Cambridge O Level Physics 5054

Past-paper question bank with official mark schemes

Paper 1 (multiple choice) and Paper 2 (theory) · May/June 2004 to May/June 2026 · by syllabus subsection

What is in this file

- 2709 entries on 4358 pages: 1723 Paper 1 questions and 986 Paper 2 entries (551 original Paper 2 questions).
- Only questions on the selected syllabus sections: 1.1 to 1.7 (pressure, 1.8, is not included), 3.1 to 3.4 and 4.2.
- Only papers that have an official mark scheme in the repository (156 question papers). Questions of papers without a mark scheme are listed in the exclusions file and are not shown.

How an entry is built

- An entry is one original question. It shows the WHOLE question as printed (introduction, diagrams, tables, graphs, data, answer spaces and every part), followed at once by the WHOLE official mark scheme of that question.
- Paper 2: the question comes first, one picture for each source page it occupies; its mark scheme starts on a new page. A question with EITHER / OR alternatives keeps both alternatives and both schemes.
- Paper 1: the question and the row of the published answer key for that question are on the same page.
- A Paper 2 question that tests several subsections appears once in each of them. The header names the parts that belong to the subsection and their marks, out of the marks of the question (EITHER / OR question: the marks with one alternative).

Order

- Syllabus chapter and subsection; then year (oldest first); session (May/June, then October/November); variant (1, then 2); paper number; original question number. Before 2010 there is one variant.

Reading a page

- Grey bars are added by this bank: the header (subsection, year, session, paper, variant, question, record number), the line above each picture (source file name and source page number) and the footer (record ID and page number).
- Everything else is a picture of the original PDF at its original size. Page numbers, copyright footers, blank pages and the shared lined answer pages of the older papers are not reproduced. (In the Paper 2 papers of 2004 to 2007 the page number stands on the line of the examiner label at the top of the page and is inside the picture.)
- The repository stores the source PDFs with inverted colours; the pictures here show the restored original pages (black on white).

Companion files

- physics_5054_question_bank_tracker (every entry and every excluded question), physics_5054_question_bank_exclusions (what is left out and why) and physics_5054_question_bank_audit (counts and checks).
- The subsection of each question was assigned in this project by reading the question together with its mark scheme. It is not an official classification by Cambridge.

Contents

Syllabus subsection	Paper 1	Paper 2	entries	page
1 Motion, forces and energy				
1.1 Physical quantities and measurement techniques	156	26	182	3
1.2 Motion	100	76	176	230
1.3 Mass and weight	73	39	112	538
1.4 Density	57	35	92	718
1.5 Forces				
1.5.1 Balanced and unbalanced forces	85	62	147	868
1.5.2 Friction	66	45	111	1119
1.5.3 Elastic deformation	62	22	84	1310
1.5.4 Circular motion	52	15	67	1433
1.5.5 Turning effect of forces	59	37	96	1526
1.5.6 Centre of gravity	41	14	55	1680
1.6 Momentum	8	11	19	1760
1.7 Energy, work and power				
1.7.1 Energy	68	83	151	1799
1.7.2 Work	38	30	68	2086
1.7.3 Energy resources	27	17	44	2200
1.7.4 Efficiency	41	29	70	2273
1.7.5 Power	60	48	108	2390
3 Waves				
3.1 General properties of waves	89	53	142	2574
3.2 Light				
3.2.1 Reflection of light	55	9	64	2799
3.2.2 Refraction of light	76	37	113	2878
3.2.3 Thin lenses	71	22	93	3053
3.2.4 Dispersion of light	29	13	42	3180
3.3 Electromagnetic spectrum	53	46	99	3244
3.4 Sound	115	38	153	3411
4 Electricity and magnetism				
4.2 Electrical quantities				
4.2.1 Electrical charge	90	29	119	3624
4.2.2 Electrical current	35	40	75	3784
4.2.3 Electromotive force and potential difference	44	29	73	3933
4.2.4 Resistance	73	81	154	4063

Total: 2709 entries (1723 Paper 1, 986 Paper 2) on 4358 pages.

Chapter 1 Motion, forces and energy

1.1 Physical quantities and measurement techniques

182 entries: 156 Paper 1 questions, 26 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 2 of 20

1 A student has been asked to determine, as accurately as possible, the volume of a piece of wire. The wire is about 80 cm long and about 0.2 cm in diameter.

Which measuring instruments should the student use?

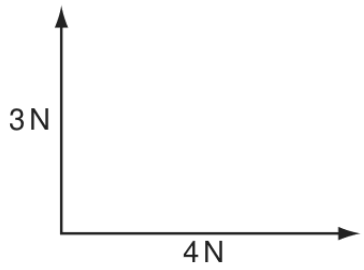
	length	diameter
A	metre rule	micrometer
B	metre rule	vernier callipers
C	micrometer	vernier callipers
D	vernier callipers	micrometer

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 1

1 **A**

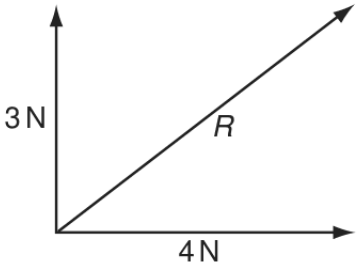
QUESTION · 5054_s04_qp_1.pdf · source page 5 of 20

10 Forces of 3 N and 4 N are acting as shown in the diagram.

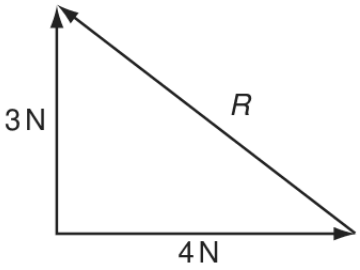


Which diagram may be used to find the resultant *R* of these two forces?

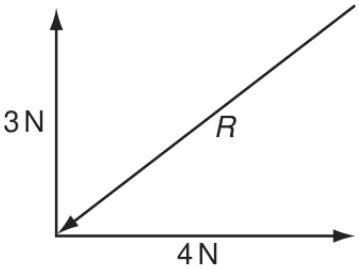
A



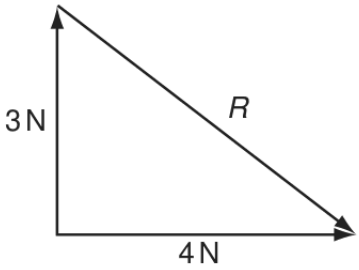
B



C



D

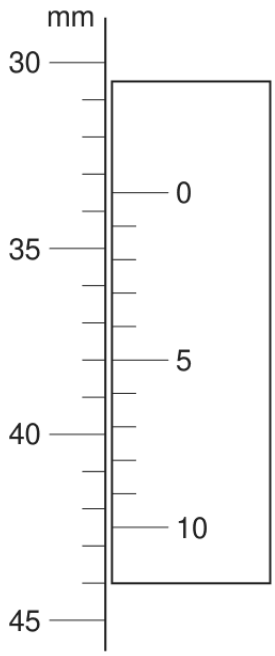


MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 10

10 **A**

QUESTION · 5054_w04_qp_1.pdf · source page 2 of 20

1 The diagram shows part of a vernier scale.



What is the correct reading?

A 30.5 mm **B** 33.5 mm **C** 38.0 mm **D** 42.5 mm

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 1

1 **B**

QUESTION · 5054_s05_qp_1.pdf · source page 2 of 20

1 Which instrument is used to measure the internal diameter of a pipe with a single measurement?

A manometer

B measuring cylinder

C micrometer

D vernier calipers

MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 1

1	**D**

QUESTION · 5054_s05_qp_1.pdf · source page 4 of 20

9 Which is the correct statement about force and velocity?

A Force and velocity are both scalars.

B Force and velocity are both vectors.

C Force is a scalar, velocity is a vector.

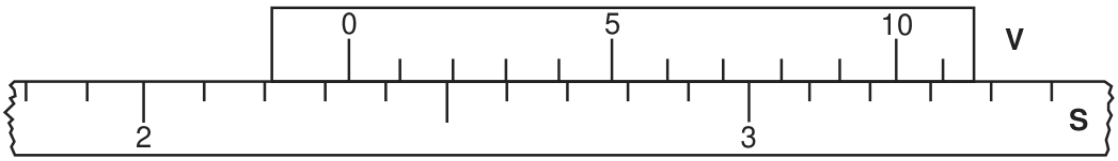
D Force is a vector, velocity is a scalar.

MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 9

9	**B**

QUESTION · 5054_w05_qp_1.pdf · source page 2 of 20

1 The diagram shows a vernier **V** placed against a scale **S**.



What is the vernier reading?

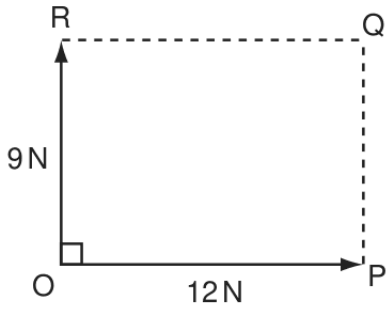
A 2.23 **B** 2.26 **C** 2.33 **D** 2.36

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 1

1 **C**

QUESTION · 5054_w05_qp_1.pdf · source page 6 of 20

10 Two forces act at right angles at a point O as shown.



What is the resultant of the forces?

	magnitude	direction
A	15 N	OQ
B	15 N	PR
C	21 N	OQ
D	21 N	PR

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 10

10 **A**

QUESTION · 5054_s06_qp_1.pdf · source page 2 of 20

3 The diagram shows a vernier scale.

What is the reading on the vernier scale?

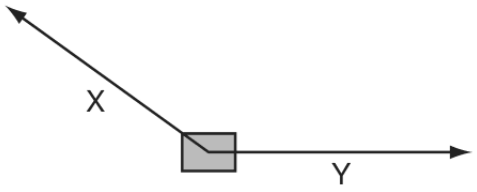
A 6.50 cm **B** 6.55 cm **C** 7.00 cm **D** 7.05 cm

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 3

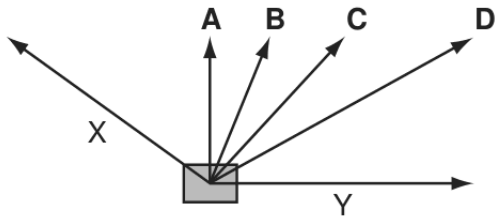
3 **B**

QUESTION · 5054_s06_qp_1.pdf · source page 3 of 20

4 Forces X and Y act on a block in the directions shown on the scale diagram.



In which direction is the resultant force acting?

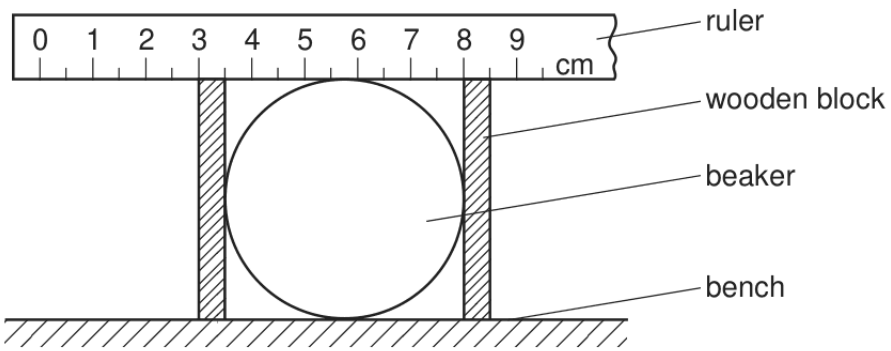


MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 4

4 **B**

QUESTION · 5054_w06_qp_1.pdf · source page 2 of 16

1 The diagram shows one method of measuring the diameter of a beaker.



What is the diameter of the beaker?

A 4.5 cm **B** 5.0 cm **C** 5.5 cm **D** 8.0 cm

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 1

1	**A**

QUESTION · 5054_w06_qp_1.pdf · source page 2 of 16

2 A student studies some equations.

power = work / time

force = mass × acceleration

velocity = displacement / time

How many vector quantities are contained in the equations?

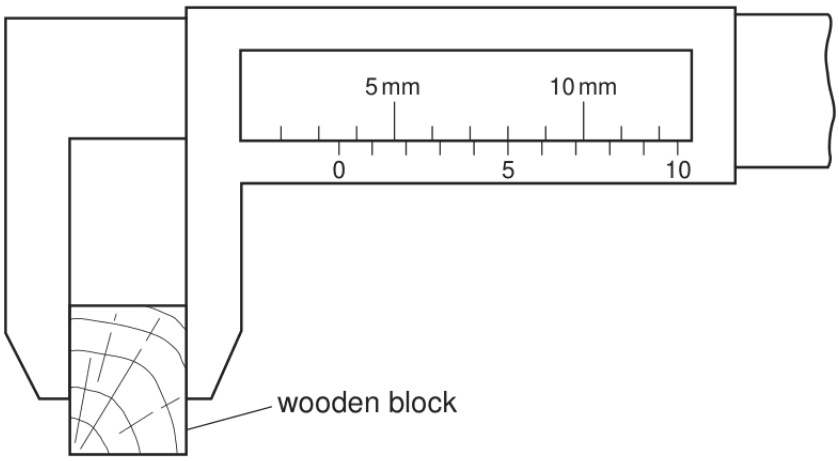
A 1 **B** 2 **C** 3 **D** 4

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 2

2 **D**

QUESTION · 5054_s07_qp_1.pdf · source page 2 of 20

1 The width of a wooden block is measured using vernier calipers.



What is the width of the block?

A 3.5 mm **B** 5.3 mm **C** 8.0 mm **D** 8.5 mm

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 1

1	**A**

QUESTION · 5054_s07_qp_1.pdf · source page 2 of 20

2 Which of the following correctly lists one scalar and one vector quantity?

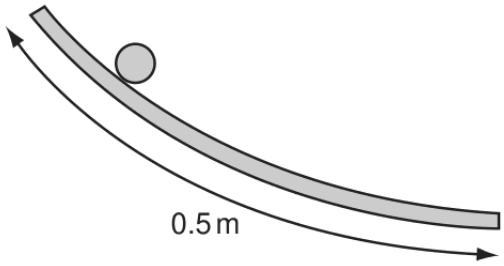
	scalar quantity	vector quantity
A	displacement	work
B	energy	force
C	force	acceleration
D	velocity	mass

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 2

2 **B**

QUESTION · 5054_w07_qp_1.pdf · source page 2 of 20

1 In an experiment, a ball is rolled down a curved track that is about half a metre long.



Which measuring device should be used to measure the length accurately?

A metre rule

B micrometer

C tape measure

D vernier calipers

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 1

1	**C**

QUESTION · 5054_w07_qp_1.pdf · source page 2 of 20

2 Forces of 30 N and 50 N act on the same body, but in different directions.

Which value could **not** be the resultant force on the body?

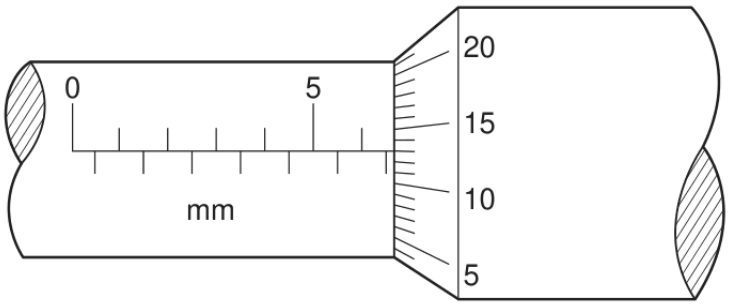
A 10 N **B** 30 N **C** 50 N **D** 70 N

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 2

2	**A**

QUESTION · 5054_s08_qp_1.pdf · source page 2 of 20

1 What is the reading on this micrometer?



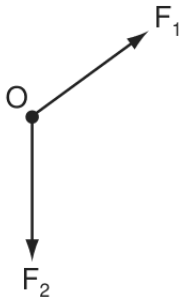
A 5.43 mm **B** 6.63 mm **C** 7.30 mm **D** 8.13 mm

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 1

1	**B**

QUESTION · 5054_s08_qp_1.pdf · source page 2 of 20

2 Two forces F_1 and F_2 act on an object O in the directions shown.



What is the direction of the resultant force?

A **B** **C** **D**

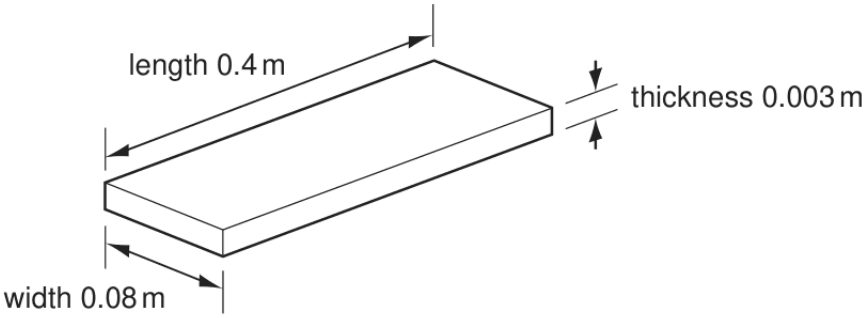
MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 2

2 **D**

QUESTION · 5054_w08_qp_1.pdf · source page 2 of 20

1 A manufacturer needs to measure accurately the dimensions of a wooden floor tile.

The approximate dimensions of the tile are shown.



Which instruments measure each of these dimensions accurately?

	length	thickness	width
A	metre rule	micrometer	vernier calipers
B	metre rule	vernier calipers	micrometer
C	micrometer	metre rule	vernier calipers
D	vernier calipers	micrometer	metre rule

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 1

1 **A**

Section A

For Examiner's Use

Answer **all** the questions in this section.

1 A microphone has a weight *W* of 6.0 N. It is suspended by wire X from the ceiling in a radio studio.

Fig. 1.1 shows the microphone held in the correct position by a horizontal wire Y.

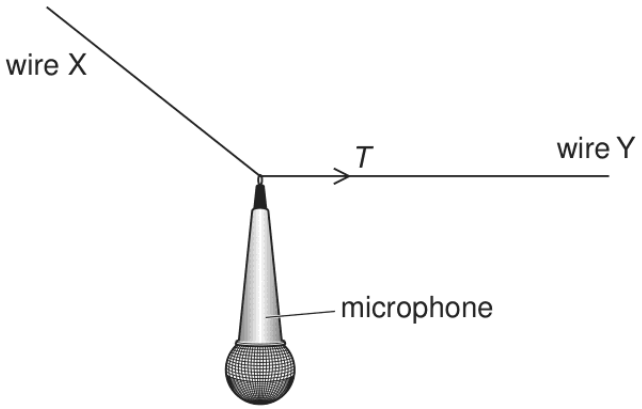


Fig. 1.1

(a) The tension *T* in wire Y is 8.0 N. Use a vector diagram to determine the magnitude and the direction of the resultant of *W* and *T.*

magnitude = ...

direction = ...

[4]

(b) The microphone is at rest. State the tension in wire X.

tension in X = ... [1]

Section A

1	**(a)**	diagram of two forces **and** resultant	B1	
		W / 6(N) **and** *T* / 8(N) marked on perp. forces **or** scale given	B1	
		10(.0 ±0.2) N	B1	
		35–39° from *T*/Y/horizontal or 51–55° from *W*/vertical **and** correct resultant	B1	
	(b)	10(.0) N **or** e.c.f.	B1	[5]

QUESTION · 5054_s09_qp_1.pdf · source page 2 of 20

1 Which instrument is most easily used to measure the internal diameter of a pipe?

A manometer

B measuring cylinder

C micrometer

D vernier calipers

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 1

1	**D**

QUESTION · 5054_s09_qp_1.pdf · source page 2 of 20

2 Which statement about scalars and vectors is correct?

A A scalar has direction but no size.

B A scalar has size but no direction.

C A vector has direction but no size.

D A vector has size but no direction.

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 2

2	**B**

QUESTION · 5054_w09_qp_1.pdf · source page 2 of 20

1 Vernier calipers read to one tenth of a millimetre.

Which reading shows this precision?

A 3.3 cm **B** 3.31 cm **C** 3.310 cm **D** 3.312 cm

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 1

1	**B**

QUESTION · 5054_w09_qp_1.pdf · source page 2 of 20

2 Which list contains only scalar quantities?

A acceleration, displacement, mass

B acceleration, distance, speed

C displacement, mass, velocity

D distance, mass, speed

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 2

2	**D**

QUESTION · 5054_s10_qp_11.pdf · source page 2 of 20

1 Power is measured in watts.

What is the correct symbol for millions of watts?

A mw **B** mW **C** Mw **D** MW

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 1

1	**D**

QUESTION · 5054_s10_qp_11.pdf · source page 2 of 20

2 The following statements are about motion.

1 A plane flies due East for 600 km.

2 A runner's average speed in a race around a track is 5 m/s.

3 A snail crawls at 3 mm/s in a straight line towards a lettuce.

4 A tourist travels 500 km on a journey.

Which statements describe vector quantities?

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 2 and 4

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 2

2 **B**

QUESTION · 5054_s10_qp_12.pdf · source page 2 of 20

1 The following statements are about motion.

1 A plane flies due East for 600 km.

2 A runner's average speed in a race around a track is 5 m / s.

3 A snail crawls at 3 mm / s in a straight line towards a lettuce.

4 A tourist travels 500 km on a journey.

Which statements describe vector quantities?

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 2 and 4

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 1

1	**B**

QUESTION · 5054_s10_qp_12.pdf · source page 2 of 20

2 Power is measured in watts.

What is the correct symbol for millions of watts?

A mw **B** mW **C** Mw **D** MW

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 2

2	**D**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 A journey consists of two displacements: the first is 500 m in a northerly direction and the second is 200 m in an easterly direction.

(a) In the space below draw, to scale, a vector diagram of these displacements.

State the scale of your diagram.
On your diagram, show the two displacements and the resultant displacement.
Determine the size (magnitude) and direction of the resultant displacement.

scale = ...

size = ...

direction = ...
[3]

(b) Another journey covers a distance of 700 m. Describe how it is possible that this journey has no resultant displacement.

.....

.....[1]

Section A

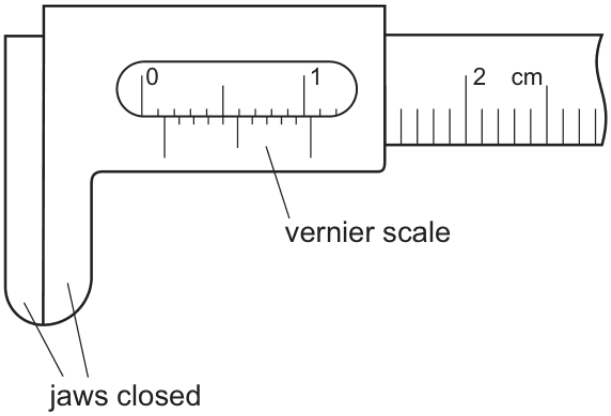
1 **(a)** components shown on correct diagram with correct resultant (i.e. towards NE) and a scale given B1
540 (±10) m B1
22º ± 3° E of N with correct diagonal B1 [3]

(b) idea that ends at start, returns in opposite direction B1 [1]

[Total: 4]

QUESTION · 5054_w10_qp_11.pdf · source page 2 of 20

1 Vernier calipers are shown with the jaws closed.



What is the zero error?

A 0.04 cm **B** 0.05 cm **C** 0.14 cm **D** 0.15 cm

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 1

1	**C**

QUESTION · 5054_w10_qp_11.pdf · source page 2 of 20

2 Which list contains only scalar quantities?

A acceleration, displacement, velocity

B distance, force, speed

C force, length, time

D length, mass, speed

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 2

2	**D**

QUESTION · 5054_w10_qp_12.pdf · source page 2 of 16

1 Which list contains only scalar quantities?

A acceleration, displacement, velocity

B distance, force, speed

C force, length, time

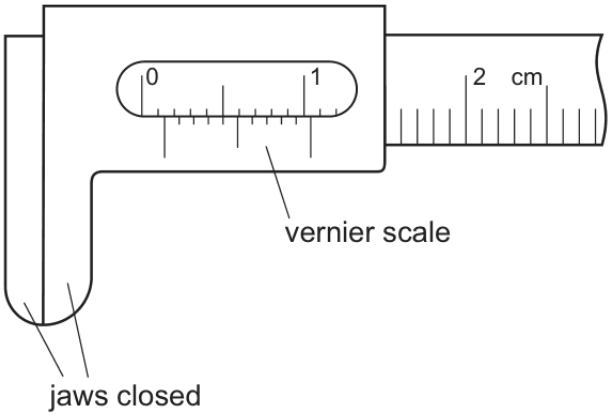
D length, mass, speed

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 1

1	**D**

QUESTION · 5054_w10_qp_12.pdf · source page 3 of 16

4 Vernier calipers are shown with the jaws closed.



What is the zero error?

A 0.04 cm **B** 0.05 cm **C** 0.14 cm **D** 0.15 cm

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 4

4 **C**

For Examiner's Use

10 A bus breaks down on a road with a 10° upward slope. The passengers get out and push the bus to the top of the slope at a constant speed. Fig. 10.1 shows the passengers exerting a force on the bus parallel to the line of the slope.

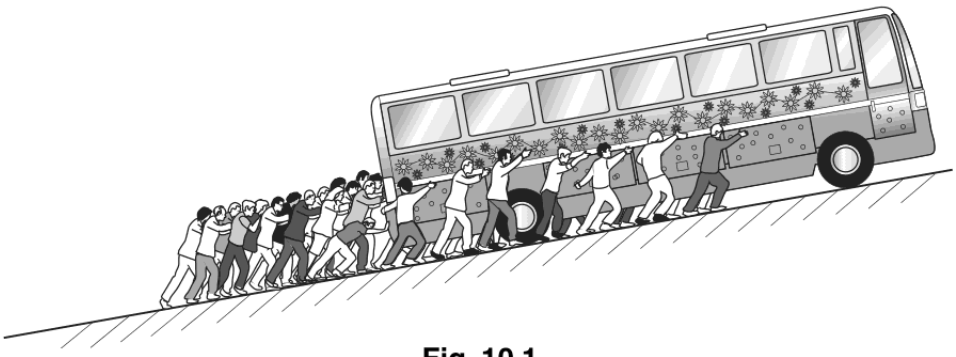


Fig. 10.1

(a) The total mass of the bus is 3200 kg.

(i) Calculate the weight of the bus.

weight = ...[1]

(ii) On Fig. 10.1, draw **two** arrows to show the direction of the weight of the bus and the direction of the force exerted on the bus by the passengers. [1]

(iii) The total force exerted by the passengers on the bus is 17000 N along the line of the slope. Use a graphical method to determine the size and direction of the resultant of this force and the weight of the bus. State the scale used.

scale = ...

size of resultant = ...

direction of resultant = ...

[4]

For Examiner's Use

(iv) The bus travels at a constant speed. State the size of the resultant of all the forces acting on the bus.

.....[1]

(b) As the passengers push the bus up the slope, work is done against two forces. State the names of these two forces and, for each force, state the form of energy produced as work is done.

force 1 .....

form of energy 1 .....

force 2 .....

form of energy 2 .....

[4]

(c) When the bus reaches the top of the slope, it stops and the passengers get back in.

(i) On the axes of Fig. 10.2, sketch the distance-time graph for the bus from a time when the passengers are pushing the bus at constant speed until they get back into the stationary bus.

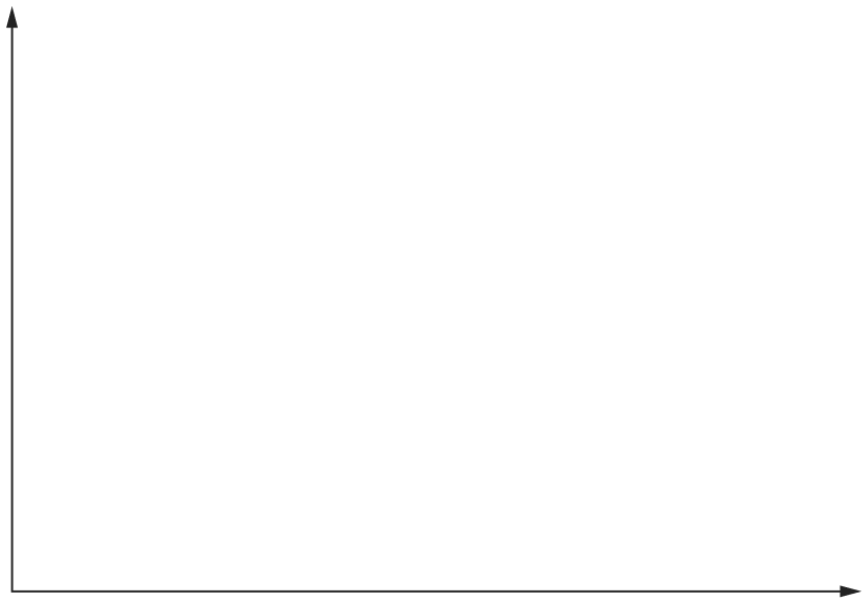


Fig. 10.2

[3]

(ii) State how the speed of a moving object may be obtained from an accurately drawn distance-time graph.

.....

.....[1]

10	**(a)**	**(i)**	32 000 N	B1	
		(ii)	two arrows/lines in correct direction by eye	B1	
		(iii)	scale given	B1	
			two arrows/lines **and** correct resultant drawn	B1	
			32.0 → 35.0 kN (2/3 sig. fig. only)	B1	
			58.5 → 61.5° to horizontal (2/3 sig. fig. only; don't penalise twice)	B1	
		(iv)	zero/no force/0	B1	[7]
	(b)		weight/gravitational force/gravitational attraction (**not** gravity)	B1	
			higher in gravitational field **or** (to gravitational) potential energy	B1	
			friction/air resistance	B1	
			heat/thermal/internal energy	B1	[4]
	(c)	**(i)**	labelled axes **and** correct way round ($x \rightarrow t$)	B1	
			straight line of positive slope	B1	
			followed only by horizontal line (**ign** curve at junction)	B1	
		(ii)	distance travelled/time taken (from points) **or** calculate the gradient	B1	[4]

QUESTION · 5054_s11_qp_11.pdf · source page 2 of 16

1 A plumber measures, as **accurately** as possible, the length and internal diameter of a straight copper pipe.

The length is approximately 80 cm and the internal diameter is approximately 2 cm.

What is the best combination of instruments for the plumber to use?

	internal diameter	length
A	rule	rule
B	rule	tape
C	vernier calipers	rule
D	vernier calipers	tape

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 1

1 **C**

QUESTION · 5054_s11_qp_11.pdf · source page 2 of 16

2 What is the correct unit for the quantity shown?

	quantity	unit
A	electromotive force (e.m.f.)	N
B	latent heat	J
C	pressure	kg/m^3
D	weight	kg

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 2

2 **B**

QUESTION · 5054_s11_qp_12.pdf · source page 2 of 16

1 What is the correct unit for the quantity shown?

	quantity	unit
A	electromotive force (e.m.f.)	N
B	latent heat	J
C	pressure	kg/m^3
D	weight	kg

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 1

1 **B**

QUESTION · 5054_s11_qp_12.pdf · source page 2 of 16

2 A plumber measures, as **accurately** as possible, the length and internal diameter of a straight copper pipe.

The length is approximately 80 cm and the internal diameter is approximately 2 cm.

What is the best combination of instruments for the plumber to use?

	internal diameter	length
A	rule	rule
B	rule	tape
C	vernier calipers	rule
D	vernier calipers	tape

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 2

2 **C**

QUESTION · 5054_w11_qp_11.pdf · source page 2 of 20

1 The diameter and the length of a thin wire, approximately 1 m in length, are measured as accurately as possible.

What are the best instruments to use?

	diameter	length
A	micrometer	rule
B	micrometer	vernier calipers
C	rule	tape
D	vernier calipers	rule

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 1

1 **A**

QUESTION · 5054_w11_qp_11.pdf · source page 2 of 20

2 A quantity is quoted as having a value of 6.2 ms.

In what units is it measured?

A metres

B metres per second

C microseconds

D milliseconds

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 2

2	**D**

QUESTION · 5054_w11_qp_12.pdf · source page 4 of 20

6 The diameter and the length of a thin wire, approximately 1 m in length, are measured as accurately as possible.

What are the best instruments to use?

	diameter	length
A	micrometer	rule
B	micrometer	vernier calipers
C	rule	tape
D	vernier calipers	rule

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 6

6 **A**

QUESTION · 5054_w11_qp_12.pdf · source page 4 of 20

7 A quantity is quoted as having a value of 6.2 ms.

In what units is it measured?

A metres

B metres per second

C microseconds

D milliseconds

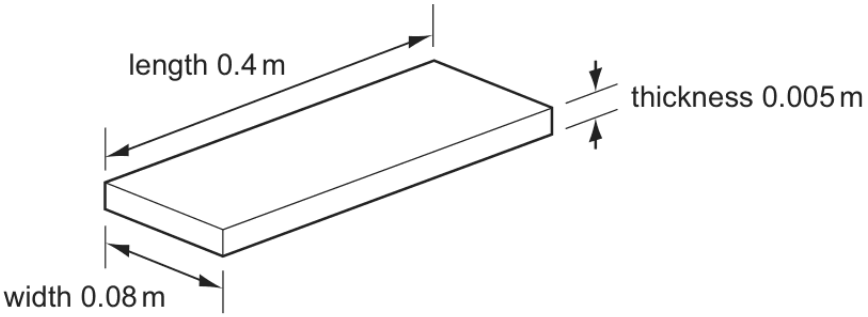
MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 7

7	**D**

QUESTION · 5054_s12_qp_11.pdf · source page 2 of 16

1 A manufacturer measures accurately the dimensions of a wooden floor tile.

The approximate dimensions of the tile are shown.



Which instruments are used to measure accurately each of these dimensions?

	length	thickness	width
A	metre rule	micrometer	vernier calipers
B	metre rule	vernier calipers	micrometer
C	micrometer	metre rule	vernier calipers
D	vernier calipers	micrometer	metre rule

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 1

1 **A**

QUESTION · 5054_s12_qp_11.pdf · source page 2 of 16

2 Which pair of quantities includes one scalar and one vector?

A	mass	time
B	temperature	time
C	temperature	velocity
D	velocity	weight

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 2

2	**C**

QUESTION · 5054_s12_qp_12.pdf · source page 2 of 16

1 A reel of copper wire is labelled 'length 30 m' and 'diameter 2 mm'. A student calculates the volume of the copper wire.

Which instruments does he use to measure accurately the length and the diameter of the wire?

	length	diameter
A	rule	calipers
B	rule	micrometer
C	tape	calipers
D	tape	micrometer

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 1

1 **D**

QUESTION · 5054_s12_qp_12.pdf · source page 2 of 16

2 Which row correctly shows examples of a vector quantity and a scalar quantity?

	vector	scalar
A	area	force
B	mass	density
C	velocity	acceleration
D	weight	volume

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 2

2 **D**

QUESTION · 5054_w12_qp_11.pdf · source page 2 of 16

1 Vernier calipers read to one tenth of a millimetre.

Which reading is given to this precision?

A 3.3 cm **B** 3.31 cm **C** 3.310 cm **D** 3.312 cm

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 1

1	**B**

QUESTION · 5054_w12_qp_11.pdf · source page 2 of 16

2 Velocity is given by the change in displacement divided by the change in time.

How many vector quantities appear in this statement?

A 0 **B** 1 **C** 2 **D** 3

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 2

2	**C**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 A parachutist of total weight 950 N falls vertically at a constant velocity of 6.5 m/s.

(a) State the size and direction of the resistive (drag) force acting on the parachutist.

size of resistive force: .....

direction of resistive force: .....
[2]

(b) At a certain height, the wind starts to blow and the parachutist moves horizontally at a velocity of 4.5 m/s as he continues to fall.

Use a graphical method to add together the horizontal and the vertical velocities to determine the size of the resultant velocity of the parachutist. Label the velocities and state the scale that you use.

size of velocity = ...

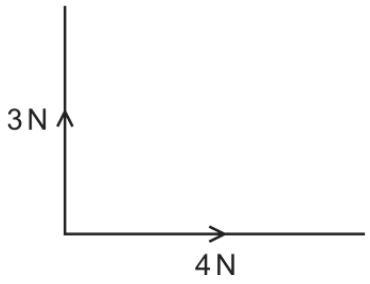
scale = ...
[3]

Section A

1	**(a)**	950 N	B1	
		upwards	B1	
	(b)	correct rectangle **and** diagonal **and** at least one velocity labelled **or** correct triangle and at least one velocity labelled (either way round)	B1	
		from 7.8(0000) to 8.0(0000) m / s (inclusive)	B1	
		scale stated	B1	[5]

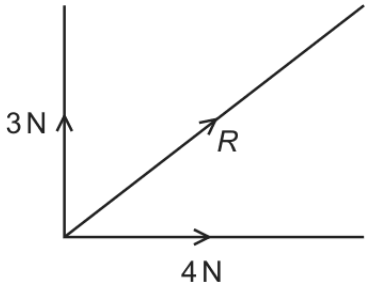
QUESTION · 5054_s13_qp_11.pdf · source page 2 of 20

1 Forces of 3 N and 4 N act as shown in the diagram.

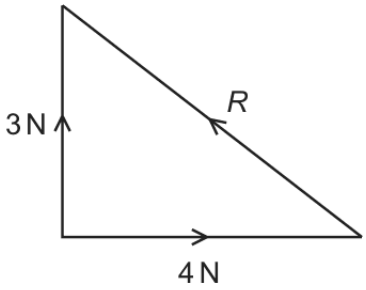


Which diagram shows the resultant *R* of these two forces?

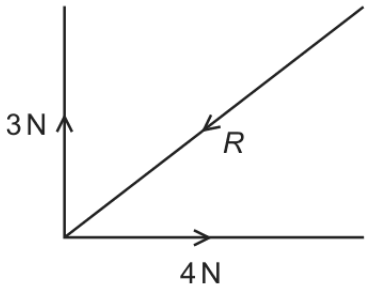
A



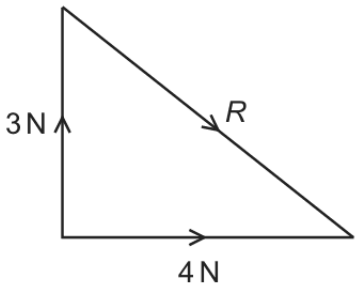
B



C



D

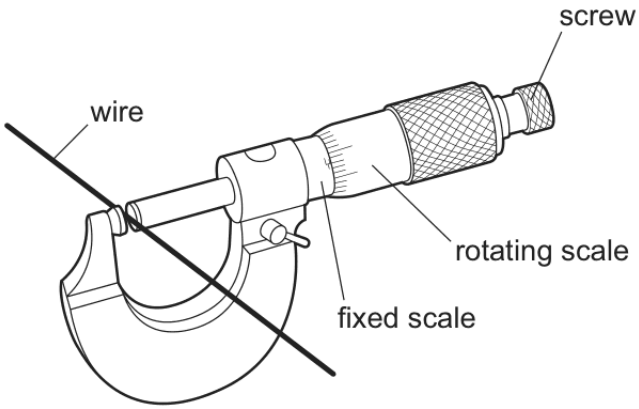


MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 1

1	**A**

QUESTION · 5054_s13_qp_11.pdf · source page 3 of 20

2 A micrometer is used to measure the diameter of a uniform wire.



What is done to obtain an accurate answer?

A Find the reading and add or subtract the zero error.

B Make the micrometer horizontal.

C Subtract the fixed scale reading from the rotating scale reading.

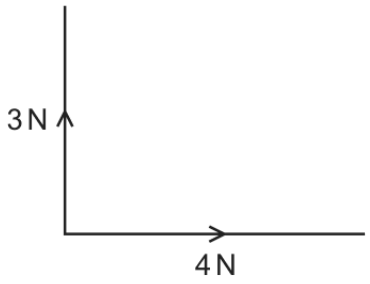
D Subtract the rotating scale reading from the fixed scale reading.

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 2

2	**A**

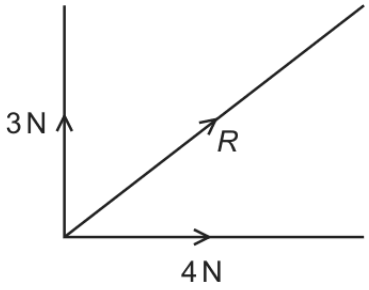
QUESTION · 5054_s13_qp_12.pdf · source page 2 of 16

1 Forces of 3 N and 4 N act as shown in the diagram.

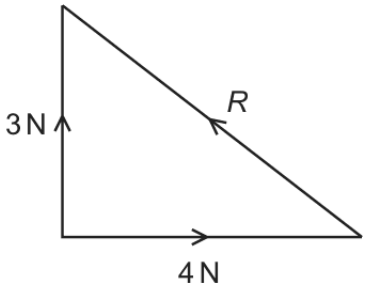


Which diagram shows the resultant *R* of these two forces?

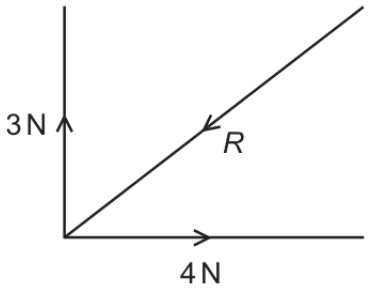
A



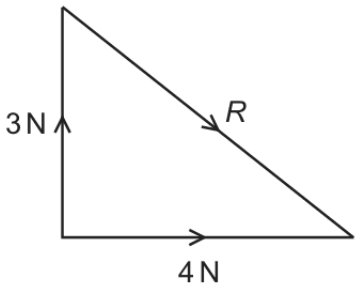
B



C



D



MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 1

1	**A**

QUESTION · 5054_s13_qp_12.pdf · source page 2 of 16

2 Before marking the finishing line on a running track, a groundsman measures out its 100 m length.

Which instrument is the most appropriate for this purpose?

A measuring tape

B metre rule

C 30 cm ruler

D micrometer

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 2

2	**A**

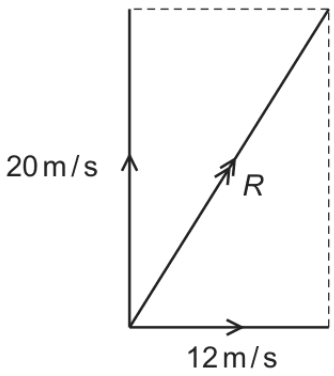
QUESTION · 5054_w13_qp_11.pdf · source page 2 of 20

1 When there is no wind, the engines of an airship push it due north at 20 m / s.

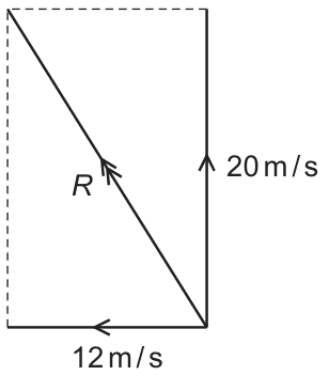
The wind is blowing from the west at 12 m / s.

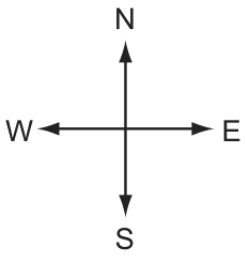
Which vector diagram correctly shows how the resultant velocity *R* of the airship is obtained?

A

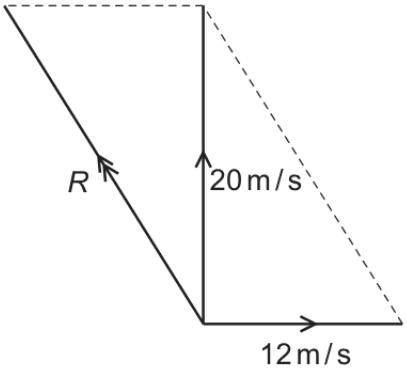


B

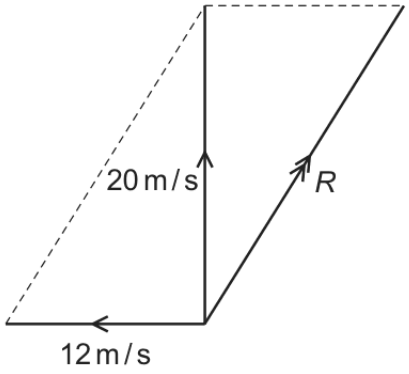




C



D

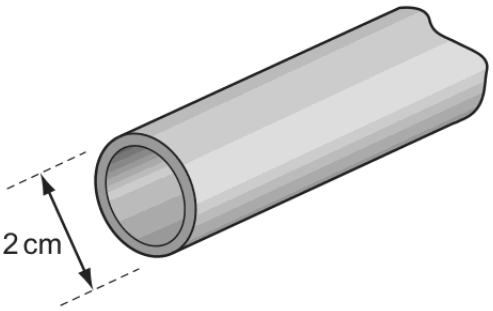


MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 1

1	**A**

QUESTION · 5054_w13_qp_11.pdf · source page 3 of 20

2 A length of copper pipe, of uniform cross-section and several metres long, carries water to a tap.



Measurements are taken to determine accurately the volume of copper in the pipe.

Which instruments are used?

A calipers and micrometer

B micrometer and rule

C rule and tape

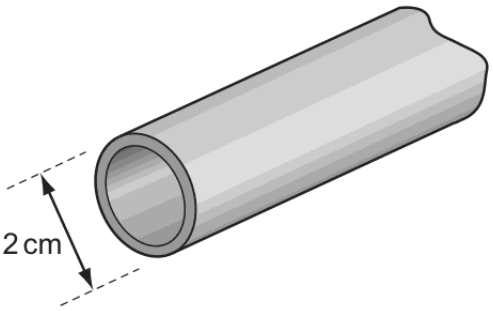
D tape and calipers

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 2

2 **D**

QUESTION · 5054_w13_qp_12.pdf · source page 2 of 20

1 A length of copper pipe, of uniform cross-section and several metres long, carries water to a tap.



Measurements are taken to determine accurately the volume of copper in the pipe.

Which instruments are used?

A calipers and micrometer

B micrometer and rule

C rule and tape

D tape and calipers

MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 1

1	**D**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 A set of traffic lights hangs from the end of a metal cable. A horizontal chain pulls the traffic lights to the right so that they are above the middle of the road. Fig. 1.1 shows the metal cable inclined to the vertical.

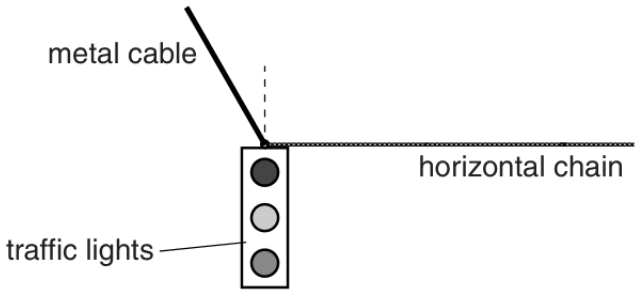


Fig. 1.1

The weight of the traffic lights is 240 N.

(a) Two of the forces on the traffic lights are the tension in the horizontal chain and the weight of the traffic lights.

On Fig. 1.1, mark

(i) an arrow that represents the tension in the horizontal chain, [1]

(ii) an arrow that represents the weight of the traffic lights. [1]

(b) The tension in the horizontal chain is 140 N. Use a scale diagram to determine the size of the resultant of the weight and the tension in the chain. State the scale used for the diagram.

scale = ...

resultant force = ...
[3]

Section A

1	**(a)**	**(i)**	arrow(head) on chain pointing to the right	B1	
		(ii)	vertical arrow downwards and part of arrow touching **or** within rectangle of lights **or** direction of arrow in **(i) and (ii)** correct (by eye)	B1	
	(b)		scale given (must have unit of cm:N **or** cm/N **or** N:cm **or** N/cm)	B1	
			correct triangle **or** rectangle (might be implied) and correct resultant (compulsory e.c.f. from **(i) or (ii)**: i.e. correct diagonal according to candidate's **(i)** and **(ii)**)	B1	
			272 ≤ candidate's value ≤ 283 N	B1	[5]

QUESTION · 5054_s14_qp_11.pdf · source page 2 of 16

1 A workman measures, as **accurately** as possible, the length and internal diameter of a straight copper pipe.

The length is approximately 600 cm and the internal diameter is approximately 2 cm.

What is the best combination of instruments for the workman to use?

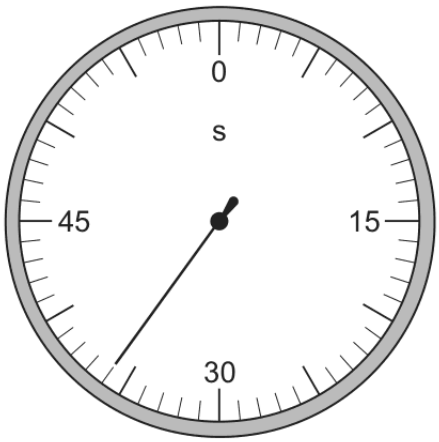
	internal diameter	length
A	ruler	ruler
B	ruler	tape
C	vernier calipers	ruler
D	vernier calipers	tape

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 1

1 **D**

QUESTION · 5054_s14_qp_11.pdf · source page 2 of 16

2 The diagram shows a stopwatch.



What is the reading on the stopwatch?

A 30.6 s **B** 33.0 s **C** 36.0 s **D** 36.6 s

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 2

2 **C**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A small boat travels with a velocity of 4.0 m/s due north in an area of still water.

(a) State what is meant by *velocity*.

.....

.....

..... [1]

(b) The boat travels from the area of still water into an area where the velocity of the water is 2.0 m/s towards the north-east, as shown in Fig. 1.1.

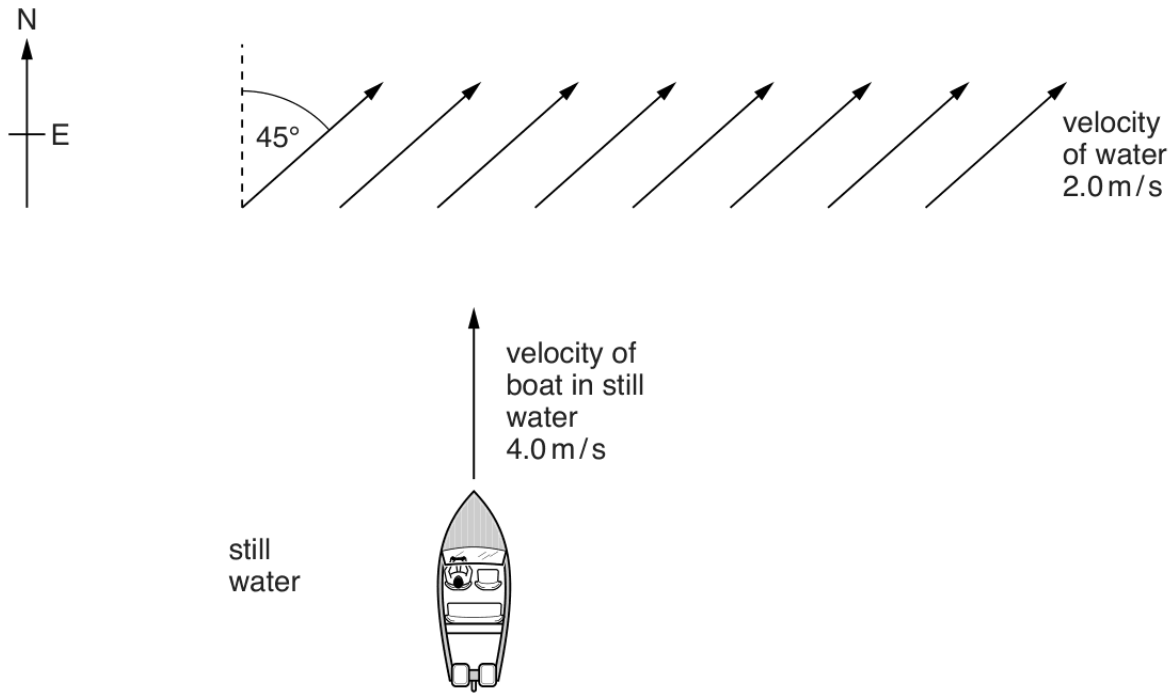


Fig. 1.1 (not to scale)

Combining the initial velocity of the boat with the velocity of the water gives the resultant velocity of the boat.

In the space on the page opposite, draw a vector diagram to show the resultant velocity.

Use your diagram to find the size and direction of the resultant velocity.

size of resultant velocity = ...

direction of resultant velocity = ...

[4]

Section A

1 **(a)** speed and direction **or** (change in) distance per sec / unit time + direction
or (change in) displacement per sec / unit time [B1]

(b) two vectors at 45° and one twice the other by eye or labelled 2 (m / s) and 4 (m / s) [B1]
correct resultant shown for two labelled vectors at any angle with directions shown [B1]
resultant speed 5.6 ± 0.2 m / s unit needed [B1]
direction (0)15° **or** N15 °E, etc. **or** any clear direction expressed
or angle mentioned on answer line **and** shown on diagram [B1]

[5]

QUESTION · 5054_s14_qp_12.pdf · source page 2 of 16

1 Each row contains a vector and a scalar.

In which row is the size of the vector equal to the size of the scalar?

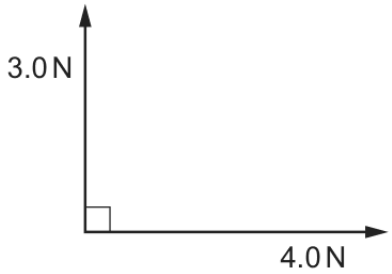
	vector	scalar
A	displacement of a car	speed of the car
B	velocity of a car	distance travelled by the car
C	velocity of a car	speed of the car
D	weight of a car	mass of the car

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 1

1 **C**

QUESTION · 5054_s14_qp_12.pdf · source page 2 of 16

2 What is the size of the resultant of the two forces shown in the diagram?



A 1.0 N **B** 3.5 N **C** 5.0 N **D** 7.0 N

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 2

2 **C**

QUESTION · 5054_s14_qp_12.pdf · source page 2 of 16

3 A student measures, as **accurately** as possible, the length and internal diameter of a straight glass tube.

The length is approximately 25 cm and the internal diameter is approximately 2 cm.

What is the best combination of instruments for the student to use?

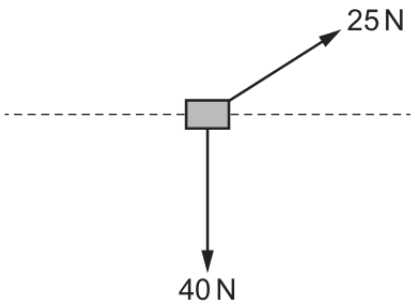
	internal diameter	length
A	ruler	micrometer
B	ruler	ruler
C	vernier calipers	micrometer
D	vernier calipers	ruler

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 3

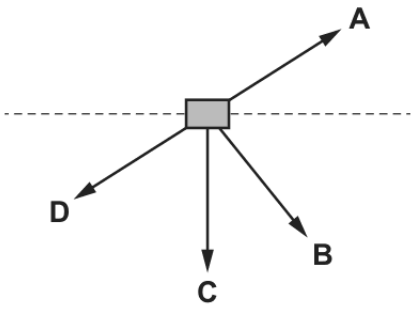
3 **D**

QUESTION · 5054_w14_qp_11.pdf · source page 2 of 20

1 Forces of 25 N and 40 N act on an object in the directions shown.



Which arrow shows the direction of the resultant force on the object?



MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 1

1	**B**

QUESTION · 5054_w14_qp_11.pdf · source page 2 of 20

2 Which device can be used to measure the thickness of a single sheet of paper?

A a metre rule

B a micrometer

C a plastic ruler

D a measuring tape

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 2

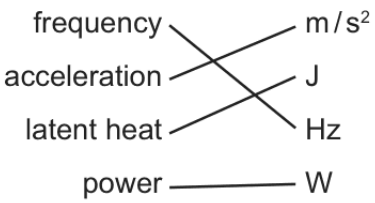
2	**B**

QUESTION · 5054_w14_qp_11.pdf · source page 3 of 20

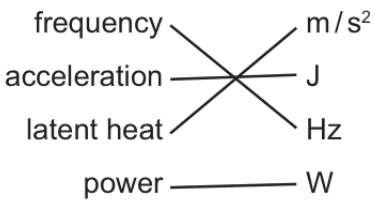
3 In a test, four students linked the quantities on the left with their units on the right.

Which student matched them all correctly?

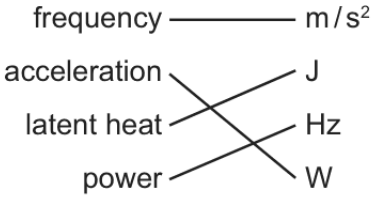
A



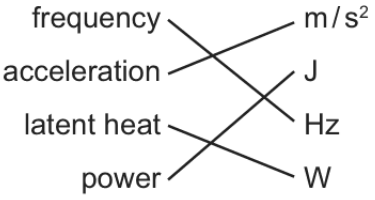
B



C



D



MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 3

3 **A**

3 Fig. 3.1 shows a helium cylinder being used at a funfair to inflate balloons that float in the air.

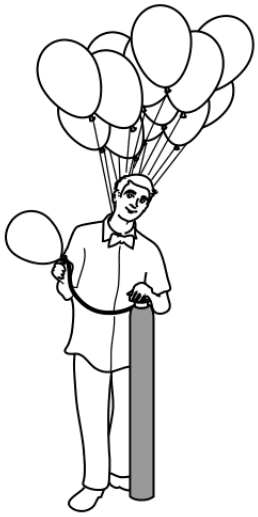


Fig. 3.1

Initially, all the helium is stored in the cylinder at a pressure of 2.0×10^7 Pa.

When no more balloons can be filled, all the helium occupies a total volume of 1.8 m^3 and is at a pressure of 1.0×10^5 Pa. The temperature of the helium has not changed.

(a) Calculate, stating the formula used, the initial volume of the helium in the cylinder.

volume = ...[3]

(b) The balloon seller takes the cylinder back to the helium supplier and it is refilled. The mass of helium in the cylinder is 0.30 kg.

(i) Calculate the density of helium in the cylinder.

density = ...[2]

(ii) The mass of the empty cylinder is 16 kg.

Suggest why weighing the cylinder is not an accurate method of determining the amount of helium supplied.

.....

.....[1]

3	**(a)**	$V_1 = p_2V_2/p_1$ **or** $p \propto 1/V$	B1	
		$1.0 \times 10^5 \times (1.8/2.0) \times 10^7$	C1	
		$9.0 \times 10^{-3}\,m^3$ **or** $9000\,cm^3$	A1	
	(b)	**(i)** $(\rho =)\ m/V$ **or** $(0.30/9.0) \times 10^{-3}$	C1	
		$33(.3333)\,kg/m^3$ **or** $0.033(3333)\,g/cm^3$	A1	
		(ii) helium mass / weight small (fraction of total / mass of air included) **or** this includes the weight of the cylinder	B1	**[6]**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 **(a)** Explain what is meant by *uniform acceleration*.

.....

.....

.....[2]

(b) Acceleration is a vector quantity.

State how a vector quantity differs from a scalar quantity.

.....

.....[1]

(c) A rock from space is travelling in a straight line at high speed when it enters the Earth's atmosphere. Fig. 9.1 is the speed-time graph for the rock from time $t = 0$ to time $t = 50$ s.

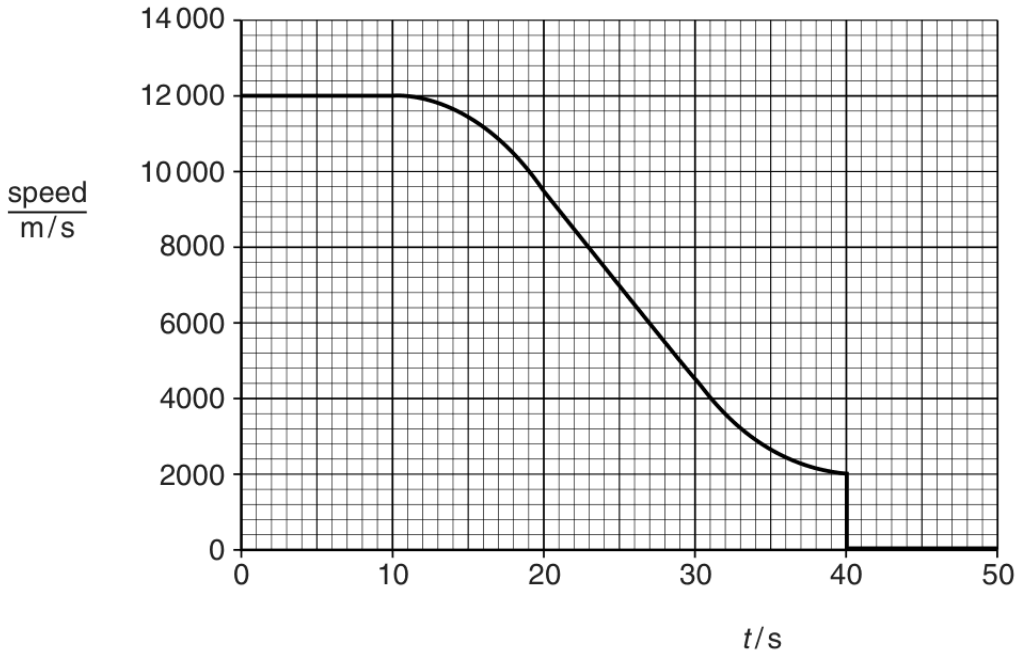


Fig. 9.1

(i) On Fig. 9.1, mark

- a letter X, where the rock is moving with a constant speed,
- a letter Y, where the rock has a uniform deceleration,
- a letter Z, where the rock has a non-uniform deceleration.

[3]

(ii) At time $t = 25\,s$, the mass of the rock is 8.4 kg. For the time $t = 25\,s$, determine

1. the size of the acceleration of the rock,

acceleration = ...[3]

2. the size of the resultant force on the rock.

force = ...[2]

(iii) Fig. 9.2 shows the rock at $t = 25\,s$.

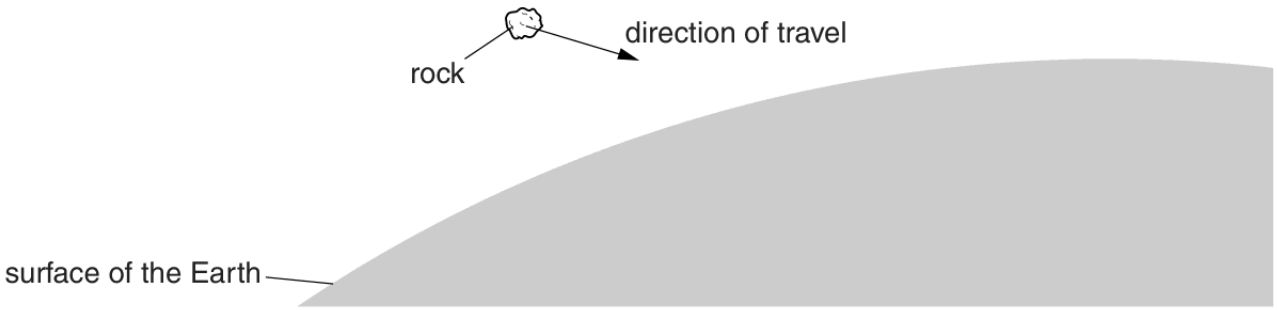


Fig. 9.2 (not to scale)

1. On Fig. 9.2, draw and label two arrows to show the directions of the gravitational force *F* and the air resistance *R* acting on the rock. [2]

2. Suggest why the size of the air resistance changes as the rock travels through the Earth's atmosphere.

.....

.....[1]

(iv) Suggest what happens to the rock at $t = 40\,s$.

.....

.....[1]

Section B

9	**(a)**	changing speed / velocity	C1	
		change in speed / velocity / time constant **or** (v–u) / t constant **or** constant / equal rate of change of speed / velocity	A1	**[2]**
	(b)	(a vector quantity has) direction	B1	**[1]**
	(c) **(i)**	**1.** X between $t \geq 0$ and $t < 10$ s	B1	
		2. Y between $t > 20$ s and $t < 30$ s	B1	
		3. Z between $t > 10$ s and $t < 20$ s **or** between $t > 30$ s and $t < 40$ s	B1	
	(ii)	**1. two** speed values from graph between 15 and 35 s (±1 mm)	C1	
		two corresponding time values from graph between 15 and 35 s (±1 mm) **or** (a =)$\Delta v / t$	C1	
		500 m / s^2	A1	
		2. (F =) ma **or** 8.4 × 500	C1	
		4200 N	A1	
	(iii)	**1.** arrow labelled F perpendicular to surface of Earth	B1	
		arrow labelled R opposite to direction of travel (by eye) from rock	B1	
		2. speed changes **or** density / pressure of air changes **or** cross-sectional area (of rock) changes	B1	
	(iv)	it hits the ground / surface of the earth **or** stops **or** speed is zero	B1	**[12]**
				[Total: 15]

QUESTION · 5054_s15_qp_11.pdf · source page 2 of 16

1 Is mass a scalar or a vector, and is acceleration a scalar or a vector?

	mass	acceleration
A	scalar	scalar
B	scalar	vector
C	vector	scalar
D	vector	vector

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 1

1 **B**

QUESTION · 5054_s15_qp_11.pdf · source page 2 of 16

2 The diameter and the length of a thin wire, approximately 50 cm in length, are measured as precisely as possible.

What are the best instruments to use?

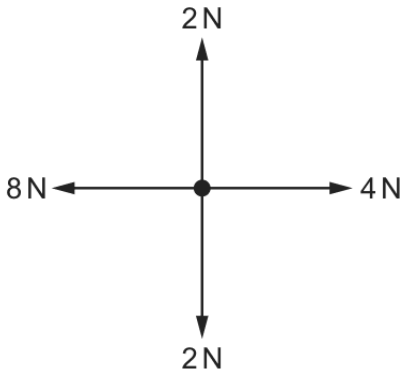
	diameter	length
A	micrometer	rule
B	micrometer	vernier calipers
C	rule	tape
D	vernier calipers	rule

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 2

2 **A**

QUESTION · 5054_s15_qp_11.pdf · source page 4 of 16

6 Four forces act at a point as shown.



What is the size of the resultant force?

A 0 N **B** 4 N **C** 6 N **D** 8 N

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 6

6 **B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a satellite in orbit around the Earth.

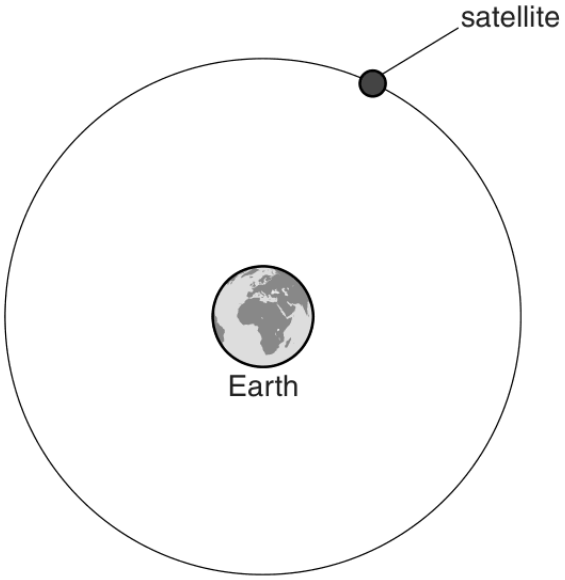


Fig. 9.1 (not to scale)

(a) The satellite travels at a constant speed in a circular orbit.

(i) Underline the quantities in the list below that are scalars.

acceleration force mass speed velocity [2]

(ii) The velocity of the satellite changes, but its speed is constant.

1. State what is meant by *velocity*. .....

.....

.....[1]

2. Explain why the velocity changes. .....

.....

.....[1]

(iii) Explain what makes this satellite move in an orbit that is circular.

.....

.....

.....

.....[2]

(b) The satellite is placed into orbit by a rocket. Fig. 9.2 shows the rocket as it takes off.

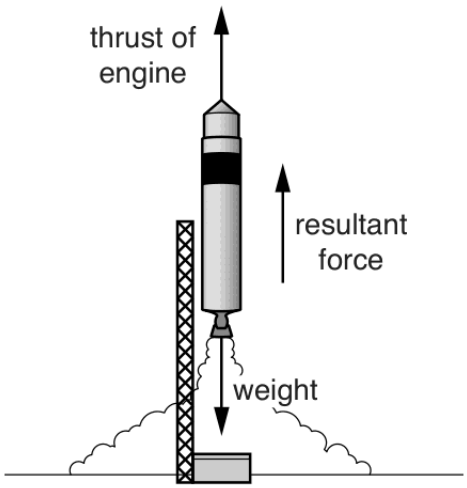


Fig. 9.2

The rocket and fuel have a total mass of 40 000 kg and a total weight of 400 000 N. The resultant force acting upwards on the rocket is 50 000 N.

(i) Calculate the thrust produced by the rocket engine.

thrust = ...[1]

(ii) Calculate the acceleration of the rocket.

acceleration = ...[2]

(c) In the first four minutes after take-off, the acceleration of the rocket is uniform.

(i) State what is meant by a *uniform acceleration*.

.....

.....

.....[2]

(ii) Fig. 9.3 describes the motion of the rocket in the first 12 minutes.

time / minutes	motion of the rocket
0 to 4	uniform acceleration
4 to 6	increasing acceleration
6 to 10	decreasing acceleration
10 to 12	constant speed

Fig. 9.3

On Fig. 9.4, sketch the speed-time graph of the rocket for the first 12 minutes.

You do not need to give values for the speed.

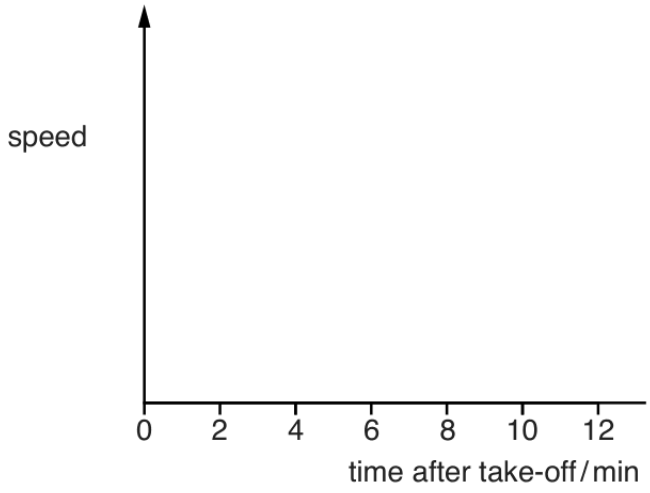


Fig. 9.4

[3]

(iii) State how the speed-time graph in **(ii)** can be used to find the distance travelled by the rocket.

.....

.....[1]

9	**(a)**	**(i)**	speed and mass	B2
		(ii)	**1** speed and direction **or** distance / time and direction **or** displacement / time	B1
			2 direction changes	B1
		(iii)	force of gravity from / towards Earth	B1
			force is centripetal **or** at right angles to motion / velocity	B1
	(b)	**(i)**	450 000 N	B1
		(ii)	(a =) F/m or 50 000 / 40 000	C1
			1.25 m / s^2	A1
	(c)	**(i)**	same change in velocity / speed	M1
			in same time period	A1
		(ii)	start at origin and straight line for first 4 minutes	B1
			gradient increases at first after 4 and then decreases	B1
			constant speed from 10 minutes until 12 minutes	B1
		(iii)	area **under** graph	B1

QUESTION · 5054_s15_qp_12.pdf · source page 2 of 16

1 Which quantity is a scalar?

A acceleration

B force

C temperature

D velocity

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 1

1	**C**

QUESTION · 5054_s15_qp_12.pdf · source page 2 of 16

2 The diameter and the length of a thin wire, approximately 50 cm in length, are measured as precisely as possible.

What are the best instruments to use?

	diameter	length
A	micrometer	rule
B	micrometer	vernier calipers
C	rule	tape
D	vernier calipers	rule

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 2

2 **A**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 The apparatus shown in Fig. 1.1 is used to measure the extension of a spring.

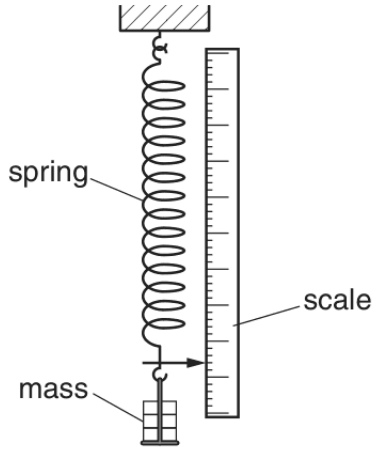


Fig. 1.1

(a) Explain how the mass causes a force on the spring.

.....

..... [1]

(b) The force on the spring is a vector quantity. State what is meant by a *vector* quantity.

.....

..... [1]

(c) Fig. 1.2 shows a graph of the length of the spring plotted against the force on the spring, for forces between 0 and 10 N.

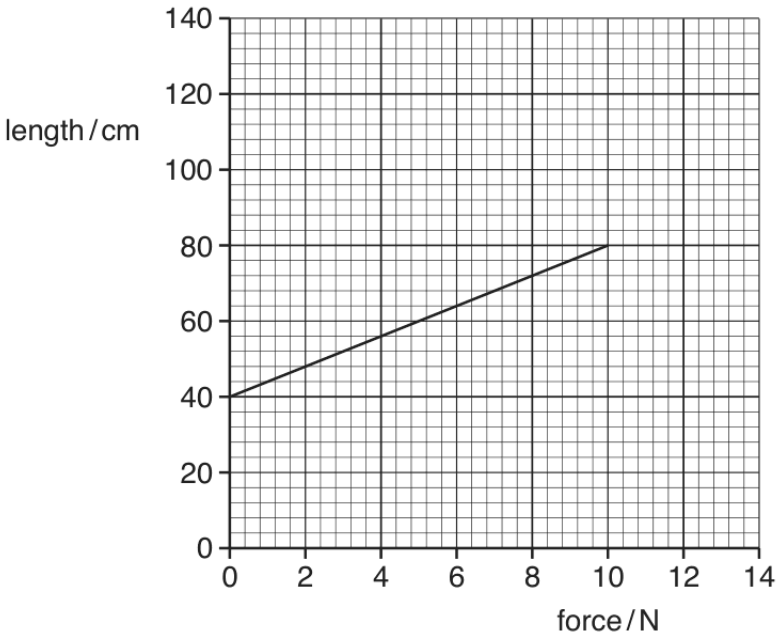


Fig. 1.2

(i) **1.** State a formula that relates the unstretched length l_0 of the spring, the stretched length l of the spring and the extension e of the spring.

..... [1]

2. A mass produces a force of 9.0 N on the spring. Determine the extension of the spring caused by this mass.

extension = ... [1]

(ii) The limit of proportionality of the spring is reached when the force is 10 N.

The spring is easier to stretch after the limit of proportionality.

On Fig. 1.2, continue the line to suggest how the length changes when the force is greater than 10 N. [1]

1 **(a)** weight (pulls spring down and causes tension) B1
or force/pull of **gravity**
or mass is in gravitational field

(b) has a direction B1

(c) **(i)** **1** $l = l_0 + e$ **or** $l_0 = l - e$ **or** $e = l - l_0$ B1

2 36 cm B1

(ii) curve upwards after 10 N B1

2 Fig. 2.1 shows two engineers measuring the length of a wall made from concrete.

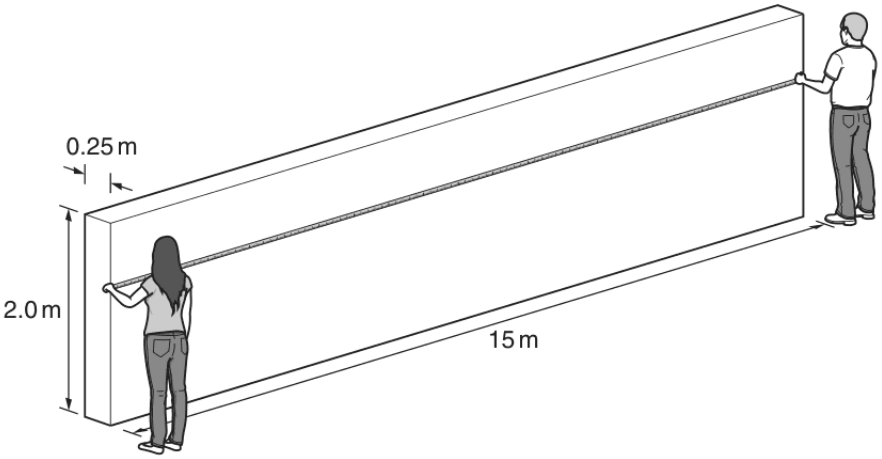


Fig. 2.1 (not to scale)

The wall is 2.0 m high, 15 m long and 0.25 m thick.

The weight of the wall is 180 000 N and the mass of the wall is 18 000 kg.

(a) The engineers measure the length of the wall in one single measurement.

State the name of the measuring instrument they use.

..... [1]

(b) The engineers state that the density of the concrete affects the pressure exerted by the wall on the ground but that the length of the wall does not affect this pressure.

(i) Define *density*.

.....

..... [1]

(ii) Calculate the density of the concrete.

density = ... [2]

(iii) Calculate the pressure exerted by the wall on the ground.

pressure = ... [2]

(iv) Without further calculation, explain why doubling the length of the wall does not change the pressure that the wall exerts on the ground.

.....

.....

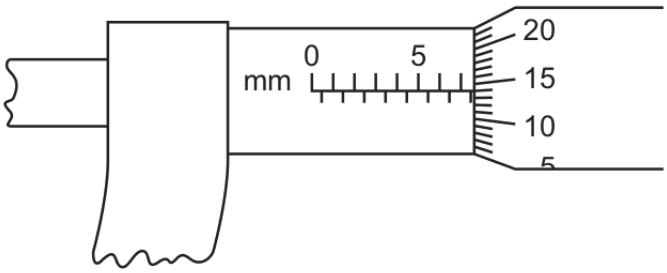
.....

..... [1]

2	**(a)**	tape	B1
	(b) **(i)**	mass ÷ volume **or** mass per unit volume	B1
	(ii)	(V=) 15 × 0.25 × 2 or 7.5 seen	C1
		2400 kg / m^3	A1
	(iii)	(A=) 15 × 0.25 **or** 3.75 **or** (P=) F / A	C1
		or (P=) 18 0000 / A **or** (P=) *dgh* / ρgh seen	
		48 000 Pa	A1
	(iv)	(length doubles) so both area **and** weight / force double	B1
		or area **and** force / weight both increase / large**r** (in proportion)	
		or height **and** density the same (in $P = dgh$)	

QUESTION · 5054_w15_qp_11.pdf · source page 2 of 16

2 The diagram shows a micrometer scale.



Which reading is shown?

A 5.64 mm **B** 7.14 mm **C** 7.16 mm **D** 7.64 mm

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 2

2 **D**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Force is a vector quantity and mass is a scalar quantity.

(a) (i) State how a vector quantity differs from a scalar quantity.

.....

..... [1]

(ii) State one other vector quantity and one other scalar quantity.

vector quantity .....

scalar quantity .....

[2]

(b) An aeroplane flies through the air in a straight line, with its engines at full power. The weight of the aeroplane is 160 kN. There is also a force of 320 kN acting upwards on the aeroplane at 30° to the horizontal.

In the space below, use a graphical method to determine the size and the direction of the resultant of these two forces. Use a scale of 1.0 cm : 40 kN.

size of resultant force = ...

direction of resultant force = ...

[4]

(c) An aeroplane is travelling in a straight line at a constant height. The aeroplane accelerates.

(i) State the energy change that is taking place as the aeroplane accelerates.

.....

.....

..... [2]

(ii) As the aeroplane accelerates, the pilot keeps the engines at constant power. The aeroplane eventually reaches a constant speed.

1. Explain why the aeroplane stops accelerating and reaches a constant speed.

.....

.....

.....

.....

.....

..... [3]

2. When travelling at a constant speed, the pilot directs the aeroplane into a tight turn that follows a circular path.

Explain why the aeroplane is accelerating even though its speed is not changing.

.....

.....

.....

.....

..... [3]

Section B

9	**(a)**	**(i)**	(vector) has direction **or** scalar does not have direction **or** (vectors) may cancel **or** scalars cannot cancel	B1	
		(ii)	one vector quantity e.g.: displacement; velocity, acceleration	B1	
			one scalar quantity e.g.: distance, length, speed, time, current, resistance	B1	[3]
	(b)		downward weight arrow of length 3.9–4.1 cm	B1	
			correct triangle/parallelogram drawn **and** correct diagonal clear	B1	
			270–285 kN	B1	
			horizontal (±3°)	B1	
	(c)	**(i)**	from chemical / fuel energy	B1	
			to kinetic (and thermal energy) (**not** with any intermediate)	B1	
		(ii)	**air** resistance / friction / drag	B1	
			air resistance / friction / drag increases **or** resultant force decreases **or** acceleration decreases	B1	
			resultant force is zero **or** (air) resistance / friction equals thrust	B1	
			direction of motion is changing	B1	
			velocity is vector **or** has a direction	B1	
			(acceleration depends on) changing velocity **or** **resultant** force towards centre (of circle) **or centripetal** force	B1	[12]
					[15]

QUESTION · 5054_w15_qp_12.pdf · source page 2 of 16

1 Which is a scalar quantity?

A mass

B force

C velocity

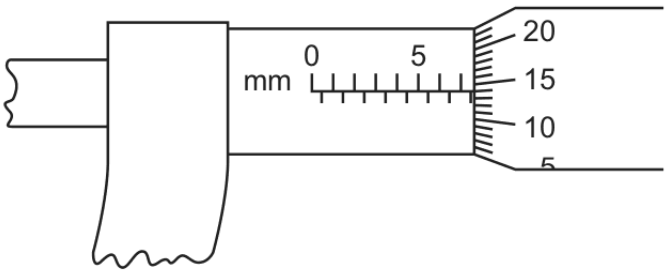
D weight

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 1

1	**A**

QUESTION · 5054_w15_qp_12.pdf · source page 2 of 16

3 The diagram shows a micrometer scale.



Which reading is shown?

A 5.64 mm **B** 7.14 mm **C** 7.16 mm **D** 7.64 mm

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 3

3 **D**

3 A very deep tank is used when training sailors to escape from submarines. The tank is cylindrical and open to the air at the top. Fig. 3.1 shows the tank.

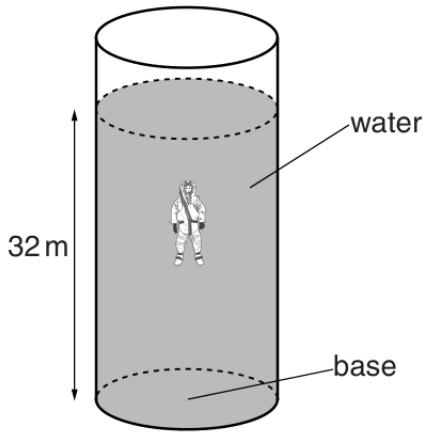


Fig. 3.1 (not to scale)

The area of the base of the tank is $45\,m^2$ and the tank is filled with water to a depth of 32 m.

(a) The density of water is $1000\,kg/m^3$ and the gravitational field strength g is $10\,N/kg$.

Calculate the pressure, due to the water, on the base of the tank.

pressure = ... [2]

(b) The pressure in the water at the base of the tank is $4.2 \times 10^5\,Pa$.

(i) Explain why the pressure in the water at the base of the tank differs from the value calculated in **(a)**.

.....

.....

..... [1]

(ii) Calculate the force exerted on the base of the tank.

force = ... [2]

(c) Force is a vector quantity and pressure is a scalar quantity.

State how a vector quantity differs from a scalar quantity.

.....

..... [1]

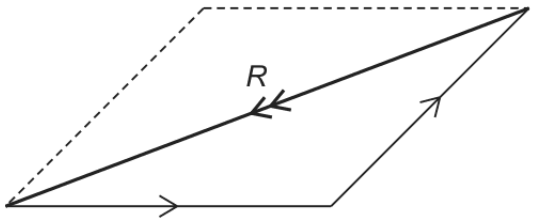
3	**(a)**	$(p =)h\rho g$ **or** $32 \times 1000 \times 10$	C1	
		3.2×10^5 Pa	A1	
	(b)	**(i)** atmospheric pressure (is also acting on the surface of the water)	B1	
		(ii) $(F =)pA$ **or** $4.2 \times 10^5 \times 45$ **or** $3.2 \times 10^5 \times 45$ **or** 1.44×10^7	C1	
		$1.9 / 1.89 \times 10^7$ N	A1	
	(c)	(vector) has a direction **or** scalar does not have a direction **or** (vectors) may cancel **or** scalars cannot cancel	B1	[6]

QUESTION · 5054_s16_qp_11.pdf · source page 2 of 16

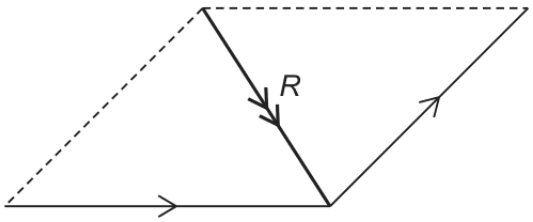
1 The diagram shows arrows representing two vector quantities.

Which diagram shows the resultant *R* of these two vectors?

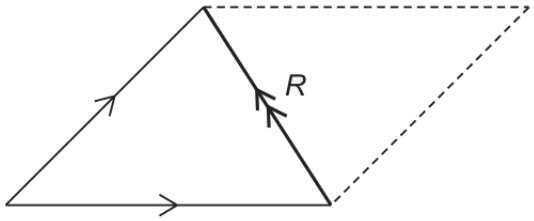
A



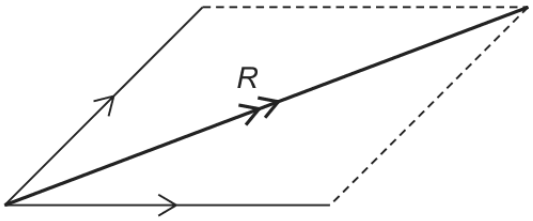
B



C



D



MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 1

1	**D**

QUESTION · 5054_s16_qp_11.pdf · source page 2 of 16

2 Which set of quantities are all vectors?

A acceleration, displacement, velocity

B chemical energy, mass, power

C extension, force, gravitational potential energy

D weight, kinetic energy, work

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 2

2	**A**

QUESTION · 5054_s16_qp_11.pdf · source page 2 of 16

3 A student determines the circumference of a golf ball.

Which instrument gives a reading that is the circumference of the golf ball?

A calipers

B micrometer

C rule

D tape

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 3

3	**D**

3 Fig. 3.1 shows a thin sheet of plastic. A student tries to measure the thickness of the sheet with a ruler, but the sheet is too thin to measure accurately.

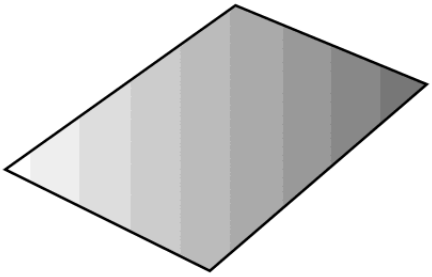


Fig. 3.1 (not to scale)

The student measures the mass of the sheet and obtains the value 0.12 g.

(a) (i) State what is meant by *mass*.

.....

.....[1]

(ii) The student is told that the density of the plastic is 0.91 g/cm^3.

Calculate the volume of the plastic sheet.

volume = ...[2]

(iii) The student measures the length and width of the sheet. The readings obtained are:

length of sheet = 3.0 cm
width of sheet = 2.0 cm

Calculate the thickness of the sheet.

thickness = ... [2]

(b) State a measuring instrument that can be used to measure the thickness of the sheet accurately.

.....[1]

3	**(a)**	**(i)**	amount of matter / substance / material **or** the ability of an object to resist a change in its state of motion (when a force is applied)	B1
		(ii)	(V=) M / D in any form numerical or algebraic	C1
			0.13(19) cm^3	A1
		(iii)	V / ($l \times$ w) in any form numerical or algebraic	C1
			0.022 cm	A1
	(b)		micrometer (screw gauge) **or** calipers	B1

QUESTION · 5054_s16_qp_12.pdf · source page 2 of 16

1 A student determines the circumference of a golf ball.

Which instrument gives a reading that is the circumference of the golf ball?

A calipers

B micrometer

C rule

D tape

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 1

1	**D**

QUESTION · 5054_s16_qp_12.pdf · source page 4 of 16

7 Which quantity is a vector?

A energy

B force

C speed

D time

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 7

7	**B**

QUESTION · 5054_w16_qp_11.pdf · source page 2 of 20

1 A length of copper wire is labelled ‘length 30 m’ and ‘diameter 0.50 mm’.

Which instruments are most suitable to measure accurately the length and the diameter of the wire?

	length	diameter
A	rule	calipers
B	rule	micrometer
C	tape	calipers
D	tape	micrometer

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 1

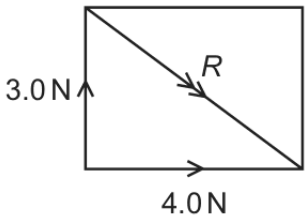
1 **D**

QUESTION · 5054_w16_qp_11.pdf · source page 4 of 20

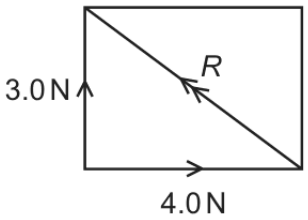
7 Two forces of 3.0 N and 4.0 N act at right-angles to each other.

Which diagram shows the resultant *R* of these forces?

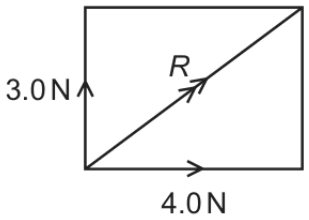
A



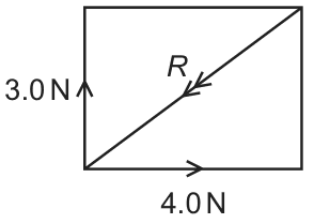
B



C



D



MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 7

7	**C**

QUESTION · 5054_w16_qp_12.pdf · source page 2 of 16

1 A length of copper wire is labelled ‘length 30 m’ and ‘diameter 0.50 mm’.

Which instruments are most suitable to measure accurately the length and the diameter of the wire?

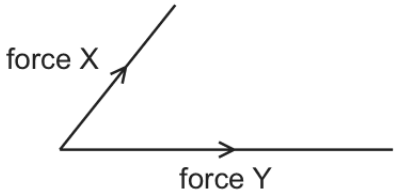
	length	diameter
A	rule	calipers
B	rule	micrometer
C	tape	calipers
D	tape	micrometer

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 1

1 **D**

QUESTION · 5054_w16_qp_12.pdf · source page 5 of 16

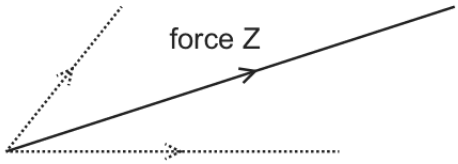
10 Two forces, X and Y, act on an object and produce a resultant force. The scale diagram represents the sizes and directions of forces X and Y.



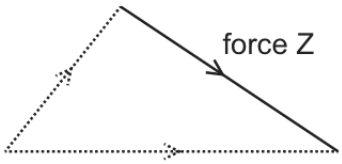
Force Z balances the resultant force due to X and Y and keeps the object stationary.

Which arrow represents force Z?

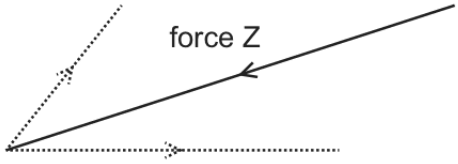
A



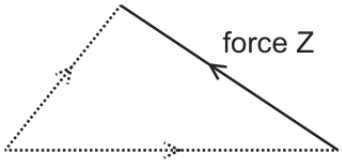
B



C



D



MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 10

10 **C**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 A spring is taken from a drawer in a physics laboratory.

(a) Fig. 8.1 is the extension-load graph for the spring.

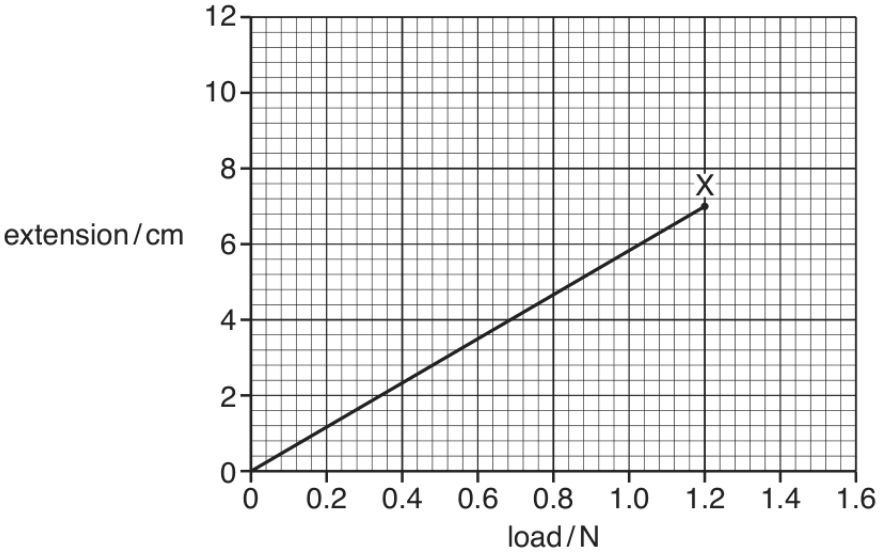


Fig. 8.1

(i) Using Fig. 8.1, determine the load on the spring when its extension is 4.9 cm.

load = ...[1]

(ii) The point X on Fig. 8.1 is the limit of proportionality for the spring.

On Fig. 8.1, sketch a line to show how the extension-load graph might continue after X. [2]

(b) A rectangular, plastic block is suspended from the lower end of the spring. Fig. 8.2 shows that the bottom 3.5 cm of the plastic block is in the water in a beaker.

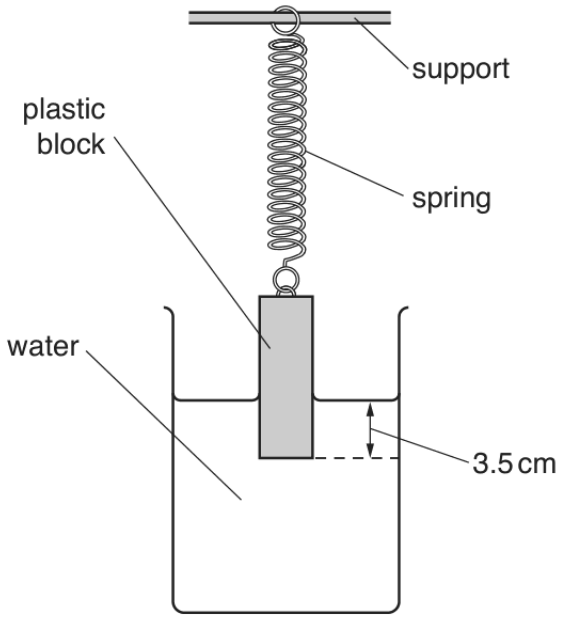


Fig. 8.2 (not to scale)

The density of water is 1000 kg / m^3 and the gravitational field strength g is 10 N / kg. The base of the plastic block is a square with sides of 4.0 cm.

(i) Calculate the pressure due to the water at a depth of 3.5 cm below the surface of the water.

pressure = ...[3]

(ii) Calculate the upward force on the base of the block due to the pressure of the water.

force = ...[2]

(iii) In Fig. 8.2, the extension of the spring is 4.9 cm.

Using the answer from **(a)(i)** and the answer from **(b)(ii)**, determine the weight of the block.

weight = ...[1]

(c) In the equipment shown in Fig. 8.2, the pressure of the atmosphere is transmitted through the water. The total pressure in the water at a depth of 3.5 cm is larger than the pressure due to the water alone.

(i) Explain why, in this experiment, atmospheric pressure does not affect the position of the plastic block in the water.

.....

.....

.....

.....[2]

(ii) Force is a vector quantity and pressure is a scalar quantity.

State how a *vector* quantity differs from a *scalar* quantity.

.....

.....[1]

(d) The beaker in Fig. 8.2 is gradually moved upwards so that more of the plastic block is in the water.

Explain what happens to the forces that act on the block.

.....

.....

.....

.....

.....

.....

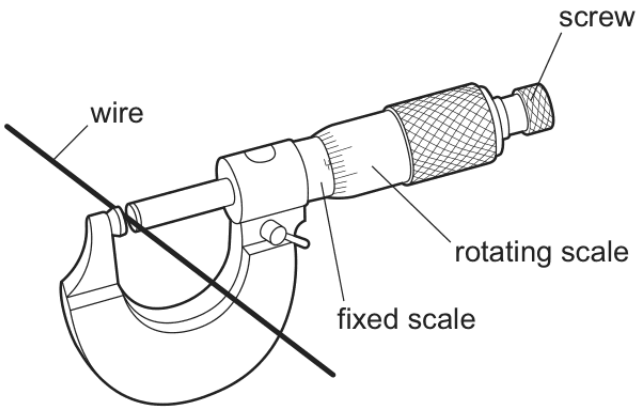
.....[3]

Section B

8	**(a)**	**(i)**	0.83 – 0.86 N	B1	
		(ii)	line curved	B1	
			line (curved) upwards	B1	[3]
	(b)	**(i)**	(P =)$h\rho g$	C1	
			0.035 × 1000 × 10 **or** 3.5 × 1000 × 10 **or** 35 × 1000 ×10	C1	
			350 Pa	A1	
		(ii)	(F =)PA **or** 350 × 0.0016 **or** 350 × 16 **or** 5600	C1	
			0.56 N	A1	
		(iii)	1.4 N **or (a)(i)** + **(b)(ii)** calculated	B1	[6]
	(c)	**(i)**	(atmospheric pressure) exerts a downward force / pressure	B1	
			(on top of the block)		
			(cancels out the) extra upward force / pressure	B1	
		(ii)	(vector) has direction (in addition to magnitude)	B1	[3]
	(d)	any **three** lines from force due to water increases force due to spring decreases increased pressure (at base) they add to give a constant value / weight of block **or** total force constant		B3	[3]
					[15]

QUESTION · 5054_s17_qp_11.pdf · source page 2 of 16

1 A micrometer is used to measure the diameter of a uniform wire.



What is done to obtain an accurate answer?

A Make the micrometer horizontal and then use the scales to find the reading.

B Subtract the fixed-scale reading from the rotating-scale reading.

C Subtract the rotating-scale reading from the fixed-scale reading.

D Use the scales to find the reading and add or subtract any zero error.

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 1

1	D	**1**

QUESTION · 5054_s17_qp_11.pdf · source page 2 of 16

2 A car's acceleration and maximum speed are improved by using an engine of smaller mass and greater driving force.

How many of the underlined quantities are vectors?

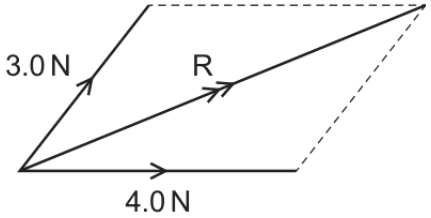
A 1 **B** 2 **C** 3 **D** 4

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 2

2	B	**1**

QUESTION · 5054_s17_qp_11.pdf · source page 2 of 16

3 The diagram shows the resultant R of a 3.0 N force and a 4.0 N force acting at a point.



The angle between the 3.0 N force and the 4.0 N force can be any value from 0° to 90°.

Which value of R is **not** possible?

A 4.0 N **B** 5.0 N **C** 6.0 N **D** 7.0 N

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 3

3	A	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 2 of 20

1 A car accelerates at $5.0\,m/s^2$ along a straight, horizontal road and reaches a velocity of $20\,m/s$ in a time of $4.0\,s$.

During this time, its total displacement is $40\,m$.

Which quantity is a scalar?

A a displacement of $40\,m$

B a time of $4.0\,s$

C a velocity of $20\,m/s$

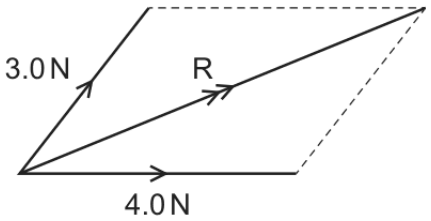
D an acceleration of $5.0\,m/s^2$

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 2 of 20

2 The diagram shows the resultant R of a 3.0 N force and a 4.0 N force acting at a point.



The angle between the 3.0 N force and the 4.0 N force can be any value from 0° to 90°.

Which value of R is **not** possible?

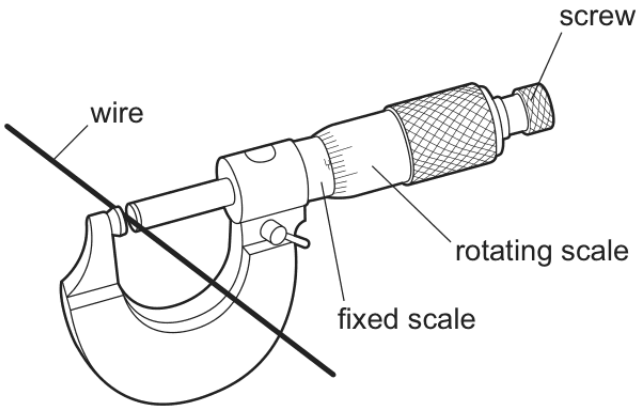
A 4.0 N **B** 5.0 N **C** 6.0 N **D** 7.0 N

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 2

2	A	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 2 of 20

3 A micrometer is used to measure the diameter of a uniform wire.



What is done to obtain an accurate answer?

A Make the micrometer horizontal and then use the scales to find the reading.

B Subtract the fixed-scale reading from the rotating-scale reading.

C Subtract the rotating-scale reading from the fixed-scale reading.

D Use the scales to find the reading and add or subtract any zero error.

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

2 A swing is made by tying rope loosely to the branch of a tree, as shown in Fig. 2.1.

A child swings backwards and forwards several times, starting at the highest point A.

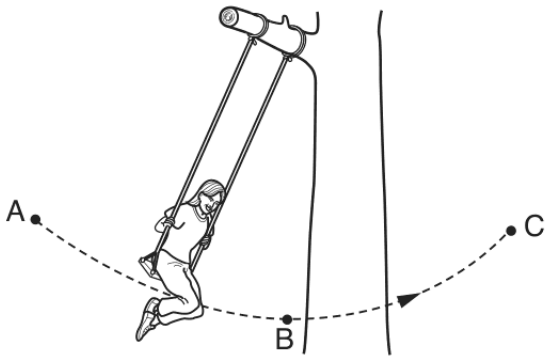


Fig. 2.1

(a) Explain how another child can obtain an accurate measurement of the time for one complete swing.

.....

.....

.....

.....

.....[2]

(b) When the child moves from A to B, she falls a vertical distance of 0.60 m. She loses 240 J of gravitational potential energy. The gravitational field strength g is 10 N/kg.

(i) Calculate the mass of the child.

mass = ...[2]

(ii) Suggest two reasons why her kinetic energy at B is not equal to 240 J.

1. .....

.....

2. .....

.....
[2]

Question	Answer	Marks
2(a)	use of stopwatch or electronic timer	**B1**
	time at least 5 swings **and** divide by number of swings **or** use of fiducial mark **or** definition of one swing clear e.g. A to C to A or from A and back to A	**B1**
2(b)(i)	(m=) P.E. / gh **or** 240 / 10 × 0.6	**C1**
	40 kg	**A1**
2(b)(ii)	air resistance **or** friction (with air or rope and tree)	**B1**
	heat produced / work done (in / against air or friction) **or** effect of wind **or** work done by arms / legs	**B1**

QUESTION · 5054_w17_qp_11.pdf · source page 2 of 20

1 Which list contains only scalar quantities?

A acceleration, displacement, velocity

B distance, force, speed

C force, length, time

D length, mass, speed

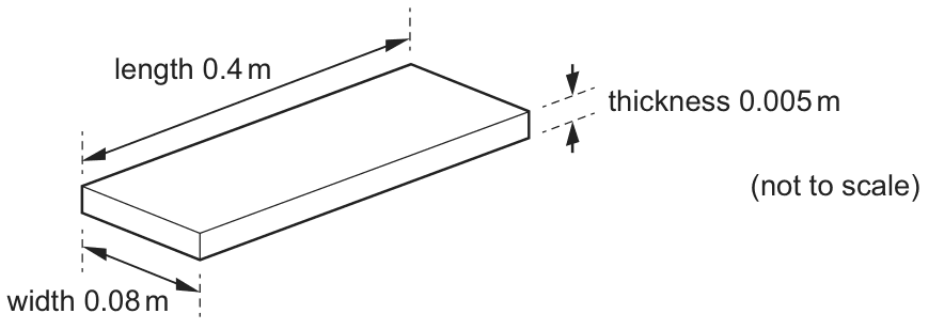
MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 1

1	D	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 2 of 20

2 A manufacturer measures the three dimensions of a wooden floor tile using three different instruments.

The approximate dimensions of the tile are shown.



Which instruments are used to measure accurately each of these dimensions?

	length	thickness	width
A	metre rule	micrometer	calipers
B	metre rule	calipers	micrometer
C	micrometer	metre rule	calipers
D	calipers	micrometer	metre rule

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 2

2	A	**1**

QUESTION · 5054_w17_qp_12.pdf · source page 2 of 16

1 Which list contains only scalar quantities?

A acceleration, displacement, velocity

B distance, force, speed

C force, length, time

D length, mass, speed

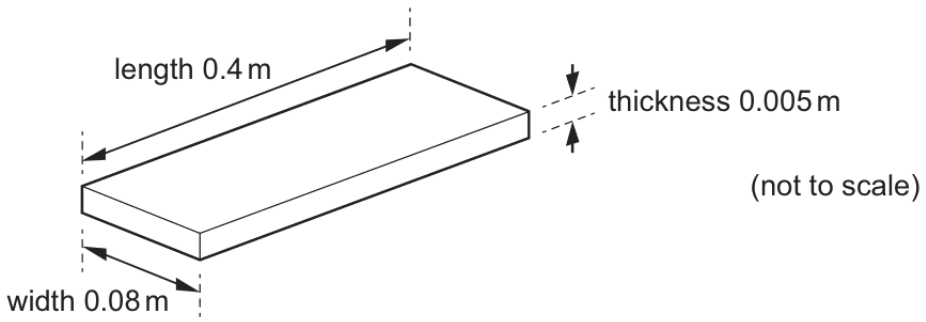
MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 1

1	D	**1**

QUESTION · 5054_w17_qp_12.pdf · source page 2 of 16

2 A manufacturer measures the three dimensions of a wooden floor tile using three different instruments.

The approximate dimensions of the tile are shown.



Which instruments are used to measure accurately each of these dimensions?

	length	thickness	width
A	metre rule	micrometer	calipers
B	metre rule	calipers	micrometer
C	micrometer	metre rule	calipers
D	calipers	micrometer	metre rule

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 2

2	A	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Two small tugboats are pulling a large ship in a harbour. Fig. 1.1 represents the view from above and shows the directions of the forces on the ship.

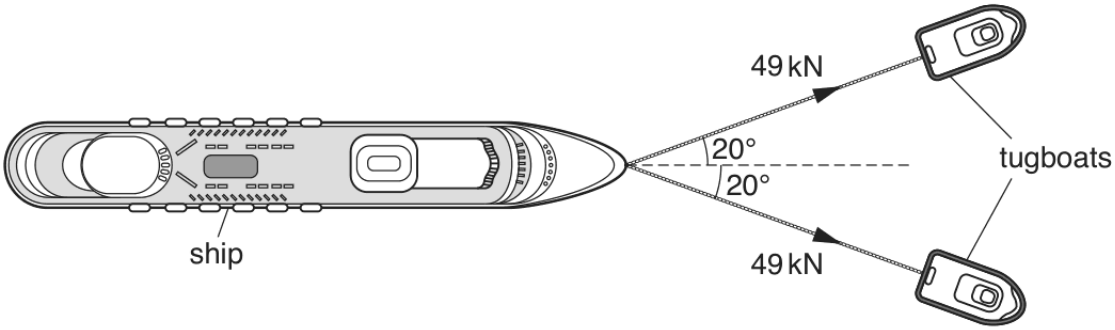


Fig. 1.1 (not to scale)

Each of the tugboats is exerting a force of 49 kN on the ship.

(a) Determine by a graphical method the resultant of these two forces and state the scale used.

scale .....

resultant = ...
[3]

(b) The engines of the ship are not operating and the water in the harbour is stationary. The ship is moving in a straight line in the direction of the total force exerted by the tugboats. It is travelling at a constant speed.

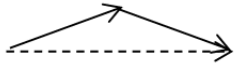
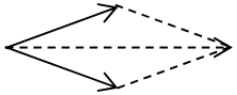
Explain, in terms of the forces acting, why the ship is moving in a straight line at constant speed.

.....

.....

.....

.....[2]

Question	Answer	Marks
1(a)	scale at least: 1.0 cm : 10 kN	**B1**
	correct triangle **or** correct parallelogram **and** resultant, i.e.  **or** 	**B1**
	90 kN ⩽ resultant ⩽ 94 kN	**B1**
1(b)	no <u>resultant</u> force **or** balanced forces **or** all forces cancel	**B1**
	(force of tugboats cancelled) by drag / water resistance / water friction etc.	**B1**

QUESTION · 5054_s18_qp_11.pdf · source page 2 of 20

2 A student wishes to measure directly the circumference of a football.

Which is the most suitable instrument to use?

A calipers

B a measuring tape

C a micrometer

D a ruler

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 2

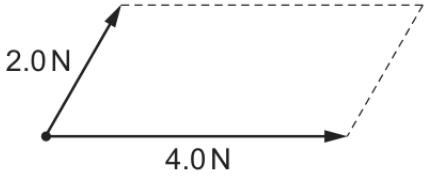
2	B	**1**

QUESTION · 5054_s18_qp_11.pdf · source page 5 of 20

11 Forces of 4.0 N and 2.0 N act at a point.

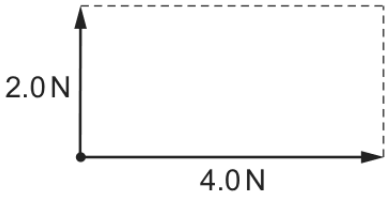
Which scale diagram shows the forces that have a resultant of 4.0 N?

A



B

C



D

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 11

11	B	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 2 of 16

1 Which is a vector quantity?

A a mass of 2.0 kg

B a temperature of –10 °C

C a weight of 15 N

D an average speed of 20 m / s

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 1

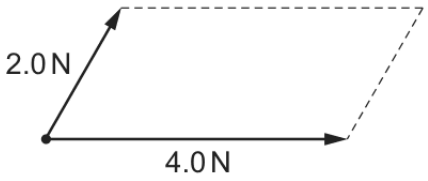
1	C	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 2 of 16

2 Forces of 4.0 N and 2.0 N act at a point.

Which scale diagram shows the forces that have a resultant of 4.0 N?

A



B

C

D

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 2

2	B	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 2 of 16

3 What is the name and value of the unit of power written as mW?

	name	value
A	megawatt	10^{-3} W
B	megawatt	10^{6} W
C	milliwatt	10^{-3} W
D	milliwatt	10^{6} W

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 3

3	C	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 3 of 16

4 Micrometers, metre rules, tapes and calipers are used for measuring lengths.

Which row identifies the most suitable device for accurately measuring the stated length?

	length	measuring device
A	0.15 mm	micrometer
B	0.50 mm	metre rule
C	0.15 m	tape
D	0.50 m	calipers

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 4

4	A	**1**

QUESTION · 5054_w18_qp_11.pdf · source page 2 of 16

1 What is a possible mass for a normal adult person?

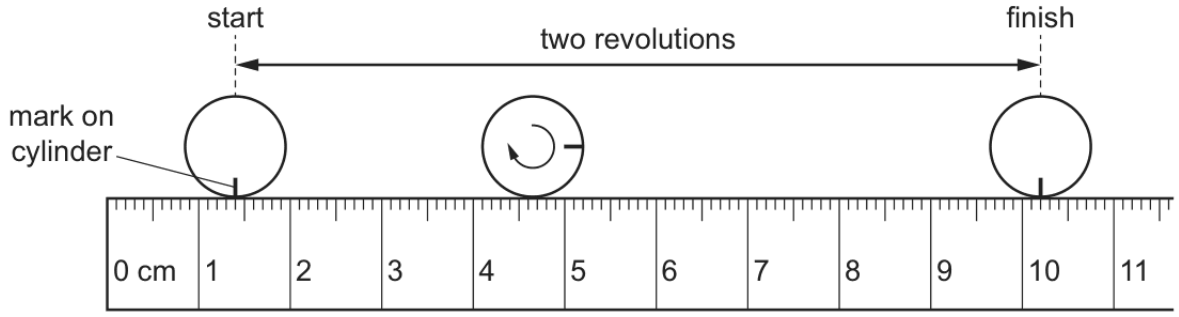
A 7.5 kg **B** 75 kg **C** 750 kg **D** 7500 kg

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

QUESTION · 5054_w18_qp_11.pdf · source page 2 of 16

2 A small cylinder is rolled along a ruler and completes two revolutions.



The circumference is the distance around the outside of a circle.

What is the circumference of the cylinder?

A 4.4 cm **B** 5.2 cm **C** 8.8 cm **D** 10.2 cm

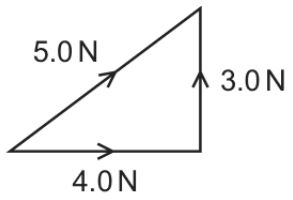
MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 2

2	A	**1**

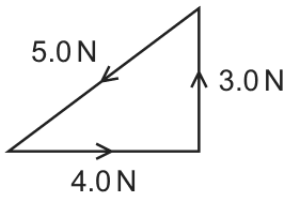
QUESTION · 5054_w18_qp_11.pdf · source page 4 of 16

8 Which diagram shows the addition of the 4.0 N and the 3.0 N forces?

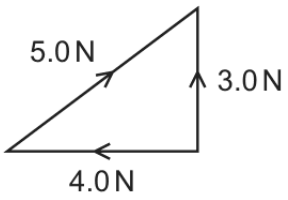
A



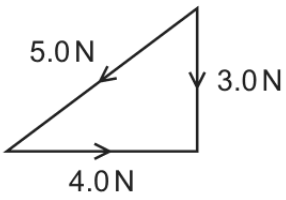
B



C



D

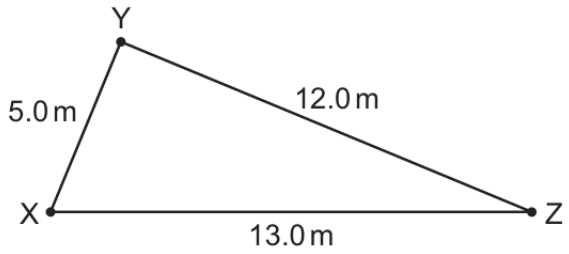


MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 8

8	A	**1**

QUESTION · 5054_w18_qp_12.pdf · source page 5 of 16

9 Paths are laid as shown between points X, Y and Z.



A person walks along the paths from X to Y to Z and then back to X.

What is the value of the total displacement and of the total distance travelled?

	total displacement / m	total distance travelled / m
A	0	0
B	0	30
C	30	0
D	30	30

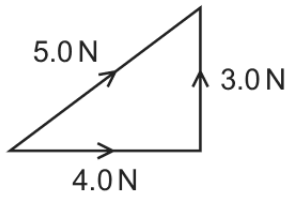
MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 9

9	B	**1**

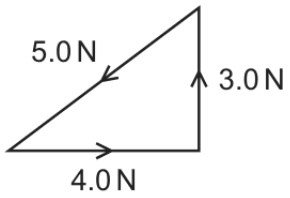
QUESTION · 5054_w18_qp_12.pdf · source page 5 of 16

10 Which diagram shows the addition of the 4.0 N and the 3.0 N forces?

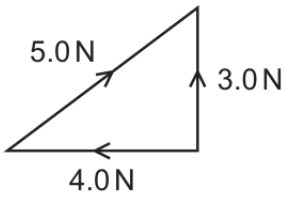
A



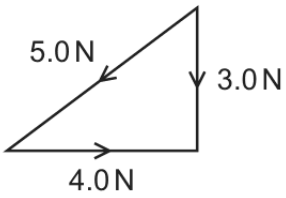
B



C



D



MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 10

10	A	**1**

QUESTION · 5054_s19_qp_11.pdf · source page 2 of 20

1 Which quantities are both vectors?

A acceleration and length

B distance and velocity

C length and distance

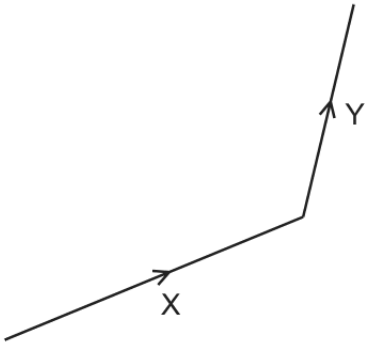
D velocity and acceleration

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 1

1	D	**1**

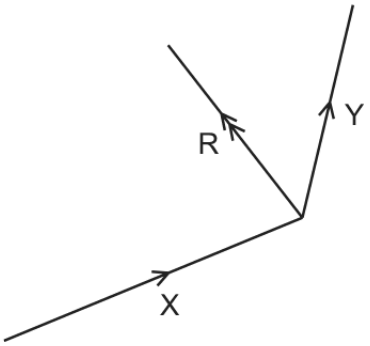
QUESTION · 5054_s19_qp_11.pdf · source page 2 of 20

2 Two forces X and Y act as shown.

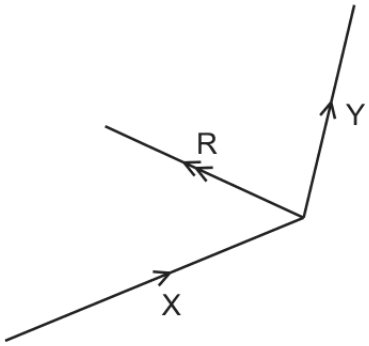


Which diagram shows the resultant force R of X and Y?

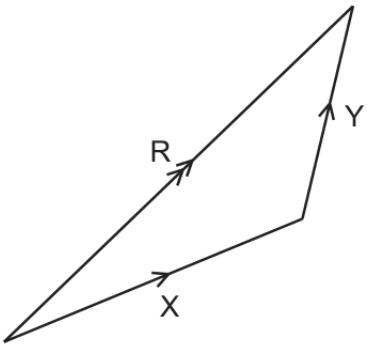
A



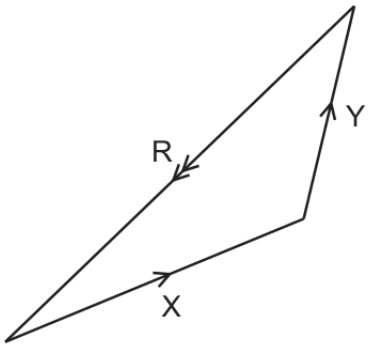
B



C



D



MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 2

2	C	**1**

QUESTION · 5054_s19_qp_11.pdf · source page 2 of 20

3 Which reading is given to one tenth of a millimetre?

A 3.3 cm **B** 3.31 cm **C** 3.310 cm **D** 3.312 cm

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 3

3	B	**1**

QUESTION · 5054_s19_qp_11.pdf · source page 3 of 20

4 The magnitudes of three different electric charges are given below.

What is the correct order of size, from largest to smallest?

A $1\,\text{mC} \rightarrow 1\,\text{MC} \rightarrow 1\,\text{kC}$

B $1\,\text{MC} \rightarrow 1\,\text{mC} \rightarrow 1\,\text{kC}$

C $1\,\text{MC} \rightarrow 1\,\text{kC} \rightarrow 1\,\text{mC}$

D $1\,\text{kC} \rightarrow 1\,\text{mC} \rightarrow 1\,\text{MC}$

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 4

4	C	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 2 of 16

1 Which reading is given to one tenth of a millimetre?

A 3.3 cm **B** 3.31 cm **C** 3.310 cm **D** 3.312 cm

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a map of the route taken by a car as it travels from town A to town B.

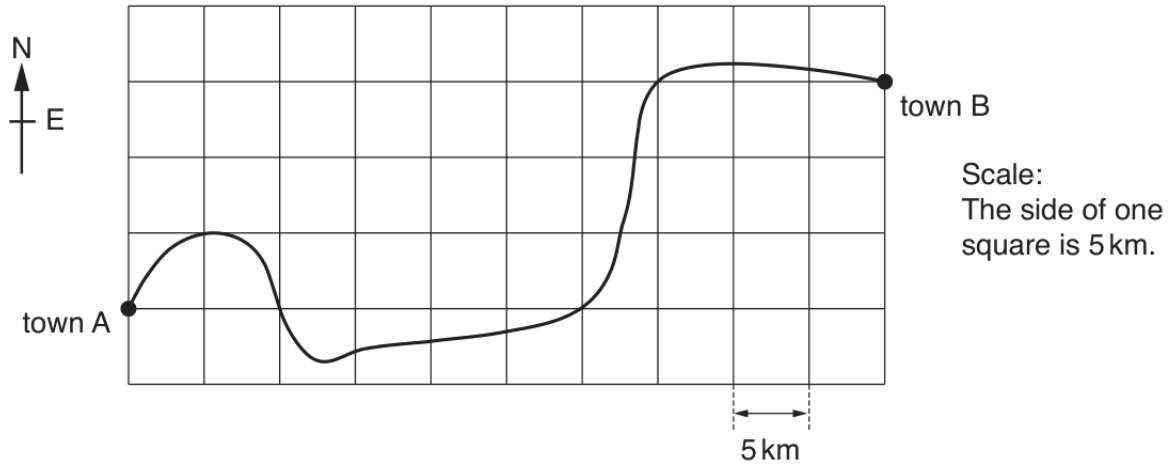


Fig. 9.1

The journey involves a distance and a displacement.

(a) (i) Using the scale on Fig. 9.1, determine the displacement in travelling from A to B.

size of displacement = ...

direction of displacement = ...
[3]

(ii) State how *distance* differs from *displacement*.

.....

.....

..... [1]

(iii) In some parts of the journey, the car is travelling at a constant speed but is also accelerating.

Explain how this is possible.

.....

.....

..... [2]

(b) At town B the car makes an emergency stop.

The total stopping distance of the car is the thinking distance added to the braking distance.

(i) Describe what is meant by the *thinking distance*.

.....

..... [1]

(ii) The brakes on the car work badly. This increases the braking distance.

State **two** other factors that increase the braking distance.

1. .....

.....

2. .....

.....
[2]

(c) The mass of the car and driver is 1200 kg. Just before the brakes are applied, the speed of the car is 30 m/s.

(i) Calculate the kinetic energy of the car and driver just before the brakes are applied.

kinetic energy = ... [3]

(ii) The braking force that stops the car is 1800 N.

Calculate the deceleration of the car.

deceleration = ... [2]

(iii) Explain why the temperature of the brakes increases during braking.

.....

..... [1]

[Total: 15]

Question	Answer	Marks
9(a)(i)	10.2–10.4 (cm) **or** 50 (km) **or** 15 (km) seen	**C1**
	51–53 km	**A1**
	direction (0)72–74° **or** N 72–74 °E etc.	**B1**
9(a)(ii)	(distance) has no direction **or (**distance) is a scalar	**B1**
9(a)(iii)	changes direction **or** goes round a corner	**B1**
	velocity changes **or** velocity depends on direction **or** acceleration / force is towards the centre of circle / centripetal	**B1**

Question	Answer	Marks
9(b)(i)	distance travelled before brakes applied (after seeing emergency)	**B1**
9(b)(ii)	any 2 of • poor tyre surface • poor road conditions, e.g. rain, ice • high speed (of car) **or** good aerodynamic car shape **or** tail wind • large mass / weight of car • driver applies less force to pedal / less braking force	**B1B1**
9(c)(i)	(KE =) ½mv^2 algebraic or numerical	**C1**
	½ × 1200 × 30^2	**C1**
	540 000 J	**A1**
9(c)(ii)	(a=) F / m algebraic or numerical	**C1**
	1.5 m / s^2	**A1**
9(c)(iii)	friction (produces thermal energy) **or** kinetic energy changes to heat **or** work done against friction	**B1**

QUESTION · 5054_w19_qp_11.pdf · source page 2 of 20

1 Which word is the name of a scalar quantity?

A acceleration

B force

C mass

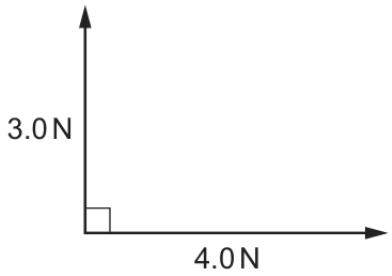
D velocity

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 1

1	C	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 2 of 20

2 What is the size of the resultant of the two forces shown in the diagram?



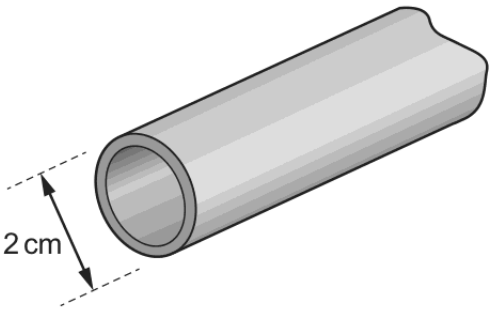
A 1.0 N **B** 3.5 N **C** 5.0 N **D** 7.0 N

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 2

2	C	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 2 of 20

3 A length of copper pipe, of uniform cross-section and several metres long, carries water to a tap.



Measurements are taken to determine accurately the volume of copper in the pipe.

Which instruments are used?

A micrometer and rule

B micrometer and calipers

C rule and tape

D tape and calipers

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 3 of 20

4 What is the correct unit for the quantity shown?

	quantity	unit
A	electromotive force (e.m.f.)	N
B	latent heat	J
C	pressure	kg/m^3
D	weight	kg

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 2 of 16

1 Which word is the name of a vector quantity?

A density

B displacement

C energy

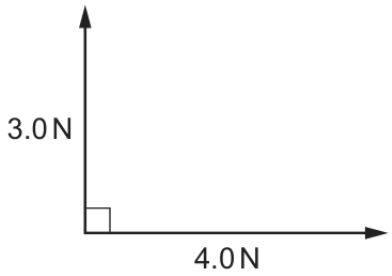
D speed

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 2 of 16

2 What is the size of the resultant of the two forces shown in the diagram?



A 1.0 N **B** 3.5 N **C** 5.0 N **D** 7.0 N

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 2

2	C	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 2 of 16

3 What is the correct unit for the quantity shown?

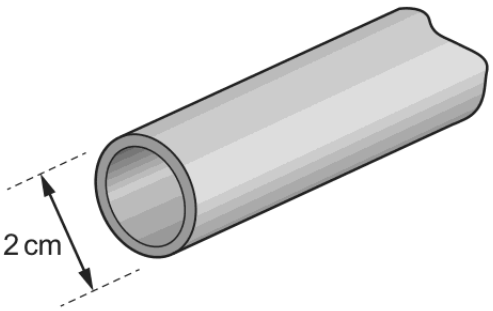
	quantity	unit
A	electromotive force (e.m.f.)	N
B	latent heat	J
C	pressure	kg/m^3
D	weight	kg

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 3

3	B	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 3 of 16

4 A length of copper pipe, of uniform cross-section and several metres long, carries water to a tap.



Measurements are taken to determine accurately the volume of copper in the pipe.

Which instruments are used?

A micrometer and rule

B micrometer and calipers

C rule and tape

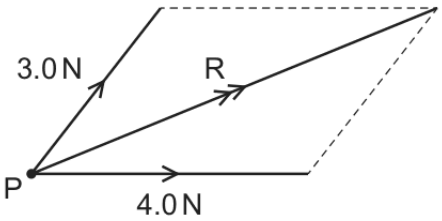
D tape and calipers

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 4

4	D	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 2 of 16

1 The diagram shows the resultant R of a 3.0 N force and a 4.0 N force that act at a point P.



The angle between the 3.0 N force and the 4.0 N force can be any value from 0° to 90°.

Which value of R is **not** possible?

A 4.0 N **B** 5.0 N **C** 6.0 N **D** 7.0 N

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 1

1	A	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 2 of 16

2 What is measured using a micrometer?

A area

B current

C length

D mass

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 2

2	C	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 2 of 16

1 Which quantity is a vector?

A acceleration

B distance

C speed

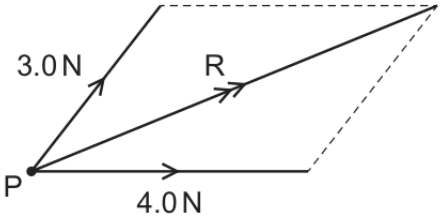
D time

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 1

1	A	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 2 of 16

2 The diagram shows the resultant R of a 3.0 N force and a 4.0 N force that act at a point P.



The angle between the 3.0 N force and the 4.0 N force can be any value from 0° to 90°.

Which value of R is **not** possible?

A 4.0 N **B** 5.0 N **C** 6.0 N **D** 7.0 N

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 2

2	A	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 2 of 16

3 What is measured using a micrometer?

A area

B current

C length

D mass

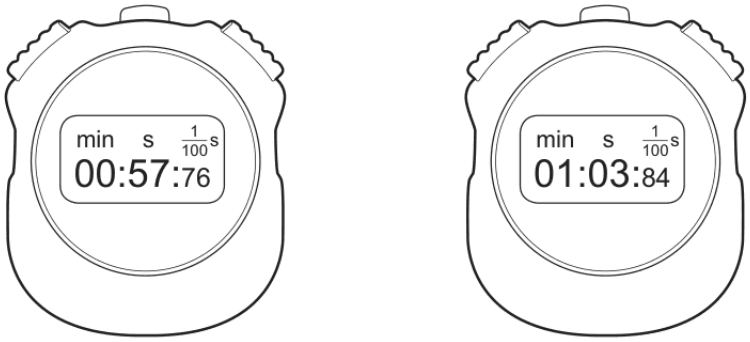
MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 3

3	C	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 2 of 16

4 Stop-watches are used to time the runners in a race.

The stop-watches show the times recorded for the winner and another runner.



winner other runner

What is the difference in time between the winner and the other runner?

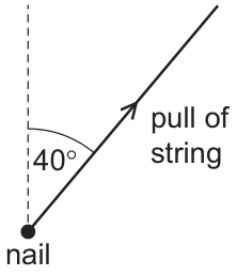
A 0.4608 s **B** 6.08 s **C** 46.08 s **D** 608 s

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 2 of 16

1 A heavy nail is fixed firmly to a wall. It is pulled by a string at 40° to the vertical. The nail does not move.



Three forces act on the nail:

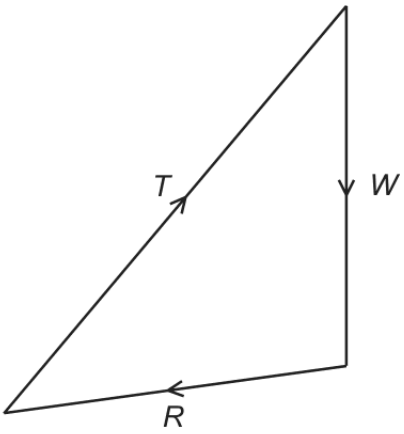
its weight *W*,

the tension *T* in the string,

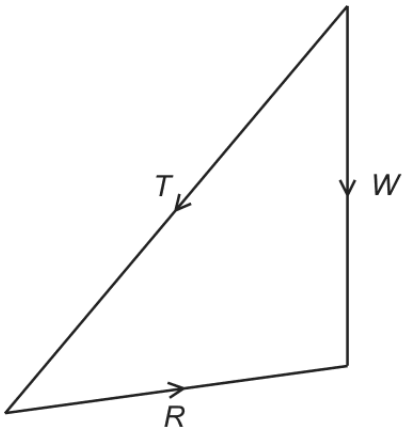
the force *R* exerted by the wall.

Which diagram, drawn to scale, represents the three forces?

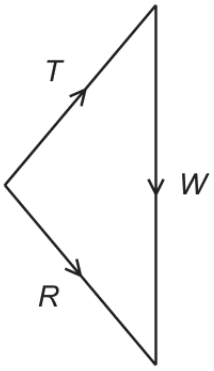
A



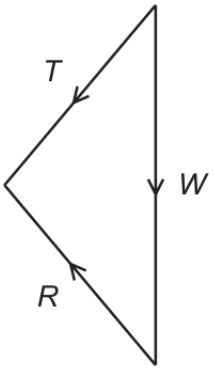
B



C



D



MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 1

1	A	**1**

2 Fig. 2.1 shows a satellite moving at a constant speed in a circular orbit around the Earth.

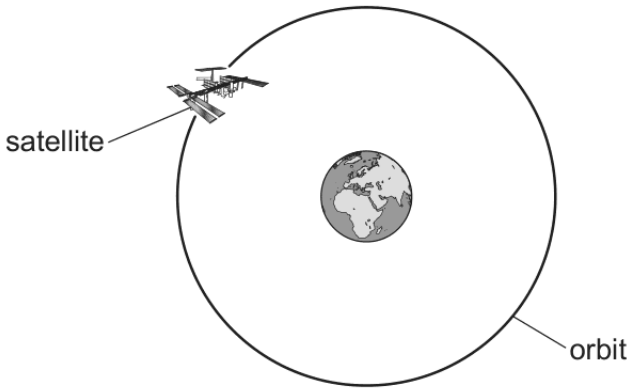


Fig. 2.1 (not to scale)

Speed is a scalar quantity but velocity is a vector quantity.

(a) State how a scalar quantity differs from a vector quantity.

.....

..... [1]

(b) Underline every vector quantity in the list.

distance **displacement** **force** **length** **mass** **time**

[1]

(c) There is a resultant force acting on the satellite in Fig. 2.1.

(i) Explain how the motion of the satellite shows that a resultant force is acting on it.

.....

.....

..... [2]

(ii) State the cause of this force.

.....

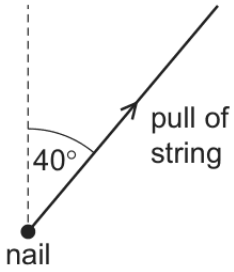
..... [1]

[Total: 5]

Question	Answer	Marks
2(a)	it / a scalar does not have a direction	**B1**
2(b)	*displacement* **and** *force* underlined	**B1**
2(c)(i)	direction (of motion) is changing	**B1**
	velocity is changing (with time) **or** it is accelerating (which requires a resultant force)	**B1**
2(c)(ii)	gravitational attraction / field of the earth	**B1**

QUESTION · 5054_w20_qp_12.pdf · source page 2 of 16

1 A heavy nail is fixed firmly to a wall. It is pulled by a string at 40° to the vertical. The nail does not move.



Three forces act on the nail:

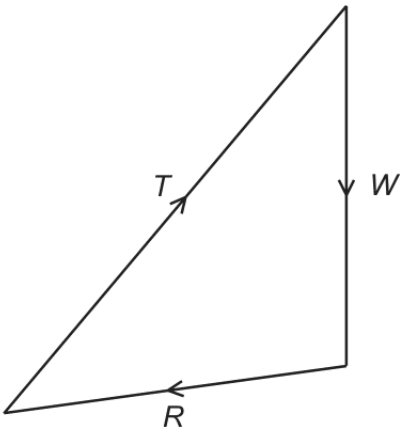
its weight *W*,

the tension *T* in the string,

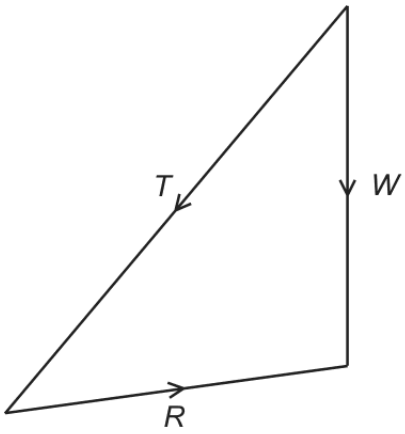
the force *R* exerted by the wall.

Which diagram, drawn to scale, represents the three forces?

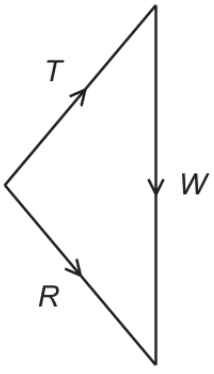
A



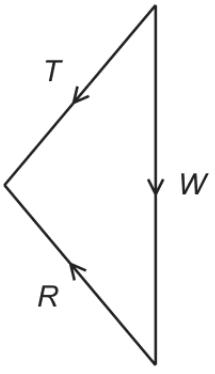
B



C



D



MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 1

1	A	**1**

QUESTION · 5054_s21_qp_11.pdf · source page 2 of 16

1 A list of various quantities is shown.

acceleration

displacement

force

length

mass

velocity

How many of these quantities are vectors?

A 2 **B** 3 **C** 4 **D** 5

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 1

1	C	**1**

QUESTION · 5054_s21_qp_11.pdf · source page 2 of 16

2 A student determines the circumference of a football.

Which instrument gives a reading that is the circumference of the football?

A calipers

B micrometer

C rule

D tape

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 2

2	D	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 An aircraft flies at a constant height.

Air drag and the force from the aircraft's engines together produce a force on the aircraft of 36 kN due north, as shown in Fig. 1.1.

The wind produces a force of 12 kN towards the east.

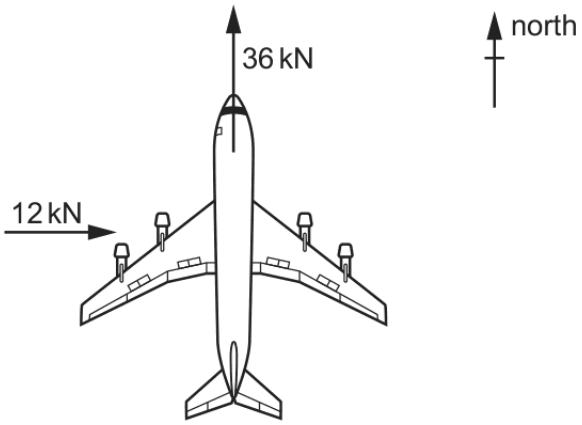


Fig. 1.1 (not to scale)

(a) Draw a scale drawing to show the resultant force acting on the aircraft.

Use your drawing to determine the size of the resultant force and the angle between the resultant force and north.

size of resultant force = ...

angle = ...
[3]

(b) The acceleration of the aircraft is uniform.

(i) Describe how a uniform acceleration differs from a non-uniform acceleration.

.....

.....

.....

..... [2]

(ii) The mass of the aircraft is 60 000 kg.

Calculate the acceleration of the aircraft.

acceleration = ... [2]

[Total: 7]

Question	Answer	Marks
1(a)	scale vector diagram showing 12 N, 36 N and correct resultant force in correct direction	**B1**
	38 ± 1 kN	**B1**
	018 ± 3°	**B1**
1(b)(i)	(only in uniform acceleration) same increase / change in speed / velocity **or** has a constant acceleration **or** speed-time graph is a straight line	**B1**
	(only in uniform acceleration) same increase / change in speed / velocity in same time (period) **or** same rate of change of speed / velocity (with time) **or** change in speed / velocity is proportional to time	**B1**
1(b)(ii)	(a =) F / m in any way, algebraic or numerical	**C1**
	0.63–0.66 m / s^2	**A1**

QUESTION · 5054_s21_qp_12.pdf · source page 2 of 16

1 A student determines the circumference of a football.

Which instrument gives a reading that is the circumference of the football?

A calipers

B micrometer

C rule

D tape

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 1

1	D	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 **(a)** **(i)** State the difference between a scalar quantity and a vector quantity.

.....

.....

..... [1]

(ii) State **two** examples of each type of quantity.

scalar quantity	**vector quantity**
1	1 .....
2	2 .....

[2]

(b) Fig. 1.1 shows the direction and size of two vectors P and Q.

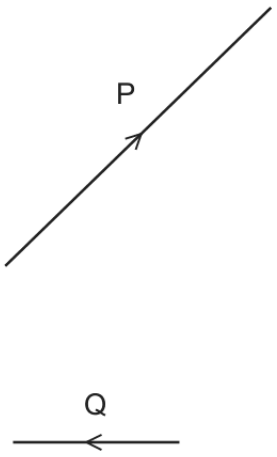


Fig. 1.1

In the space next to Fig. 1.1, draw a labelled vector diagram to show the resultant vector obtained by adding vector P to vector Q.

Draw vector P, vector Q and the resultant vector to the same scale as in Fig. 1.1.

[2]

[Total: 5]

Question	Answer	Marks
1(a)(i)	vector quantity has direction / scalar quantity does not have direction	**B1**
1(a)(ii)	any two scalar quantities, e.g. mass, distance / length, time, speed	**B1**
	any two vector quantities, e.g. velocity, acceleration, displacement, force / weight	**B1**
1(b)	P and Q drawn in correct directions to meet at a point with correct size	**B1**
	triangle or parallelogram shown with all three vectors in correct directions **and** the resultant clear (e.g. as another vector)	**B1**

QUESTION · 5054_w21_qp_11.pdf · source page 2 of 20

1 Which quantity is a vector?

A speed

B force

C mass

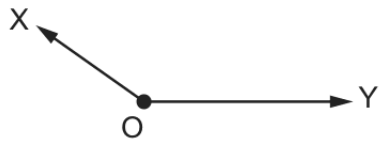
D distance

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 2 of 20

2 Two forces, X and Y, act upon an object O. The arrows represent the magnitudes and directions of the forces.



Which arrow shows the direction of the resultant force?

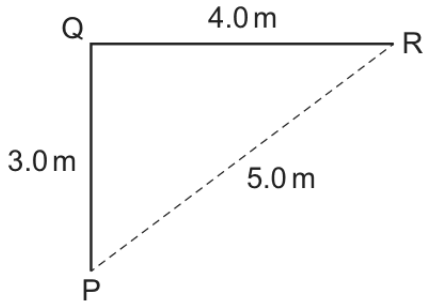
A **B** **C** **D**

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 2

2	A	**1**

QUESTION · 5054_w21_qp_12.pdf · source page 2 of 16

1 A boy starts at P and walks 3.0 m due north from P to Q and then 4.0 m due east from Q to R.



What is the shortest distance that he must now walk to have an overall displacement of zero?

A 3.0 m **B** 4.0 m **C** 5.0 m **D** 7.0 m

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 1

1	C	**1**

QUESTION · 5054_w21_qp_12.pdf · source page 2 of 16

2 A student investigates the motion of a ball falling through the air.

Which quantity is a vector?

A the diameter of the ball

B the gravitational force on the ball

C the distance from which the ball is dropped

D the speed at which the ball hits the ground

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 2

2	B	**1**

2 Force is a vector quantity.

(a) State what is meant by a *vector*.

.....

..... [1]

(b) A swimmer reaches the end wall of a swimming pool and turns around under the water.

Fig. 2.1 shows the swimmer immediately after turning around.

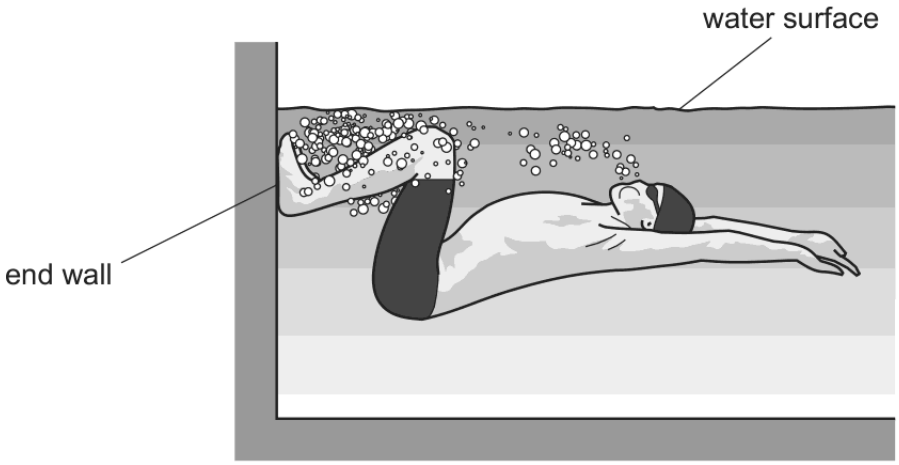


Fig. 2.1

(i) The swimmer pushes against the end wall of the pool with his legs.

Explain, in terms of Newton's third law, why the swimmer accelerates away from the end wall.

.....

.....

.....

..... [3]

(ii) While swimming, there is a constant forward force on the swimmer. His speed increases until eventually he reaches a constant speed.

Explain why he reaches a constant speed.

.....

.....

.....

..... [3]

[Total: 7]

MARK SCHEME · 5054_w21_ms_22.pdf · source page 5 of 11 · Question 2

Question	Answer	Marks
2(a)	(a vector quantity has) magnitude **and** direction	**B1**
2(b)(i)	the wall exerts a force on swimmer	**B1**
	opposite in direction (to force of swimmer on wall) **or** to the right	**B1**
	equal in size (to force of swimmer on wall) **or** (resultant) force causes acceleration **or** force on object Y due to X is equal and opposite to force on object X due to Y	**B1**

MARK SCHEME · 5054_w21_ms_22.pdf · source page 6 of 11 · Question 2

Question	Answer	Marks
2(b)(ii)	(water) resistance / resistive force / friction (on swimmer)	**B1**
	(water) resistance / backward force / resistive force / friction / opposing force increases (with speed / time)	**B1**
	(at constant speed) forward force is equal to resistive / backward force **or** no resultant force **or** forces balance	**B1**

QUESTION · 5054_s22_qp_11.pdf · source page 2 of 16

1 A length of copper wire is labelled: length 0.50 m and diameter 0.50 mm.

Which instruments are most suitable to measure accurately the length and the diameter of the wire?

	length	diameter
A	metre rule	metre rule
B	metre rule	micrometer
C	calipers	metre rule
D	calipers	micrometer

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 Fig. 7.1 shows a toy helicopter. It can hover and travel through the air.

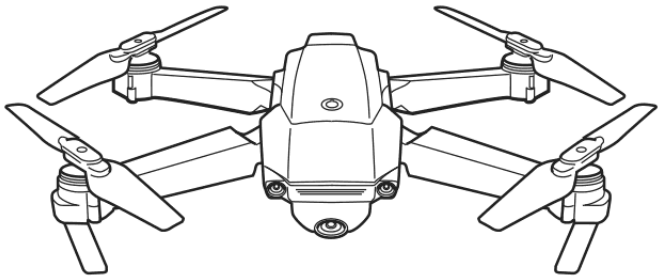


Fig. 7.1

A student flies the toy helicopter on a journey from A to B to C to D at a constant height.

Fig. 7.2 is a scale drawing of the path of the helicopter, viewed from above.

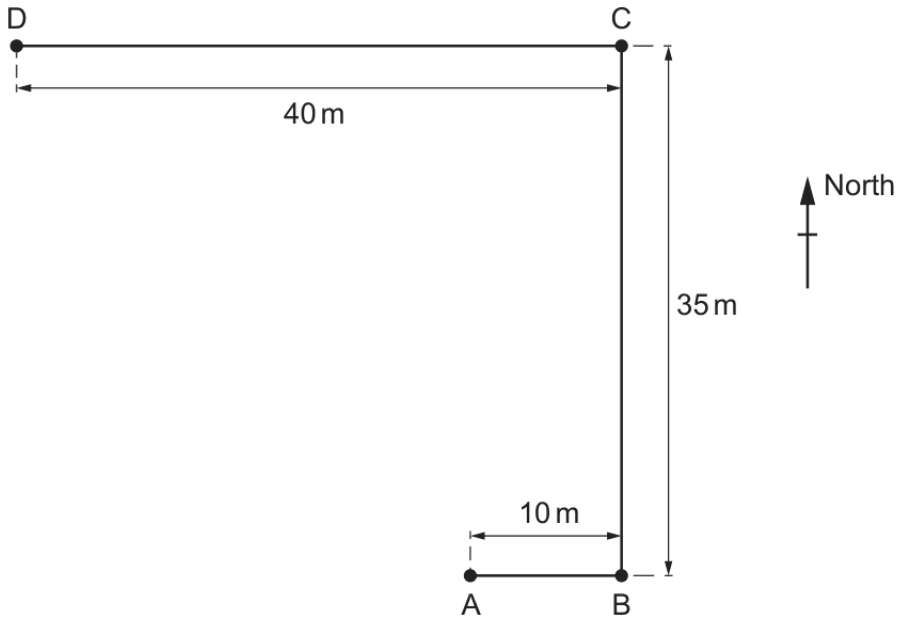


Fig. 7.2 (to scale)

(a) (i) Determine the total distance travelled by the toy helicopter.

..... [1]

(ii) The toy helicopter makes the journey in 40 s.

Calculate its average speed during the journey.

speed = ... [2]

(b) (i) State the difference between distance and displacement.

.....

..... [1]

(ii) Determine the scale used to construct Fig. 7.2 and complete the sentence.

1 cm on the diagram represents ..... on the path. [1]

(iii) Using Fig. 7.2, determine the size of the displacement of point D from point A.

Show your working.

size of displacement = ... [2]

(iv) Determine the angle between North and the direction of the displacement of point D from point A.

angle = ... [1]

(v) State what is meant by velocity.

..... [1]

(vi) Another toy helicopter flies directly from point A to point D in 40 s.

Explain why the magnitude of the velocity of this toy helicopter is smaller than the answer in **(a)(ii)**.

.....

..... [1]

(c) When the toy helicopter hovers at D, its motor fails and it falls. It reaches terminal velocity as it falls.

Explain, in terms of the forces and acceleration, what happens as the helicopter falls and reaches terminal velocity.

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 15]

Question 8 begins over the page.

MARK SCHEME · 5054_s22_ms_21.pdf · source page 8 of 11 · Question 7

Question	Answer	Marks
7(a)(i)	85 m	**B1**
7(a)(ii)	(speed =) d / t in any form algebraic or numerical	**C1**
	2.1 m / s	**A1**
7(b)(i)	displacement involves direction **or** is a vector	**B1**
7(b)(ii)	5 m shown on dotted line	**B1**

MARK SCHEME · 5054_s22_ms_21.pdf · source page 9 of 11 · Question 7

Question	Answer		Marks
7(b)(iii)	length of diagonal 9.2 (cm)	$\sqrt{30 \times 30 + 35 \times 35}$	**B1**
	46 m	46 m	**B1**
7(b)(iv)	39–42° **or** 318–320°		**B1**
7(b)(v)	(average) velocity = (change in) displacement / time **or** (average) velocity is displacement per unit time **or** rate of change of displacement		**B1**
7(b)(vi)	(size of) displacement < distance in **(a)(ii)** or distance covered is less (than 85 m / in **(a)(ii)**)		**B1**
7(c)	air resistance **and** weight / gravitational force mentioned		**B1**
	as speed increases air resistance increases		**B1**
	<u>resultant / net</u> force decreases **or** acceleration decreases		**B1**
	(eventually) air resistance = weight (at terminal velocity) / resultant force is zero		**B1**
	(at terminal velocity / eventually) zero acceleration / constant speed		**B1**

QUESTION · 5054_s22_qp_12.pdf · source page 2 of 20

1 A force of 3.0 N and a force of 4.0 N act on an object.

What is the maximum possible resultant of these two forces?

A 1.0 N **B** 5.0 N **C** 7.0 N **D** 12 N

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 1

1	C	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 2 of 20

2 A length of copper wire is labelled: length 0.50 m and diameter 0.50 mm.

Which instruments are most suitable to measure accurately the length and the diameter of the wire?

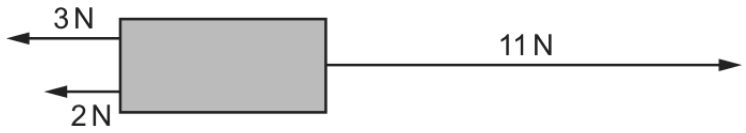
	length	diameter
A	metre rule	metre rule
B	metre rule	micrometer
C	calipers	metre rule
D	calipers	micrometer

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 2

2	B	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 2 of 16

1 The diagram shows three forces acting on a block. The resultant force is 6 N to the right.



Which additional force produces a resultant force of 3 N to the left?

A 3 N to the left

B 9 N to the left

C 6 N to the right

D 13 N to the right

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 2 of 16

2 Velocity is given by the change in displacement divided by the change in time.

How many vector quantities appear in this statement?

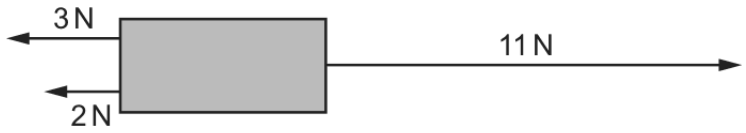
A 0 **B** 1 **C** 2 **D** 3

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 2

2	C	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 2 of 16

1 The diagram shows three forces acting on a block. The resultant force is 6 N to the right.



Which additional force produces a resultant force of 3 N to the left?

A 3 N to the left

B 9 N to the left

C 6 N to the right

D 13 N to the right

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 2 of 16

2 A teacher measures the length of her classroom.

What is the most appropriate instrument to use?

A a 30 cm ruler

B a caliper

C a micrometer

D a tape

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 2

2	D	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 2 of 16

3 Which value is one-thousandth of a metre?

A 0.0001 cm **B** 0.001 cm **C** 0.01 cm **D** 0.1 cm

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

QUESTION · 5054_s23_qp_11.pdf · source page 2 of 20

1 A student wishes to measure directly the circumference of a football.

What is the most suitable instrument to use?

A a clock only

B a measuring tape only

C a micrometer only

D a ruler only

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

QUESTION · 5054_s23_qp_11.pdf · source page 2 of 20

2 Which equation contains two vector quantities?

A $\text{acceleration} = \dfrac{\text{change in velocity}}{\text{time taken}}$

B $\text{average speed} = \dfrac{\text{distance travelled}}{\text{time taken}}$

C $\text{density} = \dfrac{\text{mass}}{\text{volume}}$

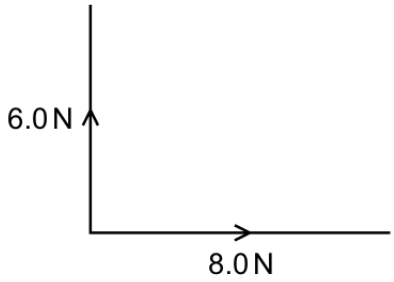
D $\text{volume} = \text{length} \times \text{width} \times \text{height}$

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 2

2	A	**1**

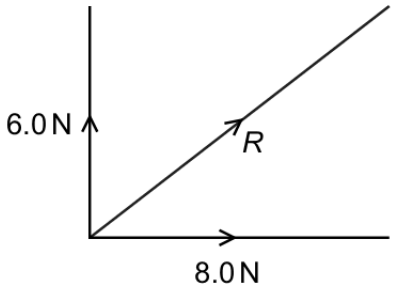
QUESTION · 5054_s23_qp_11.pdf · source page 3 of 20

3 Forces of 6.0 N and 8.0 N act as shown.

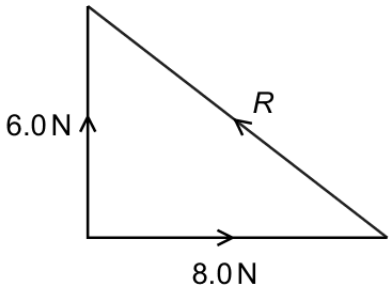


Which diagram shows the resultant *R* of these two forces?

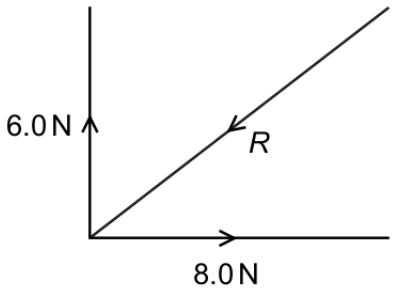
A



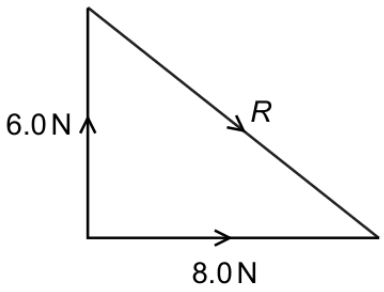
B



C



D

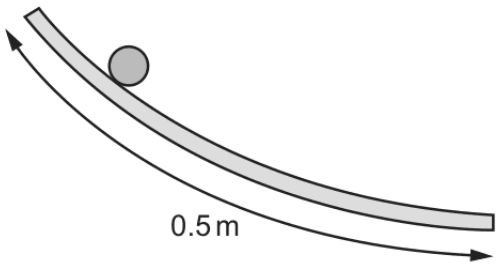


MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 3

3	A	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 2 of 16

1 In an experiment, a ball is rolled down a curved track that is about half a metre long.



Which measuring device is used to measure the length accurately?

A metre rule

B micrometer

C stop-watch

D tape measure

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 1

1	D	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 2 of 16

2 Which row describes acceleration, displacement, distance and speed?

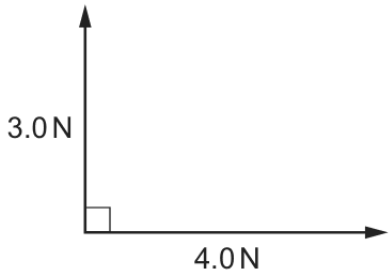
	acceleration	displacement	distance	speed
A	scalar	scalar	vector	scalar
B	scalar	vector	scalar	vector
C	vector	scalar	vector	vector
D	vector	vector	scalar	scalar

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 2

2	D	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 2 of 16

3 What is the size of the resultant of the two forces shown?



A 1.0 N **B** 3.5 N **C** 5.0 N **D** 7.0 N

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 3

3	C	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 2 of 16

1 Which description is of a scalar quantity?

A It has a direction only.

B It has direction and unit only.

C It has direction, magnitude and unit.

D It has magnitude and unit only.

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 1

1	D	**1**

9 The planet Venus orbits the Sun at a constant speed of 3.5×10^4 m/s and takes a time T_V to complete one orbit.

(a) Venus is always 1.1×10^{11} m from the Sun.

Calculate T_V.

T_V = ... s [2]

(b) Explain why speed is a scalar quantity.

.....

..... [1]

(c) Velocity is a vector quantity.

(i) Explain what happens to the velocity of Venus as it orbits the Sun.

.....

.....

.....

..... [2]

(ii) There is a resultant force on Venus as it orbits the Sun. The force is perpendicular to the direction of the motion of Venus.

State why this force is needed.

.....

..... [1]

(iii) Fig. 9.1 shows the orbit of Venus around the Sun.

On Fig. 9.1, draw an arrow to show the resultant force on Venus.

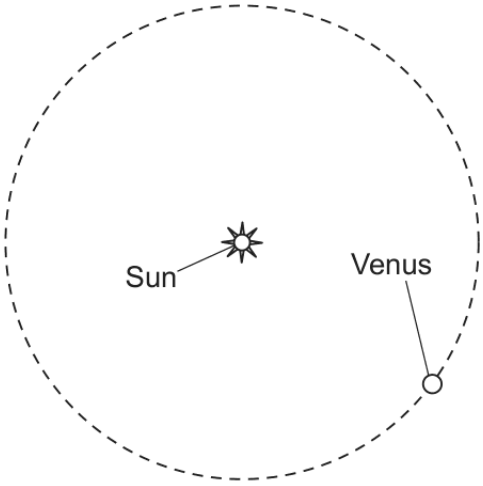


Fig. 9.1

[1]

(iv) State what causes the force acting on Venus.

.....

.....

..... [2]

(d) One planet in the Solar System is closer to the Sun than Venus.

(i) State the name of this planet.

..... [1]

(ii) Compare the time that this planet takes to complete one orbit of the Sun with T_V and explain the difference.

.....

.....

.....

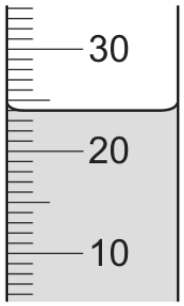
..... [2]

[Total: 12]

Question	Answer	Marks
9(a)	(T =) $2\pi r / v$ **or** $2\pi \times 1.1 \times 10^{11} / 3.5 \times 10^{4}$	**C1**
	2.0×10^{7} (s)	**A1**
9(b)	it / speed does not have a direction	**B1**
9(c)(i)	it / velocity changes	**M1**
	(because) its direction changes	**A1**
9(c)(ii)	to change the velocity **or** to keep Venus in a circular orbit **or** to accelerate Venus towards the centre of the orbit / Sun	**B1**
9(c)(iii)	arrow towards Sun **and** on Venus	**B1**
9(c)(iv)	it / Venus is in the gravitational field **or** the gravitational attraction	**M1**
	of the Sun	**A1**
9(d)(i)	Mercury	**B1**
9(d)(ii)	time taken is less **and** orbital speed is greater / circumference of orbit is less / closer to the Sun	**B1**
	circumference of orbit is less **and** orbital speed is greater.	**B1**

QUESTION · 5054_w23_qp_12.pdf · source page 2 of 20

1 The diagram shows the level of liquid in a measuring cylinder. The scale is marked in cm^3.



What is the volume of the liquid?

A $20.4\,cm^3$ **B** $24.0\,cm^3$ **C** $24.5\,cm^3$ **D** $25.0\,cm^3$

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

1 Velocity is a vector quantity but speed is a scalar quantity.

(a) State how a vector quantity differs from a scalar quantity.

.....

..... [1]

(b) Underline **all** the vector quantities in the list.

acceleration distance energy mass momentum temperature [1]

(c) Fig. 1.1 shows a passenger running across a large ship at right angles to the direction in which the ship is moving.

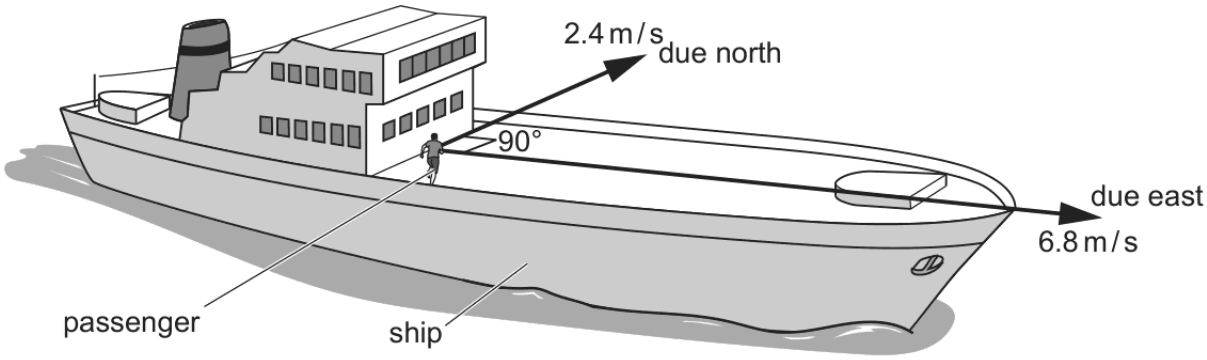


Fig. 1.1 (not to scale)

The passenger runs across the ship at 2.4 m/s due north.

The large ship is travelling due east at 6.8 m/s.

Determine, either graphically or by calculation, the resultant velocity of the passenger. Give the direction in which the passenger is moving relative to due north.

Show how the answer is obtained.

velocity = ... m/s at ..... ° from due north [4]

[Total: 6]

Question	Answer	Marks
1(a)	(a vector quantity) / it has a direction	**B1**
1(b)	**acceleration** and **momentum** underlined **and** no others	**B1**
1(c)	**speed:** (v =) $\sqrt{6.82 + 2.42}$ **or** $v^2 = 6.8^2 + 2.4^2$ **or** correct triangle / rectangle drawn / intersecting arcs (by eye)	**B1**
	speed: 7.1 – 7.3 (m / s)	**B1**
	angle: (θ =) $\tan^{-1}(6.8 / 2.4)$ **or** $\tan\theta = 6.8 / 2.4$ **or** 18 – 21(°) **or** **or** correct triangle / rectangle drawn / intersecting arcs **and** scale used $\geqslant$ 1 cm: 1 m / s	**B1**
	angle: 69 – 72(°)	**B1**

QUESTION · 5054_s24_qp_11.pdf · source page 2 of 16

2 Which quantity is a vector?

A distance

B force

C mass

D speed

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 2

2	B	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 2 of 16

1 What is measured using a micrometer?

A area

B current

C length

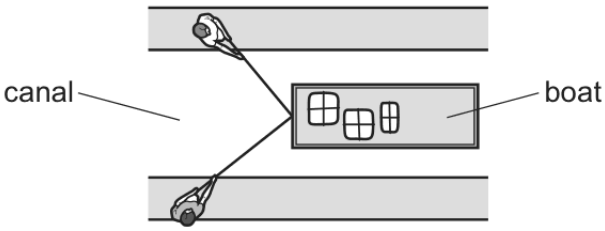
D mass

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 1

1	C	**1**

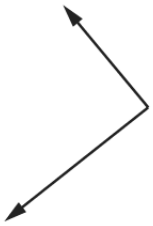
QUESTION · 5054_s24_qp_12.pdf · source page 2 of 16

2 Two people pull on ropes to move a boat along a canal.



The boat moves at a constant velocity.

The vector diagram for the tension in the ropes is shown.



Which vector is the drag acting on the boat?

A **B** **C** **D**

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 2

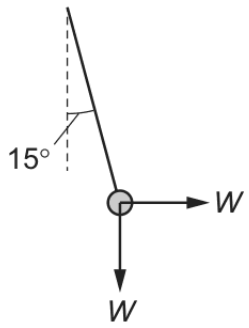
2	A	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 2 of 16

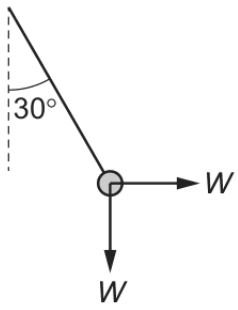
3 The bob of a simple pendulum is pulled horizontally by a force that is equal to weight *W* of the bob.

Which diagram shows the angle between the pendulum thread and the vertical?

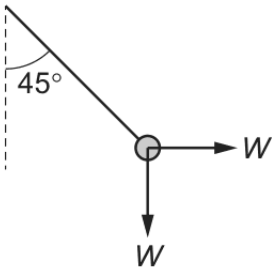
A



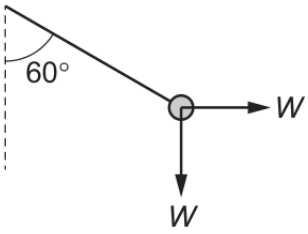
B



C



D

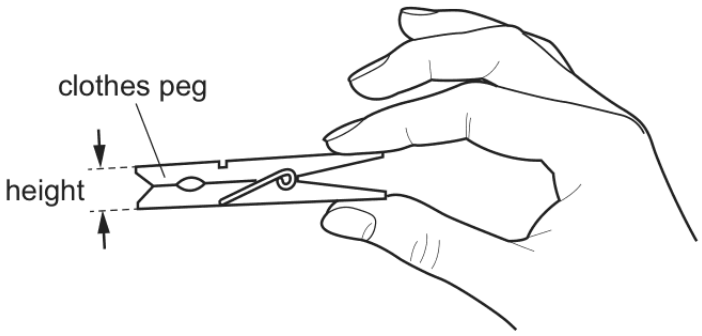


MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 3

3	C	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 2 of 16

1 The diagram shows a clothes peg.



Which device is suitable for obtaining an accurate value for the height of the clothes peg?

A a tape

B a metre rule

C a micrometer

D a measuring cylinder

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 1

1	C	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 2 of 16

2 Which set of quantities are all vectors?

A acceleration, displacement, velocity

B chemical energy, mass, power

C extension, force, gravitational potential energy

D weight, kinetic energy, work

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 2

2	A	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 2 of 20

1 Which quantity has both magnitude and direction, and is it a scalar quantity or a vector quantity?

	quantity with magnitude and direction	scalar quantity or vector quantity
A	mass	scalar
B	mass	vector
C	weight	scalar
D	weight	vector

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 1

1	D	**1**

1 Fig. 1.1 shows a skydiver falling vertically through the air.

Fig. 1.1

In the first part of the fall, her speed increases and her acceleration decreases.

In the second part of the fall, her speed is constant.

(a) On Fig. 1.2 sketch the speed–time graph for the skydiver.

On your graph, mark **one** point where the speed of the skydiver is increasing with an A and **one** point where the speed of the skydiver is constant with a B.

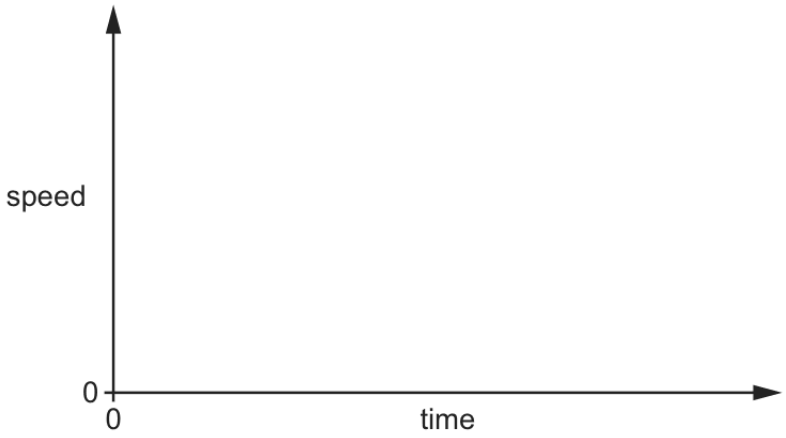


Fig. 1.2

[2]

(b) Explain how the graph shows that the acceleration decreases as the speed increases.

.....

.....

..... [1]

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(c) During the first part of the fall, there is a resultant vertical force acting downwards on the skydiver.

(i) One of the vertical forces acting on the skydiver is her weight.

State the name of the other vertical force that acts on the skydiver.

..... [1]

(ii) Explain why the resultant vertical force eventually becomes zero.

.....

.....

.....

..... [2]

(d) At one instant, the vertical force on the skydiver is 400 N downwards.

At the same instant, the wind causes an additional horizontal force of 100 N to the right to act on the skydiver.

Draw a vector diagram to determine the resultant of the 400 N vertical force and the 100 N horizontal force. Place an arrow on all of the forces to show their directions.

Determine the magnitude (size) of this resultant force and its direction to the vertical.

magnitude of resultant = ... N

direction = ...° to vertical
[3]

[Total: 9]

Question	Answer	Marks
1(a)	curve upwards (at start) with decreasing gradient labelled A	**B1**
	horizontal line labelled B	**B1**
1(b)	gradient / slope decreases **or** increase in speed for same time less at later times	**B1**
1(c)(i)	air resistance / drag	**B1**
1(c)(ii)	air resistance / upwards force increases (with increased speed)	**B1**
	air resistance / upwards force equals / balances weight	**B1**
1(d)	vector diagram showing 400 N and 100 N force at right angles with correct resultant and all directions shown	**B1**
	410 (N)	**B1**
	12–16°	**B1**

QUESTION · 5054_s25_qp_12.pdf · source page 2 of 16

1 Which list contains **only** vector quantities?

A mass, distance, weight

B momentum, gravitational field strength, displacement

C energy, force, velocity

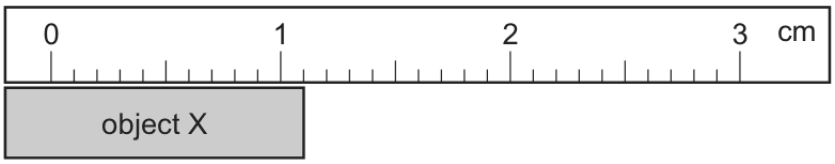
D acceleration, time, speed

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 1

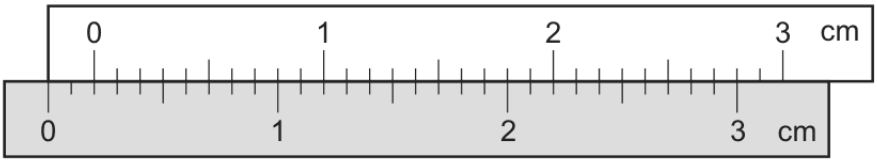
1	B	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 2 of 16

1 Object X is placed against a ruler.



This ruler is placed against another ruler to measure the end correction.



What is the length of the object X?

A 0.9 cm **B** 1.1 cm **C** 1.3 cm **D** 1.5 cm

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 1

1	C	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 2 of 16

2 Quantities can be described as scalar or vector.

Which row shows scalar quantities and vector quantities?

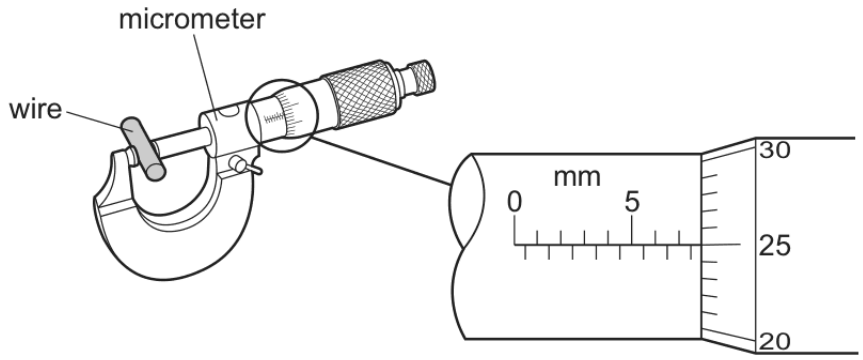
	scalar quantities	vector quantities
A	displacement, force, temperature	distance, weight, mass
B	displacement, mass, force	distance, weight, temperature
C	distance, force, weight	displacement, mass, temperature
D	distance, mass, temperature	displacement, force, weight

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 2

2	D	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 2 of 16

1 A micrometer is used to measure the diameter of a piece of wire.



What is the reading shown on the micrometer?

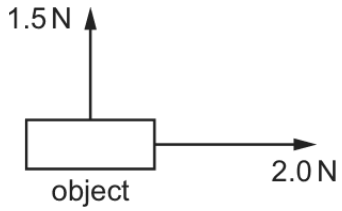
A 7.25 mm **B** 7.75 mm **C** 10.25 mm **D** 25.77 mm

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 2 of 16

2 Two forces act on an object. 1.5 N acts vertically upwards and 2.0 N acts horizontally.



What is the magnitude of the resultant force?

A 0.50 N **B** 2.5 N **C** 3.0 N **D** 3.5 N

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 2

2	B	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 2 of 16

1 A car travels along a road at x, where x is 15 m / s due north.

Which term describes x?

A acceleration

B scalar

C speed

D vector

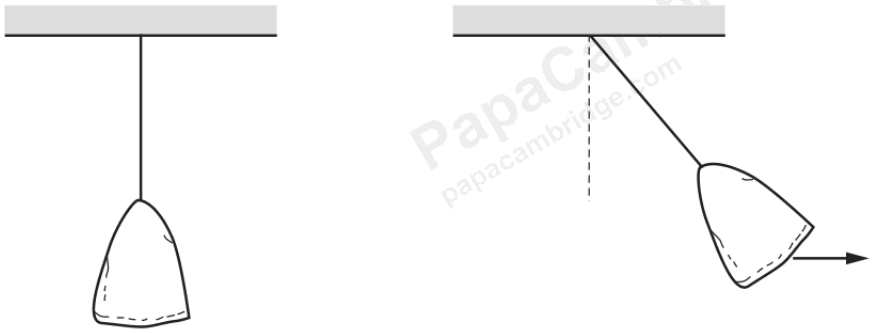
MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 1

1	D	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 2 of 16

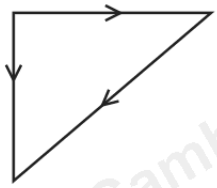
2 A heavy sack is suspended from a ceiling by a rope.

The sack is then pulled to one side and held stationary.

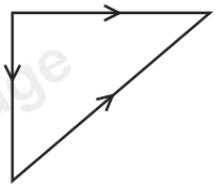


Which vector diagram represents the forces acting on the sack in its new position?

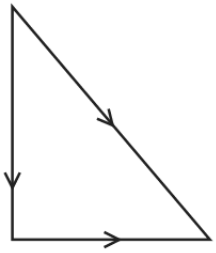
A



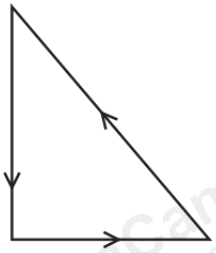
B



C



D



MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 2

2	D	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 2 of 16

1 What is the most accurate method to determine the period of an oscillating pendulum?

A Use a digital timer to measure the time for 1 period. Repeat this 10 times and calculate the average.

B Count the number of complete oscillations in 10 seconds. Divide by 10.

C Use a digital timer to measure the time for 1 period.

D Use a digital timer to measure the time for 10 complete oscillations. Divide by 10.

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 1

1	D	**1**

Chapter 1 Motion, forces and energy

1.2 Motion

176 entries: 100 Paper 1 questions, 76 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 2 of 20

2 Which quantity X is calculated using this equation?

$$X = \frac{\text{change in velocity}}{\text{time taken}}$$

A acceleration

B average velocity

C distance travelled

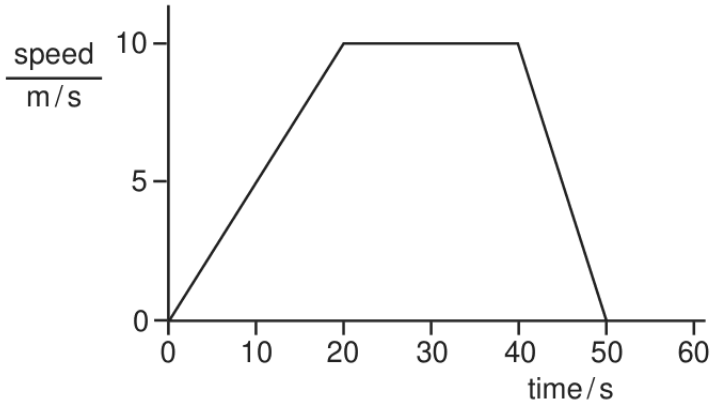
D speed

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 2

2	**A**

QUESTION · 5054_s04_qp_1.pdf · source page 2 of 20

3 The graph shows the movement of a car over a period of 50 s.



What was the distance travelled by the car while its speed was increasing?

A 10 m **B** 20 m **C** 100 m **D** 200 m

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3 **C**

For Examiner's Use

4 Fig. 4.1 shows a student clapping in front of a vertical wall. The wall reflects the sound.

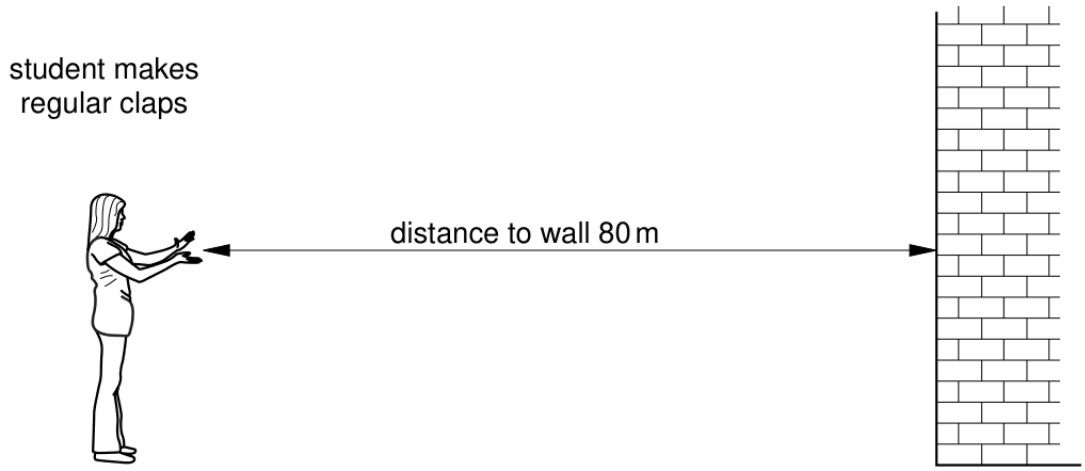


Fig. 4.1

The student changes the number of claps made in 1 minute until the reflection of each clap returns to her at exactly the same time as she makes the next clap.

The speed of sound in air is 330 m/s.

(a) Explain what is meant by *speed*.

.....

.....[1]

(b) Calculate the time between claps.

time = …….. [3]

(c) Calculate the number of claps in 1 minute.

number of claps = …….. [2]

4	**(a)**	distance traveled per unit time **or** in one second / distance ÷ time		
		or rate of change of distance		B1
	(b)	s = d/t in any algebraic or numerical form		C1
		any doubling of distance or final time		C1
		0.48 s	(allow 0.24s 2/3 accept 0.5s)	A1
	(c)	60/0.48 (5)		C1
		123.75 accept 120, 123, 124	(ecf **(b)**)	A1
			Total	**[6]**

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows a simplified speed-time graph for a train that travels between two stations.

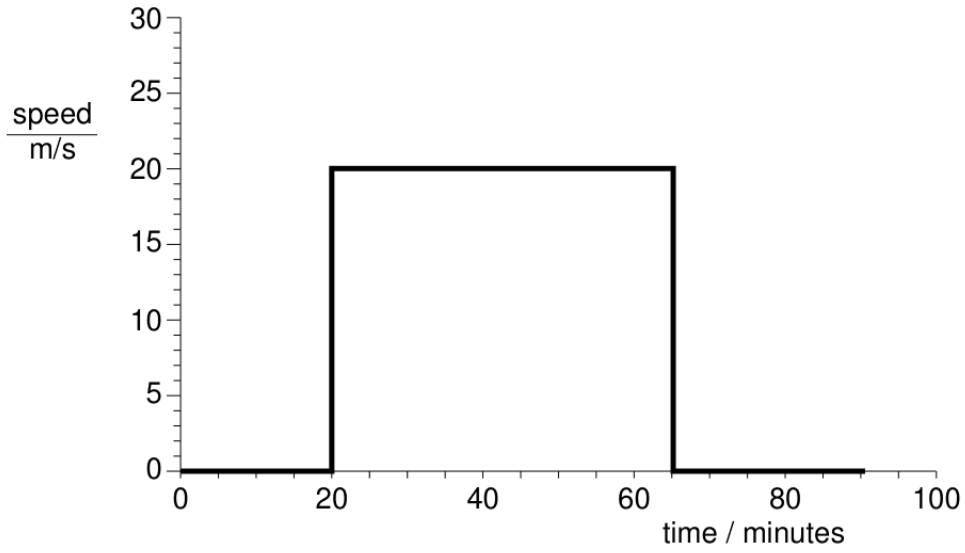


Fig. 1.1

(a) Describe the motion of the train.

.....

.....

.....

.....

.....[2]

(b) Calculate the distance travelled by the train between the two stations.

distance = …….. [2]

(c) Another train travels between the same two stations on a parallel track. This train travels at a constant speed. It starts its journey at time $t = 0$ and finishes at $t = 90$ minutes.

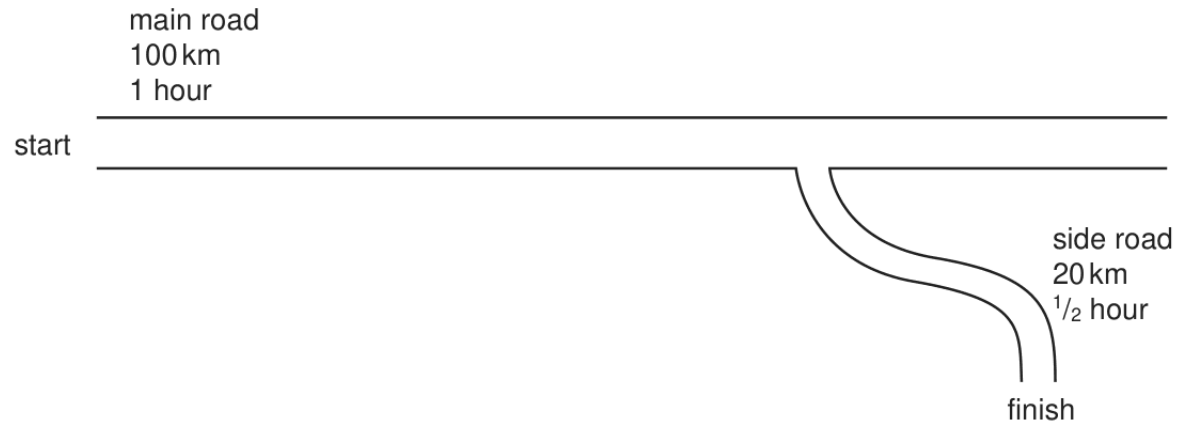
On Fig. 1.1, draw the speed-time graph for this train. [2]

Section A

1	**(a)**	**speed** uniform or 20 m/s	**B1**	
		stationary/not moving till 20 minutes or after 65 minutes **or** moves for 45 minutes	**B1**	
		(**not** if inconsistent; all times ±2 min; **ignore** acceleration/deceleration periods)		
	(b)	d = st any algebraic **or** area calculated		
		or 20 x 45, 20 x 90, 20 x 45 x 60, 20 x 90 x 60	**C1**	
		54 000 m **or** 54 km	**A1**	
	(c)	any constant speed from **0 to 90** minutes (may stop at 90 or go down to axis)	**M1**	
		10 m/s (no e.c.f. b)	**A1**	**6**

QUESTION · 5054_s05_qp_1.pdf · source page 2 of 20

3 A car takes 1 hour to travel 100 km along a main road and then ½ hour to travel 20 km along a side road.



What is the average speed of the car for the whole journey?

A 60 km / h

B 70 km / h

C 80 km / h

D 100 km / h

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3	**C**

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 represents the motion of Earth and the planet Venus around the Sun. The orbits shown are circles.

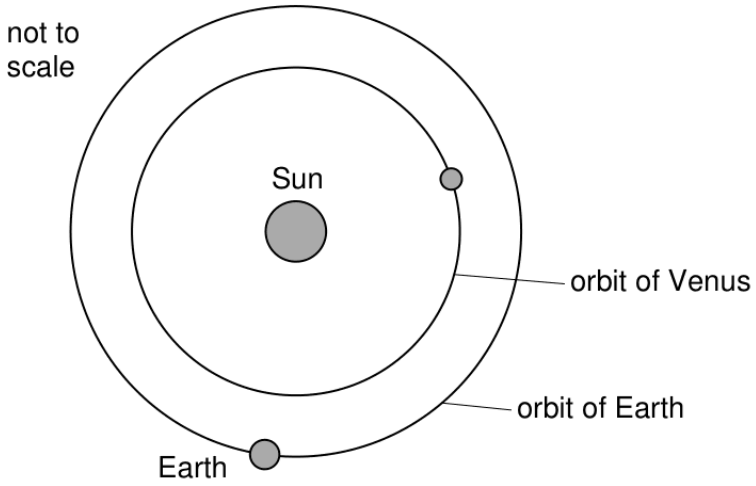


Fig. 1.1

(a) On Fig. 1.1, draw an arrow to show the direction of the force exerted by the Sun on the Earth. [1]

(b) Information about Earth and Venus is given in the table.

planet	time for one orbit in (Earth) years	radius of orbit / million km	circumference of orbit / million km
Venus	0.7	108	679
Earth	1.0	150	942

(i) Use the information in the table to show that Venus has a greater speed than Earth.

.....

.....

.....

..... [2]

(ii) As Earth and Venus move in their orbits, the distance between them changes. Calculate the largest possible distance between them.

distance = ... million km [1]

Section A

1	**(a)**	arrow from Earth to Sun (by eye would pass through Sun)	**B1**	
	(b) (i)	use of circumference/time **or** s=d/t **or** radius/t	**C1**	
		two speeds **clearly** found using circumference e.g. 970 and 942 (allow conversion to other units)	**A1**	
	(ii)	258 (million km)	**B1**	**4**

Section B

Answer **two** questions from this section.

Write your answers on the separate answer paper provided.

9 A train travels from one station to the next. It starts from rest at time $t = 0$ and accelerates uniformly for the first 20 s. At $t = 20$ s it reaches its top speed of 25 m/s. It then travels at this speed for a further 30 s before decelerating uniformly to rest. The total time for the journey is 60 s.

(a) **(i)** Sketch a speed-time graph for the motion of the train. Do **not** use graph paper. Put the speed of the train on the *y*-axis and time along the *x*-axis. [3]

(ii) Write down the equation, **in words**, that relates acceleration, time and change in velocity. [1]

(iii) Explain what is meant by a *uniform* acceleration. [1]

(iv) Use your graph to calculate the deceleration of the train as it comes to rest. [2]

(b) Several forces act on the train when it is moving.

(i) Name the horizontal and vertical forces that act on the train and give the direction of each force. [4]

(ii) Explain whether the horizontal forces are balanced or unbalanced,

1. when the train accelerates,

2. when the train travels at constant speed,

3. when the train decelerates.

You may draw diagrams to help your explanations. [3]

(c) A second train has a non-uniform acceleration. Sketch a speed-time graph showing a non-uniform acceleration. Do **not** use graph paper. [1]

Section B

9 **(a) (i)** y axis labelled speed or m/s **and** x axis labelled time or s **B1**
straight line from 0,0 to t = 20, speed = 25 **B1**
uniform speed from t = 20 to 50 **and** uniform deceleration from t = 50 to 60 **B1**

(ii) acceleration = change in velocity/time or per unit time
or rate of change of velocity with time
accept equation but must be written in words or defined symbols **B1**

(iii) constant increase in speed/velocity in 1sec/ /same time interval
or rate of change of speed/velocity constant **or** Δv proportional to time
or acceleration constant **with time** **B1**

(iv) 25/10 e.c.f. time interval from graph **C1**
2.5 m/s^2 accept -ve **A1**

(b) (i) weight/gravitational force (accept gravity) **downwards**
normal/reaction/contact force/force from ground **upwards**
air resistance/drag **or** friction (due to air) **backwards or** opposite to train (direction)
braking force **or** friction **or** resistive force **backwards or** same direction as air drag
tractive **or** thrust **or** driving force **or** force of engine **forwards**
ANY 4 **B4**
accept from diagram (-1 each wrong force more than 4)

(ii) 1. unbalanced **since** forward force > backwards force **or** resultant/net **forward** force **B1**
2. balanced **since** forward force = backwards force **or** forces cancel **or** zero resultant **B1**
3. unbalanced **since** backwards force > forwards force **or** only backwards force **or** resultant/net backwards force **B1**
accept sizes of forces from lengths of arrows on diagram

(c) sketch graph with **axes labelled** and non straight line **B1**

QUESTION · 5054_w05_qp_1.pdf · source page 3 of 20

3 What **must** change when a body is accelerating?

A the force acting on the body

B the mass of the body

C the speed of the body

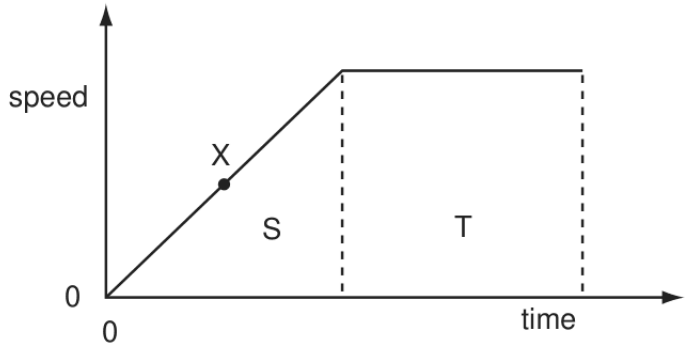
D the velocity of the body

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 3

3	**D**

QUESTION · 5054_w05_qp_1.pdf · source page 3 of 20

4 The graph illustrates the motion of an object.



Which feature of the graph represents the distance travelled by the object whilst moving at a constant speed?

A area S

B area S + area T

C area T

D the gradient at point X

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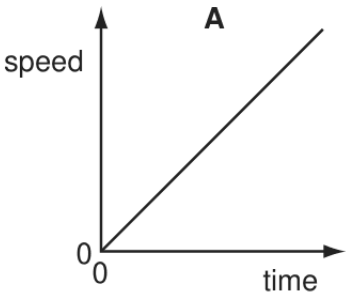
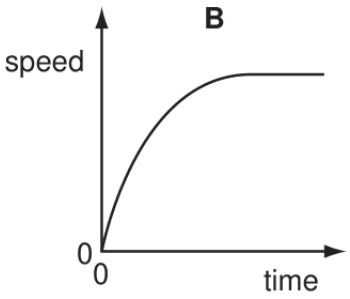
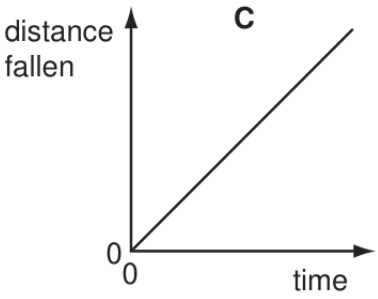
4 **C**

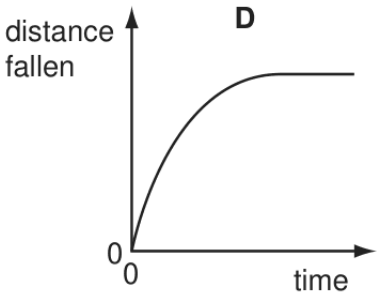
QUESTION · 5054_s06_qp_1.pdf · source page 2 of 20

1 Which graph shows the motion of a heavy, steel ball falling from a height of 2 m?









MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 1

1	**A**

For Examiner's Use

2 A cyclist starts from rest. He accelerates and then travels at a constant speed. At 12 s, the cyclist applies the brakes and slows down. Photographs are taken of the cyclist at 4 s intervals. Fig. 2.1 shows the results.

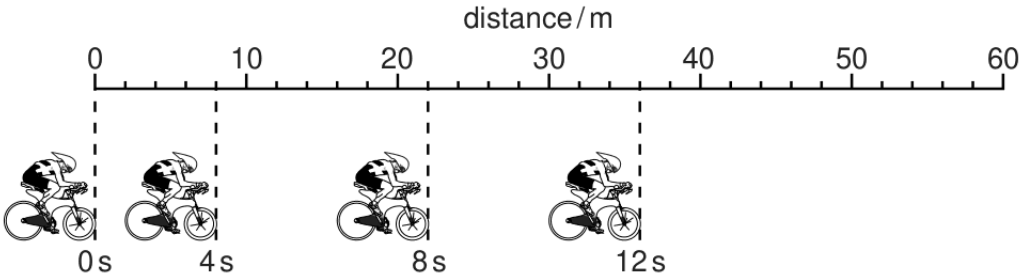


Fig. 2.1

(a) On Fig. 2.1, draw a possible position of the front wheel of the cycle at 16 s. [1]

(b) On Fig. 2.2, plot a distance-time graph of the cyclist for the first 16 s. [4]

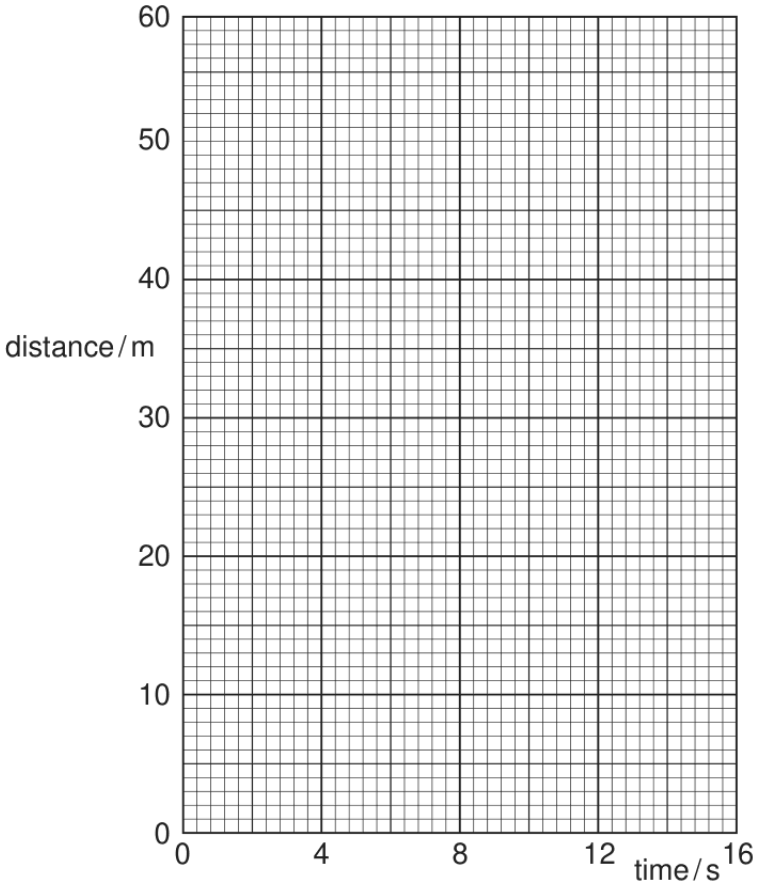


Fig. 2.2

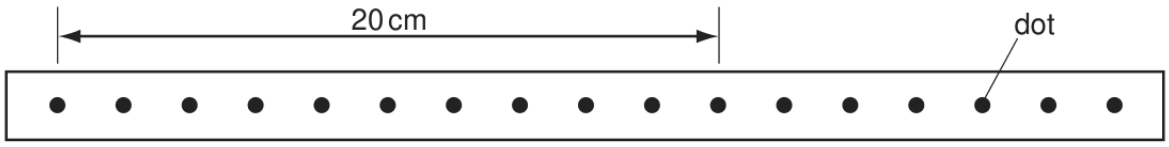
(c) Calculate the average speed of the cyclist during the first 12 s.

average speed = ... [2]

2	any position before 50 m	B1	
(b)	points plotted correctly at 4,8 and 12 s (± ½ square)	B1	
	from origin to 4 sec **curve** drawn	B1	
	from 4 to 12 sec straight line positive gradient	B1	
	from 12 to 16 sec gradient decreases (but not –ve)	B1	
(c)	speed/time	C1	
	3 m/s	A1	**[7]**

QUESTION · 5054_w06_qp_1.pdf · source page 3 of 16

5 The diagram shows a strip of paper tape that has been pulled under a vibrating arm by an object moving at constant speed. The arm was vibrating regularly, making 50 dots per second.



What was the speed of the object?

A 2.0 cm / s **B** 5.0 cm / s **C** 100 cm / s **D** 200 cm / s

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5	**C**

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 represents the motion of a car along a straight road. As the car approaches a small town, it slows down. The car travels at a constant speed from the start of the town to the end of the town. After passing through the town, the car speeds up.

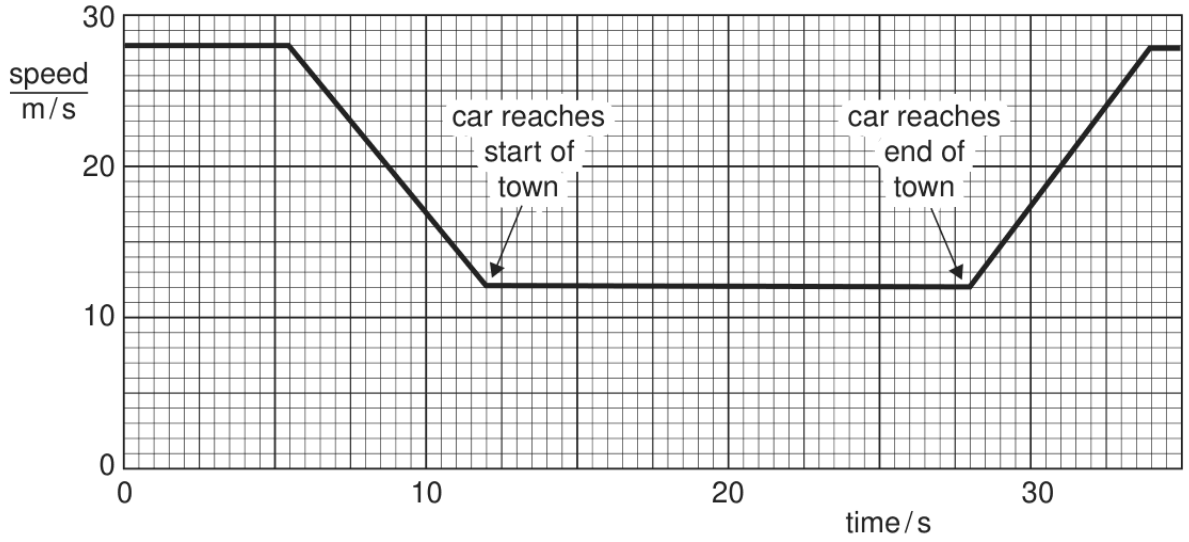

Fig. 1.1

(a) **(i)** Determine the speed of the car in the town.

speed = ... [1]

(ii) Determine the time taken by the car to pass through the town.

time taken = ... [1]

(iii) Calculate the distance travelled by the car in the town.

distance = ... [1]

(b) The car accelerates after passing through the town.

Calculate the acceleration. Give the unit of your answer.

acceleration = ... [3]

Section A

1	**(a)**	**(i)**	12 m/s	B1
		(ii)	16 s	B1
		(iii)	192 m **or** (i) × (ii)	B1
	(b)		a = (v-u)/t in any format e.g. numerical (allow 4 clearly attributable wrong numbers) **or** gradient of v-t/the graph	C1
			2.7 (2/3 sig. fig. only, do not accept fraction, **cao**)	A1
			m/s^2	B1 6

For Examiner's Use

2 Fig. 2.1 shows apparatus that demonstrates how a coin and a piece of paper fall from rest.

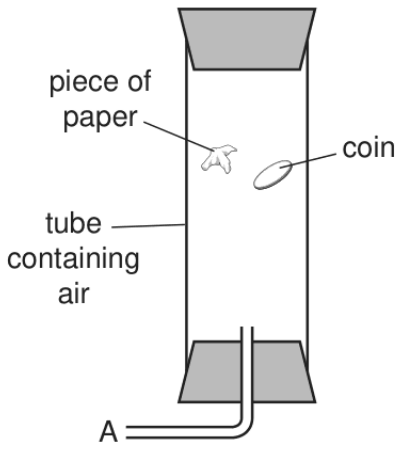

Fig. 2.1

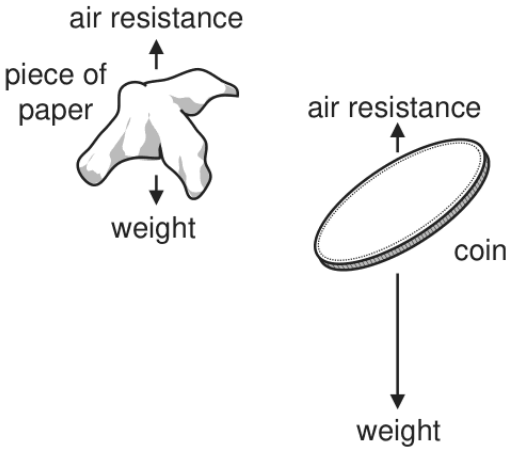

Fig. 2.2

At the positions shown in Fig. 2.1, the coin and paper are falling through air in the tube. The forces on them are shown in Fig. 2.2. The length of an arrow indicates the size of each force.

(a) State the initial value of the acceleration of the coin as it falls.

..... [1]

(b) Explain how Fig. 2.2 shows that

(i) the paper falls with constant speed,

.....

.....

(ii) the coin accelerates.

.....

.....
[3]

(c) A vacuum pump is connected to A and the air in the tube is removed.

The coin and paper fall differently in a vacuum from the way they fall in air.

State two of these differences.

1. .....

.....

2. .....

..... [2]

2	**(a)**		9.8 **or** 10 **or** 9.83 to 9.79. (m/s^2), ignore wrong unit		B1
	(b)	**(i)**	air resistance balances/equals/is same as weight (accept gravity) no resultant force **or** upwards force = downwards force		
		(ii)	weight larger than air resistance (accept gravity) resultant force (down**) or** downwards force greater **or** upwards force less	ANY 3	B3
	(c)		coin and/or paper fall faster **or** hit base sooner coin and/or paper accelerate at g coin falls with paper **or** at same rate **or** same av. speed **or** same acceleration **or** hit bottom together **or** at same time (NOT fall at same speed/same time)	ANY 2 lines	B2 6

QUESTION · 5054_s07_qp_1.pdf · source page 2 of 20

4 A car is brought to rest in 5 s from a speed of 10 m / s.

What is the average deceleration of the car?

A $0.5\,m/s^2$ **B** $2\,m/s^2$ **C** $15\,m/s^2$ **D** $50\,m/s^2$

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4 **B**

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Two athletes, A and B, run a 100 m race. At time $t = 0$, a gun is fired to start the race.

Fig. 1.1 shows the distance-time graph for the two athletes.

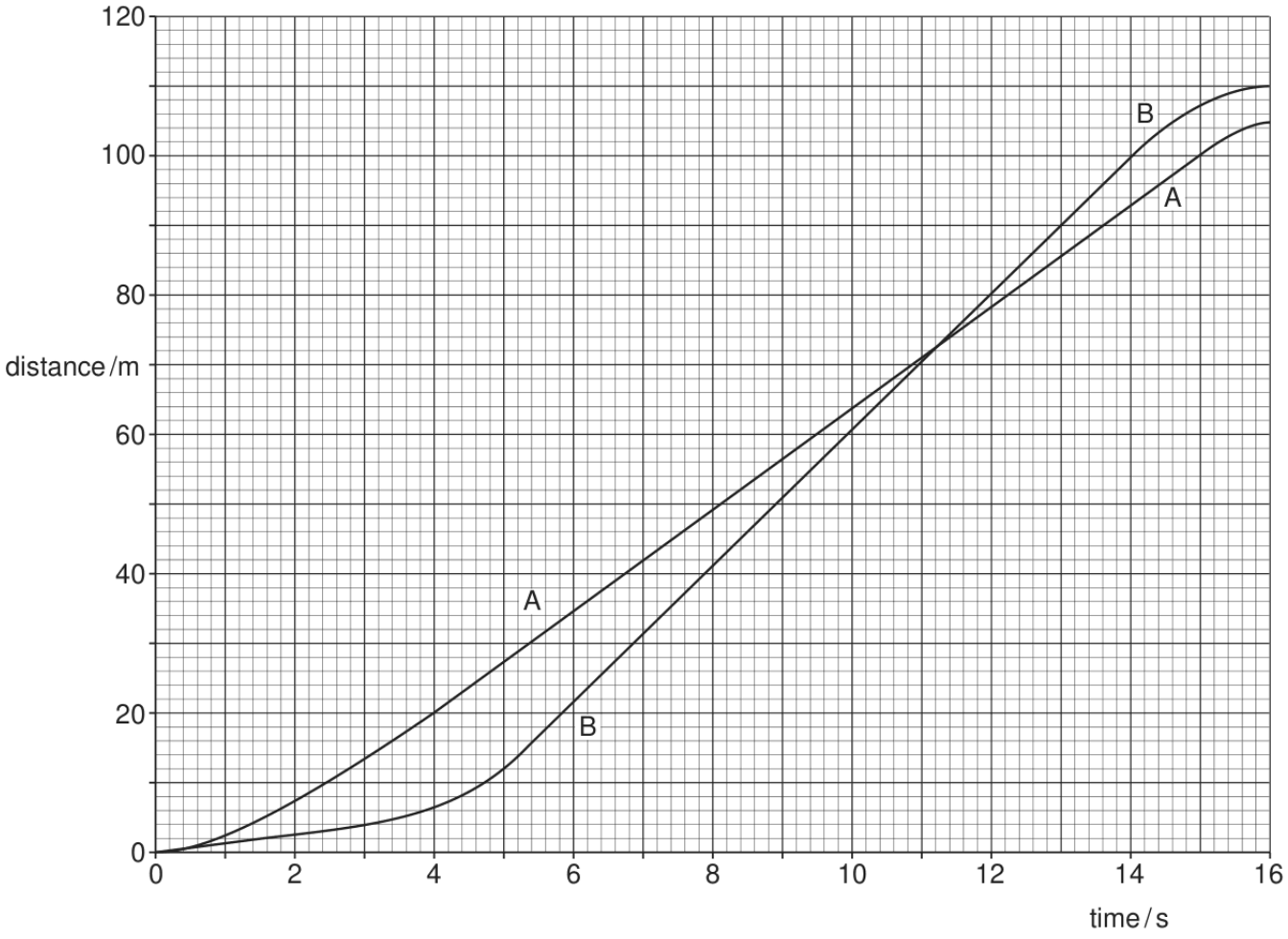


Fig. 1.1

(a) Describe the motion of athlete A during the first 8 s of the race.

.....

.....

.....

.....

..... [2]

For Examiner's Use

(b) State the distance between the two athletes as the winner passes the 100 m mark.

..... [1]

(c) Calculate the speed of athlete A between $t = 4$ s and $t = 15$ s.

speed = ... [2]

Section A

1	**(a)**	accelerates or speed increases **from rest/for 2-4s/for 8-20m**	B1	
		then a constant/uniform speed **or** velocity	B1	
	(b)	between 7 and 8 m	B1	
	(c)	distance 80 (+ 2) or s= d/t in any algebraic or numerical form	C1	
		7.3 **or** 7.27 **or** 7.273 m/s	A1	[5]

Section B

Answer **two** questions from this section.

Use the lined page provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 Fig. 9.1 shows the horizontal forces acting on a moving car.

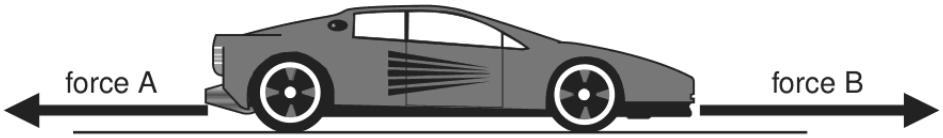


Fig. 9.1

(a) Compare the sizes and directions of the two forces when the car is

(i) moving along a straight road at constant speed, [1]

(ii) accelerating. [1]

(b) Suggest the direction of the resultant force on the car when the car turns a corner at constant speed. [1]

(c) Fig. 9.2 shows the speed-time graph for the first 24 s of the motion of the car along a straight road.

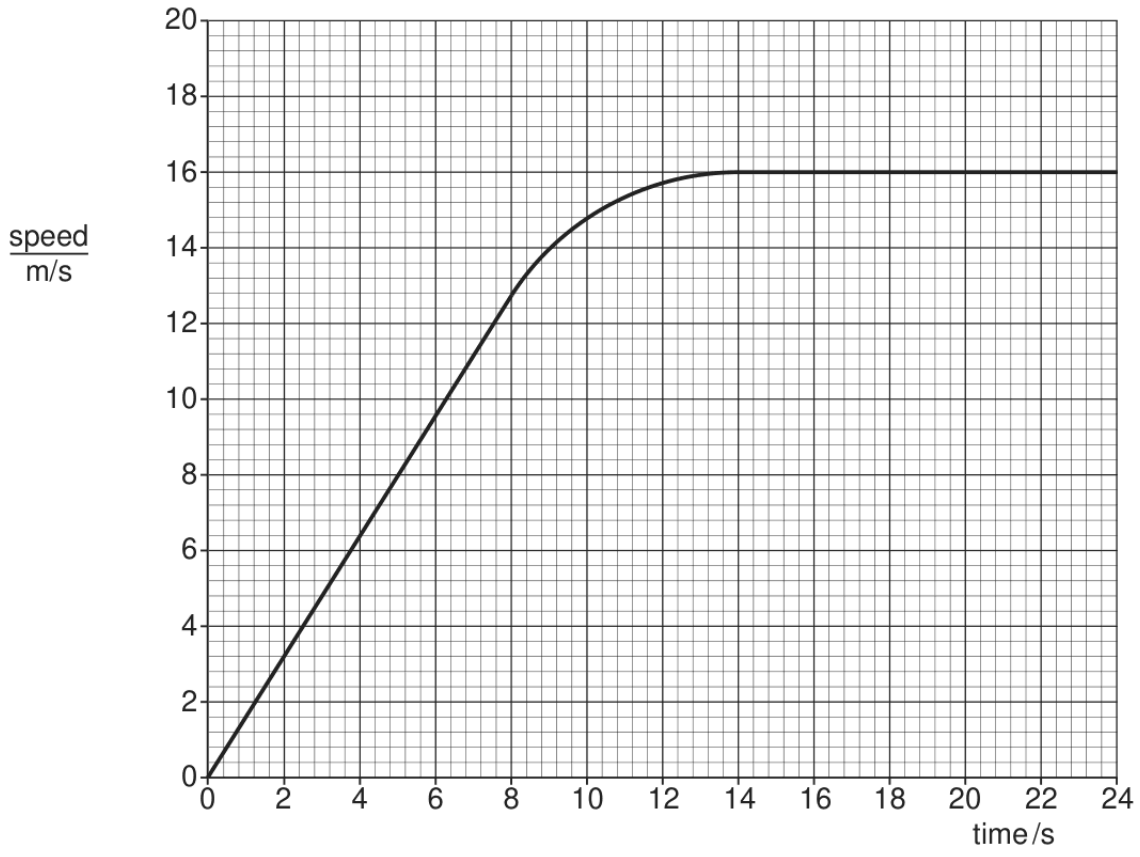


Fig. 9.2

(i) During the motion there is a period of uniform acceleration. State both the start and finish times of this period. [1]

(ii) State the main energy changes that occur in the 24 s. [3]

(iii) Calculate the acceleration of the car during the first 5 s. State clearly the equation that you use. [3]

(iv) Determine the distance travelled in the first 5 s. [3]

(d) The car is stopped by applying the brakes. Various factors can affect the distance travelled by the car during the time that the brakes are applied. Apart from the force applied by the brakes, state **two** of these factors. [2]

Section B

9	**(a)**	**(i)**	*A* = *B* (assume opposite direction and co-linear)	B1	
		(ii)	B > A (assume opposite direction and co-linear) maximum of 1 mark if directions wrong	B1	[2]
	(b)		towards **centre** of circle/corner	B1	[1]
	(c)	**(i)**	0 **and** 8–9 s	B1	
		(ii)	chemical (potential energy) (**accept** electrical if electrical car clear) **to** kinetic energy **or** K.E. increases thermal energy/heat/internal energy produced Max 2/3 if clear error	B1 B1 B1	
		(iii)	acceleration = (*v*–*u*)/t **or** gradient (7.8 to) 8/5 (**accept** any corresponding period e.g. 8s 12.6–12.8, 6s 9.4–9.6) 1.6 m/s^2 (**accept** 1.56–1.60)	C1 C1 A1	
		(iv)	area under graph **or** average speed 4 (m/s) **or** ½ 5 speed used in **(iii)** at 5 s ½ x (7.8 to) 8 x 5 20 m (**accept** 19.5–20; **ecf speed used in (iii)** at 5 s)	C1 C1 A1	[10]
	(d)		speed of car/friction with road (accept slippery road or ice or water or oil on road)/ friction in engine/tyre condition or area or pressure/air resistance/wind speed or direction/mass or inertia **of car or passengers**/slope of road	B2	[2]

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 A parachutist jumps from an aircraft. Some time later, the parachute opens.

Fig. 1.1 is a graph of the vertical speed of the parachutist plotted against time *t*.

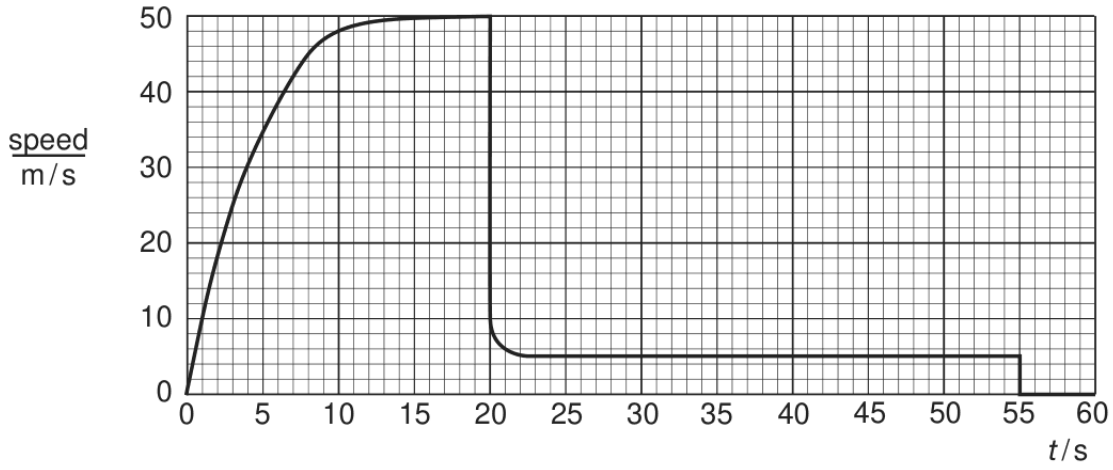


Fig. 1.1

(a) State what happens at $t = 20$ s and $t = 55$ s.

at 20 s .....

at 55 s ..... [1]

(b) Describe the motion of the parachutist between $t = 0$ and $t = 20$ s.

.....

.....

..... [2]

(c) Explain, in terms of the forces acting, why the speed of the parachutist is constant between $t = 25$ s and t = 55 s.

.....

.....

..... [2]

(d) Calculate the distance travelled by the parachutist between $t = 25$ s and $t = 55$ s.

distance = ... [2]

Section A

1 **(a)** parachute opens or speed drops from (50 to 5 m/s) **or** decelerates (**e.g.** uniformly) **and** lands/hits ground **or** speed becomes 0 or stops (**e.g.** decelerates) B1

(b) accelerates **or** speed increases (**not** increasing acceleration) B1
acceleration decreases (to 0) **or** speed becomes constant B1

(c) forces balance/cancel **or** no resultant **or** equal and opposite (**not just** forces equal) B1
weight/gravity **and** air resistance/drag mentioned (**not** upthrust/friction) B1

(d) (d =) st **or** s=d/t or any speed x any time **or** area under graph C1
150 m A1 [7]

Section B

Answer **two** questions from this section.

Use the separate sheets available from the Supervisor.

9 Many cars are fitted with an air-bag, as shown in Fig. 9.1. In a collision, the air-bag inflates and reduces the effect of the impact between the passenger and the dashboard.

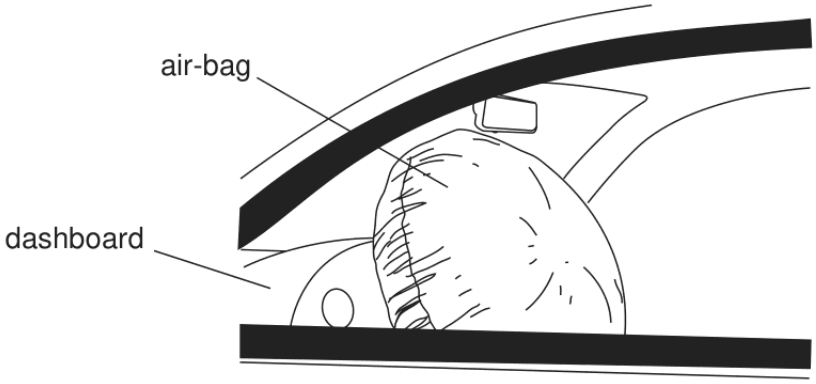


Fig. 9.1

(a) In a test of the air-bag, a heavy ball is used instead of the passenger.

The car is travelling at 14 m/s when it hits a wall. The air-bag inflates and the ball takes 3.0 s to come to rest. The ball has mass 5.0 kg.

(i) Calculate the average deceleration of the ball. [3]

(ii) Calculate the average force exerted on the ball. [2]

(iii) Using ideas about acceleration, explain how the air-bag reduces the force on the ball during the test. [2]

(b) If there was no air-bag, a large pressure would be exerted on the ball at the point where it hits the dashboard.

(i) Define *pressure*. [1]

(ii) The inflated air-bag reduces the pressure exerted on the ball. State **two** reasons why the pressure is reduced. [2]

(c) Compressed gas from a small cylinder inflates the air-bag. The cylinder contains a volume of 600 cm^3 of gas at a pressure of 1.4×10^7 Pa. The cylinder and the inflated air-bag have a volume of 30 000 cm^3.

(i) Calculate the pressure of the gas in the inflated air-bag, assuming that the temperature is constant. [3]

(ii) The pressure inside the cylinder decreases as the air-bag is inflated. Explain, using ideas about molecules, why the pressure decreases. [2]

Section B

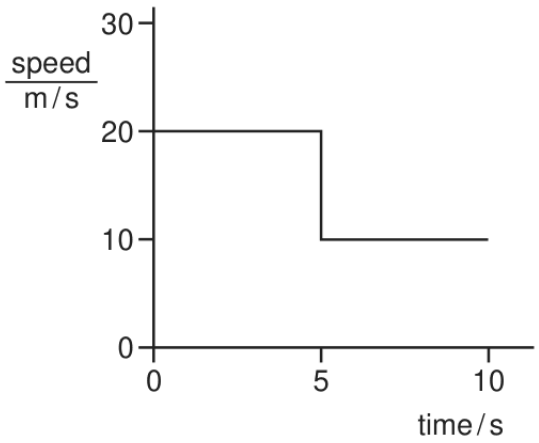
9	**(a)**	**(i)**	(acc =) (v-u)/t	C1	
			14/3	C1	
			4.7 m/s^2 (penalise halving to 2.35 m/s^2, accept 2 or more sig figs **not** fractions)	A1	
		(ii)	F = ma or 5 x **(i)**	C1	
			23 N (penalise **second** halving to 5.75 N, ecf **(i)** acc. 2 or more sig figs **not** fractions)	A1	
		(iii)	longer time of impact/slows down ball gradually/stops the ball more slowly	B1	
			less acceleration	B1	[7]
	(b)	**(i)**	force / area **or** F/A (acc. force on unit area **not** force on an area; N/m^2)	B1	
		(ii)	larger area	B1	
			smaller force	B1	[3]
	(c)	**(i)**	$P_1V_1 = P_2V_2$ **or** PV = constant	C1	
			1.4 x 10^7 x 600 = P x 30000 **or** 1.4 x 10^7 x 600/30000	C1	
			280 000 Pa	A1	
		(ii)	molecules hit sides (of cylinder) (**not** each other)	B1	
			molecules leave cylinder **or** fewer in cylinder **or** enter air bag	B1	[5]

QUESTION · 5054_s08_qp_1.pdf · source page 3 of 20

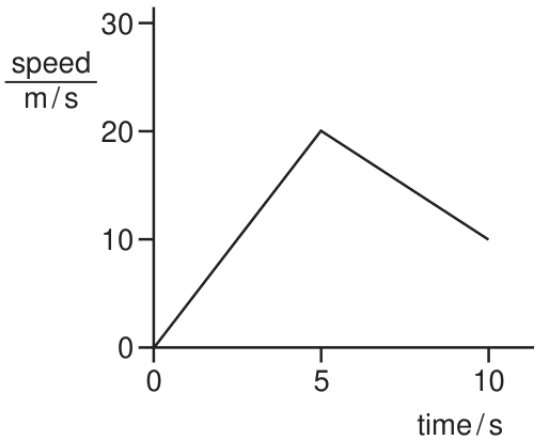
4 A skier is travelling downhill. The acceleration on hard snow is $4\,m/s^2$ and on soft snow is $2\,m/s^2$.

Which graph shows the motion of the skier when moving from hard snow to soft snow?

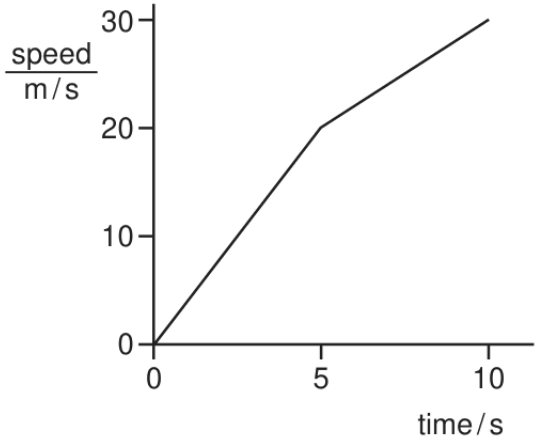
A



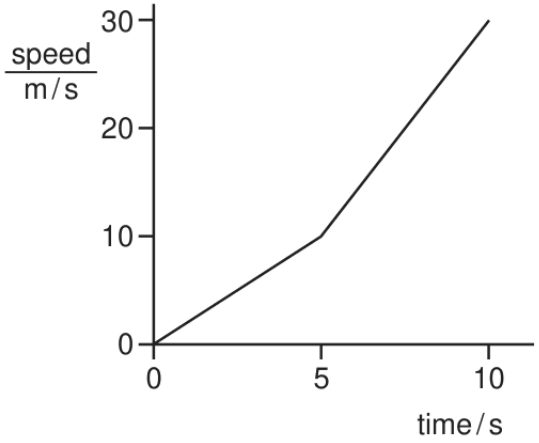
B



C



D



MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 4

4 **C**

Section B

Answer **two** questions from this section.

Use the lined pages provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 When a car driver sees an emergency ahead, he applies the brakes. During his reaction time the car travels at a steady speed and covers a distance known as the thinking distance. The braking distance is the distance travelled by the car after the brakes are applied.

(a) State the energy change that occurs as the car brakes. [2]

(b) Fig. 9.1 shows the speed-time graph of a car.
The driver sees the emergency at time $t = 0$.
The total mass of the car is 800 kg.

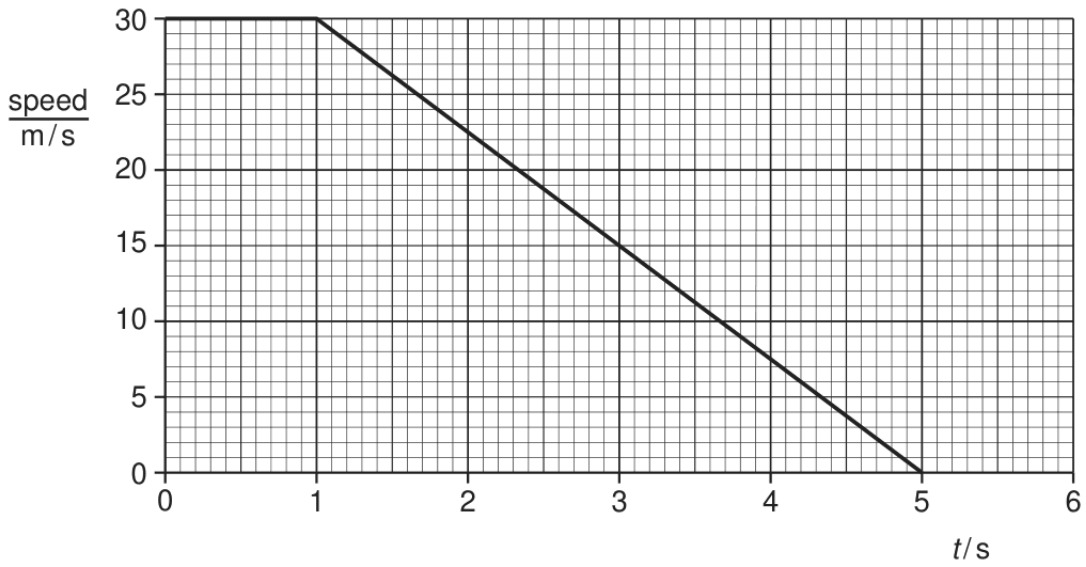


Fig. 9.1

Determine

(i) the thinking distance, [1]

(ii) the braking distance, [2]

(iii) the deceleration of the car during braking, [2]

(iv) the force provided by the brakes. [2]

(c) Using ideas about friction and deceleration, state and explain how the braking distance is affected by

(i) using new tyres rather than badly worn tyres, [2]

(ii) the car skidding on a wet road, [2]

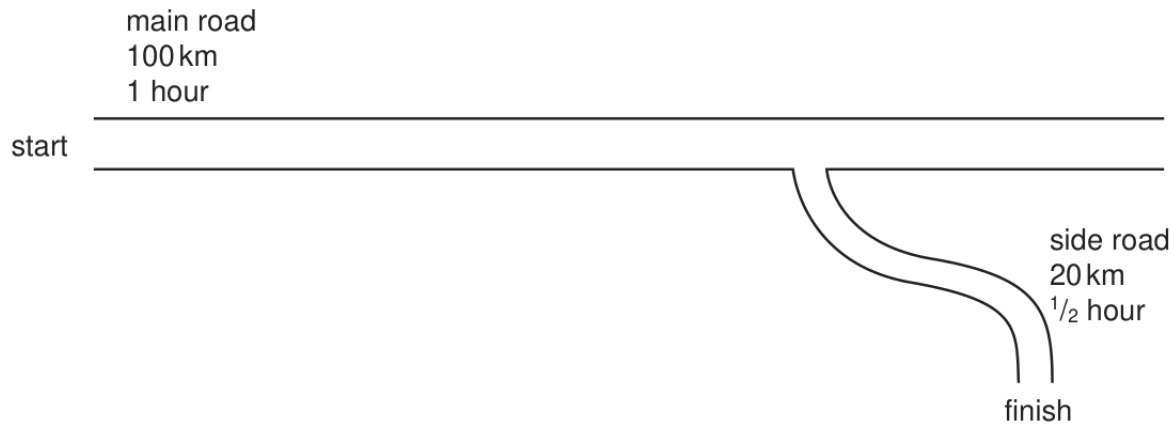
(iii) the car carrying a heavy load of passengers. [2]

Section B

9	**(a)**	K.E. (at start) to heat (+ sound)	B1 B1
			[Total: 2]
	(b) **(i)**	30 m cao	B1
	(ii)	area under graph **or** average speed × time **or** (u + v).t / 2 **or** 30 × 4/2 60 m	C1 A1
	(iii)	(v–u)/t **or** $v = u + at$ **or** 30/4 **or** gradient **or** rise/run 7.5 (±0.1) m/s^2	C1 A1
	(iv)	$F = ma$ **or** 800 × **(iii)** 6000 N ecf **(iii)**	C1 A1
			[Total: 7]
	(c) **(i)**	more friction/grip/traction **or** more deceleration **or** decelerates faster **or** decelerates in less time less (braking) **distance**	 B1 B1
	(ii)	less friction **or** less deceleration **or** decelerates slower/longer more (braking) **distance**	B1 B1
	(iii)	less deceleration **or** decelerates slower/longer more **distance**	B1 B1
			[Total: 6]

QUESTION · 5054_w08_qp_1.pdf · source page 2 of 20

2 A car takes 1 hour to travel 100 km along a main road and then ½ hour to travel 20 km along a side road.



What is the average speed of the car for the whole journey?

A 60 km / h **B** 70 km / h **C** 80 km / h **D** 100 km / h

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 2

2 **C**

For Examiner's Use

4 A passenger aeroplane accelerates from rest along a runway. It accelerates at a uniform rate for 35 s. At this point it reaches a speed of 84 m / s and then takes off.

(a) (i) Calculate the acceleration of the aeroplane along the runway.

acceleration = ... [2]

(ii) Sketch a speed-time graph for the aeroplane as it travels along the runway.

[3]

(b) Two **horizontal** forces act on the aeroplane as it accelerates along the runway.

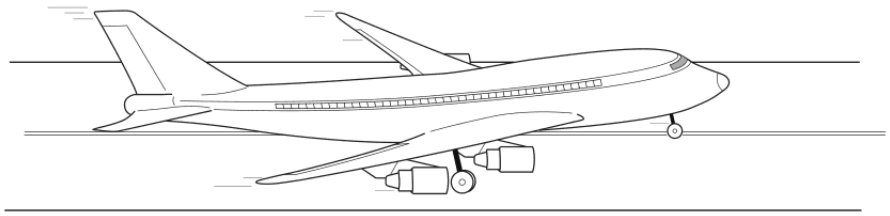


Fig. 4.1

(i) On Fig. 4.1, draw arrows to show the directions and the relative magnitudes of these forces. [1]

(ii) State what causes the smaller of these two forces.

..... [1]

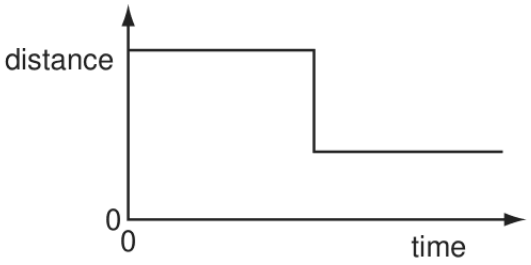
4	**(a)**	**(i)**	(a = Δ)v/t **or** 84/35	C1	
			2.4 m/s²	A1	
		(ii)	speed **and** time axes correct **and** labelled	B1	
			straight line of positive gradient through origin	B1	
			84 (m/s) **and** 35 (s) marked	B1	
	(b)	**(i)**	two arrows with forward force > backward force	B1	
		(ii)	air/wind resistance **or** friction **or** drag	B1	[7]

QUESTION · 5054_s09_qp_1.pdf · source page 2 of 20

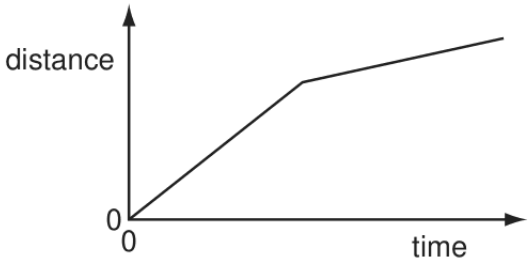
3 A free-fall parachutist falls at a constant speed. He then opens his parachute and continues to fall to Earth at a lower, constant speed.

Which diagram shows how the distance fallen by the parachutist varies with time?

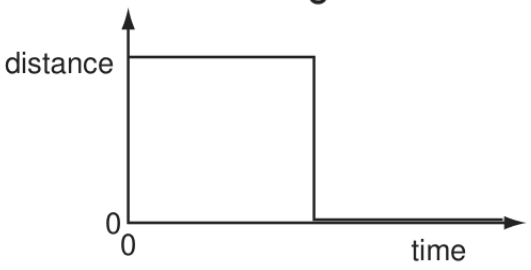
A



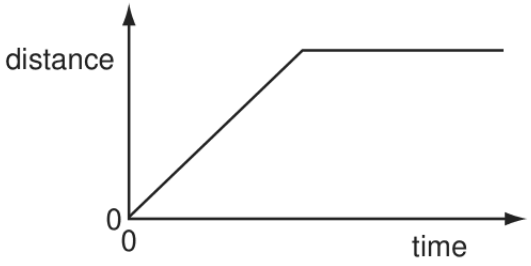
B



C



D

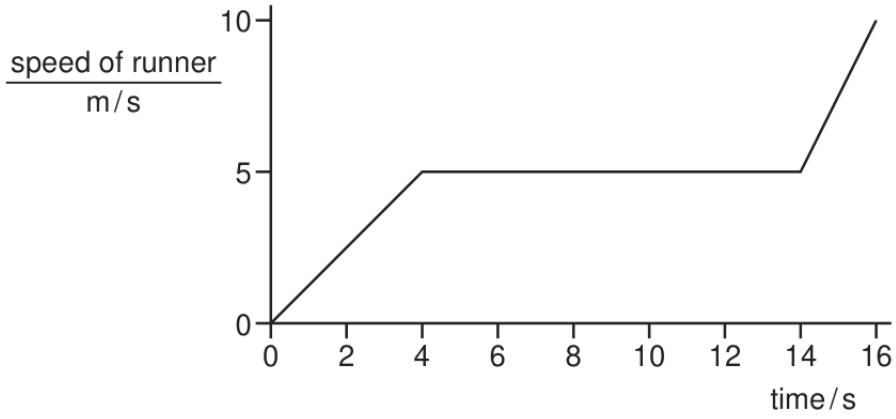


MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 3

3 **B**

QUESTION · 5054_s09_qp_1.pdf · source page 3 of 20

4 The graph shows the speed of a runner during a race.



What is the distance travelled by the runner during the race?

A 50 m **B** 65 m **C** 75 m **D** 90 m

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 4

4 **C**

Section A

For Examiner's Use

Answer **all** the questions in this section.

1 A piece of paper falls from 4.0 m above the ground.
Fig. 1.1 shows how the height h above the ground varies with the time t.

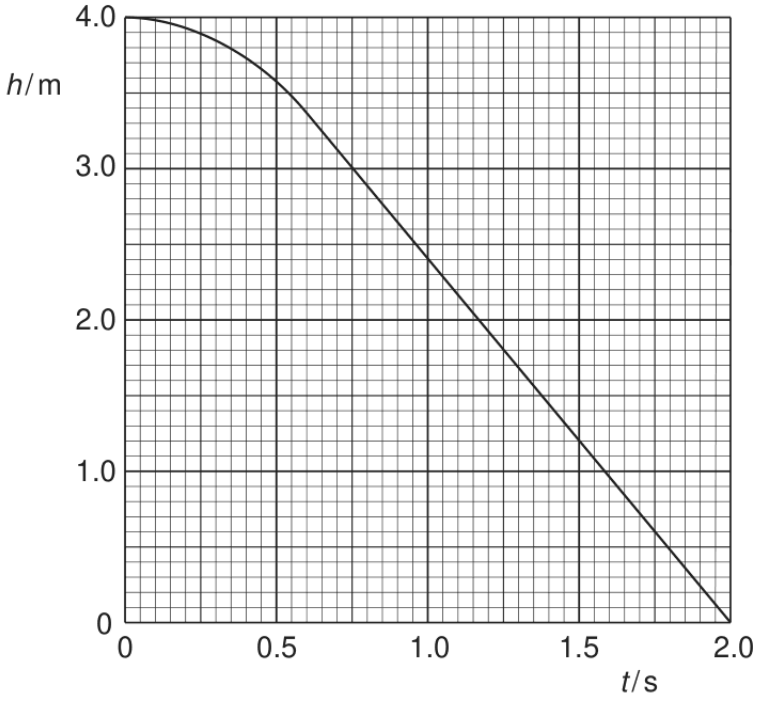


Fig. 1.1

(a) State what happens to the speed of the paper as it falls.

.....

.....

..... [2]

(b) Calculate the speed of the paper at time $t = 1.5$ s.

speed = ... [2]

(c) As the paper falls, energy changes from one form to another.
State the main energy change between $t = 1.0$ s and $t = 2.0$ s.

.....

.....

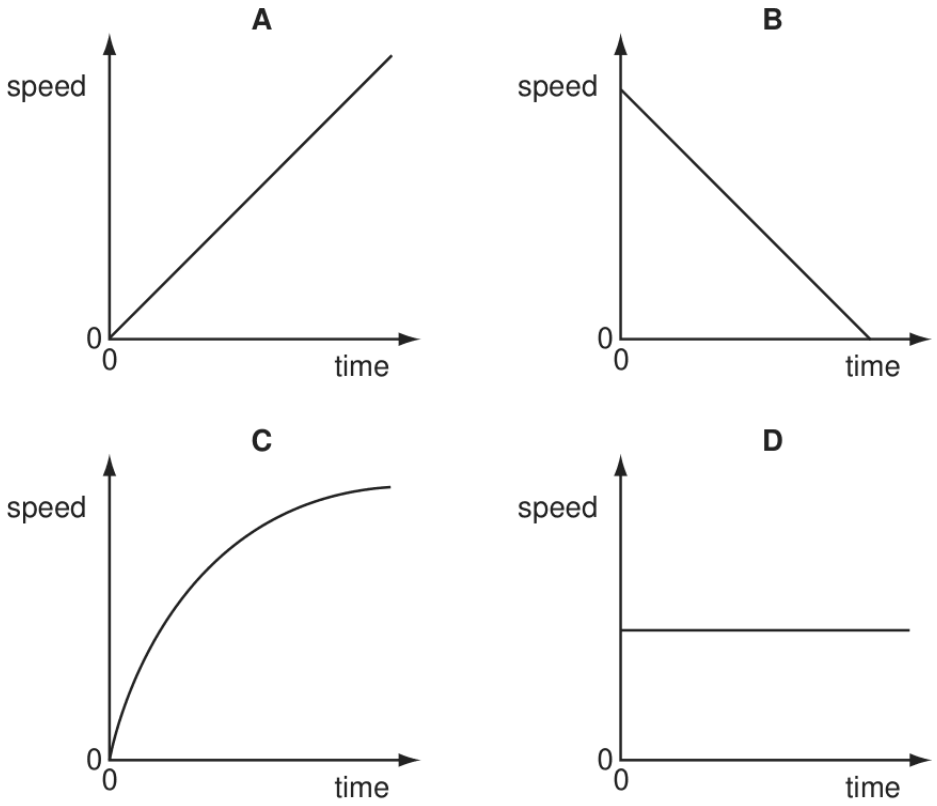
..... [2]

Section A

1	**(a)**	(speed) increases or (paper) accelerates	B1	
		(speed) becomes constant/uniform or acceleration zero (after 0.5 s)	B1	
	(b)	any clear change in distance/time or 1.87 (m/s) (allow 1.9)	C1	
		2.3–2.5 m/s	A1	
	(c)	PE at beginning of a change	B1	
		heat/internal energy/thermal energy at end of a change/K.E. of air	B1	[6]

QUESTION · 5054_w09_qp_1.pdf · source page 2 of 20

3 Which speed-time graph shows an object moving with non-uniform acceleration?

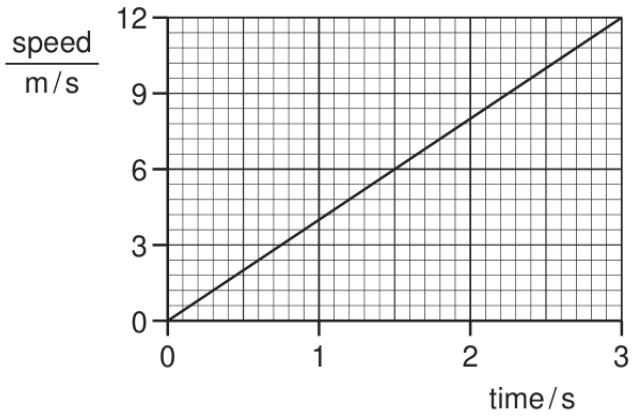


MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 3

3 **C**

QUESTION · 5054_w09_qp_1.pdf · source page 3 of 20

4 The graph shows the speed of a car as it moves from rest.



What is the average speed of the car during the first 3 s?

A 4 m/s **B** 6 m/s **C** 18 m/s **D** 36 m/s

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 4

4	**B**

QUESTION · 5054_s10_qp_11.pdf · source page 2 of 20

3 A student measures the speed of a trolley. At one instant, the speed of the trolley is 1.0 m / s and two seconds later the speed is 4.0 m / s.

What is the acceleration of the trolley?

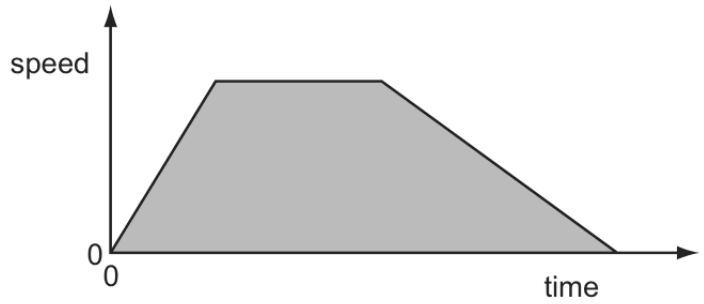
A $1.5\,m/s^2$ **B** $2.0\,m/s^2$ **C** $2.5\,m/s^2$ **D** $5.0\,m/s^2$

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 3

3 **A**

QUESTION · 5054_s10_qp_11.pdf · source page 2 of 20

4 The speed-time graph shows the movement of a car.



What does the shaded area of the graph represent?

A the average acceleration of the car

B the average speed of the car

C the total distance travelled by the car

D the total travelling time of the car

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 4

4 **C**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the speed-time graph of a ball. The ball is dropped at time $t = 0$.

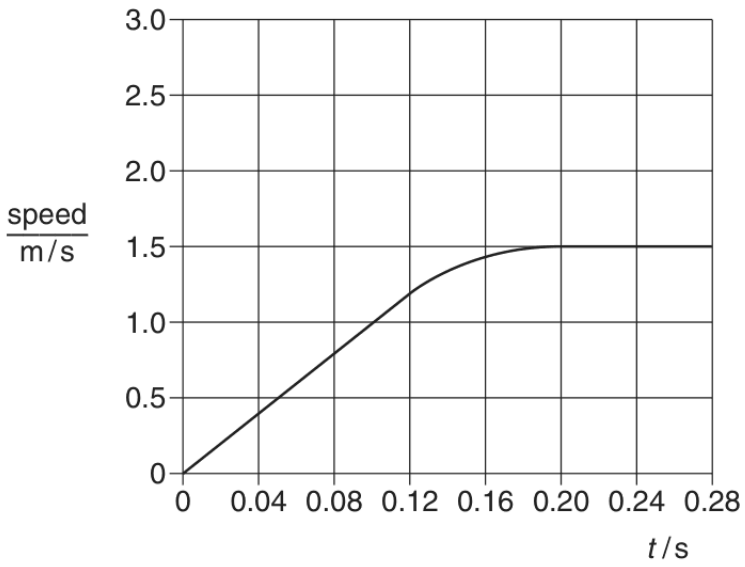


Fig. 1.1

(a) After $t = 0.20$ s, the ball falls at a constant speed.

Explain, using ideas about forces, why the speed of the ball is constant after time $t = 0.20$ s.

.....

.....

.....[1]

(b) At $t = 0$, a different ball is dropped from rest. Until $t = 0.20$ s, this ball has a constant acceleration equal to the acceleration of free-fall. After $t = 0.20$ s, its acceleration decreases.

(i) State the value of the acceleration of free-fall.

.....[1]

(ii) Determine the speed of the second ball at $t = 0.20$ s.

speed = ...[2]

(iii) On Fig. 1.1, draw the speed-time graph for the second ball from $t = 0$ to $t = 0.28$ s. [2]

Section A

1	**(a)**	forces balance/cancel **or** no net force **or** upward force = downward force **or** weight = air resistance/drag/air friction	B1	[1]
	(b) **(i)**	9.8–10 m/s^2	B1	[1]
	(ii)	$a = v(-u)/t$ algebraic or numerical	C1	
		2(.0) m/s ecf **(i)**	A1	[2]
	(iii)	straight line from (0,0) to (0.2,2) ecf **(ii)**	B1	
		after 0.2 s, decreasing but not negative gradient	B1	[2]

QUESTION · 5054_s10_qp_12.pdf · source page 3 of 20

4 A student measures the speed of a trolley. At one instant, the speed of the trolley is 1.0 m / s and two seconds later the speed is 4.0 m / s.

What is the acceleration of the trolley?

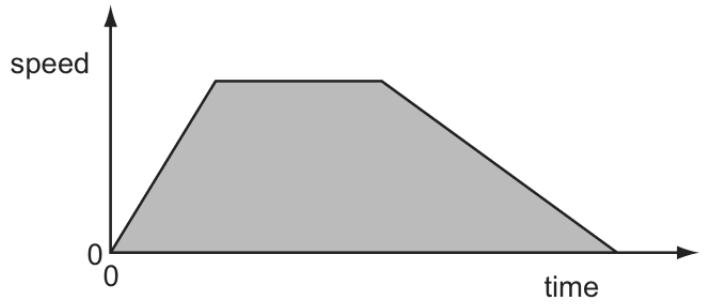
A $1.5\,m/s^2$ **B** $2.0\,m/s^2$ **C** $2.5\,m/s^2$ **D** $5.0\,m/s^2$

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 4

4 **A**

QUESTION · 5054_s10_qp_12.pdf · source page 3 of 20

5 The speed-time graph shows the movement of a car.



What does the shaded area of the graph represent?

A the average acceleration of the car

B the average speed of the car

C the total distance travelled by the car

D the total travelling time of the car

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 5

5 **C**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a car braking on a road and coming to rest.

Fig. 9.1

(a) Explain what is meant by

(i) the *thinking distance*,

.....

.....[1]

(ii) the *braking distance*.

.....

.....[1]

(b) An engineer conducts a test on the car and finds that the braking distance is greater when the car is fully loaded than when it is unloaded.

(i) Apart from the road conditions, state what must be kept the same in the test.

.....[1]

(ii) Explain why the car has a greater braking distance when fully loaded.

.....

.....[1]

(c) State and explain how one road condition affects the braking distance of the car. Use ideas about friction in your answer.

.....

.....[2]

(d) Explain how wider tyres affect the pressure of the car on the surface of the road.

.....

.....[1]

For Examiner's Use

(e) The car has a total mass of 900 kg and is travelling at 20 m/s. At time $t = 0$, the driver sees an accident ahead. He applies the brakes at $t = 0.60$ s to stop the car. After the brakes are applied, the car comes to rest in a further 4.0 s.

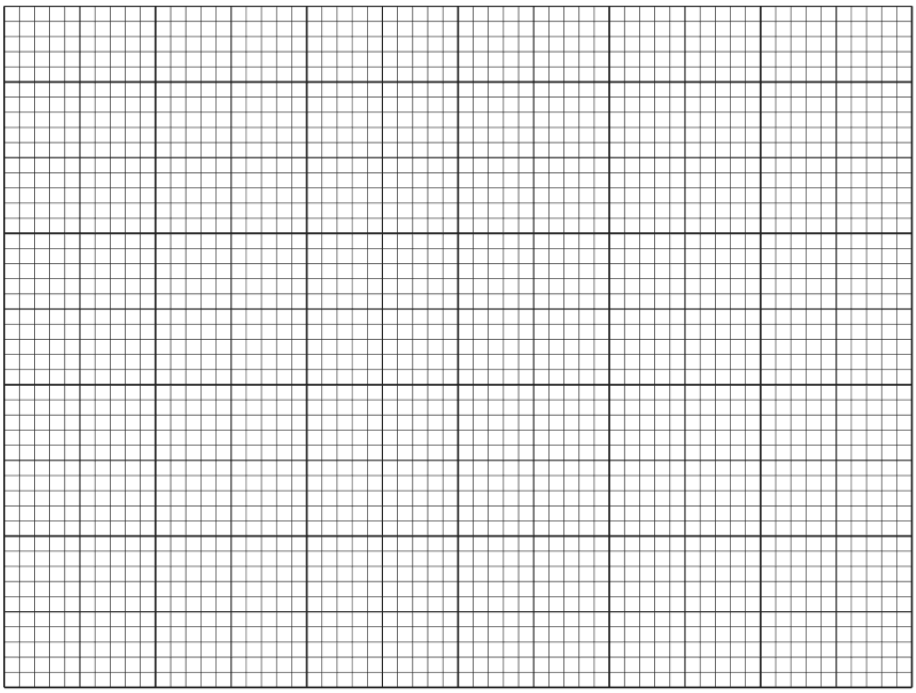
(i) Calculate the deceleration of the car as it brakes.

deceleration = ...[2]

(ii) Calculate the braking force acting on the car.

force = ...[2]

(iii) On Fig. 9.2, draw a speed-time graph for the car as it brakes.



[3]

Fig. 9.2

(iv) State how your graph in **(iii)** can be used to find the total distance travelled by the car.

.....

.....[1]

Section B

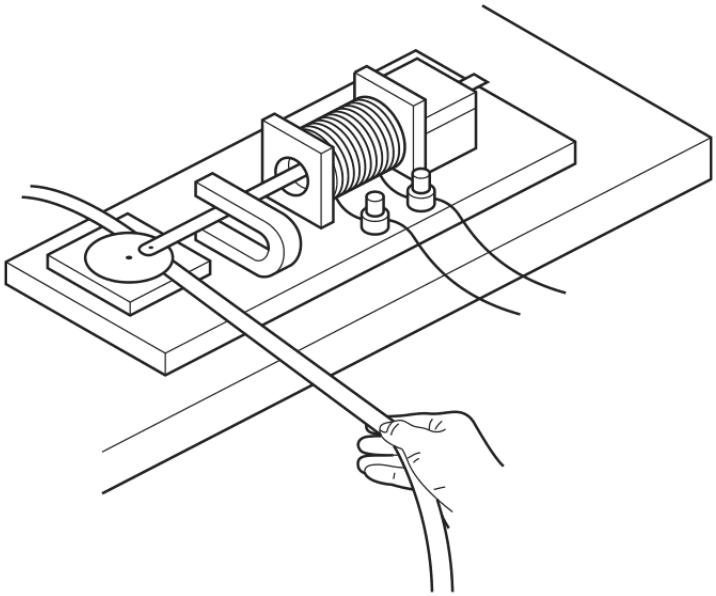
9	**(a)**	**(i)**	distance travelled while thinking/in reaction time or before braking starts	B1	[1]
		(ii)	distance travelled while brakes applied/car decelerates	B1	[1]
	(b)	**(i)**	speed (of cars) or same force/pressure on pedal or same braking force or same tyres or condition of brakes	B1	[1]
		(ii)	greater inertia/kinetic energy/momentum or smaller deceleration/acceleration	B1	[1]
	(c)		any road condition, e.g. icy, wet, poor surface, slippery/smooth/rough surface and its correct effect on distance	B1	
			correct explanation that refers to friction e.g. more friction when dry	B1	[2]
	(d)		pressure low(er) (with larger area)	B1	[1]
	(e)	**(i)**	$a = v/t$ any algebraic or numerical value e.g. 20/4; 20/3.4; 20/4.6; 20/0.6	C1	
			5(.0) m/s^2	A1	[2]
		(ii)	$F = ma$ algebraic or numerical e.g. 900 × **(i)**	C1	
			4500 N e.c.f. **(i)**	A1	[2]
		(iii)	correct axes labelled with quantity and/or unit	B1	
			horizontal line at 20 m/s from 0 to 0.6 s	B1	
			straight line from end of horizontal section or from (0.6,20) to (4.6,0) or (4,0)	B1	[3]
		(iv)	area (under graph or of trapezium)	B1	[1]

[Total: 15]

QUESTION · 5054_w10_qp_11.pdf · source page 3 of 20

3 A student pulls a piece of tape through a ticker-tape timer. Every 0.02 s, the timer prints a dot on the tape.

First the tape is pulled quickly, then slowly, then quickly again.



Which piece of tape does the student obtain?

A

B

C

D

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 3

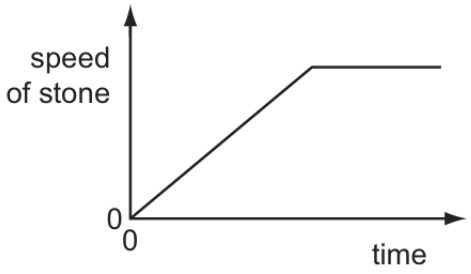
3 **B**

QUESTION · 5054_w10_qp_11.pdf · source page 4 of 20

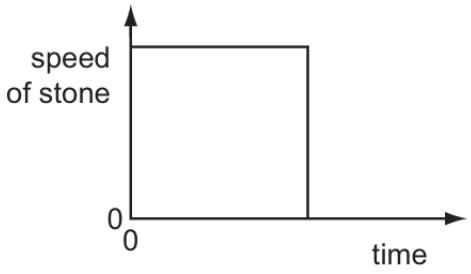
4 A small stone is dropped from the top of a ladder, falls and hits the ground. It does not rebound.

Which speed-time graph is correct?

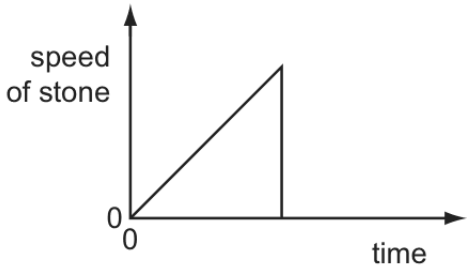
A



B



C



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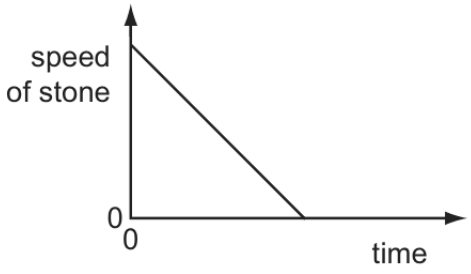


MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 4

4 **C**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 A window cleaner drops a sponge from the balcony of a hotel at time $t = 0$. Fig. 1.1 is the speed-time graph for the motion of the sponge.

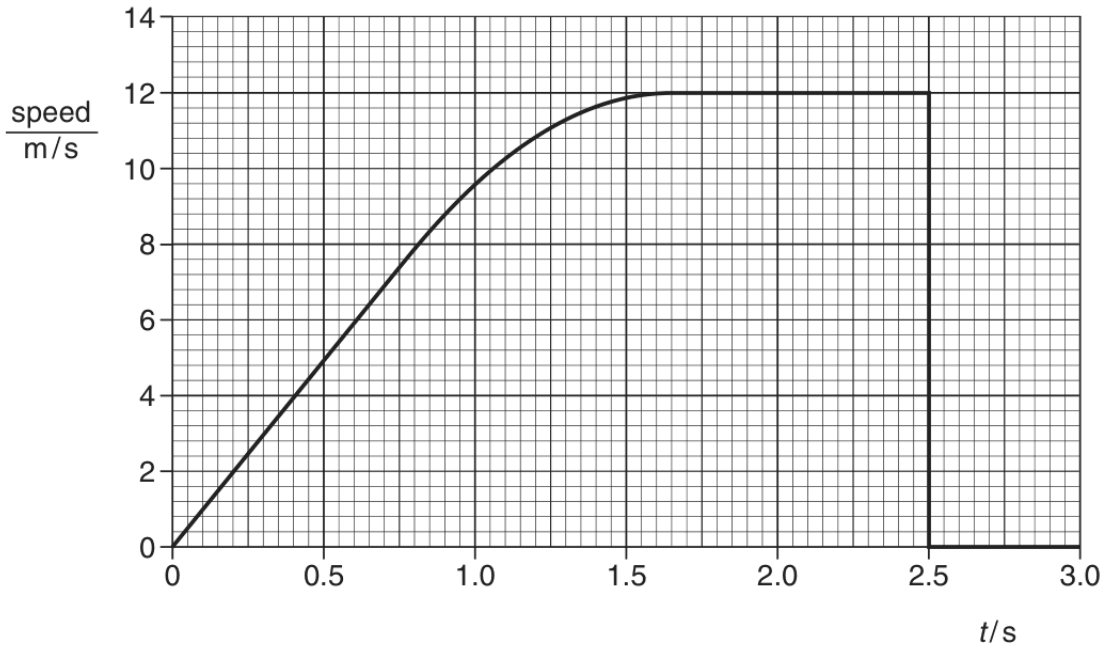


Fig. 1.1

(a) State a value for t when the sponge is

(i) moving with a uniform speed,

..... [1]

(ii) accelerating at a non-uniform rate,

..... [1]

(iii) decelerating.

..... [1]

(b) Calculate the distance travelled by the sponge between $t = 0$ and $t = 0.75$ s.

distance = ... [2]

Section A

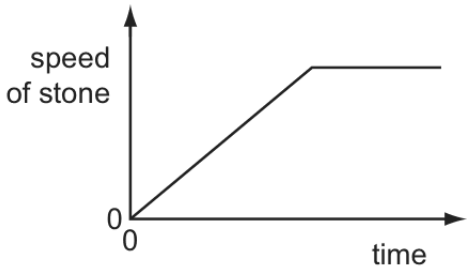
1	(a)	(i)	any **one** time between 1.60 and 2.50 s **or** range of correct values	B1	[1]
		(ii)	any **one** time between 0.75 and 1.65 s **or** range of correct values	B1	[1]
		(iii)	2.5(0) s	B1	[1]
	(b)		area (under graph) **or** ½bh **or** ½gt^2 **or** ½ × 0.75 × (7.3 to 7.5)	C1	
			2.7(375) to 2.8(125) m	A1	[2]

QUESTION · 5054_w10_qp_12.pdf · source page 2 of 16

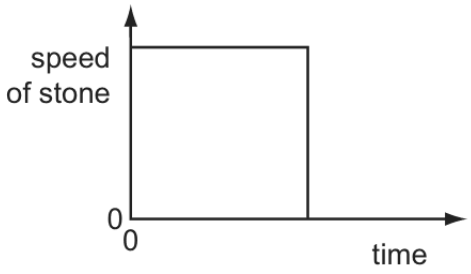
2 A small stone is dropped from the top of a ladder, falls and hits the ground. It does not rebound.

Which speed-time graph is correct?

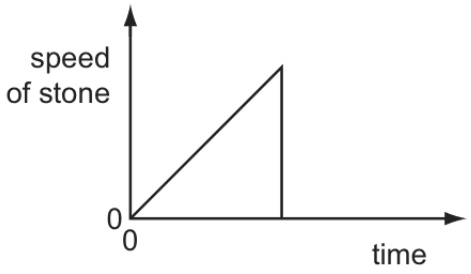
A



B



C



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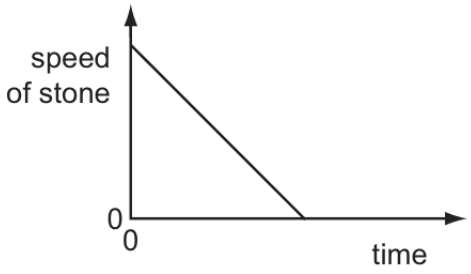


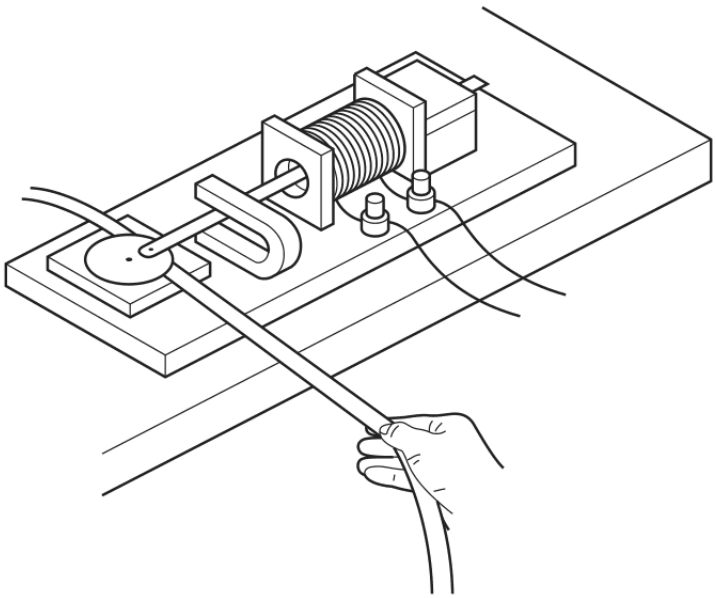
MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 2

2 **C**

QUESTION · 5054_w10_qp_12.pdf · source page 3 of 16

5 A student pulls a piece of tape through a ticker-tape timer. Every 0.02 s, the timer prints a dot on the tape.

First the tape is pulled quickly, then slowly, then quickly again.



Which piece of tape does the student obtain?

A

B

C

D

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5	**B**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a cricket ball as it comes into contact with a cricket bat.

Fig. 1.1

The cricket ball has a mass of 0.16 kg and it hits the bat with a speed of 25 m/s. After being in contact with the bat for 0.0013 s, the ball rebounds with a speed of 22 m/s in the direction exactly opposite to its original direction.

(a) State the difference between *speed* and *velocity*.

.....

.....[1]

(b) Calculate

(i) the change in velocity of the cricket ball,

velocity change = ...[1]

(ii) the average acceleration of the ball whilst it is in contact with the bat,

acceleration = ...[2]

(iii) the average force exerted on the ball by the bat.

force = ...[2]

Section A

1 **(a)** velocity has a direction/is a vector **or** speed does not have a direction/is not a vector **or** displacement/time **and** distance/time
(**ign** speed is a scalar) B1

(b) **(i)** (–) 47 m/s B1

(ii) (a =) v/t **or** 47/0.0013 C1
(–) 3.6(1538 etc.) × 10^4 m/s^2 A1

(iii) (F =) ma **or** 0.16 × 3.6 × 10^4 C1
(–) 5.8(**or** 5.78461 etc.) × 10^3 N A1 [6]

10 A bus breaks down on a road with a 10° upward slope. The passengers get out and push the bus to the top of the slope at a constant speed. Fig. 10.1 shows the passengers exerting a force on the bus parallel to the line of the slope.

For Examiner's Use

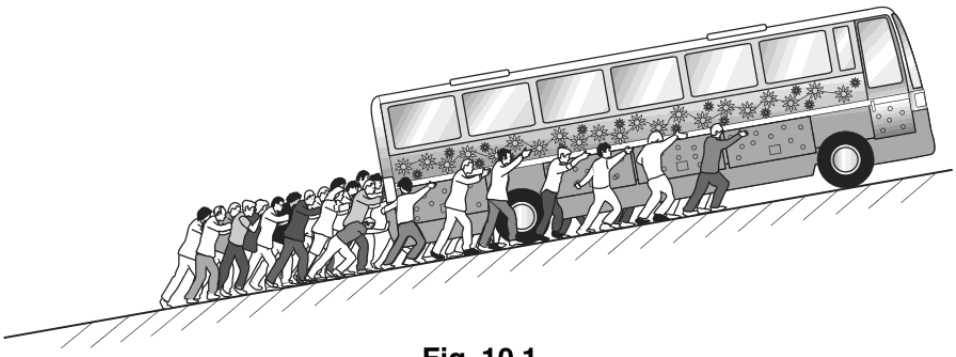


Fig. 10.1

(a) The total mass of the bus is 3200 kg.

(i) Calculate the weight of the bus.

weight = ...[1]

(ii) On Fig. 10.1, draw **two** arrows to show the direction of the weight of the bus and the direction of the force exerted on the bus by the passengers. [1]

(iii) The total force exerted by the passengers on the bus is 17000 N along the line of the slope. Use a graphical method to determine the size and direction of the resultant of this force and the weight of the bus. State the scale used.

scale = ...

size of resultant = ...

direction of resultant = ...

[4]

For Examiner's Use

(iv) The bus travels at a constant speed. State the size of the resultant of all the forces acting on the bus.

.....[1]

(b) As the passengers push the bus up the slope, work is done against two forces. State the names of these two forces and, for each force, state the form of energy produced as work is done.

force 1 .....

form of energy 1 .....

force 2 .....

form of energy 2 .....
[4]

(c) When the bus reaches the top of the slope, it stops and the passengers get back in.

(i) On the axes of Fig. 10.2, sketch the distance-time graph for the bus from a time when the passengers are pushing the bus at constant speed until they get back into the stationary bus.

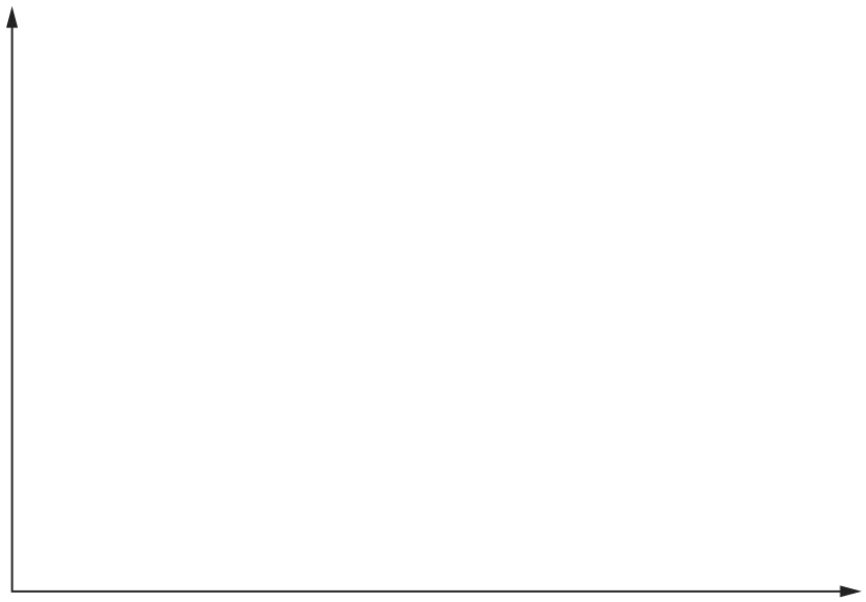


Fig. 10.2

[3]

(ii) State how the speed of a moving object may be obtained from an accurately drawn distance-time graph.

.....

.....[1]

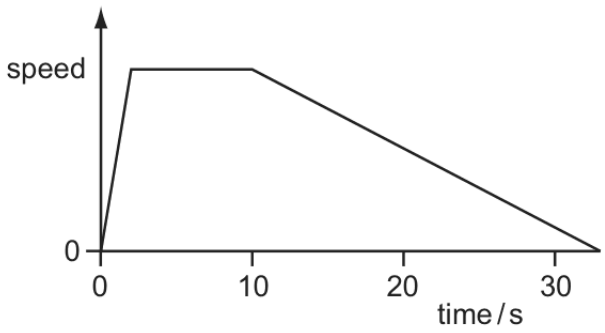
10	**(a)**	**(i)**	32 000 N	B1	
		(ii)	two arrows/lines in correct direction by eye	B1	
		(iii)	scale given	B1	
			two arrows/lines **and** correct resultant drawn	B1	
			32.0 → 35.0 kN (2/3 sig. fig. only)	B1	
			58.5 → 61.5° to horizontal (2/3 sig. fig. only; don't penalise twice)	B1	
		(iv)	zero/no force/0	B1	[7]
	(b)		weight/gravitational force/gravitational attraction (**not** gravity)	B1	
			higher in gravitational field **or** (to gravitational) potential energy	B1	
			friction/air resistance	B1	
			heat/thermal/internal energy	B1	[4]
	(c)	**(i)**	labelled axes **and** correct way round ($x \rightarrow t$)	B1	
			straight line of positive slope	B1	
			followed only by horizontal line (**ign** curve at junction)	B1	
		(ii)	distance travelled/time taken (from points) **or** calculate the gradient	B1	[4]

QUESTION · 5054_s11_qp_11.pdf · source page 3 of 16

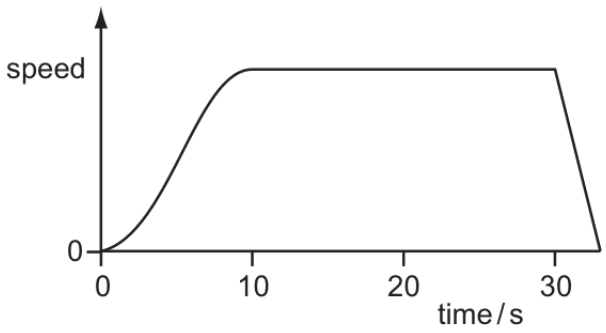
4 A car accelerates from traffic lights for 10 s. It stays at a steady speed for 20 s and then brakes to a stop in 3 s.

Which graph shows the journey?

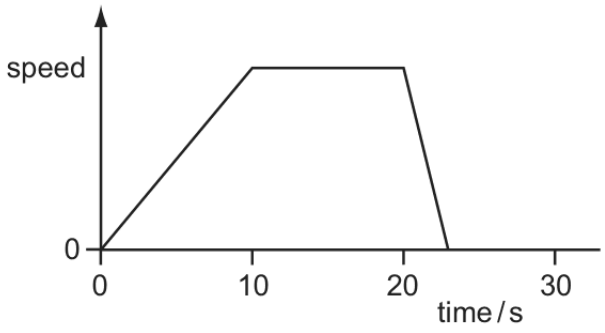
A



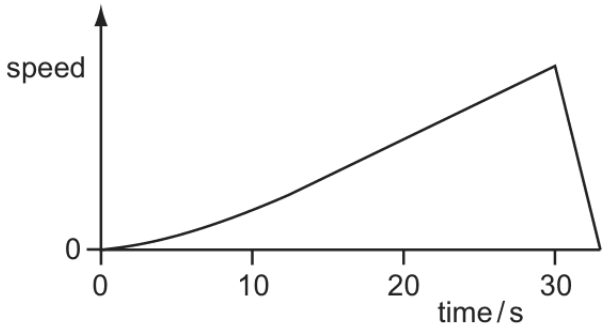
B



C



D



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4 **B**

QUESTION · 5054_s11_qp_11.pdf · source page 3 of 16

5 Which vehicle has an acceleration of $5\,m/s^2$?

A a bicycle, when its speed changes from rest to 2.5 m/s in 2 s

B a car, when its speed changes from rest to 15 m/s in 5 s

C a lorry, when its speed changes from rest to 20 m/s in 15 s

D a motorbike, when its speed changes from rest to 50 m/s in 10 s

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 5

5	**D**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the speed-time graph for a car.

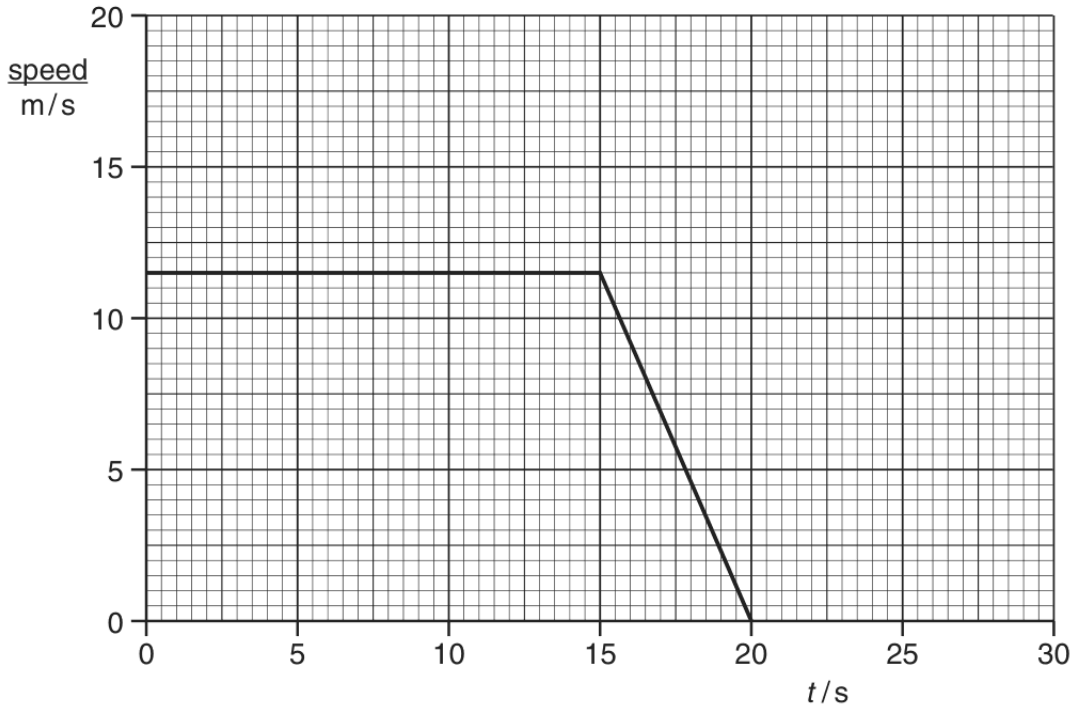


Fig. 1.1

At time $t = 15$ s, the brakes are applied and the car comes to rest with uniform deceleration.

(a) (i) State the speed of the car just before the brakes are applied.

speed = ... [1]

(ii) Explain what is meant by *uniform deceleration*.

.....

.....

..... [2]

(b) A lorry travels at a constant speed of 18 m/s for 15 s. At time $t = 15$ s, the brakes are applied and the lorry slows down with the same deceleration as the car.

(i) On Fig. 1.1, draw the speed-time graph for the lorry. [2]

(ii) Explain how your graph shows that, while braking, the lorry travels further than the car.

.....

..... [1]

Section A

1	**(a)**	**(i)**	11.5 m/s	B1	
		(ii)	decrease in speed // negative gradient	C1	
			equal changes/decreases in speed in the same time // const. neg. grad. on v-t graph	A1	
	(b)	**(i)**	flat line at 18 m/s from t = 0 to 15	B1	
			constant slope downwards parallel to initial line (by eye)	B1	
		(ii)	greater area **under** graph // higher **initial/average** speed	B1	[6]

QUESTION · 5054_s11_qp_12.pdf · source page 3 of 16

4 Which vehicle has an acceleration of $5\,m/s^2$?

A a bicycle, when its speed changes from rest to 2.5 m/s in 2 s

B a car, when its speed changes from rest to 15 m/s in 5 s

C a lorry, when its speed changes from rest to 20 m/s in 15 s

D a motorbike, when its speed changes from rest to 50 m/s in 10 s

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 4

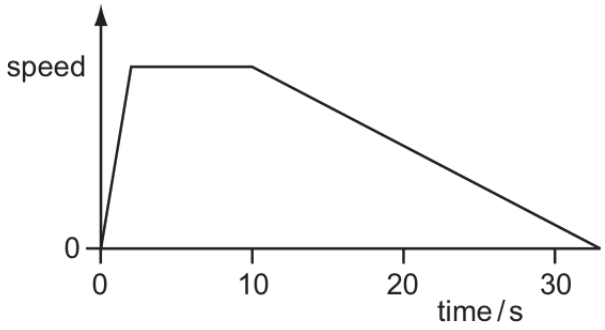
4	**D**

QUESTION · 5054_s11_qp_12.pdf · source page 3 of 16

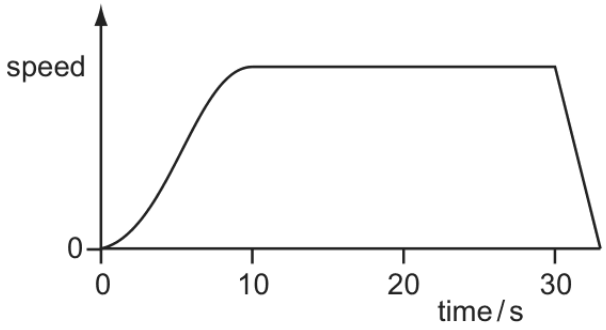
5 A car accelerates from traffic lights for 10 s. It stays at a steady speed for 20 s and then brakes to a stop in 3 s.

Which graph shows the journey?

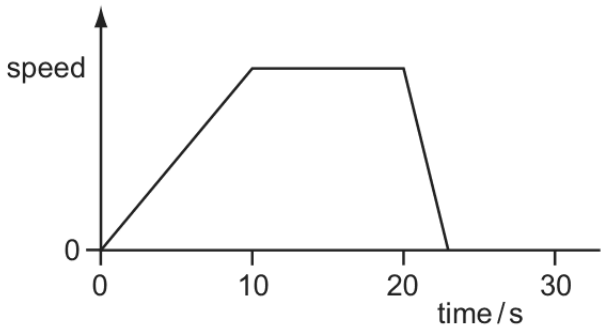
A



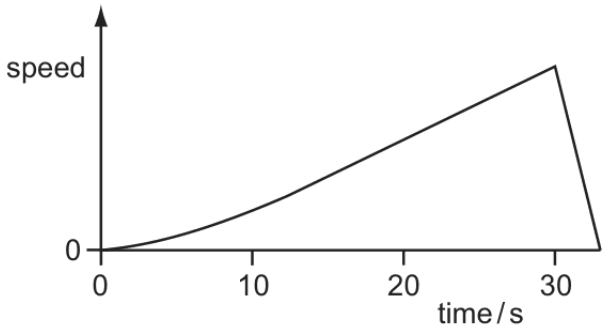
B



C



D



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5 **B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a sky-diver falling vertically.

Fig. 9.1

The sky-diver starts from rest at time $t = 0$.
His acceleration is non-uniform until he reaches a steady speed of 50 m/s at $t = 10$ s.
He opens his parachute at $t = 20$ s and decelerates until $t = 25$ s.
From $t = 25$ s he falls at a steady speed of 5 m/s.

(a) (i) On Fig. 9.2, draw the speed-time graph for the sky-diver. [4]

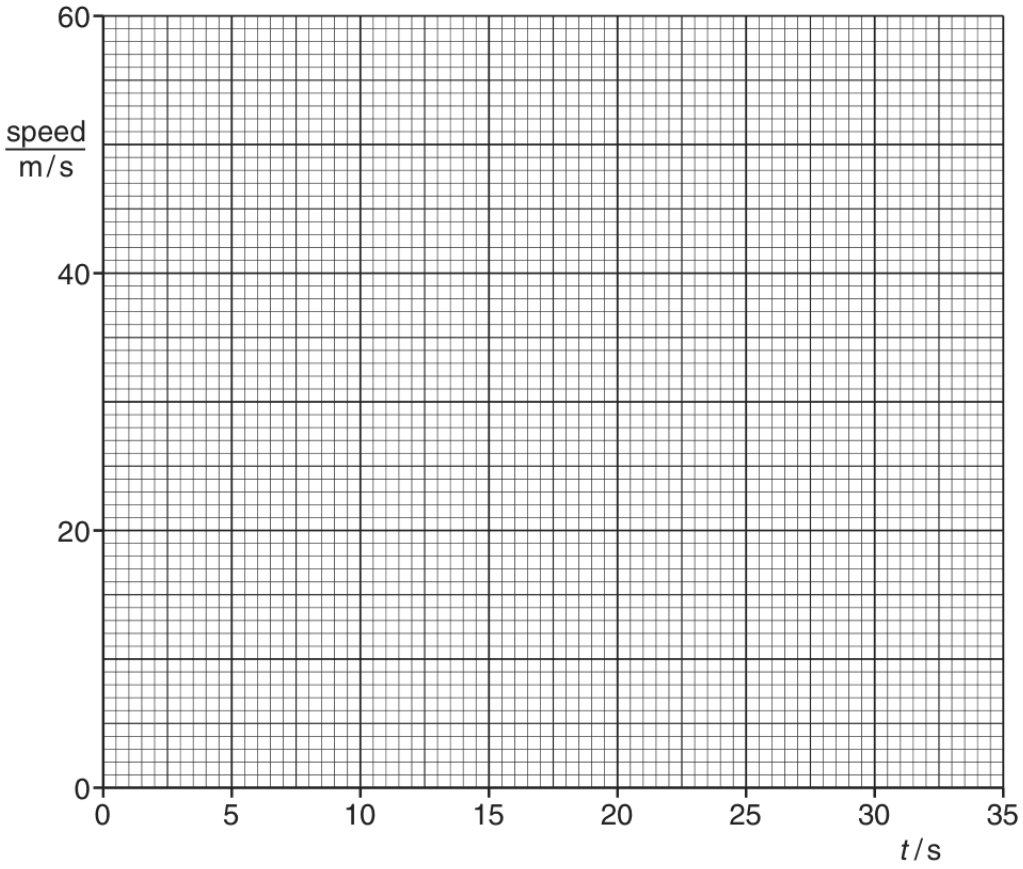


Fig. 9.2

(ii) State how your graph shows that the acceleration is non-uniform between $t = 0$ and $t = 10$ s.

.....

.....[1]

(b) Explain, in detail, why **after** the sky-diver opens his parachute, he decelerates and eventually reaches a steady speed.

.....

.....

.....

.....

.....

.....[4]

(c) For the time interval between t = 10 s and t = 20 s, calculate the distance that the sky-diver falls.

distance = ...[1]

(d) The mass of the sky-diver is 60 kg.

For the time interval between t = 20 s and t = 25 s,

(i) calculate the average deceleration of the sky-diver,

average deceleration = ...[2]

(ii) calculate the average resultant force acting on the sky-diver,

force = ...[2]

(iii) state how your graph in Fig. 9.2 may be used to obtain the distance that the sky-diver falls.

.....

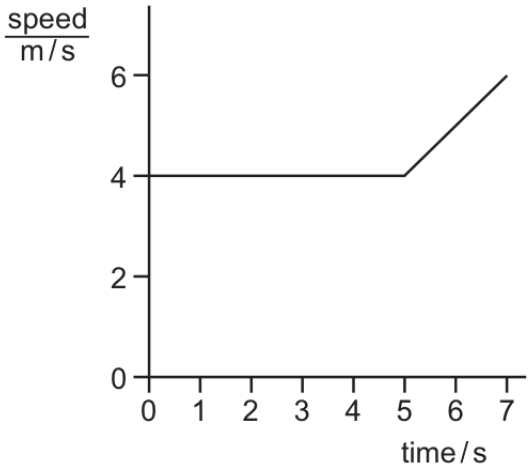
.....[1]

Section B

9	**(a)**	**(i)**	curve with decreasing gradient from origin to 50 m/s at 10 s	B1
			constant speed from 10 to 20 s	B1
			decrease to 5 m/s at 25 s	B1
			constant speed from 25 s until at least 30 s	B1
		(ii)	gradient/slope not constant/decreases // graph curves // graph not a (straight) line // increase (in speed) per second/unit time not equal	B1
	(b)		any mention of **air** resistance/drag/upward force	B1
			(initially) force upwards larger than force downwards // resultant force upwards	B1
			air resistance decreases (with fall in speed)	B1
			(at constant speed) air resistance/friction/drag equals weight // forces (up and down) balance // zero resultant force	B1
	(c)		500 m	B1
	(d)	**(i)**	(a =) $\frac{v-u}{t}$ in any numerical or algebraic form, e.g. 45/5	C1
			9(.0) m/s^2 ecf **(a)(i)**	A1
		(ii)	($F = ma$) in any numerical or algebraic form, e.g. 60 × 9 ecf **(i)**	C1
			540 N	A1
		(iii)	area **under** graph/line/curve	B1 [15]

QUESTION · 5054_w11_qp_11.pdf · source page 2 of 20

3 The graph shows part of a journey made by a cyclist.



How far did the cyclist travel in 7 s?

A 28 m **B** 30 m **C** 32 m **D** 42 m

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3 **B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 is the speed-time graph for a rocket from the moment that the fuel starts to burn at time $t = 0$.

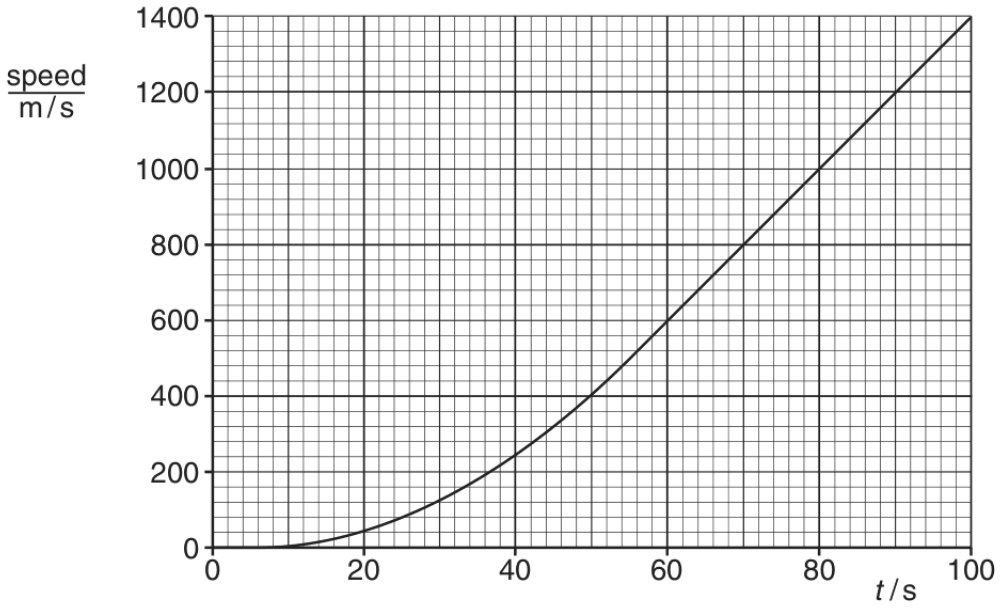


Fig. 9.1

(a) State the principal energy changes taking place as the rocket accelerates upwards.

.....

.....

.....

.....[4]

(b) **(i)** State the size of the acceleration of the rocket at $t = 0$.

.....[1]

(ii) State what happens to the acceleration of the rocket between $t = 5\,s$ and $t = 80\,s$.

.....

.....[2]

(iii) Calculate the acceleration of the rocket at $t = 80\,s$.

acceleration = ...[2]

(iv) The total mass of the rocket at $t = 80\,s$ is $1.6 \times 10^6\,kg$. Calculate the resultant force on the rocket at this time.

resultant force = ...[2]

(v) The total weight of the rocket at $t = 80\,s$ is $1.6 \times 10^7\,N$. Calculate the upward force on the rocket at this time, caused by the burning fuel.

upward force = ...[1]

(c) As the rocket burns fuel, it ejects hot gas downwards.

(i) State *Newton's third law of motion*.

.....

.....[1]

(ii) Explain how Newton's third law of motion applies to the upward force on the rocket and to the force on the hot gas.

.....

.....

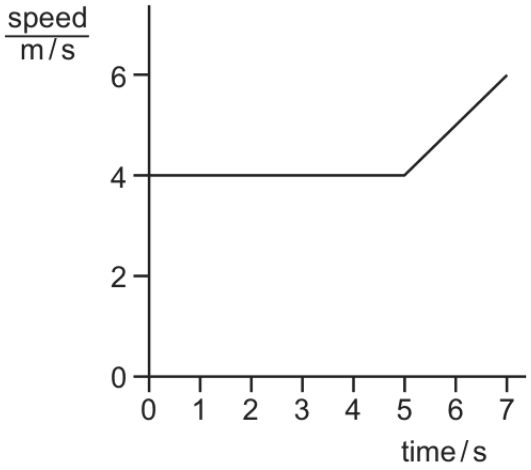
.....[2]

Section B

9	(a)	**at start:** chemical (potential) energy	B1	
		at end: PE/GPE/gravitational energy	B1	
		at end: heat/thermal/internal energy	B1	
		at end: KE **or** intermediate KE from chemical energy	B1	
	(b) (i)	0	B1	
	(ii)	it increases	B1	
		to constant value	B1	
	(iii)	gradient **or** $(v–u)/t$ **or** (1400-600)/40 **or** other correct numbers	C1	
		$20\,m/s^2$	A1	
	(iv)	$(F =)\ ma$ **or** $1.6 \times 10^6 \times 20$	C1	
		$3.2 \times 10^7\,N$	A1	
	(v)	$4.8 \times 10^7\,N$	B1	
	(c) (i)	to every action there is an equal and opposite reaction **or** forces act in pairs of equal size **and** in opposite directions/on different bodies	B1	
	(ii)	downward force on gas	B1	
		equal and opposite to upward/**(b)(v)** force (on rocket)	B1	[15]

QUESTION · 5054_w11_qp_12.pdf · source page 3 of 20

3 The graph shows part of a journey made by a cyclist.



How far did the cyclist travel in 7 s?

A 28 m **B** 30 m **C** 32 m **D** 42 m

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 3

3 **B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 is the speed-time graph for a racing car of total mass 650 kg as it sets off from rest at the start of a race.

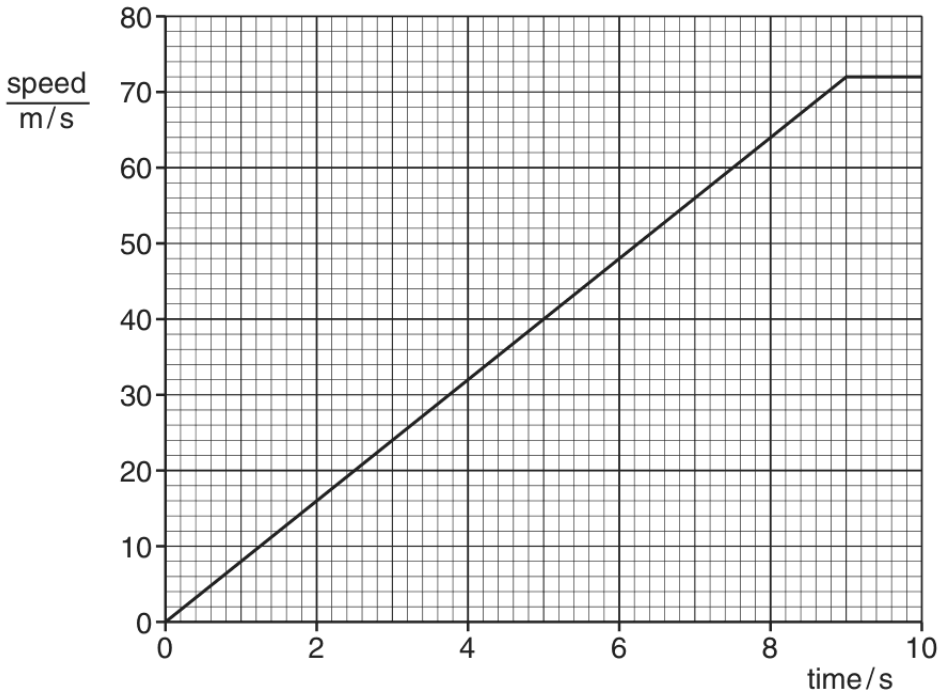


Fig. 9.1

The car travels in a straight line until time $t = 9.0$ s.

(a) State the speed of the car at time $t = 9.0$ s.

speed = ...[1]

(b) Calculate, for the car between times $t = 0$ and $t = 9.0$ s,

(i) the distance travelled,

distance = ...[2]

(ii) the acceleration of the car,

acceleration = ...[2]

(iii) the resultant force acting on the car.

force = ...[2]

(c) The acceleration of the car is constant between $t = 0$ and $t = 9.0$ s.

Suggest and explain why the driving force on the car must increase to keep the acceleration constant.

.....

.....

.....

.....[3]

(d) After $t = 9.0$ s, the car starts to turn a corner and follows a circular path at constant speed.

(i) Explain why the car is accelerating even though its speed is constant.

.....

.....

.....[2]

(ii) State the direction of this acceleration.

.....[1]

(iii) State and explain what causes the car to accelerate as it turns the corner.

.....

.....[2]

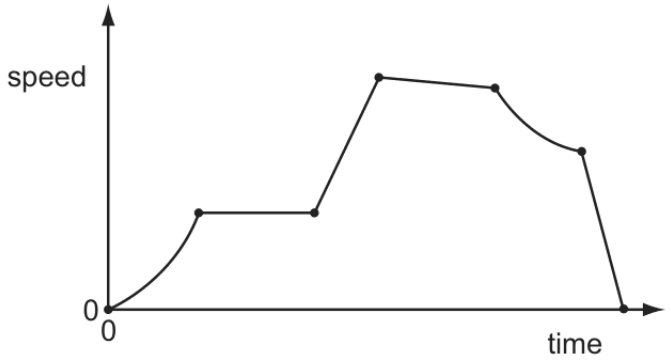
Section B

9	**(a)**	72 m / s			B1	
	(b)	**(i)**	area (under graph) **or** ½ base × height **or** ½*vt* **or** ½ × 9 × 72		C1	
			320/324 m		A1	
		(ii)	**change** in velocity/time **or** $\Delta v/t$ **or** 72/9		C1	
			8(.0) m / s^2		A1	
		(iii)	(*F* =) *ma* **or** 650 × 8.0		C1	
			5.2 × 10^3 N		A1	
	(c)		friction **or** air/wind resistance **or** drag		M1	
			increases as speed increases		A1	
			resultant/net/unbalanced force remains constant		B1	
	(d)	**(i)**	**direction** (of car/motion/speed/velocity) changes		B1	
			(therefore) velocity changes		B1	
		(ii)	towards centre (of circle)/centripetal		B1	
		(iii)	friction **with ground**	**OR** banking of track	B1	
			mention wheels/tyres	reaction force (acts towards centre)	B1	[15]

QUESTION · 5054_s12_qp_11.pdf · source page 2 of 16

3 The speed-time graph represents the journey of a car.

The dots separate different sections of the journey. There are six different sections.



How many sections represent the car moving with non-uniform acceleration?

A 0 **B** 1 **C** 2 **D** 3

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 3

3 **C**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 An astronaut standing on the Moon throws a stone vertically upwards. The stone leaves his hand at time $t = 0$. The line on Fig. 1.1 shows how the velocity v of the stone varies with time t until $t = 2.0\,s$.

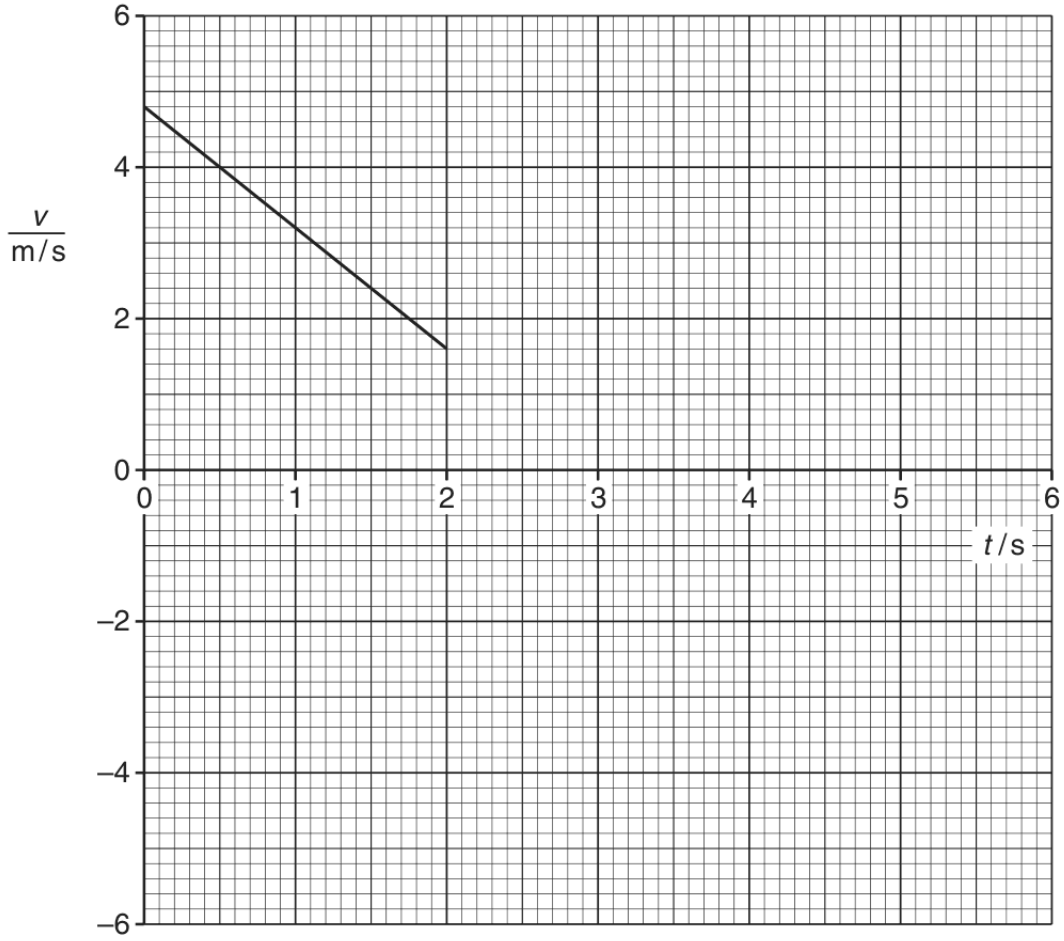


Fig. 1.1

(a) After rising, the stone falls. The astronaut catches the stone at $t = 6.0\,s$.
There is no air resistance on the Moon.

(i) Complete Fig. 1.1 until $t = 6.0\,s$. [1]

(ii) State the value of t when the stone is at its highest point.

.....[1]

(b) Calculate the acceleration of the stone between $t = 0$ and $t = 2.0$ s.

acceleration = ...[2]

(c) A stone is thrown vertically upwards on the Earth with the same initial velocity.
State two ways in which the velocity-time graph for this stone differs from Fig. 1.1.

1. .....

2. .....
[2]

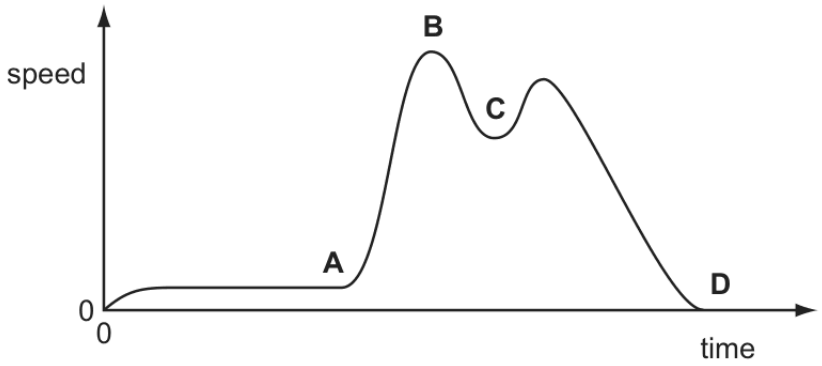
Section A

1	**(a)**	**(i)**	straight line continues to 6 ± 0.2 s	B1	
		(ii)	3(.0) s OR the time on Fig. 1.1 when $v = 0$	B1	
	(b)		(a =) ($v - u$)/t in any form numerical or algebraic	C1	
			(–)1.6 m / s^2	A1	
	(c)		any TWO lines: (at first) graph steeper/higher acceleration/deceleration caught sooner/shorter time to maximum graph curves (due to air resistance)	B2	[6]

QUESTION · 5054_s12_qp_12.pdf · source page 2 of 16

3 A cyclist travels along a hilly road without using the pedals or brakes. Air resistance and friction are negligible. The speed / time graph of the cyclist is shown.

At which point did he reach the bottom of the first hill?



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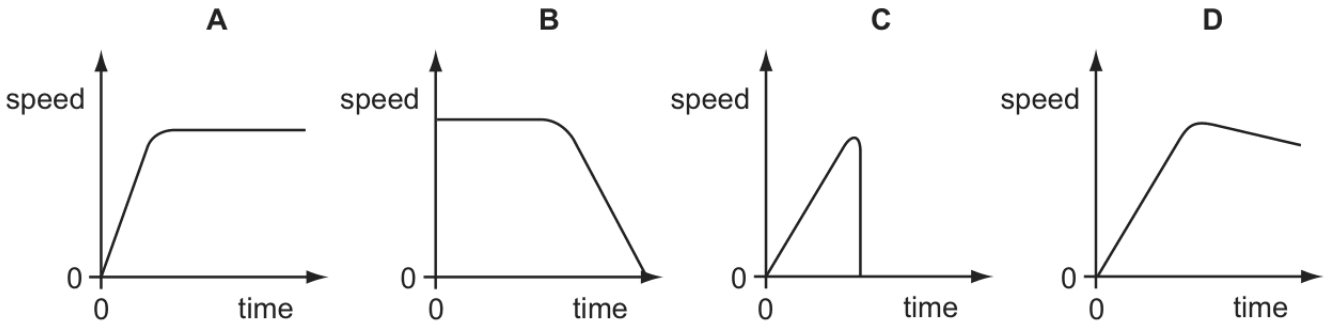
3	**B**

QUESTION · 5054_w12_qp_11.pdf · source page 2 of 16

3 A ball starts to roll down a steep slope and then along rough horizontal ground.

Which graph best shows the **speed** of the ball?



MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 3

3 **D**

QUESTION · 5054_w12_qp_11.pdf · source page 2 of 16

4 A car is stationary at a set of traffic lights.

When the lights turn green, the car begins to move and continues to speed up until it reaches the maximum speed allowed. It continues to travel at this constant speed for the rest of the journey.

What happens to the acceleration and to the velocity of the car during this journey?

A Both the acceleration and the velocity change.

B Only the acceleration changes.

C Only the velocity changes.

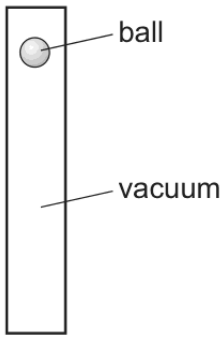
D Neither the acceleration nor the velocity changes.

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 4

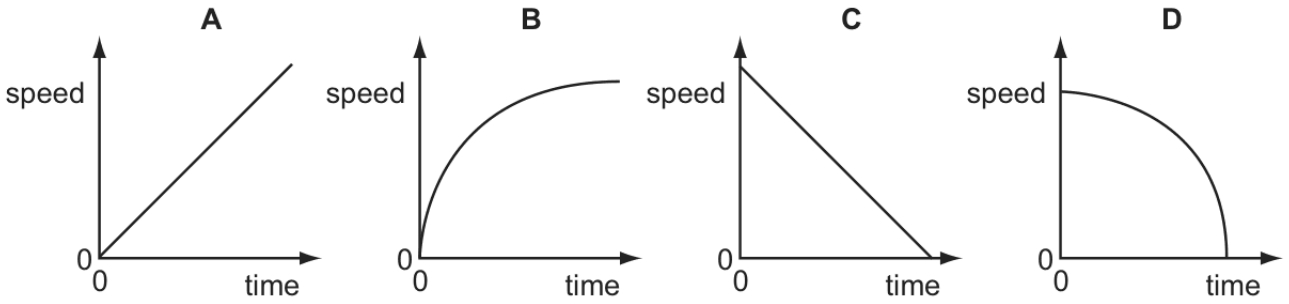
4	**A**

QUESTION · 5054_w12_qp_12.pdf · source page 2 of 16

3 A table-tennis ball is released from the top of an evacuated tube.



Which graph shows how the speed of the ball changes with time as it falls to the bottom of the tube?

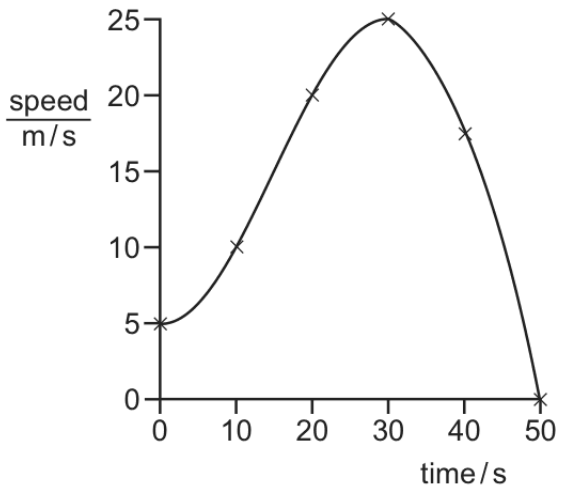


MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 3

3 **A**

QUESTION · 5054_w12_qp_12.pdf · source page 3 of 16

4 The speed-time graph for a car is shown.



What is the acceleration of the car at 30 s?

A 0 **B** $\frac{25-5}{30}$ m/s^2 **C** $\frac{25}{30}$ m/s^2 **D** $\frac{25}{50}$ m/s^2

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 4

4	**A**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a diver working below the surface of a lake. The density of the water in the lake is 1000 kg/m^3, the atmospheric pressure at the surface is 1.0×10^5 Pa and the gravitational field strength is 10 N/kg.

Fig. 9.1

The diver inflates a balloon with air at a depth of 15 m and attaches the balloon to a tray of objects.

(a) Calculate

(i) the pressure due to 15 m of water,

pressure = ...[2]

(ii) the total pressure at 15 m below the surface of the lake.

total pressure = ...[1]

(b) The air in the balloon occupies a volume of 0.048 m^3 at the pressure calculated in **(a)(ii)**. The diver releases the tray and the balloon, and they begin to rise. The temperature of the air in the balloon does not change.

(i) Calculate the volume occupied by the air in the balloon at atmospheric pressure.

volume = ...[2]

For Examiner's Use

(ii) The pressure of the air inside the balloon is less at the surface than at a depth of 15 m. Explain, in terms of the air molecules inside the balloon, why the pressure is less.

.....

.....

.....

.....[3]

(c) State one difference between the arrangement of the molecules of water in the lake and the molecules of air in the balloon.

.....

.....[1]

(d) When the diver releases the tray, the balloon accelerates upwards and reaches a constant speed before it arrives at the surface.

(i) Explain how the forces acting on the balloon cause it to behave in this way.

.....

.....

.....

.....[3]

(ii) On the axes in Fig. 9.2, sketch the distance-time graph for the balloon as it travels 15 m to the surface.

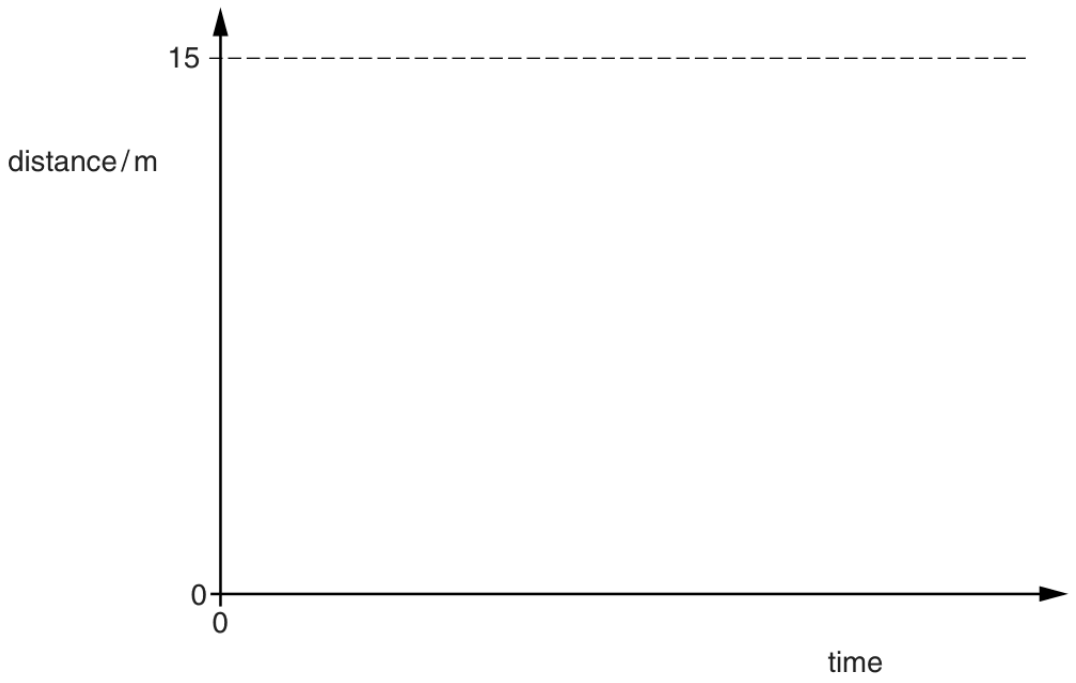


Fig. 9.2 [3]

Section B

9	**(a)**	**(i)**	$(p =)\rho hg$ **or** 1000 × 15 × 10	C1	
			1.5×10^5 Pa	A1	
		(ii)	2.5×10^5 Pa	B1	[3]
	(b)	**(i)**	$p_1V_1 = p_2V_2$ **or** 250 000 × 0.048 = 100 000 × V_2	C1	
			0.12 m^3	A1	
		(ii)	molecules/particles: further apart **or** their speed is unchanged	B1	
			(molecular) collisions with balloon/walls/unit area	B1	
			less frequent collisions (**not** if force/violence of each collision less)	B1	[5]
	(c)		water molecules: close(r)/move in clusters/move within the liquid **or** air molecules: far/further apart/move individually/move throughout container	B1	[1]
	(d)	**(i)**	net/resultant/unbalanced force upwards (at first) **or** upwards force greater	B1	
			friction/resistance/drag/downward force increases		
			(until) downward force = upward force/forces balance/no resultant force	B3	
		(ii)	starts from **marked** (0,0) **or** initial gradient = 0	B1	
			increasing gradient initially	B1	
			constant gradient (must be greater than zero) finally	B1	[6]

[Total: 15]

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows how the length of a spring varies as the force applied to it increases.

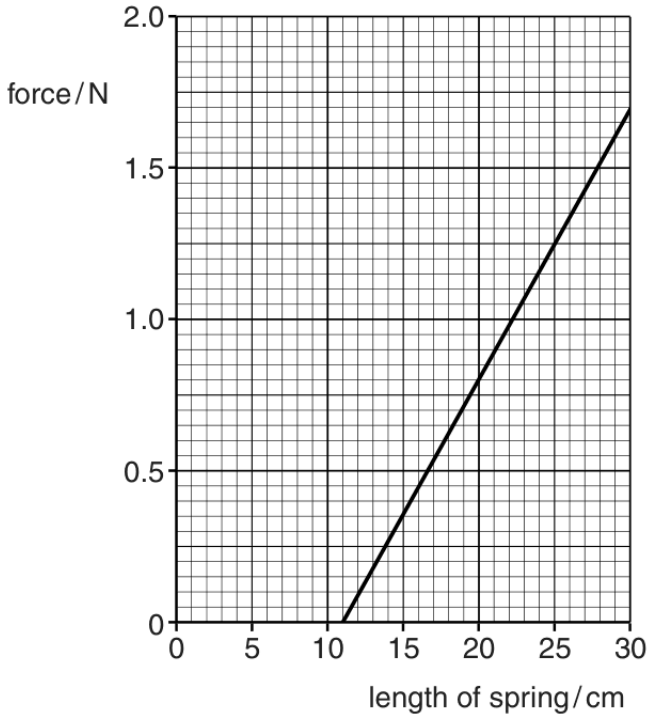


Fig. 1.1

(a) Determine the length of the unstretched spring.

length = ...[1]

(b) Explain how the graph shows that the limit of proportionality is not reached.

.....

.....[1]

For Examiner's Use

(c) The spring is attached to a mass M of 0.20 kg and placed on a frictionless surface, as shown in Fig. 1.2.

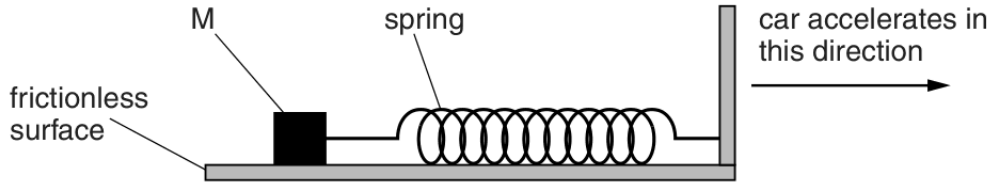


Fig. 1.2 (not to scale)

The apparatus is placed on the floor of a car.

When the car accelerates uniformly in the direction shown, the spring extends.

(i) State what is meant by a *uniform acceleration*.

.....

.....

.....[2]

(ii) The extension of the spring is 9.0 cm.

Using Fig. 1.1, determine

1. the horizontal force on M,

force = ...[1]

2. the acceleration of M.

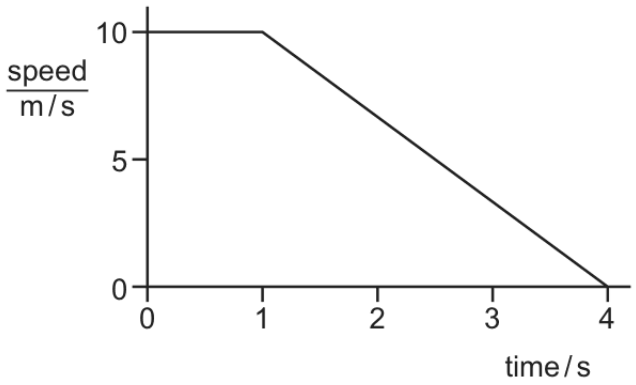
acceleration = ...[2]

Section A

1	**(a)**	11 cm	B1	
	(b)	(graph is) a straight line/linear **or** has constant gradient **or** not curved	B1	
	(c) **(i)**	change in speed/velocity	M1	
		same change in uniform/same time **or** in 1 s	A1	
	(ii) **1.**	0.8 N	B1	
	2.	(a=) F/m algebraic or numerical e.g. $F = ma$; 0.8/0.2	C1	
		(ecf **1.** but **not** if F = 0)		
		4(.0) m/s^2	A1	[7]

QUESTION · 5054_s13_qp_12.pdf · source page 3 of 16

4 The diagram shows the speed-time graph of the motion of a car for four seconds.



What is the distance travelled by the car in the four seconds?

A 15 m **B** 25 m **C** 30 m **D** 40 m

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4 **B**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

Answer only one of the two alternative parts in **Question 8**.

1 A ball rolls down a slope, as shown in Fig. 1.1.

The metre rule shows the position of the ball at times $t = 0$, 1.0 s, 2.0 s and 3.0 s.

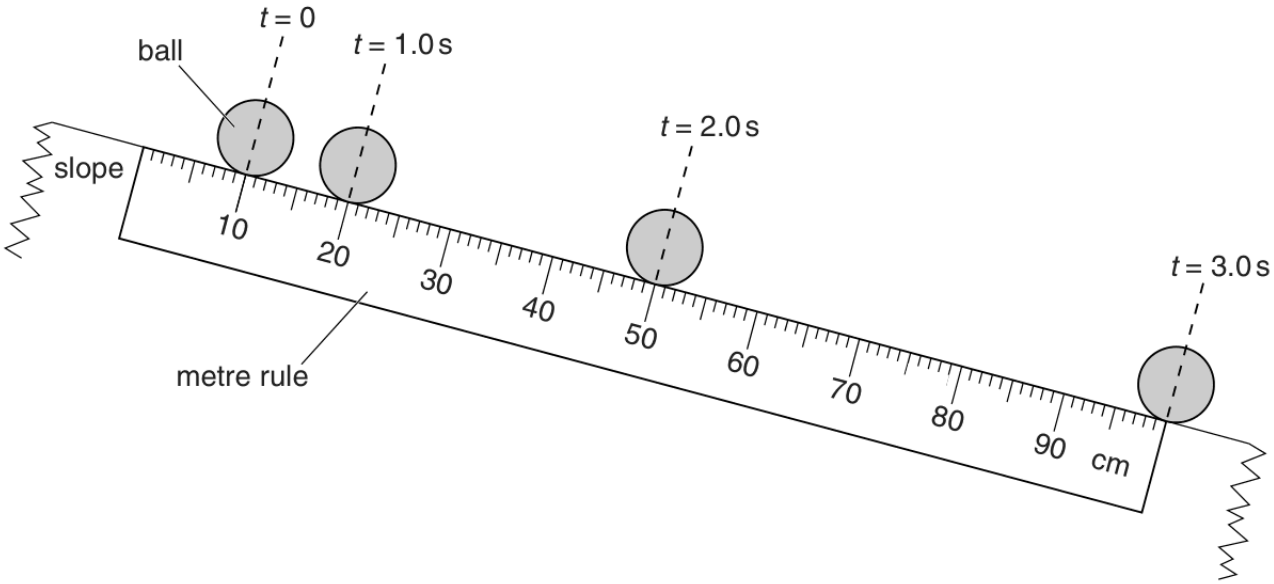


Fig. 1.1

(a) Explain how Fig. 1.1 shows that the ball is accelerating.

.....

..... [1]

(b) Calculate the average speed of the ball between $t = 1.0$ s and 3.0 s.

average speed = ... [2]

(c) Two of the forces that act on the ball are air resistance and weight.

State what, if anything, happens to these forces as the ball accelerates.

air resistance: .....

weight: .....
[2]

(d) Explain why, if the slope is long enough, the ball eventually reaches a constant speed.

.....

..... [1]

Section A

1	**(a)**	travels further in each second / in same time / between images	B1	
	(b)	(*s*=) *d*/*t* in any form algebraic or numerical	C1	
		40 cm/s; 0.4(0) m/s	A1	
	(c)	air resistance increases	B1	
		weight constant	B1	
	(d)	forces balance /cancel **or** no resultant/net force **or** resultant of any two forces equal and opposite to third	B1	[6]

QUESTION · 5054_w13_qp_11.pdf · source page 3 of 20

3 An object moves from P to Q in 10 s with uniform acceleration.

velocity at P = 5 m / s

velocity at Q = 12 m / s

What is the acceleration?

A $0.5\,m/s^2$ **B** $0.7\,m/s^2$ **C** $1.2\,m/s^2$ **D** $1.7\,m/s^2$

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 3

3	**B**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a skydiver, of mass 70 kg, falling towards the Earth at constant speed, a long time after jumping from an aeroplane.

Fig. 9.1

At time $t = 0$, he receives a radio signal. He opens his parachute 12 s later. Fig. 9.2 is the speed-time graph for the skydiver.

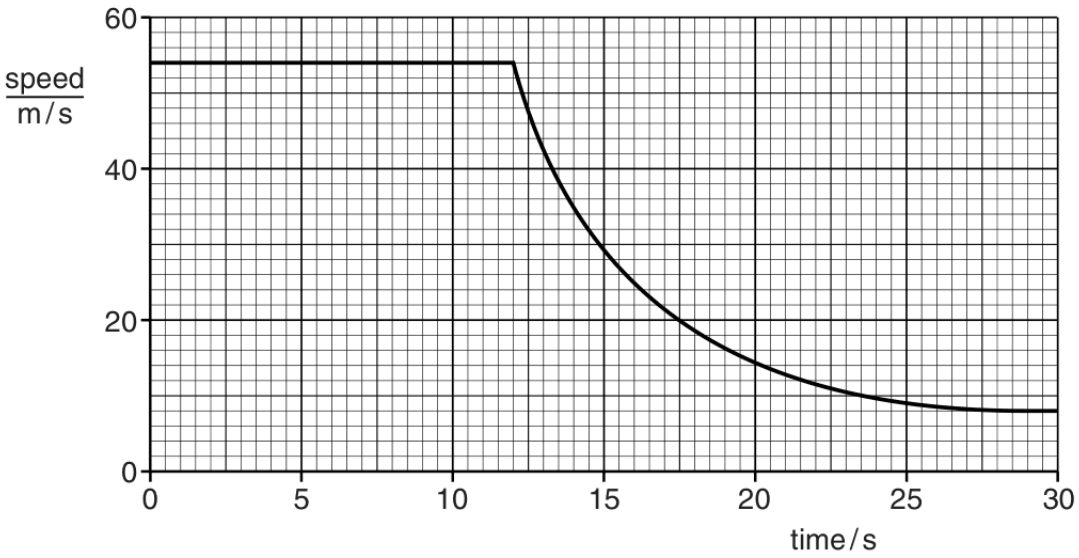


Fig. 9.2

(a) State the difference between *speed* and *velocity*.

.....

.....[1]

(b) The gravitational field strength g is 10 N/kg.

(i) Calculate the weight of the skydiver.

weight = ...[1]

For Examiner's Use

(ii) State the size of the air resistance acting on the skydiver between $t = 0$ and $t = 12$ s.

air resistance = ...[1]

(c) For the period between $t = 0$ and $t = 12$ s, determine

(i) the speed of the skydiver,

speed = ...[1]

(ii) the distance fallen by the skydiver,

distance = ...[2]

(iii) the change in the gravitational potential energy of the skydiver.

change in energy = ...[2]

(d) As he falls at constant speed, his gravitational potential energy changes into another form. State the name of this other form of energy.

.....[1]

(e) **(i)** State and explain what happens to the air resistance as the skydiver opens his parachute.

.....

.....

.....[2]

(ii) State and explain the effect on the motion of the skydiver of opening the parachute.

.....

.....

.....[2]

(f) By $t = 15$ s, his parachute is fully open.

State and explain what happens to the air resistance after $t = 15$ s.

.....

.....

.....[2]

Section B

9	**(a)**	speed does not have direction **and** velocity does or speed = distance/time **and** velocity = displacement/time or speed is a scalar **and** velocity is a vector	B1	[1]
	(b) **(i)**	700 N	B1	
	(ii)	700 N	B1	[2]
	(c) **(i)**	54 m / s	B1	
	(ii)	(height/distance =) area (under graph) **or** (x =)vt **or** 54 × 12 648/650 m	C1 A1	
	(iii)	(GPE =) mgh **or** 70 × 10 × 648 4.5/4.54/4.536 × 10^5 J	C1 A1	[5]
	(d)	(becomes) heat/thermal energy/internal energy (**not** kinetic energy (of skydiver) unless qualified as KE of air)	B1	[1]
	(e) **(i)**	(air resistance) increases larger area of parachute	B1 B1	
	(ii)	(skydiver) decelerates/slows down (**not** rises up) net upward force	B1 B1	[4]
	(f)	air resistance decreases speed decreases	B1 B1	[2]

[Total: 15]

QUESTION · 5054_w13_qp_12.pdf · source page 2 of 20

2 An object moves from P to Q in 10 s with uniform acceleration.

velocity at P = 5 m / s

velocity at Q = 12 m / s

What is the acceleration?

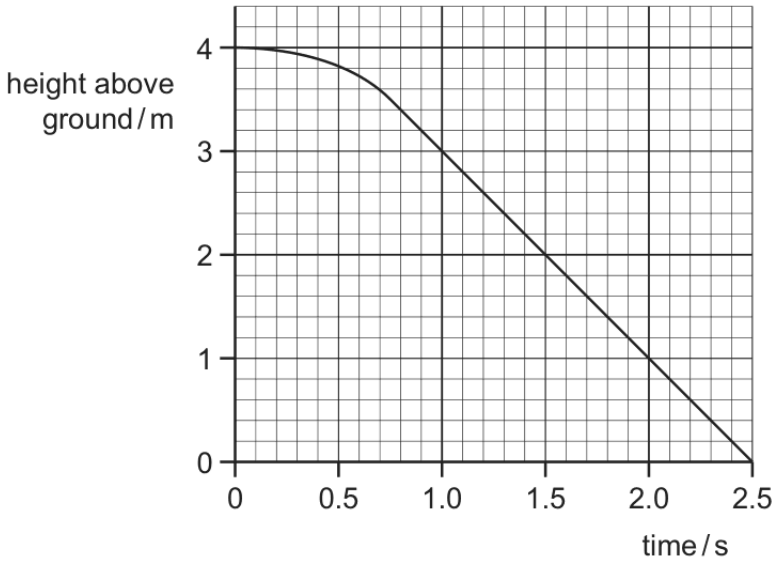
A $0.5\,m/s^2$ **B** $0.7\,m/s^2$ **C** $1.2\,m/s^2$ **D** $1.7\,m/s^2$

MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 2

2	**B**

QUESTION · 5054_w13_qp_12.pdf · source page 3 of 20

3 The graph shows how the height of an object above the ground changes with time.



What is the terminal velocity?

A 1.0 m/s **B** 1.3 m/s **C** 1.6 m/s **D** 2.0 m/s

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3 **D**

For Examiner's Use

10 Fig. 10.1 shows an electric train of mass 5.5×10^5 kg.

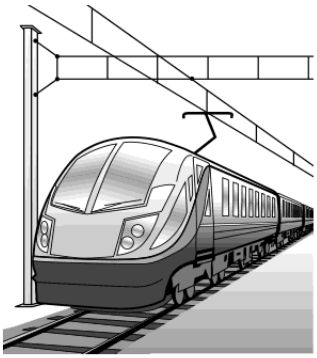


Fig. 10.1

The train is initially at rest. The electric motor exerts a constant force and the train accelerates. Its acceleration decreases until the train reaches a speed of 40 m/s.

The train then continues at this constant speed.

(a) (i) On the axes in Fig. 10.2, sketch a speed-time graph for the train.

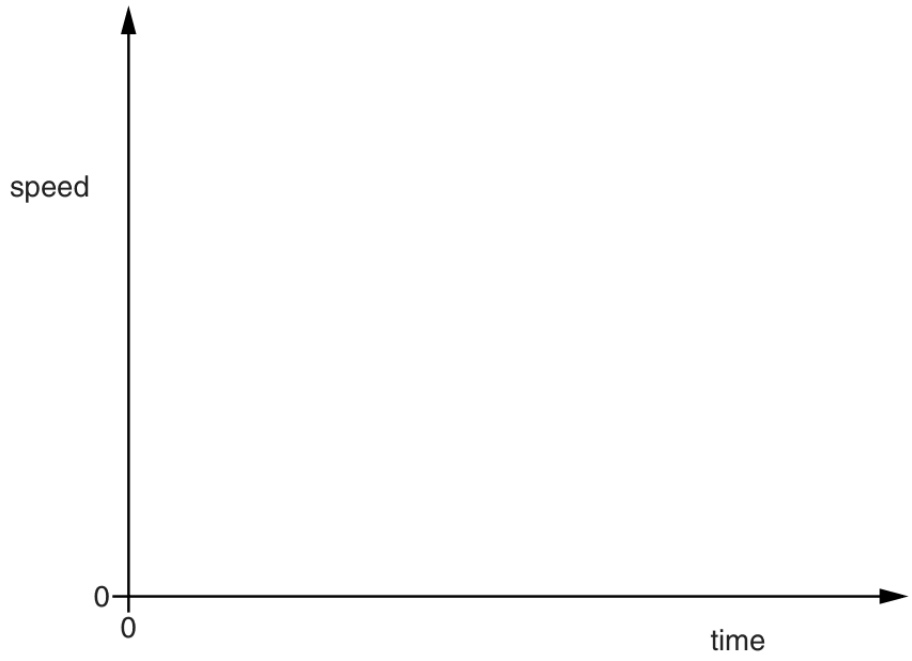


Fig. 10.2

[3]

(ii) Describe how the speed-time graph for the train is used to determine the distance travelled by the train when it is accelerating.

.....

.....

.....[2]

For Examiner's Use

(b) Explain, in terms of the forces acting,

(i) why the acceleration of the train decreases,

.....

.....

.....[2]

(ii) why the train eventually reaches constant speed.

.....

.....[1]

(c) **(i)** Calculate the kinetic energy of the train when it is travelling at a speed of 40 m/s.

kinetic energy = ...[3]

(ii) As the train accelerates to 40 m/s, electrical energy supplied to the motor is converted into kinetic energy of the train. The efficiency of this process is 0.40 (40 %).

Calculate the electrical energy supplied to the motor.

energy = ...[2]

(iii) The electrical energy is generated in an oil-fired power station. Chemical energy in oil is converted into the electrical energy supplied to the motor.

Suggest two places where energy is lost as heat in this process.

1. .....

2. .....
[2]

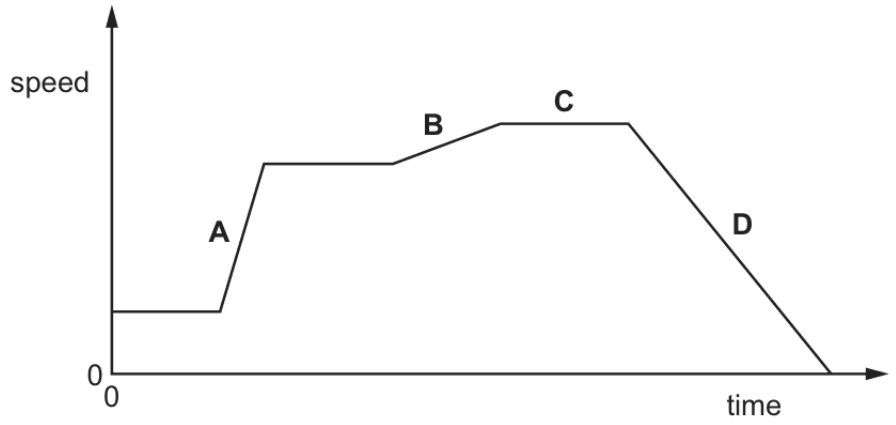
10	**(a)**	**(i)**	start at origin **and** not horizontal	B1	
			gradient (gradually) decreasing (**ignore** sudden decrease) (not if part of curve above horizontal section)	B1	
			final horizontal section (≥ 1 cm) (not if v is shown as ≠ 40 m/s)	B1	
		(ii)	area **under** the graph **or** count squares **under** graph between t = 0 and horizontal section **or** when speed is changing **or** calculate equivalent distance to 1 cm^2 (after counting squares)	M1 A1	[5]
	(b)	**(i)**	friction/air resistance increases (as speed increases)	B1	
			resultant force decreases (**not** if driving force decreases)	B1	
		(ii)	(air resistance increases until) net force becomes zero **or** forces balance **or** air resistance and driving/forward force are in equilibrium/balanced/equal	B1	[3]
	(c)	**(i)**	(KE =) $\frac{1}{2}mv^2$	C1	
			$\frac{1}{2} \times 5.5 \times 10^5 \times 40^2$	C1	
			4.4×10^8 J	A1	
		(ii)	(total energy input =) useful energy output efficiency **or** efficiency = useful power output/total power input **or** $4.4 \times 10^8/0.40$	C1	
			1.1×10^9 J	A1	
		(iii)	two valid examples e.g. furnace/boiler/turbines/generator/coils/cooling water/cooling towers/heat exchanger/transformer/chimney/waste gases/ transmission cables/lines/wires (**ignore** power station/all parts of motor)	B2	[7]

[Total: 15]

QUESTION · 5054_s14_qp_11.pdf · source page 2 of 16

3 The graph shows how the speed of a car travelling in a straight line changes with time.

Which section shows the largest acceleration?



MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 3

3 **A**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A small boat travels with a velocity of 4.0 m/s due north in an area of still water.

(a) State what is meant by *velocity*.

.....

.....

..... [1]

(b) The boat travels from the area of still water into an area where the velocity of the water is 2.0 m/s towards the north-east, as shown in Fig. 1.1.

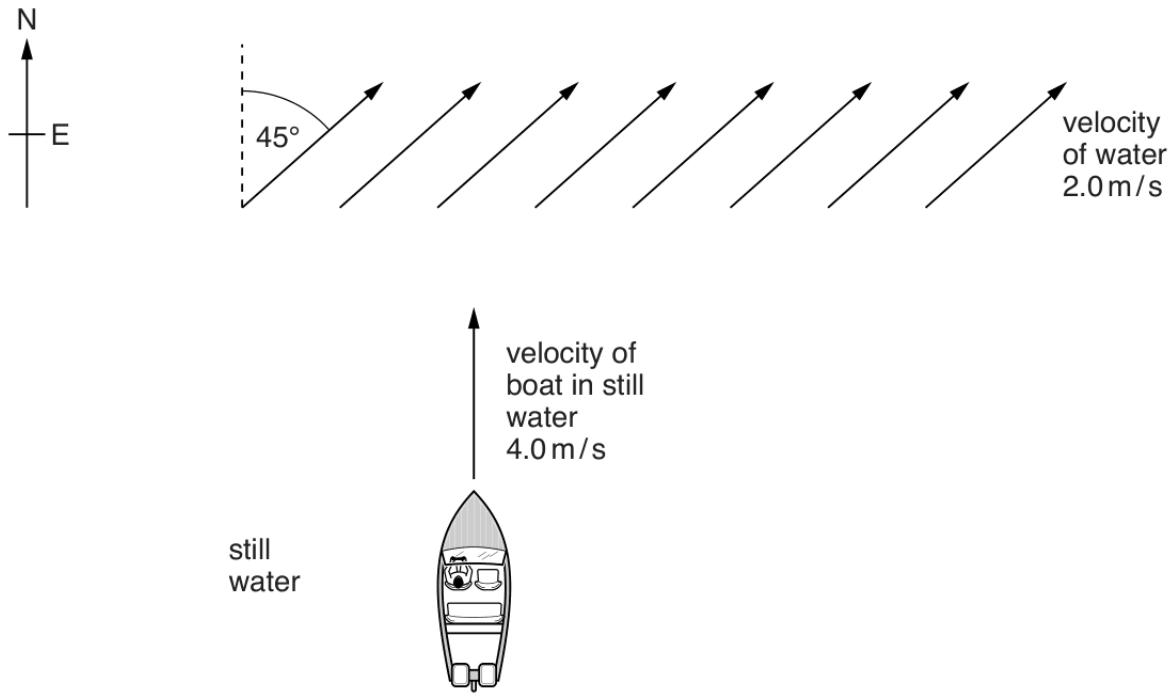


Fig. 1.1 (not to scale)

Combining the initial velocity of the boat with the velocity of the water gives the resultant velocity of the boat.

In the space on the page opposite, draw a vector diagram to show the resultant velocity.

Use your diagram to find the size and direction of the resultant velocity.

size of resultant velocity = ...

direction of resultant velocity = ...

[4]

Section A

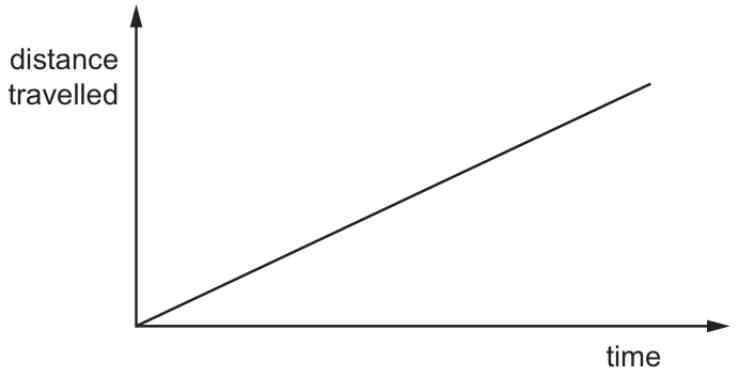
1 **(a)** speed and direction **or** (change in) distance per sec / unit time + direction
or (change in) displacement per sec / unit time [B1]

(b) two vectors at 45° and one twice the other by eye or labelled 2 (m / s) and 4 (m / s) [B1]
correct resultant shown for two labelled vectors at any angle with directions shown [B1]
resultant speed 5.6 ± 0.2 m / s unit needed [B1]
direction (0)15° **or** N15 °E, etc. **or** any clear direction expressed
or angle mentioned on answer line **and** shown on diagram [B1]

[5]

QUESTION · 5054_s14_qp_12.pdf · source page 3 of 16

6 The distance travelled by a car is increasing uniformly as it is driven along a straight road up a hill.



Which quantity for the car is constant but not zero?

A acceleration

B gravitational potential energy

C kinetic energy

D resultant force

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6	**C**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 A children's ride consists of a steel cable that runs between two posts of different heights, as shown in Fig. 9.1.

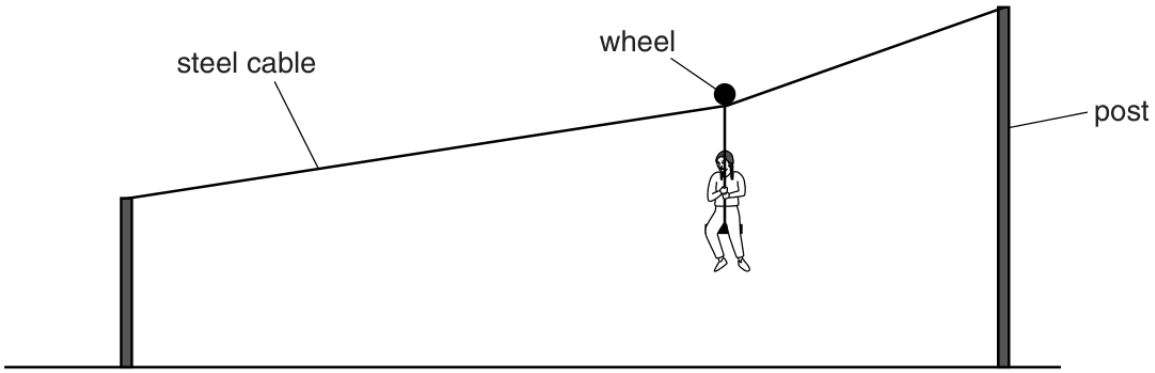


Fig. 9.1

A girl starts and finishes the ride at rest. Her horizontal motion can be taken as
- an initial uniform acceleration for 3.0 s, followed by
- a constant speed of 2.4 m/s for a further 5.0 s and
- a final uniform deceleration that lasts for 1.0 s.

(a) On Fig. 9.2, draw a speed-time graph of the horizontal motion.

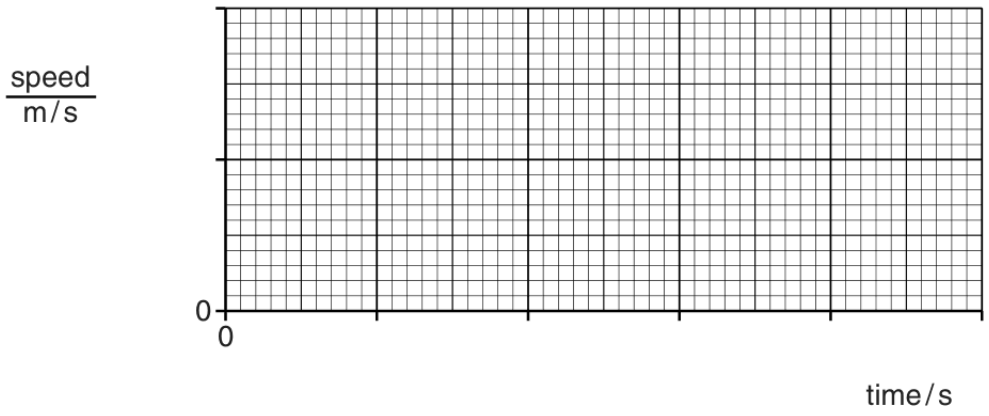


Fig. 9.2 [3]

(b) Explain what is meant by *uniform acceleration*.

.....

.....

..... [2]

(c) The final deceleration is larger in size than the initial acceleration.

Explain how the data shows this.

.....

..... [1]

(d) Calculate the horizontal distance travelled by the girl in the first 8.0 s.

distance = ... [3]

(e) **(i)** The girl has a mass of 30 kg and falls a vertical distance of 1.6 m during the ride.

The gravitational field strength g is 10 N / kg.

Calculate the decrease in gravitational potential energy of the girl.

decrease in potential energy = ... [2]

(ii) The gain in kinetic energy of the girl is less than the decrease in her potential energy. Suggest one reason for this.

.....

..... [1]

(f) A group of pupils make measurements to show that the girl's speed is constant during the middle section of the ride.

Suggest what measurements are made and how they show that the speed is constant.

.....

.....

.....

.....

.....

..... [3]

9 **(a)** straight line from (0, 0) to (3, 2.4) [B1]
horizontal line from 3 s to 8 s [B1]
straight line from **end of a horizontal line** to zero in 1 s [B1]

(b) constant / same increase in velocity **or constant** change in velocity [C1]
constant / same increase in velocity per sec / unit time [A1]

(c) occurs in a short(er) time
or acceleration took 3 s and deceleration took 1 s [B1]

(d) (d =) speed × time numerical or algebraic **or** area under graph clear [C1]
1.2 × 3 **or** 3.6 (m) **or** 2.4 × 5 **or** 12 seen [C1]
15.6 m **or** 16 m [A1]

(e) **(i)** mgh seen in any algebraic or numerical form, e.g. 30 × 10 × 1.6 [C1]
480 J [A1]

(ii) heat or thermal energy or sound produced
or work done against friction / air resistance [B1]

(f) at least two distances and corresponding times mentioned [C1]

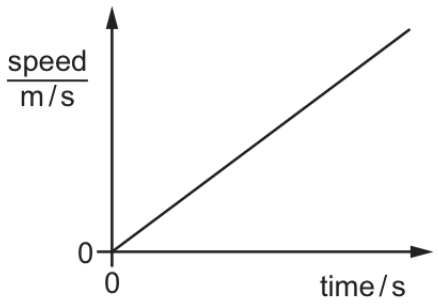
how the **actual measurement** is made, e.g. **(any one from)**
- make mark on ground every second and **measure** distances
- note video position every sec and use a scale to **find** distances
- make mark on ground every meter and **measure / take** time as girl passes [A1]

how constant speed is proved using measurement, e.g. **(any one from)**
- same distance between each position for the same time interval
- same time interval for equal distances
- $\Delta d / \Delta t$ constant **or** slope of distance-time graph constant [B1]

[15]

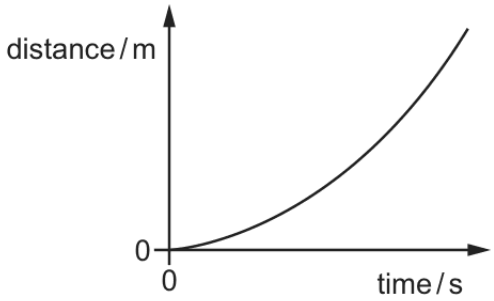
QUESTION · 5054_w14_qp_11.pdf · source page 4 of 20

4 The speed-time graph represents a short journey.

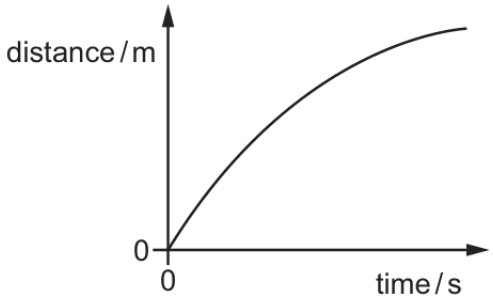


Which distance-time graph represents the same journey?

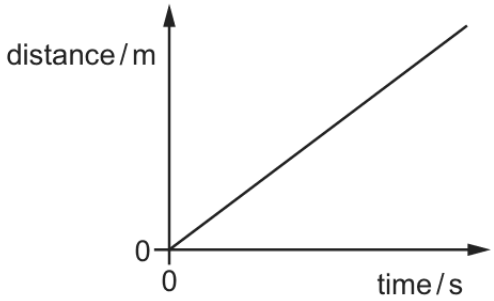
A



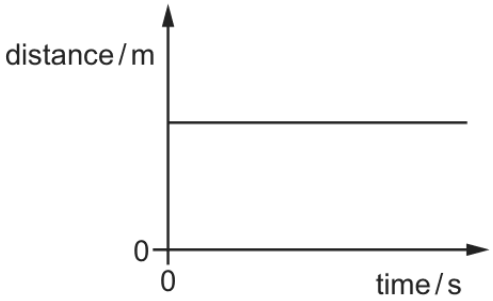
B



C



D



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4 **A**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a motorcycle during a race.

Fig. 1.1

The motorcycle accelerates along a straight section of the track from a speed of 40 m/s to maximum speed.

Fig. 1.2 is the speed-time graph for the motorcycle along the straight section of the track.

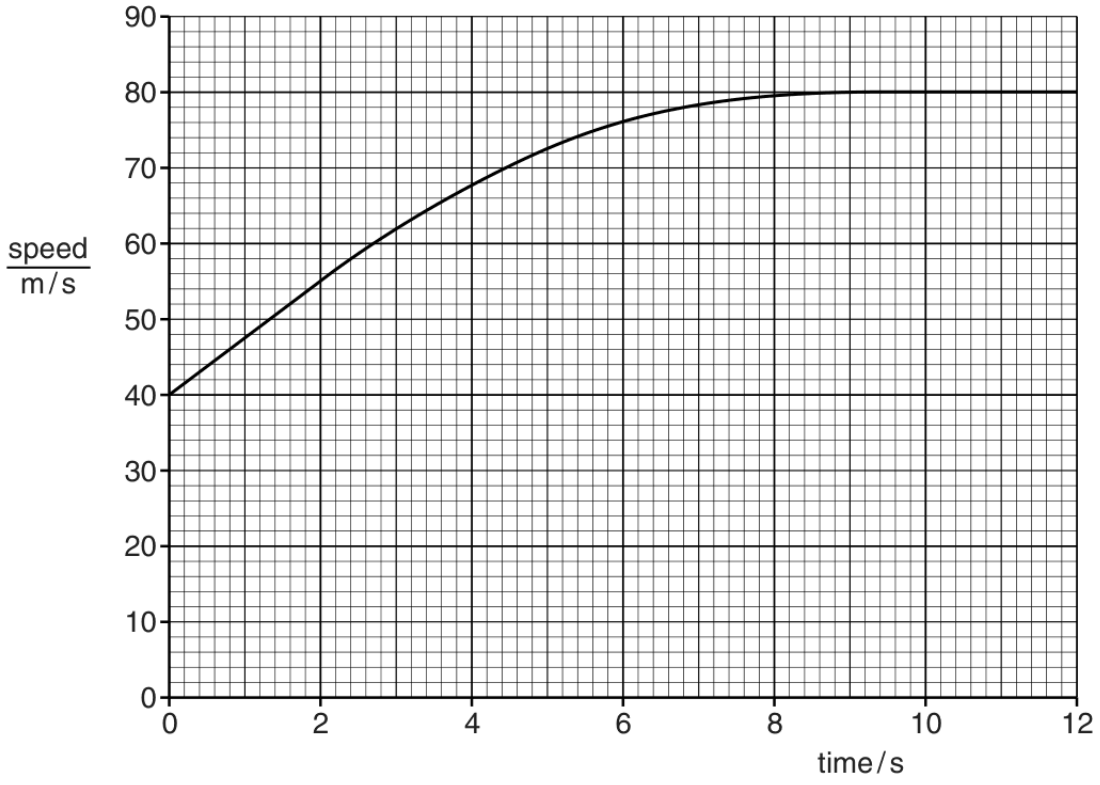


Fig. 1.2

The mass of the motorcycle is 180 kg.

(a) For the time 0 to 2.0 s, determine

(i) the acceleration of the motorcycle,

acceleration = ...[2]

(ii) the resultant force acting on the motorcycle.

force = ...[2]

(b) The driving force acting on the motorcycle remains constant throughout the 12 s spent on the straight section of track.

(i) Using Fig. 1.2, describe how the acceleration of the motorcycle changes during this time.

.....

.....[1]

(ii) Explain, in terms of the forces acting, why the acceleration changes in this way.

.....

.....

.....

.....[3]

Section A

1	**(a)**	**(i)**	(a =) $(v-u)/t$ **or** $\Delta v/t$ **or** (55–40)/2 **or** equivalent values from graph	C1	
			$7.5\,m/s^2$	A1	
		(ii)	(F =) ma **or** 180×7.5	C1	
			1300 / 1350 / 1400 N	A1	
	(b)	**(i)**	(acceleration) decreases (to zero)	B1	
		(ii)	air resistance / friction / drag mentioned	B1	
			air resistance / friction / drag increases (with speed) **or** resultant force decreases (with speed)	B1	
			(finally) (air) resistance = driving force **or** resultant is zero	B1	**[8]**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 **(a)** Explain what is meant by *uniform acceleration*.

.....

.....

.....[2]

(b) Acceleration is a vector quantity.

State how a vector quantity differs from a scalar quantity.

.....

.....[1]

(c) A rock from space is travelling in a straight line at high speed when it enters the Earth's atmosphere. Fig. 9.1 is the speed-time graph for the rock from time $t = 0$ to time $t = 50$ s.

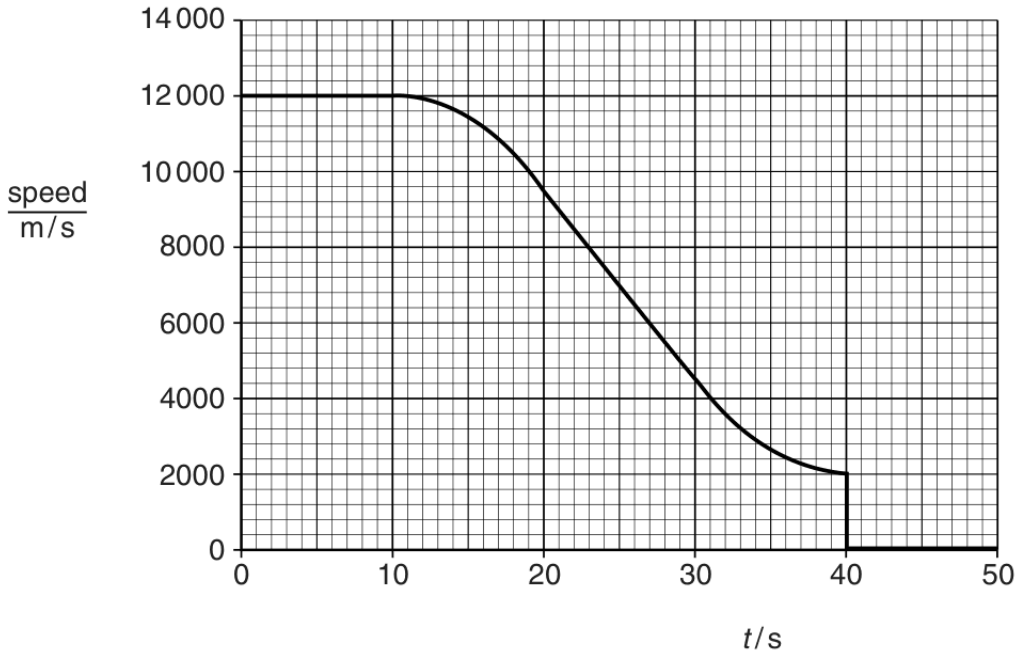


Fig. 9.1

(i) On Fig. 9.1, mark

- a letter X, where the rock is moving with a constant speed,
- a letter Y, where the rock has a uniform deceleration,
- a letter Z, where the rock has a non-uniform deceleration.

[3]

(ii) At time $t = 25\,s$, the mass of the rock is 8.4 kg. For the time $t = 25\,s$, determine

1. the size of the acceleration of the rock,

acceleration = ...[3]

2. the size of the resultant force on the rock.

force = ...[2]

(iii) Fig. 9.2 shows the rock at $t = 25\,s$.

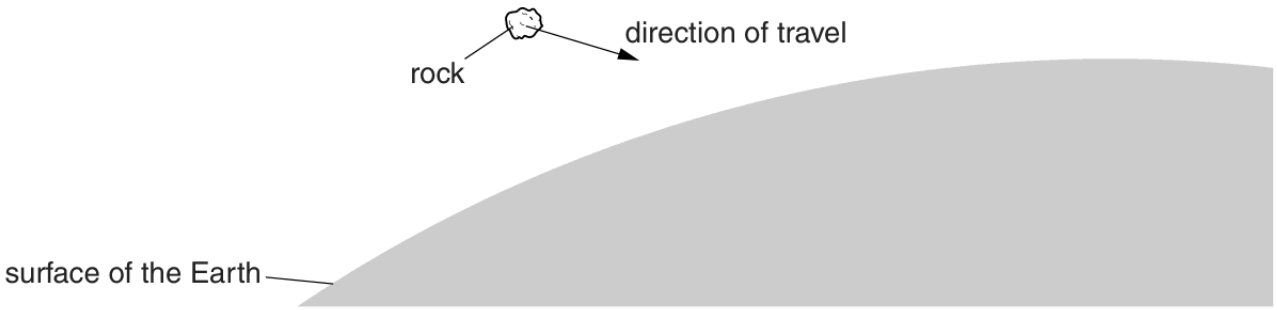


Fig. 9.2 (not to scale)

1. On Fig. 9.2, draw and label two arrows to show the directions of the gravitational force *F* and the air resistance *R* acting on the rock. [2]

2. Suggest why the size of the air resistance changes as the rock travels through the Earth's atmosphere.

.....

.....[1]

(iv) Suggest what happens to the rock at $t = 40\,s$.

.....

.....[1]

Section B

9	**(a)**	changing speed / velocity		C1	
		change in speed / velocity / time constant **or** (v–u) / t constant **or** constant / equal rate of change of speed / velocity		A1	**[2]**
	(b)	(a vector quantity has) direction		B1	**[1]**
	(c)	**(i)**	**1.** X between $t \geq 0$ and $t < 10$ s	B1	
			2. Y between $t > 20$ s and $t < 30$ s	B1	
			3. Z between $t > 10$ s and $t < 20$ s **or** between $t > 30$ s and $t < 40$ s	B1	
		(ii)	**1. two** speed values from graph between 15 and 35 s (±1 mm)	C1	
			two corresponding time values from graph between 15 and 35 s (±1 mm) **or** (a =)$\Delta v / t$	C1	
			500 m / s^2	A1	
			2. (F =) ma **or** 8.4 × 500	C1	
			4200 N	A1	
		(iii)	**1.** arrow labelled F perpendicular to surface of Earth	B1	
			arrow labelled R opposite to direction of travel (by eye) from rock	B1	
			2. speed changes **or** density / pressure of air changes **or** cross-sectional area (of rock) changes	B1	
		(iv)	it hits the ground / surface of the earth **or** stops **or** speed is zero	B1	**[12]**
					[Total: 15]

QUESTION · 5054_s15_qp_11.pdf · source page 2 of 16

3 A student walks at a constant speed. He takes 100 s to walk 160 paces. The length of each pace is 0.80 m.

How far does the student walk in 50 s?

A 64 m **B** 80 m **C** 128 m **D** 256 m

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 3

3	**A**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the distance-time graph for two cyclists A and B. They start a 500 m race together but finish the race at different times.

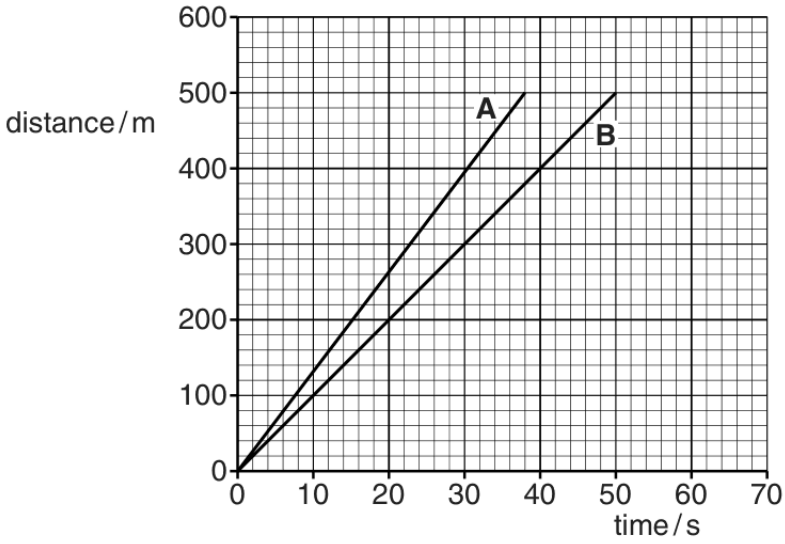


Fig. 1.1

(a) Use Fig. 1.1 to determine

(i) the distance between A and B at time $t = 20$ s,

.....[1]

(ii) the difference in the time taken by A and B for the race.

.....[1]

(b) Cyclist C starts the race at the same time as A and B and covers the first 200 m of the race at a constant speed of 5.0 m / s. He then accelerates and finishes the race at $t = 60$ s.

(i) On Fig. 1.1, draw the distance-time graph for cyclist C. [2]

(ii) Calculate the average speed of cyclist C for the whole race.

speed = ...[2]

1	**(a)**	**(i)**	60 m	B1
		(ii)	12 s	B1
	(b)	**(i)**	straight line from origin to 200 m at 40 s	B1
			any line straight or curved from (40,200) to (60,500)	B1
		(ii)	$s = d / t$ or 500 / 60	C1
			8.3 m / s	A1

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a satellite in orbit around the Earth.

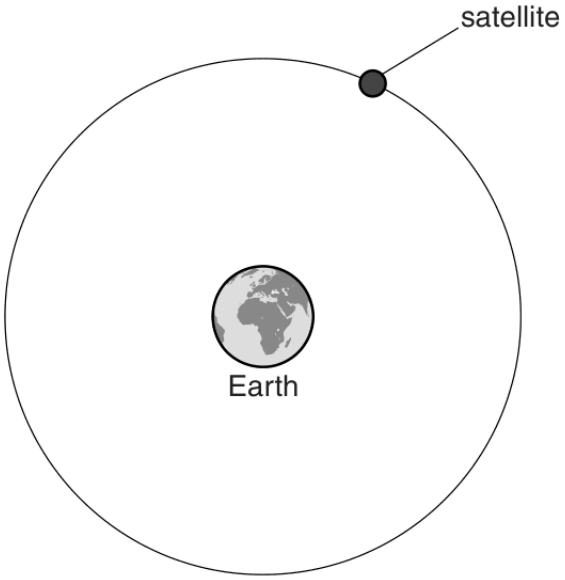


Fig. 9.1 (not to scale)

(a) The satellite travels at a constant speed in a circular orbit.

(i) Underline the quantities in the list below that are scalars.

acceleration force mass speed velocity [2]

(ii) The velocity of the satellite changes, but its speed is constant.

1. State what is meant by *velocity*. .....

.....

.....[1]

2. Explain why the velocity changes. .....

.....

.....[1]

(iii) Explain what makes this satellite move in an orbit that is circular.

.....

.....

.....

.....[2]

(b) The satellite is placed into orbit by a rocket. Fig. 9.2 shows the rocket as it takes off.

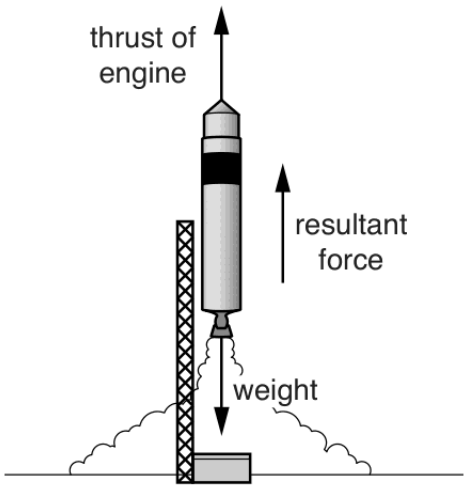


Fig. 9.2

The rocket and fuel have a total mass of 40 000 kg and a total weight of 400 000 N. The resultant force acting upwards on the rocket is 50 000 N.

(i) Calculate the thrust produced by the rocket engine.

thrust = ...[1]

(ii) Calculate the acceleration of the rocket.

acceleration = ...[2]

(c) In the first four minutes after take-off, the acceleration of the rocket is uniform.

(i) State what is meant by a *uniform acceleration*.

.....

.....

.....[2]

(ii) Fig. 9.3 describes the motion of the rocket in the first 12 minutes.

time / minutes	motion of the rocket
0 to 4	uniform acceleration
4 to 6	increasing acceleration
6 to 10	decreasing acceleration
10 to 12	constant speed

Fig. 9.3

On Fig. 9.4, sketch the speed-time graph of the rocket for the first 12 minutes.

You do not need to give values for the speed.

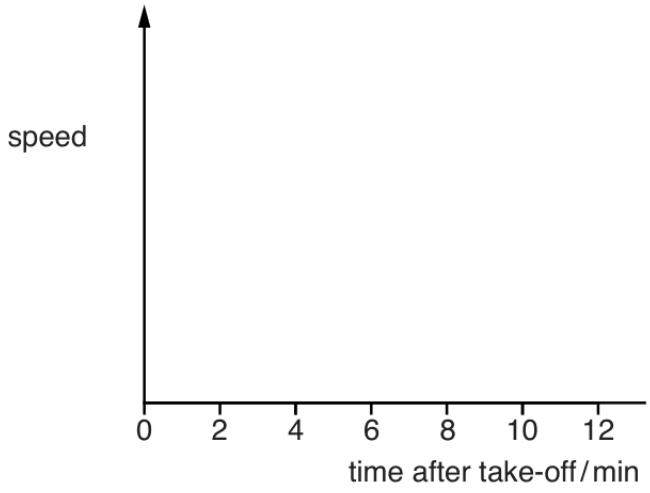


Fig. 9.4

[3]

(iii) State how the speed-time graph in **(ii)** can be used to find the distance travelled by the rocket.

.....

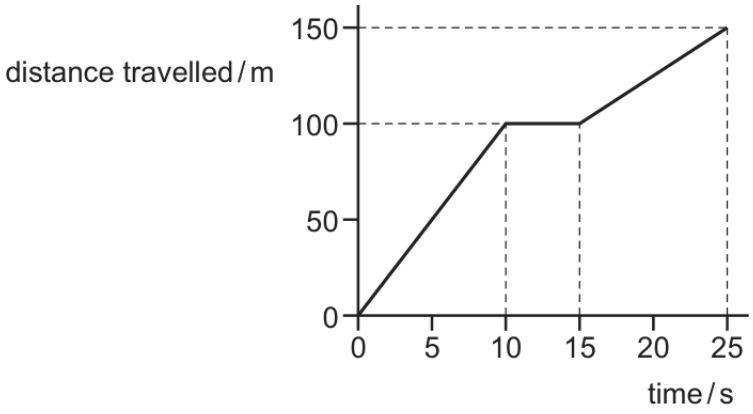
.....[1]

9	**(a)**	**(i)**	speed and mass	B2
		(ii)	**1** speed and direction **or** distance / time and direction **or** displacement / time	B1
			2 direction changes	B1
		(iii)	force of gravity from / towards Earth	B1
			force is centripetal **or** at right angles to motion / velocity	B1
	(b)	**(i)**	450 000 N	B1
		(ii)	(a =) F/m or 50 000 / 40 000	C1
			1.25 m / s^2	A1
	(c)	**(i)**	same change in velocity / speed	M1
			in same time period	A1
		(ii)	start at origin and straight line for first 4 minutes	B1
			gradient increases at first after 4 and then decreases	B1
			constant speed from 10 minutes until 12 minutes	B1
		(iii)	area **under** graph	B1

QUESTION · 5054_s15_qp_12.pdf · source page 2 of 16

3 A cyclist takes a ride lasting 25 s.

The diagram shows how her distance travelled from the starting position varies with time.



What is her average speed for the whole ride?

A 6.0 m/s **B** 7.5 m/s **C** 10.0 m/s **D** 11.0 m/s

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3 **A**

QUESTION · 5054_s15_qp_12.pdf · source page 3 of 16

4 A car begins to move. It speeds up until it reaches a constant speed. It continues to travel at this constant speed for the rest of the journey.

What happens to the acceleration and what happens to the velocity of the car during the journey?

A Both the acceleration and the velocity change.

B Only the acceleration changes.

C Only the velocity changes.

D Neither the acceleration nor the velocity changes.

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4	**A**

11 A bungee jumper falls from a bridge above a river, as shown in Fig. 11.1.

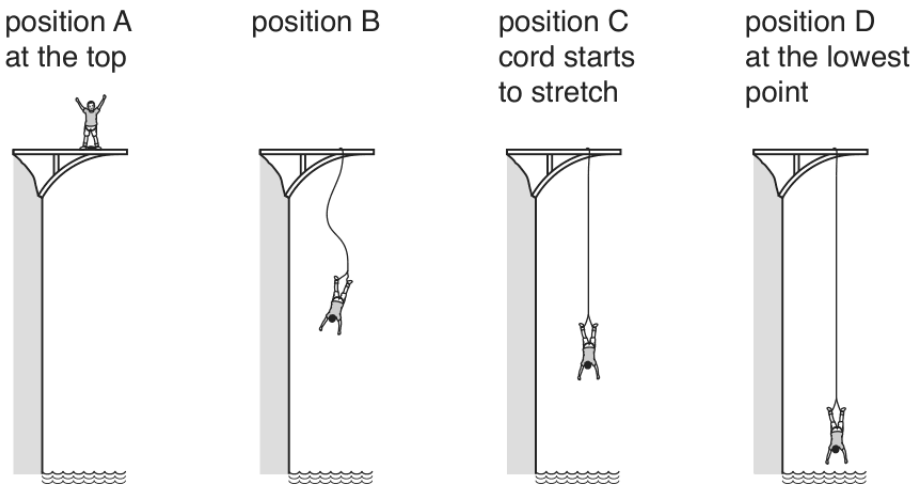


Fig. 11.1 (not to scale)

The man starts from position A in Fig. 11.1. The elastic cord starts to stretch at position C and he stops for the first time at position D. He continues to rise and fall.

Fig. 11.2 shows how the velocity of the man varies with time *t*.

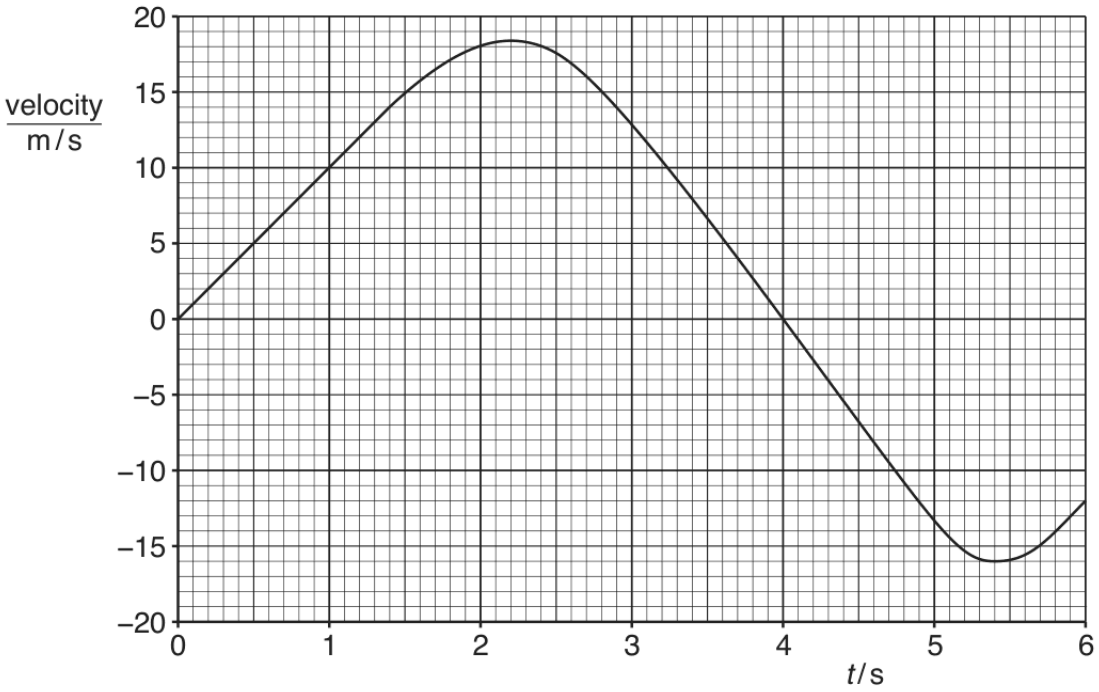


Fig. 11.2

(a) (i) State what is meant by *velocity*.

.....

..... [2]

(ii) State the difference between a positive velocity and a negative velocity.

.....

..... [1]

(iii) In the first 1.4 s the acceleration is uniform.

1. Using values from Fig. 11.2, determine the acceleration of the man in the first 1.4 s.

acceleration = ... [3]

2. Comment on your value of acceleration.

.....

..... [1]

(iv) 1. State the value of t when the man is at position D.

..... [1]

2. Explain, in terms of the forces acting, why the man is accelerating upwards at D.

.....

.....

.....

.....

..... [3]

(b) Fig. 11.3 shows the values for the gravitational potential energy of the man, the kinetic energy of the man and the elastic potential energy in the cord at A, C and D.

You may ignore the effect of air resistance in this question.

	gravitational potential energy/J	kinetic energy/J	elastic potential energy/J
position A	20000	0	0
position C	15000		0
position D	0	0	

Fig. 11.3

(i) Complete Fig. 11.3 to show the kinetic energy of the man at C and the elastic potential energy in the cord at D. [2]

(ii) The man has a mass of 50 kg.

The gravitational field strength g is 10 N/kg.

Using values from Fig. 11.3, calculate the vertical distance between A and C.

distance = ... [2]

11 (a) (i) distance (travelled) per second **or** speed — C1
distance (travelled) per second / speed in a given direction — A1
or displacement / time
or change in displacement per unit time
or displacement (travelled/covered) per unit time
or rate of change of displacement

(ii) opposite direction — B1

(iii) 1 value seen for v and corresponding value of t — C1
$0 < t \leqslant 1.4$ and $0 < v \leqslant 14$
(a=) v-u / t algebraic or numerical equation — C1
$10\,m/s^2$ — A1

2 sensible comment — A1

(iv) 1 4(.0 s) — B1

2 weight **or** force due to gravity mentioned (at D) — B1
mention of — B1
- upwards force (on man) from cord
- tension / elastic force from cord (on man)

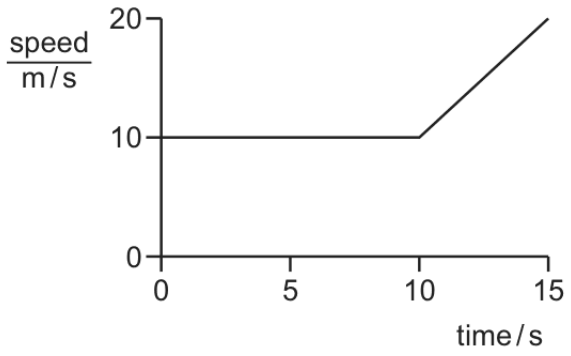
force in cord / upward force / tension greater than downwards force — B1
or resultant force upwards

(b) (i) 5000 — B1
20 000 — B1

(ii) (h =) PE / mg **or** $5000 = 50 \times 10 \times h$ — C1
10 m — A1

QUESTION · 5054_w15_qp_11.pdf · source page 2 of 16

3 An object travels at a constant speed of 10 m/s for 10 s. During the next 5 s, it accelerates uniformly to 20 m/s.



What is the total distance travelled by the object?

A 150 m **B** 175 m **C** 200 m **D** 300 m

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 3

3 **B**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows an archer firing an arrow at a target during a competition.

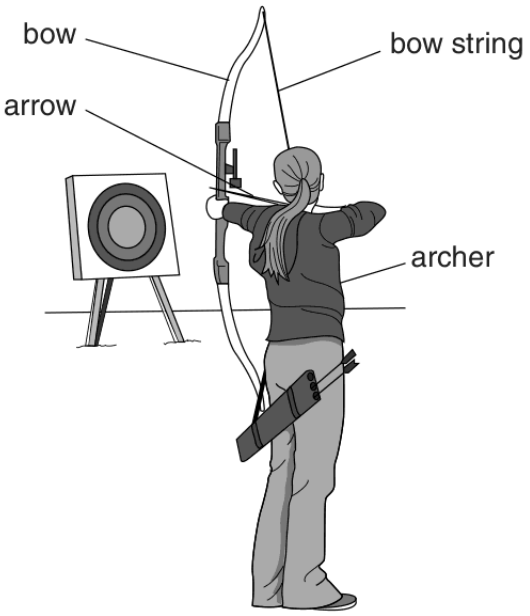


Fig. 1.1

The arrow has a mass of 0.018 kg and is initially at rest. The arrow leaves the bow 0.011 s after the bow string is released. When it leaves the bow, the velocity of the arrow is 95 m/s.

(a) Calculate

(i) the average acceleration of the arrow during the 0.011 s,

average acceleration = ... [2]

(ii) the average force exerted on the arrow during the 0.011 s.

average force = ... [2]

(b) In practice, the force exerted on the arrow gradually decreases to zero during firing.

On Fig. 1.2, sketch a possible speed-time graph for the arrow as it is being fired.

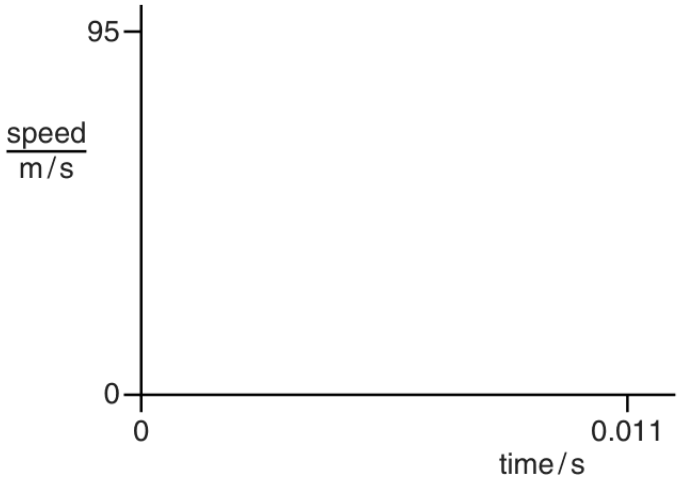


Fig. 1.2

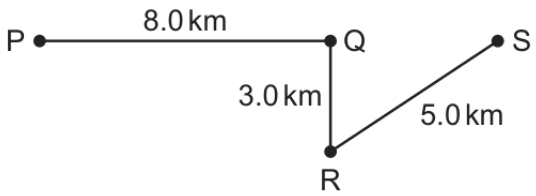
[2]

Section A

1	**(a)**	**(i)**	$(a =)\Delta v / t$ **or** $95 / 0.011$	C1	
			$8.6(3636) \times 10^3\ \text{m/s}^2$	A1	
		(ii)	$(F =)ma$ **or** $0.018 \times 8.63 \times 10^3$	C1	
			150 / 155(.4545) / 160 N	A1	
	(b)		line from (0, 0) to (0.011, 95) with decreasing gradient	B1	
			(becomes) horizontal at (0.011, 95)	B1	[6]

QUESTION · 5054_w15_qp_12.pdf · source page 3 of 16

4 A lorry takes 15 minutes to travel along the path PQRS.



What is the average speed of the lorry?

A 4.0 km / h **B** 22 km / h **C** 48 km / h **D** 64 km / h

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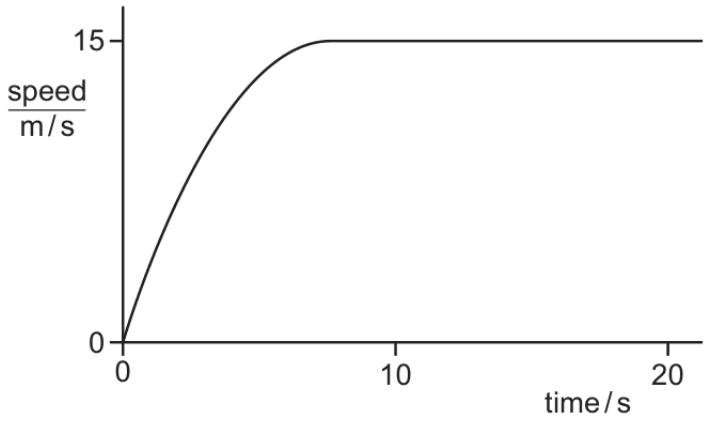
4 **D**

QUESTION · 5054_s16_qp_11.pdf · source page 3 of 16

4 The graph shows how the speed of a car varies with time.



Which statement about the acceleration of the car between 10 s and 20 s is correct?

A The acceleration decreases.

B The acceleration increases.

C The acceleration is constant, but not zero.

D The acceleration is zero.

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4 **D**

12 The overall stopping distance of a cyclist is made up of two parts, as shown in Fig. 12.1:

- the distance the cyclist travels during the reaction time of the cyclist (the thinking distance)
- the distance the cyclist travels after the brakes are applied (the braking distance)

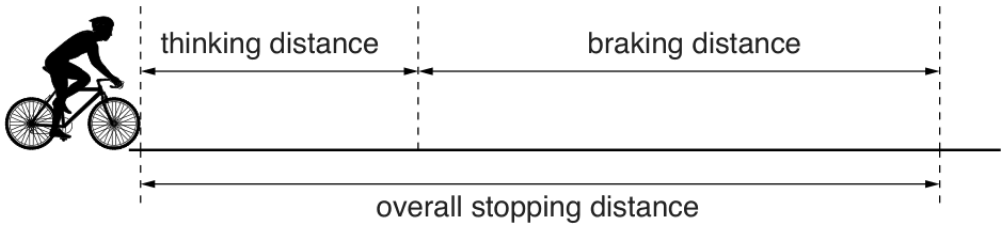


Fig. 12.1 (not to scale)

(a) State the energy change that occurs during braking.

.....

.....[2]

(b) A ball rolls in front of a cyclist at time $t = 0$ and the cyclist brakes and comes to rest.

Fig. 12.2 shows the speed-time graph for the cyclist.

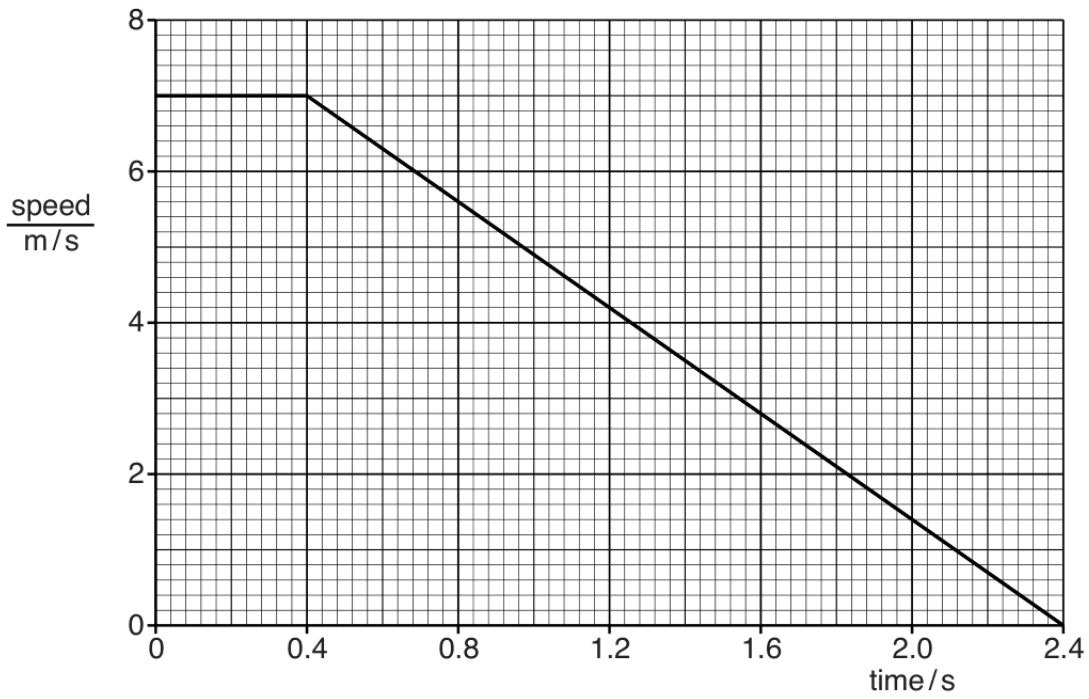


Fig. 12.2

(i) Using Fig. 12.2, determine the reaction time of the cyclist.

reaction time = ...[1]

(ii) Using Fig. 12.2, calculate the thinking distance.

thinking distance = ...[2]

(iii) State how the braking distance is found using Fig. 12.2.

.....

.....[1]

(iv) On another occasion, the same cyclist travels at an initial speed of 5.0 m/s. A ball rolls in front of the cycle at time $t = 0$.

The cyclist has the same reaction time and the deceleration of the cycle is the same as in Fig. 12.2.

On Fig. 12.2 draw the new speed-time graph for the cyclist. [2]

(v) The braking distance is longer when the cyclist stops on a wet road.

Explain why.

.....

.....

.....

.....

.....

.....[3]

(c) Fig. 12.3 shows the hydraulic braking system of the cycle.

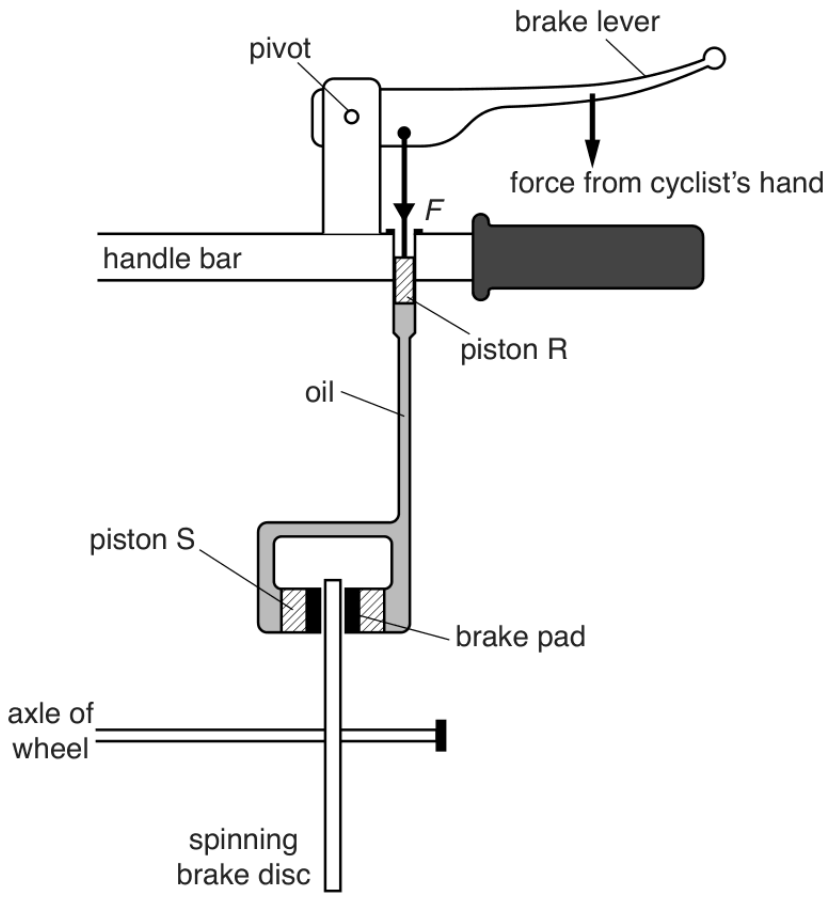


Fig. 12.3 (not to scale)

(i) The cyclist applies a force on the brake lever. This increases the pressure in the oil by 1.2×10^6 Pa.

The cross-sectional area of piston R is 5.0×10^{-5} m^2.

Calculate the force *F* applied by the brake lever to piston R.

F = ... [2]

(ii) The force applied to each of the brake pads is larger than *F*.

Explain why.

.....

.....

.....[2]

12	**(a)**	kinetic energy at start	B1
		thermal energy / heat energy / internal energy at end	B1
	(b) **(i)**	0.4(0) s	B1
	(ii)	(d=) s × t numerical or algebraic	C1
		2.8 m	A1
	(iii)	area under graph (between 0.4 and 2.4 s) **or** time (difference) × <u>average</u> speed **or** ½ × time (difference) × initial speed	B1
	(iv)	horizontal line from (0,5) to (0.4,5)	B1
		line showing braking with same gradient as original line	B1
	(v)	less friction	B1
		less deceleration **or** graph less steep **or** less <u>force</u> <u>backwards</u> / less <u>force</u> <u>opposing motion</u> **or** same KE lost / work done by friction	B1
		longer time to stop **or** larger area under (speed-time) graph **or** work = force x distance applied correctly	B1
	(c) **(i)**	(F=) P × A numerical or algebraic	C1
		60 N	A1
	(ii)	same pressure	B1
		larger area (of S / brake pads)	B1

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the speed-time graph for a car travelling along a horizontal road.

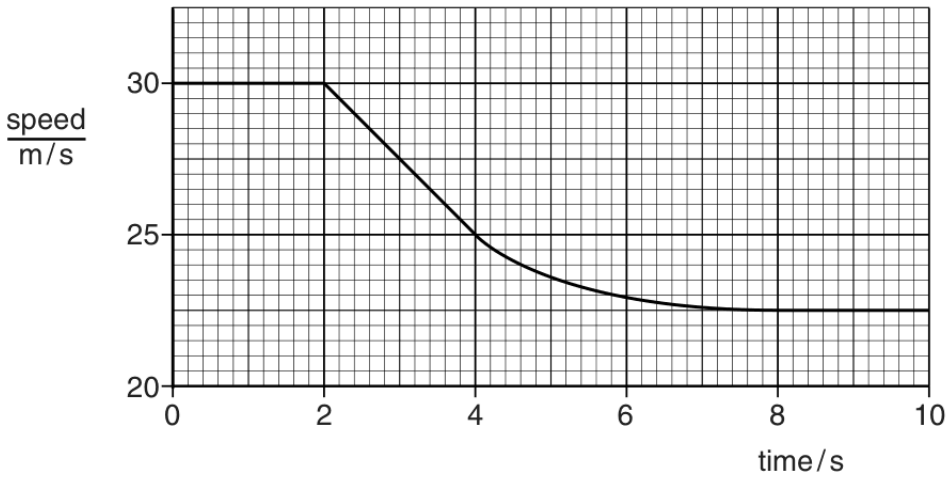


Fig. 1.1

(a) On Fig. 1.1, mark and label a point where the car has a non-uniform deceleration. [1]

(b) Calculate the deceleration of the car at $t = 3.0\,s$.

deceleration = ...[2]

(c) Explain, in terms of the horizontal forces that act on the car, why its speed is constant at $t = 1.0\,s$.

.....

.....

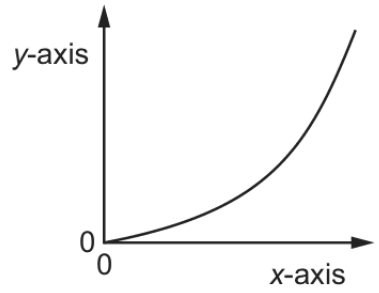
.....

.....[2]

1	**(a)**	mark at a time <u>between</u> 4.0 and 7.5 seconds	B1
	(b)	(a =) (v – u) / t numerical or algebraic	C1
		or (a =) gradient of graph stated	
		$2.5\,m/s^2$	A1
	(c)	forward force and backward force clear	B1
		equal forces (in horizontal direction)	B1
		or no resultant force	
		or forces cancel / balance / in equilibrium	

QUESTION · 5054_w16_qp_11.pdf · source page 3 of 20

4 A car has stopped at a red light. When the light changes to green, the car starts moving with a constant acceleration. The graph represents this motion.



Which quantity is plotted on the *x*-axis and which quantity is plotted on the *y*-axis?

	x-axis	*y*-axis
A	distance	time
B	speed	time
C	time	distance
D	time	speed

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4 **C**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A car of mass 800 kg is travelling at a speed of 25 m/s along a straight, horizontal road.

(a) State how *velocity* differs from *speed*.

.....

.....[1]

(b) The driver presses the accelerator and the speed of the car starts to increase from 25 m/s. The car accelerates at 1.5 m/s^2.

(i) Calculate the resultant force acting on the car as it accelerates.

force = ...[2]

(ii) Explain why the forward force acting on the car due to the engine is greater than the value in **(b)(i)**.

.....

.....

.....

.....[2]

(iii) Determine the speed of the car 4.0 s after it starts to accelerate.

speed = ...[2]

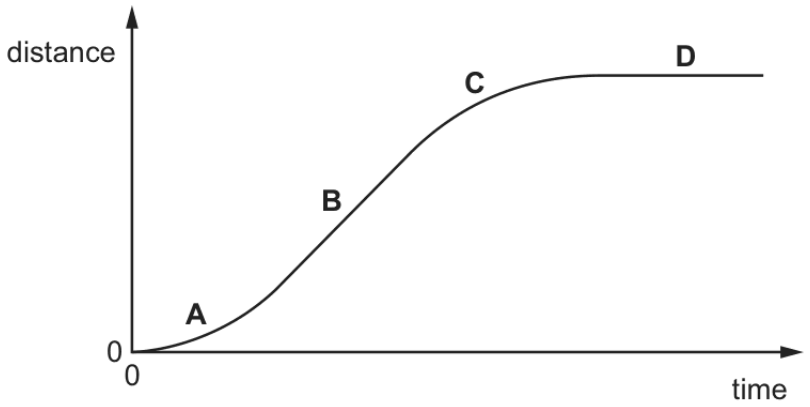
Section A

1	**(a)**	velocity / it has a direction / is a vector	B1	
	(b) **(i)**	(F =)ma **or** 800 × 1.5	C1	
		1200 N	A1	
	(ii)	friction / air resistance acts on car	B1	
		opposes force due to engine	B1	
	(iii)	(Δv =)at **or** 1.5 × 4.0 **or** 6.0	C1	
		31 m / s	A1	[7]

QUESTION · 5054_w16_qp_12.pdf · source page 3 of 16

4 The graph shows a distance-time graph for a car travelling in a straight line.

In which region is the car decelerating?



MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 4

4 **C**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A skier sets off from rest and accelerates uniformly at 3.4 m/s^2 in a straight line for 5.0 s.

(a) Calculate the speed of the skier after 5.0 s.

speed = ...[2]

(b) At 5.0 s, the skier stops accelerating and travels on for a further 10.0 s at a constant speed.

(i) State the size of the resultant force acting on the skier during these 10.0 s.

.....[1]

(ii) On Fig. 1.1, sketch a speed-time graph for the skier during the whole 15.0 s. [3]

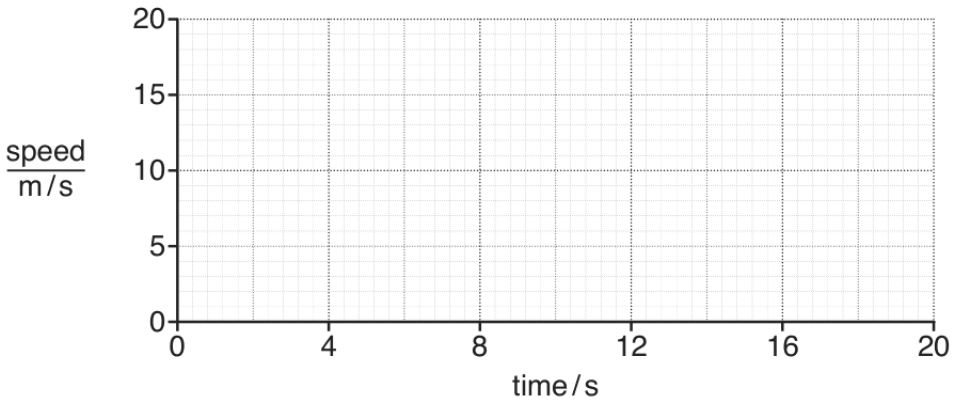


Fig. 1.1

(iii) State how the distance travelled by the skier can be determined using the speed-time graph.

.....

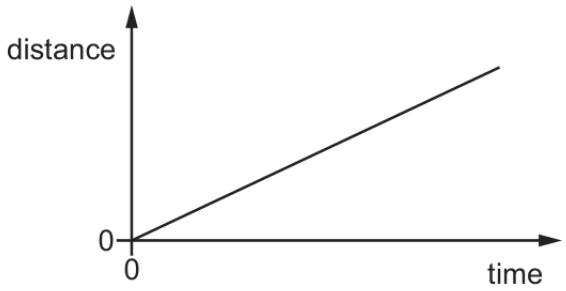
.....[1]

Section A

1	**(a)**	$(v = u +)at$ **or** 3.4×5.0	C1	
		17 m / s	A1	
	(b)	**(i)** 0 **or** zero **or** no resultant force	B1	
		(ii) straight line of positive gradient from (0, 0)	B1	
		horizontal line at $v > 0$ **and** after initial acceleration	B1	
		straight line from (0, 0) to (5.0, 17) **and** straight line from (5.0, 17) to at least (15.0, 17)	B1	
		(iii) calculate the area **under** the graph **or** area of trapezium	B1	[7]

QUESTION · 5054_s17_qp_11.pdf · source page 3 of 16

4 The diagram shows the distance-time graph of a car.



The car is travelling along a straight road up a hill.

Which quantity for the car is constant and greater than zero?

A acceleration

B gravitational potential energy

C kinetic energy

D resultant force

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4	C	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the directions of four forces acting on a racing car as it travels in a horizontal straight line.

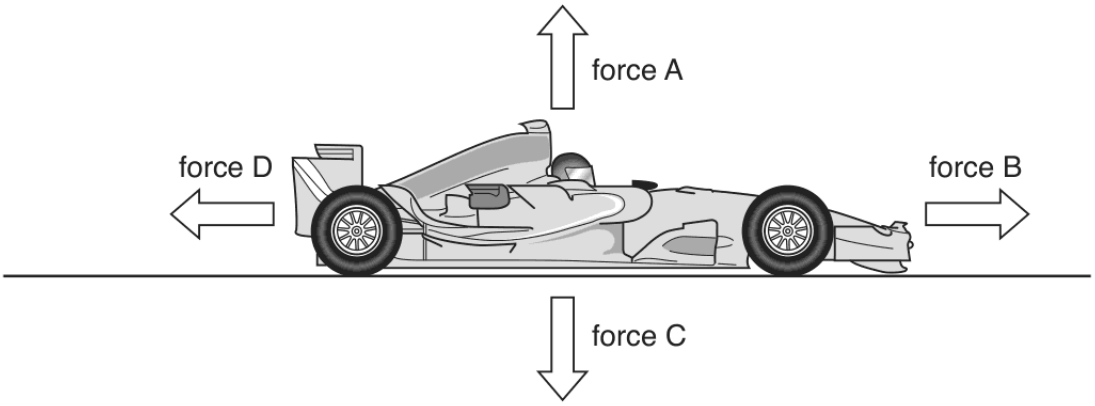


Fig. 1.1

(a) Draw a line from each box on the left to the correct description of each force.

force A	driving force
force B	contact or normal reaction force
force C	air resistance and friction
force D	force of gravity

[1]

(b) The table shows the sizes of the forces acting on the car at one time.

force A/N	force B/N	force C/N	force D/N
8000	1000	8000	600

The gravitational field strength g is 10 N/kg.

Calculate

(i) the mass of the car,

mass = ...[1]

(ii) the resultant force on the car,

resultant force = ...[1]

(iii) the acceleration of the car.

acceleration = ...[2]

(c) At another time, the car is travelling at speed u. It then accelerates for 5.0 s with an acceleration of 1.6 m/s^2, and reaches a speed of 20 m/s.

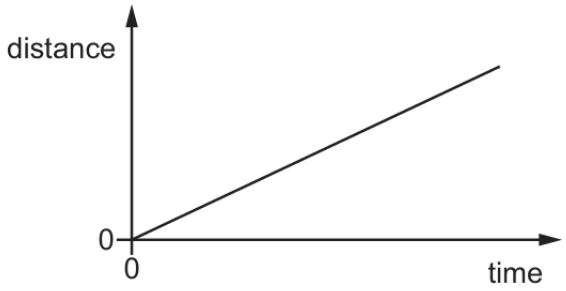
Calculate the value of u.

u = ...[2]

Question	Answer	Marks
1(a)	A – contact or reaction force B – driving force C – force of gravity D – air resistance and friction	**B1**
1(b)(i)	800 kg	**B1**
1(b)(ii)	400 N	**B1**
1(b)(iii)	(a =) F / m formula or with numbers	**C1**
	0.50 m / s^2	**A1**
1(c)	change in v = at in words or numbers	**C1**
	12 m / s	**A1**

QUESTION · 5054_s17_qp_12.pdf · source page 3 of 20

4 The diagram shows the distance-time graph of a car.



The car is travelling along a straight road up a hill.

Which quantity for the car is constant and greater than zero?

A acceleration

B gravitational potential energy

C kinetic energy

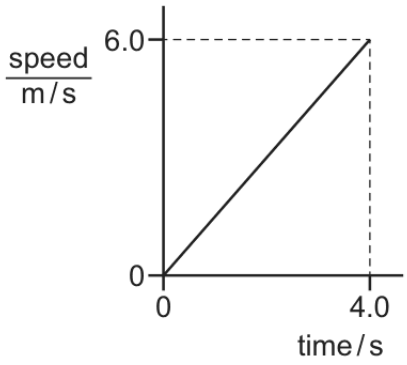
D resultant force

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 4

4	C	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 3 of 20

5 The diagram shows a speed-time graph for an object moving with uniform acceleration.



What is the distance travelled in the first 4.0 s?

A 0.67 m **B** 1.5 m **C** 12 m **D** 24 m

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5	C	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A car accelerates from rest in a straight line. During the first 14 s, the acceleration is uniform and the car reaches a speed of 25 m/s.

(a) (i) Calculate the acceleration of the car.

acceleration = ...[2]

(ii) After the first 14 s, the speed of the car continues to increase but the acceleration decreases. From 70 s to 80 s after the start, the car moves at a constant speed of 55 m/s.

On Fig. 1.1, draw a possible speed-time graph for the car.

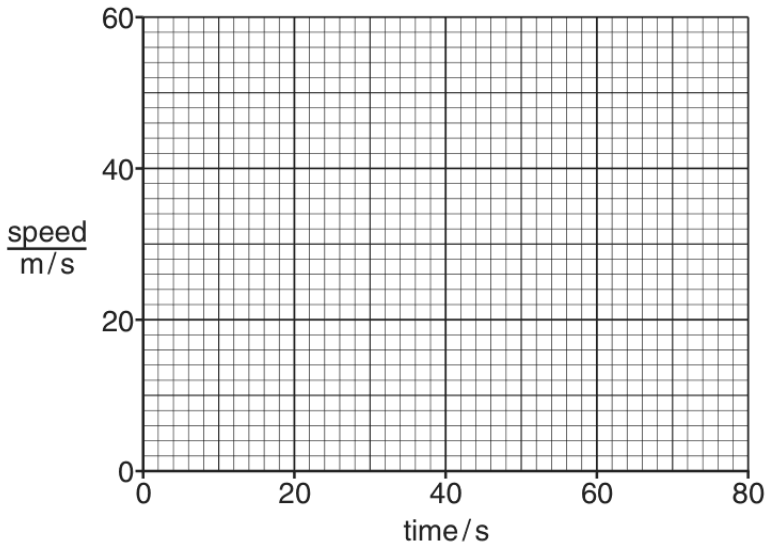


Fig. 1.1

[2]

(b) At a later time, the driver applies the brakes to stop. He is wearing a seat belt and slows down in his seat. A bag on the seat next to him slides forwards, across the seat towards the front of the car.

Using ideas about the forces acting, explain why the driver slows down but the bag slides forwards.

.....

.....

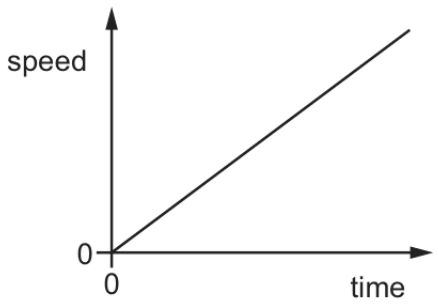
.....

.....[3]

Question	Answer	Marks
1(a)((i)	(a=) v(–u) / t **or** 25 / 14	**C1**
	1.8 m / s^2	**A1**
1(a)(ii)	initial straight line from (0,0) to (14,25)	**B1**
	gradient of line decreases after 14 s **and** flat from (70,55) to (80,55)	**B1**
1(b)	force backwards on driver / car B1	**any 3**
	force produced by seat belt / steering wheel **or** friction with seat / friction between tyres and road B1	
	no / small (backward) force / friction on bag B1	**B3**
	(mass of bag) resists change (from state of motion) **or** carries on in straight line **or** has constant velocity **or** (bag) has inertia B1	

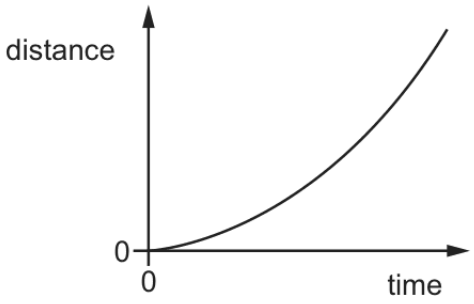
QUESTION · 5054_w17_qp_11.pdf · source page 3 of 20

3 The speed-time graph represents a short journey.

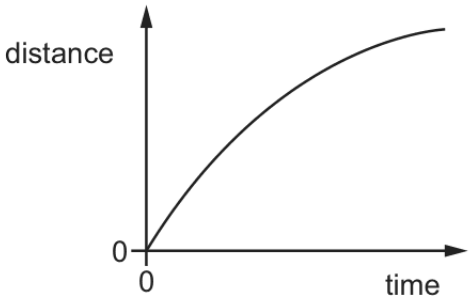


Which distance-time graph represents the same journey?

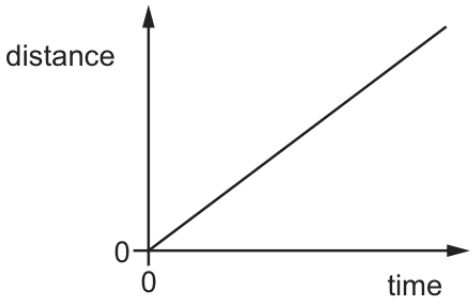
A



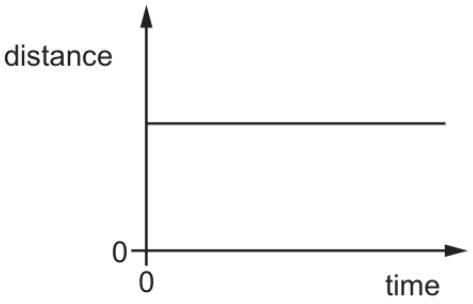
B



C



D



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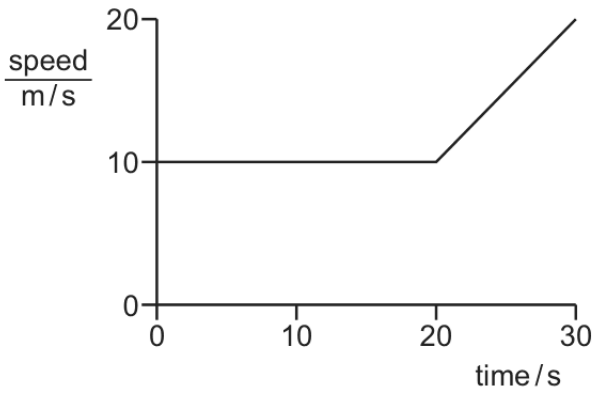
3	A	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 4 of 20

4 An object travels for 20 s with a constant speed of 10 m / s. For the next 10 s, it accelerates uniformly to 20 m / s.



What is the total distance travelled by the object in the 30 s?

A 300 m **B** 350 m **C** 400 m **D** 450 m

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4	B	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a large container ship travelling at constant speed in a straight line.

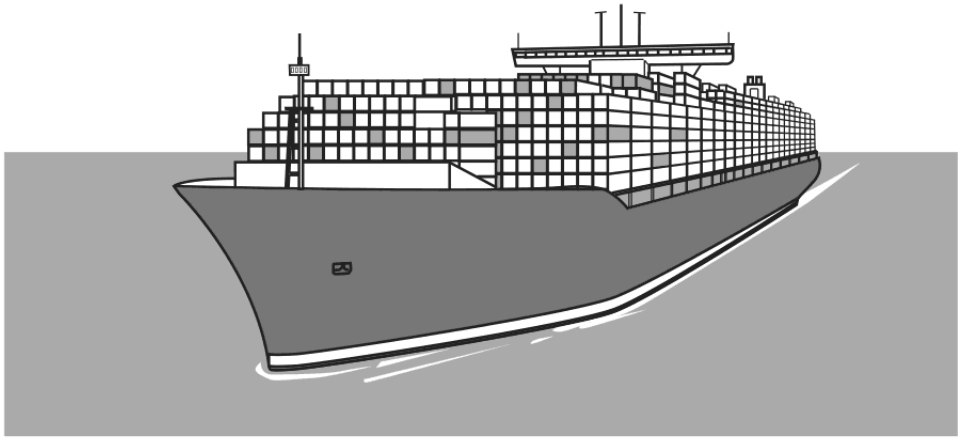


Fig. 9.1

The resistive force acting on the ship is 2.8×10^6 N.

(a) The speed of the ship is 9.7 m/s.

(i) Calculate the work done against the resistive force on the ship in 1.0 s.

work done = ... [2]

(ii) The engines are powered by oil.

State the energy transfer that is taking place when the ship is travelling at constant speed.

.....

..... [2]

(b) The mass of the ship is 2.2×10^8 kg. The engines are switched off and the resistive force causes the ship to decelerate.

(i) Calculate the initial deceleration of the ship.

deceleration = ... [2]

(ii) As the speed of the ship decreases, its deceleration changes.

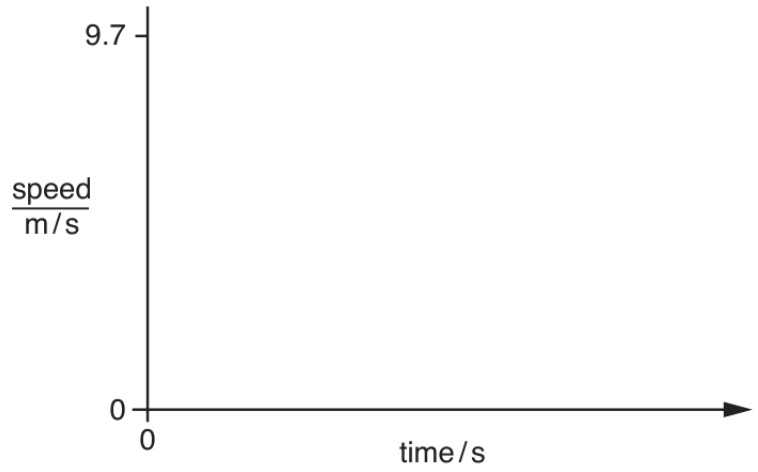
1. Suggest and explain how the deceleration changes.

.....

.....

..... [2]

2. On Fig. 9.2, sketch a possible speed-time graph for the ship as it decelerates to rest.



[2]

Fig. 9.2

3. Explain how the distance travelled by the ship may be determined from the speed-time graph.

.....

.....

..... [2]

(c) When the ship is travelling at a different speed, energy is being supplied to the engines at a rate of 33 MJ/s. The efficiency of the engines is 0.36 (36%).

(i) State a relationship that defines *efficiency*.

.....

..... [1]

(ii) Calculate the rate at which energy is wasted in the engines.

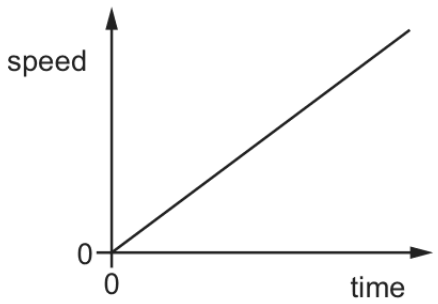
rate at which energy is wasted = ... [2]

Section B

Question	Answer	Marks
9(a)(i)	(*WD*) = *Fx* **or** $2.8 \times 10^6 \times 9.7$	**C1**
	2.7×10^7 J	**A1**
9(a)(ii)	from chemical (potential energy)	**B1**
	to thermal (energy) / heat	**B1**
9(b)(i)	(*a* =)*F* / *m* **or** $2.8 \times 10^6 / 2.2 \times 10^8$	**C1**
	0.013 m / s^2	**A1**
9(b)(ii)	**1** deceleration / it decreases	**B1**
	resistive force decreases **or** resistive force depends on speed	**B1**
	2 curve/line from 9.7 to zero **and** gradient negative (**allow** zero at end)	**B1**
	magnitude of gradient decreasing	**B1**
	3 area mentioned	**B1**
	area <u>under</u> line / curve **or** convert cm^2 (of graph paper) to distance **or** in terms of the scales	**B1**
9(c)(i)	(efficiency =) useful energy output / total energy input **or** useful power output / total power input	**B1**
9(c)(ii)	33×0.64 **or** 12 MJ / s	**C1**
	21 MJ / s	**A1**

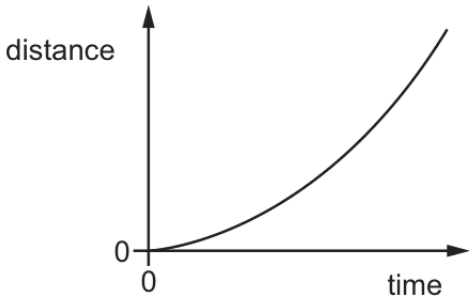
QUESTION · 5054_w17_qp_12.pdf · source page 3 of 16

3 The speed-time graph represents a short journey.

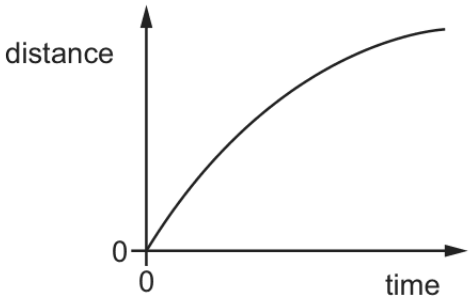


Which distance-time graph represents the same journey?

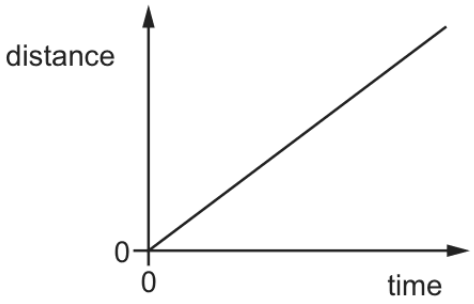
A



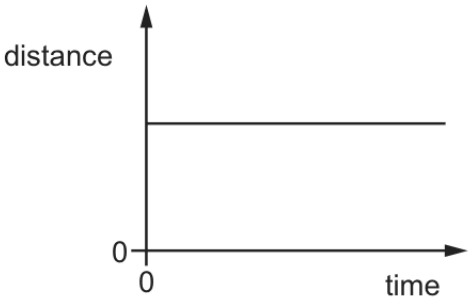
B



C



D



MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 3

3	A	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows an astronaut in space near to a space station in orbit above the Earth.

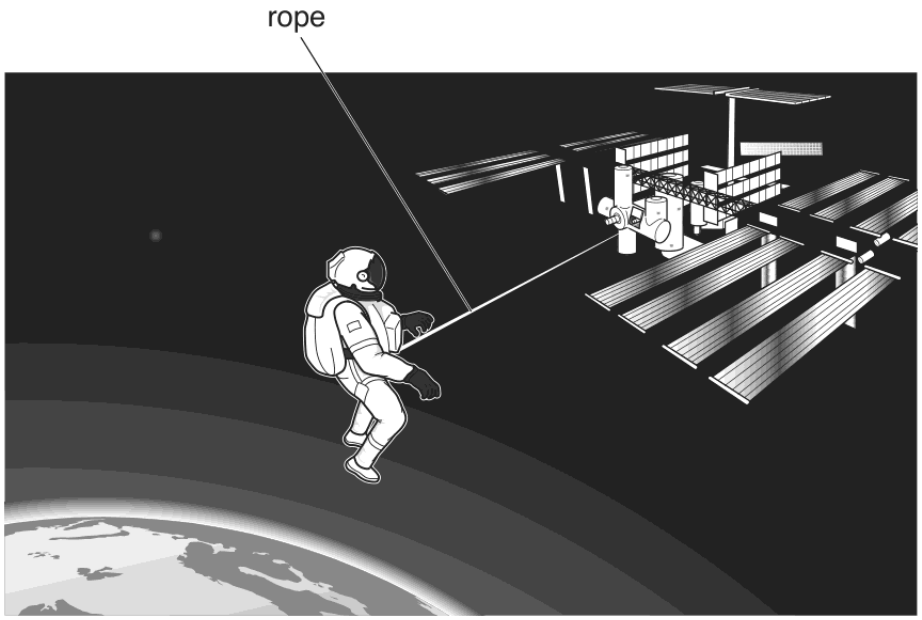


Fig. 9.1

The total mass of the astronaut and all his equipment is 160 kg.

The astronaut is initially at rest relative to the space station but he is then pulled towards the space station by the rope. His acceleration towards the space station is 0.35 m/s^2 for 1.2 s.

(a) Calculate

(i) the resultant force that causes a mass of 160 kg to accelerate at 0.35 m/s^2,

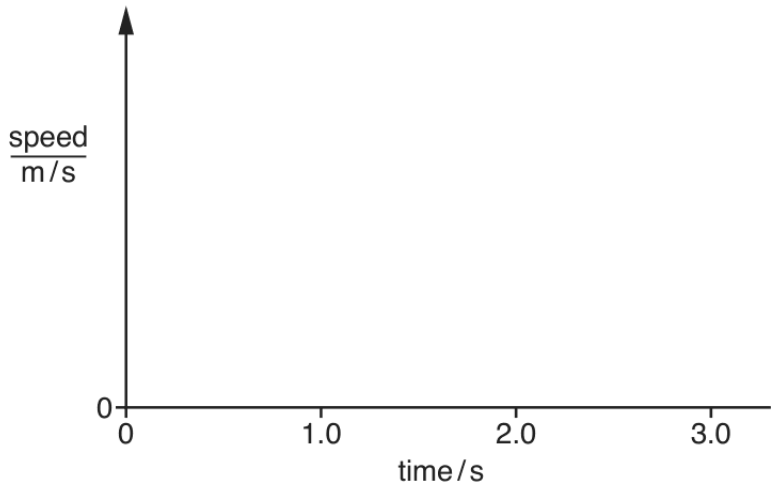
resultant force = ...[2]

(ii) the speed at which the astronaut is travelling after 1.2 s.

speed = ...[2]

(b) The resultant force on the astronaut is constant for 1.2 s, but then it decreases to zero instantly as the rope becomes slack.

(i) On Fig. 9.2, sketch the speed-time graph for the astronaut for the first 3.0 s of his motion.



[2]

Fig. 9.2

(ii) Describe how the distance moved by the astronaut during the first 3.0 s may be found using the speed-time graph.

.....

.....

.....[2]

(c) The space station is travelling at a constant speed in a circular orbit around the Earth as shown in Fig. 9.3.

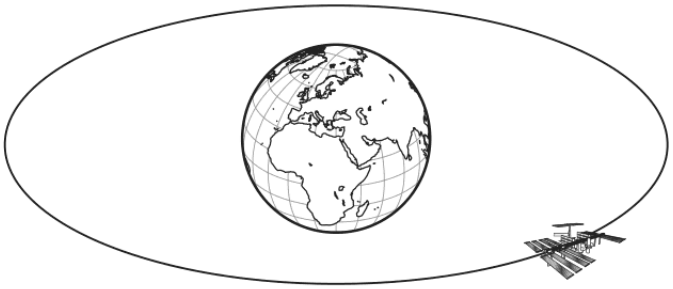


Fig. 9.3

(i) A force acts on the space station to keep it in this orbit.

1. On Fig. 9.3, draw an arrow to show the direction of this force. [1]

2. Explain what causes this force.

.....

.....

.....[2]

(ii) State what is meant by the *velocity* of an object.

.....

.....

.....[2]

(iii) State and explain what happens during the orbit to

1. the velocity of the space station,

.....

.....

.....[1]

2. the kinetic energy of the space station.

.....

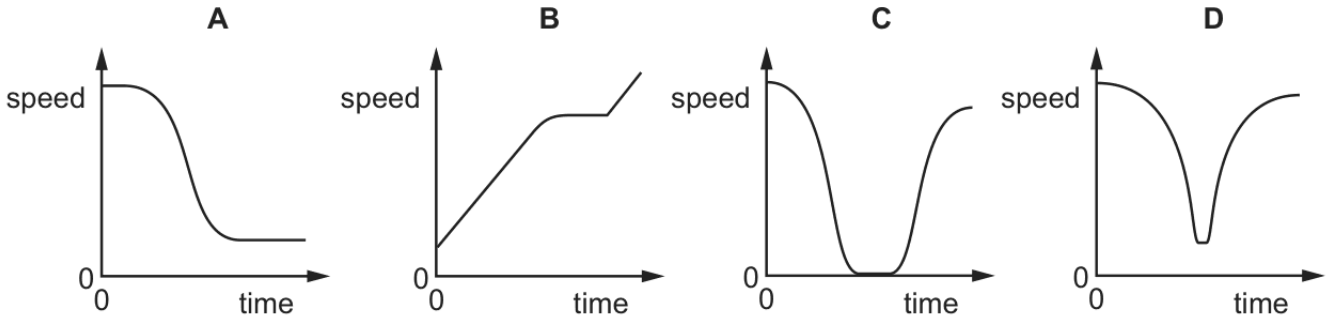
.....

.....[1]

Question	Answer	Marks
9(a)(i)	(F =) ma **or** 160 × 0.35	**C1**
	56 N	**A1**
9(a)(ii)	(v = u +) at **or** 0.35 × 1.2	**C1**
	0. 42 m / s	**A1**
9(b)(i)	straight line of positive gradient from origin to t = 1.2 s	**B1**
	horizontal line after t = 1.2 s	**B1**
9(b)(ii)	area mentioned	**B1**
	area under the line **or** convert cm^2 (of graph paper) to distance **or** in terms of the scales	**B1**
9(c)(i)	**1** arrow from space-station towards centre of Earth	**B1**
	2 gravitational field / attraction	**B1**
	of Earth	**B1**
9(c)(ii)	**two** separate approaches	
	I change of displacement	**M1**
	per unit time **or** divided by time	**A1**
	or **II** rate of change of distance **or** distance moved per unit time	**M1**
	in specified direction	**A1**
9(c)(iii)	**1** (it / velocity) changes **and** (because its) direction changes / force perpendicular to velocity	**B1**
	2 (remains) constant **and** depends on speed / g.p.e. remains constant / (k.e.) is a scalar quantity / speed is constant	**B1**

QUESTION · 5054_s18_qp_11.pdf · source page 3 of 20

4 Which speed-time graph represents the motion of a railway train making a short stop at a station?



MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 4

4	C	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A cyclist travels along a straight, horizontal road, as shown in Fig. 1.1.

Fig. 1.1

(a) Describe how the average speed of the cyclist can be measured.

In your account, suggest what equipment is used, the measurements that are made and how the average speed is calculated.

.....

.....

.....

.....

.....

.....[3]

(b) For part of the journey, the cyclist has a constant, negative acceleration.

(i) State what is meant by a *negative acceleration*.

.....

.....[1]

(ii) The size of the acceleration is $1.8\,m/s^2$.

Calculate the change in speed of the cyclist in a time of 2.5 s.

change in speed = ...[2]

Question	Answer	Marks
1(a)	distance measured (between 2 marks) with metre rule or tape measure	**B1**
	time measured (between 2 marks) with stopwatch / stopclock / timer	**B1**
	(average speed =) (total) distance / time	**B1**
1(b)(i)	decrease in speed **or** deceleration	**B1**
1(b)(ii)	(Δv =) a $\times$ t in any form numerical or algebraic	**C1**
	4.5 m / s	**A1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A student drops a small metal object into a cylinder of oil. The object falls alongside a vertical ruler and a camera records its position at 1.0 s intervals, as shown in Fig. 1.1.

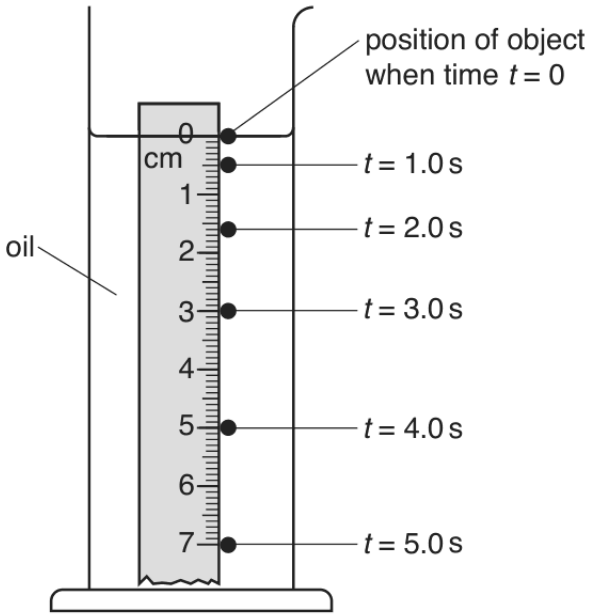


Fig. 1.1

(a) On the grid in Fig. 1.2, plot a distance-time graph for the object.

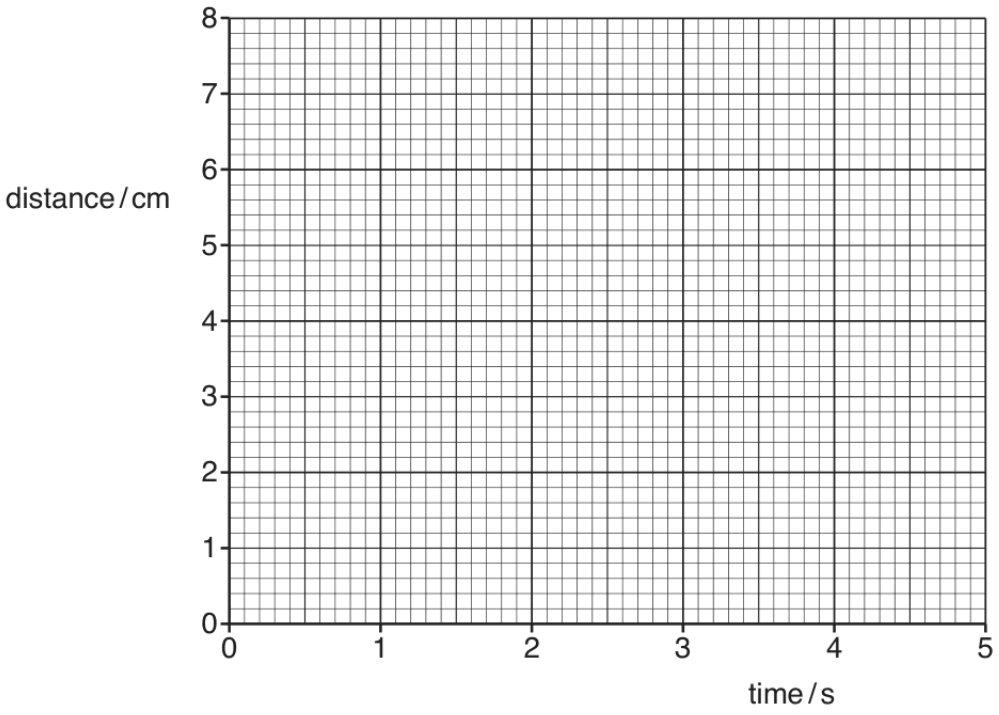


Fig. 1.2

[2]

(b) The object has a constant speed for some of the time.

(i) State and explain how a distance-time graph shows that speed is constant.

.....

.....

.....

.....[2]

(ii) Explain, in terms of the forces acting on the object, how it is able to fall at a constant speed.

.....

.....

.....

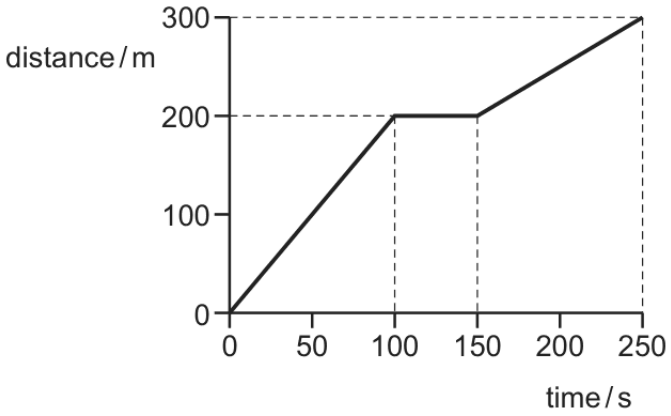
.....[2]

Question	Answer	Marks
1(a)	points plotted correctly at (1,0.5) (2,1.6) (3,3) (4,5) and (5,7)	**B1**
	smooth curve from origin	**B1**
1(b)(i)	straight line **or** gradient / slope constant	**B1**
	travels equal distances in same time **or** gradient equals speed	**B1**
1(b)(ii)	weight **and** force (upwards) from oil / liquid	**B1**
	forces balanced / cancel / are equal (and opposite) **or** no resultant (force)	**B1**

QUESTION · 5054_w18_qp_11.pdf · source page 3 of 16

4 A cyclist takes a ride lasting 250 s.

The graph shows how the distance from the starting position varies with time.



What is his average speed for the whole journey?

A 1.0 m / s **B** 1.2 m / s **C** 1.5 m / s **D** 2.0 m / s

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A girl of mass 35 kg, on a bicycle, accelerates from rest and travels down a slope in a straight line. The girl does not use the pedals.

Fig. 1.1 shows that the gradient of the slope is constant.

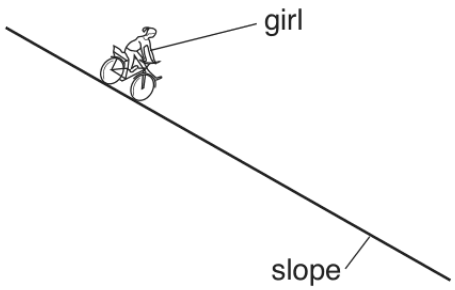


Fig. 1.1

(a) Calculate the resultant force on the girl when she is accelerating at 2.6 m/s^2.

resultant force = ... [2]

(b) At first, her acceleration is constant. At time t_1, her acceleration starts to decrease gradually until she is travelling at a constant speed in a straight line.

(i) On Fig. 1.2, sketch a speed-time graph for the girl from when she starts moving until she is travelling at a constant speed. [2]

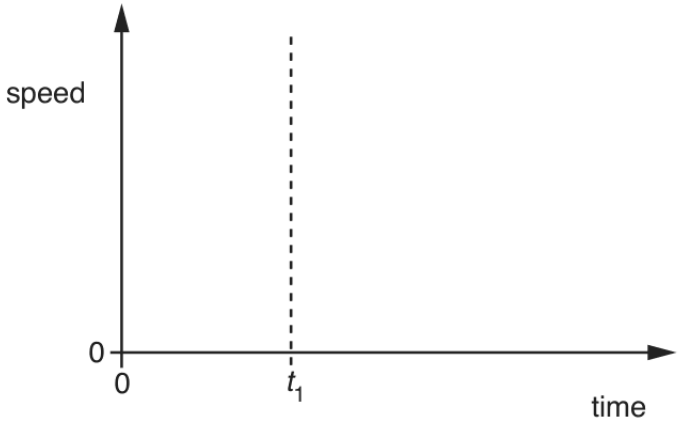


Fig. 1.2

(ii) State how the distance travelled is found from a speed-time graph.

.....

..... [1]

Question	Answer	Marks
1(a)	(F =) ma **or** 35 × 2.6	**C1**
	91 N	**A1**
1(b)(i)	straight line from origin to dashed line / t_1	**B1**
	gradually decreasing gradient until the line is horizontal	**B1**
1(b)(ii)	area under the line **or** area between line and x-axis	**B1**

QUESTION · 5054_w18_qp_12.pdf · source page 2 of 16

2 The table shows how the speeds of four bodies, **A**, **B**, **C** and **D**, change with time.

Which body has an acceleration that is **not** constant?

time / s	speed of **A** / m/s	speed of **B** / m/s	speed of **C** / m/s	speed of **D** / m/s
0	0	0	0	5.5
1	1.0	2.0	3.0	6.5
2	3.0	4.0	6.0	7.5
3	6.0	6.0	9.0	8.5

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2	A	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a satellite travelling at a constant speed in a circular orbit around the Earth.

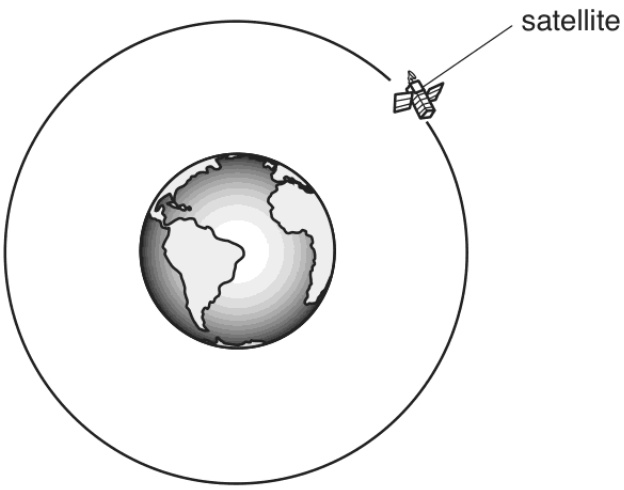


Fig. 1.1 (not to scale)

(a) State how *speed* differs from *velocity*.

.....

..... [1]

(b) As it orbits the Earth, the satellite is experiencing an acceleration.

(i) Explain, in terms of its velocity, why the satellite is accelerating.

.....

.....

..... [2]

(ii) On Fig. 1.1, draw an arrow, starting on the satellite, to show the direction of the satellite's acceleration. [1]

(c) As the satellite orbits the Earth, it experiences a force due to gravitational attraction.

State and explain whether this force does work on the satellite and state whether the energy of the satellite is affected.

.....

.....

.....

..... [2]

Question	Answer	Marks
1(a)	it / speed does **not** have a direction / is a scalar quantity	**B1**
1(b)(i)	direction (of velocity) changes (as it moves around the Earth)	**B1**
	its velocity changes with time (this is an acceleration)	**B1**
1(b)(ii)	arrow from satellite towards (centre of) Earth	**B1**
1(c)	**no** work done **and** force perpendicular to motion / no movement in direction of force	**B1**
	(kinetic and gravitational potential) energy remains constant / no effect	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 During a game of cricket, a player hits a ball with a bat. The ball then travels vertically upwards, as shown in Fig. 8.1.

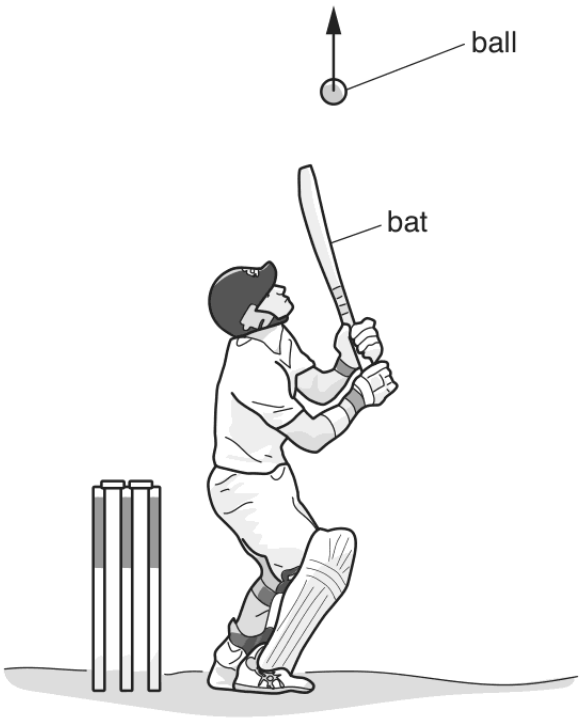


Fig. 8.1

The speed of the ball as it leaves the bat is 8.7 m/s.

(a) The mass of the ball is 0.16 kg.

(i) **1.** State what is meant by *mass*.

.....

..... [1]

2. State the name of a measuring instrument that is used to determine mass.

.....

..... [1]

(ii) Calculate the kinetic energy of the ball as it leaves the bat.

kinetic energy = ... [3]

(b) Fig. 8.2 is the velocity-time graph for the ball.

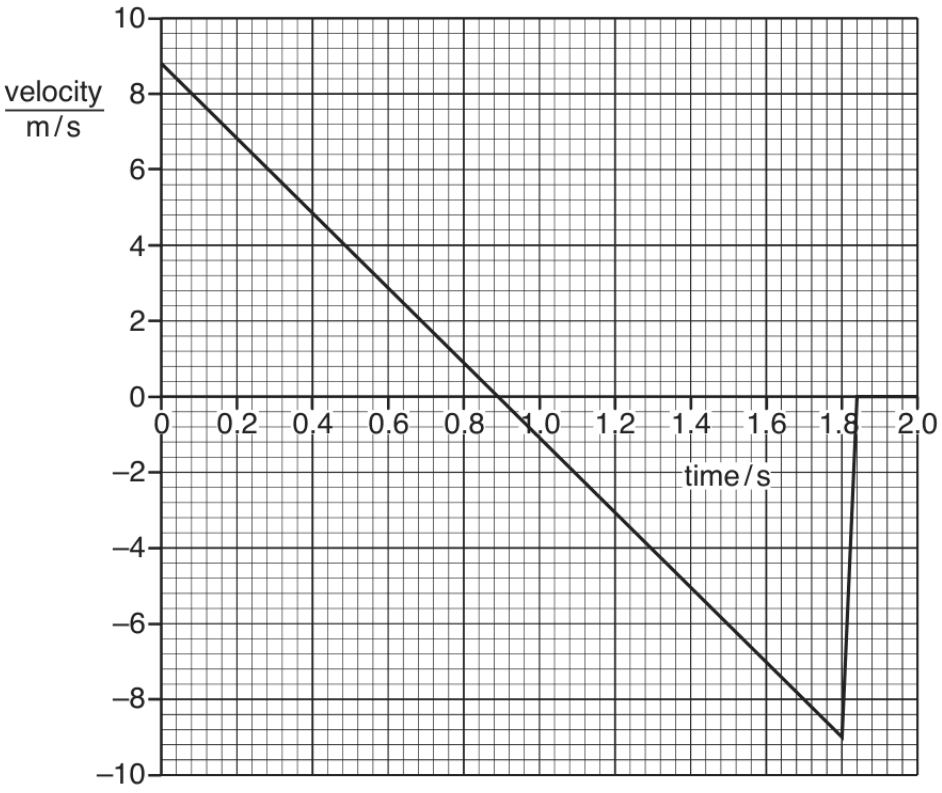


Fig. 8.2

(i) Fig. 8.2 shows that immediately after being hit, the ball has a negative acceleration.

1. State the name given to a negative acceleration.

..... [1]

2. State how the graph shows that the ball has a negative acceleration.

.....

..... [1]

(ii) Using Fig. 8.2, determine

1. the time at which the ball stops moving upwards,

time = ... [1]

2. the distance travelled before the ball stops moving upwards.

distance = ... [2]

(iii) Suggest why the graph, in Fig. 8.2, is almost vertical when time = 1.84 s.

.....

..... [1]

(iv) Fig. 8.2 shows that the ball remains stationary after a time of 1.84 s.

Describe the energy change that is occurring at a time of 1.84 s.

.....

.....

..... [2]

(c) For this ball moving in the way shown, the effect of air resistance is negligible.

When a ball with a much smaller mass is hit vertically upwards at the same speed, air resistance produces a noticeable effect on its motion.

Suggest two ways in which the velocity-time graph for the ball with a smaller mass differs from Fig. 8.2.

1. .....

.....

2. .....

.....
[2]

MARK SCHEME · 5054_w18_ms_22.pdf · source page 7 of 10 · Question 8

Question	Answer	Marks
8(a)(i)	1 quantity of matter (in a body)	**B1**
	2 balance **or** scales	**B1**
8(a)(ii)	(*k.e.* =) $\frac{1}{2}mv^2$	**C1**
	$\frac{1}{2} \times 0.16 \times 8.7^2$	**C1**
	6.1 J	**A1**
8(b)(i)	1 deceleration **or** retardation	**B1**
	2 negative gradient **or** line slopes downwards (left to right)	**B1**
8(b)(ii)	1 0.88 s ⩽ time ⩽ 0.90 s	**B1**
	2 area **or** counting squares **or** $\frac{1}{2}bh$ in some form	**C1**
	3.7 s ⩽ distance ⩽ 4.1 m	**A1**
8(b)(iii)	ball hits ground **or** short time for deceleration **or** large force **or** ground is hard	**B1**
8(b)(iv)	internal / thermal energy (of ball and ground) has increased	**B1**
	(internal energy) from kinetic energy	**B1**

MARK SCHEME · 5054_w18_ms_22.pdf · source page 8 of 10 · Question 8

Question	Answer	Marks
8(c)	any **two** from: smaller time to drop to zero velocity / hit ground line not straight **or** velocity does not change uniformly **or** gradient not constant smaller area under (first part of) graph **or** less distance travelled slower final velocity initial downward gradient steeper	**B2**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the distance-time graph for a journey made by a cyclist between town A and town B.

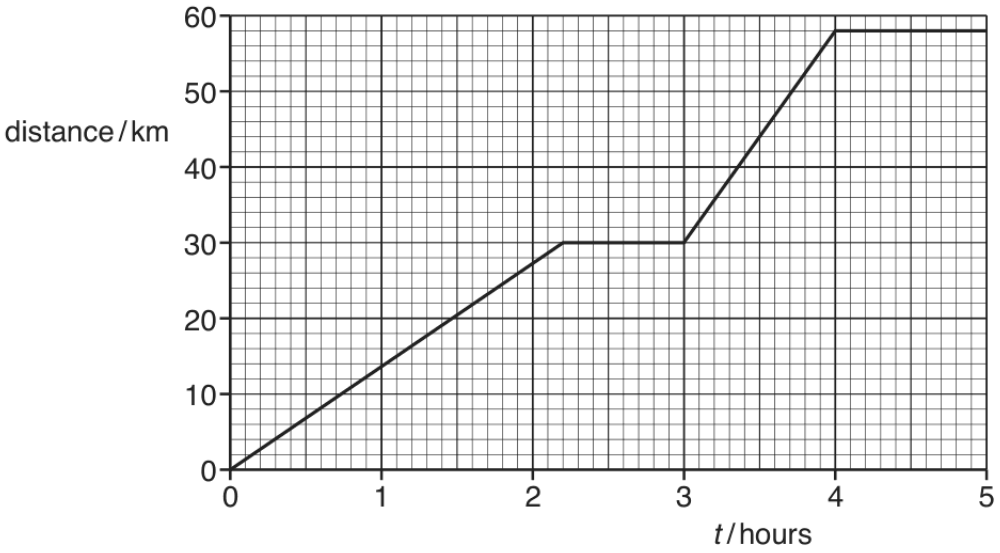


Fig. 1.1

The cyclist leaves town A at time $t = 0$ and arrives at town B at $t = 4.0$ hours.

(a) (i) Determine the distance between the two towns.

distance = ... [1]

(ii) Determine the time for which the cyclist rests during the journey, before he reaches town B.

time = ... [1]

(b) The speed of the cyclist near the end of the journey is greater than the speed at the beginning.

State how the graph shows this.

.....

..... [1]

(c) Calculate the average speed of the cyclist for the journey from A to B.

average speed = ... [2]

[Total: 5]

Question	Answer	Marks
1(a)(i)	58 km	**B1**
1(a)(ii)	0.8(0) hr **or** 48 min	**B1**
1(b)	steep(er) / larg(er) slope / gradient **or** travels a larger distance in the same / unit time **or** takes a shorter time to travel the same distance **or** covers more distance in less time	**B1**
1(c)	(s =) d / t numerical or algebraic; 58 / 4	**C1**
	15 km / hr	**A1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a parachutist falling vertically towards the ground.

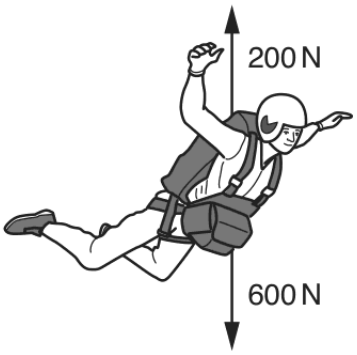


Fig. 9.1

The mass of the parachutist is 60 kg and his weight is 600 N.

(a) State what is meant by:

(i) *mass*

.....

..... [1]

(ii) *weight*.

.....

..... [1]

(b) Calculate the acceleration of the parachutist when the air resistance is 200 N.

acceleration = ... [3]

(c) The parachutist falls from rest at time $t = 0$.

He accelerates non-uniformly until he reaches a terminal velocity of 44 m/s at $t = 30$ s.

At $t = 40$ s the parachute opens. He decelerates for 4.0 s, reaching a smaller terminal velocity of 5.0 m/s.

He then falls at this speed until he lands on the ground at $t = 200$ s.

(i) On Fig. 9.2, draw the speed–time graph for the parachutist.

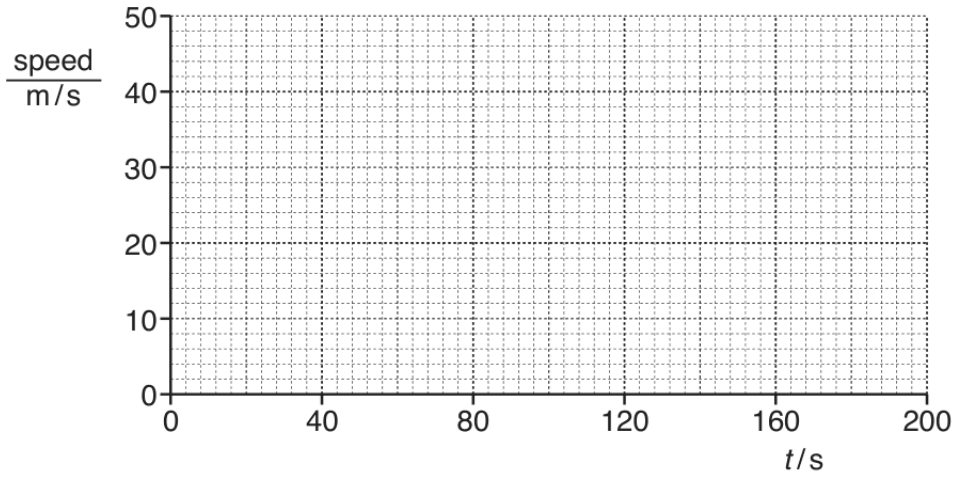


Fig. 9.2

[4]

(ii) Calculate the average deceleration of the parachutist between $t = 40\,s$ and $t = 44\,s$.

deceleration = ... [2]

(d) Explain, in terms of the forces involved, why:

(i) the parachutist reaches a terminal velocity at $t = 30\,s$

.....

..... [1]

(ii) the parachutist decelerates when the parachute opens

.....

.....

..... [2]

(iii) the second terminal velocity, when the parachute is open, is smaller than the first.

.....

..... [1]

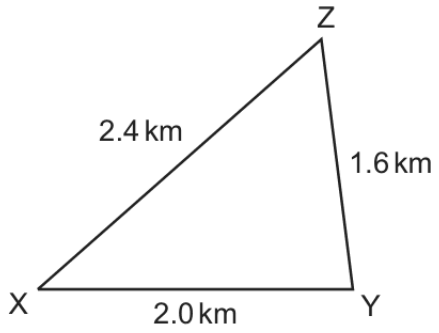
[Total: 15]

Section B

Question	Answer	Marks
9(a)(i)	amount / measure / quantity of matter / substance (in a body) **or** (property of a body) that resists (change in) motion	**B1**
9(a)(ii)	force / pull of gravity **or** in a gravitational field	**B1**
9(b)	(a =) F / m algebraic or numerical **or** 400 (N) seen	**C1**
	400 / 60	**C1**
	6.7 m/s^2	**A1**
9(c)(i)	initial curve of correct shape from origin at 0 to 30 s	**B1**
	horizontal line at 44 m/s from 30 s to 40 s	**B1**
	speed drops from 44 to 5 m/s in a time of 4 s	**B1**
	constant speed of 5 m/s until 200 s (and then to zero)	**B1**
9(c)(ii)	(a =) (v – u) / t algebraic or numerical	**C1**
	9.7 **or** 9.8 m/s^2	**A1**
9(d)(i)	upwards force / air resistance = weight / downward force	**B1**
9(d)(ii)	increase in air resistance / upwards force (from parachute)	**B1**
	upwards / force / air resistance larger than weight / downwards force **or** resultant force upwards	**B1**
9(d)(iii)	air resistance decreases (as speed reduces) until it equals weight **or** (with parachute open) air resistance = weight at lower speed	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 2 of 16

2 A woman runs 2.0 km from X to Y in 20 minutes and then rests at Y for 10 minutes. She then takes 10 minutes to run 1.6 km from Y to Z.



What is the size of her average velocity for the journey from X to Z?

A 0.060 km / minute

B 0.080 km / minute

C 0.090 km / minute

D 0.130 km / minute

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2	A	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 2 of 16

3 A cyclist travelling in a straight line at 8.0 m / s accelerates to 12 m / s in a time of 6.0 s.

Which expression gives the cyclist's acceleration?

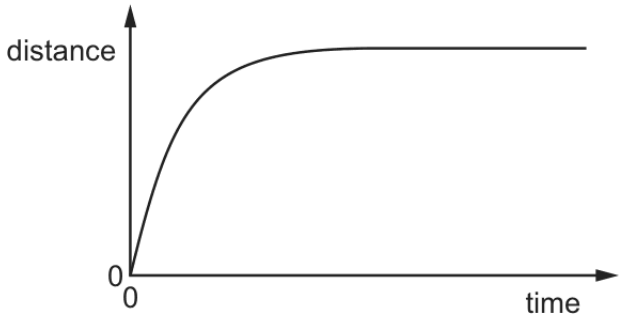
A $\frac{6}{12-8}$ **B** $\frac{8}{12-6}$ **C** $\frac{12-6}{8}$ **D** $\frac{12-8}{6}$

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 3 of 16

4 The diagram shows the distance–time graph for a moving object.



What is the moving object?

A a ball thrown vertically upwards and falling back to the thrower

B a car, starting from rest, speeding up as it moves away from a traffic signal

C a rock dropped from a high cliff, falling into the sea below

D a train braking to a halt as it stops at a station

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 4

4	D	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a map of the route taken by a car as it travels from town A to town B.

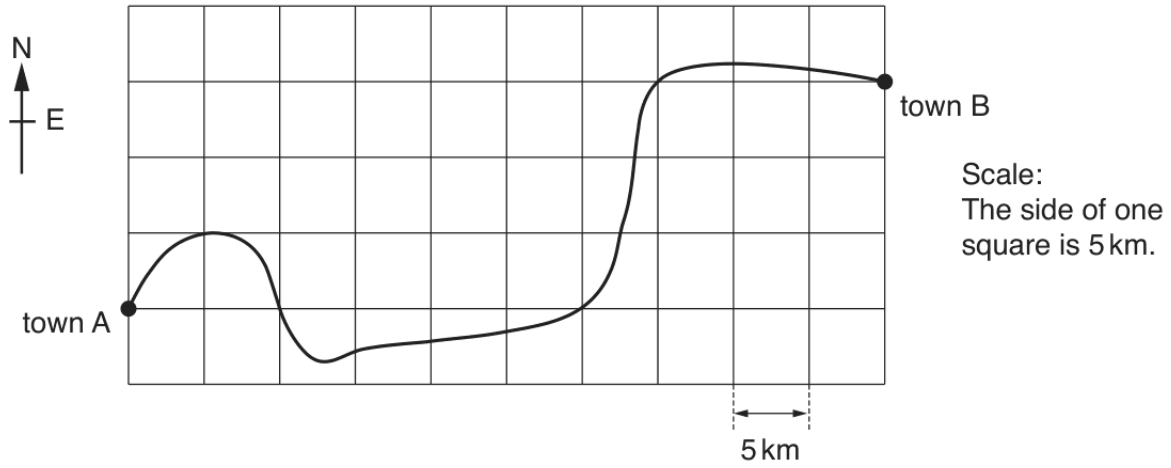


Fig. 9.1

The journey involves a distance and a displacement.

(a) (i) Using the scale on Fig. 9.1, determine the displacement in travelling from A to B.

size of displacement = ...

direction of displacement = ...
[3]

(ii) State how *distance* differs from *displacement*.

.....

.....

..... [1]

(iii) In some parts of the journey, the car is travelling at a constant speed but is also accelerating.

Explain how this is possible.

.....

.....

..... [2]

(b) At town B the car makes an emergency stop.

The total stopping distance of the car is the thinking distance added to the braking distance.

(i) Describe what is meant by the *thinking distance*.

.....

..... [1]

(ii) The brakes on the car work badly. This increases the braking distance.

State **two** other factors that increase the braking distance.

1. .....

.....

2. .....

.....
[2]

(c) The mass of the car and driver is 1200 kg. Just before the brakes are applied, the speed of the car is 30 m/s.

(i) Calculate the kinetic energy of the car and driver just before the brakes are applied.

kinetic energy = ... [3]

(ii) The braking force that stops the car is 1800 N.

Calculate the deceleration of the car.

deceleration = ... [2]

(iii) Explain why the temperature of the brakes increases during braking.

.....

..... [1]

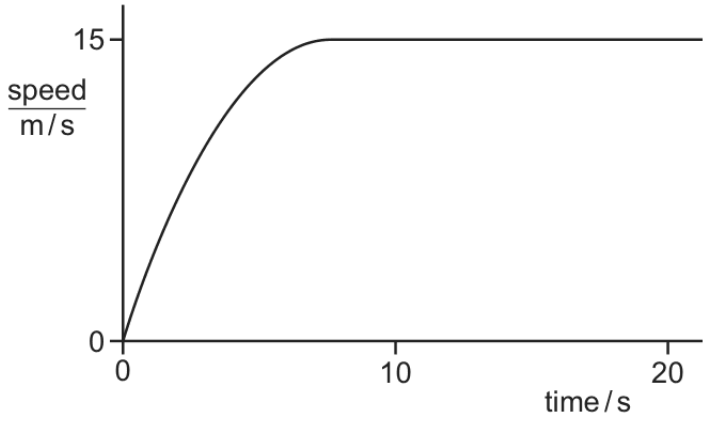
[Total: 15]

Question	Answer	Marks
9(a)(i)	10.2–10.4 (cm) **or** 50 (km) **or** 15 (km) seen	**C1**
	51–53 km	**A1**
	direction (0)72–74° **or** N 72–74 °E etc.	**B1**
9(a)(ii)	(distance) has no direction **or (**distance) is a scalar	**B1**
9(a)(iii)	changes direction **or** goes round a corner	**B1**
	velocity changes **or** velocity depends on direction **or** acceleration / force is towards the centre of circle / centripetal	**B1**

Question	Answer	Marks
9(b)(i)	distance travelled before brakes applied (after seeing emergency)	**B1**
9(b)(ii)	any 2 of • poor tyre surface • poor road conditions, e.g. rain, ice • high speed (of car) **or** good aerodynamic car shape **or** tail wind • large mass / weight of car • driver applies less force to pedal / less braking force	**B1B1**
9(c)(i)	(KE =) ½mv^2 algebraic or numerical	**C1**
	½ × 1200 × 30^2	**C1**
	540 000 J	**A1**
9(c)(ii)	(a=) F / m algebraic or numerical	**C1**
	1.5 m / s^2	**A1**
9(c)(iii)	friction (produces thermal energy) **or** kinetic energy changes to heat **or** work done against friction	**B1**

QUESTION · 5054_w19_qp_11.pdf · source page 3 of 20

5 The graph shows how the speed of a car varies with time.



Which statement about the acceleration of the car between the times 10 s and 20 s is correct?

A The acceleration decreases.

B The acceleration increases.

C The acceleration is constant, but not zero.

D The acceleration is zero.

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 5

5	D	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 6 of 20

10 There is no atmosphere on the Moon.

Two metal spheres of identical volume but different mass are at rest at the same height above the ground on the Moon. They are dropped at the same time.

How do the spheres move after release?

A The sphere with the larger mass has a smaller acceleration.

B The sphere with the larger mass hits the ground first.

C The sphere with the smaller mass has a smaller acceleration.

D The two spheres hit the ground together.

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 10

10	D	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 3 of 16

5 Part of a speed–time graph for two runners is shown. They are running along the same track.

Which statement **must** be correct?

A Both runners started at the same moment.

B Runner 1 has a greater acceleration than runner 2.

C Runner 1 is moving faster than runner 2.

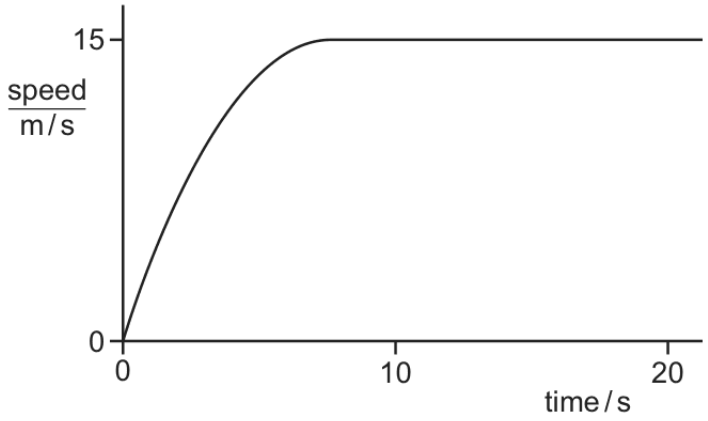
D The distance between the two runners stays constant.

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 5

5	C	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 4 of 16

6 The graph shows how the speed of a car varies with time.



Which statement about the acceleration of the car between the times 10 s and 20 s is correct?

A The acceleration decreases.

B The acceleration increases.

C The acceleration is constant, but not zero.

D The acceleration is zero.

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 4 of 16

7 In which descent is the acceleration equal to the acceleration of free fall *g* at all times?

A a bungee jumper leaping from a bridge

B a feather falling in a vertical tube that contains a vacuum

C a hailstone travelling to Earth at terminal velocity

D a sky-diver dropping from an aircraft towards the ground

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 7

7	B	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 There is no atmosphere on the Moon.

An astronaut on the Moon drops a feather and a hammer from the same height at the same time. They both accelerate downwards at $1.6\,m/s^2$ and they hit the ground at the same time.

(a) The weight of the hammer is much larger than that of the feather.

Explain, in terms of their weights and masses, why their accelerations are equal.

.....

.....

.....

.....

.....

..... [3]

(b) Both the feather and the hammer take 1.5 s to fall to the ground from rest.

(i) Calculate the speed of the objects as they hit the ground.

speed = ... [2]

(ii) On Fig. 1.1, draw the speed-time graph for the fall. At the correct position on the *y*-axis, write the value of the speed at time $t = 1.5$ s.

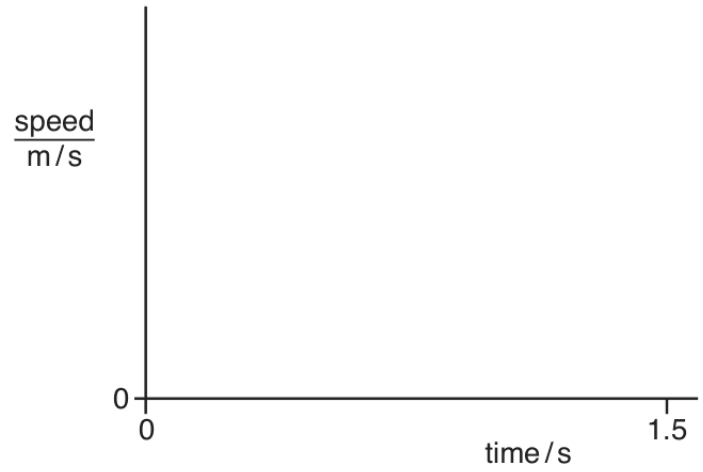


Fig. 1.1

[1]

(iii) Using the speed-time graph in **(b)(ii)**, determine the height from which the objects are dropped.

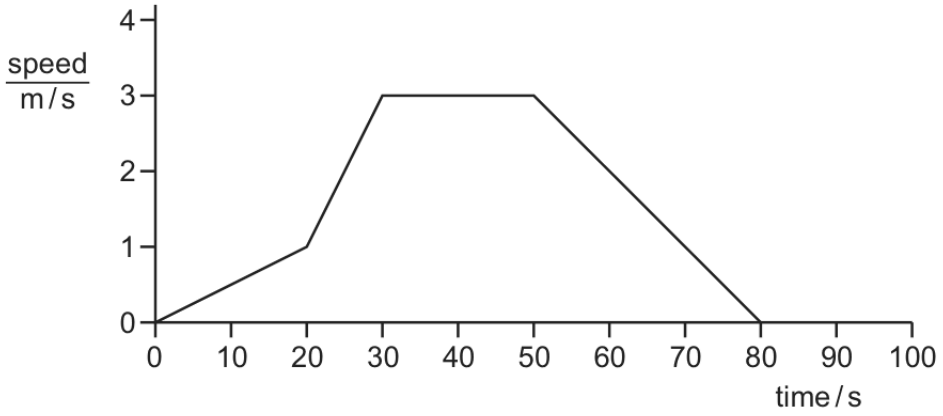
height = ... [2]

[Total: 8]

Question	Answer	Marks
1(a)	weight is directly proportional to mass **or** $W = mg$	**B1**
	a larger force acts on the more massive body **or** $a = F / m$ **or** $a = W / m$	**B1**
	mass cancels **or** acceleration inversely proportional to mass	**B1**
1(b)(i)	($\Delta v =$) at **or** 1.6×1.5	**C1**
	2.4 m / s	**A1**
1(b)(ii)	2.4 marked on y-axis **and** straight line from (0, 0) to (1.5, 2.4)	**B1**
1(b)(iii)	$\frac{1}{2} \times 2.4 \times 1.5$ **or** area under the graph mentioned / attempted	**C1**
	1.8 m	**A1**

QUESTION · 5054_s20_qp_11.pdf · source page 2 of 16

3 The graph shows a short journey. It is a speed–time graph.



What is the greatest speed reached?

A 1 m/s **B** 2 m/s **C** 3 m/s **D** 4 m/s

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 3

3	C	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 3 of 16

5 Two cameras are a known distance apart. The exact time that a vehicle passes each of the cameras is recorded.

What can be obtained from the information?

A average acceleration of the vehicle

B average speed of the vehicle

C maximum acceleration of the vehicle

D maximum speed of the vehicle

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 5

5	B	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows part of the speed–time graph for an athlete in a race.

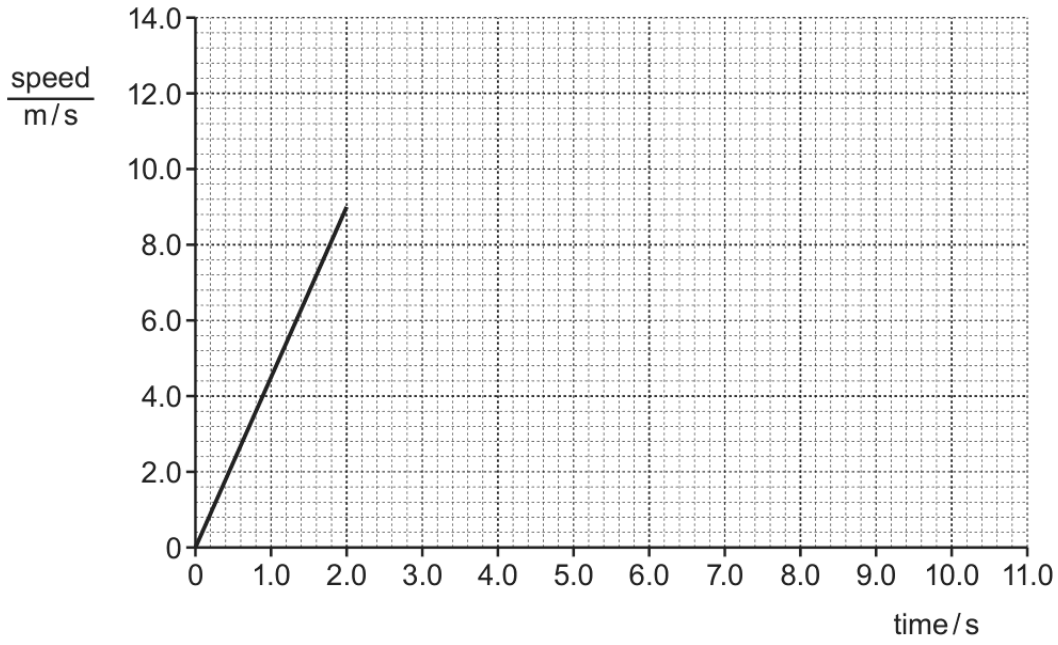


Fig. 1.1

(a) During the race, the acceleration of the athlete is uniform in the first 2.0 s.

State how the graph shows that the acceleration is uniform.

.....

..... [1]

(b) Determine the distance travelled by the athlete in the first 2.0 s.

distance = ... [2]

(c) During the rest of the race:
- from 2.0 s to 5.5 s, the acceleration of the athlete decreases
- at 5.5 s, the athlete reaches a maximum speed of 12 m/s
- from 5.5 s to 8.0 s, the athlete travels at a speed of 12 m/s
- from 8.0 s to 11.0 s, the athlete decelerates, finishing the race at a speed of 10 m/s.

On Fig. 1.1, complete the speed–time graph for times between 2.0 and 11.0 s. [3]

[Total: 6]

Question	Answer	Marks
1(a)	constant gradient	**B1**
1(b)	use of area (under graph) **or** $\frac{1}{2}$vt **or** average speed 4.5 (m/s)	**C1**
	9(.0) m	**A1**
1(c)	line of decreasing slope after 2 s	**B1**
	horizontal line at 12 m/s from 5.5 to 8 s	**B1**
	line, curved or straight, from (8, 12) to (11,10)	**B1**

QUESTION · 5054_w20_qp_11.pdf · source page 3 of 16

2 A car begins to move. It speeds up until it reaches a constant speed. It continues to travel at this constant speed for the rest of the journey.

What happens to the acceleration and what happens to the velocity of the car during the journey?

A Both the acceleration and the velocity change.

B Only the acceleration changes.

C Only the velocity changes.

D Neither the acceleration nor the velocity changes.

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 2

2	A	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 is the distance–time graph for a skydiver who jumps from a balloon at time $t = 0$.

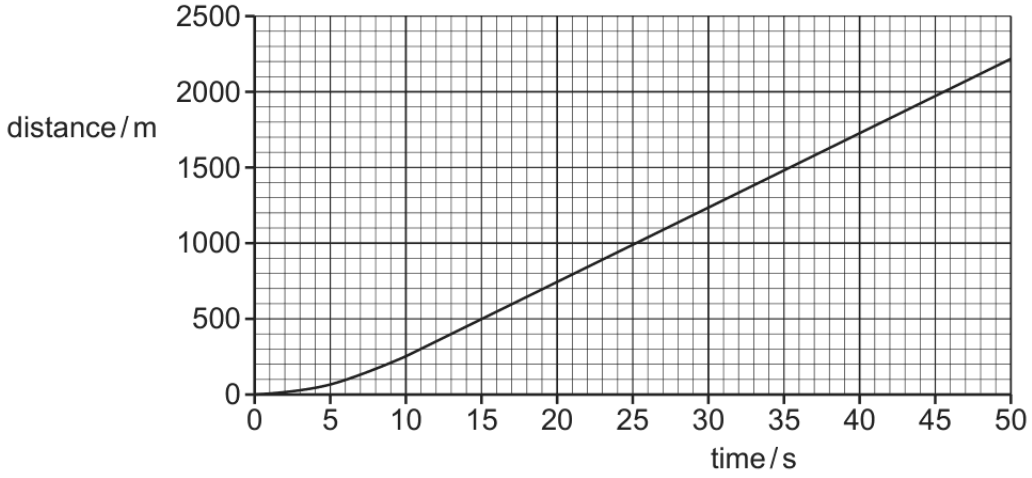


Fig. 1.1

(a) The first part of the graph shows the motion of the skydiver from when he jumps until he reaches terminal velocity.

(i) Describe the motion of the skydiver between $t = 0$ and $t = 20$ s.

.....

.....

..... [2]

(ii) Explain the motion of the skydiver between $t = 0$ and $t = 20$ s in terms of the forces acting on him.

.....

.....

.....

..... [3]

(b) Using Fig. 1.1, determine the terminal velocity of the skydiver.

On Fig. 1.1, indicate any values used for your calculation.

terminal velocity = ... [3]

[Total: 8]

Question	Answer	Marks
1(a)(i)	accelerating **or** increasing speed	**B1**
	decreasing acceleration **or** speed increasing at a decreasing rate **or** reaches a constant speed	**B1**
1(a)(ii)	initially acceleration is due to force of gravity / weight (only) **or** force of gravity / weight larger than air resistance from (t = 0 to ≈ 7 s)	**B1**
	air resistance increases (with speed)	**B1**
	resultant force decreases / becomes zero **or** forces balance **or** air resistance equals weight	**B1**
1(b)	two different pairs of co-ordinates from straight-line section of graph seen	**C1**
	correct use of co-ordinates in a division	**C1**
	48 – 52 m / s	**A1**

QUESTION · 5054_w20_qp_12.pdf · source page 3 of 16

2 A car begins to move. It speeds up until it reaches a constant speed. It continues to travel at this constant speed for the rest of the journey.

What happens to the acceleration and what happens to the velocity of the car during the journey?

A Both the acceleration and the velocity change.

B Only the acceleration changes.

C Only the velocity changes.

D Neither the acceleration nor the velocity changes.

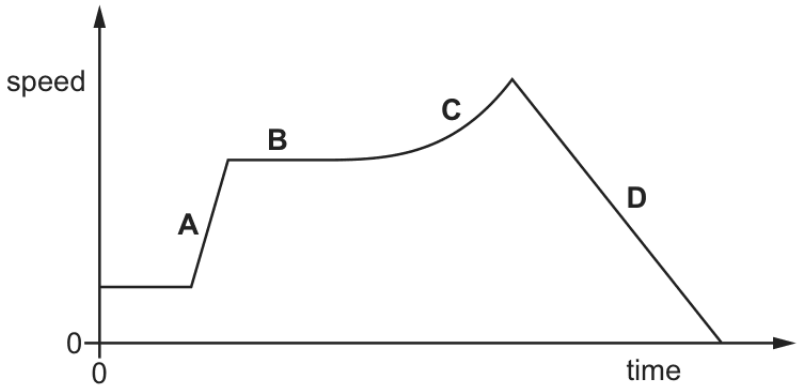
MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 2

2	A	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 3 of 16

3 The speed–time graph for a car's journey is shown.

During which part of the journey is the car moving with non-uniform acceleration?



MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 3

3	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 A bus leaves a bus-stop at time $t = 0$ and travels along a horizontal road until it reaches a second bus-stop. Fig. 7.1 is the distance-time graph for the bus between $t = 0$ and $t = 60$ s.

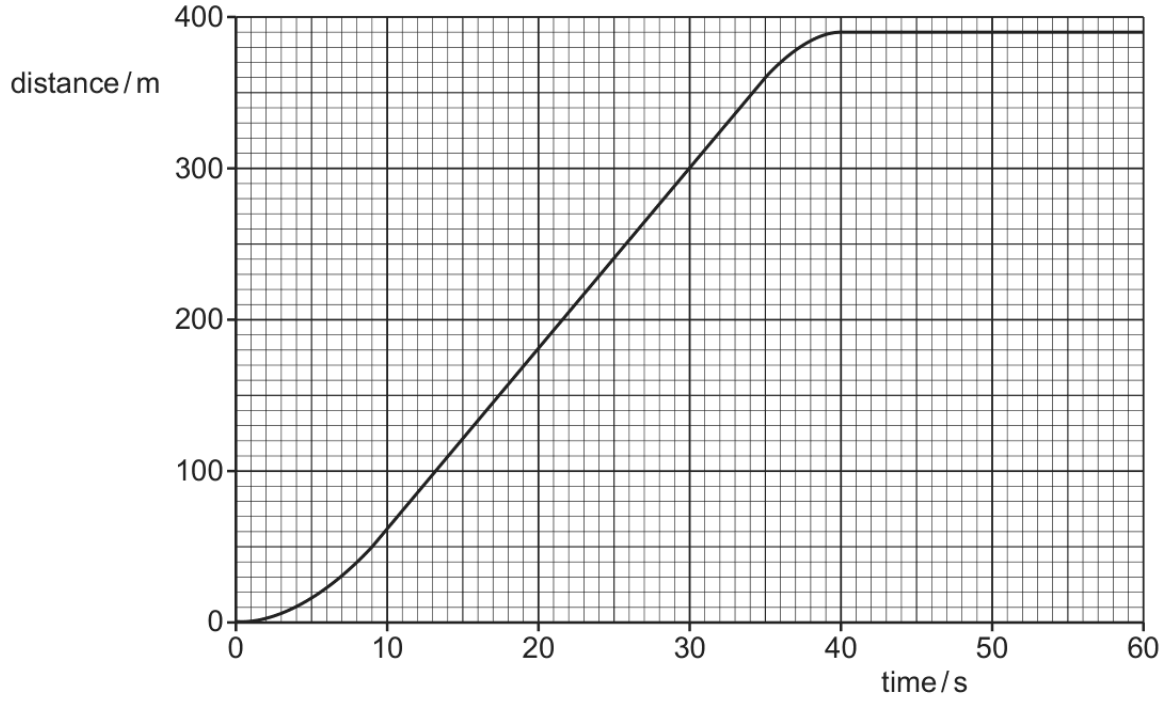


Fig. 7.1

The road on which the bus is travelling is straight except for a short, curved section. The bus travels around this circular curve between $t = 21$ s and $t = 24$ s.

(a) Describe how the motion of the bus between $t = 0$ and $t = 10$ s differs from its motion between $t = 35$ s and $t = 40$ s.

.....

.....

.....

.....

..... [3]

(b) Determine:

(i) the maximum speed of the bus during these 60 s

maximum speed = ... [3]

(ii) the average speed of the bus between leaving the first bus-stop and arriving at the second bus-stop.

average speed = ... [2]

(c) **(i)** State how *velocity* differs from *speed*.

.....

..... [1]

(ii) There are **three** periods during the 60 s when there is a non-zero resultant force acting on the bus.

Complete the statements to indicate these three time periods and state the direction of the resultant force in that period.

1. Between *t* = ... and *t* = ... the direction of the resultant force is

.....

2. Between *t* = ... and *t* = ... the direction of the resultant force is

.....

3. Between *t* = ... and *t* = ... the direction of the resultant force is

.....
[4]

(d) During the journey, the air resistance acting on the bus varies.

(i) State why the air resistance changes during the journey.

.....

..... [1]

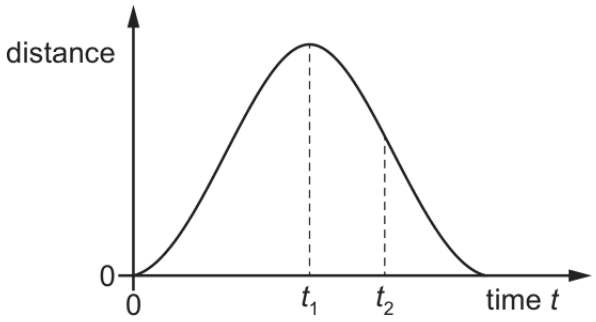
(ii) On Fig. 7.1, mark and label with an M a time when the air resistance is a maximum value.
[1]

[Total: 15]

Question	Answer	Marks
7(a)	(between t = 0 and t = 10 s) it is accelerating **or** speed / velocity increasing	**B1**
	between t = 35 s and t = 40 s, it is decelerating **or** speed / velocity decreasing	**B1**
	magnitude of acceleration less than magnitude of deceleration **or** between t = 0 and t = 10 s starts from rest / zero speed **or** between t = 35ms and t = 40 s it finishes at rest / with zero speed	**B1**
7(b)(i)	two pairs of co-ordinates from the straight-line section	**C1**
	appropriate division using candidate's coordinates	**C1**
	11.7 ⩽ speed ⩽ 12.0 m / s	**A1**
7(b)(ii)	total distance / total time **or** (distance =) 390 (m) **and** 39 (s) ⩽ (time =) ⩽ 40 (s)	**C1**
	9.7 to 10.0 m / s	**A1**
7(c)(i)	it / velocity has a direction **or** is a vector **or** depends on displacement (rather than distance)	**B1**
7(c)(ii)	**answers need not be in this order** 1 0 **and** 10 (s) **and** forwards / in the direction of movement	**B1**
	2 21 (s) **and** 24 (s)	**B1**
	perpendicular to the motion / to the direction of movement	**B1**
	3 35 (s) **and** 40 (s) **and** backwards / opposite to the direction of movement	**B1**
7(d)(i)	speed (of bus) varies / changes / decreases / increases	**B1**
7(d)(ii)	any time on Fig. 1.1 from 10 s ⩽ t ⩽ 35 s labelled M	**B1**

QUESTION · 5054_s21_qp_11.pdf · source page 2 of 16

3 A train sets off from a station at time $t = 0$. The graph shows how the distance between the train and the station varies with time.



Which statement about the movement of the train between time t_1 and t_2 is correct?

A Its speed is decreasing and it is moving away from the station.

B Its speed is decreasing and it is moving towards the station.

C Its speed is increasing and it is moving away from the station.

D Its speed is increasing and it is moving towards the station.

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 An aircraft flies at a constant height.

Air drag and the force from the aircraft's engines together produce a force on the aircraft of 36 kN due north, as shown in Fig. 1.1.

The wind produces a force of 12 kN towards the east.

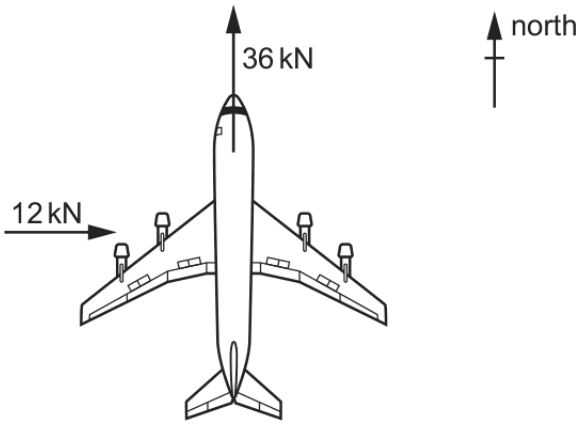


Fig. 1.1 (not to scale)

(a) Draw a scale drawing to show the resultant force acting on the aircraft.

Use your drawing to determine the size of the resultant force and the angle between the resultant force and north.

size of resultant force = ...

angle = ...
[3]

(b) The acceleration of the aircraft is uniform.

(i) Describe how a uniform acceleration differs from a non-uniform acceleration.

.....

.....

.....

..... [2]

(ii) The mass of the aircraft is 60 000 kg.

Calculate the acceleration of the aircraft.

acceleration = ... [2]

[Total: 7]

Question	Answer	Marks
1(a)	scale vector diagram showing 12 N, 36 N and correct resultant force in correct direction	**B1**
	38 ± 1 kN	**B1**
	018 ± 3°	**B1**
1(b)(i)	(only in uniform acceleration) same increase / change in speed / velocity **or** has a constant acceleration **or** speed-time graph is a straight line	**B1**
	(only in uniform acceleration) same increase / change in speed / velocity in same time (period) **or** same rate of change of speed / velocity (with time) **or** change in speed / velocity is proportional to time	**B1**
1(b)(ii)	(a =) F / m in any way, algebraic or numerical	**C1**
	0.63–0.66 m / s^2	**A1**

2 A car approaches a set of traffic lights. The lights change to red at time $t = 0$.

Fig. 2.1 shows how the speed of the car changes with time.

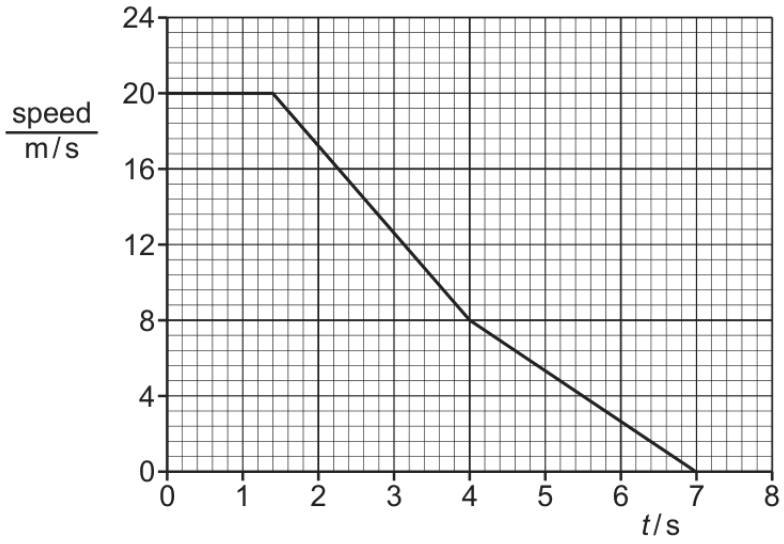


Fig. 2.1

(a) The car starts to slow down a short time after the lights change to red.

Determine the time between the lights changing to red and the car starting to slow down.

..... [1]

(b) **(i)** State what is meant by *uniform acceleration*.

.....

..... [1]

(ii) State how Fig. 2.1 shows that the deceleration of the car between $t = 2$ s and $t = 7$ s is *non-uniform*.

.....

..... [1]

(c) Determine the distance the car travels from the moment the car starts to slow down until it stops.

distance = ... [3]

[Total: 6]

Question	Answer	Marks
2(a)	1.4 s	**B1**
2(b)(i)	same change in speed / velocity in the same time	**B1**
2(b)(ii)	slope / gradient changes	**B1**
2(c)	any clear statement or any use of area under graph **or** $d = s \times t$ **or** an (average) speed × time	**C1**
	calculation of any area between 1.4 and 7.0 s on Fig.2.1 e.g. 14×2.6; 36.4; 4×3; 12; $½ \times 12 \times 2.6$; 15.6; 2.6×8; 20.8 (m) seen	**C1**
	48 m	**A1**

QUESTION · 5054_w21_qp_11.pdf · source page 2 of 20

3 Two identical objects begin to fall from rest.

One object falls from 200 m above the Earth's surface and the other falls from 200 m above the Moon's surface. One second after they have started to fall, both objects are still accelerating.

There is no atmosphere on the Moon and the weight of each object is constant.

Which row describes the motion of both objects at this time?

	acceleration of object falling above the Earth's surface	acceleration of object falling above the Moon's surface
A	decreasing	constant
B	constant	constant
C	increasing	decreasing
D	increasing	increasing

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 3

3	A	**1**

2 Fig. 2.1 shows a man of mass 80 kg standing in a lift (elevator).

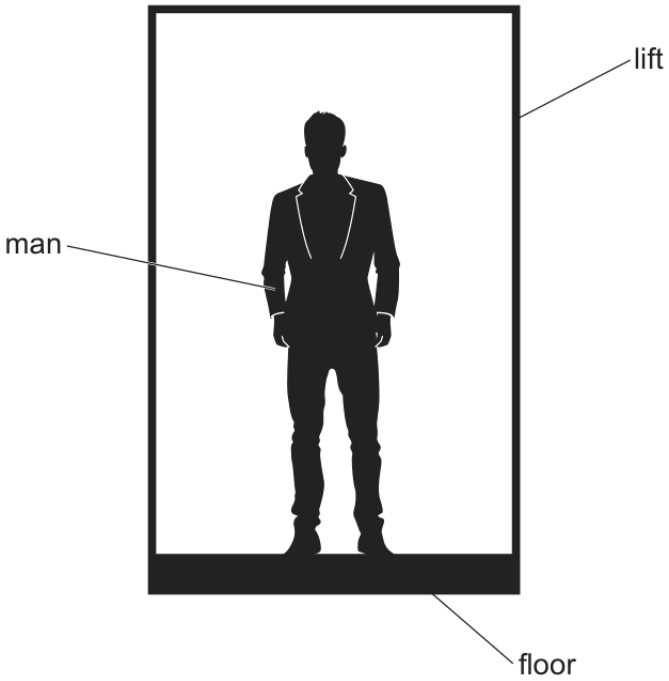


Fig. 2.1

The gravitational field strength g is 10 N / kg.

(a) Calculate the weight of the man.

weight = ... [1]

(b) The lift accelerates upwards uniformly at 0.50 m / s^2.

Calculate:

(i) the resultant upward force on the man

resultant force = ... [2]

(ii) the force exerted on the man by the floor of the lift.

force = ... [1]

(c) The lift accelerates upwards uniformly from rest for 4.0 s and then decelerates uniformly to rest in 4.0 s. The total distance travelled is 8.0 m.

On Fig. 2.2, sketch the distance-time graph for this journey.

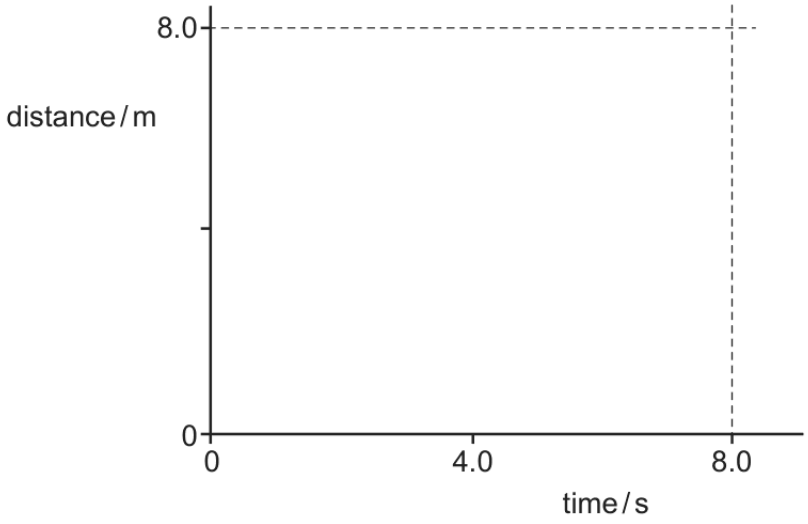


Fig. 2.2

[3]

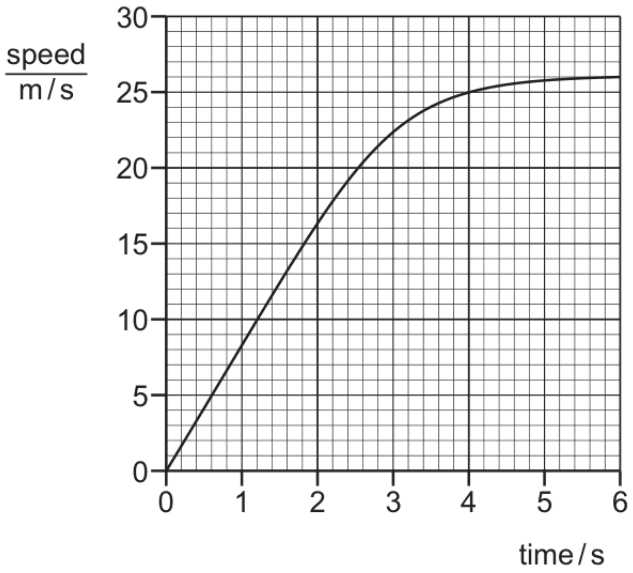
[Total: 7]

Question	Answer	Marks
2(a)	800 N **c.a.o.**	**B1**
2(b)(i)	(F =) ma **or** 80×0.50	**C1**
	40 N	**A1**
2(b)(ii)	840 N	**B1**
2(c)	increasing gradient from origin to t = 4.0 s	**B1**
	decreasing gradient from end of first curve	**B1**
	line horizontal at origin **or** at t = 8.0 s (and beyond)	**B1**

QUESTION · 5054_w21_qp_12.pdf · source page 2 of 16

3 A skier slides down a slope.

The diagram shows how his speed varies with time.



What is his average acceleration during the 6.0 s?

A 2.2 m/s^2 **B** 4.3 m/s^2 **C** 8.0 m/s^2 **D** 13.0 m/s^2

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 3

3	B	**1**

5 Fig. 5.1 shows a simple d.c. motor used in a toy car.

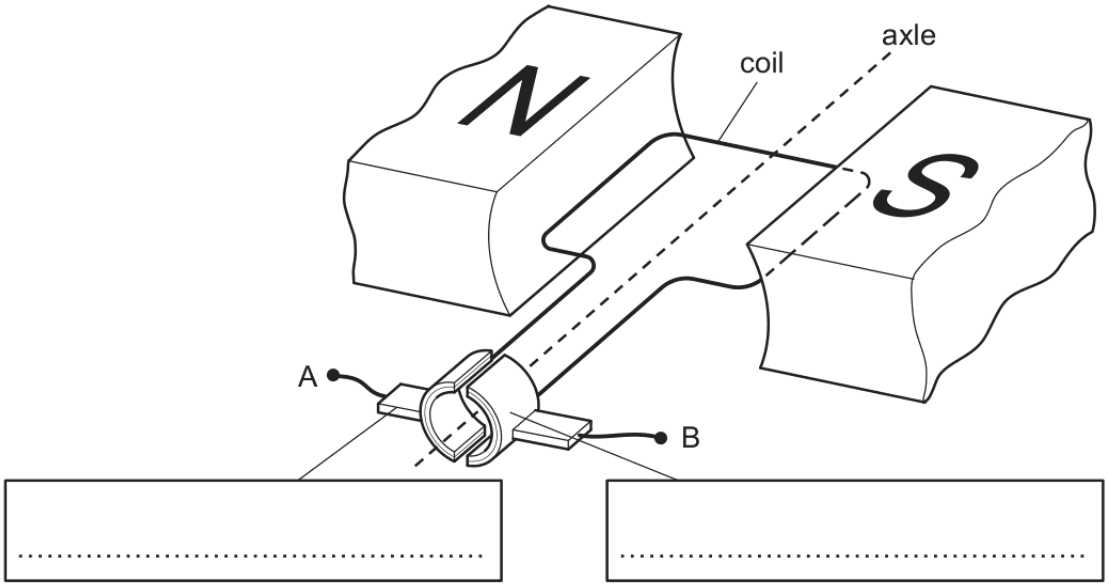


Fig. 5.1

Terminals A and B are connected to a battery and the motor rotates.

(a) On Fig. 5.1, add labels inside the boxes to identify the parts of the motor indicated. [2]

(b) State what happens to the rotation of the coil when:

(i) the number of turns on the coil is increased

..... [1]

(ii) the magnetic field between the poles of the magnet is reversed.

..... [1]

(c) The power supply to the motor is switched on and off at a steady rate.

Fig. 5.2 shows how the speed of the toy car varies with time as a result of the power supply being switched on and off.

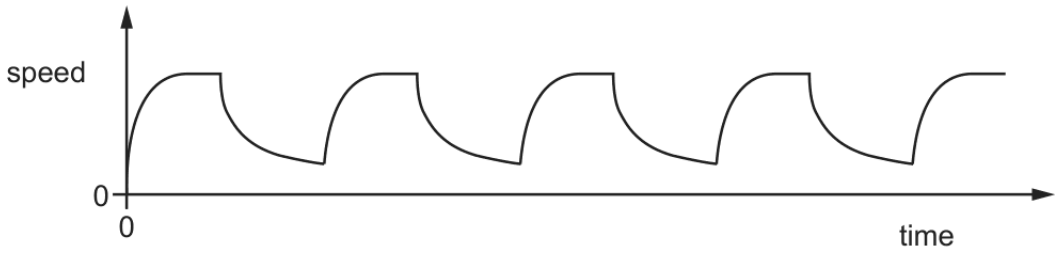


Fig. 5.2

(i) Describe the motion of the toy car.

.....

.....

.....

..... [2]

(ii) The voltage supplied to the motor is switched on for a longer period of time and off for a longer period of time, at a steady rate.

Suggest what happens to the motion of the toy car.

.....

.....

..... [1]

[Total: 7]

Question	Answer	Marks
5(a)	brush / carbon	**B1**
	split-ring **or** commutator	**B1**
5(b)(i)	faster **or** greater turning effect	**B1**
5(b)(ii)	goes backwards / turns backwards / reverses	**B1**
5(c)(i)	speeds up **and** slows down (when turned on / off)	**B1**
	reaches a steady speed / terminal speed / maximum speed **or** slows down but never to rest **or** cycle continuously repeats	**B1**
5(c)(ii)	same top / constant speed **or** constant speed / maximum speed for a longer time **or** drops to a lower speed / comes to rest	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 Fig. 7.1 shows a toy helicopter. It can hover and travel through the air.

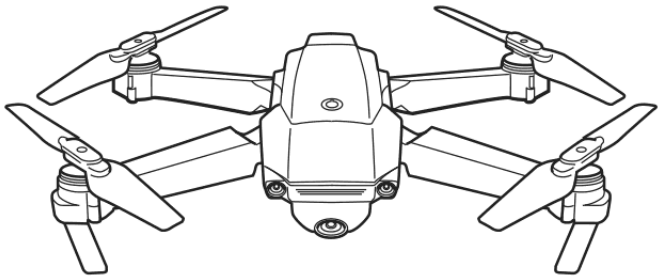


Fig. 7.1

A student flies the toy helicopter on a journey from A to B to C to D at a constant height.

Fig. 7.2 is a scale drawing of the path of the helicopter, viewed from above.

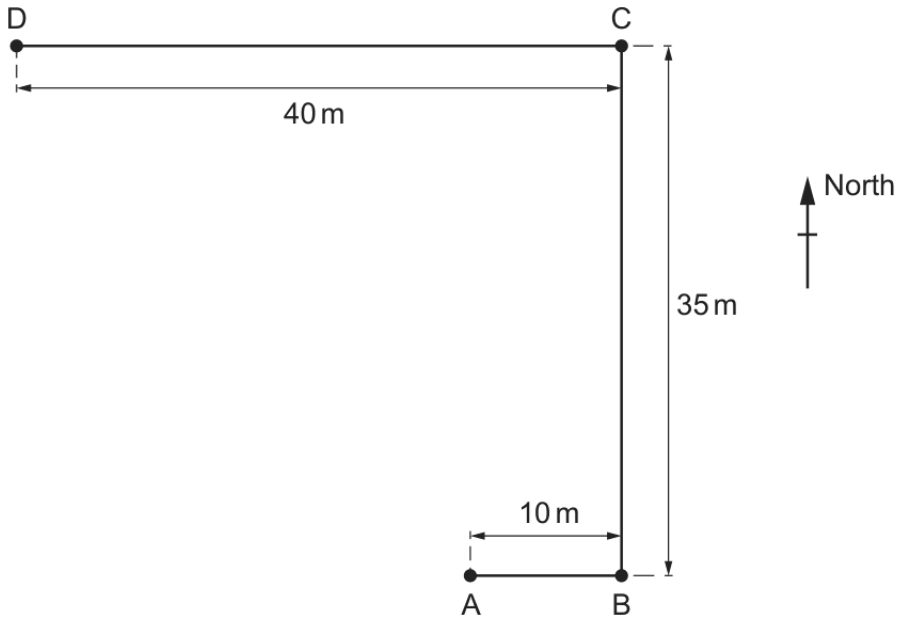


Fig. 7.2 (to scale)

(a) (i) Determine the total distance travelled by the toy helicopter.

..... [1]

(ii) The toy helicopter makes the journey in 40 s.

Calculate its average speed during the journey.

speed = ... [2]

(b) (i) State the difference between distance and displacement.

.....

..... [1]

(ii) Determine the scale used to construct Fig. 7.2 and complete the sentence.

1 cm on the diagram represents ..... on the path. [1]

(iii) Using Fig. 7.2, determine the size of the displacement of point D from point A.

Show your working.

size of displacement = ... [2]

(iv) Determine the angle between North and the direction of the displacement of point D from point A.

angle = ... [1]

(v) State what is meant by velocity.

..... [1]

(vi) Another toy helicopter flies directly from point A to point D in 40 s.

Explain why the magnitude of the velocity of this toy helicopter is smaller than the answer in **(a)(ii)**.

.....

..... [1]

(c) When the toy helicopter hovers at D, its motor fails and it falls. It reaches terminal velocity as it falls.

Explain, in terms of the forces and acceleration, what happens as the helicopter falls and reaches terminal velocity.

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 15]

Question 8 begins over the page.

MARK SCHEME · 5054_s22_ms_21.pdf · source page 8 of 11 · Question 7

Question	Answer	Marks
7(a)(i)	85 m	**B1**
7(a)(ii)	(speed =) d / t in any form algebraic or numerical	**C1**
	2.1 m / s	**A1**
7(b)(i)	displacement involves direction **or** is a vector	**B1**
7(b)(ii)	5 m shown on dotted line	**B1**

MARK SCHEME · 5054_s22_ms_21.pdf · source page 9 of 11 · Question 7

Question	Answer	Marks
7(b)(iii)	length of diagonal 9.2 (cm) $\sqrt{30 \times 30 + 35 \times 35}$	**B1**
	46 m 46 m	**B1**
7(b)(iv)	39–42° **or** 318–320°	**B1**
7(b)(v)	(average) velocity = (change in) displacement / time **or** (average) velocity is displacement per unit time **or** rate of change of displacement	**B1**
7(b)(vi)	(size of) displacement < distance in **(a)(ii)** or distance covered is less (than 85 m / in **(a)(ii)**)	**B1**
7(c)	air resistance **and** weight / gravitational force mentioned	**B1**
	as speed increases air resistance increases	**B1**
	resultant / net force decreases **or** acceleration decreases	**B1**
	(eventually) air resistance = weight (at terminal velocity) / resultant force is zero	**B1**
	(at terminal velocity / eventually) zero acceleration / constant speed	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 Fig. 7.1 shows the speed–time graph for a car travelling on a straight horizontal road.

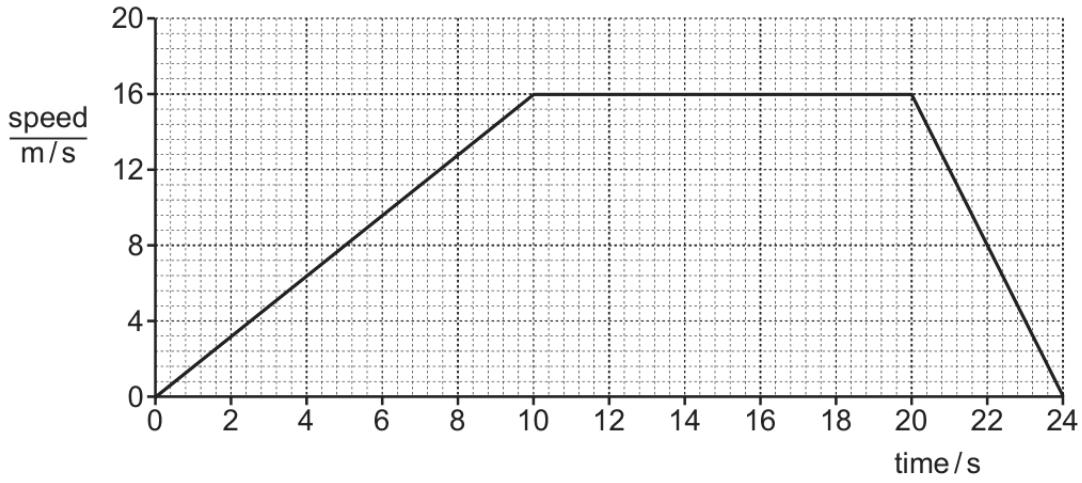


Fig. 7.1

(a) (i) Describe the motion of the car.

.....

.....

.....

.....

..... [3]

(ii) Using Fig. 7.1, calculate the distance travelled by the car during the 24 s of its motion.

Show your working.

distance = ... [3]

(iii) Calculate the average speed of the car during its motion.

average speed = ... [2]

(iv) A second car travels at a steady speed. It travels the same distance as the first car in the 24 s of the journey.

On Fig. 7.1, draw the speed–time graph for the second car. [2]

(b) The thinking distance is the distance travelled by a car between the time that a hazard is seen and the time that the brakes are applied.

The braking distance is the distance travelled while the car slows down to rest.

Table 7.1 shows the thinking and braking distances for an alert driver when the car travels at different speeds.

Table 7.1

$\frac{\text{speed}}{\text{km/h}}$	thinking distance / m	braking distance / m
20	9	2
40	18	9
60		20
80	36	36
100	45	56

(i) Complete Table 7.1. [1]

(ii) The time it takes for the driver to react to the hazard is constant at different speeds.

Explain how the table shows this.

.....

.....

.....

..... [2]

(iii) State what happens to the thinking distance and the braking distance when the driver is tired.

thinking distance .....

braking distance .....
[2]

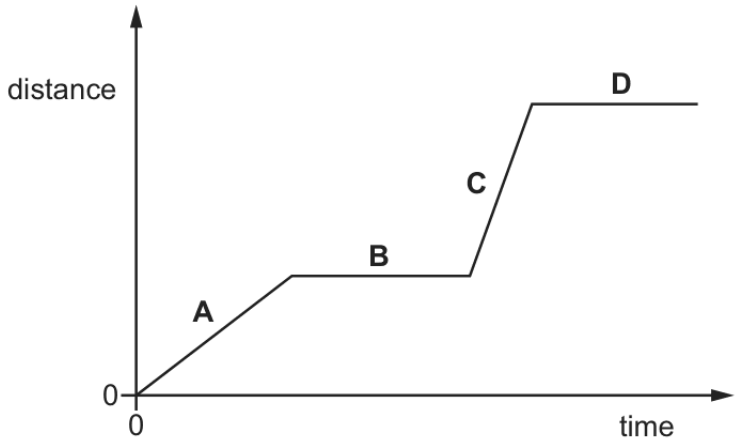
[Total: 15]

Question	Answer	Marks
7(a)(i)	acceleration constant / for 10 s **or** acceleration 1.6 m / s^2 **or** speeds up for 10 s / until it reaches 16 m / s	**B1**
	constant / uniform speed for 10 s / from 10 to 20 s / of 16 m / s **or** zero acceleration for 10 s / from 10 to 20 s	**B1**
	decelerates constantly / for 4 s / from 20 s to 24 s / until 24 s **or** deceleration 4 m / s^2 **or** acceleration −4 m / s^2	**B1**

Question	Answer	Marks
7(a)(ii)	area under curve **or** formula for area of a triangle / trapezium **seen** **or** working shown used	**B1**
	(accelerating) 80 (m) **or** (at constant speed) 160 (m) **or** (decelerating) 32 (m) seen	**C1**
	(total distance) 270 **or** 272 m	**A1**
7(a)(iii)	(average speed) = total distance / (total) time algebraic or numerical	**C1**
	11 m / s	**A1**
7(a)(iv)	horizontal line drawn from 0 to 24 s at any level **or** horizontal line at the value calculated in **7(a)(iii)** for at least 16 s	**B1**
	horizontal line from 0 to 24 s at the value calculated in **7(a)(iii)**	**B1**
7(b)(i)	27	**B1**
7(b)(ii)	(statement that thinking) distance is proportional to speed **or** thinking distance increases uniformly with speed **or** time = (thinking) distance / speed and stated to be equal / not affected **or** ratio of (thinking) distance to speed is constant **or** (thinking) distance and speed both increase by same ratio **or** same change in (thinking) distance for equal changes in speed	**C1**
	at least two values thinking distance ÷ speed (or inverse) calculated **or** all values of thinking distance/speed stated to be 0.45 (or inverse 2.2) **or** proportionality shown by at least two values **or** double the speed doubles the (thinking) distance **or** each 20 (km / h) / same increase in speed covers an extra 9 (m) distance	**A1**
7(b)(iii)	*thinking distances* increases	**B1**
	braking distances no change	**B1**

QUESTION · 5054_w22_qp_11.pdf · source page 2 of 16

4 Which part of the distance–time graph shows the fastest constant speed?



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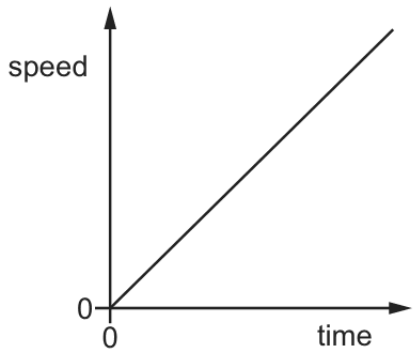
4	C	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 3 of 16

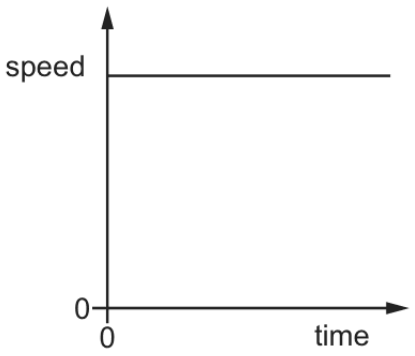
5 An object is accelerating along a straight, horizontal track. The acceleration is uniform.

Which speed–time graph represents the motion of the object?

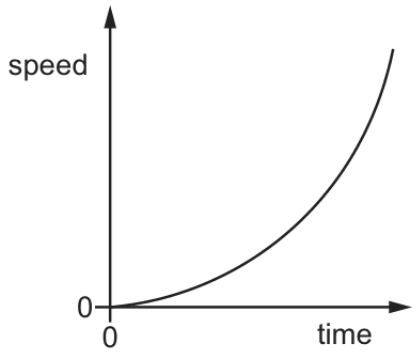
A



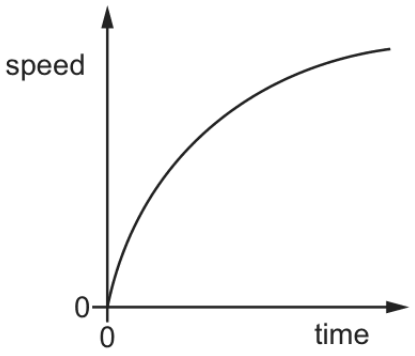
B



C



D



MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 5

5	A	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 Fig. 7.1 shows a child sitting on a sledge on a snow-covered hill of constant slope.

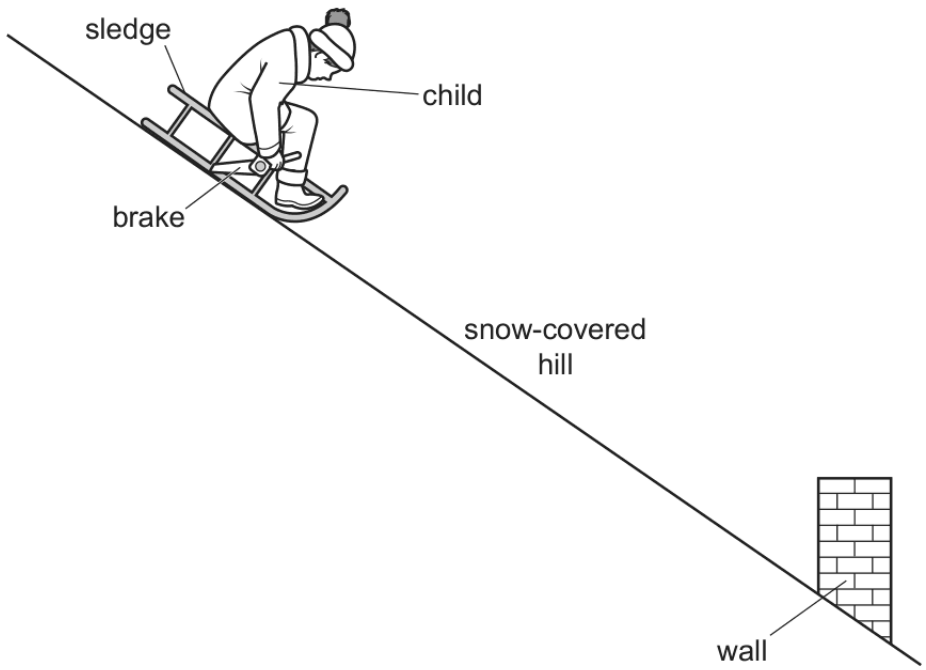


Fig. 7.1 (not to scale)

At time $t = 0$, the child and the sledge begin to move down the hill in a straight line.

When the child sees a wall ahead, he applies the brake.

The child and sledge continue to travel in a straight line until they come to a stop before hitting the wall.

Fig. 7.2 is the speed-time graph for the journey.

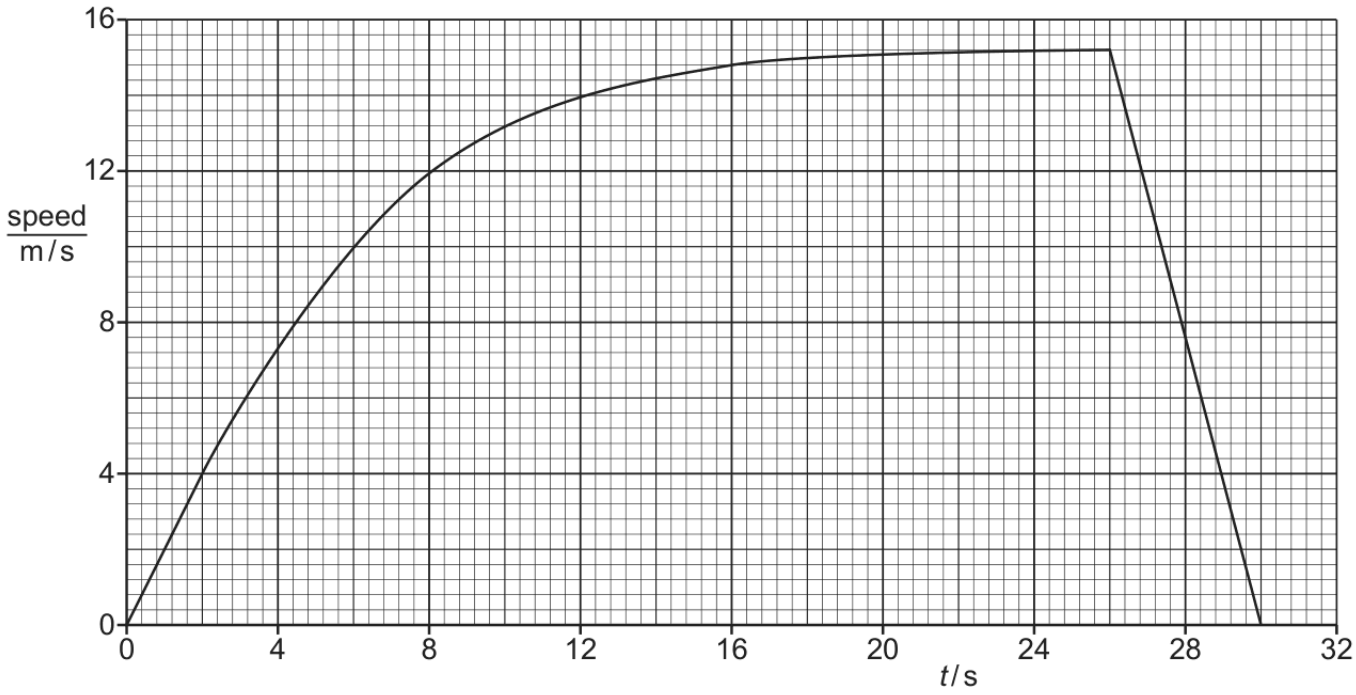


Fig. 7.2

The brake is applied at $t = 26$ s.

(a) Fig 7.2 shows how the speed of the child and sledge varies over the whole of the journey.

Explain why, between $t = 0$ and $t = 26$ s, the speed varies in the way shown by the graph.

.....

.....

.....

.....

.....

..... [3]

(b) At $t = 26$ s, the front of the sledge is 35 m from the wall.

Determine the distance between the front of the sledge and the wall when the sledge stops.

distance = ... [3]

(c) At $t = 26$ s, the child and sledge begin to decelerate.

(i) Determine the size of the deceleration.

deceleration = ... [3]

(ii) The mass of the child is 46 kg and the mass of the sledge is 9.0 kg.

Calculate the resultant force on the child and sledge as they decelerate.

resultant force = ... [2]

(iii) State the energy transfer that is taking place as the child and sledge decelerate.

.....

.....

..... [2]

(d) At $t = 26$ s, when the brake is first applied, the child jerks forwards on the sledge.

Explain why.

.....

.....

..... [2]

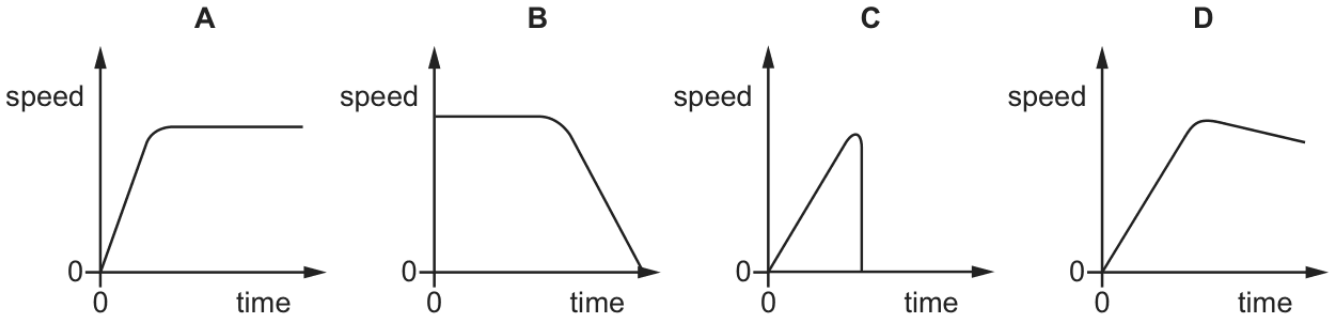
[Total: 15]

Question	Answer	Marks
7(a)	acceleration / increase in speed **and** gravity / weight mentioned	**B1**
	air resistance / friction increases (with speed)	**B1**
	terminal / constant velocity reached **or** air resistance (becomes) equal to accelerating force	**B1**
7(b)	(distance travelled =) area under the graph **or** (x =) ½(u + v)t **or** ½ × 15.2 × 4.0	**C1**
	35 – (½ × 15.2 × 4.0) **or** 30.4 **or** 30	**C1**
	4.6 m	**A1**
7(c)(i)	(magnitude of deceleration =) gradient **or** Dv / Dt **or** 15.2 / 4.0 **or** equivalent numbers (from graph) in same ratio	**C1**
	15.2 / 4.0 **or** equivalent numbers in same ratio	**C1**
	3.8 m / s^2	**A1**
7(c)(ii)	(F =) ma **or** (46 + 9.0) × 3.8 **or** 55 × 3.8	**C1**
	210 N	**A1**
7(c)(iii)	<u>from</u> kinetic energy (and from gravitational potential energy)	**B1**
	<u>to</u> thermal energy **and** no intermediate energy	**B1**
7(d)	due to his mass **or** mass mentioned	**B1**
	(his) mass resists a change to its state of motion	**B1**

QUESTION · 5054_w22_qp_12.pdf · source page 3 of 16

5 A ball starts from rest and rolls down a steep slope. The ball then rolls along rough horizontal ground.

Which graph shows the speed of the ball at different times?



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5	D	**1**

Section A

Answer **all** questions in this section. Answer in the spaces provided.

1 A train travels along a straight horizontal track. At time $t = 0$, the train passes through station P at constant speed without stopping.

The driver applies the brakes 70 s before reaching station Q. The train decelerates.

Fig. 1.1 is the speed–time graph for the train from $t = 0$ until it stops at station Q.

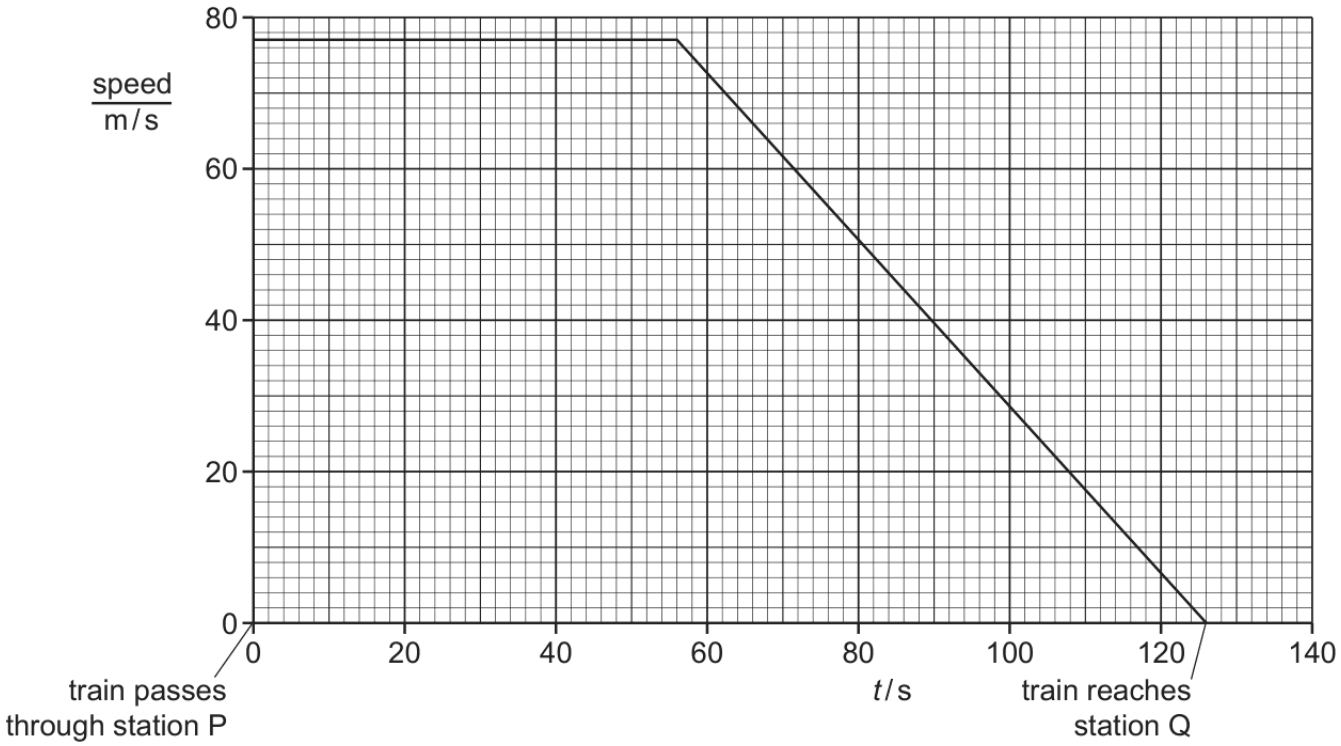


Fig. 1.1

(a) Using Fig. 1.1, determine the distance between station P and station Q.

distance = ... [3]

(b) The mass of the train is 3.8×10^5 kg.

(i) Determine the deceleration of the train in the 70 s before it stops at station Q.

deceleration = ... [2]

(ii) Calculate the resultant force on the train as it decelerates.

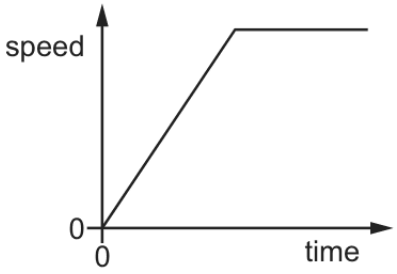
resultant force = ... [2]

[Total: 7]

Question	Answer	Marks
1(a)	(distance =) area (under graph) **or** 77 × 56 **or** ½ × 77 × 70 **or** 4300 **or** 2700 **or** ½ × 77 × (56 + 126)	**C1**
	77 × 56 + (½ × 77 × 70) **or** 4300 + 2700 **or** ½ × 77 × (56 + 126)	**C1**
	7000 m **or** 7.0×10^3 m **or** 7.0 km	**A1**
1(b)(i)	(a =) $\Delta v / \Delta t$ **or** 77 / 70 (etc.)	**C1**
	1.1 m / s^2	**A1**
1(b)(ii)	(F =) ma **or** $3.8 \times 10^5 \times 1.1$	**C1**
	4.2×10^5 N	**A1**

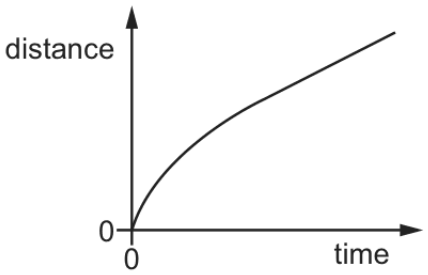
QUESTION · 5054_s23_qp_11.pdf · source page 4 of 20

4 A speed–time graph for the journey of a car is shown.

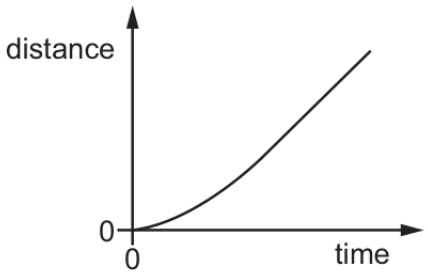


What is the distance–time graph for the journey?

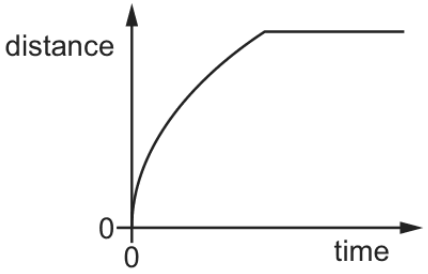
A



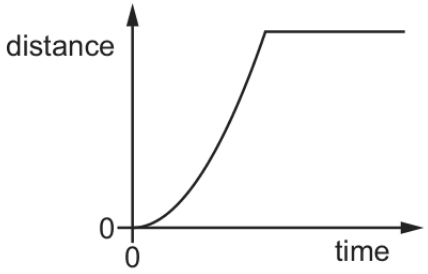
B



C



D



MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

1 An aircraft pulls a glider along a runway as shown in Fig. 1.1.

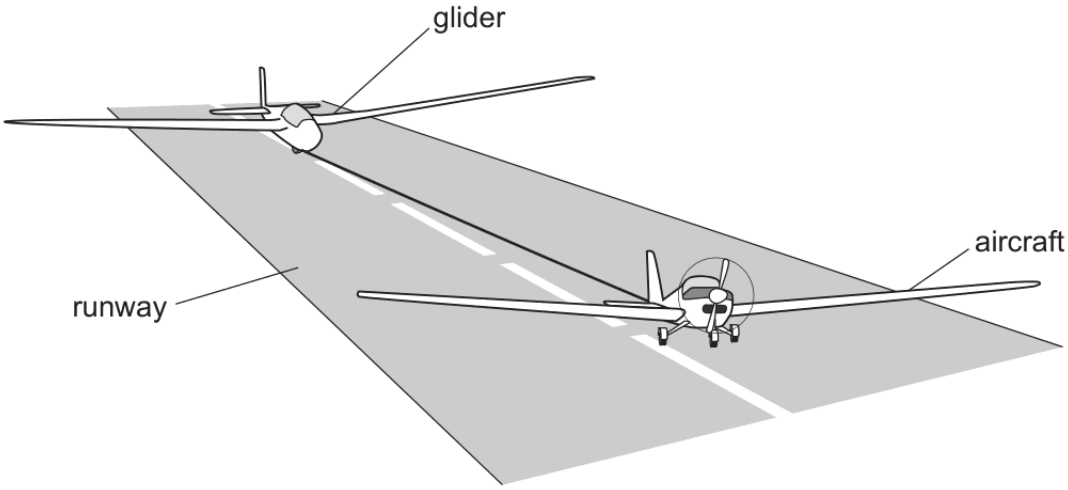


Fig. 1.1

Fig. 1.2 shows the speed of the glider during the first 12 s of the motion.

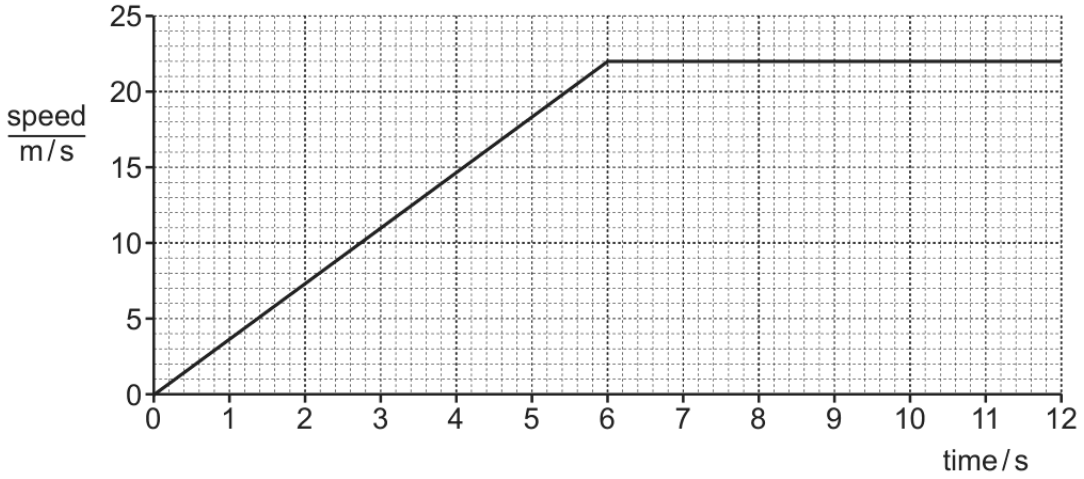


Fig. 1.2

(a) Describe the motion of the glider in the first 12 s.

.....

.....

..... [2]

(b) In the first 6.0 s of the motion, there is a resultant force of 1800 N on the glider.

Using the increase in speed in the first 6.0 s, calculate the mass of the glider.

mass = ... kg [3]

(c) Determine the distance travelled by the glider in the first 6.0 s of its motion.

distance = ... m [2]

(d) The glider has no engine and stays in the air with the use of convection currents.

(i) State what is meant by a 'convection current'.

.....

.....

..... [1]

(ii) Suggest how the convection current that supports the glider is formed.

.....

.....

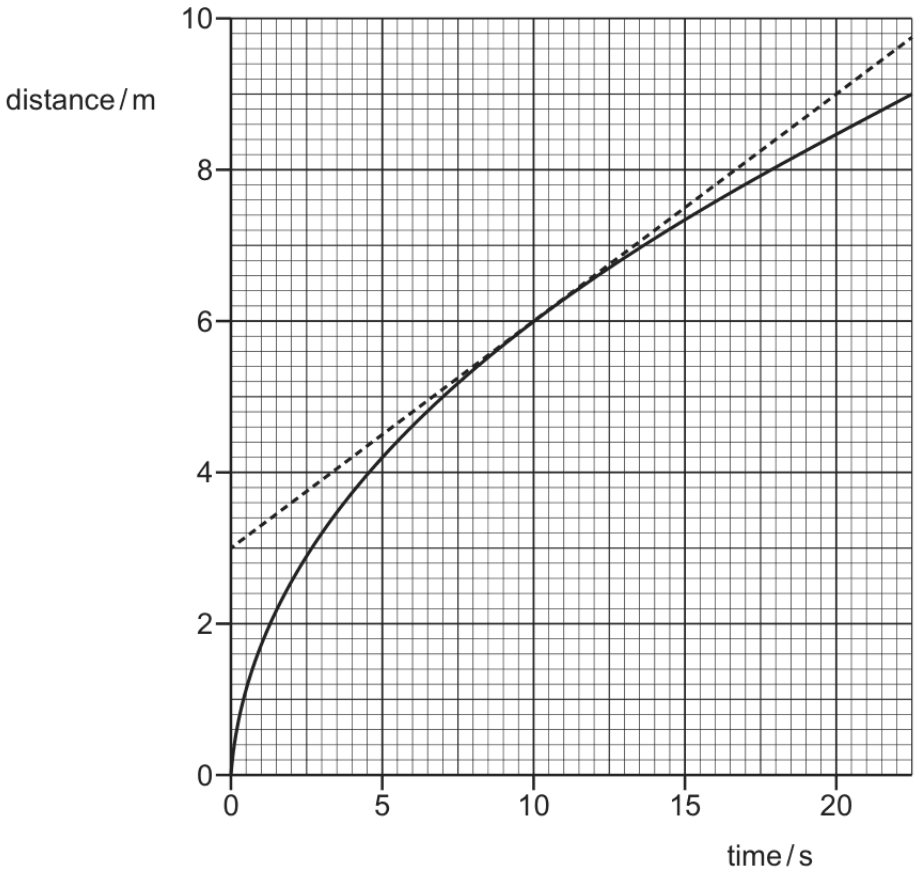
..... [1]

[Total: 9]

Question	Answer	Marks
1(a)	uniform / constant acceleration (for 6 s) / acceleration of 3.7 m / s^2	**B1**
	after 6(.0 s) constant speed / zero acceleration	**B1**
1(b)	(a =) (v–u) / t in any form numerical or algebraic **or** 3.7 (m / s^2) seen	**C1**
	(m =) F / a **or** (m =) Ft / v in any form numerical or algebraic	**C1**
	490 (kg)	**A1**
1(c)	area under graph **or** average speed × time numerical or algebraic	**C1**
	66 (m)	**A1**
1(d)(i)	rising of hot air / hot fluid (and falling of cold air) **or** circulation of air / fluid with temperature differences	**B1**
1(d)(ii)	high temperature spot on Earth, e.g. from a chimney, dark patch in sunlight, above a town / land / surface; Sun heats surface **or** (hot) less dense air rises upwards	**B1**

QUESTION · 5054_s23_qp_12.pdf · source page 3 of 16

4 The curved line on the distance–time graph shows the motion of a toy car. The straight line is the tangent to the curve at 10 s.



What is the speed of the toy car at 10 s?

A 0.3 m / s **B** 0.6 m / s **C** 1.7 m / s **D** 3.3 m / s

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 4

4	A	**1**

1 Fig. 1.1 shows the speed–time graph for a car travelling on a straight horizontal road.

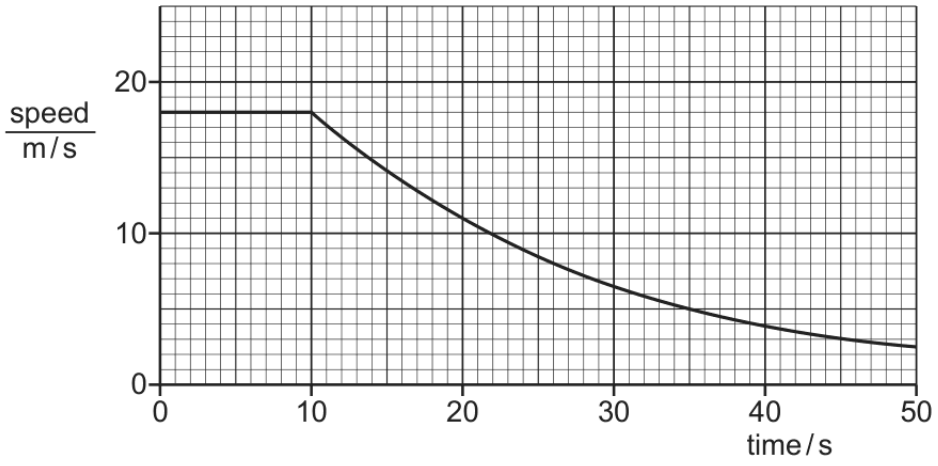


Fig. 1.1

(a) Describe the motion of the car shown in Fig. 1.1.

.....

.....

.....

..... [2]

(b) At time t = 10 s the engine of the car is switched off. The brakes are not applied.

(i) Name **two** forces that act on the car to cause the change in motion after t = 10 s.

1 .....

2 .....

[1]

(ii) Suggest why Fig. 1.1 is a curve after t = 10 s.

.....

.....

..... [1]

(c) Between $t = 10\,s$ and $t = 20\,s$ the speed of the car changes from $18\,m/s$ to $11\,m/s$.

The mass of the car is $1200\,kg$.

(i) Calculate the change in momentum of the car in this time.

Give the unit of your answer

momentum change = ... unit ..... [2]

(ii) Calculate the average resultant force exerted on the car during this time.

average resultant force = ... N [2]

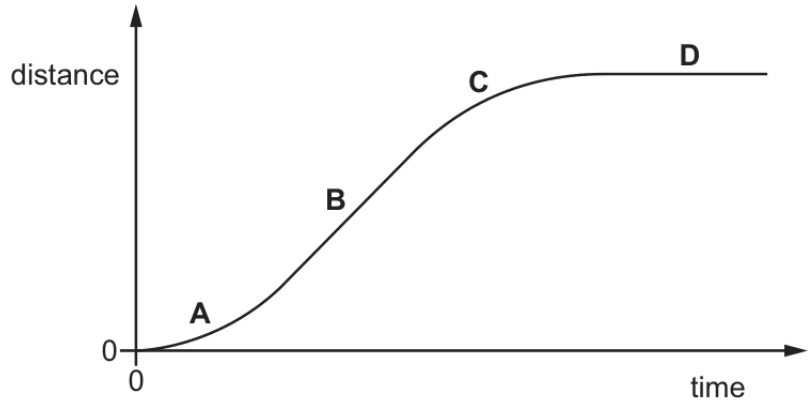
[Total: 8]

Question	Answer	Marks
1(a)	speed is constant for 10 s / at first	**B1**
	deceleration • has (average) value 0.39 m / s^2 • lasts for 40 s • is non-uniform	**B1**
1(b)(i)	friction (with road) **and** air resistance **or** (air) drag	**B1**
1(b)(ii)	resistive force(s) change / decrease (as speed decreases) **or** (graph shows) deceleration is not constant / decreases **or** non-uniform decrease in speed	**B1**
1(c)(i)	(momentum =) *mv* in any form algebraic or numerical	**C1**
	8400 **and** kg m / s **or** Ns	**A1**
1(c)(ii)	(F =) $\Delta mv / t$ **or** $F = ma$ in any form algebraic or numerical	**C1**
	840 (N)	**A1**

QUESTION · 5054_w23_qp_11.pdf · source page 2 of 16

2 The diagram shows a distance–time graph for a car travelling in a straight line.

In which region is the car decelerating?



MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 2

2	C	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 2 of 16

4 A car begins to move. It speeds up until it reaches a constant speed.

What happens to the acceleration and what happens to the velocity of the car?

A Both the acceleration and the velocity change.

B Only the acceleration changes.

C Only the velocity changes.

D Neither the acceleration nor the velocity changes.

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 4

4	A	**1**

2 A lorry (truck) is at rest on a straight, horizontal road.

A constant driving force of 5.4×10^4 N acts on the lorry and it accelerates forwards.

(a) Define what is meant by 'acceleration'.

.....

.....

..... [2]

(b) As the lorry moves forwards, work is done on it by the driving force.

(i) State the equation that defines work done.

Refer to the direction of the force exerted.

.....

..... [1]

(ii) Calculate the work done on the lorry by the driving force when the lorry travels a distance of 280 m.

work done = ... J [2]

(iii) At one point on the road, the kinetic energy of the lorry is 3.2×10^6 J and its speed is 12 m/s.

Calculate the mass of the lorry.

mass = ... kg [3]

[Total: 8]

MARK SCHEME · 5054_w23_ms_21.pdf · source page 6 of 11 · Question 2

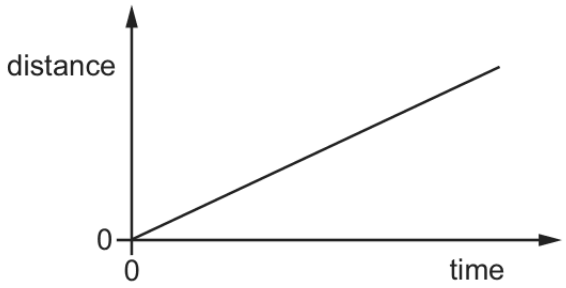
Question	Answer	Marks
2(a)	change / increase of velocity (with time)	**B1**
	change / increase of velocity per unit time **or** rate of change / increase of velocity	**B1**
2(b)(i)	force × distance moved in the direction of the force / forwards	**B1**
2(b)(ii)	(work done =) $5.4 \times 10^4 \times 280$	**C1**
	1.5×10^7 (J)	**A1**

MARK SCHEME · 5054_w23_ms_21.pdf · source page 7 of 11 · Question 2

Question	Answer	Marks
2(b)(iii)	$E_k = ½mv^2$ **or** (m =) $2 \times E_k / v^2$	**C1**
	(m =) $2 \times 3.2 \times 10^6 / 12^2$ **or** $2 \times 3.2 \times 10^6 / 144$	**C1**
	4.4×10^4 (kg)	**A1**

QUESTION · 5054_w23_qp_12.pdf · source page 2 of 20

2 The diagram shows the distance–time graph of a car.



The car is travelling along a straight road up a hill.

Which quantity for the car is constant and greater than zero?

A acceleration

B displacement

C gravitational potential energy

D kinetic energy

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 2

2	D	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 2 of 20

3 A distance–time graph and a speed–time graph show the motion of an object.

How are the speed and the acceleration of the object determined from these graphs?

	speed	acceleration
A	area under the distance–time graph	area under the speed–time graph
B	area under the distance–time graph	gradient of the speed–time graph
C	gradient of the distance–time graph	area under the speed–time graph
D	gradient of the distance–time graph	gradient of the speed–time graph

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 3 of 16

3 The table shows the distance travelled by a car in each 2 s time interval during a 12 s period of its journey.

time interval / s	0–2	2–4	4–6	6–8	8–10	10–12
distance travelled in time interval / m	10	10	11	13	16	23

Which statement describes the motion of the car?

A moving with constant speed from 0–4 s, followed by non-uniform acceleration

B moving with constant speed from 0–4 s, then uniform acceleration

C at rest from 0–4 s, then non-uniform acceleration

D at rest from 0–4 s, then uniform acceleration

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 3

3	A	**1**

1 A ball is released from rest at point A and moves along a smooth track ABCDE as shown in Fig. 1.1. The ball is shown at point A and as it passes point B.

The ball is always in contact with the track and air resistance is negligible.

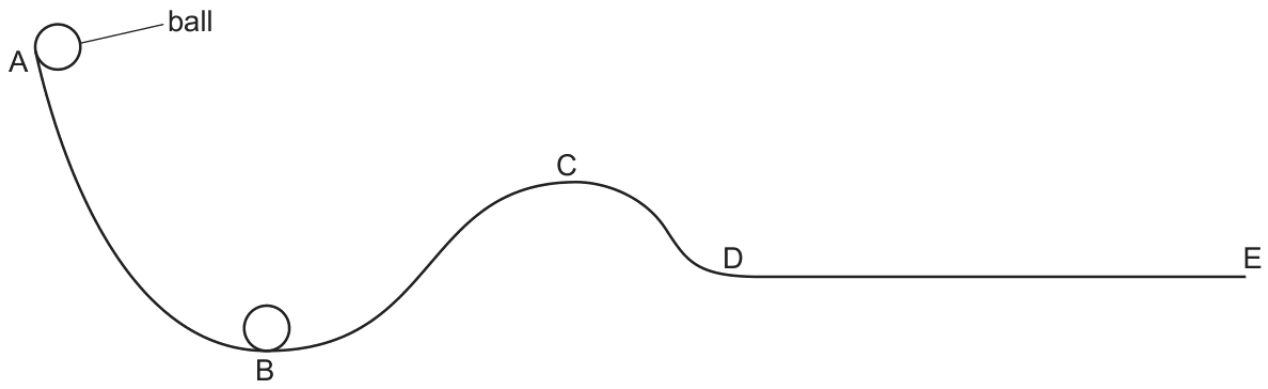


Fig. 1.1

Fig. 1.2 shows the distance−time graph for the ball as it moves from A to E.

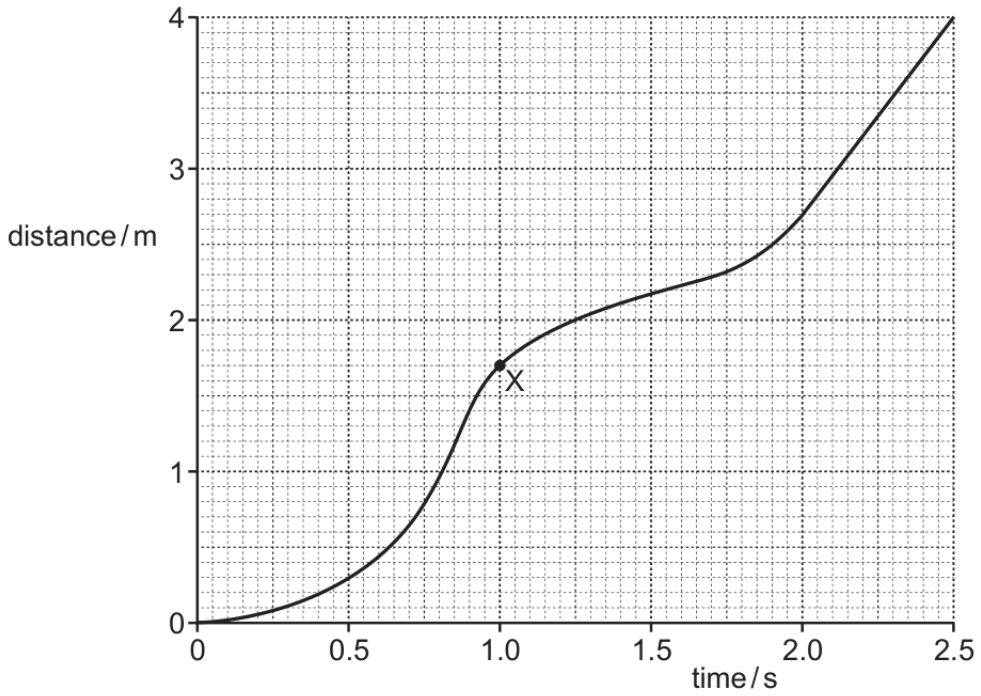


Fig. 1.2

(a) (i) On Fig. 1.2, mark:

- with the letter P **one** point where the ball accelerates
- with the letter Q **one** point where the ball has constant speed.

[2]

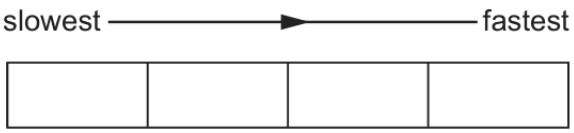
(ii) Determine the speed of the ball at point X on Fig. 1.2.

Show your working.

speed = ... m/s [3]

(b) The speeds of the ball at A, B, C and D are v_A, v_B, v_C and v_D respectively.

Arrange these four speeds from slowest to fastest.



[1]

(c) Work is done to transfer energy between energy stores as the ball moves from A to B.

Name the force involved in the work done and describe the energy transfer.

.....

.....

.....

..... [3]

(d) The track at point B is circular in shape.

On Fig. 1.1, draw an arrow to show the direction of the resultant force on the ball at point B.

[1]

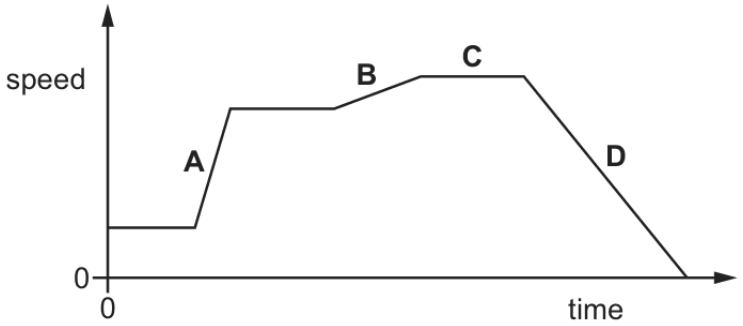
[Total: 10]

Question	Answer	Marks
1(a)(i)	P marked in region t = 0–0.8 s **or** in region 1.6–2.0 s	**B1**
	Q marked between 2.0 and 2.5 s	**B1**
1(a)(ii)	tangent drawn at X	**B1**
	(speed =) d/t in any form numerical or algebraic	**C1**
	1.9–2.4 m/s	**A1**
1(b)	$v_A < v_C < v_D < v_B$	**B1**
1(c)	(work done by) gravity / weight / gravitational attraction	**B1**
	from gravitational potential energy (store)	**B1**
	to kinetic energy (store)	**B1**
1(d)	arrow upwards at D	**B1**

QUESTION · 5054_s24_qp_12.pdf · source page 2 of 16

3 The graph shows how the speed of a car travelling in a straight line changes with time.

Which section shows the largest acceleration?



MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 3

3	A	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 3 of 16

4 A man walks along a path from X to Y. The diagram shows the path from above.

The man measures the distance he walks and the time taken.

Which quantity can be calculated using this data only?

A acceleration

B average speed

C average velocity

D power

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

1 Fig. 1.1 shows two trolleys. On the front of trolley A, there is a wooden rod. Trolley B is initially at rest.

As trolley A moves towards the right, the rod enters the modelling clay. Trolley A slows down and trolley B starts moving.

The trolleys then stick together and continue moving towards the right.

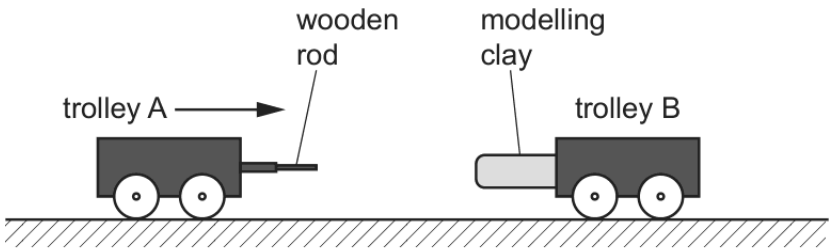


Fig. 1.1

Fig. 1.2 shows the speed−time graph for the two trolleys.

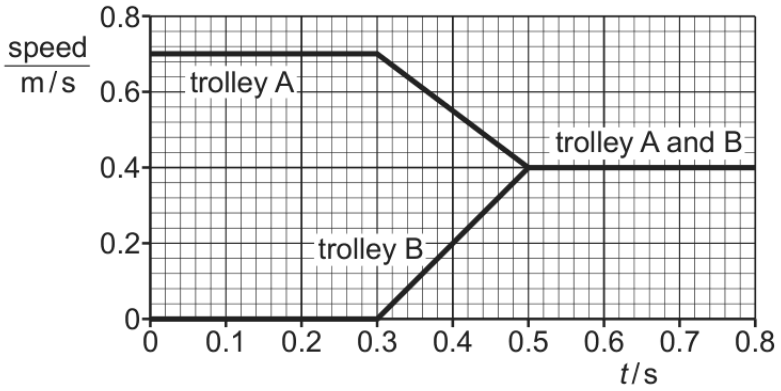


Fig. 1.2

The trolleys start to collide at time t = 0.30 s. At t = 0.50 s, the trolleys are moving at the same speed.

(a) (i) State how Fig. 1.2 shows that, during the collision, trolley B has a uniform acceleration.

.....

..... [1]

(ii) Describe how the graph in Fig. 1.2 shows that the magnitude (size) of the acceleration of trolley B is larger than the magnitude of the deceleration of trolley A.

.....

..... [1]

(iii) Calculate the acceleration of trolley B when $t = 0.40\,s$.

acceleration = ... m/s^2 [2]

(b) The mass of trolley A = 0.80 kg. The mass of trolley B = 0.60 kg.

Show that momentum is conserved in the collision.

[2]

(c) In another collision between the same trolleys, the rod and modelling clay are not present. Trolley A hits trolley B with the same initial speed.

Explain why the force between the trolleys is larger in this collision.

.....

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
1(a)(i)	straight line (on speed-time graph)	**B1**
1(a)(ii)	gradient of line steeper (for B) **or** larger change in speed (in same time)	**B1**
1(a)(iii)	(*a* =) change in speed / time in any form	**C1**
	2(.0) m / s^2	**A1**
1(b)	(p =) mv **or** (initial) momentum of A = 0.56 (kg m / s)	**B1**
	and final momentum of A and B = 1.4 × 0.4 = 0.56	**B1**
1(c)	Any two out of three: • contact surfaces are more rigid / less pliable **aaw** • decreases the time of contact **or** final velocity of B is larger (than 0.4 m / s) / final velocity of A is smaller • increases acceleration **or** force = (mv - mu) / t **or** impulse / t	**B2**

QUESTION · 5054_w24_qp_11.pdf · source page 2 of 16

4 A student takes 20 minutes to walk from X to Y at a speed of 3.0 km / h.

She turns 90° at Y and walks at a speed of 4.0 km / h for 15 minutes to reach Z.

What is her average speed from X to Z?

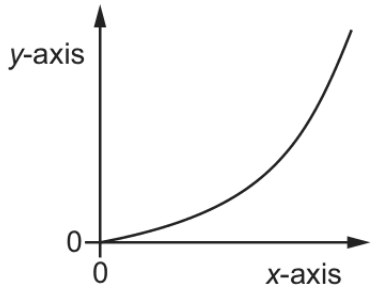
A 2.4 km / h **B** 3.4 km / h **C** 3.5 km / h **D** 5.0 km / h

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 3 of 16

5 A car is at rest by a red traffic light. When the light changes to green, the car starts moving. The graph represents the motion of the car.



The car has a constant acceleration.

Which quantity is plotted on the *x*-axis and which quantity is plotted on the *y*-axis?

	x-axis	*y*-axis
A	distance	time
B	speed	time
C	time	distance
D	time	speed

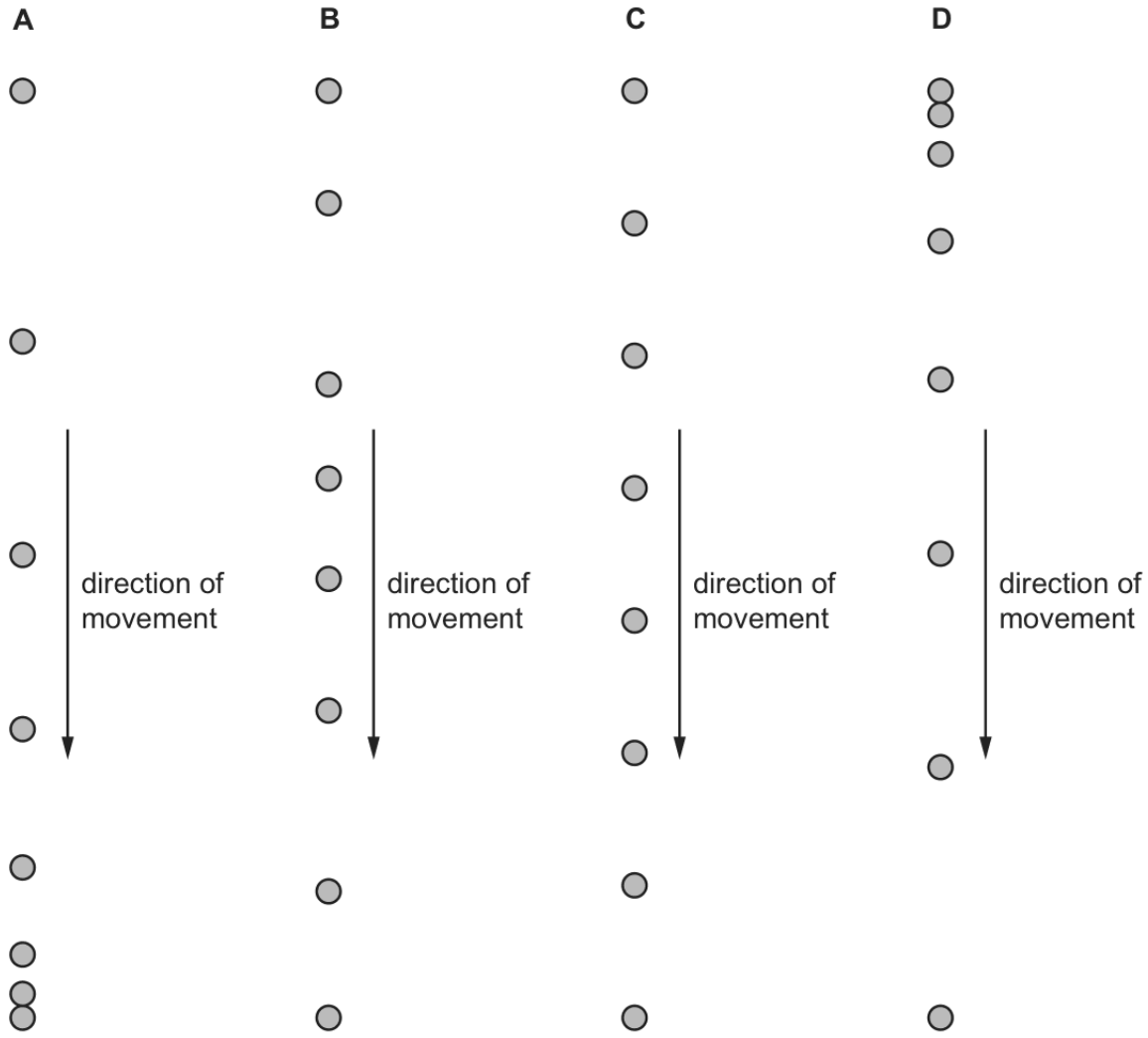
MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 5

5	C	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 4 of 16

6 An object falls without air resistance. Its position at equal time intervals is shown by a sequence of dots.

Which sequence of dots is correct?



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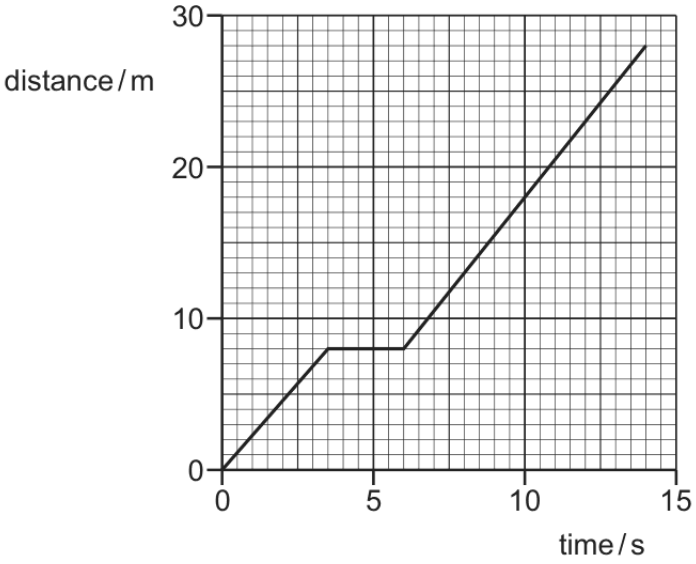
6	D	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 2 of 16

3 The graph shows the motion of a cyclist during a ride that lasts for 14 s.



What is the average speed of the cyclist for the ride?

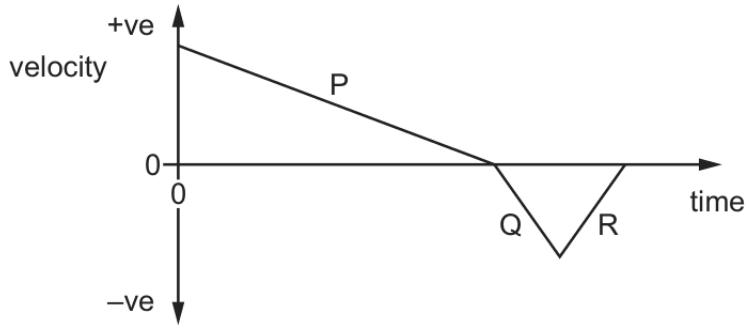
A 2.0 m / s **B** 2.3 m / s **C** 2.4 m / s **D** 2.5 m / s

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 3

3	A	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 3 of 16

4 The diagram shows how the velocity of a train varies with time.



Which sections of the graph show where the magnitude of the velocity of the train is decreasing?

A P and Q **B** P and R **C** P only **D** R only

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

9 The Earth has a radius of 6.4×10^6 m and rotates on its axis once in every 24 hours.

(a) Fig. 9.1 shows that as the Earth rotates, objects on the surface of the Earth move in circular paths around the axis of rotation.

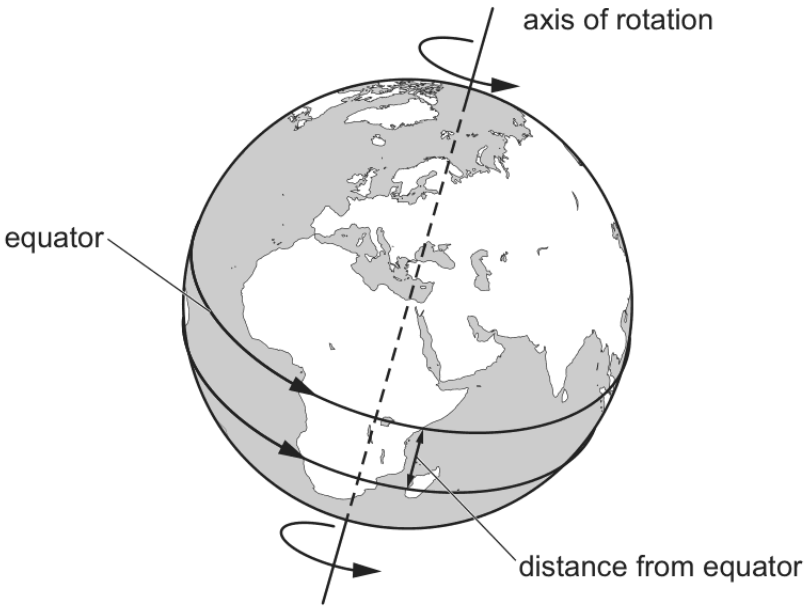


Fig. 9.1

(i) Calculate the speed of an object at the equator as it moves in a circular path around the axis of rotation.

speed = ... m/s [3]

(ii) Explain why the speed around the axis of rotation of an object on the Earth decreases as the distance of the object from the equator increases.

.....

.....

..... [2]

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(b) The Earth moves around the Sun because a force acts on the Earth.

(i) State what provides this force.

.....

.....

..... [2]

(ii) The orbit of the Earth around the Sun is an ellipse.

Describe what happens to the magnitude of the force that acts on the Earth as it moves in this elliptical orbit.

.....

.....

.....

..... [2]

(iii) Explain why the Earth travels the slowest when it is at its furthest distance from the Sun.

.....

.....

.....

..... [2]

[Total: 11]

MARK SCHEME · 5054_w24_ms_22.pdf · source page 9 of 10 · Question 9

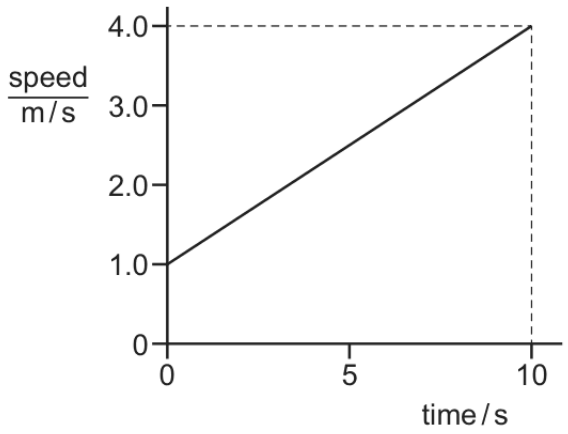
Question	Answer	Marks
9(a)(i)	(v =) s / t **or** any relevant distance / any relevant time	**C1**
	(v =) $2\pi \times 6.4 \times 10^6$ / (24×3600)	**C1**
	470 (m / s)	**A1**

MARK SCHEME · 5054_w24_ms_22.pdf · source page 10 of 10 · Question 9

Question	Answer	Marks
9(a)(ii)	(circumference of) path decreases (with distance from Equator)	**B1**
	rotation period / time for rotation remains constant	**B1**
9(b)(i)	gravitational attraction / field	**B1**
	due to Sun	**B1**
9(b)(ii)	increases **and** decreases **or** fluctuates **or** varies **or** not constant	**B1**
	decreases when moving away from Sun **or** increases when moving towards Sun	**B1**
9(b)(iii)	(at greatest distance) the gravitational potential energy is the greatest	**B1**
	(hence) kinetic energy the smallest (and so speed is the smallest)	**B1**

QUESTION · 5054_s25_qp_11.pdf · source page 2 of 20

2 The speed–time graph shows part of the journey of a bicycle for a time of 10 s.



What is the distance that the bicycle travels in the 10 s?

A 20 m **B** 25 m **C** 30 m **D** 40 m

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 2

2	B	**1**

1 Fig. 1.1 shows a skydiver falling vertically through the air.

Fig. 1.1

In the first part of the fall, her speed increases and her acceleration decreases.

In the second part of the fall, her speed is constant.

(a) On Fig. 1.2 sketch the speed–time graph for the skydiver.

On your graph, mark **one** point where the speed of the skydiver is increasing with an A and **one** point where the speed of the skydiver is constant with a B.

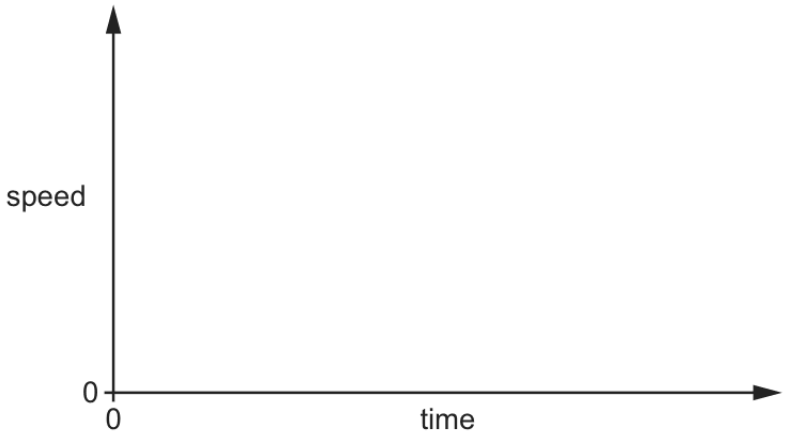


Fig. 1.2

[2]

(b) Explain how the graph shows that the acceleration decreases as the speed increases.

.....

.....

..... [1]

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(c) During the first part of the fall, there is a resultant vertical force acting downwards on the skydiver.

(i) One of the vertical forces acting on the skydiver is her weight.

State the name of the other vertical force that acts on the skydiver.

..... [1]

(ii) Explain why the resultant vertical force eventually becomes zero.

.....

.....

.....

..... [2]

(d) At one instant, the vertical force on the skydiver is 400 N downwards.

At the same instant, the wind causes an additional horizontal force of 100 N to the right to act on the skydiver.

Draw a vector diagram to determine the resultant of the 400 N vertical force and the 100 N horizontal force. Place an arrow on all of the forces to show their directions.

Determine the magnitude (size) of this resultant force and its direction to the vertical.

magnitude of resultant = ... N

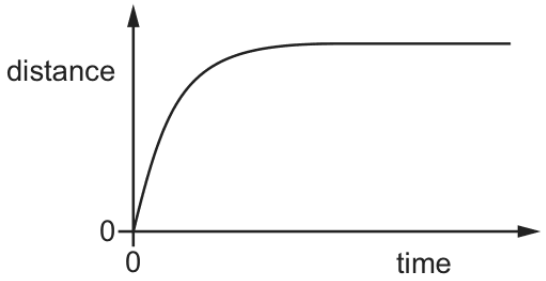
direction = ...° to vertical
[3]

[Total: 9]

Question	Answer	Marks
1(a)	curve upwards (at start) with decreasing gradient labelled A	**B1**
	horizontal line labelled B	**B1**
1(b)	gradient / slope decreases **or** increase in speed for same time less at later times	**B1**
1(c)(i)	air resistance / drag	**B1**
1(c)(ii)	air resistance / upwards force increases (with increased speed)	**B1**
	air resistance / upwards force equals / balances weight	**B1**
1(d)	vector diagram showing 400 N and 100 N force at right angles with correct resultant and all directions shown	**B1**
	410 (N)	**B1**
	12–16°	**B1**

QUESTION · 5054_s25_qp_12.pdf · source page 2 of 16

2 The diagram shows the distance–time graph for a moving object.



What is a description of the moving object?

A a ball thrown vertically upwards and falling back to the thrower

B a car starting from rest and speeding up as it moves away from a traffic signal

C a rock dropped from a high cliff and falling into the sea below

D a train braking to a halt as it stops at a station

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 2

2	D	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 4 of 16

6 There is no atmosphere on the Moon.

Two metal spheres of identical volume but different mass are at rest at the same height above the ground on the Moon. The two spheres are dropped at the same time.

How do the spheres move after they are dropped?

A The sphere with the larger mass has a smaller acceleration.

B The sphere with the larger mass hits the ground first.

C The sphere with the smaller mass has a smaller acceleration.

D The two spheres hit the ground together.

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

1 A ball is dropped by the side of a vertical scale that is marked in centimetres.

A video recording of the fall shows the position of the ball on the scale every 0.10 s.

The distance the ball falls is shown by the position on the scale.

Table 1.1 shows the results.

Table 1.1

time / s	0	0.10	0.20	0.30	0.40	0.50
distance / cm	0	5	20	44	78	123

(a) (i) On Fig. 1.1, plot the distance–time graph for the ball. Draw a suitable line.

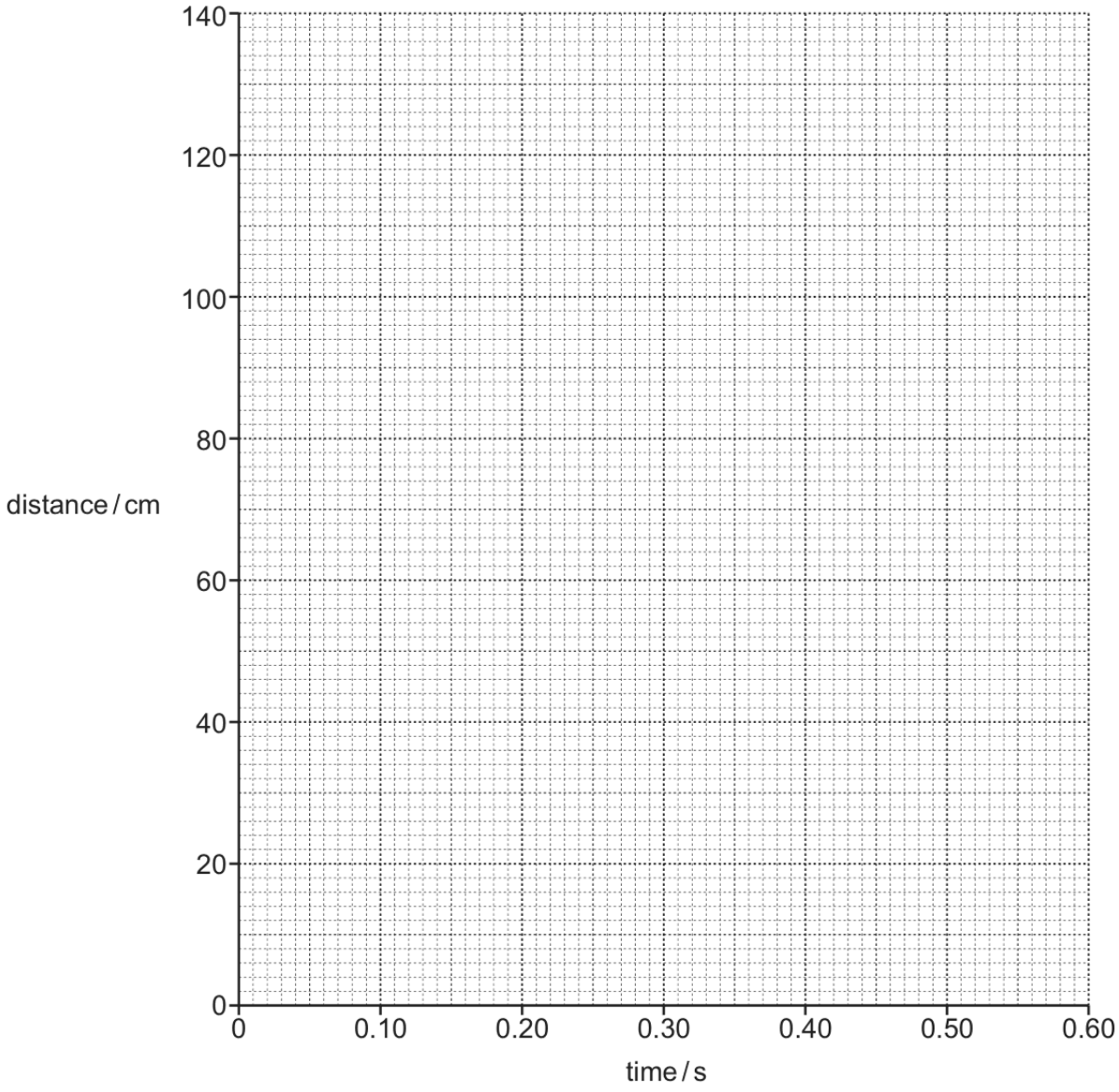


Fig. 1.1

[2]

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(ii) Calculate the average speed of the ball between time $t = 0$ and time $t = 0.40\,s$.

speed = ... cm / s [2]

(iii) Describe how to use the graph in Fig. 1.1 to determine the speed of the ball at time $t = 0.40\,s$.

.....

.....

..... [2]

(b) As the ball falls, the acceleration of the ball decreases and eventually becomes zero before it reaches the ground.

(i) Explain, in terms of the forces acting, why the acceleration decreases, and why it eventually becomes zero.

.....

.....

.....

.....

.....

..... [3]

(ii) Describe the appearance of a distance–time graph when the acceleration of the ball is zero.

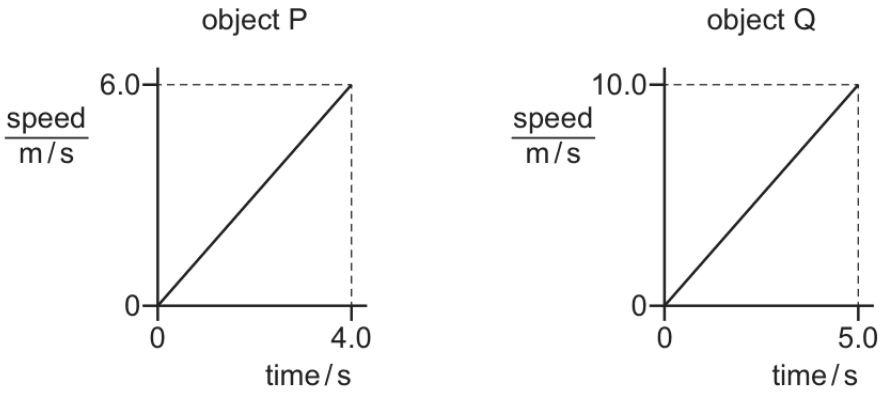
..... [1]

[Total: 10]

Question	Answer	Marks
1(a)(i)	points plotted to within ½ a small square	**B1**
	smooth curve drawn	**B1**
1(a)(ii)	(average speed =) d / t in any form	**C1**
	195 (cm / s)	**A1**
1(a)(iii)	draw a tangent (at t = 0.40 s)	**B1**
	find the gradient / slope (of the graph) **or** the gradient of the tangent	**B1**
1(b)(i)	mention of air resistance / resistive force	**B1**
	resultant force decreases **or** air resistance / upwards force increases (as speed increases) / becomes high(er)	**B1**
	(eventually) air resistance = weight (and acceleration is zero) **or** air resistance balances / cancels weight	**B1**
1(b)(ii)	straight line (of positive) **or** (line of) constant gradient / slope	**B1**

QUESTION · 5054_w25_qp_11.pdf · source page 3 of 16

3 The diagram shows the speed–time graphs for two objects moving with uniform acceleration.



A student uses the graphs to make two conclusions.

1 The distance travelled by object P is 26 m less than the distance travelled by object Q.

2 The acceleration of object P is 0.50 m / s^2 less than the acceleration of object Q.

Which conclusions are correct?

A neither 1 nor 2

B 1 only

C 2 only

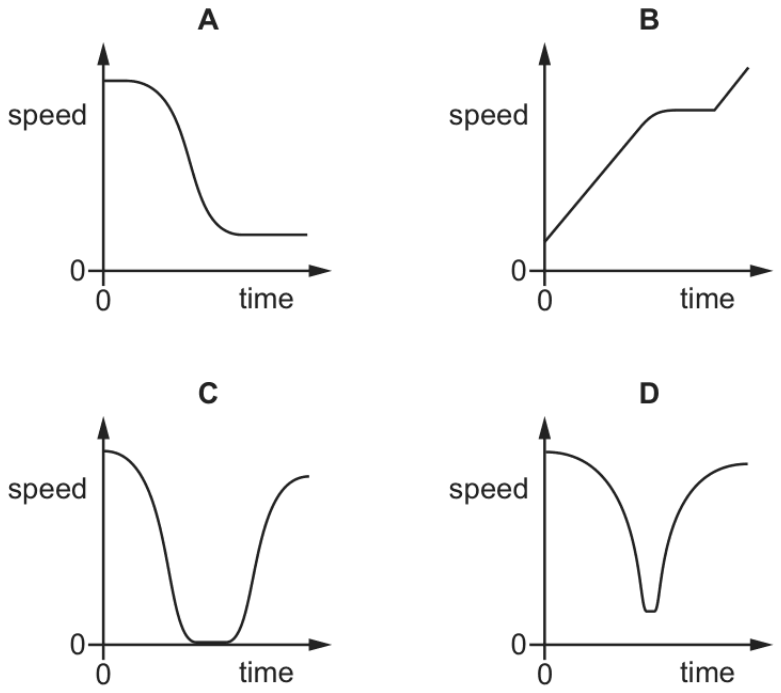
D both 1 and 2

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 3

3	C	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 3 of 16

4 Which speed–time graph represents the motion of a railway train making a short stop at a station?



MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 4

4	C	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 2 of 16

3 Which equation defines acceleration?

A $\frac{\text{change in distance}}{\text{time taken}}$

B $\frac{\text{change in distance in a fixed direction}}{\text{time taken}}$

C $\frac{\text{change in speed}}{\text{time taken}}$

D $\frac{\text{change in velocity}}{\text{time taken}}$

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

1 A car travels along a straight horizontal road, initially at a constant speed. The speed–time graph in Fig. 1.1 shows the motion of the car from time $t = 0$.

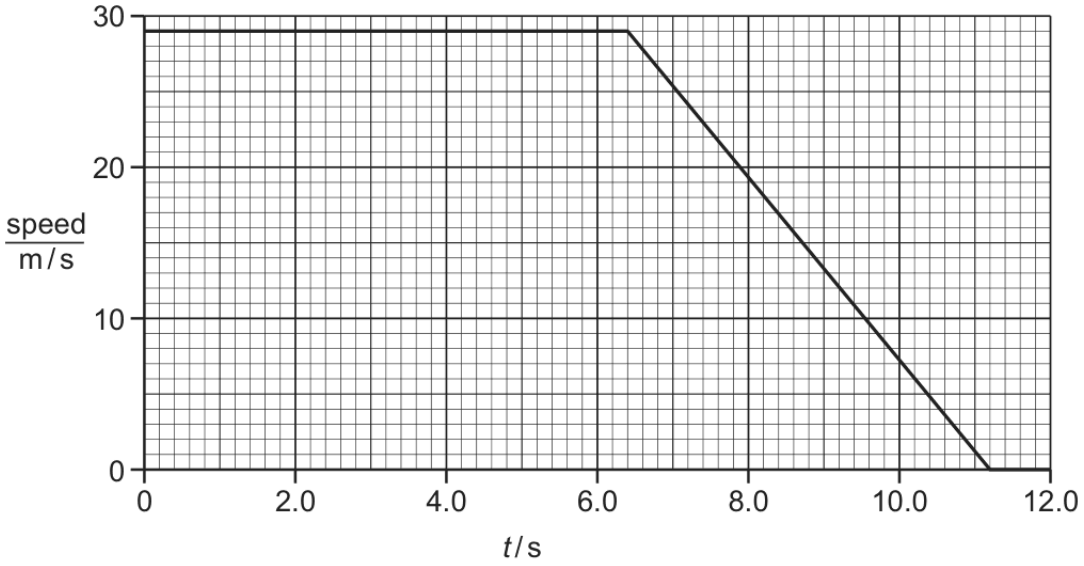


Fig. 1.1

At time $t = 5.7$ s, the driver sees an obstruction in the road.

The driver applies the brakes a short time later, and the car begins to slow down.

(a) Using Fig. 1.1, determine:

(i) the speed of the car at time $t = 0$

speed = ... m/s [1]

(ii) the distance travelled by the car between the driver seeing the obstruction at $t = 5.7$ s and the car beginning to slow down

distance = ... m [2]

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(iii) the distance travelled by the car between $t = 0$ and $t = 11.2\,s$.

distance = ... m [3]

(b) The speed of the car affects both the thinking distance and the braking distance.

State **one** factor that affects **only** the braking distance of the car.

Explain how a change in this factor increases the braking distance.

factor .....

.....

explanation .....

.....

.....

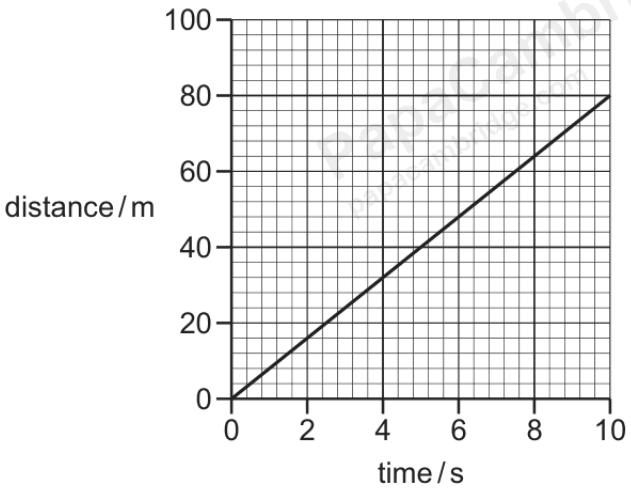
[3]

[Total: 9]

Question	Answer	Marks
1(a)(i)	29 (m / s)	**B1**
1(a)(ii)	(s =) vt **or** 29 × (6.4 – 5.7)	**C1**
	20 (m)	**A1**
1(a)(iii)	(distance travelled =) area under graph **or** ½bh **or** ½(w_1 + w_2)h	**C1**
	½ × 29 × (11.2 – 6.4) **or** ½ × 29 × 4.8 **or** 69.6 (m) **or** ½ × 29 × (11.2 + 6.4)	**C1**
	260 (m)	**A1**
1(b)	any **one** factor from: • (mass) of load / car • (condition / wear on) tyres • (state of) road • brakes • reference to friction	**B1**
	any **one** matching explanation from: • heavy load • worn tyres • slippery / smooth road reference to reduced friction	**B1**
	reduced deceleration **or** increases the time taken (to stop)	**B1**

QUESTION · 5054_s26_qp_11.pdf · source page 3 of 16

3 The graph is the distance–time graph for the motion of a cyclist.



Which statement describes the motion of the cyclist?

A The cyclist is at rest.

B The cyclist is moving with constant speed.

C The cyclist is accelerating.

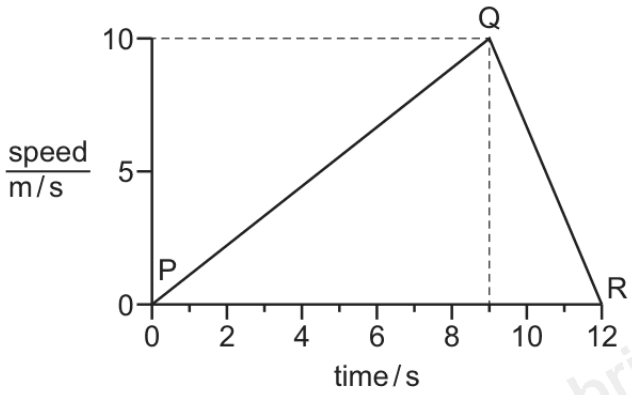
D The cyclist is decelerating.

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 3

3	B	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 3 of 16

4 The speed–time graph shows the motion of a lift working in a tall building.



At which of the points P, Q and R is the lift at rest?

A P, Q and R **B** P and Q only **C** P and R only **D** Q and R only

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 4

4	C	**1**

1 A bus starts a journey with a uniform acceleration for the first 20 s.

(a) (i) State what is meant by 'uniform acceleration'.

.....

..... [1]

(ii) The bus starts from rest at time $t = 0$ and reaches a speed of 16 m/s at $t = 20$ s.

Calculate the acceleration of the bus.

acceleration = ...m/s^2 [2]

(b) At $t = 20$ s, the acceleration of the bus changes.

- From $t = 20$ s to $t = 100$ s, the acceleration of the bus is zero.
- From $t = 100$ s to $t = 120$ s, the bus decelerates with an acceleration of −0.30 m/s^2.

On Figure 1.1, plot a speed–time graph for the first 120 s of the journey.

Use the space provided for any calculations.

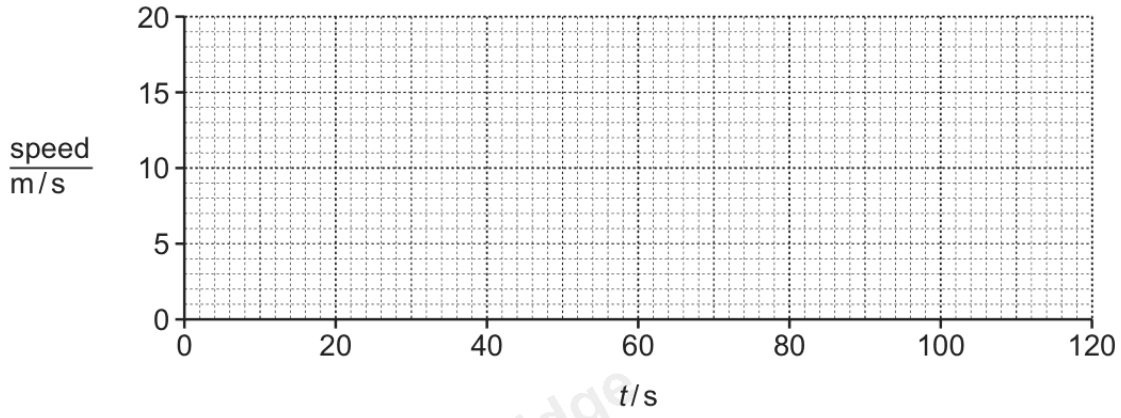


Figure 1.1

[3]

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(c) Explain, by referring to the relative magnitudes and directions of the forces on the bus, why the bus initially accelerates and later decelerates.

.....

.....

.....

..... [2]

[Total: 8]

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Question	Answer	Marks	Guidance
1(a)(i)	same change in speed / velocity in same time / unit time / in 1 s	**1**	**A** any named time period; same increase in speed over time; straight line on speed-time graph; rate of change in velocity is constant; increasing speed at a constant rate **I** formula alone; speed increases steadily; accelerates at a constant rate
1(a)(ii)	(a =) $v \div t$ in any form 0.8 (m/s^2)	**1**	
1(b)	straight line from (0,0) to (20,16) horizontal line from (20,16) to (100,16) straight line from (100,16) with a decrease of 6 m/s	**1**	**A** within ½ square penalise wrong value plotted for 16 once only; **ecf** from a misplotted value of speed or time at one point each time
1(c)	*acceleration / t = 20 to 100 (s):* forwards force larger than backwards force *deceleration / t = 100 (s) onwards:* resistive force acts alone **or** backwards force > forwards force	**1**	throughout **A** driving force / thrust for forwards force **A** resistive force / force from brakes / air resistance / resistance with road for backwards force; mention of resultant force forwards **bod** force of acceleration is forwards force **I** other regions / just 'forces are unbalanced' / 'resultant force acts' just 'backwards force increases'

QUESTION · 5054_s26_qp_12.pdf · source page 2 of 16

2 Which feature of a graph represents total distance travelled?

A the area under a distance–time graph

B the area under a speed–time graph

C the gradient of a distance–time graph

D the gradient of a speed–time graph

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 2

2	B	**1**

1 An athlete and a cyclist both travel along the same straight road in the same direction.

- The athlete and the cyclist start next to each other at time $t = 0$.
- The athlete runs at a steady speed of 8.0 m/s from $t = 0$ to $t = 6.0$ s.
- The cyclist accelerates uniformly from rest at $t = 0$ to a speed of 18 m/s at $t = 6.0$ s.

(a) On Figure 1.1, draw speed–time graphs for the runner and for the cyclist.

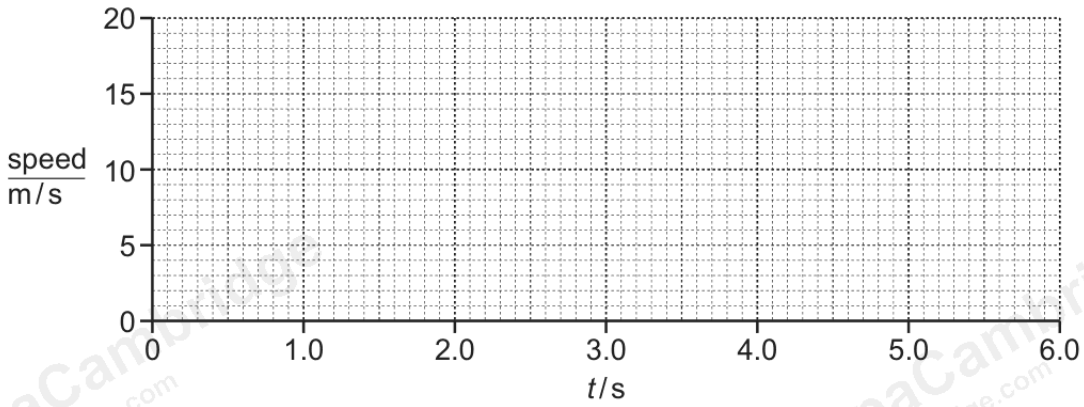

Figure 1.1

[2]

(b) Calculate the acceleration of the cyclist. Give the unit of your answer.

acceleration = ... unit ..... [3]

(c) Show, by calculation, that the athlete has travelled a greater distance than the cyclist at $t = 5.0$ s.

[2]

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(d) State Newton's first law of motion, and explain how it applies to the motion of the athlete.

.....

.....

.....

..... [2]

[Total: 9]

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Question	Answer	Marks	Guidance
1(a)	horizontal line from (0, 8) to (6, 8)	**B1**	all points ± ½ small square throughout
	straight line from (0, 0) to (6, 18)	**B1**	labels not needed
1(b)	(a =) v (- u) / t in any form algebraic or numerical	**C1**	**A** numerical values for cyclist or runner or ± ½ graph small square
	3(.0)	**A1**	**ecf as always, here** from an incorrect graph; **accept** –3 remember to round answer to 2 s.f., so e.g. **A** 2.95 but **not** 2.9;
	m / s^2	**B1**	**ignore** N / kg as the unit
1(c)	*working must be shown and final value calculated*:		**expected** values 40 (m) and 37.5 or 38 (m)
	(runner) 5 × 8 **and** 40	**B1**	**cao** correct answer only
	(cyclist = area under graph) ½ ×15 (±0.5) × 5 **and** 37.5 or 38	**B1**	**ecf** from graph with ±0.5 on y–axis for cyclist graph **max 1** if no working, i.e. if only 40 and 37.5 or 38 seen or wrong units for distance clearly shown, e.g. m^2; m / s **SC** answers using 6 s as time (48 m and 54 m) 1 mark even with no working

Question	Answer	Marks	Guidance
1(d)	simple statement of 1st law: an object/athlete remains at rest / uniform motion / constant motion / keeps moving unless a force acts	**C1**	i**gnore** just 'remains at rest'; F=ma **must** mention the word 'force' **A** a force causes movement / motion / speed / velocity to change
	complete statement and correctly applied to athlete: athlete moves at constant velocity / constant speed (in a straight line) / uniform motion / zero acceleration unless acted upon by a net / resultant / unbalanced force **OR** constant velocity / speed in a constant direction / uniform motion / no acceleration means no resultant force on athlete	**A1**	look for • uniform motion / uniform speed etc ignore at rest • unless / no resultant / net / unbalanced force • applied correctly to athlete (e.g. mentions athlete's constant speed or describes forces on athlete) **I** constant motion/remains at rest / external force in this mark

Chapter 1 Motion, forces and energy

1.3 Mass and weight

112 entries: 73 Paper 1 questions, 39 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 2 of 20

4 Which relationship defines gravitational field strength?

A mass x 10

B mass x weight

C mass / weight

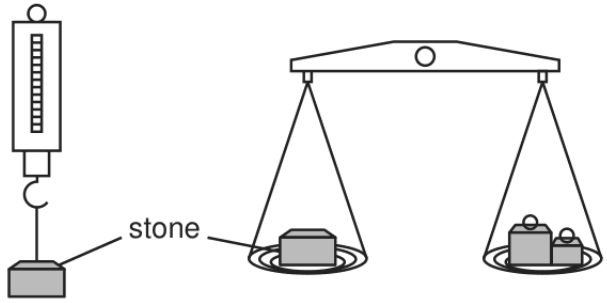
D weight / mass

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 4

4	**D**

QUESTION · 5054_w04_qp_1.pdf · source page 4 of 20

6 A stone is weighed using a newtonmeter (spring balance) and a pair of scales (pan balance).



newtonmeter scales

This experiment is repeated on the Moon.

Are the readings for each balance the same or different when taken on Earth and on the Moon?

	reading on newtonmeter	reading on scales
A	different	different
B	different	same
C	same	different
D	same	same

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 6

6 **B**

QUESTION · 5054_s05_qp_1.pdf · source page 3 of 20

4 At a point on the surface of the Earth, the gravitational field strength is 9.8 N / kg.

Which pair of values for mass and weight are correct for an object placed at this point?

	mass / kg	weight / N
A	9.8	10
B	10	9.8
C	10	98
D	98	10

MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 4

4 **C**

QUESTION · 5054_w05_qp_1.pdf · source page 4 of 20

6 An object that has a mass of 15 kg on the Earth is taken to the Moon.

The gravitational field strength on the Earth is 10 N / kg and on the Moon is 1.6 N / kg.

What are the mass and the weight of the object on the Moon?

	mass / kg	weight / N
A	15	24
B	15	150
C	24	15
D	150	24

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 6

6 **A**

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows a 0.4 kg mass hanging at rest from a spring.

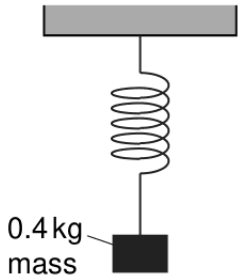


Fig. 1.1

(a) State what is meant by the *mass* of an object.

.....

.....

..... [1]

(b) **(i)** On Fig. 1.1, draw an arrow showing the direction and the line of action for each of the two forces that act on the mass. Write the name of the force next to each arrow. [2]

(ii) The gravitational field strength is 10 N/kg. Calculate the size of each of the two forces acting on the mass.

first force = ... second force = ... [2]

(c) The mass is pulled downwards. State and explain what happens to the upward force.

.....

.....

.....

.....

..... [2]

Section A

1	**(a)**		amount of matter/substance **or** ability to resist motion (accept Force/acc)	**B1**	
	(b)	**(i)**	downwards force labelled weight/gravity continuation of vertical line	**B1**	
			upwards force (labelled tension) along vertical line	**B1**	
			(allow 1 mark if both labels correct but lines of action not accurate, e.g. displaced to side)		
		(ii)	4 N	**B1**	
			4 N or same as other	**B1**	
	(c)		upwards force/tension increases	**B1**	
			spring stretched/longer	**B1**	**7**

QUESTION · 5054_s06_qp_1.pdf · source page 3 of 20

6 The inertia of a body is its resistance to changes in motion.

Which property is a measure of the body's inertia?

A its density

B its mass

C the height of its sides

D the size of its base

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 6

6	**B**

10 Fig.10.1 shows a section through a very old bicycle tyre that is made of solid rubber.

Fig. 10.1

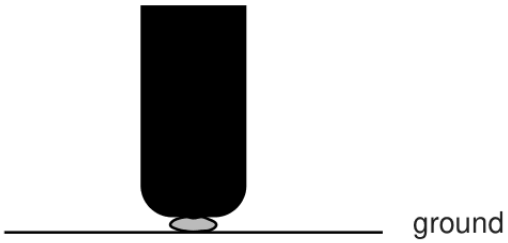


Fig. 10.2

(a) The solid rubber tyre exerts a pressure on the ground.

The unit of pressure is the pascal (Pa).

(i) Define the pascal. [2]

(ii) The mass of the bicycle and rider is 60 kg and the total area of the tyres in contact with the road is $2.4 \times 10^{-3}\,m^2$. The gravitational field strength is 10 N/kg.

Calculate the pressure exerted on the road. [3]

(iii) One of the solid rubber tyres goes over a small stone, as shown in Fig. 10.2. Describe and explain how the pressure exerted on the ground changes. [2]

(b) Fig. 10.3 shows a modern tyre that contains air. A heavy person sits on the bicycle and the shape of the tyre changes, as shown in Fig. 10.4.

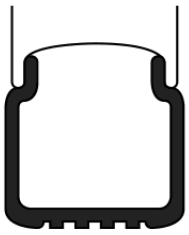


Fig. 10.3

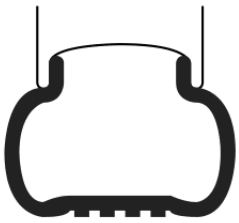


Fig. 10.4

The table contains information about the tyre in Fig. 10.3 and Fig. 10.4.

	tyre in Fig. 10.3	tyre in Fig. 10.4
air pressure in tyre / Pa	1.9×10^5	2.1×10^5
volume of air / m^3	0.016	V

(i) Calculate the volume V of the air in the tyre in Fig. 10.4. State one assumption that you make in your calculation. [4]

(ii) Another way to change the pressure in the tyre is to raise the temperature.

State what happens to the pressure in the tyre when the temperature rises and explain your answer using ideas about molecules. [4]

10	**(a)**	**(i)**	Force/area	C1	
			(force of) 1N acting on (area of) $1m^2$	A1	
		(ii)	600 (N) seen	C1	
			force/area in any form e.g. $600/2.4 \times 10^{-3}$	C1	
			2.5×10^5 Pa	A1	
		(iii)	increases	B1	
			less surface area (in contact with ground)	B1	**[7]**
	(b)	**(i)**	PV = constant in any form	C1	
			$1.9 \times 10^5 . 0.016 = 2.1 \times 10^5 . V$	C1	
			$0.014(476)\ m^3$	A1	
			temperature **or** amount of gas constant	B1	
		(ii)	(pressure) increases	B1	
			speed/K.E. of molecules increases	B1	
			(molecules) hit walls harder or with more force	B1	
			hit more often /more frequently	B1	**[8] (15)**

QUESTION · 5054_w07_qp_1.pdf · source page 3 of 20

6 A body of mass 10 kg falling freely in the gravitational field close to the Moon's surface has an acceleration of $1.6\,m/s^2$.

What is the gravitational field strength on the Moon?

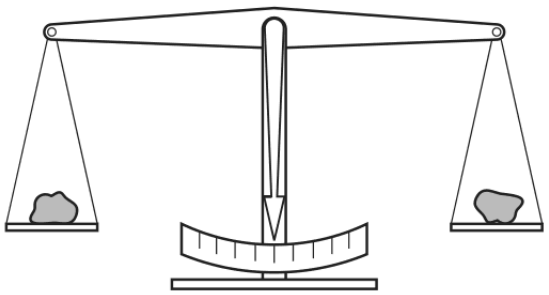
A 0 N/kg **B** 1.6 N/kg **C** 10 N/kg **D** 16 N/kg

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 6

6	**B**

QUESTION · 5054_w08_qp_1.pdf · source page 4 of 20

5 The diagram shows two objects on a beam balance in equilibrium.



Which need not be the same?

A the masses of the two objects

B the moments about the pivot of the two objects

C the volumes of the two objects

D the weights of the two objects

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 5

5	**C**

10 A space research organisation plans to send astronauts to Mars to examine rocks on its surface. The organisation will produce a report containing information about conditions on Mars.

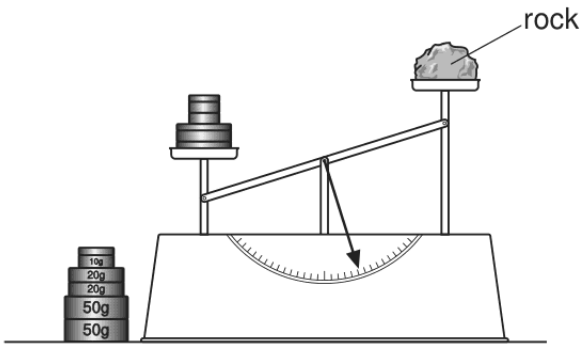
(a) (i) The gravitational field strength on the surface of Mars is 3.7 N/kg. Calculate the weight, on Mars, of a rock of mass 0.50 kg. [2]

(ii) A rock dropped on Mars falls to the surface. State the acceleration of the falling rock. Assume that there is no air resistance on Mars. [1]

(iii) Calculate the kinetic energy of a 0.50 kg rock as it strikes the surface at a speed of 3.2 m/s. [3]

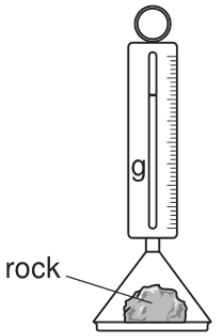
(b) While still on the surface of Mars, the astronauts will measure the mass of each rock collected. Fig. 10.1 shows two devices for measuring mass.

device A



A lever arm balance and a set of brass discs. The mass of each disc is accurately determined on Earth and the value is marked on it.

device B



A spring balance. The spring balance is accurately calibrated on Earth and the mass values are marked on the vertical scale.

Fig. 10.1

(i) When the two devices are used on Mars, they will give different readings for the mass of the same rock. Explain why. [3]

(ii) State which device will give the correct reading for the mass. [1]

(c) The astronauts will also determine the density of each rock.

(i) Apart from the mass of the rock, state what other information is needed in order to calculate the density of the rock. [1]

(ii) Describe a method for determining the density of a small, irregularly shaped rock of known mass *m*. [4]

10 (a) (i) (W =)mg **or** 0.5 × 3.7 C1
1.8/1.85/1.9 N A1

(ii) 3.7 m/s^2 **not** N/kg B1

(iii) (KE =) ½mv^2 C1
½ × 0.50 × 3.2^2 C1
2.6 **or** 2.56 J A1 [6]

(b) (i) A compares/measures (unknown and known) masses/amount of matter B1
B measures/is dependent on weight/force of gravity (and hence mass obtained) B1
Mars weights/forces of gravity are less than/different from (Earth) B1

(ii) A **or** lever arm balance **or** balance with discs B1 [4]

(c) (i) volume B1

(ii)

Either:	**Or:**
record value of water in measuring cylinder (**not** beaker)	water/liquid in displacement/eureka can
insert rock	full to overflowing
record new value	immerse rock
subtract (to obtain volume) **or** measure rise)	(new) overflow in measuring cylinder (**not** beaker is volume)
m/volume **or** *m*/subtraction (max 4)	*m*/volume **or** *m*/subtraction (max 4)

B4 [5]

[Total: 15]

QUESTION · 5054_s10_qp_11.pdf · source page 4 of 20

7 A passenger is sitting in an aeroplane, which takes off and climbs to 10 000 m.

During this time, what happens to the mass and to the weight of the passenger?

	mass	weight
A	decreases	decreases
B	increases	increases
C	unchanged	decreases
D	unchanged	increases

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 7

7 **C**

QUESTION · 5054_s10_qp_12.pdf · source page 3 of 20

6 A passenger is sitting in an aeroplane, which takes off and climbs to 10 000 m.

During this time, what happens to the mass and to the weight of the passenger?

	mass	weight
A	decreases	decreases
B	increases	increases
C	unchanged	decreases
D	unchanged	increases

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 6

6 **C**

QUESTION · 5054_w10_qp_11.pdf · source page 6 of 20

7 A person of weight 600 N at the bottom of a mountain climbs to the top. The gravitational field strength changes from 10.00 N / kg at the bottom to 9.97 N / kg at the top. His mass is unchanged as he climbs.

What are his mass and his weight at the top of the mountain?

	mass at top of mountain / kg	weight at top of mountain / N
A	60.0	598
B	60.0	600
C	60.1	598
D	60.1	600

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 7

7 **A**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

8 A car has a gas-filled shock absorber for each of its four wheels. Fig. 8.1 shows one of these shock absorbers.

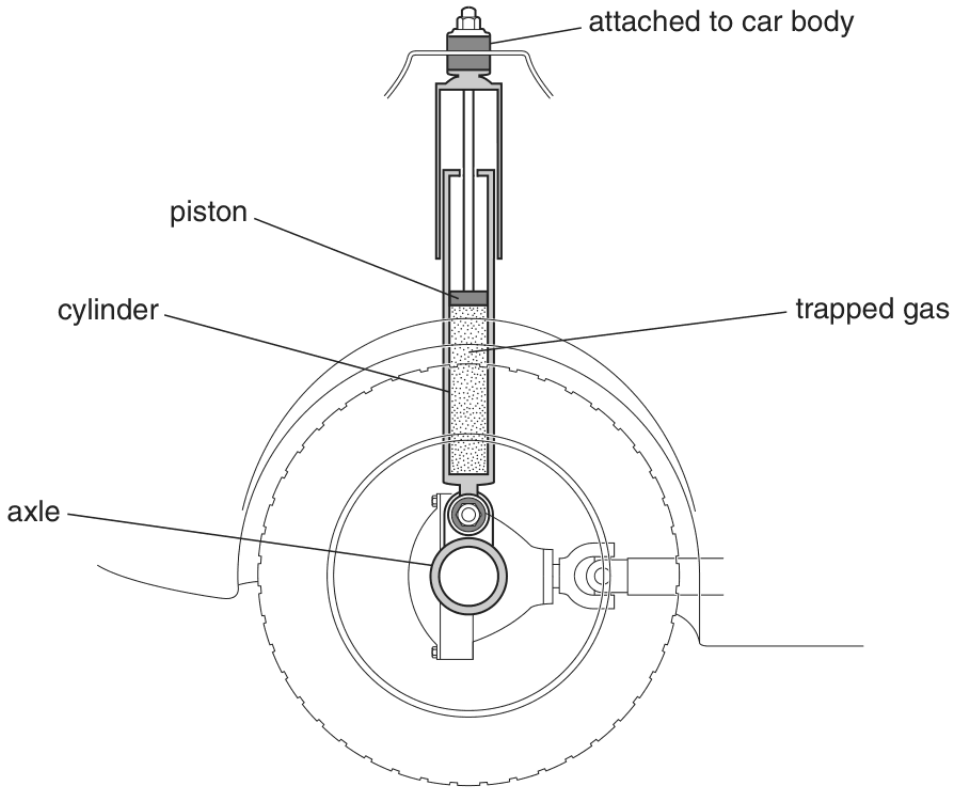


Fig. 8.1

The axles are attached to the cylinders. The body of the car is supported by the four pistons, which can move up and down inside the cylinders. Each piston has a cross-sectional area of 35 cm^2.

(a) A driver of mass 70 kg gets into the car. Calculate

(i) the driver's weight,

weight = ... [2]

(ii) the average increase in the pressure of the gas in the cylinders.

pressure increase = ... [3]

For Examiner's Use

(b) (i) Explain in detail how the molecules of the gas trapped inside the cylinder by the piston exert a pressure on the cylinder.

.....

.....

.....

.....

.....

.....

..... [4]

(ii) The temperature of the trapped gas remains constant. Explain why the pressure of the gas increases as the piston is moved further into the cylinder.

.....

.....

..... [2]

(c) At the end of a long journey, the temperature of the trapped gas in the shock absorbers has increased substantially.

(i) State what happens to the molecules of the trapped gas due to the temperature increase.

.....

..... [1]

(ii) State and explain the effect of the increased temperature of the gas on the height of the car body above the road surface.

.....

.....

.....

.....

..... [3]

Section B

8	**(a)**	**(i)**	(*W* =) *mg* **or** 70 × 10 **or** 70 × 9.8(1) etc.	C1	
			700(.0) N	A1	[2]
		(ii)	(*P* =) *F*/*A* **or** 700/35	C1	
			700/(35 × 4) **or** 700/0.0035 **or** 700/(0.0035 × 4)	C1	
			50 000 Pa **or** 50.0 kPa **or** 5.0 N/cm^2	A1	[3]
	(b)	**(i)**	molecules/atoms/particles move **or** collide	B1	
			molecules/atoms/particles collide with cylinder/walls	M1	
			exert force on walls (as they collide)	A1	
			spread out effect (of forces) is pressure **or** (force)/m^2 **or** similar	B1	[4]
		(ii)	molecules/atoms/particles closer/denser/more in given volume	C1	
			more collisions per (unit) area/m^2 **or** per (unit) time/s (**not** faster)	A1	[2]
	(c)	**(i)**	speed (of molecules/atoms/particles) increases/k.e. increases	B1	[1]
		(ii)	car (body) higher (off the ground)	B1	
			collisions more violent **or** gas in cylinder expanded	B1	
			fewer collisions of molecules/atoms/particles needed **or** pressure rises **initially**	B1	[3]

QUESTION · 5054_w10_qp_12.pdf · source page 5 of 16

9 A person of weight 600 N at the bottom of a mountain climbs to the top. The gravitational field strength changes from 10.00 N / kg at the bottom to 9.97 N / kg at the top. His mass is unchanged as he climbs.

What are his mass and his weight at the top of the mountain?

	mass at top of mountain / kg	weight at top of mountain / N
A	60.0	598
B	60.0	600
C	60.1	598
D	60.1	600

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 9

9 **A**

10 A bus breaks down on a road with a 10° upward slope. The passengers get out and push the bus to the top of the slope at a constant speed. Fig. 10.1 shows the passengers exerting a force on the bus parallel to the line of the slope.

For Examiner's Use

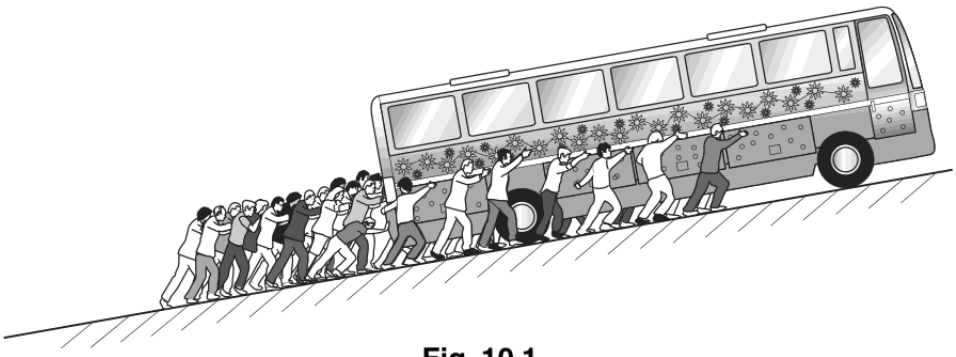


Fig. 10.1

(a) The total mass of the bus is 3200 kg.

(i) Calculate the weight of the bus.

weight = ...[1]

(ii) On Fig. 10.1, draw **two** arrows to show the direction of the weight of the bus and the direction of the force exerted on the bus by the passengers. [1]

(iii) The total force exerted by the passengers on the bus is 17000 N along the line of the slope. Use a graphical method to determine the size and direction of the resultant of this force and the weight of the bus. State the scale used.

scale = ...

size of resultant = ...

direction of resultant = ...
[4]

For Examiner's Use

(iv) The bus travels at a constant speed. State the size of the resultant of all the forces acting on the bus.

.....[1]

(b) As the passengers push the bus up the slope, work is done against two forces. State the names of these two forces and, for each force, state the form of energy produced as work is done.

force 1 .....

form of energy 1 .....

force 2 .....

form of energy 2 .....
[4]

(c) When the bus reaches the top of the slope, it stops and the passengers get back in.

(i) On the axes of Fig. 10.2, sketch the distance-time graph for the bus from a time when the passengers are pushing the bus at constant speed until they get back into the stationary bus.

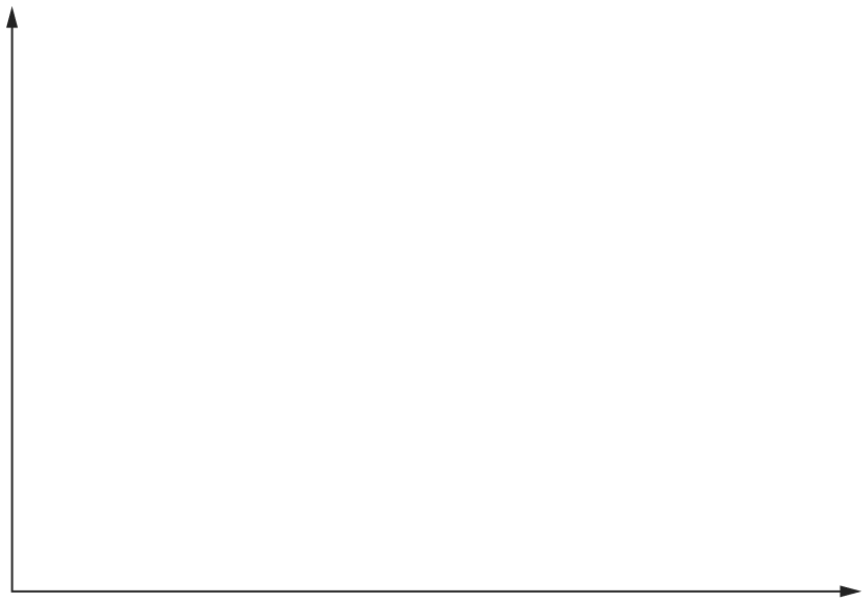


Fig. 10.2

[3]

(ii) State how the speed of a moving object may be obtained from an accurately drawn distance-time graph.

.....

.....[1]

10	**(a)**	**(i)**	32 000 N	B1	
		(ii)	two arrows/lines in correct direction by eye	B1	
		(iii)	scale given	B1	
			two arrows/lines **and** correct resultant drawn	B1	
			32.0 → 35.0 kN (2/3 sig. fig. only)	B1	
			58.5 → 61.5° to horizontal (2/3 sig. fig. only; don't penalise twice)	B1	
		(iv)	zero/no force/0	B1	[7]
	(b)		weight/gravitational force/gravitational attraction (**not** gravity)	B1	
			higher in gravitational field **or** (to gravitational) potential energy	B1	
			friction/air resistance	B1	
			heat/thermal/internal energy	B1	[4]
	(c)	**(i)**	labelled axes **and** correct way round ($x \rightarrow t$)	B1	
			straight line of positive slope	B1	
			followed only by horizontal line (**ign** curve at junction)	B1	
		(ii)	distance travelled/time taken (from points) **or** calculate the gradient	B1	[4]

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A student wishes to find the density of a stone. He uses a measuring cylinder and a spring balance with a scale marked in newtons. The measuring cylinder, spring balance and stone are shown in Fig. 1.1.

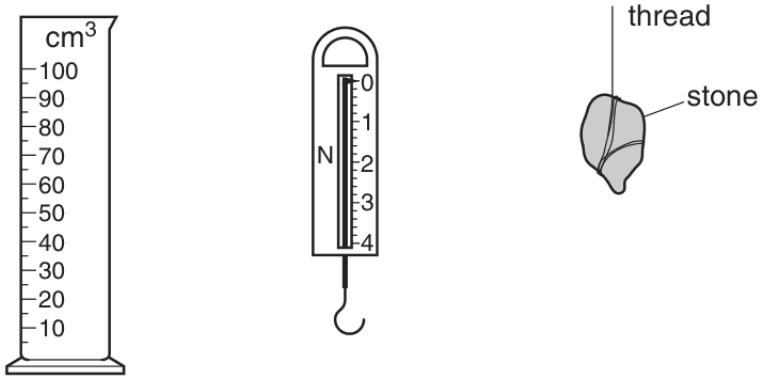


Fig. 1.1

The student knows that the gravitational field strength is 10 N/kg.

(a) Describe how the student uses the spring balance to find the mass of the stone.

.....

.....

.....[2]

(b) Describe how the student uses the measuring cylinder to find the volume of the stone.

.....

.....

.....[2]

(c) The mass of the stone is 150 g and its volume is 70 cm^3.
Calculate the density of the stone.

density of stone = ...[1]

(d) The stone is taken to another place, where the gravitational field strength is less than 10 N/kg. State how this affects the mass and the weight of the stone.

mass .....

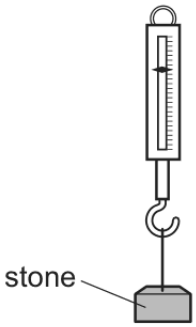
weight .....
[1]

Section A

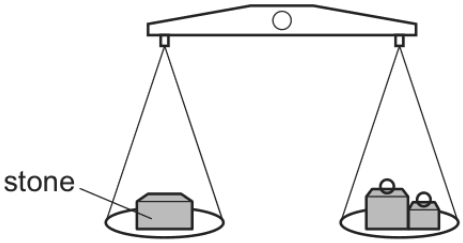
1	**(a)**	(uses spring balance) for a reading/value // finds weight/force of gravity	B1	
		divides reading/weight by 10/g // uses $W = mg$	B1	
	(b)	reading (of measuring cylinder) taken with liquid/water (alone) // initial volume mentioned // fill to certain level	C1	
		measure increase/change when stone (totally) immersed/in cylinder	A1	
	(c)	2.1 or 2.14 g/cm^3 // 2142.86 kg/m^3 // 0.00214286 kg/cm^3	B1	
	(d)	mass unchanged **and** weight less	B1	[6]

QUESTION · 5054_w11_qp_11.pdf · source page 4 of 20

7 The weight of a stone is found using a newton meter, and its mass is found using a pan balance.



newton meter



pan balance

The experiment is carried out on the Earth and on the Moon.

For each meter, is its reading the same or different on the Earth and on the Moon?

	reading on newton meter	reading on pan balance
A	different	different
B	different	same
C	same	different
D	same	same

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 7

7 **B**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows an ice cube at 0 °C.

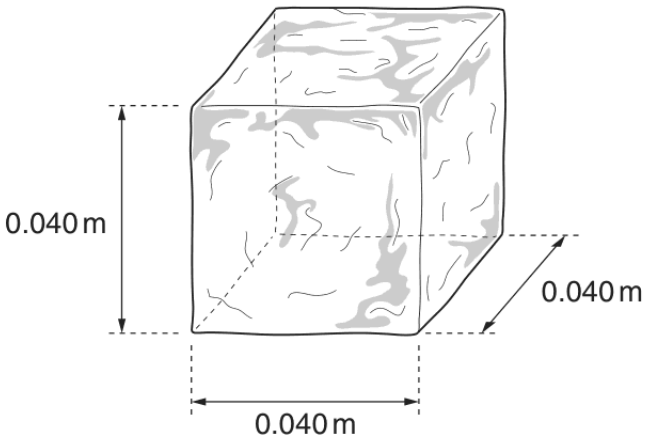


Fig. 1.1

The sides of the cube are of length 0.040 m. Ice at 0 °C has a density of 920 kg/m^3.

(a) Calculate

(i) the mass of the ice cube,

mass = ...[3]

(ii) the weight of the ice cube.

weight = ...[1]

(b) The specific latent heat of fusion of ice is 3.4×10^5 J/kg. Calculate the thermal energy (heat) absorbed by the ice cube as it melts.

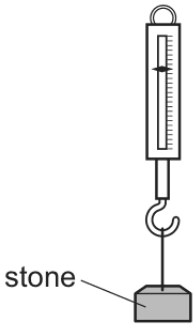
thermal energy = ...[2]

Section A

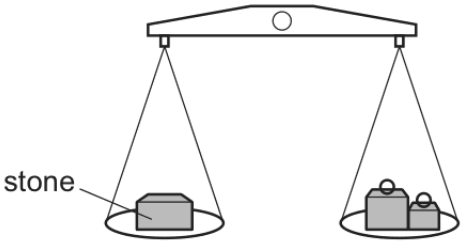
1	**(a)**	**(i)**	(V =) 64 **or** 6.4×10^{-5} **or** 4^3 **or** 0.04^3	C1	
			(m =) ρV **or** $920 \times 6.4 \times 10^{-5}$ **or** 920×0.04^3	C1	
			0.059 kg **or** 59 g **or** 0.05888 kg	A1	
		(ii)	0.59 N **or** 0.5888 N	B1	
	(b)		(Q =) ml **or** $0.059 \times 3.4 \times 10^5$	C1	
			$2.0(0) \times 10^4$ / $2.0(1) \times 10^4$ / 2.006×10^4 J	A1	[6]

QUESTION · 5054_w11_qp_12.pdf · source page 4 of 20

5 The weight of a stone is found using a newton meter, and its mass is found using a pan balance.



newton meter



pan balance

The experiment is carried out on the Earth and on the Moon.

For each meter, is its reading the same or different on the Earth and on the Moon?

	reading on newton meter	reading on pan balance
A	different	different
B	different	same
C	same	different
D	same	same

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 5

5 **B**

2 Fig. 2.1 shows a skier of mass 85 kg skiing down a very steep slope.

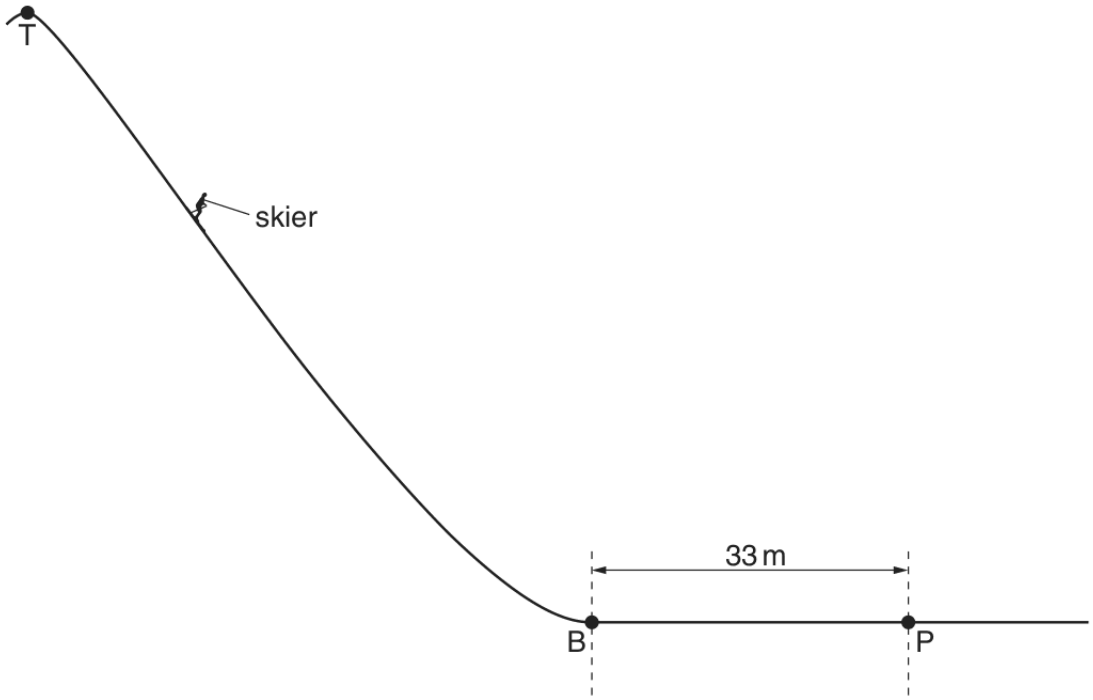


Fig. 2.1 (not to scale)

The skier starts from rest at the top T of the slope. The force of gravity accelerates him down the slope.
When he reaches the bottom B of the slope, his kinetic energy is 5.5×10^4 J.

(a) The gravitational field strength is 10 N/kg. Calculate

(i) the weight of the skier,

weight = ...[1]

(ii) the minimum possible difference in height between T and B.

height difference = ...[2]

(b) At B, the skier digs his skis into the snow and stops at the point P after travelling 33 m horizontally.

Calculate the average horizontal force that acts on the skier between B and P.

force = ...[2]

2	**(a)**	**(i)**	850 N	B1	
		(ii)	KE = PE/*mgh* **or** *mgh* = 5.5×10^4	C1	
			65/64.7(0588) m	A1	
	(b)		WD = *Fx* **or** KE/*x* **or** $5.5 \times 10^4/33$ **or** *v* = 35(.97) **and** *a* = 19(.60) **and** *F* = *ma*	C1	
			1700/1670/1667/1666.7 N	A1	[5]

QUESTION · 5054_s12_qp_11.pdf · source page 4 of 16

10 A block of metal is taken from the Earth to the Moon. Which property of the block changes?

A density

B mass

C volume

D weight

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 10

10	**D**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows apparatus used to obtain the readings for a graph of force against extension for a spring.

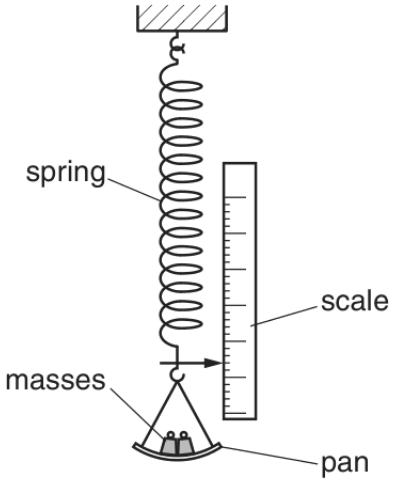


Fig. 1.1

The masses added to the pan produce a force that stretches the spring.

(a) (i) State what is meant by the *mass* of a body.

.....

.....[1]

(ii) Describe how the scale is used to find the extension of the spring.

.....

.....

.....[1]

(b) Fig. 1.2 shows the force-extension graphs for two different springs.

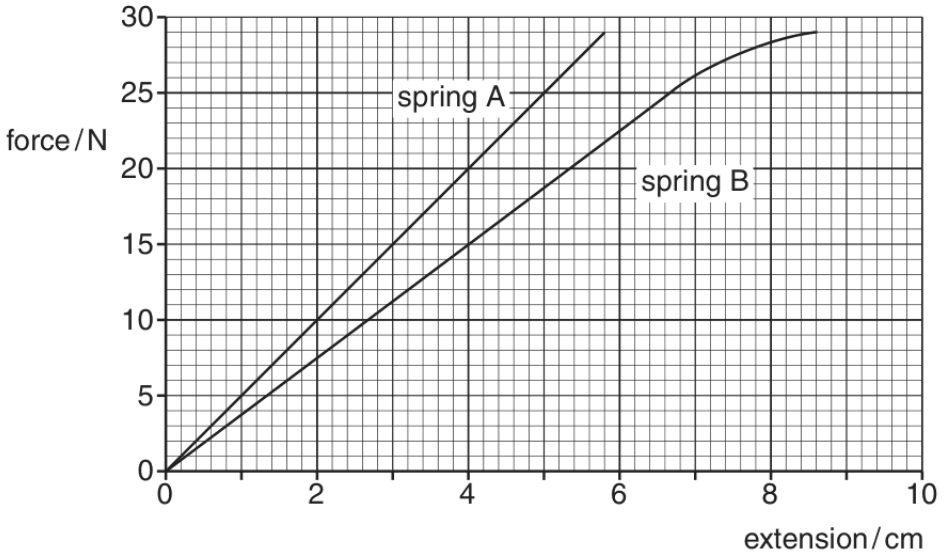


Fig. 1.2

(i) A student states that spring B is easier to stretch than spring A.

Use values from Fig. 1.2 to explain what the student means.

.....

.....

.....[1]

(ii) When a force of 25 N is applied, spring B reaches its limit of proportionality but spring A does not. Explain how Fig. 1.2 shows this.

.....

.....

.....[1]

(iii) The same force is applied to each spring.

Using Fig. 1.2, determine the force that produces an extension of spring B that is 1.0 cm greater than the extension of spring A.

force = ...[1]

Section A

1	**(a)**	**(i)**	(amount of) matter/material/substance it contains	B1	
		(ii)	**use of scale** and subtraction/difference/increase in lengths/readings OR read distance between two marks **on the scale** with different masses	B1	
	(b)	**(i)**	two force **values** with $F_A > F_B$ for the same extension OR two extension **values** with $e_B > e_A$ for the same force/at maximum	B1	
		(ii)	idea that A is a straight line and B is not OR gradient constant in A but not in B OR same increase in F every cm for A but not B	B1	
		(iii)	15 N	B1	[5]

For Examiner's Use

2 A pupil hangs a load of 45 N from a length of copper wire. The wire extends by 2.5 mm and does not exceed the limit of proportionality.

(a) (i) The gravitational field strength is 10 N/kg. Calculate the mass of the 45 N load.

mass = ...[1]

(ii) Use the grid in Fig. 2.1 to plot the extension-load graph for this wire for a load between 0 and 45 N.

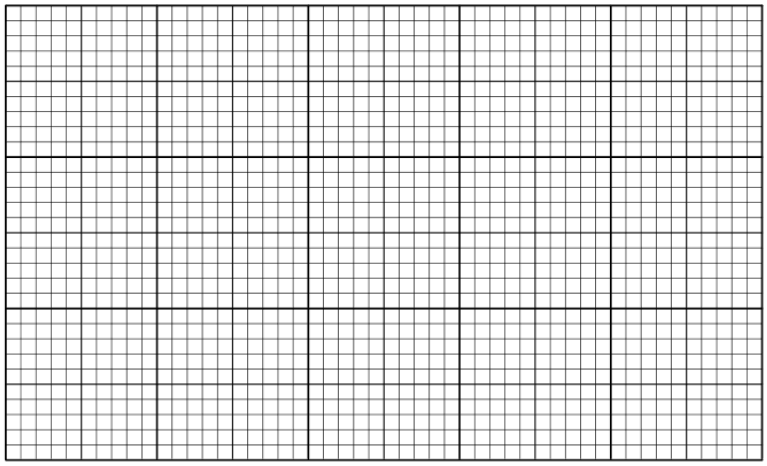


Fig. 2.1 [3]

(b) Use the graph plotted in Fig. 2.1 to determine the load needed to produce an extension of 1.3 mm.

load = ...[1]

2	**(a)**	**(i)**	4.5 kg	B1	
		(ii)	axes labelled with quantity **and** unit	B1	
			linear scale	B1	
			straight line from clear (0,0) to correct point	B1	
	(b)		answer from candidate's line	B1	[5]

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a conveyor belt carrying suitcases into an aeroplane.

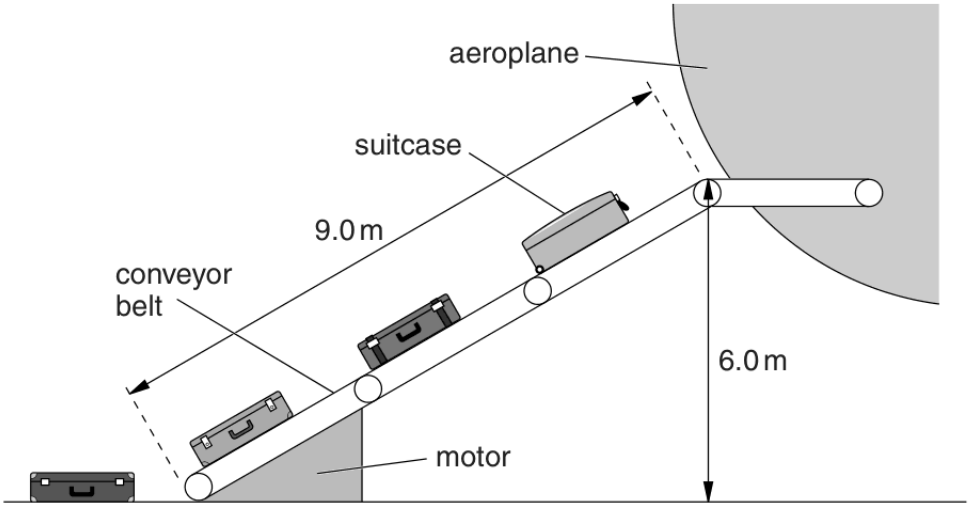


Fig. 9.1 (not to scale)

An electric motor drives the conveyor belt.

(a) A suitcase of mass 20 kg is lifted from the ground into the aeroplane.

(i) Explain what is meant by *mass* and by *weight*.

.....

.....

.....[2]

(ii) The gravitational field strength g is 10 N/kg.
Calculate the increase in the gravitational potential energy of the suitcase.

increase in potential energy = ...[2]

(iii) The suitcase takes 12 s to travel 9.0 m along the conveyor belt.
Calculate the kinetic energy of the suitcase.

kinetic energy = ...[3]

For Examiner's Use

(iv) The electric motor is powered by an input voltage of 230 V and a current of 1.5 A. Calculate the electrical energy input to the motor in 12 s.

energy = ...[2]

(v) **1.** State the principle of the conservation of energy.

.....

.....

.....[2]

2. Explain how this principle applies to the lifting of the suitcases into the aeroplane.

.....

.....

.....[1]

(b) Some of the electrical energy for the motor comes from renewable energy sources.

(i) Explain what is meant by a *renewable energy source*.

.....

.....

.....[1]

(ii) State two renewable energy sources.

1. .....

2. .....
[2]

Section B

9 (a) (i) mass is the amount of matter/substance **or** to resist (change in) motion **or** (measurement of)inertia B1
weight is the pull/force of gravity **or** pull of Earth B1
accept *mg* where *g* stated as 10 (N/kg) or grav. field strength or acc. due to gravity
ignore *mg* where *g* is gravity or grav. force or undefined

(ii) *mgh* algebraic or numerical C1
1200 J A1

(iii) ½ mv^2 algebraic or numerical C1
(speed =) 9/12 **or** 0.75 seen A1
5.6(25) J A1

(iv) (*E*=) *VIt* algebraic or numerical C1
4100 J **or** 4140 J A1

(v) **1.** energy can neither be created or destroyed/lost B1
(but) may change form / be transferred B1

2. electrical energy changes to P.E. (and K.E. and heat/work against friction; **ign**. mechanical energy) B1 [12]

(b) (i) will not run out **or** infinite **or** being replaced (**allow** does not finish/always available)
(**ign.** cannot be reused/recycled) B1

(ii) wind, tidal, solar/Sun, geothermal, hydroelectric, biomass, waves, wood (**not** nuclear) B2 [3]
(**allow** biogas/biofuel e.g. cane into petrol, dung into gas etc.) (**ign.** tidal waves)

[Total: 15]

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a flask, partly full of a liquid and partly full of air.

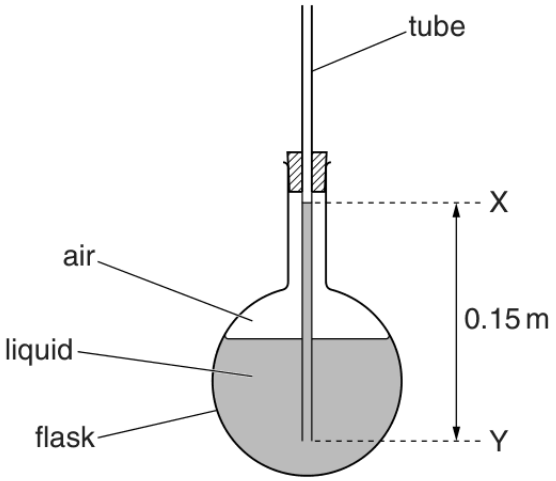


Fig. 9.1

When the flask is heated, the pressure of the air inside the flask increases and the liquid rises up the tube.

(a) Explain, using ideas about molecules, why heating the air inside the flask causes the pressure to increase.

.....

.....

..... [3]

(b) A teacher uses the flask as a thermometer. He marks the liquid level at 0 °C and at 100 °C.

(i) Describe how the teacher obtains these temperatures.

.....

.....

.....

..... [2]

(ii) State two ways in which the thermometer can be made more sensitive.

1. .....

2. .....

[2]

For Examiner's Use

(iii) The thermometer has a non-linear scale.

Explain what is meant by a *non-linear* scale on a thermometer.

.....

.....

..... [2]

(c) The inner cross-sectional area of the tube is $5.0 \times 10^{-5}\,m^2$.
The density of liquid in the tube is $1200\,kg/m^3$.
The distance between X and Y is 0.15 m.
The gravitational field strength g is 10 N/kg.

Calculate

(i) the mass of liquid in the tube,

mass = ... [2]

(ii) the weight of liquid in the tube,

weight = ... [1]

(iii) the pressure at the bottom of the tube caused by the liquid column between X and Y.

pressure = ... [2]

(d) When the flask is completely full of liquid, the liquid moves up the tube a smaller distance for the same temperature rise. State why.

.....

..... [1]

Section B

9	**(a)**	(air) molecules hit walls / liquid (surface)	B1	
		(air) molecules move fast(er) /great(er) kinetic energy	B1	
		(air) molecules hit more often/more frequently/greater rate / harder / more force **or** (liquid) molecules evaporate	B1	[3]
	(b) **(i)**	(flask) in (pure) melting ice (and water)	B1	
		(flask) in (pure) boiling water / above boiling water (at one atmosphere)	B1	
	(ii)	thin(ner) tube **or** large(r) flask **or** more air/less liquid **or** use liquid that expands more (1 mark for each)	B2	
	(iii)	divisions not equally spaced **or** scale not uniform/not proportional	C1	
		different distance (along scale) for same temperature rise **or** different change in temperature for same distance (along scale)	A1	[6]
	(c) **(i)**	(M=) $d \times V$ in any form **or** $1200 \times 5 \times 10^{-5} \times 0.15$	C1	
		$9(.0) \times 10^{-3}$ kg; 0.009(0) kg	A1	
	(ii)	0.09(0) N ecf **(i)**	B1	
	(iii)	(P=) hdg in any form **or** (P=) F/A in any form	C1	
		1800 Pa	A1	[5]
	(d)	liquids expand less (than air) **or** great(er) forces between liquid molecules	B1	[1]

[Total: 15]

QUESTION · 5054_w13_qp_11.pdf · source page 4 of 20

6 A spacecraft travels from the Earth to the Moon.

At a certain point in the journey, it has zero weight.

Why is the weight zero at this point?

A The gravitational fields of the Earth and the Moon cancel each other.

B The spacecraft leaves the Earth's gravitational field.

C The spacecraft stops moving.

D There is no air resistance on the spacecraft.

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 6

6	**A**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a skydiver, of mass 70 kg, falling towards the Earth at constant speed, a long time after jumping from an aeroplane.

Fig. 9.1

At time $t = 0$, he receives a radio signal. He opens his parachute 12 s later. Fig. 9.2 is the speed-time graph for the skydiver.

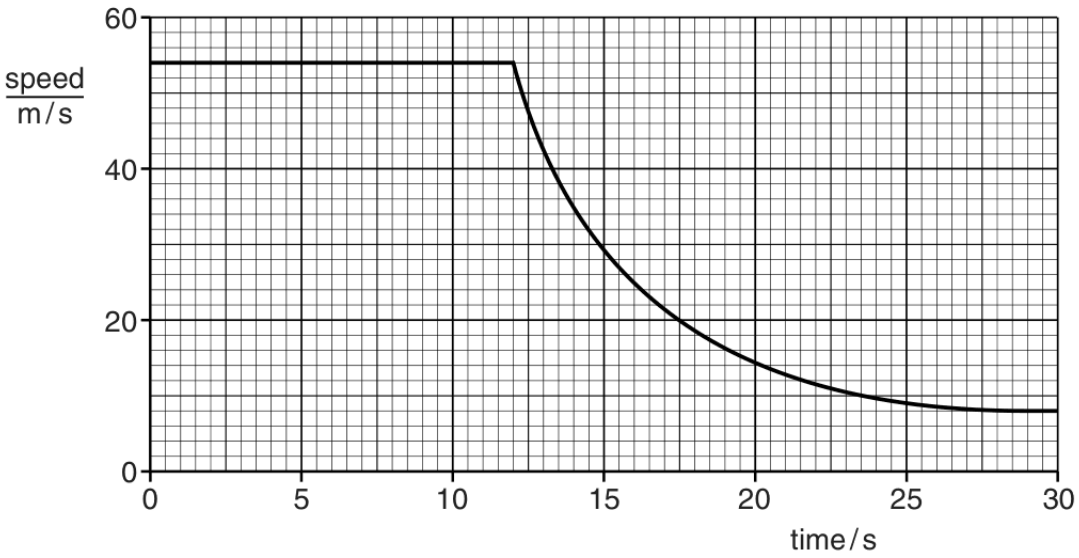


Fig. 9.2

(a) State the difference between *speed* and *velocity*.

.....

.....[1]

(b) The gravitational field strength g is 10 N/kg.

(i) Calculate the weight of the skydiver.

weight = ...[1]

For Examiner's Use

(ii) State the size of the air resistance acting on the skydiver between $t = 0$ and $t = 12$ s.

air resistance = ...[1]

(c) For the period between $t = 0$ and $t = 12$ s, determine

(i) the speed of the skydiver,

speed = ...[1]

(ii) the distance fallen by the skydiver,

distance = ...[2]

(iii) the change in the gravitational potential energy of the skydiver.

change in energy = ...[2]

(d) As he falls at constant speed, his gravitational potential energy changes into another form. State the name of this other form of energy.

.....[1]

(e) **(i)** State and explain what happens to the air resistance as the skydiver opens his parachute.

.....

.....

.....[2]

(ii) State and explain the effect on the motion of the skydiver of opening the parachute.

.....

.....

.....[2]

(f) By $t = 15$ s, his parachute is fully open.

State and explain what happens to the air resistance after $t = 15$ s.

.....

.....

.....[2]

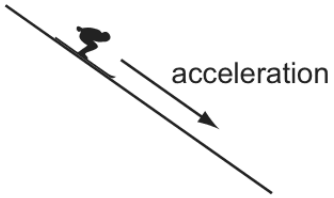
Section B

9	**(a)**	speed does not have direction **and** velocity does or speed = distance/time **and** velocity = displacement/time or speed is a scalar **and** velocity is a vector	B1	[1]
	(b) **(i)**	700 N	B1	
	(ii)	700 N	B1	[2]
	(c) **(i)**	54 m / s	B1	
	(ii)	(height/distance =) area (under graph) **or** (x =)vt **or** 54 × 12 648/650 m	C1 A1	
	(iii)	(GPE =) mgh **or** 70 × 10 × 648 4.5/4.54/4.536 × 10^5 J	C1 A1	[5]
	(d)	(becomes) heat/thermal energy/internal energy (**not** kinetic energy (of skydiver) unless qualified as KE of air)	B1	[1]
	(e) **(i)**	(air resistance) increases larger area of parachute	B1 B1	
	(ii)	(skydiver) decelerates/slows down (**not** rises up) net upward force	B1 B1	[4]
	(f)	air resistance decreases speed decreases	B1 B1	[2]

[Total: 15]

QUESTION · 5054_w13_qp_12.pdf · source page 3 of 20

4 A skier of mass 80 kg accelerates down a slope at $2\,m/s^2$.



The gravitational field strength g is 10 N / kg.

What is the weight of the skier and the resultant force on the skier?

	weight / N	resultant force / N
A	8	168
B	80	240
C	800	160
D	800	640

MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 4

4 **C**

QUESTION · 5054_w13_qp_12.pdf · source page 4 of 20

6 The weight of a cylinder on the Moon is less than its weight on the Earth.

How do the gravitational field strength and the mass of the cylinder on the Moon compare with their values on the Earth?

	gravitational field strength on Moon	mass on Moon
A	the same	less
B	the same	the same
C	less	the same
D	zero	zero

MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 6

6 **C**

QUESTION · 5054_s14_qp_11.pdf · source page 4 of 16

8 Four objects are situated in places with different gravitational field strengths.

Which object has the greatest weight?

	$\frac{\text{mass}}{\text{kg}}$	$\frac{\text{gravitational field strength}}{\text{N/kg}}$
A	3.0	10.4
B	3.5	9.5
C	4.0	10.2
D	4.5	9.0

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 8

8 **C**

QUESTION · 5054_s14_qp_12.pdf · source page 4 of 16

7 Four rocks on different planets have masses and weights as shown.

Which planet has the greatest gravitational field strength?

	mass / kg	weight / N
A	2.0	14
B	2.5	20
C	3.0	21
D	3.5	19

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 7

7 **B**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a lorry accelerating in a straight line along a horizontal road.

Fig. 1.1

(a) The driving force on the lorry in the forward direction is *D* and the total backward force on the lorry is *B*.

(i) State and explain whether *D* or *B* is the larger force.

.....

..... [1]

(ii) Suggest one possible cause of the backward force *B*.

..... [1]

(b) The weight of the lorry is 300 000 N.

The gravitational field strength *g* is 10 N/kg.

(i) Calculate the mass of the lorry.

mass = ... [1]

(ii) The resultant force on the lorry is 15 000 N. Calculate the acceleration of the lorry.

acceleration = ... [2]

(c) Later, the lorry turns a corner at constant speed.

Explain why the lorry accelerates even though the speed is constant.

.....

..... [1]

1 **(a)** **(i)** *D* **and either** lorry accelerates (forward) **or** resultant force is forward [B1]

(ii) air resistance **or** (air) drag **or** friction (between tyres and road) [B1]

(b) **(i)** 30 000 kg [B1]

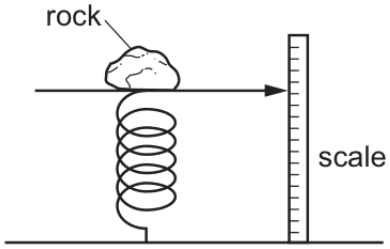
(ii) (a=) F / m algebraic in any form or numerical [C1]
0.5(0) m / s^2 [A1]

(c) direction or velocity is changing **or** acceleration or force is sideways or towards centre (of circle) [B1]

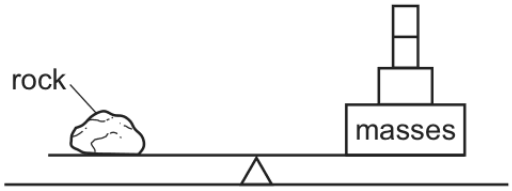
[6]

QUESTION · 5054_w14_qp_11.pdf · source page 6 of 20

6 A scientist places a rock on a spring balance. She then places the same rock on a beam balance.



spring balance



beam balance

At the North Pole, the gravitational field strength is greater than at the Equator.

She performs the experiment at the North Pole and at the Equator.

How do the readings at the North Pole compare with those at the Equator?

	scale reading on spring balance	masses needed on beam balance
A	different at North Pole	different at North Pole
B	different at North Pole	same at North Pole
C	same at North Pole	different at North Pole
D	same at North Pole	same at North Pole

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 6

6 **B**

2 A student places a metre rule on the edge of a triangular prism. The prism is used as a pivot and the rule balances about the 50 cm mark.

The student then places a block of wood at the 10 cm mark on the rule and an empty measuring cylinder at the 80 cm mark. The rule is still balanced. Fig. 2.1 shows the arrangement.

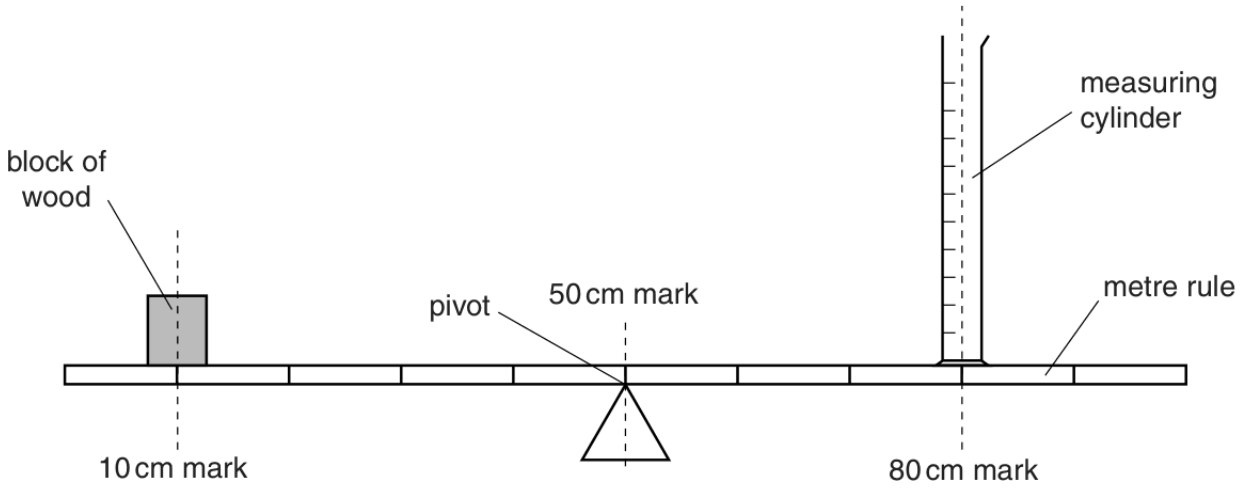


Fig. 2.1 (not to scale)

The student now places a weight of 0.39 N on top of the block of wood. She then starts to pour oil into the measuring cylinder. The rule balances again when there is 60 cm^3 of oil in the measuring cylinder.

The gravitational field strength *g* is 10 N/kg.

(a) Calculate

(i) the weight of the oil in the cylinder,

weight = ...[2]

(ii) the mass of the oil in the cylinder.

mass = ...[1]

(b) Calculate the density of the oil.

density = ...[2]

2 **(a)** **(i)** $F_1 \times d_1 = F_2 \times d_2$ **or** $(0.39 \times 0.40)/0.30$ C1
$0.52\,\text{N}$ A1

(ii) $0.052\,\text{kg}$ **or** $52\,\text{g}$ B1

(b) $(\rho =)\ m/V$ **or** $52/60$ **or** $0.052/0.000\,060$ **or** $0.052/60$ B1
$870/867/866.7\,\text{kg/m}^3$ **or** $0.87\,\text{g/cm}^3$ **or** $8.7 \times 10^{-4}\,\text{kg/cm}^3$ etc. B1 **[5]**

QUESTION · 5054_w14_qp_12.pdf · source page 2 of 16

4 A boy stands on some bathroom scales. The reading on the scales is 50 kg.

What is the mass and what is the weight of the boy?

	mass	weight
A	50 kg	50 N
B	50 kg	500 N
C	5.0 N	50 kg
D	50 N	5.0 kg

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 4

4 **B**

QUESTION · 5054_s15_qp_11.pdf · source page 3 of 16

4 The mass of a stone is found on Earth using a pan balance. The weight of the stone is found using a newton meter.

Are the readings the same or different on the Moon?

	reading on pan balance	reading on newton meter
A	different	different
B	same	different
C	different	same
D	same	same

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 4

4 **B**

QUESTION · 5054_s15_qp_11.pdf · source page 3 of 16

5 The table shows the weights and masses of four objects on different planets.

On which planet is the gravitational field strength the largest?

	weight / N	mass / kg
A	2.0	20
B	4.0	30
C	6.0	40
D	8.0	50

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 5

5 **D**

QUESTION · 5054_s15_qp_12.pdf · source page 4 of 16

8 The mass of a stone is found on Earth using a pan balance. The weight of the stone is found using a newton meter.

Are the readings the same or different on the Moon?

	reading on pan balance	reading on newton meter
A	different	different
B	same	different
C	different	same
D	same	same

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 8

8 **B**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 The apparatus shown in Fig. 1.1 is used to measure the extension of a spring.

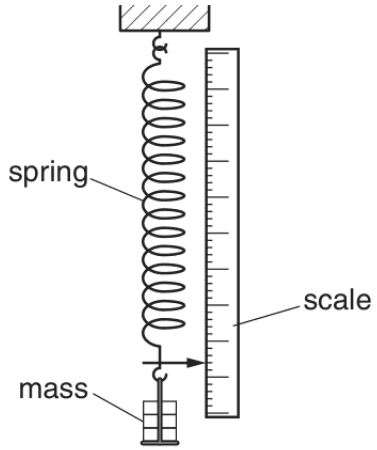


Fig. 1.1

(a) Explain how the mass causes a force on the spring.

.....

..... [1]

(b) The force on the spring is a vector quantity. State what is meant by a *vector* quantity.

.....

..... [1]

(c) Fig. 1.2 shows a graph of the length of the spring plotted against the force on the spring, for forces between 0 and 10 N.

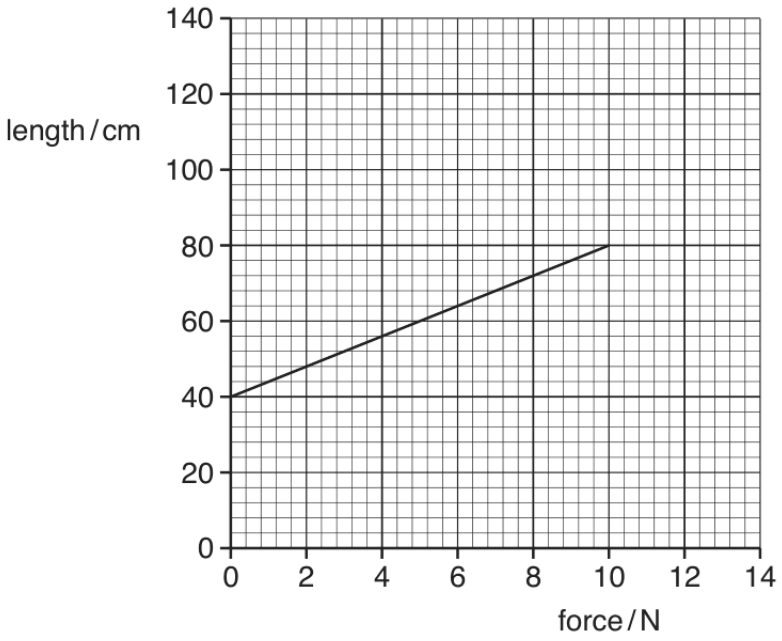


Fig. 1.2

(i) **1.** State a formula that relates the unstretched length l_0 of the spring, the stretched length l of the spring and the extension e of the spring.

..... [1]

2. A mass produces a force of 9.0 N on the spring. Determine the extension of the spring caused by this mass.

extension = ... [1]

(ii) The limit of proportionality of the spring is reached when the force is 10 N.

The spring is easier to stretch after the limit of proportionality.

On Fig. 1.2, continue the line to suggest how the length changes when the force is greater than 10 N. [1]

1	**(a)**	weight (pulls spring down and causes tension) **or** force/pull of **gravity** **or** mass is in gravitational field		B1
	(b)	has a direction		B1
	(c)	**(i)** **1**	$l = l_0 + e$ **or** $l_0 = l - e$ **or** $e = l - l_0$	B1
		2	36 cm	B1
		(ii)	curve upwards after 10 N	B1

QUESTION · 5054_w15_qp_11.pdf · source page 4 of 16

7 Which statement about mass is correct?

A Mass is density divided by volume.

B Mass is the amount of space occupied by a body.

C Mass is weight multiplied by the gravitational field strength.

D Mass resists a change in motion.

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 7

7	**D**

4 Fig. 4.1 shows a screwdriver of mass 64 g resting in equilibrium on a pivot.

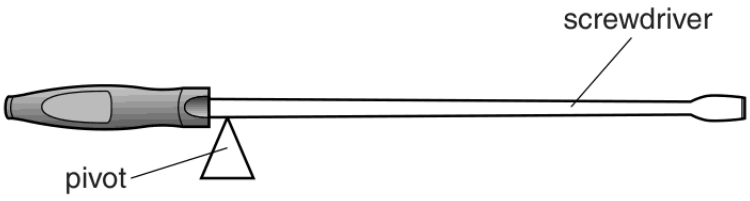


Fig. 4.1

(a) On Fig. 4.1, mark and label with a C, the centre of mass of the screwdriver. [1]

(b) The gravitational field strength is 10 N/kg.

(i) Calculate the weight of the screwdriver.

weight = ... [1]

(ii) On Fig. 4.1, draw an arrow labelled *W* to represent the weight of the screwdriver. [1]

(c) State two conditions that apply when an object is in equilibrium.

1. .....

.....

.....

2. .....

.....

.....

[2]

4 **(a)** (point) C immediately above tip of pivot (and in middle(vertically) of screwdriver (±1 mm)) B1

(b) **(i)** 0.64 N B1

(ii) arrow *W* vertically downwards through candidate's C or pivot B1

(c) no resultant force **or** upward force = downward force **or** force left = force right B1
no resultant moment (of force) **or** clockwise moment = anticlockwise moment B1 [5]

QUESTION · 5054_w15_qp_12.pdf · source page 4 of 16

9 A student produces some revision notes on gravity, mass and weight.

Which statement is **not** correct?

A A gravitational field is a region in which a mass experiences a gravitational force.

B Gravitational field strength has the unit of N / kg.

C Mass is a measure of the amount of matter.

D Mass is equal to weight multiplied by gravitational field strength.

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 9

9	**D**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Aeroplanes fly at high altitudes where the temperature is well below 0 °C. Ice that forms on an aeroplane can fall to earth and strike the ground.

Fig. 9.1 shows a block of ice falling from an aeroplane as it approaches an airport.

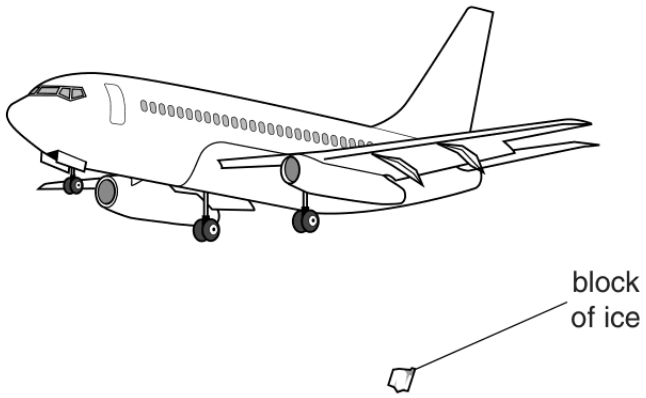


Fig. 9.1 (not to scale)

The mass of the falling block of ice is 1.2 kg and the gravitational field strength g is 10 N/kg.

(a) Calculate the weight of the block of ice.

weight = ... [1]

(b) Fig. 9.2 is the speed-time graph for the block of ice as it falls to the ground.

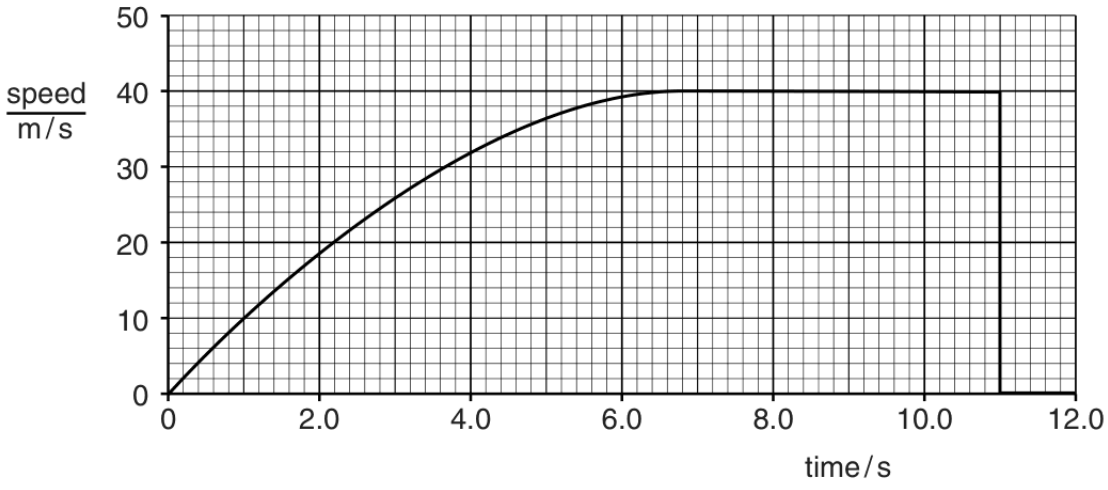


Fig. 9.2

At first, the acceleration of the block of ice is equal to the acceleration of free-fall. The acceleration of the block then decreases to zero as the block reaches terminal velocity.

(i) As the block of ice falls, the force F of air resistance acting on the block changes.

1. State the value of F at time = 0.

F = ... [1]

2. State the value of F at time = 10.0 s.

F = ... [1]

3. Explain why F changes.

.....

.....

..... [1]

(ii) State the energy change that takes place when the block is falling at terminal velocity.

.....

..... [1]

(iii) Using information from Fig. 9.2, determine the maximum kinetic energy of the block when it is falling to the ground with terminal velocity.

maximum kinetic energy = ... [3]

(c) The block strikes the ground and it stops moving. This impact causes some of the ice to melt.

(i) The specific latent heat of fusion of ice is 330 J / g.

Calculate the maximum possible mass of ice that melts as a result of the impact.

mass = ... [2]

(ii) In practice, the mass of ice that melts on impact is less than the value calculated in **(c)(i)**.

Suggest two reasons for this.

1. .....

.....

2. .....

..... [2]

(d) As the solid ice melts, it changes into liquid water.

Describe, in terms of molecules, how ice differs from liquid water.

.....

.....

.....

.....

.....

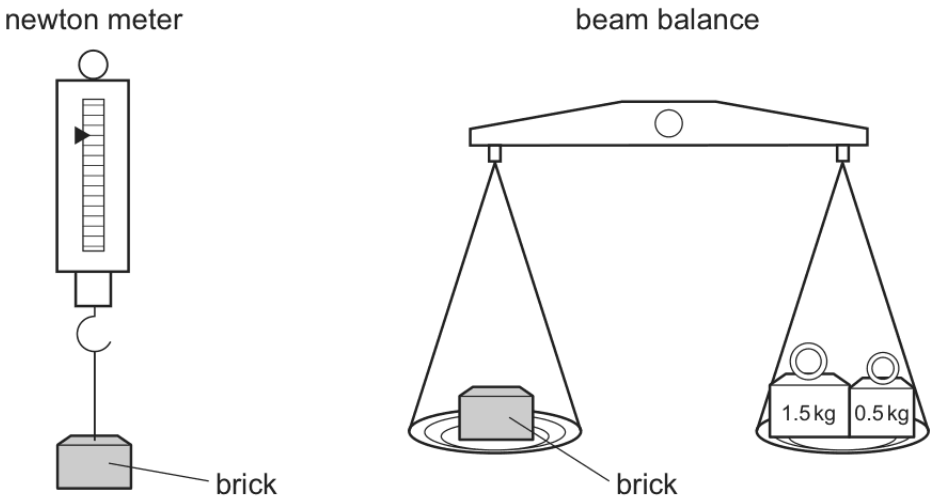
..... [3]

Section B

9	**(a)**	12 N	B1	[1]
	(b) **(i)**	0 **or** zero	B1	
		12 N **or** it is equal to the weight	B1	
		(*F* increases) as the speed increases	B1	
	(ii)	(gravitational) potential to thermal energy **or** to k.e. **of air**	B1	
	(iii)	(KE =)$\frac{1}{2}mv^2$	C1	
		$\frac{1}{2} \times 1.2 \times 40^2$	C1	
		960 J	A1	[7]
	(c) **(i)**	(m =)E / l_f **or** Q / l_f **or** 960 / 330	C1	
		2.9 / 2.91 g **or** 2.9 / 2.91 $\times 10^{-3}$ kg	A1	
	(ii)	any **two** from: ice is below 0 °C thermal energy transferred / lost (to ground / air) work done compressing / compacting the ground	B2	[4]
	(d)	any **three** from: molecules fixed in position **or** water molecules move around molecules vibrate **or** water molecules do not vibrate molecules in regular lattice **or** water molecules placed randomly (interatomic) forces between ice molecules larger ice molecules further apart	B3	[3]
				[15]

QUESTION · 5054_s16_qp_11.pdf · source page 4 of 16

7 A brick is placed on a newton meter and then on a beam balance.



What is measured by each instrument?

	newton meter	beam balance
A	mass	mass
B	mass	weight
C	weight	mass
D	weight	weight

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 7

7 **C**

2 A student hangs various masses from the end of a spring and determines the extension produced. Fig. 2.1 shows the spring hanging vertically on its own and with an unknown mass X at one end.

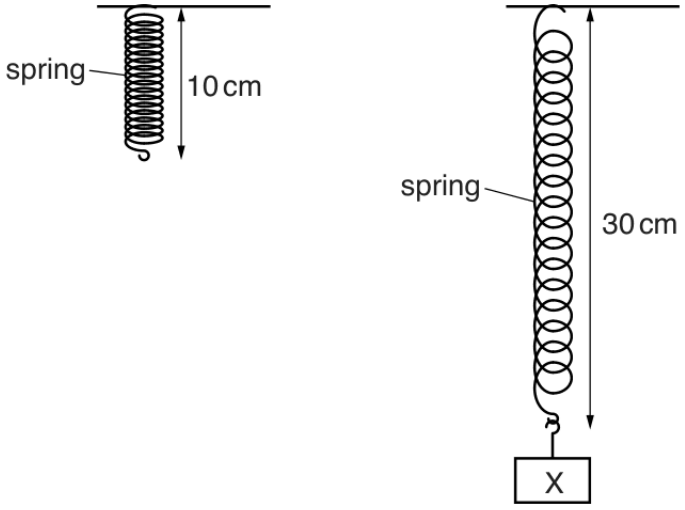


Fig. 2.1 (not to scale)

(a) The student plots a graph of the extension of the spring against the mass hanging on the spring. Fig. 2.2 shows the student's graph.

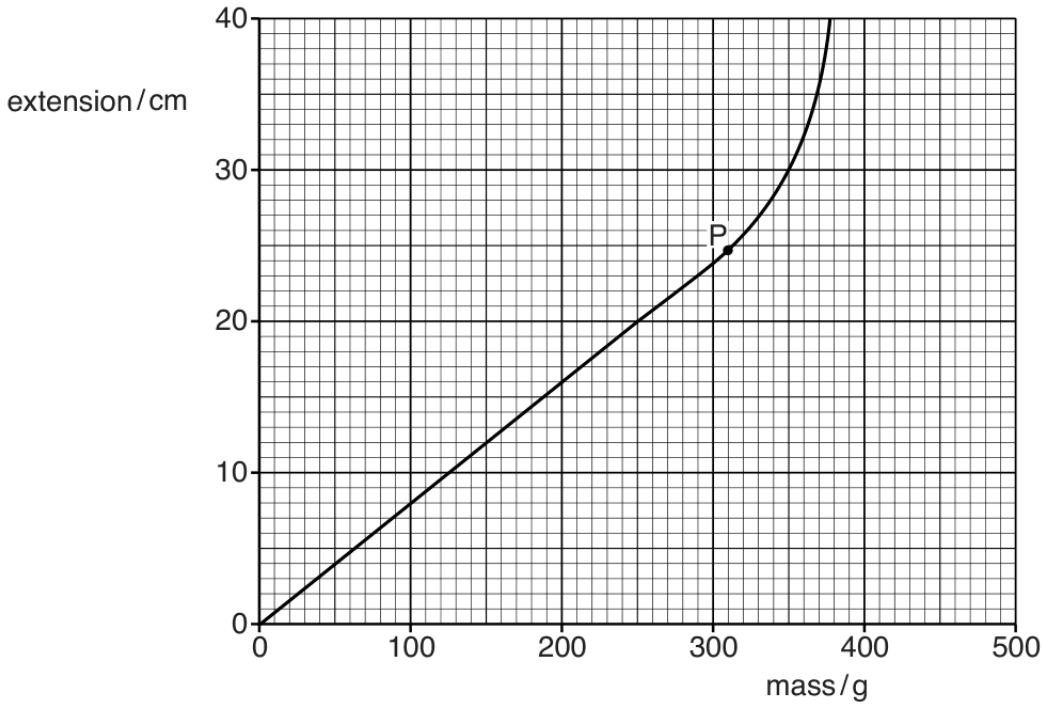


Fig. 2.2

(i) At point P on the graph, the line begins to curve.

State the name of point P.

.....[1]

(ii) Using Fig. 2.1 and Fig. 2.2, determine the mass X.

mass = ... [1]

(iii) The gravitational field strength g is 10 N/kg.

Calculate the weight of X.

weight = ... [1]

(b) An identical spring is used with the original spring, as shown in Fig. 2.3.

Together they support the mass X.

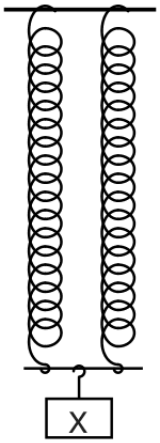


Fig. 2.3 (not to scale)

State and explain how the extension in Fig. 2.3 compares with the extension in Fig. 2.1.

.....

.....

.....

.....[2]

2	**(a)**	**(i)**	limit of proportionality	B1
		(ii)	250 g	B1
		(iii)	2.5 N	B1
	(b)		half the extension / 10 cm	B1
			each / both / another spring shares / distributes the weight / mass **or** both springs bear / carry the load	B1

3 Fig. 3.1 shows a thin sheet of plastic. A student tries to measure the thickness of the sheet with a ruler, but the sheet is too thin to measure accurately.

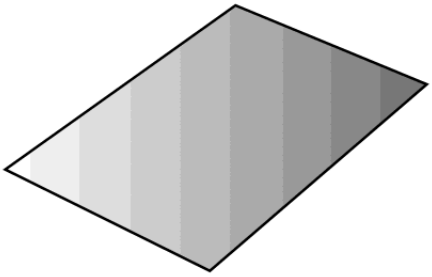


Fig. 3.1 (not to scale)

The student measures the mass of the sheet and obtains the value 0.12 g.

(a) (i) State what is meant by *mass*.

.....

.....[1]

(ii) The student is told that the density of the plastic is 0.91 g/cm^3.

Calculate the volume of the plastic sheet.

volume = ...[2]

(iii) The student measures the length and width of the sheet. The readings obtained are:

length of sheet = 3.0 cm
width of sheet = 2.0 cm

Calculate the thickness of the sheet.

thickness = ... [2]

(b) State a measuring instrument that can be used to measure the thickness of the sheet accurately.

.....[1]

3	**(a)**	**(i)**	amount of matter / substance / material **or** the ability of an object to resist a change in its state of motion (when a force is applied)	B1
		(ii)	(V=) M / D in any form numerical or algebraic	C1
			0.13(19) cm^3	A1
		(iii)	V / ($l \times$ w) in any form numerical or algebraic	C1
			0.022 cm	A1
	(b)		micrometer (screw gauge) **or** calipers	B1

QUESTION · 5054_s16_qp_12.pdf · source page 2 of 16

4 Which property of a body resists change from a state of rest or of motion?

A density

B mass

C volume

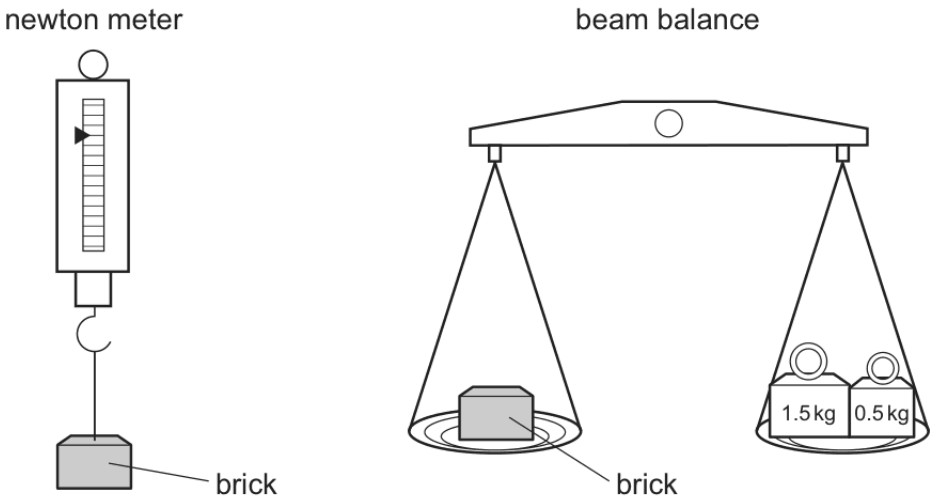
D weight

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 4

4	**B**

QUESTION · 5054_s16_qp_12.pdf · source page 3 of 16

5 A brick is placed on a newton meter and then on a beam balance.



What is measured by each instrument?

	newton meter	beam balance
A	mass	mass
B	mass	weight
C	weight	mass
D	weight	weight

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 5

5 **C**

QUESTION · 5054_w16_qp_11.pdf · source page 3 of 20

5 Which statement about mass and weight is correct?

A A mass experiences a weight due to a gravitational field.

B Mass and weight are different types of force.

C Mass and weight have the same unit.

D When an object expands, its mass changes but its weight stays the same.

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 5

5	**A**

2 Fig. 2.1 shows a bed that folds away against a wall during the day.

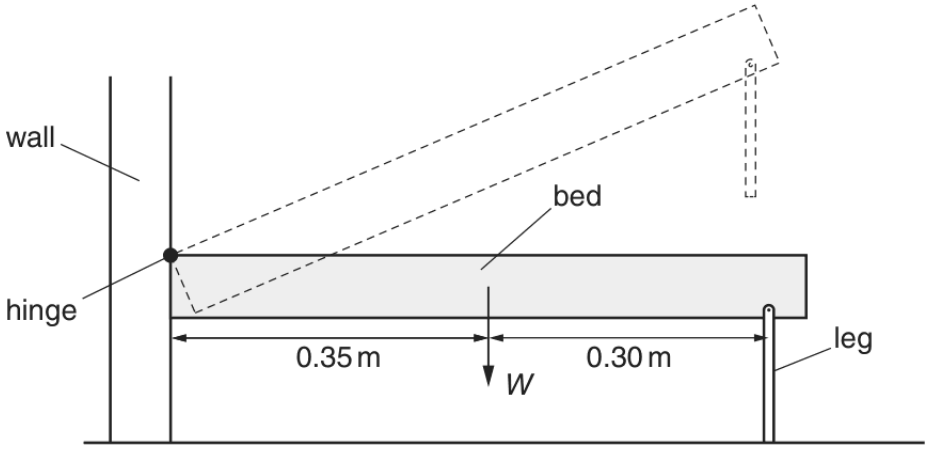


Fig. 2.1 (not to scale)

When it is horizontal, the bed is supported on one side by a hinge and on the other side by two legs. The weight W of the bed acts through its centre of mass, which is at a horizontal distance of 0.35 m from the hinge.

(a) The mass of the bed is 26 kg. The gravitational field strength g is 10 N/kg.

Calculate the weight W of the bed.

W = ...[1]

(b) (i) State the principle of moments.

.....

.....

.....

.....[2]

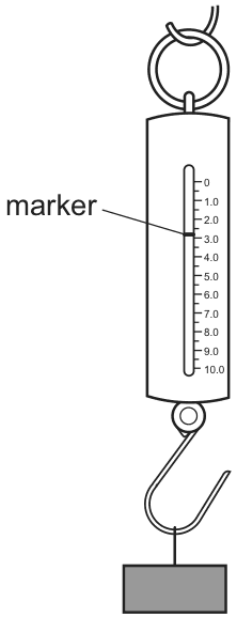
(ii) Determine the size of the total upward force exerted on the bed by the two legs when the bed is horizontal.

total force = ...[3]

2	**(a)**	260 N	B1	
	(b)	**(i)** for a body in equilibrium	B1	
		(total) clockwise moment = (total) anticlockwise moment	B1	
		(ii) $F_1d_1 = F_2d_2$ **or** 260×0.35 **or** 91 **or** $F \times 0.65$	C1	
		$260 \times 0.35 = F \times 0.65$ **or** $260 \times 0.35/0.65$ **or** $91 = F \times 0.65$ **or** $91/0.65$	C1	
		140 N	A1	[6]

QUESTION · 5054_w16_qp_12.pdf · source page 4 of 16

7 A block is hung on a spring balance. The marker inside the balance is pulled down by the block.



What can the position of the marker be used to determine?

A the mass of the block in kg

B the mass of the block in N

C the moment caused by the block in N m

D the weight of the block in kg

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 7

7 **A**

3 When the lid of a freezer is opened, it pivots about the hinge at the back of the freezer. The handle is at the front.

Fig. 3.1 is a side view of the freezer.

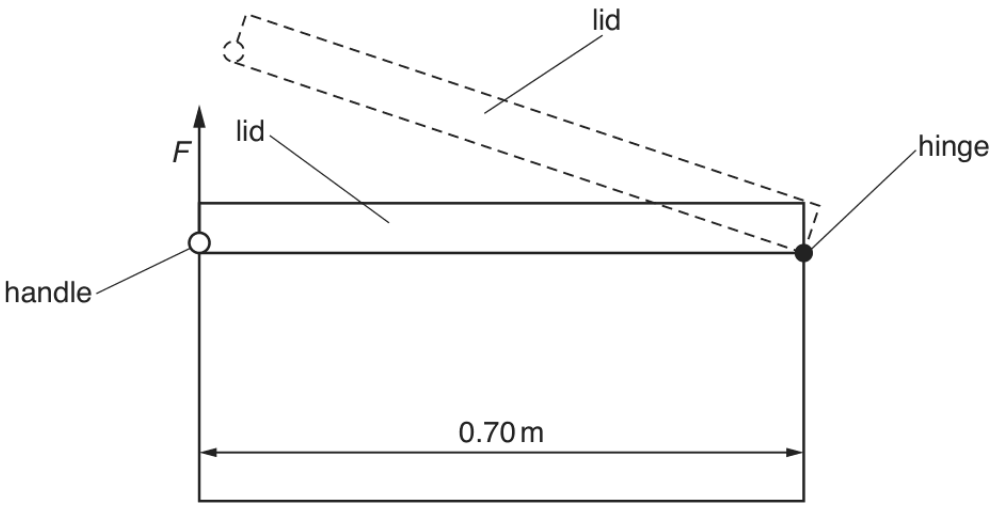


Fig. 3.1

The handle is 0.70 m from the hinge. The lid has a mass of 2.0 kg.

(a) The gravitational field strength g is 10 N / kg.

(i) Calculate the weight of the lid.

weight = ...[1]

(ii) The lid is uniform and its centre of mass is at its centre. The weight of the lid produces a moment about the hinge.

1. Calculate the moment about the hinge when the lid is closed.

moment = ...[2]

2. The moment required to open the lid is greater than the value calculated in **(a)(ii)1.**

Suggest one reason for this.

.....

.....[1]

(b) The lid is closed. To open the lid, a force *F* is applied to the handle as shown in Fig. 3.1.

The direction of *F* is vertically upwards and *F* is the smallest possible force that opens the lid.

A force on the handle in any other direction must be larger than *F* in order to open the lid. Explain why.

.....

.....

.....[1]

3 (a) (i) 20 N B1

(ii) 1. (Γ =)Fd **or** 20 × 0.35 **or** 20 × 0.70 **or** 14 C1
7.0 N m A1

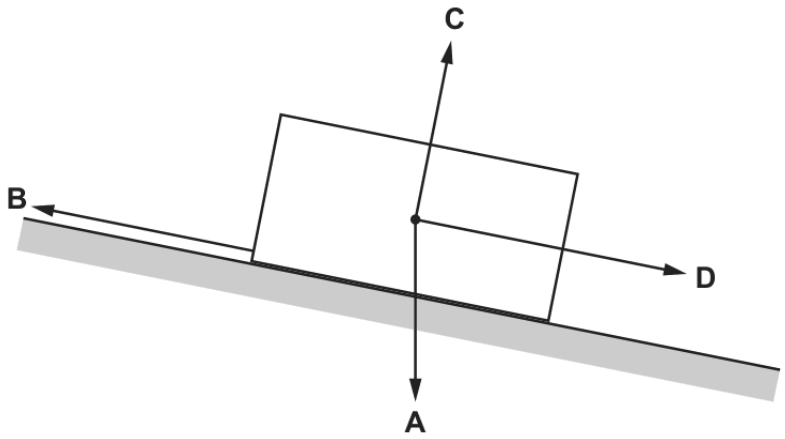
2. friction (at hinge / seal) **or** air resistance **or** to cause an initial acceleration B1

(b) (for other directions) **perpendicular** distance is less B1 [5]

QUESTION · 5054_s17_qp_11.pdf · source page 5 of 16

9 The diagram shows a block of wood resting on a sloping board.

Which arrow shows the direction of the gravitational force acting on the block?



MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 9

9	A	**1**

QUESTION · 5054_s17_qp_11.pdf · source page 6 of 16

10 An astronaut travels to the International Space Station.

Which row describes how his mass and his weight compare with their sizes on Earth?

	mass	weight
A	different	different
B	different	the same
C	the same	different
D	the same	the same

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 10

10	C	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the directions of four forces acting on a racing car as it travels in a horizontal straight line.

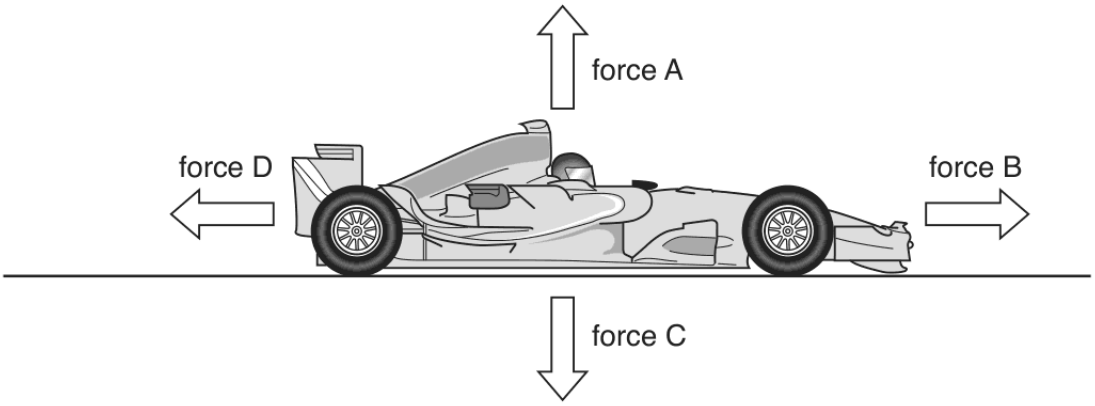


Fig. 1.1

(a) Draw a line from each box on the left to the correct description of each force.

force A	driving force
force B	contact or normal reaction force
force C	air resistance and friction
force D	force of gravity

[1]

(b) The table shows the sizes of the forces acting on the car at one time.

force A/N	force B/N	force C/N	force D/N
8000	1000	8000	600

The gravitational field strength g is 10 N/kg.

Calculate

(i) the mass of the car,

mass = ...[1]

(ii) the resultant force on the car,

resultant force = ...[1]

(iii) the acceleration of the car.

acceleration = ...[2]

(c) At another time, the car is travelling at speed u. It then accelerates for 5.0 s with an acceleration of 1.6 m/s^2, and reaches a speed of 20 m/s.

Calculate the value of u.

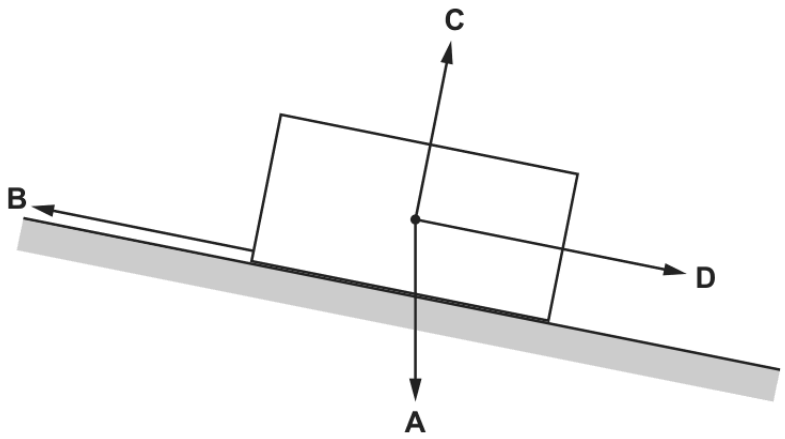
u = ...[2]

Question	Answer	Marks
1(a)	A – contact or reaction force B – driving force C – force of gravity D – air resistance and friction	**B1**
1(b)(i)	800 kg	**B1**
1(b)(ii)	400 N	**B1**
1(b)(iii)	(a =) F / m formula or with numbers	**C1**
	0.50 m / s^2	**A1**
1(c)	change in v = at in words or numbers	**C1**
	12 m / s	**A1**

QUESTION · 5054_s17_qp_12.pdf · source page 5 of 20

9 The diagram shows a block of wood resting on a sloping board.

Which arrow shows the direction of the gravitational force acting on the block?



MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 9

9	A	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 5 of 20

10 On the Earth's surface, the gravitational field strength is 10 N / kg.

On the surface of Mars, the gravitational field strength is 3.8 N / kg.

A robot vehicle has a weight of 2000 N on Earth.

What is the weight of the robot vehicle on Mars?

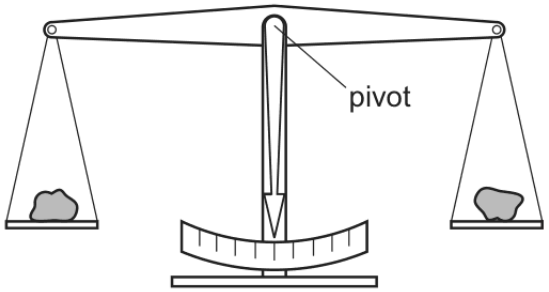
A 200 N **B** 760 N **C** 7600 N **D** 76 000 N

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 10

10	B	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 5 of 20

6 The diagram shows two objects on a beam balance.



The beam balance is in equilibrium.

Which quantities may be different?

A the masses of the two objects

B the moments about the pivot of the two objects

C the volumes of the two objects

D the weights of the two objects

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 6

6	C	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

Answer only one of the two alternative parts in **Question 6**.

1 Fig. 1.1 shows a painter standing on a wooden plank, directly above the right-hand support.

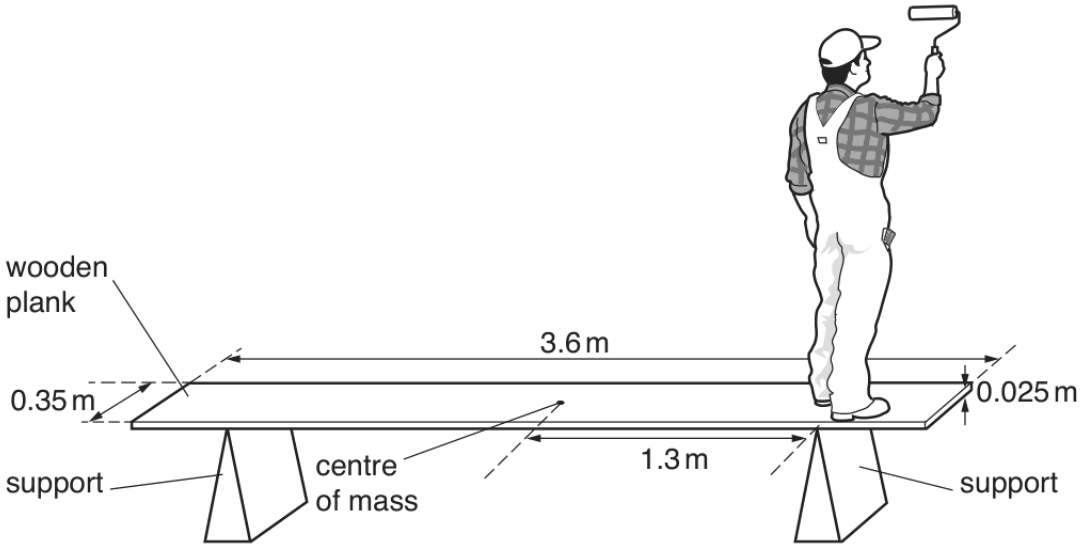


Fig. 1.1

The plank has length 3.6 m, width 0.35 m and thickness 0.025 m.

The gravitational field strength g is 10 N/kg and the mass of the plank is 23 kg.

(a) Calculate the density of the wood from which the plank is made.

density = ... [2]

(b) The centre of mass of the plank is in the middle of the plank at a distance of 1.3 m from each of the supports.

Calculate

(i) the weight of the plank,

weight = ... [1]

(ii) the moment of the plank about the right-hand support.

moment = ... [2]

(c) The painter moves further to the right along the plank and the plank rotates about the right-hand support.

Explain why the plank rotates.

.....

.....

.....

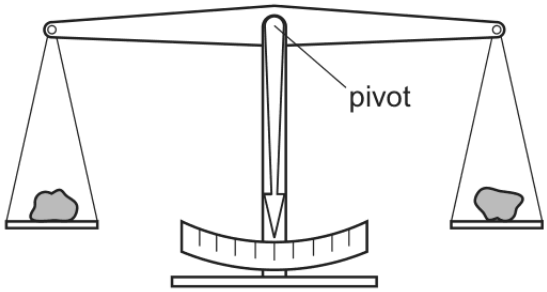
..... [1]

Section A

Question	Answer	Marks
1(a)	(ρ =)m / V **or** 23 / (3.6 × 0.35 × 0.025) **or** 23 / 0.0315	**C1**
	730 kg / m^3	**A1**
1(b)(i)	230 N	**B1**
1(b)(ii)	(Γ =)$Wx_{\perp r}$ **or** 230 × 1.3	**C1**
	300 Nm	**A1**
1(c)	moment of painter / clockwise moment (about support) is greater than / not equal to / different from moment of plank / anticlockwise moment	**B1**

QUESTION · 5054_w17_qp_12.pdf · source page 4 of 16

5 The diagram shows two objects on a beam balance.



The beam balance is in equilibrium.

Which quantities may be different?

A the masses of the two objects

B the moments about the pivot of the two objects

C the volumes of the two objects

D the weights of the two objects

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 5

5	C	**1**

2 A force applied to a solid object may cause it to accelerate so that its velocity changes.

(a) State two other properties of the object which may change when a force is applied.

1. .....

2. .....

[2]

(b) A spring has a mass of 0.012 kg.

(i) The gravitational field strength g is 10 N/kg.

Calculate the weight of the spring.

weight = ...[1]

(ii) The spring is suspended vertically and a load is attached to its lower end. The spring extends by 2.7 cm and does not pass the limit of proportionality.

Fig. 2.1 shows the spring and load attached to the lower end of a second identical spring.

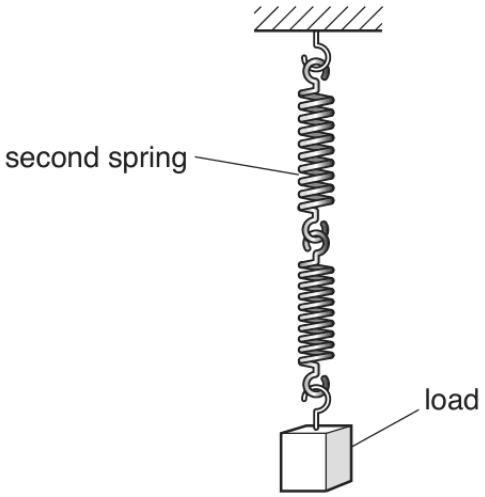


Fig. 2.1

Suggest one reason why the extension of the second spring differs from 2.7 cm.

.....

.....

.....[1]

(iii) Explain what is meant by *limit of proportionality*.

.....

.....

.....[1]

(c) The load in Fig. 2.1 is pulled down below its equilibrium position.

(i) State the form of energy stored in the stretched springs.

.....[1]

(ii) The load is released and it moves upwards and downwards. The distance travelled in each movement decreases until the load stops moving.

Explain why the load stops moving.

.....

.....

.....[1]

Question	Answer	Marks
2(a)	shape	**B1**
	size / volume / density	**B1**
2(b)(i)	0.12 N	**B1**
2(b)(ii)	weight of first spring (increases extension) **or** weight of first spring (takes spring beyond the limit of proportionality)	**B1**
2(b)(iii)	where / point / load / extension / limit beyond which the extension is not directly proportional to load / extension-load graph is not straight	**B1**
2(c)(i)	elastic potential energy	**B1**
2(c)(ii)	friction / air resistance mentioned **or** thermal energy / heat produced	**B1**

QUESTION · 5054_s18_qp_11.pdf · source page 3 of 20

6 Which piece of equipment is used to measure mass?

A balance

B manometer

C measuring cylinder

D newton meter

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 6

6	A	**1**

QUESTION · 5054_s18_qp_11.pdf · source page 3 of 20

7 A body of mass 10 kg falling freely in the gravitational field close to the Moon's surface has an acceleration of 1.6 m/s^2.

What is the gravitational field strength on the Moon?

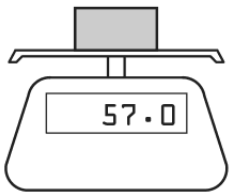
A 0 N/kg **B** 1.6 N/kg **C** 10 N/kg **D** 16 N/kg

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 7

7	B	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 4 of 16

8 A block of metal is placed on an electronic balance to record its mass.



What is the unit of the reading on the electronic balance and what is the unit of weight?

	unit on electronic balance	unit of weight
A	g	g
B	g	N
C	N	g
D	N	N

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 8

8	B	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 5 of 16

9 A body of mass 10 kg falling freely in the gravitational field close to the Moon's surface has an acceleration of 1.6 m/s^2.

What is the gravitational field strength on the Moon?

A 0 N/kg **B** 1.6 N/kg **C** 10 N/kg **D** 16 N/kg

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 9

9	B	**1**

QUESTION · 5054_w18_qp_11.pdf · source page 4 of 16

6 What happens to an object when it is moved to a location where the gravitational field strength is slightly greater?

A Its density decreases.

B Its mass decreases.

C Its weight increases.

D Its volume increases.

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 6

6	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 A bucket of water is pulled up out of a well using a rope. Fig. 8.1 shows the rope winding on to a cylinder as the handle is turned.

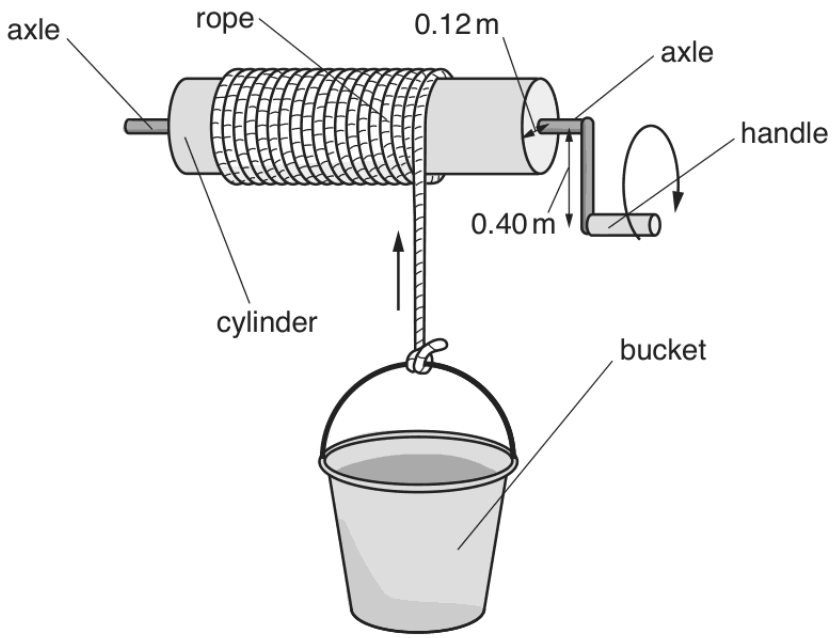


Fig. 8.1

The empty bucket has a mass of 1.0 kg.

(a) Complete the sentences that describe mass by filling in the gaps.

The mass of a body is a measure of the amount of .....

in the body. It resists a change in the state of ..... of

the body. [2]

(b) When the bucket is full, it contains $2.4 \times 10^{-2}\,m^3$ of water.

The gravitational field strength g is equal to 10 N/kg.

(i) Explain what is meant by a *gravitational field*.

.....

..... [1]

(ii) The density of water is 1000 kg / m^3.

Determine the total weight of the bucket and the water.

weight = ... [3]

(c) The radius of the cylinder is 0.12 m and the handle is 0.40 m from the axle of the cylinder. The weight of the bucket and the water produce a moment that acts on the cylinder.

(i) Calculate this moment.

moment = ... [2]

(ii) Calculate the minimum force on the handle that balances this moment.

force = ... [1]

(d) A farmer pulls the bucket of water up at a constant speed. He needs to exert a force on the handle that is greater than that calculated in **(c)(ii)**. He notices that there is a slight increase in temperature where the axle is in contact with the frame holding it.

(i) Suggest two reasons why the force exerted is greater than the value calculated.

1. .....

.....

2. .....

.....
[2]

(ii) State the energy changes that are taking place as the bucket is being lifted at a constant speed.

.....

.....

..... [3]

(e) State what can be deduced about the forces acting on the bucket when it is travelling upwards at a constant speed.

.....

..... [1]

Question	Answer	Marks
8(a)	substance **or** matter	**B1**
	rest **or** motion	**B1**
8(b)(i)	(a region of space) where a mass experiences a force (due to gravitational attraction)	**B1**
8(b)(ii)	(m =) ρV **or** $1000 \times 2.4 \times 10^{-2}$ **or** 24 (kg)	**C1**
	25 (kg) **or** 240 (N)	**C1**
	250 N	**A1**
8(c)(i)	(Γ =) Fx **or** 250×0.12	**C1**
	30 N m	**A1**
8(c)(ii)	75 N	**B1**
8(d)(i)	any **two** from: friction (between axle and cylinder) weight of rope force not perpendicular (to radius) rope wraps over itself (and increases moment)	**B2**

Question	Answer	Marks
8(d)(ii)	from chemical (energy) **or** chemical (energy) as first term	**B1**
	to gravitational potential (energy) **or** to thermal / heat (energy) as last term	**B1**
	to thermal / internal (energy) / heat **and** thermal / internal (energy) / heat	**B1**
8(e)	no resultant force **or** forces balance / cancel **or** tension equals weight (of bucket) **or** upwards force equals downward force	**B1**

QUESTION · 5054_w18_qp_12.pdf · source page 3 of 16

5 Which piece of apparatus may be used to compare the masses of two objects?

A balance

B manometer

C measuring cylinder

D micrometer

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 5

5	A	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 During a game of cricket, a player hits a ball with a bat. The ball then travels vertically upwards, as shown in Fig. 8.1.

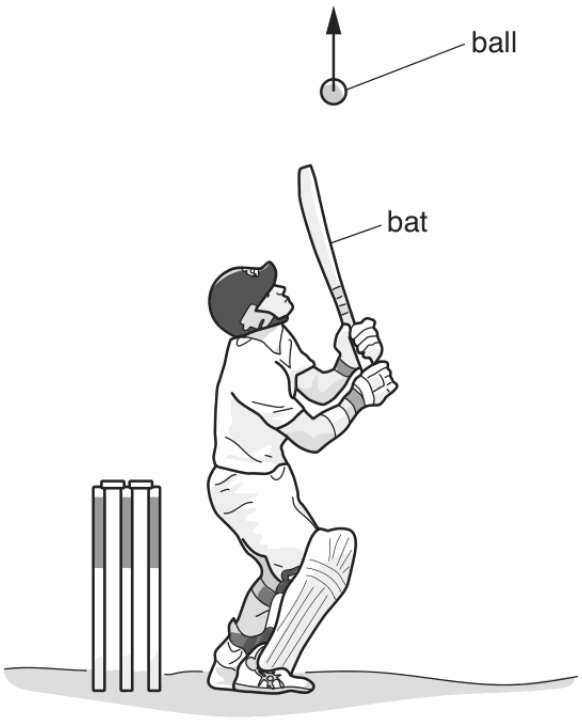


Fig. 8.1

The speed of the ball as it leaves the bat is 8.7 m/s.

(a) The mass of the ball is 0.16 kg.

(i) **1.** State what is meant by *mass*.

.....

..... [1]

2. State the name of a measuring instrument that is used to determine mass.

.....

..... [1]

(ii) Calculate the kinetic energy of the ball as it leaves the bat.

kinetic energy = ... [3]

(b) Fig. 8.2 is the velocity-time graph for the ball.

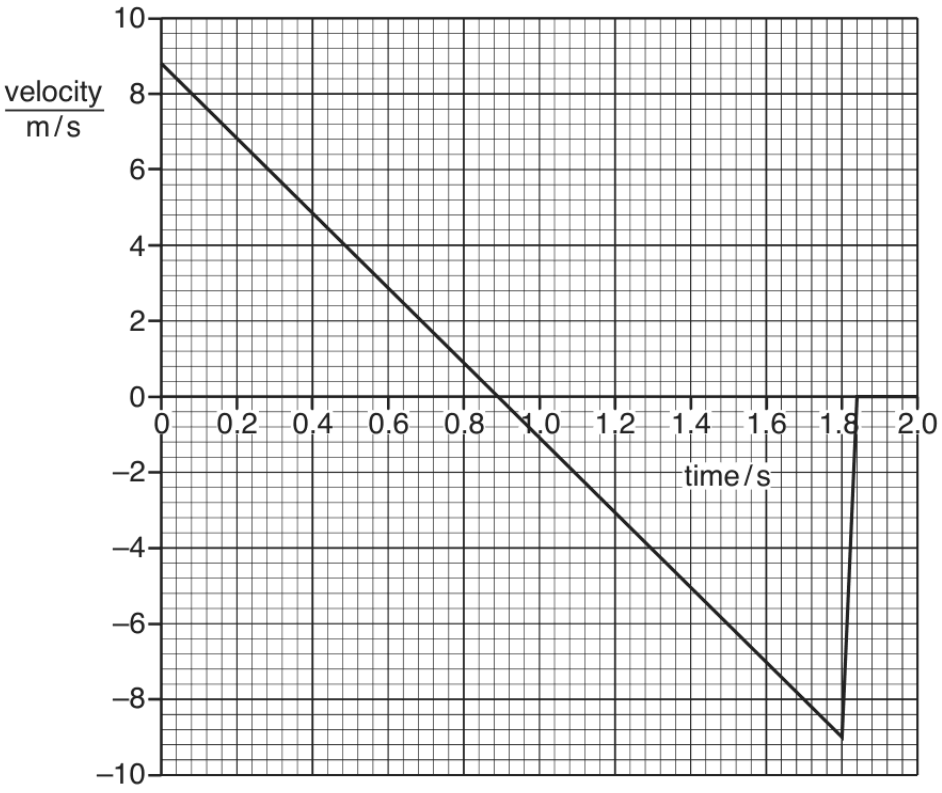


Fig. 8.2

(i) Fig. 8.2 shows that immediately after being hit, the ball has a negative acceleration.

1. State the name given to a negative acceleration.

..... [1]

2. State how the graph shows that the ball has a negative acceleration.

.....

..... [1]

(ii) Using Fig. 8.2, determine

1. the time at which the ball stops moving upwards,

time = ... [1]

2. the distance travelled before the ball stops moving upwards.

distance = ... [2]

(iii) Suggest why the graph, in Fig. 8.2, is almost vertical when time = 1.84 s.

.....

..... [1]

(iv) Fig. 8.2 shows that the ball remains stationary after a time of 1.84 s.

Describe the energy change that is occurring at a time of 1.84 s.

.....

.....

..... [2]

(c) For this ball moving in the way shown, the effect of air resistance is negligible.

When a ball with a much smaller mass is hit vertically upwards at the same speed, air resistance produces a noticeable effect on its motion.

Suggest two ways in which the velocity-time graph for the ball with a smaller mass differs from Fig. 8.2.

1. .....

.....

2. .....

.....
[2]

MARK SCHEME · 5054_w18_ms_22.pdf · source page 7 of 10 · Question 8

Question	Answer	Marks
8(a)(i)	1 quantity of matter (in a body)	**B1**
	2 balance **or** scales	**B1**
8(a)(ii)	(*k.e.* =) $\frac{1}{2}mv^2$	**C1**
	$\frac{1}{2} \times 0.16 \times 8.7^2$	**C1**
	6.1 J	**A1**
8(b)(i)	1 deceleration **or** retardation	**B1**
	2 negative gradient **or** line slopes downwards (left to right)	**B1**
8(b)(ii)	1 0.88 s ⩽ time ⩽ 0.90 s	**B1**
	2 area **or** counting squares **or** $\frac{1}{2}bh$ in some form	**C1**
	3.7 s ⩽ distance ⩽ 4.1 m	**A1**
8(b)(iii)	ball hits ground **or** short time for deceleration **or** large force **or** ground is hard	**B1**
8(b)(iv)	internal / thermal energy (of ball and ground) has increased	**B1**
	(internal energy) from kinetic energy	**B1**

MARK SCHEME · 5054_w18_ms_22.pdf · source page 8 of 10 · Question 8

Question	Answer	Marks
8(c)	any **two** from: smaller time to drop to zero velocity / hit ground line not straight **or** velocity does not change uniformly **or** gradient not constant smaller area under (first part of) graph **or** less distance travelled slower final velocity initial downward gradient steeper	**B2**

QUESTION · 5054_s19_qp_11.pdf · source page 4 of 20

8 Four objects of different masses are situated in places with different gravitational field strengths.

Which object has the greatest weight?

	mass / kg	$\frac{\text{gravitational field strength}}{\text{N/kg}}$
A	3.0	10.4
B	3.5	9.5
C	4.0	10.2
D	4.5	9.0

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 8

8	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a parachutist falling vertically towards the ground.

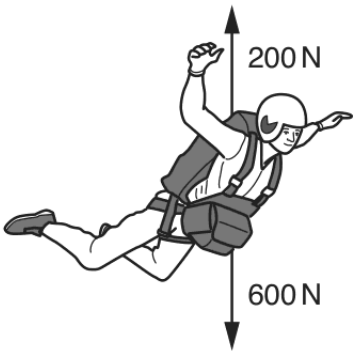


Fig. 9.1

The mass of the parachutist is 60 kg and his weight is 600 N.

(a) State what is meant by:

(i) *mass*

.....

..... [1]

(ii) *weight*.

.....

..... [1]

(b) Calculate the acceleration of the parachutist when the air resistance is 200 N.

acceleration = ... [3]

(c) The parachutist falls from rest at time $t = 0$.

He accelerates non-uniformly until he reaches a terminal velocity of 44 m/s at $t = 30$ s.

At $t = 40$ s the parachute opens. He decelerates for 4.0 s, reaching a smaller terminal velocity of 5.0 m/s.

He then falls at this speed until he lands on the ground at $t = 200$ s.

(i) On Fig. 9.2, draw the speed–time graph for the parachutist.

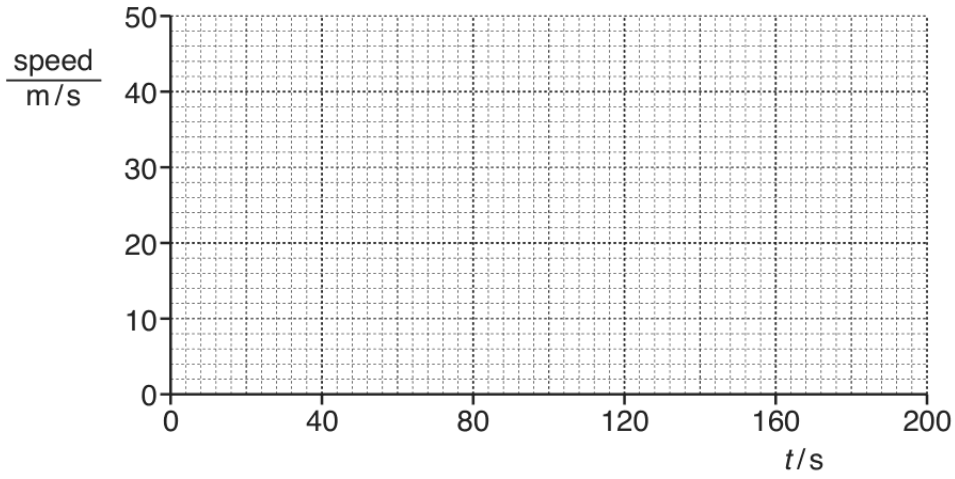


Fig. 9.2

[4]

(ii) Calculate the average deceleration of the parachutist between $t = 40$ s and $t = 44$ s.

deceleration = ... [2]

(d) Explain, in terms of the forces involved, why:

(i) the parachutist reaches a terminal velocity at $t = 30$ s

.....

..... [1]

(ii) the parachutist decelerates when the parachute opens

.....

.....

..... [2]

(iii) the second terminal velocity, when the parachute is open, is smaller than the first.

.....

..... [1]

[Total: 15]

Section B

Question	Answer	Marks
9(a)(i)	amount / measure / quantity of matter / substance (in a body) **or** (property of a body) that resists (change in) motion	**B1**
9(a)(ii)	force / pull of gravity **or** in a gravitational field	**B1**
9(b)	(a =) F / m algebraic or numerical **or** 400 (N) seen	**C1**
	400 / 60	**C1**
	6.7 m/s^2	**A1**
9(c)(i)	initial curve of correct shape from origin at 0 to 30 s	**B1**
	horizontal line at 44 m/s from 30 s to 40 s	**B1**
	speed drops from 44 to 5 m/s in a time of 4 s	**B1**
	constant speed of 5 m/s until 200 s (and then to zero)	**B1**
9(c)(ii)	(a =) (v – u) / t algebraic or numerical	**C1**
	9.7 **or** 9.8 m/s^2	**A1**
9(d)(i)	upwards force / air resistance = weight / downward force	**B1**
9(d)(ii)	increase in air resistance / upwards force (from parachute)	**B1**
	upwards / force / air resistance larger than weight / downwards force **or** resultant force upwards	**B1**
9(d)(iii)	air resistance decreases (as speed reduces) until it equals weight **or** (with parachute open) air resistance = weight at lower speed	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 4 of 16

8 The mass of object P is greater than the mass of object Q.

The objects contain different amounts of matter and have a different resistance to change of motion.

Which row is correct?

	greater amount of matter	greater resistance to change of motion
A	P	P
B	P	Q
C	Q	P
D	Q	Q

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 8

8	A	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 4 of 16

9 Four objects of different masses are situated in places with different gravitational field strengths.

Which object has the greatest weight?

	mass / kg	$\frac{\text{gravitational field strength}}{\text{N/kg}}$
A	3.0	10.4
B	3.5	9.5
C	4.0	10.2
D	4.5	9.0

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 9

9	C	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 There is no atmosphere on the Moon.

An astronaut on the Moon drops a feather and a hammer from the same height at the same time. They both accelerate downwards at $1.6\,m/s^2$ and they hit the ground at the same time.

(a) The weight of the hammer is much larger than that of the feather.

Explain, in terms of their weights and masses, why their accelerations are equal.

.....

.....

.....

.....

.....

..... [3]

(b) Both the feather and the hammer take 1.5 s to fall to the ground from rest.

(i) Calculate the speed of the objects as they hit the ground.

speed = ... [2]

(ii) On Fig. 1.1, draw the speed-time graph for the fall. At the correct position on the *y*-axis, write the value of the speed at time $t = 1.5$ s.

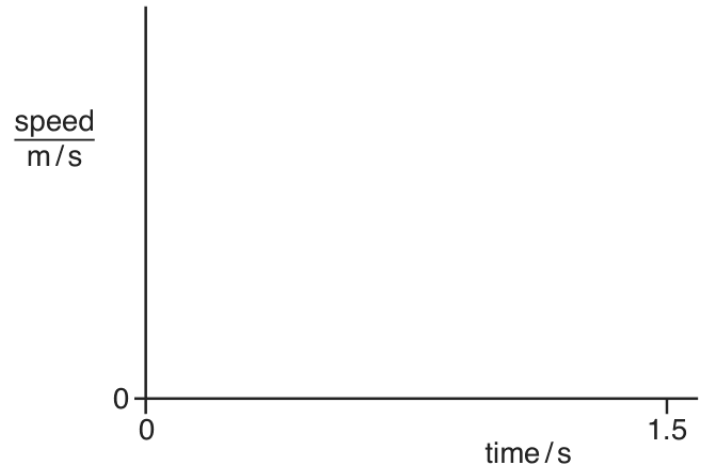


Fig. 1.1

[1]

(iii) Using the speed-time graph in **(b)(ii)**, determine the height from which the objects are dropped.

height = ... [2]

[Total: 8]

Question	Answer	Marks
1(a)	weight is directly proportional to mass **or** $W = mg$	**B1**
	a larger force acts on the more massive body **or** $a = F / m$ **or** $a = W / m$	**B1**
	mass cancels **or** acceleration inversely proportional to mass	**B1**
1(b)(i)	($\Delta v =$) at **or** 1.6×1.5	**C1**
	2.4 m / s	**A1**
1(b)(ii)	2.4 marked on y-axis **and** straight line from (0, 0) to (1.5, 2.4)	**B1**
1(b)(iii)	$\frac{1}{2} \times 2.4 \times 1.5$ **or** area under the graph mentioned / attempted	**C1**
	1.8 m	**A1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 Fig. 8.1 shows an elastic rope (bungee).

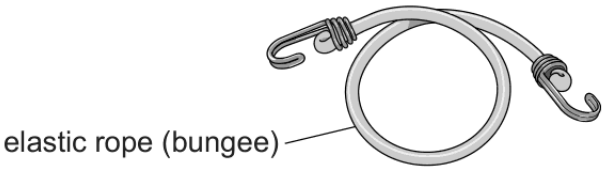


Fig. 8.1

The unstretched length of the rope is 0.80 m.

A student uses the hook at one end to suspend the rope from a shelf and hangs an empty paint can of mass 0.70 kg from the other end. The rope stretches to a length of 0.97 m but does not exceed its limit of proportionality.

(a) Explain what is meant by *limit of proportionality*.

.....

.....

..... [2]

(b) State what is meant by:

(i) *mass* ..... [1]

(ii) *weight.* .....

..... [2]

(c) The gravitational field strength is 10 N/kg.

(i) Calculate the weight of the empty paint can.

weight = ... [1]

(ii) For the rope, calculate a value for $\frac{\text{load}}{\text{extension}}$.

$\frac{\text{load}}{\text{extension}}$ = ... [1]

(d) The student repeatedly pours small volumes of paint into the can.

When there is $2.5 \times 10^{-3}\,m^3$ of paint in the can, the total length of the rope is 1.70 m. This is where the rope reaches the limit of proportionality.

(i) Determine the mass of the paint in the can.

mass = ... [3]

(ii) The student continues to pour paint into the can until there is a volume of $5.0 \times 10^{-3}\,m^3$ in it. He notices that the elastic rope becomes harder to stretch after the limit of proportionality.

He plots a graph of the total length of the rope against the volume of paint added.

On Fig. 8.2, sketch the length-volume graph for the range 0 to $5.0 \times 10^{-3}\,m^3$.

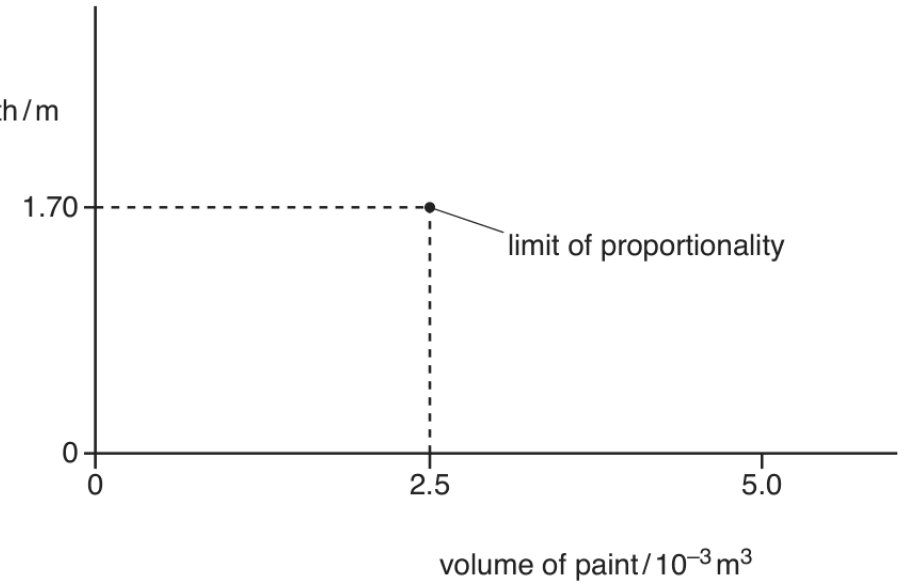


Fig. 8.2

[3]

(e) The student suddenly removes the can from the end of the stretched rope and, as it contracts, the rope jumps into the air.

State the energy change that is taking place in the rope as it contracts and jumps into the air.

.....

..... [2]

[Total: 15]

MARK SCHEME · 5054_w19_ms_22.pdf · source page 7 of 10 · Question 8

Question	Answer	Marks
8(a)	where / when extension is not directly proportional to load / tension / force (applied)	**C1**
	point where extension stops being directly proportional to load / tension / force (applied)	**A1**
8(b)(i)	the amount matter in a body **or** (a measure of) the inertia of a body	**B1**
8(b)(ii)	force (on a body)	**B1**
	in a gravitational field	**B1**
8(c)(i)	7.0 N **c.a.o.**	**B1**
8(c)(ii)	41 N / m **or** 0.41 N / cm	**B1**

MARK SCHEME · 5054_w19_ms_22.pdf · source page 8 of 10 · Question 8

Question	Answer	Marks
8(d)(i)	two separate approaches **I** / **II**	
8(d)(i)**I**	0.90	**C1**
	41×0.90 **or** 37 **or** 3.7 **or** $3.7 - 0.70$	**C1**
	3.0 kg	**A1**
	or	
8(d)(i)**II**	$1.7 - 0.97$ **or** 0.73	**C1**
	$41 \times (1.7 - 0.97)$ **or** 41×0.73 **or** 30	**C1**
	3.0 kg	**A1**
8(d)(ii)	straight line to marked point	**B1**
	0 < location of *y*-intercept < 1.70 m (intercept need not be labelled)	**B1**
	curve of changing / changed gradient beyond marked point	**B1**
8(e)	from elastic (potential energy) **or** strain (energy)	**B1**
	to kinetic (energy) **or** to gravitational (potential energy)	**B1**

QUESTION · 5054_s20_qp_11.pdf · source page 4 of 16

9 Four objects of different masses are on different planets.

The weight of each object on its planet is determined.

Which object is on the planet with the smallest gravitational field strength?

	mass / kg	weight / N
A	5	125
B	15	150
C	20	220
D	25	225

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 9

9	D	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 4 of 16

10 Which physical quantity does **not** change when a piece of copper is heated?

A density

B mass

C temperature

D volume

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 10

10	B	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 4 of 16

9 An object weighs 40 N on Earth, where the gravitational field strength is 10 N / kg. The object is taken to a planet where the gravitational field strength is 4.0 N / kg.

Which row is correct?

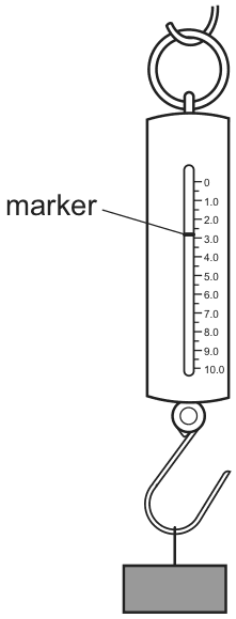
	mass of object on the planet / kg	weight of object on the planet / N
A	4.0	1.0
B	4.0	16
C	400	100
D	400	1600

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 9

9	B	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 4 of 16

6 A block is hung on a spring balance. The marker inside the balance is pulled down by the block.



What can the position of the marker be used to determine?

A the mass of the block in kg

B the mass of the block in N

C the moment caused by the block in N

D the weight of the block in kg

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 6

6	A	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 4 of 16

8 Which statement about mass and weight is correct?

A Mass is a scientific term that means the same as weight.

B The mass of an object on Earth is 10 times its weight.

C Weight is a scalar quantity, mass is a vector quantity.

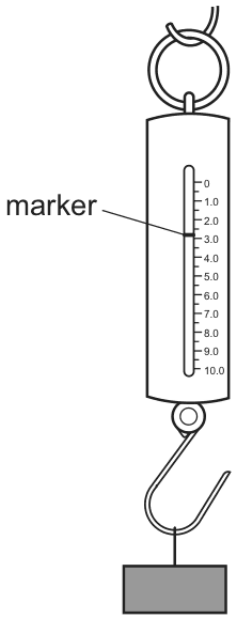
D Weight is the force of gravity pulling on a mass.

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 8

8	D	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 4 of 16

7 A block is hung on a spring balance. The marker inside the balance is pulled down by the block.



What can the position of the marker be used to determine?

A the mass of the block in kg

B the mass of the block in N

C the moment caused by the block in N

D the weight of the block in kg

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 7

7	A	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 5 of 16

8 There is an astronaut on a planet which has a gravitational field strength one fifth of the value on Earth.

He hangs samples of rock from a spring balance designed for use on Earth. Each sample has a mass of 0.1 kg.

The spring balance reads 1.0 kg.

How many samples of rock are there?

A 2 **B** 10 **C** 25 **D** 50

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 8

8	D	**1**

QUESTION · 5054_s21_qp_11.pdf · source page 4 of 16

6 Which row shows the mass and the weight of an object on the Earth's surface?

[gravitational field strength g = 10 N / kg]

	mass / kg	weight / N
A	2	0.20
B	2	10
C	5	5.0
D	5	50

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 4 of 16

6 The gravitational field strength in space is smaller than on the Earth's surface.

A rocket is used to launch a satellite from the Earth's surface into space.

How are the mass and the weight of the satellite affected as the satellite moves away from the surface of the Earth and into space?

A Both the mass and the weight are unaffected.

B The mass decreases and the weight decreases.

C The mass increases and the weight is unaffected.

D The mass is unaffected and the weight decreases.

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

3 Fig. 3.1 shows a small brick hanging from a newton meter.

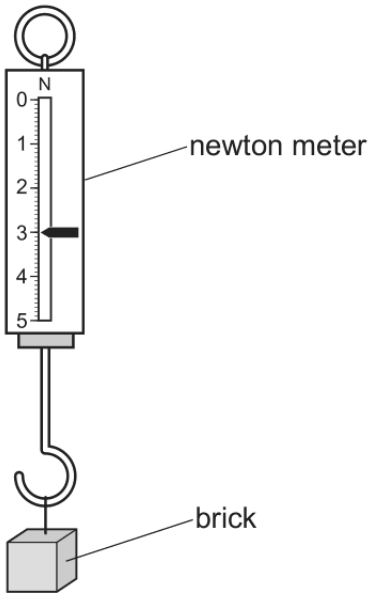


Fig. 3.1

The reading on the newton meter is 3.0 N.

(a) Describe how the reading on the newton meter is used to find the mass of the brick.

.....

..... [1]

(b) The same brick and newton meter are used in the apparatus shown in Fig. 3.2. The meter rule is pivoted at its centre and is balanced. The reading on the newton meter is not shown.

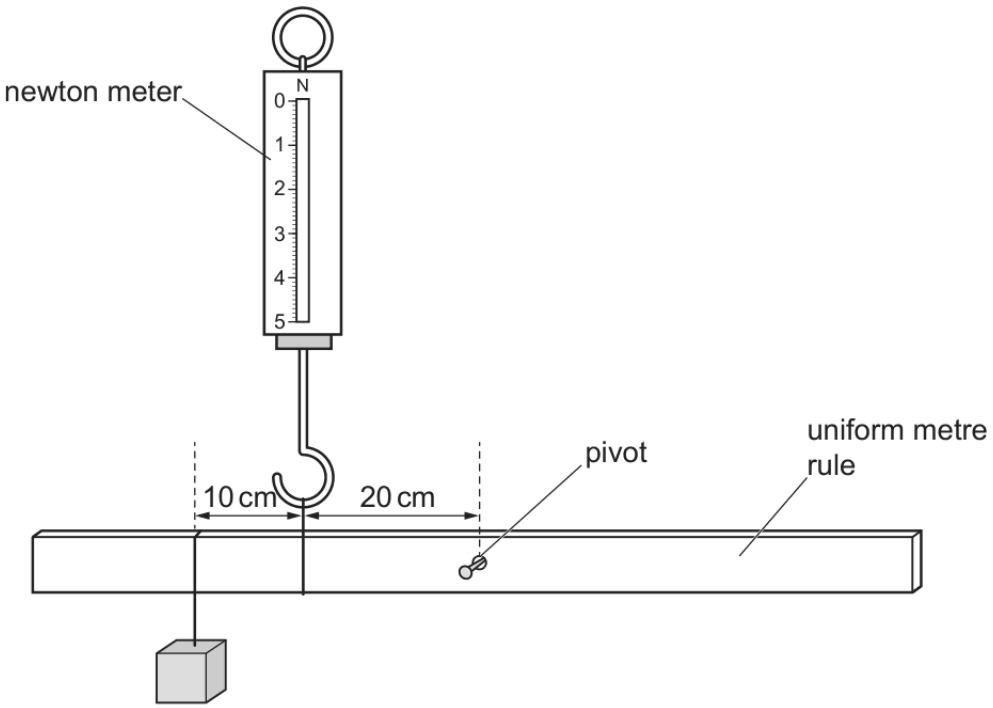


Fig. 3.2

(i) State the *principle of moments* for a body in equilibrium.

.....

.....

..... [1]

(ii) Determine the reading on the newton meter shown in Fig. 3.2.

reading = ... [2]

(c) A beaker of water is placed so that the brick is partly submerged in the water, as shown in Fig. 3.3. The apparatus is adjusted to keep the rule horizontal.

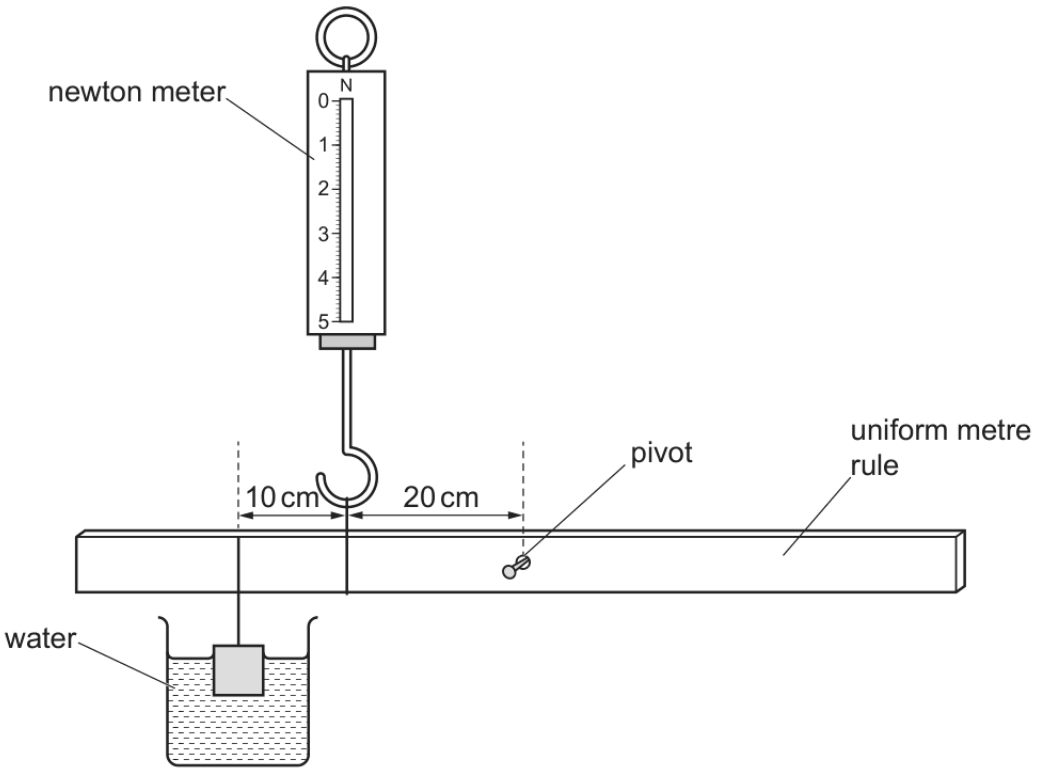


Fig. 3.3

Suggest why the reading on the newton meter is less than your answer in **(b)(ii)**.

.....

.....

..... [2]

[Total: 6]

Question	Answer	Marks
3(a)	(mass =) meter reading / *g* meter reading / gravitational field (strength meter reading / acceleration due to gravity	**B1**
3(b)(i)	(sum of) clockwise moments = (sum of) anticlockwise moments (about any point)	**B1**

Question	Answer	Marks
3(b)(ii)	any attempt at using moments, e.g. $3 \times 30 = F \times 20$	**C1**
	4.5 N	**A1**
3(c)	force upwards (on brick due to water) **or** force due to water (on brick) **or** force downwards (on brick) is reduced / counteracted (by water) **or** force exerted by brick (on rule) reduced **or** force / tension in string reduced	**B1**
	(force caused by) pressure (of water) **or** reduced anticlockwise moment	**B1**

2 Fig. 2.1 shows a man of mass 80 kg standing in a lift (elevator).

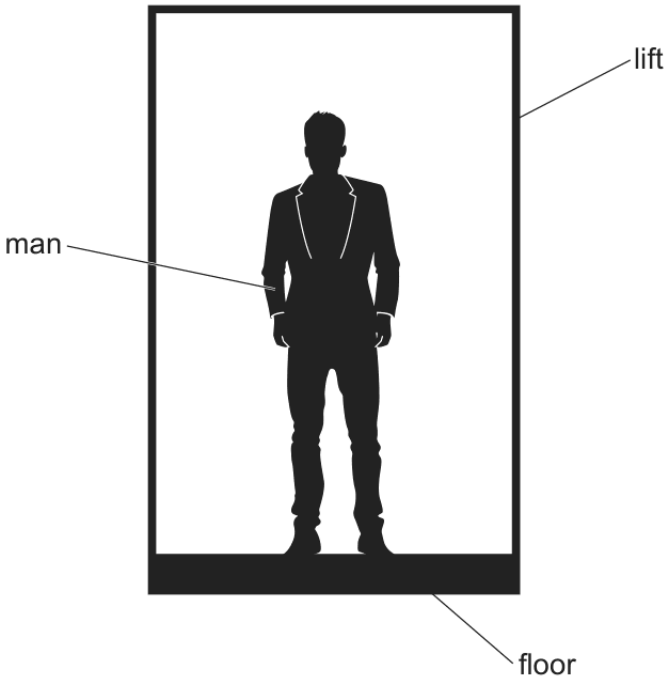


Fig. 2.1

The gravitational field strength g is 10 N / kg.

(a) Calculate the weight of the man.

weight = ... [1]

(b) The lift accelerates upwards uniformly at 0.50 m / s^2.

Calculate:

(i) the resultant upward force on the man

resultant force = ... [2]

(ii) the force exerted on the man by the floor of the lift.

force = ... [1]

(c) The lift accelerates upwards uniformly from rest for 4.0 s and then decelerates uniformly to rest in 4.0 s. The total distance travelled is 8.0 m.

On Fig. 2.2, sketch the distance-time graph for this journey.

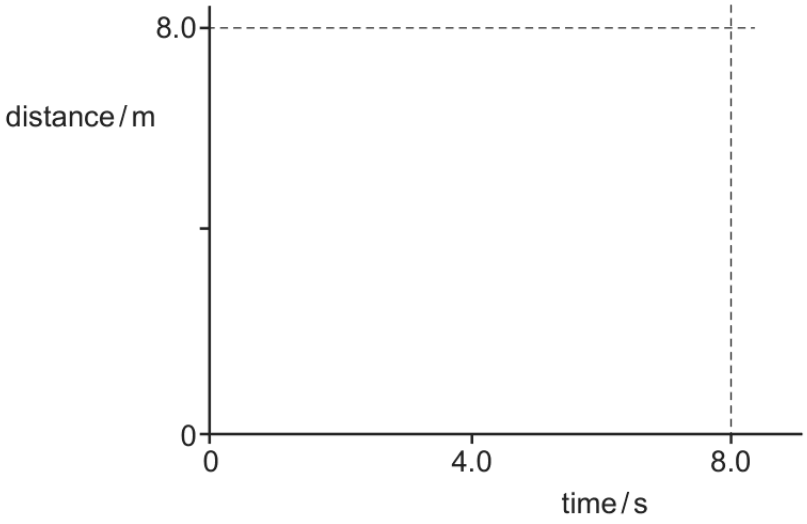


Fig. 2.2

[3]

[Total: 7]

Question	Answer	Marks
2(a)	800 N **c.a.o.**	**B1**
2(b)(i)	(F =) ma **or** 80×0.50	**C1**
	40 N	**A1**
2(b)(ii)	840 N	**B1**
2(c)	increasing gradient from origin to t = 4.0 s	**B1**
	decreasing gradient from end of first curve	**B1**
	line horizontal at origin **or** at t = 8.0 s (and beyond)	**B1**

QUESTION · 5054_w21_qp_12.pdf · source page 3 of 16

6 An object in a laboratory is used to determine the gravitational field strength g.

Which quantity is equal to g?

A the weight of the object

B the mass of the object

C the $\frac{\text{weight}}{\text{mass}}$ of the object

D the $\frac{\text{mass}}{\text{weight}}$ of the object

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 6

6	C	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a jug of liquid and an empty measuring cylinder that is on an electronic balance.

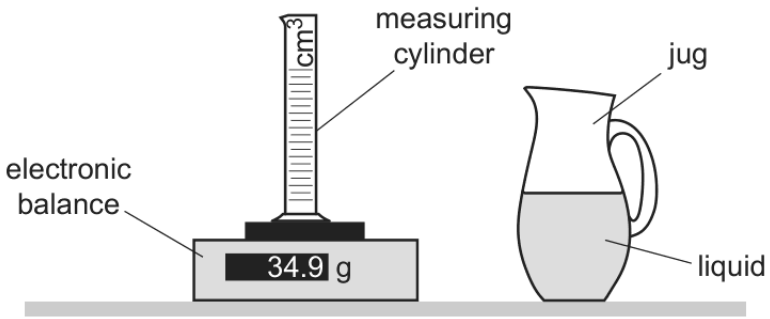


Fig. 1.1

(a) The electronic balance uses the weight of the measuring cylinder to determine its mass.

Weight and mass have different units.

State **two** other ways in which weight differs from mass.

1. .....

.....

2. .....

.....

[2]

(b) Describe how to determine the density of the liquid using the apparatus shown in Fig. 1.1.

.....

.....

.....

.....

.....

..... [4]

(c) The density of the liquid is 780 kg / m^3.

Determine the volume of 0.65 kg of the liquid.

volume = ... [2]

[Total: 8]

Question	Answer	Marks
1(a)	any **two** from: weight / it is a vector / has a direction weight / it is a force weight / it depends on location / gravitational field strength weight / it is measured with a spring balance (not a beam balance) / electronic balance / top-pan balance / newton meter	**B2**
1(b)	(pour liquid into) measuring cylinder **and** read / find the volume **or** zero the reading on the balance	**B1**
	take reading on balance **or** find the mass (of full measuring cylinder) **or** (pour liquid into) measuring cylinder **and** read / find the volume	**B1**
	subtract original reading / 34.9 g (from reading on balance) **or** take reading on balance **or** find the mass (of full measuring cylinder)	**B1**
	divide difference between readings by volume **or** divide (final) reading on balance by volume	**B1**
1(c)	(ρ =) 0.65 / 780	**C1**
	8.3×10^{-4} m^3 **or** 830 cm^3	**A1**

8 The turning effect of a force is measured by its moment.

(a) Fig. 8.1 shows a force *F* acting on an object at point P. The object is free to rotate about an axis at X that is perpendicular to the page.

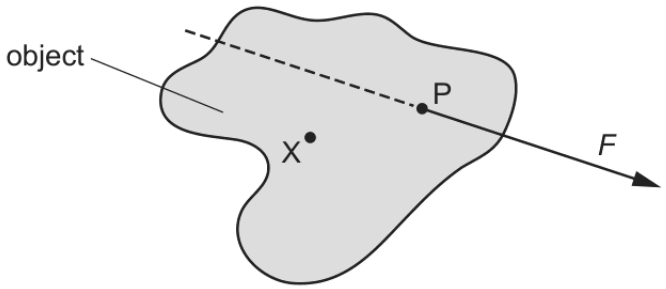


Fig. 8.1

Write down an expression for the moment of *F* about the axis at X. Draw on Fig. 8.1 to show what is meant by any other term used in your expression.

.....

..... [2]

(b) **(i)** State the principle of moments.

.....

.....

..... [2]

(ii) Describe an experiment to verify the principle of moments. Include a diagram to help the description.

.....

.....

.....

.....

.....

..... [3]

(c) A worker carries a ladder on his shoulder. His shoulder acts as a pivot. Fig. 8.2 shows that the ladder is horizontal.

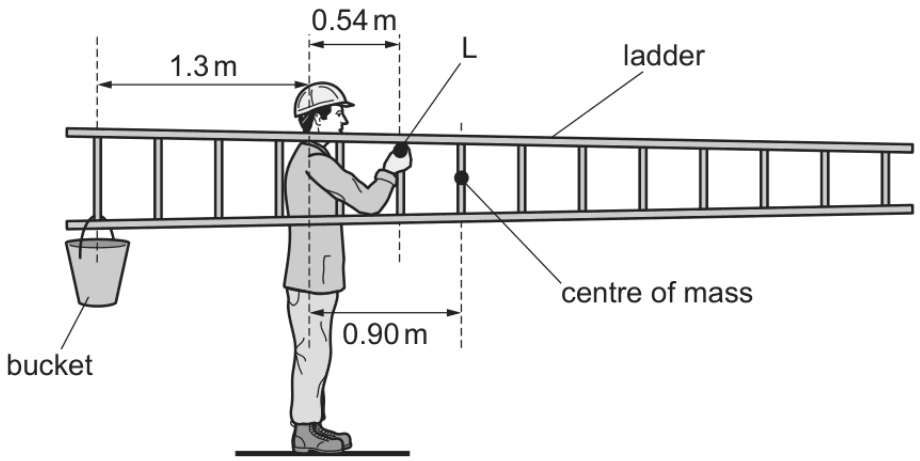


Fig. 8.2

The ladder is wider at one end than at the other end. The mass of the ladder is 8.0 kg.

(i) The gravitational field strength is 10 N / kg.

Calculate the weight of the ladder.

weight = ... [1]

(ii) State what is meant by *centre of mass*.

.....

..... [1]

(iii) The centre of mass of the ladder is not halfway along its length.

State what this shows about the ladder.

.....

..... [1]

(iv) The centre of mass of the ladder is a horizontal distance of 0.90 m from the worker's shoulder.

Calculate the moment about the worker's shoulder of the weight of the ladder.

moment = ... [2]

(v) A bucket of weight 87 N is suspended from the ladder at a horizontal distance of 1.3 m from the worker's shoulder.

The worker keeps the ladder horizontal by exerting a vertical force at point L. L is a horizontal distance of 0.54 m from his shoulder.

Determine the size and direction of the force exerted at L.

size of force = ...

direction = ...
[3]

[Total: 15]

MARK SCHEME · 5054_w21_ms_22.pdf · source page 9 of 11 · Question 8

Question	Answer	Marks
8(a)	$F \times$ distance / displacement	**M1**
	perpendicular distance to line of action of force shown on Fig. 7.1	**A1**
8(b)(i)	for an object in equilibrium	**B1**
	(sum of) clockwise moments is equal to (sum of) anticlockwise moments **or** no resultant moment	**B1**
8(b)(ii)	equipment shown in diagram (e.g. rule, two loads, pivot)	**B1**
	what is done to achieve balance (e.g. balance rule on pivot, one load each side of pivot)	**B1**
	how principle is verified (e.g. calculate two opposite moments and find they are equal)	**B1**
8(c)(i)	80 N	**B1**
8(c)(ii)	(point) where the (whole) weight of an object acts / seems to act **or** point where the object balances	**B1**

MARK SCHEME · 5054_w21_ms_22.pdf · source page 10 of 11 · Question 8

Question	Answer	Marks
8(c)(iii)	mass not uniformly distributed along length **or** more mass at the one end	**B1**
8(c)(iv)	(Γ=) 80×0.90	**C1**
	72 N m	**A1**
8(c)(v)	moment / 0.54 **or** 209 (N) **or** 41.1 / 0.54	**C1**
	76 N	**A1**
	downwards	**B1**

QUESTION · 5054_s22_qp_11.pdf · source page 3 of 16

4 The mass m of an object is related to its weight W and the gravitational field strength g.

Which equation shows the relationship?

A $m = W \times g$ **B** $m = \frac{W}{g}$ **C** $m = \frac{g}{W}$ **D** $m = W - g$

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a waterskier pulled horizontally across the water surface.

Fig. 1.1

The mass of the waterskier is 60 kg. The gravitational field strength g is 10 N/kg.

(a) Calculate the weight of the waterskier.

weight = ... [1]

(b) Mass and weight have different units.

State **two** other differences between mass and weight.

.....

.....

.....

..... [2]

(c) A boat pulls the waterskier to the right with a horizontal force of 50 N. The waterskier travels at a constant speed.

(i) Explain, in terms of the horizontal forces acting, why the speed is constant.

.....

..... [2]

(ii) The horizontal force acting to the right increases from 50 N to 70 N. The sizes of any other forces are unchanged.

Calculate the acceleration of the waterskier.

acceleration = ... [3]

[Total: 8]

Question	Answer	Marks
1(a)	600 N	**B1**
1(b)	ANY 2 separate bullet points from: • weight is a vector **/** weight has direction / mass has no direction / mass is a scalar • mass is the amount of matter / substance • weight is a force / weight is mass × gravitational field strength • weight varies with position (of object) / gravitational field (strength) / mass does not vary with position • mass resists change of motion / has inertia	**B2**
1(c)(i)	friction (with water) **or** drag **or** air / water resistance mentioned	**B1**
	(forces) balance / cancel **or** no resultant force	**B1**
1(c)(ii)	(resultant force) 20 (N) or 70 – 50 seen	**C1**
	(*a* =) *F* / *m* algebraic or numerical in any form	**C1**
	0.33 m / s^2	**A1**

QUESTION · 5054_w22_qp_11.pdf · source page 4 of 16

9 X is the result of Y divided by Z.

What are X, Y and Z?

	X	Y	Z
A	mass	weight	gravitational field strength
B	weight	mass	gravitational field strength
C	mass	gravitational field strength	weight
D	weight	gravitational field strength	mass

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 9

9	A	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 4 of 16

9 A student writes two statements about mass and weight.

1 Mass is the property of a body which resists change in its state of rest or motion.

2 Weight is the amount of substance in a body.

Which statements are correct?

A statement 1 only

B statement 2 only

C statement 1 and statement 2

D neither statement 1 nor statement 2

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 9

9	A	**1**

QUESTION · 5054_s23_qp_11.pdf · source page 4 of 20

5 The gravitational field strength on Earth is 9.8 N / kg.

The gravitational field strength on Mars is 3.7 N / kg.

The difference between the weight of an object on Earth and the weight of the same object on Mars is 25 N.

What is the mass of the object?

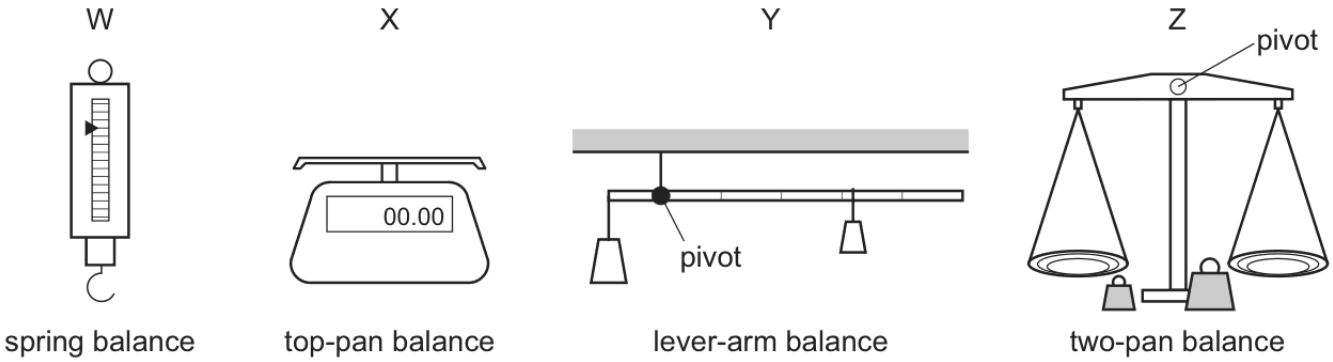
A 1.9 kg **B** 2.6 kg **C** 4.1 kg **D** 6.8 kg

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 5

5	C	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 3 of 16

5 The diagrams show four balances, W, X, Y and Z.



spring balance | top-pan balance | lever-arm balance | two-pan balance

The scales of all the balances are calibrated on the Earth to measure mass.

Which balances also measure mass correctly when used on the Moon?

A W and X **B** W and Z **C** X and Y **D** Y and Z

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 5

5	D	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 4 of 16

6 A spacecraft carries a probe to Mars where the gravitational field strength is 3.7 N / kg. The weight of the probe on Mars is 370 N.

The gravitational field strength on Earth is 9.8 N / kg.

What are the mass and the weight of the probe on Earth?

	mass / kg	weight / N
A	100	370
B	100	980
C	370	98
D	370	3700

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 6

6	B	**1**

3 A plane, irregular lamina has a mass of 50 g. It hangs from a nail that passes through a small hole H near to the edge of the lamina.

The nail acts as a pivot and the lamina can swing about it. The lamina is held in the position shown in Fig. 3.1, a small distance above a horizontal bench.

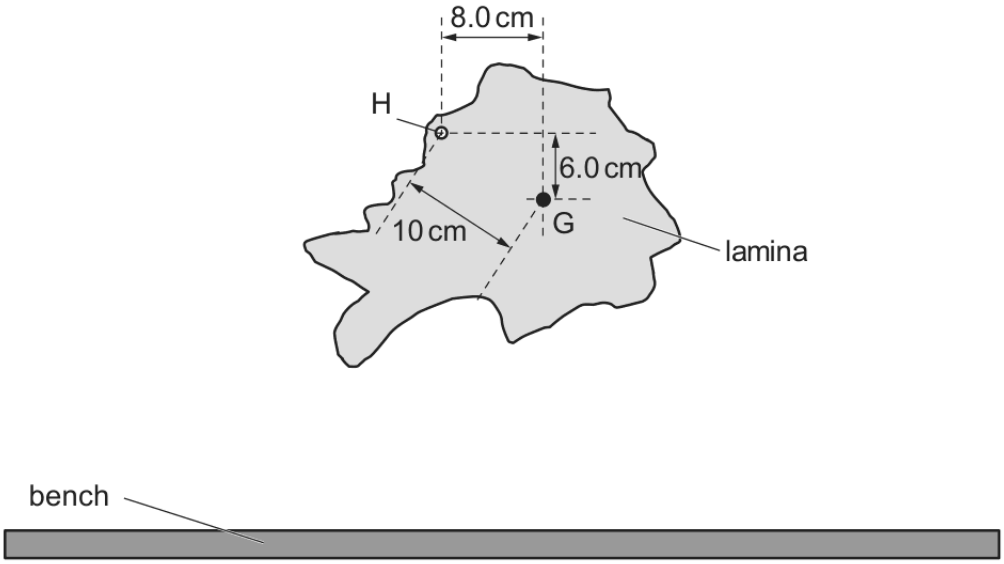


Fig. 3.1 (not to scale)

The centre of gravity of the lamina is at G.

(a) The weight of the lamina is a force that acts downwards.

(i) Explain why the lamina experiences this downward force.

.....

..... [1]

(ii) Calculate the weight of the lamina on Earth.

weight = ... N [2]

(b) The weight produces a moment about the pivot through H.

(i) Using Fig. 3.1, calculate the moment of the weight about the pivot.

moment = ... N m [2]

(ii) The lamina is now released from the position shown in Fig. 3.1.

Describe what happens to the lamina from the time it is released until it stops moving.

.....

.....

.....

..... [3]

[Total: 8]

Question	Answer	Marks
3(a)(i)	it / lamina is in the gravitational field (of Earth)	**B1**
3(a)(ii)	($F =$) mg **or** 0.050×9.8 **or** 50×9.8 **or** 490	**C1**
	0.49 (N)	**A1**
3(b)(i)	($M =$) $F \times x_{\perp r}$ **or** 0.49×8.0 **or** 0.49×0.080	**C1**
	0.039 (N m)	**A1**
3(b)(ii)	it / lamina moves clockwise (about the pivot)	**B1**
	it speeds up **and** then slows down **or** it oscillates (with decreasing amplitude) **or** overshoots	**B1**
	(stops with) G vertically below H	**B1**

8 Table 8.1 shows data about three planets, Mercury, Venus and Earth.

Table 8.1

	time for one orbit of Sun / days	distance from Sun / km	average density / kg / m^3	gravitational field strength at surface / N / kg
Mercury	88	5.8×10^7	5400	3.7
Venus	220	1.1×10^8	5200	8.9
Earth	365	1.5×10^8	5500	9.8

(a) Fig. 8.1 shows these planets in alignment with the Sun. They rotate around the Sun in the direction shown.

Mark and label on Fig. 8.1 the positions of the three planets 110 days after the position shown in Fig. 8.1.

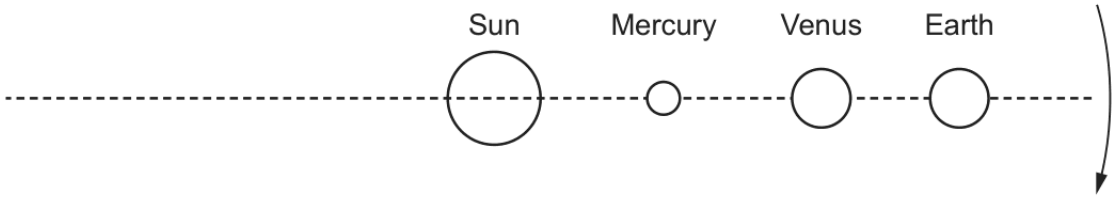


Fig. 8.1 (not to scale)

[3]

(b) (i) Each of the three planets has a similar average density.

Suggest why the gravitational field strength at the surface of Mercury is much smaller than at the surface of Venus.

.....

..... [1]

(ii) An object has a weight of 37 N on the surface of Mercury.

Calculate its weight on the surface of the Earth. Show your working.

weight = ... N [2]

(c) State what is meant by a 'moon'.

..... [1]

(d) The Sun will eventually run out of hydrogen.

Describe what happens to the Sun when the hydrogen runs out.

.....

.....

.....

.....

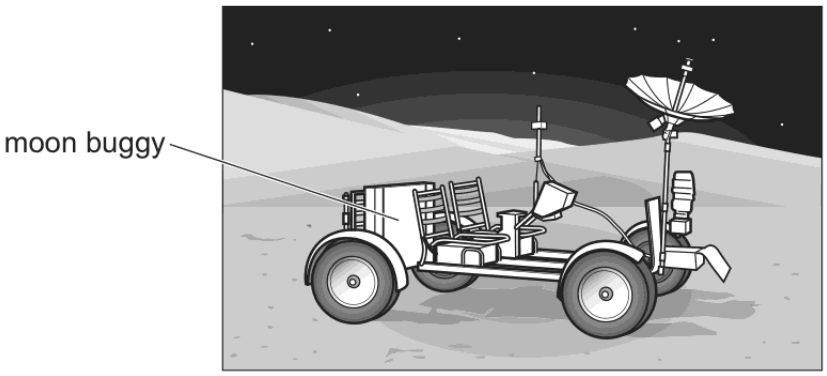
..... [3]

[Total: 10]

Question	Answer	Marks
8(a)	Venus on left of Sun on dashed line	**B1**
	Earth approximately 30% round orbit (106°)	**B1**
	Mercury (1=) 1/4 round orbit i.e. approx. vertically below Sun	**B1**
8(b)(i)	smaller mass (of mercury)	**B1**
8(b)(ii)	(mass =) 10 (kg) clearly seen	**C1**
	98 (N)	**A1**
8(c)	a (large, natural) object that orbits a planet	**B1**
8(d)	becomes a red giant / increase in size	**B1**
	forms a planetary nebula	**B1**
	becomes a white dwarf	**B1**

QUESTION · 5054_s24_qp_12.pdf · source page 3 of 16

6 The diagram shows a moon buggy used by astronauts.



The mass of the moon buggy on the Earth is 210 kg. The gravitational field strength on the Moon is $\frac{1}{6}$ of that on the Earth.

What is the weight of the moon buggy on the Moon?

A zero **B** 35 N **C** 210 N **D** 340 N

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 2 of 16

2 Which statement about mass is correct?

A Mass is a scalar and its unit is the kilogram.

B Mass is a scalar and its unit is the newton.

C Mass is a vector and its unit is the kilogram.

D Mass is a vector and its unit is the newton.

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 2

2	A	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 4 of 16

8 A car has a weight of 8000 N.

What is the mass of the car?

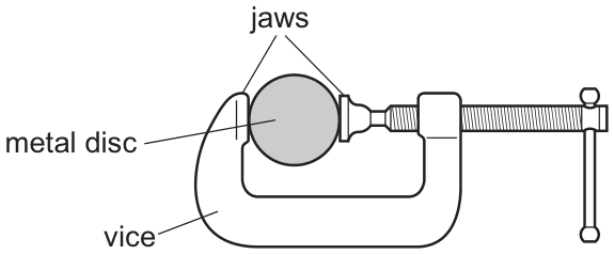
A 810 kg **B** 820 kg **C** 8000 kg **D** 78 000 kg

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 8

8	B	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 3 of 16

5 A metal disc is squeezed in the jaws of a vice.



Which property of the metal disc is unchanged when it is squeezed?

A mass

B radius

C distance between its atoms

D forces between its atoms

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 5

5	A	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 3 of 16

6 The gravitational field strength in space is smaller than on the Earth's surface.

How are the mass and the weight of a satellite affected as the satellite moves away from the surface of the Earth and into space?

A Both the mass and the weight are unaffected.

B The mass decreases and the weight decreases.

C The mass increases and the weight is unaffected.

D The mass is unaffected and the weight decreases.

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 5 of 20

10 The mass of object P is greater than the mass of object Q.

The objects contain different amounts of matter and have a different resistance to change of motion.

Which row is correct?

	greater amount of matter	greater resistance to change of motion
A	P	P
B	P	Q
C	Q	P
D	Q	Q

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 10

10	A	**1**

2 **(a)** State the principle of moments.

.....

.....

..... [2]

(b) An airline passenger wishes to check the weight of a suitcase that he carries as hand luggage onto an aeroplane.

He uses a uniform plank of wood, pivoted at its centre, as shown in Fig. 2.1.

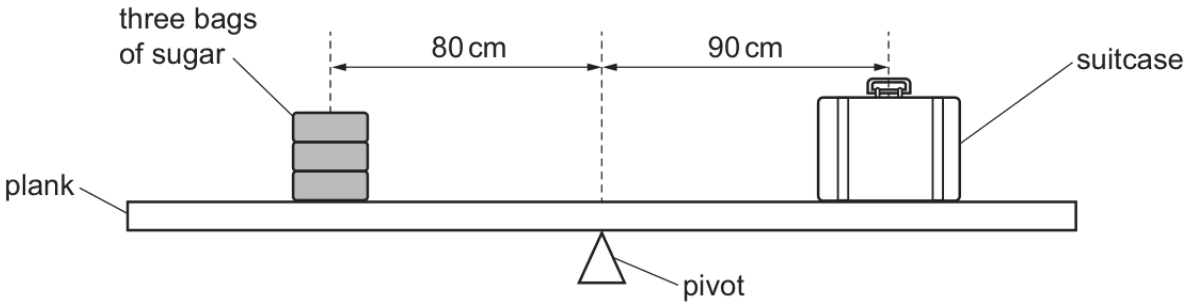


Fig. 2.1

The plank is balanced with the suitcase on one side of the pivot and three bags of sugar, each of mass 2.0 kg, on the other side.

The distances are shown on Fig. 2.1.

(i) Calculate the weight of **one** bag of sugar.

weight = ... N [1]

(ii) The maximum weight of a suitcase that can be carried onto the aeroplane is 67 N.

Determine whether the weight of the suitcase exceeds the maximum allowed.

Show a calculation in your answer.

.....

..... [2]

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(c) A tall bus is tested for stability.

Fig. 2.2 shows the bus on a slope.

The centre of gravity of the bus is marked.

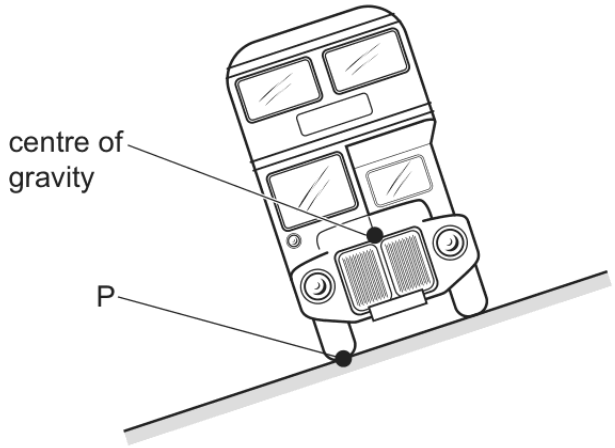

Fig. 2.2

(i) State what is meant by the 'centre of gravity' of an object.

.....

..... [1]

(ii) When the slope is made very steep, the bus falls over by rotating about point P.

Explain why the bus falls over.

.....

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
2(a)	(total) clockwise moments = anticlockwise moments	**M1**
	in equilibrium	**A1**
2(b)(i)	19.6 or 20 N	**B1**
2(b)(ii)	80 × 3 × **(b)(i)** = 90 × weight	**B1**
	52 (N) **and** weight of suitcase does not exceed maximum	**B1**
2(c)(i)	(place) where weight of object (can be taken to) acts	**B1**
2(c)(ii)	centre of gravity falls outside base / to the left of P	**B1**
	weight causes anticlockwise moment / turning effect	**B1**

2 In a factory, an electric motor is used to lift the boxes shown in Fig. 2.1.

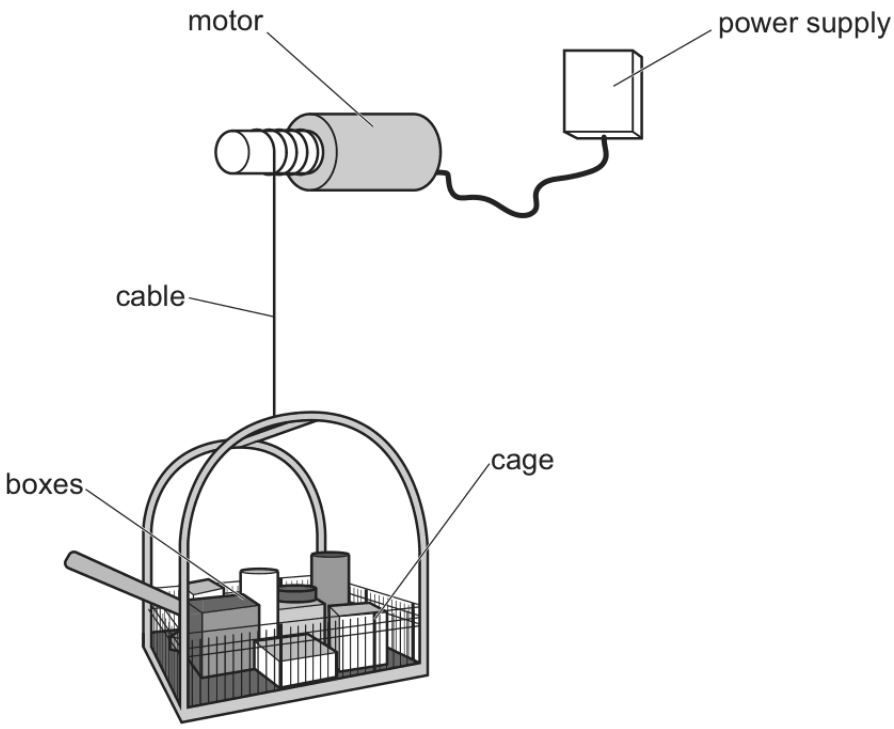

Fig. 2.1

The boxes are placed in a cage and the motor winds a cable around a cylinder to lift the cage and boxes.

The total mass of the boxes is 37 kg and they are lifted vertically through 13 m to a storage area.

(a) Calculate:

(i) the total weight of the boxes

weight = ... N [1]

(ii) the useful work done in lifting the boxes to the storage area.

useful work done = ... J [2]

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(b) The e.m.f. of the power supply for the motor is 230 V. The current in the motor is 5.6 A.

It takes 11 s for the motor to lift the cage and boxes to the storage area.

Calculate:

(i) the total energy transferred electrically to the motor in 11 s

energy = ... J [2]

(ii) the percentage efficiency of the lifting process.

efficiency = ...% [1]

(c) Some of the energy transferred by the motor during the lifting process is not transferred usefully.

State **two** reasons why the efficiency of the lifting process is less than 100%.

1 .....

.....

2 .....

.....

[2]

[Total: 8]

MARK SCHEME · 5054_w25_ms_21.pdf · source page 8 of 14 · Question 2

Question	Answer	Marks
2(a)(i)	360 (N)	**B1**
2(a)(ii)	(work done =) Fx **or** 360×13	**C1**
	4700 (J)	**A1**
2(b)(i)	(energy =) VIt **or** $230 \times 5.6 \times 11$	**C1**
	1.4×10^4 (J)	**A1**
2(b)(ii)	0.33 **or** 33%	**B1**

MARK SCHEME · 5054_w25_ms_21.pdf · source page 9 of 14 · Question 2

Question	Answer	Marks
2(c)	any **two** from: (work done / GPE) to lift the cage / cable (energy transferred) thermally in motor (coils) (work done) overcoming friction / air resistance	**B2**

QUESTION · 5054_w25_qp_12.pdf · source page 2 of 16

4 The acceleration of free fall on the surface of the Moon is $\frac{1}{6}$ the acceleration of free fall on the surface of the Earth.

What is the approximate weight of a 1 kg mass on the Moon?

A 0.98 N **B** 1.6 N **C** 9.8 N **D** 16 N

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 3 of 16

6 Which row shows a definition of the mass of an object and an instrument used to measure mass?

	definition of mass	measuring instrument
A	force experienced by an object	electronic balance
B	force experienced by an object	force meter
C	quantity of matter in an object	electronic balance
D	quantity of matter in an object	force meter

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 6

6	C	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 4 of 16

6 An object of mass 5.4 kg has a weight of 42 N on planet P.

The object is moved to Earth.

Which statement is correct?

A The gravitational field strength on planet P is 7.8 N / kg.

B The mass of the object increases.

C The weight of the object decreases.

D The weight of the object on Earth is 42 N.

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 6

6	A	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 2 of 16

3 Which statement about mass is correct?

A Mass is density divided by volume.

B Mass is the volume of space occupied by a body.

C Mass is weight multiplied by the gravitational field strength.

D Mass resists a change in motion.

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 2 of 16

4 Which unit is a unit of weight?

A kg **B** kg / m **C** N **D** N m

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 4

4	C	**1**

Chapter 1 Motion, forces and energy

1.4 Density

92 entries: 57 Paper 1 questions, 35 Paper 2 questions

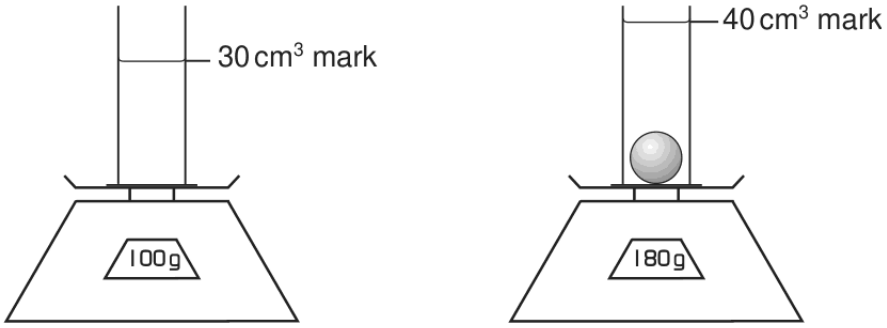
Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 3 of 20

5 A measuring cylinder containing some water stands on a scale pan. A solid ball is lowered into the water.

The water level rises from the 30 cm^3 mark to the 40 cm^3 mark.

The scale reading increases from 100 g to 180 g.



What is the density of the material of the ball?

A 2.0 g / cm^3 **B** 4.5 g / cm^3 **C** 8.0 g / cm^3 **D** 18 g / cm^3

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 5

5	**C**

10 Fig. 10.1 is a full-scale drawing of a small aluminium bar.

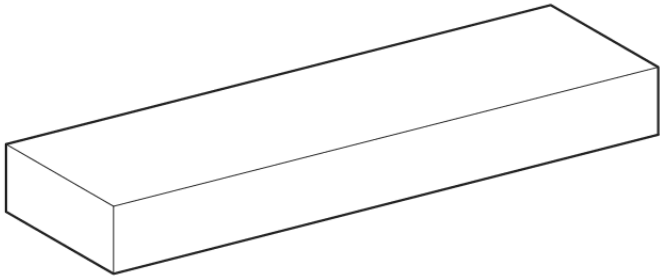


Fig. 10.1

(a) Describe how the density of aluminium may be determined using the bar. In your account

(i) describe the readings to be taken,

(ii) state the instruments used to make the readings,

(iii) describe how the density is calculated from the readings.

[6]

(b) The bar is placed in a small furnace.
Fig. 10.2 shows how the temperature of the bar varies with time *t*.

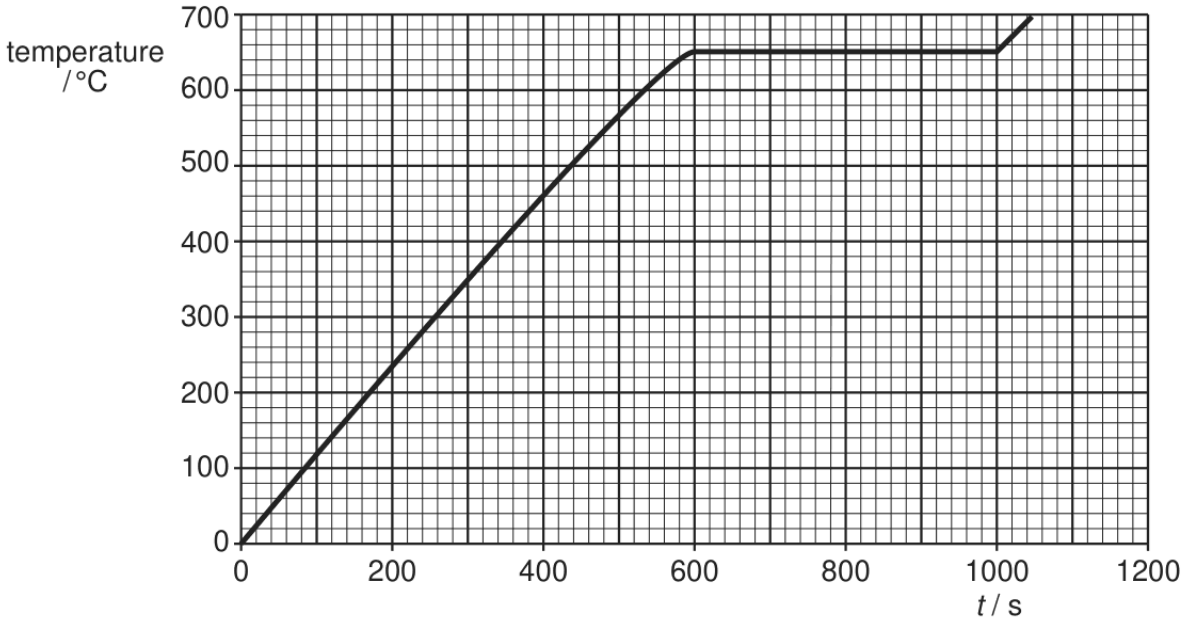


Fig. 10.2

(i) State what happens to the bar between $t = 600$ s and 1000 s.

(ii) Explain what effect the energy supplied to the bar has on its molecules between $t = 0$ and 1000 s.

(iii) The mass of the bar is 0.030 kg and the specific heat capacity of aluminium is 880 J/(kg °C). Calculate the energy supplied to the bar between $t = 0$ and 600 s.

(iv) Between $t = 600$ s and 1000 s the furnace supplies 30 joules of energy per second to the bar. Calculate the specific latent heat of fusion of aluminium.

[9]

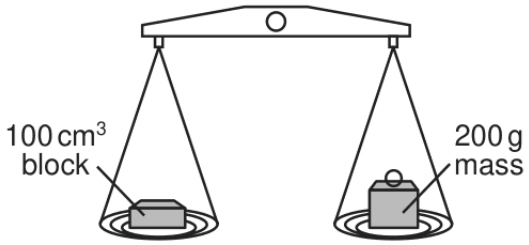
10	**(a)**		mass of bar (measured)	M1
			using (top-pan) balance / spring balance / scales etc.	A1
			length, breadth and height measured	
			or volume water + bar measured **or** displacement can (full) with water	M1
			volume = length x breadth x height	
			or subtract volume water alone **or** collect water displaced	A1
			using ruler / calipers / micrometer **or** measuring cylinder	A1
			density = mass / volume	B1
	(b)	**(i)**	melts / changes state / becomes liquid	B1
		(ii)	(initial) increase in vibration / K.E. of molecules (to 600s)	B1
			then later / after 600s or on melting	
			bonds broken (accept molecules break free / overcome attraction / not fixed in place)	B1
		(iii)	E = mc (Δ)T algebraic form seen	C1
			645 – 655 (°C) seen)	C1
			17 160 J (allow 1700, 17200, 20000)	A1
		(iv)	30*400 or 12 000 (J) seen)	C1
			E = mL any algebraic form seen **or** 12 000/0.3	C1
			40 000 J/kg	A1
			Total	**[15]**

QUESTION · 5054_w04_qp_1.pdf · source page 5 of 20

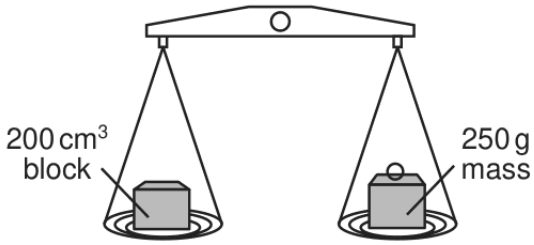
7 Four blocks, each made from a different material, are placed on scales and balanced as shown in the diagrams below.

In which diagram does the block have the greatest density?

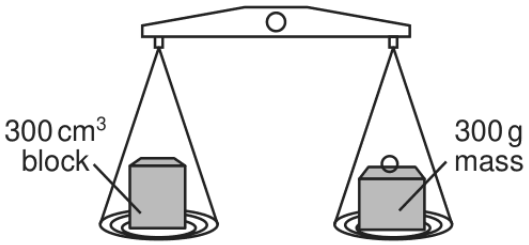
A



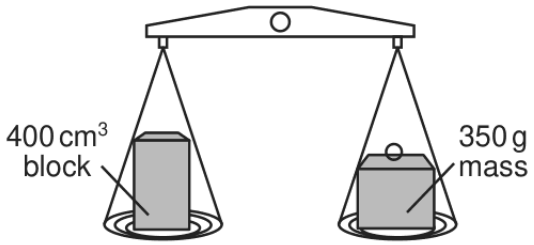
B



C



D



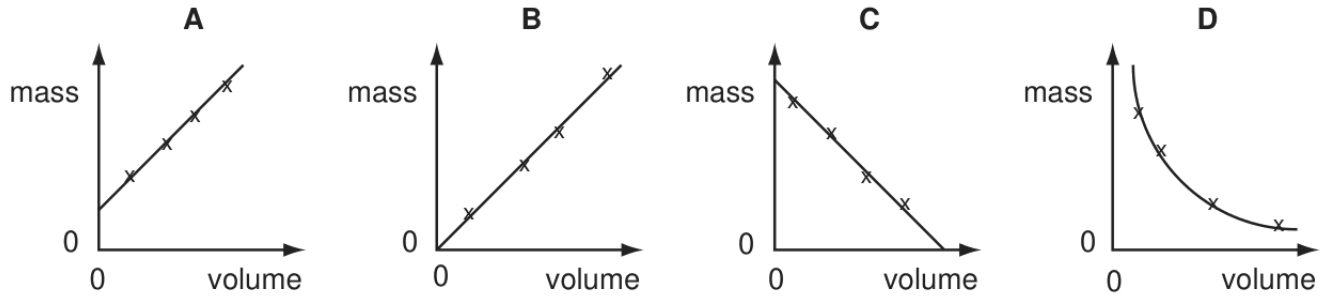
MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 7

7	**A**

QUESTION · 5054_s05_qp_1.pdf · source page 3 of 20

5 Some students measure the masses and the volumes of different sized samples of a type of wood.

Which graph shows their results?



MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 5

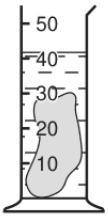
5 **B**

QUESTION · 5054_w05_qp_1.pdf · source page 4 of 20

7 A student is trying to find the density of a stone, but he has mixed up the instruction cards.

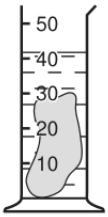
Find the mass of the stone using a mass balance

Card 1



Read the new level of the liquid in the measuring cylinder

Card 2

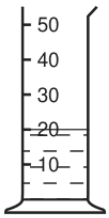


Put the stone into the liquid

Card 3

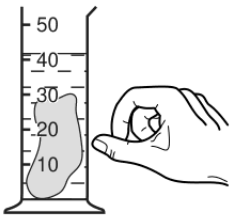
Divide the mass of the stone by its volume to find the density

Card 4



Put some liquid into a measuring cylinder and read the level of the liquid

Card 5



Flick the cylinder to get rid of any air bubbles

Card 6

Subtract the original volume of liquid from the volume of liquid plus the stone

Card 7

What order should the cards be in?

A 5 → 3 → 6 → 2 → 1 → 4 → 7

B 1 → 5 → 3 → 6 → 2 → 7 → 4

C 5 → 6 → 3 → 2 → 1 → 7 → 4

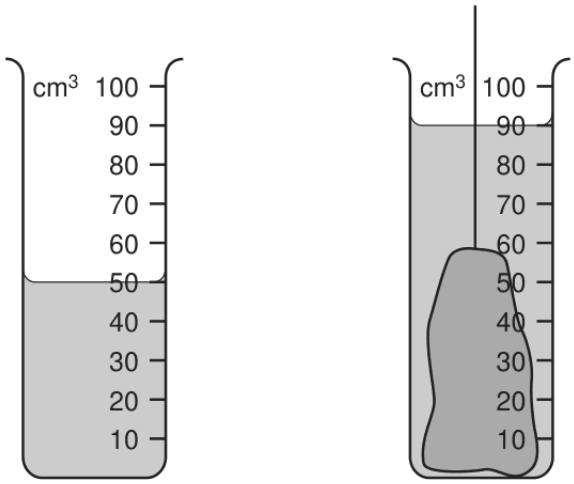
D 1 → 4 → 5 → 3 → 6 → 2 → 7

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 7

7 **B**

QUESTION · 5054_s06_qp_1.pdf · source page 4 of 20

7 An object of mass 100 g is immersed in water as shown in the diagram.



What is the density of the material from which the object is made?

A 0.4 g / cm^3

B 0.9 g / cm^3

C 1.1 g / cm^3

D 2.5 g / cm^3

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 7

7	**D**

Section B

Answer **two** questions from this section.

Use the separate answer sheets available from the Supervisor.

9 A student has an open tank for storing water outside her house. The tank is black and is in direct sunlight.

She notices that the level of water inside the tank slowly decreases as water evaporates.

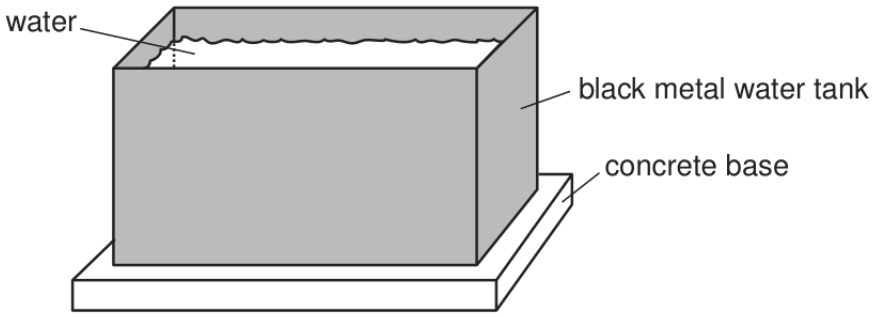


Fig. 9.1

She investigates whether some changes, shown in the table, will affect the amount of water evaporated each day.

change 1	a heater is used to increase the temperature of the water
change 2	a sheet of plastic is used to cover half the surface of the water
change 3	a fan is used to blow air over the top of the tank
change 4	the outside of the tank is painted white

Only one change is made at a time. All other factors are kept constant.

(a) State whether each change will increase, leave unchanged or decrease the amount of water evaporated each day. Explain each of your answers. [8]

(b) She notices that the water level in the tank falls by 0.005 m in 40 000 s.

The cross-sectional area of the tank is 3.0 m^2. The density of water is 1000 kg/m^3.

The specific latent heat of vaporisation of water is 2.2×10^6 J/kg.

Calculate

(i) the volume of water that evaporates, [1]

(ii) the mass of water that evaporates, [2]

(iii) the energy required to evaporate the water, [2]

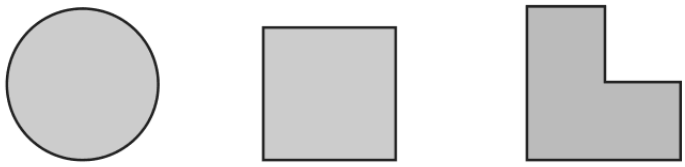
(iv) the average rate at which energy is used to evaporate the water. [2]

Section B

9	(a)	**change 1** increases evaporation	B1	
		water molecules have more K.E./move faster/more have enough energy	B1	
		change 2 decreases evaporation	B1	
		less surface for molecules to escape (through)	B1	
		change 3 increases evaporation	B1	
		liquid molecules leaving surface removed by collisions with air molecules	B1	
		change 4 decreases evaporation	B1	
		heat/light/infra-red reflected by white surface **or** tank cooler	B1	**[8]**
	(b)	**(i)** $0.015\ m^3$	B1	
		(ii) M = D. V in any form	C1	
		15 kg (ecf **(i)**)	A1	
		(iii) m .L	C1	
		3.3×10^7 J (ecf **(ii)**)	A1	
		(iv) energy/time in any form	C1	
		825 J/s **or** W (ecf **(iii)**)	A1	**[7] (15)**

QUESTION · 5054_w06_qp_1.pdf · source page 3 of 16

6 Three objects are cut from the same sheet of steel. They are different shapes but they all have the same mass.



disc square L-shape

Which object has the greatest density?

A the disc

B the L-shape

C the square

D they all have the same density

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 6

6	**D**

QUESTION · 5054_s07_qp_1.pdf · source page 3 of 20

6 A student does an experiment to estimate the density of an irregularly-shaped stone.

Which items of equipment are needed?

A a balance and a measuring cylinder containing water

B a balance and a ruler

C a ruler and a measuring cylinder containing water

D only a measuring cylinder containing water

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 6

6	**A**

For Examiner's Use

2 A student measures the mass and the volume of four samples of rock A, B, C and D. The results are shown in Fig. 2.1.

	A	B	C	D
mass/g	101	202	448	4508
volume/cm^3	22	44	80	978

Fig. 2.1

(a) (i) Describe in detail how a measuring cylinder is used to find the volume of rock A.

.....

.....

.....

.....

..... [2]

(ii) Explain why the volume of rock D cannot be found with an ordinary laboratory measuring cylinder.

.....

..... [1]

(b) Calculate the density of rock A.

density = ... [2]

(c) Three of the rocks are made from the same material.

State and explain which of the rocks is made from a different material.

.....

.....

.....

..... [2]

2 **(a)** **(i)** take reading of liquid before rock placed in **or** pour in a known/specified volume
or fill eureka can to spout/overflowing B1
take reading with rock and subtract **or** add rock **and** measure overflow B1

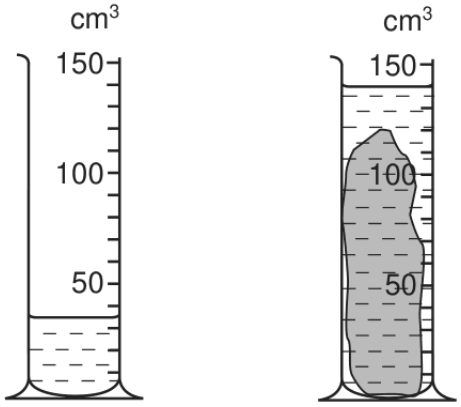
(ii) will not fit in **or** volume too large B1

(b) (d =) m/v **or** 101/22 C1
4.59 g/cm^3 A1

(c) C B1
mass/volume **or** density different **or** mass not proportional to volume B1 [7]

QUESTION · 5054_s08_qp_1.pdf · source page 4 of 20

6 A lump of metal has a mass of 210 g. It is lowered into a measuring cylinder containing water. The level of the water rises from 35 cm^3 to 140 cm^3.



What is the density of the metal?

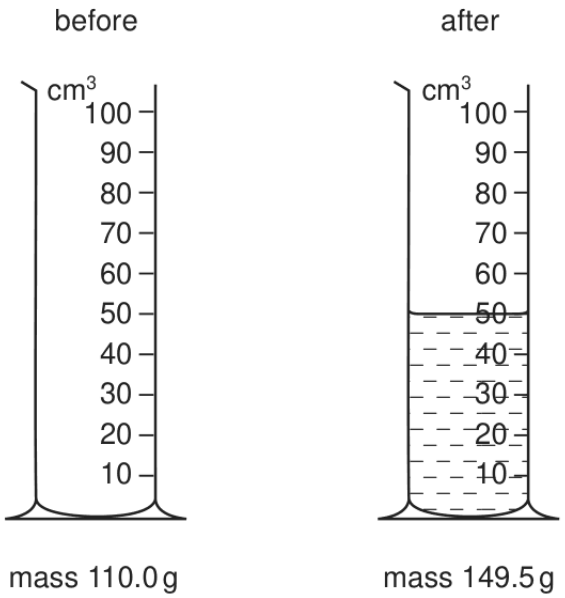
A 0.67 g / cm^3 **B** 1.5 g / cm^3 **C** 2.0 g / cm^3 **D** 6.0 g / cm^3

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 6

6	**C**

QUESTION · 5054_w08_qp_1.pdf · source page 4 of 20

6 The mass of a measuring cylinder is measured before and after pouring a liquid into it.



mass 110.0 g mass 149.5 g

What is the density of the liquid?

A $0.79\,g/cm^3$ **B** $1.3\,g/cm^3$ **C** $1.4\,g/cm^3$ **D** $2.2\,g/cm^3$

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 6

6	**A**

QUESTION · 5054_s09_qp_1.pdf · source page 4 of 20

7 A measuring cylinder contains 118 cm^3 of water. When a small object is fully immersed in the water, the reading goes up to 132 cm^3. The object has a mass of 42 g.

What is the density of the object?

A $\frac{14}{42}\,g/cm^3$ **B** $\frac{42}{14}\,g/cm^3$ **C** $\frac{42}{118}\,g/cm^3$ **D** $\frac{132}{42}\,g/cm^3$

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 7

7	**B**

10 Fig. 10.1 shows the horizontal forces as a cyclist travels forwards.

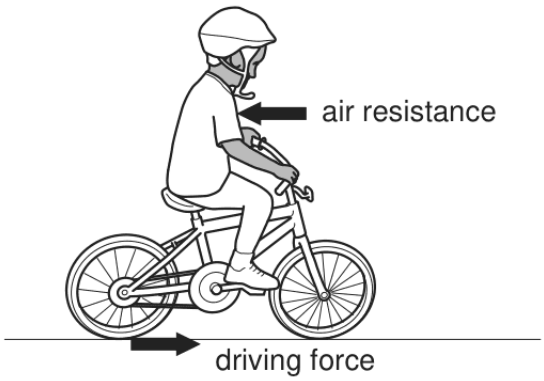


Fig. 10.1

The cyclist produces the driving force that acts on the back wheel.
In this question, you may ignore any frictional force acting on the front wheel.

(a) The bicycle accelerates until a constant speed is reached.

(i) Describe how the size of the air resistance changes during this time. [2]

(ii) Compare the sizes of the two horizontal forces when the bicycle is accelerating. [1]

(b) The total mass of the bicycle and the cyclist is 75 kg. At one instant, the speed of the bicycle is 4.0 m/s, the driving force is 30 N and the air resistance is 20 N.

Calculate

(i) the total kinetic energy of the bicycle and the cyclist, [3]

(ii) the acceleration of the bicycle and the cyclist. [2]

(c) As the bicycle moves, energy is transmitted from the pedals to the back wheel.

Fig. 10.2 shows what happens to the energy input to the pedals.

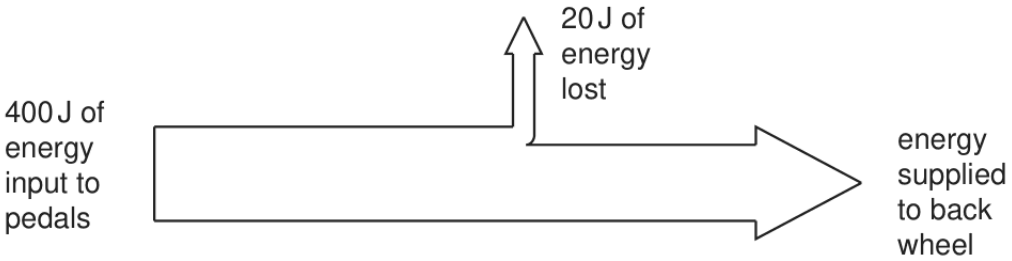


Fig. 10.2

(i) As energy is transmitted to the back wheel, some is lost. Explain how this happens. [2]

(ii) Calculate the efficiency of the bicycle in transmitting energy from the pedals to the back wheel. [2]

(d) Some bicycles are made from low density materials. Explain why this is an advantage. [3]

10 (a) (i) air resistance increases (as speed increases) B1
(at constant speed) becomes equal to driving force/applied force etc. B1

(ii) driving force (forward force) larger (than air resistance/backwards force) B1

(b) (i) (E =) ½ mv^2 algebraic formula C1
½ × 75 × 4^2 C1
600 J A1

(ii) (a =) F/m algebraic seen or 10 (N) used as force C1
0.13 m/s^2 A1

(c) (i) friction (in chain/axles) or rubbing of surfaces B1
heat or thermal energy produced B1

(ii) (efficiency = useful) energy output/energy input algebraic or numerical or 380 seen C1
0.95 or 95% A1

(d) lower mass/weight of cycle B1

less force needed	same acceleration/get up hill/to stop cycle/lift cycle	
greater acceleration/easier to acc.	for same force or $F = ma$ quoted	
less energy/work (input)	to go uphill/due to less friction/½ mv^2 less	
less kinetic energy	less stopping distance/less force to stop	
more efficient/less energy wasted	going uphill/less friction	
less friction	easier turn handlebars/higher (top) speed	
less pressure (on ground)	sinks less into ground	M1A1 [15]

QUESTION · 5054_w09_qp_1.pdf · source page 4 of 20

7 A room measures 4.0 m × 3.0 m × 2.0 m. The density of the air in the room is 1.3 kg / m^3.

What is the mass of air in the room?

A 0.054 kg **B** 18 kg **C** 24 kg **D** 31 kg

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 7

7	**D**

10 A space research organisation plans to send astronauts to Mars to examine rocks on its surface. The organisation will produce a report containing information about conditions on Mars.

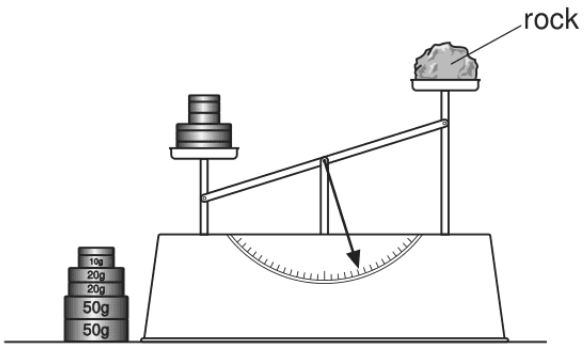
(a) (i) The gravitational field strength on the surface of Mars is 3.7 N/kg. Calculate the weight, on Mars, of a rock of mass 0.50 kg. [2]

(ii) A rock dropped on Mars falls to the surface. State the acceleration of the falling rock. Assume that there is no air resistance on Mars. [1]

(iii) Calculate the kinetic energy of a 0.50 kg rock as it strikes the surface at a speed of 3.2 m/s. [3]

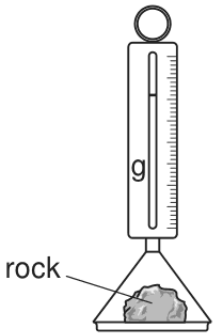
(b) While still on the surface of Mars, the astronauts will measure the mass of each rock collected. Fig. 10.1 shows two devices for measuring mass.

device A



A lever arm balance and a set of brass discs. The mass of each disc is accurately determined on Earth and the value is marked on it.

device B



A spring balance. The spring balance is accurately calibrated on Earth and the mass values are marked on the vertical scale.

Fig. 10.1

(i) When the two devices are used on Mars, they will give different readings for the mass of the same rock. Explain why. [3]

(ii) State which device will give the correct reading for the mass. [1]

(c) The astronauts will also determine the density of each rock.

(i) Apart from the mass of the rock, state what other information is needed in order to calculate the density of the rock. [1]

(ii) Describe a method for determining the density of a small, irregularly shaped rock of known mass *m*. [4]

10	**(a)**	**(i)**	(W =)mg **or** 0.5 × 3.7	C1	
			1.8/1.85/1.9 N	A1	
		(ii)	3.7 m/s^2 **not** N/kg	B1	
		(iii)	(KE =) ½mv^2	C1	
			½ × 0.50 × 3.2^2	C1	
			2.6 **or** 2.56 J	A1	[6]
	(b)	**(i)**	A compares/measures (unknown and known) masses/amount of matter	B1	
			B measures/is dependent on weight/force of gravity (and hence mass obtained)	B1	
			Mars weights/forces of gravity are less than/different from (Earth)	B1	
		(ii)	A **or** lever arm balance **or** balance with discs	B1	[4]

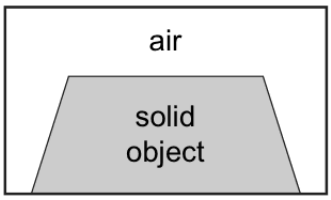
	(c)	**(i)**	volume	B1	
		(ii)	**Either:**	**Or:**	
			record value of water in measuring cylinder (**not** beaker)	water/liquid in displacement/eureka can	
			insert rock	full to overflowing	
			record new value	immerse rock	
			subtract (to obtain volume) **or** measure rise)	(new) overflow in measuring cylinder (**not** beaker is volume)	
			m/volume **or** *m*/subtraction (max 4)	*m*/volume **or** *m*/subtraction (max 4)	B4 [5]

[Total: 15]

QUESTION · 5054_w10_qp_11.pdf · source page 6 of 20

8 A box has an internal volume of 1000 cm^3. When a solid object is placed in the closed box, the volume of air in the box is 520 cm^3.

The density of the object is 8.00 g/cm^3.



What is the mass of the object?

A 60.0 g **B** 3840 g **C** 4160 g **D** 8000 g

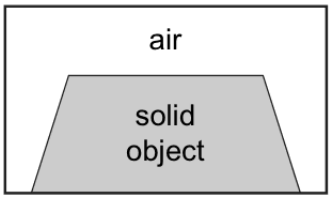
MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 8

8	**B**

QUESTION · 5054_w10_qp_12.pdf · source page 6 of 16

11 A box has an internal volume of 1000 cm^3. When a solid object is placed in the closed box, the volume of air in the box is 520 cm^3.

The density of the object is 8.00 g / cm^3.



What is the mass of the object?

A 60.0 g **B** 3840 g **C** 4160 g **D** 8000 g

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 11

11	**B**

For Examiner's Use

2 A U-shaped tube, of constant cross-sectional area, contains some water of density 1000 kg/m^3. Oil that does not mix with water is then poured into the right-hand side of the tube. Fig. 2.1 shows the levels of the water and the oil when equilibrium is reached.

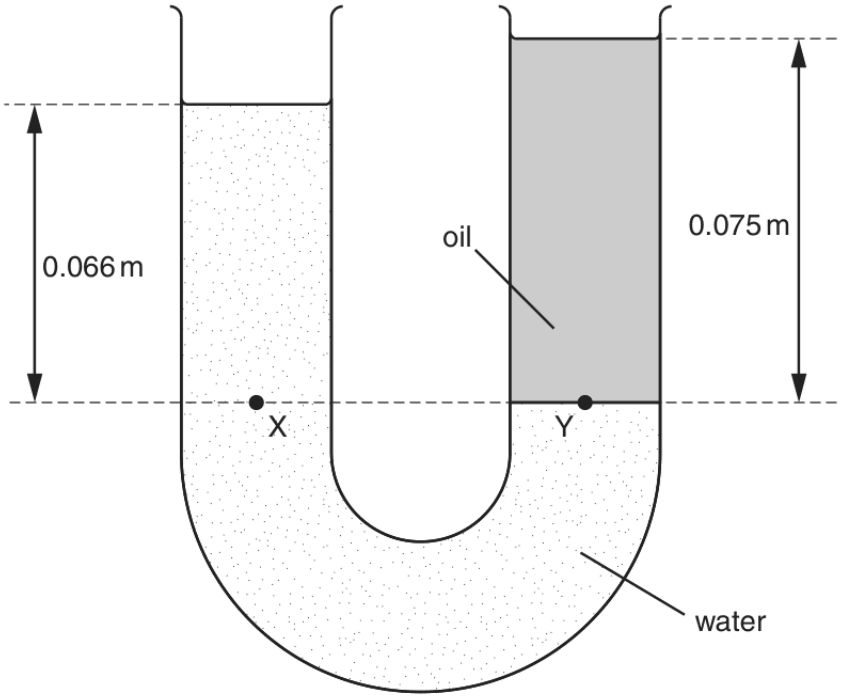


Fig. 2.1

Points X and Y are at the same horizontal level. X is 0.066 m below the top surface of the water. Y is 0.075 m below the top surface of the oil.

(a) State two quantities that influence the pressure beneath the surface of a liquid.

1. .....

2. .....

[2]

(b) The cross-sectional area of the tube is 5.0×10^{-4} m^2.

(i) Calculate the mass of water above the level of X.

mass = ...[2]

(ii) The pressure caused by 0.066 m of water at X is equal to that caused by 0.075 m of the oil at Y.

Determine the density of the oil.

density = ...[2]

2 **(a)** any **two** points:
depth/height;
density (of liquid);
atmospheric pressure;
g/gravitational field strength/acceleration of free-fall (**not** gravity) B2

(b) **(i)** (m =) ρV **or** $5.0 \times 10^{-4} \times 0.066 \times 1000$ **or** $3.3 \times 10^{-5} \times 1000$ C1
0.033 kg (**not** factor of 10 caused by omitted density) A1

(ii) **mass of oil** = 0.033 (kg)/mass of water **above X**
or $1000 \times 0.066/0.075$ **or** $0.033/(5.0 \times 10^{-4} \times 0.075)$
or $0.033/(3.75 \times 10^{-5})$ **or** inversely proportional to height C1
880 kg/m^3 A1 [6]

QUESTION · 5054_s11_qp_11.pdf · source page 4 of 16

7 A student collects stones and finds their density.

Which apparatus is needed to measure the mass and the volume of the stones?

	mass	volume
A	newton meter	measuring cylinder and water
B	newton meter	ruler and calipers
C	top-pan balance	measuring cylinder and water
D	top-pan balance	ruler and calipers

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 7

7 **C**

QUESTION · 5054_s11_qp_11.pdf · source page 7 of 16

16 A balloon filled with air is gently heated.

What happens to the mass and the density of the air inside the balloon?

	mass	density
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 16

16 **C**

QUESTION · 5054_s11_qp_12.pdf · source page 4 of 16

8 A student collects stones and finds their density.

Which apparatus is needed to measure the mass and the volume of the stones?

	mass	volume
A	newton meter	measuring cylinder and water
B	newton meter	ruler and calipers
C	top-pan balance	measuring cylinder and water
D	top-pan balance	ruler and calipers

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 8

8 **C**

QUESTION · 5054_s11_qp_12.pdf · source page 7 of 16

15 A balloon filled with air is gently heated.

What happens to the mass and the density of the air inside the balloon?

	mass	density
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 15

15 **C**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A student wishes to find the density of a stone. He uses a measuring cylinder and a spring balance with a scale marked in newtons. The measuring cylinder, spring balance and stone are shown in Fig. 1.1.

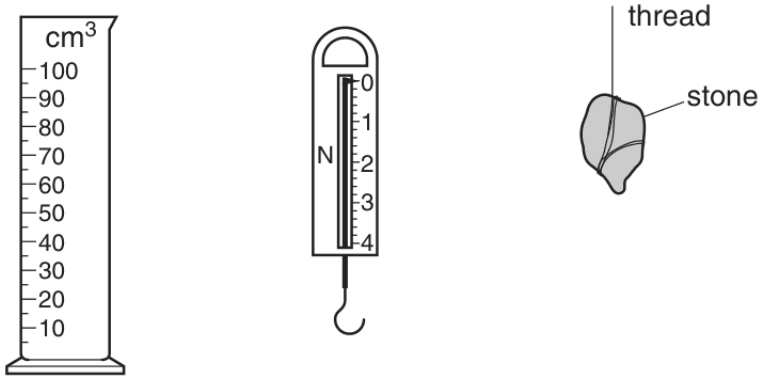


Fig. 1.1

The student knows that the gravitational field strength is 10 N/kg.

(a) Describe how the student uses the spring balance to find the mass of the stone.

.....

.....

.....[2]

(b) Describe how the student uses the measuring cylinder to find the volume of the stone.

.....

.....

.....[2]

(c) The mass of the stone is 150 g and its volume is 70 cm^3.
Calculate the density of the stone.

density of stone = ...[1]

(d) The stone is taken to another place, where the gravitational field strength is less than 10 N/kg. State how this affects the mass and the weight of the stone.

mass .....

weight .....
[1]

Section A

1	**(a)**	(uses spring balance) for a reading/value // finds weight/force of gravity	B1	
		divides reading/weight by 10/g // uses $W = mg$	B1	
	(b)	reading (of measuring cylinder) taken with liquid/water (alone) // initial volume mentioned // fill to certain level	C1	
		measure increase/change when stone (totally) immersed/in cylinder	A1	
	(c)	2.1 or 2.14 g/cm^3 // 2142.86 kg/m^3 // 0.00214286 kg/cm^3	B1	
	(d)	mass unchanged **and** weight less	B1	[6]

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows an ice cube at 0 °C.

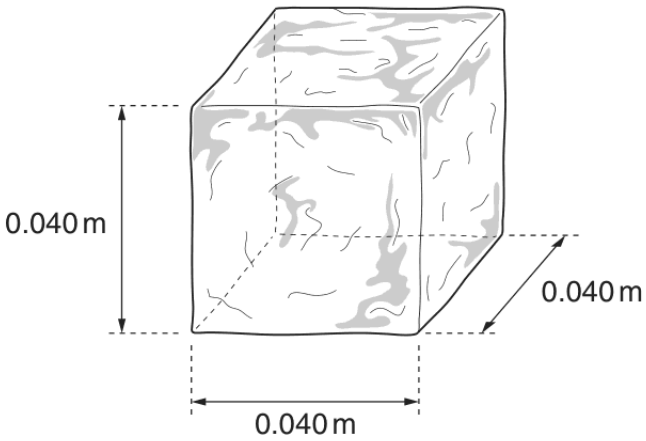


Fig. 1.1

The sides of the cube are of length 0.040 m. Ice at 0 °C has a density of 920 kg/m^3.

(a) Calculate

(i) the mass of the ice cube,

mass = ...[3]

(ii) the weight of the ice cube.

weight = ...[1]

(b) The specific latent heat of fusion of ice is 3.4×10^5 J/kg. Calculate the thermal energy (heat) absorbed by the ice cube as it melts.

thermal energy = ...[2]

Section A

1	**(a)**	**(i)**	(V =) 64 **or** 6.4×10^{-5} **or** 4^3 **or** 0.04^3	C1	
			(m =) ρV **or** $920 \times 6.4 \times 10^{-5}$ **or** 920×0.04^3	C1	
			0.059 kg **or** 59 g **or** 0.05888 kg	A1	
		(ii)	0.59 N **or** 0.5888 N	B1	
	(b)		(Q =) ml **or** $0.059 \times 3.4 \times 10^5$	C1	
			$2.0(0) \times 10^4$ / $2.0(1) \times 10^4$ / 2.006×10^4 J	A1	[6]

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A builder needs to determine the density of a solid cube of wood.

He places the 50 cm mark of a uniform metre rule on a pivot, so that the rule balances.

He then places the cube on the rule with its centre of gravity directly above the 75 cm mark. A mass of 0.050 kg is moved along the rule until balance is restored. This is shown in Fig. 1.1.

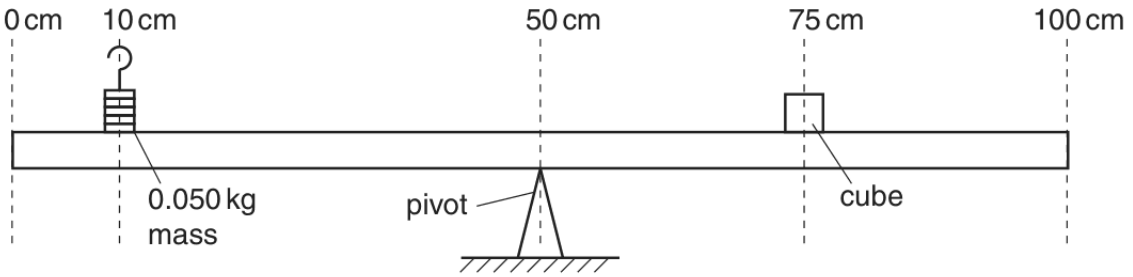


Fig. 1.1 (not to scale)

The rule is balanced when the 0.050 kg mass is at the 10 cm mark.

(a) Calculate the mass of the cube.

mass = ...[3]

(b) The cube has a volume of $1.6 \times 10^{-4}\,m^3$. Determine the density of the wood.

density = ... [2]

Section A

1	**(a)**	$m_1(g)x_1$ **or** $m_2(g)x_2$ **or** 2 **or** one of these in numbers **or** 40 **and** 25 seen	C1	
		$0.050 \times (10) \times 40 = m_2 \times (10) \times 25$		
		or anticlockwise **moment =**clockwise **moment**	C1	
		0.080 kg **or** 80 g	A1	
	(b)	$(\rho/d =)\ m/V$ **or** $0.08/1.6 \times 10^{-4}$	C1	
		$500\,kg/m^3$ **or** $0.50\,g/cm^3$	A1	[5]

QUESTION · 5054_s12_qp_12.pdf · source page 5 of 16

10 The mass of a paper-clip is 0.50 g and the density of its material is 8.0 g / cm^3. The total volume of a number of clips is 20 cm^3.

How many paper-clips are there?

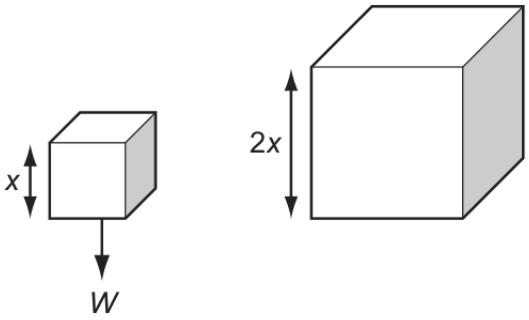
A 80 **B** 160 **C** 240 **D** 320

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 10

10 **D**

QUESTION · 5054_w12_qp_11.pdf · source page 5 of 16

10 The diagram shows two cubes made from the same material. One cube has sides that are twice as long as the sides of the other cube.



The weight of the small cube is *W*.

What is the weight of the larger cube?

A $2W$ **B** $4W$ **C** $8W$ **D** $16W$

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 10

10 **C**

4 A petrol tanker has mass 2800 kg when empty and holds 30 m^3 of petrol when full. The density of petrol is 740 kg/m^3.

For Examiner's Use

(a) Calculate the total mass of the tanker when full of petrol.

mass = ... [2]

(b) The full tanker is travelling along a road when the driver observes an obstruction and applies the brakes. A resultant force of 30 000 N acts on the tanker.

Calculate the deceleration of the tanker.

deceleration = ... [2]

4	**(a)**	(m=)ρV **or** 740 × 30 or 22 000/22 200	C1	
		25 000/2.5 × 10^4 kg (allow 24 800 from 22 000)	A1	
	(b)	(a =)F/m **or** 30 000/25 000	C1	
		(–)1.2 m / s^2	A1	[4]

QUESTION · 5054_w12_qp_12.pdf · source page 2 of 16

1 The level of water in a measuring cylinder is 75 cm^3. A stone of volume 20 cm^3 is lowered into the water.

What is the new reading of the water level?

A 20 cm^3 **B** 55 cm^3 **C** 75 cm^3 **D** 95 cm^3

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 1

1	**D**

QUESTION · 5054_w12_qp_12.pdf · source page 5 of 16

10 The mass and the volume of a bar made from metal X are measured.

The masses and volumes of four other bars are measured.

Which bar is made from a metal with a density that is double that of X?

	mass compared with X	volume compared with X
A	double	half
B	half	same
C	same	double
D	same	half

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 10

10 **D**

QUESTION · 5054_s13_qp_11.pdf · source page 4 of 20

6 Paper is sold in packets labelled $80\,g/m^2$. This means that a sheet of paper of area $10\,000\,cm^2$ has a mass of 80 g.

The thickness of each sheet is 0.11 mm.

What is the density of the paper?

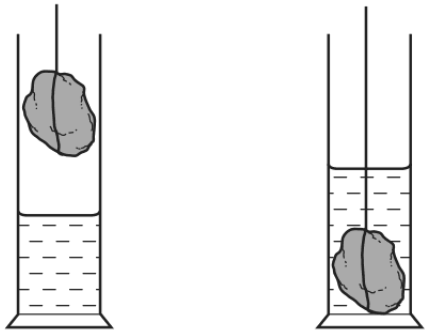
A $0.073\,g/cm^3$ **B** $0.088\,g/cm^3$ **C** $0.73\,g/cm^3$ **D** $0.88\,g/cm^3$

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 6

6	**C**

QUESTION · 5054_s13_qp_12.pdf · source page 4 of 16

8 A measuring cylinder contains 20 cm^3 of water. A stone is placed in the water and the water level rises to 38 cm^3.



What is the volume of the stone?

A 18 cm^3 **B** 29 cm^3 **C** 38 cm^3 **D** 58 cm^3

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 8

8	**A**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a flask, partly full of a liquid and partly full of air.

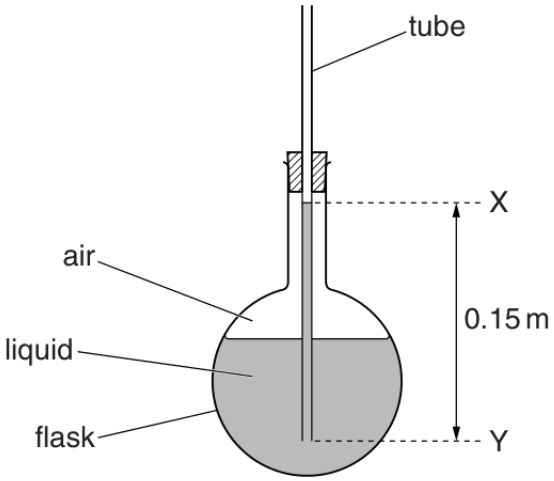


Fig. 9.1

When the flask is heated, the pressure of the air inside the flask increases and the liquid rises up the tube.

(a) Explain, using ideas about molecules, why heating the air inside the flask causes the pressure to increase.

.....

.....

..... [3]

(b) A teacher uses the flask as a thermometer. He marks the liquid level at 0 °C and at 100 °C.

(i) Describe how the teacher obtains these temperatures.

.....

.....

.....

..... [2]

(ii) State two ways in which the thermometer can be made more sensitive.

1. .....

2. .....

[2]

For Examiner's Use

(iii) The thermometer has a non-linear scale.

Explain what is meant by a *non-linear* scale on a thermometer.

.....

.....

..... [2]

(c) The inner cross-sectional area of the tube is $5.0 \times 10^{-5}\,m^2$.
The density of liquid in the tube is $1200\,kg/m^3$.
The distance between X and Y is 0.15 m.
The gravitational field strength g is 10 N/kg.

Calculate

(i) the mass of liquid in the tube,

mass = ... [2]

(ii) the weight of liquid in the tube,

weight = ... [1]

(iii) the pressure at the bottom of the tube caused by the liquid column between X and Y.

pressure = ... [2]

(d) When the flask is completely full of liquid, the liquid moves up the tube a smaller distance for the same temperature rise. State why.

.....

..... [1]

Section B

9	**(a)**	(air) molecules hit walls / liquid (surface)	B1	
		(air) molecules move fast(er) /great(er) kinetic energy	B1	
		(air) molecules hit more often/more frequently/greater rate / harder / more force **or** (liquid) molecules evaporate	B1	[3]
	(b) **(i)**	(flask) in (pure) melting ice (and water)	B1	
		(flask) in (pure) boiling water / above boiling water (at one atmosphere)	B1	
	(ii)	thin(ner) tube **or** large(r) flask **or** more air/less liquid **or** use liquid that expands more (1 mark for each)	B2	
	(iii)	divisions not equally spaced **or** scale not uniform/not proportional	C1	
		different distance (along scale) for same temperature rise **or** different change in temperature for same distance (along scale)	A1	[6]
	(c) **(i)**	(M=) $d \times V$ in any form **or** $1200 \times 5 \times 10^{-5} \times 0.15$	C1	
		$9(.0) \times 10^{-3}$ kg; 0.009(0) kg	A1	
	(ii)	0.09(0) N ecf **(i)**	B1	
	(iii)	(P=) hdg in any form **or** (P=) F/A in any form	C1	
		1800 Pa	A1	[5]
	(d)	liquids expand less (than air) **or** great(er) forces between liquid molecules	B1	[1]

[Total: 15]

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 A swimming pool is filled with 450 m^3 of water at 12 °C.

The density of water is 1000 kg/m^3 and its specific heat capacity is 4.2 J/(g °C).

(a) Calculate the mass of water in the swimming pool.

mass = ...[2]

(b) **(i)** Calculate the thermal energy (heat) needed to raise the temperature of the water to 27 °C.

thermal energy = ...[3]

(ii) Suggest one reason why the thermal energy supplied to the water by the heating system, when raising the temperature of the water to 27 °C, differs from the value calculated in **(i)**.

.....

.....[1]

Section A

1	**(a)**	(m =) ρV **or** 1000 × 450	C1	
		4.5×10^5 kg	A1	
	(b)	**(i)** (Q =) $mc\Delta T$ **or** $4.5 \times 10^5 \times 4.2 \times 15$ **or** 4200 **and** 15/(27–12)	C1	
		$4.5 \times 10^5 \times 4200 \times 15$ **or** $2.8(35) \times 10^7$	C1	
		$2.8(35) \times 10^{10}$ J	A1	
		(ii) thermal/internal energy/heat lost **or** gained by something specific (e.g. air/pool walls/tiles etc.) **or** heat lost by evaporation	B1	[6]

For Examiner's Use

2 Fig. 2.1 shows a boy moving a water container in a wheelbarrow.

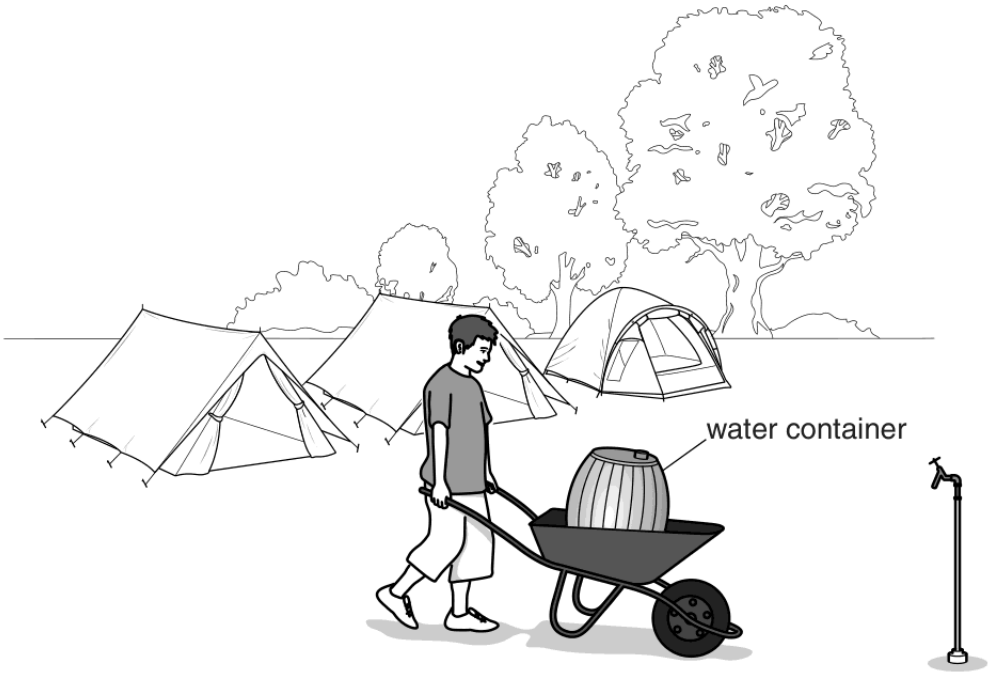


Fig. 2.1

The container has a volume of 0.15 m^3 and is filled with water of density 1000 kg/m^3.

(a) Calculate the mass of water in the container when it is full.

mass = ...[2]

(b) It is harder to stop the wheelbarrow when the container is full than when it is empty.

Explain this.

.....

.....

.....[2]

2	**(a)**	(m =) ρV **or** 1000 × 0.150	C1	
		150 kg	A1	
	(b)	(when full) greater mass **or** greater momentum	B1	
		more inertia **or** mass resists change in state of motion **or** small(er) deceleration (for same force) **or** large(r) force for same deceleration (rate of decrease of momentum for deceleration)	B1	
		or		
		greater kinetic energy	(B1)	
		more work done in same distance (to stop)	(B1)	[4]

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 A garden pond contains a small fountain. An electric pump in the water causes the water to rise above the surface of the pond, as shown in Fig. 9.1.

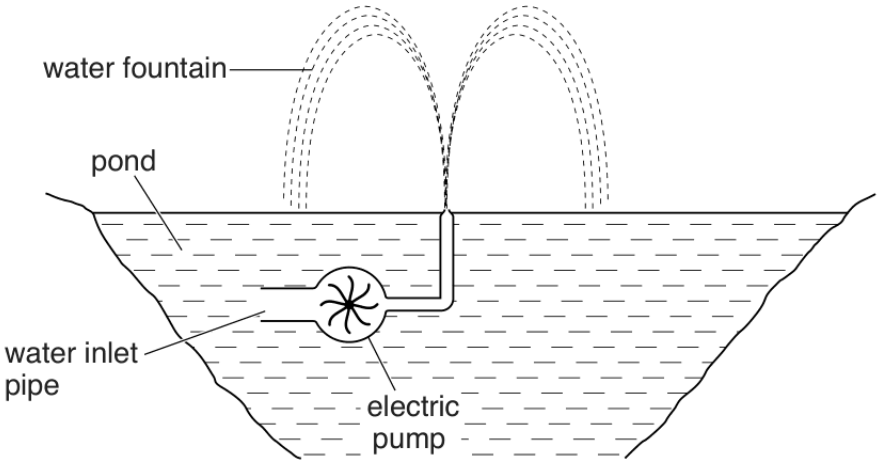


Fig. 9.1

(a) The pressure of the water in the pond increases with depth.

(i) Explain what is meant by *pressure*.

.....

..... [1]

(ii) Explain why the pressure below the surface of the water increases with depth.

.....

..... [2]

(iii) State the unit of pressure.

..... [1]

(b) Describe the energy changes that occur within the pump.

.....

.....

..... [3]

(c) A volume of 0.78 m^3 of water passes through the pump in one hour. The density of water is 1000 kg/m^3.

(i) Calculate the mass of water that passes through the pump in one hour.

mass = ... [2]

(ii) The water rises 1.5 m. The gravitational field strength g is 10 N/kg.

Calculate the useful work done in one hour raising the water to the top of the fountain.

work done = ... [2]

(iii) Calculate the minimum power output of the pump.

power = ... [2]

(d) Describe an experiment to check that the density of water is 1000 kg/m^3.

.....

.....

.....

.....

..... [2]

SECTION B

9 (a) (i) force per unit area **or** force divided by area **or** force on 1 m^2 **or** force on unit area [B1]

(ii) more water above **or** more force from water **or** more atoms / molecules above [C1]

larger weight of water above [A1]

(iii) Pa **or** N/m^2 [B1]

(b) electrical energy at start of process [B1]
kinetic energy of water / turbine/blade produced [B1]
some heat energy / thermal energy / internal energy produce [B1]

(c) (i) (M=) D × V in any algebraic or numerical form [C1]
780 kg [A1]

(ii) (W=) mgh or Fd in any algebraic or numerical form [C1]
11700 J **or** 12000 J [A1]

(iii) (P=) W / t in any algebraic or numerical form **or** 195(W) [C1]
3.25 **or** 3.2 **or** 3.3 W [A1]

(d) find difference in mass of container, with and without water [B1]
measuring cylinder or similar instrument used to find volume **and** density checked [B1]

other methods are possible, e.g. use of calibrated hydrometer

[15]

QUESTION · 5054_s14_qp_12.pdf · source page 4 of 16

8 A stone has a mass of 390 g and a density of 2.7 g / cm^3.

Cooking oil has a density of 0.90 g / cm^3.

Which mass of oil has the same volume as the stone?

A 130 g **B** 160 g **C** 900 g **D** 1200 g

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 8

8	**A**

2 A student places a metre rule on the edge of a triangular prism. The prism is used as a pivot and the rule balances about the 50 cm mark.

The student then places a block of wood at the 10 cm mark on the rule and an empty measuring cylinder at the 80 cm mark. The rule is still balanced. Fig. 2.1 shows the arrangement.

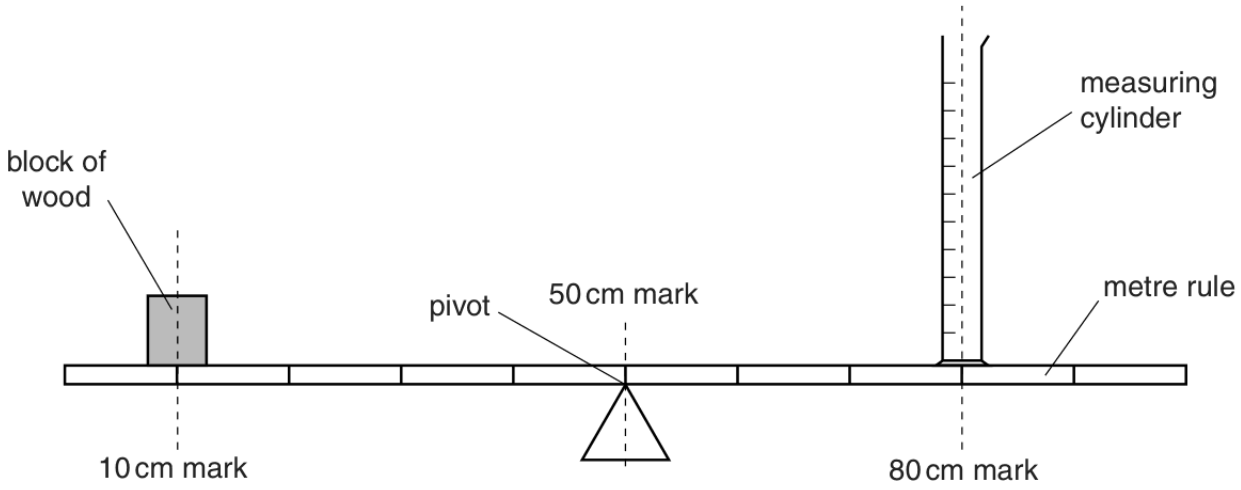


Fig. 2.1 (not to scale)

The student now places a weight of 0.39 N on top of the block of wood. She then starts to pour oil into the measuring cylinder. The rule balances again when there is 60 cm^3 of oil in the measuring cylinder.

The gravitational field strength g is 10 N/kg.

(a) Calculate

(i) the weight of the oil in the cylinder,

weight = ...[2]

(ii) the mass of the oil in the cylinder.

mass = ...[1]

(b) Calculate the density of the oil.

density = ...[2]

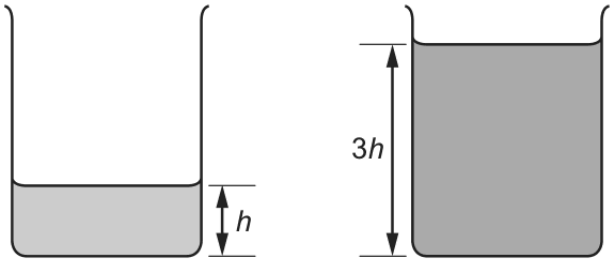
2 **(a)** **(i)** $F_1 \times d_1 = F_2 \times d_2$ **or** $(0.39 \times 0.40)/0.30$ C1
$0.52\,\text{N}$ A1

(ii) $0.052\,\text{kg}$ **or** $52\,\text{g}$ B1

(b) $(\rho =)\ m/V$ **or** $52/60$ **or** $0.052/0.000\,060$ **or** $0.052/60$ B1
$870/867/866.7\,\text{kg/m}^3$ **or** $0.87\,\text{g/cm}^3$ **or** $8.7 \times 10^{-4}\,\text{kg/cm}^3$ etc. B1 **[5]**

QUESTION · 5054_w14_qp_12.pdf · source page 3 of 16

5 Two identical beakers contain the same mass of liquid. There is a different liquid in each beaker.



Liquid Q has a density ρ.

What is the density of liquid P?

A $\frac{\rho}{3}$ **B** ρ **C** 3ρ **D** 9ρ

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 5

5 **C**

3 Fig. 3.1 shows a helium cylinder being used at a funfair to inflate balloons that float in the air.

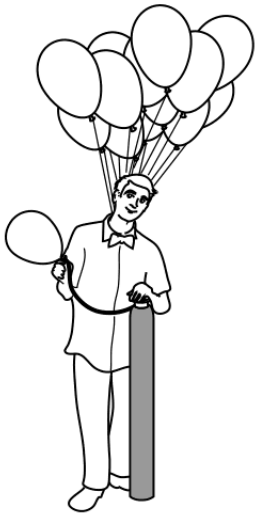


Fig. 3.1

Initially, all the helium is stored in the cylinder at a pressure of 2.0×10^7 Pa.

When no more balloons can be filled, all the helium occupies a total volume of $1.8\,m^3$ and is at a pressure of 1.0×10^5 Pa. The temperature of the helium has not changed.

(a) Calculate, stating the formula used, the initial volume of the helium in the cylinder.

volume = ...[3]

(b) The balloon seller takes the cylinder back to the helium supplier and it is refilled. The mass of helium in the cylinder is 0.30 kg.

(i) Calculate the density of helium in the cylinder.

density = ...[2]

(ii) The mass of the empty cylinder is 16 kg.

Suggest why weighing the cylinder is not an accurate method of determining the amount of helium supplied.

.....

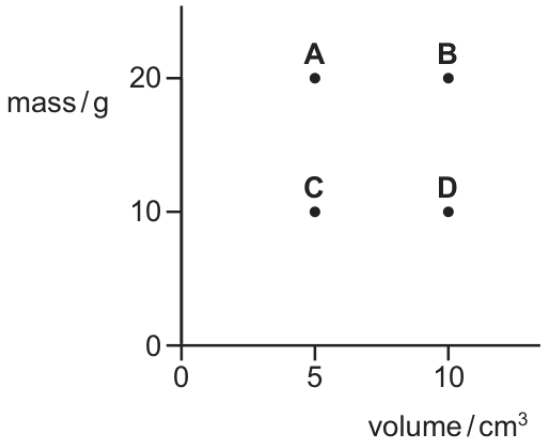
.....[1]

3	**(a)**	$V_1 = p_2V_2/p_1$ **or** $p \propto 1/V$	B1	
		$1.0 \times 10^5 \times (1.8/2.0) \times 10^7$	C1	
		$9.0 \times 10^{-3}\,m^3$ **or** $9000\,cm^3$	A1	
	(b)	**(i)** $(\rho =)\ m/V$ **or** $(0.30/9.0) \times 10^{-3}$	C1	
		$33(.3333)\,kg/m^3$ **or** $0.033(3333)\,g/cm^3$	A1	
		(ii) helium mass / weight small (fraction of total / mass of air included) **or** this includes the weight of the cylinder	B1	**[6]**

QUESTION · 5054_s15_qp_11.pdf · source page 4 of 16

8 The mass and volume of four different objects are plotted as shown.

Which object has the smallest density?



MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 8

8 **D**

2 Fig. 2.1 shows two engineers measuring the length of a wall made from concrete.

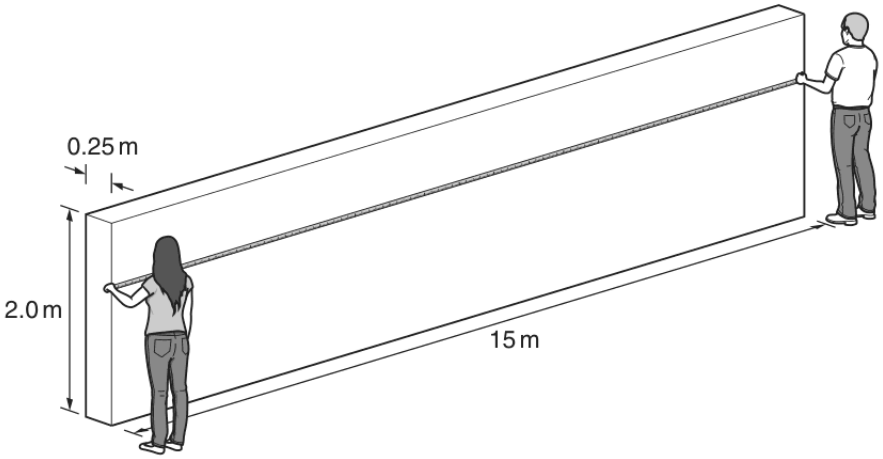


Fig. 2.1 (not to scale)

The wall is 2.0 m high, 15 m long and 0.25 m thick.

The weight of the wall is 180 000 N and the mass of the wall is 18 000 kg.

(a) The engineers measure the length of the wall in one single measurement.

State the name of the measuring instrument they use.

..... [1]

(b) The engineers state that the density of the concrete affects the pressure exerted by the wall on the ground but that the length of the wall does not affect this pressure.

(i) Define *density*.

.....

..... [1]

(ii) Calculate the density of the concrete.

density = ... [2]

(iii) Calculate the pressure exerted by the wall on the ground.

pressure = ... [2]

(iv) Without further calculation, explain why doubling the length of the wall does not change the pressure that the wall exerts on the ground.

.....

.....

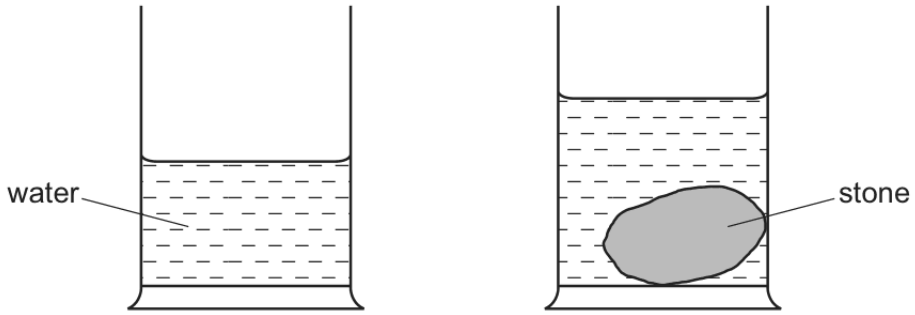
.....

..... [1]

2	**(a)**	tape	B1
	(b) **(i)**	mass ÷ volume **or** mass per unit volume	B1
	(ii)	(V=) 15 × 0.25 × 2 or 7.5 seen	C1
		2400 kg / m^3	A1
	(iii)	(A=) 15 × 0.25 **or** 3.75 **or** (P=) F / A	C1
		or (P=) 18 0000 / A **or** (P=) *dgh* / ρgh seen	
		48 000 Pa	A1
	(iv)	(length doubles) so both area **and** weight / force double	B1
		or area **and** force / weight both increase / large**r** (in proportion)	
		or height **and** density the same (in $P = dgh$)	

QUESTION · 5054_w15_qp_11.pdf · source page 2 of 16

1 During an experiment to find the density of a stone, the stone is lowered into a measuring cylinder partly filled with water.



Which statement is correct?

A The difference between the readings gives the density of the stone.

B The difference between the readings gives the volume of the stone.

C The final reading gives the density of the stone.

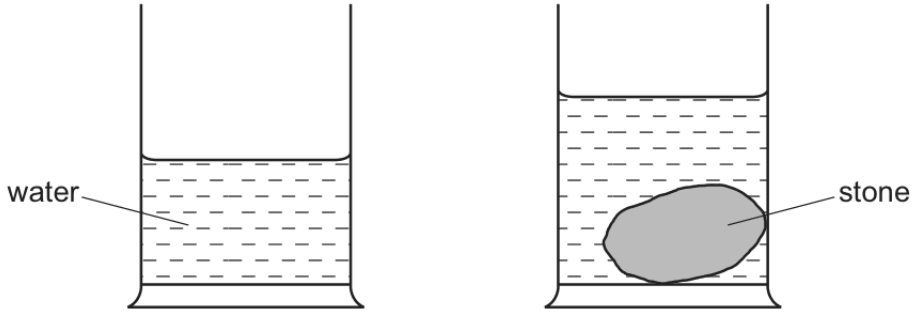
D The final reading gives the volume of the stone.

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 1

1	**B**

QUESTION · 5054_w15_qp_12.pdf · source page 2 of 16

2 During an experiment to find the density of a stone, the stone is lowered into a measuring cylinder partly filled with water.



Which statement is correct?

A The difference between the readings gives the density of the stone.

B The difference between the readings gives the volume of the stone.

C The final reading gives the density of the stone.

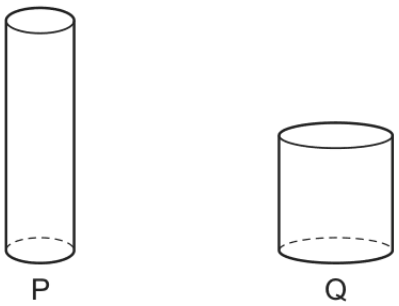
D The final reading gives the volume of the stone.

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 2

2 **B**

QUESTION · 5054_w15_qp_12.pdf · source page 5 of 16

10 Two cylinders P and Q are made of copper.



The height of P is twice the height of Q. The diameter of P is half the diameter of Q.

Which statement is correct?

A The density of cylinder P is four times that of cylinder Q.

B The density of cylinder P is twice that of cylinder Q.

C The density of cylinder P is equal to that of cylinder Q.

D The density of cylinder P is half that of cylinder Q.

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 10

10	**C**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Water is transported to a village in a tank pulled by a tractor.

Fig. 1.1 shows the tank being pulled by a tractor.

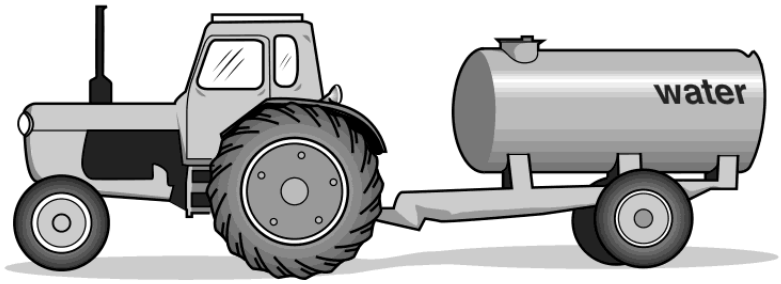


Fig. 1.1

The combined mass of the tractor and the tank is 4100 kg when the tank is empty and 6500 kg when the tank is full of water.

(a) The density of water is 1000 kg / m^3.

Calculate the volume of water in the tank when it is full.

volume = ... [2]

(b) At the start of the journey, the tractor and tank accelerate from rest along a straight, horizontal road. As their speed increases, one form of energy is decreasing.

(i) State the name of the form of energy that is decreasing.

..... [1]

(ii) Explain what happens to this energy.

.....

.....

.....

..... [2]

(c) The village is located on a mountain at a vertical height of 850 m above the water supply. The gravitational field strength g is 10 N/kg.

Calculate the gravitational potential energy gained by the water as it is transported from the supply to the village.

energy gained = ... [2]

Section A

1	**(a)**	$(V =)\, m/\rho$ **or** 10 600 / 6500 / 4100 / 2400 ÷ 1000 **or** 6.5 **or** 4.1	C1	
		2.4 m^3 / 2.4 × 10^6 cm^3	A1	
	(b)	**(i)** fuel / chemical (potential energy)	B1	
		(ii) some to heat / thermal (energy)	B1	
		some to kinetic (energy of air **or** tractor)	B1	
	(c)	$(GPE =)\, mgh$ **or** 2400 × 10 × 850	C1	
		2.0 / 2.04 × 10^7 J	A1	[7]

3 Fig. 3.1 shows a thin sheet of plastic. A student tries to measure the thickness of the sheet with a ruler, but the sheet is too thin to measure accurately.

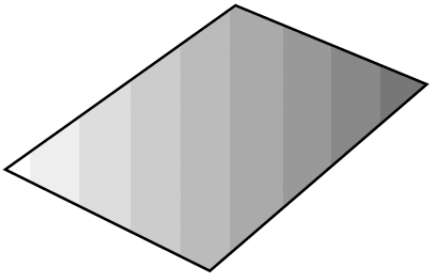


Fig. 3.1 (not to scale)

The student measures the mass of the sheet and obtains the value 0.12 g.

(a) (i) State what is meant by *mass*.

.....

.....[1]

(ii) The student is told that the density of the plastic is 0.91 g/cm^3.

Calculate the volume of the plastic sheet.

volume = ...[2]

(iii) The student measures the length and width of the sheet. The readings obtained are:

length of sheet = 3.0 cm
width of sheet = 2.0 cm

Calculate the thickness of the sheet.

thickness = ... [2]

(b) State a measuring instrument that can be used to measure the thickness of the sheet accurately.

.....[1]

3	**(a)**	**(i)**	amount of matter / substance / material **or** the ability of an object to resist a change in its state of motion (when a force is applied)	B1
		(ii)	(V=) M / D in any form numerical or algebraic	C1
			$0.13(19)\,cm^3$	A1
		(iii)	V / ($l \times$ w) in any form numerical or algebraic	C1
			0.022 cm	A1
	(b)		micrometer (screw gauge) **or** calipers	B1

10 A particular type of cooking oil is a liquid at room temperature.

(a) At room temperature, air is easily compressed but the oil cannot be compressed.

Explain, in terms of molecules,

(i) why the density of the oil is much greater than that of a gas,

.....

.....

.....[1]

(ii) why the oil cannot be compressed.

.....

.....

.....[1]

(b) Some of the oil is poured into a metal pan. The oil in the pan has a mass of 680 g.

(i) The density of the oil is 0.85 g / cm^3.

Calculate the volume of oil in the pan.

volume = ...[2]

(ii) The pan is placed on an electric hotplate, as shown in Fig. 10.1.

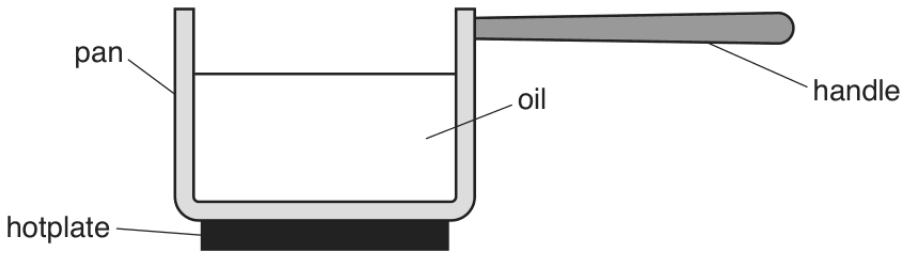


Fig. 10.1

1. Explain, in terms of molecules or free electrons, how thermal energy is conducted through the metal base of the pan.

.....

.....

.....

.....

.....

.....[3]

2. Explain how thermal energy spreads to all the oil in the pan.

.....

.....

.....

.....

.....

.....[3]

3. Suggest and explain a suitable material for the handle of the pan.

.....

.....

.....[1]

(c) The specific heat capacity of the oil is 2.0 J/(g °C). The oil is heated from 20 °C to its boiling point of 260 °C.

(i) State what is meant by *boiling point*.

.....

.....[1]

(ii) Calculate the thermal energy needed to heat the 680 g of oil to its boiling point.

thermal energy = ...[2]

(iii) Suggest one reason why the energy output from the hotplate, when heating the oil, is larger than the value calculated in **(c)(ii)**.

.....

.....

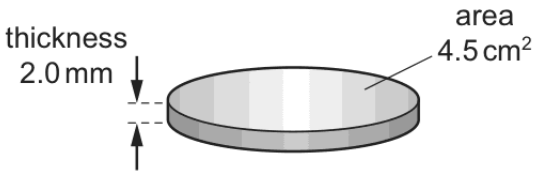
.....[1]

10	**(a)**	**(i)**	molecules / they close together **or** small gaps between molecules		B1	
		(ii)	molecules / they exert large (repulsive) forces		B1	[2]
	(b)	**(i)**	(V =)m/ρ **or** 680 / 0.85		C1	
			800 cm^3 **or** 8.0×10^{-4} m^3		A1	
		(ii)	**1** molecules <u>vibrate</u>	molecules <u>vibrate</u>	B1	
			collide with neighbours **or**	collide with electrons	B1	
			transfer vibration / energy	electrons travel through metal	B1	
			2 heated / hot oil expands / less dense		B1	
			rises		B1	
			convection current/circulation established		B1	
			3 any suitable named insulator **and** it is a poor conductor		B1	[9]
	(c)	**(i)**	<u>temperature</u> at which (liquid) vaporises / becomes gas / steam		B1	
		(ii)	(Q =)$mc\,\Delta T$ **or** $680 \times 2.0 \times (260 - 20)$ **or** $680 \times 2.0 \times 240$		C1	
			3.3×10^5 J		A1	
		(iii)	heat supplied to <u>pan</u> **or** heat lost to <u>air / surroundings</u>		B1	[4]
						[15]

QUESTION · 5054_w16_qp_12.pdf · source page 4 of 16

8 The density of gold is $19\,g/cm^3$.

A gold coin is 2.0 mm thick. Each face of the coin has an area of $4.5\,cm^2$.



What is the mass of the coin?

A 2.1 g **B** 17 g **C** 21 g **D** 170 g

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 8

8	**B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Different energy sources are used to generate electricity.

(a) Energy sources are renewable or non-renewable.

(i) Nuclear energy is described as a non-renewable source.

Explain what is meant by a *non-renewable* energy source.

.....

.....[1]

(ii) Four of the energy sources used are:

hydroelectric **oil** **geothermal** **wind**

Write the name of these energy sources in the correct column of the table below.

non-renewable	renewable and caused by energy from the Sun	renewable and not caused by energy from the Sun

[3]

(iii) State one way in which using nuclear energy is better for the environment than using oil.

.....

.....[1]

(iv) State one way in which using nuclear energy is worse for the environment than using oil.

.....

.....[1]

(b) Fig. 9.1 is a block diagram of a power station that produces electrical energy from oil.

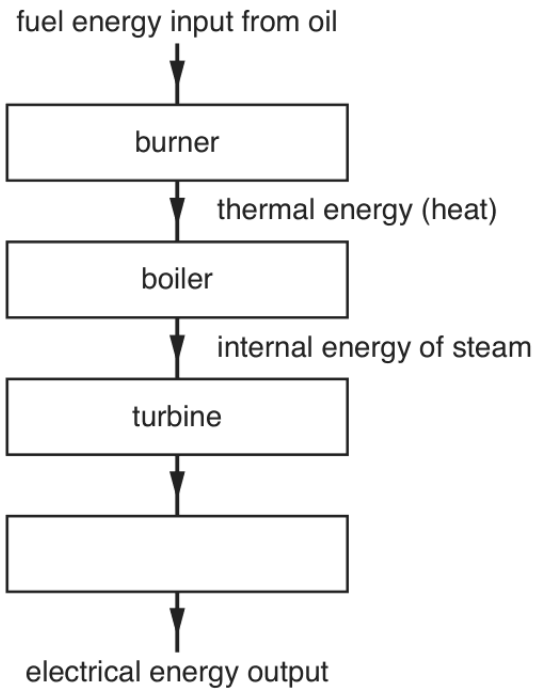


Fig. 9.1

(i) Write the name of the missing part of the power station in the empty box on Fig. 9.1. [1]

(ii) State the form of energy that the turbine possesses.

.....[1]

(iii) A small boiler in the power station contains 24 m^3 of water at 30 °C. High pressure in the boiler increases the boiling point of water to 120 °C.

Thermal energy supplied to the boiler is used to heat the water from 30 °C to 120 °C and then to turn it all to steam at 120 °C.

The density of water is 1000 kg/m^3.
The specific heat capacity of water is 4200 J/(kg °C).
The specific latent heat of vaporisation of water is 2.3×10^6 J/kg.

1. Calculate the mass of water in the boiler.

mass = ...[1]

2. Calculate the total thermal energy (heat) supplied to the boiler.

thermal energy = ...[4]

(iv) The electrical energy output from the power station is transmitted over long distances at a high voltage.

Explain why electricity is transmitted at a high voltage.

.....

.....

.....

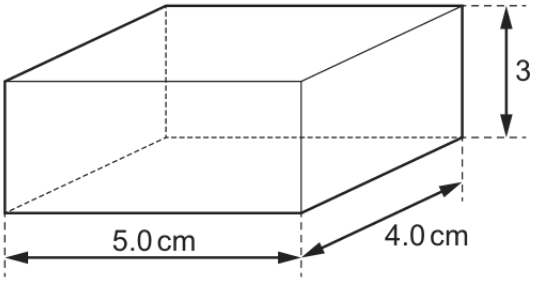
.....[2]

Section B

Question	Answer	Marks
9(a)(i)	not being replaced **or** will run out	**B1**
9(a)(ii)	only oil in 1st column	**B1**
	only wind and hydroelectric in 2nd column	**B1**
	geothermal in 3rd column	**B1**
9(a)(iii)	less greenhouse gases / global warming / acid rain / **or** toxic gases / oil spills **and** how they affect a named organism / ecosystem	**B1**
9(a)(iv)	nuclear waste / radioactive waste causes (storage) problems **or** explosion / melt down / leak emits radioactivity	**B1**
9(b)(i)	generator (and transformer)	**B1**
9(b)(ii)	kinetic (energy)	**B1**
9(b)(iii)	1 24 000 kg	**B1**
	2 formula (E) = mL **or** (E **=**) mcT seen	**C1**
	$24\,000 \times 2.3 \times 10^6$ **or** $5.5(2) \times 10^{10}$ (J)	**C1**
	$24\,000 \times 4200 \times 90$ **or** $9.0(72) \times 10^9$ (J)	**C1**
	6.4×10^{10} J	**A1**
9(b)(iv)	low current	**B1**
	less energy / heat loss (in resistance or cables) **or** thinner wires can be used	**B1**

QUESTION · 5054_s17_qp_12.pdf · source page 5 of 20

11 The diagram shows a rectangular block.



(not to scale)

The density of the block is $2.5\,g/cm^3$.

What is the mass of the block?

A 18 g **B** 24 g **C** 50 g **D** 150 g

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11	D	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

Answer only one of the two alternative parts in **Question 6**.

1 Fig. 1.1 shows a painter standing on a wooden plank, directly above the right-hand support.

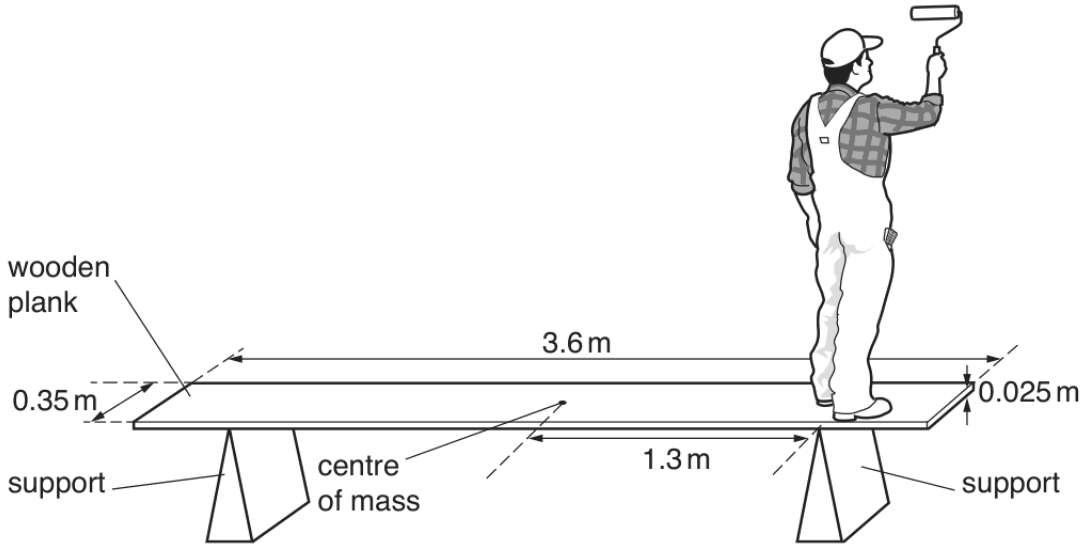


Fig. 1.1

The plank has length 3.6 m, width 0.35 m and thickness 0.025 m.

The gravitational field strength *g* is 10 N/kg and the mass of the plank is 23 kg.

(a) Calculate the density of the wood from which the plank is made.

density = ... [2]

(b) The centre of mass of the plank is in the middle of the plank at a distance of 1.3 m from each of the supports.

Calculate

(i) the weight of the plank,

weight = ... [1]

(ii) the moment of the plank about the right-hand support.

moment = ... [2]

(c) The painter moves further to the right along the plank and the plank rotates about the right-hand support.

Explain why the plank rotates.

.....

.....

.....

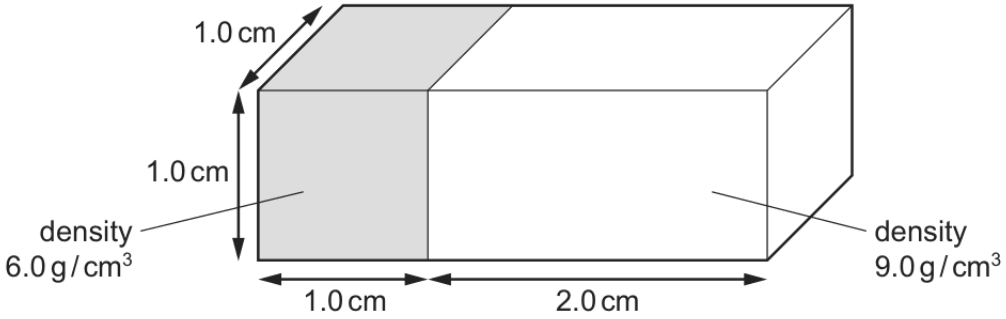
..... [1]

Section A

Question	Answer	Marks
1(a)	(ρ =)m / V **or** 23 / (3.6 × 0.35 × 0.025) **or** 23 / 0.0315	**C1**
	730 kg / m^3	**A1**
1(b)(i)	230 N	**B1**
1(b)(ii)	(Γ =)$Wx_{\perp r}$ **or** 230 × 1.3	**C1**
	300 Nm	**A1**
1(c)	moment of painter / clockwise moment (about support) is greater than / not equal to / different from moment of plank / anticlockwise moment	**B1**

QUESTION · 5054_w17_qp_12.pdf · source page 4 of 16

6 Two blocks are joined together.



One block has a density of $6.0\,g/cm^3$ and the other has a density of $9.0\,g/cm^3$.

What is the overall density of the two blocks joined together?

A $7.0\,g/cm^3$ **B** $7.5\,g/cm^3$ **C** $8.0\,g/cm^3$ **D** $15\,g/cm^3$

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 6

6	C	**1**

10 A boiler containing water is plugged into a 230 V mains supply. The heater in the boiler is operating at full power.

(a) The power used by the heater is 2.8 kW.

(i) Calculate the current in the heater.

current = ...[3]

(ii) The wiring in the cable from the socket and in the boiler remains safe and does not overheat provided that the current is less than 20 A.

Suggest a suitable fuse rating for the circuit.

.....[1]

(b) The specific heat capacity of water is 4200 J/(kg °C). The heater is used to increase the temperature of 6.3 kg of water.

(i) In a certain time, the temperature of the water increases from 23 °C to 49 °C.

Calculate the change in the internal energy of the water.

change in internal energy = ...[2]

(ii) The heater supplies thermal energy to the water at a constant rate. It takes more time for the temperature of the water to increase from 49 °C to 75 °C than from 23 °C to 49 °C. In both cases, the increase in temperature is the same.

Suggest why it takes more time for the temperature to increase from 49 °C to 75 °C.

.....

.....

.....[1]

(c) Fig. 10.1 shows the heater at the bottom of the water in the boiler.

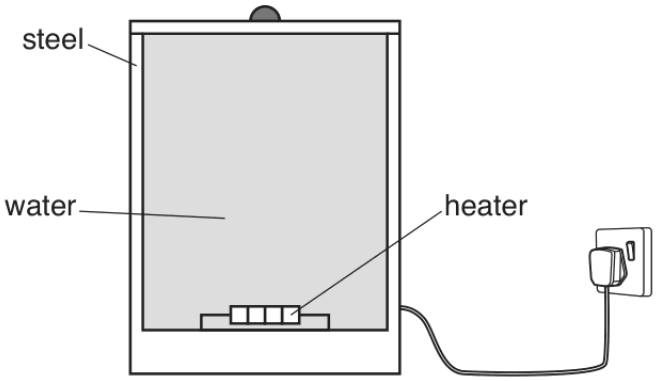


Fig. 10.1

(i) Explain how thermal energy is transferred from the heater throughout the water.

.....

.....

.....

.....

.....[3]

(ii) The part of the boiler that holds the water is made from steel.

Describe, in terms of particles, how thermal energy passes through the steel.

.....

.....

.....

.....

.....[3]

(iii) State and explain, in terms of particles, what happens to the density of the steel as its temperature increases.

.....

.....

.....

.....[2]

Question	Answer	Marks
10(a)(i)	(I =) P / V **or** 2.8 / 230 **or** 2800 / 230	**C1**
	0.012 **or** 12	**C1**
	12 A	**A1**
10(a)(ii)	12 A < integral number of amperes < 20 A	**B1**
10(b)(i)	(Q =) $m\Delta Tc$ **or** 6.3 × (49 – 23) × 4200 **or** 6.3 × 26 × 4200	**C1**
	6.9×10^5 J	**A1**
10(b)(ii)	more thermal energy is lost (per unit time)	**B1**

Question	Answer	Marks
10(c)(i)	(when heated) water expands / volume of water increases / water becomes less dense	**B1**
	heated water / water with more energetic particles rises	**B1**
	cooler / denser water sinks **or** circulation **or** convection current	**B1**
10(c)(ii)	**two** separate approaches **I** / **II**	
	I atoms / molecules / ions / particles vibrate (more violently)	**B1**
	collide with neighbouring atoms/molecules / ions / particles	**B1**
	energy / heat / vibration passed on (to neighbour)	**B1**
	or **II** particles / atoms / molecules / ions vibrate (more violently)	**B1**
	atoms / molecules / ions / particles strike electrons	**B1**
	electrons travel through transporting energy / heat	**B1**
10(c)(iii)	particles / molecules / atoms move apart (on average)	**B1**
	volume increases **and** density decreases	**B1**

10 (a) (i) Describe two differences between boiling and evaporation.

1. .....

.....

.....

2. .....

.....

.....

[4]

(ii) Explain, using ideas about molecules, why thermal energy is needed to boil a liquid.

.....

.....

.....

.....[2]

(b) In one type of bathroom shower, cold water passes through a metal pipe which contains an electric heater.

The cold water is heated and emerges from the shower head.

The temperature of the cold water before heating is measured and the hot water emerging from the shower in 1.0 minute is collected in a container.

Measurements and other data are:

temperature of water before heating = 16 °C
temperature of water after heating = 37 °C
volume of water collected in 1.0 minute = $4.6 \times 10^{-3}\,m^3$
specific heat capacity of water = 4200 J / (kg °C)
density of water = 1000 kg / m^3

(i) Calculate the mass of water leaving the shower in 1.0 s.

mass = ...[2]

(ii) Calculate the thermal energy (heat) gained by the water in 1.0 s.

energy = ...[3]

(iii) To calculate the efficiency of the shower, the value in **(b)(ii)** and the value of one other quantity are needed.

1. State what other quantity is needed.

.....

.....

2. Describe how these two values are used to calculate the efficiency of the shower.

.....

.....

.....
[2]

(iv) The metal pipe that contains the electric heater is earthed.

Explain why this is necessary.

.....

.....

.....

.....[2]

Question	Answer	Marks
10(a)(i)	*Pair 1* boiling occurs at one temperature / constant temperature evaporation occurs at all / any temperature	**B1**
	Pair 2 boiling occurs within / throughout liquid or causes bubbles evaporation occurs at surface	**B1**
	Pair 3 boiling requires a heat source / affected by rate of heating evaporation causes cooling / affected by wind etc.	**B1**
	Pair 4 temperature remains constant during boiling evaporation temperature drops / liquid cools	**B1**
10(a)(ii)	(thermal energy) used to break bonds **or** work done against intermolecular forces	**B1**
	molecules move further apart **or** P.E. increases **or** (molecules) push back the atmosphere	**B1**

Question	Answer	Marks
10(b)(i)	(M=) DV in any form algebraic or numerical	**C1**
	0.077 kg / s **or** 0.077 kg	**A1**
10(b)(ii)	(E=) mcT in any form algebraic or numerical	**C1**
	37 – 16 or 21 (°C) seen	**C1**
	6800 J	**A1**
10(b)(iii)	1input / electrical energy or power mentioned	**B1**
10(b)(iii)	2 (efficiency =) output (energy / power) ÷ input (energy / power)	**B1**
10(b)(iv)	avoid electrocution / shock / current through body **or** avoid pipe being live **or** to allow a current / charges to earth / ground **or** fuse blows to stop electrocution	**B1**
	when live / heater touches metal / water	**B1**

3 Equal volumes of metal, water and air are heated from 20 °C to 80 °C.

(a) State whether the metal, the water or the air expands the most.

.....[1]

(b) Fig. 3.1 is a diagram of the metal at 20 °C. The metal is a large, square sheet with a square hole at its centre. The sheet lies flat on a table.

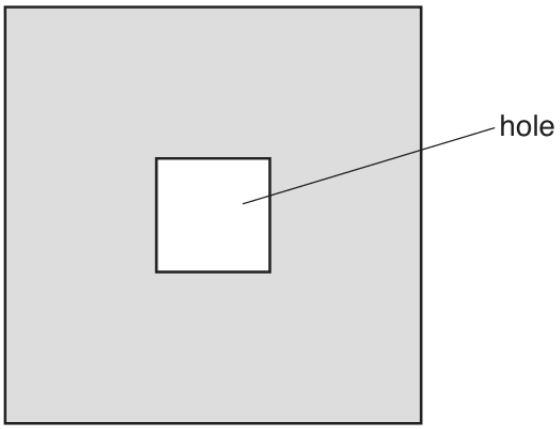


Fig. 3.1

On Fig. 3.1, draw the outline of the sheet and of the hole after the metal is heated. Make sure that the expansion is noticeable. [2]

(c) The metal has a mass of 5.0 kg and a density of 7.5×10^3 kg / m^3.

(i) Define *density*.

.....

.....[1]

(ii) Calculate the volume of the metal.

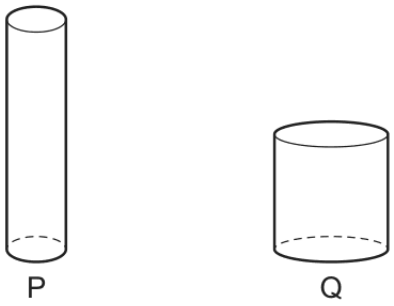
Give your answer to a suitable number of significant figures.

volume = ... [2]

Question	Answer	Marks
3(a)	the air	**B1**
3(b)	square larger	**B1**
	hole larger	**B1**
3(c)(i)	mass divided by volume **or** mass per unit volume	**B1**
3(c)(ii)	m / d **or** (V=) m / d **or** $5 / 7.5 \times 10^{3}$ i.e. rearrangement algebraic or numerical to show V as the subject	**C1**
	6.7×10^{-4} m^3 **or** 6.67×10^{-4} m^3 c.a.o.	**A1**

QUESTION · 5054_w18_qp_11.pdf · source page 4 of 16

7 Two cylinders P and Q are made of copper.



The height of P is twice the height of Q. The diameter of P is half the diameter of Q.

Which statement is correct?

A The density of cylinder P is four times that of cylinder Q.

B The density of cylinder P is twice that of cylinder Q.

C The density of cylinder P is equal to that of cylinder Q.

D The density of cylinder P is half that of cylinder Q.

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 A bucket of water is pulled up out of a well using a rope. Fig. 8.1 shows the rope winding on to a cylinder as the handle is turned.

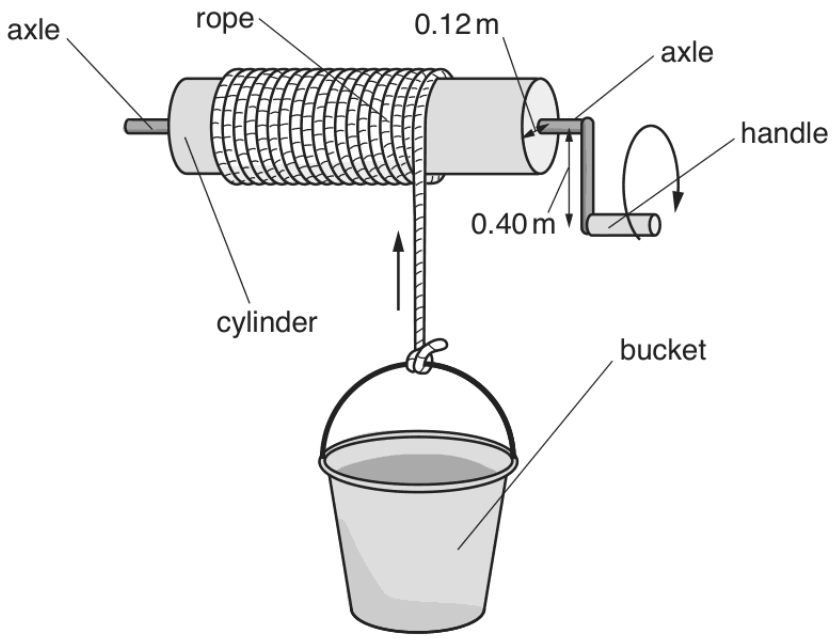


Fig. 8.1

The empty bucket has a mass of 1.0 kg.

(a) Complete the sentences that describe mass by filling in the gaps.

The mass of a body is a measure of the amount of .....

in the body. It resists a change in the state of ..... of

the body. [2]

(b) When the bucket is full, it contains $2.4 \times 10^{-2}\,m^3$ of water.

The gravitational field strength g is equal to 10 N/kg.

(i) Explain what is meant by a *gravitational field*.

.....

..... [1]

(ii) The density of water is 1000 kg / m^3.

Determine the total weight of the bucket and the water.

weight = ... [3]

(c) The radius of the cylinder is 0.12 m and the handle is 0.40 m from the axle of the cylinder. The weight of the bucket and the water produce a moment that acts on the cylinder.

(i) Calculate this moment.

moment = ... [2]

(ii) Calculate the minimum force on the handle that balances this moment.

force = ... [1]

(d) A farmer pulls the bucket of water up at a constant speed. He needs to exert a force on the handle that is greater than that calculated in **(c)(ii)**. He notices that there is a slight increase in temperature where the axle is in contact with the frame holding it.

(i) Suggest two reasons why the force exerted is greater than the value calculated.

1. .....

.....

2. .....

.....
[2]

(ii) State the energy changes that are taking place as the bucket is being lifted at a constant speed.

.....

.....

..... [3]

(e) State what can be deduced about the forces acting on the bucket when it is travelling upwards at a constant speed.

.....

..... [1]

MARK SCHEME · 5054_w18_ms_21.pdf · source page 7 of 10 · Question 8

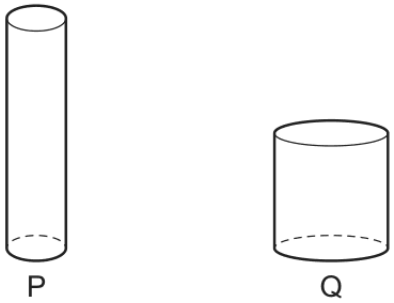
Question	Answer	Marks
8(a)	substance **or** matter	**B1**
	rest **or** motion	**B1**
8(b)(i)	(a region of space) where a mass experiences a force (due to gravitational attraction)	**B1**
8(b)(ii)	(m =) ρV **or** $1000 \times 2.4 \times 10^{-2}$ **or** 24 (kg)	**C1**
	25 (kg) **or** 240 (N)	**C1**
	250 N	**A1**
8(c)(i)	(Γ =) Fx **or** 250×0.12	**C1**
	30 N m	**A1**
8(c)(ii)	75 N	**B1**
8(d)(i)	any **two** from: friction (between axle and cylinder) weight of rope force not perpendicular (to radius) rope wraps over itself (and increases moment)	**B2**

MARK SCHEME · 5054_w18_ms_21.pdf · source page 8 of 10 · Question 8

Question	Answer	Marks
8(d)(ii)	from chemical (energy) **or** chemical (energy) as first term	**B1**
	to gravitational potential (energy) **or** to thermal / heat (energy) as last term	**B1**
	to thermal / internal (energy) / heat **and** thermal / internal (energy) / heat	**B1**
8(e)	no resultant force **or** forces balance / cancel **or** tension equals weight (of bucket) **or** upwards force equals downward force	**B1**

QUESTION · 5054_w18_qp_12.pdf · source page 4 of 16

7 Two cylinders P and Q are made of copper.



The height of P is twice the height of Q. The diameter of P is half the diameter of Q.

Which statement is correct?

A The density of cylinder P is four times that of cylinder Q.

B The density of cylinder P is twice that of cylinder Q.

C The density of cylinder P is equal to that of cylinder Q.

D The density of cylinder P is half that of cylinder Q.

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A student records the mass and volume of two irregular objects, one made of iron and the other of copper. The student uses the equipment shown in Fig. 1.1.

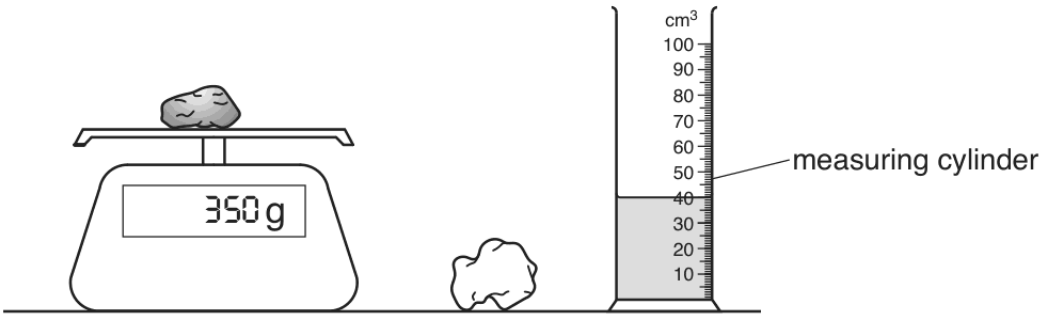


Fig. 1.1

Table 1.1 shows the results obtained.

Table 1.1

	mass / g	volume / cm^3
iron	400	51
copper	350	39

(a) Describe how to determine the volume of an irregular object with the measuring cylinder.

.....

.....

.....

.....

..... [2]

(b) Calculate the density of iron.

density = ... [2]

(c) A third object, which is also made of copper, has the same volume as the iron object.

Calculate the mass of this copper object.

mass = ... [2]

(d) State and explain what happens to the density of the iron object when it is heated.

.....

.....

..... [1]

[Total: 7]

Question	Answer	Marks
1(a)	reading / volume with water / liquid alone **or** use displacement can filled (to spout)	**B1**
	subtracted from reading (of volume) with object (submerged) in cylinder **or** place rock in displacement can and measure volume of overflow with measuring cylinder	**B1**
1(b)	(d =) M / V algebraic or numerical	**C1**
	7.8 g / cm^3	**A1**
1(c)	350 × 51 / 39 **or** (density of Cu =) 350 / 39 **or** or 9.0 (g / cm^3) seen	**C1**
	460 g	**A1**
1(d)	(density) decreases **and** volume increases / object expands **or** molecular explanation of expansion	**B1**

5 An electronic balance is calibrated to display the weight of an object that is placed on its top plate.

Fig. 5.1 shows the balance with an empty measuring cylinder on the top plate.

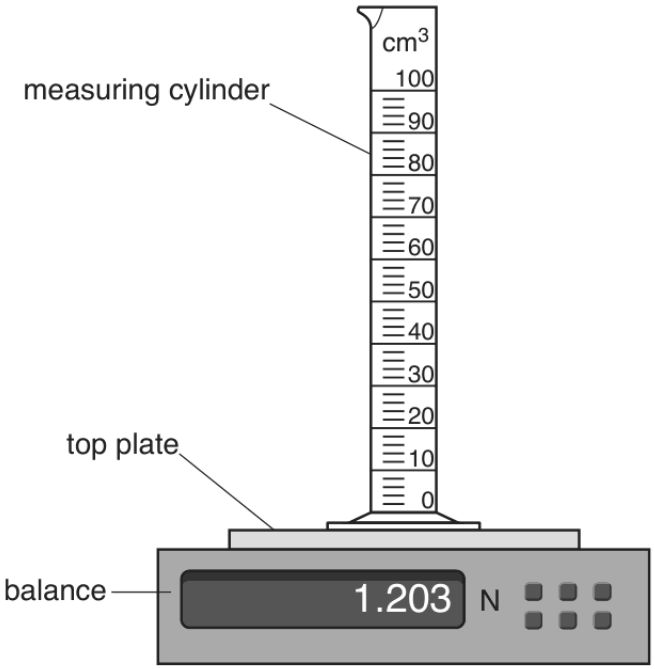


Fig. 5.1

A student pours a small quantity of oil into the measuring cylinder and records the volume and the new reading on the balance.

Fig. 5.2 shows how the reading on the balance varies with the volume of oil in the measuring cylinder.

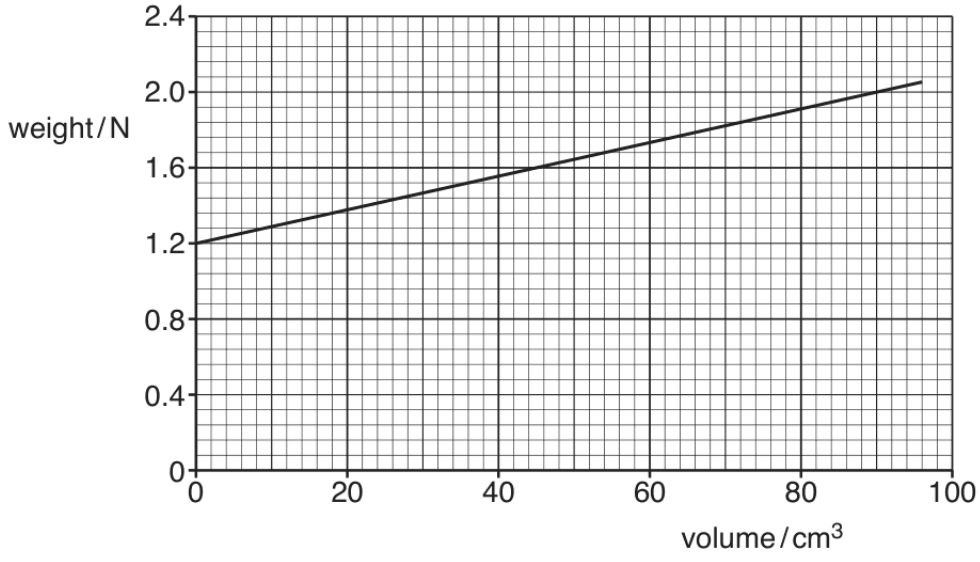


Fig. 5.2

The gravitational field strength g is equal to 10 N/kg.

Using Fig. 5.2, determine the density of the oil.

density = ... [4]

MARK SCHEME · 5054_w19_ms_21.pdf · source page 6 of 10 · Question 5

Question	Answer	Marks
5	two pairs of results from the line **and** separated by $\geqslant 50\ cm^3$	**B1**
	or (ρ) = m / V use of results to obtain gradient **or** any mass / corresponding volume	**C1**
	$W = mg$ **or** (ρ =) gradient / g **or** $0.00881\ (N / cm^3) \leqslant \rho \leqslant 0.00895\ (N / cm^3)$	**C1**
	$0.000881\ kg / cm^3 \leqslant \rho \leqslant 0.000895\ kg / cm^3$ **or** $0.881\ g / cm^3 \leqslant \rho \leqslant 0.895\ g / cm^3$	**A1**

QUESTION · 5054_s20_qp_11.pdf · source page 4 of 16

11 A single metal bolt has a mass of 34 g.

Three of the bolts are immersed in a measuring cylinder that contains 30 cm^3 of water.

The reading on the measuring cylinder rises to 42 cm^3.

What is the density of the metal?

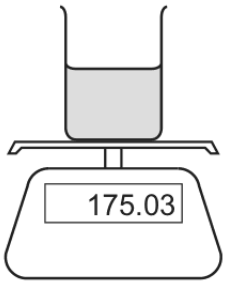
A 0.81 g/cm^3 **B** 2.8 g/cm^3 **C** 7.5 g/cm^3 **D** 8.5 g/cm^3

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 11

11	D	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 4 of 16

10 The diagram shows the equipment used to determine the density of a liquid.



Which equation is used to calculate the density of the liquid?

A $\frac{\text{mass of beaker and liquid} - \text{mass of beaker}}{\text{volume of liquid}}$

B $\frac{\text{mass of beaker and liquid} - \text{mass of liquid}}{\text{volume of liquid}}$

C $\frac{\text{volume of liquid}}{\text{mass of beaker and liquid} - \text{mass of beaker}}$

D $\frac{\text{volume of liquid}}{\text{mass of beaker and liquid} - \text{mass of liquid}}$

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 10

10	A	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 5 of 16

9 Which expression is used to calculate density?

A mass × volume

B $\frac{\text{mass}}{\text{volume}}$

C volume + mass

D $\frac{\text{volume}}{\text{mass}}$

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 9

9	B	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A glass beaker has a mass of 50 g. A liquid of density 1.8 g / cm^3 is poured into the beaker until it reaches the 200 cm^3 mark.

(a) Calculate the total mass of the beaker and its contents.

mass = ... [3]

(b) The centre of mass of a metre rule is at the 50 cm mark.

(i) State what is meant by *centre of mass*.

.....

..... [1]

(ii) The metre rule is placed on a pivot. The tip of the pivot is under the 80 cm mark on the rule.

The beaker with its contents is then placed at different positions along the rule until the rule is balanced.

Fig. 1.1 shows the arrangement with the rule balanced.

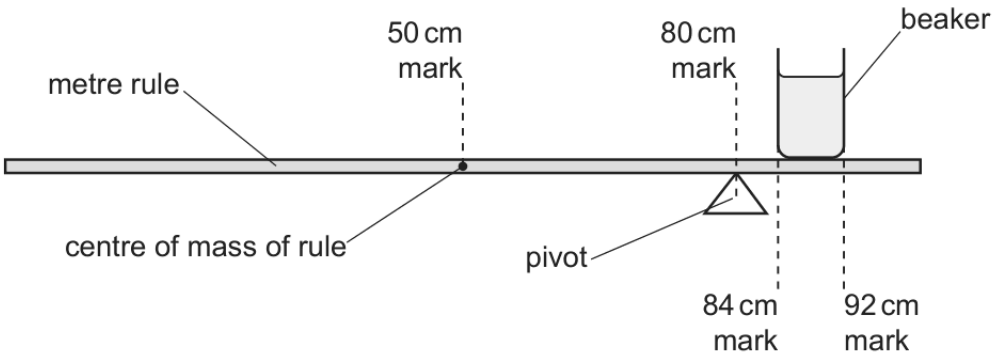


Fig. 1.1

One side of the beaker is at the 84 cm mark and the other side is at the 92 cm mark.

Calculate the mass of the rule.

mass = ... [3]

[Total: 7]

Question	Answer	Marks
1(a)	(m =) ρV **or** 1.8 × 200 **or** 360	**C1**
	(1.8 × 200) + 50 **or** 360 + 50 **or** 360 **or** candidate's mass + 50	**C1**
	410 g **or** 0.41 kg	**A1**
1(b)(i)	(place / region) where the mass appears to be concentrated **or** where the object balances	**B1**
1(b)(ii)	88 (cm) seen **or** (84 + 92) / 2 seen **or** 30 (cm) seen **or** 8(.0 cm) seen **or** 164 (g)	**C1**
	$m_1gx_1 = m_2gx_2$ **or** $F_1x_1 = F_2x_2$ **or** $W_1x_1 = W_2x_2$ **or** (m_1 =) m_2x_2 / x_1 **or** 410 × (88 – 80) / 30 **or** 164 (g)	**C1**
	110 g **or** 0.11 kg	**A1**

QUESTION · 5054_s21_qp_11.pdf · source page 4 of 16

7 Water is added to a measuring cylinder containing 100 cm^3 of liquid paraffin.

(The density of paraffin is 0.80 g/cm^3 and that of the water is 1.0 g/cm^3.)

As the water is added, the level of the paraffin rises to 150 cm^3. The paraffin and water do not mix.

What finally is the total mass of liquid in the measuring cylinder?

A 130 g **B** 140 g **C** 167 g **D** 175 g

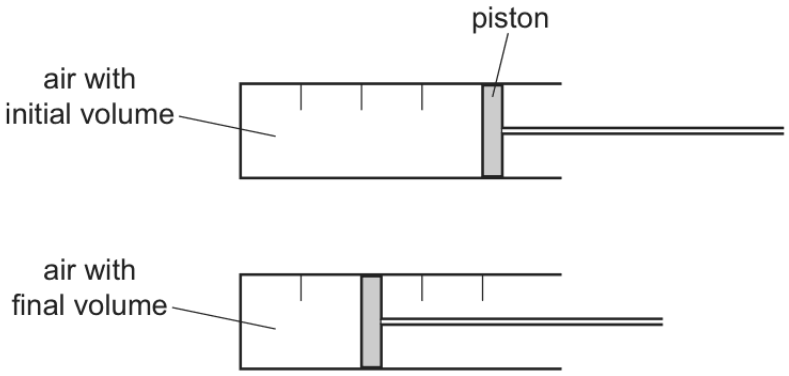
MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 7

7	A	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 4 of 16

7 Air is trapped in a cylinder by a piston.

The piston is moved inwards so that the volume of the air reduces.



The density of the air in the syringe at the initial volume is $0.0012\,g/cm^3$.

What is the density of the air in the syringe at the final volume?

A $0.0006\,g/cm^3$

B $0.0012\,g/cm^3$

C $0.0024\,g/cm^3$

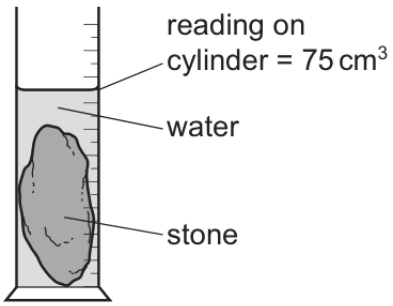
D $0.0048\,g/cm^3$

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 3 of 20

6 A stone is placed in a measuring cylinder of water, as shown.



The volume of water in the cylinder before adding the stone is 15 cm^3 and the stone has a mass of 90 g.

What is the density of the stone?

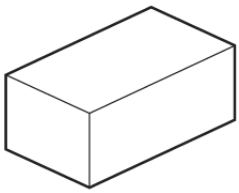
A 1.0 g / cm^3 **B** 1.2 g / cm^3 **C** 1.5 g / cm^3 **D** 6.0 g / cm^3

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 6

6	C	**1**

QUESTION · 5054_w21_qp_12.pdf · source page 4 of 16

7 A student takes measurements to find the density of a regularly shaped block.



Which formula is used to calculate the density of the block?

A $\text{density} = \frac{\text{mass}}{\text{volume}}$

B $\text{density} = \frac{\text{volume}}{\text{mass}}$

C $\text{density} = \frac{\text{weight}}{\text{area}}$

D $\text{density} = \frac{\text{weight}}{\text{volume}}$

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 7

7	A	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a jug of liquid and an empty measuring cylinder that is on an electronic balance.

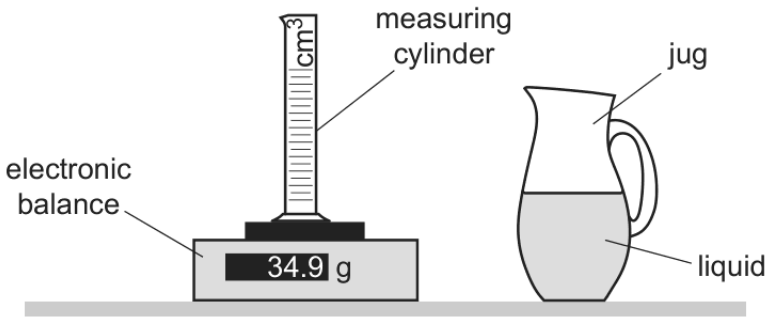


Fig. 1.1

(a) The electronic balance uses the weight of the measuring cylinder to determine its mass.

Weight and mass have different units.

State **two** other ways in which weight differs from mass.

1. .....

.....

2. .....

.....

[2]

(b) Describe how to determine the density of the liquid using the apparatus shown in Fig. 1.1.

.....

.....

.....

.....

.....

..... [4]

(c) The density of the liquid is 780 kg / m^3.

Determine the volume of 0.65 kg of the liquid.

volume = ... [2]

[Total: 8]

Question	Answer	Marks
1(a)	any **two** from: weight / it is a vector / has a direction weight / it is a force weight / it depends on location / gravitational field strength weight / it is measured with a spring balance (not a beam balance) / electronic balance / top-pan balance / newton meter	**B2**
1(b)	(pour liquid into) measuring cylinder **and** read / find the volume **or** zero the reading on the balance	**B1**
	take reading on balance **or** find the mass (of full measuring cylinder) **or** (pour liquid into) measuring cylinder **and** read / find the volume	**B1**
	subtract original reading / 34.9 g (from reading on balance) **or** take reading on balance **or** find the mass (of full measuring cylinder)	**B1**
	divide difference between readings by volume **or** divide (final) reading on balance by volume	**B1**
1(c)	(ρ =) 0.65 / 780	**C1**
	8.3×10^{-4} m^3 **or** 830 cm^3	**A1**

QUESTION · 5054_s22_qp_11.pdf · source page 3 of 16

5 A plastic cube, of density ρ, is compressed so that each of its sides becomes one-half of its original length.

What is the new density of the material of the cube?

A 0.125ρ **B** 0.50ρ **C** 2.0ρ **D** 8.0ρ

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 5

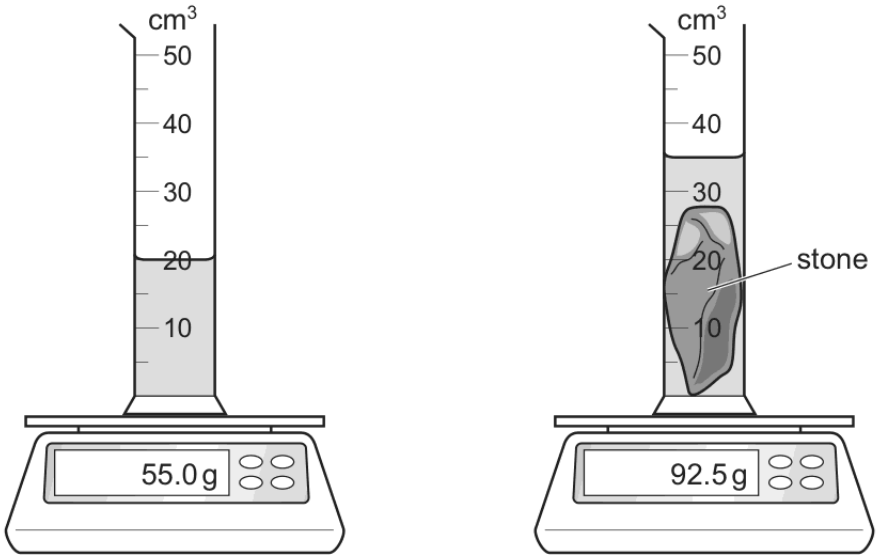
5	D	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 3 of 20

6 A measuring cylinder containing water is placed on a balance.

A stone is placed into the water.

The diagram shows the readings on the balance and on the measuring cylinder.



What is the density of the stone?

A $1.1\,g/cm^3$ **B** $1.5\,g/cm^3$ **C** $2.5\,g/cm^3$ **D** $2.6\,g/cm^3$

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 6

6	C	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 5 of 16

11 A student finds the density of an irregularly shaped object. He chooses his equipment from this list.

1 stop-watch

2 measuring cylinder

3 balance

Which equipment does the student need to use?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 11

11	D	**1**

2 The foundations that support a building are long concrete cylinders that are pointed at one end. A pile-driver is a machine that forces the pointed concrete cylinders into the ground.

Fig. 2.1 shows a pile-driver.

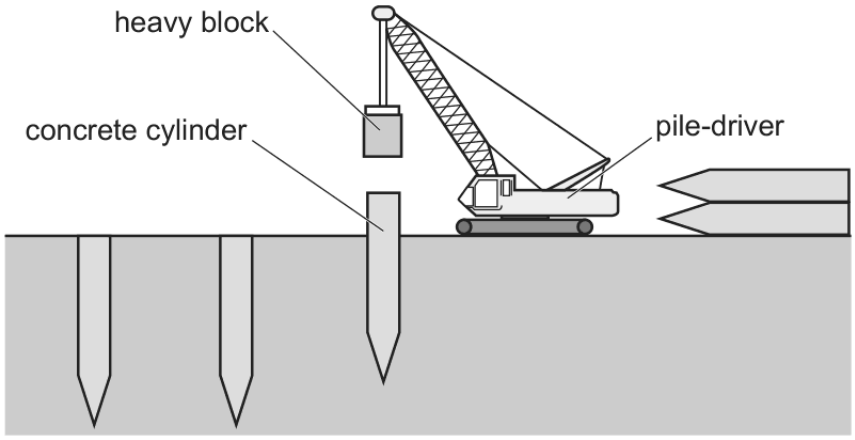


Fig. 2.1

A heavy block of mass 2.9×10^4 kg is lifted into the air then dropped onto the top of a concrete cylinder. This forces the cylinder into the ground.

(a) Fig. 2.2 shows the heavy block.

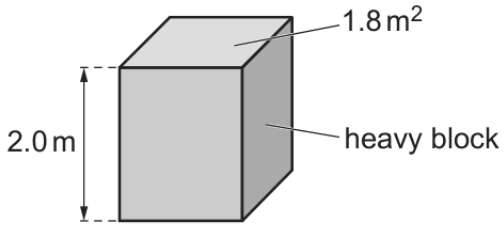


Fig. 2.2

The block is 2.0 m tall and has a cross-sectional area of 1.8 m^2.

Calculate the density of the material used to make the block.

density = ... [2]

(b) The pile-driver lifts the block from the top of a concrete cylinder, through a height of 0.80 m.

The gravitational field strength g is 10 N / kg.

(i) Calculate the gravitational potential energy gained by the block.

gravitational potential energy = ... [2]

(ii) The block is then dropped from rest onto the top of the concrete cylinder.

Calculate the speed of the block just before it hits the concrete cylinder.

speed = ... [3]

[Total: 7]

Question	Answer	Marks
2(a)	($\rho =$) m / V **or** ($\rho =$) m / hA **or** $2.9 \times 10^4 / 3.6$ **or** $2.9 \times 10^4 / (2.0 \times 1.8)$	**C1**
	8.1×10^3 kg / m^3	**A1**
2(b)(i)	(g.p.e. =) $mg\Delta h$ **or** $2.9 \times 10^4 \times 10 \times 0.80$	**C1**
	2.3×10^5 J	**A1**
2(b)(ii)	k.e. / ½mv^2 = g.p.e. / $mg\Delta h$ / 2.3×10^5 **or** ($v =$) $\sqrt{2gDh}$ **or** $\sqrt{2 \times 10 \times 0.80}$	**C1**
	($v =$) $\sqrt{2gDh}$ **or** $\sqrt{2 \times 10 \times 0.80}$	**C1**
	4.0 m / s	**A1**

QUESTION · 5054_s23_qp_11.pdf · source page 5 of 20

6 A student determines the density of a solid metal cube.

Which pieces of equipment does the student use?

A a balance and a force meter

B a balance and a ruler

C a micrometer and a measuring cylinder

D a ruler and a measuring cylinder

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 6

6	B	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 4 of 16

7 A student determines the density of a liquid.

Which equipment does the student use?

A a balance and a ruler

B a balance and a measuring cylinder

C a beaker and a measuring cylinder

D a ruler and a beaker

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 7

7	B	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 3 of 20

4 A student finds that the mass of a rock is 65 g.

She pours 30 cm^3 of water into a measuring cylinder.

She places the rock in the water so that the water covers the rock completely. The water level rises to 55 cm^3.

What is the density of the rock?

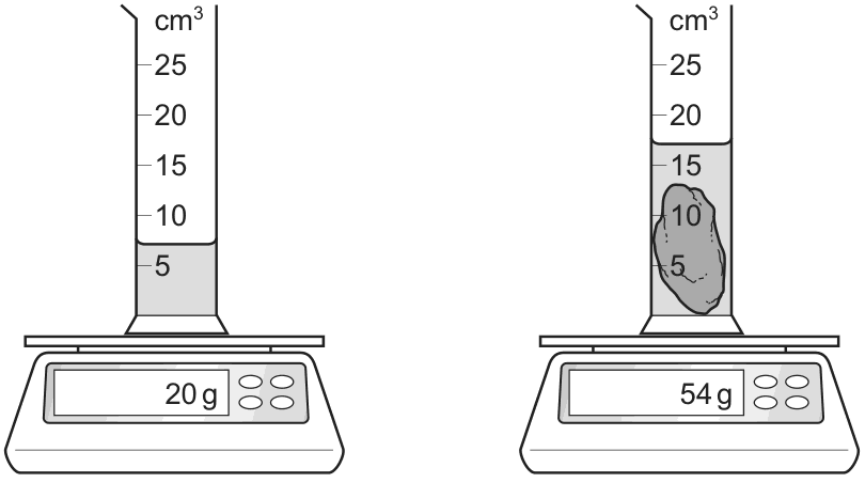
A 0.38 g/cm^3 **B** 1.2 g/cm^3 **C** 2.2 g/cm^3 **D** 2.6 g/cm^3

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 4

4	D	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 2 of 16

1 A measuring cylinder containing water is placed on a top-pan balance. An object is placed into the measuring cylinder.



What is the mass and what is the volume of the object?

	mass / g	volume / cm^3
A	10	7
B	10	34
C	34	10
D	34	17

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 1

1	C	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 3 of 16

4 Two cubes, X and Y, have the same mass, but the length of a side of X is twice that of Y.

What is the value of the ratio $\frac{\text{density of X}}{\text{density of Y}}$?

A 0.125 **B** 1.0 **C** 4.0 **D** 8.0

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 4

4	A	**1**

2 Fig. 2.1 shows a small swimming pool containing water.

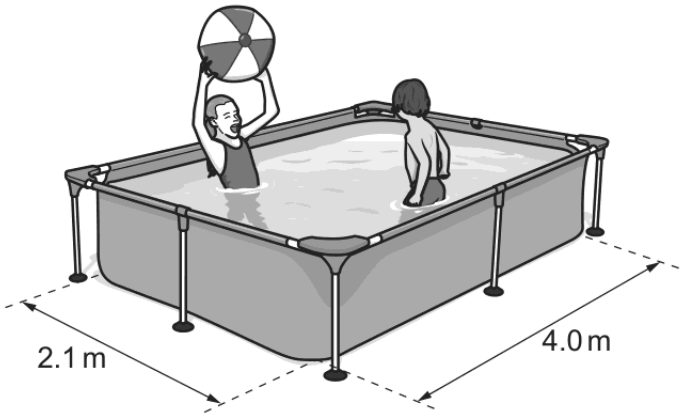


Fig. 2.1

The depth of water in the pool is 0.80 m. The density of water is 1000 kg / m^3.

(a) (i) Show that the mass of water in the pool is approximately 6700 kg.

[2]

(ii) Define 'pressure'.

.....

..... [1]

(iii) Calculate the pressure on the base of the pool due to the water.

pressure = ... Pa [2]

(iv) The water in the pool is initially at a temperature of 10 °C.

The temperature rises when 5.1×10^8 J of energy is transferred to the water.

The specific heat capacity of water is 4200 J / (kg °C).

Calculate the final temperature of the water.

temperature = ... °C [3]

(b) (i) Explain, in terms of the movement of particles, how evaporation causes cooling.

.....

.....

.....

..... [2]

(ii) Changes to factors in the environment of the swimming pool can cause an increase or decrease in the amount of evaporation from the surface of the water.

State **two** changes to environmental factors that **increase** the amount of evaporation from the surface of the water.

1 .....

2 .....
[1]

[Total: 11]

MARK SCHEME · 5054_s24_ms_22.pdf · source page 5 of 11 · Question 2

Question	Answer	Marks
2(a)(i)	(V =) length × breadth × height in any form	**C1**
	6.7(2) × 1000 **clearly seen** or 6720 (kg) seen	**A1**
2(a)(ii)	force per unit area	**B1**
2(a)(iii)	Either (P =) dgh in any form, e.g.1000 × 9.8 x 0.8 Or (area of base) 2.1 × 4.0 **or** 8.4 (m^2)	**C1**
	7800 (Pa)	**A1**

MARK SCHEME · 5054_s24_ms_22.pdf · source page 6 of 11 · Question 2

Question	Answer	Marks
2(a)(iv)	(ΔT =) E / mc in any form	**C1**
	18 (°C)	**C1**
	28 (°C)	**A1**
2(b)(i)	fastest moving / more energetic particles escape	**B1**
	leaving slowest / less energetic particles	**B1**
2(b)(ii)	ANY 2 from higher temperature / wind / low humidity	**B1**

1 A trolley of mass 0.20 kg is at rest on a frictionless surface.

A block of wood is attached to the trolley.

The volume of the block of wood is 0.0012 m^3, and the density of the wood is 650 kg/m^3.

(a) Calculate the combined mass of the trolley and block.

combined mass = ... kg [2]

(b) Fig. 1.1 shows that a student attaches a spring of spring constant 14 N/m to the front of the trolley.

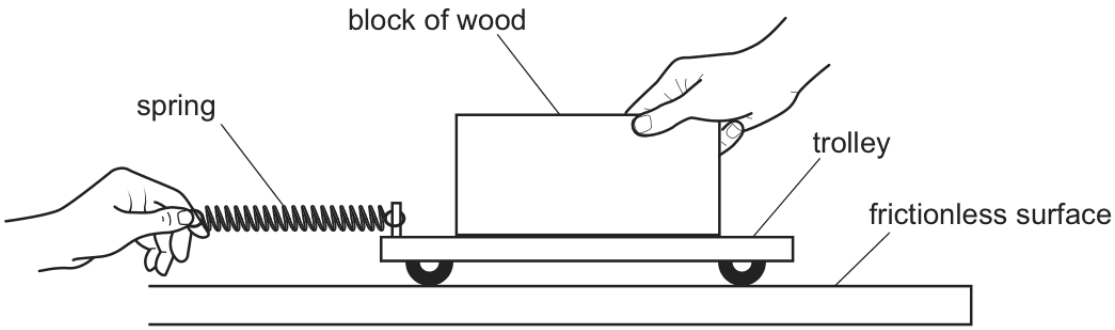


Fig. 1.1

The student holds the trolley and block and stretches the spring. He releases the trolley and block. The trolley and block accelerate from rest.

As the trolley and block accelerate, the student keeps the extension of the spring at 0.035 m.

(i) Calculate the force exerted by the spring on the trolley.

force = ... N [2]

(ii) The trolley and block are pulled a distance of 0.86 m by the spring.

Calculate the work done on the trolley.

work done = ... J [2]

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(iii) Explain why the power transferred to the trolley and block increases as the speed increases.

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
1(a)	($m =$) ρV **or** 650×0.0012 **or** 0.78	**C1**
	(combined mass =) 0.98 (kg)	**A1**
1(b)(i)	($F =$) kx **or** 14×0.035	**C1**
	0.49 (N)	**A1**
1(b)(ii)	($W =$) Fx **or** 0.49×0.86	**C1**
	0.42 (J)	**A1**
1(b)(iii)	the distance moved / work done (in one second / unit time / a given time) increases (as the speed increases)	**B1**
	the work done in one second / unit time increases	**B1**

QUESTION · 5054_w24_qp_12.pdf · source page 4 of 16

7 A material has a density of $2.4\,g/cm^3$.

What is the density in kg/m^3?

A $0.0024\,kg/m^3$

B $0.24\,kg/m^3$

C $24\,kg/m^3$

D $2400\,kg/m^3$

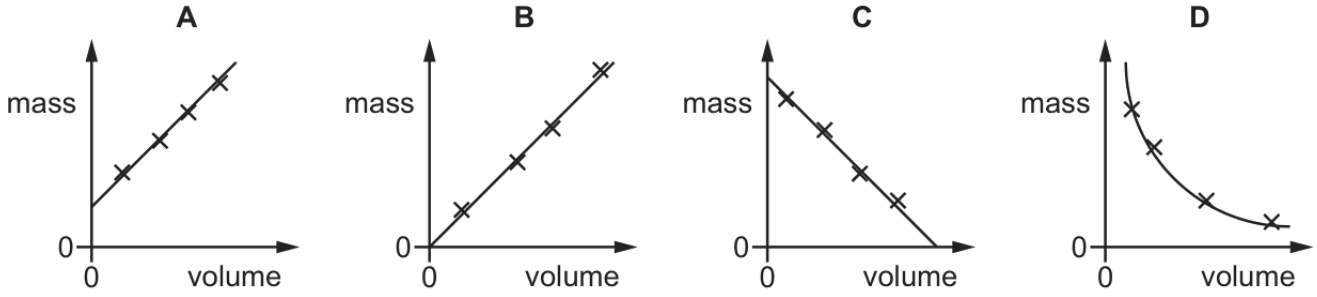
MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 7

7	D	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 3 of 20

5 Some students measure the masses and the volumes of differently sized samples of a type of wood.

Which graph shows their results?



MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 5

5	B	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 4 of 16

7 A student determines the density of a small rock.

The weight of the small rock is 1.2 N.

She pours 100 cm^3 of water into a measuring cylinder.

She carefully submerges the rock in the 100 cm^3 of water. The new reading on the measuring cylinder is 147 cm^3.

What is the density of the rock?

A 0.026 N/cm^3 **B** 0.83 g/cm^3 **C** 1.8 N/cm^3 **D** 2.6 g/cm^3

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 7

7	D	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 4 of 16

7 A student measures the density of an irregularly shaped stone.

Which items of equipment are needed?

A a balance and a measuring cylinder containing water

B a balance and a ruler

C a ruler and a measuring cylinder containing water

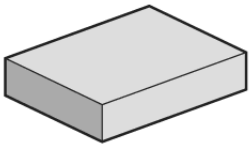
D a measuring cylinder containing water only

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 7

7	A	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 6 of 16

11 The diagram shows a solid metal block.



Which equation is used to calculate the density of the metal from which the block is made?

A density = mass × volume

B $\text{density} = \frac{\text{mass}}{\text{volume}}$

C $\text{density} = \frac{\text{weight}}{\text{area}}$

D $\text{density} = \frac{\text{weight}}{\text{volume}}$

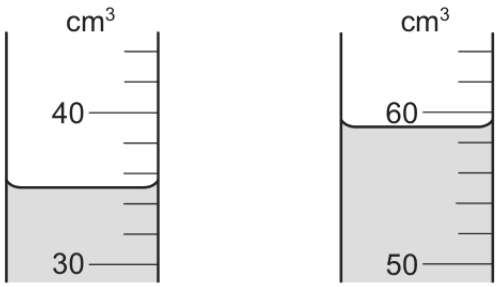
MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 11

11	B	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 3 of 16

7 A quantity of water is poured into a measuring cylinder. A small piece of rock is then added carefully.

The diagrams show the water levels and the measuring cylinder scales.



What are the correct values for the volumes of water and rock?

	volume of water / cm^3	volume of rock / cm^3
A	32.5	22.0
B	32.5	54.5
C	35.0	24.0
D	35.0	59.0

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

3 Fig. 3.1 shows a firefighter directing a jet of water at a wall.

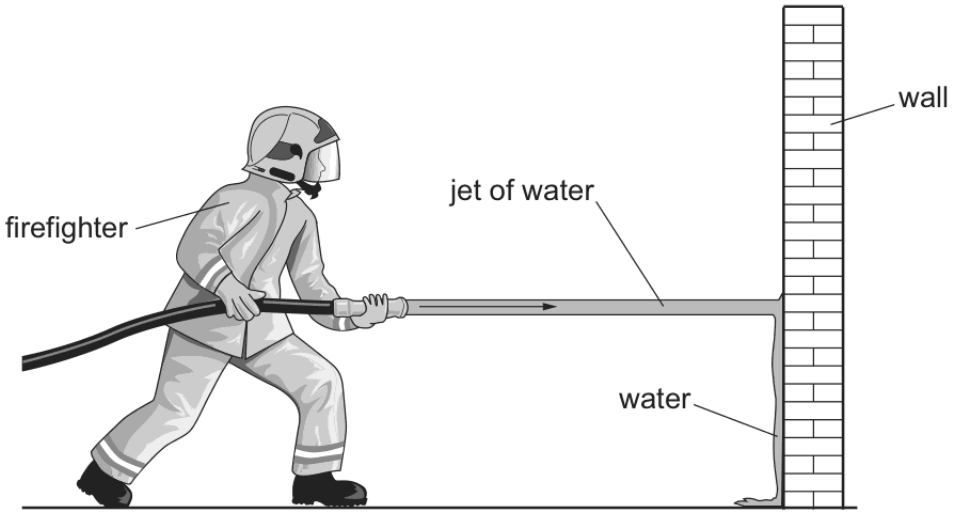


Fig. 3.1

In 1.0 s, a volume of $7.5 \times 10^{-3}\,m^3$ of water hits the wall horizontally, at a speed of 24 m/s.

The density of water is $1000\,kg/m^3$.

(a) Calculate:

(i) the mass of the water that hits the wall in 1.0 s

mass = ... kg [2]

(ii) the horizontal momentum of the water that hits the wall in 1.0 s.

momentum = ... kg m/s [2]

(b) The horizontal momentum of the water decreases to zero when the water hits the wall. None of the water bounces back from the wall.

Explain why the momentum calculated in **(a)(ii)** is equal in size to the horizontal force exerted on the water by the wall.

.....

.....

..... [2]

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(c) Explain, in terms of Newton's third law, why there is a horizontal force exerted on the wall by the water.

.....

.....

.....

.....

..... [2]

(d) The pump that forces the water to flow through the hose is adjusted, and the speed of the water leaving the hose doubles.

Explain why the force exerted by the water on the wall increases by a factor of 4.

.....

.....

..... [1]

[Total: 9]

MARK SCHEME · 5054_w25_ms_22.pdf · source page 9 of 14 · Question 3

Question	Answer	Marks
3(a)(i)	(m =) ρV **or** $1000 \times 7.5 \times 10^{-3}$	**C1**
	7.5 (kg)	**A1**
3(a)(ii)	(p =) mv **or** 7.5×24	**C1**
	180 (kg m / s)	**A1**
3(b)	force needed to change momentum **or** answer to **(a)(ii)** is change in momentum	**C1**
	force = $\frac{\text{change in momentum}}{\text{time taken}}$ **or** the momentum of the water that hits the wall in 1.0 s is the rate of change of momentum	**A1**

MARK SCHEME · 5054_w25_ms_22.pdf · source page 10 of 14 · Question 3

Question	Answer	Marks
3(c)	Newton's third law quoted **or** when object A exerts a force on object B, then object B exerts an equal and opposite force on object A **or** forces (only) occur in pairs	**B1**
	the wall exerts a force on the water	**B1**
3(d)	mass (flowing through pipe / hitting the wall in 1.0 s) doubles **and** momentum (water hitting wall in 1.0 s) increases by a factor of 4	**B1**

9 Table 9.1 gives data about some of the planets in the Solar system.

Table 9.1

planet	distance from Sun / km	orbital period / days	mass of planet / kg	average density / kg / m^3
Mercury	5.8×10^7	88	3.3×10^{23}	5400
Venus	1.1×10^8	220	4.9×10^{24}	5200
Earth	1.5×10^8	370	6.0×10^{24}	5500
Mars	2.3×10^8	690	6.4×10^{23}	3900
Jupiter	7.8×10^8	4 300	1.9×10^{27}	1300
Saturn	1.4×10^9	11 000	5.7×10^{26}	690

The orbital period is the time taken by a planet to make one orbit around the Sun.

(a) (i) The distance between Earth and Venus is not constant.

Explain why.

You may draw a diagram if you wish.

.....

.....

.....

..... [2]

(ii) Use Table 9.1 to determine the smallest distance between Earth and Venus.

smallest distance = ... $\times 10^7$ km [1]

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(b) A student suggests that the larger the mass of the planet, the greater is its orbital period.

Using data from Table 9.1, suggest whether the student is correct or not.

Explain your answer.

.....

.....

.....

..... [2]

(c) Using data from Table 9.1, calculate the volume of the Earth.

volume = ...m^3 [2]

(d) A student uses data from Table 9.1 to find the orbital speed of a planet.

The student uses:

- the distance R of the planet from the Sun
- the orbital period T of the planet.

State the equation involving R and T that is used to find the orbital speed of the planet

orbital speed =

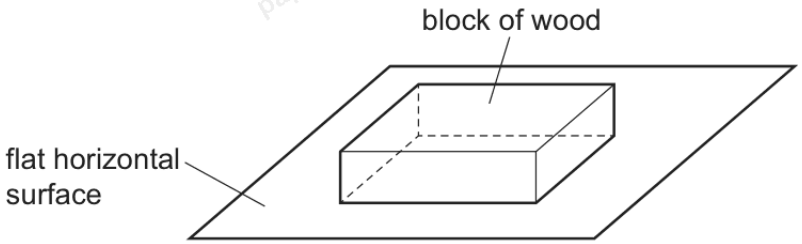
[1]

[Total: 8]

Question	Answer	Marks	Guidance
9(a)(i)	planets take different times to orbit the Sun / move at different (angular) speeds sometimes they are on the same side / sometimes on different sides of the Sun / do not always align or WTTE	**1**	different speeds / rates around Sun **not** Venus has larger orbital period than Earth, Earth moves faster than Venus **A** the orbits are elliptical / not circular; **accept** from diagrams with planets labelled
9(a)(ii)	4(.0)	**1**	**A** 4.(0) $\times 10^7$ with extra $\times 10^7$ km
9(b)	Incorrect **and** because Mars and Venus, or Mars and Earth, or Saturn and Jupiter do not follow the relationship Mars has less mass than Earth or Venus and a longer orbital period **or** Jupiter has larger mass than Saturn but a smaller orbital period	**1**	only one pair from Mars / Venus or Mars / Earth or Saturn / Jupiter needs to be mentioned **or** Jupiter has largest mass but not the largest period **A** values quoted **A** one of the pairs from first mark explained.
9(c)	(V =) M/d in any form 1.1×10^{21} (m^3)	**1**	1.0909
9(d)	orbital speed = $2\pi R / T$	**1**	

QUESTION · 5054_s26_qp_12.pdf · source page 3 of 16

5 The diagram shows a block of wood at rest on a flat horizontal surface.



Which equation is used to calculate the density of the wood?

A density = mass of block × area of bottom surface of block

B $\text{density} = \dfrac{\text{mass of block}}{\text{area of bottom surface of block}}$

C density = mass of block × volume of block

D $\text{density} = \dfrac{\text{mass of block}}{\text{volume of block}}$

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 5

5	D	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 3 of 16

6 A stone of mass 400 g is lowered into a measuring cylinder containing water.

The water level rises from 300 cm^3 to 500 cm^3.

What is the density of the stone?

A 0.50 g/cm^3 **B** 0.80 g/cm^3 **C** 1.33 g/cm^3 **D** 2.0 g/cm^3

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

Chapter 1 Motion, forces and energy

1.5 Forces

1.5.1 Balanced and unbalanced forces

147 entries: 85 Paper 1 questions, 62 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 3 of 20

6 Which is a statement of Newton's third law of motion?

A Every force causes a reaction.

B If there is no resultant force on a body then there is no acceleration.

C The forces acting on a body are always equal and opposite.

D To every action there is an equal but opposite reaction.

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 6

6	**D**

11 Fig. 11.1 shows a children's ride. A carriage containing children is pulled up the slope by a motor. The carriage stops at A and then runs down through B, C and D without further input of energy. Between D and E the carriage turns through a bend at constant speed, as shown in Fig. 11.2. At E, brakes are applied and the carriage slows to a stop at F. The height of the ride is 30 m at A and 10 m at C.

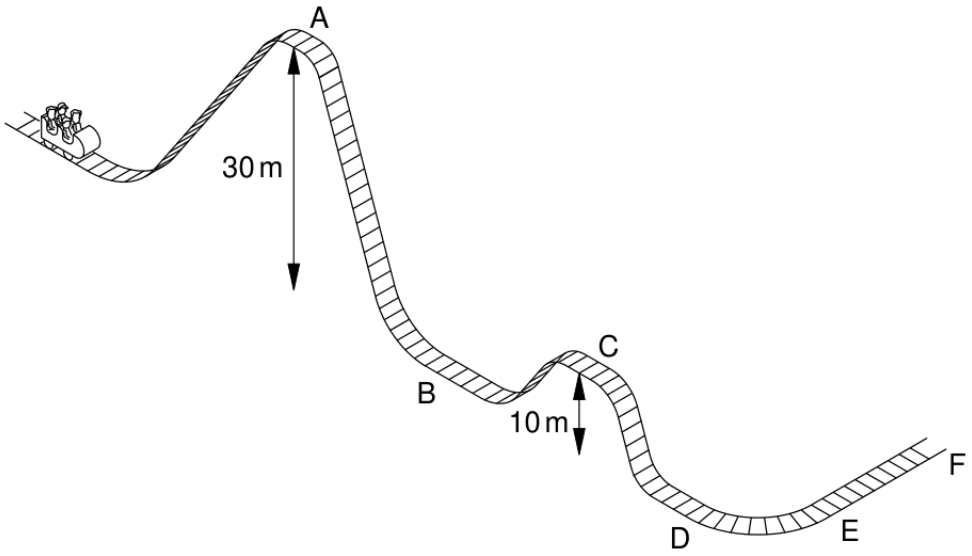


Fig. 11.1

The mass of the carriage and children is 500 kg.
Take the gravitational field strength as 10 N/kg.

(a) **(i)** Discuss the energy changes that occur in the ride from A to D.

(ii) Calculate the maximum potential energy of the carriage and children.

(iii) Assuming that there is no friction between A and C, determine the kinetic energy of the carriage and children at C. Show your working.

[9]

(b)

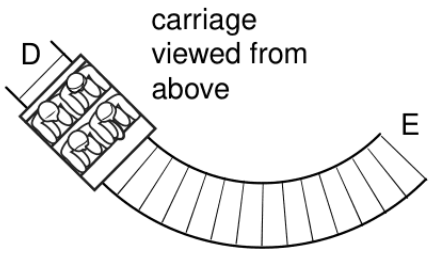


Fig. 11.2

Between D and E, the carriage goes round part of a horizontal circle at constant speed. During this time the velocity of the carriage changes.

(i) Explain how the carriage can have a constant speed but a changing velocity.

(ii) State the direction of the force that acts on the carriage to make it move round the curve.

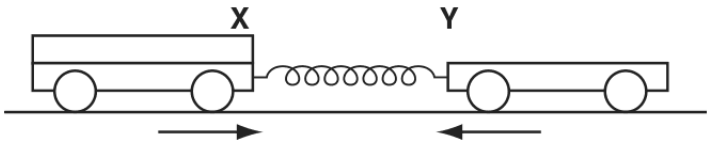
[3]

(c) Between E and F, a frictional force of 3000 N acts to slow the carriage. Calculate the deceleration of the carriage. [3]

11	**(a)**	**(i)**	P.E. decreases (A to B or C to D or downhill or initially)		B1
			K.E. gained (P.E. → K.E….2)		B1
			K.E. to P.E. **change** must be clear and from B to C or uphill		B1
		(ii)	mgh algebraic form seen		C1
			500*10*30		C1
			150 000 J		A1
		(iii)	conservation of energy cited **or** clear that loss of P.E. has become K.E.		C1
			500*10*20 or 500*10*10 or 50 000 seen		C1
			100 000 J (allow g=9.8)		A1
	(b)	**(i)**	velocity involves direction **or** is a vector (speed does not)		B1
			direction (of carriage) changes / carriage turns (accept on diagram)		
		(ii)	force towards centre (of curve) / inwards (accept centripetal)		B1
	(c)		F = ma in any algebraic form or 3000 = 500a		C1
			3000/500		C1
			6(.0) m/s^2		A1
				Total	**[15]**

QUESTION · 5054_w04_qp_1.pdf · source page 3 of 20

4 Trolley **X** and trolley **Y** are joined by a stretched spring. Trolley **X** has twice the mass of trolley **Y**.



When the trolleys are released, the acceleration of **X** is $2\,m/s^2$ to the right.

What is the initial acceleration of trolley **Y** to the left?

A $1\,m/s^2$ **B** $2\,m/s^2$ **C** $3\,m/s^2$ **D** $4\,m/s^2$

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 4

4 **D**

Section B

Answer **two** questions from this section.

Write your answers on the separate answer paper provided.

9 A train travels from one station to the next. It starts from rest at time $t = 0$ and accelerates uniformly for the first 20 s. At $t = 20$ s it reaches its top speed of 25 m/s. It then travels at this speed for a further 30 s before decelerating uniformly to rest. The total time for the journey is 60 s.

(a) **(i)** Sketch a speed-time graph for the motion of the train. Do **not** use graph paper. Put the speed of the train on the *y*-axis and time along the *x*-axis. [3]

(ii) Write down the equation, **in words**, that relates acceleration, time and change in velocity. [1]

(iii) Explain what is meant by a *uniform* acceleration. [1]

(iv) Use your graph to calculate the deceleration of the train as it comes to rest. [2]

(b) Several forces act on the train when it is moving.

(i) Name the horizontal and vertical forces that act on the train and give the direction of each force. [4]

(ii) Explain whether the horizontal forces are balanced or unbalanced,

1. when the train accelerates,

2. when the train travels at constant speed,

3. when the train decelerates.

You may draw diagrams to help your explanations. [3]

(c) A second train has a non-uniform acceleration. Sketch a speed-time graph showing a non-uniform acceleration. Do **not** use graph paper. [1]

Section B

9 **(a) (i)** y axis labelled speed or m/s **and** x axis labelled time or s **B1**
straight line from 0,0 to t = 20, speed = 25 **B1**
uniform speed from t = 20 to 50 **and** uniform deceleration from t = 50 to 60 **B1**

(ii) acceleration = change in velocity/time or per unit time
or rate of change of velocity with time
accept equation but must be written in words or defined symbols **B1**

(iii) constant increase in speed/velocity in 1sec/ /same time interval
or rate of change of speed/velocity constant **or** Δv proportional to time
or acceleration constant **with time** **B1**

(iv) 25/10 e.c.f. time interval from graph **C1**
2.5 m/s^2 accept -ve **A1**

(b) (i) weight/gravitational force (accept gravity) **downwards**
normal/reaction/contact force/force from ground **upwards**
air resistance/drag **or** friction (due to air) **backwards or** opposite to train (direction)
braking force **or** friction **or** resistive force **backwards or** same direction as air drag
tractive **or** thrust **or** driving force **or** force of engine **forwards**
ANY 4 **B4**
accept from diagram (-1 each wrong force more than 4)

(ii) 1. unbalanced **since** forward force > backwards force **or** resultant/net **forward** force **B1**
2. balanced **since** forward force = backwards force **or** forces cancel **or** zero resultant **B1**
3. unbalanced **since** backwards force > forwards force **or** only backwards force **or** resultant/net backwards force **B1**
accept sizes of forces from lengths of arrows on diagram

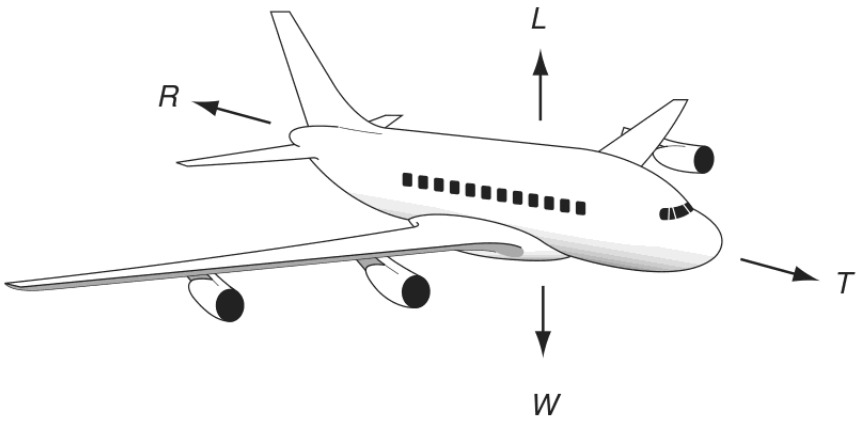
(c) sketch graph with **axes labelled** and non straight line **B1**

QUESTION · 5054_w05_qp_1.pdf · source page 2 of 20

2 An aircraft, flying at a constant height, is gaining speed.

The four forces acting are

L	lift due to the wings
R	air resistance
T	the thrust due to the engines
W	the weight



What is correct?

	vertical forces	horizontal forces
A	$L = W$	$T = R$
B	$L > W$	$T > R$
C	$L = W$	$T > R$
D	$L > W$	$T = R$

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 2

2 **C**

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows a 0.4 kg mass hanging at rest from a spring.

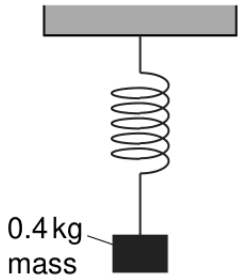


Fig. 1.1

(a) State what is meant by the *mass* of an object.

.....

.....

..... [1]

(b) **(i)** On Fig. 1.1, draw an arrow showing the direction and the line of action for each of the two forces that act on the mass. Write the name of the force next to each arrow. [2]

(ii) The gravitational field strength is 10 N/kg. Calculate the size of each of the two forces acting on the mass.

first force = ... second force = ... [2]

(c) The mass is pulled downwards. State and explain what happens to the upward force.

.....

.....

.....

.....

..... [2]

Section A

1	(a)		amount of matter/substance **or** ability to resist motion (accept Force/acc)	**B1**	
	(b)	(i)	downwards force labelled weight/gravity continuation of vertical line	**B1**	
			upwards force (labelled tension) along vertical line	**B1**	
			(allow 1 mark if both labels correct but lines of action not accurate, e.g. displaced to side)		
		(ii)	4 N	**B1**	
			4 N or same as other	**B1**	
	(c)		upwards force/tension increases	**B1**	
			spring stretched/longer	**B1**	**7**

QUESTION · 5054_s06_qp_1.pdf · source page 2 of 20

2 A force of 20 N pushes an object of mass 5.0 kg along a rough horizontal surface where the frictional force is 5.0 N.

What is the acceleration of the object?

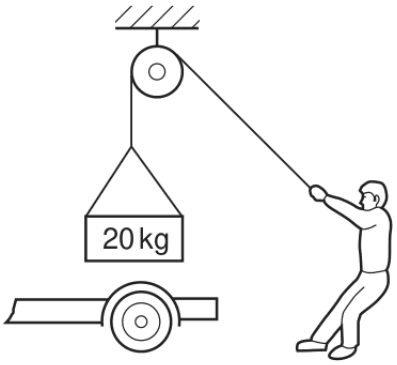
A $1.0\,m/s^2$ **B** $2.0\,m/s^2$ **C** $3.0\,m/s^2$ **D** $4.0\,m/s^2$

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 2

2	**C**

QUESTION · 5054_w06_qp_1.pdf · source page 2 of 16

3 A person just supports a mass of 20 kg suspended from a rope.



What is the resultant force acting on the mass?

A 0 N **B** 10 N **C** 20 N **D** 200 N

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 3

3	**A**

QUESTION · 5054_w06_qp_1.pdf · source page 3 of 16

4 When a block of wood of mass 2 kg is pushed along the horizontal flat surface of a bench, the friction force is 4 N.

When the block is pushed along the bench with a force of 10 N, it moves with a constant

A speed of 3 m / s.

B speed of 5 m / s.

C acceleration of 3 m / s^2.

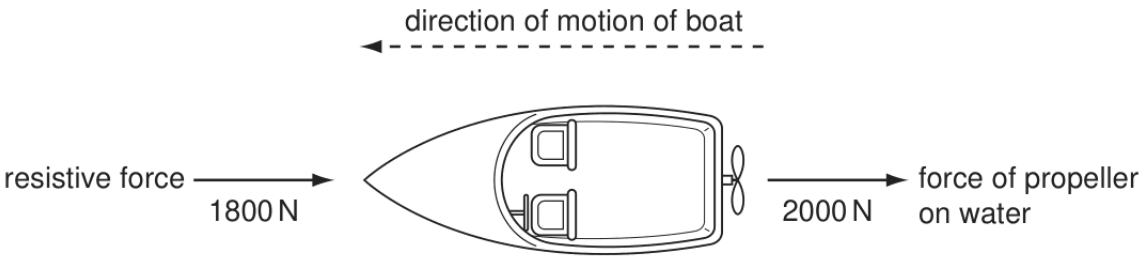
D acceleration of 5 m / s^2.

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 4

4	**C**

QUESTION · 5054_s07_qp_1.pdf · source page 3 of 20

5 The propeller on a boat pushes water backwards with a force of 2000 N. The boat moves through the water against a total resistive force of 1800 N.



According to Newton's third law, what is the forward force on the propeller due to the water?

A 3800 N **B** 2000 N **C** 1800 N **D** 200 N

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 5

5	**B**

Section B

Answer **two** questions from this section.

Use the lined page provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 Fig. 9.1 shows the horizontal forces acting on a moving car.

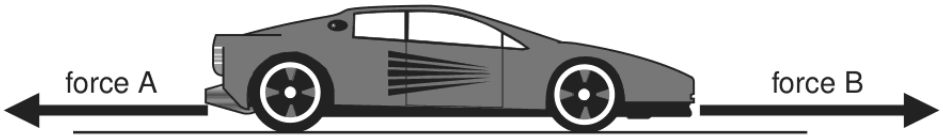


Fig. 9.1

(a) Compare the sizes and directions of the two forces when the car is

(i) moving along a straight road at constant speed, [1]

(ii) accelerating. [1]

(b) Suggest the direction of the resultant force on the car when the car turns a corner at constant speed. [1]

(c) Fig. 9.2 shows the speed-time graph for the first 24 s of the motion of the car along a straight road.

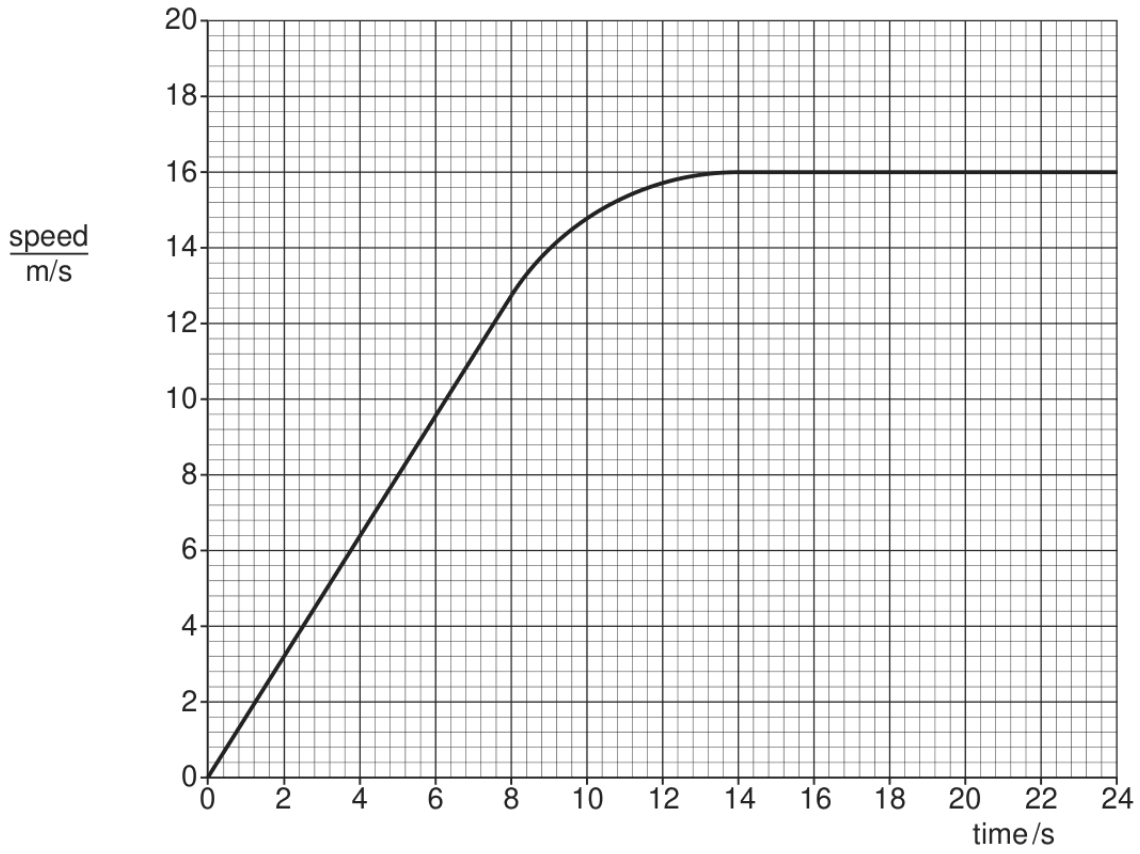


Fig. 9.2

(i) During the motion there is a period of uniform acceleration. State both the start and finish times of this period. [1]

(ii) State the main energy changes that occur in the 24 s. [3]

(iii) Calculate the acceleration of the car during the first 5 s. State clearly the equation that you use. [3]

(iv) Determine the distance travelled in the first 5 s. [3]

(d) The car is stopped by applying the brakes. Various factors can affect the distance travelled by the car during the time that the brakes are applied. Apart from the force applied by the brakes, state **two** of these factors. [2]

Section B

9	**(a)**	**(i)**	*A* = *B* (assume opposite direction and co-linear)	B1	
		(ii)	B > A (assume opposite direction and co-linear) maximum of 1 mark if directions wrong	B1	[2]
	(b)		towards **centre** of circle/corner	B1	[1]
	(c)	**(i)**	0 **and** 8–9 s	B1	
		(ii)	chemical (potential energy) (**accept** electrical if electrical car clear) **to** kinetic energy **or** K.E. increases thermal energy/heat/internal energy produced Max 2/3 if clear error	B1 B1 B1	
		(iii)	acceleration = (*v*–*u*)/t **or** gradient (7.8 to) 8/5 (**accept** any corresponding period e.g. 8s 12.6–12.8, 6s 9.4–9.6) 1.6 m/s^2 (**accept** 1.56–1.60)	C1 C1 A1	
		(iv)	area under graph **or** average speed 4 (m/s) **or** ½ 5 speed used in **(iii)** at 5 s ½ x (7.8 to) 8 x 5 20 m (**accept** 19.5–20; **ecf speed used in (iii)** at 5 s)	C1 C1 A1	[10]
	(d)		speed of car/friction with road (accept slippery road or ice or water or oil on road)/ friction in engine/tyre condition or area or pressure/air resistance/wind speed or direction/mass or inertia **of car or passengers**/slope of road	B2	[2]

Section B

Answer **two** questions from this section.

Use the separate sheets available from the Supervisor.

9 Many cars are fitted with an air-bag, as shown in Fig. 9.1. In a collision, the air-bag inflates and reduces the effect of the impact between the passenger and the dashboard.

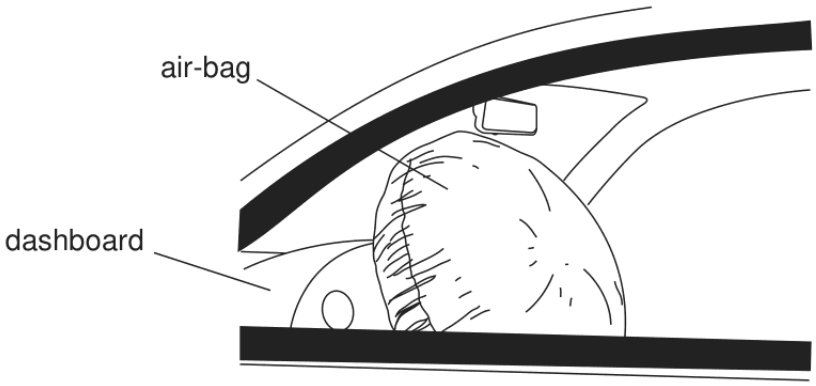


Fig. 9.1

(a) In a test of the air-bag, a heavy ball is used instead of the passenger.

The car is travelling at 14 m/s when it hits a wall. The air-bag inflates and the ball takes 3.0 s to come to rest. The ball has mass 5.0 kg.

(i) Calculate the average deceleration of the ball. [3]

(ii) Calculate the average force exerted on the ball. [2]

(iii) Using ideas about acceleration, explain how the air-bag reduces the force on the ball during the test. [2]

(b) If there was no air-bag, a large pressure would be exerted on the ball at the point where it hits the dashboard.

(i) Define *pressure*. [1]

(ii) The inflated air-bag reduces the pressure exerted on the ball. State **two** reasons why the pressure is reduced. [2]

(c) Compressed gas from a small cylinder inflates the air-bag. The cylinder contains a volume of 600 cm^3 of gas at a pressure of 1.4×10^7 Pa. The cylinder and the inflated air-bag have a volume of 30 000 cm^3.

(i) Calculate the pressure of the gas in the inflated air-bag, assuming that the temperature is constant. [3]

(ii) The pressure inside the cylinder decreases as the air-bag is inflated. Explain, using ideas about molecules, why the pressure decreases. [2]

Section B

9	**(a)**	**(i)**	(acc =) (v-u)/t	C1	
			14/3	C1	
			4.7 m/s^2 (penalise halving to 2.35 m/s^2, accept 2 or more sig figs **not** fractions)	A1	
		(ii)	F = ma or 5 x **(i)**	C1	
			23 N (penalise **second** halving to 5.75 N, ecf **(i)** acc. 2 or more sig figs **not** fractions)	A1	
		(iii)	longer time of impact/slows down ball gradually/stops the ball more slowly	B1	
			less acceleration	B1	[7]
	(b)	**(i)**	force / area **or** F/A (acc. force on unit area **not** force on an area; N/m^2)	B1	
		(ii)	larger area	B1	
			smaller force	B1	[3]
	(c)	**(i)**	$P_1V_1 = P_2V_2$ **or** PV = constant	C1	
			1.4 x 10^7 x 600 = P x 30000 **or** 1.4 x 10^7 x 600/30000	C1	
			280 000 Pa	A1	
		(ii)	molecules hit sides (of cylinder) (**not** each other)	B1	
			molecules leave cylinder **or** fewer in cylinder **or** enter air bag	B1	[5]

For Examiner's Use

2 Fig. 2.1 shows a device for punching holes in a piece of paper. A person applies a force *F* at the end of the arm. Just before the hole is made in the paper, the arm is at rest.

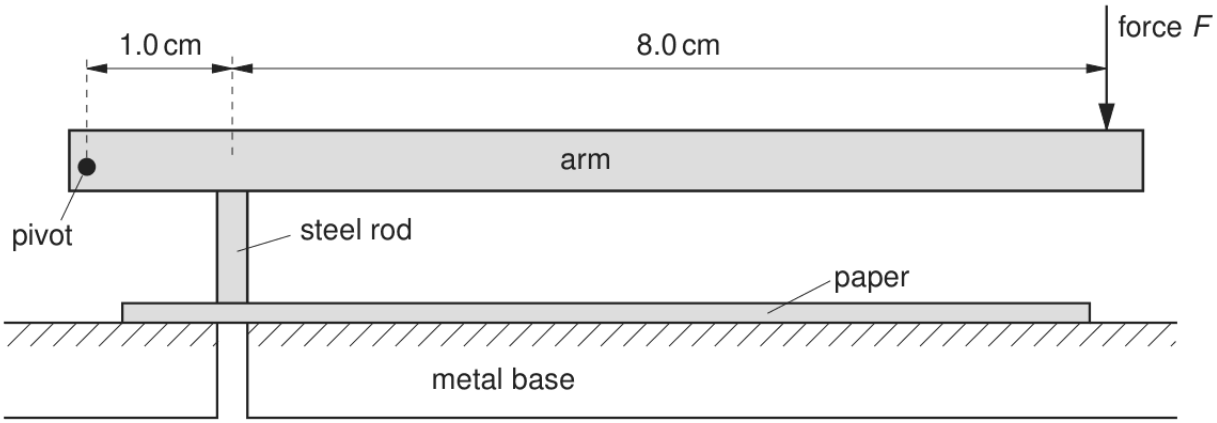


Fig. 2.1

(a) Just before the hole is made, the force upwards on the steel rod is 7.2 N.

Calculate the value of *F*. Use the distances marked on Fig. 2.1.

F = ... [2]

(b) The steel rod pushes the paper into the hole in the metal base with a force of 7.2 N. The end of the steel rod has an area of $2.8 \times 10^{-5}\,m^2$.

Calculate the pressure exerted on the paper.
State the formula that you use in your calculation.

pressure = ... [2]

(c) The force downwards on the paper due to the rod and the force upwards on the rod due to the paper are related by Newton's third law. State Newton's third law.

.....

.....

..... [1]

2 **(a)** any attempt at a moment calculation, e.g. any $F_1d_1 = F_2d_2$ seen, or answer 0.9 N C1
0.8(0) N A1

(b) P = F/A **formula** stated B1
2.6×10^5 Pa (2.571×10^5 Pa) B1

(c) action and reaction are equal and opposite **or** every force has an equal and opposite force **or** force on body A is equal and opposite to force on body B B1

[Total: 5]

Section B

Answer **two** questions from this section.

Use the lined pages provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 When a car driver sees an emergency ahead, he applies the brakes. During his reaction time the car travels at a steady speed and covers a distance known as the thinking distance. The braking distance is the distance travelled by the car after the brakes are applied.

(a) State the energy change that occurs as the car brakes. [2]

(b) Fig. 9.1 shows the speed-time graph of a car.
The driver sees the emergency at time $t = 0$.
The total mass of the car is 800 kg.

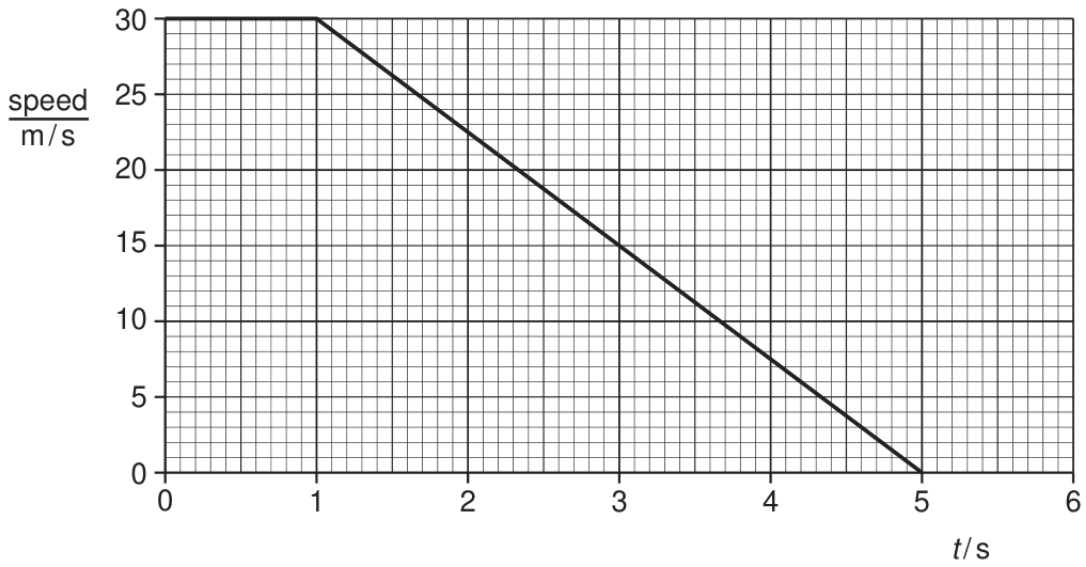


Fig. 9.1

Determine

(i) the thinking distance, [1]

(ii) the braking distance, [2]

(iii) the deceleration of the car during braking, [2]

(iv) the force provided by the brakes. [2]

(c) Using ideas about friction and deceleration, state and explain how the braking distance is affected by

(i) using new tyres rather than badly worn tyres, [2]

(ii) the car skidding on a wet road, [2]

(iii) the car carrying a heavy load of passengers. [2]

Section B

9	**(a)**	K.E. (at start) to heat (+ sound)	B1 B1
			[Total: 2]
	(b) **(i)**	30 m cao	B1
	(ii)	area under graph **or** average speed × time **or** (u + v).t / 2 **or** 30 × 4/2 60 m	C1 A1
	(iii)	$(v–u)/t$ **or** $v = u + at$ **or** 30/4 **or** gradient **or** rise/run 7.5 (±0.1) m/s^2	C1 A1
	(iv)	$F = ma$ **or** 800 × **(iii)** 6000 N ecf **(iii)**	C1 A1
			[Total: 7]
	(c) **(i)**	more friction/grip/traction **or** more deceleration **or** decelerates faster **or** decelerates in less time less (braking) **distance**	 B1 B1
	(ii)	less friction **or** less deceleration **or** decelerates slower/longer more (braking) **distance**	B1 B1
	(iii)	less deceleration **or** decelerates slower/longer more **distance**	B1 B1
			[Total: 6]

QUESTION · 5054_w08_qp_1.pdf · source page 3 of 20

3 How is the motion of a body affected by balanced and unbalanced forces acting on it?

	balanced forces	unbalanced forces
A	velocity changes	velocity changes
B	velocity changes	velocity constant
C	velocity constant	velocity changes
D	velocity constant	velocity constant

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 3

3 **C**

Section A

For Examiner's Use

Answer **all** the questions in this section.

1 A microphone has a weight *W* of 6.0 N. It is suspended by wire X from the ceiling in a radio studio.

Fig. 1.1 shows the microphone held in the correct position by a horizontal wire Y.

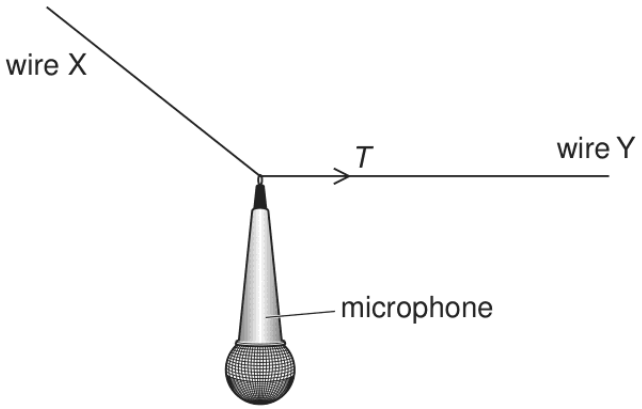


Fig. 1.1

(a) The tension *T* in wire Y is 8.0 N. Use a vector diagram to determine the magnitude and the direction of the resultant of *W* and *T.*

magnitude = ...

direction = ...
[4]

(b) The microphone is at rest. State the tension in wire X.

tension in X = ... [1]

Section A

1	**(a)**	diagram of two forces **and** resultant	B1	
		W / 6(N) **and** *T* / 8(N) marked on perp. forces **or** scale given	B1	
		10(.0 ±0.2) N	B1	
		35–39° from *T*/Y/horizontal or 51–55° from *W*/vertical **and** correct resultant	B1	
	(b)	10(.0) N **or** e.c.f.	B1	[5]

For Examiner's Use

4 A passenger aeroplane accelerates from rest along a runway. It accelerates at a uniform rate for 35 s. At this point it reaches a speed of 84 m / s and then takes off.

(a) (i) Calculate the acceleration of the aeroplane along the runway.

acceleration = ... [2]

(ii) Sketch a speed-time graph for the aeroplane as it travels along the runway.

[3]

(b) Two **horizontal** forces act on the aeroplane as it accelerates along the runway.

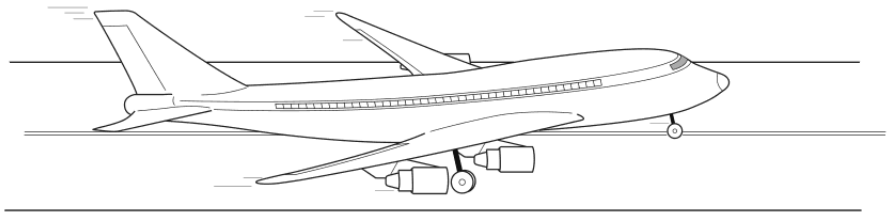


Fig. 4.1

(i) On Fig. 4.1, draw arrows to show the directions and the relative magnitudes of these forces. [1]

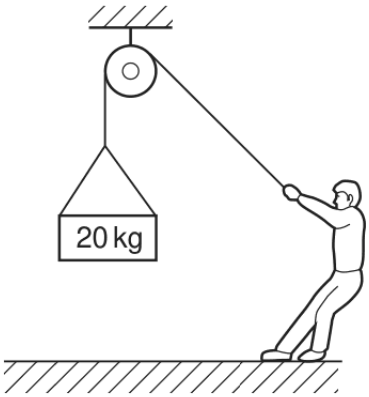
(ii) State what causes the smaller of these two forces.

..... [1]

4	**(a)**	**(i)**	(a = Δ)v/t **or** 84/35	C1	
			2.4 m/s²	A1	
		(ii)	speed **and** time axes correct **and** labelled	B1	
			straight line of positive gradient through origin	B1	
			84 (m/s) **and** 35 (s) marked	B1	
	(b)	**(i)**	two arrows with forward force > backward force	B1	
		(ii)	air/wind resistance **or** friction **or** drag	B1	[7]

QUESTION · 5054_s09_qp_1.pdf · source page 3 of 20

5 A person just supports a mass of 20 kg suspended from a rope.



What is the resultant force acting on the mass?

A 0 N **B** 10 N **C** 20 N **D** 200 N

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 5

5	**A**

10 Fig. 10.1 shows the horizontal forces as a cyclist travels forwards.

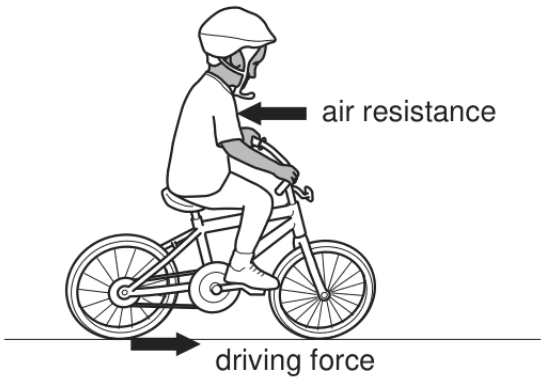


Fig. 10.1

The cyclist produces the driving force that acts on the back wheel.
In this question, you may ignore any frictional force acting on the front wheel.

(a) The bicycle accelerates until a constant speed is reached.

(i) Describe how the size of the air resistance changes during this time. [2]

(ii) Compare the sizes of the two horizontal forces when the bicycle is accelerating. [1]

(b) The total mass of the bicycle and the cyclist is 75 kg. At one instant, the speed of the bicycle is 4.0 m/s, the driving force is 30 N and the air resistance is 20 N.

Calculate

(i) the total kinetic energy of the bicycle and the cyclist, [3]

(ii) the acceleration of the bicycle and the cyclist. [2]

(c) As the bicycle moves, energy is transmitted from the pedals to the back wheel.

Fig. 10.2 shows what happens to the energy input to the pedals.

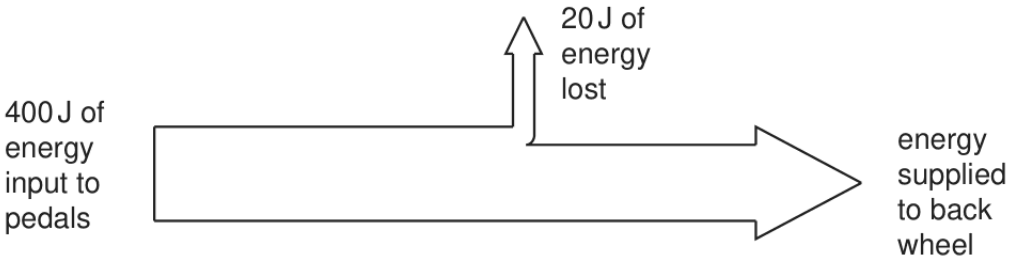


Fig. 10.2

(i) As energy is transmitted to the back wheel, some is lost. Explain how this happens. [2]

(ii) Calculate the efficiency of the bicycle in transmitting energy from the pedals to the back wheel. [2]

(d) Some bicycles are made from low density materials. Explain why this is an advantage. [3]

10 (a) (i) air resistance increases (as speed increases) B1
(at constant speed) becomes equal to driving force/applied force etc. B1

(ii) driving force (forward force) larger (than air resistance/backwards force) B1

(b) (i) (E =) ½ mv^2 algebraic formula C1
½ × 75 × 4^2 C1
600 J A1

(ii) (a =) F/m algebraic seen or 10 (N) used as force C1
0.13 m/s^2 A1

(c) (i) friction (in chain/axles) or rubbing of surfaces B1
heat or thermal energy produced B1

(ii) (efficiency = useful) energy output/energy input algebraic or numerical or 380 seen C1
0.95 or 95% A1

(d) lower mass/weight of cycle B1

less force needed	same acceleration/get up hill/to stop cycle/lift cycle	
greater acceleration/easier to acc.	for same force or $F = ma$ quoted	
less energy/work (input)	to go uphill/due to less friction/½ mv^2 less	
less kinetic energy	less stopping distance/less force to stop	
more efficient/less energy wasted	going uphill/less friction	
less friction	easier turn handlebars/higher (top) speed	
less pressure (on ground)	sinks less into ground	M1A1 [15]

QUESTION · 5054_w09_qp_1.pdf · source page 3 of 20

5 A tractor pulls a trailer at a constant speed.

The tractor exerts a forward force of 1600 N on the trailer.

What is the force exerted by the trailer on the tractor?

A 0 N

B 1600 N backwards

C 1600 N forwards

D 3200 N forwards

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 5

5	**B**

For Examiner's Use

2 Fig. 2.1 shows a motor used to operate a lift (elevator). There is a man in the lift.

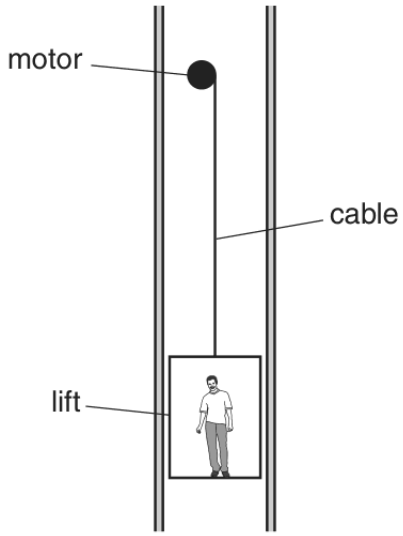


Fig. 2.1 (not to scale)

The input power to the motor is 6200 watts.

(a) Define the *watt*.

.....

.....[2]

(b) The lift and the man have a total mass of 580 kg. The lift moves up a distance of 12 m in 15 s.

(i) Calculate the tension in the cable when the lift is moving at constant speed.

tension = ...[1]

(ii) Calculate the increase in potential energy of the lift and man.

potential energy = ...[2]

(iii) Calculate the efficiency of the motor.

efficiency = ...[2]

2 **(a)** energy/time C1
1 joule in one second A1 [2]

(b) **(i)** 5800 N or 5684 N or 5700 N B1 [1]

(ii) *mgh* algebraic, words or numerical (i.e. 580 ×10 × 12) C1
69 600 J or 70 000 J or 68 208 J or 68 000 J A1 [2]

(iii) (efficiency =) output power or energy/input power or energy algebraic or numerical or 93 000 seen or 4 640 seen C1
0.75 or 75% (accept 0.748) e.c.f. from (ii) A1 [2]

[Total: 7]

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a car braking on a road and coming to rest.

Fig. 9.1

(a) Explain what is meant by

(i) the *thinking distance*,

.....

.....[1]

(ii) the *braking distance*.

.....

.....[1]

(b) An engineer conducts a test on the car and finds that the braking distance is greater when the car is fully loaded than when it is unloaded.

(i) Apart from the road conditions, state what must be kept the same in the test.

.....[1]

(ii) Explain why the car has a greater braking distance when fully loaded.

.....

.....[1]

(c) State and explain how one road condition affects the braking distance of the car. Use ideas about friction in your answer.

.....

.....[2]

(d) Explain how wider tyres affect the pressure of the car on the surface of the road.

.....

.....[1]

For Examiner's Use

(e) The car has a total mass of 900 kg and is travelling at 20 m/s. At time $t = 0$, the driver sees an accident ahead. He applies the brakes at $t = 0.60$ s to stop the car. After the brakes are applied, the car comes to rest in a further 4.0 s.

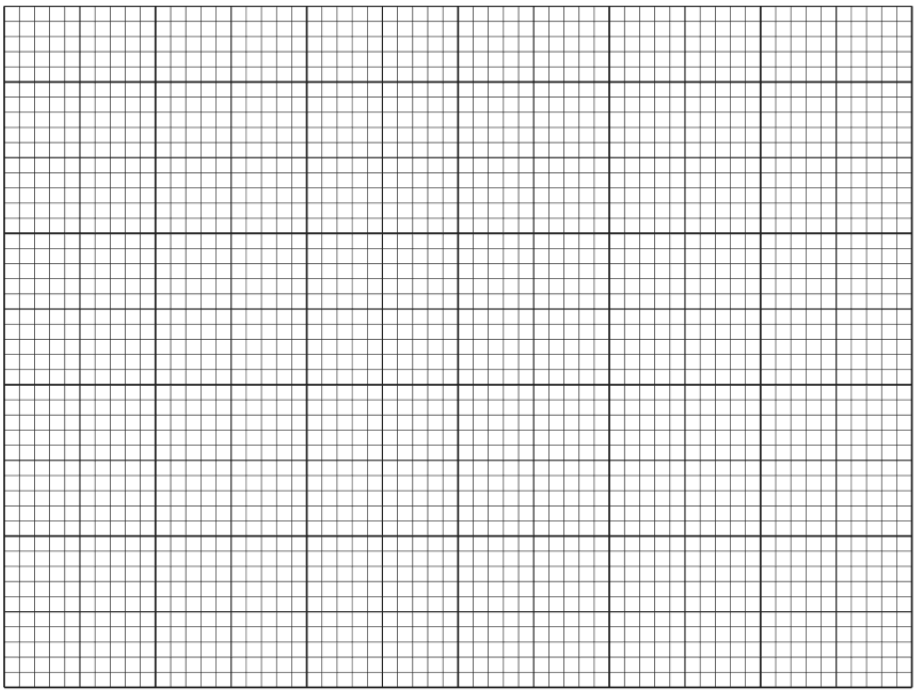
(i) Calculate the deceleration of the car as it brakes.

deceleration = ...[2]

(ii) Calculate the braking force acting on the car.

force = ...[2]

(iii) On Fig. 9.2, draw a speed-time graph for the car as it brakes.



[3]

Fig. 9.2

(iv) State how your graph in **(iii)** can be used to find the total distance travelled by the car.

.....

.....[1]

Section B

9	**(a)**	**(i)**	distance travelled while thinking/in reaction time or before braking starts	B1	[1]
		(ii)	distance travelled while brakes applied/car decelerates	B1	[1]
	(b)	**(i)**	speed (of cars) or same force/pressure on pedal or same braking force or same tyres or condition of brakes	B1	[1]
		(ii)	greater inertia/kinetic energy/momentum or smaller deceleration/acceleration	B1	[1]
	(c)		any road condition, e.g. icy, wet, poor surface, slippery/smooth/rough surface and its correct effect on distance	B1	
			correct explanation that refers to friction e.g. more friction when dry	B1	[2]
	(d)		pressure low(er) (with larger area)	B1	[1]
	(e)	**(i)**	$a = v/t$ any algebraic or numerical value e.g. 20/4; 20/3.4; 20/4.6; 20/0.6	C1	
			5(.0) m/s^2	A1	[2]
		(ii)	$F = ma$ algebraic or numerical e.g. 900 × **(i)**	C1	
			4500 N e.c.f. **(i)**	A1	[2]
		(iii)	correct axes labelled with quantity and/or unit	B1	
			horizontal line at 20 m/s from 0 to 0.6 s	B1	
			straight line from end of horizontal section or from (0.6,20) to (4.6,0) or (4,0)	B1	[3]
		(iv)	area (under graph or of trapezium)	B1	[1]

[Total: 15]

QUESTION · 5054_w10_qp_11.pdf · source page 5 of 20

5 An aircraft, flying at a constant height, is gaining speed.

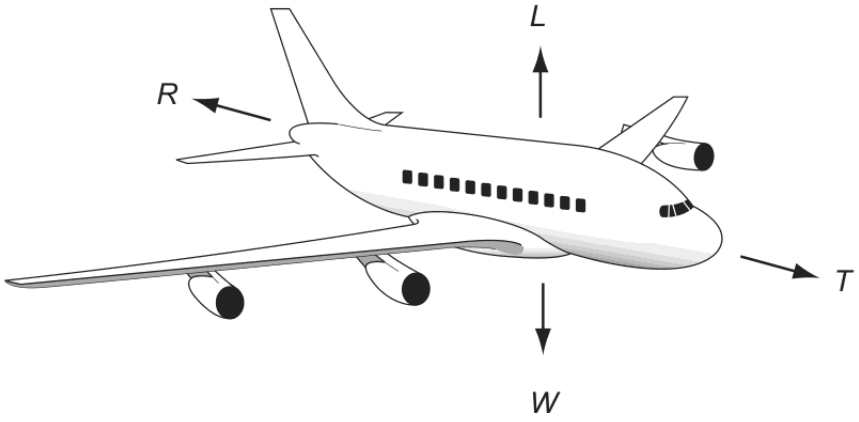
The four forces acting are

L lift due to the wings

R air resistance

T the thrust due to the engines

W the weight



Which row is correct?

	vertical forces	horizontal forces
A	$L = W$	$T = R$
B	$L > W$	$T > R$
C	$L = W$	$T > R$
D	$L > W$	$T = R$

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 5

5 **C**

2 Fig. 2.1 shows the planet Venus orbiting the Sun in a circular orbit at constant speed.

For Examiner's Use

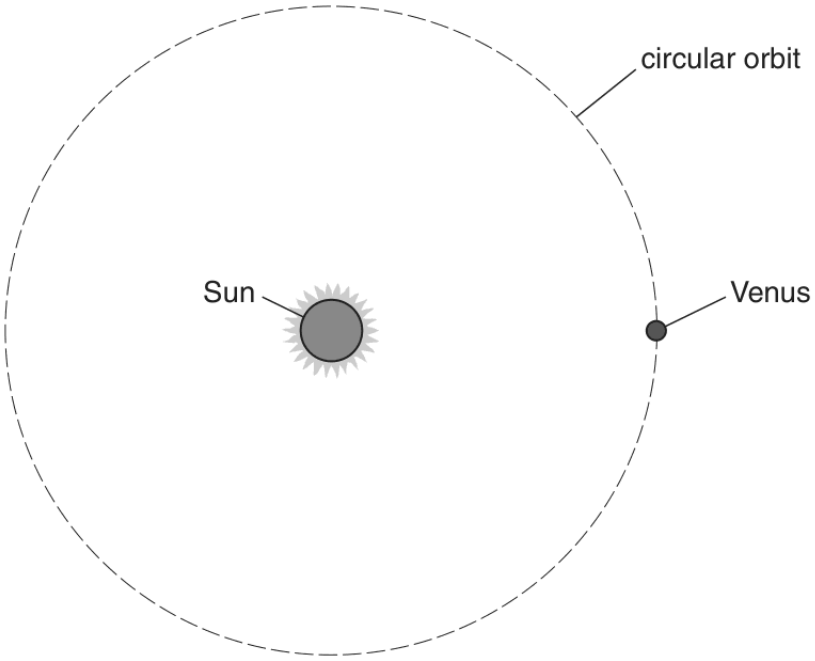


Fig. 2.1 (not to scale)

A force acting on Venus keeps it moving in a circle.

(a) State the name of this force and the object that causes it.

.....

..... [2]

(b) **(i)** On Fig. 2.1, draw an arrow through Venus that represents the direction of its acceleration. [1]

(ii) The size of this acceleration is $9.7 \times 10^{-3}\,m/s^2$ and the mass of Venus is $4.9 \times 10^{24}\,kg$. Calculate the size of the force acting on the planet.

force = ... [2]

(c) As time passes, the planet Venus moves a considerable distance around the circular orbit. State why no work is done by the force acting on the planet as it moves.

.....

.....

..... [2]

2	**(a)**	gravitational/centripetal (pull/attraction)	B1	
		of the **Sun**	B1	[2]
	(b) **(i)**	arrow touching Venus towards centre/left (must pass through Sun if extended)	B1	[1]
	(ii)	(F =) ma **or** $4.9 \times 10^{24} \times 9.7 \times 10^{-3}$	C1	
		$4.8(4.753) \times 10^{22}$ N	A1	[2]
	(c)	direction of movement ⊥ direction of force	C1	
		force/it is perpendicular/at right angles to distance moved **or** does not move **any** distance in direction of force	A1	[2]

QUESTION · 5054_w10_qp_12.pdf · source page 4 of 16

6 An aircraft, flying at a constant height, is gaining speed.

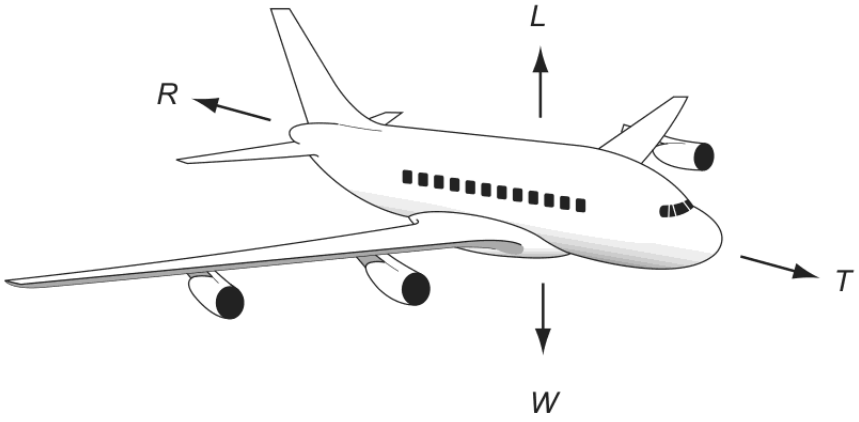
The four forces acting are

L lift due to the wings

R air resistance

T the thrust due to the engines

W the weight



Which row is correct?

	vertical forces	horizontal forces
A	$L = W$	$T = R$
B	$L > W$	$T > R$
C	$L = W$	$T > R$
D	$L > W$	$T = R$

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 6

6 **C**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a cricket ball as it comes into contact with a cricket bat.

Fig. 1.1

The cricket ball has a mass of 0.16 kg and it hits the bat with a speed of 25 m/s. After being in contact with the bat for 0.0013 s, the ball rebounds with a speed of 22 m/s in the direction exactly opposite to its original direction.

(a) State the difference between *speed* and *velocity*.

.....

.....[1]

(b) Calculate

(i) the change in velocity of the cricket ball,

velocity change = ...[1]

(ii) the average acceleration of the ball whilst it is in contact with the bat,

acceleration = ...[2]

(iii) the average force exerted on the ball by the bat.

force = ...[2]

Section A

1	**(a)**	velocity has a direction/is a vector **or** speed does not have a direction/is not a vector **or** displacement/time **and** distance/time (**ign** speed is a scalar)	B1	
	(b) **(i)**	(–) 47 m/s	B1	
	(ii)	(a =) v/t **or** 47/0.0013	C1	
		(–) 3.6(1538 etc.) × 10^4 m/s^2	A1	
	(iii)	(F =) ma **or** 0.16 × 3.6 × 10^4	C1	
		(–) 5.8(**or** 5.78461 etc.) × 10^3 N	A1	[6]

10 A bus breaks down on a road with a 10° upward slope. The passengers get out and push the bus to the top of the slope at a constant speed. Fig. 10.1 shows the passengers exerting a force on the bus parallel to the line of the slope.

For Examiner's Use

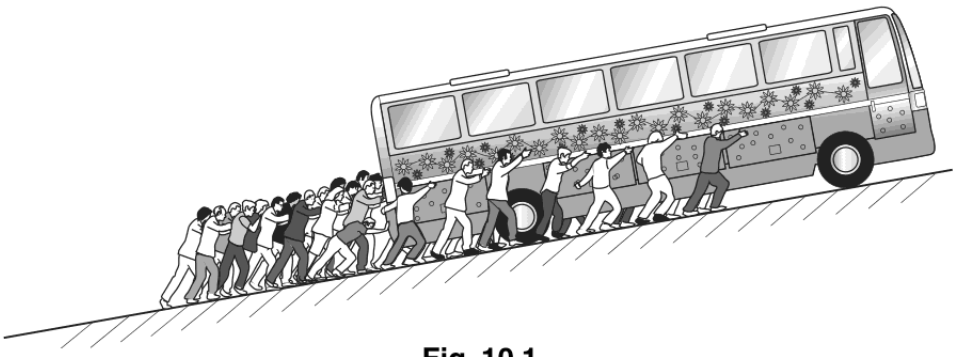


Fig. 10.1

(a) The total mass of the bus is 3200 kg.

(i) Calculate the weight of the bus.

weight = ...[1]

(ii) On Fig. 10.1, draw **two** arrows to show the direction of the weight of the bus and the direction of the force exerted on the bus by the passengers. [1]

(iii) The total force exerted by the passengers on the bus is 17000 N along the line of the slope. Use a graphical method to determine the size and direction of the resultant of this force and the weight of the bus. State the scale used.

scale = ...

size of resultant = ...

direction of resultant = ...

[4]

For Examiner's Use

(iv) The bus travels at a constant speed. State the size of the resultant of all the forces acting on the bus.

.....[1]

(b) As the passengers push the bus up the slope, work is done against two forces. State the names of these two forces and, for each force, state the form of energy produced as work is done.

force 1 .....

form of energy 1 .....

force 2 .....

form of energy 2 .....
[4]

(c) When the bus reaches the top of the slope, it stops and the passengers get back in.

(i) On the axes of Fig. 10.2, sketch the distance-time graph for the bus from a time when the passengers are pushing the bus at constant speed until they get back into the stationary bus.

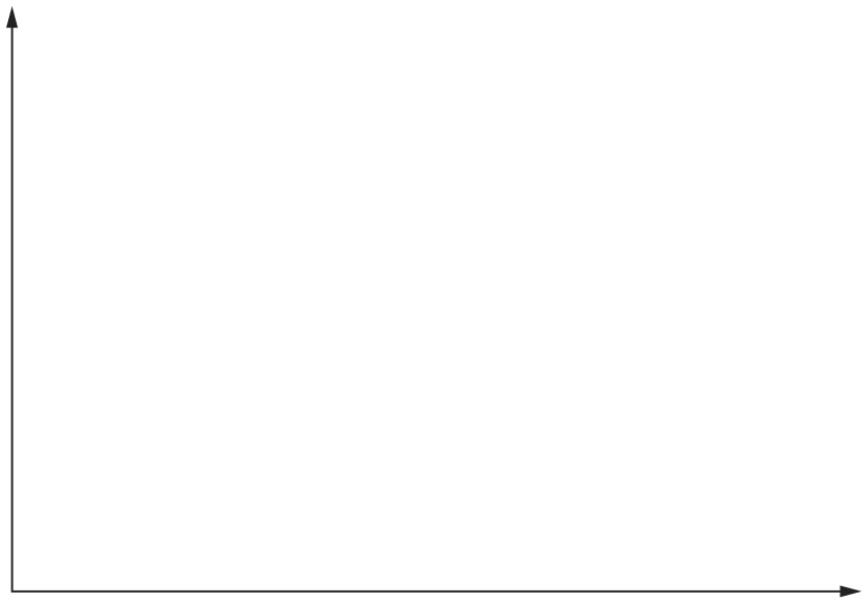


Fig. 10.2

[3]

(ii) State how the speed of a moving object may be obtained from an accurately drawn distance-time graph.

.....

.....[1]

10	**(a)**	**(i)**	32 000 N	B1	
		(ii)	two arrows/lines in correct direction by eye	B1	
		(iii)	scale given	B1	
			two arrows/lines **and** correct resultant drawn	B1	
			32.0 → 35.0 kN (2/3 sig. fig. only)	B1	
			58.5 → 61.5° to horizontal (2/3 sig. fig. only; don't penalise twice)	B1	
		(iv)	zero/no force/0	B1	[7]
	(b)		weight/gravitational force/gravitational attraction (**not** gravity)	B1	
			higher in gravitational field **or** (to gravitational) potential energy	B1	
			friction/air resistance	B1	
			heat/thermal/internal energy	B1	[4]
	(c)	**(i)**	labelled axes **and** correct way round ($x \rightarrow t$)	B1	
			straight line of positive slope	B1	
			followed only by horizontal line (**ign** curve at junction)	B1	
		(ii)	distance travelled/time taken (from points) **or** calculate the gradient	B1	[4]

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a sky-diver falling vertically.

Fig. 9.1

The sky-diver starts from rest at time $t = 0$.
His acceleration is non-uniform until he reaches a steady speed of 50 m/s at $t = 10$ s.
He opens his parachute at $t = 20$ s and decelerates until $t = 25$ s.
From $t = 25$ s he falls at a steady speed of 5 m/s.

(a) (i) On Fig. 9.2, draw the speed-time graph for the sky-diver. [4]

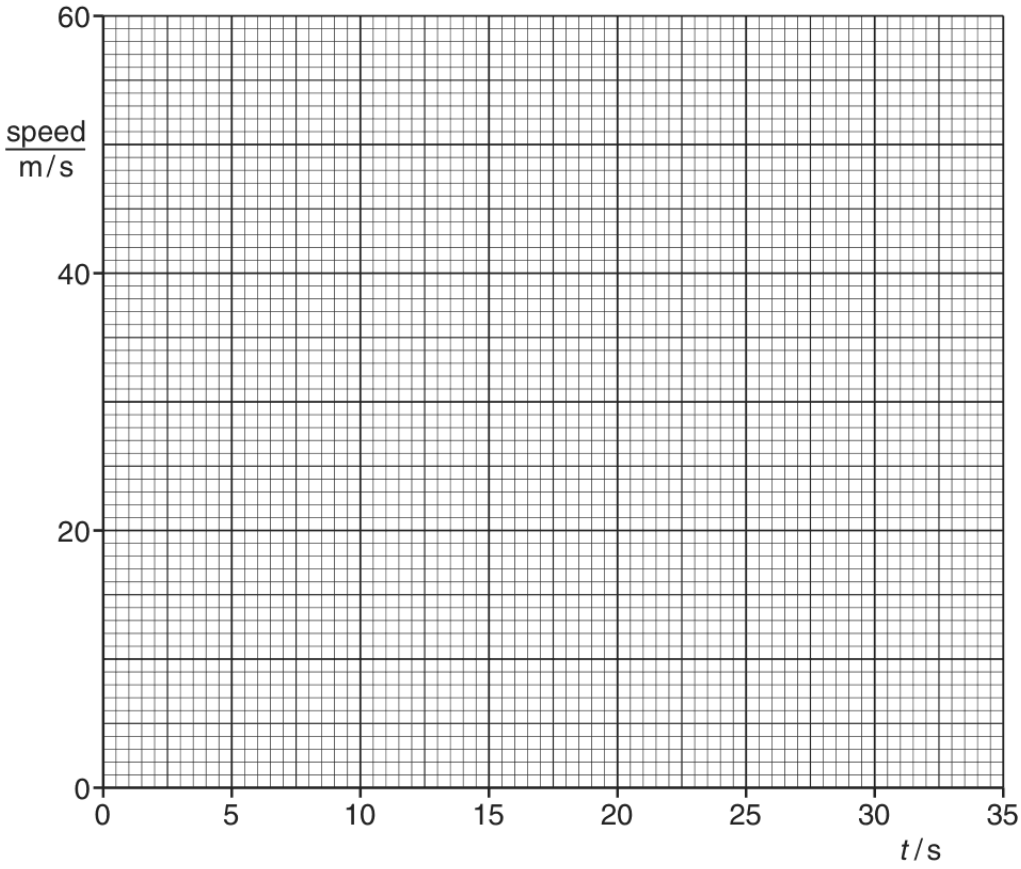


Fig. 9.2

(ii) State how your graph shows that the acceleration is non-uniform between $t = 0$ and $t = 10$ s.

.....

.....[1]

(b) Explain, in detail, why **after** the sky-diver opens his parachute, he decelerates and eventually reaches a steady speed.

.....

.....

.....

.....

.....

.....[4]

(c) For the time interval between t = 10 s and t = 20 s, calculate the distance that the sky-diver falls.

distance = ...[1]

(d) The mass of the sky-diver is 60 kg.

For the time interval between t = 20 s and t = 25 s,

(i) calculate the average deceleration of the sky-diver,

average deceleration = ...[2]

(ii) calculate the average resultant force acting on the sky-diver,

force = ...[2]

(iii) state how your graph in Fig. 9.2 may be used to obtain the distance that the sky-diver falls.

.....

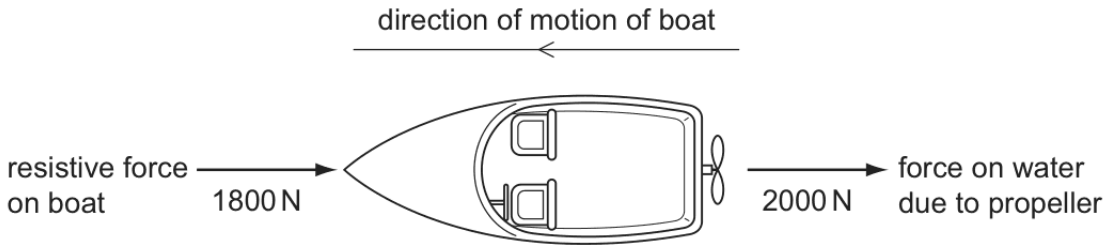
.....[1]

Section B

9	**(a)**	**(i)**	curve with decreasing gradient from origin to 50 m/s at 10 s	B1
			constant speed from 10 to 20 s	B1
			decrease to 5 m/s at 25 s	B1
			constant speed from 25 s until at least 30 s	B1
		(ii)	gradient/slope not constant/decreases // graph curves // graph not a (straight) line // increase (in speed) per second/unit time not equal	B1
	(b)		any mention of **air** resistance/drag/upward force	B1
			(initially) force upwards larger than force downwards // resultant force upwards	B1
			air resistance decreases (with fall in speed)	B1
			(at constant speed) air resistance/friction/drag equals weight // forces (up and down) balance // zero resultant force	B1
	(c)		500 m	B1
	(d)	**(i)**	(a =) $\frac{v-u}{t}$ in any numerical or algebraic form, e.g. 45/5	C1
			9(.0) m/s^2 ecf **(a)(i)**	A1
		(ii)	($F = ma$) in any numerical or algebraic form, e.g. 60 × 9 ecf **(i)**	C1
			540 N	A1
		(iii)	area **under** graph/line/curve	B1 [15]

QUESTION · 5054_w11_qp_11.pdf · source page 3 of 20

4 The propeller on a boat pushes water backwards with a force of 2000 N. The boat moves through the water against a total resistive force of 1800 N.



According to Newton's third law, what is the forward force on the propeller due to the water?

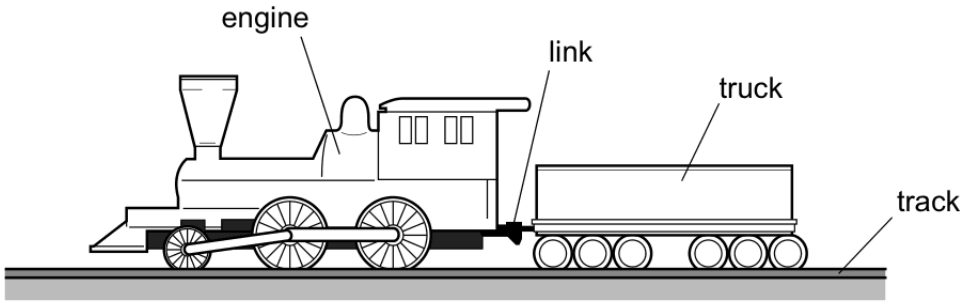
A 200 N **B** 1800 N **C** 2000 N **D** 3800 N

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 4

4	**C**

QUESTION · 5054_w11_qp_11.pdf · source page 4 of 20

6 An engine pulls a truck at constant speed on a level track.



The link between the truck and the engine breaks. The driving force on the engine remains constant.

What effect does this have on the truck and on the engine?

	truck	engine
A	slows down	speed stays constant
B	slows down	speeds up
C	stops immediately	speed stays constant
D	stops immediately	speeds up

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 6

6 **B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 is the speed-time graph for a rocket from the moment that the fuel starts to burn at time $t = 0$.

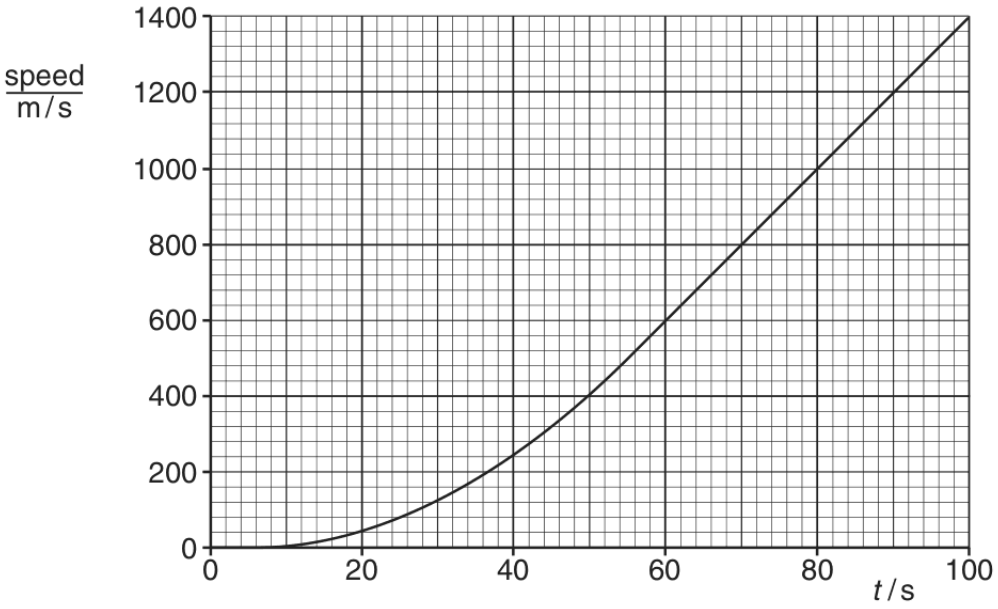


Fig. 9.1

(a) State the principal energy changes taking place as the rocket accelerates upwards.

.....

.....

.....

.....[4]

(b) **(i)** State the size of the acceleration of the rocket at $t = 0$.

.....[1]

(ii) State what happens to the acceleration of the rocket between $t = 5\,s$ and $t = 80\,s$.

.....

.....[2]

(iii) Calculate the acceleration of the rocket at $t = 80\,s$.

acceleration = ...[2]

(iv) The total mass of the rocket at $t = 80\,s$ is $1.6 \times 10^6\,kg$. Calculate the resultant force on the rocket at this time.

resultant force = ...[2]

(v) The total weight of the rocket at $t = 80\,s$ is $1.6 \times 10^7\,N$. Calculate the upward force on the rocket at this time, caused by the burning fuel.

upward force = ...[1]

(c) As the rocket burns fuel, it ejects hot gas downwards.

(i) State *Newton's third law of motion*.

.....

.....[1]

(ii) Explain how Newton's third law of motion applies to the upward force on the rocket and to the force on the hot gas.

.....

.....

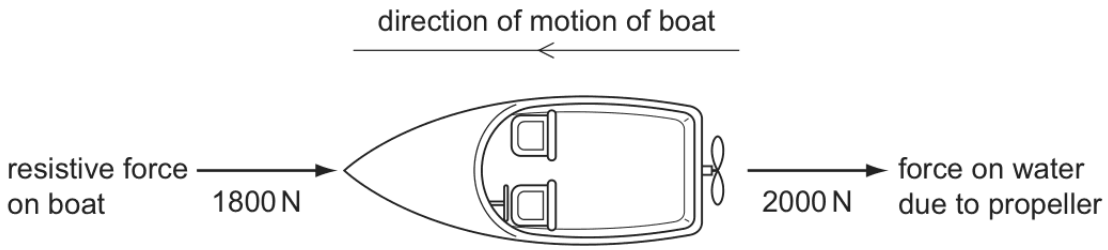
.....[2]

Section B

9	(a)	**at start:** chemical (potential) energy	B1	
		at end: PE/GPE/gravitational energy	B1	
		at end: heat/thermal/internal energy	B1	
		at end: KE **or** intermediate KE from chemical energy	B1	
	(b) (i)	0	B1	
	(ii)	it increases	B1	
		to constant value	B1	
	(iii)	gradient **or** $(v–u)/t$ **or** (1400-600)/40 **or** other correct numbers	C1	
		$20\,m/s^2$	A1	
	(iv)	$(F =)\ ma$ **or** $1.6 \times 10^6 \times 20$	C1	
		$3.2 \times 10^7\,N$	A1	
	(v)	$4.8 \times 10^7\,N$	B1	
	(c) (i)	to every action there is an equal and opposite reaction **or** forces act in pairs of equal size **and** in opposite directions/on different bodies	B1	
	(ii)	downward force on gas	B1	
		equal and opposite to upward/**(b)(v)** force (on rocket)	B1	[15]

QUESTION · 5054_w11_qp_12.pdf · source page 2 of 20

2 The propeller on a boat pushes water backwards with a force of 2000 N. The boat moves through the water against a total resistive force of 1800 N.



According to Newton's third law, what is the forward force on the propeller due to the water?

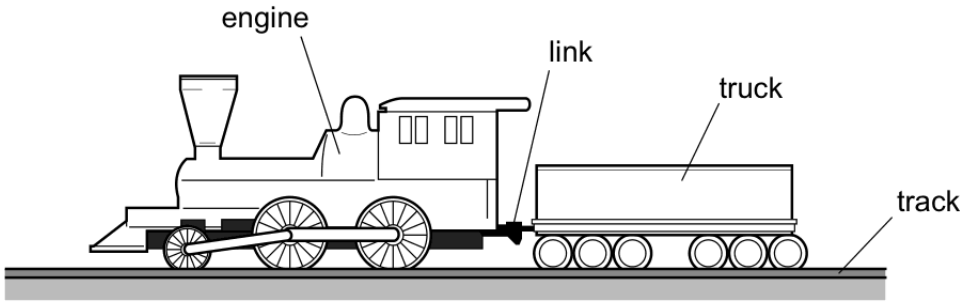
A 200 N **B** 1800 N **C** 2000 N **D** 3800 N

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 2

2	**C**

QUESTION · 5054_w11_qp_12.pdf · source page 3 of 20

4 An engine pulls a truck at constant speed on a level track.



The link between the truck and the engine breaks. The driving force on the engine remains constant.

What effect does this have on the truck and on the engine?

	truck	engine
A	slows down	speed stays constant
B	slows down	speeds up
C	stops immediately	speed stays constant
D	stops immediately	speeds up

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 4

4 **B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 is the speed-time graph for a racing car of total mass 650 kg as it sets off from rest at the start of a race.

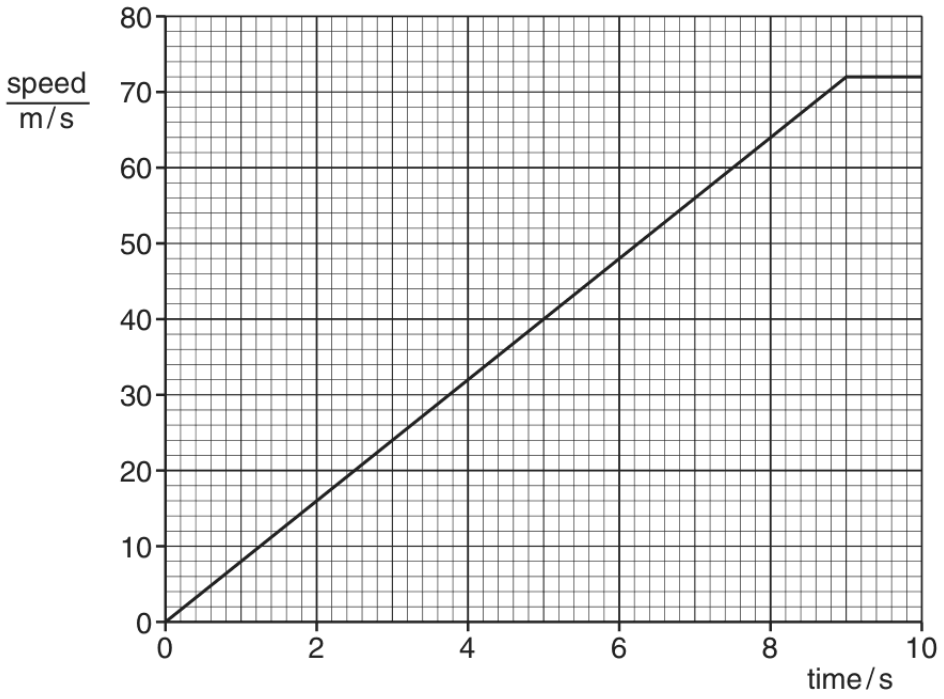


Fig. 9.1

The car travels in a straight line until time $t = 9.0$ s.

(a) State the speed of the car at time $t = 9.0$ s.

speed = ...[1]

(b) Calculate, for the car between times $t = 0$ and $t = 9.0$ s,

(i) the distance travelled,

distance = ...[2]

(ii) the acceleration of the car,

acceleration = ...[2]

(iii) the resultant force acting on the car.

force = ...[2]

(c) The acceleration of the car is constant between $t = 0$ and $t = 9.0$ s.

Suggest and explain why the driving force on the car must increase to keep the acceleration constant.

.....

.....

.....

.....[3]

(d) After $t = 9.0$ s, the car starts to turn a corner and follows a circular path at constant speed.

(i) Explain why the car is accelerating even though its speed is constant.

.....

.....

.....[2]

(ii) State the direction of this acceleration.

.....[1]

(iii) State and explain what causes the car to accelerate as it turns the corner.

.....

.....[2]

Section B

9	**(a)**	72 m / s			B1	
	(b)	**(i)**	area (under graph) **or** ½ base × height **or** ½*vt* **or** ½ × 9 × 72		C1	
			320/324 m		A1	
		(ii)	**change** in velocity/time **or** $\Delta v/t$ **or** 72/9		C1	
			8(.0) m / s^2		A1	
		(iii)	(*F* =) *ma* **or** 650 × 8.0		C1	
			5.2 × 10^3 N		A1	
	(c)	friction **or** air/wind resistance **or** drag			M1	
		increases as speed increases			A1	
		resultant/net/unbalanced force remains constant			B1	
	(d)	**(i)**	**direction** (of car/motion/speed/velocity) changes		B1	
			(therefore) velocity changes		B1	
		(ii)	towards centre (of circle)/centripetal		B1	
		(iii)	friction **with ground**	**OR** banking of track	B1	
			mention wheels/tyres	reaction force (acts towards centre)	B1	[15]

QUESTION · 5054_s12_qp_11.pdf · source page 3 of 16

5 A hard stone hits the ground and comes to rest almost immediately.

As the stone hits the ground, what is the direction and the size of the force acting on the ground?

A downwards and equal to the weight of the stone

B downwards and larger than the weight of the stone

C upwards and equal to the weight of the stone

D upwards and larger than the weight of the stone

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 5

5	**B**

QUESTION · 5054_s12_qp_11.pdf · source page 3 of 16

6 A car of mass 1500 kg is towing a trailer of mass 1100 kg along a level road. The acceleration of the car is $1.30\,m/s^2$.

Ignoring friction and air resistance, what is the driving force on the car?

A 1430 N **B** 1950 N **C** 2000 N **D** 3380 N

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 6

6	**D**

2 Fig. 2.1 shows a windsurfer, sail and board.

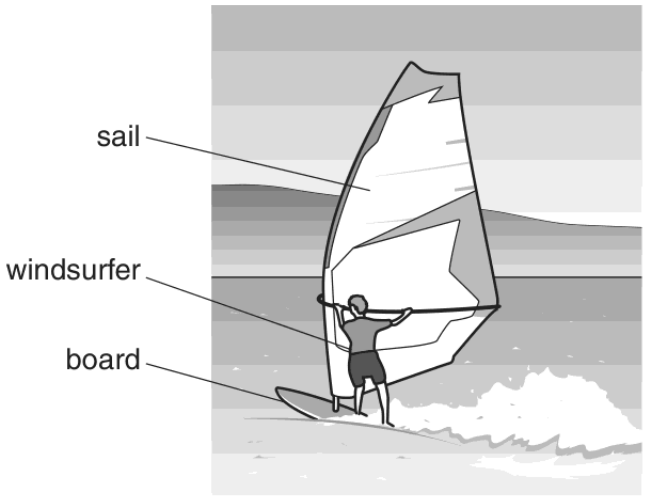


Fig. 2.1

The total mass of the windsurfer, sail and board is 90 kg. They travel at a constant speed of 5.0 m/s.

(a) Calculate the total kinetic energy of the windsurfer, sail and board.

kinetic energy = ...[2]

(b) The wind exerts a forward force on the sail.

(i) Explain, in terms of the forces acting, why the board travels at a constant speed.

.....

.....

.....[2]

(ii) The wind does work on the sail. Explain, in terms of the conservation of energy, why the total kinetic energy does not increase.

.....

.....

.....[1]

2 **(a)** (K.E.) = ½ mv^2; ½ × 90 × 5^2 C1
1125 J **or** 1100 J A1

(b) **(i)** no resultant force; forwards force = backwards force/drag/friction (ignore air resistance) B1
water resistance/**water** drag mentioned/**water** friction OR sail exerts force on board B1

(ii) heat produced OR equal to work done against backwards force/drag/friction B1 [5]

2 **(a)** State what is meant by *friction*.

.....

.....[2]

(b) Fig. 2.1 shows two horizontal forces that act on a car. Force B is caused by air resistance and friction.

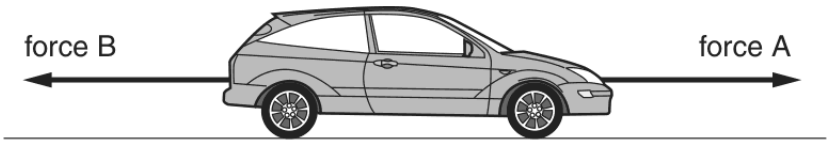


Fig. 2.1

The car is travelling along a straight level road.

(i) The forward force A and the backward force B are equal.
Describe the motion of the car.

.....

.....[1]

(ii) The mass of the car is 800 kg.
Force A increases to 5000 N. This causes the car to accelerate initially at 1.5 m/s^2.
Calculate the size of force B.

force B = ...[3]

(iii) Force A remains constant at 5000 N. Explain why the acceleration decreases as the car travels along the level road.

.....

.....[2]

(c) A car of mass 800 kg climbs a hill and rises a vertical distance of 200 m.
Calculate the gain in potential energy of the car. (The gravitational field strength *g* is 10 N/kg.)

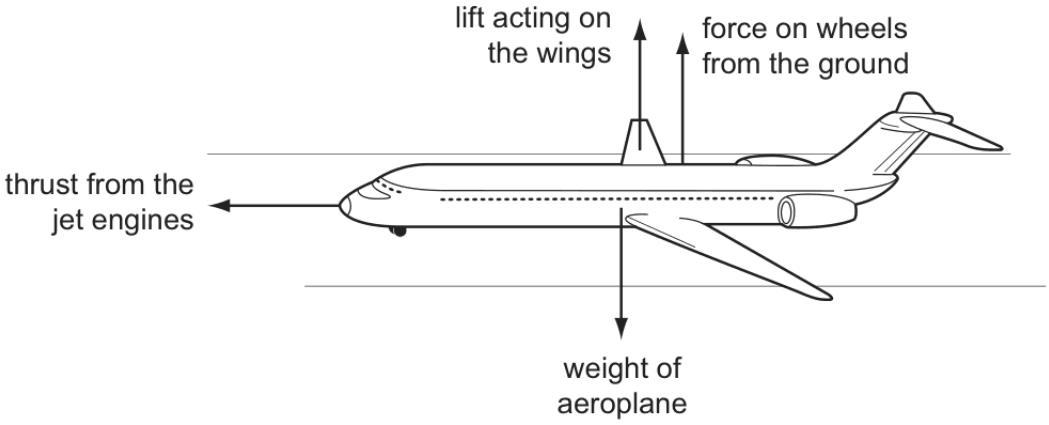
gain in potential energy = ...[2]

2	**(a)**	a force	B1	
		when objects slide over/rub one another OR opposes (relative) **motion**/**movement**	B1	
	(b) **(i)**	constant/uniform speed OR constant/uniform velocity OR zero acceleration	B1	
	(ii)	(*F* =) *ma* seen in any form numerical or algebraic	C1	
		1200 (N) OR 6200 N seen	C1	
		3800 N	A1	
	(iii)	Force B increases OR backwards force/resistance/friction/drag increases	M1	
		as speed/velocity increases	A1	
	(c)	(PE =) *mgh* in any form numerical or algebraic	C1	
		1 600 000 J	A1	[10]

QUESTION · 5054_w12_qp_11.pdf · source page 3 of 16

5 An aeroplane is half way along a runway before it takes off.

The arrows show the directions of four forces acting on the aeroplane.



The drag forces on the aeroplane are negligible.

The gravitational field strength g is 10 N / kg.

The acceleration of the aeroplane is 2 m / s^2.

What is the largest force?

A the force on the wheels from the ground

B the lift acting on the wings

C the thrust from the jet engines

D the weight of the aeroplane

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 5

5	**D**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 A parachutist of total weight 950 N falls vertically at a constant velocity of 6.5 m/s.

(a) State the size and direction of the resistive (drag) force acting on the parachutist.

size of resistive force: .....

direction of resistive force: .....
[2]

(b) At a certain height, the wind starts to blow and the parachutist moves horizontally at a velocity of 4.5 m/s as he continues to fall.

Use a graphical method to add together the horizontal and the vertical velocities to determine the size of the resultant velocity of the parachutist. Label the velocities and state the scale that you use.

size of velocity = ...

scale = ...
[3]

Section A

1	**(a)**	950 N	B1	
		upwards	B1	
	(b)	correct rectangle **and** diagonal **and** at least one velocity labelled **or** correct triangle and at least one velocity labelled (either way round)	B1	
		from 7.8(0000) to 8.0(0000) m / s (inclusive)	B1	
		scale stated	B1	[5]

For Examiner's Use

4 A petrol tanker has mass 2800 kg when empty and holds 30 m^3 of petrol when full. The density of petrol is 740 kg/m^3.

(a) Calculate the total mass of the tanker when full of petrol.

mass = ... [2]

(b) The full tanker is travelling along a road when the driver observes an obstruction and applies the brakes. A resultant force of 30 000 N acts on the tanker.

Calculate the deceleration of the tanker.

deceleration = ... [2]

4	**(a)**	$(m=)\rho V$ **or** 740 × 30 or 22 000/22 200	C1	
		25 000/2.5 × 10^4 kg (allow 24 800 from 22 000)	A1	
	(b)	$(a=)F/m$ **or** 30 000/25 000	C1	
		$(-)1.2\,m/s^2$	A1	[4]

QUESTION · 5054_w12_qp_12.pdf · source page 2 of 16

2 The diagram shows three forces acting on a block.

Which additional force will produce a resultant force of 3 N to the left?

A 3 N to the left

B 6 N to the right

C 9 N to the left

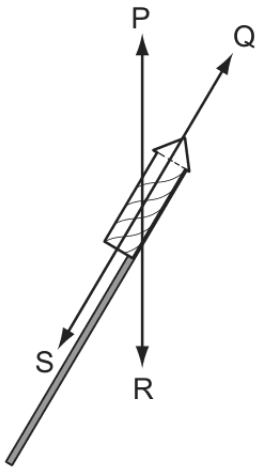
D 13 N to the right

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 2

2	**C**

QUESTION · 5054_w12_qp_12.pdf · source page 3 of 16

5 The diagram shows a firework rocket.



As the rocket flies through the air, three forces act on it. These forces are weight, thrust and air resistance.

What are the three forces?

	thrust	air resistance	weight
A	P	R	S
B	P	S	R
C	Q	R	S
D	Q	S	R

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 5

5 **D**

QUESTION · 5054_w12_qp_12.pdf · source page 4 of 16

6 An object has a mass of 15 kg. It is pushed horizontally by a force of 40 N. The frictional force is 10 N.

What is the acceleration of the object?

A $0.50\,m/s^2$ **B** $1.5\,m/s^2$ **C** $2.0\,m/s^2$ **D** $2.7\,m/s^2$

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 6

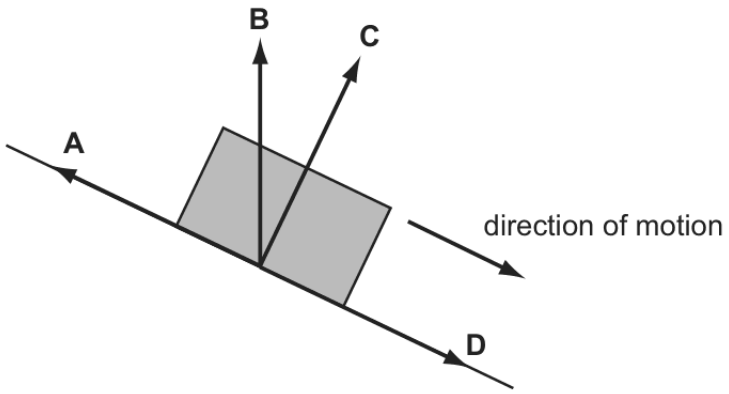
6	**C**

QUESTION · 5054_s13_qp_11.pdf · source page 4 of 20

5 A body slides down a frictionless slope as shown.

As the body presses on the surface, the surface pushes back on the body. The force of the surface on the body is sometimes called the reaction force.

In which direction does the reaction force act?



MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 5

5 **C**

QUESTION · 5054_s13_qp_12.pdf · source page 3 of 16

5 A car of mass 1500 kg travels along a horizontal road.

It accelerates steadily from 10 m/s to 25 m/s in 5.0 s.

What is the force needed to produce this acceleration?

A 300 N **B** 500 N **C** 4500 N **D** 7500 N

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 5

5	**C**

QUESTION · 5054_s13_qp_12.pdf · source page 4 of 16

6 Balanced forces are acting on a moving body.

What happens to the direction of movement and to the speed of the body?

	direction of movement	speed
A	changes	changes
B	changes	does not change
C	does not change	changes
D	does not change	does not change

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 6

6 **D**

For Examiner's Use

2 **(a)** Explain what is meant by the *moment of a force*.

.....

.....

..... [2]

(b) Fig. 2.1 shows a system for raising a heavy piece of metal into a vertical position.

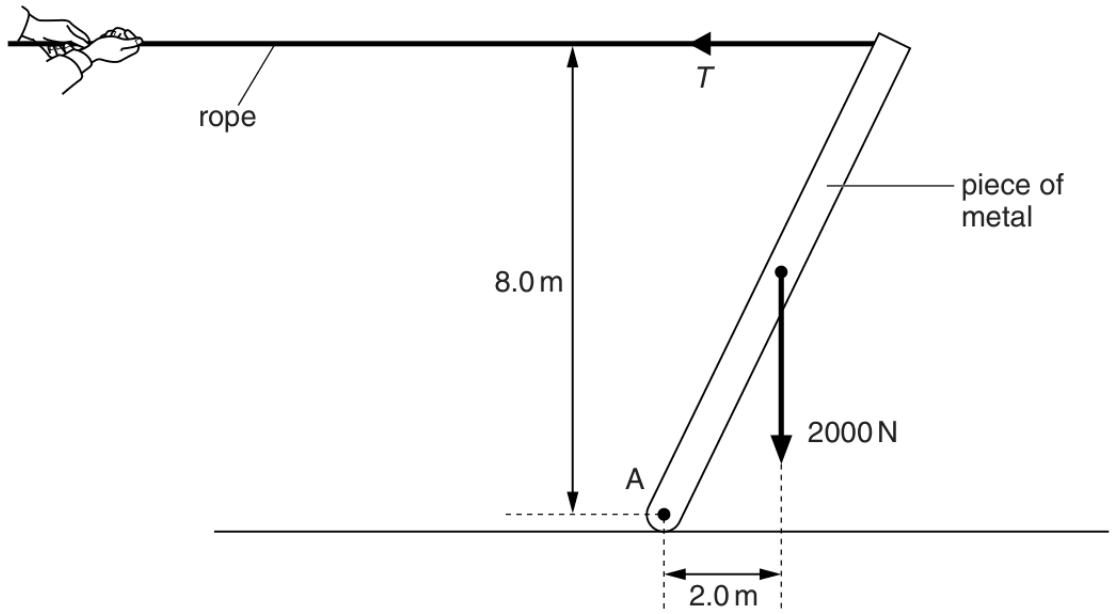


Fig. 2.1 (not to scale)

A man pulls on the rope with a horizontal force *T*. The piece of metal has a weight of 2000 N and is freely pivoted at A. The system is in equilibrium.

(i) By taking moments about A, calculate *T*.

T = ... [2]

(ii) The force *T* and the force that the rope exerts on the man are related by Newton's third law. State the relationship between these forces.

.....

.....

..... [2]

2	**(a)**	force × distance perpendicular distance	M1 A1	
	(b) **(i)**	$T \times 8$ or 2000×2 seen 500 N	C1 A1	
	(ii)	(two forces) equal (in magnitude) (two forces) opposite (in direction)	B1 B1	[6]

QUESTION · 5054_w13_qp_11.pdf · source page 3 of 20

4 A student kicks a ball horizontally along the ground. As he does so, his foot applies a force to the ball.

At the same time, the ball applies a force to his foot.

How do these forces compare?

A The force on the foot is equal to the force on the ball and in the opposite direction.

B The force on the foot is equal to the force on the ball and in the same direction.

C The force on the foot is smaller than the force on the ball and in the opposite direction.

D The force on the foot is smaller than the force on the ball and in the same direction.

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 4

4	**A**

2 Fig. 2.1 shows a boy moving a water container in a wheelbarrow.

For Examiner's Use

Fig. 2.1

The container has a volume of 0.15 m^3 and is filled with water of density 1000 kg/m^3.

(a) Calculate the mass of water in the container when it is full.

mass = ...[2]

(b) It is harder to stop the wheelbarrow when the container is full than when it is empty.

Explain this.

.....

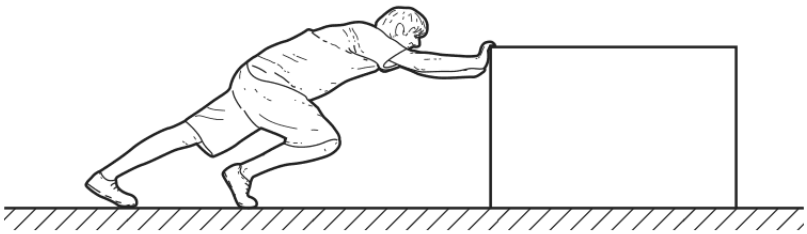
.....

.....[2]

2	**(a)**	(m =) ρV **or** 1000 × 0.150	C1	
		150 kg	A1	
	(b)	(when full) greater mass **or** greater momentum	B1	
		more inertia **or** mass resists change in state of motion **or** small(er) deceleration (for same force) **or** large(r) force for same deceleration (rate of decrease of momentum for deceleration)	B1	
		or		
		greater kinetic energy	(B1)	
		more work done in same distance (to stop)	(B1)	[4]

QUESTION · 5054_s14_qp_11.pdf · source page 3 of 16

5 A man pushes a heavy box along the ground.



A force acts between the man's hands and the box.

Another force acts between the man's feet and the floor.

In which direction do these forces act on the man?

	force on man's hands	force on man's feet
A	towards the left	towards the left
B	towards the left	towards the right
C	towards the right	towards the left
D	towards the right	towards the right

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 5

5 **B**

QUESTION · 5054_s14_qp_11.pdf · source page 3 of 16

6 In which situation is the resultant force on the body equal to zero?

A a car turning a corner at constant speed

B a rock falling freely on the Moon

C a train going up a straight hill at constant speed in a straight line

D an aircraft accelerating along a runway in a straight line

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 6

6	**C**

QUESTION · 5054_s14_qp_11.pdf · source page 5 of 16

11 A force is applied to a body.

Which property of the body **cannot** be changed by the force?

A its mass

B its shape

C its size

D its velocity

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 11

11	**A**

QUESTION · 5054_s14_qp_12.pdf · source page 5 of 16

11 A force is applied to a body.

Which property of the body **cannot** be changed by the force?

A its mass

B its shape

C its size

D its velocity

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 11

11	**A**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a lorry accelerating in a straight line along a horizontal road.

Fig. 1.1

(a) The driving force on the lorry in the forward direction is *D* and the total backward force on the lorry is *B*.

(i) State and explain whether *D* or *B* is the larger force.

.....

..... [1]

(ii) Suggest one possible cause of the backward force *B*.

..... [1]

(b) The weight of the lorry is 300 000 N.

The gravitational field strength *g* is 10 N/kg.

(i) Calculate the mass of the lorry.

mass = ... [1]

(ii) The resultant force on the lorry is 15 000 N. Calculate the acceleration of the lorry.

acceleration = ... [2]

(c) Later, the lorry turns a corner at constant speed.

Explain why the lorry accelerates even though the speed is constant.

.....

..... [1]

1 **(a)** **(i)** *D* **and either** lorry accelerates (forward) **or** resultant force is forward [B1]

(ii) air resistance **or** (air) drag **or** friction (between tyres and road) [B1]

(b) **(i)** 30 000 kg [B1]

(ii) (a=) F / m algebraic in any form or numerical [C1]
0.5(0) m / s^2 [A1]

(c) direction or velocity is changing **or** acceleration or force is sideways or towards centre (of circle) [B1]

[6]

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a motorcycle during a race.

Fig. 1.1

The motorcycle accelerates along a straight section of the track from a speed of 40 m/s to maximum speed.

Fig. 1.2 is the speed-time graph for the motorcycle along the straight section of the track.

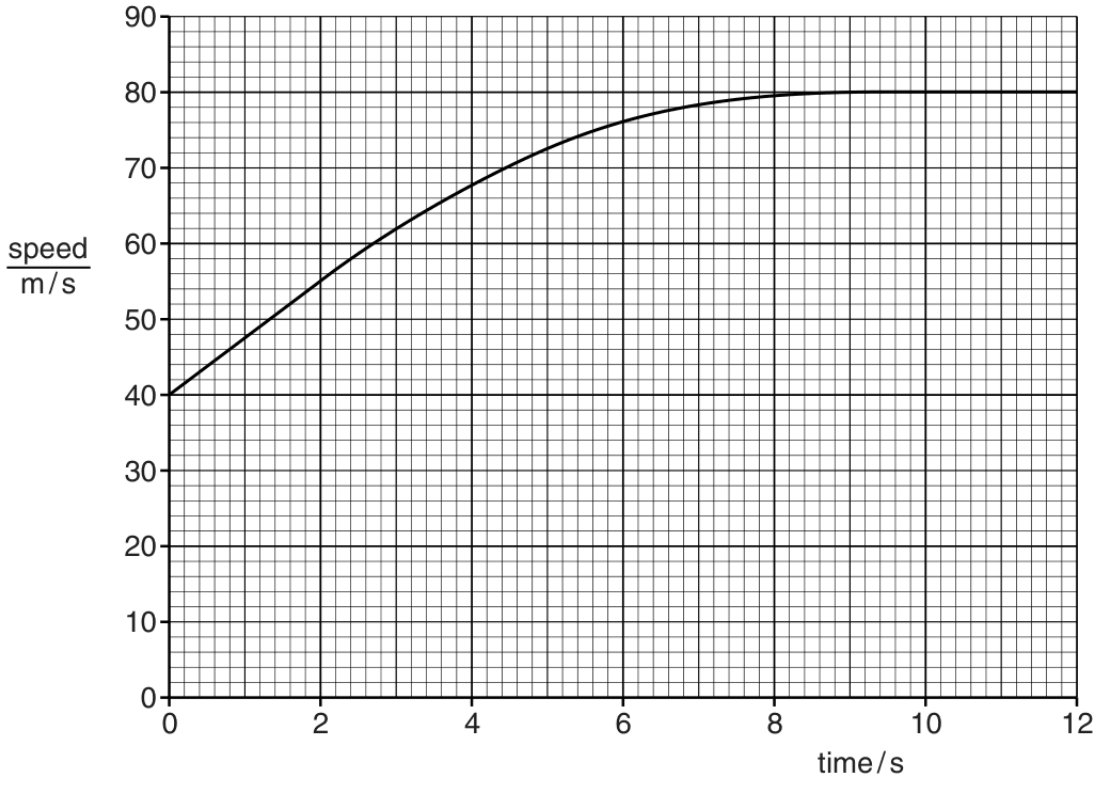


Fig. 1.2

The mass of the motorcycle is 180 kg.

(a) For the time 0 to 2.0 s, determine

(i) the acceleration of the motorcycle,

acceleration = ...[2]

(ii) the resultant force acting on the motorcycle.

force = ...[2]

(b) The driving force acting on the motorcycle remains constant throughout the 12 s spent on the straight section of track.

(i) Using Fig. 1.2, describe how the acceleration of the motorcycle changes during this time.

.....

.....[1]

(ii) Explain, in terms of the forces acting, why the acceleration changes in this way.

.....

.....

.....

.....[3]

Section A

1	**(a)**	**(i)**	(a =) $(v-u)/t$ **or** $\Delta v/t$ **or** (55–40)/2 **or** equivalent values from graph	C1	
			$7.5\,m/s^2$	A1	
		(ii)	(F =) ma **or** 180×7.5	C1	
			1300 / 1350 / 1400 N	A1	
	(b)	**(i)**	(acceleration) decreases (to zero)	B1	
		(ii)	air resistance / friction / drag mentioned	B1	
			air resistance / friction / drag increases (with speed) **or** resultant force decreases (with speed)	B1	
			(finally) (air) resistance = driving force **or** resultant is zero	B1	**[8]**

QUESTION · 5054_s15_qp_11.pdf · source page 5 of 16

9 Newton's third law involves two quantities which are equal in size and opposite in direction.

What is the unit for these two quantities?

A J

B m/s^2

C N

D W

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 9

9	**C**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a satellite in orbit around the Earth.

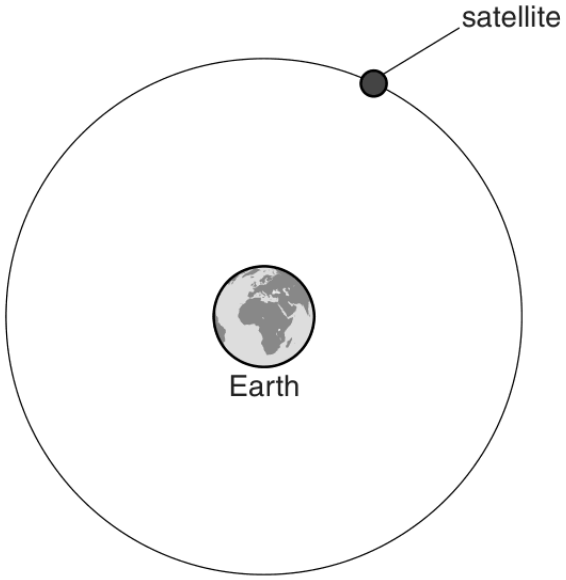


Fig. 9.1 (not to scale)

(a) The satellite travels at a constant speed in a circular orbit.

(i) Underline the quantities in the list below that are scalars.

acceleration force mass speed velocity [2]

(ii) The velocity of the satellite changes, but its speed is constant.

1. State what is meant by *velocity*. .....

.....

.....[1]

2. Explain why the velocity changes. .....

.....

.....[1]

(iii) Explain what makes this satellite move in an orbit that is circular.

.....

.....

.....

.....[2]

(b) The satellite is placed into orbit by a rocket. Fig. 9.2 shows the rocket as it takes off.

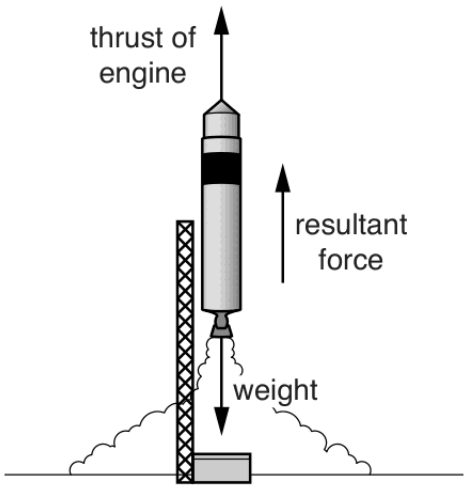


Fig. 9.2

The rocket and fuel have a total mass of 40 000 kg and a total weight of 400 000 N. The resultant force acting upwards on the rocket is 50 000 N.

(i) Calculate the thrust produced by the rocket engine.

thrust = ...[1]

(ii) Calculate the acceleration of the rocket.

acceleration = ...[2]

(c) In the first four minutes after take-off, the acceleration of the rocket is uniform.

(i) State what is meant by a *uniform acceleration*.

.....

.....

.....[2]

(ii) Fig. 9.3 describes the motion of the rocket in the first 12 minutes.

time / minutes	motion of the rocket
0 to 4	uniform acceleration
4 to 6	increasing acceleration
6 to 10	decreasing acceleration
10 to 12	constant speed

Fig. 9.3

On Fig. 9.4, sketch the speed-time graph of the rocket for the first 12 minutes.

You do not need to give values for the speed.

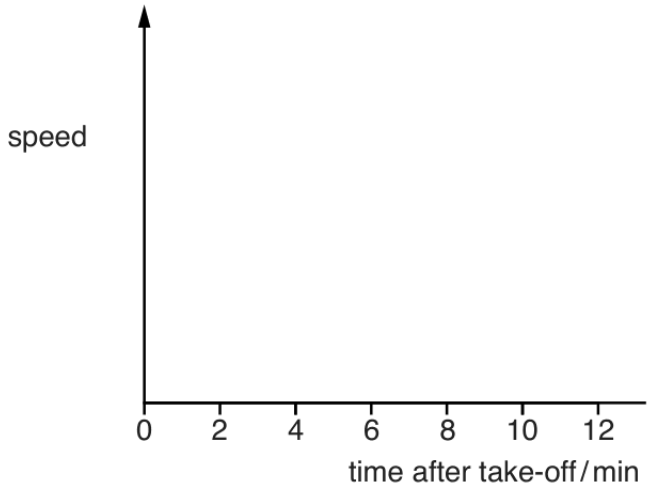


Fig. 9.4

[3]

(iii) State how the speed-time graph in **(ii)** can be used to find the distance travelled by the rocket.

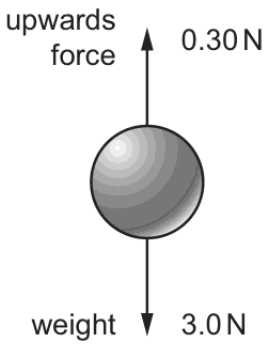
.....

.....[1]

9	**(a)**	**(i)**	speed and mass	B2
		(ii)	**1** speed and direction **or** distance / time and direction **or** displacement / time	B1
			2 direction changes	B1
		(iii)	force of gravity from / towards Earth	B1
			force is centripetal **or** at right angles to motion / velocity	B1
	(b)	**(i)**	450 000 N	B1
		(ii)	(a =) F/m or 50 000 / 40 000	C1
			1.25 m / s^2	A1
	(c)	**(i)**	same change in velocity / speed	M1
			in same time period	A1
		(ii)	start at origin and straight line for first 4 minutes	B1
			gradient increases at first after 4 and then decreases	B1
			constant speed from 10 minutes until 12 minutes	B1
		(iii)	area **under** graph	B1

QUESTION · 5054_s15_qp_12.pdf · source page 3 of 16

5 A metal ball of mass 0.30 kg and weight 3.0 N is held so that it is below the surface of oil.
It experiences an upwards force of 0.30 N.



When the ball is released, what is its initial acceleration?

A $1.0\,m/s^2$ **B** $9.0\,m/s^2$ **C** $10\,m/s^2$ **D** $11\,m/s^2$

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 5

5 **B**

11 A bungee jumper falls from a bridge above a river, as shown in Fig. 11.1.

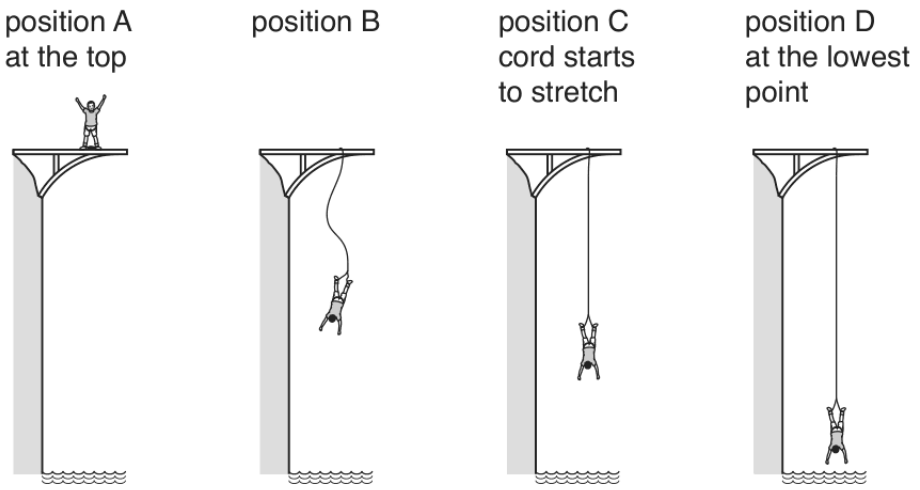


Fig. 11.1 (not to scale)

The man starts from position A in Fig. 11.1. The elastic cord starts to stretch at position C and he stops for the first time at position D. He continues to rise and fall.

Fig. 11.2 shows how the velocity of the man varies with time *t*.

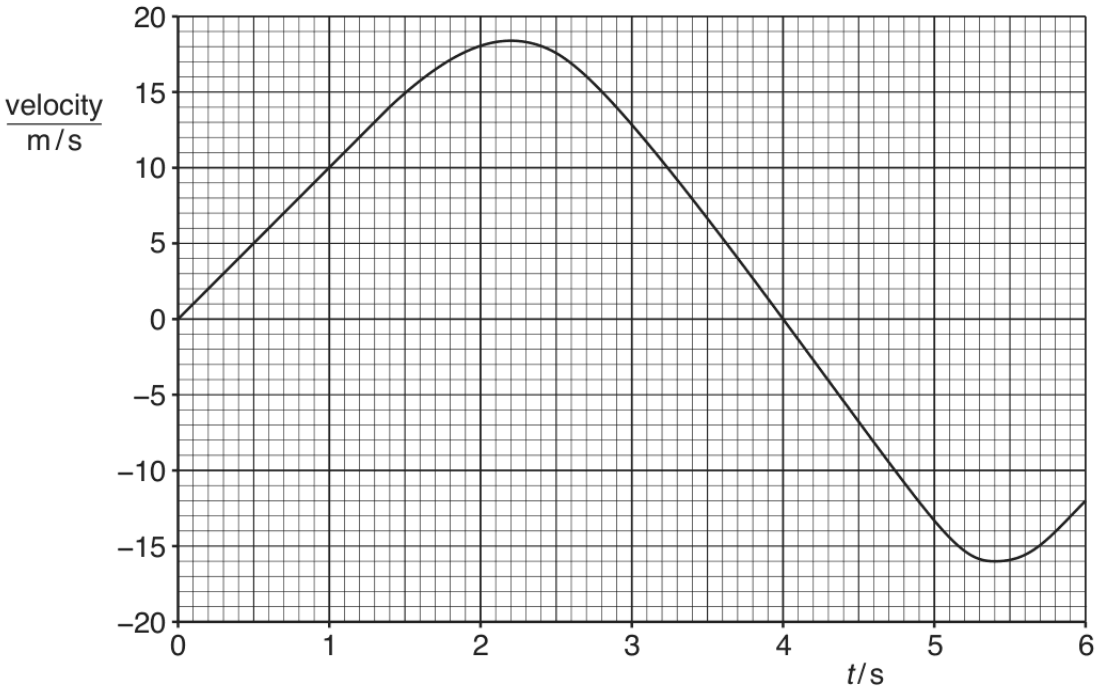


Fig. 11.2

(a) (i) State what is meant by *velocity*.

.....

..... [2]

(ii) State the difference between a positive velocity and a negative velocity.

.....

..... [1]

(iii) In the first 1.4 s the acceleration is uniform.

1. Using values from Fig. 11.2, determine the acceleration of the man in the first 1.4 s.

acceleration = ... [3]

2. Comment on your value of acceleration.

.....

..... [1]

(iv) 1. State the value of t when the man is at position D.

..... [1]

2. Explain, in terms of the forces acting, why the man is accelerating upwards at D.

.....

.....

.....

.....

..... [3]

(b) Fig. 11.3 shows the values for the gravitational potential energy of the man, the kinetic energy of the man and the elastic potential energy in the cord at A, C and D.

You may ignore the effect of air resistance in this question.

	gravitational potential energy/J	kinetic energy/J	elastic potential energy/J
position A	20000	0	0
position C	15000		0
position D	0	0	

Fig. 11.3

(i) Complete Fig. 11.3 to show the kinetic energy of the man at C and the elastic potential energy in the cord at D. [2]

(ii) The man has a mass of 50 kg.

The gravitational field strength g is 10 N/kg.

Using values from Fig. 11.3, calculate the vertical distance between A and C.

distance = ... [2]

11	**(a)**	**(i)**	distance (travelled) per second **or** speed	C1
			distance (travelled) per second / speed in a given direction **or** displacement / time **or** change in displacement per unit time **or** displacement (travelled/covered) per unit time **or** rate of change of displacement	A1
		(ii)	opposite direction	B1
		(iii)	**1** value seen for v and corresponding value of t $0 < t \leqslant 1.4$ and $0 < v \leqslant 14$	C1
			(a=) v-u / t algebraic or numerical equation	C1
			$10\,m/s^2$	A1
			2 sensible comment	A1
		(iv)	**1** 4(.0 s)	B1
			2 weight **or** force due to gravity mentioned (at D)	B1
			mention of • upwards force (on man) from cord • tension / elastic force from cord (on man)	B1
			force in cord / upward force / tension greater than downwards force **or** resultant force upwards	B1
	(b)	**(i)**	5000	B1
			20 000	B1
		(ii)	(h =) PE / mg **or** $5000 = 50 \times 10 \times h$	C1
			10 m	A1

QUESTION · 5054_w15_qp_11.pdf · source page 3 of 16

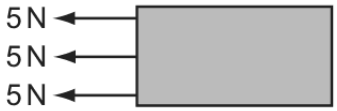
4 Which object has the largest resultant force?

A

B

C

D



MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 4

4	**C**

QUESTION · 5054_w15_qp_11.pdf · source page 4 of 16

9 A force acts on a body.

Which properties of the body may all be changed by the force?

A mass, shape and size

B mass, shape and velocity

C mass, size and velocity

D shape, size and velocity

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 9

9	**D**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows an archer firing an arrow at a target during a competition.

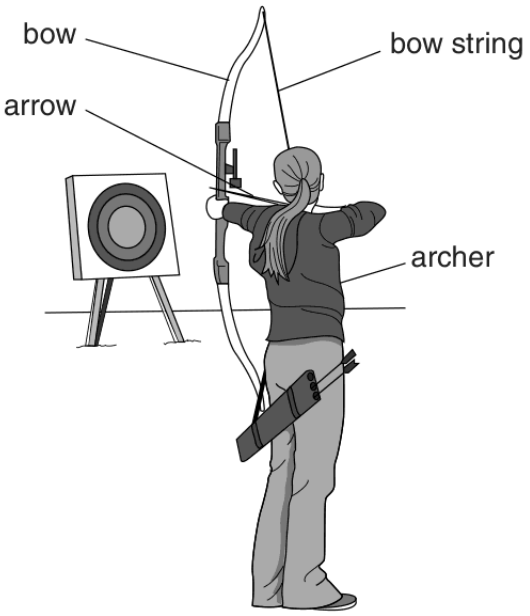


Fig. 1.1

The arrow has a mass of 0.018 kg and is initially at rest. The arrow leaves the bow 0.011 s after the bow string is released. When it leaves the bow, the velocity of the arrow is 95 m/s.

(a) Calculate

(i) the average acceleration of the arrow during the 0.011 s,

average acceleration = ... [2]

(ii) the average force exerted on the arrow during the 0.011 s.

average force = ... [2]

(b) In practice, the force exerted on the arrow gradually decreases to zero during firing.

On Fig. 1.2, sketch a possible speed-time graph for the arrow as it is being fired.

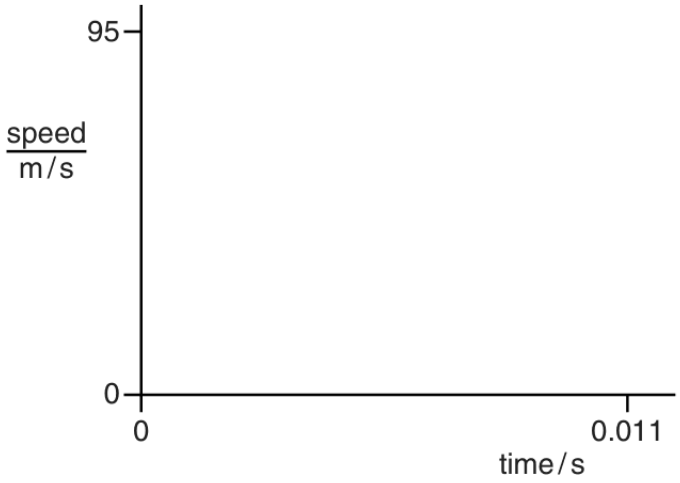


Fig. 1.2

[2]

Section A

1	**(a)**	**(i)**	$(a =)\Delta v / t$ **or** $95 / 0.011$	C1	
			$8.6(3636) \times 10^3\ \text{m/s}^2$	A1	
		(ii)	$(F =)ma$ **or** $0.018 \times 8.63 \times 10^3$	C1	
			150 / 155(.4545) / 160 N	A1	
	(b)		line from (0, 0) to (0.011, 95) with decreasing gradient	B1	
			(becomes) horizontal at (0.011, 95)	B1	[6]

QUESTION · 5054_w15_qp_12.pdf · source page 3 of 16

5 An apple of mass 0.15 kg and weight 1.5 N falls from a tree. At one point during its fall, the air resistance on the apple is 0.60 N upwards.

What is the acceleration of the apple at this point?

A $4.0\,m/s^2$ **B** $6.0\,m/s^2$ **C** $10\,m/s^2$ **D** $14\,m/s^2$

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 5

5	**B**

QUESTION · 5054_w15_qp_12.pdf · source page 3 of 16

7 Four of the gravitational forces that act between bodies in the Solar System are described below.

P the force on the Moon due to the Earth

Q the force on the Earth due to the Sun

R the force on the Earth due to the Moon

S the force on the Moon due to the Sun

Which two forces are a Newton's third law pair (action and reaction)?

A P and Q **B** P and R **C** Q and S **D** R and S

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 7

7 **B**

2 **(a)** The surface of a running track is made of rubber. A heavy trolley is pulled on to the track and it exerts a large force on the rubber track.

State two effects that this force has on the rubber.

1. .....

2. .....
[2]

(b) A spring is suspended from a support with a small pan attached to its lower end. Masses are added to the pan until the spring is extended well beyond the limit of proportionality.

Fig. 2.1 shows this apparatus.

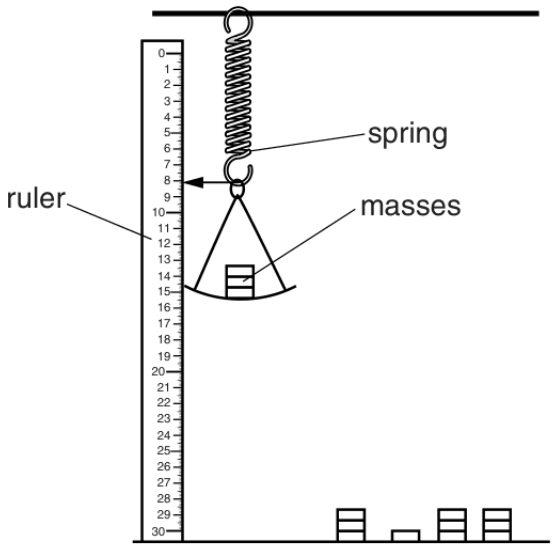


Fig. 2.1

A ruler is used when determining the extension of the spring.

(i) On Fig. 2.2, sketch the extension-load graph for the spring and label the limit of proportionality P. [2]

Fig. 2.2

(ii) The masses are then removed and the extension of the spring decreases.

Suggest what is observed when all the masses are removed.

.....

..... [1]

2	**(a)**	any **two** from different lines of: distort / stretch / change in shape / squeezed / bends / deforms compresses / change in size / volume / density / depth / height change in temperature / gets hot(ter) / generates heat	B2	
	(b) **(i)**	straight line from origin	B1	
		upward curve labelled / clear from limit of proportionality	B1	
	(ii)	permanent extension **or** spring is longer than it was originally	B1	[5]

QUESTION · 5054_s16_qp_11.pdf · source page 3 of 16

5 A man pulls a sledge of mass 25 kg across level ground with a horizontal force of 60 N. A constant force of friction of 20 N acts on the sledge.

What is the acceleration of the sledge?

A $0.63\,m/s^2$ **B** $1.6\,m/s^2$ **C** $2.4\,m/s^2$ **D** $3.2\,m/s^2$

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 5

5 **B**

QUESTION · 5054_s16_qp_11.pdf · source page 4 of 16

8 A force acts on a body.

Which list contains only quantities that can be changed by the force?

A mass, shape, velocity

B mass, shape, volume

C mass, velocity, volume

D shape, velocity, volume

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 8

8	**D**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

Answer only one of the two alternative parts in **Question 9**.

1 A sky-diver jumps from a stationary balloon. His initial downwards acceleration is 10 m/s^2.

Fig. 1.1 shows the directions of the air resistance and the weight of the sky-diver.

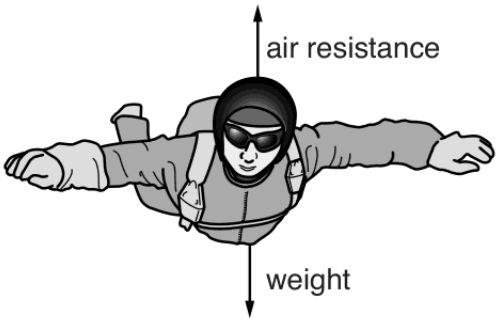


Fig. 1.1

The mass of the sky-diver is 60 kg and his weight is 600 N.

(a) Explain, using ideas about the forces, why his initial downwards acceleration is 10 m/s^2.

.....

.....

.....

.....

.....[2]

(b) When the parachute opens, the sky-diver experiences an upward acceleration for a short time.

Explain why.

.....

.....

.....[1]

(c) The total force of air resistance on the sky-diver and open parachute changes with their speed, as shown in Fig. 1.2.

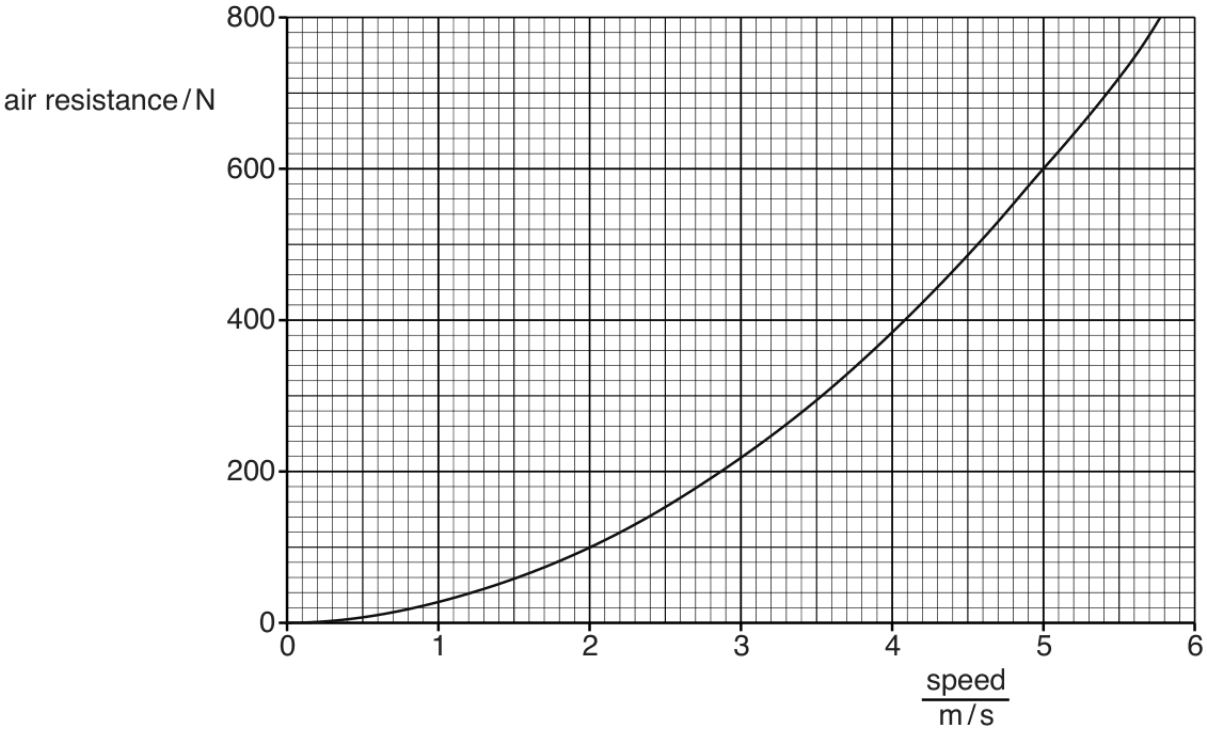


Fig. 1.2

After the parachute opens, the sky-diver slows down to a terminal velocity.

The weight of the sky-diver is 600 N.

(i) Using Fig. 1.2, determine the terminal velocity of the sky-diver.

terminal velocity = ...[1]

(ii) Calculate the resultant force on the sky-diver when his speed is 5.5 m / s.

resultant force = ...[1]

1	**(a)**	air resistance is zero **or** no air resistance acts (at first) **or** weight <u>much larger</u> than air resistance	B1
		(a=) F / m **or** weight / mass **or** 600 / 60 **or** weight is 10 times mass	B1
	(b)	air resistance / upwards force is larger than weight / 600 N / downwards force	B1
	(c) **(i)**	5(.0) m / s	B1
	(ii)	120 N	B1

QUESTION · 5054_s16_qp_12.pdf · source page 2 of 16

2 A man pulls a sledge of mass 25 kg across level ground with a horizontal force of 60 N. A constant force of friction of 20 N acts on the sledge.

What is the acceleration of the sledge?

A $0.63\,m/s^2$ **B** $1.6\,m/s^2$ **C** $2.4\,m/s^2$ **D** $3.2\,m/s^2$

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 2

2	**B**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the speed-time graph for a car travelling along a horizontal road.

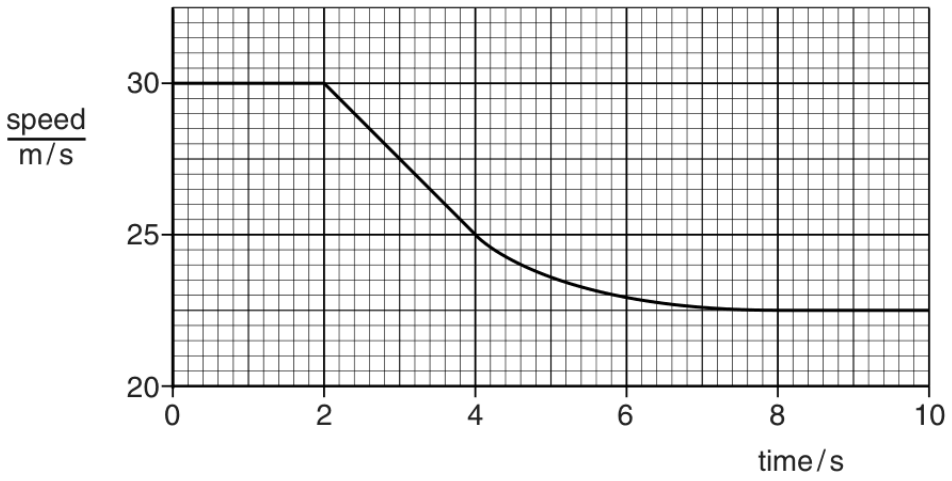


Fig. 1.1

(a) On Fig. 1.1, mark and label a point where the car has a non-uniform deceleration. [1]

(b) Calculate the deceleration of the car at $t = 3.0\,s$.

deceleration = ...[2]

(c) Explain, in terms of the horizontal forces that act on the car, why its speed is constant at $t = 1.0\,s$.

.....

.....

.....

.....[2]

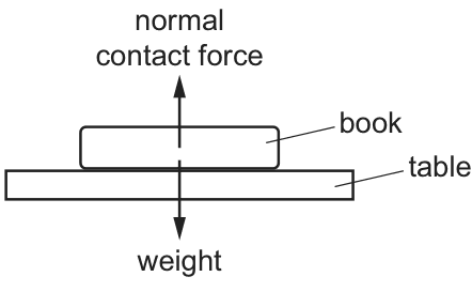
1	**(a)**	mark at a time <u>between</u> 4.0 and 7.5 seconds	B1
	(b)	(a =) (v – u) / t numerical or algebraic	C1
		or (a =) gradient of graph stated	
		$2.5\,m/s^2$	A1
	(c)	forward force and backward force clear	B1
		equal forces (in horizontal direction)	B1
		or no resultant force	
		or forces cancel / balance / in equilibrium	

QUESTION · 5054_w16_qp_11.pdf · source page 4 of 20

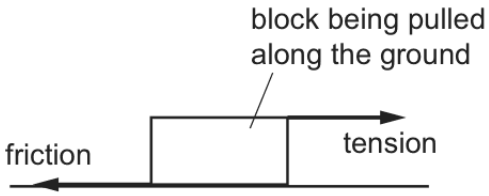
6 Each diagram shows two forces that are equal in magnitude.

Which diagram shows two forces that are a Newton's third law pair of forces?

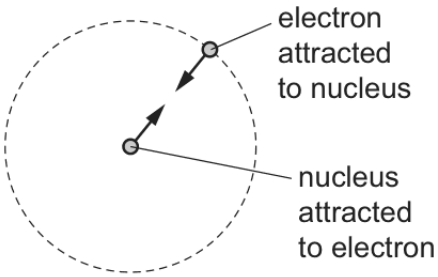
A



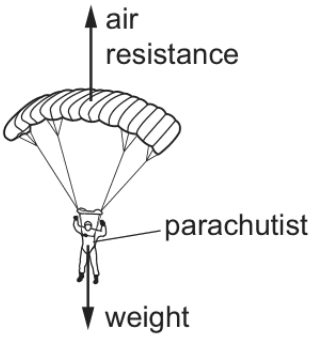
B



C



D



MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 6

6	**C**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A car of mass 800 kg is travelling at a speed of 25 m/s along a straight, horizontal road.

(a) State how *velocity* differs from *speed*.

.....

.....[1]

(b) The driver presses the accelerator and the speed of the car starts to increase from 25 m/s. The car accelerates at 1.5 m/s^2.

(i) Calculate the resultant force acting on the car as it accelerates.

force = ...[2]

(ii) Explain why the forward force acting on the car due to the engine is greater than the value in **(b)(i)**.

.....

.....

.....

.....[2]

(iii) Determine the speed of the car 4.0 s after it starts to accelerate.

speed = ...[2]

Section A

1	**(a)**	velocity / it has a direction / is a vector	B1	
	(b) **(i)**	(F =)ma **or** 800 × 1.5	C1	
		1200 N	A1	
	(ii)	friction / air resistance acts on car	B1	
		opposes force due to engine	B1	
	(iii)	(Δv =)at **or** 1.5 × 4.0 **or** 6.0	C1	
		31 m / s	A1	[7]

QUESTION · 5054_w16_qp_12.pdf · source page 2 of 16

3 Two aluminium spheres P and Q, of masses 0.50 kg and 1.0 kg respectively, are released at the same time from the same height.

Why do they have the same acceleration?

A The density of P is equal to the density of Q.

B The force on P is equal to the force on Q.

C The force on Q is double the force on P.

D The gravitational field strength for Q is double that for P.

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 3

3 **C**

QUESTION · 5054_w16_qp_12.pdf · source page 3 of 16

5 A force of 20 N acts on an object of mass 5.0 kg in the forward direction. A second force also acts on the object. The forward acceleration of the object is 3.0 m / s^2.

What is the size and direction of the second force?

	size of force / N	direction of force
A	5.0	backwards
B	15	backwards
C	15	forwards
D	35	forwards

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 5

5 **A**

QUESTION · 5054_w16_qp_12.pdf · source page 5 of 16

9 A man with an open parachute is falling to Earth at constant speed. The following forces are acting.

P the upward force of the parachute on the man.

Q the upward force of the man on the Earth.

R the downward force of the Earth on the parachute.

S the downward force of the man on the parachute.

Which two forces are a Newton's third law pair?

A P and Q **B** P and R **C** P and S **D** Q and R

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 9

9	**C**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A skier sets off from rest and accelerates uniformly at 3.4 m/s^2 in a straight line for 5.0 s.

(a) Calculate the speed of the skier after 5.0 s.

speed = ...[2]

(b) At 5.0 s, the skier stops accelerating and travels on for a further 10.0 s at a constant speed.

(i) State the size of the resultant force acting on the skier during these 10.0 s.

.....[1]

(ii) On Fig. 1.1, sketch a speed-time graph for the skier during the whole 15.0 s. [3]

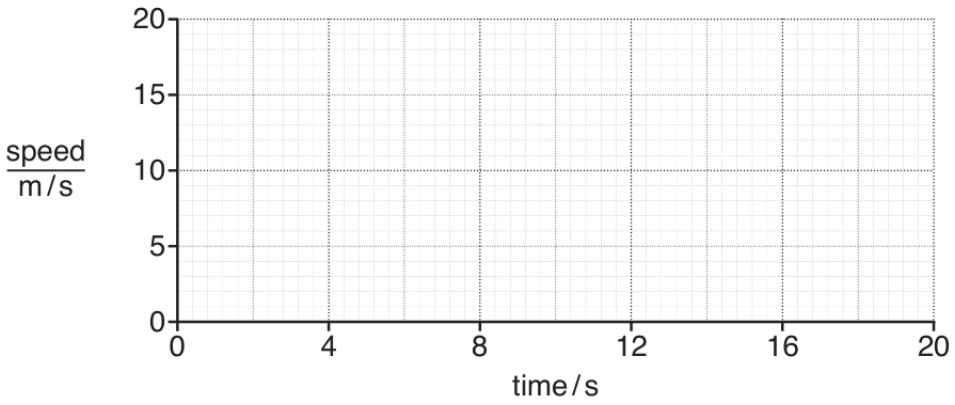


Fig. 1.1

(iii) State how the distance travelled by the skier can be determined using the speed-time graph.

.....

.....[1]

Section A

1	**(a)**	$(v = u +)at$ **or** 3.4×5.0	C1	
		17 m / s	A1	
	(b)	**(i)** 0 **or** zero **or** no resultant force	B1	
		(ii) straight line of positive gradient from (0, 0)	B1	
		horizontal line at $v > 0$ **and** after initial acceleration	B1	
		straight line from (0, 0) to (5.0, 17) **and** straight line from (5.0, 17) to at least (15.0, 17)	B1	
		(iii) calculate the area **under** the graph **or** area of trapezium	B1	[7]

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 A spring is taken from a drawer in a physics laboratory.

(a) Fig. 8.1 is the extension-load graph for the spring.

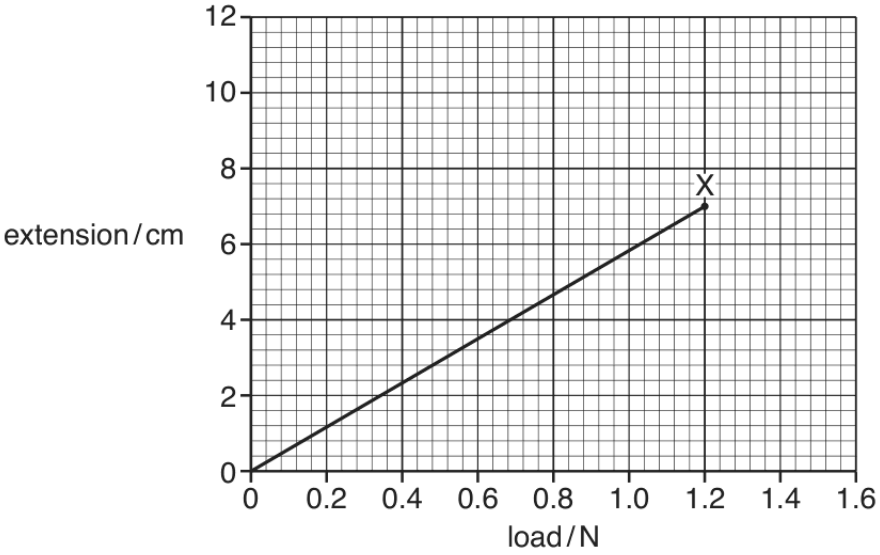


Fig. 8.1

(i) Using Fig. 8.1, determine the load on the spring when its extension is 4.9 cm.

load = ...[1]

(ii) The point X on Fig. 8.1 is the limit of proportionality for the spring.

On Fig. 8.1, sketch a line to show how the extension-load graph might continue after X. [2]

(b) A rectangular, plastic block is suspended from the lower end of the spring. Fig. 8.2 shows that the bottom 3.5 cm of the plastic block is in the water in a beaker.

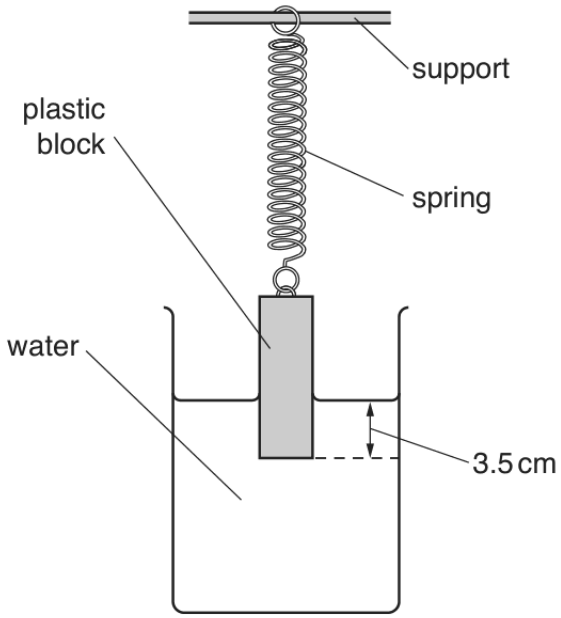


Fig. 8.2 (not to scale)

The density of water is 1000 kg / m^3 and the gravitational field strength g is 10 N / kg. The base of the plastic block is a square with sides of 4.0 cm.

(i) Calculate the pressure due to the water at a depth of 3.5 cm below the surface of the water.

pressure = ...[3]

(ii) Calculate the upward force on the base of the block due to the pressure of the water.

force = ...[2]

(iii) In Fig. 8.2, the extension of the spring is 4.9 cm.

Using the answer from **(a)(i)** and the answer from **(b)(ii)**, determine the weight of the block.

weight = ...[1]

(c) In the equipment shown in Fig. 8.2, the pressure of the atmosphere is transmitted through the water. The total pressure in the water at a depth of 3.5 cm is larger than the pressure due to the water alone.

(i) Explain why, in this experiment, atmospheric pressure does not affect the position of the plastic block in the water.

.....

.....

.....

.....[2]

(ii) Force is a vector quantity and pressure is a scalar quantity.

State how a *vector* quantity differs from a *scalar* quantity.

.....

.....[1]

(d) The beaker in Fig. 8.2 is gradually moved upwards so that more of the plastic block is in the water.

Explain what happens to the forces that act on the block.

.....

.....

.....

.....

.....

.....

.....[3]

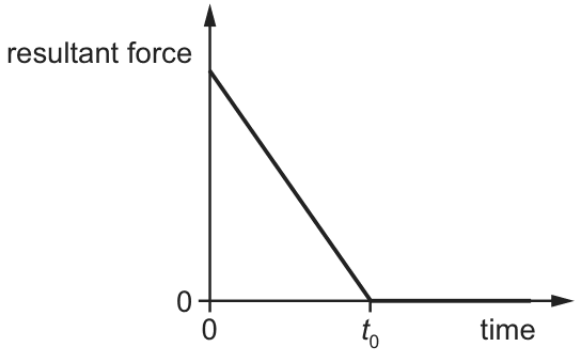
Section B

8	**(a)**	**(i)**	0.83 – 0.86 N	B1	
		(ii)	line curved	B1	
			line (curved) upwards	B1	[3]
	(b)	**(i)**	(P =)$h\rho g$	C1	
			0.035 × 1000 × 10 **or** 3.5 × 1000 × 10 **or** 35 × 1000 ×10	C1	
			350 Pa	A1	
		(ii)	(F =)PA **or** 350 × 0.0016 **or** 350 × 16 **or** 5600	C1	
			0.56 N	A1	
		(iii)	1.4 N **or (a)(i)** + **(b)(ii)** calculated	B1	[6]
	(c)	**(i)**	(atmospheric pressure) exerts a downward force / pressure (on top of the block)	B1	
			(cancels out the) extra upward force / pressure	B1	
		(ii)	(vector) has direction (in addition to magnitude)	B1	[3]
	(d)	any **three** lines from force due to water increases force due to spring decreases increased pressure (at base) they add to give a constant value / weight of block **or** total force constant		B3	[3]
					[15]

QUESTION · 5054_s17_qp_11.pdf · source page 4 of 16

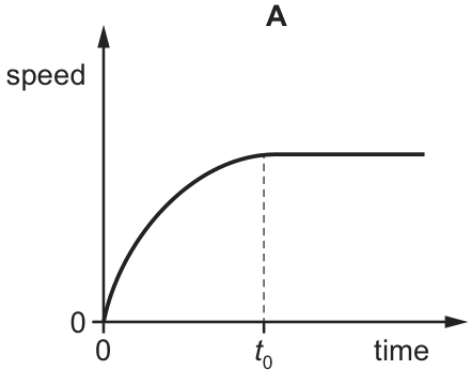
5 A resultant force acts on an object and causes it to move in a straight line.

The graph shows how the resultant force varies with time.

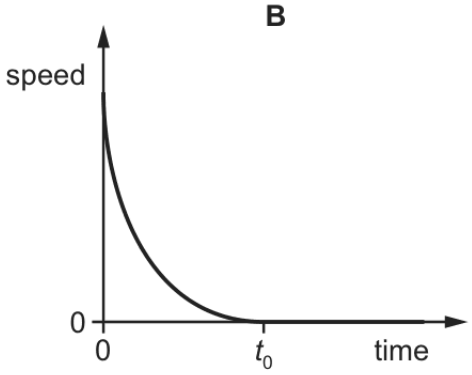


Which graph is the speed-time graph for the object?

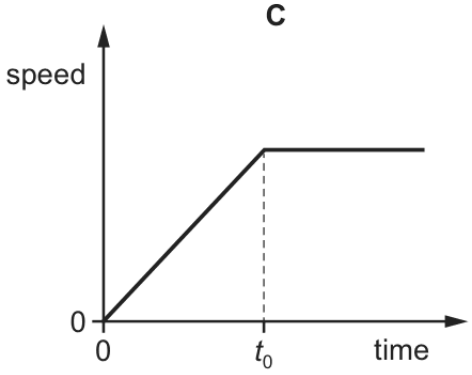
A



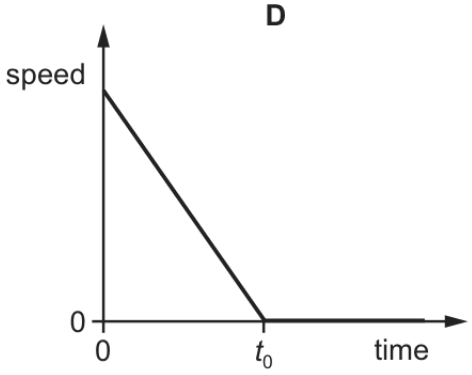
B



C



D



MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 5

5	A	**1**

QUESTION · 5054_s17_qp_11.pdf · source page 5 of 16

7 A satellite is orbiting the Earth in a circular orbit.

Which two quantities are always in the same direction as each other?

A the acceleration of the satellite and the displacement of the satellite

B the acceleration of the satellite and the velocity of the satellite

C the resultant force on the satellite and the acceleration of the satellite

D the resultant force on the satellite and the velocity of the satellite

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the directions of four forces acting on a racing car as it travels in a horizontal straight line.

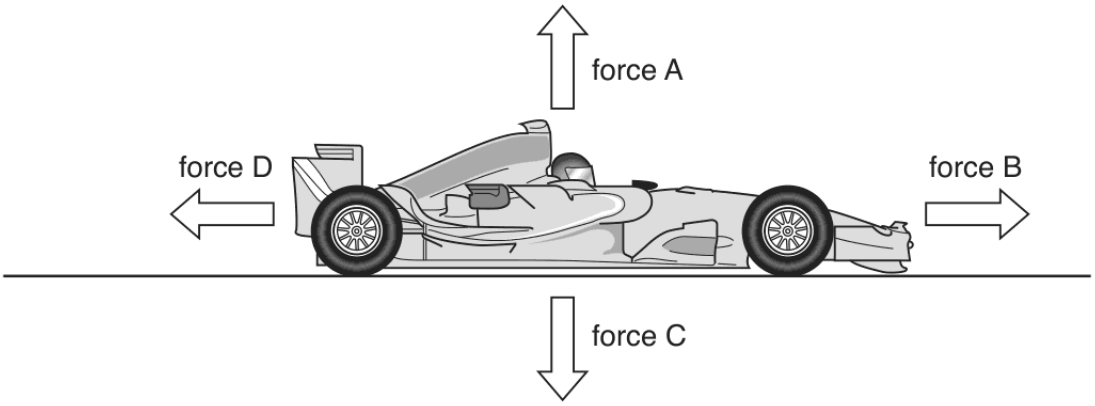


Fig. 1.1

(a) Draw a line from each box on the left to the correct description of each force.

force A	driving force
force B	contact or normal reaction force
force C	air resistance and friction
force D	force of gravity

[1]

(b) The table shows the sizes of the forces acting on the car at one time.

force A/N	force B/N	force C/N	force D/N
8000	1000	8000	600

The gravitational field strength g is 10 N/kg.

Calculate

(i) the mass of the car,

mass = ...[1]

(ii) the resultant force on the car,

resultant force = ...[1]

(iii) the acceleration of the car.

acceleration = ...[2]

(c) At another time, the car is travelling at speed u. It then accelerates for 5.0 s with an acceleration of 1.6 m/s^2, and reaches a speed of 20 m/s.

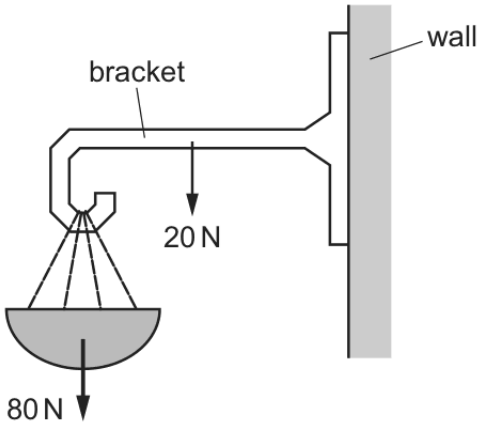
Calculate the value of u.

u = ...[2]

Question	Answer	Marks
1(a)	A – contact or reaction force B – driving force C – force of gravity D – air resistance and friction	**B1**
1(b)(i)	800 kg	**B1**
1(b)(ii)	400 N	**B1**
1(b)(iii)	(a =) F / m formula or with numbers	**C1**
	0.50 m / s^2	**A1**
1(c)	change in v = at in words or numbers	**C1**
	12 m / s	**A1**

QUESTION · 5054_s17_qp_12.pdf · source page 4 of 20

6 A hanging basket is fixed to a wall by a bracket.



The weight of the basket is 80 N. The weight of the bracket is 20 N.

What is the size of the upwards force exerted on the bracket by the wall?

A 60 N **B** 70 N **C** 90 N **D** 100 N

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 4 of 20

7 A train of mass 240 000 kg is travelling at a speed of 60 m/s. The brakes are applied and it decelerates for 10 minutes until it comes to rest.

What is the average resultant force?

A 24 000 N **B** 40 000 N **C** 480 000 N **D** 720 000 N

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 7

7	A	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A car accelerates from rest in a straight line. During the first 14 s, the acceleration is uniform and the car reaches a speed of 25 m/s.

(a) (i) Calculate the acceleration of the car.

acceleration = ...[2]

(ii) After the first 14 s, the speed of the car continues to increase but the acceleration decreases. From 70 s to 80 s after the start, the car moves at a constant speed of 55 m/s.

On Fig. 1.1, draw a possible speed-time graph for the car.

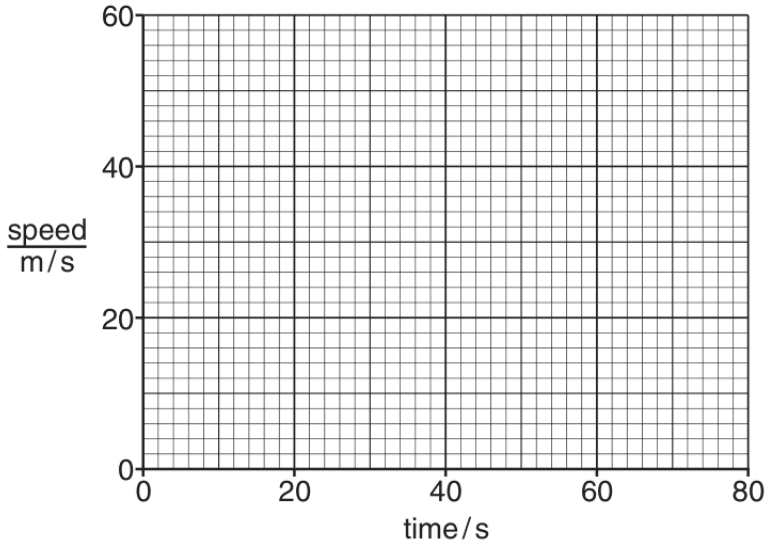


Fig. 1.1

[2]

(b) At a later time, the driver applies the brakes to stop. He is wearing a seat belt and slows down in his seat. A bag on the seat next to him slides forwards, across the seat towards the front of the car.

Using ideas about the forces acting, explain why the driver slows down but the bag slides forwards.

.....

.....

.....

.....[3]

Question	Answer	Marks
1(a)((i)	(a=) v(–u) / t **or** 25 / 14	**C1**
	1.8 m / s^2	**A1**
1(a)(ii)	initial straight line from (0,0) to (14,25)	**B1**
	gradient of line decreases after 14 s **and** flat from (70,55) to (80,55)	**B1**
1(b)	force backwards on driver / car B1	**any 3**
	force produced by seat belt / steering wheel **or** friction with seat / friction between tyres and road B1	
	no / small (backward) force / friction on bag B1	**B3**
	(mass of bag) resists change (from state of motion) **or** carries on in straight line **or** has constant velocity **or** (bag) has inertia B1	

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a large container ship travelling at constant speed in a straight line.

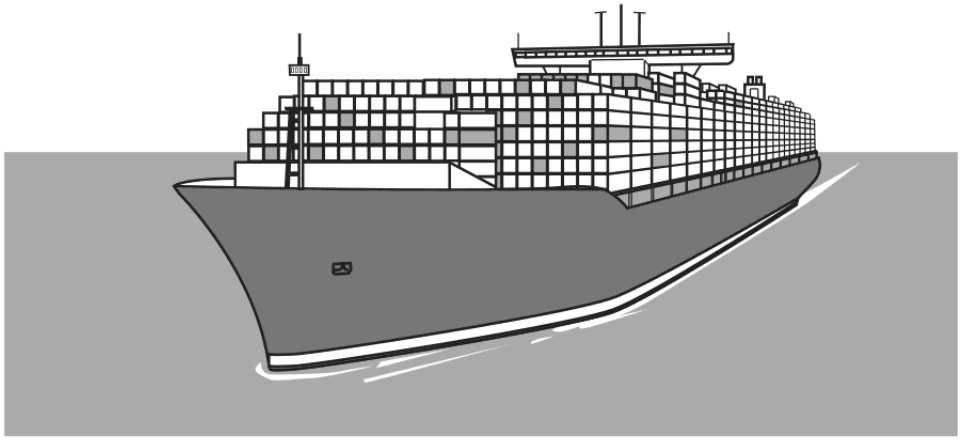


Fig. 9.1

The resistive force acting on the ship is 2.8×10^6 N.

(a) The speed of the ship is 9.7 m/s.

(i) Calculate the work done against the resistive force on the ship in 1.0 s.

work done = ... [2]

(ii) The engines are powered by oil.

State the energy transfer that is taking place when the ship is travelling at constant speed.

.....

..... [2]

(b) The mass of the ship is 2.2×10^8 kg. The engines are switched off and the resistive force causes the ship to decelerate.

(i) Calculate the initial deceleration of the ship.

deceleration = ... [2]

(ii) As the speed of the ship decreases, its deceleration changes.

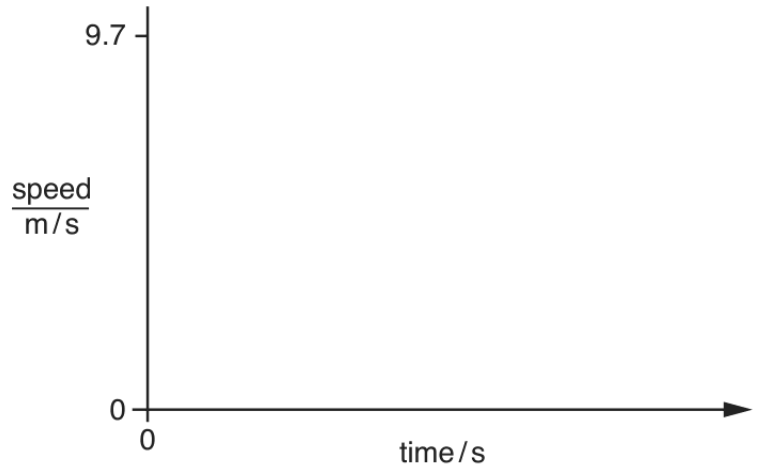
1. Suggest and explain how the deceleration changes.

.....

.....

..... [2]

2. On Fig. 9.2, sketch a possible speed-time graph for the ship as it decelerates to rest.



[2]

Fig. 9.2

3. Explain how the distance travelled by the ship may be determined from the speed-time graph.

.....

.....

..... [2]

(c) When the ship is travelling at a different speed, energy is being supplied to the engines at a rate of 33 MJ/s. The efficiency of the engines is 0.36 (36%).

(i) State a relationship that defines *efficiency*.

.....

..... [1]

(ii) Calculate the rate at which energy is wasted in the engines.

rate at which energy is wasted = ... [2]

Section B

Question	Answer	Marks
9(a)(i)	(WD) = Fx **or** $2.8 \times 10^6 \times 9.7$	**C1**
	2.7×10^7 J	**A1**
9(a)(ii)	from chemical (potential energy)	**B1**
	to thermal (energy) / heat	**B1**
9(b)(i)	(a =)F / m **or** $2.8 \times 10^6 / 2.2 \times 10^8$	**C1**
	0.013 m / s^2	**A1**
9(b)(ii)	**1** deceleration / it decreases	**B1**
	resistive force decreases **or** resistive force depends on speed	**B1**
	2 curve/line from 9.7 to zero **and** gradient negative (**allow** zero at end)	**B1**
	magnitude of gradient decreasing	**B1**
	3 area mentioned	**B1**
	area <u>under</u> line / curve **or** convert cm^2 (of graph paper) to distance **or** in terms of the scales	**B1**
9(c)(i)	(efficiency =) useful energy output / total energy input **or** useful power output / total power input	**B1**
9(c)(ii)	33×0.64 **or** 12 MJ / s	**C1**
	21 MJ / s	**A1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Two small tugboats are pulling a large ship in a harbour. Fig. 1.1 represents the view from above and shows the directions of the forces on the ship.

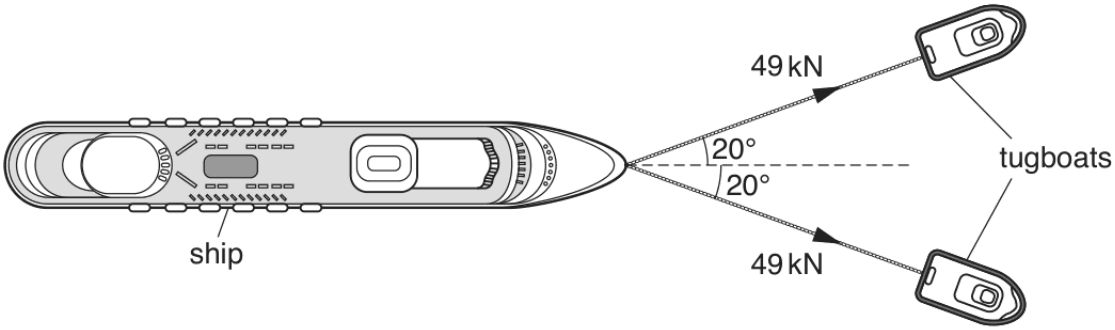


Fig. 1.1 (not to scale)

Each of the tugboats is exerting a force of 49 kN on the ship.

(a) Determine by a graphical method the resultant of these two forces and state the scale used.

scale .....

resultant = ...
[3]

(b) The engines of the ship are not operating and the water in the harbour is stationary. The ship is moving in a straight line in the direction of the total force exerted by the tugboats. It is travelling at a constant speed.

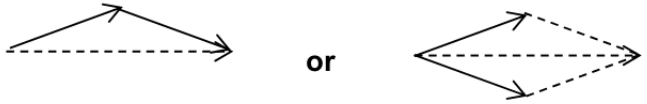
Explain, in terms of the forces acting, why the ship is moving in a straight line at constant speed.

.....

.....

.....

.....[2]

Question	Answer	Marks
1(a)	scale at least: 1.0 cm : 10 kN	**B1**
	correct triangle **or** correct parallelogram **and** resultant, i.e. 	**B1**
	90 kN ⩽ resultant ⩽ 94 kN	**B1**
1(b)	no <u>resultant</u> force **or** balanced forces **or** all forces cancel	**B1**
	(force of tugboats cancelled) by drag / water resistance / water friction etc.	**B1**

2 A force applied to a solid object may cause it to accelerate so that its velocity changes.

(a) State two other properties of the object which may change when a force is applied.

1. .....

2. .....

[2]

(b) A spring has a mass of 0.012 kg.

(i) The gravitational field strength *g* is 10 N/kg.

Calculate the weight of the spring.

weight = ...[1]

(ii) The spring is suspended vertically and a load is attached to its lower end. The spring extends by 2.7 cm and does not pass the limit of proportionality.

Fig. 2.1 shows the spring and load attached to the lower end of a second identical spring.

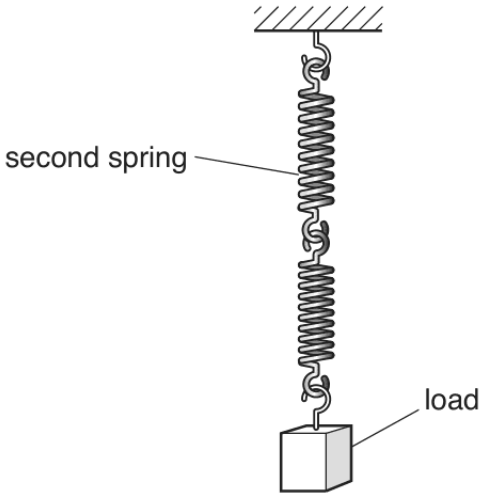


Fig. 2.1

Suggest one reason why the extension of the second spring differs from 2.7 cm.

.....

.....

.....[1]

(iii) Explain what is meant by *limit of proportionality*.

.....

.....

.....[1]

(c) The load in Fig. 2.1 is pulled down below its equilibrium position.

(i) State the form of energy stored in the stretched springs.

.....[1]

(ii) The load is released and it moves upwards and downwards. The distance travelled in each movement decreases until the load stops moving.

Explain why the load stops moving.

.....

.....

.....[1]

Question	Answer	Marks
2(a)	shape	**B1**
	size / volume / density	**B1**
2(b)(i)	0.12 N	**B1**
2(b)(ii)	weight of first spring (increases extension) **or** weight of first spring (takes spring beyond the limit of proportionality)	**B1**
2(b)(iii)	where / point / load / extension / limit beyond which the extension is not directly proportional to load / extension-load graph is not straight	**B1**
2(c)(i)	elastic potential energy	**B1**
2(c)(ii)	friction / air resistance mentioned **or** thermal energy / heat produced	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows an astronaut in space near to a space station in orbit above the Earth.

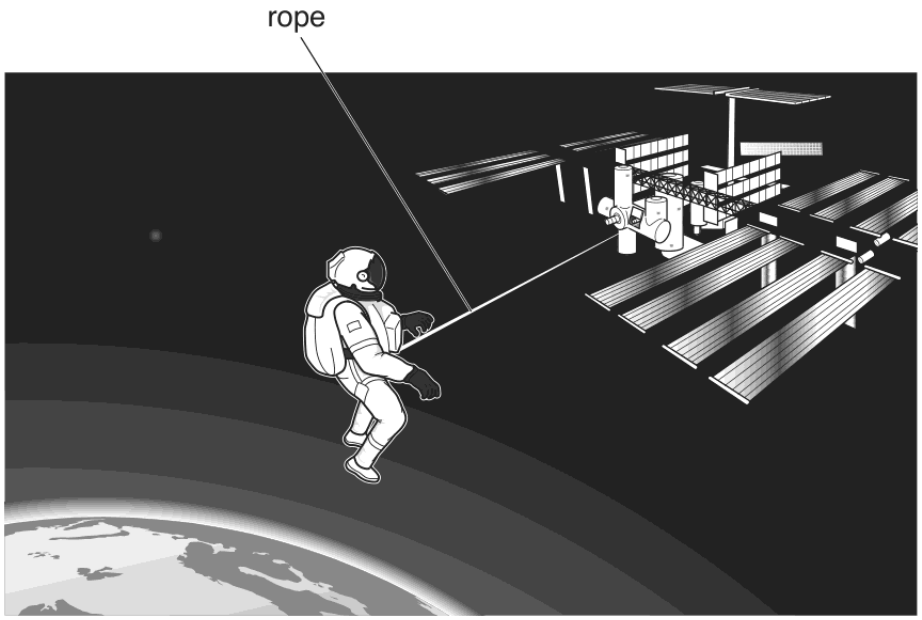


Fig. 9.1

The total mass of the astronaut and all his equipment is 160 kg.

The astronaut is initially at rest relative to the space station but he is then pulled towards the space station by the rope. His acceleration towards the space station is 0.35 m/s^2 for 1.2 s.

(a) Calculate

(i) the resultant force that causes a mass of 160 kg to accelerate at 0.35 m/s^2,

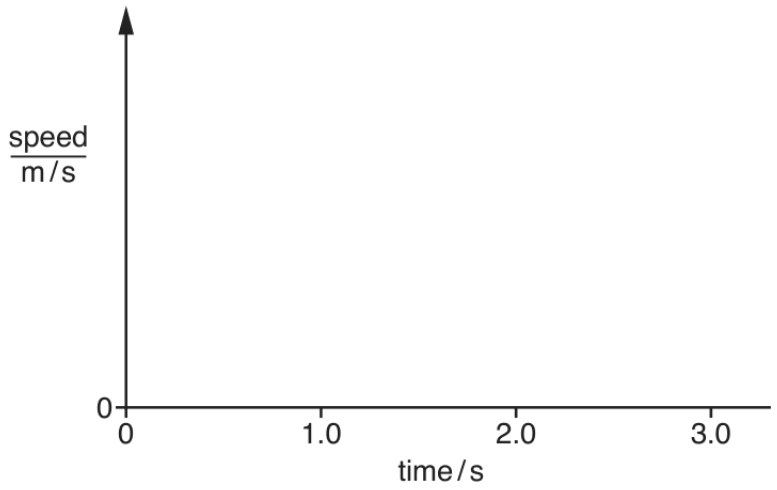
resultant force = ...[2]

(ii) the speed at which the astronaut is travelling after 1.2 s.

speed = ...[2]

(b) The resultant force on the astronaut is constant for 1.2 s, but then it decreases to zero instantly as the rope becomes slack.

(i) On Fig. 9.2, sketch the speed-time graph for the astronaut for the first 3.0 s of his motion.



[2]

Fig. 9.2

(ii) Describe how the distance moved by the astronaut during the first 3.0 s may be found using the speed-time graph.

.....

.....

.....[2]

(c) The space station is travelling at a constant speed in a circular orbit around the Earth as shown in Fig. 9.3.

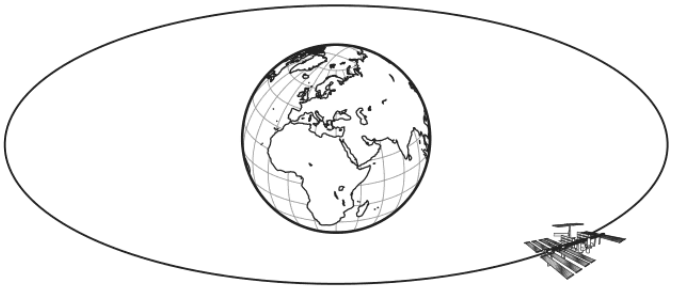


Fig. 9.3

(i) A force acts on the space station to keep it in this orbit.

1. On Fig. 9.3, draw an arrow to show the direction of this force. [1]

2. Explain what causes this force.

.....

.....

.....[2]

(ii) State what is meant by the *velocity* of an object.

.....

.....

.....[2]

(iii) State and explain what happens during the orbit to

1. the velocity of the space station,

.....

.....

.....[1]

2. the kinetic energy of the space station.

.....

.....

.....[1]

Question	Answer	Marks
9(a)(i)	(F =) ma **or** 160 × 0.35	**C1**
	56 N	**A1**
9(a)(ii)	(v = u +) at **or** 0.35 × 1.2	**C1**
	0. 42 m / s	**A1**
9(b)(i)	straight line of positive gradient from origin to t = 1.2 s	**B1**
	horizontal line after t = 1.2 s	**B1**
9(b)(ii)	area mentioned	**B1**
	area under the line **or** convert cm^2 (of graph paper) to distance **or** in terms of the scales	**B1**
9(c)(i)	**1** arrow from space-station towards centre of Earth	**B1**
	2 gravitational field / attraction	**B1**
	of Earth	**B1**
9(c)(ii)	**two** separate approaches	
	I change of displacement	**M1**
	per unit time **or** divided by time	**A1**
	or **II** rate of change of distance **or** distance moved per unit time	**M1**
	in specified direction	**A1**
9(c)(iii)	**1** (it / velocity) changes **and** (because its) direction changes / force perpendicular to velocity	**B1**
	2 (remains) constant **and** depends on speed / g.p.e. remains constant / (k.e.) is a scalar quantity / speed is constant	**B1**

2 Fig. 2.1 shows a man pushing a heavy box with a force *P*. A frictional force *F* acts in a horizontal direction.

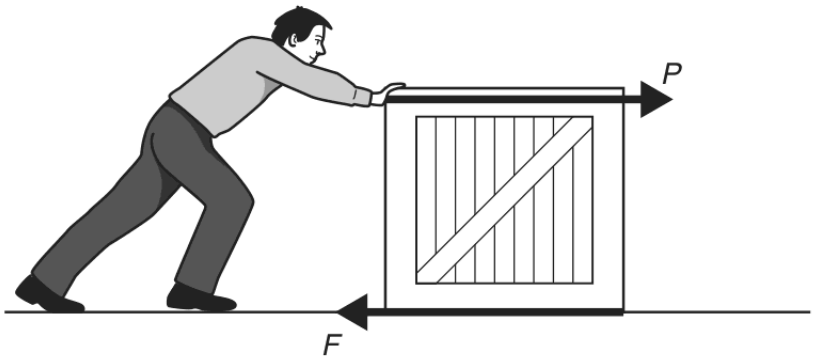


Fig. 2.1

(a) The forces on the box are *balanced* and the box is stationary.

(i) State what is meant by *balanced forces*.

.....

.....[1]

(ii) The frictional force in Fig. 2.1 does not produce any heating effect.

State what must happen for the frictional force to produce heating.

.....

.....[1]

(iii) Apart from being stationary, describe one other possible state of motion of the box when the forces are balanced.

.....

.....[1]

(b) When $P = 100\,\text{N}$ and $F = 85\,\text{N}$, the box accelerates. The mass of the box is 25 kg.

Calculate the acceleration of the box.

acceleration = ...[3]

Question	Answer	Marks
2(a)(i)	equal and opposite forces **or** no resultant force	**B1**
2(a)(ii)	box moves / slides	**B1**
2(a)(iii)	constant velocity **or** constant speed and direction	**B1**
2(b)	(resultant force) 15 (N) seen	**C1**
	(a=) F / m in any form numerical or algebraic	**C1**
	0.60 m / s^2	**A1**

QUESTION · 5054_s18_qp_12.pdf · source page 5 of 16

11 A force of 1600 N accelerates a car, of mass 800 kg, from rest.

What is the car's acceleration and its velocity after 4.0 s?

	acceleration / m/s^2	velocity / m/s
A	0.50	0.13
B	0.50	0.50
C	2.0	2.0
D	2.0	8.0

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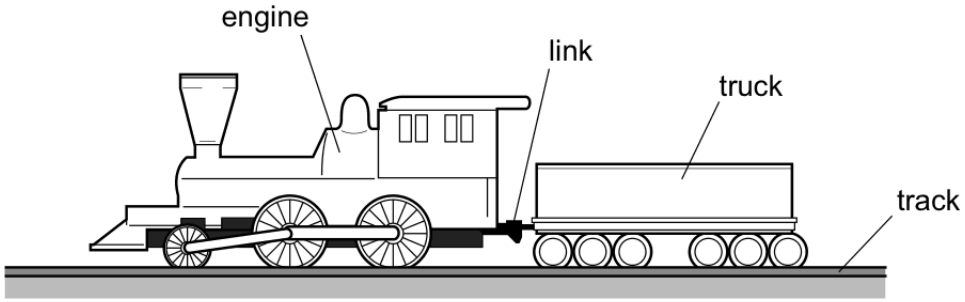
11	D	**1**

QUESTION · 5054_w18_qp_11.pdf · source page 3 of 16

5 An engine pulls a truck at constant speed on a level track.



The link between the engine and the truck breaks. The driving force on the engine remains constant.

What effect does this have on the engine and on the truck?

	engine	truck
A	speed stays constant	slows down
B	speeds up	slows down
C	speed stays constant	stops immediately
D	speeds up	stops immediately

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 5

5	B	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A girl of mass 35 kg, on a bicycle, accelerates from rest and travels down a slope in a straight line. The girl does not use the pedals.

Fig. 1.1 shows that the gradient of the slope is constant.

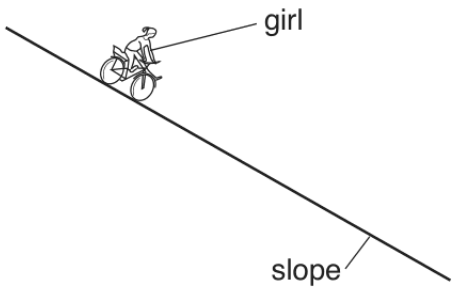


Fig. 1.1

(a) Calculate the resultant force on the girl when she is accelerating at 2.6 m/s^2.

resultant force = ... [2]

(b) At first, her acceleration is constant. At time t_1, her acceleration starts to decrease gradually until she is travelling at a constant speed in a straight line.

(i) On Fig. 1.2, sketch a speed-time graph for the girl from when she starts moving until she is travelling at a constant speed. [2]

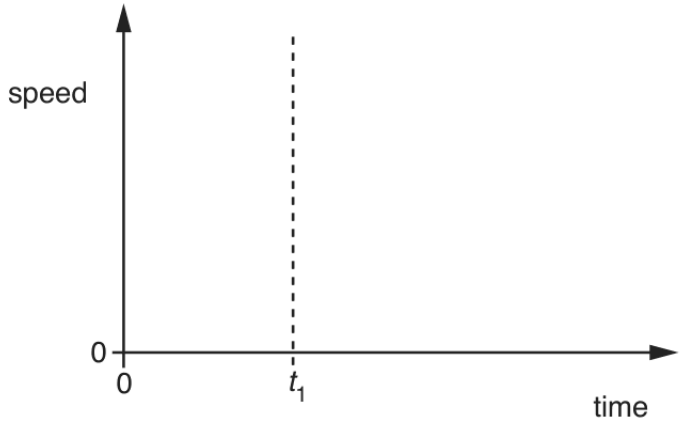


Fig. 1.2

(ii) State how the distance travelled is found from a speed-time graph.

.....

..... [1]

Question	Answer	Marks
1(a)	(F =) ma **or** 35 × 2.6	**C1**
	91 N	**A1**
1(b)(i)	straight line from origin to dashed line / t_1	**B1**
	gradually decreasing gradient until the line is horizontal	**B1**
1(b)(ii)	area under the line **or** area between line and x-axis	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 A bucket of water is pulled up out of a well using a rope. Fig. 8.1 shows the rope winding on to a cylinder as the handle is turned.

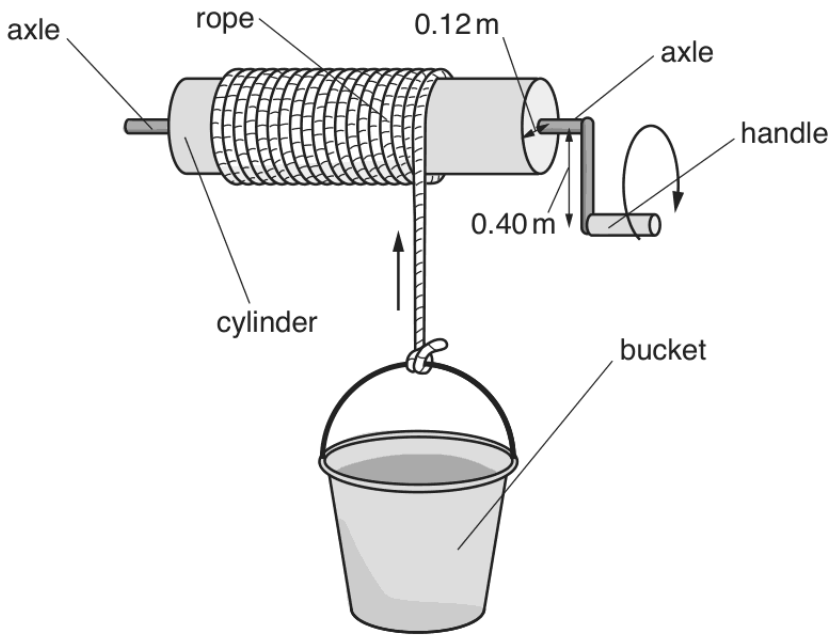


Fig. 8.1

The empty bucket has a mass of 1.0 kg.

(a) Complete the sentences that describe mass by filling in the gaps.

The mass of a body is a measure of the amount of .....

in the body. It resists a change in the state of ..... of

the body. [2]

(b) When the bucket is full, it contains $2.4 \times 10^{-2}\,m^3$ of water.

The gravitational field strength g is equal to 10 N/kg.

(i) Explain what is meant by a *gravitational field*.

.....

..... [1]

(ii) The density of water is 1000 kg / m^3.

Determine the total weight of the bucket and the water.

weight = ... [3]

(c) The radius of the cylinder is 0.12 m and the handle is 0.40 m from the axle of the cylinder. The weight of the bucket and the water produce a moment that acts on the cylinder.

(i) Calculate this moment.

moment = ... [2]

(ii) Calculate the minimum force on the handle that balances this moment.

force = ... [1]

(d) A farmer pulls the bucket of water up at a constant speed. He needs to exert a force on the handle that is greater than that calculated in **(c)(ii)**. He notices that there is a slight increase in temperature where the axle is in contact with the frame holding it.

(i) Suggest two reasons why the force exerted is greater than the value calculated.

1. .....

.....

2. .....

.....
[2]

(ii) State the energy changes that are taking place as the bucket is being lifted at a constant speed.

.....

.....

..... [3]

(e) State what can be deduced about the forces acting on the bucket when it is travelling upwards at a constant speed.

.....

..... [1]

Question	Answer	Marks
8(a)	substance **or** matter	**B1**
	rest **or** motion	**B1**
8(b)(i)	(a region of space) where a mass experiences a force (due to gravitational attraction)	**B1**
8(b)(ii)	(m =) ρV **or** $1000 \times 2.4 \times 10^{-2}$ **or** 24 (kg)	**C1**
	25 (kg) **or** 240 (N)	**C1**
	250 N	**A1**
8(c)(i)	(Γ =) Fx **or** 250×0.12	**C1**
	30 N m	**A1**
8(c)(ii)	75 N	**B1**
8(d)(i)	any **two** from: friction (between axle and cylinder) weight of rope force not perpendicular (to radius) rope wraps over itself (and increases moment)	**B2**

Question	Answer	Marks
8(d)(ii)	from chemical (energy) **or** chemical (energy) as first term	**B1**
	to gravitational potential (energy) **or** to thermal / heat (energy) as last term	**B1**
	to thermal / internal (energy) / heat **and** thermal / internal (energy) / heat	**B1**
8(e)	no resultant force **or** forces balance / cancel **or** tension equals weight (of bucket) **or** upwards force equals downward force	**B1**

QUESTION · 5054_w18_qp_12.pdf · source page 2 of 16

1 An apple falls from a tree.

Which row describes the forces acting on the apple and the motion of the apple as it falls?

	forces acting on the apple	motion of the apple
A	balanced	acceleration
B	balanced	constant speed
C	unbalanced	acceleration
D	unbalanced	constant speed

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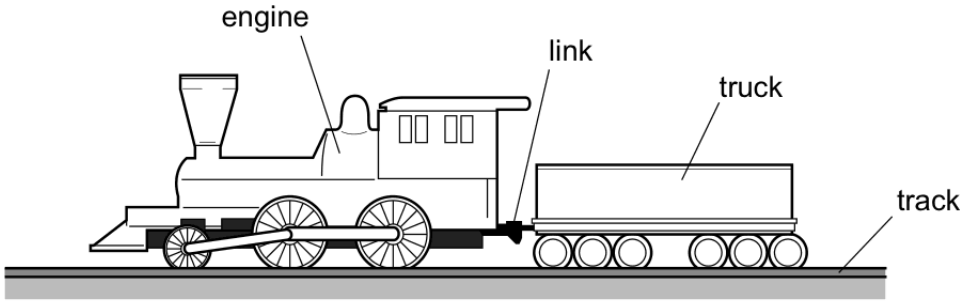
1	C	**1**

QUESTION · 5054_w18_qp_12.pdf · source page 2 of 16

3 An engine pulls a truck at constant speed on a level track.



The link between the engine and the truck breaks. The driving force on the engine remains constant.

What effect does this have on the engine and on the truck?

	engine	truck
A	speed stays constant	slows down
B	speeds up	slows down
C	speed stays constant	stops immediately
D	speeds up	stops immediately

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3	B	**1**

QUESTION · 5054_s19_qp_11.pdf · source page 3 of 20

5 Four of the gravitational forces that act between bodies in the Solar System are listed below.

P the force on the Moon due to the Earth

Q the force on the Earth due to the Sun

R the force on the Earth due to the Moon

S the force on the Moon due to the Sun

Which two forces are a Newton's third law pair (action and reaction)?

A P and Q **B** P and R **C** Q and S **D** R and S

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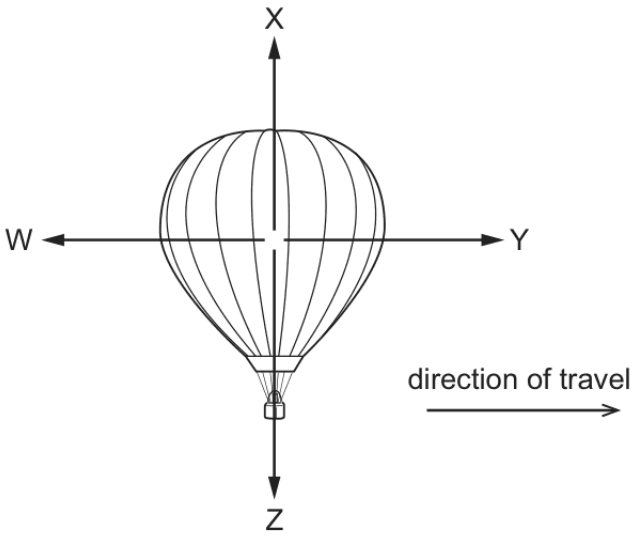
5	B	**1**

QUESTION · 5054_s19_qp_11.pdf · source page 3 of 20

6 A hot-air balloon is travelling at constant velocity and is at a constant height above the ground.

The diagram shows the only four forces acting on the balloon.



Which statement is correct?

A W and Y are equal, X and Z are equal.

B W and Y are equal, X is greater than Z.

C X and Z are equal, Y is greater than W.

D Y is greater than W, X is greater than Z.

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 6

6	A	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a parachutist falling vertically towards the ground.

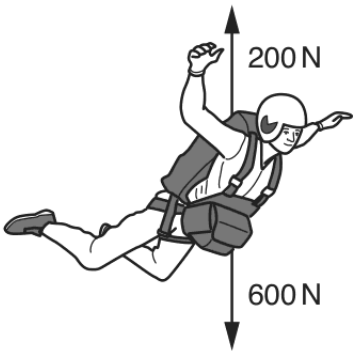


Fig. 9.1

The mass of the parachutist is 60 kg and his weight is 600 N.

(a) State what is meant by:

(i) *mass*

.....

..... [1]

(ii) *weight*.

.....

..... [1]

(b) Calculate the acceleration of the parachutist when the air resistance is 200 N.

acceleration = ... [3]

(c) The parachutist falls from rest at time $t = 0$.

He accelerates non-uniformly until he reaches a terminal velocity of 44 m/s at $t = 30$ s.

At $t = 40$ s the parachute opens. He decelerates for 4.0 s, reaching a smaller terminal velocity of 5.0 m/s.

He then falls at this speed until he lands on the ground at $t = 200$ s.

(i) On Fig. 9.2, draw the speed–time graph for the parachutist.

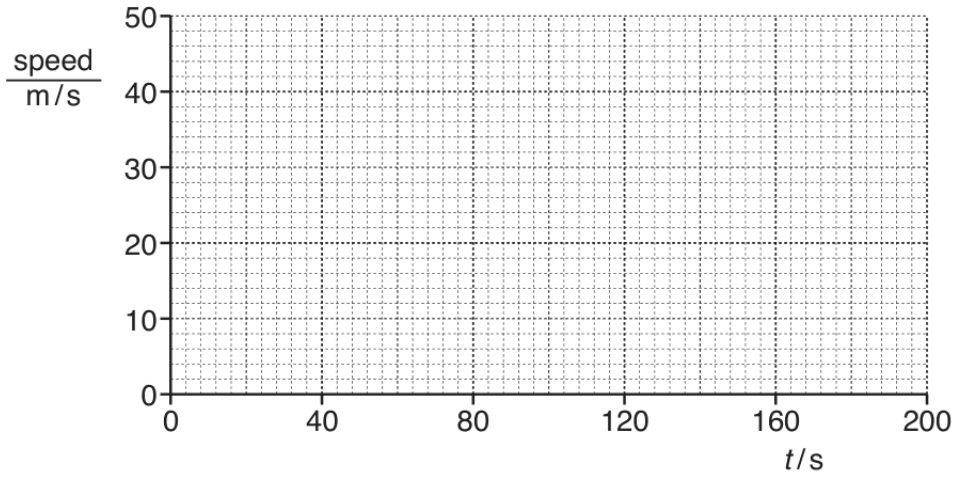


Fig. 9.2

[4]

(ii) Calculate the average deceleration of the parachutist between $t = 40$ s and $t = 44$ s.

deceleration = ... [2]

(d) Explain, in terms of the forces involved, why:

(i) the parachutist reaches a terminal velocity at $t = 30$ s

.....

..... [1]

(ii) the parachutist decelerates when the parachute opens

.....

.....

..... [2]

(iii) the second terminal velocity, when the parachute is open, is smaller than the first.

.....

..... [1]

[Total: 15]

Section B

Question	Answer	Marks
9(a)(i)	amount / measure / quantity of matter / substance (in a body) **or** (property of a body) that resists (change in) motion	**B1**
9(a)(ii)	force / pull of gravity **or** in a gravitational field	**B1**
9(b)	(a =) F / m algebraic or numerical **or** 400 (N) seen	**C1**
	400 / 60	**C1**
	6.7 m/s^2	**A1**
9(c)(i)	initial curve of correct shape from origin at 0 to 30 s	**B1**
	horizontal line at 44 m/s from 30 s to 40 s	**B1**
	speed drops from 44 to 5 m/s in a time of 4 s	**B1**
	constant speed of 5 m/s until 200 s (and then to zero)	**B1**
9(c)(ii)	(a =) (v – u) / t algebraic or numerical	**C1**
	9.7 **or** 9.8 m/s^2	**A1**
9(d)(i)	upwards force / air resistance = weight / downward force	**B1**
9(d)(ii)	increase in air resistance / upwards force (from parachute)	**B1**
	upwards / force / air resistance larger than weight / downwards force **or** resultant force upwards	**B1**
9(d)(iii)	air resistance decreases (as speed reduces) until it equals weight **or** (with parachute open) air resistance = weight at lower speed	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 3 of 16

5 Four of the gravitational forces that act between bodies in the Solar System are listed below.

P the force on the Moon due to the Earth

Q the force on the Earth due to the Sun

R the force on the Earth due to the Moon

S the force on the Moon due to the Sun

Which two forces are a Newton's third law pair (action and reaction)?

A P and Q **B** P and R **C** Q and S **D** R and S

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 5

5	B	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a map of the route taken by a car as it travels from town A to town B.

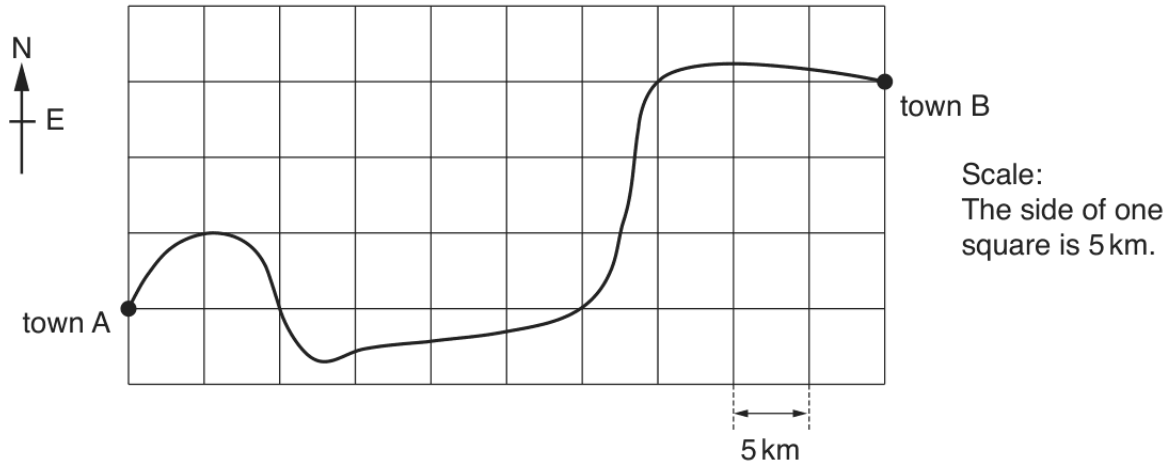


Fig. 9.1

The journey involves a distance and a displacement.

(a) (i) Using the scale on Fig. 9.1, determine the displacement in travelling from A to B.

size of displacement = ...

direction of displacement = ...
[3]

(ii) State how *distance* differs from *displacement*.

.....

.....

..... [1]

(iii) In some parts of the journey, the car is travelling at a constant speed but is also accelerating.

Explain how this is possible.

.....

.....

..... [2]

(b) At town B the car makes an emergency stop.

The total stopping distance of the car is the thinking distance added to the braking distance.

(i) Describe what is meant by the *thinking distance*.

.....

..... [1]

(ii) The brakes on the car work badly. This increases the braking distance.

State **two** other factors that increase the braking distance.

1. .....

.....

2. .....

.....
[2]

(c) The mass of the car and driver is 1200 kg. Just before the brakes are applied, the speed of the car is 30 m/s.

(i) Calculate the kinetic energy of the car and driver just before the brakes are applied.

kinetic energy = ... [3]

(ii) The braking force that stops the car is 1800 N.

Calculate the deceleration of the car.

deceleration = ... [2]

(iii) Explain why the temperature of the brakes increases during braking.

.....

..... [1]

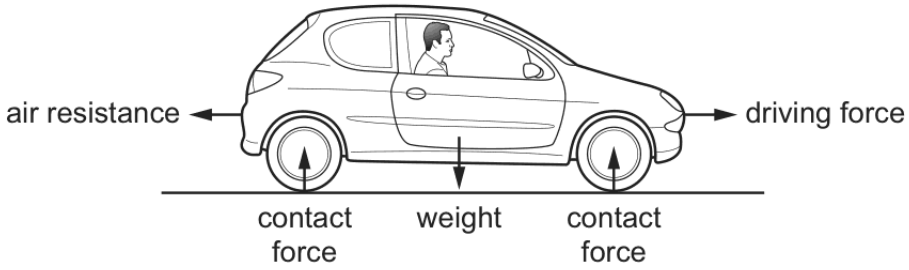
[Total: 15]

Question	Answer	Marks
9(a)(i)	10.2–10.4 (cm) **or** 50 (km) **or** 15 (km) seen	**C1**
	51–53 km	**A1**
	direction (0)72–74° **or** N 72–74 °E etc.	**B1**
9(a)(ii)	(distance) has no direction **or (**distance) is a scalar	**B1**
9(a)(iii)	changes direction **or** goes round a corner	**B1**
	velocity changes **or** velocity depends on direction **or** acceleration / force is towards the centre of circle / centripetal	**B1**

Question	Answer	Marks
9(b)(i)	distance travelled before brakes applied (after seeing emergency)	**B1**
9(b)(ii)	any 2 of • poor tyre surface • poor road conditions, e.g. rain, ice • high speed (of car) **or** good aerodynamic car shape **or** tail wind • large mass / weight of car • driver applies less force to pedal / less braking force	**B1B1**
9(c)(i)	(KE =) ½mv^2 algebraic or numerical	**C1**
	½ × 1200 × 30^2	**C1**
	540 000 J	**A1**
9(c)(ii)	(a=) F / m algebraic or numerical	**C1**
	1.5 m / s^2	**A1**
9(c)(iii)	friction (produces thermal energy) **or** kinetic energy changes to heat **or** work done against friction	**B1**

QUESTION · 5054_w19_qp_11.pdf · source page 4 of 20

6 A car is accelerating along a straight, horizontal road. The diagram shows forces acting on the car.



Which forces are balanced?

A contact forces and air resistance

B contact forces and weight

C driving force and air resistance

D driving force and weight

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 6

6	B	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 4 of 20

7 A single force is applied to a body.

What **cannot** happen?

A a change in the direction of the moving body

B a change in the mass of the body initially at rest

C a change in the position of the body initially at rest

D a change in the speed of the moving body

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 7

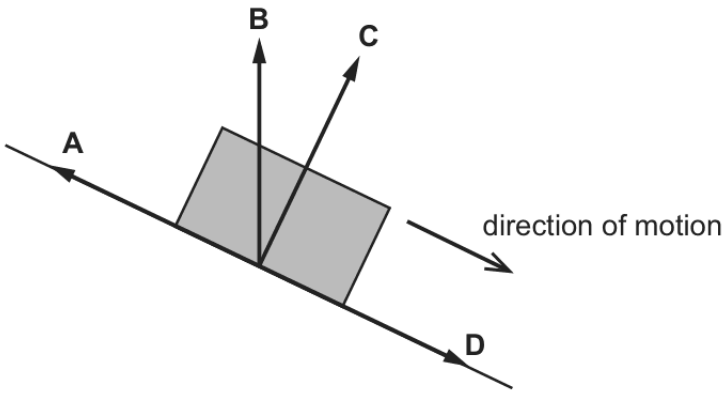
7	B	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 5 of 20

9 A body slides down a frictionless slope, as shown.

As the body presses on the surface, the surface pushes back on the body.

In which direction does the surface push back on the body?



MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 9

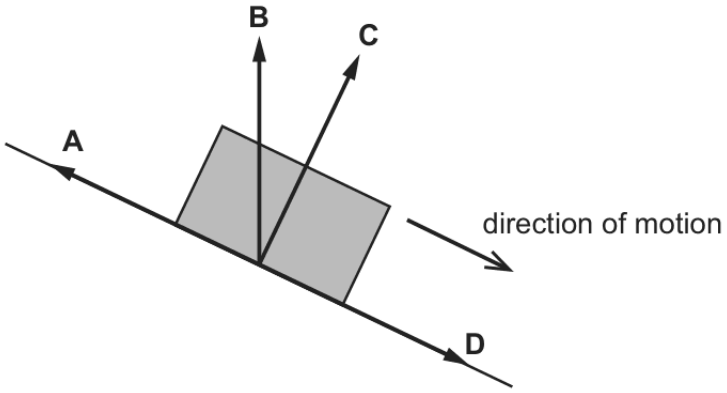
9	C	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 4 of 16

8 A body slides down a frictionless slope, as shown.

As the body presses on the surface, the surface pushes back on the body.

In which direction does the surface push back on the body?



MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 8

8	C	**1**

9 Fig. 9.1 shows a boiling liquid at its boiling point, trapped in a cylinder by a piston.

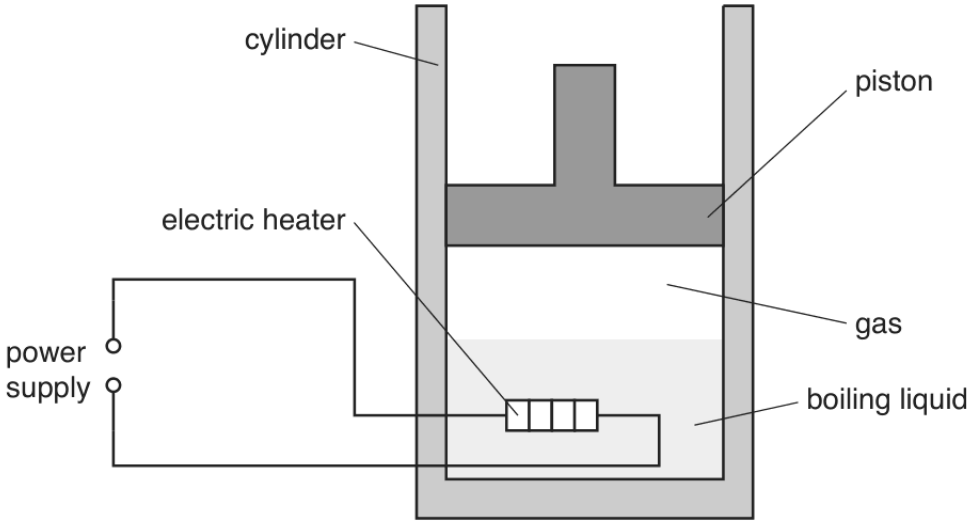


Fig. 9.1

There is an electric heater in the liquid which is connected to a power supply. The thermal energy produced by the heater gradually turns the boiling liquid into a gas.

(a) State what is meant by *boiling point*.

.....

..... [2]

(b) Explain, in terms of molecules, why it is necessary to supply energy in order to turn a liquid at its boiling point into a gas.

.....

.....

.....

..... [3]

(c) In order to determine the power of the electric heater, two meters are connected into the circuit. One meter measures the current in the heater and the other meter measures the potential difference (p.d.) across it.

On the circuit in Fig. 9.1, draw symbols to show the two meters used and where they are connected. [2]

(d) The current in the heater is 2.0 A and the p.d. across it is 6.0 V.

(i) Calculate the power of the heater.

power = ... [2]

(ii) Calculate the thermal energy supplied to the liquid by the heater in 1.0 minute.

energy = ... [2]

(iii) The specific latent heat of vaporisation of the liquid is 9.0×10^5 J/kg.

Calculate the mass of liquid that vaporises every minute.

mass = ... [2]

(e) The piston is free to move in the cylinder. As the liquid boils, the piston is pushed upwards in the cylinder at a constant speed. The volume occupied by the gas just above the liquid increases.

Discuss whether the upward force on the piston changes as the piston moves upwards at constant speed.

.....

.....

.....

..... [2]

[Total: 15]

Question	Answer	Marks
9(a)	temperature	**B1**
	at which a liquid becomes a gas / vapour	**B1**
9(b)	forces (between the molecules of the liquid) / bonds mentioned	**B1**
	move molecules apart **or** overcome forces (between the molecules)	**B1**
	work done against forces **or** gains (molecular) potential energy	**B1**
9(c)	ammeter symbol in series	**B1**
	voltmeter symbol across the heater / power supply	**B1**
9(d)(i)	(P =) VI **or** 6.0×2.0	**C1**
	12 W	**A1**
9(d)(ii)	(E =) Pt **or** 12×60 **or** (E =) VIt **or** $6.0 \times 2.0 \times 60$	**C1**
	720 J	**A1**
9(d)(iii)	(m =) Q / l_v **or** $720 / 9.0 \times 10^5$	**C1**
	8.0×10^{-4} kg **or** 0.80 g	**A1**
9(e)	(it / upward force) does not change	**B1**
	always equal to weight of piston + force due to atmospheric pressure **or** no resultant force **or** piston is not accelerating	**B1**

QUESTION · 5054_s20_qp_11.pdf · source page 2 of 16

4 A car of weight 11 000 N moves with constant velocity along a horizontal road. A driving force of 5000 N acts on the car.

What is the force opposing the motion of the car?

A 5000 N **B** 6000 N **C** 11 000 N **D** 16 000 N

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 4

4	A	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 3 of 16

5 A man with an open parachute falls to Earth at constant speed. The following forces act:

P the upward force of the parachute on the man

Q the upward force of the man on the Earth

R the downward force of the Earth on the parachute

S the downward force of the man on the parachute

Which two forces are a Newton's third law pair?

A P and Q **B** P and R **C** P and S **D** Q and R

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 5

5	C	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 3 of 16

6 A car of weight 11 000 N moves with constant velocity along a horizontal road. A driving force of 5000 N acts on the car.

What is the force opposing the motion of the car?

A 5000 N **B** 6000 N **C** 11 000 N **D** 16 000 N

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 6

6	A	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 3 of 16

7 A man with an open parachute falls to Earth at constant speed. The following forces act:

P the upward force of the parachute on the man

Q the upward force of the man on the Earth

R the downward force of the Earth on the parachute

S the downward force of the man on the parachute

Which two forces are a Newton's third law pair?

A P and Q **B** P and R **C** P and S **D** Q and R

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 3 of 16

3 A car of mass 800 kg has a forward acceleration of 2.5 m / s^2. A frictional force of 1200 N opposes the motion of the car.

What is the driving force due to the engine of the car?

A 800 N **B** 1200 N **C** 2000 N **D** 3200 N

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 5 of 16

11 Which list contains only quantities that can be changed by a force?

A mass, shape, velocity

B mass, shape, volume

C mass, velocity, volume

D shape, velocity, volume

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 11

11	D	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 3 of 16

5 A car of mass 800 kg has a forward acceleration of 2.5 m / s^2. A frictional force of 1200 N opposes the motion of the car.

What is the driving force due to the engine of the car?

A 800 N **B** 1200 N **C** 2000 N **D** 3200 N

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 5

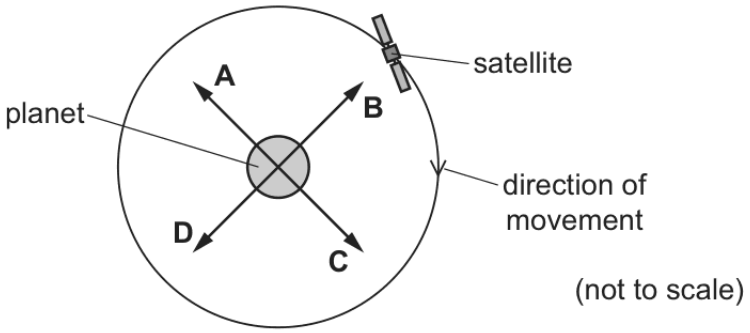
5	D	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 4 of 16

6 A satellite orbits a planet in a circular path as shown. It has constant speed.

There is a force on the planet due to the satellite.

In which direction is the force on the planet?



MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 6

6	B	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 5 of 16

10 Which list contains only quantities that can be changed by a force?

A mass, shape, velocity

B mass, shape, volume

C mass, velocity, volume

D shape, velocity, volume

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 10

10	D	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 A bus leaves a bus-stop at time $t = 0$ and travels along a horizontal road until it reaches a second bus-stop. Fig. 7.1 is the distance-time graph for the bus between $t = 0$ and $t = 60$ s.

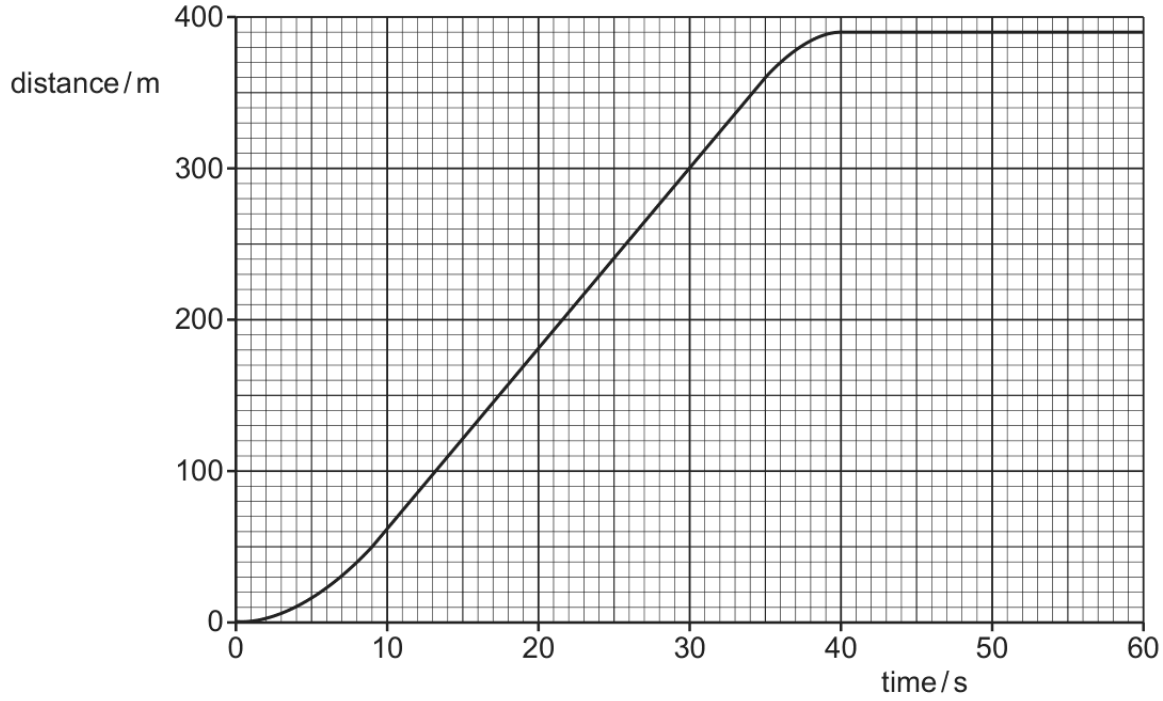


Fig. 7.1

The road on which the bus is travelling is straight except for a short, curved section. The bus travels around this circular curve between $t = 21$ s and $t = 24$ s.

(a) Describe how the motion of the bus between $t = 0$ and $t = 10$ s differs from its motion between $t = 35$ s and $t = 40$ s.

.....

.....

.....

.....

..... [3]

(b) Determine:

(i) the maximum speed of the bus during these 60 s

maximum speed = ... [3]

(ii) the average speed of the bus between leaving the first bus-stop and arriving at the second bus-stop.

average speed = ... [2]

(c) **(i)** State how *velocity* differs from *speed*.

.....

..... [1]

(ii) There are **three** periods during the 60 s when there is a non-zero resultant force acting on the bus.

Complete the statements to indicate these three time periods and state the direction of the resultant force in that period.

1. Between *t* = ... and *t* = ... the direction of the resultant force is

.....

2. Between *t* = ... and *t* = ... the direction of the resultant force is

.....

3. Between *t* = ... and *t* = ... the direction of the resultant force is

.....
[4]

(d) During the journey, the air resistance acting on the bus varies.

(i) State why the air resistance changes during the journey.

.....

..... [1]

(ii) On Fig. 7.1, mark and label with an M a time when the air resistance is a maximum value.
[1]

[Total: 15]

Question	Answer	Marks
7(a)	(between t = 0 and t = 10 s) it is accelerating **or** speed / velocity increasing	**B1**
	between t = 35 s and t = 40 s, it is decelerating **or** speed / velocity decreasing	**B1**
	magnitude of acceleration less than magnitude of deceleration **or** between t = 0 and t = 10 s starts from rest / zero speed **or** between t = 35ms and t = 40 s it finishes at rest / with zero speed	**B1**
7(b)(i)	two pairs of co-ordinates from the straight-line section	**C1**
	appropriate division using candidate's coordinates	**C1**
	11.7 ⩽ speed ⩽ 12.0 m / s	**A1**
7(b)(ii)	total distance / total time **or** (distance =) 390 (m) **and** 39 (s) ⩽ (time =) ⩽ 40 (s)	**C1**
	9.7 to 10.0 m / s	**A1**
7(c)(i)	it / velocity has a direction **or** is a vector **or** depends on displacement (rather than distance)	**B1**
7(c)(ii)	**answers need not be in this order** 1 0 **and** 10 (s) **and** forwards / in the direction of movement	**B1**
	2 21 (s) **and** 24 (s)	**B1**
	perpendicular to the motion / to the direction of movement	**B1**
	3 35 (s) **and** 40 (s) **and** backwards / opposite to the direction of movement	**B1**
7(d)(i)	speed (of bus) varies / changes / decreases / increases	**B1**
7(d)(ii)	any time on Fig. 1.1 from 10 s ⩽ t ⩽ 35 s labelled M	**B1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 An aircraft flies at a constant height.

Air drag and the force from the aircraft's engines together produce a force on the aircraft of 36 kN due north, as shown in Fig. 1.1.

The wind produces a force of 12 kN towards the east.

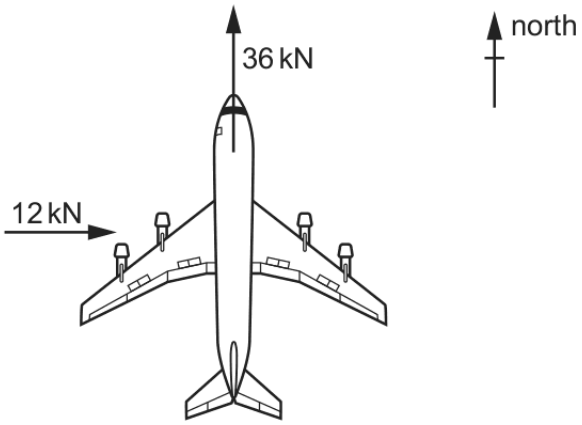


Fig. 1.1 (not to scale)

(a) Draw a scale drawing to show the resultant force acting on the aircraft.

Use your drawing to determine the size of the resultant force and the angle between the resultant force and north.

size of resultant force = ...

angle = ...
[3]

(b) The acceleration of the aircraft is uniform.

(i) Describe how a uniform acceleration differs from a non-uniform acceleration.

.....

.....

.....

..... [2]

(ii) The mass of the aircraft is 60 000 kg.

Calculate the acceleration of the aircraft.

acceleration = ... [2]

[Total: 7]

Question	Answer	Marks
1(a)	scale vector diagram showing 12 N, 36 N and correct resultant force in correct direction	**B1**
	38 ± 1 kN	**B1**
	018 ± 3°	**B1**
1(b)(i)	(only in uniform acceleration) same increase / change in speed / velocity **or** has a constant acceleration **or** speed-time graph is a straight line	**B1**
	(only in uniform acceleration) same increase / change in speed / velocity in same time (period) **or** same rate of change of speed / velocity (with time) **or** change in speed / velocity is proportional to time	**B1**
1(b)(ii)	(a =) F / m in any way, algebraic or numerical	**C1**
	0.63–0.66 m / s^2	**A1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 Fig. 8.1 shows a stationary horse and its rider, about to jump over two fences.

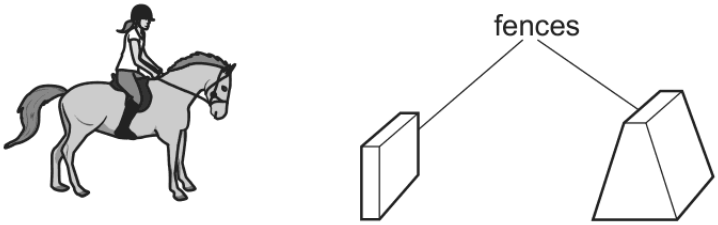


Fig. 8.1

(a) Fig. 8.2 shows a side view of the horse.

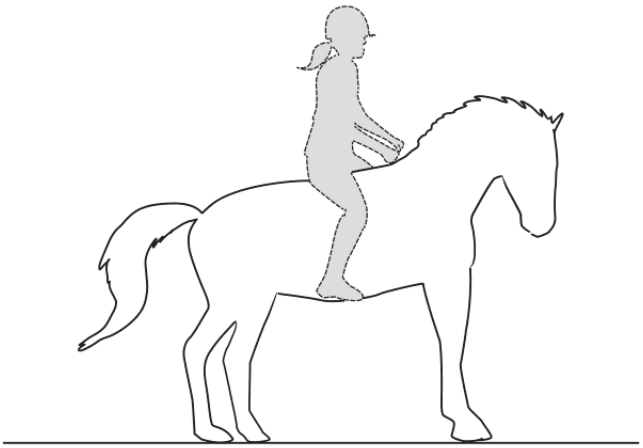


Fig. 8.2

(i) On Fig. 8.2, draw and label the forces acting on the horse.

Include the force that the rider exerts on the horse. Label this force *F*. [3]

(ii) Explain how Newton's third law applies to force *F*.

.....

.....

..... [2]

(b) Fig. 8.3 shows a side view of the two fences. They both have the same height and a uniform density.

Fig. 8.3

(i) On each fence in Fig. 8.3, mark with a cross the centre of mass. [2]

(ii) Explain why a wider base makes the fence more stable.

.....

.....

..... [2]

(c) The total mass of the horse and rider is 520 kg.

(i) As they approach a fence, the horse and rider have a total kinetic energy of 4000 J.

Calculate their speed.

speed = ... [3]

(ii) The centre of mass of the horse and rider is 1.4 m above the ground.

The maximum potential energy gained by the horse and rider as they jump over the fence is 3000 J.

Calculate the maximum height above the ground of the centre of mass during the jump.

The gravitational field strength g = 10 N / kg.

height = ... [3]

[Total: 15]

Question	Answer	Marks
8(a)(i)	weight of horse in mid-section of horse **and** labelled	**B1**
	force upwards on at least two legs **and** labelled	**B1**
	force downwards on saddle area labelled *F*	**B1**
8(a)(ii)	forces equal **and** opposite	**B1**
	acts on rider (upwards from horse)	**B1**
8(b)(i)	cross at centre of left-hand fence	**B1**
	cross at centre of right-hand fence **and** lower than for right-hand fence	**B1**

Question	Answer	Marks
8(b)(ii)	does not topple / falls back as fence tilts / rotates / turns / changes angle **or** turns to larger angle / needs more force before toppling **or** centre of mass is lower	**B1**
	(low angle of tilt) centre of mass still inside / above base **or** weight turns fence back if tilted	**B1**
8(c)(i)	K.E. = $\frac{1}{2}mv^2$ in any form algebraic or numerical	**C1**
	$v^2 = 2 \times 4000 / 520$ **or** $v = \sqrt{\frac{2\times4000}{520}}$	**C1**
	3.9 m / s	**A1**
8(c)(ii)	(*h* =) P.E. / *mg* in any form algebraic or numerical	**C1**
	3000 / (10 × 520) **or** 0.58 m	**C1**
	2.0 m	**A1**

2 Fig. 2.1 shows a man of mass 80 kg standing in a lift (elevator).

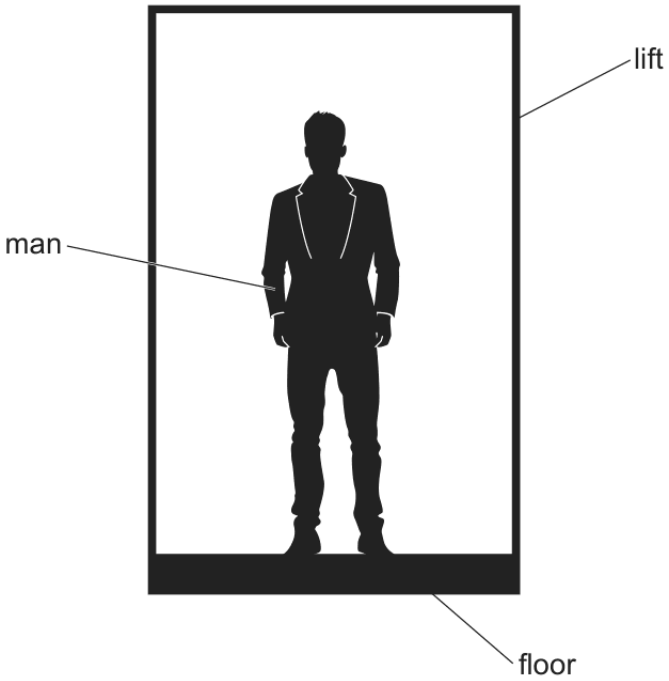


Fig. 2.1

The gravitational field strength g is 10 N / kg.

(a) Calculate the weight of the man.

weight = ... [1]

(b) The lift accelerates upwards uniformly at 0.50 m / s^2.

Calculate:

(i) the resultant upward force on the man

resultant force = ... [2]

(ii) the force exerted on the man by the floor of the lift.

force = ... [1]

(c) The lift accelerates upwards uniformly from rest for 4.0 s and then decelerates uniformly to rest in 4.0 s. The total distance travelled is 8.0 m.

On Fig. 2.2, sketch the distance-time graph for this journey.

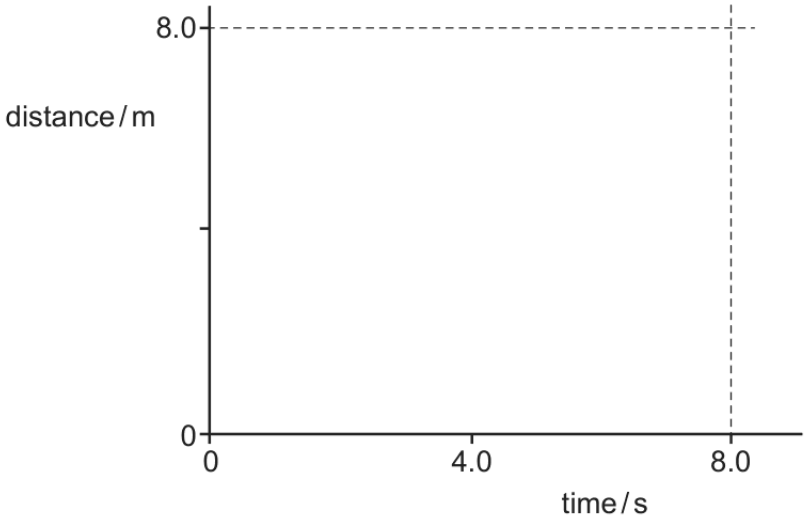


Fig. 2.2

[3]

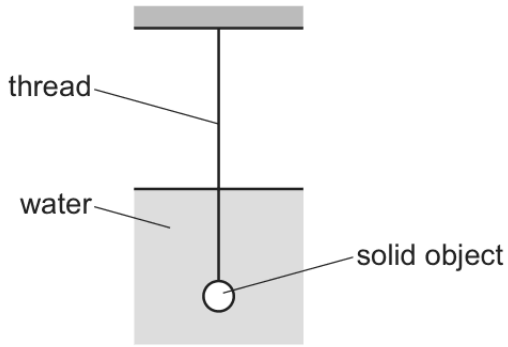
[Total: 7]

Question	Answer	Marks
2(a)	800 N **c.a.o.**	**B1**
2(b)(i)	(F =) ma **or** 80×0.50	**C1**
	40 N	**A1**
2(b)(ii)	840 N	**B1**
2(c)	increasing gradient from origin to t = 4.0 s	**B1**
	decreasing gradient from end of first curve	**B1**
	line horizontal at origin **or** at t = 8.0 s (and beyond)	**B1**

QUESTION · 5054_w21_qp_12.pdf · source page 3 of 16

4 A solid object, immersed in water, hangs from an elastic thread. Three forces act on the object: its weight *W*, the tension in the thread *T*, and a force *F* from the water.

The force *F* acts upwards.



Which equation is correct when the object is stationary?

A $F + W = 0$

B $F - T = 0$

C $F - W - T = 0$

D $F - W + T = 0$

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 4

4	D	**1**

2 Force is a vector quantity.

(a) State what is meant by a *vector*.

.....

..... [1]

(b) A swimmer reaches the end wall of a swimming pool and turns around under the water.

Fig. 2.1 shows the swimmer immediately after turning around.

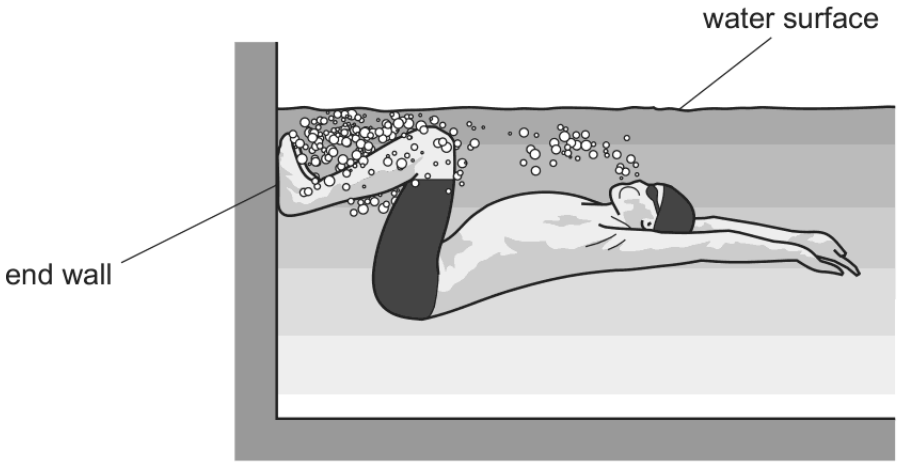


Fig. 2.1

(i) The swimmer pushes against the end wall of the pool with his legs.

Explain, in terms of Newton's third law, why the swimmer accelerates away from the end wall.

.....

.....

.....

..... [3]

(ii) While swimming, there is a constant forward force on the swimmer. His speed increases until eventually he reaches a constant speed.

Explain why he reaches a constant speed.

.....

.....

.....

..... [3]

[Total: 7]

MARK SCHEME · 5054_w21_ms_22.pdf · source page 5 of 11 · Question 2

Question	Answer	Marks
2(a)	(a vector quantity has) magnitude **and** direction	**B1**
2(b)(i)	the wall exerts a force on swimmer	**B1**
	opposite in direction (to force of swimmer on wall) **or** to the right	**B1**
	equal in size (to force of swimmer on wall) **or** (resultant) force causes acceleration **or** force on object Y due to X is equal and opposite to force on object X due to Y	**B1**

MARK SCHEME · 5054_w21_ms_22.pdf · source page 6 of 11 · Question 2

Question	Answer	Marks
2(b)(ii)	(water) resistance / resistive force / friction (on swimmer)	**B1**
	(water) resistance / backward force / resistive force / friction / opposing force increases (with speed / time)	**B1**
	(at constant speed) forward force is equal to resistive / backward force **or** no resultant force **or** forces balance	**B1**

QUESTION · 5054_s22_qp_12.pdf · source page 2 of 20

4 A 60 kg passenger enters a stationary lift. The gravitational field strength g is 10 N / kg.

How much force does the floor of the lift exert on the passenger when the lift accelerates upwards at 2.0 m / s^2?

A 120 N **B** 480 N **C** 600 N **D** 720 N

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 4

4	D	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a waterskier pulled horizontally across the water surface.

Fig. 1.1

The mass of the waterskier is 60 kg. The gravitational field strength g is 10 N/kg.

(a) Calculate the weight of the waterskier.

weight = ... [1]

(b) Mass and weight have different units.

State **two** other differences between mass and weight.

.....

.....

.....

..... [2]

(c) A boat pulls the waterskier to the right with a horizontal force of 50 N. The waterskier travels at a constant speed.

(i) Explain, in terms of the horizontal forces acting, why the speed is constant.

.....

..... [2]

(ii) The horizontal force acting to the right increases from 50 N to 70 N. The sizes of any other forces are unchanged.

Calculate the acceleration of the waterskier.

acceleration = ... [3]

[Total: 8]

Question	Answer	Marks
1(a)	600 N	**B1**
1(b)	ANY 2 separate bullet points from: • weight is a vector **/** weight has direction / mass has no direction / mass is a scalar • mass is the amount of matter / substance • weight is a force / weight is mass × gravitational field strength • weight varies with position (of object) / gravitational field (strength) / mass does not vary with position • mass resists change of motion / has inertia	**B2**
1(c)(i)	friction (with water) **or** drag **or** air / water resistance mentioned	**B1**
	(forces) balance / cancel **or** no resultant force	**B1**
1(c)(ii)	(resultant force) 20 (N) or 70 – 50 seen	**C1**
	(*a* =) *F* / *m* algebraic or numerical in any form	**C1**
	0.33 m / s^2	**A1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 Fig. 7.1 shows a child sitting on a sledge on a snow-covered hill of constant slope.

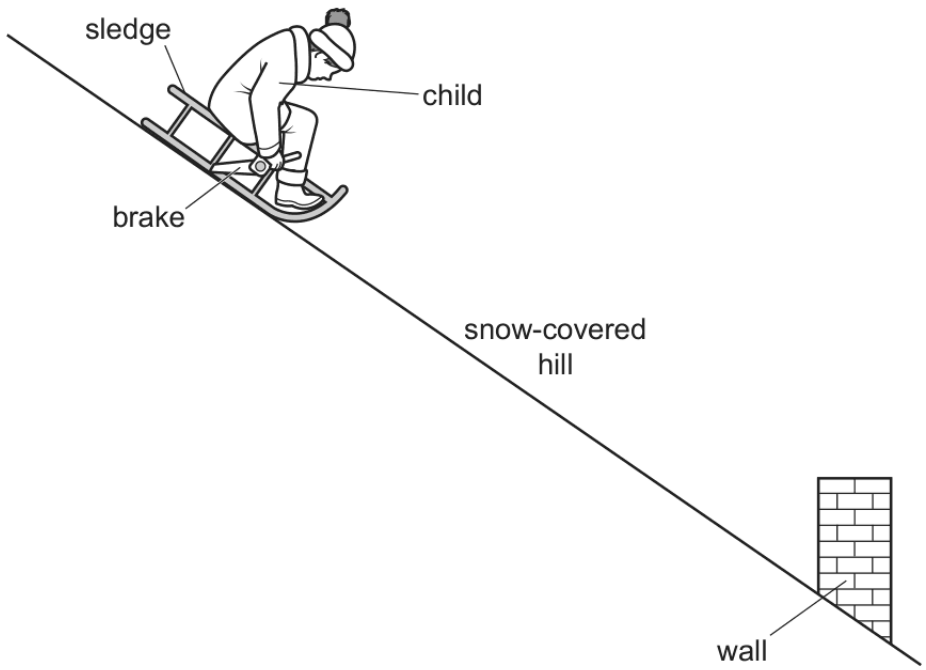


Fig. 7.1 (not to scale)

At time $t = 0$, the child and the sledge begin to move down the hill in a straight line.

When the child sees a wall ahead, he applies the brake.

The child and sledge continue to travel in a straight line until they come to a stop before hitting the wall.

Fig. 7.2 is the speed-time graph for the journey.

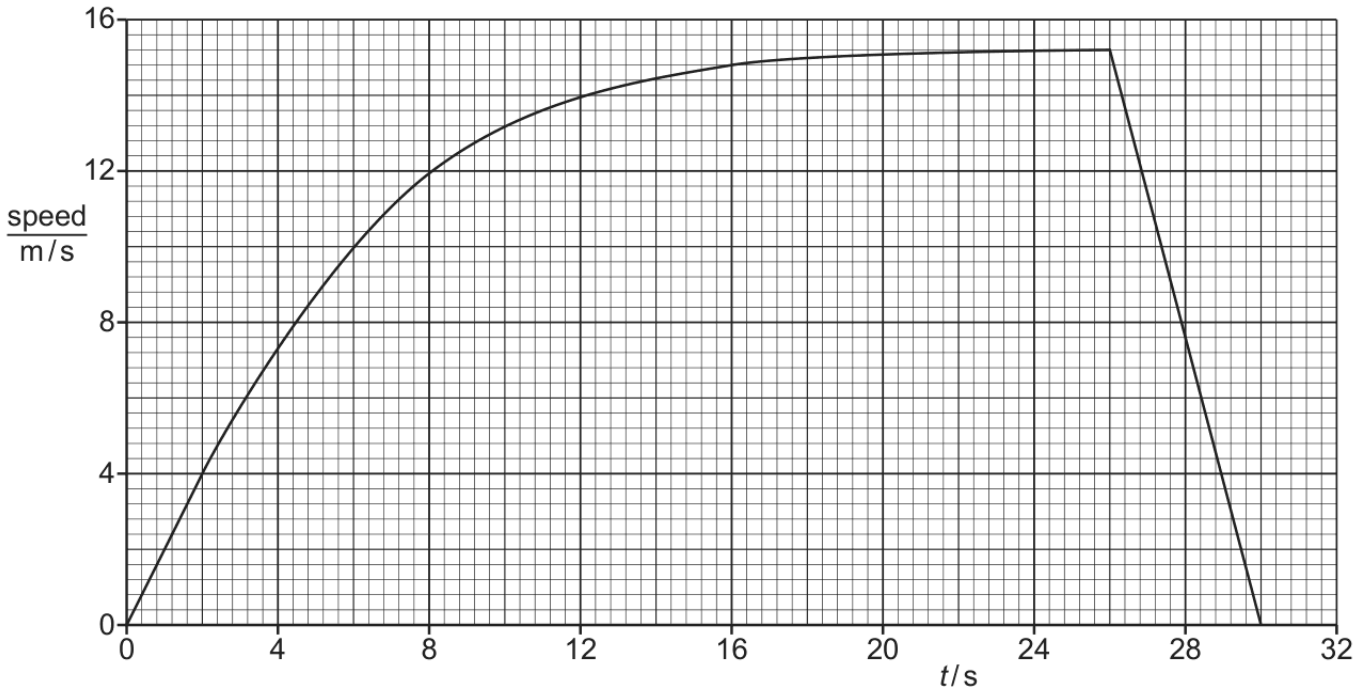


Fig. 7.2

The brake is applied at $t = 26$ s.

(a) Fig 7.2 shows how the speed of the child and sledge varies over the whole of the journey.

Explain why, between $t = 0$ and $t = 26$ s, the speed varies in the way shown by the graph.

.....

.....

.....

.....

.....

..... [3]

(b) At $t = 26$ s, the front of the sledge is 35 m from the wall.

Determine the distance between the front of the sledge and the wall when the sledge stops.

distance = ... [3]

(c) At $t = 26$ s, the child and sledge begin to decelerate.

(i) Determine the size of the deceleration.

deceleration = ... [3]

(ii) The mass of the child is 46 kg and the mass of the sledge is 9.0 kg.

Calculate the resultant force on the child and sledge as they decelerate.

resultant force = ... [2]

(iii) State the energy transfer that is taking place as the child and sledge decelerate.

.....

.....

..... [2]

(d) At $t = 26$ s, when the brake is first applied, the child jerks forwards on the sledge.

Explain why.

.....

.....

..... [2]

[Total: 15]

Question	Answer	Marks
7(a)	acceleration / increase in speed **and** gravity / weight mentioned	**B1**
	air resistance / friction increases (with speed)	**B1**
	terminal / constant velocity reached **or** air resistance (becomes) equal to accelerating force	**B1**
7(b)	(distance travelled =) area under the graph **or** (x =) ½(u + v)t **or** ½ × 15.2 × 4.0	**C1**
	35 – (½ × 15.2 × 4.0) **or** 30.4 **or** 30	**C1**
	4.6 m	**A1**
7(c)(i)	(magnitude of deceleration =) gradient **or** Dv / Dt **or** 15.2 / 4.0 **or** equivalent numbers (from graph) in same ratio	**C1**
	15.2 / 4.0 **or** equivalent numbers in same ratio	**C1**
	3.8 m / s^2	**A1**
7(c)(ii)	(F =) ma **or** (46 + 9.0) × 3.8 **or** 55 × 3.8	**C1**
	210 N	**A1**
7(c)(iii)	<u>from</u> kinetic energy (and from gravitational potential energy)	**B1**
	<u>to</u> thermal energy **and** no intermediate energy	**B1**
7(d)	due to his mass **or** mass mentioned	**B1**
	(his) mass resists a change to its state of motion	**B1**

Section A

Answer **all** questions in this section. Answer in the spaces provided.

1 A train travels along a straight horizontal track. At time $t = 0$, the train passes through station P at constant speed without stopping.

The driver applies the brakes 70 s before reaching station Q. The train decelerates.

Fig. 1.1 is the speed–time graph for the train from $t = 0$ until it stops at station Q.

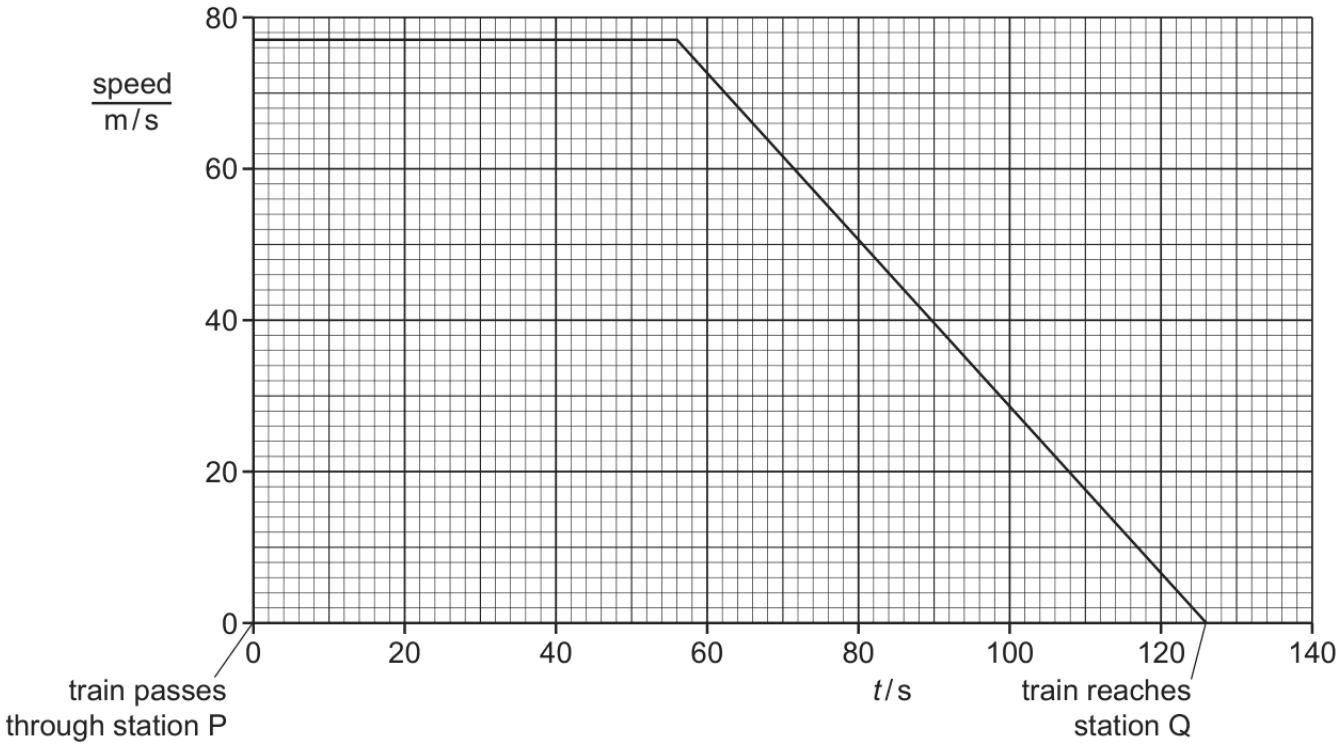


Fig. 1.1

(a) Using Fig. 1.1, determine the distance between station P and station Q.

distance = ... [3]

(b) The mass of the train is 3.8×10^5 kg.

(i) Determine the deceleration of the train in the 70 s before it stops at station Q.

deceleration = ... [2]

(ii) Calculate the resultant force on the train as it decelerates.

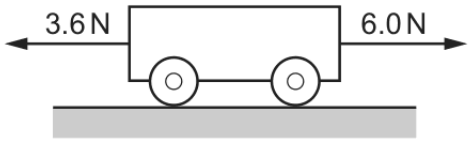
resultant force = ... [2]

[Total: 7]

Question	Answer	Marks
1(a)	(distance =) area (under graph) **or** 77 × 56 **or** ½ × 77 × 70 **or** 4300 **or** 2700 **or** ½ × 77 × (56 + 126)	**C1**
	77 × 56 + (½ × 77 × 70) **or** 4300 + 2700 **or** ½ × 77 × (56 + 126)	**C1**
	7000 m **or** 7.0×10^3 m **or** 7.0 km	**A1**
1(b)(i)	(a =) $\Delta v / \Delta t$ **or** 77 / 70 (etc.)	**C1**
	1.1 m / s^2	**A1**
1(b)(ii)	(F =) ma **or** $3.8 \times 10^5 \times 1.1$	**C1**
	4.2×10^5 N	**A1**

QUESTION · 5054_s23_qp_11.pdf · source page 5 of 20

7 The diagram shows the only two horizontal forces acting on a trolley that is accelerating at $0.12\,m/s^2$.



What is the mass of the trolley?

A 0.050 kg **B** 0.13 kg **C** 20 kg **D** 80 kg

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

1 An aircraft pulls a glider along a runway as shown in Fig. 1.1.

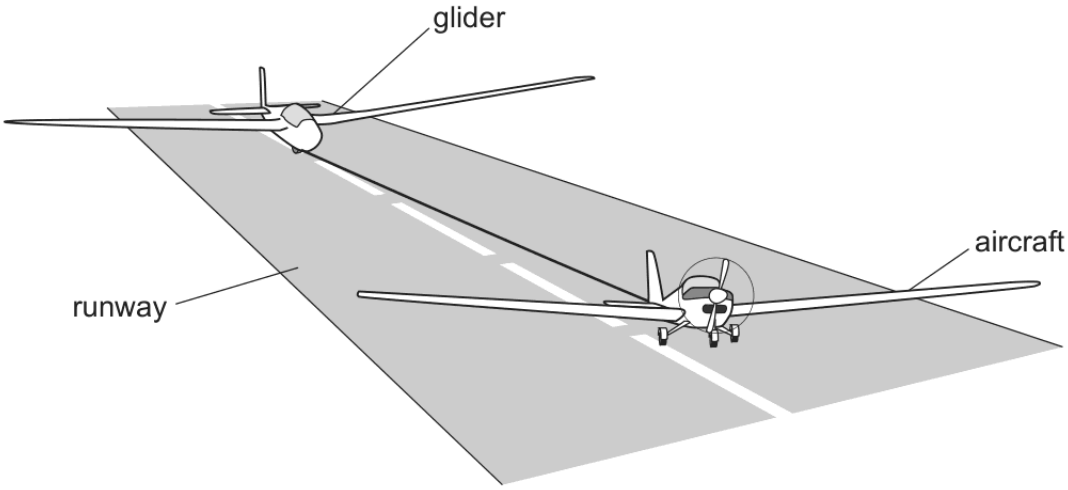


Fig. 1.1

Fig. 1.2 shows the speed of the glider during the first 12 s of the motion.

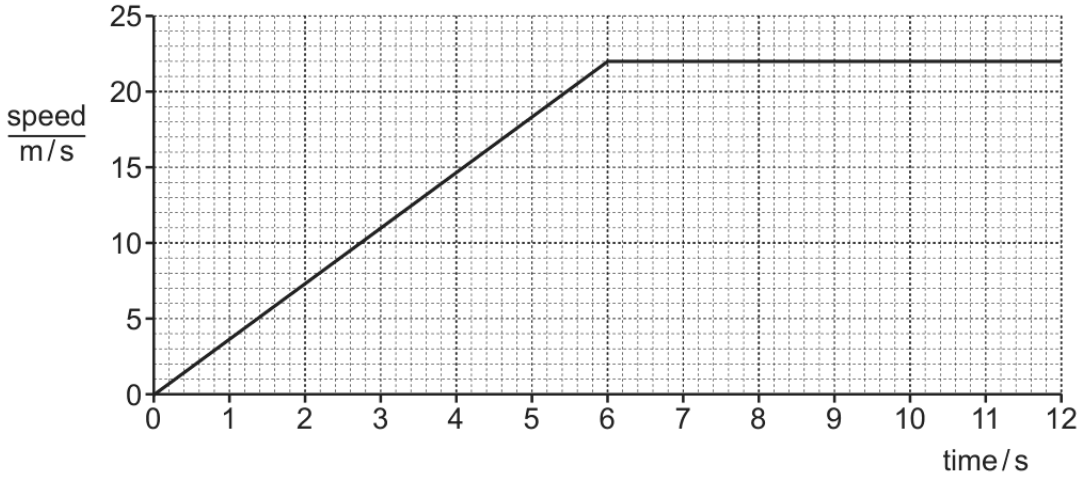


Fig. 1.2

(a) Describe the motion of the glider in the first 12 s.

.....

.....

..... [2]

(b) In the first 6.0 s of the motion, there is a resultant force of 1800 N on the glider.

Using the increase in speed in the first 6.0 s, calculate the mass of the glider.

mass = ... kg [3]

(c) Determine the distance travelled by the glider in the first 6.0 s of its motion.

distance = ... m [2]

(d) The glider has no engine and stays in the air with the use of convection currents.

(i) State what is meant by a 'convection current'.

.....

.....

..... [1]

(ii) Suggest how the convection current that supports the glider is formed.

.....

.....

..... [1]

[Total: 9]

Question	Answer	Marks
1(a)	uniform / constant acceleration (for 6 s) / acceleration of 3.7 m / s^2	**B1**
	after 6(.0 s) constant speed / zero acceleration	**B1**
1(b)	(a =) (v–u) / t in any form numerical or algebraic **or** 3.7 (m / s^2) seen	**C1**
	(m =) F / a **or** (m =) Ft / v in any form numerical or algebraic	**C1**
	490 (kg)	**A1**
1(c)	area under graph **or** average speed × time numerical or algebraic	**C1**
	66 (m)	**A1**
1(d)(i)	rising of hot air / hot fluid (and falling of cold air) **or** circulation of air / fluid with temperature differences	**B1**
1(d)(ii)	high temperature spot on Earth, e.g. from a chimney, dark patch in sunlight, above a town / land / surface; Sun heats surface **or** (hot) less dense air rises upwards	**B1**

QUESTION · 5054_s23_qp_12.pdf · source page 4 of 16

8 Newton's third law describes a pair of forces.

Which row shows whether the two forces are of the same type and whether they act on the same object?

	the types of forces in the pair	the objects on which they act
A	different	different
B	different	same
C	same	different
D	same	same

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 8

8	C	**1**

1 Fig. 1.1 shows a toy water rocket.

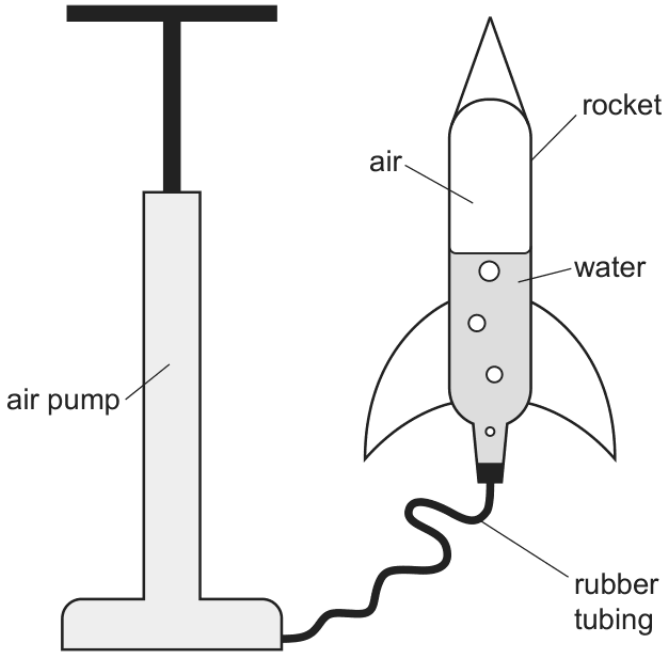


Fig. 1.1

The rocket is half-filled with water and connected by rubber tubing to a pump.

(a) Air is then pumped into the rocket so that the pressure of the air inside the rocket increases.

(i) Explain, in terms of particles, why the air in the rocket exerts a pressure on the walls of the rocket.

.....

.....

.....

..... [3]

(ii) The temperature of the air in the rocket remains constant as air is pumped into the rocket.

Explain why increasing the number of air particles in the rocket increases the pressure of the air in the rocket.

.....

..... [1]

(b) When the air pressure in the rocket is high, the rubber tubing is removed from the rocket.

The water in the rocket is expelled downwards.

Explain, in terms of momentum, why the rocket begins to accelerate upwards.

.....

.....

.....

..... [3]

(c) As the water is expelled from the rocket, the upward acceleration of the rocket changes.

Suggest **two** reasons for the change in the acceleration of the rocket.

1 .....

.....

2 .....

.....

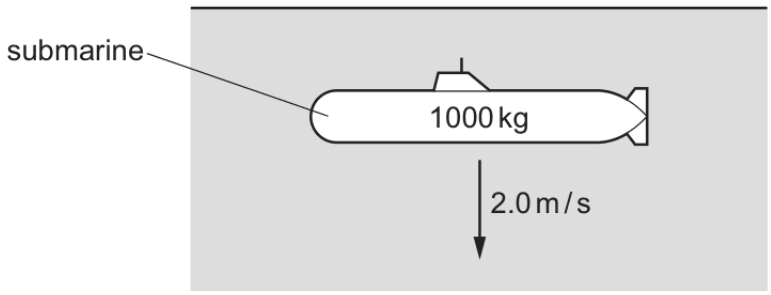
[2]

[Total: 9]

Question	Answer	Marks
1(a)(i)	particles move / collide / hit	**B1**
	particles collide with (inner) walls (of rocket)	**B1**
	force exerted by collisions over / on an area	**B1**
1(a)(ii)	more collisions per unit time	**B1**
1(b)	(expelled) water gains / has (downward) momentum	**B1**
	momentum conserved	**B1**
	rocket gains (upward) momentum (and accelerates)	**B1**
1(c)	any **two** from: mass (of water and rocket) decreases / changes pressure (of air) decreases (air) resistance (acts) gravitational field strength decreases (with height) less water expelled per second / water expelled at slower speed	**B2**

QUESTION · 5054_w23_qp_12.pdf · source page 3 of 20

5 A small submarine of mass 1000 kg sinks in water with a constant speed of 2.0 m/s.



What is the resultant force exerted on the submarine as it sinks?

A 0

B 500 N downwards

C 2000 N downwards

D 9800 N downwards

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 5

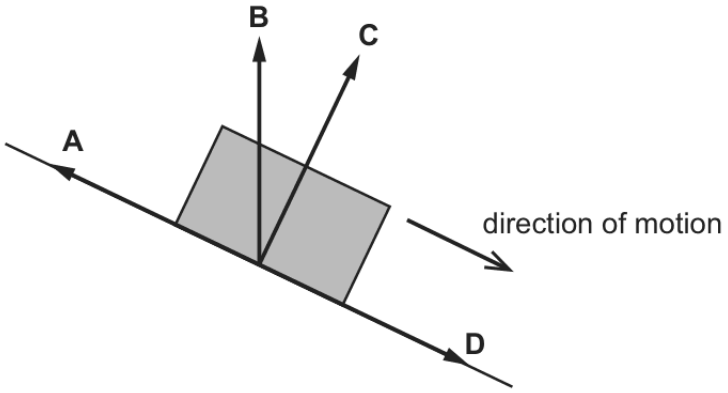
5	A	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 3 of 16

5 An object slides down a frictionless slope as shown.

As the object presses on the surface, the surface pushes back on the object.

In which direction does the surface push back on the object?



MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 5

5	C	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 3 of 16

5 Which property of an object determines its resistance to a change from its state of rest or motion?

A its mass

B its shape

C its surface area

D its volume

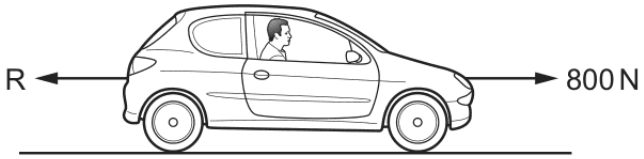
MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 5

5	A	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 4 of 16

7 A car travels along a road. The force on the car due to the engine is 800 N.

The motion of the car depends on the value of the total resistive force R.



Which row shows the motion of the car for the given value of R?

	value of resistive force R / N	motion
A	500	deceleration
B	800	acceleration
C	900	deceleration
D	1000	acceleration

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 4 of 16

8 Four of the gravitational forces that act between objects in the Solar System are listed.

P the force on the Moon due to the Earth

Q the force on the Earth due to the Sun

R the force on the Earth due to the Moon

S the force on the Moon due to the Sun

Which two forces are a Newton's third law pair?

A P and Q **B** P and R **C** Q and S **D** R and S

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 8

8	B	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 5 of 16

9 A lorry is travelling along a straight, horizontal road. The constant driving force on the lorry is 80 kN and the total resistive force is 20 kN.

Which row describes the motion of the lorry and gives the resultant force acting on it?

	motion of lorry	resultant force / kN
A	acceleration	60
B	acceleration	100
C	deceleration	60
D	deceleration	100

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 9

9	A	**1**

1 A jet ski is a type of boat that carries one or two people and travels at high speed on water.

Fig. 1.1 shows a student riding on a jet ski.

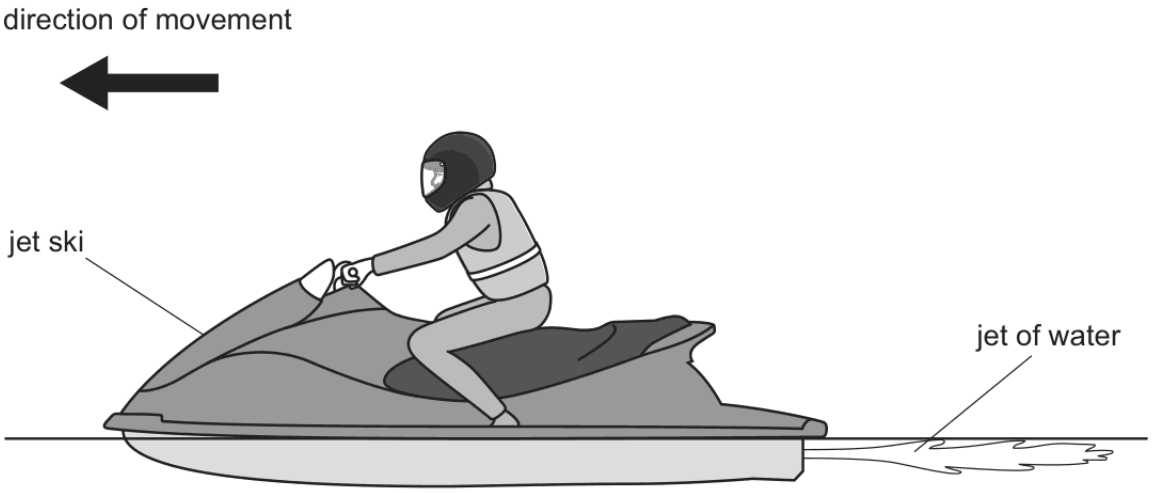


Fig. 1.1

A high-speed jet of water is forced backwards out of the back of the jet ski by a pump inside the jet ski.

(a) The pump increases the momentum of the water that is forced backwards out of the back of the jet ski.

(i) Complete the word equation to show the relationship between the resultant force on an object and the change in momentum of the object.

resultant force =

[1]

(ii) In 2.0 s, the pump increases the backwards speed of 180 kg of water by 30 m/s.

Calculate the backwards force exerted on the water.

force = ... N [2]

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DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) (i) Using Newton's third law of motion, explain why there is a forwards force on the jet ski.

.....

.....

.....

..... [2]

(ii) The student has a mass of 70 kg and the jet ski has a mass of 280 kg.

Use your answer from **(a)(ii)** to determine the acceleration of the student and jet ski when no resistive forces are acting.

acceleration = ... m/s^2 [2]

(c) The jet ski reaches a speed of 20 m/s.

Calculate the total kinetic energy of the student and jet ski at this speed.

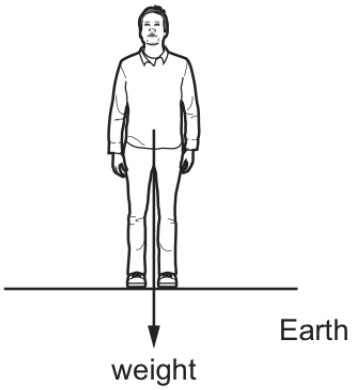
kinetic energy = ... J [3]

[Total: 10]

Question	Answer	Marks
1(a)(i)	resultant force = increase in momentum / unit time **or** resultant force = rate of increase in momentum	**B1**
1(a)(ii)	(F =) 180 × 30 / 2.0 **or** 180 × 15	**C1**
	2700 (N)	**A1**
1(b)(i)	pump **/** jet ski / engine exerts force on water	**B1**
	water exerts an opposite force on the pump / jet ski / engine	**B1**
1(b)(ii)	(a =) F / m **or** 2700 / (70 + 280) **or** 2700 / 350	**C1**
	7.7 (m / s^2)	**A1**
1(c)	(K.E. =) ½ mv^2	**C1**
	½ × (70 + 280) × 20^2 **or** ½ × 350 × 20^2	**C1**
	70 000 (J)	**A1**

QUESTION · 5054_w25_qp_11.pdf · source page 5 of 16

9 A man stands at rest on the surface of the Earth.



The weight of the man is the gravitational force that acts on him.

According to Newton's third law, the weight of the man is one force in a pair of forces.

Which force is the other force in the pair?

A a force downwards on the Earth

B a force downwards on the man

C a force upwards on the Earth

D a force upwards on the man

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 9

9	C	**1**

9 An object moves through space. It is a large distance from any other object.

(a) There are no external forces acting on the object.

Describe **two** features of the motion of the object that are a consequence of Newton's first law.

1 .....

.....

2 .....

.....

[2]

(b) Planet X and Venus are two of the planets in the Solar System.

Fig. 9.1 shows the orbit of Venus and the orbit of planet X. Both orbits are approximately circular.

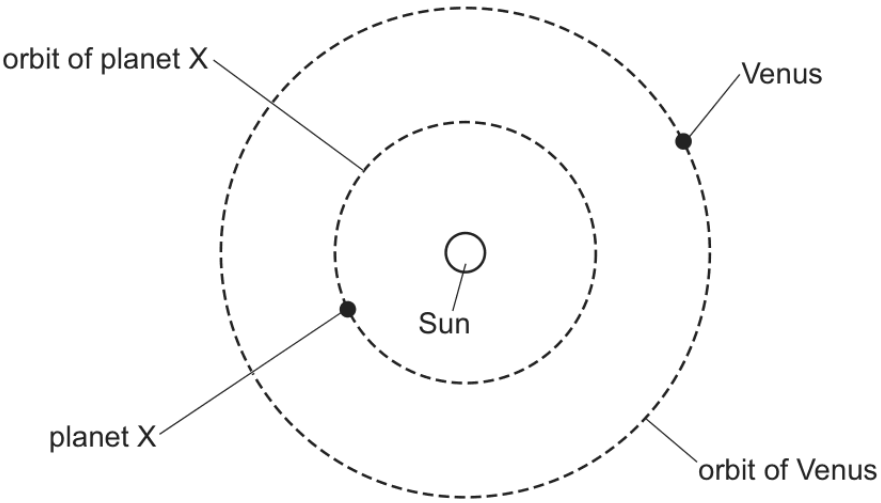


Fig. 9.1

Venus experiences an external force as it orbits the Sun.

(i) State why a force must act on Venus to keep it following an approximately circular orbit.

.....

..... [1]

(ii) State what causes this force, and state the direction of the force.

cause .....

.....

direction .....

[3]

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(iii) The average radius of the orbit of Venus is 1.1×10^{11} m. It takes Venus 5400 hours to complete one orbit of the Sun.

Calculate the average orbital speed of Venus around the Sun.

speed = ... m/s [2]

(iv) Planet X is closer to the Sun than Venus.

State the name of planet X and state how its orbital speed around the Sun compares with the orbital speed of Venus.

name .....

speed .....

.....
[2]

[Total: 10]

MARK SCHEME · 5054_w25_ms_21.pdf · source page 13 of 14 · Question 9

Question	Answer	Marks
9(a)	speed is constant	**B1**
	(motion is) in a straight line	**B1**
9(b)(i)	to change the direction of the motion / velocity (of Venus) / otherwise it would move in a straight line	**B1**
9(b)(ii)	gravitational field / gravitational attraction / gravity	**B1**
	(gravitational field) of Sun	**B1**
	towards centre of circle / Sun / inward	**B1**

MARK SCHEME · 5054_w25_ms_21.pdf · source page 14 of 14 · Question 9

Question	Answer	Marks
9(b)(iii)	(speed =) $\frac{2\pi r}{T}$ **or** $\frac{(2\times\pi)\times 1.1\times 1011}{5400\times(3600)}$	**C1**
	3.6×10^{4} (m / s)	**A1**
9(b)(iv)	Mercury	**B1**
	(its speed is) greater (than that of Venus)	**B1**

QUESTION · 5054_w25_qp_12.pdf · source page 4 of 16

8 Which list contains only quantities that can be changed by a force?

A mass, shape, velocity

B mass, shape, volume

C mass, velocity, volume

D shape, velocity, volume

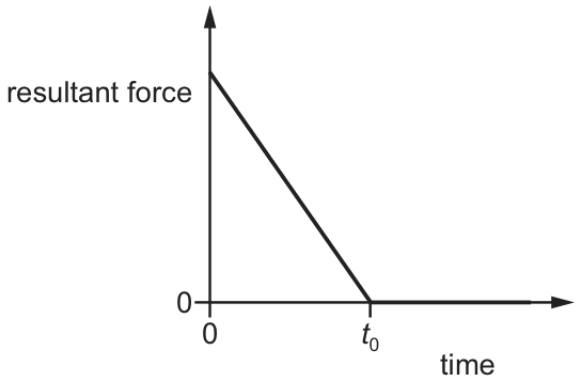
MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 8

8	D	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 4 of 16

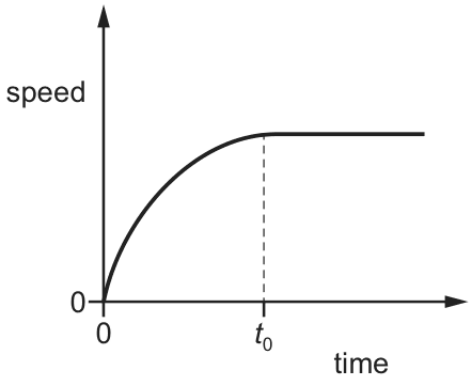
9 A resultant force acts on an object and causes it to move in a straight line.

The graph shows how the resultant force varies with time.

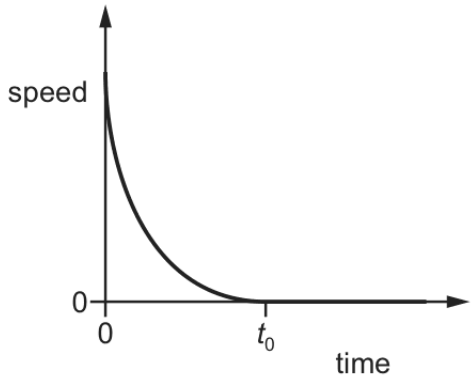


Which graph is the speed–time graph for the object?

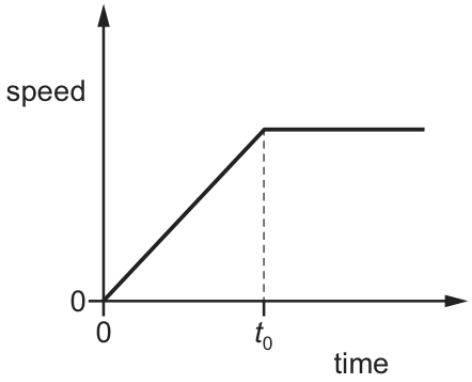
A



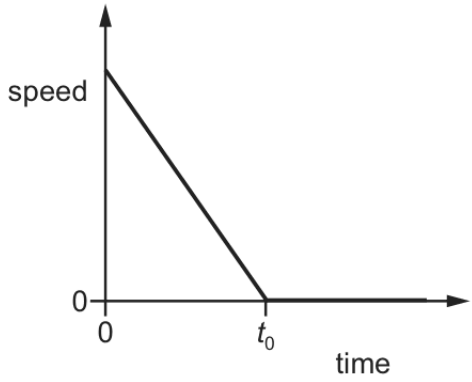
B



C



D



MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 9

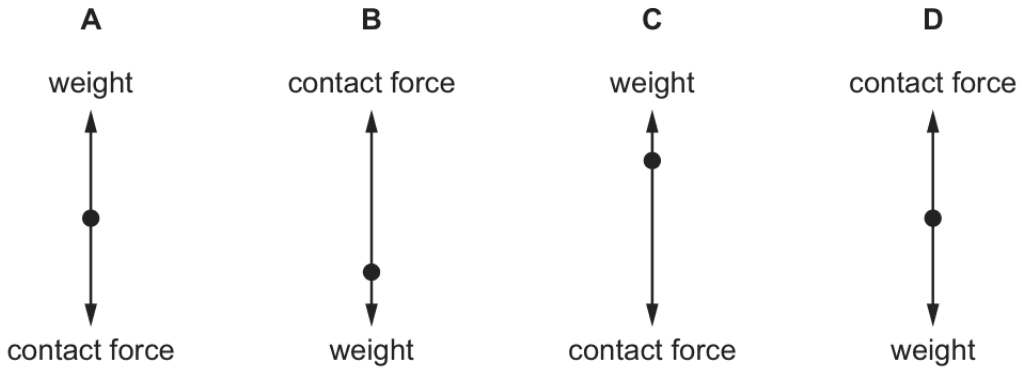
9	A	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 5 of 16

10 An object is at rest on a table.

The length of the arrows in the free-body diagrams represent the magnitude and the direction of the forces acting on the object.

Which diagram shows the free-body diagram for the object?



MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 10

10	D	**1**

3 Fig. 3.1 shows a firefighter directing a jet of water at a wall.

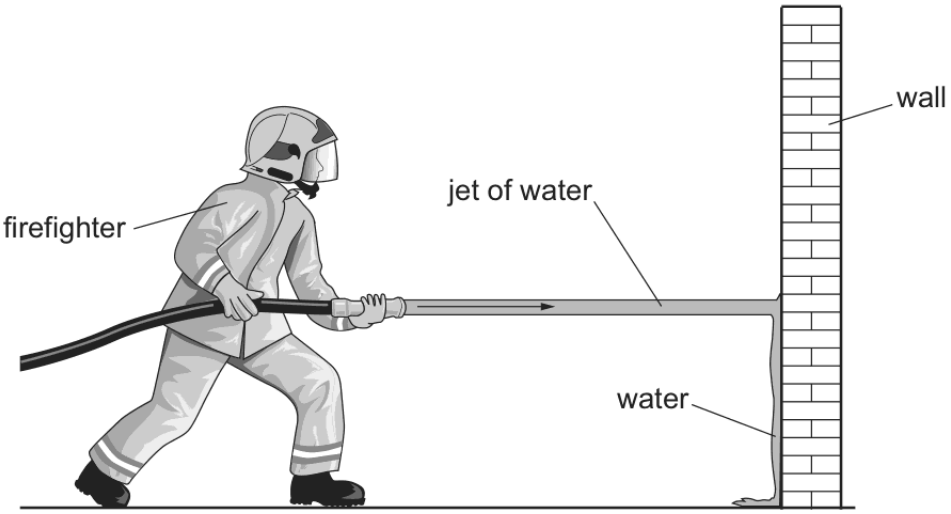


Fig. 3.1

In 1.0 s, a volume of $7.5 \times 10^{-3}\,m^3$ of water hits the wall horizontally, at a speed of 24 m/s.

The density of water is $1000\,kg/m^3$.

(a) Calculate:

(i) the mass of the water that hits the wall in 1.0 s

mass = ... kg [2]

(ii) the horizontal momentum of the water that hits the wall in 1.0 s.

momentum = ... kg m/s [2]

(b) The horizontal momentum of the water decreases to zero when the water hits the wall. None of the water bounces back from the wall.

Explain why the momentum calculated in **(a)(ii)** is equal in size to the horizontal force exerted on the water by the wall.

.....

.....

..... [2]

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(c) Explain, in terms of Newton's third law, why there is a horizontal force exerted on the wall by the water.

.....

.....

.....

.....

..... [2]

(d) The pump that forces the water to flow through the hose is adjusted, and the speed of the water leaving the hose doubles.

Explain why the force exerted by the water on the wall increases by a factor of 4.

.....

.....

..... [1]

[Total: 9]

MARK SCHEME · 5054_w25_ms_22.pdf · source page 9 of 14 · Question 3

Question	Answer	Marks
3(a)(i)	(m =) ρV **or** $1000 \times 7.5 \times 10^{-3}$	**C1**
	7.5 (kg)	**A1**
3(a)(ii)	(p =) mv **or** 7.5×24	**C1**
	180 (kg m / s)	**A1**
3(b)	force needed to change momentum **or** answer to **(a)(ii)** is change in momentum	**C1**
	force = $\frac{\text{change in momentum}}{\text{time taken}}$ **or** the momentum of the water that hits the wall in 1.0 s is the rate of change of momentum	**A1**

MARK SCHEME · 5054_w25_ms_22.pdf · source page 10 of 14 · Question 3

Question	Answer	Marks
3(c)	Newton's third law quoted **or** when object A exerts a force on object B, then object B exerts an equal and opposite force on object A **or** forces (only) occur in pairs	**B1**
	the wall exerts a force on the water	**B1**
3(d)	mass (flowing through pipe / hitting the wall in 1.0 s) doubles **and** momentum (water hitting wall in 1.0 s) increases by a factor of 4	**B1**

QUESTION · 5054_s26_qp_11.pdf · source page 4 of 16

5 Which term is used to describe the property of resisting a change in the state of rest or motion of an object?

A inertia

B moment

C momentum

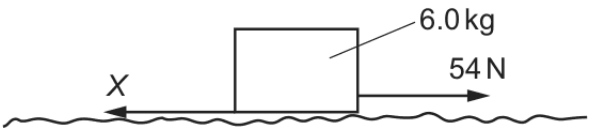
D resultant force

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 5

5	A	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 5 of 16

8 A block of mass 6.0 kg is pulled across a rough surface by a 54 N force. Force *X* also acts on the block but in the opposite direction.



The acceleration of the block is 6.0 m / s^2.

What is the magnitude of *X*?

A 9.0 N **B** 18 N **C** 36 N **D** 54 N

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 8

8	B	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 5 of 16

9 Force F_1 and force F_2 are a Newton's third law pair of forces.

Force F_1 is a gravitational force that acts on object O.

Which type of force is force F_2, and what is the object that it acts on?

A F_2 is a gravitational force that acts on O.

B F_2 is a gravitational force that acts on a different object.

C F_2 is a force of a different type that acts on O.

D F_2 is a force of a different type that acts on a different object.

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 9

9	B	**1**

1 A bus starts a journey with a uniform acceleration for the first 20 s.

(a) (i) State what is meant by 'uniform acceleration'.

.....

..... [1]

(ii) The bus starts from rest at time $t = 0$ and reaches a speed of 16 m/s at $t = 20$ s.

Calculate the acceleration of the bus.

acceleration = ...m/s^2 [2]

(b) At $t = 20$ s, the acceleration of the bus changes.

- From $t = 20$ s to $t = 100$ s, the acceleration of the bus is zero.
- From $t = 100$ s to $t = 120$ s, the bus decelerates with an acceleration of −0.30 m/s^2.

On Figure 1.1, plot a speed–time graph for the first 120 s of the journey.

Use the space provided for any calculations.

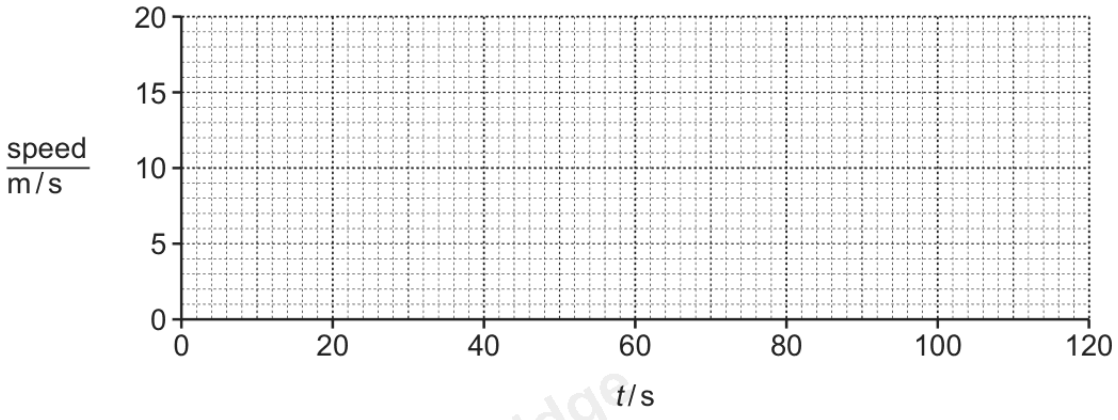


Figure 1.1

[3]

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(c) Explain, by referring to the relative magnitudes and directions of the forces on the bus, why the bus initially accelerates and later decelerates.

.....

.....

.....

..... [2]

[Total: 8]

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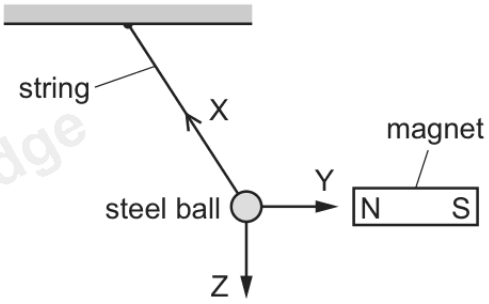
Question	Answer	Marks	Guidance
1(a)(i)	same change in speed / velocity in same time / unit time / in 1 s	**1**	**A** any named time period; same increase in speed over time; straight line on speed-time graph; rate of change in velocity is constant; increasing speed at a constant rate **I** formula alone; speed increases steadily; accelerates at a constant rate
1(a)(ii)	(a =) $v \div t$ in any form 0.8 (m/s^2)	**1**	
1(b)	straight line from (0,0) to (20,16) horizontal line from (20,16) to (100,16) straight line from (100,16) with a decrease of 6 m/s	**1**	**A** within ½ square penalise wrong value plotted for 16 once only; **ecf** from a misplotted value of speed or time at one point each time
1(c)	*acceleration / t = 20 to 100 (s):* forwards force larger than backwards force *deceleration / t = 100 (s) onwards:* resistive force acts alone **or** backwards force > forwards force	**1**	throughout **A** driving force / thrust for forwards force **A** resistive force / force from brakes / air resistance / resistance with road for backwards force; mention of resultant force forwards **bod** force of acceleration is forwards force **I** other regions / just 'forces are unbalanced' / 'resultant force acts' just 'backwards force increases'

QUESTION · 5054_s26_qp_12.pdf · source page 4 of 16

7 A steel ball is suspended from the end of a length of string.

A magnet is brought near to the steel ball. Three forces, X, Y and Z, act on the steel ball.



What are the names of forces X, Y and Z?

	X	Y	Z
A	contact force	magnetic force	friction
B	contact force	electrostatic force	weight
C	tension	magnetic force	weight
D	tension	electrostatic force	friction

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

1 An athlete and a cyclist both travel along the same straight road in the same direction.

- The athlete and the cyclist start next to each other at time $t = 0$.
- The athlete runs at a steady speed of 8.0 m/s from $t = 0$ to $t = 6.0$ s.
- The cyclist accelerates uniformly from rest at $t = 0$ to a speed of 18 m/s at $t = 6.0$ s.

(a) On Figure 1.1, draw speed–time graphs for the runner and for the cyclist.

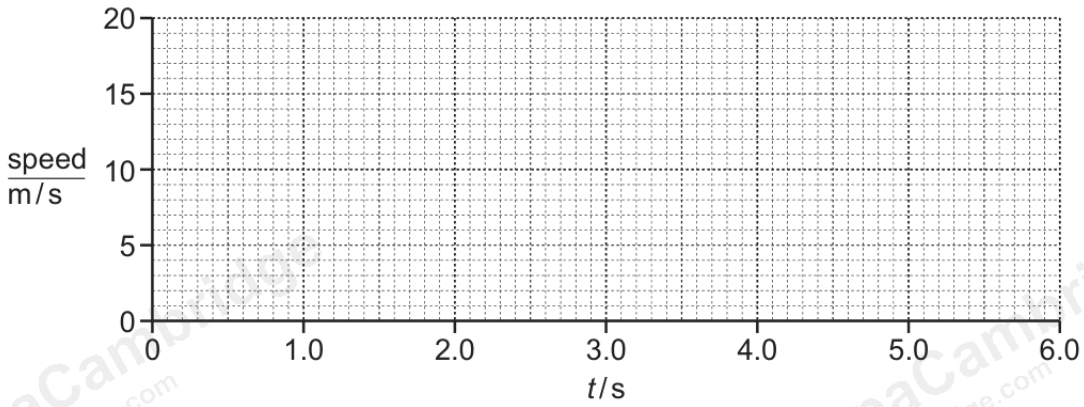


Figure 1.1

[2]

(b) Calculate the acceleration of the cyclist. Give the unit of your answer.

acceleration = ... unit ..... [3]

(c) Show, by calculation, that the athlete has travelled a greater distance than the cyclist at $t = 5.0$ s.

[2]

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(d) State Newton's first law of motion, and explain how it applies to the motion of the athlete.

.....

.....

.....

..... [2]

[Total: 9]

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Question	Answer	Marks	Guidance
1(a)	horizontal line from (0, 8) to (6, 8)	**B1**	all points ± ½ small square throughout
	straight line from (0, 0) to (6, 18)	**B1**	labels not needed
1(b)	(a =) v (- u) / t in any form algebraic or numerical	**C1**	**A** numerical values for cyclist or runner or ± ½ graph small square
	3(.0)	**A1**	**ecf as always, here** from an incorrect graph; **accept** –3 remember to round answer to 2 s.f., so e.g. **A** 2.95 but **not** 2.9;
	m / s^2	**B1**	**ignore** N / kg as the unit
1(c)	*working must be shown and final value calculated*:		**expected** values 40 (m) and 37.5 or 38 (m)
	(runner) 5 × 8 **and** 40	**B1**	**cao** correct answer only
	(cyclist = area under graph) ½ ×15 (±0.5) × 5 **and** 37.5 or 38	**B1**	**ecf** from graph with ±0.5 on y–axis for cyclist graph **max 1** if no working, i.e. if only 40 and 37.5 or 38 seen or wrong units for distance clearly shown, e.g. m^2; m / s **SC** answers using 6 s as time (48 m and 54 m) 1 mark even with no working

Question	Answer	Marks	Guidance
1(d)	simple statement of 1st law: an object/athlete remains at rest / uniform motion / constant motion / keeps moving unless a force acts	**C1**	i**gnore** just 'remains at rest'; F=ma **must** mention the word 'force' **A** a force causes movement / motion / speed / velocity to change
	complete statement and correctly applied to athlete: athlete moves at constant velocity / constant speed (in a straight line) / uniform motion / zero acceleration unless acted upon by a net / resultant / unbalanced force **OR** constant velocity / speed in a constant direction / uniform motion / no acceleration means no resultant force on athlete	**A1**	look for • uniform motion / uniform speed etc ignore at rest • unless / no resultant / net / unbalanced force • applied correctly to athlete (e.g. mentions athlete's constant speed or describes forces on athlete) **I** constant motion/remains at rest / external force in this mark

Chapter 1 Motion, forces and energy

1.5 Forces

1.5.2 Friction

111 entries: 66 Paper 1 questions, 45 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 3 of 20

7 A car is travelling at constant speed along a road and drives over a large patch of oil. The driver applies the brakes to stop the car.

Compared to braking on a dry road, what may happen?

A The car slows down more quickly because of the greater friction between the tyres and the road.

B The car speeds up at first because of the reduced friction between the tyres and the road.

C The car takes longer to slow down because of the reduced friction between the tyres and the road.

D The car takes longer to slow down because the thinking distance of the driver is greater.

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 7

7	**C**

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows a free-fall parachutist falling vertically downwards. Fig. 1.2 shows how the speed of the parachutist varies with time.

Fig. 1.1

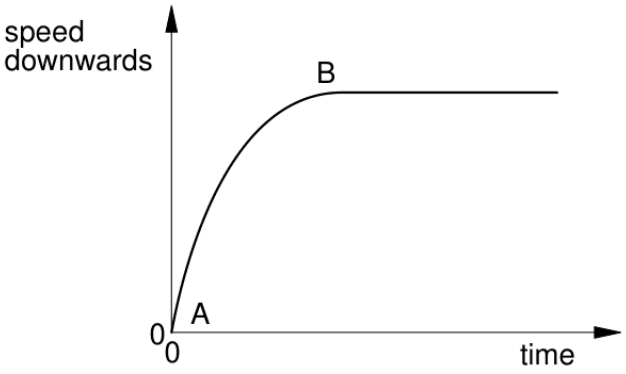


Fig. 1.2

(a) (i) State the name of the downward force acting on the parachutist.

.....

(ii) State the name of one upward force acting on the parachutist.

.....
[2]

(b) (i) State the initial value of the acceleration of the parachutist. Give the unit of your answer.

.....

(ii) Explain why the acceleration decreases from A to B.

.....

.....

.....

(iii) Explain why the parachutist falls at a constant speed after B.

.....

.....

.....
[4]

Section A

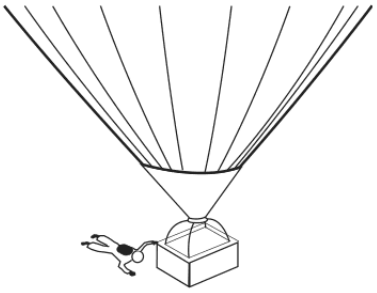
1	**(a)**	**(i)**	weight / gravity / gravitational (force)		B1
		(ii)	**air / wind** resistance or drag or friction / upthrust		B1
	(b)	**(i)**	9.8 or 10 m/s^2 or N/kg		B1
		(ii)	air resistance increases (**not** if parachute open)		B2
			less resultant force **or** sensible statement about upwards force	C1	
			e.g. resistance opposes gravity or decreases acc.	C1	
			slope of line decreases	C1	
		(iii)	air resistance = weight / no **resultant / net / overall** force / downwards		
			force balances upwards force		B1
				Total	**[6]**

QUESTION · 5054_w04_qp_1.pdf · source page 2 of 20

2 The diagrams show a parachutist in four positions after she jumps from a high balloon.

At which position does she have terminal velocity?

A not yet moving



B accelerating

C not accelerating

D not moving

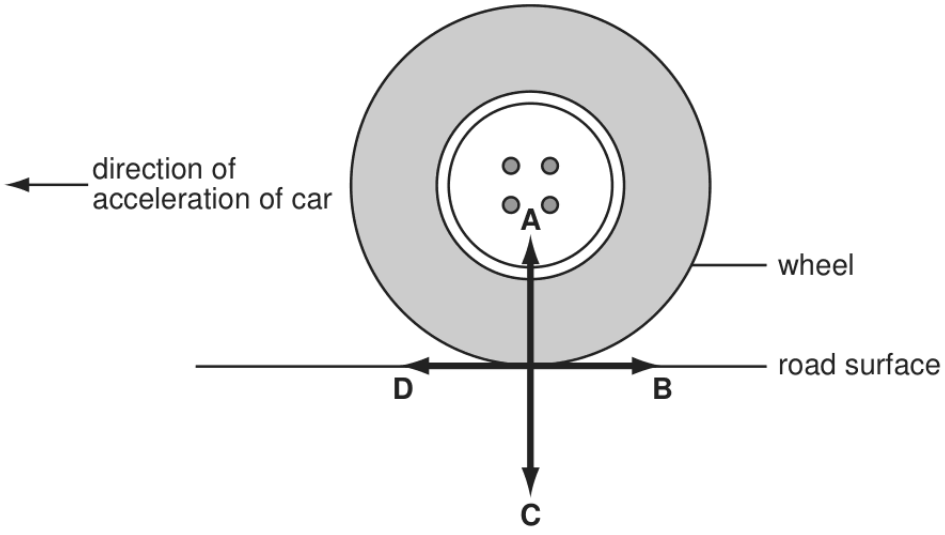
MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 2

2	**C**

QUESTION · 5054_w04_qp_1.pdf · source page 4 of 20

5 The wheel of a moving car is driven by the engine. The car is accelerating in the direction shown.

In which direction does the frictional force act on the wheel?



MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 5

5	**D**

For Examiner's Use

3 A pole-vaulter runs along a track, reaching a maximum speed of 8.4 m/s. At the end of the track, he places a pole into the ground as shown in Fig. 3.1, and uses the pole to push himself vertically upwards.

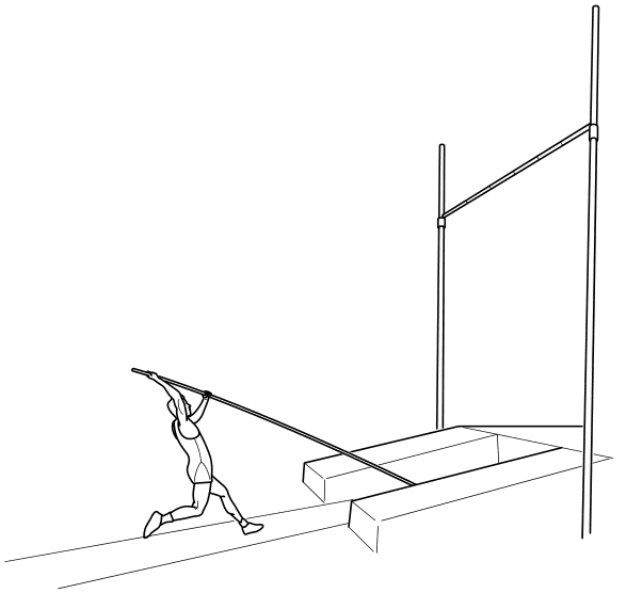


Fig. 3.1

(a) When the pole-vaulter runs along the track, there is a constant forward force on him of 320 N and a backwards resistive force that varies with his speed as shown in Fig. 3.2.

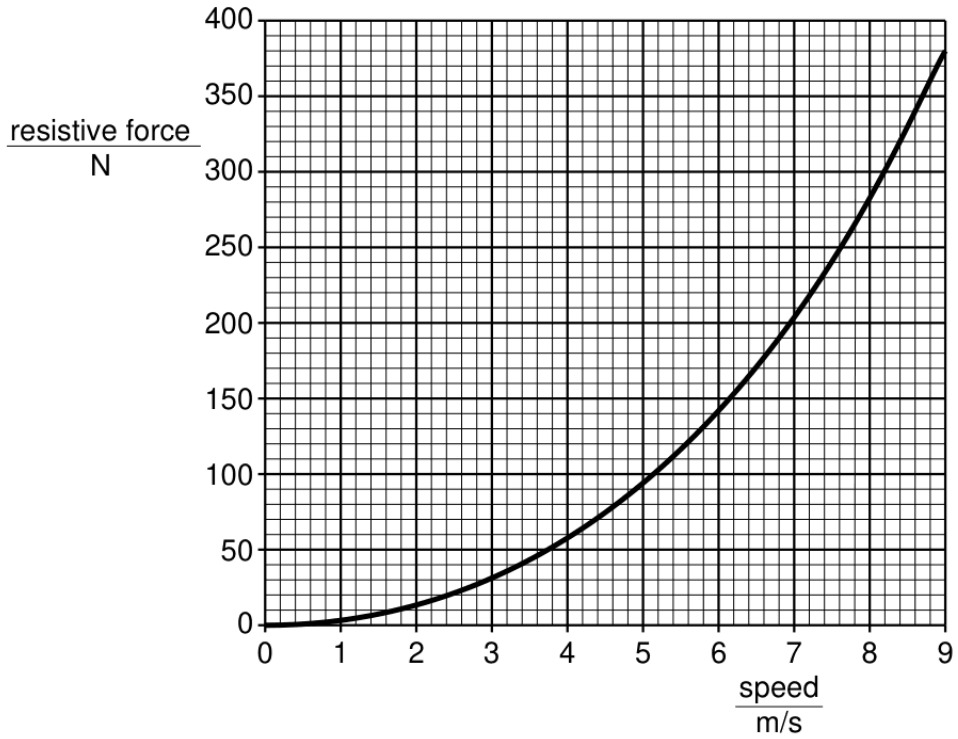


Fig. 3.2

Explain why the maximum speed that he can reach is 8.4 m/s.

.....

.....[1]

For Examiner's Use

(b) The mass of the pole-vaulter is 60 kg.

(i) Calculate the maximum kinetic energy of the pole-vaulter as he runs along the track. State clearly the formula that you use.

kinetic energy = ...

(ii) The pole is used to convert all this kinetic energy into gravitational potential energy. Calculate the height through which the pole-vaulter rises. Give your answer to an appropriate number of significant figures. The gravitational field strength is 10 N/kg.

height = ...
[6]

3 **(a)** (at 8.4 m/s) resistive force = 320 N/forward force **or** no resultant **or** forces cancel/balance

or if forward force > resistive force then runner accelerates

or if forward force < resistive force then runner decelerates **B1**

(**not** resistive force a maximum, **accept** backwards force = resistive force)

(b) **(i)** **½ mv²** **B1**

½ x 60 x 8.4^2 **C1**

2100 J (accept 2120, 2117, 2116.8) **A1**

(ii) mgh **C1**

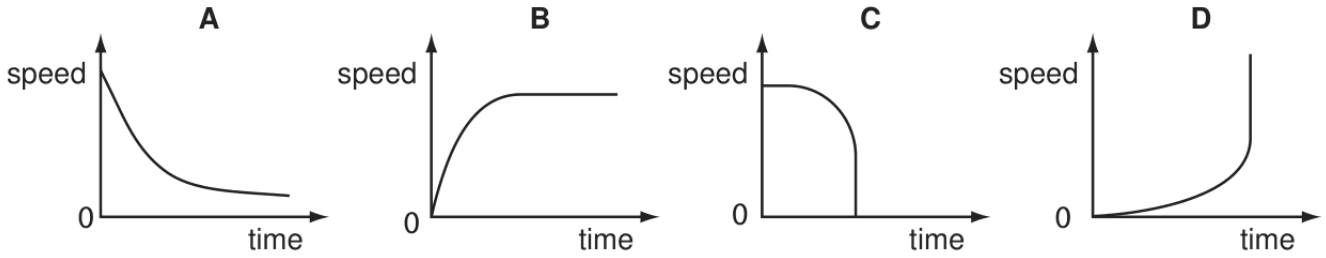
2117/60 x 10 or h = P.E. or K.E./mg e.c.f. **(i)** **C1**

3.5 m

(accept 3.50, 3.52, 3.53 i.e. **2 or 3 significant figures** only) **A1** **7**

QUESTION · 5054_s05_qp_1.pdf · source page 2 of 20

2 Which graph represents the motion of a body falling vertically that reaches a terminal velocity?



MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 2

2	**B**

For Examiner's Use

2 Fig. 2.1 shows apparatus that demonstrates how a coin and a piece of paper fall from rest.

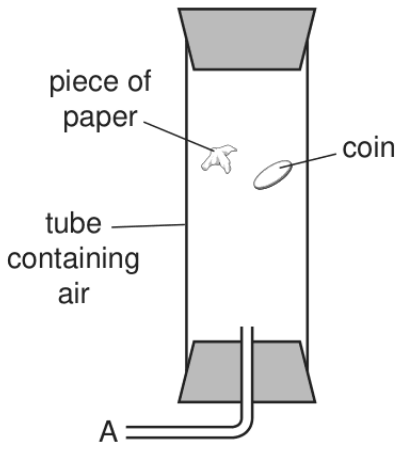


Fig. 2.1

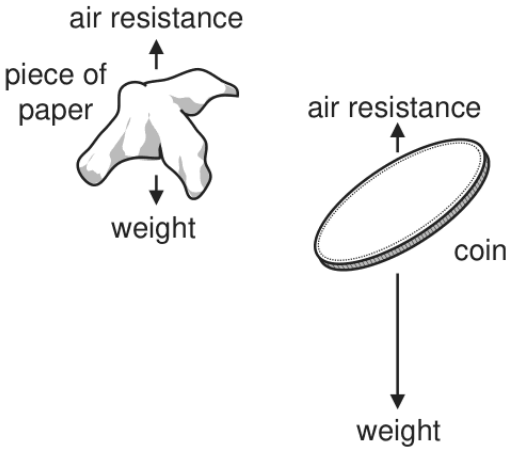


Fig. 2.2

At the positions shown in Fig. 2.1, the coin and paper are falling through air in the tube. The forces on them are shown in Fig. 2.2. The length of an arrow indicates the size of each force.

(a) State the initial value of the acceleration of the coin as it falls.

..... [1]

(b) Explain how Fig. 2.2 shows that

(i) the paper falls with constant speed,

.....

.....

(ii) the coin accelerates.

.....

.....
[3]

(c) A vacuum pump is connected to A and the air in the tube is removed.

The coin and paper fall differently in a vacuum from the way they fall in air.

State two of these differences.

1. .....

.....

2. .....

..... [2]

2	**(a)**		9.8 **or** 10 **or** 9.83 to 9.79. (m/s^2), ignore wrong unit		B1
	(b)	**(i)**	air resistance balances/equals/is same as weight (accept gravity) no resultant force **or** upwards force = downwards force		
		(ii)	weight larger than air resistance (accept gravity) resultant force (down**) or** downwards force greater **or** upwards force less	ANY 3	B3
	(c)		coin and/or paper fall faster **or** hit base sooner coin and/or paper accelerate at g coin falls with paper **or** at same rate **or** same av. speed **or** same acceleration **or** hit bottom together **or** at same time (NOT fall at same speed/same time)	ANY 2 lines	B2 6

QUESTION · 5054_s07_qp_1.pdf · source page 2 of 20

3 A student drops a table-tennis ball in air.

What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 3

3 **C**

Section B

Answer **two** questions from this section.

Use the lined page provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 Fig. 9.1 shows the horizontal forces acting on a moving car.

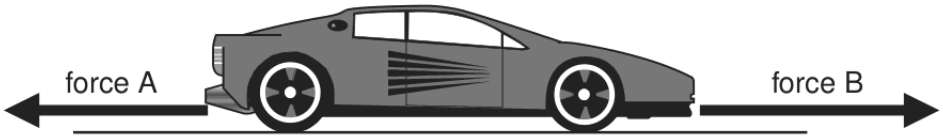


Fig. 9.1

(a) Compare the sizes and directions of the two forces when the car is

(i) moving along a straight road at constant speed, [1]

(ii) accelerating. [1]

(b) Suggest the direction of the resultant force on the car when the car turns a corner at constant speed. [1]

(c) Fig. 9.2 shows the speed-time graph for the first 24 s of the motion of the car along a straight road.

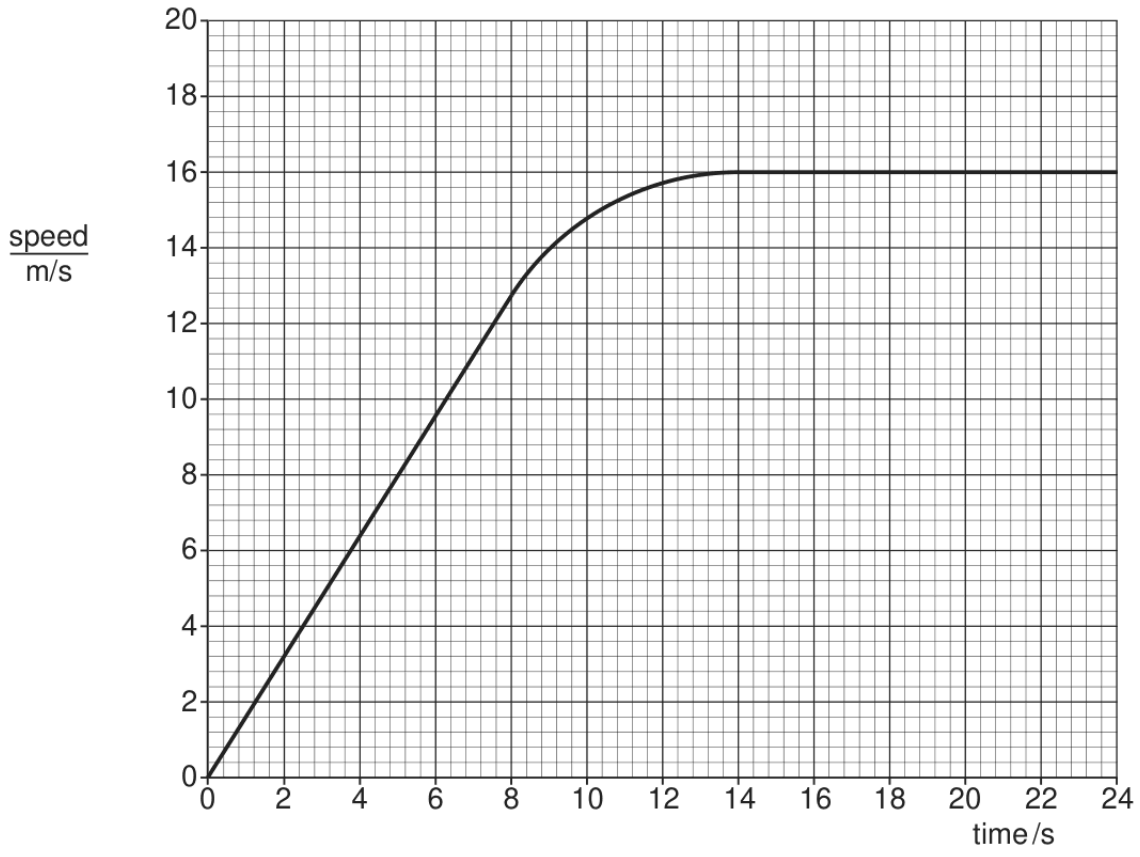


Fig. 9.2

(i) During the motion there is a period of uniform acceleration. State both the start and finish times of this period. [1]

(ii) State the main energy changes that occur in the 24 s. [3]

(iii) Calculate the acceleration of the car during the first 5 s. State clearly the equation that you use. [3]

(iv) Determine the distance travelled in the first 5 s. [3]

(d) The car is stopped by applying the brakes. Various factors can affect the distance travelled by the car during the time that the brakes are applied. Apart from the force applied by the brakes, state **two** of these factors. [2]

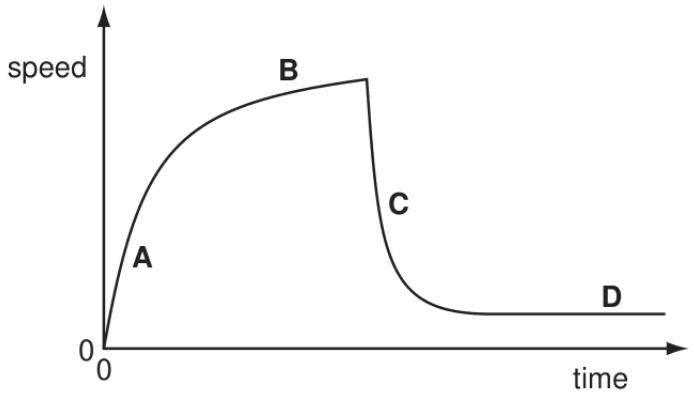
Section B

9	**(a)**	**(i)**	*A* = *B* (assume opposite direction and co-linear)	B1	
		(ii)	B > A (assume opposite direction and co-linear) maximum of 1 mark if directions wrong	B1	[2]
	(b)		towards **centre** of circle/corner	B1	[1]
	(c)	**(i)**	0 **and** 8–9 s	B1	
		(ii)	chemical (potential energy) (**accept** electrical if electrical car clear) **to** kinetic energy **or** K.E. increases thermal energy/heat/internal energy produced Max 2/3 if clear error	B1 B1 B1	
		(iii)	acceleration = (*v*–*u*)/t **or** gradient (7.8 to) 8/5 (**accept** any corresponding period e.g. 8s 12.6–12.8, 6s 9.4–9.6) 1.6 m/s^2 (**accept** 1.56–1.60)	C1 C1 A1	
		(iv)	area under graph **or** average speed 4 (m/s) **or** ½ 5 speed used in **(iii)** at 5 s ½ x (7.8 to) 8 x 5 20 m (**accept** 19.5–20; **ecf speed used in (iii)** at 5 s)	C1 C1 A1	[10]
	(d)		speed of car/friction with road (accept slippery road or ice or water or oil on road)/ friction in engine/tyre condition or area or pressure/air resistance/wind speed or direction/mass or inertia **of car or passengers**/slope of road	B2	[2]

QUESTION · 5054_w07_qp_1.pdf · source page 2 of 20

3 The speed-time graph for a falling skydiver is shown below. The skydiver alters his fall first by spreading his arms and legs and then by using a parachute.

Which part of the graph shows the diver falling with terminal velocity?

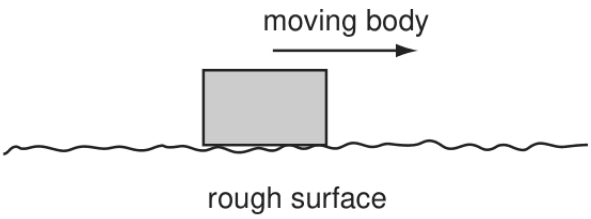


MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 3

3 **D**

QUESTION · 5054_w07_qp_1.pdf · source page 3 of 20

4 When a body moves across a rough surface, a frictional force is produced.



Which statement about this force is **always** true?

A It acts in the direction of the motion.

B It is equal in value to the force producing the motion.

C It makes the body recoil in the opposite direction after stopping it.

D It opposes the motion across the surface.

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 4

4	**D**

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 A parachutist jumps from an aircraft. Some time later, the parachute opens.

Fig. 1.1 is a graph of the vertical speed of the parachutist plotted against time *t*.

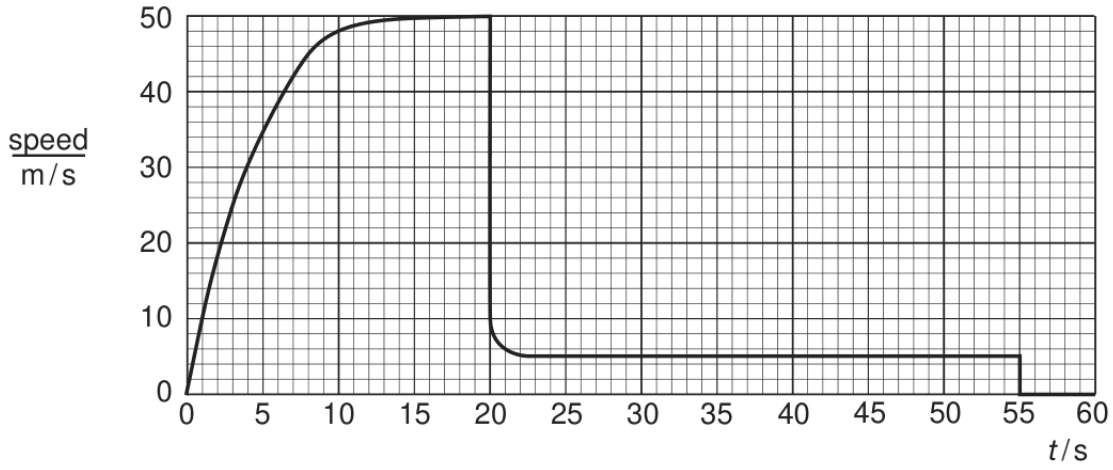


Fig. 1.1

(a) State what happens at $t = 20$ s and $t = 55$ s.

at 20 s .....

at 55 s ..... [1]

(b) Describe the motion of the parachutist between $t = 0$ and $t = 20$ s.

.....

.....

..... [2]

(c) Explain, in terms of the forces acting, why the speed of the parachutist is constant between $t = 25$ s and t = 55 s.

.....

.....

..... [2]

(d) Calculate the distance travelled by the parachutist between $t = 25$ s and $t = 55$ s.

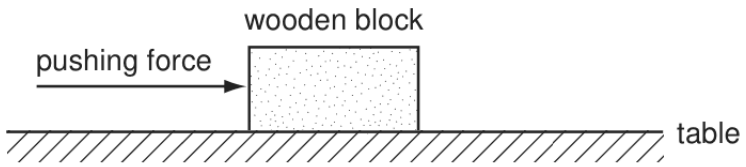
distance = ... [2]

Section A

1	**(a)**	parachute opens or speed drops from (50 to 5 m/s) **or** decelerates (**e.g.** uniformly) **and** lands/hits ground **or** speed becomes 0 or stops (**e.g.** decelerates)	B1	
	(b)	accelerates **or** speed increases (**not** increasing acceleration)	B1	
		acceleration decreases (to 0) **or** speed becomes constant	B1	
	(c)	forces balance/cancel **or** no resultant **or** equal and opposite (**not just** forces equal)	B1	
		weight/gravity **and** air resistance/drag mentioned (**not** upthrust/friction)	B1	
	(d)	(d =) st **or** s=d/t or any speed x any time **or** area under graph	C1	
		150 m	A1	[7]

QUESTION · 5054_s08_qp_1.pdf · source page 2 of 20

3 A wooden block is pushed across a table at constant speed.



Which statement is correct?

A The frictional force increases as the block moves at constant speed.

B The frictional force is equal and opposite to the pushing force.

C The frictional force is greater than the pushing force.

D The frictional force is less than the pushing force.

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 3

3	**B**

Section B

Answer **two** questions from this section.

Use the lined pages provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 When a car driver sees an emergency ahead, he applies the brakes. During his reaction time the car travels at a steady speed and covers a distance known as the thinking distance. The braking distance is the distance travelled by the car after the brakes are applied.

(a) State the energy change that occurs as the car brakes. [2]

(b) Fig. 9.1 shows the speed-time graph of a car.
The driver sees the emergency at time $t = 0$.
The total mass of the car is 800 kg.

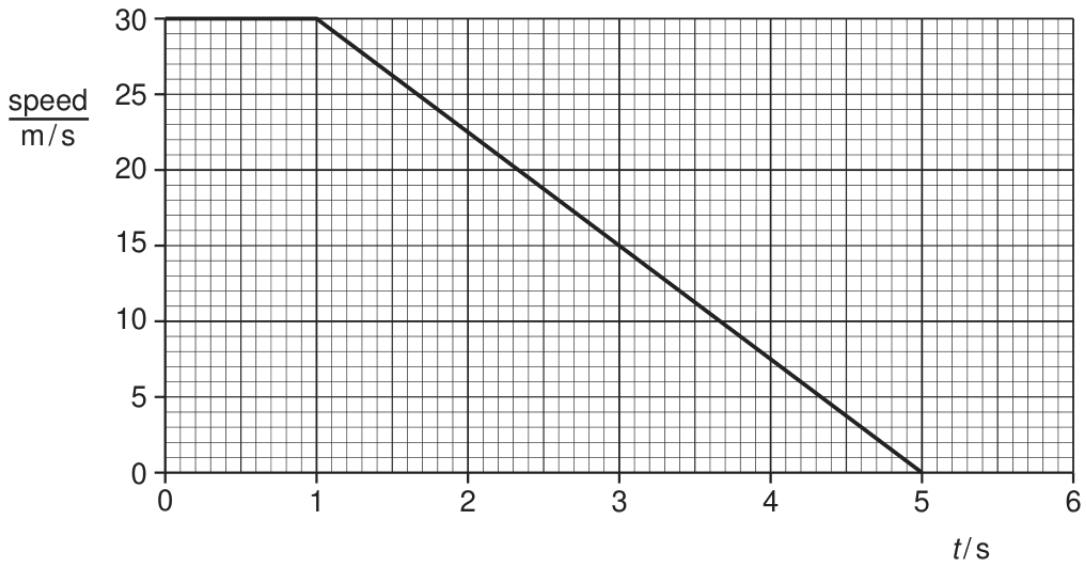


Fig. 9.1

Determine

(i) the thinking distance, [1]

(ii) the braking distance, [2]

(iii) the deceleration of the car during braking, [2]

(iv) the force provided by the brakes. [2]

(c) Using ideas about friction and deceleration, state and explain how the braking distance is affected by

(i) using new tyres rather than badly worn tyres, [2]

(ii) the car skidding on a wet road, [2]

(iii) the car carrying a heavy load of passengers. [2]

Section B

9	**(a)**	K.E. (at start) to heat (+ sound)	B1 B1
			[Total: 2]
	(b) **(i)**	30 m cao	B1
	(ii)	area under graph **or** average speed × time **or** (u + v).t / 2 **or** 30 × 4/2 60 m	C1 A1
	(iii)	$(v–u)/t$ **or** $v = u + at$ **or** 30/4 **or** gradient **or** rise/run 7.5 (±0.1) m/s^2	C1 A1
	(iv)	$F = ma$ **or** 800 × **(iii)** 6000 N ecf **(iii)**	C1 A1
			[Total: 7]
	(c) **(i)**	more friction/grip/traction **or** more deceleration **or** decelerates faster **or** decelerates in less time less (braking) **distance**	 B1 B1
	(ii)	less friction **or** less deceleration **or** decelerates slower/longer more (braking) **distance**	B1 B1
	(iii)	less deceleration **or** decelerates slower/longer more **distance**	B1 B1
			[Total: 6]

For Examiner's Use

4 A passenger aeroplane accelerates from rest along a runway. It accelerates at a uniform rate for 35 s. At this point it reaches a speed of 84 m / s and then takes off.

(a) (i) Calculate the acceleration of the aeroplane along the runway.

acceleration = ... [2]

(ii) Sketch a speed-time graph for the aeroplane as it travels along the runway.

[3]

(b) Two **horizontal** forces act on the aeroplane as it accelerates along the runway.

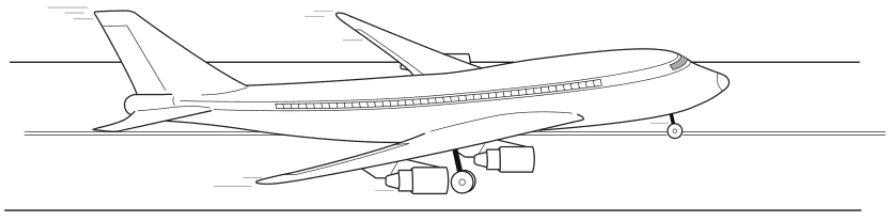


Fig. 4.1

(i) On Fig. 4.1, draw arrows to show the directions and the relative magnitudes of these forces. [1]

(ii) State what causes the smaller of these two forces.

..... [1]

4	**(a)**	**(i)**	$(a = \Delta)v/t$ **or** 84/35	C1	
			2.4 m/s²	A1	
		(ii)	speed **and** time axes correct **and** labelled	B1	
			straight line of positive gradient through origin	B1	
			84 (m/s) **and** 35 (s) marked	B1	
	(b)	**(i)**	two arrows with forward force > backward force	B1	
		(ii)	air/wind resistance **or** friction **or** drag	B1	[7]

10 Fig. 10.1 shows the horizontal forces as a cyclist travels forwards.

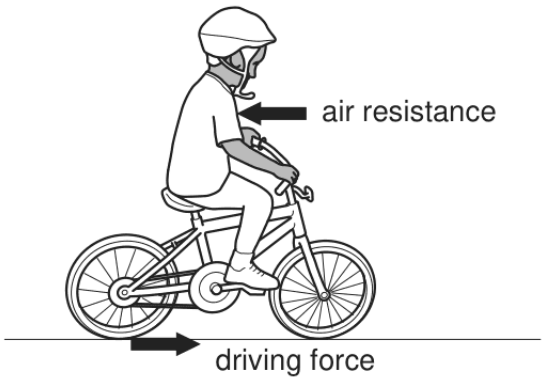


Fig. 10.1

The cyclist produces the driving force that acts on the back wheel.
In this question, you may ignore any frictional force acting on the front wheel.

(a) The bicycle accelerates until a constant speed is reached.

(i) Describe how the size of the air resistance changes during this time. [2]

(ii) Compare the sizes of the two horizontal forces when the bicycle is accelerating. [1]

(b) The total mass of the bicycle and the cyclist is 75 kg. At one instant, the speed of the bicycle is 4.0 m/s, the driving force is 30 N and the air resistance is 20 N.

Calculate

(i) the total kinetic energy of the bicycle and the cyclist, [3]

(ii) the acceleration of the bicycle and the cyclist. [2]

(c) As the bicycle moves, energy is transmitted from the pedals to the back wheel.

Fig. 10.2 shows what happens to the energy input to the pedals.

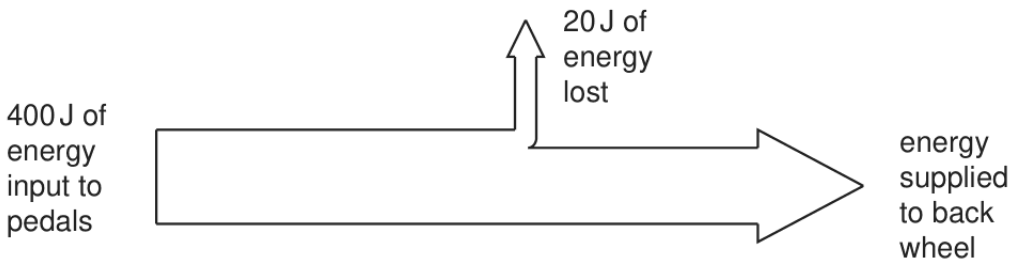


Fig. 10.2

(i) As energy is transmitted to the back wheel, some is lost. Explain how this happens. [2]

(ii) Calculate the efficiency of the bicycle in transmitting energy from the pedals to the back wheel. [2]

(d) Some bicycles are made from low density materials. Explain why this is an advantage. [3]

10 (a) (i) air resistance increases (as speed increases) B1
(at constant speed) becomes equal to driving force/applied force etc. B1

(ii) driving force (forward force) larger (than air resistance/backwards force) B1

(b) (i) (E =) ½ mv^2 algebraic formula C1
½ × 75 × 4^2 C1
600 J A1

(ii) (a =) F/m algebraic seen or 10 (N) used as force C1
0.13 m/s^2 A1

(c) (i) friction (in chain/axles) or rubbing of surfaces B1
heat or thermal energy produced B1

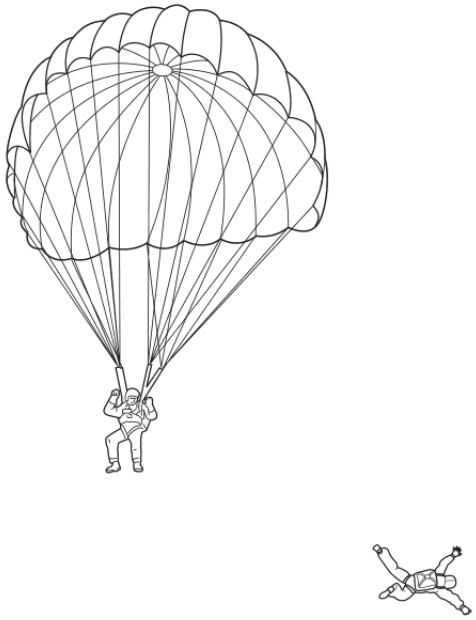
(ii) (efficiency = useful) energy output/energy input algebraic or numerical or 380 seen C1
0.95 or 95% A1

(d) lower mass/weight of cycle B1

less force needed	same acceleration/get up hill/to stop cycle/lift cycle	
greater acceleration/easier to acc.	for same force or $F = ma$ quoted	
less energy/work (input)	to go uphill/due to less friction/½ mv^2 less	
less kinetic energy	less stopping distance/less force to stop	
more efficient/less energy wasted	going uphill/less friction	
less friction	easier turn handlebars/higher (top) speed	
less pressure (on ground)	sinks less into ground	M1A1 [15]

QUESTION · 5054_s10_qp_11.pdf · source page 3 of 20

5 Two men jump out of an aeroplane at the same time. One of the men opens his parachute and the other man remains in free-fall.



Why is the man in free-fall moving faster than the parachutist?

A The man in free-fall experiences greater air resistance.

B The man in free-fall has a greater mass.

C The parachutist experiences greater air resistance.

D The parachutist has not reached terminal velocity.

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 5

5 **C**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the speed-time graph of a ball. The ball is dropped at time $t = 0$.

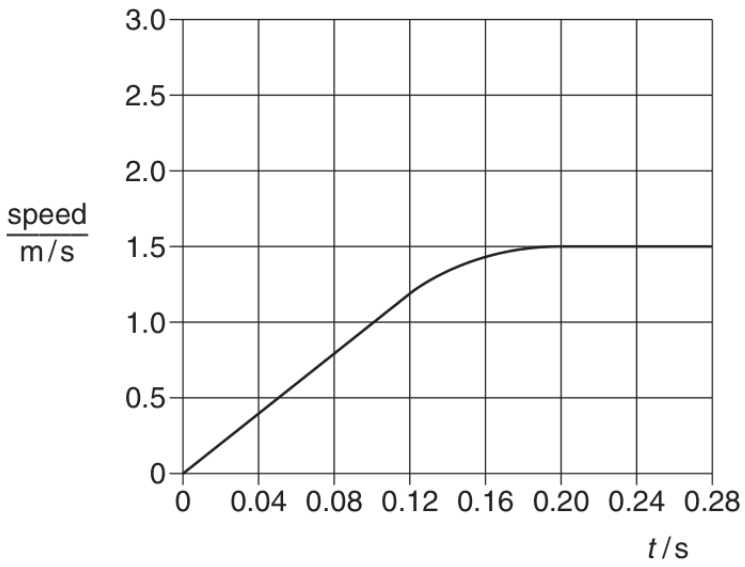


Fig. 1.1

(a) After $t = 0.20$ s, the ball falls at a constant speed.

Explain, using ideas about forces, why the speed of the ball is constant after time $t = 0.20$ s.

.....

.....

.....[1]

(b) At $t = 0$, a different ball is dropped from rest. Until $t = 0.20$ s, this ball has a constant acceleration equal to the acceleration of free-fall. After $t = 0.20$ s, its acceleration decreases.

(i) State the value of the acceleration of free-fall.

.....[1]

(ii) Determine the speed of the second ball at $t = 0.20$ s.

speed = ...[2]

(iii) On Fig. 1.1, draw the speed-time graph for the second ball from $t = 0$ to $t = 0.28$ s. [2]

Section A

1	**(a)**	forces balance/cancel **or** no net force **or** upward force = downward force **or** weight = air resistance/drag/air friction	B1	[1]
	(b) **(i)**	9.8–10 m/s^2	B1	[1]
	(ii)	$a = v(-u)/t$ algebraic or numerical	C1	
		2(.0) m/s ecf **(i)**	A1	[2]
	(iii)	straight line from (0,0) to (0.2,2) ecf **(ii)**	B1	
		after 0.2 s, decreasing but not negative gradient	B1	[2]

For Examiner's Use

3 Fig. 3.1 shows a student rubbing her hands together.

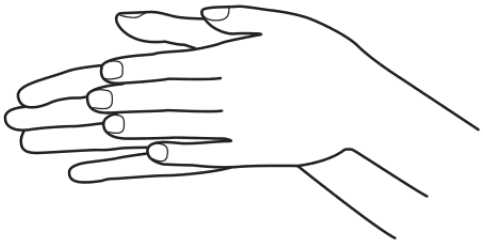


Fig. 3.1

(a) State the main energy conversion that causes the hands to become warm.

.....

.....[2]

(b) State why the hands become even warmer if they are pressed harder together when rubbing.

.....

.....[1]

(c) The average force used to slide one hand along the other is 1.2 N.

In each movement, one hand moves 0.080 m. The other hand remains stationary.

(i) Calculate the number of movements needed for 2.0 J of work to be done.

number = ...[3]

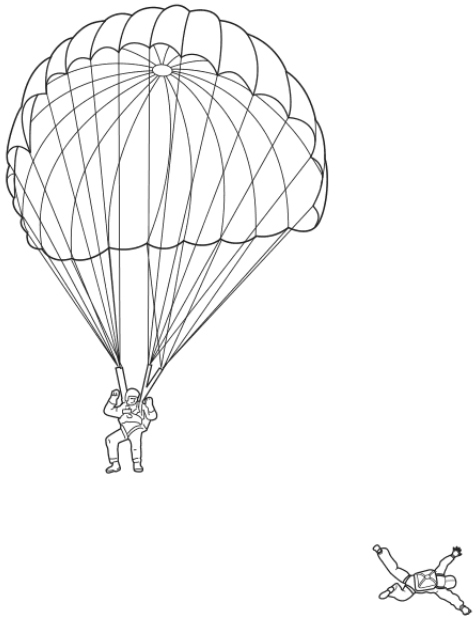
(ii) Each movement takes 0.20 s. Calculate the average power developed.

power = ...[2]

3	**(a)**	chemical energy to **or** K.E to	B1	
		heat/thermal energy/internal energy – at end	B1	[2]
	(b)	friction/resistive force **increases**	B1	[1]
	(c)	**(i)** work = force × distance in words, number or symbols	C1	
		1.2 × 0.08 **or** 0.096 **or** 20 **or** 0.208 seen	C1	
		20.8 **or** 21	A1	[3]
		(ii) power = work/time **or** energy/time, numerical or algebraic – may use values in **(i)**		
		accept 2/0.2 or 10 **W**	C1	
		0.48 W (ecf **(i)** including power of ten error in **(i)**)	A1	[2]

QUESTION · 5054_s10_qp_12.pdf · source page 2 of 20

3 Two men jump out of an aeroplane at the same time. One of the men opens his parachute and the other man remains in free-fall.



Why is the man in free-fall moving faster than the parachutist?

A The man in free-fall experiences greater air resistance.

B The man in free-fall has a greater mass.

C The parachutist experiences greater air resistance.

D The parachutist has not reached terminal velocity.

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 3

3 **C**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a car braking on a road and coming to rest.

Fig. 9.1

(a) Explain what is meant by

(i) the *thinking distance*,

.....

.....[1]

(ii) the *braking distance*.

.....

.....[1]

(b) An engineer conducts a test on the car and finds that the braking distance is greater when the car is fully loaded than when it is unloaded.

(i) Apart from the road conditions, state what must be kept the same in the test.

.....[1]

(ii) Explain why the car has a greater braking distance when fully loaded.

.....

.....[1]

(c) State and explain how one road condition affects the braking distance of the car. Use ideas about friction in your answer.

.....

.....[2]

(d) Explain how wider tyres affect the pressure of the car on the surface of the road.

.....

.....[1]

For Examiner's Use

(e) The car has a total mass of 900 kg and is travelling at 20 m/s. At time $t = 0$, the driver sees an accident ahead. He applies the brakes at $t = 0.60$ s to stop the car. After the brakes are applied, the car comes to rest in a further 4.0 s.

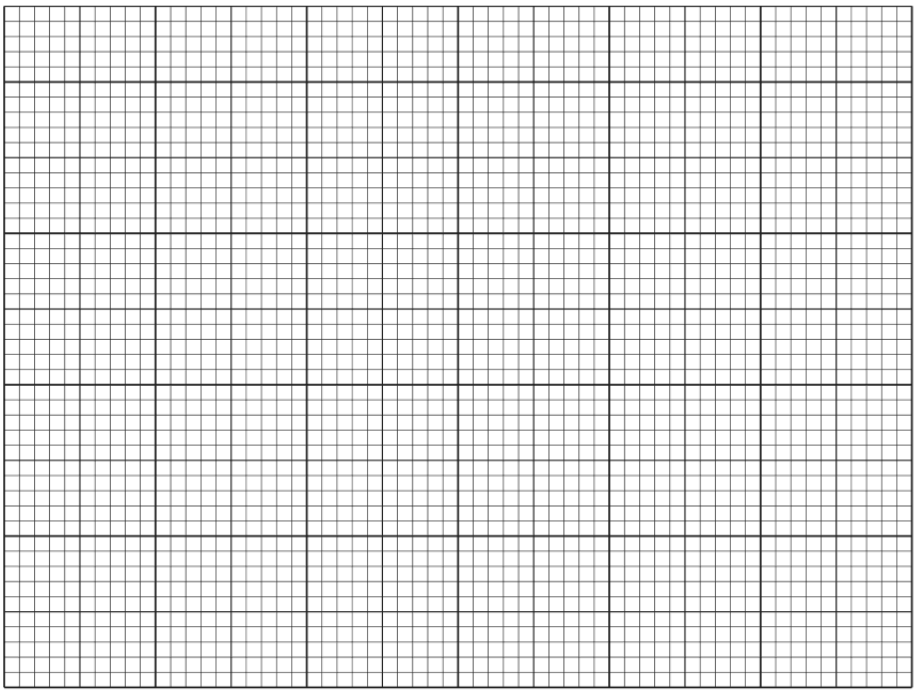
(i) Calculate the deceleration of the car as it brakes.

deceleration = ...[2]

(ii) Calculate the braking force acting on the car.

force = ...[2]

(iii) On Fig. 9.2, draw a speed-time graph for the car as it brakes.



[3]

Fig. 9.2

(iv) State how your graph in **(iii)** can be used to find the total distance travelled by the car.

.....

.....[1]

Section B

9	**(a)**	**(i)**	distance travelled while thinking/in reaction time or before braking starts	B1	[1]
		(ii)	distance travelled while brakes applied/car decelerates	B1	[1]
	(b)	**(i)**	speed (of cars) or same force/pressure on pedal or same braking force or same tyres or condition of brakes	B1	[1]
		(ii)	greater inertia/kinetic energy/momentum or smaller deceleration/acceleration	B1	[1]
	(c)		any road condition, e.g. icy, wet, poor surface, slippery/smooth/rough surface and its correct effect on distance	B1	
			correct explanation that refers to friction e.g. more friction when dry	B1	[2]
	(d)		pressure low(er) (with larger area)	B1	[1]
	(e)	**(i)**	$a = v/t$ any algebraic or numerical value e.g. 20/4; 20/3.4; 20/4.6; 20/0.6	C1	
			5(.0) m/s^2	A1	[2]
		(ii)	$F = ma$ algebraic or numerical e.g. 900 × **(i)**	C1	
			4500 N e.c.f. **(i)**	A1	[2]
		(iii)	correct axes labelled with quantity and/or unit	B1	
			horizontal line at 20 m/s from 0 to 0.6 s	B1	
			straight line from end of horizontal section or from (0.6,20) to (4.6,0) or (4,0)	B1	[3]
		(iv)	area (under graph or of trapezium)	B1	[1]

[Total: 15]

QUESTION · 5054_w10_qp_11.pdf · source page 5 of 20

6 A car travels along a road. The driver stops the car by pushing his foot down on the brake pedal.

What does **not** change if he pushes harder on the brake pedal?

A the braking distance

B the braking force

C the stopping distance

D the thinking distance

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 6

6	**D**

QUESTION · 5054_w10_qp_12.pdf · source page 2 of 16

3 A car travels along a road. The driver stops the car by pushing his foot down on the brake pedal.

What does **not** change if he pushes harder on the brake pedal?

A the braking distance

B the braking force

C the stopping distance

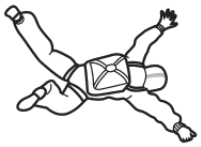
D the thinking distance

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 3

3	**D**

QUESTION · 5054_s11_qp_11.pdf · source page 2 of 16

3 A skydiver falls from rest through the air and reaches terminal velocity.



What is the acceleration of the skydiver during his fall?

A constant at $0\,m/s^2$

B constant at $10\,m/s^2$

C starting at $0\,m/s^2$ and increasing to $10\,m/s^2$

D starting at $10\,m/s^2$ and decreasing to $0\,m/s^2$

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 3

3	**D**

2 Fig. 2.1 shows a sky-diver falling vertically downwards at terminal velocity.

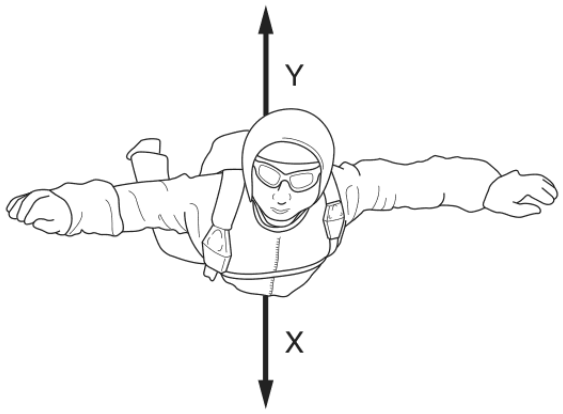


Fig. 2.1

Arrows X and Y show the two main forces acting on the sky-diver.

(a) (i) State the name of force X and the name of force Y.

X .....

Y .....
[1]

(ii) Explain why force Y acts upwards.

.....

..... [1]

(b) When the sky-diver first started to fall, forces X and Y were unbalanced.

(i) Describe and explain the effect of the unbalanced forces on the motion of the sky-diver.

.....

.....

..... [2]

(ii) State what happened to the size of force X and the size of force Y as the sky-diver fell and reached terminal velocity.

.....

..... [2]

2 **(a)** **(i)** X weight // (force of) gravity // gravitational (force)
and Y **air** resistance // (air) drag // wind resistance // **air** friction B1

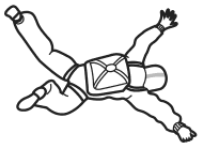
(ii) (Y) opposes motion // diver moves down // air molecules hit faster from below B1

(b) **(i)** accelerates // falls faster // speed/velocity greater B1
X is larger than Y // resultant/net force downwards B1

(ii) **two** of:
X stays the same // Y increases // forces become equal (and opposite) B2 [6]

QUESTION · 5054_s11_qp_12.pdf · source page 3 of 16

6 A skydiver falls from rest through the air and reaches terminal velocity.



What is the acceleration of the skydiver during his fall?

A constant at $0\,m/s^2$

B constant at $10\,m/s^2$

C starting at $0\,m/s^2$ and increasing to $10\,m/s^2$

D starting at $10\,m/s^2$ and decreasing to $0\,m/s^2$

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 6

6	**D**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a sky-diver falling vertically.

Fig. 9.1

The sky-diver starts from rest at time $t = 0$.
His acceleration is non-uniform until he reaches a steady speed of 50 m/s at $t = 10$ s.
He opens his parachute at $t = 20$ s and decelerates until $t = 25$ s.
From $t = 25$ s he falls at a steady speed of 5 m/s.

(a) (i) On Fig. 9.2, draw the speed-time graph for the sky-diver. [4]

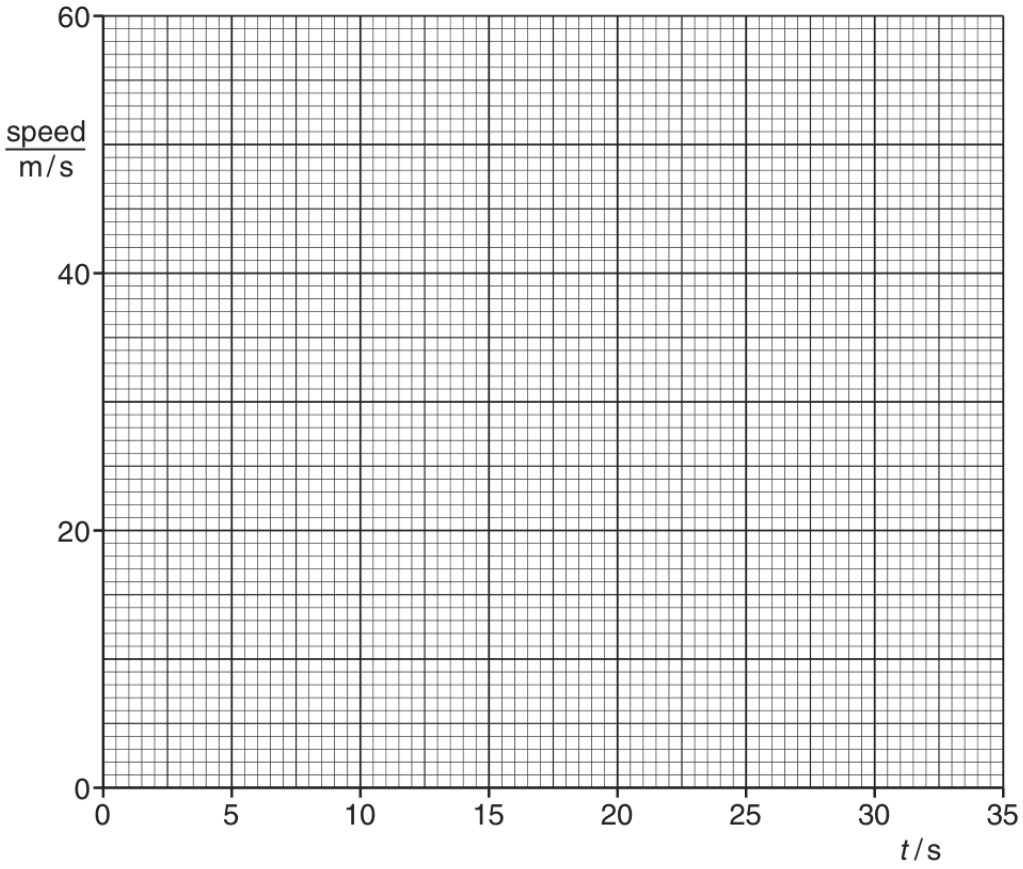


Fig. 9.2

(ii) State how your graph shows that the acceleration is non-uniform between $t = 0$ and $t = 10$ s.

.....

.....[1]

(b) Explain, in detail, why **after** the sky-diver opens his parachute, he decelerates and eventually reaches a steady speed.

.....

.....

.....

.....

.....

.....[4]

(c) For the time interval between t = 10 s and t = 20 s, calculate the distance that the sky-diver falls.

distance = ...[1]

(d) The mass of the sky-diver is 60 kg.

For the time interval between t = 20 s and t = 25 s,

(i) calculate the average deceleration of the sky-diver,

average deceleration = ...[2]

(ii) calculate the average resultant force acting on the sky-diver,

force = ...[2]

(iii) state how your graph in Fig. 9.2 may be used to obtain the distance that the sky-diver falls.

.....

.....[1]

Section B

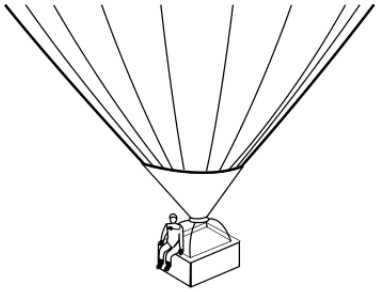
9	**(a)**	**(i)**	curve with decreasing gradient from origin to 50 m/s at 10 s	B1
			constant speed from 10 to 20 s	B1
			decrease to 5 m/s at 25 s	B1
			constant speed from 25 s until at least 30 s	B1
		(ii)	gradient/slope not constant/decreases // graph curves // graph not a (straight) line // increase (in speed) per second/unit time not equal	B1
	(b)		any mention of **air** resistance/drag/upward force	B1
			(initially) force upwards larger than force downwards // resultant force upwards	B1
			air resistance decreases (with fall in speed)	B1
			(at constant speed) air resistance/friction/drag equals weight // forces (up and down) balance // zero resultant force	B1
	(c)		500 m	B1
	(d)	**(i)**	(a =) $\frac{v-u}{t}$ in any numerical or algebraic form, e.g. 45/5	C1
			9(.0) m/s^2 ecf **(a)(i)**	A1
		(ii)	($F = ma$) in any numerical or algebraic form, e.g. 60 × 9 ecf **(i)**	C1
			540 N	A1
		(iii)	area **under** graph/line/curve	B1 [15]

QUESTION · 5054_w11_qp_11.pdf · source page 3 of 20

5 The diagrams show a parachutist in four positions after she jumps from a high balloon.

At which position does she have terminal velocity?

A not yet moving



B accelerating

C not accelerating

D not moving

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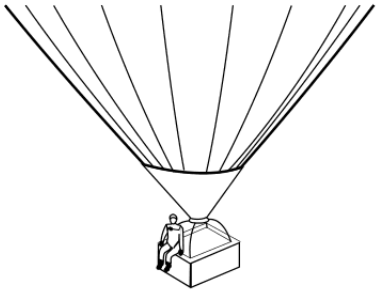
5	**C**

QUESTION · 5054_w11_qp_12.pdf · source page 2 of 20

1 The diagrams show a parachutist in four positions after she jumps from a high balloon.

At which position does she have terminal velocity?

A not yet moving



B accelerating

C not accelerating

D not moving

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 1

1	**C**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 is the speed-time graph for a racing car of total mass 650 kg as it sets off from rest at the start of a race.

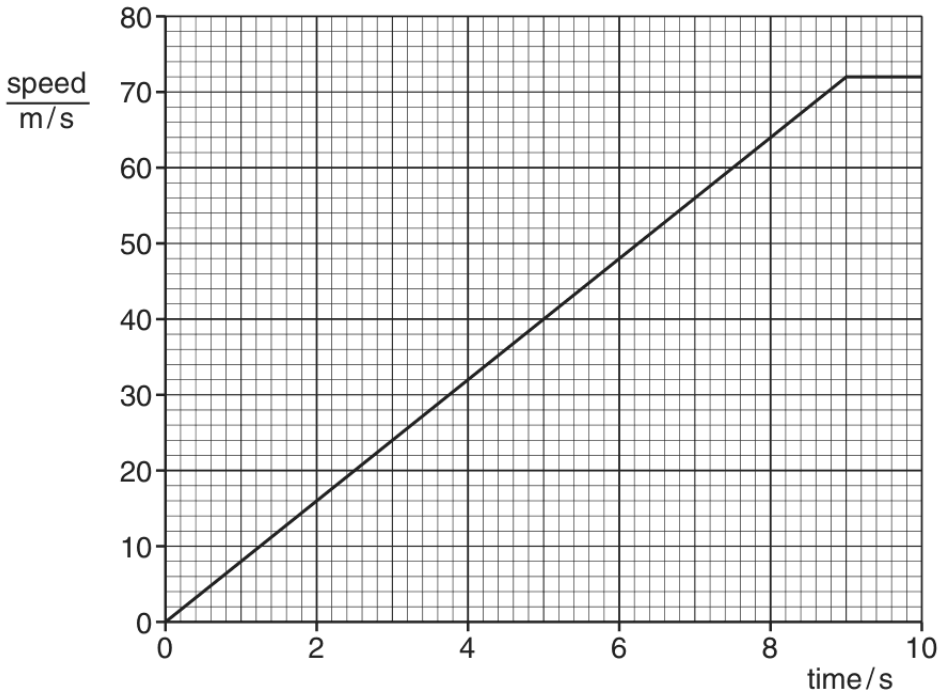


Fig. 9.1

The car travels in a straight line until time $t = 9.0$ s.

(a) State the speed of the car at time $t = 9.0$ s.

speed = ...[1]

(b) Calculate, for the car between times $t = 0$ and $t = 9.0$ s,

(i) the distance travelled,

distance = ...[2]

(ii) the acceleration of the car,

acceleration = ...[2]

(iii) the resultant force acting on the car.

force = ...[2]

(c) The acceleration of the car is constant between $t = 0$ and $t = 9.0$ s.

Suggest and explain why the driving force on the car must increase to keep the acceleration constant.

.....

.....

.....

.....[3]

(d) After $t = 9.0$ s, the car starts to turn a corner and follows a circular path at constant speed.

(i) Explain why the car is accelerating even though its speed is constant.

.....

.....

.....[2]

(ii) State the direction of this acceleration.

.....[1]

(iii) State and explain what causes the car to accelerate as it turns the corner.

.....

.....[2]

Section B

9	**(a)**	72 m / s			B1	
	(b)	**(i)**	area (under graph) **or** ½ base × height **or** ½*vt* **or** ½ × 9 × 72		C1	
			320/324 m		A1	
		(ii)	**change** in velocity/time **or** $\Delta v/t$ **or** 72/9		C1	
			8(.0) m / s^2		A1	
		(iii)	(*F* =) *ma* **or** 650 × 8.0		C1	
			5.2 × 10^3 N		A1	
	(c)	friction **or** air/wind resistance **or** drag			M1	
		increases as speed increases			A1	
		resultant/net/unbalanced force remains constant			B1	
	(d)	**(i)**	**direction** (of car/motion/speed/velocity) changes		B1	
			(therefore) velocity changes		B1	
		(ii)	towards centre (of circle)/centripetal		B1	
		(iii)	friction **with ground**	**OR** banking of track	B1	
			mention wheels/tyres	reaction force (acts towards centre)	B1	[15]

QUESTION · 5054_s12_qp_11.pdf · source page 3 of 16

4 A steel ball is released just below the surface of thick oil in a cylinder.

During the first few centimetres of travel, what is the acceleration of the ball?

A constant and equal to $10\,m/s^2$

B constant but less than $10\,m/s^2$

C decreasing

D increasing

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 4

4	**C**

QUESTION · 5054_s12_qp_12.pdf · source page 3 of 16

4 A student drops a table-tennis ball in air.

What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

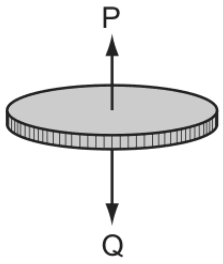
MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 4

4 **C**

QUESTION · 5054_s12_qp_12.pdf · source page 3 of 16

5 A coin falls through the air from rest, and eventually reaches a constant speed.

Two forces P and Q act on the coin.



What happens to the force P and to the resultant force acting on the coin before it reaches constant speed?

	force P	resultant force
A	decreases	increases
B	decreases	decreases
C	increases	decreases
D	increases	increases

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 5

5 **C**

2 **(a)** State what is meant by *friction*.

.....

.....[2]

(b) Fig. 2.1 shows two horizontal forces that act on a car. Force B is caused by air resistance and friction.

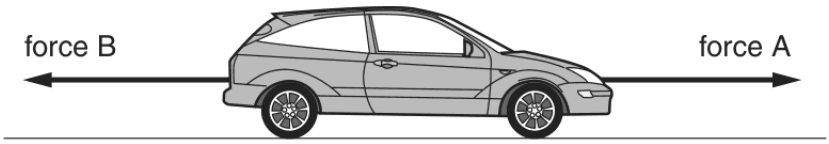


Fig. 2.1

The car is travelling along a straight level road.

(i) The forward force A and the backward force B are equal.
Describe the motion of the car.

.....

.....[1]

(ii) The mass of the car is 800 kg.
Force A increases to 5000 N. This causes the car to accelerate initially at 1.5 m/s^2.
Calculate the size of force B.

force B = ...[3]

(iii) Force A remains constant at 5000 N. Explain why the acceleration decreases as the car travels along the level road.

.....

.....[2]

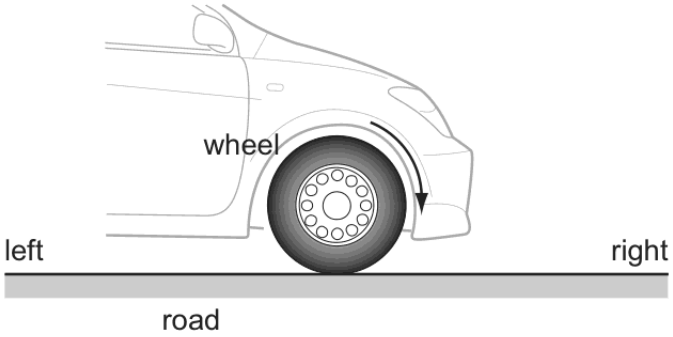
(c) A car of mass 800 kg climbs a hill and rises a vertical distance of 200 m.
Calculate the gain in potential energy of the car. (The gravitational field strength *g* is 10 N/kg.)

gain in potential energy = ...[2]

2	**(a)**	a force	B1	
		when objects slide over/rub one another OR opposes (relative) **motion**/**movement**	B1	
	(b)	**(i)** constant/uniform speed OR constant/uniform velocity OR zero acceleration	B1	
		(ii) (*F* =) *ma* seen in any form numerical or algebraic	C1	
		1200 (N) OR 6200 N seen	C1	
		3800 N	A1	
		(iii) Force B increases OR backwards force/resistance/friction/drag increases	M1	
		as speed/velocity increases	A1	
	(c)	(PE =) *mgh* in any form numerical or algebraic	C1	
		1 600 000 J	A1	[10]

QUESTION · 5054_w12_qp_11.pdf · source page 4 of 16

6 The front wheel of a car is turned in a clockwise direction by the engine as the car accelerates towards the right, as shown in the diagram. There is a force of friction between the wheel and the road.



What are the directions of the frictional forces on the wheel of the car and on the road?

	direction of frictional force on wheel	direction of frictional force on road
A	to the left	to the left
B	to the left	to the right
C	to the right	to the left
D	to the right	to the right

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 6

6 **C**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a diver working below the surface of a lake. The density of the water in the lake is 1000 kg/m^3, the atmospheric pressure at the surface is 1.0×10^5 Pa and the gravitational field strength is 10 N/kg.

Fig. 9.1

The diver inflates a balloon with air at a depth of 15 m and attaches the balloon to a tray of objects.

(a) Calculate

(i) the pressure due to 15 m of water,

pressure = ...[2]

(ii) the total pressure at 15 m below the surface of the lake.

total pressure = ...[1]

(b) The air in the balloon occupies a volume of 0.048 m^3 at the pressure calculated in **(a)(ii)**. The diver releases the tray and the balloon, and they begin to rise. The temperature of the air in the balloon does not change.

(i) Calculate the volume occupied by the air in the balloon at atmospheric pressure.

volume = ...[2]

For Examiner's Use

(ii) The pressure of the air inside the balloon is less at the surface than at a depth of 15 m. Explain, in terms of the air molecules inside the balloon, why the pressure is less.

.....

.....

.....

.....[3]

(c) State one difference between the arrangement of the molecules of water in the lake and the molecules of air in the balloon.

.....

.....[1]

(d) When the diver releases the tray, the balloon accelerates upwards and reaches a constant speed before it arrives at the surface.

(i) Explain how the forces acting on the balloon cause it to behave in this way.

.....

.....

.....

.....[3]

(ii) On the axes in Fig. 9.2, sketch the distance-time graph for the balloon as it travels 15 m to the surface.

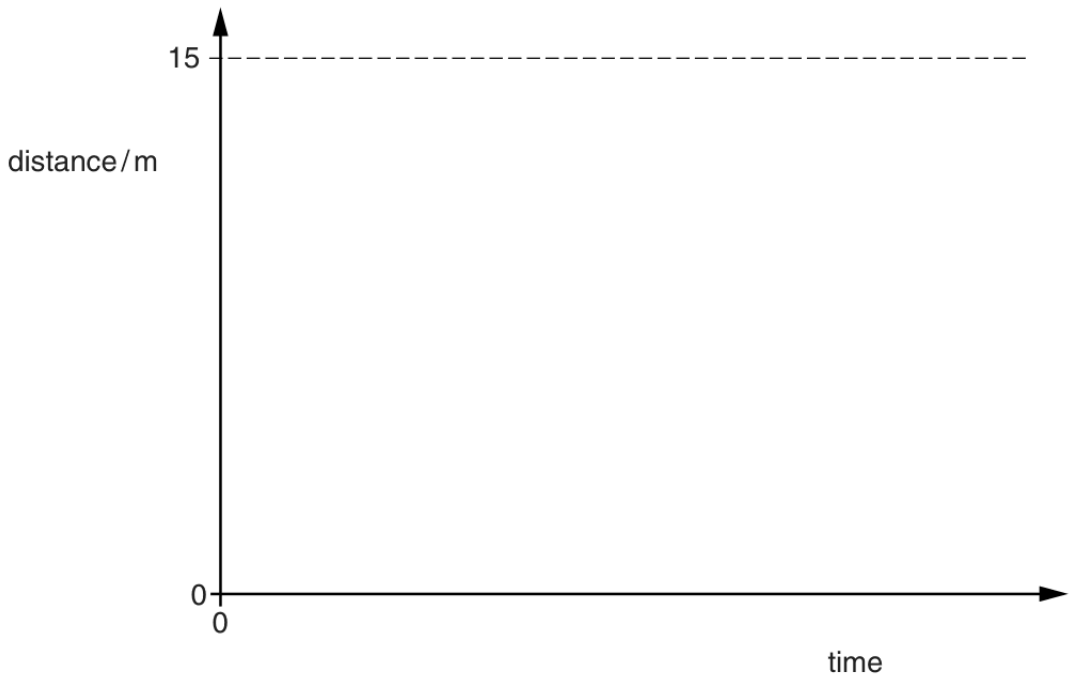


Fig. 9.2 [3]

Section B

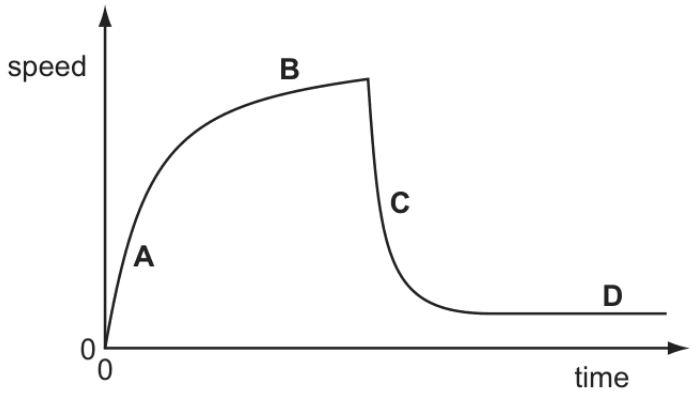
9	**(a)**	**(i)**	$(p =)\rho hg$ **or** 1000 × 15 × 10	C1	
			1.5×10^5 Pa	A1	
		(ii)	2.5×10^5 Pa	B1	[3]
	(b)	**(i)**	$p_1V_1 = p_2V_2$ **or** 250 000 × 0.048 = 100 000 × V_2	C1	
			0.12 m^3	A1	
		(ii)	molecules/particles: further apart **or** their speed is unchanged	B1	
			(molecular) collisions with balloon/walls/unit area	B1	
			less frequent collisions (**not** if force/violence of each collision less)	B1	[5]
	(c)		water molecules: close(r)/move in clusters/move within the liquid **or** air molecules: far/further apart/move individually/move throughout container	B1	[1]
	(d)	**(i)**	net/resultant/unbalanced force upwards (at first) **or** upwards force greater	B1	
			friction/resistance/drag/downward force increases		
			(until) downward force = upward force/forces balance/no resultant force	B3	
		(ii)	starts from **marked** (0,0) **or** initial gradient = 0	B1	
			increasing gradient initially	B1	
			constant gradient (must be greater than zero) finally	B1	[6]

[Total: 15]

QUESTION · 5054_s13_qp_11.pdf · source page 3 of 20

3 The speed-time graph for a falling skydiver is shown below. As he falls, the skydiver spreads out his arms and legs and then opens his parachute.

Which part of the graph shows the skydiver falling with terminal velocity?



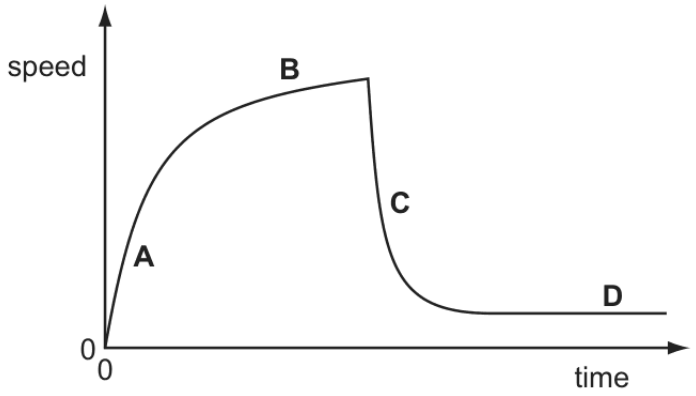
MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 3

3 **D**

QUESTION · 5054_s13_qp_12.pdf · source page 3 of 16

3 The speed-time graph for a falling skydiver is shown below. As he falls, the skydiver spreads out his arms and legs and then opens his parachute.

Which part of the graph shows the skydiver falling with terminal velocity?



MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 3

3	**D**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

Answer only one of the two alternative parts in **Question 8**.

1 A ball rolls down a slope, as shown in Fig. 1.1.

The metre rule shows the position of the ball at times $t = 0$, 1.0 s, 2.0 s and 3.0 s.

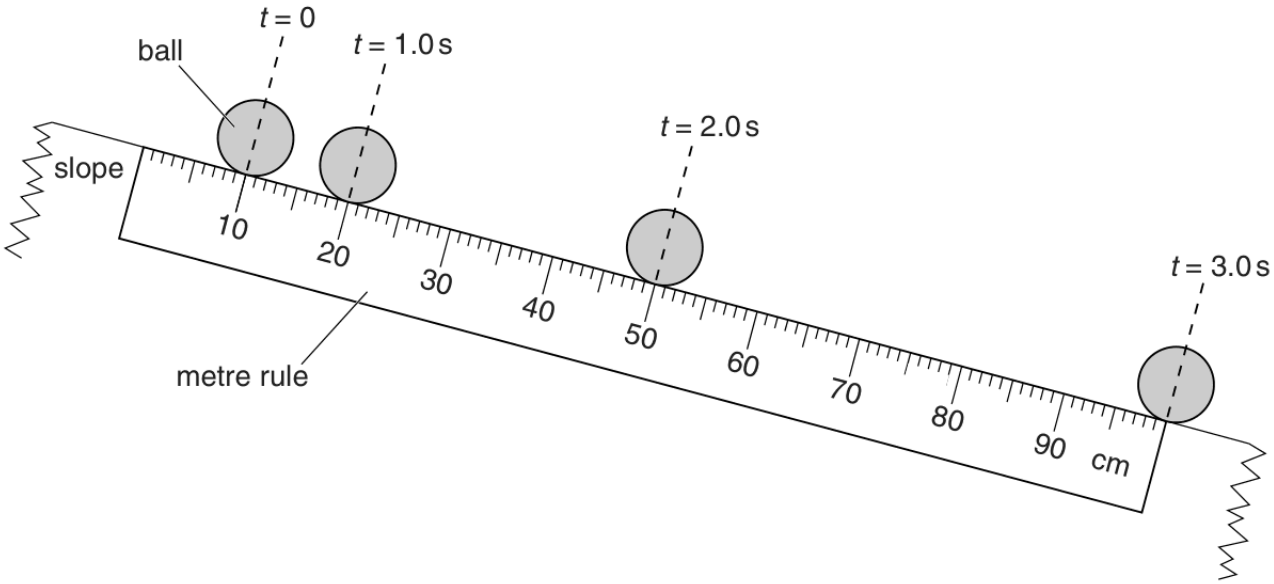


Fig. 1.1

(a) Explain how Fig. 1.1 shows that the ball is accelerating.

.....

..... [1]

(b) Calculate the average speed of the ball between $t = 1.0$ s and 3.0 s.

average speed = ... [2]

(c) Two of the forces that act on the ball are air resistance and weight.

State what, if anything, happens to these forces as the ball accelerates.

air resistance: .....

weight: .....
[2]

(d) Explain why, if the slope is long enough, the ball eventually reaches a constant speed.

.....

..... [1]

Section A

1	**(a)**	travels further in each second / in same time / between images	B1	
	(b)	(s=) d/t in any form algebraic or numerical	C1	
		40 cm/s; 0.4(0) m/s	A1	
	(c)	air resistance increases	B1	
		weight constant	B1	
	(d)	forces balance /cancel **or** no resultant/net force **or** resultant of any two forces equal and opposite to third	B1	[6]

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a skydiver, of mass 70 kg, falling towards the Earth at constant speed, a long time after jumping from an aeroplane.

Fig. 9.1

At time $t = 0$, he receives a radio signal. He opens his parachute 12 s later. Fig. 9.2 is the speed-time graph for the skydiver.

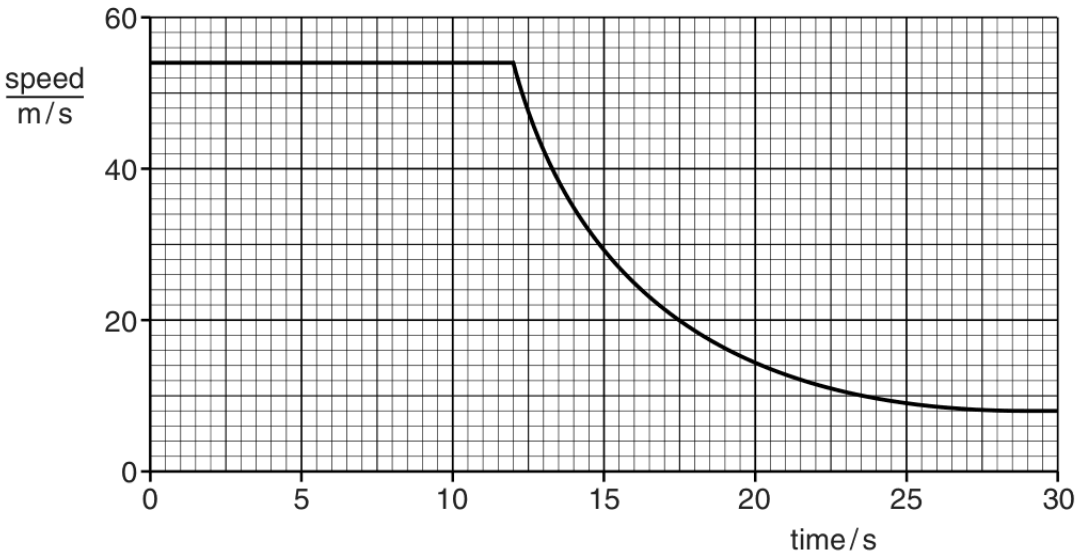


Fig. 9.2

(a) State the difference between *speed* and *velocity*.

.....

.....[1]

(b) The gravitational field strength g is 10 N/kg.

(i) Calculate the weight of the skydiver.

weight = ...[1]

For Examiner's Use

(ii) State the size of the air resistance acting on the skydiver between $t = 0$ and $t = 12\,s$.

air resistance = ...[1]

(c) For the period between $t = 0$ and $t = 12\,s$, determine

(i) the speed of the skydiver,

speed = ...[1]

(ii) the distance fallen by the skydiver,

distance = ...[2]

(iii) the change in the gravitational potential energy of the skydiver.

change in energy = ...[2]

(d) As he falls at constant speed, his gravitational potential energy changes into another form. State the name of this other form of energy.

.....[1]

(e) **(i)** State and explain what happens to the air resistance as the skydiver opens his parachute.

.....

.....

.....[2]

(ii) State and explain the effect on the motion of the skydiver of opening the parachute.

.....

.....

.....[2]

(f) By $t = 15\,s$, his parachute is fully open.

State and explain what happens to the air resistance after $t = 15\,s$.

.....

.....

.....[2]

Section B

9	**(a)**	speed does not have direction **and** velocity does or speed = distance/time **and** velocity = displacement/time or speed is a scalar **and** velocity is a vector	B1	[1]
	(b) **(i)**	700 N	B1	
	(ii)	700 N	B1	[2]
	(c) **(i)**	54 m / s	B1	
	(ii)	(height/distance =) area (under graph) **or** (x =)vt **or** 54 × 12 648/650 m	C1 A1	
	(iii)	(GPE =) mgh **or** 70 × 10 × 648 4.5/4.54/4.536 × 10^5 J	C1 A1	[5]
	(d)	(becomes) heat/thermal energy/internal energy (**not** kinetic energy (of skydiver) unless qualified as KE of air)	B1	[1]
	(e) **(i)**	(air resistance) increases larger area of parachute	B1 B1	
	(ii)	(skydiver) decelerates/slows down (**not** rises up) net upward force	B1 B1	[4]
	(f)	air resistance decreases speed decreases	B1 B1	[2]

[Total: 15]

10 Fig. 10.1 shows an electric train of mass 5.5×10^5 kg.

For Examiner's Use

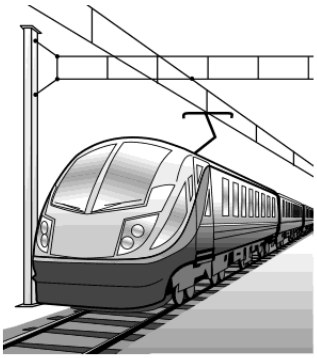


Fig. 10.1

The train is initially at rest. The electric motor exerts a constant force and the train accelerates. Its acceleration decreases until the train reaches a speed of 40 m/s.

The train then continues at this constant speed.

(a) (i) On the axes in Fig. 10.2, sketch a speed-time graph for the train.

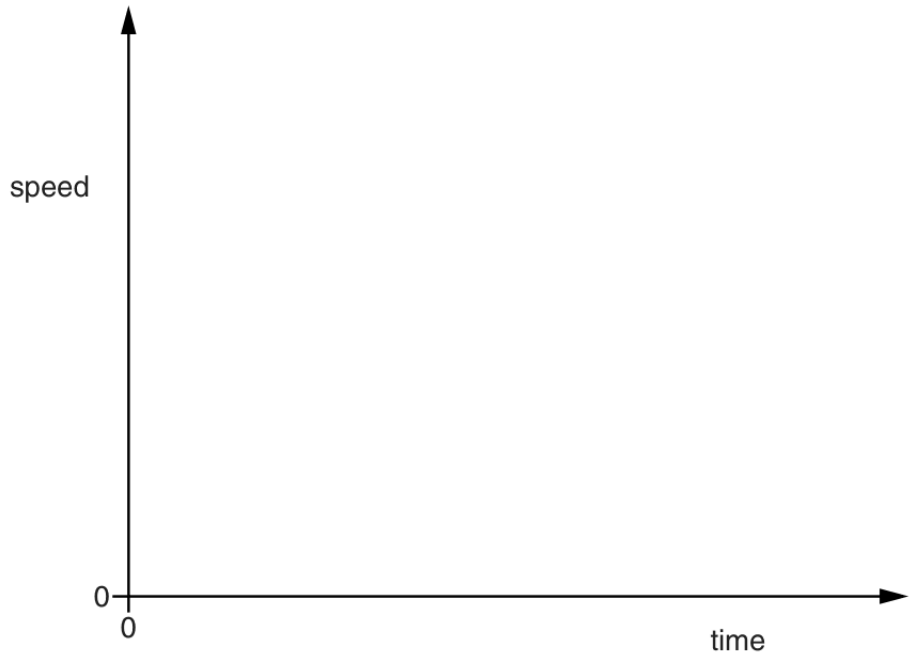


Fig. 10.2

[3]

(ii) Describe how the speed-time graph for the train is used to determine the distance travelled by the train when it is accelerating.

.....

.....

.....[2]

(b) Explain, in terms of the forces acting,

For Examiner's Use

(i) why the acceleration of the train decreases,

.....

.....

.....[2]

(ii) why the train eventually reaches constant speed.

.....

.....[1]

(c) **(i)** Calculate the kinetic energy of the train when it is travelling at a speed of 40 m/s.

kinetic energy = ...[3]

(ii) As the train accelerates to 40 m/s, electrical energy supplied to the motor is converted into kinetic energy of the train. The efficiency of this process is 0.40 (40 %).

Calculate the electrical energy supplied to the motor.

energy = ...[2]

(iii) The electrical energy is generated in an oil-fired power station. Chemical energy in oil is converted into the electrical energy supplied to the motor.

Suggest two places where energy is lost as heat in this process.

1. .....

2. .....
[2]

10	**(a)**	**(i)**	start at origin **and** not horizontal	B1	
			gradient (gradually) decreasing (**ignore** sudden decrease) (not if part of curve above horizontal section)	B1	
			final horizontal section (≥ 1 cm) (not if v is shown as ≠ 40 m/s)	B1	
		(ii)	area **under** the graph **or** count squares **under** graph between t = 0 and horizontal section **or** when speed is changing **or** calculate equivalent distance to 1 cm^2 (after counting squares)	M1 A1	[5]
	(b)	**(i)**	friction/air resistance increases (as speed increases)	B1	
			resultant force decreases (**not** if driving force decreases)	B1	
		(ii)	(air resistance increases until) net force becomes zero **or** forces balance **or** air resistance and driving/forward force are in equilibrium/balanced/equal	B1	[3]
	(c)	**(i)**	(KE =) $\frac{1}{2}mv^2$	C1	
			$\frac{1}{2} \times 5.5 \times 10^5 \times 40^2$	C1	
			4.4×10^8 J	A1	
		(ii)	(total energy input =) useful energy output efficiency **or** efficiency = useful power output/total power input **or** $4.4 \times 10^8/0.40$	C1	
			1.1×10^9 J	A1	
		(iii)	two valid examples e.g. furnace/boiler/turbines/generator/coils/cooling water/cooling towers/heat exchanger/transformer/chimney/waste gases/ transmission cables/lines/wires (**ignore** power station/all parts of motor)	B2	[7]

[Total: 15]

QUESTION · 5054_s14_qp_11.pdf · source page 3 of 16

4 An object falls from rest through the air and the air resistance on it increases. The object reaches terminal velocity.

Which quantity **decreases** until its terminal velocity is reached?

A acceleration

B kinetic energy

C speed

D weight

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 4

4	**A**

QUESTION · 5054_s14_qp_12.pdf · source page 2 of 16

4 An object falls from rest through the air. Its velocity increases until it reaches terminal velocity.

Which quantity **increases** until its terminal velocity is reached?

A acceleration

B air resistance

C resultant force

D weight

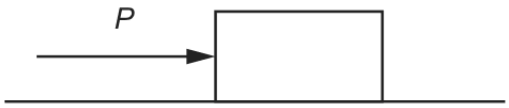
MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 4

4	**B**

QUESTION · 5054_s14_qp_12.pdf · source page 3 of 16

5 The diagram shows a block of stone on a rough horizontal surface.

Force *P* acts on the block as shown.



The block is at rest. A frictional force *F* acts on the block.

Which row shows the direction and size of *F*?

	direction of *F*	size of *F*
A	to the left	more than *P*
B	to the right	more than *P*
C	to the left	same as *P*
D	to the right	same as *P*

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 5

5 **C**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a lorry accelerating in a straight line along a horizontal road.

Fig. 1.1

(a) The driving force on the lorry in the forward direction is *D* and the total backward force on the lorry is *B*.

(i) State and explain whether *D* or *B* is the larger force.

.....

..... [1]

(ii) Suggest one possible cause of the backward force *B*.

..... [1]

(b) The weight of the lorry is 300 000 N.

The gravitational field strength *g* is 10 N/kg.

(i) Calculate the mass of the lorry.

mass = ... [1]

(ii) The resultant force on the lorry is 15 000 N. Calculate the acceleration of the lorry.

acceleration = ... [2]

(c) Later, the lorry turns a corner at constant speed.

Explain why the lorry accelerates even though the speed is constant.

.....

..... [1]

1 **(a)** **(i)** *D* **and either** lorry accelerates (forward) **or** resultant force is forward [B1]

(ii) air resistance **or** (air) drag **or** friction (between tyres and road) [B1]

(b) **(i)** 30 000 kg [B1]

(ii) (a=) F / m algebraic in any form or numerical [C1]
0.5(0) m / s^2 [A1]

(c) direction or velocity is changing **or** acceleration or force is sideways or towards centre (of circle) [B1]

[6]

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a motorcycle during a race.

Fig. 1.1

The motorcycle accelerates along a straight section of the track from a speed of 40 m/s to maximum speed.

Fig. 1.2 is the speed-time graph for the motorcycle along the straight section of the track.

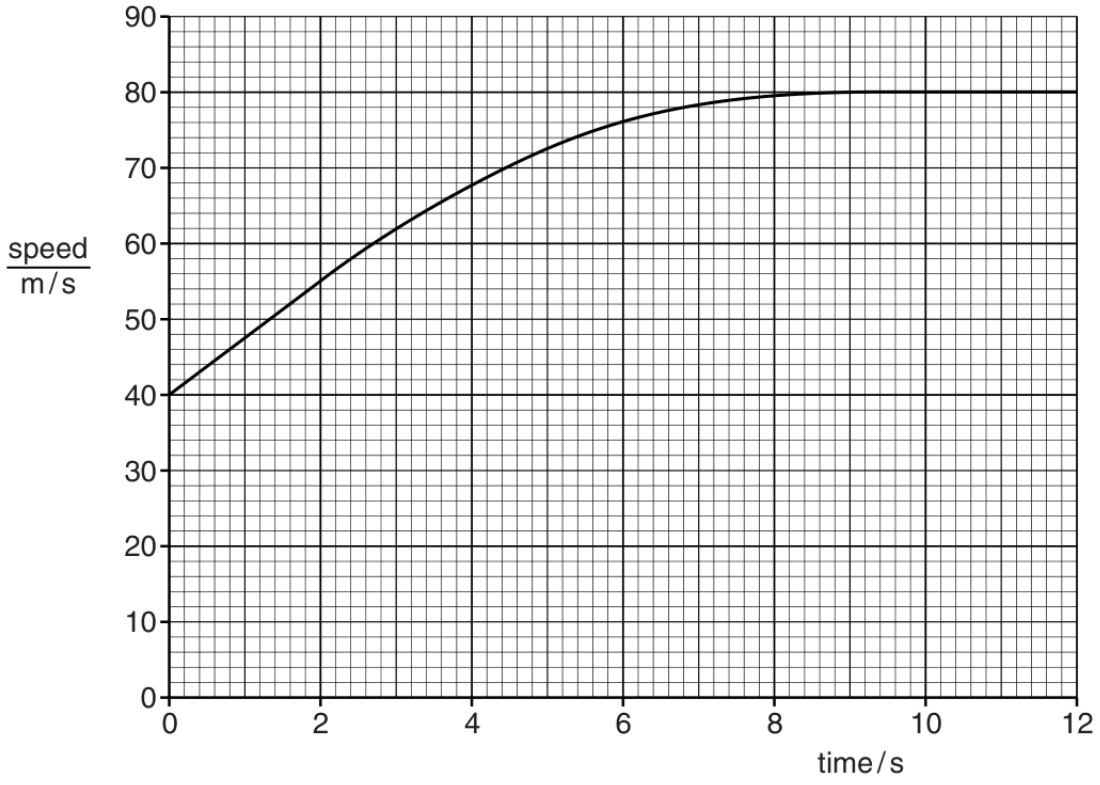


Fig. 1.2

The mass of the motorcycle is 180 kg.

(a) For the time 0 to 2.0 s, determine

(i) the acceleration of the motorcycle,

acceleration = ...[2]

(ii) the resultant force acting on the motorcycle.

force = ...[2]

(b) The driving force acting on the motorcycle remains constant throughout the 12 s spent on the straight section of track.

(i) Using Fig. 1.2, describe how the acceleration of the motorcycle changes during this time.

.....

.....[1]

(ii) Explain, in terms of the forces acting, why the acceleration changes in this way.

.....

.....

.....

.....[3]

Section A

1	(a)	(i)	(a =) $(v-u)/t$ **or** $\Delta v/t$ **or** (55–40)/2 **or** equivalent values from graph	C1	
			7.5 m/s^2	A1	
		(ii)	(F =) ma **or** 180 × 7.5	C1	
			1300 / 1350 / 1400 N	A1	
	(b)	(i)	(acceleration) decreases (to zero)	B1	
		(ii)	air resistance / friction / drag mentioned	B1	
			air resistance / friction / drag increases (with speed) **or** resultant force decreases (with speed)	B1	
			(finally) (air) resistance = driving force **or** resultant is zero	B1	**[8]**

QUESTION · 5054_w14_qp_12.pdf · source page 2 of 16

1 When a heavy coin falls a short distance towards the ground it does **not** reach terminal velocity.

Why is this?

A The coin has not hit the ground.

B The weight of the coin equals the air resistance.

C The weight of the coin increases as air resistance increases.

D The weight of the coin is always more than air resistance.

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 1

1	**D**

QUESTION · 5054_w14_qp_12.pdf · source page 2 of 16

2 A car travels along a road at 50 km / h.

The driver applies the same braking force at the same place on a day when the surface is dry and then on a day when the road is wet.

On the **wet** surface, how many of these distances are greater than on the dry surface?

braking distance

stopping distance

thinking distance

A 0 **B** 1 **C** 2 **D** 3

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 2

2 **C**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 **(a)** Explain what is meant by *uniform acceleration*.

.....

.....

.....[2]

(b) Acceleration is a vector quantity.

State how a vector quantity differs from a scalar quantity.

.....

.....[1]

(c) A rock from space is travelling in a straight line at high speed when it enters the Earth's atmosphere. Fig. 9.1 is the speed-time graph for the rock from time $t = 0$ to time $t = 50$ s.

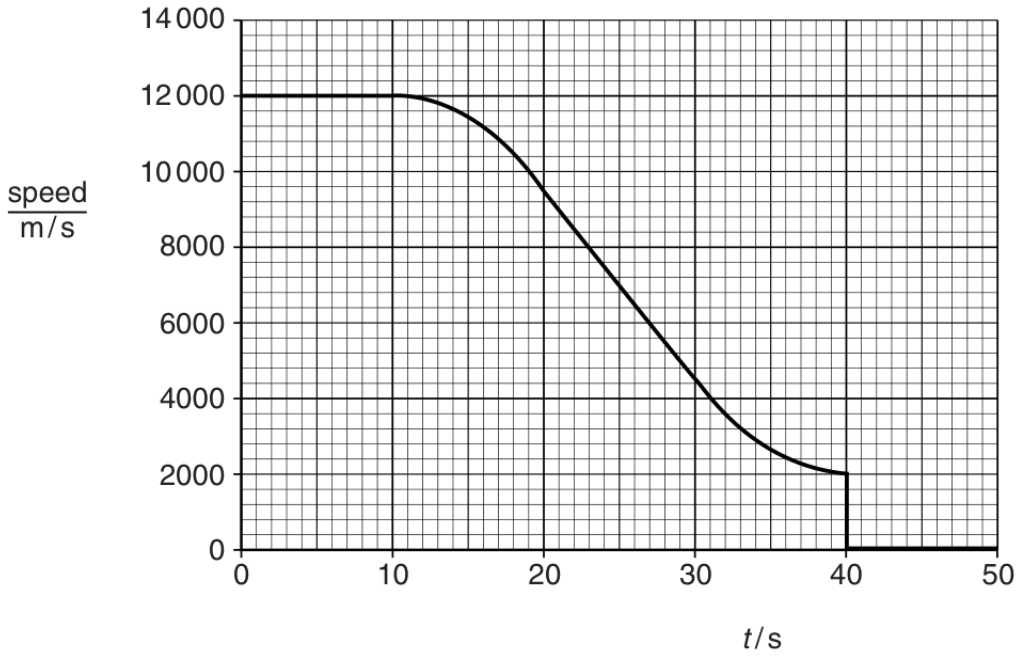


Fig. 9.1

(i) On Fig. 9.1, mark

- a letter X, where the rock is moving with a constant speed,
- a letter Y, where the rock has a uniform deceleration,
- a letter Z, where the rock has a non-uniform deceleration.

[3]

(ii) At time $t = 25\,s$, the mass of the rock is 8.4 kg. For the time $t = 25\,s$, determine

1. the size of the acceleration of the rock,

acceleration = ...[3]

2. the size of the resultant force on the rock.

force = ...[2]

(iii) Fig. 9.2 shows the rock at $t = 25\,s$.

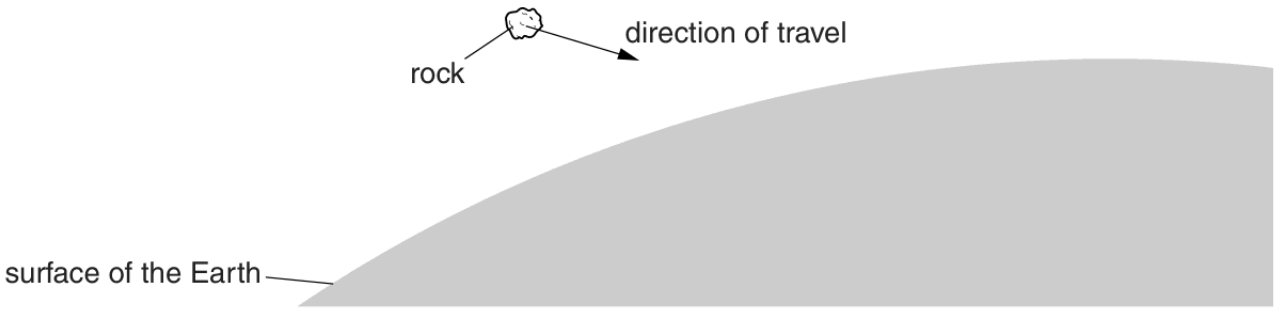


Fig. 9.2 (not to scale)

1. On Fig. 9.2, draw and label two arrows to show the directions of the gravitational force *F* and the air resistance *R* acting on the rock. [2]

2. Suggest why the size of the air resistance changes as the rock travels through the Earth's atmosphere.

.....

.....[1]

(iv) Suggest what happens to the rock at $t = 40\,s$.

.....

.....[1]

Section B

9	**(a)**	changing speed / velocity	C1	
		change in speed / velocity / time constant **or** (v–u) / t constant **or** constant / equal rate of change of speed / velocity	A1	**[2]**
	(b)	(a vector quantity has) direction	B1	**[1]**
	(c)	**(i)** **1.** X between $t \geq 0$ and $t < 10$ s	B1	
		2. Y between $t > 20$ s and $t < 30$ s	B1	
		3. Z between $t > 10$ s and $t < 20$ s **or** between $t > 30$ s and $t < 40$ s	B1	
		(ii) **1.** **two** speed values from graph between 15 and 35 s (±1 mm)	C1	
		two corresponding time values from graph between 15 and 35 s (±1 mm) **or** (a =)$\Delta v / t$	C1	
		500 m / s^2	A1	
		2. (F =) ma **or** 8.4 × 500	C1	
		4200 N	A1	
		(iii) **1.** arrow labelled F perpendicular to surface of Earth	B1	
		arrow labelled R opposite to direction of travel (by eye) from rock	B1	
		2. speed changes **or** density / pressure of air changes **or** cross-sectional area (of rock) changes	B1	
		(iv) it hits the ground / surface of the earth **or** stops **or** speed is zero	B1	**[12]**
				[Total: 15]

QUESTION · 5054_s15_qp_11.pdf · source page 4 of 16

7 A student drops, from rest, a table-tennis ball in air.

What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 7

7	**C**

QUESTION · 5054_s15_qp_12.pdf · source page 3 of 16

6 A student drops, from rest, a table-tennis ball in air.

What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 6

6 **C**

QUESTION · 5054_w15_qp_11.pdf · source page 3 of 16

5 In which example does friction act in the direction of forward motion of the object on which it acts?

A on a suitcase accelerating on a conveyor belt

B on sandpaper rubbing a wooden block

C on the skis of a skier accelerating downhill on smooth snow

D on the tyres of a braking car

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 5

5	**A**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Force is a vector quantity and mass is a scalar quantity.

(a) (i) State how a vector quantity differs from a scalar quantity.

.....

..... [1]

(ii) State one other vector quantity and one other scalar quantity.

vector quantity .....

scalar quantity .....
[2]

(b) An aeroplane flies through the air in a straight line, with its engines at full power. The weight of the aeroplane is 160 kN. There is also a force of 320 kN acting upwards on the aeroplane at 30° to the horizontal.

In the space below, use a graphical method to determine the size and the direction of the resultant of these two forces. Use a scale of 1.0 cm : 40 kN.

size of resultant force = ...

direction of resultant force = ...
[4]

(c) An aeroplane is travelling in a straight line at a constant height. The aeroplane accelerates.

(i) State the energy change that is taking place as the aeroplane accelerates.

.....

.....

..... [2]

(ii) As the aeroplane accelerates, the pilot keeps the engines at constant power. The aeroplane eventually reaches a constant speed.

1. Explain why the aeroplane stops accelerating and reaches a constant speed.

.....

.....

.....

.....

.....

..... [3]

2. When travelling at a constant speed, the pilot directs the aeroplane into a tight turn that follows a circular path.

Explain why the aeroplane is accelerating even though its speed is not changing.

.....

.....

.....

.....

..... [3]

Section B

9	(a)	(i)	(vector) has direction **or** scalar does not have direction **or** (vectors) may cancel **or** scalars cannot cancel	B1	
		(ii)	one vector quantity e.g.: displacement; velocity, acceleration	B1	
			one scalar quantity e.g.: distance, length, speed, time, current, resistance	B1	[3]
	(b)		downward weight arrow of length 3.9–4.1 cm	B1	
			correct triangle/parallelogram drawn **and** correct diagonal clear	B1	
			270–285 kN	B1	
			horizontal (±3°)	B1	
	(c)	(i)	from chemical / fuel energy	B1	
			to kinetic (and thermal energy) (**not** with any intermediate)	B1	
		(ii)	**air** resistance / friction / drag	B1	
			air resistance / friction / drag increases **or** resultant force decreases **or** acceleration decreases	B1	
			resultant force is zero **or** (air) resistance / friction equals thrust	B1	
			direction of motion is changing	B1	
			velocity is vector **or** has a direction	B1	
			(acceleration depends on) changing velocity **or** **resultant** force towards centre (of circle) **or centripetal** force	B1	[12]
					[15]

QUESTION · 5054_w15_qp_12.pdf · source page 3 of 16

6 At time t_1, a stone is dropped from a stationary balloon. The stone reaches terminal velocity at time t_2.

Which row gives the acceleration of the stone at time t_1 and at time t_2?

	acceleration at t_1 / m/s^2	acceleration at t_2 / m/s^2
A	0	0
B	10	0
C	0	10
D	10	10

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 6

6 **B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Aeroplanes fly at high altitudes where the temperature is well below 0 °C. Ice that forms on an aeroplane can fall to earth and strike the ground.

Fig. 9.1 shows a block of ice falling from an aeroplane as it approaches an airport.

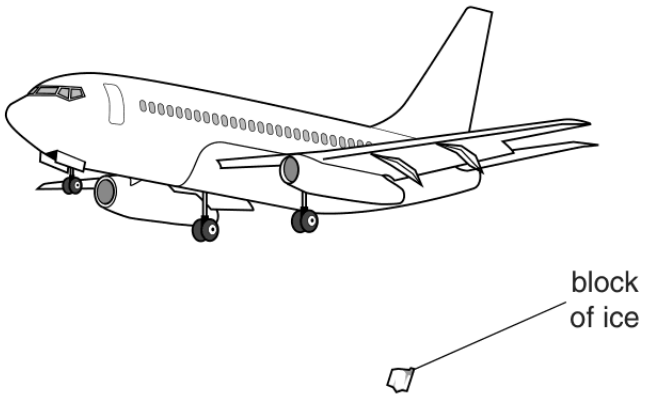


Fig. 9.1 (not to scale)

The mass of the falling block of ice is 1.2 kg and the gravitational field strength g is 10 N/kg.

(a) Calculate the weight of the block of ice.

weight = ... [1]

(b) Fig. 9.2 is the speed-time graph for the block of ice as it falls to the ground.

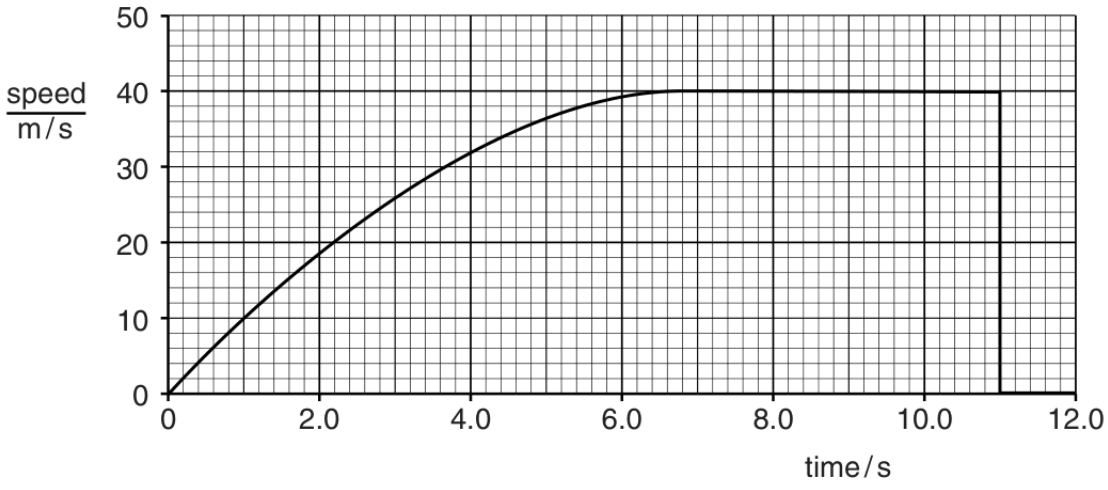


Fig. 9.2

At first, the acceleration of the block of ice is equal to the acceleration of free-fall. The acceleration of the block then decreases to zero as the block reaches terminal velocity.

(i) As the block of ice falls, the force F of air resistance acting on the block changes.

1. State the value of F at time = 0.

$F =$ ..... [1]

2. State the value of F at time = 10.0 s.

$F =$ ..... [1]

3. Explain why F changes.

.....

.....

..... [1]

(ii) State the energy change that takes place when the block is falling at terminal velocity.

.....

..... [1]

(iii) Using information from Fig. 9.2, determine the maximum kinetic energy of the block when it is falling to the ground with terminal velocity.

maximum kinetic energy = ... [3]

(c) The block strikes the ground and it stops moving. This impact causes some of the ice to melt.

(i) The specific latent heat of fusion of ice is 330 J / g.

Calculate the maximum possible mass of ice that melts as a result of the impact.

mass = ... [2]

(ii) In practice, the mass of ice that melts on impact is less than the value calculated in **(c)(i)**.

Suggest two reasons for this.

1. .....

.....

2. .....

..... [2]

(d) As the solid ice melts, it changes into liquid water.

Describe, in terms of molecules, how ice differs from liquid water.

.....

.....

.....

.....

.....

..... [3]

Section B

9	**(a)**	12 N	B1	[1]
	(b) **(i)**	0 **or** zero	B1	
		12 N **or** it is equal to the weight	B1	
		(*F* increases) as the speed increases	B1	
	(ii)	(gravitational) potential to thermal energy **or** to k.e. **of air**	B1	
	(iii)	(KE =)$\frac{1}{2}mv^2$	C1	
		$\frac{1}{2} \times 1.2 \times 40^2$	C1	
		960 J	A1	[7]
	(c) **(i)**	(m =)E / l_f **or** Q / l_f **or** 960 / 330	C1	
		2.9 / 2.91 g **or** 2.9 / 2.91 $\times 10^{-3}$ kg	A1	
	(ii)	any **two** from: ice is below 0 °C thermal energy transferred / lost (to ground / air) work done compressing / compacting the ground	B2	[4]
	(d)	any **three** from: molecules fixed in position **or** water molecules move around molecules vibrate **or** water molecules do not vibrate molecules in regular lattice **or** water molecules placed randomly (interatomic) forces between ice molecules larger ice molecules further apart	B3	[3]
				[15]

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

Answer only one of the two alternative parts in **Question 9**.

1 A sky-diver jumps from a stationary balloon. His initial downwards acceleration is $10\,m/s^2$.

Fig. 1.1 shows the directions of the air resistance and the weight of the sky-diver.

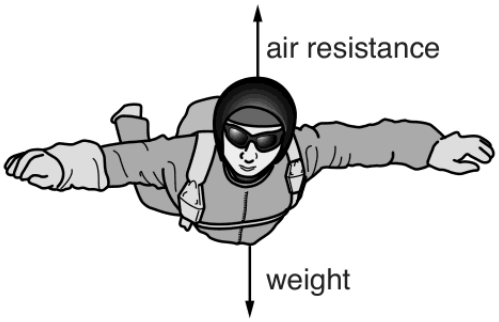


Fig. 1.1

The mass of the sky-diver is 60 kg and his weight is 600 N.

(a) Explain, using ideas about the forces, why his initial downwards acceleration is $10\,m/s^2$.

.....

.....

.....

.....

.....[2]

(b) When the parachute opens, the sky-diver experiences an upward acceleration for a short time.

Explain why.

.....

.....

.....[1]

(c) The total force of air resistance on the sky-diver and open parachute changes with their speed, as shown in Fig. 1.2.

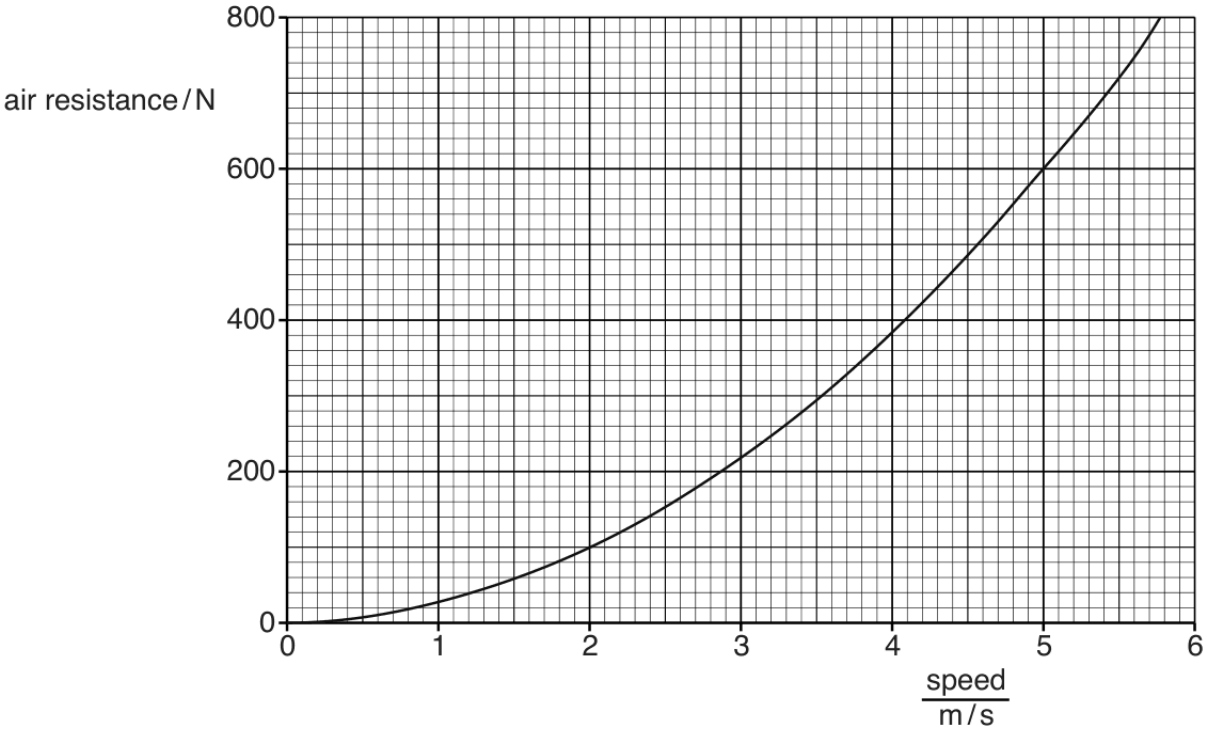


Fig. 1.2

After the parachute opens, the sky-diver slows down to a terminal velocity.

The weight of the sky-diver is 600 N.

(i) Using Fig. 1.2, determine the terminal velocity of the sky-diver.

terminal velocity = ...[1]

(ii) Calculate the resultant force on the sky-diver when his speed is 5.5 m / s.

resultant force = ...[1]

1 **(a)** air resistance is zero B1
or no air resistance acts (at first)
or weight much larger than air resistance

(a=) F / m **or** weight / mass **or** 600 / 60 B1
or weight is 10 times mass

(b) air resistance / upwards force is larger than weight / 600 N / downwards force B1

(c) **(i)** 5(.0) m / s B1

(ii) 120 N B1

12 The overall stopping distance of a cyclist is made up of two parts, as shown in Fig. 12.1:

- the distance the cyclist travels during the reaction time of the cyclist (the thinking distance)
- the distance the cyclist travels after the brakes are applied (the braking distance)

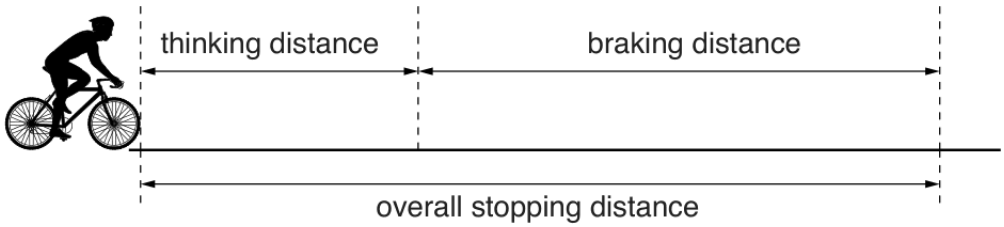


Fig. 12.1 (not to scale)

(a) State the energy change that occurs during braking.

.....

.....[2]

(b) A ball rolls in front of a cyclist at time $t = 0$ and the cyclist brakes and comes to rest.

Fig. 12.2 shows the speed-time graph for the cyclist.

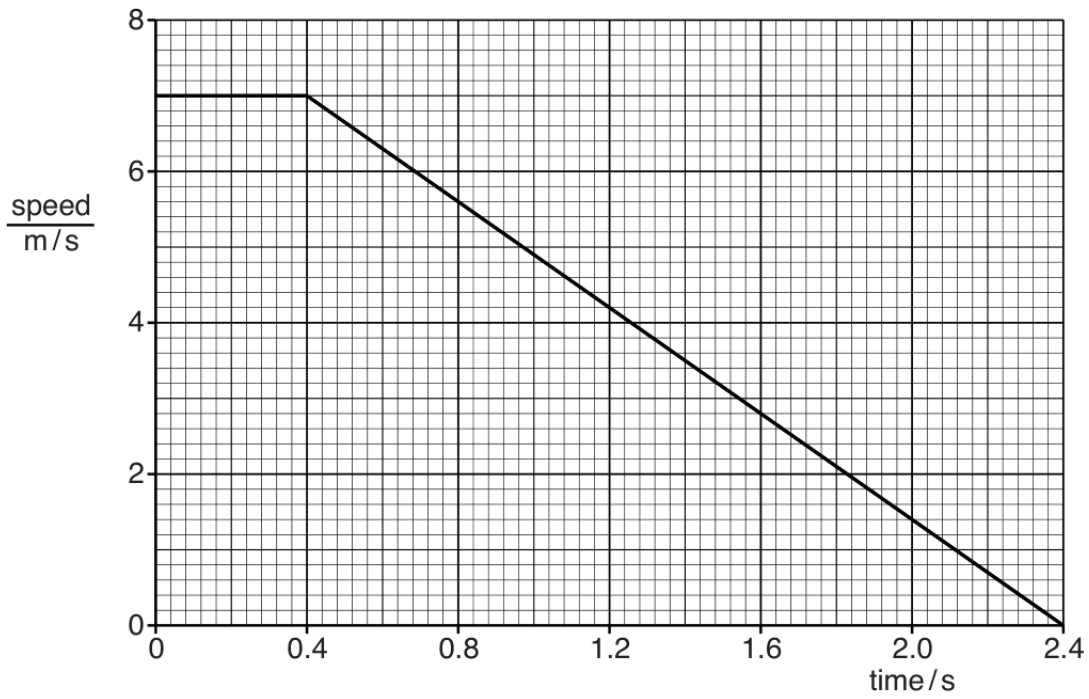


Fig. 12.2

(i) Using Fig. 12.2, determine the reaction time of the cyclist.

reaction time = ...[1]

(ii) Using Fig. 12.2, calculate the thinking distance.

thinking distance = ...[2]

(iii) State how the braking distance is found using Fig. 12.2.

.....

.....[1]

(iv) On another occasion, the same cyclist travels at an initial speed of 5.0 m/s. A ball rolls in front of the cycle at time $t = 0$.

The cyclist has the same reaction time and the deceleration of the cycle is the same as in Fig. 12.2.

On Fig. 12.2 draw the new speed-time graph for the cyclist. [2]

(v) The braking distance is longer when the cyclist stops on a wet road.

Explain why.

.....

.....

.....

.....

.....

.....[3]

(c) Fig. 12.3 shows the hydraulic braking system of the cycle.

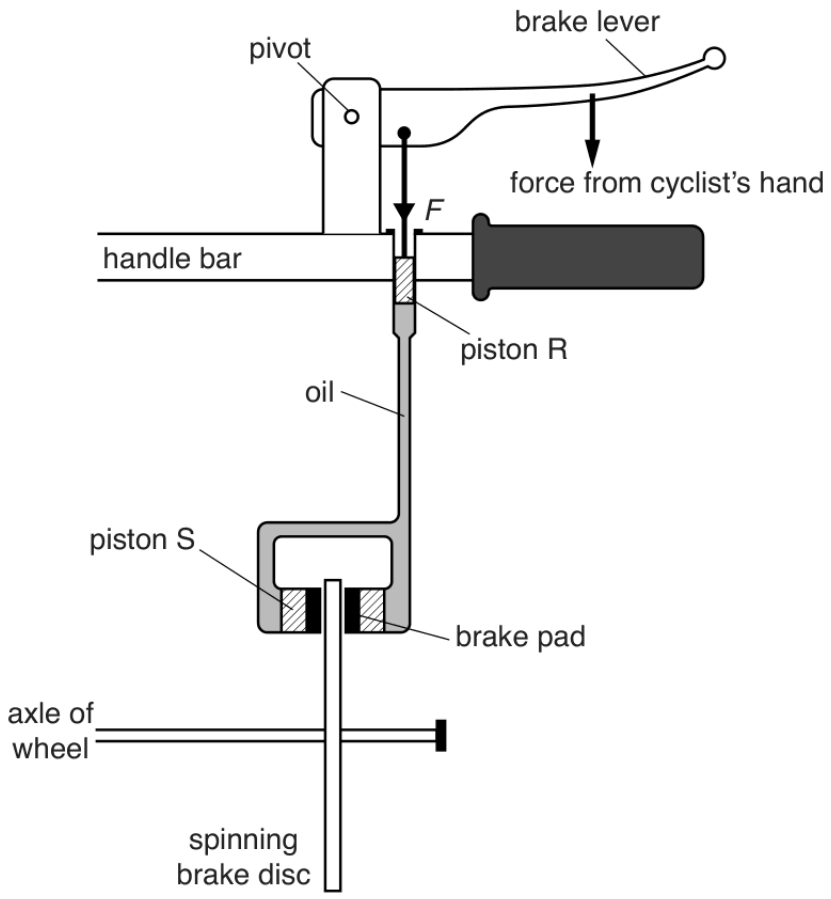


Fig. 12.3 (not to scale)

(i) The cyclist applies a force on the brake lever. This increases the pressure in the oil by 1.2×10^{6} Pa.

The cross-sectional area of piston R is 5.0×10^{-5} m^2.

Calculate the force *F* applied by the brake lever to piston R.

F = ... [2]

(ii) The force applied to each of the brake pads is larger than *F*.

Explain why.

.....

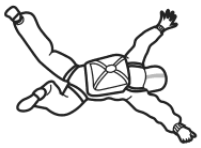
.....

.....[2]

12	**(a)**	kinetic energy at start	B1
		thermal energy / heat energy / internal energy at end	B1
	(b) **(i)**	0.4(0) s	B1
	(ii)	(d=) s × t numerical or algebraic	C1
		2.8 m	A1
	(iii)	area under graph (between 0.4 and 2.4 s)	B1
		or time (difference) × <u>average</u> speed	
		or ½ × time (difference) × initial speed	
	(iv)	horizontal line from (0,5) to (0.4,5)	B1
		line showing braking with same gradient as original line	B1
	(v)	less friction	B1
		less deceleration **or** graph less steep	B1
		or less <u>force</u> <u>backwards</u> / less <u>force</u> <u>opposing motion</u>	
		or same KE lost / work done by friction	
		longer time to stop	B1
		or larger area under (speed-time) graph	
		or work = force x distance applied correctly	
	(c) **(i)**	(F=) P × A numerical or algebraic	C1
		60 N	A1
	(ii)	same pressure	B1
		larger area (of S / brake pads)	B1

QUESTION · 5054_w16_qp_11.pdf · source page 2 of 20

2 A skydiver falls from rest through the air and reaches terminal velocity.



What is the acceleration of the skydiver during his fall?

A constant at $0\,m/s^2$

B constant at $10\,m/s^2$

C starting at $0\,m/s^2$ and increasing to $10\,m/s^2$

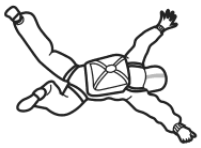
D starting at $10\,m/s^2$ and decreasing to $0\,m/s^2$

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 2

2	**D**

QUESTION · 5054_w16_qp_12.pdf · source page 2 of 16

2 A skydiver falls from rest through the air and reaches terminal velocity.



What is the acceleration of the skydiver during his fall?

A constant at $0\,m/s^2$

B constant at $10\,m/s^2$

C starting at $0\,m/s^2$ and increasing to $10\,m/s^2$

D starting at $10\,m/s^2$ and decreasing to $0\,m/s^2$

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 2

2	**D**

QUESTION · 5054_s17_qp_11.pdf · source page 4 of 16

6 A skydiver jumps from an aeroplane. After a few seconds, he reaches a terminal velocity without opening his parachute.

Why does he reach terminal velocity?

A Air resistance becomes greater than his weight and slows him down.

B Air resistance decreases and he speeds up.

C Air resistance increases and balances his weight so that his acceleration is zero.

D His weight decreases and balances the air resistance.

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 6

6	C	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 4 of 20

5 A skydiver falls at terminal velocity. He then opens his parachute.

Which row gives the direction of the resultant force on the skydiver and the direction of the acceleration of the skydiver immediately after the parachute opens?

	resultant force	acceleration
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 5

5	D	**1**

QUESTION · 5054_w17_qp_12.pdf · source page 3 of 16

4 A free-fall skydiver jumps from a plane. As he falls there is a force acting upwards and a force acting downwards on his body. These produce a resultant force.

Before he reaches terminal velocity, how do the sizes of the forces change?

	downward force	upward force	resultant force
A	decreases	decreases	stays the same
B	increases	stays the same	decreases
C	stays the same	increases	decreases
D	stays the same	increases	increases

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 4

4	C	**1**

QUESTION · 5054_s18_qp_11.pdf · source page 3 of 20

5 The minimum braking distance for a car is tested on a dry road.

The test is then repeated on a wet road.

What happens to the braking distance and to the frictional force between the tyres and the road?

	braking distance	frictional force
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 5

5	C	**1**

2 Fig. 2.1 shows a man pushing a heavy box with a force *P*. A frictional force *F* acts in a horizontal direction.

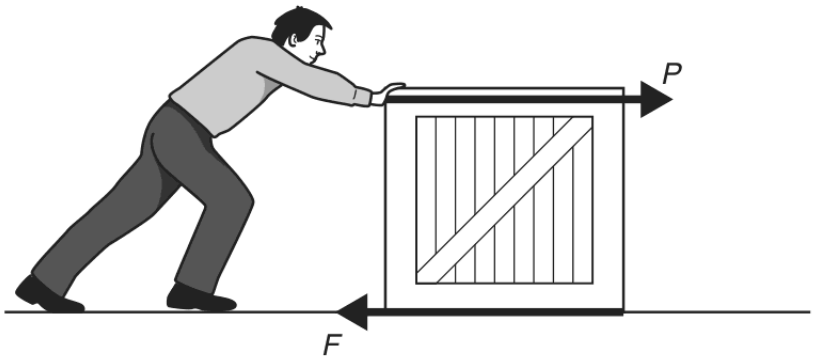


Fig. 2.1

(a) The forces on the box are *balanced* and the box is stationary.

(i) State what is meant by *balanced forces*.

.....

.....[1]

(ii) The frictional force in Fig. 2.1 does not produce any heating effect.

State what must happen for the frictional force to produce heating.

.....

.....[1]

(iii) Apart from being stationary, describe one other possible state of motion of the box when the forces are balanced.

.....

.....[1]

(b) When $P = 100\,\text{N}$ and $F = 85\,\text{N}$, the box accelerates. The mass of the box is 25 kg.

Calculate the acceleration of the box.

acceleration = ...[3]

Question	Answer	Marks
2(a)(i)	equal and opposite forces **or** no resultant force	**B1**
2(a)(ii)	box moves / slides	**B1**
2(a)(iii)	constant velocity **or** constant speed and direction	**B1**
2(b)	(resultant force) 15 (N) seen	**C1**
	(a=) F / m in any form numerical or algebraic	**C1**
	0.60 m / s^2	**A1**

QUESTION · 5054_s18_qp_12.pdf · source page 3 of 16

6 Which forces act on a skydiver who is falling at terminal velocity?

A air resistance and weight

B air resistance only

C weight only

D no forces act

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 6

6	A	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A student drops a small metal object into a cylinder of oil. The object falls alongside a vertical ruler and a camera records its position at 1.0 s intervals, as shown in Fig. 1.1.

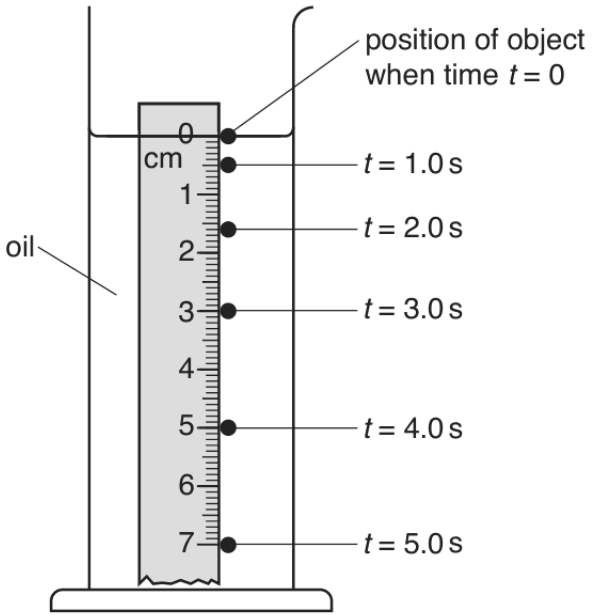


Fig. 1.1

(a) On the grid in Fig. 1.2, plot a distance-time graph for the object.

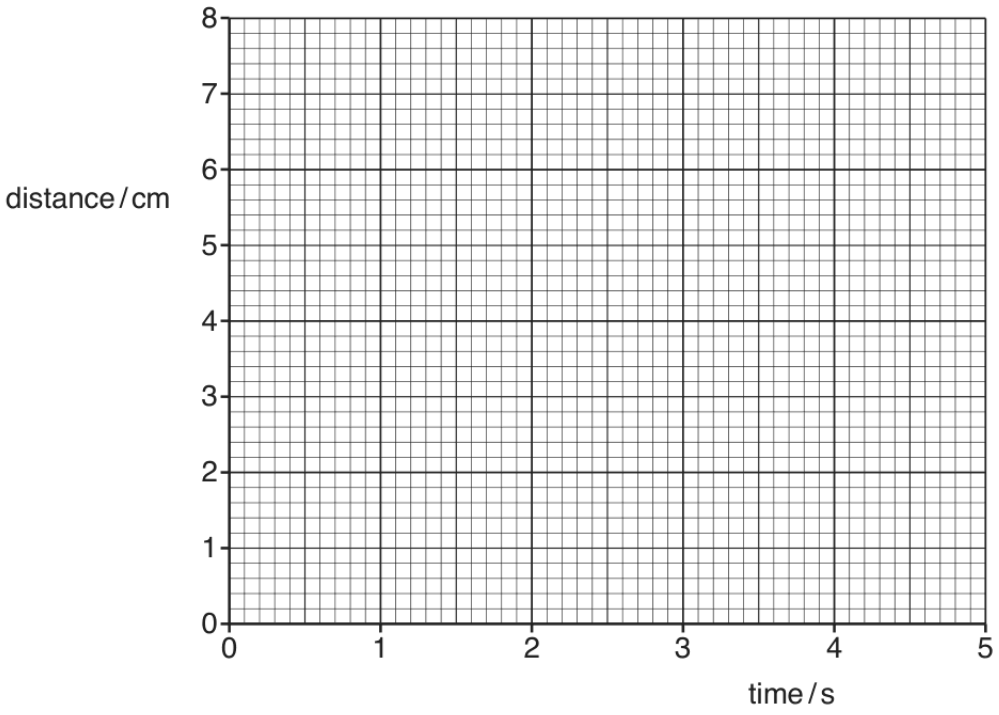


Fig. 1.2

[2]

(b) The object has a constant speed for some of the time.

(i) State and explain how a distance-time graph shows that speed is constant.

.....

.....

.....

.....[2]

(ii) Explain, in terms of the forces acting on the object, how it is able to fall at a constant speed.

.....

.....

.....

.....[2]

Question	Answer	Marks
1(a)	points plotted correctly at (1,0.5) (2,1.6) (3,3) (4,5) and (5,7)	**B1**
	smooth curve from origin	**B1**
1(b)(i)	straight line **or** gradient / slope constant	**B1**
	travels equal distances in same time **or** gradient equals speed	**B1**
1(b)(ii)	weight **and** force (upwards) from oil / liquid	**B1**
	forces balanced / cancel / are equal (and opposite) **or** no resultant (force)	**B1**

QUESTION · 5054_w18_qp_11.pdf · source page 2 of 16

3 A skydiver is falling at terminal velocity.

Which row describes the acceleration of the skydiver and the velocity of the skydiver?

	acceleration of the skydiver	velocity of the skydiver
A	downwards	constant
B	downwards	zero
C	zero	constant
D	zero	zero

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 3

3	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 During a game of cricket, a player hits a ball with a bat. The ball then travels vertically upwards, as shown in Fig. 8.1.

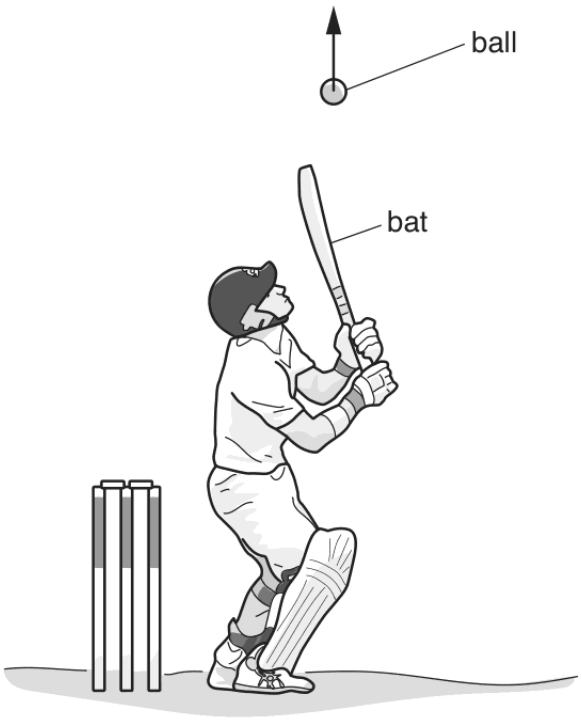


Fig. 8.1

The speed of the ball as it leaves the bat is 8.7 m/s.

(a) The mass of the ball is 0.16 kg.

(i) **1.** State what is meant by *mass*.

.....

..... [1]

2. State the name of a measuring instrument that is used to determine mass.

.....

..... [1]

(ii) Calculate the kinetic energy of the ball as it leaves the bat.

kinetic energy = ... [3]

(b) Fig. 8.2 is the velocity-time graph for the ball.

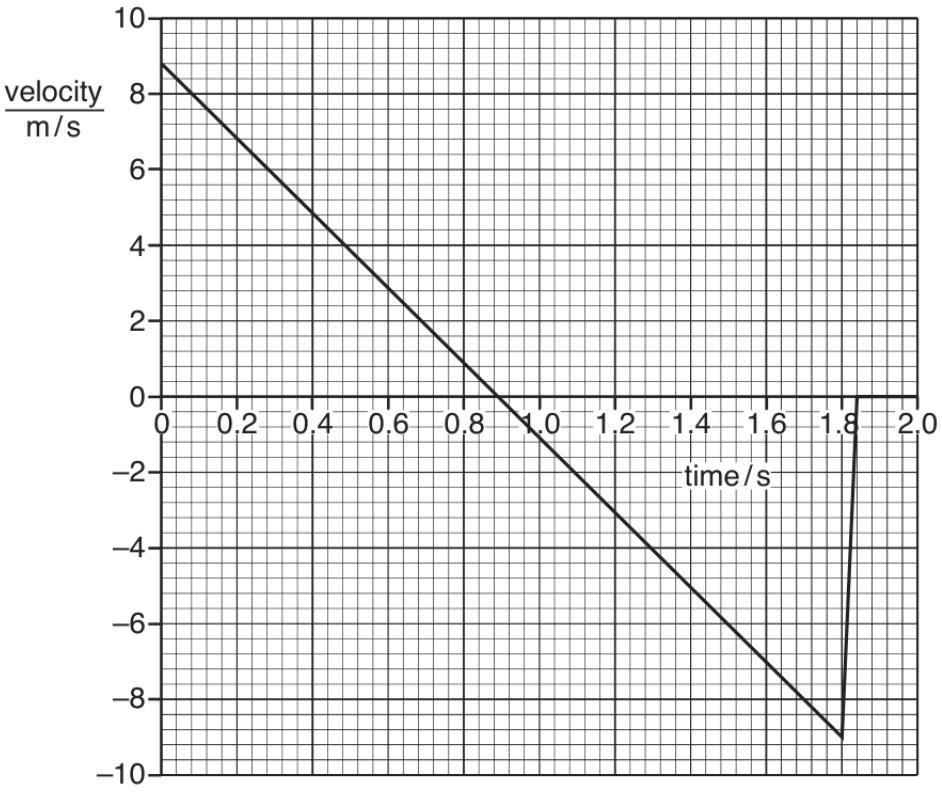


Fig. 8.2

(i) Fig. 8.2 shows that immediately after being hit, the ball has a negative acceleration.

1. State the name given to a negative acceleration.

..... [1]

2. State how the graph shows that the ball has a negative acceleration.

.....

..... [1]

(ii) Using Fig. 8.2, determine

1. the time at which the ball stops moving upwards,

time = ... [1]

2. the distance travelled before the ball stops moving upwards.

distance = ... [2]

(iii) Suggest why the graph, in Fig. 8.2, is almost vertical when time = 1.84 s.

.....

..... [1]

(iv) Fig. 8.2 shows that the ball remains stationary after a time of 1.84 s.

Describe the energy change that is occurring at a time of 1.84 s.

.....

.....

..... [2]

(c) For this ball moving in the way shown, the effect of air resistance is negligible.

When a ball with a much smaller mass is hit vertically upwards at the same speed, air resistance produces a noticeable effect on its motion.

Suggest two ways in which the velocity-time graph for the ball with a smaller mass differs from Fig. 8.2.

1. .....

.....

2. .....

.....
[2]

MARK SCHEME · 5054_w18_ms_22.pdf · source page 7 of 10 · Question 8

Question	Answer	Marks
8(a)(i)	1 quantity of matter (in a body)	**B1**
	2 balance **or** scales	**B1**
8(a)(ii)	(*k.e.* =) $\frac{1}{2}mv^2$	**C1**
	$\frac{1}{2} \times 0.16 \times 8.7^2$	**C1**
	6.1 J	**A1**
8(b)(i)	1 deceleration **or** retardation	**B1**
	2 negative gradient **or** line slopes downwards (left to right)	**B1**
8(b)(ii)	1 0.88 s ⩽ time ⩽ 0.90 s	**B1**
	2 area **or** counting squares **or** $\frac{1}{2}bh$ in some form	**C1**
	3.7 s ⩽ distance ⩽ 4.1 m	**A1**
8(b)(iii)	ball hits ground **or** short time for deceleration **or** large force **or** ground is hard	**B1**
8(b)(iv)	internal / thermal energy (of ball and ground) has increased	**B1**
	(internal energy) from kinetic energy	**B1**

MARK SCHEME · 5054_w18_ms_22.pdf · source page 8 of 10 · Question 8

Question	Answer	Marks
8(c)	any **two** from: smaller time to drop to zero velocity / hit ground line not straight **or** velocity does not change uniformly **or** gradient not constant smaller area under (first part of) graph **or** less distance travelled slower final velocity initial downward gradient steeper	**B2**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a parachutist falling vertically towards the ground.

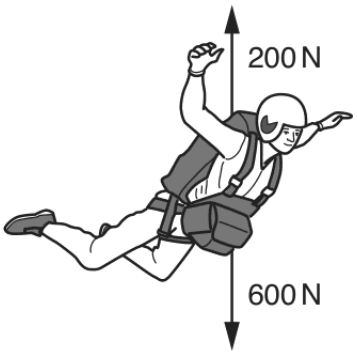


Fig. 9.1

The mass of the parachutist is 60 kg and his weight is 600 N.

(a) State what is meant by:

(i) *mass*

.....

..... [1]

(ii) *weight*.

.....

..... [1]

(b) Calculate the acceleration of the parachutist when the air resistance is 200 N.

acceleration = ... [3]

(c) The parachutist falls from rest at time $t = 0$.

He accelerates non-uniformly until he reaches a terminal velocity of 44 m/s at $t = 30$ s.

At $t = 40$ s the parachute opens. He decelerates for 4.0 s, reaching a smaller terminal velocity of 5.0 m/s.

He then falls at this speed until he lands on the ground at $t = 200$ s.

(i) On Fig. 9.2, draw the speed–time graph for the parachutist.

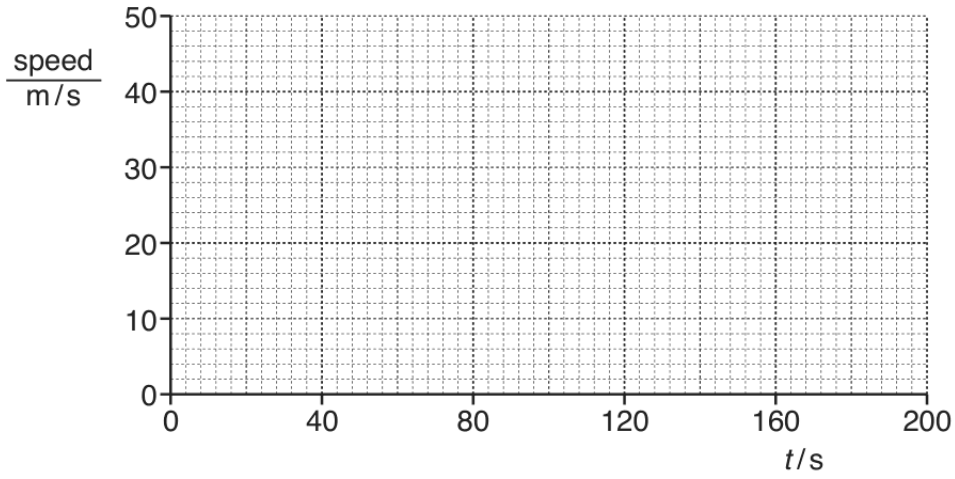


Fig. 9.2

[4]

(ii) Calculate the average deceleration of the parachutist between $t = 40\,s$ and $t = 44\,s$.

deceleration = ... [2]

(d) Explain, in terms of the forces involved, why:

(i) the parachutist reaches a terminal velocity at $t = 30\,s$

.....

..... [1]

(ii) the parachutist decelerates when the parachute opens

.....

.....

..... [2]

(iii) the second terminal velocity, when the parachute is open, is smaller than the first.

.....

..... [1]

[Total: 15]

Section B

Question	Answer	Marks
9(a)(i)	amount / measure / quantity of matter / substance (in a body) **or** (property of a body) that resists (change in) motion	**B1**
9(a)(ii)	force / pull of gravity **or** in a gravitational field	**B1**
9(b)	(a =) F / m algebraic or numerical **or** 400 (N) seen	**C1**
	400 / 60	**C1**
	6.7 m/s^2	**A1**
9(c)(i)	initial curve of correct shape from origin at 0 to 30 s	**B1**
	horizontal line at 44 m/s from 30 s to 40 s	**B1**
	speed drops from 44 to 5 m/s in a time of 4 s	**B1**
	constant speed of 5 m/s until 200 s (and then to zero)	**B1**
9(c)(ii)	(a =) (v – u) / t algebraic or numerical	**C1**
	9.7 **or** 9.8 m/s^2	**A1**
9(d)(i)	upwards force / air resistance = weight / downward force	**B1**
9(d)(ii)	increase in air resistance / upwards force (from parachute)	**B1**
	upwards / force / air resistance larger than weight / downwards force **or** resultant force upwards	**B1**
9(d)(iii)	air resistance decreases (as speed reduces) until it equals weight **or** (with parachute open) air resistance = weight at lower speed	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 3 of 16

6 A parachutist of mass 60 kg falls at a steady speed of 10 m/s. The gravitational field strength g is 10 N/kg.

What is the upward force acting on the parachutist?

A 0 **B** 6.0 N **C** 60 N **D** 600 N

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 6

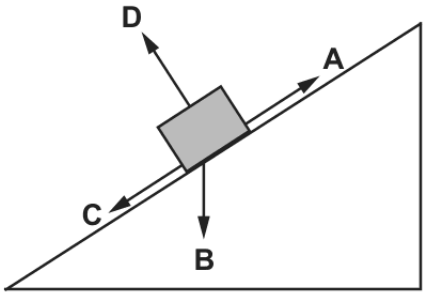
6	D	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 4 of 16

7 A force moves a box up a rough slope.

The diagram shows four forces acting on the box.

Which is the force due to friction on the box?



MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a map of the route taken by a car as it travels from town A to town B.

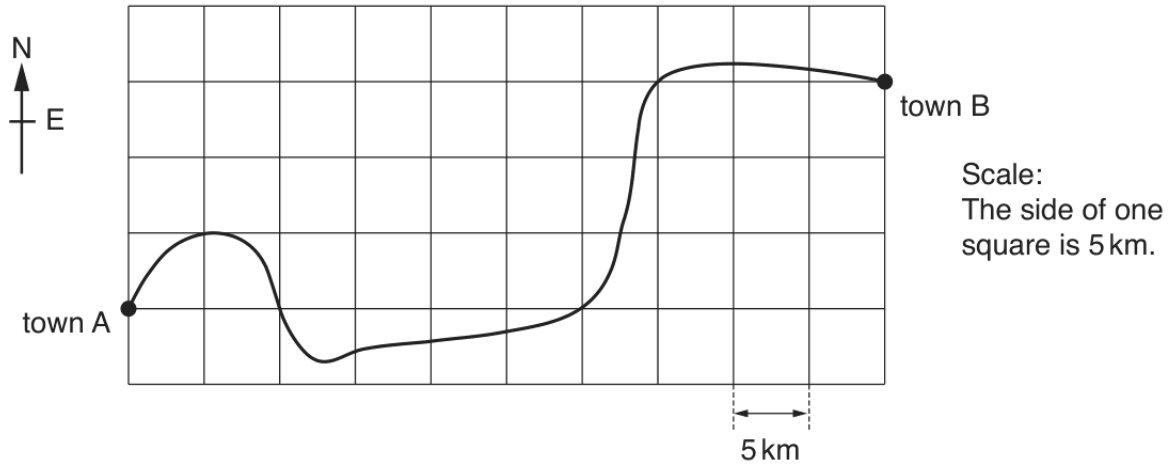


Fig. 9.1

The journey involves a distance and a displacement.

(a) (i) Using the scale on Fig. 9.1, determine the displacement in travelling from A to B.

size of displacement = ...

direction of displacement = ...
[3]

(ii) State how *distance* differs from *displacement*.

.....

.....

..... [1]

(iii) In some parts of the journey, the car is travelling at a constant speed but is also accelerating.

Explain how this is possible.

.....

.....

..... [2]

(b) At town B the car makes an emergency stop.

The total stopping distance of the car is the thinking distance added to the braking distance.

(i) Describe what is meant by the *thinking distance*.

.....

..... [1]

(ii) The brakes on the car work badly. This increases the braking distance.

State **two** other factors that increase the braking distance.

1. .....

.....

2. .....

.....
[2]

(c) The mass of the car and driver is 1200 kg. Just before the brakes are applied, the speed of the car is 30 m/s.

(i) Calculate the kinetic energy of the car and driver just before the brakes are applied.

kinetic energy = ... [3]

(ii) The braking force that stops the car is 1800 N.

Calculate the deceleration of the car.

deceleration = ... [2]

(iii) Explain why the temperature of the brakes increases during braking.

.....

..... [1]

[Total: 15]

Question	Answer	Marks
9(a)(i)	10.2–10.4 (cm) **or** 50 (km) **or** 15 (km) seen	**C1**
	51–53 km	**A1**
	direction (0)72–74° **or** N 72–74 °E etc.	**B1**
9(a)(ii)	(distance) has no direction **or (**distance) is a scalar	**B1**
9(a)(iii)	changes direction **or** goes round a corner	**B1**
	velocity changes **or** velocity depends on direction **or** acceleration / force is towards the centre of circle / centripetal	**B1**

Question	Answer	Marks
9(b)(i)	distance travelled before brakes applied (after seeing emergency)	**B1**
9(b)(ii)	any 2 of • poor tyre surface • poor road conditions, e.g. rain, ice • high speed (of car) **or** good aerodynamic car shape **or** tail wind • large mass / weight of car • driver applies less force to pedal / less braking force	**B1B1**
9(c)(i)	(KE =) ½mv^2 algebraic or numerical	**C1**
	½ × 1200 × 30^2	**C1**
	540 000 J	**A1**
9(c)(ii)	(a=) F / m algebraic or numerical	**C1**
	1.5 m / s^2	**A1**
9(c)(iii)	friction (produces thermal energy) **or** kinetic energy changes to heat **or** work done against friction	**B1**

2 A small coin of mass *m* is initially at rest. It is dropped from a height *h* above ground level. The coin has a speed *v* as it hits the ground.

The gravitational field strength *g* is equal to 10 N/kg.

(a) State an expression for:

- the gravitational potential energy of the coin at a height *h* above ground level

  gravitational potential energy = ...

- the kinetic energy of the coin when travelling at a speed *v*.

  kinetic energy = ...

[1]

(b) The coin is dropped from the top of a building of height 380 m.

Determine the speed of the coin as it hits the ground. You may ignore air resistance.

speed = ... [3]

(c) When air resistance is negligible, a heavier coin hits the ground at the same speed as a lighter coin when they are both dropped from the same height.

(i) Explain why.

.....

.....

..... [1]

(ii) When air resistance acts, coins of different masses do **not** hit the ground at the same speed when they are dropped from the same height.

Explain why.

.....

.....

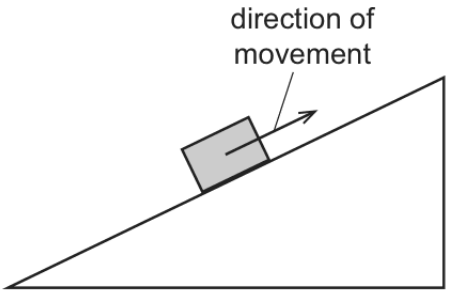
..... [2]

[Total: 7]

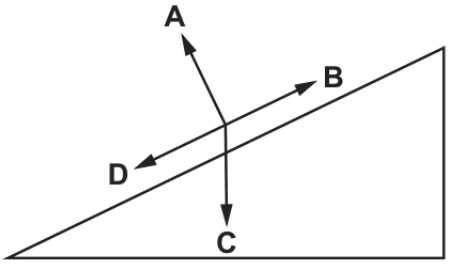
Question	Answer	Marks
2(a)	mgh **and** $\frac{1}{2}mv^2$	**B1**
2(b)	$\frac{1}{2}mv^2 = mgh$ **or** $v^2 = 2gh$ **or** $v^2 = 2 \times 10 \times 380$ **or** $v^2 = 7600$	**C1**
	$(v =)\sqrt{2gh}$ **or** $\sqrt{2 \times 10 \times 380}$	**C1**
	87 m / s	**A1**
2(c)(i)	both the g.p.e. and the k.e. are proportional to mass **or** mass cancels out in the equation **or** they accelerate at the same rate / g	**B1**
2(c)(ii)	air resistance (force) is a smaller proportion of weight (for heavier coin)	**B1**
	acceleration of heavier coin greater / less affected by air resistance	**B1**

QUESTION · 5054_s20_qp_11.pdf · source page 3 of 16

6 A box is pulled up a rough slope, as shown.



In which direction does friction act on the box?



MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the thinking distance and the braking distance for a car being driven along a dry road and along a wet road at the same speed.

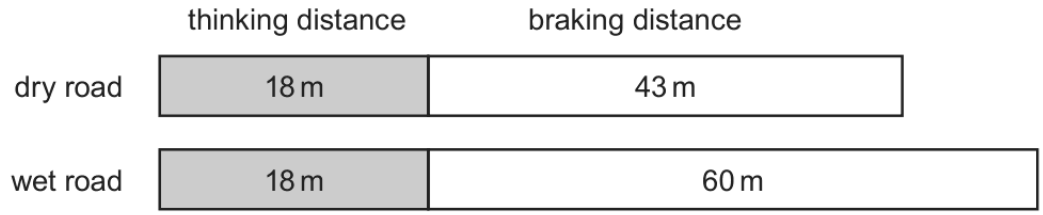


Fig. 1.1

(a) Calculate the total stopping distance for the car on the wet road.

distance = ... [1]

(b) Complete the sentence.

The thinking distance is the distance travelled between seeing a hazard and .....

..... [1]

(c) **(i)** Suggest why the thinking distance is the same on both roads.

.....

.....

..... [1]

(ii) Explain why the braking distance is larger when the road is wet.

.....

.....

.....

..... [2]

[Total: 5]

Question	Answer	Marks
1(a)	78 m	**B1**
1(b)	(starting to) apply the brakes	**B1**
1(c)(i)	time taken to think / react is the same (and speed is the same)	**B1**
1(c)(ii)	friction is lower **or** less backwards / resistant force **or** car slides or skids / tyres do not grip road / car aquaplanes	**B1**
	takes longer time to stop **or** acceleration is less	**B1**

2 Fig. 2.1 shows the thinking distance and the braking distance for a car driven at 100 km / h.

The car has old, smooth tyres.

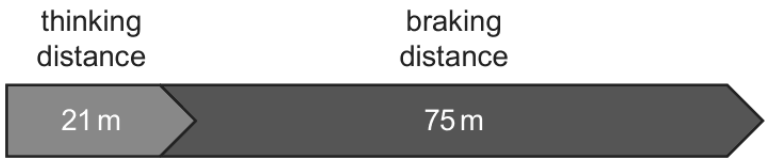


Fig. 2.1

(a) Calculate the total stopping distance for the car.

stopping distance = ... [1]

(b) The car is now fitted with new tyres.

At a speed greater than 100 km / h, the total stopping distance is the same as in **(a)**.

(i) State and explain the effect that the increase in speed and the use of new tyres have on the thinking distance.

effect .....

explanation .....

.....

..... [2]

(ii) State and explain the effect that the increase in speed and the new tyres have on the braking distance.

effect .....

explanation .....

.....

..... [2]

[Total: 5]

Question	Answer	Marks
2(a)	96 m	**B1**
2(b)(i)	increases	**B1**
	time unchanged but speed faster	**B1**
2(b)(ii)	decreases	**B1**
	(force of) friction larger	**B1**

QUESTION · 5054_w20_qp_11.pdf · source page 3 of 16

4 A parachutist falling at a steady speed opens her parachute.

Which row is correct for the direction of the resultant force and for the direction of the acceleration of the parachutist just after her parachute opens?

	resultant force direction	acceleration direction
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 4

4	D	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 is the distance–time graph for a skydiver who jumps from a balloon at time $t = 0$.

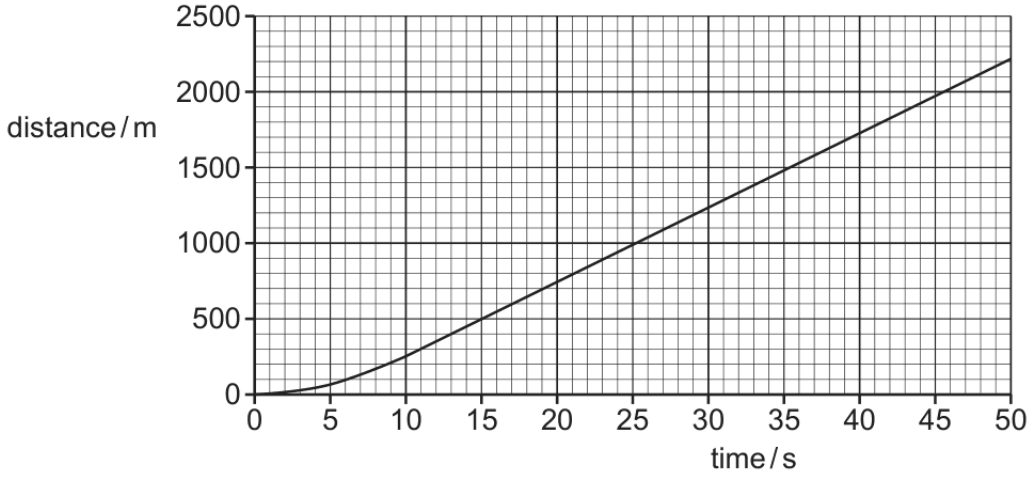


Fig. 1.1

(a) The first part of the graph shows the motion of the skydiver from when he jumps until he reaches terminal velocity.

(i) Describe the motion of the skydiver between $t = 0$ and $t = 20$ s.

.....

.....

..... [2]

(ii) Explain the motion of the skydiver between $t = 0$ and $t = 20$ s in terms of the forces acting on him.

.....

.....

.....

..... [3]

(b) Using Fig. 1.1, determine the terminal velocity of the skydiver.

On Fig. 1.1, indicate any values used for your calculation.

terminal velocity = ... [3]

[Total: 8]

Question	Answer	Marks
1(a)(i)	accelerating **or** increasing speed	**B1**
	decreasing acceleration **or** speed increasing at a decreasing rate **or** reaches a constant speed	**B1**
1(a)(ii)	initially acceleration is due to force of gravity / weight (only) **or** force of gravity / weight larger than air resistance from (t = 0 to ≈ 7 s)	**B1**
	air resistance increases (with speed)	**B1**
	resultant force decreases / becomes zero **or** forces balance **or** air resistance equals weight	**B1**
1(b)	two different pairs of co-ordinates from straight-line section of graph seen	**C1**
	correct use of co-ordinates in a division	**C1**
	48 – 52 m / s	**A1**

QUESTION · 5054_w20_qp_12.pdf · source page 3 of 16

4 A car is travelling forwards at high speed. The brakes are applied and the car skids along the road surface.

In which direction and where does the friction force act?

A backwards on the tyres

B forwards on the tyres

C upwards on the tyres

D downwards on the road

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 4

4	A	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 A bus leaves a bus-stop at time $t = 0$ and travels along a horizontal road until it reaches a second bus-stop. Fig. 7.1 is the distance-time graph for the bus between $t = 0$ and $t = 60$ s.

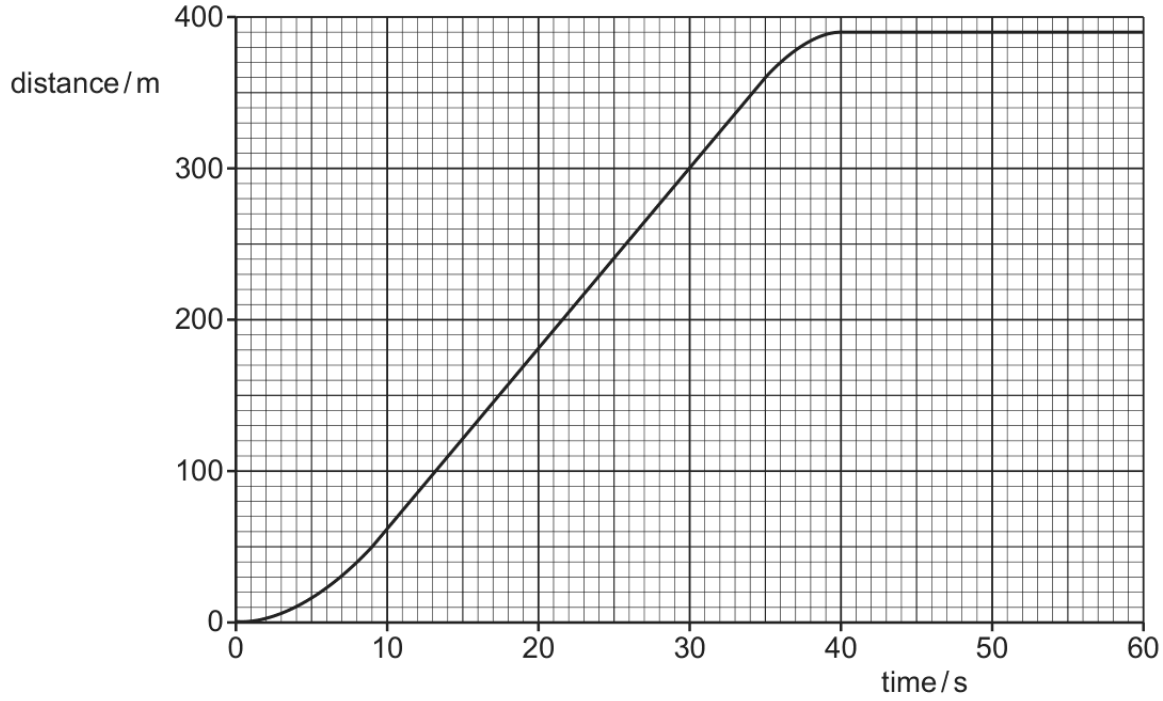


Fig. 7.1

The road on which the bus is travelling is straight except for a short, curved section. The bus travels around this circular curve between $t = 21$ s and $t = 24$ s.

(a) Describe how the motion of the bus between $t = 0$ and $t = 10$ s differs from its motion between $t = 35$ s and $t = 40$ s.

.....

.....

.....

.....

..... [3]

(b) Determine:

(i) the maximum speed of the bus during these 60 s

maximum speed = ... [3]

(ii) the average speed of the bus between leaving the first bus-stop and arriving at the second bus-stop.

average speed = ... [2]

(c) **(i)** State how *velocity* differs from *speed*.

.....

..... [1]

(ii) There are **three** periods during the 60 s when there is a non-zero resultant force acting on the bus.

Complete the statements to indicate these three time periods and state the direction of the resultant force in that period.

1. Between *t* = ... and *t* = ... the direction of the resultant force is

.....

2. Between *t* = ... and *t* = ... the direction of the resultant force is

.....

3. Between *t* = ... and *t* = ... the direction of the resultant force is

.....
[4]

(d) During the journey, the air resistance acting on the bus varies.

(i) State why the air resistance changes during the journey.

.....

..... [1]

(ii) On Fig. 7.1, mark and label with an M a time when the air resistance is a maximum value.
[1]

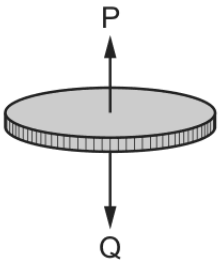
[Total: 15]

Question	Answer	Marks
7(a)	(between t = 0 and t = 10 s) it is accelerating **or** speed / velocity increasing	**B1**
	between t = 35 s and t = 40 s, it is decelerating **or** speed / velocity decreasing	**B1**
	magnitude of acceleration less than magnitude of deceleration **or** between t = 0 and t = 10 s starts from rest / zero speed **or** between t = 35ms and t = 40 s it finishes at rest / with zero speed	**B1**
7(b)(i)	two pairs of co-ordinates from the straight-line section	**C1**
	appropriate division using candidate's coordinates	**C1**
	11.7 ⩽ speed ⩽ 12.0 m / s	**A1**
7(b)(ii)	total distance / total time **or** (distance =) 390 (m) **and** 39 (s) ⩽ (time =) ⩽ 40 (s)	**C1**
	9.7 to 10.0 m / s	**A1**
7(c)(i)	it / velocity has a direction **or** is a vector **or** depends on displacement (rather than distance)	**B1**
7(c)(ii)	**answers need not be in this order** 1 0 **and** 10 (s) **and** forwards / in the direction of movement	**B1**
	2 21 (s) **and** 24 (s)	**B1**
	perpendicular to the motion / to the direction of movement	**B1**
	3 35 (s) **and** 40 (s) **and** backwards / opposite to the direction of movement	**B1**
7(d)(i)	speed (of bus) varies / changes / decreases / increases	**B1**
7(d)(ii)	any time on Fig. 1.1 from 10 s ⩽ t ⩽ 35 s labelled M	**B1**

QUESTION · 5054_s21_qp_11.pdf · source page 3 of 16

4 A coin falls from rest through the air and eventually reaches a constant speed.

There is a resultant force acting on the coin due to the two forces P and Q shown in the diagram.



What happens to force P and what happens to the resultant force before the coin reaches constant speed?

	force P	resultant force
A	decreases	increases
B	decreases	decreases
C	increases	decreases
D	increases	increases

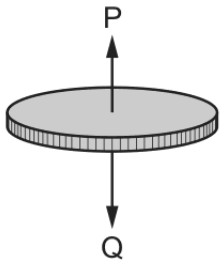
MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 4

4	C	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 2 of 16

2 A coin falls from rest through the air and eventually reaches a constant speed.

There is a resultant force acting on the coin due to the two forces P and Q shown in the diagram.



What happens to force P and what happens to the resultant force before the coin reaches constant speed?

	force P	resultant force
A	decreases	increases
B	decreases	decreases
C	increases	decreases
D	increases	increases

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 2

2	C	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 3 of 16

4 A student lists three changes that affect the stopping distance of a car.

1 increasing the braking force

2 increasing the friction between the tyres and the road

3 increasing the speed of the car

Which change or changes increase the stopping distance?

A 1 and 2 **B** 1 and 3 **C** 1 only **D** 3 only

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 4

4	D	**1**

2 Force is a vector quantity.

(a) State what is meant by a *vector*.

.....

..... [1]

(b) A swimmer reaches the end wall of a swimming pool and turns around under the water.

Fig. 2.1 shows the swimmer immediately after turning around.

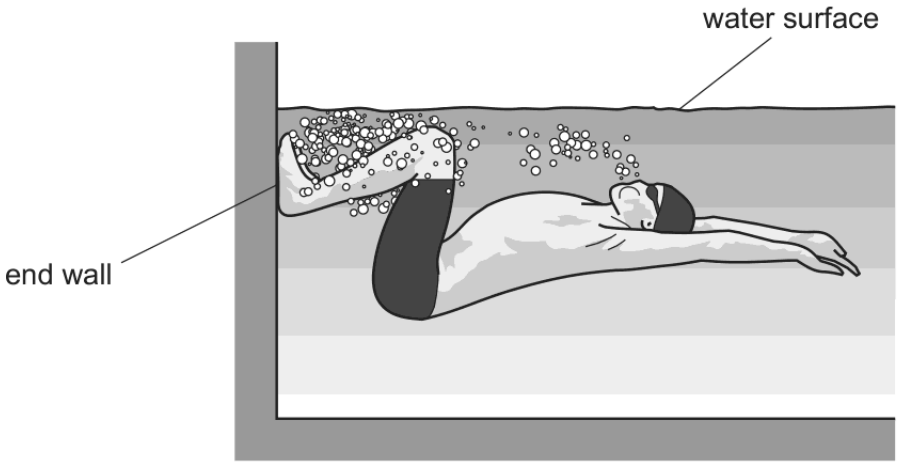


Fig. 2.1

(i) The swimmer pushes against the end wall of the pool with his legs.

Explain, in terms of Newton's third law, why the swimmer accelerates away from the end wall.

.....

.....

.....

..... [3]

(ii) While swimming, there is a constant forward force on the swimmer. His speed increases until eventually he reaches a constant speed.

Explain why he reaches a constant speed.

.....

.....

.....

..... [3]

[Total: 7]

MARK SCHEME · 5054_w21_ms_22.pdf · source page 5 of 11 · Question 2

Question	Answer	Marks
2(a)	(a vector quantity has) magnitude **and** direction	**B1**
2(b)(i)	the wall exerts a force on swimmer	**B1**
	opposite in direction (to force of swimmer on wall) **or** to the right	**B1**
	equal in size (to force of swimmer on wall) **or** (resultant) force causes acceleration **or** force on object Y due to X is equal and opposite to force on object X due to Y	**B1**

MARK SCHEME · 5054_w21_ms_22.pdf · source page 6 of 11 · Question 2

Question	Answer	Marks
2(b)(ii)	(water) resistance / resistive force / friction (on swimmer)	**B1**
	(water) resistance / backward force / resistive force / friction / opposing force increases (with speed / time)	**B1**
	(at constant speed) forward force is equal to resistive / backward force **or** no resultant force **or** forces balance	**B1**

QUESTION · 5054_s22_qp_11.pdf · source page 2 of 16

2 A driver drives a car at uniform velocity along a road.

The driver sees a hazard and applies the brakes.

What is the car's motion as it moves through the thinking distance and as it moves through the braking distance?

	motion through the thinking distance	motion through the braking distance
A	deceleration	deceleration
B	deceleration	uniform velocity
C	uniform velocity	deceleration
D	uniform velocity	uniform velocity

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 2

2	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 Fig. 7.1 shows a toy helicopter. It can hover and travel through the air.

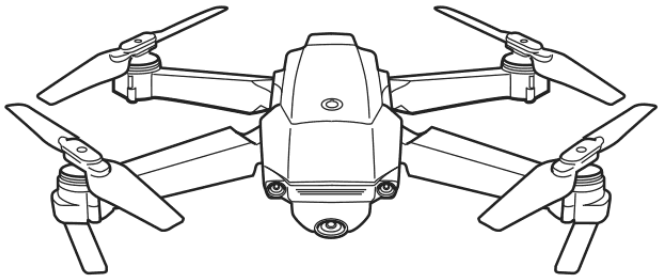


Fig. 7.1

A student flies the toy helicopter on a journey from A to B to C to D at a constant height.

Fig. 7.2 is a scale drawing of the path of the helicopter, viewed from above.

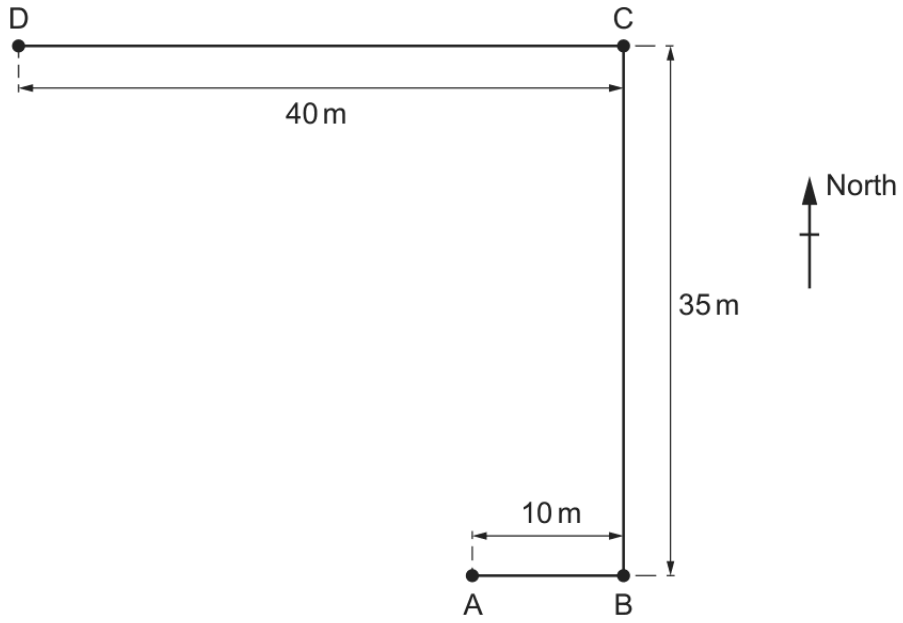


Fig. 7.2 (to scale)

(a) (i) Determine the total distance travelled by the toy helicopter.

..... [1]

(ii) The toy helicopter makes the journey in 40 s.

Calculate its average speed during the journey.

speed = ... [2]

(b) (i) State the difference between distance and displacement.

.....

..... [1]

(ii) Determine the scale used to construct Fig. 7.2 and complete the sentence.

1 cm on the diagram represents ..... on the path. [1]

(iii) Using Fig. 7.2, determine the size of the displacement of point D from point A.

Show your working.

size of displacement = ... [2]

(iv) Determine the angle between North and the direction of the displacement of point D from point A.

angle = ... [1]

(v) State what is meant by velocity.

..... [1]

(vi) Another toy helicopter flies directly from point A to point D in 40 s.

Explain why the magnitude of the velocity of this toy helicopter is smaller than the answer in **(a)(ii)**.

.....

..... [1]

(c) When the toy helicopter hovers at D, its motor fails and it falls. It reaches terminal velocity as it falls.

Explain, in terms of the forces and acceleration, what happens as the helicopter falls and reaches terminal velocity.

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 15]

Question 8 begins over the page.

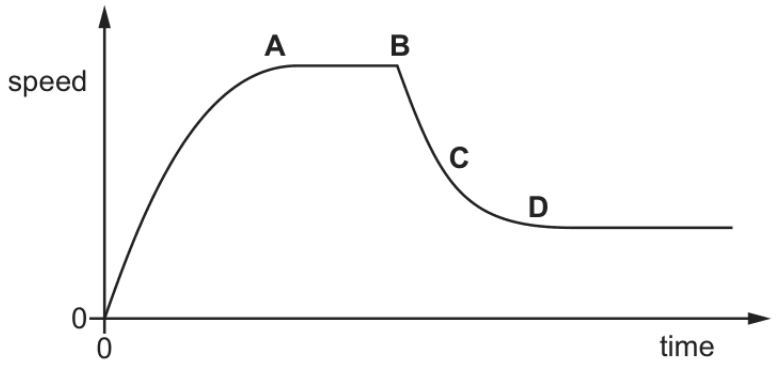
Question	Answer	Marks
7(a)(i)	85 m	**B1**
7(a)(ii)	(speed =) d / t in any form algebraic or numerical	**C1**
	2.1 m / s	**A1**
7(b)(i)	displacement involves direction **or** is a vector	**B1**
7(b)(ii)	5 m shown on dotted line	**B1**

Question	Answer	Marks
7(b)(iii)	length of diagonal 9.2 (cm) $\sqrt{30 \times 30 + 35 \times 35}$	**B1**
	46 m 46 m	**B1**
7(b)(iv)	39–42° **or** 318–320°	**B1**
7(b)(v)	(average) velocity = (change in) displacement / time **or** (average) velocity is displacement per unit time **or** rate of change of displacement	**B1**
7(b)(vi)	(size of) displacement < distance in **(a)(ii)** or distance covered is less (than 85 m / in **(a)(ii)**)	**B1**
7(c)	air resistance **and** weight / gravitational force mentioned	**B1**
	as speed increases air resistance increases	**B1**
	resultant / net force decreases **or** acceleration decreases	**B1**
	(eventually) air resistance = weight (at terminal velocity) / resultant force is zero	**B1**
	(at terminal velocity / eventually) zero acceleration / constant speed	**B1**

QUESTION · 5054_s22_qp_12.pdf · source page 2 of 20

3 The graph shows the speed–time graph for a parachutist who jumps from a plane but does not open his parachute immediately.

At which point does he open his parachute?



MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 3

3	B	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 Fig. 7.1 shows the speed–time graph for a car travelling on a straight horizontal road.

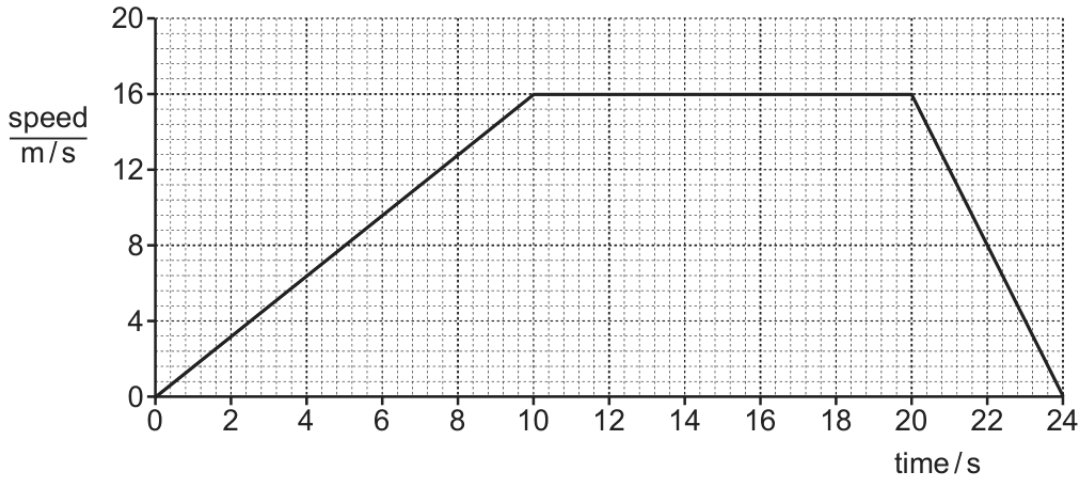


Fig. 7.1

(a) (i) Describe the motion of the car.

.....

.....

.....

.....

..... [3]

(ii) Using Fig. 7.1, calculate the distance travelled by the car during the 24 s of its motion.

Show your working.

distance = ... [3]

(iii) Calculate the average speed of the car during its motion.

average speed = ... [2]

(iv) A second car travels at a steady speed. It travels the same distance as the first car in the 24 s of the journey.

On Fig. 7.1, draw the speed–time graph for the second car. [2]

(b) The thinking distance is the distance travelled by a car between the time that a hazard is seen and the time that the brakes are applied.

The braking distance is the distance travelled while the car slows down to rest.

Table 7.1 shows the thinking and braking distances for an alert driver when the car travels at different speeds.

Table 7.1

$\frac{\text{speed}}{\text{km/h}}$	thinking distance / m	braking distance / m
20	9	2
40	18	9
60		20
80	36	36
100	45	56

(i) Complete Table 7.1. [1]

(ii) The time it takes for the driver to react to the hazard is constant at different speeds.

Explain how the table shows this.

.....

.....

.....

..... [2]

(iii) State what happens to the thinking distance and the braking distance when the driver is tired.

thinking distance .....

braking distance .....
[2]

[Total: 15]

Question	Answer	Marks
7(a)(i)	acceleration constant / for 10 s **or** acceleration 1.6 m / s^2 **or** speeds up for 10 s / until it reaches 16 m / s	**B1**
	constant / uniform speed for 10 s / from 10 to 20 s / of 16 m / s **or** zero acceleration for 10 s / from 10 to 20 s	**B1**
	decelerates constantly / for 4 s / from 20 s to 24 s / until 24 s **or** deceleration 4 m / s^2 **or** acceleration −4 m / s^2	**B1**

Question	Answer	Marks
7(a)(ii)	area under curve **or** formula for area of a triangle / trapezium **seen** **or** working shown used	**B1**
	(accelerating) 80 (m) **or** (at constant speed) 160 (m) **or** (decelerating) 32 (m) seen	**C1**
	(total distance) 270 **or** 272 m	**A1**
7(a)(iii)	(average speed) = total distance / (total) time algebraic or numerical	**C1**
	11 m / s	**A1**
7(a)(iv)	horizontal line drawn from 0 to 24 s at any level **or** horizontal line at the value calculated in **7(a)(iii)** for at least 16 s	**B1**
	horizontal line from 0 to 24 s at the value calculated in **7(a)(iii)**	**B1**
7(b)(i)	27	**B1**
7(b)(ii)	(statement that thinking) distance is proportional to speed **or** thinking distance increases uniformly with speed **or** time = (thinking) distance / speed and stated to be equal / not affected **or** ratio of (thinking) distance to speed is constant **or** (thinking) distance and speed both increase by same ratio **or** same change in (thinking) distance for equal changes in speed	**C1**
	at least two values thinking distance ÷ speed (or inverse) calculated **or** all values of thinking distance/speed stated to be 0.45 (or inverse 2.2) **or** proportionality shown by at least two values **or** double the speed doubles the (thinking) distance **or** each 20 (km / h) / same increase in speed covers an extra 9 (m) distance	**A1**
7(b)(iii)	*thinking distances* increases	**B1**
	braking distances no change	**B1**

QUESTION · 5054_w22_qp_11.pdf · source page 3 of 16

6 What causes the frictional effects experienced by a cyclist to increase?

A using a lighter bicycle

B travelling at greater speed

C a smoother tyre surface

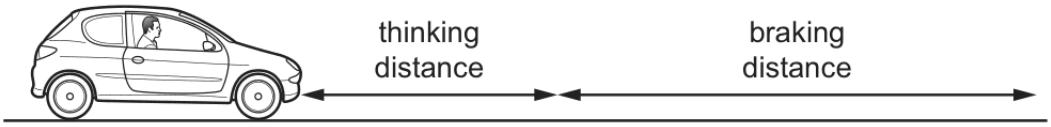
D a road surface covered in oil

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 6

6	B	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 3 of 16

7 The diagram shows a car at the moment the driver sees a hazard ahead.



Which expression is equal to the stopping distance?

A braking distance + thinking distance

B braking distance – thinking distance

C braking distance × thinking distance

D braking distance ÷ thinking distance

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 7

7	A	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 Fig. 7.1 shows a child sitting on a sledge on a snow-covered hill of constant slope.

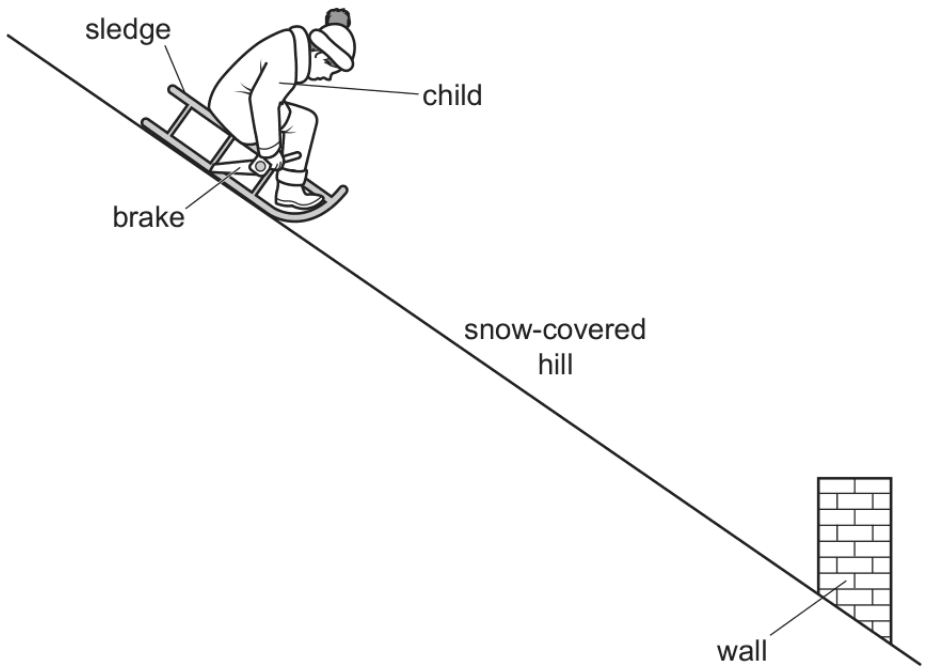


Fig. 7.1 (not to scale)

At time $t = 0$, the child and the sledge begin to move down the hill in a straight line.

When the child sees a wall ahead, he applies the brake.

The child and sledge continue to travel in a straight line until they come to a stop before hitting the wall.

Fig. 7.2 is the speed-time graph for the journey.

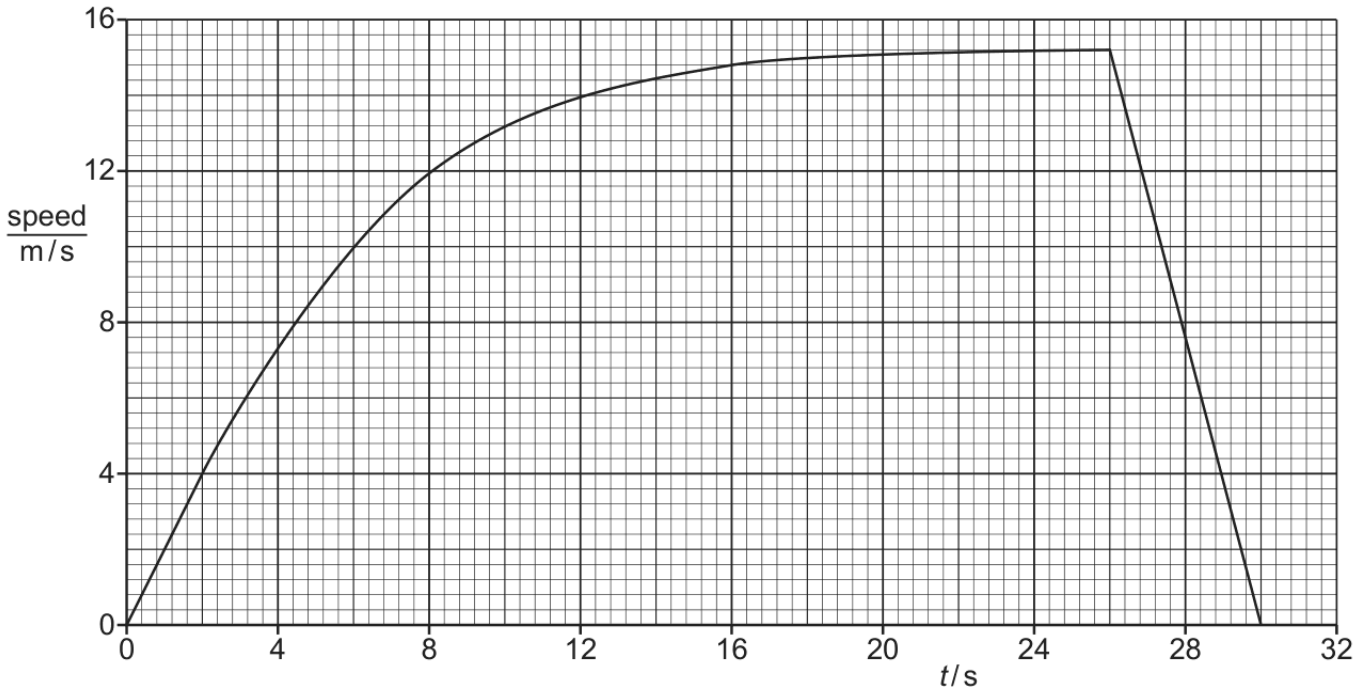


Fig. 7.2

The brake is applied at $t = 26\,s$.

(a) Fig 7.2 shows how the speed of the child and sledge varies over the whole of the journey.

Explain why, between $t = 0$ and $t = 26\,s$, the speed varies in the way shown by the graph.

.....

.....

.....

.....

.....

..... [3]

(b) At $t = 26\,s$, the front of the sledge is 35 m from the wall.

Determine the distance between the front of the sledge and the wall when the sledge stops.

distance = ... [3]

(c) At $t = 26$ s, the child and sledge begin to decelerate.

(i) Determine the size of the deceleration.

deceleration = ... [3]

(ii) The mass of the child is 46 kg and the mass of the sledge is 9.0 kg.

Calculate the resultant force on the child and sledge as they decelerate.

resultant force = ... [2]

(iii) State the energy transfer that is taking place as the child and sledge decelerate.

.....

.....

..... [2]

(d) At $t = 26$ s, when the brake is first applied, the child jerks forwards on the sledge.

Explain why.

.....

.....

..... [2]

[Total: 15]

Question	Answer	Marks
7(a)	acceleration / increase in speed **and** gravity / weight mentioned	**B1**
	air resistance / friction increases (with speed)	**B1**
	terminal / constant velocity reached **or** air resistance (becomes) equal to accelerating force	**B1**
7(b)	(distance travelled =) area under the graph **or** (x =) ½(u + v)t **or** ½ × 15.2 × 4.0	**C1**
	35 – (½ × 15.2 × 4.0) **or** 30.4 **or** 30	**C1**
	4.6 m	**A1**
7(c)(i)	(magnitude of deceleration =) gradient **or** Dv / Dt **or** 15.2 / 4.0 **or** equivalent numbers (from graph) in same ratio	**C1**
	15.2 / 4.0 **or** equivalent numbers in same ratio	**C1**
	3.8 m / s^2	**A1**
7(c)(ii)	(F =) ma **or** (46 + 9.0) × 3.8 **or** 55 × 3.8	**C1**
	210 N	**A1**
7(c)(iii)	<u>from</u> kinetic energy (and from gravitational potential energy)	**B1**
	<u>to</u> thermal energy **and** no intermediate energy	**B1**
7(d)	due to his mass **or** mass mentioned	**B1**
	(his) mass resists a change to its state of motion	**B1**

QUESTION · 5054_w22_qp_12.pdf · source page 3 of 16

7 A car travels at 60 km / h on a straight road. The road is dry. The driver applies the brakes suddenly. The table shows the thinking distance, the braking distance and the stopping distance.

thinking distance / m	braking distance / m	stopping distance / m
27	22	49

The same car is driven by the same driver at 60 km / h on the same road when it is wet. The driver applies the brakes suddenly again.

What is the effect of the wet road on the thinking distance, the braking distance and the stopping distance?

	thinking distance	braking distance	stopping distance
A	no change	increases	increases
B	no change	no change	no change
C	increases	increases	increases
D	increases	no change	increases

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 7

7	A	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 5 of 16

9 The stopping distance of a car is the sum of the thinking distance and the braking distance.

Which factors affect the braking distance?

	speed of car	tiredness of driver	condition of road
A	✓	✗	✗
B	✓	✓	✓
C	✓	✗	✓
D	✗	✓	✓

key

✓ = affects braking distance

✗ = does not affect braking distance

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 9

9	C	**1**

1 Fig. 1.1 shows the speed–time graph for a car travelling on a straight horizontal road.

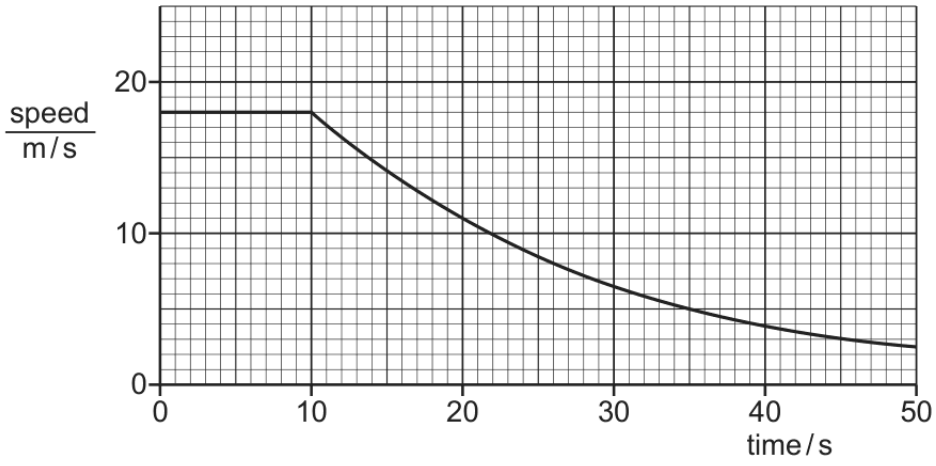


Fig. 1.1

(a) Describe the motion of the car shown in Fig. 1.1.

.....

.....

.....

..... [2]

(b) At time t = 10 s the engine of the car is switched off. The brakes are not applied.

(i) Name **two** forces that act on the car to cause the change in motion after t = 10 s.

1 .....

2 .....

[1]

(ii) Suggest why Fig. 1.1 is a curve after t = 10 s.

.....

.....

..... [1]

(c) Between $t = 10\,s$ and $t = 20\,s$ the speed of the car changes from $18\,m/s$ to $11\,m/s$.

The mass of the car is $1200\,kg$.

(i) Calculate the change in momentum of the car in this time.

Give the unit of your answer

momentum change = ... unit ..... [2]

(ii) Calculate the average resultant force exerted on the car during this time.

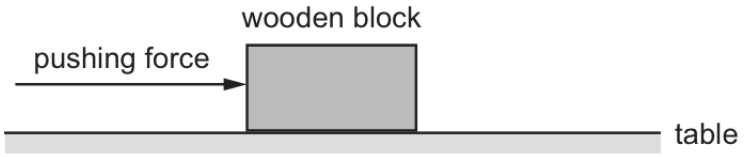
average resultant force = ... N [2]

[Total: 8]

Question	Answer	Marks
1(a)	speed is constant for 10 s / at first	**B1**
	deceleration • has (average) value 0.39 m / s^2 • lasts for 40 s • is non-uniform	**B1**
1(b)(i)	friction (with road) **and** air resistance **or** (air) drag	**B1**
1(b)(ii)	resistive force(s) change / decrease (as speed decreases) **or** (graph shows) deceleration is not constant / decreases **or** non-uniform decrease in speed	**B1**
1(c)(i)	(momentum =) *mv* in any form algebraic or numerical	**C1**
	8400 **and** kg m / s **or** Ns	**A1**
1(c)(ii)	(F =) $\Delta mv / t$ **or** $F = ma$ in any form algebraic or numerical	**C1**
	840 (N)	**A1**

QUESTION · 5054_w23_qp_11.pdf · source page 2 of 16

3 A wooden block is pushed across a table at constant speed.



Which statement is correct?

A The frictional force increases as the block moves at constant speed.

B The frictional force is equal and opposite to the pushing force.

C The frictional force is greater than the pushing force.

D The frictional force is less than the pushing force.

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 3

3	B	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 3 of 20

6 A student drops, from rest, a light ball in air. The ball experiences some air resistance as it falls.

What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 6

6	C	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 4 of 20

7 The stopping distance of a car travelling at a certain speed is measured in four different situations.

- on a dry road with new tyres
- on a dry road with old, worn tyres
- on a wet road with new tyres
- on a wet road with old, worn tyres

The results are shown.

	W	X	Y	Z
stopping distance / m	12	16	17	40

Which two results are possible measurements on the dry road with old, worn tyres?

A W and X **B** X and Y **C** X and Z **D** Y and Z

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 7

7	B	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 4 of 16

6 A car, travelling on a straight road, brakes and comes to a stop.

Which row gives the expression for calculating the thinking distance and gives a variable that affects thinking distance?

	expression for calculating thinking distance	variable that affects thinking distance
A	braking distance + stopping distance	load carried
B	braking distance + stopping distance	speed of car
C	stopping distance – braking distance	load carried
D	stopping distance – braking distance	speed of car

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 6

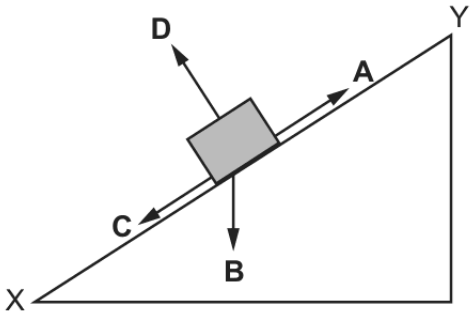
6	D	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 4 of 16

9 A box is moved up a rough slope from X to Y.

The diagram shows four forces acting on the box.

Which force is the force due to friction on the box?



MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 9

9	C	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 5 of 16

10 As a cyclist uses the brakes on a bicycle, energy is transferred from the kinetic store to the thermal store.

What causes the transfer of energy?

A friction

B mass

C thrust

D weight

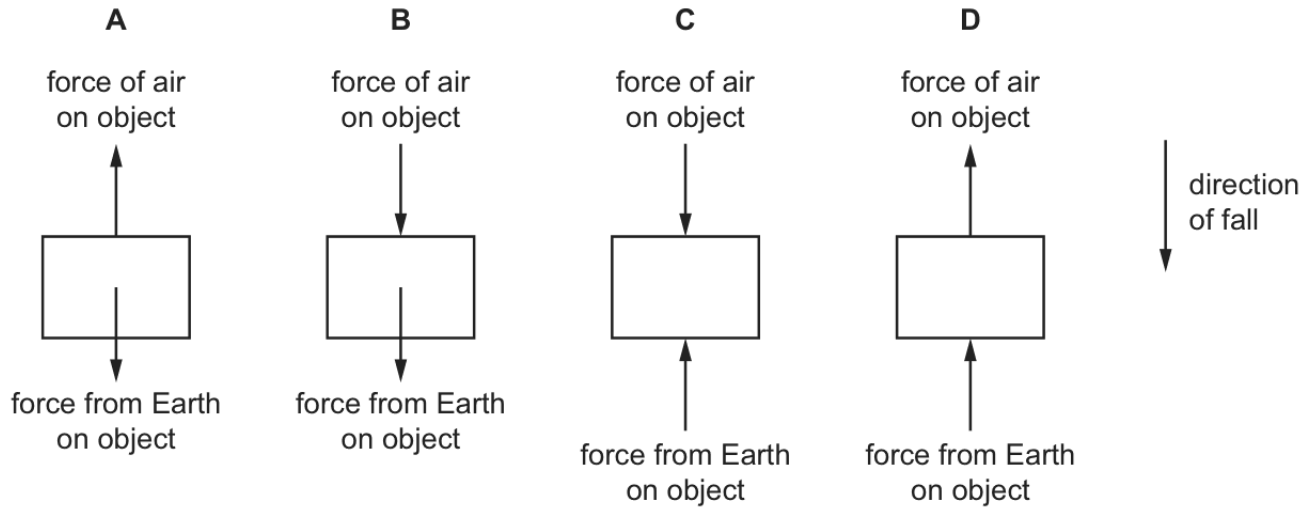
MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 10

10	A	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 4 of 16

8 An object falls vertically through the air.

Which diagram represents the forces acting on the object?



MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 8

8	A	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 2 of 20

3 Which statement explains why a heavy coin that falls through a short distance towards the ground does **not** reach terminal velocity?

A The coin has **not** hit the ground.

B The weight of the coin equals the air resistance.

C The weight of the coin increases as the air resistance increases.

D The weight of the coin is always more than the air resistance.

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

1 Fig. 1.1 shows a skydiver falling vertically through the air.

Fig. 1.1

In the first part of the fall, her speed increases and her acceleration decreases.

In the second part of the fall, her speed is constant.

(a) On Fig. 1.2 sketch the speed–time graph for the skydiver.

On your graph, mark **one** point where the speed of the skydiver is increasing with an A and **one** point where the speed of the skydiver is constant with a B.

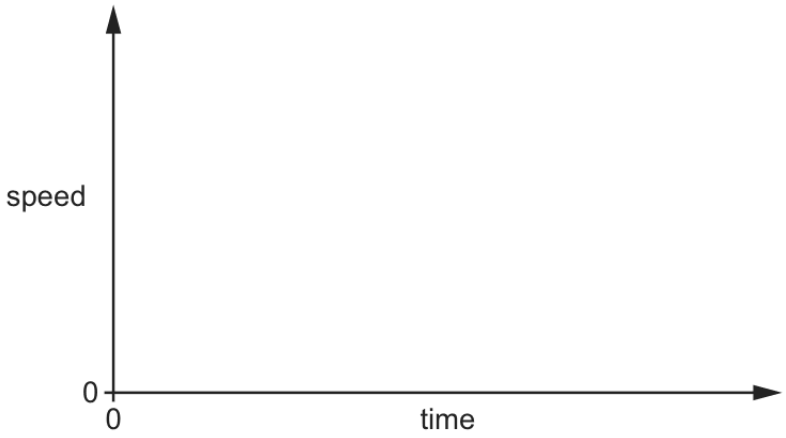


Fig. 1.2

[2]

(b) Explain how the graph shows that the acceleration decreases as the speed increases.

.....

.....

..... [1]

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(c) During the first part of the fall, there is a resultant vertical force acting downwards on the skydiver.

(i) One of the vertical forces acting on the skydiver is her weight.

State the name of the other vertical force that acts on the skydiver.

..... [1]

(ii) Explain why the resultant vertical force eventually becomes zero.

.....

.....

.....

..... [2]

(d) At one instant, the vertical force on the skydiver is 400 N downwards.

At the same instant, the wind causes an additional horizontal force of 100 N to the right to act on the skydiver.

Draw a vector diagram to determine the resultant of the 400 N vertical force and the 100 N horizontal force. Place an arrow on all of the forces to show their directions.

Determine the magnitude (size) of this resultant force and its direction to the vertical.

magnitude of resultant = ... N

direction = ...° to vertical
[3]

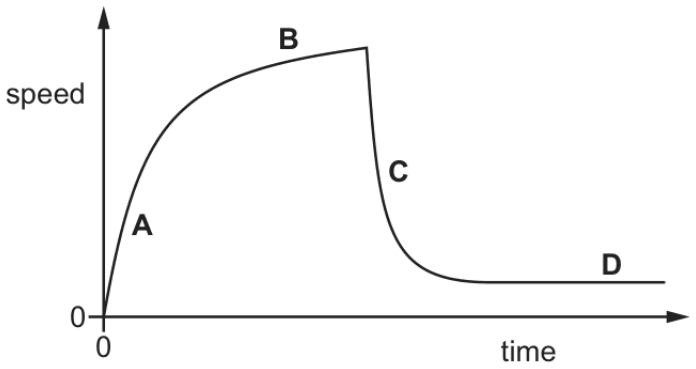
[Total: 9]

Question	Answer	Marks
1(a)	curve upwards (at start) with decreasing gradient labelled A	**B1**
	horizontal line labelled B	**B1**
1(b)	gradient / slope decreases **or** increase in speed for same time less at later times	**B1**
1(c)(i)	air resistance / drag	**B1**
1(c)(ii)	air resistance / upwards force increases (with increased speed)	**B1**
	air resistance / upwards force equals / balances weight	**B1**
1(d)	vector diagram showing 400 N and 100 N force at right angles with correct resultant and all directions shown	**B1**
	410 (N)	**B1**
	12–16°	**B1**

QUESTION · 5054_s25_qp_12.pdf · source page 2 of 16

3 The speed–time graph for a falling skydiver is shown. As he falls, the skydiver spreads out his arms and legs and then opens his parachute.

Which part of the graph shows the skydiver falling with terminal velocity?



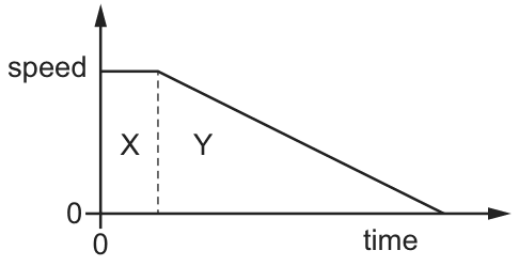
MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 3 of 16

4 A car driver travelling at constant speed in a straight line sees a hazard in the road ahead at time = 0.

The speed–time graph shows how the speed of the car changes from time = 0.



Three students suggest what area X and area Y represent.

student 1 Area X represents the thinking distance.

student 2 Area Y represents the stopping distance.

student 3 Area X + area Y represents the stopping distance.

Which students are correct?

A 1 and 2 **B** 1 and 3 **C** 1 only **D** 2 only

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

1 A ball is dropped by the side of a vertical scale that is marked in centimetres.

A video recording of the fall shows the position of the ball on the scale every 0.10 s.

The distance the ball falls is shown by the position on the scale.

Table 1.1 shows the results.

Table 1.1

time / s	0	0.10	0.20	0.30	0.40	0.50
distance / cm	0	5	20	44	78	123

(a) (i) On Fig. 1.1, plot the distance–time graph for the ball. Draw a suitable line.

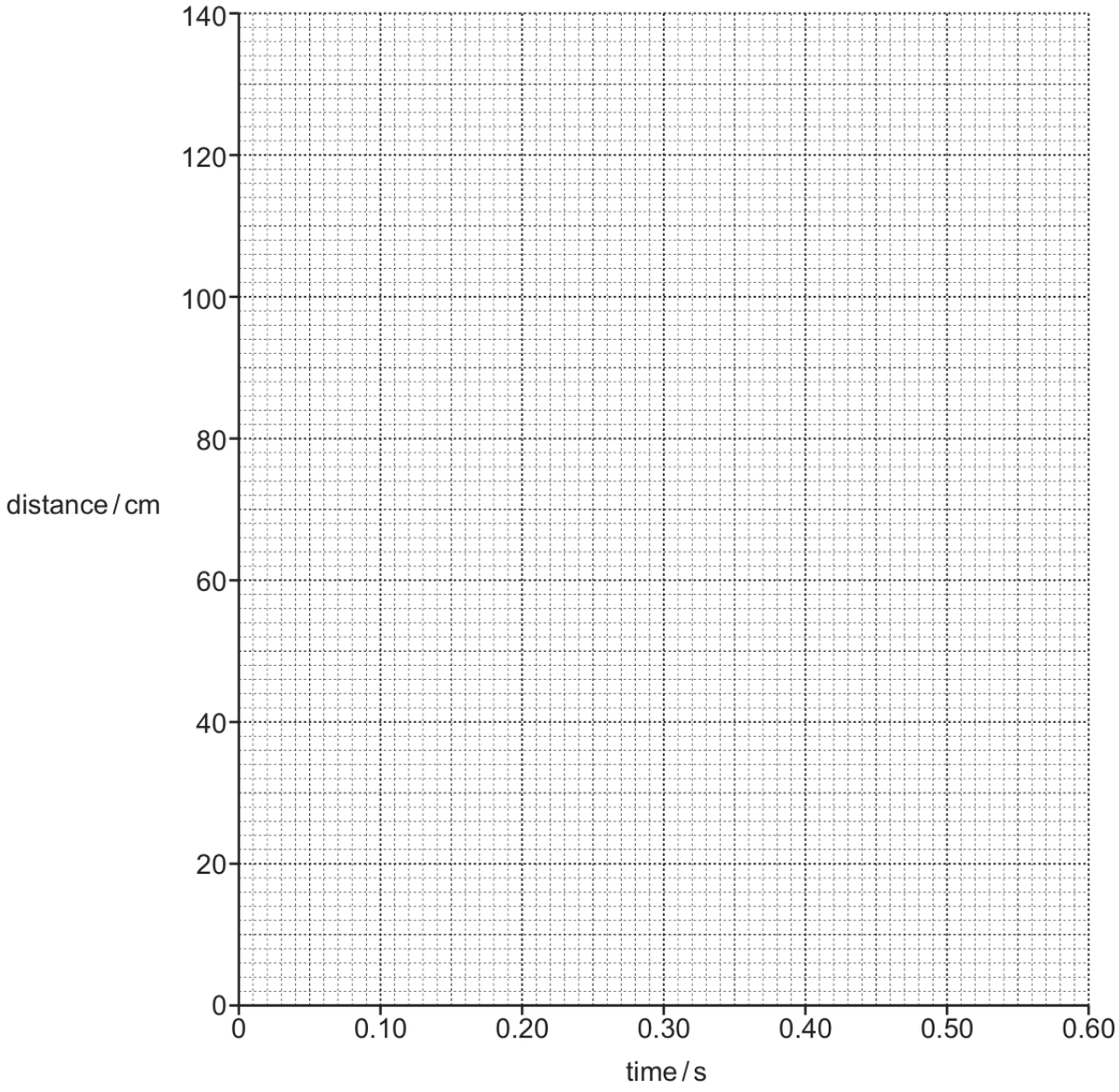


Fig. 1.1

[2]

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(ii) Calculate the average speed of the ball between time $t = 0$ and time $t = 0.40\,s$.

speed = ... cm / s [2]

(iii) Describe how to use the graph in Fig. 1.1 to determine the speed of the ball at time $t = 0.40\,s$.

.....

.....

..... [2]

(b) As the ball falls, the acceleration of the ball decreases and eventually becomes zero before it reaches the ground.

(i) Explain, in terms of the forces acting, why the acceleration decreases, and why it eventually becomes zero.

.....

.....

.....

.....

.....

..... [3]

(ii) Describe the appearance of a distance–time graph when the acceleration of the ball is zero.

..... [1]

[Total: 10]

Question	Answer	Marks
1(a)(i)	points plotted to within ½ a small square	**B1**
	smooth curve drawn	**B1**
1(a)(ii)	(average speed =) d / t in any form	**C1**
	195 (cm / s)	**A1**
1(a)(iii)	draw a tangent (at t = 0.40 s)	**B1**
	find the gradient / slope (of the graph) **or** the gradient of the tangent	**B1**
1(b)(i)	mention of air resistance / resistive force	**B1**
	resultant force decreases **or** air resistance / upwards force increases (as speed increases) / becomes high(er)	**B1**
	(eventually) air resistance = weight (and acceleration is zero) **or** air resistance balances / cancels weight	**B1**
1(b)(ii)	straight line (of positive) **or** (line of) constant gradient / slope	**B1**

QUESTION · 5054_w25_qp_11.pdf · source page 4 of 16

6 A parachutist falls towards the ground at constant speed with a fully open parachute.

Which statement is correct?

A There are **no** forces acting on the parachutist.

B The upward force on the parachute is equal to the weight of the parachutist.

C The upward force on the parachute is greater than the weight of the parachutist.

D The upward force on the parachute is less than the weight of the parachutist.

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 6

6	B	**1**

1 A car travels along a straight horizontal road, initially at a constant speed. The speed–time graph in Fig. 1.1 shows the motion of the car from time $t = 0$.

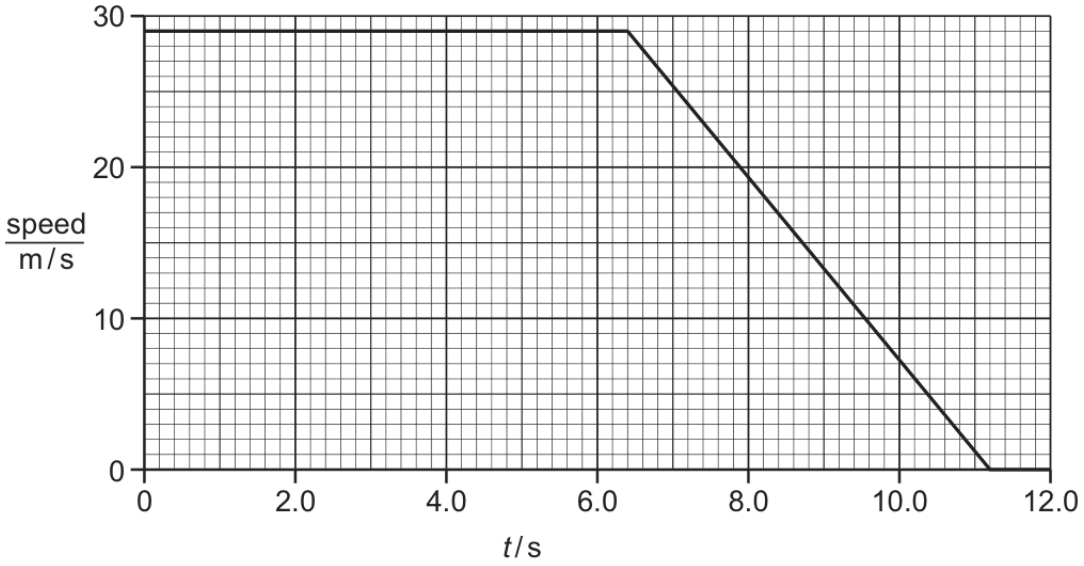

Fig. 1.1

At time $t = 5.7$ s, the driver sees an obstruction in the road.

The driver applies the brakes a short time later, and the car begins to slow down.

(a) Using Fig. 1.1, determine:

(i) the speed of the car at time $t = 0$

speed = ... m/s [1]

(ii) the distance travelled by the car between the driver seeing the obstruction at $t = 5.7$ s and the car beginning to slow down

distance = ... m [2]

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(iii) the distance travelled by the car between $t = 0$ and $t = 11.2$ s.

distance = ... m [3]

(b) The speed of the car affects both the thinking distance and the braking distance.

State **one** factor that affects **only** the braking distance of the car.

Explain how a change in this factor increases the braking distance.

factor .....

.....

explanation .....

.....

.....

[3]

[Total: 9]

Question	Answer	Marks
1(a)(i)	29 (m / s)	**B1**
1(a)(ii)	(s =) vt **or** 29 × (6.4 – 5.7)	**C1**
	20 (m)	**A1**
1(a)(iii)	(distance travelled =) area under graph **or** ½bh **or** ½(w_1 + w_2)h	**C1**
	½ × 29 × (11.2 – 6.4) **or** ½ × 29 × 4.8 **or** 69.6 (m) **or** ½ × 29 × (11.2 + 6.4)	**C1**
	260 (m)	**A1**
1(b)	any **one** factor from: • (mass) of load / car • (condition / wear on) tyres • (state of) road • brakes • reference to friction	**B1**
	any **one** matching explanation from: • heavy load • worn tyres • slippery / smooth road reference to reduced friction	**B1**
	reduced deceleration **or** increases the time taken (to stop)	**B1**

2 A spherical container is used to carry measuring instruments to the bottom of a lake.

Fig. 2.1 shows the container held at rest just below the water surface of a lake.

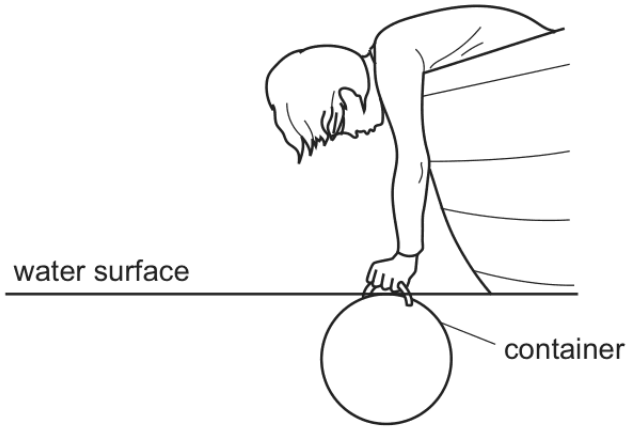

Fig. 2.1 (not to scale)

The container is released and falls through the water to the bottom of the lake.

(a) The container reaches terminal velocity before it hits the bottom of the lake.

Explain, in terms of the forces on the container, the motion of the container through the water until it reaches terminal velocity.

.....

.....

.....

.....

.....

..... [3]

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(b) The container falls through a vertical distance of 32 m before it hits the bottom of the lake.

The container and instruments have a total mass of 74 kg.

(i) Calculate the energy transferred from the gravitational potential energy store of the container.

energy transferred = ... J [2]

(ii) State the principle of conservation of energy.

.....

.....

..... [2]

(iii) Work is done to transfer energy from the gravitational potential energy store as the container falls at terminal velocity.

Describe, in terms of the work done, how the energy is transferred from the gravitational potential energy store to **one** other energy store in the water.

.....

.....

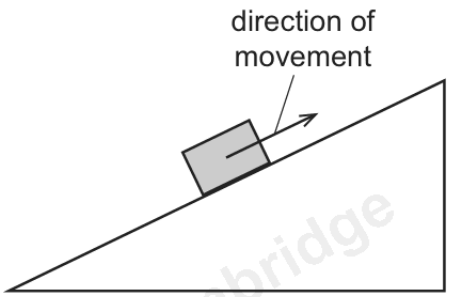
..... [2]

[Total: 9]

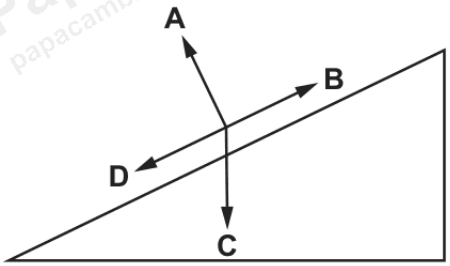
Question	Answer	Marks
2(a)	(initially its weight causes) acceleration **or** speeds up **or** gains kinetic energy	**B1**
	friction / drag / resistance / resistive force / viscous force increases (as the speed increases) **or** resultant force decreases	**B1**
	acceleration decreases to zero **or** friction / drag / resistance / resistive force / viscous force is (eventually) equal to weight **or** forces balance **or** resultant force = 0	**B1**
2(b)(i)	(ΔE_P =) $mg\Delta h$ **or** $74.0 \times 9.8 \times 32$	**C1**
	2.3×10^4 (J)	**A1**
2(b)(ii)	energy can be neither created **and** nor destroyed	**B1**
	energy can be transferred from one energy store to another	**B1**
2(b)(iii)	(energy transferred) by doing work against friction / drag / resistance / resistive force	**B1**
	to internal (energy) store	**B1**

QUESTION · 5054_s26_qp_11.pdf · source page 4 of 16

7 A box is pulled up a slope as shown. There is friction between the box and the slope.



In which direction does friction act on the box?



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MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 7

7	D	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 4 of 16

8 A skydiver jumps from an aeroplane.

After a few seconds, he reaches terminal velocity without opening his parachute.

Why does he reach terminal velocity?

A Air resistance becomes greater than his weight and slows him down.

B Air resistance decreases and he speeds up.

C Air resistance increases and balances his weight so that his acceleration is zero.

D His weight decreases and balances the air resistance.

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 8

8	C	**1**

Chapter 1 Motion, forces and energy

1.5 Forces

1.5.3 Elastic deformation

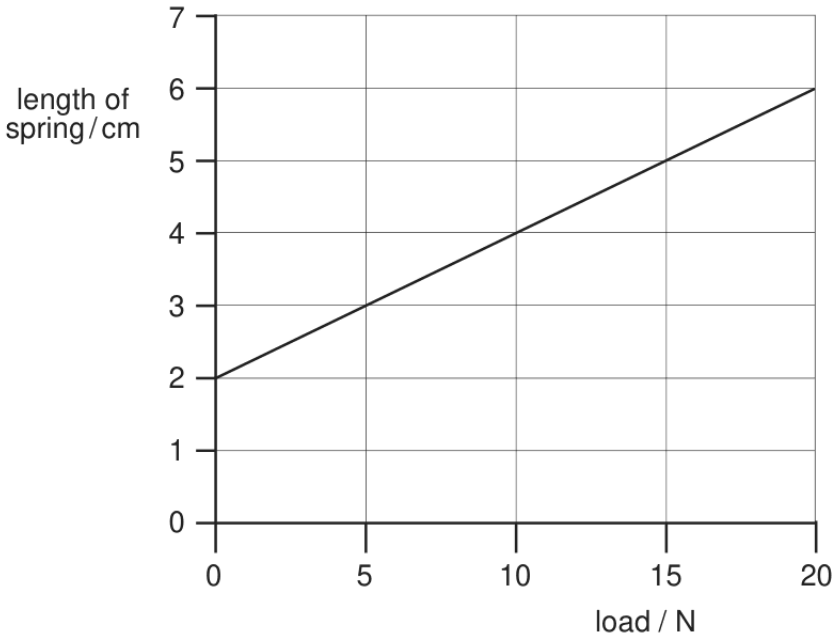
84 entries: 62 Paper 1 questions, 22 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 4 of 20

9 A spring balance is calibrated to give readings in newtons.

The graph shows how the length of the spring varies with the load.



A load causes the spring of the balance to extend by 3 cm.

What is the balance reading?

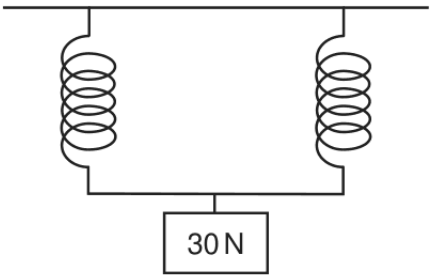
A 3 N **B** 5 N **C** 10 N **D** 15 N

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 9

9	**D**

QUESTION · 5054_w04_qp_1.pdf · source page 6 of 20

10 A spring extends by 4 cm when 10 N is suspended from it. Two of these springs are used as shown to carry a 30 N load.



What is the extension of each spring?

A 4 cm **B** 6 cm **C** 8 cm **D** 12 cm

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 10

10 **B**

QUESTION · 5054_s05_qp_1.pdf · source page 4 of 20

8 The table shows how the extension of a spring varies with load.

load / N	0	2	4	6	8	10	12	14	16
extension / cm	0	3	6	9	12	15	20	27	38

Between which two loads would you find the limit of proportionality?

A 0 N and 2 N

B 8 N and 10 N

C 10 N and 12 N

D 14 N and 16 N

MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 8

8 **C**

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows a 0.4 kg mass hanging at rest from a spring.

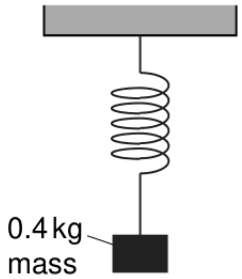


Fig. 1.1

(a) State what is meant by the *mass* of an object.

.....

.....

..... [1]

(b) **(i)** On Fig. 1.1, draw an arrow showing the direction and the line of action for each of the two forces that act on the mass. Write the name of the force next to each arrow. [2]

(ii) The gravitational field strength is 10 N/kg. Calculate the size of each of the two forces acting on the mass.

first force = ... second force = ... [2]

(c) The mass is pulled downwards. State and explain what happens to the upward force.

.....

.....

.....

.....

..... [2]

Section A

1	**(a)**		amount of matter/substance **or** ability to resist motion (accept Force/acc)	**B1**	
	(b)	**(i)**	downwards force labelled weight/gravity continuation of vertical line	**B1**	
			upwards force (labelled tension) along vertical line	**B1**	
			(allow 1 mark if both labels correct but lines of action not accurate, e.g. displaced to side)		
		(ii)	4 N	**B1**	
			4 N or same as other	**B1**	
	(c)		upwards force/tension increases	**B1**	
			spring stretched/longer	**B1**	**7**

QUESTION · 5054_s06_qp_1.pdf · source page 5 of 20

10 A metal wire, initially 1.000 m long, extends by 4 mm when a load of 2 N is added to it.

What will the length of the wire be if a further 3 N is added, assuming it does not extend beyond its limit of proportionality?

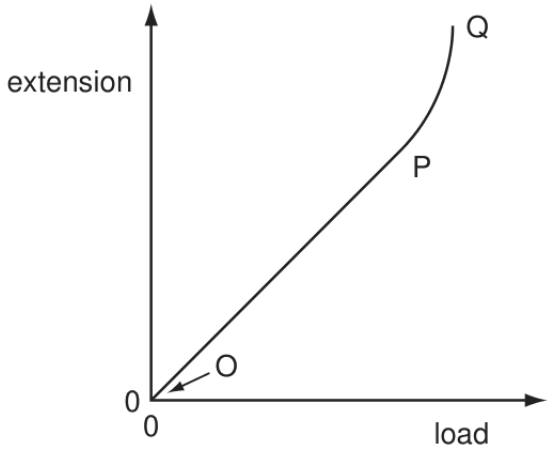
A 1.060 m **B** 1.080 m **C** 1.010 m **D** 1.012 m

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 10

10	**C**

QUESTION · 5054_w06_qp_1.pdf · source page 4 of 16

7 Which part of the graph shows the limit of proportionality for an elastic solid?



A O **B** OP **C** P **D** PQ

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 7

7 **C**

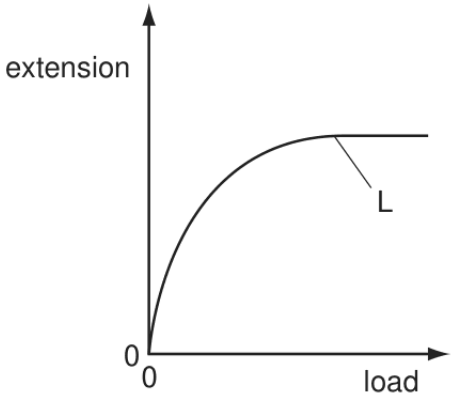
QUESTION · 5054_s07_qp_1.pdf · source page 5 of 20

8 A steel spring is stretched by a load. The load is increased.

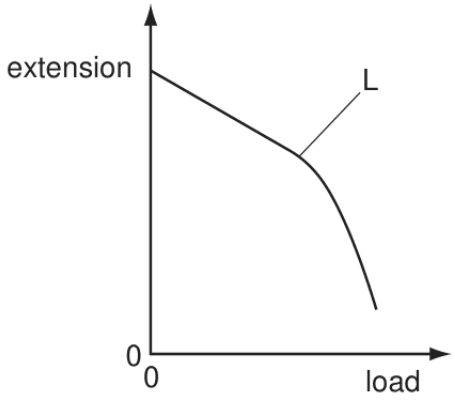
At first the extension is proportional to the load. The spring reaches its limit of proportionality at L.

Which is the correct graph of the extension against load for the spring?

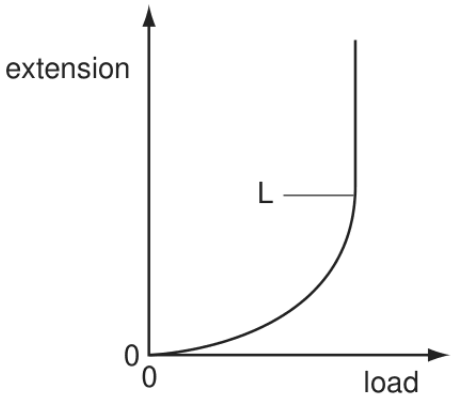
A



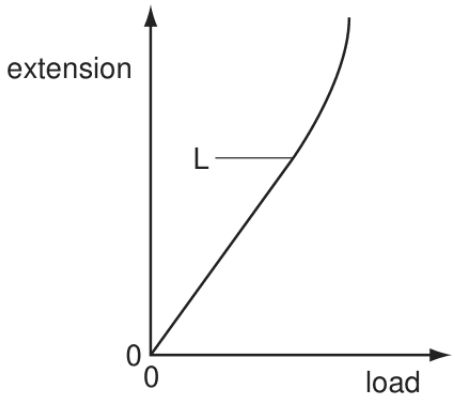
B



C



D



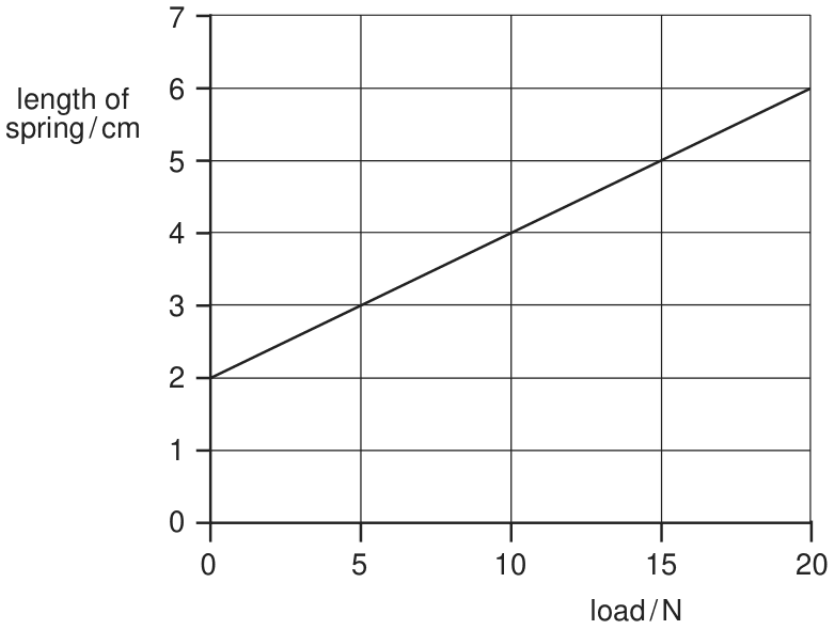
MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 8

8	**D**

QUESTION · 5054_w07_qp_1.pdf · source page 4 of 20

8 A spring balance is calibrated to give readings in newtons.

The graph shows how the length of the spring varies with the load.



A load causes the spring of the balance to extend by 3 cm.

What is the balance reading?

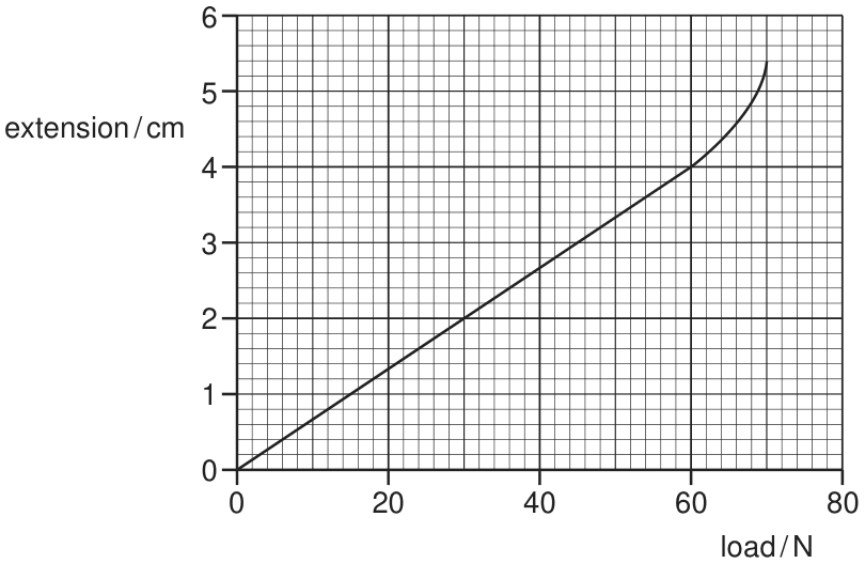
A 3 N **B** 5 N **C** 10 N **D** 15 N

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 8

8 **D**

QUESTION · 5054_s08_qp_1.pdf · source page 4 of 20

7 An extension-load graph for a wire is shown.



What is the load at the limit of proportionality for the wire?

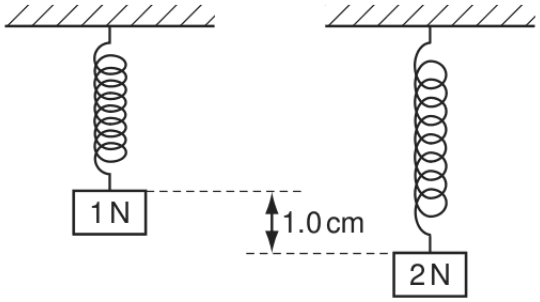
A 4 N **B** 15 N **C** 60 N **D** 70 N

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 7

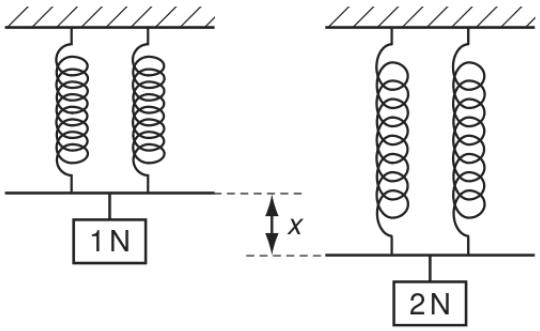
7	**C**

QUESTION · 5054_w08_qp_1.pdf · source page 6 of 20

9 A single spring is loaded with a 1 N weight. The load is then increased to 2 N and the extension increases by 1.0 cm, as shown.



Two springs that are identical to the first one are put side by side. They are connected at both ends, and a 1 N weight is hung on them. The load is then increased to 2 N.



What is the new increase *x* in the extension?

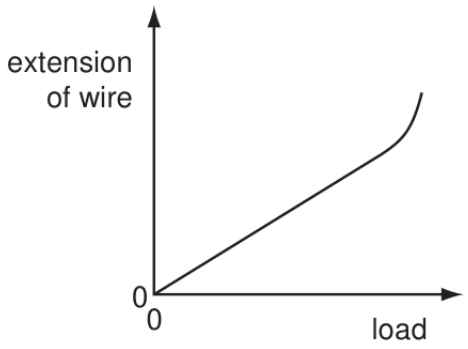
A 0.5 cm **B** 1.0 cm **C** 2.0 cm **D** 3.0 cm

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 9

9 **A**

QUESTION · 5054_s09_qp_1.pdf · source page 6 of 20

9 The graph shows the extension of a piece of copper wire as the load on it is increased.



What does the graph show?

A At a certain load the wire becomes easier to extend.

B At a certain load the wire becomes harder to extend.

C The load and extension are directly proportional for any load.

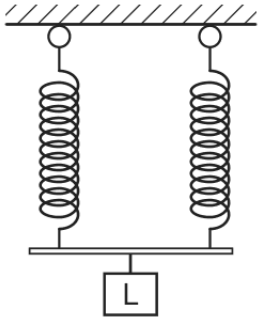
D The load and extension are inversely proportional for any load.

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 9

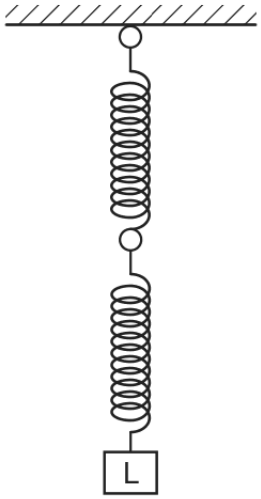
9	**A**

QUESTION · 5054_w09_qp_1.pdf · source page 5 of 20

9 A load L is suspended from two springs that are in parallel. The extension of each spring is x.



The springs are then arranged to hang vertically, one below the other.



In this new arrangement, what is the total extension of the two springs?

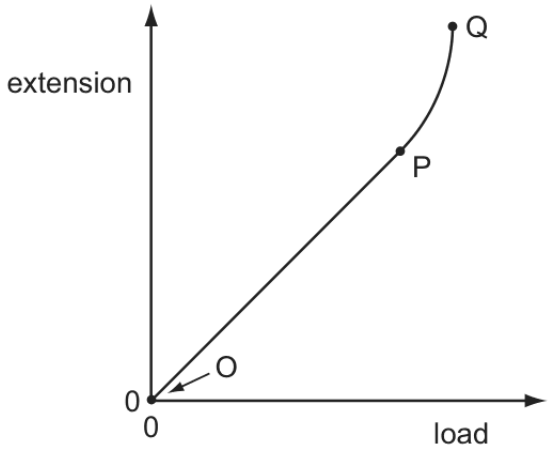
A $\frac{1}{2}x$ **B** x **C** $2x$ **D** $4x$

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 9

9	**D**

QUESTION · 5054_s10_qp_11.pdf · source page 4 of 20

9 Which part of the graph shows the limit of proportionality for an elastic solid?



A O **B** OP **C** P **D** PQ

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 9

9 **C**

For Examiner's Use

2 Fig. 2.1 shows a weight of 4.0 N attached to a spring.

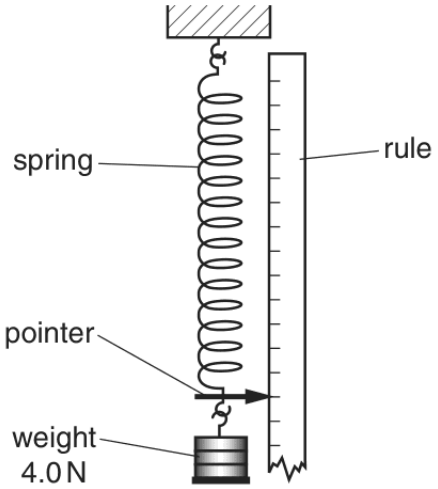


Fig. 2.1

The unstretched length of the spring is 8.0 cm. With the 4.0 N weight attached to the spring, the length is 14.0 cm. The spring is within its limit of proportionality.

(a) State what is meant by the *limit of proportionality* of a spring.

.....

.....

.....[1]

(b) The 4.0 N weight is replaced with a 2.0 N weight. Calculate the new length of the spring.

new length = ...[2]

(c) Describe how the apparatus in Fig. 2.1 is used to obtain readings to plot an extension-load graph.

.....

.....

.....

.....[2]

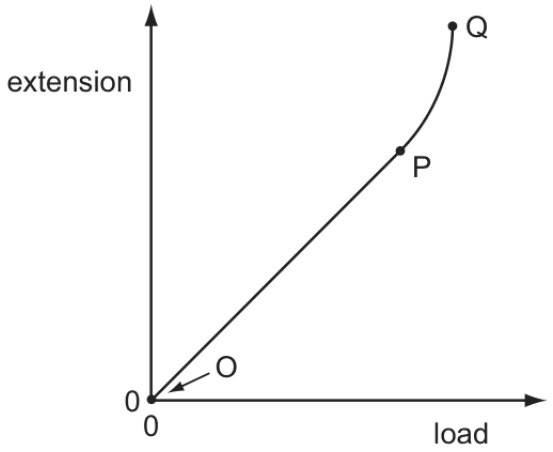
2 **(a)** **where** extension/stretching stops being proportional to force/load/weight/mass
or extension/load = constant
or point where length or extension against load graph curves C1 [1]

(b) 4 = k 6 **or** 4/6 **or** 6/4 **or** 6 × 2/4 **or** 3 (cm) seen C1
11 cm A1 [2]

(c) different weights/masses/load **and** measure new length B1
how extension is found e.g. reading on scale for loaded spring subtracted from reading with no load/mass/original B1 [2]

QUESTION · 5054_s10_qp_12.pdf · source page 5 of 20

10 Which part of the graph shows the limit of proportionality for an elastic solid?



A O **B** OP **C** P **D** PQ

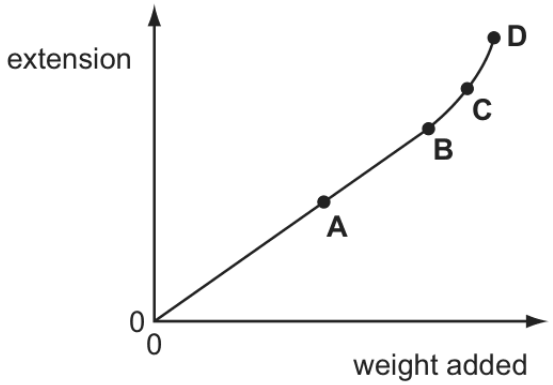
MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 10

10 **C**

QUESTION · 5054_w11_qp_11.pdf · source page 5 of 20

8 The extension of a spring is measured as weights are added. The graph shows the results.

Which point is the spring's limit of proportionality?



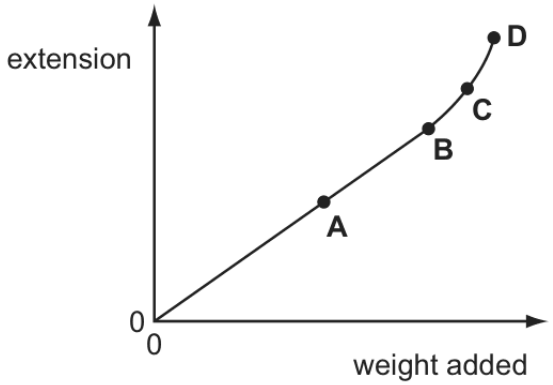
MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 8

8 **B**

QUESTION · 5054_w11_qp_12.pdf · source page 6 of 20

11 The extension of a spring is measured as weights are added. The graph shows the results.

Which point is the spring's limit of proportionality?

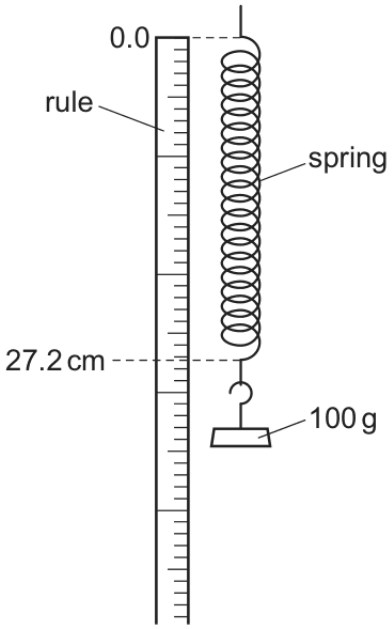


MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 11

11 **B**

QUESTION · 5054_s12_qp_11.pdf · source page 4 of 16

9 A 100 g mass is suspended from a spring next to a vertical metre rule. The top of the spring is level with the 0.0 cm mark. The bottom of the spring is level with the 27.2 cm mark.



The 100 g mass is replaced with a 600 g mass. The length of the spring is now 89.7 cm. The spring has not reached the limit of proportionality.

The 600 g mass is replaced with a 200 g mass. What is the length of the spring?

A 29.9 cm **B** 33.4 cm **C** 39.7 cm **D** 54.4 cm

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 9

9	**C**

3 A student measures the pressure inside a bicycle tyre using a pressure gauge he has constructed. Fig. 3.1 shows the apparatus he uses. The piston and rod move along the smooth cylindrical tube.

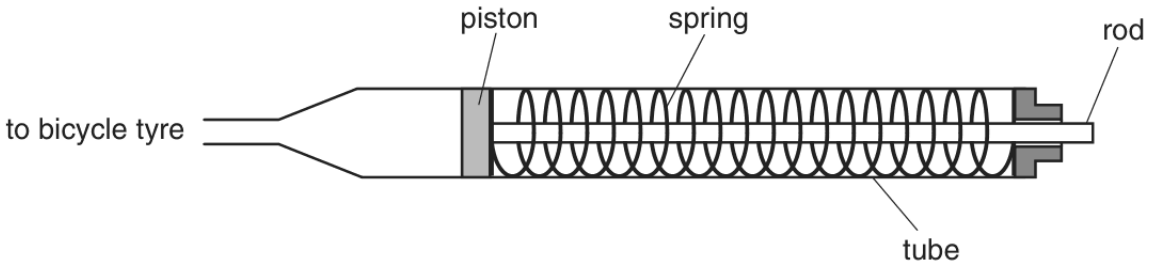


Fig. 3.1

The change in length of the spring is proportional to the force applied to the spring. A force of 2.8 N compresses the spring by 2.0 cm.

The student connects the pressure gauge to the tyre. Air from the tyre exerts a force on the piston. As the piston moves, the spring is compressed. The piston moves 10 cm to the right and then stops.

(a) Calculate the force exerted on the piston by the spring.

force = ...[1]

(b) The cross-sectional area of the piston is $3.0 \times 10^{-5}\,m^2$. Calculate the pressure of the air inside the tyre.

pressure = ...[2]

(c) Suggest one change to the apparatus so that the piston moves a smaller distance when measuring the same pressure.

.....[1]

(d) Each time that the pressure gauge is used, the pressure in the tyre falls slightly. The temperature stays constant. Describe, using ideas about molecules, why the pressure falls.

.....

.....

.....

.....[2]

3 **(a)** 14 N B1

(b) (P =) F/A algebraically in symbols or words in any form; $14/3.0 \times 10^{-5}$ C1
4.67×10^5 Pa; 4.7×10^5 Pa ecf **(a)** A1

(c) stiffer/stronger spring; piston has less area/diameter; smaller piston (and tube) B1

(d) molecules/particles/atoms collide with tyre/walls/piston B1
air/molecules enter gauge; fewer molecules in the tyre; fewer hits/sec;
less frequent hits; volume increases B1 [6]

QUESTION · 5054_s12_qp_12.pdf · source page 4 of 16

9 A metal wire, of initial length 1000 mm, extends by 4 mm when a load of 2 N is added to it.

What is the length of the wire when a further 3 N is added, assuming that the wire does not extend beyond the limit of proportionality?

A 1006 mm **B** 1008 mm **C** 1010 mm **D** 1012 mm

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 9

9	**C**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows apparatus used to obtain the readings for a graph of force against extension for a spring.

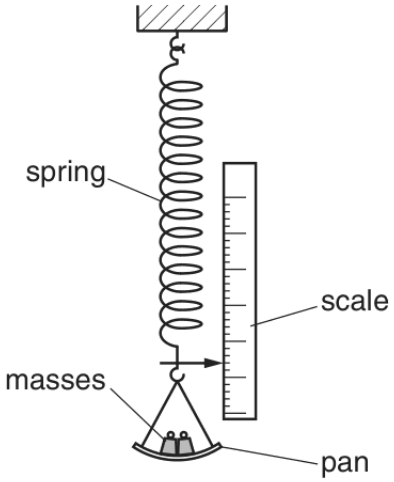


Fig. 1.1

The masses added to the pan produce a force that stretches the spring.

(a) (i) State what is meant by the *mass* of a body.

.....

.....[1]

(ii) Describe how the scale is used to find the extension of the spring.

.....

.....

.....[1]

(b) Fig. 1.2 shows the force-extension graphs for two different springs.

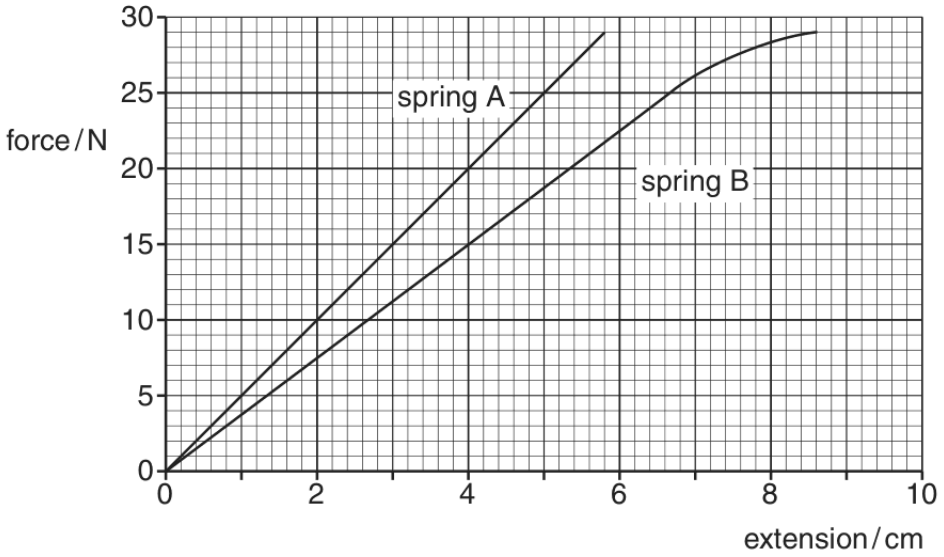


Fig. 1.2

(i) A student states that spring B is easier to stretch than spring A.

Use values from Fig. 1.2 to explain what the student means.

.....

.....

.....[1]

(ii) When a force of 25 N is applied, spring B reaches its limit of proportionality but spring A does not. Explain how Fig. 1.2 shows this.

.....

.....

.....[1]

(iii) The same force is applied to each spring.

Using Fig. 1.2, determine the force that produces an extension of spring B that is 1.0 cm greater than the extension of spring A.

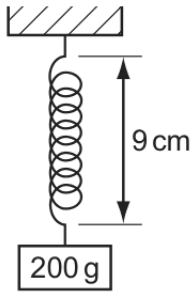
force = ...[1]

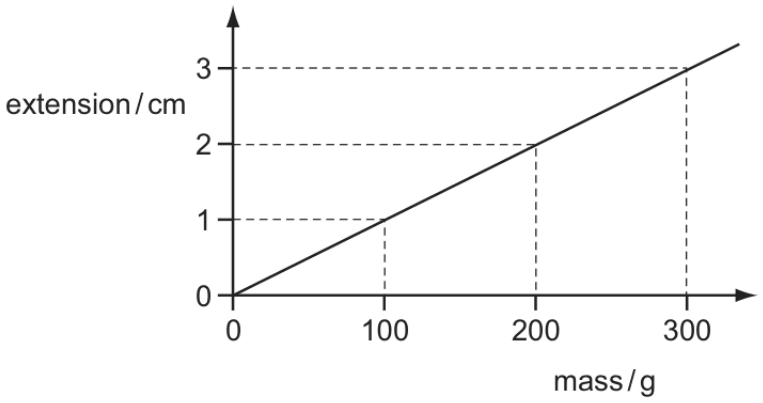
Section A

1	**(a)**	**(i)**	(amount of) matter/material/substance it contains	B1	
		(ii)	**use of scale** and subtraction/difference/increase in lengths/readings OR read distance between two marks **on the scale** with different masses	B1	
	(b)	**(i)**	two force **values** with $F_A > F_B$ for the same extension OR two extension **values** with $e_B > e_A$ for the same force/at maximum	B1	
		(ii)	idea that A is a straight line and B is not OR gradient constant in A but not in B OR same increase in F every cm for A but not B	B1	
		(iii)	15 N	B1	[5]

QUESTION · 5054_w12_qp_11.pdf · source page 5 of 16

9 A spring has a length of 9 cm when loaded with a 200 g mass. The extension-mass graph for the spring is shown.





The 200 g mass is replaced with a 100 g mass. What is the new length of the spring?

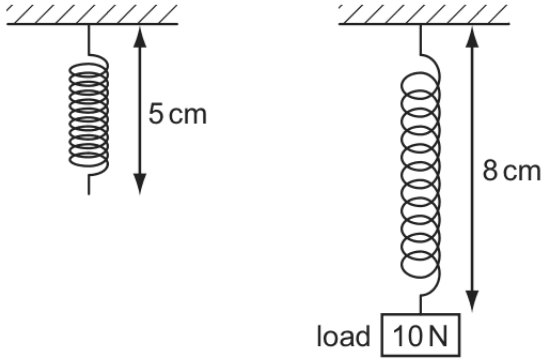
A 7 cm **B** 8 cm **C** 9 cm **D** 10 cm

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 9

9	**B**

QUESTION · 5054_w12_qp_12.pdf · source page 5 of 16

9 The diagram shows how the length of a spring changes when a load of 10 N is hung on it.



The 10 N load is replaced by a 20 N load. What is the new length of the spring?

A 6 cm **B** 11 cm **C** 14 cm **D** 16 cm

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 9

9	**B**

For Examiner's Use

2 A pupil hangs a load of 45 N from a length of copper wire. The wire extends by 2.5 mm and does not exceed the limit of proportionality.

(a) (i) The gravitational field strength is 10 N/kg. Calculate the mass of the 45 N load.

mass = ...[1]

(ii) Use the grid in Fig. 2.1 to plot the extension-load graph for this wire for a load between 0 and 45 N.

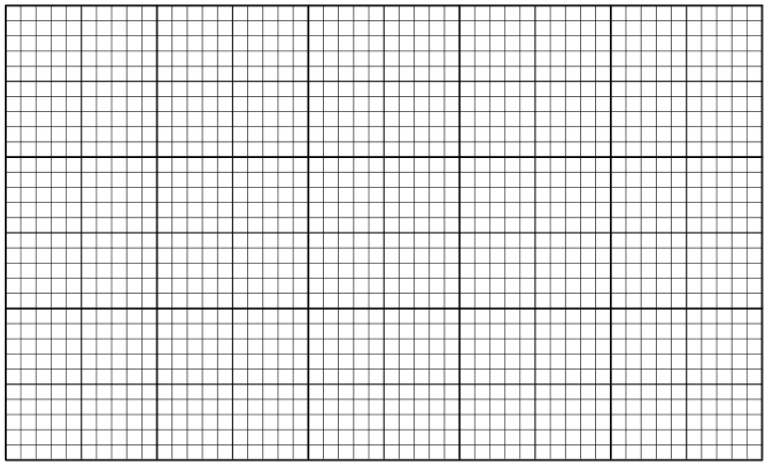


Fig. 2.1 [3]

(b) Use the graph plotted in Fig. 2.1 to determine the load needed to produce an extension of 1.3 mm.

load = ...[1]

2	**(a)**	**(i)**	4.5 kg	B1	
		(ii)	axes labelled with quantity **and** unit	B1	
			linear scale	B1	
			straight line from clear (0,0) to correct point	B1	
	(b)		answer from candidate's line	B1	[5]

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows how the length of a spring varies as the force applied to it increases.

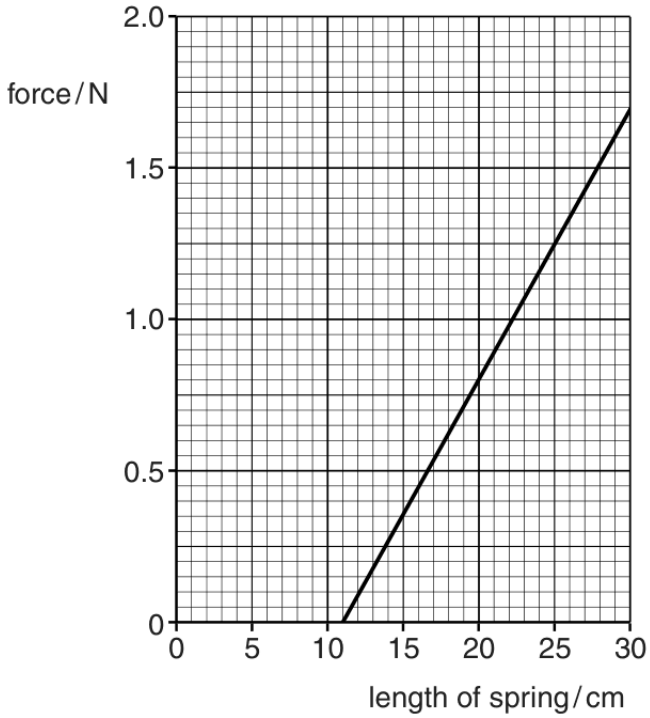


Fig. 1.1

(a) Determine the length of the unstretched spring.

length = ...[1]

(b) Explain how the graph shows that the limit of proportionality is not reached.

.....

.....[1]

For Examiner's Use

(c) The spring is attached to a mass M of 0.20 kg and placed on a frictionless surface, as shown in Fig. 1.2.

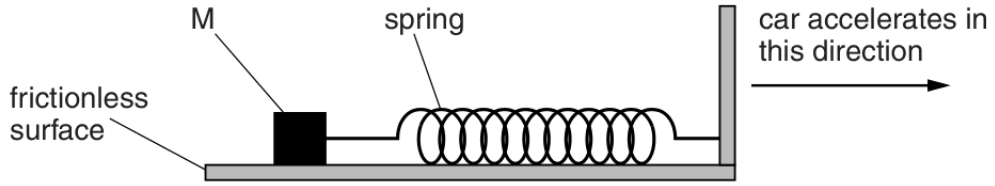


Fig. 1.2 (not to scale)

The apparatus is placed on the floor of a car.

When the car accelerates uniformly in the direction shown, the spring extends.

(i) State what is meant by a *uniform acceleration*.

.....

.....

.....[2]

(ii) The extension of the spring is 9.0 cm.

Using Fig. 1.1, determine

1. the horizontal force on M,

force = ...[1]

2. the acceleration of M.

acceleration = ...[2]

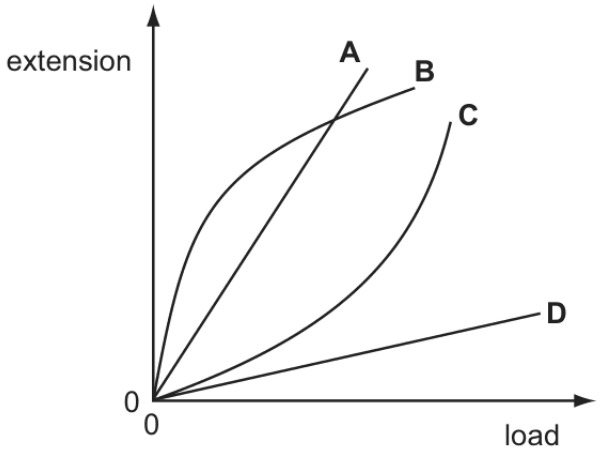
Section A

1	**(a)**	11 cm	B1	
	(b)	(graph is) a straight line/linear **or** has constant gradient **or** not curved	B1	
	(c) **(i)**	change in speed/velocity	M1	
		same change in uniform/same time **or** in 1 s	A1	
	(ii) **1.**	0.8 N	B1	
	2.	(a=) F/m algebraic or numerical e.g. $F = ma$; 0.8/0.2	C1	
		(ecf **1.** but **not** if $F = 0$)		
		4(.0) m/s^2	A1	[7]

QUESTION · 5054_s13_qp_12.pdf · source page 5 of 16

10 The graph shows extension-load curves for four fibres.

Which fibre is the hardest to stretch over the range of loads shown?



MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 10

10 **D**

3 Fig. 3.1 shows a man using a chest expander to increase the strength of his arms.

For Examiner's Use

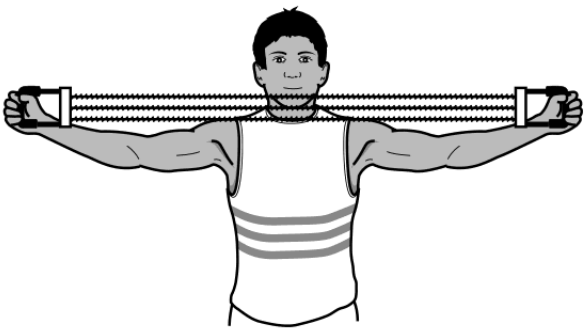


Fig. 3.1

The chest expander may be considered to be a single spring.

Fig. 3.2 shows the force needed to produce different extensions of the spring.

force / N	extension / cm
0	0
60	10
120	20
180	30

Fig. 3.2

(a) The man increases the force on the spring from 0 to 180 N.
The spring extends by 30 cm and the average force exerted during this process is 90 N.

(i) Calculate the work done on the spring.

work = ... [2]

(ii) Twenty extensions are made in 1.0 minute.

Calculate the power used to extend the spring.

power = ... [2]

(b) The force at the limit of proportionality of the spring is 800 N.

For Examiner's Use

(i) Calculate the extension of the spring when the force is 800 N.

extension = ... [2]

(ii) The force on the spring is increased from 800 N to 860 N.

Suggest what happens to the extension.

.....

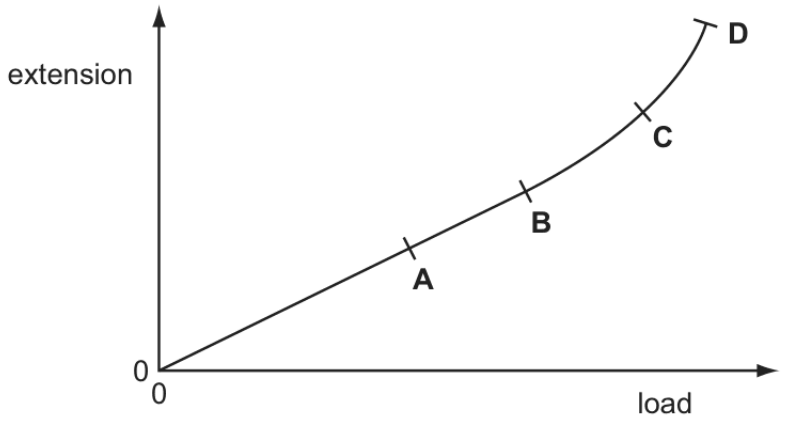
..... [1]

3	**(a)**	**(i)**	(W=) Fd **or** 90×0.3 **or** 90×30	C1	
			27 J	A1	
		(ii)	(P=) W/t **or** Fd/t **or** $27(\times 20)/60$ **or** $27/3$	C1	
			9(.0) W	A1	
	(b)	**(i)**	$800 \times 30/180$ **or** 800/6 **or** 6 seen **or** proportionality clearly used	C1	
			133 **or** 130 cm	A1	
		(ii)	extension more than 143 cm **or** (extra) extension > 10 cm **or** (some) extension permanent	B1	[7]

QUESTION · 5054_w13_qp_11.pdf · source page 5 of 20

8 An extension-load graph is plotted to show the result of increasing the load on a spring.

Which point marks the limit of proportionality for this spring?



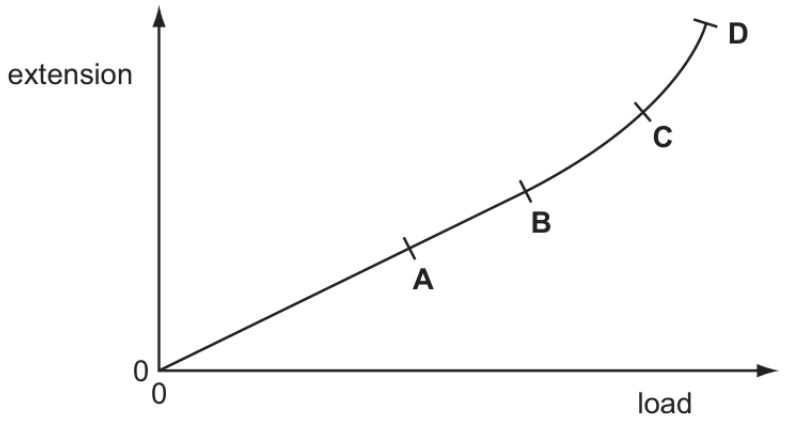
MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 8

8	**B**

QUESTION · 5054_w13_qp_12.pdf · source page 6 of 20

9 An extension-load graph is plotted to show the result of increasing the load on a spring.

Which point marks the limit of proportionality for this spring?

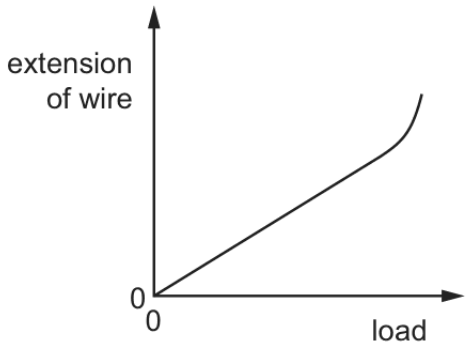


MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 9

9 **B**

QUESTION · 5054_s14_qp_11.pdf · source page 5 of 16

12 The graph shows the extension of a piece of copper wire as the load on it is increased.



What does the graph show?

A At a certain load the wire becomes easier to extend.

B At a certain load the wire becomes harder to extend.

C The load and extension are directly proportional for all loads.

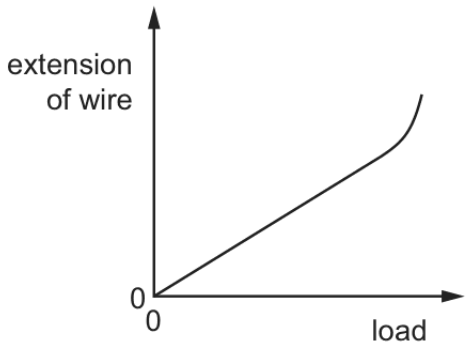
D The load and extension are inversely proportional for all loads.

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 12

12	**A**

QUESTION · 5054_s14_qp_12.pdf · source page 5 of 16

12 The graph shows the extension of a piece of copper wire as the load on it is increased.



What does the graph show?

A At a certain load the wire becomes easier to extend.

B At a certain load the wire becomes harder to extend.

C The load and extension are directly proportional for all loads.

D The load and extension are inversely proportional for all loads.

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 12

12	**A**

QUESTION · 5054_w14_qp_11.pdf · source page 6 of 20

8 Which object behaves as an elastic solid as it deforms?

A a bullet as it hits a solid metal wall

B a car damaged in a collision

C a piece of metal cut by a saw

D a football as it is kicked

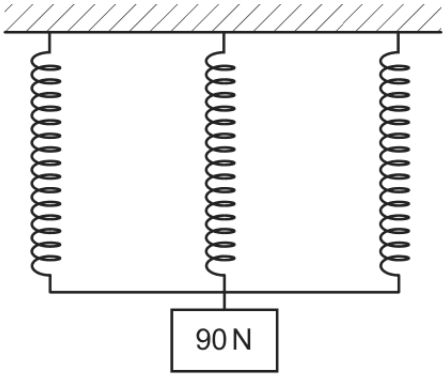
MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 8

8	**D**

QUESTION · 5054_w14_qp_11.pdf · source page 7 of 20

9 A spring extends by 6.0 cm when a 15 N weight is suspended from its lower end.

Three of these springs are used as shown to support a 90 N weight.



What is the extension of each spring?

A 2.0 cm **B** 12 cm **C** 36 cm **D** 110 cm

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 9

9 **B**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A student suspends a spring from a clamp stand. He measures the unstretched length of the spring with no load attached. He then attaches different loads to the lower end of the spring and measures the new lengths of the spring.

Fig. 1.1 shows how the total length of the spring depends on the load attached.

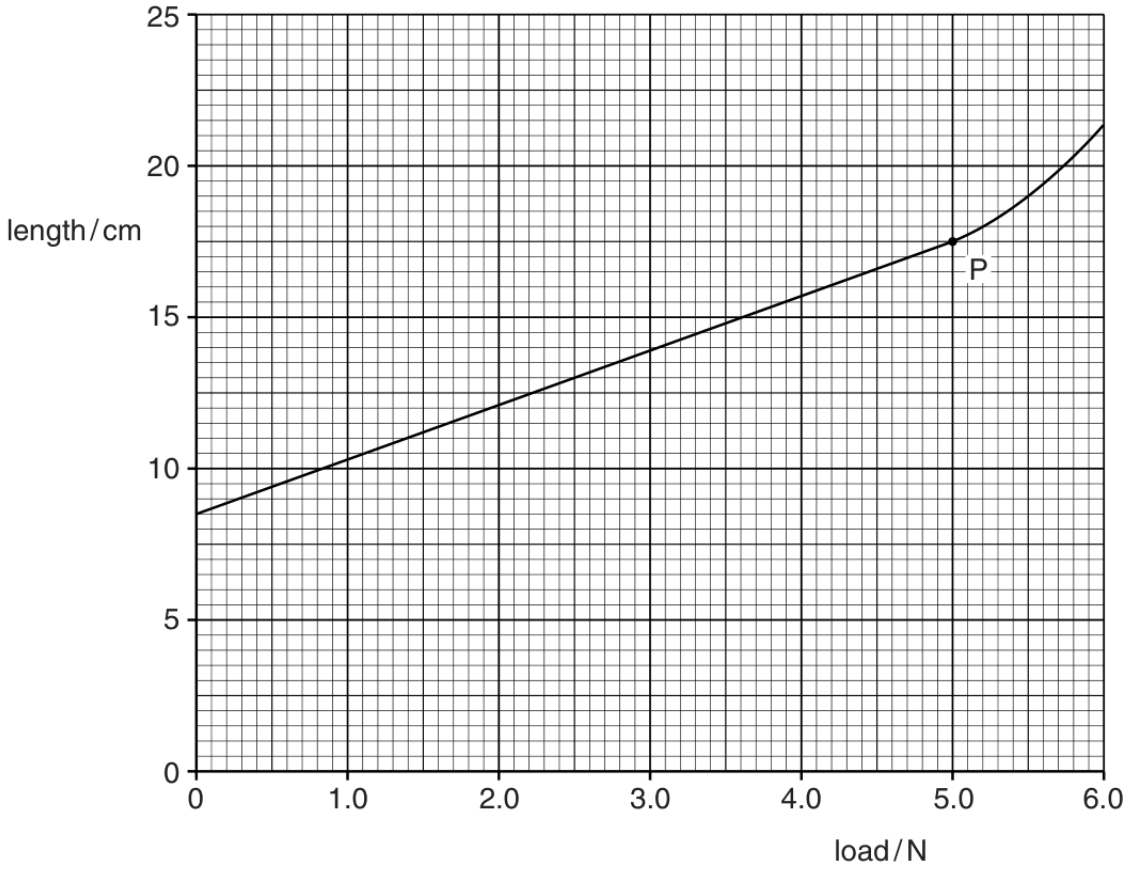


Fig. 1.1

(a) At point P on Fig. 1.1, the line stops being straight and begins to curve.

State the name of point P.

.....[1]

(b) **(i)** Using Fig. 1.1, determine the unstretched length of the spring.

unstretched length = ...[1]

(ii) Calculate the extension of the spring for a load of 4.0 N.

extension = ...[1]

(c) The student attaches a small block of wood to an identical spring. The extension of the spring is 2.7 cm. The gravitational field strength g is 10 N/kg.

Calculate the mass of the block of wood.

mass = ...[2]

Section A

1	**(a)**	limit of proportionality (<u>not</u> breaking point)	B1	
	(b) **(i)**	8.5 cm **cao**	B1	
	(ii)	7.1 – 7.3 cm	B1	
	(c)	4.0 × (2.7 / 7.2) **or** 5.0 × (2.7 / 9.0) **or** 1.5 (N) **or** read from graph **or** 11.2 (cm)	C1	
		0.148 – 0.152 kg **or** 148 – 152 g	A1	**[5]**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 The apparatus shown in Fig. 1.1 is used to measure the extension of a spring.

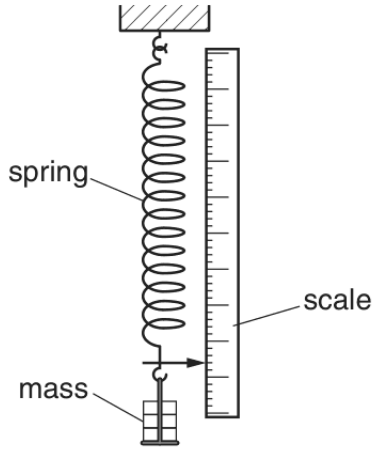


Fig. 1.1

(a) Explain how the mass causes a force on the spring.

.....

..... [1]

(b) The force on the spring is a vector quantity. State what is meant by a *vector* quantity.

.....

..... [1]

(c) Fig. 1.2 shows a graph of the length of the spring plotted against the force on the spring, for forces between 0 and 10 N.

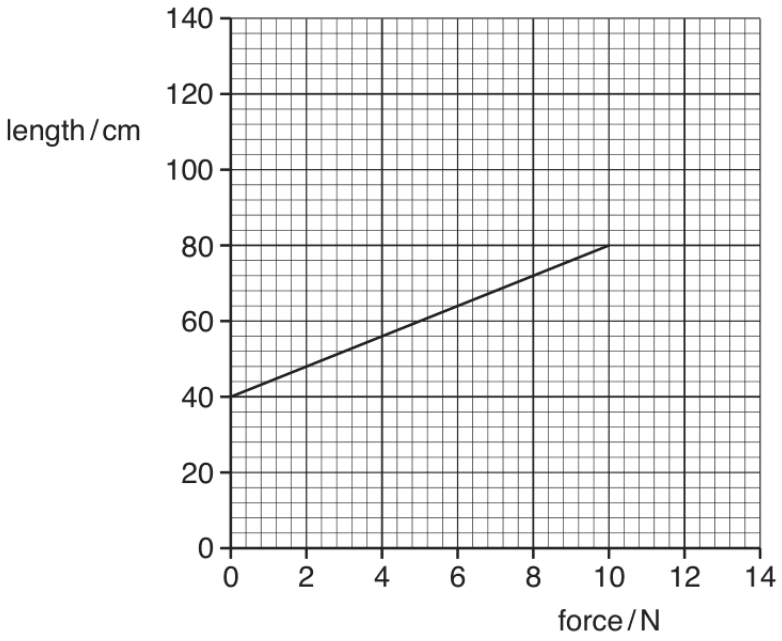


Fig. 1.2

(i) **1.** State a formula that relates the unstretched length l_0 of the spring, the stretched length l of the spring and the extension e of the spring.

..... [1]

2. A mass produces a force of 9.0 N on the spring. Determine the extension of the spring caused by this mass.

extension = ... [1]

(ii) The limit of proportionality of the spring is reached when the force is 10 N.

The spring is easier to stretch after the limit of proportionality.

On Fig. 1.2, continue the line to suggest how the length changes when the force is greater than 10 N. [1]

1 **(a)** weight (pulls spring down and causes tension) B1
or force/pull of **gravity**
or mass is in gravitational field

(b) has a direction B1

(c) **(i)** **1** $l = l_0 + e$ **or** $l_0 = l - e$ **or** $e = l - l_0$ B1

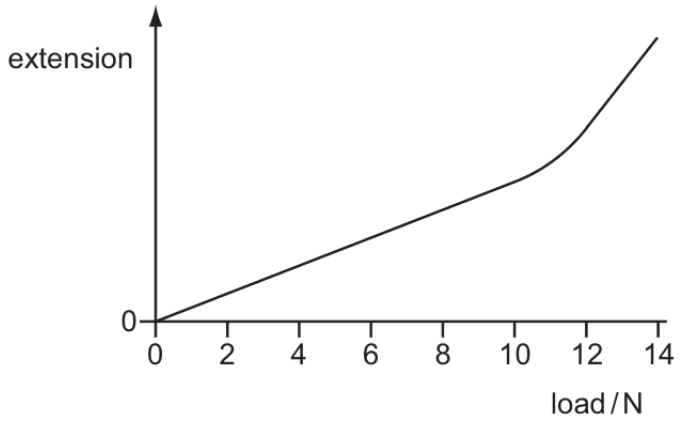
2 36 cm B1

(ii) curve upwards after 10 N B1

QUESTION · 5054_w15_qp_11.pdf · source page 5 of 16

10 Some students plan to use a spring to make a spring balance with a linear scale.

The graph shows how the extension of their spring changes with the load on it.



What is the total range of a balance with a linear scale using this spring?

A 0 N to 10 N **B** 0 N to 12 N **C** 10 N to 12 N **D** 10 N to 14 N

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 10

10	**A**

2 **(a)** The surface of a running track is made of rubber. A heavy trolley is pulled on to the track and it exerts a large force on the rubber track.

State two effects that this force has on the rubber.

1. .....

2. .....
[2]

(b) A spring is suspended from a support with a small pan attached to its lower end. Masses are added to the pan until the spring is extended well beyond the limit of proportionality.

Fig. 2.1 shows this apparatus.

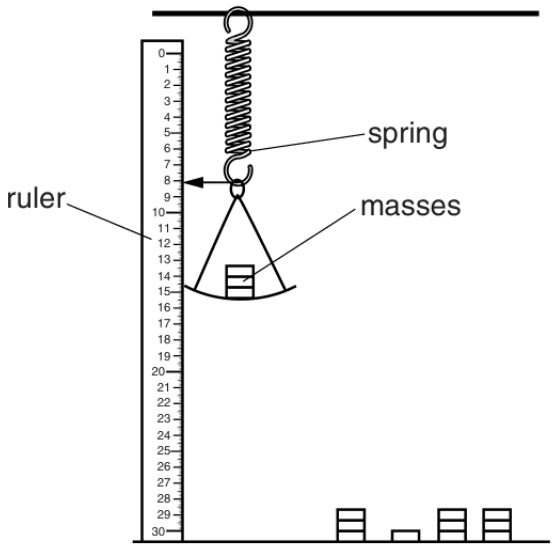


Fig. 2.1

A ruler is used when determining the extension of the spring.

(i) On Fig. 2.2, sketch the extension-load graph for the spring and label the limit of proportionality P. [2]

Fig. 2.2

(ii) The masses are then removed and the extension of the spring decreases.

Suggest what is observed when all the masses are removed.

.....

..... [1]

2 **(a)** any **two** from different lines of:
distort / stretch / change in shape / squeezed / bends / deforms
compresses / change in size / volume / density / depth / height
change in temperature / gets hot(ter) / generates heat B2

(b) **(i)** straight line from origin B1
upward curve labelled / clear from limit of proportionality B1

(ii) permanent extension **or** spring is longer than it was originally B1 [5]

2 A student hangs various masses from the end of a spring and determines the extension produced. Fig. 2.1 shows the spring hanging vertically on its own and with an unknown mass X at one end.

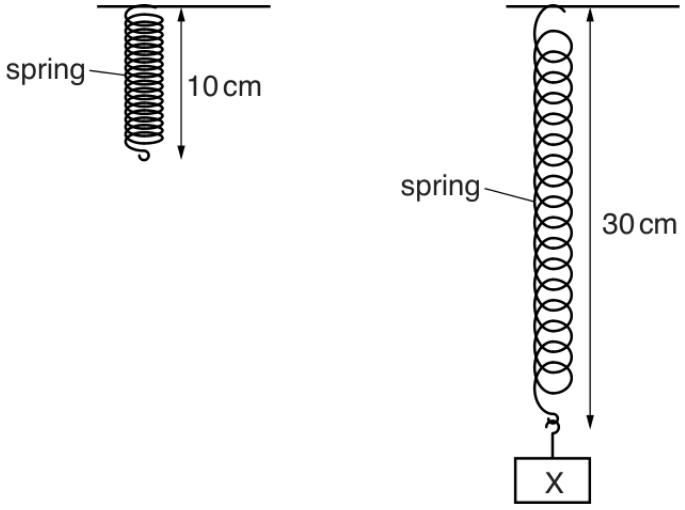


Fig. 2.1 (not to scale)

(a) The student plots a graph of the extension of the spring against the mass hanging on the spring. Fig. 2.2 shows the student's graph.

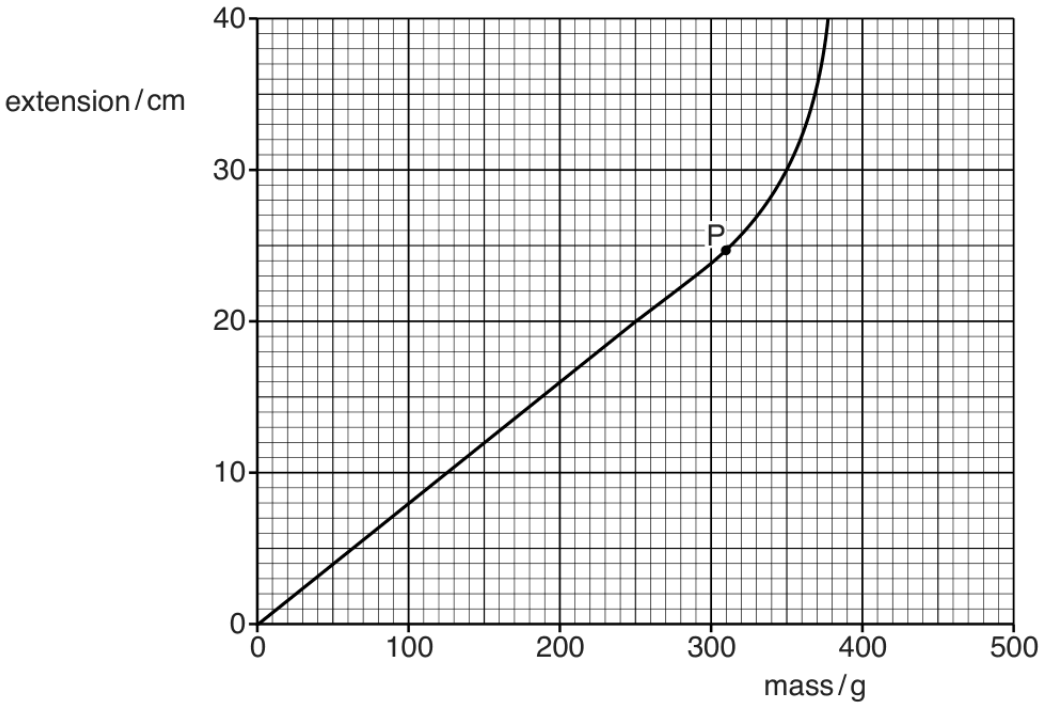


Fig. 2.2

(i) At point P on the graph, the line begins to curve.

State the name of point P.

.....[1]

(ii) Using Fig. 2.1 and Fig. 2.2, determine the mass X.

mass = ... [1]

(iii) The gravitational field strength g is 10 N/kg.

Calculate the weight of X.

weight = ... [1]

(b) An identical spring is used with the original spring, as shown in Fig. 2.3.

Together they support the mass X.

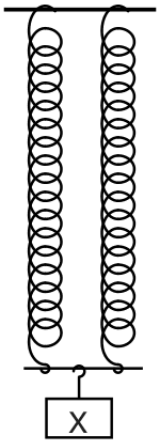


Fig. 2.3 (not to scale)

State and explain how the extension in Fig. 2.3 compares with the extension in Fig. 2.1.

.....

.....

.....

.....[2]

2	**(a)**	**(i)**	limit of proportionality	B1
		(ii)	250 g	B1
		(iii)	2.5 N	B1
	(b)		half the extension / 10 cm	B1
			each / both / another spring shares / distributes the weight / mass **or** both springs bear / carry the load	B1

3 A student has three springs A, B and C. He measures the length of each spring, in turn, when different weights are placed on the end of each spring.

His table of results is shown in Fig. 3.1.

weight/N	length of spring A/cm	length of spring B/cm	length of spring C/cm
1.0	6.1	8.6	9.7
1.5	6.9	9.5	10.5
2.0	7.7	10.4	11.3
2.5	8.5	11.2	13.1
3.0	9.3	12.1	16.9

Fig. 3.1

(a) (i) State which spring has been stretched past the limit of proportionality.

.....

(ii) Using data from Fig. 3.1, explain how you obtained your answer to **(a)(i)**.

.....

.....

.....

.....

[2]

(iii) Calculate the unstretched length of spring A.

unstretched length = ...[1]

(b) Describe how the student can use spring A to determine the mass of a small rock.

.....

.....

.....

.....

.....[2]

3 (a) (i) C M1

(ii) **data** quoted to prove stretches more at end A1
or extensions / changes in length increase / are not the same (at higher loads)

(iii) 4.5 cm B1

(b) (tie rock to spring A) B1
- find weight / force / newtons using length or extension **and** graph **or** match readings (in table)
- find known weight / mass / force / N that gives same extension of spring
- use of proportionality with length or extension
- extension (in cm) / 1.6

(mass =) weight / g B1
or weight / gravitational field (strength)

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 A spring is taken from a drawer in a physics laboratory.

(a) Fig. 8.1 is the extension-load graph for the spring.

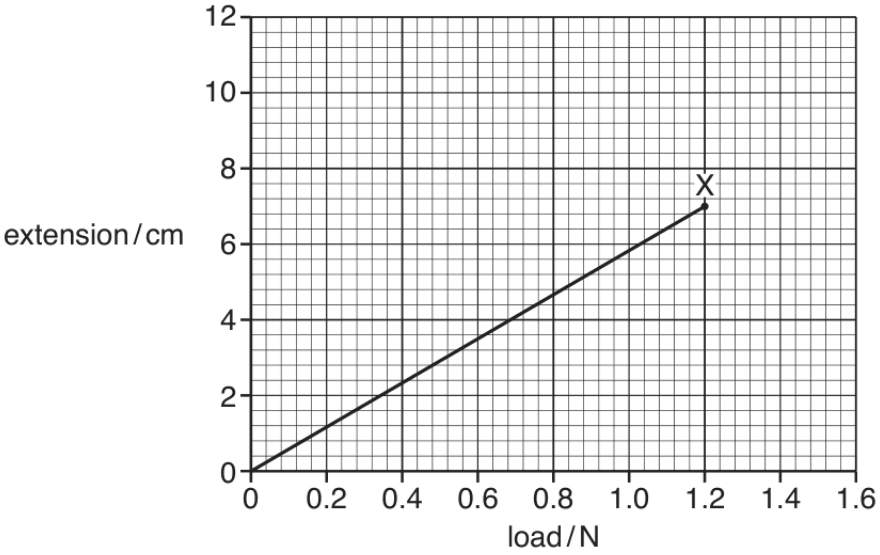


Fig. 8.1

(i) Using Fig. 8.1, determine the load on the spring when its extension is 4.9 cm.

load = ...[1]

(ii) The point X on Fig. 8.1 is the limit of proportionality for the spring.

On Fig. 8.1, sketch a line to show how the extension-load graph might continue after X. [2]

(b) A rectangular, plastic block is suspended from the lower end of the spring. Fig. 8.2 shows that the bottom 3.5 cm of the plastic block is in the water in a beaker.

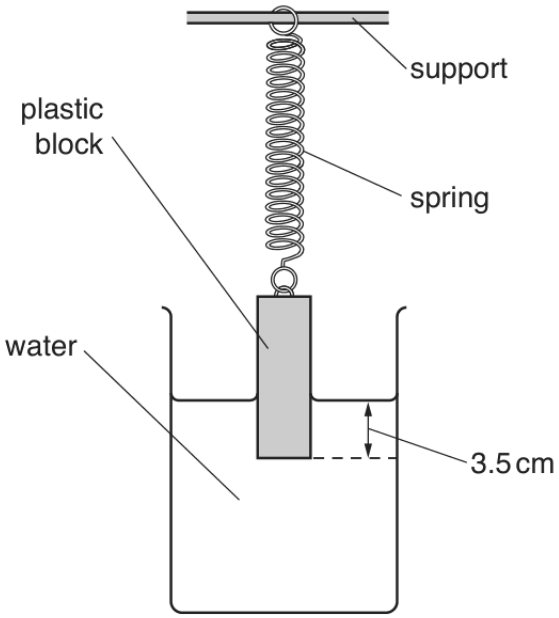


Fig. 8.2 (not to scale)

The density of water is 1000 kg / m^3 and the gravitational field strength g is 10 N / kg. The base of the plastic block is a square with sides of 4.0 cm.

(i) Calculate the pressure due to the water at a depth of 3.5 cm below the surface of the water.

pressure = ...[3]

(ii) Calculate the upward force on the base of the block due to the pressure of the water.

force = ...[2]

(iii) In Fig. 8.2, the extension of the spring is 4.9 cm.

Using the answer from **(a)(i)** and the answer from **(b)(ii)**, determine the weight of the block.

weight = ...[1]

(c) In the equipment shown in Fig. 8.2, the pressure of the atmosphere is transmitted through the water. The total pressure in the water at a depth of 3.5 cm is larger than the pressure due to the water alone.

(i) Explain why, in this experiment, atmospheric pressure does not affect the position of the plastic block in the water.

.....

.....

.....

.....[2]

(ii) Force is a vector quantity and pressure is a scalar quantity.

State how a *vector* quantity differs from a *scalar* quantity.

.....

.....[1]

(d) The beaker in Fig. 8.2 is gradually moved upwards so that more of the plastic block is in the water.

Explain what happens to the forces that act on the block.

.....

.....

.....

.....

.....

.....

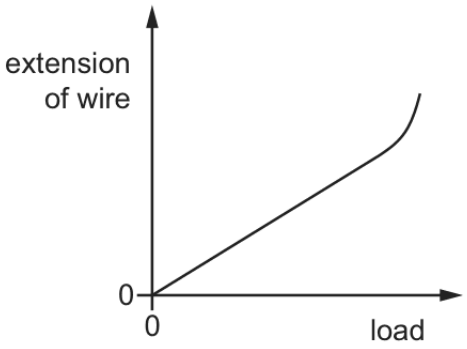
.....[3]

Section B

8	**(a)**	**(i)**	0.83 – 0.86 N	B1	
		(ii)	line curved	B1	
			line (curved) upwards	B1	[3]
	(b)	**(i)**	(P =)$h\rho g$	C1	
			0.035 × 1000 × 10 **or** 3.5 × 1000 × 10 **or** 35 × 1000 ×10	C1	
			350 Pa	A1	
		(ii)	(F =)PA **or** 350 × 0.0016 **or** 350 × 16 **or** 5600	C1	
			0.56 N	A1	
		(iii)	1.4 N **or (a)(i)** + **(b)(ii)** calculated	B1	[6]
	(c)	**(i)**	(atmospheric pressure) exerts a downward force / pressure (on top of the block)	B1	
			(cancels out the) extra upward force / pressure	B1	
		(ii)	(vector) has direction (in addition to magnitude)	B1	[3]
	(d)	any **three** lines from force due to water increases force due to spring decreases increased pressure (at base) they add to give a constant value / weight of block **or** total force constant		B3	[3]
					[15]

QUESTION · 5054_s17_qp_11.pdf · source page 6 of 16

11 The graph shows the extension of a piece of copper wire as the load on it is increased.



What does the graph show?

A At a certain load, the wire becomes easier to extend.

B At a certain load, the wire becomes harder to extend.

C The load and the extension are directly proportional for all loads.

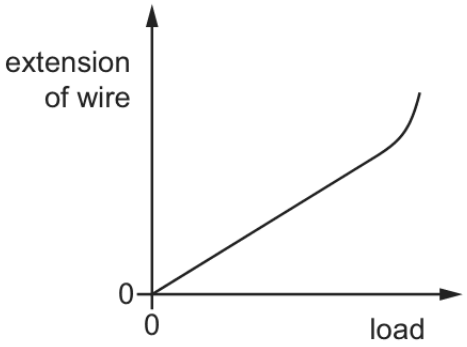
D The load and the extension are inversely proportional for all loads.

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 11

11	A	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 6 of 20

12 The graph shows the extension of a piece of copper wire as the load on it is increased.



What does the graph show?

A At a certain load, the wire becomes easier to extend.

B At a certain load, the wire becomes harder to extend.

C The load and the extension are directly proportional for all loads.

D The load and the extension are inversely proportional for all loads.

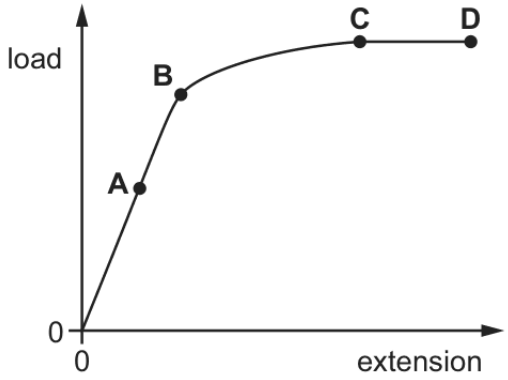
MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 12

12	A	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 6 of 20

13 The diagram shows a load-extension graph for a wire.

Which point is the limit of proportionality?



MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 13

13	B	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 5 of 20

7 A metal wire is suspended from a support and loads of different mass are attached to the lower end.

The table shows how the extension of the wire depends on the mass of the load.

mass / kg	0	2.0	4.0	6.0	8.0	10.0
extension / mm	0	0.35	0.70	1.05	1.50	2.05

Between which values of mass does the limit of proportionality lie?

A 0 and 2.0 kg

B 4.0 kg and 6.0 kg

C 6.0 kg and 8.0 kg

D 8.0 kg and 10.0 kg

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

2 A force applied to a solid object may cause it to accelerate so that its velocity changes.

(a) State two other properties of the object which may change when a force is applied.

1. .....

2. .....

[2]

(b) A spring has a mass of 0.012 kg.

(i) The gravitational field strength g is 10 N/kg.

Calculate the weight of the spring.

weight = ...[1]

(ii) The spring is suspended vertically and a load is attached to its lower end. The spring extends by 2.7 cm and does not pass the limit of proportionality.

Fig. 2.1 shows the spring and load attached to the lower end of a second identical spring.

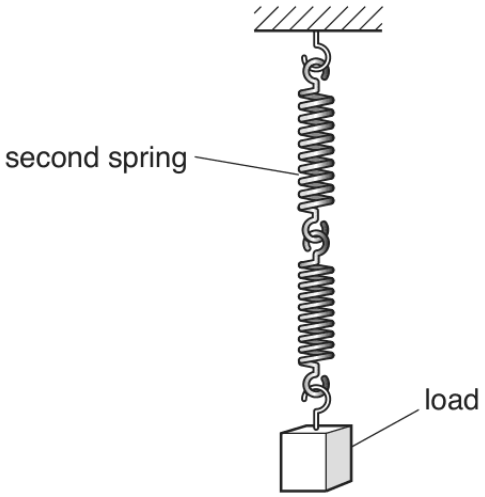


Fig. 2.1

Suggest one reason why the extension of the second spring differs from 2.7 cm.

.....

.....

.....[1]

(iii) Explain what is meant by *limit of proportionality*.

.....

.....

.....[1]

(c) The load in Fig. 2.1 is pulled down below its equilibrium position.

(i) State the form of energy stored in the stretched springs.

.....[1]

(ii) The load is released and it moves upwards and downwards. The distance travelled in each movement decreases until the load stops moving.

Explain why the load stops moving.

.....

.....

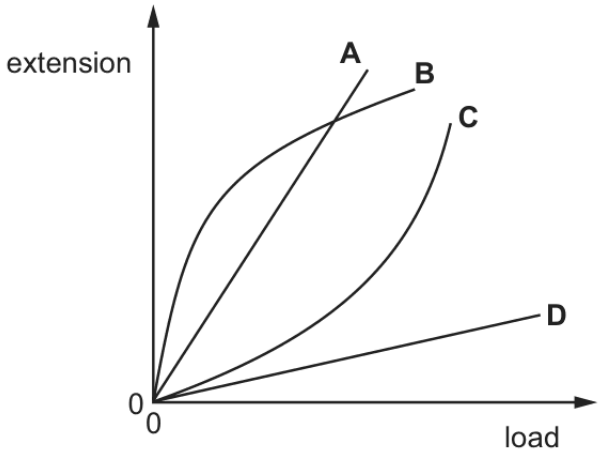
.....[1]

Question	Answer	Marks
2(a)	shape	**B1**
	size / volume / density	**B1**
2(b)(i)	0.12 N	**B1**
2(b)(ii)	weight of first spring (increases extension) **or** weight of first spring (takes spring beyond the limit of proportionality)	**B1**
2(b)(iii)	where / point / load / extension / limit beyond which the extension is not directly proportional to load / extension-load graph is not straight	**B1**
2(c)(i)	elastic potential energy	**B1**
2(c)(ii)	friction / air resistance mentioned **or** thermal energy / heat produced	**B1**

QUESTION · 5054_s18_qp_11.pdf · source page 4 of 20

8 The graph shows extension-load curves for four fibres.

Which fibre is the most difficult to stretch over the range of loads shown?



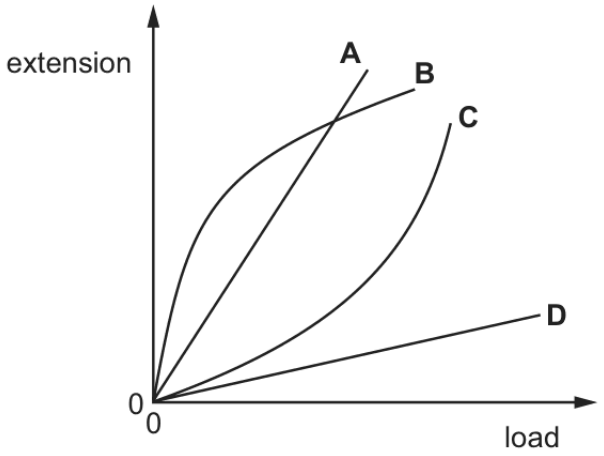
MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 8

8	D	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 6 of 16

12 The graph shows extension-load curves for four fibres.

Which fibre is the most difficult to stretch over the range of loads shown?



MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 12

12	D	**1**

2 In an experiment, a student takes measurements and determines the extension of a spring for different loads. The apparatus is shown in Fig. 2.1.

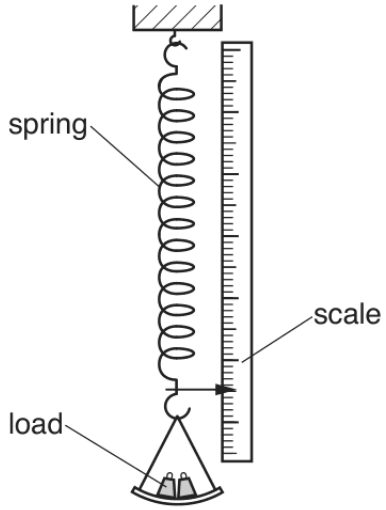


Fig. 2.1

The table in Fig. 2.2 shows the results.

load/N	0	1.0	2.0	3.0	4.0	5.0
length of spring/mm	200	235	270	305	350	420
extension/mm	0	35	70	105		

Fig. 2.2

(a) Only some of the extensions are shown in the table.

Complete the table to show all of the extensions. [1]

(b) Calculate the load that produces an extension of 49 mm.

load = ...[2]

(c) The student pulls a load downwards from position A to position B and holds it fixed at position B, as shown in Fig. 2.3.

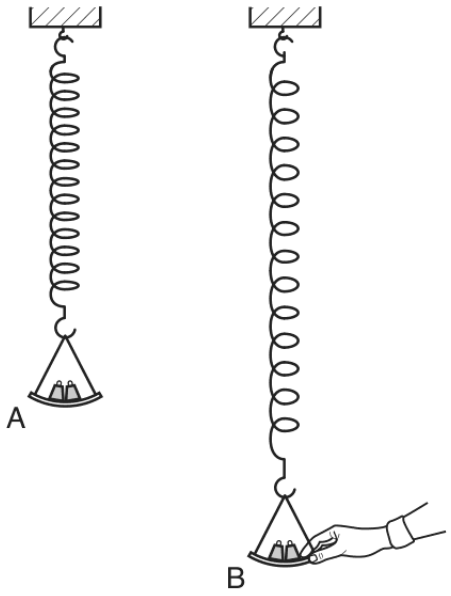


Fig. 2.3

The load is stationary at A and at B. The load has no kinetic energy at either point.

(i) Place ticks (✓) in the boxes to show how the value of each of the forms of energy compares at A and B.

	equal at A and B	larger at A	larger at B
energy stored in spring	☐	☐	☐
gravitational potential energy of load	☐	☐	☐

[1]

(ii) Work is done by the student to pull the load down. The law of conservation of energy states that energy cannot be created and cannot be destroyed.

Explain how this principle applies in this case.

.....

.....

.....

.....[2]

Question	Answer	Marks
2(a)	150 and 220 in table	**B1**
2(b)	any use of proportionality	**C1**
	1.4 N	**A1**
2(c)(i)	1st line larger at B **and** 2nd line larger at A	**B1**
2(c)(ii)	student loses energy / chemical energy decreases **or** (gravitational) potential energy decreases **and** spring energy increases	**B1**
	idea that increase in spring energy > change / decrease in (grav) PE (so student provides work needed)	**B1**

3 A student uses the apparatus shown in Fig. 3.1 to obtain the extension-load graph of a spring.

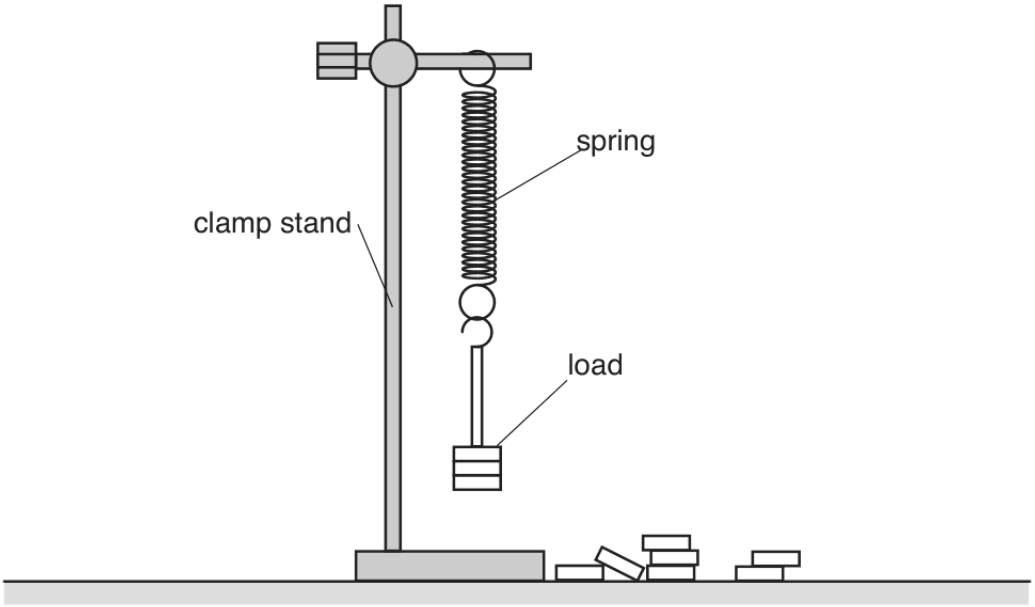


Fig. 3.1 (not to scale)

The weights of the loads are known. The student suspends loads of increasing size from the spring and determines the different extensions produced.

Fig. 3.2 is the graph the student obtains.

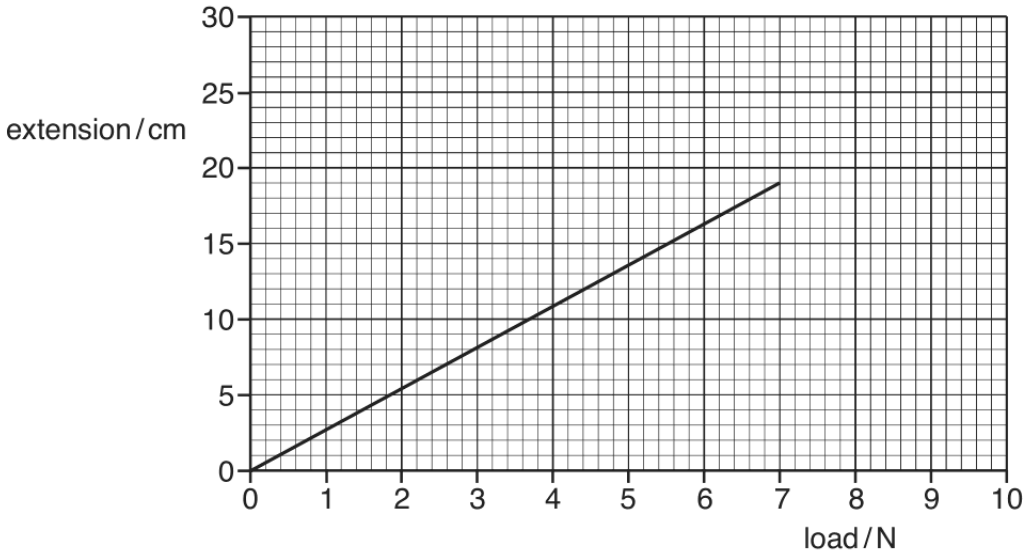


Fig. 3.2

The spring reaches its limit of proportionality when the load is 7.0 N.

(a) On Fig. 3.2, sketch the shape of the graph for loads greater than 7.0 N. [1]

(b) All the loads are removed from the spring. A block of wood is suspended from the end of the spring. The length of the spring increases from 4.9 cm to 13.4 cm.

Determine the weight of the wooden block.

weight = ... [2]

(c) The student takes a second spring that is identical to the one in Fig. 3.1 and hangs it next to the first spring.

Fig. 3.3 shows a light rod passing through the rings at the lower ends of the springs.

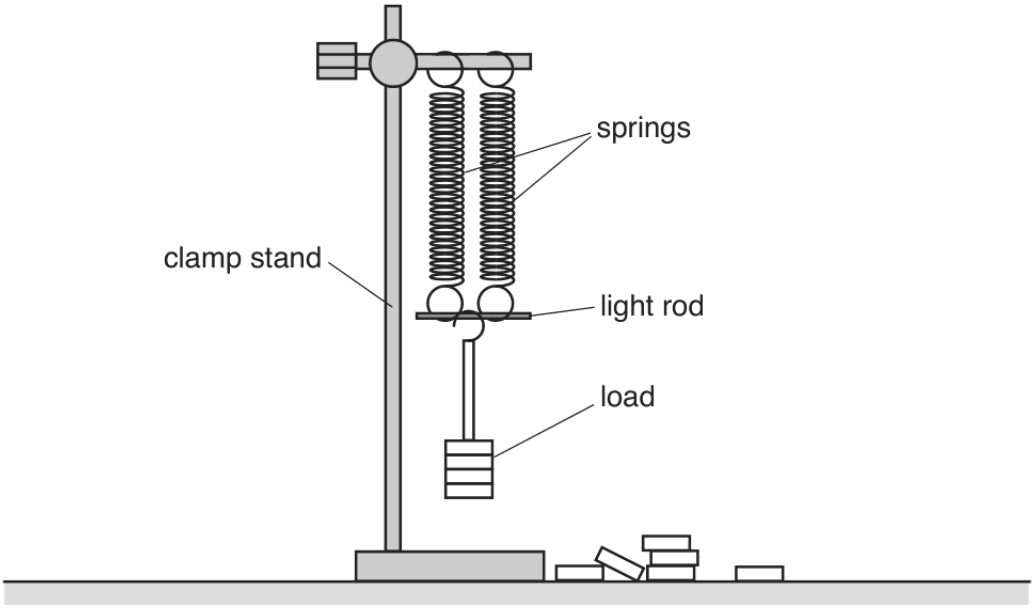


Fig. 3.3 (not to scale)

The student repeats the experiment and gradually increases the load to 9.0 N.

State and explain whether the limit of proportionality is reached.

.....

.....

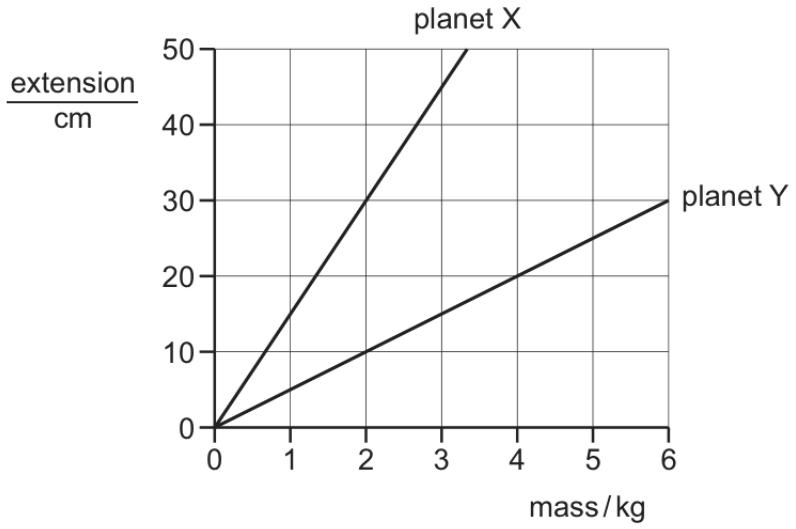
.....

..... [2]

Question	Answer	Marks
3(a)	line continues from (7.0, 0.19) **and** curves upwards	**B1**
3(b)	(extension =) 8.5 (cm)	**C1**
	3.1–3.2 N	**A1**
3(c)	(limit of proportionality is) not reached **and** load shared / distributed equally **or** tension in each spring 4.5 N **or** limit of proportionality now 14 N	**B1**
	or tension in each spring halved **or** tension in each spring < 7.0 N	**B1**

QUESTION · 5054_w18_qp_12.pdf · source page 3 of 16

6 The graph shows how the extension of a spring changes with the masses suspended from it when the spring is on planet X and when the spring is on planet Y.



Which conclusion can be drawn from these graphs?

A It is not possible to compare the gravitational field strengths on planets X and Y.

B The gravitational field strength on planet X is equal to the gravitational field strength on planet Y.

C The gravitational field strength on planet X is one third of the gravitational field strength on planet Y.

D The gravitational field strength on planet X is three times the gravitational field strength on planet Y.

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

3 A large spring is stretched by an athlete to increase the strength of his arms.

The athlete extends the spring, as shown in Fig. 3.1, and then releases it gently. He extends and releases the spring several times during a period of 60 s.

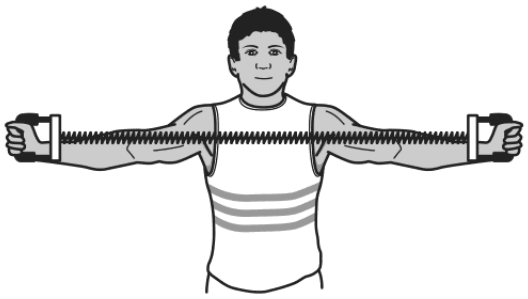


Fig. 3.1

(a) State the form of energy stored in the stretched spring and the form of energy stored in the muscles of the athlete.

form of energy in spring .....

form of energy in muscles .....
[2]

(b) Suggest what happens if he extends the spring beyond the limit of proportionality.

.....

..... [1]

(c) During one extension of the spring, its length increases from 70 cm to 93 cm. The average force exerted by the athlete is 400 N.

(i) Calculate the work done by the athlete in one extension.

work done = ... [2]

(ii) Describe how the athlete can use the value obtained in **(i)** to determine his average power during the 60 s.

.....

.....

.....

..... [2]

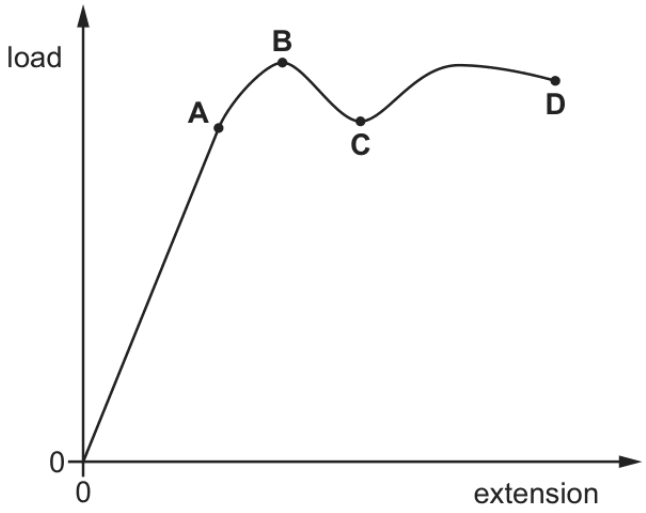
[Total: 7]

Question	Answer	Marks
3(a)	*in spring:* elastic or strain (potential energy)	**B1**
	in muscles: chemical (potential energy)	**B1**
3(b)	spring extends a different distance for the same (extra) load **or** load is no longer proportional to extension **or** becomes easier to pull	**B1**
3(c)(i)	(W =) Fd numerical or algebraic; 400×0.23	**C1**
	92 J	**A1**
3(c)(ii)	measure / determine the no. of times he extends the spring in 60 s	**B1**
	power = (total) work / time explained	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 5 of 16

12 The diagram shows a graph of load against extension for a material.

At which point is the limit of proportionality?



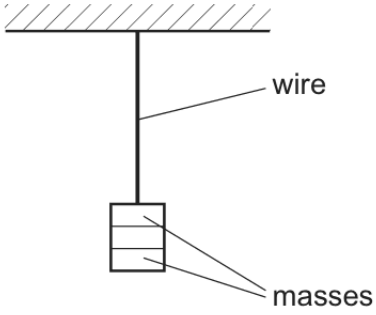
MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 12

12	A	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 6 of 16

13 A length of metal wire is made from a radioactive isotope of the metal.

The diagram shows masses suspended from the end of the wire.



The masses are removed from the wire.

Which property changes?

A half-life of the isotope

B length of the wire

C mass of the wire

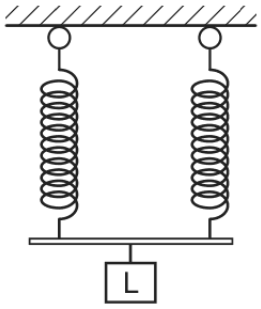
D melting point of the metal

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 13

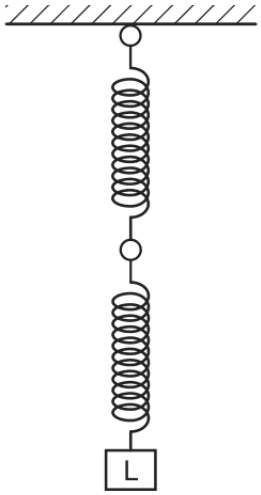
13	B	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 7 of 20

13 A load L is suspended from two springs that are in parallel. The extension of each spring is x.



The springs are then arranged to hang vertically, one below the other.



In this new arrangement, what is the total extension of the two springs?

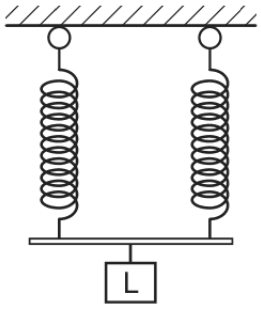
A $\frac{1}{2}x$ **B** x **C** $2x$ **D** $4x$

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 13

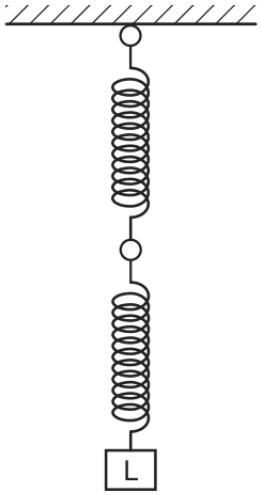
13	D	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 6 of 16

11 A load L is suspended from two springs that are in parallel. The extension of each spring is x.



The springs are then arranged to hang vertically, one below the other.



In this new arrangement, what is the total extension of the two springs?

A $\frac{1}{2}x$ **B** x **C** $2x$ **D** $4x$

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 11

11	D	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 Fig. 8.1 shows an elastic rope (bungee).

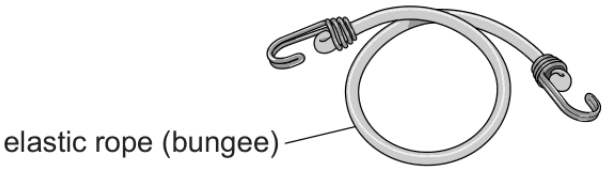


Fig. 8.1

The unstretched length of the rope is 0.80 m.

A student uses the hook at one end to suspend the rope from a shelf and hangs an empty paint can of mass 0.70 kg from the other end. The rope stretches to a length of 0.97 m but does not exceed its limit of proportionality.

(a) Explain what is meant by *limit of proportionality*.

.....

.....

..... [2]

(b) State what is meant by:

(i) *mass* ..... [1]

(ii) *weight.* .....

..... [2]

(c) The gravitational field strength is 10 N/kg.

(i) Calculate the weight of the empty paint can.

weight = ... [1]

(ii) For the rope, calculate a value for $\frac{\text{load}}{\text{extension}}$.

$\frac{\text{load}}{\text{extension}}$ = ... [1]

(d) The student repeatedly pours small volumes of paint into the can.

When there is $2.5 \times 10^{-3}\,m^3$ of paint in the can, the total length of the rope is 1.70 m. This is where the rope reaches the limit of proportionality.

(i) Determine the mass of the paint in the can.

mass = ... [3]

(ii) The student continues to pour paint into the can until there is a volume of $5.0 \times 10^{-3}\,m^3$ in it. He notices that the elastic rope becomes harder to stretch after the limit of proportionality.

He plots a graph of the total length of the rope against the volume of paint added.

On Fig. 8.2, sketch the length-volume graph for the range 0 to $5.0 \times 10^{-3}\,m^3$.

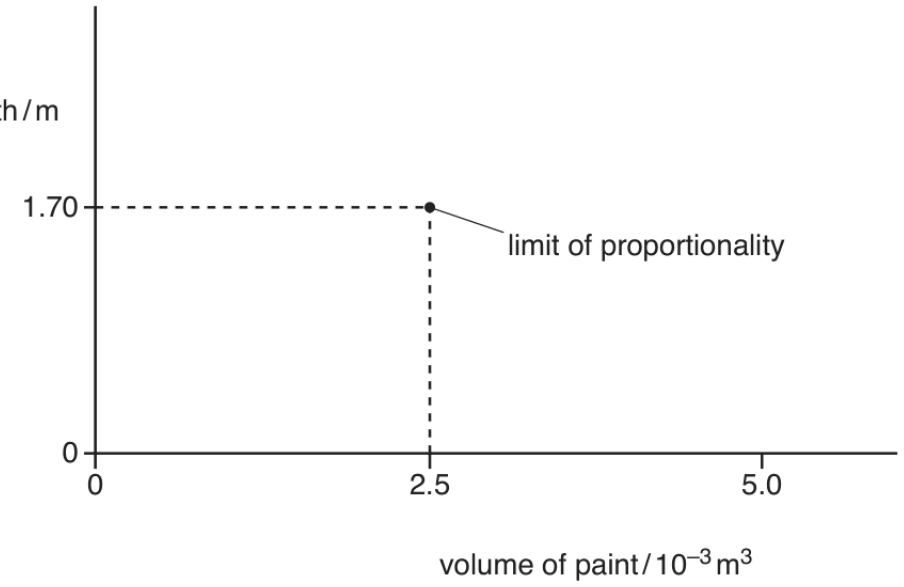


Fig. 8.2

[3]

(e) The student suddenly removes the can from the end of the stretched rope and, as it contracts, the rope jumps into the air.

State the energy change that is taking place in the rope as it contracts and jumps into the air.

.....

..... [2]

[Total: 15]

MARK SCHEME · 5054_w19_ms_22.pdf · source page 7 of 10 · Question 8

Question	Answer	Marks
8(a)	where / when extension is not directly proportional to load / tension / force (applied)	**C1**
	point where extension stops being directly proportional to load / tension / force (applied)	**A1**
8(b)(i)	the amount matter in a body **or** (a measure of) the inertia of a body	**B1**
8(b)(ii)	force (on a body)	**B1**
	in a gravitational field	**B1**
8(c)(i)	7.0 N **c.a.o.**	**B1**
8(c)(ii)	41 N / m **or** 0.41 N / cm	**B1**

MARK SCHEME · 5054_w19_ms_22.pdf · source page 8 of 10 · Question 8

Question	Answer	Marks
8(d)(i)	two separate approaches **I** / **II**	
8(d)(i)**I**	0.90	**C1**
	41×0.90 **or** 37 **or** 3.7 **or** $3.7 - 0.70$	**C1**
	3.0 kg	**A1**
	or	
8(d)(i)**II**	$1.7 - 0.97$ **or** 0.73	**C1**
	$41 \times (1.7 - 0.97)$ **or** 41×0.73 **or** 30	**C1**
	3.0 kg	**A1**
8(d)(ii)	straight line to marked point	**B1**
	0 < location of *y*-intercept < 1.70 m (intercept need not be labelled)	**B1**
	curve of changing / changed gradient beyond marked point	**B1**
8(e)	from elastic (potential energy) **or** strain (energy)	**B1**
	to kinetic (energy) **or** to gravitational (potential energy)	**B1**

3 A student suspends a spring from a support. He attaches different loads to the lower end of the spring. For each load attached, he measures the extension of the spring.

Fig. 3.1 shows the extension–load graph obtained.

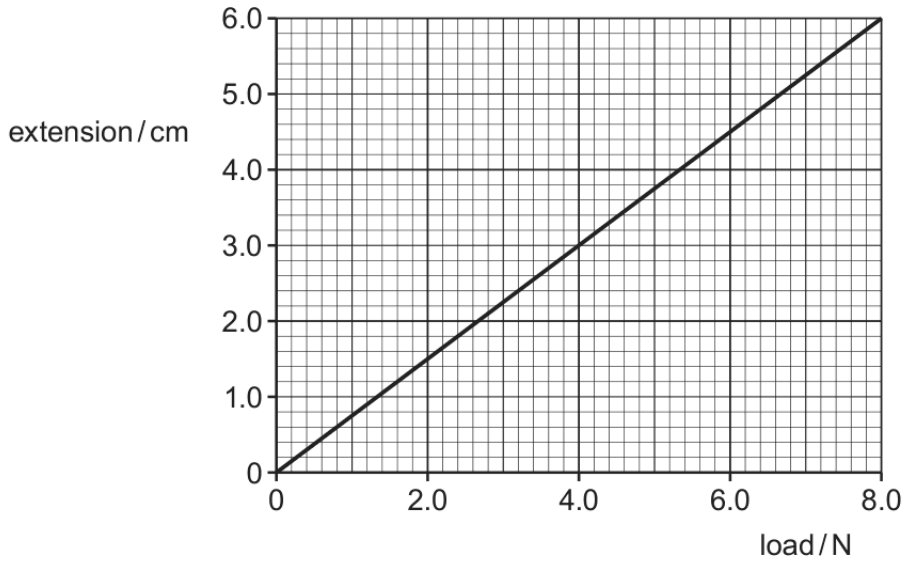


Fig. 3.1

(a) The extension of the spring is directly proportional to the load.

State the **two** features of the graph that show this.

1. .....

2. .....

[2]

(b) Describe how the student can show that the spring reaches its limit of proportionality at 8.0 N.

.....

.....

.....

.....

.....

..... [2]

(c) The spring is used in a simple device known as an accelerometer, shown in Fig. 3.2.

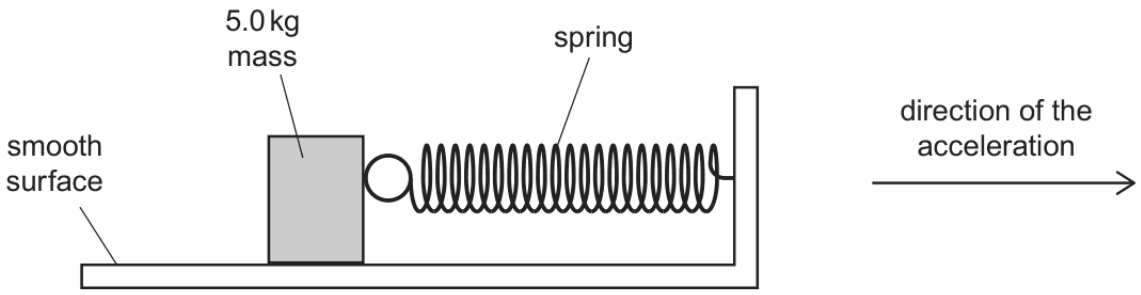


Fig. 3.2

The spring is fixed at one end and attached to a mass of 5.0 kg at the other end.

The mass rests on a smooth, horizontal surface. There is no friction between the mass and the surface.

The whole device is placed in a car.

The car accelerates and the student notices that the extension of the spring is 3.0 cm.

Using Fig. 3.1, calculate the acceleration of the car.

acceleration = ... [3]

[Total: 7]

Question	Answer	Marks
3(a)	straight line	**B1**
	through the origin	**B1**
3(b)	add loads above 8 N	**B1**
	increase in length same increase in load is different **or** graph does not continue as a straight line	**B1**
3(c)	(force of) 4(.0 N) seen	**C1**
	(a =) F/m in any form algebraic or numerical	**C1**
	0.8(0) m/s^2	**A1**

QUESTION · 5054_s20_qp_12.pdf · source page 5 of 16

13 An unstretched elastic cord of length 15.0 cm stretches by 4.0 cm when a weight of 6.0 N is suspended from it. The cord does not exceed the limit of proportionality.

What is the total length of the cord when the weight is 3.0 N?

A 2.0 cm **B** 7.5 cm **C** 9.5 cm **D** 17.0 cm

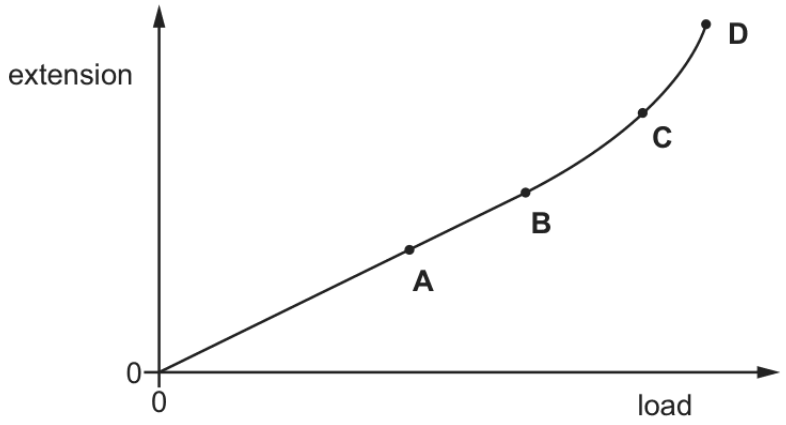
MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 13

13	D	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 6 of 16

12 An extension–load graph is plotted for a spring.

Which point marks the limit of proportionality for this spring?



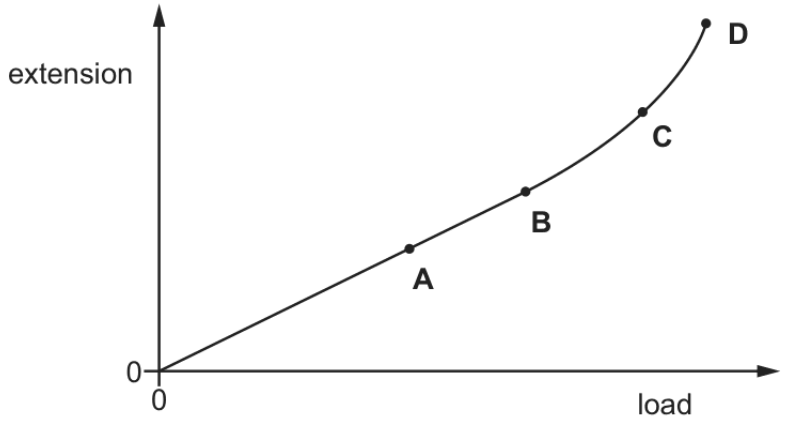
MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 12

12	B	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 5 of 16

11 An extension–load graph is plotted for a spring.

Which point marks the limit of proportionality for this spring?

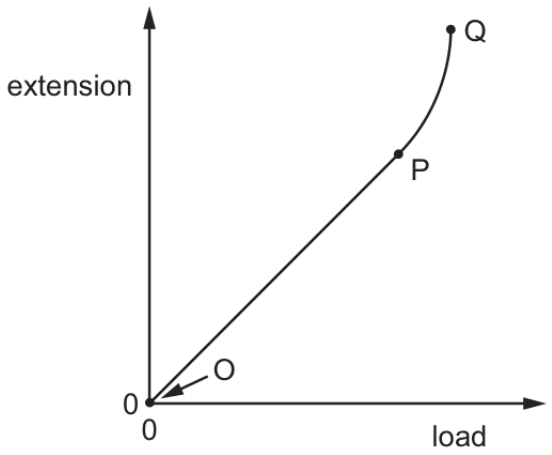


MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 11

11	B	**1**

QUESTION · 5054_s21_qp_11.pdf · source page 5 of 16

9 Where on the graph is the limit of proportionality for an elastic solid?



A between O and P

B at P

C between P and Q

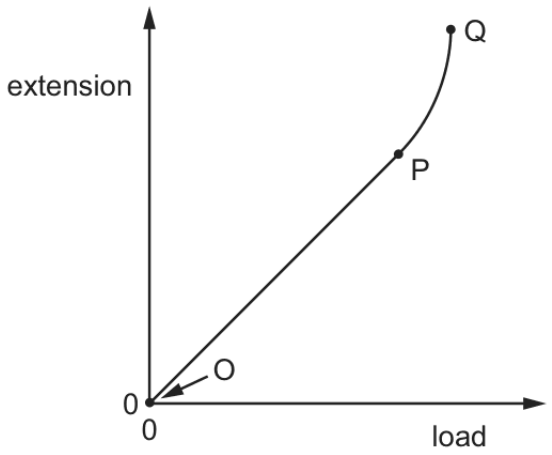
D at Q

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 9

9	B	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 6 of 16

10 Where on the graph is the limit of proportionality for an elastic solid?



A between O and P

B at P

C between P and Q

D at Q

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 10

10	B	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 6 of 16

11 A weight of 2.0 N is hung from a spring. The extension produced is 6.0 cm. The 2.0 N weight is removed and an 8.0 N weight is hung from the spring. The spring does not pass its limit of proportionality.

What is the new extension of the spring?

A 6.0 cm **B** 24 cm **C** 36 cm **D** 48 cm

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 11

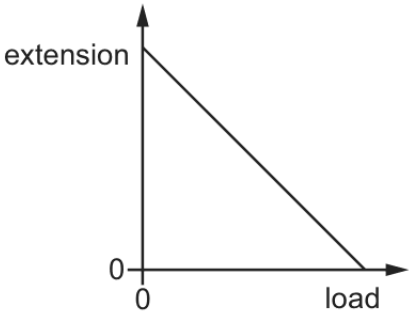
11	B	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 4 of 20

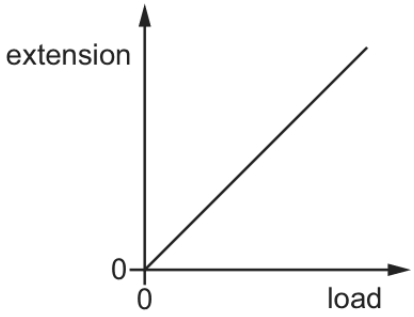
7 The extension of a spring is measured for various loads. The limit of proportionality is not reached.

Which graph shows the extension–load graph for the spring?

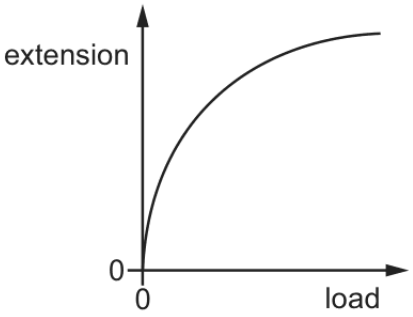
A



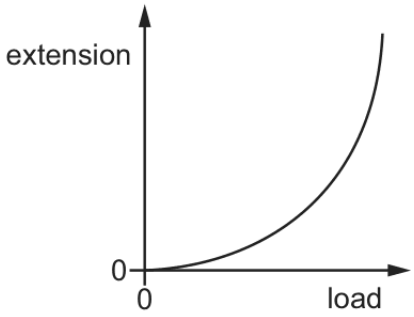
B



C



D



MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 7

7	B	**1**

QUESTION · 5054_w21_qp_12.pdf · source page 4 of 16

9 A metal wire is stretched by a force *F* up to the limit of proportionality.

Which statement describes what happens when *F* is increased slightly?

A The wire breaks.

B The wire continues to extend but not in direct proportion to *F*.

C The wire continues to extend in direct proportion to *F*.

D The wire does not extend any further.

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 9

9	B	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a model of the human arm. The rubber band represents the muscle that moves part of the arm XY up.

A mass is suspended from XY, as shown in Fig. 1.2. The weight of section XY is negligible and the model is at rest.

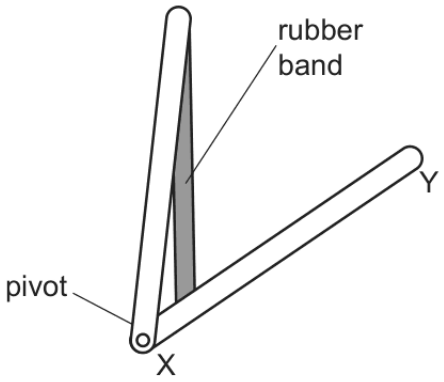


Fig. 1.1

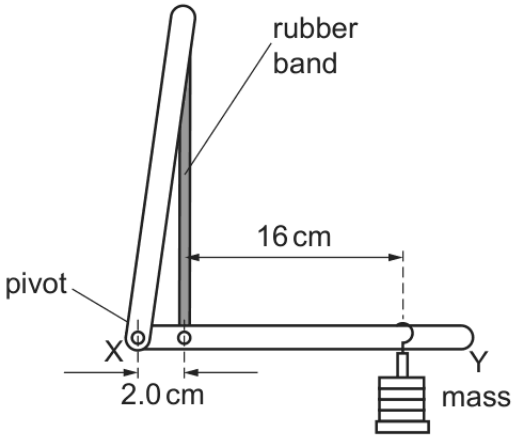


Fig. 1.2 (not to scale)

(a) (i) State **two** ways in which the dimensions of the rubber band change as the mass is added to section XY.

.....

..... [2]

(ii) State the form of energy stored in the stretched rubber band.

..... [1]

(b) (i) State the principle of moments.

.....

.....

.....

..... [2]

(ii) Explain why the force that the rubber band exerts on section XY is larger than the weight of the mass.

.....

.....

..... [1]

(iii) The mass suspended from section XY in Fig. 1.2 has a weight of 4.0 N.

Calculate the force that the rubber band exerts on section XY.

force = ... [2]

(iv) Explain how your answer to **(b)(iii)** is different if the weight of section XY is **not** negligible.

.....

.....

..... [1]

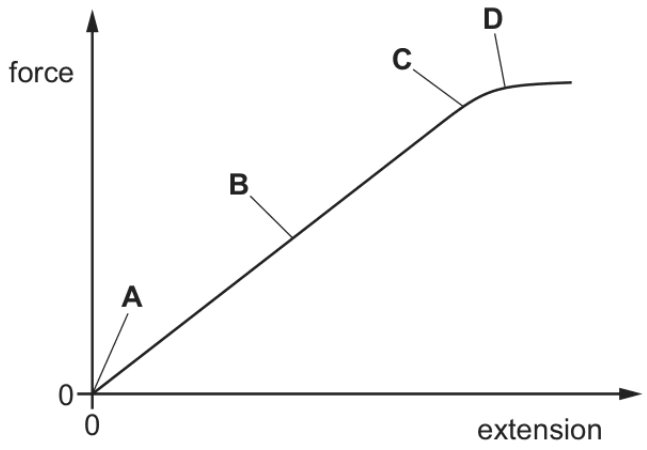
[Total: 9]

Question	Answer	Marks
1(a)(i)	longer	**B1**
	thinner **or** smaller (cross-sectional) area	**B1**
1(a)(ii)	elastic potential **or** strain (energy)	**B1**
1(b)(i)	sum of / total / net / resultant moment is zero	**M1**
	when (the object is) in equilibrium	**A1**
1(b)(ii)	distance of rubber band to pivot is smaller than the distance of the mass (to pivot)	**B1**
1(b)(iii)	$4 \times 18 = F \times 2$	**C1**
	36 N	**A1**
1(b)(iv)	greater **and** the weight (of XY) contributes to the (clockwise) moment/turning effect **or** the moment of the rubber band force is larger	**B1**

QUESTION · 5054_s22_qp_12.pdf · source page 5 of 20

9 The graph shows how the extension of a spring depends on the force applied.

Which point is the limit of proportionality?



MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 9

9	C	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 5 of 20

10 An elastic spring has an unstretched length of 30 cm.

A load of 6.0 N is hung from the spring and the length of the spring is now 66 cm.

The 6.0 N load is removed and the spring returns to its original length. A load of 2.0 N is now hung from the spring.

What is the new length of the spring?

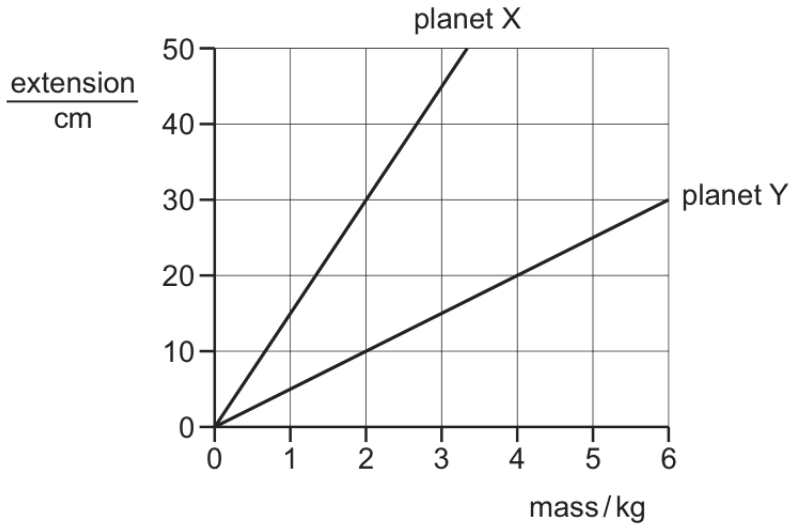
A 22 cm **B** 40 cm **C** 42 cm **D** 52 cm

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 10

10	C	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 5 of 16

10 The graph shows how the extension of a spring changes with the masses suspended from it when the spring is on planet X and when the spring is on planet Y.



Which conclusion can be drawn from these graphs?

A It is not possible to compare the gravitational field strengths on planets X and Y.

B The gravitational field strength on planet X is equal to the gravitational field strength on planet Y.

C The gravitational field strength on planet X is one third of the gravitational field strength on planet Y.

D The gravitational field strength on planet X is three times the gravitational field strength on planet Y.

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 10

10	D	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a steel spring.

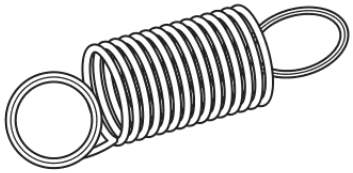


Fig. 1.1

(a) Describe an experimental procedure for obtaining the results needed to plot an extension–load graph for this spring.

.....

.....

.....

.....

.....

..... [3]

(b) In an experiment to obtain results for an extension–load graph, a spring is stretched beyond its limit of proportionality.

(i) On Fig. 1.2, sketch the extension–load graph for this spring.

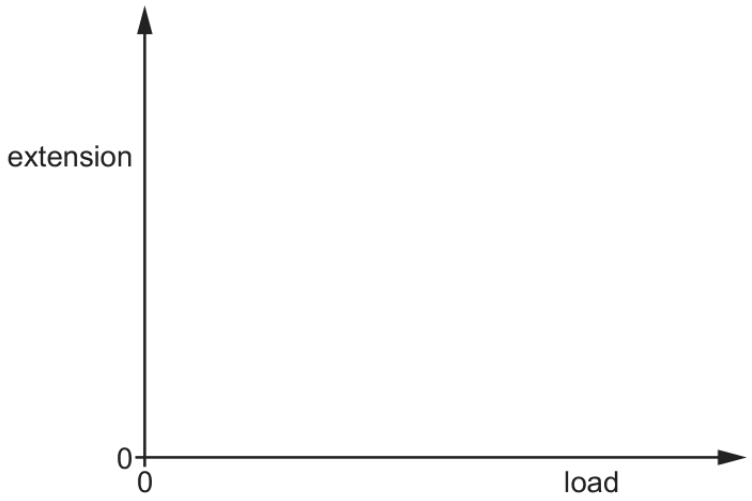


Fig. 1.2

[2]

(ii) On your graph in Fig. 1.2, mark and label with a P the limit of proportionality. [1]

(c) The limit of proportionality for this spring occurs at a load of 8.5 N.

The extension of the spring is equal to 0.014 m when the load is equal to 3.5 N.

Calculate the extension for a load of 5.5 N.

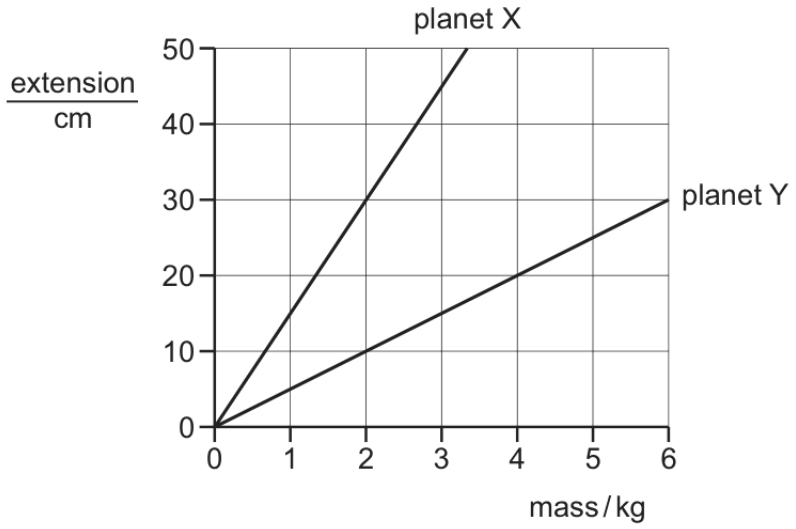
extension = ... [2]

[Total: 8]

Question	Answer	Marks
1(a)	measure length / attach marker	**B1**
	attach load **and** measure new length **and** calculate extension **or** attach load **and** read extension (using marker) against zeroed rule	**B1**
	obtain a new result for different load **or** use of rule(r) **or** calculation of extension explained	**B1**
1(b)(i)	straight line of positive gradient from origin **and** for at least half of the drawn line	**B1**
	curve of increasing gradient starting at end of straight line **and** never becomes vertical	**B1**
1(b)(ii)	P marked where the gradient starts to change	**B1**
1(c)	(x =) 0.014 × 5.5 / 3.5 **or** extension is directly proportional to load	**C1**
	0.022 m	**A1**

QUESTION · 5054_w22_qp_12.pdf · source page 5 of 16

10 The graph shows how the extension of a spring changes with the masses suspended from it when the spring is on planet X and when the spring is on planet Y.



Which conclusion can be drawn from these graphs?

A It is not possible to compare the gravitational field strengths on planets X and Y.

B The gravitational field strength on planet X is equal to the gravitational field strength on planet Y.

C The gravitational field strength on planet X is one third of the gravitational field strength on planet Y.

D The gravitational field strength on planet X is three times the gravitational field strength on planet Y.

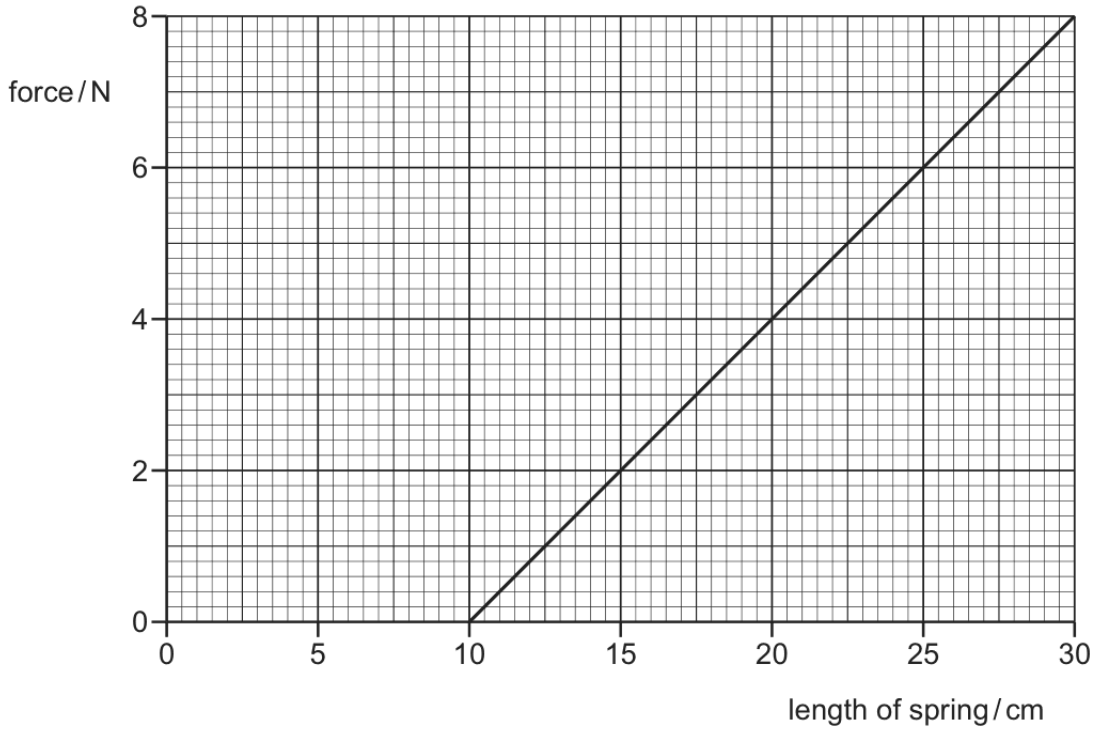
MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 10

10	D	**1**

QUESTION · 5054_s23_qp_11.pdf · source page 5 of 20

8 A spring is stretched to different lengths by a variable force.

The graph shows how the size of the force depends on the length to which the spring is stretched.



What is the spring constant *k* of the spring?

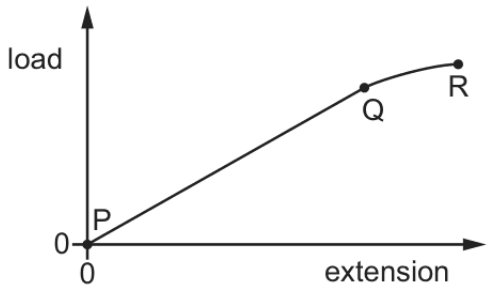
A 0.27 N / cm **B** 0.40 N / cm **C** 2.5 N / cm **D** 3.8 N / cm

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 8

8	B	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 5 of 16

10 A load–extension graph for a spring is shown.



Which point is the limit of proportionality for the spring and what is the spring constant of the spring equal to?

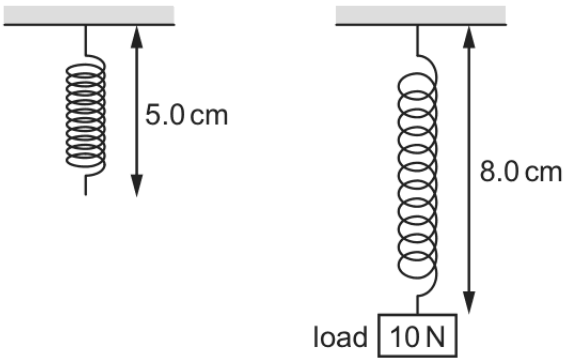
	limit of proportionality	spring constant is equal to
A	Q	the gradient of PQ
B	Q	the gradient of QR
C	R	the gradient of PQ
D	R	the gradient of QR

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 10

10	A	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 3 of 16

5 The diagram shows how the length of a spring changes when a load of 10 N is suspended from it.



The 10 N load is replaced by a 20 N load.

What is the new length of the spring?

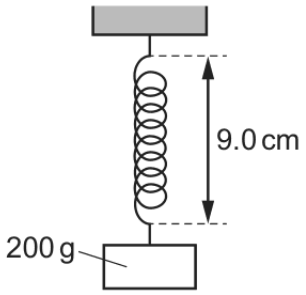
A 6.0 cm **B** 11 cm **C** 14 cm **D** 16 cm

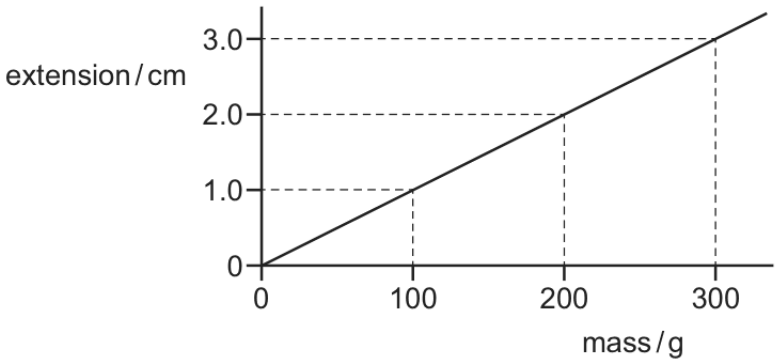
MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 5

5	B	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 4 of 20

8 A spring has a length of 9.0 cm when loaded with a 200 g mass. The extension–mass graph for the spring is shown.





The 200 g mass is replaced with a 100 g mass.

What is the new length of the spring?

A 7.0 cm **B** 8.0 cm **C** 9.0 cm **D** 10 cm

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 8

8	B	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 4 of 16

7 An unstretched spring of length 15.0 cm stretches by 4.0 cm when a mass of weight 6.0 N is suspended from it. The spring does not exceed the limit of proportionality.

What is the total length of the spring when the weight of the suspended mass is 3.0 N?

A 2.0 cm **B** 7.5 cm **C** 9.5 cm **D** 17.0 cm

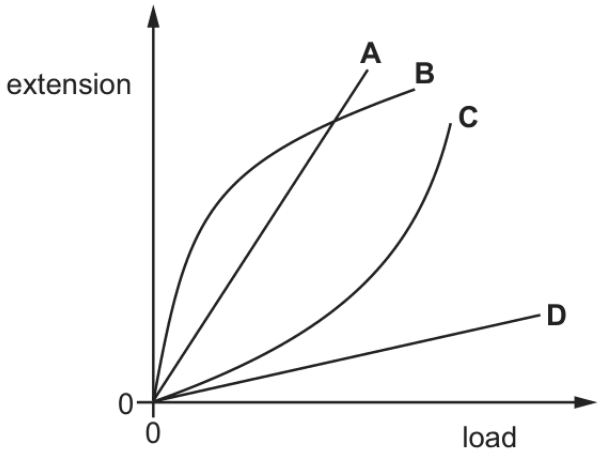
MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 7

7	D	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 5 of 16

10 The graph shows how the extension of four different threads depends on the load attached.

Which thread is the most difficult to stretch over the range of loads shown?



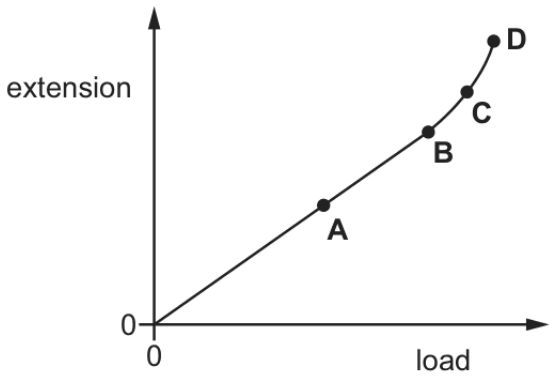
MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 10

10	D	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 5 of 16

11 The extension of a spring is measured as the load stretching it is increased. The graph shows the results.

Which point is the spring's limit of proportionality?



MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 11

11	B	**1**

1 A trolley of mass 0.20 kg is at rest on a frictionless surface.

A block of wood is attached to the trolley.

The volume of the block of wood is 0.0012 m^3, and the density of the wood is 650 kg/m^3.

(a) Calculate the combined mass of the trolley and block.

combined mass = ... kg [2]

(b) Fig. 1.1 shows that a student attaches a spring of spring constant 14 N/m to the front of the trolley.

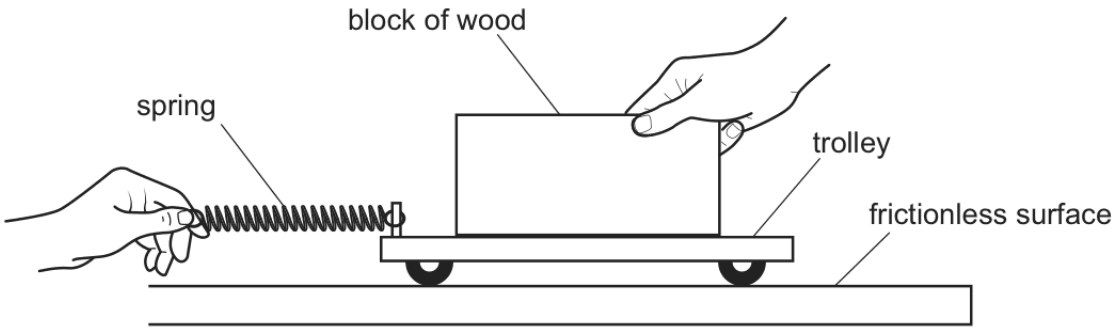


Fig. 1.1

The student holds the trolley and block and stretches the spring. He releases the trolley and block. The trolley and block accelerate from rest.

As the trolley and block accelerate, the student keeps the extension of the spring at 0.035 m.

(i) Calculate the force exerted by the spring on the trolley.

force = ... N [2]

(ii) The trolley and block are pulled a distance of 0.86 m by the spring.

Calculate the work done on the trolley.

work done = ... J [2]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS M

(iii) Explain why the power transferred to the trolley and block increases as the speed increases.

.....

.....

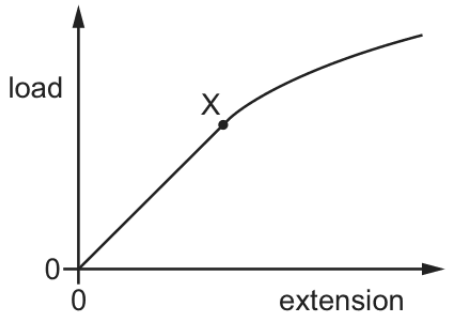
..... [2]

[Total: 8]

Question	Answer	Marks
1(a)	($m =$) ρV **or** 650×0.0012 **or** 0.78	**C1**
	(combined mass =) 0.98 (kg)	**A1**
1(b)(i)	($F =$) kx **or** 14×0.035	**C1**
	0.49 (N)	**A1**
1(b)(ii)	($W =$) Fx **or** 0.49×0.86	**C1**
	0.42 (J)	**A1**
1(b)(iii)	the distance moved / work done (in one second / unit time / a given time) increases (as the speed increases)	**B1**
	the work done in one second / unit time increases	**B1**

QUESTION · 5054_s25_qp_11.pdf · source page 3 of 20

4 The diagram shows the load–extension graph for a spring.



What is the name given to point X on the graph?

A limit of force

B limit of proportionality

C limit of the extension

D limit of the spring

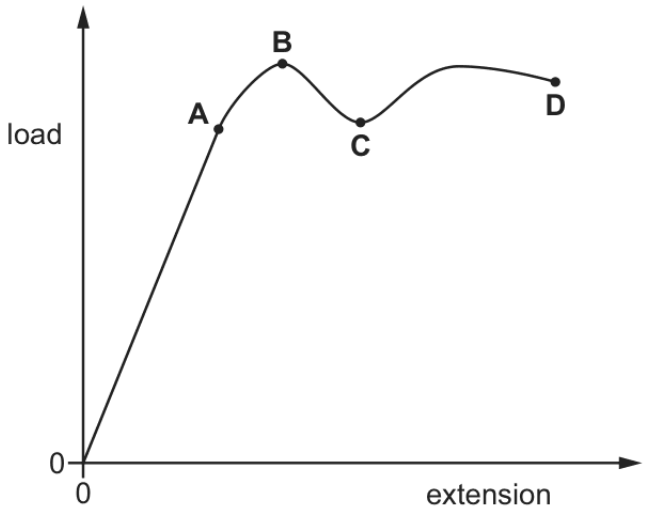
MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 3 of 16

5 The diagram shows a graph of load against extension for a material.

At which point is the limit of proportionality?



MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 5

5	A	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 5 of 16

8 The table shows how the extension of a spring varies with load.

load / N	0	2	4	6	8	10	12	14	16
extension / cm	0	3	6	9	12	15	20	27	38

Between which two loads is the limit of proportionality?

A 0 N and 2 N

B 8 N and 10 N

C 10 N and 12 N

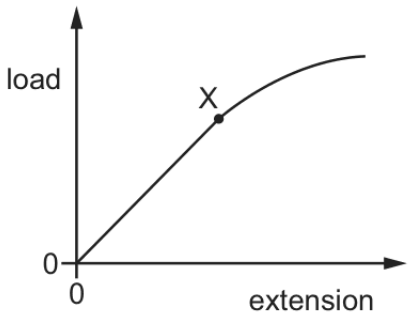
D 14 N and 16 N

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 8

8	C	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 5 of 16

11 The load–extension graph for a spring is shown.



What does point X show?

A the elastic deformation

B the limit of proportionality

C the maximum extension

D the spring constant

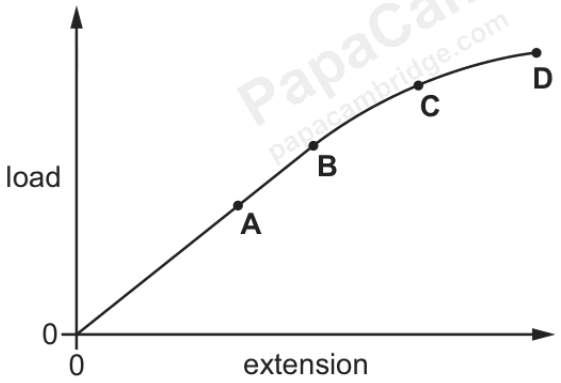
MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 11

11	B	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 5 of 16

10 A load–extension graph is plotted for a spring.

Which point marks the limit of proportionality for this spring?



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MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 10

10	B	**1**

Chapter 1 Motion, forces and energy

1.5 Forces

1.5.4 Circular motion

67 entries: 52 Paper 1 questions, 15 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

11 Fig. 11.1 shows a children's ride. A carriage containing children is pulled up the slope by a motor. The carriage stops at A and then runs down through B, C and D without further input of energy. Between D and E the carriage turns through a bend at constant speed, as shown in Fig. 11.2. At E, brakes are applied and the carriage slows to a stop at F. The height of the ride is 30 m at A and 10 m at C.

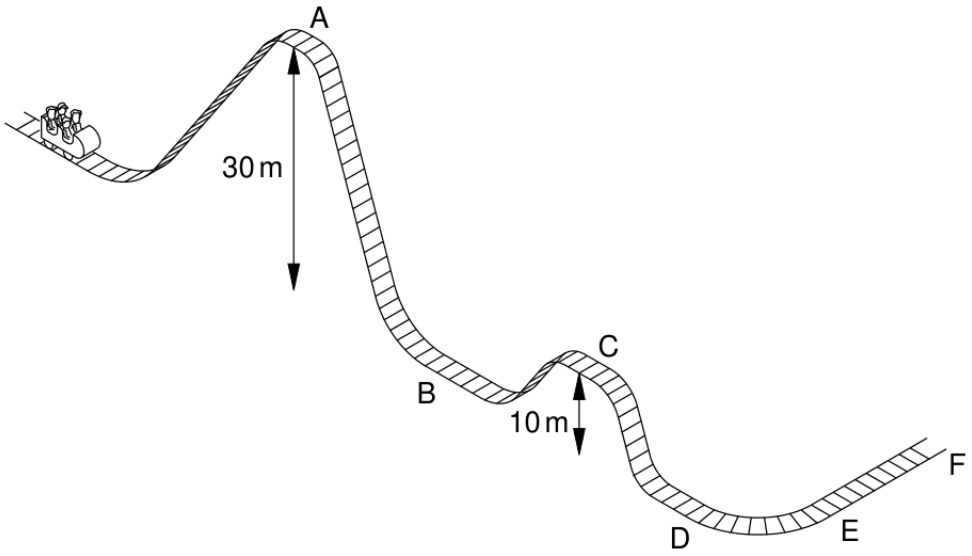


Fig. 11.1

The mass of the carriage and children is 500 kg.
Take the gravitational field strength as 10 N/kg.

(a) **(i)** Discuss the energy changes that occur in the ride from A to D.

(ii) Calculate the maximum potential energy of the carriage and children.

(iii) Assuming that there is no friction between A and C, determine the kinetic energy of the carriage and children at C. Show your working.

[9]

(b)

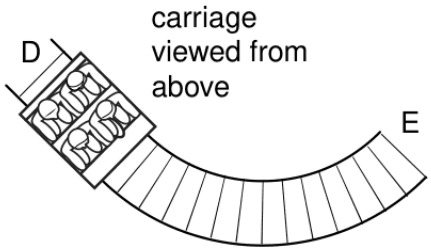


Fig. 11.2

Between D and E, the carriage goes round part of a horizontal circle at constant speed. During this time the velocity of the carriage changes.

(i) Explain how the carriage can have a constant speed but a changing velocity.

(ii) State the direction of the force that acts on the carriage to make it move round the curve.

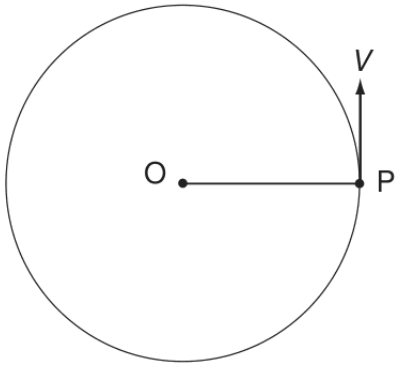
[3]

(c) Between E and F, a frictional force of 3000 N acts to slow the carriage. Calculate the deceleration of the carriage. [3]

11	**(a)**	**(i)**	P.E. decreases (A to B or C to D or downhill or initially)		B1
			K.E. gained (P.E. → K.E….2)		B1
			K.E. to P.E. **change** must be clear and from B to C or uphill		B1
		(ii)	mgh algebraic form seen		C1
			500*10*30		C1
			150 000 J		A1
		(iii)	conservation of energy cited **or** clear that loss of P.E. has become K.E.		C1
			500*10*20 or 500*10*10 or 50 000 seen		C1
			100 000 J (allow g=9.8)		A1
	(b)	**(i)**	velocity involves direction **or** is a vector (speed does not)		B1
			direction (of carriage) changes / carriage turns (accept on diagram)		
		(ii)	force towards centre (of curve) / inwards (accept centripetal)		B1
	(c)		F = ma in any algebraic form or 3000 = 500a		C1
			3000/500		C1
			6(.0) m/s^2		A1
				Total	**[15]**

QUESTION · 5054_w04_qp_1.pdf · source page 3 of 20

3 A particle P is moving in a horizontal circle about O. It moves at constant speed *V*.



Which statement is true?

A A force of constant size is acting in the direction of *V*.

B A force of constant size is acting towards O.

C The force on P varies in size as it moves around the circle.

D There are no forces acting on P.

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 3

3	**B**

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 represents the motion of Earth and the planet Venus around the Sun. The orbits shown are circles.

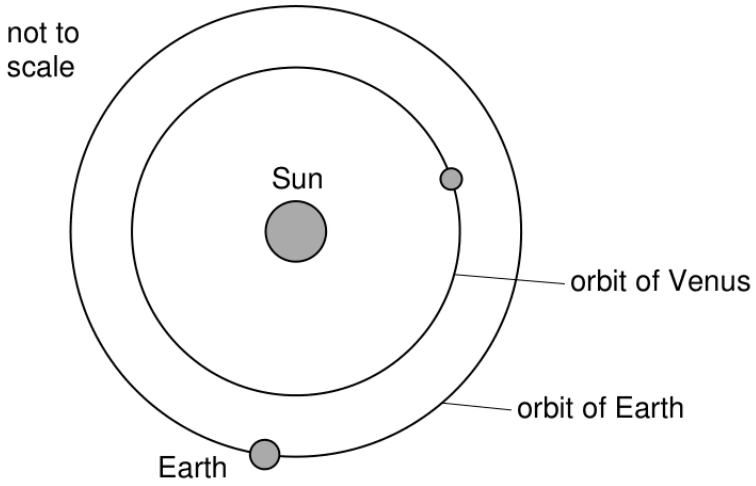


Fig. 1.1

(a) On Fig. 1.1, draw an arrow to show the direction of the force exerted by the Sun on the Earth. [1]

(b) Information about Earth and Venus is given in the table.

planet	time for one orbit in (Earth) years	radius of orbit / million km	circumference of orbit / million km
Venus	0.7	108	679
Earth	1.0	150	942

(i) Use the information in the table to show that Venus has a greater speed than Earth.

.....

.....

.....

..... [2]

(ii) As Earth and Venus move in their orbits, the distance between them changes. Calculate the largest possible distance between them.

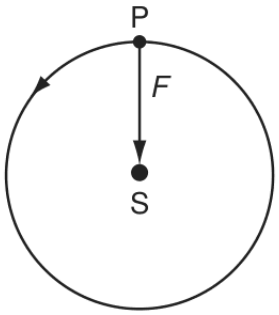
distance = ... million km [1]

Section A

1	**(a)**	arrow from Earth to Sun (by eye would pass through Sun)	**B1**	
	(b) (i)	use of circumference/time **or** s=d/t **or** radius/t	**C1**	
		two speeds **clearly** found using circumference e.g. 970 and 942 (allow conversion to other units)	**A1**	
	(ii)	258 (million km)	**B1**	**4**

QUESTION · 5054_w05_qp_1.pdf · source page 3 of 20

5 A body P moves in a circle around a point S. A force *F* keeps it moving in the circle.



What happens if the force *F* suddenly disappears?

A P moves directly towards S.

B P moves in a circle closer to S.

C P moves away from S in a curved path.

D P goes off in a straight line.

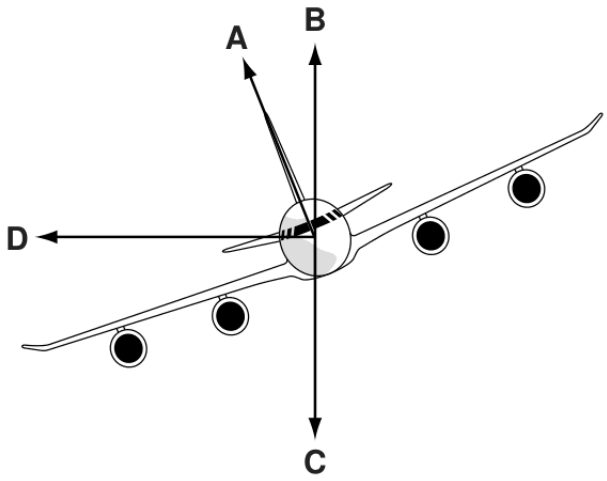
MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 5

5	**D**

QUESTION · 5054_s06_qp_1.pdf · source page 3 of 20

5 The diagram shows an aeroplane turning in a horizontal circle at constant speed.

In which direction is there a resultant force?



MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 5

5	**D**

Section B

Answer **two** questions from this section.

Use the lined page provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 Fig. 9.1 shows the horizontal forces acting on a moving car.

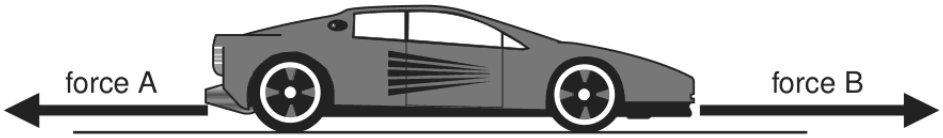


Fig. 9.1

(a) Compare the sizes and directions of the two forces when the car is

(i) moving along a straight road at constant speed, [1]

(ii) accelerating. [1]

(b) Suggest the direction of the resultant force on the car when the car turns a corner at constant speed. [1]

(c) Fig. 9.2 shows the speed-time graph for the first 24 s of the motion of the car along a straight road.

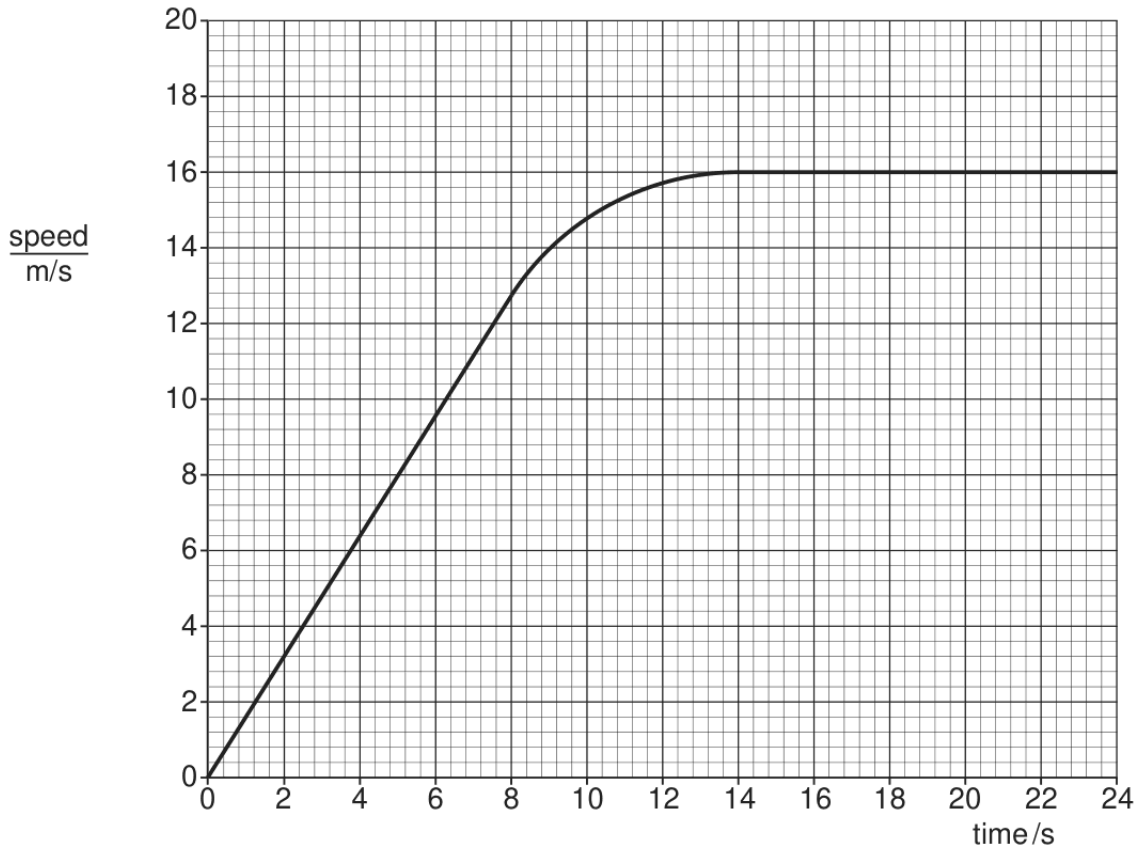


Fig. 9.2

(i) During the motion there is a period of uniform acceleration. State both the start and finish times of this period. [1]

(ii) State the main energy changes that occur in the 24 s. [3]

(iii) Calculate the acceleration of the car during the first 5 s. State clearly the equation that you use. [3]

(iv) Determine the distance travelled in the first 5 s. [3]

(d) The car is stopped by applying the brakes. Various factors can affect the distance travelled by the car during the time that the brakes are applied. Apart from the force applied by the brakes, state **two** of these factors. [2]

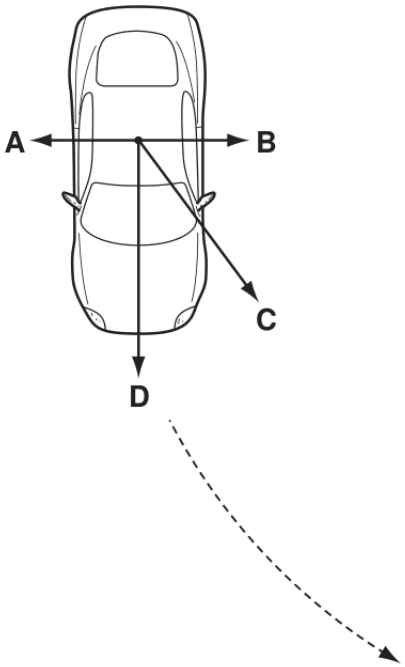
Section B

9	**(a)**	**(i)**	*A* = *B* (assume opposite direction and co-linear)	B1	
		(ii)	B > A (assume opposite direction and co-linear) maximum of 1 mark if directions wrong	B1	[2]
	(b)		towards **centre** of circle/corner	B1	[1]
	(c)	**(i)**	0 **and** 8–9 s	B1	
		(ii)	chemical (potential energy) (**accept** electrical if electrical car clear) **to** kinetic energy **or** K.E. increases thermal energy/heat/internal energy produced Max 2/3 if clear error	B1 B1 B1	
		(iii)	acceleration = (*v*–*u*)/t **or** gradient (7.8 to) 8/5 (**accept** any corresponding period e.g. 8s 12.6–12.8, 6s 9.4–9.6) 1.6 m/s^2 (**accept** 1.56–1.60)	C1 C1 A1	
		(iv)	area under graph **or** average speed 4 (m/s) **or** ½ 5 speed used in **(iii)** at 5 s ½ x (7.8 to) 8 x 5 20 m (**accept** 19.5–20; **ecf speed used in (iii)** at 5 s)	C1 C1 A1	[10]
	(d)		speed of car/friction with road (accept slippery road or ice or water or oil on road)/ friction in engine/tyre condition or area or pressure/air resistance/wind speed or direction/mass or inertia **of car or passengers**/slope of road	B2	[2]

QUESTION · 5054_w07_qp_1.pdf · source page 3 of 20

5 A car moves in a circle at a constant speed.

What is the direction of the resultant force acting on the car?



MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 5

5 **B**

QUESTION · 5054_s08_qp_1.pdf · source page 3 of 20

5 What keeps an electron moving in a circle around the nucleus of an atom?

A a gravitational force away from the nucleus

B a gravitational force towards the nucleus

C an electrostatic force away from the nucleus

D an electrostatic force towards the nucleus

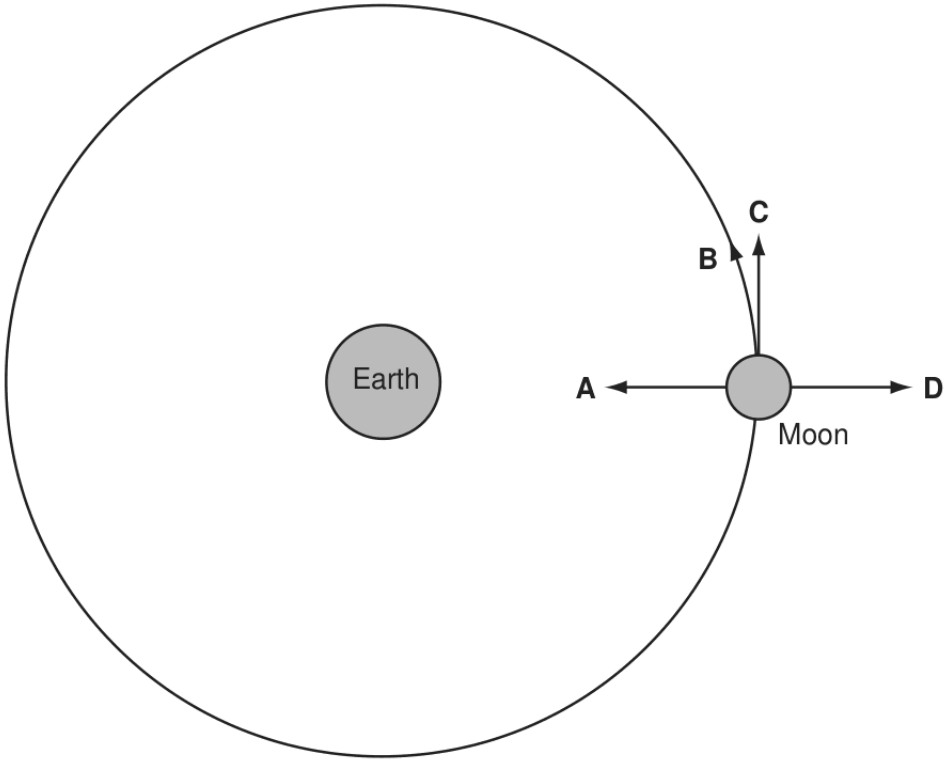
MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 5

5	**D**

QUESTION · 5054_w08_qp_1.pdf · source page 3 of 20

4 The diagram represents the Moon in its orbit around the Earth.

Which arrow represents the direction of the resultant force acting on the Moon at the instant shown?

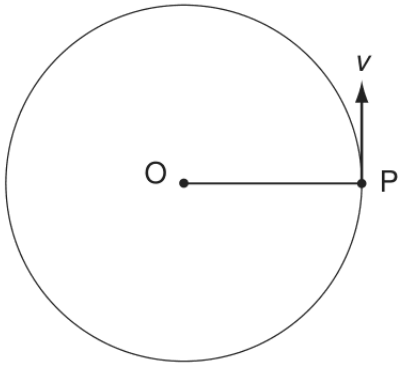


MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 4

4 **A**

QUESTION · 5054_s09_qp_1.pdf · source page 4 of 20

6 A particle P is moving in a horizontal circle about O. P moves at constant speed *v*.



Which statement is true?

A A force of constant magnitude is acting in the same direction as *v*.

B A force of constant magnitude is acting towards O.

C The force on P varies in size as it moves around the circle.

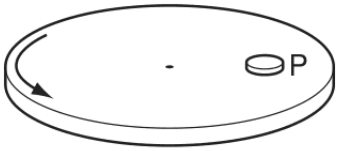
D There is no resultant force acting on P.

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 6

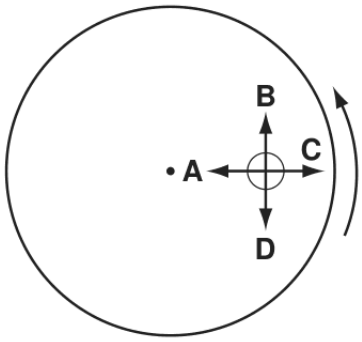
6	**B**

QUESTION · 5054_w09_qp_1.pdf · source page 3 of 20

6 A turntable rotates at constant speed. A coin is placed on the turntable at P. The friction force between the coin and the turntable keeps the coin in the same position on the turntable.



In which direction does the friction force act?



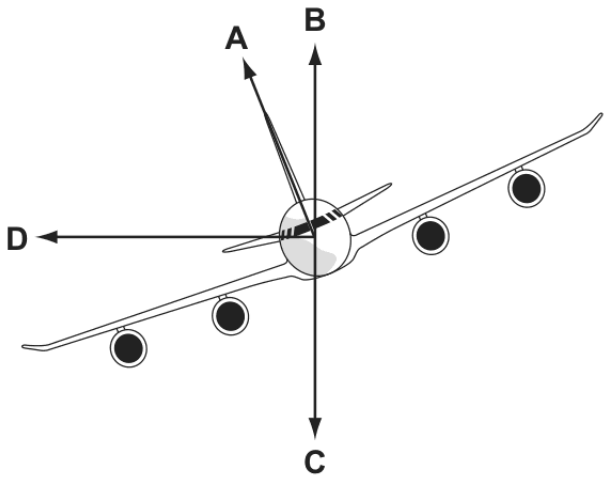
MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 6

6 **A**

QUESTION · 5054_s10_qp_11.pdf · source page 3 of 20

6 The diagram shows an aeroplane turning in a horizontal circle at constant speed.

In which direction is there a resultant force?



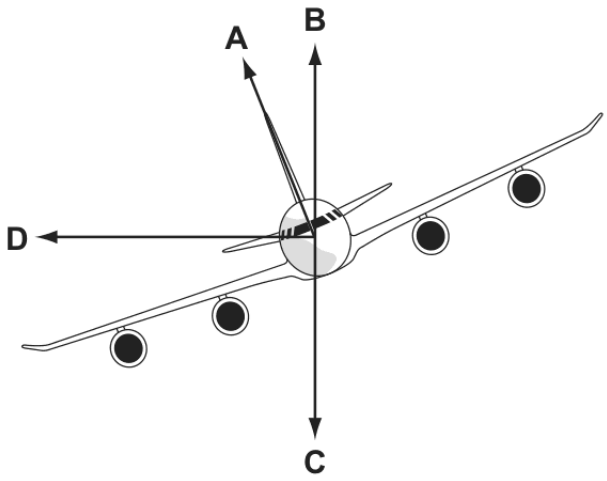
MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 6

6 **D**

QUESTION · 5054_s10_qp_12.pdf · source page 4 of 20

8 The diagram shows an aeroplane turning in a horizontal circle at constant speed.

In which direction is there a resultant force?



MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 8

8	**D**

For Examiner's Use

2 Fig. 2.1 shows the planet Venus orbiting the Sun in a circular orbit at constant speed.

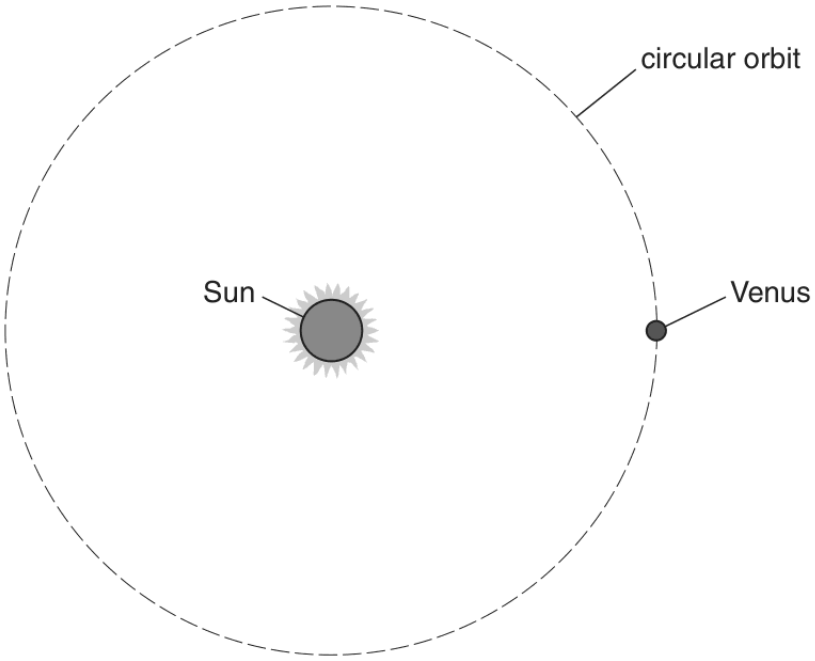


Fig. 2.1 (not to scale)

A force acting on Venus keeps it moving in a circle.

(a) State the name of this force and the object that causes it.

.....

..... [2]

(b) **(i)** On Fig. 2.1, draw an arrow through Venus that represents the direction of its acceleration. [1]

(ii) The size of this acceleration is $9.7 \times 10^{-3}\,m/s^2$ and the mass of Venus is $4.9 \times 10^{24}\,kg$. Calculate the size of the force acting on the planet.

force = ... [2]

(c) As time passes, the planet Venus moves a considerable distance around the circular orbit. State why no work is done by the force acting on the planet as it moves.

.....

.....

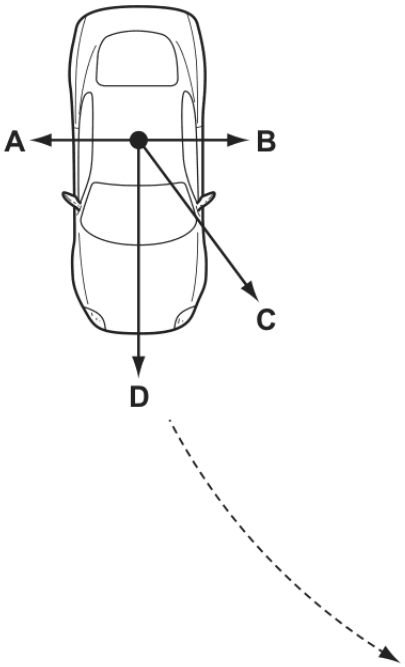
..... [2]

2	**(a)**	gravitational/centripetal (pull/attraction)	B1	
		of the **Sun**	B1	[2]
	(b) **(i)**	arrow touching Venus towards centre/left (must pass through Sun if extended)	B1	[1]
	(ii)	(F =) ma **or** $4.9 \times 10^{24} \times 9.7 \times 10^{-3}$	C1	
		$4.8(4.753) \times 10^{22}$ N	A1	[2]
	(c)	direction of movement ⊥ direction of force	C1	
		force/it is perpendicular/at right angles to distance moved **or** does not move **any** distance in direction of force	A1	[2]

QUESTION · 5054_s11_qp_11.pdf · source page 4 of 16

6 A car moves in a circle at constant speed.

What is the direction of the resultant force acting on the car?



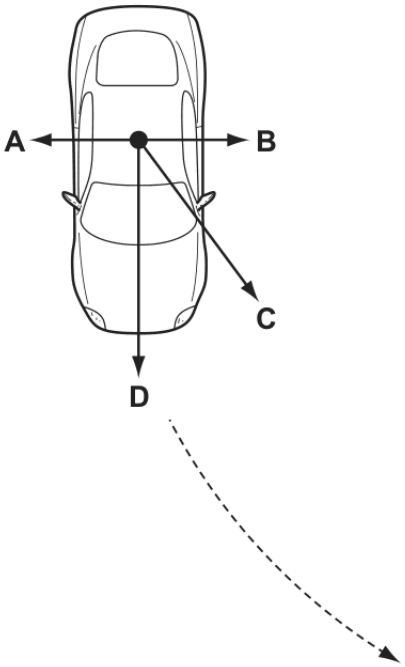
MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 6

6	**B**

QUESTION · 5054_s11_qp_12.pdf · source page 2 of 16

3 A car moves in a circle at constant speed.

What is the direction of the resultant force acting on the car?



MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 3

3 **B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 is the speed-time graph for a racing car of total mass 650 kg as it sets off from rest at the start of a race.

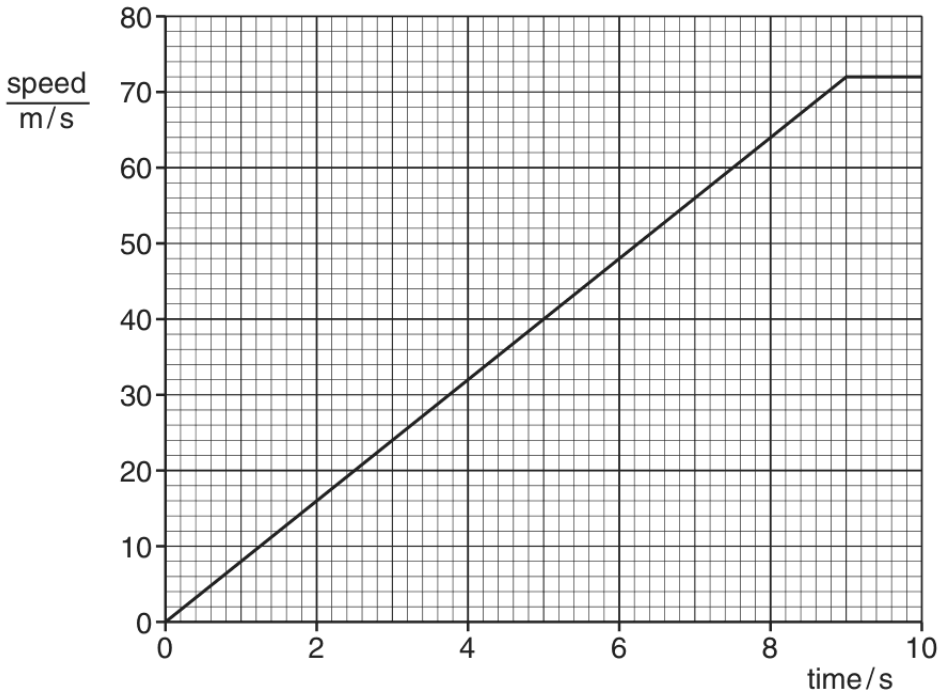


Fig. 9.1

The car travels in a straight line until time $t = 9.0$ s.

(a) State the speed of the car at time $t = 9.0$ s.

speed = ...[1]

(b) Calculate, for the car between times $t = 0$ and $t = 9.0$ s,

(i) the distance travelled,

distance = ...[2]

(ii) the acceleration of the car,

acceleration = ...[2]

(iii) the resultant force acting on the car.

force = ...[2]

(c) The acceleration of the car is constant between $t = 0$ and $t = 9.0$ s.

Suggest and explain why the driving force on the car must increase to keep the acceleration constant.

.....

.....

.....

.....[3]

(d) After $t = 9.0$ s, the car starts to turn a corner and follows a circular path at constant speed.

(i) Explain why the car is accelerating even though its speed is constant.

.....

.....

.....[2]

(ii) State the direction of this acceleration.

.....[1]

(iii) State and explain what causes the car to accelerate as it turns the corner.

.....

.....[2]

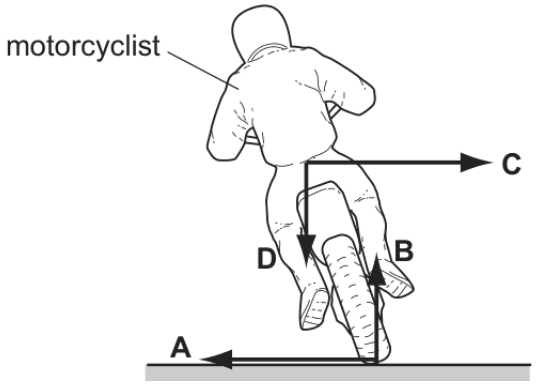
Section B

9	**(a)**	72 m / s			B1	
	(b)	**(i)**	area (under graph) **or** ½ base × height **or** ½*vt* **or** ½ × 9 × 72		C1	
			320/324 m		A1	
		(ii)	**change** in velocity/time **or** $\Delta v/t$ **or** 72/9		C1	
			8(.0) m / s^2		A1	
		(iii)	(*F* =) *ma* **or** 650 × 8.0		C1	
			5.2 × 10^3 N		A1	
	(c)	friction **or** air/wind resistance **or** drag			M1	
		increases as speed increases			A1	
		resultant/net/unbalanced force remains constant			B1	
	(d)	**(i)**	**direction** (of car/motion/speed/velocity) changes		B1	
			(therefore) velocity changes		B1	
		(ii)	towards centre (of circle)/centripetal		B1	
		(iii)	friction **with ground**	**OR** banking of track	B1	
			mention wheels/tyres	reaction force (acts towards centre)	B1	[15]

QUESTION · 5054_s12_qp_12.pdf · source page 4 of 16

6 The diagram shows a motorcyclist leaning over in order to move around a corner.

Which force causes him to move around the corner?

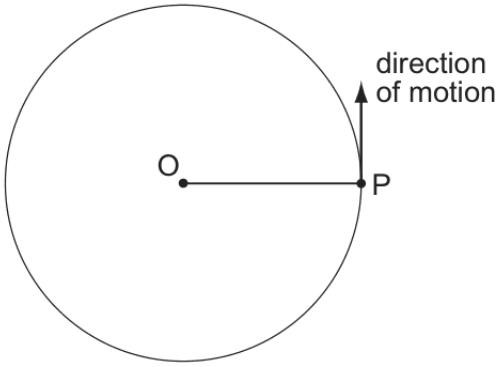


MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 6

6 **A**

QUESTION · 5054_s13_qp_11.pdf · source page 4 of 20

4 A particle P is moving in a horizontal circle about O. P moves at constant speed.



Which statement is true?

A A force of constant size acts on P in the direction of motion.

B A force of constant size acts on P towards O.

C The force on P varies in size as it moves around the circle.

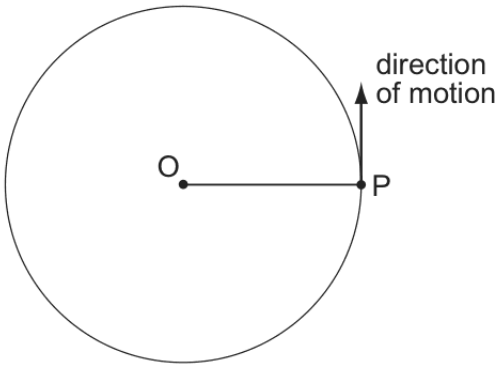
D There is no resultant force acting on P.

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 4

4	**B**

QUESTION · 5054_s13_qp_12.pdf · source page 4 of 16

7 A particle P is moving in a horizontal circle about O. P moves at constant speed.



Which statement is true?

A A force of constant size acts on P in the direction of motion.

B A force of constant size acts on P towards O.

C The force on P varies in size as it moves around the circle.

D There is no resultant force acting on P.

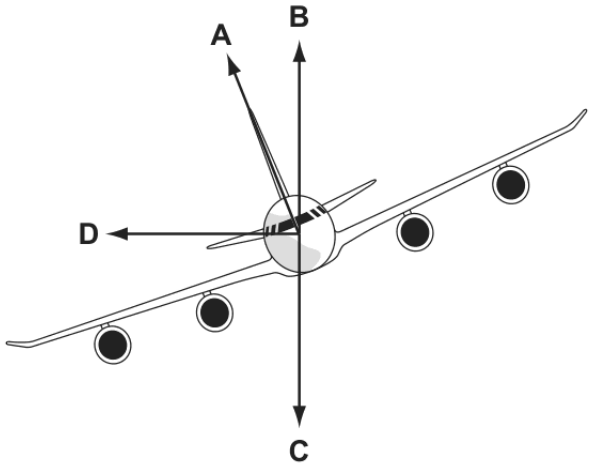
MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 7

7	**B**

QUESTION · 5054_w13_qp_11.pdf · source page 4 of 20

5 The diagram shows an aeroplane turning in a horizontal circle at constant speed.

In which direction is there a resultant force?



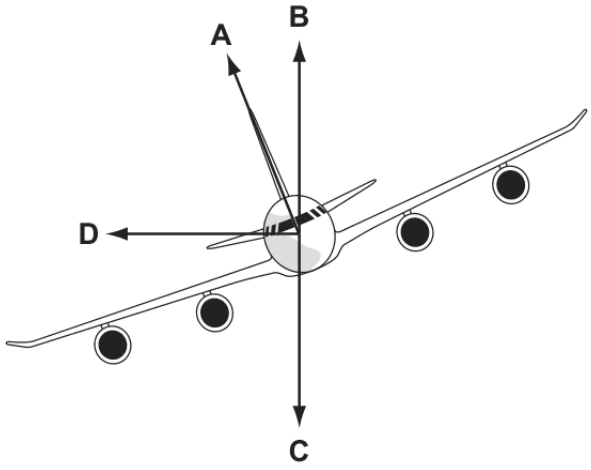
MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 5

5 **D**

QUESTION · 5054_w13_qp_12.pdf · source page 4 of 20

5 The diagram shows an aeroplane turning in a horizontal circle at constant speed.

In which direction is there a resultant force?



MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 5

5 **D**

QUESTION · 5054_s14_qp_11.pdf · source page 4 of 16

7 A satellite is orbiting the Earth.

What is the direction of the force on the satellite causing this circular motion?

A away from the centre of the Earth

B in the direction of motion of the satellite

C in the opposite direction to the motion of the satellite

D towards the centre of the Earth

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 7

7	**D**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a lorry accelerating in a straight line along a horizontal road.

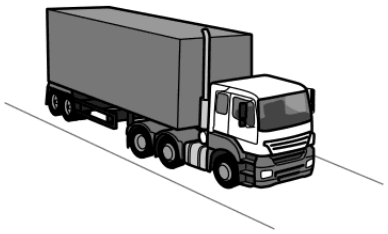


Fig. 1.1

(a) The driving force on the lorry in the forward direction is *D* and the total backward force on the lorry is *B*.

(i) State and explain whether *D* or *B* is the larger force.

.....

..... [1]

(ii) Suggest one possible cause of the backward force *B*.

..... [1]

(b) The weight of the lorry is 300 000 N.

The gravitational field strength *g* is 10 N/kg.

(i) Calculate the mass of the lorry.

mass = ... [1]

(ii) The resultant force on the lorry is 15 000 N. Calculate the acceleration of the lorry.

acceleration = ... [2]

(c) Later, the lorry turns a corner at constant speed.

Explain why the lorry accelerates even though the speed is constant.

.....

..... [1]

1 **(a)** **(i)** *D* **and either** lorry accelerates (forward) **or** resultant force is forward [B1]

(ii) air resistance **or** (air) drag **or** friction (between tyres and road) [B1]

(b) **(i)** 30 000 kg [B1]

(ii) (a=) F / m algebraic in any form or numerical [C1]
0.5(0) m / s^2 [A1]

(c) direction or velocity is changing **or** acceleration or force is sideways or towards centre (of circle) [B1]

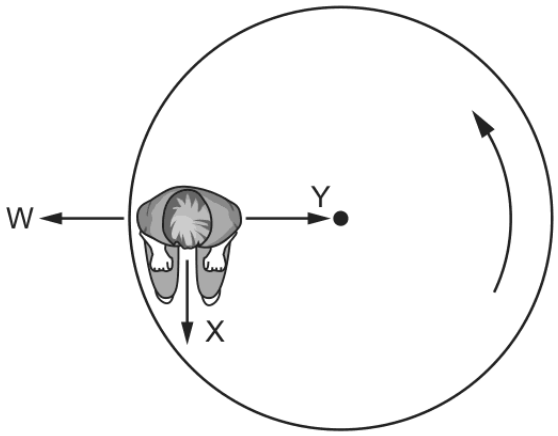
[6]

QUESTION · 5054_w14_qp_11.pdf · source page 5 of 20

5 A boy sits on a playground roundabout (carousel).

The roundabout carries the boy in a horizontal, anticlockwise circle at a constant speed.

The diagram shows the view of the roundabout from above.



What describes the resultant force on the boy?

A The resultant force is zero.

B The resultant force is in direction W.

C The resultant force is in direction X.

D The resultant force is in direction Y.

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 5

5	**D**

QUESTION · 5054_w14_qp_12.pdf · source page 2 of 16

3 In a model of an atom, electrons move in circular orbits around a nucleus.

Which statement about the electrons is correct?

A The electrostatic force on the electrons is away from the nucleus.

B The acceleration of the electrons is towards the central nucleus.

C The speed of the electrons varies continuously.

D The velocity of the electrons remains constant.

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 3

3	**B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a satellite in orbit around the Earth.

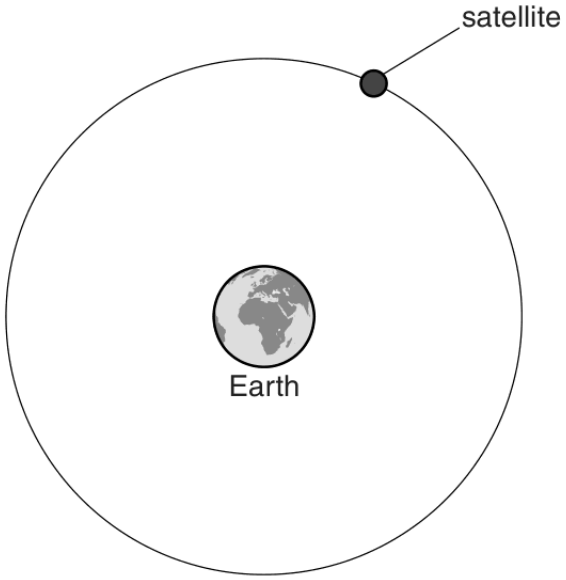


Fig. 9.1 (not to scale)

(a) The satellite travels at a constant speed in a circular orbit.

(i) Underline the quantities in the list below that are scalars.

acceleration force mass speed velocity [2]

(ii) The velocity of the satellite changes, but its speed is constant.

1. State what is meant by *velocity*. .....

.....

.....[1]

2. Explain why the velocity changes. .....

.....

.....[1]

(iii) Explain what makes this satellite move in an orbit that is circular.

.....

.....

.....

.....[2]

(b) The satellite is placed into orbit by a rocket. Fig. 9.2 shows the rocket as it takes off.

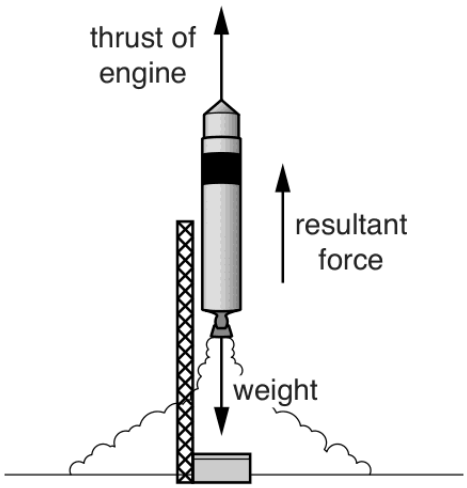


Fig. 9.2

The rocket and fuel have a total mass of 40 000 kg and a total weight of 400 000 N. The resultant force acting upwards on the rocket is 50 000 N.

(i) Calculate the thrust produced by the rocket engine.

thrust = ...[1]

(ii) Calculate the acceleration of the rocket.

acceleration = ...[2]

(c) In the first four minutes after take-off, the acceleration of the rocket is uniform.

(i) State what is meant by a *uniform acceleration*.

.....

.....

.....[2]

(ii) Fig. 9.3 describes the motion of the rocket in the first 12 minutes.

time / minutes	motion of the rocket
0 to 4	uniform acceleration
4 to 6	increasing acceleration
6 to 10	decreasing acceleration
10 to 12	constant speed

Fig. 9.3

On Fig. 9.4, sketch the speed-time graph of the rocket for the first 12 minutes.

You do not need to give values for the speed.

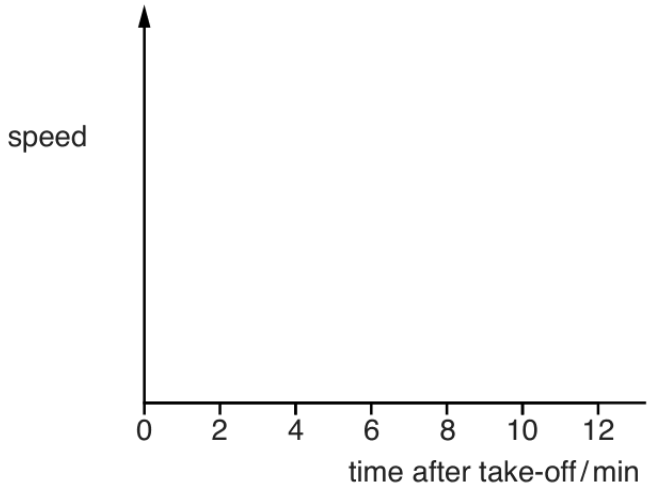


Fig. 9.4

[3]

(iii) State how the speed-time graph in **(ii)** can be used to find the distance travelled by the rocket.

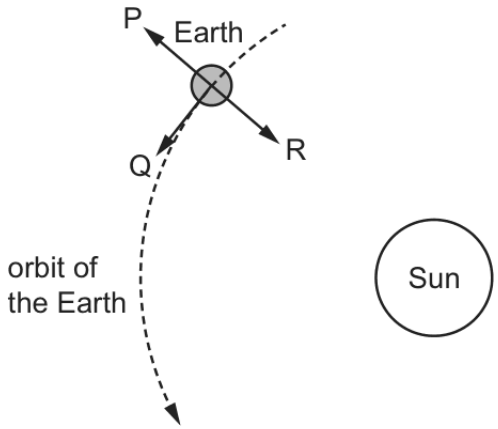
.....

.....[1]

9	**(a)**	**(i)**	speed and mass	B2
		(ii)	**1** speed and direction **or** distance / time and direction **or** displacement / time	B1
			2 direction changes	B1
		(iii)	force of gravity from / towards Earth	B1
			force is centripetal **or** at right angles to motion / velocity	B1
	(b)	**(i)**	450 000 N	B1
		(ii)	(a =) F/m or 50 000 / 40 000	C1
			1.25 m / s^2	A1
	(c)	**(i)**	same change in velocity / speed	M1
			in same time period	A1
		(ii)	start at origin and straight line for first 4 minutes	B1
			gradient increases at first after 4 and then decreases	B1
			constant speed from 10 minutes until 12 minutes	B1
		(iii)	area **under** graph	B1

QUESTION · 5054_s15_qp_12.pdf · source page 4 of 16

7 The Earth travels in a circular orbit around the Sun at constant speed.



Which arrows show the direction of the acceleration of the Earth and the direction of the velocity of the Earth?

	direction of acceleration	direction of velocity
A	P	Q
B	Q	P
C	Q	R
D	R	Q

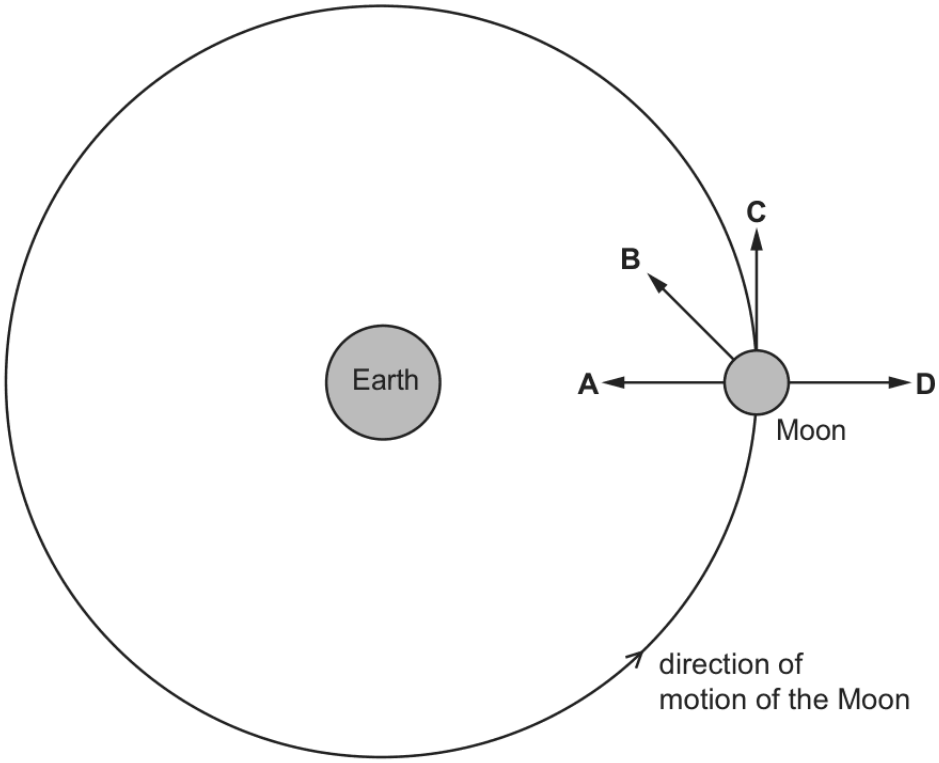
MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 7

7 **D**

QUESTION · 5054_w15_qp_11.pdf · source page 3 of 16

6 The diagram represents the Moon in its orbit around the Earth.

Which arrow represents the direction of the resultant force acting on the Moon at the instant shown?



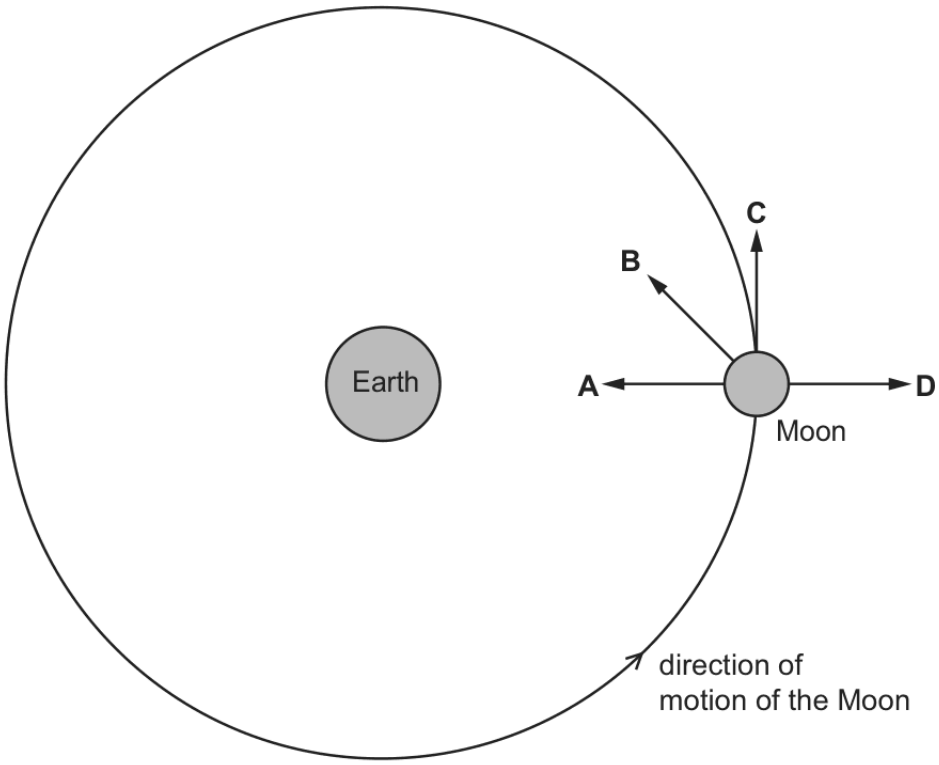
MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 6

6 **A**

QUESTION · 5054_w15_qp_12.pdf · source page 4 of 16

8 The diagram represents the Moon in its orbit around the Earth.

Which arrow represents the direction of the resultant force acting on the Moon at the instant shown?



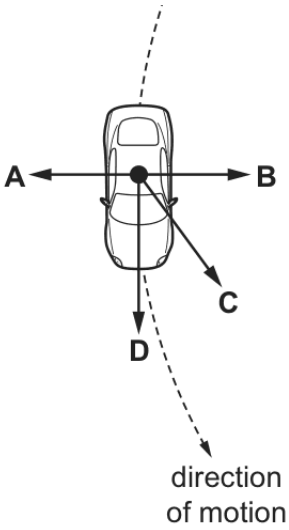
MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 8

8	**A**

QUESTION · 5054_s16_qp_11.pdf · source page 3 of 16

6 A car moves in a circle at constant speed.

What is the direction of the resultant force acting on the car?



× centre of circle turned by car

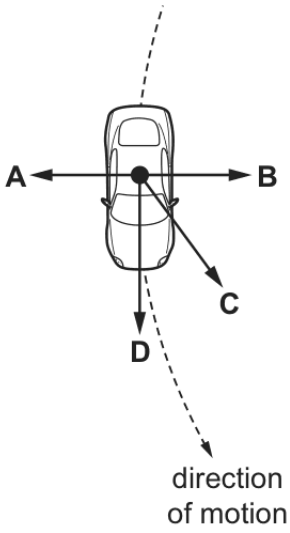
MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 6

6 **B**

QUESTION · 5054_s16_qp_12.pdf · source page 2 of 16

3 A car moves in a circle at constant speed.

What is the direction of the resultant force acting on the car?



× centre of circle turned by car

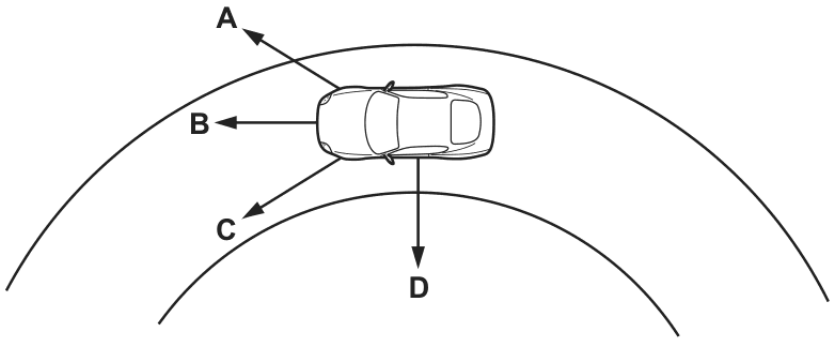
MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 3

3 **B**

QUESTION · 5054_w16_qp_11.pdf · source page 2 of 20

3 The diagram shows a car going around a circular track at constant speed.

Which arrow shows the direction of the resultant force on the car?



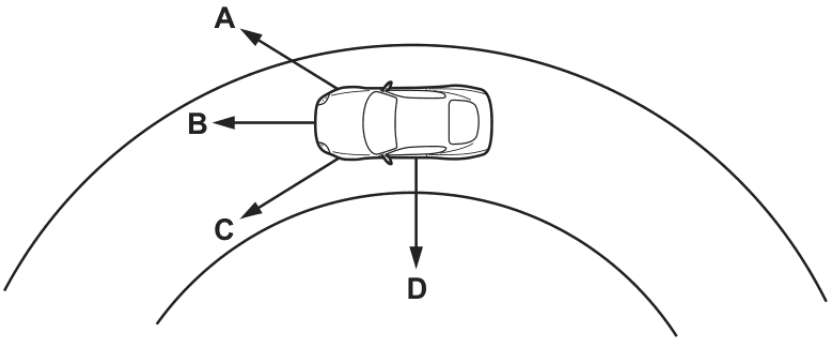
MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 3

3 **D**

QUESTION · 5054_w16_qp_12.pdf · source page 3 of 16

6 The diagram shows a car going around a circular track at constant speed.

Which arrow shows the direction of the resultant force on the car?



MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 6

6	**D**

2 A skateboarder of mass 45 kg is at the top of a ramp.

Fig. 2.1 shows the skateboarder and the ramp.

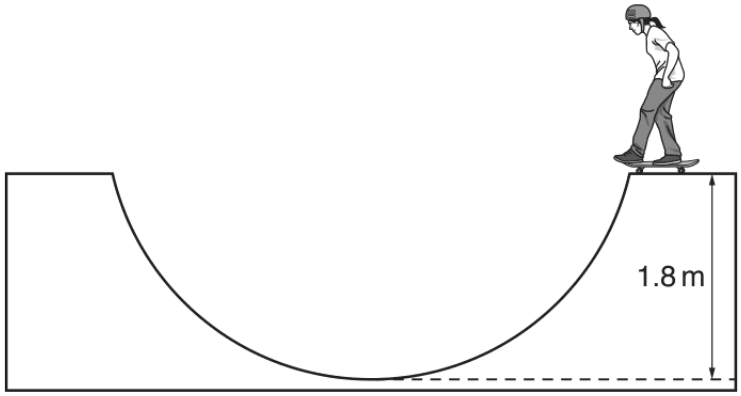


Fig. 2.1 (not to scale)

(a) The skateboarder moves off and she descends vertically through 1.8 m. The gravitational field strength g is 10 N/kg.

(i) Calculate the change in gravitational potential energy of the skateboarder.

energy change = ...[2]

(ii) State the name of two forms of energy that increase as she descends.

1. .....

2. .....
[2]

(b) At the lowest point on the ramp, the skateboarder is travelling at a constant speed along a path which is part of a vertical circle.

State

(i) the direction of the resultant force on the skateboarder at this point,

.....[1]

(ii) how the weight of the skateboarder compares with the upward force that the ramp exerts on her.

.....

.....

.....[1]

2	**(a)**	**(i)**	(GPE =)*mgh* **or** 45 × 10 × 1.8	B1	
			810 J	B1	
		(ii)	kinetic either order thermal / internal / heat / sound } either order	B1 B1	
	(b)	**(i)**	upwards / centripetal / towards centre (of circle)	B1	
		(ii)	it / weight less (than normal contact force) **or** upward force greater	B1	[6]

QUESTION · 5054_s17_qp_11.pdf · source page 5 of 16

8 A satellite is in a circular orbit around a planet.

Which statement is correct?

A Its acceleration is constant in direction but not in size.

B Its acceleration is constant in size but not in direction.

C Its gravitational potential energy varies.

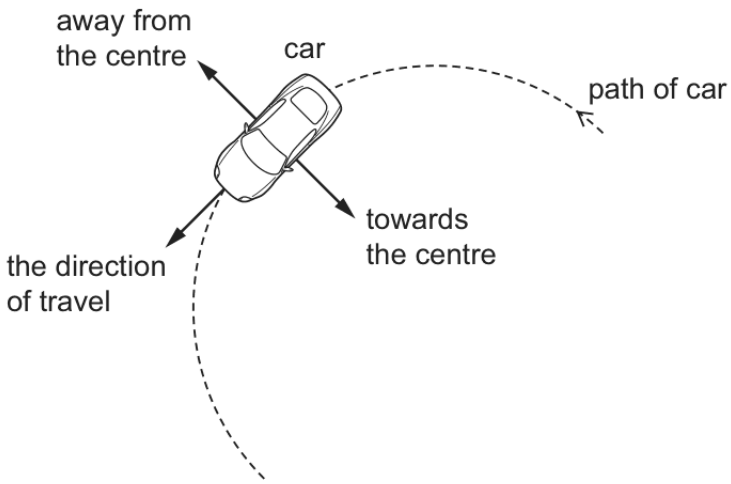
D Its velocity is constant.

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 8

8	B	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 4 of 20

8 A car travels at a constant speed along a circular, horizontal path.



Which statement describes the forces acting on the car?

A They are balanced as the car is moving at constant speed.

B They are unbalanced with a resultant in the direction away from the centre.

C They are unbalanced with a resultant in the direction of travel of the car.

D They are unbalanced with a resultant in the direction towards the centre.

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 8

8	D	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows an astronaut in space near to a space station in orbit above the Earth.

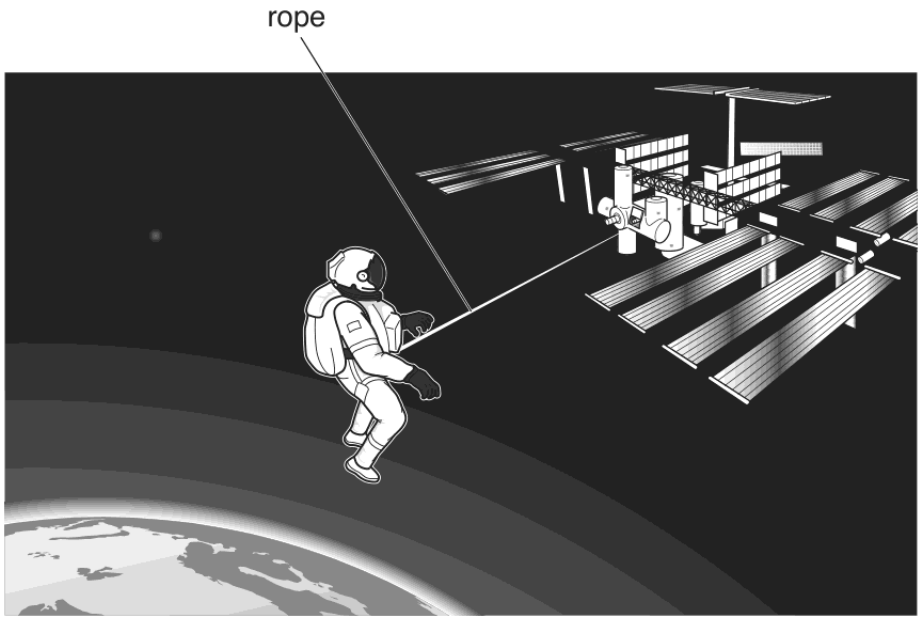


Fig. 9.1

The total mass of the astronaut and all his equipment is 160 kg.

The astronaut is initially at rest relative to the space station but he is then pulled towards the space station by the rope. His acceleration towards the space station is 0.35 m/s^2 for 1.2 s.

(a) Calculate

(i) the resultant force that causes a mass of 160 kg to accelerate at 0.35 m/s^2,

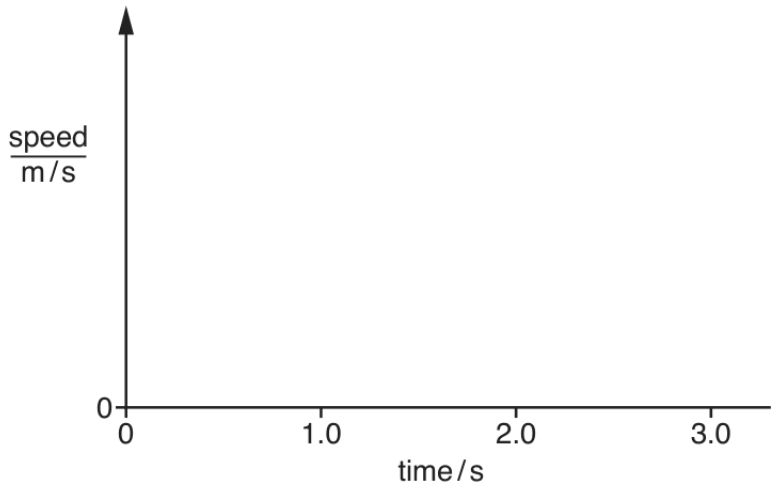
resultant force = ...[2]

(ii) the speed at which the astronaut is travelling after 1.2 s.

speed = ...[2]

(b) The resultant force on the astronaut is constant for 1.2 s, but then it decreases to zero instantly as the rope becomes slack.

(i) On Fig. 9.2, sketch the speed-time graph for the astronaut for the first 3.0 s of his motion.



[2]

Fig. 9.2

(ii) Describe how the distance moved by the astronaut during the first 3.0 s may be found using the speed-time graph.

.....

.....

.....[2]

(c) The space station is travelling at a constant speed in a circular orbit around the Earth as shown in Fig. 9.3.

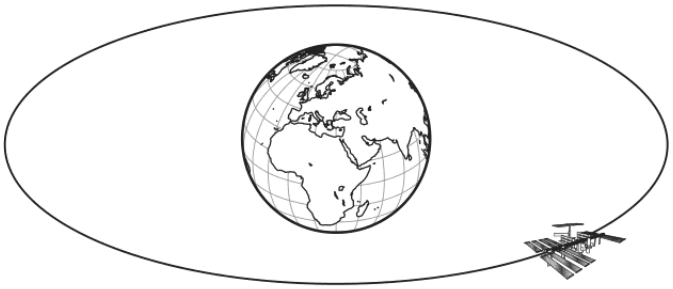


Fig. 9.3

(i) A force acts on the space station to keep it in this orbit.

1. On Fig. 9.3, draw an arrow to show the direction of this force. [1]

2. Explain what causes this force.

.....

.....

.....[2]

(ii) State what is meant by the *velocity* of an object.

.....

.....

.....[2]

(iii) State and explain what happens during the orbit to

1. the velocity of the space station,

.....

.....

.....[1]

2. the kinetic energy of the space station.

.....

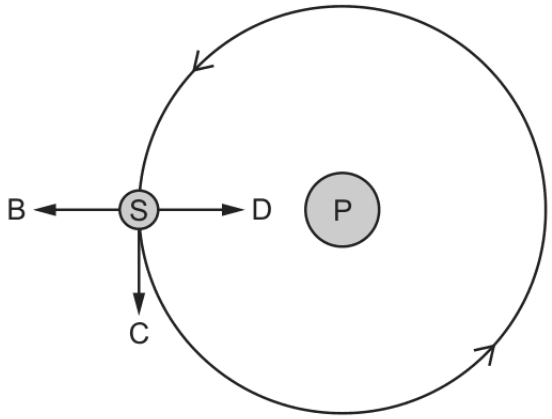
.....

.....[1]

Question	Answer	Marks
9(a)(i)	(F =) ma **or** 160 × 0.35	**C1**
	56 N	**A1**
9(a)(ii)	(v = u +) at **or** 0.35 × 1.2	**C1**
	0. 42 m / s	**A1**
9(b)(i)	straight line of positive gradient from origin to t = 1.2 s	**B1**
	horizontal line after t = 1.2 s	**B1**
9(b)(ii)	area mentioned	**B1**
	area under the line **or** convert cm^2 (of graph paper) to distance **or** in terms of the scales	**B1**
9(c)(i)	**1** arrow from space-station towards centre of Earth	**B1**
	2 gravitational field / attraction	**B1**
	of Earth	**B1**
9(c)(ii)	**two** separate approaches	
	I change of displacement	**M1**
	per unit time **or** divided by time	**A1**
	or **II** rate of change of distance **or** distance moved per unit time	**M1**
	in specified direction	**A1**
9(c)(iii)	**1** (it / velocity) changes **and** (because its) direction changes / force perpendicular to velocity	**B1**
	2 (remains) constant **and** depends on speed / g.p.e. remains constant / (k.e.) is a scalar quantity / speed is constant	**B1**

QUESTION · 5054_s18_qp_11.pdf · source page 2 of 20

3 The diagram shows a satellite S travelling at a constant speed in a circular orbit around a planet P.



Which statement is correct?

A The resultant force on the satellite is zero.

B The resultant force on the satellite is in direction B.

C The resultant force on the satellite is in direction C.

D The resultant force on the satellite is in direction D.

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 3

3	D	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 3 of 16

5 The planets in the Solar System orbit the Sun.

Which statement is correct?

A There is a force on each planet away from the Sun.

B There is a force on each planet in the direction in which it travels.

C There is a force on each planet opposite to the direction in which it travels.

D There is a force on each planet towards the Sun.

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 5

5	D	**1**

QUESTION · 5054_w18_qp_12.pdf · source page 3 of 16

4 On which car is there a resultant force?

A a car moving along a straight horizontal road at constant speed

B a car moving around a bend at constant speed

C a car moving uphill at constant velocity

D a car that is stationary

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 4

4	B	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a satellite travelling at a constant speed in a circular orbit around the Earth.

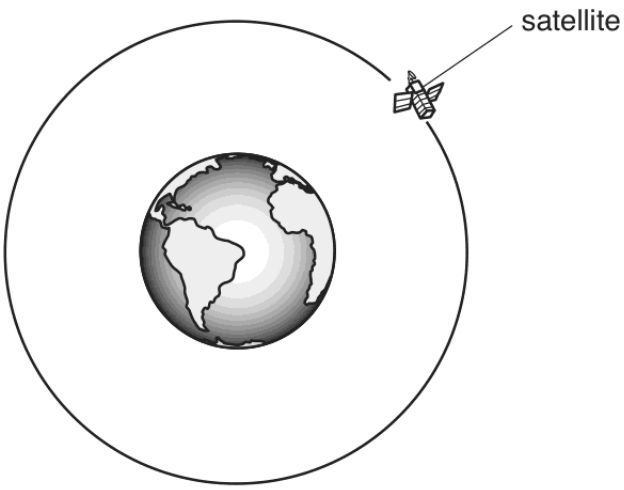


Fig. 1.1 (not to scale)

(a) State how *speed* differs from *velocity*.

.....

..... [1]

(b) As it orbits the Earth, the satellite is experiencing an acceleration.

(i) Explain, in terms of its velocity, why the satellite is accelerating.

.....

.....

..... [2]

(ii) On Fig. 1.1, draw an arrow, starting on the satellite, to show the direction of the satellite's acceleration. [1]

(c) As the satellite orbits the Earth, it experiences a force due to gravitational attraction.

State and explain whether this force does work on the satellite and state whether the energy of the satellite is affected.

.....

.....

.....

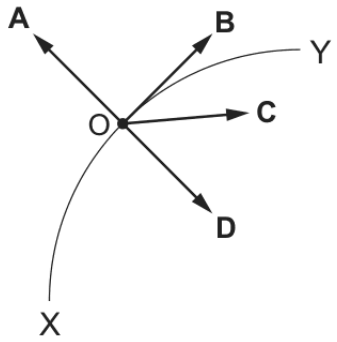
..... [2]

Question	Answer	Marks
1(a)	it / speed does **not** have a direction / is a scalar quantity	**B1**
1(b)(i)	direction (of velocity) changes (as it moves around the Earth)	**B1**
	its velocity changes with time (this is an acceleration)	**B1**
1(b)(ii)	arrow from satellite towards (centre of) Earth	**B1**
1(c)	**no** work done **and** force perpendicular to motion / no movement in direction of force	**B1**
	(kinetic and gravitational potential) energy remains constant / no effect	**B1**

QUESTION · 5054_s19_qp_11.pdf · source page 4 of 20

7 The diagram represents an object O moving from X to Y along a circular path at constant speed.

What is the direction of the resultant force on O in the position shown?

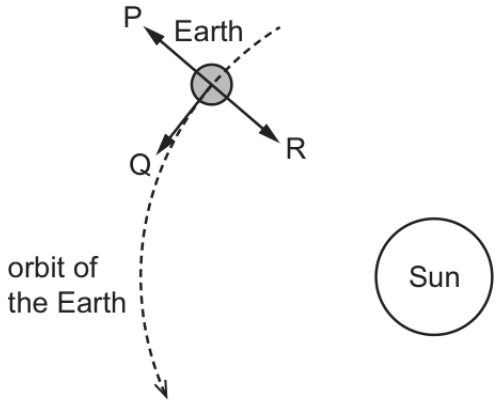


MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 7

7	D	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 5 of 20

8 The Earth travels at constant speed in a circular orbit around the Sun.



Which arrows show the direction of the acceleration of the Earth and the direction of the velocity of the Earth?

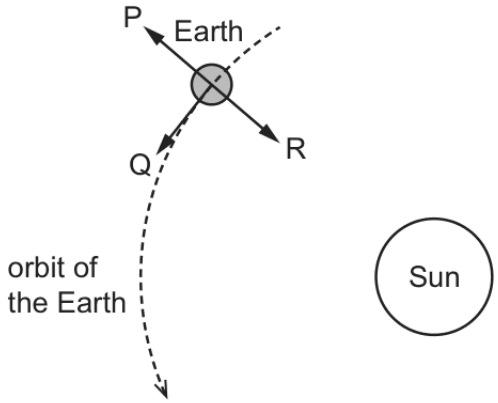
	direction of acceleration	direction of velocity
A	P	Q
B	Q	P
C	Q	R
D	R	Q

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 8

8	D	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 5 of 16

9 The Earth travels at constant speed in a circular orbit around the Sun.



Which arrows show the direction of the acceleration of the Earth and the direction of the velocity of the Earth?

	direction of acceleration	direction of velocity
A	P	Q
B	Q	P
C	Q	R
D	R	Q

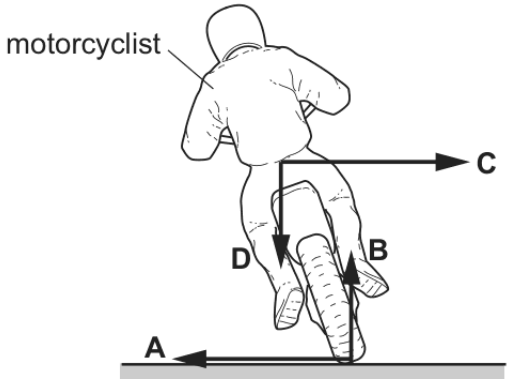
MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 9

9	D	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 3 of 16

7 The diagram shows a motorcyclist leaning over in order to turn the corner to the left.

Which force causes him to turn?



left right

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 7

7	A	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 4 of 16

8 Which type of force causes the Earth to orbit the Sun?

A elastic

B electrostatic

C gravitational

D magnetic

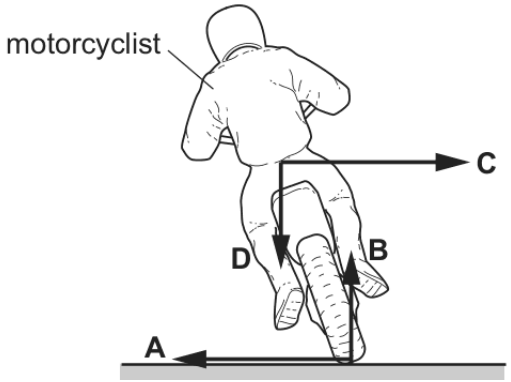
MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 8

8	C	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 3 of 16

8 The diagram shows a motorcyclist leaning over in order to turn the corner to the left.

Which force causes him to turn?



MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 8

8	A	**1**

2 Fig. 2.1 shows a satellite moving at a constant speed in a circular orbit around the Earth.

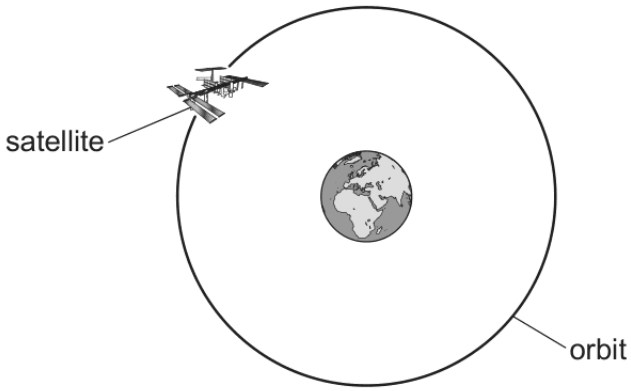


Fig. 2.1 (not to scale)

Speed is a scalar quantity but velocity is a vector quantity.

(a) State how a scalar quantity differs from a vector quantity.

.....

..... [1]

(b) Underline every vector quantity in the list.

distance **displacement** **force** **length** **mass** **time**

[1]

(c) There is a resultant force acting on the satellite in Fig. 2.1.

(i) Explain how the motion of the satellite shows that a resultant force is acting on it.

.....

.....

..... [2]

(ii) State the cause of this force.

.....

..... [1]

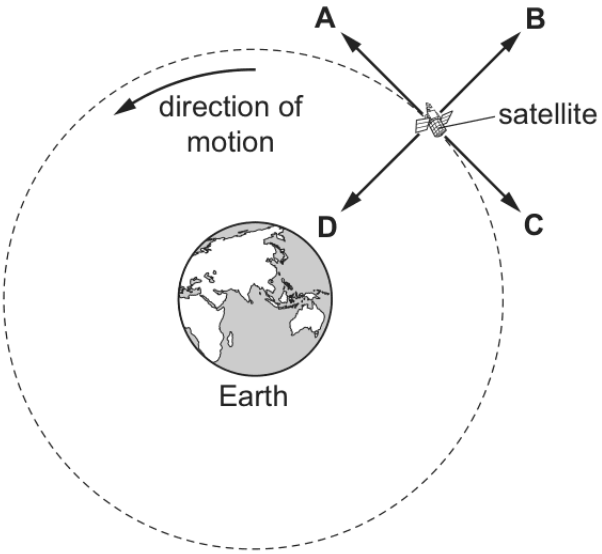
[Total: 5]

Question	Answer	Marks
2(a)	it / a scalar does not have a direction	**B1**
2(b)	*displacement* **and** *force* underlined	**B1**
2(c)(i)	direction (of motion) is changing	**B1**
	velocity is changing (with time) **or** it is accelerating (which requires a resultant force)	**B1**
2(c)(ii)	gravitational attraction / field of the earth	**B1**

QUESTION · 5054_s21_qp_11.pdf · source page 3 of 16

5 A satellite is shown moving around the Earth in a circular path at a constant speed.

Which arrow shows the direction of the force on the satellite?



not to scale

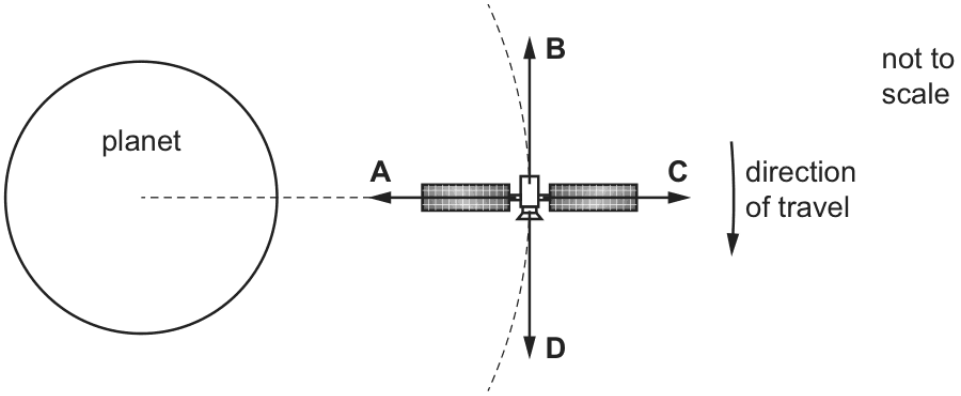
MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 5

5	D	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 3 of 16

5 The diagram shows a satellite orbiting a planet at a steady speed.

In which direction does the resultant force act on the satellite?



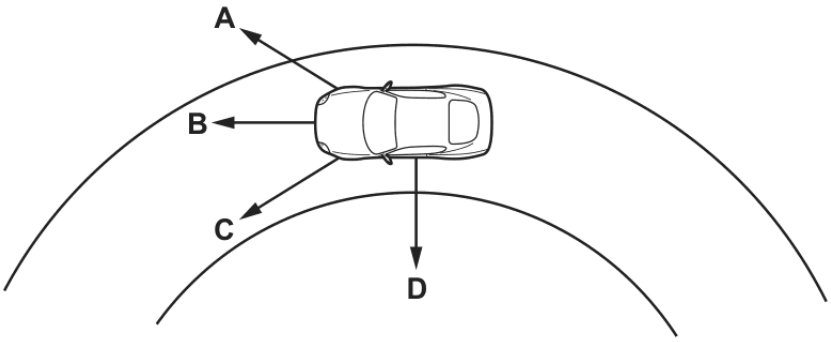
MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 5

5	A	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 3 of 20

4 The diagram shows a car going around a circular track at constant speed.

Which arrow shows the direction of the resultant force on the car?



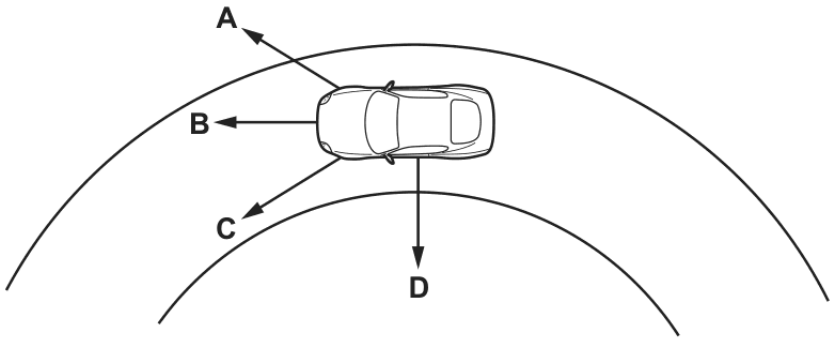
MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 4

4	D	**1**

QUESTION · 5054_w21_qp_12.pdf · source page 3 of 16

5 The diagram shows a car going around a circular track at constant speed.

Which arrow shows the direction of the resultant force on the car?



MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 5

5	D	**1**

6 Isotope X is radioactive. It decays by alpha-particle emission to a stable isotope.

(a) State how a nucleus of X changes when it emits an alpha-particle.

.....

.....

..... [2]

(b) There is a radiation detector in a laboratory where there are no radioactive samples.

The detector is switched on and shows an average count rate of 22 counts / minute.

(i) State why the radiation detector shows a count rate.

.....

..... [1]

(ii) A sample of isotope X is placed 2 cm from the detector and the reading displayed is 8000 counts / minute.

The sample is moved a distance of 10 cm from the detector. The reading returns to an average value of 22 counts / minute.

Explain why the reading returns to the original value.

.....

.....

..... [2]

(c) An alpha-particle passes into a region where there is a magnetic field. In the magnetic field, a force acts on the alpha-particle so that it follows a circular path. Fig. 6.1 shows that the particle passes through point J.

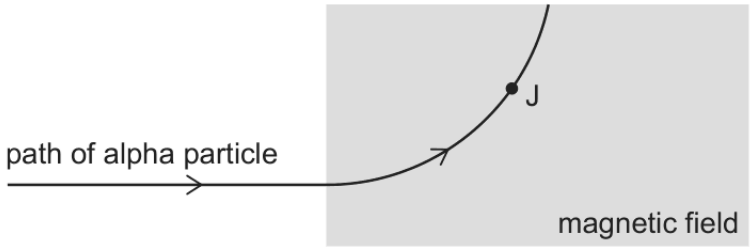


Fig. 6.1

(i) On Fig. 6.1, draw an arrow through point J to show the direction of the force on the alpha-particle at J. [1]

(ii) Determine the direction of the magnetic field and mark a tick in the box (✓) that indicates this direction.

- ☐ to the left
- ☐ to the right
- ☐ towards the top of the page
- ☐ towards the bottom of the page
- ☐ into the page
- ☐ out of the page

[1]

(iii) Explain whether this force does work on the alpha-particle as the particle moves along the circular path.

.....

.....

..... [2]

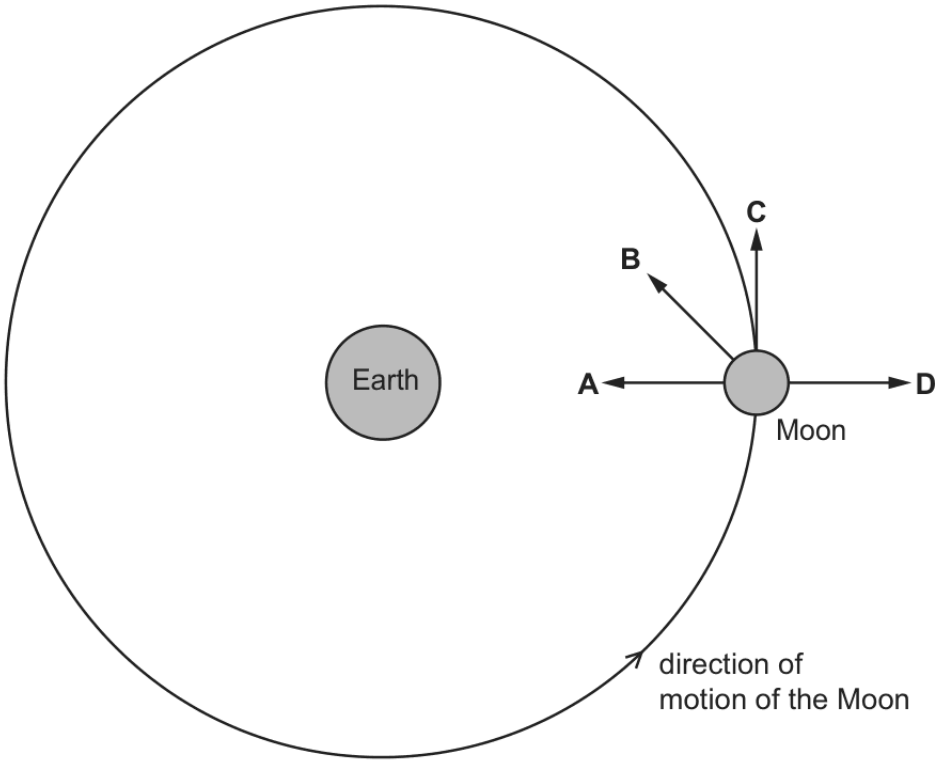
[Total: 9]

Question	Answer	Marks
6(a)	any **two** from: number of protons decreases by two number of neutrons decreases by two **or** number of nucleons decreases by four	**B2**
6(b)(i)	background radiation (detected) **or** source of background radiation mentioned (e.g. cosmic rays; radiation in the air)	**B1**
6(b)(ii)	alpha-particles are absorbed by the air (between the source and detector)	**B1**
	all the alpha-particles / radiation are absorbed (by the air between the source and detector) **or** alpha-particles travel less than 10 cm (in air)	**B1**
6(c)(i)	arrow through J towards centre of circular path	**B1**
6(c)(ii)	into the page (i.e. fifth box ticked)	**B1**
6(c)(iii)	force perpendicular to direction of motion **or** no component of velocity along line of action of force **or** work done = force × distance moved in direction of motion	**M1**
	no work done	**A1**

QUESTION · 5054_s22_qp_11.pdf · source page 3 of 16

3 The diagram represents the Moon in its orbit around the Earth.

Which arrow represents the direction of the resultant force acting on the Moon at the instant shown?



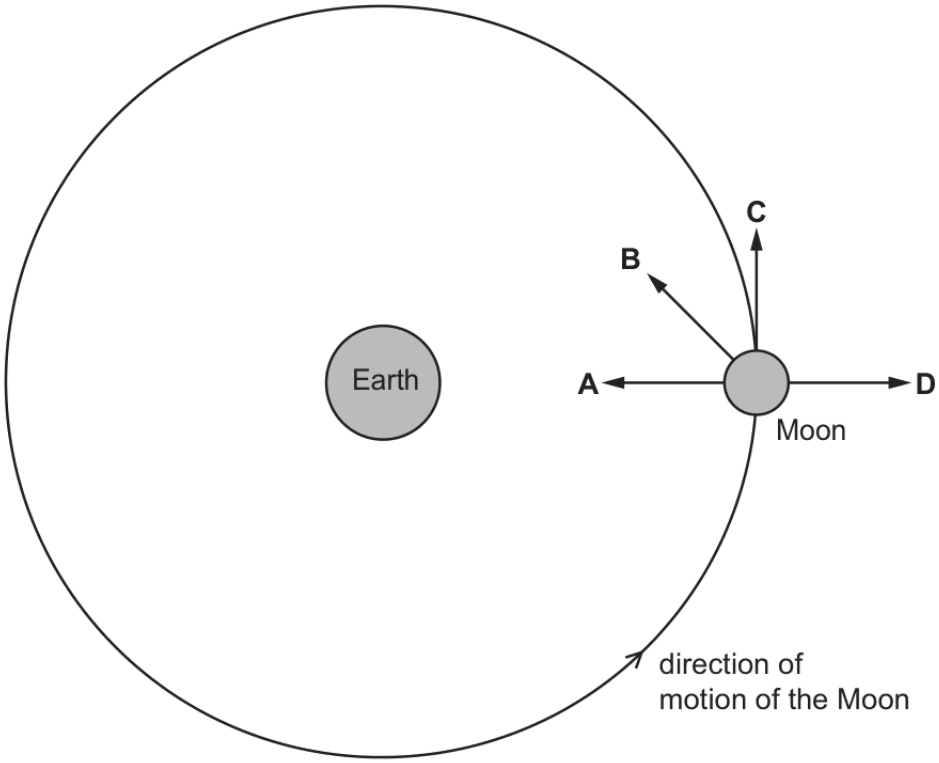
MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 3

3	A	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 3 of 20

5 The diagram represents the Moon in its orbit around the Earth.

Which arrow represents the direction of the resultant force acting on the Moon at the instant shown?



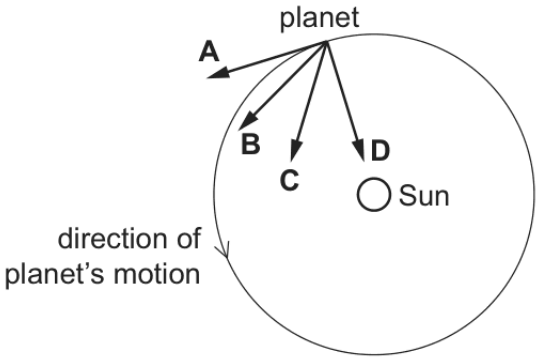
MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 5

5	A	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 4 of 16

8 A planet orbits the Sun. The planet moves in a circular path at constant speed, but is accelerating because it is changing direction.

In which direction does the resultant force act to cause this acceleration?



MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 8

8	D	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 4 of 16

8 A car travels on a horizontal road around a bend at constant speed.

What is the direction of the resultant force on the car?

A backwards

B forwards

C towards the centre of the bend

D towards the road

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 8

8	C	**1**

9 The planet Venus orbits the Sun at a constant speed of 3.5×10^4 m/s and takes a time T_V to complete one orbit.

(a) Venus is always 1.1×10^{11} m from the Sun.

Calculate T_V.

T_V = ... s [2]

(b) Explain why speed is a scalar quantity.

.....

..... [1]

(c) Velocity is a vector quantity.

(i) Explain what happens to the velocity of Venus as it orbits the Sun.

.....

.....

.....

..... [2]

(ii) There is a resultant force on Venus as it orbits the Sun. The force is perpendicular to the direction of the motion of Venus.

State why this force is needed.

.....

..... [1]

(iii) Fig. 9.1 shows the orbit of Venus around the Sun.

On Fig. 9.1, draw an arrow to show the resultant force on Venus.

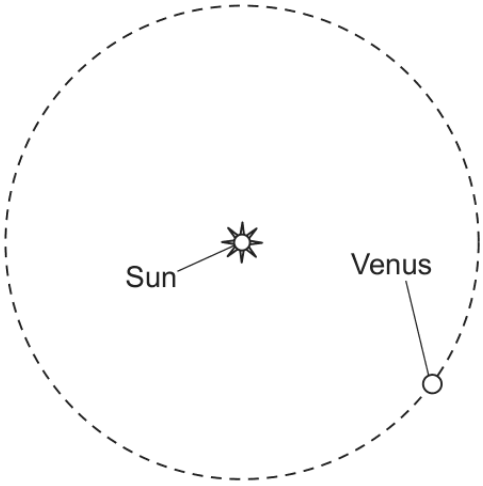


Fig. 9.1

[1]

(iv) State what causes the force acting on Venus.

.....

.....

..... [2]

(d) One planet in the Solar System is closer to the Sun than Venus.

(i) State the name of this planet.

..... [1]

(ii) Compare the time that this planet takes to complete one orbit of the Sun with T_V and explain the difference.

.....

.....

.....

..... [2]

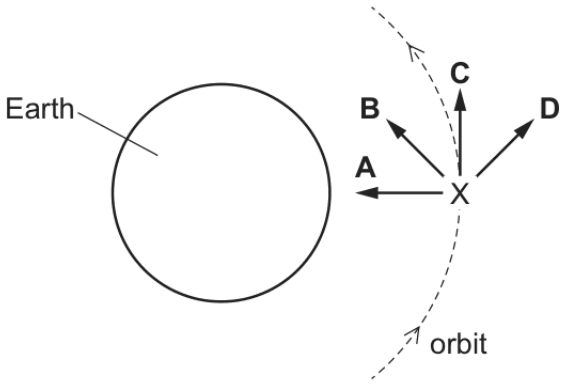
[Total: 12]

Question	Answer	Marks
9(a)	(T =) $2\pi r / v$ **or** $2\pi \times 1.1 \times 10^{11} / 3.5 \times 10^{4}$	**C1**
	2.0×10^{7} (s)	**A1**
9(b)	it / speed does not have a direction	**B1**
9(c)(i)	it / velocity changes	**M1**
	(because) its direction changes	**A1**
9(c)(ii)	to change the velocity **or** to keep Venus in a circular orbit **or** to accelerate Venus towards the centre of the orbit / Sun	**B1**
9(c)(iii)	arrow towards Sun **and** on Venus	**B1**
9(c)(iv)	it / Venus is in the gravitational field **or** the gravitational attraction	**M1**
	of the Sun	**A1**
9(d)(i)	Mercury	**B1**
9(d)(ii)	time taken is less **and** orbital speed is greater / circumference of orbit is less / closer to the Sun	**B1**
	circumference of orbit is less **and** orbital speed is greater.	**B1**

QUESTION · 5054_w23_qp_12.pdf · source page 5 of 20

9 The International Space Station orbits the Earth.

Which arrow shows the direction of the force on the International Space Station when it is at point X?



MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 9

9	A	**1**

1 A ball is released from rest at point A and moves along a smooth track ABCDE as shown in Fig. 1.1. The ball is shown at point A and as it passes point B.

The ball is always in contact with the track and air resistance is negligible.

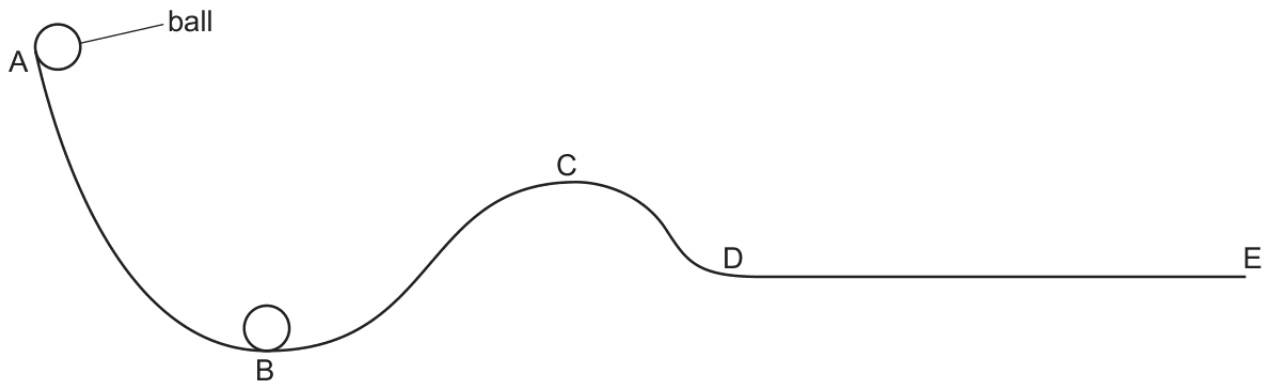


Fig. 1.1

Fig. 1.2 shows the distance−time graph for the ball as it moves from A to E.

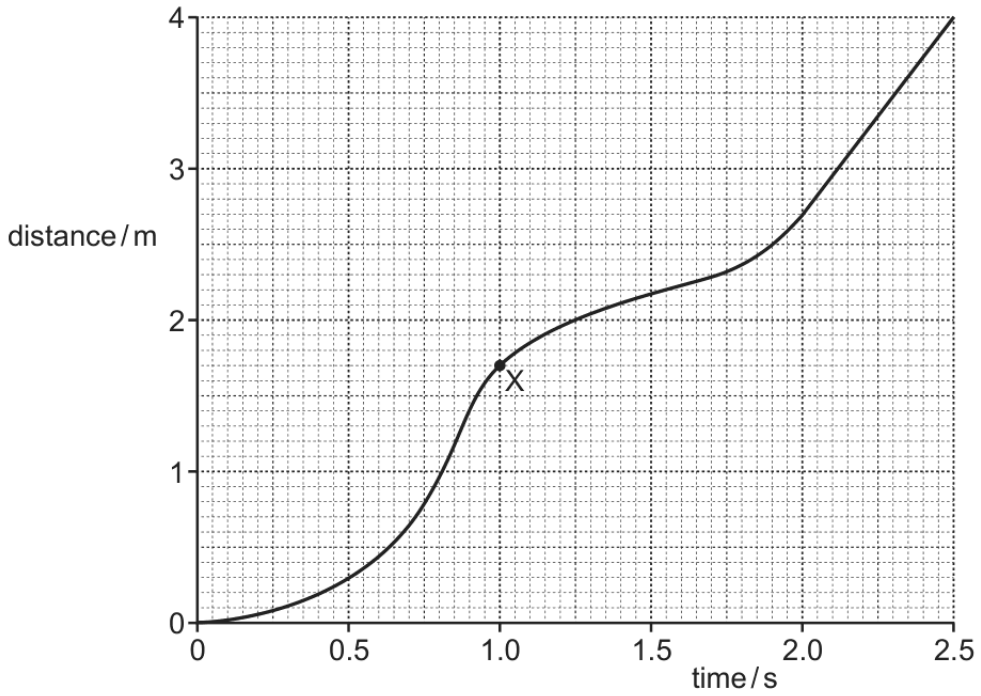


Fig. 1.2

(a) (i) On Fig. 1.2, mark:

- with the letter P **one** point where the ball accelerates
- with the letter Q **one** point where the ball has constant speed.

[2]

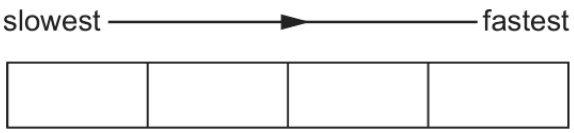
(ii) Determine the speed of the ball at point X on Fig. 1.2.

Show your working.

speed = ... m/s [3]

(b) The speeds of the ball at A, B, C and D are v_A, v_B, v_C and v_D respectively.

Arrange these four speeds from slowest to fastest.



[1]

(c) Work is done to transfer energy between energy stores as the ball moves from A to B.

Name the force involved in the work done and describe the energy transfer.

.....

.....

.....

..... [3]

(d) The track at point B is circular in shape.

On Fig. 1.1, draw an arrow to show the direction of the resultant force on the ball at point B.

[1]

[Total: 10]

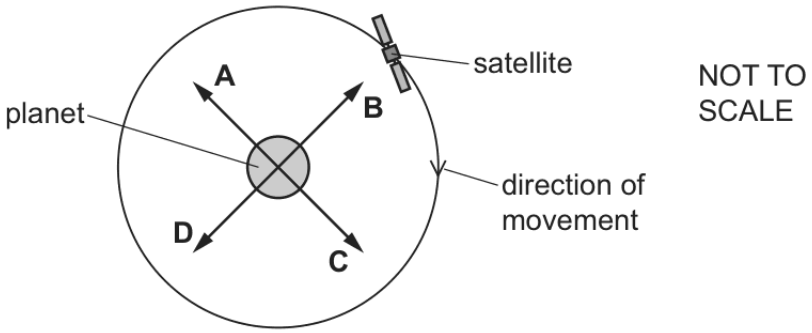
Question	Answer	Marks
1(a)(i)	P marked in region t = 0–0.8 s **or** in region 1.6–2.0 s	**B1**
	Q marked between 2.0 and 2.5 s	**B1**
1(a)(ii)	tangent drawn at X	**B1**
	(speed =) d/t in any form numerical or algebraic	**C1**
	1.9–2.4 m/s	**A1**
1(b)	$v_A < v_C < v_D < v_B$	**B1**
1(c)	(work done by) gravity / weight / gravitational attraction	**B1**
	from gravitational potential energy (store)	**B1**
	to kinetic energy (store)	**B1**
1(d)	arrow upwards at D	**B1**

QUESTION · 5054_s24_qp_12.pdf · source page 5 of 16

11 A satellite orbits a planet in a circular path as shown. It has constant speed.

There is a force on the satellite due to the planet.

In which direction is the force on the satellite when it is in the position shown?



MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 11

11	D	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 5 of 16

12 A satellite orbits the Earth.

What is the direction of the force on the satellite causing this circular motion?

A away from the centre of the Earth

B in the direction of motion of the satellite

C in the opposite direction to the motion of the satellite

D towards the centre of the Earth

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 12

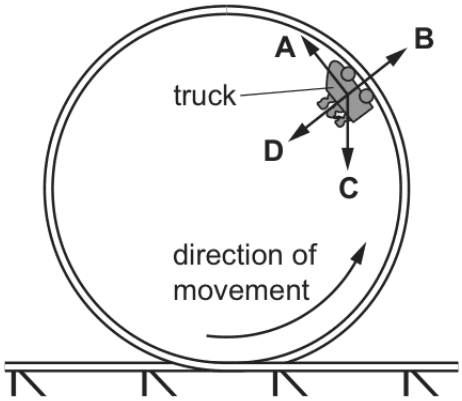
12	D	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 4 of 20

6 The diagram shows a section of a roller coaster ride that is a circle arranged vertically.

The truck is moving at constant speed at the position shown.

Which arrow shows the direction of the resultant force acting on the truck?



MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 16 of 16

38 The planets in the Solar System orbit the Sun.

Which statement is correct?

A There is a force on each planet away from the Sun.

B There is a force on each planet in the direction in which it travels.

C There is a force on each planet opposite to the direction in which it travels.

D There is a force on each planet towards the Sun.

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 3 of 3 · row for Question 38

38	D	**1**

9 An object moves through space. It is a large distance from any other object.

(a) There are no external forces acting on the object.

Describe **two** features of the motion of the object that are a consequence of Newton's first law.

1 .....

.....

2 .....

.....

[2]

(b) Planet X and Venus are two of the planets in the Solar System.

Fig. 9.1 shows the orbit of Venus and the orbit of planet X. Both orbits are approximately circular.

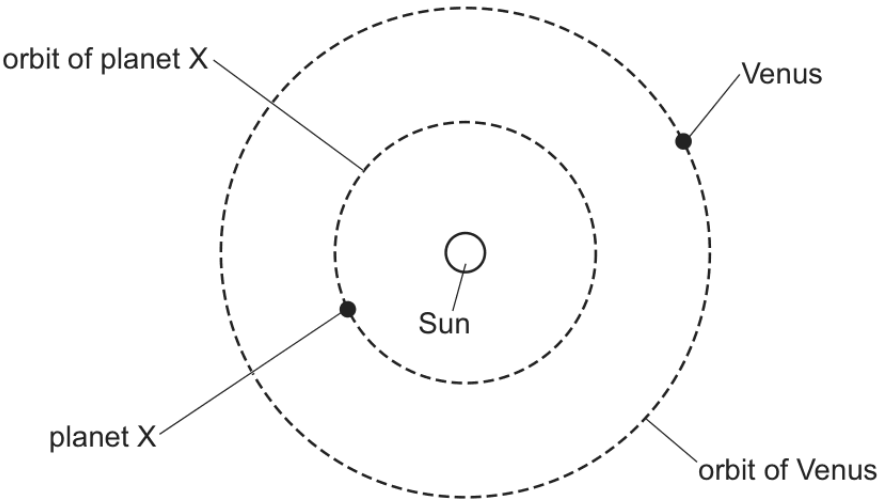


Fig. 9.1

Venus experiences an external force as it orbits the Sun.

(i) State why a force must act on Venus to keep it following an approximately circular orbit.

.....

..... [1]

(ii) State what causes this force, and state the direction of the force.

cause .....

.....

direction .....

[3]

DO NOT WRITE IN THIS MARGIN
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DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(iii) The average radius of the orbit of Venus is 1.1×10^{11} m. It takes Venus 5400 hours to complete one orbit of the Sun.

Calculate the average orbital speed of Venus around the Sun.

speed = ... m/s [2]

(iv) Planet X is closer to the Sun than Venus.

State the name of planet X and state how its orbital speed around the Sun compares with the orbital speed of Venus.

name .....

speed .....

.....
[2]

[Total: 10]

MARK SCHEME · 5054_w25_ms_21.pdf · source page 13 of 14 · Question 9

Question	Answer	Marks
9(a)	speed is constant	**B1**
	(motion is) in a straight line	**B1**
9(b)(i)	to change the direction of the motion / velocity (of Venus) / otherwise it would move in a straight line	**B1**
9(b)(ii)	gravitational field / gravitational attraction / gravity	**B1**
	(gravitational field) of Sun	**B1**
	towards centre of circle / Sun / inward	**B1**

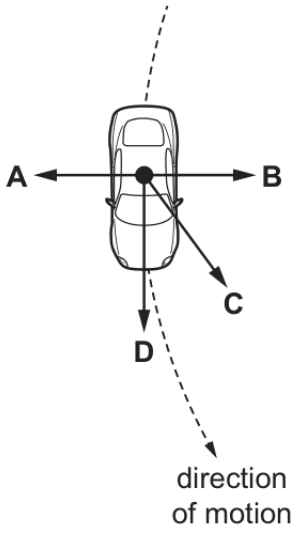
MARK SCHEME · 5054_w25_ms_21.pdf · source page 14 of 14 · Question 9

Question	Answer	Marks
9(b)(iii)	(speed =) $\frac{2\pi r}{T}$ **or** $\frac{(2\times\pi)\times 1.1\times 1011}{5400\times(3600)}$	**C1**
	3.6×10^{4} (m / s)	**A1**
9(b)(iv)	Mercury	**B1**
	(its speed is) greater (than that of Venus)	**B1**

QUESTION · 5054_w25_qp_12.pdf · source page 6 of 16

12 A car moves in a circle at constant speed.

What is the direction of the resultant force acting on the car?



× centre of circle turned by car

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 12

12	B	**1**

Chapter 1 Motion, forces and energy

1.5 Forces

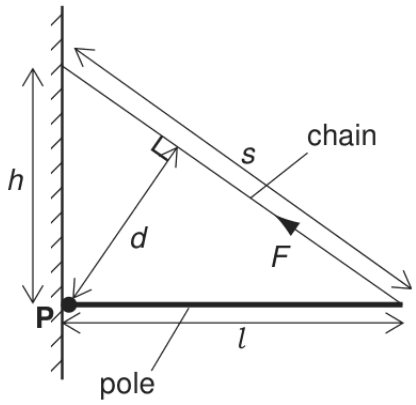
1.5.5 Turning effect of forces

96 entries: 59 Paper 1 questions, 37 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 4 of 20

8 A horizontal pole is attached to the side of a building. There is a pivot **P** at the wall and a chain is connected from the end of the pole to a point higher up the wall.



There is a tension force *F* in the chain.

What is the moment of the force *F* about the pivot **P**?

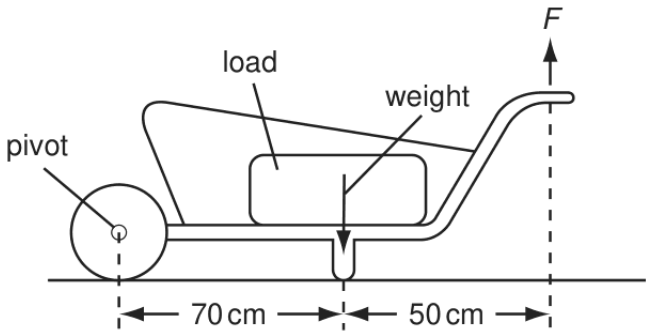
A $F \times d$ **B** $F \times h$ **C** $F \times l$ **D** $F \times s$

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 8

8	**A**

QUESTION · 5054_w04_qp_1.pdf · source page 5 of 20

8 A load is to be moved using a wheelbarrow. The total mass of the load and wheelbarrow is 60 kg. The gravitational field strength is 10 N / kg.



What is the size of force *F* needed just to lift the loaded wheelbarrow?

A 350 N **B** 430 N **C** 600 N **D** 840 N

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 8

8 **A**

9 Fig. 9.1 is a diagram of a simple d.c. motor.

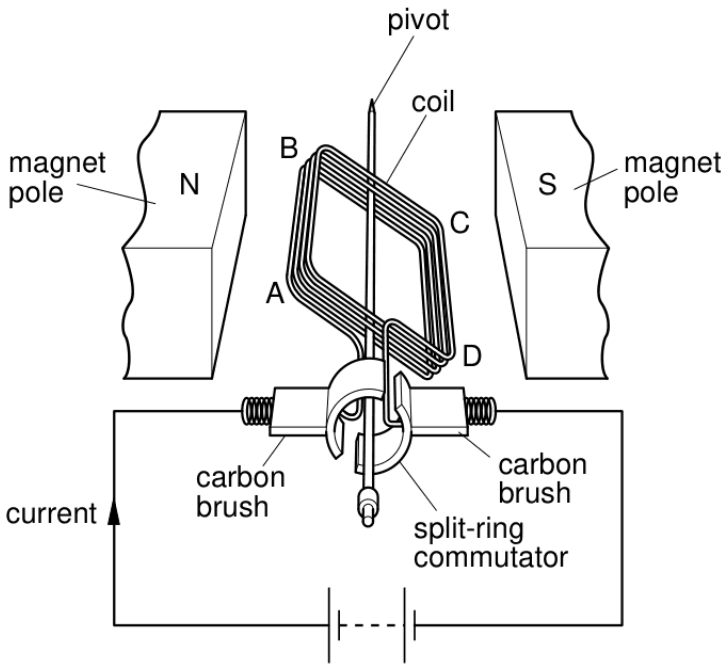


Fig. 9.1

The gap between the two halves of the split-ring commutator is so wide that a carbon brush can only touch one half of the split-ring at any time. This protects the circuit. It also means that sometimes the motor will not start when switched on.

The coil is rotated by vertical forces that act downwards on side AB and upwards on side CD. The current causes a constant force of 3.0 N on each side. The moment created by these forces varies as the coil turns. The moment is a maximum when the coil is horizontal.

The distances AD and BC are both 0.065 m.

(a) Explain why

(i) the carbon brushes must **not** be allowed to touch both halves of the split-ring at the same time,

(ii) sometimes the motor does **not** start when switched on, even if there is no friction.

[3]

(b) (i) Define the *moment of a force*.

(ii) Calculate the value of the maximum moment created on the coil.

(iii) Explain why the moment is a maximum when the coil is horizontal.

(iv) Sketch a graph to suggest how the moment acting on the coil varies with time as the coil rotates at constant speed. Label each axis. On the time axis, mark clearly the time taken for one revolution of the coil.

[7]

(c) (i) State the measurements you would make in order to find the electrical power input to the motor and state how your measurements are used to find the electrical power.

(ii) Draw a diagram of the circuit you would use to make these measurements.

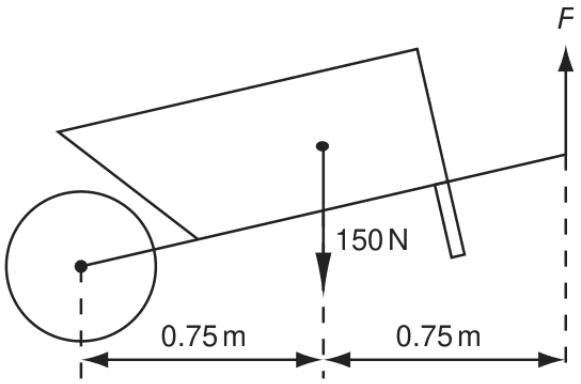
[5]

9	**(a)**	**(i)**	low resistance **or** short circuit **or** large current (in battery) **or** no current in coil	**B1**	
		(ii)	brushes touch gaps or no contact with ring **or** coil vertical	**B1**	
			no current **or** open circuit **or** no forces **or** no moment	**B1**	
	(b)	**(i)**	force x distance	**M1**	
			perpendicular distance	**A1**	

		(ii)	3 x 0.065 or 3 x 0.065/2	**C1**	
			0.195 Nm		
			(accept 0.19 or 0.20; 0.39 or 0.0975 Nm…C1)	**A1**	
		(iii)	large (perpendicular) distance (between forces/axis when coil horizontal)	**B1**	
		(iv)	axes labelled **and** graph any repeating shape with same sign	**B1**	
			(**not** a sine wave either side of axis)		
			1 revolution correct on time axis	**B1**	
			(should be between three maxima/minima if graph always same sign, if graph goes either side of axis e.g. sine wave, award mark for period of wave drawn)		
	(c)	**(i)**	voltage (p.d.) (across motor)	**B1**	
			current (through motor)	**B1**	
			power = VI	**B1**	
		(ii)	correct series circuit with ammeter, cell etc., any symbol for motor		
			(**accept** lamp labelled motor **condone** V in series)	**B1**	
			voltmeter in parallel with motor		
			(or cell if no extra resistor)	**B1**	**15**

QUESTION · 5054_s05_qp_1.pdf · source page 3 of 20

6 The diagram shows a wheelbarrow and its load, which have a total weight of 150 N. This is supported by a vertical force *F* at the ends of the handles.



What is the value of *F*?

A 75 N **B** 150 N **C** 225 N **D** 300 N

MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 6

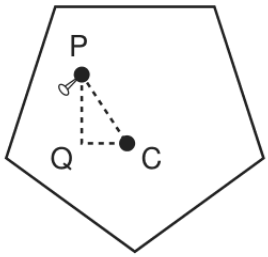
6	**A**

QUESTION · 5054_w05_qp_1.pdf · source page 5 of 20

9 A plane lamina is freely suspended from point P.

The weight of the lamina is 2.0 N and the centre of mass is at C.

PC = 0.50 m
PQ = 0.40 m
QC = 0.30 m



lamina

The lamina is displaced to the position shown. What is the moment that will cause the lamina to swing?

A 0.60 N m clockwise

B 0.80 N m anticlockwise

C 1.0 N m clockwise

D 1.0 N m anticlockwise

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 9

9	**A**

QUESTION · 5054_s06_qp_1.pdf · source page 4 of 20

8 If a nut and bolt are difficult to undo, it may be easier to turn the nut by using a longer spanner.

This is because the longer spanner gives

A a larger turning moment.

B a smaller turning moment.

C less friction.

D more friction.

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 8

8	**A**

Section B

Answer **two** questions from this section.

Use the separate answer sheets available from the Supervisor.

9 Fig. 9.1 shows a thin sheet of metal suspended from a hole in one corner at A. The weight of the metal is 0.10 N and the centre of mass is at B. The diagram is drawn full scale.

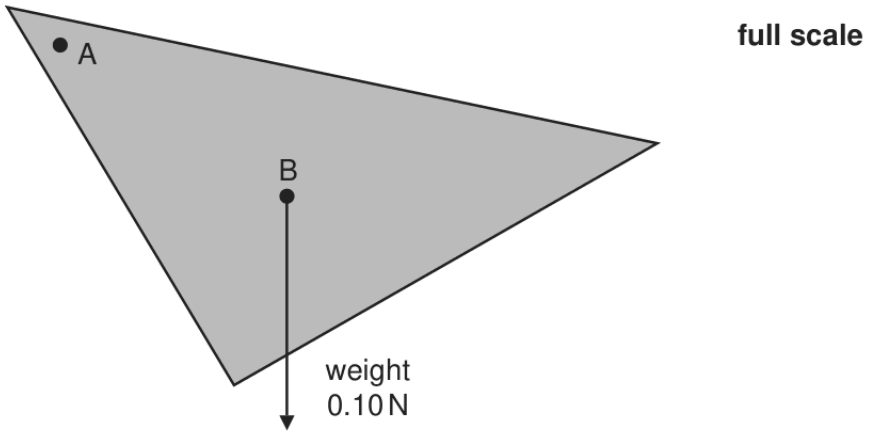


Fig. 9.1

(a) Describe in detail how you would experimentally determine the position of the centre of mass of the sheet of metal. [6]

(b) The sheet turns because of the moment of the weight about point A.

(i) Define what is meant by the *moment of a force*. [2]

(ii) Using a distance measured on Fig. 9.1, calculate the moment of the weight about point A. State clearly which distance you measured and give the unit of your final answer. [3]

(c) Fig. 9.2 and Fig. 9.3 show a thick piece of wood with one corner on a table.

Fig. 9.4 shows the same piece of wood balanced on the table. B is the centre of mass.

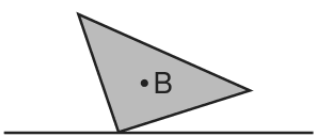


Fig. 9.2

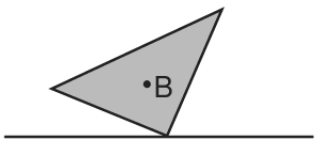


Fig. 9.3

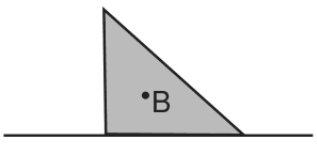


Fig. 9.4

(i) Explain why in Fig. 9.2 the piece of wood falls to the right and in Fig. 9.3 it falls to the left. [2]

(ii) Explain why the piece of wood in Fig. 9.4 does not fall over. [1]

(iii) Suggest how the thickness of the wood in Fig. 9.4 affects its stability. [1]

Section B

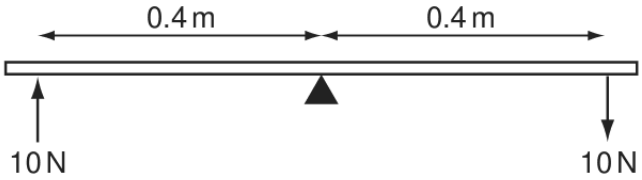
9	**(a)**	set wood swinging/let metal pivot **or** fall **OR** allow to come to rest use of plumb line **from hole** mark line along plumb line (on metal) hang from another hole line intersection is centre of mass hang from 3rd hole	balance on sort of edge clearly a sharp edge mark line of edge repeat in new position intersection is centre of mass repeat for 3rd position	
			OR balance on point sharp (compass) point move till balanced	
		ANY 6 consistent lines max.	point is centre of mass	B6
	(b) **(i)**	force × distance		**M**1
		perpendicular (accept symbol) distance **or** shortest distance to line of action of force		A1
	(ii)	correct perpendicular distance (2.9 – 3.1 cm)		B1
		worked out value of: 0.1 × distance reading		B1
		Ncm (or Nm if conversion of distance to m clear)		B1
	(c) **(i)**	moment **or** turning effect of weight		C1
		anticlockwise and clockwise moment **or** weight to right **and** left of corner		A1
	(ii)	moments balance/cancel **or** weight inside base		B1
	(iii)	thicker more stable/thinner less stable		B1 15

QUESTION · 5054_s07_qp_1.pdf · source page 4 of 20

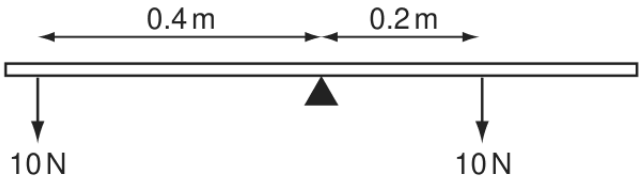
7 Forces are applied to a uniform beam pivoted at its centre.

Which beam is balanced?

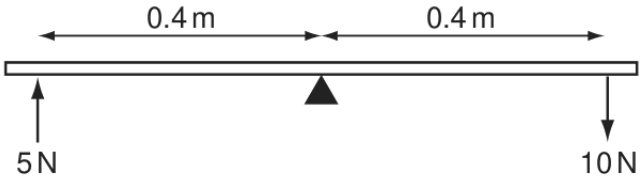
A



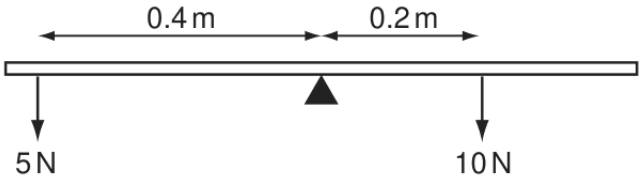
B



C



D

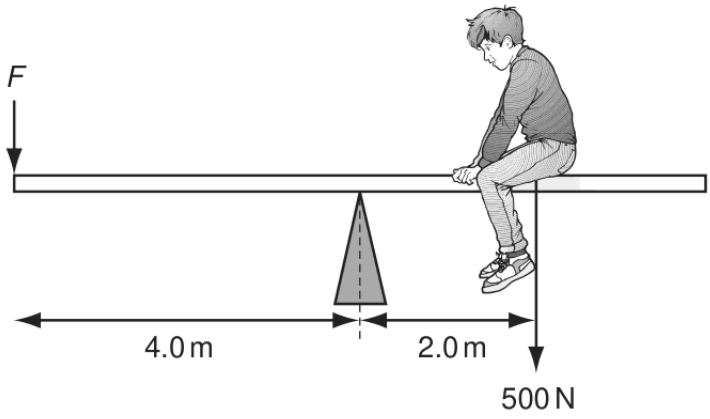


MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 7

7	**D**

QUESTION · 5054_w07_qp_1.pdf · source page 4 of 20

7 The diagram shows a boy of weight 500 N sitting on a see-saw. He sits 2.0 m from the pivot.



What is the force *F* needed to balance the see-saw?

A 250 N **B** 750 N **C** 1000 N **D** 3000 N

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 7

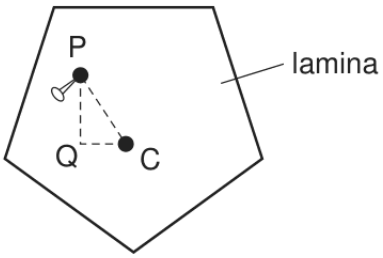
7	**A**

QUESTION · 5054_s08_qp_1.pdf · source page 5 of 20

8 A flat lamina is freely suspended from point P.

The weight of the lamina is 2.0 N and the centre of mass is at C.

PC = 0.50 m
PQ = 0.40 m
QC = 0.30 m



The lamina is displaced to the position shown.

What is the moment that will cause the lamina to swing?

A 0.60 N m clockwise

B 0.80 N m anticlockwise

C 1.0 N m clockwise

D 1.0 N m anticlockwise

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 8

8	**A**

For Examiner's Use

2 Fig. 2.1 shows a device for punching holes in a piece of paper. A person applies a force *F* at the end of the arm. Just before the hole is made in the paper, the arm is at rest.

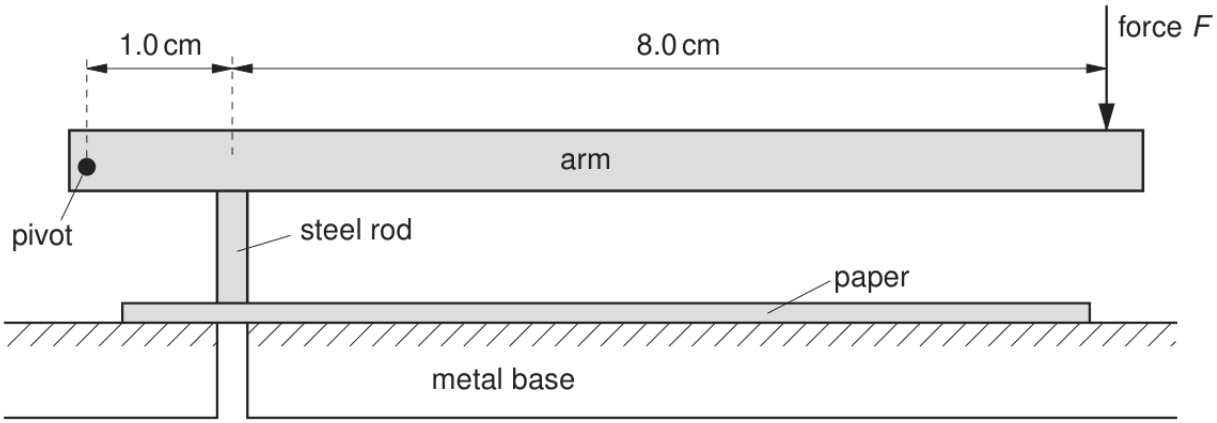


Fig. 2.1

(a) Just before the hole is made, the force upwards on the steel rod is 7.2 N.

Calculate the value of *F*. Use the distances marked on Fig. 2.1.

F = ... [2]

(b) The steel rod pushes the paper into the hole in the metal base with a force of 7.2 N. The end of the steel rod has an area of $2.8 \times 10^{-5}\,m^2$.

Calculate the pressure exerted on the paper.
State the formula that you use in your calculation.

pressure = ... [2]

(c) The force downwards on the paper due to the rod and the force upwards on the rod due to the paper are related by Newton's third law. State Newton's third law.

.....

.....

..... [1]

2 **(a)** any attempt at a moment calculation, e.g. any $F_1d_1 = F_2d_2$ seen, or answer 0.9 N C1
0.8(0) N A1

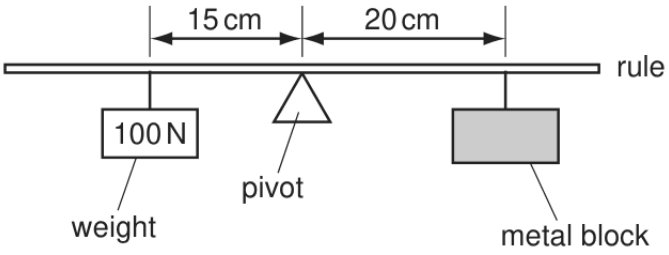
(b) P = F/A **formula** stated B1
2.6×10^5 Pa (2.571×10^5 Pa) B1

(c) action and reaction are equal and opposite **or** every force has an equal and opposite force **or** force on body A is equal and opposite to force on body B B1

[Total: 5]

QUESTION · 5054_w08_qp_1.pdf · source page 5 of 20

7 The diagram shows a uniform half-metre rule balanced at its mid-point.



What is the weight of the metal block?

A 50 N **B** 75 N **C** 100 N **D** 150 N

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 7

7	**B**

For Examiner's Use

2 Fig. 2.1 shows a wooden walking-stick that has a metal head and a rubber foot. It balances on a pencil placed 0.50 m from its rubber foot.

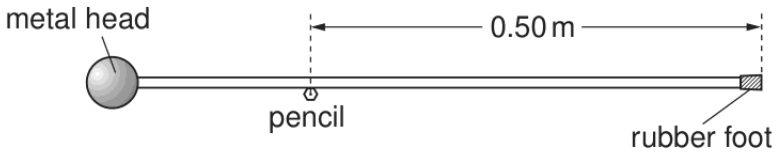


Fig. 2.1

(a) State the distance between the centre of mass of the walking-stick and the end of the rubber foot.

distance = ... [1]

(b) The pencil is moved along the walking-stick towards its rubber foot. State and explain the motion of the walking-stick.

.....

.....

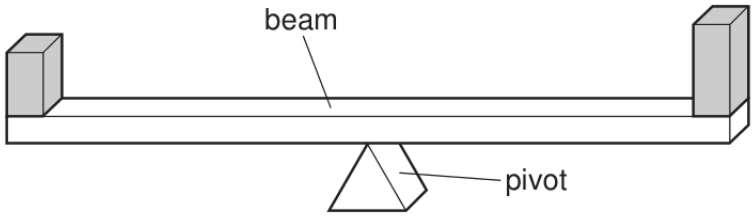
.....

..... [3]

2	**(a)**	0.5(0) m	B1	
	(b)	rotates/tilts/unbalanced/one side down/one side up	C1	
		rotates anticlockwise/down on left **or** head down **or** foot up	A1	
		(net) anticlockwise moment **or** moment on left > moment on right **or** weight/CM on left of pivot	B1	[4]

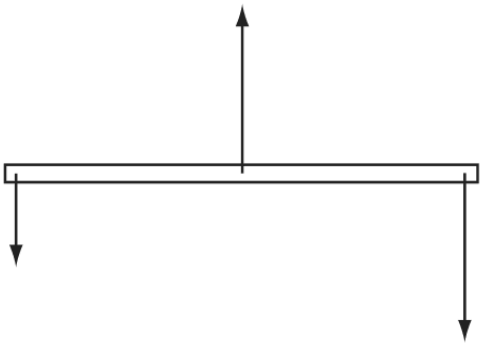
QUESTION · 5054_s09_qp_1.pdf · source page 5 of 20

8 Two blocks are placed on a beam which balances on a pivot at its centre. The weight of the beam is negligible.

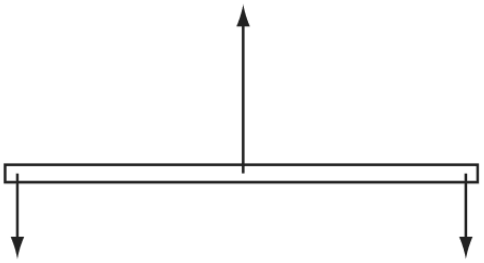


Which diagram shows the forces acting on the beam?
(The length of each arrow represents the size of a force.)

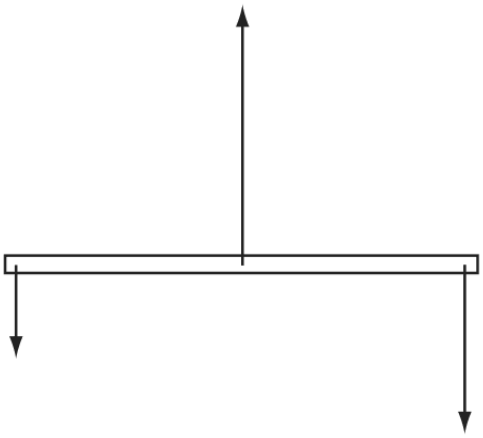
A



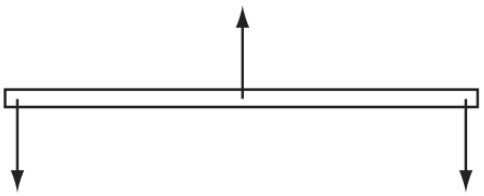
B



C



D



MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 8

8	**B**

For Examiner's Use

7 Fig. 7.1 shows a rectangular concrete slab of weight 18 000 N. It rests on a brick wall and is the roof of a bus shelter. The concrete slab is 3.0 m wide.

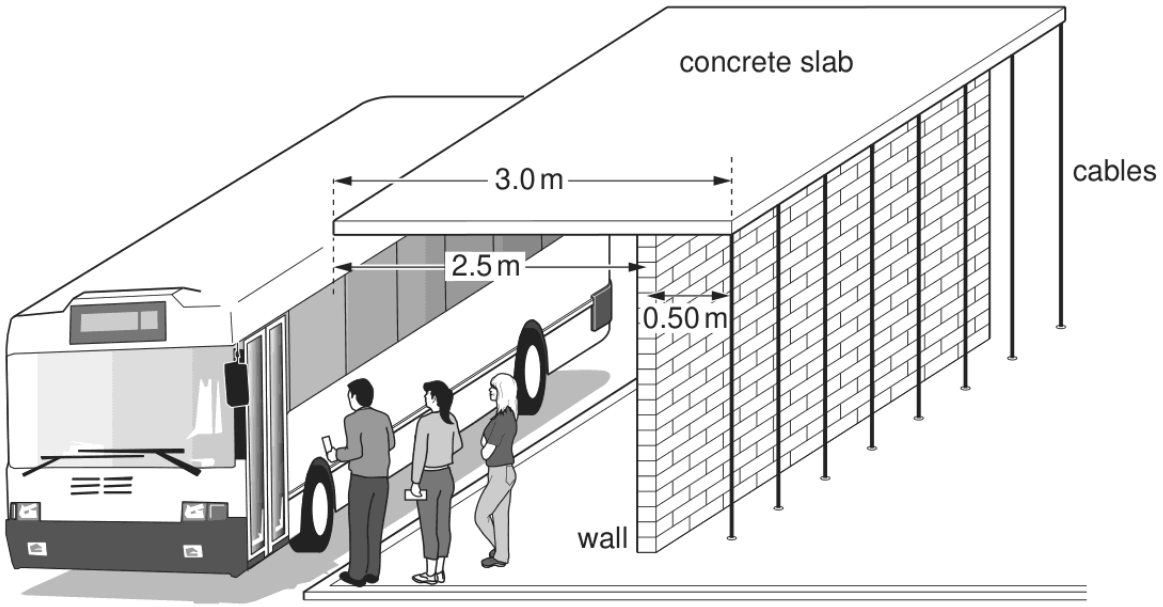


Fig. 7.1

The wall is 2.5 m from the front of the concrete slab and 0.50 m from the back. The cables behind the shelter pull downwards and stop the slab toppling forwards.

(a) The concrete slab is of uniform thickness and density. Determine the perpendicular distance between the wall and the centre of mass of the slab.

distance = ...[1]

(b) **(i)** State the principle of moments.

.....

.....

.....[2]

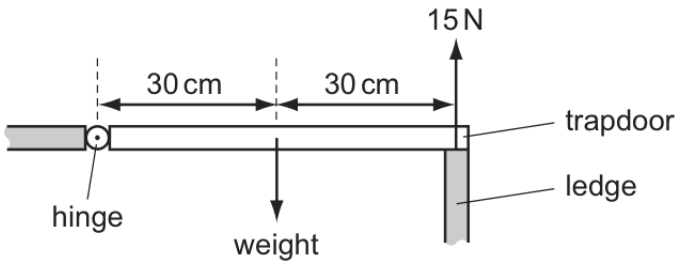
(ii) Calculate the total downward force exerted by the cables on the slab.

force = ...[2]

7	**(a)**	1.(0) m	B1	
	(b)	**(i)** (for an object in) equilibrium/balance	B1	
		$W_1x = W_2y$ (clear) **or** anticlockwise moment/torque/turning force = clockwise moment/torque/turning force	B1	
		(ii) 18 000 × 1. 0 = T × 0.5	C1	
		36 000 N	A1	[5]

QUESTION · 5054_s10_qp_11.pdf · source page 4 of 20

8 A wooden trapdoor is hinged along one side and, when closed, is supported on the other side by a ledge.



When the trapdoor is closed, the ledge exerts an upward force of 15 N on the trapdoor. The gravitational field strength is 10 N / kg.

What is the mass of the trapdoor?

A 1.5 kg **B** 3.0 kg **C** 30 kg **D** 150 kg

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 8

8	**B**

10 Fig. 10.1 shows a car braking system. The brake fluid is an oily liquid.

For Examiner's Use

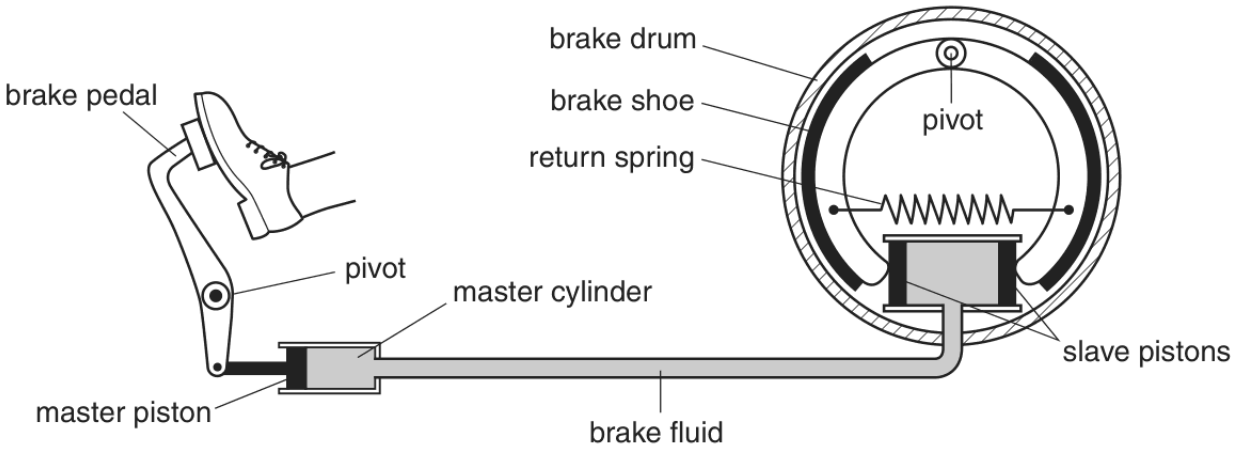


Fig. 10.1

The brake drum rotates with the wheel of the car.

(a) Explain how pushing the brake pedal makes the brake shoes rub against the drum.

.....

.....[2]

(b) The cross-sectional area of the master piston is 2.0 cm^2. A force of 140 N is applied to the master piston.

(i) Calculate the pressure created in the brake fluid by the master piston.

pressure = ... N/cm^2 [2]

(ii) The cross-sectional area of each slave piston is 2.8 cm^2.

Calculate the force exerted on each slave piston by the brake fluid.

force = ...[2]

(iii) The force exerted on the master piston is greater than the force applied by the foot on the brake pedal. Using the principle of moments, explain this.

.....

.....

.....[2]

For Examiner's Use

(c) Fig. 10.2 shows a master cylinder sealed at one end. Instead of brake fluid, the cylinder contains air.

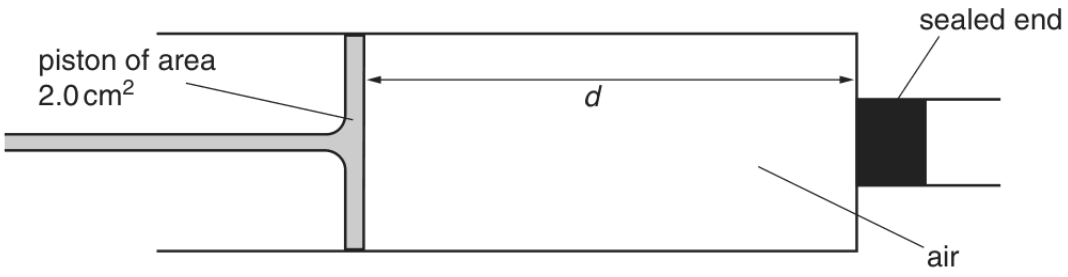


Fig. 10.2

When a force is applied to the piston, the length *d* changes from 6.0 cm to 4.0 cm. The pressure of the air increases but the temperature stays constant.

(i) Describe how the molecules of the air exert a pressure.

.....

.....[1]

(ii) Explain why the pressure increases even though the temperature stays constant.

.....

.....

.....[2]

(iii) The initial pressure of the air inside the cylinder is 1.0×10^5 Pa.

Calculate the final pressure of the air. State the formula that you use in your calculation.

pressure = ...[3]

(d) Air bubbles form in the brake fluid of Fig. 10.1. State the effect this has on the braking system.

.....

.....

.....[1]

10 (a) (master cylinder creates) **pressure** in brake **fluid or** pressure from master piston transmitted to slave piston B1
fluid/pressure produces **force/push** (not press) (on slave piston) **or** force from master piston transmitted (to slave piston) B1 [2]

(b) (i) P = F/A **or** 140/2.0 C1
70 (N/cm^2) A1 [2]

(ii) 70 × 2.8 C1
200 N **accept** 196 N ecf **(i)** A1 [2]

(iii) distance foot to pivot larger than piston to pivot B1
force × distance constant B1 [2]

(c) (i) molecules hit against walls/piston (**ignore** hit each other) B1 [1]

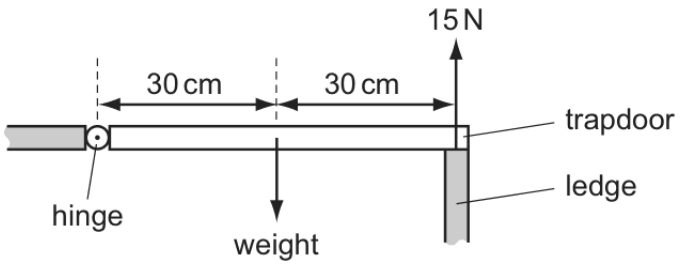
(ii) hit more often/more frequently (**accept** hit each other more often) B1
smaller volume **or** molecules closer/less space B1 [2]

(iii) $P_1V_1 = P_2V_2$ **or** PV = constant B1
$1 \times 10^5 \times 6$ (×2) = P × 4 × (2) C1
1.5×10^5 Pa A1 [3]

(d) air/bubbles compress/reduce in volume **or** brakes pushed further/spongy B1 [1]
ignore: efficiency; less pressure; less force transmitted

QUESTION · 5054_s10_qp_12.pdf · source page 4 of 20

7 A wooden trapdoor is hinged along one side and, when closed, is supported on the other side by a ledge.



When the trapdoor is closed, the ledge exerts an upward force of 15 N on the trapdoor. The gravitational field strength is 10 N / kg.

What is the mass of the trapdoor?

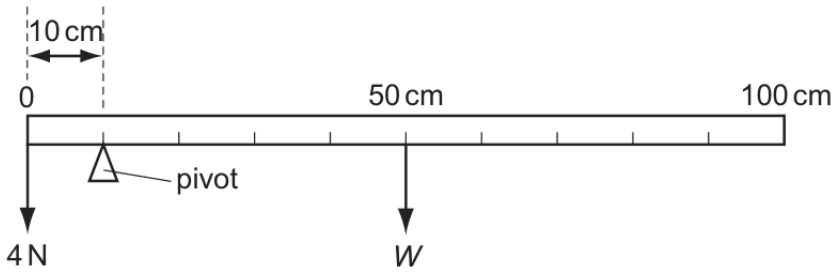
A 1.5 kg **B** 3.0 kg **C** 30 kg **D** 150 kg

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 7

7	**B**

QUESTION · 5054_w10_qp_11.pdf · source page 7 of 20

10 A uniform metre rule is balanced by a 4 N weight as shown in the diagram.



What is the weight *W* of the metre rule?

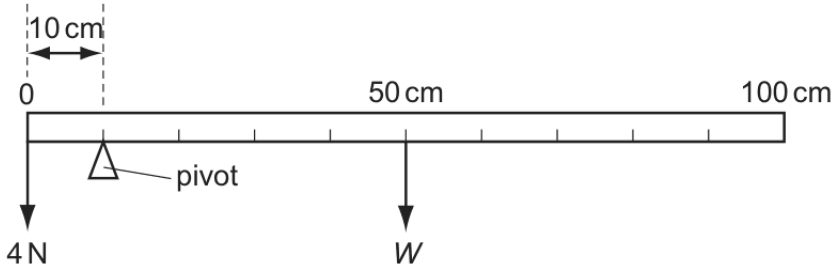
A 1 N **B** 4 N **C** 16 N **D** 40 N

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 10

10 **A**

QUESTION · 5054_w10_qp_12.pdf · source page 6 of 16

12 A uniform metre rule is balanced by a 4 N weight as shown in the diagram.



What is the weight *W* of the metre rule?

A 1 N **B** 4 N **C** 16 N **D** 40 N

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 12

12	**A**

For Examiner's Use

3 Fig. 3.1 shows a firefighter of total weight 840 N in equilibrium at the top of a ladder that is pivoted at point P.

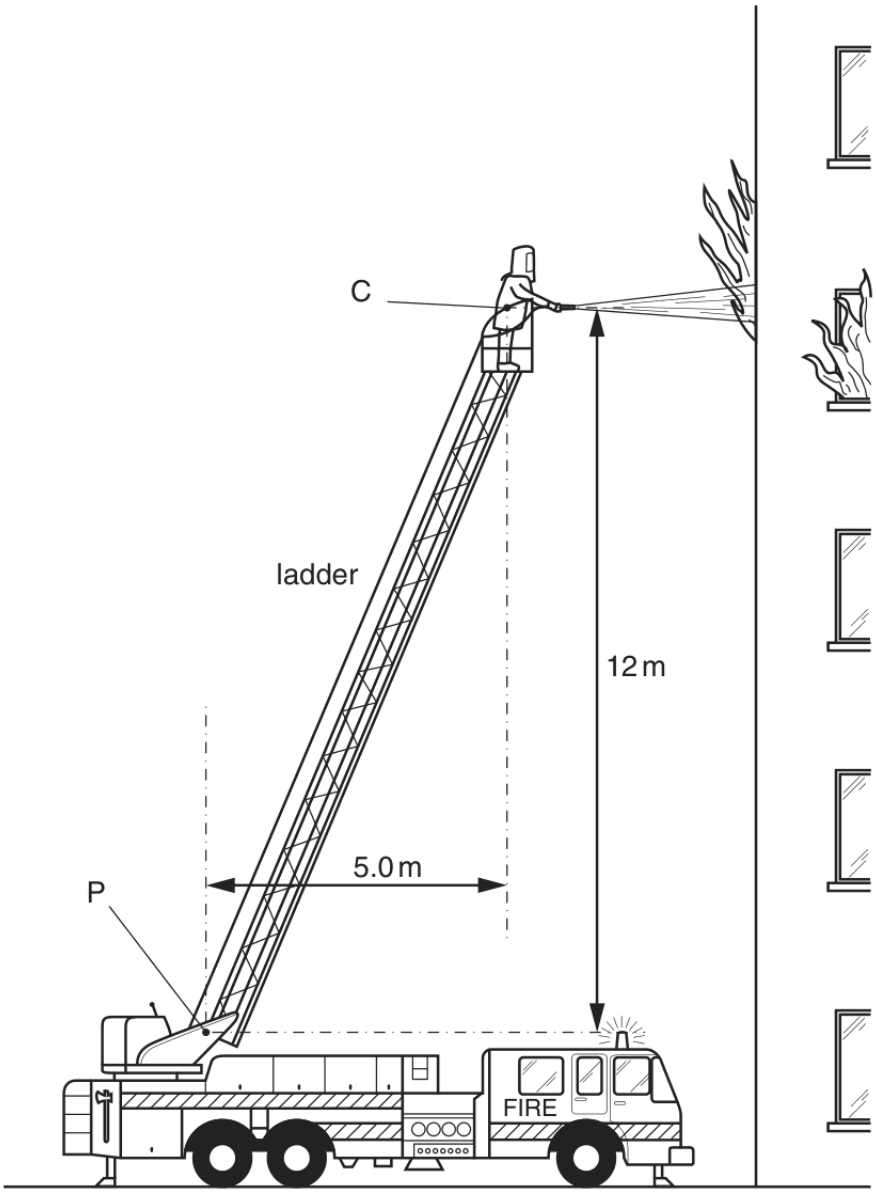


Fig. 3.1

The ladder leans towards a burning building at an angle such that the centre of gravity C of the firefighter is 12 m above and 5.0 m to the right of P. The firefighter holds a hose that directs a high-speed jet of water horizontally into a burning building.

(a) (i) Calculate the moment *M* of the firefighter's weight about P.

moment = ...[2]

For Examiner's Use

(ii) The jet of water causes a horizontal force *R* on the firefighter that acts towards the left, through C. This opposes the turning effect of his weight. Calculate the size of *R* that, on its own, ensures that *M* is exactly cancelled.

force = ...[1]

(iii) Suggest a third force that has a turning effect about P on the ladder.

.....

.....[1]

(b) Fig. 3.2 shows the firefighter.

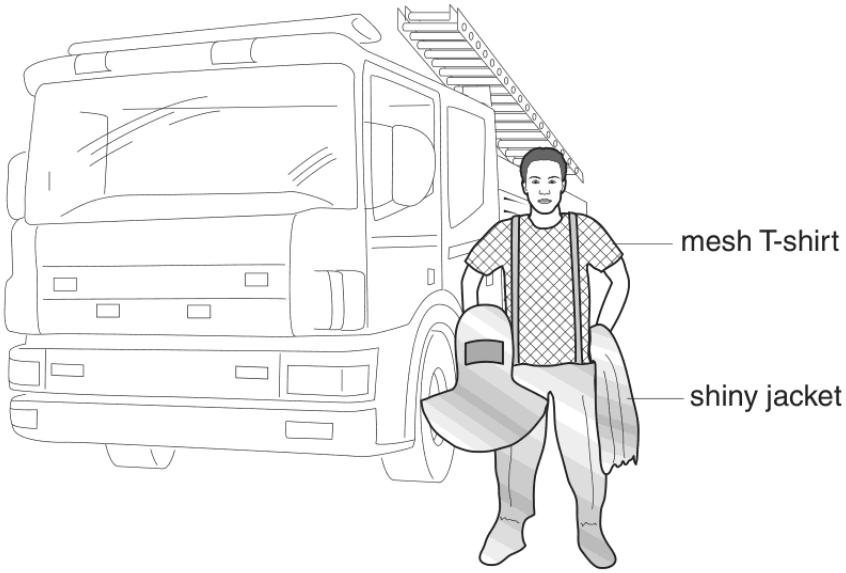


Fig. 3.2

The jacket of his protection suit has a shiny, silver-coloured outer surface. Underneath it he wears a loosely-woven mesh T-shirt (string vest).

Explain how wearing the shiny jacket and the mesh T-shirt helps to keep the firefighter cool when he is close to a source of intense heat.

.....

.....

.....

.....

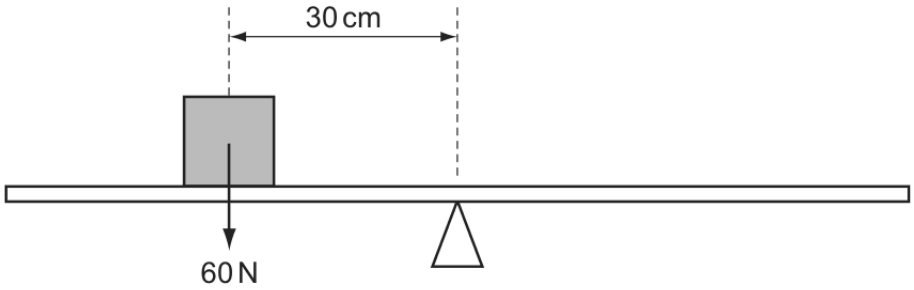
.....

.....[4]

3	**(a)**	**(i)**	(*M* =) force × **perpendicular** distance **or** 840 × 5 (formula mark can be scored if not given in **3(a)(ii)**) 4200 N m	C1 A1	
		(ii)	350 N **or (a)(i)**/12 **and** calculated	B1	
		(iii)	weight of ladder/hose **or** friction at P/pivot/axle (**not** air resistance; **ign**. friction)	B1	
	(b)		any **four** lines: (mesh) traps air **air** poor conductor/good insulator **or** convection prevented (shiny surface) reflects/(good) reflector of IR/radiation/heat (shiny surface) does not absorb/poor absorber of IR/radiation/heat (**not** with radiator/emitter/conductor) less heat transmitted/to **firefighter**	B4	[8]

QUESTION · 5054_s11_qp_11.pdf · source page 5 of 16

8 A uniform beam is balanced at its midpoint. An object is placed on the beam, as shown.



Which force will rebalance the beam?

A 30 N acting upwards, 60 cm to the left of the midpoint

B 30 N acting upwards, 60 cm to the right of the midpoint

C 45 N acting downwards, 45 cm to the right of the midpoint

D 90 N acting downwards, 20 cm to the left of the midpoint

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 8

8	**A**

8 **(a)** A wire carrying a current in a magnetic field experiences a force due to the current. On Fig. 8.1, insert the words **current**, **field** and **force** in the boxes to show the relative directions of the current, the magnetic field and the force.

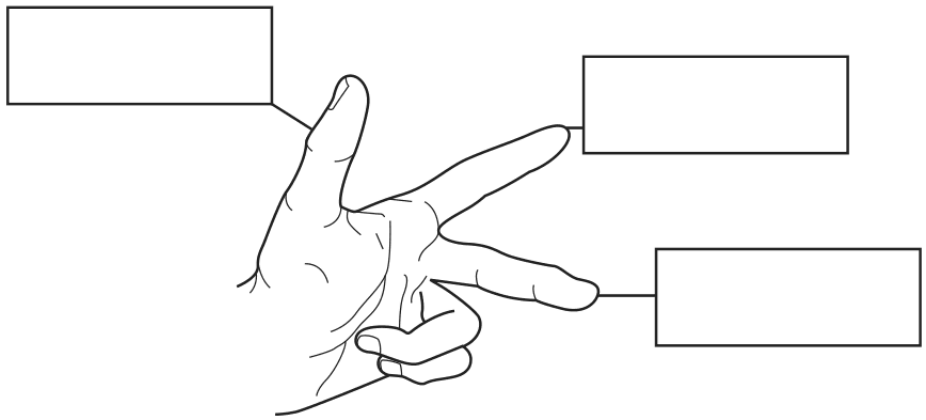


Fig. 8.1

[1]

(b) Fig. 8.2 shows a current-carrying coil ABCD in a magnetic field.

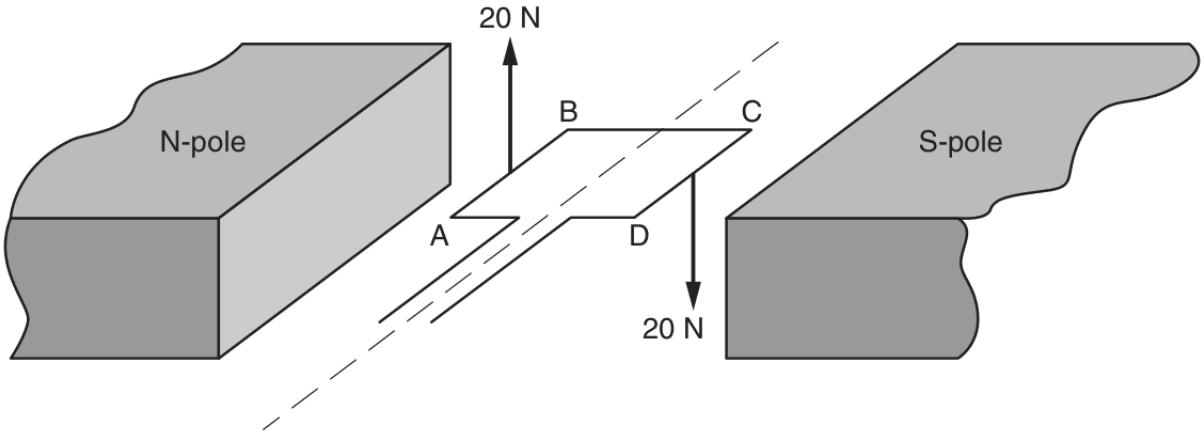


Fig. 8.2

Each side of the coil is 4.0 cm in length. The force on AB is 20 N and the force on CD is 20 N.

(i) Calculate the total moment caused by these forces.

moment = ... [2]

(ii) The moment is increased by using a stronger magnetic field.

State two other ways to increase the moment.

1. .....

2. .....

[2]

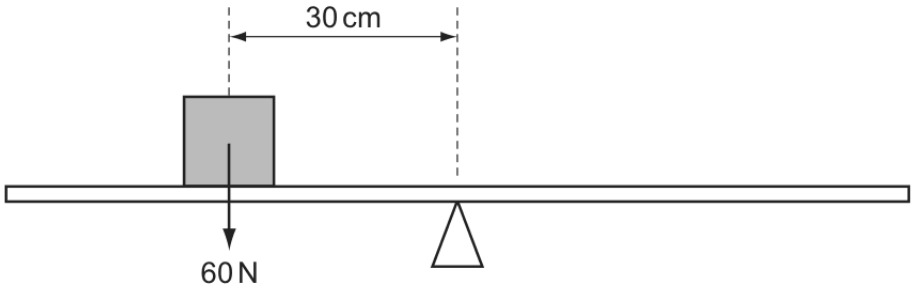
8 **(a)** all three correct: force, field, current B1

(b) **(i)** *Fd* // 20 × 4; 20 × 2; 20 × 0.04; 20 × 0.02; 20 × 4 × 2; 160 C1
80 N cm, 0.8 N m A1

(ii) more turns (on coil)
use soft iron
more current
increase AB or CD
increase BC or AD ANY 2 B1 [5]

QUESTION · 5054_s11_qp_12.pdf · source page 4 of 16

7 A uniform beam is balanced at its midpoint. An object is placed on the beam, as shown.



Which force will rebalance the beam?

A 30 N acting upwards, 60 cm to the left of the midpoint

B 30 N acting upwards, 60 cm to the right of the midpoint

C 45 N acting downwards, 45 cm to the right of the midpoint

D 90 N acting downwards, 20 cm to the left of the midpoint

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 7

7	**A**

11 A crankshaft is a shaped metal bar that is part of a car engine. It is free to rotate about an axis, as shown in Fig. 11.1.

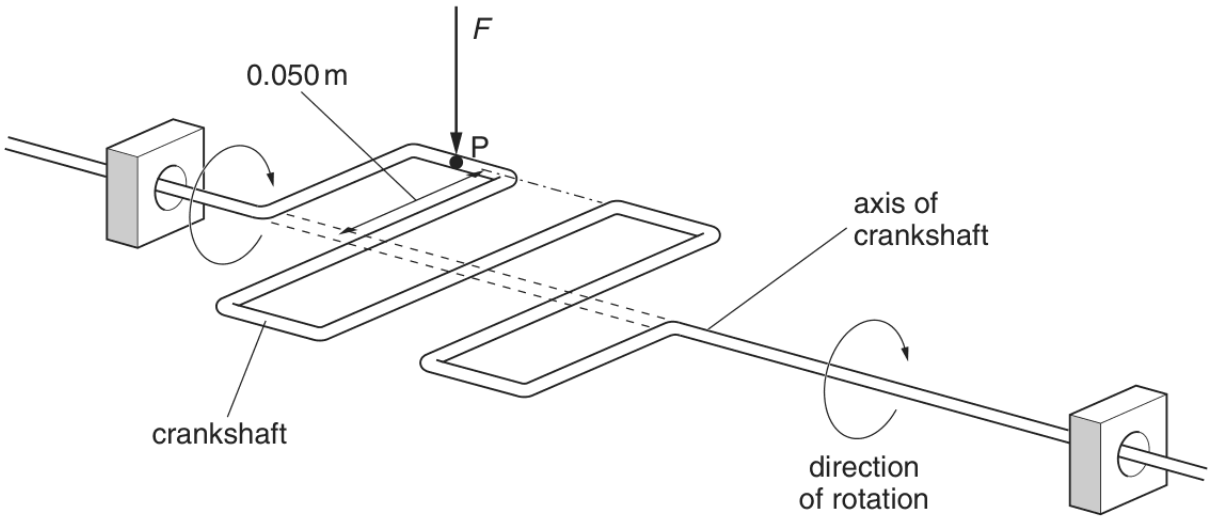


Fig. 11.1

When the crankshaft is horizontal, a vertical force *F* of 8200 N acts downwards on the crankshaft at P. This causes the crankshaft to rotate. The distance between P and the axis of the crankshaft is 0.050 m.

(a) (i) State what is meant by the *moment* of a force.

.....

.....[2]

(ii) The crankshaft is horizontal. Calculate the moment of *F* about the axis of the crankshaft.

moment = ...[2]

(iii) The size and direction of the vertical force *F*, acting on the crankshaft at P, remain constant. Explain why the moment of *F* decreases as the crankshaft rotates through a small angle.

.....

.....[1]

(b) The force *F* on the crankshaft at P is exerted by a metal rod connected to a piston. The piston traps high-pressure gas in a cylinder, as shown in Fig. 11.2.

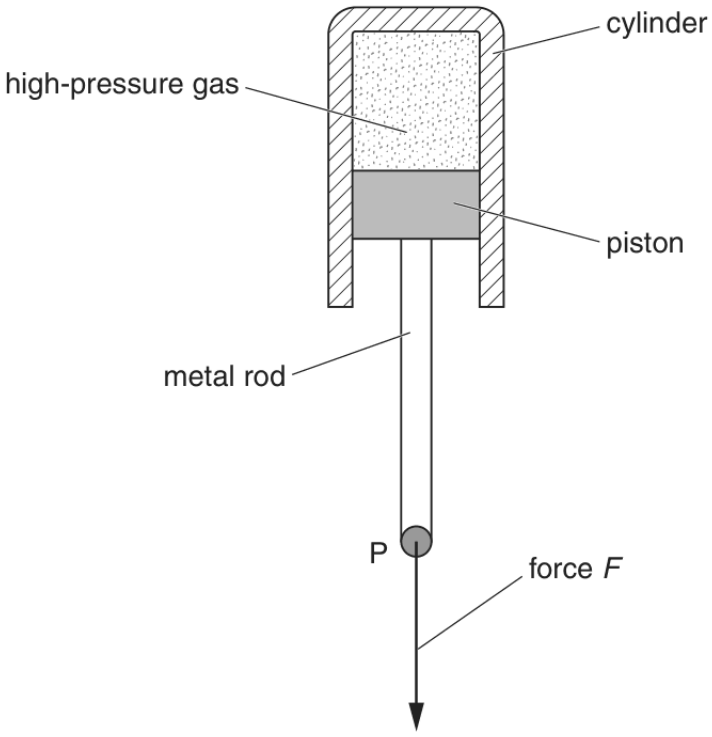


Fig. 11.2

The force *F*, of 8200 N, acts because the high-pressure gas pushes down on the piston. The air pressure outside the cylinder is 1.0×10^5 Pa and the cross-sectional area of the piston is $0.0067\,m^2$.

(i) Calculate the minimum value of the pressure of the gas in the cylinder.

pressure = ...[3]

(ii) Suggest why, in practice, the pressure of the gas in the cylinder is greater than the value calculated in **(i)**.

.....

.....

.....[2]

Question 11 continues on the next page.

(c) The piston moves in the cylinder and the gas expands. State and explain what happens to the force *F*.

.....[1]

(d) Explain, in molecular terms, why heating a gas at constant volume causes its pressure to increase.

.....

.....

.....

.....

.....[4]

11	**(a)**	**(i)**	force × distance force × perpendicular distance (from the axis)	C1 A1	
		(ii)	8200 × 0.05 410 N m	C1 A1	
		(iii)	(perpendicular) distance reduced/force not perpendicular/only a component of the force is perpendicular	B1	
	(b)	**(i)**	(P =) F/A **or** 8200/0.0067 $1.2(23881) \times 10^6$ $1.3(23881) \times 10^6$ Pa	C1 C1 A1	
		(ii)	friction exerts opposing force **or** between piston **and** cylinder	M1 A1	
	(c)		pressure decreases **or** F decreases (no contradiction)	B1	
	(d)		any **four** lines: molecules collide with/hit walls molecules move faster/kinetic energy increases molecules collide harder (with walls) molecules collide more frequently (with walls) greater force/impulse/momentum change (on walls)	B4	[15]

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A builder needs to determine the density of a solid cube of wood.

He places the 50 cm mark of a uniform metre rule on a pivot, so that the rule balances.

He then places the cube on the rule with its centre of gravity directly above the 75 cm mark. A mass of 0.050 kg is moved along the rule until balance is restored. This is shown in Fig. 1.1.

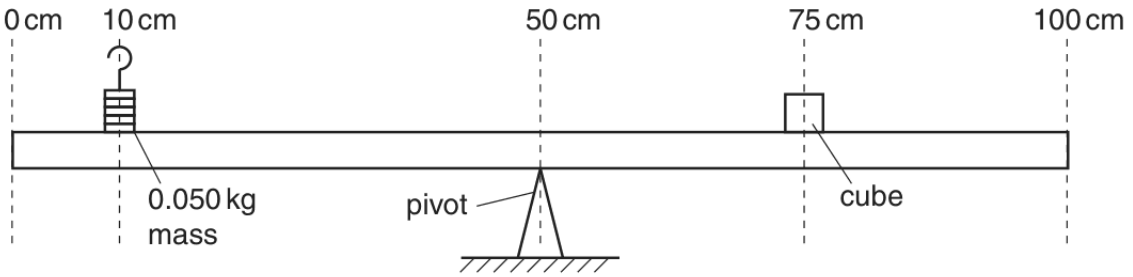


Fig. 1.1 (not to scale)

The rule is balanced when the 0.050 kg mass is at the 10 cm mark.

(a) Calculate the mass of the cube.

mass = ...[3]

(b) The cube has a volume of $1.6 \times 10^{-4}\,m^3$. Determine the density of the wood.

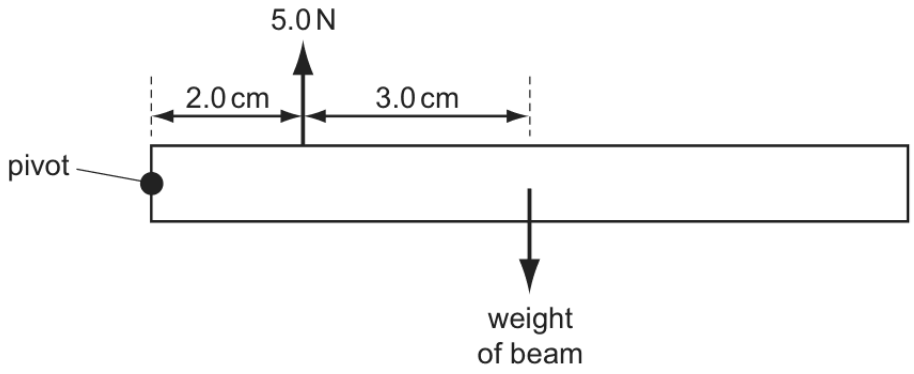
density = ... [2]

Section A

1	**(a)**	$m_1(g)x_1$ **or** $m_2(g)x_2$ **or** 2 **or** one of these in numbers **or** 40 **and** 25 seen	C1	
		$0.050 \times (10) \times 40 = m_2 \times (10) \times 25$		
		or anticlockwise **moment =**clockwise **moment**	C1	
		0.080 kg **or** 80 g	A1	
	(b)	$(\rho/d =)\ m/V$ **or** $0.08/1.6 \times 10^{-4}$	C1	
		500 kg / m^3 **or** 0.50 g / cm^3	A1	[5]

QUESTION · 5054_s12_qp_11.pdf · source page 3 of 16

7 A beam pivoted at one end has a force of 5.0 N acting vertically upwards on it as shown. The beam is in equilibrium.



What is the weight of the beam?

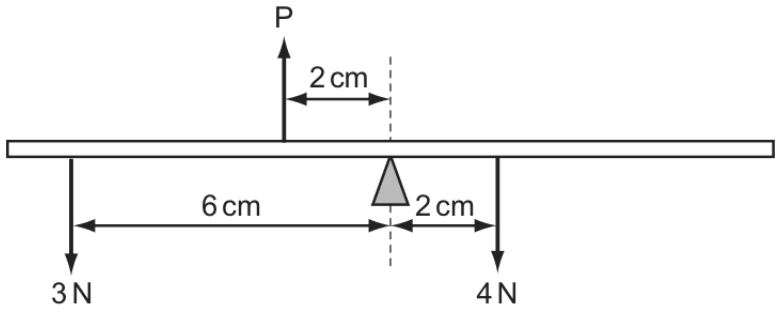
A 2.0 N **B** 3.0 N **C** 3.3 N **D** 5.0 N

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 7

7	**A**

QUESTION · 5054_s12_qp_12.pdf · source page 4 of 16

7 The diagram shows a uniform balanced beam, pivoted about its centre.



What is the value of force P?

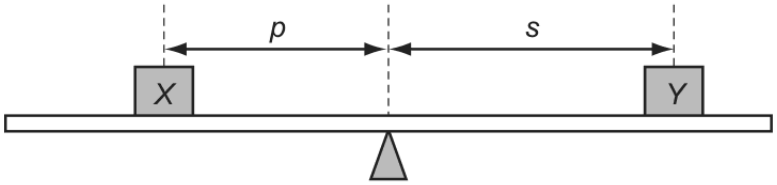
A 5 N **B** 7 N **C** 10 N **D** 13 N

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 7

7	**A**

QUESTION · 5054_w12_qp_11.pdf · source page 4 of 16

7 Masses *X* and *Y* are placed on opposite sides of the centre of a uniform metre rule, which is pivoted at its centre.



Which combination of masses and distances balances the rule?

	mass / g		distance / cm	
	X	*Y*	*p*	*s*
A	200	200	5	10
B	200	300	10	15
C	400	300	12	16
D	500	200	15	30

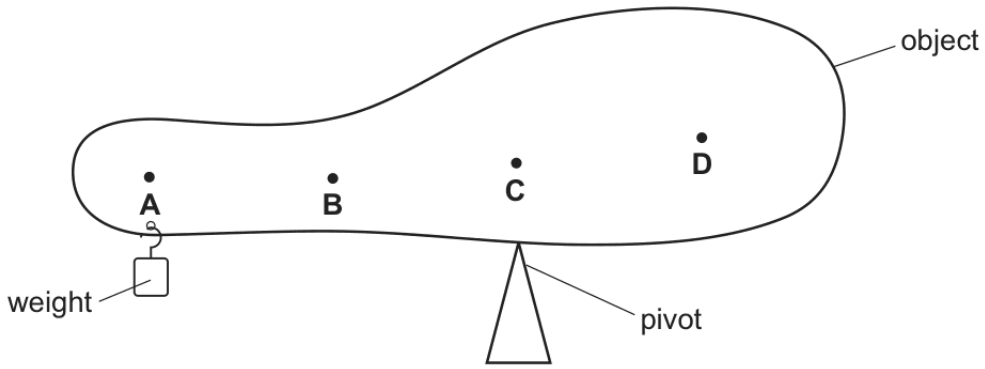
MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 7

7 **C**

QUESTION · 5054_w12_qp_11.pdf · source page 5 of 16

8 A student balances a non-uniform object on a pivot. To do this, a weight is suspended near the left-hand end of the object.

Where is the centre of mass of the object?

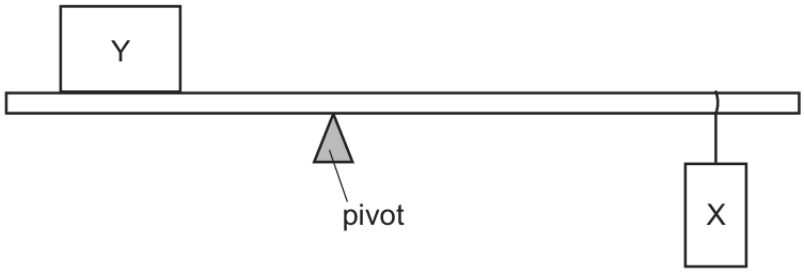


MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 8

8 **D**

QUESTION · 5054_w12_qp_12.pdf · source page 4 of 16

7 An object Y is in a fixed position on a rod. A weight X is moved and the position of a pivot is adjusted until the rod balances on the pivot, as shown.



The experiment is repeated in a region where the gravitational field strength is lower.

What is done to keep the rod balanced?

	pivot	X
A	move left	no movement
B	move right	move left
C	no movement	move right
D	no movement	no movement

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 7

7 **D**

Section A

For Examiner's Use

Answer **all** the questions in this section. Answer in the spaces provided.

1 **(a)** Describe an experiment to verify *the principle of moments*. You may include a diagram in your answer.

.....

.....

.....

.....

.....[4]

(b) Fig. 1.1 shows a spanner tightening a nut.

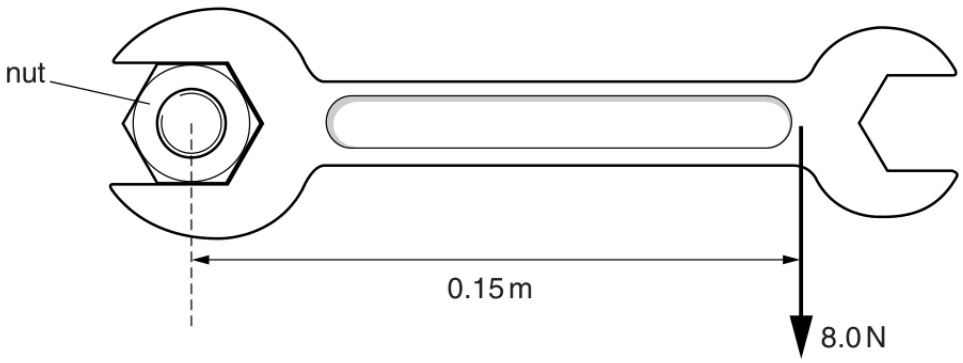


Fig. 1.1

A force of 8.0 N is applied to the spanner at a perpendicular distance of 0.15 m from the centre of the nut.

Calculate the moment of the force acting on the nut.

moment = ...[2]

Section A

1	**(a)**	appropriate apparatus e.g. ruler, weights, fulcrum	B1	
		action e.g. balance weights on each side	B1	
		one of: force/mass × distance **or** calculate moment	B1	
		vary **or** repeat	B1	
	(b)	$F \times d$ **or** 8.0 × 0.15	C1	
		1.2 N m (**not** J)	A1	[6]

For Examiner's Use

2 Fig. 2.1 shows apparatus used to investigate the turning effect of a force.

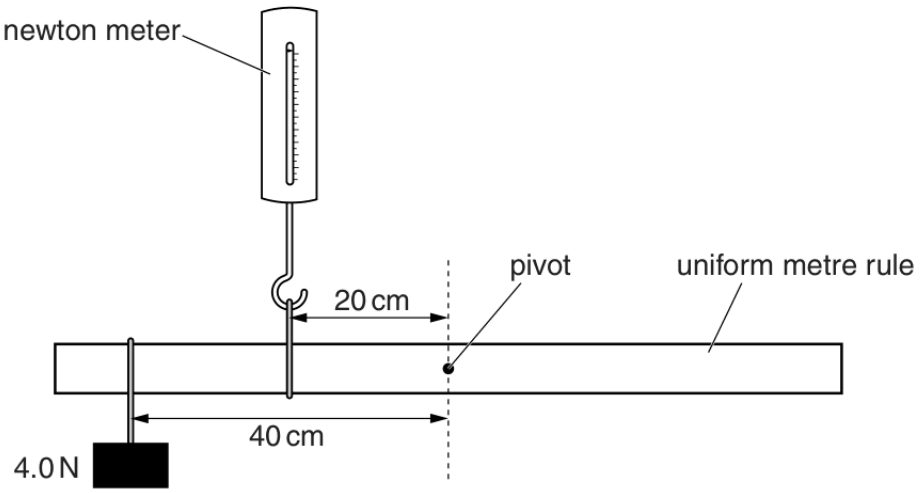


Fig. 2.1

The uniform metre rule is freely pivoted at its centre.

The newton meter is 20 cm from the pivot and a 4.0 N weight is 40 cm from the pivot.

The metre rule is in equilibrium.

(a) State the principle of moments for a body in equilibrium.

.....

.....

.....[1]

(b) Calculate the reading on the newton meter.

reading = ...[2]

(c) The weight of the metre rule is 1.2 N.

Calculate the size of the force exerted on the metre rule by the pivot.

force = ...[2]

2	**(a)**	total/resultant **moment** zero **or** (sum of) clockwise = anticlockwise **moment**	B1	
	(b)	F_1d_1 **or** F_2d_2 seen in any form	C1	
		8(.0) N	A1	
	(c)	4 + 1.2 or 5.2 seen	C1	
		2.8 N ecf **(b)** i.e. accept 5.2 – **(b)** or **(b)** – 5.2	A1	[5]

For Examiner's Use

2 **(a)** Explain what is meant by the *moment of a force*.

.....

.....

..... [2]

(b) Fig. 2.1 shows a system for raising a heavy piece of metal into a vertical position.

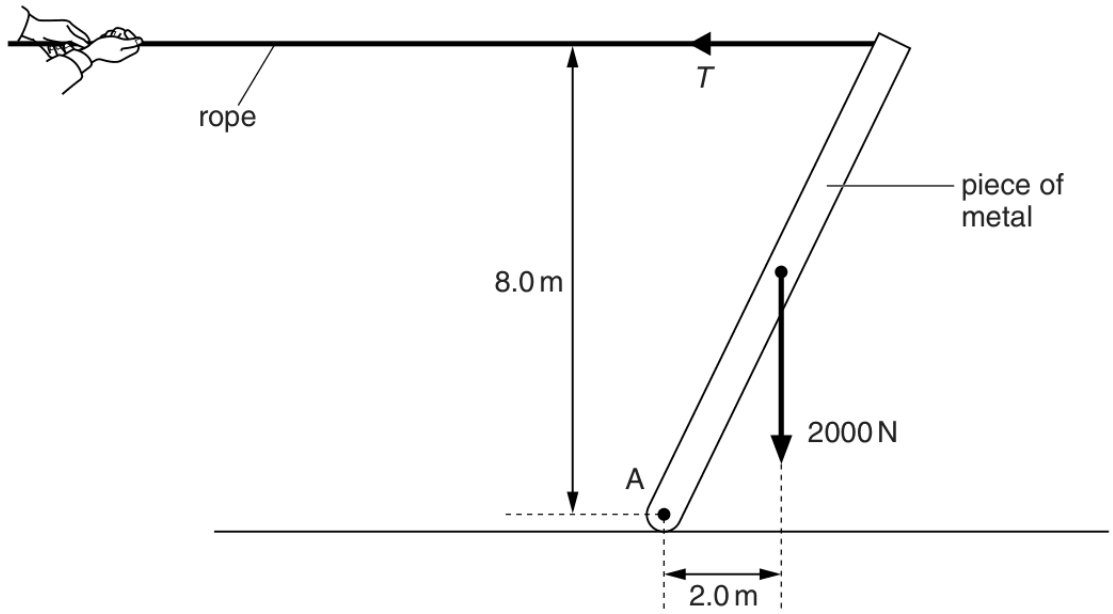


Fig. 2.1 (not to scale)

A man pulls on the rope with a horizontal force *T*. The piece of metal has a weight of 2000 N and is freely pivoted at A. The system is in equilibrium.

(i) By taking moments about A, calculate *T*.

T = ... [2]

(ii) The force *T* and the force that the rope exerts on the man are related by Newton's third law. State the relationship between these forces.

.....

.....

..... [2]

2	**(a)**	force × distance perpendicular distance	M1 A1	
	(b) **(i)**	$T \times 8$ or 2000×2 seen 500 N	C1 A1	
	(ii)	(two forces) equal (in magnitude) (two forces) opposite (in direction)	B1 B1	[6]

For Examiner's Use

2 A girl of weight 550 N is playing on a see-saw with her brother. Fig. 2.1 shows her brother of weight *W* sitting 1.1 m to the right of the balance point.

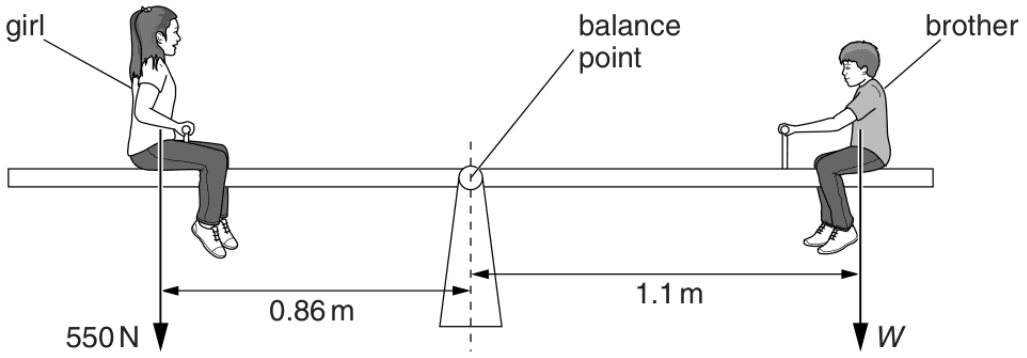


Fig. 2.1 (not to scale)

The see-saw is balanced when the girl sits 0.86 m to the left of the balance point.

(a) Calculate *W*.

W = ...[2]

(b) The girl and her brother slide equal distances along the see-saw away from each other.

Describe and explain what happens.

.....

.....

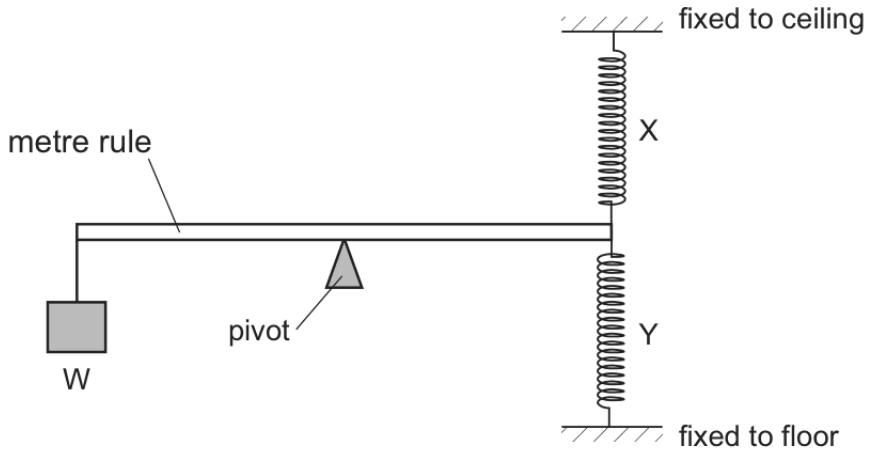
.....

.....[3]

2	**(a)**	$F_1x_1 = F_2x_2$ **or** 550 × (0.86 or 86)/(1.1 or 110)	C1	
		430 N	A1	
	(b)	**both** moments increase	C1	
		girl's moment increases more **or** girl's moment > brother's **or** anticlockwise moment greater	A1	
		see-saw tips down on girl's side	B1	[5]

QUESTION · 5054_w13_qp_12.pdf · source page 5 of 20

7 Two stretched springs X and Y are attached to one end of a metre rule as shown. A weight W is hung from the other end. A pivot is at the centre of the rule.



The weight W is moved towards the pivot.

How does the extension of each spring change?

	spring X	spring Y
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 7

7 **C**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 **(a)** State what is meant by the *moment of a force*.

.....

.....[2]

(b) The anchor of a sailing ship has a mass of 350 kg. Six sailors raise the anchor from the sea-bed by turning a large axle. They push the handles attached to the axle and it rotates. The anchor is on the end of a chain that winds on to the rotating axle.

Fig. 9.1 shows the sailors lifting the anchor.

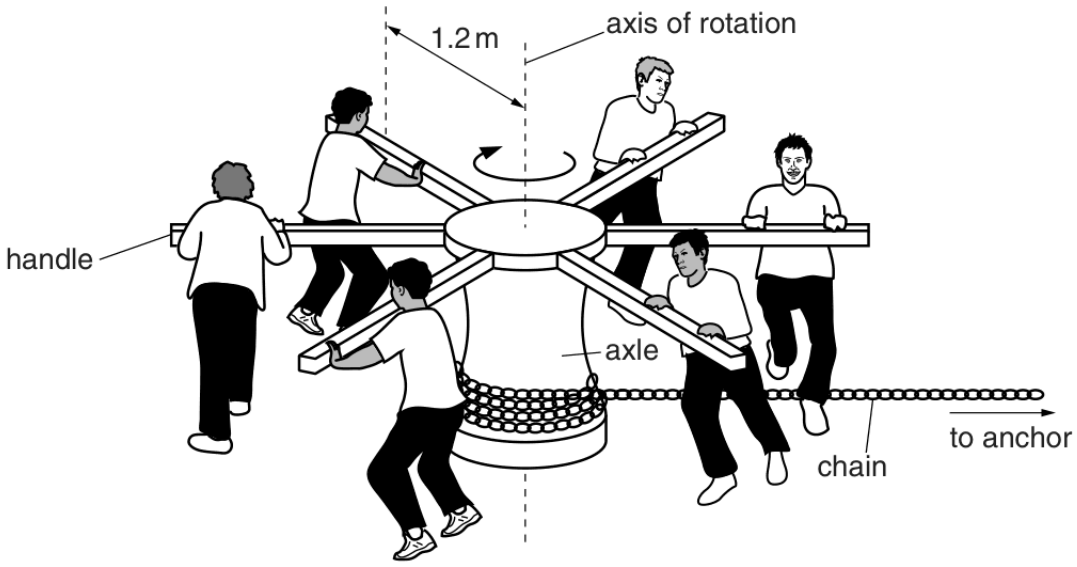


Fig. 9.1

Each of the sailors exerts a force of 750 N on his handle at a distance of 1.2 m from the axis of rotation. The axle rotates through one complete revolution and the anchor is lifted through a distance of 160 cm.

The gravitational field strength g is 10 N/kg.

(i) Calculate

1. the total moment exerted on the axle by the six sailors,

moment = ...[2]

2. the gravitational potential energy gained by the anchor as the axle rotates through one complete revolution.

For Examiner's Use

energy = ...[3]

(ii) The work done on the axle by the sailors is very much larger than the gravitational potential energy gained by the anchor.

State and explain how energy is wasted.

.....

.....

.....[2]

(iii) Explain why the power produced by the sailors is larger when the anchor is lifted at a faster rate.

.....

.....[2]

(c) Describe, with the aid of a labelled diagram, how to verify the principle of moments.

.....

.....

.....

.....

.....

.....[4]

Section B

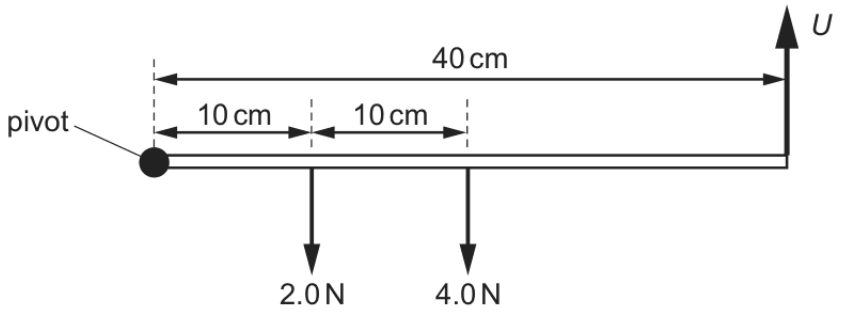
9	**(a)**	force × distance **or** $F \times d$ with F and d defined **or** $F \times d_{perp}$		C1	
		force × perpendicular distance **or** $F \times d_{perp}$ with F and d_{perp} defined		A1	[2]
	(b)	**(i)**	**1.** 6 × 750 × 1.2 **or** 750 × 1.2 **or** 900	C1	
			5400 N m	A1	
			2. *mgh* **or** 350 × 10 × 160 **or** 350 × 10 × 1.6	C1	
			350 × 10 × 1.6 **or** 5.6 × 105	C1	
			5600 J	A1	
		(ii)	friction at axle/boat **or** drag due to water **or** chain lifted also	B1	
			heat produced (**ignore** in sailors) **or** work done against friction/drag **or** work done raising chain	B1	
		(iii)	same amount of work done **or** $P = E/t$ **or** $P = WD/t$	B1	
			in less time **or** power inversely proportional to time (**ignore** faster rate)	B1	[9]
	(c)	clear/labelled diagram with ruler, fulcrum, at least two weights		B1	
		any three of the following points made in words: balance ruler (on its own) place weights on ruler so it balances clockwise and anticlockwise moments equal **or** net moment zero repeat (apply similar principles to other possible methods)		B3	[4]

[Total: 15]

QUESTION · 5054_s14_qp_11.pdf · source page 4 of 16

9 A beam of length 40 cm is pivoted at one end.

The weight of the beam is 4.0 N and acts at a point 20 cm from the pivot. A 2.0 N weight hangs 10 cm from the pivot.



An upward force *U* is needed to keep the beam horizontal.

What is the size of *U*?

A 0.5 N **B** 1.5 N **C** 2.5 N **D** 6.0 N

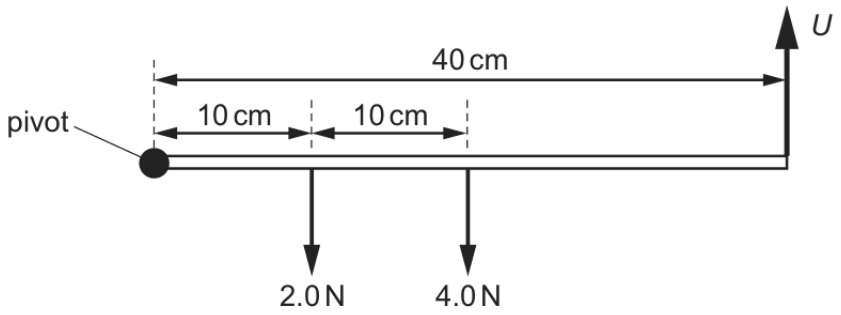
MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 9

9	**C**

QUESTION · 5054_s14_qp_12.pdf · source page 4 of 16

9 A beam of length 40 cm is pivoted at one end.

The weight of the beam is 4.0 N and acts at a point 20 cm from the pivot. A 2.0 N weight hangs 10 cm from the pivot.



An upward force *U* is needed to keep the beam horizontal.

What is the size of *U*?

A 0.5 N **B** 1.5 N **C** 2.5 N **D** 6.0 N

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 9

9	**C**

7 Fig. 7.1 shows a compass needle near a bar magnet. Magnetic poles are shown on the compass needle and on the magnet.

A finger stops the compass needle from turning.

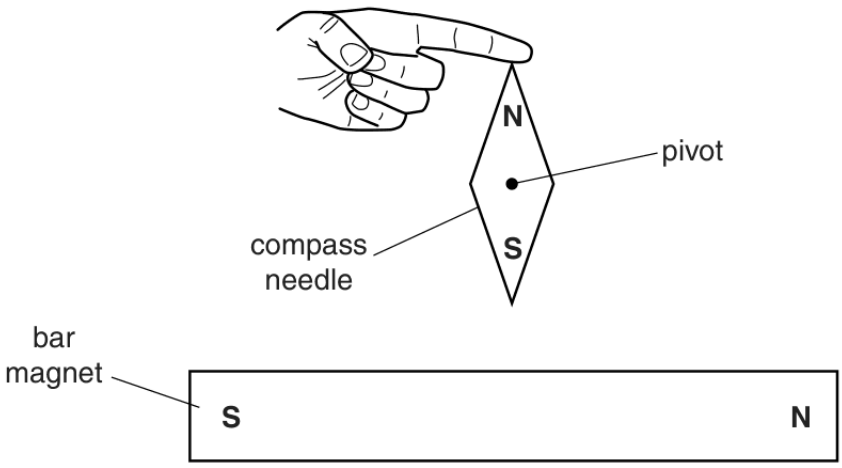


Fig. 7.1 (not to scale)

(a) (i) The magnet causes a force on the S-pole of the compass needle.

On Fig. 7.1, draw an arrow from the S-pole of the compass needle to show the direction of this force. [1]

(ii) Explain why the compass needle turns when the finger is removed.

.....

..... [1]

(b) A small compass is used to plot the magnetic field lines of the magnet.

Describe how the compass is used to plot magnetic field lines on a piece of paper.

.....

.....

.....

.....

..... [3]

7 (a) (i) **horizontal** arrow to right (by eye) [B1]

(ii) forces / resultant causes moment or (turns because) force is not at pivot [B1]

(b) mark made at one end/pole/direction of compass (on paper) [B1]
move compass so that other end of compass is on mark **and** remark [B1]
join marks made as compass moved on in some way (to draw line) [B1]

[5]

2 A student places a metre rule on the edge of a triangular prism. The prism is used as a pivot and the rule balances about the 50 cm mark.

The student then places a block of wood at the 10 cm mark on the rule and an empty measuring cylinder at the 80 cm mark. The rule is still balanced. Fig. 2.1 shows the arrangement.

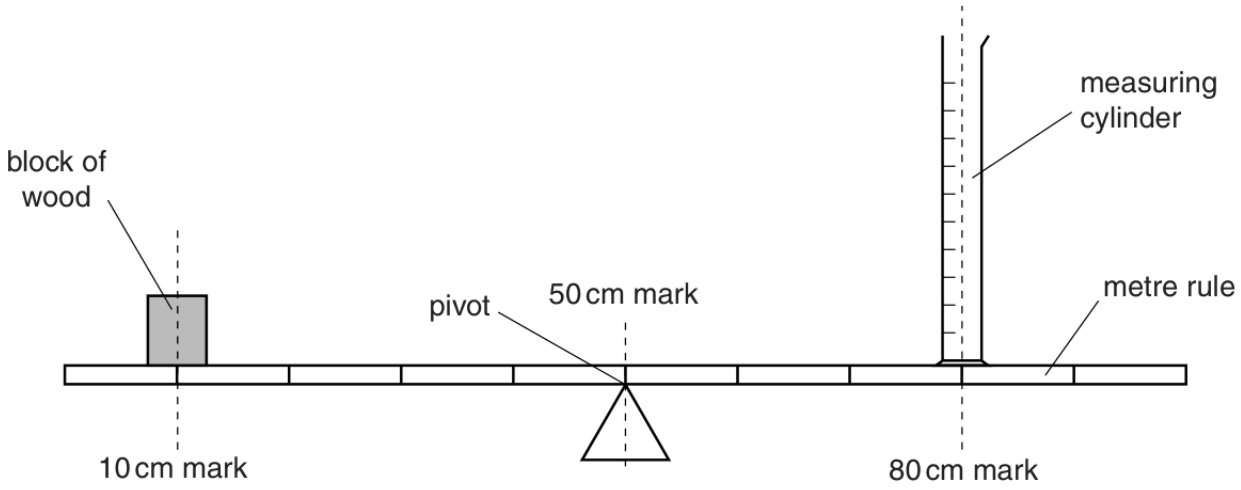


Fig. 2.1 (not to scale)

The student now places a weight of 0.39 N on top of the block of wood. She then starts to pour oil into the measuring cylinder. The rule balances again when there is 60 cm^3 of oil in the measuring cylinder.

The gravitational field strength g is 10 N/kg.

(a) Calculate

(i) the weight of the oil in the cylinder,

weight = ...[2]

(ii) the mass of the oil in the cylinder.

mass = ...[1]

(b) Calculate the density of the oil.

density = ...[2]

2	**(a)**	**(i)**	$F_1 \times d_1 = F_2 \times d_2$ **or** $(0.39 \times 0.40)/0.30$	C1	
			0.52 N	A1	
		(ii)	0.052 kg **or** 52 g	B1	
	(b)		$(\rho =)\ m/V$ **or** $52/60$ **or** $0.052/0.000\ 060$ **or** $0.052/60$	B1	
			$870/867/866.7\ \text{kg/m}^3$ **or** $0.87\ \text{g/cm}^3$ **or** $8.7 \times 10^{-4}\ \text{kg/cm}^3$ etc.	B1	**[5]**

2 In hospitals, doctors and nurses operate taps with their elbows in order to avoid contamination.

Fig. 2.1 shows a hospital tap with a long handle.

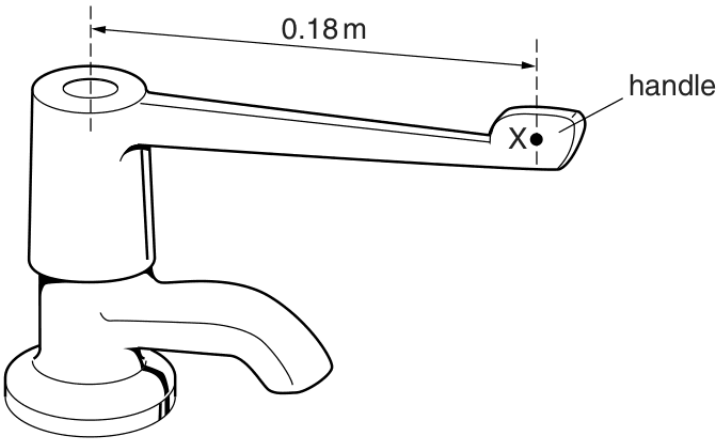


Fig. 2.1

(a) A nurse applies a force of 2.5 N at a point X on the handle, 0.18 m from the axis of the tap.

(i) Calculate the maximum moment about the axis that this force can produce.

moment = ...[2]

(ii) The moment produced by the nurse is less than this maximum value.

Suggest one reason why this is so.

.....

.....[1]

(b) State how the force needed to operate the tap is affected by the length of the handle.

.....

.....[1]

2	**(a)**	**(i)**	Fd **or** 2.5×0.18	C1	
			0.45 N m	A1	
		(ii)	force not applied at right angles to the tap	B1	
	(b)		long(er) distance needs small(er) force (for same moment) **or** inversely related/proportional	B1	**[4]**

2 Fig. 2.1 shows a student doing a press-up. A total force *F* acts upwards on his hands. There is also a force upwards on his toes.

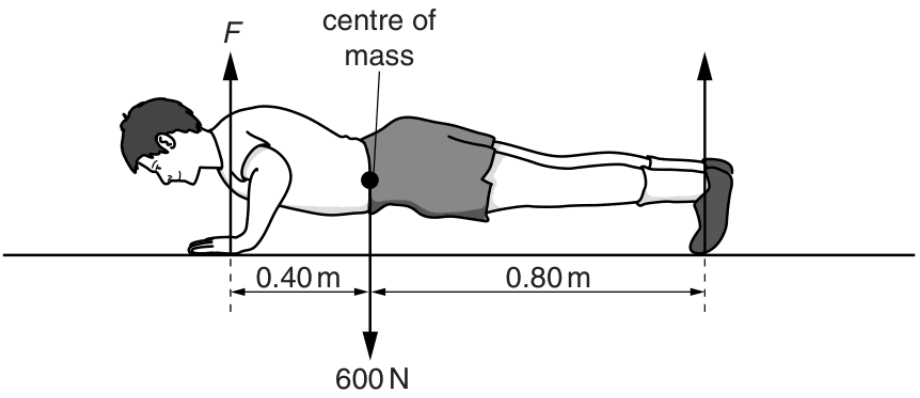


Fig. 2.1 (not to scale)

The weight of the student is 600 N and this force acts downwards from his centre of mass.

(a) (i) Explain why the student does work as his body rises from the ground.

.....

.....[1]

(ii) State the form of energy that the student uses to do this work.

.....[1]

(b) At the position shown in Fig. 2.1, the student is stationary.
The weight of the student causes a moment about his toes.

(i) Calculate the moment of the weight of the student about his toes.

moment = ...[1]

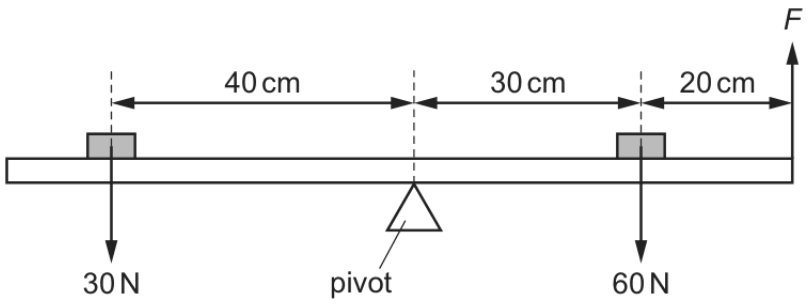
(ii) Calculate the value of the force *F*.

F = ...[2]

2	**(a)**	**(i)**	force moves through a distance (in same direction)	B1
		(ii)	chemical (potential) energy	B1
	(b)	**(i)**	480 Nm	B1
		(ii)	attempt to apply moments with two forces and distances	C1
			400 N	A1

QUESTION · 5054_s15_qp_12.pdf · source page 5 of 16

9 A uniform beam is pivoted at its centre. Two weights are placed on the beam in the positions shown and the beam is balanced by an upward force *F*.



What is the size of *F*?

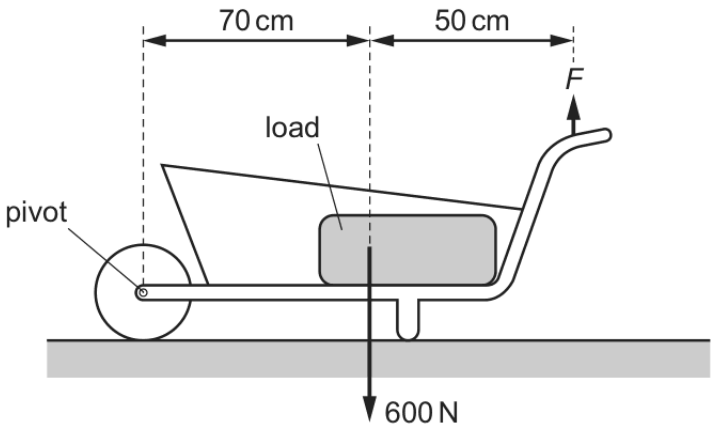
A 6 N **B** 12 N **C** 30 N **D** 60 N

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 9

9	**B**

QUESTION · 5054_w15_qp_11.pdf · source page 4 of 16

8 The total weight of the load and the wheelbarrow shown is 600 N.



What is the size of force *F* needed just to lift the loaded wheelbarrow?

A 350 N **B** 430 N **C** 600 N **D** 840 N

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 8

8	**A**

4 Fig. 4.1 shows a screwdriver of mass 64 g resting in equilibrium on a pivot.

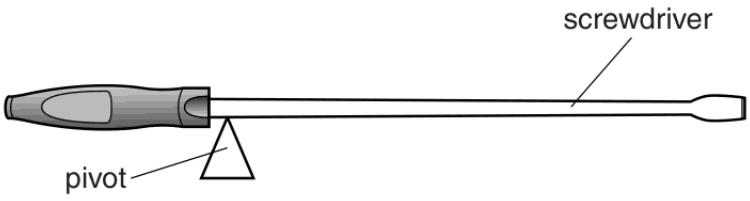


Fig. 4.1

(a) On Fig. 4.1, mark and label with a C, the centre of mass of the screwdriver. [1]

(b) The gravitational field strength is 10 N/kg.

(i) Calculate the weight of the screwdriver.

weight = ... [1]

(ii) On Fig. 4.1, draw an arrow labelled *W* to represent the weight of the screwdriver. [1]

(c) State two conditions that apply when an object is in equilibrium.

1. .....

.....

.....

2. .....

.....

.....

[2]

4 **(a)** (point) C immediately above tip of pivot (and in middle(vertically) of screwdriver (±1 mm)) B1

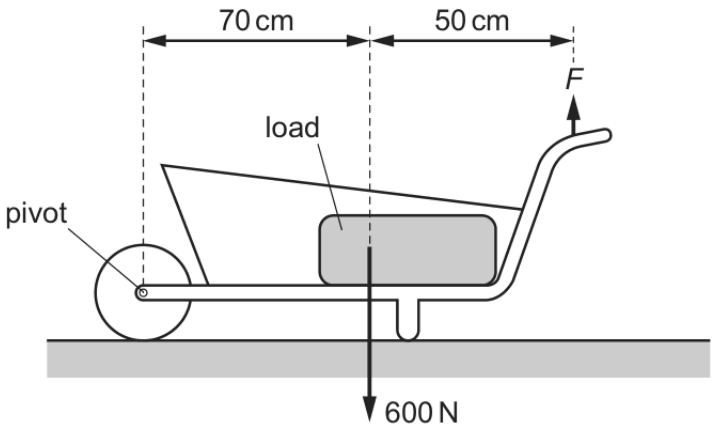
(b) **(i)** 0.64 N B1

(ii) arrow *W* vertically downwards through candidate's C or pivot B1

(c) no resultant force **or** upward force = downward force **or** force left = force right B1
no resultant moment (of force) **or** clockwise moment = anticlockwise moment B1 [5]

QUESTION · 5054_w15_qp_12.pdf · source page 5 of 16

11 The total weight of the load and the wheelbarrow shown is 600 N.



What is the size of force *F* needed just to lift the loaded wheelbarrow?

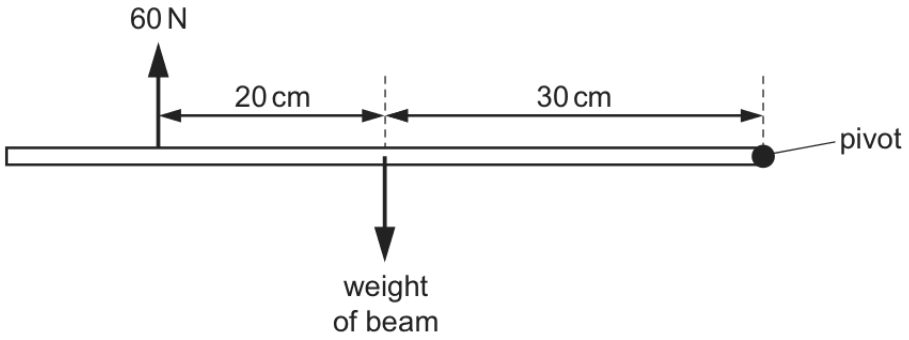
A 350 N **B** 430 N **C** 600 N **D** 840 N

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 11

11 **A**

QUESTION · 5054_s16_qp_11.pdf · source page 5 of 16

9 A uniform horizontal beam, pivoted at its right-hand end, is in equilibrium. A force of 60 N acts vertically upwards on the beam as shown.



What is the weight of the beam?

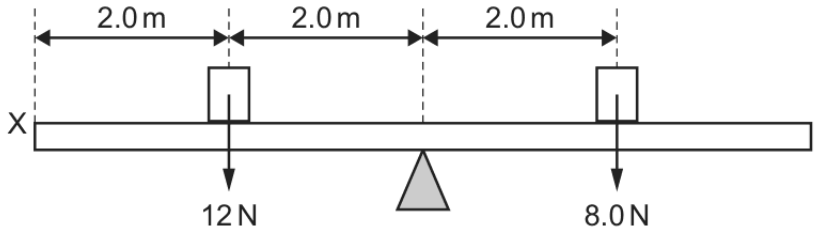
A 36 N **B** 40 N **C** 90 N **D** 100 N

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 9

9	**D**

QUESTION · 5054_s16_qp_12.pdf · source page 3 of 16

6 A uniform plank is pivoted at its mid-point. Two weights are added to the plank, one weight on each side of the pivot in the positions shown.



A vertical force is applied at point X to balance the plank.

What is the size and direction of this force?

	size / N	direction
A	2.0	downwards
B	2.0	upwards
C	4.0	downwards
D	4.0	upwards

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 6

6 **B**

2 Fig. 2.1 shows a bed that folds away against a wall during the day.

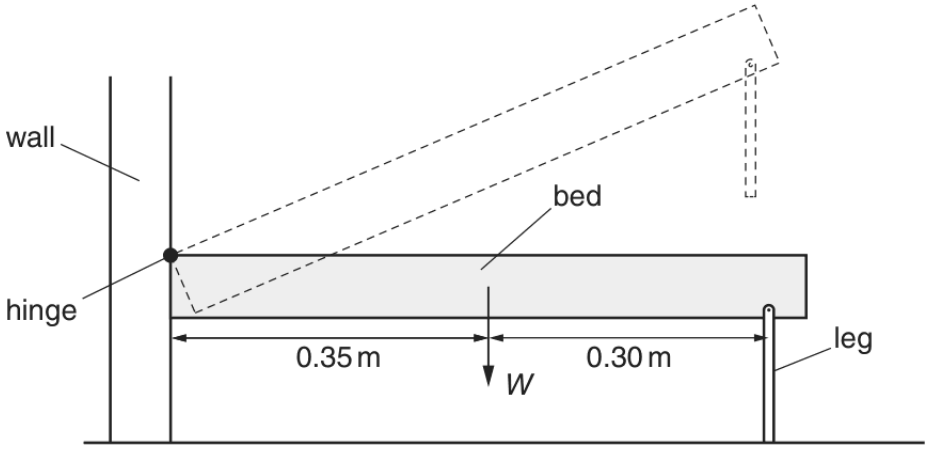


Fig. 2.1 (not to scale)

When it is horizontal, the bed is supported on one side by a hinge and on the other side by two legs. The weight W of the bed acts through its centre of mass, which is at a horizontal distance of 0.35 m from the hinge.

(a) The mass of the bed is 26 kg. The gravitational field strength g is 10 N/kg.

Calculate the weight W of the bed.

W = ...[1]

(b) (i) State the principle of moments.

.....

.....

.....

.....[2]

(ii) Determine the size of the total upward force exerted on the bed by the two legs when the bed is horizontal.

total force = ...[3]

2	**(a)**	260 N	B1	
	(b)	**(i)** for a body in equilibrium	B1	
		(total) clockwise moment = (total) anticlockwise moment	B1	
		(ii) $F_1d_1 = F_2d_2$ **or** 260×0.35 **or** 91 **or** $F \times 0.65$	C1	
		$260 \times 0.35 = F \times 0.65$ **or** $260 \times 0.35/0.65$ **or** $91 = F \times 0.65$ **or** $91/0.65$	C1	
		140 N	A1	[6]

3 When the lid of a freezer is opened, it pivots about the hinge at the back of the freezer. The handle is at the front.

Fig. 3.1 is a side view of the freezer.

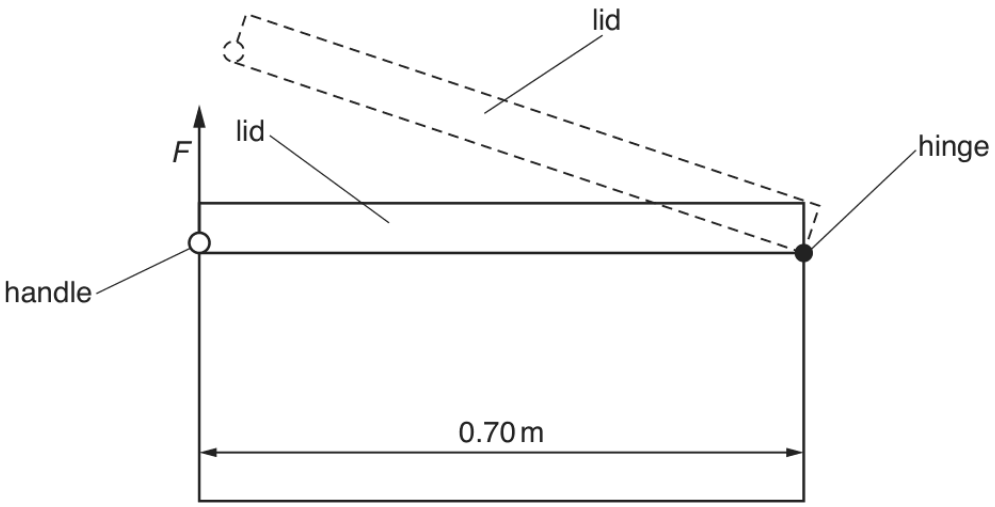


Fig. 3.1

The handle is 0.70 m from the hinge. The lid has a mass of 2.0 kg.

(a) The gravitational field strength g is 10 N / kg.

(i) Calculate the weight of the lid.

weight = ...[1]

(ii) The lid is uniform and its centre of mass is at its centre. The weight of the lid produces a moment about the hinge.

1. Calculate the moment about the hinge when the lid is closed.

moment = ...[2]

2. The moment required to open the lid is greater than the value calculated in **(a)(ii)1.**

Suggest one reason for this.

.....

.....[1]

(b) The lid is closed. To open the lid, a force *F* is applied to the handle as shown in Fig. 3.1.

The direction of *F* is vertically upwards and *F* is the smallest possible force that opens the lid.

A force on the handle in any other direction must be larger than *F* in order to open the lid. Explain why.

.....

.....

.....[1]

3 (a) (i) 20 N B1

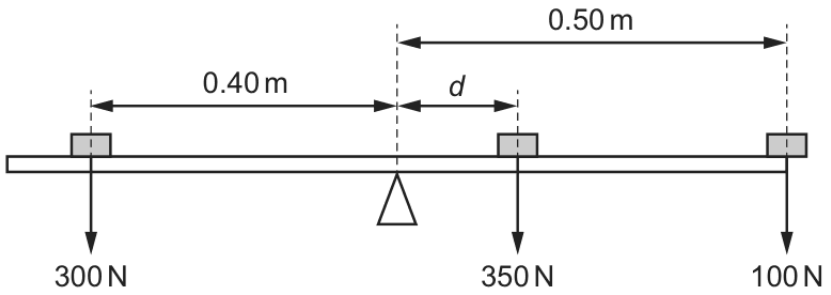
(ii) 1. ($\Gamma =$)Fd **or** 20×0.35 **or** 20×0.70 **or** 14 C1
7.0 N m A1

2. friction (at hinge / seal) **or** air resistance **or** to cause an initial acceleration B1

(b) (for other directions) **perpendicular** distance is less B1 [5]

QUESTION · 5054_s17_qp_11.pdf · source page 6 of 16

12 A uniform beam is pivoted at its centre. The beam is balanced by three weights in the positions shown.



What is the length *d*?

A 0.020 m **B** 0.050 m **C** 0.20 m **D** 0.48 m

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 12

12	C	**1**

3 Fig. 3.1 shows the brake pedal of a car which is connected to a brake cylinder.

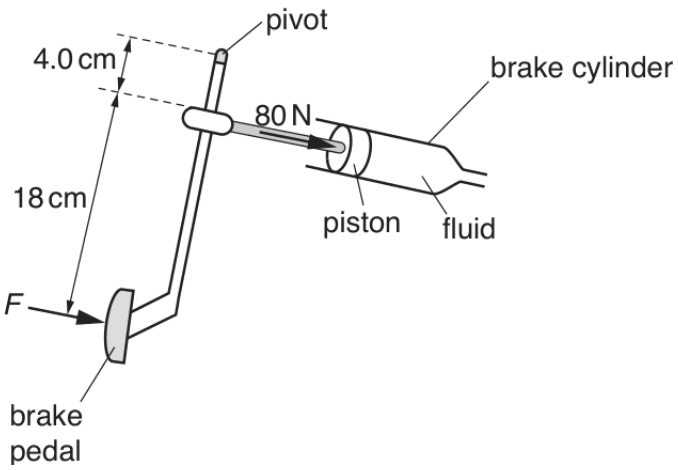


Fig. 3.1 (not to scale)

The brake is pressed with a force *F*. This force produces a moment about the pivot.

Pressing the brake causes a force of 80 N to act on the piston.

(a) Define the term *moment*.

.....

.....[2]

(b) Calculate the force *F* applied to the brake pedal.

F = ...[2]

(c) The cross-sectional area of the piston is 0.0012 m^2.

Calculate the pressure exerted by the brake piston on the fluid.

pressure = ...[2]

Question	Answer	Marks
3(a)	force × distance **ignore** force into distance	**C1**
	force × **perpendicular** distance (from line of action to point / pivot)	**A1**
3(b)	any moment calculation seen, e.g. F × 22 = 80 × 4	**C1**
	15 N	**A1**
3(c)	(P=) force / area **or** 80 / 0.0012	**C1**
	6.7×10^4 Pa	**A1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

Answer only one of the two alternative parts in **Question 6**.

1 Fig. 1.1 shows a painter standing on a wooden plank, directly above the right-hand support.

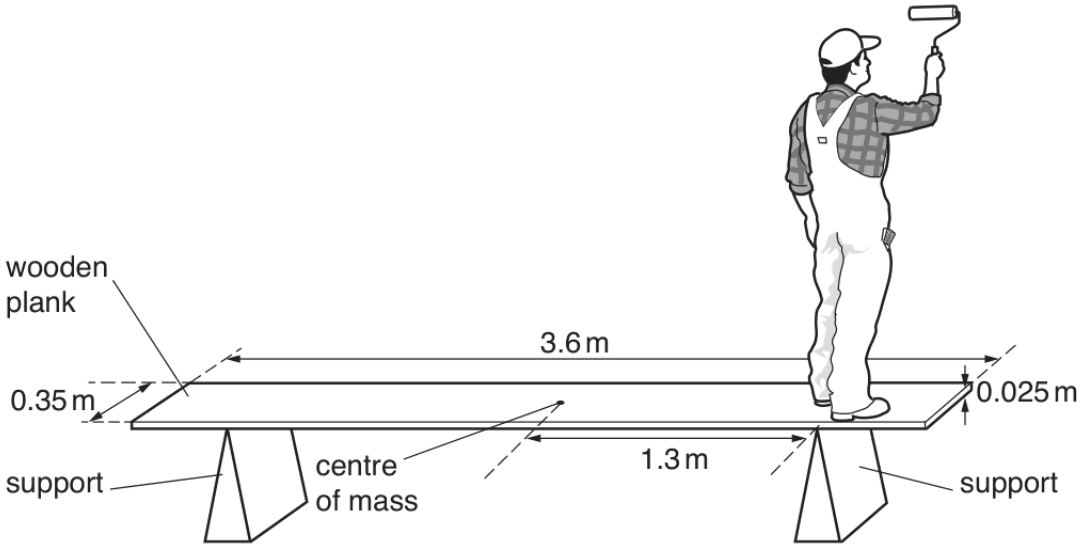


Fig. 1.1

The plank has length 3.6 m, width 0.35 m and thickness 0.025 m.

The gravitational field strength *g* is 10 N/kg and the mass of the plank is 23 kg.

(a) Calculate the density of the wood from which the plank is made.

density = ... [2]

(b) The centre of mass of the plank is in the middle of the plank at a distance of 1.3 m from each of the supports.

Calculate

(i) the weight of the plank,

weight = ... [1]

(ii) the moment of the plank about the right-hand support.

moment = ... [2]

(c) The painter moves further to the right along the plank and the plank rotates about the right-hand support.

Explain why the plank rotates.

.....

.....

.....

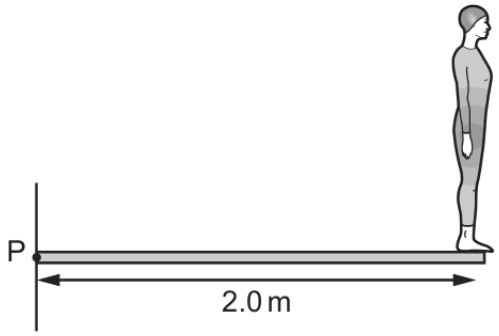
..... [1]

Section A

Question	Answer	Marks
1(a)	(ρ =)m / V **or** 23 / (3.6 × 0.35 × 0.025) **or** 23 / 0.0315	**C1**
	730 kg / m^3	**A1**
1(b)(i)	230 N	**B1**
1(b)(ii)	(Γ =)$Wx_{\perp r}$ **or** 230 × 1.3	**C1**
	300 Nm	**A1**
1(c)	moment of painter / clockwise moment (about support) is greater than / not equal to / different from moment of plank / anticlockwise moment	**B1**

QUESTION · 5054_w17_qp_12.pdf · source page 5 of 16

7 A diver of weight 500 N stands at the end of a springboard that is 2.0 m long and is fixed at point P.



The springboard has a weight of 500 N. The centre of mass of the springboard is in the centre of the board.

What is the total moment about point P of the diver and the board?

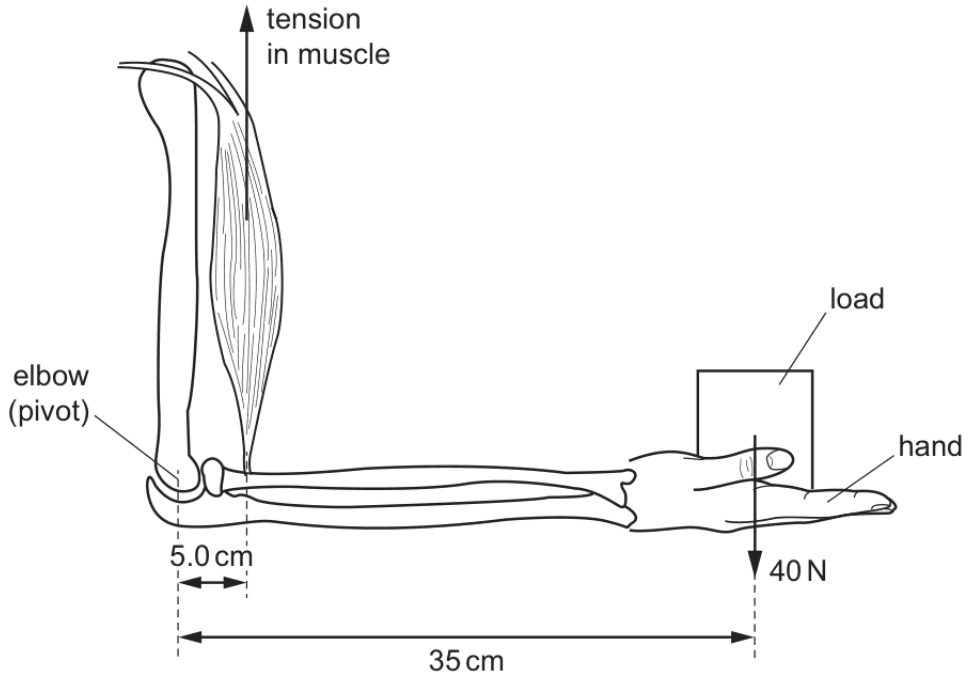
A 500 N m **B** 750 N m **C** 1000 N m **D** 1500 N m

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 7

7	D	**1**

QUESTION · 5054_s18_qp_11.pdf · source page 4 of 20

9 The diagram shows a muscle and bones in a person's arm. The hand holds a load of weight 40 N. The elbow acts as a pivot and the tension in the muscle keeps the lower part of the arm horizontal.



What is the tension in the muscle due to the load?

A 200 N **B** 240 N **C** 280 N **D** 1400 N

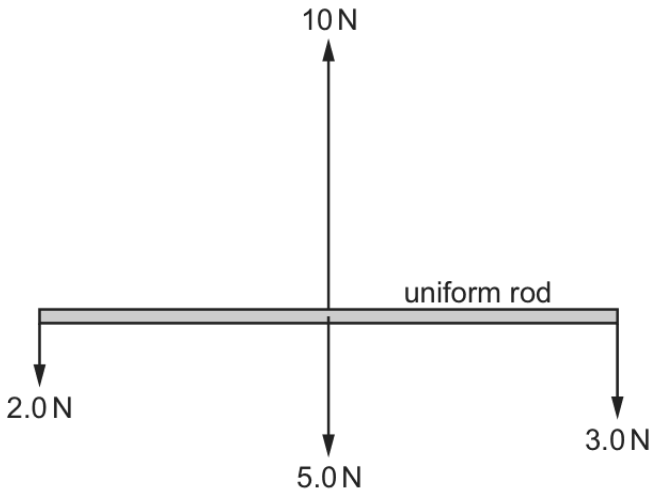
MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 9

9	C	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 4 of 16

7 A uniform rod of weight 5.0 N is held initially at rest.

The diagram shows the forces acting on the rod when it is released.



What happens to the rod when it is released?

A It does not move.

B It moves to the right.

C It moves upwards.

D It starts to rotate.

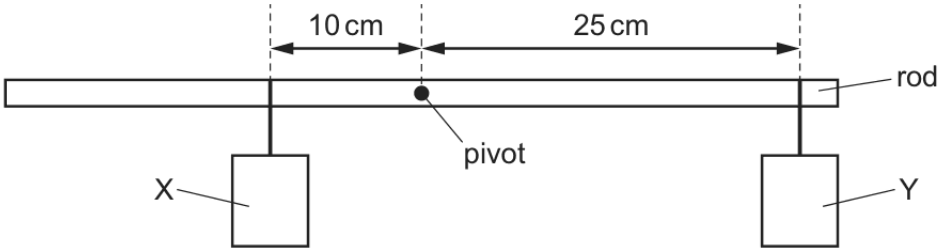
MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 7

7	D	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 5 of 16

10 Two objects X and Y are suspended from a uniform rod, pivoted at its centre.

The rod is in equilibrium.



Which statement about X and Y is correct?

A The mass of X is 0.4 times the mass of Y.

B The mass of X is 2.5 times the mass of Y.

C The mass of X is 3.5 times the mass of Y.

D The mass of X is equal to the mass of Y.

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 10

10	B	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 A bucket of water is pulled up out of a well using a rope. Fig. 8.1 shows the rope winding on to a cylinder as the handle is turned.

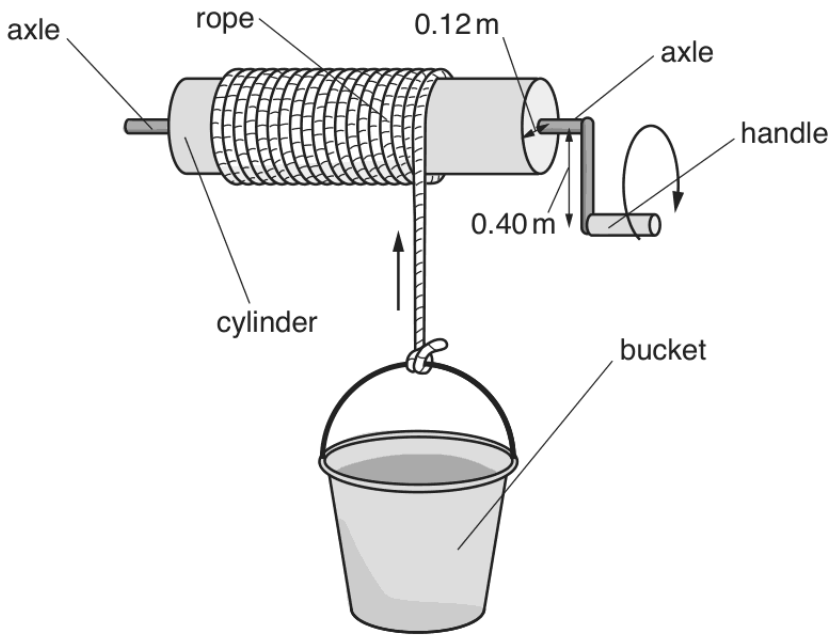


Fig. 8.1

The empty bucket has a mass of 1.0 kg.

(a) Complete the sentences that describe mass by filling in the gaps.

The mass of a body is a measure of the amount of .....

in the body. It resists a change in the state of ..... of

the body. [2]

(b) When the bucket is full, it contains $2.4 \times 10^{-2}\,m^3$ of water.

The gravitational field strength g is equal to 10 N/kg.

(i) Explain what is meant by a *gravitational field*.

.....

..... [1]

(ii) The density of water is 1000 kg / m^3.

Determine the total weight of the bucket and the water.

weight = ... [3]

(c) The radius of the cylinder is 0.12 m and the handle is 0.40 m from the axle of the cylinder. The weight of the bucket and the water produce a moment that acts on the cylinder.

(i) Calculate this moment.

moment = ... [2]

(ii) Calculate the minimum force on the handle that balances this moment.

force = ... [1]

(d) A farmer pulls the bucket of water up at a constant speed. He needs to exert a force on the handle that is greater than that calculated in **(c)(ii)**. He notices that there is a slight increase in temperature where the axle is in contact with the frame holding it.

(i) Suggest two reasons why the force exerted is greater than the value calculated.

1. .....

.....

2. .....

.....
[2]

(ii) State the energy changes that are taking place as the bucket is being lifted at a constant speed.

.....

.....

..... [3]

(e) State what can be deduced about the forces acting on the bucket when it is travelling upwards at a constant speed.

.....

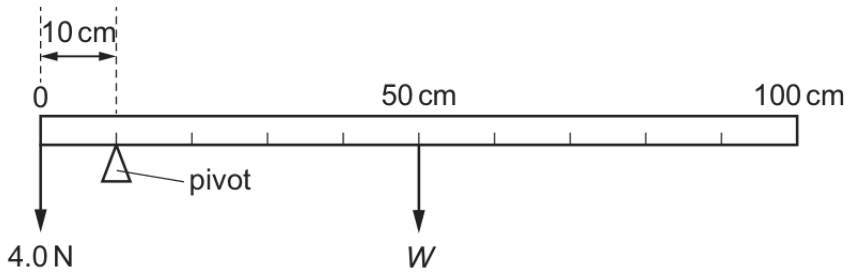
..... [1]

Question	Answer	Marks
8(a)	substance **or** matter	**B1**
	rest **or** motion	**B1**
8(b)(i)	(a region of space) where a mass experiences a force (due to gravitational attraction)	**B1**
8(b)(ii)	(m =) ρV **or** $1000 \times 2.4 \times 10^{-2}$ **or** 24 (kg)	**C1**
	25 (kg) **or** 240 (N)	**C1**
	250 N	**A1**
8(c)(i)	(Γ =) Fx **or** 250×0.12	**C1**
	30 N m	**A1**
8(c)(ii)	75 N	**B1**
8(d)(i)	any **two** from: friction (between axle and cylinder) weight of rope force not perpendicular (to radius) rope wraps over itself (and increases moment)	**B2**

Question	Answer	Marks
8(d)(ii)	from chemical (energy) **or** chemical (energy) as first term	**B1**
	to gravitational potential (energy) **or** to thermal / heat (energy) as last term	**B1**
	to thermal / internal (energy) / heat **and** thermal / internal (energy) / heat	**B1**
8(e)	no resultant force **or** forces balance / cancel **or** tension equals weight (of bucket) **or** upwards force equals downward force	**B1**

QUESTION · 5054_s19_qp_11.pdf · source page 4 of 20

9 A uniform metre rule is balanced by a 4.0 N weight as shown.



What is the weight *W* of the metre rule?

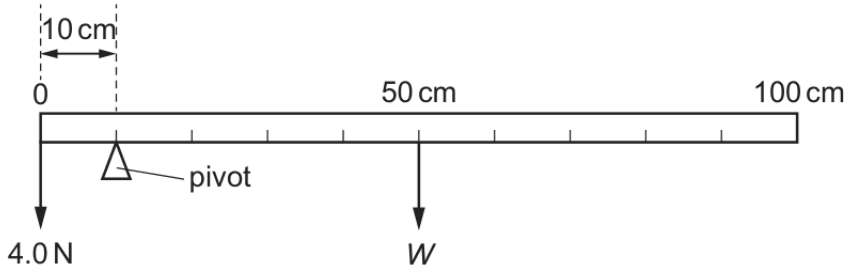
A 1.0 N **B** 4.0 N **C** 16 N **D** 40 N

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 9

9	A	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 5 of 16

10 A uniform metre rule is balanced by a 4.0 N weight as shown.



What is the weight *W* of the metre rule?

A 1.0 N **B** 4.0 N **C** 16 N **D** 40 N

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 10

10	A	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 6 of 20

11 What is an expression for the moment of a force about a pivot?

A distance the force moves from the pivot ÷ force

B force × distance the force moves from the pivot

C force × perpendicular distance of the force from the pivot

D force ÷ perpendicular distance of the force from the pivot

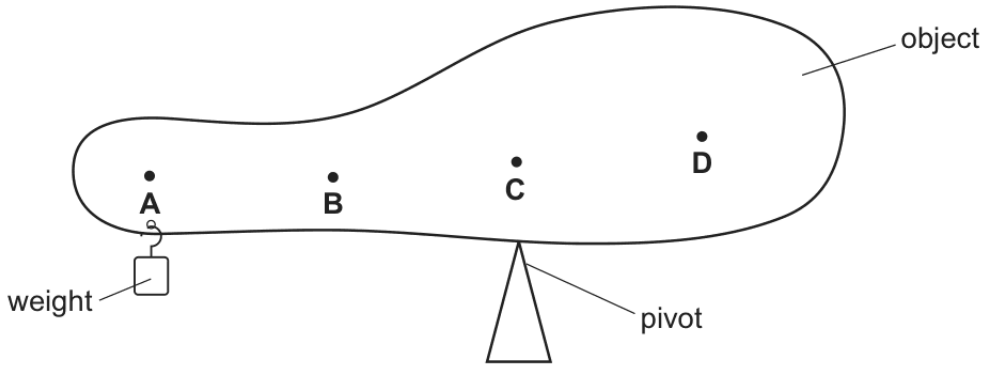
MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 11

11	C	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 6 of 20

12 A student balances a non-uniform object on a pivot. To do this, a weight is suspended near the left-hand end of the object, as shown.

Where is the centre of mass of the object?



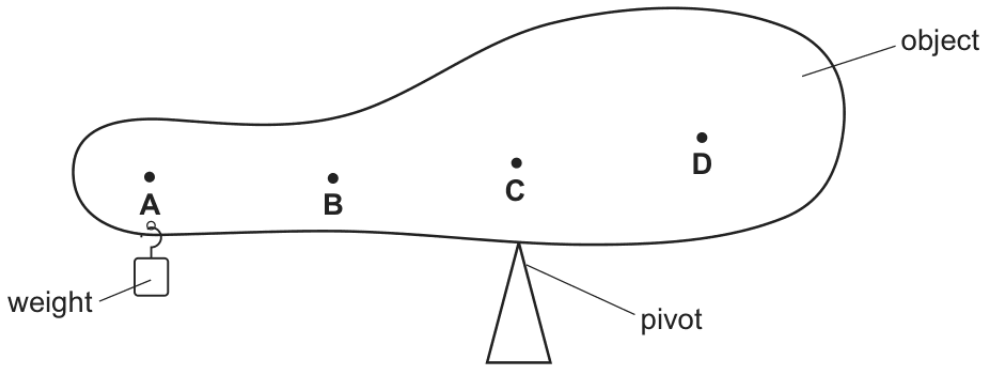
MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 12

12	D	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 5 of 16

10 A student balances a non-uniform object on a pivot. To do this, a weight is suspended near the left-hand end of the object, as shown.

Where is the centre of mass of the object?



MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 10

10	D	**1**

7 Fig. 7.1 shows a horizontal, rectangular coil ABCD placed between a magnetic N-pole and an S-pole.

There is a current in the coil.

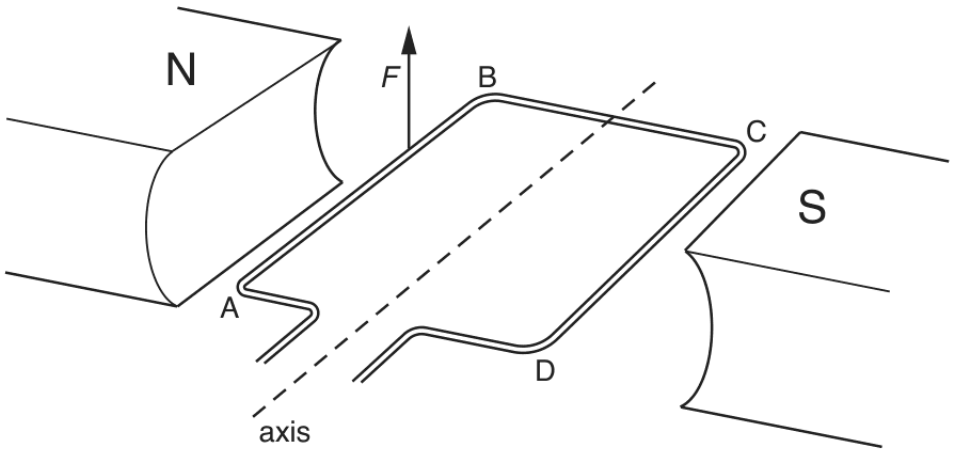


Fig. 7.1

An upward force F of size 9.6×10^{-3} N acts on side AB of the coil.

(a) (i) Explain why there is a force on side AB.

.....

..... [1]

(ii) Determine the direction of the current in AB and state how the direction is deduced.

.....

.....

..... [2]

(b) Both side AB and side CD of the coil are 2.5 cm from the axis.

Determine the total moment acting on the coil.

moment = ... [2]

(c) The coil in Fig. 7.1 is part of a direct current (d.c.) motor.

(i) State the name of the device that connects the coil of a d.c. motor to the electricity supply.

..... [1]

(ii) State **one** change to the arrangement in Fig. 7.1 that produces a greater turning effect in coil ABCD.

..... [1]

[Total: 7]

Section B begins over the page

MARK SCHEME · 5054_w19_ms_22.pdf · source page 6 of 10 · Question 7

Question	Answer	Marks
7(a)(i)	current **and** magnetic field / magnetic flux (experiences a force) **or** the two magnetic fields interact	**B1**
7(a)(ii)	three separate approaches **I** / **II** / **III**	
7(a)(ii)**I**	B to A	**B1**
	left-hand rule mentioned / described	**B1**
	or	

MARK SCHEME · 5054_w19_ms_22.pdf · source page 7 of 10 · Question 7

Question	Answer	Marks
7(a)(ii)**II**	B to A	**B1**
	top face of coil attracted to S-pole **or** top face acts as an N-pole	**B1**
	or	
7(a)(ii)**III**	B to A	**B1**
	more field lines under AB **or** fewer lines above AB	**B1**
7(b)	(Γ=) Fx **or** $2 \times 9.6 \times 10^{-3} \times 0.025$ **or** $9.6 \times 10^{-3} \times 0.025$ **or** 2.4×10^{-4} (N m) **or** 0.024 (N cm)	**C1**
	4.8×10^{-4} N m **or** 0.048 N cm	**A1**
7(c)(i)	split-ring commutator **or** (carbon) brushes	**B1**
7(c)(ii)	(soft-)iron core **or** stronger magnet(s) **or** larger current **or** more turns **or** smaller resistance **or** greater electromotive force (e.m.f.)	**B1**

3 Fig. 3.1 shows part of a hydraulic press that is used to compress waste paper into a brick for burning.

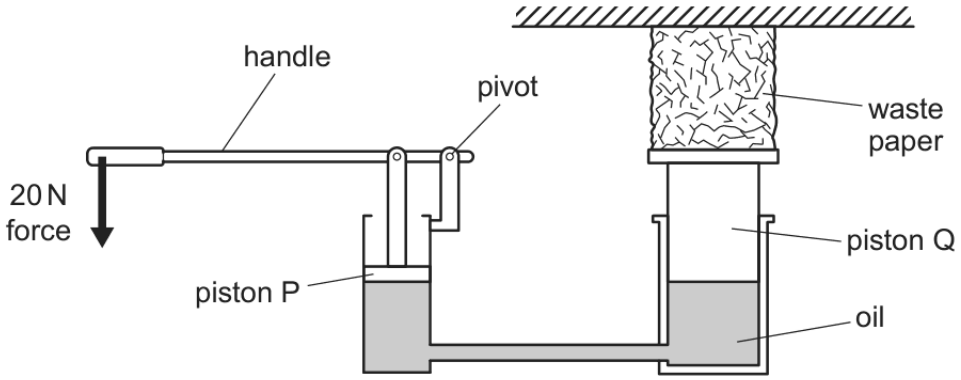


Fig. 3.1

A force of 20 N is exerted downwards on the end of the handle.

(a) The force on the handle creates a moment about the pivot.

(i) Define the *moment of a force*.

.....

.....

.....

..... [2]

(ii) Explain why the force exerted on piston P is greater than the force exerted on the handle.

.....

.....

..... [1]

(b) Explain how the hydraulic press enables a greater force to be exerted on piston Q than is exerted on piston P.

.....

.....

.....

.....

..... [2]

(c) In moving the handle downwards, the 20 N force moves through a distance of 0.60 m and piston Q rises by 0.020 m.

The force exerted by piston Q on the paper is 400 N.

Calculate:

(i) the work done in moving the handle downwards

work = ... [2]

(ii) the efficiency of the hydraulic press.

efficiency = ... [2]

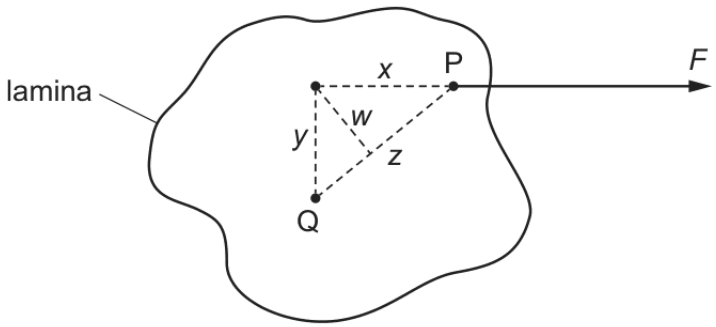
[Total: 9]

Question	Answer	Marks
3(a)(i)	force × distance	**C1**
	force × perpendicular distance	**A1**
3(a)(ii)	smaller distance to pivot (than from P to pivot)	**B1**

Question	Answer	Marks
3(b)	same / equal pressure on piston P and piston Q	**B1**
	larger area of piston Q (than piston P)	**B1**
3(c)(i)	(W = Fd) numerical or algebraic	**C1**
	12 J	**A1**
3(c)(ii)	(efficiency) = work output / work input numerical or algebraic, e.g. 400 × 0.02 / 12	**C1**
	0.67 or 67%	**A1**

QUESTION · 5054_w20_qp_11.pdf · source page 5 of 16

10 A length of thread is attached to a lamina at point P, as shown in the diagram.



The lamina is free to rotate about point Q.

The tension in the thread is *F*.

What is the moment of *F* about Q?

A *Fw* **B** *Fx* **C** *Fy* **D** *Fz*

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 10

10	C	**1**

3 Fig. 3.1 shows a door and an automatic door-closer viewed from above.

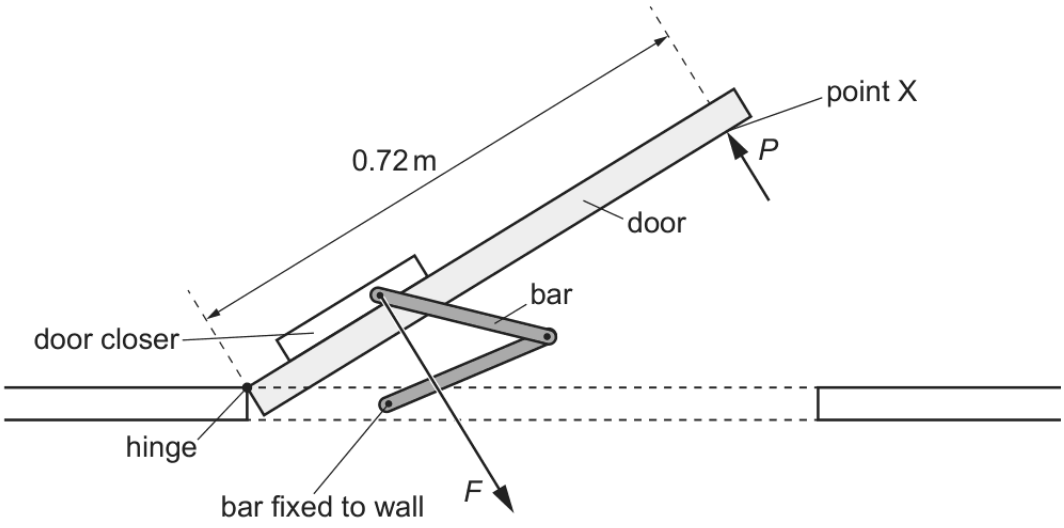


Fig. 3.1

When the door opens and closes, the hinge acts as a pivot.

A girl opens the door by exerting a force *P* at point X.

Force *P* is perpendicular to the surface of the door.

(a) Fig. 3.1 shows that point X is a distance of 0.72 m along the front of the door from the hinge. The force *P* is 25 N.

(i) Calculate the moment of force *P* about the hinge.

moment of force = ... [2]

(ii) The door rotates about the hinge by 90°. The circumference of a circle of radius 0.72 m is 4.5 m.

Calculate the work done on the door by force *P*.

work done = ... [2]

(b) As the door opens, there is a force *F* on the door in the direction shown in Fig. 3.1.

Although force *F* is larger than force *P*, the door rotates about the hinge.

Explain why.

.....

.....

.....

..... [2]

[Total: 6]

MARK SCHEME · 5054_w20_ms_21.pdf · source page 6 of 10 · Question 3

Question	Answer	Marks
3(a)(i)	(Γ =) $Fx_{\perp r}$ **or** 25×0.72	**C1**
	18 N m	**A1**
3(a)(ii)	(WD =) $Fx_{\parallel}$ **or** $F\pi r / 2$ or $25 \times 4.5 / 4$ **or** 25×1.13	**B1**
	28 J	**B1**
3(b)	moment due to force *P* is greater	**B1**
	distance of X to hinge is greater than distance from where force *F* acts	**B1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A glass beaker has a mass of 50 g. A liquid of density 1.8 g / cm^3 is poured into the beaker until it reaches the 200 cm^3 mark.

(a) Calculate the total mass of the beaker and its contents.

mass = ... [3]

(b) The centre of mass of a metre rule is at the 50 cm mark.

(i) State what is meant by *centre of mass*.

.....

..... [1]

(ii) The metre rule is placed on a pivot. The tip of the pivot is under the 80 cm mark on the rule.

The beaker with its contents is then placed at different positions along the rule until the rule is balanced.

Fig. 1.1 shows the arrangement with the rule balanced.

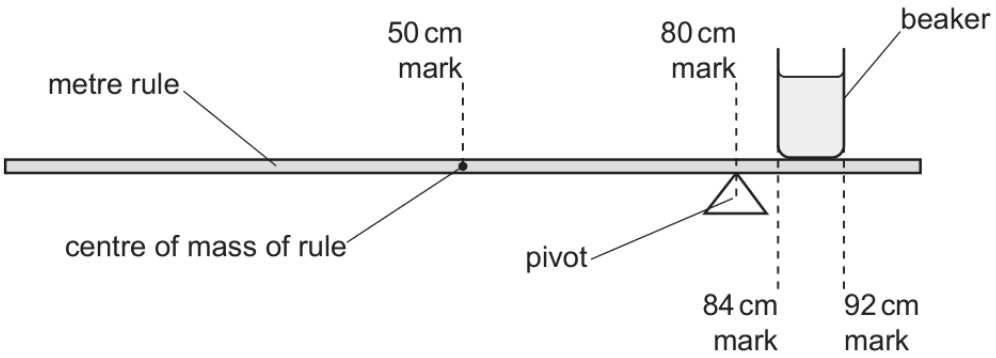


Fig. 1.1

One side of the beaker is at the 84 cm mark and the other side is at the 92 cm mark.

Calculate the mass of the rule.

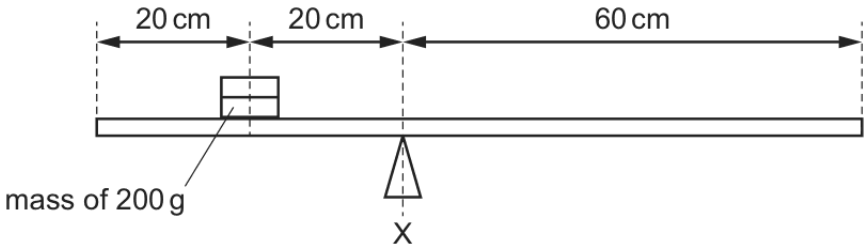
mass = ... [3]

[Total: 7]

Question	Answer	Marks
1(a)	(m =) ρV **or** 1.8×200 **or** 360	**C1**
	$(1.8 \times 200) + 50$ **or** $360 + 50$ **or** 360 **or** candidate's mass + 50	**C1**
	410 g **or** 0.41 kg	**A1**
1(b)(i)	(place / region) where the mass appears to be concentrated **or** where the object balances	**B1**
1(b)(ii)	88 (cm) seen **or** $(84 + 92) / 2$ seen **or** 30 (cm) seen **or** 8(.0 cm) seen **or** 164 (g)	**C1**
	$m_1gx_1 = m_2gx_2$ **or** $F_1x_1 = F_2x_2$ **or** $W_1x_1 = W_2x_2$ **or** (m_1 =) m_2x_2 / x_1 **or** $410 \times (88 - 80) / 30$ **or** 164 (g)	**C1**
	110 g **or** 0.11 kg	**A1**

QUESTION · 5054_s21_qp_11.pdf · source page 4 of 16

8 A horizontal beam is pivoted at X. A mass of 200 g rests on the beam as shown. The centre of mass of the beam is 50 cm from the right-hand end of the beam.



The beam is balanced.

What is the mass of the beam?

A 80 g **B** 100 g **C** 400 g **D** 800 g

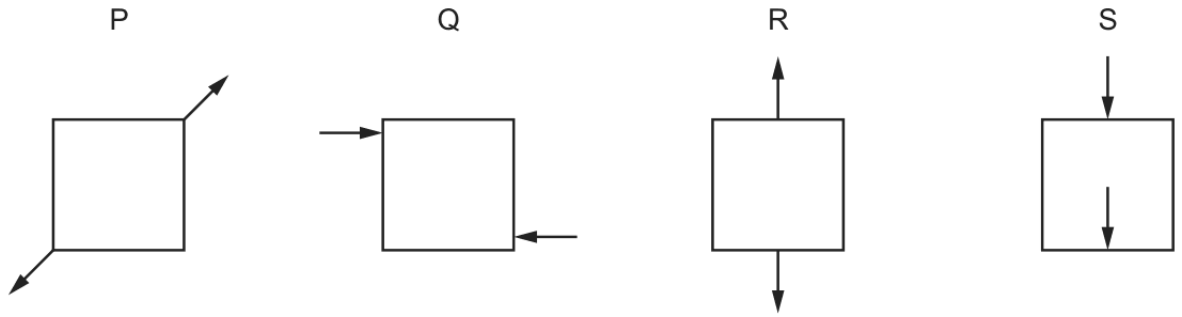
MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 8

8	C	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 3 of 16

3 The diagrams show the forces acting on four identical solid blocks.

Each arrow represents a force of 20 N.



Which blocks are in equilibrium?

A P and R only

B Q and S only

C P, Q and R only

D P, Q, R and S

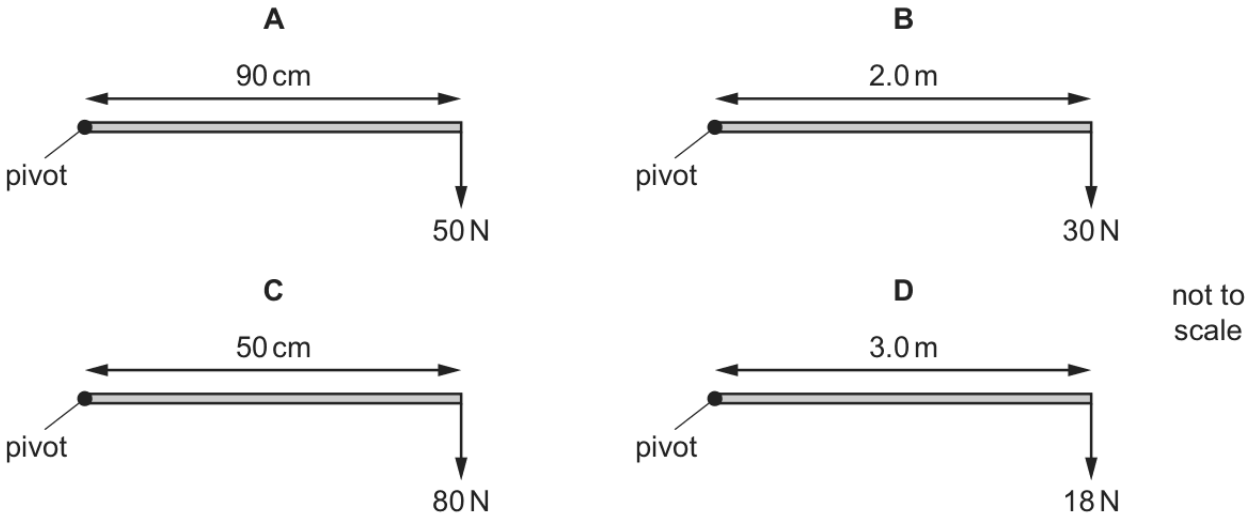
MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 3

3	A	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 5 of 16

8 Some forces are applied at different distances from a pivot.

Which diagram shows the force that produces the largest moment about the pivot?



MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 8

8	B	**1**

3 Fig. 3.1 shows a small brick hanging from a newton meter.

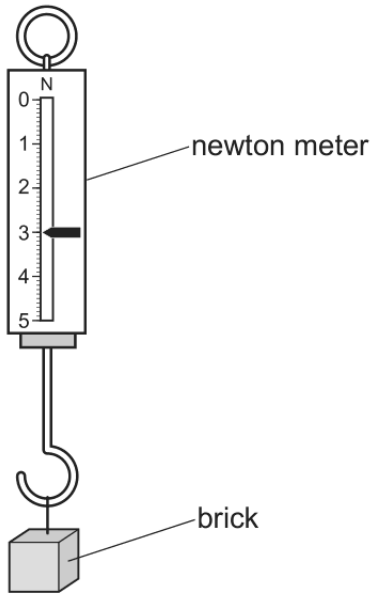


Fig. 3.1

The reading on the newton meter is 3.0 N.

(a) Describe how the reading on the newton meter is used to find the mass of the brick.

.....

..... [1]

(b) The same brick and newton meter are used in the apparatus shown in Fig. 3.2. The meter rule is pivoted at its centre and is balanced. The reading on the newton meter is not shown.

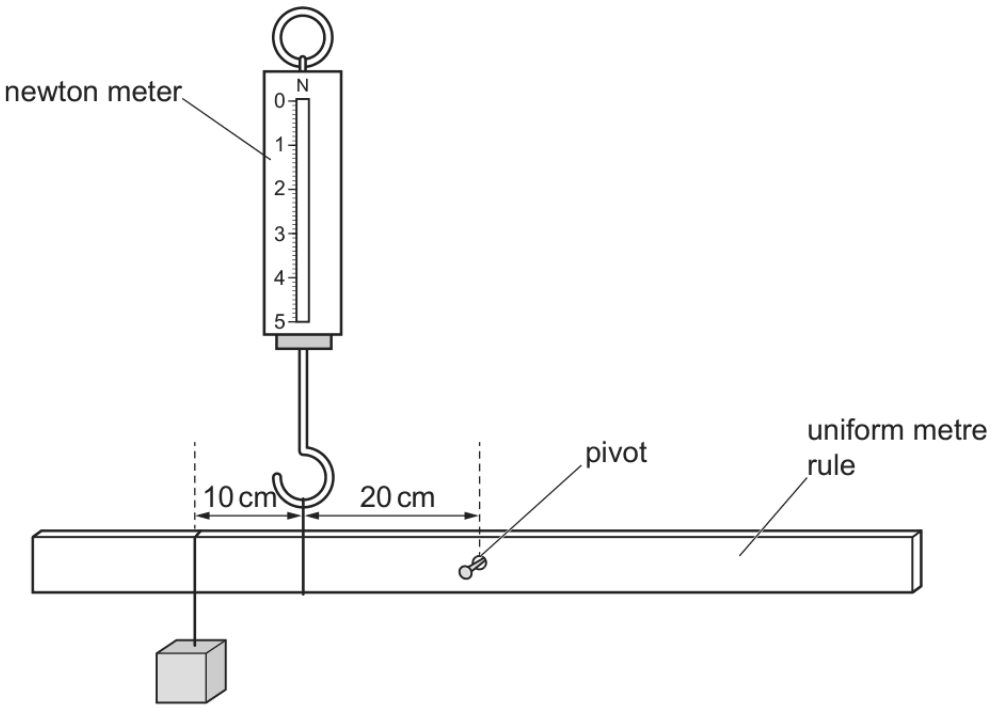


Fig. 3.2

(i) State the *principle of moments* for a body in equilibrium.

.....

.....

..... [1]

(ii) Determine the reading on the newton meter shown in Fig. 3.2.

reading = ... [2]

(c) A beaker of water is placed so that the brick is partly submerged in the water, as shown in Fig. 3.3. The apparatus is adjusted to keep the rule horizontal.

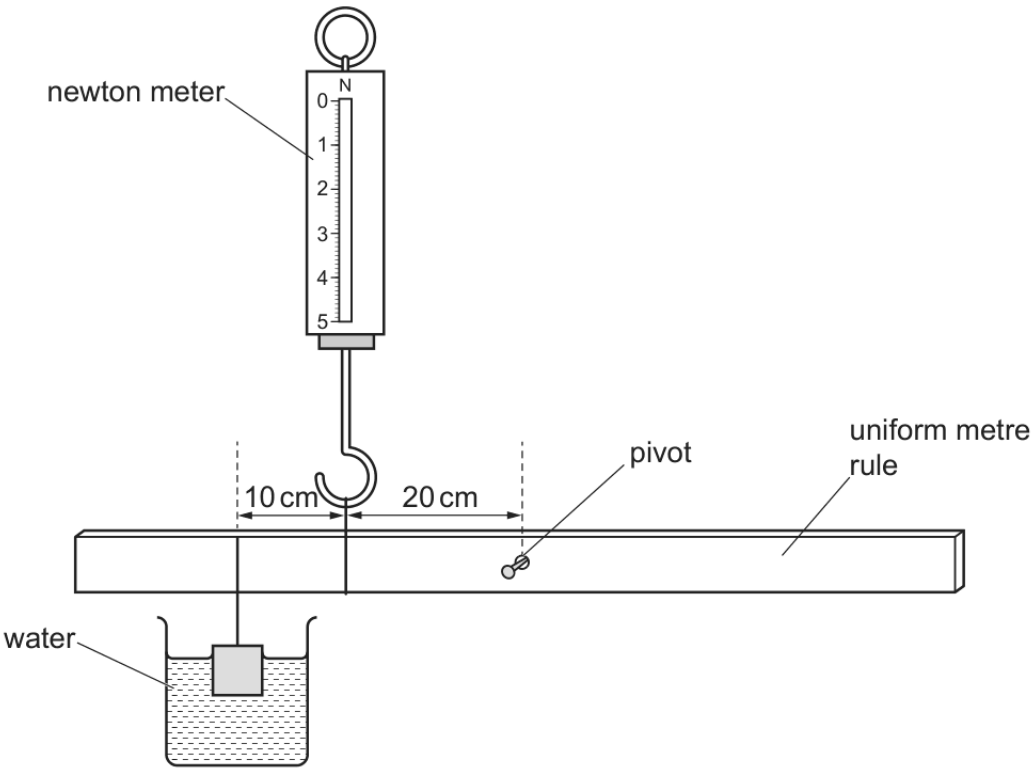


Fig. 3.3

Suggest why the reading on the newton meter is less than your answer in **(b)(ii)**.

.....

.....

..... [2]

[Total: 6]

Question	Answer	Marks
3(a)	(mass =) meter reading / *g* meter reading / gravitational field (strength meter reading / acceleration due to gravity	**B1**
3(b)(i)	(sum of) clockwise moments = (sum of) anticlockwise moments (about any point)	**B1**

Question	Answer	Marks
3(b)(ii)	any attempt at using moments, e.g. $3 \times 30 = F \times 20$	**C1**
	4.5 N	**A1**
3(c)	force upwards (on brick due to water) **or** force due to water (on brick) **or** force downwards (on brick) is reduced / counteracted (by water) **or** force exerted by brick (on rule) reduced **or** force / tension in string reduced	**B1**
	(force caused by) pressure (of water) **or** reduced anticlockwise moment	**B1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a wooden bench of weight 2000 N.

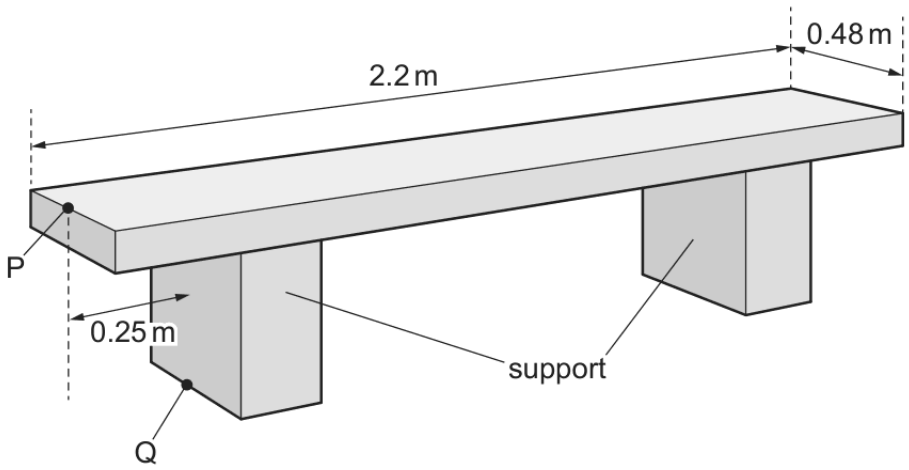


Fig. 1.1

(a) Each of the two supports has an area of 0.040 m^2 in contact with the ground.

Calculate the pressure on the ground due to the bench.

pressure = ... [2]

(b) The centre of mass of the bench is 1.1 m from the left-hand end of the bench and 0.24 m from the front.

(i) Suggest **one** reason why the centre of mass is in this position.

.....

.....

..... [2]

(ii) There is a force exerted vertically downwards from the point P shown in Fig. 1.1.

Calculate the maximum force that can be exerted vertically downwards at P without the bench rotating about the point Q shown in Fig. 1.1.

maximum force = ... [3]

[Total: 7]

Question	Answer	Marks
1(a)	(p =) F / A or 2000 / (2 × 0.040) **or** 50 000 (P–a)	**C1**
	25 000 Pa	**A1**
1(b)(i)	symmetrical left-to-right **or** symmetrical front-to-back **or** bench is stable	**B1**
	because centre of mass is in middle (horizontally) **or** centre of mass is within the base	**B1**
1(b)(ii)	(moment =) Fx **or** Wy **or** 2000 × (1.1 – 0.25) **or** 2000 × 0.85 clockwise moment = anticlockwise moment **or** Fx = Wy **or**	**C1**
	Fx = 2000 × (1.1 – 0.25) **or** Fx = 2000 × 0.85 **or** (F =)2000 × 0.85 / 0.25	**C1**
	6800 N	**A1**

8 The turning effect of a force is measured by its moment.

(a) Fig. 8.1 shows a force *F* acting on an object at point P. The object is free to rotate about an axis at X that is perpendicular to the page.

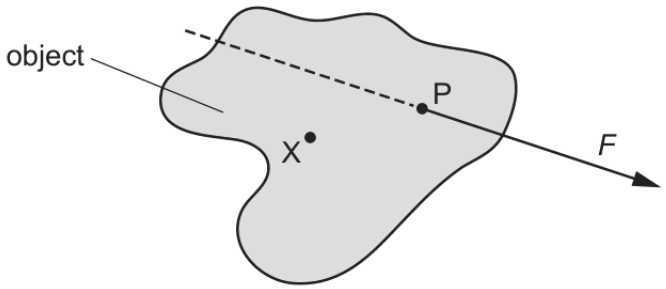


Fig. 8.1

Write down an expression for the moment of *F* about the axis at X. Draw on Fig. 8.1 to show what is meant by any other term used in your expression.

.....

..... [2]

(b) **(i)** State the principle of moments.

.....

.....

..... [2]

(ii) Describe an experiment to verify the principle of moments. Include a diagram to help the description.

.....

.....

.....

.....

.....

..... [3]

(c) A worker carries a ladder on his shoulder. His shoulder acts as a pivot. Fig. 8.2 shows that the ladder is horizontal.

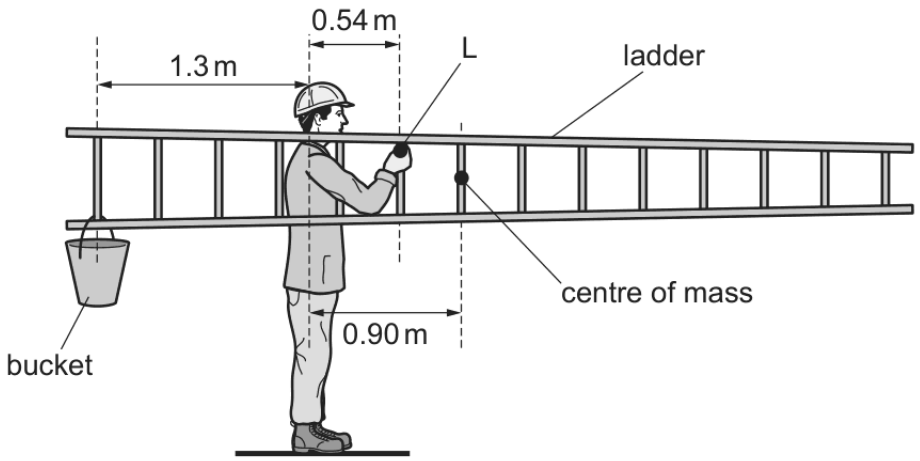


Fig. 8.2

The ladder is wider at one end than at the other end. The mass of the ladder is 8.0 kg.

(i) The gravitational field strength is 10 N / kg.

Calculate the weight of the ladder.

weight = ... [1]

(ii) State what is meant by *centre of mass*.

.....

..... [1]

(iii) The centre of mass of the ladder is not halfway along its length.

State what this shows about the ladder.

.....

..... [1]

(iv) The centre of mass of the ladder is a horizontal distance of 0.90 m from the worker's shoulder.

Calculate the moment about the worker's shoulder of the weight of the ladder.

moment = ... [2]

(v) A bucket of weight 87 N is suspended from the ladder at a horizontal distance of 1.3 m from the worker's shoulder.

The worker keeps the ladder horizontal by exerting a vertical force at point L. L is a horizontal distance of 0.54 m from his shoulder.

Determine the size and direction of the force exerted at L.

size of force = ...

direction = ...
[3]

[Total: 15]

MARK SCHEME · 5054_w21_ms_22.pdf · source page 9 of 11 · Question 8

Question	Answer	Marks
8(a)	$F \times$ distance / displacement	**M1**
	perpendicular distance to line of action of force shown on Fig. 7.1	**A1**
8(b)(i)	for an object in equilibrium	**B1**
	(sum of) clockwise moments is equal to (sum of) anticlockwise moments **or** no resultant moment	**B1**
8(b)(ii)	equipment shown in diagram (e.g. rule, two loads, pivot)	**B1**
	what is done to achieve balance (e.g. balance rule on pivot, one load each side of pivot)	**B1**
	how principle is verified (e.g. calculate two opposite moments and find they are equal)	**B1**
8(c)(i)	80 N	**B1**
8(c)(ii)	(point) where the (whole) weight of an object acts / seems to act **or** point where the object balances	**B1**

MARK SCHEME · 5054_w21_ms_22.pdf · source page 10 of 11 · Question 8

Question	Answer	Marks
8(c)(iii)	mass not uniformly distributed along length **or** more mass at the one end	**B1**
8(c)(iv)	(Γ=) 80×0.90	**C1**
	72 N m	**A1**
8(c)(v)	moment / 0.54 **or** 209 (N) **or** 41.1 / 0.54	**C1**
	76 N	**A1**
	downwards	**B1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a model of the human arm. The rubber band represents the muscle that moves part of the arm XY up.

A mass is suspended from XY, as shown in Fig. 1.2. The weight of section XY is negligible and the model is at rest.

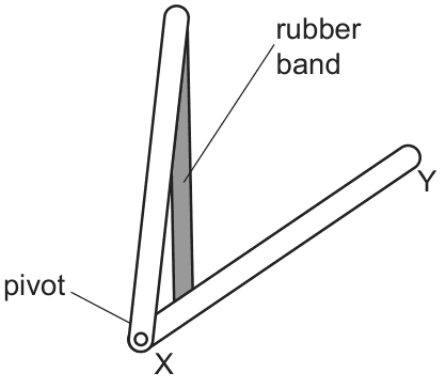


Fig. 1.1

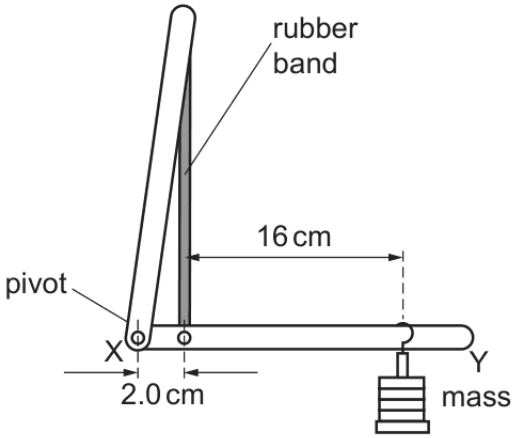


Fig. 1.2 (not to scale)

(a) (i) State **two** ways in which the dimensions of the rubber band change as the mass is added to section XY.

.....

..... [2]

(ii) State the form of energy stored in the stretched rubber band.

..... [1]

(b) (i) State the principle of moments.

.....

.....

.....

..... [2]

(ii) Explain why the force that the rubber band exerts on section XY is larger than the weight of the mass.

.....

.....

..... [1]

(iii) The mass suspended from section XY in Fig. 1.2 has a weight of 4.0 N.

Calculate the force that the rubber band exerts on section XY.

force = ... [2]

(iv) Explain how your answer to **(b)(iii)** is different if the weight of section XY is **not** negligible.

.....

.....

..... [1]

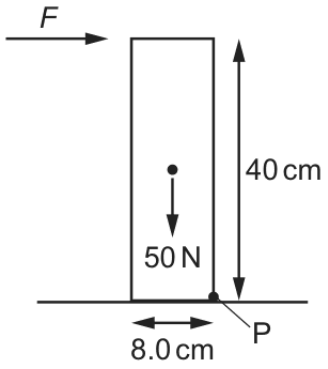
[Total: 9]

Question	Answer	Marks
1(a)(i)	longer	**B1**
	thinner **or** smaller (cross-sectional) area	**B1**
1(a)(ii)	elastic potential **or** strain (energy)	**B1**
1(b)(i)	sum of / total / net / resultant moment is zero	**M1**
	when (the object is) in equilibrium	**A1**
1(b)(ii)	distance of rubber band to pivot is smaller than the distance of the mass (to pivot)	**B1**
1(b)(iii)	$4 \times 18 = F \times 2$	**C1**
	36 N	**A1**
1(b)(iv)	greater **and** the weight (of XY) contributes to the (clockwise) moment/turning effect **or** the moment of the rubber band force is larger	**B1**

QUESTION · 5054_s22_qp_12.pdf · source page 4 of 20

7 The diagram shows a uniform solid rectangular block of weight 50 N that is pivoted about point P.

The height of the block is 40 cm. The base of the block is 8.0 cm wide.



Which horizontal force *F* just makes the block start to rotate about P?

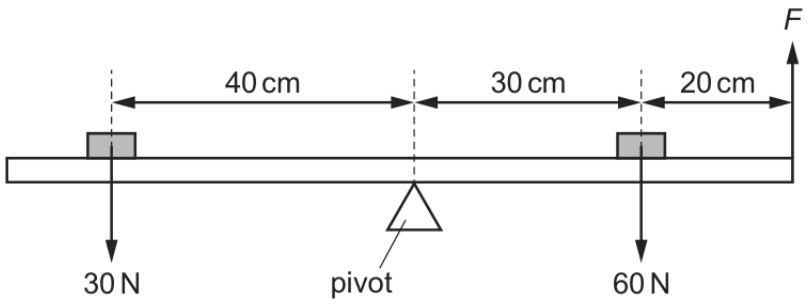
A 2.5 N **B** 5.0 N **C** 10 N **D** 160 N

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 7

7	B	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 5 of 16

11 A uniform beam is pivoted at its centre. Two weights are placed on the beam in the positions shown and the beam is balanced by an upward force *F*.



What is the size of *F*?

A 6.0 N **B** 12 N **C** 30 N **D** 60 N

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 11

11	B	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 3 of 16

6 A block of wood is placed on a table. External forces act on the block but the block remains stationary.

A student suggests three conditions for the block to remain stationary.

condition P The resultant force on the block **must** equal zero.

condition Q The resultant moment on the block **must** equal zero.

condition R The external forces **must** act through the centre of mass of the block.

Which conditions are necessary for the block to remain stationary?

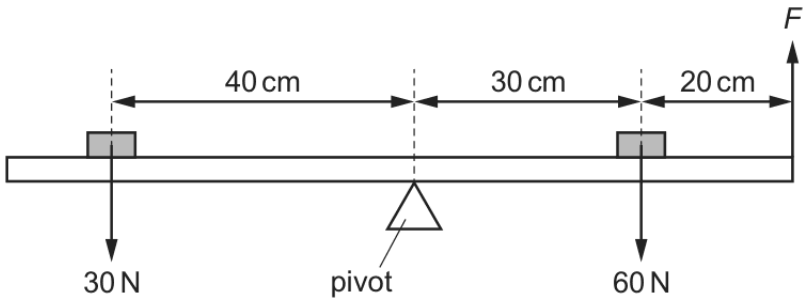
A P and Q only **B** P and R only **C** Q and R only **D** P, Q and R

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 6

6	A	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 6 of 16

12 A uniform beam is pivoted at its centre. Two weights are placed on the beam in the positions shown and the beam is balanced by an upward force *F*.



What is the size of *F*?

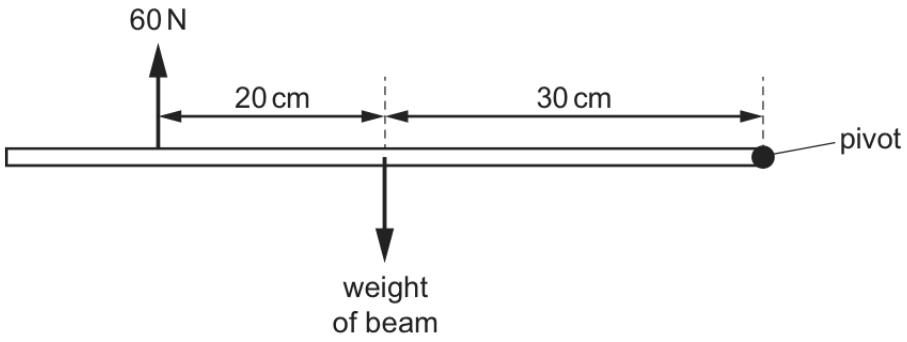
A 6.0 N **B** 12 N **C** 30 N **D** 60 N

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 12

12	B	**1**

QUESTION · 5054_s23_qp_11.pdf · source page 6 of 20

9 A uniform horizontal beam, pivoted at its right-hand end, is in equilibrium. A force of 60 N acts vertically upwards on the beam as shown.



What is the weight of the beam?

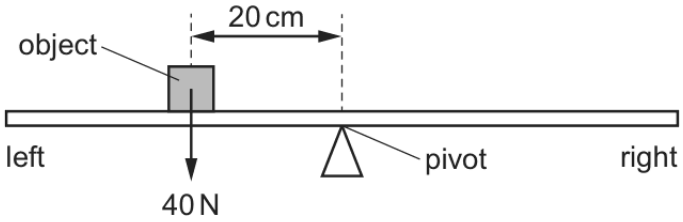
A 36 N **B** 40 N **C** 90 N **D** 100 N

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 9

9	D	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 6 of 16

11 A uniform beam is pivoted at its midpoint. An object weighing 40 N is placed on the beam 20 cm to the left of the pivot as shown.



Which force and position balances the system?

A 20 N acting downwards, 40 cm to the right of the pivot

B 20 N acting upwards, 40 cm to the right of the pivot

C 50 N acting downwards, 10 cm to the left of the pivot

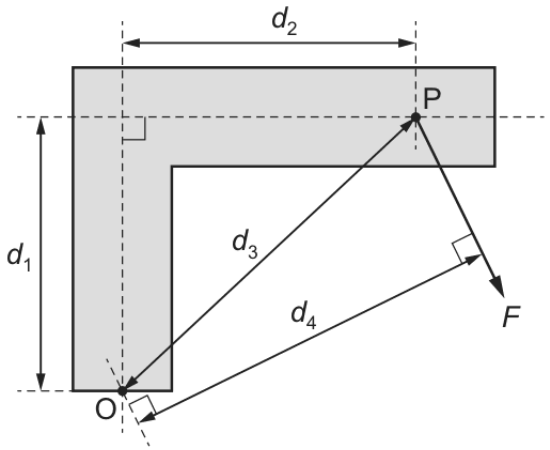
D 50 N acting upwards, 10 cm to the left of the pivot

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 11

11	A	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 3 of 16

6 A force *F* causes a moment about point O on an L-shaped bar. The force *F* acts at point P.



What is the moment of *F* about O?

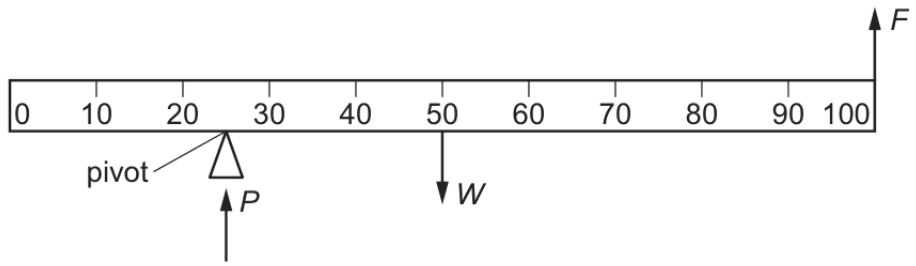
A Fd_1 **B** Fd_2 **C** Fd_3 **D** Fd_4

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 6

6	D	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 4 of 16

7 A uniform metre rule of weight W is pivoted at the 25 cm mark and held horizontal by a force F applied upwards at the 100 cm mark. The rule is supported by a vertical force P acting at the pivot.



What is the magnitude of P?

A $\frac{1}{3}W$ **B** $\frac{1}{2}W$ **C** $\frac{2}{3}W$ **D** $\frac{4}{3}W$

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 7

7	C	**1**

3 A plane, irregular lamina has a mass of 50 g. It hangs from a nail that passes through a small hole H near to the edge of the lamina.

The nail acts as a pivot and the lamina can swing about it. The lamina is held in the position shown in Fig. 3.1, a small distance above a horizontal bench.

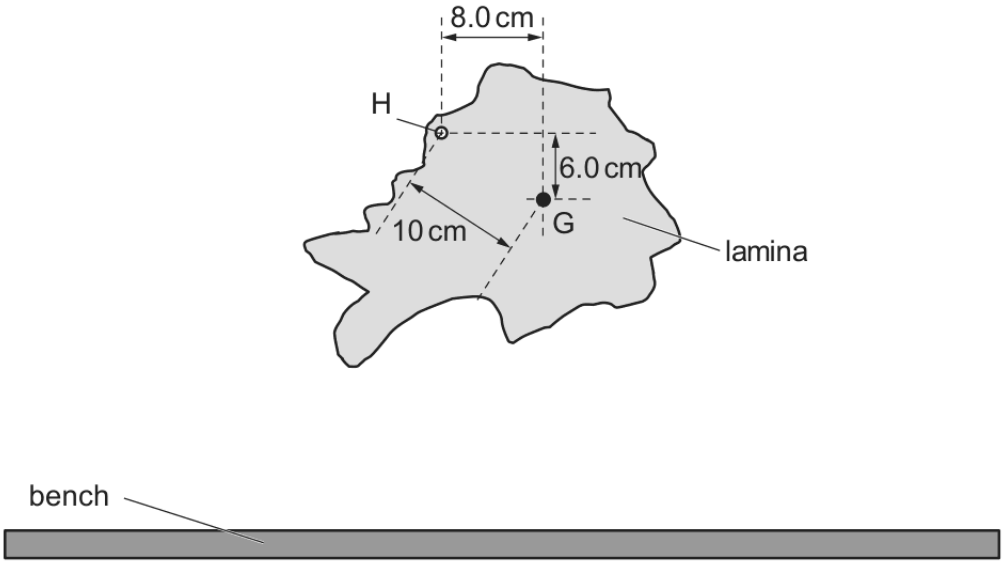


Fig. 3.1 (not to scale)

The centre of gravity of the lamina is at G.

(a) The weight of the lamina is a force that acts downwards.

(i) Explain why the lamina experiences this downward force.

.....

..... [1]

(ii) Calculate the weight of the lamina on Earth.

weight = ... N [2]

(b) The weight produces a moment about the pivot through H.

(i) Using Fig. 3.1, calculate the moment of the weight about the pivot.

moment = ... N m [2]

(ii) The lamina is now released from the position shown in Fig. 3.1.

Describe what happens to the lamina from the time it is released until it stops moving.

.....

.....

.....

..... [3]

[Total: 8]

Question	Answer	Marks
3(a)(i)	it / lamina is in the gravitational field (of Earth)	**B1**
3(a)(ii)	($F =$) mg **or** 0.050×9.8 **or** 50×9.8 **or** 490	**C1**
	0.49 (N)	**A1**
3(b)(i)	($M =$) $F \times x_{\perp r}$ **or** 0.49×8.0 **or** 0.49×0.080	**C1**
	0.039 (N m)	**A1**
3(b)(ii)	it / lamina moves clockwise (about the pivot)	**B1**
	it speeds up **and** then slows down **or** it oscillates (with decreasing amplitude) **or** overshoots	**B1**
	(stops with) G vertically below H	**B1**

QUESTION · 5054_s24_qp_11.pdf · source page 4 of 16

8 Which statements describing the moment of a force about a pivot are correct?

1 The moment of a force is a measure of the turning effect of the force.

2 The moment of a force is equal to the force × perpendicular distance from the pivot.

3 The moment of a force is equal to the force × pressure on the pivot.

A 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

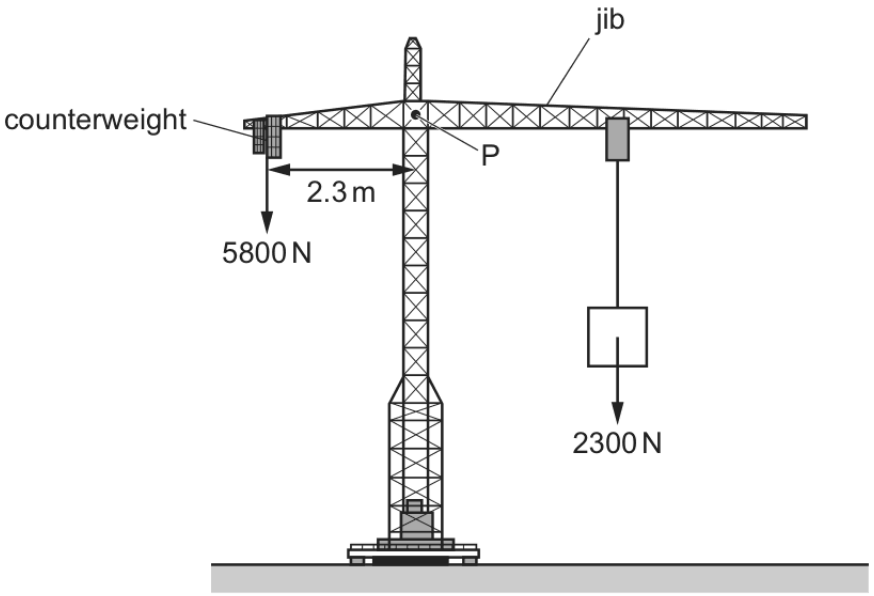
MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 8

8	A	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 6 of 16

12 A crane has a 5800 N counterweight positioned 2.3 m from the tower along a horizontal jib.

The centre of gravity P of the crane jib is marked.



What is the horizontal distance between the 2300 N load and P so that there is no moment about P?

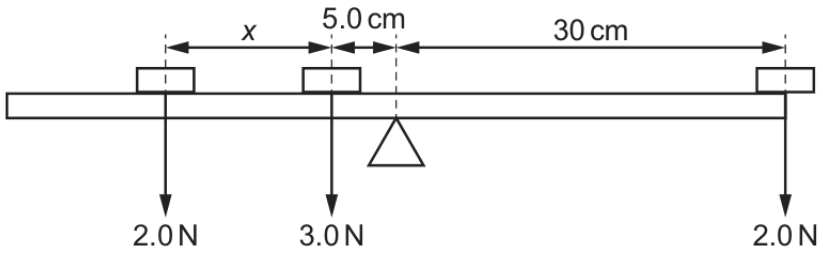
A 0.91 m **B** 3.5 m **C** 5.8 m **D** 8.1 m

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 12

12	C	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 6 of 16

13 A uniform beam is pivoted at its centre and balanced by three weights as shown.



What is distance *x*?

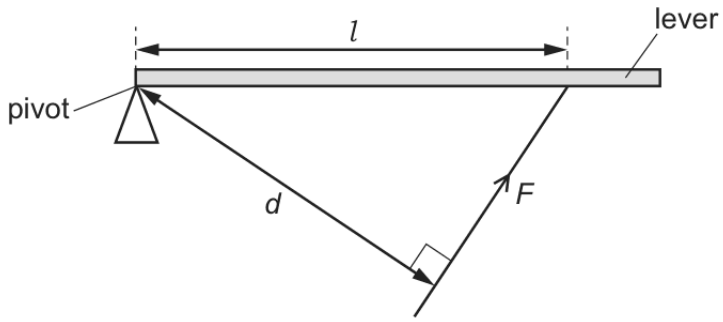
A 17.5 cm **B** 22.5 cm **C** 35.0 cm **D** 37.5 cm

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 13

13	A	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 4 of 16

9 A force *F* is applied to a lever at distance l from the pivot as shown.



Length *d* is the perpendicular distance between the force and the pivot.

Which expression is the moment of force *F* about the pivot?

A $\frac{F}{d}$ **B** $\frac{F}{l}$ **C** $F \times d$ **D** $F \times l$

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 9

9	C	**1**

2 **(a)** State the principle of moments.

.....

.....

..... [2]

(b) An airline passenger wishes to check the weight of a suitcase that he carries as hand luggage onto an aeroplane.

He uses a uniform plank of wood, pivoted at its centre, as shown in Fig. 2.1.

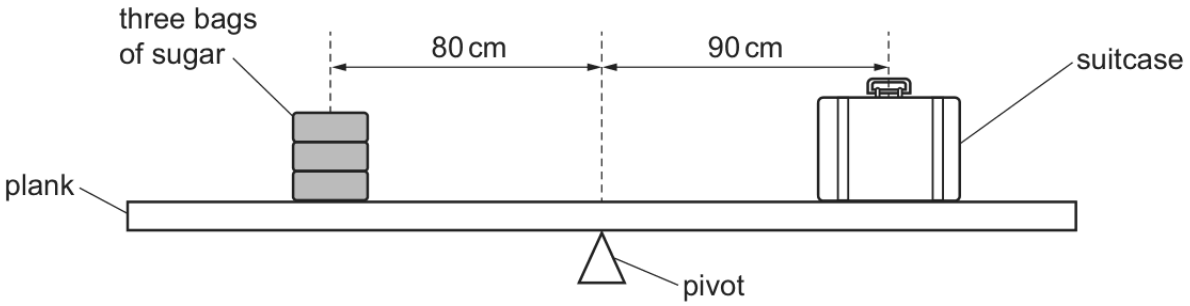


Fig. 2.1

The plank is balanced with the suitcase on one side of the pivot and three bags of sugar, each of mass 2.0 kg, on the other side.

The distances are shown on Fig. 2.1.

(i) Calculate the weight of **one** bag of sugar.

weight = ... N [1]

(ii) The maximum weight of a suitcase that can be carried onto the aeroplane is 67 N.

Determine whether the weight of the suitcase exceeds the maximum allowed.

Show a calculation in your answer.

.....

..... [2]

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(c) A tall bus is tested for stability.

Fig. 2.2 shows the bus on a slope.

The centre of gravity of the bus is marked.

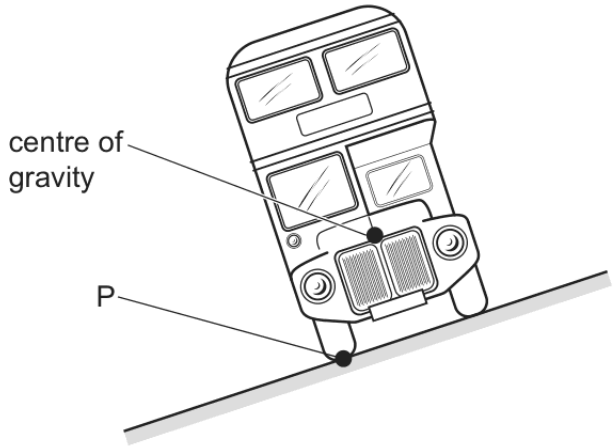

Fig. 2.2

(i) State what is meant by the 'centre of gravity' of an object.

.....

..... [1]

(ii) When the slope is made very steep, the bus falls over by rotating about point P.

Explain why the bus falls over.

.....

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
2(a)	(total) clockwise moments = anticlockwise moments	**M1**
	in equilibrium	**A1**
2(b)(i)	19.6 or 20 N	**B1**
2(b)(ii)	80 × 3 × **(b)(i)** = 90 × weight	**B1**
	52 (N) **and** weight of suitcase does not exceed maximum	**B1**
2(c)(i)	(place) where weight of object (can be taken to) acts	**B1**
2(c)(ii)	centre of gravity falls outside base / to the left of P	**B1**
	weight causes anticlockwise moment / turning effect	**B1**

7 Fig. 7.1 shows a simple d.c. motor. The poles at the ends of the magnets are not labelled.

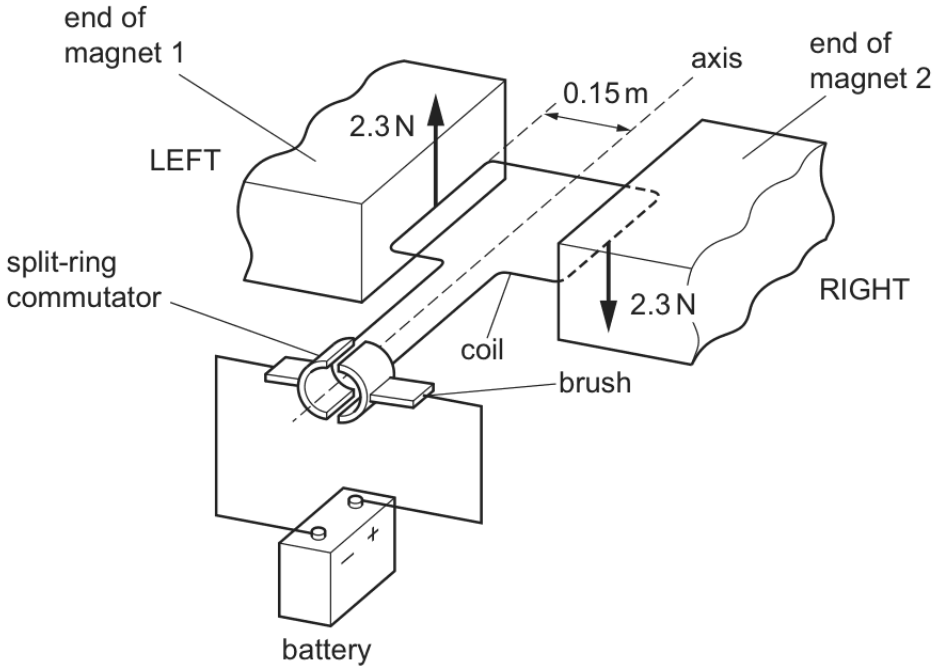

Fig. 7.1

(a) State the purpose of the split-ring commutator and the brushes.

split-ring commutator .....

.....

.....

brushes .....

.....

[3]

(b) There is an upward force shown on the left side of the coil.

There is a magnetic field in the space between the ends of the magnets.

State the direction of this magnetic field.

Explain how you determined this direction.

direction .....

explanation .....

.....

.....

.....

[3]

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(c) State what happens to the rotation of the coil if the magnetic field and the current in the coil are both reversed at the same time.

.....

..... [1]

(d) In Fig. 7.1, the coil is horizontal. The force upward on the left side and the force downward on the right side are each 2.3 N.

The distance between the sides of the coil and the axis is 0.15 m.

(i) Calculate the total moment exerted on the coil by these forces.

Give the unit of your answer.

moment = ...

unit = ...
[3]

(ii) Explain why the turning effect becomes smaller as the coil becomes more vertical.

.....

.....

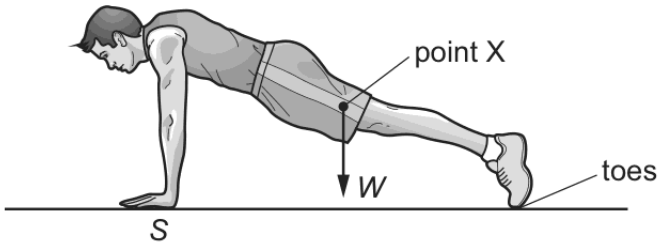
..... [1]

[Total: 11]

Question	Answer	Marks
7(a)	*split ring commutator*: to reverse the current (in the coil)	**B1**
	reverse current / connections every half turn / twice in one turn / every 180° / when coil is vertical	**B1**
	brushes: to connect coil / commutator or (split) rings to battery / (external) circuit **or** to provide current to coil, commutator or (split) rings **or** to avoid wires (to coil) tangling / twisting	**B1**
7(b)	*direction*: (from left) to right / from magnet 1 to magnet 2	**B1**
	explanation: current is from plus to minus / towards front / out of plane / page / outwards / top to bottom of wire / downwards along wire / towards the split ring on the left	**B1**
	left-hand rule mentioned and explained using two of current, field or motion / force	**B1**
7(c)	no change	**B1**
7(d)(i)	(moment =) force × distance in any form	**C1**
	0.69	**A1**
	Nm	**B1**
7(d)(ii)	(perpendicular) distance between (line of action of) forces decreases **or** (sides of) coil enter weaker magnetic field **or** coil leaves magnetic field	**B1**

QUESTION · 5054_w25_qp_11.pdf · source page 4 of 16

5 The diagram shows an athlete doing press-ups on a floor. The athlete is using both hands. The area of contact of each hand with the floor is *S*. The athlete's toes act as a pivot during each press-up.



The athlete's weight *W* acts through point X. The distance between his toes and point X is three quarters of the distance between his toes and his shoulders.

What is the pressure exerted on the palm of each of the athlete's hands?

A $\frac{W}{8S}$ **B** $\frac{3W}{8S}$ **C** $\frac{W}{4S}$ **D** $\frac{3W}{4S}$

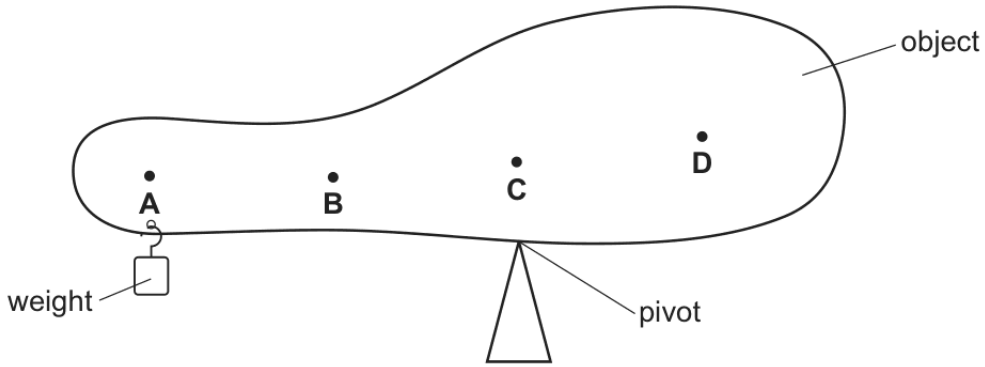
MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 5

5	B	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 7 of 16

12 A student balances a non-uniform object on a pivot. To do this, a weight is suspended near the left-hand end of the object, as shown.

Where is the centre of gravity of the object?



MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 12

12	D	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 6 of 16

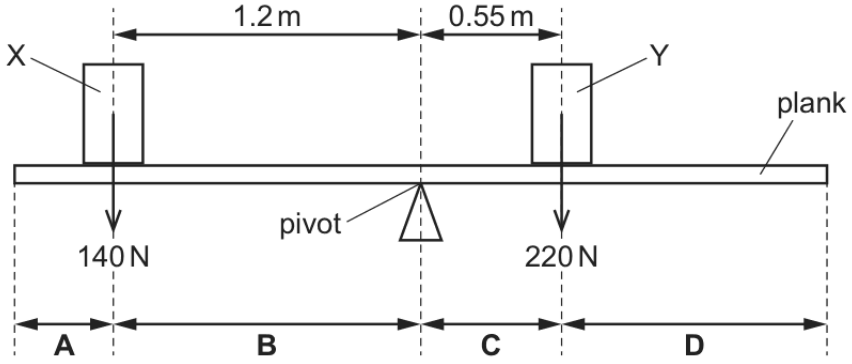
13 A long plank is pivoted at its centre of gravity.

Two objects, X and Y, are then placed on the plank.

X has a weight of 140 N and is 1.2 m to the left of the pivot. Y has a weight of 220 N and is 0.55 m to the right of the pivot.

A third object of weight 110 N is placed on the plank so that the plank is in equilibrium.

On which part of the plank is the object placed?



MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 13

13	C	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 6 of 16

11 Which quantity is a measure of the turning effect of a force?

A the direction of a force

B the moment of a force

C the momentum of a force

D the perpendicular distance of a force

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 11

11	B	**1**

2 A heavy stone is placed on a plank of wood to make a diving board as shown in Figure 2.1.

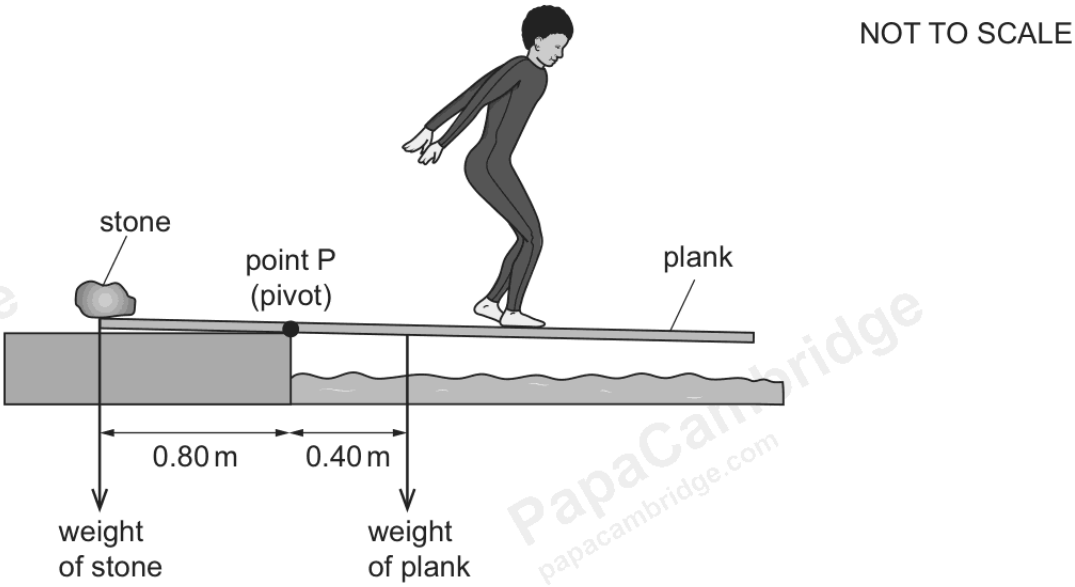


Figure 2.1

The point P is a pivot.

The weight of the stone is 650 N, and the weight of the plank is 200 N.

(a) State the principle of moments.

.....

.....

..... [2]

(b) Calculate the moment of the weight of the plank about P.

moment = ... N m [2]

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(c) A student of mass 40 kg walks to the right along the plank until the plank tips into the water.

Calculate the horizontal distance between point P and the student when the plank tips into the water.

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distance = ... m [3]

[Total: 7]

Question	Answer	Marks	Guidance
2(a)	(total) clockwise moment equals (total) anti–clockwise moment	**M1**	**A** CW = ACW or similar as long as moment is mentioned
	when the object is stationary / not turning / in equilibrium / balanced	**A1**	

Question	Answer	Marks	Guidance
2(b)	(moment =) force × distance in any form algebraic or numerical	**C1**	**ignore** $F_1d_1 = F_2d_2$ algebraic equation **A** moment = F_1d_1 ; **A** any force (200 or 650) multiplied by any distance (0.4, 0.8 or 1.2) even in an $F_1d_1 = F_2d_2$ equation
	80 (Nm)	**A1**	
2(c)	(weight of student =) 40 × 9.8 or 392 (N) seen	**C1**	allow 10 for 9.8
	moment equation seen with $F_1d_1 = F_2d_2$ (+ 80)	**C1**	e.g. 650×0.8 = 40 × 9.8 × d (+ value from **(b)**); e.g. 392d ⩾520 **A** = or > in a moment equation
	1.1(2) m	**A1**	**I** distances greater than 1.12 m on answer line

Chapter 1 Motion, forces and energy

1.5 Forces

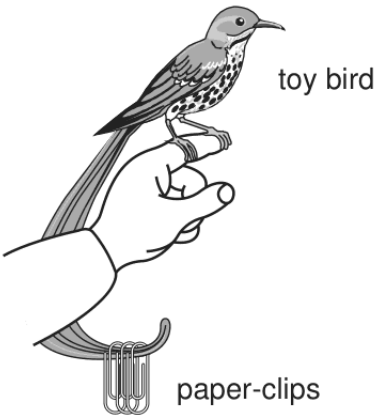
1.5.6 Centre of gravity

55 entries: 41 Paper 1 questions, 14 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_w04_qp_1.pdf · source page 6 of 20

9 A girl uses paper-clips to balance a toy bird on her finger as shown.



What is the effect of the paper-clips?

A They help to raise the centre of mass above her finger.

B They help to raise the centre of mass to her finger.

C They help to lower the centre of mass below her finger.

D They do not affect the centre of mass but increase the weight.

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 9

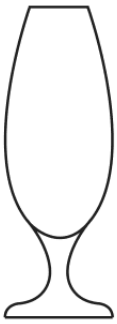
9	**C**

QUESTION · 5054_s05_qp_1.pdf · source page 4 of 20

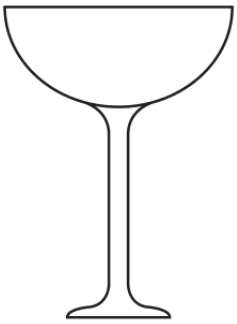
7 The diagrams show the cross-sections of different glasses.

Which one is the least stable when filled with a liquid?

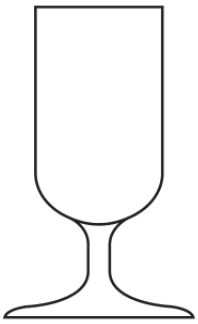
A



B



C



D

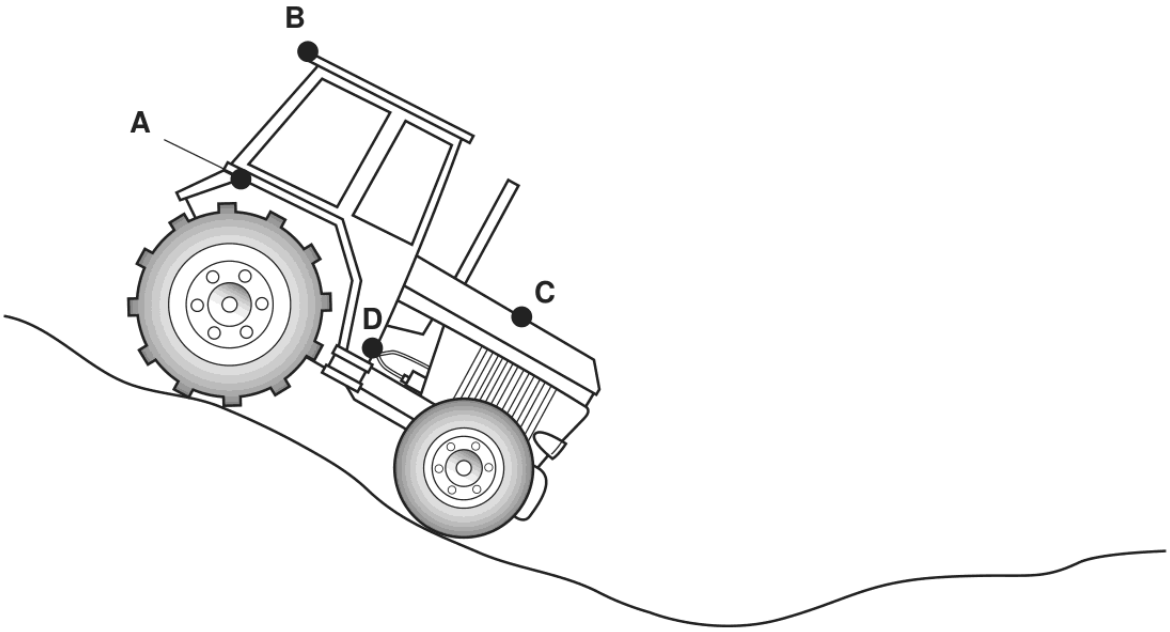
MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 7

7	**B**

QUESTION · 5054_w05_qp_1.pdf · source page 5 of 20

8 A tractor is being used on rough ground.

What is the safest position for its centre of mass?



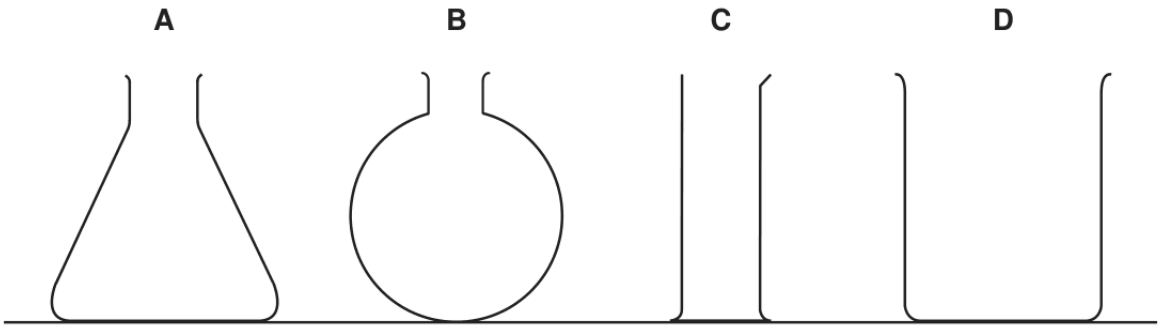
MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 8

8 **D**

QUESTION · 5054_s06_qp_1.pdf · source page 4 of 20

9 Some containers are made from thin glass.

Which empty container is the **most** stable?



MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 9

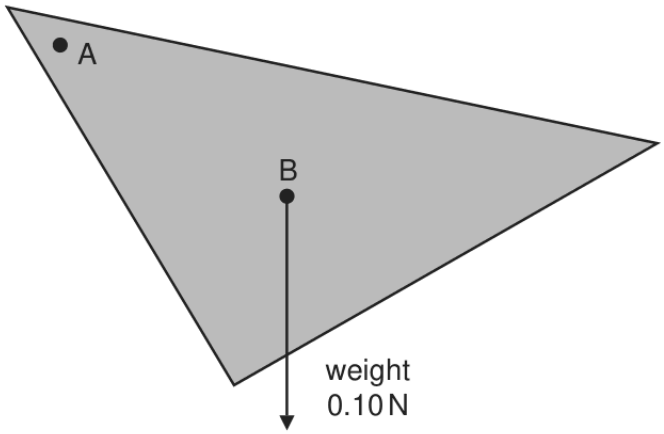
9 **A**

Section B

Answer **two** questions from this section.

Use the separate answer sheets available from the Supervisor.

9 Fig. 9.1 shows a thin sheet of metal suspended from a hole in one corner at A. The weight of the metal is 0.10 N and the centre of mass is at B. The diagram is drawn full scale.



full scale

Fig. 9.1

(a) Describe in detail how you would experimentally determine the position of the centre of mass of the sheet of metal. [6]

(b) The sheet turns because of the moment of the weight about point A.

(i) Define what is meant by the *moment of a force*. [2]

(ii) Using a distance measured on Fig. 9.1, calculate the moment of the weight about point A. State clearly which distance you measured and give the unit of your final answer. [3]

(c) Fig. 9.2 and Fig. 9.3 show a thick piece of wood with one corner on a table.

Fig. 9.4 shows the same piece of wood balanced on the table. B is the centre of mass.

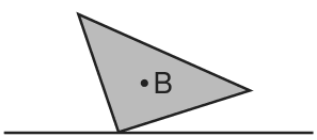


Fig. 9.2

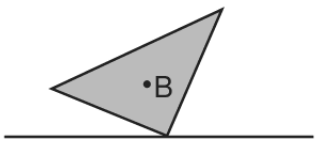


Fig. 9.3

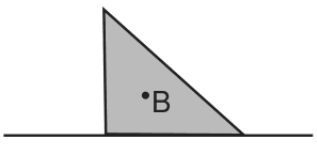


Fig. 9.4

(i) Explain why in Fig. 9.2 the piece of wood falls to the right and in Fig. 9.3 it falls to the left. [2]

(ii) Explain why the piece of wood in Fig. 9.4 does not fall over. [1]

(iii) Suggest how the thickness of the wood in Fig. 9.4 affects its stability. [1]

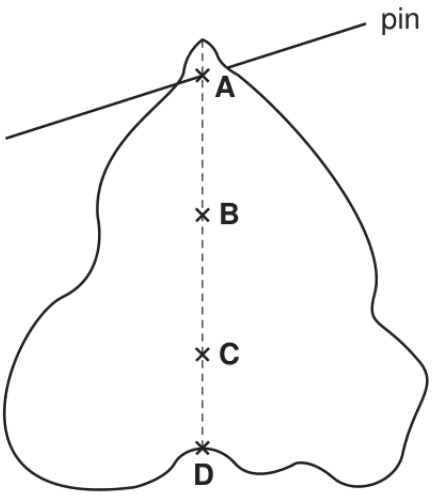
Section B

9	**(a)**	set wood swinging/let metal pivot **or** fall **OR**		balance on sort of edge	
		allow to come to rest		clearly a sharp edge	
		use of plumb line **from hole**		mark line of edge	
		mark line along plumb line (on metal)		repeat in new position	
		hang from another hole		intersection is centre of mass	
		line intersection is centre of mass		repeat for 3rd position	
		hang from 3rd hole			
			OR	balance on point	
				sharp (compass) point	
				move till balanced	
		ANY 6 consistent lines max.		point is centre of mass	B6
	(b) **(i)**	force × distance			**M**1
		perpendicular (accept symbol) distance **or** shortest distance to line of action of force			A1
	(ii)	correct perpendicular distance (2.9 – 3.1 cm)			B1
		worked out value of: 0.1 × distance reading			B1
		Ncm (or Nm if conversion of distance to m clear)			B1
	(c) **(i)**	moment **or** turning effect of weight			C1
		anticlockwise and clockwise moment **or** weight to right **and** left of corner			A1
	(ii)	moments balance/cancel **or** weight inside base			B1
	(iii)	thicker more stable/thinner less stable			B1 15

QUESTION · 5054_w08_qp_1.pdf · source page 5 of 20

8 A piece of uniform card is suspended freely from a horizontal pin.

At which of the points shown is its centre of gravity?



MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 8

8 **C**

For Examiner's Use

2 Fig. 2.1 shows a wooden walking-stick that has a metal head and a rubber foot. It balances on a pencil placed 0.50 m from its rubber foot.

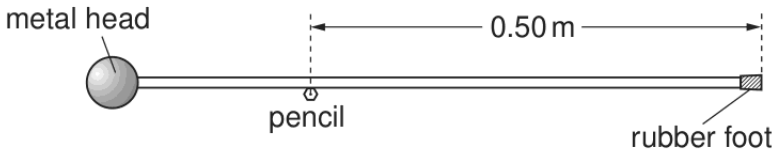


Fig. 2.1

(a) State the distance between the centre of mass of the walking-stick and the end of the rubber foot.

distance = ... [1]

(b) The pencil is moved along the walking-stick towards its rubber foot. State and explain the motion of the walking-stick.

.....

.....

.....

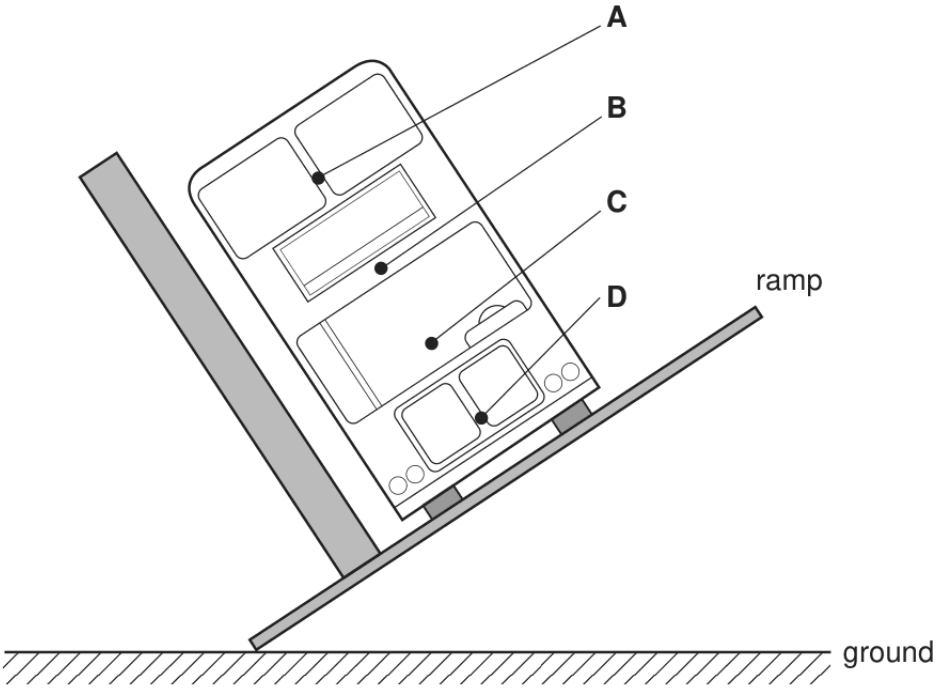
..... [3]

2	**(a)**	0.5(0) m	B1	
	(b)	rotates/tilts/unbalanced/one side down/one side up	C1	
		rotates anticlockwise/down on left **or** head down **or** foot up	A1	
		(net) anticlockwise moment **or** moment on left > moment on right **or** weight/CM on left of pivot	B1	[4]

QUESTION · 5054_w09_qp_1.pdf · source page 4 of 20

8 The stability of a bus is tested by tilting it on a ramp. The diagram shows a bus that is just about to topple over.

Where is the centre of mass of the bus?



MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 8

8 **C**

For Examiner's Use

7 Fig. 7.1 shows a rectangular concrete slab of weight 18000 N. It rests on a brick wall and is the roof of a bus shelter. The concrete slab is 3.0 m wide.

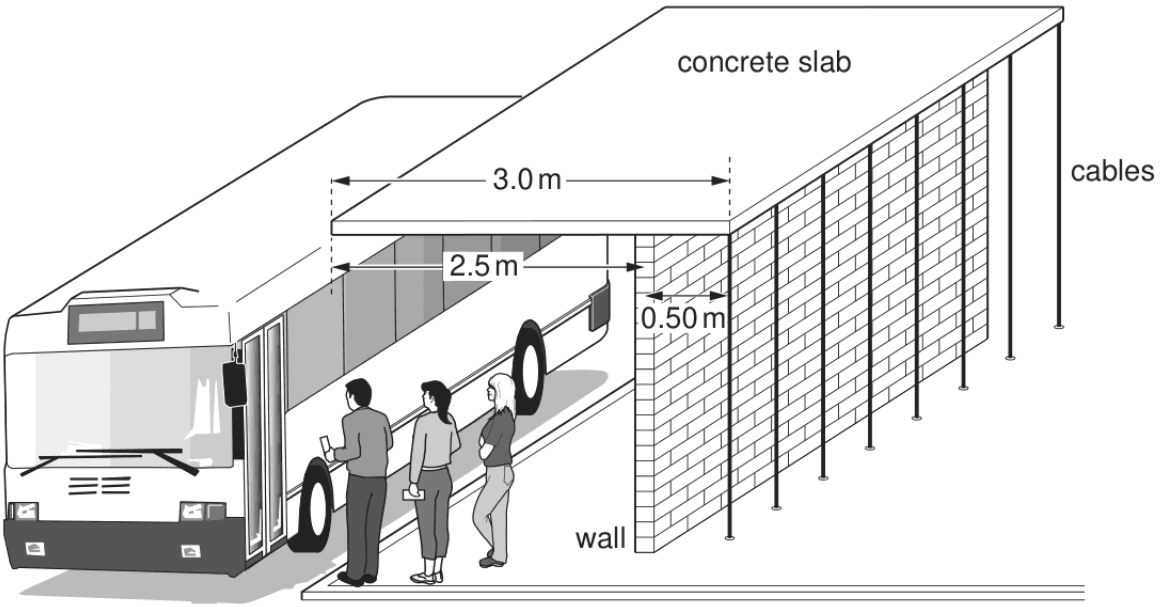


Fig. 7.1

The wall is 2.5 m from the front of the concrete slab and 0.50 m from the back. The cables behind the shelter pull downwards and stop the slab toppling forwards.

(a) The concrete slab is of uniform thickness and density. Determine the perpendicular distance between the wall and the centre of mass of the slab.

distance = ...[1]

(b) **(i)** State the principle of moments.

.....

.....

.....[2]

(ii) Calculate the total downward force exerted by the cables on the slab.

force = ...[2]

7	**(a)**	1.(0) m	B1	
	(b)	**(i)** (for an object in) equilibrium/balance	B1	
		$W_1x = W_2y$ (clear) **or** anticlockwise moment/torque/turning force = clockwise moment/torque/turning force	B1	
		(ii) 18 000 × 1. 0 = T × 0.5	C1	
		36 000 N	A1	[5]

QUESTION · 5054_w10_qp_11.pdf · source page 6 of 20

9 What affects the stability of an object?

A only its base area

B only its base area and the location of its centre of mass

C only its weight and its base area

D only the location of its centre of mass

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 9

9	**B**

QUESTION · 5054_w10_qp_12.pdf · source page 5 of 16

8 What affects the stability of an object?

A only its base area

B only its base area and the location of its centre of mass

C only its weight and its base area

D only the location of its centre of mass

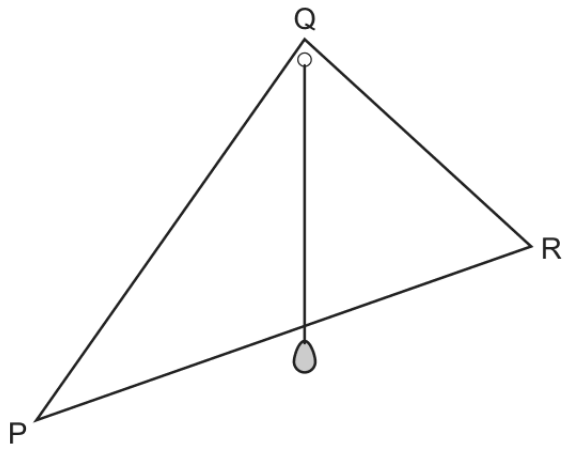
MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 8

8	**B**

QUESTION · 5054_s11_qp_11.pdf · source page 5 of 16

9 A student finds the centre of mass of a triangular lamina PQR.

He drills a small hole at Q. He suspends the lamina from a pin through the hole at Q so that the lamina swings freely. He then hangs a plumb-line from the pin at Q, as shown. He marks the position of the plumb-line on the lamina.



To determine the location of the centre of mass, the student then repeats the experiment but with one change.

What is the change?

A He suspends the lamina from the hole at Q, with R on the left and P on the right.

B He suspends the lamina from a pin through a hole at R.

C He uses a heavier weight on the plumb-line.

D He uses a longer plumb-line.

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 9

9 **B**

QUESTION · 5054_s11_qp_12.pdf · source page 5 of 16

9 A student finds the centre of mass of a triangular lamina PQR.

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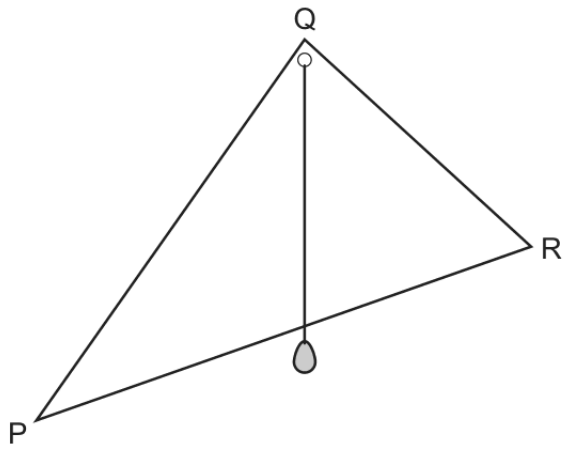

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What is the change?

A He suspends the lamina from the hole at Q, with R on the left and P on the right.

B He suspends the lamina from a pin through a hole at R.

C He uses a heavier weight on the plumb-line.

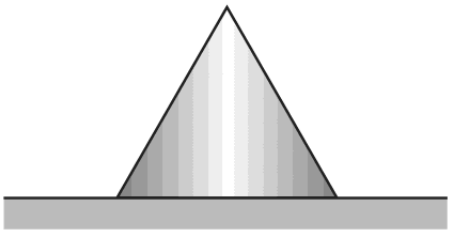
D He uses a longer plumb-line.

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 9

9	**B**

QUESTION · 5054_w11_qp_11.pdf · source page 5 of 20

9 A metal cone with a circular base is placed on a flat surface.



The stability of the cone depends on

A its weight only.

B the diameter of its base and the position of its centre of mass.

C the diameter of its base only.

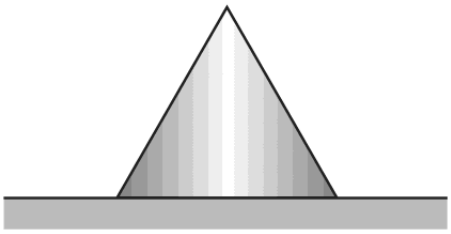
D the position of its centre of mass only.

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 9

9	**B**

QUESTION · 5054_w11_qp_12.pdf · source page 6 of 20

12 A metal cone with a circular base is placed on a flat surface.



The stability of the cone depends on

A its weight only.

B the diameter of its base and the position of its centre of mass.

C the diameter of its base only.

D the position of its centre of mass only.

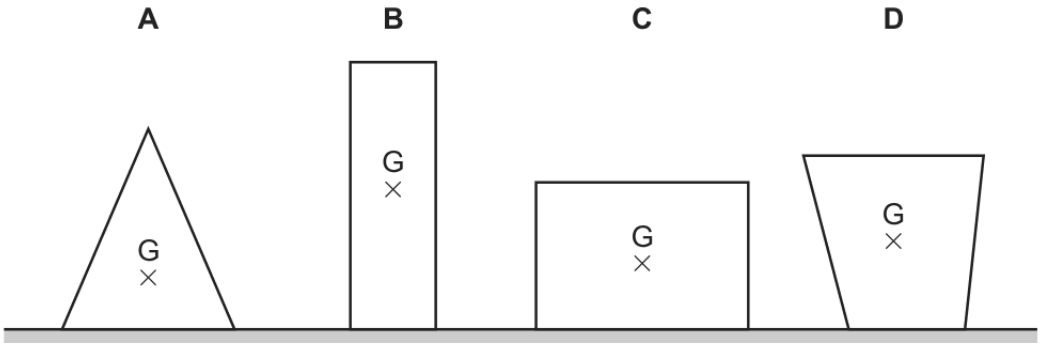
MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 12

12	**B**

QUESTION · 5054_s12_qp_11.pdf · source page 4 of 16

8 Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

Which object is the least stable?



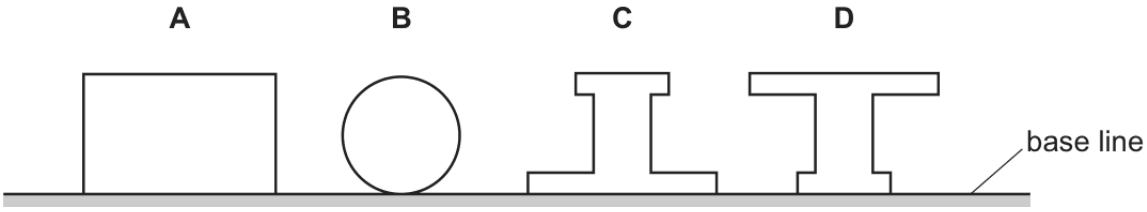
MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 8

8 **B**

QUESTION · 5054_s12_qp_12.pdf · source page 4 of 16

8 The diagram shows four shapes, cut from the same piece of card.

Which shape has its centre of mass nearest to the base line?



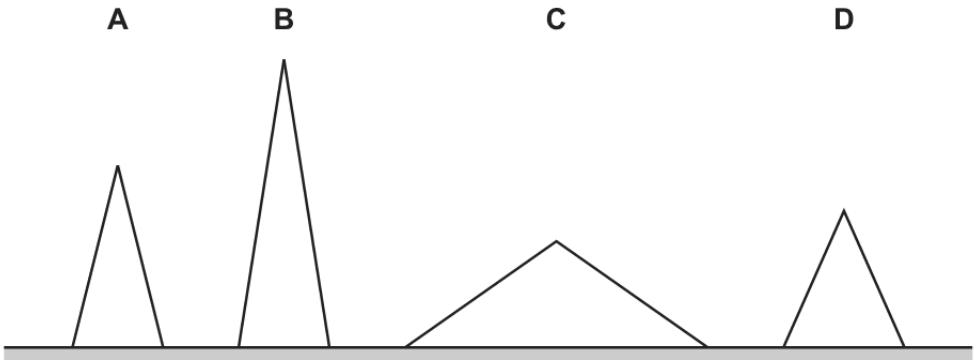
MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 8

8	**C**

QUESTION · 5054_w12_qp_12.pdf · source page 4 of 16

8 Four solid uniform cones have equal weight. They are placed on a bench as shown in the scale diagram.

Which cone is the most stable?

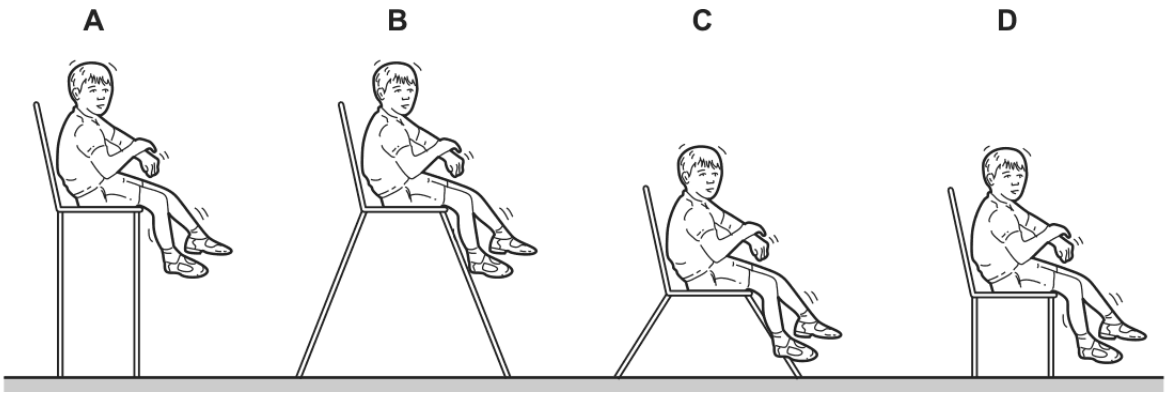


MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 8

8 **C**

QUESTION · 5054_s13_qp_11.pdf · source page 5 of 20

7 Which chair is the **least** stable if the child moves?

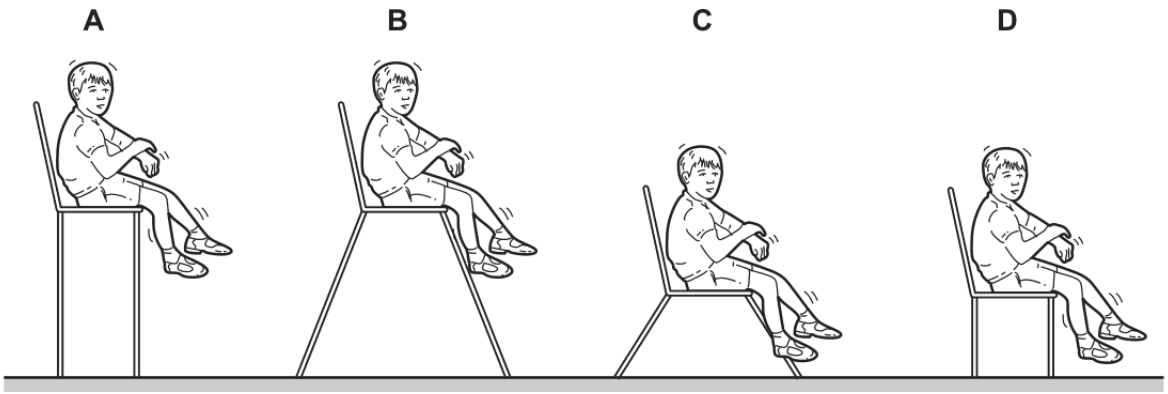


MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 7

7	**A**

QUESTION · 5054_s13_qp_12.pdf · source page 5 of 16

9 Which chair is the **least** stable if the child moves?



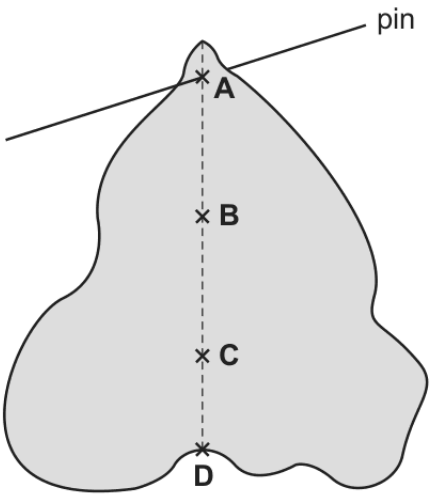
MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 9

9	**A**

QUESTION · 5054_w13_qp_11.pdf · source page 4 of 20

7 A piece of uniform card is suspended freely from a horizontal pin.

Which point is its centre of mass?



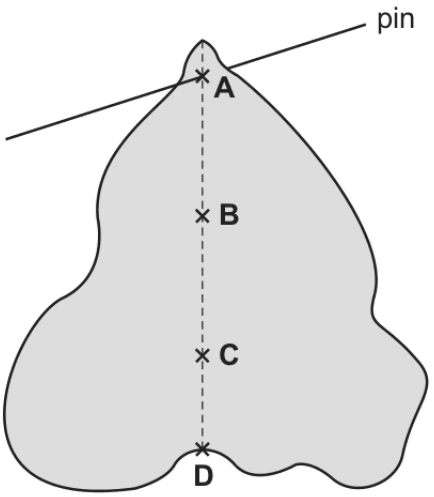
MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 7

7 **C**

QUESTION · 5054_w13_qp_12.pdf · source page 5 of 20

8 A piece of uniform card is suspended freely from a horizontal pin.

Which point is its centre of mass?



MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 8

8 **C**

QUESTION · 5054_s14_qp_11.pdf · source page 5 of 16

10 A car is designed to be stable.

To achieve good stability, where is the centre of mass of the car?

A above the front wheels

B above the rear wheels

C as high in the car as possible

D as low in the car as possible

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 10

10	**D**

QUESTION · 5054_s14_qp_12.pdf · source page 4 of 16

10 A man uses clay to make a pot. He wants the pot to be as stable as possible when placed on a flat surface.

Which two features of the pot must the man consider?

A the area of the base and the height of the centre of gravity

B the density of the clay and the area of the base

C the density of the clay and the height of the centre of gravity

D the weight and the height of the centre of gravity

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 10

10	**A**

QUESTION · 5054_w14_qp_11.pdf · source page 6 of 20

7 What affects the stability of an object?

A only its base area and the location of its centre of mass

B only its weight and its base area

C only the location of its centre of mass

D only its weight

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 7

7	**A**

QUESTION · 5054_w14_qp_12.pdf · source page 3 of 16

6 What affects the stability of an object?

A only its base area and the location of its centre of mass

B only its weight and its base area

C only the location of its centre of mass

D only its weight

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 6

6	**A**

QUESTION · 5054_s15_qp_11.pdf · source page 5 of 16

10 When a car turns a corner at speed, it risks toppling over. Two factors affecting the stability of a car are the height of its centre of mass and the distance between its front wheels.

Which factors make the car most stable?

	centre of mass	distance between front wheels
A	high	small
B	high	large
C	low	small
D	low	large

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 10

10 **D**

4 Fig. 4.1 shows a screwdriver of mass 64 g resting in equilibrium on a pivot.

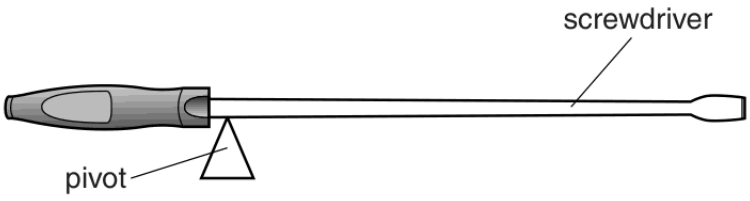


Fig. 4.1

(a) On Fig. 4.1, mark and label with a C, the centre of mass of the screwdriver. [1]

(b) The gravitational field strength is 10 N/kg.

(i) Calculate the weight of the screwdriver.

weight = ... [1]

(ii) On Fig. 4.1, draw an arrow labelled *W* to represent the weight of the screwdriver. [1]

(c) State two conditions that apply when an object is in equilibrium.

1. .....

.....

.....

2. .....

.....

.....

[2]

4 **(a)** (point) C immediately above tip of pivot (and in middle(vertically) of screwdriver (±1 mm)) B1

(b) **(i)** 0.64 N B1

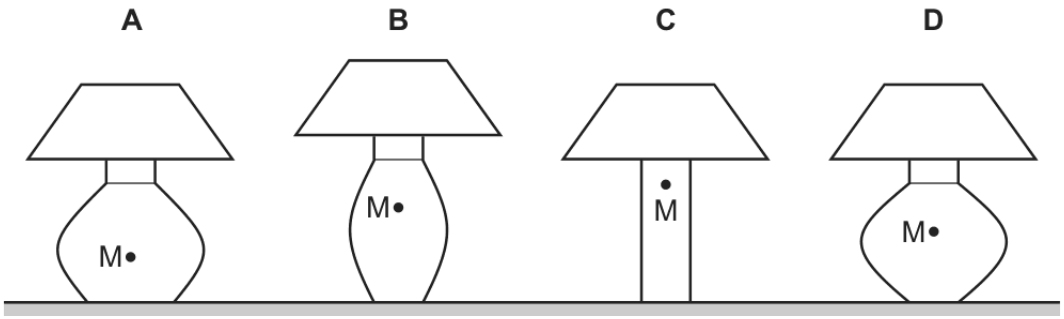
(ii) arrow *W* vertically downwards through candidate's C or pivot B1

(c) no resultant force **or** upward force = downward force **or** force left = force right B1
no resultant moment (of force) **or** clockwise moment = anticlockwise moment B1 [5]

QUESTION · 5054_s16_qp_11.pdf · source page 5 of 16

10 Four table lamps are shown along with the position M of the centre of mass in each case.

Which lamp is the most stable?



MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 10

10	**A**

2 Fig. 2.1 shows a student sitting on a chair. Fig. 2.2 shows the same student with his chair tilted backwards slightly.

Fig. 2.1

Fig. 2.2

(a) State and explain how the pressure of the chair on the floor differs in the two positions.

.....

.....

.....

.....[2]

(b) The chair and student fall over if the chair is tilted backwards more than in Fig. 2.2.

Explain why.

.....

.....

.....

.....

.....[2]

Question	Answer	Marks
2(a)	greater pressure in Fig. 2.2 **and** smaller area (in contact) in Fig. 2.2 **or** more force through the back legs	**B1**
	pressure is inversely proportional to area **or** pressure is force / area	**B1**
2(b)	Any 2 of • centre of mass (on or) outside base / chair / leg • weight acts on left of / outside chair leg	**B1**
	• (anticlockwise) moment (created)	**B1**

3 A solid, rectangular concrete block is lying horizontally on flat ground with one of its largest sides in contact with the ground. Fig. 3.1 shows the dimensions of the block.

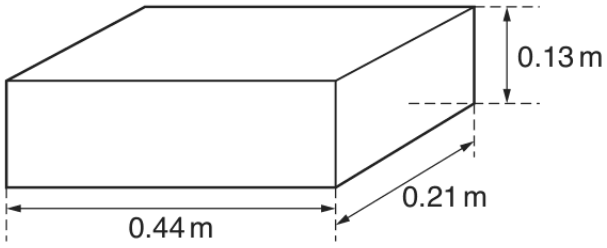


Fig. 3.1

The weight of the block is 240 N.

(a) Calculate the pressure on the ground caused by the block.

pressure = ...[2]

(b) State why the total pressure on the ground underneath the block is larger than the value obtained in **(a)**.

.....

.....[1]

(c) The block is of uniform density.

(i) State the height of the centre of mass of the block above the ground.

height = ...[1]

(ii) The block is rotated about its lower, left-hand edge so that it comes to rest on one of its smallest sides.

This is shown in Fig. 3.2.

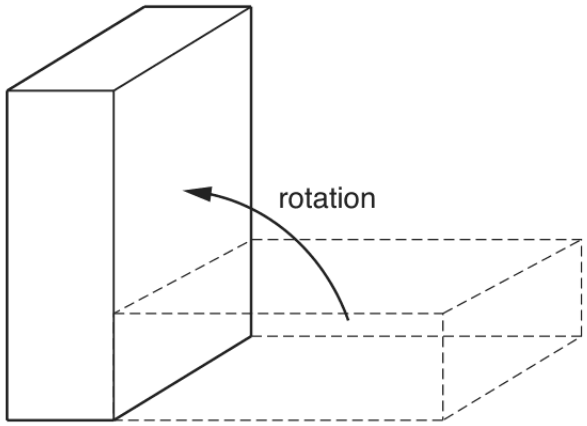


Fig. 3.2

Explain why work is done as the block is rotated.

.....

.....

.....

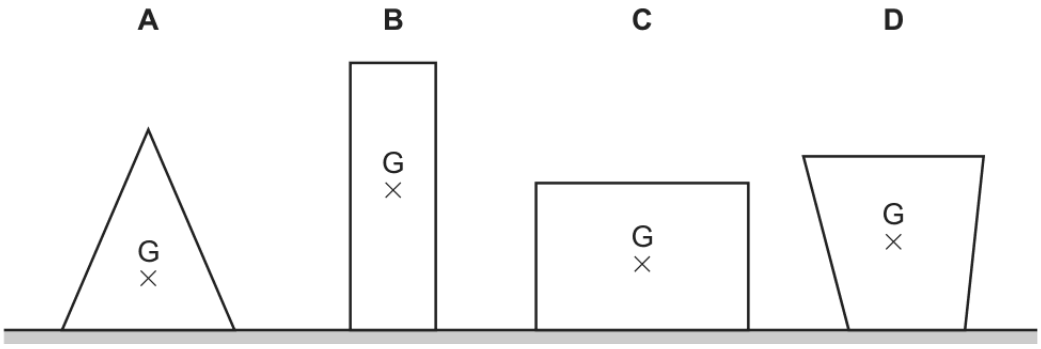
.....[2]

Question	Answer	Marks
3(a)	(P =) F / A **or** 240 / (0.44 × 0.21) **or** 240 / 0.092(4)	**C1**
	2.6×10^3 Pa	**A1**
3(b)	atmospheric pressure (is acting on the block / ground)	**B1**
3(c)(i)	0.065 m **or** 6.5 cm **or** 65 mm	**B1**
3(c)(ii)	**two** separate approaches **I** / **II**	
	I centre of mass rises	**B1**
	gravitational potential energy gained	**B1**
	or **II** block has weight	**B1**
	force **moves or** something moves in direction of force **or** force makes block move / rotate	**B1**

QUESTION · 5054_s18_qp_11.pdf · source page 5 of 20

10 Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

Which object is the least stable?



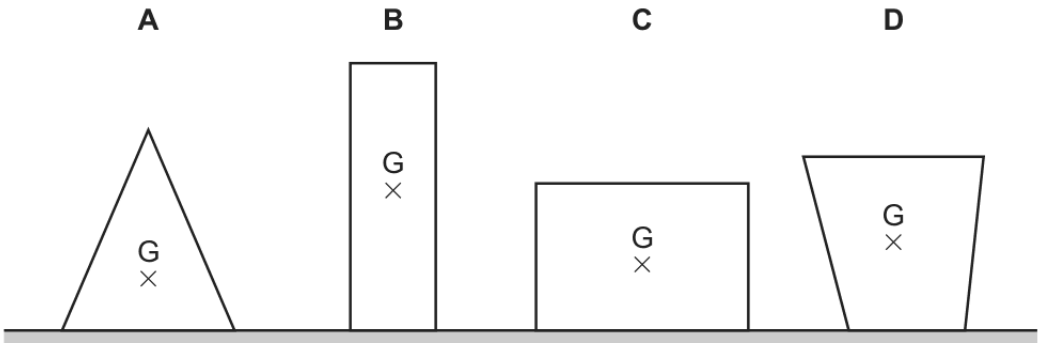
MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 10

10	B	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 6 of 16

13 Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

Which object is the least stable?



MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 13

13	B	**1**

2 A sports car is designed to be very stable when turning a corner at high speed.

Fig. 2.1 shows the position of the centre of mass of the car.

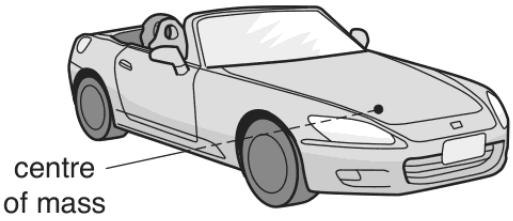


Fig. 2.1

(a) State what is meant by *centre of mass*.

.....

..... [1]

(b) State two features of the design that make the car in Fig. 2.1 stable.

1. .....

2. .....
[2]

(c) Fig. 2.2 shows an irregularly shaped piece of card.

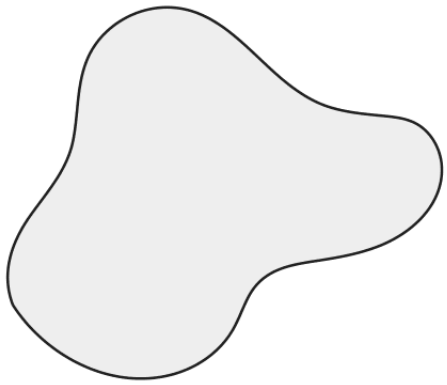


Fig. 2.2

Describe how to determine the position of the centre of mass of the piece of card.

You may draw on Fig. 2.2 or in the blank space.

.....

.....

.....

.....

.....

.....

.....

..... [3]

Question	Answer	Marks
2(a)	(place where entire) mass (seems) to be located	**B1**
2(b)	wide base / base area large	**B1**
	low centre of mass	**B1**
2(c)	suspend lamina next to plumb line / mass on string	**B1**
	mark vertical line on lamina / line along string / plumb line	**B1**
	repeat from another point **and** centre of mass at intersection of lines	**B1**

QUESTION · 5054_s19_qp_11.pdf · source page 4 of 20

10 Which statement about centre of mass is correct?

A Objects with a centre of mass at the same height are less stable when the base is larger.

B Objects with a centre of mass at the same height are more stable when the base is larger.

C Objects with higher centres of mass and smaller bases are more stable.

D Objects with identical bases are more stable when the centre of mass is higher.

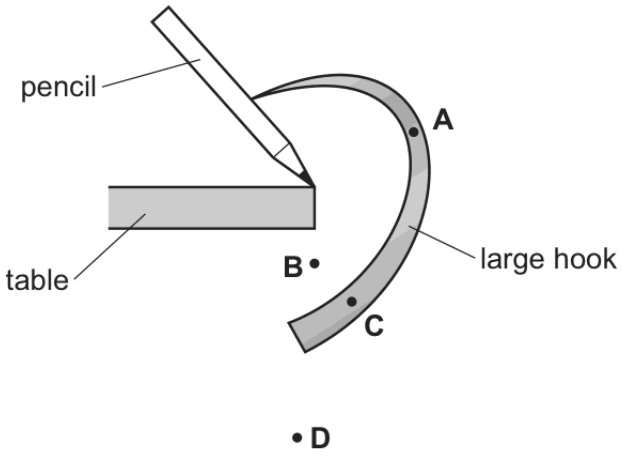
MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 10

10	B	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 5 of 16

11 A large hook sticks into a pencil. The hook and pencil are balanced on the edge of a table, as shown.

Where is the centre of mass of the hook and pencil?

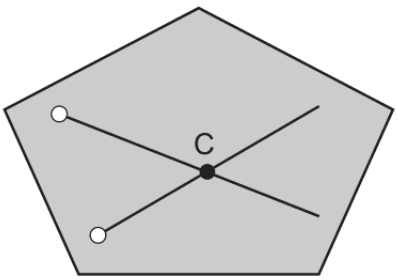


MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 11

11	B	**1**

2 A student performs an experiment to mark the centre of mass C on a thin piece of card. There are two holes in the card.

Fig. 2.1 shows the card and two lines that the student draws on the card.



key
○ = hole
C = centre of mass

Fig. 2.1

(a) Describe a method used to draw these two lines in their correct positions on the card.

Make clear what extra apparatus is needed. You may draw a diagram, if you wish.

.....

.....

.....

.....

.....

.....

..... [3]

(b) The student holds the card loosely between her fingers. The card is vertical, resting with its lower edge on a bench as shown in Fig. 2.2.

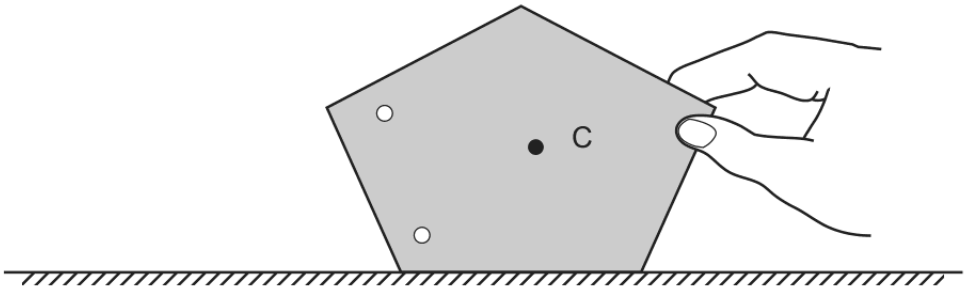


Fig. 2.2

The card is tilted slightly, as shown in Fig. 2.3, and then released.

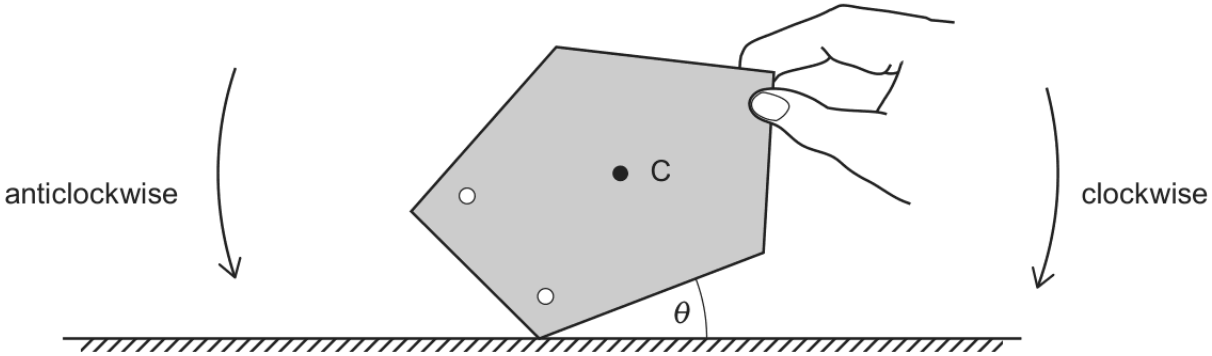


Fig. 2.3

When angle θ is small, the card falls clockwise, back to the position shown in Fig. 2.2.

(i) Explain why the card falls anticlockwise when θ is large.

.....

.....

.....

..... [2]

(ii) State one change to the card that makes it more stable.

.....

..... [1]

[Total: 6]

Question	Answer	Marks
2(a)	Hang card from pin / nail though hole **and** wait until stationary	**B1**
	using plumb line from hole mark line (on card)	**B1**
	repeat with another hole	**B1**
2(b)(i)	centre of mass / C falls outside base	**B1**
	weight creates moment / turning effect	**B1**
2(b)(ii)	lower centre of gravity **or** wider / longer base	**C1**

QUESTION · 5054_s20_qp_12.pdf · source page 4 of 16

11 Which single item can be used to find the centre of mass of a plane lamina of irregular shape?

A a balance

B a measuring tape

C a micrometer

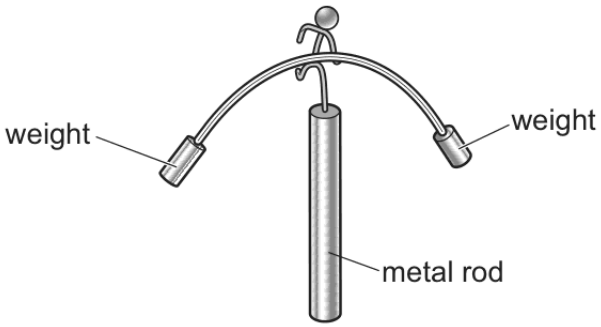
D a vertical pin

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 11

11	D	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 5 of 16

12 A student makes a model toy which balances on the end of a metal rod.



Why are the weights needed?

A They increase the pressure on the rod.

B They increase the weight of the model.

C They lower the centre of mass below the top of the rod.

D They raise the centre of mass above the top of the rod.

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 12

12	C	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 A glass beaker has a mass of 50 g. A liquid of density 1.8 g / cm^3 is poured into the beaker until it reaches the 200 cm^3 mark.

(a) Calculate the total mass of the beaker and its contents.

mass = ... [3]

(b) The centre of mass of a metre rule is at the 50 cm mark.

(i) State what is meant by *centre of mass*.

.....

..... [1]

(ii) The metre rule is placed on a pivot. The tip of the pivot is under the 80 cm mark on the rule.

The beaker with its contents is then placed at different positions along the rule until the rule is balanced.

Fig. 1.1 shows the arrangement with the rule balanced.

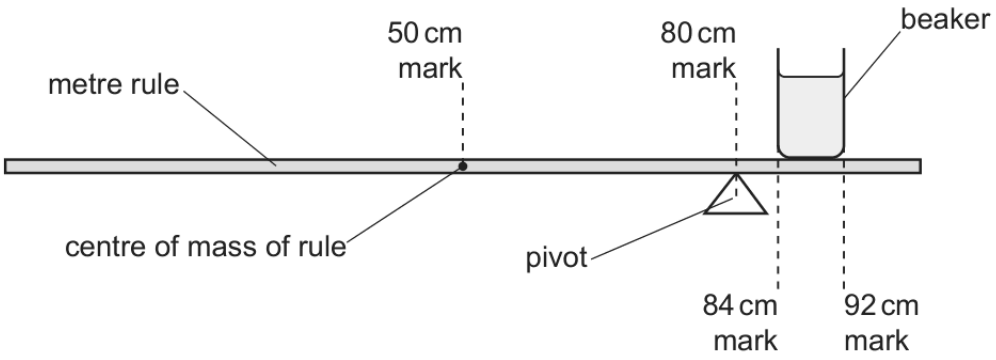


Fig. 1.1

One side of the beaker is at the 84 cm mark and the other side is at the 92 cm mark.

Calculate the mass of the rule.

mass = ... [3]

[Total: 7]

Question	Answer	Marks
1(a)	(m =) ρV **or** 1.8×200 **or** 360	**C1**
	$(1.8 \times 200) + 50$ **or** $360 + 50$ **or** 360 **or** candidate's mass + 50	**C1**
	410 g **or** 0.41 kg	**A1**
1(b)(i)	(place / region) where the mass appears to be concentrated **or** where the object balances	**B1**
1(b)(ii)	88 (cm) seen **or** $(84 + 92) / 2$ seen **or** 30 (cm) seen **or** 8(.0 cm) seen **or** 164 (g)	**C1**
	$m_1gx_1 = m_2gx_2$ **or** $F_1x_1 = F_2x_2$ **or** $W_1x_1 = W_2x_2$ **or** (m_1 =) m_2x_2 / x_1 **or** $410 \times (88 - 80) / 30$ **or** 164 (g)	**C1**
	110 g **or** 0.11 kg	**A1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 Fig. 8.1 shows a stationary horse and its rider, about to jump over two fences.

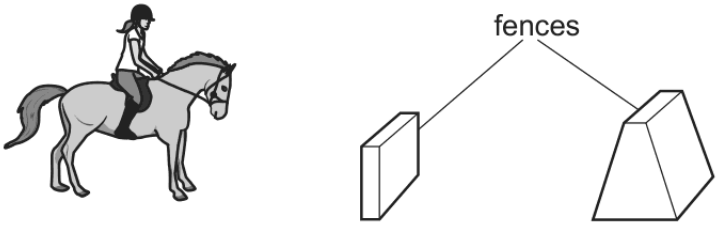


Fig. 8.1

(a) Fig. 8.2 shows a side view of the horse.

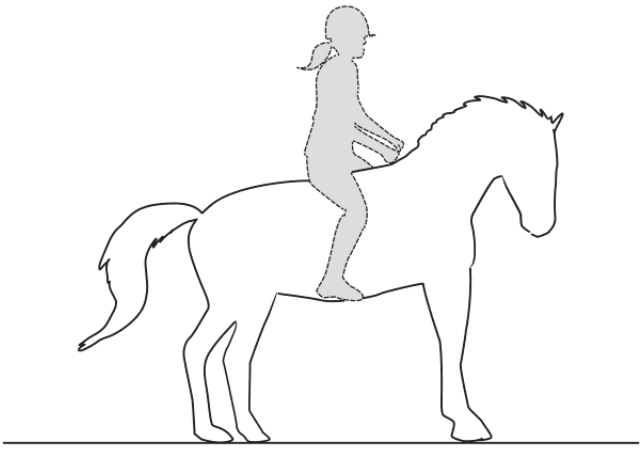


Fig. 8.2

(i) On Fig. 8.2, draw and label the forces acting on the horse.

Include the force that the rider exerts on the horse. Label this force *F*. [3]

(ii) Explain how Newton's third law applies to force *F*.

.....

.....

..... [2]

(b) Fig. 8.3 shows a side view of the two fences. They both have the same height and a uniform density.

Fig. 8.3

(i) On each fence in Fig. 8.3, mark with a cross the centre of mass. [2]

(ii) Explain why a wider base makes the fence more stable.

.....

.....

..... [2]

(c) The total mass of the horse and rider is 520 kg.

(i) As they approach a fence, the horse and rider have a total kinetic energy of 4000 J.

Calculate their speed.

speed = ... [3]

(ii) The centre of mass of the horse and rider is 1.4 m above the ground.

The maximum potential energy gained by the horse and rider as they jump over the fence is 3000 J.

Calculate the maximum height above the ground of the centre of mass during the jump.

The gravitational field strength g = 10 N / kg.

height = ... [3]

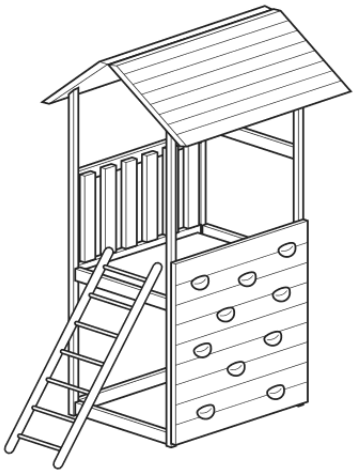
[Total: 15]

Question	Answer	Marks
8(a)(i)	weight of horse in mid-section of horse **and** labelled	**B1**
	force upwards on at least two legs **and** labelled	**B1**
	force downwards on saddle area labelled *F*	**B1**
8(a)(ii)	forces equal **and** opposite	**B1**
	acts on rider (upwards from horse)	**B1**
8(b)(i)	cross at centre of left-hand fence	**B1**
	cross at centre of right-hand fence **and** lower than for right-hand fence	**B1**

Question	Answer	Marks
8(b)(ii)	does not topple / falls back as fence tilts / rotates / turns / changes angle **or** turns to larger angle / needs more force before toppling **or** centre of mass is lower	**B1**
	(low angle of tilt) centre of mass still inside / above base **or** weight turns fence back if tilted	**B1**
8(c)(i)	K.E. = $\frac{1}{2}mv^2$ in any form algebraic or numerical	**C1**
	$v^2 = 2 \times 4000 / 520$ **or** $v = \sqrt{\frac{2\times4000}{520}}$	**C1**
	3.9 m / s	**A1**
8(c)(ii)	(h =) P.E. / mg in any form algebraic or numerical	**C1**
	3000 / (10 × 520) **or** 0.58 m	**C1**
	2.0 m	**A1**

QUESTION · 5054_s21_qp_12.pdf · source page 5 of 16

9 The diagram shows a children's wooden play tower.



Which change to the tower makes it more stable?

A making the tower narrower

B making the tower taller

C lowering the centre of mass

D raising the centre of mass

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 9

9	C	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a wooden bench of weight 2000 N.

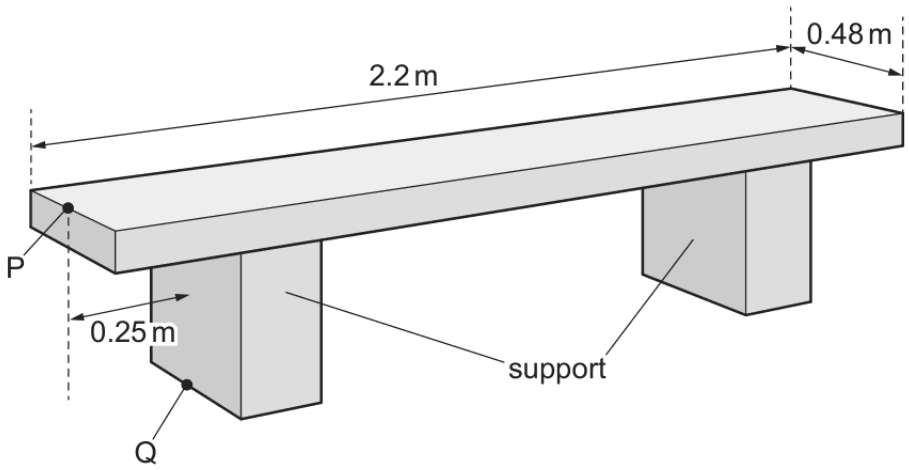


Fig. 1.1

(a) Each of the two supports has an area of 0.040 m^2 in contact with the ground.

Calculate the pressure on the ground due to the bench.

pressure = ... [2]

(b) The centre of mass of the bench is 1.1 m from the left-hand end of the bench and 0.24 m from the front.

(i) Suggest **one** reason why the centre of mass is in this position.

.....

.....

..... [2]

(ii) There is a force exerted vertically downwards from the point P shown in Fig. 1.1.

Calculate the maximum force that can be exerted vertically downwards at P without the bench rotating about the point Q shown in Fig. 1.1.

maximum force = ... [3]

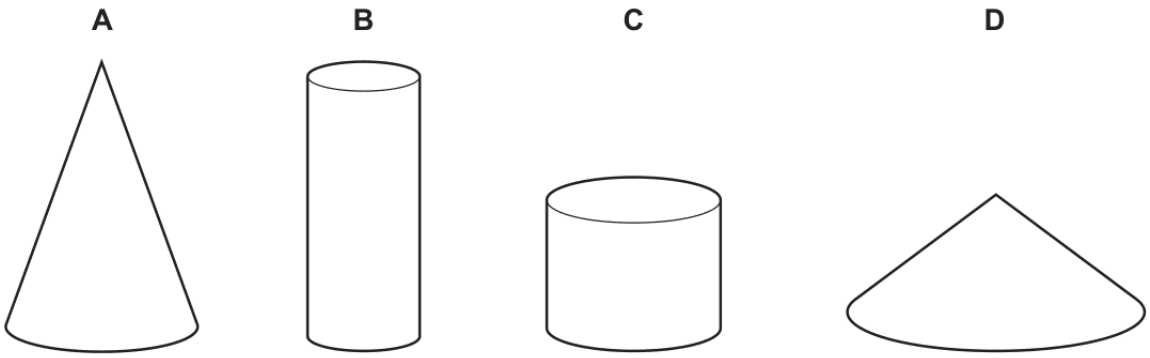
[Total: 7]

Question	Answer	Marks
1(a)	(p =) F / A or 2000 / (2 × 0.040) **or** 50 000 (P–a)	**C1**
	25 000 Pa	**A1**
1(b)(i)	symmetrical left-to-right **or** symmetrical front-to-back **or** bench is stable	**B1**
	because centre of mass is in middle (horizontally) **or** centre of mass is within the base	**B1**
1(b)(ii)	(moment =) Fx **or** Wy **or** 2000 × (1.1 – 0.25) **or** 2000 × 0.85 clockwise moment = anticlockwise moment **or** Fx = Wy **or**	**C1**
	Fx = 2000 × (1.1 – 0.25) **or** Fx = 2000 × 0.85 **or** (F =)2000 × 0.85 / 0.25	**C1**
	6800 N	**A1**

QUESTION · 5054_w21_qp_12.pdf · source page 4 of 16

8 Four solid objects are made from the same material and have equal volumes.

Which object is the most stable?



MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 8

8	D	**1**

8 The turning effect of a force is measured by its moment.

(a) Fig. 8.1 shows a force *F* acting on an object at point P. The object is free to rotate about an axis at X that is perpendicular to the page.

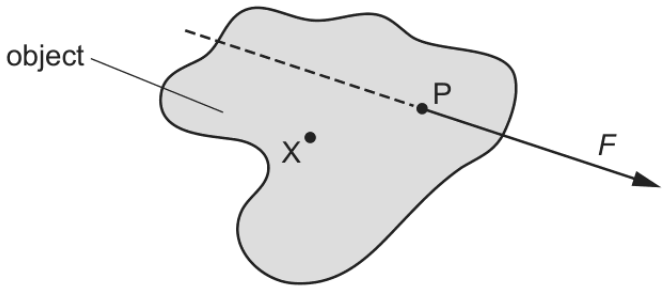


Fig. 8.1

Write down an expression for the moment of *F* about the axis at X. Draw on Fig. 8.1 to show what is meant by any other term used in your expression.

.....

..... [2]

(b) **(i)** State the principle of moments.

.....

.....

..... [2]

(ii) Describe an experiment to verify the principle of moments. Include a diagram to help the description.

.....

.....

.....

.....

.....

..... [3]

(c) A worker carries a ladder on his shoulder. His shoulder acts as a pivot. Fig. 8.2 shows that the ladder is horizontal.

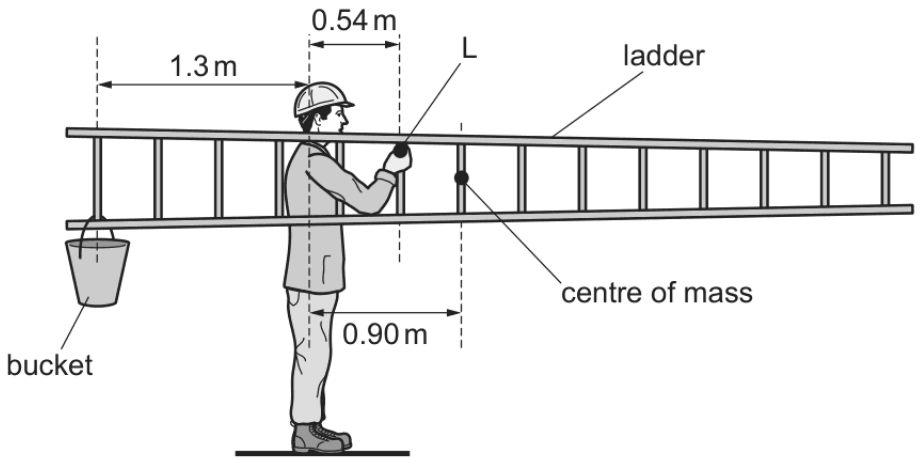


Fig. 8.2

The ladder is wider at one end than at the other end. The mass of the ladder is 8.0 kg.

(i) The gravitational field strength is 10 N / kg.

Calculate the weight of the ladder.

weight = ... [1]

(ii) State what is meant by *centre of mass*.

.....

..... [1]

(iii) The centre of mass of the ladder is not halfway along its length.

State what this shows about the ladder.

.....

..... [1]

(iv) The centre of mass of the ladder is a horizontal distance of 0.90 m from the worker's shoulder.

Calculate the moment about the worker's shoulder of the weight of the ladder.

moment = ... [2]

(v) A bucket of weight 87 N is suspended from the ladder at a horizontal distance of 1.3 m from the worker's shoulder.

The worker keeps the ladder horizontal by exerting a vertical force at point L. L is a horizontal distance of 0.54 m from his shoulder.

Determine the size and direction of the force exerted at L.

size of force = ...

direction = ...
[3]

[Total: 15]

MARK SCHEME · 5054_w21_ms_22.pdf · source page 9 of 11 · Question 8

Question	Answer	Marks
8(a)	$F \times$ distance / displacement	**M1**
	perpendicular distance to line of action of force shown on Fig. 7.1	**A1**
8(b)(i)	for an object in equilibrium	**B1**
	(sum of) clockwise moments is equal to (sum of) anticlockwise moments **or** no resultant moment	**B1**
8(b)(ii)	equipment shown in diagram (e.g. rule, two loads, pivot)	**B1**
	what is done to achieve balance (e.g. balance rule on pivot, one load each side of pivot)	**B1**
	how principle is verified (e.g. calculate two opposite moments and find they are equal)	**B1**
8(c)(i)	80 N	**B1**
8(c)(ii)	(point) where the (whole) weight of an object acts / seems to act **or** point where the object balances	**B1**

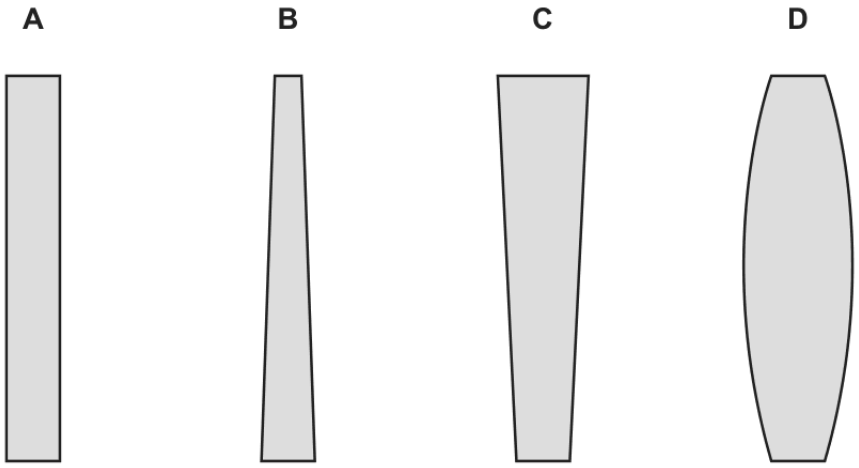
MARK SCHEME · 5054_w21_ms_22.pdf · source page 10 of 11 · Question 8

Question	Answer	Marks
8(c)(iii)	mass not uniformly distributed along length **or** more mass at the one end	**B1**
8(c)(iv)	(Γ=) 80×0.90	**C1**
	72 N m	**A1**
8(c)(v)	moment / 0.54 **or** 209 (N) **or** 41.1 / 0.54	**C1**
	76 N	**A1**
	downwards	**B1**

QUESTION · 5054_s22_qp_11.pdf · source page 4 of 16

6 Four glass objects have square bases of equal area.

Which object is the least stable?

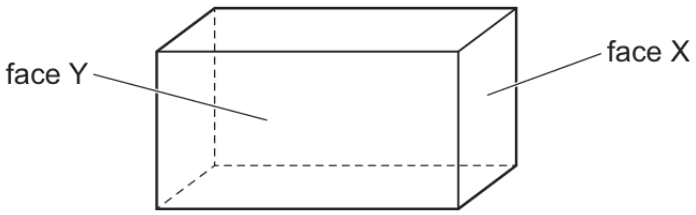


MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 6

6	C	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 4 of 20

8 The centre of mass of a solid rectangular block is at its centre. A small heavy weight is available.



In which arrangement is the centre of mass the lowest?

A with face X on a table

B with face Y on a table

C with face X on a table and the heavy weight attached centrally on top of the block

D with face Y on a table and the heavy weight attached centrally on top of the block

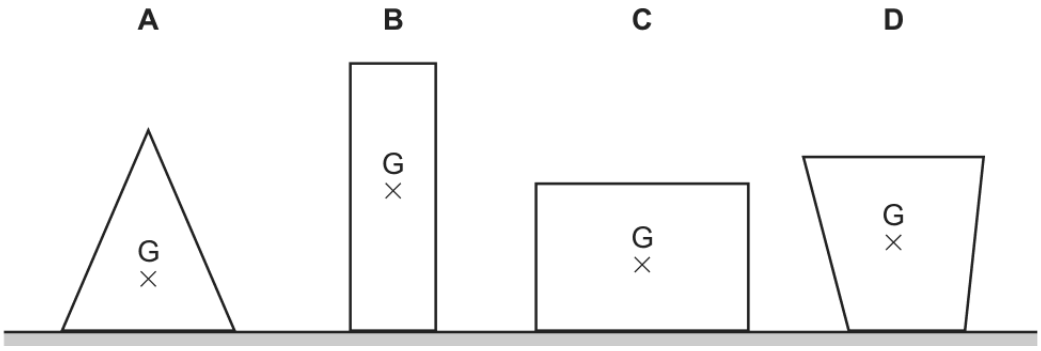
MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 8

8	B	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 6 of 16

12 Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

Which object is the least stable?



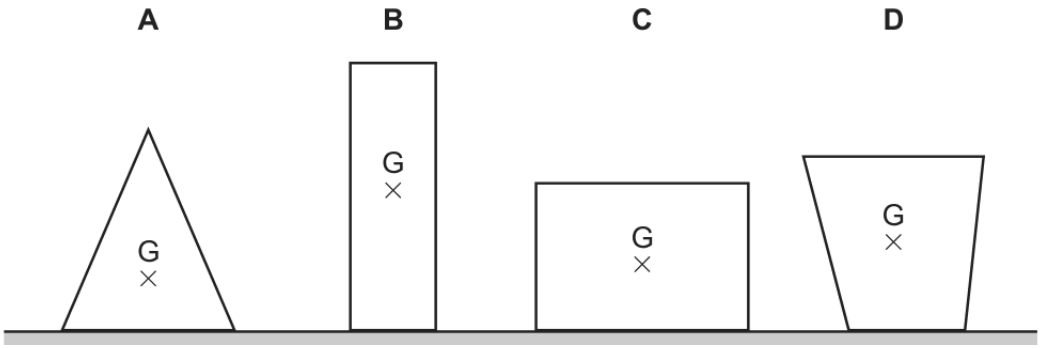
MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 12

12	B	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 6 of 16

13 Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

Which object is the least stable?



MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 13

13	B	**1**

3 A plane, irregular lamina has a mass of 50 g. It hangs from a nail that passes through a small hole H near to the edge of the lamina.

The nail acts as a pivot and the lamina can swing about it. The lamina is held in the position shown in Fig. 3.1, a small distance above a horizontal bench.

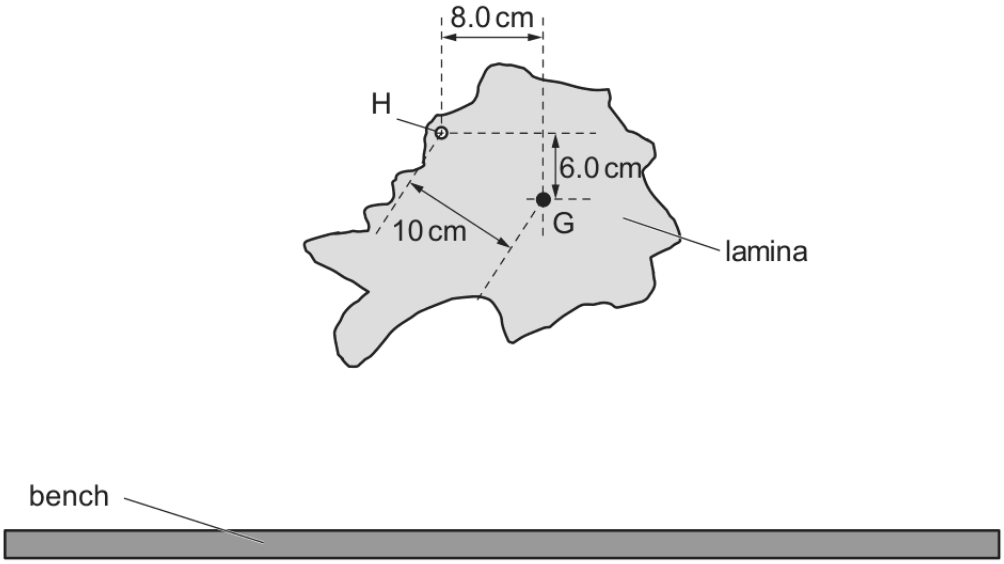


Fig. 3.1 (not to scale)

The centre of gravity of the lamina is at G.

(a) The weight of the lamina is a force that acts downwards.

(i) Explain why the lamina experiences this downward force.

.....

..... [1]

(ii) Calculate the weight of the lamina on Earth.

weight = ... N [2]

(b) The weight produces a moment about the pivot through H.

(i) Using Fig. 3.1, calculate the moment of the weight about the pivot.

moment = ... N m [2]

(ii) The lamina is now released from the position shown in Fig. 3.1.

Describe what happens to the lamina from the time it is released until it stops moving.

.....

.....

.....

..... [3]

[Total: 8]

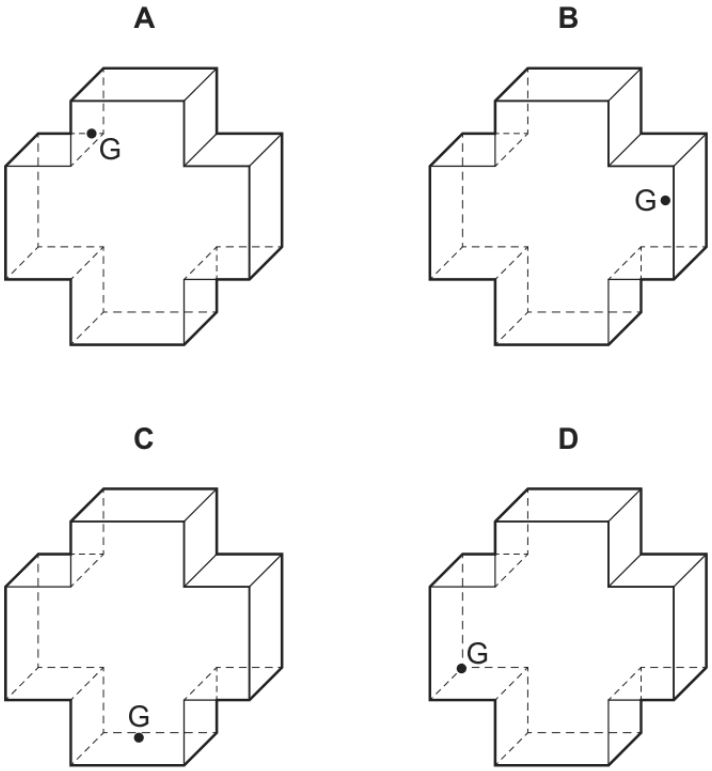
Question	Answer	Marks
3(a)(i)	it / lamina is in the gravitational field (of Earth)	**B1**
3(a)(ii)	($F =$) mg **or** 0.050×9.8 **or** 50×9.8 **or** 490	**C1**
	0.49 (N)	**A1**
3(b)(i)	($M =$) $F \times x_{\perp r}$ **or** 0.49×8.0 **or** 0.49×0.080	**C1**
	0.039 (N m)	**A1**
3(b)(ii)	it / lamina moves clockwise (about the pivot)	**B1**
	it speeds up **and** then slows down **or** it oscillates (with decreasing amplitude) **or** overshoots	**B1**
	(stops with) G vertically below H	**B1**

QUESTION · 5054_s24_qp_11.pdf · source page 5 of 16

9 Four objects have identical shapes but different distributions of mass.

The centre of gravity of each object is shown by the dot marked G.

Which object is the most stable?



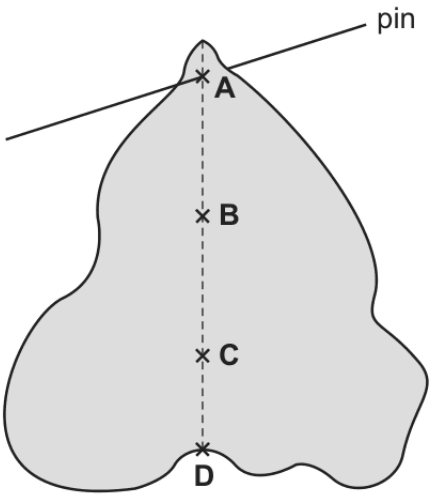
MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 9

9	C	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 6 of 16

14 A piece of card of uniform thickness and density is suspended freely from a horizontal pin. The diagram shows the card at rest.

Which cross is the centre of gravity of the card?



MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 14

14	C	**1**

2 **(a)** State the principle of moments.

.....

.....

..... [2]

(b) An airline passenger wishes to check the weight of a suitcase that he carries as hand luggage onto an aeroplane.

He uses a uniform plank of wood, pivoted at its centre, as shown in Fig. 2.1.

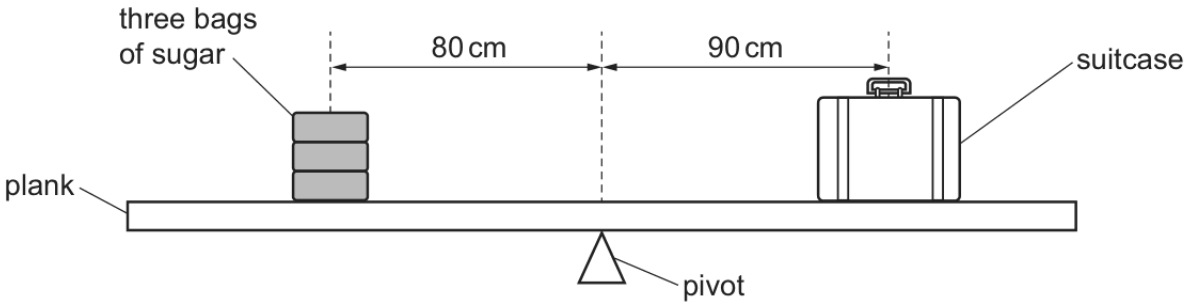


Fig. 2.1

The plank is balanced with the suitcase on one side of the pivot and three bags of sugar, each of mass 2.0 kg, on the other side.

The distances are shown on Fig. 2.1.

(i) Calculate the weight of **one** bag of sugar.

weight = ... N [1]

(ii) The maximum weight of a suitcase that can be carried onto the aeroplane is 67 N.

Determine whether the weight of the suitcase exceeds the maximum allowed.

Show a calculation in your answer.

.....

..... [2]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(c) A tall bus is tested for stability.

Fig. 2.2 shows the bus on a slope.

The centre of gravity of the bus is marked.

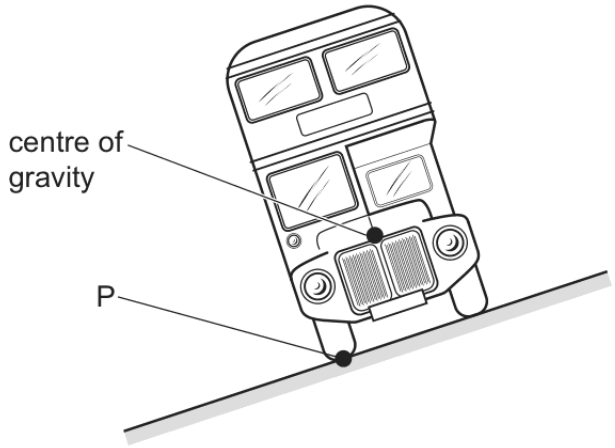


Fig. 2.2

(i) State what is meant by the 'centre of gravity' of an object.

.....

..... [1]

(ii) When the slope is made very steep, the bus falls over by rotating about point P.

Explain why the bus falls over.

.....

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
2(a)	(total) clockwise moments = anticlockwise moments	**M1**
	in equilibrium	**A1**
2(b)(i)	19.6 or 20 N	**B1**
2(b)(ii)	80 × 3 × **(b)(i)** = 90 × weight	**B1**
	52 (N) **and** weight of suitcase does not exceed maximum	**B1**
2(c)(i)	(place) where weight of object (can be taken to) acts	**B1**
2(c)(ii)	centre of gravity falls outside base / to the left of P	**B1**
	weight causes anticlockwise moment / turning effect	**B1**

QUESTION · 5054_w25_qp_12.pdf · source page 6 of 16

14 Which statement about centre of gravity is correct?

A Objects with a centre of gravity at the same height are less stable when the base is larger.

B Objects with a centre of gravity at the same height are more stable when the base is larger.

C Objects with higher centres of gravity and smaller bases are more stable.

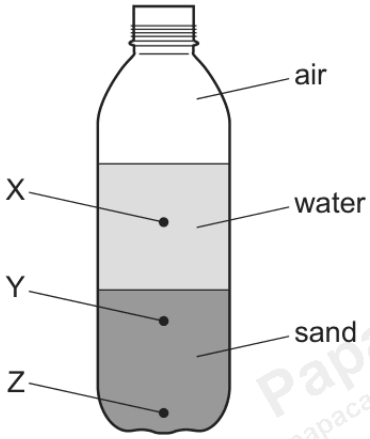
D Objects with identical bases are more stable when the centre of gravity is higher.

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 14

14	B	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 5 of 16

9 The diagram shows a bottle containing air, water and sand.



More sand is added to the bottle. This affects the position of the centre of gravity of the bottle and its contents.

How might the centre of gravity move?

A from X towards Y

B from Y towards X

C from Y towards Z

D from Z towards Y

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 9

9	B	**1**

Chapter 1 Motion, forces and energy

1.6 Momentum

19 entries: 8 Paper 1 questions, 11 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s23_qp_11.pdf · source page 6 of 20

10 A constant frictional force acts on a moving object for a short amount of time causing the momentum of the object to decrease.

Which expression is used to calculate the time for which the force acts?

A $\text{time} = \dfrac{\text{change in momentum}}{\text{force}}$

B $\text{time} = \text{change in momentum} \times \text{force}$

C $\text{time} = \dfrac{\text{momentum}}{\text{force}}$

D $\text{time} = \text{momentum} \times \text{force}$

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 10

10	A	**1**

2 In a safety test, a car of mass 1100 kg travels at a speed of 10 m/s and collides with a stationary van of mass 3000 kg.

After the collision the car and the van move together with a velocity *v*.

Fig. 2.1 shows the car and van before and after the collision.

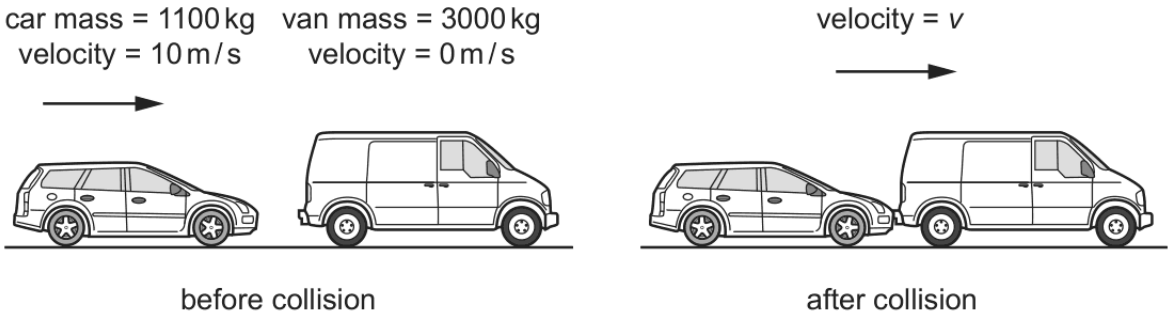


Fig. 2.1

The total momentum of the car and van is conserved during the collision.

(a) (i) Define 'momentum'.

.....

..... [1]

(ii) State the unit of momentum.

..... [1]

(b) Calculate the velocity *v* of the car and van after the collision.

v = ... m/s [2]

(c) (i) Calculate the total kinetic energy of the car and van after the collision.

kinetic energy = ... J [2]

(ii) State the transfer of energy that occurs in the collision.

.....

..... [1]

[Total: 7]

Question	Answer	Marks
2(a)(i)	the product of mass and velocity	**B1**
2(a)(ii)	kg m / s **or** Ns	**B1**
2(b)	1100 × 10 = 4100 × v	**C1**
	2.7 (m / s)	**A1**
2(c)(i)	(*KE* =) ½ mv^2 algebraic or numerical for the car	**C1**
	15 000 (J)	**A1**
2(c)(ii)	(kinetic energy) becomes heat / thermal energy / internal energy (of vehicles / air) **or** kinetic energy (of car) goes to kinetic energy **of van**	**B1**

1 Fig. 1.1 shows the speed–time graph for a car travelling on a straight horizontal road.

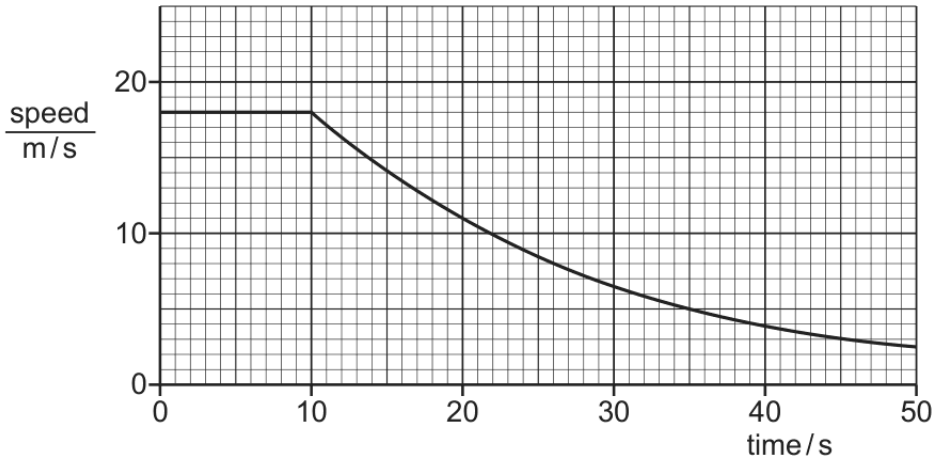


Fig. 1.1

(a) Describe the motion of the car shown in Fig. 1.1.

.....

.....

.....

..... [2]

(b) At time t = 10 s the engine of the car is switched off. The brakes are not applied.

(i) Name **two** forces that act on the car to cause the change in motion after t = 10 s.

1 .....

2 .....

[1]

(ii) Suggest why Fig. 1.1 is a curve after t = 10 s.

.....

.....

..... [1]

(c) Between $t = 10\,s$ and $t = 20\,s$ the speed of the car changes from $18\,m/s$ to $11\,m/s$.

The mass of the car is $1200\,kg$.

(i) Calculate the change in momentum of the car in this time.

Give the unit of your answer

momentum change = ... unit ..... [2]

(ii) Calculate the average resultant force exerted on the car during this time.

average resultant force = ... N [2]

[Total: 8]

Question	Answer	Marks
1(a)	speed is constant for 10 s / at first	**B1**
	deceleration • has (average) value 0.39 m / s^2 • lasts for 40 s • is non-uniform	**B1**
1(b)(i)	friction (with road) **and** air resistance **or** (air) drag	**B1**
1(b)(ii)	resistive force(s) change / decrease (as speed decreases) **or** (graph shows) deceleration is not constant / decreases **or** non-uniform decrease in speed	**B1**
1(c)(i)	(momentum =) *mv* in any form algebraic or numerical	**C1**
	8400 **and** kg m / s **or** Ns	**A1**
1(c)(ii)	(F =) $\Delta mv / t$ **or** $F = ma$ in any form algebraic or numerical	**C1**
	840 (N)	**A1**

1 Fig. 1.1 shows a toy water rocket.

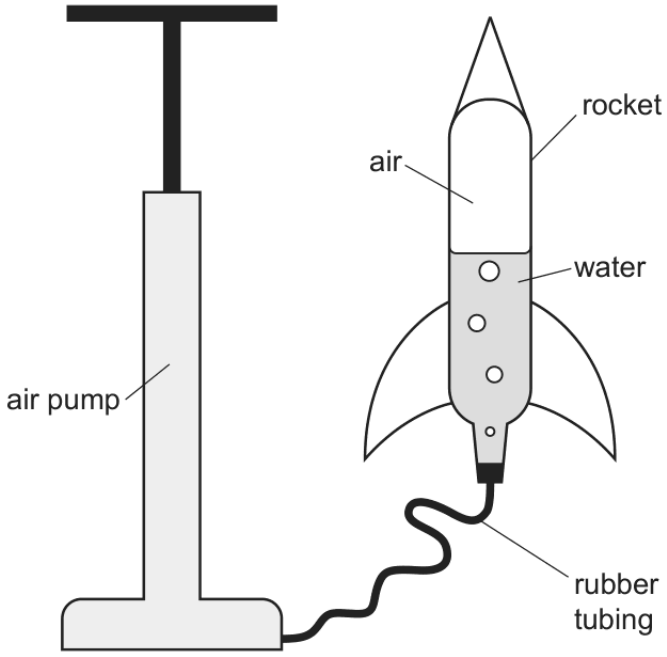


Fig. 1.1

The rocket is half-filled with water and connected by rubber tubing to a pump.

(a) Air is then pumped into the rocket so that the pressure of the air inside the rocket increases.

(i) Explain, in terms of particles, why the air in the rocket exerts a pressure on the walls of the rocket.

.....

.....

.....

..... [3]

(ii) The temperature of the air in the rocket remains constant as air is pumped into the rocket.

Explain why increasing the number of air particles in the rocket increases the pressure of the air in the rocket.

.....

..... [1]

(b) When the air pressure in the rocket is high, the rubber tubing is removed from the rocket.

The water in the rocket is expelled downwards.

Explain, in terms of momentum, why the rocket begins to accelerate upwards.

.....

.....

.....

..... [3]

(c) As the water is expelled from the rocket, the upward acceleration of the rocket changes.

Suggest **two** reasons for the change in the acceleration of the rocket.

1 .....

.....

2 .....

.....

[2]

[Total: 9]

Question	Answer	Marks
1(a)(i)	particles move / collide / hit	**B1**
	particles collide with (inner) walls (of rocket)	**B1**
	force exerted by collisions over / on an area	**B1**
1(a)(ii)	more collisions per unit time	**B1**
1(b)	(expelled) water gains / has (downward) momentum	**B1**
	momentum conserved	**B1**
	rocket gains (upward) momentum (and accelerates)	**B1**
1(c)	any **two** from: mass (of water and rocket) decreases / changes pressure (of air) decreases (air) resistance (acts) gravitational field strength decreases (with height) less water expelled per second / water expelled at slower speed	**B2**

QUESTION · 5054_w23_qp_12.pdf · source page 5 of 20

10 Which expression is equal to impulse?

A change in momentum ÷ time

B change in momentum

C force × distance moved in the direction of the force

D force ÷ area

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 10

10	B	**1**

2 A spacecraft of mass 300 kg is moving in a straight line in space, at a speed of 8000 m/s.

(a) Calculate the momentum of the spacecraft.

momentum = ... kg m/s [2]

(b) The fuel on the spacecraft explodes and the spacecraft splits into two parts. The direction in which the parts move does not change.

(i) After the explosion, the speed of the front part increases to 9000 m/s. It has a mass of 150 kg.

Calculate the speed of the rear part after the explosion.

speed = ... m/s [3]

(ii) The total kinetic energy of the two parts after the explosion is greater than the original kinetic energy of the spacecraft.

State the energy transfer that occurs in the explosion.

.....

..... [1]

(iii) The explosion lasts for 0.20 s.

Calculate the average force on the front part during this time.

force = ... N [3]

[Total: 9]

Question	Answer	Marks
2(a)	(p =) mv **or** 300 × 8000	**C1**
	2.4×10^6 (kg m / s)	**A1**
2(b)(i)	momentum conserved **or** $mv = m_1v_1 + m_2v_2$	**C1**
	(v_2 =) $(2.4 \times 10^6 - (150 \times 9000)) \div 150$ **or** (v_2 =) (1 050 000) / 150	**C1**
	7000 (m / s)	**A1**
2(b)(ii)	from energy stored as chemical (potential) energy	**B1**
2(b)(iii)	Δp **or** J **or** I = 150 × (9000 – 8000) **or** 150 000 **or** (a =) (9000 – 8000) ÷ 0.20 **or** (a =) 5000	**C1**
	(F =) $\Delta p / (\Delta)t$ **or** (F =) $I / (\Delta)t$ **or** (F =) ma **or** 150 × (9000 – 8000) / 0.20 **or** 150 × 1000 / 0.20	**C1**
	7.5×10^5 (N)	**A1**

2 A small aircraft takes off from the horizontal deck of a ship.

Before taking off, the aircraft is held in place by a holdback bar.

When the holdback bar is released, the aircraft is pulled along the deck by a steam-powered piston as shown in Fig. 2.1.

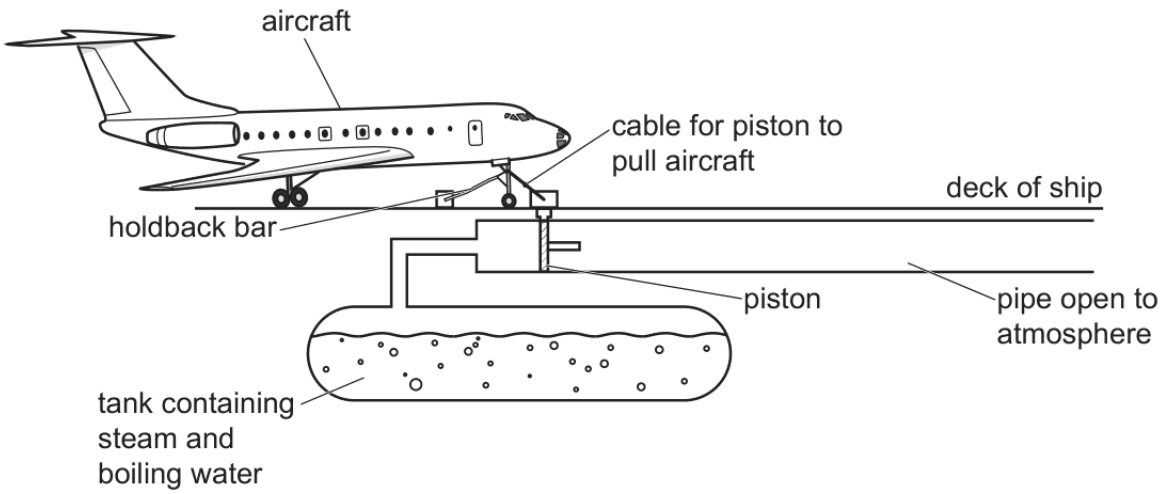


Fig. 2.1 (not to scale)

The steam exerts a high pressure on the piston.

(a) (i) Explain, using ideas about particles, how the steam creates a pressure on the piston.

.....

.....

.....

..... [2]

(ii) Explain why the pressure on the piston increases as the temperature of the steam increases.

.....

.....

.....

..... [2]

(b) When the pressure is high enough, the holdback bar is released. The steam pushes the piston along the pipe shown in Fig. 2.1.

The piston has a cross-sectional area of $0.30\,m^2$.

The pressure of the steam in the tank is $2.1 \times 10^6\,Pa$.

Atmospheric pressure is $1.0 \times 10^5\,Pa$.

Determine the resultant force on the piston caused by the pressure difference.

Show your working.

force = ... N [2]

(c) The force calculated in **(b)** causes the aircraft to accelerate to a maximum speed of $28\,m/s$ from rest.

The mass of the aircraft is $3.0 \times 10^4\,kg$.

(i) Calculate the momentum of the aircraft at the maximum speed.

momentum = ... kg m/s [2]

(ii) Calculate the time taken for the aircraft to reach the maximum speed.

time = ... s [2]

[Total: 10]

MARK SCHEME · 5054_s24_ms_21.pdf · source page 5 of 10 · Question 2

Question	Answer	Marks
2(a)(i)	particles hit piston / sides of container	**B1**
	each hit creates a small force **or** force on unit area is pressure **or** particles momentum changes / direction changes **or** many hits per second	**B1**
2(a)(ii)	particles move faster / have higher kinetic energy / more momentum	**B1**
	more hits / sec. / more frequent hits / larger momentum change per second / larger force on each hit	**B1**

MARK SCHEME · 5054_s24_ms_21.pdf · source page 6 of 10 · Question 2

Question	Answer	Marks
2(b)	(force =) P × A in any form numerical or algebraic with any pressure	**B1**
	6.0×10^5 (N)	**B1**
2(c)(i)	(momentum =) mv in any form numerical or algebraic	**C1**
	8.4×10^5 (kg m/s)	**A1**
2(c)(ii)	(time =) mv ($-mu$) / F **or** (t =) v ($-u$) / a **and** $F = ma$	**C1**
	1.4 s	**A1**

QUESTION · 5054_s24_qp_12.pdf · source page 6 of 16

13 A car of mass 750 kg travels 400 m at 25 m / s. It then accelerates to 35 m / s and travels a further 400 m.

What is the change in the momentum of the car due to acceleration?

A 7500 kg m / s

B 24 000 kg m / s

C 45 000 kg m / s

D 75 000 kg m / s

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 13

13	A	**1**

1 Fig. 1.1 shows two trolleys. On the front of trolley A, there is a wooden rod. Trolley B is initially at rest.

As trolley A moves towards the right, the rod enters the modelling clay. Trolley A slows down and trolley B starts moving.

The trolleys then stick together and continue moving towards the right.

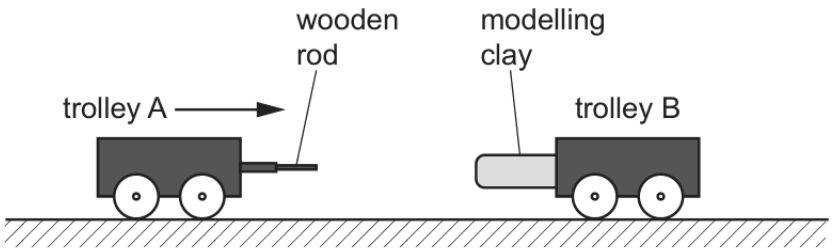


Fig. 1.1

Fig. 1.2 shows the speed−time graph for the two trolleys.

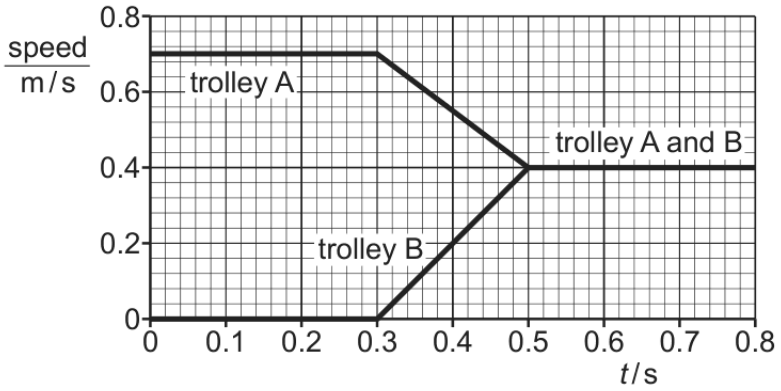


Fig. 1.2

The trolleys start to collide at time t = 0.30 s. At t = 0.50 s, the trolleys are moving at the same speed.

(a) (i) State how Fig. 1.2 shows that, during the collision, trolley B has a uniform acceleration.

.....

..... [1]

(ii) Describe how the graph in Fig. 1.2 shows that the magnitude (size) of the acceleration of trolley B is larger than the magnitude of the deceleration of trolley A.

.....

..... [1]

(iii) Calculate the acceleration of trolley B when $t = 0.40\,s$.

acceleration = ... m/s^2 [2]

(b) The mass of trolley A = 0.80 kg. The mass of trolley B = 0.60 kg.

Show that momentum is conserved in the collision.

[2]

(c) In another collision between the same trolleys, the rod and modelling clay are not present. Trolley A hits trolley B with the same initial speed.

Explain why the force between the trolleys is larger in this collision.

.....

.....

.....

..... [2]

[Total: 8]

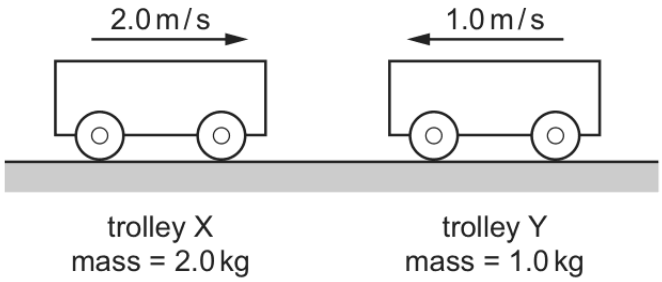
Question	Answer	Marks
1(a)(i)	straight line (on speed-time graph)	**B1**
1(a)(ii)	gradient of line steeper (for B) **or** larger change in speed (in same time)	**B1**
1(a)(iii)	(*a* =) change in speed / time in any form	**C1**
	2(.0) m / s^2	**A1**
1(b)	(p =) mv **or** (initial) momentum of A = 0.56 (kg m / s)	**B1**
	and final momentum of A and B = 1.4 × 0.4 = 0.56	**B1**
1(c)	Any two out of three: • contact surfaces are more rigid / less pliable **aaw** • decreases the time of contact **or** final velocity of B is larger (than 0.4 m / s) / final velocity of A is smaller • increases acceleration **or** force = (mv - mu) / t **or** impulse / t	**B2**

QUESTION · 5054_w24_qp_11.pdf · source page 7 of 16

15 Trolley X of mass 2.0 kg is moving to the right at 2.0 m/s.

Trolley Y of mass 1.0 kg is moving to the left at 1.0 m/s.

The trolleys collide and stick together.



The collision takes a total time of 0.20 s.

What is the average force acting on trolley X during the collision?

A 3.3 N **B** 10 N **C** 15 N **D** 25 N

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 15

15	B	**1**

1 A jet ski is a type of boat that carries one or two people and travels at high speed on water.

Fig. 1.1 shows a student riding on a jet ski.

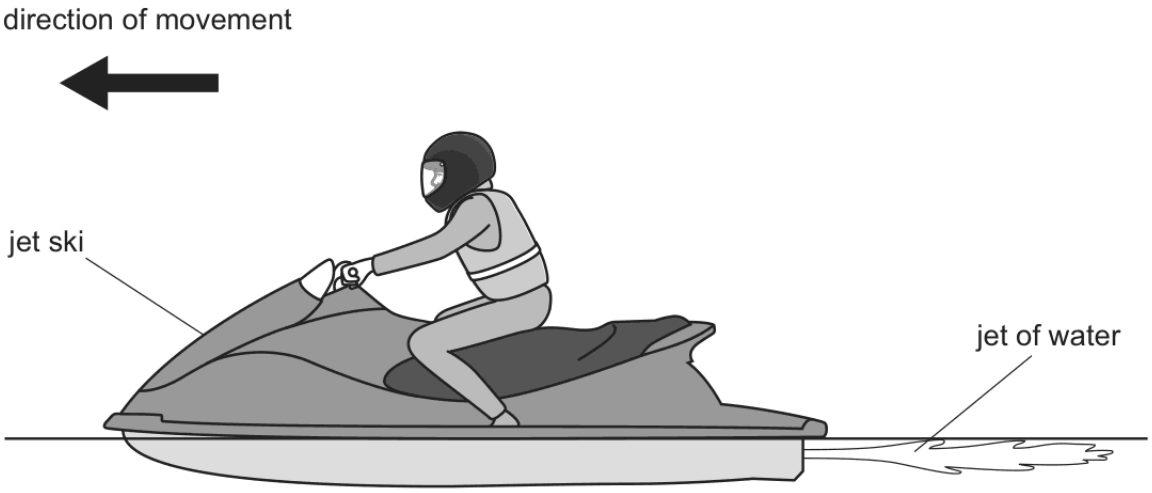


Fig. 1.1

A high-speed jet of water is forced backwards out of the back of the jet ski by a pump inside the jet ski.

(a) The pump increases the momentum of the water that is forced backwards out of the back of the jet ski.

(i) Complete the word equation to show the relationship between the resultant force on an object and the change in momentum of the object.

resultant force =

[1]

(ii) In 2.0 s, the pump increases the backwards speed of 180 kg of water by 30 m/s.

Calculate the backwards force exerted on the water.

force = ... N [2]

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(b) (i) Using Newton's third law of motion, explain why there is a forwards force on the jet ski.

.....

.....

.....

..... [2]

(ii) The student has a mass of 70 kg and the jet ski has a mass of 280 kg.

Use your answer from **(a)(ii)** to determine the acceleration of the student and jet ski when no resistive forces are acting.

acceleration = ... m/s^2 [2]

(c) The jet ski reaches a speed of 20 m/s.

Calculate the total kinetic energy of the student and jet ski at this speed.

kinetic energy = ... J [3]

[Total: 10]

Question	Answer	Marks
1(a)(i)	resultant force = increase in momentum / unit time **or** resultant force = rate of increase in momentum	**B1**
1(a)(ii)	(F =) 180 × 30 / 2.0 **or** 180 × 15	**C1**
	2700 (N)	**A1**
1(b)(i)	pump **/** jet ski / engine exerts force on water	**B1**
	water exerts an opposite force on the pump / jet ski / engine	**B1**
1(b)(ii)	(a =) F / m **or** 2700 / (70 + 280) **or** 2700 / 350	**C1**
	7.7 (m / s^2)	**A1**
1(c)	(K.E. =) ½ mv^2	**C1**
	½ × (70 + 280) × 20^2 **or** ½ × 350 × 20^2	**C1**
	70 000 (J)	**A1**

QUESTION · 5054_s25_qp_11.pdf · source page 4 of 20

7 What is a unit for momentum?

A kgm/s **B** kgm/s^2 **C** Nm **D** N/m^2

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 7

7	A	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 4 of 20

8 In a safety test, a car collides with a concrete block.

The impulse exerted by the block on the car is 14 000 N s.

The collision with the block lasts for 70 ms.

What is the force exerted on the car?

A 200 N **B** 14 000 N **C** 200 000 N **D** 980 000 N

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 8

8	C	**1**

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3 Fig. 3.1 shows a flat tyre on a car. The tyre needs to be inflated.

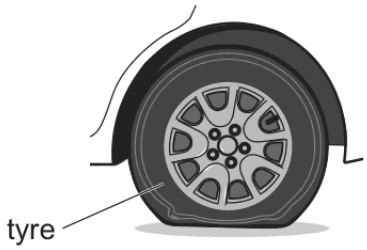

Fig. 3.1

(a) The force exerted by the flat tyre on the road is 1350 N.

The area of the flat tyre in contact with the road is 0.0070 m^2.

(i) Calculate the pressure exerted by the flat tyre on the road.

pressure = ... Pa [2]

(ii) Air is pumped into the tyre to inflate it. This increases the pressure inside. The force exerted by the tyre on the road does not change.

State how the area of the tyre in contact with the road changes.

..... [1]

(b) Inside the inflated tyre, a particle of air collides against the wall of the tyre.

(i) Define 'momentum'.

..... [1]

(ii) Using ideas about momentum, explain why there is a force on the wall of the tyre as the particle collides with it.

.....

.....

..... [2]

(c) As the car moves, the temperature of the air inside the tyre increases. The increase in temperature causes an increase in the pressure of the air.

Using ideas about particles, explain why the pressure increases.

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
3(a)(i)	(P =) force / area in any form	**C1**
	1.9×10^5 (Pa)	**A1**
3(a)(ii)	(increasing pressure) decreases the area	**B1**
3(b)(i)	mass (of the particle) × velocity (of the particle)	**B1**
3(b)(ii)	*on collision*: change of momentum / direction / velocity **or** *on collision:* momentum transferred to wall	**B1**
	force related to change in momentum ÷ time **or** force of equal size on tyre / wall and on particle (Newton's 3rd law)	**B1**
3(c)	(particles have) high(er) speed / velocity / momentum / kinetic energy	**B1**
	more collisions per second **or** hit (wall) hard(er) **or** larger change in momentum per sec	**B1**

1 A car is at rest on a straight horizontal road.

At time $t = 0$, the car begins to accelerate.

Fig. 1.1 shows how the momentum of the car varies with time.

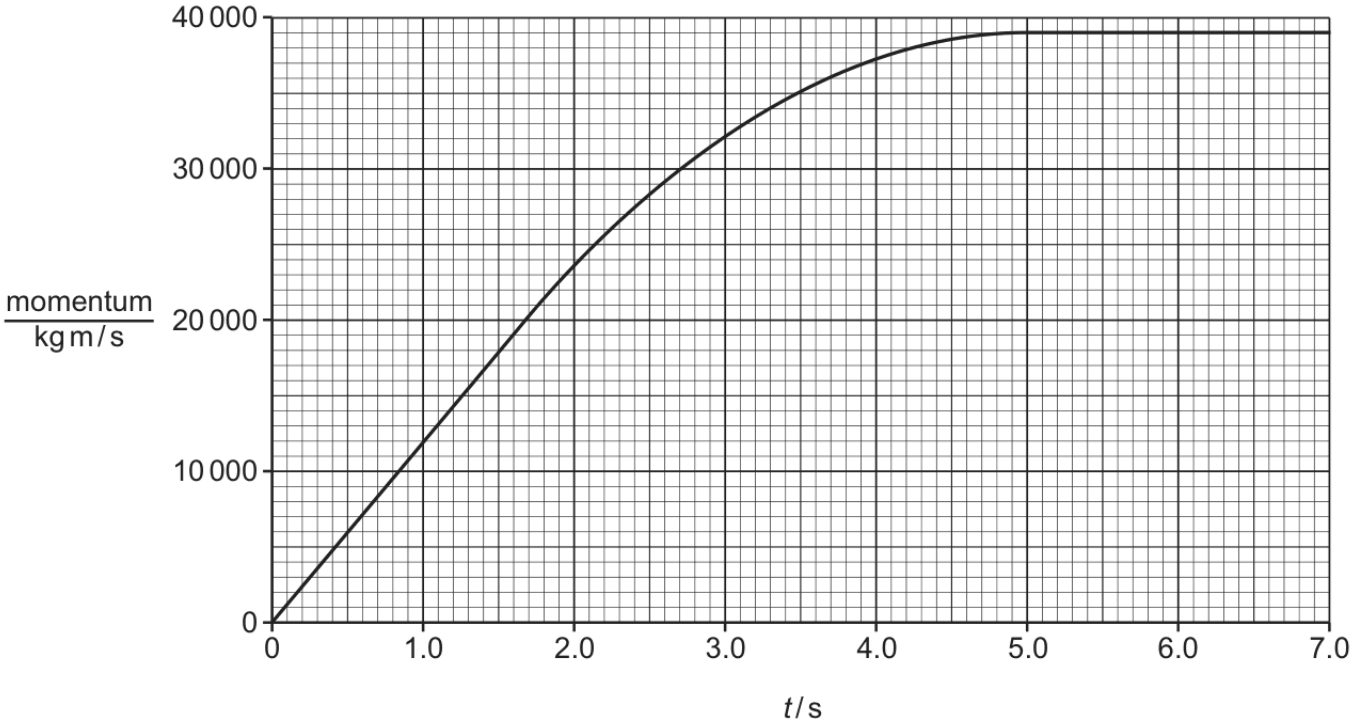

Fig. 1.1

(a) Use Fig. 1.1 to determine the resultant force acting on the car during the first 1.5 s of its motion.

resultant force = ... N
[3]

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(b) The car has mass m.

(i) When the momentum of the car is p, its kinetic energy is E_k.

Show that $\frac{p^2}{2m} = E_k$.

[2]

(ii) At $t = 6.0\,s$, the kinetic energy of the car is $5.4 \times 10^5\,J$.

Use Fig. 1.1 to determine the momentum p of the car at $t = 6.0\,s$, and use the equation in **(b)(i)** to determine the mass m of the car.

Show your working.

p = ... kg m/s

m = ... kg
[3]

[Total: 8]

Question	Answer	Marks
1(a)	(change in momentum / momentum =) 18 000 (kg m / s)	**C1**
	(force =) $\frac{\text{change in momentum}}{\text{time taken}}$ **or** $\frac{18\,000}{1.5}$	**C1**
	12 000 (N)	**A1**
1(b)(i)	$\left(\frac{p^2}{2m} =\right) \frac{m^2v^2}{2m}$	**B1**
	shown to be $\frac{1}{2}mv^2$ (and equal to E_k / kinetic energy) $E_k = \frac{1}{2}mv^2$ seen	**B1**
1(b)(ii)	39 000 seen	**B1**
	$(m =) \frac{39\,000^2}{2 \times 5.4 \times 105}$	**C1**
	1400 (kg)	**A1**

3 Fig. 3.1 shows a firefighter directing a jet of water at a wall.

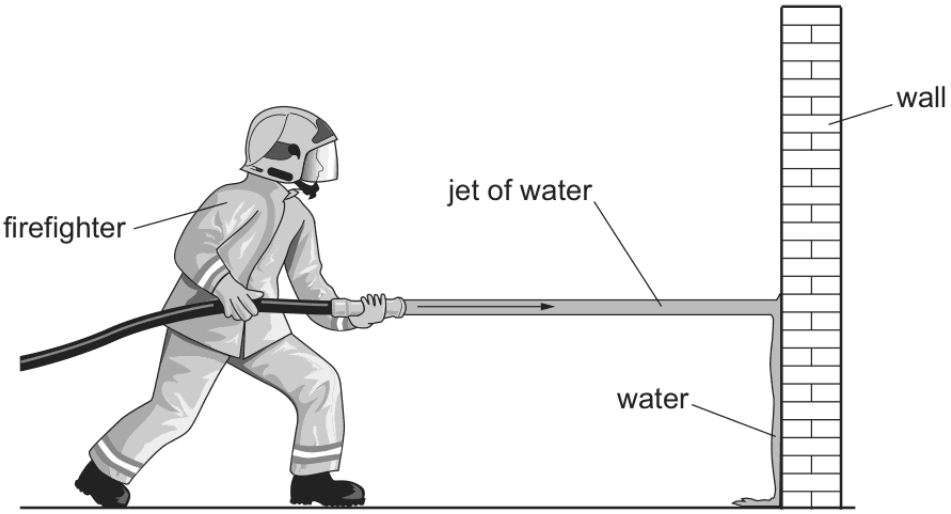

Fig. 3.1

In 1.0 s, a volume of $7.5 \times 10^{-3}\,m^3$ of water hits the wall horizontally, at a speed of 24 m/s.

The density of water is $1000\,kg/m^3$.

(a) Calculate:

(i) the mass of the water that hits the wall in 1.0 s

mass = ... kg [2]

(ii) the horizontal momentum of the water that hits the wall in 1.0 s.

momentum = ... kg m/s [2]

(b) The horizontal momentum of the water decreases to zero when the water hits the wall. None of the water bounces back from the wall.

Explain why the momentum calculated in **(a)(ii)** is equal in size to the horizontal force exerted on the water by the wall.

.....

.....

..... [2]

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(c) Explain, in terms of Newton's third law, why there is a horizontal force exerted on the wall by the water.

.....

.....

.....

.....

..... [2]

(d) The pump that forces the water to flow through the hose is adjusted, and the speed of the water leaving the hose doubles.

Explain why the force exerted by the water on the wall increases by a factor of 4.

.....

.....

..... [1]

[Total: 9]

MARK SCHEME · 5054_w25_ms_22.pdf · source page 9 of 14 · Question 3

Question	Answer	Marks
3(a)(i)	(m =) ρV **or** $1000 \times 7.5 \times 10^{-3}$	**C1**
	7.5 (kg)	**A1**
3(a)(ii)	(p =) mv **or** 7.5×24	**C1**
	180 (kg m / s)	**A1**
3(b)	force needed to change momentum **or** answer to **(a)(ii)** is change in momentum	**C1**
	force = $\frac{\text{change in momentum}}{\text{time taken}}$ **or** the momentum of the water that hits the wall in 1.0 s is the rate of change of momentum	**A1**

MARK SCHEME · 5054_w25_ms_22.pdf · source page 10 of 14 · Question 3

Question	Answer	Marks
3(c)	Newton's third law quoted **or** when object A exerts a force on object B, then object B exerts an equal and opposite force on object A **or** forces (only) occur in pairs	**B1**
	the wall exerts a force on the water	**B1**
3(d)	mass (flowing through pipe / hitting the wall in 1.0 s) doubles **and** momentum (water hitting wall in 1.0 s) increases by a factor of 4	**B1**

QUESTION · 5054_s26_qp_11.pdf · source page 6 of 16

12 Quantity 1 is equal to the change in quantity 2 per unit time.

What are quantity 1 and quantity 2?

	quantity 1	quantity 2
A	force	momentum
B	impulse	acceleration
C	momentum	velocity
D	velocity	acceleration

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 12

12	A	**1**

3 A syringe containing air is connected to a pressure sensor in a school laboratory as shown in Figure 3.1.

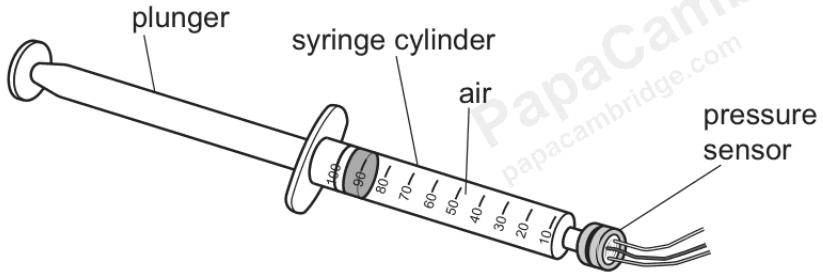

Figure 3.1

The pressure of the air in the syringe cylinder is 1.0×10^5 Pa. There is no friction between the plunger and the walls of the syringe cylinder.

(a) (i) The internal cross-sectional area of the syringe cylinder is $1.2 \times 10^{-4}\,m^2$.

Calculate the force exerted on the plunger by the air inside the syringe cylinder.

force = ... N [2]

(ii) The plunger does not move, even though the force in **3(a)(i)** acts on the plunger.

Explain why the plunger does not move.

.....

..... [1]

(iii) An air particle has a mass of 4.7×10^{-26} kg and moves with a speed of 500 m/s.

Calculate the momentum of the air particle. Give the unit of your answer.

momentum = ... unit ..... [2]

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(iv) Explain, using ideas about particles, how air exerts a large pressure on the wall of the syringe cylinder, even though the momentum of each particle is very small.

.....

.....

.....

..... [2]

(b) The volume of air in the syringe cylinder is initially 90 cm^3.

The plunger is pushed inwards so that the volume of air decreases by 40 cm^3. The temperature of the air in the syringe cylinder remains unchanged.

Calculate the new pressure of the air in the syringe cylinder.

pressure = ...Pa [3]

[Total: 10]

Question	Answer	Marks	Guidance
3(a)(i)	(F =) PA numerical or algebraic 12 N	**1**	
3(a)(ii)	(air) pressure/force outside syringe = (air) pressure / force inside syringe **or** atmospheric pressure (outside syringe) pushes to the right / inwards /produces zero resultant force	**1**	**I** pressure in syringe less than atmospheric pressure; comments about friction
3(a)(iii)	2.4×10^{-23} kg m / s **or** N s	**1**	**A** 2.35, 2.3×10^{-23} **I** formula mv **not** kg/m/s
3(a)(iv)	many particles in gas many hits per second / many hits in unit time / frequent hits	**1**	**A** atoms and molecules for particles; large number of collisions **I** no of particles increase **A** high(er) frequency of collisions; hit often **I** many hits etc. on each other; hit rapidly
3(b)	$p_1V_1 = p_2V_2$ numerical or algebraic $1 \times 10^5 \times 90 = p \times 50$ 1.8×10^5 (Pa)	**1**	numerical calculation could involve 40 $2.2(5) \times 10^5$ gets 2 marks

QUESTION · 5054_s26_qp_12.pdf · source page 5 of 16

10 What is the unit of impulse?

A kg **B** $kg\,m/s^2$ **C** N/s **D** Ns

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 10

10	D	**1**

Chapter 1 Motion, forces and energy

1.7 Energy, work and power

1.7.1 Energy

151 entries: 68 Paper 1 questions, 83 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

11 Fig. 11.1 shows a children's ride. A carriage containing children is pulled up the slope by a motor. The carriage stops at A and then runs down through B, C and D without further input of energy. Between D and E the carriage turns through a bend at constant speed, as shown in Fig. 11.2. At E, brakes are applied and the carriage slows to a stop at F. The height of the ride is 30 m at A and 10 m at C.

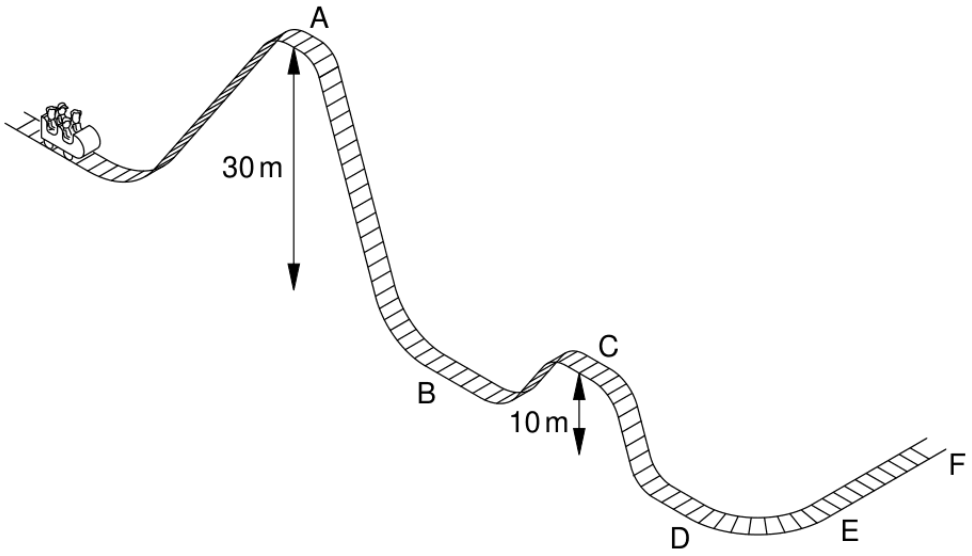


Fig. 11.1

The mass of the carriage and children is 500 kg.
Take the gravitational field strength as 10 N/kg.

(a) **(i)** Discuss the energy changes that occur in the ride from A to D.

(ii) Calculate the maximum potential energy of the carriage and children.

(iii) Assuming that there is no friction between A and C, determine the kinetic energy of the carriage and children at C. Show your working.

[9]

(b)

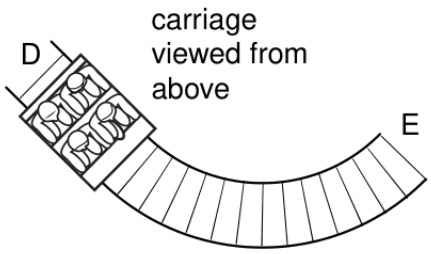


Fig. 11.2

Between D and E, the carriage goes round part of a horizontal circle at constant speed. During this time the velocity of the carriage changes.

(i) Explain how the carriage can have a constant speed but a changing velocity.

(ii) State the direction of the force that acts on the carriage to make it move round the curve.

[3]

(c) Between E and F, a frictional force of 3000 N acts to slow the carriage. Calculate the deceleration of the carriage. [3]

11	**(a)**	**(i)**	P.E. decreases (A to B or C to D or downhill or initially)		B1
			K.E. gained (P.E. → K.E….2)		B1
			K.E. to P.E. **change** must be clear and from B to C or uphill		B1
		(ii)	mgh algebraic form seen		C1
			500*10*30		C1
			150 000 J		A1
		(iii)	conservation of energy cited **or** clear that loss of P.E. has become K.E.		C1
			500*10*20 or 500*10*10 or 50 000 seen		C1
			100 000 J (allow g=9.8)		A1
	(b)	**(i)**	velocity involves direction **or** is a vector (speed does not)		B1
			direction (of carriage) changes / carriage turns (accept on diagram)		
		(ii)	force towards centre (of curve) / inwards (accept centripetal)		B1
	(c)		F = ma in any algebraic form or 3000 = 500a		C1
			3000/500		C1
			6(.0) m/s^2		A1
				Total	**[15]**

For Examiner's Use

3 A pole-vaulter runs along a track, reaching a maximum speed of 8.4 m/s. At the end of the track, he places a pole into the ground as shown in Fig. 3.1, and uses the pole to push himself vertically upwards.

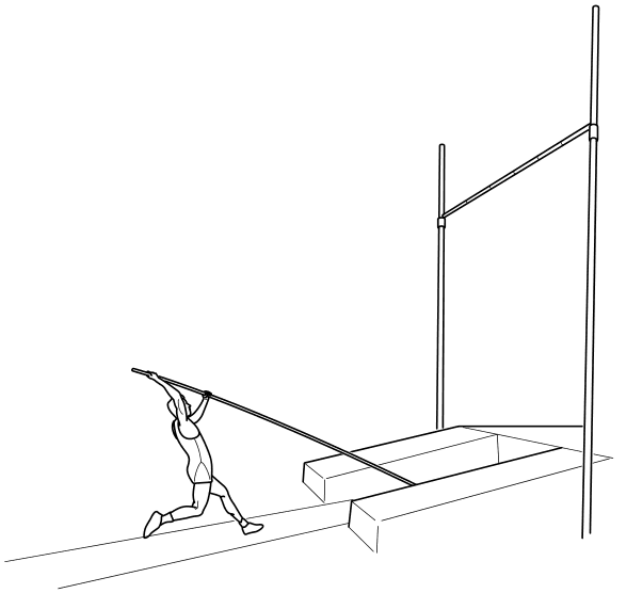


Fig. 3.1

(a) When the pole-vaulter runs along the track, there is a constant forward force on him of 320 N and a backwards resistive force that varies with his speed as shown in Fig. 3.2.

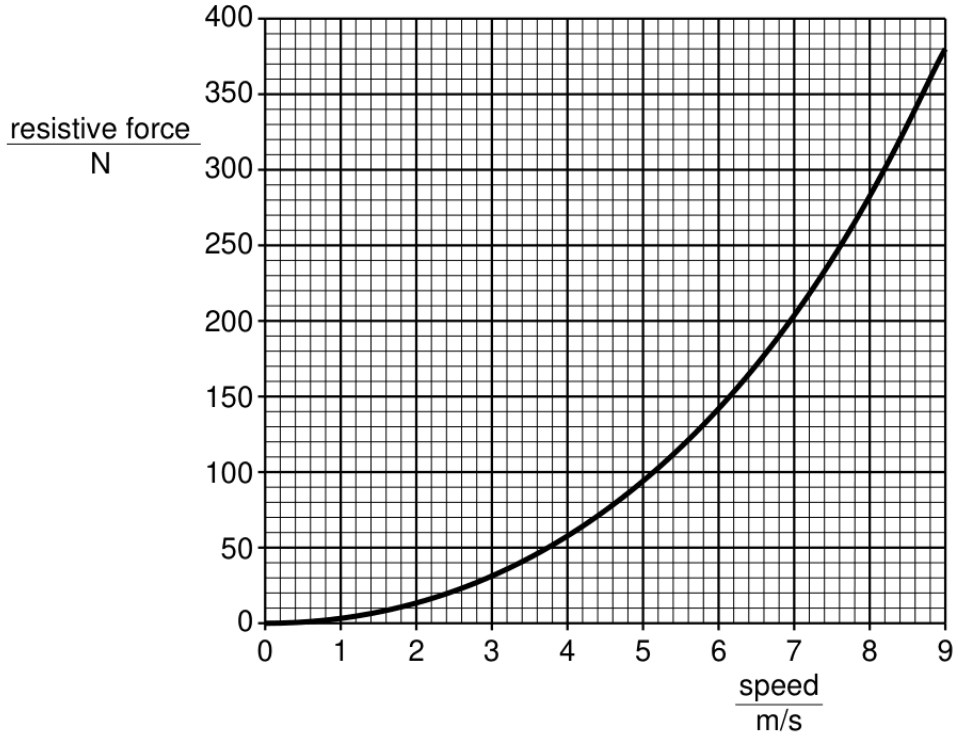


Fig. 3.2

Explain why the maximum speed that he can reach is 8.4 m/s.

.....

.....[1]

For Examiner's Use

(b) The mass of the pole-vaulter is 60 kg.

(i) Calculate the maximum kinetic energy of the pole-vaulter as he runs along the track. State clearly the formula that you use.

kinetic energy = ...

(ii) The pole is used to convert all this kinetic energy into gravitational potential energy. Calculate the height through which the pole-vaulter rises. Give your answer to an appropriate number of significant figures. The gravitational field strength is 10 N/kg.

height = ...
[6]

3 **(a)** (at 8.4 m/s) resistive force = 320 N/forward force **or** no resultant **or** forces cancel/balance

or if forward force > resistive force then runner accelerates

or if forward force < resistive force then runner decelerates **B1**

(**not** resistive force a maximum, **accept** backwards force = resistive force)

(b) **(i)** **½ mv^2** **B1**

½ x 60 x 8.4^2 **C1**

2100 J (accept 2120, 2117, 2116.8) **A1**

(ii) mgh **C1**

2117/60 x 10 or h = P.E. or K.E./mg e.c.f. **(i)** **C1**

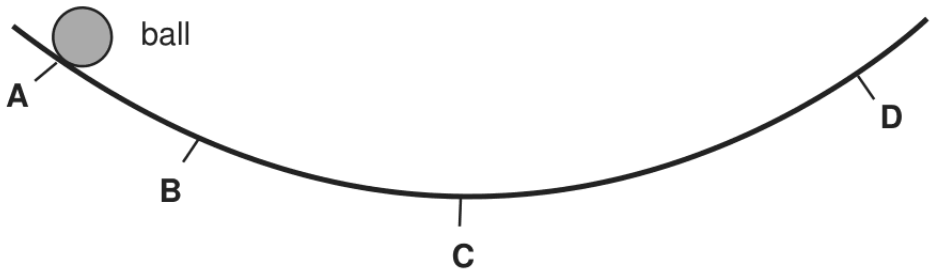
3.5 m

(accept 3.50, 3.52, 3.53 i.e. **2 or 3 significant figures** only) **A1** **7**

QUESTION · 5054_s05_qp_1.pdf · source page 5 of 20

10 The diagram shows a curved curtain rail that has a steel ball rolling on it. The ball is released at point **A**.

At which point does the ball have maximum kinetic energy?



MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 10

10 **C**

For Examiner's Use

4 Fig. 4.1 shows a hydroelectric power station. Water from the lake is used to produce electricity in the turbine house.

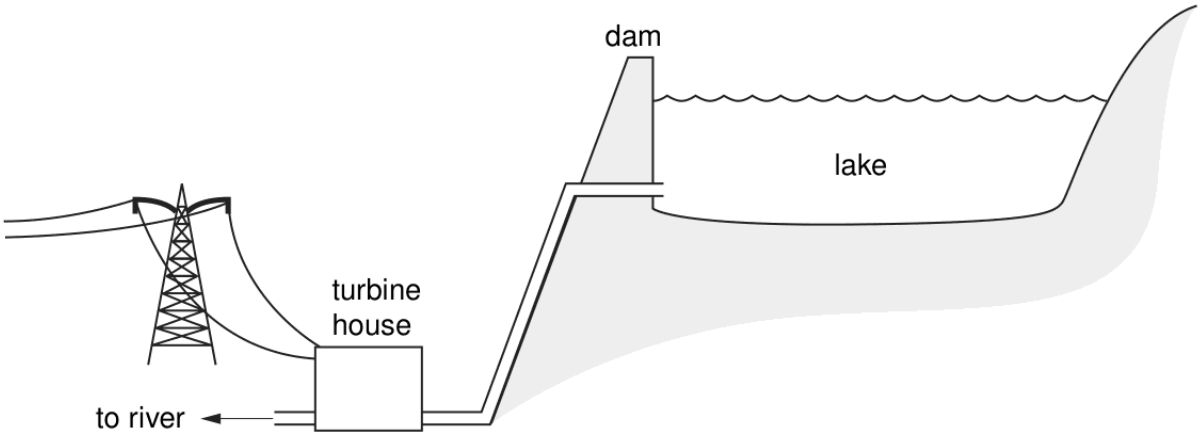


Fig. 4.1

(a) State where the water in Fig. 4.1 has the least potential energy.

..... [1]

(b) In 30 minutes, the water loses 5.0×10^9 J of energy and 4.5×10^9 J of electrical energy is produced in the turbine house.

(i) Calculate the efficiency of the energy conversion.

efficiency = ... [1]

(ii) Calculate, in watts, the electrical power output from the turbine house.

power = ... W [3]

(c) Some power stations burn coal to produce the same electrical power output. State one advantage of the hydroelectric power station.

.....

.....

..... [1]

(d) State one harmful effect that the hydroelectric power station may have on the environment.

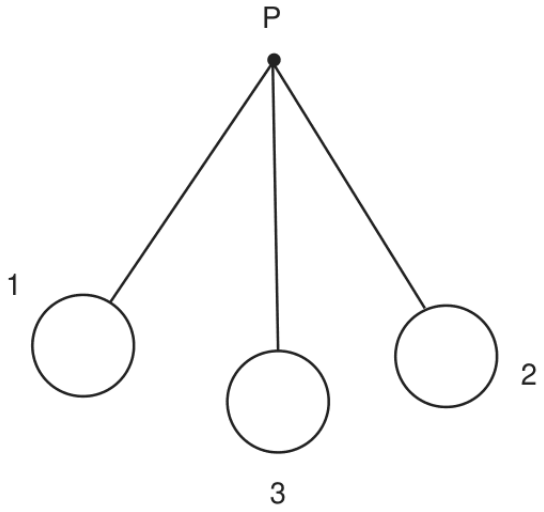
.....

..... [1]

4	**(a)**	in river/(emerging from or entering) turbine house	**B1**
	(b) **(i)**	0.9 **or** 90% **or** 0.47 **or** 47% (penalise unit error)	**B1**
	(ii)	P = E/t in symbols **or** any energy/any time	**C1**
		30 x 60 **or** 1800 seen	**C1**
		2.5×10^6 (W)	
		(150 or 2.78MW score 2/3)	**A1**
	(c)	any sensible suggestion e.g. no costs for water/energy supply **or** less pollution (accept coal produces smoke/dust/harmful gases/CO_2 **or** no need to transport coal **or** renewable **or** rapid response to power demand **or** less heat produced/more efficient	**B1**
	(d)	any sensible suggestion e.g. flooding **or** fish unable to pass **or** turbines kill fish **or** destroy habitats **or** less land **or** uses up large space **or** fells trees **or** unsightly/destroys scenery **or** lake/river silt up **or** more rain/evaporation	**B1** **7**

QUESTION · 5054_w05_qp_1.pdf · source page 6 of 20

11 A mass hangs on a string fixed at point P. It starts from position 1 and swings to the furthest position on the opposite side, position 2. It then oscillates several times with decreasing amplitude before ending at position 3.



Where does the ball have the most kinetic energy?

A at position 1

B at position 2

C the first time at position 3

D the last time at position 3

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 11

11	**C**

For Examiner's Use

5 This question is about the sound produced by a mobile phone and the energy changes while it is operating.

(a) The sound produced by a ringing phone consists of two notes, one after the other.

Fig. 5.1 shows the trace on an oscilloscope screen produced by the first of the notes.

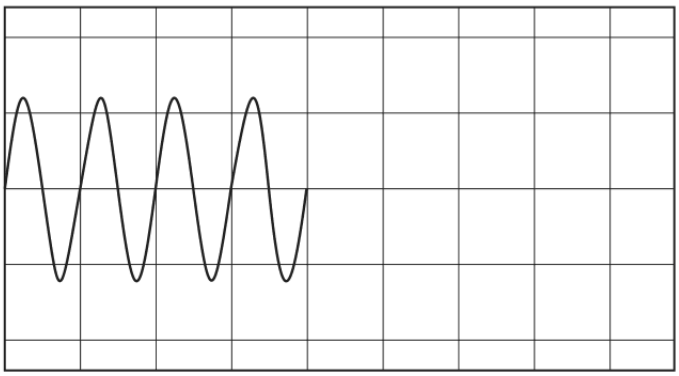


Fig. 5.1

The second note is louder and has a higher pitch.

(i) On Fig. 5.1, continue the trace to show what happens when the second note is sounding. [1]

(ii) Explain the differences between the two traces.

.....

.....

.....

.....

..... [2]

(b) The mobile phone contains a rechargeable battery.

Energy is stored inside the battery as it is charged.

Describe the useful energy change that takes place inside the battery as it is charged from the mains.

.....

.....

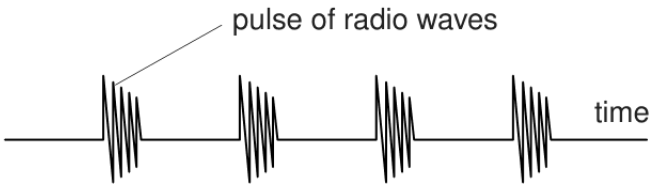
..... [2]

5	**(a)**	**(i)**	diagram with larger amplitude and shorter “wavelength”	**B1**	
		(ii)	louder means larger amplitude/height	**B1**	
			higher pitch means higher frequency/more waves on screen/shorter wavelength	**B1**	
	(b)		electrical at start	**B1**	
			chemical at end and a clear transformation without errors (allow 1 mark for chemical to electrical)	**B1**	**5**

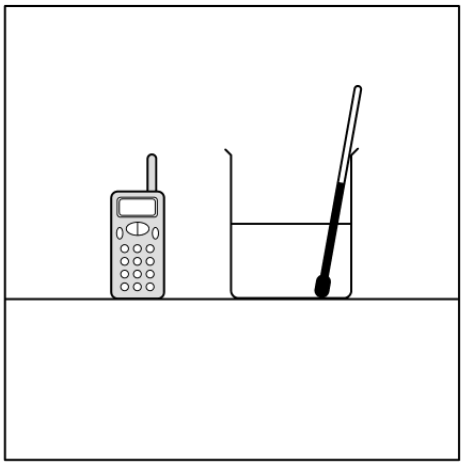
10 A student uses a mobile phone. He is concerned that the energy of the radio waves may cause a temperature rise in his brain. To investigate this effect, he calculates and tries to measure the heating effect of the phone on a nearby glass beaker of water.

Information from manufacturer

constant input power 0.20 W



The mobile phone emits pulses of radio waves.
Each pulse of waves has energy 0.00012 J.
There are 216 pulses of waves in 1.0 s



The phone is used for 360 s when next to a glass beaker containing 50 g of water.

(a) Calculate

(i) the number of pulses of radio waves produced during the phone call, [2]

(ii) the total energy of the radio waves emitted during the phone call, [2]

(iii) the maximum temperature rise produced in 50 g of water if all of the energy calculated in **(ii)** is absorbed by the water. The specific heat capacity of water is 4.2 J/(g °C). [3]

(b) The manufacturer gives the input power of the phone as 0.20 W.

(i) Calculate the energy supplied by the battery when the phone is used for 360 s. [2]

(ii) Calculate the fraction of the energy supplied by the battery that is converted into radio wave energy during the phone call. [1]

(c) The student uses the mercury-in-glass thermometer shown in Fig. 10.1.

He does not detect any temperature rise in the water in the beaker when the phone is used.

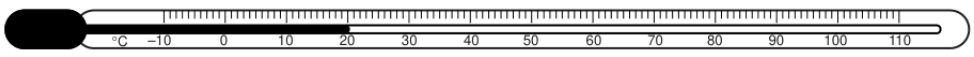


Fig. 10.1

(i) Describe how you would check the 0 °C and 100 °C points on the thermometer. [2]

(ii) Explain why the thermometer is not sensitive enough to detect the temperature rise. [1]

(iii) State and explain one change that will make a mercury-in-glass thermometer more sensitive. [2]

10	**(a)**	**(i)**	360 x 216	**C1**
			77 800 (no sig fig penalty)	**A1**
		(ii)	77 800 x 0.00012	**C1**
			9.33 J ecf **(i)**	**A1**
		(iii)	E = mcΔT in any form, algebraic or numerical	**B1**
			9.33/(50 x 4.2)	**C1**
			0.044 ºC ecf **(ii)**	**A1**
	(b)	**(i)**	E=Pt in any form, algebraic or numerical	**C1**
			72 J	**A1**
		(ii)	0.13 (accept 13%) no s.f. penalty	**A1**
	(c)	**(i)**	(pure) melting ice for 0° C	**B1**
			(pure) boiling water/steam above boiling water (at 1 atmosphere) for 100° C	**B1**
		(ii)	each division on thermometer is too small described in some way e.g. does not expand far up tube (**not** bore too thin, not enough mercury)	**B1**
		(iii)	**change** use more mercury **or** use smaller bore	**M1**
			reason more expansion **or** further distance up tube (for same expansion)	**A1**

QUESTION · 5054_s06_qp_1.pdf · source page 5 of 20

11 In a hydroelectric power station, water from a reservoir falls down a long pipe before entering the turbines. The turbines then turn the generator.

What is the **overall** energy conversion?

A electrical energy into kinetic energy

B electrical energy into potential energy

C kinetic energy into chemical energy

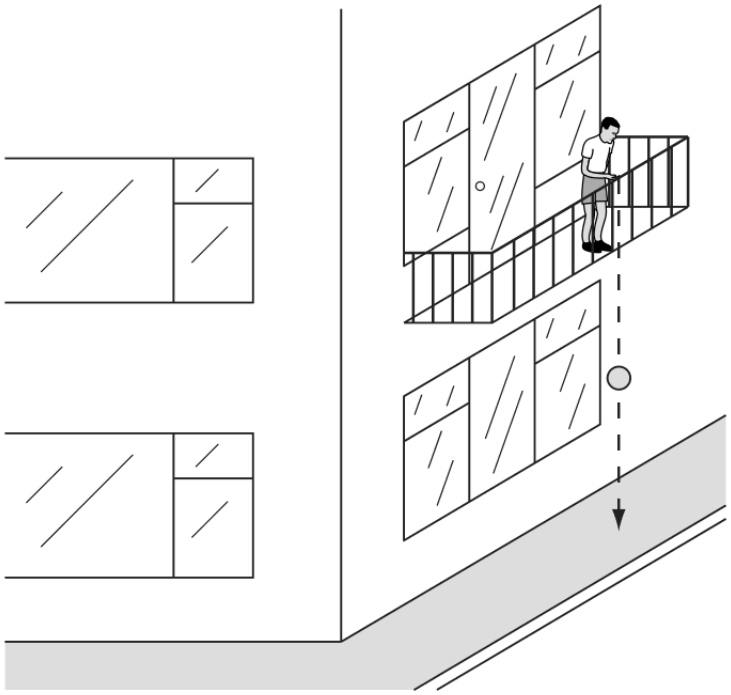
D potential energy into electrical energy

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 11

11	**D**

QUESTION · 5054_w06_qp_1.pdf · source page 4 of 16

8 A young child holds a ball over the edge of a balcony. The ball has gravitational potential energy. The ball is then released. It falls onto a concrete path below, and bounces back up.



Which sequence represents, in the correct order, the transformations of the gravitational potential energy after the ball is released?

A → elastic potential energy → kinetic energy → chemical potential energy

B → elastic potential energy → kinetic energy → gravitational potential energy

C → kinetic energy → elastic potential energy → kinetic energy

D → kinetic energy → gravitational potential energy → kinetic energy

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 8

8	**C**

For Examiner's Use

6 Fig. 6.1 shows a device used to generate electricity. Water entering at the top turns the wheel. A generator connected to the wheel produces electric current.

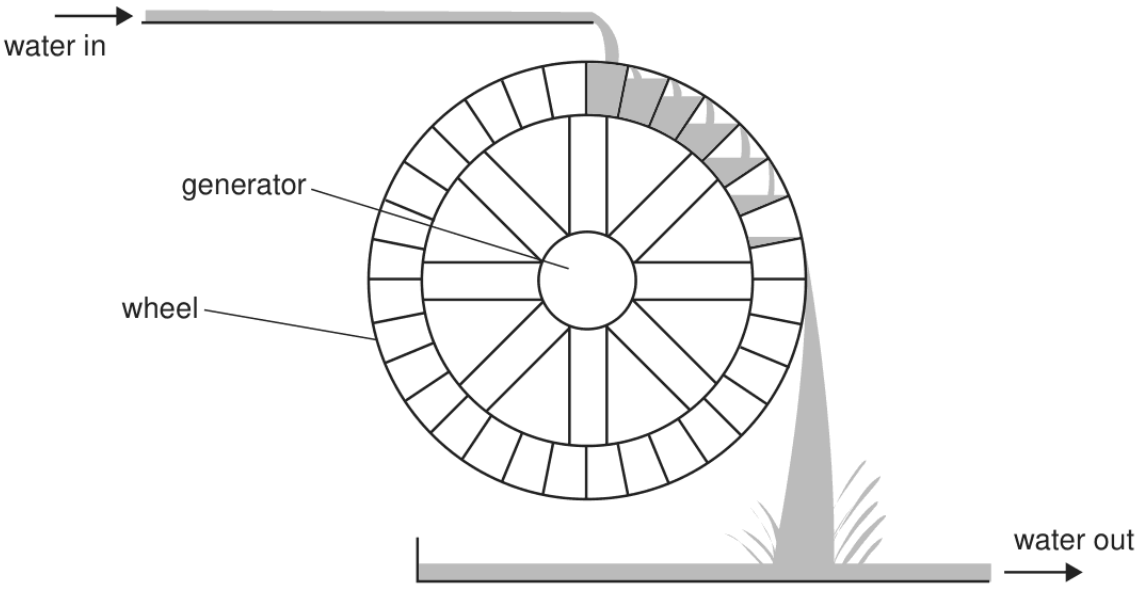


Fig. 6.1

(a) State the two main energy changes that take place in this device as water starts to flow.

1. .....

2. ..... [2]

(b) The change in the potential energy of the water every second is 2000 J. The electrical energy output is 1200 J every second.

Calculate the efficiency of the process.

efficiency = ... [2]

(c) State two ways in which the device wastes energy.

1. .....

2. ..... [2]

6	**(a)**	P.E. (of water) to K.E. (of wheel or water**)/**K.E. (of wheel) to electrical energy/ KE **of water** to KE **of wheel** /PE to electrical energy ANY 2 (-1 each clearly wrong answer beyond 2)	B2
	(b)	1200/2000 **or** energy output/ energy input **or** power output/power input (NOT output/input)	C1
		0.60 **or** 60% (NOT fractions; 0.6 YES)	A1
	(c)	friction in wheel or generator (bearings/axle) **or** water out has K.E. **or** produces heat in windings/in resistance **or** heat (in bearings) due to friction (ignore sound) ANY 2 (-1 each clearly wrong answer beyond 2)	B2 6

11 Stars that are being formed emit infra-red radiation. Some of this radiation is received by a telescope that orbits the Earth. Microwave signals from the telescope are sent to the Earth's surface, as shown in Fig. 11.1.

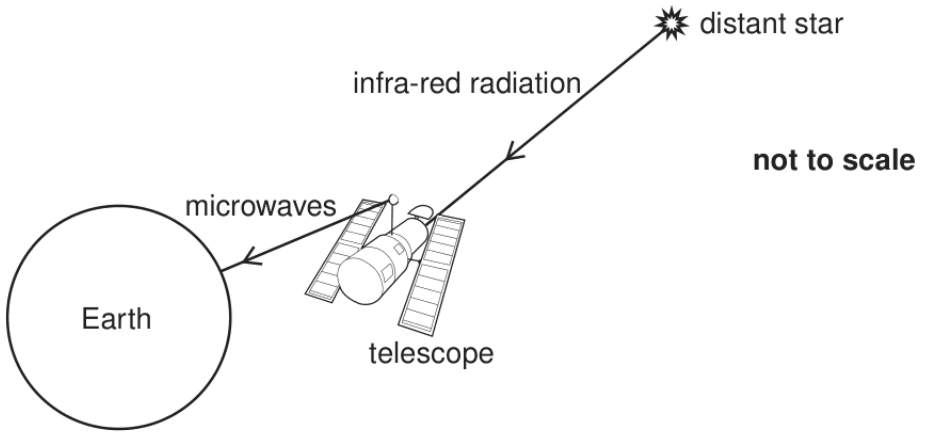


Fig. 11.1

(a) **(i)** Microwaves travel at a speed of 300 000 km/s. The telescope is 600 km above the Earth's surface. Calculate the time for a microwave signal from the telescope to reach the Earth's surface. [3]

(ii) Infra-red and microwave radiation are both part of the electromagnetic spectrum.

State **two** other similarities and give **one** difference between infra-red and microwave radiation. [3]

(b) As clouds of gas and dust come together to form a star, there is an energy change similar to the energy change as a ball falls to the ground.

The temperature in the clouds of dust and gas becomes so high that nuclear fusion occurs.

In one such reaction, two isotopes of hydrogen, ${}^{2}_{1}\mathrm{H}$ and ${}^{3}_{1}\mathrm{H}$, fuse together.

(i) Explain why the gas and dust come together. [1]

(ii) State the energy change that causes the rise in temperature as the gas and dust come together. [2]

(iii) Explain why high temperatures are needed for nuclear fusion to occur. [2]

(iv) Describe the structure of a nucleus of ${}^{3}_{1}\mathrm{H}$. [2]

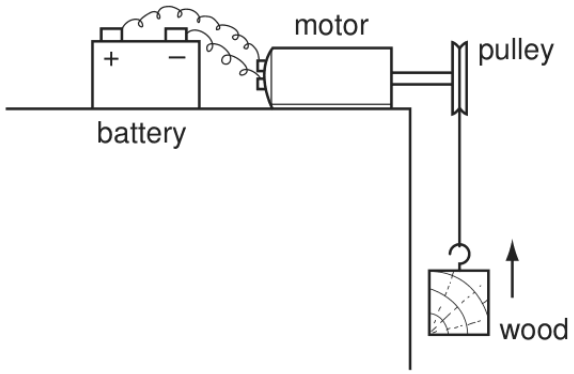
(v) State which element is created by the fusion of two hydrogen nuclei. [1]

(vi) State **one** effect of nuclear fusion on a star. [1]

11 (a)	**(i)**	d = speed × time in any format		C1
		600/300 000 **or** 600 000/300 000 000		C1
		0.002 s		A1
	(ii)	**similarities:**		
		same speed (in vacuum)		
		travel in a vacuum		
		travel in straight lines		
		refract/reflect/diffract/interfere		
		carry energy		
		transverse/polarisable ANY 2 (-1 each clearly wrong answer beyond 2)		B2
		(NOT both obey c = fλ/waves/invisible/undeflected by magnetic/electric field)		
		differences:		
		wavelength		
		frequency		
		microwave received by aerials ANY 1 line		B1
		(wavelength of IR different YES; wavelength of IR longer NO)		
(b)	**(i)**	gravity		B1
	b	potential energy to kinetic energy		B1
		kinetic energy to heat/thermal energy		B1
		OR potential energy to heat/thermal energy	**OR**	B2
		-1 each clearly wrong answer beyond 2		
	(iii)	nuclei repel **or** nuclei are positive		B1
		nuclei need high speed/ K.E. (so high temperature)		B1
	(iv)	1 proton **or** proton number = 1		B1
		2 neutrons **or** neutron number = 2 (electron(s) **max 1**)		B1
	(v)	He **or** helium		B1
	(vi)	energy/heat produced **or** raises temperature **or** becomes hot **or** causes star to expand **or** counters gravitational collapse **or** loses mass		B1 15

QUESTION · 5054_s07_qp_1.pdf · source page 6 of 20

9 The diagram shows a battery-operated motor lifting a block of wood at constant speed.



What is the overall energy change taking place?

A chemical → gravitational potential

B gravitational potential → electrical

C gravitational potential → kinetic

D kinetic → gravitational potential

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 9

9	**A**

Section B

Answer **two** questions from this section.

Use the lined page provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 Fig. 9.1 shows the horizontal forces acting on a moving car.

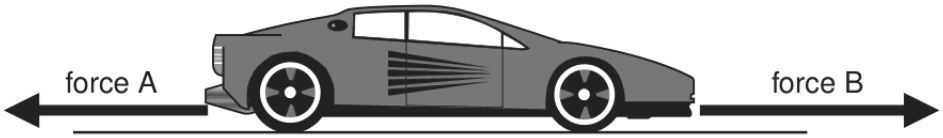


Fig. 9.1

(a) Compare the sizes and directions of the two forces when the car is

(i) moving along a straight road at constant speed, [1]

(ii) accelerating. [1]

(b) Suggest the direction of the resultant force on the car when the car turns a corner at constant speed. [1]

(c) Fig. 9.2 shows the speed-time graph for the first 24 s of the motion of the car along a straight road.

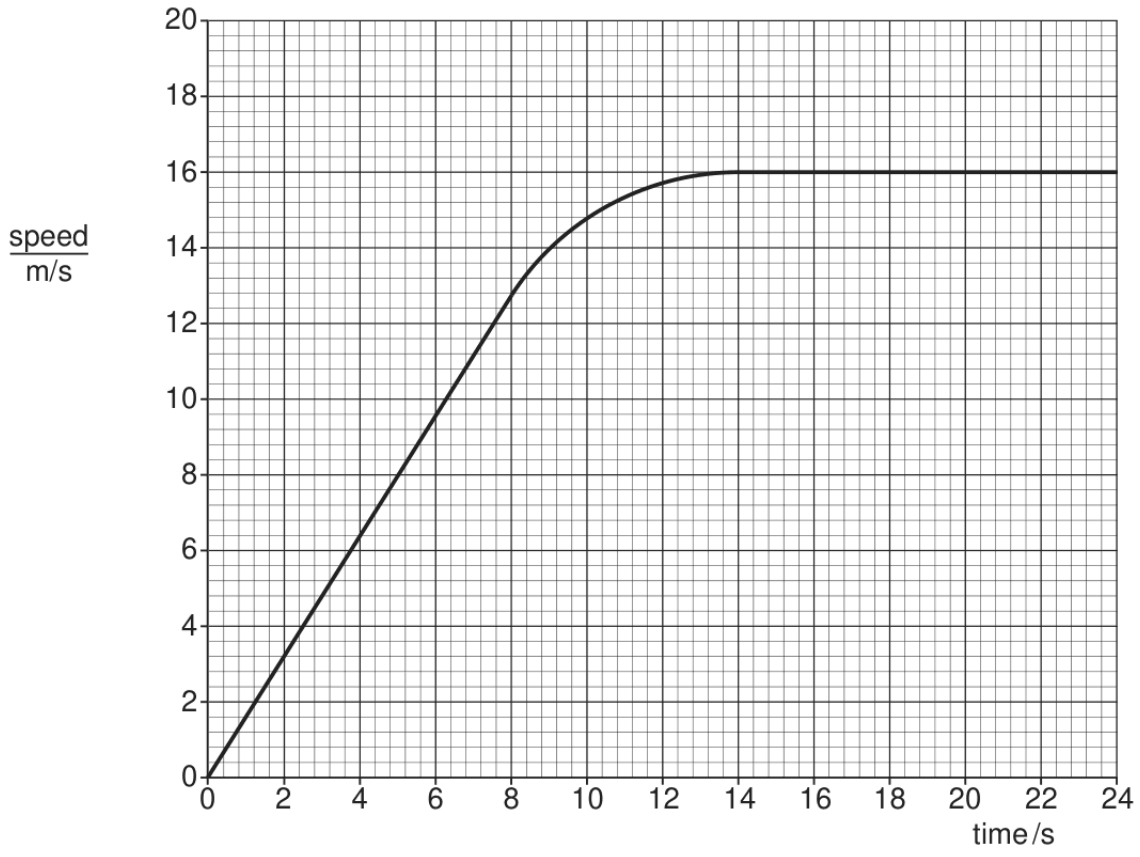


Fig. 9.2

(i) During the motion there is a period of uniform acceleration. State both the start and finish times of this period. [1]

(ii) State the main energy changes that occur in the 24 s. [3]

(iii) Calculate the acceleration of the car during the first 5 s. State clearly the equation that you use. [3]

(iv) Determine the distance travelled in the first 5 s. [3]

(d) The car is stopped by applying the brakes. Various factors can affect the distance travelled by the car during the time that the brakes are applied. Apart from the force applied by the brakes, state **two** of these factors. [2]

Section B

9	**(a)**	**(i)**	*A* = *B* (assume opposite direction and co-linear)	B1	
		(ii)	B > A (assume opposite direction and co-linear) maximum of 1 mark if directions wrong	B1	[2]
	(b)		towards **centre** of circle/corner	B1	[1]
	(c)	**(i)**	0 **and** 8–9 s	B1	
		(ii)	chemical (potential energy) (**accept** electrical if electrical car clear) **to** kinetic energy **or** K.E. increases thermal energy/heat/internal energy produced Max 2/3 if clear error	B1 B1 B1	
		(iii)	acceleration = (*v*–*u*)/t **or** gradient (7.8 to) 8/5 (**accept** any corresponding period e.g. 8s 12.6–12.8, 6s 9.4–9.6) 1.6 m/s^2 (**accept** 1.56–1.60)	C1 C1 A1	
		(iv)	area under graph **or** average speed 4 (m/s) **or** ½ 5 speed used in **(iii)** at 5 s ½ x (7.8 to) 8 x 5 20 m (**accept** 19.5–20; **ecf speed used in (iii)** at 5 s)	C1 C1 A1	[10]
	(d)		speed of car/friction with road (accept slippery road or ice or water or oil on road)/ friction in engine/tyre condition or area or pressure/air resistance/wind speed or direction/mass or inertia **of car or passengers**/slope of road	B2	[2]

QUESTION · 5054_w07_qp_1.pdf · source page 5 of 20

9 A 2 kg mass is moving at constant speed.

The kinetic energy of the mass is 400 J.

What is the speed of the mass?

A 0.4 m/s **B** 20 m/s **C** 200 m/s **D** 400 m/s

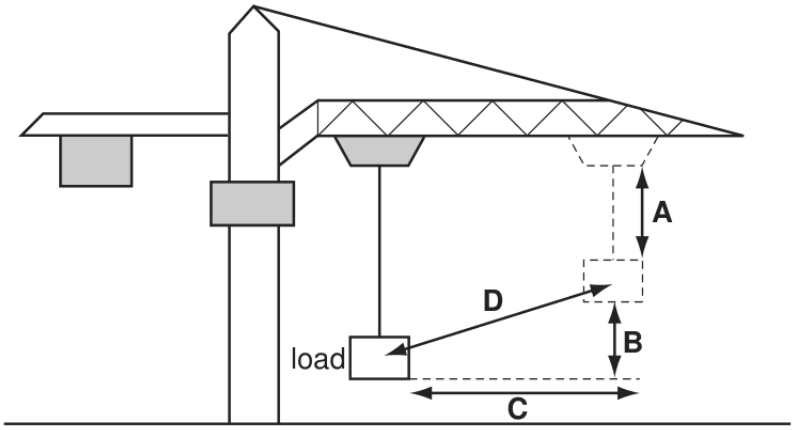
MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 9

9	**B**

QUESTION · 5054_w07_qp_1.pdf · source page 5 of 20

10 A crane moves its load diagonally, as shown.

By what distance is the weight of the load multiplied in order to calculate the increase in gravitational potential energy of the load?

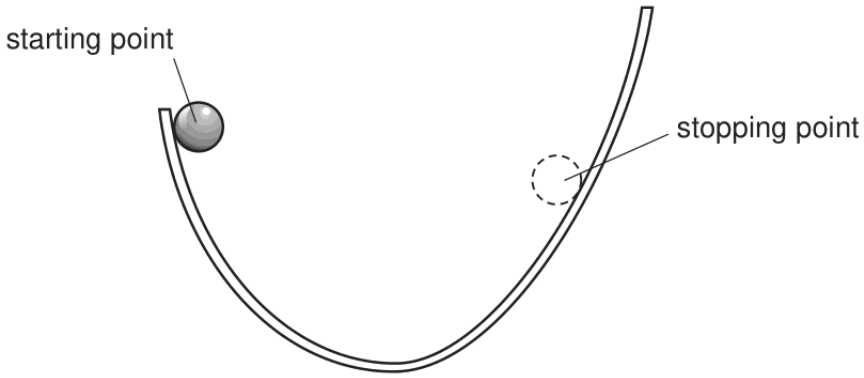


MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 10

10 **B**

QUESTION · 5054_s08_qp_1.pdf · source page 5 of 20

9 A ball is held at rest on one side of a curved track.



The ball is released. It rolls down one side of the track and part of the way up the other side. It then stops, before rolling back down again. The height of the stopping point is less than that of the starting point.

What is the sequence of energy changes between starting and stopping for the first time?

A potential energy → kinetic energy → potential energy + heat

B potential energy → kinetic energy → heat → potential energy

C potential energy → heat → kinetic energy → potential energy

D potential energy → kinetic energy + heat → potential energy + heat

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 9

9	**D**

Section B

Answer **two** questions from this section.

Use the lined pages provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 When a car driver sees an emergency ahead, he applies the brakes. During his reaction time the car travels at a steady speed and covers a distance known as the thinking distance. The braking distance is the distance travelled by the car after the brakes are applied.

(a) State the energy change that occurs as the car brakes. [2]

(b) Fig. 9.1 shows the speed-time graph of a car.
The driver sees the emergency at time $t = 0$.
The total mass of the car is 800 kg.

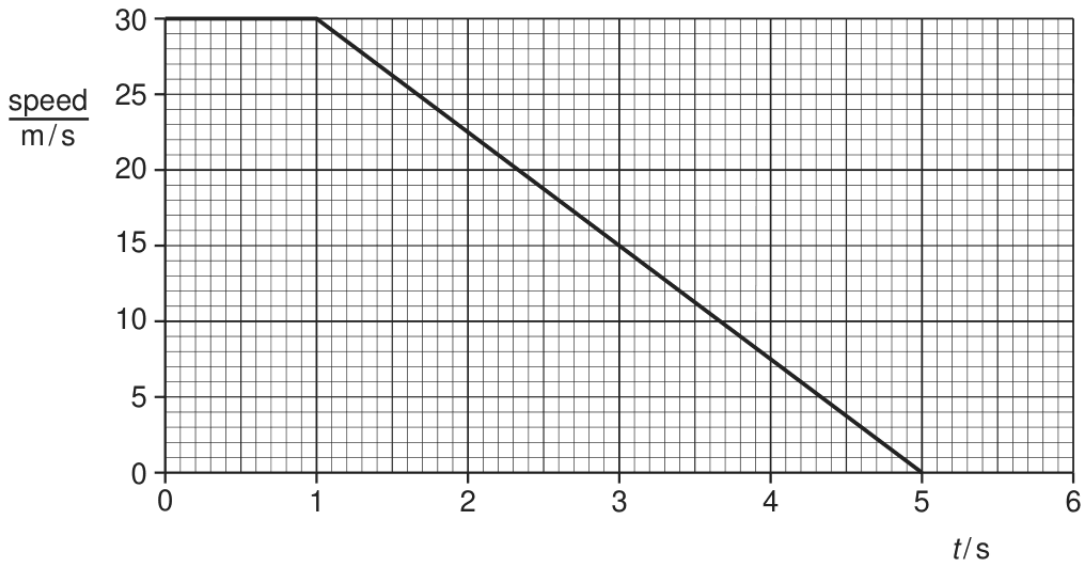


Fig. 9.1

Determine

(i) the thinking distance, [1]

(ii) the braking distance, [2]

(iii) the deceleration of the car during braking, [2]

(iv) the force provided by the brakes. [2]

(c) Using ideas about friction and deceleration, state and explain how the braking distance is affected by

(i) using new tyres rather than badly worn tyres, [2]

(ii) the car skidding on a wet road, [2]

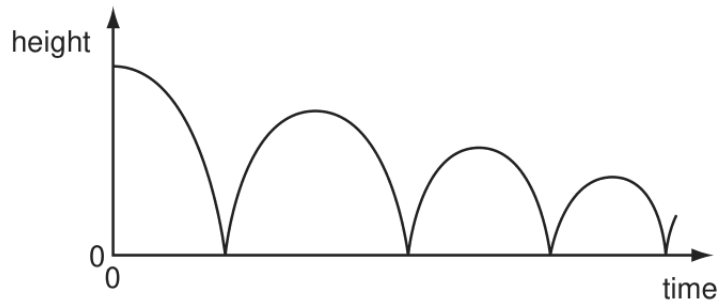
(iii) the car carrying a heavy load of passengers. [2]

Section B

9	**(a)**	K.E. (at start) to heat (+ sound)	B1 B1
			[Total: 2]
	(b) **(i)**	30 m cao	B1
	(ii)	area under graph **or** average speed × time **or** (u + v).t / 2 **or** 30 × 4/2 60 m	C1 A1
	(iii)	$(v–u)/t$ **or** $v = u + at$ **or** 30/4 **or** gradient **or** rise/run 7.5 (±0.1) m/s^2	C1 A1
	(iv)	$F = ma$ **or** 800 × **(iii)** 6000 N ecf **(iii)**	C1 A1
			[Total: 7]
	(c) **(i)**	more friction/grip/traction **or** more deceleration **or** decelerates faster **or** decelerates in less time less (braking) **distance**	 B1 B1
	(ii)	less friction **or** less deceleration **or** decelerates slower/longer more (braking) **distance**	B1 B1
	(iii)	less deceleration **or** decelerates slower/longer more **distance**	B1 B1
			[Total: 6]

QUESTION · 5054_w08_qp_1.pdf · source page 6 of 20

10 The graph shows how the height above the ground of a bouncing ball changes with time.



Which statement explains why the height of each peak decreases with time?

A Kinetic energy is converted to potential energy at each bounce.

B Kinetic energy is converted to thermal energy at each bounce.

C The ball gains energy on impact with the floor.

D The ball is wearing out.

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 10

10	**B**

For Examiner's Use

3 A river flows over a cliff, producing a waterfall. The water, at a temperature of 7.2 °C, falls 700 m into a pool. The gravitational field strength is 10 N / kg.

(a) Calculate the change in the gravitational potential energy of each kilogram of water due to its fall.

energy change = ... [2]

(b) Assume that all of this energy is converted into thermal energy (heat) in the water in the pool. Calculate the temperature of this water when it is in the pool. The specific heat capacity of water is 4200 J / (kg °C).

temperature = ... [3]

3	**(a)**	*mgh* **or** *F* × *d* **or** 10 × 700	C1	
		(–)7000 J	A1	
	(b)	*Q/E/H* = *mc*Δ*T* **or** (Δ*T* =) 7000/(1) × 4200	C1	
		1.7 **or** 1.67 **or** 5.5	C1	
		8.9 °C e.c.f. **(a)**	A1	[5]

Section A

For Examiner's Use

Answer **all** the questions in this section.

1 A piece of paper falls from 4.0 m above the ground.
Fig. 1.1 shows how the height h above the ground varies with the time t.

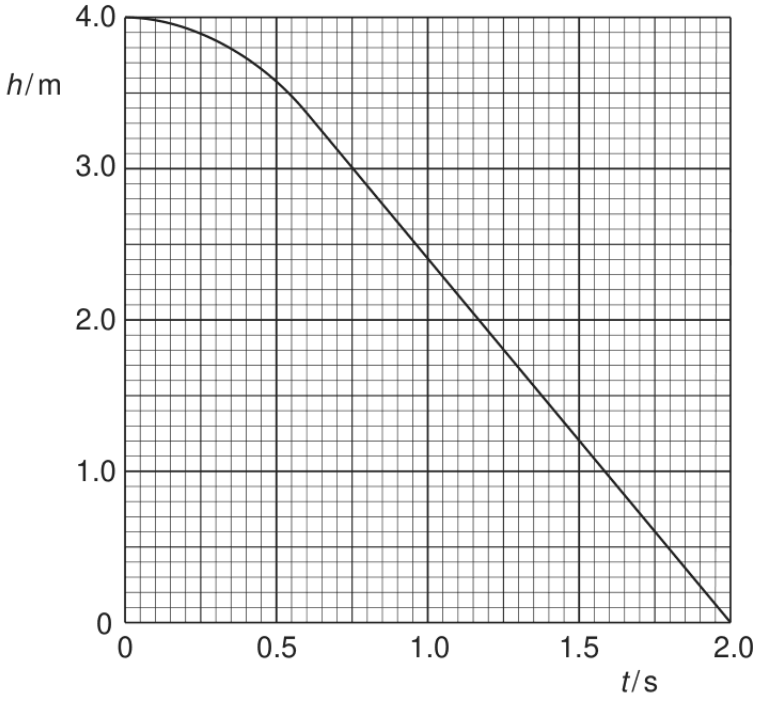


Fig. 1.1

(a) State what happens to the speed of the paper as it falls.

.....

.....

..... [2]

(b) Calculate the speed of the paper at time $t = 1.5$ s.

speed = ... [2]

(c) As the paper falls, energy changes from one form to another.
State the main energy change between $t = 1.0$ s and $t = 2.0$ s.

.....

.....

..... [2]

Section A

1	**(a)**	(speed) increases or (paper) accelerates	B1	
		(speed) becomes constant/uniform or acceleration zero (after 0.5 s)	B1	
	(b)	any clear change in distance/time or 1.87 (m/s) (allow 1.9)	C1	
		2.3–2.5 m/s	A1	
	(c)	PE at beginning of a change	B1	
		heat/internal energy/thermal energy at end of a change/K.E. of air	B1	[6]

10 Fig. 10.1 shows the horizontal forces as a cyclist travels forwards.

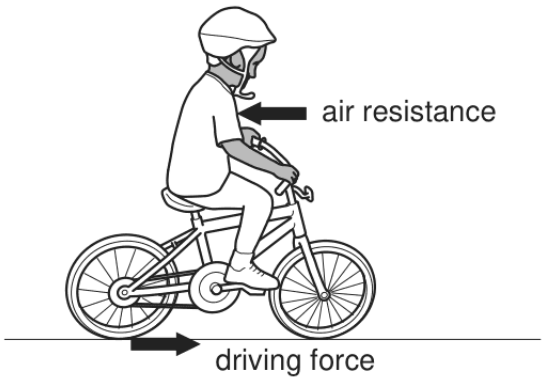


Fig. 10.1

The cyclist produces the driving force that acts on the back wheel.
In this question, you may ignore any frictional force acting on the front wheel.

(a) The bicycle accelerates until a constant speed is reached.

(i) Describe how the size of the air resistance changes during this time. [2]

(ii) Compare the sizes of the two horizontal forces when the bicycle is accelerating. [1]

(b) The total mass of the bicycle and the cyclist is 75 kg. At one instant, the speed of the bicycle is 4.0 m/s, the driving force is 30 N and the air resistance is 20 N.

Calculate

(i) the total kinetic energy of the bicycle and the cyclist, [3]

(ii) the acceleration of the bicycle and the cyclist. [2]

(c) As the bicycle moves, energy is transmitted from the pedals to the back wheel.

Fig. 10.2 shows what happens to the energy input to the pedals.

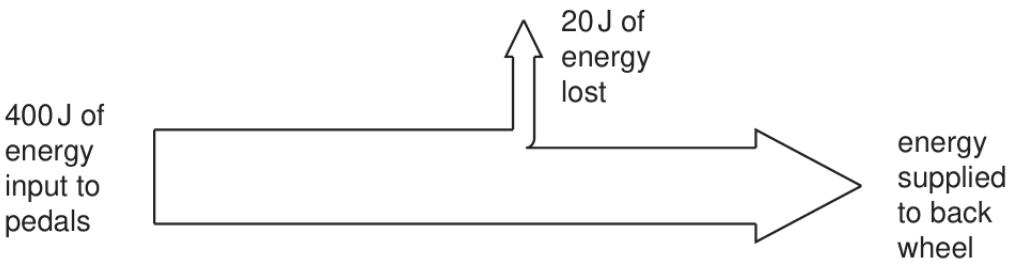


Fig. 10.2

(i) As energy is transmitted to the back wheel, some is lost. Explain how this happens. [2]

(ii) Calculate the efficiency of the bicycle in transmitting energy from the pedals to the back wheel. [2]

(d) Some bicycles are made from low density materials. Explain why this is an advantage. [3]

10 (a) (i) air resistance increases (as speed increases) B1
(at constant speed) becomes equal to driving force/applied force etc. B1

(ii) driving force (forward force) larger (than air resistance/backwards force) B1

(b) (i) (E =) ½ mv^2 algebraic formula C1
½ × 75 × 4^2 C1
600 J A1

(ii) (a =) F/m algebraic seen or 10 (N) used as force C1
0.13 m/s^2 A1

(c) (i) friction (in chain/axles) or rubbing of surfaces B1
heat or thermal energy produced B1

(ii) (efficiency = useful) energy output/energy input algebraic or numerical or 380 seen C1
0.95 or 95% A1

(d) lower mass/weight of cycle B1

less force needed	same acceleration/get up hill/to stop cycle/lift cycle	
greater acceleration/easier to acc.	for same force or $F = ma$ quoted	
less energy/work (input)	to go uphill/due to less friction/½ mv^2 less	
less kinetic energy	less stopping distance/less force to stop	
more efficient/less energy wasted	going uphill/less friction	
less friction	easier turn handlebars/higher (top) speed	
less pressure (on ground)	sinks less into ground	M1A1 [15]

10 A space research organisation plans to send astronauts to Mars to examine rocks on its surface. The organisation will produce a report containing information about conditions on Mars.

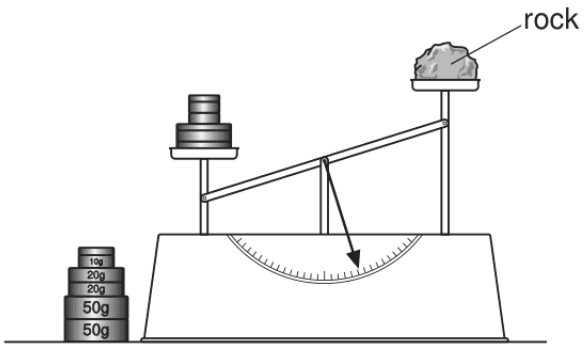
(a) (i) The gravitational field strength on the surface of Mars is 3.7 N/kg. Calculate the weight, on Mars, of a rock of mass 0.50 kg. [2]

(ii) A rock dropped on Mars falls to the surface. State the acceleration of the falling rock. Assume that there is no air resistance on Mars. [1]

(iii) Calculate the kinetic energy of a 0.50 kg rock as it strikes the surface at a speed of 3.2 m/s. [3]

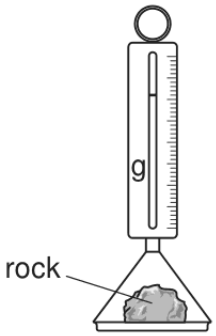
(b) While still on the surface of Mars, the astronauts will measure the mass of each rock collected. Fig. 10.1 shows two devices for measuring mass.

device A



A lever arm balance and a set of brass discs. The mass of each disc is accurately determined on Earth and the value is marked on it.

device B



A spring balance. The spring balance is accurately calibrated on Earth and the mass values are marked on the vertical scale.

Fig. 10.1

(i) When the two devices are used on Mars, they will give different readings for the mass of the same rock. Explain why. [3]

(ii) State which device will give the correct reading for the mass. [1]

(c) The astronauts will also determine the density of each rock.

(i) Apart from the mass of the rock, state what other information is needed in order to calculate the density of the rock. [1]

(ii) Describe a method for determining the density of a small, irregularly shaped rock of known mass *m*. [4]

10	**(a)**	**(i)**	(W =)mg **or** 0.5 × 3.7	C1	
			1.8/1.85/1.9 N	A1	
		(ii)	3.7 m/s^2 **not** N/kg	B1	
		(iii)	(KE =) ½mv^2	C1	
			½ × 0.50 × 3.2^2	C1	
			2.6 **or** 2.56 J	A1	[6]
	(b)	**(i)**	A compares/measures (unknown and known) masses/amount of matter	B1	
			B measures/is dependent on weight/force of gravity (and hence mass obtained)	B1	
			Mars weights/forces of gravity are less than/different from (Earth)	B1	
		(ii)	A **or** lever arm balance **or** balance with discs	B1	[4]

	(c)	**(i)**	volume	B1	
		(ii)	**Either:**	**Or:**	
			record value of water in measuring cylinder (**not** beaker)	water/liquid in displacement/eureka can	
			insert rock	full to overflowing	
			record new value	immerse rock	
			subtract (to obtain volume) **or** measure rise)	(new) overflow in measuring cylinder (**not** beaker is volume)	
			m/volume **or** *m*/subtraction (max 4)	*m*/volume **or** *m*/subtraction (max 4)	B4 [5]

[Total: 15]

QUESTION · 5054_s10_qp_11.pdf · source page 6 of 20

12 A parachutist has opened his parachute and is falling to Earth at constant speed.

What is the principal energy conversion taking place as he falls?

A kinetic energy → potential energy

B kinetic energy → thermal energy (heat)

C potential energy → kinetic energy

D potential energy → thermal energy (heat)

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 12

12	**D**

For Examiner's Use

3 Fig. 3.1 shows a student rubbing her hands together.

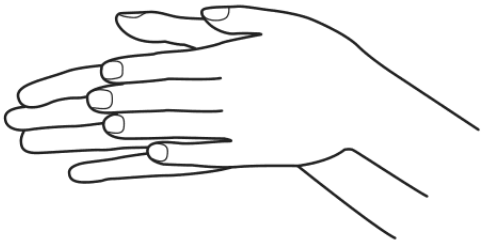


Fig. 3.1

(a) State the main energy conversion that causes the hands to become warm.

.....

.....[2]

(b) State why the hands become even warmer if they are pressed harder together when rubbing.

.....

.....[1]

(c) The average force used to slide one hand along the other is 1.2 N.

In each movement, one hand moves 0.080 m. The other hand remains stationary.

(i) Calculate the number of movements needed for 2.0 J of work to be done.

number = ...[3]

(ii) Each movement takes 0.20 s. Calculate the average power developed.

power = ...[2]

3	**(a)**	chemical energy to **or** K.E to	B1	
		heat/thermal energy/internal energy – at end	B1	[2]
	(b)	friction/resistive force **increases**	B1	[1]
	(c) **(i)**	work = force × distance in words, number or symbols	C1	
		1.2 × 0.08 **or** 0.096 **or** 20 **or** 0.208 seen	C1	
		20.8 **or** 21	A1	[3]
	(ii)	power = work/time **or** energy/time, numerical or algebraic – may use values in **(i)**		
		accept 2/0.2 or 10 **W**	C1	
		0.48 W (ecf **(i)** including power of ten error in **(i)**)	A1	[2]

QUESTION · 5054_s10_qp_12.pdf · source page 6 of 20

13 A parachutist has opened his parachute and is falling to Earth at constant speed.

What is the principal energy conversion taking place as he falls?

A kinetic energy → potential energy

B kinetic energy → thermal energy (heat)

C potential energy → kinetic energy

D potential energy → thermal energy (heat)

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 13

13	**D**

For Examiner's Use

2 Fig. 2.1 shows a motor used to operate a lift (elevator). There is a man in the lift.

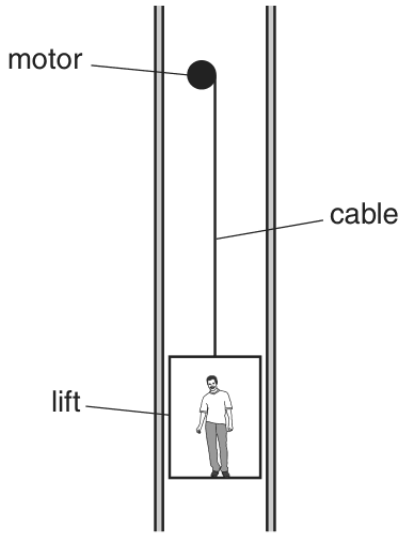


Fig. 2.1 (not to scale)

The input power to the motor is 6200 watts.

(a) Define the *watt*.

.....

.....[2]

(b) The lift and the man have a total mass of 580 kg. The lift moves up a distance of 12 m in 15 s.

(i) Calculate the tension in the cable when the lift is moving at constant speed.

tension = ...[1]

(ii) Calculate the increase in potential energy of the lift and man.

potential energy = ...[2]

(iii) Calculate the efficiency of the motor.

efficiency = ...[2]

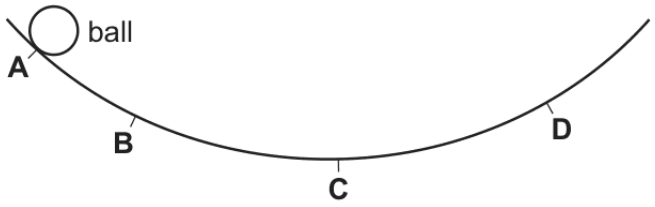
2	**(a)**	energy/time	C1	
		1 joule in one second	A1	[2]
	(b)	**(i)** 5800 N or 5684 N or 5700 N	B1	[1]
		(ii) *mgh* algebraic, words or numerical (i.e. 580 ×10 × 12)	C1	
		69 600 J or 70 000 J or 68 208 J or 68 000 J	A1	[2]
		(iii) (efficiency =) output power or energy/input power or energy algebraic or numerical or 93 000 seen or 4 640 seen	C1	
		0.75 or 75% (accept 0.748) e.c.f. from (ii)	A1	[2]

[Total: 7]

QUESTION · 5054_w10_qp_11.pdf · source page 8 of 20

13 The diagram shows a curved track. A ball is released from the position shown.

At which point does the ball have the maximum gravitational potential energy?



MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 13

13	**A**

For Examiner's Use

3 **(a)** State the *principle of conservation of energy*.

.....

.....

..... [2]

(b) Fig. 3.1 shows a miner using a head torch powered by a battery pack. The battery pack contains the cells that power the filament lamp in the head torch.

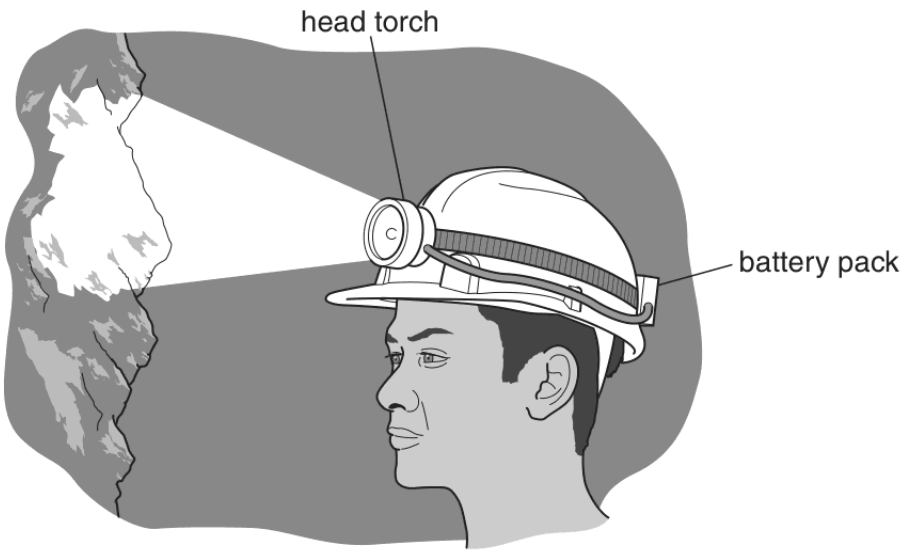


Fig. 3.1

(i) State the main energy change that takes place in the cells.

.....

..... [2]

(ii) State two different forms of energy that are produced in the filament lamp.

1. .....

2. .....

[2]

(c) An identical battery pack powers a different head torch that uses light-emitting diodes (LEDs) instead of the filament lamp. The two torches are equally bright but the efficiency of the torch using the LEDs is much higher.

State how the amounts of the different forms of energy produced in the LEDs compare with the amounts produced in the filament lamp.

.....

.....

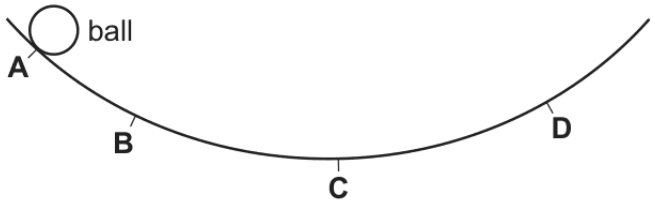
..... [2]

3	**(a)**	energy cannot be created/destroyed (nb. only one required)		B1	
		energy cannot be destroyed or created (i.e. the other one as well) **or** (merely) transformed **or** total energy in an isolated system is constant		B1	[2]
	(b) **(i)**	chemical (potential) **at beginning**		B1	
		to electrical (and heat) **at end**	others present: max 1	B1	[2]
	(ii)	light		B1	
		heat/thermal/internal	others present: max 1	B1	[2]
	(c)	less heat; same light		B2	
		or less chemical/electrical; less heat		B2	
		or less chemical/electrical; same light		B2	[2]

QUESTION · 5054_w10_qp_12.pdf · source page 8 of 16

20 The diagram shows a curved track. A ball is released from the position shown.

At which point does the ball have the maximum gravitational potential energy?



MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 20

20	**A**

QUESTION · 5054_s11_qp_11.pdf · source page 6 of 16

12 The diagram represents parts of a power station.

What is the order of the energy changes taking place?

A chemical → heat → kinetic → electrical

B chemical → kinetic → heat → electrical

C heat → chemical → kinetic → electrical

D kinetic → chemical → heat → electrical

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 12

12	**A**

QUESTION · 5054_s11_qp_12.pdf · source page 5 of 16

10 The diagram represents parts of a power station.

What is the order of the energy changes taking place?

A chemical → heat → kinetic → electrical

B chemical → kinetic → heat → electrical

C heat → chemical → kinetic → electrical

D kinetic → chemical → heat → electrical

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 10

10	**A**

2 Fig. 2.1 illustrates the journey of a cyclist from point A to point B. Points A and B are at the same height.

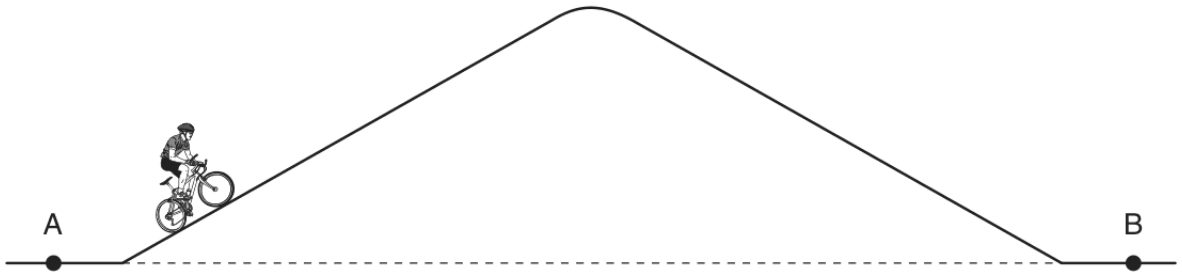


Fig. 2.1

The cyclist starts from rest at A and pedals up and over a hill. Near the bottom of the hill, she starts to brake and comes to rest at B.

(a) Describe the energy changes that take place as she pedals up the hill at constant speed.

.....

.....

.....[3]

(b) Explain how the law of conservation of energy applies to the complete journey from A to B.

.....

.....

.....[1]

(c) At one point in the journey, the gravitational potential energy of the cyclist has increased by 5400 J. The mass of the cyclist is 60 kg. The gravitational field strength is 10 N/kg.

Calculate the height above A of the cyclist at this point.

height above A = ...[2]

2	**(a)**	chemical (potential) energy at start	B1	
		gravitational/potential energy increases	B1	
		thermal energy/heat/internal energy produced	B1	
	(b)	energy not created/lost/destroyed // energy **only** changes form // total energy constant **and** at least one attempt to explain a conversion **in the journey** // all ends up as heat	B1	
	(c)	(h =) PE/mg numerical or algebraic seen, e.g. 5400/10 × 60	C1	
		9(.0) m	A1	[6]

11 **(a)** In an experiment to measure the specific heat capacity of water, an electric heater heats water in a glass beaker. The temperature of the water is measured at regular intervals of time. Fig. 11.1 shows how the temperature varies with time *t*.

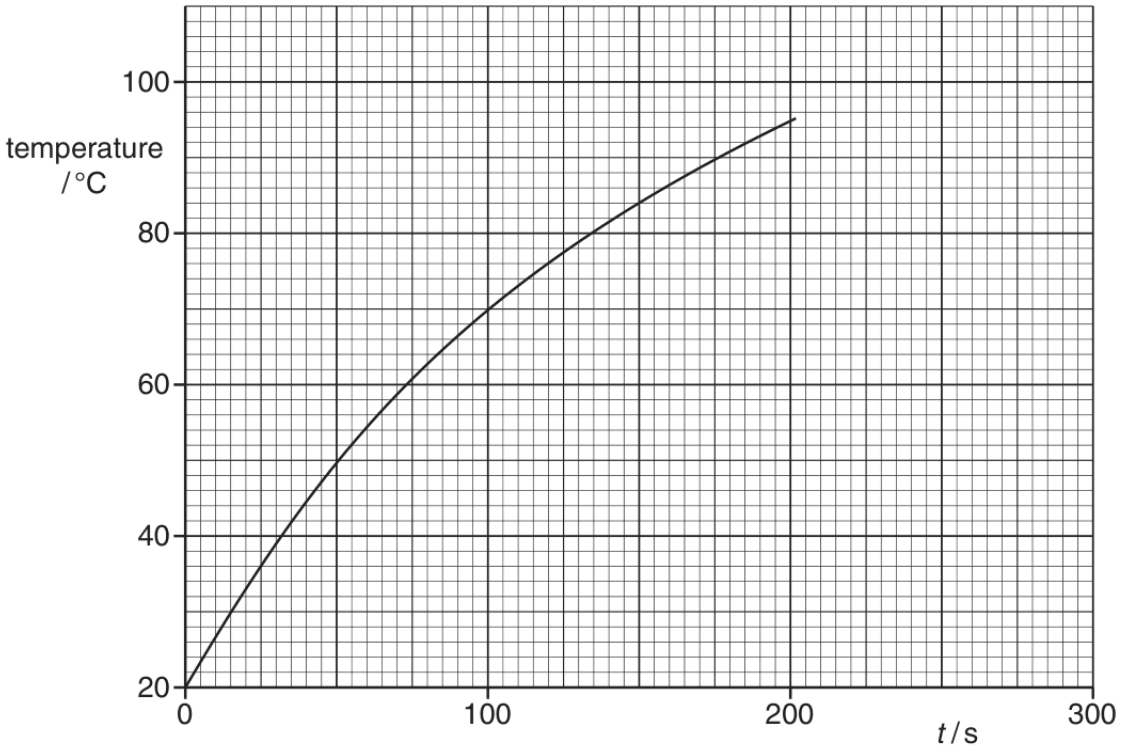


Fig. 11.1

(i) Use Fig. 11.1 to determine the change in temperature between

$t = 0$ and $t = 100$ s,

change = ...

$t = 100$ s and $t = 200$ s.

change = ...
[1]

(ii) State and explain why the values in **(i)** are different.

.....

.....

.....[2]

(iii) Describe and explain what happens to the water if the heating is continued.

.....

.....

.....[2]

(b) (i) The experiment in **(a)** is repeated using 72 g of water. The heater supplies 7400 J of thermal energy (heat) to the water and the temperature rise of the water is 23 °C.

Calculate the specific heat capacity of water.

specific heat capacity = ...[2]

(ii) A bullet of mass 72 g is fired from a gun at a speed of 450 m / s.

Calculate the kinetic energy of the bullet.

energy = ...[3]

(iii) The amount of internal energy gained by the water and the amount of kinetic energy gained by the bullet are approximately equal.

Describe the change in the motion of the molecules of the water and of the molecules of the bullet that this addition of energy has caused.

water: .....

.....

bullet: .....

.....

[3]

(c) A thermocouple is used in the experiment in **(a)**. In the space below, draw a labelled diagram of a thermocouple thermometer. Show clearly the part of the thermocouple that is placed in the water in this experiment.

[2]

11 (a) (i) 50°C **and** 24/25°C B1

(ii) heat loss or evaporation mentioned // molecules escape // liquid to vapour C1
more heat loss or **more** evaporation etc. **because** temperature is higher A1

(iii) temperature becomes 100°C // reaches boiling point // temperature becomes steady B1
water boils // water turns to steam/gas // energy loss = energy gain B1

(b) (i) (c =) $E/m\Delta T$ in any form, numerical or algebraic, e.g. 7400/72 × 23 C1
4.5 or 4.47 or 4.4686 J/(g °C) // 4468.6 J/(kg °C) A1

(ii) (E =) ½ mv^2 algebraic only C1
½ . 0.072. 450^2 C1
7300 J // 7290 J (7 290 000 (J) alone gets 2/3) A1

(iii) *water molecules* move/vibrate fast(er)/(more) vigorously B1
water molecules **random** motion // move (more) throughout liquid/all directions // slide over each other // move in convection // hit more often // move further apart B1
bullet molecules motion in one direction/away from gun/towards target/ all have same (increase in) speed B1

(c) two different metals at any junction/two outside wires if three used B1
joined wires connected to meter/voltmeter/ammeter/galvanometer B1 [15]

QUESTION · 5054_w11_qp_11.pdf · source page 6 of 20

11 Coal is burned as fuel to heat water in a boiler, producing steam. The steam drives a turbine, which is connected to an electric generator.

In which order do the major energy transformations take place?

A chemical energy → heat energy → electrical energy → kinetic energy

B chemical energy → heat energy → kinetic energy → electrical energy

C heat energy → chemical energy → electrical energy → kinetic energy

D heat energy → chemical energy → kinetic energy → electrical energy

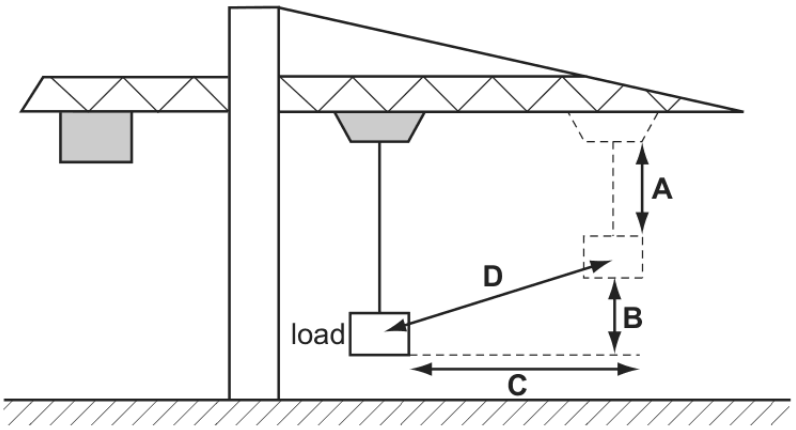
MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 11

11	**B**

QUESTION · 5054_w11_qp_11.pdf · source page 6 of 20

12 A crane moves its load diagonally, as shown.

By which distance is the weight of the load multiplied to calculate the change in gravitational potential energy of the load?



MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 12

12 **B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 is the speed-time graph for a rocket from the moment that the fuel starts to burn at time $t = 0$.

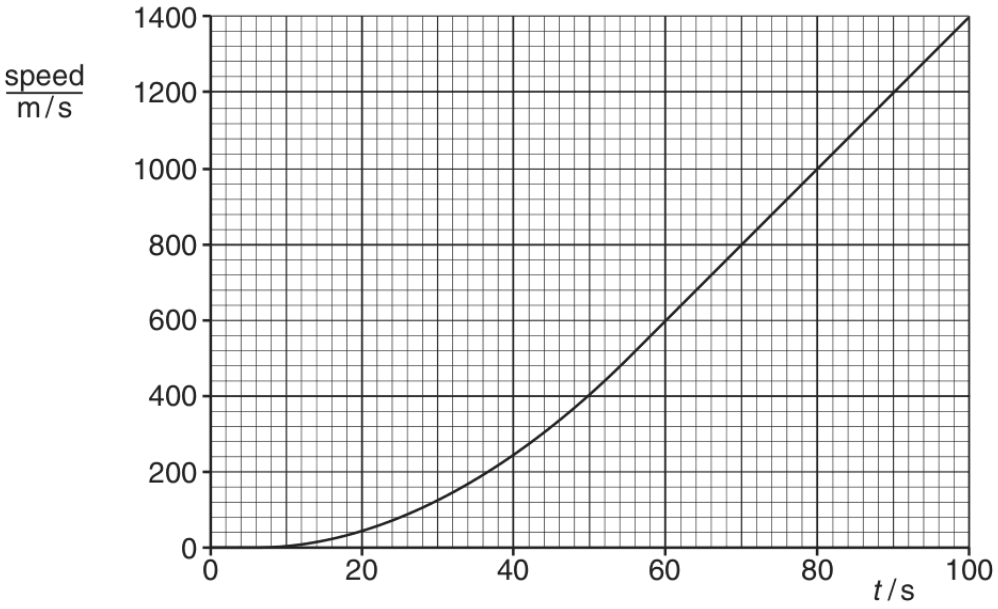


Fig. 9.1

(a) State the principal energy changes taking place as the rocket accelerates upwards.

.....

.....

.....

.....[4]

(b) **(i)** State the size of the acceleration of the rocket at $t = 0$.

.....[1]

(ii) State what happens to the acceleration of the rocket between $t = 5\,s$ and $t = 80\,s$.

.....

.....[2]

(iii) Calculate the acceleration of the rocket at $t = 80\,s$.

acceleration = ...[2]

(iv) The total mass of the rocket at $t = 80\,s$ is $1.6 \times 10^6\,kg$. Calculate the resultant force on the rocket at this time.

resultant force = ...[2]

(v) The total weight of the rocket at $t = 80\,s$ is $1.6 \times 10^7\,N$. Calculate the upward force on the rocket at this time, caused by the burning fuel.

upward force = ...[1]

(c) As the rocket burns fuel, it ejects hot gas downwards.

(i) State *Newton's third law of motion*.

.....

.....[1]

(ii) Explain how Newton's third law of motion applies to the upward force on the rocket and to the force on the hot gas.

.....

.....

.....[2]

Section B

9	(a)	**at start:** chemical (potential) energy	B1	
		at end: PE/GPE/gravitational energy	B1	
		at end: heat/thermal/internal energy	B1	
		at end: KE **or** intermediate KE from chemical energy	B1	
	(b) (i)	0	B1	
	(ii)	it increases	B1	
		to constant value	B1	
	(iii)	gradient **or** $(v–u)/t$ **or** (1400-600)/40 **or** other correct numbers	C1	
		$20\,m/s^2$	A1	
	(iv)	$(F =)\ ma$ **or** $1.6 \times 10^6 \times 20$	C1	
		$3.2 \times 10^7\,N$	A1	
	(v)	$4.8 \times 10^7\,N$	B1	
	(c) (i)	to every action there is an equal and opposite reaction **or** forces act in pairs of equal size **and** in opposite directions/on different bodies	B1	
	(ii)	downward force on gas	B1	
		equal and opposite to upward/**(b)(v)** force (on rocket)	B1	[15]

QUESTION · 5054_w11_qp_12.pdf · source page 5 of 20

9 Coal is burned as fuel to heat water in a boiler, producing steam. The steam drives a turbine, which is connected to an electric generator.

In which order do the major energy transformations take place?

A chemical energy → heat energy → electrical energy → kinetic energy

B chemical energy → heat energy → kinetic energy → electrical energy

C heat energy → chemical energy → electrical energy → kinetic energy

D heat energy → chemical energy → kinetic energy → electrical energy

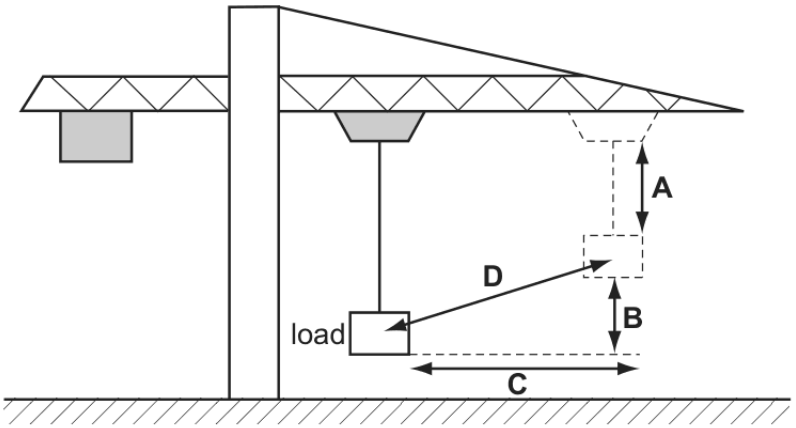
MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 9

9	**B**

QUESTION · 5054_w11_qp_12.pdf · source page 5 of 20

10 A crane moves its load diagonally, as shown.

By which distance is the weight of the load multiplied to calculate the change in gravitational potential energy of the load?



MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 10

10 **B**

2 Fig. 2.1 shows a skier of mass 85 kg skiing down a very steep slope.

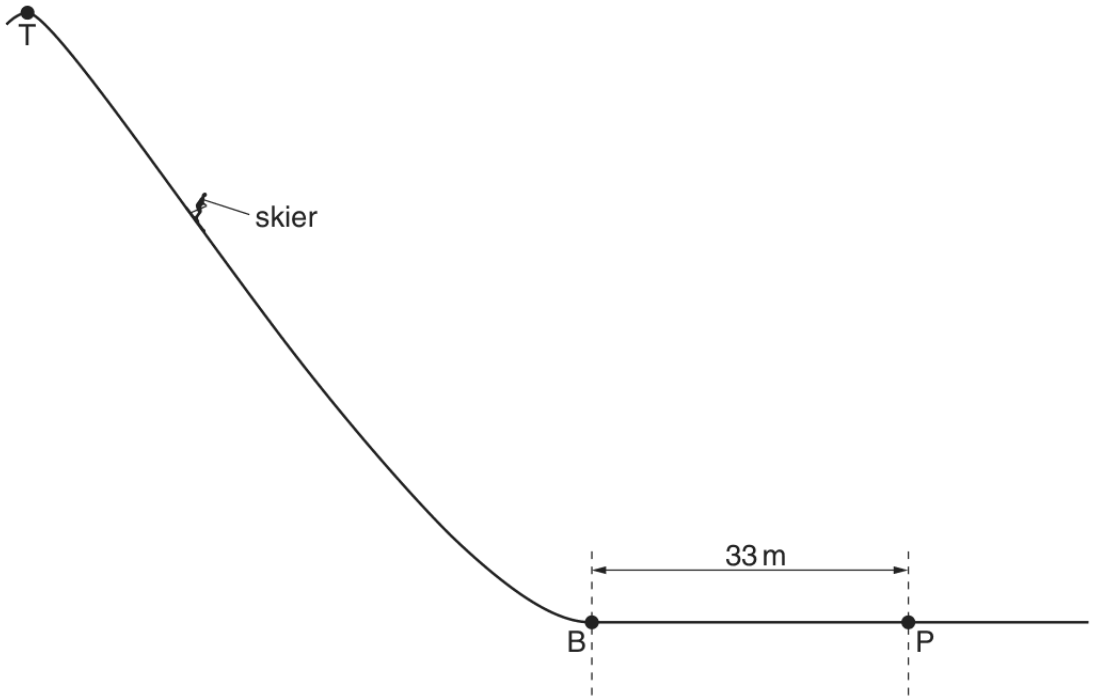


Fig. 2.1 (not to scale)

The skier starts from rest at the top T of the slope. The force of gravity accelerates him down the slope.
When he reaches the bottom B of the slope, his kinetic energy is 5.5×10^4 J.

(a) The gravitational field strength is 10 N/kg. Calculate

(i) the weight of the skier,

weight = ...[1]

(ii) the minimum possible difference in height between T and B.

height difference = ...[2]

(b) At B, the skier digs his skis into the snow and stops at the point P after travelling 33 m horizontally.

Calculate the average horizontal force that acts on the skier between B and P.

force = ...[2]

2	**(a)**	**(i)**	850 N	B1	
		(ii)	KE = PE/*mgh* **or** *mgh* = 5.5×10^4	C1	
			65/64.7(0588) m	A1	
	(b)		WD = *Fx* **or** KE/*x* **or** $5.5 \times 10^4/33$ **or** *v* = 35(.97) **and** *a* = 19(.60) **and** *F* = *ma*	C1	
			1700/1670/1667/1666.7 N	A1	[5]

10 A lead bullet of mass 1.9 g is fired from a rifle in a sports club. The bullet misses the target and embeds itself in a wall behind the target. The bullet melts as it is stopped by the wall. The specific latent heat of fusion of lead is 2.2×10^4 J/kg.

(a) State what is meant by *melting point*.

.....

.....[1]

(b) **(i)** Calculate the energy required to melt the bullet, at its melting point, without raising its temperature.

energy = ...[3]

(ii) Assume that the energy that melts the bullet is equal to its kinetic energy just before it strikes the wall. Calculate the speed of the bullet just before it strikes the wall.

speed = ...[3]

(iii) Suggest two reasons why the speed of the bullet as it leaves the rifle is greater than the value calculated in **(ii)**.

1. .....

2. .....
[2]

(c) Describe how the molecular structure of the lead changes as it melts.

.....

.....

.....

.....

.....[3]

(d) On another occasion, lead bullets of twice the mass are used. One of these heavier bullets hits the wall with the speed calculated in **(b)(ii)**. State and explain whether this bullet melts as it is stopped by the wall.

.....

.....

.....

.....[3]

10 (a) **temperature** where: liquid and solid may exist together **or** solid turns to liquid — B1

(b) (i) (E =) ml — C1
$0.0019 \times 2.2 \times 10^4$ **or** $1.9 \times 2.2 \times 10^4$ **or** 41 800 **or** 42 000 — C1
42 (41.8) J — A1

(ii) $½mv^2$ **or** ½ × 0.0019 × v^2 **or** ½ × 1.9 × v^2 — C1
(v^2 =) 44 000 **or** 44 — C1
210 (209.761 etc.) m / s — A1

(iii) any **two** of:
heat lost to wall
heat to raise bullet to m.p.
air resistance/**air friction** reduces energy/speed/velocity **or** work done against **air resistance**/**air friction** (in air/as bullet travels) — B2

(c) any **three** of:
molecules become further apart
molecules become randomly positioned/less ordered
molecules moving throughout liquid/in clusters/were fixed/free to move/slide over each other
bonds broken/overcome/weaker **or** forces reduced — B3

(d) twice the energy needed
(bullets have) twice the KE

they melt — A1

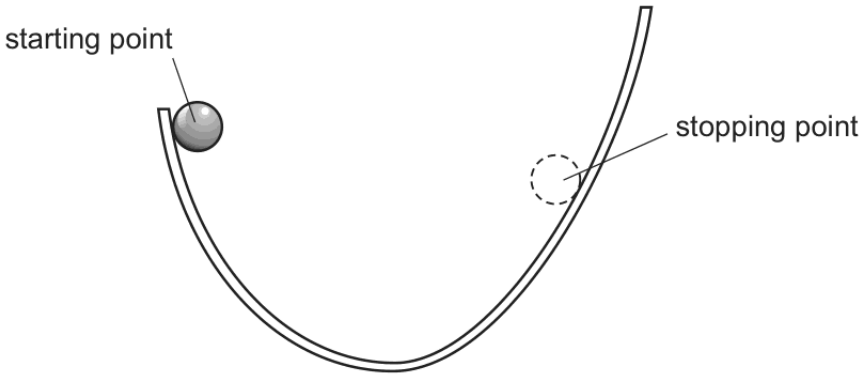
OR

$ml = ½mv^2$ — M1
m cancels **or** mass irrelevant **or** w.t.t.e. **or** calculation — M1

[15]

QUESTION · 5054_s12_qp_11.pdf · source page 6 of 16

13 A ball is held at rest on one side of a curved track.



The ball is released. It rolls down one side of the track and part of the way up the other side. It then stops, before rolling back down again. The height of the stopping point is less than that of the starting point.

What is the sequence of energy changes between starting and stopping for the first time?

A potential energy → kinetic energy → potential energy

B potential energy → kinetic energy → heat → potential energy

C potential energy → heat → kinetic energy → potential energy

D potential energy → kinetic energy + heat → potential energy + heat

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 13

13	**D**

2 Fig. 2.1 shows a windsurfer, sail and board.

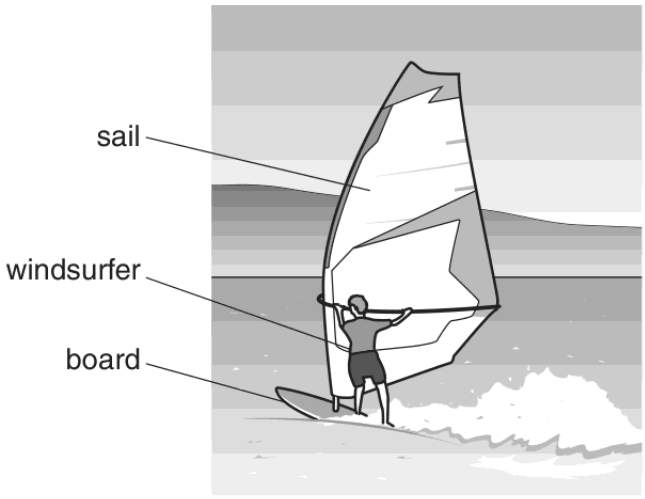


Fig. 2.1

The total mass of the windsurfer, sail and board is 90 kg. They travel at a constant speed of 5.0 m/s.

(a) Calculate the total kinetic energy of the windsurfer, sail and board.

kinetic energy = ...[2]

(b) The wind exerts a forward force on the sail.

(i) Explain, in terms of the forces acting, why the board travels at a constant speed.

.....

.....

.....[2]

(ii) The wind does work on the sail. Explain, in terms of the conservation of energy, why the total kinetic energy does not increase.

.....

.....

.....[1]

2 **(a)** (K.E.) = ½ mv^2; ½ × 90 × 5^2 C1
1125 J **or** 1100 J A1

(b) **(i)** no resultant force; forwards force = backwards force/drag/friction (ignore air resistance) B1
water resistance/**water** drag mentioned/**water** friction OR sail exerts force on board B1

(ii) heat produced OR equal to work done against backwards force/drag/friction B1 [5]

QUESTION · 5054_s12_qp_12.pdf · source page 6 of 16

13 Two major components of a coal-fired power station are a turbine and a generator.

What are the output forms of energy from the turbine and from the generator?

	turbine	generator
A	electrical	electrical
B	electrical	kinetic
C	heat	kinetic
D	kinetic	electrical

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 13

13 **D**

2 **(a)** State what is meant by *friction*.

.....

.....[2]

(b) Fig. 2.1 shows two horizontal forces that act on a car. Force B is caused by air resistance and friction.

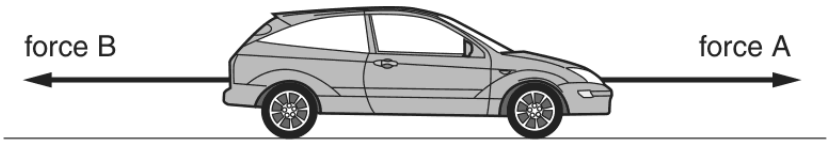


Fig. 2.1

The car is travelling along a straight level road.

(i) The forward force A and the backward force B are equal.
Describe the motion of the car.

.....

.....[1]

(ii) The mass of the car is 800 kg.
Force A increases to 5000 N. This causes the car to accelerate initially at 1.5 m/s^2.
Calculate the size of force B.

force B = ...[3]

(iii) Force A remains constant at 5000 N. Explain why the acceleration decreases as the car travels along the level road.

.....

.....[2]

(c) A car of mass 800 kg climbs a hill and rises a vertical distance of 200 m.
Calculate the gain in potential energy of the car. (The gravitational field strength *g* is 10 N/kg.)

gain in potential energy = ...[2]

2	**(a)**	a force	B1	
		when objects slide over/rub one another OR opposes (relative) **motion**/**movement**	B1	
	(b)	**(i)** constant/uniform speed OR constant/uniform velocity OR zero acceleration	B1	
		(ii) (*F* =) *ma* seen in any form numerical or algebraic	C1	
		1200 (N) OR 6200 N seen	C1	
		3800 N	A1	
		(iii) Force B increases OR backwards force/resistance/friction/drag increases	M1	
		as speed/velocity increases	A1	
	(c)	(PE =) *mgh* in any form numerical or algebraic	C1	
		1 600 000 J	A1	[10]

For Examiner's Use

3 A man of mass 75 kg falls from a platform high above a lake.
Fig. 3.1a shows the man tied to the platform by a long elastic rope (bungee).
Fig. 3.1b shows the man when he has fallen 20 m. After this point the rope begins to stretch.
Fig. 3.1c shows the man at 25 m below the platform where he is first stopped by the rope.

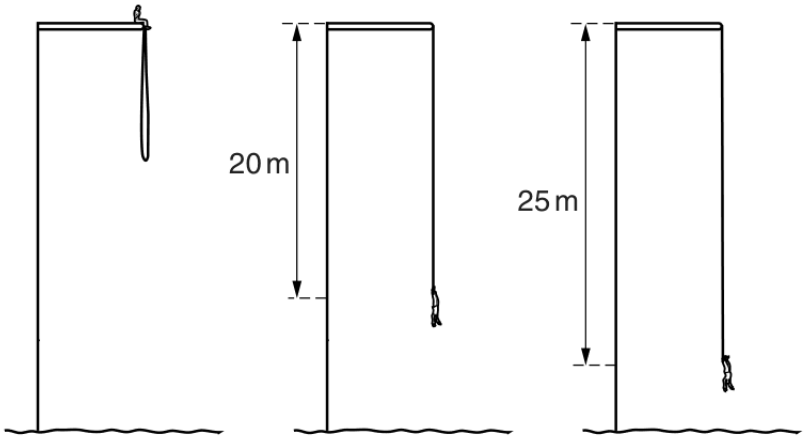


Fig. 3.1a **Fig. 3.1b** **Fig. 3.1c**

(a) As the man falls, his gravitational potential energy changes.

(i) The gravitational field strength is 10 N/kg. Calculate the change in his gravitational potential energy as he falls through 20 m.

change in potential energy = ...[2]

(ii) When he is 20 m below the platform, his kinetic energy is equal to the change in his gravitational potential energy calculated in **(i)**. Calculate his speed at this point.

speed = ...[3]

(b) State the energy changes that take place as he falls from 20 m to 25 m below the platform. Ignore the effect of air resistance.

.....

.....

.....[3]

3	**(a)**	**(i)**	(PE =)mgh **or** 75 × 10 × 20	C1	
			1.5×10^4 J	A1	
		(ii)	$\frac{1}{2}mv^2$ **or** $\frac{1}{2}75v^2$	C1	
			$v^2 = 400$ (if this is seen it scores the first 2 marks)	C1	
			$v = 20$ m / s	A1	
	(b)		(G)PE **at start**	B1	
			KE **at start**	B1	
			to elastic/strain/clear equivalent /EPE **at end**		
			(**not** stretch energy; any intermediate energy –1)	B1	[8]

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a conveyor belt carrying suitcases into an aeroplane.

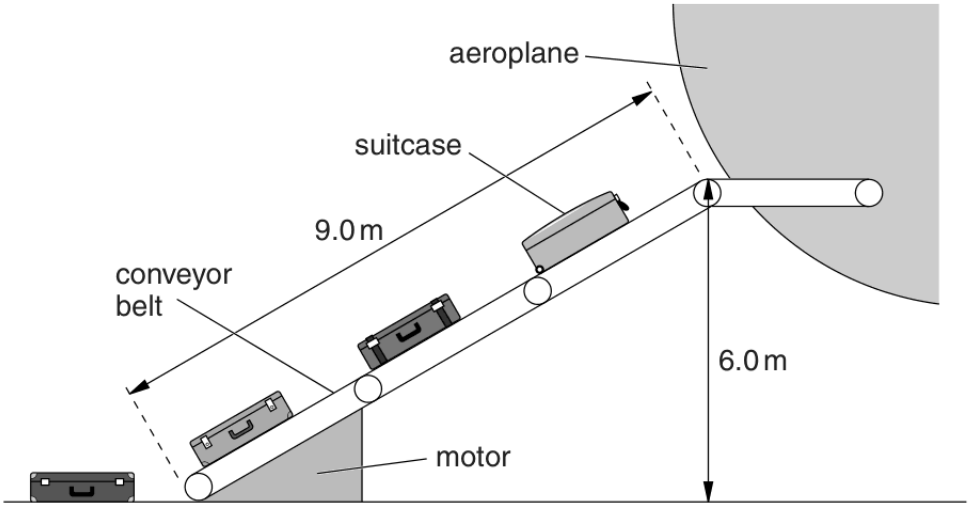


Fig. 9.1 (not to scale)

An electric motor drives the conveyor belt.

(a) A suitcase of mass 20 kg is lifted from the ground into the aeroplane.

(i) Explain what is meant by *mass* and by *weight*.

.....

.....

.....[2]

(ii) The gravitational field strength g is 10 N/kg.
Calculate the increase in the gravitational potential energy of the suitcase.

increase in potential energy = ...[2]

(iii) The suitcase takes 12 s to travel 9.0 m along the conveyor belt.
Calculate the kinetic energy of the suitcase.

kinetic energy = ...[3]

For Examiner's Use

(iv) The electric motor is powered by an input voltage of 230 V and a current of 1.5 A. Calculate the electrical energy input to the motor in 12 s.

energy = ...[2]

(v) **1.** State the principle of the conservation of energy.

.....

.....

.....[2]

2. Explain how this principle applies to the lifting of the suitcases into the aeroplane.

.....

.....

.....[1]

(b) Some of the electrical energy for the motor comes from renewable energy sources.

(i) Explain what is meant by a *renewable energy source*.

.....

.....

.....[1]

(ii) State two renewable energy sources.

1. .....

2. .....
[2]

Section B

9 (a) (i) mass is the amount of matter/substance **or** to resist (change in) motion **or** (measurement of)inertia B1
weight is the pull/force of gravity **or** pull of Earth B1
accept *mg* where *g* stated as 10 (N/kg) or grav. field strength or acc. due to gravity
ignore *mg* where *g* is gravity or grav. force or undefined

(ii) *mgh* algebraic or numerical C1
1200 J A1

(iii) ½ mv^2 algebraic or numerical C1
(speed =) 9/12 **or** 0.75 seen A1
5.6(25) J A1

(iv) (*E*=) *VIt* algebraic or numerical C1
4100 J **or** 4140 J A1

(v) **1.** energy can neither be created or destroyed/lost B1
(but) may change form / be transferred B1

2. electrical energy changes to P.E. (and K.E. and heat/work against friction; **ign**. mechanical energy) B1 [12]

(b) (i) will not run out **or** infinite **or** being replaced (**allow** does not finish/always available)
(**ign.** cannot be reused/recycled) B1

(ii) wind, tidal, solar/Sun, geothermal, hydroelectric, biomass, waves, wood (**not** nuclear) B2 [3]
(**allow** biogas/biofuel e.g. cane into petrol, dung into gas etc.) (**ign.** tidal waves)

[Total: 15]

QUESTION · 5054_s13_qp_12.pdf · source page 6 of 16

13 A swimmer dives into a very deep pool at high speed. He slows down as he moves towards the bottom of the pool.

What is the overall energy transformation as the diver moves downwards through the water?

A gravitational potential energy → kinetic energy + thermal energy (heat)

B gravitational potential energy → kinetic energy → thermal energy (heat)

C kinetic energy + gravitational potential energy → thermal energy (heat)

D kinetic energy → gravitational potential energy + thermal energy (heat)

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 13

13	**C**

QUESTION · 5054_s13_qp_12.pdf · source page 6 of 16

14 A lorry of mass 10 000 kg takes 5000 kg of sand to the top of a hill 50 m high, unloads the sand and then returns to the bottom of the hill.

The gravitational field strength is 10 N / kg.

What is the overall gain in potential energy?

A 250 000 J **B** 750 000 J **C** 2 500 000 J **D** 7 500 000 J

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 14

14	**C**

QUESTION · 5054_w13_qp_11.pdf · source page 6 of 20

11 Three objects P, Q and R have different masses and different speeds as shown in the table.

	$\frac{\text{mass}}{\text{kg}}$	$\frac{\text{speed}}{\text{m/s}}$
P	1	3
Q	2	2
R	5	1

What is the order of increasing kinetic energy (smallest first) of the objects?

A $P \rightarrow Q \rightarrow R$

B $P \rightarrow R \rightarrow Q$

C $R \rightarrow P \rightarrow Q$

D $R \rightarrow Q \rightarrow P$

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 11

11 **D**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a skydiver, of mass 70 kg, falling towards the Earth at constant speed, a long time after jumping from an aeroplane.

Fig. 9.1

At time $t = 0$, he receives a radio signal. He opens his parachute 12 s later. Fig. 9.2 is the speed-time graph for the skydiver.

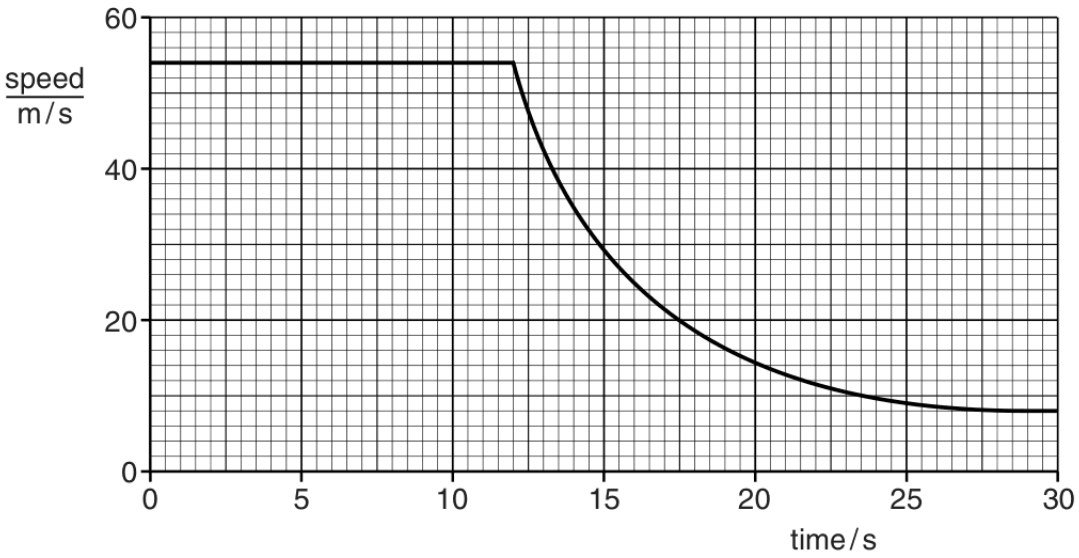


Fig. 9.2

(a) State the difference between *speed* and *velocity*.

.....

.....[1]

(b) The gravitational field strength g is 10 N/kg.

(i) Calculate the weight of the skydiver.

weight = ...[1]

For Examiner's Use

(ii) State the size of the air resistance acting on the skydiver between $t = 0$ and $t = 12$ s.

air resistance = ...[1]

(c) For the period between $t = 0$ and $t = 12$ s, determine

(i) the speed of the skydiver,

speed = ...[1]

(ii) the distance fallen by the skydiver,

distance = ...[2]

(iii) the change in the gravitational potential energy of the skydiver.

change in energy = ...[2]

(d) As he falls at constant speed, his gravitational potential energy changes into another form. State the name of this other form of energy.

.....[1]

(e) **(i)** State and explain what happens to the air resistance as the skydiver opens his parachute.

.....

.....

.....[2]

(ii) State and explain the effect on the motion of the skydiver of opening the parachute.

.....

.....

.....[2]

(f) By $t = 15$ s, his parachute is fully open.

State and explain what happens to the air resistance after $t = 15$ s.

.....

.....

.....[2]

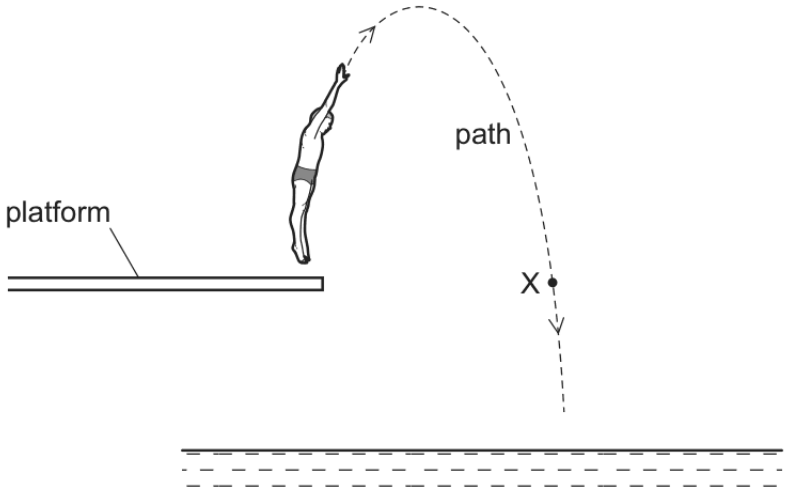
Section B

9	**(a)**	speed does not have direction **and** velocity does or speed = distance/time **and** velocity = displacement/time or speed is a scalar **and** velocity is a vector	B1	[1]
	(b) **(i)**	700 N	B1	
	(ii)	700 N	B1	[2]
	(c) **(i)**	54 m / s	B1	
	(ii)	(height/distance =) area (under graph) **or** (x =)vt **or** 54 × 12 648/650 m	C1 A1	
	(iii)	(GPE =) mgh **or** 70 × 10 × 648 4.5/4.54/4.536 × 10^5 J	C1 A1	[5]
	(d)	(becomes) heat/thermal energy/internal energy (**not** kinetic energy (of skydiver) unless qualified as KE of air)	B1	[1]
	(e) **(i)**	(air resistance) increases larger area of parachute	B1 B1	
	(ii)	(skydiver) decelerates/slows down (**not** rises up) net upward force	B1 B1	[4]
	(f)	air resistance decreases speed decreases	B1 B1	[2]

[Total: 15]

QUESTION · 5054_w13_qp_12.pdf · source page 8 of 20

14 The diagram shows the path of a diver after leaving a platform and before entering the water in a swimming pool.



The gravitational potential energy of the diver is zero when he is at the surface of the water.

Which statement about the diver's energy along the path is correct?

A At point X he has only gravitational potential energy.

B At point X he has only kinetic energy.

C His gravitational potential energy is always more than his kinetic energy.

D The sum of his gravitational potential energy and kinetic energy is constant.

MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 14

14	**D**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 **(a)** State what is meant by the *moment of a force*.

.....

.....[2]

(b) The anchor of a sailing ship has a mass of 350 kg. Six sailors raise the anchor from the sea-bed by turning a large axle. They push the handles attached to the axle and it rotates. The anchor is on the end of a chain that winds on to the rotating axle.

Fig. 9.1 shows the sailors lifting the anchor.

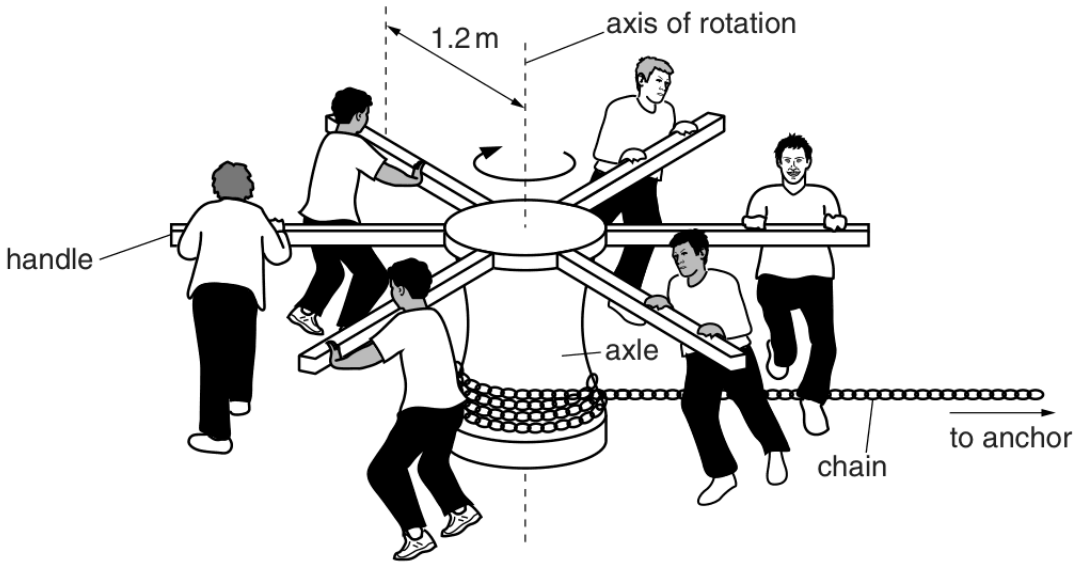


Fig. 9.1

Each of the sailors exerts a force of 750 N on his handle at a distance of 1.2 m from the axis of rotation. The axle rotates through one complete revolution and the anchor is lifted through a distance of 160 cm.

The gravitational field strength g is 10 N/kg.

(i) Calculate

1. the total moment exerted on the axle by the six sailors,

moment = ...[2]

For Examiner's Use

2. the gravitational potential energy gained by the anchor as the axle rotates through one complete revolution.

energy = ...[3]

(ii) The work done on the axle by the sailors is very much larger than the gravitational potential energy gained by the anchor.

State and explain how energy is wasted.

.....

.....

.....[2]

(iii) Explain why the power produced by the sailors is larger when the anchor is lifted at a faster rate.

.....

.....[2]

(c) Describe, with the aid of a labelled diagram, how to verify the principle of moments.

.....

.....

.....

.....

.....

.....[4]

Section B

9	**(a)**	force × distance **or** $F \times d$ with F and d defined **or** $F \times d_{perp}$	C1	
		force × perpendicular distance **or** $F \times d_{perp}$ with F and d_{perp} defined	A1	[2]
	(b) **(i)**	**1.** 6 × 750 × 1.2 **or** 750 × 1.2 **or** 900	C1	
		5400 N m	A1	
		2. *mgh* **or** 350 × 10 × 160 **or** 350 × 10 × 1.6	C1	
		350 × 10 × 1.6 **or** 5.6 × 105	C1	
		5600 J	A1	
	(ii)	friction at axle/boat **or** drag due to water **or** chain lifted also	B1	
		heat produced (**ignore** in sailors) **or** work done against friction/drag **or** work done raising chain	B1	
	(iii)	same amount of work done **or** $P = E/t$ **or** $P = WD/t$	B1	
		in less time **or** power inversely proportional to time (**ignore** faster rate)	B1	[9]
	(c)	clear/labelled diagram with ruler, fulcrum, at least two weights	B1	
		any three of the following points made in words: balance ruler (on its own) place weights on ruler so it balances clockwise and anticlockwise moments equal **or** net moment zero repeat (apply similar principles to other possible methods)	B3	[4]

[Total: 15]

For Examiner's Use

10 Fig. 10.1 shows an electric train of mass 5.5×10^5 kg.

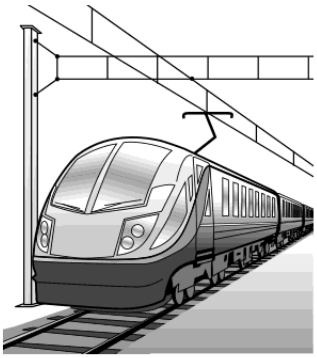


Fig. 10.1

The train is initially at rest. The electric motor exerts a constant force and the train accelerates. Its acceleration decreases until the train reaches a speed of 40 m/s.

The train then continues at this constant speed.

(a) (i) On the axes in Fig. 10.2, sketch a speed-time graph for the train.

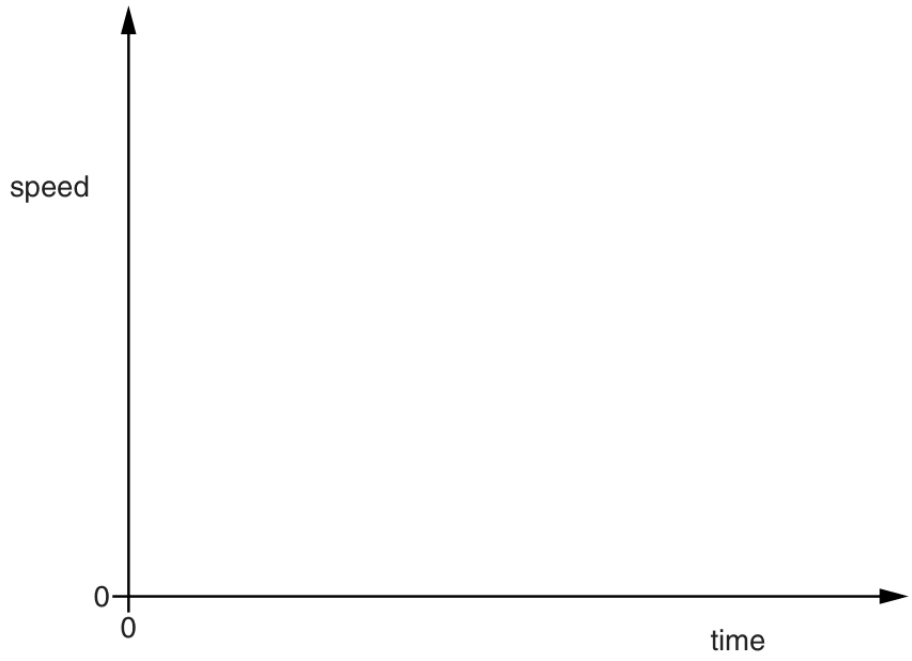


Fig. 10.2

[3]

(ii) Describe how the speed-time graph for the train is used to determine the distance travelled by the train when it is accelerating.

.....

.....

.....[2]

For Examiner's Use

(b) Explain, in terms of the forces acting,

(i) why the acceleration of the train decreases,

.....

.....

.....[2]

(ii) why the train eventually reaches constant speed.

.....

.....[1]

(c) **(i)** Calculate the kinetic energy of the train when it is travelling at a speed of 40 m/s.

kinetic energy = ...[3]

(ii) As the train accelerates to 40 m/s, electrical energy supplied to the motor is converted into kinetic energy of the train. The efficiency of this process is 0.40 (40%).

Calculate the electrical energy supplied to the motor.

energy = ...[2]

(iii) The electrical energy is generated in an oil-fired power station. Chemical energy in oil is converted into the electrical energy supplied to the motor.

Suggest two places where energy is lost as heat in this process.

1. .....

2. .....
[2]

10	**(a)**	**(i)**	start at origin **and** not horizontal	B1	
			gradient (gradually) decreasing (**ignore** sudden decrease) (not if part of curve above horizontal section)	B1	
			final horizontal section (≥ 1 cm) (not if v is shown as ≠ 40 m/s)	B1	
		(ii)	area **under** the graph **or** count squares **under** graph	M1	
			between t = 0 and horizontal section **or** when speed is changing **or** calculate equivalent distance to 1 cm^2 (after counting squares)	A1	[5]
	(b)	**(i)**	friction/air resistance increases (as speed increases)	B1	
			resultant force decreases (**not** if driving force decreases)	B1	
		(ii)	(air resistance increases until) net force becomes zero **or** forces balance **or** air resistance and driving/forward force are in equilibrium/balanced/equal	B1	[3]
	(c)	**(i)**	(KE =) $\frac{1}{2}mv^2$	C1	
			$\frac{1}{2} \times 5.5 \times 10^5 \times 40^2$	C1	
			4.4×10^8 J	A1	
		(ii)	(total energy input =) useful energy output efficiency **or** efficiency = useful power output/total power input **or** $4.4 \times 10^8/0.40$	C1	
			1.1×10^9 J	A1	
		(iii)	two valid examples e.g. furnace/boiler/turbines/generator/coils/cooling water/cooling towers/heat exchanger/transformer/chimney/waste gases/ transmission cables/lines/wires (**ignore** power station/all parts of motor)	B2	[7]

[Total: 15]

QUESTION · 5054_s14_qp_11.pdf · source page 6 of 16

17 A javelin has a mass of 0.80 kg.

The javelin is released with 250 J of kinetic energy.

What is the speed of the javelin?

A 13 m/s **B** 18 m/s **C** 25 m/s **D** 630 m/s

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 17

17	**C**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 A garden pond contains a small fountain. An electric pump in the water causes the water to rise above the surface of the pond, as shown in Fig. 9.1.

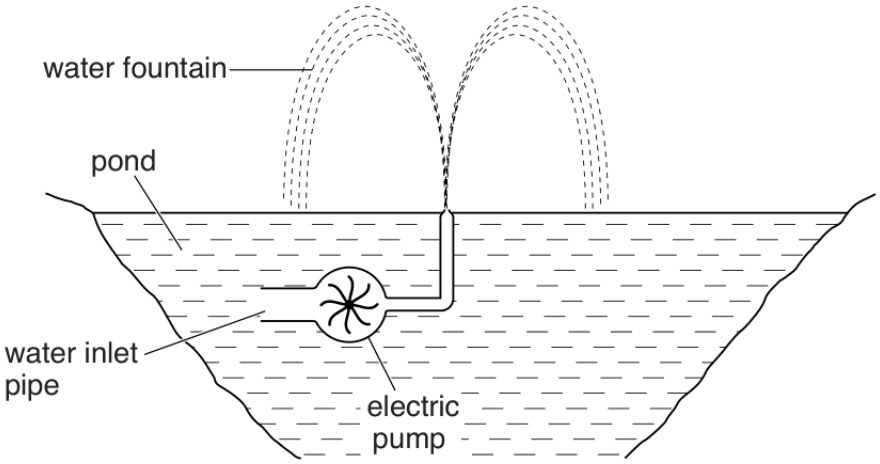


Fig. 9.1

(a) The pressure of the water in the pond increases with depth.

(i) Explain what is meant by *pressure*.

.....

..... [1]

(ii) Explain why the pressure below the surface of the water increases with depth.

.....

..... [2]

(iii) State the unit of pressure.

..... [1]

(b) Describe the energy changes that occur within the pump.

.....

.....

..... [3]

(c) A volume of 0.78 m^3 of water passes through the pump in one hour. The density of water is 1000 kg/m^3.

(i) Calculate the mass of water that passes through the pump in one hour.

mass = ... [2]

(ii) The water rises 1.5 m. The gravitational field strength g is 10 N/kg.

Calculate the useful work done in one hour raising the water to the top of the fountain.

work done = ... [2]

(iii) Calculate the minimum power output of the pump.

power = ... [2]

(d) Describe an experiment to check that the density of water is 1000 kg/m^3.

.....

.....

.....

.....

..... [2]

SECTION B

9 (a) (i) force per unit area **or** force divided by area **or** force on 1 m^2 **or** force on unit area [B1]

(ii) more water above **or** more force from water **or** more atoms / molecules above [C1]

larger weight of water above [A1]

(iii) Pa **or** N / m^2 [B1]

(b) electrical energy at start of process [B1]
kinetic energy of water / turbine/blade produced [B1]
some heat energy / thermal energy / internal energy produce [B1]

(c) (i) (M=) D × V in any algebraic or numerical form [C1]
780 kg [A1]

(ii) (W=) mgh or Fd in any algebraic or numerical form [C1]
11700 J **or** 12000 J [A1]

(iii) (P=) W / t in any algebraic or numerical form **or** 195(W) [C1]
3.25 **or** 3.2 **or** 3.3 W [A1]

(d) find difference in mass of container, with and without water [B1]
measuring cylinder or similar instrument used to find volume **and** density checked [B1]

other methods are possible, e.g. use of calibrated hydrometer

[15]

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 A children's ride consists of a steel cable that runs between two posts of different heights, as shown in Fig. 9.1.

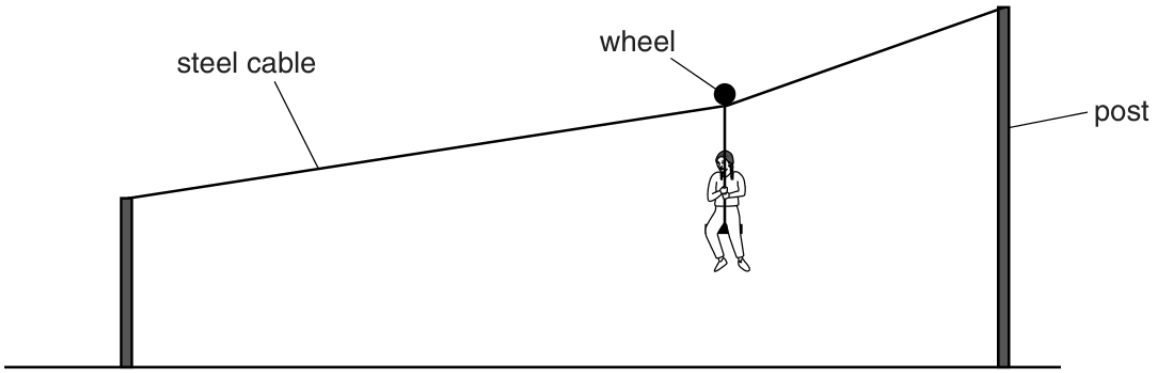


Fig. 9.1

A girl starts and finishes the ride at rest. Her horizontal motion can be taken as
- an initial uniform acceleration for 3.0 s, followed by
- a constant speed of 2.4 m/s for a further 5.0 s and
- a final uniform deceleration that lasts for 1.0 s.

(a) On Fig. 9.2, draw a speed-time graph of the horizontal motion.

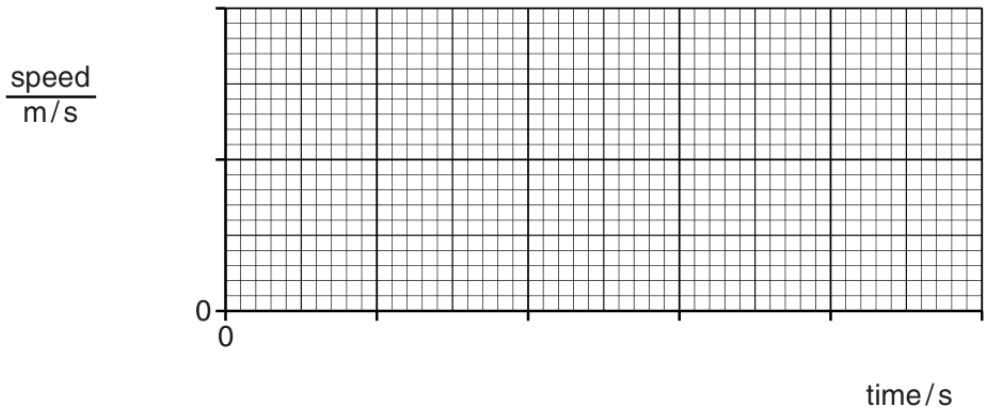


Fig. 9.2 [3]

(b) Explain what is meant by *uniform acceleration*.

.....

.....

..... [2]

(c) The final deceleration is larger in size than the initial acceleration.

Explain how the data shows this.

.....

..... [1]

(d) Calculate the horizontal distance travelled by the girl in the first 8.0 s.

distance = ... [3]

(e) **(i)** The girl has a mass of 30 kg and falls a vertical distance of 1.6 m during the ride.

The gravitational field strength g is 10 N / kg.

Calculate the decrease in gravitational potential energy of the girl.

decrease in potential energy = ... [2]

(ii) The gain in kinetic energy of the girl is less than the decrease in her potential energy. Suggest one reason for this.

.....

..... [1]

(f) A group of pupils make measurements to show that the girl's speed is constant during the middle section of the ride.

Suggest what measurements are made and how they show that the speed is constant.

.....

.....

.....

.....

.....

..... [3]

9 **(a)** straight line from (0, 0) to (3, 2.4) [B1]
horizontal line from 3 s to 8 s [B1]
straight line from **end of a horizontal line** to zero in 1 s [B1]

(b) constant / same increase in velocity **or constant** change in velocity [C1]
constant / same increase in velocity per sec / unit time [A1]

(c) occurs in a short(er) time
or acceleration took 3 s and deceleration took 1 s [B1]

(d) (d =) speed × time numerical or algebraic **or** area under graph clear [C1]
1.2 × 3 **or** 3.6 (m) **or** 2.4 × 5 **or** 12 seen [C1]
15.6 m **or** 16 m [A1]

(e) **(i)** mgh seen in any algebraic or numerical form, e.g. 30 × 10 × 1.6 [C1]
480 J [A1]

(ii) heat or thermal energy or sound produced
or work done against friction / air resistance [B1]

(f) at least two distances and corresponding times mentioned [C1]

how the **actual measurement** is made, e.g. **(any one from)**
- make mark on ground every second and **measure** distances
- note video position every sec and use a scale to **find** distances
- make mark on ground every meter and **measure / take** time as girl passes [A1]

how constant speed is proved using measurement, e.g. **(any one from)**
- same distance between each position for the same time interval
- same time interval for equal distances
- $\Delta d / \Delta t$ constant **or** slope of distance-time graph constant [B1]

[15]

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows the dam and reservoir of a hydroelectric power station.

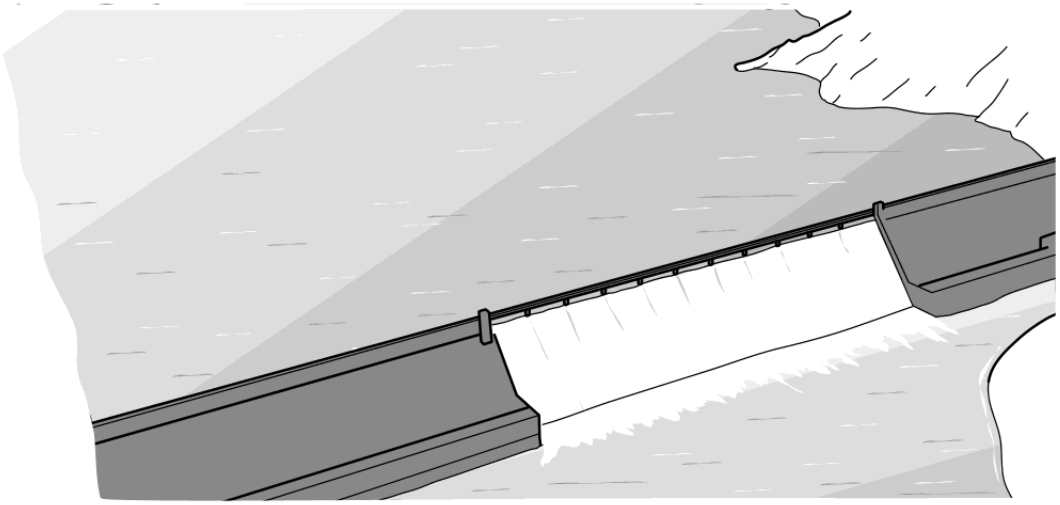


Fig. 9.1

A hydroelectric power station uses a renewable energy source.

(a) State two other renewable energy sources.

1. .....

2. .....
[2]

(b) When the power station operates at full capacity, the electrical power output is 6.8×10^9W.

(i) **1.** Calculate the electrical energy output of the power station if it operates at full capacity for one year.

energy = ...[1]

2. Suggest why, in practice, the power station does not operate at full capacity throughout the year.

.....

.....[1]

(ii) The water surface in the reservoir that supplies the hydroelectric power station is at a vertical height of 170 m above the turbines. In one hour, 1.6×10^{10} kg of water flows from the reservoir through the turbines. The gravitational field strength g is 10 N / kg.

1. Calculate the gravitational potential energy that is converted into other forms of energy in one hour.

energy = ...[2]

2. Calculate the efficiency of the power station when operating at full capacity.

efficiency = ...[2]

3. Suggest two reasons why the efficiency of the power station is less than 1.0 (100%).

1. .....

2. .....
[2]

(c) Before the electrical energy is transmitted along power lines, the voltage is increased to a very large value.

(i) State and explain why a very high voltage is used to transmit electrical energy over long distances.

.....

.....

.....

.....[3]

(ii) State the name of the device that is used to increase the voltage.

.....[1]

(iii) Explain why the generators in the power station must be a.c. (alternating current) generators.

.....

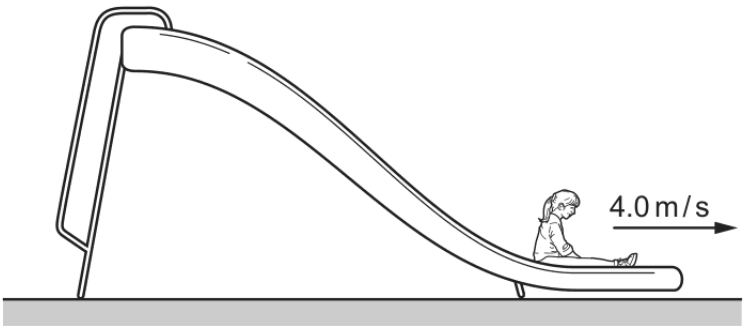
.....[1]

Section B

9	**(a)**	any **two** from: biomass / wood; geothermal power; solar power; tidal power; wave power; wind power		B2	**[2]**
	(b) **(i)**	**1.**	$2.1(4) \times 10^{17}$ J (**allow** $2.1(5) \times 10^{17}$ J if candidate uses 365.24 / 5)	B1	
		2.	any **one** from: not enough water (to maintain maximum flow); rainfall varies (during the year); periods of low demand	B1	
	(ii)	**1.**	(GPE =) *mgh* **or** $1.6 \times 10^{10} \times 10 \times 170$	C1	
			$2.7(2) \times 10^{13}$ J	A1	
		2.	$2.7(2) \times 10^{13} / 3600$ **or** $6.8 \times 10^{9} \times 3600$ **or** $6.8 \times 10^{9} / 7.5(55) \times 10^{9}$ **or** $2.4(48) \times 10^{13} / 2.7(2) \times 10^{13}$	C1	
			0.90 **or** 90%	A1	
		3.	any **two** from: friction (of water) with pipe / turbine /; viscosity of water; friction at bearings; resistance / heat in the wires; KE of water leaving turbine	B2	**[8]**
	(c) **(i)**		less energy lost / wasted **or** more efficient	B1	
			(for a given power) a high voltage results in a small(er) current	B1	
			less heat generated in wires **or** I^2R **or** less resistive losses (**not** if <u>changed</u> resistance mentioned)	B1	
	(ii)		transformer	B1	
	(iii)		transformers only work with an a.c. supply	B1	**[5]**
					[Total: 15]

QUESTION · 5054_w14_qp_12.pdf · source page 4 of 16

9 A child of mass 30 kg is moving at a speed of 4.0 m/s when she reaches the bottom of a slide.



What is her kinetic energy?

A 60 J **B** 120 J **C** 240 J **D** 480 J

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 9

9 **C**

4 Fig. 4.1 shows a black car going up a hill on a bright, sunny day.

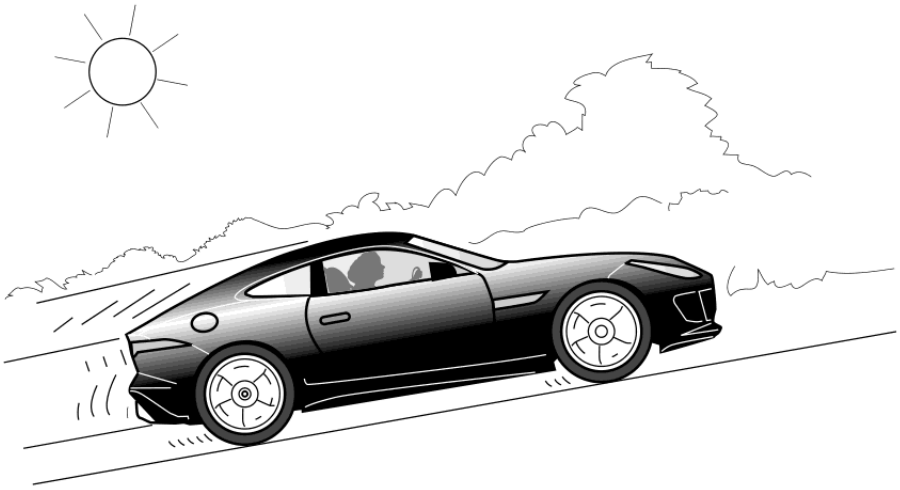


Fig. 4.1

(a) State

(i) one way in which the car is gaining thermal energy,

.....

.....[1]

(ii) one way in which the car is losing thermal energy.

.....[1]

(b) The car accelerates up the hill. In addition to the changes in the thermal energy of the car, there are other energy changes taking place.

Describe the other energy changes occurring as the car moves.

.....

.....

.....

.....[3]

4	**(a)**	**(i)**	heat gained from burning fuel / combustion **or** friction between moving parts / with air / road **or** from (radiation of) Sun	B1	
		(ii)	heat lost to air/surroundings **or** by convection (currents) **or** exhaust/hot gases/fumes **or** from exhaust **or** heat emitted (by hot car) **or** by radiation	B1	
	(b)		at start chemical energy decreases **or** at start chemical energy to	B1	
			gravitational / potential energy (of car) increases **or** at end of process	B1	
			kinetic energy (of car **or** air) increases	B1	**[5]**

QUESTION · 5054_s15_qp_11.pdf · source page 5 of 16

11 A cyclist travels along a horizontal track at constant speed.

The work done by the cyclist is equal to

A the change in kinetic energy.

B the force of air resistance.

C the force of friction in the bicycle.

D the thermal energy (heat) produced.

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 11

11	**D**

QUESTION · 5054_s15_qp_11.pdf · source page 5 of 16

12 A 2.0 kg mass has 300 J of kinetic energy.

What is the speed of the mass?

A 8.7 m/s **B** 12 m/s **C** 17 m/s **D** 24 m/s

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 12

12	**C**

2 Fig. 2.1 shows a student doing a press-up. A total force *F* acts upwards on his hands. There is also a force upwards on his toes.

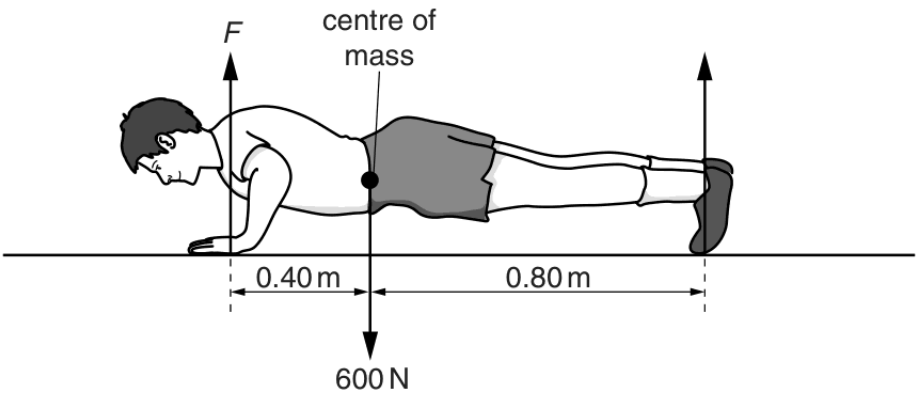


Fig. 2.1 (not to scale)

The weight of the student is 600 N and this force acts downwards from his centre of mass.

(a) **(i)** Explain why the student does work as his body rises from the ground.

.....

.....[1]

(ii) State the form of energy that the student uses to do this work.

.....[1]

(b) At the position shown in Fig. 2.1, the student is stationary.
The weight of the student causes a moment about his toes.

(i) Calculate the moment of the weight of the student about his toes.

moment = ...[1]

(ii) Calculate the value of the force *F*.

F = ...[2]

2	**(a)**	**(i)**	force moves through a distance (in same direction)	B1
		(ii)	chemical (potential) energy	B1
	(b)	**(i)**	480 Nm	B1
		(ii)	attempt to apply moments with two forces and distances	C1
			400 N	A1

QUESTION · 5054_s15_qp_12.pdf · source page 6 of 16

12 A rocket of total mass M is travelling at a speed v. The engine of the rocket is fired and fuel is used up. The mass of the rocket decreases to $M/2$ and its speed increases to $2v$.

What happens to the kinetic energy of the rocket?

A it doubles

B it halves

C it increases by a factor of four

D it stays the same

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 12

12	**A**

11 A bungee jumper falls from a bridge above a river, as shown in Fig. 11.1.

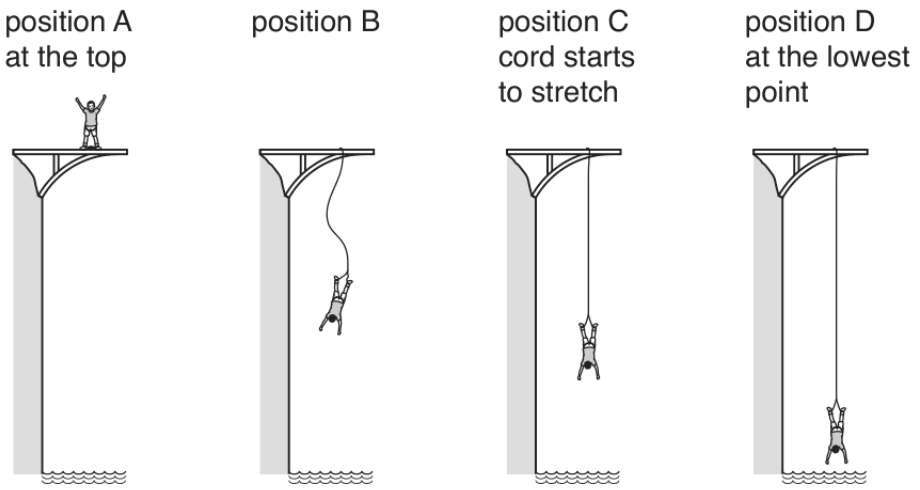


Fig. 11.1 (not to scale)

The man starts from position A in Fig. 11.1. The elastic cord starts to stretch at position C and he stops for the first time at position D. He continues to rise and fall.

Fig. 11.2 shows how the velocity of the man varies with time *t*.

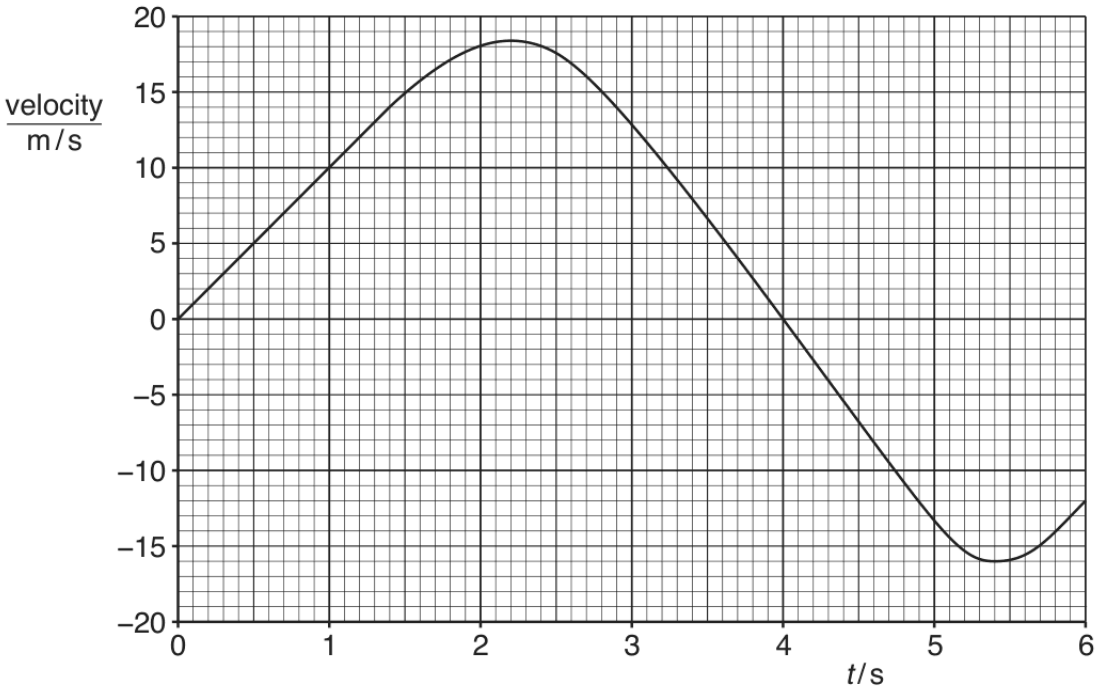


Fig. 11.2

(a) (i) State what is meant by *velocity*.

.....

..... [2]

(ii) State the difference between a positive velocity and a negative velocity.

.....

..... [1]

(iii) In the first 1.4 s the acceleration is uniform.

1. Using values from Fig. 11.2, determine the acceleration of the man in the first 1.4 s.

acceleration = ... [3]

2. Comment on your value of acceleration.

.....

..... [1]

(iv) 1. State the value of *t* when the man is at position D.

..... [1]

2. Explain, in terms of the forces acting, why the man is accelerating upwards at D.

.....

.....

.....

.....

..... [3]

(b) Fig. 11.3 shows the values for the gravitational potential energy of the man, the kinetic energy of the man and the elastic potential energy in the cord at A, C and D.

You may ignore the effect of air resistance in this question.

	gravitational potential energy/J	kinetic energy/J	elastic potential energy/J
position A	20000	0	0
position C	15000		0
position D	0	0	

Fig. 11.3

(i) Complete Fig. 11.3 to show the kinetic energy of the man at C and the elastic potential energy in the cord at D. [2]

(ii) The man has a mass of 50 kg.

The gravitational field strength g is 10 N/kg.

Using values from Fig. 11.3, calculate the vertical distance between A and C.

distance = ... [2]

11 (a) (i) distance (travelled) per second **or** speed — C1
distance (travelled) per second / speed in a given direction — A1
or displacement / time
or change in displacement per unit time
or displacement (travelled/covered) per unit time
or rate of change of displacement

(ii) opposite direction — B1

(iii) 1 value seen for v and corresponding value of t — C1
$0 < t \leqslant 1.4$ and $0 < v \leqslant 14$
(a=) v-u / t algebraic or numerical equation — C1
$10\,m/s^2$ — A1

2 sensible comment — A1

(iv) 1 4(.0 s) — B1

2 weight **or** force due to gravity mentioned (at D) — B1
mention of — B1
- upwards force (on man) from cord
- tension / elastic force from cord (on man)

force in cord / upward force / tension greater than downwards force — B1
or resultant force upwards

(b) (i) 5000 — B1
20 000 — B1

(ii) (h =) PE / mg **or** 5000 **=** 50 × 10 × h — C1
10 m — A1

3 A lorry of mass 4.4×10^4 kg travels along a straight, horizontal road at 20 m/s.

(a) Calculate the kinetic energy of the lorry.

kinetic energy = ... [3]

(b) The lorry driver sees an obstruction ahead and applies the brakes. The lorry slows down and stops. The lorry stops a distance of 40 m from where the driver first applies the brakes.

As it slows down, the kinetic energy of the lorry decreases to zero. The kinetic energy decreases as work is done by the braking force exerted on the lorry.

(i) State the formula that relates the work done to the braking force.

.....

.....

..... [1]

(ii) Calculate the braking force exerted on the lorry.

force = ... [2]

3	**(a)**	(k.e. =)$\frac{1}{2}mv^2$	C1	
		$\frac{1}{2} \times 4.4 \times 10^4 \times 20^2$	C1	
		8.8×10^6 J	A1	
	(b)	**(i)** $WD = F \times x_{(\|\|)}$ **or** force × distance (parallel to / in direction of force)	B1	
		(ii) $8.8 \times 10^6 / 40$ **or** a = (–)5.0 (m / s^2) **or** t = 4.0 (s) seen	C1	
		2.2×10^5 N	A1	[6]

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Force is a vector quantity and mass is a scalar quantity.

(a) (i) State how a vector quantity differs from a scalar quantity.

.....

..... [1]

(ii) State one other vector quantity and one other scalar quantity.

vector quantity .....

scalar quantity .....
[2]

(b) An aeroplane flies through the air in a straight line, with its engines at full power. The weight of the aeroplane is 160 kN. There is also a force of 320 kN acting upwards on the aeroplane at 30° to the horizontal.

In the space below, use a graphical method to determine the size and the direction of the resultant of these two forces. Use a scale of 1.0 cm : 40 kN.

size of resultant force = ...

direction of resultant force = ...
[4]

(c) An aeroplane is travelling in a straight line at a constant height. The aeroplane accelerates.

(i) State the energy change that is taking place as the aeroplane accelerates.

.....

.....

..... [2]

(ii) As the aeroplane accelerates, the pilot keeps the engines at constant power. The aeroplane eventually reaches a constant speed.

1. Explain why the aeroplane stops accelerating and reaches a constant speed.

.....

.....

.....

.....

.....

..... [3]

2. When travelling at a constant speed, the pilot directs the aeroplane into a tight turn that follows a circular path.

Explain why the aeroplane is accelerating even though its speed is not changing.

.....

.....

.....

.....

..... [3]

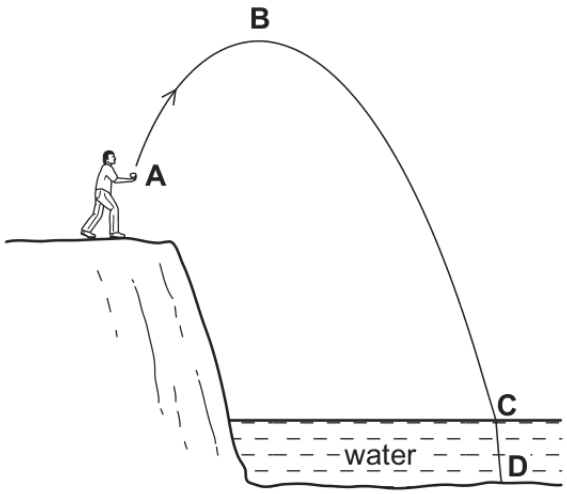
Section B

9	**(a)**	**(i)**	(vector) has direction **or** scalar does not have direction **or** (vectors) may cancel **or** scalars cannot cancel	B1	
		(ii)	one vector quantity e.g.: displacement; velocity, acceleration	B1	
			one scalar quantity e.g.: distance, length, speed, time, current, resistance	B1	[3]
	(b)		downward weight arrow of length 3.9–4.1 cm	B1	
			correct triangle/parallelogram drawn **and** correct diagonal clear	B1	
			270–285 kN	B1	
			horizontal (±3°)	B1	
	(c)	**(i)**	from chemical / fuel energy	B1	
			to kinetic (and thermal energy) (**not** with any intermediate)	B1	
		(ii)	**air** resistance / friction / drag	B1	
			air resistance / friction / drag increases **or** resultant force decreases **or** acceleration decreases	B1	
			resultant force is zero **or** (air) resistance / friction equals thrust	B1	
			direction of motion is changing	B1	
			velocity is vector **or** has a direction	B1	
			(acceleration depends on) changing velocity **or** **resultant** force towards centre (of circle) **or centripetal** force	B1	[12]
					[15]

QUESTION · 5054_w15_qp_12.pdf · source page 7 of 16

14 A person throws a stone so that it follows the path shown in the diagram.

In which position does the stone have the most gravitational potential energy?



MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 14

14 **B**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Water is transported to a village in a tank pulled by a tractor.

Fig. 1.1 shows the tank being pulled by a tractor.

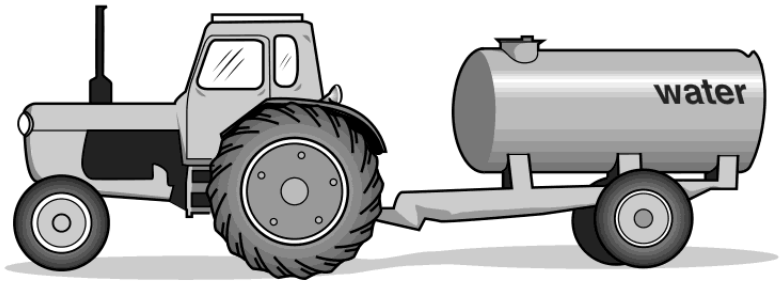


Fig. 1.1

The combined mass of the tractor and the tank is 4100 kg when the tank is empty and 6500 kg when the tank is full of water.

(a) The density of water is 1000 kg / m^3.

Calculate the volume of water in the tank when it is full.

volume = ... [2]

(b) At the start of the journey, the tractor and tank accelerate from rest along a straight, horizontal road. As their speed increases, one form of energy is decreasing.

(i) State the name of the form of energy that is decreasing.

..... [1]

(ii) Explain what happens to this energy.

.....

.....

.....

..... [2]

(c) The village is located on a mountain at a vertical height of 850 m above the water supply. The gravitational field strength g is 10 N/kg.

Calculate the gravitational potential energy gained by the water as it is transported from the supply to the village.

energy gained = ... [2]

Section A

1	**(a)**	$(V =)\ m/\rho$ **or** 10 600 / 6500 / 4100 / 2400 ÷ 1000 **or** 6.5 **or** 4.1	C1	
		2.4 m^3 / 2.4 × 10^6 cm^3	A1	
	(b)	**(i)** fuel / chemical (potential energy)	B1	
		(ii) some to heat / thermal (energy)	B1	
		some to kinetic (energy of air **or** tractor)	B1	
	(c)	$(GPE =)\ mgh$ **or** 2400 × 10 × 850	C1	
		2.0 / 2.04 × 10^7 J	A1	[7]

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Aeroplanes fly at high altitudes where the temperature is well below 0 °C. Ice that forms on an aeroplane can fall to earth and strike the ground.

Fig. 9.1 shows a block of ice falling from an aeroplane as it approaches an airport.

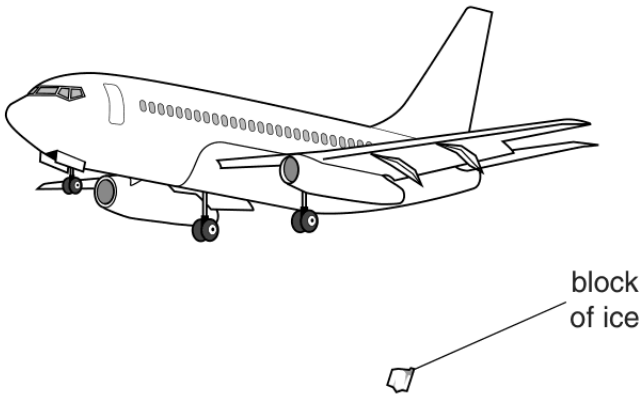


Fig. 9.1 (not to scale)

The mass of the falling block of ice is 1.2 kg and the gravitational field strength g is 10 N/kg.

(a) Calculate the weight of the block of ice.

weight = ... [1]

(b) Fig. 9.2 is the speed-time graph for the block of ice as it falls to the ground.

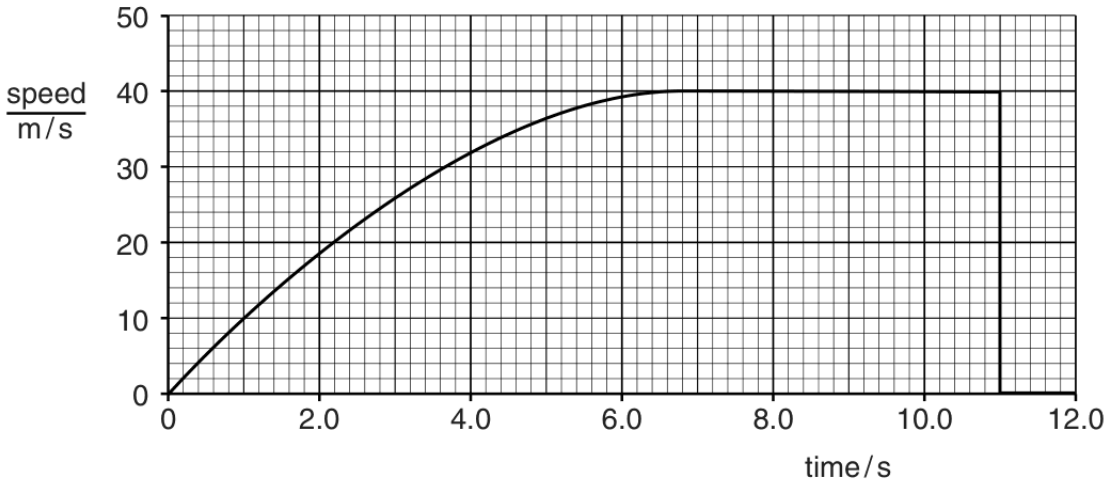


Fig. 9.2

At first, the acceleration of the block of ice is equal to the acceleration of free-fall. The acceleration of the block then decreases to zero as the block reaches terminal velocity.

(i) As the block of ice falls, the force F of air resistance acting on the block changes.

1. State the value of F at time = 0.

F = ... [1]

2. State the value of F at time = 10.0 s.

F = ... [1]

3. Explain why F changes.

.....

.....

..... [1]

(ii) State the energy change that takes place when the block is falling at terminal velocity.

.....

..... [1]

(iii) Using information from Fig. 9.2, determine the maximum kinetic energy of the block when it is falling to the ground with terminal velocity.

maximum kinetic energy = ... [3]

(c) The block strikes the ground and it stops moving. This impact causes some of the ice to melt.

(i) The specific latent heat of fusion of ice is 330 J / g.

Calculate the maximum possible mass of ice that melts as a result of the impact.

mass = ... [2]

(ii) In practice, the mass of ice that melts on impact is less than the value calculated in **(c)(i)**.

Suggest two reasons for this.

1. .....

.....

2. .....

.....

[2]

(d) As the solid ice melts, it changes into liquid water.

Describe, in terms of molecules, how ice differs from liquid water.

.....

.....

.....

.....

.....

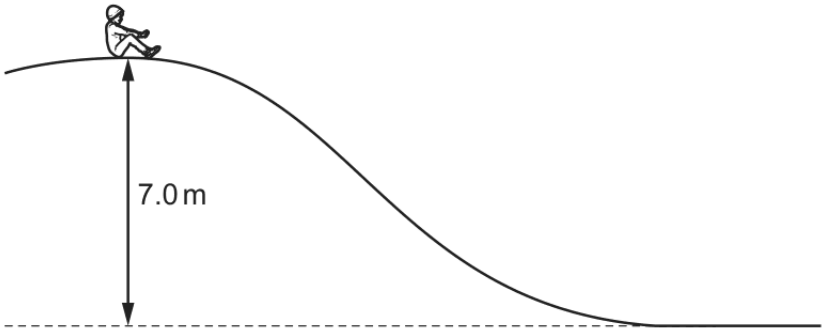
..... [3]

Section B

9	**(a)**	12 N	B1	[1]
	(b) **(i)**	0 **or** zero	B1	
		12 N **or** it is equal to the weight	B1	
		(*F* increases) as the speed increases	B1	
	(ii)	(gravitational) potential to thermal energy **or** to k.e. **of air**	B1	
	(iii)	(KE =)$\frac{1}{2}mv^2$	C1	
		$\frac{1}{2} \times 1.2 \times 40^2$	C1	
		960 J	A1	[7]
	(c) **(i)**	(m =)E / l_f **or** Q / l_f **or** 960 / 330	C1	
		2.9 / 2.91 g **or** 2.9 / 2.91 $\times 10^{-3}$ kg	A1	
	(ii)	any **two** from: ice is below 0 °C thermal energy transferred / lost (to ground / air) work done compressing / compacting the ground	B2	[4]
	(d)	any **three** from: molecules fixed in position **or** water molecules move around molecules vibrate **or** water molecules do not vibrate molecules in regular lattice **or** water molecules placed randomly (interatomic) forces between ice molecules larger ice molecules further apart	B3	[3]
				[15]

QUESTION · 5054_s16_qp_11.pdf · source page 6 of 16

11 A child slides down a slide.



The weight of the child is 250 N. The height of the slide is 7.0 m. The work done against friction as the child travels down the slide is 1300 J.

What is the change in gravitational potential energy and what is the final kinetic energy of the child?

	change in gravitational potential energy	final kinetic energy
A	1750	450
B	1750	1750
C	17 500	16 200
D	17 500	17 500

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 11

11 **A**

12 The overall stopping distance of a cyclist is made up of two parts, as shown in Fig. 12.1:

- the distance the cyclist travels during the reaction time of the cyclist (the thinking distance)
- the distance the cyclist travels after the brakes are applied (the braking distance)

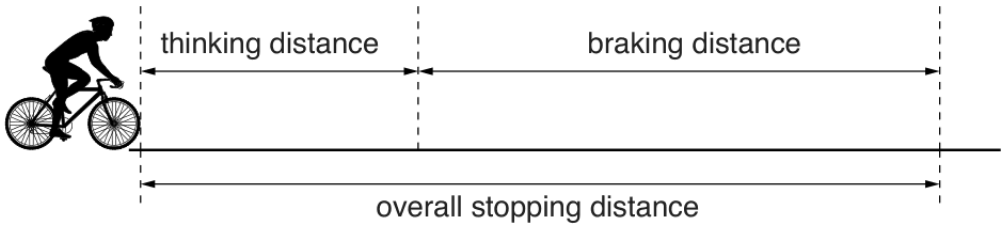


Fig. 12.1 (not to scale)

(a) State the energy change that occurs during braking.

.....

.....[2]

(b) A ball rolls in front of a cyclist at time $t = 0$ and the cyclist brakes and comes to rest.

Fig. 12.2 shows the speed-time graph for the cyclist.

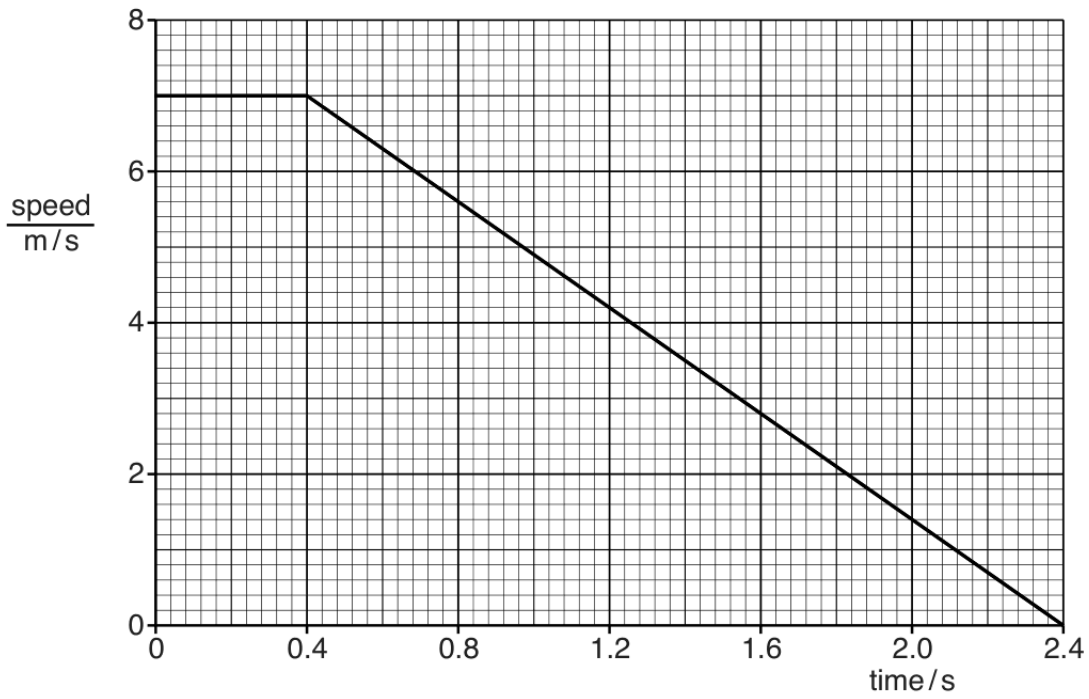


Fig. 12.2

(i) Using Fig. 12.2, determine the reaction time of the cyclist.

reaction time = ...[1]

(ii) Using Fig. 12.2, calculate the thinking distance.

thinking distance = ...[2]

(iii) State how the braking distance is found using Fig. 12.2.

.....

.....[1]

(iv) On another occasion, the same cyclist travels at an initial speed of 5.0 m/s. A ball rolls in front of the cycle at time $t = 0$.

The cyclist has the same reaction time and the deceleration of the cycle is the same as in Fig. 12.2.

On Fig. 12.2 draw the new speed-time graph for the cyclist. [2]

(v) The braking distance is longer when the cyclist stops on a wet road.

Explain why.

.....

.....

.....

.....

.....

.....[3]

(c) Fig. 12.3 shows the hydraulic braking system of the cycle.

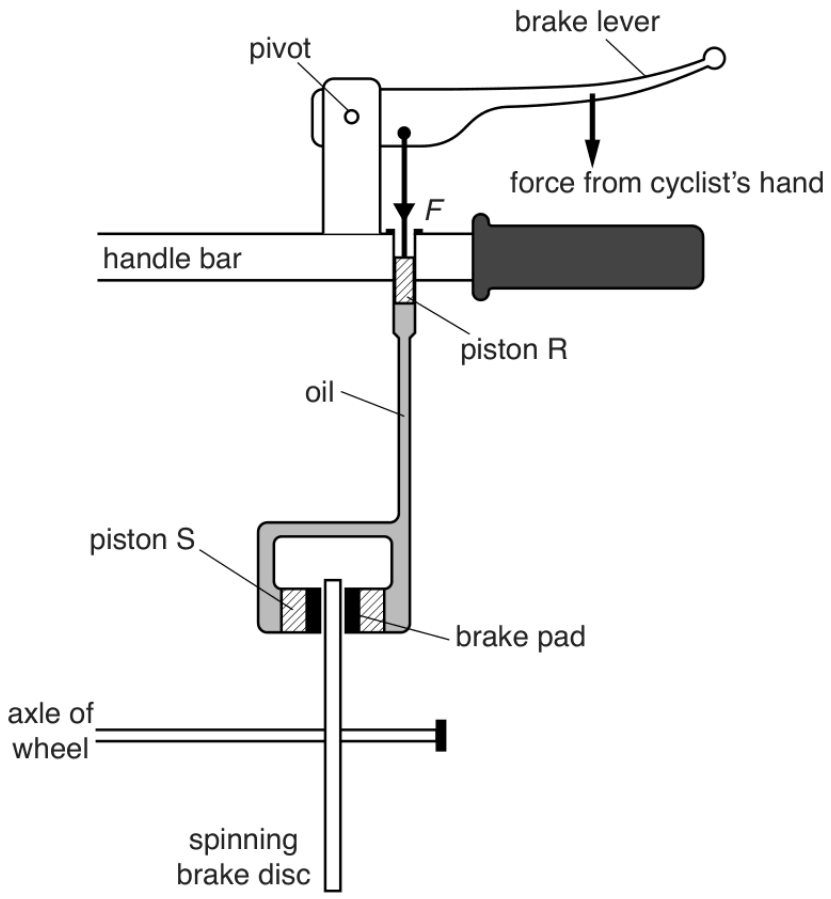


Fig. 12.3 (not to scale)

(i) The cyclist applies a force on the brake lever. This increases the pressure in the oil by 1.2×10^6 Pa.

The cross-sectional area of piston R is 5.0×10^{-5} m^2.

Calculate the force *F* applied by the brake lever to piston R.

F = ... [2]

(ii) The force applied to each of the brake pads is larger than *F*.

Explain why.

.....

.....

.....[2]

12	**(a)**	kinetic energy at start	B1
		thermal energy / heat energy / internal energy at end	B1
	(b) **(i)**	0.4(0) s	B1
	(ii)	(d=) s × t numerical or algebraic	C1
		2.8 m	A1
	(iii)	area under graph (between 0.4 and 2.4 s) **or** time (difference) × <u>average</u> speed **or** ½ × time (difference) × initial speed	B1
	(iv)	horizontal line from (0,5) to (0.4,5)	B1
		line showing braking with same gradient as original line	B1
	(v)	less friction	B1
		less deceleration **or** graph less steep **or** less <u>force</u> <u>backwards</u> / less <u>force</u> <u>opposing motion</u> **or** same KE lost / work done by friction	B1
		longer time to stop **or** larger area under (speed-time) graph **or** work = force x distance applied correctly	B1
	(c) **(i)**	(F=) P × A numerical or algebraic	C1
		60 N	A1
	(ii)	same pressure	B1
		larger area (of S / brake pads)	B1

2 Fig. 2.1 shows an electric motor lifting a load.

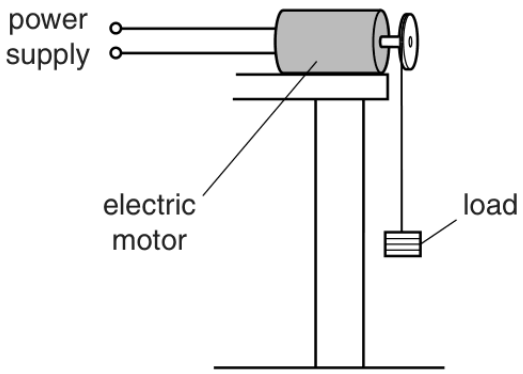


Fig. 2.1

The load, of weight 5.0 N, is raised through a vertical height of 3.5 m at a constant speed. The efficiency of the electric motor is 0.65 (65%).

(a) Calculate the increase in gravitational potential energy of the load.

gravitational potential energy = ...[2]

(b) (i) State the formula that relates efficiency, energy input and useful energy output.

.....

.....[1]

(ii) Calculate the energy input to the motor.

energy input = ...[2]

(c) Suggest one reason why the efficiency of the motor is less than 1.0 (100%).

.....

.....[1]

2 **(a)** (PE =) mgh **or** Fd **or** 5 × 3.5 C1
17.5 J **or** 17 J **or** 18 J A1

(b) **(i)** (efficiency = useful) energy output / energy input B1
in any form but all three quantities must be mentioned
if efficiency is not the subject of the equation

(ii) 17.5 / 0.65 **or** 17.5 / 65 C1
or 0.65 / 65 = **(a)** / energy input
26.9 J **or** 27 J A1

(c) due to friction (in bearings of motor) B1
or due to (electrical) resistance (in motor)
or air resistance acts
or thermal energy / heat produced / lost (in resistance of motor / due to friction)

QUESTION · 5054_w16_qp_11.pdf · source page 5 of 20

9 Three objects P, Q and R have different masses and different speeds, as shown in the table.

	mass / kg	speed / m/s
P	1.0	3.0
Q	2.0	2.0
R	5.0	1.0

What is the order of increasing kinetic energy (smallest first) of the objects?

A $P \rightarrow Q \rightarrow R$

B $P \rightarrow R \rightarrow Q$

C $R \rightarrow P \rightarrow Q$

D $R \rightarrow Q \rightarrow P$

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 9

9 **D**

QUESTION · 5054_w16_qp_11.pdf · source page 6 of 20

10 A weightlifter lifts some masses through a height of 1.8 m.

The gravitational field strength *g* is 10 N / kg.

During the lift, the gravitational potential energy of the masses increases by 4500 J.

What is the total mass lifted?

A 25 kg **B** 250 kg **C** 450 kg **D** 810 kg

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 10

10	**B**

3 A large power station burns oil to heat water in a boiler.

(a) State the name of the form of energy stored in the oil before it is burnt.

.....[1]

(b) (i) State and explain whether oil is a renewable or a non-renewable energy source.

.....

.....

.....[1]

(ii) Describe one environmental issue caused by burning oil to generate electricity in a power station.

.....

.....[1]

(c) (i) State what is meant by *efficiency*.

.....

.....[1]

(ii) The efficiency of the power station is 0.38 (38%). Its electrical power output is 1.9×10^9 W.

Calculate

1. the input power to the power station,

input power = ...[2]

2. the energy wasted in 2.0 hours.

energy wasted = ...[2]

3	**(a)**		chemical (potential energy)	B1	
	(b)	**(i)**	non-renewable **and** oil / it is not replaced / will run out	B1	
		(ii)	acid rain **or** produces CO_2 **or** warms lakes / rivers / sea **or** global warming **or** greenhouse effect	B1	
	(c)	**(i)**	<u>useful</u> energy output / (total) energy input **or** power for energy twice	B1	
		(ii)	**1** $1.9 \times 10^9/0.38$ **or** $1.9 \times 10^9 \times 100/38$	C1	
			5.0×10^9 W	A1	
			2 (E =)Pt **or** $0.62 \times 5.0 \times 10^9 \times 2.0$ ($\times$ 3600) **or** $(5.0 - 1.9) \times 10^9$ etc.	C1	
			2.2×10^{13} J	A1	[8]

QUESTION · 5054_w16_qp_12.pdf · source page 6 of 16

11 Three objects P, Q and R have different masses and different speeds, as shown in the table.

	mass / kg	speed / m/s
P	1.0	3.0
Q	2.0	2.0
R	5.0	1.0

What is the order of increasing kinetic energy (smallest first) of the objects?

A $P \rightarrow Q \rightarrow R$

B $P \rightarrow R \rightarrow Q$

C $R \rightarrow P \rightarrow Q$

D $R \rightarrow Q \rightarrow P$

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 11

11 **D**

2 A skateboarder of mass 45 kg is at the top of a ramp.

Fig. 2.1 shows the skateboarder and the ramp.

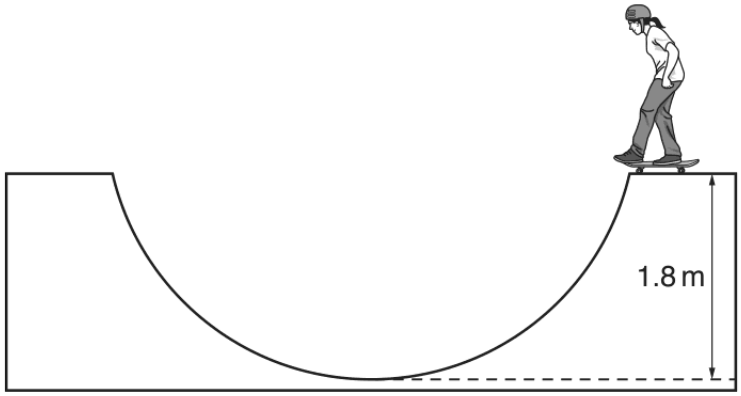


Fig. 2.1 (not to scale)

(a) The skateboarder moves off and she descends vertically through 1.8 m. The gravitational field strength g is 10 N/kg.

(i) Calculate the change in gravitational potential energy of the skateboarder.

energy change = ...[2]

(ii) State the name of two forms of energy that increase as she descends.

1. .....

2. .....
[2]

(b) At the lowest point on the ramp, the skateboarder is travelling at a constant speed along a path which is part of a vertical circle.

State

(i) the direction of the resultant force on the skateboarder at this point,

.....[1]

(ii) how the weight of the skateboarder compares with the upward force that the ramp exerts on her.

.....

.....

.....[1]

2	**(a)**	**(i)**	(GPE =)*mgh* **or** 45 × 10 × 1.8	B1	
			810 J	B1	
		(ii)	kinetic either order thermal / internal / heat / sound } either order	B1 B1	
	(b)	**(i)**	upwards / centripetal / towards centre (of circle)	B1	
		(ii)	it / weight less (than normal contact force) **or** upward force greater	B1	[6]

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Different energy sources are used to generate electricity.

(a) Energy sources are renewable or non-renewable.

(i) Nuclear energy is described as a non-renewable source.

Explain what is meant by a *non-renewable* energy source.

.....

.....[1]

(ii) Four of the energy sources used are:

hydroelectric **oil** **geothermal** **wind**

Write the name of these energy sources in the correct column of the table below.

non-renewable	renewable and caused by energy from the Sun	renewable and not caused by energy from the Sun

[3]

(iii) State one way in which using nuclear energy is better for the environment than using oil.

.....

.....[1]

(iv) State one way in which using nuclear energy is worse for the environment than using oil.

.....

.....[1]

(b) Fig. 9.1 is a block diagram of a power station that produces electrical energy from oil.

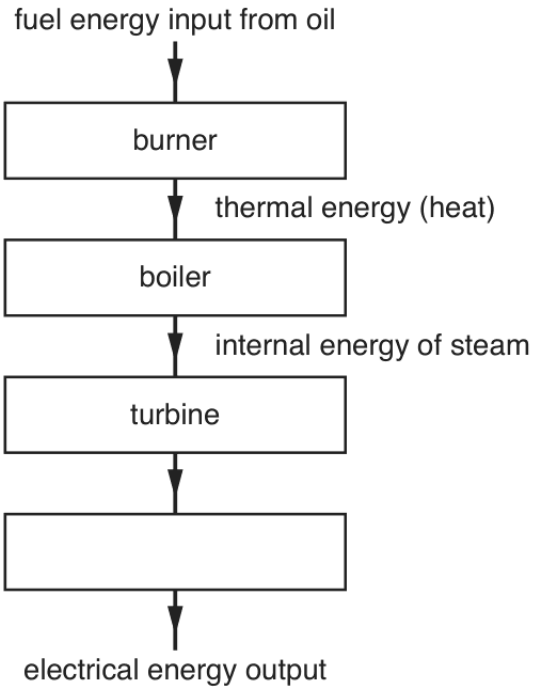


Fig. 9.1

(i) Write the name of the missing part of the power station in the empty box on Fig. 9.1. [1]

(ii) State the form of energy that the turbine possesses.

.....[1]

(iii) A small boiler in the power station contains 24 m^3 of water at 30 °C. High pressure in the boiler increases the boiling point of water to 120 °C.

Thermal energy supplied to the boiler is used to heat the water from 30 °C to 120 °C and then to turn it all to steam at 120 °C.

The density of water is 1000 kg / m^3.
The specific heat capacity of water is 4200 J / (kg °C).
The specific latent heat of vaporisation of water is 2.3×10^6 J / kg.

1. Calculate the mass of water in the boiler.

mass = ...[1]

2. Calculate the total thermal energy (heat) supplied to the boiler.

thermal energy = ...[4]

(iv) The electrical energy output from the power station is transmitted over long distances at a high voltage.

Explain why electricity is transmitted at a high voltage.

.....

.....

.....

.....[2]

Section B

Question	Answer	Marks
9(a)(i)	not being replaced **or** will run out	**B1**
9(a)(ii)	only oil in 1st column	**B1**
	only wind and hydroelectric in 2nd column	**B1**
	geothermal in 3rd column	**B1**
9(a)(iii)	less greenhouse gases / global warming / acid rain / **or** toxic gases / oil spills **and** how they affect a named organism / ecosystem	**B1**
9(a)(iv)	nuclear waste / radioactive waste causes (storage) problems **or** explosion / melt down / leak emits radioactivity	**B1**
9(b)(i)	generator (and transformer)	**B1**
9(b)(ii)	kinetic (energy)	**B1**
9(b)(iii)	1 24 000 kg	**B1**
	2 formula (E) = mL **or** (E **=**) mcT seen	**C1**
	$24\,000 \times 2.3 \times 10^6$ **or** $5.5(2) \times 10^{10}$ (J)	**C1**
	$24\,000 \times 4200 \times 90$ **or** $9.0(72) \times 10^9$ (J)	**C1**
	6.4×10^{10} J	**A1**
9(b)(iv)	low current	**B1**
	less energy / heat loss (in resistance or cables) **or** thinner wires can be used	**B1**

2 A swing is made by tying rope loosely to the branch of a tree, as shown in Fig. 2.1.

A child swings backwards and forwards several times, starting at the highest point A.

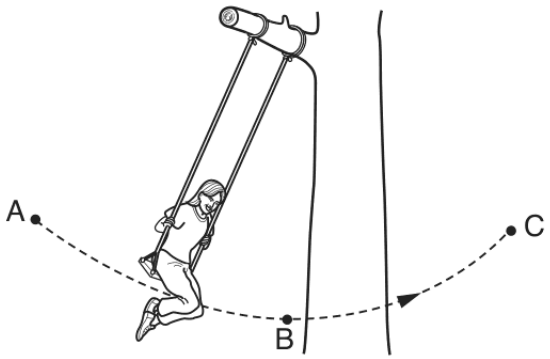


Fig. 2.1

(a) Explain how another child can obtain an accurate measurement of the time for one complete swing.

.....

.....

.....

.....

.....[2]

(b) When the child moves from A to B, she falls a vertical distance of 0.60 m. She loses 240 J of gravitational potential energy. The gravitational field strength g is 10 N/kg.

(i) Calculate the mass of the child.

mass = ...[2]

(ii) Suggest two reasons why her kinetic energy at B is not equal to 240 J.

1. .....

.....

2. .....

.....

[2]

Question	Answer	Marks
2(a)	use of stopwatch or electronic timer	**B1**
	time at least 5 swings **and** divide by number of swings **or** use of fiducial mark **or** definition of one swing clear e.g. A to C to A or from A and back to A	**B1**
2(b)(i)	(m=) P.E. / gh **or** 240 / 10 × 0.6	**C1**
	40 kg	**A1**
2(b)(ii)	air resistance **or** friction (with air or rope and tree)	**B1**
	heat produced / work done (in / against air or friction) **or** effect of wind **or** work done by arms / legs	**B1**

QUESTION · 5054_w17_qp_11.pdf · source page 6 of 20

9 An astronaut on the Moon drops a tool box of mass 3.0 kg. It falls from rest and its kinetic energy as it hits the surface is 0.96 J.

At which speed does the tool box hit the surface of the Moon?

A 0.46 m/s **B** 0.57 m/s **C** 0.64 m/s **D** 0.80 m/s

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 9

9	D	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a large container ship travelling at constant speed in a straight line.

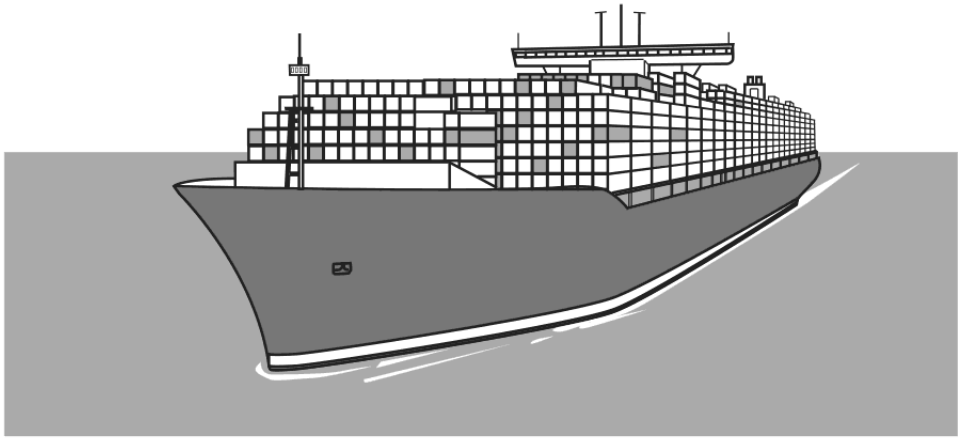


Fig. 9.1

The resistive force acting on the ship is 2.8×10^6 N.

(a) The speed of the ship is 9.7 m/s.

(i) Calculate the work done against the resistive force on the ship in 1.0 s.

work done = ... [2]

(ii) The engines are powered by oil.

State the energy transfer that is taking place when the ship is travelling at constant speed.

.....

..... [2]

(b) The mass of the ship is 2.2×10^8 kg. The engines are switched off and the resistive force causes the ship to decelerate.

(i) Calculate the initial deceleration of the ship.

deceleration = ... [2]

(ii) As the speed of the ship decreases, its deceleration changes.

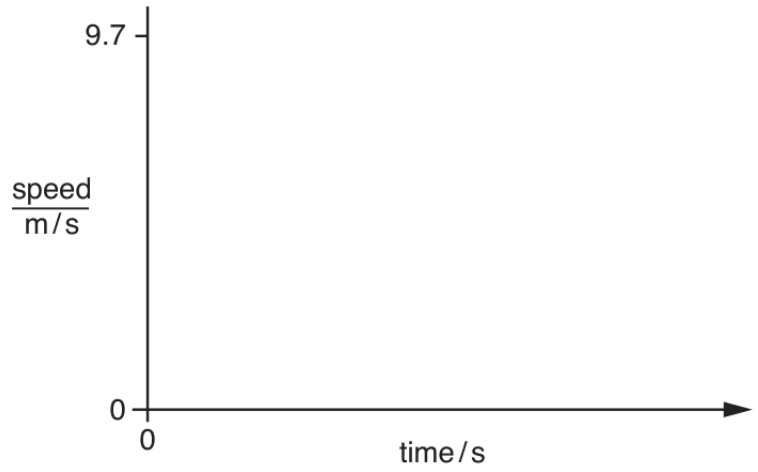
1. Suggest and explain how the deceleration changes.

.....

.....

..... [2]

2. On Fig. 9.2, sketch a possible speed-time graph for the ship as it decelerates to rest.



[2]

Fig. 9.2

3. Explain how the distance travelled by the ship may be determined from the speed-time graph.

.....

.....

..... [2]

(c) When the ship is travelling at a different speed, energy is being supplied to the engines at a rate of 33 MJ/s. The efficiency of the engines is 0.36 (36%).

(i) State a relationship that defines *efficiency*.

.....

..... [1]

(ii) Calculate the rate at which energy is wasted in the engines.

rate at which energy is wasted = ... [2]

Section B

Question	Answer	Marks
9(a)(i)	(*WD*) = *Fx* **or** $2.8 \times 10^6 \times 9.7$	**C1**
	2.7×10^7 J	**A1**
9(a)(ii)	from chemical (potential energy)	**B1**
	to thermal (energy) / heat	**B1**
9(b)(i)	(*a* =)*F* / *m* **or** $2.8 \times 10^6 / 2.2 \times 10^8$	**C1**
	0.013 m / s^2	**A1**
9(b)(ii)	**1** deceleration / it decreases	**B1**
	resistive force decreases **or** resistive force depends on speed	**B1**
	2 curve/line from 9.7 to zero **and** gradient negative (**allow** zero at end)	**B1**
	magnitude of gradient decreasing	**B1**
	3 area mentioned	**B1**
	area <u>under</u> line / curve **or** convert cm^2 (of graph paper) to distance **or** in terms of the scales	**B1**
9(c)(i)	(efficiency =) useful energy output / total energy input **or** useful power output / total power input	**B1**
9(c)(ii)	33×0.64 **or** 12 MJ / s	**C1**
	21 MJ / s	**A1**

QUESTION · 5054_w17_qp_12.pdf · source page 5 of 16

8 A tennis ball of mass 56 g is travelling at 1500 metres / minute.

Which expression is equal to the kinetic energy, in joules, of the tennis ball?

A $\frac{1}{2} \times 0.056 \times (25)^2$

B $\frac{1}{2} \times 0.056 \times (1500)^2$

C $\frac{1}{2} \times 56 \times (25)^2$

D $\frac{1}{2} \times 56 \times (1500)^2$

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 8

8	A	**1**

2 A force applied to a solid object may cause it to accelerate so that its velocity changes.

(a) State two other properties of the object which may change when a force is applied.

1. .....

2. .....

[2]

(b) A spring has a mass of 0.012 kg.

(i) The gravitational field strength g is 10 N/kg.

Calculate the weight of the spring.

weight = ...[1]

(ii) The spring is suspended vertically and a load is attached to its lower end. The spring extends by 2.7 cm and does not pass the limit of proportionality.

Fig. 2.1 shows the spring and load attached to the lower end of a second identical spring.

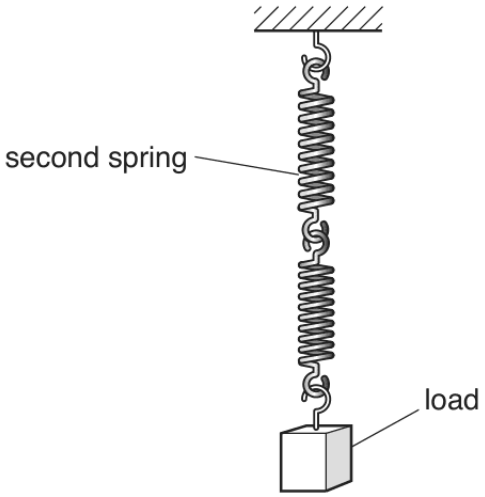


Fig. 2.1

Suggest one reason why the extension of the second spring differs from 2.7 cm.

.....

.....

.....[1]

(iii) Explain what is meant by *limit of proportionality*.

.....

.....

.....[1]

(c) The load in Fig. 2.1 is pulled down below its equilibrium position.

(i) State the form of energy stored in the stretched springs.

.....[1]

(ii) The load is released and it moves upwards and downwards. The distance travelled in each movement decreases until the load stops moving.

Explain why the load stops moving.

.....

.....

.....[1]

Question	Answer	Marks
2(a)	shape	**B1**
	size / volume / density	**B1**
2(b)(i)	0.12 N	**B1**
2(b)(ii)	weight of first spring (increases extension) **or** weight of first spring (takes spring beyond the limit of proportionality)	**B1**
2(b)(iii)	where / point / load / extension / limit beyond which the extension is not directly proportional to load / extension-load graph is not straight	**B1**
2(c)(i)	elastic potential energy	**B1**
2(c)(ii)	friction / air resistance mentioned **or** thermal energy / heat produced	**B1**

4 A woman does work as she raises a flag by pulling downwards on a rope. This is shown in Fig. 4.1.

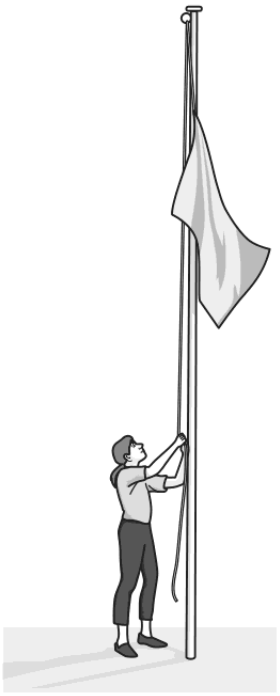


Fig. 4.1

(a) Define what is meant by *work done*.

.....

.....[2]

(b) The mass of the flag is 0.35 kg and the flag is raised through a height of 6.0 m.

The gravitational field strength g is 10 N/kg.

Calculate the increase in gravitational potential energy of the flag.

gravitational potential energy = ...[2]

(c) Suggest and explain a reason why the work done on the rope is larger than the value calculated in **(b)**.

.....

.....

.....

.....[2]

Question	Answer	Marks
4(a)	force × distance (moved)	**C1**
	force × distance <u>moved</u> in direction of force	**A1**
4(b)	mgh in any form numerical or algebraic	**C1**
	21 J	**A1**
4(c)	larger force is needed (by woman / on rope)	**B1**
	(because of either) • friction / heat produced • <u>weight</u> of rope / her arms • flag / rope gains K.E. • flag accelerates / gains speed	**B1**

2 In an experiment, a student takes measurements and determines the extension of a spring for different loads. The apparatus is shown in Fig. 2.1.

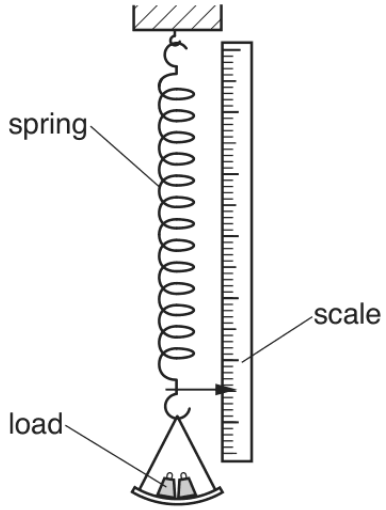


Fig. 2.1

The table in Fig. 2.2 shows the results.

load/N	0	1.0	2.0	3.0	4.0	5.0
length of spring/mm	200	235	270	305	350	420
extension/mm	0	35	70	105		

Fig. 2.2

(a) Only some of the extensions are shown in the table.

Complete the table to show all of the extensions. [1]

(b) Calculate the load that produces an extension of 49 mm.

load = ...[2]

(c) The student pulls a load downwards from position A to position B and holds it fixed at position B, as shown in Fig. 2.3.

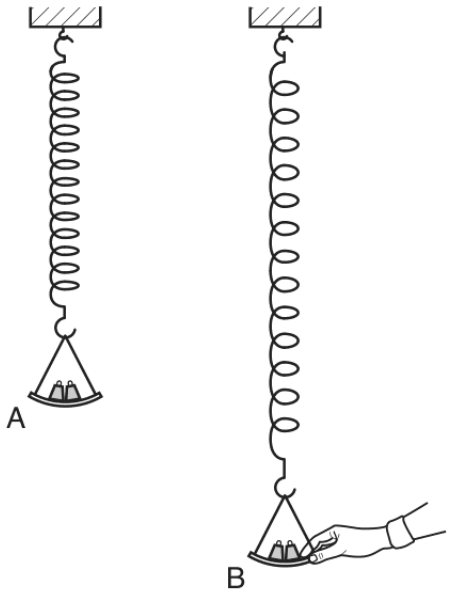


Fig. 2.3

The load is stationary at A and at B. The load has no kinetic energy at either point.

(i) Place ticks (✓) in the boxes to show how the value of each of the forms of energy compares at A and B.

	equal at A and B	larger at A	larger at B
energy stored in spring	☐	☐	☐
gravitational potential energy of load	☐	☐	☐

[1]

(ii) Work is done by the student to pull the load down. The law of conservation of energy states that energy cannot be created and cannot be destroyed.

Explain how this principle applies in this case.

.....

.....

.....

.....[2]

Question	Answer	Marks
2(a)	150 and 220 in table	**B1**
2(b)	any use of proportionality	**C1**
	1.4 N	**A1**
2(c)(i)	1st line larger at B **and** 2nd line larger at A	**B1**
2(c)(ii)	student loses energy / chemical energy decreases **or** (gravitational) potential energy decreases **and** spring energy increases	**B1**
	idea that increase in spring energy > change / decrease in (grav) PE (so student provides work needed)	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Wind energy is a renewable energy source. A wind turbine and generator convert energy in the wind to electrical energy in a generator. Fig. 9.1 shows some wind turbines.

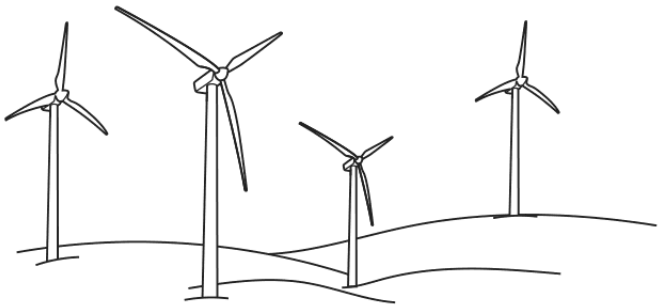


Fig. 9.1

(a) (i) State what is meant by a *renewable energy source*.

.....

.....

.....[1]

(ii) State the name of **one other** renewable energy source.

.....

.....[1]

(iii) Some methods of generating electricity cause much more global warming than wind energy.

1. State **one** method of generating electricity that causes a large amount of global warming.

.....[1]

2. Describe, in outline, how the method you have chosen causes a significant amount of global warming.

.....

.....

.....[1]

(b) During a 30 minute period, a mass of 4.2×10^7 kg of air enters the turbine with a speed of 15 m/s.

(i) Calculate the kinetic energy of the air that enters the turbine in 30 minutes.

energy = ...[2]

(ii) The electrical power output of the generator connected to the turbine is 8.4×10^5 W.

Calculate the electrical energy output from the turbine in 30 minutes.

energy output = ...[2]

(iii) Calculate the efficiency of the turbine and generator in converting the kinetic energy of the air to electrical energy.

efficiency = ...[2]

(c) The generator contains a coil that rotates between the poles of a magnet.

The coil rotates 50 times in one second and produces an alternating voltage output with a maximum value of 500 V.

(i) On the grid in Fig. 9.2, sketch a voltage–time graph showing the output voltage.

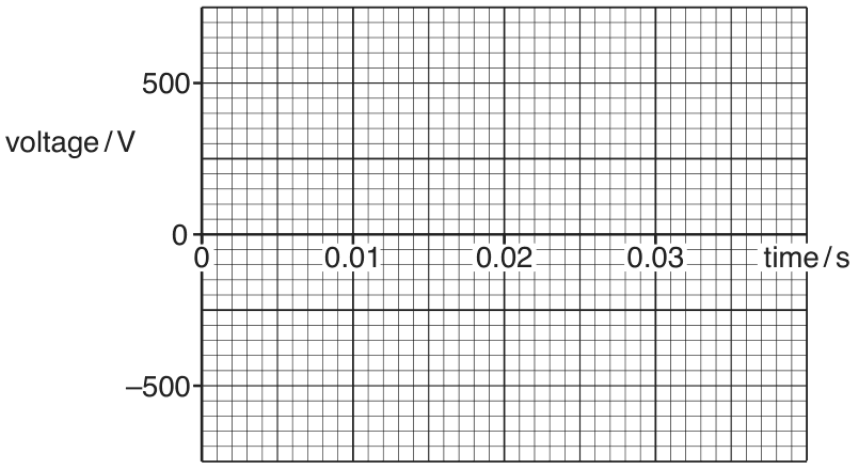


Fig. 9.2

[2]

(ii) The output voltage from the generator is stepped up by a transformer. The electrical energy then passes along a transmission line to a distant house.

Explain why

1. a high voltage is used for the transmission of electrical energy,

.....

.....

.....

.....[2]

2. a transformer is used to connect the transmission line to the house.

.....

.....

.....[1]

Question	Answer	Marks
9(a)(i)	will not run out **or** infinite **or** being replaced	**B1**
9(a)(ii)	solar / Sun, tidal, geothermal, biomass, hydroelectric, water waves	**B1**
9(a)(iii)	1 (burning) use of a fossil fuel, e.g. coal, oil, natural gas etc.	**B1**
	2 produces carbon dioxide **or** greenhouse gases (absorb infra-red from Earth)	**B1**
9(b)(i)	(E=) ½mv^2 seen in any form algebraic or numerical	**C1**
	4.7×10^9 J	**A1**
9(b)(ii)	(E=)Pt seen in any form algebraic or numerical	**C1**
	1.5×10^9 J	**A1**
9(b)(iii)	(efficiency=) (energy) output / (energy) input in any form	**C1**
	0.32 **or** 32%	**A1**
9(c)(i)	all peaks and troughs at 500 and -500 to within one small square **and** roughly sinusoidal	**B1**
	one wave takes 0.02 s within one small square	**B1**
9(c)(ii)	1 less energy loss / heat loss / power loss / more efficient	**B1**
	low(er) currents (in line) **or** thin(ner) wires can be used	**B1**
	2 step down transformer **or** voltage reduced (to 240 V) **or** otherwise voltage dangerous	**B1**

4 A rock from space is travelling at high speed towards the Earth.

(a) Complete the equation for the kinetic energy of an object travelling with a speed v and state what any other terms in the equation represent.

kinetic energy =

[1]

(b) The mass of the rock is 0.60 kg. When it hits the Earth, its average temperature increases by 25 °C.

(i) The specific heat capacity of the rock is 560 J / (kg °C).

Calculate the increase in the internal energy of the rock.

increase in internal energy = ... [2]

(ii) Use the answer from **(b)(i)** to determine the minimum possible speed of the rock when it hits the Earth.

minimum speed = ... [2]

(iii) Suggest one reason why the speed determined in **(b)(ii)** is less than the actual speed of the rock when it hits the Earth.

.....

..... [1]

Question	Answer	Marks
4(a)	(kinetic energy =) $\frac{1}{2}mv^2$ **and** m is the mass of the object	**B1**
4(b)(i)	(ΔQ =) $mc\Delta T$ **or** $0.60 \times 560 \times 25$	**C1**
	8.4×10^3 J	**A1**
4(b)(ii)	$v^2 = 2\Delta Q / m$ **or** $8.4 \times 10^3 = \frac{1}{2} \times 0.60 \times v^2$ **or** (v^2 =) 2.8×10^4 (m^2 / s^2)	**C1**
	167 m / s	**A1**
4(b)(iii)	some internal / thermal energy lost (to ground / air) **or** work done to shatter rock **or** rock bounces **or** energy used to compress ground **or** rock melts	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 A bucket of water is pulled up out of a well using a rope. Fig. 8.1 shows the rope winding on to a cylinder as the handle is turned.

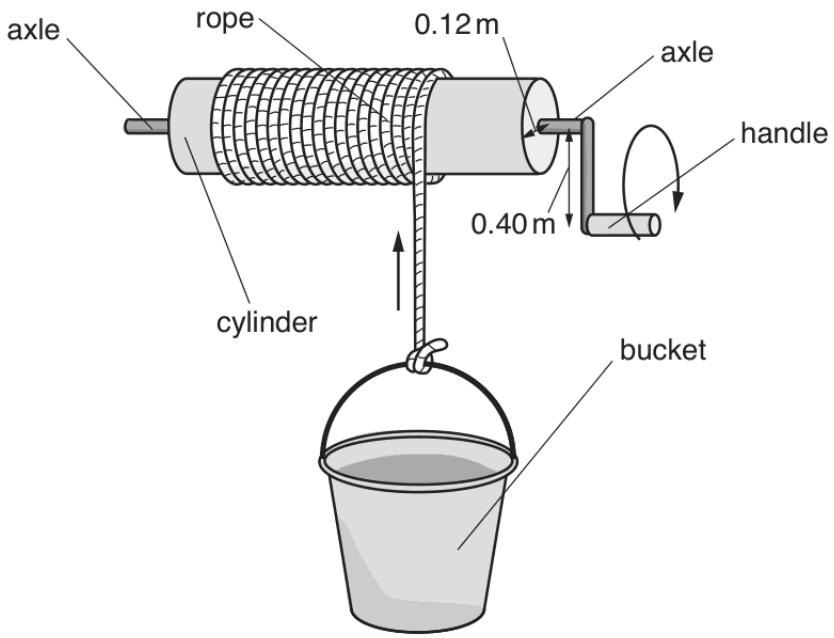


Fig. 8.1

The empty bucket has a mass of 1.0 kg.

(a) Complete the sentences that describe mass by filling in the gaps.

The mass of a body is a measure of the amount of .....

in the body. It resists a change in the state of ..... of

the body. [2]

(b) When the bucket is full, it contains $2.4 \times 10^{-2}\,m^3$ of water.

The gravitational field strength g is equal to 10 N/kg.

(i) Explain what is meant by a *gravitational field*.

.....

..... [1]

(ii) The density of water is 1000 kg / m^3.

Determine the total weight of the bucket and the water.

weight = ... [3]

(c) The radius of the cylinder is 0.12 m and the handle is 0.40 m from the axle of the cylinder. The weight of the bucket and the water produce a moment that acts on the cylinder.

(i) Calculate this moment.

moment = ... [2]

(ii) Calculate the minimum force on the handle that balances this moment.

force = ... [1]

(d) A farmer pulls the bucket of water up at a constant speed. He needs to exert a force on the handle that is greater than that calculated in **(c)(ii)**. He notices that there is a slight increase in temperature where the axle is in contact with the frame holding it.

(i) Suggest two reasons why the force exerted is greater than the value calculated.

1. .....

.....

2. .....

.....
[2]

(ii) State the energy changes that are taking place as the bucket is being lifted at a constant speed.

.....

.....

..... [3]

(e) State what can be deduced about the forces acting on the bucket when it is travelling upwards at a constant speed.

.....

..... [1]

Question	Answer	Marks
8(a)	substance **or** matter	**B1**
	rest **or** motion	**B1**
8(b)(i)	(a region of space) where a mass experiences a force (due to gravitational attraction)	**B1**
8(b)(ii)	(m =) ρV **or** $1000 \times 2.4 \times 10^{-2}$ **or** 24 (kg)	**C1**
	25 (kg) **or** 240 (N)	**C1**
	250 N	**A1**
8(c)(i)	(Γ =) Fx **or** 250×0.12	**C1**
	30 N m	**A1**
8(c)(ii)	75 N	**B1**
8(d)(i)	any **two** from: friction (between axle and cylinder) weight of rope force not perpendicular (to radius) rope wraps over itself (and increases moment)	**B2**

Question	Answer	Marks
8(d)(ii)	from chemical (energy) **or** chemical (energy) as first term	**B1**
	to gravitational potential (energy) **or** to thermal / heat (energy) as last term	**B1**
	to thermal / internal (energy) / heat **and** thermal / internal (energy) / heat	**B1**
8(e)	no resultant force **or** forces balance / cancel **or** tension equals weight (of bucket) **or** upwards force equals downward force	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 During a game of cricket, a player hits a ball with a bat. The ball then travels vertically upwards, as shown in Fig. 8.1.

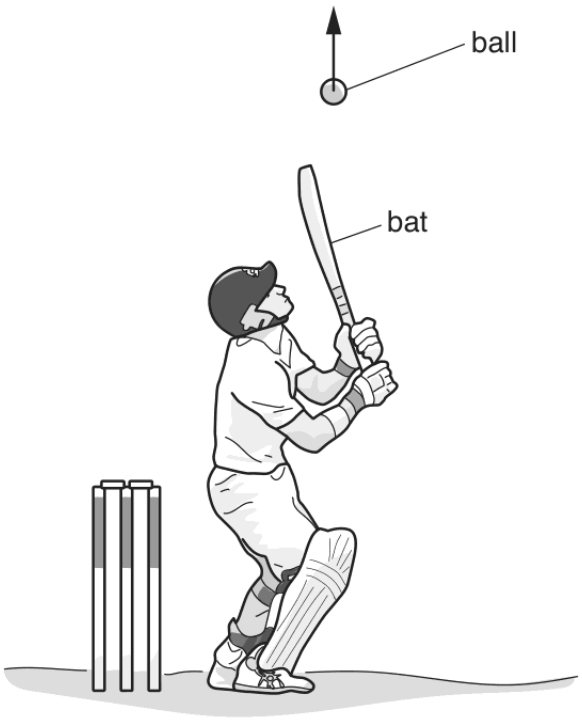


Fig. 8.1

The speed of the ball as it leaves the bat is 8.7 m/s.

(a) The mass of the ball is 0.16 kg.

(i) **1.** State what is meant by *mass*.

.....

..... [1]

2. State the name of a measuring instrument that is used to determine mass.

.....

..... [1]

(ii) Calculate the kinetic energy of the ball as it leaves the bat.

kinetic energy = ... [3]

(b) Fig. 8.2 is the velocity-time graph for the ball.

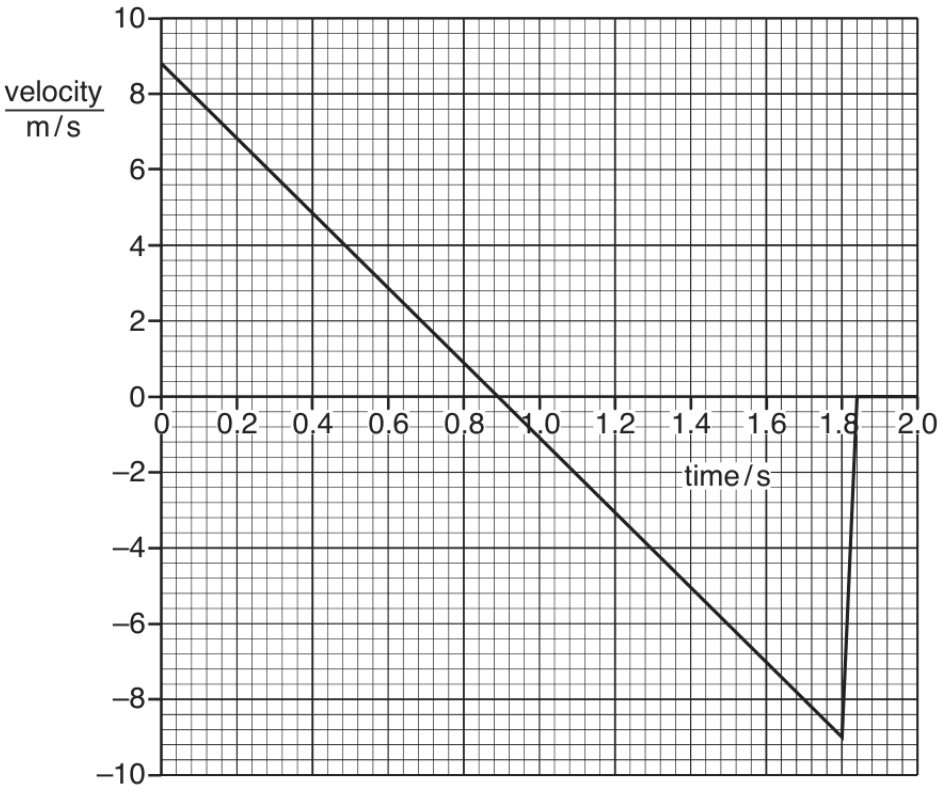


Fig. 8.2

(i) Fig. 8.2 shows that immediately after being hit, the ball has a negative acceleration.

1. State the name given to a negative acceleration.

..... [1]

2. State how the graph shows that the ball has a negative acceleration.

.....

..... [1]

(ii) Using Fig. 8.2, determine

1. the time at which the ball stops moving upwards,

time = ... [1]

2. the distance travelled before the ball stops moving upwards.

distance = ... [2]

(iii) Suggest why the graph, in Fig. 8.2, is almost vertical when time = 1.84 s.

.....

..... [1]

(iv) Fig. 8.2 shows that the ball remains stationary after a time of 1.84 s.

Describe the energy change that is occurring at a time of 1.84 s.

.....

.....

..... [2]

(c) For this ball moving in the way shown, the effect of air resistance is negligible.

When a ball with a much smaller mass is hit vertically upwards at the same speed, air resistance produces a noticeable effect on its motion.

Suggest two ways in which the velocity-time graph for the ball with a smaller mass differs from Fig. 8.2.

1. .....

.....

2. .....

.....
[2]

MARK SCHEME · 5054_w18_ms_22.pdf · source page 7 of 10 · Question 8

Question	Answer	Marks
8(a)(i)	1 quantity of matter (in a body)	**B1**
	2 balance **or** scales	**B1**
8(a)(ii)	(*k.e.* =) $\frac{1}{2}mv^2$	**C1**
	$\frac{1}{2} \times 0.16 \times 8.7^2$	**C1**
	6.1 J	**A1**
8(b)(i)	1 deceleration **or** retardation	**B1**
	2 negative gradient **or** line slopes downwards (left to right)	**B1**
8(b)(ii)	1 0.88 s ⩽ time ⩽ 0.90 s	**B1**
	2 area **or** counting squares **or** $\frac{1}{2}bh$ in some form	**C1**
	3.7 s ⩽ distance ⩽ 4.1 m	**A1**
8(b)(iii)	ball hits ground **or** short time for deceleration **or** large force **or** ground is hard	**B1**
8(b)(iv)	internal / thermal energy (of ball and ground) has increased	**B1**
	(internal energy) from kinetic energy	**B1**

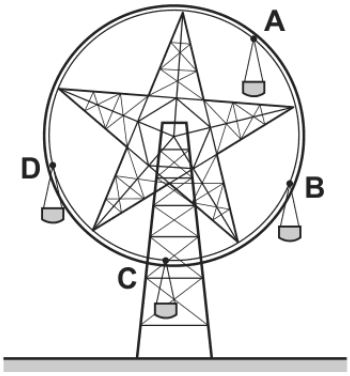
MARK SCHEME · 5054_w18_ms_22.pdf · source page 8 of 10 · Question 8

Question	Answer	Marks
8(c)	any **two** from: smaller time to drop to zero velocity / hit ground line not straight **or** velocity does not change uniformly **or** gradient not constant smaller area under (first part of) graph **or** less distance travelled slower final velocity initial downward gradient steeper	**B2**

QUESTION · 5054_s19_qp_11.pdf · source page 6 of 20

14 The diagram shows a stationary fairground ride with four chairs of equal mass.

Which chair has the most gravitational potential energy?



MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 14

14	A	**1**

3 A large spring is stretched by an athlete to increase the strength of his arms.

The athlete extends the spring, as shown in Fig. 3.1, and then releases it gently. He extends and releases the spring several times during a period of 60 s.

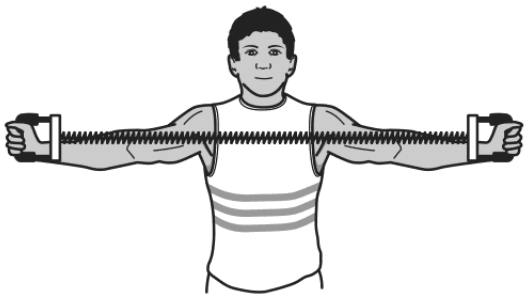


Fig. 3.1

(a) State the form of energy stored in the stretched spring and the form of energy stored in the muscles of the athlete.

form of energy in spring .....

form of energy in muscles .....
[2]

(b) Suggest what happens if he extends the spring beyond the limit of proportionality.

.....

..... [1]

(c) During one extension of the spring, its length increases from 70 cm to 93 cm. The average force exerted by the athlete is 400 N.

(i) Calculate the work done by the athlete in one extension.

work done = ... [2]

(ii) Describe how the athlete can use the value obtained in **(i)** to determine his average power during the 60 s.

.....

.....

.....

..... [2]

[Total: 7]

Question	Answer	Marks
3(a)	*in spring:* elastic or strain (potential energy)	**B1**
	in muscles: chemical (potential energy)	**B1**
3(b)	spring extends a different distance for the same (extra) load **or** load is no longer proportional to extension **or** becomes easier to pull	**B1**
3(c)(i)	(W =) Fd numerical or algebraic; 400×0.23	**C1**
	92 J	**A1**
3(c)(ii)	measure / determine the no. of times he extends the spring in 60 s	**B1**
	power = (total) work / time explained	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 7 of 16

16 An object is lifted.

Which expression gives the increase in gravitational potential energy of the object?

A mass multiplied by its velocity

B mass multiplied by its acceleration

C the rate at which work is done

D weight multiplied by the increase in height

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 16

16	D	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a map of the route taken by a car as it travels from town A to town B.

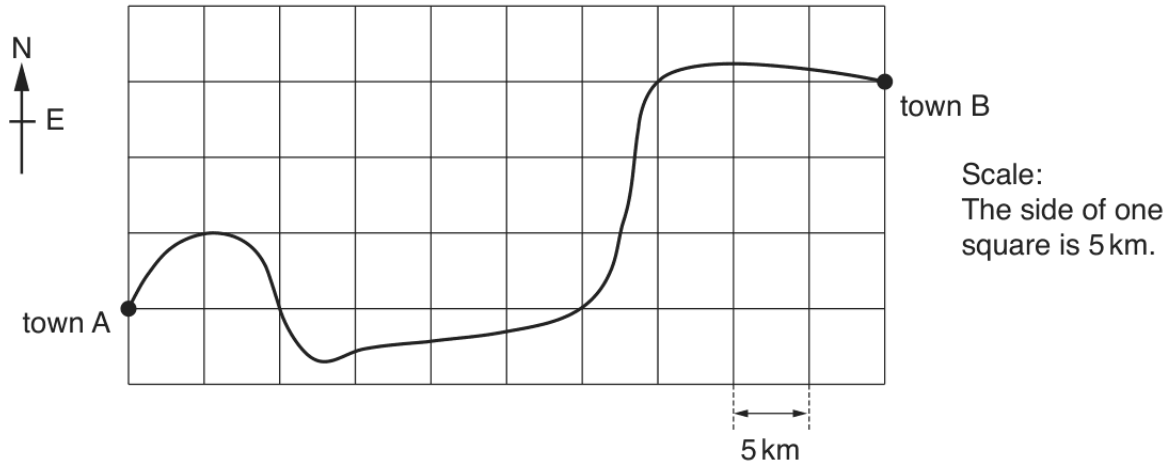


Fig. 9.1

The journey involves a distance and a displacement.

(a) (i) Using the scale on Fig. 9.1, determine the displacement in travelling from A to B.

size of displacement = ...

direction of displacement = ...
[3]

(ii) State how *distance* differs from *displacement*.

.....

.....

..... [1]

(iii) In some parts of the journey, the car is travelling at a constant speed but is also accelerating.

Explain how this is possible.

.....

.....

..... [2]

(b) At town B the car makes an emergency stop.

The total stopping distance of the car is the thinking distance added to the braking distance.

(i) Describe what is meant by the *thinking distance*.

.....

..... [1]

(ii) The brakes on the car work badly. This increases the braking distance.

State **two** other factors that increase the braking distance.

1. .....

.....

2. .....

.....
[2]

(c) The mass of the car and driver is 1200 kg. Just before the brakes are applied, the speed of the car is 30 m/s.

(i) Calculate the kinetic energy of the car and driver just before the brakes are applied.

kinetic energy = ... [3]

(ii) The braking force that stops the car is 1800 N.

Calculate the deceleration of the car.

deceleration = ... [2]

(iii) Explain why the temperature of the brakes increases during braking.

.....

..... [1]

[Total: 15]

Question	Answer	Marks
9(a)(i)	10.2–10.4 (cm) **or** 50 (km) **or** 15 (km) seen	**C1**
	51–53 km	**A1**
	direction (0)72–74° **or** N 72–74 °E etc.	**B1**
9(a)(ii)	(distance) has no direction **or (**distance) is a scalar	**B1**
9(a)(iii)	changes direction **or** goes round a corner	**B1**
	velocity changes **or** velocity depends on direction **or** acceleration / force is towards the centre of circle / centripetal	**B1**

Question	Answer	Marks
9(b)(i)	distance travelled before brakes applied (after seeing emergency)	**B1**
9(b)(ii)	any 2 of • poor tyre surface • poor road conditions, e.g. rain, ice • high speed (of car) **or** good aerodynamic car shape **or** tail wind • large mass / weight of car • driver applies less force to pedal / less braking force	**B1B1**
9(c)(i)	(KE =) ½mv^2 algebraic or numerical	**C1**
	½ × 1200 × 30^2	**C1**
	540 000 J	**A1**
9(c)(ii)	(a=) F / m algebraic or numerical	**C1**
	1.5 m / s^2	**A1**
9(c)(iii)	friction (produces thermal energy) **or** kinetic energy changes to heat **or** work done against friction	**B1**

2 A small coin of mass *m* is initially at rest. It is dropped from a height *h* above ground level. The coin has a speed *v* as it hits the ground.

The gravitational field strength *g* is equal to 10 N/kg.

(a) State an expression for:

- the gravitational potential energy of the coin at a height *h* above ground level

  gravitational potential energy = ...

- the kinetic energy of the coin when travelling at a speed *v*.

  kinetic energy = ...

[1]

(b) The coin is dropped from the top of a building of height 380 m.

Determine the speed of the coin as it hits the ground. You may ignore air resistance.

speed = ... [3]

(c) When air resistance is negligible, a heavier coin hits the ground at the same speed as a lighter coin when they are both dropped from the same height.

(i) Explain why.

.....

.....

..... [1]

(ii) When air resistance acts, coins of different masses do **not** hit the ground at the same speed when they are dropped from the same height.

Explain why.

.....

.....

..... [2]

[Total: 7]

Question	Answer	Marks
2(a)	mgh **and** $\frac{1}{2}mv^2$	**B1**
2(b)	$\frac{1}{2}mv^2 = mgh$ **or** $v^2 = 2gh$ **or** $v^2 = 2 \times 10 \times 380$ **or** $v^2 = 7600$	**C1**
	$(v =)\sqrt{2gh}$ **or** $\sqrt{2 \times 10 \times 380}$	**C1**
	87 m / s	**A1**
2(c)(i)	both the g.p.e. and the k.e. are proportional to mass **or** mass cancels out in the equation **or** they accelerate at the same rate / g	**B1**
2(c)(ii)	air resistance (force) is a smaller proportion of weight (for heavier coin)	**B1**
	acceleration of heavier coin greater / less affected by air resistance	**B1**

QUESTION · 5054_w19_qp_12.pdf · source page 7 of 16

14 An object X has mass m and velocity v.

A second object Y has mass $2m$ and velocity $2v$.

What is the value of the ratio $\frac{\text{kinetic energy of Y}}{\text{kinetic energy of X}}$?

A 1 **B** 2 **C** 4 **D** 8

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 14

14	D	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 Fig. 8.1 shows an elastic rope (bungee).

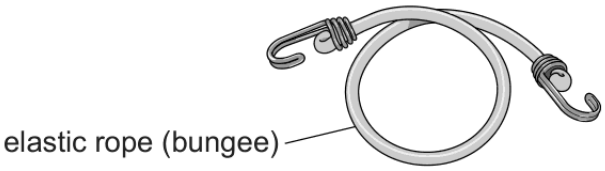


Fig. 8.1

The unstretched length of the rope is 0.80 m.

A student uses the hook at one end to suspend the rope from a shelf and hangs an empty paint can of mass 0.70 kg from the other end. The rope stretches to a length of 0.97 m but does not exceed its limit of proportionality.

(a) Explain what is meant by *limit of proportionality*.

.....

.....

..... [2]

(b) State what is meant by:

(i) *mass* ..... [1]

(ii) *weight.* .....

..... [2]

(c) The gravitational field strength is 10 N/kg.

(i) Calculate the weight of the empty paint can.

weight = ... [1]

(ii) For the rope, calculate a value for $\frac{\text{load}}{\text{extension}}$.

$\frac{\text{load}}{\text{extension}}$ = ... [1]

(d) The student repeatedly pours small volumes of paint into the can.

When there is $2.5 \times 10^{-3}\,m^3$ of paint in the can, the total length of the rope is 1.70 m. This is where the rope reaches the limit of proportionality.

(i) Determine the mass of the paint in the can.

mass = ... [3]

(ii) The student continues to pour paint into the can until there is a volume of $5.0 \times 10^{-3}\,m^3$ in it. He notices that the elastic rope becomes harder to stretch after the limit of proportionality.

He plots a graph of the total length of the rope against the volume of paint added.

On Fig. 8.2, sketch the length-volume graph for the range 0 to $5.0 \times 10^{-3}\,m^3$.

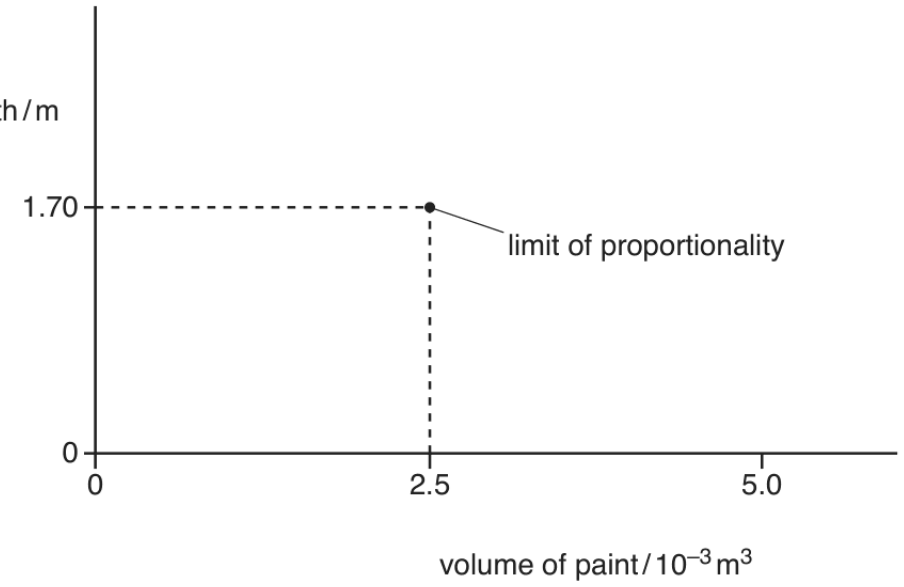


Fig. 8.2

[3]

(e) The student suddenly removes the can from the end of the stretched rope and, as it contracts, the rope jumps into the air.

State the energy change that is taking place in the rope as it contracts and jumps into the air.

.....

..... [2]

[Total: 15]

MARK SCHEME · 5054_w19_ms_22.pdf · source page 7 of 10 · Question 8

Question	Answer	Marks
8(a)	where / when extension is not directly proportional to load / tension / force (applied)	**C1**
	point where extension stops being directly proportional to load / tension / force (applied)	**A1**
8(b)(i)	the amount matter in a body **or** (a measure of) the inertia of a body	**B1**
8(b)(ii)	force (on a body)	**B1**
	in a gravitational field	**B1**
8(c)(i)	7.0 N **c.a.o.**	**B1**
8(c)(ii)	41 N / m **or** 0.41 N / cm	**B1**

MARK SCHEME · 5054_w19_ms_22.pdf · source page 8 of 10 · Question 8

Question	Answer	Marks
8(d)(i)	two separate approaches **I** / **II**	
8(d)(i)**I**	0.90	**C1**
	41×0.90 **or** 37 **or** 3.7 **or** $3.7 - 0.70$	**C1**
	3.0 kg	**A1**
	or	
8(d)(i)**II**	$1.7 - 0.97$ **or** 0.73	**C1**
	$41 \times (1.7 - 0.97)$ **or** 41×0.73 **or** 30	**C1**
	3.0 kg	**A1**
8(d)(ii)	straight line to marked point	**B1**
	0 < location of *y*-intercept < 1.70 m (intercept need not be labelled)	**B1**
	curve of changing / changed gradient beyond marked point	**B1**
8(e)	from elastic (potential energy) **or** strain (energy)	**B1**
	to kinetic (energy) **or** to gravitational (potential energy)	**B1**

QUESTION · 5054_s20_qp_11.pdf · source page 5 of 16

12 A rocket of mass *M* when empty carries a mass *M* of fuel. The rocket and fuel travel at speed *v*. The engine of the rocket is fired and all of the fuel is expelled. The speed of the rocket increases to 2*v*.

What happens to the kinetic energy of the rocket?

A It doubles.

B It halves.

C It increases by a factor of four.

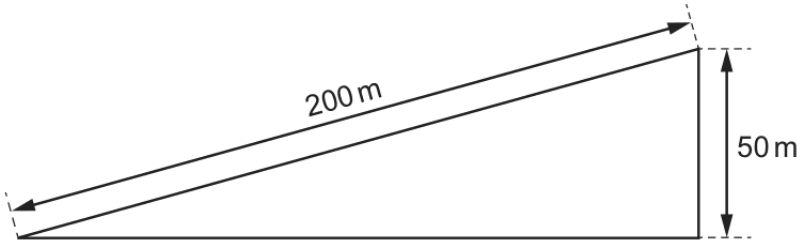
D It stays the same.

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 12

12	C	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 5 of 16

13 A car of mass 1000 kg is driven 200 m up an incline so that it rises 50 m vertically.



The acceleration of free fall g is $10\,m/s^2$.

What is the gain in gravitational potential energy?

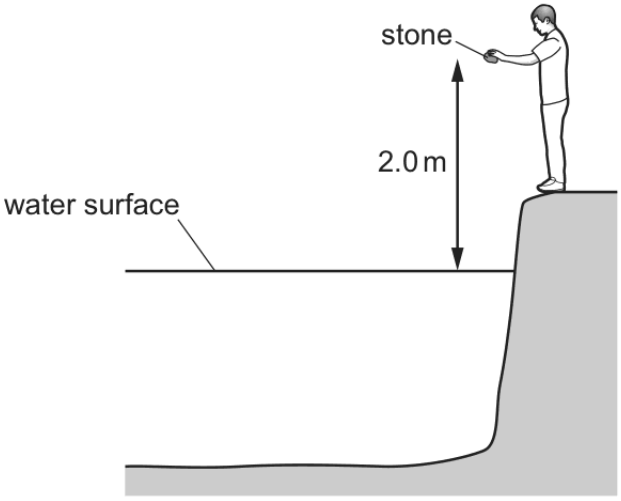
A 5000 J **B** 200 000 J **C** 500 000 J **D** 2 000 000 J

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 13

13	C	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 7 of 16

17 A boy is standing by the side of a lake.



The boy drops a heavy stone from a height of 2.0 m above the water surface.

The acceleration due to gravity is $10\,m/s^2$.

What is the speed of the stone when it hits the surface of the water?

A 4.5 m/s **B** 6.3 m/s **C** 20 m/s **D** 40 m/s

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 17

17	B	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 Fig. 8.1 shows a lamp from a car. It contains two metal filaments.

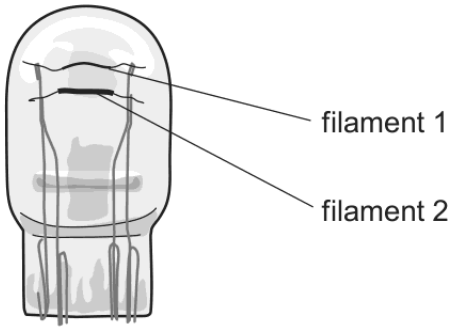
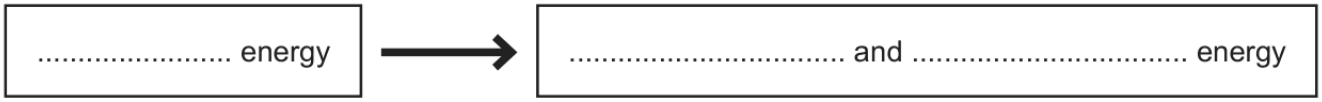


Fig. 8.1

(a) (i) Complete the boxes to describe the transfer of energy that takes place when the lamp is switched on.



[3]

(ii) The efficiency of the metal filament lamp is less than 10%.

State what is meant by *efficiency*.

.....

..... [2]

(b) The two filaments are usually connected in parallel to a car battery.

A student investigates what happens when the filaments are connected in series, rather than in parallel. He uses the same battery for the investigation.

State whether the current, the voltage across each filament and the total power produced *increases*, *decreases* or *stays the same* when the two filaments are connected in series.

current .....

voltage .....

power .....

[2]

(c) Fig. 8.2 shows the current–voltage graph for the two filaments.

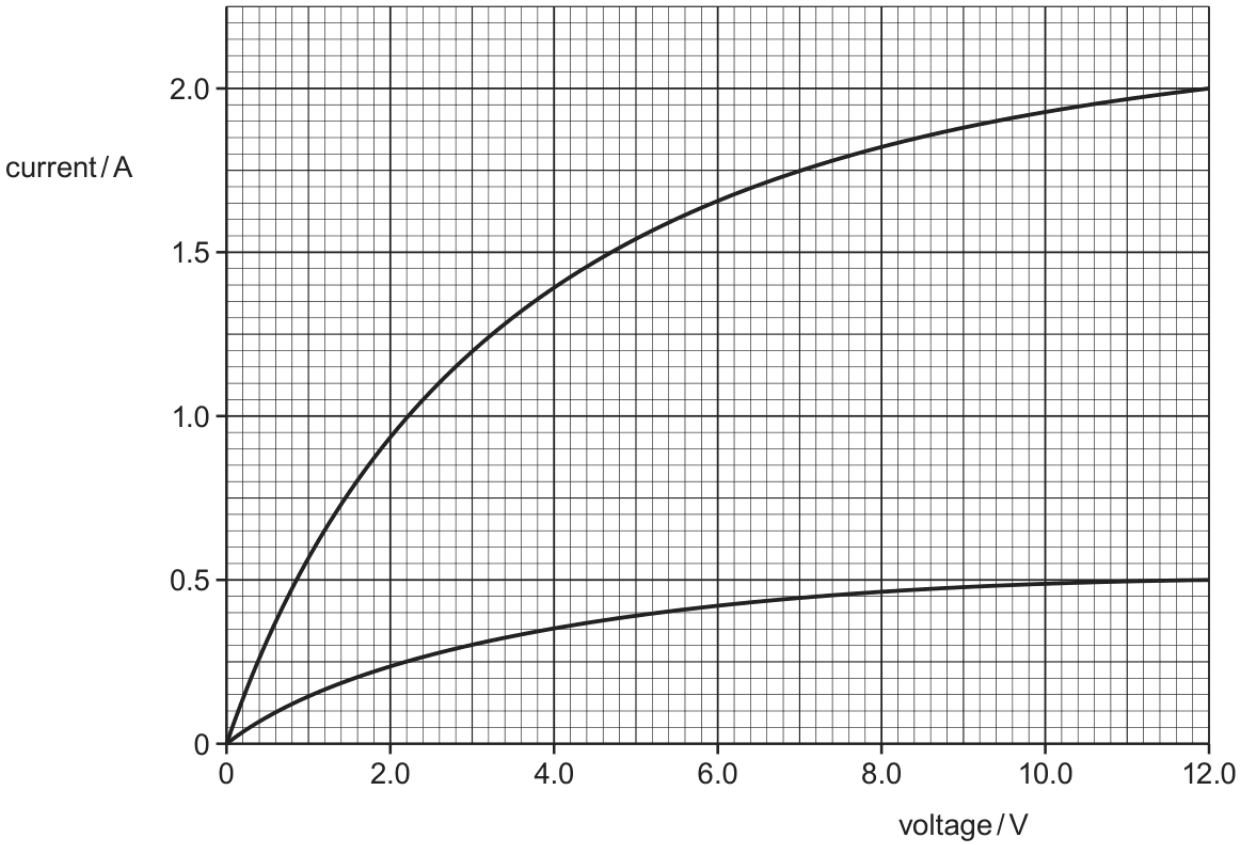


Fig. 8.2

(i) Calculate the total resistance of the two filaments when they are connected in parallel to a voltage of 12 V.

resistance = ... [3]

(ii) The two filaments are made from the same type of metal and have the same length, when uncoiled. They both operate at the same temperature.

Suggest why one filament has a resistance that is greater than that of the other filament.

.....

..... [1]

(d) Fig. 8.3 shows a relay used to switch on a car headlamp.

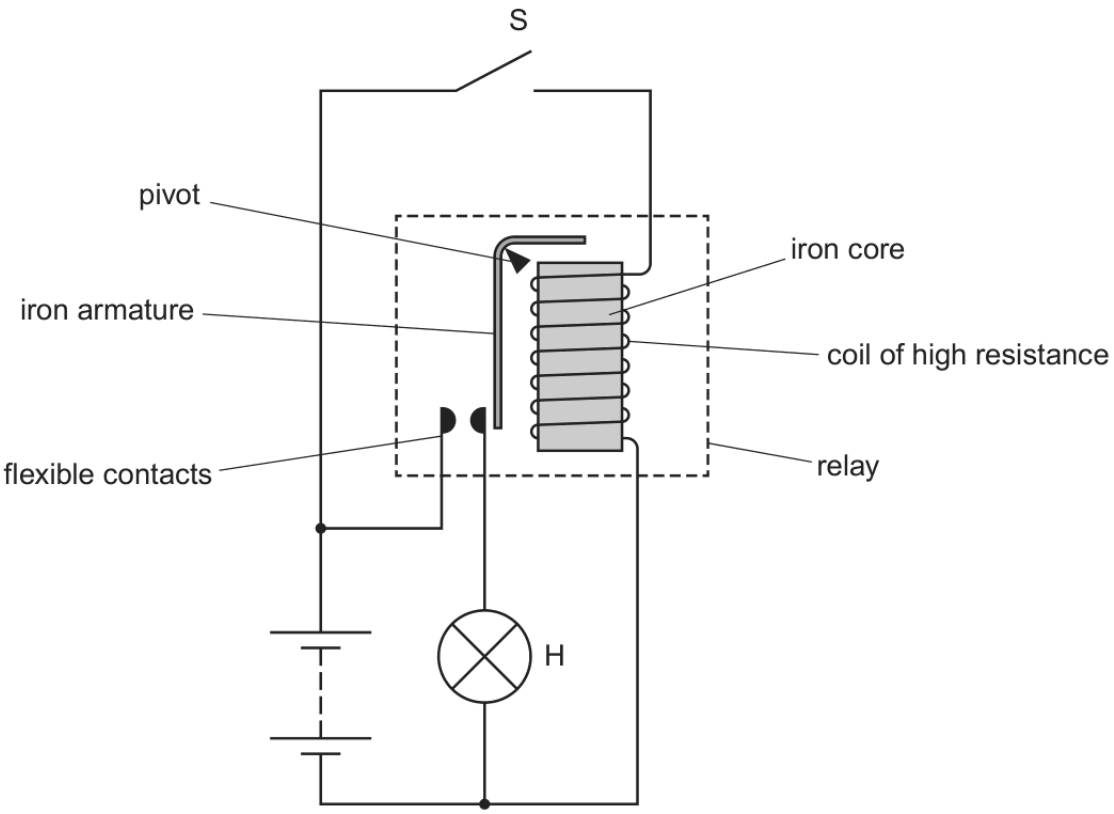


Fig. 8.3

Explain why headlamp H lights up when switch S is closed.

.....

.....

.....

.....

.....

..... [4]

[Total: 15]

MARK SCHEME · 5054_s20_ms_21.pdf · source page 8 of 11 · Question 8

Question	Answer	Marks
8(a)(i)	electrical in first box	**B1**
	heat	**B1**
	light	**B1**
8(a)(ii)	energy output / energy input	**C1**
	useful output energy / total energy input	**A1**
8(b)	two decrease	**C1**
	all three decrease	**A1**
8(c)(i)	(R =) V / I in any form numerical or algebraic	**C1**
	total current = 2.5 A **or** both resistances 24 (Ω) and 6(.0 Ω)	**C1**
	4.8 Ω	**A1**

MARK SCHEME · 5054_s20_ms_21.pdf · source page 9 of 11 · Question 8

Question	Answer	Marks
8(c)(ii)	smaller area / diameter / radius	**B1**
8(d)	current in the coil	**B1**
	coil / core becomes magnetised	**B1**
	attracts iron armature	**B1**
	armature turns **and** contacts close	**B1**

10 Fig. 10.1 shows a motor lifting a mass. Fig. 10.2 shows part of the circuit diagram of the connections to the motor.

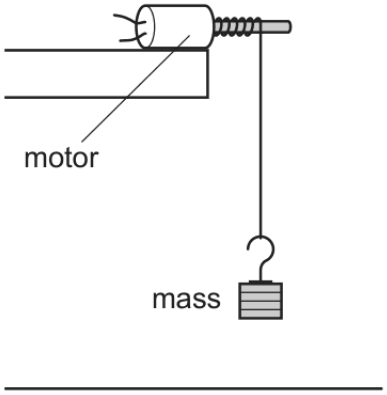


Fig. 10.1

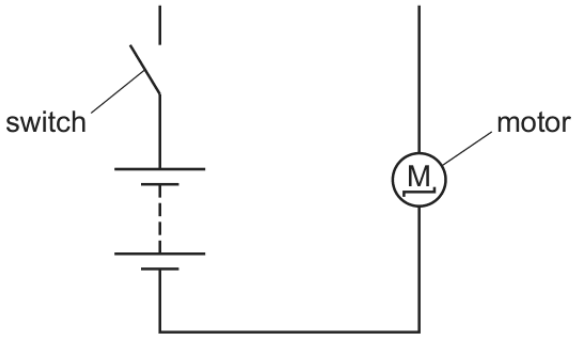


Fig. 10.2

(a) The current in the motor is 1.5A and the voltage supplied by the battery is 8.0 V.

(i) Complete the circuit diagram in Fig. 10.2 to show an ammeter and a voltmeter in the correct positions to take these measurements while the motor is working. [2]

(ii) The motor takes 4.0 s to lift the mass.

Calculate the electrical energy transferred to the motor in this time.

energy = ... [2]

(iii) The motor lifts the 150 g mass through a height of 80 cm in the 4.0 s.

Calculate the gravitational potential energy gained by the mass.

The gravitational field strength g = 10 N / kg.

gravitational potential energy = ... [3]

(iv) State **two** reasons why the gravitational potential energy gained by the mass is less than the electrical energy supplied to the motor.

1. .....

2. .....

[2]

(b) Fig. 10.3 shows the structure of the motor.

When the mass reaches the top of its motion, the switch is opened. This disconnects the battery and causes the mass to fall. The coil turns as the mass falls.

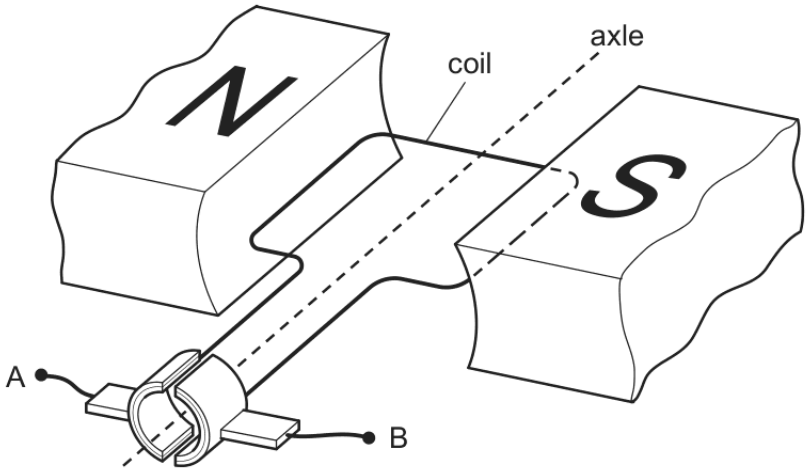


Fig. 10.3

As the coil turns, a small voltage is produced.

(i) Explain why a voltage is produced as the coil turns.

.....

.....

..... [3]

(ii) As the mass falls, a student connects a wire between the points A and B shown in Fig. 10.3.

He notices that the mass takes a longer time to fall when the wire is connected.

The student suggests that this is an example of Lenz's law.

State Lenz's law and suggest why the mass takes longer to fall.

.....

.....

.....

.....

.....

..... [3]

[Total: 15]

MARK SCHEME · 5054_s20_ms_21.pdf · source page 10 of 11 · Question 10

Question	Answer	Marks
10(a)(i)	ammeter in series connecting switch with motor	**B1**
	voltmeter across motor or battery	**B1**
10(a)(ii)	(E =) VIt in any form numerical or algebraic **or** P =VI and P = E / t in any form	**C1**
	48 J	**A1**
10(a)(iii)	(E = mgh) in any form numerical or algebraic	**C1**
	$0.15 \times 0.8(0) \times 10$	**C1**
	1.2 J	**A1**
10(a)(iv)	ANY two of • motor / mass has kinetic energy • heat generated in motor / wires • heat generated in air due to drag / resistance • work done to lift string	**B2**

MARK SCHEME · 5054_s20_ms_21.pdf · source page 11 of 11 · Question 10

Question	Answer	Marks
10(b)(i)	mention of magnetic field (between N and S poles)	**B1**
	coil cuts lines of magnetic flux / magnetic field lines **or** flux / field in coil changes	**B1**
	induction	**B1**
10(b)(ii)	***statement of Lenz's law***, e.g. (induced) current flows in a direction to oppose the change producing it	**B1**
	coil becomes a magnet **or** current flowing in coil produces force in field	**M1**
	poles of coil attracted to opposite poles on magnet **or** force up on a side as side moves down	**A1**

QUESTION · 5054_s20_qp_12.pdf · source page 6 of 16

16 A rocket of mass *M* when empty carries a mass *M* of fuel. The rocket and fuel travel at speed *v*. The engine of the rocket is fired and all of the fuel is expelled. The speed of the rocket increases to 2*v*.

What happens to the kinetic energy of the rocket?

A It doubles.

B It halves.

C It increases by a factor of four.

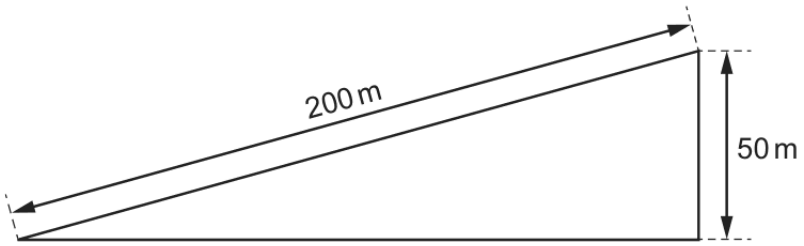
D It stays the same.

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 16

16	C	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 6 of 16

17 A car of mass 1000 kg is driven 200 m up an incline so that it rises 50 m vertically.



The acceleration of free fall g is $10\,m/s^2$.

What is the gain in gravitational potential energy?

A 5000 J **B** 200 000 J **C** 500 000 J **D** 2 000 000 J

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 17

17	C	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 7 of 16

20 In which situation is there **no** transfer of energy?

A a car battery turning the starter motor

B a car moving along a horizontal road at constant speed

C a solar panel warming water

D a spacecraft orbiting the earth at constant height

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 20

20	D	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 6 of 16

14 There is a current in a resistor.

Which energy transfer takes place?

A electrical → chemical

B electrical → internal

C kinetic → electrical

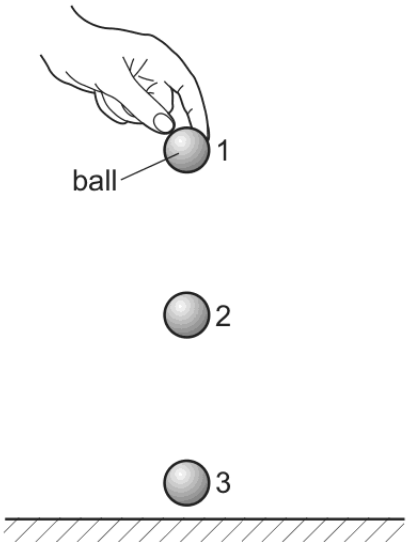
D potential → electrical

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 14

14	B	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 7 of 16

15 A ball is dropped from a height as shown.



Ignoring the effects of air resistance, which statement about the total energy of the ball is correct?

A It is the same at all points.

B It is greatest at point 1.

C It is greatest at point 2.

D It is greatest at point 3.

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 15

15	A	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 7 of 16

17 The table shows some data from a high-jump competition.

Which athlete jumps the highest?

	weight / N	increase in gravitational potential energy / J
A	600	1320
B	700	1610
C	800	1760
D	900	1800

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 17

17	B	**1**

7 Nuclear fusion is a reaction that occurs in the Sun and other stars.

(a) Explain what is meant by *nuclear fusion*.

.....

.....

..... [2]

(b) Describe the nuclear fusion reaction that takes place in the Sun.

.....

.....

..... [1]

(c) A star forms when the temperature of a large cloud of gas in space increases as the cloud collapses inwards.

(i) State the energy transfers that occur as the cloud collapses.

.....

.....

..... [2]

(ii) The temperature increase starts a nuclear fusion reaction.

Explain why the temperature of the star eventually reaches a steady value.

.....

.....

..... [2]

[Total: 7]

MARK SCHEME · 5054_w20_ms_21.pdf · source page 7 of 10 · Question 7

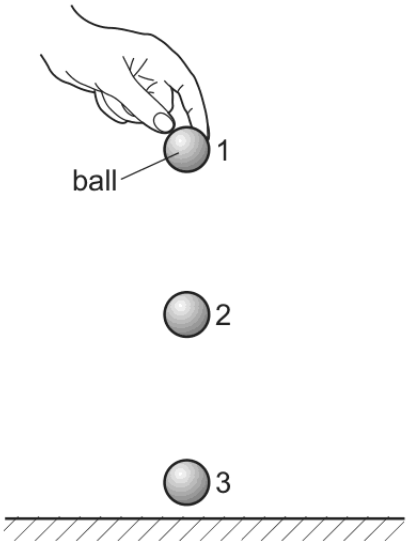
Question	Answer	Marks
7(a)	joining together of (smaller) nuclei / hydrogen nuclei	**B1**
	releasing (large amounts of) energy	**B1**
7(b)	hydrogen reacts to produce helium	**B1**

MARK SCHEME · 5054_w20_ms_21.pdf · source page 8 of 10 · Question 7

Question	Answer	Marks
7(c)(i)	gravitational potential energy to internal energy / kinetic energy	**C1**
	gravitational potential energy to kinetic energy to internal energy / light	**A1**
7(c)(ii)	energy emitted by (infrared / electromagnetic) radiation	**B1**
	energy loss is equal to / balances energy from fusion reaction	**B1**

QUESTION · 5054_w20_qp_12.pdf · source page 6 of 16

13 A ball is dropped from a height as shown.



Ignoring the effects of air resistance, which statement about the total energy of the ball is correct?

A It is the same at all points.

B It is greatest at point 1.

C It is greatest at point 2.

D It is greatest at point 3.

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 13

13	A	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 11 of 16

29 What is stored in a battery and what is its unit?

	quantity	unit
A	current	A
B	current	A s
C	energy	J
D	energy	J / s

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 3 of 3 · row for Question 29

29	C	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 13 of 16

35 What changes kinetic (movement) energy into electrical energy?

A a car battery

B an electric fire

C an electric motor

D a wind turbine

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 3 of 3 · row for Question 35

35	D	**1**

2 The total mass of a wheelbarrow and its load is 90 kg. A worker pushes the wheelbarrow and load up a plank of length 2.0 m on to a platform, as shown in Fig. 2.1. The platform is at a height of 0.60 m above the ground.

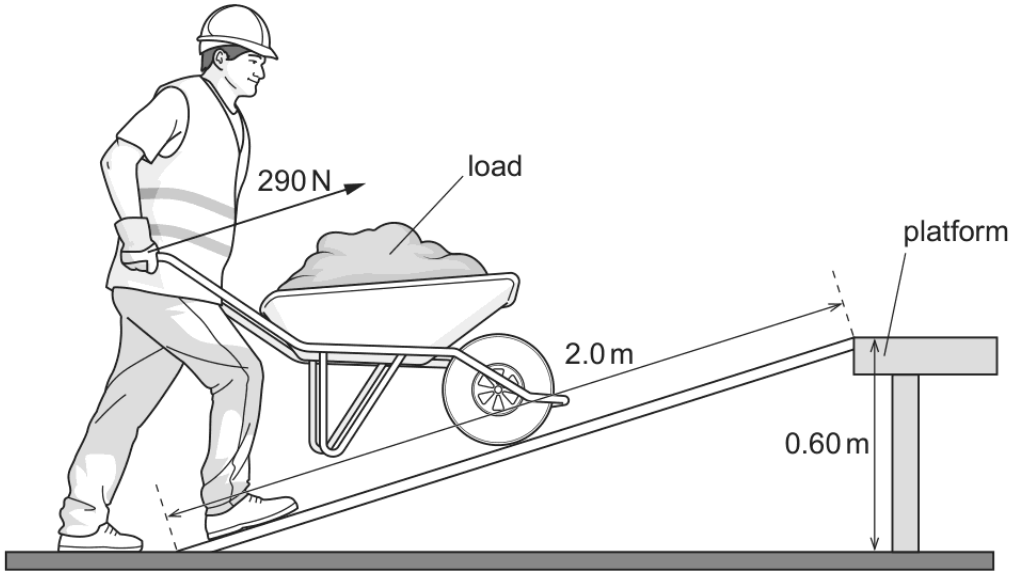


Fig. 2.1

The worker exerts a force of 290 N on the wheelbarrow in the direction in which the wheelbarrow moves.

(a) The gravitational field strength g is equal to 10 N / kg.

(i) Calculate the gravitational potential energy gained by the wheelbarrow and its load.

energy = ... [2]

(ii) The worker pushes the wheelbarrow 2.0 m along the plank.

Calculate the work done on the wheelbarrow by the worker.

work done = ... [2]

(iii) Suggest one reason why the answer to **(a)(ii)** is not equal to the answer to **(a)(i)**.

.....

..... [1]

(b) The worker finds that walking up the plank pushing the wheelbarrow is even more inefficient than the answers in **(a)** suggest.

(i) State what is meant by *efficiency*.

.....

..... [1]

(ii) Suggest **one** reason why this method of lifting the load onto the platform is so inefficient.

.....

..... [1]

[Total: 7]

Question	Answer	Marks
2(a)(i)	(g.p.e. =) *mgh* **or** $90 \times 10 \times 0.60$	**C1**
	540 J	**A1**
2(a)(ii)	(W.D. =) *Fx* **or** 290×2.0	**C1**
	580 J	**A1**
2(a)(ii)	friction (between wheel and axle) **or** friction (between tyre / wheel and plank) **or** (work done against) friction	**B1**
2(b)(i)	$\frac{\text{useful energy output}}{\text{(total) energy input}}$ ($\times$ 100%) **or** $\frac{\text{useful power output}}{\text{(total) power input}}$ ($\times$ 100%)	**B1**
2(b)(ii)	wheelbarrow / worker has to be lifted up **ignore** references to friction / air resistance	**B1**

QUESTION · 5054_s21_qp_11.pdf · source page 13 of 16

33 Which energy transfer takes place in an electric kettle?

A chemical to electrical

B electrical to heat

C electrical to chemical

D heat to electrical

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 3 of 3 · row for Question 33

33	B	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 Fig. 8.1 shows a stationary horse and its rider, about to jump over two fences.

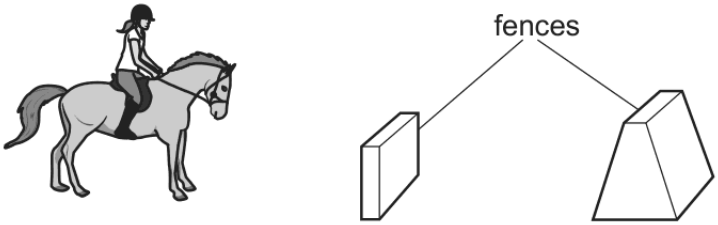


Fig. 8.1

(a) Fig. 8.2 shows a side view of the horse.

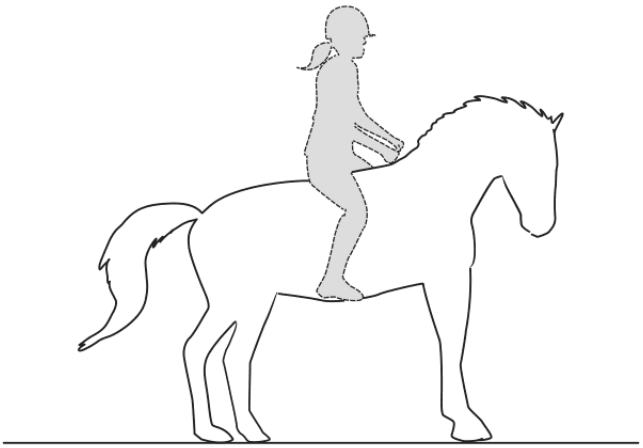


Fig. 8.2

(i) On Fig. 8.2, draw and label the forces acting on the horse.

Include the force that the rider exerts on the horse. Label this force *F*. [3]

(ii) Explain how Newton's third law applies to force *F*.

.....

.....

..... [2]

(b) Fig. 8.3 shows a side view of the two fences. They both have the same height and a uniform density.

Fig. 8.3

(i) On each fence in Fig. 8.3, mark with a cross the centre of mass. [2]

(ii) Explain why a wider base makes the fence more stable.

.....

.....

..... [2]

(c) The total mass of the horse and rider is 520 kg.

(i) As they approach a fence, the horse and rider have a total kinetic energy of 4000 J.

Calculate their speed.

speed = ... [3]

(ii) The centre of mass of the horse and rider is 1.4 m above the ground.

The maximum potential energy gained by the horse and rider as they jump over the fence is 3000 J.

Calculate the maximum height above the ground of the centre of mass during the jump.

The gravitational field strength g = 10 N / kg.

height = ... [3]

[Total: 15]

MARK SCHEME · 5054_s21_ms_21.pdf · source page 9 of 12 · Question 8

Question	Answer	Marks
8(a)(i)	weight of horse in mid-section of horse **and** labelled	**B1**
	force upwards on at least two legs **and** labelled	**B1**
	force downwards on saddle area labelled *F*	**B1**
8(a)(ii)	forces equal **and** opposite	**B1**
	acts on rider (upwards from horse)	**B1**
8(b)(i)	cross at centre of left-hand fence	**B1**
	cross at centre of right-hand fence **and** lower than for right-hand fence	**B1**

MARK SCHEME · 5054_s21_ms_21.pdf · source page 10 of 12 · Question 8

Question	Answer	Marks
8(b)(ii)	does not topple / falls back as fence tilts / rotates / turns / changes angle **or** turns to larger angle / needs more force before toppling **or** centre of mass is lower	**B1**
	(low angle of tilt) centre of mass still inside / above base **or** weight turns fence back if tilted	**B1**
8(c)(i)	K.E. = $\frac{1}{2}mv^2$ in any form algebraic or numerical	**C1**
	$v^2 = 2 \times 4000 / 520$ **or** $v = \sqrt{\frac{2\times4000}{520}}$	**C1**
	3.9 m / s	**A1**
8(c)(ii)	(*h* =) P.E. / *mg* in any form algebraic or numerical	**C1**
	3000 / (10 × 520) **or** 0.58 m	**C1**
	2.0 m	**A1**

QUESTION · 5054_s21_qp_12.pdf · source page 7 of 16

16 A girl lifts an object of mass 1.0 kg from the floor and puts it on a ledge 2.0 m above the floor.

How much potential energy is gained by the object?

[gravitational field strength g = 10 N / kg]

A 2.0 J **B** 5.0 J **C** 10 J **D** 20 J

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 16

16	D	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 **(a)** Fig. 9.1 represents the conversion of energy in a mobile phone.

battery mobile phone

Fig. 9.1

(i) State the form of energy stored in the battery.

..... [1]

(ii) State the useful form of output energy from the battery.

..... [1]

(iii) State **two** useful forms of output energy from the mobile phone.

..... [2]

(b) When the battery is in use, the average current is 1.3 mA.

Determine the charge that passes through the battery in a time of 2.0 minutes.

charge = ... [2]

(c) A student estimates the thermal energy produced in the battery when the mobile phone is used and uses this value to estimate the efficiency of the battery.

He measures a temperature rise of 5.0 °C within the battery in a period of time when the useful energy output from the battery is 5200 J.

The mass of the battery is 110 g and its specific heat capacity is 830 J / (kg °C).

(i) Calculate the thermal energy (heat) needed to raise the temperature of the battery by 5.0 °C.

thermal energy = ... [3]

(ii) State what is meant by *efficiency*.

.....

..... [2]

(iii) Calculate the efficiency of the battery.

efficiency = ... [2]

(iv) Suggest and explain **one** reason why the value for the efficiency calculated in **(iii)** is larger than the actual efficiency of the battery.

.....

.....

.....

..... [2]

[Total: 15]

MARK SCHEME · 5054_s21_ms_22.pdf · source page 13 of 16 · Question 9

Question	Answer	Marks
9(a)(i)	chemical	**B1**
9(a)(ii)	electrical	**B1**
9(a)(iii)	ANY 2 from • light • sound • radio / microwave / electromagnetic (waves)	**B2**
9(b)	(*Q* =) *It* in any form	**C1**
	0.16 C	**A1**

MARK SCHEME · 5054_s21_ms_22.pdf · source page 14 of 16 · Question 9

Question	Answer	Marks
9(c)(i)	(*E* =) *mcT* numerical or algebraic in any form	**C1**
	$0.11 \times 830 \times 5$	**C1**
	460 J	**A1**
9(c)(ii)	energy output / energy input	**C1**
	useful energy output / (total) energy input	**A1**
9(c)(iii)	(energy input =) 460 + 5200 **or** 5660 (J) seen **or** 5200 ÷ (energy in **9(c)(i)** + 5200)	**C1**
	92% **or** 0.92	**A1**
9(c)(iv)	(thermal) energy lost / heat lost (from battery to environment)	**B1**
	energy input larger than calculated / 5660 (J) **or** temperature rise should be larger / is smaller than it should be **or** thermal energy / heat produced larger than 460 J / calculated	**B1**

4 In a coal-fired power station, coal is burnt and electrical energy is produced.

(a) State the form of energy in the coal that is transferred to electrical energy.

..... [1]

(b) (i) State the principle of the conservation of energy.

.....

.....

.....

..... [2]

(ii) The quantity of electrical energy produced by the power station is much less than the quantity of energy in the coal that is burnt to generate the electricity.

Explain why this is so.

.....

.....

.....

..... [2]

(c) State **one** environmental consequence of generating electricity in coal-fired power stations.

.....

..... [1]

[Total: 6]

Question	Answer	Marks
4(a)	chemical (energy)	**B1**
4(b)(i)	energy cannot be created **and** energy cannot be destroyed	**B1**
	energy can be transferred from one form into another	**B1**
4(b)(ii)	some chemical energy is transferred to thermal energy (sound and light)	**B1**
	thermal energy lost to air / water / buildings / equipment	**B1**
4(c)	any **one** from: carbon dioxide produced; carbon monoxide produced; global warming; greenhouse gases nitrous oxide produced; sulfur dioxide produced; acid rain; rise in sea-level	**B1**

QUESTION · 5054_s22_qp_11.pdf · source page 4 of 16

8 A stone falls at terminal velocity in air.

What is equal to the change in gravitational potential energy of the stone?

A the change in kinetic energy

B the change in temperature of the stone

C the friction between the stone and the air

D the thermal energy produced

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 8

8	D	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a model of the human arm. The rubber band represents the muscle that moves part of the arm XY up.

A mass is suspended from XY, as shown in Fig. 1.2. The weight of section XY is negligible and the model is at rest.

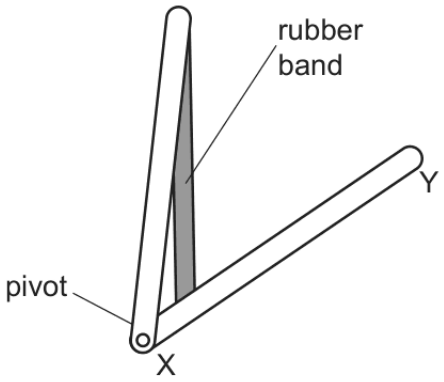


Fig. 1.1

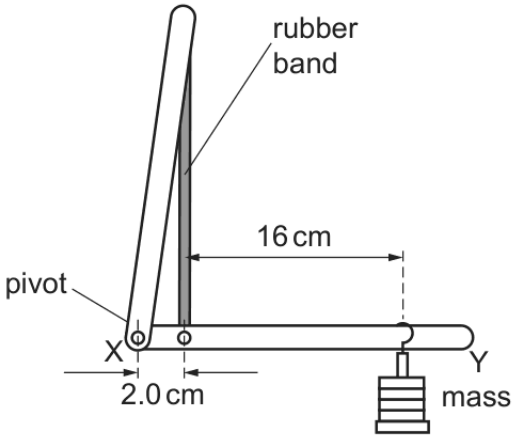


Fig. 1.2 (not to scale)

(a) (i) State **two** ways in which the dimensions of the rubber band change as the mass is added to section XY.

.....

..... [2]

(ii) State the form of energy stored in the stretched rubber band.

..... [1]

(b) (i) State the principle of moments.

.....

.....

.....

..... [2]

(ii) Explain why the force that the rubber band exerts on section XY is larger than the weight of the mass.

.....

.....

..... [1]

(iii) The mass suspended from section XY in Fig. 1.2 has a weight of 4.0 N.

Calculate the force that the rubber band exerts on section XY.

force = ... [2]

(iv) Explain how your answer to **(b)(iii)** is different if the weight of section XY is **not** negligible.

.....

.....

..... [1]

[Total: 9]

Question	Answer	Marks
1(a)(i)	longer	**B1**
	thinner **or** smaller (cross-sectional) area	**B1**
1(a)(ii)	elastic potential **or** strain (energy)	**B1**
1(b)(i)	sum of / total / net / resultant moment is zero	**M1**
	when (the object is) in equilibrium	**A1**
1(b)(ii)	distance of rubber band to pivot is smaller than the distance of the mass (to pivot)	**B1**
1(b)(iii)	$4 \times 18 = F \times 2$	**C1**
	36 N	**A1**
1(b)(iv)	greater **and** the weight (of XY) contributes to the (clockwise) moment/turning effect **or** the moment of the rubber band force is larger	**B1**

2 Fig. 2.1 shows a wind turbine.

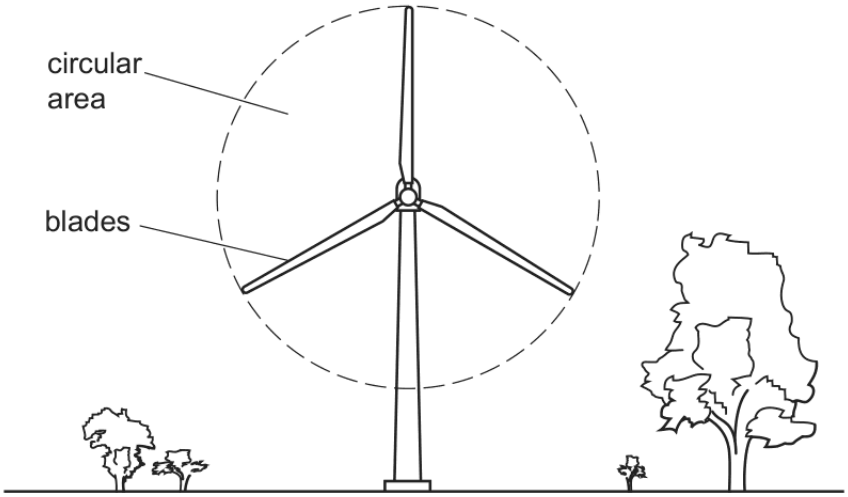


Fig. 2.1

(a) The wind blows directly towards the turbine with a speed of 12 m / s.

In one second, 60 000 kg of air passes through the circular area swept out by the blades.

Calculate the kinetic energy of this mass of air.

kinetic energy = ... [3]

(b) A wind turbine releases no carbon dioxide into the atmosphere while generating electricity. Compared with a coal-fired power station, this is an advantage of using a wind turbine.

(i) State **one** reason why it is important to reduce the amount of carbon dioxide produced.

.....

..... [1]

(ii) Suggest **one** advantage of using a coal-fired power station compared with a wind turbine.

.....

..... [1]

(c) A coal-fired power station releases 0.96 kg of carbon dioxide when it generates 1.0 kW h of electrical energy.

(i) Define the kilowatt-hour (kW h).

.....

..... [1]

(ii) Calculate the mass of carbon dioxide saved when the wind turbine has a power output of 2000 kW and operates for 12 hours.

mass = ... [1]

(d) Wind energy is a form of renewable energy.

State the name of **one** other form of renewable energy.

..... [1]

[Total: 8]

MARK SCHEME · 5054_s22_ms_21.pdf · source page 5 of 11 · Question 2

Question	Answer	Marks
2(a)	(K.E. =) ½mv^2 numerical or algebraic	**C1**
	$0.5 \times 60\,000 \times 12^2$	**C1**
	4 300 000 J	**A1**
2(b)(i)	reduces global warming	**B1**
2(b)(ii)	wind not always blowing **or** wind not reliable / not available in some places **or** smaller land area (for same power) **or** less harm to birds **or** less visual effect **or** no interference with tv / radar signals	**B1**

MARK SCHEME · 5054_s22_ms_21.pdf · source page 6 of 11 · Question 2

Question	Answer	Marks
2(c)(i)	(amount of energy of a device of power) 1 kW used for 1 hour	**B1**
2(c)(ii)	23 000 kg	**B1**
2(d)	solar, tidal, geothermal, hydro(electric), biomass, wave	**B1**

2 Fig. 2.1 shows a long cardboard tube, sealed at both ends, which contains many small pieces of metal.

Fig. 2.1

The tube is turned vertically so that the pieces of metal fall from one end to the other.
The temperature of the pieces increases as a result of the fall.

During the fall, the gravitational potential energy of the metal pieces is transferred to other forms of energy.

(a) State **two** forms of energy to which the gravitational potential energy is transferred.

.....

..... [2]

(b) The pieces fall an average distance of 1.2 m during one turn. The total mass of the metal pieces is 150 g.

(i) Calculate the loss in the gravitational potential energy of the pieces as they fall once.

The gravitational field strength g is 10 N / kg.

loss in gravitational potential energy = ... [2]

(ii) A student turns the tube quickly. After the small metal pieces have fallen from one end to the other 80 times, their temperature has increased by 7.0 °C.

Determine the specific heat capacity of the metal.

specific heat capacity = ... [3]

(iii) The student repeats the experiment, turning the tube more slowly.

Suggest why a different temperature increase is obtained.

.....

..... [1]

[Total: 8]

Question	Answer	Marks
2(a)	ANY 2 from: • kinetic energy • heat / thermal energy / internal energy • sound	**B2**
2(b)(i)	(PE =) *m g h* algebraic or numerical in any form with any mass or distance	**C1**
	1.8 J	**A1**

Question	Answer	Marks
2(b)(ii)	(shc =) $E / m \times T$ algebraic or numerical in any form	**C1**
	$80 \times 1.8 / (0.15 \times 7.0)$ **or** 1.7 **or** 0.0017	**C1**
	140 J / (kg °C)	**A1**
2(b)(iii)	any 1 of • (more time for) heat / energy <u>loss</u> • more time to cool down • masses do not fall full distance / masses slide along tube • thermal energy / temperature (rise) shared with tube • tube heats up • friction <u>with tube</u> (increases) • less kinetic energy	**B1**

2 A child's toy consists of a flexible track and a model car.

Fig. 2.1 shows a diagram of the toy.

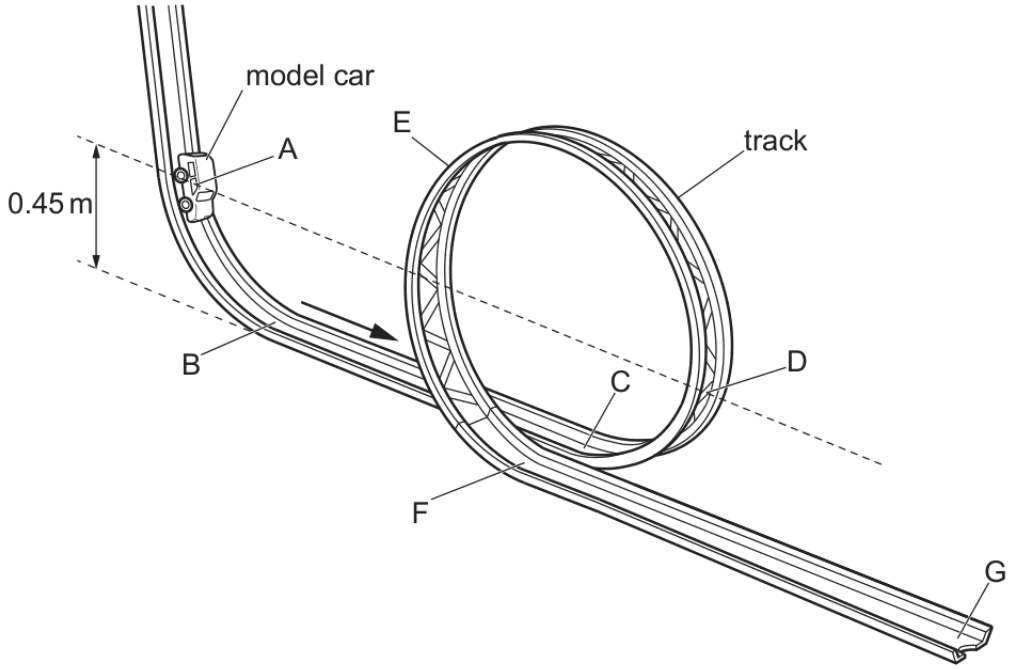


Fig. 2.1

The child first holds the car stationary at point A which is 0.45 m above the horizontal sections of track BC and FG. The mass of the car is 0.12 kg.

The child then releases the car which travels towards point B. Both air resistance and friction between the car and the track are negligible.

The gravitational field strength g is 10 N / kg.

(a) Calculate the change in gravitational potential energy (g.p.e.) of the car as it travels from A to B.

change in g.p.e. = ... [2]

(b) Calculate the speed of the car when it reaches B.

speed = ... [3]

(c) After releasing it, the child expects the car to follow the track along the route ABCDEFG. In fact, the model car does not reach F.

(i) Explain, in terms of energy, why the car does not go past D, which is also 0.45 m above the horizontal track.

.....

..... [1]

(ii) Immediately after being released at A, the car travels to B, to C and then to D.

Describe the motion of the car after it reaches D.

.....

..... [1]

[Total: 7]

Question	Answer	Marks
2(a)	(change in g.p.e.=) mgh **or** $0.12 \times 10 \times 0.45$	**C1**
	0.54 J	**A1**
2(b)	k.e. = $\frac{1}{2}mv^2$ **or** $0.54 = \frac{1}{2} \times 0.12 \times v^2$	**C1**
	(v =) $\sqrt{2gh}$ **or** $\sqrt{2\ 10\ \ 0.45}$ **or** $v^2 = 2 \times 0.54 / 0.12$ **or** $v^2 = 9$	**C1**
	3.0 m / s	**A1**
2(c)(i)	not enough (kinetic) energy (at C) to travel beyond D / to top of circle **or** more g.p.e. needed originally to travel beyond D / to top of circle **or** g.p.e. gained cannot be greater than original g.p.e. (transferred to k.e)	**B1**
2(c)(ii)	it travels back down to C **and** returns to A	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

7 Fig. 7.1 shows a child sitting on a sledge on a snow-covered hill of constant slope.

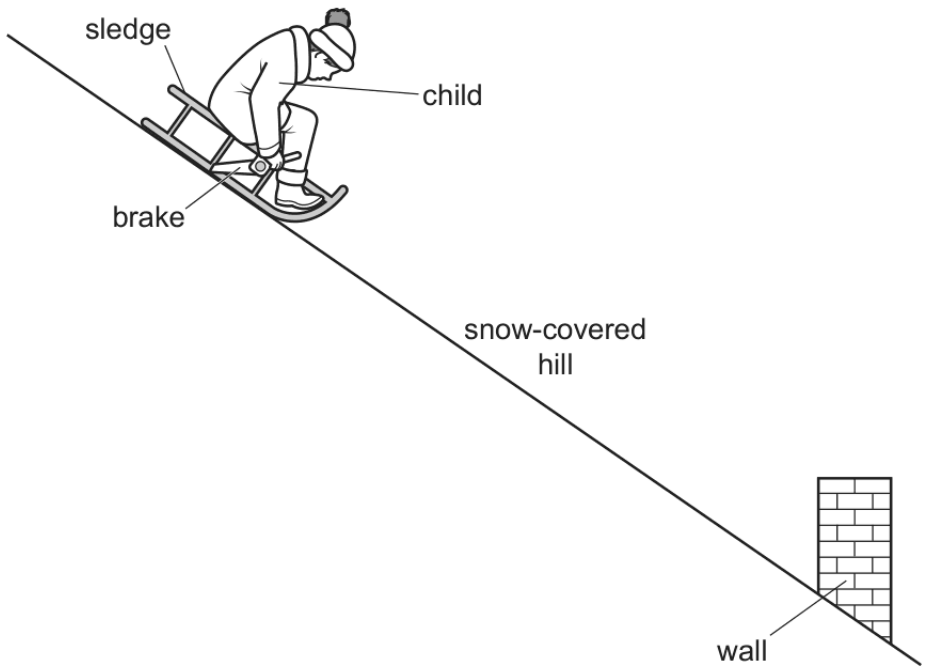


Fig. 7.1 (not to scale)

At time $t = 0$, the child and the sledge begin to move down the hill in a straight line.

When the child sees a wall ahead, he applies the brake.

The child and sledge continue to travel in a straight line until they come to a stop before hitting the wall.

Fig. 7.2 is the speed-time graph for the journey.

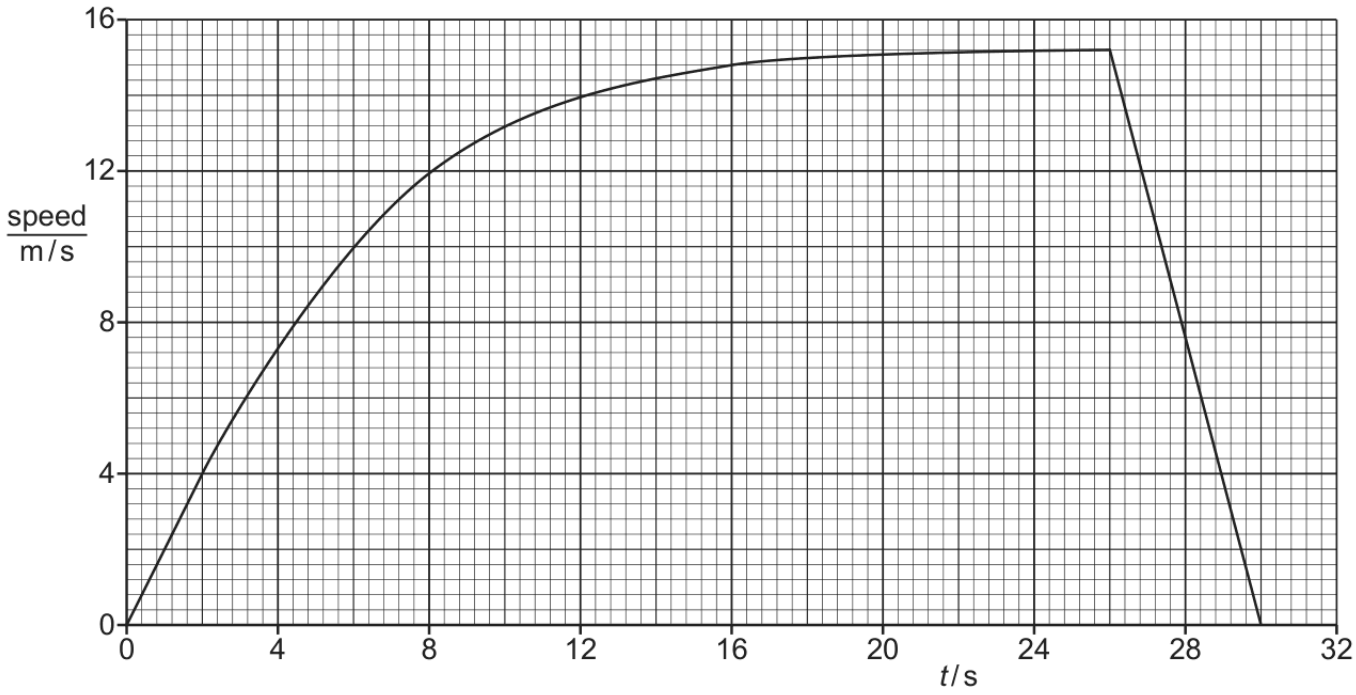


Fig. 7.2

The brake is applied at $t = 26$ s.

(a) Fig 7.2 shows how the speed of the child and sledge varies over the whole of the journey.

Explain why, between $t = 0$ and $t = 26$ s, the speed varies in the way shown by the graph.

.....

.....

.....

.....

.....

..... [3]

(b) At $t = 26$ s, the front of the sledge is 35 m from the wall.

Determine the distance between the front of the sledge and the wall when the sledge stops.

distance = ... [3]

(c) At $t = 26$ s, the child and sledge begin to decelerate.

(i) Determine the size of the deceleration.

deceleration = ... [3]

(ii) The mass of the child is 46 kg and the mass of the sledge is 9.0 kg.

Calculate the resultant force on the child and sledge as they decelerate.

resultant force = ... [2]

(iii) State the energy transfer that is taking place as the child and sledge decelerate.

.....

.....

..... [2]

(d) At $t = 26$ s, when the brake is first applied, the child jerks forwards on the sledge.

Explain why.

.....

.....

..... [2]

[Total: 15]

Question	Answer	Marks
7(a)	acceleration / increase in speed **and** gravity / weight mentioned	**B1**
	air resistance / friction increases (with speed)	**B1**
	terminal / constant velocity reached **or** air resistance (becomes) equal to accelerating force	**B1**
7(b)	(distance travelled =) area under the graph **or** (x =) ½(u + v)t **or** ½ × 15.2 × 4.0	**C1**
	35 – (½ × 15.2 × 4.0) **or** 30.4 **or** 30	**C1**
	4.6 m	**A1**
7(c)(i)	(magnitude of deceleration =) gradient **or** Dv / Dt **or** 15.2 / 4.0 **or** equivalent numbers (from graph) in same ratio	**C1**
	15.2 / 4.0 **or** equivalent numbers in same ratio	**C1**
	3.8 m / s^2	**A1**
7(c)(ii)	(F =) ma **or** (46 + 9.0) × 3.8 **or** 55 × 3.8	**C1**
	210 N	**A1**
7(c)(iii)	<u>from</u> kinetic energy (and from gravitational potential energy)	**B1**
	<u>to</u> thermal energy **and** no intermediate energy	**B1**
7(d)	due to his mass **or** mass mentioned	**B1**
	(his) mass resists a change to its state of motion	**B1**

QUESTION · 5054_w22_qp_12.pdf · source page 14 of 16

33 Which device is used to convert electrical energy to kinetic energy?

A iron

B kettle

C lamp

D motor

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 3 of 3 · row for Question 33

33	D	**1**

2 The foundations that support a building are long concrete cylinders that are pointed at one end. A pile-driver is a machine that forces the pointed concrete cylinders into the ground.

Fig. 2.1 shows a pile-driver.

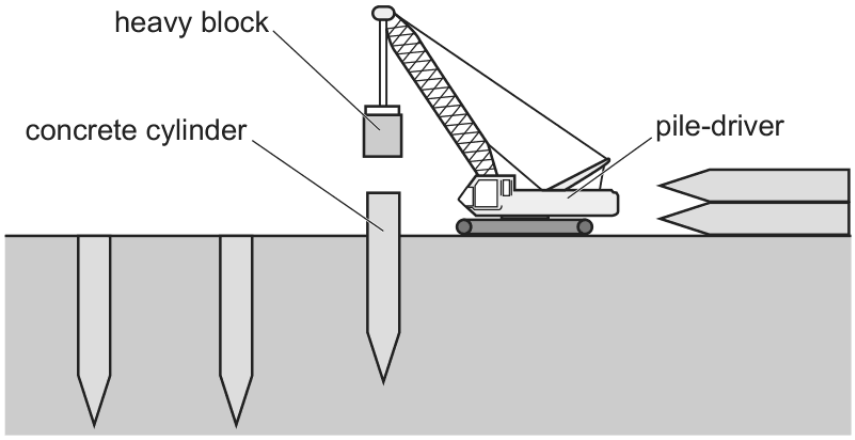


Fig. 2.1

A heavy block of mass 2.9×10^4 kg is lifted into the air then dropped onto the top of a concrete cylinder. This forces the cylinder into the ground.

(a) Fig. 2.2 shows the heavy block.

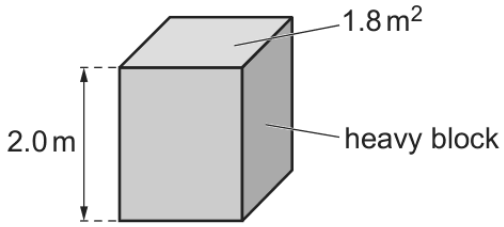


Fig. 2.2

The block is 2.0 m tall and has a cross-sectional area of 1.8 m^2.

Calculate the density of the material used to make the block.

density = ... [2]

(b) The pile-driver lifts the block from the top of a concrete cylinder, through a height of 0.80 m.

The gravitational field strength g is 10 N / kg.

(i) Calculate the gravitational potential energy gained by the block.

gravitational potential energy = ... [2]

(ii) The block is then dropped from rest onto the top of the concrete cylinder.

Calculate the speed of the block just before it hits the concrete cylinder.

speed = ... [3]

[Total: 7]

Question	Answer	Marks
2(a)	$(\rho =)\ m / V$ **or** $(\rho =)\ m / hA$ **or** $2.9 \times 10^4 / 3.6$ **or** $2.9 \times 10^4 / (2.0 \times 1.8)$	**C1**
	$8.1 \times 10^3\ \text{kg / m}^3$	**A1**
2(b)(i)	(g.p.e. =) $mg\Delta h$ **or** $2.9 \times 10^4 \times 10 \times 0.80$	**C1**
	$2.3 \times 10^5\ \text{J}$	**A1**
2(b)(ii)	k.e. / ½mv^2 = g.p.e. / $mg\Delta h$ / 2.3×10^5 **or** $(v =)\ \sqrt{2gDh}$ **or** $\sqrt{2 \times 10 \times 0.80}$	**C1**
	$(v =)\ \sqrt{2gDh}$ **or** $\sqrt{2 \times 10 \times 0.80}$	**C1**
	4.0 m / s	**A1**

3 Fig. 3.1 shows a man standing underneath an outdoor heater on a cold evening.

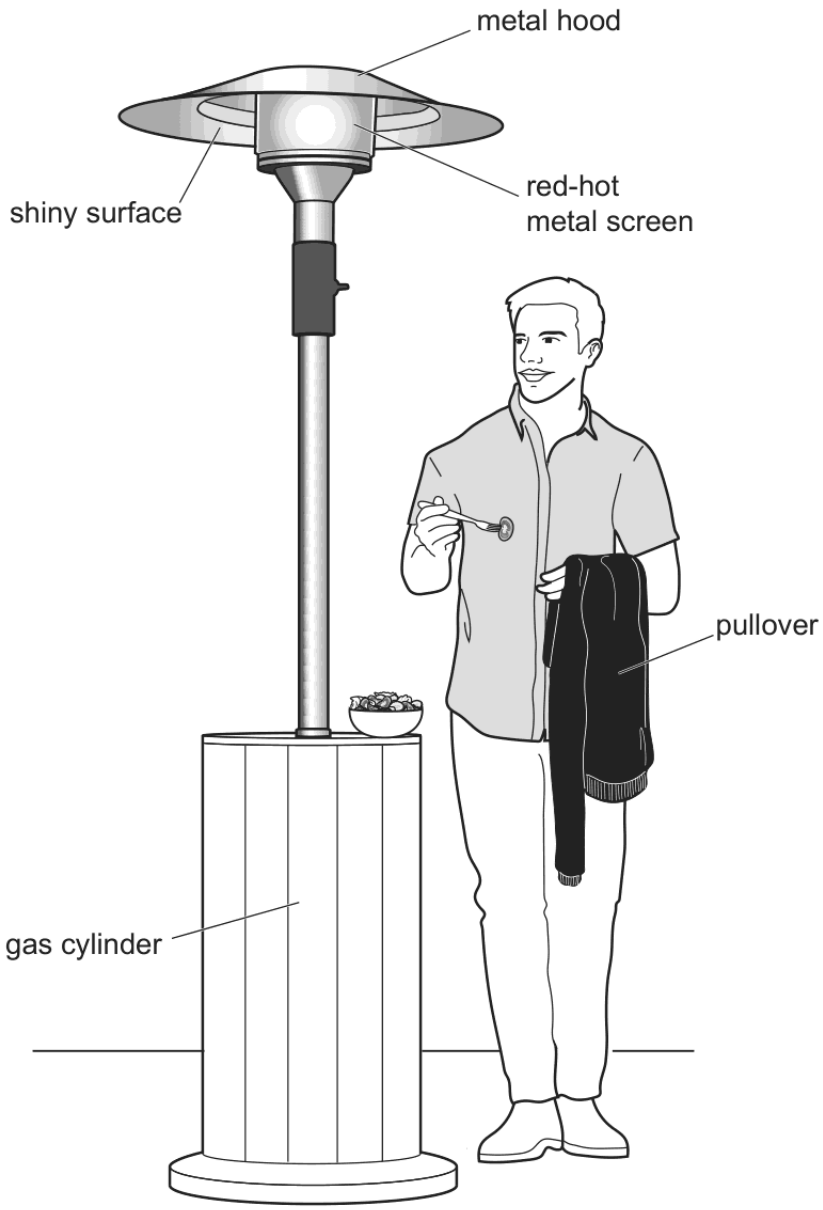


Fig. 3.1

Gas in the cylinder at the base of the heater is the fuel for the heater. When the heater is operating, the gas travels to the top of the heater where it burns.

(a) State the form of energy stored in the gas that is transferred by the heater.

..... [1]

(b) A metal screen surrounding the burning gas is heated by the burning gas until it is red-hot. The hot metal screen warms the man who is standing underneath it.

(i) Describe how thermal energy in the red-hot metal screen is transferred to the man and how it warms him.

.....

.....

.....

.....

..... [3]

(ii) At the top of the heater is a metal hood that has a shiny lower surface.

Explain why this makes the energy transfer from the metal screen more efficient.

.....

.....

..... [2]

(iii) The air temperature decreases and the man puts on a black pullover.

Explain **one** way in which wearing the black pullover helps to keep the man warm.

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
3(a)	chemical potential energy **and** no other energy	**B1**
3(b)(i)	radiation **or** infrared mentioned	**B1**
	radiation **and** infrared (travel to the man)	**B1**
	energy / infrared / radiation <u>absorbed</u> by the man / pullover **or** travels in straight line **or** travels at the speed of light **or** increases (man's) internal energy	**B1**
3(b)(ii)	(shiny surface) <u>reflects</u> radiation (to man)	**B1**
	less (thermal) energy escapes **or** more (thermal) energy reaches / absorbed by man	**B1**
3(b)(iii)	**EITHER** black (surface) is good absorber / poor reflector of infrared radiation	**B1**
	more (thermal) energy absorbed **or** more thermal energy transferred to man	**B1**
	OR trapped air **and** reduced convection / air poor conductor / air is a good insulator	**(B1)**
	less (thermal) energy lost	**(B1)**

QUESTION · 5054_s23_qp_11.pdf · source page 7 of 20

12 The diagram shows a burning candle.

Which energy store is associated with the burning candle?

A chemical

B elastic

C electrostatic

D nuclear

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 12

12	A	**1**

2 In a safety test, a car of mass 1100 kg travels at a speed of 10 m/s and collides with a stationary van of mass 3000 kg.

After the collision the car and the van move together with a velocity *v*.

Fig. 2.1 shows the car and van before and after the collision.

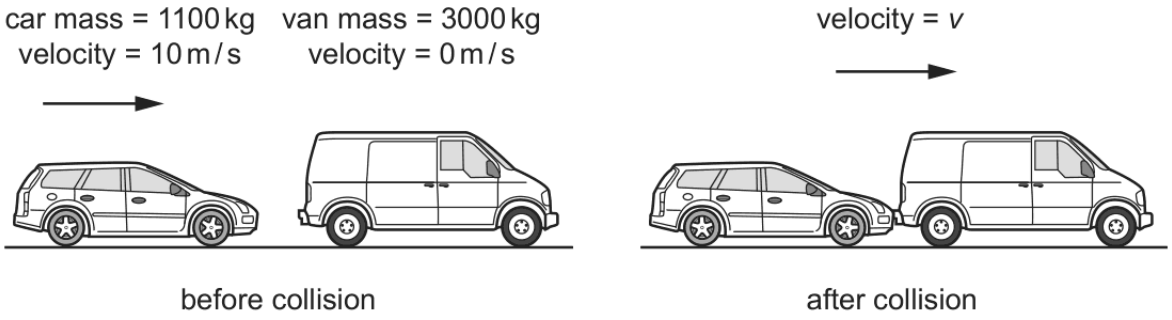


Fig. 2.1

The total momentum of the car and van is conserved during the collision.

(a) (i) Define 'momentum'.

.....

..... [1]

(ii) State the unit of momentum.

..... [1]

(b) Calculate the velocity *v* of the car and van after the collision.

v = ... m/s [2]

(c) (i) Calculate the total kinetic energy of the car and van after the collision.

kinetic energy = ... J [2]

(ii) State the transfer of energy that occurs in the collision.

.....

..... [1]

[Total: 7]

Question	Answer	Marks
2(a)(i)	the product of mass and velocity	**B1**
2(a)(ii)	kg m / s **or** Ns	**B1**
2(b)	1100 × 10 = 4100 × v	**C1**
	2.7 (m / s)	**A1**
2(c)(i)	(*KE* =) ½ mv^2 algebraic or numerical for the car	**C1**
	15 000 (J)	**A1**
2(c)(ii)	(kinetic energy) becomes heat / thermal energy / internal energy (of vehicles / air) **or** kinetic energy (of car) goes to kinetic energy **of van**	**B1**

QUESTION · 5054_s23_qp_12.pdf · source page 6 of 16

13 A battery is connected in a circuit to light a lamp.

What is the energy store in the battery?

A chemical

B electrostatic

C gravitational

D kinetic

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 13

13	A	**1**

2 Fig. 2.2 shows a rider on an electric scooter.

Fig. 2.2

The scooter contains a battery and a motor to drive the back wheel.

(a) (i) State the name of the energy store in the battery.

..... [1]

(ii) Describe, in terms of work done, the stages of energy transfer from the energy store in the battery to the kinetic energy of the scooter.

.....

.....

.....

..... [2]

(b) The total mass of the scooter and the rider is 70 kg.

Calculate the total kinetic energy of the rider and scooter when the scooter has a speed of 4.0 m/s.

kinetic energy = ... J [2]

(c) The battery is marked 'energy capacity 0.35 kilowatt-hour (kW h)'.

(i) Define what is meant by a kilowatt-hour.

.....

..... [1]

(ii) The scooter stops working because the battery is totally discharged (flat). This means that there is no more energy stored in the battery.

The battery is then recharged using a 70 W power supply.

Calculate the time taken to fully recharge the battery.

time = ... hours [2]

[Total: 8]

MARK SCHEME · 5054_s23_ms_22.pdf · source page 6 of 12 · Question 2

Question	Answer	Marks
2(a)(i)	chemical (potential energy)	**B1**
2(a)(ii)	electrical work / (electrical) current **or** electrons (made to) move	**B1**
	mechanical) work is done by the motor / by or on the (back) wheel	**B1**
2(b)	(KE =) ½ mv^2 in any form	**C1**
	560 (J)	**A1**
2(c)(i)	(energy provided if power is) 1 kW for 1 hour / 1000 W for 1 hour	**B1**

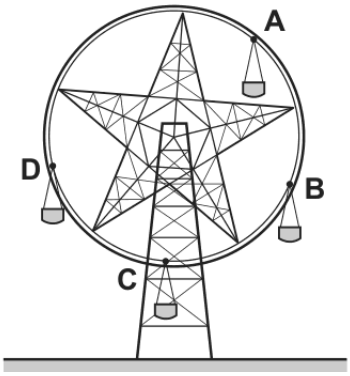
MARK SCHEME · 5054_s23_ms_22.pdf · source page 7 of 12 · Question 2

Question	Answer	Marks
2(c)(ii)	*(t =) E / P* **or** 0.35 / 0.07 **or** 350 / 70	**C1**
	5(.0 hours)	**A1**

QUESTION · 5054_w23_qp_11.pdf · source page 4 of 16

8 The diagram shows a stationary fairground ride with four chairs of equal mass.

Which chair has the most gravitational potential energy?



MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 8

8	A	**1**

2 A lorry (truck) is at rest on a straight, horizontal road.

A constant driving force of 5.4×10^4 N acts on the lorry and it accelerates forwards.

(a) Define what is meant by 'acceleration'.

.....

.....

..... [2]

(b) As the lorry moves forwards, work is done on it by the driving force.

(i) State the equation that defines work done.

Refer to the direction of the force exerted.

.....

..... [1]

(ii) Calculate the work done on the lorry by the driving force when the lorry travels a distance of 280 m.

work done = ... J [2]

(iii) At one point on the road, the kinetic energy of the lorry is 3.2×10^6 J and its speed is 12 m/s.

Calculate the mass of the lorry.

mass = ... kg [3]

[Total: 8]

MARK SCHEME · 5054_w23_ms_21.pdf · source page 6 of 11 · Question 2

Question	Answer	Marks
2(a)	change / increase of velocity (with time)	**B1**
	change / increase of velocity per unit time **or** rate of change / increase of velocity	**B1**
2(b)(i)	force × distance moved in the direction of the force / forwards	**B1**
2(b)(ii)	(work done =) $5.4 \times 10^4 \times 280$	**C1**
	1.5×10^7 (J)	**A1**

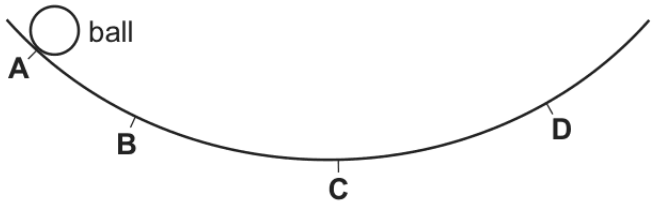
MARK SCHEME · 5054_w23_ms_21.pdf · source page 7 of 11 · Question 2

Question	Answer	Marks
2(b)(iii)	$E_k = ½mv^2$ **or** (m =) $2 \times E_k / v^2$	**C1**
	(m =) $2 \times 3.2 \times 10^6 / 12^2$ **or** $2 \times 3.2 \times 10^6 / 144$	**C1**
	4.4×10^4 (kg)	**A1**

QUESTION · 5054_w23_qp_12.pdf · source page 5 of 20

12 The diagram shows a curved track. A ball is released from the position shown.

At which point is the gravitational potential energy stored in the ball the maximum?



MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 12

12	A	**1**

2 A spacecraft of mass 300 kg is moving in a straight line in space, at a speed of 8000 m/s.

(a) Calculate the momentum of the spacecraft.

momentum = ... kg m/s [2]

(b) The fuel on the spacecraft explodes and the spacecraft splits into two parts. The direction in which the parts move does not change.

(i) After the explosion, the speed of the front part increases to 9000 m/s. It has a mass of 150 kg.

Calculate the speed of the rear part after the explosion.

speed = ... m/s [3]

(ii) The total kinetic energy of the two parts after the explosion is greater than the original kinetic energy of the spacecraft.

State the energy transfer that occurs in the explosion.

.....

..... [1]

(iii) The explosion lasts for 0.20 s.

Calculate the average force on the front part during this time.

force = ... N [3]

[Total: 9]

Question	Answer	Marks
2(a)	(p =) mv **or** 300 × 8000	**C1**
	2.4×10^6 (kg m / s)	**A1**
2(b)(i)	momentum conserved **or** $mv = m_1v_1 + m_2v_2$	**C1**
	(v_2 =) (2.4 × 10^6 – (150 × 9000)) ÷ 150 **or** (v_2 =) (1 050 000) / 150	**C1**
	7000 (m / s)	**A1**
2(b)(ii)	<u>from</u> energy stored as chemical (potential) energy	**B1**
2(b)(iii)	Δp **or** J **or** I = 150 × (9000 – 8000) **or** 150 000 **or** (a =) (9000 – 8000) ÷ 0.20 **or** (a =) 5000	**C1**
	(F =) $\Delta p / (\Delta)t$ **or** (F =) $I / (\Delta)t$ **or** (F =) ma **or** 150 × (9000 – 8000) / 0.20 **or** 150 × 1000 / 0.20	**C1**
	7.5×10^5 (N)	**A1**

9 In space, an interstellar cloud of dust and gas collapses to form a protostar. The cloud contains hydrogen.

(a) Describe the energy transfers that take place as the cloud collapses and forms the protostar.

.....

.....

..... [2]

(b) The collapse results in a nuclear reaction in the protostar that involves the isotope hydrogen-3 ($^{3}_{1}H$).

The isotope hydrogen-3 ($^{3}_{1}H$) is radioactive. It decays by beta particle emission.

(i) State how the nuclei of isotopes of the same element are different.

.....

..... [1]

(ii) Explain why hydrogen-3 cannot decay by alpha particle emission.

.....

..... [1]

(iii) Complete the equation for the decay of hydrogen-3 to an isotope of helium (He).

$$^{3}_{1}H \longrightarrow \begin{matrix}\cdots\cdots\\ \cdots\cdots\end{matrix}\beta \; + \; \begin{matrix}\cdots\cdots\\ \cdots\cdots\end{matrix}He$$

[2]

(iv) Explain how the nuclear reaction in the protostar stops further collapse as the protostar becomes a stable star.

.....

.....

.....

.....

..... [3]

(c) The cloud of dust and gas that collapses also contains atoms of the heaviest elements.

State which part of the life cycle of a star is responsible for the production of the heaviest elements.

.....

..... [1]

[Total: 10]

MARK SCHEME · 5054_w23_ms_22.pdf · source page 10 of 11 · Question 9

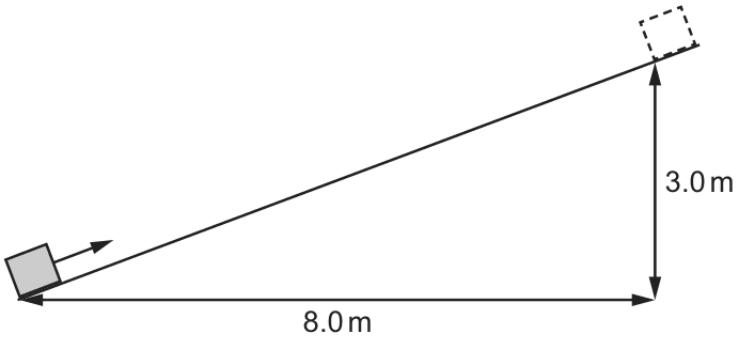
Question	Answer	Marks
9(a)	any **two** from: from gravitational potential energy **or** gravitational potential energy to .... **or** gravitational potential energy decreases kinetic energy mentioned to internal energy **or** internal energy increases	**B2**
9(b)(i)	different number of neutrons (in the nucleus)	**B1**
9(b)(ii)	(its nucleus only contains) only has one proton (alpha–particles contain two protons) **or** not enough protons	**B1**
9(b)(iii)	${}^{0}_{-1}(\beta)$	**B1**
	${}^{3}_{2}(\mathrm{He})$	**B1**

MARK SCHEME · 5054_w23_ms_22.pdf · source page 11 of 11 · Question 9

Question	Answer	Marks
9(b)(iv)	(nuclear reaction produces) a high temperature (in core)	**B1**
	outward force (due to high temperature / nuclear reaction) **or** force due to high temperature / nuclear reaction **or** radiation pressure	**B1**
	balances (gravitational) force (inwards)	**B1**
9(c)	(red supergiant explodes as) supernova (explosion)	**B1**

QUESTION · 5054_s24_qp_11.pdf · source page 5 of 16

10 An object of mass 5.0 kg is pushed up a slope at a steady speed of 2.0 m / s.



Which calculation gives the change in gravitational potential energy from the bottom to the top?

A $0.50 \times 5.0 \times 2.0$

B $5.0 \times 9.8 \times 3.0$

C $0.50 \times 5.0 \times 2.0^2$

D $5.0 \times 9.8 \times 8.0$

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 10

10	B	**1**

1 A ball is released from rest at point A and moves along a smooth track ABCDE as shown in Fig. 1.1. The ball is shown at point A and as it passes point B.

The ball is always in contact with the track and air resistance is negligible.

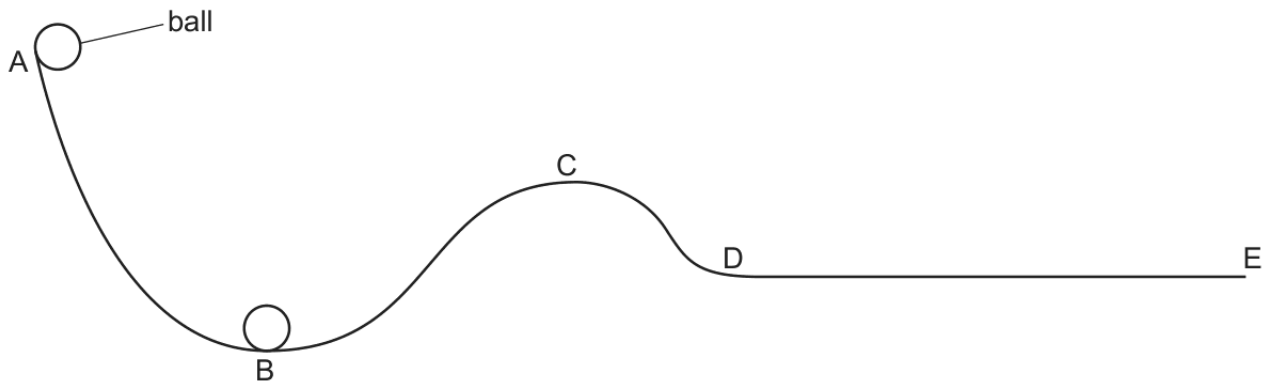


Fig. 1.1

Fig. 1.2 shows the distance−time graph for the ball as it moves from A to E.

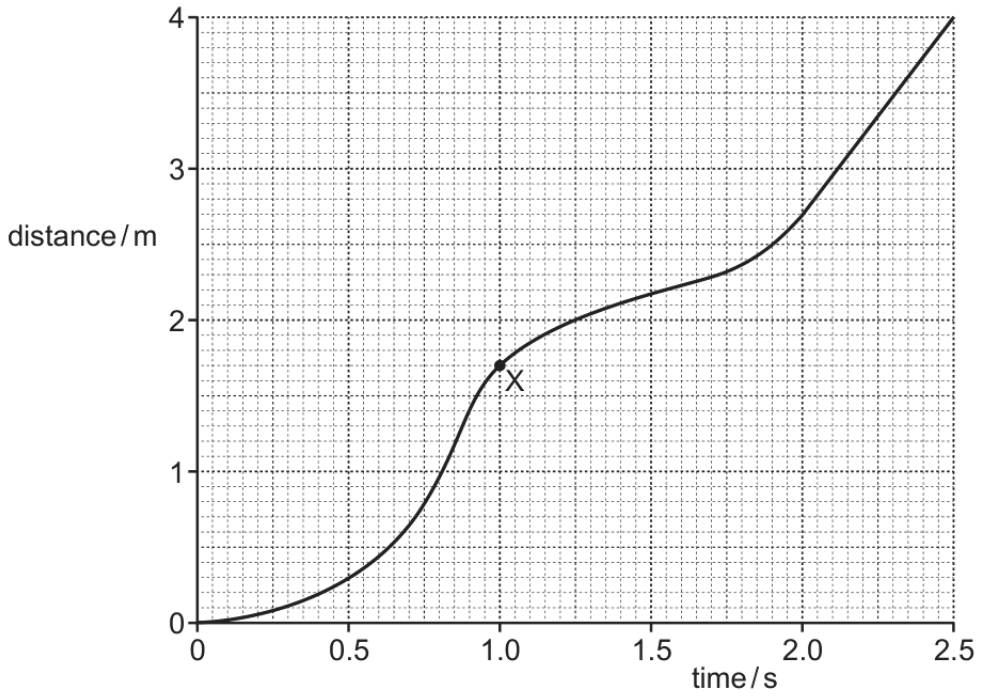


Fig. 1.2

(a) (i) On Fig. 1.2, mark:

- with the letter P **one** point where the ball accelerates
- with the letter Q **one** point where the ball has constant speed.

[2]

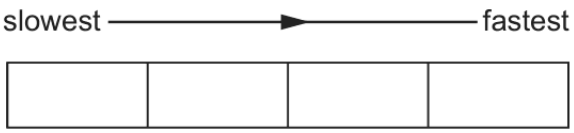
(ii) Determine the speed of the ball at point X on Fig. 1.2.

Show your working.

speed = ... m/s [3]

(b) The speeds of the ball at A, B, C and D are v_A, v_B, v_C and v_D respectively.

Arrange these four speeds from slowest to fastest.



[1]

(c) Work is done to transfer energy between energy stores as the ball moves from A to B.

Name the force involved in the work done and describe the energy transfer.

.....

.....

.....

..... [3]

(d) The track at point B is circular in shape.

On Fig. 1.1, draw an arrow to show the direction of the resultant force on the ball at point B.

[1]

[Total: 10]

Question	Answer	Marks
1(a)(i)	P marked in region t = 0–0.8 s **or** in region 1.6–2.0 s	**B1**
	Q marked between 2.0 and 2.5 s	**B1**
1(a)(ii)	tangent drawn at X	**B1**
	(speed =) d/t in any form numerical or algebraic	**C1**
	1.9–2.4 m/s	**A1**
1(b)	$v_A < v_C < v_D < v_B$	**B1**
1(c)	(work done by) gravity / weight / gravitational attraction	**B1**
	from gravitational potential energy (store)	**B1**
	to kinetic energy (store)	**B1**
1(d)	arrow upwards at D	**B1**

QUESTION · 5054_s24_qp_12.pdf · source page 6 of 16

14 A ball is dropped from rest at the top of a building. Air resistance is negligible.

The velocity of the ball is 14 m / s when it hits the ground.

What is the height of the building?

A 2.9 m **B** 10 m **C** 20 m **D** 40 m

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 14

14	B	**1**

3 Fig. 3.1 shows a solar-powered charger connected to a cell phone (mobile phone).

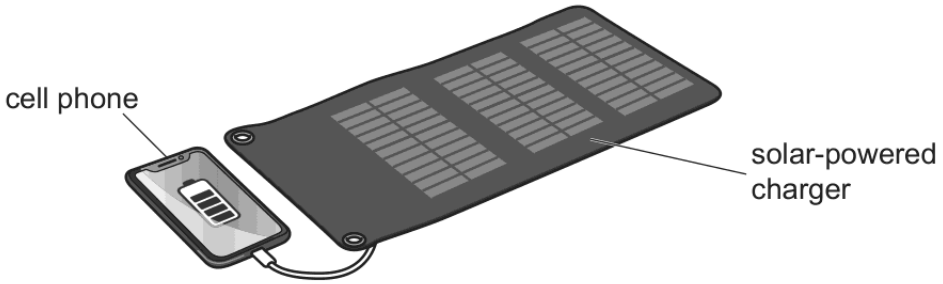


Fig. 3.1

The battery inside the cell phone is charged by the solar-powered charger.

(a) (i) Complete Fig. 3.2 to show the useful transfer of energy from the Sun to the battery.

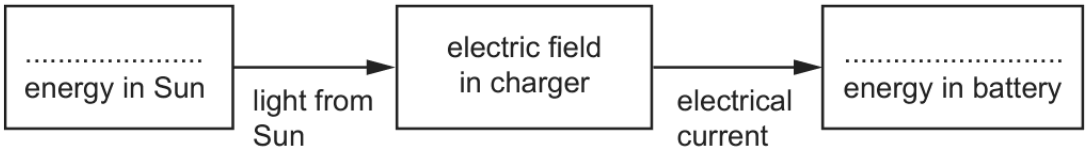


Fig. 3.2

[2]

(ii) Explain why the battery takes a long time to charge on a cloudy day.

.....

..... [1]

(b) After use, the outside surface of the cell phone is warm. When switched off, the cell phone cools down.

Name and describe the **three** processes by which thermal energy is transferred as the cell phone cools down.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) It takes 4.5 hours to charge the battery with an average current of 300 mA.

Calculate the quantity of charge that enters the battery. Give the unit of your answer.

charge = ... unit ..... [3]

[Total: 9]

Question	Answer	Marks
3(a)(i)	nuclear (energy in Sun)	**B1**
	chemical (energy in battery)	**B1**
3(a)(ii)	less / no light / energy (reaching the charger) **or** less / no electrical current (produced)	**B1**
3(b)	conduction **and** energy passing from particle to particle in solid / from outside particles to air particles	**B1**
	convection **and** in air / hot air rises	**B1**
	radiation **and** emitted from surface	**B1**
3(c)	(Q =) It in any form	**C1**
	4 900	**A1**
	C **or** coulomb	**B1**

QUESTION · 5054_w24_qp_11.pdf · source page 7 of 16

16 An object of mass 5.0 kg is moving at a constant speed. It has 10 J of kinetic energy.

What is the speed of the object?

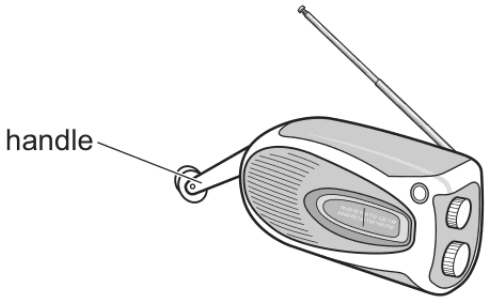
A 0.25 m/s **B** 0.5 m/s **C** 2.0 m/s **D** 4.0 m/s

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 16

16	C	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 7 of 16

17 A wind-up radio is used in areas where mains electricity is not available.



A person winds a spring in the radio using the handle. When the spring unwinds, it makes a small coil rotate in a magnetic field.

Which energy transfer is **not** useful for this radio?

A elastic potential energy → kinetic energy

B kinetic energy → thermal energy

C chemical energy → elastic potential energy

D kinetic energy → sound energy

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 17

17	B	**1**

2 A student makes a ball from modelling clay. The ball has a mass of 0.20 kg.

She drops the ball from a height of 40 m above the ground.

(a) (i) Calculate the initial gravitational potential energy of the ball.

gravitational potential energy = ... J [2]

(ii) Calculate the speed of the ball immediately before the ball hits the ground.

Ignore the effect of air resistance.

speed = ... m/s [3]

(b) When the ball hits the ground, it stops moving.

The kinetic energy of the ball is transferred to internal energy.

The temperature of the ball increases.

(i) The specific heat capacity of modelling clay is 1400 J/(kg °C).

Calculate the maximum possible temperature increase of the ball. Show your working.

temperature increase = ... °C [2]

(ii) Suggest **one** reason why the actual temperature increase of the ball may be smaller than the value calculated in **(b)(i)**.

.....

.....

..... [1]

[Total: 8]

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Question	Answer	Marks
2(a)(i)	(G.P.E.=) mgh **or** $0.20 \times 9.8 \times 40$	**C1**
	78 (J)	**A1**
2(a)(ii)	(K.E. =) $\frac{1}{2}mv^2$ or $\frac{1}{2} \times 0.2 \times v^2$	**C1**
	(v =) $\sqrt{2gh}$ **or** $\sqrt{2 \times 9.8 \times 40}$ **or** $\sqrt{784}$	**C1**
	28 (m / s)	**A1**
2(b)(i)	($\Delta\theta$ =) $\Delta E / mc$ **or** $78.4 / (0.20 \times 1400)$ **or** gh / c **or** $9.8 \times 40 / 1400$	**C1**
	0.28 (°C)	**A1**
2(b)(ii)	some (thermal) energy is transferred to the ground **or** work is done deforming the modelling clay **or** air resistance / work done means that speed / k.e. is less than the calculated value	**B1**

QUESTION · 5054_w24_qp_12.pdf · source page 12 of 16

32 How is energy transferred within a kettle that is powered by the mains supply?

A by electrical work done to chemical energy

B by electrical work done to thermal energy

C by mechanical work done to chemical energy

D by mechanical work done to thermal energy

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 3 of 3 · row for Question 32

32	B	**1**

1 A jet ski is a type of boat that carries one or two people and travels at high speed on water.

Fig. 1.1 shows a student riding on a jet ski.

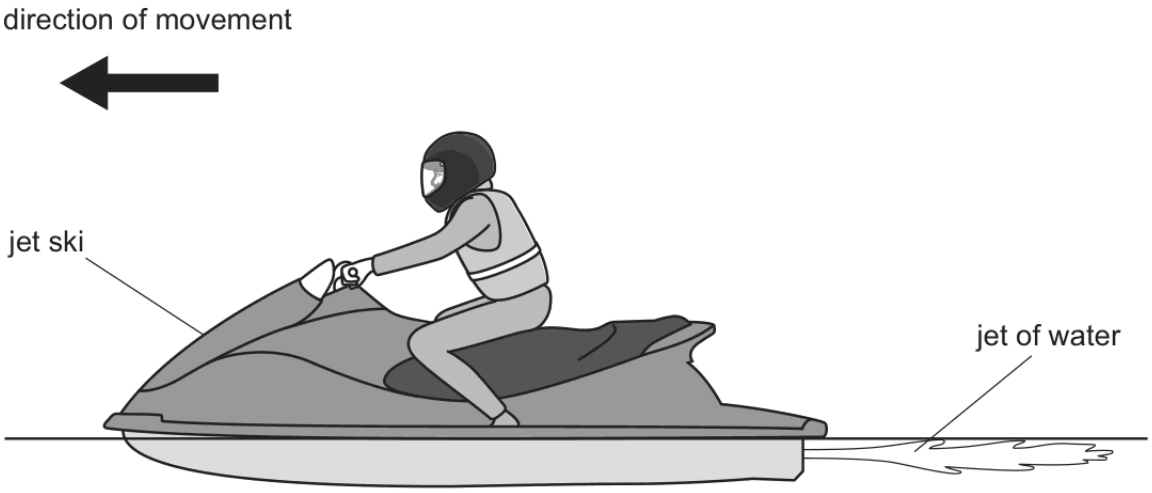

Fig. 1.1

A high-speed jet of water is forced backwards out of the back of the jet ski by a pump inside the jet ski.

(a) The pump increases the momentum of the water that is forced backwards out of the back of the jet ski.

(i) Complete the word equation to show the relationship between the resultant force on an object and the change in momentum of the object.

resultant force =

[1]

(ii) In 2.0 s, the pump increases the backwards speed of 180 kg of water by 30 m/s.

Calculate the backwards force exerted on the water.

force = ... N [2]

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(b) (i) Using Newton's third law of motion, explain why there is a forwards force on the jet ski.

.....

.....

.....

..... [2]

(ii) The student has a mass of 70 kg and the jet ski has a mass of 280 kg.

Use your answer from **(a)(ii)** to determine the acceleration of the student and jet ski when no resistive forces are acting.

acceleration = ... m/s^2 [2]

(c) The jet ski reaches a speed of 20 m/s.

Calculate the total kinetic energy of the student and jet ski at this speed.

kinetic energy = ... J [3]

[Total: 10]

Question	Answer	Marks
1(a)(i)	resultant force = increase in momentum / unit time **or** resultant force = rate of increase in momentum	**B1**
1(a)(ii)	(F =) 180 × 30 / 2.0 **or** 180 × 15	**C1**
	2700 (N)	**A1**
1(b)(i)	pump **/** jet ski / engine exerts force on water	**B1**
	water exerts an opposite force on the pump / jet ski / engine	**B1**
1(b)(ii)	(a =) F / m **or** 2700 / (70 + 280) **or** 2700 / 350	**C1**
	7.7 (m / s^2)	**A1**
1(c)	(K.E. =) ½ mv^2	**C1**
	½ × (70 + 280) × 20^2 **or** ½ × 350 × 20^2	**C1**
	70 000 (J)	**A1**

3 A battery, a pulley and a motor are used to lift a load as shown in Fig. 3.1.

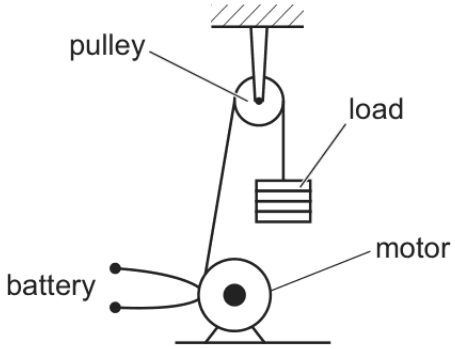

Fig. 3.1

(a) Describe the transfers between energy stores that occur as the load is lifted.

.....

.....

.....

..... [3]

(b) The efficiency of the motor, pulley and load system is less than 100%.

(i) By comparing the input energy and the useful output energy, explain why the efficiency is less than 100%.

.....

..... [1]

(ii) Explain how the principle of conservation of energy applies in lifting the load.

.....

.....

..... [2]

(c) The input power to the motor is 15 W. The motor is used for 20 s. The efficiency of the motor is 60%.

Calculate the energy supplied to the load.

energy = ... J [2]

[Total: 8]

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Question	Answer	Marks
3(a)	*Any three from*: • reduction in chemical energy store in battery • electrical work done by motor • mechanical work done on weight • increase in gravitational energy store of weight • increase in thermal energy (in motor or air) • electrical heating (by current)	**B3**
3(b)(i)	useful or output energy < input energy	**B1**
3(b)(ii)	total energy is constant	**B1**
	energy from power supply = energy gained by weight + thermal energy (to air / due to friction)	**B1**
3(c)	(*E* =) power × time in any form	**C1**
	180 J	**A1**

QUESTION · 5054_s25_qp_12.pdf · source page 4 of 16

8 A rock of mass 20 kg is travelling in space at a speed of 6.0 m / s.

What is its kinetic energy?

A 60 J **B** 120 J **C** 360 J **D** 720 J

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 8

8	C	**1**

2 Fig. 2.1 shows a fork-lift truck used to lift a load.

Fig. 2.1

When lifting the load, the electric motor on the fork-lift truck has a useful output power of 600 W.

(a) Define 'power'.

.....

..... [1]

(b) The efficiency of the motor is 70%.

Calculate the input power to the motor.

input power = ...W [2]

(c) The fork-lift truck lifts a load with mass of 50 kg through a vertical distance of 2.3 m.

(i) Calculate the work done in lifting the load.

work done = ... J [2]

(ii) Calculate the time taken to lift the load.

time taken = ... s [2]

(d) Describe how energy is transferred to thermal energy as the load is lifted by the fork-lift truck.

.....

..... [1]

[Total: 8]

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Question	Answer	Marks
2(a)	work done or energy per unit time	**B1**
2(b)	(input power =) power output / efficiency in any form	**C1**
	860 (W)	**A1**
2(c)(i)	(work =) force × distance **or** *mgh*	**C1**
	1100 (J)	**A1**
2(c)(ii)	(time =) work ÷ (output) power	**C1**
	1.9 or 1.8 s	**A1**
2(d)	*Any one of*: • friction / air resistance • electrical heating	**B1**

QUESTION · 5054_w25_qp_12.pdf · source page 7 of 16

15 Energy is stored in batteries, fuel and food.

What is the name of this energy store?

A chemical

B electrostatic

C nuclear

D thermal

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 15

15	A	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 7 of 16

16 A rocket has mass *M* when empty. The rocket carries an additional mass *M* of fuel.

The rocket and fuel travel at a speed *v*. When the engine of the rocket is fired, all of the fuel is expelled and the speed of the rocket increases to 2*v*.

What happens to the kinetic energy of the **rocket**?

A It doubles.

B It halves.

C It increases by a factor of four.

D It stays the same.

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 16

16	C	**1**

2 A spherical container is used to carry measuring instruments to the bottom of a lake.

Fig. 2.1 shows the container held at rest just below the water surface of a lake.

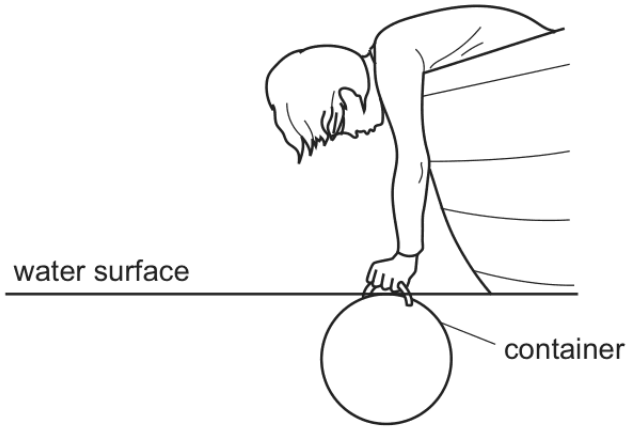

Fig. 2.1 (not to scale)

The container is released and falls through the water to the bottom of the lake.

(a) The container reaches terminal velocity before it hits the bottom of the lake.

Explain, in terms of the forces on the container, the motion of the container through the water until it reaches terminal velocity.

.....

.....

.....

.....

.....

..... [3]

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(b) The container falls through a vertical distance of 32 m before it hits the bottom of the lake.

The container and instruments have a total mass of 74 kg.

(i) Calculate the energy transferred from the gravitational potential energy store of the container.

energy transferred = ... J [2]

(ii) State the principle of conservation of energy.

.....

.....

..... [2]

(iii) Work is done to transfer energy from the gravitational potential energy store as the container falls at terminal velocity.

Describe, in terms of the work done, how the energy is transferred from the gravitational potential energy store to **one** other energy store in the water.

.....

.....

..... [2]

[Total: 9]

Question	Answer	Marks
2(a)	(initially its weight causes) acceleration **or** speeds up **or** gains kinetic energy	**B1**
	friction / drag / resistance / resistive force / viscous force increases (as the speed increases) **or** resultant force decreases	**B1**
	acceleration decreases to zero **or** friction / drag / resistance / resistive force / viscous force is (eventually) equal to weight **or** forces balance **or** resultant force = 0	**B1**
2(b)(i)	(ΔE_P =) $mg\Delta h$ **or** $74.0 \times 9.8 \times 32$	**C1**
	2.3×10^4 (J)	**A1**
2(b)(ii)	energy can be neither created **and** nor destroyed	**B1**
	energy can be transferred from one energy store to another	**B1**
2(b)(iii)	(energy transferred) by doing work against friction / drag / resistance / resistive force	**B1**
	to internal (energy) store	**B1**

2 Figure 2.1 shows a waterfall.

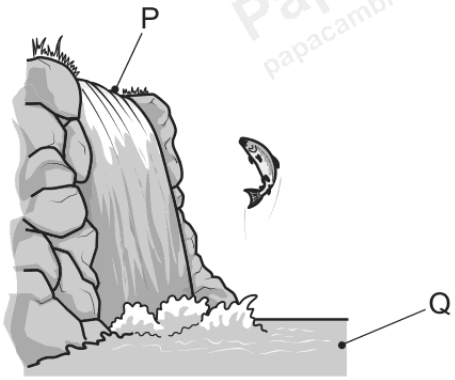

Figure 2.1

(a) The speed of the water flow at points P and Q is the same.

(i) Describe the transfers of energy that occur as the water moves from point P to point Q.

.....

.....

.....

..... [2]

(ii) The particles of water move at random as well as moving with the water flow past point P and point Q.

Compare the temperature of the water at point P with the temperature of the water at point Q, and compare the average speed of the random movement of particles at point P with the average speed of the random movement of the particles at point Q.

the temperature at point P compared with point Q is .....

the average speed of particles at point P compared with point Q is .....
[1]

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(b) A fish of mass 2.5 kg rises vertically, leaving the water at a speed of 8.0 m/s.

(i) Calculate the kinetic energy of the fish as it leaves the water.

kinetic energy = ... J [3]

(ii) Calculate the maximum height above the surface of the water that the fish reaches.

height = ... m [2]

[Total: 8]

Question	Answer	Marks	Guidance
2(a)(i)	gravitational (potential) energy to kinetic energy (kinetic energy) to internal energy	**1**	**A** GPE to KE at Q; a change is needed **A** KE to heat energy / thermal energy **I** back to GPE at bottom of fall (in splashes); energy to sound
2(a)(ii)	lower / slower / smaller/ less for both temperature and average speed	**1**	**A** colder
2(b)(i)	(K.E. =) $\frac{1}{2}mv^2$ algebraic $\frac{1}{2} \times 2.5 \times 8^2$ numerical 80 (J)	**1**	Award of this mark also earns 1st mark. **not** 2500
2(b)(ii)	GPE or **(a)(iii)** = *mgh* numerical or algebraic 3.3 (m)	**1**	**A** use of distance = $v^2 / 2g$ 3.265 accept 3.2 if 10 N/kg used

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Chapter 1 Motion, forces and energy

1.7 Energy, work and power

1.7.2 Work

68 entries: 38 Paper 1 questions, 30 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

For Examiner's Use

2 Fig. 2.1 shows a stationary piston in a cylinder.

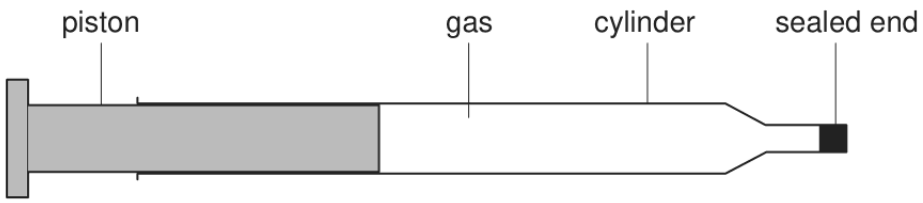


Fig. 2.1

(a) The gas in the cylinder exerts the same pressure on the piston as it does on the sealed end. The sealed end has a smaller cross-sectional area.

Use ideas about molecules to explain why the pressures are the same.

.....

.....

.....

.....

..... [2]

(b) The piston is pushed inwards and the temperature of the gas stays constant.

(i) The piston moves 0.10 m. The average force exerted on the piston to compress the gas is 23 N.

Calculate the work done. State the formula that you use.

work done = ... [2]

(ii) The gas in the cylinder starts at a pressure of 1.0×10^5 Pa and has a volume of 100 cm^3. The volume of the gas decreases to 80 cm^3.

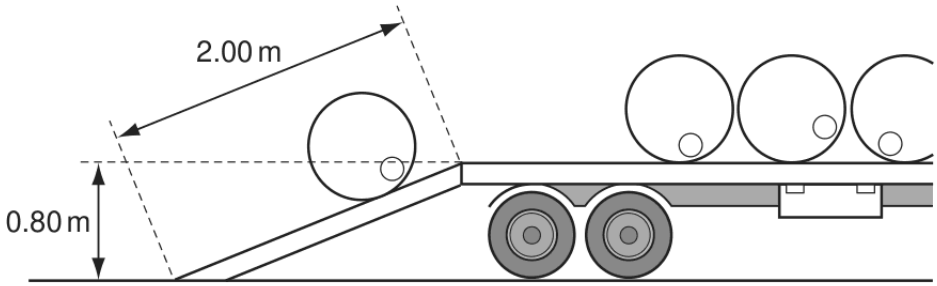
Calculate the final pressure of the gas. State the formula that you use.

pressure = ... [3]

2	**(a)**		molecules hit piston/end/walls	**C1**	
			same number molecules hit equal (unit) areas of piston and end or more molecules hit piston but area is larger	**A1**	
	(b)	**(i)**	F x d formula	**B1**	
			2.3 J c.a.o.	**B1**	
		(ii)	PV = constant or $P_1 V_1 = P_2 V_2$ formula	**B1**	
			$1.0 \times 10^5 . 100 = P . 80$	**C1**	
			1.25×10^5 Pa	**A1**	**7**

QUESTION · 5054_s08_qp_1.pdf · source page 6 of 20

10 A workman rolls a barrel of weight 2000 N up a plank of length 2.00 m and on to a lorry. The back of the lorry is 0.80 m above the horizontal surface of the road.



What is the work done on the barrel against gravity?

A 1000 J **B** 1600 J **C** 2500 J **D** 4000 J

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 10

10	**B**

2 Fig. 2.1 shows a boat stationary in still water in a narrow tunnel.

For Examiner's Use

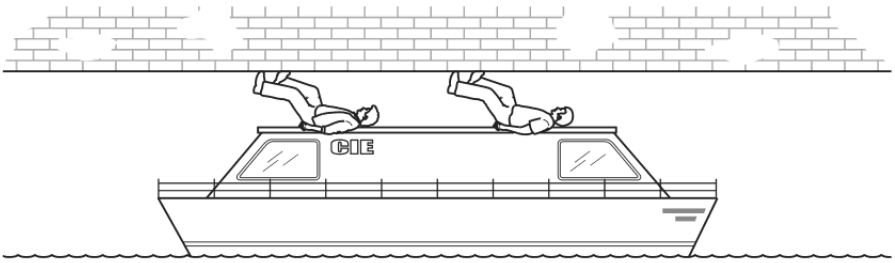


Fig. 2.1

To move the boat, two men lie on the top of it and push against the tunnel roof with their legs. They exert a total forwards force of 1680 N on the boat.

(a) The men push the boat 50.0 m to the end of the tunnel. Calculate the work done on the boat by the men.

work done = ...[2]

(b) Explain why the kinetic energy of the boat at the end of the tunnel is less than the work done on the boat by the men.

.....

.....

.....[2]

2	**(a)**	(WD =)Fx **or** 1680 × 50	C1	
		84 000 J	A1	
	(b)	friction/drag/resistance **of water/air**	B1	
		work done against friction/resistance/drag **or** energy lost due to friction/resistance/ drag **or** energy lost as heat/internal/thermal	B1	[4]

For Examiner's Use

3 Fig. 3.1 shows a student rubbing her hands together.

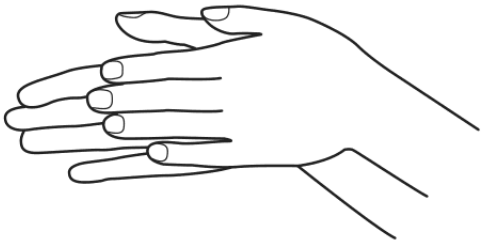


Fig. 3.1

(a) State the main energy conversion that causes the hands to become warm.

.....

.....[2]

(b) State why the hands become even warmer if they are pressed harder together when rubbing.

.....

.....[1]

(c) The average force used to slide one hand along the other is 1.2 N.

In each movement, one hand moves 0.080 m. The other hand remains stationary.

(i) Calculate the number of movements needed for 2.0 J of work to be done.

number = ...[3]

(ii) Each movement takes 0.20 s. Calculate the average power developed.

power = ...[2]

3	**(a)**	chemical energy to **or** K.E to	B1	
		heat/thermal energy/internal energy – at end	B1	[2]
	(b)	friction/resistive force **increases**	B1	[1]
	(c)	**(i)** work = force × distance in words, number or symbols	C1	
		1.2 × 0.08 **or** 0.096 **or** 20 **or** 0.208 seen	C1	
		20.8 **or** 21	A1	[3]
		(ii) power = work/time **or** energy/time, numerical or algebraic – may use values in **(i)**		
		accept 2/0.2 or 10 **W**	C1	
		0.48 W (ecf **(i)** including power of ten error in **(i)**)	A1	[2]

For Examiner's Use

2 Fig. 2.1 shows the planet Venus orbiting the Sun in a circular orbit at constant speed.

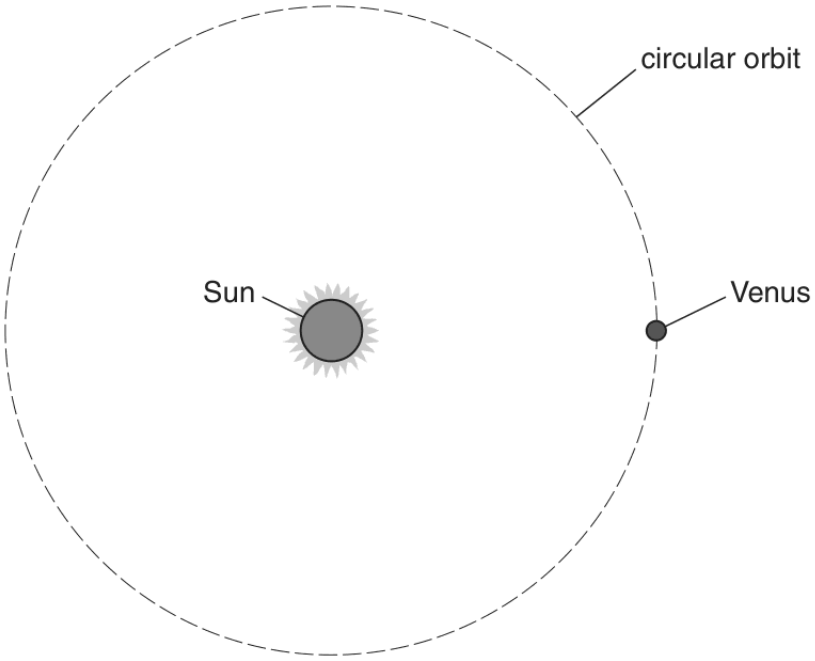


Fig. 2.1 (not to scale)

A force acting on Venus keeps it moving in a circle.

(a) State the name of this force and the object that causes it.

.....

..... [2]

(b) **(i)** On Fig. 2.1, draw an arrow through Venus that represents the direction of its acceleration. [1]

(ii) The size of this acceleration is $9.7 \times 10^{-3}\,m/s^2$ and the mass of Venus is $4.9 \times 10^{24}\,kg$. Calculate the size of the force acting on the planet.

force = ... [2]

(c) As time passes, the planet Venus moves a considerable distance around the circular orbit. State why no work is done by the force acting on the planet as it moves.

.....

.....

..... [2]

2	**(a)**	gravitational/centripetal (pull/attraction)	B1	
		of the **Sun**	B1	[2]
	(b) **(i)**	arrow touching Venus towards centre/left (must pass through Sun if extended)	B1	[1]
	(ii)	(F =) ma **or** $4.9 \times 10^{24} \times 9.7 \times 10^{-3}$	C1	
		$4.8(4.753) \times 10^{22}$ N	A1	[2]
	(c)	direction of movement ⊥ direction of force	C1	
		force/it is perpendicular/at right angles to distance moved **or** does not move **any** distance in direction of force	A1	[2]

10 A bus breaks down on a road with a 10° upward slope. The passengers get out and push the bus to the top of the slope at a constant speed. Fig. 10.1 shows the passengers exerting a force on the bus parallel to the line of the slope.

For Examiner's Use

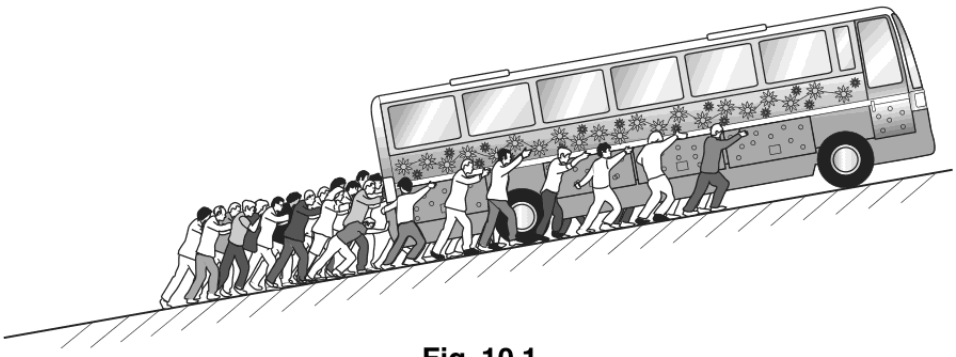


Fig. 10.1

(a) The total mass of the bus is 3200 kg.

(i) Calculate the weight of the bus.

weight = ...[1]

(ii) On Fig. 10.1, draw **two** arrows to show the direction of the weight of the bus and the direction of the force exerted on the bus by the passengers. [1]

(iii) The total force exerted by the passengers on the bus is 17000 N along the line of the slope. Use a graphical method to determine the size and direction of the resultant of this force and the weight of the bus. State the scale used.

scale = ...

size of resultant = ...

direction of resultant = ...

[4]

For Examiner's Use

(iv) The bus travels at a constant speed. State the size of the resultant of all the forces acting on the bus.

.....[1]

(b) As the passengers push the bus up the slope, work is done against two forces. State the names of these two forces and, for each force, state the form of energy produced as work is done.

force 1 .....

form of energy 1 .....

force 2 .....

form of energy 2 .....

[4]

(c) When the bus reaches the top of the slope, it stops and the passengers get back in.

(i) On the axes of Fig. 10.2, sketch the distance-time graph for the bus from a time when the passengers are pushing the bus at constant speed until they get back into the stationary bus.

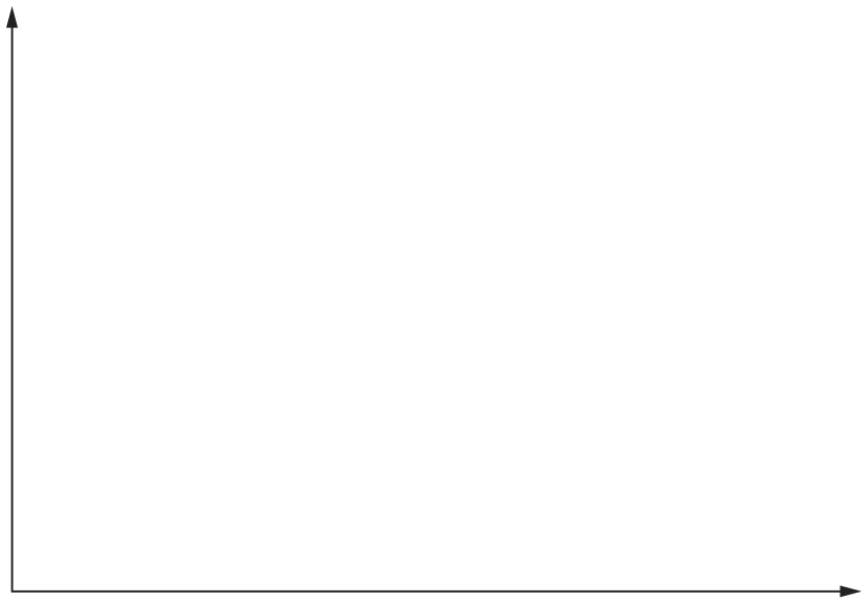


Fig. 10.2

[3]

(ii) State how the speed of a moving object may be obtained from an accurately drawn distance-time graph.

.....

.....[1]

10	**(a)**	**(i)**	32 000 N	B1	
		(ii)	two arrows/lines in correct direction by eye	B1	
		(iii)	scale given	B1	
			two arrows/lines **and** correct resultant drawn	B1	
			32.0 → 35.0 kN (2/3 sig. fig. only)	B1	
			58.5 → 61.5° to horizontal (2/3 sig. fig. only; don't penalise twice)	B1	
		(iv)	zero/no force/0	B1	[7]
	(b)		weight/gravitational force/gravitational attraction (**not** gravity)	B1	
			higher in gravitational field **or** (to gravitational) potential energy	B1	
			friction/air resistance	B1	
			heat/thermal/internal energy	B1	[4]
	(c)	**(i)**	labelled axes **and** correct way round ($x \rightarrow t$)	B1	
			straight line of positive slope	B1	
			followed only by horizontal line (**ign** curve at junction)	B1	
		(ii)	distance travelled/time taken (from points) **or** calculate the gradient	B1	[4]

2 Fig. 2.1 shows an elevator (lift) that is supported by a cable.

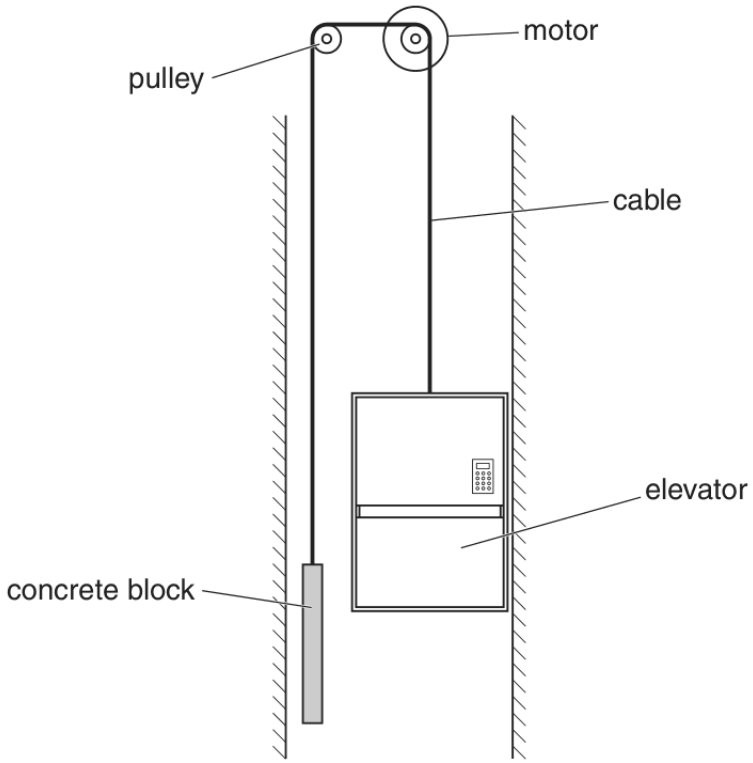


Fig. 2.1 (not to scale)

The cable passes around the axle of an electric motor and over a pulley. The cable is then attached to a concrete block that has the same mass as the empty elevator. As the elevator moves up, the concrete block moves down.

(a) Explain why very little work is done by the electric motor when lifting up the empty elevator.

.....

.....

.....[2]

(b) Several people of total weight 4900 N are travelling upwards at constant speed in the elevator. They rise through a distance of 24 m in 28 s.

Calculate the minimum power output of the electric motor.

power = ...[3]

2	**(a)**	work is done by the (falling) block **or** block loses (G)PE **or** energy transferred from block to elevator **or** forces balance	B1	
		(WD by falling block) raises the elevator **or** converted to GPE **of elevator or** (WD against) friction/air resistance **or** WD to accelerate elevator	B1	
	(b)	(WD =) $F \times d$ **or** 4900 × 24 **or** 117 600 **or** (P =) E/t	C1	
		4900 × 24/28 **or** 117 600/28	C1	
		4.2×10^3 W **or** 4.2 kW	A1	[5]

2 Fig. 2.1 shows a skier of mass 85 kg skiing down a very steep slope.

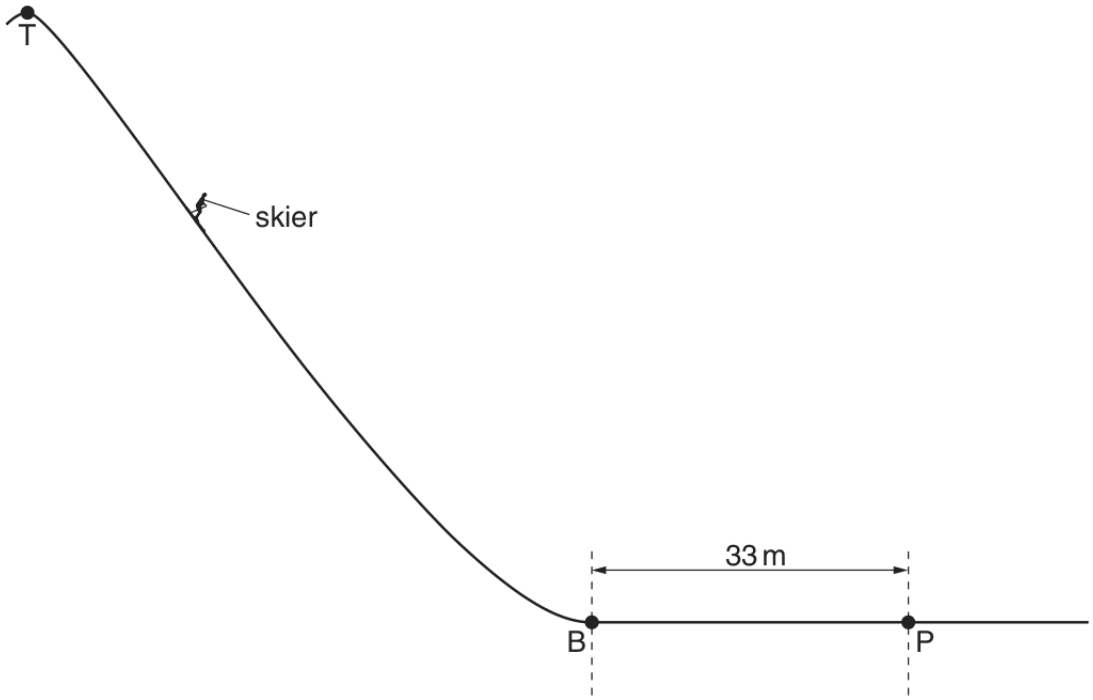


Fig. 2.1 (not to scale)

The skier starts from rest at the top T of the slope. The force of gravity accelerates him down the slope.
When he reaches the bottom B of the slope, his kinetic energy is 5.5×10^4 J.

(a) The gravitational field strength is 10 N/kg. Calculate

(i) the weight of the skier,

weight = ...[1]

(ii) the minimum possible difference in height between T and B.

height difference = ...[2]

(b) At B, the skier digs his skis into the snow and stops at the point P after travelling 33 m horizontally.

Calculate the average horizontal force that acts on the skier between B and P.

force = ...[2]

2	**(a)**	**(i)**	850 N	B1	
		(ii)	KE = PE/*mgh* **or** *mgh* = 5.5×10^4	C1	
			65/64.7(0588) m	A1	
	(b)		WD = *Fx* **or** KE/*x* **or** $5.5 \times 10^4/33$ **or** *v* = 35(.97) **and** *a* = 19(.60) **and** *F* = *ma*	C1	
			1700/1670/1667/1666.7 N	A1	[5]

3 Fig. 3.1 shows an electric motor being used on a building site to lift bricks.

For Examiner's Use

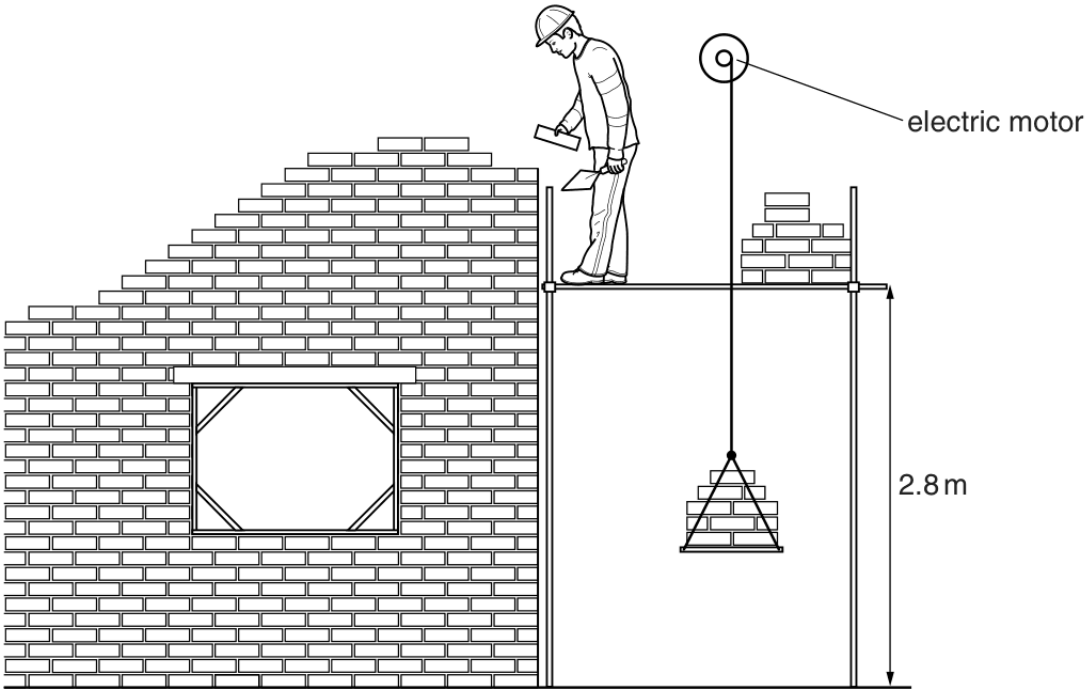


Fig. 3.1

The gravitational field strength is 10 N / kg. The motor lifts bricks of total mass 54 kg through a height of 2.8 m in 3.0 s.

(a) Calculate

(i) the work done in lifting the bricks,

work done = ... [2]

(ii) the useful output power of the motor.

power = ... [2]

(b) Suggest two reasons why the electrical power supplied to the motor is greater than its useful output power.

1. .....

2. .....
[2]

For Examiner's Use

(c) (i) Fig. 3.2 shows a battery and an electric motor. The battery is used to power the motor. Complete Fig. 3.2 to show the circuit diagram of a circuit that is used to determine the power supplied to the electric motor.

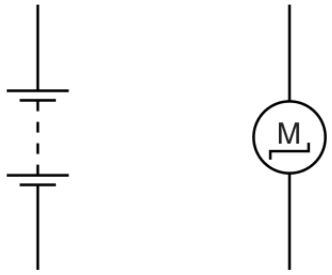


Fig. 3.2 [2]

(ii) State how the power supplied is calculated.

.....

..... [1]

3	**(a)**	**(i)**	(WD =)*mgh* **or** 54 × 10 × 2.8	C1	
			1500/1510/1512 J	A1	
		(ii)	(*P* =) WD/*t* or *E*/*t* **or** 1500/3 **or** 1510/3 **or** 1512/3	C1	
			500/503/504 W	A1	
	(b)		any **two** of: also lifting board/rope heat in motor/wires/cable friction with something named e.g. axle/spindle/air	B2	
	(c)	**(i)**	power supply, motor and ammeter in series (**ignore** series voltmeter and other components)	B1	
			voltmeter to measure voltage across motor	B1	
		(ii)	current (reading) × voltage (reading) or VI	B1	[9]

QUESTION · 5054_w12_qp_12.pdf · source page 6 of 16

14 An object of mass 5 kg is carried a distance of 4 m across a room at a constant height above the floor.

What is the work done on the object?

A 0 **B** 20 J **C** 50 J **D** 200 J

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 14

14	**A**

For Examiner's Use

4 Fig. 4.1 shows gas trapped in a cylinder by a piston.

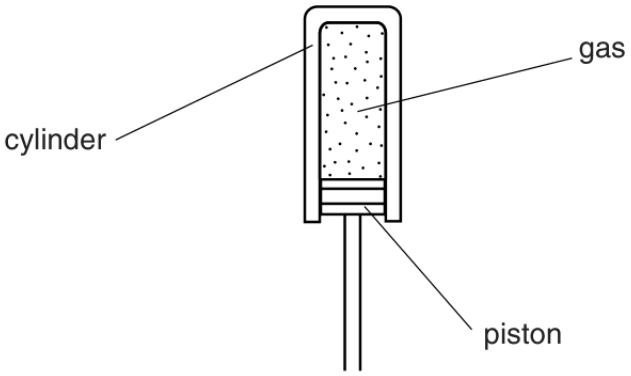


Fig. 4.1

The piston has a cross-sectional area of $0.0050\,m^2$. It moves upwards, through a distance of 0.074 m, compressing the gas. During this compression, the average pressure of the gas is 4.6×10^5 Pa.

(a) Calculate

(i) the average force exerted on the piston by the gas during compression,

force = ...[2]

(ii) the work done on the gas during compression.

work done = ...[2]

(b) The gas in the cylinder has a heat capacity of 0.27 J/°C and heats up as it is compressed.

(i) Calculate the maximum possible temperature rise of the gas.

temperature rise = ...[2]

(ii) Suggest a reason why the actual temperature rise of the gas is less than the value calculated in **(i)**.

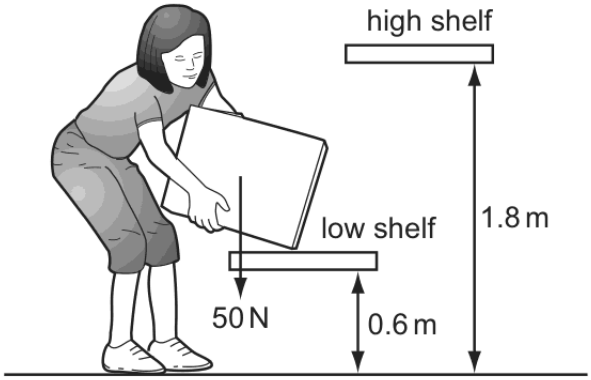
.....

.....[1]

4	**(a)**	**(i)**	$(F =)PA$ **or** $4.6 \times 10^5 \times 0.005$		C1	
			2300 N		A1	
		(ii)	$(WD =)F \times d$ **or** 2300×0.074		C1	
			170(.2) J		A1	
	(b)	**(i)**	$(\Delta T =)Q/C$ **or** 170/0.27		C1	
			629.6(2)/630(.370) °C	(° is **not** correct)	A1	
		(ii)	thermal energy/heat lost to cylinder/environment/atmosphere (**not** just 'lost') **or** work done against/heat lost due to friction		B1	[7]

QUESTION · 5054_s13_qp_11.pdf · source page 6 of 20

11 The diagram shows a girl lifting a box of weight 50 N from a low shelf to a high shelf.



How much work is done on the box?

A 30 J **B** 60 J **C** 90 J **D** 120 J

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 11

11	**B**

For Examiner's Use

3 Fig. 3.1 shows a man using a chest expander to increase the strength of his arms.

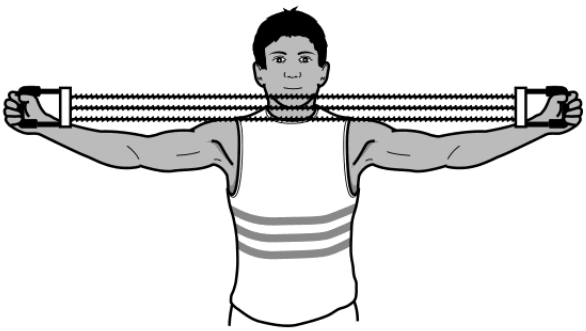


Fig. 3.1

The chest expander may be considered to be a single spring.

Fig. 3.2 shows the force needed to produce different extensions of the spring.

force / N	extension / cm
0	0
60	10
120	20
180	30

Fig. 3.2

(a) The man increases the force on the spring from 0 to 180 N.
The spring extends by 30 cm and the average force exerted during this process is 90 N.

(i) Calculate the work done on the spring.

work = ... [2]

(ii) Twenty extensions are made in 1.0 minute.

Calculate the power used to extend the spring.

power = ... [2]

For Examiner's Use

(b) The force at the limit of proportionality of the spring is 800 N.

(i) Calculate the extension of the spring when the force is 800 N.

extension = ... [2]

(ii) The force on the spring is increased from 800 N to 860 N.

Suggest what happens to the extension.

.....

..... [1]

3	**(a)**	**(i)**	(*W*=) *Fd* **or** 90 × 0.3 **or** 90 × 30	C1	
			27 J	A1	
		(ii)	(*P*=) *W*/*t* **or** *Fd*/*t* **or** 27(× 20)/60 **or** 27/3	C1	
			9(.0) W	A1	
	(b)	**(i)**	800 × 30/180 **or** 800/6 **or** 6 seen **or** proportionality clearly used	C1	
			133 **or** 130 cm	A1	
		(ii)	extension more than 143 cm **or** (extra) extension > 10 cm **or** (some) extension permanent	B1	[7]

For Examiner's Use

3 A farmer uses a hydraulic system to operate machinery that is pulled behind a tractor.

Two cylinders and the flexible pipe that joins them contain oil. Two pistons keep the oil in the cylinders. The arrangement is shown in Fig. 3.1.

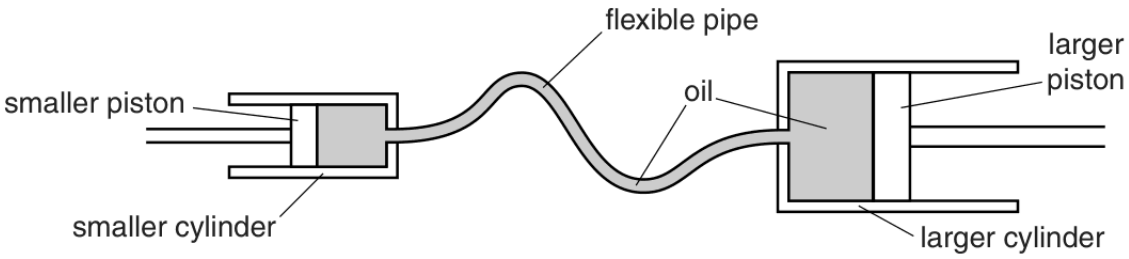


Fig. 3.1

The cross-sectional area of the smaller cylinder is 0.048 m^2.
The cross-sectional area of the larger cylinder is 0.14 m^2.

The smaller piston exerts a force of 12 000 N on the oil.

(a) Calculate

(i) the pressure in the oil,

pressure = ...[2]

(ii) the force exerted by the oil on the larger piston.

force = ...[1]

(b) Suggest why the resultant force on the larger piston is less than the value obtained in **(a)(ii)**.

.....

.....[1]

(c) The smaller piston moves a distance of 0.065 m into the cylinder.

Calculate the work done on the oil by the smaller piston.

work done = ...[2]

(d) Suggest one advantage of using oil rather than air in the system.

.....[1]

3 **(a)** **(i)** (p =) F/A **or** 12 000/0.048 **or** 12 000/0.14
or (in **(ii)**) (F =) pA **or** 2.5 × 105 × 0.14 C1
2.5×10^5 Pa A1

(ii) 35 000 N A1

(b) atmospheric pressure **or** friction (between cylinder and piston/oil)
(accept bubbles (of air) in oil **or** viscosity of oil) B1

(c) (W.D. =) $F \times d$ **or** 12 000 × 0.065 **or** 35 000 × 0.065 **or** 2275 C1
780 J A1

(d) (liquids) incompressible **or** air spongy **or** (oil) lubricates the system **or** (oil) reduces friction
(**ignore** density references, **ignore** oil compresses less) B1 [7]

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 A garden pond contains a small fountain. An electric pump in the water causes the water to rise above the surface of the pond, as shown in Fig. 9.1.

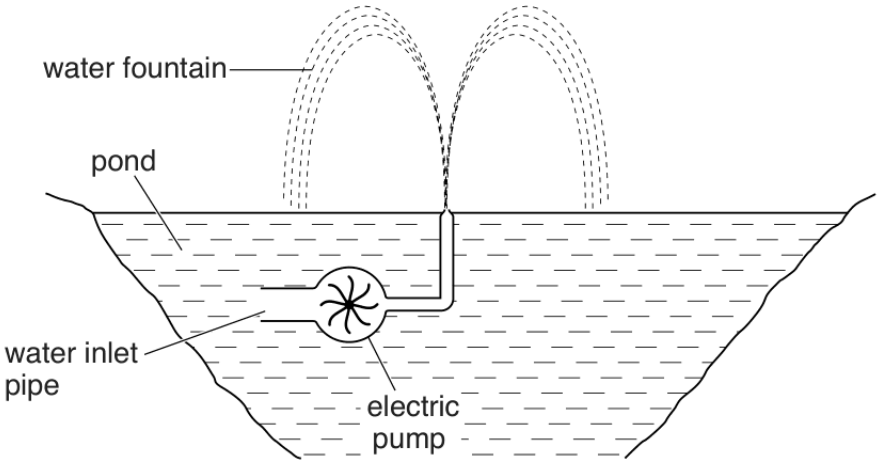


Fig. 9.1

(a) The pressure of the water in the pond increases with depth.

(i) Explain what is meant by *pressure*.

.....

..... [1]

(ii) Explain why the pressure below the surface of the water increases with depth.

.....

..... [2]

(iii) State the unit of pressure.

..... [1]

(b) Describe the energy changes that occur within the pump.

.....

.....

..... [3]

(c) A volume of 0.78 m^3 of water passes through the pump in one hour. The density of water is 1000 kg/m^3.

(i) Calculate the mass of water that passes through the pump in one hour.

mass = ... [2]

(ii) The water rises 1.5 m. The gravitational field strength g is 10 N/kg.

Calculate the useful work done in one hour raising the water to the top of the fountain.

work done = ... [2]

(iii) Calculate the minimum power output of the pump.

power = ... [2]

(d) Describe an experiment to check that the density of water is 1000 kg/m^3.

.....

.....

.....

.....

..... [2]

SECTION B

9 (a) (i) force per unit area **or** force divided by area **or** force on 1 m^2 **or** force on unit area [B1]

(ii) more water above **or** more force from water **or** more atoms / molecules above [C1]

larger weight of water above [A1]

(iii) Pa **or** N / m^2 [B1]

(b) electrical energy at start of process [B1]
kinetic energy of water / turbine/blade produced [B1]
some heat energy / thermal energy / internal energy produce [B1]

(c) (i) (M=) D × V in any algebraic or numerical form [C1]
780 kg [A1]

(ii) (W=) mgh or Fd in any algebraic or numerical form [C1]
11700 J **or** 12000 J [A1]

(iii) (P=) W / t in any algebraic or numerical form **or** 195(W) [C1]
3.25 **or** 3.2 **or** 3.3 W [A1]

(d) find difference in mass of container, with and without water [B1]
measuring cylinder or similar instrument used to find volume **and** density checked [B1]

other methods are possible, e.g. use of calibrated hydrometer

[15]

QUESTION · 5054_w14_qp_11.pdf · source page 9 of 20

13 A student uses a newton meter to pull an object horizontally along rough ground. The student multiplies the newton meter reading by the distance moved by the object.

What does this calculation give?

A the efficiency of the process

B the kinetic energy of the object

C the power generated

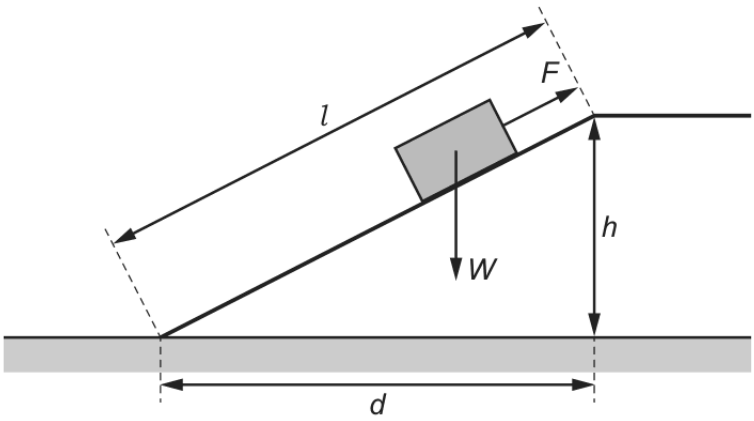
D the work done on the object

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 13

13	**D**

QUESTION · 5054_w14_qp_12.pdf · source page 5 of 16

10 A constant force *F* pulls a block of weight *W* up the slope shown.



How much work is done in pulling the block up the slope?

A $F \times h$ **B** $F \times l$ **C** $W \times d$ **D** $W \times l$

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 10

10 **B**

2 Fig. 2.1 shows a student doing a press-up. A total force *F* acts upwards on his hands. There is also a force upwards on his toes.

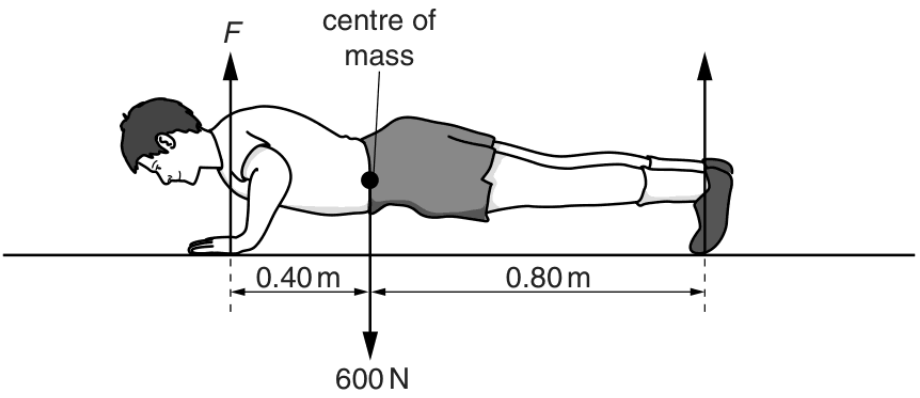


Fig. 2.1 (not to scale)

The weight of the student is 600 N and this force acts downwards from his centre of mass.

(a) **(i)** Explain why the student does work as his body rises from the ground.

.....

.....[1]

(ii) State the form of energy that the student uses to do this work.

.....[1]

(b) At the position shown in Fig. 2.1, the student is stationary.
The weight of the student causes a moment about his toes.

(i) Calculate the moment of the weight of the student about his toes.

moment = ...[1]

(ii) Calculate the value of the force *F*.

F = ...[2]

2	**(a)**	**(i)**	force moves through a distance (in same direction)	B1
		(ii)	chemical (potential) energy	B1
	(b)	**(i)**	480 Nm	B1
		(ii)	attempt to apply moments with two forces and distances	C1
			400 N	A1

3 A lorry of mass 4.4×10^4 kg travels along a straight, horizontal road at 20 m/s.

(a) Calculate the kinetic energy of the lorry.

kinetic energy = ... [3]

(b) The lorry driver sees an obstruction ahead and applies the brakes. The lorry slows down and stops. The lorry stops a distance of 40 m from where the driver first applies the brakes.

As it slows down, the kinetic energy of the lorry decreases to zero. The kinetic energy decreases as work is done by the braking force exerted on the lorry.

(i) State the formula that relates the work done to the braking force.

.....

.....

..... [1]

(ii) Calculate the braking force exerted on the lorry.

force = ... [2]

3	**(a)**	(k.e. =)$\frac{1}{2}mv^2$	C1	
		$\frac{1}{2} \times 4.4 \times 10^4 \times 20^2$	C1	
		8.8×10^6 J	A1	
	(b)	**(i)** $WD = F \times x_{(\|\|)}$ **or** force × distance (parallel to / in direction of force)	B1	
		(ii) $8.8 \times 10^6 / 40$ **or** $a = (-)5.0$ (m / s^2) **or** t = 4.0 (s) seen	C1	
		2.2×10^5 N	A1	[6]

QUESTION · 5054_w15_qp_12.pdf · source page 7 of 16

15 A piston of area $10\,cm^2$ is pushed slowly into a very large cylinder containing gas at a pressure of $10\,N/cm^2$. The pressure of the gas remains constant as the piston moves a distance of 0.10 m.

What is the force of the gas on the piston and what is the work done by the piston on the gas?

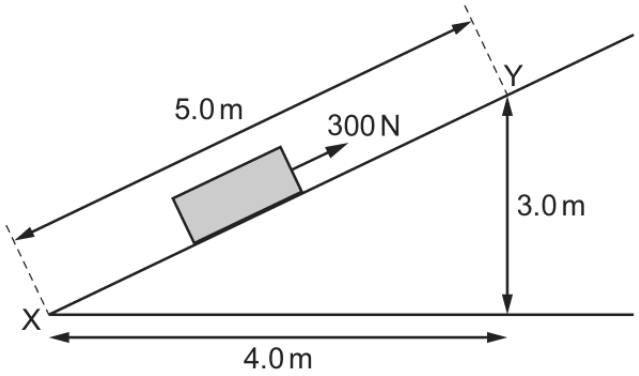
	force / N	work done / J
A	1.0	0.1
B	1.0	10
C	100	10
D	100	1000

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 15

15 **C**

QUESTION · 5054_s16_qp_11.pdf · source page 7 of 16

14 A 300 N force is applied to a box to move it up a ramp, as shown.



How much work is done by the force when moving the box from X to Y?

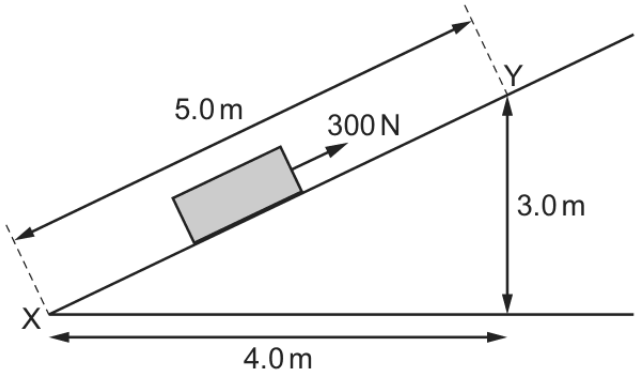
A 900 J **B** 1200 J **C** 1500 J **D** 3000 J

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 14

14	**C**

QUESTION · 5054_s16_qp_12.pdf · source page 4 of 16

9 A 300 N force is applied to a box to move it up a ramp, as shown.



How much work is done by the force when moving the box from X to Y?

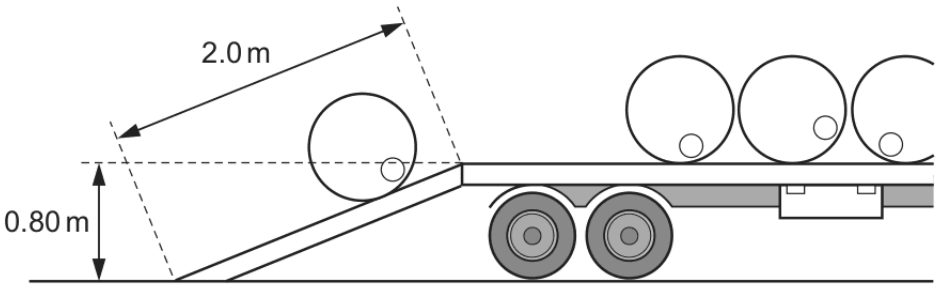
A 900 J **B** 1200 J **C** 1500 J **D** 3000 J

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 9

9	**C**

QUESTION · 5054_w16_qp_11.pdf · source page 5 of 20

8 A workman rolls a barrel of weight 2000 N up a plank of length 2.0 m and onto a lorry. The back of the lorry is 0.80 m above the horizontal surface of the road.



What is the work done on the barrel against gravity?

A 1000 J **B** 1600 J **C** 2500 J **D** 4000 J

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 8

8	**B**

QUESTION · 5054_w16_qp_12.pdf · source page 6 of 16

13 Which force does no work?

A the force from a car's engine that causes the car to accelerate

B the force that causes a box to slide over a rough horizontal surface

C the force that causes a satellite to move in a circular orbit

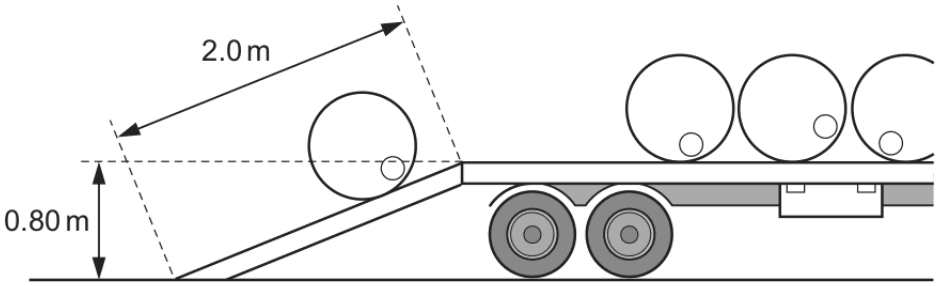
D the force that causes a spring to stretch

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 13

13	**C**

QUESTION · 5054_w16_qp_12.pdf · source page 6 of 16

14 A workman rolls a barrel of weight 2000 N up a plank of length 2.0 m and onto a lorry. The back of the lorry is 0.80 m above the horizontal surface of the road.



What is the work done on the barrel against gravity?

A 1000 J **B** 1600 J **C** 2500 J **D** 4000 J

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 14

14 **B**

QUESTION · 5054_s17_qp_11.pdf · source page 7 of 16

13 The work done by a force on a body is calculated by multiplying the force by a quantity.

Which quantity?

A the distance travelled in the direction of the force

B the distance travelled perpendicular to the direction of the force

C the speed in the direction perpendicular to the force

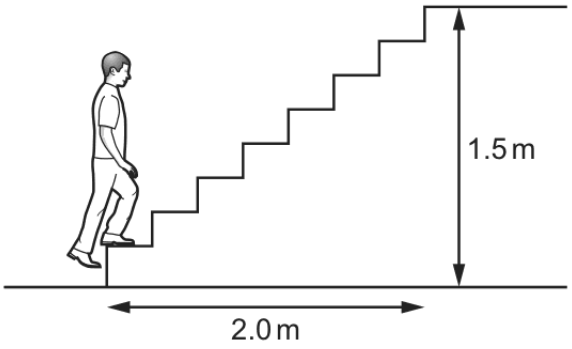
D the velocity in the direction of the force

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 13

13	A	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 7 of 20

15 A student of mass 60 kg climbs some steps. He travels a horizontal distance of 2.0 m and a vertical distance of 1.5 m. The gravitational field strength *g* is 10 N / kg.



(not to scale)

What is the work done against the force of gravity?

A 90 J **B** 120 J **C** 900 J **D** 1200 J

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 15

15	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a large container ship travelling at constant speed in a straight line.

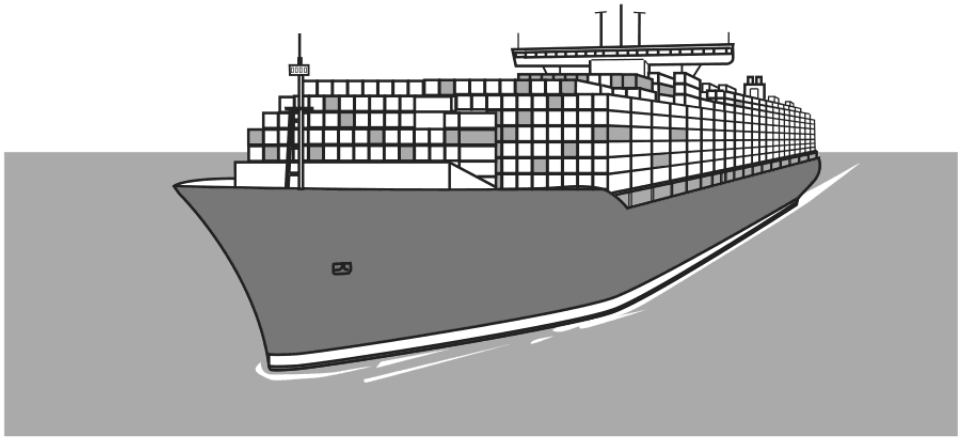


Fig. 9.1

The resistive force acting on the ship is 2.8×10^6 N.

(a) The speed of the ship is 9.7 m/s.

(i) Calculate the work done against the resistive force on the ship in 1.0 s.

work done = ... [2]

(ii) The engines are powered by oil.

State the energy transfer that is taking place when the ship is travelling at constant speed.

.....

..... [2]

(b) The mass of the ship is 2.2×10^8 kg. The engines are switched off and the resistive force causes the ship to decelerate.

(i) Calculate the initial deceleration of the ship.

deceleration = ... [2]

(ii) As the speed of the ship decreases, its deceleration changes.

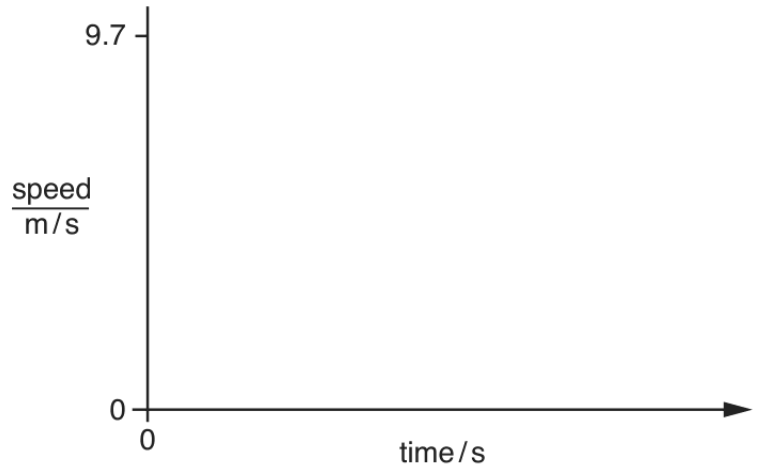
1. Suggest and explain how the deceleration changes.

.....

.....

..... [2]

2. On Fig. 9.2, sketch a possible speed-time graph for the ship as it decelerates to rest.



[2]

Fig. 9.2

3. Explain how the distance travelled by the ship may be determined from the speed-time graph.

.....

.....

..... [2]

(c) When the ship is travelling at a different speed, energy is being supplied to the engines at a rate of 33 MJ/s. The efficiency of the engines is 0.36 (36%).

(i) State a relationship that defines *efficiency*.

.....

..... [1]

(ii) Calculate the rate at which energy is wasted in the engines.

rate at which energy is wasted = ... [2]

Section B

Question	Answer	Marks
9(a)(i)	(WD) = Fx **or** $2.8 \times 10^6 \times 9.7$	**C1**
	2.7×10^7 J	**A1**
9(a)(ii)	from chemical (potential energy)	**B1**
	to thermal (energy) / heat	**B1**
9(b)(i)	(a =)F / m **or** $2.8 \times 10^6 / 2.2 \times 10^8$	**C1**
	0.013 m / s^2	**A1**
9(b)(ii)	**1** deceleration / it decreases	**B1**
	resistive force decreases **or** resistive force depends on speed	**B1**
	2 curve/line from 9.7 to zero **and** gradient negative (**allow** zero at end)	**B1**
	magnitude of gradient decreasing	**B1**
	3 area mentioned	**B1**
	area <u>under</u> line / curve **or** convert cm^2 (of graph paper) to distance **or** in terms of the scales	**B1**
9(c)(i)	(efficiency =) useful energy output / total energy input **or** useful power output / total power input	**B1**
9(c)(ii)	33×0.64 **or** 12 MJ / s	**C1**
	21 MJ / s	**A1**

3 A solid, rectangular concrete block is lying horizontally on flat ground with one of its largest sides in contact with the ground. Fig. 3.1 shows the dimensions of the block.

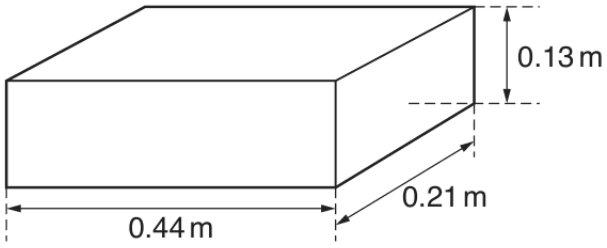


Fig. 3.1

The weight of the block is 240 N.

(a) Calculate the pressure on the ground caused by the block.

pressure = ...[2]

(b) State why the total pressure on the ground underneath the block is larger than the value obtained in **(a)**.

.....

.....[1]

(c) The block is of uniform density.

(i) State the height of the centre of mass of the block above the ground.

height = ...[1]

(ii) The block is rotated about its lower, left-hand edge so that it comes to rest on one of its smallest sides.

This is shown in Fig. 3.2.

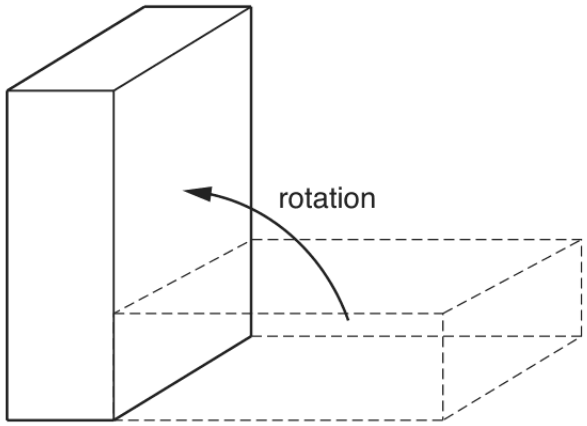


Fig. 3.2

Explain why work is done as the block is rotated.

.....

.....

.....

.....[2]

Question	Answer	Marks
3(a)	(P =) F / A **or** 240 / (0.44 × 0.21) **or** 240 / 0.092(4)	**C1**
	2.6×10^3 Pa	**A1**
3(b)	atmospheric pressure (is acting on the block / ground)	**B1**
3(c)(i)	0.065 m **or** 6.5 cm **or** 65 mm	**B1**
3(c)(ii)	**two** separate approaches **I** / **II**	
	I centre of mass rises	**B1**
	gravitational potential energy gained	**B1**
	or **II** block has weight	**B1**
	force **moves or** something moves in direction of force **or** force makes block move / rotate	**B1**

4 A woman does work as she raises a flag by pulling downwards on a rope. This is shown in Fig. 4.1.

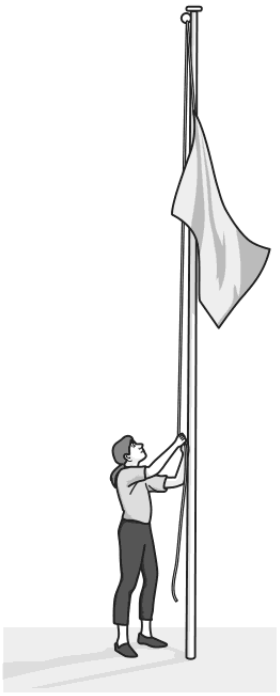


Fig. 4.1

(a) Define what is meant by *work done*.

.....

.....[2]

(b) The mass of the flag is 0.35 kg and the flag is raised through a height of 6.0 m.

The gravitational field strength g is 10 N / kg.

Calculate the increase in gravitational potential energy of the flag.

gravitational potential energy = ...[2]

(c) Suggest and explain a reason why the work done on the rope is larger than the value calculated in **(b)**.

.....

.....

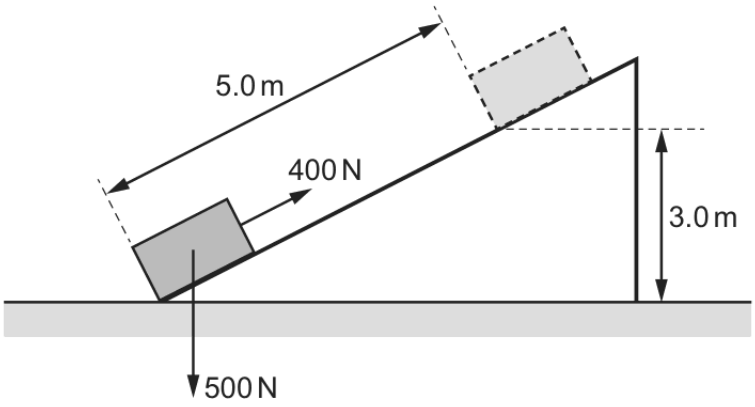
.....

.....[2]

Question	Answer	Marks
4(a)	force × distance (moved)	**C1**
	force × distance <u>moved</u> in direction of force	**A1**
4(b)	mgh in any form numerical or algebraic	**C1**
	21 J	**A1**
4(c)	larger force is needed (by woman / on rope)	**B1**
	(because of either) • friction / heat produced • <u>weight</u> of rope / her arms • flag / rope gains K.E. • flag accelerates / gains speed	**B1**

QUESTION · 5054_s18_qp_12.pdf · source page 7 of 16

15 Work is done when a force of 400 N pulls a crate of weight 500 N at a constant speed along a ramp, as shown.



Part of the work done increases the gravitational potential energy *E* of the crate and the rest is work done *W* against friction.

What are the values of *E* and *W*?

	E / J	*W* / J
A	1500	500
B	1500	2000
C	2000	2500
D	3500	500

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 15

15	A	**1**

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows a satellite travelling at a constant speed in a circular orbit around the Earth.

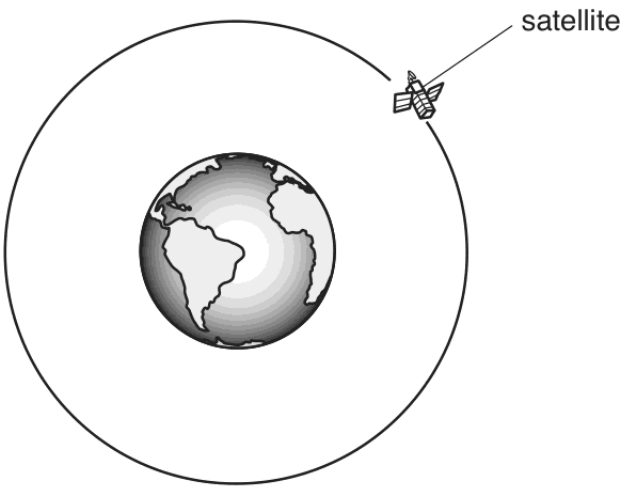


Fig. 1.1 (not to scale)

(a) State how *speed* differs from *velocity*.

.....

..... [1]

(b) As it orbits the Earth, the satellite is experiencing an acceleration.

(i) Explain, in terms of its velocity, why the satellite is accelerating.

.....

.....

..... [2]

(ii) On Fig. 1.1, draw an arrow, starting on the satellite, to show the direction of the satellite's acceleration. [1]

(c) As the satellite orbits the Earth, it experiences a force due to gravitational attraction.

State and explain whether this force does work on the satellite and state whether the energy of the satellite is affected.

.....

.....

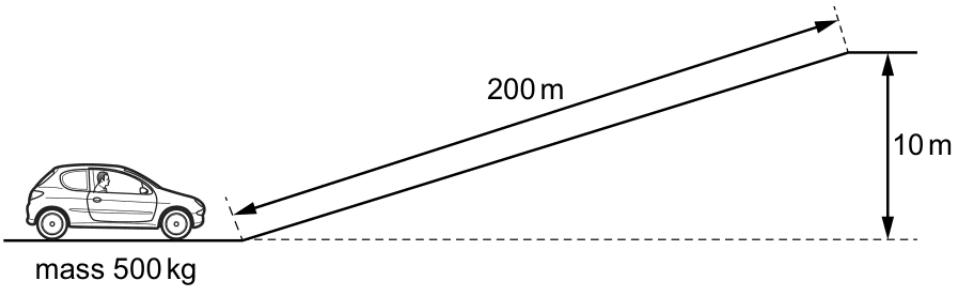
.....

..... [2]

Question	Answer	Marks
1(a)	it / speed does **not** have a direction / is a scalar quantity	**B1**
1(b)(i)	direction (of velocity) changes (as it moves around the Earth)	**B1**
	its velocity changes with time (this is an acceleration)	**B1**
1(b)(ii)	arrow from satellite towards (centre of) Earth	**B1**
1(c)	**no** work done **and** force perpendicular to motion / no movement in direction of force	**B1**
	(kinetic and gravitational potential) energy remains constant / no effect	**B1**

QUESTION · 5054_s19_qp_11.pdf · source page 6 of 20

15 The diagram shows a small car of mass 500 kg approaching a hill. It moves up the hill with constant speed.



The gravitational field strength g is 10 N / kg.

Ignoring friction, how much work is done in moving the car up the hill?

A 5.0×10^3 J **B** 5.0×10^4 J **C** 1.0×10^5 J **D** 1.0×10^6 J

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 15

15	B	**1**

3 A large spring is stretched by an athlete to increase the strength of his arms.

The athlete extends the spring, as shown in Fig. 3.1, and then releases it gently. He extends and releases the spring several times during a period of 60 s.

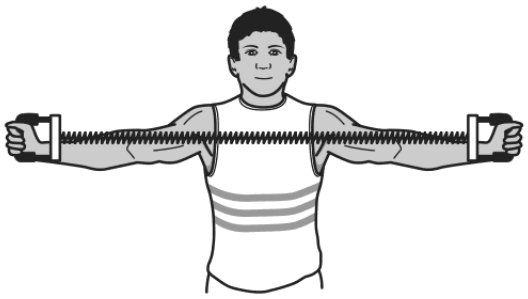


Fig. 3.1

(a) State the form of energy stored in the stretched spring and the form of energy stored in the muscles of the athlete.

form of energy in spring .....

form of energy in muscles .....
[2]

(b) Suggest what happens if he extends the spring beyond the limit of proportionality.

.....

..... [1]

(c) During one extension of the spring, its length increases from 70 cm to 93 cm. The average force exerted by the athlete is 400 N.

(i) Calculate the work done by the athlete in one extension.

work done = ... [2]

(ii) Describe how the athlete can use the value obtained in **(i)** to determine his average power during the 60 s.

.....

.....

.....

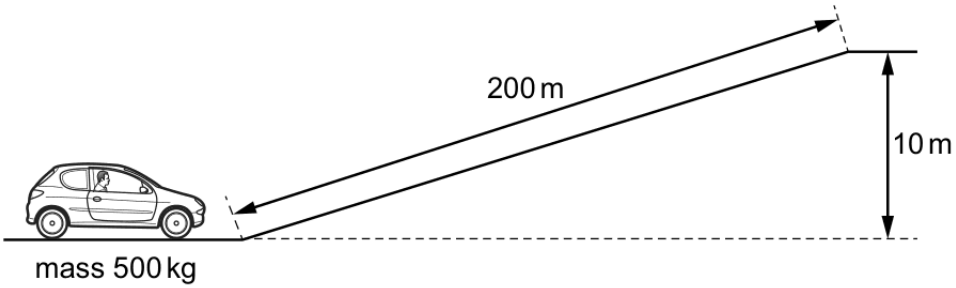
..... [2]

[Total: 7]

Question	Answer	Marks
3(a)	*in spring:* elastic or strain (potential energy)	**B1**
	in muscles: chemical (potential energy)	**B1**
3(b)	spring extends a different distance for the same (extra) load **or** load is no longer proportional to extension **or** becomes easier to pull	**B1**
3(c)(i)	(W =) Fd numerical or algebraic; 400×0.23	**C1**
	92 J	**A1**
3(c)(ii)	measure / determine the no. of times he extends the spring in 60 s	**B1**
	power = (total) work / time explained	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 7 of 16

17 The diagram shows a small car of mass 500 kg approaching a hill. It moves up the hill with constant speed.



The gravitational field strength g is 10 N / kg.

Ignoring friction, how much work is done in moving the car up the hill?

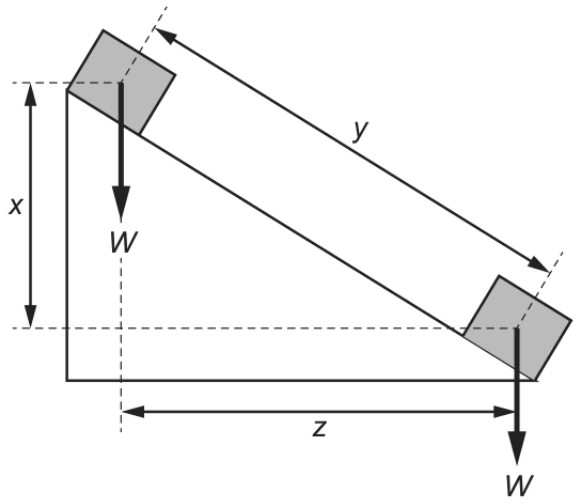
A 5.0×10^3 J **B** 5.0×10^4 J **C** 1.0×10^5 J **D** 1.0×10^6 J

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 17

17	B	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 9 of 20

17 A heavy box of weight W slides down a frictionless slope.



What is the work done by the weight W in moving from top to bottom?

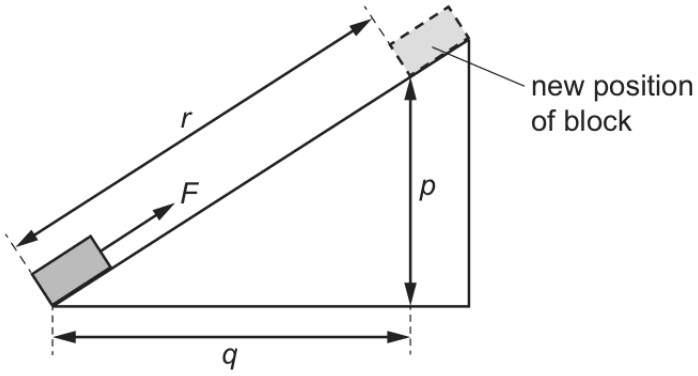
A Wx **B** Wy **C** $W(y - z)$ **D** $W(x + z)$

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 17

17	A	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 7 of 16

19 The diagram shows a block being pulled up a slope by a force *F*.



The block reaches the new position at the top of the slope.

What is the work done by force *F* in moving the block to its new position?

A $F \times r$ **B** $F \times p$ **C** $F(q + p)$ **D** $F(r + p)$

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 19

19	A	**1**

3 Fig. 3.1 shows part of a hydraulic press that is used to compress waste paper into a brick for burning.

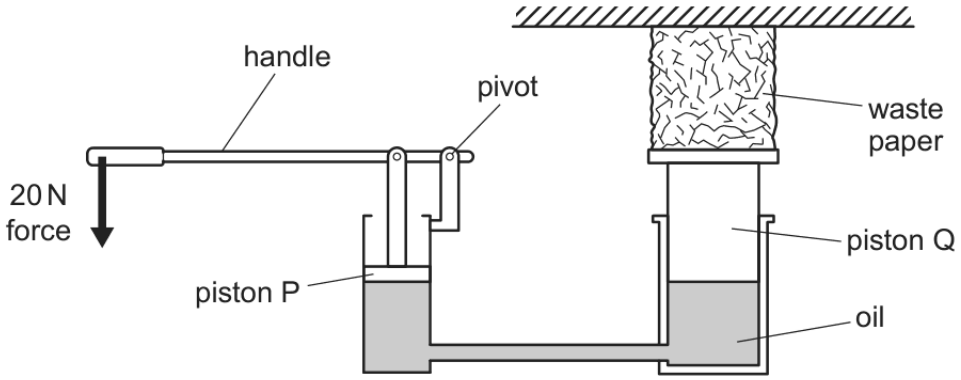


Fig. 3.1

A force of 20 N is exerted downwards on the end of the handle.

(a) The force on the handle creates a moment about the pivot.

(i) Define the *moment of a force*.

.....

.....

.....

..... [2]

(ii) Explain why the force exerted on piston P is greater than the force exerted on the handle.

.....

.....

..... [1]

(b) Explain how the hydraulic press enables a greater force to be exerted on piston Q than is exerted on piston P.

.....

.....

.....

.....

..... [2]

(c) In moving the handle downwards, the 20 N force moves through a distance of 0.60 m and piston Q rises by 0.020 m.

The force exerted by piston Q on the paper is 400 N.

Calculate:

(i) the work done in moving the handle downwards

work = ... [2]

(ii) the efficiency of the hydraulic press.

efficiency = ... [2]

[Total: 9]

MARK SCHEME · 5054_s20_ms_22.pdf · source page 7 of 13 · Question 3

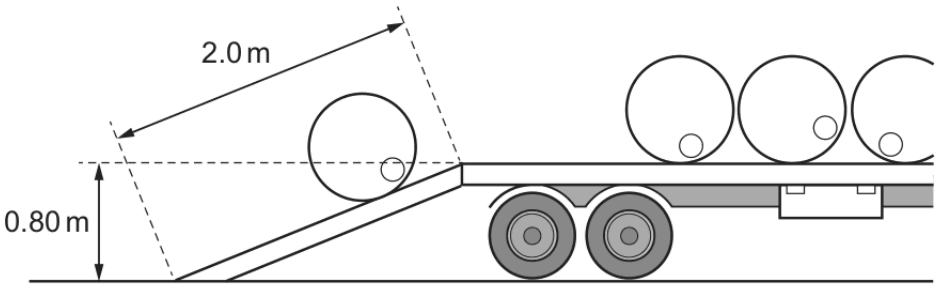
Question	Answer	Marks
3(a)(i)	force × distance	**C1**
	force × perpendicular distance	**A1**
3(a)(ii)	smaller distance to pivot (than from P to pivot)	**B1**

MARK SCHEME · 5054_s20_ms_22.pdf · source page 8 of 13 · Question 3

Question	Answer	Marks
3(b)	same / equal pressure on piston P and piston Q	**B1**
	larger area of piston Q (than piston P)	**B1**
3(c)(i)	(W = Fd) numerical or algebraic	**C1**
	12 J	**A1**
3(c)(ii)	(efficiency) = work output / work input numerical or algebraic, e.g. 400 × 0.02 / 12	**C1**
	0.67 or 67%	**A1**

QUESTION · 5054_w20_qp_11.pdf · source page 8 of 16

18 A workman rolls a barrel of weight 2000 N up a plank of length 2.0 m and onto a lorry. The back of the lorry is 0.80 m above the horizontal surface of the road.



What is the work done on the barrel against gravity?

A 1000 J **B** 1600 J **C** 2500 J **D** 4000 J

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 18

18	B	**1**

3 Fig. 3.1 shows a door and an automatic door-closer viewed from above.

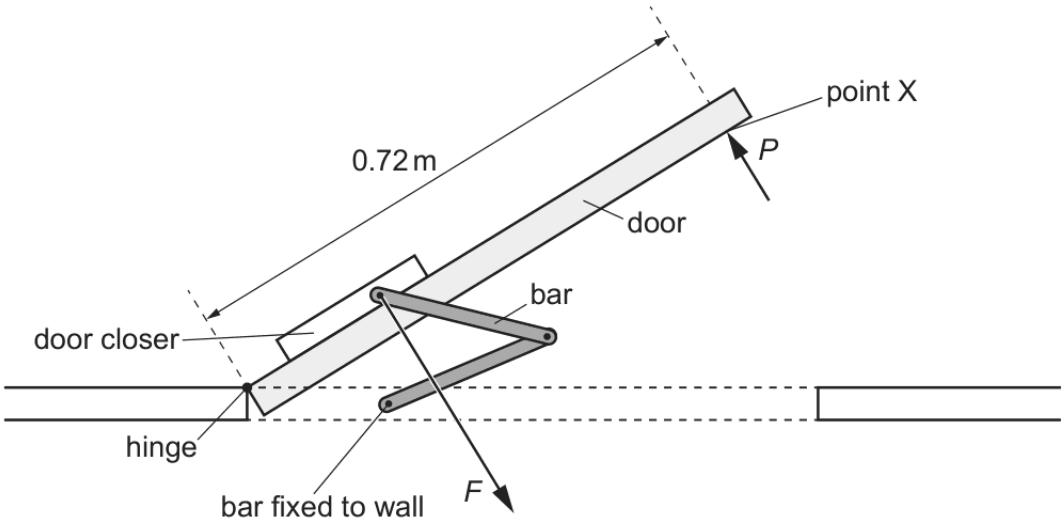


Fig. 3.1

When the door opens and closes, the hinge acts as a pivot.

A girl opens the door by exerting a force *P* at point X.

Force *P* is perpendicular to the surface of the door.

(a) Fig. 3.1 shows that point X is a distance of 0.72 m along the front of the door from the hinge. The force *P* is 25 N.

(i) Calculate the moment of force *P* about the hinge.

moment of force = ... [2]

(ii) The door rotates about the hinge by 90°. The circumference of a circle of radius 0.72 m is 4.5 m.

Calculate the work done on the door by force *P*.

work done = ... [2]

(b) As the door opens, there is a force *F* on the door in the direction shown in Fig. 3.1.

Although force *F* is larger than force *P*, the door rotates about the hinge.

Explain why.

.....

.....

.....

..... [2]

[Total: 6]

MARK SCHEME · 5054_w20_ms_21.pdf · source page 6 of 10 · Question 3

Question	Answer	Marks
3(a)(i)	(Γ =) $Fx_{\perp r}$ **or** 25×0.72	**C1**
	18 N m	**A1**
3(a)(ii)	(WD =) $Fx_{\parallel}$ **or** $F\pi r / 2$ or $25 \times 4.5 / 4$ **or** 25×1.13	**B1**
	28 J	**B1**
3(b)	moment due to force *P* is greater	**B1**
	distance of X to hinge is greater than distance from where force *F* acts	**B1**

QUESTION · 5054_w20_qp_12.pdf · source page 7 of 16

14 A student uses an average force of F to push a box a distance of d across a horizontal table. The mass of the box is m and the gravitational field strength is g.

How is the work done by the student pushing the box calculated?

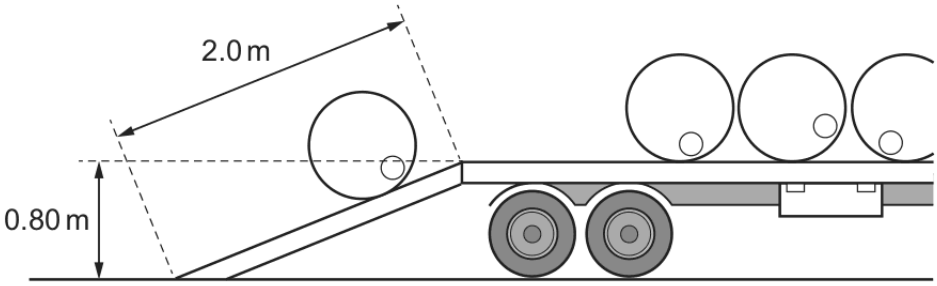
A $F \times d$ **B** $\frac{F \times d}{g}$ **C** $m \times d$ **D** $m \times g \times d$

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 14

14	A	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 7 of 16

15 A workman rolls a barrel of weight 2000 N up a plank of length 2.0 m and onto a lorry. The back of the lorry is 0.80 m above the horizontal surface of the road.



What is the work done on the barrel against gravity?

A 1000 J **B** 1600 J **C** 2500 J **D** 4000 J

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 15

15	B	**1**

2 The total mass of a wheelbarrow and its load is 90 kg. A worker pushes the wheelbarrow and load up a plank of length 2.0 m on to a platform, as shown in Fig. 2.1. The platform is at a height of 0.60 m above the ground.

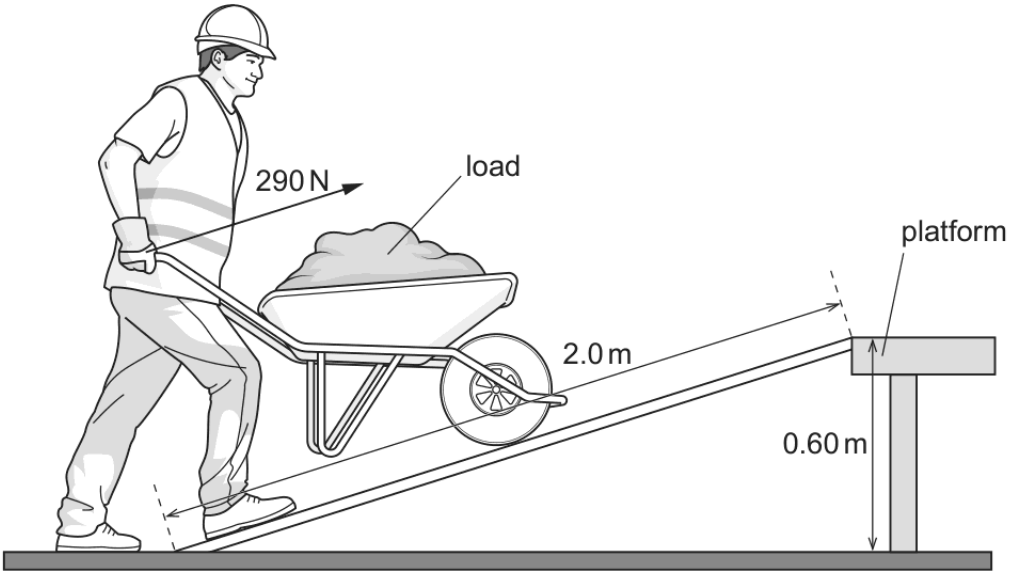


Fig. 2.1

The worker exerts a force of 290 N on the wheelbarrow in the direction in which the wheelbarrow moves.

(a) The gravitational field strength *g* is equal to 10 N / kg.

(i) Calculate the gravitational potential energy gained by the wheelbarrow and its load.

energy = ... [2]

(ii) The worker pushes the wheelbarrow 2.0 m along the plank.

Calculate the work done on the wheelbarrow by the worker.

work done = ... [2]

(iii) Suggest one reason why the answer to **(a)(ii)** is not equal to the answer to **(a)(i)**.

.....

..... [1]

(b) The worker finds that walking up the plank pushing the wheelbarrow is even more inefficient than the answers in **(a)** suggest.

(i) State what is meant by *efficiency*.

.....

..... [1]

(ii) Suggest **one** reason why this method of lifting the load onto the platform is so inefficient.

.....

..... [1]

[Total: 7]

Question	Answer	Marks
2(a)(i)	(g.p.e. =) *mgh* **or** $90 \times 10 \times 0.60$	**C1**
	540 J	**A1**
2(a)(ii)	(W.D. =) *Fx* **or** 290×2.0	**C1**
	580 J	**A1**
2(a)(ii)	friction (between wheel and axle) **or** friction (between tyre / wheel and plank) **or** (work done against) friction	**B1**
2(b)(i)	$\frac{\text{useful energy output}}{\text{(total) energy input}}$ ($\times$ 100%) **or** $\frac{\text{useful power output}}{\text{(total) power input}}$ ($\times$ 100%)	**B1**
2(b)(ii)	wheelbarrow / worker has to be lifted up **ignore** references to friction / air resistance	**B1**

QUESTION · 5054_s21_qp_11.pdf · source page 6 of 16

13 The work done by a force *F* on a body is calculated by multiplying *F* by a quantity *q*.

What is *q*?

A the distance travelled in the direction of the force

B the distance travelled perpendicular to the direction of the force

C the velocity in the direction of the force

D the velocity in the direction perpendicular to the force

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 13

13	A	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 7 of 16

15 The work done by a force *F* on a body is calculated by multiplying *F* by a quantity *q*.

What is *q*?

A the distance travelled in the direction of the force

B the distance travelled perpendicular to the direction of the force

C the velocity in the direction of the force

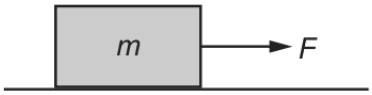
D the velocity in the direction perpendicular to the force

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 15

15	A	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 5 of 20

11 The diagram shows a block of mass *m* pulled in a straight line along a horizontal surface by a force *F*.



The block moves a distance *d* in time *t*. The average speed at which the block moves is *v*.

Which two quantities must be known to calculate the work done?

A *F* and *d** **B** *F* and *m** **C** *F* and *t** **D** *F* and *v*

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 11

11	A	**1**

QUESTION · 5054_w21_qp_12.pdf · source page 6 of 16

12 A 60 kg ice-skater, travelling initially at 3.0 m / s, glides across a horizontal icy surface. The skater brakes and comes to rest after 5.0 m.

The work done against the braking force is the change in the kinetic energy of the skater.

What is the braking force?

A 18 N **B** 27 N **C** 54 N **D** 108 N

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 12

12	C	**1**

6 Isotope X is radioactive. It decays by alpha-particle emission to a stable isotope.

(a) State how a nucleus of X changes when it emits an alpha-particle.

.....

.....

..... [2]

(b) There is a radiation detector in a laboratory where there are no radioactive samples.

The detector is switched on and shows an average count rate of 22 counts / minute.

(i) State why the radiation detector shows a count rate.

.....

..... [1]

(ii) A sample of isotope X is placed 2 cm from the detector and the reading displayed is 8000 counts / minute.

The sample is moved a distance of 10 cm from the detector. The reading returns to an average value of 22 counts / minute.

Explain why the reading returns to the original value.

.....

.....

..... [2]

(c) An alpha-particle passes into a region where there is a magnetic field. In the magnetic field, a force acts on the alpha-particle so that it follows a circular path. Fig. 6.1 shows that the particle passes through point J.

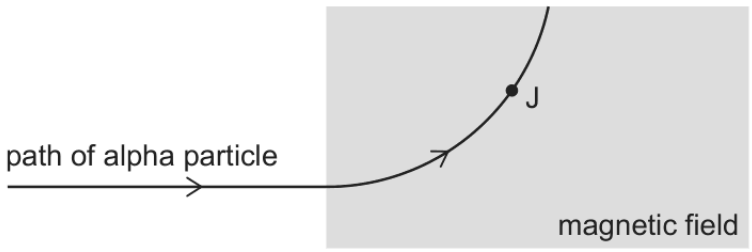


Fig. 6.1

(i) On Fig. 6.1, draw an arrow through point J to show the direction of the force on the alpha-particle at J. [1]

(ii) Determine the direction of the magnetic field and mark a tick in the box (✓) that indicates this direction.

- ☐ to the left
- ☐ to the right
- ☐ towards the top of the page
- ☐ towards the bottom of the page
- ☐ into the page
- ☐ out of the page

[1]

(iii) Explain whether this force does work on the alpha-particle as the particle moves along the circular path.

.....

.....

..... [2]

[Total: 9]

Question	Answer	Marks
6(a)	any **two** from: number of protons decreases by two number of neutrons decreases by two **or** number of nucleons decreases by four	**B2**
6(b)(i)	background radiation (detected) **or** source of background radiation mentioned (e.g. cosmic rays; radiation in the air)	**B1**
6(b)(ii)	alpha-particles are absorbed by the air (between the source and detector)	**B1**
	all the alpha-particles / radiation are absorbed (by the air between the source and detector) **or** alpha-particles travel less than 10 cm (in air)	**B1**
6(c)(i)	arrow through J towards centre of circular path	**B1**
6(c)(ii)	into the page (i.e. fifth box ticked)	**B1**
6(c)(iii)	force perpendicular to direction of motion **or** no component of velocity along line of action of force **or** work done = force × distance moved in direction of motion	**M1**
	no work done	**A1**

QUESTION · 5054_w22_qp_11.pdf · source page 2 of 16

3 A car of mass 1000 kg is travelling down a steep hill. The brakes fail and the driver uses a horizontal sand-filled safety road to stop the car.

The car enters the sand at a speed of 10 m/s and experiences a constant stopping force of 2500 N.

How far does the car travel in the sand before coming to rest?

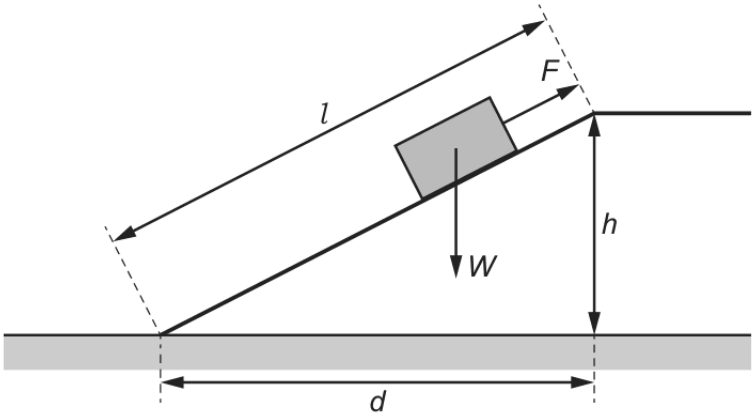
A 2.0 m **B** 4.0 m **C** 20 m **D** 40 m

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 3

3	C	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 7 of 16

15 A constant force F pulls a block of weight W up the slope shown.



How much work is done by F in pulling the block up the slope?

A $F \times h$ **B** $F \times l$ **C** $W \times d$ **D** $W \times l$

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 15

15	B	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 2 of 16

4 A car of mass 1000 kg is travelling down a steep hill. The brakes fail and the driver uses a horizontal sand-filled safety road to stop the car.

The car enters the sand at a speed of 10 m/s and experiences a constant stopping force of 2500 N.

How far does the car travel in the sand before coming to rest?

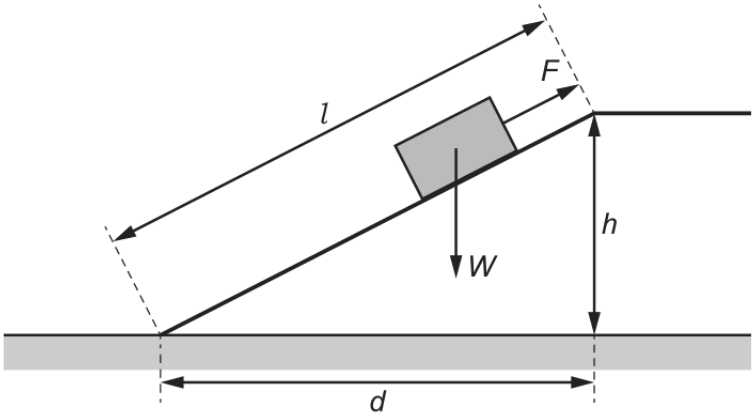
A 2.0 m **B** 4.0 m **C** 20 m **D** 40 m

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 4

4	C	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 7 of 16

16 A constant force *F* pulls a block of weight *W* up the slope shown.



How much work is done by *F* in pulling the block up the slope?

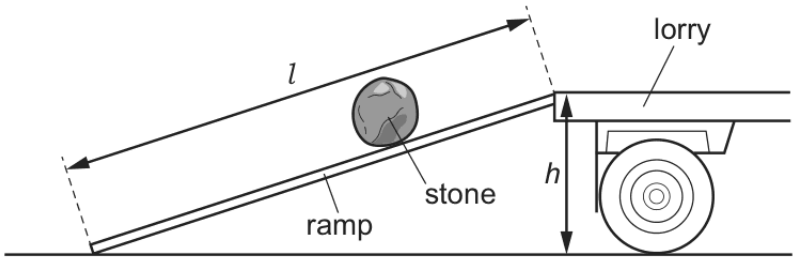
A $F \times h$ **B** $F \times l$ **C** $W \times d$ **D** $W \times l$

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 16

16	B	**1**

QUESTION · 5054_s23_qp_11.pdf · source page 6 of 20

11 A large stone is rolled onto a lorry using a ramp.



The length of the ramp is l. The stone is lifted a height of h.

The mass of the stone is m. The weight of the stone is W.

Which expression is equal to the useful work done on the stone?

A $m \times h$ **B** $W \times h$ **C** $m \times l$ **D** $W \times l$

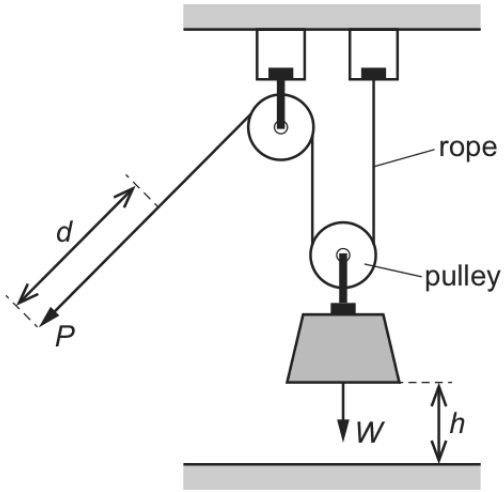
MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 11

11	B	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 6 of 16

12 A worker pulls a rope with a force *P* through a distance *d* in the direction of the force as shown.

This causes a block of weight *W* to move vertically upwards through a height *h*.



How much work is done by the worker?

A *Ph* **B** *Pd* **C** (*P* + *W*)*h* **D** (*P* + *W*)*d*

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 12

12	B	**1**

2 Fig. 2.2 shows a rider on an electric scooter.

Fig. 2.2

The scooter contains a battery and a motor to drive the back wheel.

(a) (i) State the name of the energy store in the battery.

..... [1]

(ii) Describe, in terms of work done, the stages of energy transfer from the energy store in the battery to the kinetic energy of the scooter.

.....

.....

.....

..... [2]

(b) The total mass of the scooter and the rider is 70 kg.

Calculate the total kinetic energy of the rider and scooter when the scooter has a speed of 4.0 m/s.

kinetic energy = ... J [2]

(c) The battery is marked 'energy capacity 0.35 kilowatt-hour (kW h)'.

(i) Define what is meant by a kilowatt-hour.

.....

..... [1]

(ii) The scooter stops working because the battery is totally discharged (flat). This means that there is no more energy stored in the battery.

The battery is then recharged using a 70 W power supply.

Calculate the time taken to fully recharge the battery.

time = ... hours [2]

[Total: 8]

MARK SCHEME · 5054_s23_ms_22.pdf · source page 6 of 12 · Question 2

Question	Answer	Marks
2(a)(i)	chemical (potential energy)	**B1**
2(a)(ii)	electrical work / (electrical) current **or** electrons (made to) move	**B1**
	mechanical) work is done by the motor / by or on the (back) wheel	**B1**
2(b)	(KE =) ½ mv^2 in any form	**C1**
	560 (J)	**A1**
2(c)(i)	(energy provided if power is) 1 kW for 1 hour / 1000 W for 1 hour	**B1**

MARK SCHEME · 5054_s23_ms_22.pdf · source page 7 of 12 · Question 2

Question	Answer	Marks
2(c)(ii)	*(t =) E / P* **or** 0.35 / 0.07 **or** 350 / 70	**C1**
	5(.0 hours)	**A1**

2 A lorry (truck) is at rest on a straight, horizontal road.

A constant driving force of 5.4×10^4 N acts on the lorry and it accelerates forwards.

(a) Define what is meant by 'acceleration'.

.....

.....

..... [2]

(b) As the lorry moves forwards, work is done on it by the driving force.

(i) State the equation that defines work done.

Refer to the direction of the force exerted.

.....

..... [1]

(ii) Calculate the work done on the lorry by the driving force when the lorry travels a distance of 280 m.

work done = ... J [2]

(iii) At one point on the road, the kinetic energy of the lorry is 3.2×10^6 J and its speed is 12 m/s.

Calculate the mass of the lorry.

mass = ... kg [3]

[Total: 8]

MARK SCHEME · 5054_w23_ms_21.pdf · source page 6 of 11 · Question 2

Question	Answer	Marks
2(a)	change / increase of velocity (with time)	**B1**
	change / increase of velocity per unit time **or** rate of change / increase of velocity	**B1**
2(b)(i)	force × distance moved in the direction of the force / forwards	**B1**
2(b)(ii)	(work done =) $5.4 \times 10^4 \times 280$	**C1**
	1.5×10^7 (J)	**A1**

MARK SCHEME · 5054_w23_ms_21.pdf · source page 7 of 11 · Question 2

Question	Answer	Marks
2(b)(iii)	$E_k = \frac{1}{2}mv^2$ **or** (m =) $2 \times E_k / v^2$	**C1**
	(m =) $2 \times 3.2 \times 10^6 / 12^2$ **or** $2 \times 3.2 \times 10^6 / 144$	**C1**
	4.4×10^4 (kg)	**A1**

QUESTION · 5054_w23_qp_12.pdf · source page 5 of 20

11 A box with mass 4.0 kg is pulled from rest along a horizontal floor by a constant force *F*.

A constant frictional force of 1.0 N acts on the box.

When the box has travelled 1.0 m its velocity is 1.5 m / s.

The kinetic energy of the box is equal to the work done on the box.

What is force *F*?

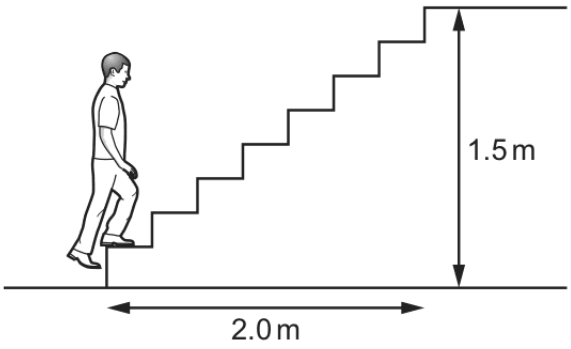
A 2.0 N **B** 3.5 N **C** 4.0 N **D** 5.5 N

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 11

11	D	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 6 of 16

11 A student of mass 60 kg climbs some steps. He travels a horizontal distance of 2.0 m and a vertical distance of 1.5 m.



NOT TO SCALE

What is the work done against the force of gravity?

A 88 J **B** 118 J **C** 880 J **D** 1200 J

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 11

11	C	**1**

1 A ball is released from rest at point A and moves along a smooth track ABCDE as shown in Fig. 1.1. The ball is shown at point A and as it passes point B.

The ball is always in contact with the track and air resistance is negligible.

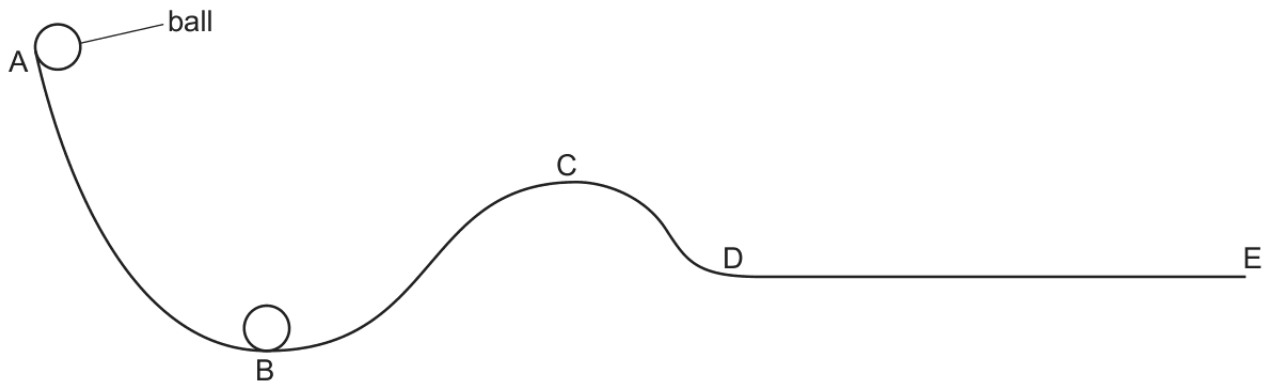


Fig. 1.1

Fig. 1.2 shows the distance−time graph for the ball as it moves from A to E.

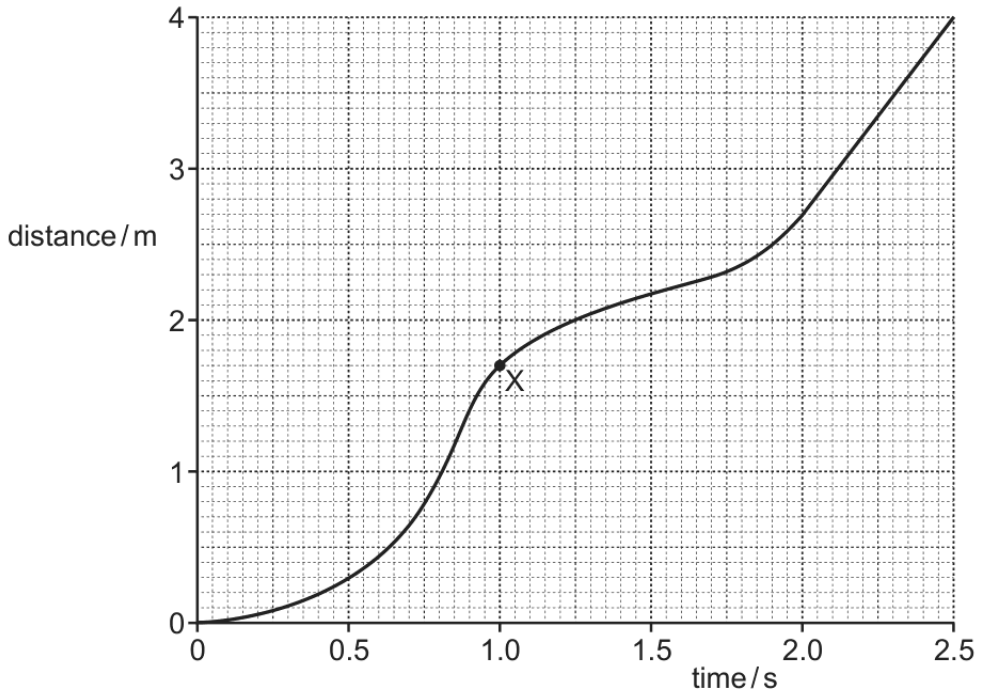


Fig. 1.2

(a) (i) On Fig. 1.2, mark:

- with the letter P **one** point where the ball accelerates
- with the letter Q **one** point where the ball has constant speed.

[2]

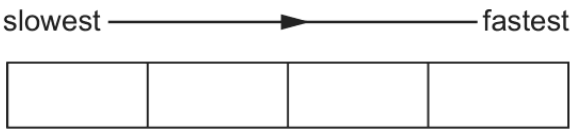
(ii) Determine the speed of the ball at point X on Fig. 1.2.

Show your working.

speed = ... m/s [3]

(b) The speeds of the ball at A, B, C and D are v_A, v_B, v_C and v_D respectively.

Arrange these four speeds from slowest to fastest.



[1]

(c) Work is done to transfer energy between energy stores as the ball moves from A to B.

Name the force involved in the work done and describe the energy transfer.

.....

.....

.....

..... [3]

(d) The track at point B is circular in shape.

On Fig. 1.1, draw an arrow to show the direction of the resultant force on the ball at point B.

[1]

[Total: 10]

Question	Answer	Marks
1(a)(i)	P marked in region t = 0–0.8 s **or** in region 1.6–2.0 s	**B1**
	Q marked between 2.0 and 2.5 s	**B1**
1(a)(ii)	tangent drawn at X	**B1**
	(speed =) d/t in any form numerical or algebraic	**C1**
	1.9–2.4 m/s	**A1**
1(b)	$v_A < v_C < v_D < v_B$	**B1**
1(c)	(work done by) gravity / weight / gravitational attraction	**B1**
	from gravitational potential energy (store)	**B1**
	to kinetic energy (store)	**B1**
1(d)	arrow upwards at D	**B1**

1 A trolley of mass 0.20 kg is at rest on a frictionless surface.

A block of wood is attached to the trolley.

The volume of the block of wood is 0.0012 m^3, and the density of the wood is 650 kg/m^3.

(a) Calculate the combined mass of the trolley and block.

combined mass = ... kg [2]

(b) Fig. 1.1 shows that a student attaches a spring of spring constant 14 N/m to the front of the trolley.

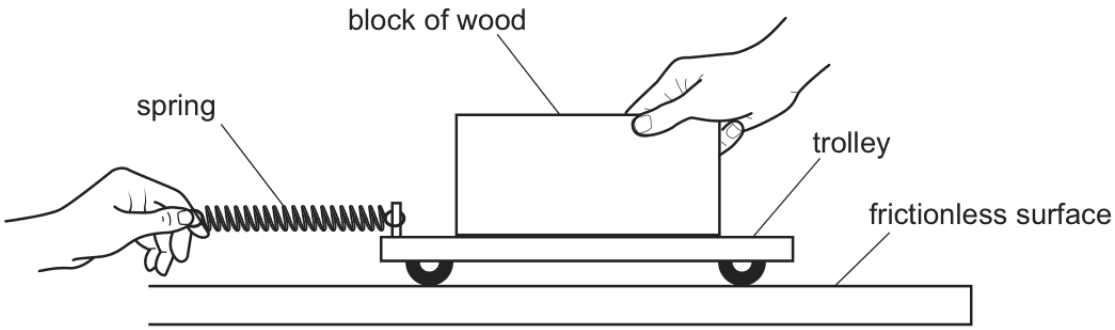


Fig. 1.1

The student holds the trolley and block and stretches the spring. He releases the trolley and block. The trolley and block accelerate from rest.

As the trolley and block accelerate, the student keeps the extension of the spring at 0.035 m.

(i) Calculate the force exerted by the spring on the trolley.

force = ... N [2]

(ii) The trolley and block are pulled a distance of 0.86 m by the spring.

Calculate the work done on the trolley.

work done = ... J [2]

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(iii) Explain why the power transferred to the trolley and block increases as the speed increases.

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
1(a)	($m =$) ρV **or** 650×0.0012 **or** 0.78	**C1**
	(combined mass =) 0.98 (kg)	**A1**
1(b)(i)	($F =$) kx **or** 14×0.035	**C1**
	0.49 (N)	**A1**
1(b)(ii)	($W =$) Fx **or** 0.49×0.86	**C1**
	0.42 (J)	**A1**
1(b)(iii)	the distance moved / work done (in one second / unit time / a given time) increases (as the speed increases)	**B1**
	the work done in one second / unit time increases	**B1**

QUESTION · 5054_w24_qp_12.pdf · source page 5 of 16

10 A rocket is launched vertically upwards by a constant force from its engines.

The data for the launch is shown.

weight of rocket	1.9 MN
average air resistance	0.10 MN
force from engine	3.5 MN

How much of the work done on the rocket is used to increase its speed in the first 100 m?

A 150 MJ **B** 170 MJ **C** 350 MJ **D** 550 MJ

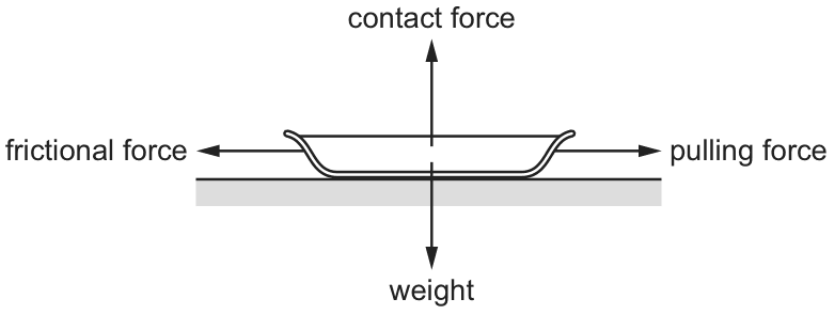
MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 10

10	A	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 5 of 20

9 An object pulled along horizontal ground accelerates to the right.

The diagram shows the directions of the forces acting on the object.



Which force is multiplied by the distance moved to calculate the work done to increase the kinetic energy store?

A contact force

B frictional force

C pulling force

D weight

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 9

9	C	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 5 of 16

10 Which row gives the definition of work done and the unit of work done?

	definition of work done	unit of work done
A	the energy transferred per unit time	J
B	the energy transferred per unit time	N / m
C	the force multiplied by the distance moved in the direction of the force	J
D	the force multiplied by the distance moved in the direction of the force	N / m

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 10

10	C	**1**

2 Fig. 2.1 shows a fork-lift truck used to lift a load.

Fig. 2.1

When lifting the load, the electric motor on the fork-lift truck has a useful output power of 600 W.

(a) Define 'power'.

.....

..... [1]

(b) The efficiency of the motor is 70%.

Calculate the input power to the motor.

input power = ...W [2]

(c) The fork-lift truck lifts a load with mass of 50 kg through a vertical distance of 2.3 m.

(i) Calculate the work done in lifting the load.

work done = ... J [2]

(ii) Calculate the time taken to lift the load.

time taken = ... s [2]

(d) Describe how energy is transferred to thermal energy as the load is lifted by the fork-lift truck.

.....

..... [1]

[Total: 8]

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Question	Answer	Marks
2(a)	work done or energy per unit time	**B1**
2(b)	(input power =) power output / efficiency in any form	**C1**
	860 (W)	**A1**
2(c)(i)	(work =) force × distance **or** *mgh*	**C1**
	1100 (J)	**A1**
2(c)(ii)	(time =) work ÷ (output) power	**C1**
	1.9 or 1.8 s	**A1**
2(d)	*Any one of*: • friction / air resistance • electrical heating	**B1**

2 In a factory, an electric motor is used to lift the boxes shown in Fig. 2.1.

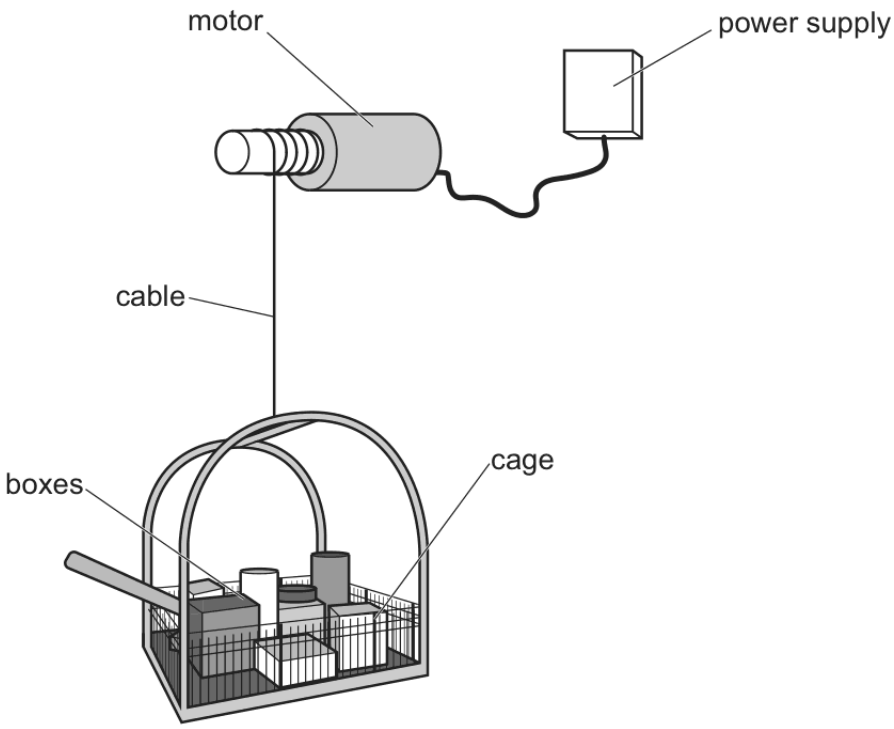

Fig. 2.1

The boxes are placed in a cage and the motor winds a cable around a cylinder to lift the cage and boxes.

The total mass of the boxes is 37 kg and they are lifted vertically through 13 m to a storage area.

(a) Calculate:

(i) the total weight of the boxes

weight = ... N [1]

(ii) the useful work done in lifting the boxes to the storage area.

useful work done = ... J [2]

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(b) The e.m.f. of the power supply for the motor is 230 V. The current in the motor is 5.6 A.

It takes 11 s for the motor to lift the cage and boxes to the storage area.

Calculate:

(i) the total energy transferred electrically to the motor in 11 s

energy = ... J [2]

(ii) the percentage efficiency of the lifting process.

efficiency = ...% [1]

(c) Some of the energy transferred by the motor during the lifting process is not transferred usefully.

State **two** reasons why the efficiency of the lifting process is less than 100%.

1 .....

.....

2 .....

.....

[2]

[Total: 8]

MARK SCHEME · 5054_w25_ms_21.pdf · source page 8 of 14 · Question 2

Question	Answer	Marks
2(a)(i)	360 (N)	**B1**
2(a)(ii)	(work done =) Fx **or** 360×13	**C1**
	4700 (J)	**A1**
2(b)(i)	(energy =) VIt **or** $230 \times 5.6 \times 11$	**C1**
	1.4×10^4 (J)	**A1**
2(b)(ii)	0.33 **or** 33%	**B1**

MARK SCHEME · 5054_w25_ms_21.pdf · source page 9 of 14 · Question 2

Question	Answer	Marks
2(c)	any **two** from: (work done / GPE) to lift the cage / cable (energy transferred) thermally in motor (coils) (work done) overcoming friction / air resistance	**B2**

2 A spherical container is used to carry measuring instruments to the bottom of a lake.

Fig. 2.1 shows the container held at rest just below the water surface of a lake.

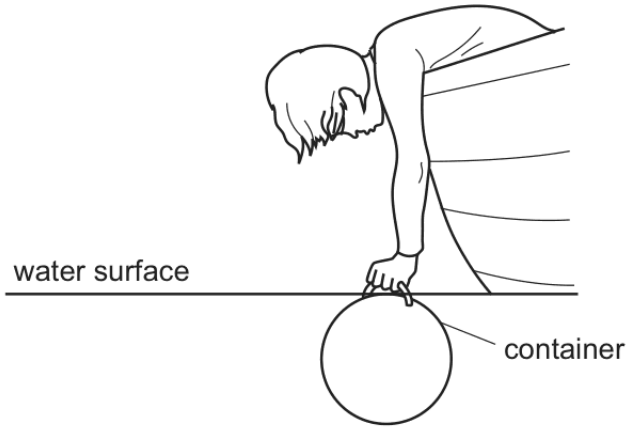

Fig. 2.1 (not to scale)

The container is released and falls through the water to the bottom of the lake.

(a) The container reaches terminal velocity before it hits the bottom of the lake.

Explain, in terms of the forces on the container, the motion of the container through the water until it reaches terminal velocity.

.....

.....

.....

.....

.....

..... [3]

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(b) The container falls through a vertical distance of 32 m before it hits the bottom of the lake.

The container and instruments have a total mass of 74 kg.

(i) Calculate the energy transferred from the gravitational potential energy store of the container.

energy transferred = ... J [2]

(ii) State the principle of conservation of energy.

.....

.....

..... [2]

(iii) Work is done to transfer energy from the gravitational potential energy store as the container falls at terminal velocity.

Describe, in terms of the work done, how the energy is transferred from the gravitational potential energy store to **one** other energy store in the water.

.....

.....

..... [2]

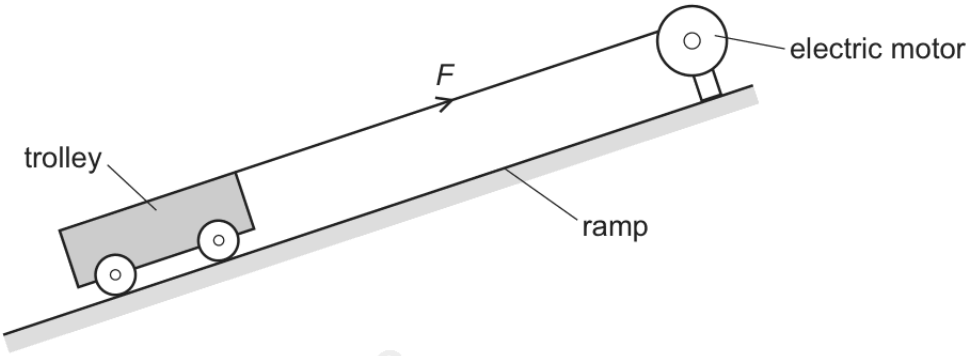
[Total: 9]

Question	Answer	Marks
2(a)	(initially its weight causes) acceleration **or** speeds up **or** gains kinetic energy	**B1**
	friction / drag / resistance / resistive force / viscous force increases (as the speed increases) **or** resultant force decreases	**B1**
	acceleration decreases to zero **or** friction / drag / resistance / resistive force / viscous force is (eventually) equal to weight **or** forces balance **or** resultant force = 0	**B1**
2(b)(i)	(ΔE_P =) $mg\Delta h$ **or** $74.0 \times 9.8 \times 32$	**C1**
	2.3×10^4 (J)	**A1**
2(b)(ii)	energy can be neither created **and** nor destroyed	**B1**
	energy can be transferred from one energy store to another	**B1**
2(b)(iii)	(energy transferred) by doing work against friction / drag / resistance / resistive force	**B1**
	to internal (energy) store	**B1**

QUESTION · 5054_s26_qp_12.pdf · source page 6 of 16

11 An electric motor produces a force *F* to pull a trolley up a ramp.

There is no friction or air resistance acting.



Which quantity is equal to the work done by the force *F*?

A the acceleration of the trolley

B the energy given to the trolley

C the power of the motor

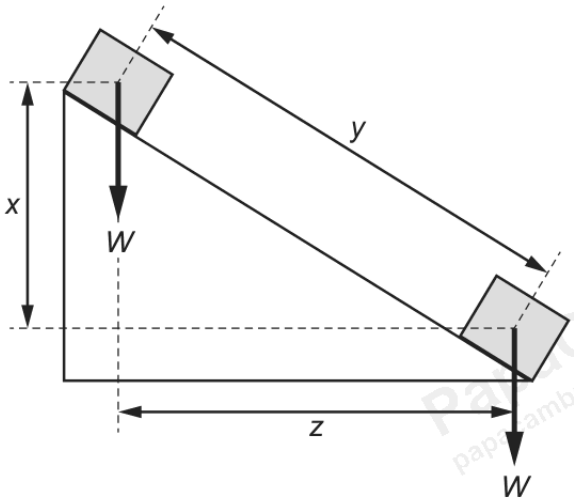
D the weight of the trolley

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 11

11	B	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 6 of 16

12 A heavy box of weight W slides down a frictionless slope.



What is the work done by the weight W in moving from top to bottom?

A Wx **B** Wy **C** $W(y - z)$ **D** $W(x + z)$

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 12

12	A	**1**

Chapter 1 Motion, forces and energy

1.7 Energy, work and power

1.7.3 Energy resources

44 entries: 27 Paper 1 questions, 17 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 6 of 20

13 Which represents the main energy changes that take place in a coal-fired power station?

A chemical → heat → kinetic → electrical

B chemical → heat → light → electrical

C chemical → kinetic → electrical → potential

D kinetic → heat → light → electrical

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 13

13	**A**

QUESTION · 5054_w04_qp_1.pdf · source page 6 of 20

11 Hydroelectric, tidal and fossil fuels are three sources of energy.

Which of these are renewable energy sources?

	hydroelectric	tidal	fossil fuels
A	no	yes	yes
B	no	no	yes
C	yes	no	no
D	yes	yes	no

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 11

11 **D**

For Examiner's Use

4 Fig. 4.1 shows a hydroelectric power station. Water from the lake is used to produce electricity in the turbine house.

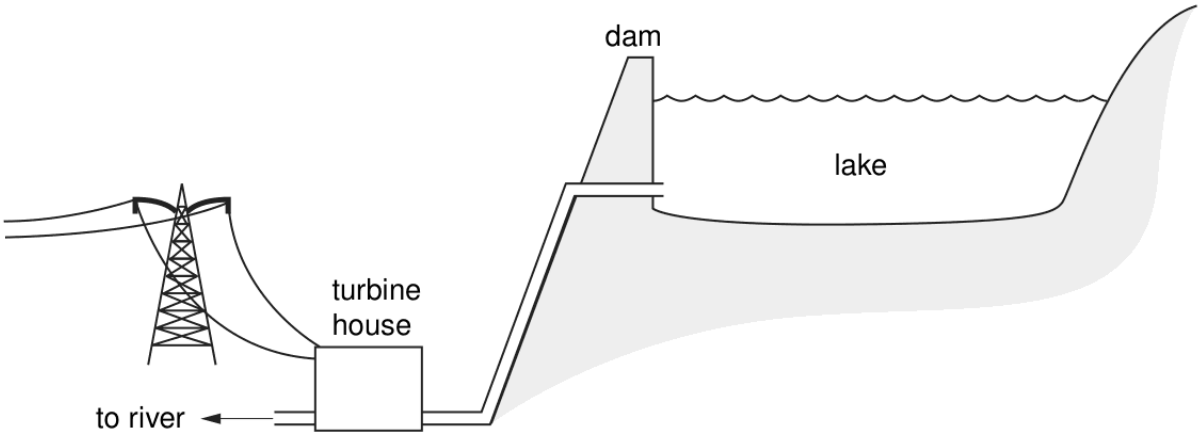


Fig. 4.1

(a) State where the water in Fig. 4.1 has the least potential energy.

..... [1]

(b) In 30 minutes, the water loses 5.0×10^9 J of energy and 4.5×10^9 J of electrical energy is produced in the turbine house.

(i) Calculate the efficiency of the energy conversion.

efficiency = ... [1]

(ii) Calculate, in watts, the electrical power output from the turbine house.

power = ... W [3]

(c) Some power stations burn coal to produce the same electrical power output. State one advantage of the hydroelectric power station.

.....

.....

..... [1]

(d) State one harmful effect that the hydroelectric power station may have on the environment.

.....

..... [1]

4	**(a)**	in river/(emerging from or entering) turbine house	**B1**
	(b) **(i)**	0.9 **or** 90% **or** 0.47 **or** 47% (penalise unit error)	**B1**
	(ii)	P = E/t in symbols **or** any energy/any time	**C1**
		30 x 60 **or** 1800 seen	**C1**
		2.5×10^6 (W)	
		(150 or 2.78MW score 2/3)	**A1**
	(c)	any sensible suggestion e.g. no costs for water/energy supply **or** less pollution (accept coal produces smoke/dust/harmful gases/CO_2 **or** no need to transport coal **or** renewable **or** rapid response to power demand **or** less heat produced/more efficient	**B1**
	(d)	any sensible suggestion e.g. flooding **or** fish unable to pass **or** turbines kill fish **or** destroy habitats **or** less land **or** uses up large space **or** fells trees **or** unsightly/destroys scenery **or** lake/river silt up **or** more rain/evaporation	**B1** **7**

For Examiner's Use

3 One type of renewable energy source is shown in Fig. 3.1.

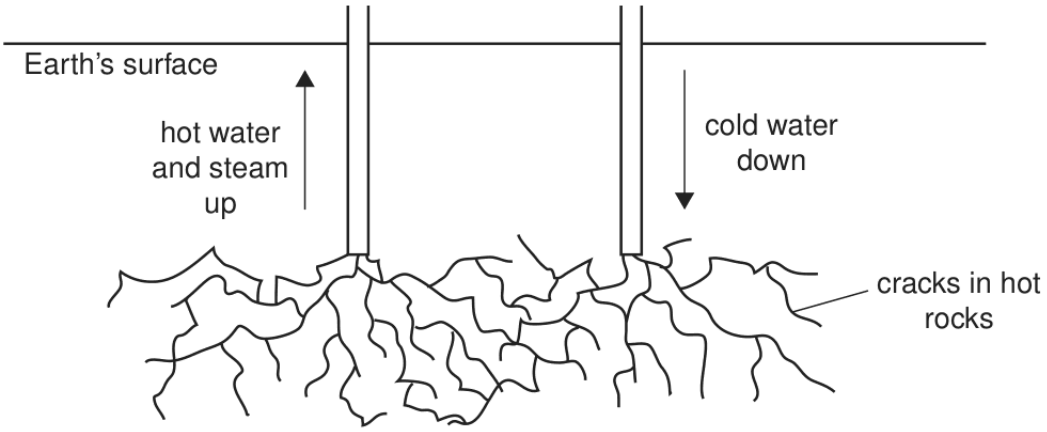


Fig. 3.1

(a) (i) State the name of the renewable energy source shown in Fig. 3.1.

..... [1]

(ii) State what is meant by a *renewable* energy source.

.....

.....

..... [1]

(b) 1000 kg of cold water at a temperature of 20 °C is pumped down to the hot rocks. 100 kg of water returns as steam and 900 kg as hot water, both at a temperature of 100 °C.

The specific heat capacity of water is 4200 J/(kg °C) and the specific latent heat of vaporisation of water is 2.3×10^6 J/kg.

Calculate

(i) the energy needed to heat 1000 kg of water from 20 °C to 100 °C,

energy = ... [2]

(ii) the energy needed to produce 100 kg of steam from water that is already at 100 °C.

energy = ... [2]

3	**(a)**	**(i)**	geothermal		B1
		(ii)	will not run out **or** infinite **or** being replaced	(**not** can be used again/recycled)	B1
	(b)	**(i)**	(E =) mcT **or** 1000 x 4200 x 80 **or** whole equation rearranged		C1
			3.36×10^8 J		A1
		(ii)	(E=) mL **or** 100 x 2.3 x 10^6 **or** whole equation rearranged		C1
			2.3×10^8 J		A1 [6]

Section A

For Examiner's Use

Answer **all** the questions in this section.

1 Fig. 1.1 is a block diagram of a power station.

The four boxes represent different parts of the power station. The first box is labelled boiler.

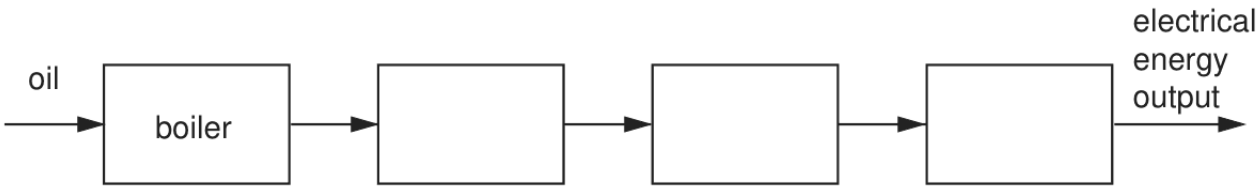


Fig. 1.1

Each of the other three boxes should contain **one** of the labels from the following list.

generator, motor, transformer, turbine, solar panel

(a) On Fig. 1.1, label the boxes using words from the list. [2]

(b) State **one** environmental problem caused by burning oil to produce electricity.

.....

..... [1]

(c) Oil is a non-renewable energy source.

(i) State why oil is described as a non-renewable energy source.

.....

.....

..... [1]

(ii) State **one** renewable energy source.

.....

..... [1]

Section A

1 **(a)** turbine in first box **or** transformer in third box C1
turbine, generator, transformer A1

(b) pollution (e.g. smoke, fumes, **toxic** gases e.g. CO, SO_2 **not** ozone layer affected), global warming, greenhouse effect, acid rain B1

(c) **(i)** cannot be replaced, not being renewed/made, will run out, many years to form, finite (**not** cannot be used again/reused/recycled) B1

(ii) solar/Sun, wind, tidal, geothermal, biomass, hydro-electric, wave B1

[Total: 5]

3 Fig. 3.1 shows part of a small hydroelectric system that generates electricity from moving water.

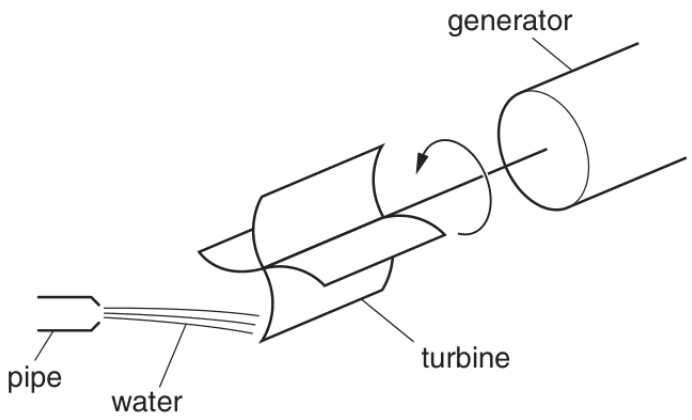


Fig. 3.1

(a) Every minute, water with kinetic energy 14 000 J emerges from the pipe. The water turns a turbine that is connected to the generator.
Half of the kinetic energy of the water is given to the generator.

(i) Calculate the power input to the generator.

power = ... [3]

(ii) Suggest a reason why only some of the kinetic energy of the water is given to the generator.

.....

..... [1]

(b) Hydroelectric systems use sources of renewable energy.

(i) State what is meant by *renewable energy*.

.....

..... [1]

(ii) State one source of **non**-renewable energy.

..... [1]

3	**(a)**	**(i)**	7000 (J) seen **or** 50% **used** somewhere	C1	
			(*P* =) *E*/*t* // 14000/*t* // 7000/*t* seen // 7000 J/minute // 420 000 J/hour	C1	
			120 W	A1	
		(ii)	water after hitting turbine still moves // has KE/energy/velocity // energy lost due to friction // friction **and** location // heat/internal energy **and** location // water misses turbine	B1	
	(b)	**(i)**	can be replaced/made // will not run out	B1	
		(ii)	coal, oil, gas, peat, nuclear, uranium (**not** solar)	B1	[6]

QUESTION · 5054_w12_qp_11.pdf · source page 6 of 16

13 Which is a non-renewable source of energy?

A a battery in a watch

B a hydroelectric power station

C a solar panel

D a wind turbine

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 13

13	**A**

QUESTION · 5054_w12_qp_12.pdf · source page 6 of 16

13 Hydroelectric, tidal and fossil fuel power stations generate electrical energy.

Do these use renewable sources of energy?

	hydroelectric	tidal	fossil fuel
A	no	yes	yes
B	no	no	yes
C	yes	no	no
D	yes	yes	no

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 13

13 **D**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a conveyor belt carrying suitcases into an aeroplane.

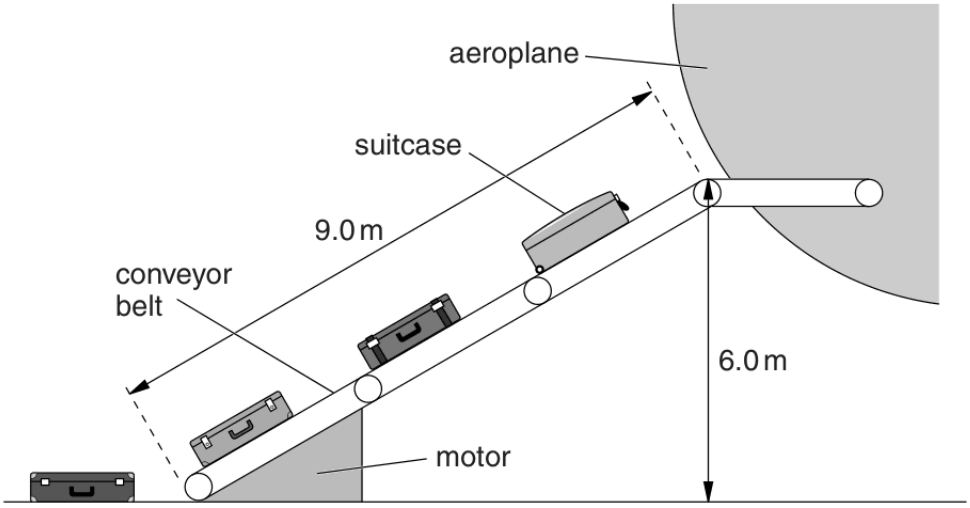


Fig. 9.1 (not to scale)

An electric motor drives the conveyor belt.

(a) A suitcase of mass 20 kg is lifted from the ground into the aeroplane.

(i) Explain what is meant by *mass* and by *weight*.

.....

.....

.....[2]

(ii) The gravitational field strength g is 10 N/kg.
Calculate the increase in the gravitational potential energy of the suitcase.

increase in potential energy = ...[2]

(iii) The suitcase takes 12 s to travel 9.0 m along the conveyor belt.
Calculate the kinetic energy of the suitcase.

kinetic energy = ...[3]

For Examiner's Use

(iv) The electric motor is powered by an input voltage of 230 V and a current of 1.5 A. Calculate the electrical energy input to the motor in 12 s.

energy = ...[2]

(v) **1.** State the principle of the conservation of energy.

.....

.....

.....[2]

2. Explain how this principle applies to the lifting of the suitcases into the aeroplane.

.....

.....

.....[1]

(b) Some of the electrical energy for the motor comes from renewable energy sources.

(i) Explain what is meant by a *renewable energy source*.

.....

.....

.....[1]

(ii) State two renewable energy sources.

1. .....

2. .....
[2]

Section B

9 (a) (i) mass is the amount of matter/substance **or** to resist (change in) motion **or** (measurement of)inertia B1
weight is the pull/force of gravity **or** pull of Earth B1
accept *mg* where *g* stated as 10 (N/kg) or grav. field strength or acc. due to gravity
ignore *mg* where *g* is gravity or grav. force or undefined

(ii) *mgh* algebraic or numerical C1
1200 J A1

(iii) ½ mv^2 algebraic or numerical C1
(speed =) 9/12 **or** 0.75 seen A1
5.6(25) J A1

(iv) (*E*=) *VIt* algebraic or numerical C1
4100 J **or** 4140 J A1

(v) **1.** energy can neither be created or destroyed/lost B1
(but) may change form / be transferred B1

2. electrical energy changes to P.E. (and K.E. and heat/work against friction; **ign**. mechanical energy) B1 [12]

(b) (i) will not run out **or** infinite **or** being replaced (**allow** does not finish/always available)
(**ign.** cannot be reused/recycled) B1

(ii) wind, tidal, solar/Sun, geothermal, hydroelectric, biomass, waves, wood (**not** nuclear) B2 [3]
(**allow** biogas/biofuel e.g. cane into petrol, dung into gas etc.) (**ign.** tidal waves)

[Total: 15]

QUESTION · 5054_w13_qp_11.pdf · source page 6 of 20

12 Which energy resource comes from hot rocks beneath the Earth's surface?

A geothermal energy

B hydroelectric energy

C solar energy

D wind energy

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 12

12	**A**

For Examiner's Use

10 Fig. 10.1 shows an electric train of mass 5.5×10^5 kg.

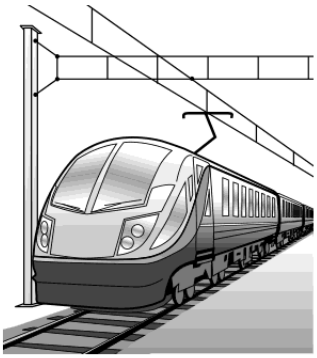


Fig. 10.1

The train is initially at rest. The electric motor exerts a constant force and the train accelerates. Its acceleration decreases until the train reaches a speed of 40 m/s.

The train then continues at this constant speed.

(a) (i) On the axes in Fig. 10.2, sketch a speed-time graph for the train.

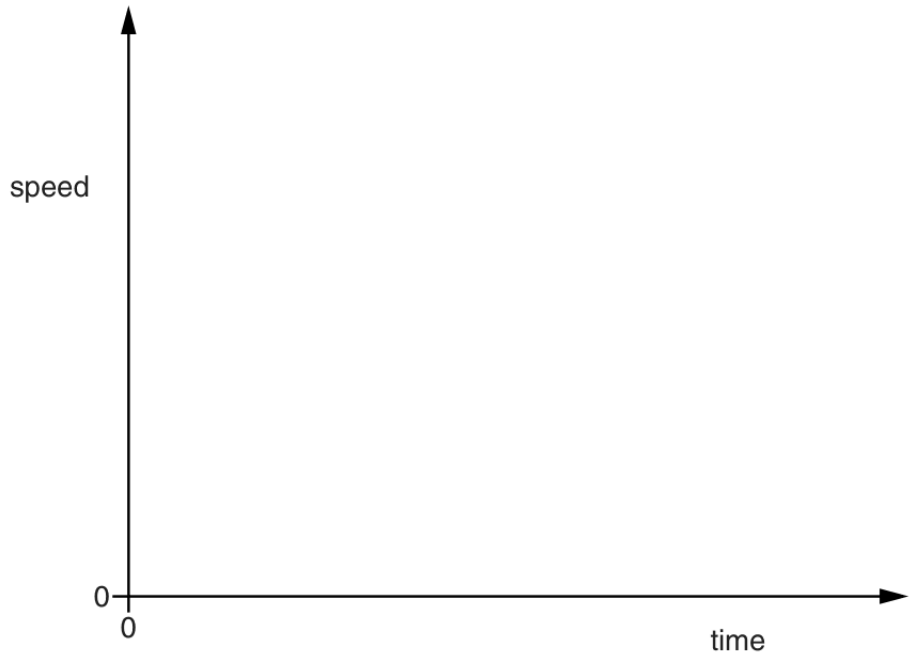


Fig. 10.2

[3]

(ii) Describe how the speed-time graph for the train is used to determine the distance travelled by the train when it is accelerating.

.....

.....

.....[2]

For Examiner's Use

(b) Explain, in terms of the forces acting,

(i) why the acceleration of the train decreases,

.....

.....

.....[2]

(ii) why the train eventually reaches constant speed.

.....

.....[1]

(c) **(i)** Calculate the kinetic energy of the train when it is travelling at a speed of 40 m/s.

kinetic energy = ...[3]

(ii) As the train accelerates to 40 m/s, electrical energy supplied to the motor is converted into kinetic energy of the train. The efficiency of this process is 0.40 (40 %).

Calculate the electrical energy supplied to the motor.

energy = ...[2]

(iii) The electrical energy is generated in an oil-fired power station. Chemical energy in oil is converted into the electrical energy supplied to the motor.

Suggest two places where energy is lost as heat in this process.

1. .....

2. .....
[2]

10	**(a)**	**(i)**	start at origin **and** not horizontal	B1	
			gradient (gradually) decreasing (**ignore** sudden decrease) (not if part of curve above horizontal section)	B1	
			final horizontal section (≥ 1 cm) (not if v is shown as ≠ 40 m/s)	B1	
		(ii)	area **under** the graph **or** count squares **under** graph	M1	
			between t = 0 and horizontal section **or** when speed is changing **or** calculate equivalent distance to 1 cm^2 (after counting squares)	A1	[5]
	(b)	**(i)**	friction/air resistance increases (as speed increases)	B1	
			resultant force decreases (**not** if driving force decreases)	B1	
		(ii)	(air resistance increases until) net force becomes zero **or** forces balance **or** air resistance and driving/forward force are in equilibrium/balanced/equal	B1	[3]
	(c)	**(i)**	(KE =) $\frac{1}{2}mv^2$	C1	
			$\frac{1}{2} \times 5.5 \times 10^5 \times 40^2$	C1	
			4.4×10^8 J	A1	
		(ii)	(total energy input =) useful energy output efficiency **or** efficiency = useful power output/total power input **or** $4.4 \times 10^8/0.40$	C1	
			1.1×10^9 J	A1	
		(iii)	two valid examples e.g. furnace/boiler/turbines/generator/coils/cooling water/cooling towers/heat exchanger/transformer/chimney/waste gases/ transmission cables/lines/wires (**ignore** power station/all parts of motor)	B2	[7]

[Total: 15]

QUESTION · 5054_s14_qp_12.pdf · source page 7 of 16

16 Which source releases carbon dioxide, a greenhouse gas, when generating electricity?

A fossil fuels

B geothermal

C hydroelectric

D nuclear

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 16

16	**A**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows the dam and reservoir of a hydroelectric power station.

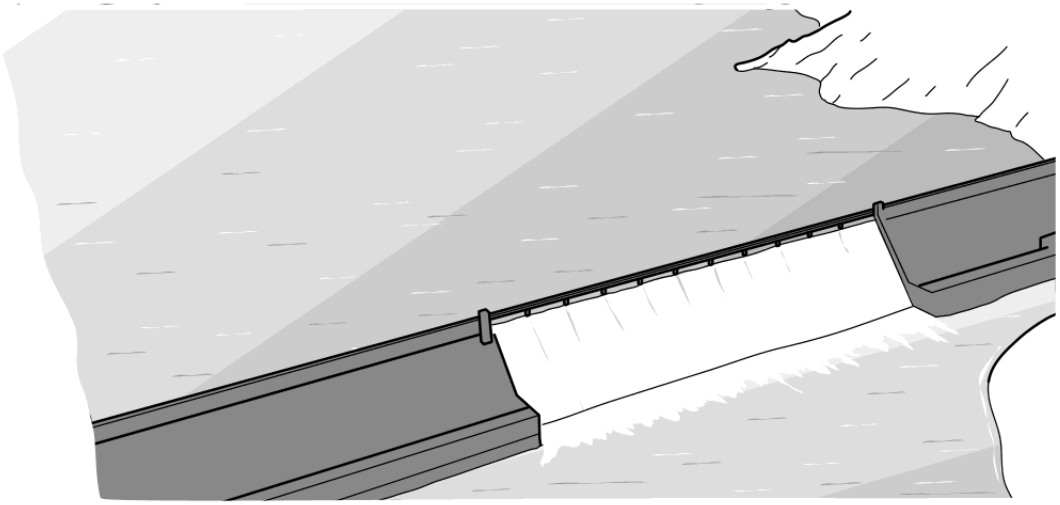


Fig. 9.1

A hydroelectric power station uses a renewable energy source.

(a) State two other renewable energy sources.

1. .....

2. .....
[2]

(b) When the power station operates at full capacity, the electrical power output is 6.8×10^9W.

(i) **1.** Calculate the electrical energy output of the power station if it operates at full capacity for one year.

energy = ...[1]

2. Suggest why, in practice, the power station does not operate at full capacity throughout the year.

.....

.....[1]

(ii) The water surface in the reservoir that supplies the hydroelectric power station is at a vertical height of 170 m above the turbines. In one hour, 1.6×10^{10} kg of water flows from the reservoir through the turbines. The gravitational field strength g is 10 N/kg.

1. Calculate the gravitational potential energy that is converted into other forms of energy in one hour.

energy = ...[2]

2. Calculate the efficiency of the power station when operating at full capacity.

efficiency = ...[2]

3. Suggest two reasons why the efficiency of the power station is less than 1.0 (100%).

1. .....

2. .....
[2]

(c) Before the electrical energy is transmitted along power lines, the voltage is increased to a very large value.

(i) State and explain why a very high voltage is used to transmit electrical energy over long distances.

.....

.....

.....

.....[3]

(ii) State the name of the device that is used to increase the voltage.

.....[1]

(iii) Explain why the generators in the power station must be a.c. (alternating current) generators.

.....

.....[1]

Section B

9	**(a)**	any **two** from: biomass / wood; geothermal power; solar power; tidal power; wave power; wind power		B2	**[2]**
	(b) **(i)**	**1.**	$2.1(4) \times 10^{17}$ J (**allow** $2.1(5) \times 10^{17}$ J if candidate uses 365.24 / 5)	B1	
		2.	any **one** from: not enough water (to maintain maximum flow); rainfall varies (during the year); periods of low demand	B1	
	(ii)	**1.**	(GPE =) *mgh* **or** $1.6 \times 10^{10} \times 10 \times 170$	C1	
			$2.7(2) \times 10^{13}$ J	A1	
		2.	$2.7(2) \times 10^{13} / 3600$ **or** $6.8 \times 10^{9} \times 3600$ **or** $6.8 \times 10^{9} / 7.5(55) \times 10^{9}$ **or** $2.4(48) \times 10^{13} / 2.7(2) \times 10^{13}$	C1	
			0.90 **or** 90%	A1	
		3.	any **two** from: friction (of water) with pipe / turbine /; viscosity of water; friction at bearings; resistance / heat in the wires; KE of water leaving turbine	B2	**[8]**
	(c) **(i)**		less energy lost / wasted **or** more efficient	B1	
			(for a given power) a high voltage results in a small(er) current	B1	
			less heat generated in wires **or** I^2R **or** less resistive losses (**not** if <u>changed</u> resistance mentioned)	B1	
	(ii)		transformer	B1	
	(iii)		transformers only work with an a.c. supply	B1	**[5]**
					[Total: 15]

QUESTION · 5054_s16_qp_11.pdf · source page 6 of 16

12 What uses non-renewable energy?

A a geothermal heating system

B a nuclear power station

C a solar panel

D a wind turbine

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 12

12	**B**

3 A large power station burns oil to heat water in a boiler.

(a) State the name of the form of energy stored in the oil before it is burnt.

.....[1]

(b) (i) State and explain whether oil is a renewable or a non-renewable energy source.

.....

.....

.....[1]

(ii) Describe one environmental issue caused by burning oil to generate electricity in a power station.

.....

.....[1]

(c) (i) State what is meant by *efficiency*.

.....

.....[1]

(ii) The efficiency of the power station is 0.38 (38%). Its electrical power output is 1.9×10^9 W.

Calculate

1. the input power to the power station,

input power = ...[2]

2. the energy wasted in 2.0 hours.

energy wasted = ...[2]

3	**(a)**	chemical (potential energy)		B1	
	(b)	**(i)**	non-renewable **and** oil / it is not replaced / will run out	B1	
		(ii)	acid rain **or** produces CO_2 **or** warms lakes / rivers / sea **or** global warming **or** greenhouse effect	B1	
	(c)	**(i)**	<u>useful</u> energy output / (total) energy input **or** power for energy twice	B1	
		(ii)	**1** $1.9 \times 10^9/0.38$ **or** $1.9 \times 10^9 \times 100/38$	C1	
			5.0×10^9 W	A1	
			2 (E =)Pt **or** $0.62 \times 5.0 \times 10^9 \times 2.0$ ($\times$ 3600) **or** $(5.0 - 1.9) \times 10^9$ etc.	C1	
			2.2×10^{13} J	A1	[8]

QUESTION · 5054_w16_qp_12.pdf · source page 6 of 16

12 Which power stations all rely on renewable sources of energy?

A hydroelectric, oil-fired, solar power stations

B hydroelectric, solar, wind power stations

C hydroelectric, oil-fired, wind power stations

D oil-fired, wind, solar power stations

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 12

12	**B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Different energy sources are used to generate electricity.

(a) Energy sources are renewable or non-renewable.

(i) Nuclear energy is described as a non-renewable source.

Explain what is meant by a *non-renewable* energy source.

.....

.....[1]

(ii) Four of the energy sources used are:

hydroelectric **oil** **geothermal** **wind**

Write the name of these energy sources in the correct column of the table below.

non-renewable	renewable and caused by energy from the Sun	renewable and not caused by energy from the Sun

[3]

(iii) State one way in which using nuclear energy is better for the environment than using oil.

.....

.....[1]

(iv) State one way in which using nuclear energy is worse for the environment than using oil.

.....

.....[1]

(b) Fig. 9.1 is a block diagram of a power station that produces electrical energy from oil.

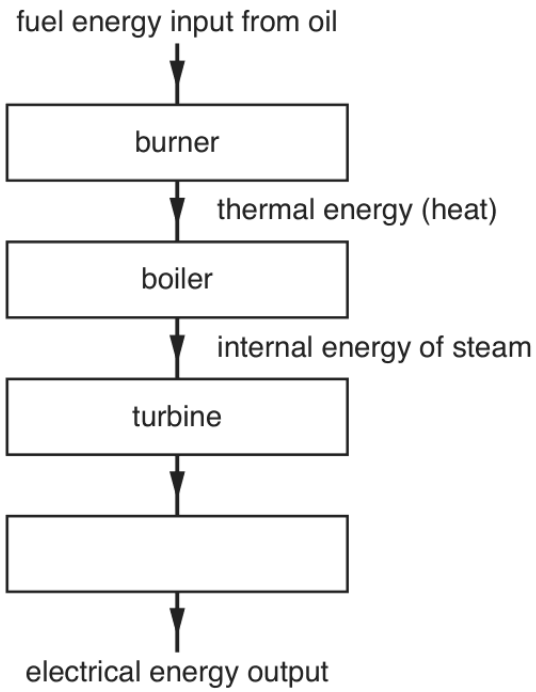


Fig. 9.1

(i) Write the name of the missing part of the power station in the empty box on Fig. 9.1. [1]

(ii) State the form of energy that the turbine possesses.

.....[1]

(iii) A small boiler in the power station contains 24 m^3 of water at 30 °C. High pressure in the boiler increases the boiling point of water to 120 °C.

Thermal energy supplied to the boiler is used to heat the water from 30 °C to 120 °C and then to turn it all to steam at 120 °C.

The density of water is 1000 kg/m^3.
The specific heat capacity of water is 4200 J/(kg °C).
The specific latent heat of vaporisation of water is 2.3×10^6 J/kg.

1. Calculate the mass of water in the boiler.

mass = ...[1]

2. Calculate the total thermal energy (heat) supplied to the boiler.

thermal energy = ...[4]

(iv) The electrical energy output from the power station is transmitted over long distances at a high voltage.

Explain why electricity is transmitted at a high voltage.

.....

.....

.....

.....[2]

Section B

Question	Answer	Marks
9(a)(i)	not being replaced **or** will run out	**B1**
9(a)(ii)	only oil in 1st column	**B1**
	only wind and hydroelectric in 2nd column	**B1**
	geothermal in 3rd column	**B1**
9(a)(iii)	less greenhouse gases / global warming / acid rain / **or** toxic gases / oil spills **and** how they affect a named organism / ecosystem	**B1**
9(a)(iv)	nuclear waste / radioactive waste causes (storage) problems **or** explosion / melt down / leak emits radioactivity	**B1**
9(b)(i)	generator (and transformer)	**B1**
9(b)(ii)	kinetic (energy)	**B1**
9(b)(iii)	1 24 000 kg	**B1**
	2 formula (E) = mL **or** (E **=**) mcT seen	**C1**
	$24\,000 \times 2.3 \times 10^6$ **or** $5.5(2) \times 10^{10}$ (J)	**C1**
	$24\,000 \times 4200 \times 90$ **or** $9.0(72) \times 10^9$ (J)	**C1**
	6.4×10^{10} J	**A1**
9(b)(iv)	low current	**B1**
	less energy / heat loss (in resistance or cables) **or** thinner wires can be used	**B1**

QUESTION · 5054_s17_qp_12.pdf · source page 7 of 20

16 Which energy source is non-renewable?

A nuclear energy

B solar energy

C tidal energy

D wind energy

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 16

16	A	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 7 of 20

17 In a coal-fired power station, the coal is burnt and thermal energy (heat) is produced. The thermal energy is used to produce electrical energy.

In which order does the energy pass through parts of the power station?

A boiler → generator → turbine

B boiler → turbine → generator

C turbine → boiler → generator

D turbine → generator → boiler

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 17

17	B	**1**

4 A 75 W filament lamp in an office is accidentally left switched on for 63 hours when the office is closed.

(a) (i) Calculate the electrical energy that is supplied to this lamp in this time.

electrical energy = ...[3]

(ii) The efficiency of the lamp is 0.12 (12%).

Calculate the energy released by the lamp in the form of light.

light energy = ...[1]

(b) The power station that generates the electricity supplied to the office uses oil as its fuel.

(i) State and explain whether oil is a renewable or a non-renewable source of energy.

.....

.....

.....[1]

(ii) Discuss one environmental consequence of using oil as a source of energy.

.....

.....

.....[2]

Question	Answer	Marks
4(a)(i)	(E =) Pt **or** 75×63 **or** $75 \times 63 \times 60$	**C1**
	$75 \times 63 \times 60 \times 60$ **or** 2.835×10^5 **or** 0.075×63	**C1**
	1.7×10^7 J	**A1**
4(a)(ii)	2.0×10^6 J	**B1**
4(b)(i)	non-renewable **and** gets used up / not replaced / will run out / finite	**B1**
4(b)(ii)	**three** separate approaches **I** / **II** / **III**	
	I greenhouse gases / CO / CO_2 emitted	**B1**
	global warming / <u>increase</u> greenhouse effect / ice caps melt	**B1**
	or II acid rain / NOx / SO_2 emitted	**B1**
	breathing difficulty / damage to buildings / acidify lakes **or** damages / kills plants / sea-life / animals / fish	**B1**
	or III oil spills	**B1**
	damages / kills plants / sea-life / animals / fish	**B1**

QUESTION · 5054_s18_qp_11.pdf · source page 6 of 20

13 How does an oil-fired power station differ from a nuclear power station?

A Gases emitted by hot fuel are emitted into the atmosphere.

B Steam is produced in a boiler using hot fuel.

C The hot steam is used to turn a turbine.

D Turbines are used to drive an electric generator.

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 13

13	A	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 6 of 16

14 Which power station produces carbon dioxide when operating?

A gas-fired power station

B geothermal power station

C nuclear power station

D wind power station

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 14

14	A	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Wind energy is a renewable energy source. A wind turbine and generator convert energy in the wind to electrical energy in a generator. Fig. 9.1 shows some wind turbines.

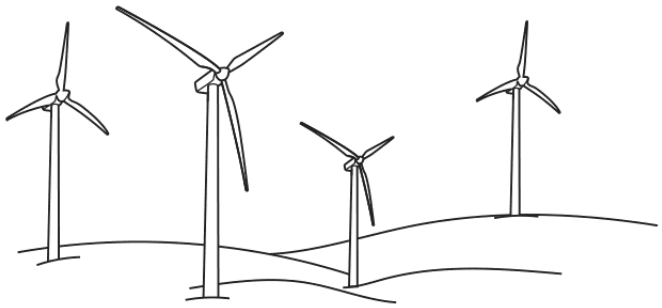


Fig. 9.1

(a) (i) State what is meant by a *renewable energy source*.

.....

.....

.....[1]

(ii) State the name of **one other** renewable energy source.

.....

.....[1]

(iii) Some methods of generating electricity cause much more global warming than wind energy.

1. State **one** method of generating electricity that causes a large amount of global warming.

.....[1]

2. Describe, in outline, how the method you have chosen causes a significant amount of global warming.

.....

.....

.....[1]

(b) During a 30 minute period, a mass of 4.2×10^7 kg of air enters the turbine with a speed of 15 m/s.

(i) Calculate the kinetic energy of the air that enters the turbine in 30 minutes.

energy = ...[2]

(ii) The electrical power output of the generator connected to the turbine is 8.4×10^5 W.

Calculate the electrical energy output from the turbine in 30 minutes.

energy output = ...[2]

(iii) Calculate the efficiency of the turbine and generator in converting the kinetic energy of the air to electrical energy.

efficiency = ...[2]

(c) The generator contains a coil that rotates between the poles of a magnet.

The coil rotates 50 times in one second and produces an alternating voltage output with a maximum value of 500 V.

(i) On the grid in Fig. 9.2, sketch a voltage–time graph showing the output voltage.

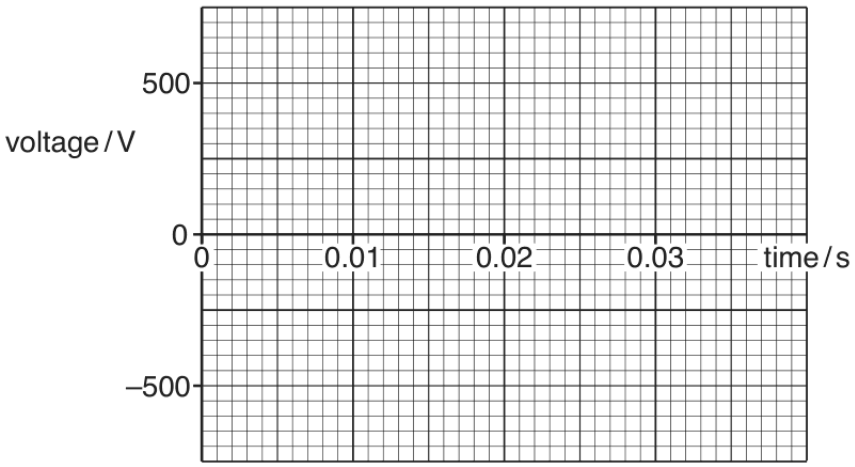


Fig. 9.2

[2]

(ii) The output voltage from the generator is stepped up by a transformer. The electrical energy then passes along a transmission line to a distant house.

Explain why

1. a high voltage is used for the transmission of electrical energy,

.....

.....

.....

.....[2]

2. a transformer is used to connect the transmission line to the house.

.....

.....

.....[1]

Question	Answer	Marks
9(a)(i)	will not run out **or** infinite **or** being replaced	**B1**
9(a)(ii)	solar / Sun, tidal, geothermal, biomass, hydroelectric, water waves	**B1**
9(a)(iii)	1 (burning) use of a fossil fuel, e.g. coal, oil, natural gas etc.	**B1**
	2 produces carbon dioxide **or** greenhouse gases (absorb infra-red from Earth)	**B1**
9(b)(i)	(E=) ½mv^2 seen in any form algebraic or numerical	**C1**
	4.7×10^9 J	**A1**
9(b)(ii)	(E=)Pt seen in any form algebraic or numerical	**C1**
	1.5×10^9 J	**A1**
9(b)(iii)	(efficiency=) (energy) output / (energy) input in any form	**C1**
	0.32 **or** 32%	**A1**
9(c)(i)	all peaks and troughs at 500 and -500 to within one small square **and** roughly sinusoidal	**B1**
	one wave takes 0.02 s within one small square	**B1**
9(c)(ii)	1 less energy loss / heat loss / power loss / more efficient	**B1**
	low(er) currents (in line) **or** thin(ner) wires can be used	**B1**
	2 step down transformer **or** voltage reduced (to 240 V) **or** otherwise voltage dangerous	**B1**

10 **(a)** Two neutral atoms are atoms of different isotopes of the same element.

State

(i) one similarity in the structures of these two atoms,

..... [1]

(ii) one difference between the structures of these two atoms.

..... [1]

(b) The isotope plutonium-239 decays by the emission of an alpha-particle (symbol α) to an isotope of uranium (symbol U). The half-life for this decay is 2.4×10^4 years.

(i) Complete the equation for this decay.

$$^{239}_{94}\text{Pu} \longrightarrow {}^{\cdots\cdots}_{\cdots\cdots}\text{U} + {}^{\cdots\cdots}_{\cdots\cdots}\alpha$$

[3]

(ii) The count rate measured due to a pure sample of plutonium-239 is 6400 counts / second.

Calculate the count rate after a period of 1.2×10^5 years has elapsed.

count rate = ... [3]

(iii) Fig. 10.1 shows an alpha-particle entering a uniform magnetic field.

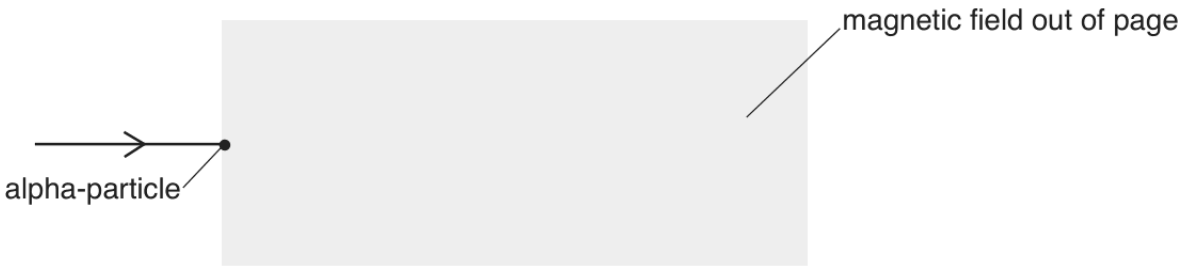


Fig. 10.1

1. On Fig. 10.1, draw the path of the alpha-particle in the magnetic field. [2]

2. Explain why the alpha-particle follows the path you drew in **(b)(iii)1**.

.....

..... [1]

(c) In a nuclear reactor at a power station, plutonium-239 is made to undergo nuclear fission.

(i) State what is meant by *nuclear fission*.

.....

..... [1]

(ii) State what happens to a plutonium-239 nucleus in the nuclear reactor to cause fission.

.....

..... [1]

(iii) State one advantage of generating electricity in a nuclear power station rather than using

1. an oil-fired power station,

.....

..... [1]

2. an array of wind turbines.

.....

..... [1]

MARK SCHEME · 5054_w18_ms_21.pdf · source page 9 of 10 · Question 10

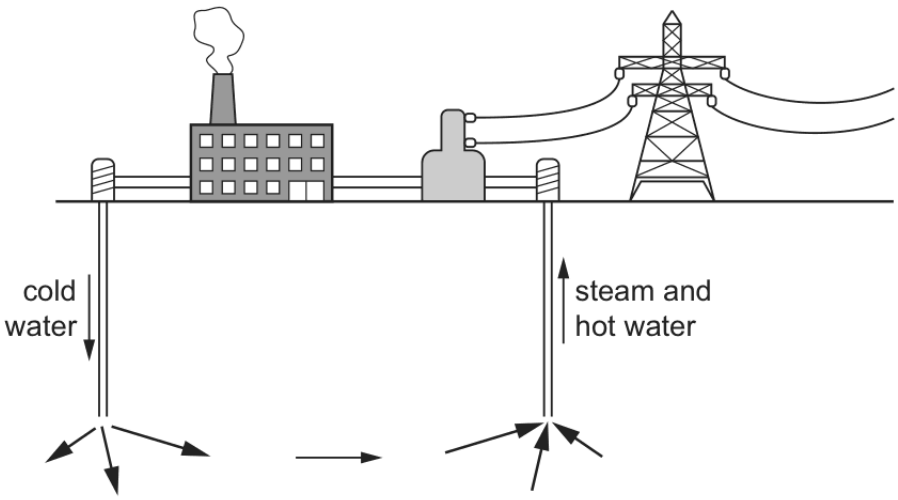
Question	Answer	Marks
10(a)(i)	same number of protons **or** same number of electrons	**B1**
10(a)(ii)	different number of neutrons	**B1**
10(b)(i)	$^{4}_{2}\alpha$	**B1**
	$^{\cdots\cdots}_{92}U$	**B1**
	$^{235}_{\cdots\cdots}U$	**B1**
10(b)(ii)	(no. of half-lives =) 1.2×10^5 / 2.4×10^4 **or** 5 (half-lives)	**C1**
	relevant halving **or** $1 / 2^5$ **or** 1 / 32 **or** 6400 / 32	**C1**
	200 counts / second	**A1**

MARK SCHEME · 5054_w18_ms_21.pdf · source page 10 of 10 · Question 10

Question	Answer	Marks
10(b)(iii)	1 curving (only) downwards with correct curvature	**M1**
	no straight section (by eye) anywhere in magnetic field	**A1**
	2 (Fleming's) left-hand rule **or** motor rule **or** rule described **or** moving charge is a current	**B1**
10(c)(i)	splitting of a <u>nucleus</u>	**B1**
10(c)(ii)	neutron fired at a plutonium nucleus **or** plutonium nucleus absorbs neutron	**B1**
10(c)(iii)	1 no CO_2 / SO_2 / nitrogen oxides / greenhouse gases emitted **or** do not contribute to global warming / acid rain	**B1**
	2 reliable / not intermittent **or** smaller land area	**B1**

QUESTION · 5054_w18_qp_12.pdf · source page 6 of 16

11 The diagram represents a geothermal power station.



Which useful energy transformation is taking place?

A electrical energy → potential energy

B electrical energy → thermal energy

C potential energy → electrical energy

D thermal energy → electrical energy

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 11

11	D	**1**

2 **(a)** Some sources of energy, used to produce electricity, may run out in the future.

Underline the **two** sources of energy that are likely to run out first, before the others.

coal oil solar tides wave wind

[1]

(b) Fig. 2.1 is a simplified diagram of a geothermal power station.

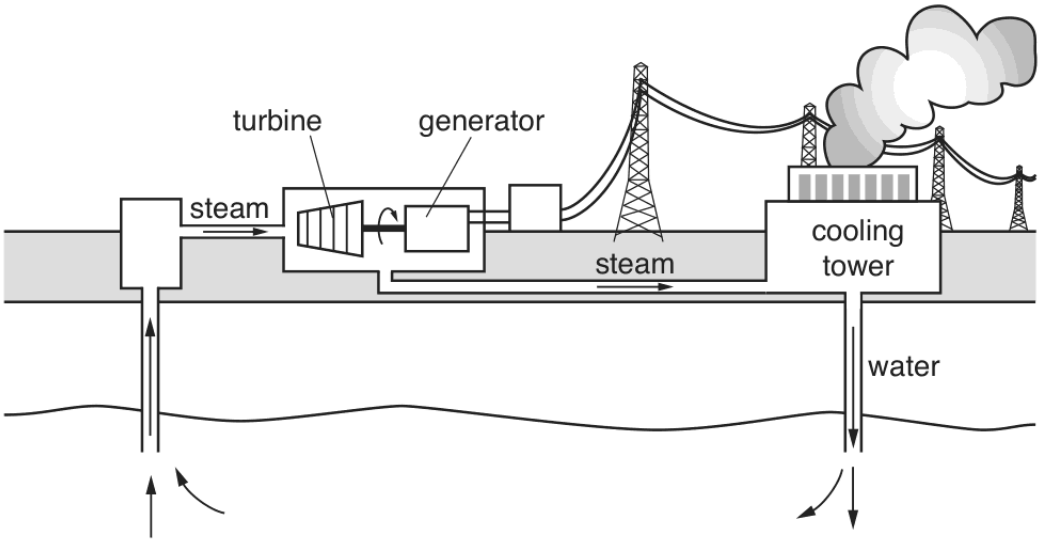


Fig. 2.1

(i) Thermal energy is given to the water as it passes through rocks underground.

Describe the process by which this thermal energy causes the turbine to turn.

.....

.....

.....

..... [2]

(ii) Water from the cooling tower at 30 °C passes into the ground and is heated to 160 °C.

The specific heat capacity of water is 4200 J/(kg °C).

Calculate the thermal energy (heat) needed to heat 90 kg of water from 30 °C to 160 °C.

thermal energy = ... [2]

[Total: 5]

Question	Answer	Marks
2(a)	coal and oil underlined	**B1**
2(b)(i)	water boils or turns to steam	**B1**
	pressure / force of steam turns turbine **or** steam at high pressure / pressure difference (across turbine)	**B1**
2(b)(ii)	(Q =) mcT numerical or algebraic	**B1**
	4.9×10^7 J	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 7 of 16

18 Which method of producing electricity does **not** use an a.c. generator?

A geothermal power station

B solar panels

C wave-energy generators

D wind turbine

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 18

18	B	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 7 of 16

19 The main stages in the operation of a coal-fired power station are listed. They are **not** in the correct order.

1 The turbine turns a generator.

2 Water in the boiler becomes hot.

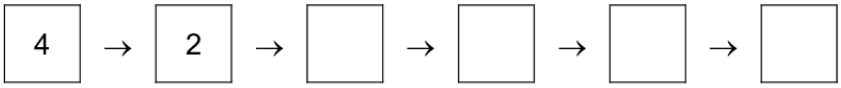
3 Steam turns a turbine.

4 Burning coal produces thermal energy.

5 Electromagnetic induction produces electrical energy.

6 Steam is produced.

The flow chart shows the first two stages.



What is the correct order for the remaining stages?

A 6 → 3 → 1 → 5

B 6 → 1 → 3 → 5

C 1 → 6 → 3 → 5

D 6 → 3 → 5 → 1

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 19

19	A	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 7 of 16

16 Which source of electrical energy is **not** renewable?

A solar cells

B hydroelectric generators

C wind turbines

D nuclear reactors

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 16

16	D	**1**

4 In a coal-fired power station, coal is burnt and electrical energy is produced.

(a) State the form of energy in the coal that is transferred to electrical energy.

..... [1]

(b) (i) State the principle of the conservation of energy.

.....

.....

.....

..... [2]

(ii) The quantity of electrical energy produced by the power station is much less than the quantity of energy in the coal that is burnt to generate the electricity.

Explain why this is so.

.....

.....

.....

..... [2]

(c) State **one** environmental consequence of generating electricity in coal-fired power stations.

.....

..... [1]

[Total: 6]

Question	Answer	Marks
4(a)	chemical (energy)	**B1**
4(b)(i)	energy cannot be created **and** energy cannot be destroyed	**B1**
	energy can be transferred from one form into another	**B1**
4(b)(ii)	some chemical energy is transferred to thermal energy (sound and light)	**B1**
	thermal energy lost to air / water / buildings / equipment	**B1**
4(c)	any **one** from: carbon dioxide produced; carbon monoxide produced; global warming; greenhouse gases nitrous oxide produced; sulfur dioxide produced; acid rain; rise in sea-level	**B1**

QUESTION · 5054_s22_qp_11.pdf · source page 5 of 16

9 Water in a river is used to turn a turbine as the water flows downstream.

How is this type of power source described?

A hydroelectric

B tidal

C geothermal

D nuclear

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 9

9	A	**1**

2 Fig. 2.1 shows a wind turbine.

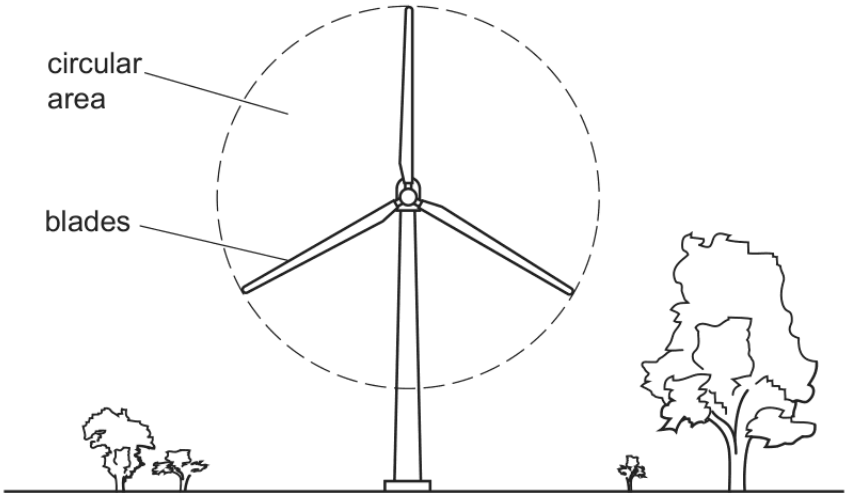


Fig. 2.1

(a) The wind blows directly towards the turbine with a speed of 12 m / s.

In one second, 60 000 kg of air passes through the circular area swept out by the blades.

Calculate the kinetic energy of this mass of air.

kinetic energy = ... [3]

(b) A wind turbine releases no carbon dioxide into the atmosphere while generating electricity. Compared with a coal-fired power station, this is an advantage of using a wind turbine.

(i) State **one** reason why it is important to reduce the amount of carbon dioxide produced.

.....

..... [1]

(ii) Suggest **one** advantage of using a coal-fired power station compared with a wind turbine.

.....

..... [1]

(c) A coal-fired power station releases 0.96 kg of carbon dioxide when it generates 1.0 kW h of electrical energy.

(i) Define the kilowatt-hour (kW h).

.....

..... [1]

(ii) Calculate the mass of carbon dioxide saved when the wind turbine has a power output of 2000 kW and operates for 12 hours.

mass = ... [1]

(d) Wind energy is a form of renewable energy.

State the name of **one** other form of renewable energy.

..... [1]

[Total: 8]

MARK SCHEME · 5054_s22_ms_21.pdf · source page 5 of 11 · Question 2

Question	Answer	Marks
2(a)	(K.E. =) ½mv^2 numerical or algebraic	**C1**
	$0.5 \times 60\,000 \times 12^2$	**C1**
	4 300 000 J	**A1**
2(b)(i)	reduces global warming	**B1**
2(b)(ii)	wind not always blowing **or** wind not reliable / not available in some places **or** smaller land area (for same power) **or** less harm to birds **or** less visual effect **or** no interference with tv / radar signals	**B1**

MARK SCHEME · 5054_s22_ms_21.pdf · source page 6 of 11 · Question 2

Question	Answer	Marks
2(c)(i)	(amount of energy of a device of power) 1 kW used for 1 hour	**B1**
2(c)(ii)	23 000 kg	**B1**
2(d)	solar, tidal, geothermal, hydro(electric), biomass, wave	**B1**

QUESTION · 5054_s22_qp_12.pdf · source page 7 of 20

13 Which device provides a continuous, steady energy output for the longest time?

A a nuclear reactor

B a solar panel

C a 1.5 V cell

D a wind generator

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 13

13	A	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 4 of 16

9 Hydroelectric and tidal power stations generate electrical energy.

Do these use renewable sources of energy?

	hydroelectric	tidal
A	no	no
B	no	yes
C	yes	no
D	yes	yes

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 9

9	D	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 5 of 16

10 Which energy resource is correctly described?

	energy resource	renewable	availability
A	biofuels	yes	at all times
B	nuclear	yes	intermittent
C	solar radiation	no	at all times
D	wind	no	intermittent

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 10

10	A	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 6 of 20

13 How does an oil-fired power station differ from a nuclear power station?

A Gases emitted by hot fuel are emitted into the atmosphere.

B Steam is produced in a boiler using hot fuel.

C The hot steam is used to turn a turbine.

D Turbines are used to drive an electric generator.

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 13

13	A	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 6 of 16

12 Which method of producing electricity does **not** involve a turbine?

A hydroelectric power station

B nuclear power station

C solar cells

D wind generator

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 12

12	C	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 7 of 16

15 Which energy source is available constantly over a 24-hour period?

A natural gas

B solar cells

C tidal

D wind

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 15

15	A	**1**

3 Fig. 3.1 shows a solar-powered charger connected to a cell phone (mobile phone).

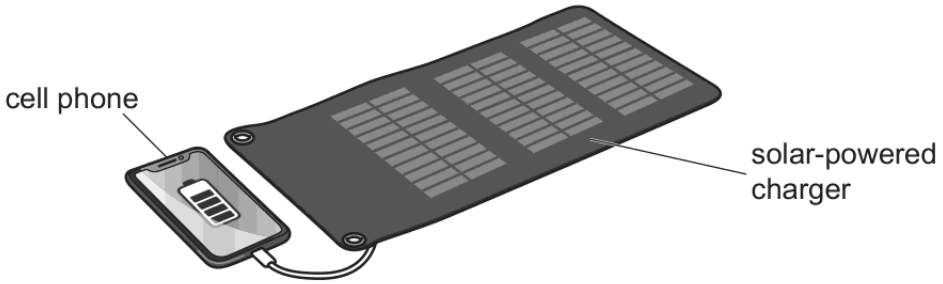


Fig. 3.1

The battery inside the cell phone is charged by the solar-powered charger.

(a) **(i)** Complete Fig. 3.2 to show the useful transfer of energy from the Sun to the battery.

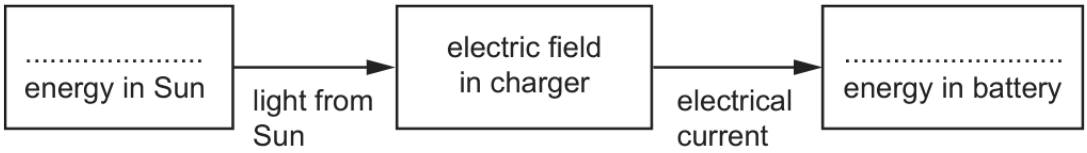


Fig. 3.2

[2]

(ii) Explain why the battery takes a long time to charge on a cloudy day.

.....

..... [1]

(b) After use, the outside surface of the cell phone is warm. When switched off, the cell phone cools down.

Name and describe the **three** processes by which thermal energy is transferred as the cell phone cools down.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) It takes 4.5 hours to charge the battery with an average current of 300 mA.

Calculate the quantity of charge that enters the battery. Give the unit of your answer.

charge = ... unit ..... [3]

[Total: 9]

Question	Answer	Marks
3(a)(i)	nuclear (energy in Sun)	**B1**
	chemical (energy in battery)	**B1**
3(a)(ii)	less / no light / energy (reaching the charger) **or** less / no electrical current (produced)	**B1**
3(b)	conduction **and** energy passing from particle to particle in solid / from outside particles to air particles	**B1**
	convection **and** in air / hot air rises	**B1**
	radiation **and** emitted from surface	**B1**
3(c)	(Q =) It in any form	**C1**
	4 900	**A1**
	C **or** coulomb	**B1**

QUESTION · 5054_s25_qp_12.pdf · source page 4 of 16

9 Which row shows an advantage and a disadvantage of using nuclear fuel to generate electrical power?

	advantage	disadvantage
A	does **not** produce carbon dioxide	produces radioactive waste
B	does **not** produce carbon dioxide	only works in some weather conditions
C	energy source is renewable	produces radioactive waste
D	energy source is renewable	only works in some weather conditions

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 9

9	A	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 7 of 16

13 Seven energy sources are listed.

fossil fuel **hydroelectric** **solar** **nuclear** **geothermal** **wind** **tides**

How many of these energy sources are renewable energy sources?

A 3 **B** 4 **C** 5 **D** 6

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 13

13	C	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 6 of 16

14 Why are fossil fuels described as non-renewable?

A Burning fossil fuels badly affects the environment.

B Fossil fuels are being used up faster than they are replaced.

C Fossil fuels are only found in some countries.

D The energy in fossil fuels comes initially from the Sun.

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 14

14	B	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 7 of 16

13 In a fossil fuel power station, chemical energy stored in the fossil fuel is used to generate electricity.

Which stage is **not** involved in the generation of electricity in a nuclear power station?

A Fuel is burned in a furnace.

B Steam is used to turn a turbine.

C The turbine is used to turn a generator.

D Water is heated in a boiler to make steam.

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 13

13	A	**1**

3 Figure 3.1 shows a dam used as part of a hydroelectric power station.

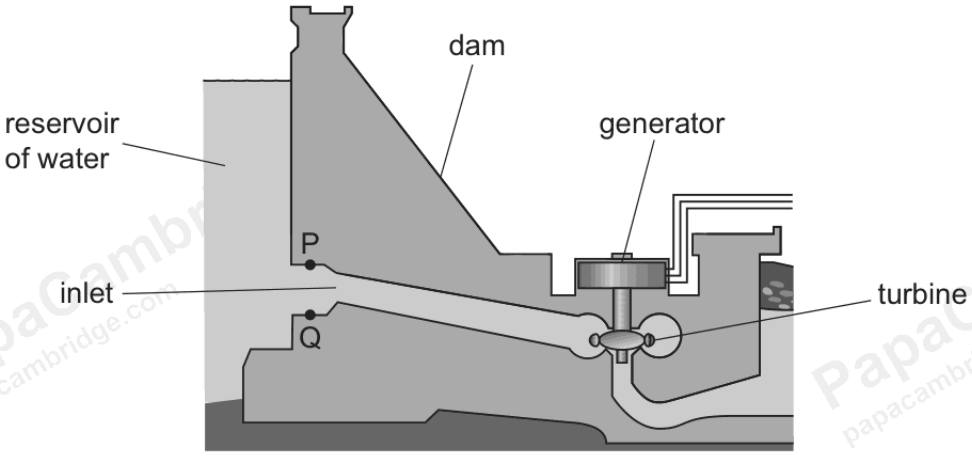


Figure 3. 1

Water flows to the turbine through an inlet in the dam. The inlet is shown in Figure 3.1.

Points P and Q are on the horizontal surfaces next to the inlet.

(a) (i) State **two** factors that determine the size of the pressure at P.

.....

..... [1]

(ii) The pressure in the water causes a force on the horizontal surfaces at P and Q.

State the directions of the forces due to the water on the surfaces at P and Q.

direction at P .....

direction at Q .....

[1]

(b) This hydroelectric energy resource is a renewable source of energy.

State the name of **one** other renewable energy source.

..... [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
OT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(c) The power output from the generator is 2.5 MW.

(i) Calculate the energy **output** from the generator in 30 minutes.

energy output = ... J [2]

(ii) The efficiency of the energy transfer in the generator is 95%.

Calculate the power **input** to the generator.

PapaCambridge
papacambridge.com
PapaCambridge
papacambridge.com

power input = ... MW [2]

[Total: 7]

Question	Answer	Marks	Guidance
3(a)(i)	At least 2 from • height / depth (of water) / quantity or amount of water in dam • density (of water / liquid) • acceleration of gravity / gravitational field strength • atmospheric pressure • force (on surface / inlet) • area (of surface / inlet)	**B1**	**I** thickness; area of dam
3(a)(ii)	at P – upwards / towards P at Q – downwards / towards Q	**B1**	**I** top, bottom; north, south both needed
3(b)	ANY one from geothermal, tidal, solar, wind, biomass, wave (energy)	**B1**	**A** sunlight; solar panels; solar cell; wind turbines; biogas ; solar power; wood from a managed source **I** just 'wood' **not** nuclear

Question	Answer	Marks	Guidance
3(c)(i)	(E =) $P \times t$ in any form algebraic or numerical	**C1**	**A** 2.5 ($\times 10^6$)$\times$ 30; 2.5$\times$30$\times$60; 4.5 $\times$ 10^n; P=W / t
	4.5×10^9 (J)	**A1**	
3(c)(ii)	efficiency = (power) output / (power) input in any form	**C1**	**A** 0.95 = 2.5 / P_{IN}; (useful) energy output / energy input e.g. 0.95 =**3(c)(i)** / input in any form
	2.6(3 MW)	**A1**	**A** 2.6×10^6 with unit MW only 1 mark

Chapter 1 Motion, forces and energy

1.7 Energy, work and power

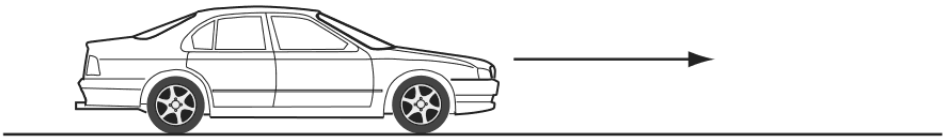
1.7.4 Efficiency

70 entries: 41 Paper 1 questions, 29 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 6 of 20

12 A car is driven along a level road. The total energy input from the petrol is 60 kJ, and the car wastes 45 kJ of energy.



What is the efficiency of the car?

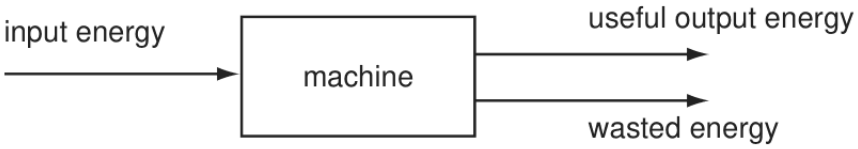
A 15 % **B** 25 % **C** 45 % **D** 75 %

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 12

12	**B**

QUESTION · 5054_w04_qp_1.pdf · source page 7 of 20

12 The diagram shows the energy transfer through a machine.



The machine is 50 % efficient.

Which is correct?

A input energy = useful output energy

B useful output energy = input energy + wasted energy

C wasted energy = input energy + useful output energy

D wasted energy = useful output energy

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 12

12 **D**

For Examiner's Use

4 Fig. 4.1 shows a hydroelectric power station. Water from the lake is used to produce electricity in the turbine house.

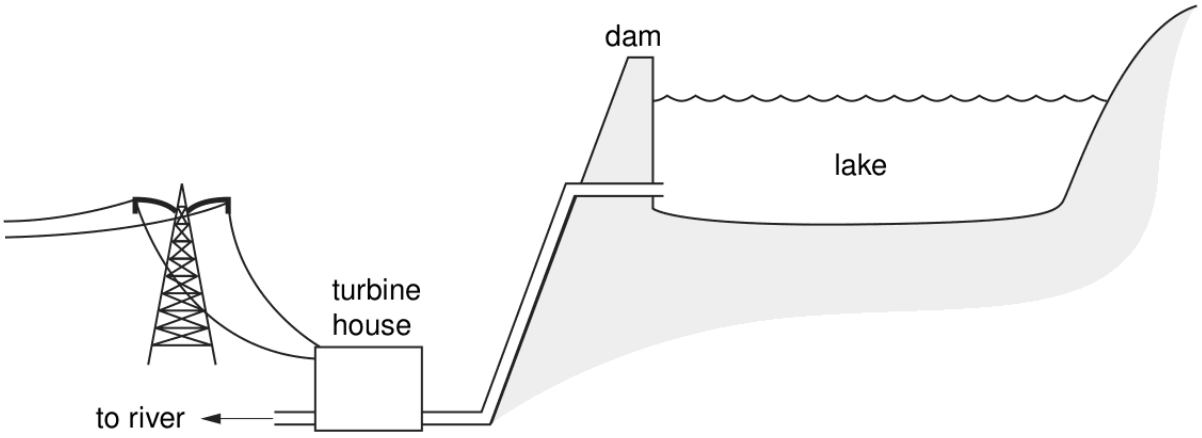


Fig. 4.1

(a) State where the water in Fig. 4.1 has the least potential energy.

..... [1]

(b) In 30 minutes, the water loses 5.0×10^9 J of energy and 4.5×10^9 J of electrical energy is produced in the turbine house.

(i) Calculate the efficiency of the energy conversion.

efficiency = ... [1]

(ii) Calculate, in watts, the electrical power output from the turbine house.

power = ... W [3]

(c) Some power stations burn coal to produce the same electrical power output. State one advantage of the hydroelectric power station.

.....

.....

..... [1]

(d) State one harmful effect that the hydroelectric power station may have on the environment.

.....

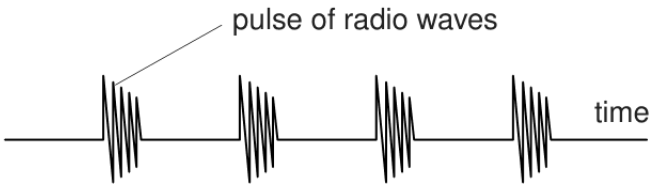
..... [1]

4	**(a)**	in river/(emerging from or entering) turbine house	**B1**
	(b) **(i)**	0.9 **or** 90% **or** 0.47 **or** 47% (penalise unit error)	**B1**
	(ii)	P = E/t in symbols **or** any energy/any time	**C1**
		30 x 60 **or** 1800 seen	**C1**
		2.5×10^6 (W)	
		(150 or 2.78MW score 2/3)	**A1**
	(c)	any sensible suggestion e.g. no costs for water/energy supply **or** less pollution (accept coal produces smoke/dust/harmful gases/CO_2 **or** no need to transport coal **or** renewable **or** rapid response to power demand **or** less heat produced/more efficient	**B1**
	(d)	any sensible suggestion e.g. flooding **or** fish unable to pass **or** turbines kill fish **or** destroy habitats **or** less land **or** uses up large space **or** fells trees **or** unsightly/destroys scenery **or** lake/river silt up **or** more rain/evaporation	**B1** **7**

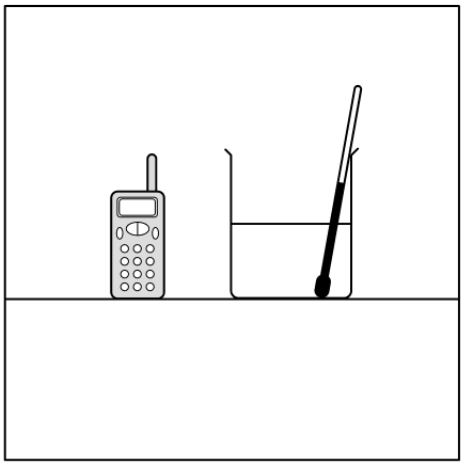
10 A student uses a mobile phone. He is concerned that the energy of the radio waves may cause a temperature rise in his brain. To investigate this effect, he calculates and tries to measure the heating effect of the phone on a nearby glass beaker of water.

Information from manufacturer

constant input power 0.20 W



The mobile phone emits pulses of radio waves.
Each pulse of waves has energy 0.00012 J.
There are 216 pulses of waves in 1.0 s



The phone is used for 360 s when next to a glass beaker containing 50 g of water.

(a) Calculate

(i) the number of pulses of radio waves produced during the phone call, [2]

(ii) the total energy of the radio waves emitted during the phone call, [2]

(iii) the maximum temperature rise produced in 50 g of water if all of the energy calculated in **(ii)** is absorbed by the water. The specific heat capacity of water is 4.2 J/(g °C). [3]

(b) The manufacturer gives the input power of the phone as 0.20 W.

(i) Calculate the energy supplied by the battery when the phone is used for 360 s. [2]

(ii) Calculate the fraction of the energy supplied by the battery that is converted into radio wave energy during the phone call. [1]

(c) The student uses the mercury-in-glass thermometer shown in Fig. 10.1.

He does not detect any temperature rise in the water in the beaker when the phone is used.

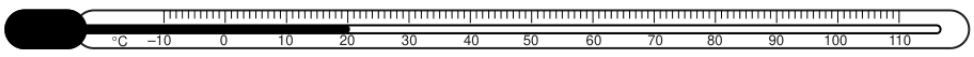


Fig. 10.1

(i) Describe how you would check the 0 °C and 100 °C points on the thermometer. [2]

(ii) Explain why the thermometer is not sensitive enough to detect the temperature rise. [1]

(iii) State and explain one change that will make a mercury-in-glass thermometer more sensitive. [2]

10	**(a)**	**(i)**	360 x 216	**C1**
			77 800 (no sig fig penalty)	**A1**
		(ii)	77 800 x 0.00012	**C1**
			9.33 J ecf **(i)**	**A1**
		(iii)	E = mcΔT in any form, algebraic or numerical	**B1**
			9.33/(50 x 4.2)	**C1**
			0.044 ºC ecf **(ii)**	**A1**
	(b)	**(i)**	E=Pt in any form, algebraic or numerical	**C1**
			72 J	**A1**
		(ii)	0.13 (accept 13%) no s.f. penalty	**A1**
	(c)	**(i)**	(pure) melting ice for 0° C	**B1**
			(pure) boiling water/steam above boiling water (at 1 atmosphere) for 100° C	**B1**
		(ii)	each division on thermometer is too small described in some way e.g. does not expand far up tube (**not** bore too thin, not enough mercury)	**B1**
		(iii)	**change** use more mercury **or** use smaller bore	**M1**
			reason more expansion **or** further distance up tube (for same expansion)	**A1**

QUESTION · 5054_s06_qp_1.pdf · source page 5 of 20

13 The input power to a motor is 300 W. In 20 s it lifts a load of 400 N through a height of 6.0 m.

What is the efficiency of the motor?

A 12 % **B** 25 % **C** 40 % **D** 75 %

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 13

13	**C**

For Examiner's Use

6 Fig. 6.1 shows a device used to generate electricity. Water entering at the top turns the wheel. A generator connected to the wheel produces electric current.

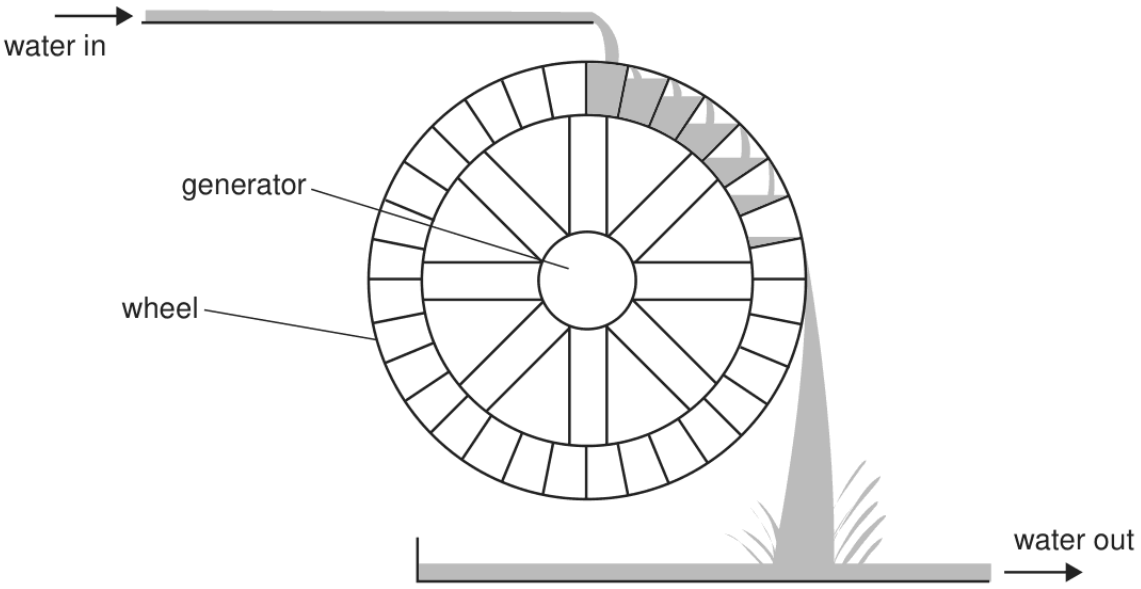


Fig. 6.1

(a) State the two main energy changes that take place in this device as water starts to flow.

1. .....

2. ..... [2]

(b) The change in the potential energy of the water every second is 2000 J. The electrical energy output is 1200 J every second.

Calculate the efficiency of the process.

efficiency = ... [2]

(c) State two ways in which the device wastes energy.

1. .....

2. ..... [2]

6	**(a)**	P.E. (of water) to K.E. (of wheel or water**)/**K.E. (of wheel) to electrical energy/ KE **of water** to KE **of wheel** /PE to electrical energy ANY 2 (-1 each clearly wrong answer beyond 2)	B2
	(b)	1200/2000 **or** energy output/ energy input **or** power output/power input (NOT output/input)	C1
		0.60 **or** 60% (NOT fractions; 0.6 YES)	A1
	(c)	friction in wheel or generator (bearings/axle) **or** water out has K.E. **or** produces heat in windings/in resistance **or** heat (in bearings) due to friction (ignore sound) ANY 2 (-1 each clearly wrong answer beyond 2)	B2 6

QUESTION · 5054_s07_qp_1.pdf · source page 6 of 20

11 Which formula gives the efficiency of an energy conversion?

A efficiency = total energy input - useful energy output

B efficiency = useful energy output × total energy input

C $\text{efficiency} = \dfrac{\text{useful energy output}}{\text{total energy input}}$

D $\text{efficiency} = \dfrac{\text{total energy input}}{\text{useful energy output}}$

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 11

11	**C**

QUESTION · 5054_w07_qp_1.pdf · source page 5 of 20

11 The efficiency of an electrical generator is 65 %.

Which useful output can be expected if the energy input to the generator is 12 kJ?

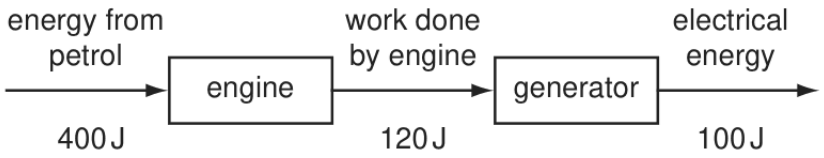
A 4.2 kJ **B** 7.8 kJ **C** 19 kJ **D** 780 kJ

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 11

11	**B**

QUESTION · 5054_s08_qp_1.pdf · source page 6 of 20

11 Energy from petrol is used to operate an engine. The engine drives a generator, which produces electrical energy.



What is the overall efficiency of the process?

A 25 % **B** 30 % **C** 55 % **D** 83 %

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 11

11	**A**

QUESTION · 5054_w08_qp_1.pdf · source page 7 of 20

12 A crane lifts a weight of 1000 N through a vertical height of 30 m.

It uses 60 000 J of energy.

What is the efficiency of the crane?

A 20 % **B** 30 % **C** 40 % **D** 50 %

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 12

12	**D**

QUESTION · 5054_s09_qp_1.pdf · source page 7 of 20

13 A crane lifts a load of 1000 N through a vertical height of 3.0 m in 10 s. The input power to the crane is 500 J / s.

What is the efficiency of the crane?

A 0.17 **B** 0.50 **C** 0.60 **D** 0.67

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 13

13	**C**

10 Fig. 10.1 shows the horizontal forces as a cyclist travels forwards.

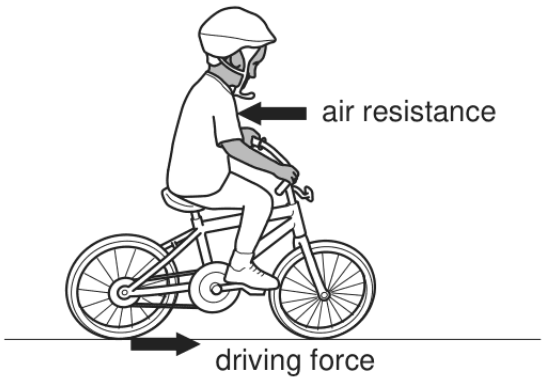


Fig. 10.1

The cyclist produces the driving force that acts on the back wheel.
In this question, you may ignore any frictional force acting on the front wheel.

(a) The bicycle accelerates until a constant speed is reached.

(i) Describe how the size of the air resistance changes during this time. [2]

(ii) Compare the sizes of the two horizontal forces when the bicycle is accelerating. [1]

(b) The total mass of the bicycle and the cyclist is 75 kg. At one instant, the speed of the bicycle is 4.0 m/s, the driving force is 30 N and the air resistance is 20 N.

Calculate

(i) the total kinetic energy of the bicycle and the cyclist, [3]

(ii) the acceleration of the bicycle and the cyclist. [2]

(c) As the bicycle moves, energy is transmitted from the pedals to the back wheel.

Fig. 10.2 shows what happens to the energy input to the pedals.

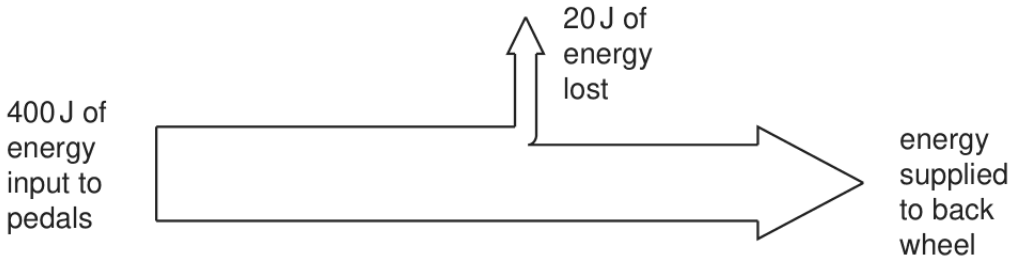


Fig. 10.2

(i) As energy is transmitted to the back wheel, some is lost. Explain how this happens. [2]

(ii) Calculate the efficiency of the bicycle in transmitting energy from the pedals to the back wheel. [2]

(d) Some bicycles are made from low density materials. Explain why this is an advantage. [3]

10 (a) (i) air resistance increases (as speed increases) B1
(at constant speed) becomes equal to driving force/applied force etc. B1

(ii) driving force (forward force) larger (than air resistance/backwards force) B1

(b) (i) (E =) ½ mv^2 algebraic formula C1
½ × 75 × 4^2 C1
600 J A1

(ii) (a =) F/m algebraic seen or 10 (N) used as force C1
0.13 m/s^2 A1

(c) (i) friction (in chain/axles) or rubbing of surfaces B1
heat or thermal energy produced B1

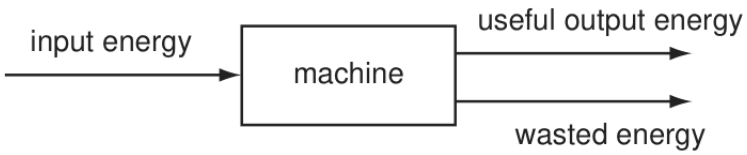
(ii) (efficiency = useful) energy output/energy input algebraic or numerical or 380 seen C1
0.95 or 95% A1

(d) lower mass/weight of cycle B1

less force needed	same acceleration/get up hill/to stop cycle/lift cycle	
greater acceleration/easier to acc.	for same force or $F = ma$ quoted	
less energy/work (input)	to go uphill/due to less friction/½ mv^2 less	
less kinetic energy	less stopping distance/less force to stop	
more efficient/less energy wasted	going uphill/less friction	
less friction	easier turn handlebars/higher (top) speed	
less pressure (on ground)	sinks less into ground	M1A1 [15]

QUESTION · 5054_w09_qp_1.pdf · source page 7 of 20

12 The diagram shows energy transfer through a machine.



What is the efficiency of the machine?

A $\frac{\text{input energy}}{\text{useful output energy}}$

B $\frac{\text{useful output energy}}{\text{input energy}}$

C $\frac{\text{useful output energy}}{\text{wasted energy}}$

D $\frac{\text{wasted energy}}{\text{input energy}}$

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 12

12	**B**

For Examiner's Use

2 Fig. 2.1 shows a motor used to operate a lift (elevator). There is a man in the lift.

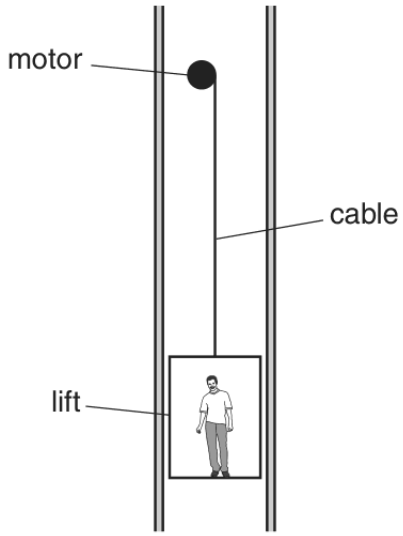


Fig. 2.1 (not to scale)

The input power to the motor is 6200 watts.

(a) Define the *watt*.

.....

.....[2]

(b) The lift and the man have a total mass of 580 kg. The lift moves up a distance of 12 m in 15 s.

(i) Calculate the tension in the cable when the lift is moving at constant speed.

tension = ...[1]

(ii) Calculate the increase in potential energy of the lift and man.

potential energy = ...[2]

(iii) Calculate the efficiency of the motor.

efficiency = ...[2]

2 **(a)** energy/time C1
1 joule in one second A1 [2]

(b) **(i)** 5800 N or 5684 N or 5700 N B1 [1]

(ii) *mgh* algebraic, words or numerical (i.e. 580 ×10 × 12) C1
69 600 J or 70 000 J or 68 208 J or 68 000 J A1 [2]

(iii) (efficiency =) output power or energy/input power or energy algebraic or numerical or 93 000 seen or 4 640 seen C1
0.75 or 75% (accept 0.748) e.c.f. from (ii) A1 [2]

[Total: 7]

QUESTION · 5054_w10_qp_11.pdf · source page 8 of 20

15 The energy output of a generator depends on its efficiency and the energy input.

Which set of values is correct?

	efficiency	energy input /MJ	energy output /MJ
A	0.1	50	25
B	0.2	30	6
C	0.4	120	30
D	0.6	60	24

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 15

15 **B**

For Examiner's Use

3 **(a)** State the *principle of conservation of energy*.

.....

.....

..... [2]

(b) Fig. 3.1 shows a miner using a head torch powered by a battery pack. The battery pack contains the cells that power the filament lamp in the head torch.

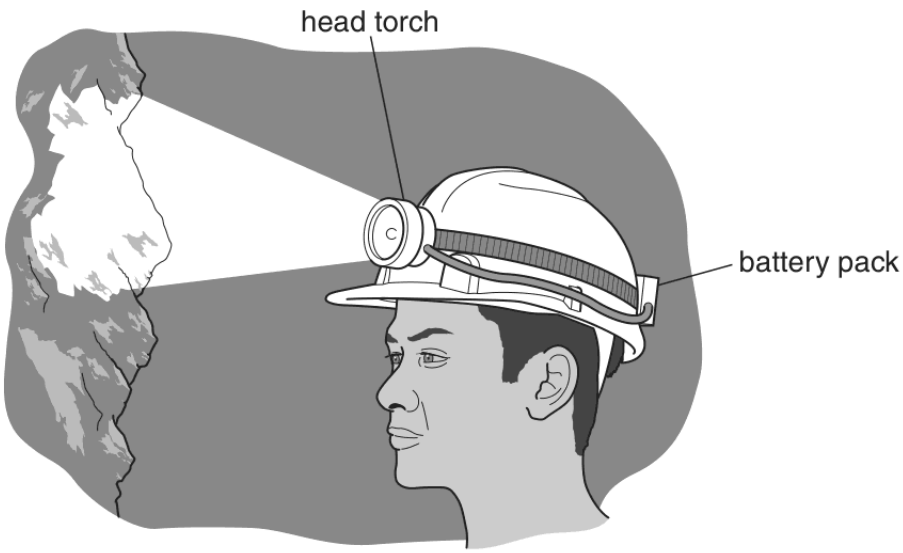


Fig. 3.1

(i) State the main energy change that takes place in the cells.

.....

..... [2]

(ii) State two different forms of energy that are produced in the filament lamp.

1. .....

2. .....

[2]

(c) An identical battery pack powers a different head torch that uses light-emitting diodes (LEDs) instead of the filament lamp. The two torches are equally bright but the efficiency of the torch using the LEDs is much higher.

State how the amounts of the different forms of energy produced in the LEDs compare with the amounts produced in the filament lamp.

.....

.....

..... [2]

3	**(a)**	energy cannot be created/destroyed (nb. only one required)		B1	
		energy cannot be destroyed or created (i.e. the other one as well) **or** (merely) transformed **or** total energy in an isolated system is constant		B1	[2]
	(b) **(i)**	chemical (potential) **at beginning**		B1	
		to electrical (and heat) **at end**	others present: max 1	B1	[2]
	(ii)	light		B1	
		heat/thermal/internal	others present: max 1	B1	[2]
	(c)	less heat; same light		B2	
		or less chemical/electrical; less heat		B2	
		or less chemical/electrical; same light		B2	[2]

QUESTION · 5054_w10_qp_12.pdf · source page 8 of 16

18 The energy output of a generator depends on its efficiency and the energy input.

Which set of values is correct?

	efficiency	energy input / MJ	energy output / MJ
A	0.1	50	25
B	0.2	30	6
C	0.4	120	30
D	0.6	60	24

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 18

18 **B**

QUESTION · 5054_s11_qp_11.pdf · source page 7 of 16

14 A crane lifts a weight of 1000 N through a vertical height of 30 m.

It uses 60 000 J of energy.

What is the efficiency of the crane?

A 20 % **B** 30 % **C** 50 % **D** 200 %

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 14

14	**C**

3 Fig. 3.1 shows part of a small hydroelectric system that generates electricity from moving water.

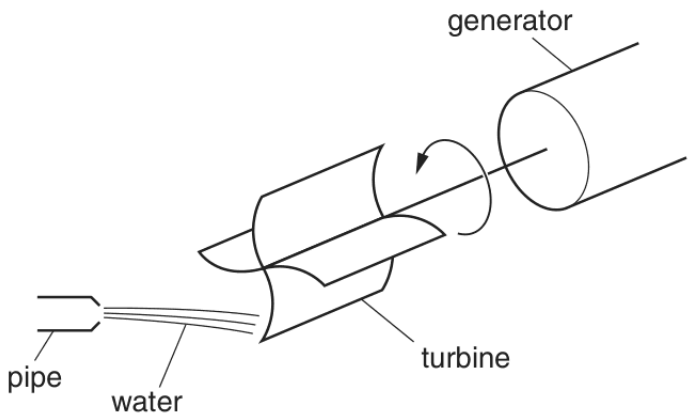


Fig. 3.1

(a) Every minute, water with kinetic energy 14 000 J emerges from the pipe. The water turns a turbine that is connected to the generator.
Half of the kinetic energy of the water is given to the generator.

(i) Calculate the power input to the generator.

power = ... [3]

(ii) Suggest a reason why only some of the kinetic energy of the water is given to the generator.

.....

..... [1]

(b) Hydroelectric systems use sources of renewable energy.

(i) State what is meant by *renewable energy*.

.....

..... [1]

(ii) State one source of **non**-renewable energy.

..... [1]

3	**(a)**	**(i)**	7000 (J) seen **or** 50% **used** somewhere	C1	
			(*P* =) *E*/*t* // 14000/*t* // 7000/*t* seen // 7000 J/minute // 420 000 J/hour	C1	
			120 W	A1	
		(ii)	water after hitting turbine still moves // has KE/energy/velocity // energy lost due to friction // friction **and** location // heat/internal energy **and** location // water misses turbine	B1	
	(b)	**(i)**	can be replaced/made // will not run out	B1	
		(ii)	coal, oil, gas, peat, nuclear, uranium (**not** solar)	B1	[6]

QUESTION · 5054_s11_qp_12.pdf · source page 7 of 16

14 A crane lifts a weight of 1000 N through a vertical height of 30 m.

It uses 60 000 J of energy.

What is the efficiency of the crane?

A 20 % **B** 30 % **C** 50 % **D** 200 %

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 14

14	**C**

QUESTION · 5054_w11_qp_11.pdf · source page 5 of 20

10 The efficiency of an electrical generator is 65 %.

The energy input to the generator is 12 kJ.

What is the useful energy output?

A 4.2 kJ **B** 5.4 kJ **C** 7.8 kJ **D** 780 kJ

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 10

10	**C**

QUESTION · 5054_w11_qp_12.pdf · source page 5 of 20

8 The efficiency of an electrical generator is 65 %.

The energy input to the generator is 12 kJ.

What is the useful energy output?

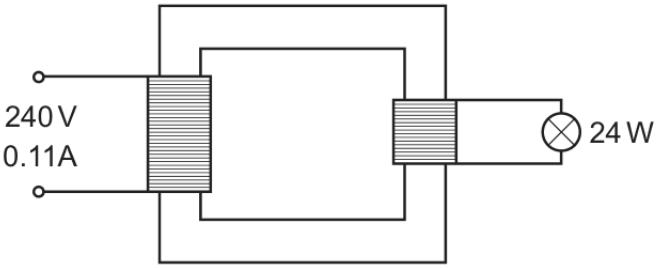
A 4.2 kJ **B** 5.4 kJ **C** 7.8 kJ **D** 780 kJ

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 8

8	**C**

QUESTION · 5054_s12_qp_11.pdf · source page 6 of 16

14 A transformer connected to a 240 V mains supply is used to light a 24 W lamp.



The input current to the transformer is 0.11 A and the input voltage is 240 V. The useful output power of the transformer is 24 W.

What is the efficiency of the transformer?

A 0.10 **B** 0.91 **C** 1.1 **D** 2.6

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 14

14	**B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1a shows a room heater. Fig. 9.1b is a diagram of the electric circuit of the heater.

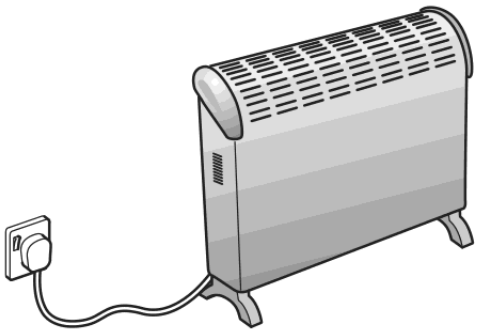


Fig. 9.1a

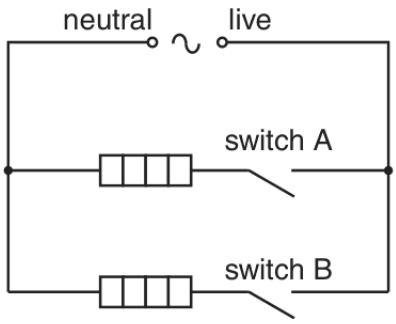


Fig. 9.1b

The fuse has not been drawn on the circuit diagram in Fig. 9.1b.

(a) (i) On Fig. 9.1b, draw the symbol for a fuse in the correct position. [2]

(ii) State the part of the room heater to which the earth wire is connected.

.....[1]

(iii) The earth wire reduces the chance of an electric shock if a fault develops in the room heater.

1. State one fault that causes an electric shock when a person uses the room heater without an earth connection.

.....

.....[1]

2. Explain how using an earth connection prevents an electric shock.

.....

.....

.....[2]

(b) (i) This type of room heater is very efficient. Explain what this means.

.....

.....[1]

(ii) The room heater is a convector heater. Describe and explain how thermal energy (heat) passes around a room by convection.

.....

.....

.....

.....

.....[3]

(c) Fig. 9.2 shows the power output of the room heater when each switch is closed.

	power / W
switch A only closed	600
switch B only closed	
both switches closed	2100

Fig. 9.2

(i) Determine the power output of the room heater when only switch B is closed.

power output = ... [1]

(ii) The room heater is used with both switches closed for 2.5 hours.

Calculate the energy output of the room heater

1. in kilowatt-hours,

energy = ... kW h [2]

2. in joules.

energy = ... J [2]

Section B

9	**(a)**	**(i)**	fuse symbol correct	B1	
			in live wire before junction of two elements	B1	
		(ii)	the (metal) case/outside	B1	
		(iii)	**1.** live wire touches case; live touches person	B1	
			2. current goes to earth; current does not go through the person	B1	
			fuse blows	B1	
	(b)	**(i)**	most of the energy output is useful/heat; little energy is wasted;	B1	
		(ii)	hot air rises (**not** heat rises)	B1	
			density of hot air is lower	B1	
			convection current mentioned OR hot air rises and cold air falls	B1	
	(c)	**(i)**	1500 W	B1	
		(ii)	**1.** conversion to kW seen on any power; 2.1 (kW) seen	C1	
			5.25; 5.2; 5.3 (kW h)	A1	
			2. $E = P \times t$ in any form, algebraic or using any power or time e.g. 600 × 2.5, 600 × 150	C1	
			1.89×10^7 (J) OR 3.6×10^6 × **(c)(ii)1.**	A1	[15]

QUESTION · 5054_s12_qp_12.pdf · source page 6 of 16

14 What is efficiency?

A $\frac{\text{total energy input}}{\text{useful energy output}}$

B $\frac{\text{total power input}}{\text{useful energy output}}$

C $\frac{\text{useful energy output}}{\text{total energy input}}$

D $\frac{\text{useful power output}}{\text{total energy input}}$

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 14

14	**C**

3 Fig. 3.1 shows an electric motor being used on a building site to lift bricks.

For Examiner's Use

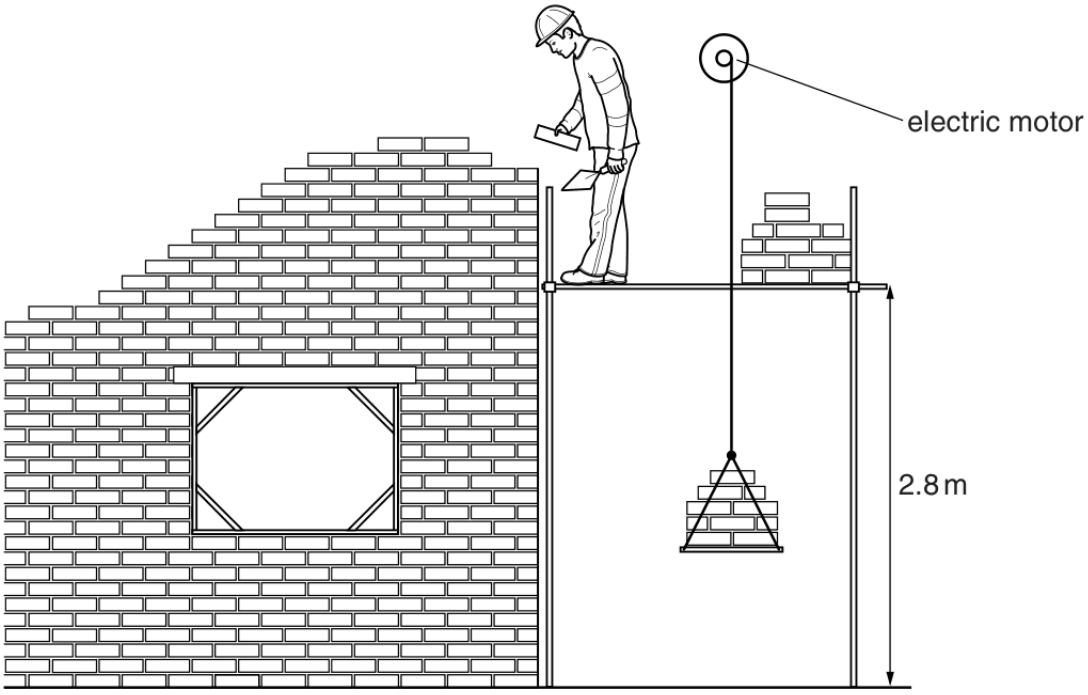


Fig. 3.1

The gravitational field strength is 10 N / kg. The motor lifts bricks of total mass 54 kg through a height of 2.8 m in 3.0 s.

(a) Calculate

(i) the work done in lifting the bricks,

work done = ... [2]

(ii) the useful output power of the motor.

power = ... [2]

(b) Suggest two reasons why the electrical power supplied to the motor is greater than its useful output power.

1. .....

2. .....
[2]

For Examiner's Use

(c) (i) Fig. 3.2 shows a battery and an electric motor. The battery is used to power the motor. Complete Fig. 3.2 to show the circuit diagram of a circuit that is used to determine the power supplied to the electric motor.

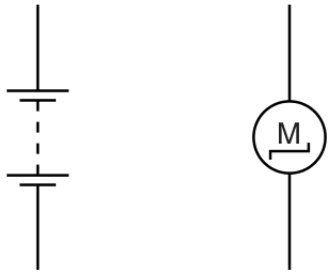


Fig. 3.2 [2]

(ii) State how the power supplied is calculated.

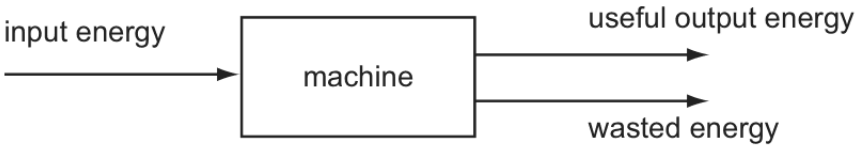
.....

..... [1]

3	**(a)**	**(i)**	(WD =)*mgh* **or** 54 × 10 × 2.8	C1	
			1500/1510/1512 J	A1	
		(ii)	(*P* =) WD/*t* or *E*/*t* **or** 1500/3 **or** 1510/3 **or** 1512/3	C1	
			500/503/504 W	A1	
	(b)		any **two** of: also lifting board/rope heat in motor/wires/cable friction with something named e.g. axle/spindle/air	B2	
	(c)	**(i)**	power supply, motor and ammeter in series (**ignore** series voltmeter and other components)	B1	
			voltmeter to measure voltage across motor	B1	
		(ii)	current (reading) × voltage (reading) or VI	B1	[9]

QUESTION · 5054_s13_qp_11.pdf · source page 6 of 20

12 The diagram shows the energy transfer through a machine.



The machine is 50% efficient.

Which equation is correct?

A input energy = useful output energy

B useful output energy = input energy + wasted energy

C wasted energy = input energy + useful output energy

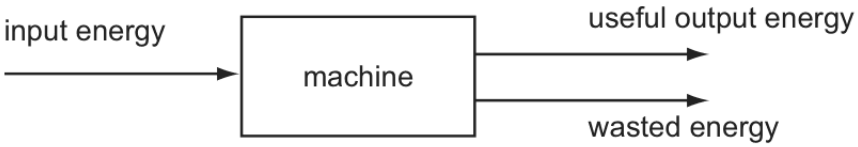
D wasted energy = useful output energy

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 12

12	**D**

QUESTION · 5054_s13_qp_12.pdf · source page 7 of 16

15 The diagram shows the energy transfer through a machine.



The machine is 50% efficient.

Which equation is correct?

A input energy = useful output energy

B useful output energy = input energy + wasted energy

C wasted energy = input energy + useful output energy

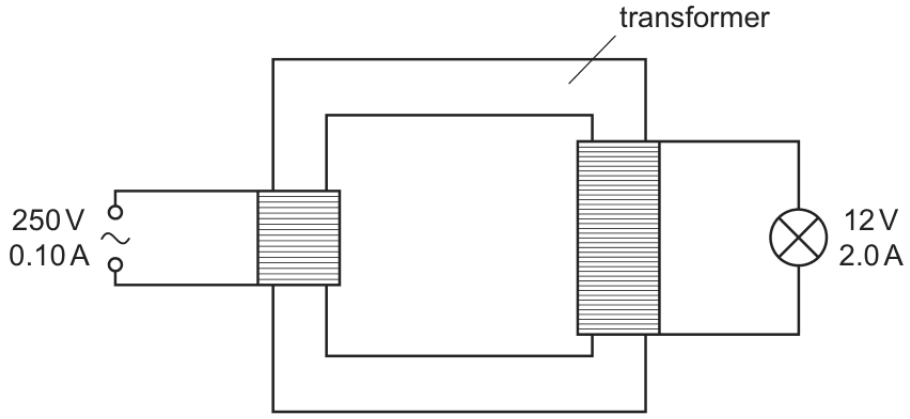
D wasted energy = useful output energy

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 15

15 **D**

QUESTION · 5054_w13_qp_11.pdf · source page 14 of 20

31 A transformer is used to operate a 12 V lamp from a 250 V mains supply.



The mains current is 0.10 A. The current in the lamp is 2.0 A.

What is the efficiency of the transformer?

A 0.048 **B** 0.050 **C** 0.96 **D** 1.04

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 31

31	**C**

QUESTION · 5054_w13_qp_12.pdf · source page 7 of 20

13 When exposed to light, a solar cell generates electrical energy.

Two solar cells of equal area are tested. Each cell is exposed to sunlight of different brightness for the same time.

Which statement could apply to the more efficient cell?

A It generates less electrical energy from light of greater brightness.

B It generates less electrical energy from light of the same brightness.

C It generates the same electrical energy from light of greater brightness.

D It generates the same electrical energy from light of lower brightness.

MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 13

13	**D**

For Examiner's Use

10 Fig. 10.1 shows an electric train of mass 5.5×10^5 kg.

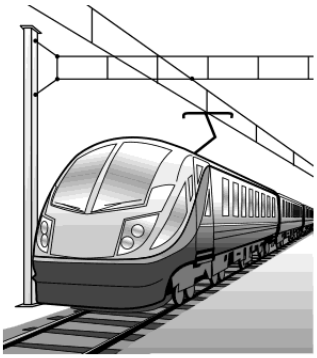


Fig. 10.1

The train is initially at rest. The electric motor exerts a constant force and the train accelerates. Its acceleration decreases until the train reaches a speed of 40 m/s.

The train then continues at this constant speed.

(a) (i) On the axes in Fig. 10.2, sketch a speed-time graph for the train.

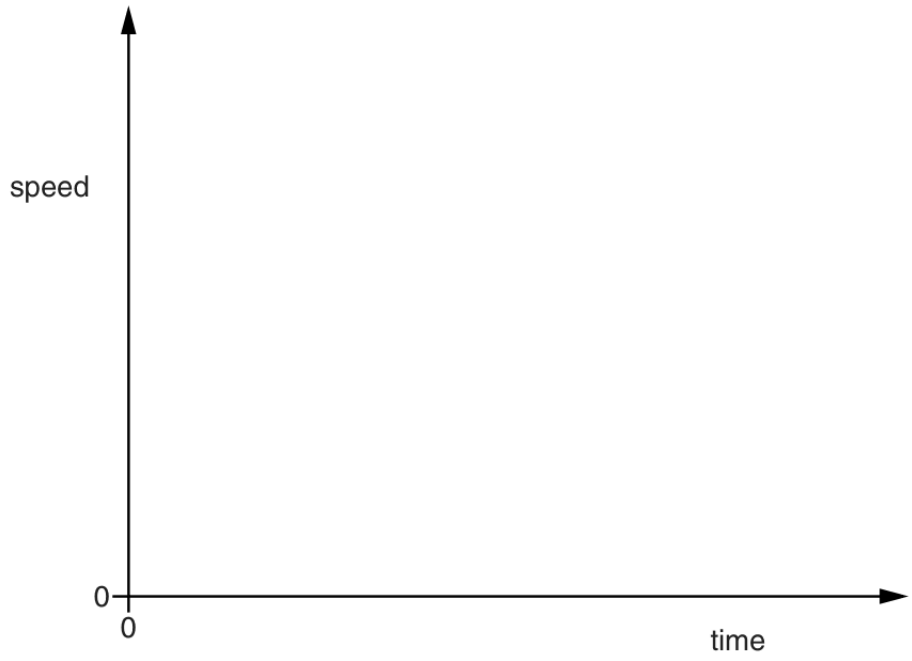


Fig. 10.2

[3]

(ii) Describe how the speed-time graph for the train is used to determine the distance travelled by the train when it is accelerating.

.....

.....

.....[2]

For Examiner's Use

(b) Explain, in terms of the forces acting,

(i) why the acceleration of the train decreases,

.....

.....

.....[2]

(ii) why the train eventually reaches constant speed.

.....

.....[1]

(c) **(i)** Calculate the kinetic energy of the train when it is travelling at a speed of 40 m/s.

kinetic energy = ...[3]

(ii) As the train accelerates to 40 m/s, electrical energy supplied to the motor is converted into kinetic energy of the train. The efficiency of this process is 0.40 (40%).

Calculate the electrical energy supplied to the motor.

energy = ...[2]

(iii) The electrical energy is generated in an oil-fired power station. Chemical energy in oil is converted into the electrical energy supplied to the motor.

Suggest two places where energy is lost as heat in this process.

1. .....

2. .....
[2]

10	**(a)**	**(i)**	start at origin **and** not horizontal	B1	
			gradient (gradually) decreasing (**ignore** sudden decrease) (not if part of curve above horizontal section)	B1	
			final horizontal section (≥ 1 cm) (not if v is shown as ≠ 40 m/s)	B1	
		(ii)	area **under** the graph **or** count squares **under** graph between t = 0 and horizontal section **or** when speed is changing **or** calculate equivalent distance to 1 cm^2 (after counting squares)	M1 A1	[5]
	(b)	**(i)**	friction/air resistance increases (as speed increases)	B1	
			resultant force decreases (**not** if driving force decreases)	B1	
		(ii)	(air resistance increases until) net force becomes zero **or** forces balance **or** air resistance and driving/forward force are in equilibrium/balanced/equal	B1	[3]
	(c)	**(i)**	(KE =) $\frac{1}{2}mv^2$	C1	
			$\frac{1}{2} \times 5.5 \times 10^5 \times 40^2$	C1	
			4.4×10^8 J	A1	
		(ii)	(total energy input =) useful energy output efficiency **or** efficiency = useful power output/total power input **or** $4.4 \times 10^8/0.40$	C1	
			1.1×10^9 J	A1	
		(iii)	two valid examples e.g. furnace/boiler/turbines/generator/coils/cooling water/cooling towers/heat exchanger/transformer/chimney/waste gases/ transmission cables/lines/wires (**ignore** power station/all parts of motor)	B2	[7]

[Total: 15]

2 Fig. 2.1 shows part of a hydraulic jack used to lift the front of a car.

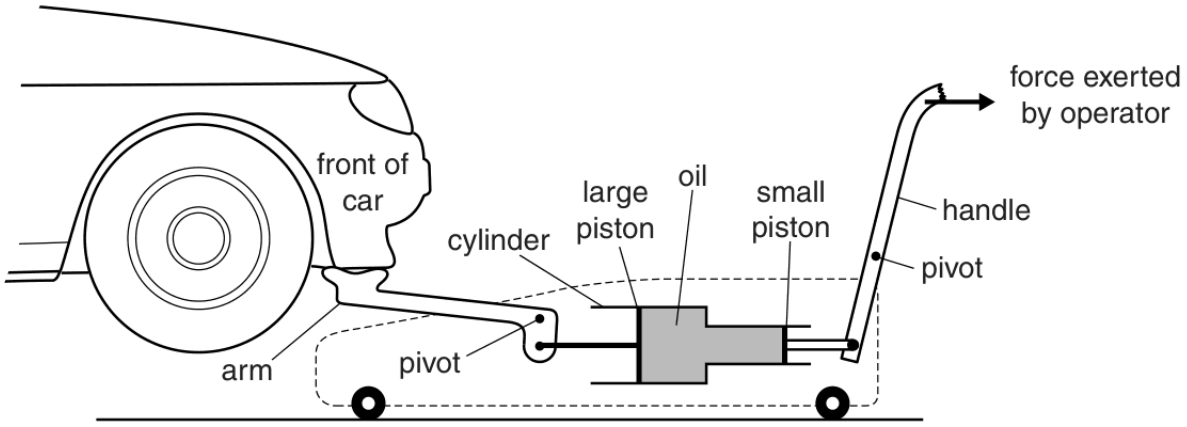


Fig. 2.1 (not to scale)

The operator pulls the handle and causes a force of 50 N to act on the small piston. The force exerted by the oil on the large piston increases by *F*. The large piston moves and rotates the arm about the pivot. This raises the front of the car.

The cross-sectional area of the small piston is 1.5 cm^2.
The cross-sectional area of the large piston is 5.0 cm^2.

(a) Calculate

(i) the pressure in the oil caused by the force on the small piston,

pressure = ... [2]

(ii) the value of *F*.

F = ... [1]

(b) Explain why the large piston moves through a shorter distance than the small piston.

.....

.....

..... [1]

(c) The efficiency of the jack is 75%. Explain what is meant by *efficiency*.

.....

..... [2]

2 **(a)** **(i)** (P=) F / A algebraic in any form or numerical [C1]
$33\,N/cm^2$ **or** $3.3 \times 10^5\,N/m^2$ [A1]

(ii) 170 N **or** 167 N **or** 166.7 **N or (i)** × 5 with unit [B1]

(b) volume (of oil) remains the same
or oil passes from small(er) to large(r) area
or work = Fd **and** force large so distance small [B1]

(c) output ÷ input **or** fraction or percentage of work mentioned [C1]
complete definition, e.g. **useful** work obtained ÷ (total) work put in [A1]

[6]

3 Two different kettles are used to heat water, as shown in Fig. 3.1.

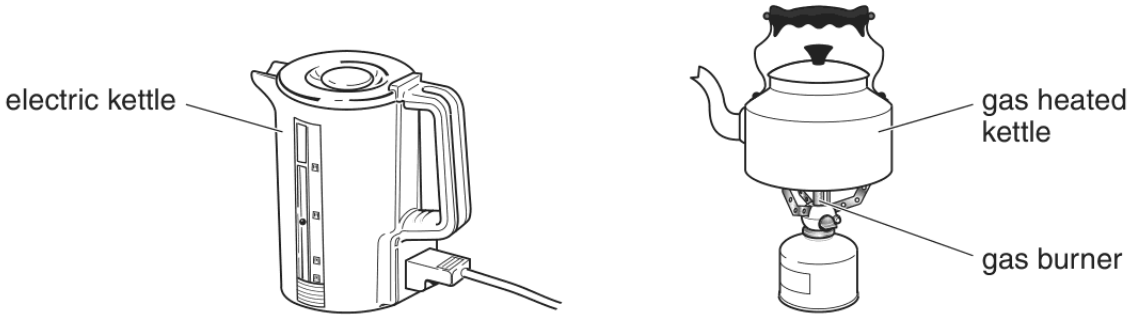


Fig. 3.1

Data for the two kettles is shown in Fig. 3.2.

	energy supplied to the kettle in one minute/J	thermal energy (heat) supplied by the kettle to the water in one minute/J
electric kettle	120 000	95 000
gas heated kettle	130 000	90 000

Fig. 3.2

(a) (i) Calculate the efficiency of the electric kettle.

efficiency = ... [2]

(ii) Calculate the useful power of the gas heated kettle.

power = ... [2]

(b) Both kettles contain the same mass of water, at the same initial temperature.

State and explain which kettle brings the water to boiling point first.

.....

.....

..... [1]

(c) The boiling water produces steam at 100 °C.

State one difference between molecules in the steam and molecules in the boiling water.

.....

.....

..... [1]

3 **(a)** **(i)** (efficiency =) useful energy ÷ input energy C1
or 95 000/120 000 (×100)
0.79(17) **or** 79(.17)**%** A1

(ii) (*P*=) energy/time **or** 90 000 / 60 C1
1500 W A1

(b) electric kettle **and** B1
- **more** energy / heat **per minute** output / into water / supplied
- more power output

transfers heat / energy faster / at a faster rate

(c) steam molecules have more potential energy; further apart; smaller force / bonds between molecules; have latent heat; more random arrangement B1

QUESTION · 5054_w15_qp_11.pdf · source page 6 of 16

14 The input power to a lamp is 6.0 W. The lamp wastes 2.7 J of energy in 3.0 s.

What is the efficiency of the lamp?

A 0.15 **B** 0.45 **C** 0.55 **D** 0.85

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 14

14	**D**

2 Fig. 2.1 shows an electric motor lifting a load.

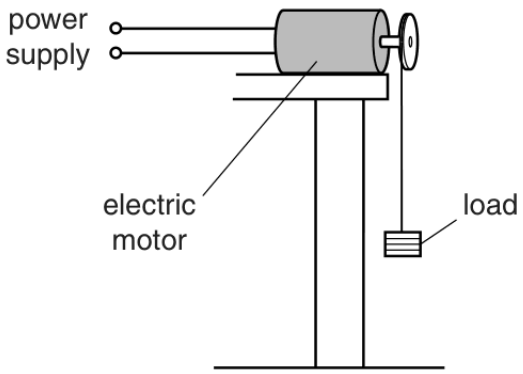


Fig. 2.1

The load, of weight 5.0 N, is raised through a vertical height of 3.5 m at a constant speed. The efficiency of the electric motor is 0.65 (65%).

(a) Calculate the increase in gravitational potential energy of the load.

gravitational potential energy = ...[2]

(b) (i) State the formula that relates efficiency, energy input and useful energy output.

.....

.....[1]

(ii) Calculate the energy input to the motor.

energy input = ...[2]

(c) Suggest one reason why the efficiency of the motor is less than 1.0 (100%).

.....

.....[1]

2 **(a)** (PE =) mgh **or** Fd **or** 5 × 3.5 C1
17.5 J **or** 17 J **or** 18 J A1

(b) **(i)** (efficiency = useful) energy output / energy input B1
in any form but all three quantities must be mentioned
if efficiency is not the subject of the equation

(ii) 17.5 / 0.65 **or** 17.5 / 65 C1
or 0.65 / 65 = **(a)** / energy input
26.9 J **or** 27 J A1

(c) due to friction (in bearings of motor) B1
or due to (electrical) resistance (in motor)
or air resistance acts
or thermal energy / heat produced / lost (in resistance of motor / due to friction)

3 A large power station burns oil to heat water in a boiler.

(a) State the name of the form of energy stored in the oil before it is burnt.

.....[1]

(b) (i) State and explain whether oil is a renewable or a non-renewable energy source.

.....

.....

.....[1]

(ii) Describe one environmental issue caused by burning oil to generate electricity in a power station.

.....

.....[1]

(c) (i) State what is meant by *efficiency*.

.....

.....[1]

(ii) The efficiency of the power station is 0.38 (38%). Its electrical power output is 1.9×10^9 W.

Calculate

1. the input power to the power station,

input power = ...[2]

2. the energy wasted in 2.0 hours.

energy wasted = ...[2]

3	**(a)**	chemical (potential energy)		B1	
	(b)	**(i)**	non-renewable **and** oil / it is not replaced / will run out	B1	
		(ii)	acid rain **or** produces CO_2 **or** warms lakes / rivers / sea **or** global warming **or** greenhouse effect	B1	
	(c)	**(i)**	<u>useful</u> energy output / (total) energy input **or** power for energy twice	B1	
		(ii)	**1** $1.9 \times 10^9/0.38$ **or** $1.9 \times 10^9 \times 100/38$	C1	
			5.0×10^9 W	A1	
			2 (E =)Pt **or** $0.62 \times 5.0 \times 10^9 \times 2.0$ ($\times$ 3600) **or** $(5.0 - 1.9) \times 10^9$ etc.	C1	
			2.2×10^{13} J	A1	[8]

QUESTION · 5054_w16_qp_12.pdf · source page 7 of 16

16 Which expression gives the efficiency of an a.c. generator?

A $\frac{\text{electrical energy output}}{\text{total energy input}}$

B $\frac{\text{electrical energy output}}{\text{thermal energy output}}$

C $\frac{\text{total energy output}}{\text{electrical energy input}}$

D $\frac{\text{thermal energy output}}{\text{total energy input}}$

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 16

16	**A**

7 Fig. 7.1 shows part of a d.c. electric motor.

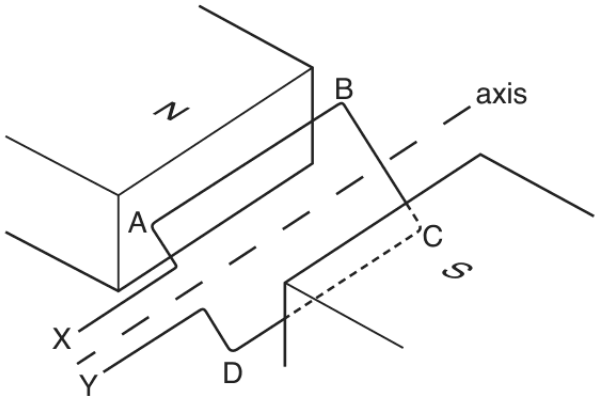


Fig. 7.1

(a) A coil of wire ABCD is placed between the poles of a magnet. Ends X and Y are connected to a commutator and a battery.

(i) State why there is a force on side AB of the coil.

.....

.....[1]

(ii) Describe how the commutator keeps the coil rotating in one direction.

.....

.....

.....

.....

.....[2]

(b) The current in the motor is 2.0 A and the battery has an electromotive force (e.m.f.) of 12 V. In a time of 8.0 s, the motor does 140 J of work.

Calculate

(i) the energy supplied to the motor in 8.0 s,

energy supplied = ...[2]

(ii) the efficiency of the motor.

efficiency = ...[1]

Question	Answer	Marks
7(a)(i)	current in coil (at right angles) in a magnetic field (of magnet) **or** left-hand rule mentioned	**B1**
7(a)(ii)	reverses / changes direction of current (in coil)	**B1**
	reverses current every half turn / when coil is vertical **or** reverses forces (on side AB / CD) **or** keeps forces in same direction for wire on one side	**B1**
7(b)(i)	(E =)VIt or 2 × 12 × 8 **or** E = Pt **and** P = VI **or** E = VQ **and** Q = It	**C1**
	190 **or** 192 J	**A1**
7(b)(ii)	73% **or** 0.73	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a large container ship travelling at constant speed in a straight line.

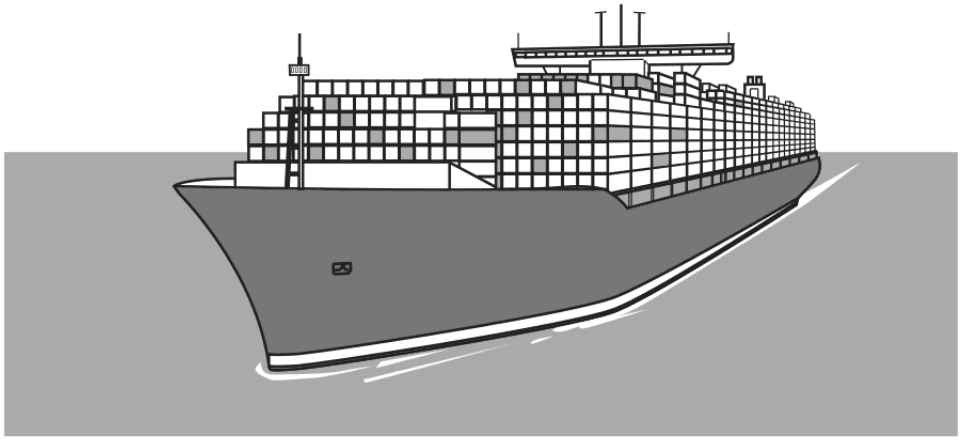


Fig. 9.1

The resistive force acting on the ship is 2.8×10^6 N.

(a) The speed of the ship is 9.7 m/s.

(i) Calculate the work done against the resistive force on the ship in 1.0 s.

work done = ... [2]

(ii) The engines are powered by oil.

State the energy transfer that is taking place when the ship is travelling at constant speed.

.....

..... [2]

(b) The mass of the ship is 2.2×10^8 kg. The engines are switched off and the resistive force causes the ship to decelerate.

(i) Calculate the initial deceleration of the ship.

deceleration = ... [2]

(ii) As the speed of the ship decreases, its deceleration changes.

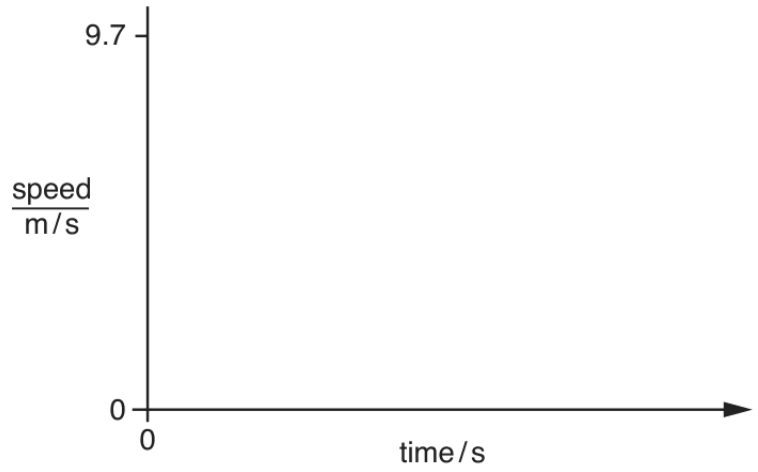
1. Suggest and explain how the deceleration changes.

.....

.....

..... [2]

2. On Fig. 9.2, sketch a possible speed-time graph for the ship as it decelerates to rest.



[2]

Fig. 9.2

3. Explain how the distance travelled by the ship may be determined from the speed-time graph.

.....

.....

..... [2]

(c) When the ship is travelling at a different speed, energy is being supplied to the engines at a rate of 33 MJ/s. The efficiency of the engines is 0.36 (36%).

(i) State a relationship that defines *efficiency*.

.....

..... [1]

(ii) Calculate the rate at which energy is wasted in the engines.

rate at which energy is wasted = ... [2]

Section B

Question	Answer	Marks
9(a)(i)	(WD) = Fx **or** $2.8 \times 10^6 \times 9.7$	**C1**
	2.7×10^7 J	**A1**
9(a)(ii)	from chemical (potential energy)	**B1**
	to thermal (energy) / heat	**B1**
9(b)(i)	(a =)F / m **or** $2.8 \times 10^6 / 2.2 \times 10^8$	**C1**
	0.013 m / s^2	**A1**
9(b)(ii)	**1** deceleration / it decreases	**B1**
	resistive force decreases **or** resistive force depends on speed	**B1**
	2 curve/line from 9.7 to zero **and** gradient negative (**allow** zero at end)	**B1**
	magnitude of gradient decreasing	**B1**
	3 area mentioned	**B1**
	area <u>under</u> line / curve **or** convert cm^2 (of graph paper) to distance **or** in terms of the scales	**B1**
9(c)(i)	(efficiency =) useful energy output / total energy input **or** useful power output / total power input	**B1**
9(c)(ii)	33×0.64 **or** 12 MJ / s	**C1**
	21 MJ / s	**A1**

4 A 75 W filament lamp in an office is accidentally left switched on for 63 hours when the office is closed.

(a) (i) Calculate the electrical energy that is supplied to this lamp in this time.

electrical energy = ...[3]

(ii) The efficiency of the lamp is 0.12 (12%).

Calculate the energy released by the lamp in the form of light.

light energy = ...[1]

(b) The power station that generates the electricity supplied to the office uses oil as its fuel.

(i) State and explain whether oil is a renewable or a non-renewable source of energy.

.....

.....

.....[1]

(ii) Discuss one environmental consequence of using oil as a source of energy.

.....

.....

.....[2]

Question	Answer	Marks
4(a)(i)	(E =) Pt **or** 75×63 **or** $75 \times 63 \times 60$	**C1**
	$75 \times 63 \times 60 \times 60$ **or** 2.835×10^5 **or** 0.075×63	**C1**
	1.7×10^7 J	**A1**
4(a)(ii)	2.0×10^6 J	**B1**
4(b)(i)	non-renewable **and** gets used up / not replaced / will run out / finite	**B1**
4(b)(ii)	**three** separate approaches **I** / **II** / **III**	
	I greenhouse gases / CO / CO_2 emitted	**B1**
	global warming / <u>increase</u> greenhouse effect / ice caps melt	**B1**
	or II acid rain / NOx / SO_2 emitted	**B1**
	breathing difficulty / damage to buildings / acidify lakes **or** damages / kills plants / sea-life / animals / fish	**B1**
	or III oil spills	**B1**
	damages / kills plants / sea-life / animals / fish	**B1**

10 (a) (i) Describe two differences between boiling and evaporation.

1. .....

.....

.....

2. .....

.....

.....

[4]

(ii) Explain, using ideas about molecules, why thermal energy is needed to boil a liquid.

.....

.....

.....

.....[2]

(b) In one type of bathroom shower, cold water passes through a metal pipe which contains an electric heater.

The cold water is heated and emerges from the shower head.

The temperature of the cold water before heating is measured and the hot water emerging from the shower in 1.0 minute is collected in a container.

Measurements and other data are:

temperature of water before heating = 16 °C
temperature of water after heating = 37 °C
volume of water collected in 1.0 minute = $4.6 \times 10^{-3}\,m^3$
specific heat capacity of water = 4200 J / (kg °C)
density of water = 1000 kg / m^3

(i) Calculate the mass of water leaving the shower in 1.0 s.

mass = ...[2]

(ii) Calculate the thermal energy (heat) gained by the water in 1.0 s.

energy = ...[3]

(iii) To calculate the efficiency of the shower, the value in **(b)(ii)** and the value of one other quantity are needed.

1. State what other quantity is needed.

.....

.....

2. Describe how these two values are used to calculate the efficiency of the shower.

.....

.....

.....
[2]

(iv) The metal pipe that contains the electric heater is earthed.

Explain why this is necessary.

.....

.....

.....

.....[2]

Question	Answer	Marks
10(a)(i)	*Pair 1* boiling occurs at one temperature / constant temperature evaporation occurs at all / any temperature	**B1**
	Pair 2 boiling occurs within / throughout liquid or causes bubbles evaporation occurs at surface	**B1**
	Pair 3 boiling requires a heat source / affected by rate of heating evaporation causes cooling / affected by wind etc.	**B1**
	Pair 4 temperature remains constant during boiling evaporation temperature drops / liquid cools	**B1**
10(a)(ii)	(thermal energy) used to break bonds **or** work done against intermolecular forces	**B1**
	molecules move further apart **or** P.E. increases **or** (molecules) push back the atmosphere	**B1**

Question	Answer	Marks
10(b)(i)	(M=) DV in any form algebraic or numerical	**C1**
	0.077 kg / s **or** 0.077 kg	**A1**
10(b)(ii)	(E=) mcT in any form algebraic or numerical	**C1**
	37 – 16 or 21 (°C) seen	**C1**
	6800 J	**A1**
10(b)(iii)	1input / electrical energy or power mentioned	**B1**
10(b)(iii)	2 (efficiency =) output (energy / power) ÷ input (energy / power)	**B1**
10(b)(iv)	avoid electrocution / shock / current through body **or** avoid pipe being live **or** to allow a current / charges to earth / ground **or** fuse blows to stop electrocution	**B1**
	when live / heater touches metal / water	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Wind energy is a renewable energy source. A wind turbine and generator convert energy in the wind to electrical energy in a generator. Fig. 9.1 shows some wind turbines.

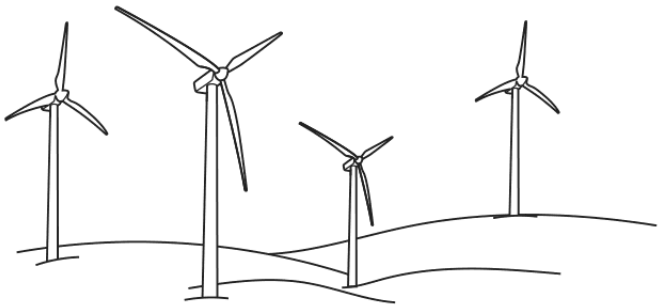


Fig. 9.1

(a) (i) State what is meant by a *renewable energy source*.

.....

.....

.....[1]

(ii) State the name of **one other** renewable energy source.

.....

.....[1]

(iii) Some methods of generating electricity cause much more global warming than wind energy.

1. State **one** method of generating electricity that causes a large amount of global warming.

.....[1]

2. Describe, in outline, how the method you have chosen causes a significant amount of global warming.

.....

.....

.....[1]

(b) During a 30 minute period, a mass of 4.2×10^7 kg of air enters the turbine with a speed of 15 m/s.

(i) Calculate the kinetic energy of the air that enters the turbine in 30 minutes.

energy = ...[2]

(ii) The electrical power output of the generator connected to the turbine is 8.4×10^5 W.

Calculate the electrical energy output from the turbine in 30 minutes.

energy output = ...[2]

(iii) Calculate the efficiency of the turbine and generator in converting the kinetic energy of the air to electrical energy.

efficiency = ...[2]

(c) The generator contains a coil that rotates between the poles of a magnet.

The coil rotates 50 times in one second and produces an alternating voltage output with a maximum value of 500 V.

(i) On the grid in Fig. 9.2, sketch a voltage–time graph showing the output voltage.

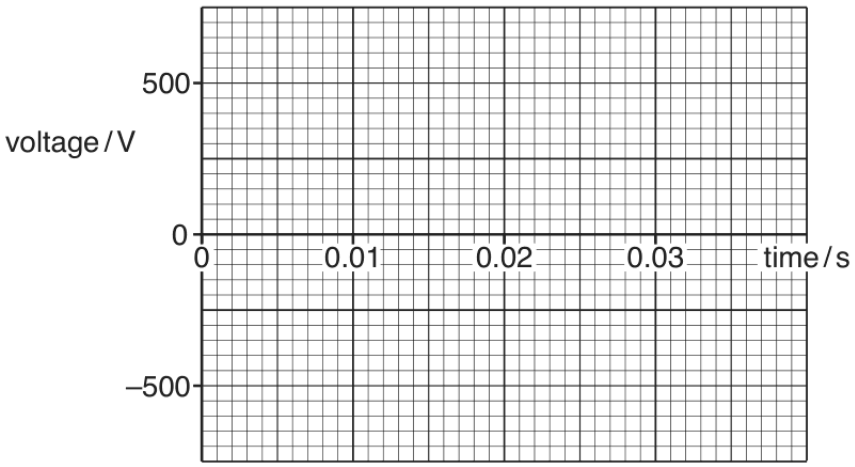


Fig. 9.2

[2]

(ii) The output voltage from the generator is stepped up by a transformer. The electrical energy then passes along a transmission line to a distant house.

Explain why

1. a high voltage is used for the transmission of electrical energy,

.....

.....

.....

.....[2]

2. a transformer is used to connect the transmission line to the house.

.....

.....

.....[1]

Question	Answer	Marks
9(a)(i)	will not run out **or** infinite **or** being replaced	**B1**
9(a)(ii)	solar / Sun, tidal, geothermal, biomass, hydroelectric, water waves	**B1**
9(a)(iii)	1 (burning) use of a fossil fuel, e.g. coal, oil, natural gas etc.	**B1**
	2 produces carbon dioxide **or** greenhouse gases (absorb infra-red from Earth)	**B1**
9(b)(i)	(E=) ½mv^2 seen in any form algebraic or numerical	**C1**
	4.7×10^9 J	**A1**
9(b)(ii)	(E=)Pt seen in any form algebraic or numerical	**C1**
	1.5×10^9 J	**A1**
9(b)(iii)	(efficiency=) (energy) output / (energy) input in any form	**C1**
	0.32 **or** 32%	**A1**
9(c)(i)	all peaks and troughs at 500 and -500 to within one small square **and** roughly sinusoidal	**B1**
	one wave takes 0.02 s within one small square	**B1**
9(c)(ii)	1 less energy loss / heat loss / power loss / more efficient	**B1**
	low(er) currents (in line) **or** thin(ner) wires can be used	**B1**
	2 step down transformer **or** voltage reduced (to 240 V) **or** otherwise voltage dangerous	**B1**

QUESTION · 5054_w18_qp_11.pdf · source page 5 of 16

9 An electric motor, connected to the mains, is used to lift bricks to the top of a building.

There is a fuse in the plug.

How can the efficiency of the motor be increased?

A Increase the friction in the motor.

B Reduce the energy losses in the motor.

C Use a fuse with a higher current rating.

D Use a fuse with a lower current rating.

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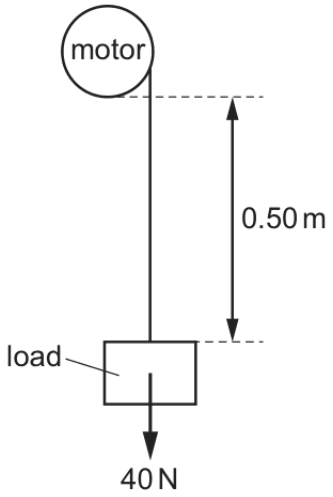
9	B	**1**

QUESTION · 5054_w18_qp_12.pdf · source page 6 of 16

12 A motor is used to lift a load 0.50 m vertically, as shown.



The load weighs 40 N. The power of the motor is 20 W and the system is 25% efficient.

How long does it take to raise the load?

A 0.040 s **B** 0.25 s **C** 4.0 s **D** 40 s

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 12

12	C	**1**

7 Fig. 7.1 shows part of a torch. The torch does not contain a battery.

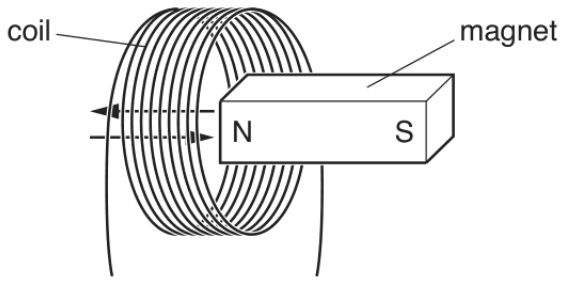


Fig. 7.1

The torch is shaken and this causes the magnet to move backwards and forwards through the coil.

(a) (i) Explain why an electromotive force (e.m.f.) is induced in the coil when the magnet moves.

.....

..... [2]

(ii) State one way to increase the e.m.f. induced.

.....

..... [1]

(b) The current in the coil is used to produce light from a light-emitting diode (LED).

(i) In the space below, draw the circuit symbol for an LED.

[1]

(ii) An LED is a *more efficient* source of light than a filament lamp.

Explain what this statement means.

.....

..... [1]

[Total: 5]

Question	Answer	Marks
7(a)(i)	magnetic field / flux mentioned	**B1**
	change in field / flux (in coil) **or** field (lines) cut coil / wire	**B1**
7(a)(ii)	more turns **or** stronger magnet **or** move magnet faster	**B1**
7(b)(i)	correct symbol	**B1**
7(b)(ii)	converts more of the input energy / power into light / useful energy / power **or** less energy / power wasted with same input energy / power **or** in same time **or** same input power / energy / current gives more light / less energy wasted / power wasted / heat / greater power out **or** less input power / energy gives same output power / energy	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 8 of 16

19 Three solar panels have a total area of 7.2 m^2. Together, they produce 1.0 kW of power.

An area of 1.0 m^2 of each panel receives 0.80 kJ of energy from the Sun in 1.0 s.

What is the efficiency of the solar panels?

A 11% **B** 17% **C** 52% **D** 80%

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 19

19	B	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 Fig. 8.1 shows a lamp from a car. It contains two metal filaments.

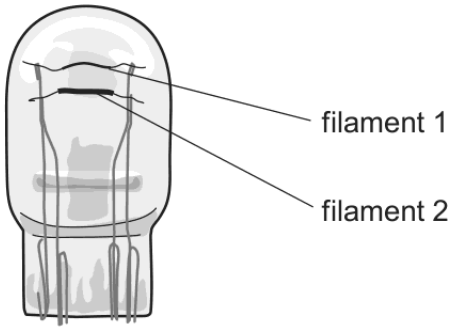


Fig. 8.1

(a) (i) Complete the boxes to describe the transfer of energy that takes place when the lamp is switched on.

..... energy	→	..... and ..... energy

[3]

(ii) The efficiency of the metal filament lamp is less than 10%.

State what is meant by *efficiency*.

.....

..... [2]

(b) The two filaments are usually connected in parallel to a car battery.

A student investigates what happens when the filaments are connected in series, rather than in parallel. He uses the same battery for the investigation.

State whether the current, the voltage across each filament and the total power produced *increases*, *decreases* or *stays the same* when the two filaments are connected in series.

current .....

voltage .....

power .....

[2]

(c) Fig. 8.2 shows the current–voltage graph for the two filaments.

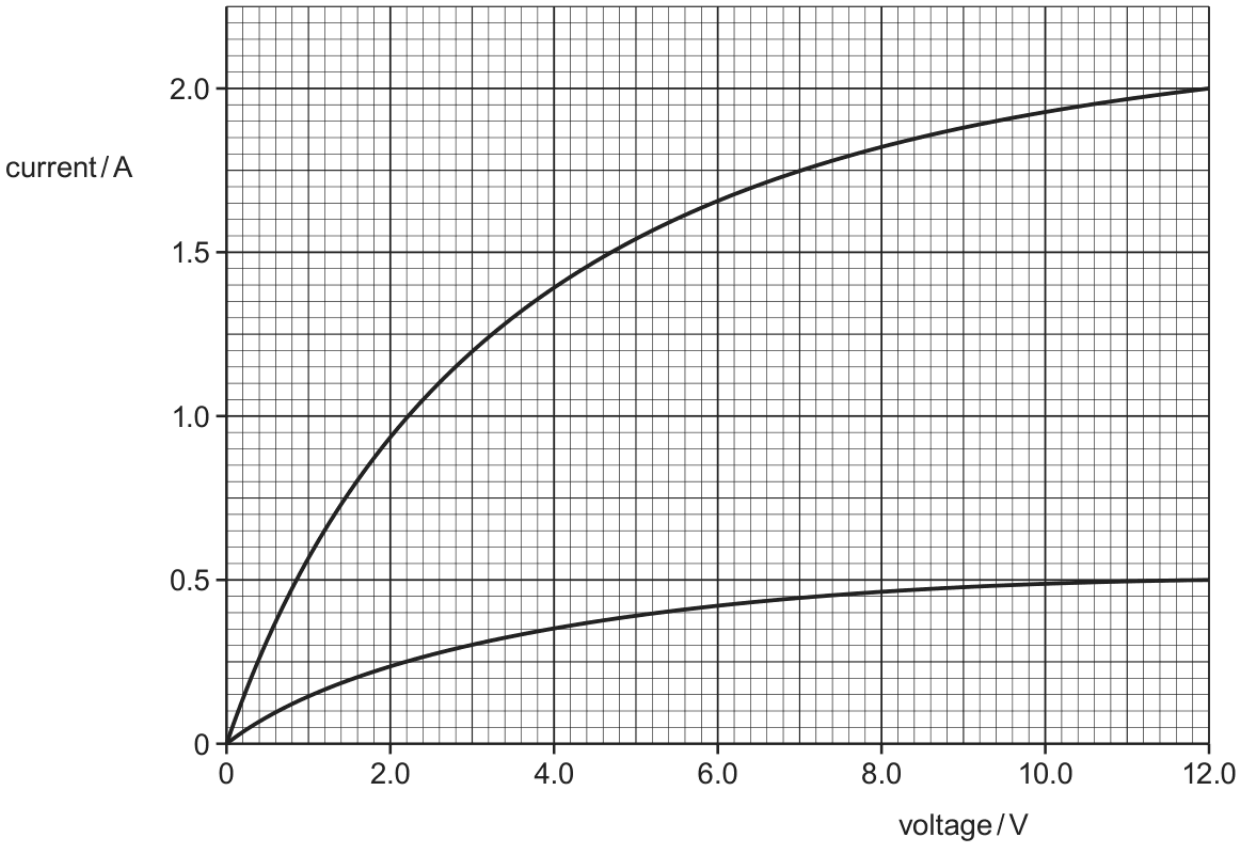


Fig. 8.2

(i) Calculate the total resistance of the two filaments when they are connected in parallel to a voltage of 12 V.

resistance = ... [3]

(ii) The two filaments are made from the same type of metal and have the same length, when uncoiled. They both operate at the same temperature.

Suggest why one filament has a resistance that is greater than that of the other filament.

.....

..... [1]

(d) Fig. 8.3 shows a relay used to switch on a car headlamp.

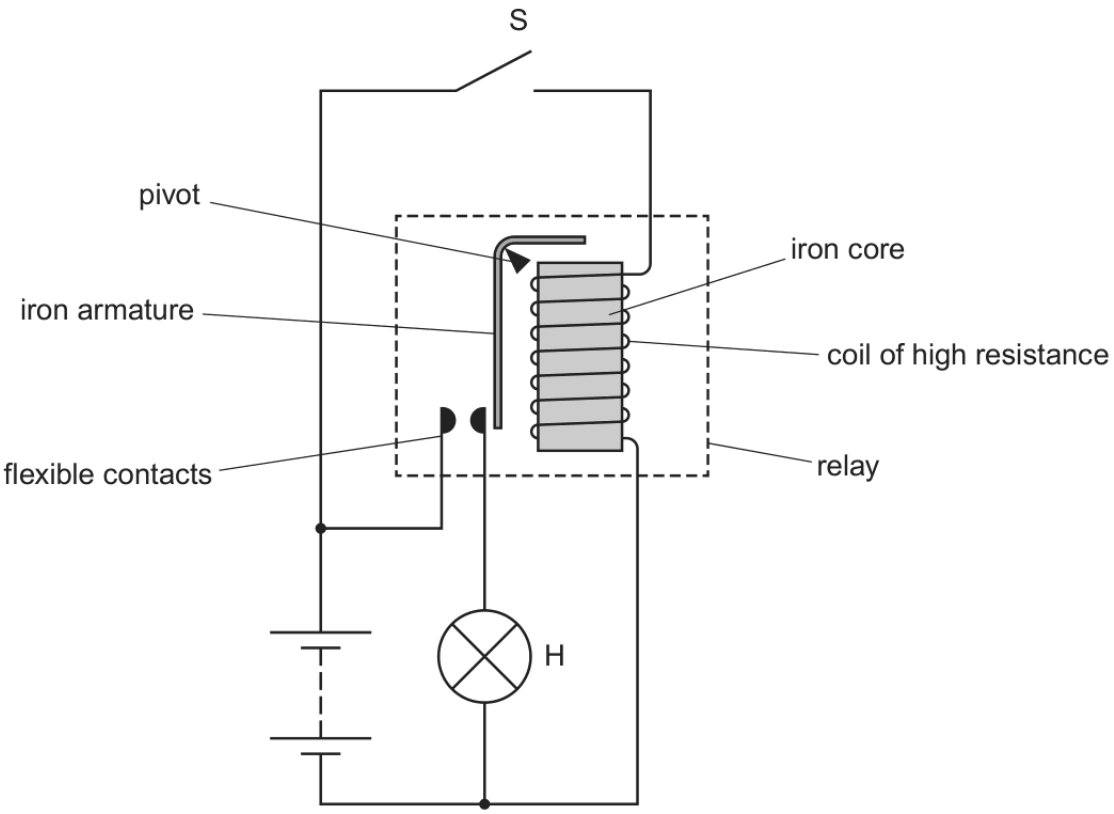


Fig. 8.3

Explain why headlamp H lights up when switch S is closed.

.....

.....

.....

.....

.....

..... [4]

[Total: 15]

MARK SCHEME · 5054_s20_ms_21.pdf · source page 8 of 11 · Question 8

Question	Answer	Marks
8(a)(i)	electrical in first box	**B1**
	heat	**B1**
	light	**B1**
8(a)(ii)	energy output / energy input	**C1**
	useful output energy / total energy input	**A1**
8(b)	two decrease	**C1**
	all three decrease	**A1**
8(c)(i)	(R =) V / I in any form numerical or algebraic	**C1**
	total current = 2.5 A **or** both resistances 24 (Ω) and 6(.0 Ω)	**C1**
	4.8 Ω	**A1**

MARK SCHEME · 5054_s20_ms_21.pdf · source page 9 of 11 · Question 8

Question	Answer	Marks
8(c)(ii)	smaller area / diameter / radius	**B1**
8(d)	current in the coil	**B1**
	coil / core becomes magnetised	**B1**
	attracts iron armature	**B1**
	armature turns **and** contacts close	**B1**

3 Fig. 3.1 shows part of a hydraulic press that is used to compress waste paper into a brick for burning.

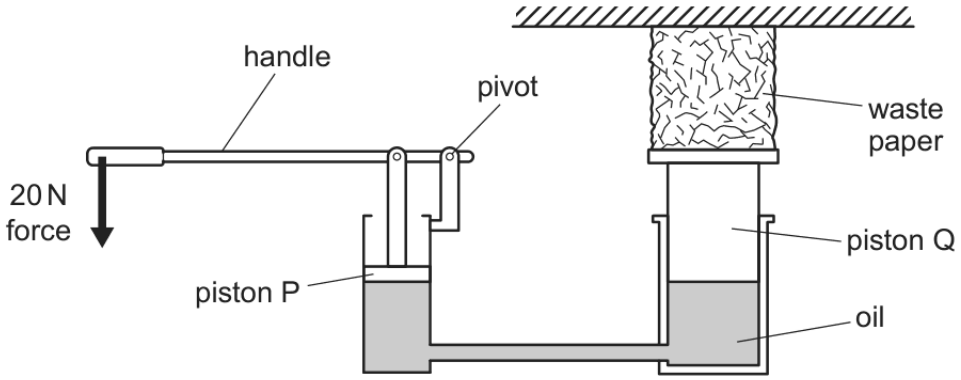


Fig. 3.1

A force of 20 N is exerted downwards on the end of the handle.

(a) The force on the handle creates a moment about the pivot.

(i) Define the *moment of a force*.

.....

.....

.....

..... [2]

(ii) Explain why the force exerted on piston P is greater than the force exerted on the handle.

.....

.....

..... [1]

(b) Explain how the hydraulic press enables a greater force to be exerted on piston Q than is exerted on piston P.

.....

.....

.....

.....

..... [2]

(c) In moving the handle downwards, the 20 N force moves through a distance of 0.60 m and piston Q rises by 0.020 m.

The force exerted by piston Q on the paper is 400 N.

Calculate:

(i) the work done in moving the handle downwards

work = ... [2]

(ii) the efficiency of the hydraulic press.

efficiency = ... [2]

[Total: 9]

MARK SCHEME · 5054_s20_ms_22.pdf · source page 7 of 13 · Question 3

Question	Answer	Marks
3(a)(i)	force × distance	**C1**
	force × perpendicular distance	**A1**
3(a)(ii)	smaller distance to pivot (than from P to pivot)	**B1**

MARK SCHEME · 5054_s20_ms_22.pdf · source page 8 of 13 · Question 3

Question	Answer	Marks
3(b)	same / equal pressure on piston P and piston Q	**B1**
	larger area of piston Q (than piston P)	**B1**
3(c)(i)	(W = Fd) numerical or algebraic	**C1**
	12 J	**A1**
3(c)(ii)	(efficiency) = work output / work input numerical or algebraic, e.g. 400 × 0.02 / 12	**C1**
	0.67 or 67%	**A1**

QUESTION · 5054_w20_qp_11.pdf · source page 8 of 16

19 Data for three types of electricity generator are shown.

	input energy / MJ	wasted energy / MJ
oil	500	300
nuclear	200	160
hydroelectric	10	1.0

Which is the least efficient generator and which is the most efficient?

	least efficient	most efficient
A	hydroelectric	nuclear
B	hydroelectric	oil
C	nuclear	hydroelectric
D	oil	hydroelectric

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 19

19	C	**1**

2 The total mass of a wheelbarrow and its load is 90 kg. A worker pushes the wheelbarrow and load up a plank of length 2.0 m on to a platform, as shown in Fig. 2.1. The platform is at a height of 0.60 m above the ground.

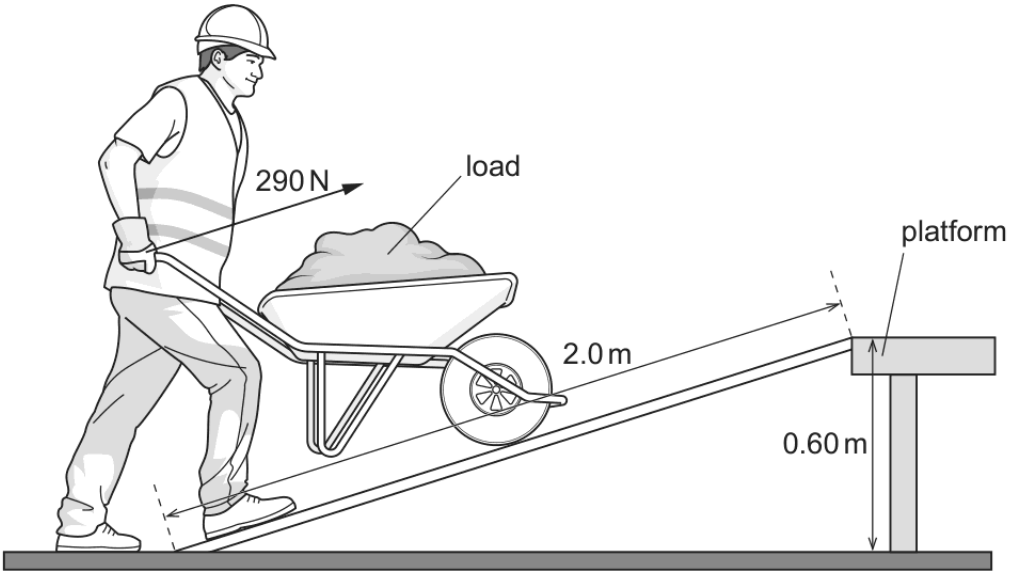


Fig. 2.1

The worker exerts a force of 290 N on the wheelbarrow in the direction in which the wheelbarrow moves.

(a) The gravitational field strength *g* is equal to 10 N / kg.

(i) Calculate the gravitational potential energy gained by the wheelbarrow and its load.

energy = ... [2]

(ii) The worker pushes the wheelbarrow 2.0 m along the plank.

Calculate the work done on the wheelbarrow by the worker.

work done = ... [2]

(iii) Suggest one reason why the answer to **(a)(ii)** is not equal to the answer to **(a)(i)**.

.....

..... [1]

(b) The worker finds that walking up the plank pushing the wheelbarrow is even more inefficient than the answers in **(a)** suggest.

(i) State what is meant by *efficiency*.

.....

..... [1]

(ii) Suggest **one** reason why this method of lifting the load onto the platform is so inefficient.

.....

..... [1]

[Total: 7]

Question	Answer	Marks
2(a)(i)	(g.p.e. =) *mgh* **or** $90 \times 10 \times 0.60$	**C1**
	540 J	**A1**
2(a)(ii)	(W.D. =) *Fx* **or** 290×2.0	**C1**
	580 J	**A1**
2(a)(ii)	friction (between wheel and axle) **or** friction (between tyre / wheel and plank) **or** (work done against) friction	**B1**
2(b)(i)	$\frac{\text{useful energy output}}{\text{(total) energy input}}$ ($\times$ 100%) **or** $\frac{\text{useful power output}}{\text{(total) power input}}$ ($\times$ 100%)	**B1**
2(b)(ii)	wheelbarrow / worker has to be lifted up **ignore** references to friction / air resistance	**B1**

QUESTION · 5054_s21_qp_11.pdf · source page 6 of 16

14 Some solar panels have a total area of 12 m^2.

Each 1.0 m^2 of the panels receives 0.85 kJ of energy from the Sun in 1.0 s.

The efficiency of the panels is 16%.

How much power do they produce?

A 1.6 kW **B** 2.2 kW **C** 64 kW **D** 160 kW

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 14

14	A	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 **(a)** Fig. 9.1 represents the conversion of energy in a mobile phone.

Fig. 9.1

(i) State the form of energy stored in the battery.

..... [1]

(ii) State the useful form of output energy from the battery.

..... [1]

(iii) State **two** useful forms of output energy from the mobile phone.

..... [2]

(b) When the battery is in use, the average current is 1.3 mA.

Determine the charge that passes through the battery in a time of 2.0 minutes.

charge = ... [2]

(c) A student estimates the thermal energy produced in the battery when the mobile phone is used and uses this value to estimate the efficiency of the battery.

He measures a temperature rise of 5.0 °C within the battery in a period of time when the useful energy output from the battery is 5200 J.

The mass of the battery is 110 g and its specific heat capacity is 830 J / (kg °C).

(i) Calculate the thermal energy (heat) needed to raise the temperature of the battery by 5.0 °C.

thermal energy = ... [3]

(ii) State what is meant by *efficiency*.

.....

..... [2]

(iii) Calculate the efficiency of the battery.

efficiency = ... [2]

(iv) Suggest and explain **one** reason why the value for the efficiency calculated in **(iii)** is larger than the actual efficiency of the battery.

.....

.....

.....

..... [2]

[Total: 15]

MARK SCHEME · 5054_s21_ms_22.pdf · source page 13 of 16 · Question 9

Question	Answer	Marks
9(a)(i)	chemical	**B1**
9(a)(ii)	electrical	**B1**
9(a)(iii)	ANY 2 from • light • sound • radio / microwave / electromagnetic (waves)	**B2**
9(b)	(*Q* =) *It* in any form	**C1**
	0.16 C	**A1**

MARK SCHEME · 5054_s21_ms_22.pdf · source page 14 of 16 · Question 9

Question	Answer	Marks
9(c)(i)	(*E* =) *mcT* numerical or algebraic in any form	**C1**
	$0.11 \times 830 \times 5$	**C1**
	460 J	**A1**
9(c)(ii)	energy output / energy input	**C1**
	useful energy output / (total) energy input	**A1**
9(c)(iii)	(energy input =) 460 + 5200 **or** 5660 (J) seen **or** 5200 ÷ (energy in **9(c)(i)** + 5200)	**C1**
	92% **or** 0.92	**A1**
9(c)(iv)	(thermal) energy lost / heat lost (from battery to environment)	**B1**
	energy input larger than calculated / 5660 (J) **or** temperature rise should be larger / is smaller than it should be **or** thermal energy / heat produced larger than 460 J / calculated	**B1**

QUESTION · 5054_w21_qp_12.pdf · source page 6 of 16

13 Which equation is used to find the efficiency of an energy transfer?

A $\text{efficiency} = \dfrac{\text{useful energy output}}{\text{total energy input}}$

B $\text{efficiency} = \dfrac{\text{total energy input}}{\text{useful energy output}}$

C $\text{efficiency} = \dfrac{\text{energy converted to the required form}}{\text{total power input}}$

D efficiency = (energy converted to the required form) × (total energy input)

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 13

13	A	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 6 of 16

11 The total energy incident in 1.0 s on a group of solar panels is 120 J / m^2.

The group of solar panels converts 720 J of light energy to 120 J of useful electrical energy in 1.0 s.

What is the total surface area of the panels and what is the efficiency of the system?

	total surface area of solar panels / m^2	efficiency / %
A	6.0	17
B	0.17	6.0
C	6.0	6.0
D	0.17	17

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 11

11	A	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 8 of 20

15 The electrical energy supplied to a kettle is 600 kJ.

Of this energy, 45 kJ is transferred to the surroundings and 15 kJ is used to heat the casing of the kettle.

The remaining energy is used to heat the water.

What is the efficiency of the kettle?

A 0.10 **B** 0.90 **C** 0.925 **D** 0.975

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 15

15	B	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 7 of 16

16 X and Y are both 40 W, 240 V lamps. Lamp X is more efficient than lamp Y.

Which statement correctly compares X with Y?

A X is more expensive to run.

B X has a lower electrical resistance.

C X produces less light energy.

D X produces less thermal energy.

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 16

16	D	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 7 of 16

17 The input power to a motor is 12 W.

The motor wastes 590 J of energy in 1.0 min.

What is the efficiency of the motor?

A 18% **B** 22% **C** 55% **D** 82%

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 17

17	A	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 7 of 16

18 Electric motors have an efficiency of about 90% when used in an electric train.

Which forms of wasted energy are produced?

A sound only

B thermal only

C thermal and chemical

D thermal and sound

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 18

18	D	**1**

8 A filament lamp is arranged above a shiny metal surface, as shown in Fig. 8.1.

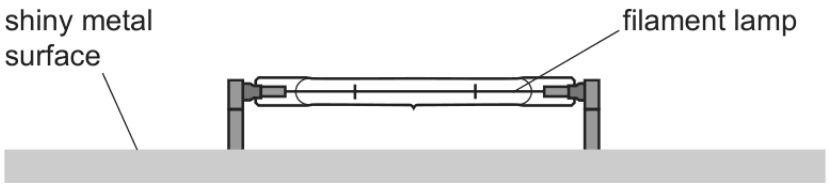


Fig. 8.1

A student reads in a textbook that a light-emitting diode (LED) is more efficient than a filament lamp.

(a) (i) State what is meant by 'efficiency'.

.....

.....

..... [1]

(ii) Suggest why the efficiency of a filament lamp is very low.

.....

.....

..... [1]

(b) The student considers replacing the filament lamp shown in Fig. 8.1 with an LED of the same brightness.

Data about the filament lamp and a suitable LED are shown in Table 8.1.

Table 8.1

	input power / W	energy efficiency
filament lamp	120	6.2%
LED	15	–

The LED emits the same amount of visible light as the filament lamp.

Using this information and the data in Table 8.1, determine the efficiency of the LED.

efficiency = ... % [2]

(c) The filament lamp is connected to the live and neutral wires in the mains supply.

The earth wire in the mains supply is connected to the shiny metal surface shown in Fig. 8.1.

There is a fuse in the live wire.

By accident, the live wire touches the shiny metal surface.

(i) Describe what then happens.

.....

.....

.....

..... [2]

(ii) In another similar lamp, the fuse is wrongly connected into the earth wire.

Explain why a person is **not** protected when the live wire touches the shiny metal surface.

.....

.....

.....

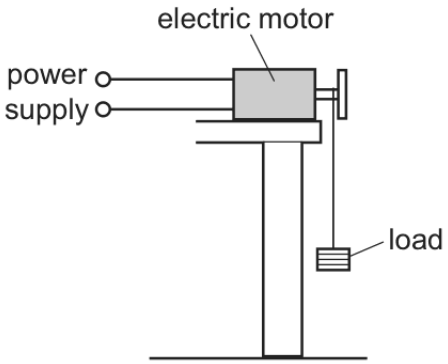
..... [2]

[Total: 8]

Question	Answer	Marks
8(a)(i)	useful energy output / (total) energy input	**B1**
8(a)(ii)	heat / thermal energy is produced **or** energy is wasted	**B1**
8(b)	(useful output power =) 120 × 0.062 ; 7.4 (4 W)	**C1**
	50% **or** 0.50	**A1**
8(c)(i)	ANY 2 from • current flows to earth / ground • fuse blows / melts • stops current / breaks the circuit/not live	**B2**
8(c)(ii)	ANY 2 from • fuse (still) melts • lamp appears off • metal case still live / can cause electrocution or shock	**B2**

QUESTION · 5054_w23_qp_12.pdf · source page 6 of 20

14 An electric motor is used to lift a load at a constant speed.



The gravitational potential energy (GPE) stored in the load increases.

The moving load also has stored kinetic energy (KE).

Which expression is used to calculate the efficiency of the motor when the load is being lifted at a constant rate?

A $\text{efficiency} = \dfrac{\text{GPE gained per second}}{\text{KE gained per second}}$

B $\text{efficiency} = \dfrac{\text{GPE gained per second}}{\text{power supplied}}$

C $\text{efficiency} = \dfrac{\text{power supplied}}{\text{GPE gained per second}}$

D $\text{efficiency} = \dfrac{\text{power supplied}}{\text{KE gained per second}}$

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 14

14	B	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 6 of 16

13 The input power to a lamp is 6.0 W. The lamp wastes 2.7 J of energy in 3.0 s.

What is the efficiency of the lamp?

A 0.15 **B** 0.45 **C** 0.55 **D** 0.85

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 13

13	D	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 7 of 16

16 A 15 W lamp is turned on for 30 minutes. It wastes 7000 J of energy.

What is the efficiency of the lamp?

A 0.26 **B** 0.35 **C** 0.59 **D** 0.74

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 16

16	D	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 8 of 16

18 Energy is transferred in a filament lamp at a rate of 60 W. The lamp wastes energy at a rate of 46 W.

Which ratio is used to calculate the efficiency of the lamp?

A $\frac{14}{74}$ **B** $\frac{14}{60}$ **C** $\frac{14}{46}$ **D** $\frac{46}{60}$

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 18

18	B	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 5 of 16

11 Which equation is used to calculate the efficiency of an energy transfer?

A $\text{efficiency} = \dfrac{\text{useful energy output}}{\text{total energy input}}$

B $\text{efficiency} = \dfrac{\text{total energy input}}{\text{useful energy output}}$

C $\text{efficiency} = \dfrac{\text{total energy output}}{\text{total power input}}$

D efficiency = (useful energy output) × (total energy input)

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 11

11	A	**1**

3 A battery, a pulley and a motor are used to lift a load as shown in Fig. 3.1.

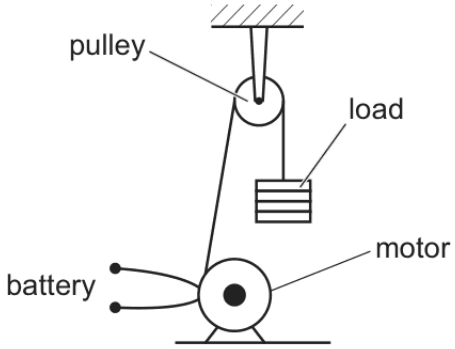


Fig. 3.1

(a) Describe the transfers between energy stores that occur as the load is lifted.

.....

.....

.....

..... [3]

(b) The efficiency of the motor, pulley and load system is less than 100%.

(i) By comparing the input energy and the useful output energy, explain why the efficiency is less than 100%.

.....

..... [1]

(ii) Explain how the principle of conservation of energy applies in lifting the load.

.....

.....

..... [2]

(c) The input power to the motor is 15 W. The motor is used for 20 s. The efficiency of the motor is 60%.

Calculate the energy supplied to the load.

energy = ... J [2]

[Total: 8]

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Question	Answer	Marks
3(a)	*Any three from*: • reduction in chemical energy store in battery • electrical work done by motor • mechanical work done on weight • increase in gravitational energy store of weight • increase in thermal energy (in motor or air) • electrical heating (by current)	**B3**
3(b)(i)	useful or output energy < input energy	**B1**
3(b)(ii)	total energy is constant	**B1**
	energy from power supply = energy gained by weight + thermal energy (to air / due to friction)	**B1**
3(c)	(E =) power × time in any form	**C1**
	180 J	**A1**

2 Fig. 2.1 shows a fork-lift truck used to lift a load.

Fig. 2.1

When lifting the load, the electric motor on the fork-lift truck has a useful output power of 600 W.

(a) Define 'power'.

.....

..... [1]

(b) The efficiency of the motor is 70%.

Calculate the input power to the motor.

input power = ...W [2]

(c) The fork-lift truck lifts a load with mass of 50 kg through a vertical distance of 2.3 m.

(i) Calculate the work done in lifting the load.

work done = ... J [2]

(ii) Calculate the time taken to lift the load.

time taken = ... s [2]

(d) Describe how energy is transferred to thermal energy as the load is lifted by the fork-lift truck.

.....

..... [1]

[Total: 8]

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Question	Answer	Marks
2(a)	work done or energy per unit time	**B1**
2(b)	(input power =) power output / efficiency in any form	**C1**
	860 (W)	**A1**
2(c)(i)	(work =) force × distance **or** *mgh*	**C1**
	1100 (J)	**A1**
2(c)(ii)	(time =) work ÷ (output) power	**C1**
	1.9 or 1.8 s	**A1**
2(d)	*Any one of*: • friction / air resistance • electrical heating	**B1**

2 In a factory, an electric motor is used to lift the boxes shown in Fig. 2.1.

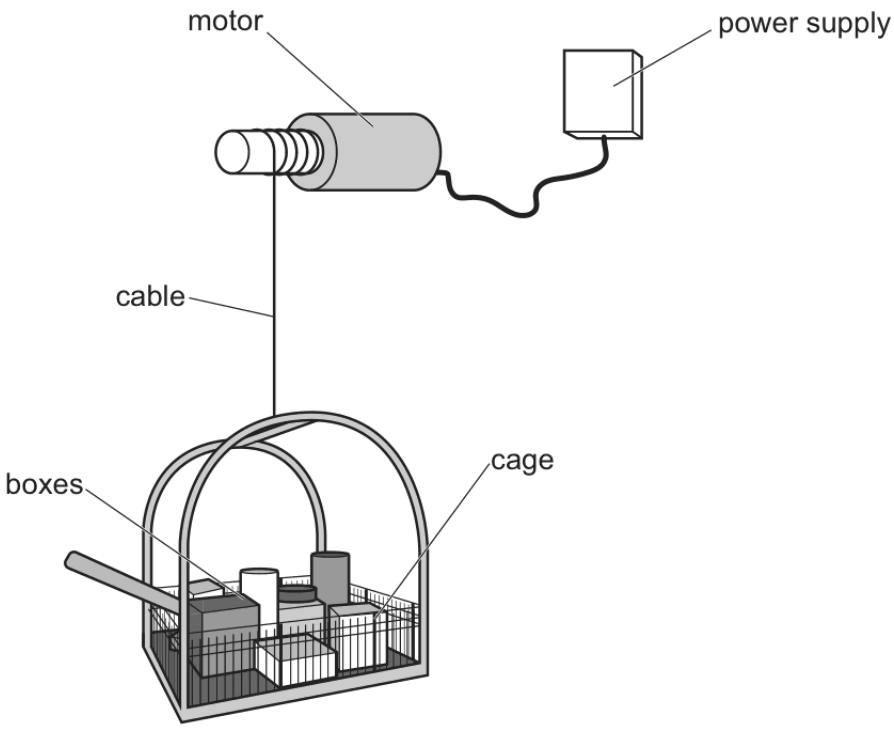

Fig. 2.1

The boxes are placed in a cage and the motor winds a cable around a cylinder to lift the cage and boxes.

The total mass of the boxes is 37 kg and they are lifted vertically through 13 m to a storage area.

(a) Calculate:

(i) the total weight of the boxes

weight = ... N [1]

(ii) the useful work done in lifting the boxes to the storage area.

useful work done = ... J [2]

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(b) The e.m.f. of the power supply for the motor is 230 V. The current in the motor is 5.6 A.

It takes 11 s for the motor to lift the cage and boxes to the storage area.

Calculate:

(i) the total energy transferred electrically to the motor in 11 s

energy = ... J [2]

(ii) the percentage efficiency of the lifting process.

efficiency = ...% [1]

(c) Some of the energy transferred by the motor during the lifting process is not transferred usefully.

State **two** reasons why the efficiency of the lifting process is less than 100%.

1 .....

.....

2 .....

.....

[2]

[Total: 8]

MARK SCHEME · 5054_w25_ms_21.pdf · source page 8 of 14 · Question 2

Question	Answer	Marks
2(a)(i)	360 (N)	**B1**
2(a)(ii)	(work done =) Fx **or** 360×13	**C1**
	4700 (J)	**A1**
2(b)(i)	(energy =) VIt **or** $230 \times 5.6 \times 11$	**C1**
	1.4×10^4 (J)	**A1**
2(b)(ii)	0.33 **or** 33%	**B1**

MARK SCHEME · 5054_w25_ms_21.pdf · source page 9 of 14 · Question 2

Question	Answer	Marks
2(c)	any **two** from: (work done / GPE) to lift the cage / cable (energy transferred) thermally in motor (coils) (work done) overcoming friction / air resistance	**B2**

QUESTION · 5054_w25_qp_12.pdf · source page 7 of 16

17 What is the definition of efficiency?

A $\text{efficiency} = \dfrac{\text{useful energy input}}{\text{total energy output}}$

B $\text{efficiency} = \dfrac{\text{total energy input}}{\text{total energy output}}$

C $\text{efficiency} = \dfrac{\text{useful energy output}}{\text{total energy input}}$

D $\text{efficiency} = \dfrac{\text{total energy output}}{\text{total energy input}}$

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 17

17	C	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 6 of 16

13 A microwave oven has an efficiency of 60%.

The useful power output of the microwave oven is 800 W.

What is the total power input to the microwave oven?

A 13 W **B** 75 W **C** 1300 W **D** 48 000 W

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 13

13	C	**1**

3 Figure 3.1 shows a dam used as part of a hydroelectric power station.

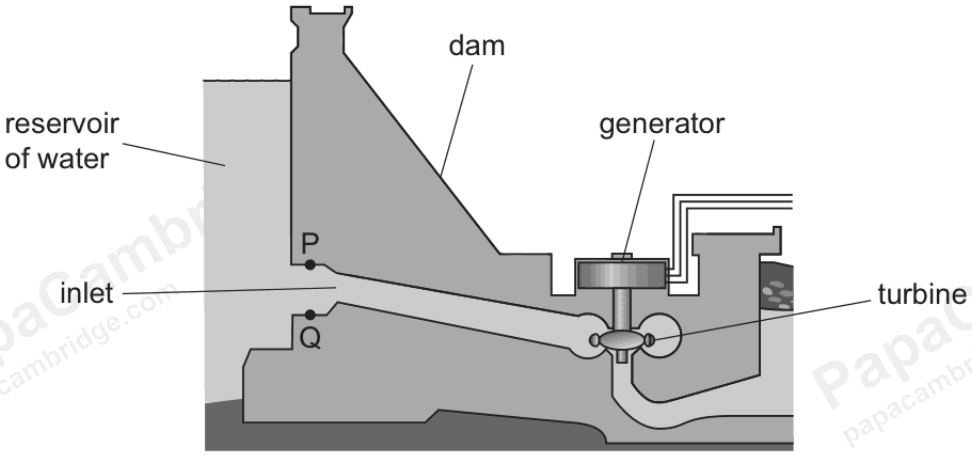


Figure 3. 1

Water flows to the turbine through an inlet in the dam. The inlet is shown in Figure 3.1.

Points P and Q are on the horizontal surfaces next to the inlet.

(a) (i) State **two** factors that determine the size of the pressure at P.

.....

..... [1]

(ii) The pressure in the water causes a force on the horizontal surfaces at P and Q.

State the directions of the forces due to the water on the surfaces at P and Q.

direction at P .....

direction at Q .....

[1]

(b) This hydroelectric energy resource is a renewable source of energy.

State the name of **one** other renewable energy source.

..... [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
OT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(c) The power output from the generator is 2.5 MW.

(i) Calculate the energy **output** from the generator in 30 minutes.

energy output = ... J [2]

(ii) The efficiency of the energy transfer in the generator is 95%.

Calculate the power **input** to the generator.

PapaCambridge
papacambridge.com
PapaCambridge
papacambridge.com

power input = ... MW [2]

[Total: 7]

MARK SCHEME · 5054_s26_ms_22.pdf · source page 9 of 18 · Question 3

Question	Answer	Marks	Guidance
3(a)(i)	At least 2 from • height / depth (of water) / quantity or amount of water in dam • density (of water / liquid) • acceleration of gravity / gravitational field strength • atmospheric pressure • force (on surface / inlet) • area (of surface / inlet)	**B1**	**I** thickness; area of dam
3(a)(ii)	at P – upwards / towards P at Q – downwards / towards Q	**B1**	**I** top, bottom; north, south both needed
3(b)	ANY one from geothermal, tidal, solar, wind, biomass, wave (energy)	**B1**	**A** sunlight; solar panels; solar cell; wind turbines; biogas ; solar power; wood from a managed source **I** just 'wood' **not** nuclear

MARK SCHEME · 5054_s26_ms_22.pdf · source page 10 of 18 · Question 3

Question	Answer	Marks	Guidance
3(c)(i)	(E =) $P \times t$ in any form algebraic or numerical	**C1**	**A** 2.5 ($\times 10^6$)$\times$ 30; 2.5$\times$30$\times$60; 4.5 $\times$ 10^n; P=W / t
	4.5×10^9 (J)	**A1**	
3(c)(ii)	efficiency = (power) output / (power) input in any form	**C1**	**A** 0.95 = 2.5 / P_{IN}; (useful) energy output / energy input e.g. 0.95 =**3(c)(i)** / input in any form
	2.6(3 MW)	**A1**	**A** 2.6×10^6 with unit MW only 1 mark

Chapter 1 Motion, forces and energy

1.7 Energy, work and power

1.7.5 Power

108 entries: 60 Paper 1 questions, 48 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

10 Fig. 10.1 is a full-scale drawing of a small aluminium bar.

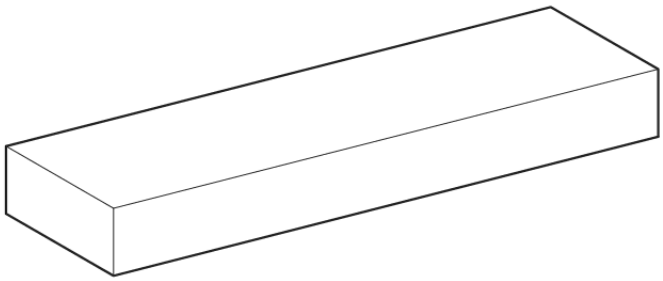


Fig. 10.1

(a) Describe how the density of aluminium may be determined using the bar. In your account

(i) describe the readings to be taken,

(ii) state the instruments used to make the readings,

(iii) describe how the density is calculated from the readings.

[6]

(b) The bar is placed in a small furnace.
Fig. 10.2 shows how the temperature of the bar varies with time *t*.

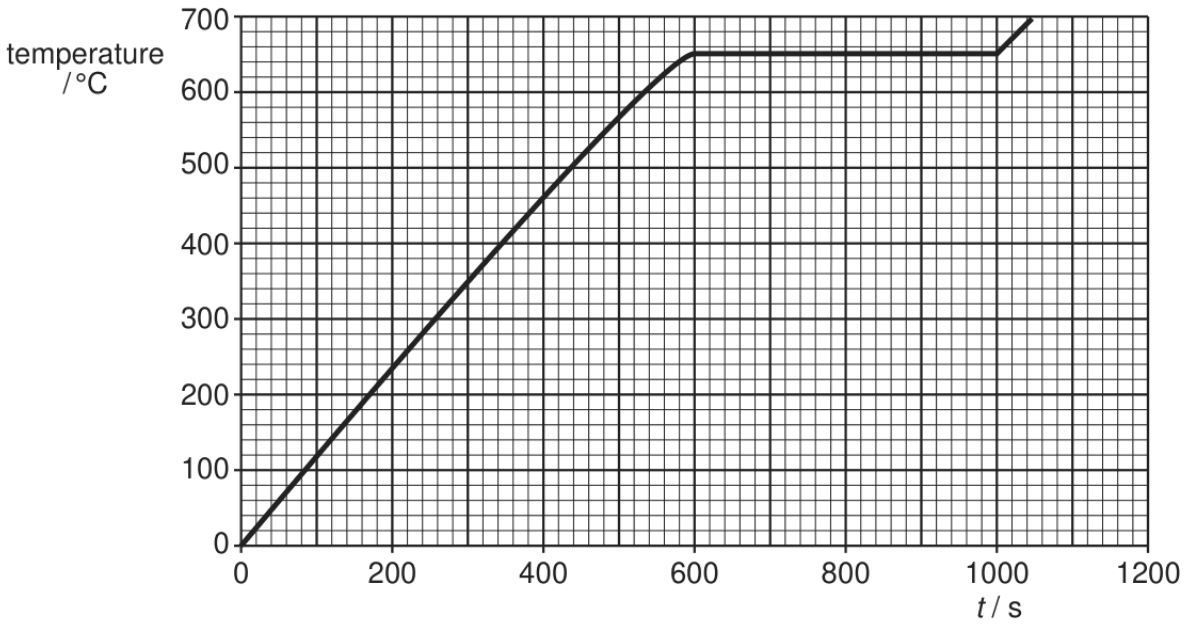


Fig. 10.2

(i) State what happens to the bar between $t = 600$ s and 1000 s.

(ii) Explain what effect the energy supplied to the bar has on its molecules between $t = 0$ and 1000 s.

(iii) The mass of the bar is 0.030 kg and the specific heat capacity of aluminium is 880 J/(kg °C). Calculate the energy supplied to the bar between $t = 0$ and 600 s.

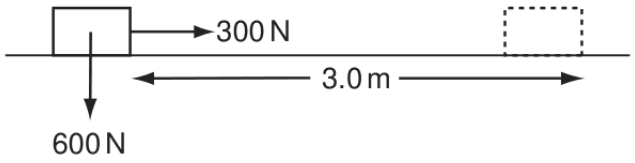
(iv) Between $t = 600$ s and 1000 s the furnace supplies 30 joules of energy per second to the bar. Calculate the specific latent heat of fusion of aluminium.

[9]

10	**(a)**		mass of bar (measured)	M1
			using (top-pan) balance / spring balance / scales etc.	A1
			length, breadth and height measured	
			or volume water + bar measured **or** displacement can (full) with water	M1
			volume = length x breadth x height	
			or subtract volume water alone **or** collect water displaced	A1
			using ruler / calipers / micrometer **or** measuring cylinder	A1
			density = mass / volume	B1
	(b)	**(i)**	melts / changes state / becomes liquid	B1
		(ii)	(initial) increase in vibration / K.E. of molecules (to 600s)	B1
			then later / after 600s or on melting	
			bonds broken (accept molecules break free / overcome attraction / not fixed in place)	B1
		(iii)	E = mc (Δ)T algebraic form seen	C1
			645 – 655 (°C) seen)	C1
			17 160 J (allow 1700, 17200, 20000)	A1
		(iv)	30*400 or 12 000 (J) seen)	C1
			E = mL any algebraic form seen **or** 12 000/0.3	C1
			40 000 J/kg	A1
			Total	**[15]**

QUESTION · 5054_w04_qp_1.pdf · source page 7 of 20

13 When a 300 N force is applied to a box weighing 600 N, the box moves 3.0 m horizontally in 20 s.



What is the average power?

A 45 W **B** 90 W **C** 900 W **D** 1800 W

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 13

13	**A**

For Examiner's Use

4 Fig. 4.1 shows a hydroelectric power station. Water from the lake is used to produce electricity in the turbine house.

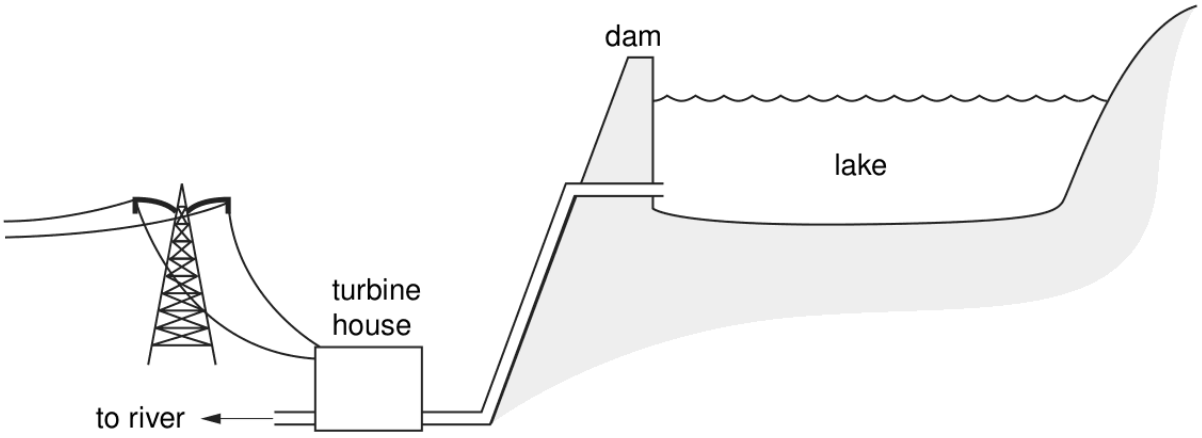


Fig. 4.1

(a) State where the water in Fig. 4.1 has the least potential energy.

..... [1]

(b) In 30 minutes, the water loses 5.0×10^9 J of energy and 4.5×10^9 J of electrical energy is produced in the turbine house.

(i) Calculate the efficiency of the energy conversion.

efficiency = ... [1]

(ii) Calculate, in watts, the electrical power output from the turbine house.

power = ... W [3]

(c) Some power stations burn coal to produce the same electrical power output. State one advantage of the hydroelectric power station.

.....

.....

..... [1]

(d) State one harmful effect that the hydroelectric power station may have on the environment.

.....

..... [1]

4	**(a)**	in river/(emerging from or entering) turbine house	**B1**
	(b) **(i)**	0.9 **or** 90% **or** 0.47 **or** 47% (penalise unit error)	**B1**
	(ii)	P = E/t in symbols **or** any energy/any time	**C1**
		30 x 60 **or** 1800 seen	**C1**
		2.5×10^6 (W)	
		(150 or 2.78MW score 2/3)	**A1**
	(c)	any sensible suggestion e.g. no costs for water/energy supply **or** less pollution (accept coal produces smoke/dust/harmful gases/CO_2 **or** no need to transport coal **or** renewable **or** rapid response to power demand **or** less heat produced/more efficient	**B1**
	(d)	any sensible suggestion e.g. flooding **or** fish unable to pass **or** turbines kill fish **or** destroy habitats **or** less land **or** uses up large space **or** fells trees **or** unsightly/destroys scenery **or** lake/river silt up **or** more rain/evaporation	**B1** **7**

For Examiner's Use

6 The table gives information about two household appliances.

appliance	mains supply voltage / V	current through appliance / A	power / W	power / kW	time used per day / h	energy used per day / kW h
television	240	1.20	288	0.288	2.50	0.720
water heater	240	12.6			0.50	

(a) Write the missing values in the empty spaces in the table. [3]

(b) Why is more power needed for the water heater than for the television?

.....

.....

..... [1]

(c) The water heater is connected to the mains supply. Explain why using a 3 A fuse would **not** be suitable.

.....

.....

.....

.....

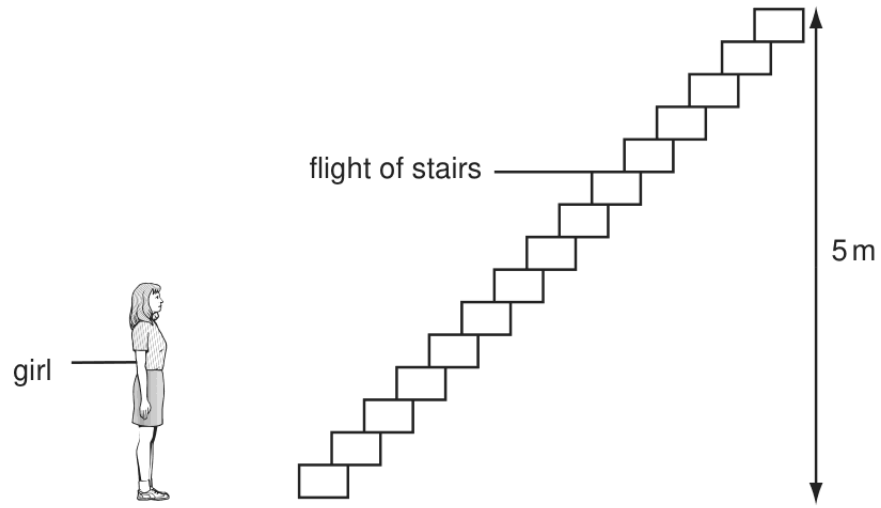
.....

..... [2]

6	**(a)**	3024	**B1**
		3.024 (or1/1000 of previous answer)	**B1**
		1.512 (or ½ of previous answer)	**B1**
	(b)	smaller resistance **accept** more current	**B1**
	(c)	heater uses more than 3A **accept** current 12.6A	**B1**
		causes fuse to melt/blow/burn/break	**B1** **6**

QUESTION · 5054_w05_qp_1.pdf · source page 7 of 20

12 A girl of weight 500 N runs up a flight of stairs in 10 seconds. The vertical height of the stairs is 5 m.



What is the average power developed by the girl?

A 50 W **B** 100 W **C** 250 W **D** 1000 W

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 12

12	**C**

QUESTION · 5054_w05_qp_1.pdf · source page 14 of 20

30 A combined bathroom unit of a heater and a lamp is controlled by one switch. The unit contains a 2 kW heater and a 100 W lamp. In one week, the lamp uses 1 kW h of electrical energy.

How much electrical energy is used by the heater alone?

A 2 kW h **B** 4 kW h **C** 10 kW h **D** 20 kW h

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 30

30	**D**

Section B

Answer **two** questions from this section.

Use the lined pages provided and, if necessary, continue on the separate sheets available from the Supervisor.

9 Fig. 9.1 shows an electric kettle.

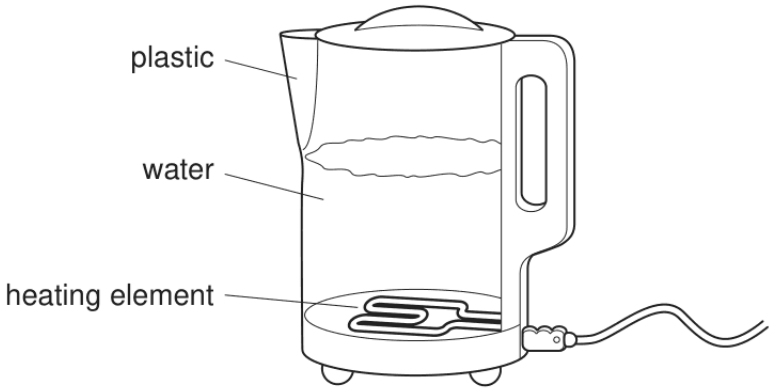


Fig. 9.1

(a) The body of the kettle is plastic but the outside casing of the heating element is metal.

(i) Explain why a fuse is included in the circuit and explain what happens when the fuse blows. [3]

(ii) Explain why the metal casing of the heating element is connected to earth. [2]

(b) The electrical power input to the kettle is 2000 W and the kettle is used for 6 minutes (0.1 hour).

Calculate

(i) the electrical energy, in J, supplied to the kettle. State clearly the formula that relates power, energy and time. [3]

(ii) the electrical energy, in kW h, supplied to the kettle. [2]

(iii) the cost of using the kettle if 1 kW h costs 8 cents. [2]

(c) When the kettle is switched off, the water cools down. Explain, in molecular terms, how evaporation causes a loss of energy from the water. [3]

Section B

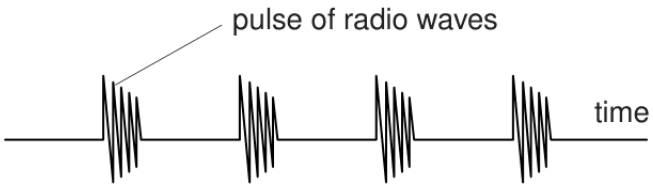
9	**(a)**	**(i)**	protects the circuit **or** stops a fire (if) current is too large fuse melts stops current/breaks circuit	not a surge of current/power **any 3 lines**	 **B3**
		(ii)	heating element fault allows water to conduct (electricity) (with earth connected, if a fault) current flows to earth no current (through water) to person no (electric) shock	 **any 2 lines**	 **B2**
	(b)	**(i)**	P x t seen in any form 2000 x 360 720 000 (J)		**B1** **C1** **A1**
		(ii)	conversion of 2000 W to 2 kW 0.2 (kWh)		**C1** **A1**
		(iii)	0.2 x 8 1.6 c		**C1** **A1**

	(c)		fast/energetic molecules escape remaining molecules slower/less K.E molecules separate (molecular) bonds are broken escaping molecules have greater P.E.	**B1** **B1** **B1** **B1** **B1**	 **any 3** **B3**

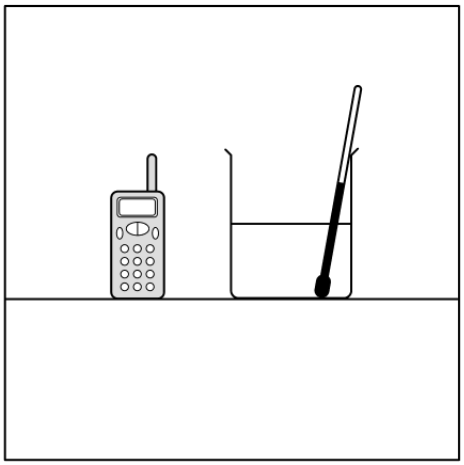
10 A student uses a mobile phone. He is concerned that the energy of the radio waves may cause a temperature rise in his brain. To investigate this effect, he calculates and tries to measure the heating effect of the phone on a nearby glass beaker of water.

Information from manufacturer

constant input power 0.20 W



The mobile phone emits pulses of radio waves.
Each pulse of waves has energy 0.00012 J.
There are 216 pulses of waves in 1.0 s



The phone is used for 360 s when next to a glass beaker containing 50 g of water.

(a) Calculate

(i) the number of pulses of radio waves produced during the phone call, [2]

(ii) the total energy of the radio waves emitted during the phone call, [2]

(iii) the maximum temperature rise produced in 50 g of water if all of the energy calculated in **(ii)** is absorbed by the water. The specific heat capacity of water is 4.2 J/(g °C). [3]

(b) The manufacturer gives the input power of the phone as 0.20 W.

(i) Calculate the energy supplied by the battery when the phone is used for 360 s. [2]

(ii) Calculate the fraction of the energy supplied by the battery that is converted into radio wave energy during the phone call. [1]

(c) The student uses the mercury-in-glass thermometer shown in Fig. 10.1.

He does not detect any temperature rise in the water in the beaker when the phone is used.

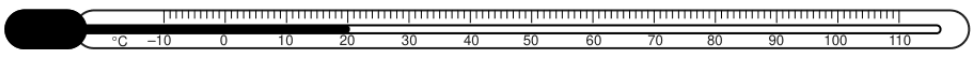


Fig. 10.1

(i) Describe how you would check the 0 °C and 100 °C points on the thermometer. [2]

(ii) Explain why the thermometer is not sensitive enough to detect the temperature rise. [1]

(iii) State and explain one change that will make a mercury-in-glass thermometer more sensitive. [2]

10	**(a)**	**(i)**	360 x 216	**C1**
			77 800 (no sig fig penalty)	**A1**
		(ii)	77 800 x 0.00012	**C1**
			9.33 J ecf **(i)**	**A1**
		(iii)	E = mcΔT in any form, algebraic or numerical	**B1**
			9.33/(50 x 4.2)	**C1**
			0.044 ºC ecf **(ii)**	**A1**
	(b)	**(i)**	E=Pt in any form, algebraic or numerical	**C1**
			72 J	**A1**
		(ii)	0.13 (accept 13%) no s.f. penalty	**A1**
	(c)	**(i)**	(pure) melting ice for 0° C	**B1**
			(pure) boiling water/steam above boiling water (at 1 atmosphere) for 100° C	**B1**
		(ii)	each division on thermometer is too small described in some way e.g. does not expand far up tube (**not** bore too thin, not enough mercury)	**B1**
		(iii)	**change** use more mercury **or** use smaller bore	**M1**
			reason more expansion **or** further distance up tube (for same expansion)	**A1**

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 A student carries out an experiment to measure the power he produces. He runs up a flight of stairs, as shown in Fig. 1.1.

Fig. 1.1

The student takes readings to calculate his power.

(a) (i) List all of the readings he must take.

.....

.....

.....

..... [3]

(ii) State one precaution he should take to make sure one of his readings is accurate.

.....

..... [1]

(b) Write down all of the equations he must use to calculate his power from the readings.

You may use symbols or words in your equations.

.....

.....

.....

.....

..... [2]

Section A

1	**(a)**	**(i)**	mass or weight	B1	
			time (to run up steps) **or** speed	B1	
			height **or** number of steps and height of each	B1	
		(ii)	mass – ensure balance reads zero without person time – have two timers (or repeat) **or** use stopclock to better than 0.1 s height – use callipers for step height **or** ensure rule vertical **or** tape taut ANY 1 sensible comment	B1	
	(b)		work = force x distance **or** power = work/time **or** K.E. = ½ mv^2	C1	
			mass x *g* x height/time **or** weight x height /time (accept distance for height for both marks only if clear in **(a)(ii)**	A1	**[6]**

Section B

Answer **two** questions from this section.

Use the separate answer sheets available from the Supervisor.

9 A student has an open tank for storing water outside her house. The tank is black and is in direct sunlight.

She notices that the level of water inside the tank slowly decreases as water evaporates.

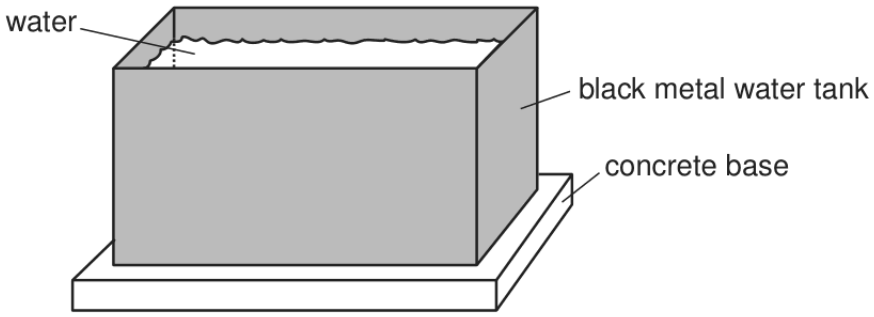


Fig. 9.1

She investigates whether some changes, shown in the table, will affect the amount of water evaporated each day.

change 1	a heater is used to increase the temperature of the water
change 2	a sheet of plastic is used to cover half the surface of the water
change 3	a fan is used to blow air over the top of the tank
change 4	the outside of the tank is painted white

Only one change is made at a time. All other factors are kept constant.

(a) State whether each change will increase, leave unchanged or decrease the amount of water evaporated each day. Explain each of your answers. [8]

(b) She notices that the water level in the tank falls by 0.005 m in 40 000 s.

The cross-sectional area of the tank is 3.0 m^2. The density of water is 1000 kg/m^3.

The specific latent heat of vaporisation of water is 2.2×10^6 J/kg.

Calculate

(i) the volume of water that evaporates, [1]

(ii) the mass of water that evaporates, [2]

(iii) the energy required to evaporate the water, [2]

(iv) the average rate at which energy is used to evaporate the water. [2]

Section B

9	(a)	**change 1** increases evaporation	B1	
		water molecules have more K.E./move faster/more have enough energy	B1	
		change 2 decreases evaporation	B1	
		less surface for molecules to escape (through)	B1	
		change 3 increases evaporation	B1	
		liquid molecules leaving surface removed by collisions with air molecules	B1	
		change 4 decreases evaporation	B1	
		heat/light/infra-red reflected by white surface **or** tank cooler	B1	**[8]**
	(b)	**(i)** $0.015\ m^3$	B1	
		(ii) M = D. V in any form	C1	
		15 kg (ecf **(i)**)	A1	
		(iii) m .L	C1	
		3.3×10^7 J (ecf **(ii)**)	A1	
		(iv) energy/time in any form	C1	
		825 J/s **or** W (ecf **(iii)**)	A1	**[7] (15)**

QUESTION · 5054_w06_qp_1.pdf · source page 11 of 16

29 Which of the following would cost the **least** if operated from the same voltage supply?

A a 5000 W electric cooker used for 1 minute

B a 1000 W electric fire used for 10 minutes

C a 500 W electric iron used for 1 hour

D a 100 W lamp used for 1 day

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 29

29	**A**

10 Fig. 10.1 shows a cable containing three wires coloured brown, blue and yellow/green, and a mains plug with the cover removed.

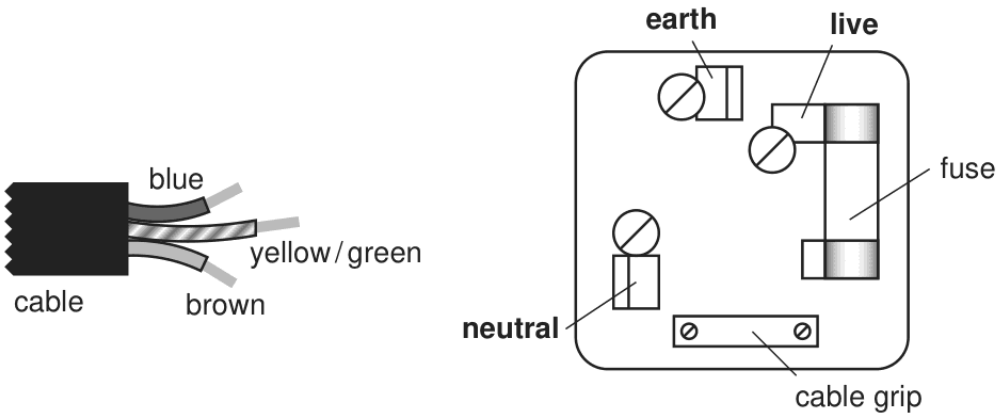


Fig. 10.1

(a) Describe how to connect the cable and the three wires correctly and safely to the plug. [4]

(b) The table lamp shown in Fig. 10.2 is made from plastic. It has only two wires in the cable to connect it to the plug.

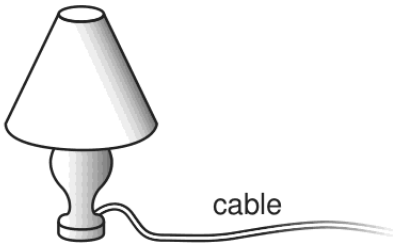


Fig. 10.2

The lamp has a power rating of 100 W and is used with a 230 V supply.

(i) Which wire, earth, live or neutral, is **not** needed in the cable for the lamp? [1]

(ii) Explain why the lamp is safe to use even though it has only two wires in the cable. [2]

(iii) Explain what is meant by a *power rating of 100 W*. [2]

(iv) Calculate the value of the fuse that should be used in the plug for this lamp. [3]

(v) Calculate the electrical energy supplied to the lamp in 30 minutes. [3]

10 (a)		yellow/green to earth blue to neutral **and** brown to live tighten terminal screws cable (outer cover) under grip no bare metal on wires earth wire longest put cover back on ANY 4 (-1 each clearly wrong answer beyond 4)		B4
(b)	**(i)**	earth		B1
	(ii)	plastic/lamp/cover/base made from insulator/does not conduct electricity		B1
		doubly insulated **or** plastic/lamp/cover/base cannot be live **or** cannot electrocute/shock		B1
	(iii)	100 J (100 J/s first mark only)		B1
		(electrical)(energy) used/transformed/converted/delivered/arrives **per second**		B1
	(iv)	P = VI (in any form numerical or algebraic)		C1
		0.43(48) (accept 1 sig.fig.)		A1
		Fuse: 0.5/1.0/2.0/3.0 A		B1
	(v)	VIt **or** Pt (in any form numerical or algebraic)		C1
		30 × 60 **or** 1800 (s) seen		C1
		180 000 J (3000 J 2/3; 0.05 kWh 3/3)		A1 15

QUESTION · 5054_s07_qp_1.pdf · source page 6 of 20

10 Four students exercise in a gym.

Which student does the most work?

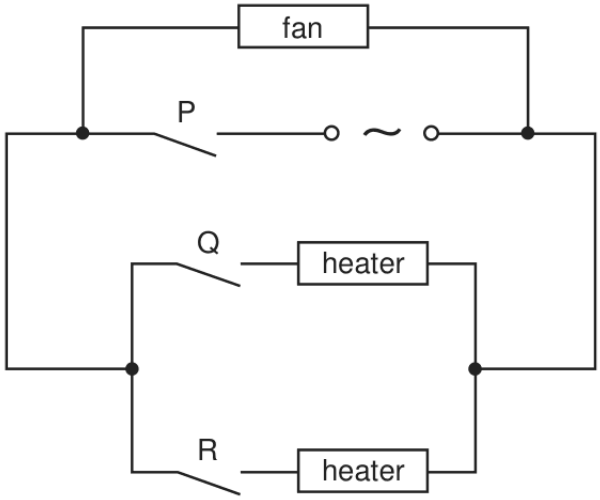
	exercise time / s	power developed / W
A	50	250
B	100	150
C	200	200
D	250	30

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 10

10 **C**

QUESTION · 5054_s07_qp_1.pdf · source page 15 of 20

31 The diagram shows the circuit for a hair-dryer.



The fan has a power rating of 0.1 kW and the heaters each have a rating of 0.4 kW. The cost of electricity is 8 cents/kW h.

What is the cost of running the dryer for two hours with switches P and Q closed and switch R open?

A 1.6 cents **B** 3.2 cents **C** 6.4 cents **D** 8.0 cents

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 31

31	**D**

10 A farmer connects a house to the mains electricity. The house is a long way from the nearest 230 V mains electricity supply.

Fig. 10.1 shows the mains supply connected to the house.

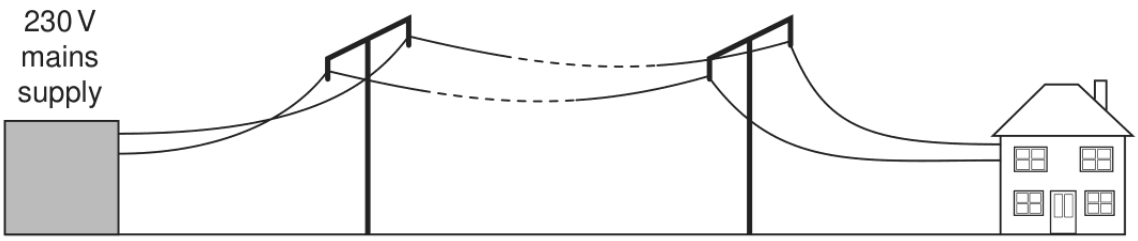


Fig. 10.1

(a) The farmer uses 230 V lamps in the house but they do not light up at full brightness. Explain why the lamps are dim. [2]

(b) The farmer adds transformers, as shown in Fig. 10.2.

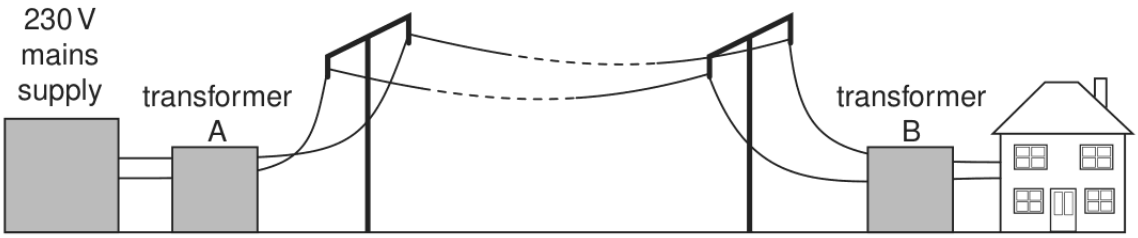


Fig. 10.2

The lamps in the distant house light up at full brightness.

Explain why the lamps are now bright. [3]

(c) **(i)** Describe the structure of a transformer. You may draw a diagram to help your explanation. [3]

(ii) Explain in detail how a transformer produces an output voltage. [3]

(d) The 230 V mains supply provides 690 W of power to transformer A in Fig. 10.2.

(i) Calculate the current supplied to the transformer. [2]

(ii) Calculate the energy supplied to the transformer in 10 minutes. Give your answer in joules. [2]

10	**(a)**	resistance of cables		B1	
		power/energy/heat loss **or** voltage drop **or** current low	in cables/wires clear	B1	[2]
	(b)	low(er) current in line **or** less voltage drop/power/heat/energy loss		B1	
		A steps voltage up or increases voltage or reduces current		B1	
		B steps voltage down or decreases voltage or increases current		B1	[3]
	(c) **(i)**	two coils (no label needed)		M1	
		coils labelled/described primary/input **and** secondary/output **or** insulated **or** copper		A1	
		coils on complete (soft) iron (core) (accept from labelled diagram or description)		B1	
	(ii)	alternating/changing **current** input		B1	
		(alternating) magnetic field (produced in core **or** coil)		B1	
		induced e.m.f./voltage/current (in secondary coil)		B1	[6]
	(d) **(i)**	$I = P/V$ algebraic or numerical		C1	
		3 A		A1	
	(ii)	$E = VIt$ **or** Pt algebraic or numerical **or** 600 (s) used		C1	
		414 000 (J) **or** 414kJ **or** 410 000 (J)		A1	[4]

QUESTION · 5054_w08_qp_1.pdf · source page 7 of 20

13 A rock climber of weight 600 N climbs up a rock face of vertical height 300 m in 3600 s.

What is the average power she generates against gravity during this time?

A 0.020 W **B** 50 W **C** 1800 W **D** 7200 W

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 13

13	**B**

QUESTION · 5054_w08_qp_1.pdf · source page 14 of 20

31 An electric heater is rated at 3 kW. Electrical energy costs 20 cents per kW h.

What is the cost of using the heater for five hours?

A 12 cents **B** 60 cents **C** 100 cents **D** 300 cents

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 31

31	**D**

For Examiner's Use

3 Fig. 3.1 shows apparatus to measure the specific latent heat of fusion of water.
In this question, you may ignore heat transfer to the ice from the room.

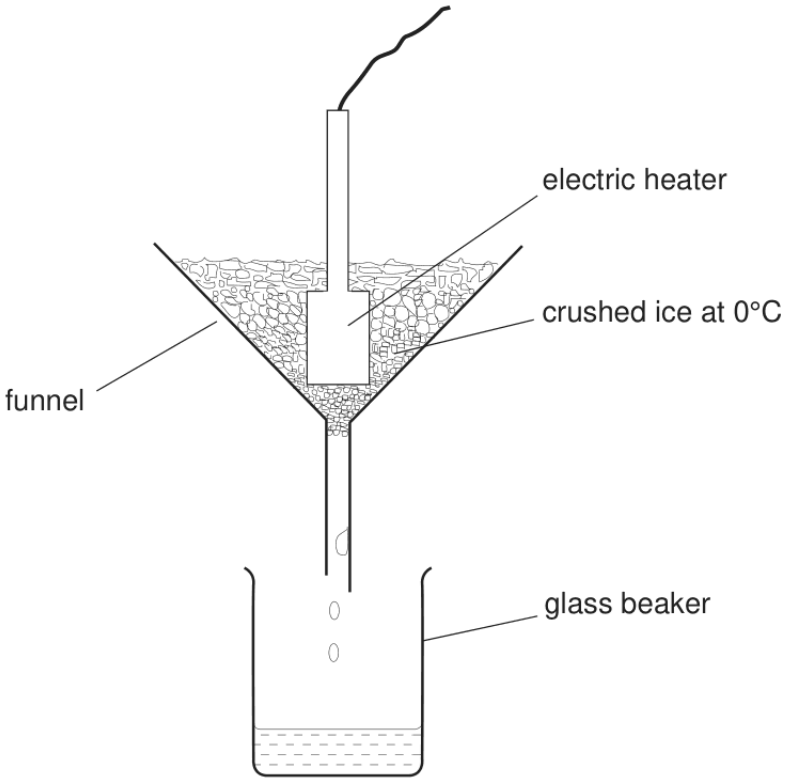


Fig. 3.1

The heater is switched on and water drips into the beaker at a constant rate.
In 2.0 minutes, 31 g of water drips into the beaker. The power of the heater is 85 W.

(a) Calculate the amount of electrical energy supplied to the heater in 2.0 minutes.

energy = ... [2]

(b) Use your answer to **(a)** to calculate the specific latent heat of fusion of water.

specific latent heat = ... [2]

(c) In another experiment using the same heater, ice colder than 0 °C is used.
State why less water drips into the beaker in 2.0 minutes.

.....

.....

..... [1]

3 **(a)** (E =) $P.t$ in any algebraic form or 85 × 120 or 85 × 2 or 170 C1
10200 J or 2.8×10^{-3} kW h A1

(b) (H =) mL seen in any algebraic form or **(a)**/31 or **(a)**/0.031 C1
330 or 329 J/g or 3.29×10^{5} J/kg ecf **(a)** A1

(c) heat/time needed to warm ice to 0°C/melting point/freezing point B1 [5]

QUESTION · 5054_w09_qp_1.pdf · source page 7 of 20

13 A man weighs 600 N. He runs up a staircase of total height 4.0 metres in 3.0 seconds.

How much useful power is needed to do this?

A 450 W **B** 800 W **C** 2400 W **D** 7200 W

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 13

13	**B**

QUESTION · 5054_w09_qp_1.pdf · source page 14 of 20

30 Which costs the **most** if operated from the same mains supply?

A a 5000 W electric cooker used for 1 minute

B a 1000 W electric fire used for 10 minutes

C a 500 W electric iron used for 1 hour

D a 100 W lamp used for 1 day

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 30

30	**D**

Section B

Answer **two** questions from this section.

Use the lined pages provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 Fig. 9.1 shows an electric boiler in a school kitchen.

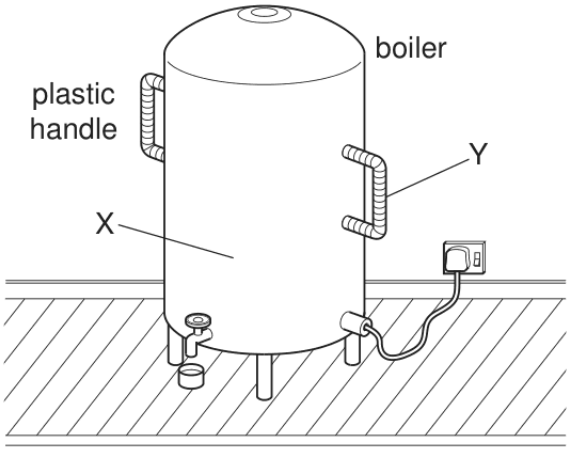


Fig. 9.1

The boiler contains 35 kg of water at 22 °C. The specific heat capacity of water is 4200 J / (kg °C).

(a) (i) Calculate the thermal energy (heat) needed to raise the temperature of the water from 22 °C to its boiling point. [3]

(ii) The water in the boiler is heated with a 2600 W immersion heater. Calculate the minimum time for the heater to bring the water to its boiling point. [2]

(iii) Suggest **one** reason why the actual time is greater than the time calculated in **(ii)**. [1]

(b) (i) The immersion heater is placed in the water at the bottom of the boiler. Explain in detail how this ensures that the thermal energy (heat) is transferred throughout the water. [4]

(ii) The boiler is made of steel and has two large plastic handles. When the water is boiling, the steel surface at X is hot while the plastic handle at Y is cool. Explain why. [2]

(c) Before the water reaches boiling point, water vapour is seen escaping from the boiler.

(i) State the name of the process that produces this water vapour. [1]

(ii) State **two** differences between this process and boiling. [2]

Section B

9	**(a)**	**(i)**	100 – 22 **or** 78		C1	
			(Q =)mcΔT **or** 35 × 4200 × 78		C1	
			1.1/1.1466/1.15 × 10^7 J		A1	
		(ii)	(t =)E/P **or** P = E/t **or** 1.15 × 10^7/2600		C1	
			4.4/4.41/4.42 × 10^3 s		A1	
		(iii)	heat escapes/lost (to kitchen) **or** heat to heat the boiler/heater **or** not all heat ends up in water **or** heat to cause evaporation **or** used as latent heat (**not** heat wasted)		B1	[6]
	(b)	**(i)**	hot/warm **water** expands (**not** molecules expand)		B1	
			density (of hot/warm water) decreases		B1	
			hot/warm water rises		B1	
			convection current/circulation **or** cold water sinks		B1	
			mixes water (max 4)		B1	
		(ii)	metal/steel is (good) conductor/poor insulator **or** plastic is poor conductor/ insulator		B1	
			more heat transferred through steel/less through plastic **or** heat transferred more quickly through steel/less quickly through plastic		B1	[6]
	(c)	**(i)**	evaporation **OR** condensation		B1	
		(ii)	any **two** points			
			only occurs at surface	boiling needs heat/ condensation releases heat	B1	
			occurs at any temperature			
			produces cooling	boiling: liquid to gas/ condensation: gas to liquid	B1	
			no bubbles	B2		[3]

[Total: 15]

QUESTION · 5054_s10_qp_11.pdf · source page 6 of 20

13 A small emergency generator supplies 432 000 000 J of electrical energy in twenty-four hours.

What is the average power output of the generator?

A 5000 W

B 300 000 W

C 18 000 000 W

D 432 000 000 W

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 13

13	**A**

QUESTION · 5054_s10_qp_11.pdf · source page 13 of 20

30 Which quantity is measured in kilowatt-hours?

A charge

B current

C energy

D power

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 30

30	**C**

For Examiner's Use

3 Fig. 3.1 shows a student rubbing her hands together.

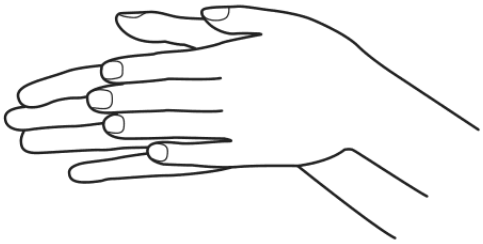


Fig. 3.1

(a) State the main energy conversion that causes the hands to become warm.

.....

.....[2]

(b) State why the hands become even warmer if they are pressed harder together when rubbing.

.....

.....[1]

(c) The average force used to slide one hand along the other is 1.2 N.

In each movement, one hand moves 0.080 m. The other hand remains stationary.

(i) Calculate the number of movements needed for 2.0 J of work to be done.

number = ...[3]

(ii) Each movement takes 0.20 s. Calculate the average power developed.

power = ...[2]

3	**(a)**	chemical energy to **or** K.E to	B1	
		heat/thermal energy/internal energy – at end	B1	[2]
	(b)	friction/resistive force **increases**	B1	[1]
	(c)	**(i)** work = force × distance in words, number or symbols	C1	
		1.2 × 0.08 **or** 0.096 **or** 20 **or** 0.208 seen	C1	
		20.8 **or** 21	A1	[3]
		(ii) power = work/time **or** energy/time, numerical or algebraic – may use values in **(i)**		
		accept 2/0.2 or 10 **W**	C1	
		0.48 W (ecf **(i)** including power of ten error in **(i)**)	A1	[2]

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a laboratory thermometer.

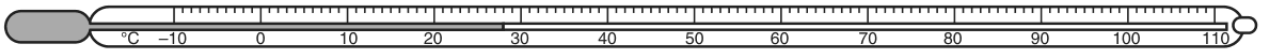


Fig. 9.1

(a) (i) State the range of the thermometer.

.....[1]

(ii) State one change in the design of the thermometer to increase its range.

.....

.....[1]

(b) (i) Describe how the behaviour of a more sensitive thermometer is different from a less sensitive thermometer.

.....

.....[1]

(ii) State one change in the design of the thermometer to make it more sensitive.

.....

.....[1]

(c) Describe how a clinical thermometer differs from a laboratory thermometer. A diagram may be included in your answer.

.....

.....

.....

.....[3]

For Examiner's Use

(d) (i) In the space below, draw a labelled diagram of a thermocouple thermometer.

[2]

(ii) State **two** reasons why a thermocouple thermometer is sometimes a better choice than a laboratory thermometer.

.....

.....

.....[2]

(e) Fig. 9.2 shows a copper block of mass 1.8 kg with two holes in the top. An 80 W heater is placed in one hole and a thermometer in the other.

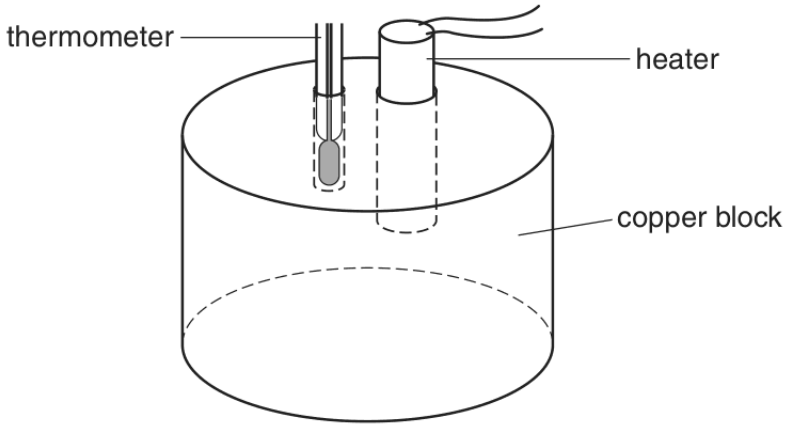


Fig. 9.2

The heater is switched on for 5.0 minutes. Assume that no energy is lost from the block.

(i) Calculate the energy supplied to the block.

energy supplied = ...[2]

(ii) The specific heat capacity of copper is 390 J/(kg °C). Calculate the rise in temperature of the block.

temperature rise = ...[2]

Section B

9	**(a)**	**(i)**	120°C **or** –10°C to 110°C	B1	[1]
		(ii)	longer thermometer **or** wider bore **or** less mercury **or** smaller bulb **not** change liquid	B1	[1]
	(b)	**(i)**	measures small(er) change in temperature **or** small(er) range for same distance **or** large(r) expansion for (same) temperature rise	B1	[1]
		(ii)	larger bulb **or** more liquid **or** narrower bore/tube **or** use liquid that expands more	B1	[1]
	(c)		constriction/narrowing (**accept** 1st and 3rd marks on diagram) mercury/thread breaks at constriction (on cooling) **or** thermometer is a "maximum" thermometer range different more sensitive/divisions further apart triangular cross-section/acts as lens thin(ner) bulb (quick response to temperature change) ANY 3 lines	B3	[3]
	(d)	**(i)**	two different metals joined	M1	
			connected to meter/ammeter/galvanometer/voltmeter	A1	[2]
		(ii)	low/high temperatures **or** greater range responds quickly/measures rapidly changing temperatures measures temperature at a point electronic output more robust measures temperatures at a distance (**not** more sensitive) ANY 2 lines	B2	[2]
	(e)	**(i)**	(Q =) Pt **or** 80 × 5 × 60 **or** 80 × 5 **or** 400 **or** 300 (s) seen	C1	
			24000 J cao	A1	[2]
		(ii)	(Q =) $mc\Delta T$ **or** 1.8 × 390 × T = 24000 in any form ecf **(i)**	C1	
			34°C (**accept** 34.188, 34.18, 34.19, 34.2)	A1	[2]

QUESTION · 5054_s10_qp_12.pdf · source page 6 of 20

12 A small emergency generator supplies 432 000 000 J of electrical energy in twenty-four hours.

What is the average power output of the generator?

A 5000 W

B 300 000 W

C 18 000 000 W

D 432 000 000 W

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 12

12	**A**

QUESTION · 5054_s10_qp_12.pdf · source page 12 of 20

28 Which quantity is measured in kilowatt-hours?

A charge

B current

C energy

D power

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 28

28	**C**

For Examiner's Use

2 Fig. 2.1 shows a motor used to operate a lift (elevator). There is a man in the lift.

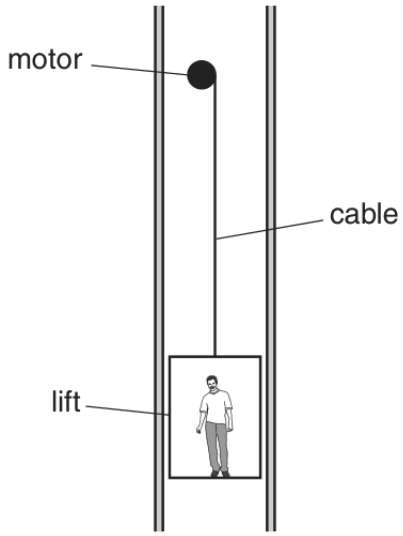


Fig. 2.1 (not to scale)

The input power to the motor is 6200 watts.

(a) Define the *watt*.

.....

.....[2]

(b) The lift and the man have a total mass of 580 kg. The lift moves up a distance of 12 m in 15 s.

(i) Calculate the tension in the cable when the lift is moving at constant speed.

tension = ...[1]

(ii) Calculate the increase in potential energy of the lift and man.

potential energy = ...[2]

(iii) Calculate the efficiency of the motor.

efficiency = ...[2]

2	**(a)**	energy/time	C1	
		1 joule in one second	A1	[2]
	(b)	**(i)** 5800 N or 5684 N or 5700 N	B1	[1]
		(ii) *mgh* algebraic, words or numerical (i.e. 580 ×10 × 12)	C1	
		69 600 J or 70 000 J or 68 208 J or 68 000 J	A1	[2]
		(iii) (efficiency =) output power or energy/input power or energy algebraic or numerical or 93 000 seen or 4 640 seen	C1	
		0.75 or 75% (accept 0.748) e.c.f. from (ii)	A1	[2]

[Total: 7]

For Examiner's Use

5 A motor car has a 12 V electrical supply. Fig. 5.1 shows a heating element in the rear window. This heater uses a current of 35 A from the supply.

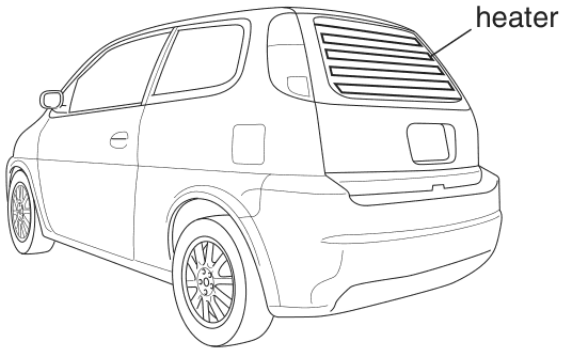


Fig. 5.1

(a) Calculate

(i) the power developed by the heater,

power = ... [2]

(ii) the thermal energy (heat) generated by the heater in 2.0 minutes.

thermal energy = ... [3]

(b) One cold morning, the rear window of this car is covered in ice. Ice has a specific latent heat of fusion of 330 J/g.

(i) Calculate the maximum mass of ice that this heater melts in 2.0 minutes.

mass = ... [2]

(ii) State two reasons why the actual quantity of ice melted in 2.0 minutes may be less than the value calculated.

1. .....

2. .....

[2]

5	**(a)**	**(i)**	(P =) VI **or** 12 × 35	C1	
			420 W **or** J/s	A1	[2]
		(ii)	(Q =) Pt **or** VIt **or** 12 × 35 × 2 **or** 420/ecf **(i)** × 2	C1	
			12 × 35 × 120 **or** 420/ecf **(i)** × 120	C1	
			$5.0(4) \times 10^4$ J	A1	[3]
	(b)	**(i)**	(m =) Q/l_f **or** $5.04 \times 10^4/330$	C1	
			150/153/152.7272) g **or** 152 g from 5.0×10^4 J	A1	[2]
		(ii)	heat lost to **glass/air/wires/water/surroundings**		
			(i.e. specified heat loss)	B1	
			ice below 0°C	B1	[2]

11 Fig. 11.1 shows a workman using a cordless electric drill.

For Examiner's Use

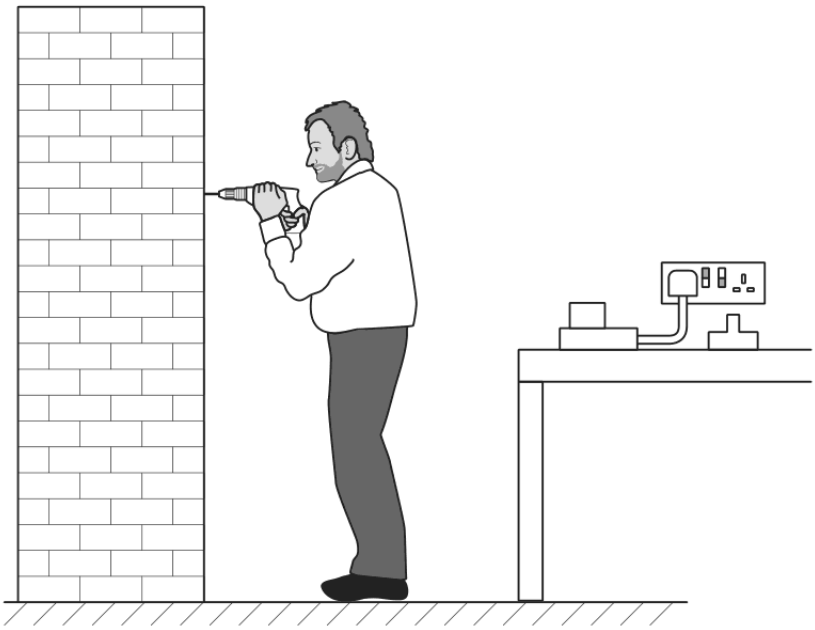


Fig. 11.1

The motor of the drill is powered by a rechargeable battery with an electromotive force (e.m.f.) of 18 V. When the drill is used, the power supplied to the motor is 450 W.

(a) Explain what is meant by an *e.m.f. of 18 V*.

.....

.....

.....[2]

(b) The workman uses the drill for 90 minutes.

Calculate

(i) the electrical energy supplied to the motor,

energy = ...[3]

(ii) the charge that the battery supplies.

charge = ...[2]

For Examiner's Use

(c) After 90 minutes of use, the battery is flat. It is connected to a charger and is recharged. The charger includes a transformer that produces a 23V a.c. output from a 230V a.c. mains supply.

(i) Draw a labelled diagram to show the structure of the transformer.

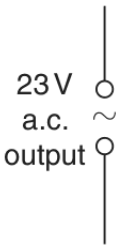
[2]

(ii) State how the transformer ensures that the a.c. output has a value of 23V when the input is the 230V a.c. mains supply.

.....

.....[1]

(iii) The 23V a.c. output of the transformer is used in a rectifying circuit to produce a direct current that recharges the battery. Complete the circuit diagram of the circuit.



[2]

(d) State and explain **one** advantage of using an alternating current (a.c.) for long-distance transmission of electrical power.

.....

.....

.....

.....[3]

11	**(a)**	energy released/unit charge **or** power released/unit current		C1	
		18 J/C **or 18** W/A		A1	[2]
	(b)	**(i)** (*t* =) 5400 **or** 60 × 90 **or** 1.5 **or** 90/60 **or** (*E* =) *Pt* **or** 450 × 90 450 × 60 × 90 **or** 450 × 5400 **or** 4.0/4.05/4.1 × 10^4 **or** 0.45 × 1.5		B1	
			or 0.45 × 90/60 **or** 450 × 1.5 **or** 450 × 90/60	C1	
			2.4(3) × 10^6 J **or** 0.675 kWh	A1	
		(ii) (*Q* =) *E*/emf (**ign.** emf = *E*/*Q*) **OR** (*I* =) 25 (A) **or** 25 × 5400			
			or 2.4(3) × 10^6/18 **or** 25 × 60 × 90	C1	
			1.3/1.35/1.4 × 10^5 C	A1	[5]
	(c)	**(i)** laminated/iron core		B1	
			two coils on core	B1	
		(ii) turns ratio = 10:1 (may be shown on diagram)		B1	
		(iii) diode symbol		B1	
			symbol for battery/cell (**allow** either polarity w.r.t. diode) **and** complete circuit	B1	[5]

(d)	can be transformed/operate transformer/voltage can be changed		B1	
	high voltage/low current transmission (possible) **or** changing magnetic field		B1	
	less energy/power loss **or** less heating (in wires) **or** thinner wires		B1	[3]

MARKING SCHEME CODE:

3 Fig. 3.1 shows part of a small hydroelectric system that generates electricity from moving water.

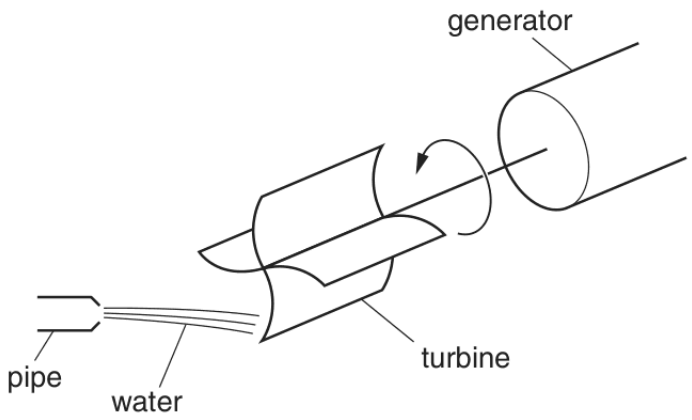


Fig. 3.1

(a) Every minute, water with kinetic energy 14 000 J emerges from the pipe. The water turns a turbine that is connected to the generator.
Half of the kinetic energy of the water is given to the generator.

(i) Calculate the power input to the generator.

power = ... [3]

(ii) Suggest a reason why only some of the kinetic energy of the water is given to the generator.

.....

..... [1]

(b) Hydroelectric systems use sources of renewable energy.

(i) State what is meant by *renewable energy*.

.....

..... [1]

(ii) State one source of **non**-renewable energy.

..... [1]

3	**(a)**	**(i)**	7000 (J) seen **or** 50% **used** somewhere	C1	
			(*P* =) *E*/*t* // 14000/*t* // 7000/*t* seen // 7000 J/minute // 420 000 J/hour	C1	
			120 W	A1	
		(ii)	water after hitting turbine still moves // has KE/energy/velocity // energy lost due to friction // friction **and** location // heat/internal energy **and** location // water misses turbine	B1	
	(b)	**(i)**	can be replaced/made // will not run out	B1	
		(ii)	coal, oil, gas, peat, nuclear, uranium (**not** solar)	B1	[6]

6 The cable from the mains plug to a washing machine contains a live wire, a neutral wire and an earth wire. The earth wire is connected to the metal case of the washing machine.

(a) Explain how connecting the earth wire to the metal case makes the washing machine safer.

.....

.....

.....[2]

(b) When in use, the average input power to the washing machine is 500 W.

Calculate the number of kW h of energy used by the washing machine in 45 minutes of use.

number of kW h = ...[2]

6 **(a)** no shock // no electrocution B1
(if) case becomes live // live touches case B1

(b) correct conversion to kW, 0.5 seen // conversion to hours // 0.75 // $\frac{45}{60}$ // (E =) $P \times t$ C1
0.375 // 0.38 // 0.37 (kW h) A1 [4]

2 Fig. 2.1 shows an elevator (lift) that is supported by a cable.

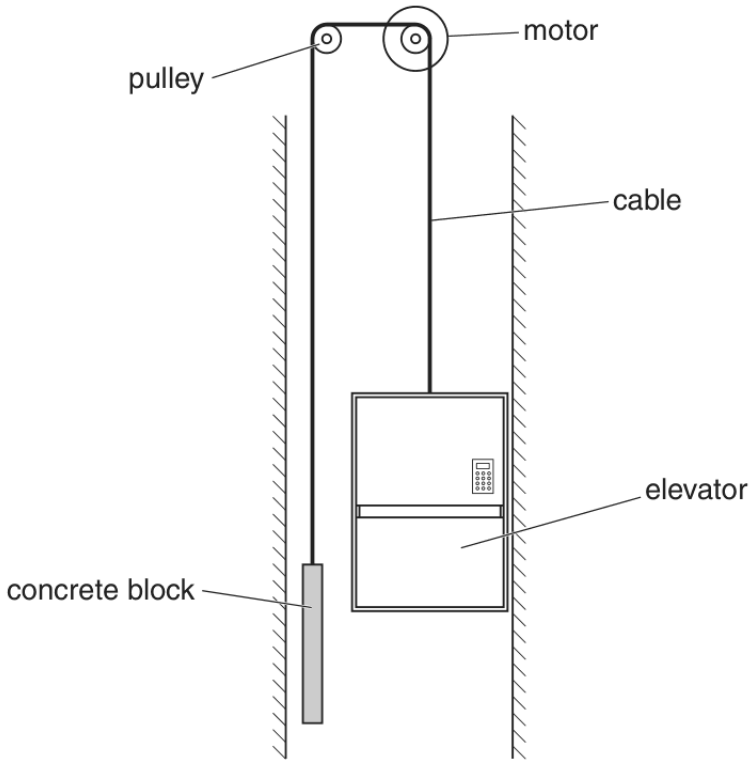


Fig. 2.1 (not to scale)

The cable passes around the axle of an electric motor and over a pulley. The cable is then attached to a concrete block that has the same mass as the empty elevator. As the elevator moves up, the concrete block moves down.

(a) Explain why very little work is done by the electric motor when lifting up the empty elevator.

.....

.....

.....[2]

(b) Several people of total weight 4900 N are travelling upwards at constant speed in the elevator. They rise through a distance of 24 m in 28 s.

Calculate the minimum power output of the electric motor.

power = ...[3]

2	**(a)**	work is done by the (falling) block **or** block loses (G)PE **or** energy transferred from block to elevator **or** forces balance	B1	
		(WD by falling block) raises the elevator **or** converted to GPE **of elevator or** (WD against) friction/air resistance **or** WD to accelerate elevator	B1	
	(b)	(WD =) $F \times d$ **or** 4900 × 24 **or** 117 600 **or** (P =) E/t	C1	
		4900 × 24/28 **or** 117 600/28	C1	
		4.2×10^3 W **or** 4.2 kW	A1	[5]

QUESTION · 5054_s12_qp_11.pdf · source page 12 of 16

31 A lamp connected to a 12 V supply converts energy at a rate of 36 W.

How much energy will be converted in 10 s?

A 30 J **B** 36 J **C** 120 J **D** 360 J

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 31

31	**D**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1a shows a room heater. Fig. 9.1b is a diagram of the electric circuit of the heater.

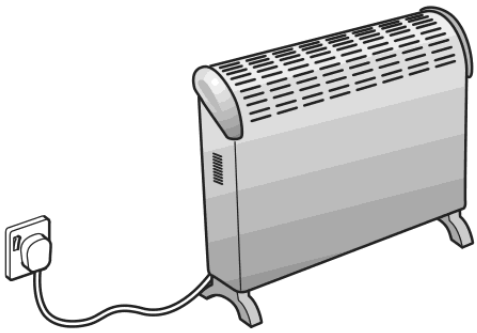


Fig. 9.1a

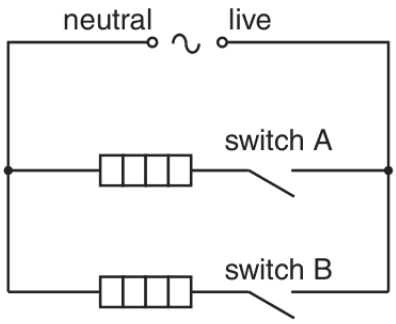


Fig. 9.1b

The fuse has not been drawn on the circuit diagram in Fig. 9.1b.

(a) (i) On Fig. 9.1b, draw the symbol for a fuse in the correct position. [2]

(ii) State the part of the room heater to which the earth wire is connected.

.....[1]

(iii) The earth wire reduces the chance of an electric shock if a fault develops in the room heater.

1. State one fault that causes an electric shock when a person uses the room heater without an earth connection.

.....

.....[1]

2. Explain how using an earth connection prevents an electric shock.

.....

.....

.....[2]

(b) (i) This type of room heater is very efficient. Explain what this means.

.....

.....[1]

(ii) The room heater is a convector heater. Describe and explain how thermal energy (heat) passes around a room by convection.

.....

.....

.....

.....

.....[3]

(c) Fig. 9.2 shows the power output of the room heater when each switch is closed.

	power / W
switch A only closed	600
switch B only closed	
both switches closed	2100

Fig. 9.2

(i) Determine the power output of the room heater when only switch B is closed.

power output = ... [1]

(ii) The room heater is used with both switches closed for 2.5 hours.

Calculate the energy output of the room heater

1. in kilowatt-hours,

energy = ... kW h [2]

2. in joules.

energy = ... J [2]

Section B

9	**(a)**	**(i)**	fuse symbol correct	B1	
			in live wire before junction of two elements	B1	
		(ii)	the (metal) case/outside	B1	
		(iii)	**1.** live wire touches case; live touches person	B1	
			2. current goes to earth; current does not go through the person	B1	
			fuse blows	B1	
	(b)	**(i)**	most of the energy output is useful/heat; little energy is wasted;	B1	
		(ii)	hot air rises (**not** heat rises)	B1	
			density of hot air is lower	B1	
			convection current mentioned OR hot air rises and cold air falls	B1	
	(c)	**(i)**	1500 W	B1	
		(ii)	**1.** conversion to kW seen on any power; 2.1 (kW) seen	C1	
			5.25; 5.2; 5.3 (kW h)	A1	
			2. $E = P \times t$ in any form, algebraic or using any power or time e.g. 600 × 2.5, 600 × 150	C1	
			1.89×10^7 (J) OR 3.6×10^6 × **(c)(ii)1.**	A1	[15]

3 Fig. 3.1 shows the plan of a bedroom and part of the main room of a house. Other rooms are not shown.

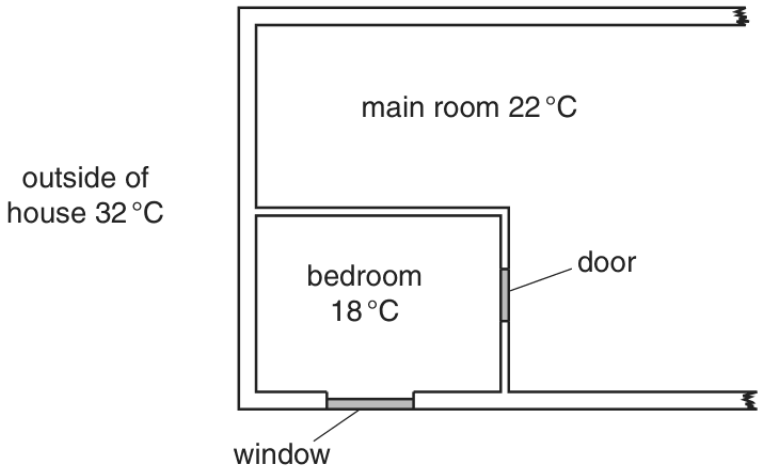


Fig. 3.1

The temperatures of the main room, the bedroom and the outside of the house are shown on Fig. 3.1.

Fig. 3.2 shows all the thermal energy (heat) inputs to the bedroom in one hour.

thermal energy input to bedroom	
through door and walls from main room	50 000 J
through walls from outside of house	2 000 000 J
through window	1 000 000 J
from person sleeping in bedroom	250 000 J

Fig. 3.2

(a) Suggest why more thermal energy enters the bedroom from the outside of the house than from the main room.

.....[1]

(b) An air conditioner keeps the temperature constant in the bedroom by removing thermal energy.

(i) Calculate the total thermal energy that the air conditioner removes from the bedroom in one hour.

thermal energy = ...[1]

(ii) The electrical power input to the air conditioner is 300W.
Calculate the electrical energy input into the air conditioner in 1 hour.

energy = ...[2]

(c) The air conditioner cools the air at the top of the room. This causes a convection current in the room.

Explain how the cold air gives rise to the convection current.

.....

.....

.....[3]

3	**(a)**	large(r) temperature difference (between bedroom and outside) OR outside is hot(ter than main room)	B1	
	(b) **(i)**	3 300 000 J(/hour)	B1	
	(ii)	(E =) $P \times t$ in any form; $300 \times 60 \times 60$ 1.08×106 J; 1.1×106 J OR 0.3 **kWh**	C1 A1	
	(c)	cold air sinks	B1	
		(cold air has a) high(er) density or contracts	B1	
		hot air rises OR hot air has a low(er) density OR (hot) air comes in to replace cold air	B1	[7]

QUESTION · 5054_w12_qp_11.pdf · source page 6 of 16

14 A builder lifts eight slabs from the ground on to the back of a lorry 1.5 m high.

The total time taken is 48 s and each slab weighs 20 N.

How much useful power does the builder produce?

A 5 W **B** 40 W **C** 240 W **D** 320 W

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 14

14 **A**

QUESTION · 5054_w12_qp_11.pdf · source page 13 of 16

31 A lamp connected to a 12 V supply converts energy at a rate of 36 W.

How much energy will be converted in 10 s?

A 30 J **B** 36 J **C** 120 J **D** 360 J

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 31

31	**D**

3 Fig. 3.1 shows an electric motor being used on a building site to lift bricks.

For Examiner's Use

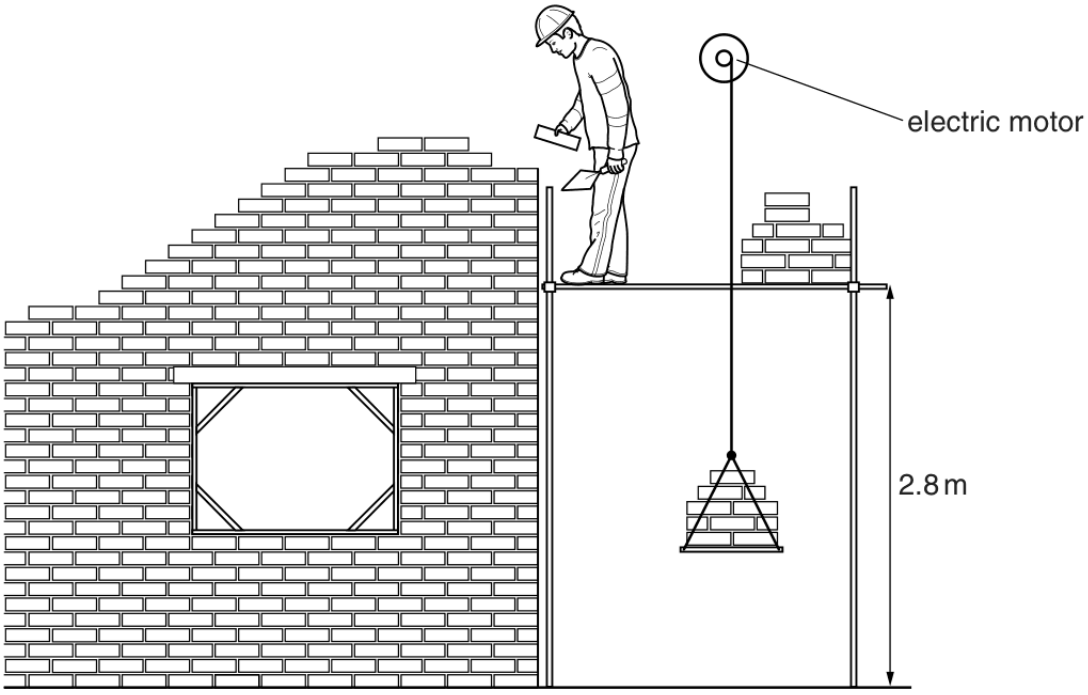


Fig. 3.1

The gravitational field strength is 10 N / kg. The motor lifts bricks of total mass 54 kg through a height of 2.8 m in 3.0 s.

(a) Calculate

(i) the work done in lifting the bricks,

work done = ... [2]

(ii) the useful output power of the motor.

power = ... [2]

(b) Suggest two reasons why the electrical power supplied to the motor is greater than its useful output power.

1. .....

2. .....
[2]

For Examiner's Use

(c) (i) Fig. 3.2 shows a battery and an electric motor. The battery is used to power the motor. Complete Fig. 3.2 to show the circuit diagram of a circuit that is used to determine the power supplied to the electric motor.

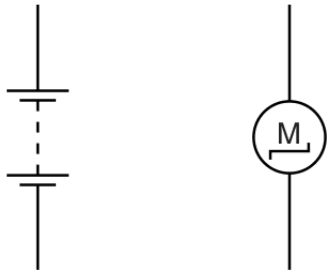


Fig. 3.2 [2]

(ii) State how the power supplied is calculated.

.....

..... [1]

3	**(a)**	**(i)**	(WD =)*mgh* **or** 54 × 10 × 2.8	C1	
			1500/1510/1512 J	A1	
		(ii)	(*P* =) WD/*t* or *E*/*t* **or** 1500/3 **or** 1510/3 **or** 1512/3	C1	
			500/503/504 W	A1	
	(b)		any **two** of: also lifting board/rope heat in motor/wires/cable friction with something named e.g. axle/spindle/air	B2	
	(c)	**(i)**	power supply, motor and ammeter in series (**ignore** series voltmeter and other components)	B1	
			voltmeter to measure voltage across motor	B1	
		(ii)	current (reading) × voltage (reading) or VI	B1	[9]

For Examiner's Use

10 Fig. 10.1 shows a domestic heater that is used to heat a room. It contains 16 large bricks each of mass 7.5 kg.

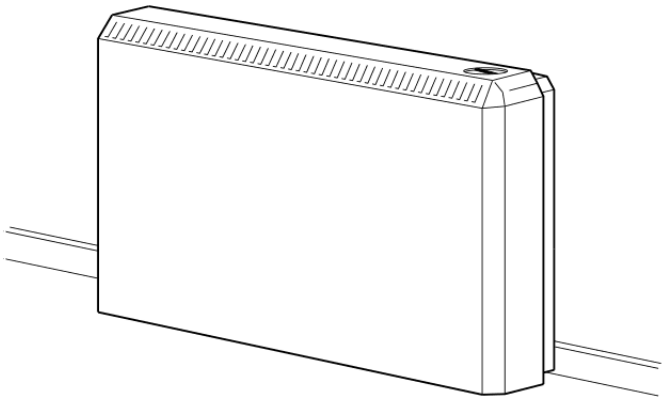


Fig. 10.1

During the night, the bricks are heated from a temperature of 17 °C to 96 °C. The bricks are made from a material that has a specific heat capacity of 2300 J/(kg °C).

(a) Calculate the thermal energy (heat) supplied to the bricks.

thermal energy = ... [3]

(b) During the day, the bricks gradually cool and the stored thermal energy is released to the room. After 7.0 hours, the bricks have cooled to 17 °C.

(i) Calculate the average rate of release of thermal energy to the room during these 7.0 hours.

rate = ... [2]

(ii) At the beginning of the day, the heater releases thermal energy at a greater rate than later in the day.

Suggest why.

.....

.....

..... [2]

For Examiner's Use

(c) Fig. 10.2 is a diagram of the heater standing on the floor of the room and placed next to one of the walls.

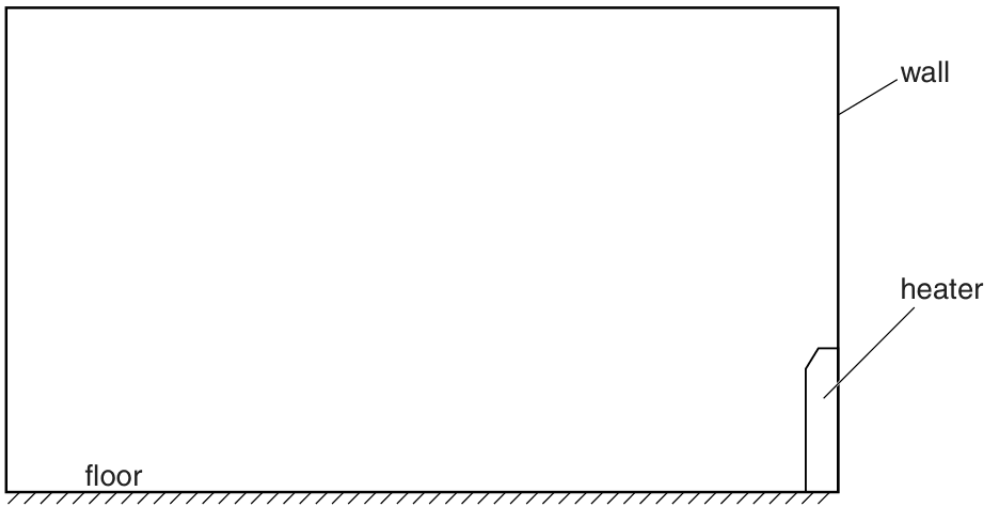


Fig. 10.2

Describe, and explain in detail, how the thermal energy is transferred throughout the room. You may draw on Fig. 10.2.

.....

.....

.....

.....

.....

..... [5]

(d) State one method of thermal insulation that is used to keep a room warm, and explain why it is effective.

.....

.....

.....

.....

..... [3]

10	**(a)**	16 × 7.5 **or** 120 **or** 96–17 **or** 79			C1
		(Q =)$mc\Delta T$ **or** 120 × 2300 × 79			C1
		2.2(2.1804) × 10^7 J			A1 [3]
	(b)	**(i)** 2.2 × 10^7/7 **or** 2.2 × 10^7/ (7 × 60) **or** 2.2 × 10^7/ (7 × 3600)			C1
		3.1 × 10^6 J / h or 5.2 × 10^4 J / min or 870 J / s or W			A1
		(ii) (heater/bricks) hot(ter) (**not** room cooler)			B1
		great(er) temperature difference (between heater and room)			B1 [4]
	(c)	air (next to heater) gets hot or conduction through metal/casing			B1
		expands or radiation or IR (radiation)			B1
		less dense			B1
		rises			B1
		circulation or convection current or arrows on Fig. 10.2			B1 [5]
	(d)	double glazing/cavity walls/ceiling tiles/carpet/curtains/loft insulation etc.	**or**	shiny foil	B1
		traps air		radiation reflected	M1
		air is poor conductor/convection prevented		IR radiation/ back into room	A1 [3]

[Total: 15]

QUESTION · 5054_w12_qp_12.pdf · source page 11 of 16

32 The cost of electricity is 16 c per kW h.

What is the cost of using a 3 kW heater for 4 hours?

A $\frac{3 \times 4}{16}$ c **B** $\frac{3 \times 16}{4}$ c **C** $4 \times 3 \times 16$ c **D** $4 \times 3 \times 60 \times 16$ c

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 32

32	**C**

QUESTION · 5054_s13_qp_11.pdf · source page 8 of 20

15 Using an electric kettle, 200 g of water at 100 °C is converted into steam at 100 °C in 300 seconds.

The specific latent heat of steam is 2250 J / g.

What is the average electrical power used?

A $\frac{2250}{300 \times 200}$ W

B $\frac{200 \times 2250}{300}$ W

C $\frac{300 \times 2250}{200}$ W

D $200 \times 300 \times 2250$ W

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 15

15	**B**

QUESTION · 5054_s13_qp_12.pdf · source page 8 of 16

19 Using an electric kettle, 200 g of water at 100 °C is converted into steam at 100 °C in 300 seconds.

The specific latent heat of steam is 2250 J / g.

What is the average electrical power used?

A $\frac{2250}{300 \times 200}$ W

B $\frac{200 \times 2250}{300}$ W

C $\frac{300 \times 2250}{200}$ W

D $200 \times 300 \times 2250$ W

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 19

19	**B**

For Examiner's Use

3 Fig. 3.1 shows a man using a chest expander to increase the strength of his arms.

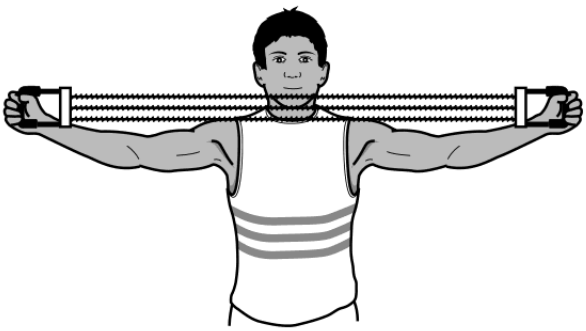


Fig. 3.1

The chest expander may be considered to be a single spring.

Fig. 3.2 shows the force needed to produce different extensions of the spring.

force / N	extension / cm
0	0
60	10
120	20
180	30

Fig. 3.2

(a) The man increases the force on the spring from 0 to 180 N.
The spring extends by 30 cm and the average force exerted during this process is 90 N.

(i) Calculate the work done on the spring.

work = ... [2]

(ii) Twenty extensions are made in 1.0 minute.

Calculate the power used to extend the spring.

power = ... [2]

For Examiner's Use

(b) The force at the limit of proportionality of the spring is 800 N.

(i) Calculate the extension of the spring when the force is 800 N.

extension = ... [2]

(ii) The force on the spring is increased from 800 N to 860 N.

Suggest what happens to the extension.

.....

..... [1]

3	**(a)**	**(i)**	(*W*=) *Fd* **or** 90 × 0.3 **or** 90 × 30	C1	
			27 J	A1	
		(ii)	(*P*=) *W*/*t* **or** *Fd*/*t* **or** 27(× 20)/60 **or** 27/3	C1	
			9(.0) W	A1	
	(b)	**(i)**	800 × 30/180 **or** 800/6 **or** 6 seen **or** proportionality clearly used	C1	
			133 **or** 130 cm	A1	
		(ii)	extension more than 143 cm **or** (extra) extension > 10 cm **or** (some) extension permanent	B1	[7]

11 A student places a small electrical heater inside a cup of water, as shown in Fig. 11.1.

For Examiner's Use

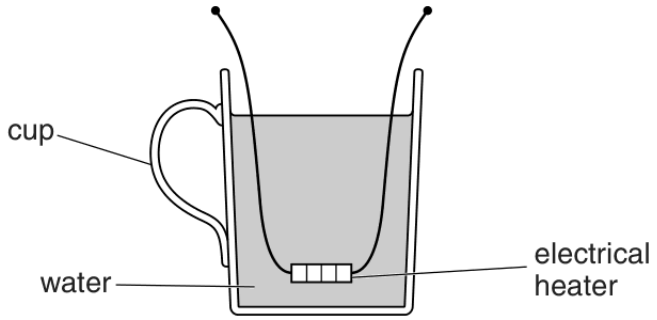


Fig. 11.1

The student determines the electrical power of the heater.

(a) In the space above the cup on Fig. 11.1, draw the electrical circuit that the student uses. Include an ammeter, a voltmeter and a power supply. [2]

(b) The voltage of the power supply is 12 V and the current is 4.2 A.

(i) Calculate the electrical power input to the heater.

power = ... [2]

(ii) Calculate the energy input to the heater in 8.0 minutes. Give your answer in kW h.

energy = ... kW h [3]

For Examiner's Use

(c) During heating, the student notices that some of the water evaporates from the cup.

(i) Describe, using ideas about molecules of water, what happens during evaporation.

.....

.....

..... [2]

(ii) The student finds that the amount of evaporation increases when the temperature of the water is higher.

State and explain one other change that increases the amount of evaporation.

.....

.....

..... [2]

(iii) State two differences between evaporation and boiling.

1. .....

.....

2. .....

.....

[2]

(d) The student turns off the power supply and the water cools.

Describe and explain how convection in the air causes the water to cool.

.....

.....

.....

..... [2]

MARK SCHEME · 5054_s13_ms_22.pdf · source page 6 of 7 · Question 11

11 (a) power supply with ammeter and heater in series B1
voltmeter in parallel with heater/ power supply B1 [2]

(b) (i) (*P*=)*VI* in any form C1
or 4.2 × 12

50(.4) W A1

(ii) (*E*=) *Pt* i.e. any power × any time e.g. 50(.4) × 8 C1

8/60 C1
or 0.13(3) seen
or division by 1000 seen anywhere

0.0067(2) (kWh) A1 [5]

MARK SCHEME · 5054_s13_ms_22.pdf · source page 7 of 7 · Question 11

(c) (i) molecules escape (from surface/leave water) / become gas or vapour / break bonds C1

fast(er) moving / high energy/ energetic molecules escape A1

(ii)

change **M1**	**explanation** **A1**
wind / draught / breeze	wind knocks molecules away
or larger surface area	more chance/possibility of escape/more space to escape **or** more molecules come to/near/at surface
or decrease humidity / drier air	fewer molecules return/from air
or decrease atmospheric pressure	fewer air molecules to hit during escape

(iii) evaporation occurs at surface **and** boiling inside liquid/bubbles
evaporation occurs at any temperature (accept room temperature)
and boiling occurs at boiling point/100°C/ fixed / specific temperature
evaporation increased by draughts/higher temp/more area **and** boiling is not
OR increase in pressure stops boiling but only reduces evaporation
any two B2 [6]

(d) water heats air (by conduction) B1
or water loses heat/energy (to cup or air)
or air gains heat/energy (from water)

hot / heated air / particles rise B1 [2]
or cold air / particles sink
or hot air is less dense
or cold air is more dense

[Total: 15]

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 **(a)** State what is meant by the *moment of a force*.

.....

.....[2]

(b) The anchor of a sailing ship has a mass of 350 kg. Six sailors raise the anchor from the sea-bed by turning a large axle. They push the handles attached to the axle and it rotates. The anchor is on the end of a chain that winds on to the rotating axle.

Fig. 9.1 shows the sailors lifting the anchor.

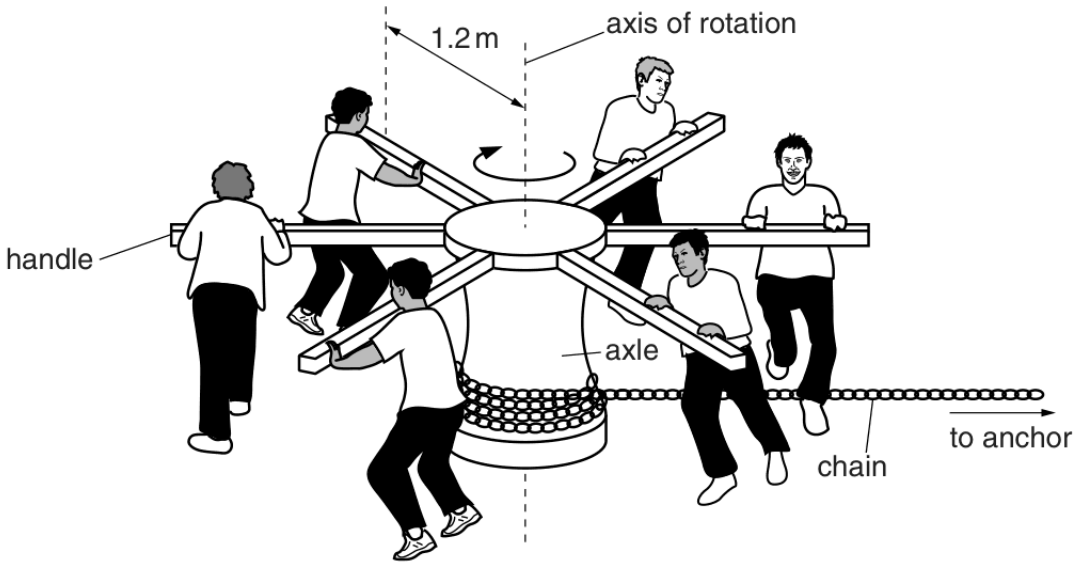


Fig. 9.1

Each of the sailors exerts a force of 750 N on his handle at a distance of 1.2 m from the axis of rotation. The axle rotates through one complete revolution and the anchor is lifted through a distance of 160 cm.

The gravitational field strength g is 10 N/kg.

(i) Calculate

1. the total moment exerted on the axle by the six sailors,

moment = ...[2]

For Examiner's Use

2. the gravitational potential energy gained by the anchor as the axle rotates through one complete revolution.

energy = ...[3]

(ii) The work done on the axle by the sailors is very much larger than the gravitational potential energy gained by the anchor.

State and explain how energy is wasted.

.....

.....

.....[2]

(iii) Explain why the power produced by the sailors is larger when the anchor is lifted at a faster rate.

.....

.....[2]

(c) Describe, with the aid of a labelled diagram, how to verify the principle of moments.

.....

.....

.....

.....

.....

.....[4]

Section B

9	**(a)**	force × distance **or** $F \times d$ with F and d defined **or** $F \times d_{perp}$	C1	
		force × perpendicular distance **or** $F \times d_{perp}$ with F and d_{perp} defined	A1	[2]
	(b) **(i)**	**1.** 6 × 750 × 1.2 **or** 750 × 1.2 **or** 900	C1	
		5400 N m	A1	
		2. *mgh* **or** 350 × 10 × 160 **or** 350 × 10 × 1.6	C1	
		350 × 10 × 1.6 **or** 5.6 × 105	C1	
		5600 J	A1	
	(ii)	friction at axle/boat **or** drag due to water **or** chain lifted also	B1	
		heat produced (**ignore** in sailors) **or** work done against friction/drag **or** work done raising chain	B1	
	(iii)	same amount of work done **or** $P = E/t$ **or** $P = WD/t$	B1	
		in less time **or** power inversely proportional to time (**ignore** faster rate)	B1	[9]
	(c)	clear/labelled diagram with ruler, fulcrum, at least two weights	B1	
		any three of the following points made in words: balance ruler (on its own) place weights on ruler so it balances clockwise and anticlockwise moments equal **or** net moment zero repeat (apply similar principles to other possible methods)	B3	[4]

[Total: 15]

QUESTION · 5054_s14_qp_11.pdf · source page 6 of 16

16 A crane lifts a load of 6000 N through a vertical distance of 15 m in 30 s.

What is the average useful power during this operation?

A 200 W **B** 400 W **C** 3000 W **D** 12 000 W

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 16

16 **C**

QUESTION · 5054_s14_qp_11.pdf · source page 12 of 16

32 One kilowatt-hour of energy costs $0.24.

How much does it cost to run a 2 kW heater for three hours?

A $0.24 **B** $0.48 **C** $0.72 **D** $1.44

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 32

32	**D**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 A garden pond contains a small fountain. An electric pump in the water causes the water to rise above the surface of the pond, as shown in Fig. 9.1.

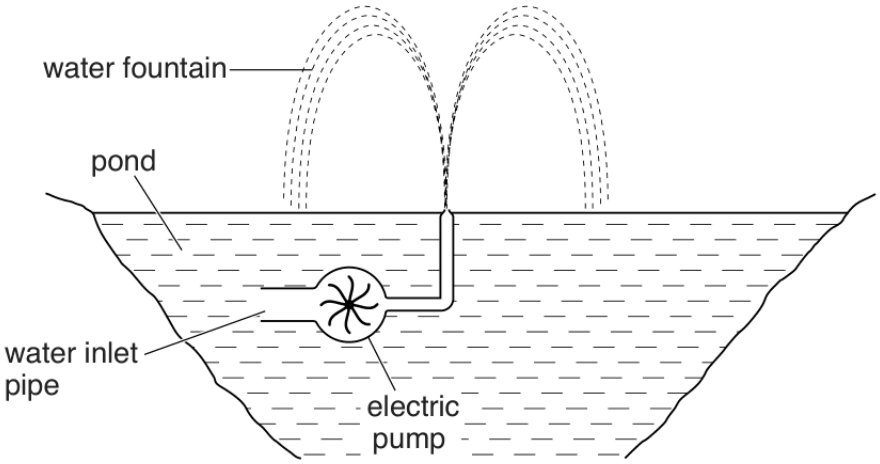


Fig. 9.1

(a) The pressure of the water in the pond increases with depth.

(i) Explain what is meant by *pressure*.

.....

..... [1]

(ii) Explain why the pressure below the surface of the water increases with depth.

.....

..... [2]

(iii) State the unit of pressure.

..... [1]

(b) Describe the energy changes that occur within the pump.

.....

.....

..... [3]

(c) A volume of 0.78 m^3 of water passes through the pump in one hour. The density of water is 1000 kg/m^3.

(i) Calculate the mass of water that passes through the pump in one hour.

mass = ... [2]

(ii) The water rises 1.5 m. The gravitational field strength g is 10 N/kg.

Calculate the useful work done in one hour raising the water to the top of the fountain.

work done = ... [2]

(iii) Calculate the minimum power output of the pump.

power = ... [2]

(d) Describe an experiment to check that the density of water is 1000 kg/m^3.

.....

.....

.....

.....

..... [2]

SECTION B

9 (a) (i) force per unit area **or** force divided by area **or** force on 1 m^2 **or** force on unit area [B1]

(ii) more water above **or** more force from water **or** more atoms / molecules above [C1]

larger weight of water above [A1]

(iii) Pa **or** N / m^2 [B1]

(b) electrical energy at start of process [B1]
kinetic energy of water / turbine/blade produced [B1]
some heat energy / thermal energy / internal energy produce [B1]

(c) (i) (M=) D × V in any algebraic or numerical form [C1]
780 kg [A1]

(ii) (W=) mgh or Fd in any algebraic or numerical form [C1]
11700 J **or** 12000 J [A1]

(iii) (P=) W / t in any algebraic or numerical form **or** 195(W) [C1]
3.25 **or** 3.2 **or** 3.3 W [A1]

(d) find difference in mass of container, with and without water [B1]
measuring cylinder or similar instrument used to find volume **and** density checked [B1]

other methods are possible, e.g. use of calibrated hydrometer

[15]

QUESTION · 5054_s14_qp_12.pdf · source page 7 of 16

18 A crane lifts a load of 6000 N through a vertical distance of 15 m in 30 s.

What is the average useful power during this operation?

A 200 W **B** 400 W **C** 3000 W **D** 12 000 W

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 18

18	**C**

QUESTION · 5054_w14_qp_11.pdf · source page 9 of 20

14 A builder carrying six bricks climbs a ladder. Each brick weighs 25 N and the builder climbs a vertical distance of 4.0 m in 8.0 s.

What is the average power used to raise the bricks?

A 13 W **B** 75 W **C** 750 W **D** 4800 W

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 14

14	**B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows the dam and reservoir of a hydroelectric power station.

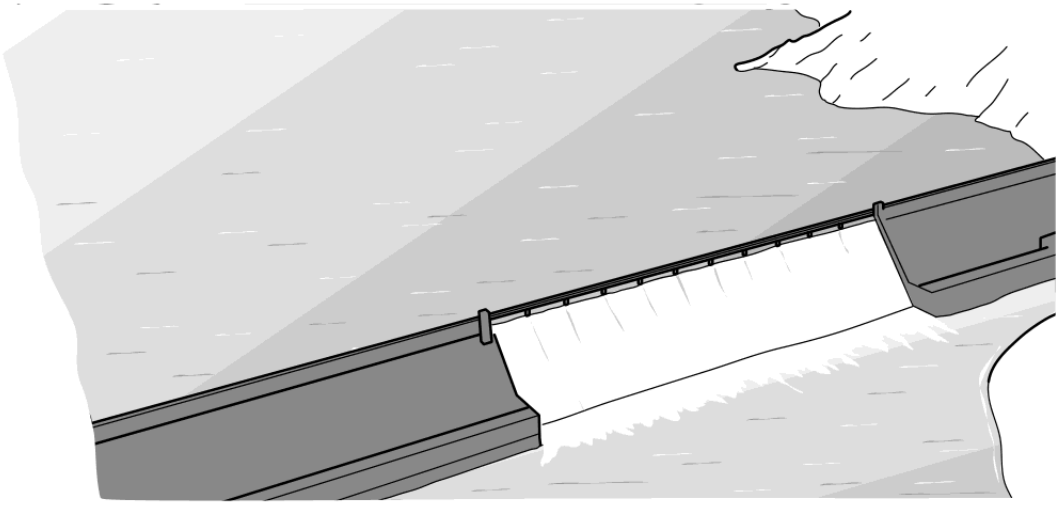


Fig. 9.1

A hydroelectric power station uses a renewable energy source.

(a) State two other renewable energy sources.

1. .....

2. .....
[2]

(b) When the power station operates at full capacity, the electrical power output is 6.8×10^9W.

(i) **1.** Calculate the electrical energy output of the power station if it operates at full capacity for one year.

energy = ...[1]

2. Suggest why, in practice, the power station does not operate at full capacity throughout the year.

.....

.....[1]

(ii) The water surface in the reservoir that supplies the hydroelectric power station is at a vertical height of 170 m above the turbines. In one hour, 1.6×10^{10} kg of water flows from the reservoir through the turbines. The gravitational field strength g is 10 N/kg.

1. Calculate the gravitational potential energy that is converted into other forms of energy in one hour.

energy = ...[2]

2. Calculate the efficiency of the power station when operating at full capacity.

efficiency = ...[2]

3. Suggest two reasons why the efficiency of the power station is less than 1.0 (100%).

1. .....

2. .....
[2]

(c) Before the electrical energy is transmitted along power lines, the voltage is increased to a very large value.

(i) State and explain why a very high voltage is used to transmit electrical energy over long distances.

.....

.....

.....

.....[3]

(ii) State the name of the device that is used to increase the voltage.

.....[1]

(iii) Explain why the generators in the power station must be a.c. (alternating current) generators.

.....

.....[1]

Section B

9	**(a)**	any **two** from: biomass / wood; geothermal power; solar power; tidal power; wave power; wind power		B2	**[2]**
	(b) **(i)**	**1.**	$2.1(4) \times 10^{17}$ J (**allow** $2.1(5) \times 10^{17}$ J if candidate uses 365.24 / 5)	B1	
		2.	any **one** from: not enough water (to maintain maximum flow); rainfall varies (during the year); periods of low demand	B1	
	(ii)	**1.**	(GPE =) *mgh* **or** $1.6 \times 10^{10} \times 10 \times 170$	C1	
			$2.7(2) \times 10^{13}$ J	A1	
		2.	$2.7(2) \times 10^{13} / 3600$ **or** $6.8 \times 10^{9} \times 3600$ **or** $6.8 \times 10^{9} / 7.5(55) \times 10^{9}$ **or** $2.4(48) \times 10^{13} / 2.7(2) \times 10^{13}$	C1	
			0.90 **or** 90%	A1	
		3.	any **two** from: friction (of water) with pipe / turbine /; viscosity of water; friction at bearings; resistance / heat in the wires; KE of water leaving turbine	B2	**[8]**
	(c) **(i)**		less energy lost / wasted **or** more efficient	B1	
			(for a given power) a high voltage results in a small(er) current	B1	
			less heat generated in wires **or** I^2R **or** less resistive losses (**not** if changed resistance mentioned)	B1	
	(ii)		transformer	B1	
	(iii)		transformers only work with an a.c. supply	B1	**[5]**
					[Total: 15]

QUESTION · 5054_w14_qp_12.pdf · source page 11 of 16

29 Which unit measures the energy input to an electrical appliance?

A ampere

B kilowatt-hour

C volt

D watt

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 29

29	**B**

10 **(a)** A student makes a diagram of the molecules in a liquid, as shown in Fig. 10.1.

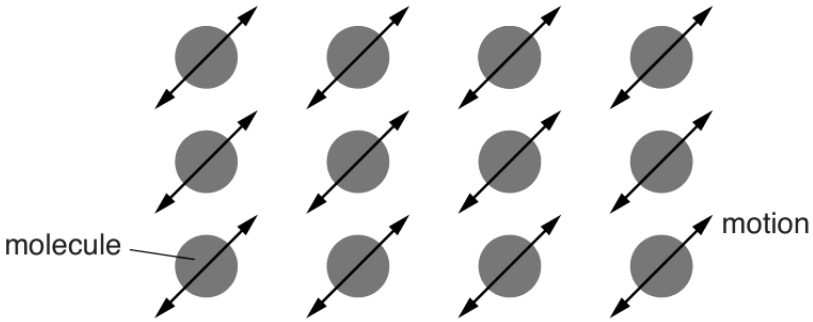


Fig. 10.1

Suggest two ways in which the student's diagram does not show the positions and motion of the molecules accurately.

1. .....

.....

2. .....

.....[2]

(b) Some of the molecules of a liquid evaporate.

(i) Describe what happens during evaporation.

.....

.....[1]

(ii) Explain why evaporation has a cooling effect on the liquid.

.....

.....

.....

.....[2]

(c) The beaker shown in Fig. 10.2 contains pure water at a temperature of 80 °C. A thermometer, initially at room temperature, is placed in the air just above the water.
The reading on the thermometer starts to increase slowly.

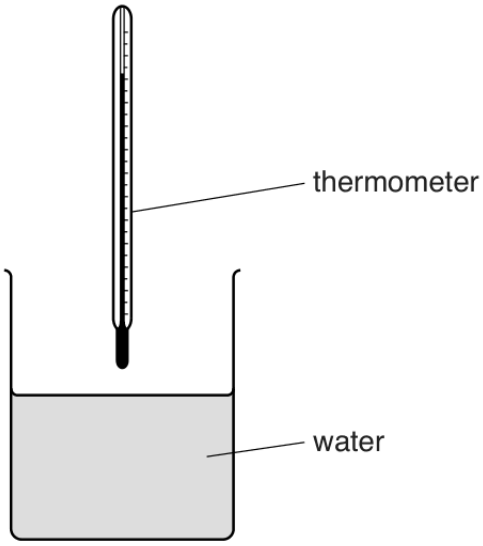


Fig. 10.2

(i) Explain how convection in the air causes the reading on the thermometer to increase.

.....

.....[1]

(ii) A heater is placed underneath the beaker and the water boils vigorously to produce steam. The thermometer reading increases quickly.

Explain how the water in the gas state gives thermal energy (heat) to the thermometer.

.....

.....

.....

.....[2]

(iii) The heater placed underneath the beaker has a power of 200 W.

The mass of water in the beaker is initially 100 g.

The water in the beaker is heated for a time of 120 s. During this time, the temperature of the water in the beaker rises from 80 °C to 100 °C and then the water boils. At the end of this time, the mass of water in the beaker is 95 g.

The specific heat capacity of water is 4.2 J / (g °C).
The specific latent heat of vaporisation of water is 2250 J / g.

Calculate the thermal energy (heat)

1. produced by the heater in 120 s,

thermal energy = ...[2]

2. used to warm the water from 80°C to 100°C,

thermal energy = ...[2]

3. used to reduce the mass of water in the beaker,

thermal energy = ...[2]

4. produced by the heater that is not transferred to the water.

thermal energy = ...[1]

MARK SCHEME · 5054_s15_ms_21.pdf · source page 4 of 5 · Question 10

10	**(a)**	(liquid) molecules not arranged (so) regularly	B1
		(liquid) molecules not vibrating / moving in same direction **or** do not have same speed	B1
	(b) **(i)**	molecules / liquid escape (from surface) / break bonds	B1
	(ii)	fast moving / more energetic molecules evaporate / escape	B1
		leaving slow molecules **or** molecules with less **kinetic** energy (on average)	B1
	(c) **(i)**	hot air rises	B1
	(ii)	(steam) condenses or changes to liquid (on thermometer)	B1
		or heat (conducted) from hot to cold gives out latent heat (to thermometer)	B1
		or explanation involving bonds being made	

MARK SCHEME · 5054_s15_ms_21.pdf · source page 5 of 5 · Question 10

(iii)	**1**	(E =)Pt **or** 200×120	C1
		24 000 J	A1
	2	(E =) mcT or $100 \times 4.2 \times 20$	C1
		8400 J	A1
	3	(E =) mL or 5×2250	C1
		11 250 J	A1
	4	4350 J **or** 1 – (2+3)	B1

QUESTION · 5054_s15_qp_12.pdf · source page 6 of 16

13 A builder lifts eight slabs from the ground on to the back of a lorry 1.5 m high.

The total time taken is 48 s and each slab weighs 200 N.

How much useful power does the builder produce?

A 50 W **B** 400 W **C** 2400 W **D** 3200 W

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 13

13	**A**

3 Two different kettles are used to heat water, as shown in Fig. 3.1.

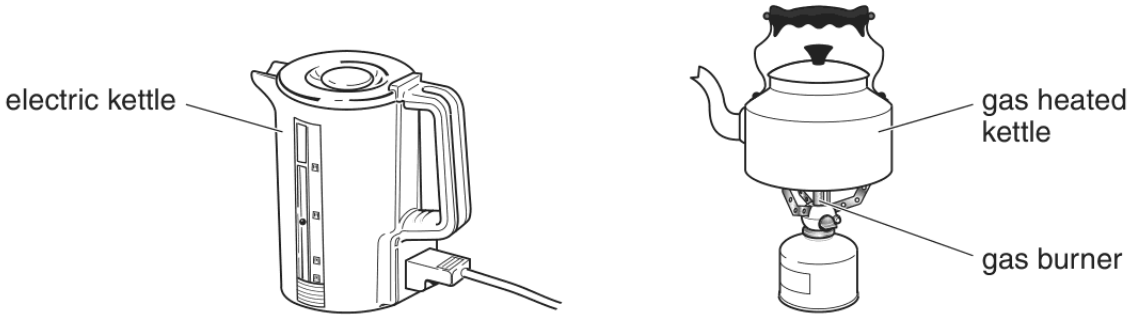


Fig. 3.1

Data for the two kettles is shown in Fig. 3.2.

	energy supplied to the kettle in one minute/J	thermal energy (heat) supplied by the kettle to the water in one minute/J
electric kettle	120 000	95 000
gas heated kettle	130 000	90 000

Fig. 3.2

(a) (i) Calculate the efficiency of the electric kettle.

efficiency = ... [2]

(ii) Calculate the useful power of the gas heated kettle.

power = ... [2]

(b) Both kettles contain the same mass of water, at the same initial temperature.

State and explain which kettle brings the water to boiling point first.

.....

.....

..... [1]

(c) The boiling water produces steam at 100 °C.

State one difference between molecules in the steam and molecules in the boiling water.

.....

.....

..... [1]

3 **(a)** **(i)** (efficiency =) useful energy ÷ input energy C1
or 95 000/120 000 (×100)
0.79(17) **or** 79(.17)**%** A1

(ii) (*P*=) energy/time **or** 90 000 / 60 C1
1500 W A1

(b) electric kettle **and** B1
- **more** energy / heat **per minute** output / into water / supplied
- more power output

transfers heat / energy faster / at a faster rate

(c) steam molecules have more potential energy; further apart; smaller force / bonds between molecules; have latent heat; more random arrangement B1

10 (a) Fig. 10.1 shows a solenoid (long coil) X connected in series with a battery, a switch S and a variable resistor (rheostat).

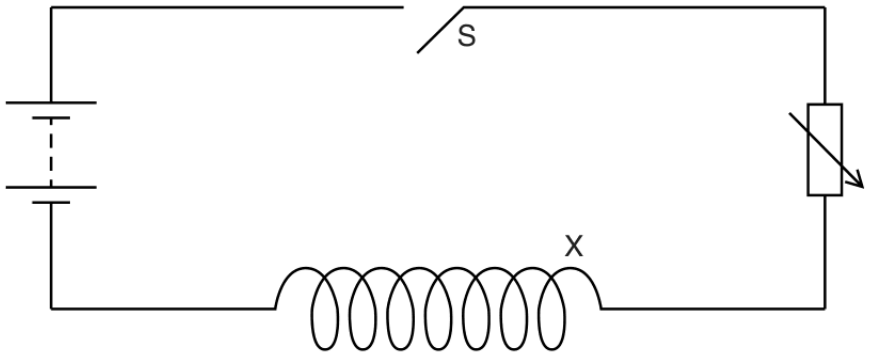


Fig. 10.1

The switch S is closed and there is a magnetic field due to the current in the solenoid.

(i) On Fig. 10.1, draw the pattern of the magnetic field in, above and below the solenoid. [3]

(ii) A second solenoid Y is connected to a sensitive centre-zero ammeter. Solenoid Y is placed in the magnetic field of X.

1. The resistance of the variable resistor is gradually decreased and the pointer of the sensitive ammeter deflects slightly to one side.

Explain why this happens.

.....

.....

.....

.....

.....

.....

..... [3]

2. The switch S is now opened.

Describe and explain what happens to the deflection on the ammeter as the switch is opened.

.....

.....

.....

.....

.....

.....

..... [3]

(b) Fig. 10.2 shows a transformer that consists of two coils wound on an iron core. The transformer is connected to an electricity transmission cable.

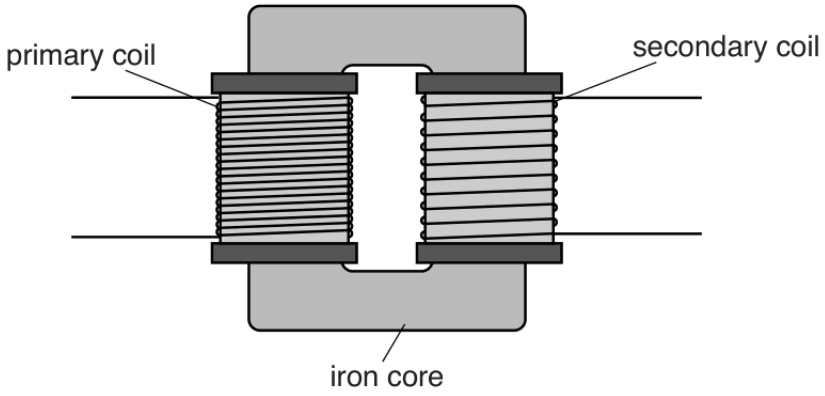


Fig. 10.2

(i) Explain the purpose of the iron core in the transformer.

.....

.....

.....

.....

.....

..... [2]

(ii) The transformer supplies electrical power to a factory at 33000V. The current supplied is 85A.

Calculate

1. the electrical power that the factory is receiving,

power = ... [2]

2. the electrical energy the factory receives in 1.0 hour.

energy = ... [2]

10	**(a)**	**(i)**	at least two straight parallel lines inside the coil	B1	
			at least two (complete) lines one above the coil and one below the coil	B1	
			third line in middle **and** evenly spaced **and** two closed loops	B1	
			(any crossings max. 2 / 3)		
		(ii)	current (in X) increases	B1	
			magnetic field becomes stronger / changes	B1	
			current / e.m.f. / voltage **induced** in Y / electromagnetic induction	B1	
			opposite deflection	B1	
			larger deflection	B1	
			magnetic field decreasing **or** quicker (rate of) change	B1	[9]
	(b)	**(i)**	to increase the strength of the magnetic field	B1	
			to direct / concentrate the magnetic field (into the secondary coil)	B1	
		(ii)	(P =)VI **or** $33\,000 \times 85$	C1	
			2.8×10^6 W **or** 2800 kW **or** 2.8 MW	A1	
		(iii)	(E =)VIt **or** $33\,000 \times 85 \times 3600$ **or** $2.8 \times 10^6 \times 3600$	C1	
			$1.0 / 1.01 / 1.008 \times 10^{10}$ J	A1	[6]
					[15]

8 A spotlight in a television studio is at full brightness. The lamp inside the spotlight is powered by a 230 V supply and the current in the filament of the lamp is 27 A.

(a) Calculate the power of the lamp.

power = ... [2]

(b) The spotlight is kept switched on at full brightness.

(i) Calculate the energy transformed by the lamp in 30 minutes.

energy = ... [1]

(ii) The cost of using one kilowatt-hour (kWh) of electricity is 23 cents.

Calculate the cost of using the spotlight for 30 minutes.

cost = ... [2]

8	**(a)**	$(P =)VI$ **or** 230×27	C1	
		6200/6210 W **or** 6.21/6.2 kW	A1	
	(b)	**(i)** $1.1/1.12/1.1178 \times 10^7$ J	B1	
		(ii) $6.21 \times 0.5 \times 23$ **or** $6.21 \times 30 \times 23$ **or** $3.1/3.105 \times 23$	C1	
		71/71.3/71.4/71.415 c **or** \$ 0.71/0.714/0.71415	A1	[5]
				[45]

11 **(a)** Induced currents can be formed in a coil of wire by electromagnetic induction.

(i) Describe how to demonstrate the formation of an induced current in a coil of wire.

Sketch and label a diagram of the arrangement of the apparatus.

.....

.....

.....[3]

(ii) State two changes to the apparatus that increase the induced current.

1. .....

2. .....
[2]

(iii) The direction of the induced current is determined by Lenz's Law.

1. State Lenz's Law.

.....

.....

.....[2]

2. Describe how Lenz's Law applies in the experiment you described in **(a)(i)**.

.....

.....

.....

.....

.....[2]

(b) An electrical generator supplies power to a distant factory.

(i) The factory receives 500 kW of electrical power at a voltage of 33 kV.

Calculate

1. the current in the wires supplying the power,

current = ...[2]

2. the electrical energy supplied to the factory in one hour.

energy = ...[2]

(ii) The 33 kV voltage supplied to the factory is very high.

Transmitting electrical energy at high voltage reduces energy losses in the transmission lines.

Explain why.

.....

.....

.....

.....

.....[2]

11	**(a)**	**(i)**	diagram showing coil of wire and either • magnet or • another coil and supply (dc and switch or ac)	B1
			coil of wire connected to an ammeter or voltmeter or cro or other method of detection, e.g. lamp	B1
			magnet or coil moved **or** change in current mentioned if another coil used	B1
		(ii)	ANY 2 from • move magnet (or coil) faster • larger current in primary (if transformer drawn) • more turns in coil • stronger magnet (if magnet drawn) • soft iron core	B2
		(iii) **1**	direction of induced current / induced emf opposes the change (that produces it)	B1 B1
		(iii) **2**	(magnetic) flux / field / poles in coil caused by movement / (induced) current in coil	B1
			statement of how opposition occurs, e.g. repulsion as magnet moves in; N pole created (by induction) at end of coil as N pole approaches	B1
	(b)	**(i)** **1**	(I=) P / V numerical or algebraic 15(.15) A	C1 A1
		2	(E=) Pt **or** VIt **or** 500(000) × 60 × 60 1.8×10^9 J **or** 500 kWh	C1 A1
		(ii)	low current P = I^2R **or** E = I^2Rt explained	B1 B1

QUESTION · 5054_s16_qp_12.pdf · source page 12 of 16

31 A 100 W lamp is switched on for five hours each day for three weeks.

The cost of one unit of electricity is $0.24.

How much does it cost to run the lamp for this time?

A $0.36 **B** $0.84 **C** $2.52 **D** $25.20

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 31

31	**C**

QUESTION · 5054_w16_qp_11.pdf · source page 6 of 20

11 Which expression is used to calculate power?

A $\frac{\text{distance}}{\text{work done}}$

B work done × distance

C $\frac{\text{work done}}{\text{time taken}}$

D work done × time taken

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 11

11	**C**

QUESTION · 5054_w16_qp_12.pdf · source page 7 of 16

15 An electric motor is used to operate a lift (elevator). The motor does 140 000 J of work in 70 s.

What is the output power of the motor?

A 0.50 milliwatts

B 2.0 watts

C 2.0 kilowatts

D 9.8 megawatts

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 15

15	**C**

QUESTION · 5054_s17_qp_11.pdf · source page 7 of 16

14 A small motor has an input power rating of 10 W and is run for 5.0 minutes.

What is the electrical energy input to the motor in this time?

A 2.0 J **B** 50 J **C** 300 J **D** 3000 J

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 14

14	D	**1**

QUESTION · 5054_s17_qp_11.pdf · source page 7 of 16

15 A small hydroelectric power station diverts water from a river. Every second, 20 kg of water flows through a pipe and falls through a vertical drop of 15 m. The efficiency of the power station is 0.60 (60%).

The gravitational field strength *g* is 10 N / kg.

What is the power output?

A 0.18 kW **B** 1.8 kW **C** 3.0 kW **D** 180 kW

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 15

15	B	**1**

6 Fig. 6.1 shows a 240 V a.c. mains supply connected to a television and two lamps.

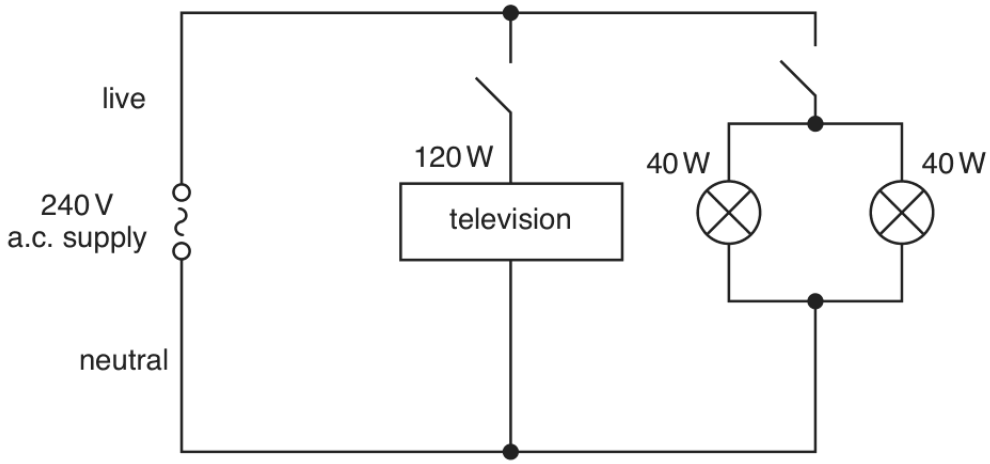


Fig. 6.1

In normal operation, the power supplied to each lamp is 40 W and the power supplied to the television is 120 W.

(a) Calculate, in normal operation,

(i) the total power supplied,

power = ...[1]

(ii) the total number of kilowatt-hours (kW h) of energy supplied to the circuit in 3.0 hours,

number of kW h = ...[2]

(iii) the current in each lamp.

current = ...[2]

(b) Explain why the switches are placed in the live wire and not in the neutral wire.

.....

.....

.....

.....[2]

Question	Answer	Marks
6(a)(i)	200 W	**B1**
6(a)(ii)	power × time **or** any numerical value of power × 3	**C1**
	0.6(0)	**A1**
6(a)(iii)	(I =) P / V algebraic or numerical	**C1**
	0.17 A	**A1**
6(b)	(if in neutral) rest of circuit / lamps / television not live / still high voltage / still 240 V **or** live has high voltage / 240 V / can cause shock	**C1**
	(if in neutral) <u>when</u> (switched) <u>off</u> circuit / lamps / television not live / still high voltage / still 240 V	**A1**

QUESTION · 5054_s17_qp_12.pdf · source page 6 of 20

14 A small motor has an input power rating of 10 W and is run for 5.0 minutes.

What is the electrical energy input to the motor in this time?

A 2.0 J **B** 50 J **C** 300 J **D** 3000 J

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 14

14	D	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 6 of 20

10 A car travels a distance of 200 m in 20 s. The engine of the car provides a driving force of 1000 N.

What is the power output of the engine?

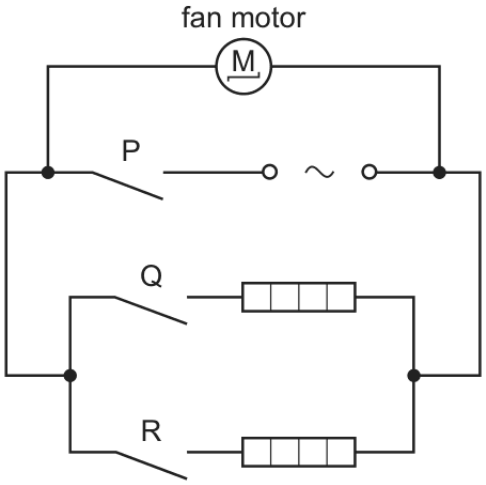
A 0.25 W **B** 4.0 W **C** 100 W **D** 10 000 W

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 10

10	D	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 15 of 20

32 The diagram shows the circuit for a hair-dryer.



The fan motor has a power rating of 0.10 kW and the heaters each have a rating of 0.40 kW. The cost of electricity is 8.0 cents / kW h.

What is the cost of running the hair-dryer for two hours with switches P and Q closed and switch R open?

A 1.6 cents **B** 4.0 cents **C** 6.4 cents **D** 8.0 cents

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 3 of 3 · row for Question 32

32	D	**1**

QUESTION · 5054_w17_qp_12.pdf · source page 5 of 16

9 In a hydroelectric power station, 4.2×10^5 kg of water passes through the turbines every second. The turbines are at a height of 50 m below the surface of the reservoir. The gravitational field strength g is 10 N / kg.

Assuming there are no energy losses, what is the power output of the power station?

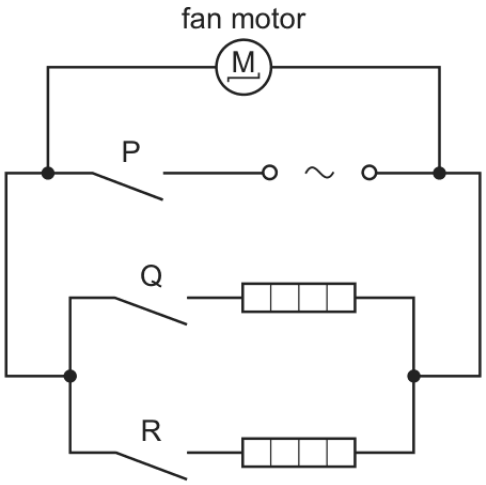
A 8.4×10^3 W **B** 8.4×10^4 W **C** 2.1×10^7 W **D** 2.1×10^8 W

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 9

9	D	**1**

QUESTION · 5054_w17_qp_12.pdf · source page 13 of 16

32 The diagram shows the circuit for a hair-dryer.



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What is the cost of running the hair-dryer for two hours with switches P and Q closed and switch R open?

A 1.6 cents **B** 4.0 cents **C** 6.4 cents **D** 8.0 cents

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 3 of 3 · row for Question 32

32	D	**1**

4 A 75 W filament lamp in an office is accidentally left switched on for 63 hours when the office is closed.

(a) (i) Calculate the electrical energy that is supplied to this lamp in this time.

electrical energy = ...[3]

(ii) The efficiency of the lamp is 0.12 (12%).

Calculate the energy released by the lamp in the form of light.

light energy = ...[1]

(b) The power station that generates the electricity supplied to the office uses oil as its fuel.

(i) State and explain whether oil is a renewable or a non-renewable source of energy.

.....

.....

.....[1]

(ii) Discuss one environmental consequence of using oil as a source of energy.

.....

.....

.....[2]

Question	Answer	Marks
4(a)(i)	(E =) Pt **or** 75×63 **or** $75 \times 63 \times 60$	**C1**
	$75 \times 63 \times 60 \times 60$ **or** 2.835×10^5 **or** 0.075×63	**C1**
	1.7×10^7 J	**A1**
4(a)(ii)	2.0×10^6 J	**B1**
4(b)(i)	non-renewable **and** gets used up / not replaced / will run out / finite	**B1**
4(b)(ii)	**three** separate approaches **I** / **II** / **III**	
	I greenhouse gases / CO / CO_2 emitted	**B1**
	global warming / <u>increase</u> greenhouse effect / ice caps melt	**B1**
	or II acid rain / NOx / SO_2 emitted	**B1**
	breathing difficulty / damage to buildings / acidify lakes **or** damages / kills plants / sea-life / animals / fish	**B1**
	or III oil spills	**B1**
	damages / kills plants / sea-life / animals / fish	**B1**

QUESTION · 5054_s18_qp_11.pdf · source page 5 of 20

12 A student calculates his power in running up a flight of stairs. He measures the vertical height of the stairs, the time taken to run up the stairs and his weight.

How does he calculate his power?

A height × time × weight

B $\frac{\text{height} \times \text{weight}}{\text{time}}$

C $\frac{\text{time} \times \text{weight}}{\text{height}}$

D $\frac{\text{weight}}{\text{height} \times \text{time}}$

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 12

12	B	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Wind energy is a renewable energy source. A wind turbine and generator convert energy in the wind to electrical energy in a generator. Fig. 9.1 shows some wind turbines.

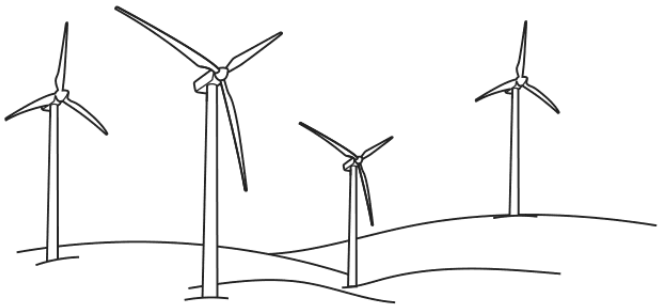


Fig. 9.1

(a) (i) State what is meant by a *renewable energy source*.

.....

.....

.....[1]

(ii) State the name of **one other** renewable energy source.

.....

.....[1]

(iii) Some methods of generating electricity cause much more global warming than wind energy.

1. State **one** method of generating electricity that causes a large amount of global warming.

.....[1]

2. Describe, in outline, how the method you have chosen causes a significant amount of global warming.

.....

.....

.....[1]

(b) During a 30 minute period, a mass of 4.2×10^7 kg of air enters the turbine with a speed of 15 m/s.

(i) Calculate the kinetic energy of the air that enters the turbine in 30 minutes.

energy = ...[2]

(ii) The electrical power output of the generator connected to the turbine is 8.4×10^5 W.

Calculate the electrical energy output from the turbine in 30 minutes.

energy output = ...[2]

(iii) Calculate the efficiency of the turbine and generator in converting the kinetic energy of the air to electrical energy.

efficiency = ...[2]

(c) The generator contains a coil that rotates between the poles of a magnet.

The coil rotates 50 times in one second and produces an alternating voltage output with a maximum value of 500 V.

(i) On the grid in Fig. 9.2, sketch a voltage–time graph showing the output voltage.

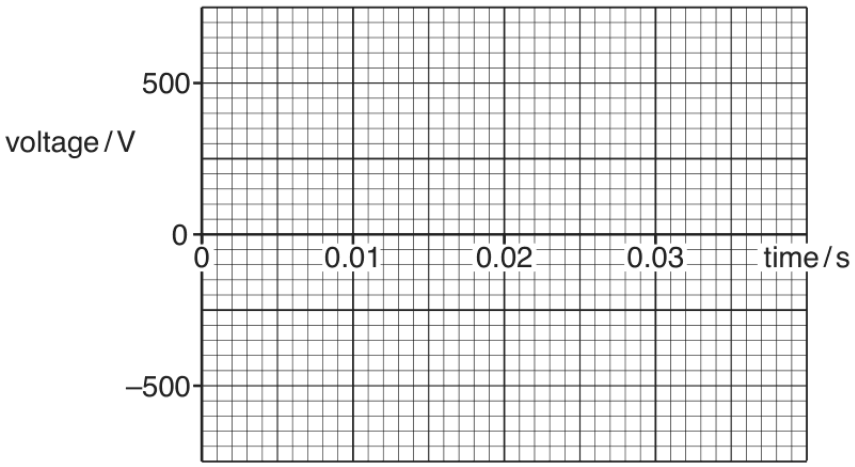


Fig. 9.2

[2]

(ii) The output voltage from the generator is stepped up by a transformer. The electrical energy then passes along a transmission line to a distant house.

Explain why

1. a high voltage is used for the transmission of electrical energy,

.....

.....

.....

.....[2]

2. a transformer is used to connect the transmission line to the house.

.....

.....

.....[1]

Question	Answer	Marks
9(a)(i)	will not run out **or** infinite **or** being replaced	**B1**
9(a)(ii)	solar / Sun, tidal, geothermal, biomass, hydroelectric, water waves	**B1**
9(a)(iii)	1 (burning) use of a fossil fuel, e.g. coal, oil, natural gas etc.	**B1**
	2 produces carbon dioxide **or** greenhouse gases (absorb infra-red from Earth)	**B1**
9(b)(i)	(E=) ½mv^2 seen in any form algebraic or numerical	**C1**
	4.7×10^9 J	**A1**
9(b)(ii)	(E=)Pt seen in any form algebraic or numerical	**C1**
	1.5×10^9 J	**A1**
9(b)(iii)	(efficiency=) (energy) output / (energy) input in any form	**C1**
	0.32 **or** 32%	**A1**
9(c)(i)	all peaks and troughs at 500 and -500 to within one small square **and** roughly sinusoidal	**B1**
	one wave takes 0.02 s within one small square	**B1**
9(c)(ii)	1 less energy loss / heat loss / power loss / more efficient	**B1**
	low(er) currents (in line) **or** thin(ner) wires can be used	**B1**
	2 step down transformer **or** voltage reduced (to 240 V) **or** otherwise voltage dangerous	**B1**

QUESTION · 5054_w18_qp_11.pdf · source page 11 of 16

31 The power of an electrical heater is 3.0 kW.

The heater is switched on for a time of 2 hours and 30 minutes.

The cost of using the heater for this time is 120 c.

What is the cost of 1.0 kW h of electrical energy?

A 16 c **B** 17 c **C** 40 c **D** 48 c

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 3 of 3 · row for Question 31

31	A	**1**

3 A large spring is stretched by an athlete to increase the strength of his arms.

The athlete extends the spring, as shown in Fig. 3.1, and then releases it gently. He extends and releases the spring several times during a period of 60 s.

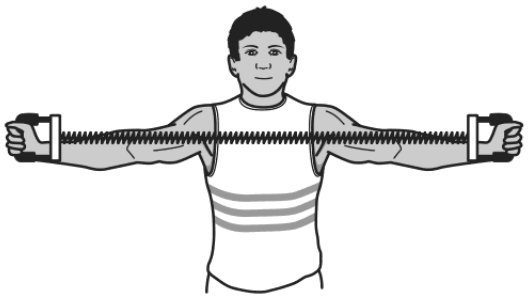


Fig. 3.1

(a) State the form of energy stored in the stretched spring and the form of energy stored in the muscles of the athlete.

form of energy in spring .....

form of energy in muscles .....
[2]

(b) Suggest what happens if he extends the spring beyond the limit of proportionality.

.....

..... [1]

(c) During one extension of the spring, its length increases from 70 cm to 93 cm. The average force exerted by the athlete is 400 N.

(i) Calculate the work done by the athlete in one extension.

work done = ... [2]

(ii) Describe how the athlete can use the value obtained in **(i)** to determine his average power during the 60 s.

.....

.....

.....

..... [2]

[Total: 7]

Question	Answer	Marks
3(a)	*in spring:* elastic or strain (potential energy)	**B1**
	in muscles: chemical (potential energy)	**B1**
3(b)	spring extends a different distance for the same (extra) load **or** load is no longer proportional to extension **or** becomes easier to pull	**B1**
3(c)(i)	(W =) Fd numerical or algebraic; 400×0.23	**C1**
	92 J	**A1**
3(c)(ii)	measure / determine the no. of times he extends the spring in 60 s	**B1**
	power = (total) work / time explained	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 8 of 16

20 A student suggests three different definitions for power.

1 energy transferred per second

2 force × distance moved

3 rate of doing work

Which definitions are correct?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 3 only

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 20

20	C	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 7 of 16

16 A student uses a newton meter to pull an object a distance *d* horizontally along a flat surface in a time of *t*. The newton meter reading is *F*.

What is given by the expression $(F \times d) / t$?

A efficiency

B energy

C power

D work

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 16

16	C	**1**

9 Fig. 9.1 shows a boiling liquid at its boiling point, trapped in a cylinder by a piston.

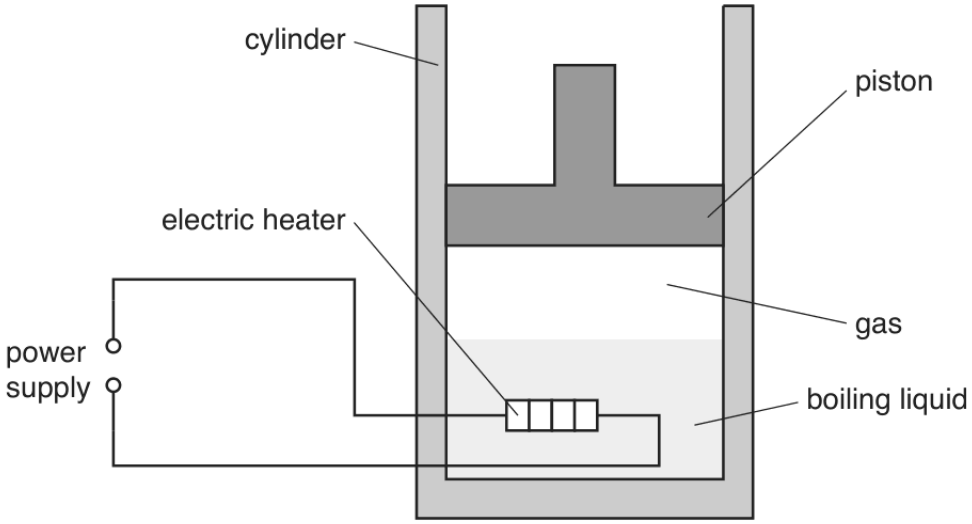


Fig. 9.1

There is an electric heater in the liquid which is connected to a power supply. The thermal energy produced by the heater gradually turns the boiling liquid into a gas.

(a) State what is meant by *boiling point*.

.....

..... [2]

(b) Explain, in terms of molecules, why it is necessary to supply energy in order to turn a liquid at its boiling point into a gas.

.....

.....

.....

..... [3]

(c) In order to determine the power of the electric heater, two meters are connected into the circuit. One meter measures the current in the heater and the other meter measures the potential difference (p.d.) across it.

On the circuit in Fig. 9.1, draw symbols to show the two meters used and where they are connected. [2]

(d) The current in the heater is 2.0 A and the p.d. across it is 6.0 V.

(i) Calculate the power of the heater.

power = ... [2]

(ii) Calculate the thermal energy supplied to the liquid by the heater in 1.0 minute.

energy = ... [2]

(iii) The specific latent heat of vaporisation of the liquid is 9.0×10^5 J/kg.

Calculate the mass of liquid that vaporises every minute.

mass = ... [2]

(e) The piston is free to move in the cylinder. As the liquid boils, the piston is pushed upwards in the cylinder at a constant speed. The volume occupied by the gas just above the liquid increases.

Discuss whether the upward force on the piston changes as the piston moves upwards at constant speed.

.....

.....

.....

..... [2]

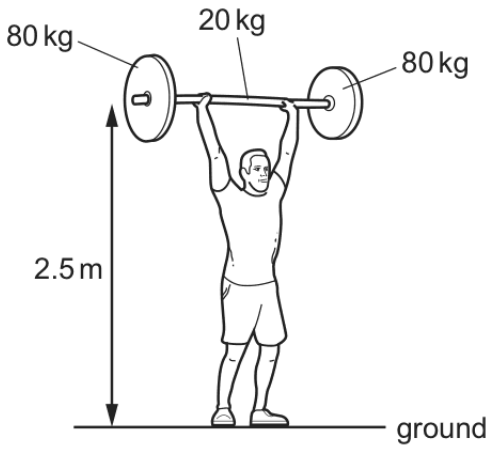
[Total: 15]

Question	Answer	Marks
9(a)	temperature	**B1**
	at which a liquid becomes a gas / vapour	**B1**
9(b)	forces (between the molecules of the liquid) / bonds mentioned	**B1**
	move molecules apart **or** overcome forces (between the molecules)	**B1**
	work done against forces **or** gains (molecular) potential energy	**B1**
9(c)	ammeter symbol in series	**B1**
	voltmeter symbol across the heater / power supply	**B1**
9(d)(i)	(P =) VI **or** 6.0×2.0	**C1**
	12 W	**A1**
9(d)(ii)	(E =) Pt **or** 12×60 **or** (E =) VIt **or** $6.0 \times 2.0 \times 60$	**C1**
	720 J	**A1**
9(d)(iii)	(m =) Q / l_v **or** $720 / 9.0 \times 10^5$	**C1**
	8.0×10^{-4} kg **or** 0.80 g	**A1**
9(e)	(it / upward force) does not change	**B1**
	always equal to weight of piston + force due to atmospheric pressure **or** no resultant force **or** piston is not accelerating	**B1**

QUESTION · 5054_s20_qp_11.pdf · source page 6 of 16

16 In a weightlifting contest, an athlete lifts a metal bar of mass 20 kg fitted with a mass of 80 kg on each end.

The lift from ground to a height of 2.5 m takes 0.50 s.



The gravitational field strength g is 10 N / kg.

What average power does the weightlifter exert in providing the gravitational potential energy during this lift?

A 0.90 kW **B** 5.0 kW **C** 8.0 kW **D** 9.0 kW

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 16

16	D	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 8 of 16

22 In an experiment to measure the power output of a small steam engine, a known load is lifted by the engine.

Which two measuring instruments are also required?

A measuring cylinder and thermometer

B measuring cylinder and metre rule

C metre rule and stop-watch

D stop-watch and thermometer

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 2 of 3 · row for Question 22

22	C	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 16 of 16

39 A student calculates the amount of energy used by an electric heater.

What is the equation for calculating the energy E in kW h?

A $E = I$ (ampere) $\times$ V (volt) $\times$ t (second)

B $E = I$ (ampere) $\times$ V (volt) $\times$ t (hour)

C $E = I$ (ampere) $\times$ V (volt) $\times$ t (second) $\div$ 1000

D $E = I$ (ampere) $\times$ V (volt) $\times$ t (hour) $\div$ 1000

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 3 of 3 · row for Question 39

39	D	**1**

9 A small glass measuring cylinder of oil is placed inside a freezer where the temperature is –18 °C. Fig. 9.1 shows how the temperature of the oil varies with time *t*.

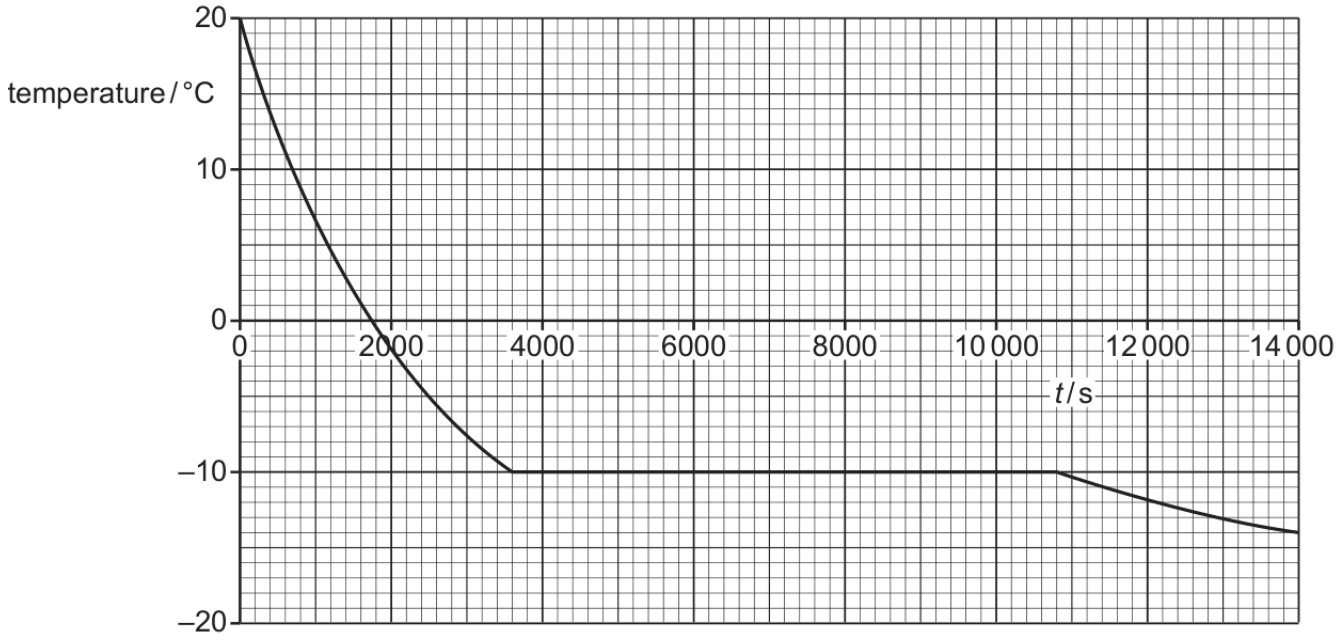


Fig. 9.1

Fig. 9.1 shows that it takes 700 s for the temperature to decrease from 20 °C to 10 °C but that it takes 1900 s to decrease from 0 °C to –10 °C.

(a) Suggest why these times are different.

.....

.....

.....

..... [3]

(b) Explain what happens to the molecules of the oil and what happens to the level of the oil in the glass measuring cylinder as the temperature decreases from 20 °C to 0 °C.

.....

.....

.....

..... [3]

(c) **(i)** Determine the melting point of the oil.

..... [1]

(ii) Explain, in terms of molecules, why the temperature of the oil does not change between $t = 3600\,\text{s}$ and $t = 10\,800\,\text{s}$.

.....

.....

.....

..... [3]

(d) There is 45 g of oil in the glass measuring cylinder and the specific latent heat of fusion (melting) of the oil is $5.7 \times 10^4\,\text{J/kg}$.

Calculate:

(i) the energy transferred from the oil between $t = 3600\,\text{s}$ and $t = 10\,800\,\text{s}$

energy = ... [2]

(ii) the average rate at which energy is transferred from the oil between $t = 3600\,\text{s}$ and $t = 10\,800\,\text{s}$.

rate = ... [1]

(e) The graph in Fig. 9.1 is steeper before the horizontal section than it is after.

Use this observation to compare the specific heat capacity of oil in the liquid and solid states and explain your reasoning.

.....

.....

.....

..... [2]

[Total: 15]

Question	answer	Marks
9(a)	(oil) hotter initially **or** cooler finally	**B1**
	temperature difference (between oil and freezer) decreases	**B1**
	loses energy faster when hotter **or** more slowly when cooler	**B1**
9(b)	molecules move more slowly **or** less kinetic energy **or** less internal energy	**B1**
	level decreases **and** oil contracts / volume of oil decreases	**B1**
	oil / liquid contracts more than glass / solid **or** molecules move closer	**B1**
9(c)(i)	–10°C	**B1**
9(c)(ii)	(attractive intermolecular) force between molecules **or** liquid solidifies	**B1**
	(intermolecular) potential energy decreases (as molecules move closer)	**B1**
	thermal energy lost (to freezer) is potential energy lost **or** latent heat lost	**B1**
9(d)(i)	(Q =) ml_f **or** $0.045 \times 5.7 \times 10^4$	**C1**
	2.6×10^3 J	**A1**
9(d)(ii)	0.36 J / s **or** 0.36 W	**B1**
9(e)	specific heat capacity of liquid oil is smaller (than of oil in solid state)	**B1**
	the temperature decreases more quickly (in liquid state)	**B1**

QUESTION · 5054_w20_qp_12.pdf · source page 7 of 16

16 What is *power*?

A force per second

B the maximum force an object exerts

C the total energy of an object

D work done per second

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 16

16	D	**1**

QUESTION · 5054_s21_qp_11.pdf · source page 12 of 16

30 A 100 W lamp is switched on for five hours each day for three weeks.

The cost of one unit of electricity is $0.24.

How much does it cost to run the lamp for this time?

A $0.36 **B** $0.84 **C** $2.52 **D** $25.20

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 3 of 3 · row for Question 30

30	C	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 12 of 16

35 A 100 W lamp is switched on for five hours each day for three weeks.

The cost of one unit of electricity is $0.24.

How much does it cost to run the lamp for this time?

A $0.36 **B** $0.84 **C** $2.52 **D** $25.20

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 3 of 3 · row for Question 35

35	C	**1**

6 The power input to a filament lamp when connected to a 12 V supply is 20 W.

(a) State what is meant by *power*.

.....

..... [1]

(b) Calculate:

(i) the current in the filament

current = ... [2]

(ii) the resistance of the filament.

resistance = ... [1]

(c) The filament in another lamp is made from thinner wire of the same length and made from the same material. Both lamps are connected to a 12 V supply.

State and explain how using thinner wire in the filament affects the power input to the lamp.

.....

.....

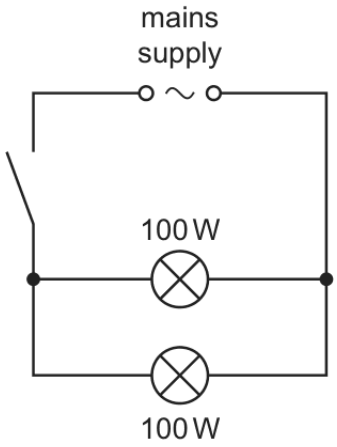
..... [2]

[Total: 6]

Question	Answer	Marks
6(a)	work done / energy (transfer) per unit time	**B1**
6(b)(i)	(I =) P / V in any form numerical or algebraic	**C1**
	1.7 A	**A1**
6(b)(ii)	6.9−7.2 Ω	**B1**
6(c)	resistance (of wire) is larger	**B1**
	power is less **and** **either** current is smaller **or** $P = V^2 / R$ ***or*** $P = VI$ **and** $V = IR$	**B1**

QUESTION · 5054_w21_qp_11.pdf · source page 15 of 20

32 Two 100 W lamps are connected in parallel to the mains supply, as shown.



How much electrical energy is supplied by the mains when the switch is closed for 36 minutes?

A 0.030 kW h **B** 0.060 kW h **C** 0.12 kW h **D** 7.2 kW h

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 3 of 3 · row for Question 32

32	C	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 15 of 20

33 The cost of electricity is 9.0 cents / kW h.

It takes 0.060 hours and costs 1.2 cents to boil the water in a kettle.

What is the power rating of the kettle?

A 0.45 kW **B** 0.65 kW **C** 2.2 kW **D** 130 kW

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 3 of 3 · row for Question 33

33	C	**1**

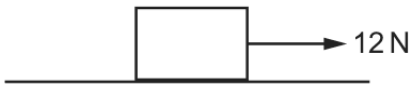
QUESTION · 5054_s22_qp_11.pdf · source page 5 of 16

10 A load is pulled by a rope attached to a motor. The resultant force exerted by the rope on the load is shown in the diagrams.

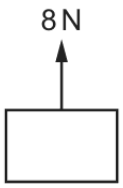
In each diagram, the load moves in the direction of the force shown and takes 10 s to travel 1.0 m.

In which diagram does the motor work with the greatest power?

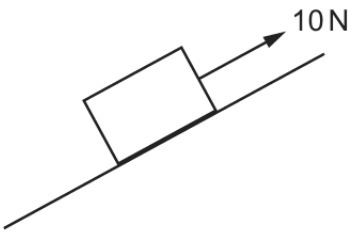
A



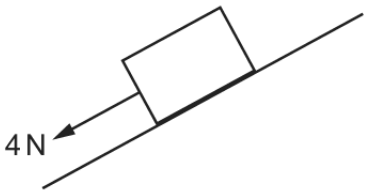
B



C



D



MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 10

10	A	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 6 of 16

12 A 1000 W, 240 V electrical device is switched on.

How much energy does it use in 5.0 minutes?

A 5000 J **B** 12 000 J **C** 72 000 J **D** 300 000 J

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 12

12	D	**1**

2 Fig. 2.1 shows a wind turbine.

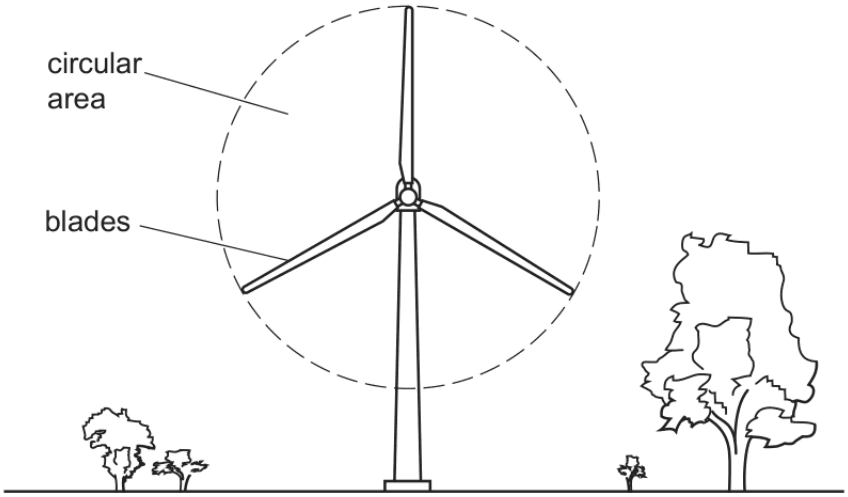


Fig. 2.1

(a) The wind blows directly towards the turbine with a speed of 12 m / s.

In one second, 60 000 kg of air passes through the circular area swept out by the blades.

Calculate the kinetic energy of this mass of air.

kinetic energy = ... [3]

(b) A wind turbine releases no carbon dioxide into the atmosphere while generating electricity. Compared with a coal-fired power station, this is an advantage of using a wind turbine.

(i) State **one** reason why it is important to reduce the amount of carbon dioxide produced.

.....

..... [1]

(ii) Suggest **one** advantage of using a coal-fired power station compared with a wind turbine.

.....

..... [1]

(c) A coal-fired power station releases 0.96 kg of carbon dioxide when it generates 1.0 kW h of electrical energy.

(i) Define the kilowatt-hour (kW h).

.....

..... [1]

(ii) Calculate the mass of carbon dioxide saved when the wind turbine has a power output of 2000 kW and operates for 12 hours.

mass = ... [1]

(d) Wind energy is a form of renewable energy.

State the name of **one** other form of renewable energy.

..... [1]

[Total: 8]

MARK SCHEME · 5054_s22_ms_21.pdf · source page 5 of 11 · Question 2

Question	Answer	Marks
2(a)	(K.E. =) ½mv^2 numerical or algebraic	**C1**
	$0.5 \times 60\,000 \times 12^2$	**C1**
	4 300 000 J	**A1**
2(b)(i)	reduces global warming	**B1**
2(b)(ii)	wind not always blowing **or** wind not reliable / not available in some places **or** smaller land area (for same power) **or** less harm to birds **or** less visual effect **or** no interference with tv / radar signals	**B1**

MARK SCHEME · 5054_s22_ms_21.pdf · source page 6 of 11 · Question 2

Question	Answer	Marks
2(c)(i)	(amount of energy of a device of power) 1 kW used for 1 hour	**B1**
2(c)(ii)	23 000 kg	**B1**
2(d)	solar, tidal, geothermal, hydro(electric), biomass, wave	**B1**

QUESTION · 5054_s22_qp_12.pdf · source page 8 of 20

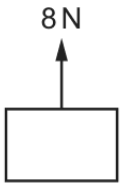
14 A load is pulled by a rope attached to a motor. The resultant force exerted by the rope on the load is shown in the diagrams.

In each diagram, the load moves in the direction of the force shown and takes 10 s to travel 1.0 m.

In which diagram does the motor work with the greatest power?

A

B



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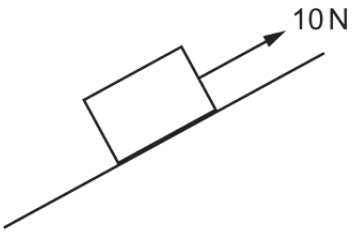

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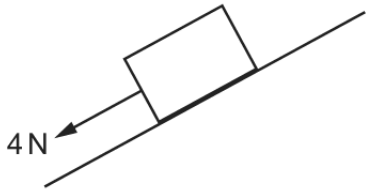

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 14

14	A	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 7 of 16

19 What is a unit of power?

A Js **B** J/s **C** Ns **D** N/s

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 19

19	B	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 12 of 16

31 The power consumption of a microwave oven is 1500 W.

The cost of 1.0 kW h of electrical energy is 24 cents.

What is the cost of using the microwave oven for 10 minutes?

A 6 cents

B 36 cents

C 60 cents

D 360 cents

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 3 of 3 · row for Question 31

31	A	**1**

2 Fig. 2.2 shows a rider on an electric scooter.

Fig. 2.2

The scooter contains a battery and a motor to drive the back wheel.

(a) (i) State the name of the energy store in the battery.

..... [1]

(ii) Describe, in terms of work done, the stages of energy transfer from the energy store in the battery to the kinetic energy of the scooter.

.....

.....

.....

..... [2]

(b) The total mass of the scooter and the rider is 70 kg.

Calculate the total kinetic energy of the rider and scooter when the scooter has a speed of 4.0 m/s.

kinetic energy = ... J [2]

(c) The battery is marked 'energy capacity 0.35 kilowatt-hour (kW h)'.

(i) Define what is meant by a kilowatt-hour.

.....

..... [1]

(ii) The scooter stops working because the battery is totally discharged (flat). This means that there is no more energy stored in the battery.

The battery is then recharged using a 70 W power supply.

Calculate the time taken to fully recharge the battery.

time = ... hours [2]

[Total: 8]

MARK SCHEME · 5054_s23_ms_22.pdf · source page 6 of 12 · Question 2

Question	Answer	Marks
2(a)(i)	chemical (potential energy)	**B1**
2(a)(ii)	electrical work / (electrical) current **or** electrons (made to) move	**B1**
	mechanical) work is done by the motor / by or on the (back) wheel	**B1**
2(b)	(KE =) ½ mv^2 in any form	**C1**
	560 (J)	**A1**
2(c)(i)	(energy provided if power is) 1 kW for 1 hour / 1000 W for 1 hour	**B1**

MARK SCHEME · 5054_s23_ms_22.pdf · source page 7 of 12 · Question 2

Question	Answer	Marks
2(c)(ii)	*(t =) E / P* **or** 0.35 / 0.07 **or** 350 / 70	**C1**
	5(.0 hours)	**A1**

QUESTION · 5054_w23_qp_11.pdf · source page 5 of 16

11 A small motor has an input power rating of 10 W and is switched on for 5.0 minutes.

What is the electrical energy input to the motor in this time?

A 2.0 J **B** 50 J **C** 300 J **D** 3000 J

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 11

11	D	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 6 of 20

15 A battery delivers 3000 J of energy to a small 2.0 W electric motor before the battery stops working.

For how long is the motor switched on?

A 1500 minutes

B 100 minutes

C 50 minutes

D 25 minutes

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 15

15	D	**1**

3 A pan containing ice at –15 °C is placed on a gas heater as shown in Fig. 3.1.

Fig. 3.1

(a) Thermal conduction occurs in the metal from which the pan is made.

By referring to particles, describe the process of thermal conduction in a metal.

.....

.....

.....

..... [2]

(b) Fig. 3.2 shows how the temperature of the contents of the pan varies with time.

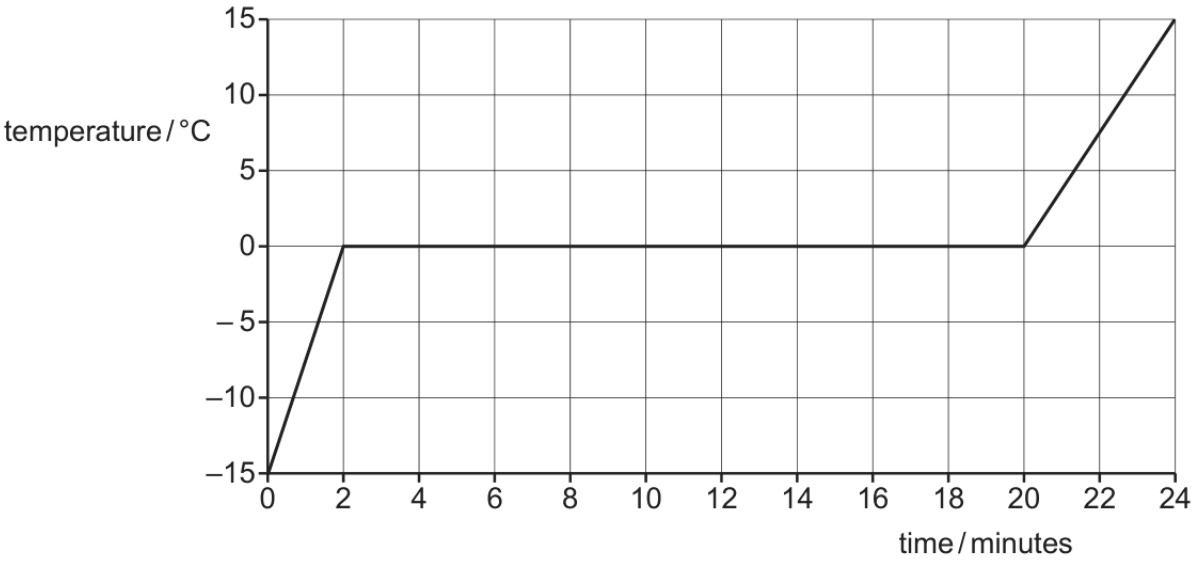

Fig. 3.2

(i) State the melting temperature of ice in kelvin.

melting temperature = ... K [1]

(ii) Explain why the temperature of the water is constant as the ice melts.

.....

.....

.....

..... [2]

(iii) The initial mass of ice in the pan is 1.5 kg and the initial temperature is –15 °C.

The specific heat capacity of ice is 2100 J / (kg °C).

Calculate the energy required to warm the ice to its melting temperature.

energy = ... J [2]

(iv) All the energy transferred to the pan comes from the heater.

Using your answer to **(iii)** and Fig. 3.2, determine the power of the heater used to warm the ice.

power = ... W [2]

(v) The graph in Fig. 3.2 has a smaller gradient when the water is liquid than when it is solid.

Suggest **one** reason why.

.....

..... [1]

[Total: 10]

MARK SCHEME · 5054_s24_ms_21.pdf · source page 6 of 10 · Question 3

Question	Answer	Marks
3(a)	free / delocalised electrons **or** electrons move	**B1**
	electrons / thermal energy move from hot area to cold area **or** electrons hit by or hit particles (of metal) **or** vibrations / collisions between particles passes on energy	**B1**
3(b)(i)	273 (K)	**B1**
3(b)(ii)	latent heat (required) **or** (particles need) more potential energy / energy supplied is being used to weaken forces between particles	**B1**
	particles further apart OR all the energy is used to weaken forces OR kinetic energy of particles stays constant	**B1**
3(b)(iii)	(Q =) mcT in any form	**C1**
	47 000 (J)	**A1**
3(b)(iv)	(P =) E / t in any form	**C1**
	390 (W)	**A1**

MARK SCHEME · 5054_s24_ms_21.pdf · source page 7 of 10 · Question 3

Question	Answer	Marks
3(b)(v)	(liquid) water has a higher specific heat capacity; water has stronger bonds	**B1**

QUESTION · 5054_s24_qp_12.pdf · source page 7 of 16

17 Which quantity is **not** measured in joules (J)?

A gravitational potential energy

B latent heat

C power

D work

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 17

17	C	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 11 of 16

31 How many kilowatt-hours of energy are used by a 1000 W heater connected to a 230 V supply for 30 minutes?

A 0.30 kW h **B** 0.50 kW h **C** 30 kW h **D** 120 kW h

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 3 of 3 · row for Question 31

31	B	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 2 of 16

1 A crane lifts a heavy load at a steady rate.

Which unit is suitable for the output power of the crane's motor?

A kN **B** kW **C** kN/s **D** kWh

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 1

1	B	**1**

1 A trolley of mass 0.20 kg is at rest on a frictionless surface.

A block of wood is attached to the trolley.

The volume of the block of wood is 0.0012 m^3, and the density of the wood is 650 kg/m^3.

(a) Calculate the combined mass of the trolley and block.

combined mass = ... kg [2]

(b) Fig. 1.1 shows that a student attaches a spring of spring constant 14 N/m to the front of the trolley.

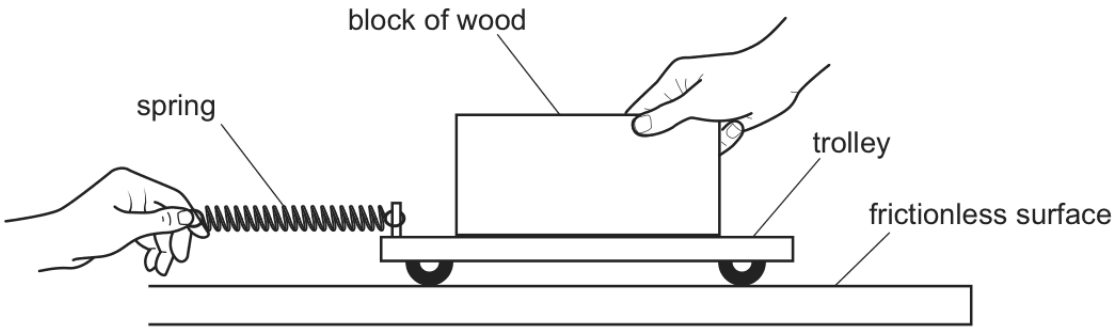

Fig. 1.1

The student holds the trolley and block and stretches the spring. He releases the trolley and block. The trolley and block accelerate from rest.

As the trolley and block accelerate, the student keeps the extension of the spring at 0.035 m.

(i) Calculate the force exerted by the spring on the trolley.

force = ... N [2]

(ii) The trolley and block are pulled a distance of 0.86 m by the spring.

Calculate the work done on the trolley.

work done = ... J [2]

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(iii) Explain why the power transferred to the trolley and block increases as the speed increases.

.....

.....

..... [2]

[Total: 8]

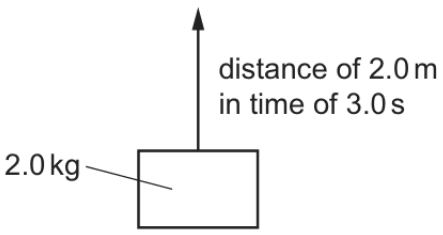
Question	Answer	Marks
1(a)	($m =$) ρV **or** 650×0.0012 **or** 0.78	**C1**
	(combined mass =) 0.98 (kg)	**A1**
1(b)(i)	($F =$) kx **or** 14×0.035	**C1**
	0.49 (N)	**A1**
1(b)(ii)	($W =$) Fx **or** 0.49×0.86	**C1**
	0.42 (J)	**A1**
1(b)(iii)	the distance moved / work done (in one second / unit time / a given time) increases (as the speed increases)	**B1**
	the work done in one second / unit time increases	**B1**

QUESTION · 5054_w24_qp_12.pdf · source page 5 of 16

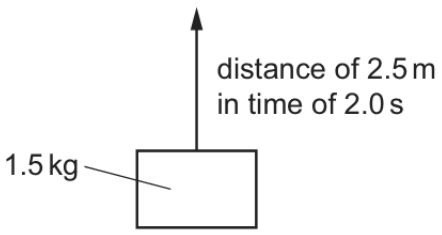
12 Four objects of different masses are lifted through different distances in different amounts of time.

In which combination of mass, distance and time is the greatest power transferred?

A



B



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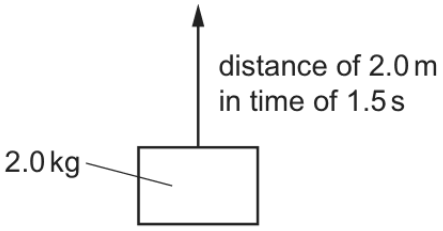

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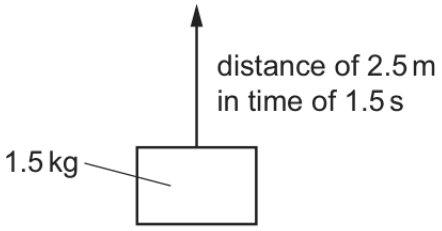

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 12

12	C	**1**

6 Fig. 6.1 shows the circuit symbol for a relay and two labelled terminals.

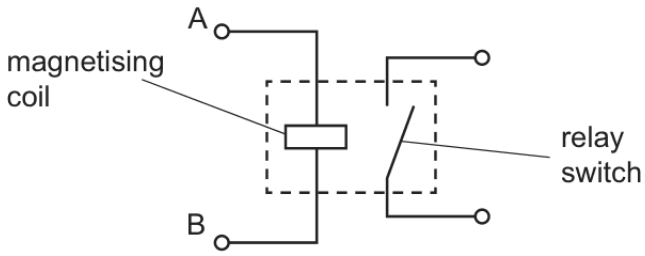

Fig. 6.1

(a) By referring to Fig. 6.1, explain the effect of a current passing from A to B.

.....

.....

.....

..... [2]

(b) Fig. 6.2 shows two circuits linked by a relay. The heater transfers energy at a rate of 40 W when it is connected to a 12 V supply.

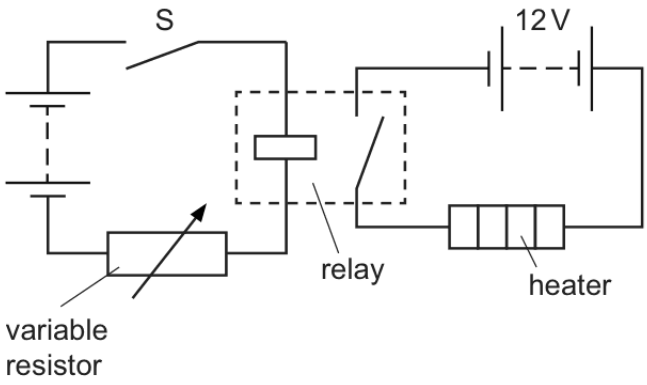

Fig. 6.2

Switch S is closed. The variable resistor is adjusted so that its resistance *R* increases from 0.

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Fig. 6.3 shows how the power transferred in the 40 W heater varies as R increases from 0. The graph shows a sharp decrease when the resistance R reaches R_H.

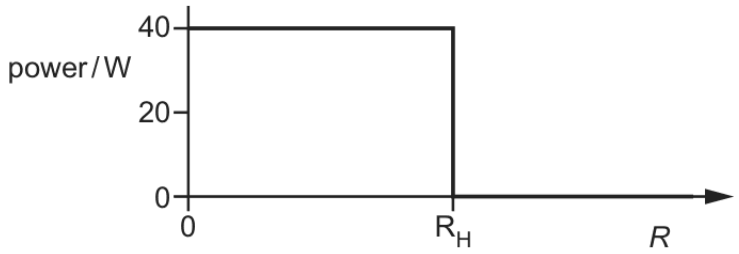

Fig. 6.3

Explain the shape of the graph in Fig. 6.3.

.....

.....

.....

.....

.....

..... [3]

(c) The 40 W heater connected to the 12 V supply is switched on.

Calculate:

(i) the energy transferred in the heater in 2.0 minutes

energy = ... J [2]

(ii) the resistance of the heater.

resistance = ... Ω [3]

[Total: 10]

MARK SCHEME · 5054_w24_ms_22.pdf · source page 7 of 10 · Question 6

Question	Answer	Marks
6(a)	current / it in AB / the (magnetising) coil produces magnetic field	**B1**
	(relay) switch (between X and Y) closes **or** switch attracted by magnetic field / coil **or** (relay) switch made of iron	**B1**

MARK SCHEME · 5054_w24_ms_22.pdf · source page 8 of 10 · Question 6

Question	Answer	Marks
6(b)	(initially) the current in the relay coil is large **or** relay switch (in relay circuit / relay) is closed **or** heater is connected to a 12 V supply **or** voltage of supply does not change **or** heater is switched on	**B1**
	(as R increases) current / magnetic field (in relay coil) decreases	**B1**
	(at $R = R_H$) relay switch (in relay coil) opens **or** magnetic field not strong enough to hold switch closed	**B1**
6(c)(i)	($E =$) Pt **or** 40×2.0 **or** $40 \times 2.0 \times 60$ **or** 80	**C1**
	4800 (J)	**A1**
6(c)(ii)	$I = P / V$ **or** $I = 40 / 12$ **or** $I = 3.3$ (A)	**C1**
	($R =$) V / I **or** $12 / 3.3$	**C1**
	3.6 (Ω)	**A1**

3 A battery, a pulley and a motor are used to lift a load as shown in Fig. 3.1.

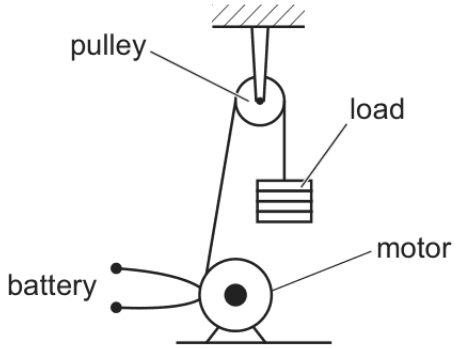

Fig. 3.1

(a) Describe the transfers between energy stores that occur as the load is lifted.

.....

.....

.....

..... [3]

(b) The efficiency of the motor, pulley and load system is less than 100%.

(i) By comparing the input energy and the useful output energy, explain why the efficiency is less than 100%.

.....

..... [1]

(ii) Explain how the principle of conservation of energy applies in lifting the load.

.....

.....

..... [2]

(c) The input power to the motor is 15 W. The motor is used for 20 s. The efficiency of the motor is 60%.

Calculate the energy supplied to the load.

energy = ... J [2]

[Total: 8]

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Question	Answer	Marks
3(a)	*Any three from*: • reduction in chemical energy store in battery • electrical work done by motor • mechanical work done on weight • increase in gravitational energy store of weight • increase in thermal energy (in motor or air) • electrical heating (by current)	**B3**
3(b)(i)	useful or output energy < input energy	**B1**
3(b)(ii)	total energy is constant	**B1**
	energy from power supply = energy gained by weight + thermal energy (to air / due to friction)	**B1**
3(c)	(E =) power × time in any form	**C1**
	180 J	**A1**

2 Fig. 2.1 shows a fork-lift truck used to lift a load.

Fig. 2.1

When lifting the load, the electric motor on the fork-lift truck has a useful output power of 600 W.

(a) Define 'power'.

.....

..... [1]

(b) The efficiency of the motor is 70%.

Calculate the input power to the motor.

input power = ...W [2]

(c) The fork-lift truck lifts a load with mass of 50 kg through a vertical distance of 2.3 m.

(i) Calculate the work done in lifting the load.

work done = ... J [2]

(ii) Calculate the time taken to lift the load.

time taken = ... s [2]

(d) Describe how energy is transferred to thermal energy as the load is lifted by the fork-lift truck.

.....

..... [1]

[Total: 8]

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Question	Answer	Marks
2(a)	work done or energy per unit time	**B1**
2(b)	(input power =) power output / efficiency in any form	**C1**
	860 (W)	**A1**
2(c)(i)	(work =) force × distance **or** *mgh*	**C1**
	1100 (J)	**A1**
2(c)(ii)	(time =) work ÷ (output) power	**C1**
	1.9 or 1.8 s	**A1**
2(d)	*Any one of*: • friction / air resistance • electrical heating	**B1**

QUESTION · 5054_w25_qp_12.pdf · source page 7 of 16

18 An object of mass m is lifted through a vertical distance d in time t.

Which equation is used to calculate the power P required to lift the object?

A $P = \frac{md}{gt}$ **B** $P = \frac{mgd}{t}$ **C** $P = \frac{dt}{mg}$ **D** $P = \frac{m}{gdt}$

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 18

18	B	**1**

3 Figure 3.1 shows a dam used as part of a hydroelectric power station.

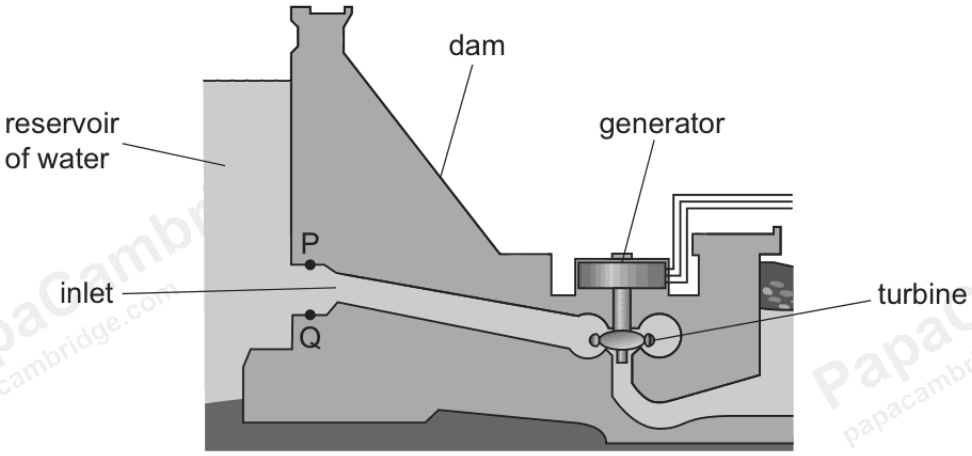

Figure 3. 1

Water flows to the turbine through an inlet in the dam. The inlet is shown in Figure 3.1.

Points P and Q are on the horizontal surfaces next to the inlet.

(a) (i) State **two** factors that determine the size of the pressure at P.

.....

..... [1]

(ii) The pressure in the water causes a force on the horizontal surfaces at P and Q.

State the directions of the forces due to the water on the surfaces at P and Q.

direction at P .....

direction at Q .....

[1]

(b) This hydroelectric energy resource is a renewable source of energy.

State the name of **one** other renewable energy source.

..... [1]

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(c) The power output from the generator is 2.5 MW.

(i) Calculate the energy **output** from the generator in 30 minutes.

energy output = ... J [2]

(ii) The efficiency of the energy transfer in the generator is 95%.

Calculate the power **input** to the generator.

PapaCambridge papacambridge.com PapaCambridge papacambridge.com

power input = ... MW [2]

[Total: 7]

MARK SCHEME · 5054_s26_ms_22.pdf · source page 9 of 18 · Question 3

Question	Answer	Marks	Guidance
3(a)(i)	At least 2 from • height / depth (of water) / quantity or amount of water in dam • density (of water / liquid) • acceleration of gravity / gravitational field strength • atmospheric pressure • force (on surface / inlet) • area (of surface / inlet)	**B1**	**I** thickness; area of dam
3(a)(ii)	at P – upwards / towards P at Q – downwards / towards Q	**B1**	**I** top, bottom; north, south both needed
3(b)	ANY one from geothermal, tidal, solar, wind, biomass, wave (energy)	**B1**	**A** sunlight; solar panels; solar cell; wind turbines; biogas ; solar power; wood from a managed source **I** just 'wood' **not** nuclear

MARK SCHEME · 5054_s26_ms_22.pdf · source page 10 of 18 · Question 3

Question	Answer	Marks	Guidance
3(c)(i)	(E =) $P \times t$ in any form algebraic or numerical	**C1**	**A** 2.5 ($\times 10^6$)$\times$ 30; 2.5$\times$30$\times$60; 4.5 $\times$ 10^n; P=W / t
	4.5×10^9 (J)	**A1**	
3(c)(ii)	efficiency = (power) output / (power) input in any form	**C1**	**A** 0.95 = 2.5 / P_{IN}; (useful) energy output / energy input e.g. 0.95 =**3(c)(i)** / input in any form
	2.6(3 MW)	**A1**	**A** 2.6×10^6 with unit MW only 1 mark

Chapter 3 Waves

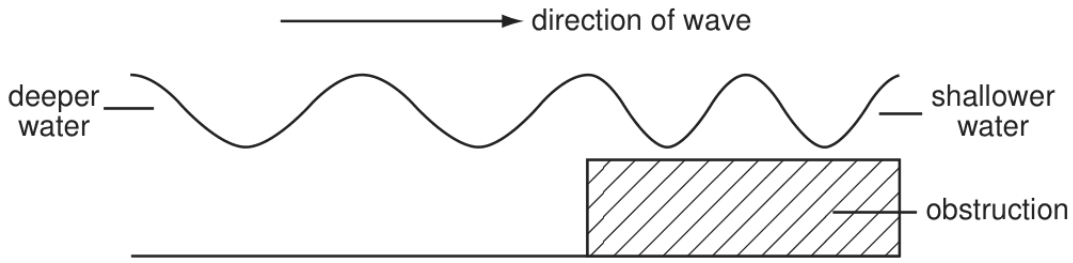
3.1 General properties of waves

142 entries: 89 Paper 1 questions, 53 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 8 of 20

19 The diagram shows a wave moving into shallower water.



The wavelength of the waves is reduced because

A both the frequency and the speed decrease.

B both the frequency and the speed increase.

C only the frequency increases.

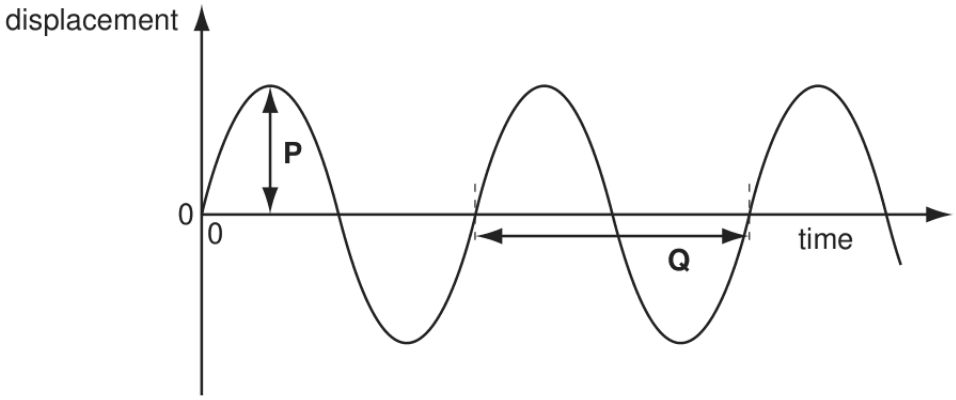
D only the speed decreases.

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 19

19 **D**

QUESTION · 5054_w04_qp_1.pdf · source page 10 of 20

22 The diagram shows a graph of wave motion.



Which quantities are shown by distances **P** and **Q**?

	P	Q
A	amplitude	period
B	amplitude	wavelength
C	half the amplitude	period
D	half the amplitude	wavelength

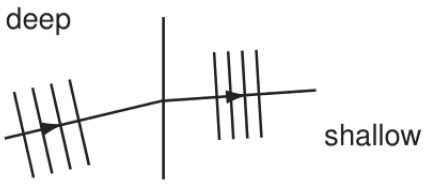
MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 22

22 **A**

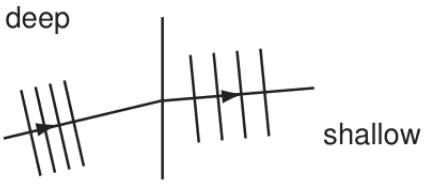
QUESTION · 5054_w04_qp_1.pdf · source page 11 of 20

23 Which diagram correctly shows water waves travelling through deep water to shallower water?

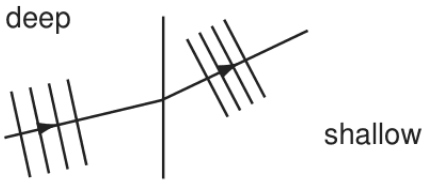
A



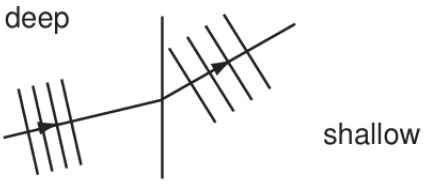
B



C



D



MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 23

23	**A**

5 Fig. 5.1 shows part of a long, thin spring used to demonstrate a transverse wave.

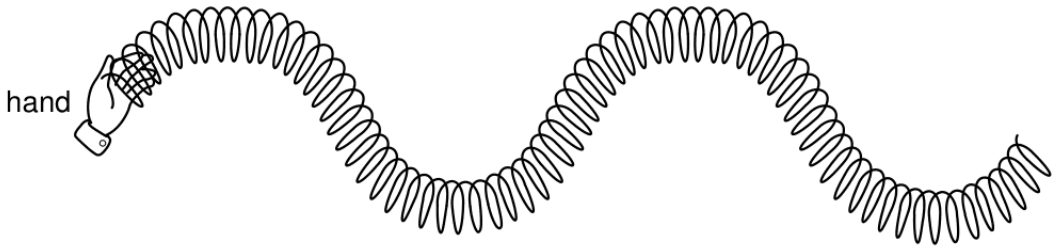


Fig. 5.1

The wave shown in Fig. 5.1 has a frequency of 4.0 Hz.

(a) (i) On Fig. 5.1, mark the direction the hand must move to make a transverse wave.

(ii) Describe how the hand must move to make a transverse wave of frequency 4.0 Hz.

.....

.....

.....
[2]

(b) The speed of the wave is 0.80 m/s. Calculate its wavelength.
State clearly the formula that you use.

wavelength = …….. [3]

(c) State what must be done to double the wavelength of the wave on the spring.

.....

.....

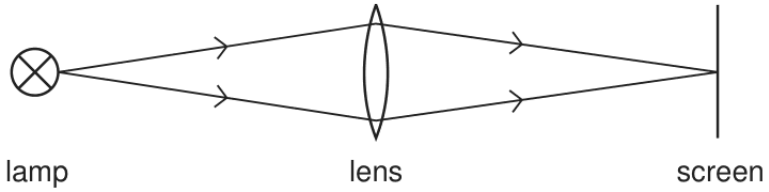
.....[1]

5	**(a)** **(i)** up and down arrow shown			
	(allow if one arrow and up/down stated in **(ii)**)		**B1**	
	(ii) 4 times in one second **or** once in 0.25 sec		**B1**	
	(b) $v = f\lambda$ in any algebraic form		**B1**	
	0.8/4		**C1**	
	0.2 m		**A1**	
	(c) halve the frequency **or** move hand once every 0.5 springs		**B1**	**6**
	(**ignore** move hand slower or at half speed or speed of wave double unless spring stretched more)			

QUESTION · 5054_s05_qp_1.pdf · source page 8 of 20

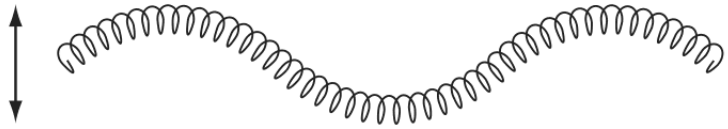
19 Which diagram shows an example of a longitudinal wave?

A light travelling from a lamp to a screen

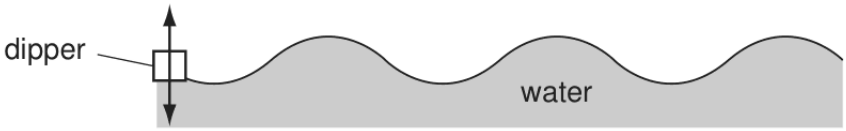


B a spring pushed backwards and forwards

C a spring pushed up and down



D a water ripple caused by a dipper moving up and down

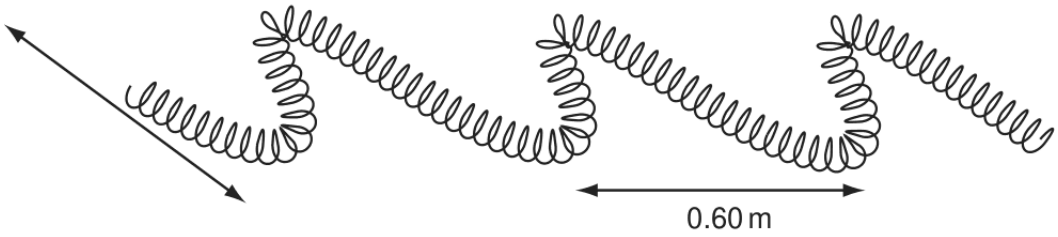


MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 19

19	**B**

QUESTION · 5054_s05_qp_1.pdf · source page 8 of 20

20 The diagram shows part of a spring that is shaken from side to side to produce a wave.



The distance between successive peaks is 0.60 m and the frequency is 2.5 Hz.

How long does it take for a wave to travel 3.0 m along the spring?

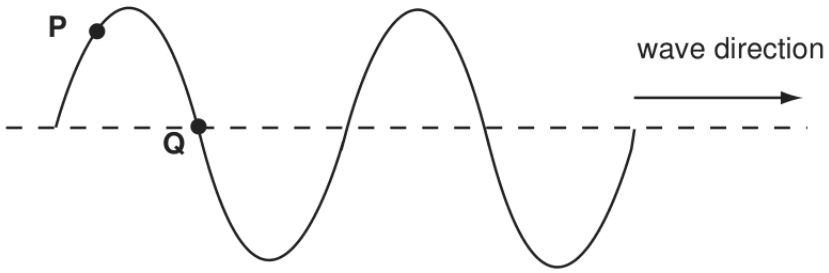
A 0.20 s **B** 0.50 s **C** 2.0 s **D** 5.0 s

MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 20

20	**C**

QUESTION · 5054_w05_qp_1.pdf · source page 11 of 20

20 The diagram shows a wave on a string with two points **P** and **Q** marked. The wave is moving in the direction shown.



What will happen next?

A **P** will move to the right.

B **P** will move up.

C **Q** will not move.

D **Q** will move up.

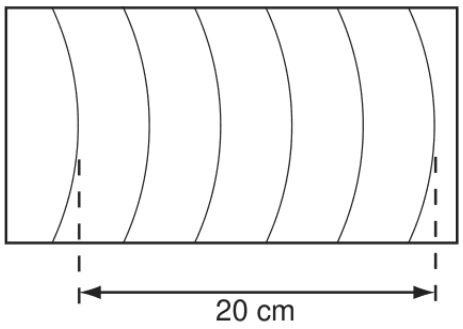
MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 20

20	**D**

QUESTION · 5054_w05_qp_1.pdf · source page 12 of 20

21 The dipper in a ripple tank vibrates at a frequency of 4.0 Hz and the resulting wave pattern is photographed.

The distance between the two crests shown is 20 cm.



What is the speed of the wave?

A 4 cm / s **B** 5 cm / s **C** 16 cm / s **D** 20 cm / s

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 21

21	**C**

3 A student produces wavefronts in a ripple tank to demonstrate refraction, as shown in Fig. 3.1. He places a sheet of glass under the water on the right-hand side of the tank. The arrows show the directions of movement of the wavefronts.

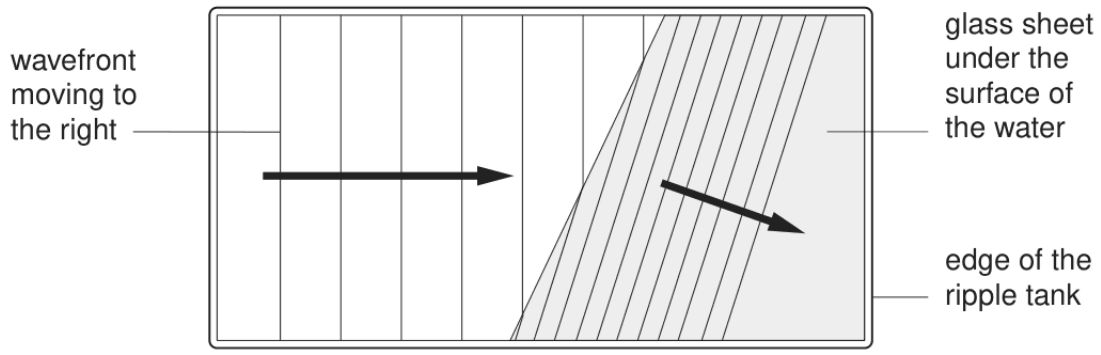


Fig. 3.1

(a) State what is meant by a *wavefront*.

.....

.....

..... [1]

(b) State what happens to each of the following quantities as the wavefronts change direction.

(i) wavelength .....

..... [1]

(ii) speed .....

..... [1]

(iii) frequency .....

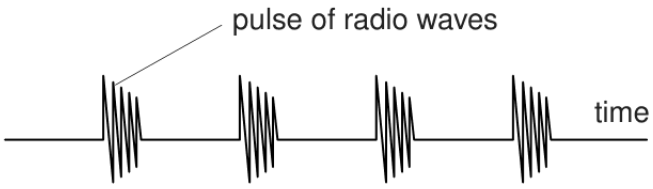
..... [1]

3	**(a)**		line joining points of same phase, e.g. line joining crests	**B1**	
	(b)	**(i)**	decreases	**B1**	
		(ii)	decreases	**B1**	
		(iii)	constant	**B1**	**4**

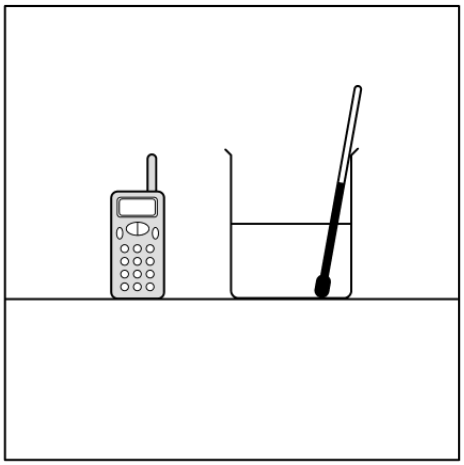
10 A student uses a mobile phone. He is concerned that the energy of the radio waves may cause a temperature rise in his brain. To investigate this effect, he calculates and tries to measure the heating effect of the phone on a nearby glass beaker of water.

Information from manufacturer

constant input power 0.20 W



The mobile phone emits pulses of radio waves.
Each pulse of waves has energy 0.00012 J.
There are 216 pulses of waves in 1.0 s



The phone is used for 360 s when next to a glass beaker containing 50 g of water.

(a) Calculate

(i) the number of pulses of radio waves produced during the phone call, [2]

(ii) the total energy of the radio waves emitted during the phone call, [2]

(iii) the maximum temperature rise produced in 50 g of water if all of the energy calculated in **(ii)** is absorbed by the water. The specific heat capacity of water is 4.2 J/(g °C). [3]

(b) The manufacturer gives the input power of the phone as 0.20 W.

(i) Calculate the energy supplied by the battery when the phone is used for 360 s. [2]

(ii) Calculate the fraction of the energy supplied by the battery that is converted into radio wave energy during the phone call. [1]

(c) The student uses the mercury-in-glass thermometer shown in Fig. 10.1.

He does not detect any temperature rise in the water in the beaker when the phone is used.

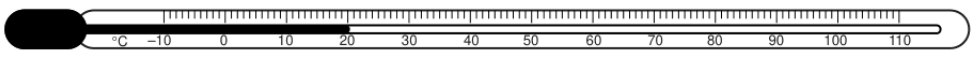


Fig. 10.1

(i) Describe how you would check the 0 °C and 100 °C points on the thermometer. [2]

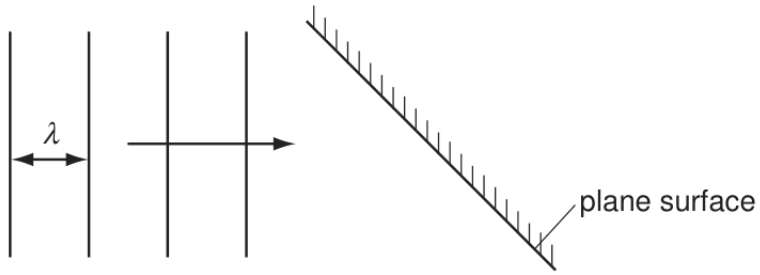
(ii) Explain why the thermometer is not sensitive enough to detect the temperature rise. [1]

(iii) State and explain one change that will make a mercury-in-glass thermometer more sensitive. [2]

10	**(a)**	**(i)**	360 x 216					**C1**
			77 800 (no sig fig penalty)					**A1**
		(ii)	77 800 x 0.00012					**C1**
			9.33 J ecf **(i)**					**A1**
		(iii)	E = mcΔT in any form, algebraic or numerical					**B1**
			9.33/(50 x 4.2)					**C1**
			0.044 ºC ecf **(ii)**					**A1**
	(b)	**(i)**	E=Pt in any form, algebraic or numerical					**C1**
			72 J					**A1**
		(ii)	0.13 (accept 13%) no s.f. penalty					**A1**
	(c)	**(i)**	(pure) melting ice for 0° C					**B1**
			(pure) boiling water/steam above boiling water (at 1 atmosphere) for 100° C					**B1**
		(ii)	each division on thermometer is too small described in some way e.g. does not expand far up tube (**not** bore too thin, not enough mercury)					**B1**
		(iii)	**change**	use more mercury	**or**	use smaller bore		**M1**
			reason	more expansion	**or**	further distance up tube (for same expansion)		**A1**

QUESTION · 5054_s06_qp_1.pdf · source page 7 of 20

21 In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



Which of the following correctly describes the waves after they are reflected from the surface?

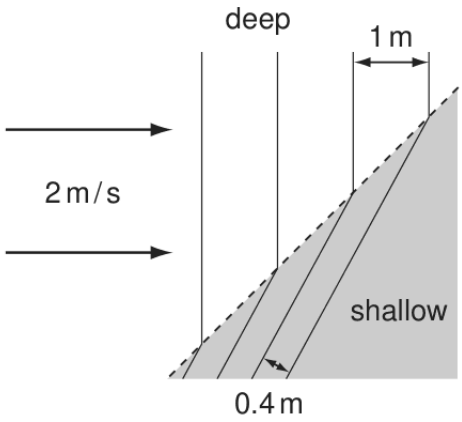
	speed of waves	wavelength λ
A	faster	shorter
B	slower	longer
C	slower	shorter
D	the same	the same

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 21

21	**D**

QUESTION · 5054_s06_qp_1.pdf · source page 8 of 20

22 Waves pass from deep water to shallow water and refraction occurs.



What is the speed of the waves in the shallow water?

A 0.2 m/s **B** 0.8 m/s **C** 2.0 m/s **D** 5.0 m/s

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 22

22	**B**

QUESTION · 5054_w06_qp_1.pdf · source page 8 of 16

19 What is meant by the term *wavefront*?

A the distance between successive peaks of a wave

B the distance between the trough and the peak of a wave

C a line joining points along the peak of a wave

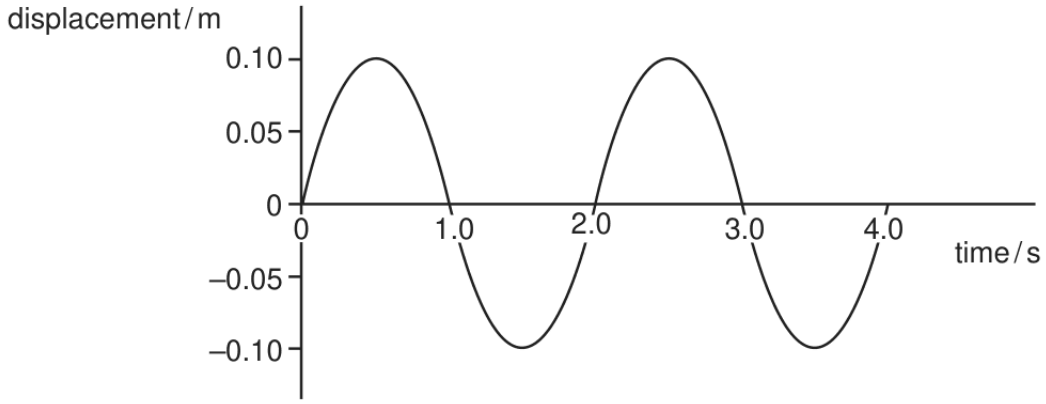
D a line joining the trough and the peak of a wave

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 19

19	**C**

QUESTION · 5054_w06_qp_1.pdf · source page 8 of 16

20 The diagram shows how displacement varies with time as a wave passes a fixed point.



What is the frequency of this wave?

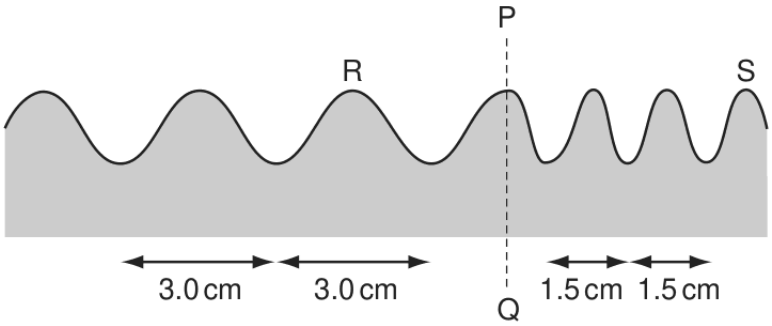
A 0.25 Hz **B** 0.50 Hz **C** 1.0 Hz **D** 2.0 Hz

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 20

20	**B**

QUESTION · 5054_s07_qp_1.pdf · source page 9 of 20

19 The diagram shows a water wave in a ripple tank.



The wave has a speed of 12 cm / s at R.

The wave crosses a boundary PQ where the distance between crests changes from 3.0 cm to 1.5 cm.

What is the speed of the wave at S?

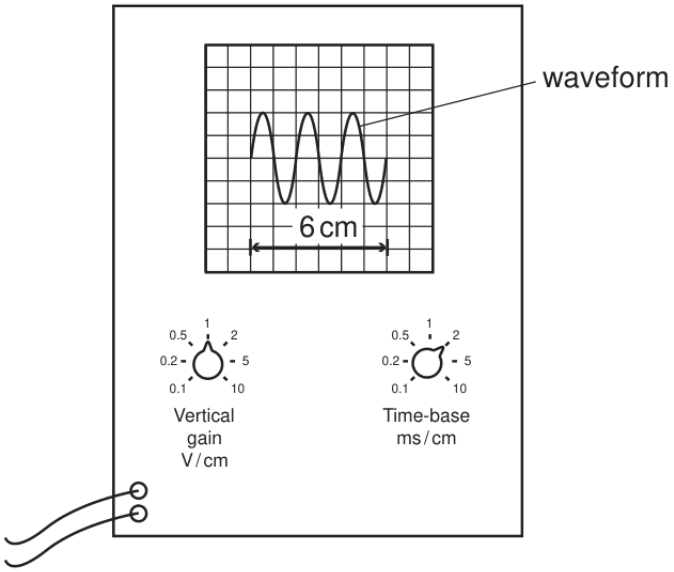
A 3.0 cm / s **B** 6.0 cm / s **C** 12 cm / s **D** 24 cm / s

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 19

19	**B**

QUESTION · 5054_s07_qp_1.pdf · source page 18 of 20

37 A waveform is displayed on a cathode-ray oscilloscope. The length of three cycles of the waveform is 6 cm. The vertical gain and the time-base settings are shown on the diagram.



What is the time taken for one cycle of the waveform?

A 1 ms **B** 2 ms **C** 4 ms **D** 6 ms

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 37

37	**C**

For Examiner's Use

4 Fig. 4.1 shows part of an optical fibre.

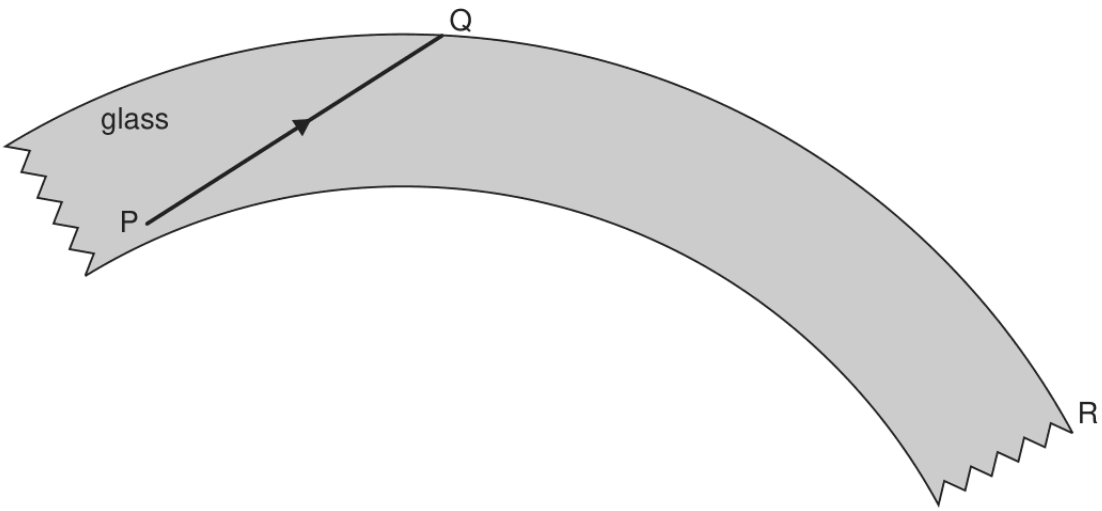


Fig. 4.1

The ray PQ undergoes total internal reflection in the optical fibre.

(a) On Fig. 4.1, continue the path of ray PQ until it reaches end R. [1]

(b) Explain what is meant by *total internal reflection*.

.....

.....

..... [1]

(c) Optical fibres are cheaper and lighter than copper wires. State one other advantage of using optical fibres rather than copper wires for telephone communications.

.....

.....

..... [1]

(d) The light in the optical fibre is travelling at a speed of 2.1×10^8 m/s and has a wavelength of 6.4×10^{-7} m.

Calculate the frequency of the light.

frequency = ... [2]

4 **(a)** reflections correct by eye B1

(b) **all** the ray **reflects** back (into the denser medium/glass)
or reflection **and** no refraction/escape into air B1

(c) more calls **or** greater bandwidth **or** more/faster data(/sec)/information **or** better quality **or** less power loss/energy loss/attenuation **or** greater distance (between repeaters) **or** harder to tap **or** less noise/interference B1

(d) $f = v/\lambda$ in any form numerical or algebraic C1
3.3×10^{14} Hz A1 [5]

For Examiner's Use

6 Fig. 6.1 shows a coil of wire wound on a cardboard tube.

There is a d.c. current in the coil. The direction of the current is shown in the key.

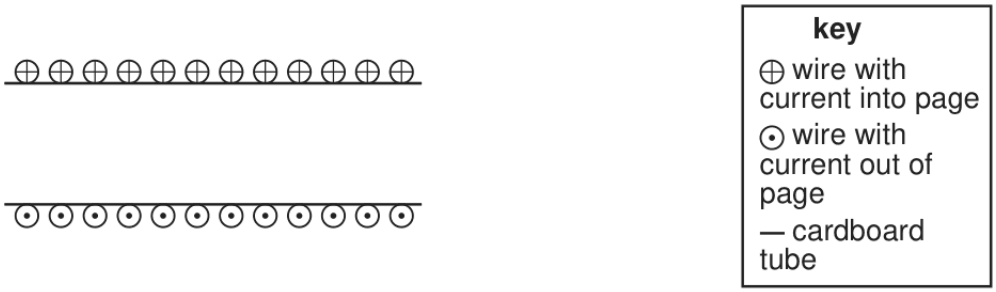


Fig. 6.1

(a) On Fig. 6.1, draw the magnetic field produced by the coil. [3]

(b) Fig. 6.2 shows a simple loudspeaker that uses the coil shown in Fig. 6.1 attached to a paper cone.

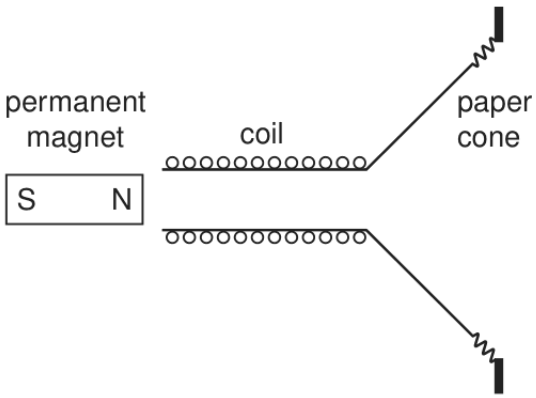


Fig. 6.2

The coil is connected to a signal generator.

There is an alternating current of frequency 100 Hz in the coil.

(i) State what is meant by a *frequency of 100 Hz*.

.....

..... [1]

For Examiner's Use

(ii) Describe and explain the movement of the coil.

.....

.....

.....

.....

.....

..... [3]

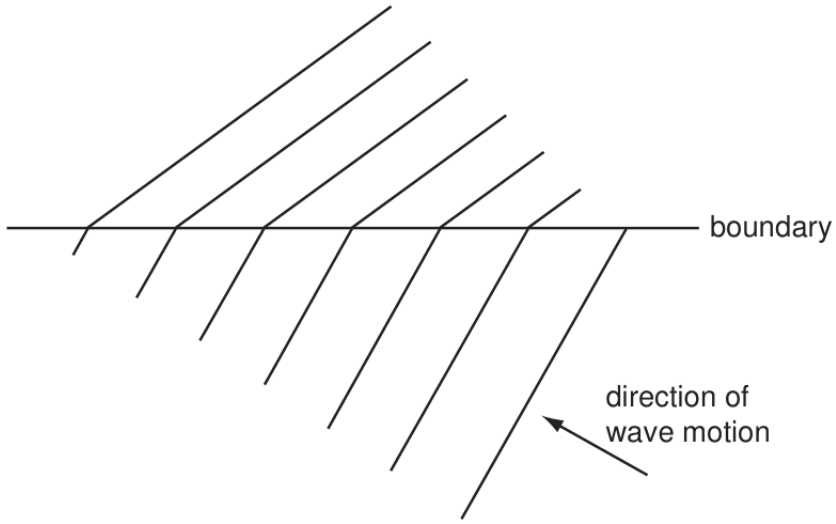
6 **(a)** at least 3 approx straight lines in middle of coil B1
one loop around top or bottom of coil B1
correct direction shown on at least one line or loop B1

(b) **(i)** 100 oscillations/cycles/waves/changes in direction (and back again) in 1 sec
or time for one oscillation etc. 0.01s B1

(ii) coil/cone moves left to right **or** back and forth **or** oscillates/vibrates **or** in and out B1
(current in) coil produces magnetic field/pole(s)
or Left Hand Rule/current in magnetic field (gives force) B1
fields/poles (of coil) oscillate/reverse B1
force (exerted by magnet) oscillates/alternates (**accept** attracts/repels) B1 ANY 2 B2 [7]

QUESTION · 5054_w07_qp_1.pdf · source page 9 of 20

19 The diagram shows the refraction of water waves as they cross a boundary in a ripple tank.



What causes this refraction?

A a change in frequency due to a change in depth

B a change in frequency due to a change in wavelength

C a change in speed due to a change in depth

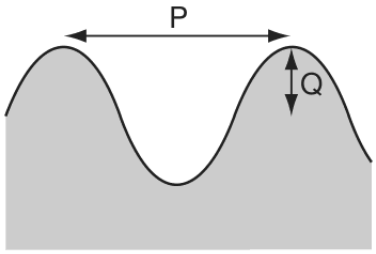
D a change in speed due to a change in frequency

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 19

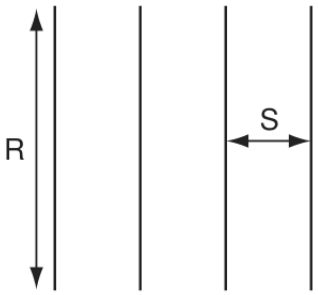
19	**C**

QUESTION · 5054_s08_qp_1.pdf · source page 9 of 20

20 The diagrams show different views of a water wave in a ripple tank.



cross-section of wave



crests seen from above

Which letters represent a wavelength and a wavefront?

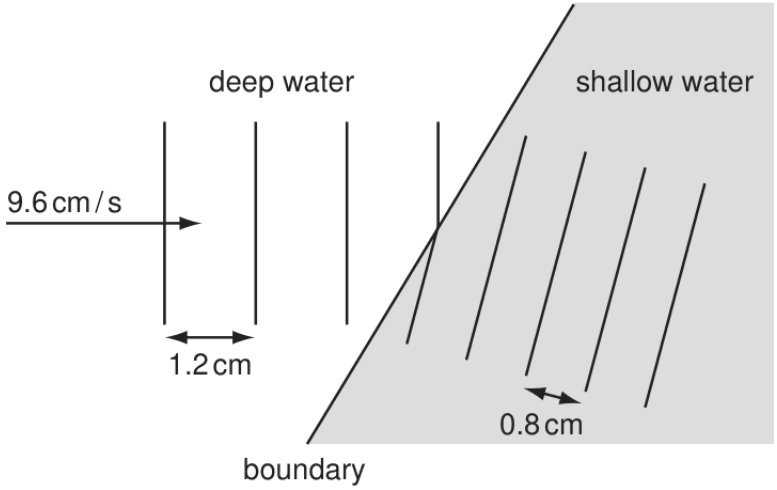
	wavelength	wavefront
A	P	R
B	P	S
C	Q	R
D	Q	S

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 20

20 **A**

QUESTION · 5054_s08_qp_1.pdf · source page 9 of 20

21 A ripple tank is used to demonstrate refraction of plane water waves.



Waves in deep water have a wavelength of 1.2 cm and a speed of 9.6 cm / s. The wavelength of the waves in shallow water is 0.8 cm.

What is the speed of the waves in the shallow water?

A 6.4 cm / s **B** 8.0 cm / s **C** 9.6 cm / s **D** 14.4 cm / s

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 21

21	**A**

10 **(a)** Water waves are transverse waves. Sound is a longitudinal wave.

(i) Describe the difference between transverse waves and longitudinal waves. In your account, draw a diagram of each type of wave. [4]

(ii) Sound contains regions of compression and regions of rarefaction. Describe what is meant by a compression and by a rarefaction. [2]

(b) A ripple tank is used to demonstrate the reflection of water waves.

(i) Draw a labelled diagram of a ripple tank. [2]

(ii) Draw a diagram showing the reflection of waves from a plane barrier in a ripple tank. [2]

(c) Fig. 10.1 shows a water wave passing a floating log. The log is stationary.

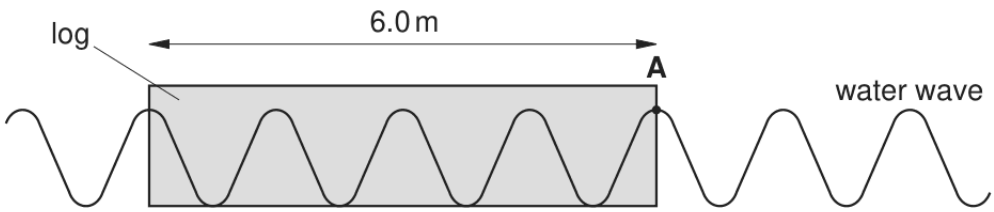


Fig. 10.1

The log is 6.0 m long and 5 complete waves take 10 seconds to pass point **A**.

Determine

(i) the wavelength of the water waves, [1]

(ii) the frequency of the water waves, [2]

(iii) the speed of the water waves. [2]

10 (a) (i) transverse-crest and troughs **and** longitudinal-compressions and rarefactions C1
transverse **vibration** at right angles **and** longitudinal along wave A1
diagram showing transverse wave at least one wavelength B1
diagram showing longitudinal wave (slinky/layers, etc.) at least one wavelength B1

(ii) high(er) pressure **or** denser **or** molecules/atoms/layers closer together B1
low(er) pressure **or** molecules, etc. further apart B1

[Total: 6]

(b) (i) tank containing water/waves and **labelled** dipper/vibrator B1
source of light (labelled or clear) and screen/paper/projected image
or stroboscope to view or illuminate B1

(ii) plane barrier (labelled or clear) + incident waves B1
reflected waves correct B1
(**accept** circular waves with correct centres 0/2 if waves go through barrier)

[Total: 4]

(c) (i) 1.5 m B1

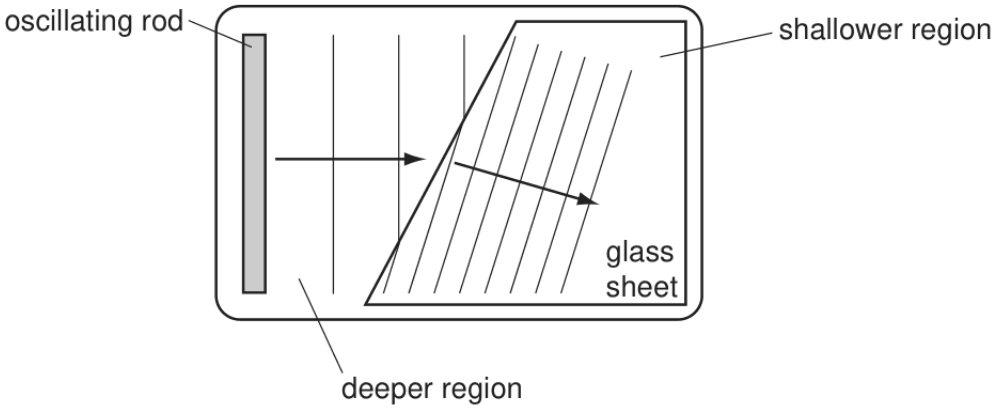
(ii) 5/10 **or** no of waves per second **or** f = 1/T C1
0.5 Hz A1

(iii) $v = f\lambda$ or **(i)** × **(ii)** allow $v = f\lambda$ anywhere in **(c)** C1
0.75 m/s ecf **(i)** and **(ii)** A1

[Total: 5]

QUESTION · 5054_w08_qp_1.pdf · source page 11 of 20

22 The diagram shows the refraction of water waves in a ripple tank. The water is shallower above the glass sheet.



When crossing into the shallower region, what is the effect on the frequency and on the speed of the waves?

	wave frequency	wave speed
A	changes	changes
B	changes	unchanged
C	unchanged	changes
D	unchanged	unchanged

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 22

22 **C**

Section B

Answer **two** questions from this section.

Use the lined pages provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 Fig. 9.1 shows one swimmer in a race starting before the signal.

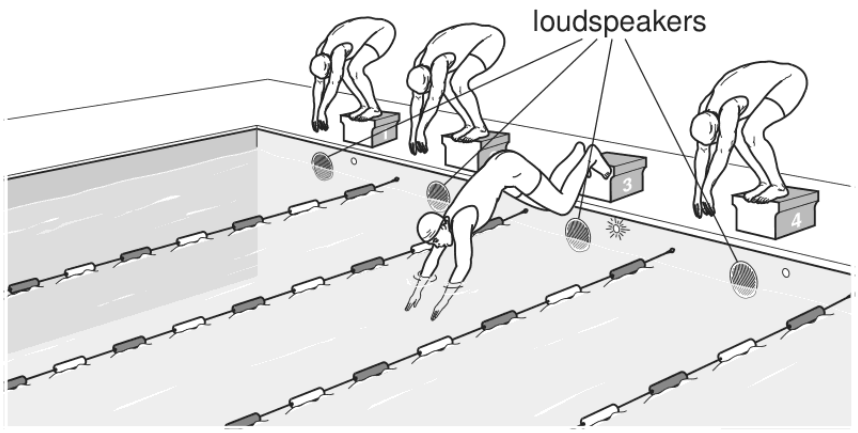


Fig. 9.1

The swimmer is called back by a loud, low-pitched sound from a loudspeaker positioned just at water level. The speed of sound in air is 330 m/s.

(a) (i) Describe how the loudspeaker causes sound to travel through the air. [3]

(ii) Explain, in terms of wave properties, what is meant by *loud* and *low-pitched*. [3]

(iii) The swimmer is 0.57 m from the loudspeaker when he hears the sound. Calculate the time taken for the sound to reach him through the air. [2]

(iv) Explain how the time taken differs when sound travels the same distance through air and through water. [2]

(b) The loudspeaker produces sound of frequency 0.20 kHz.

(i) Calculate the wavelength of this sound. [3]

(ii) Draw a diagram to show what is meant by the term *wavelength* when applied to a **longitudinal** wave such as sound. [2]

Section B

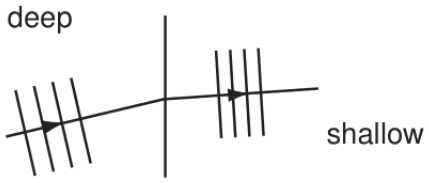
9	**(a)**	**(i)**	Any **three** lines: vibration of **cone/loudspeaker** vibration of **air/particles** (molecules) **particles/molecules** pass on vibrations/energy (to neighbours) compressions **and** rarefactions **or** longitudinal wave/movement of particle (max 3)	 B1 B1 B1 B1	
		(ii)	loud – large amplitude/max displacement low-pitched – frequency/no. of waves per sec low frequency, small frequency, etc. (long wavelength 1/2)	B1 M1 A1	
		(iii)	(t =) d/s **or** 0.57/330 0.0017 s	C1 A1	
		(iv)	speed of sound greater in water/liquid **or** v.v. less time taken in water/liquid **or** heard sooner/faster	B1 B1	[10]
	(b)	**(i)**	$v = f\lambda$ **or** 200 seen (λ =) v/f **or** 330/200 **or** 330/0.2 **or** 1650 (m) 1.6/1.65/1.7 m	C1 C1 A1	
		(ii)	attempt at compressions and rarefactions/longitudinal wave correct wavelength marked	M1 A1	[5]

[Total: 15]

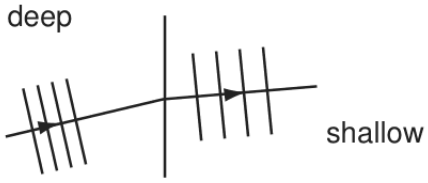
QUESTION · 5054_s09_qp_1.pdf · source page 9 of 20

19 Which diagram correctly represents water waves travelling from deep water to shallow water?

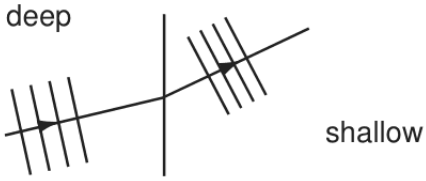
A



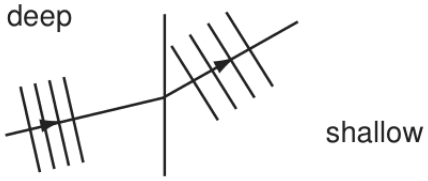
B



C



D

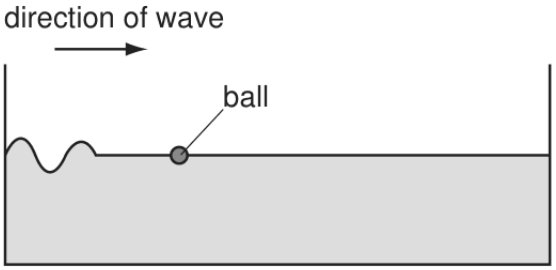


MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 19

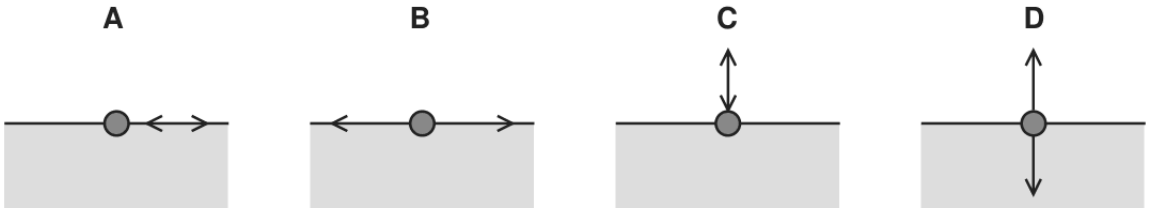
19 **A**

QUESTION · 5054_w09_qp_1.pdf · source page 10 of 20

19 The diagram shows a ball floating in a tank of water.



Which diagram shows the movement of the ball as the wave passes?



MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 19

19 **D**

5 Fig. 5.1 is a full-scale diagram that represents a sound wave of frequency 5.1 kHz in air.

For Examiner's Use

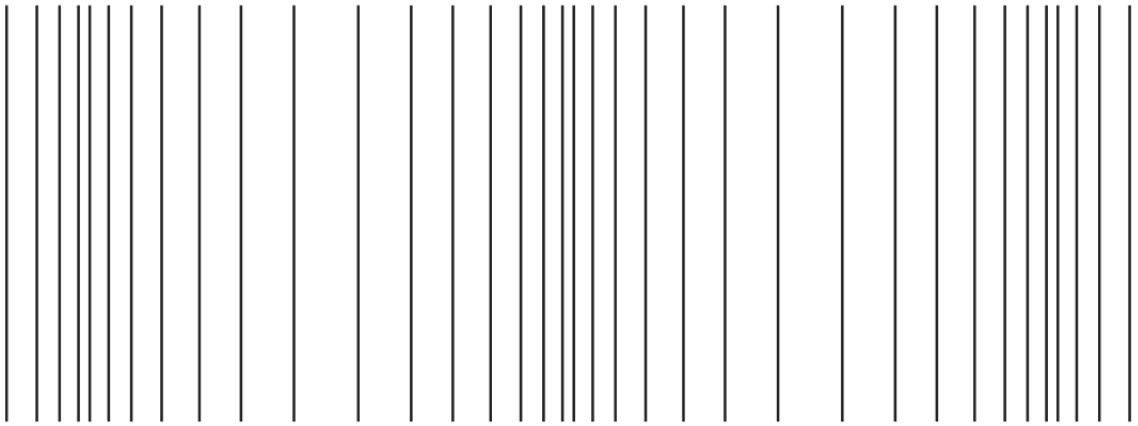


Fig. 5.1

(a) (i) On Fig. 5.1, mark the position of one compression with the letter C and of one rarefaction with the letter R. [2]

(ii) Using the full-scale diagram, measure the wavelength of this sound wave.

wavelength = ...[1]

(iii) Calculate the speed of sound in air.

speed = ...[3]

(b) A sound wave is longitudinal. Describe how a longitudinal wave differs from a transverse wave.

.....

.....

.....[2]

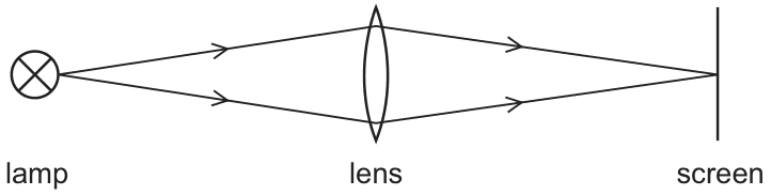
5	**(a)**	**(i)**	C in correct position i.e. gap 4, 18 **or** 32 { **allow** arrows/	B1	
			R in correct position i.e. gap 11 **or** 25 { brackets < λ/2	B1	
			OR two correct positions but R and C reversed 1/2		
		(ii)	6.2 → 6.6 cm	B1	
		(iii)	(v =)fλ **or** 5.1/5100 × 6.4/0.064 (using candidate's **5 (a) (ii)**)	C1	
			$3.16 – 3.37 \times 10^{n}$	C1	
			316 – 337 m/s	A1	

	(b)		(longitudinal wave:) vibration/oscillation direction parallel to/in same direction as wave/energy travel direction (**not** L & R)	B1	
			transverse wave: directions perpendicular **or** can be polarized (**not** up & down)	B1	[8]

QUESTION · 5054_s10_qp_11.pdf · source page 8 of 20

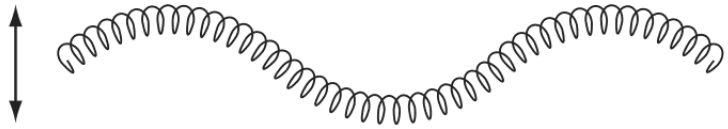
18 Which diagram shows an example of a longitudinal wave?

A light travelling from a lamp to a screen

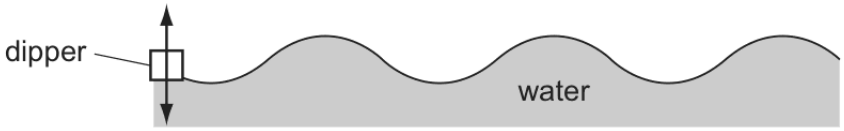


B a spring pushed backwards and forwards

C a spring pushed up and down



D a water ripple caused by a dipper moving up and down



MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 18

18	**B**

For Examiner's Use

4 Fig. 4.1 and Fig.4.2 are diagrams of a ripple tank being used to show two properties of waves.

(a) Fig. 4.1 shows wavefronts approaching a barrier in the water.

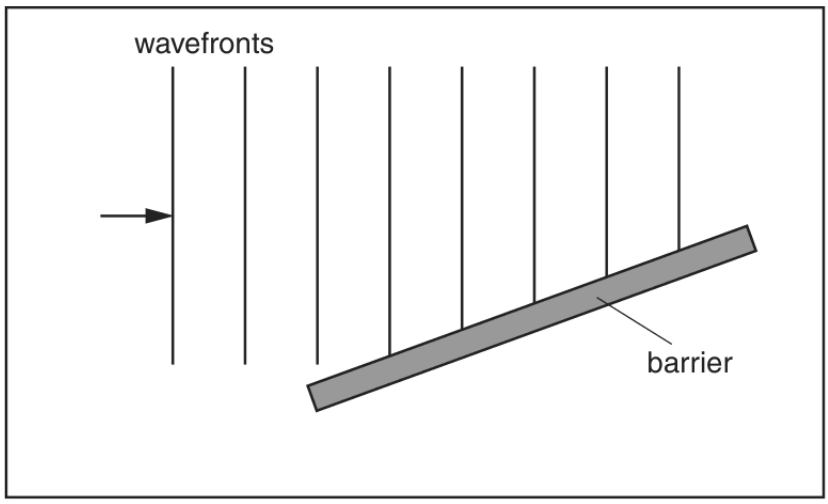


Fig. 4.1

Complete Fig. 4.1 to show the reflection of the wavefronts at the barrier. [2]

(b) Fig. 4.2 shows the wavefronts approaching shallow water above a piece of glass.

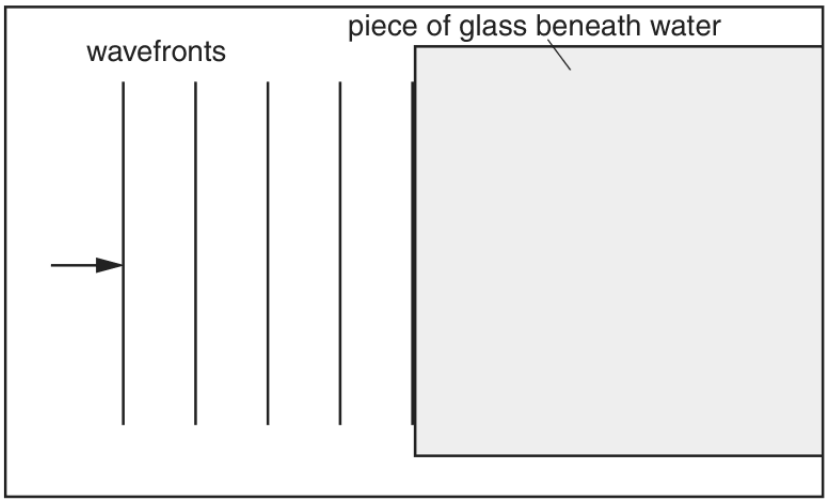


Fig. 4.2

(i) Complete Fig. 4.2 to show the wavefronts in the shallow water. [1]

(ii) As the wave passes into the shallow water, state what, if anything, happens to

1. the wave speed, ...[1]

2. the frequency. .....[1]

4 **(a)** at least 3 reflected wavefronts with same wavelength as before B1
all **and** ⩾ 2 reflected wavefronts at correct angle B1 [2]

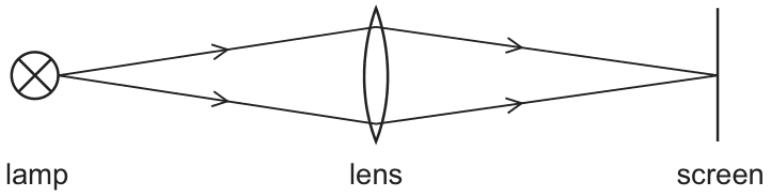
(b) **(i)** no change in direction **and** clearly smaller, approximately constant wavelength B1 [1]

(ii) **1** reduces B1
2 constant B1 [2]

QUESTION · 5054_s10_qp_12.pdf · source page 9 of 20

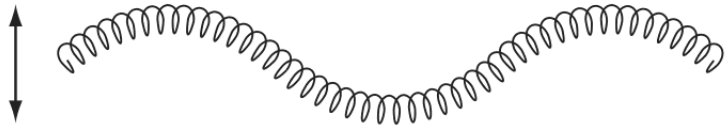
19 Which diagram shows an example of a longitudinal wave?

A light travelling from a lamp to a screen

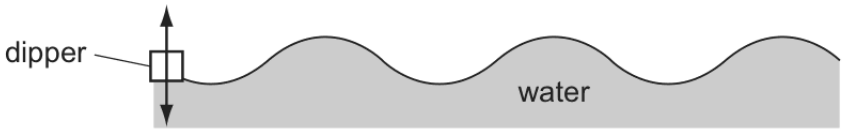


B a spring pushed backwards and forwards

C a spring pushed up and down



D a water ripple caused by a dipper moving up and down



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19	**B**

For Examiner's Use

10 (a) Describe a method for measuring the speed of sound in air. In your account, state clearly

- how the sound is made,
- what measurements are taken,
- how the result is calculated,
- one precaution to produce an accurate result.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[5]

(b) Ultrasound is used in quality control to detect cracks in metal.

Pulses of ultrasound are sent into the metal from a transmitter. A detector is placed next to the transmitter on the front surface of the metal.

Fig. 10.1 shows the oscilloscope trace of the ultrasound pulses produced if the metal contains no cracks.

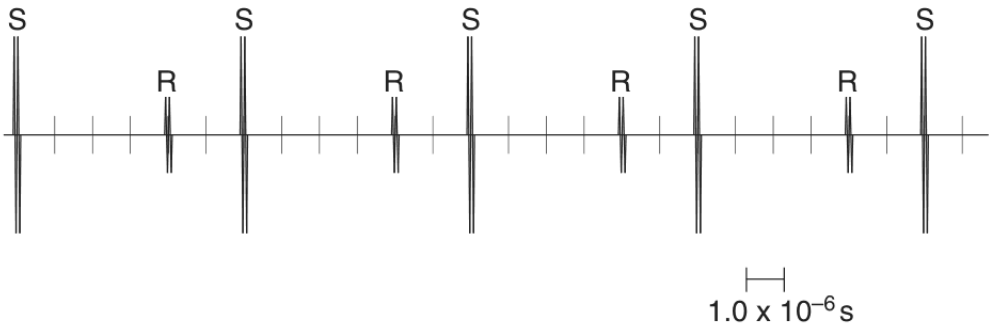


Fig. 10.1

One division along the *x*-axis represents 1.0×10^{-6} s.

Pulses labelled S are the pulses initially sent out from the transmitter. Each pulse labelled R is the reflection from the back surface of the metal of the previous pulse S.

For Examiner's Use

(i) State what is meant by *ultrasound*.

.....

.....[2]

(ii) Use Fig. 10.1 to calculate the number of pulses sent out by the source in one second.

number = ...[2]

(iii) Suggest two reasons why the amplitude of R is less than the amplitude of S.

1. .....

.....

2. .....

.....
[2]

(iv) Some time later, the piece of metal is tested again. It now has a small crack half-way between the front surface and the back surface.

On Fig. 10.1, draw the position and size of the pulses produced by this crack.

Label each of these pulses C. [2]

(v) A second beam of ultrasound has a frequency of 8.0×10^6 Hz and a speed of 4000 m/s in the metal.

Calculate the wavelength of this ultrasound in the metal.

wavelength = ...[2]

10 (a) how sound is made e.g. gun, clap hands, hit metal B1
correct measurement of time, e.g. from seeing flash to hearing sound, clap-echo B1
correct measurement of distance, e.g. gun to observer, observer to wall B1
correct calculation for measurements, e.g. d/t or $2d/t$ B1
precaution e.g. time clap on echo and time 10; ensure no wind; repeat in opposite direction; repeat and average; use large distance; use more than 200 m B1 [5]

(b) (i) (sound/wave/vibration) of high frequency or (sound that) cannot be heard C1
(frequency) above 15–20 kHz A1 [2]

(ii) $f = 1/T$ or 6×10^{-6} (s) seen or 2,3,4 pulses in 12,18,24 μs C1
1.7×10^5 allow 166667 A1 [2]

(iii) not all sound is reflected (from back surface) or some passes through the back
(some energy/sound) absorbed (by metal)
(sound/energy) spreads out/scattered/reflected in other directions/dispersed/ travels a (greater) distance any 2 lines B2 [2]

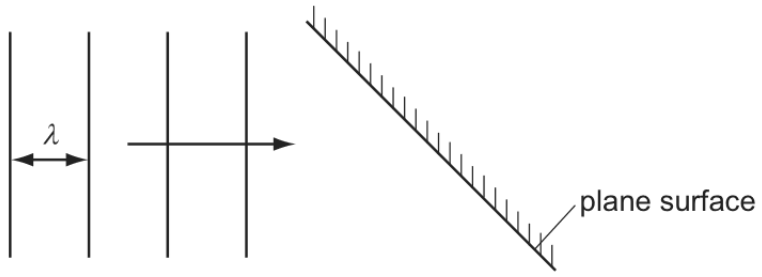
(iv) at least one pulse half way between S and R in the long gap B1
total height of pulse smaller than S and 3 or more drawn and labelled C B1 [2]

(v) $v = f\lambda$ in any algebraic or numerical form e.g. $4000/8 \times 10^6$ C1
$5(.0) \times 10^{-4}$ m A1 [2]

[Total: 15]

QUESTION · 5054_w10_qp_11.pdf · source page 10 of 20

21 In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



Which row correctly describes the waves after they are reflected from the surface?

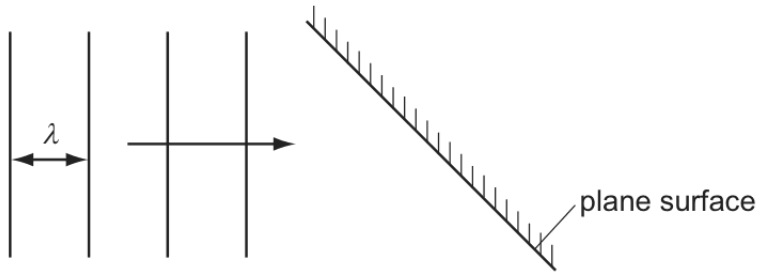
	speed of waves	wavelength λ
A	faster	shorter
B	slower	shorter
C	the same	longer
D	the same	the same

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 21

21	**D**

QUESTION · 5054_w10_qp_12.pdf · source page 9 of 16

25 In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



Which row correctly describes the waves after they are reflected from the surface?

	speed of waves	wavelength λ
A	faster	shorter
B	slower	shorter
C	the same	longer
D	the same	the same

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 25

25 **D**

For Examiner's Use

5 A microphone is connected to a cathode-ray oscilloscope (c.r.o.). A note produced by a musical instrument causes a trace on the screen. The trace is stored electronically in the c.r.o.

Fig. 5.1 shows the stored trace displayed on the screen.

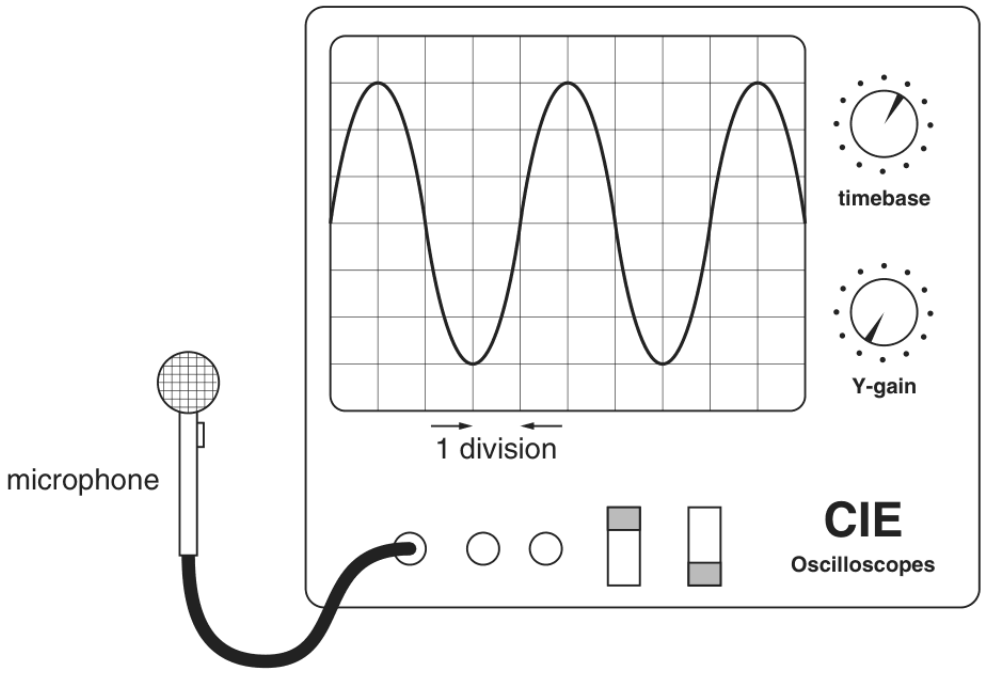


Fig. 5.1

The timebase setting on the oscilloscope is 0.20 ms / division.

(a) Determine the frequency of this note.

frequency = ...[3]

For Examiner's Use

(b) A note from a second musical instrument is played and the trace S produced by the new note is also displayed on the screen. Fig. 5.2 shows the screen displaying both the original trace and S.

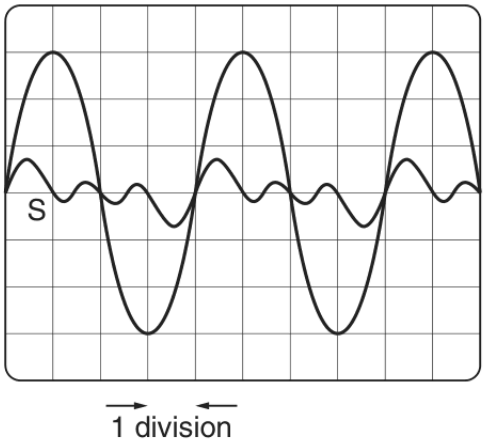


Fig. 5.2

In each of the three spaces below, describe either a similarity or a difference between the **sounds** made by the two notes.

1. .....

2. .....

3. .....

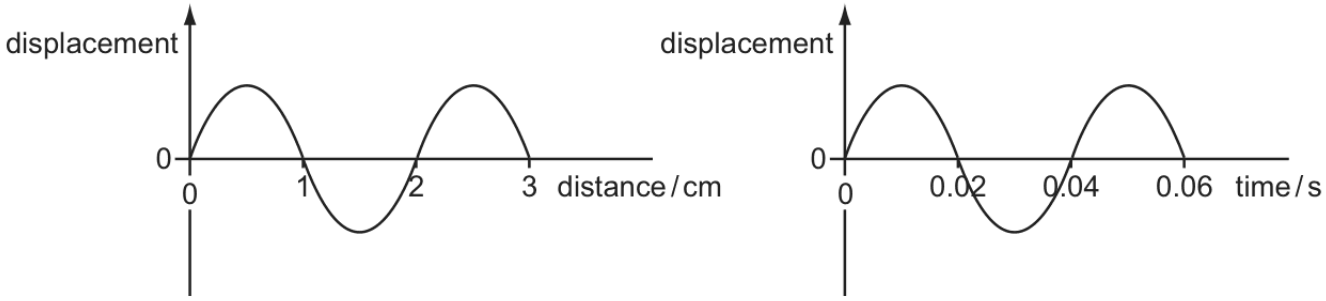
[3]

5 **(a)** 0.80 **or** 0.0008 **or** 4 × 0.20 **or** 4 × 0.0002 **or** 4 divisions C1
(f =) 1/T **or** 1.2/1.25/1.3 (Hz) C1
1200/1250/1300 Hz A1

(b) any **three** of:
equal/same pitch/frequency (**ign** wavelength)
original note louder/ S quieter/softer (**ign** amplitude)
{ different qualities/timbres/
{ more frequencies/overtones/harmonics in S B3 [6]

QUESTION · 5054_s11_qp_11.pdf · source page 8 of 16

20 The displacement-distance and displacement-time graphs are for a water wave in a ripple tank.



What is the speed of the water wave?

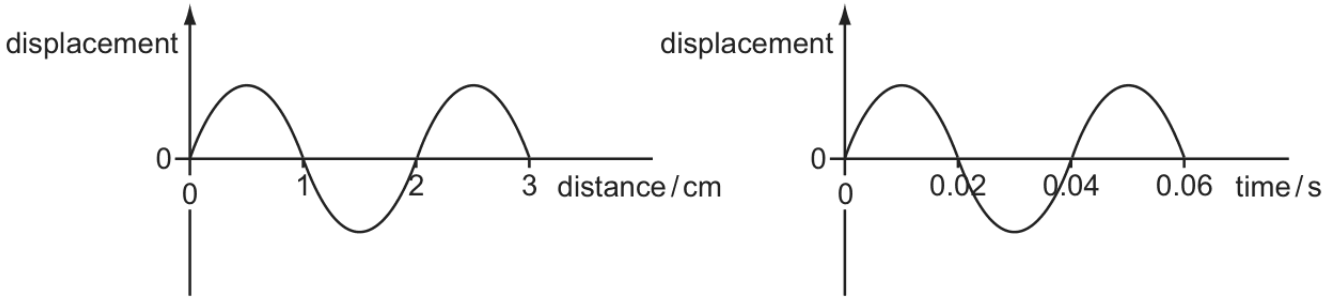
A 0.02 cm / s **B** 0.08 cm / s **C** 25 cm / s **D** 50 cm / s

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 20

20 **D**

QUESTION · 5054_s11_qp_12.pdf · source page 9 of 16

22 The displacement-distance and displacement-time graphs are for a water wave in a ripple tank.



What is the speed of the water wave?

A 0.02 cm / s **B** 0.08 cm / s **C** 25 cm / s **D** 50 cm / s

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 22

22	**D**

4 Fig. 4.1 shows circular wavefronts produced at the centre of a circular ripple tank.

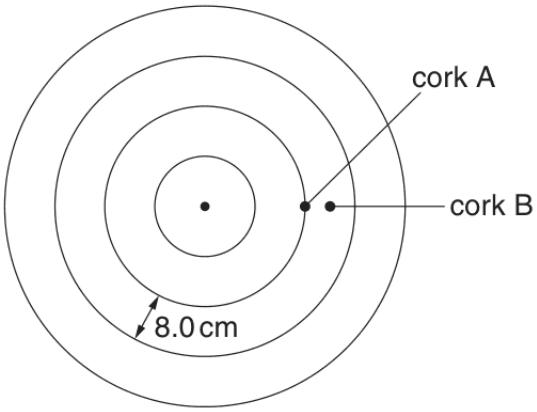


Fig. 4.1

Two corks, A and B, float on the water in the ripple tank. They move up and down on the surface of the water as the wave passes. The wavelength of the wave is 8.0 cm.

Fig. 4.2 shows how the displacement of A varies with time.

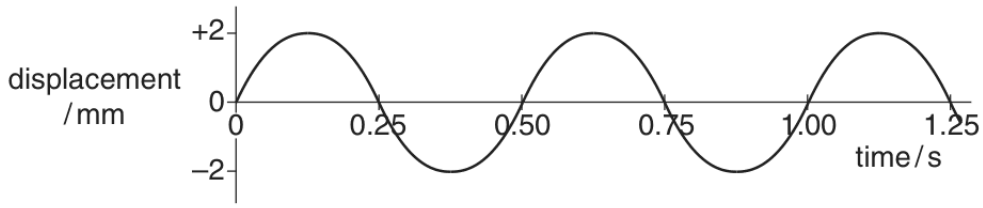


Fig. 4.2

(a) State the amplitude of the vibrations of A as the wave passes.

amplitude = ...[1]

(b) The horizontal distance between A and B is half the wavelength of the wave.

On Fig. 4.2, sketch a graph to show how the displacement of B varies with time. [2]

(c) **(i)** Use Fig. 4.2 to determine the frequency of the wave.

frequency = ...[2]

(ii) The distance between the centre of the ripple tank and its edge is 40 cm.

Determine the time taken by a wavefront to travel from the centre of the tank to the edge.

time = ...[2]

4	**(a)**	2(.0) mm	B1	
	(b)	same period (by eye), with at least one wave	B1	
		opposite phase to wave drawn	B1	
	(c)	**(i)** (f =) 1/T numerical or algebraic seen (e.g. 1/0.5) // 1 wave in 0.5 s // 2 waves in 1 s	C1	
		2(.0) Hz	A1	
		(ii) $v = f\lambda$ // 8 × 2 or 8 × **(i)** // 16 (cm / s) // 5 (wavelengths from centre to edge) // (t =) d/v	C1	
		2.5 s **ecf** from **(i)** – i.e. **accept 5/(c)(i)**	A1	[7]

4 Fig. 4.1 represents a microwave travelling in air through points A and B.

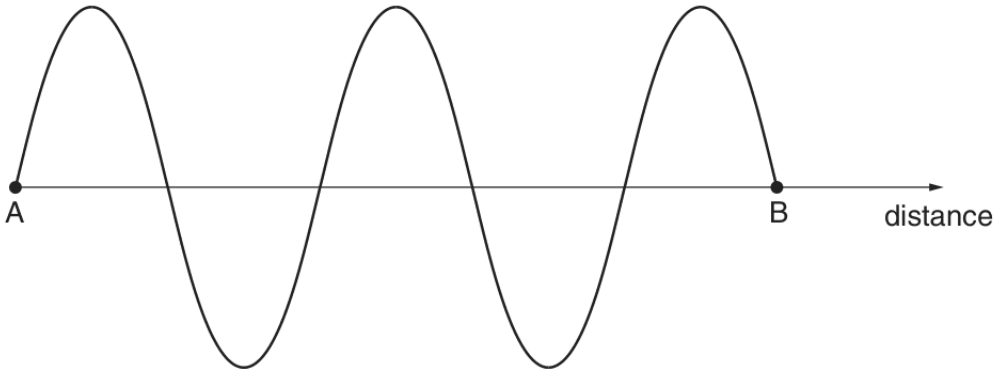


Fig. 4.1 (not to scale)

The distance between A and B is 40 cm.

(a) State the speed of microwaves in air.

speed = ...[1]

(b) Determine the wavelength of the microwave shown in Fig. 4.1.

wavelength = ...[1]

(c) Describe how microwaves are used in the transmission of television signals by satellite.

.....

.....

.....

.....[3]

(d) State two properties common to all electromagnetic waves.

1. .....

2. .....
[2]

4 **(a)** $3(.00) \times 10^8\,m/s$ B1

(b) 0.16 m **or** 16 cm B1

(c) any **three** of:
travel through space/vacuum
pass through the atmosphere/not reflected by ionosphere
encoded (with the signal)
(satellite) amplifies/boosts signal
sent to/received by satellite
transmitted/sent by satellite
transmitted/received by a (satellite) **dish** (on Earth) B3

(d) any **two** of:
same (high) speed (in air) **or** travel at speed of light
travel in vacuum/space **or** no medium needed
transfer/transmit energy
transverse (stated **or** explained)
(oscillating) magnetic **and** electric fields/waves
reflection/refraction/diffraction/interference/polarisation B2 [7]

QUESTION · 5054_s12_qp_11.pdf · source page 8 of 16

21 The speed of radio waves is *c*. A radio station transmits waves with a wavelength λ.

What is the frequency of the transmission?

A $\frac{c}{\lambda}$ **B** $\frac{\lambda}{c}$ **C** $\frac{1}{\lambda}$ **D** $c\lambda$

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 21

21	**A**

6 A sound from an electronic organ is played into a microphone which is connected to a cathode-ray oscilloscope (c.r.o.). Fig. 6.1 shows the trace on the c.r.o. produced by the sound.

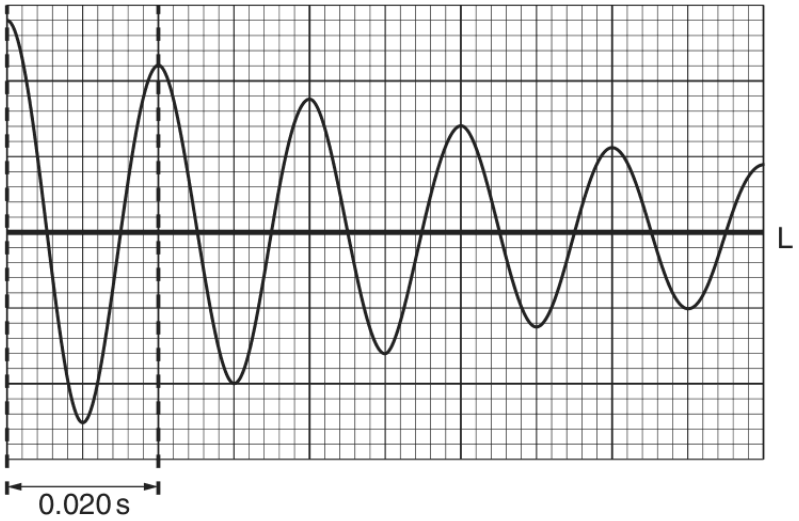


Fig. 6.1

The horizontal line marked L is the trace seen when there is no sound.

(a) (i) State how Fig. 6.1 shows that the loudness of the sound decreases with time.

.....[1]

(ii) State how Fig. 6.1 shows that the pitch of the sound is constant.

.....

.....[1]

(b) (i) State what is meant by the *frequency* of a wave.

.....

.....[1]

(ii) The time for one complete cycle of the wave in Fig. 6.1 is 0.020 s.
Calculate the frequency of the wave.

frequency = ...[2]

(iii) Determine the time taken for the amplitude of the trace in Fig. 6.1 to decrease to half its initial value.

time = ...[1]

6	**(a)**	**(i)**	amplitude decreases	B1	
		(ii)	constant frequency/time for one wave/wavelength/period	B1	
	(b)	**(i)**	number of (complete) cycles in one second	B1	
		(ii)	(f =) 1/T in any form numerical or algebraic; 1/0.02 50 Hz	C1 A1	
		(iii)	0.06 s	B1	[6]

QUESTION · 5054_s12_qp_12.pdf · source page 9 of 16

21 A wave has a frequency of 2 Hz.

How many waves are produced in one minute?

A 2×60 **B** $\frac{60}{2}$ **C** 2 **D** $\frac{2}{60}$

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 21

21	**A**

6 Fig. 6.1 shows a wave on the surface of water. The wave is travelling to the right.

Fig. 6.1 (not to scale)

(a) Describe what is meant by *wave motion*.

.....

.....

.....[2]

(b) On Fig. 6.1, draw an arrow to show the direction of the movement of a water molecule at X. [1]

(c) The frequency of the water wave is 2.0 Hz and the wavelength is 2.5 cm.

(i) Calculate the speed of the wave.

speed = ...[2]

(ii) On Fig. 6.1, mark a distance which shows how far a wavefront at X moves in 1.0 s. Label this distance *D*. [1]

6 **(a)** Any 2 of
- an oscillation/vibration/movement up and down
- carries energy
- no (net) movement of the medium/transfer of matter) B2

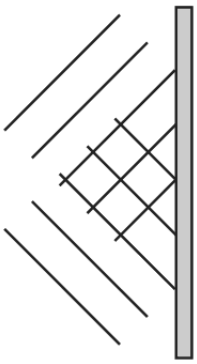
(b) arrow downwards or upwards or both B1

(c) **(i)** (v =) $f\lambda$ in any form numerical or algebraic C1
5(.0) cm/s or 0.05(0) m / s A1

(ii) line or indication **labelled D** of length 2 wavelengths B1 [6]

QUESTION · 5054_w12_qp_11.pdf · source page 9 of 16

21 The diagram shows the pattern of waves in a ripple tank.



What does the pattern show?

A waves being reflected

B waves being refracted

C waves changing frequency

D waves changing speed

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21	**A**

For Examiner's Use

5 A water wave of frequency 3.6 Hz is generated in a ripple tank. Fig. 5.1 is a scale diagram that shows the wavefronts travelling from deep water to shallow water in the ripple tank. A length of 1.0 cm on the diagram represents 8.0 cm in the ripple tank.

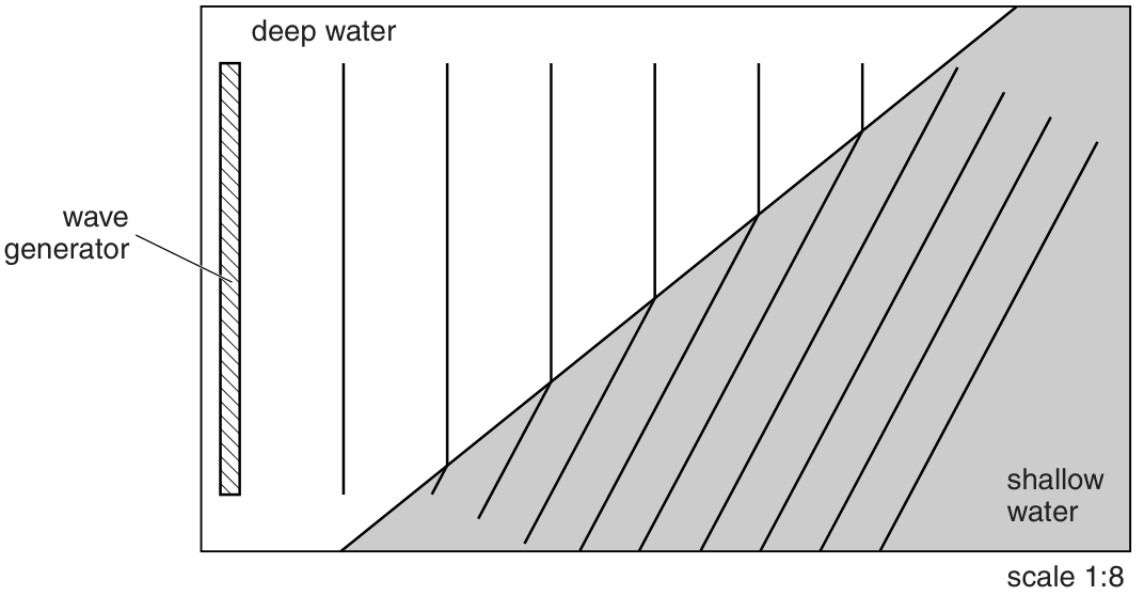


Fig. 5.1

(a) (i) Using measurements from Fig. 5.1, determine the wavelength of the wave in the deep water.

wavelength = ... [2]

(ii) Calculate the speed of the wave in the deep water.

speed = ... [2]

(b) As the wave passes into the shallow water, state what happens to

(i) the frequency,

..... [1]

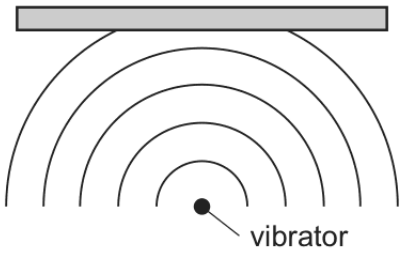
(ii) the speed.

..... [1]

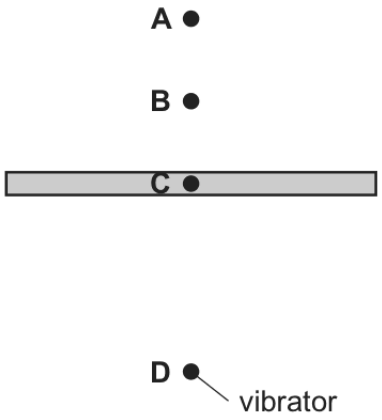
5	**(a)**	**(i)**	clear attempt at measuring more than one wavelength e.g. 6.85/5		
			or 1.30 – 1.45 cm	B1	
			10.7 – 11.3 cm	B1	
		(ii)	(v =)$f\lambda$ **or** 3.6 × **(a)(i)**	C1	
			40(39.6) cm / s	A1	
	(b)	**(i)**	stays the same	B1	
		(ii)	decreases	B1	[6]

QUESTION · 5054_w12_qp_12.pdf · source page 8 of 16

21 In a ripple tank, a vibrator produces circular wavefronts which hit a flat surface.



The reflected wavefronts are also parts of circles. Where is the centre of these circles?



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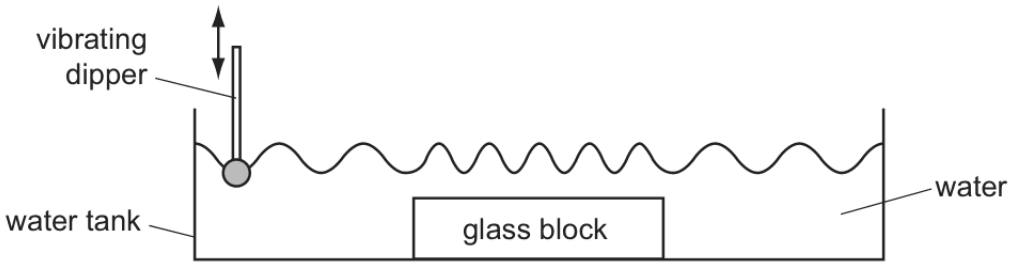
21	**A**

QUESTION · 5054_s13_qp_11.pdf · source page 8 of 20

17 A ripple tank is used to show wave behaviour. The dipper vibrates up and down at a constant frequency.



What happens to the frequency and to the speed of the wave as it reaches the glass block?

	frequency	speed
A	decreases	decreases
B	decreases	increases
C	remains the same	decreases
D	remains the same	increases

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17 **C**

QUESTION · 5054_s13_qp_11.pdf · source page 9 of 20

18 A wave has a frequency of 10 kHz.

Which pair of values of its speed and wavelength is possible?

	speed / m/s	wavelength / m
A	330	0.33
B	330	33
C	3.0×10^8	30
D	3.0×10^8	3.0×10^4

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18 **D**

QUESTION · 5054_s13_qp_12.pdf · source page 9 of 16

22 Energy can be transferred in many different ways.

In which situation is energy transferred by wave motion?

A colliding atoms in a heated copper rod

B fast-moving electrons in a cathode-ray oscilloscope

C hot water rising in a heated saucepan

D ripples passing across water in a ripple tank

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22	**D**

QUESTION · 5054_s13_qp_12.pdf · source page 9 of 16

23 A wave has a frequency of 10 kHz.

Which pair of values of its speed and wavelength is possible?

	speed / m/s	wavelength / m
A	330	0.33
B	330	33
C	3.0×10^8	30
D	3.0×10^8	3.0×10^4

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 23

23	**D**

QUESTION · 5054_s13_qp_12.pdf · source page 9 of 16

24 Which of the following travels as a longitudinal wave?

A a radio wave in air

B a sound wave in a solid

C a wave on a rope shaken from side to side

D an infra-red wave in space

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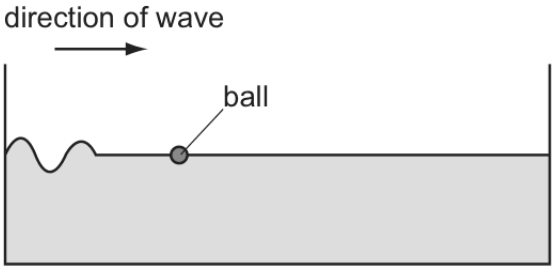
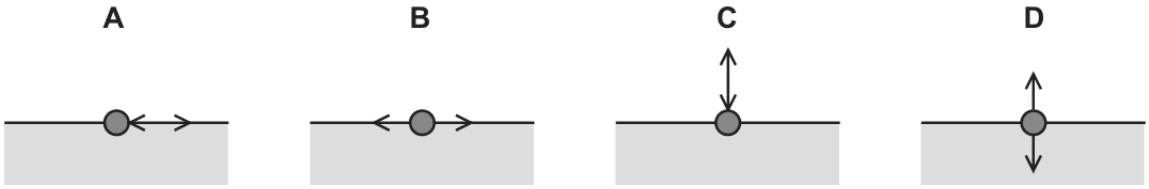
24	**B**

QUESTION · 5054_w13_qp_11.pdf · source page 9 of 20

19 The diagram shows a ball floating in a tank of water.



Which diagram shows the movement of the ball as the wave passes?



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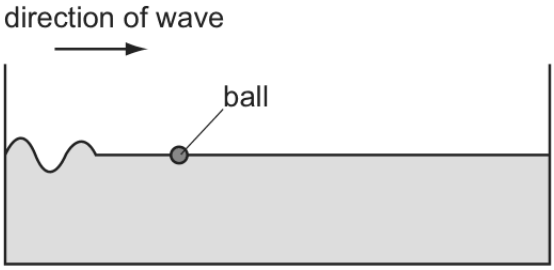
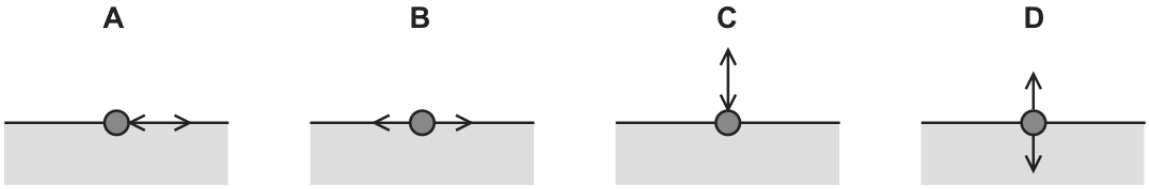
19	**D**

QUESTION · 5054_w13_qp_12.pdf · source page 11 of 20

22 The diagram shows a ball floating in a tank of water.



Which diagram shows the movement of the ball as the wave passes?



MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 22

22	**D**

For Examiner's Use

5 A physics textbook states that sound is a longitudinal pressure wave with a frequency within the audible range.

(a) Explain what is meant by a *longitudinal wave*.

.....

.....

.....[2]

(b) **(i)** State the approximate range of audible frequencies.

highest frequency: .....

lowest frequency: .....
[2]

(ii) The speed of sound in air is 330 m / s. Using your answer in **(i)**, calculate the shortest wavelength in air of sound in the audible range.

wavelength = ...[2]

5	**(a)**	transmission of energy through a medium **or** vibration **or** oscillation **or** two opposite motions (e.g. up and down) **or** compressions and rarefactions	C1	
		vibration direction parallel to energy travel/wave direction **or** similar	A1	
	(b) **(i)**	$1.5–2.5 \times 10^4$ Hz **or** 15–25 kHz cao	B1	
		15–25 Hz cao	B1	
	(ii)	(λ =) c/f **or** 330/either of candidate's frequencies	C1	
		330/candidate's higher frequency and correctly calculated with unit (candidate's higher frequency is either the one stated as the highest **or** the one that is in fact the higher)	A1	[6]

QUESTION · 5054_s14_qp_11.pdf · source page 9 of 16

23 Water waves refract at a boundary between deep water and shallow water.

What is the effect on the frequency, wavelength and speed of the waves at the boundary?

	frequency	wavelength	speed
A	changes	changes	stays the same
B	changes	stays the same	stays the same
C	stays the same	changes	changes
D	stays the same	stays the same	changes

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 23

23	**C**

10 A sound wave travels through air.

(a) Explain what is meant by the *wavelength* of a wave.

.....

..... [1]

(b) Fig. 10.1 represents air molecules in the sound wave at one instant.

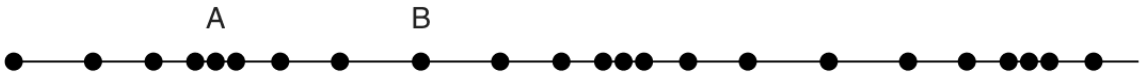


Fig. 10.1

(i) Describe the motion of a molecule of air due to the sound.

.....

..... [2]

(ii) State one difference between the motion of molecule A and the motion of molecule B.

.....

..... [1]

(c) Describe an experiment that shows that a medium is needed to transmit sound waves. Draw a labelled diagram of the apparatus.

.....

.....

.....

.....

.....

..... [4]

(d) A short pulse of sound waves produces an echo from a wall 20 m away. The echo arrives back at the source of the sound 0.12 s after the pulse is produced.

The wavelength of the sound wave is 30 mm.

(i) State what is meant by an *echo*.

.....

..... [1]

(ii) Calculate the speed of the sound.

speed = ... [2]

(iii) Calculate the frequency of the sound.

frequency = ... [3]

(iv) The sound wave produces compressions and rarefactions in the air as it passes.

Calculate the distance between a compression and the nearest rarefaction.

distance = ... [1]

10 (a) (shortest) distance between two points in phase [B1]

(b) (i) oscillate **or** vibrate **or** move closer and further [B1]
backwards and forwards **or** in direction of wave (energy) **or** longitudinal mentioned [B1]

(ii) move in opposite directions **or** when A is on right B is on left (of mean)
or A molecules next move apart **and** B next move together [B1]

(c) diagram containing sound source (bell / tuning fork, etc.) in container [B1]
vacuum / pump connected to container [B1]
remove air [M1]
sound decreases [A1]

(d) (i) sound (that returns) after a reflection [B1]

(ii) (s=) d / t in any algebraic or numerical form e.g. 20 (or 40) / 0.12 (or 0.06) [C1]
330 **or** 333 m / s [A1]

(iii) (f=) v / λ in any algebraic or numerical form e.g. 330 / 30 (or 0.03) [C1]
correct conversion of 30 mm to 0.03 m [C1]
11 100 Hz **or** 11 000 Hz [A1]

(iv) 0.015 m **or** 15 mm [B1]

[15]

QUESTION · 5054_s14_qp_12.pdf · source page 10 of 16

25 What is the frequency of a wave?

A The number of waves passing a fixed point per second.

B The number of peaks added to the number of troughs passing a fixed point per second.

C The time taken for one wave to pass a fixed point.

D The time taken for the displacement to change from maximum to minimum.

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 25

25	**A**

4 Fig. 4.1 shows part of a long rope used by a student to show a transverse wave.

Fig. 4.1

(a) On Fig. 4.1,

(i) mark the direction of movement of the student's hand, [1]

(ii) mark and label the wavelength λ of the wave, [1]

(iii) mark and label the amplitude A of the wave. [1]

(b) Describe how the frequency of the wave is found using a stopwatch.

.....

.....

.....

..... [2]

(c) Using the same rope, the student produces a wave of a longer wavelength than that shown in Fig. 4.1.

State how the student does this.

.....

..... [1]

4 **(a)** **(i)** up and down clear, e.g. by double headed arrow **or** down [B1]

(ii) any correct distance between consecutive points in phase [B1]

(iii) correct distance [B1]

(b) measure number of oscillations / count waves (passing) in a stated time or time at least one oscillation [B1]
show how to calculate number of oscillations per second [B1]

(c) **moves** (hand or rope) with slow(er) speed or rate / less frequency / less times per sec [B1]

[6]

QUESTION · 5054_w14_qp_11.pdf · source page 10 of 20

19 A wave of frequency 13 000 Hz travels 1300 m in 4.0 s.

What is the wavelength of the wave?

A 0.025 m **B** 0.40 m **C** 2.5 m **D** 40 m

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 19

19	**A**

4 Ripple tanks are often used to illustrate wave motion.

(a) Describe what is meant by *wave motion* in a ripple tank.

.....

.....

.....[2]

(b) When describing wave motion, state what is meant by

(i) *frequency*,

.....[1]

(ii) *wavelength*.

.....

.....[2]

(c) A water wave in a ripple tank strikes a barrier. Fig. 4.1 shows some wavefronts of the incident wave.

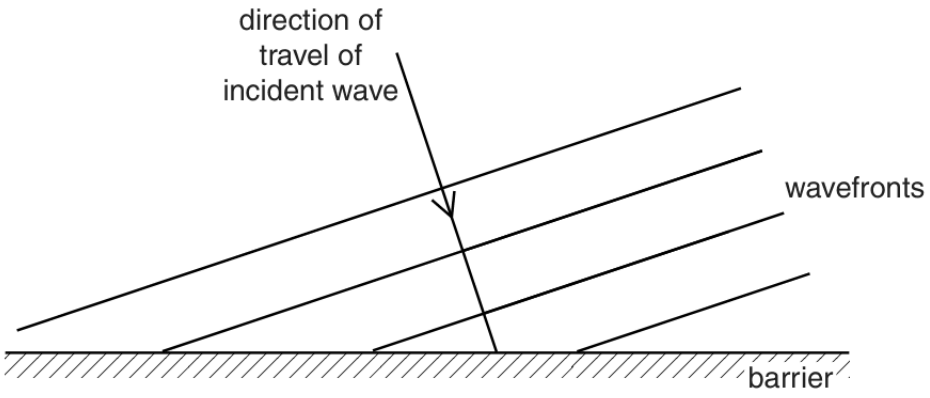


Fig. 4.1

The water wave hits the barrier and is reflected. Three of the wavefronts in Fig. 4.1 have already hit the barrier. The reflected parts of these wavefronts are not shown.

On Fig. 4.1, draw the reflected parts of these three wavefronts. [3]

4	**(a)**	any **two** from: transmission of energy without net movement of medium through vibration of particles	B2	
	(b) **(i)**	number of (complete) waves / cycles / oscillations per unit time / second	B1	
	(ii)	distance between (neighbouring) waves	C1	
		distance between (neighbouring) wavefronts / points of same phase **or** crest to crest / tough to trough distance	A1	
	(c)	three reflected wavefronts roughly correct direction	M1	
		wavelengths equal to each other and incident wavelength by eye	A1	
		reflected wavefronts joined to incident wavefronts	B1	**[8]**

QUESTION · 5054_s15_qp_11.pdf · source page 8 of 16

21 A wave in a ripple tank passes from a deeper to a shallower region and refracts.

Which wave properties decrease as the wave enters the shallow region?

A frequency only

B speed only

C frequency and wavelength

D speed and wavelength

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 21

21	**D**

4 Fig. 4.1 shows a wave on a string. The wave is travelling towards the right.

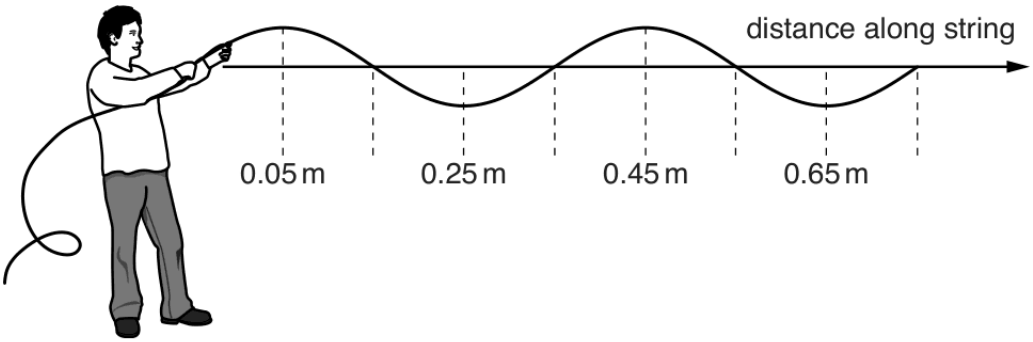


Fig. 4.1

(a) Describe the movement of the particles of the string.

.....

.....

.....

.....[2]

(b) Determine the wavelength of the wave.

wavelength = ...[1]

(c) The speed of the wave is 2.0 m/s.

(i) Calculate the frequency of the wave.

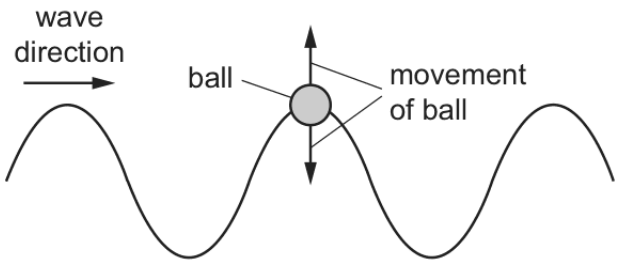
frequency = ...[2]

(ii) On Fig. 4.1, draw the string at a time 0.10 s later than the time in Fig. 4.1. [1]

4	**(a)**	oscillate / vibrate stated or described	B1
		transverse movement described	B1
	(b)	0.40 m	B1
	(c) **(i)**	$v = f\lambda$ or ($f =$) v/λ or 2 / (b)	C1
		5.0 Hz	A1
	(ii)	clear attempt to draw wave moved along 0.20 m to right	B1

QUESTION · 5054_s15_qp_12.pdf · source page 9 of 16

24 A ball floating in a ripple tank begins to move vertically up and down as a wave passes beneath it. The ball does not move horizontally.



Which statement is correct?

A Both energy and water are transferred in the wave direction.

B Energy is not transferred in the wave direction but water is.

C Energy is transferred in the wave direction but water is not.

D Neither energy nor water is transferred in the wave direction.

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 24

24	**C**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows the wavefronts of a water wave in deep water in a ripple tank.

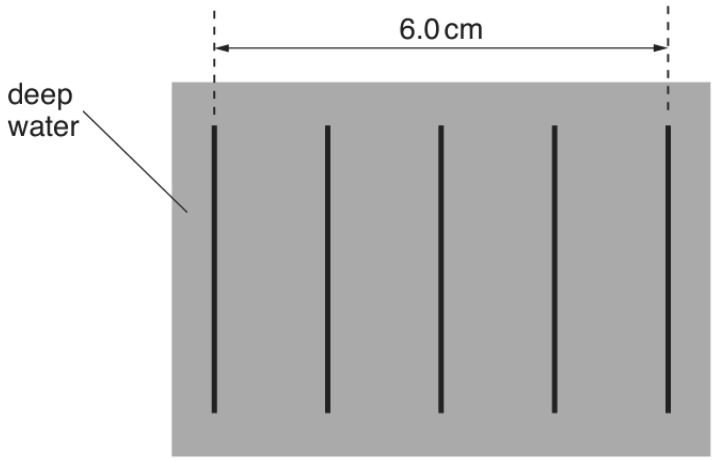


Fig. 9.1

The frequency of the wave in deep water is 5.0 Hz.

(a) Explain what is meant by the *frequency* of a wave.

.....

.....

..... [2]

(b) **(i)** Determine the wavelength of the wave in deep water.

..... [1]

(ii) Calculate the speed of the wave in deep water.

speed = ... [2]

(c) The wave passes from deep water into shallow water. The speed of the wave is less in shallower water.

(i) State and explain how this affects the wavelength of the wave.

.....

.....

..... [2]

(ii) The wave in deep water shown in Fig. 9.2 travels towards the right and enters the shallow water at an angle. The wave refracts.

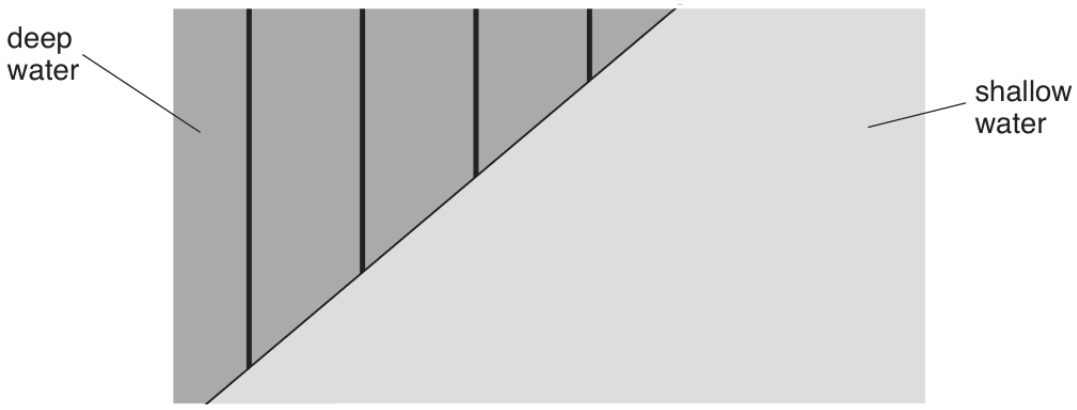


Fig. 9.2

On Fig. 9.2, draw the wavefronts in the shallow water. [3]

(d) Sound is also a wave.

(i) Describe one difference between a sound wave and a water wave.

.....

.....

.....

..... [2]

(ii) The speed of sound in carbon dioxide gas is less than the speed of sound in air. Using this information, or otherwise, describe an experiment to show the refraction of sound waves. You may include a diagram of your apparatus.

.....

.....

.....

.....

.....

..... [3]

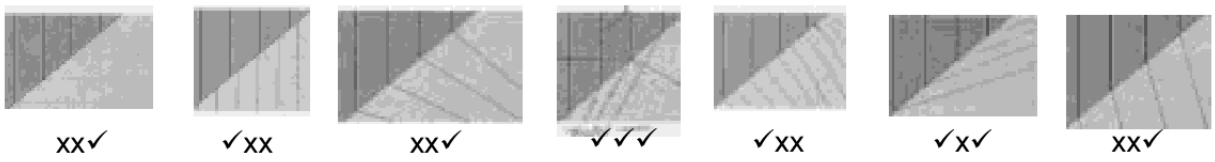
9 **(a)** number of waves (that pass a point) M1
or number of oscillations (passing a point)
in unit time **or** per second **or** in 1 second A1

(b) **(i)** 1.5 cm B1

(ii) (v =)$f\lambda$ **or** 5 × 1.5 seen C1
or (s=)d / t **and** f = 1 / t
7.5 cm / s A1

(c) **(i)** wavelength decreases B1
travels a shorter distance in the same time B1
or frequency stays the same (and $v = f\lambda$)

(ii) wavefronts with smaller wavelength (by eye) B1
smaller angle to surface (by eye) **and** slanted down B1
wavefronts join those in shallow water B1



xx✓ ✓xx xx✓ ✓✓✓ ✓xx ✓x✓ xx✓

(d) **(i)**

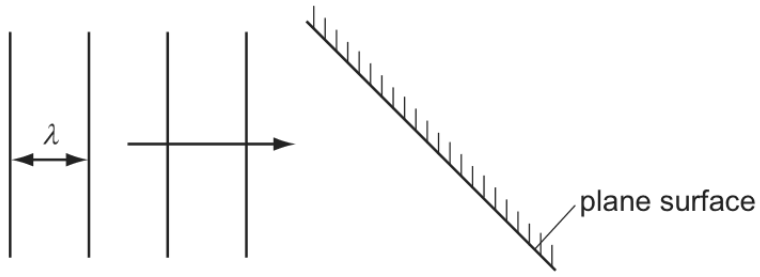
sound	water
particles / wave / source • vibrate / oscillate / move in direction of (travel of) wave / along wave • move backwards and forwards	particles / wave / source • vibrate / oscillate / move at 90° to direction of (travel of) wave • move up and down
(contains) compressions and rarefactions **or** particles come closer / further apart	(contains) crests and troughs
speed 300–330 m / s	wave slower (than sound)

B1
B1
one row only

(ii) method of generating sound, e.g. (loud) speaker (and signal generator) B1
apparatus that enables refraction clear, e.g. carbon dioxide in balloon or any shape where refraction is possible B1
method of detecting refraction, e.g. microphone and how it is used to show refraction B1

QUESTION · 5054_w15_qp_11.pdf · source page 8 of 16

19 In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



Which row correctly describes the waves after they are reflected from the surface?

	speed of waves	wavelength λ
A	larger	shorter
B	smaller	shorter
C	the same	longer
D	the same	the same

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 19

19 **D**

7 Audible sound waves and ultrasound waves both consist of compressions and rarefactions.

(a) Describe one difference between a compression and a rarefaction.

.....

.....

..... [1]

(b) In a particular medium, the speed of both audible sound and ultrasound waves is v.

(i) State the equation that relates the wavelength λ of a sound wave to its frequency f.

.....

..... [1]

(ii) Explain how, in this medium, the wavelength of an audible sound wave compares with the wavelength of an ultrasound wave.

.....

.....

.....

..... [1]

(c) Describe one use of ultrasound waves.

.....

.....

.....

.....

..... [2]

7	**(a)**	at compression: molecules closer together **or** pressure higher **or** vice versa for rarefaction		B1	
	(b)	**(i)** $v = f\lambda$ **or** in words		B1	
		(ii) larger **and** because the frequency is lower		B1	
	(c)	states one use	(e.g. prenatal scanning)	B1	
		basic idea	(e.g. ultrasound reflects off foetus)	B1	[5]

QUESTION · 5054_w15_qp_12.pdf · source page 9 of 16

21 An echo sounder produces ultrasound of frequency 24 kHz. The ultrasound travels in water at a speed of 1.5 km / s.

What is the wavelength in water of ultrasound of this frequency?

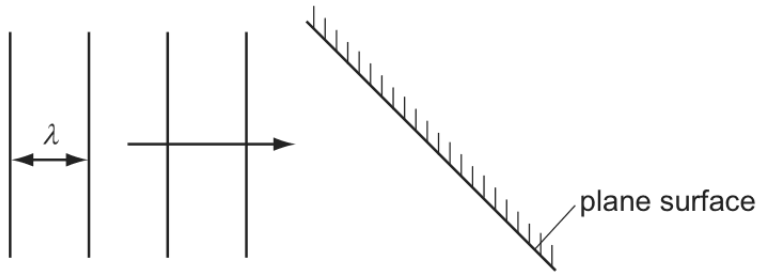
A 0.063 m **B** 16 m **C** 36 m **D** 63 m

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 21

21	**A**

QUESTION · 5054_w15_qp_12.pdf · source page 10 of 16

23 In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



Which row correctly describes the waves after they are reflected from the surface?

	speed of waves	wavelength λ
A	larger	shorter
B	smaller	shorter
C	the same	longer
D	the same	the same

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 23

23	**D**

QUESTION · 5054_s16_qp_11.pdf · source page 8 of 16

19 For a transverse wave, what is a *wavefront*?

A a line joining all points on the same crest of a wave

B a line showing the displacement of a wave

C the energy content of a wave

D the first part of a wave to reach a point

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 19

19	**A**

QUESTION · 5054_s16_qp_11.pdf · source page 8 of 16

20 A longitudinal wave passes along a spring. The coils of the spring vibrate from side to side.

The diagram shows the positions of the coils at one particular time.

Which positions are one wavelength apart?

A W and X **B** W and Z **C** X and Z **D** Y and Z

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 20

20	**D**

5 **(a)** Light is a transverse wave. The direction of vibration is perpendicular to the direction of transfer of the energy.

Complete the table of Fig. 5.1 to show the direction of vibration and the type of wave associated with a sound wave and with a wave on the surface of water in a ripple tank.

wave	direction of vibration	type of wave
light	perpendicular to the direction of transfer of the energy	transverse
sound		
water		

[2]

Fig. 5.1

(b) A wave travels along a rope. Fig. 5.2 shows how the height of one point on the rope varies with time as the wave passes.

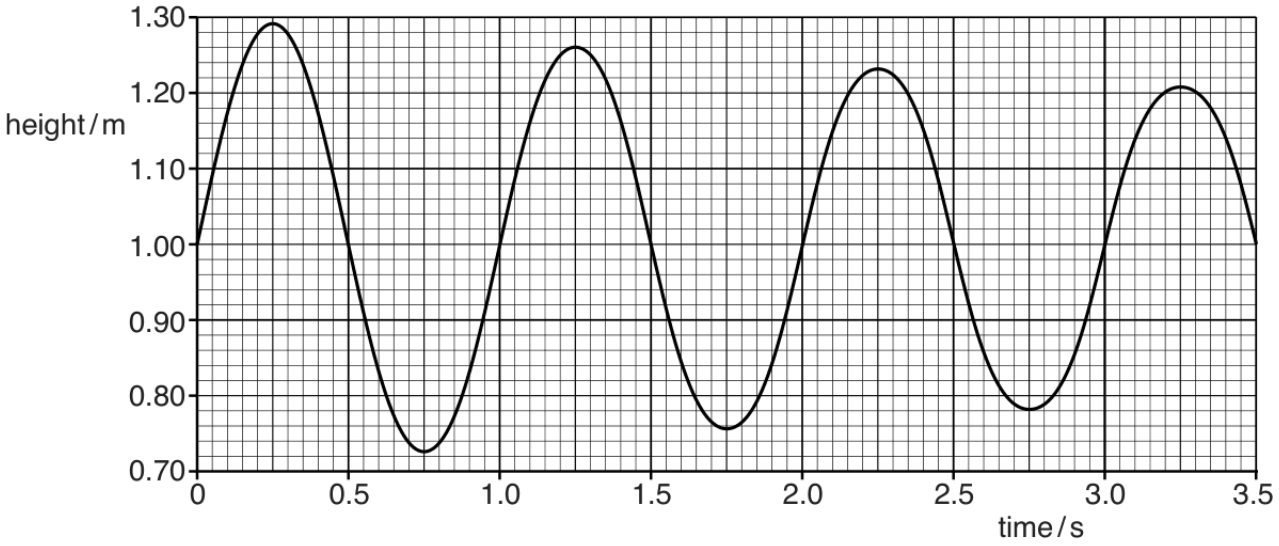


Fig. 5.2

(i) Determine the amplitude of the vibration at time = 0.25 s.

.....[1]

(ii) Describe how the graph shows that the frequency of the wave is constant.

.....

.....[1]

5 **(a)** *sound:* along or parallel (to transfer of energy or wave) **and** longitudinal B1

water: perpendicular **and** transverse B1

(b) **(i)** 0.29 – 0.28 m B1

(ii) time / period for one wave(length) / cycle constant B1
or each oscillation / cycle takes one second

7 Fig. 7.1 shows part of a loudspeaker. A coil is attached to a paper cone and is suspended freely within a cylindrical magnet.

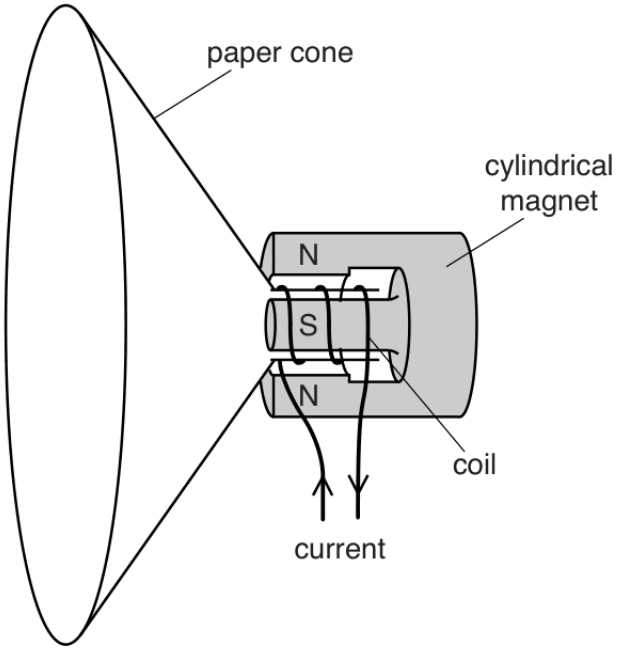


Fig. 7.1

(a) Explain why a current in the coil causes the paper cone to move.

.....

.....

.....

.....[2]

(b) Describe and explain the movement of the cone when there is an alternating current in the coil.

.....

.....

.....[2]

(c) The loudspeaker produces sound of frequency 500 Hz. The speed of sound is 320 m/s. Calculate the wavelength of the sound produced.

wavelength = ...[2]

MARK SCHEME · 5054_s16_ms_21.pdf · source page 3 of 6 · Question 7

7	**(a)**	(current in coil) creates magnetic field **or** current is at right angles to magnetic field (of permanent / cylindrical magnets)	C1
	(b)	into and out of magnet **or** left and right **o**r backwards and forwards	B1
		current is one way then reverses (so reverses force)	B1

MARK SCHEME · 5054_s16_ms_21.pdf · source page 4 of 6 · Question 7

	(c)	(λ=) v / f numerically or algebraic in any form	C1
		0.64 m	A1

QUESTION · 5054_s16_qp_12.pdf · source page 6 of 16

17 A water wave in a ripple tank refracts as it passes from deep water to shallow water.

Which properties change as the wave refracts?

A frequency and amplitude

B frequency and wavelength

C speed and frequency

D speed and wavelength

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 17

17	**D**

QUESTION · 5054_w16_qp_11.pdf · source page 9 of 20

21 A phone rings and lights up.

Which types of wave are associated with the phone's operation?

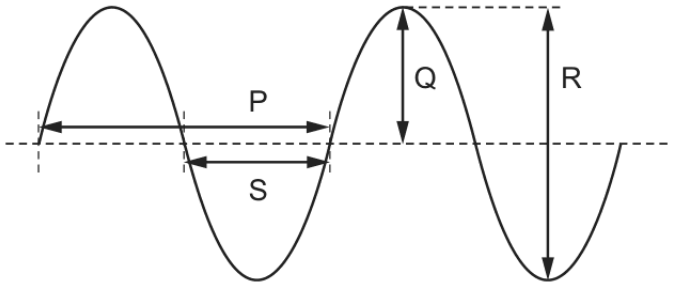
	light	sound
A	longitudinal	longitudinal
B	longitudinal	transverse
C	transverse	longitudinal
D	transverse	transverse

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 21

21 **C**

QUESTION · 5054_w16_qp_11.pdf · source page 9 of 20

22 The diagram shows the surface of water as a wave passes across a ripple tank.



Which lengths represent the amplitude and the wavelength of the wave?

	amplitude	wavelength
A	Q	P
B	Q	S
C	R	P
D	R	S

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 22

22 **A**

5 Fig. 5.1 shows a loudspeaker inside a glass bell-jar that contains air.

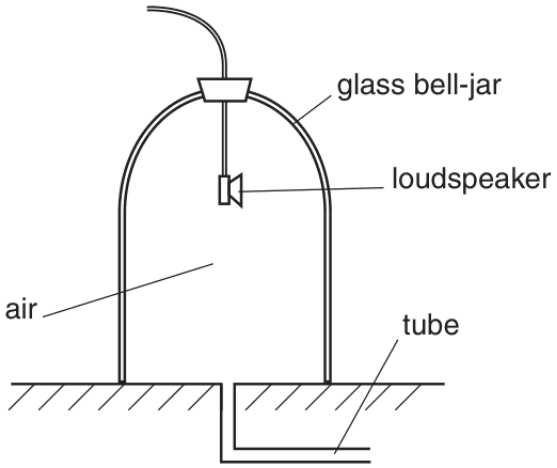


Fig. 5.1

The loudspeaker produces a sound of frequency 2200 Hz.

(a) State what is meant by the *frequency* of a sound.

.....

.....

.....[1]

(b) **(i)** The speed of sound in air is 330 m / s.

Calculate the wavelength of the sound in the air.

wavelength = ...[2]

(ii) The sound passes from the air into the glass.

State what happens to

1. the frequency of the sound,

.....

2. the speed of the sound.

.....
[1]

(c) (i) Describe

1. how sound is produced by the loudspeaker,

.....

.....

.....[1]

2. how sound is transmitted through the air.

.....

.....

.....[2]

(ii) A pump is connected to the tube at the bottom of the glass bell-jar. The pump is switched on and it removes the air from the glass bell-jar.

Explain why the volume of the sound heard outside the glass bell-jar decreases.

.....

.....

.....[1]

5	**(a)**	number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time		B1	
	(b) **(i)**	(λ =)c / f **or** 330 / 2200		C1	
		0.15 m		A1	
	(ii)	**1**	no change **and**		
		2	increases	B1	
	(c) **(i)**	**1**	loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules)	B1	
		2	compressions **and** rarefactions / molecules vibrate / longitudinal wave vibration / oscillation / energy passed on	B1 B1	
	(ii)	fewer / no molecules / particles **and** less / no energy / vibration transferred		B1	[8]

QUESTION · 5054_w16_qp_12.pdf · source page 9 of 16

25 Which statement about waves is correct?

A All electromagnetic waves are transverse waves.

B All transverse waves travel at 300 000 000 m / s.

C Longitudinal waves cannot be refracted.

D Water waves are longitudinal waves.

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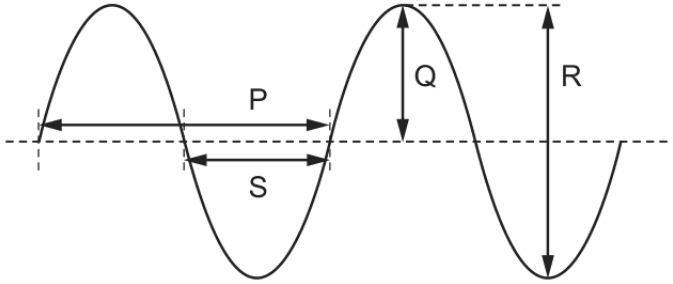
25	**A**

QUESTION · 5054_w16_qp_12.pdf · source page 10 of 16

26 The diagram shows the surface of water as a wave passes across a ripple tank.



Which lengths represent the amplitude and the wavelength of the wave?

	amplitude	wavelength
A	Q	P
B	Q	S
C	R	P
D	R	S

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 26

26 **A**

QUESTION · 5054_s17_qp_11.pdf · source page 11 of 16

26 Which waves are longitudinal?

A sound waves in water

B ultra-violet waves in air

C waves on the surface of water

D X-rays in a vacuum

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 2 of 3 · row for Question 26

26	A	**1**

10 **(a)** Fig. 10.1 shows two rays from an object that is placed in front of a plane mirror.

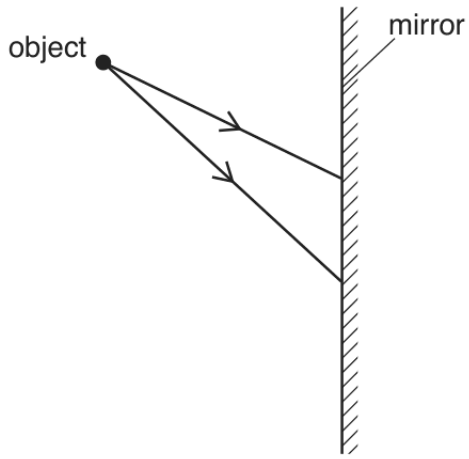


Fig. 10.1

(i) On Fig. 10.1, draw the two reflected rays and locate the position of the image. [2]

(ii) Apart from its position, state one characteristic of the image.

.....[1]

(b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

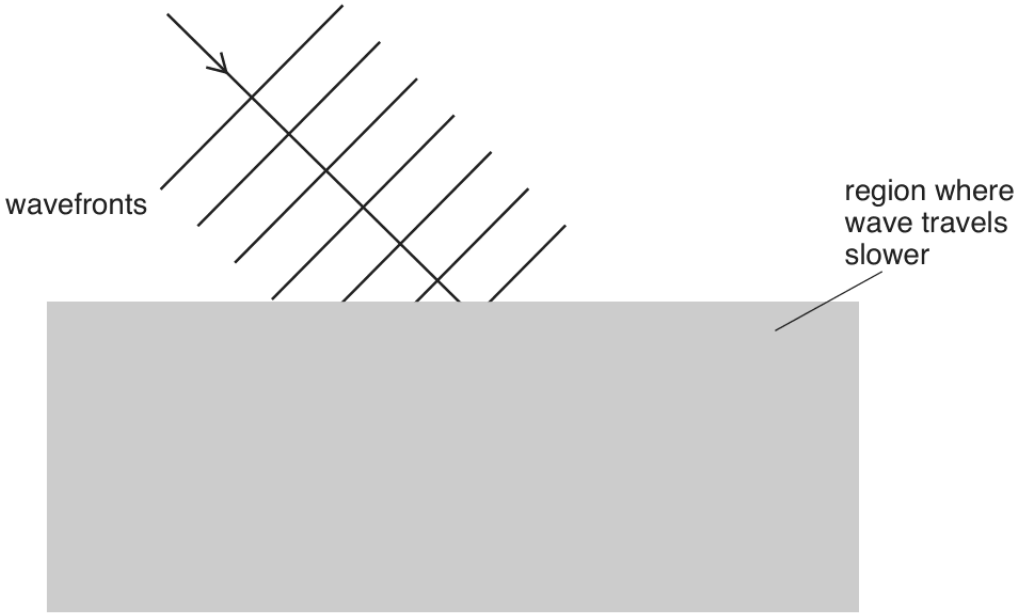


Fig. 10.2

(i) State what is meant by *wavefront*.

.....

.....[2]

(ii) Both reflection and refraction occur at the boundary.

1. On Fig. 10.2, draw the reflected wavefronts. [1]

2. On Fig. 10.2, draw the refracted wavefronts. [2]

(c) A loudspeaker produces a sound of frequency 2.0 kHz. The wavelength of this sound in air is 16 cm.

(i) Calculate the speed of sound in air.

speed = ...[2]

(ii) **1.** State the range of frequencies that can be heard by a healthy human ear.

.....[1]

2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength = ...[1]

(iii) Describe a simple experiment to show that sound waves obey the law of reflection. You may draw a diagram if you wish.

.....

.....

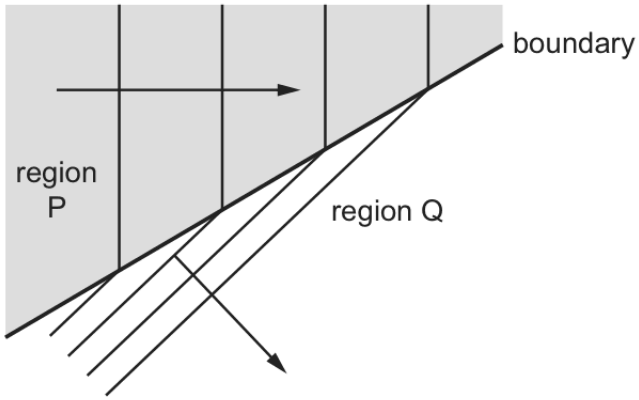
.....

.....[3]

Question	Answer	Marks
10(a)(i)	both reflected rays correct by eye	**B1**
	image in correct position shown by continuation of rays behind mirror	**B1**
10(a)(ii)	virtual **or** upright **or** laterally inverted **or** same size (as object)	**B1**
10(b)(i)	line (joining points)	**C1**
	line joining points / particles on wave with same phase **or** line joining points along a crest etc.	**A1**
10(b)(ii)1	correct angle to surface ± 7° with correct orientation **and** similar wavelength (by eye)	**B1**
10(b)(ii)2	at least two lines at smaller angle to surface with correct orientation in slower medium	**B1**
	showing smaller and constant wavelength with wavefronts deviated in correct direction	**B1**
10(c)(i)	(v =) fλ **or** 2(000) × 16	**C1**
	32 000 cm / s **or** 320 m / s	**A1**
10(c)(ii)1	a range with 15–25 Hz as the lowest frequency **and** 15–30 kHz as the highest	**B1**
10(c)(ii)2	1.6 cm (if highest frequency is 20 000 Hz)	**B1**
10(c)(iii)	tube or other method to produce (narrow) beam of sound on source and/or on detector	**B1**
	stated detector **or** stated reflector	**B1**
	detector moved to find maximum loudness **or** angles *i* and *r* measured with suitable experiment	**B1**

QUESTION · 5054_s17_qp_12.pdf · source page 12 of 20

26 The diagram shows a water wave in a ripple tank being refracted at the boundary between region P and region Q.



How does the wave in region P compare with the wave in region Q?

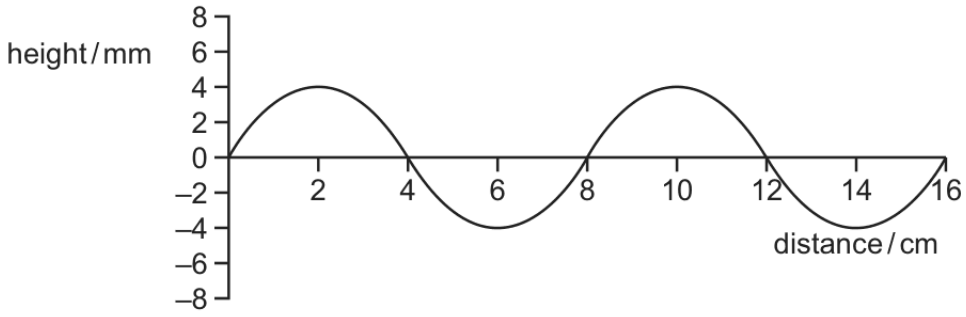
	wavelength of wave in region P	speed of wave in region P
A	larger	faster
B	larger	slower
C	smaller	faster
D	smaller	slower

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 26

26	A	**1**

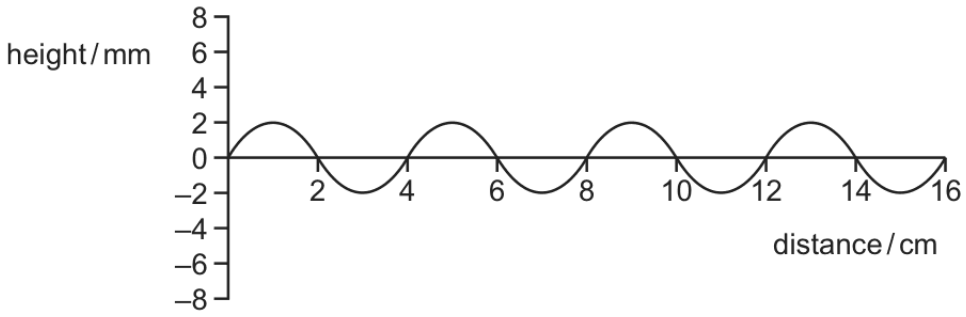
QUESTION · 5054_s17_qp_12.pdf · source page 13 of 20

27 The graph shows how the height of a water wave varies with distance along the wave.

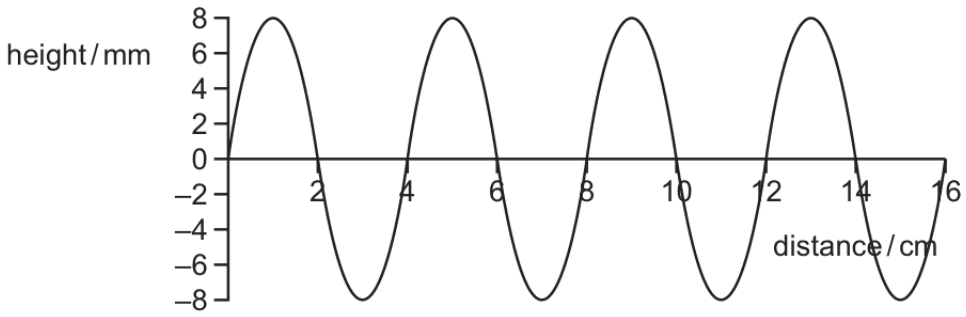


Which graph shows a wave with twice the amplitude, half the frequency, and the same speed?

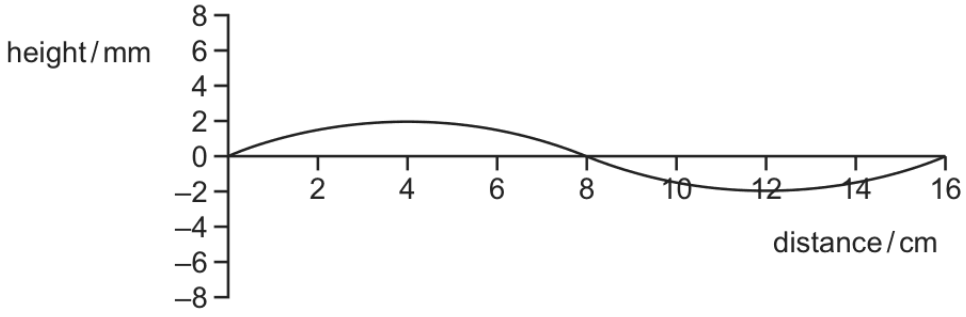
A



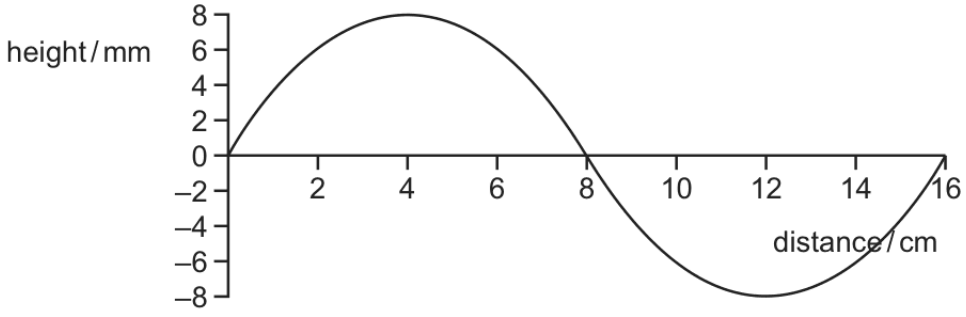
B



C



D



MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 27

27	D	**1**

6 **(a)** Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

(b) A sound wave has a frequency of 3.8 kHz and a speed of 330 m/s.

(i) Calculate the wavelength of the sound.

wavelength = ...[2]

(ii) A different sound has a frequency of 3.8 Hz.

State and explain whether a human with normal hearing is able to hear this sound.

.....

.....

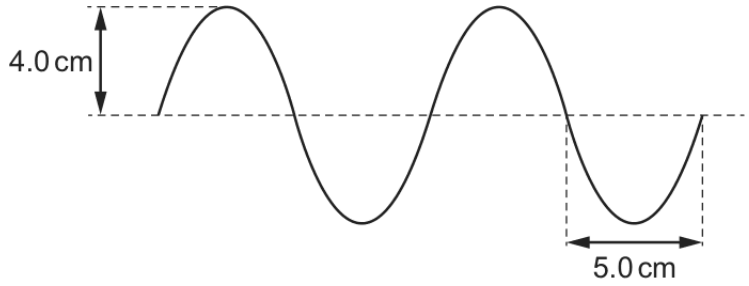
.....[1]

Question	Answer	Marks
6(a)	longitudinal - vibration / oscillation / movement to and fro **and** in direction of wave **or** has compressions and rarefactions	**B1**
	transverse – vibration / oscillation / movement up and down **and** at right angles to wave **or** has crests and troughs	**B1**
6(b)(i)	(λ =) v / f or 330 / 3800	**B1**
	0.087 m **or** 8.7 cm	**B1**
6(b)(ii)	not heard **and** because below the range of audible frequencies **or** audible range is 20 – 20 000 Hz **or** too low a pitch / frequency	**B1**

QUESTION · 5054_w17_qp_11.pdf · source page 9 of 20

18 A wave is travelling in water in a glass-sided tank.

The diagram shows a side view of the wave at one instant.



What are the amplitude and the wavelength of the wave?

	amplitude / cm	wavelength / cm
A	4.0	5.0
B	4.0	10
C	8.0	5.0
D	8.0	10

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 18

18	B	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 9 of 20

19 An earthquake wave travels through the solid surface of the Earth from east to west. The solid surface vibrates in a north-south direction.

How can the earthquake wave be described?

A electromagnetic

B longitudinal

C sound

D transverse

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 19

19	D	**1**

6 Fig. 6.1 shows a girl standing a few hundred metres in front of a large building.

Fig. 6.1 (not to scale)

The girl uses a signal generator and a loudspeaker to send a short pulse of sound towards the building. The sound has a frequency of 3700 Hz.

A short time later, the girl hears an echo.

(a) State what is meant by the term *echo*.

.....

.....[1]

(b) The pitch of the echo is the same as that of the original sound but the echo is not as loud.

State what has happened to

(i) the amplitude of the sound wave,

.....[1]

(ii) the frequency of the sound wave.

.....[1]

(c) The speed of sound in air is 330 m/s.

Calculate the wavelength of this sound.

wavelength = ...[2]

Question	Answer	Marks
6(a)	reflection of sound	**B1**
6(b)(i)	decreases	**B1**
6(b)(ii)	does not change	**B1**
6(c)	(λ =) c / f **or** 330 / 3700	**C1**
	0.089 m **or** 8.9 cm **or** 89 mm	**A1**

11 A water wave is produced in a ripple tank.

(a) (i) Draw a diagram of a ripple tank and describe how it is used to produce and observe the wave.

.....

.....

.....

.....

.....

.....

.....[4]

(ii) The water wave in the ripple tank transfers energy.

Describe how to show that there is a transfer of energy.

.....

.....

.....

.....[2]

(iii) State what is meant by a *wavefront*.

.....

.....

.....[2]

(b) Fig. 11.1 shows the variation with time *t* of the height of water at the same place in the tank for two waves P and Q. The height is sometimes above and sometimes below the mean level which is marked as 0.

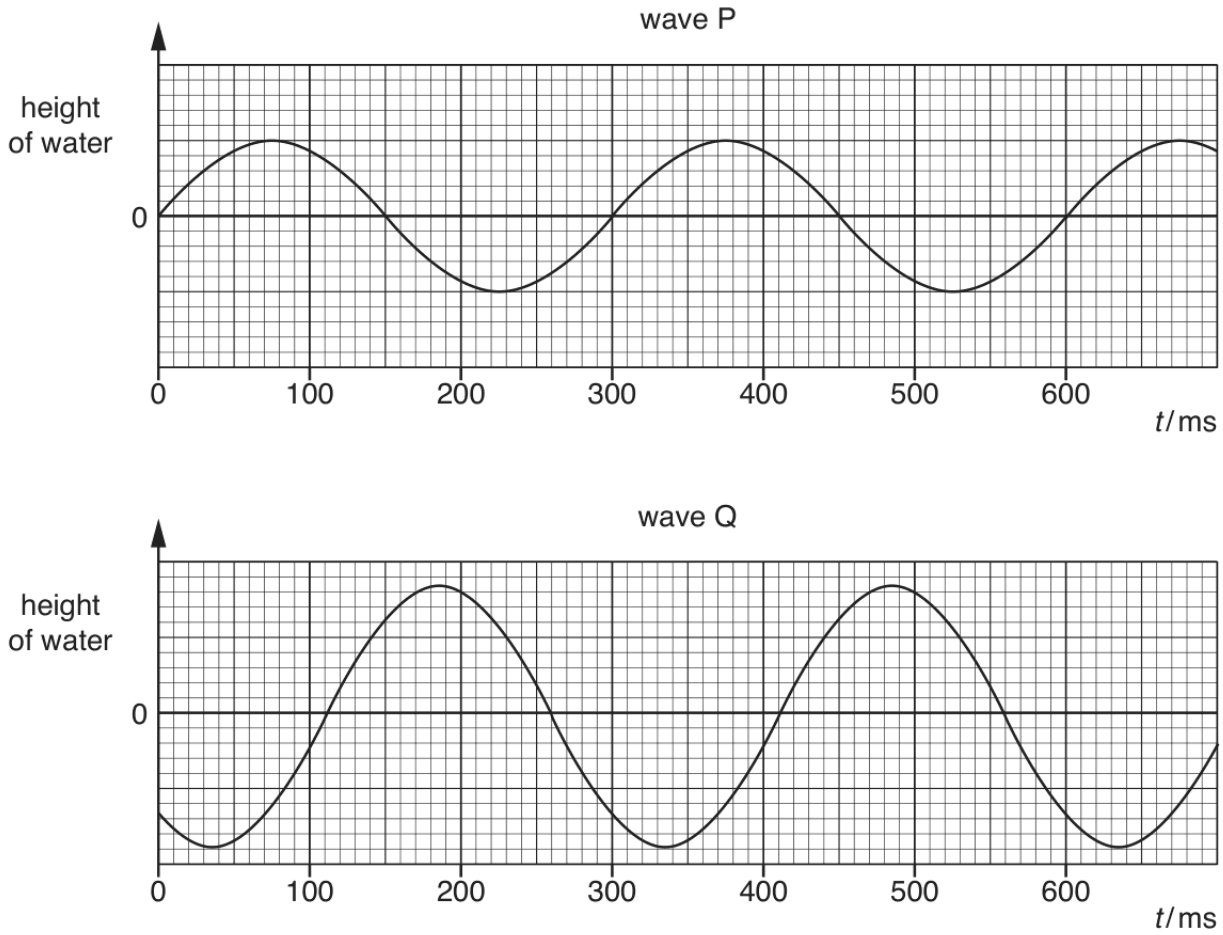


Fig. 11.1

(i) State one difference and one similarity of the two waves.

difference .....

.....

similarity .....

.....

[2]

(ii) The speed of the water wave is 0.20 m / s.

Determine

1. the frequency of wave P,

frequency = ...[3]

2. the wavelength of wave P.

wavelength = ...[2]

Question	Answer	Marks
11(a)(i)	tank with water	**B1**
	dipper (bar or point) touches or in surface of water	**B1**
	light source **/** stroboscope / video	**B1**
	dipper moves up and down (to create wave)	**B1**
11(a)(ii)	observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object	**B1**
	moves up and down	**B1**
11(a)(iii)	line joining points	**M1**
	that are at the same point in the motion	**A1**
11(b)(i)	*difference*: Q has a greater / different amplitude or phase	**B1**
	similarity: same frequency; same time for one wave (period); both transverse; oscillation up and down	**B1**
11(b)(ii)	time 300 (ms) or 0.3 (s)	**C1**
	(f=) 1 / t numerical or algebraic in any form	**C1**
	3.3 Hz	**A1**
11(b)(ii)	2 (λ =) v / f **or** v = dt – numerical or algebraic in any form	**C1**
	0.060 m	**A1**

QUESTION · 5054_s18_qp_12.pdf · source page 10 of 16

26 A communications system uses microwaves with a wavelength of 15 cm. The speed of light is 3.0×10^8 m/s.

What is the frequency of the microwaves?

A 2.0×10^7 Hz **B** 4.5×10^7 Hz **C** 2.0×10^9 Hz **D** 4.5×10^9 Hz

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 26

26	C	**1**

5 A microwave signal from a transmitter on the Earth's surface is sent up to a satellite in orbit and is then immediately transmitted back to Earth.

(a) A pulse of microwaves from the transmitter is received back at the transmitter 0.24 s later.

The speed of microwaves is 3.0×10^8 m / s.

Calculate the distance from the transmitter on the Earth to the satellite.

distance = ... [2]

(b) The microwave signal is used in a telephone system that produces sound.

Waves are described as longitudinal or transverse. Some waves are also electromagnetic.

(i) Place ticks (✓) in the table in Fig. 5.1 to show whether microwaves and sound waves are longitudinal or transverse and to show whether they are electromagnetic.

	longitudinal	transverse	electromagnetic
microwaves			
sound			

[2]

Fig. 5.1

(ii) A longitudinal wave contains a compression.

Describe what is meant by a *compression*.

.....

.....

.....[1]

Question	Answer	Marks
5(a)	(d=) s × t in any form algebraic or numerical,	**C1**
	3.6×10^7 m	**A1**
5(b)(i)	1st two columns correct (sound longitudinal and microwaves transverse)	**B1**
	3rd column correct microwaves electromagnetic	**B1**
5(b)(ii)	Layers / molecules / particles / spring coils close together **or** high(er) pressure (than atmospheric)	**B1**

QUESTION · 5054_w18_qp_11.pdf · source page 7 of 16

18 Which term is the number of crests of a wave passing a point per second?

A amplitude

B frequency

C speed

D wavelength

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 18

18	B	**1**

9 Some waves are longitudinal and some waves are transverse.

(a) State how a longitudinal wave differs from a transverse wave.

.....

..... [1]

(b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

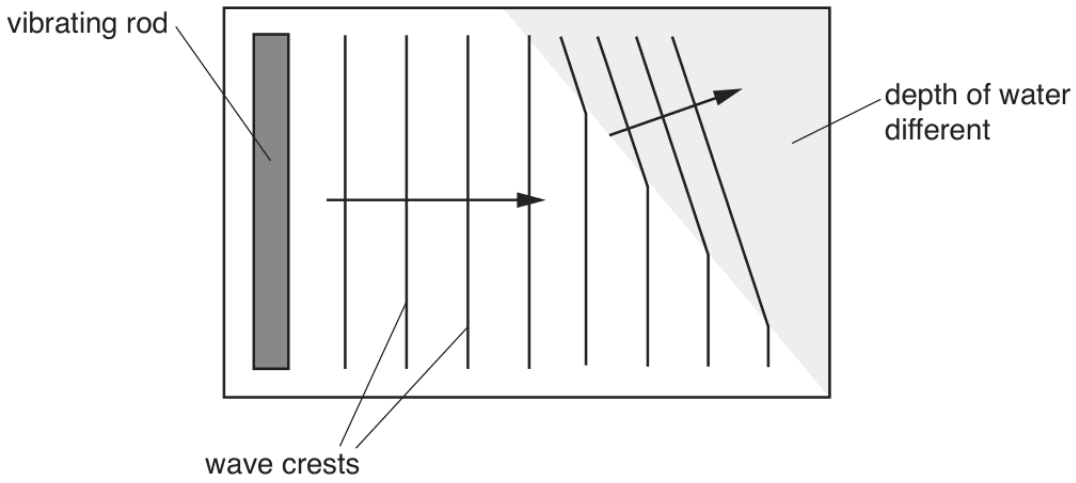


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

(i) In the left-hand section of the tank, the wavelength of the wave is 0.019 m and it travels at 0.17 m/s.

Calculate the frequency of the wave.

frequency = ... [2]

(ii) State what happens to the frequency of the wave, as it passes into the right-hand section of the tank.

..... [1]

(iii) Using Fig. 9.1, state and explain what happens to the speed of the wave as it passes into the right-hand section of the tank.

.....

.....

..... [2]

(c) Fig. 9.2 shows light, in air, striking the vertical side of a rectangular glass block at an angle of incidence of 60°.

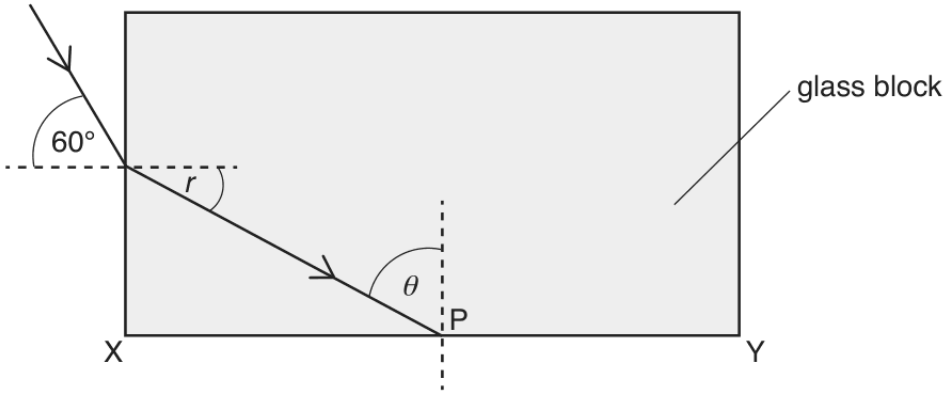


Fig. 9.2

The refractive index of the glass is 1.6. The light travels in the glass and strikes side XY at P.

(i) Underline all the terms that describe a light wave.

electromagnetic **longitudinal** **transverse**

[1]

(ii) At the point where the light enters the glass, the angle of refraction is *r*.

Calculate angle *r*.

r = ... [2]

(iii) **1.** Calculate the critical angle *c* for light travelling in the block.

c = ... [2]

2. At P, the angle θ between the ray and the normal is given by $\theta = 90° - r$.

State and explain what happens to the light when it strikes side XY.

.....

.....

..... [2]

3. On Fig. 9.2, draw the path of the light after it strikes side XY at P and the path of the light when it is again travelling in the air. [2]

Question	Answer	Marks
9(a)	the vibration direction is parallel to the wave / energy travel direction **or** longitudinal waves have compressions and rarefactions **or** transverse waves have crests and troughs **or** longitudinal waves cannot be polarised	**B1**
9(b)(i)	(f =) v / λ **or** 0.17 / 0.019	**C1**
	8.9 Hz	**A1**
9(b)(ii)	(frequency) stays constant **or** does not change	**B1**
9(b)(iii)	(speed) decreases	**B1**
	(Fig. 9.1 shows that) wavelength decreases **or** refraction is towards normal **or** $i > r$ **or** top of wave (in shaded region) lags behind bottom of wave	**B1**
9(c)(i)	electromagnetic **and** transverse underlined	**B1**
9(c)(ii)	$n = \sin(i) / \sin(r)$ **or** (r =) $\sin^{-1}(\sin(i) / n)$ **or** (r =) $\sin^{-1}(\sin(60°) / 1.6)$	**C1**
	33(°)	**A1**

Question	Answer	Marks
9(c)(iii)	1 $n = 1 / \sin(c)$ **or** (c =) $\sin^{-1}(1 / n)$ **or** $\sin^{-1}(1 / 1.6)$	**C1**
	39°	**A1**
	2 total internal reflection	**B1**
	$\theta > c$ **or** 57(°) > 39(°) **or** angle of incidence greater than the critical angle	**B1**
	3 reflected ray at P (correct angle by eye)	**B1**
	light in air at 30° (by eye) to vertical surface	**B1**

QUESTION · 5054_w18_qp_12.pdf · source page 10 of 16

22 Which statement about a water wave is correct?

A The amplitude is the vertical distance between a trough and a peak.

B The frequency is the number of troughs passing a point in one second added to the number of peaks passing a point in one second.

C The speed is the horizontal distance travelled per second by a peak.

D The wavelength is the horizontal distance between a trough and a peak.

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 22

22	C	**1**

9 Both sound and ultrasound are waves that travel in solids, liquids and gases.

(a) State how sound differs from ultrasound.

.....

..... [1]

(b) Sound and ultrasound are longitudinal waves that consist of compressions and rarefactions.

(i) Explain what is meant by a wave and describe how a longitudinal wave differs from a transverse wave.

.....

.....

.....

.....

..... [3]

(ii) Fig. 9.1 represents a longitudinal wave of frequency 25 000 Hz travelling in oil.

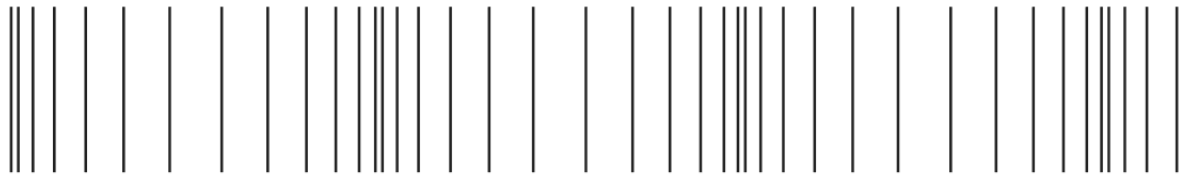


Fig. 9.1 (full scale)

1. On Fig. 9.1, mark two points at the centre of two different rarefactions and label each one R. [1]

2. On Fig. 9.1, draw a double-headed arrow to indicate a distance that is equal to one wavelength of the wave. [1]

3. Measure the length of the arrow drawn on Fig. 9.1 and use it to determine the speed of the wave in oil.

speed = ... [2]

(c) A sound wave travelling in a liquid, passes into air.

(i) State what happens to the speed of the sound wave as it enters the air.

..... [1]

(ii) The wave in the liquid travels towards the surface at an angle. Fig. 9.2 shows the centres of the compressions of the sound wave in the liquid.

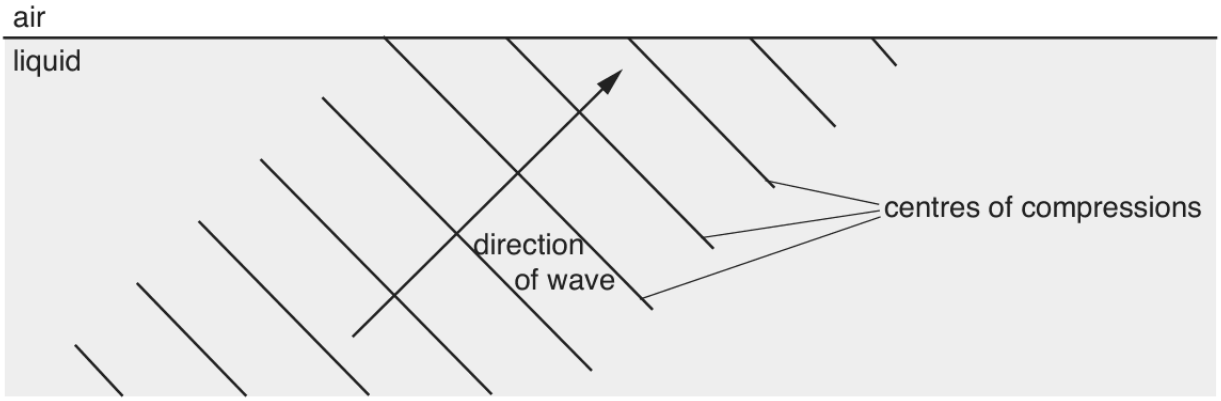


Fig. 9.2

Some compressions shown have reached the liquid-air boundary. The parts of these compressions in the air are not shown on Fig. 9.2.

On Fig. 9.2, draw the parts of these compressions that are in the air. [3]

(d) Describe, in outline, how ultrasound is used in cleaning.

.....

.....

.....

.....

..... [3]

MARK SCHEME · 5054_w18_ms_22.pdf · source page 8 of 10 · Question 9

Question	Answer	Marks
9(a)	frequency of sound wave small(er) **or** its frequency is less than 20 000 Hz	**B1**
9(b)(i)	transmission of energy	**B1**
	(through a medium) with no net movement of medium **or** by vibrating particles	**B1**
	vibrations parallel (and antiparallel) to wave / energy travel direction **or** cannot be polarised	**B1**
9(b)(ii)	1 two centres of rarefactions labelled *R*	**B1**
	2 distance from one point to adjacent identical point indicated (with double-headed arrow)	**B1**
	3. (v =) $f\lambda$ **or** 25 000 × 0.047 / 0.048 / 0.049 **or** 25 000 × 4.7 / 4.8 / 4.9 **or** 25 000 × 47 / 48 / 49	**C1**
	1200 m / s	**A1**
9(c)(i)	decreases	**B1**
9(c)(ii)	four / five straight lines in air that touch the compressions still in the liquid **and** no intermediate / extra lines between the correct lines	**B1**
	at least four compressions in air parallel to each other	**B1**
	at least four straight lines at shallower angle from horizontal **and** slope correct	**B1**

MARK SCHEME · 5054_w18_ms_22.pdf · source page 9 of 10 · Question 9

Question	Answer	Marks
9(d)	object (to be cleaned) immersed in liquid / solvent	**B1**
	object / liquid agitated / vibrated by ultrasound	**B1**
	dirt (particles) shaken off **or** dislodges / removes dirt	**B1**

QUESTION · 5054_s19_qp_11.pdf · source page 10 of 20

23 Which waves are longitudinal?

A gamma rays

B light waves

C ultrasound waves

D X-rays

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 23

23	C	**1**

10 In Fig. 10.1 the vertical lines represent the crests of a wave in a ripple tank.

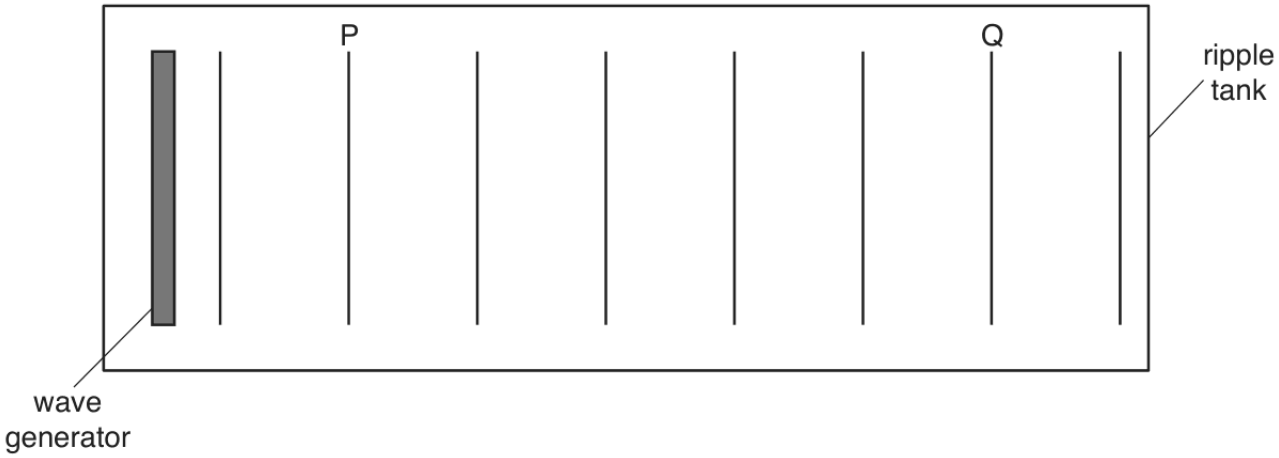


Fig. 10.1

Fig. 10.1 is drawn to scale, with 1.0 cm on the diagram representing 4.0 cm in the tank.

A crest takes 2.0 s to travel from P to Q.

(a) (i) Determine the wavelength of the wave in the ripple tank.

wavelength = ... [2]

(ii) Define the term *frequency*.

.....

..... [2]

(iii) Determine the frequency of the wave.

frequency = ... [1]

(iv) The wave generator is a bar of wood.

Describe how it is used to produce the wave shown.

.....

.....

..... [2]

(v) Some water in the ripple tank is removed. The water becomes less deep. No other change is made.

State how the change in depth affects the wavelength and the speed of the wave.

speed .....

wavelength .....
[1]

(b) A student reads in a textbook that the speed of sound in air is about 300 m / s.

He plans an experiment to check this value.

He asks another student to stand a long distance away in a large open space.

One student has a starting pistol. The pistol, when fired, produces a loud sound and a puff of smoke at the same instant.

(i) Other pieces of apparatus are needed for this experiment in order to measure the speed of sound.

Name two other pieces of apparatus that are needed.

1. .....

2. .....
[2]

(ii) Describe how all the measurements are made.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(iii) State typical values for the speed of sound in:

1. a liquid

.....

2. a solid.

.....

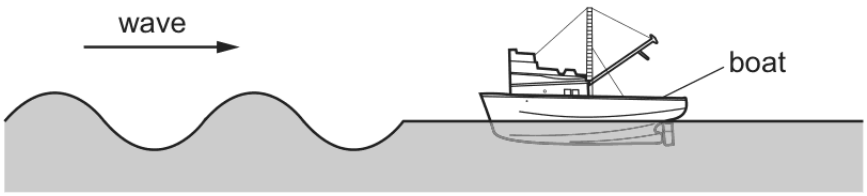
[2]

[Total: 15]

Question	Answer	Marks
10(a)(i)	1.7 (cm) seen or any multiple of wavelength measured on diagram	**C1**
	6.4–7.2 cm	**A1**
10(a)(ii)	number of waves/cycles/oscillations in one second / unit time	**C1**
	number of wavelengths/crests/troughs/ wave cycles/wave oscillations generated/made/pass a point in one second/unit time **or** number of oscillations of a source/particle in one second / unit time	**A1**
10(a)(iii)	2.5 Hz	**B1**
10(a)(iv)	moved up and down	**B1**
	regularly **or** at constant frequency / 2.5 times a second	**B1**
10(a)(v)	*speed* smaller / decreases **and** *wavelength* smaller / decreases	**B1**
10(b)(i)	tape measure **or** trundle wheel **or** microphones	**B1**
	stopwatch **or** timer	**B1**
10(b)(ii)	start timer on seeing the smoke from gun / sound picked up by microphone	**B1**
	stop timer on hearing sound	**B1**
	measure (large) distance between students	**B1**
10(b)(iii)	both between and including 1000 and 10 000 m/s	**B1**
	solid speed > liquid speed	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 9 of 16

25 A wave on water travels towards a stationary boat.



As the wave passes, the boat starts to move up and down and gains kinetic energy from the wave.

Which property of the wave changes as the wave passes the boat?

A amplitude

B frequency

C speed

D wavelength

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 25

25	A	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 9 of 16

26 Which waves are longitudinal?

A gamma rays

B light waves

C ultrasound waves

D X-rays

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 26

26	C	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 9 of 16

22 Which description defines the amplitude of a water wave in a ripple tank?

A half of the vertical distance between a trough and a peak

B the horizontal distance between adjacent peaks

C the horizontal distance between a trough and an adjacent peak

D the vertical distance between a trough and a peak

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 22

22	A	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 10 of 16

23 Here are four statements about wavefronts.

1 Wavefronts from a point source are straight lines.

2 Wavefronts from a point source are circular.

3 Wavefronts are parallel to the direction of energy movement.

4 Wavefronts are perpendicular to the direction of energy movement.

Which two statements are correct?

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 23

23	D	**1**

5 Both sound and ultrasound are longitudinal waves.

(a) Describe what is meant by *longitudinal*.

.....

.....

..... [2]

(b) Fig. 5.1 shows an ultrasound cleaner being used to clean jewellery.

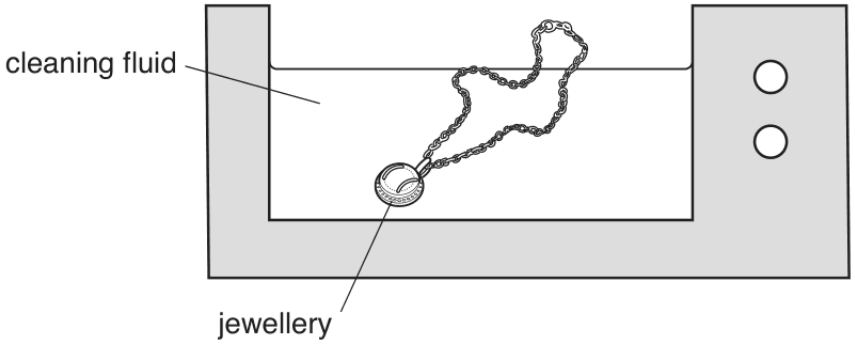


Fig. 5.1

The jewellery is lowered into the cleaning fluid and ultrasound waves of frequency 42 000 Hz are produced in the fluid.

(i) The speed of ultrasound in the fluid is 1500 m / s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = ... [2]

(ii) Describe how the ultrasound cleans the jewellery.

.....

.....

..... [2]

[Total: 6]

MARK SCHEME · 5054_w19_ms_22.pdf · source page 5 of 10 · Question 5

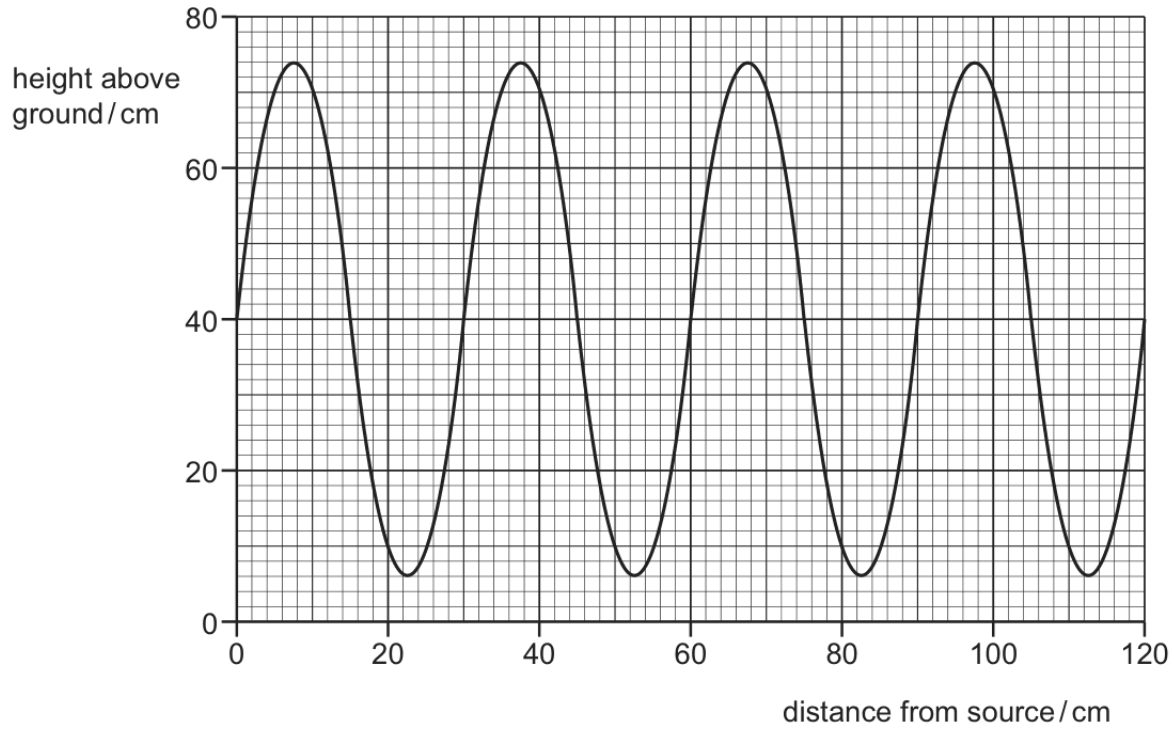
Question	Answer	Marks
5(a)	vibration / oscillation of <u>particles</u> **or** energy transfer **or** compressions and rarefactions	**B1**
	vibration / oscillation (of particles) parallel to energy travel direction / propagation direction	**B1**

MARK SCHEME · 5054_w19_ms_22.pdf · source page 6 of 10 · Question 5

Question	Answer	Marks
5(b)(i)	(λ =) v / f **or** 1500 / 42 000	**C1**
	0.036 m **or** 3.6 cm	**A1**
5(b)(ii)	(ultrasound) vibrates (the cleaning fluid / jewellery)	**B1**
	dirt shaken off	**B1**

QUESTION · 5054_s20_qp_12.pdf · source page 10 of 16

29 The diagram shows how the height above the ground of a rope transmitting a wave varies with distance from the source of vibrations at one instant in time.



What is the amplitude of the wave shown?

A 30 cm **B** 34 cm **C** 72 cm **D** 74 cm

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 3 of 3 · row for Question 29

29	B	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 10 of 16

30 A vibrator produces 12 wavelengths on the surface of water in 10 s. The spacing between the first crest and the third crest is 60 cm.

What is the speed of the wave?

A 24 cm / s **B** 25 cm / s **C** 36 cm / s **D** 72 cm / s

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 3 of 3 · row for Question 30

30	C	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 10 of 16

26 An electromagnetic wave has a speed of 3.0×10^8 m/s and a wavelength of 10 cm.

What is the frequency of the wave?

A 3.3×10^{-10} Hz

B 3.3×10^{-8} Hz

C 3.0×10^{7} Hz

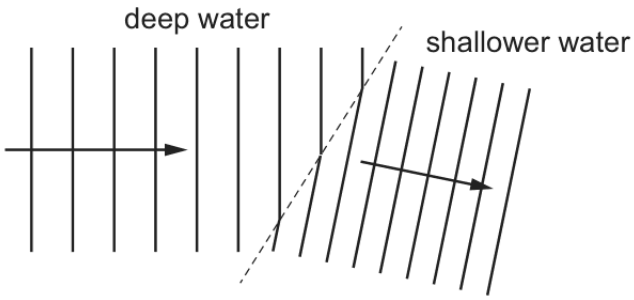
D 3.0×10^{9} Hz

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 26

26	D	**1**

QUESTION · 5054_s21_qp_11.pdf · source page 8 of 16

20 A water wave in a ripple tank refracts as it moves from deep water into shallower water.



What happens to the speed and to the frequency of the wave as it moves into shallower water?

	speed	frequency
A	decreases	stays constant
B	increases	stays constant
C	stays constant	decreases
D	stays constant	increases

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 20

20	A	**1**

4 **(a)** Fig. 4.1 shows a wave on a rope and Fig. 4.2 shows a wave on a spring. Both waves are moving in the direction shown by the arrows.

Fig. 4.1

Fig. 4.2

(i) State the name of each of the two types of wave shown.

Fig. 4.1 .....

Fig. 4.2 .....
[1]

(ii) On Fig. 4.1, draw a wave which has a smaller frequency than the wave shown. [1]

(iii) Describe the movement of point X on the spring in Fig. 4.2 as the wave passes.

.....

..... [1]

(b) Table 4.1 shows some sentences that a student writes about waves.

His teacher places a ring around each mistake.

Write a suitable correction for each mistake in Table 4.1. One has been done for you.

Table 4.1

sentence	correction
Sound travels at 3.0×10^8 m/s in air.	330
Sound with a higher pitch has a larger amplitude.	
X-rays are used for pre-natal scanning.	
In the electromagnetic spectrum, microwaves have the highest frequency.	

[3]

[Total: 6]

Question	Answer	Marks
4(a)(i)	Fig.4.1 transverse **and** Fig.4.2 longitudinal	**B1**
4(a)(ii)	wave drawn with larger wavelength	**B1**
4(a)(iii)	backwards and forwards, left and right, to and fro, vibration parallel to arrow / direction of wave (movement)	**B1**
4(b)	frequency	**B1**
	ultrasound	**B1**
	gamma	**B1**

10 **(a)** Fig. 10.1 shows a water wave moving from shallow into deep water. The wavefronts shown represent the crests of the wave.

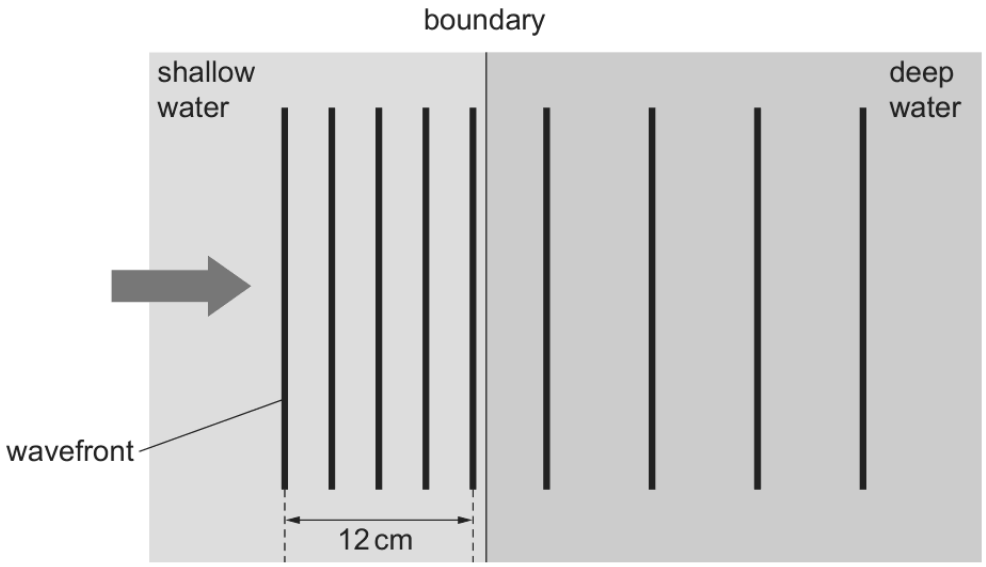


Fig. 10.1 (not to scale)

The water wave is made by dipping a wooden bar up and down in the water. The bar makes 10 complete up and down movements in 5.0 s.

(i) Determine the wavelength of the wave in the shallow water.

wavelength = ... [1]

(ii) Calculate the frequency of the wave produced.

frequency = ... [2]

(iii) Determine the speed of the water wave in the shallow water.

speed = ... [2]

(iv) On Fig. 10.2, draw lines to show what happens to the wavelength, the frequency and the speed of the water wave as it enters the deep water.

One line has been drawn for you.

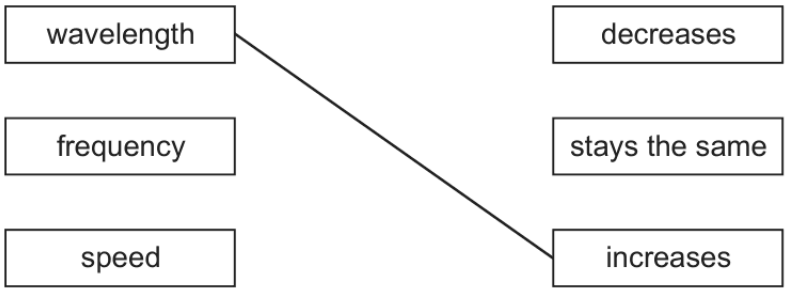


Fig. 10.2

[1]

(v) The direction of the boundary is now altered. The water wave now enters the deep water from the shallow water at an angle, as shown in Fig. 10.3.

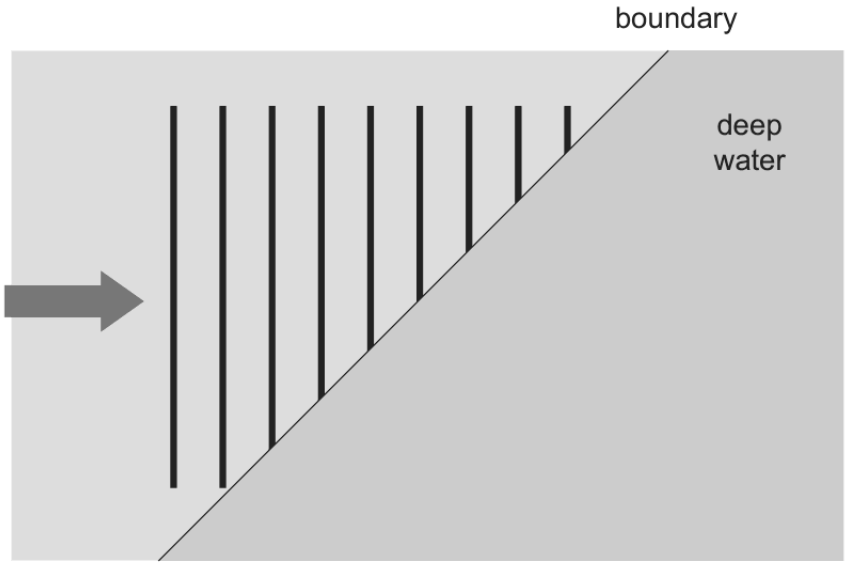


Fig. 10.3

On Fig. 10.3, draw the wavefronts in the deep water. [3]

(b) White light is made up of different colours.

(i) State the name of four of the colours in the visible spectrum and place them in order from the smallest wavelength to the largest wavelength.

smallest wavelength .....

.....

.....

largest wavelength .....

[2]

(ii) A narrow beam of white light can be split into different colours.

Fig. 10.4 shows rays of white light emitted from a lamp.

Complete Fig. 10.4 to show how a narrow beam is produced from these rays and how a spectrum is shown on the screen. Label your diagram.

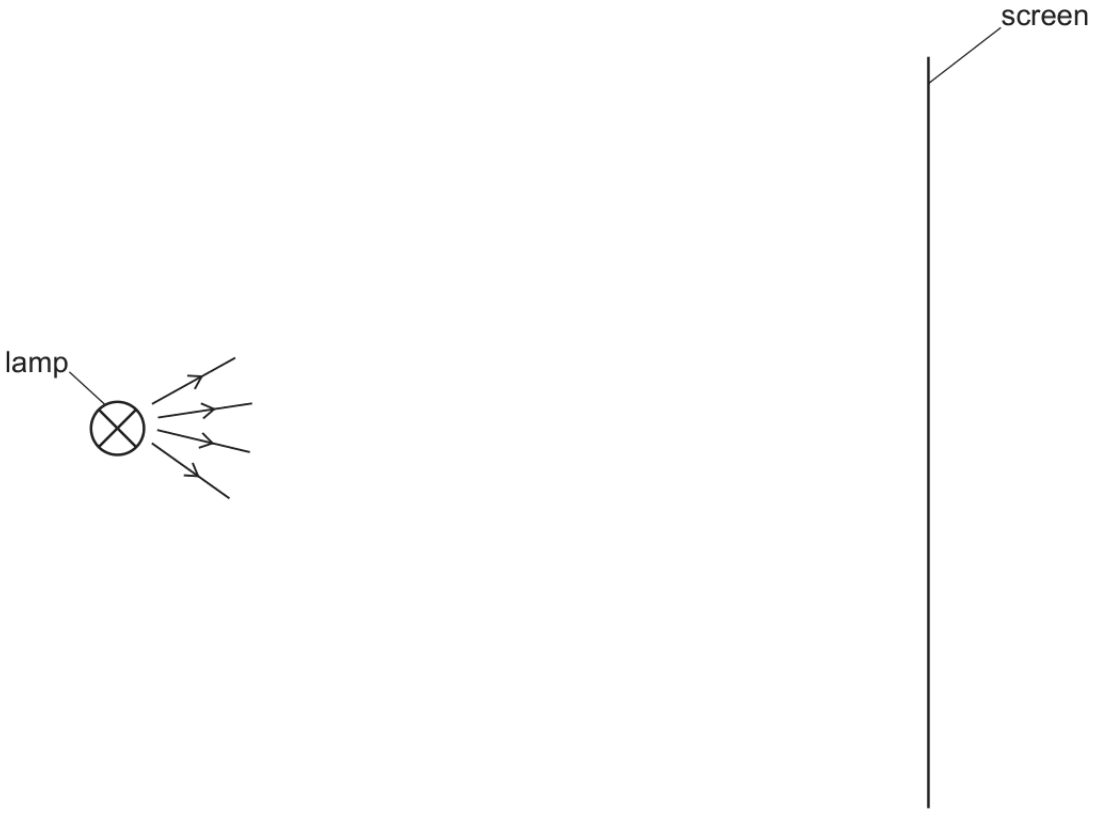


Fig. 10.4

[4]

[Total: 15]

MARK SCHEME · 5054_s21_ms_22.pdf · source page 14 of 16 · Question 10

Question	Answer	Marks
10(a)(i)	3 (.0) cm	**B1**
10(a)(ii)	number of waves per second **or** 5 / 10 **or** $f = 1 / T$	**C1**
	2(.0) Hz	**A1**
10(a)(iii)	($v =$) $f \lambda$ **or** s = d / t numerical or algebraic in any form	**C1**
	6(.0) cm / s	**A1**
10(a)(iv)	frequency stays the same **and** speed increases	**B1**

MARK SCHEME · 5054_s21_ms_22.pdf · source page 15 of 16 · Question 10

Question	Answer	Marks
10(a)(v)	⩾ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in-between	**B1**
	constant wavelength shown between all refracted wavefronts	**B1**
	all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal	**B1**
10(b)(i)	four colours from violet, indigo, blue, green, yellow, orange, red	**B1**
	correct order of four of the above colours	**B1**
10(b)(ii)	prism	**B1**
	slit (to produce narrow beam)	**B1**
	dispersion shown at first or second face, i.e. a ray becoming at least two rays at least one face	**B1**
	correct refraction on first and second faces for all rays shown	**B1**

QUESTION · 5054_w21_qp_11.pdf · source page 8 of 20

17 A water wave on a pond passes into a region where the wave travels more slowly. It then returns to a region where it travels at the original speed. Assume that the frequency of the wave stays constant.

Which side-view of the wave is correct?

A

B

C

D

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 17

17	A	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 8 of 20

18 What is meant by the *amplitude* of a wave?

A the distance between the extreme points of a particle's motion

B the maximum distance a particle moves from its equilibrium position

C the maximum energy carried by the wave

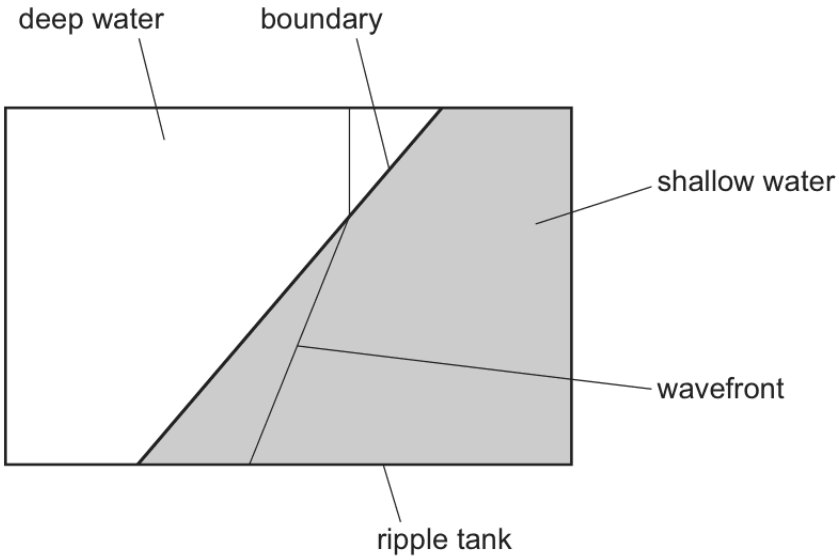
D the maximum power of the vibrations carrying the wave

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 18

18	B	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 9 of 20

19 The diagram shows one wavefront of a wave as it travels from deep water to shallow water in a ripple tank.



What happens as the wavefront moves into the shallow water?

A The speed of the wavefront increases.

B The speed of the wavefront decreases.

C The wavelength of the wave remains constant.

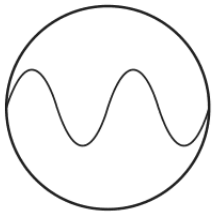
D The wavelength of the wave increases.

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 19

19	B	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 17 of 20

36 A microphone detects a musical sound and the signal is fed into an oscilloscope. The diagram shows the trace which appears on the screen.



The spot of light on the oscilloscope screen takes 1.2 ms to travel across the screen.

What is the frequency of the musical sound?

A 170 Hz **B** 330 Hz **C** 1700 Hz **D** 3300 Hz

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 3 of 3 · row for Question 36

36	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 A loudspeaker is made from a coil of wire fixed to a cardboard tube. The tube is attached to a cardboard cone.

Fig. 8.1 shows part of the arrangement of the loudspeaker.

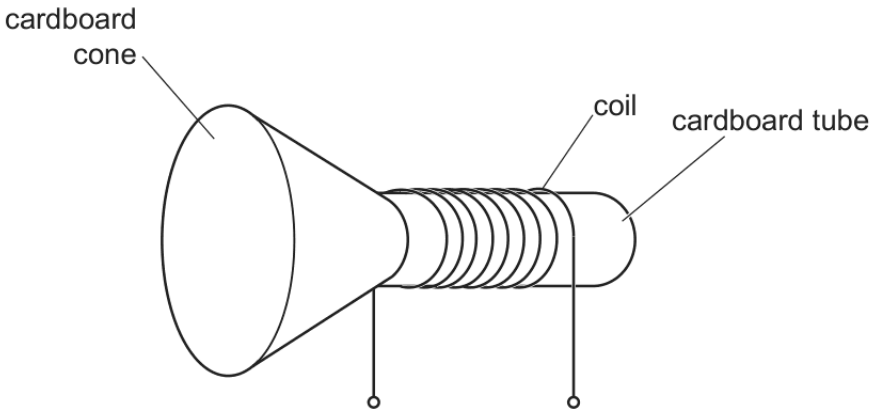


Fig. 8.1

When there is a current in the coil, the coil experiences a force.

(a) State what else is needed in a loudspeaker to make a current-carrying wire experience a force.

..... [1]

(b) A student connects the coil to the output of an alternating current (a.c.) generator. Fig. 8.2 shows how the electromotive force (e.m.f.) produced by the generator varies with time.

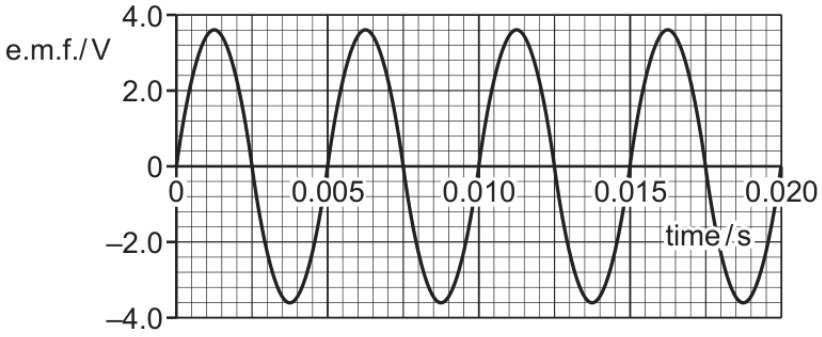


Fig. 8.2

The coil, tube and cone vibrate backwards and forwards.

(i) Explain why the e.m.f. shown in Fig. 8.2 makes the coil vibrate.

.....

.....

..... [2]

(ii) The vibrating cone produces sound in the surrounding air.

Explain, in terms of molecules, how the cone produces a sound wave that travels through the air.

.....

.....

.....

.....

..... [3]

(iii) Sound is a longitudinal wave.

Describe the difference between a longitudinal wave and a transverse wave.

.....

.....

.....

..... [2]

(iv) Using Fig. 8.2, determine the number of times that the cone reverses its direction of motion in 1.0 s.

number = ... [2]

(v) The speed of sound in air is 340 m / s.

Calculate the wavelength of the sound.

wavelength = ... [2]

(c) The student adjusts the generator so that the maximum voltage of the output is now 3.0 V. Everything else stays the same as the output shown on the graph in Fig. 8.2.

(i) Explain any effect on the loudness of the sound.

.....

..... [2]

(ii) Explain any effect on the pitch of the sound.

.....

..... [1]

[Total: 15]

Question 9 starts over the page

MARK SCHEME · 5054_w21_ms_21.pdf · source page 8 of 11 · Question 8

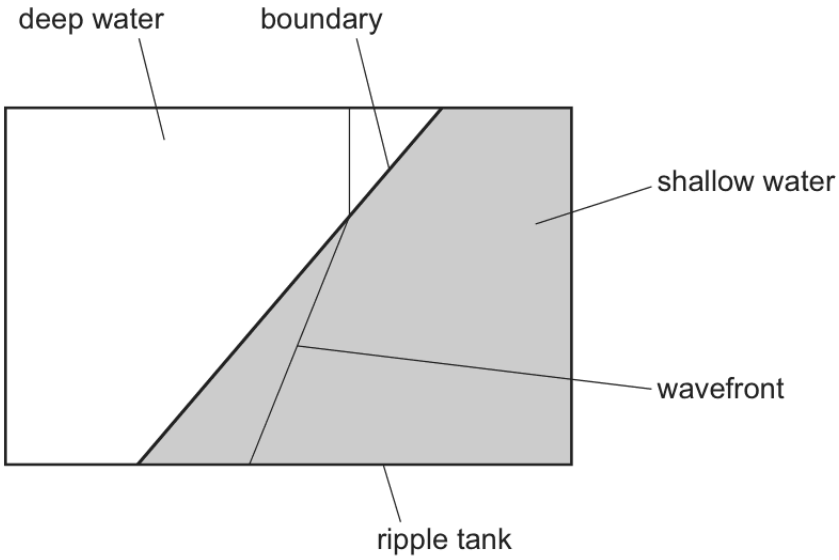
Question	Answer	Marks
8(a)(i)	magnetic field **or** magnet	**B1**
8(b)(i)	the current (in coil continually) reverses (direction) **or** the force (on the coil continually) reverses	**B1**
	the current (in the coil continually) reverses (direction) **and** the force (on the coil continually) reverses	**B1**

MARK SCHEME · 5054_w21_ms_21.pdf · source page 9 of 11 · Question 8

Question	Answer	Marks
8(b)(ii)	cone moves forwards **and** air is compressed / hits (air) molecules	**B1**
	cone moves backwards **and** rarefaction produced / molecules sucked back	**B1**
	(air) molecules vibrate **or** production of compressions and rarefactions continues	**B1**
8(b)(iii)	vibration direction is parallel to energy travel / propagation direction	**B1**
	in <u>transverse</u> wave, vibration direction is perpendicular to propagation / energy travel direction	**B1**
8(b)(iv)	(time period =) 0.0050 (s) **or** 1 / 0.0050 **or** 200 (times per second)	**C1**
	400 (times per second)	**A1**
8(b)(v)	(λ =) c / v **or** 340 / 200	**C1**
	1.7 m	**A1**
8(c)(i)	decreases / less loud / quieter	**B1**
	amplitude of sound / vibration decreases **or** (maximum) force on coil decreases	**B1**
8(c)(ii)	(pitch) stays the same / unchanged **and** no change to frequency / time period	**B1**

QUESTION · 5054_w21_qp_12.pdf · source page 9 of 16

21 The diagram shows one wavefront of a wave as it travels from deep water to shallow water in a ripple tank.



What happens as the wavefront moves into the shallow water?

A The speed of the wavefront increases.

B The speed of the wavefront decreases.

C The wavelength of the wave remains constant.

D The wavelength of the wave increases.

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 21

21	B	**1**

QUESTION · 5054_w21_qp_12.pdf · source page 9 of 16

22 Which example describes a transverse wave being refracted?

A infrared waves travelling in an optical fibre

B light waves from an object entering a magnifying glass

C ultrasound waves being used to clean jewellery

D water waves hitting a barrier

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 22

22	B	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 8 of 16

19 A wave in the sea collides with a cliff.

A crest of the wave hits the cliff once every 6.0 s.

The horizontal distance between a crest and the adjacent trough of the wave is 4.5 m.

What is the speed of the wave?

A 0.67 m/s **B** 0.75 m/s **C** 1.3 m/s **D** 1.5 m/s

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 19

19	D	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 8 of 16

20 A sound wave refracts as it passes from air to water.

Which quantities change?

A speed and frequency only

B speed and wavelength only

C wavelength and frequency only

D speed, frequency and wavelength

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 20

20	B	**1**

8 **(a)** Fig. 8.1 shows a ripple tank and the crests of the water wave that is produced in it.

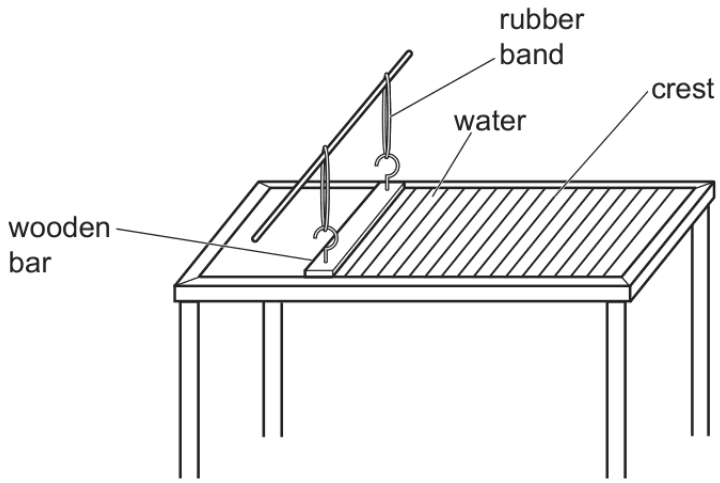


Fig. 8.1

The frequency of the water wave is 2.0 Hz and its amplitude is 3.0 mm.

(i) Calculate the number of crests produced in 1.5 s.

number = ... [1]

(ii) The height of the wave is measured from the level of the undisturbed surface of the water. The height of the wave at one point is 0 at time = 0.

On Fig. 8.2, draw a graph to show how the height of the wave at this point varies with time.

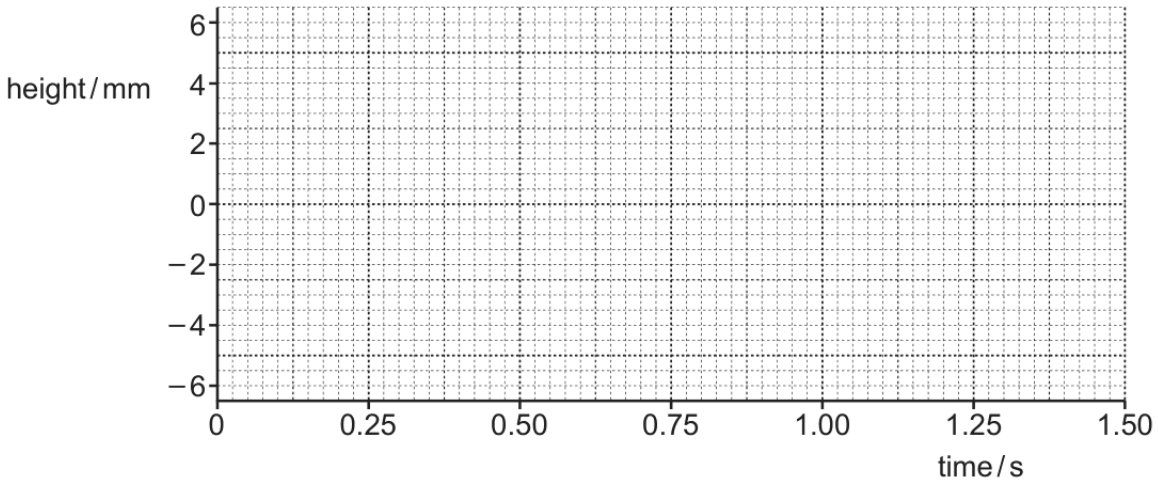


Fig. 8.2

[2]

(b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....

..... [1]

(ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed .....

wavelength .....
[2]

(c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

(i) State the additional apparatus needed to demonstrate refraction.

..... [1]

(ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.

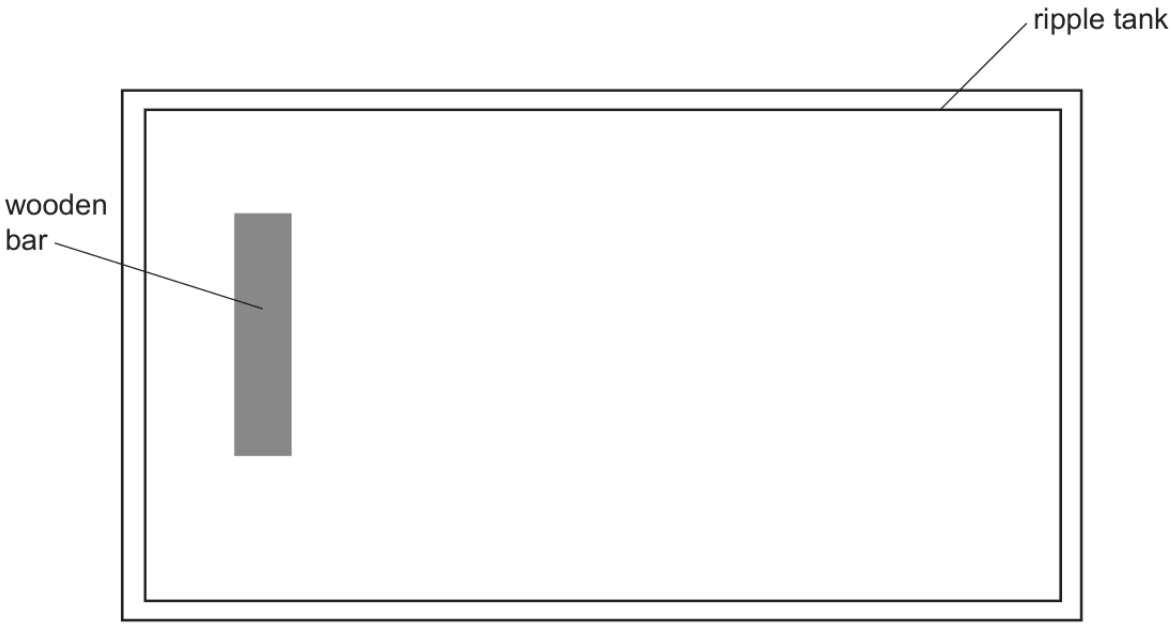


Fig. 8.3

[3]

(d) Fig. 8.4 shows a connection to the internet made from a remote station A using a satellite above the Earth's surface.

Data is sent between stations A and B using microwaves which travel to and from the satellite.

Station B is connected directly to the internet using optical fibre.

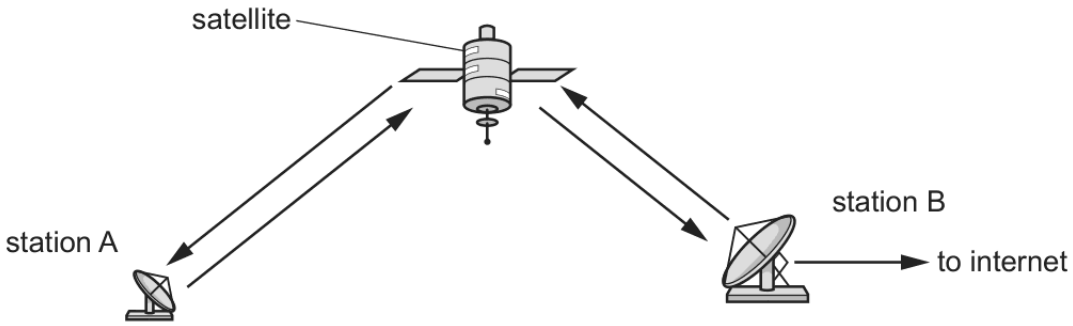


Fig. 8.4 (not to scale)

The speed of microwaves is 3.0×10^8 m/s.

Stations A and B are each 560 km from the satellite.

(i) Calculate the time taken for data to travel from A to B using microwaves.

time = ... [3]

(ii) The same data can be sent from A to B in a shorter time along the surface of the Earth using infrared waves to carry the information in optical fibres.

State **one** property that infrared waves and microwaves have in common.

.....

..... [1]

(iii) Suggest **one** other advantage of using optical fibres to connect stations A and B directly.

.....

..... [1]

[Total: 15]

Question 9 begins over the page.

Question	Answer	Marks
8(a)(i)	3	**B1**
8(a)(ii)	shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change)	**B1**
	1 wave taking 0.5 s for first wave shown within half a square **or** 1.5 s shown for the number of waves in **(a)(i)**	**B1**
8(b)(i)	move bar up and down more often (per second) / faster **or** make rubber band(s) strong(er) / short(er) / tight(er) / more in number **or** use bar of less mass / weight	**B1**
8(b)(ii)	*speed* remains the same	**B1**
	wavelength decreases	**B1**
8(c)(i)	(piece of) plastic / (piece of) glass / block / plate **or** a shallow(er) / deep(er) area / boundary	**B1**
8(c)(ii)	diagram showing within the ripple tank: • crests parallel to bar hitting boundary with a boundary labelled	**B1**
	• correct refraction of plane crests either towards or away from normal on the same side of the normal **or** no direction change if crests are initially parallel to boundary but with a wavelength change	**B1**
	• different wavelength shown before and after refraction and constant before and after (by eye)	**B1**
8(d)(i)	(t =) distance / speed algebraic or numerical seen in any form	**C1**
	2 × 560(000) seen or used **or** 0.0019 (s)	**C1**
	0.0037 s	**A1**

Question	Answer	Marks
8(d)(ii)	One of (both): • reflect / refract • have same speed (in a vacuum) • travel in a vacuum / don't need a medium • are electromagnetic • are transverse • carry energy / cause heating • have larger wavelength / smaller frequency than visible light	**B1**
8(d)(iii)	less likely to be hacked / intercepted / greater security **or** more channels / better bandwidth possible / more data **or** less likely for transmission to be interrupted **or** cost of satellites saved / satellites expensive	**B1**

QUESTION · 5054_w22_qp_11.pdf · source page 9 of 16

23 Which statement is correct?

A Both longitudinal waves and transverse waves can travel at the speed of light.

B Both longitudinal waves and transverse waves of similar wavelengths must have similar frequencies.

C Both longitudinal waves and transverse waves can travel in a vacuum.

D Both longitudinal waves and transverse waves can travel in water.

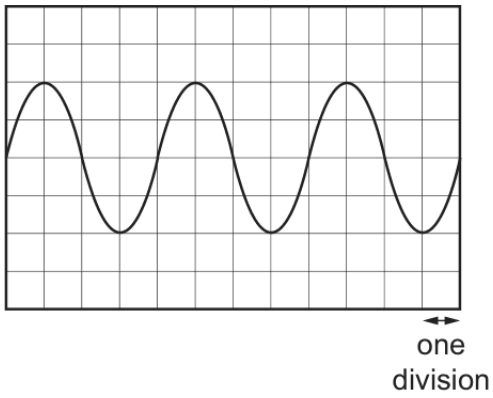
MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 23

23	D	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 14 of 16

35 The sound wave produced by a note is displayed on the screen of a cathode-ray oscilloscope.

The time-base is set at 5 ms for one division.



What is the frequency of the note?

A 20 Hz **B** 50 Hz **C** 60 Hz **D** 150 Hz

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 3 of 3 · row for Question 35

35	B	**1**

4 X-rays are transverse waves that are part of a spectrum of waves.

(a) State the name of the spectrum of waves that includes X-rays.

..... [1]

(b) **(i)** Explain what is meant by 'frequency'.

.....

.....

..... [2]

(ii) State the name of the waves in this spectrum that have the greatest frequency.

..... [1]

(c) X-rays are used in hospitals to produce an image of broken bones.

(i) Explain how this is done. You may draw a diagram if you wish.

.....

.....

.....

..... [3]

(ii) Explain why X-rays are **not** used for pre-natal scanning.

.....

..... [1]

[Total: 8]

Question	Answer	Marks
4(a)	electromagnetic (waves) **or** components of electromagnetic spectrum	**B1**
4(b)(i)	number of oscillations / cycles / wavelengths	**B1**
	per unit time	**B1**
4(b)(ii)	gamma rays	**B1**
4(c)(i)	X-rays directed towards body part with (suspected) broken bone	**B1**
	X-rays pass through flesh but not (to same extent) through bones	**B1**
	image produced by digital / electronic / CCD / CMOS detector	**B1**
4(c)(ii)	X-rays (may) cause cancer / cell mutation / birth defect / miscarriage **or** X-rays show the bone structure rather than foetal development **or** image would not be clear	**B1**

4 A sound wave, travelling in air or water, contains compressions and rarefactions.

(a) Describe an experiment to show that a medium is required to transmit sound waves. You may draw a labelled diagram if you wish.

.....

.....

.....

.....

..... [3]

(b) When a sound wave passes through air or water, the molecules of the air or water move.

Describe the motion of the molecules due to the sound wave.

.....

.....

..... [2]

(c) Describe what is meant by ‘a rarefaction’.

.....

..... [1]

(d) Fig. 4.1 shows a dolphin.

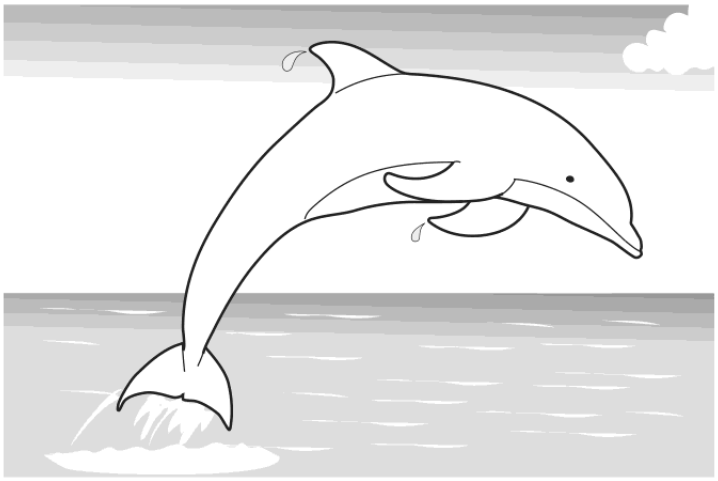


Fig. 4.1

The range of audible sound frequencies for the dolphin is 75 Hz to 140 000 Hz.

In water, the speed of sound is 1500 m / s.

Calculate the smallest wavelength of sound in water that the dolphin can hear.

wavelength = ... [2]

[Total: 8]

MARK SCHEME · 5054_w22_ms_22.pdf · source page 6 of 11 · Question 4

Question	Answer			Marks
4(a)	equipment	e.g.	(evacuable) container **and** (suspended electric) bell / source of sound **and** (air) pump	**B1**
	action	e.g.	ringing / sound (produced) / sound heard **and** evacuate air / vacuum (produced)	**B1**
	observation	e.g.	transmission / hearing of sound ceases / no sound	**B1**

MARK SCHEME · 5054_w22_ms_22.pdf · source page 7 of 11 · Question 4

Question	Answer	Marks
4(b)	(they) vibrate	**M1**
	(vibration) parallel to / in the direction of propagation / of energy travel	**A1**
4(c)	(point where) molecules are further apart (than average) **or** (point where) pressure / density is lower (than average)	**B1**
4(d)	(λ =) v / f **or** 1500 / 140 000	**C1**
	1.1×10^{-2} m **or** 1.1 cm **or** 11 mm	**A1**

7 A water wave in a ripple tank diffracts as it passes through the gap in a barrier.

Fig. 7.1 shows a drawing made by a student of the crests in the pattern.

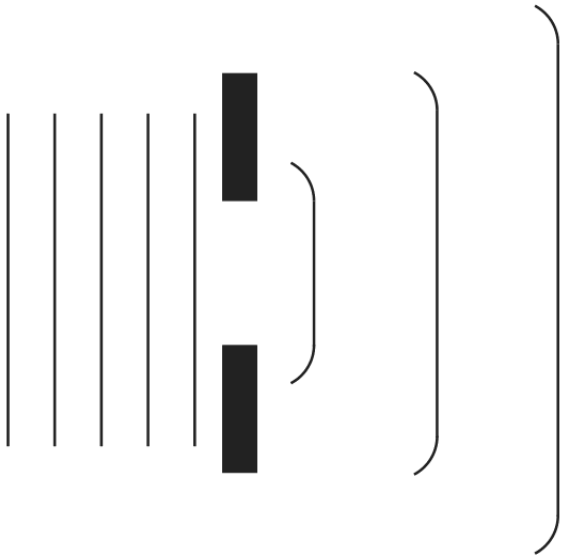


Fig. 7.1

(a) State **one** way in which the student's drawing of the crests is wrong.

.....

..... [1]

(b) The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.

Complete Fig. 7.2 with at least three crests to show the new diffraction pattern.

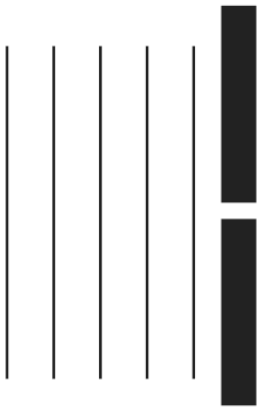


Fig. 7.2

[2]

(c) In one part of the ripple tank, the water wave has a speed of 6.0 cm/s and a wavelength of 2.0 cm.

The wave then passes into a shallower region of the tank.

The speed of the wave in the shallow region is 4.0 cm/s.

(i) Define the term 'wavelength'.

.....

..... [1]

(ii) Calculate the frequency of the wave.

frequency = ... Hz [2]

(iii) Calculate the wavelength of the wave in the shallow part of the tank.

wavelength = ... cm [1]

[Total: 7]

Question	Answer	Marks
7(a)	wavelength has altered / too large **or** crests in shadow should be bent / quarter circular above or below gap **or** vertical lines too straight on two crests	**B1**
7(b)	at least 3 semicircles centered on the gap	**B1**
	wavelength constant and the same on right and left of gap **and** some bending	**B1**
7(c)(i)	distance between two (consecutive) identical points such as two (consecutive) crests	**B1**
7(c)(ii)	$v = f\lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz)	**C1**
	3(.0) Hz)	**A1**
7(c)(iii)	1.3 (cm)	**B1**

5 Water waves are transverse waves.

(a) Underline **two** other examples of transverse waves.

seismic P-waves **seismic S-waves** **sound** **X-rays** [1]

(b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

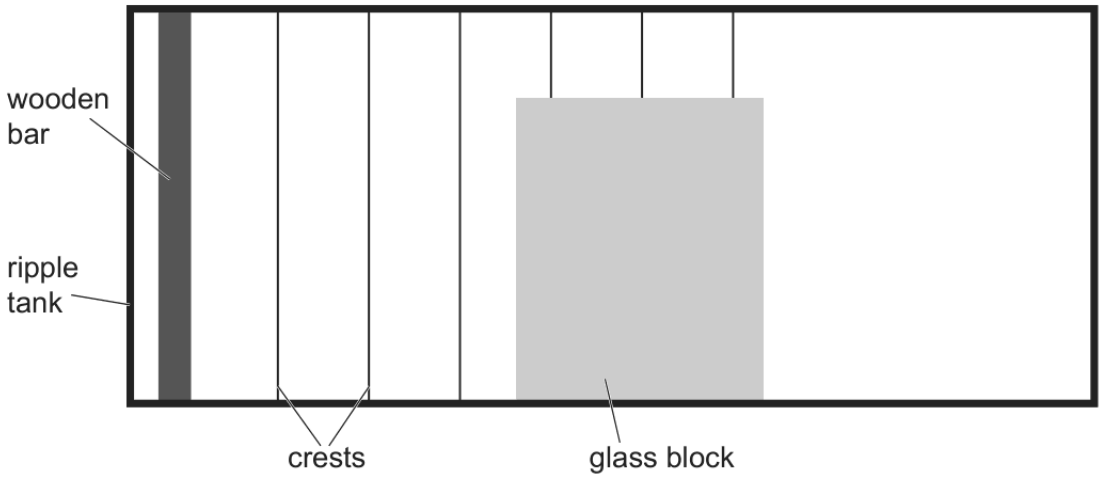


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

(i) The speed of the water wave is 27 cm / s.

Calculate the frequency and the wavelength of the wave.

frequency = ... Hz

wavelength = ... cm
[3]

(ii) The wave diffracts at the right-hand edge of the glass block.

On Fig. 5.1 draw **two** crests after they pass the glass block to show the diffraction. [2]

(iii) Describe how a wave with a smaller wavelength is made with the wooden bar.

.....

..... [1]

(iv) Describe how a decrease in wavelength affects the diffraction.

.....

..... [1]

[Total: 8]

MARK SCHEME · 5054_s23_ms_22.pdf · source page 8 of 12 · Question 5

Question	Answer	Marks
5(a)	seismic S-waves **and** X-rays	**B1**
5(b)(i)	(wavelength =) v / f in any form **or** (distance =) vt **or** $f = 1 / T$ or 1 / 0.15	**C1**
	frequency 6.7 (Hz)	**A1**
	wavelength 4.0 (cm)	**A1**
5(b)(ii)	straight wavefronts above block to the right with same wavelength	**B1**
	correct diffraction – wavefronts quarter circles with centre at top corner	**B1**
5(b)(iii)	increase frequency of movement of (wooden) bar	**B1**

MARK SCHEME · 5054_s23_ms_22.pdf · source page 9 of 12 · Question 5

Question	Answer	Marks
5(b)(iv)	less diffraction **or** less bending	**B1**

4 Red light of frequency 4.7×10^{14} Hz travels in air at a speed of 3.0×10^{8} m/s.

(a) Calculate the wavelength in air of this red light.

wavelength = ... m [3]

(b) Fig. 4.1 shows a narrow beam of this red light striking the surface of a parallel-sided glass block.

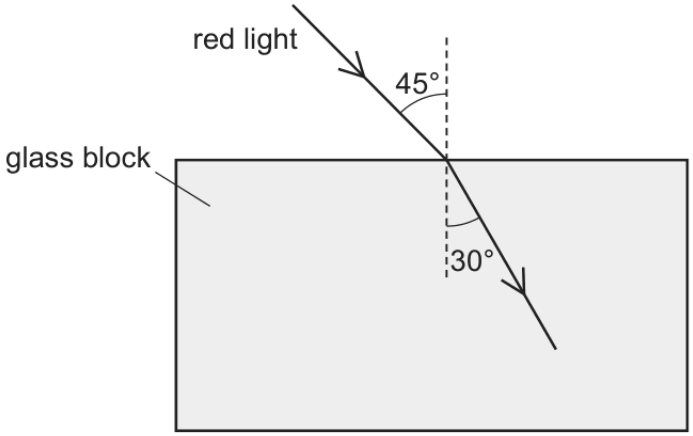


Fig. 4.1

(i) State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block.

speed .....

frequency .....

wavelength .....
[2]

(ii) Using the angles shown on Fig. 4.1, calculate the refractive index of the glass.

refractive index = ... [3]

(iii) The light continues along the path shown in Fig. 4.1 until it strikes the bottom surface of the block. Light then emerges into the air.

Draw on Fig. 4.1 to show the path taken by the light until it strikes the bottom surface and the path of the light that emerges into the air. [2]

[Total: 10]

Question	Answer	Marks
4(a)	($\lambda =$) v / f **or** $3.0 \times 10^8 / 4.7 \times 10^{14}$	**C1**
	6.4×10^{N} (m) or $3 / 4.7 \times 10^6$	**C1**
	6.4×10^{-7} (m)	**A1**
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	**B2**
4(b)(ii)	$i = 45°$ **and** $r = 30°$	**C1**
	($n =$) $\sin(i) / \sin(r)$ **or** $\sin(45°) / \sin(30°)$	**C1**
	1.4	**A1**
4(b)(iii)	straight continuation of ray in block **and** refraction away from normal in air	**B1**
	emergent ray parallel to initial incident ray	**B1**

5 Most of the energy emitted by the Sun is in three regions of the electromagnetic spectrum.

(a) The name of one of these three regions is the ultraviolet region.

(i) State the names of all three regions in order of increasing frequency.

smallest frequency .....

intermediate frequency .....

largest frequency .....

[2]

(ii) Some of the ultraviolet radiation emitted by the Sun reaches the surface of the Earth.

Give one possible damaging effect of ultraviolet radiation on the human body and the property of ultraviolet radiation that causes this damage.

damaging effect .....

.....

property .....

.....

[2]

(b) Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

.....

.....

..... [2]

[Total: 6]

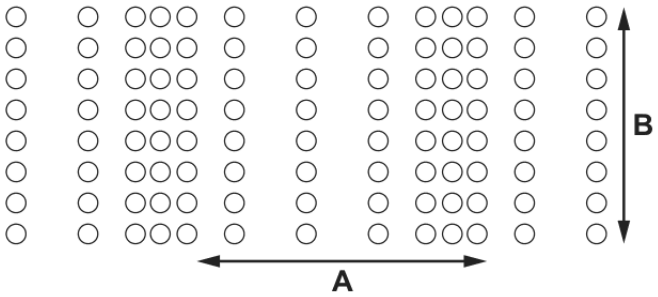
Question	Answer	Marks
5(a)(i)	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation)	**B1**
	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation) **and** in this order	**B1**
5(a)(ii)	damaging effect: (skin) cancer / cataracts	**B1**
	property: it / ultraviolet radiation is ionising	**B1**

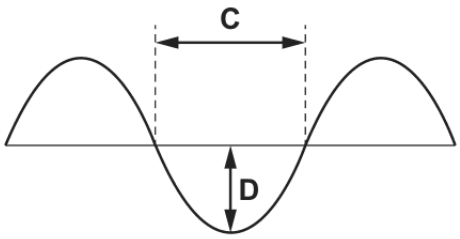
Question	Answer	Marks
5(b)	in a transverse wave, the vibration direction is perpendicular / at right angles to the propagation direction	**B1**
	in a longitudinal wave, the vibration direction is parallel to the propagation direction	**B1**

QUESTION · 5054_w23_qp_12.pdf · source page 10 of 20

24 The diagram shows two waveforms.

Which arrow shows the wavelength of a longitudinal wave?



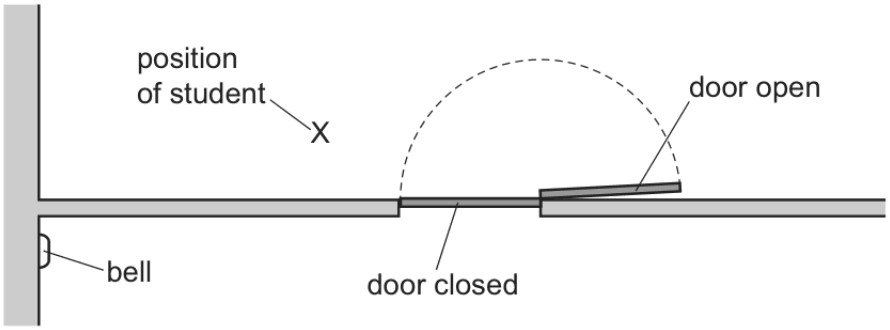


MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 24

24	A	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 11 of 20

25 A bell, on a wall outside a classroom, rings. A student is standing at point X shown on the view from above.



The door of the classroom is closed and the student in the room at X hears the bell ring.

The door is opened, and the sound heard by the student from the bell is louder.

What causes the sound reaching the student to be louder?

A diffraction

B dispersion

C reflection

D refraction

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 25

25	A	**1**

6 The loudspeaker shown in Fig. 6.1 produces a sound of frequency 800 Hz.

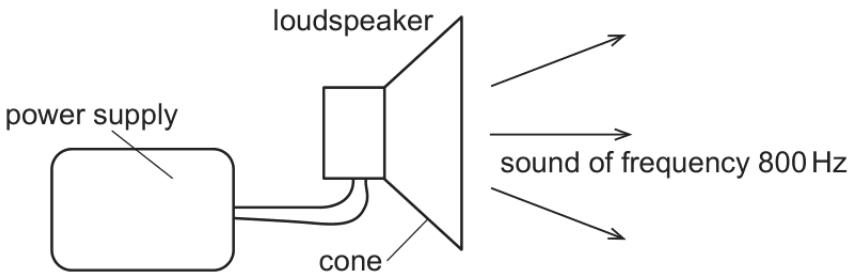


Fig. 6.1

(a) State what is meant by 'frequency'.

.....

..... [2]

(b) Explain whether the sound produced by the speaker is audible to a human with healthy hearing.

.....

..... [1]

(c) Describe the motion of the cone in Fig. 6.1 and explain how this causes the production of sound.

.....

.....

.....

.....

..... [3]

(d) The Sun is a very large and extremely violent ball of gas. Electromagnetic radiation from explosions on the surface of the Sun reaches the Earth.

Explain why sound from explosions on the surface of the Sun does **not** reach the Earth.

.....

.....

..... [2]

[Total: 8]

MARK SCHEME · 5054_w23_ms_22.pdf · source page 8 of 11 · Question 6

Question	Answer	Marks
6(a)	(frequency =) the number of wavelengths	**B1**
	(passing a point) per unit time	**B1**
6(b)	yes **and** within audible range (of human with healthy hearing)	**B1**

MARK SCHEME · 5054_w23_ms_22.pdf · source page 9 of 11 · Question 6

Question	Answer	Marks
6(c)	it / the loudspeaker / cone / coil is vibrating	**B1**
	when it / cone moves (forwards) it produces a compression / compresses (air) **or** when it moves (backwards) it produces a rarefaction / expands (air)	**B1**
	vibrations / compressions / rarefactions / energy / change in pressure / motion passed on (to neighbouring air)	**B1**
6(d)	space is a vacuum **or** no medium / air in space	**C1**
	sound cannot travel in a vacuum / space / without a medium	**A1**

QUESTION · 5054_s24_qp_11.pdf · source page 8 of 16

22 Which quantity is defined as the number of wavelengths passing a point per second?

A amplitude

B frequency

C speed

D wavelength

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 22

22	B	**1**

4 **(a)** Ultrasound of frequency 30 kHz has a wavelength of 0.011 m in air.

(i) State what is meant by 'frequency'.

.....

..... [1]

(ii) Calculate the speed of ultrasound in air.

Show your working.

speed = ... m/s [2]

(iii) State **one** use of ultrasound.

..... [1]

(b) Sound waves are longitudinal waves. Water waves are transverse waves.

(i) Describe, by referring to the movement of the particles in the wave, the difference between a longitudinal wave and a transverse wave.

You may include a labelled diagram to help your description.

.....

.....

.....

.....

..... [3]

(ii) State **one** source of waves that generates both longitudinal and transverse waves.

..... [1]

(c) Fig. 4.1 shows radio waves of long and short wavelength passing over the same hill.

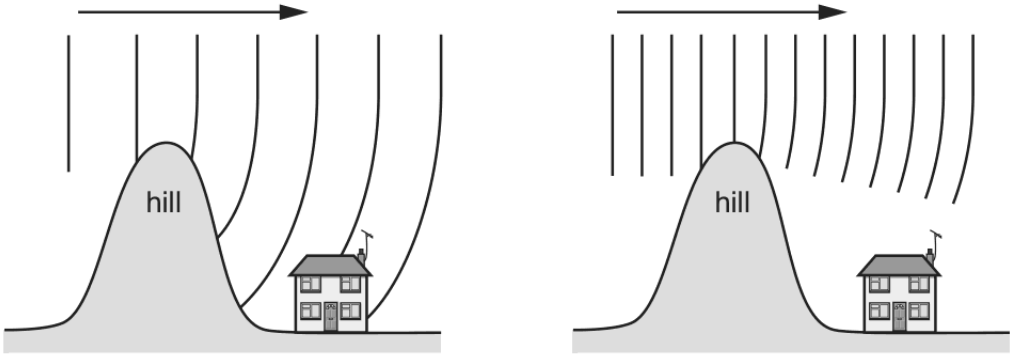


diagram A diagram B

Fig. 4.1 (not to scale)

Explain why the radio waves in diagram A reach the house but the radio waves in diagram B do not reach the house.

.....

.....

.....

..... [2]

[Total: 10]

Question	Answer	Marks
4(a)(i)	number of (complete) waves (passing a point) per second	**B1**
4(a)(ii)	(V =) $f\lambda$ in any form	**C1**
	330 m / s	**A1**
4(a)(iii)	ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance	**B1**
4(b)(i)	longitudinal wave - oscillation backwards and forwards / in direction of travel of wave	**B1**
	transverse wave - oscillation at right angles to (direction of travel of) wave	**B1**
	diagram of each type	**B1**
4(b)(ii)	earthquake (waves) or other correct source which is a single source that generates both types of wave, e.g. lightning	**B1**
4(c)	long wavelengths bend around hill **and** short wavelengths do not bend	**B1**
	(due to) diffraction	**B1**

QUESTION · 5054_w24_qp_11.pdf · source page 12 of 16

28 Which waves are longitudinal?

A sound waves in water

B ultraviolet waves in air

C waves on the surface of water

D X-rays in a vacuum

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 28

28	A	**1**

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5 Ultrasound is a longitudinal wave which cannot be heard by humans as the frequency of ultrasound is too high.

(a) Ultrasound waves **cannot** travel in a vacuum.

Explain why.

.....

.....

..... [2]

(b) Describe how a longitudinal wave differs from a transverse wave.

.....

.....

.....

..... [2]

(c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m / s.

(i) Calculate the wavelength of this ultrasound wave in soft human tissue.

wavelength = ... m [3]

(ii) The ultrasound wave passes from the soft tissue into bone where the speed of the ultrasound wave is greater than 1500 m / s.

State what happens to the frequency and to the wavelength of the ultrasound wave as it passes into the bone.

frequency .....

wavelength .....
[2]

[Total: 9]

Question	Answer	Marks
5(a)	ultrasound / sound / it is transmitted by vibrating matter / particles	**B1**
	no matter / particles in a vacuum / no medium	**B1**
5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	**B1**
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	**B1**
5(c)(i)	($\lambda =$) v / f **or** $1500 / 8.4 \times 10^{6}$	**C1**
	1.8×10^{N}	**C1**
	1.8×10^{-4} (m)	**A1**
5(c)(ii)	frequency: does not change	**B1**
	wavelength: increases	**B1**

5 **(a)** Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.

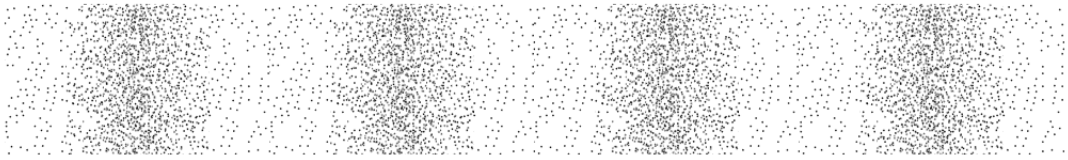

Fig. 5.1

(i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]

(ii) Describe the difference between a compression and a rarefaction.

.....

.....

..... [1]

(b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.

(i) The wooden bar makes 45 complete oscillations in 1.0 minute.

Calculate the frequency of the wave produced.

frequency = ...Hz [1]

(ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed .....

wavelength .....
[2]

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(iii) The crests of the water wave move into the shallow region shown in Fig. 5.2.

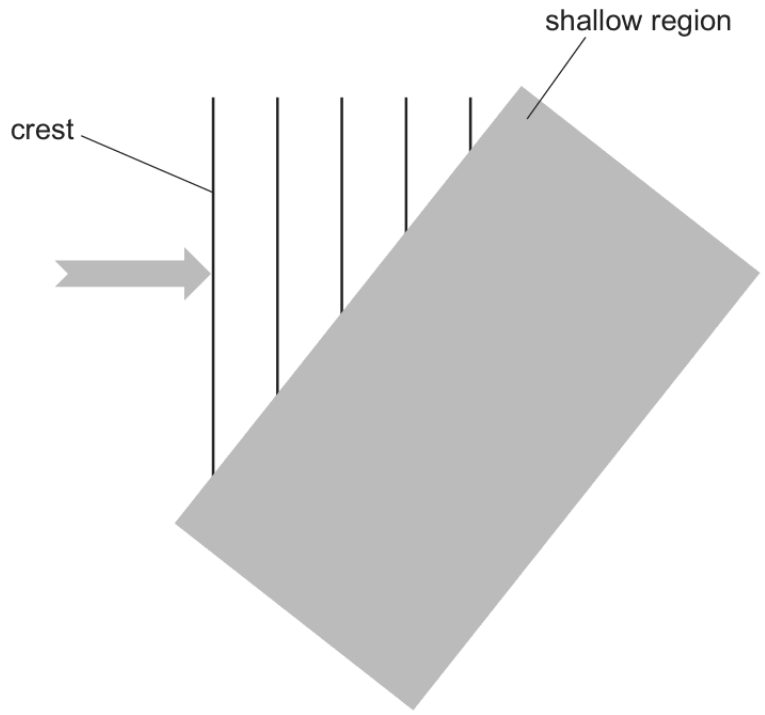


Fig. 5.2

On Fig. 5.2, draw the crests in the shallow region. [2]

(c) Describe what is meant by the diffraction of a water wave.

.....

.....

..... [1]

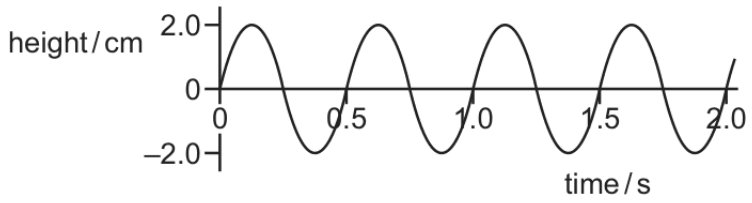
[Total: 8]

Question	Answer	Marks
5(a)(i)	C and R marked correctly	**B1**
5(a)(ii)	compressions are where pressure is high / particles close together / density is high	**B1**
5(b)(i)	0.75 (Hz)	**B1**
5(b)(ii)	*speed:* no change	**B1**
	wavelength: decreases	**B1**
5(b)(iii)	at least 3 wavefronts in the shallow region parallel to each other **and** on correct side of normal	**M1**
	refracted towards the surface on correct side of surface **and** join wavefronts shown	**A1**
5(c)	spreading out / bending at an edge / obstacle **or** spreading out after passing through a (small) gap / hole	**B1**

QUESTION · 5054_s25_qp_12.pdf · source page 8 of 16

19 A cork on the surface of water moves up and down as a wave passes across the surface.

The diagram shows how the height of the cork above the average level of the surface varies with time.



The numerical value of the wavelength of this wave is equal to the numerical value of the amplitude of this wave.

What is the speed of the wave?

A 1.0 cm / s **B** 2.0 cm / s **C** 4.0 cm / s **D** 8.0 cm / s

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 19

19	C	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 9 of 16

20 A water wave travels from deep water to shallow water in a ripple tank.

What happens to the speed and to the wavelength of the wave?

	speed of wave	wavelength
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 20

20	D	**1**

4 A sound wave from a vibrating tuning fork is displayed on the oscilloscope shown in Fig. 4.1. The sound wave is detected using a microphone.

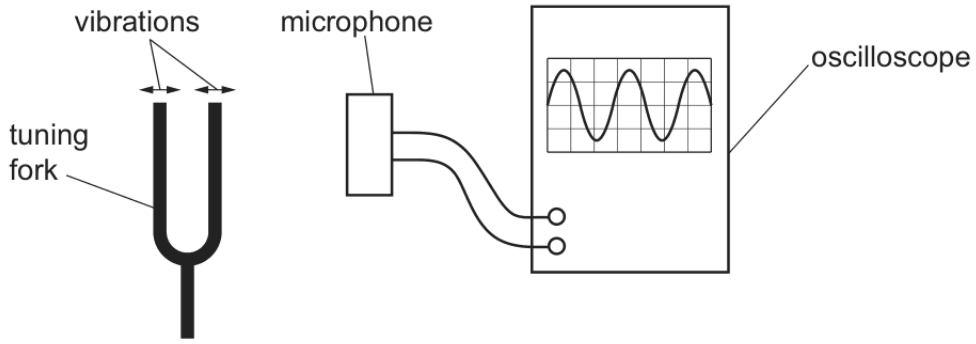


Fig. 4.1

(a) Describe the motion of the air particles as the sound wave passes from the tuning fork to the microphone.

.....

.....

..... [2]

(b) Fig. 4.2 shows the trace seen on the oscilloscope when the tuning fork vibrates.

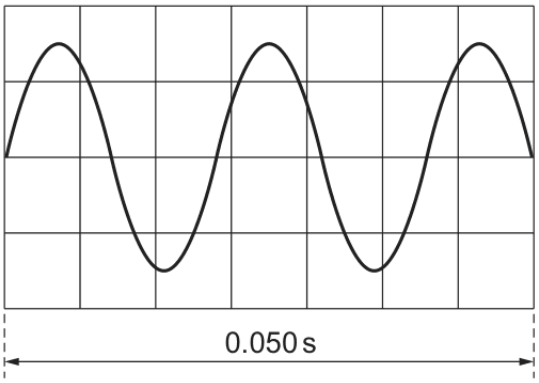


Fig. 4.2

The frequency of the sound is constant.

(i) State what is meant by 'frequency'.

.....

..... [1]

(ii) The trace shown on Fig. 4.2 is produced in 0.050 s.

Calculate the frequency of the sound wave.

frequency = ... Hz [2]

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(iii) A student near the tuning fork observes the trace and listens to the sound. He notices that the amplitude of the trace decreases.

State how the sound heard by the student changes.

..... [1]

(iv) The tuning fork is replaced by a new tuning fork that produces a sound wave of half the frequency. The controls on the oscilloscope are not changed.

On Fig. 4.3, draw the trace obtained with the new tuning fork. The trace obtained with the original tuning fork is shown.

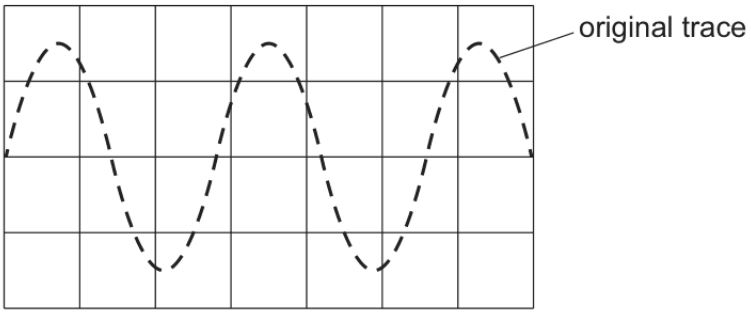

Fig. 4.3

[1]

(c) Fig. 4.4 shows three adjacent wavefronts of the sound wave as the wavefronts move towards a gap in a wall created by an open door.

Another student, standing at P on the other side of the wall, hears the sound.

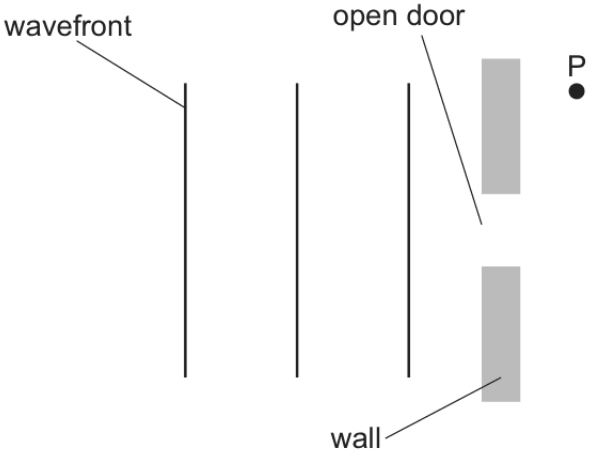

Fig. 4.4

(i) On Fig. 4.4, draw the three wavefronts after they pass through the gap. [2]

(ii) State the name of the process involved as the wavefronts pass through the gap and reach P.

..... [1]

[Total: 10]

Question	Answer	Marks
4(a)	*motion of particles:* backwards and forwards / to and fro / closer and further apart	**B1**
	(oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork **or** mention of compressions and rarefactions	**B1**
4(b)(i)	number of oscillations per second **or** number of wave(length)s (passing a point) per second	**B1**
4(b)(ii)	2.5 (waves) seen **or** (1 wave takes) 0.02 (s) **or** (f =) 1 / T or (number of) waves / time	**C1**
	50 (Hz)	**A1**
4(b)(iii)	quieter / less volume	**B1**
4(b)(iv)	half frequency shown **and** at least one complete oscillation seen	**B1**
4(c)(i)	(at least) three correctly curved wavefronts centered on gap	**B1**
	same wavelength as on left **and** some curving in correct direction	**B1**
4(c)(ii)	diffraction	**B1**

5 The microwave region is one region of the electromagnetic spectrum.

(a) Fig. 5.1 represents the electromagnetic spectrum divided into the seven main regions.

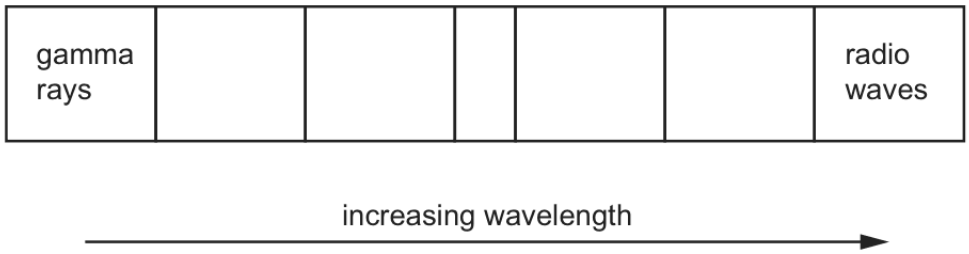

Fig. 5.1

The regions are arranged from left to right in order of increasing wavelength.

On Fig. 5.1:

- indicate the microwave region by marking it with the letter M
- indicate the ultraviolet region by marking it with the letters UV.

[1]

(b) The frequency of the microwaves used by some satellite television systems is 12 GHz.

(i) The speed of microwaves in a vacuum is 3.0×10^8 m/s.

Calculate the wavelength of these microwaves in a vacuum.

wavelength = ... m [3]

(ii) Describe how microwaves are used in satellite television systems.

.....

.....

.....

..... [2]

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(c) All microwave ovens have a switch that turns the oven off when the door is opened.

Suggest why it is important for a microwave oven to switch off when the door is opened.

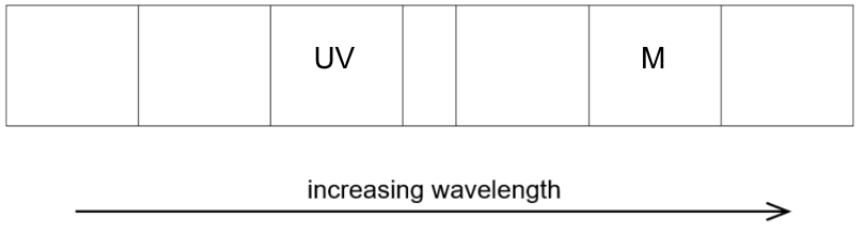
.....

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
5(a)		**B1**
5(b)(i)	$(\lambda =) \frac{c}{f}$ **or** $\frac{3.0 \times 10^{8}}{1.2 \times 10^{N}}$	**C1**
	2.5×10^{X} (m)	**C1**
	0.025 (m)	**A1**
5(b)(ii)	encoded microwave / microwave signal / microwave carrying data / microwave transmitting data (to satellite from Earth)	**B1**
	satellite transmits microwave to Earth / dish / antenna / subscriber / television	**B1**
5(c)	to prevent microwaves escaping / leaving the oven / spreading out of the device	**B1**
	microwaves / they can cause burns (in living tissue / people) / damage to tissue through heating / blisters	**B1**

QUESTION · 5054_s26_qp_11.pdf · source page 10 of 16

22 Which equation gives the speed of a wave?

A speed = amplitude ÷ frequency

B speed = amplitude × frequency

C speed = wavelength ÷ frequency

D speed = wavelength × frequency

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 22

22	D	**1**

4 **(a)** All electromagnetic waves transfer energy.

State **two** other properties that are the same for all electromagnetic waves.

1 .....

.....

2 .....

..... [2]

(b) Figure 4.1 shows an X-ray image of healthy teeth.

The detector used to create the image is placed inside the mouth, behind the teeth.

A source of X-rays is placed outside the mouth, in front of the teeth.

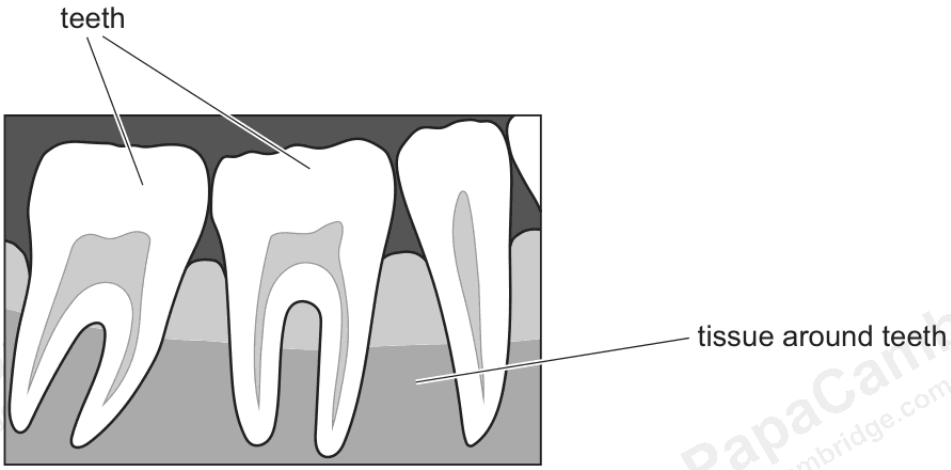


Figure 4.1

Explain why the X-rays produce an image that shows white teeth, grey tissue around the teeth and a black background.

.....

.....

.....

.....

..... [2]

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(c) Ultrasound is another type of wave used in hospitals to produce medical images.

A scanner placed on the skin of a patient emits an ultrasound wave.

(i) The ultrasound wave causes a motion of the particles in the tissue of the patient.

Describe this motion.

.....

.....

..... [2]

(ii) The ultrasound wave is reflected from a bone in the patient and arrives back at the scanner 8.0×10^{-5} s after it was emitted.

The speed of ultrasound in the patient is 1500 m / s.

Calculate the distance between the scanner and the bone.

distance = ... m [2]

(iii) Some of the ultrasound passes through the patient and emerges into the air.

The speed of ultrasound in air is smaller than the speed of ultrasound in the patient.

State and explain the change in wavelength as ultrasound travels from the patient into the air.

change in wavelength .....

explanation .....

.....

.....

.....

[2]

[Total: 10]

Question	Answer	Marks	Guidance
4(a)	Any 2 from: • can reflect / diffract / refract • travel through a vacuum / do not require a medium • transverse • are oscillating electric and magnetic fields • travel at the speed of light / same speed (in vacuum/air)	**2**	**I** no mass, no charge
4(b)	Any two from: • X rays pass through the spaces **or** X-rays make the image darker / go black • (most / all) X-rays absorbed by / cannot penetrate the teeth • X-rays partially absorbed by the tissue / some pass through tissue	**2**	X-rays reflected max 1 mark
4(c)(i)	oscillation / vibration longitudinal **or** backwards and forwards **or** in the direction of the sound / parallel to the direction of travel of the wave **or** formation of compressions and rarefactions	**1**	**A** backwards and forwards or any clear oscillation, eg up and down; left and right; to and fro **A** ultrasound longitudinal **I** left and right, up and down
4(c)(ii)	(d =) vt algebraic or numerical 0.06 (0) m	**1**	**A** ½ vt; ½ × 8.0 × 10^{-5} × 1500; 0.12 m; 12 cm
4(c)(iii)	(wavelength) decreases (in air) $v = f\lambda$ **and** frequency the same **or** each (complete) wave travels less distance (before the next arrives)	**1**	**A** frequency the same **and** $v \propto \lambda$

5 Figure 5.1 shows a ripple tank used by a student.

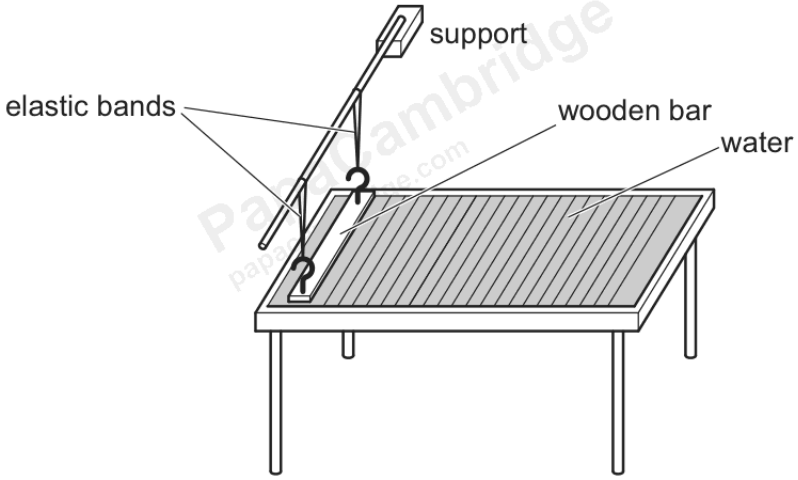

Figure 5.1

(a) (i) Describe how the student uses the apparatus to create a wave.

.....

.....

..... [1]

(ii) Describe how the student can demonstrate diffraction of the wave using the ripple tank. You should refer to any other apparatus needed.

.....

.....

.....

..... [1]

(iii) The unit of frequency is the hertz (Hz).

Describe how the student can determine the frequency of the wave in the ripple tank in Hz.

.....

.....

.....

..... [2]

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(b) Figure 5.2 shows a side view of the wave on the surface of the water. The diagram also shows the surface of the water before it is disturbed by the wave.

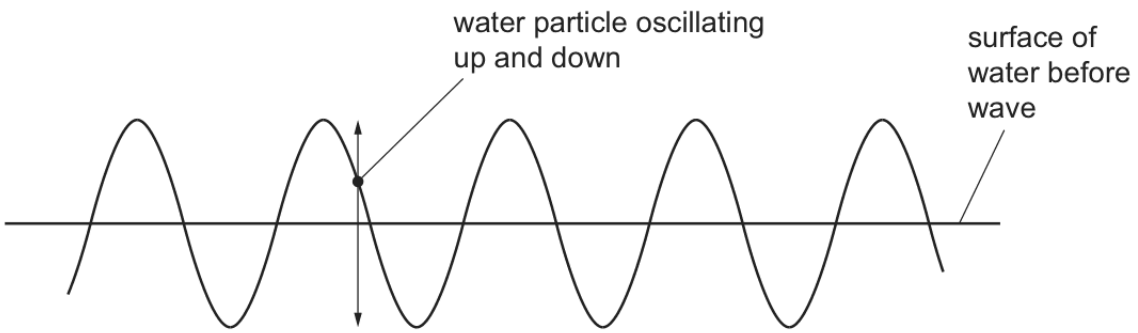


Figure 5.2

A water particle on the surface of the water oscillates up and down.

(i) On Figure 5.2, mark and label the amplitude of the wave. [1]

(ii) The frequency of the wave is 2.0 Hz.

Calculate the time taken for the water particle to move from the top of the wave to the bottom of the wave.

time = ... s [2]

(iii) Describe how the speed of the particle changes as it moves from the top of the wave to the bottom of the wave.

.....

.....

..... [2]

[Total: 9]

Question	Answer	Marks	Guidance
5(a)(i)	move the bar (up and down continuously)	**1**	**A** left and right; vibrate the bar; use bar to disturb water / bar hits water / shake the bar **I** put bar in water; move support; bar touches water; just 'stretch and release bands'
5(a)(ii)	use a (barrier with a) slit **or** (two) obstacles with a gap **or** the edge of a barrier (in the water) **or** waves bend around an obstacle	**1**	**I** just 'create a hole'; have a gap; original wooden bar acts as obstacle **NB** must mention edge or gap or slit or similar
5(a)(iii)	(count) the number of oscillations / crests (passing a point) in a (measured) time **or** measure the time for a given number of waves **or** measure / or find the time for one wave / measure or find the period divide the number of waves by the time (in seconds) **or** count the number of crests / wavefronts pass / oscillations in one second **or** 1 / time for one wave (the period)	**1**	**A (**count) number of waves in any time; time for one wave to be formed **I** $v = f$□□unless wavelength and speed measurement described; just 'find period' 1 mark for 'count no. of waves in unit time' **just** count no of waves / oscillations in one second / unit time 2 marks; if no measurement made, e.g. 'f is number of waves / oscillations per second/unit time', 1 mark
5(b)(i)	top **or** bottom of wave to the middle.	**1**	**A** at other point if clear
5(b)(ii)	(T =) 1/f **or** 0.5 (0 s) seen for time 0.25 (s)	**1**	
5(b)(iii)	increases as it moves down (to the middle) moving fastest in the middle **or** decelerates (speed decreases) after middle / near bottom **or** stops at bottom	**1**	**A** increases top to bottom for this mark; needs to indicate increase while moving down/falling; just 'increases' TV

QUESTION · 5054_s26_qp_12.pdf · source page 9 of 16

20 A wave has a frequency of 2.0 Hz.

How many waves are produced in one minute?

A 2.0×60 **B** $\frac{60}{2.0}$ **C** 2.0 **D** $\frac{2.0}{60}$

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 20

20	A	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 9 of 16

21 The frequency of a wave is 0.60 MHz and the wavelength of the wave is 0.50 km.

What is the speed of this wave?

A 0.30 m / s

B 1.2 m / s

C 1200 m / s

D 300 000 000 m / s

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 21

21	D	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 9 of 16

22 Which list consists only of transverse waves?

A electromagnetic waves, seismic S-waves, waves on the surface of water

B seismic S-waves, sound waves, electromagnetic waves

C sound waves, waves on the surface of water, seismic P-waves

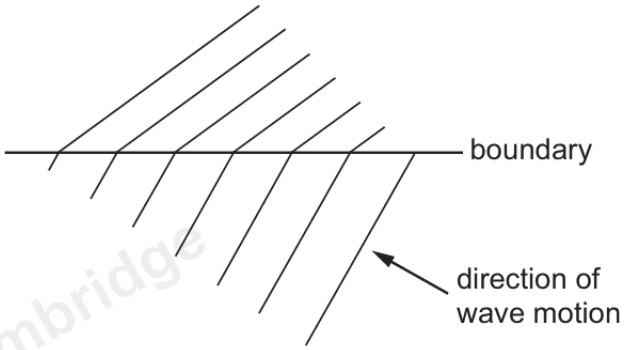
D waves on the surface of water, seismic P-waves, electromagnetic waves

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 22

22	A	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 10 of 16

23 The diagram shows the refraction of water waves as they cross a boundary in a ripple tank.



What causes this refraction?

A a change in frequency due to a change in depth

B a change in frequency due to a change in wavelength

C a change in speed due to a change in depth

D a change in speed due to a change in frequency

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 23

23	C	**1**

Chapter 3 Waves

3.2 Light

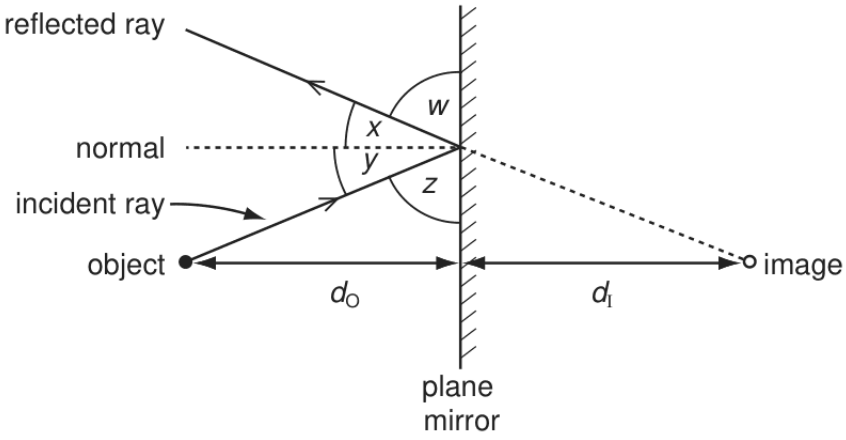
3.2.1 Reflection of light

64 entries: 55 Paper 1 questions, 9 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 9 of 20

20 An image is formed in a plane mirror.



Which statement must be correct?

	angles	distances
A	$w = y$	$d_O = d_I$
B	$w = z$	d_O is greater than d_I
C	$x = y$	$d_O = d_I$
D	$x = z$	d_O is greater than d_I

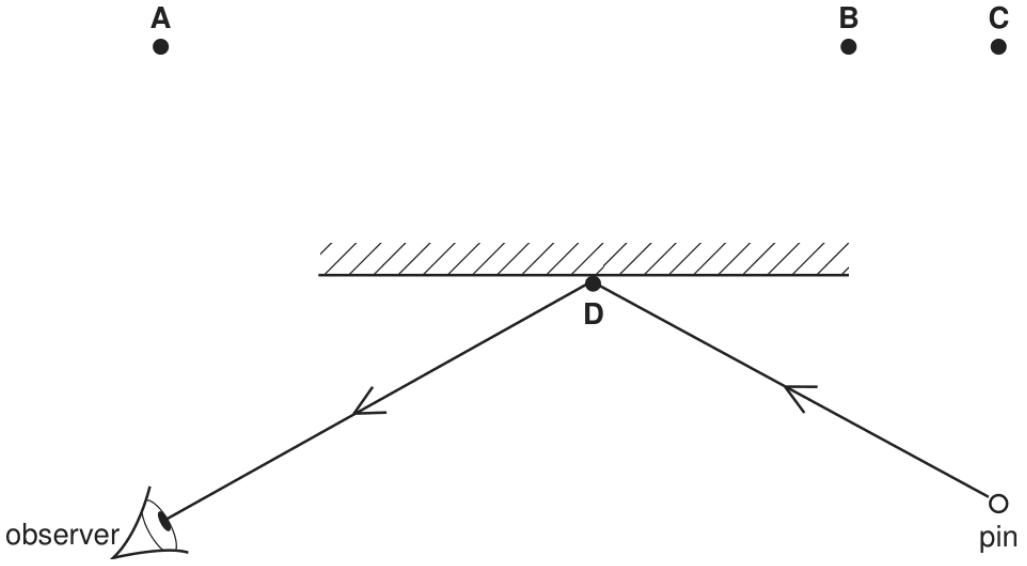
MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 20

20 **C**

QUESTION · 5054_w04_qp_1.pdf · source page 11 of 20

24 A pin is placed in front of, and to the right of, a plane mirror as shown.

Where is the image of the pin?

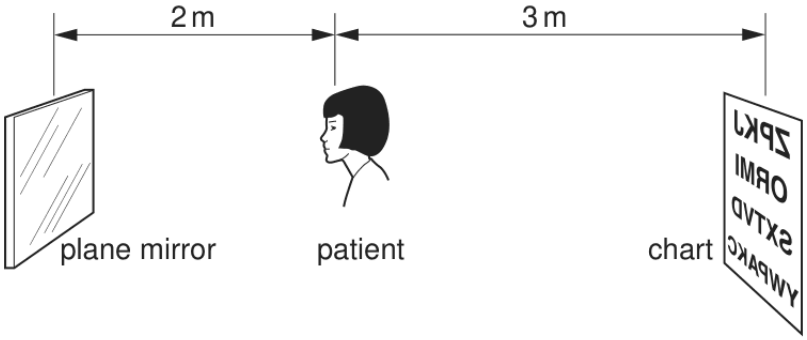


MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 24

24	C

QUESTION · 5054_s05_qp_1.pdf · source page 9 of 20

21 The diagram shows a patient having her eyes tested. A chart with letters on it is placed behind her and she sees the chart reflected in a plane mirror.



How far away from the patient is the image of the chart?

A 2 m **B** 4 m **C** 5 m **D** 7 m

MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 21

21	**D**

QUESTION · 5054_w05_qp_1.pdf · source page 12 of 20

22 Which characteristics describe an image formed in a plane mirror?

A real and inverted

B virtual and upright

C real and larger than the object

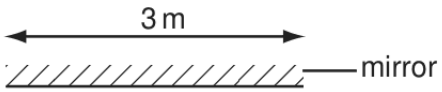
D virtual and smaller than the object

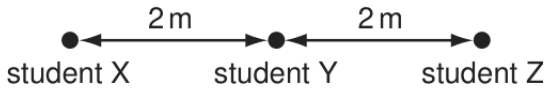
MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 22

22	**B**

QUESTION · 5054_w06_qp_1.pdf · source page 9 of 16

21 Three students stand 2 m apart in front of a plane mirror which is 3 m long.





Student Y is standing opposite the mid-point of the mirror.

How many students can see the images of the other two?

A 0 **B** 1 **C** 2 **D** 3

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 21

21	**D**

For Examiner's Use

4 **(a)** Fig. 4.1 shows a ray of light incident on a mirror at X. The incident ray makes an angle of 50° with the surface of the mirror.

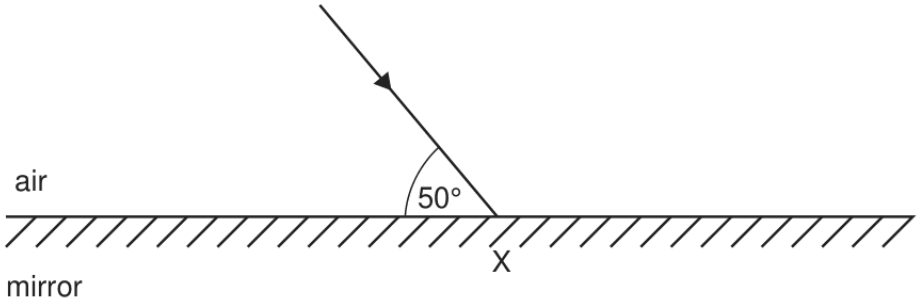


Fig. 4.1

(i) Complete Fig. 4.1 to show the normal and the reflected ray at X. [1]

(ii) State the values of

1. the angle of incidence, ... [1]

2. the angle of reflection. ..... [1]

(b) Describe with the help of a diagram how you would find the position of the image produced by a plane mirror.

.....

.....

.....

.....

.....

.....

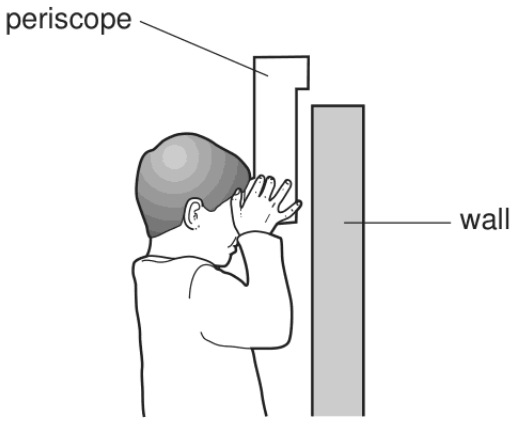
.....

..... [3]

4	**(a)**	**(i)**	reflected ray correct by eye **and** normal	B1
		(ii)	40 °	B1
			40 ° **or** same as angle of incidence	B1
	(b)		**diagram** with object, mirror, image in approx. correct position	B1
			at least 1 ray drawn from object/ray-box correctly reflecting from mirror	B1
			at least 2 rays extrapolated back to image position	B1
	OR(b)		**diagram** with object, mirror, image in approx. correct position **OR**	B1
			Use of search pin behind mirror shown/stated	B1
			no parallax used to locate image **or** described	B1 6
			(ignore arrows/do not insist on dotted lines)	

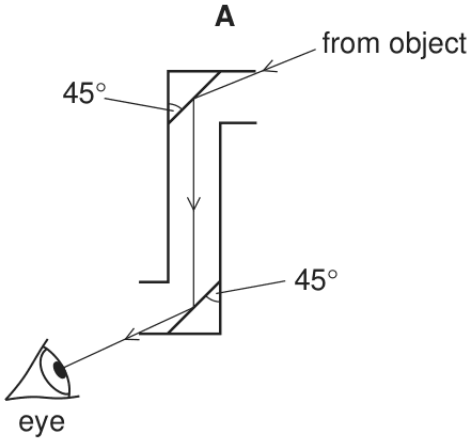
QUESTION · 5054_s07_qp_1.pdf · source page 10 of 20

20 The diagram shows a child using a periscope to look at an object on the other side of a wall.

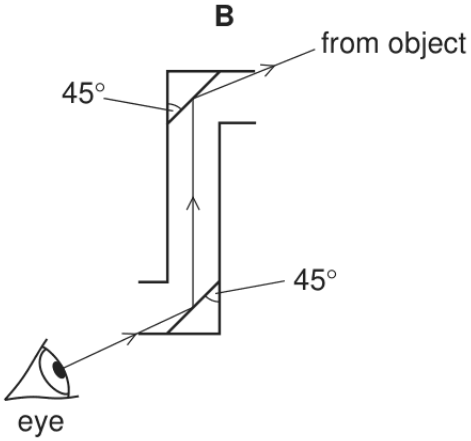


Which diagram shows a correctly drawn ray of light from the object?

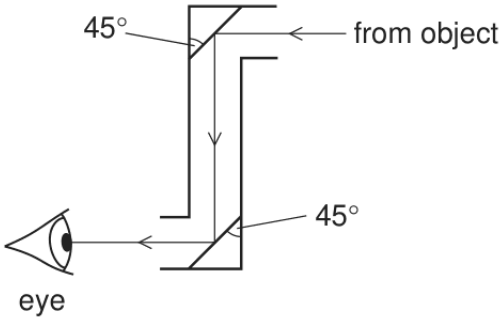
A



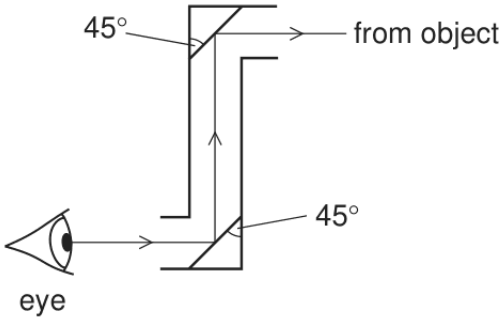
B



C



D

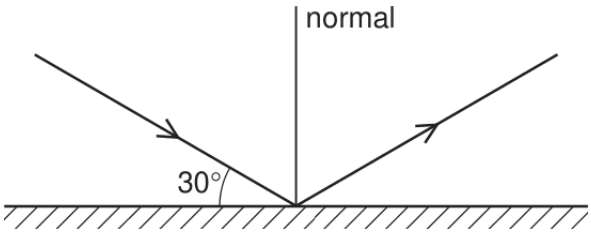


MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 20

20 **C**

QUESTION · 5054_w07_qp_1.pdf · source page 9 of 20

20 The diagram shows a ray of light reflected from a plane mirror.



What is the angle of reflection?

A 30° **B** 60° **C** 90° **D** 120°

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 20

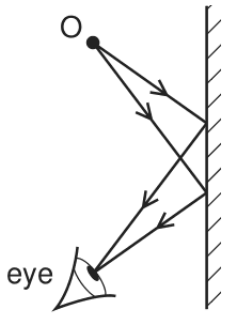
20	**B**

QUESTION · 5054_w08_qp_1.pdf · source page 12 of 20

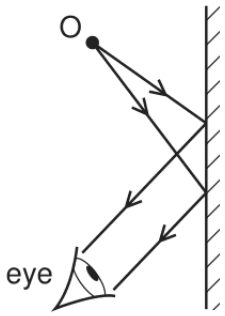
23 An eye views an object O by reflection in a plane mirror.

Which is the correct ray diagram?

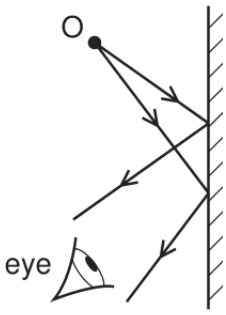
A



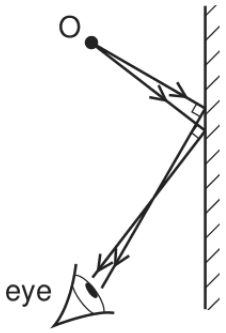
B



C



D



MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 23

23	**C**

QUESTION · 5054_s09_qp_1.pdf · source page 9 of 20

20 Which characteristics describe an image formed by a plane mirror?

A real and inverted

B virtual and upright

C real and larger than the object

D virtual and smaller than the object

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 20

20	**B**

5 Fig. 5.1 shows a man looking at his reflection in a rectangular plane mirror.

For Examiner's Use

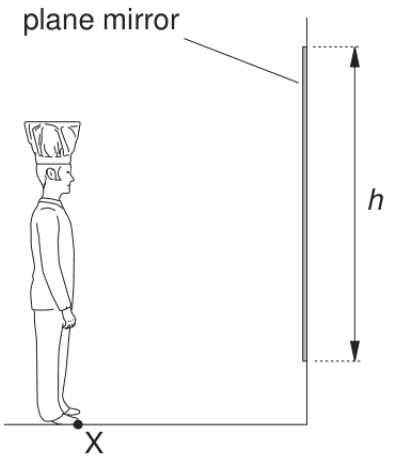


Fig. 5.1

The vertical side of the mirror has length *h*.

(a) (i) On Fig. 5.1, draw a ray of light from point X that is reflected by the mirror to the man's eye. [1]

(ii) On Fig. 5.1, mark the angle of incidence of your ray at the mirror.

Label this angle *i*. [1]

(iii) Define the *angle of incidence*.

.....

.....

..... [1]

(b) On Fig. 5.1, draw a ray of light from the top of the man's hat that is reflected by the mirror to his eye.

Use your rays to determine the smallest value of *h* that allows the man to see all of the image in the mirror, from the top of his hat to his toes.

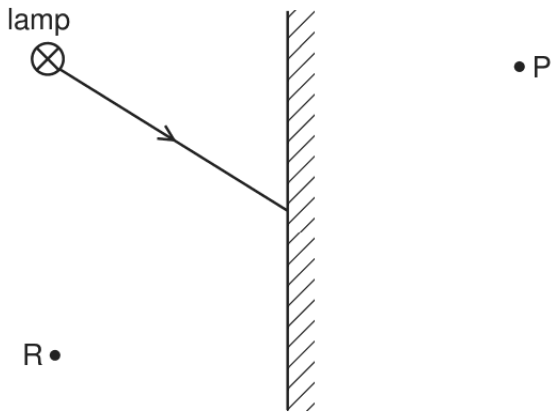
On the diagram, 1 cm represents 0.5 m.

h = ... [2]

5	**(a)**	**(i)**	correct ray	B1	
		(ii)	correct angle marked to normal	B1	
		(iii)	(the angle) between the incident ray and the normal (at the point of contact)	B1	
	(b)		correct ray from hat to eye	B1	
			0.85–1.15 m	B1	[5]

QUESTION · 5054_w09_qp_1.pdf · source page 10 of 20

20 The diagram shows a ray of light from one point on a lamp striking a plane mirror.



The image of the point on the lamp formed by the mirror is

A at P and is real.

B at P and is virtual.

C at R and is real.

D at R and is virtual.

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 20

20 **B**

For Examiner's Use

4 Fig. 4.1 shows an old coin displayed in a museum.

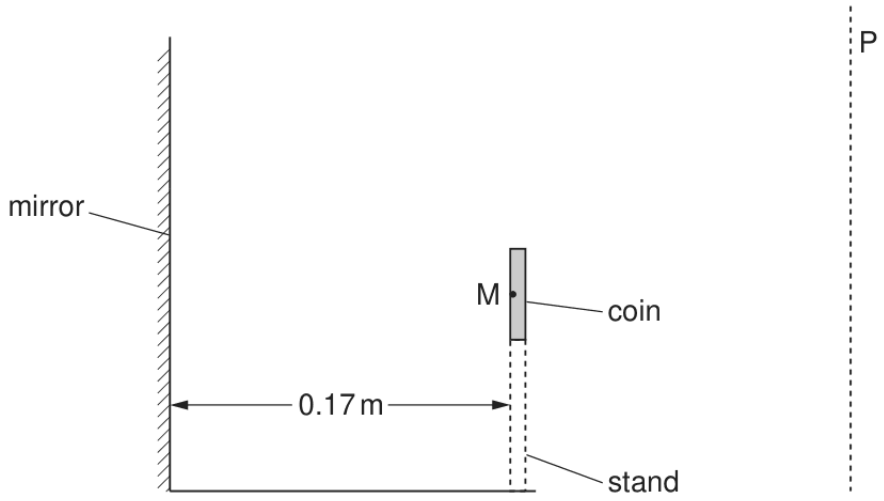


Fig. 4.1

The coin is vertical and is supported by a transparent stand. A vertical mirror 0.17 m behind the coin ensures that the back of the coin can be seen by a visitor looking from the line P.

M is a point on the back of the coin.

(a) On Fig. 4.1,

(i) draw **two** rays of light from M to show how its image is produced, [2]

(ii) label the image **I**. [1]

(b) State the distance from point M on the coin to its image.

distance = ...[1]

4	**(a)**	**(i)**	one ray from M correctly reflected – checked by eye two rays from M correctly reflected – checked by eye – **and** traced back to image	C1 A1	
		(ii)	image point **clearly** marked at intersection/correct place checked by eye	B1	
	(b)		0.34 m **cao**	B1	[4]

QUESTION · 5054_s10_qp_11.pdf · source page 9 of 20

19 A student holds a sheet of paper with letters on it facing a plane mirror.

The letters on the paper are shown.

What does the student see in the mirror?

A

B

C

D

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 19

19	**B**

QUESTION · 5054_s10_qp_12.pdf · source page 10 of 20

22 A student holds a sheet of paper with letters on it facing a plane mirror.

The letters on the paper are shown.

What does the student see in the mirror?

A

B

C

D

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 22

22	**B**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a very large plane mirror, inclined at 45° to the horizontal, beneath a pattern on the high ceiling of a hall.

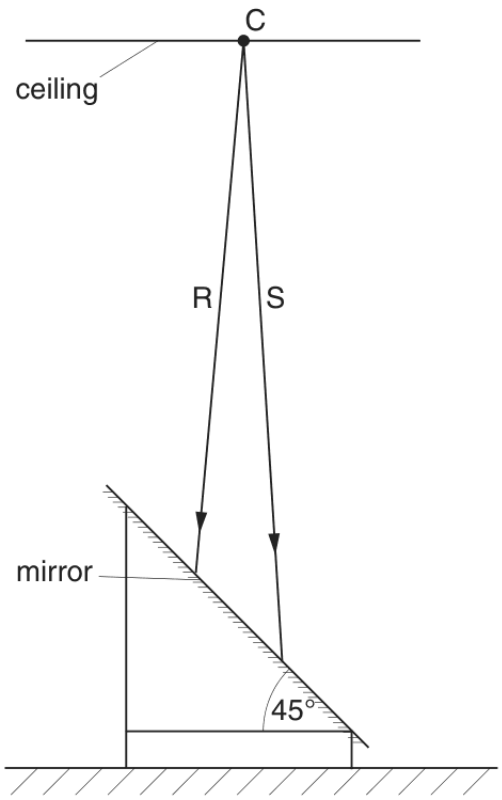


Fig. 9.1

The mirror is set on a stand at head-height immediately below the centre C of the pattern. R and S are two rays of light from C that strike the mirror.

(a) (i) On Fig. 9.1, continue the rays R and S after they strike the mirror. [1]

(ii) On Fig. 9.1, show how these rays are used to locate the image of C and mark and label the position of this image with the letter I. [2]

(iii) State two characteristics of this image.

1. .....

2. .....
[2]

(iv) Suggest how the mirror helps visitors to the hall to see the pattern on the ceiling.

.....

.....[1]

For Examiner's Use

(b) Violet light from C has a wavelength of 4.0×10^{-7} m.

(i) Calculate the frequency of this light, clearly stating the value of any constant used in the calculation.

frequency = ...[3]

(ii) State two different components of the electromagnetic spectrum that have wavelengths smaller than the wavelength of violet light.

1. .....

2. .....
[2]

(iii) **1.** Discuss a medical application of **one** of these components of the electromagnetic spectrum.

.....

.....

.....

.....

.....[3]

2. State a health risk associated with this component of the electromagnetic spectrum.

.....

.....[1]

Section B

9 (a) (i) **one** correctly reflected ray (by eye) B1

(ii) **two** reflected rays traced back to an image B1
(labelled) image in correct position (by eye) B1

(iii) any **two** of:
virtual
full size/mag = 1 **or** same distance from mirror as C
laterally inverted (**ign** upright)
dimmer B2

(iv) more comfortable/no neck strain/no need to look up/reflects to eyes B1 [6]

(b) (i) (c =) 3(.00) × 10^8 (m/s) **or** 3(.00) × 10^5 (km/s) **or** used in equation B1
(f =) c/λ **or** (3.0 × 10^8/their **stated** value/330)/4.0 × 10^{-7} C1
7.5 × 10^{14} Hz **or** correct answer from **stated** value (incl. unit)
or 8.2/8.25/8.3 × 10^8 Hz A1

(ii) any **two**:
UV(radiation); X(radiation); γ(radiation) B2

(iii) **1.**

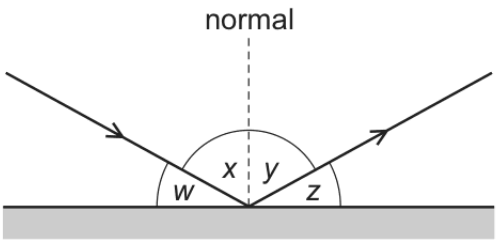
UV absorbed by skin	psoriasis destroyed	cells multiply less rapidly
X-rays absorbed by bones/not absorbed by flesh	shadow/image of bones	on film/CCD
γ-rays emitted by absorbed isotope	position/shape of organ etc. revealed	on film/CCD
tumour/cancer absorbs X/γ-ray	tumour destroyed	photons/energy/stops cells multiplying
bacteria absorb UV/X/γ-ray	Bacteria killed	sterilisation/stops bacteria multiplying

2.

UV:	X-rays:	γ-ray:	
damages eyes/skin cancer	cancer/hair loss/ radiation sickness	cancer/hair loss/ radiation sickness	B1 [9]

QUESTION · 5054_s11_qp_11.pdf · source page 9 of 16

21 A ray of light strikes a plane mirror and is reflected.



Which pair of angles **must** be equal in value?

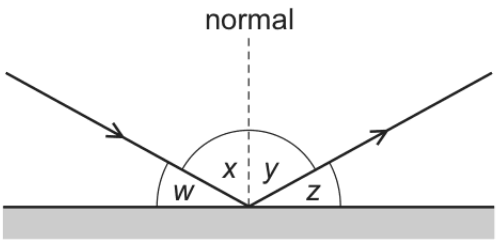
A *w* and *x* **B** *w* and *y* **C** *x* and *y* **D** *x* and *z*

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 21

21	**C**

QUESTION · 5054_s11_qp_12.pdf · source page 10 of 16

23 A ray of light strikes a plane mirror and is reflected.



Which pair of angles **must** be equal in value?

A *w* and *x** **B** *w* and *y** **C** *x* and *y** **D** *x* and *z*

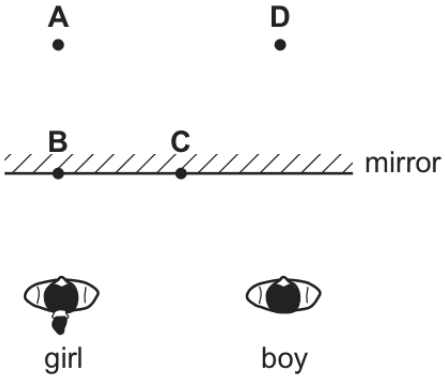
MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 23

23 **C**

QUESTION · 5054_w11_qp_11.pdf · source page 9 of 20

22 A boy stands beside a girl in front of a large plane mirror. They are both the same distance from the mirror, as shown.

Where does the boy see the girl's image?



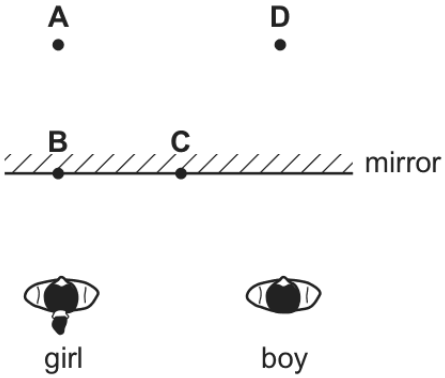
MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 22

22	**A**

QUESTION · 5054_w11_qp_12.pdf · source page 9 of 20

21 A boy stands beside a girl in front of a large plane mirror. They are both the same distance from the mirror, as shown.

Where does the boy see the girl's image?

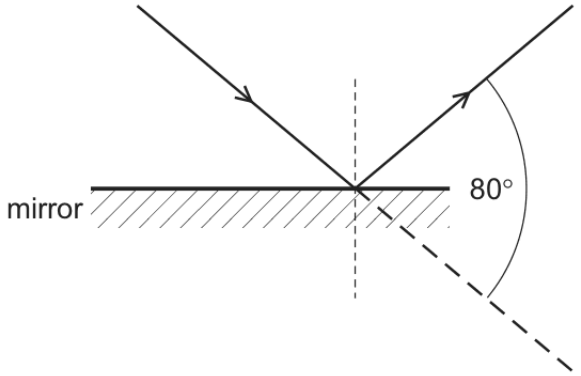


MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 21

21	**A**

QUESTION · 5054_s12_qp_11.pdf · source page 9 of 16

22 Light is incident on a mirror and is reflected as shown.



What is the angle of incidence and the angle of reflection?

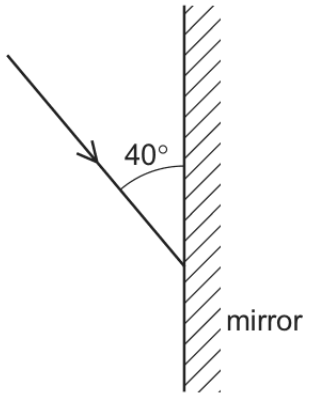
	angle of incidence / °	angle of reflection / °
A	40	40
B	40	50
C	50	40
D	50	50

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 22

22 **D**

QUESTION · 5054_s12_qp_12.pdf · source page 10 of 16

22 The diagram shows a ray of light directed at a plane mirror.



What are the angle of incidence and the angle of reflection?

	angle of incidence	angle of reflection
A	40°	40°
B	40°	50°
C	50°	40°
D	50°	50°

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 22

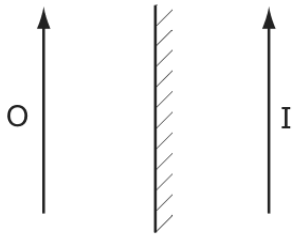
22 **D**

QUESTION · 5054_w12_qp_11.pdf · source page 9 of 16

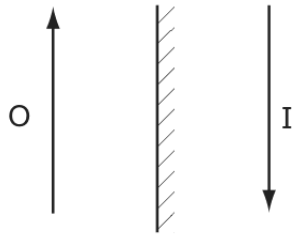
22 An object O is placed in front of a plane mirror.

Which diagram correctly represents the image I formed by the mirror?

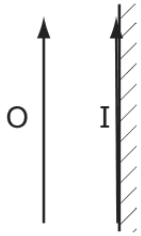
A



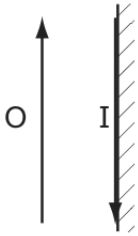
B



C



D



MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 22

22	**A**

QUESTION · 5054_w12_qp_12.pdf · source page 8 of 16

22 An object is placed in front of a plane mirror. The image produced is

A real and smaller than the object.

B real and the same size as the object.

C virtual and smaller than the object.

D virtual and the same size as the object.

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 22

22	**D**

For Examiner's Use

10 A laser produces red light of frequency 4.7×10^{14} Hz. The speed of light in glass is 2.0×10^{8} m/s.

(a) Calculate the wavelength in glass of light from this laser.

wavelength = ...[2]

(b) Describe an experiment to verify *the law of reflection* for light. You may include a diagram in your answer.

.....

.....

.....

.....

.....

.....

.....[5]

(c) Fig. 10.1 shows a ray of light travelling in an optical fibre. The ray strikes the side of the fibre at P.

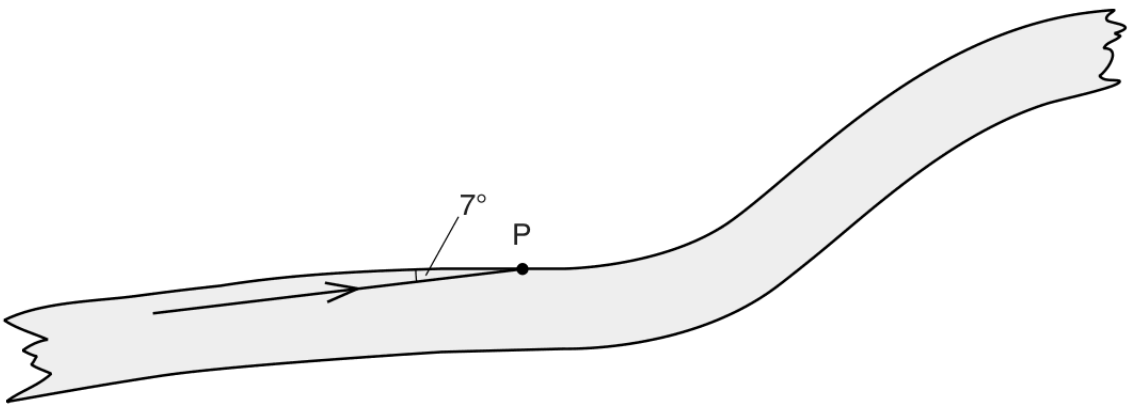


Fig. 10.1

The angle between the ray and the side of the fibre is 7°.

For Examiner's Use

(i) Determine the angle of incidence of the ray at P.

angle = ...[1]

(ii) State and explain what happens to the ray at P.

.....

.....

.....[2]

(d) A room is illuminated by wall lamps. Fig. 10.2 shows a mirror on the wall behind one of the lamps.

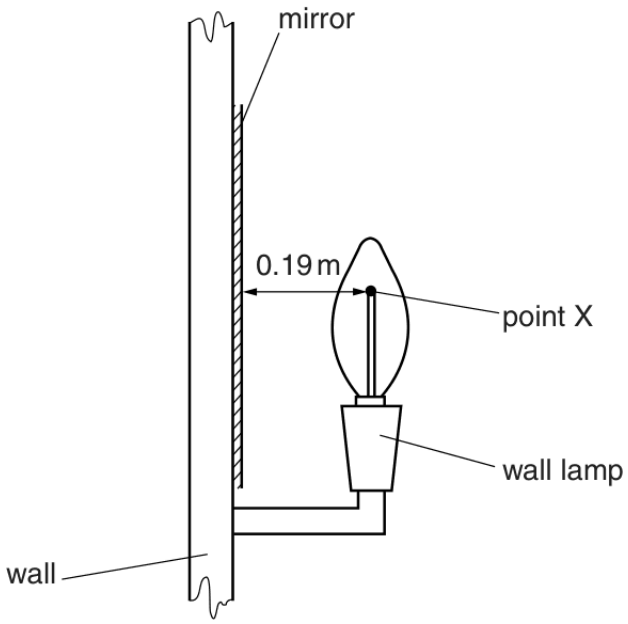


Fig. 10.2 (not to scale)

X is a point on the filament of the lamp. It is 0.19 m in front of the mirror.

(i) On Fig. 10.2, draw rays from X and locate the image of X. Label the image I. [3]

(ii) State the distance between I and the mirror.

distance = ...[1]

(iii) Suggest one advantage of placing a mirror behind the lamp in the room.

.....

.....[1]

10 (a) $(\lambda =)v/f$ **or** $2 \times 10^8/4.7 \times 10^{14}$ C1
4.3×10^{-7} m A1 [2]

(b)

raybox/light source/ laser **and** mirror		pin(s) **and** mirror	B1	
shine ray at mirror		place two pins	B1	
mark rays	**or**	two more pins in line with image	B1	
measure *i* and *r* **and** equal		measure *i* and *r* **and** equal	B1	
repeat		repeat	B1	[5]

(c) (i) 83° B1

(ii) total internal reflection **or** TIR **cao** B1
angle of incidence exceeds critical angle B1 [3]

(d) (i) (at least) one ray from X to mirror M1
(at least) **two** rays from X to mirror **and** correct reflections A1
rays traced back to marked I **or** I marked in correct place (by eye) B1

(ii) 0.19 m B1

(iii) less/no light wasted **or** hall brighter B1 [5]

[Total: 15]

QUESTION · 5054_s13_qp_11.pdf · source page 9 of 20

19 Which characteristics describe an image formed by a vertical plane mirror?

A real and inverted

B virtual and not inverted

C real and larger than the object

D virtual and smaller than the object

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 19

19	**B**

QUESTION · 5054_s13_qp_12.pdf · source page 9 of 16

25 Which characteristics describe an image formed by a vertical plane mirror?

A real and inverted

B virtual and not inverted

C real and larger than the object

D virtual and smaller than the object

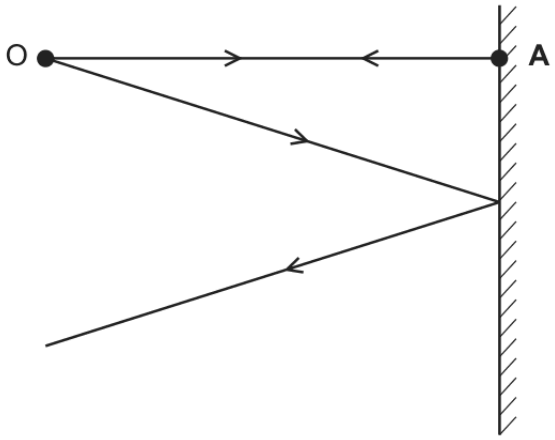
MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 25

25	**B**

QUESTION · 5054_s14_qp_11.pdf · source page 9 of 16

24 The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

At which position is the image formed?



- **B**
- **C**
- **D**

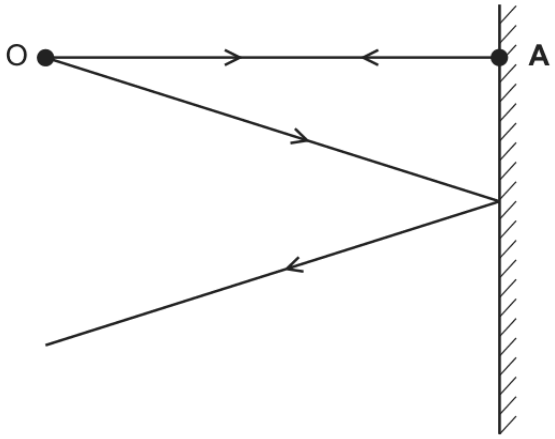
MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 24

24	**B**

QUESTION · 5054_s14_qp_12.pdf · source page 10 of 16

26 The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

At which position is the image formed?



- **B**
- **C**
- **D**

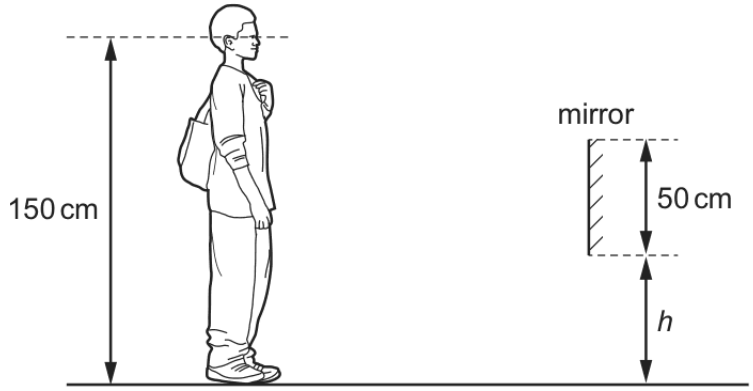
MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 26

26	**B**

QUESTION · 5054_w14_qp_11.pdf · source page 11 of 20

20 A shoe shop puts a mirror on the wall so that customers can look at their shoes.

The length of the mirror is 50 cm. A customer has eyes 150 cm above ground level.



The bottom of the mirror is at height *h* above the ground.

What is the smallest value of *h* that allows the customer to see an image of his shoes in the mirror?

A 0 **B** 25 cm **C** 50 cm **D** 75 cm

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 20

20	**B**

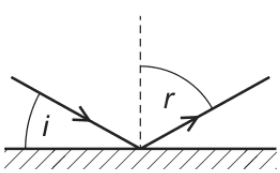
QUESTION · 5054_s16_qp_11.pdf · source page 8 of 16

21 Light is incident on a mirror. The light is reflected from the mirror.

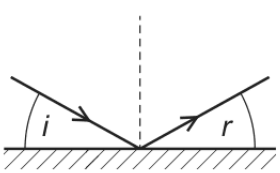
The angle of incidence is *i* and the angle of reflection is *r*.

Which diagram correctly shows *i* and *r*?

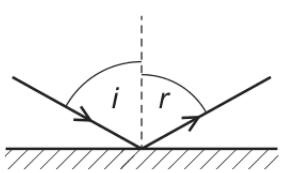
A



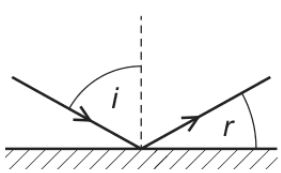
B



C



D

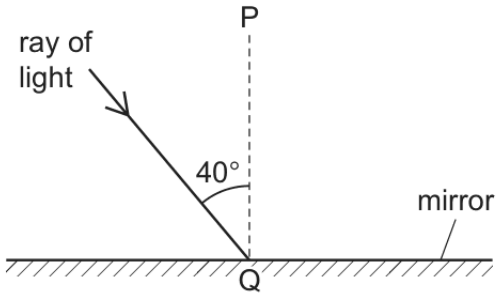


MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 21

21	C

QUESTION · 5054_s16_qp_12.pdf · source page 7 of 16

18 The diagram shows light incident on a plane mirror.



Which row gives the angle of reflection and the name of line PQ?

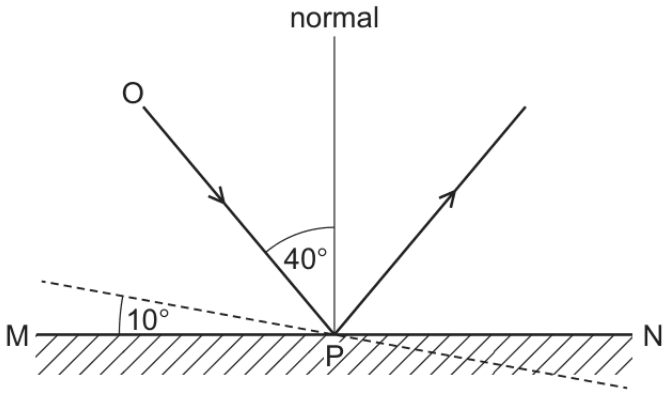
	angle of reflection	the line PQ is called the
A	40°	normal
B	40°	reflected ray
C	50°	normal
D	50°	reflected ray

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 18

18 **A**

QUESTION · 5054_w16_qp_11.pdf · source page 10 of 20

23 The angle of incidence of ray OP on the plane mirror MN is 40°.



The mirror is rotated through 10°, as shown by the dashed line. The direction of the incident ray OP does not change.

What is the new angle of incidence?

A 30° **B** 40° **C** 50° **D** 60°

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 23

23	**C**

10 **(a)** Fig. 10.1 shows two rays from an object that is placed in front of a plane mirror.

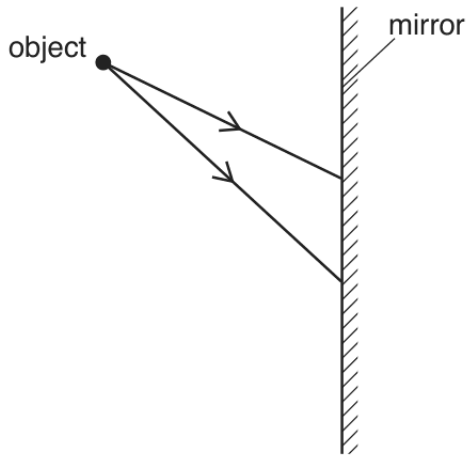


Fig. 10.1

(i) On Fig. 10.1, draw the two reflected rays and locate the position of the image. [2]

(ii) Apart from its position, state one characteristic of the image.

.....[1]

(b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

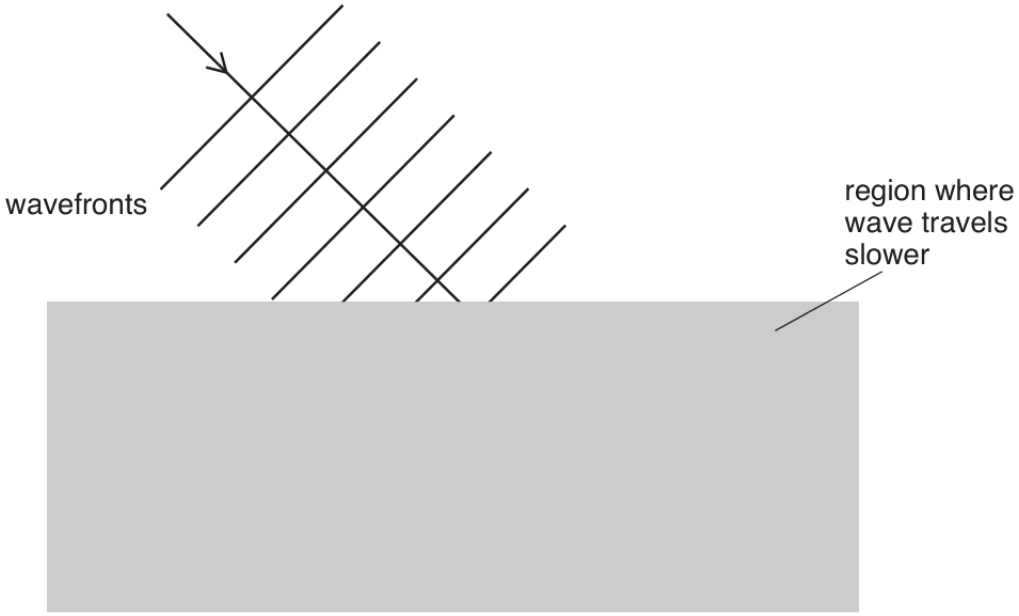


Fig. 10.2

(i) State what is meant by *wavefront*.

.....

.....[2]

(ii) Both reflection and refraction occur at the boundary.

1. On Fig. 10.2, draw the reflected wavefronts. [1]

2. On Fig. 10.2, draw the refracted wavefronts. [2]

(c) A loudspeaker produces a sound of frequency 2.0 kHz. The wavelength of this sound in air is 16 cm.

(i) Calculate the speed of sound in air.

speed = ...[2]

(ii) **1.** State the range of frequencies that can be heard by a healthy human ear.

.....[1]

2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength = ...[1]

(iii) Describe a simple experiment to show that sound waves obey the law of reflection.
You may draw a diagram if you wish.

.....

.....

.....

.....[3]

Question	Answer	Marks
10(a)(i)	both reflected rays correct by eye	**B1**
	image in correct position shown by continuation of rays behind mirror	**B1**
10(a)(ii)	virtual **or** upright **or** laterally inverted **or** same size (as object)	**B1**
10(b)(i)	line (joining points)	**C1**
	line joining points / particles on wave with same phase **or** line joining points along a crest etc.	**A1**
10(b)(ii)1	correct angle to surface ± 7° with correct orientation **and** similar wavelength (by eye)	**B1**
10(b)(ii)2	at least two lines at smaller angle to surface with correct orientation in slower medium	**B1**
	showing smaller and constant wavelength with wavefronts deviated in correct direction	**B1**
10(c)(i)	(v =) fλ **or** 2(000) × 16	**C1**
	32 000 cm / s **or** 320 m / s	**A1**
10(c)(ii)1	a range with 15–25 Hz as the lowest frequency **and** 15–30 kHz as the highest	**B1**
10(c)(ii)2	1.6 cm (if highest frequency is 20 000 Hz)	**B1**
10(c)(iii)	tube or other method to produce (narrow) beam of sound on source and/or on detector	**B1**
	stated detector **or** stated reflector	**B1**
	detector moved to find maximum loudness **or** angles *i* and *r* measured with suitable experiment	**B1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a man in a room looking into a mirror, as viewed from above.

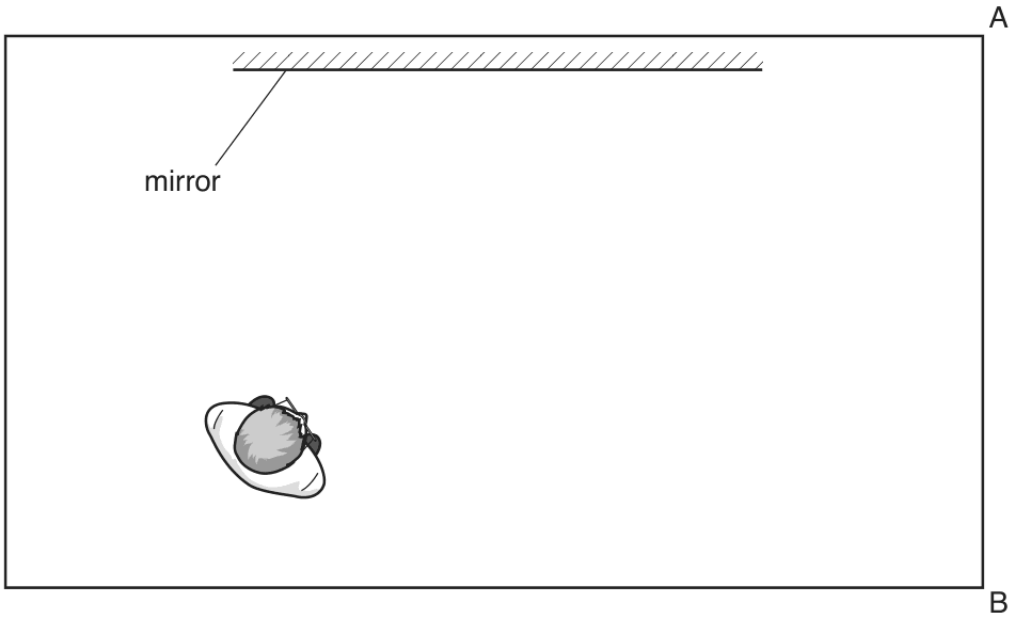


Fig. 9.1

(a) The man can see an image of part of the wall AB in the mirror.

Point C is the point on the wall closest to A that the man can see by reflection in the mirror. This point is not marked on Fig. 9.1.

(i) On Fig. 9.1, find point C and draw a ray of light from C which enables the man to see the image of C. [2]

(ii) On the ray that you have drawn on Fig. 9.1, label the angle of incidence *i* and the angle of reflection *r*. [2]

(iii) One characteristic of the image formed by the mirror is that it is virtual.

1. Explain what is meant by a *virtual* image.

.....

.....

.....[2]

2. State one other characteristic of an image formed by a plane mirror.

.....

.....[1]

(b) Light travels along optical fibres by total internal reflection.

(i) State two differences between total internal reflection and reflection in a mirror.

1. .....

.....

2. .....

.....

[2]

(ii) The critical angle for light travelling from the material of the fibre to air is 44°.

Calculate the refractive index of the material in the fibre.

refractive index = ...[2]

(iii) Light enters the optical fibre from air with an angle of incidence of 50°.

Calculate the angle of refraction.

angle of refraction = ...[2]

(iv) Optical fibres are used in telecommunications to send large amounts of data.

Pulses of light travel at almost the same speed along a fibre as electrical pulses travel along a metal wire.

State two advantages of using optical fibres rather than wires for transmitting data.

1. .....

.....

2. .....

.....

[2]

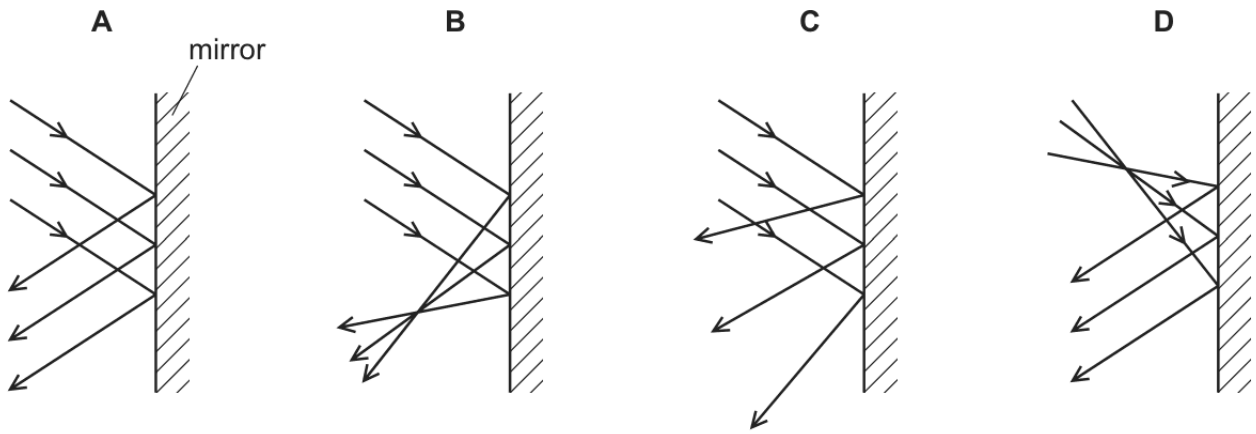
SECTION B

Question	Answer	Marks
9(a)(i)	ray from right-hand corner of mirror to eye	**B1**
	any incident and corresponding reflected ray correct by eye	**B1**
9(a)(ii)	normal drawn at any intersection of incident and reflected ray	**C1**
	both *r* and *i* labelled correctly with normal	**A1**
9(a)(iii)1	cannot be formed on a screen **or** nothing at the image (position)	**B1**
	rays do not come (all the way) from the image **or** rays only appear to come from image	**B1**
9(a)(iii)2	(same distance) behind the mirror **or** same size (as object) **or** upright / erect **or laterally** inverted	**B1**
9(b)(i)	reflection in mirror occurs at any angle **or** total internal reflection (TIR) only occurs for *i* > critical angle **or** there is no critical angle for the mirror B1	**any 2** **B2**
	TIR occurs from dense to less dense medium **or** in the dense(r) medium **or** from glass to air **or** inside / does not escape glass **or** from slow to fast (media) **or** mirror reflection from air to glass B1	
	(mirror) reflection is not total, e.g., not all reflected **or** better quality of image **or** multiple images from a mirror B1	

Question	Answer	Marks
9(b)(ii)	(n=) 1 / sinC **or** 1 / sin44	**C1**
	1.4	**A1**
9(b)(iii)	n = sini / sinr in any form, e.g. sinr = sin50 / n	**C1**
	32° – 33°	**A1**
9(b)(iv)	ANY 2 lines from • more data per second **or** per unit time • less decrease in strength / amplitude / attenuation • less heat / power produced / wasted • less need for repeating or amplification stations • less interference / noise • more secure / less chance of cross-talk • lighter / less heavy	**B1** **B1**

QUESTION · 5054_w17_qp_12.pdf · source page 9 of 16

20 Which diagram shows reflection by a plane mirror?



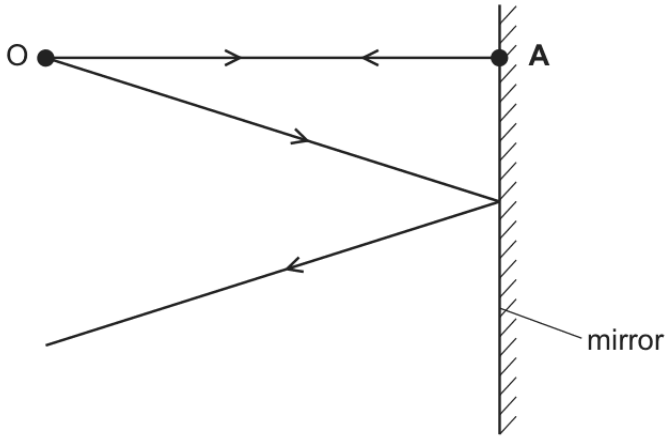
MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 20

20	A	**1**

QUESTION · 5054_s18_qp_11.pdf · source page 10 of 20

22 The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

At which position is the image formed?



- **B**
- **C**
- **D**

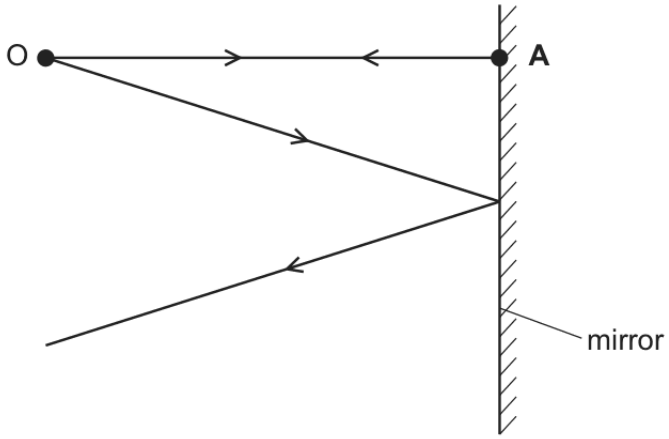
MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 22

22	B	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 11 of 16

27 The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

At which position is the image formed?



- **B**
- **C**
- **D**

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 27

27	B	**1**

QUESTION · 5054_s19_qp_11.pdf · source page 10 of 20

24 A ray of light strikes a plane mirror at an angle of incidence of 20°.

The angle of incidence is then increased by 5°.

What is the new angle between the incident ray and the reflected ray?

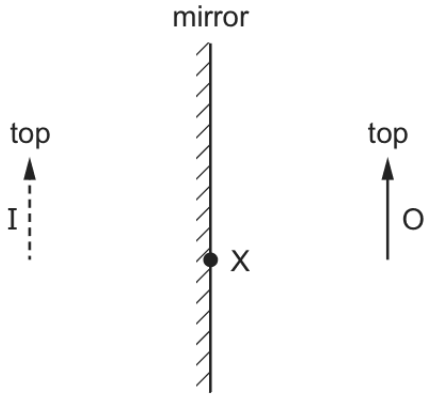
A 10° **B** 25° **C** 45° **D** 50°

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 24

24	D	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 10 of 16

24 An object O is placed in front of a plane mirror. I is the image formed.



A ray from the top of the object is incident on the mirror at X.

What happens to this ray?

A It reflects and passes through the bottom of O.

B It reflects and passes through the top of O.

C It reflects as though it came from the bottom of I.

D It reflects as though it came from the top of I.

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 24

24	D	**1**

6 A dentist uses a plane mirror to see the back of a tooth.

(a) A plane mirror produces an image of an object.

Describe the position of this image.

.....

.....

..... [2]

(b) Fig. 6.1 shows the plane mirror used by the dentist to see the point labelled X on the tooth.

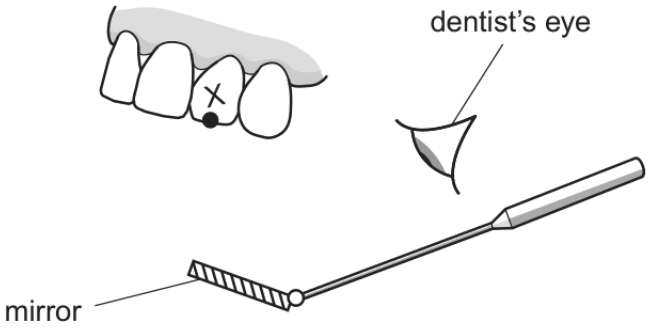


Fig. 6.1

On Fig. 6.1:

(i) mark the position of the image of X formed by the mirror [1]

(ii) draw a ray of light from X to show how the dentist can see the tooth. [2]

(c) State **one** characteristic of the image formed by the plane mirror other than its position.

.....

..... [1]

[Total: 6]

Question	Answer	Marks
6(a)	behind the mirror **or** on the opposite side (of mirror) to object	**B1**
	same distance as object (from mirror)	**B1**
6(b)(i)	image marked at correct position by eye	**B1**
6(b)(ii)	ray from tooth to mirror **and** ray from mirror to eye	**M1**
	angles of incidence and reflection equal (by eye)	**A1**
6(c)	virtual **or** same size (as object) **or** laterally inverted	**B1**

4 **(a)** Two mirrors, A and B, are inclined at an angle of 60° to each other.

Light strikes mirror A at an angle of 30°, as shown in Fig. 4.1.

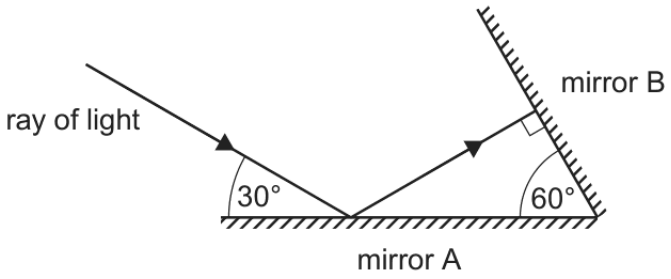


Fig. 4.1

(i) Determine the angle of incidence at mirror A.

angle of incidence at A = ... [1]

(ii) Determine the angle of incidence at mirror B.

angle of incidence at B = ... [1]

(iii) Describe the path of the reflected ray after it leaves mirror B.

.....

..... [1]

(b) A plane mirror hanging on a wall is used to form the image of an object.

State **three** characteristics of the image formed.

1. .....

2. .....

3. .....

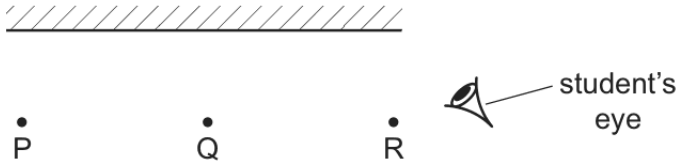
[3]

[Total: 6]

Question	Answer	Marks
4(a)(i)	60°	**B1**
4(a)(ii)	0(°)	**B1**
4(a)(iii)	back along the initial path	**B1**
4(b)	Any three of • same size (as object) • same distance (from mirror as object) • upright • virtual • laterally inverted	**B3**

QUESTION · 5054_w20_qp_11.pdf · source page 10 of 16

27 Three objects P, Q and R are placed in front of a plane mirror.



The student's eye is positioned as shown.

Which of the images of P, Q and R can the student see in the mirror?

	P	Q	R
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✗
D	✗	✗	✗

key
✓ = can see
✗ = cannot see

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 27

27	B	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 10 of 16

23 A plane mirror on a vertical wall forms an image of an object placed in front of it.

Which characteristics describe the image?

A real, inverted and smaller than the object

B real, upright and the same size as the object

C virtual, upright and smaller than the object

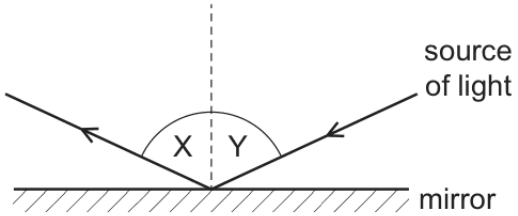
D virtual, upright and the same size as the object

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 23

23	D	**1**

QUESTION · 5054_w21_qp_12.pdf · source page 10 of 16

23 Light reflects from a plane mirror as shown.



Which row is always correct?

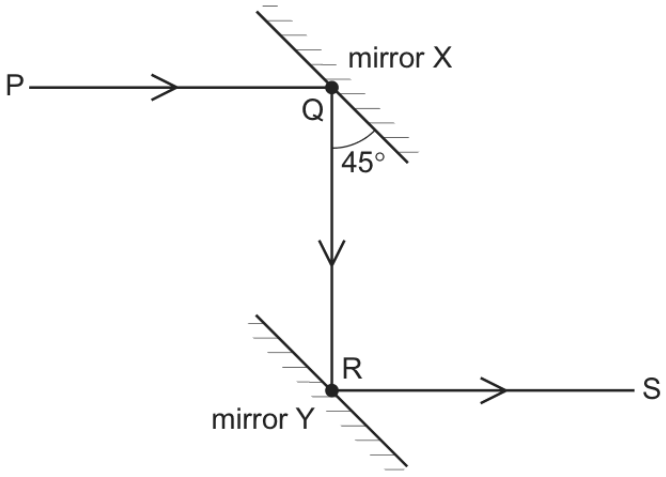
	symbol of angle X	relationship between X and Y
A	*i*	X = Y
B	*i*	X + Y = 90°
C	*r*	X = Y
D	*r*	X + Y = 90°

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 23

23	C	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 11 of 20

24 The diagram shows a ray PQ reflected by mirror X to a parallel mirror Y. The reflected ray along RS is parallel to PQ.



Which statement is correct?

A The angle between PQ and QR is 45°.

B The angle between QR and RS is 180°.

C The angle of incidence of PQ on mirror X is 60°.

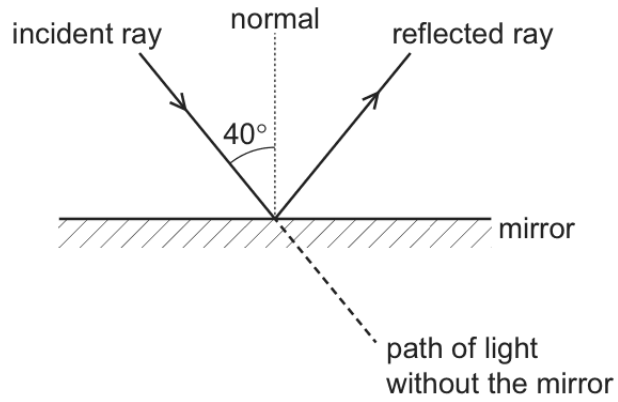
D The angle of incidence of QR on mirror Y is 45°.

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 24

24	D	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 10 of 16

24 A mirror is placed in the path of a ray of light.



Through which angle does the direction of the ray of light change?

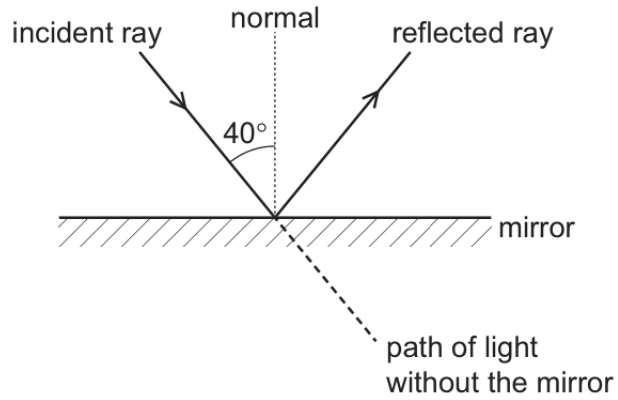
A 40° **B** 90° **C** 100° **D** 140°

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 24

24	C	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 9 of 16

22 A mirror is placed in the path of a ray of light.



Through which angle does the direction of the ray of light change?

A 40° **B** 90° **C** 100° **D** 140°

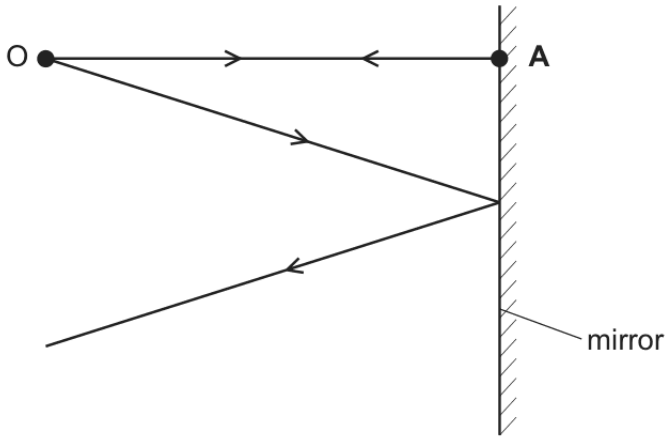
MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 22

22	C	**1**

QUESTION · 5054_s23_qp_11.pdf · source page 9 of 20

19 The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

At which position is the image formed?



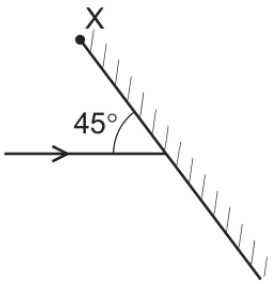
- **B**
- **C**
- **D**

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 19

19	B	**1**

QUESTION · 5054_s23_qp_11.pdf · source page 9 of 20

20 Light is incident on a mirror at an angle of 45° as shown. The mirror can be rotated about an axis into the page through point X.



The mirror is rotated until the light is reflected back along its original path.

Through which angle is the mirror rotated?

A 22.5° clockwise

B 22.5° anticlockwise

C 45° clockwise

D 45° anticlockwise

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 20

20	C	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 9 of 16

19 Which characteristics describe the image formed by a vertical plane mirror?

A real and upside down

B virtual and upright

C real and larger than the object

D virtual and smaller than the object

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 19

19	B	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 8 of 16

18 A plane mirror on a vertical wall forms an image of an object placed in front of it.

Which characteristics describe the image?

A real and inverted

B real and upright

C virtual and inverted

D virtual and upright

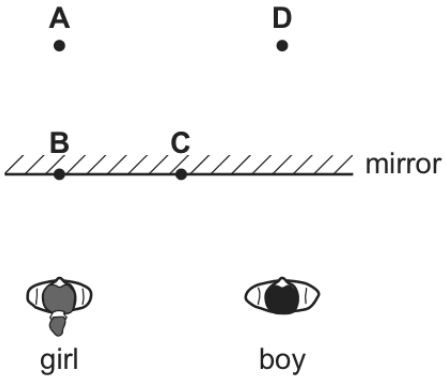
MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 18

18	D	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 11 of 20

26 A boy stands beside a girl in front of a large vertical plane mirror. They are equal distances from the mirror, as shown. The boy sees an image of the girl.

Where is the girl's image?



MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 26

26	A	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 9 of 16

23 A plane mirror forms the image of an object.

What are characteristics of the image formed?

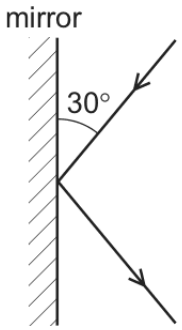
	size	type
A	same as object	real
B	same as object	virtual
C	smaller than object	real
D	smaller than object	virtual

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 23

23	B	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 8 of 16

20 The diagram shows a ray of light incident on a plane mirror.



NOT TO SCALE

What is the angle of reflection?

A 30° **B** 60° **C** 90° **D** 120°

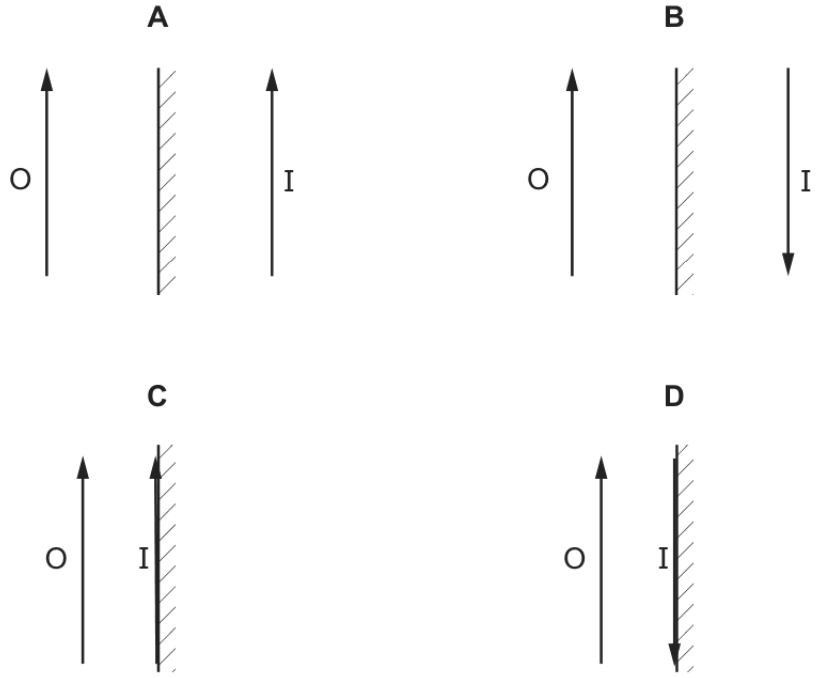
MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 20

20	B	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 8 of 16

21 An object O is placed in front of a plane mirror.

Which diagram shows the image I formed by the mirror?



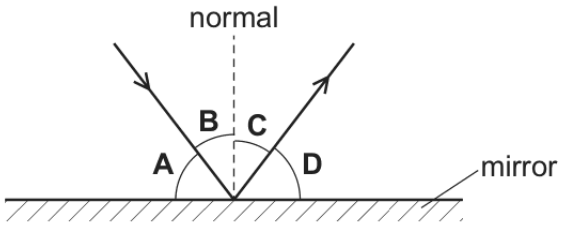
MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 21

21	A	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 8 of 20

19 The diagram shows light striking a plane mirror and reflecting.

Which angle is the angle of incidence?



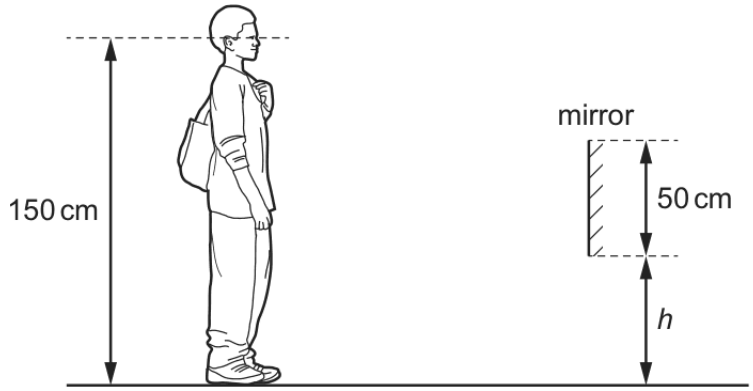
MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 19

19	B	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 9 of 20

20 A shoe shop puts a mirror on the wall so that customers can look at their shoes.

The length of the mirror is 50 cm. A customer has eyes 150 cm above ground level.



The bottom of the mirror is at height *h* above the ground.

What is the smallest value of *h* that allows the customer to see an image of his shoes in the mirror?

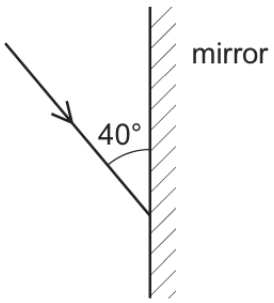
A 0 **B** 25 cm **C** 50 cm **D** 75 cm

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 20

20	B	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 9 of 16

21 The diagram shows a ray of light directed at a plane mirror.



What are the angle of incidence and the angle of reflection?

	angle of incidence / °	angle of reflection / °
A	40	40
B	40	50
C	50	40
D	50	50

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 21

21	D	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 9 of 16

19 A ray of light strikes a plane mirror at an angle of incidence of 20°.

The angle of incidence is then increased by 5°.

What is the new angle between the incident ray and the reflected ray?

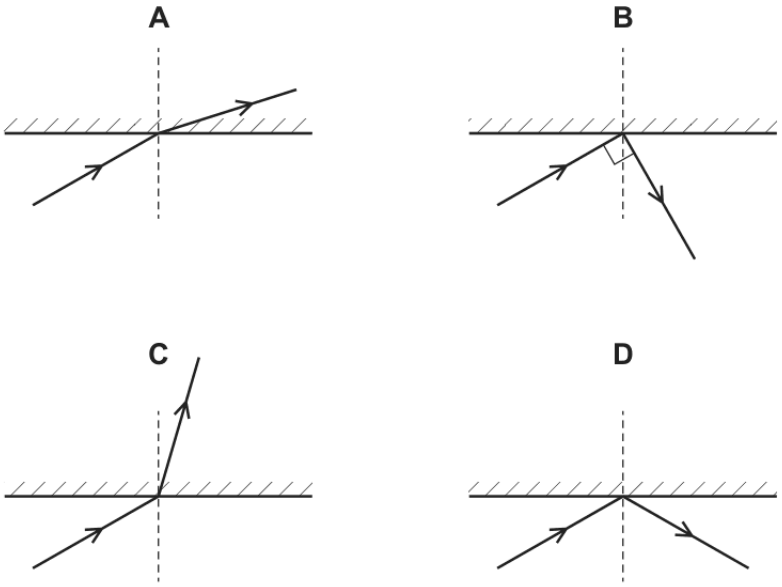
A 10° **B** 25° **C** 45° **D** 50°

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 19

19	D	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 11 of 16

28 Which diagram shows a ray of light that is reflected by a plane mirror?



MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 28

28	D	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 12 of 16

29 An object is placed in front of a plane mirror.

Which statement describes the image produced?

A real and smaller than the object

B real and the same size as the object

C virtual and smaller than the object

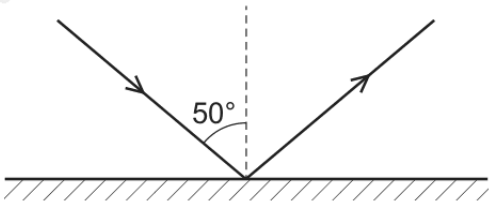
D virtual and the same size as the object

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 3 of 3 · row for Question 29

29	D	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 9 of 16

21 The ray diagram shows light reflecting off a plane mirror.



What is the angle between the incident and reflected rays?

A 40° **B** 50° **C** 80° **D** 100°

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 21

21	D	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 10 of 16

24 A woman stands in front of a plane mirror. She sees an image of herself in the mirror.

The woman moves 1.0 m further away from the mirror.

What happens to the distance between the woman and her image?

A It decreases by 1.0 m.

B It decreases by 2.0 m.

C It increases by 1.0 m.

D It increases by 2.0 m.

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 24

24	D	**1**

Chapter 3 Waves

3.2 Light

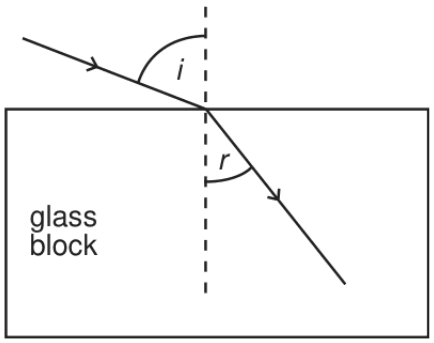
3.2.2 Refraction of light

113 entries: 76 Paper 1 questions, 37 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

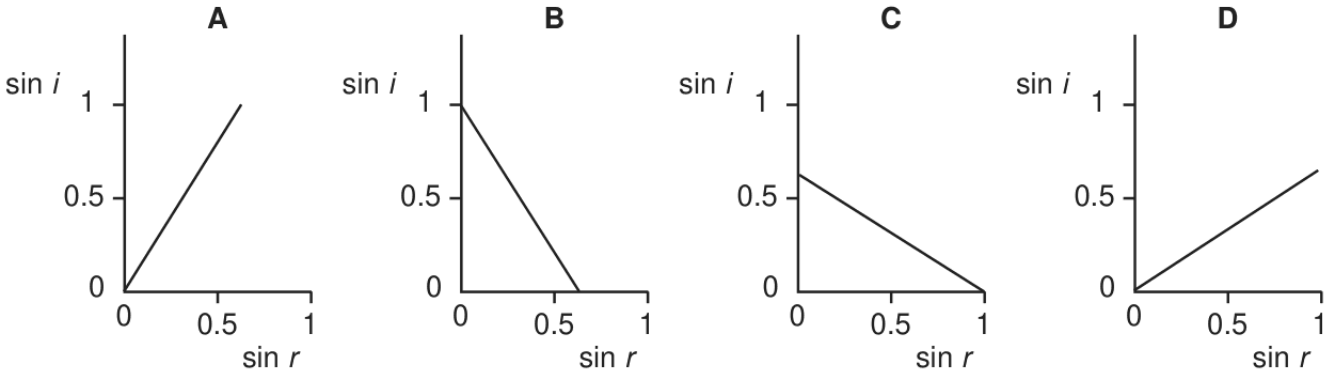
QUESTION · 5054_s04_qp_1.pdf · source page 10 of 20

21 A ray of light enters a glass block at an angle of incidence *i* producing an angle of refraction *r* in the glass.



Several different values of *i* and *r* are measured, and a graph is drawn of sin *i* against sin *r*.

Which graph is correct?



MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 21

21	**A**

For Examiner's Use

4 Fig. 4.1 shows an air bubble in water. The rays of light are incident on the air bubble.

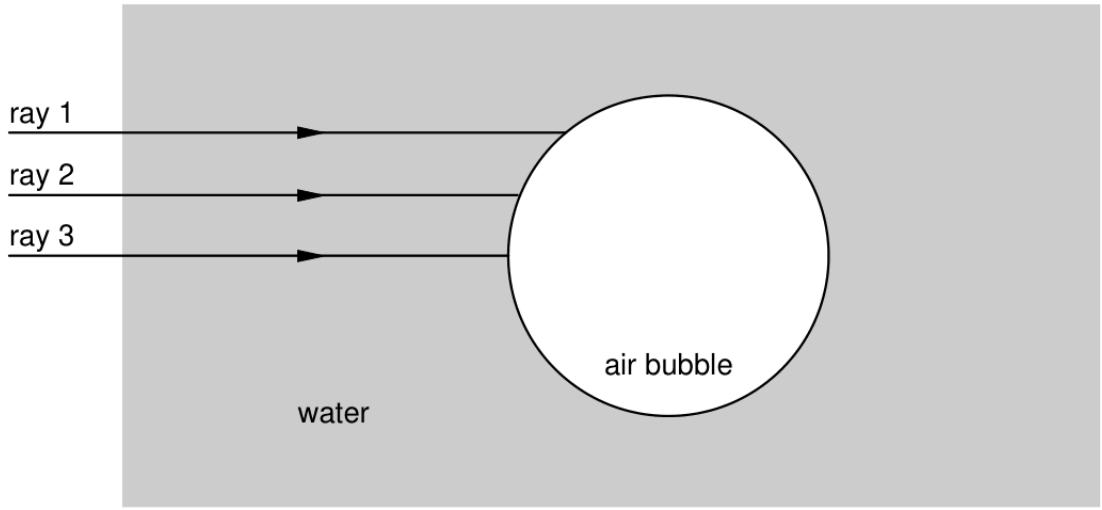


Fig. 4.1

The angle of incidence of ray 1 on the air bubble is greater than the critical angle.
The angle of incidence of ray 2 on the air bubble is less than the critical angle.
Ray 3 is perpendicular to the surface of the bubble.

The angle of incidence of ray 2 on the air bubble is 27° and the angle of refraction of ray 2 inside the air bubble is 37°.

(a) On Fig. 4.1, at the point where ray 1 meets the air bubble, mark

(i) the normal to the surface,

(ii) the angle of incidence.

[2]

(b) Complete Fig. 4.1 to show how all three rays continue after they meet the air bubble. [3]

(c) **(i)** Define what is meant by the *refractive index* of water.

.....

.....

.....

(ii) Calculate the refractive index of water.

refractive index = ...

9

[2]

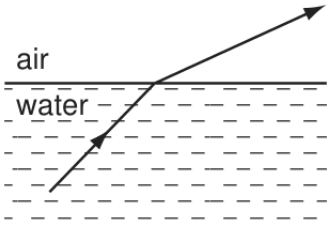
For Examiner's Use

4	**(a)**	**(i)**	correct normal (by eye to centre of circle)	**M1**	
			angle between normal and ray 1 marked	**A1**	
	(b)		ray 1 sensibly reflected **and no** refracted ray	**B1**	
			ray 2 bends upwards (ignore reflection)	**B1**	
			ray 3 undeviated (ignore all rays leaving bubble)	**B1**	
	(c)	**(i)**	sin i/sin r **or** ratio of speed in air/vacuum to speed in medium	**B1**	
			(**ignore** real/apparent depth)		
		(ii)	1.33 or 0.75		
			(accept 1.326, 1.3, 0.754, 0.8 **not** 1.325, 1, 0.76)	**B1**	**7**

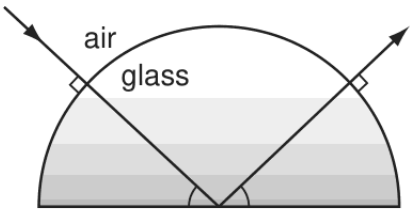
QUESTION · 5054_s05_qp_1.pdf · source page 9 of 20

22 In which diagram is the path of the light ray **not** correct?

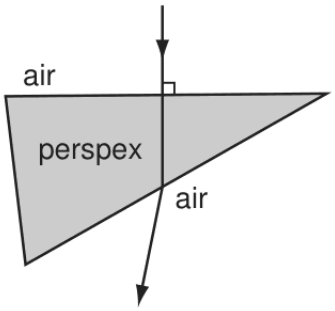
A



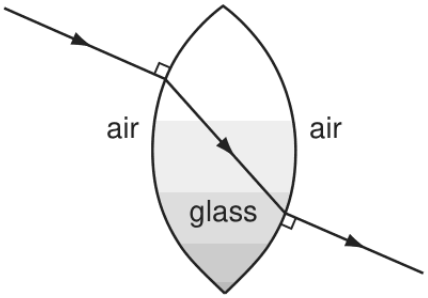
B



C



D



MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 22

22	**D**

For Examiner's Use

4 Fig. 4.1 and Fig. 4.2 show rays of light passing through the same semi-circular block of plastic.

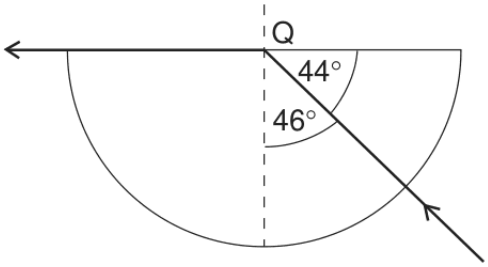


Fig. 4.1

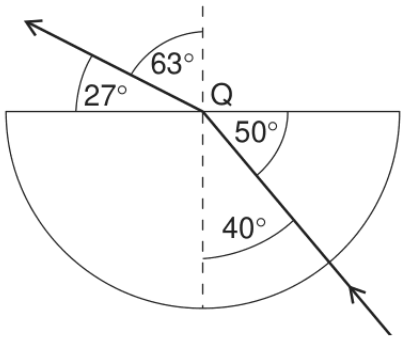


Fig. 4.2

Q is the centre of the straight side of the block.

(a) State the value of the critical angle in the plastic.

critical angle = ... [1]

(b) Explain what is meant by the *critical angle*.

.....

.....

.....

..... [2]

(c) Calculate the refractive index of the plastic. State the formula that you use.

refractive index = ... [3]

(d) Some light reflects back into the plastic at Q.

On Fig. 4.1, draw the reflected ray at Q. [1]

4	**(a)**	46 (°)	B1	
	(b)	angle of incidence	B1	
		when refracted ray is along surface **or** minimum angle of incidence for Total Internal Reflection	B1	
	(c)	sin *i*/sin *r* **or** 1/sin*C*	B1	
		sin 63/sin 40 **or** 1/sin 46	C1	
		1.39 (accept 1.3860 –1.3902)	A1	
	(d)	correct reflected ray by eye	B1	**[7]**

QUESTION · 5054_s07_qp_1.pdf · source page 10 of 20

21 What happens to light as it passes from glass into air?

A Its frequency decreases because its speed decreases.

B Its frequency increases because its speed increases.

C Its wavelength decreases because its speed decreases.

D Its wavelength increases because its speed increases.

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 21

21	**D**

For Examiner's Use

4 Fig. 4.1 shows part of an optical fibre.

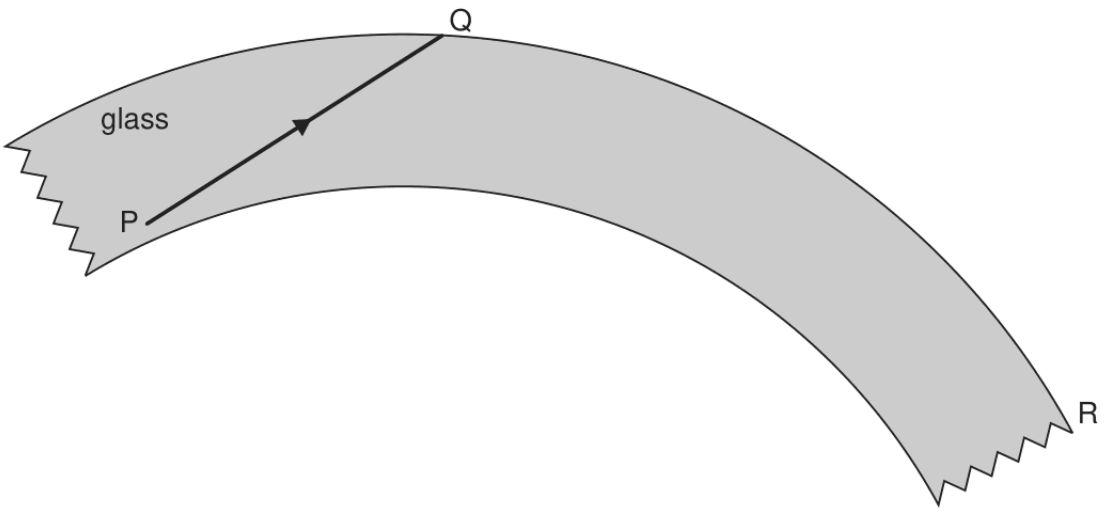


Fig. 4.1

The ray PQ undergoes total internal reflection in the optical fibre.

(a) On Fig. 4.1, continue the path of ray PQ until it reaches end R. [1]

(b) Explain what is meant by *total internal reflection*.

.....

.....

..... [1]

(c) Optical fibres are cheaper and lighter than copper wires. State one other advantage of using optical fibres rather than copper wires for telephone communications.

.....

.....

..... [1]

(d) The light in the optical fibre is travelling at a speed of 2.1×10^8 m/s and has a wavelength of 6.4×10^{-7} m.

Calculate the frequency of the light.

frequency = ... [2]

4 **(a)** reflections correct by eye B1

(b) **all** the ray **reflects** back (into the denser medium/glass)
or reflection **and** no refraction/escape into air B1

(c) more calls **or** greater bandwidth **or** more/faster data(/sec)/information **or** better quality **or** less power loss/energy loss/attenuation **or** greater distance (between repeaters) **or** harder to tap **or** less noise/interference B1

(d) $f = v/\lambda$ in any form numerical or algebraic C1
3.3×10^{14} Hz A1 [5]

11 **(a)** Fig. 11.1 shows a ray of light passing through the edge of a converging lens.

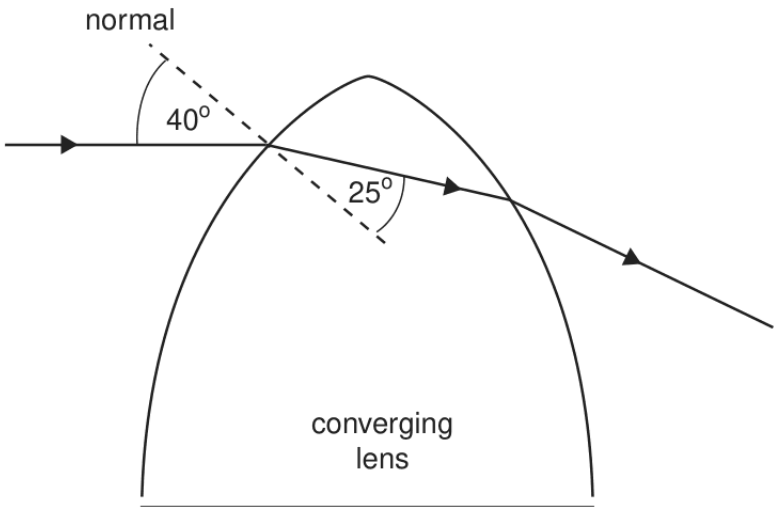


Fig. 11.1

(i) Describe what happens to the direction of the ray of light as it enters and leaves the lens. [2]

(ii) State what happens to the speed, frequency and wavelength of the light as it enters the lens. [3]

(iii) Calculate the refractive index of the glass used in the lens. [3]

(b) The focal length of the lens is 20 cm. An object is placed 50 cm from the lens and an image is formed on a screen.

(i) Explain what is meant by the *focal length* of a lens. You may draw a diagram if you wish. [2]

(ii) Draw a ray diagram to scale to show the formation of the image. [3]

(iii) The image is real. State **two** other properties of the image. [2]

11 (a) (i) (as it enters) bends towards normal B1
(as it leaves) bends away from the normal B1

(ii) speed **and** wavelength change C1
speed **and** wavelength decrease A1
frequency unaltered B1

(iii) sin(i)/sin(r) C1
sin 40°/sin 25° C1
1.5(2) (penalise °, accept 2 or more sig figs; **1.5 alone** with no working B1) A1 [8]

(b) Mark (i) and (ii) separately unless specifically referred to (i) in (ii)

(i) **Words:** distance between (principal) focus/focal point (**not** F) and lens M1
centre of lens A1
Diagram: F/(principal) focus/focal point marked **and** lens marked/curved faces/triangles
and correct arrow of some sort M1
f/FL/fl/focal length marked **and** arrow from centre of lens to F A1

(ii) diagram showing object, lens and **one** correct ray M1
second correct ray M1
correct image shown (½ < h < 1) A1

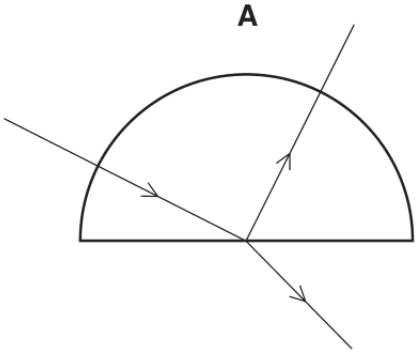
(iii) smaller / de-magnified / e.c.f **(ii)** B1
upside down B1

QUESTION · 5054_s08_qp_1.pdf · source page 10 of 20

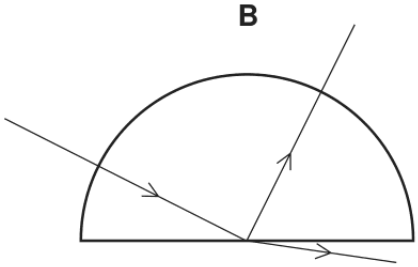
22 A ray of red light enters a semi-circular glass block normal to the curved surface.

Which diagram correctly shows the partial reflection and refraction of the ray?

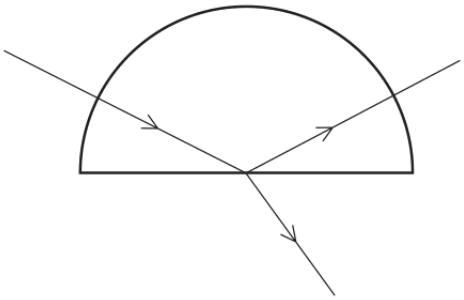
A



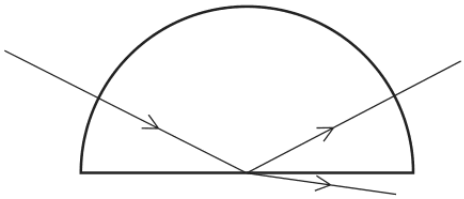
B



C



D

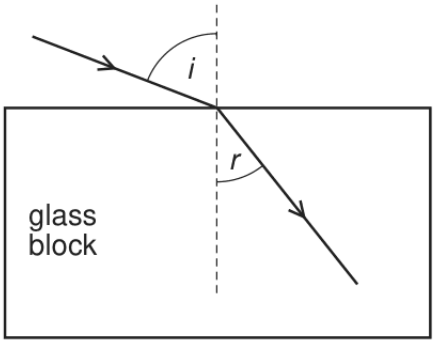


MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 22

22	**D**

QUESTION · 5054_w08_qp_1.pdf · source page 12 of 20

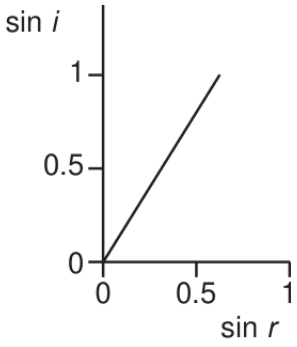
24 A ray of light enters a glass block at an angle of incidence *i* producing an angle of refraction *r* in the glass.



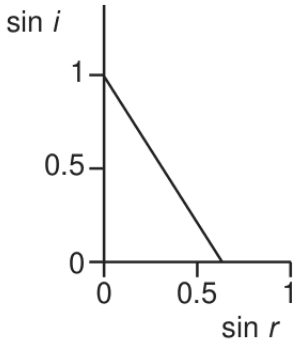
Several different values of *i* and *r* are measured, and a graph is drawn of sin *i* against sin *r*.

Which graph is correct?

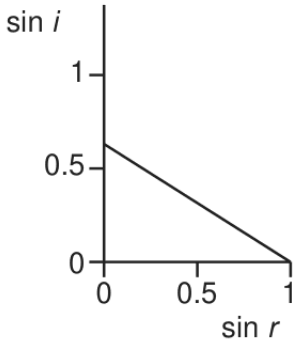
A



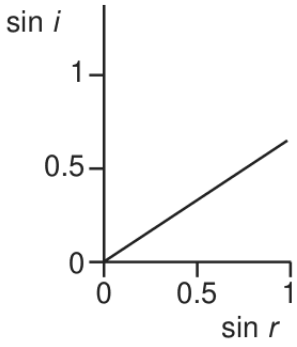
B



C



D



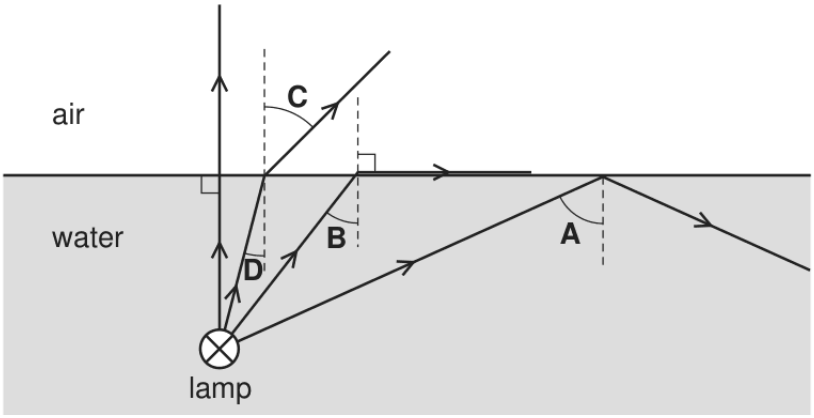
MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 24

24	**A**

QUESTION · 5054_s09_qp_1.pdf · source page 10 of 20

21 The diagram shows four rays of light from a lamp below the surface of some water.

What is the critical angle for light in water?



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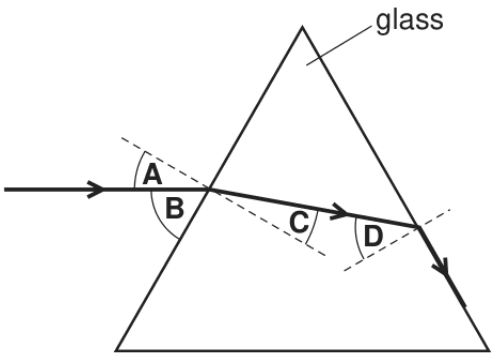
21	**B**

QUESTION · 5054_w09_qp_1.pdf · source page 11 of 20

21 The diagram shows the passage of a ray of light through a triangular glass block.

What is the critical angle of light in glass?



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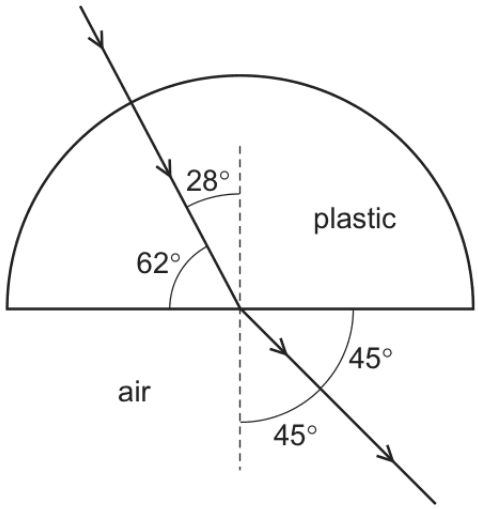
21	**D**

QUESTION · 5054_s10_qp_11.pdf · source page 9 of 20

20 A semi-circular block is made from a plastic. A ray of light passes through it at the angles shown.



To two decimal places, what is the refractive index of the plastic?

A 1.25 **B** 1.41 **C** 1.51 **D** 1.61

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 20

20 **C**

For Examiner's Use

5 Fig. 5.1 shows a ray of white light incident on a glass prism.

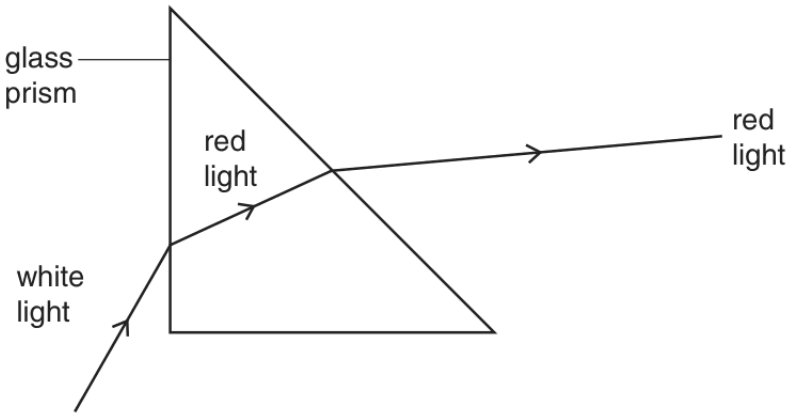


Fig. 5.1

Fig. 5.1 is not complete; it shows only the ray of red light produced from the white light.

(a) (i) On Fig. 5.1, draw the ray of blue light produced in, and beyond, the prism. [2]

(ii) State **two** colours of the spectrum found between the red and blue rays.

.....[1]

(b) Fig. 5.2 shows the same prism with the ray of white light incident at a different angle.

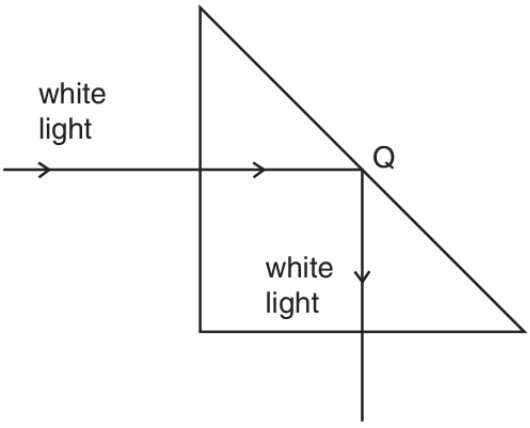


Fig. 5.2

Fig. 5.2 is complete; no spectrum is produced.

(i) Explain why no light emerges from the prism at Q.

.....

.....[1]

(ii) Explain why no spectrum is produced.

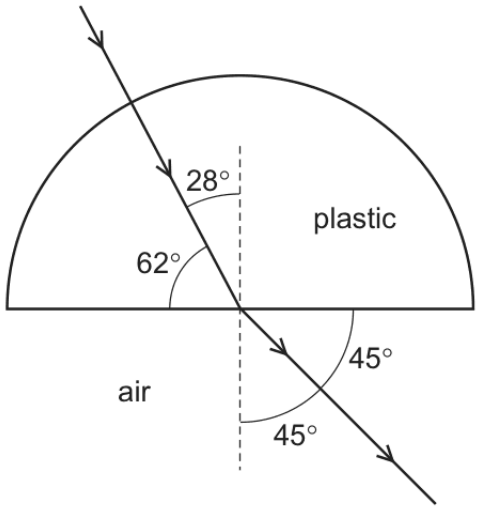
.....

.....[1]

5	**(a)**	**(i)**	correct direction of refraction at **both** faces (**not** along normal)	B1	
			blue below red **and** blue and red diverge	B1	[2]
		(ii)	any **two** from orange, yellow, green	B1	[1]
	(b)	**(i)**	**total internal reflection or** angle of incidence greater than critical angle	B1	[1]
		(ii)	all colours reflected at same angle **or** all have i = r	B1	[1]

QUESTION · 5054_s10_qp_12.pdf · source page 10 of 20

23 A semi-circular block is made from a plastic. A ray of light passes through it at the angles shown.



To two decimal places, what is the refractive index of the plastic?

A 1.25 **B** 1.41 **C** 1.51 **D** 1.61

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 23

23	**C**

For Examiner's Use

5 Fig. 5.1 shows a ray of light entering a semi-circular glass block and striking the glass surface at M, the mid-point of the straight face.

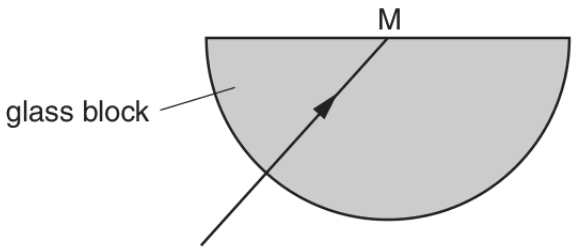


Fig. 5.1

(a) The ray of light strikes the glass surface at M with an angle of incidence *C* equal to the critical angle of light in glass.

(i) State what is meant by *critical angle*.

.....

.....[1]

(ii) On Fig. 5.1, mark and label the angle *C*. [1]

(iii) On Fig. 5.1, continue the ray of light after it strikes the glass surface at M. [1]

(b) Fig. 5.2 shows a second ray of light striking M.

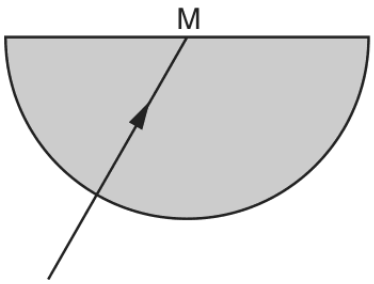


Fig. 5.2

This ray has an angle of incidence at M smaller than the critical angle.
On Fig. 5.2, continue this ray of light after it strikes the glass surface at M. [1]

(c) The refractive index of this glass is 1.5. A third ray of light enters the block from air with an angle of incidence of 50°. Calculate the angle of refraction.

angle= .....[2]

5 **(a)** **(i)** smallest angle of incidence for total internal reflection
or greatest angle of incidence that allows refraction
or angle of incidence for (refracted) ray along surface/angle of refraction 90° B1 [1]

(ii) correct angle marked to normal (by eye) B1 [1]

(iii) ray along surface or reflected ray correct (by eye) or both rays B1 [1]

(b) ray in air refracted away from normal B1 [1]

(c) refractive index = $\sin i / \sin r$ algebraic or numerical e.g. $1.5 = \sin 50 / \sin r$ C1

31° accept 30.71, 30.7 degree symbol required somewhere A1 [2]

[Total: 6]

QUESTION · 5054_w10_qp_11.pdf · source page 10 of 20

22 A ray of light strikes the surface of a glass block at an angle of incidence of 45°.

The refractive index of the glass is 1.5.

What is the angle of refraction inside the block?

A 28° **B** 30° **C** 45° **D** 67°

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 22

22	**A**

For Examiner's Use

9 **(a)** Fig. 9.1 shows a young boy lying on his back on the bottom of a swimming pool. He is holding his breath and his eyes are open. A red light is positioned on the ground at Q.

At first the boy's head is touching the pool wall. He notices that, as he slides away from the pool wall, his eye reaches a point P where he first sees the light at Q. Fig. 9.1 shows the boy in this position.

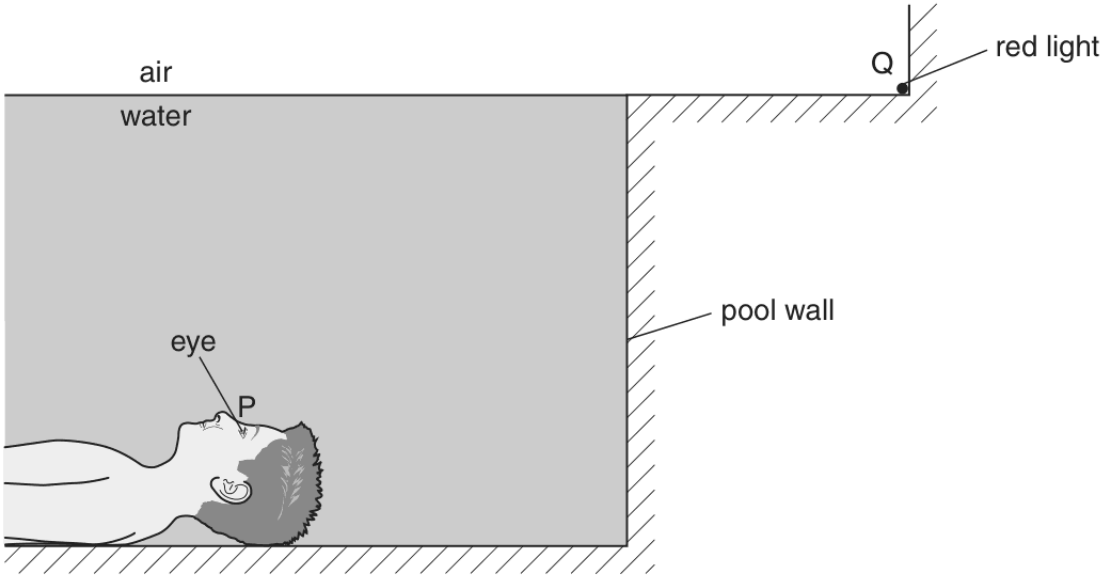


Fig. 9.1

(i) On Fig. 9.1, draw the ray of light travelling from Q to P. Mark the critical angle for light in water and label it C. [2]

(ii) Explain why the boy is unable to see the red light at Q when his eye is closer to the pool wall than P.

.....

.....

..... [2]

(iii) The critical angle is 49°. Calculate the refractive index of water.

refractive index = ... [2]

(iv) The red light is now replaced with a blue light. State the effect of this on the wavelength of the light in the air.

..... [1]

For Examiner's Use

(b) A small, very brightly illuminated display is located at the back of a projector. The projector lens produces an inverted and magnified image of the display on a white classroom wall.

Fig. 9.2 is a scale diagram showing the position and size of both the display and the image on the wall. R is a point on the display.

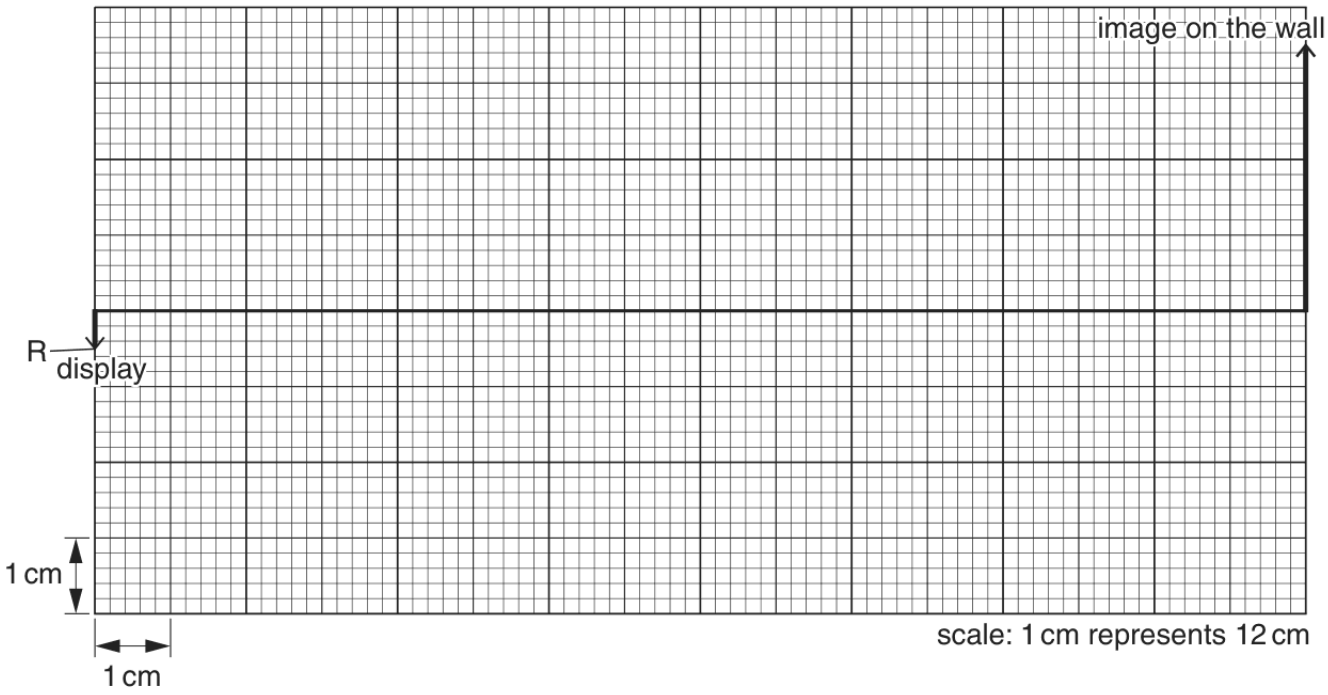


Fig. 9.2

(i) The image is inverted and magnified. State two other properties of the image.

1. .....

2. .....

[2]

(ii) On Fig. 9.2, draw the straight-line ray from R to the image. [1]

(iii) On Fig. 9.2, draw a vertical line representing the lens and label it L. [1]

(iv) A second ray from R to the image passes through a focal point (principal focus) of the lens. On Fig. 9.2, draw this ray and use it to mark this focal point. Label this focal point F. [2]

(v) Determine the focal length of the lens.

focal length = ... [2]

9	**(a)**	**(i)**	horizontal ray from Q to pool edge **and** on to P from corner	B1	
			critical angle marked *C* **or** obvious	B1	[2]
		(ii)	for $i = 90°$ **or** horizontal ray	B1	
			angle(in water) equals/cannot be less than critical/*C*	B1	[2]
		(iii)	(n =) sin i/sin r **or** 1/sin C **or** 1/n = sin C **or** sin 90°/sin 49° **or** 1/sin 49°	B1	
			1.3(2501)	B1	[2]
		(iv)	decreases	B1	[1]
	(b)	**(i)**	any **two** of: real less bright further from lens beyond 2*f*	B2	[2]
		(ii)	straight ray from R to **top** of image	B1	[1]
		(iii)	where ray crosses principal axis, vertical line (L **or** drawn lens)	B1	[1]
		(iv)	paraxial ray from R to lens refracted to top of image **or** paraxial ray from lens to top of image, traced back to R	M1	
			F marked	A1	[2]
		(v)	1.6 – 1.9 cm **or** attempt to use 1/u + 1/v	C1	
			19 – 23 cm (2 sig. fig. only)	A1	[2]

QUESTION · 5054_w10_qp_12.pdf · source page 9 of 16

23 A ray of light strikes the surface of a glass block at an angle of incidence of 45°.

The refractive index of the glass is 1.5.

What is the angle of refraction inside the block?

A 28° **B** 30° **C** 45° **D** 67°

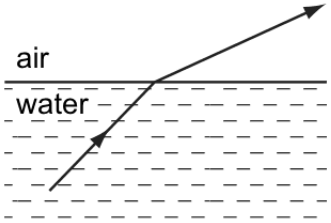
MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 23

23	**A**

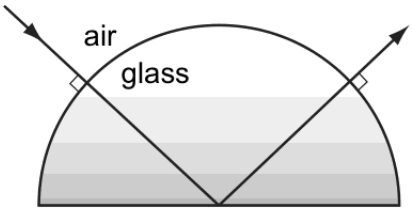
QUESTION · 5054_s11_qp_11.pdf · source page 9 of 16

22 In which diagram is the path of the light ray **not** correct?

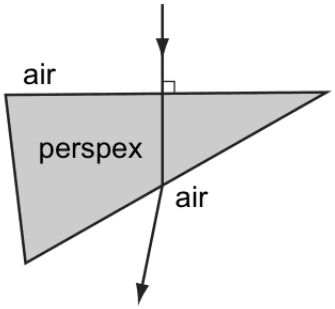
A



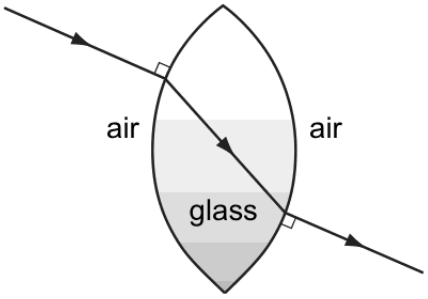
B



C



D



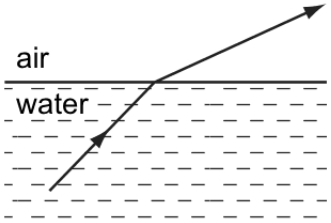
MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 22

22	**D**

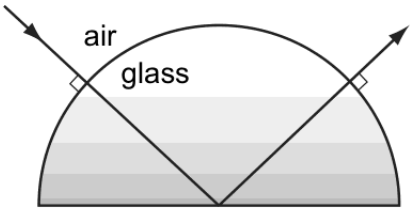
QUESTION · 5054_s11_qp_12.pdf · source page 9 of 16

21 In which diagram is the path of the light ray **not** correct?

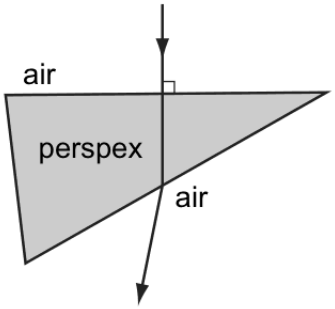
A



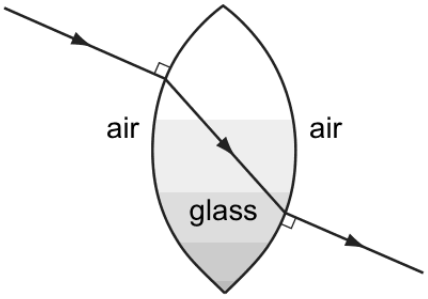
B



C



D



MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 21

21	**D**

10 (a) Describe an experiment to measure the critical angle for light in glass or perspex.

Your answer should include a labelled diagram.

.....

.....

.....

.....

.....

.....

.....

.....[5]

(b) Fig. 10.1 represents a simple camera.

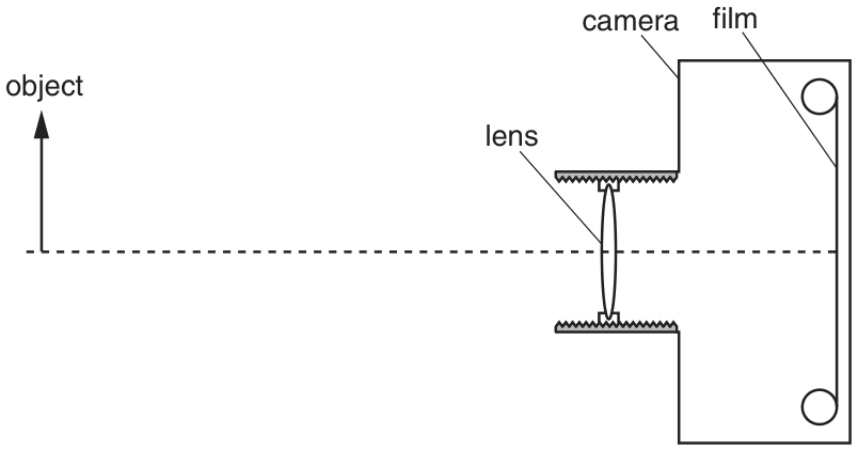


Fig. 10.1 (to scale)

(i) State the type of lens used in this simple camera.

.....[1]

(ii) Draw two rays from the top of the object to show how the image is formed on the film. Mark and label the image on the film. [3]

(iii) Define the term *linear magnification*.

.....

.....[1]

(iv) Fig. 10.1 is drawn to scale. Determine the linear magnification of the object shown in Fig. 10.1.

magnification = ...[1]

(v) Apart from its size, state one other property of the image formed by the lens.

.....[1]

(vi) Explain why, when taking photographs of other objects, it may be necessary to move the lens towards the film.

.....

.....

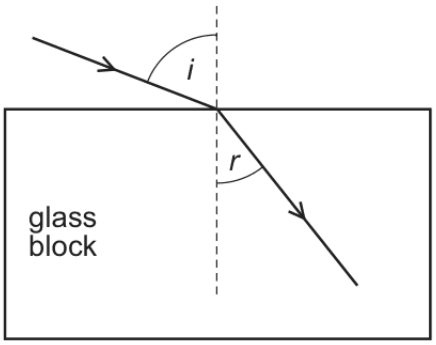
.....

.....[3]

10	**(a)**	suitable block (semicircular/rectangular/prism)	B1	
		suitable source of rays (e.g. ray box; pins on incident ray; laser **not** torch) **must** be labelled on diagram or clear in text	B1	
		diagram showing incident ray in glass/perspex (no arrow needed) **and** correct refraction out into air	B1	
		adjustment of (angle of incidence of) ray until along surface/just no longer emerges	B1	
		(measure) correct angle marked or described clearly or C marked on diagram	B1	
	(b) **(i)**	converging or convex	B1	
	(ii)	ray from top through middle of lens undeviated	B1	
		other ray from top of object to same position on film	M1	
		correct image labelled/drawn/marked	A1	
	(iii)	ratio of size/height/length/distance of image to size/height/length/distance of object	B1	
	(iv)	0.4(0) (±0.05) no ecf **(iii)**	B1	
	(v)	upside down // inverted // real // other side of lens to object // nearer lens than object	B1	
	(vi)	(otherwise) not focussed // to/adjust focus // to produce a clear/sharp image //	B1	
		(otherwise) rays do not converge on film // to converge rays onto film // image on film // object at different distance	C1	
		(otherwise) image formed in front of film // object now further	A1	[15]

QUESTION · 5054_w11_qp_11.pdf · source page 9 of 20

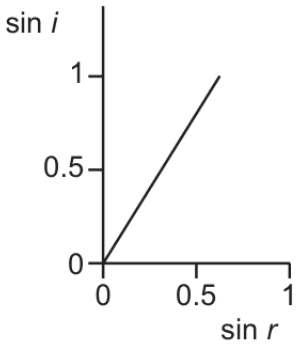
21 A ray of light enters a glass block at an angle of incidence *i*, producing an angle of refraction *r* in the glass.



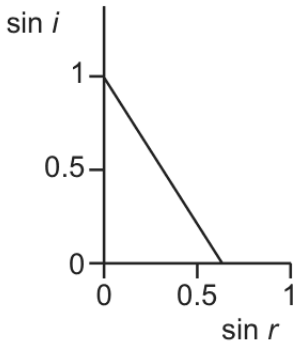
Several different values of *i* and *r* are measured, and a graph is drawn of sin *i* against sin *r*.

Which graph is correct?

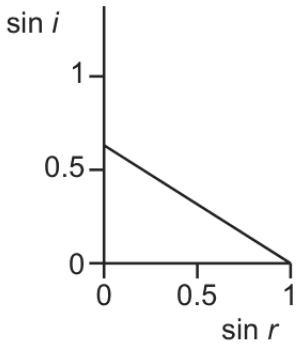
A



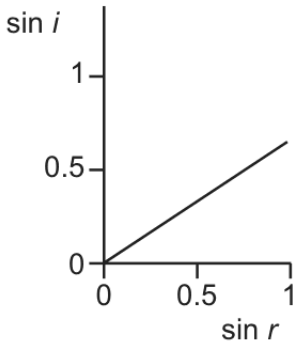
B



C



D



MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 21

21	**A**

4 Fig. 4.1 shows a glass lens in air and its two focal points F_1 and F_2.

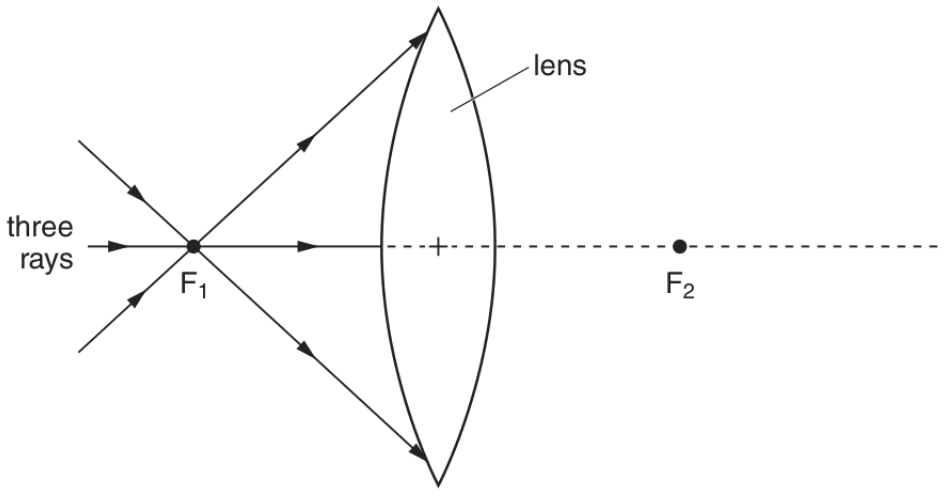


Fig. 4.1

Three rays of light pass through F_1 to the lens.

(a) On Fig. 4.1, continue the three rays through the lens and into the air. [2]

(b) State what happens to the speed of light on

(i) entering the glass lens from air,

.....[1]

(ii) leaving the lens and returning to the air.

.....[1]

(c) Light of wavelength 6.0×10^{-7} m travels in air at a speed of 3.0×10^{8} m/s.

(i) Calculate the frequency of this light.

frequency = ...[2]

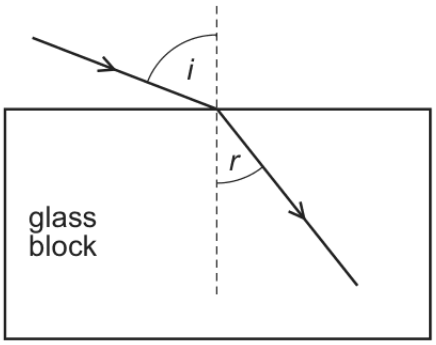
(ii) State the effect, if any, on the frequency as the light enters the glass from air.

.....[1]

4	**(a)**	one outer ray parallel to principal axis three rays parallel to the principal axis	C1 A1	
	(b) **(i)**	(speed) reduced **or** slows down	B1	
	(ii)	(speed) returns to original value/$3.0 \times 10^{8}\,m/s$	B1	
	(c) **(i)**	$(f =)\ c/\lambda$ **or** $3.0 \times 10^{8} / 6.0 \times 10^{-7}$ $5(.0) \times 10^{14}\,Hz$	C1 A1	
	(ii)	no effect/unchanged/$(f =)\ 5(.0) \times 10^{14}\,Hz$	B1	[7]

QUESTION · 5054_w11_qp_12.pdf · source page 9 of 20

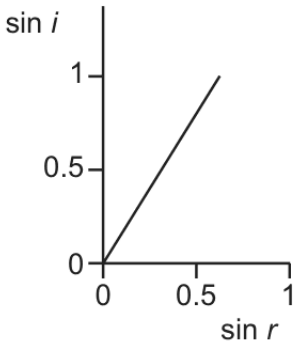
20 A ray of light enters a glass block at an angle of incidence *i*, producing an angle of refraction *r* in the glass.



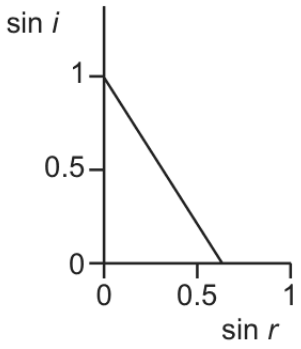
Several different values of *i* and *r* are measured, and a graph is drawn of sin *i* against sin *r*.

Which graph is correct?

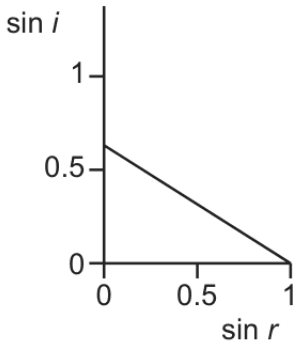
A



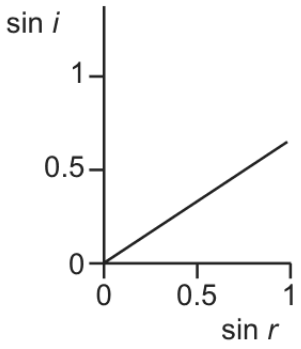
B



C



D



MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 20

20 **A**

QUESTION · 5054_s12_qp_11.pdf · source page 9 of 16

23 Light is incident on one face of a glass block at an angle of incidence of 40°. The glass block is in air.

The refractive index of the glass is 1.46.

What is the angle of refraction inside the glass block?

A 26° **B** 27° **C** 58° **D** 70°

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 23

23	**A**

10 (a) Fig. 10.1 shows the path of a ray of blue light as it passes through a glass prism.

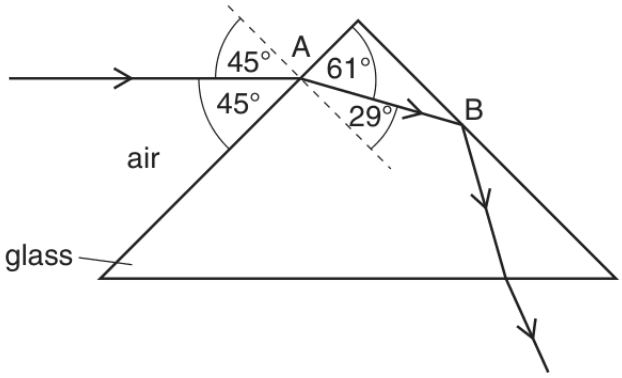


Fig. 10.1

(i) State the wave term used to describe what happens to the ray of light at A.

.....[1]

(ii) Using angles from Fig. 10.1, calculate the refractive index of the glass.

refractive index = ...[3]

(iii) Explain why the ray does not emerge from the prism at B.

.....

.....[2]

(iv) Fig. 10.2 shows a second, horizontal, ray of blue light striking the prism at point C.

On Fig. 10.2, continue the path of the second ray through and out of the glass prism. [2]

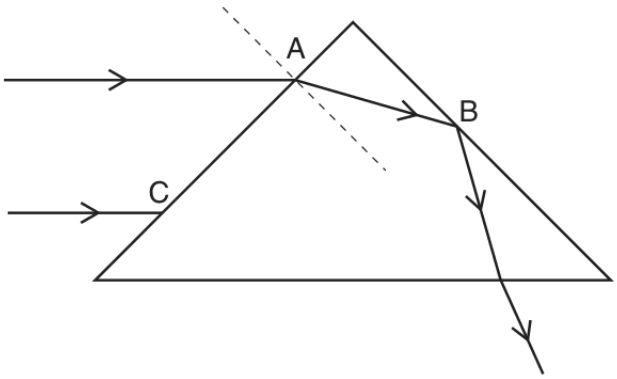


Fig. 10.2

(b) The camera lens shown in Fig. 10.3 is used to photograph the object O.

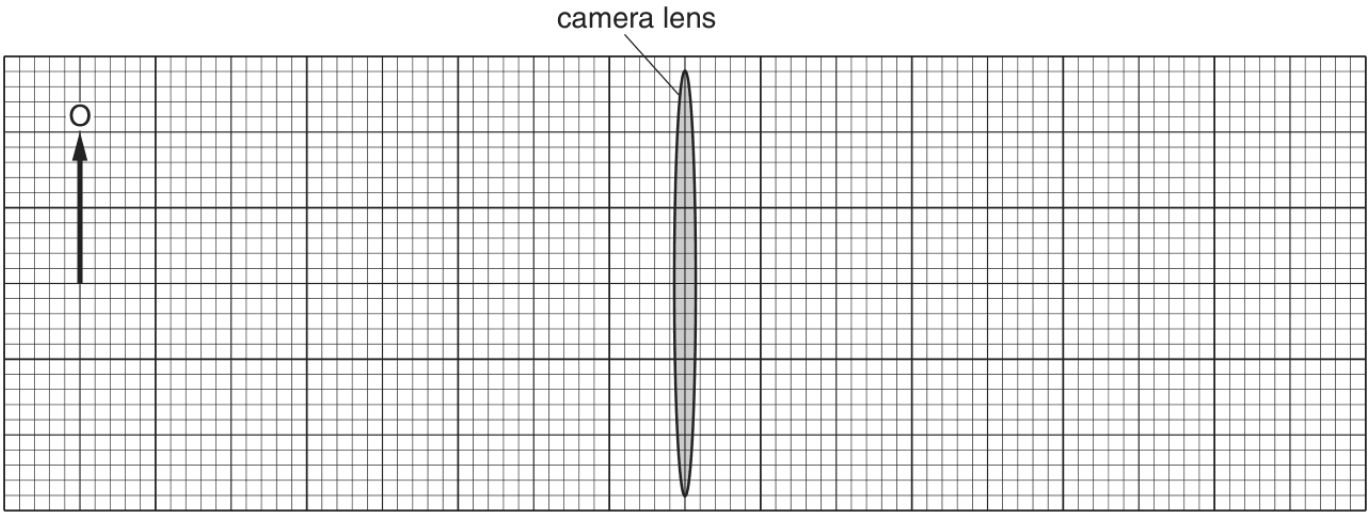


Fig. 10.3 (full scale)

The object O is 2.0 cm high and is placed 8.0 cm from the centre of the lens. The lens has a focal length of 3.0 cm.

(i) Draw rays on Fig. 10.3 to find the position and height of the image formed by the lens. Label the image I. [3]

(ii) Determine the height of the image.

.....[1]

(iii) The image formed by the lens is a real image.

1. Explain the difference between a real image and a virtual image.

.....

.....

.....[1]

2. Explain how a converging lens is used to produce and view a virtual image.

.....

.....

.....

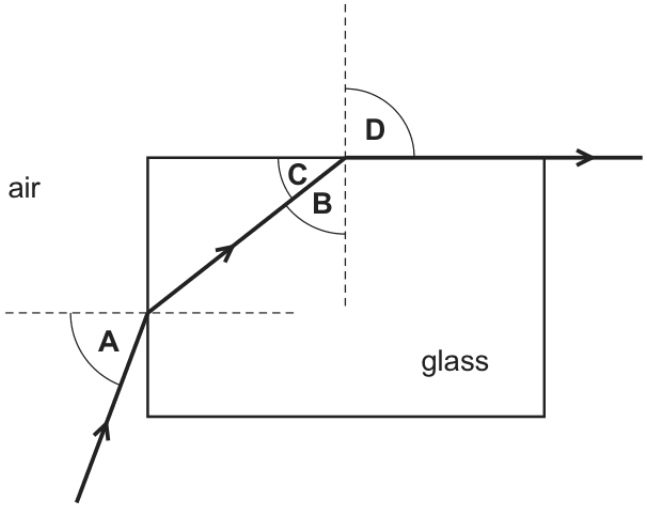
.....[2]

10	**(a)**	**(i)**	refraction	B1	
		(ii)	(n =) sin i/sin r	C1	
			sin 45°/sin 29°	C1	
			1.4585 to more than 1 sig. fig.	A1	
		(iii)	the angle of incidence/incident angle is greater than the critical angle	B1	
			total internal reflection occurs	B1	
		(iv)	correct refraction at C with ray parallel to AB	B1	
			correct reflection (and correct refraction on other face i.e. downwards)	B1	
	(b)	**(i)**	Any TWO of: undeviated ray through centre of lens ray parallel to axis through point 3 cm from lens on right after lens ray through point 3 cm to left of lens parallel to axis after lens	M2	
			rays converge and vertical image drawn and labelled I	A1	
		(ii)	1.2 ± 0.2 cm	B1	
		(iii)	**1.** real image (can be) formed on screen; virtual image not found on screen; rays converge on real image; rays do not converge on virtual image; rays only appear/seem to come from a point on virtual image	B1	
			2. place object within focal length; between lens and focal point/principal focus	B1	
			view from other side of lens; look through lens; image same side as/behind object	B1	[15]

QUESTION · 5054_s12_qp_12.pdf · source page 10 of 16

23 Light travels through a glass block as shown.

Which angle is the critical angle for light in the glass?



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23	**B**

5 Optical fibres are used to transmit telephone signals. Fig. 5.1 shows a ray of light that strikes the inside surface of an optical fibre at P.

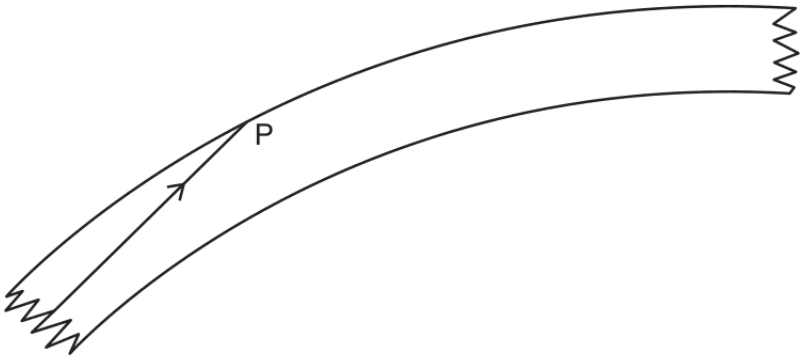


Fig. 5.1

(a) State one advantage of using optical fibres to transmit telephone signals.

.....

.....[1]

(b) **(i)** On Fig. 5.1, draw a normal at P and mark the angle of incidence with the letter *i*. [1]

(ii) State and explain what happens to the ray at P. Use the term *critical angle* in your answer.

.....

.....

.....[2]

(c) The optical fibre is made of glass of refractive index 1.5.
At the start of the optical fibre, the ray enters the glass from air.
The angle of incidence in the air is 60°.

Calculate the angle of refraction in the glass.

angle = ...[2]

5 **(a)** more telephone signals (at one time)
OR great(er) bandwidth; more data (per sec); more signals
OR faster data/information transfer
OR less attenuation; less energy/power/signal loss;
OR long(er) distance (before regeneration)
OR (more) secure
OR less noise/interference OR high(er) quality/clear(er) B1

(b) **(i)** correct normal and angle marked B1

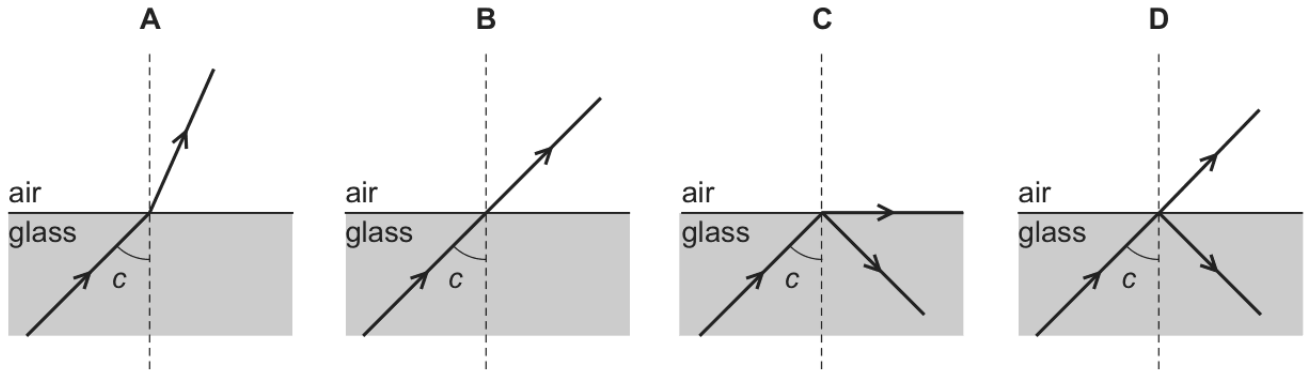
(ii) total internal reflection B1
angle of incidence is larger than critical angle B1

(c) (n =) sin i/sin r in any form numerical or algebraic C1
35(.2644)° **unit ° needed** A1 [6]

QUESTION · 5054_w12_qp_11.pdf · source page 10 of 16

23 In the following diagrams, the angle *c* is the critical angle.

Which diagram shows the correct path of the light ray?



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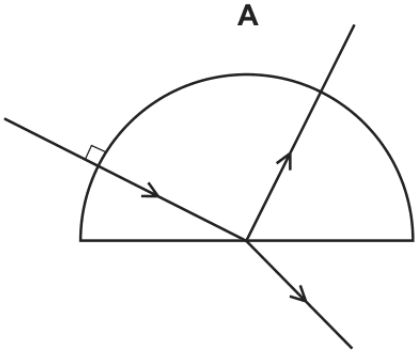
23	**C**

QUESTION · 5054_w12_qp_12.pdf · source page 9 of 16

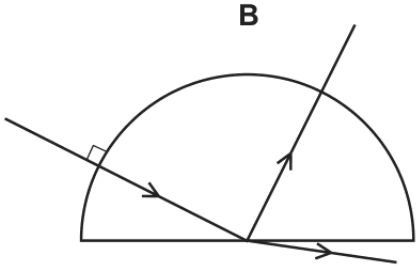
23 A ray of red light enters a semi-circular glass block normal to the curved surface.

Which diagram shows the partial reflection and refraction of the ray?

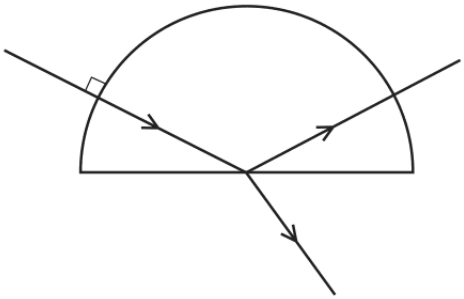
A



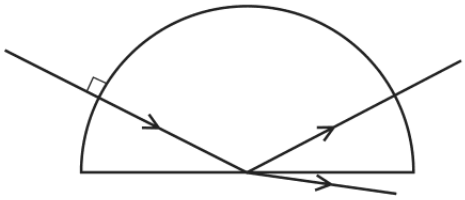
B



C



D



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23	**D**

For Examiner's Use

10 A laser produces red light of frequency 4.7×10^{14} Hz. The speed of light in glass is 2.0×10^{8} m/s.

(a) Calculate the wavelength in glass of light from this laser.

wavelength = ...[2]

(b) Describe an experiment to verify *the law of reflection* for light. You may include a diagram in your answer.

.....

.....

.....

.....

.....

.....

.....[5]

(c) Fig. 10.1 shows a ray of light travelling in an optical fibre. The ray strikes the side of the fibre at P.

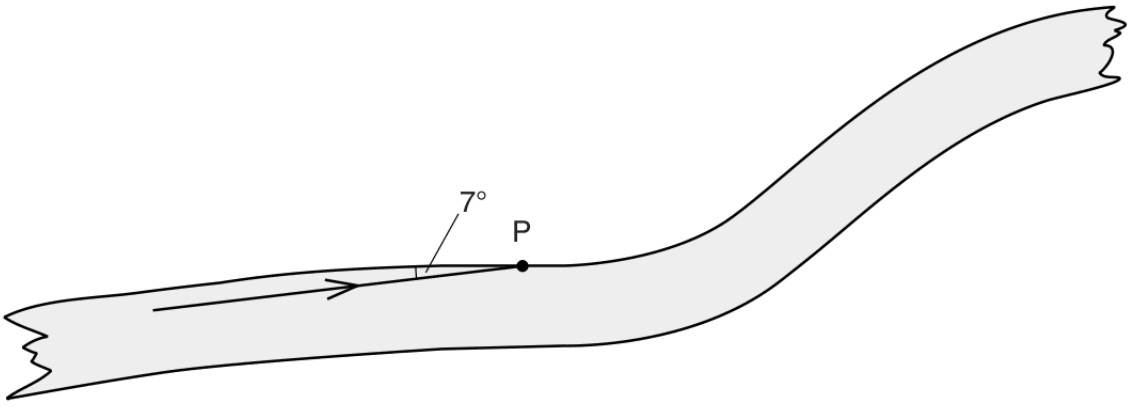


Fig. 10.1

The angle between the ray and the side of the fibre is 7°.

For Examiner's Use

(i) Determine the angle of incidence of the ray at P.

angle = ...[1]

(ii) State and explain what happens to the ray at P.

.....

.....

.....[2]

(d) A room is illuminated by wall lamps. Fig. 10.2 shows a mirror on the wall behind one of the lamps.

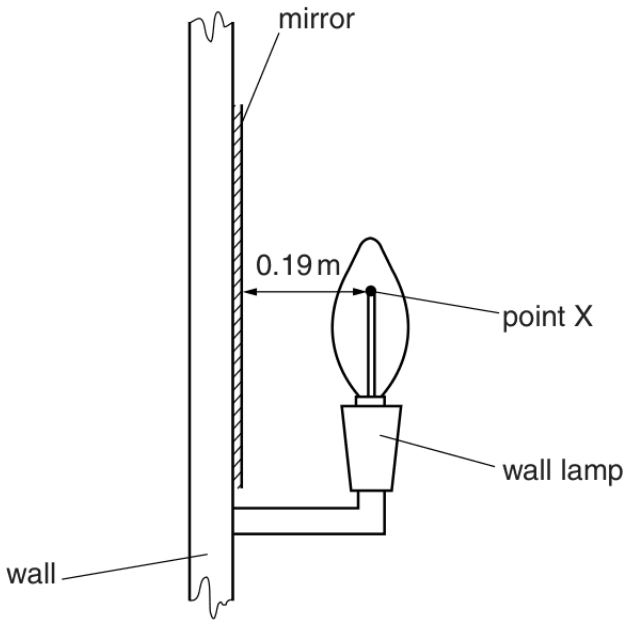


Fig. 10.2 (not to scale)

X is a point on the filament of the lamp. It is 0.19 m in front of the mirror.

(i) On Fig. 10.2, draw rays from X and locate the image of X. Label the image I. [3]

(ii) State the distance between I and the mirror.

distance = ...[1]

(iii) Suggest one advantage of placing a mirror behind the lamp in the room.

.....

.....[1]

10 (a) $(\lambda =)\, v/f$ **or** $2 \times 10^8 / 4.7 \times 10^{14}$ C1
4.3×10^{-7} m A1 [2]

(b)

raybox/light source/ laser **and** mirror		pin(s) **and** mirror	B1
shine ray at mirror		place two pins	B1
mark rays	**or**	two more pins in line with image	B1
measure *i* and *r* **and** equal		measure *i* and *r* **and** equal	B1
repeat		repeat	B1 [5]

(c) (i) 83° B1

(ii) total internal reflection **or** TIR **cao** B1
angle of incidence exceeds critical angle B1 [3]

(d) (i) (at least) one ray from X to mirror M1
(at least) **two** rays from X to mirror **and** correct reflections A1
rays traced back to marked I **or** I marked in correct place (by eye) B1

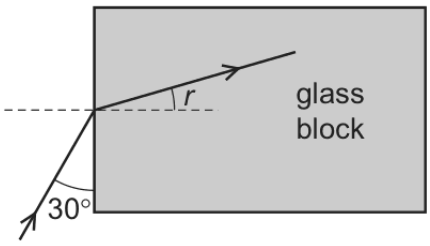
(ii) 0.19 m B1

(iii) less/no light wasted **or** hall brighter B1 [5]

[Total: 15]

QUESTION · 5054_s13_qp_11.pdf · source page 9 of 20

20 A ray of light meets the face of a glass block at an angle of 30° as shown.



The refractive index of the glass is 1.5.

What is the angle of refraction *r* inside the glass block?

A 19° **B** 20° **C** 35° **D** 40°

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 20

20	**C**

For Examiner's Use

5 Fig. 5.1 shows a ray of light that enters a semicircular glass block at A.
At B, some of the light is reflected and some light leaves the glass and travels along the surface.

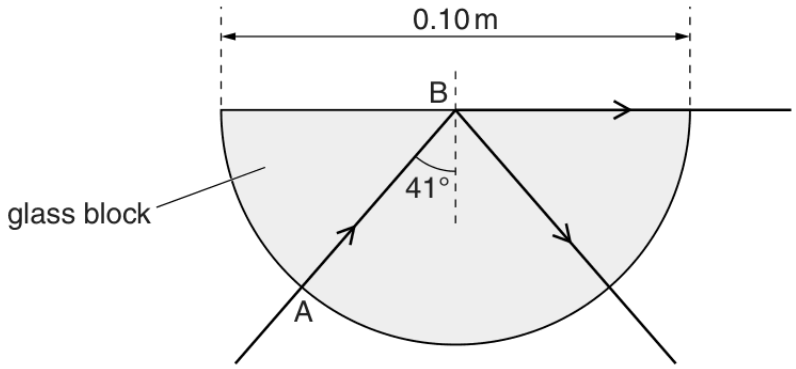


Fig. 5.1

(a) State the name of the angle of incidence marked 41°.

.....[1]

(b) Rays of light are incident at B with different angles of incidence.

(i) On Fig. 5.2a, the angle i_1 is less than 41°.
Draw the path taken by the ray of light after B.

(ii) On Fig. 5.2b, the angle i_2 is greater than 41°.
Draw the path taken by the ray of light after B.

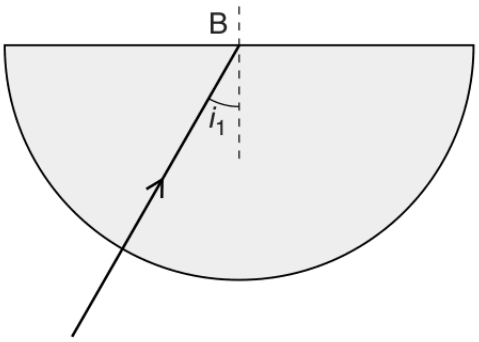


Fig. 5.2a

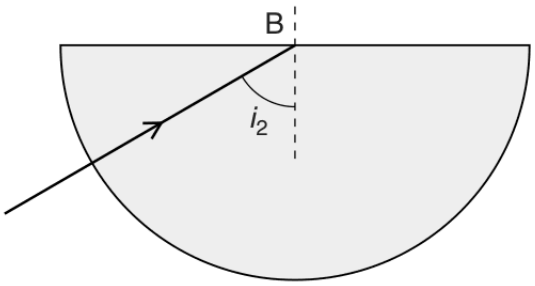


Fig. 5.2b [2]

(c) The speed of light in the glass block is 2.0×10^8 m/s.
The diameter of the glass block is 0.10 m.

Calculate the time taken for the light to travel from A to B.

time = ...[2]

5	**(a)**	critical angle	B1	
	(b)	**(i)** light refracted out into air **and** bent away from normal (ignore reflected ray)	B1	
		(ii) correct internal reflection (by eye) **and** no refracted ray (**not** at 90°)	B1	
	(c)	(t =) distance/speed in any form numerical or algebraic (e.g. d/s, s/v $10/2 \times 10^8$)	C1	
		2.5×10^{-10} s	A1	[5]

For Examiner's Use

10 A student traces the path of a ray of blue light as it enters and as it leaves a glass prism. Fig. 10.1 shows the trace obtained by the student.

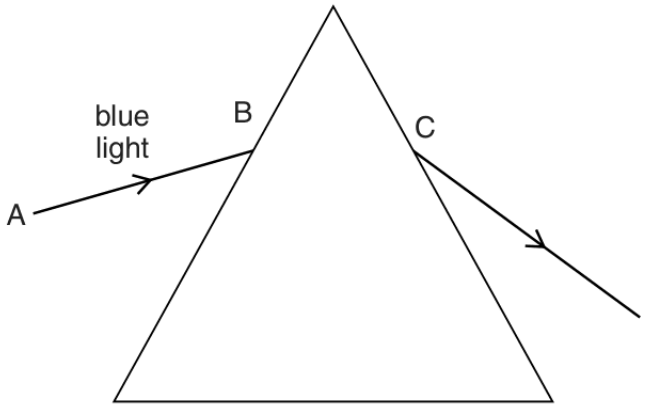


Fig. 10.1

(a) On Fig. 10.1, draw and label, at the point B, the normal, the angle of incidence *i* and the angle of refraction *r*. [3]

(b) State, in terms of the properties of light waves, why the light refracts at B.

.....

..... [1]

(c) The angle of incidence for the ray of blue light at B is 45°. The refractive index of the glass is 1.5. Calculate the angle of refraction at B.

angle of refraction = ... [3]

(d) The student performs another experiment with a ray of red light along the line AB.

On Fig. 10.1, show the path taken by this ray of light as it passes through and leaves the prism. [2]

For Examiner's Use

(e) The student performs another experiment with a semicircular glass block and a ray of white light. Fig.10.2 shows the path taken by this ray of light as it enters the glass at P until it hits the straight edge at Q.

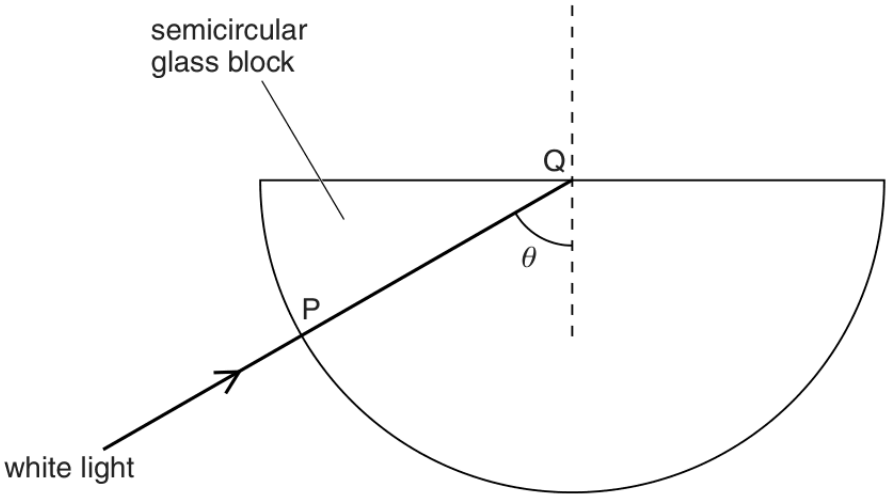


Fig. 10.2

The student finds that there is no change in direction as the ray enters the glass at P and that no light passes out of the glass at Q.

(i) Explain why the ray does not change direction at P.

.....

..... [1]

(ii) Explain why no light passes out of the glass at Q.

.....

.....

..... [2]

(iii) On Fig. 10.2, draw the complete path followed by this ray. [1]

(iv) The student directs the ray of white light into the glass along different paths, so that the angle θ is slowly reduced.

Describe what happens to the ray at Q.

.....

.....

..... [2]

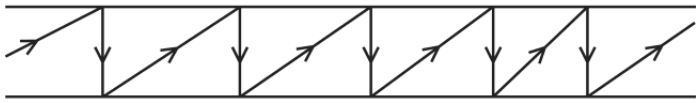
10	**(a)**	correct normal by eye	B1	
		correct angle of incidence between candidate's normal and incident ray	B1	
		correct angle of refraction marked between candidate's normal and BC	B1	[3]
	(b)	decrease / change in speed / wavelength	B1	[1]

	(c)	n = sin i/sin r seen in any form	C1	
		(sin r =) sin 45°/1.5 **or** 0.47(14) seen	C1	
		28(.1)°	C1	[3]
	(d)	refracts less at first face and on correct side of normal	B1	
		refraction at second face away from normal so that red ray and blue ray are diverging	B1	[2]
	(e) **(i)**	angle of incidence is 0 **or** ray along normal/perpendicular to glass	B1	
	(ii)	angle of incidence/θ is larger than critical angle	B1	
		total internal reflection occurs	B1	
	(iii)	reflected ray drawn correctly **and** emerging without refraction from block	B1	
	(iv)	(eventually) light emerges (into air at Q) **or** light refracts (out at Q) **or** (weak) refracted ray appears	B1	
		light emerging at Q coloured in some way **or** correct description of movement of reflected ray (as θ decreases)	B1	[6]
				[Total: 15]

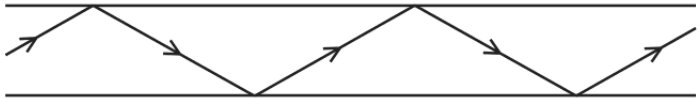
QUESTION · 5054_w13_qp_11.pdf · source page 9 of 20

20 Which diagram represents the reflection of light along an optical fibre?

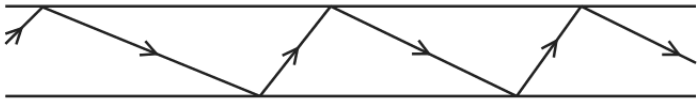
A



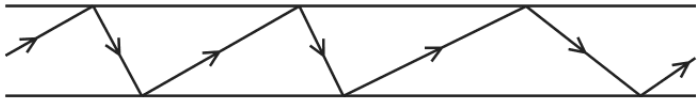
B



C



D



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20 **B**

For Examiner's Use

10 A student goes for a walk in the mountains. During a storm, she sees lightning strike a hillside in the distance. Several seconds later, she hears the thunder caused by the lightning.

(a) (i) Explain why she hears the thunder several seconds after she sees the lightning.

.....

.....[1]

(ii) The student knows the distance to the hillside. She waits for lightning to strike the hillside again. Describe how she can determine a value for the speed of sound in air.

.....

.....

.....

.....[2]

(b) After the storm, the student sees a rainbow.

(i) State the speed of light in air.

speed = ...[1]

(ii) Calculate the wavelength in air of light of frequency 7.5×10^{14} Hz.

wavelength = ...[2]

(iii) State the colours of the spectrum in order of increasing wavelength.

.....

.....[2]

For Examiner's Use

(c) In the laboratory, the student sends blue light into a glass prism placed on a sheet of paper. The arrangement is shown in Fig. 10.1.

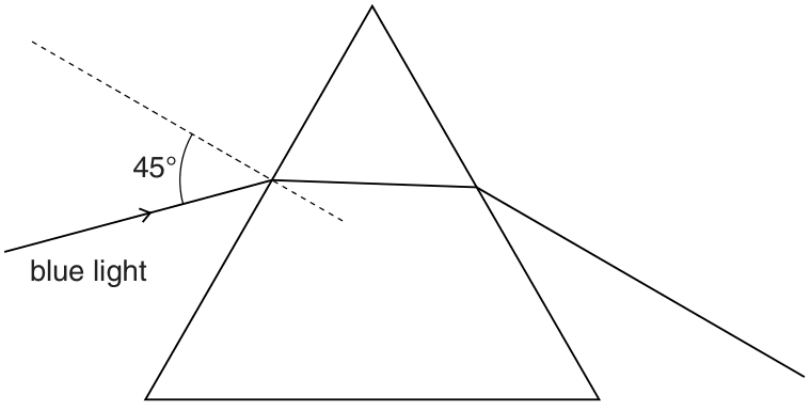


Fig. 10.1

The blue light enters the prism with an angle of incidence of 45°.

(i) On Fig. 10.1, mark the angle of refraction in the glass and label it *r*. [1]

(ii) The student wishes to determine the angle of refraction. On the sheet of paper, she draws the path the ray takes in the glass. Describe how she could do this.

.....

.....

.....[2]

(iii) **1.** State the formula that relates the angle of incidence *i*, the angle of refraction *r* and the refractive index *n*.

.....

.....[1]

2. The angle of refraction in the glass is 28°.

Calculate the refractive index of the glass.

refractive index = ...[1]

(iv) The student sends a ray of red light to the glass prism along the same path as the blue light.

On Fig. 10.1, mark the path taken by the red light after it enters the prism. [2]

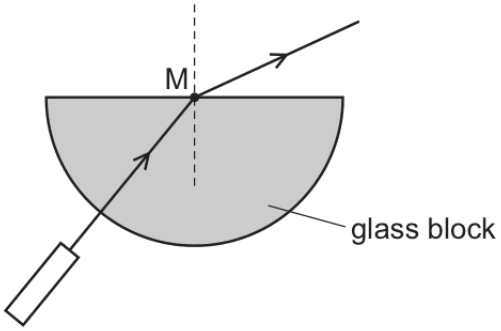
10	**(a)**	**(i)**	speed of sound is (much) less than the speed of light (accept quoted values)	B1	
		(ii)	**measure** the time delay (between the lightning and thunder)	B1	
			divide distance by time/delay	B1	[3]

	(b)	**(i)**	3.0×10^8 m / s	B1	
		(ii)	(λ =) *c*/*f* **or** $3.0 \times 10^8 / 7.5 \times 10^{14}$	C1	
			4.0×10^{-7} m	A1	
		(iii)	(in any order) blue, green, orange, red, yellow, (indigo), (violet) **or** VIBGYOR	C1	
			violet, indigo, blue, green, yellow, orange, red	A1	[5]
	(c)	**(i)**	correct angle clear/labelled *r*	B1	
		(ii)	mark/determine entrance and exit points (e.g. trace rays back to glass)	B1	
			join/draw line between entrance and exit points	B1	
		(iii)	**1.** *n* = sin *i*/sin *r*	B1	
			2. 1.5/1.51/1.506176 with no unit (**not** just 1.5 without working out)	B1	
		(iv)	correct direction of refraction at **both** faces	M1	
			completely correct (above blue)	A1	[7]

[Total: 15]

QUESTION · 5054_w13_qp_12.pdf · source page 11 of 20

23 A ray of red light from a laser passes into a semi-circular glass block.



What is shown at M?

A dispersion

B rarefaction

C reflection

D refraction

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23	**D**

QUESTION · 5054_w13_qp_12.pdf · source page 12 of 20

24 A ray of light strikes the surface of a glass block at an angle of incidence of 45°.

The refractive index of the glass is 1.8.

What is the angle of refraction inside the block?

A 23° **B** 25° **C** 45° **D** 81°

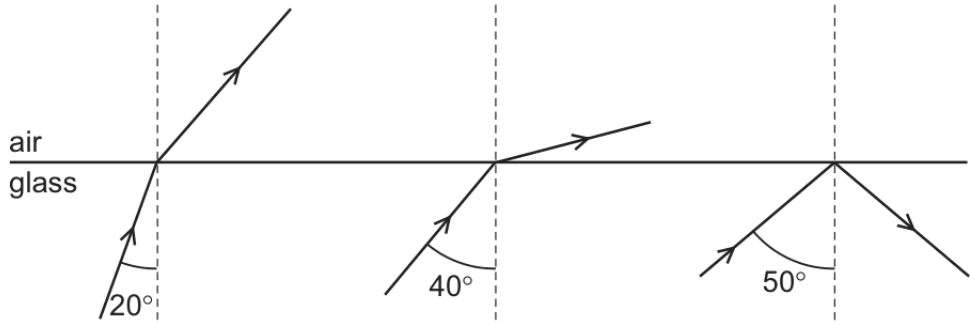
MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 24

24	**A**

QUESTION · 5054_s14_qp_11.pdf · source page 10 of 16

25 Three rays of light are incident on the boundary between a glass block and air.

The angles of incidence are different.



What is a possible critical angle for light in the glass?

A 15° **B** 30° **C** 45° **D** 60°

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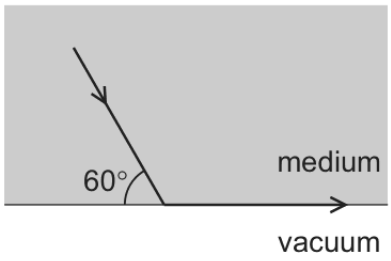
25	**C**

QUESTION · 5054_w14_qp_11.pdf · source page 11 of 20

21 The diagram shows light travelling through a medium. The light reaches the boundary with a vacuum as shown. The light emerges travelling along the surface.



What is the refractive index of the medium?

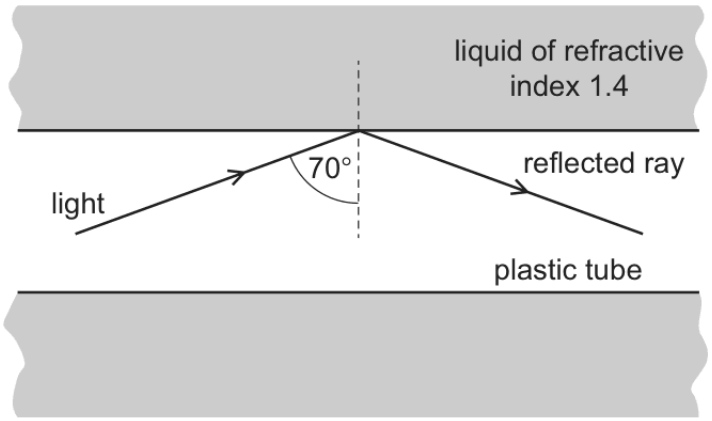
A $\frac{\sin 60°}{\sin 30°}$ **B** $\frac{\sin 60°}{\sin 90°}$ **C** $\frac{\sin 90°}{\sin 30°}$ **D** $\frac{\sin 90°}{\sin 60°}$

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 21

21	**C**

QUESTION · 5054_w14_qp_12.pdf · source page 7 of 16

16 A plastic tube is immersed in a liquid of refractive index 1.4. Light travelling in the plastic tube strikes the inside surface at an angle of incidence of 70°. The light undergoes total internal reflection.



What describes the values of the critical angle in the plastic and the refractive index of the plastic?

	critical angle in plastic	refractive index of plastic
A	greater than 70°	greater than 1.4
B	greater than 70°	less than 1.4
C	less than 70°	greater than 1.4
D	less than 70°	less than 1.4

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 16

16 **C**

10 **(a)** State the speed of light in air.

.....[1]

(b) Fig. 10.1 shows a ray of blue light passing from air into a glass block and refracting at the surface.

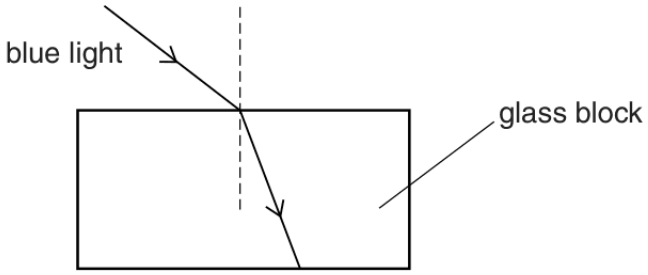


Fig. 10.1 (not to scale)

(i) As the light enters the glass, state what happens to

1. the speed of the light,

.....[1]

2. the frequency of the light,

.....[1]

3. the wavelength of the light.

.....[1]

(ii) On Fig. 10.1, mark and label the angle of incidence *i* and the angle of refraction *r*. [2]

(c) The refractive index of glass for blue light is 1.5.

(i) A ray of blue light strikes the surface of a glass block at an angle of incidence of 89°.

Calculate the angle of refraction of the light in the block.

angle of refraction = ...[3]

(ii) Explain why the angle of refraction of blue light in glass is always less than 45°.

.....

.....

.....

.....[2]

(d) Blue light, travelling in air, strikes the side of a different glass block and continues in the same direction as it enters the glass block. Fig. 10.2 shows the ray of light and the shape of the glass block. The critical angle for this glass is 42°.

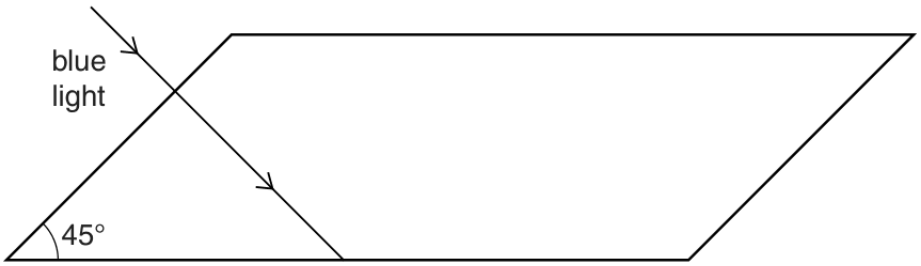


Fig. 10.2

(i) Explain why the light continues in the same direction as it enters the glass block.

.....

.....

.....[2]

(ii) On Fig. 10.2, complete the path of the light until it leaves the glass. [2]

MARK SCHEME · 5054_w14_ms_22.pdf · source page 4 of 5 · Question 10

10 (a) 3.0×10^8 m / s — B1 **[1]**

(b) (i) **1.** decreases **cao** — B1
2. no change **cao** — B1
3. decreases **cao** — B1

(ii) **1.** *i* correctly marked (to normal) — B1
2. *r* correctly marked (to normal) — B1 **[5]**

(c) (i) sin *i* / sin *r* = *n* **or** sin *i* / sin *r* = 1.5 — C1
sin 89 / sin *r* = 1.5 **or** sin 89 / 1.5 **or** 0.67(0.666565) — C1
42° **or** 41.8025° — A1

(ii) *i* equal to / close to 90° **and** *r* less than 45° — sin *i* / sin 45 = 1.5 — $\sin^{-1}(1/n) / \sin^{-1}(1/1.5)$ **and** 41.8° — B1
or **or**
i never bigger than 89° / 90° — sin *i* >1 — *r* not be more than *c* — B1 **[5]**

MARK SCHEME · 5054_w14_ms_22.pdf · source page 5 of 5 · Question 10

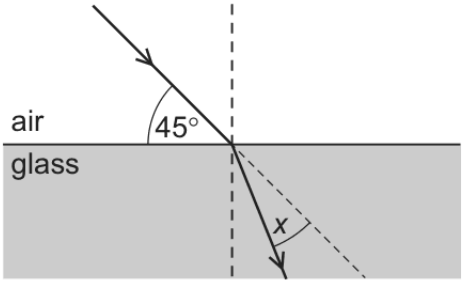
(d) (i) (sin) *i* = 0 **or** ray enters directly / along normal / perpendicular — **or** — wavefront / light hits surface all together — B1
(sin) *r* = 0 **or** no refraction — all slows down together — B1

(ii) correct reflection at bottom surface (by eye) — M1
second correct reflection at top and no refraction at either point — A1 **[4]**

[Total: 15]

QUESTION · 5054_s15_qp_11.pdf · source page 8 of 16

22 A ray of light is incident on the surface of a glass block, as shown in the diagram below.



The refractive index of the glass is 1.5.

The light ray changes direction when entering the glass.

What is the angle *x* through which the ray moves?

A 30° **B** 28° **C** 17° **D** 15°

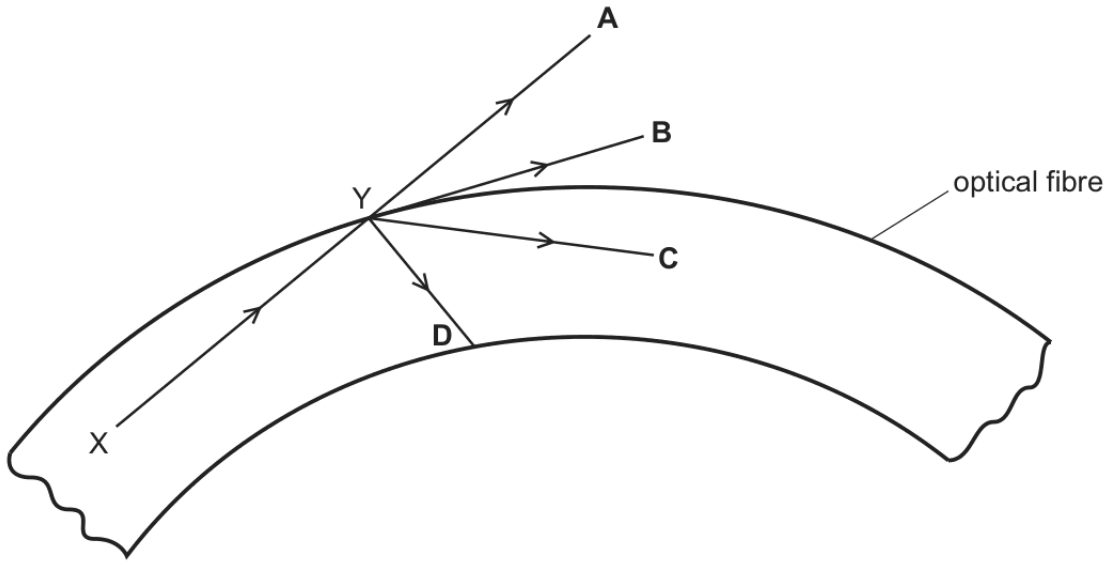
MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 22

22	**C**

QUESTION · 5054_s15_qp_11.pdf · source page 9 of 16

24 A ray of light travels from X to Y along an optical fibre. The angle of incidence at Y is greater than the critical angle.

In which direction does the ray of light travel after reaching point Y?



MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 24

24	**C**

5 Fig. 5.1 shows a ray of light entering and passing along an optical fibre.

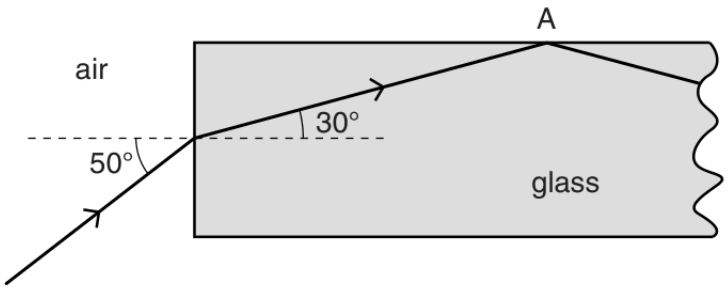


Fig. 5.1

(a) Calculate the refractive index of the glass in the optical fibre.

refractive index = ...[2]

(b) Explain why the ray of light is totally internally reflected at A.

.....

.....

.....[2]

(c) Both optical fibre and copper wire are used to transmit data.

Optical fibre is cheaper and can carry more data per second than copper wire.

State one other advantage of using optical fibre rather than copper wire to transmit data.

.....

.....[1]

MARK SCHEME · 5054_s15_ms_21.pdf · source page 2 of 5 · Question 5

5 **(a)** sin *i* / sin *r* or sin 50 / sin 30 C1
1.5(321) A1

MARK SCHEME · 5054_s15_ms_21.pdf · source page 3 of 5 · Question 5

(b) moving from more dense to less dense medium B1
or moving to lower refractive index (air)
angle of incidence is greater than critical angle B1

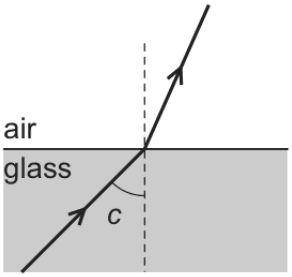
(c) less heat loss / more efficient B1
less chance of hacking / more secure / less interference
less reduction in signal / less need for boosting / larger distances possible / thinner or less bulky

QUESTION · 5054_s15_qp_12.pdf · source page 10 of 16

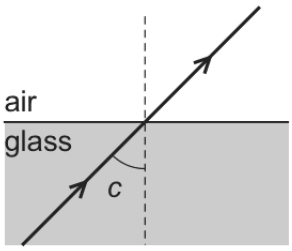
25 A ray of light in glass is incident on the surface at an angle *c*. The angle *c* is the critical angle.

Which diagram shows what happens to the light?

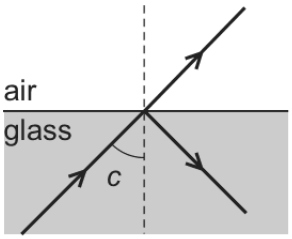
A



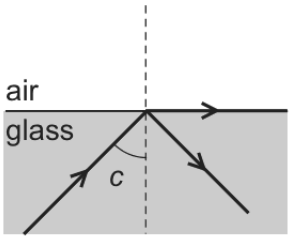
B



C



D



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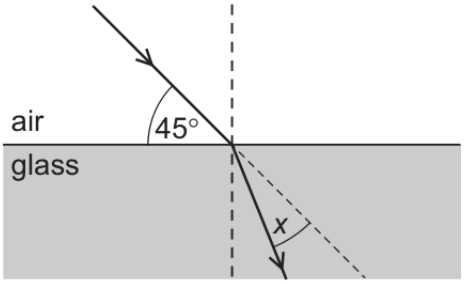
25	**D**

QUESTION · 5054_s15_qp_12.pdf · source page 10 of 16

26 A ray of light is incident on the surface of a glass block, as shown in the diagram below.



The refractive index of the glass is 1.5.

The light ray changes direction when entering the glass.

What is the angle *x* through which the ray moves?

A 30° **B** 28° **C** 17° **D** 15°

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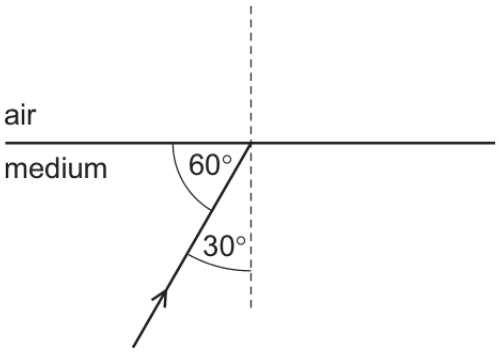
26	**C**

QUESTION · 5054_w15_qp_11.pdf · source page 9 of 16

21 A ray of light in a transparent medium of refractive index 1.8 is incident on the surface as shown. The light enters air.



What is the angle between the refracted ray and the normal in air?

A 29° **B** 33° **C** 54° **D** 64°

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21	**D**

7 Light enters a parallel-sided glass block at A. The angle between the side of the block and the ray of light in air is 35°.

The light then strikes the edge of the block at B. Fig. 7.1 shows the ray of light and the glass block.

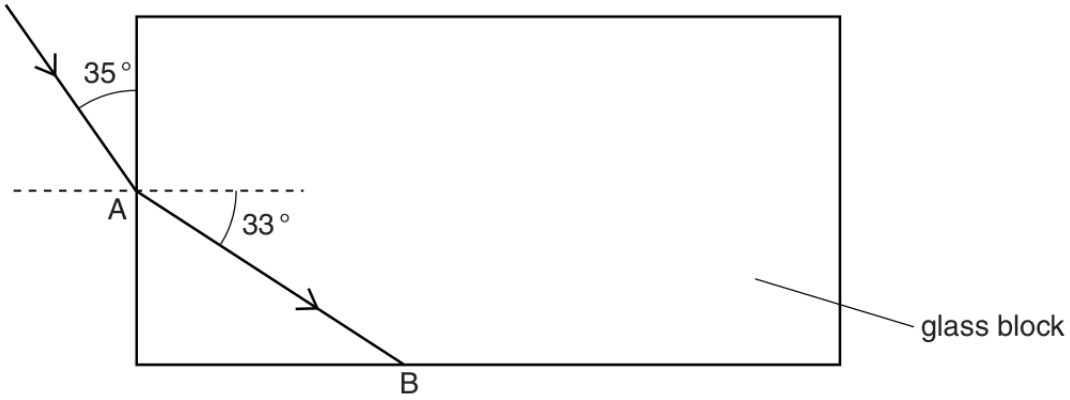


Fig. 7.1

The angle of refraction in the glass at A is 33°.

(a) Calculate the refractive index for light in the glass.

refractive index = ... [2]

(b) The light undergoes total internal reflection at B.

(i) State one condition necessary for total internal reflection to occur.

.....

.....

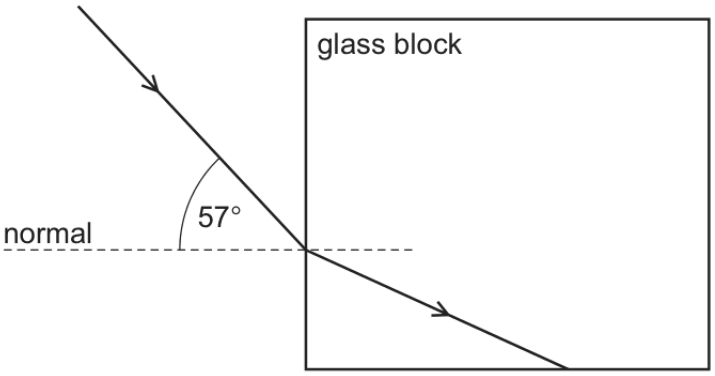
..... [1]

(ii) On Fig. 7.1, continue the ray to show the path of the light after total internal reflection at B and until it leaves the block. [1]

7	**(a)**	(n =)sin i / sin r **or** sin 55° / sin 33°	M1	
		1.5(040274)	A1	
	(b)	**(i)** angle of incidence greater than critical angle **or** denser to rarer medium	B1	
		(ii) reflected ray in correct direction (by eye) to edge of block **and** no second TIR (**ign** marked values)	B1	[4]

QUESTION · 5054_s16_qp_11.pdf · source page 9 of 16

23 Light passes from air into a glass block of refractive index 1.5, as shown.



NOT TO SCALE

What is the angle of refraction in the glass and what is the critical angle?

	angle of refraction	critical angle
A	34°	42°
B	34°	60°
C	38°	42°
D	38°	60°

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 23

23	**A**

6 **(a)** Explain what is meant by *critical angle*.

.....

.....

.....

.....[2]

(b) Diamonds are attractive because of their ability to reflect light. The critical angle for diamond in air is 24°.

(i) Calculate the refractive index of diamond.

refractive index = ...[2]

(ii) Fig. 6.1 shows two diamonds of different shapes. A ray of light enters each diamond, as shown.

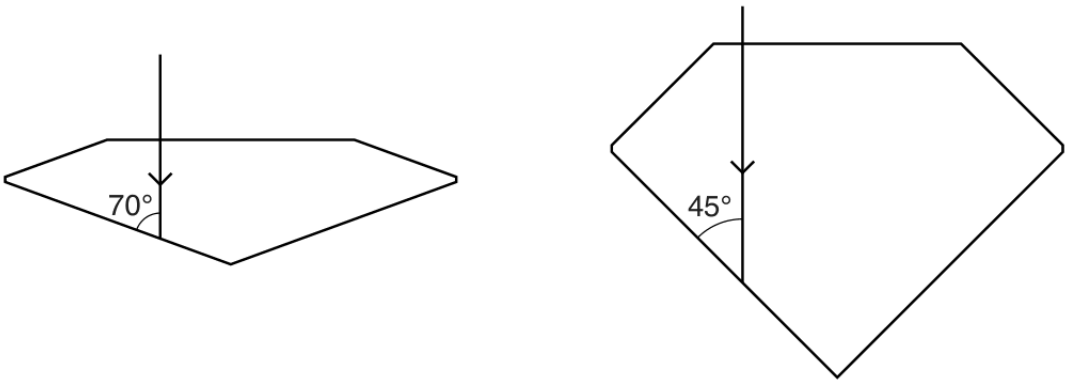


Fig. 6.1

On Fig. 6.1 continue the path of each ray of light until it emerges into the air. [3]

6 **(a)** angle of incidence B1

smallest angle for light to be totally internally reflected B1
or largest angle (of incidence) for ray to be refracted / emerge
or when light emerges along surface
or when angle of refraction is 90°

(b) **(i)** n = 1 / sinC algebraic or numerical C1
2.5 **or** 2.46 **or** 2.458(59) A1

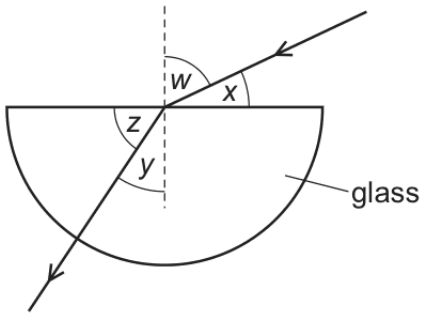
(ii) *left hand diagram* ray refracts away from normal and emerges into air at bottom left surface B1

right hand diagram reflected horizontal ray (by eye) B1

right hand diagram rest of ray completely correct to emerge into air at top face without refraction (by eye) B1

QUESTION · 5054_s16_qp_12.pdf · source page 8 of 16

21 Light passes from air into a block of glass, as shown.



Which expression is equal to the refractive index of glass?

A $\frac{\sin w}{\sin y}$ **B** $\frac{\sin w}{\sin z}$ **C** $\frac{\sin y}{\sin w}$ **D** $\frac{\sin z}{\sin x}$

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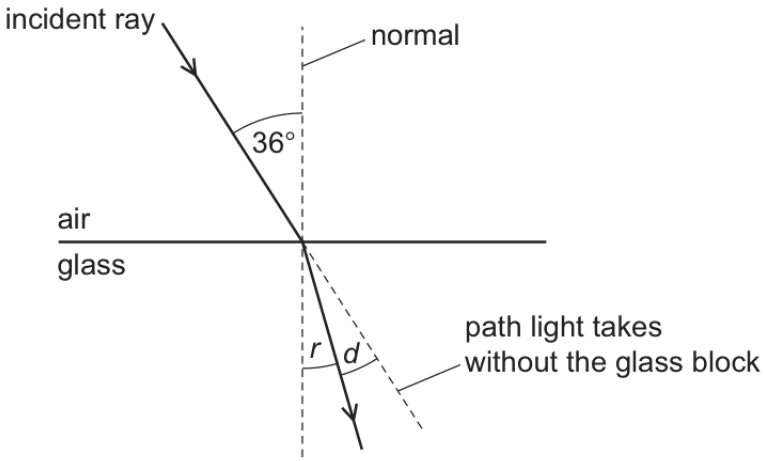
21	**A**

QUESTION · 5054_w16_qp_11.pdf · source page 10 of 20

24 A ray of light is incident on the surface of a glass block. The diagram is not drawn to scale.

The refractive index of the glass is 1.5.



The angle of refraction is *r*. The angle between the refracted ray and the path the light takes without the glass block is *d*.

What are *r* and *d*?

	r / °	*d* / °
A	23	12
B	24	12
C	23	13
D	24	13

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 24

24 **C**

4 A lamp is positioned at the bottom of a small pool of water. The critical angle for light passing from water into air is 49°.

(a) Explain what is meant by the term *critical angle*.

.....

.....

.....

.....[2]

(b) The lamp sends light towards the surface of the pool.

Fig. 4.1 shows three rays of light that are at 30°, 60° and 90° to the horizontal.

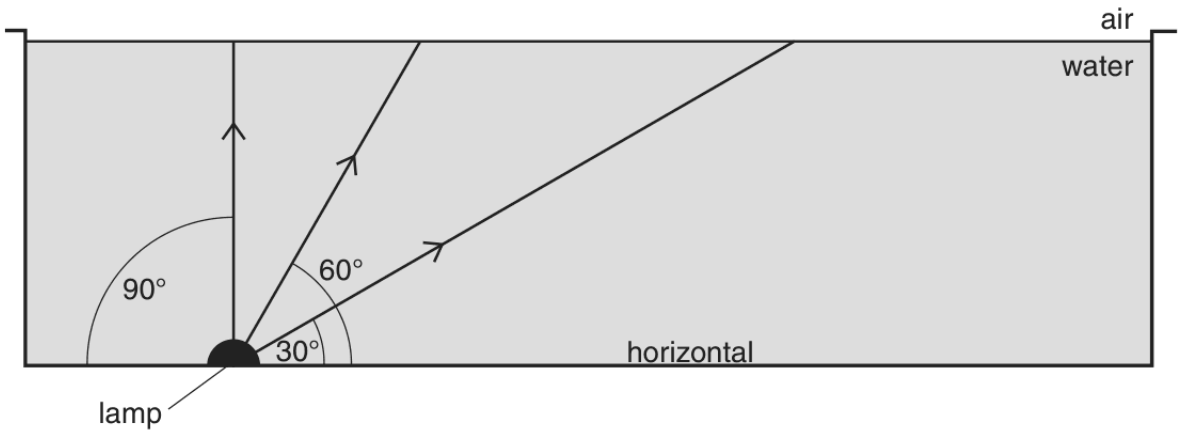


Fig. 4.1

On Fig. 4.1, draw the path taken by each of the three rays after they strike the surface of the water. [3]

4	**(a)**	smallest angle for total internal reflection **or** angle for refraction along surface	B1	
		angle of <u>incidence</u> in (optically) <u>denser</u> medium	B1	
	(b)	vertical ray continues undeviated	B1	
		second ray (60° to horizontal) refracts away from normal into the air	B1	
		third ray reflects internally **and** $i = r$ by eye **not** if any refracted ray	B1	[5]

10 A laser produces a beam of red light.

(a) The red light from the laser has a frequency of 4.3×10^{14} Hz.

(i) State the speed of light in air.

.....[1]

(ii) Calculate the wavelength of this red light in air.

wavelength = ...[2]

(b) Red light from the laser strikes one side of a glass prism at an angle of incidence *i*. The light refracts towards the normal as it enters the prism.

Fig. 10.1 shows the prism, the light and a screen.

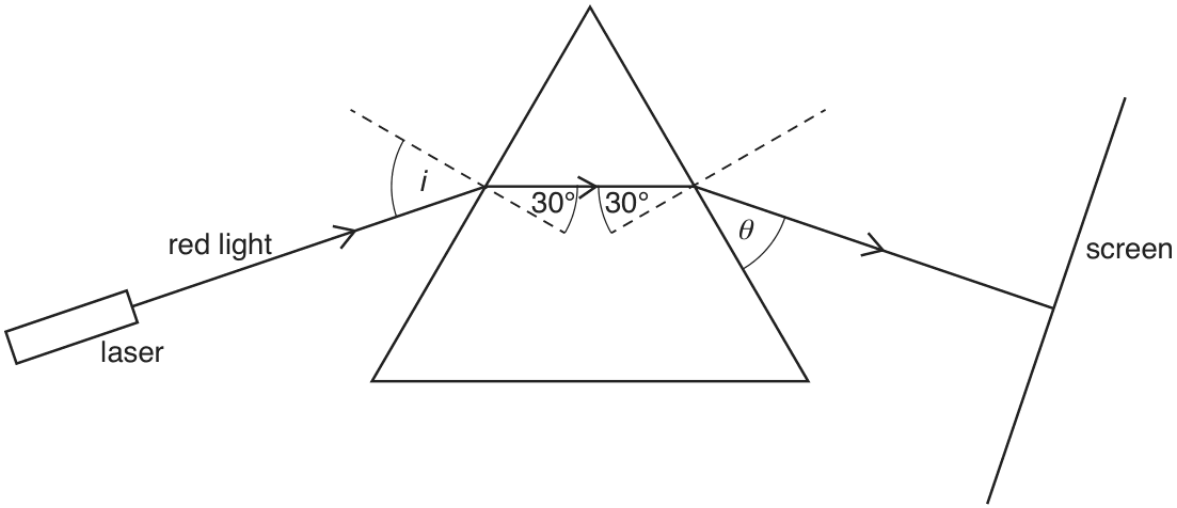


Fig. 10.1 (not to scale)

(i) State what happens to the speed of light as it enters the glass.

.....[1]

(ii) The refractive index of the glass is 1.5. The angle of refraction in the glass, where the light enters the prism, is 30°.

Calculate the angle of incidence *i*.

i = ...[2]

(iii) The light then passes back into the air and strikes the front of the white screen, as shown in Fig. 10.1.

Calculate θ, the angle between the ray in air and the side of the prism.

$\theta =$ .....[1]

(c) The laser in **(b)** is replaced with a filament lamp and a slit, as shown in Fig. 10.2.

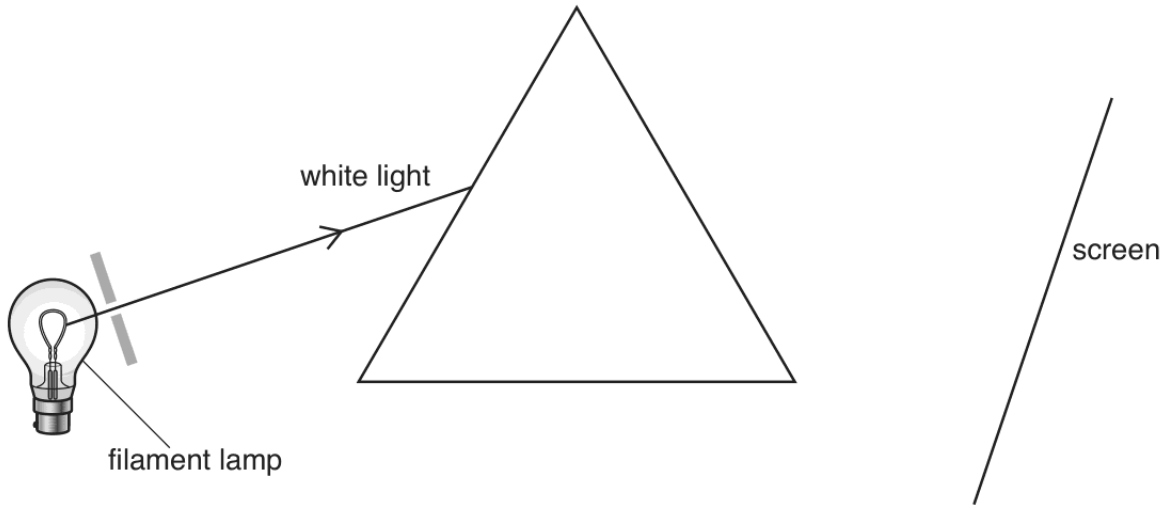


Fig. 10.2 (not to scale)

A ray of white light now strikes the prism.

(i) On Fig. 10.2, draw what happens to the light as it passes through the prism and strikes the screen. [2]

(ii) Describe what is seen on the screen.

.....

.....

.....

.....[2]

(iii) In addition to visible light, the filament lamp also emits some infra-red radiation. This infra-red radiation is able to pass through glass.

1. On the screen in Fig. 10.2, mark an X to indicate a place where infra-red radiation strikes the screen. [1]

2. Infra-red radiation is often detected by using a sensitive thermometer with a bulb that has been painted black.

Explain why the blackened bulb makes the thermometer a good detector of infra-red radiation.

.....

.....[1]

(d) Explain the role played by infra-red radiation in intruder alarms.

.....

.....

.....

.....

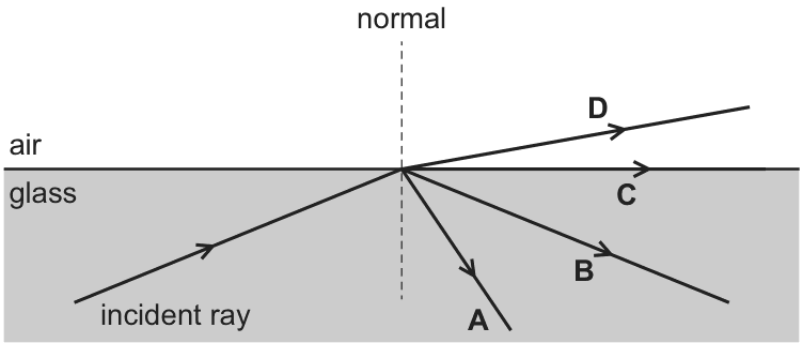
.....[2]

10	**(a)**	**(i)**	3.0×10^8 m / s		B1	
		(ii)	(λ =)c /f **or** $3.0 \times 10^8 / 4.3 \times 10^{14}$		C1	
			7.0×10^{-7} m		A1	[3]
	(b)	**(i)**	decreases		B1	
		(ii)	sin(i) = n × sin(r) **or** 1.5 × sin(30°) **or** 0.75		C1	
			49°		A1	
		(iii)	41°		B1	[4]
	(c)	**(i)**	dispersion at **both** surfaces **and** refractions in correct direction		B1	
			violet / blue light below the red light shown		B1	
		(ii)	spectrum **or** band of (continuous) colours **or** colours of rainbow		B1	
			red, orange, yellow, green, blue, (indigo, violet)		B1	
		(iii)	**1** X marked above red		B1	
			2 it is / black surfaces are good absorbers (of IR radiation)		B1	[6]
	(d)	intruder / human being emits IR	IR beam broken	IR reflected	B1	
		or	**or**			
		intruder warm **or** IR detected	does not reach detector	change detected	B1	[2]
						[15]

QUESTION · 5054_s17_qp_11.pdf · source page 11 of 16

27 Light travelling in glass is incident on a glass-air boundary. The angle of incidence of the light is greater than the critical angle.

Which arrow shows the direction of the light after it is incident on the boundary?



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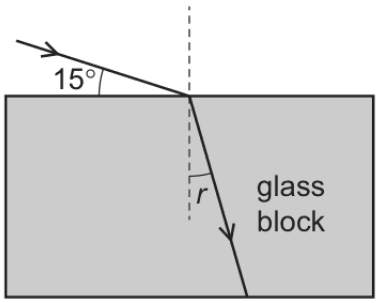
27	B	**1**

QUESTION · 5054_s17_qp_11.pdf · source page 12 of 16

28 Light strikes the top surface of a glass block at an angle of 15° as shown.



(not to scale)

The refractive index of glass is 1.5.

What is the angle of refraction *r*?

A 10° **B** 23° **C** 40° **D** 50°

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28	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a man in a room looking into a mirror, as viewed from above.

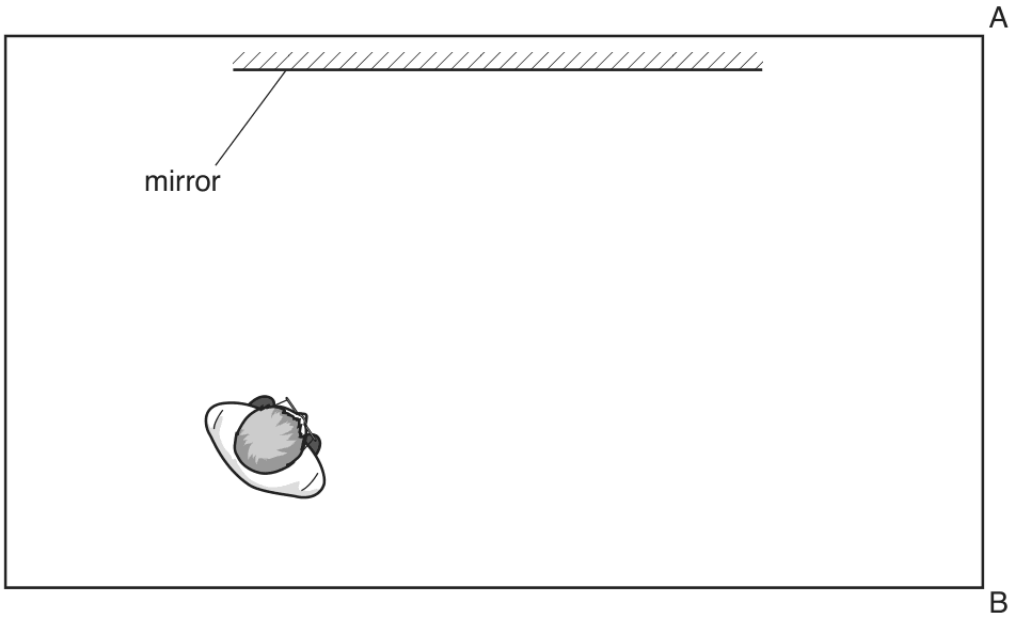


Fig. 9.1

(a) The man can see an image of part of the wall AB in the mirror.

Point C is the point on the wall closest to A that the man can see by reflection in the mirror. This point is not marked on Fig. 9.1.

(i) On Fig. 9.1, find point C and draw a ray of light from C which enables the man to see the image of C. [2]

(ii) On the ray that you have drawn on Fig. 9.1, label the angle of incidence *i* and the angle of reflection *r*. [2]

(iii) One characteristic of the image formed by the mirror is that it is virtual.

1. Explain what is meant by a *virtual* image.

.....

.....

.....[2]

2. State one other characteristic of an image formed by a plane mirror.

.....

.....[1]

(b) Light travels along optical fibres by total internal reflection.

(i) State two differences between total internal reflection and reflection in a mirror.

1. .....

.....

2. .....

.....

[2]

(ii) The critical angle for light travelling from the material of the fibre to air is 44°.

Calculate the refractive index of the material in the fibre.

refractive index = ...[2]

(iii) Light enters the optical fibre from air with an angle of incidence of 50°.

Calculate the angle of refraction.

angle of refraction = ...[2]

(iv) Optical fibres are used in telecommunications to send large amounts of data.

Pulses of light travel at almost the same speed along a fibre as electrical pulses travel along a metal wire.

State two advantages of using optical fibres rather than wires for transmitting data.

1. .....

.....

2. .....

.....

[2]

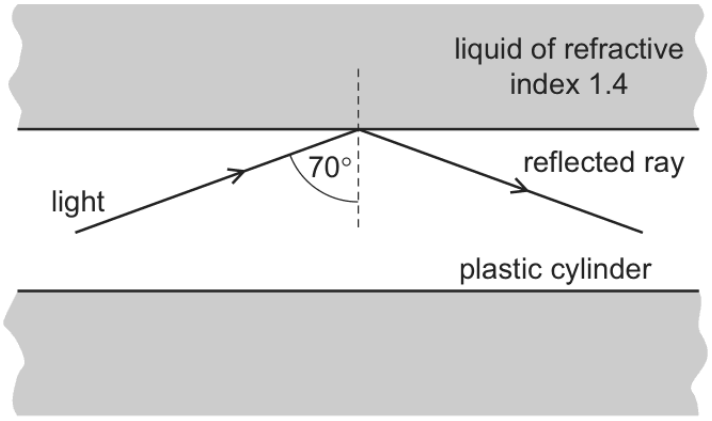
SECTION B

Question	Answer	Marks
9(a)(i)	ray from right-hand corner of mirror to eye	**B1**
	any incident and corresponding reflected ray correct by eye	**B1**
9(a)(ii)	normal drawn at any intersection of incident and reflected ray	**C1**
	both *r* and *i* labelled correctly with normal	**A1**
9(a)(iii)1	cannot be formed on a screen **or** nothing at the image (position)	**B1**
	rays do not come (all the way) from the image **or** rays only appear to come from image	**B1**
9(a)(iii)2	(same distance) behind the mirror **or** same size (as object) **or** upright / erect **or laterally** inverted	**B1**
9(b)(i)	reflection in mirror occurs at any angle **or** total internal reflection (TIR) only occurs for *i* > critical angle **or** there is no critical angle for the mirror B1	**any 2** **B2**
	TIR occurs from dense to less dense medium **or** in the dense(r) medium **or** from glass to air **or** inside / does not escape glass **or** from slow to fast (media) **or** mirror reflection from air to glass B1	
	(mirror) reflection is not total, e.g., not all reflected **or** better quality of image **or** multiple images from a mirror B1	

Question	Answer	Marks
9(b)(ii)	(n=) 1 / sinC **or** 1 / sin44	**C1**
	1.4	**A1**
9(b)(iii)	n = sini / sinr in any form, e.g. sinr = sin50 / n	**C1**
	32° – 33°	**A1**
9(b)(iv)	ANY 2 lines from • more data per second **or** per unit time • less decrease in strength / amplitude / attenuation • less heat / power produced / wasted • less need for repeating or amplification stations • less interference / noise • more secure / less chance of cross-talk • lighter / less heavy	**B1** **B1**

QUESTION · 5054_w17_qp_11.pdf · source page 10 of 20

20 A solid plastic cylinder is immersed in a liquid of refractive index 1.4. Light travelling in the plastic cylinder strikes the inside surface at an angle of incidence of 70°. The light undergoes total internal reflection.



What are the values of the critical angle in the plastic and the refractive index of the plastic?

	critical angle in plastic	refractive index of plastic
A	greater than 70°	greater than 1.4
B	greater than 70°	less than 1.4
C	less than 70°	greater than 1.4
D	less than 70°	less than 1.4

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 20

20	C	**1**

10 Fig. 10.1 represents the electromagnetic spectrum.

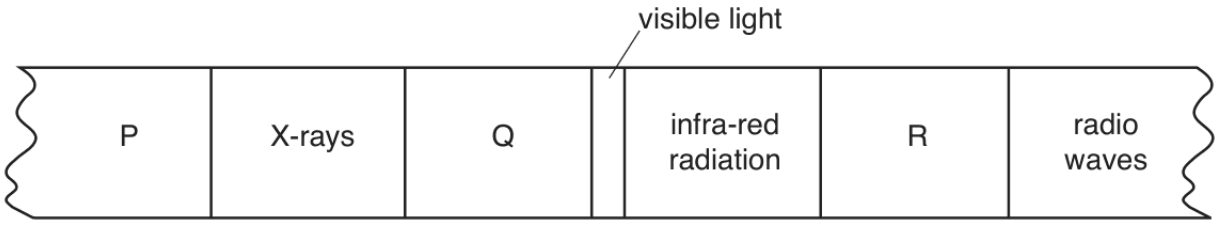


Fig. 10.1

(a) State the name of the component of the electromagnetic spectrum that is found in region

(i) P, ...

(ii) Q, ...

(iii) R. .....
[2]

(b) On Fig. 10.1, mark a tick (✓) in all of the boxes that represent the components with frequencies greater than the frequencies of visible light. [1]

(c) The speed of electromagnetic radiation in a vacuum is 3.0×10^8 m/s.

A television remote controller uses infra-red radiation of wavelength 9.4×10^{-7} m.

(i) Calculate the frequency of this radiation.

frequency = ... [3]

(ii) Explain how infra-red radiation is used in television remote controllers.

.....

.....

.....

..... [2]

(d) Fig. 10.2 shows a ray of green light in air striking the side of a glass prism.

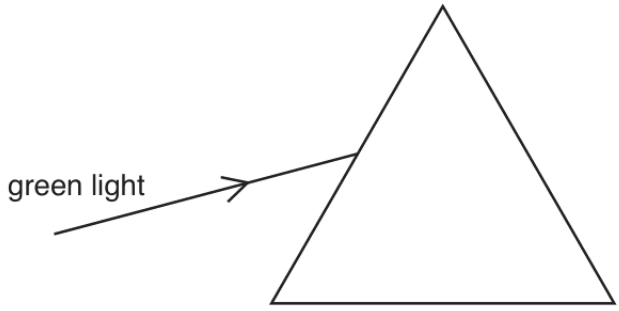


Fig. 10.2 (not to scale)

The refractive index of the glass for green light is 1.5.

(i) On Fig. 10.2,

1. draw the normal for this ray,

2. mark the angle of incidence with a letter *i*.

[1]

(ii) This angle of incidence is 57°.

Calculate the angle of refraction in the glass.

angle of refraction = ... [2]

(iii) State what happens, as the light enters the glass, to the light's

1. frequency,

.....

2. speed,

.....

3. wavelength.

.....

[2]

(iv) On Fig. 10.2, draw the path taken by the light as it passes through the glass and into the air. [2]

Question	Answer	Marks
10(a)	P - gamma(-rays) **or** γ(-rays)	
	Q - ultraviolet (radiation)	
	R - microwaves	
	any one correct	**C1**
	all three correct	**A1**
10(b)	P **and** X-rays **and** Q ticked	**B1**
10(c)(i)	(f =)c / λ **or** 3.0×10^{8} / 9.4×10^{-7}	**C1**
	3.2×10^{N}	**C1**
	3.2×10^{14} Hz	**A1**
10(c)(ii)	infra-red / radiation / signal / wave emitted by control **and** received at set	**B1**
	infra-red / radiation / signal / wave is encoded **or** is decoded	**B1**
10(d)(i)	normal indicated **and** angle of incidence indicated	**B1**
10(d)(ii)	n = sin i / sin r **or** 1.5 = sin 57(°) / sin r **or** (r =)$\sin^{-1}$(sin 57(°) / n) **or** $\sin^{-1}$(sin 57(°) / 1.5)	**C1**
	34°	**A1**
10(d)(iii)	**1** no change	**B1**
	2 decreases **and** **3** decreases	**B1**

Question	Answer	Marks
10(d)(iv)	ray in glass between normal and continuation of the incident ray	**B1**
	ray in air between continuation of the refracted ray and side of prism	**B1**

QUESTION · 5054_w17_qp_12.pdf · source page 9 of 16

21 Light is incident on a plastic block of refractive index 1.5.

The angle of incidence is 50°.

What is the angle of refraction?

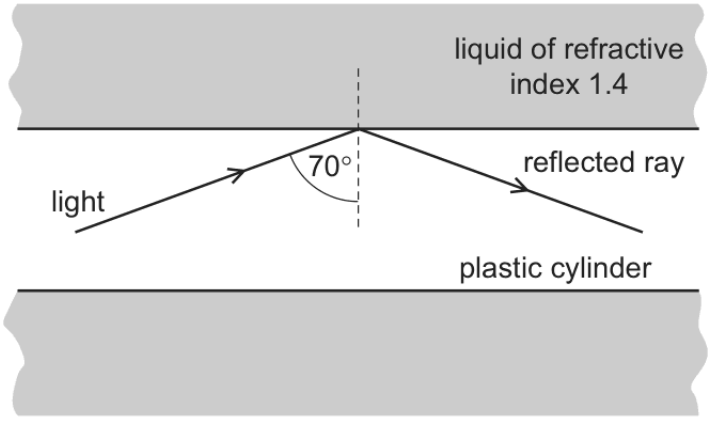
A 31° **B** 33° **C** 40° **D** 75°

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 21

21	A	**1**

QUESTION · 5054_w17_qp_12.pdf · source page 9 of 16

22 A solid plastic cylinder is immersed in a liquid of refractive index 1.4. Light travelling in the plastic cylinder strikes the inside surface at an angle of incidence of 70°. The light undergoes total internal reflection.



What are the values of the critical angle in the plastic and the refractive index of the plastic?

	critical angle in plastic	refractive index of plastic
A	greater than 70°	greater than 1.4
B	greater than 70°	less than 1.4
C	less than 70°	greater than 1.4
D	less than 70°	less than 1.4

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 22

22	C	**1**

QUESTION · 5054_s18_qp_11.pdf · source page 10 of 20

23 Which statement is correct?

A Total internal reflection only occurs when light travels from air into glass.

B The larger the refractive index of glass, the larger is the critical angle.

C When total internal reflection occurs, the angle of incidence is equal to the angle of reflection.

D When total internal reflection occurs, the angle of incidence is less than the critical angle.

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 23

23	C	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 11 of 16

29 Which device uses total internal reflection?

A magnifying glass

B optical fibre

C photographic enlarger

D projector

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 3 of 3 · row for Question 29

29	B	**1**

9 Some waves are longitudinal and some waves are transverse.

(a) State how a longitudinal wave differs from a transverse wave.

.....

..... [1]

(b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

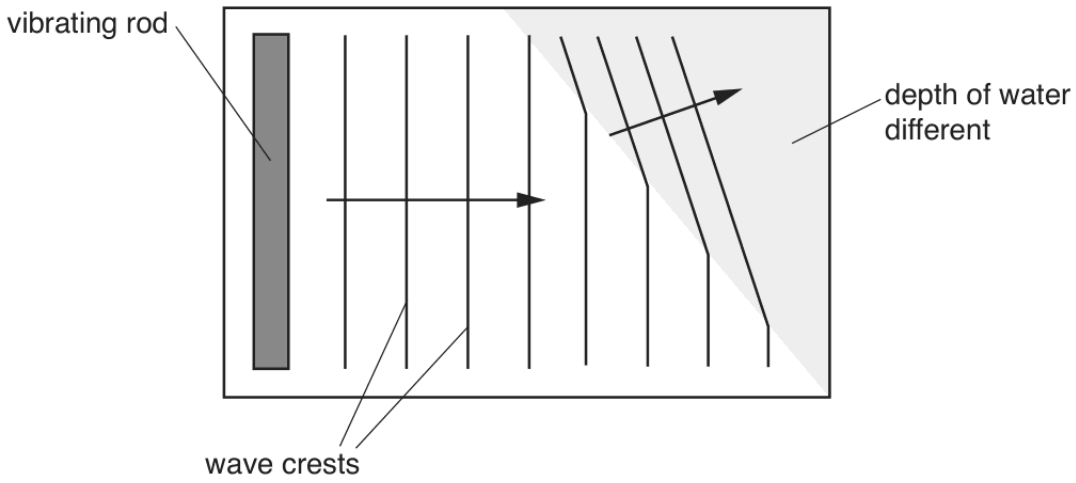


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

(i) In the left-hand section of the tank, the wavelength of the wave is 0.019 m and it travels at 0.17 m/s.

Calculate the frequency of the wave.

frequency = ... [2]

(ii) State what happens to the frequency of the wave, as it passes into the right-hand section of the tank.

..... [1]

(iii) Using Fig. 9.1, state and explain what happens to the speed of the wave as it passes into the right-hand section of the tank.

.....

.....

..... [2]

(c) Fig. 9.2 shows light, in air, striking the vertical side of a rectangular glass block at an angle of incidence of 60°.

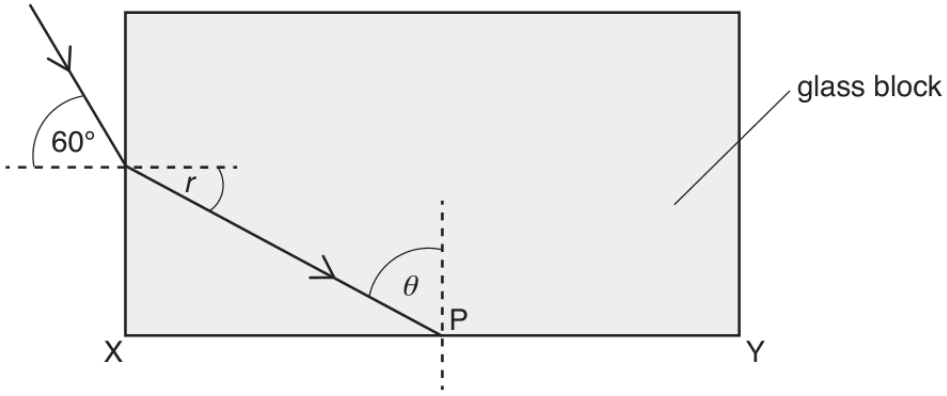


Fig. 9.2

The refractive index of the glass is 1.6. The light travels in the glass and strikes side XY at P.

(i) Underline all the terms that describe a light wave.

electromagnetic **longitudinal** **transverse**

[1]

(ii) At the point where the light enters the glass, the angle of refraction is *r*.

Calculate angle *r*.

r = ... [2]

(iii) **1.** Calculate the critical angle *c* for light travelling in the block.

c = ... [2]

2. At P, the angle θ between the ray and the normal is given by $\theta = 90° - r$.

State and explain what happens to the light when it strikes side XY.

.....

.....

..... [2]

3. On Fig. 9.2, draw the path of the light after it strikes side XY at P and the path of the light when it is again travelling in the air. [2]

Question	Answer	Marks
9(a)	the vibration direction is parallel to the wave / energy travel direction **or** longitudinal waves have compressions and rarefactions **or** transverse waves have crests and troughs **or** longitudinal waves cannot be polarised	**B1**
9(b)(i)	(f =) v / λ **or** 0.17 / 0.019	**C1**
	8.9 Hz	**A1**
9(b)(ii)	(frequency) stays constant **or** does not change	**B1**
9(b)(iii)	(speed) decreases	**B1**
	(Fig. 9.1 shows that) wavelength decreases **or** refraction is towards normal **or** $i > r$ **or** top of wave (in shaded region) lags behind bottom of wave	**B1**
9(c)(i)	electromagnetic **and** transverse underlined	**B1**
9(c)(ii)	$n = \sin(i) / \sin(r)$ **or** (r =) $\sin^{-1}(\sin(i) / n)$ **or** (r =) $\sin^{-1}(\sin(60°) / 1.6)$	**C1**
	33(°)	**A1**

Question	Answer	Marks
9(c)(iii)	1 $n = 1 / \sin(c)$ **or** (c =) $\sin^{-1}(1 / n)$ **or** $\sin^{-1}(1 / 1.6)$	**C1**
	39°	**A1**
	2 total internal reflection	**B1**
	$\theta > c$ **or** 57(°) > 39(°) **or** angle of incidence greater than the critical angle	**B1**
	3 reflected ray at P (correct angle by eye)	**B1**
	light in air at 30° (by eye) to vertical surface	**B1**

QUESTION · 5054_w18_qp_12.pdf · source page 11 of 16

25 A student reads the following in her physics book.

'The incident angle is greater than 42° which is the critical angle for glass in air.'

What is the student reading about?

A focal length of a glass lens

B reflection in a plane mirror

C refraction as light enters glass

D total internal reflection

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25	D	**1**

3 Fig. 3.1 shows three rays of light travelling in water from a light source S.

One ray of light is totally internally reflected at the boundary between water and air.

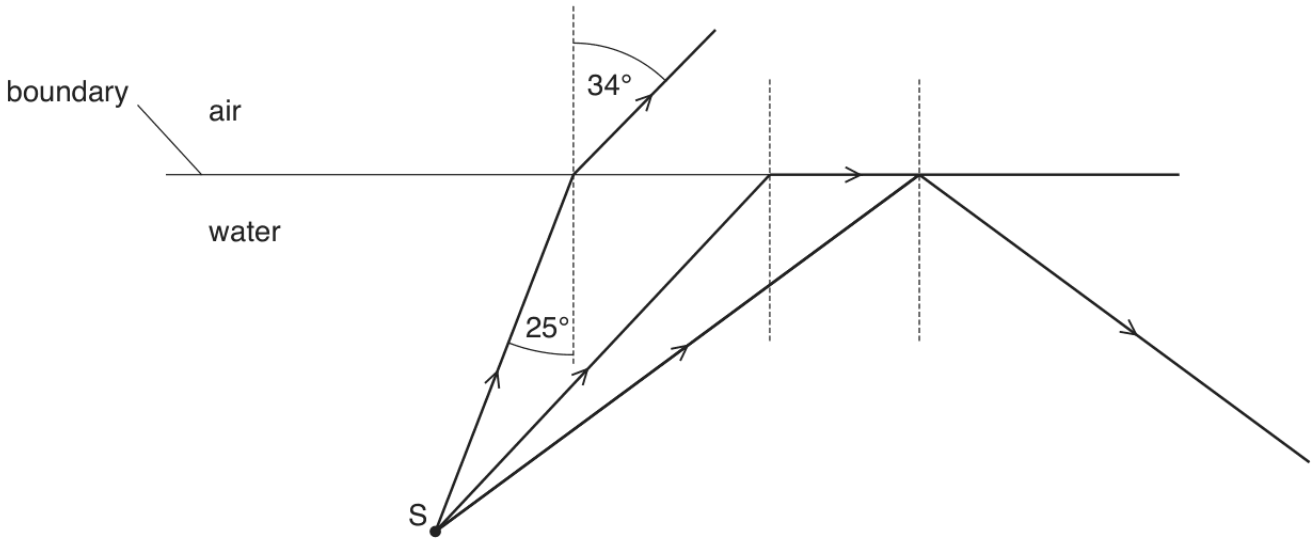


Fig. 3.1 (not to scale)

(a) On Fig. 3.1, mark the critical angle and label it *c*. [1]

(b) State why only one of the three rays is totally internally reflected.

.....

..... [1]

(c) Calculate:

(i) the refractive index for water

refractive index = ... [2]

(ii) the critical angle for the boundary between water and air.

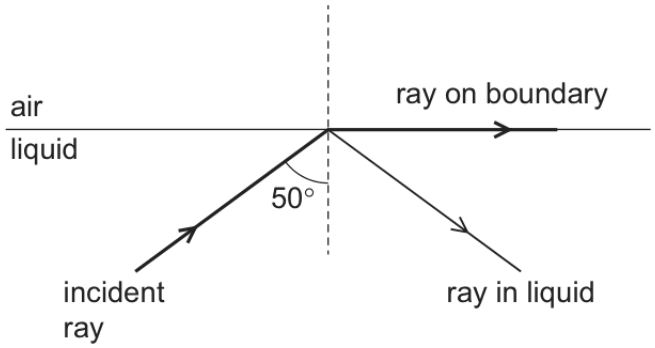
critical angle = ... [2]

[Total: 6]

Question	Answer	Marks
3(a)	middle ray angle of incidence labelled c	**B1**
3(b)	angle of incidence (of right-hand ray only) greater than the critical angle	**B1**
3(c)(i)	(n=) sin i / sinr algebraic or numerical	**C1**
	1.3	**A1**
3(c)(ii)	sin c = 1 / n algebraic or numerical	**C1**
	answer in range 49° to 50.3°	**A1**

QUESTION · 5054_w19_qp_11.pdf · source page 12 of 20

26 The diagram shows a ray of light in liquid incident on the boundary with air. Two other rays are observed. One is in the liquid and the other is in the air on the boundary.



What is the angle of refraction?

A 0° **B** 40° **C** 50° **D** 90°

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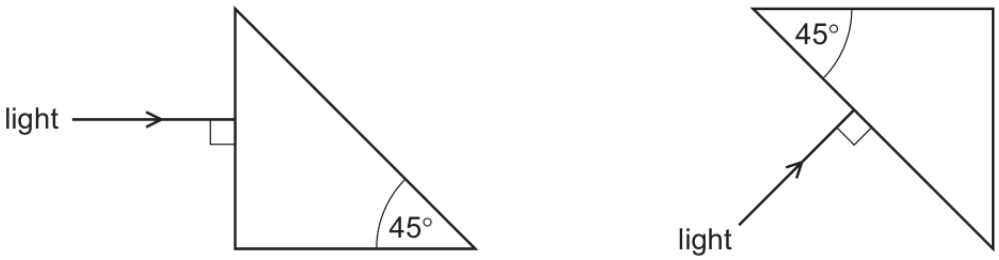
26	D	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 13 of 20

27 Light is incident at 90° on the surfaces of two glass prisms P and Q.



glass prism P glass prism Q

The critical angle for light travelling from glass into air is 42°.

Where does total internal reflection occur?

A in P and in Q

B in P only

C in Q only

D in neither P nor Q

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 27

27	A	**1**

10 Visible light is one component of the electromagnetic spectrum.

(a) State the speed of light in a vacuum.

..... [1]

(b) State the colours of the visible spectrum.

..... [1]

(c) An object O of height 3.0 cm is placed 4.0 cm from the centre of a diverging lens.

Fig. 10.1 shows the object O, the diverging lens and the two focal points (principal focuses), F_1 and F_2, of the lens.

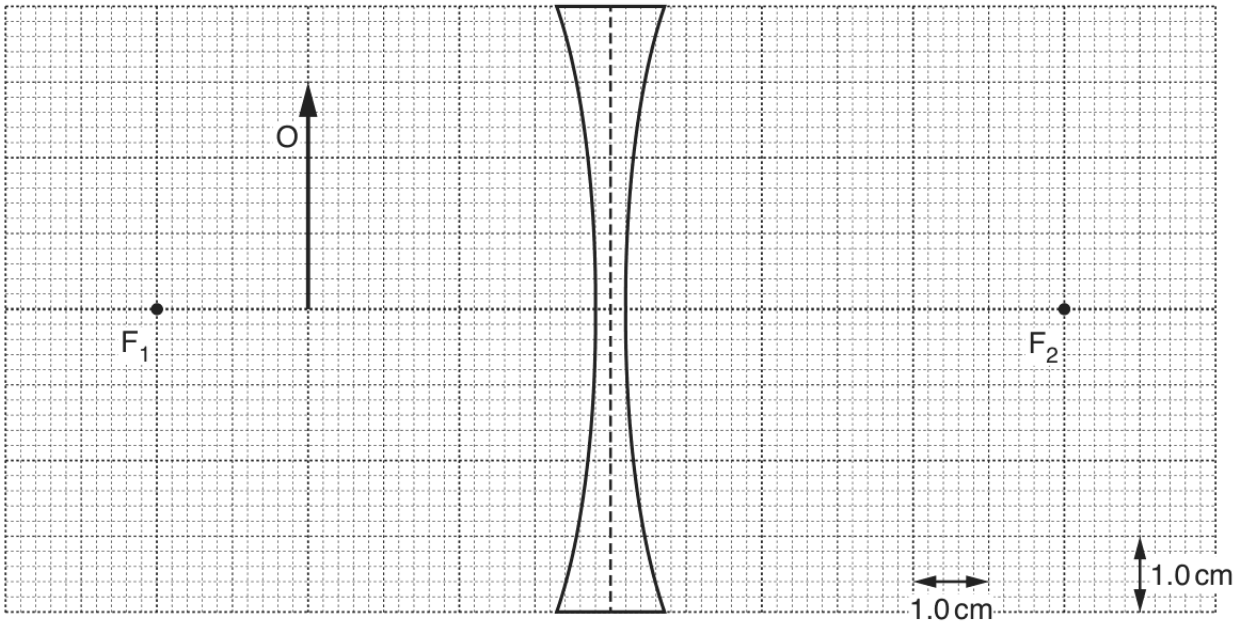


Fig. 10.1

(i) Determine the focal length of this lens.

focal length = ... [1]

(ii) The diverging lens produces an image of object O.

On Fig. 10.1, by drawing rays from the tip of O, locate this image and label it I. [4]

(iii) Determine the height of the image.

height = ... [1]

(iv) Determine the magnification of the image.

magnification = ... [1]

(v) State **two** ways in which a virtual image differs from a real image.

1. .....

.....

2. .....

..... [2]

(vi) State one use for a diverging lens.

.....

..... [1]

(d) Light passing from air into glass refracts in a similar way to a water wave passing from deep water into shallow water.

Fig. 10.2 represents light passing from air into glass at an angle to the surface.

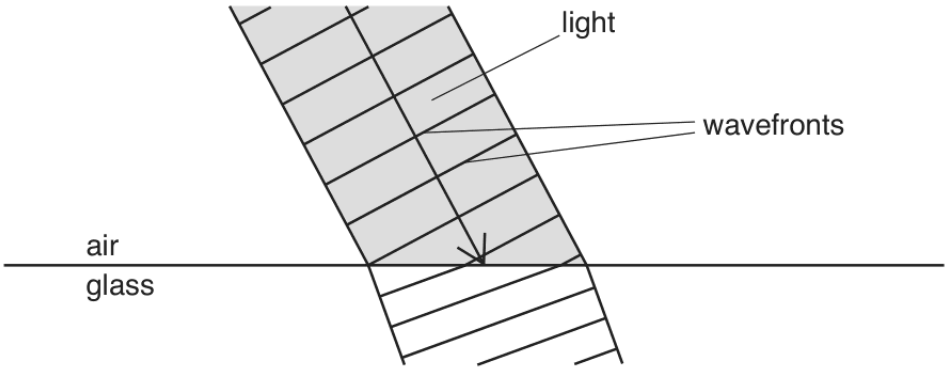


Fig. 10.2

One side of a wavefront strikes the glass before the other side.

Explain why the wavefronts change direction as the light enters the glass.

.....

.....

.....

..... [3]

[Total: 15]

MARK SCHEME · 5054_w19_ms_21.pdf · source page 9 of 10 · Question 10

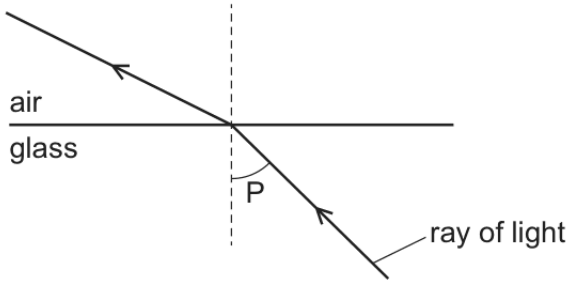
Question	Answer	Marks
10(a)	3.0×10^8 m / s	**B1**
10(b)	red orange yellow green blue indigo violet (any order)	**B1**
10(c)(i)	(–)6.(0) cm	**B1**

MARK SCHEME · 5054_w19_ms_21.pdf · source page 10 of 10 · Question 10

Question	Answer	Marks
10(c)(ii)	any **two** rays drawn from: paraxial ray that refracts and seems to come from F_1 ray through the optical centre of lens ray that aims for F_2 but refracts and emerges paraxially	**B2**
	rays traced back to point	**B1**
	(point) labelled I **and** rest of image drawn down to the principal axis	**B1**
10(c)(iii)	1.7–1.9 cm	**B1**
10(c)(iv)	candidate's **10(c)(iii)** / 3.0 evaluated	**B1**
10(c)(v)	any **two** from: a real image can be projected on to a screen light actually passes through a real image on same side (of lens) as object **or** on opposite side of mirror to object	**B2**
10(c)(vi)	correction of short-sight / myopia	**B1**
10(d)	light travels more slowly in glass **or** light changes speed	**B1**
	one side / left-hand side of wavefront slows down first	**B1**
	wavelength decreases **or** wavefront travels a shorter distance in the same time	**B1**

QUESTION · 5054_w19_qp_12.pdf · source page 11 of 16

25 The diagram shows light passing from glass into air.



What is the name of angle P?

A the angle of incidence

B the angle of reflection

C the angle of refraction

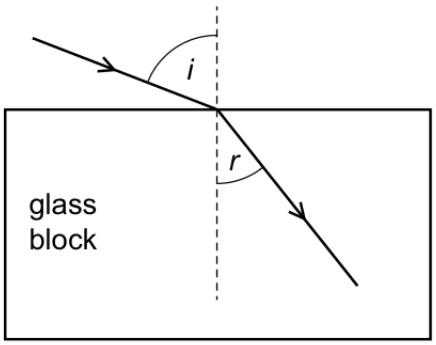
D the critical angle

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 25

25	A	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 9 of 16

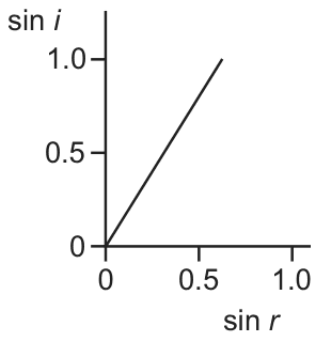
26 Light enters a glass block at an angle of incidence *i* and it produces an angle of refraction *r* in the glass.



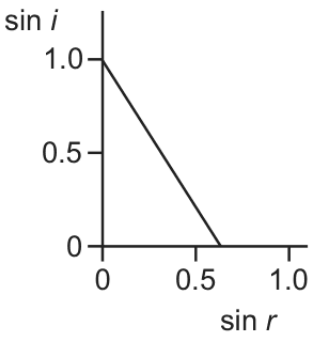
Several different values of *i* and *r* are measured, and a graph is drawn of sin *i* against sin *r*.

Which graph is correct?

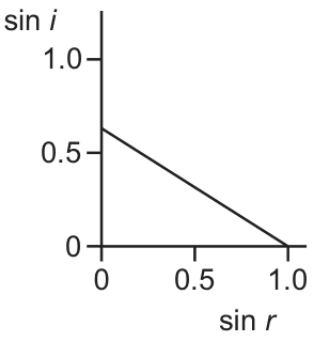
A



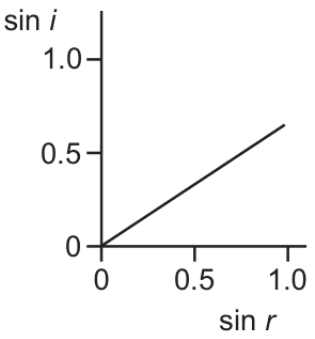
B



C



D



MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 26

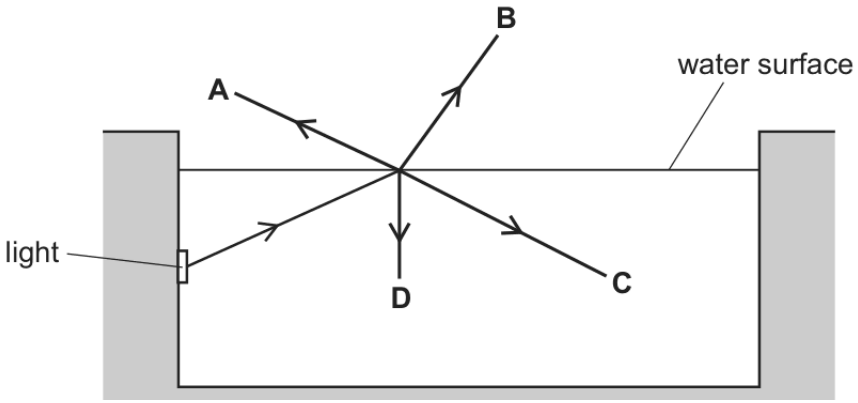
26	A	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 10 of 16

27 A swimming pool is lit by an underwater light.

A ray of light is incident on the water surface.

What is the correct path for the ray of light?



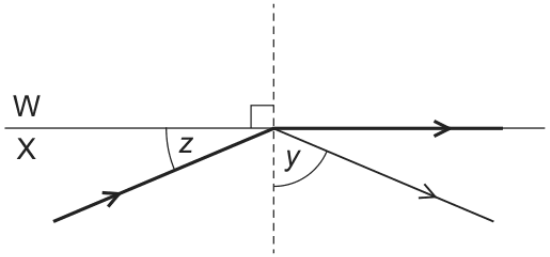
MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 27

27	C	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 11 of 16

31 The diagram shows a ray of light incident on the boundary between two mediums W and X.

The mediums have different refractive indexes.



Some light is reflected and some passes along the surface between the two mediums. Angle *y* is greater than angle *z*.

Which statement is correct?

A W has a greater refractive index than X and angle *y* is equal to the critical angle.

B W has a greater refractive index than X and angle *z* is equal to the critical angle.

C X has a greater refractive index than W and angle *y* is equal to the critical angle.

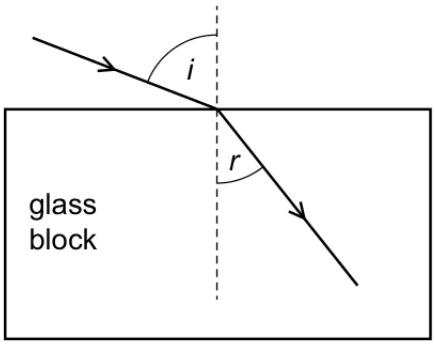
D X has a greater refractive index than W and angle *z* is equal to the critical angle.

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 3 of 3 · row for Question 31

31	C	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 11 of 16

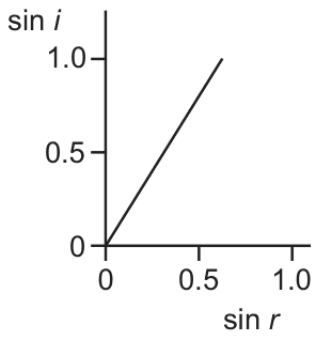
32 Light enters a glass block at an angle of incidence *i* and it produces an angle of refraction *r* in the glass.



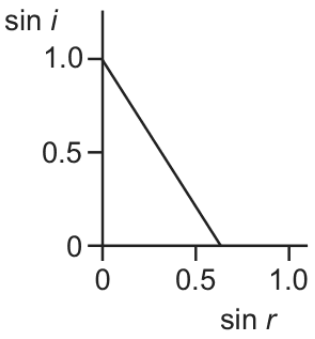
Several different values of *i* and *r* are measured, and a graph is drawn of sin *i* against sin *r*.

Which graph is correct?

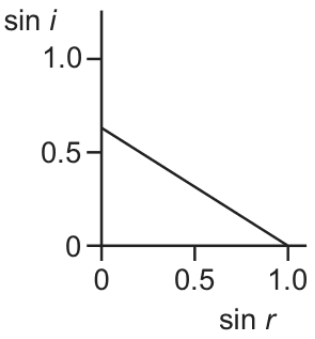
A



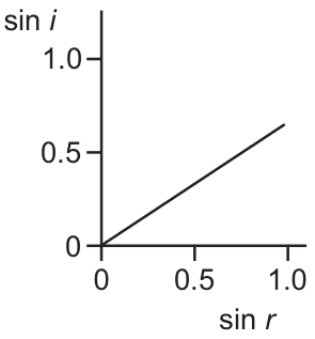
B



C



D



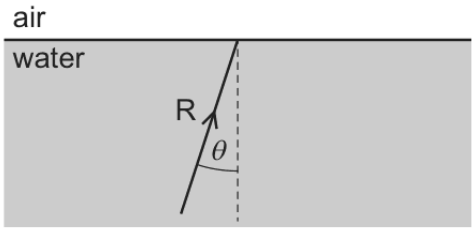
MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 3 of 3 · row for Question 32

32	A	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 11 of 16

28 A ray of light R is incident on a water-to-air surface with an angle of incidence θ.

The angle θ is less than the critical angle *c*.



Which statement describes the subsequent path of R?

A It travels back into the water with an angle of reflection equal to *c*.

B It travels back into the water with an angle of reflection greater than *c*.

C It travels into air with an angle of refraction greater than θ.

D It travels into air with an angle of refraction smaller than θ.

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 3 of 3 · row for Question 28

28	C	**1**

4 The three angles of a glass prism are 45°, 45° and 90° as shown in Fig. 4.1.

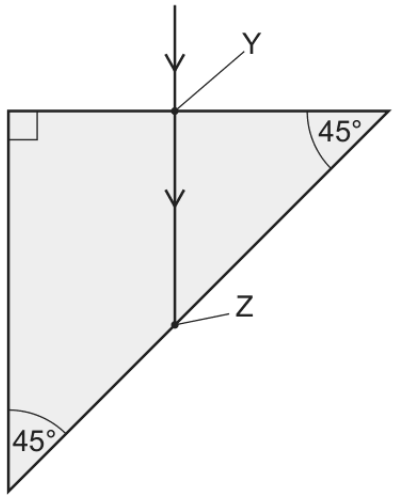


Fig. 4.1

At point Y, a ray of light of a single frequency travels in air and strikes the side of the prism at 90°. The ray passes into the glass prism.

(a) Light travels more slowly in glass than in air.

(i) State what happens to the wavelength of the light in the ray as it enters the glass.

..... [1]

(ii) State what happens to the frequency of the light in the ray as it enters the glass.

..... [1]

(b) The refractive index of glass is 1.6.

(i) Calculate the critical angle for light in glass.

critical angle = ... [2]

(ii) On Fig. 4.1, sketch the path of the light after it strikes the side of the prism at Z and after it returns to the air. [2]

[Total: 6]

Question	Answer	Marks
4(a)(i)	it decreases	**B1**
4(a)(ii)	it does not change	**B1**
4(b)(i)	(c =) $\sin^{-1}(1 / n)$ **or** $\sin^{-1}(1 / 1.6)$ **or** $\sin^{-1}(0.625)$	**C1**
	39°	**A1**
4(b)(ii)	reflected ray at Z **and** $i = r$ **and** no refracted ray	**B1**
	ray strikes vertical face perpendicularly **and** undeviated in air	**B1**

QUESTION · 5054_w20_qp_12.pdf · source page 10 of 16

24 Which statement about light passing from air to glass is correct?

A The frequency of the light waves decreases.

B The speed of the light waves decreases.

C The wavelength of the light waves increases.

D The wavelength of the light waves remains unchanged.

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 24

24	B	**1**

8 A thin converging lens is made of a transparent material of refractive index 1.4.

(a) A ray of light travelling in air strikes the surface of the lens at an angle of incidence of 55°.

(i) Calculate the angle of refraction.

angle of refraction = ... [2]

(ii) Place a tick (✓) in **one** of the boxes in the third column of Table 8.1 to indicate how the light ray deviates and what happens to the speed of the light in the ray as it enters the lens.

Table 8.1

direction of deviation	speed of light	
away from the normal	decreases	
away from the normal	does not change	
away from the normal	increases	
towards the normal	decreases	
towards the normal	does not change	
towards the normal	increases	

[1]

(iii) State what happens to the frequency of the light in the ray as it enters the glass.

..... [1]

(b) The focal length of the lens is 2.5 cm.

(i) State what is meant by *focal length*.

.....

..... [1]

(ii) Fig. 8.1 is a full-scale diagram that shows an object O of height 3.0 cm and the lens.

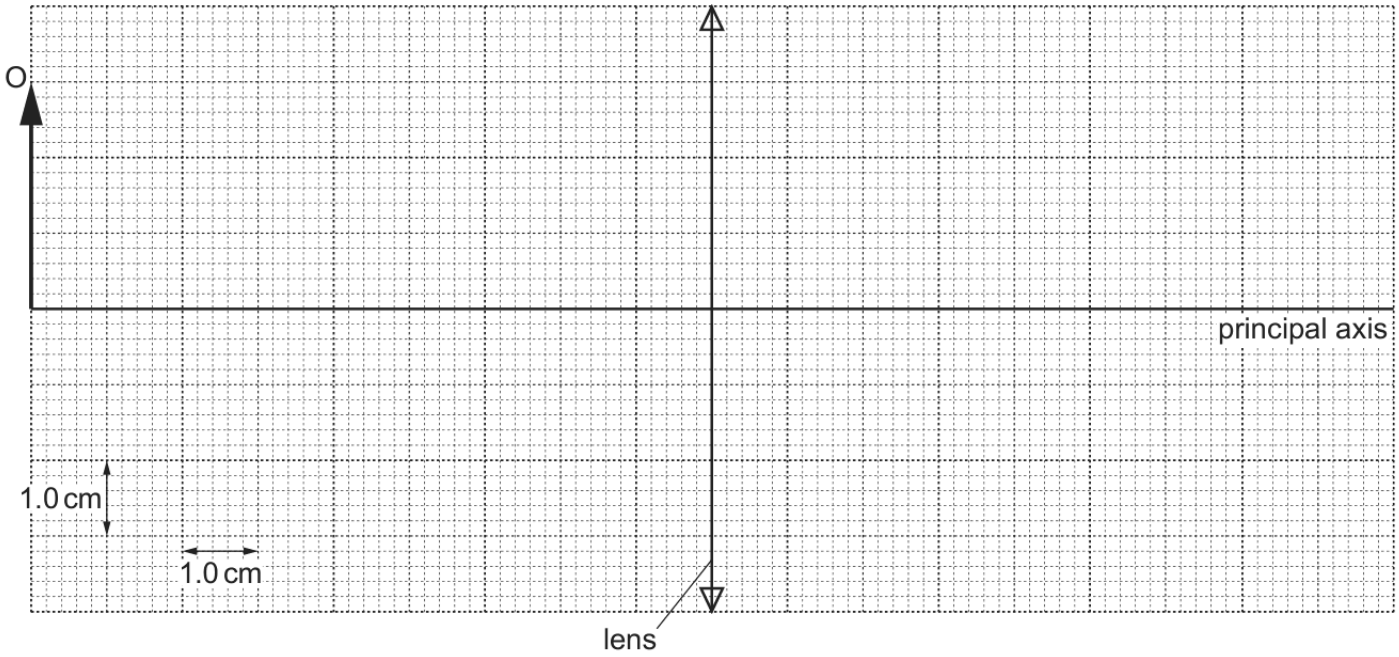


Fig. 8.1 (full scale)

By drawing on Fig. 8.1, locate and mark the image I of O. [4]

(iii) Determine the distance of I from the lens and calculate the magnification of O produced by the lens.

distance = ...

magnification = ...
[3]

(c) Describe how a converging lens is used in a camera.

.....

.....

.....

.....

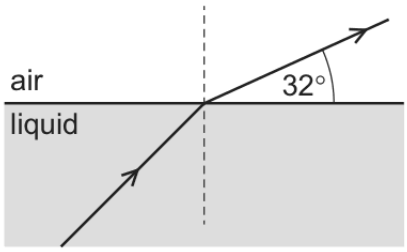
..... [3]

[Total: 15]

Question	Answer	Marks
8(a)(i)	n = sin i / sin r **or** 1.4 = sin 55° / sin r **or** (r =) $\sin^{-1}$(sin i / n) **or** (r =) $\sin^{-1}$(sin 55°/ 1.4)	**C1**
	36°	**A1**
8(a)(ii)	towards (the normal) **and** speed decreases (i.e. fourth box **only**)	**B1**
8(a)(ii)	stays constant	**B1**
8(b)(i)	distance between (one) focal point / principal focus and (the centre of) lens	**B1**
8(b)(ii)	at least one correct focal point labelled / clear / used	**B1**
	any **two** from drawn: paraxial ray from tip of O to lens and refracted through correct far F ray from tip of O through optical centre and undeviated ray from tip of O through correct near F to lens and refracted paraxially	**B2**
	I marked at intersection of correct rays	**B1**
8(b)(ii)	3.4 cm ⩽ distance ⩽ 3.5 cm	**B1**
	(magnification =) candidate's distance / 9.0 **or** candidate's height / 3.0	**C1**
	0.37 ⩽ magnification ⩽ 0.40 **c.a.o.**	**A1**
8(c)	on next page	
8(c)	any **three** from: light (from subject / object) focussed / refracted by lens by moving / sliding / adjusting lens real image (of subject / object) formed on digital light detector / film	**B3**

QUESTION · 5054_s21_qp_11.pdf · source page 8 of 16

21 Light refracts from a liquid into air as shown.



not to scale

The refractive index for light moving from air to the liquid is 1.4.

What is the angle of incidence in the liquid?

A 22° **B** 37° **C** 41° **D** 45°

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 21

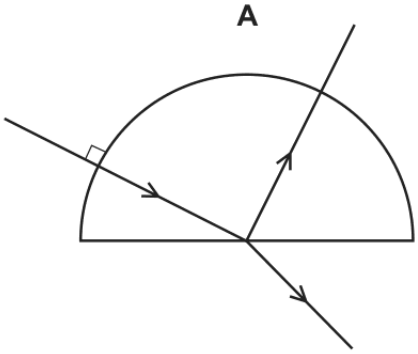
21	B	**1**

QUESTION · 5054_s21_qp_11.pdf · source page 9 of 16

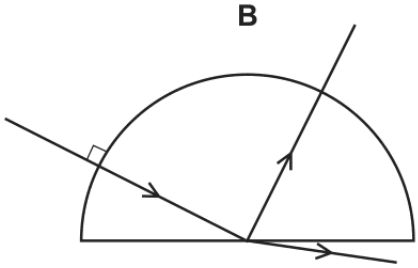
22 A ray of red light in air enters a semi-circular block.

Which diagram shows the partial reflection and the refraction of the ray?

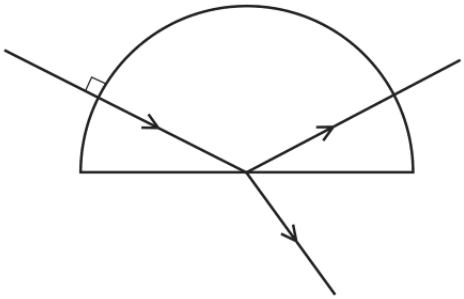
A



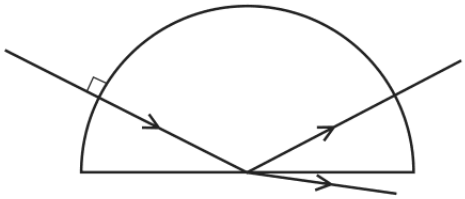
B



C



D



MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 22

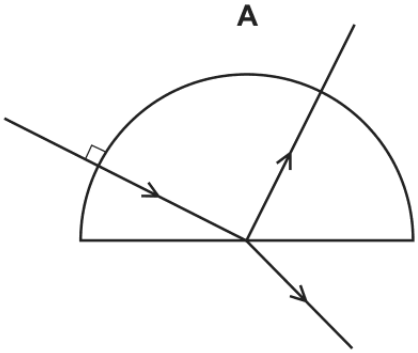
22	D	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 9 of 16

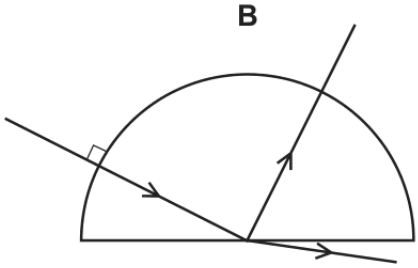
24 A ray of red light in air enters a semi-circular block.

Which diagram shows the partial reflection and the refraction of the ray?

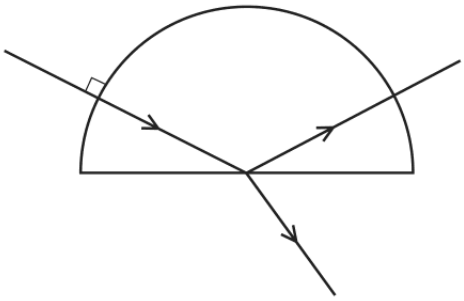
A



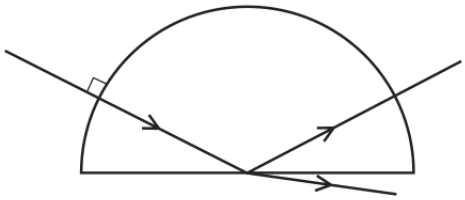
B



C



D



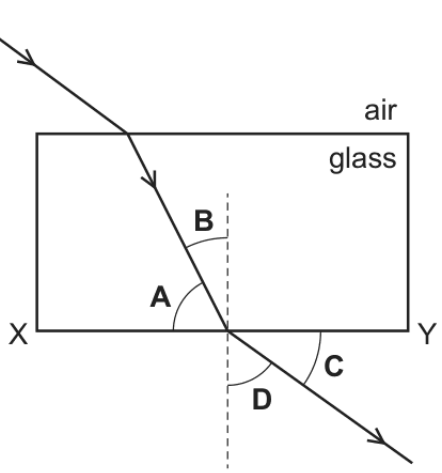
MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 24

24	D	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 9 of 20

20 A ray of light passes into a glass block. It travels through the glass block and then emerges into the air.

Which angle is the angle of refraction at the surface XY?



MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 20

20	D	**1**

3 Fig. 3.1 shows light entering a transparent block.

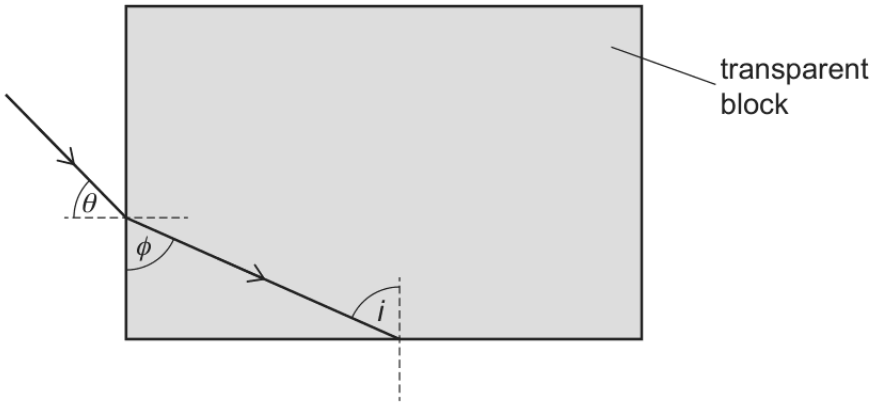


Fig. 3.1 (not to scale)

The light enters the block at an angle θ to the normal and travels through the block until it meets the bottom surface.

The angle between the ray in the block and the vertical side of the block is ϕ.

(a) Light travels more slowly in the block than in air.

(i) Explain how Fig. 3.1 shows this.

.....

.....

..... [2]

(ii) State what happens to the wavelength of the light and what happens to the frequency of the light as it enters the block.

wavelength .....

frequency .....
[2]

(b) The refractive index of the transparent material is 1.6. Angle θ is 45°.

(i) Determine angle ϕ.

ϕ = ... [3]

(ii) The angle of incidence *i* at the bottom surface is equal to ϕ and the critical angle for the material of the block in air is 39°.

Explain what happens to the light after it meets the bottom surface.

.....

.....

..... [2]

[Total: 9]

Question	Answer	Marks
3(a)(i)	it / light refracts / bends / moves towards normal **or** angle of incidence greater than angle of refraction	**B1**
	change in direction of the ray / refraction shows a change in speed (at the boundary) **or** sin(*i*) / sin(*r*) is ratio of speeds **or** top of ray slows down first **and** enters the block first	**B1**
3(a)(ii)	(wavelength) decreases	**B1**
	(frequency) does not change.	**B1**
3(b)(i)	(sin(*r*) =) sin(*i*) / *n* **or** sin(45°) / 1.6 **or** 0.71 / 1.6 **or** 0.44 **or** (*r* =) 21(°)	**C1**
	26(°)	**C1**
	64°	**A1**
3(b)(ii)	64° / angle of incidence / *i* is greater than the critical angle / 39° **or** angle of incidence = 64**°**	**B1**
	it undergoes total internal reflection	**B1**

4 Two parallel rays of light, one red and one blue, enter a glass prism.

Fig. 4.1 shows both rays of light before they enter the prism. The blue ray is also shown incident on a different side of the prism after passing through the prism.

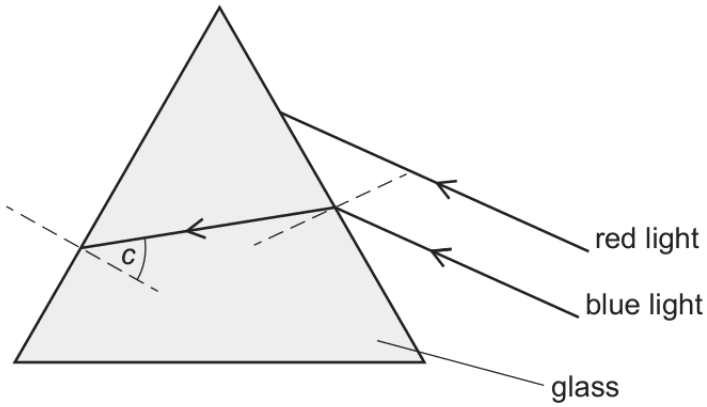


Fig. 4.1 (not to scale)

The ray of blue light strikes the left side of the prism at an angle equal to its critical angle *c*.

(a) (i) On Fig. 4.1, mark and label the angle of incidence *i* and the angle of refraction *r* for the blue light as it enters the prism. [1]

(ii) On Fig. 4.1, continue the path of the blue light after it strikes the left side of the prism. [1]

(iii) The refractive index of glass for red light is smaller than the refractive index for blue light.

On Fig. 4.1, draw the path of the red light as it travels in the prism and after it strikes the left side of the prism. [2]

(b) (i) State what is meant by the critical angle.

.....

..... [2]

(ii) The refractive index of glass for blue light is 1.5.

Calculate the critical angle *c* for blue light in glass. Show your working.

c = ... [2]

[Total: 8]

MARK SCHEME · 5054_s22_ms_21.pdf · source page 6 of 11 · Question 4

Question	Answer	Marks
4(a)(i)	correct *i* and *r* marked	**B1**
4(a)(ii)	reflection shown at approximately the correct angle **or** ray along surface	**B1**
4(a)(iii)	correct refraction as red light enters prism showing less refraction than blue light	**B1**
	correct refraction as red light emerges from left side of prism	**B1**

MARK SCHEME · 5054_s22_ms_21.pdf · source page 7 of 11 · Question 4

Question	Answer	Marks
4(b)(i)	angle of incidence	**B1**
	smallest angle when ray totally internally reflects **or** (when ray is) just internally reflected **or** when ray (refracted/emerges) along surface / at 90° to normal / angle of refraction is 90° **OR** angle of refraction when the angle of incidence is 90° B2	**B1**
4(b)(ii)	$\sin c = 1 / n$ in any form numerical or algebraic	**B1**
	42°	**B1**

4 Fig. 4.1 shows a ray of white light incident on a glass prism.

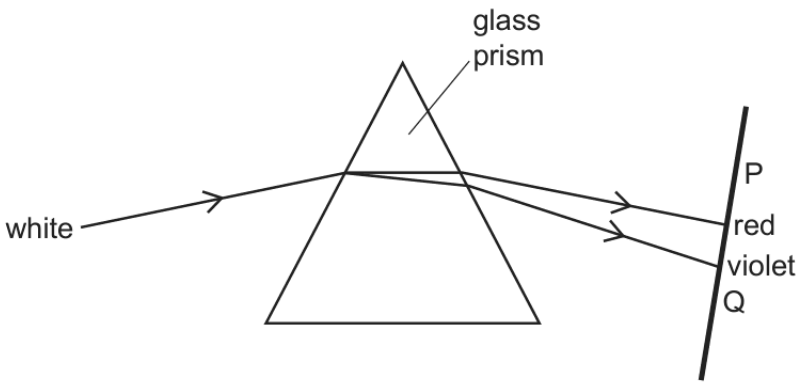


Fig. 4.1 (not to scale)

Refraction causes the white light to separate into different colours.

(a) Define the term ‘angle of refraction’.

.....

..... [2]

(b) The angle of incidence of the white light as it enters the prism is 40° and the angle of refraction for the red light is 25°.

Calculate the refractive index of the glass for red light. Show your working.

refractive index = ... [2]

(c) Using Fig. 4.1, state and explain how the refractive index for red light differs from the refractive index for violet light.

.....

.....

.....

..... [2]

(d) The source of white light used in Fig. 4.1 produces other types of electromagnetic radiation as well as visible light.

State the name of the invisible radiation found at P and the invisible radiation found at Q.

at P ..... at Q ..... [1]

[Total: 7]

Question	Answer	Marks
4(a)	clear angle is to / from the normal **or** 90°– angle between ray and surface	**C1**
	angle between/with the refracted ray / beam and the normal or vv **or** angle from the normal to the refracted ray / beam or vv	**A1**
4(b)	(refractive index =) sin*i* / sin*r* algebraic or numerical in any form	**M1**
	1.5	**A1**
4(c)	refractive index smaller (for red)	**B1**
	larger angle of refraction (for red) **or** (red) bends / refracts less (towards normal) assume comparison is for light **entering** glass at equal incident angles unless clear otherwise	**B1**
4(d)	P infrared **and Q** ultraviolet	**B1**

8 **(a)** Fig. 8.1 shows a ripple tank and the crests of the water wave that is produced in it.

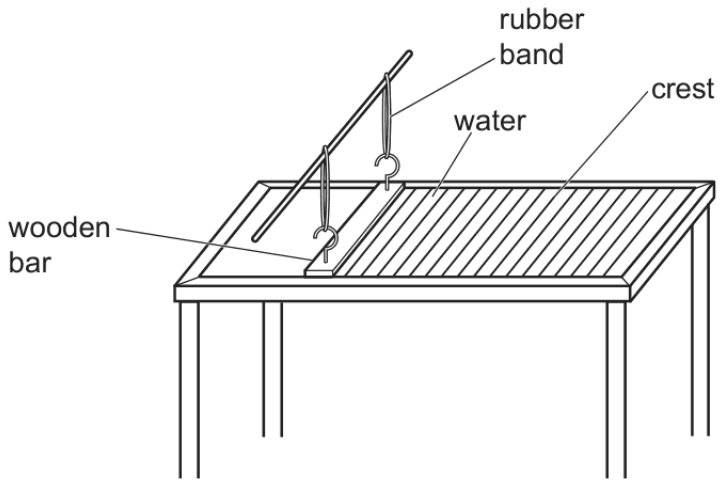


Fig. 8.1

The frequency of the water wave is 2.0 Hz and its amplitude is 3.0 mm.

(i) Calculate the number of crests produced in 1.5 s.

number = ... [1]

(ii) The height of the wave is measured from the level of the undisturbed surface of the water. The height of the wave at one point is 0 at time = 0.

On Fig. 8.2, draw a graph to show how the height of the wave at this point varies with time.

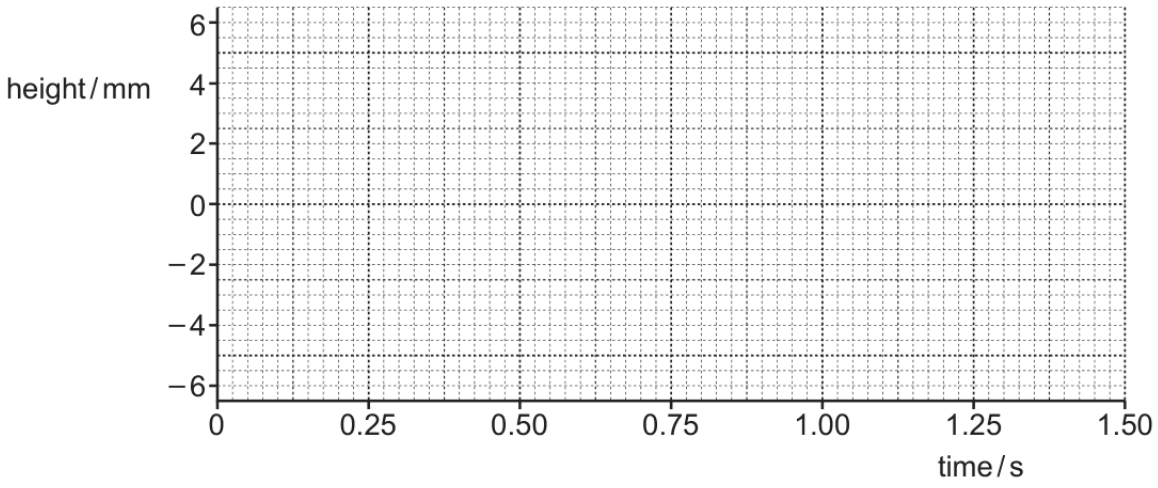


Fig. 8.2

[2]

(b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....

..... [1]

(ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed .....

wavelength .....
[2]

(c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

(i) State the additional apparatus needed to demonstrate refraction.

..... [1]

(ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.

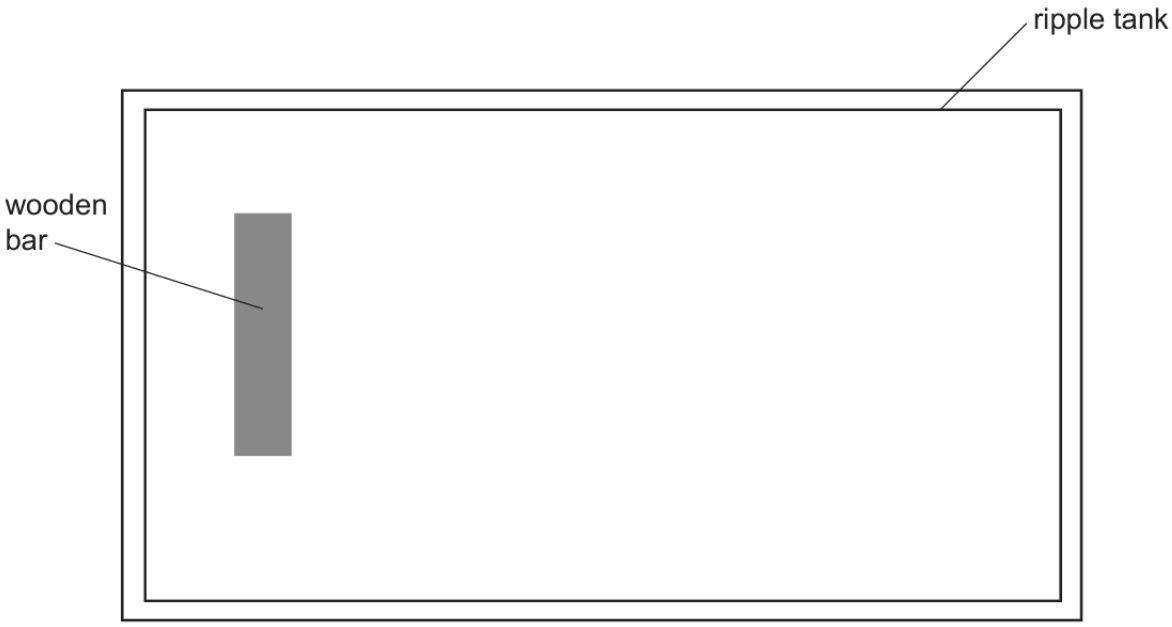


Fig. 8.3

[3]

(d) Fig. 8.4 shows a connection to the internet made from a remote station A using a satellite above the Earth's surface.

Data is sent between stations A and B using microwaves which travel to and from the satellite.

Station B is connected directly to the internet using optical fibre.

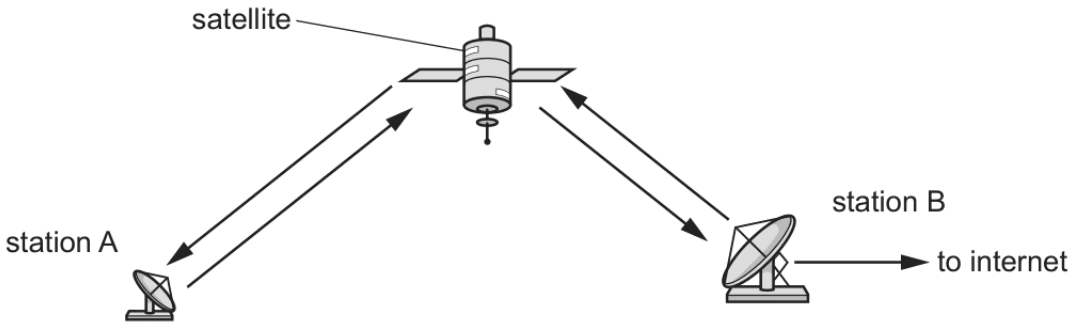


Fig. 8.4 (not to scale)

The speed of microwaves is 3.0×10^8 m/s.

Stations A and B are each 560 km from the satellite.

(i) Calculate the time taken for data to travel from A to B using microwaves.

time = ... [3]

(ii) The same data can be sent from A to B in a shorter time along the surface of the Earth using infrared waves to carry the information in optical fibres.

State **one** property that infrared waves and microwaves have in common.

.....

..... [1]

(iii) Suggest **one** other advantage of using optical fibres to connect stations A and B directly.

.....

..... [1]

[Total: 15]

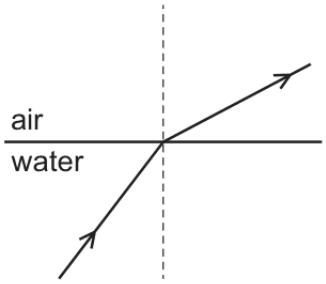
Question 9 begins over the page.

Question	Answer	Marks
8(a)(i)	3	**B1**
8(a)(ii)	shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change)	**B1**
	1 wave taking 0.5 s for first wave shown within half a square **or** 1.5 s shown for the number of waves in **(a)(i)**	**B1**
8(b)(i)	move bar up and down more often (per second) / faster **or** make rubber band(s) strong(er) / short(er) / tight(er) / more in number **or** use bar of less mass / weight	**B1**
8(b)(ii)	*speed* remains the same	**B1**
	wavelength decreases	**B1**
8(c)(i)	(piece of) plastic / (piece of) glass / block / plate **or** a shallow(er) / deep(er) area / boundary	**B1**
8(c)(ii)	diagram showing within the ripple tank: • crests parallel to bar hitting boundary with a boundary labelled	**B1**
	• correct refraction of plane crests either towards or away from normal on the same side of the normal **or** no direction change if crests are initially parallel to boundary but with a wavelength change	**B1**
	• different wavelength shown before and after refraction and constant before and after (by eye)	**B1**
8(d)(i)	(t =) distance / speed algebraic or numerical seen in any form	**C1**
	2 × 560(000) seen or used **or** 0.0019 (s)	**C1**
	0.0037 s	**A1**

Question	Answer	Marks
8(d)(ii)	One of (both): • reflect / refract • have same speed (in a vacuum) • travel in a vacuum / don't need a medium • are electromagnetic • are transverse • carry energy / cause heating • have larger wavelength / smaller frequency than visible light	**B1**
8(d)(iii)	less likely to be hacked / intercepted / greater security **or** more channels / better bandwidth possible / more data **or** less likely for transmission to be interrupted **or** cost of satellites saved / satellites expensive	**B1**

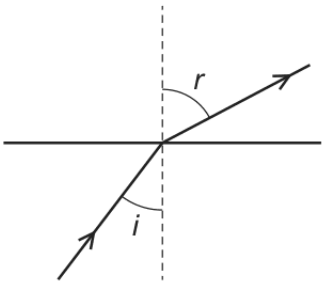
QUESTION · 5054_w22_qp_12.pdf · source page 10 of 16

23 A ray of light in water is refracted at the surface into air.

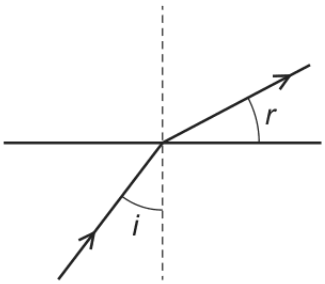


Which diagram shows the angle of incidence *i* and the angle of refraction *r*?

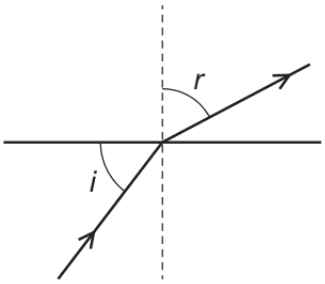
A



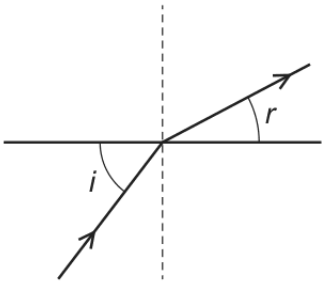
B



C



D



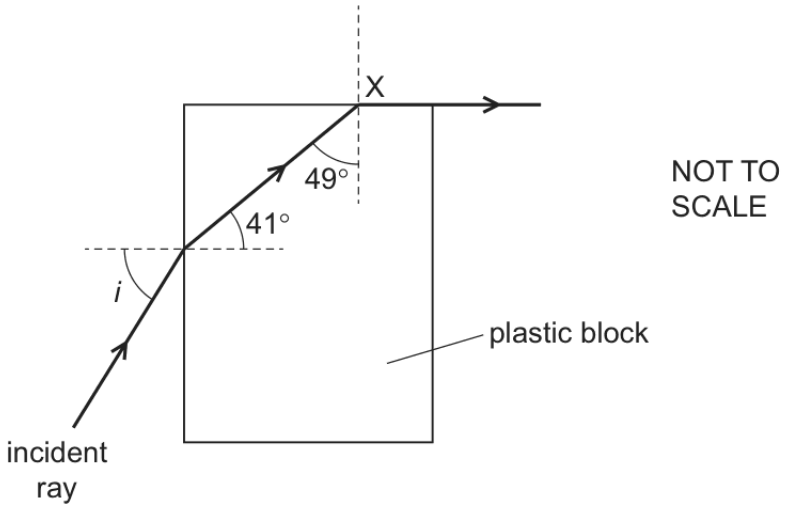
MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 23

23	A	**1**

QUESTION · 5054_s23_qp_11.pdf · source page 10 of 20

21 A ray of light is incident on a plastic block in air, at an angle of incidence *i*. The refractive index of the plastic is *n*.

The light ray refracts along the plastic-air boundary when it reaches X.



Which equation is correct?

A $\sin i = \frac{\sin 41°}{n}$

B $\sin i = \frac{n}{\sin 49°}$

C $\sin i = n \times \sin 41°$

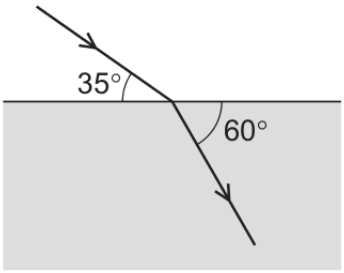
D $\sin i = n \times \sin 49°$

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 21

21	C	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 9 of 16

20 A ray of light passes into a block of transparent material as shown.



What is the refractive index of the transparent material?

A 0.66 **B** 1.15 **C** 1.64 **D** 1.83

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 20

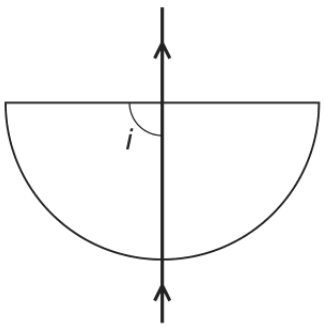
20	C	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 9 of 16

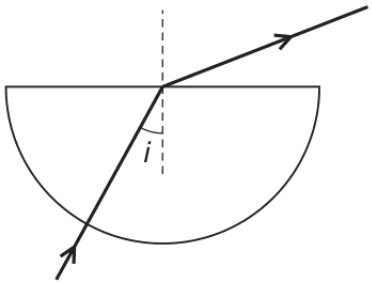
21 The diagrams show light incident on the straight edge of a semi-circular glass block after passing through it.

Which diagram shows refraction and an angle *i* equal to the critical angle of the glass?

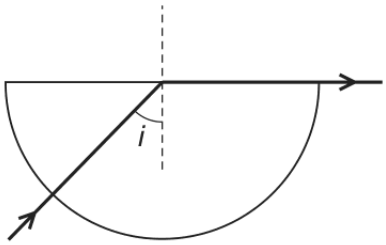
A



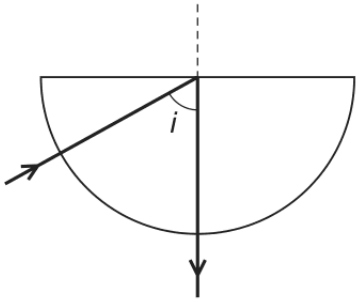
B



C



D



MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 21

21	C	**1**

7 **(a)** Ultraviolet radiation is one component of the electromagnetic spectrum.

(i) State the name of **two** components of the electromagnetic spectrum that have a smaller wavelength than ultraviolet radiation.

1 .....

2 .....
[1]

(ii) State **one** useful application of ultraviolet radiation.

..... [1]

(iii) Exposure to ultraviolet radiation from the Sun damages the eyes.

State **one** type of damage to the eye caused by ultraviolet radiation.

..... [1]

(b) Fig. 7.1 shows a ray of light. The ray passes into a semi-circular block of glass at A and leaves the glass at B, travelling along the surface to C.

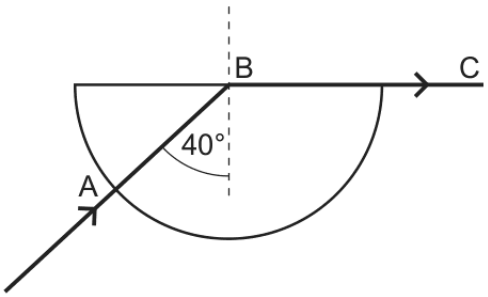


Fig. 7.1

(i) State the name given to the angle of incidence marked as 40°.

..... [1]

(ii) Calculate the refractive index of the glass.

refractive index = ... [2]

[Total: 6]

MARK SCHEME · 5054_s23_ms_22.pdf · source page 9 of 12 · Question 7

Question	Answer	Marks
7(a)(i)	X-rays **and** gamma (radiation / rays)	**B1**
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	**B1**
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	**B1**

MARK SCHEME · 5054_s23_ms_22.pdf · source page 10 of 12 · Question 7

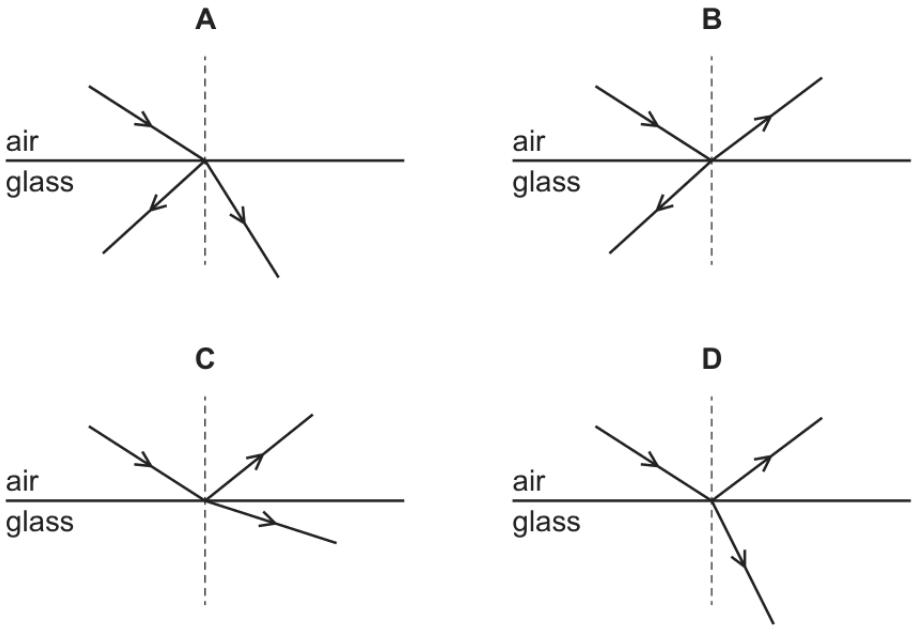
Question	Answer	Marks
7(b)(i)	critical angle	**B1**
7(b)(ii)	(n =) sin$i$ / sin$r$ **or** 1 / sin$c$	**C1**
	1.6	**A1**

QUESTION · 5054_w23_qp_11.pdf · source page 8 of 16

19 The diagrams show light travelling in air incident on the surface of a glass block.

Some light is reflected and some light is refracted.

Which diagram shows the reflection and refraction of the light?



MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 19

19	D	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 8 of 16

20 Which statement is correct?

A Total internal reflection only occurs when light travels from air into glass.

B The larger the refractive index of glass, the larger is the critical angle.

C When total internal reflection occurs, the angle of incidence is equal to the angle of reflection.

D When total internal reflection occurs, the angle of incidence is less than the critical angle.

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 20

20	C	**1**

4 Red light of frequency 4.7×10^{14} Hz travels in air at a speed of 3.0×10^{8} m/s.

(a) Calculate the wavelength in air of this red light.

wavelength = ... m [3]

(b) Fig. 4.1 shows a narrow beam of this red light striking the surface of a parallel-sided glass block.

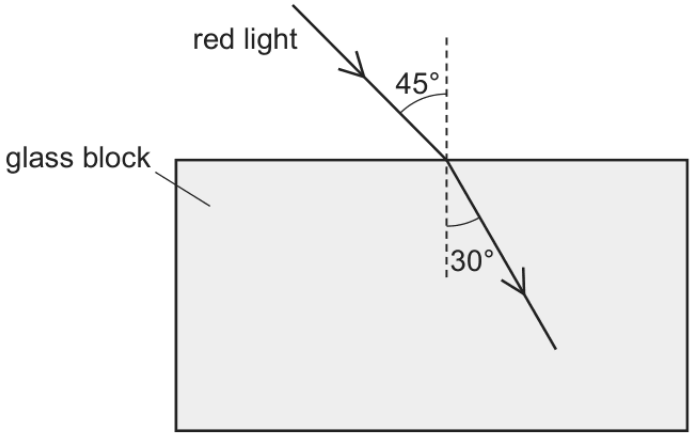


Fig. 4.1

(i) State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block.

speed .....

frequency .....

wavelength .....
[2]

(ii) Using the angles shown on Fig. 4.1, calculate the refractive index of the glass.

refractive index = ... [3]

(iii) The light continues along the path shown in Fig. 4.1 until it strikes the bottom surface of the block. Light then emerges into the air.

Draw on Fig. 4.1 to show the path taken by the light until it strikes the bottom surface and the path of the light that emerges into the air. [2]

[Total: 10]

Question	Answer	Marks
4(a)	($\lambda =$) v / f **or** $3.0 \times 10^8 / 4.7 \times 10^{14}$	**C1**
	6.4×10^{N} (m) or $3 / 4.7 \times 10^6$	**C1**
	6.4×10^{-7} (m)	**A1**
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	**B2**
4(b)(ii)	$i = 45°$ **and** $r = 30°$	**C1**
	($n =$) $\sin(i) / \sin(r)$ **or** $\sin(45°) / \sin(30°)$	**C1**
	1.4	**A1**
4(b)(iii)	straight continuation of ray in block **and** refraction away from normal in air	**B1**
	emergent ray parallel to initial incident ray	**B1**

QUESTION · 5054_w23_qp_12.pdf · source page 11 of 20

27 Light travelling in air is incident on a water surface.

The refractive index of water is 1.3 and the angle of refraction in the water is 40°.

What is the angle of incidence in the air?

A 30° **B** 50° **C** 52° **D** 57°

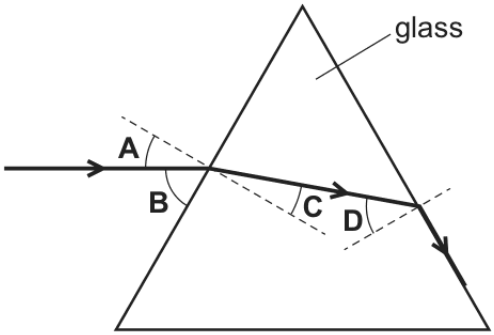
MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 27

27	D	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 12 of 20

28 The diagram shows a ray of light passing through a triangular glass block.

Which angle is the critical angle of light in glass?



MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 2 of 3 · row for Question 28

28	D	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 9 of 16

24 A ray of light strikes the surface of a glass block at an angle of incidence of 45°.

The refractive index of the glass is 1.8.

What is the angle of refraction inside the block?

A 23° **B** 25° **C** 45° **D** 81°

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 24

24	A	**1**

4 **(a)** Fig. 4.1 shows light passing through a triangular glass prism.

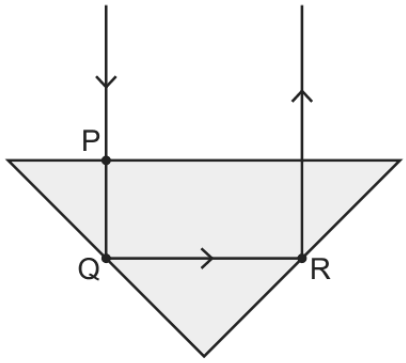


Fig. 4.1

(i) State the value of the angle of incidence at point P.

angle of incidence = ... ° [1]

(ii) Draw the normal and the angle of incidence at point R.

Label the angle of incidence. [2]

(iii) State **two** conditions needed so that no light refracts from the glass into the air at point Q.

1 .....

.....

2 .....

.....
[2]

(b) Information is sent across the internet using pulses of visible light through long, thin glass fibres and electrical signals through copper wires.

(i) State the name of **one** other type of electromagnetic radiation used to transmit information through long, thin glass fibres.

..... [1]

(ii) Suggest **two** advantages of using glass fibres rather than copper wires to transmit information from the internet.

1 .....

.....

2 .....

.....
[2]

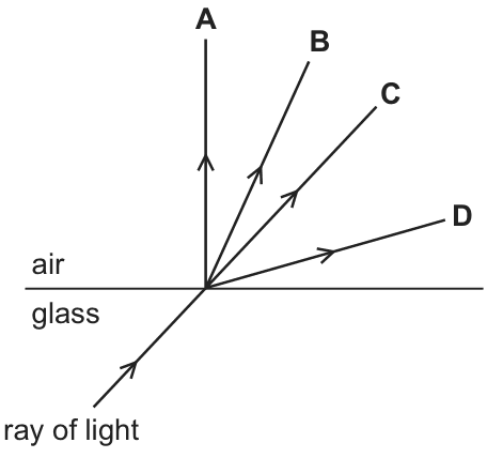
[Total: 8]

Question	Answer	Marks
4(a)(i)	0	**B1**
4(a)(ii)	normal correct by eye	**B1**
	angle of incidence correct and labelled	**B1**
4(a)(ii)	angle of incidence greater than critical angle	**B1**
	from more dense to less dense (medium) / from high to low refractive index	**B1**
4(b)(i)	infrared	**B1**
4(b)(ii)	ANY 2 from • more calls per second **or** greater bandwidth **or** faster speed • less chance of hacking • less interference • less (information) noise • longer transmission distances (before losing signal) / less attenuation • thinner **or** lighter **or** less mass/weight	**B2**

QUESTION · 5054_w24_qp_11.pdf · source page 11 of 16

25 A ray of light passes from glass to air.

Which arrow shows the direction of the ray in air?

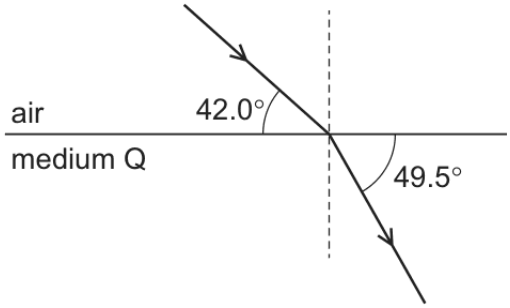


MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 25

25	D	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 9 of 16

22 The diagram shows a ray of light travelling from air to medium Q.



NOT TO SCALE

What is the critical angle for the boundary between medium Q and air?

A 57.5° **B** 58.0° **C** 60.9° **D** 61.6°

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 22

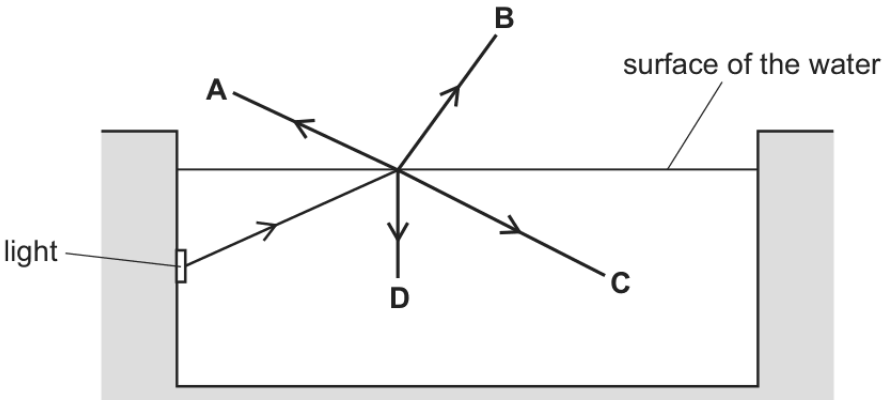
22	C	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 10 of 20

23 A swimming pool is lit by an underwater light.

A ray of light is incident on the surface of the water.

What is the path for the ray of light?



MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 23

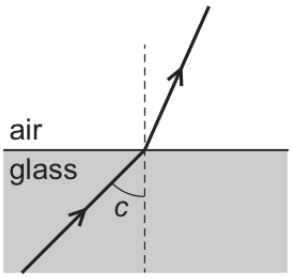
23	C	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 9 of 16

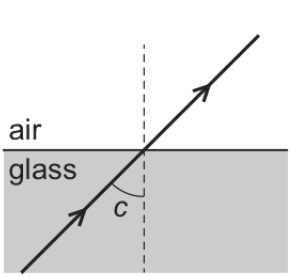
22 A ray of light in glass is incident on the surface at an angle *c*. The angle *c* is the critical angle.

Which diagram shows what happens to the light?

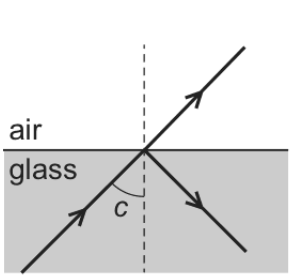
A



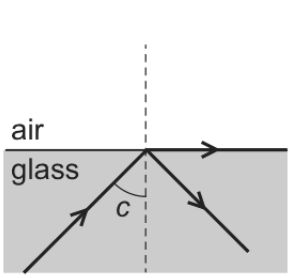
B



C



D



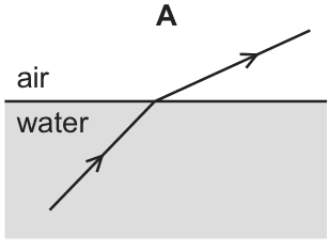
MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 22

22	D	**1**

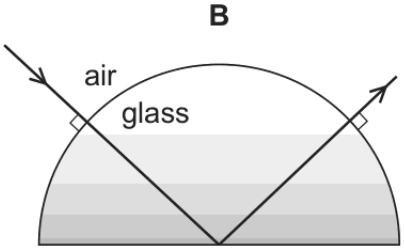
QUESTION · 5054_w25_qp_11.pdf · source page 11 of 16

22 In which diagram is the path of the light ray **not** correct?

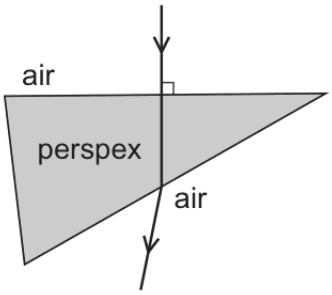
A



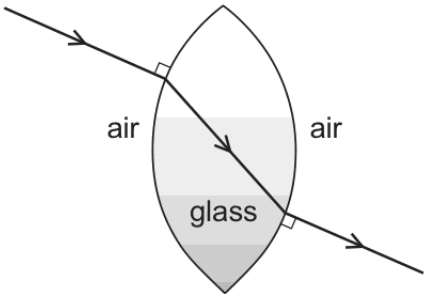
B



C



D



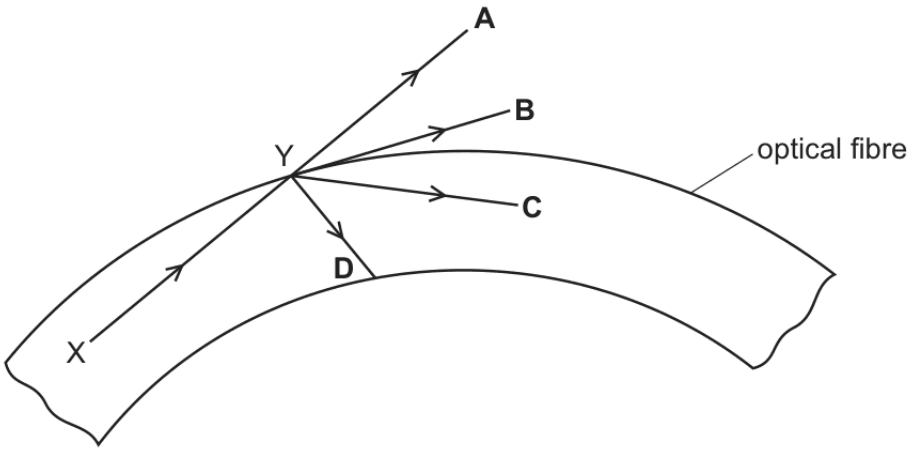
MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 22

22	D	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 11 of 16

23 A ray of light travels from X to Y along an optical fibre. The angle of incidence at Y is greater than the critical angle.

In which direction does the ray of light travel after reaching point Y?



MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 23

23	C	**1**

4 Fig. 4.1 shows a ray of light striking the left-hand surface of a thin glass converging lens.

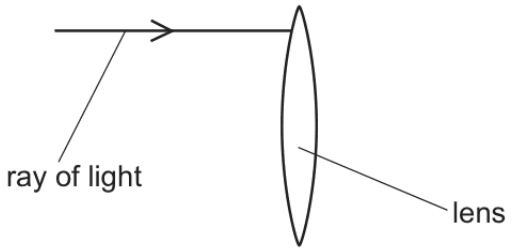


Fig. 4.1

(a) Explain why the ray refracts and changes direction as it enters the glass.

.....

.....

.....

..... [2]

(b) Explain what is meant by:

(i) the principal focus (focal point) of a lens

.....

.....

.....

..... [2]

(ii) the focal length of a lens.

.....

..... [1]

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(c) An object O of height 1.8 cm is placed at a point 2.2 cm to the left of the centre of the lens.

Fig. 4.2 is a full-scale diagram that shows the converging lens, its principal axis and its two principal focuses, F_1 and F_2.

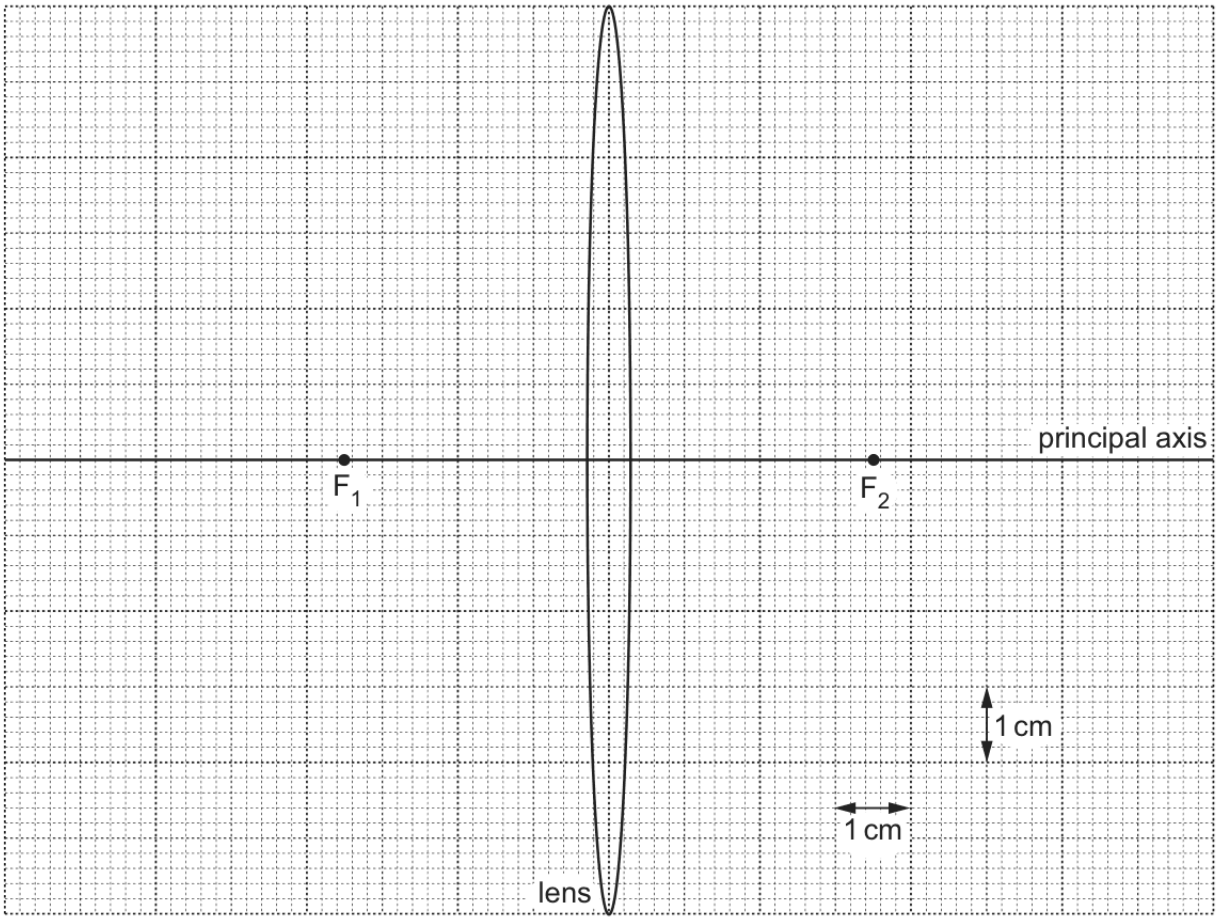


Fig. 4.2

(i) On Fig. 4.2, draw an arrow to represent the height and position of object O. The arrow should start on the principal axis. [1]

(ii) On Fig. 4.2, draw **two** rays from the tip of the arrow drawn in **(c)(i)** to locate the position of the tip of the image I of object O.

Draw an arrow to represent the image I of the object O. Label the image I. [3]

(iii) State whether the image I of the object O is a real or a virtual image, and explain your answer.

.....

.....

..... [1]

[Total: 10]

MARK SCHEME · 5054_w25_ms_21.pdf · source page 9 of 14 · Question 4

Question	Answer	Marks
4(a)	any **two** from: (in glass) light / it travels more slowly / at a different speed enters (an optically) denser medium / medium with different refractive index wavelength changes (and frequency stays the same) angle of incidence is greater than 0	**B2**

MARK SCHEME · 5054_w25_ms_21.pdf · source page 10 of 14 · Question 4

Question	Answer	Marks
4(b)(i)	(point) where parallel rays meet (after refraction)	**C1**
	(point) where rays parallel to principal axis meet (after refraction) **or** (point) where paraxial rays meet (after refraction)	**A1**
4(b)(ii)	distance between principal focus / focal point and (centre of) lens	**B1**
4(c)(i)	arrow of height 1.8 cm **and** 2.2 cm **and** on left of the centre of the lens	**B1**
4(c)(ii)	any **two** from: paraxial ray from tip of O to lens **and** from lens to F_2 straight ray through centre of lens ray that seems to come from F_1 from tip of O to lens **and** emerges paraxially from lens	**M2**
	two rays traced back to image (labelled I)	**A1**
4(c)(iii)	virtual **and** rays (merely) seem to come from I / do not pass through the image / object distance less than focal length / the image is upright	**B1**

QUESTION · 5054_w25_qp_12.pdf · source page 11 of 16

27 Light passes into a transparent material that has a refractive index n.

The angle of incidence at the surface of the transparent material is i.

Which equation is used to calculate the angle of refraction r?

A $r = n \sin i$ **B** $i = n \sin r$ **C** $\sin r = \frac{n}{\sin i}$ **D** $\sin r = \frac{\sin i}{n}$

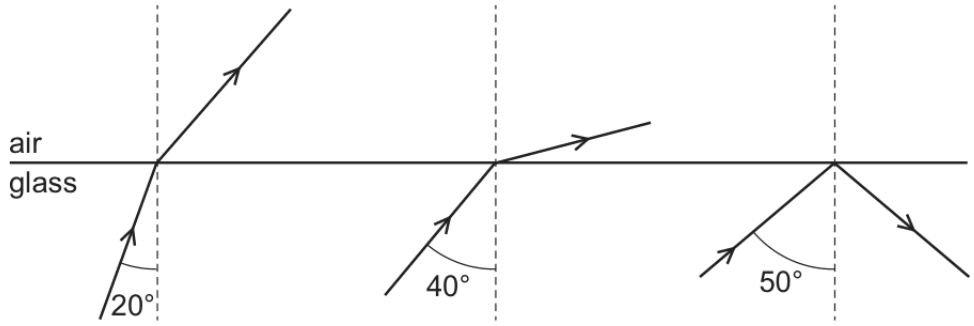
MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 27

27	D	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 12 of 16

30 Three rays of light are incident on the boundary between a glass block and air.

The angles of incidence are different.



What is a possible critical angle for light in the glass?

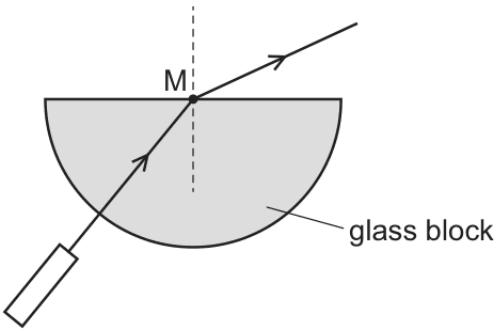
A 15° **B** 30° **C** 45° **D** 60°

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 3 of 3 · row for Question 30

30	C	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 10 of 16

23 A ray of red light from a laser passes into a semi-circular glass block.



What is shown at M?

A dispersion

B rarefaction

C reflection

D refraction

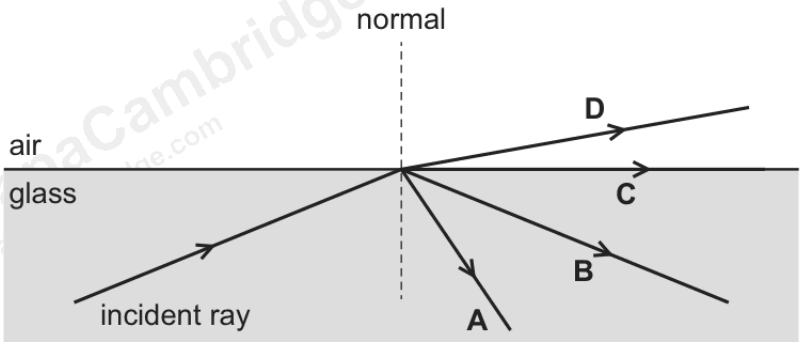
MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 23

23	D	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 10 of 16

24 Light travelling in glass is incident on a glass–air boundary. The angle of incidence of the light is greater than the critical angle.

Which arrow shows the direction of the light after it is incident on the boundary?



MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 24

24	B	1

QUESTION · 5054_s26_qp_11.pdf · source page 11 of 16

25 Three students suggest an advantage of using optical fibre cables in telecommunications rather than using copper cables.

1 Optical fibre cables are more secure.

2 Optical fibre cables transmit over greater distances without boosting the signal.

3 Optical fibre cables transmit data at a greater rate.

Which suggestions are correct?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 25

25	A	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 10 of 16

25 The refractive index of Perspex is 1.5.

What is the critical angle for light incident on the boundary between Perspex and air?

A 0.73° **B** 26° **C** 38° **D** 42°

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 25

25	D	**1**

5 A student uses a glass block and two identical glass prisms to demonstrate the action of a converging lens.

Figure 5.1 shows three parallel rays of red light incident on the glass block and the two prisms.

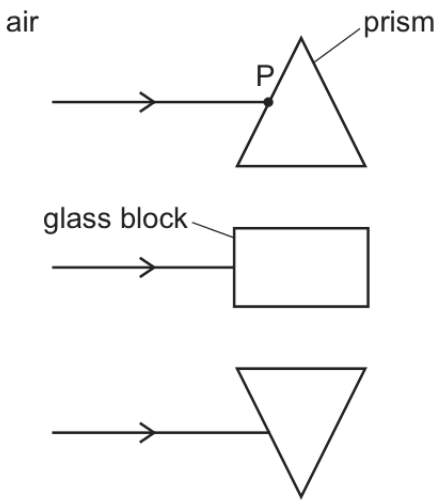


Figure 5.1

(a) (i) On Figure 5.1, draw and label the normal to the prism at point P. [1]

(ii) Complete Figure 5.1 to show how the rays of light travel through and out of both prisms and the glass block. [2]

(iii) The action of a diverging lens is also demonstrated using the two identical glass prisms and the glass block.

On Figure 5.2, draw the prisms arranged so that they demonstrate the action of a diverging lens.

Show the rays of light travelling through the prisms and the glass block.

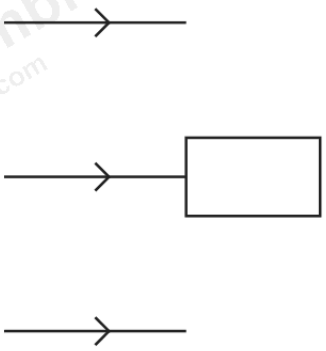


Figure 5.2

[2]

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(b) Figure 5.3 shows a converging lens, an object and two rays of light from the top of the object. The principal focus (focal point) on each side of the lens is labelled F.

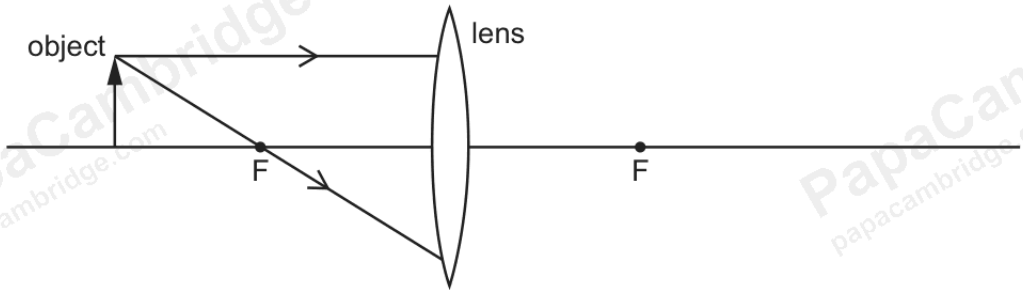


Figure 5.3

Continue the paths of both rays to show where the image of the object is formed. Draw and label the image. [2]

(c) Figure 5.4 shows the electromagnetic spectrum.

radio waves	microwaves	infrared	visible light	ultraviolet	X-rays	gamma rays

Figure 5.4

State **two** properties of electromagnetic waves that change from one end of the spectrum to the other end, and describe how these properties change from one end to the other.

1 .....

.....

2 .....

.....

[2]

[Total: 9]

5(a)(i)	correct normal	**B1**	by eye; either one side or the other of boundary or both

bod

Question	Answer	Marks	Guidance
5(a)(ii)	three correct refractions on entering and leaving the prisms	**B1**	**I** central block **con** dispersion, e.g. ray splits on refraction **allow** one wrong refraction of a ray at a boundary
	horizontal ray through and emerging from central glass block	**B1**	
5(a)(iii)	correct arrangement of prisms	**B1**	i.e. Intention that prisms inverted vertically, e.g. apex of prisms face central block
	rays diverge after prisms (top diverges up and bottom diverges down)	**B1**	**I** refraction in block and prisms**; I** ray leaving the block **not** if rays split on refraction; no prisms shown
5(b)	one correct ray; top ray passes through / to F on right of lens **or** bottom ray horizontal on right of lens	**C1**	
	both rays correct **and** image shown	**A1**	**A** point of intersection marked as I / image **A** arrow either way up as image at image position without a label
5(c)	ANY 2 from • wavelength decreases (towards the right) • frequency increases (towards the right) • damage / ionisation / energy increases (towards the right)	**B2**	**A** reverse arguments **con** speed changes assume left to right unless stated otherwise **A** just 'radio waves largest / large(r) λ'; 'γ small(est) λ' 'gamma have high(est) / high(er) frequency'; 'radio low(est) frequency' **SC** allow 1 mark for *wavelength increases and freq. decreases;* radio has largest λ **and** smallest frequency ORA for gamma

Chapter 3 Waves

3.2 Light

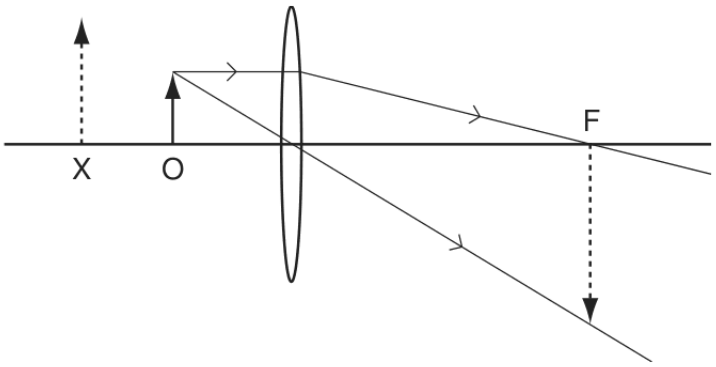
3.2.3 Thin lenses

93 entries: 71 Paper 1 questions, 22 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 10 of 20

22 A student starts to draw a ray diagram for an object at O, near a thin convex lens, but is not sure whether the image is formed at X or at F.



The correctly drawn image is

A real and formed at F.

B real and formed at X.

C virtual and formed at F.

D virtual and formed at X.

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 22

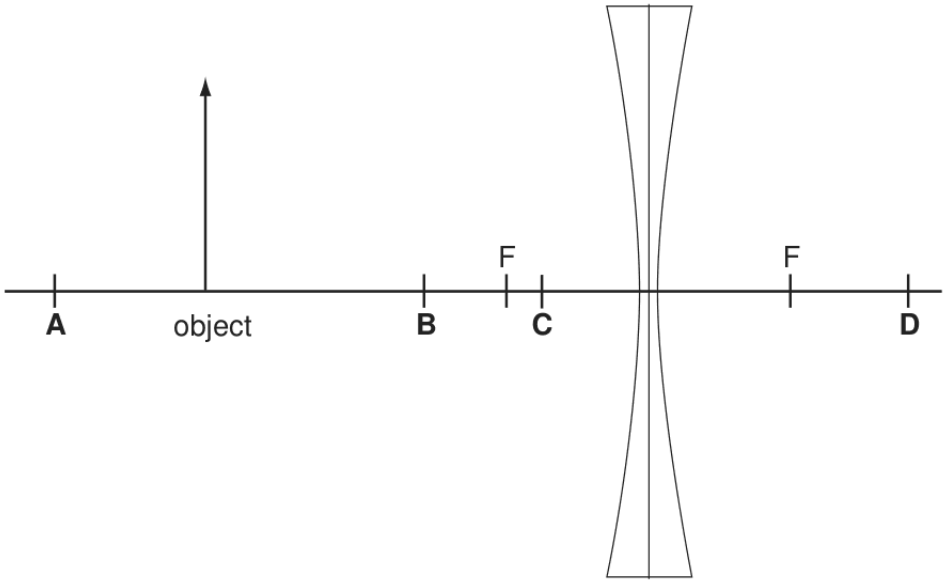
22	**D**

QUESTION · 5054_w04_qp_1.pdf · source page 12 of 20

25 An object is placed in front of a diverging lens as shown on the scale diagram.

The principal focus F is marked on each side of the lens.

At which position will the image be formed?



MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 25

25	**C**

For Examiner's Use

2 Fig. 2.1 shows the lens of a simple camera being used to photograph an object.

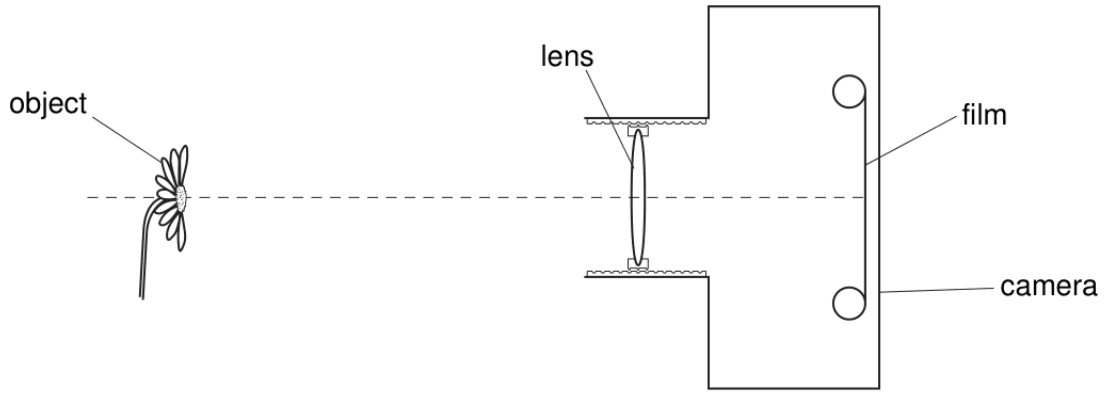


Fig. 2.1

The lens forms a focused image of the object on the film.

(a) Draw two rays from the top of the object to show how the lens forms the image. [2]

(b) The object moves closer to the camera. State how the lens is adjusted to keep the image in focus.

.....

.....

..... [1]

(c) Complete Fig. 2.2 to show how white light is split into a spectrum when it passes through a glass prism. [3]

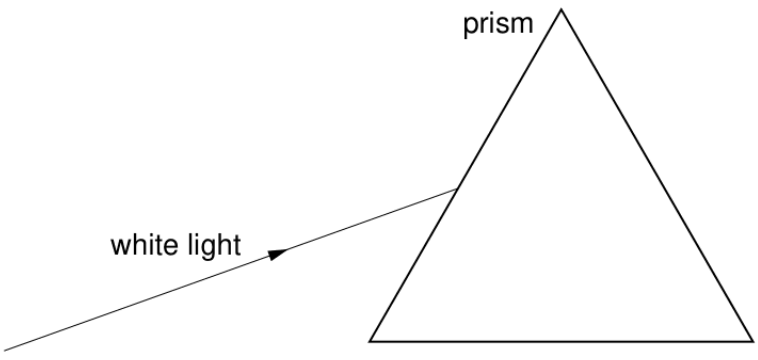
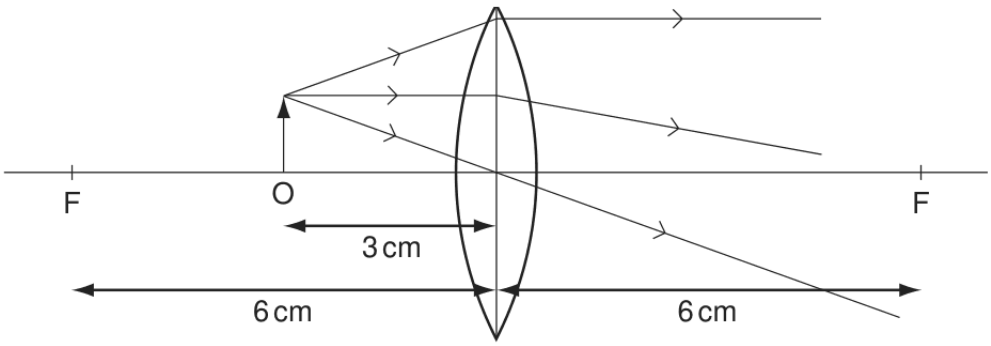


Fig. 2.2

2	**(a)**	straight line through optical centre by eye	**M1**	
		one other line from same point on object correctly to image on film	**A1**	
	(b)	move lens towards object/to left/away from film	**B1**	
	(c)	**1st and 2nd face** correct refraction for all rays shown	**B1**	
		dispersion into at least two rays at first face only	**B1**	
		colours marked on diverging rays outside prism (any 2 visible colours from spectrum, any order, accept letters)	**B1**	**6**

QUESTION · 5054_w05_qp_1.pdf · source page 12 of 20

23 The diagram shows an object O placed 3 cm away from a converging lens of focal length 6 cm.



What type of image is produced?

A real, erect and diminished

B real, inverted and magnified

C virtual, erect and magnified

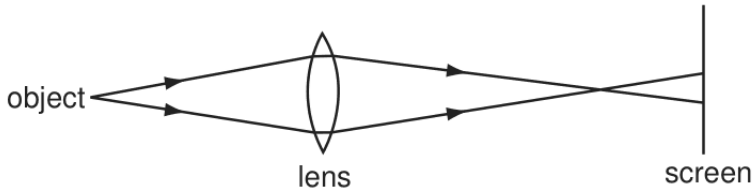
D virtual, inverted and diminished

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 23

23	**C**

QUESTION · 5054_s06_qp_1.pdf · source page 8 of 20

23 A lens forms a blurred image of an object on a screen.



How can the image be focussed on the screen?

A by moving the object away from the lens and screen

B by moving the screen away from the lens and object

C by using a brighter object at the same position

D by using a lens of longer focal length at the same position

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 23

23	**D**

QUESTION · 5054_w06_qp_1.pdf · source page 9 of 16

22 The human eye has a converging lens system that produces an image at the back of the eye.

An eye views a distant object. What type of image is produced?

A real, erect, same size

B real, inverted, diminished

C virtual, erect, diminished

D virtual, inverted, magnified

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 22

22	**B**

For Examiner's Use

5 Fig. 5.1 is drawn full scale. The focal length of the lens is 3.0 cm.

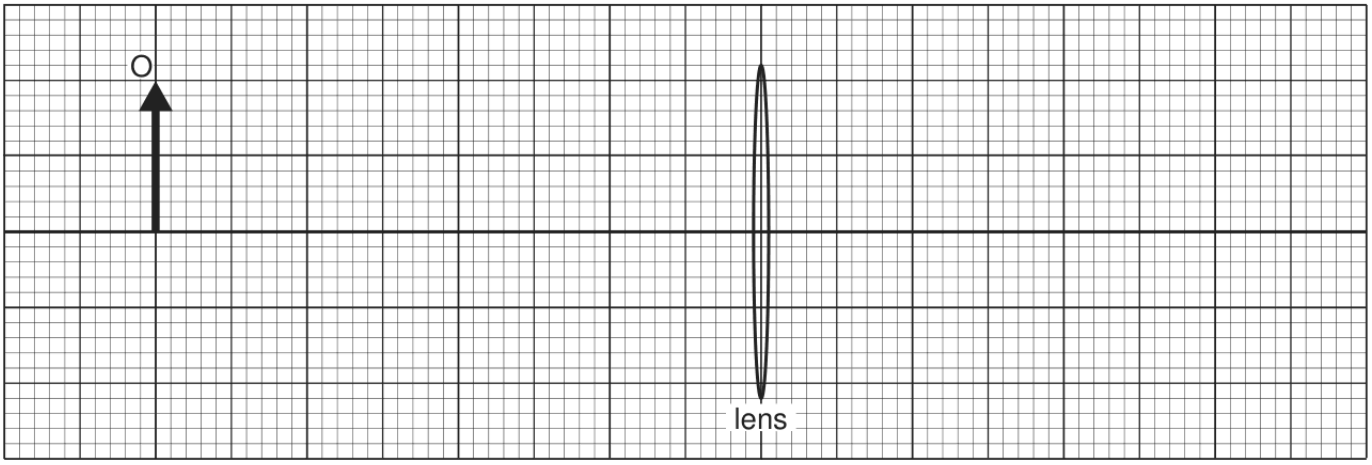


Fig. 5.1

(a) On Fig. 5.1, draw two rays from the top of the object O that meet at the image. [2]

(b) **(i)** Define the term *linear magnification*.

.....

.....

..... [1]

(ii) Determine the magnification produced by the lens in Fig. 5.1.

magnification = ... [1]

(c) Fig. 5.2 shows a normal eye viewing an object close to it. Fig. 5.3 is a long-sighted eye viewing an object at the same distance.

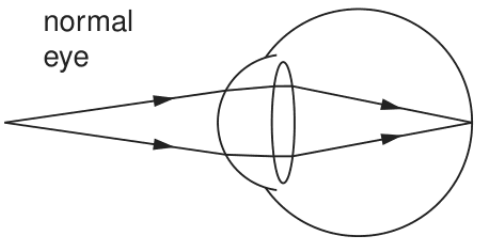


Fig. 5.2

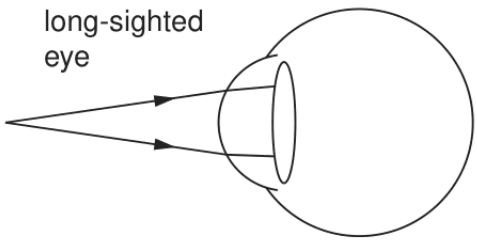


Fig. 5.3

Complete Fig. 5.3 to show the rays travelling through the eye. [1]

5 **(a)** any ray from top of object correct through lens within 1mm of optical centre or F — B1
other ray from same point correct through lens meeting 1st ray **and** none wrong — B1

(b) **(i)** image size/object size (accept image distance/object distance **or** v/u) — B1

(ii) 0.55–0.65 **ecf diagram in (a) sizes or distances** — B1

(c) rays completed to retina but would meet behind retina — B1 [5]

11 **(a)** Fig. 11.1 shows a ray of light passing through the edge of a converging lens.

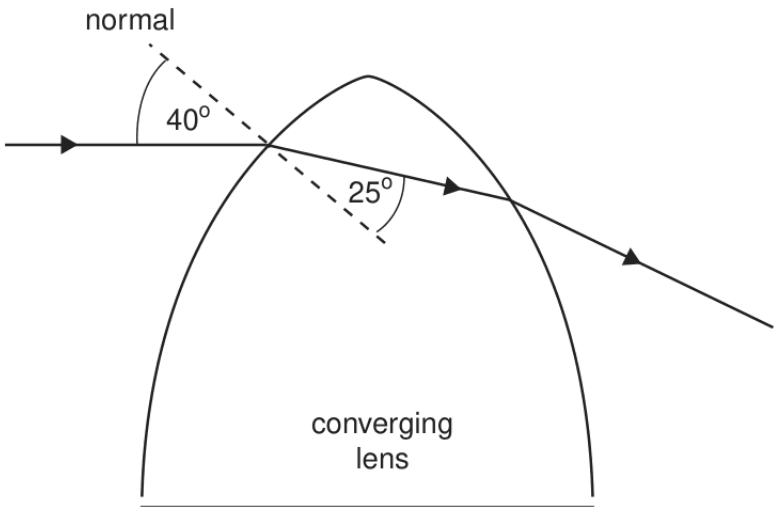


Fig. 11.1

(i) Describe what happens to the direction of the ray of light as it enters and leaves the lens. [2]

(ii) State what happens to the speed, frequency and wavelength of the light as it enters the lens. [3]

(iii) Calculate the refractive index of the glass used in the lens. [3]

(b) The focal length of the lens is 20 cm. An object is placed 50 cm from the lens and an image is formed on a screen.

(i) Explain what is meant by the *focal length* of a lens. You may draw a diagram if you wish. [2]

(ii) Draw a ray diagram to scale to show the formation of the image. [3]

(iii) The image is real. State **two** other properties of the image. [2]

11 (a) (i) (as it enters) bends towards normal B1
(as it leaves) bends away from the normal B1

(ii) speed **and** wavelength change C1
speed **and** wavelength decrease A1
frequency unaltered B1

(iii) sin(i)/sin(r) C1
sin 40°/sin 25° C1
1.5(2) (penalise °, accept 2 or more sig figs; **1.5 alone** with no working B1) A1 [8]

(b) Mark (i) and (ii) separately unless specifically referred to (i) in (ii)

(i) **Words:** distance between (principal) focus/focal point (**not** F) and lens M1
centre of lens A1
Diagram: F/(principal) focus/focal point marked **and** lens marked/curved faces/triangles
and correct arrow of some sort M1
f/FL/fl/focal length marked **and** arrow from centre of lens to F A1

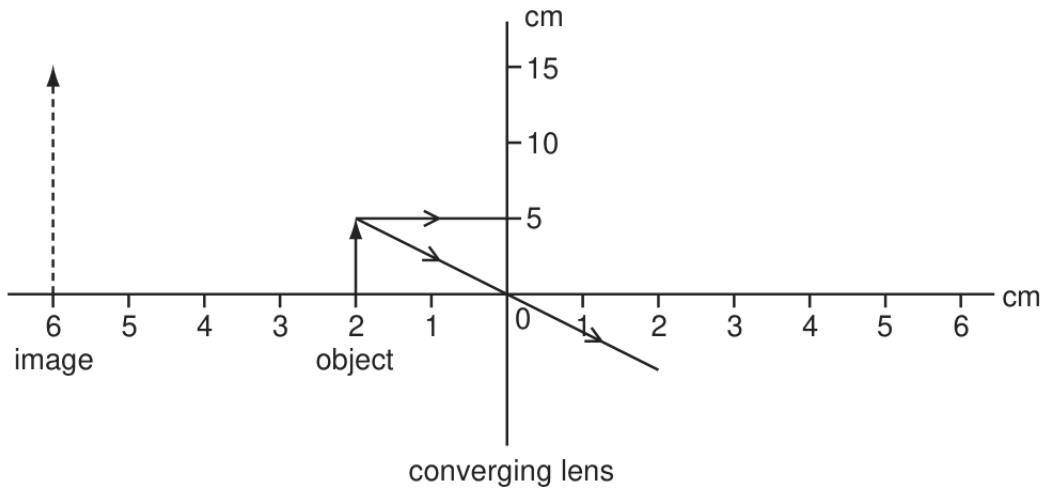
(ii) diagram showing object, lens and **one** correct ray M1
second correct ray M1
correct image shown (½ < h < 1) A1

(iii) smaller / de-magnified / e.c.f **(ii)** B1
upside down B1

QUESTION · 5054_s08_qp_1.pdf · source page 10 of 20

23 An object 5.0 cm high is placed 2.0 cm from a converging (convex) lens which is being used as a magnifying glass.

The image produced is 6.0 cm from the lens and is 15 cm high.



What is the focal length of the lens?

A 2.0 cm **B** 3.0 cm **C** 4.0 cm **D** 6.0 cm

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 23

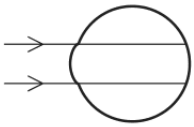
23	**B**

QUESTION · 5054_s08_qp_1.pdf · source page 11 of 20

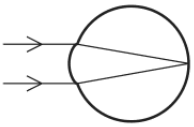
24 A man is short-sighted.

Which ray diagram shows what happens when he looks at a distant object?

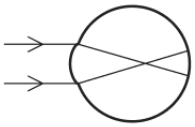
A



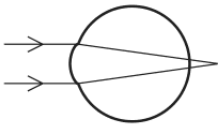
B



C



D



MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 24

24	**C**

QUESTION · 5054_w08_qp_1.pdf · source page 13 of 20

25 In a short-sighted eye, rays from distant objects are not focused on the retina.

Where are these rays focused and what type of lens is needed to correct the problem?

	where focused	lens needed
A	behind the retina	converging lens
B	behind the retina	diverging lens
C	in front of the retina	converging lens
D	in front of the retina	diverging lens

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 25

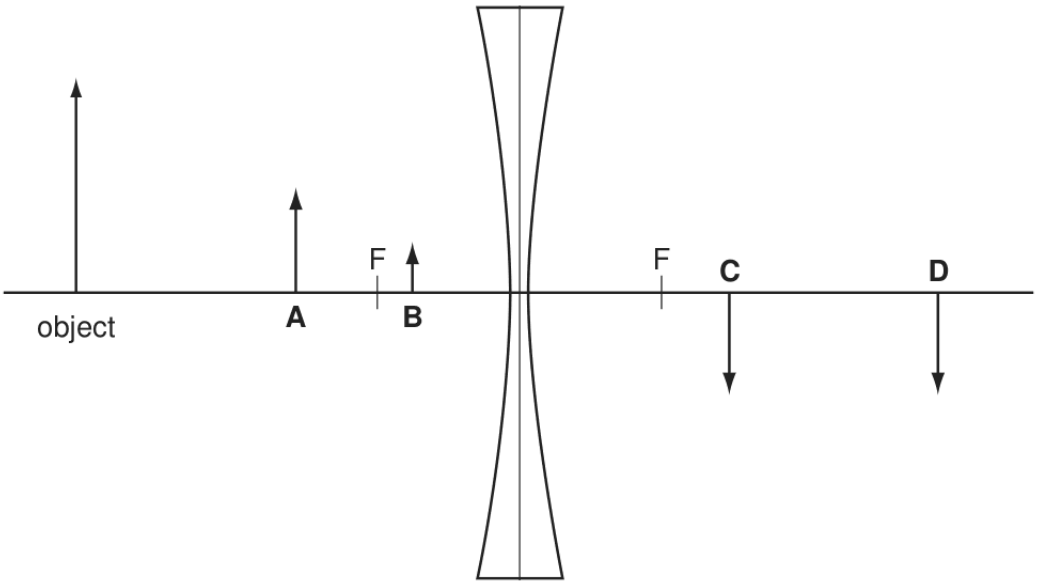
25	**D**

QUESTION · 5054_w09_qp_1.pdf · source page 11 of 20

22 An object is placed in front of a diverging lens as shown on the scale diagram.

The principal focus F is marked on each side of the lens.

What is the position of the image formed by the lens?



MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 22

22	**B**

QUESTION · 5054_s10_qp_11.pdf · source page 9 of 20

21 Convex lenses are used in cameras and as magnifying glasses.

Which types of image are formed?

	type of image in camera	type of image in magnifying glass
A	real	real
B	real	virtual
C	virtual	real
D	virtual	virtual

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 21

21	**B**

QUESTION · 5054_s10_qp_12.pdf · source page 8 of 20

18 Convex lenses are used in cameras and as magnifying glasses.

Which types of image are formed?

	type of image in camera	type of image in magnifying glass
A	real	real
B	real	virtual
C	virtual	real
D	virtual	virtual

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 18

18 **B**

QUESTION · 5054_w10_qp_11.pdf · source page 10 of 20

23 An object is viewed through a concave (diverging) lens.

What is the correct description of the image formed?

A real, inverted, magnified

B real, upright, diminished

C virtual, inverted, magnified

D virtual, upright, diminished

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 23

23	**D**

For Examiner's Use

9 **(a)** Fig. 9.1 shows a young boy lying on his back on the bottom of a swimming pool. He is holding his breath and his eyes are open. A red light is positioned on the ground at Q.

At first the boy's head is touching the pool wall. He notices that, as he slides away from the pool wall, his eye reaches a point P where he first sees the light at Q. Fig. 9.1 shows the boy in this position.

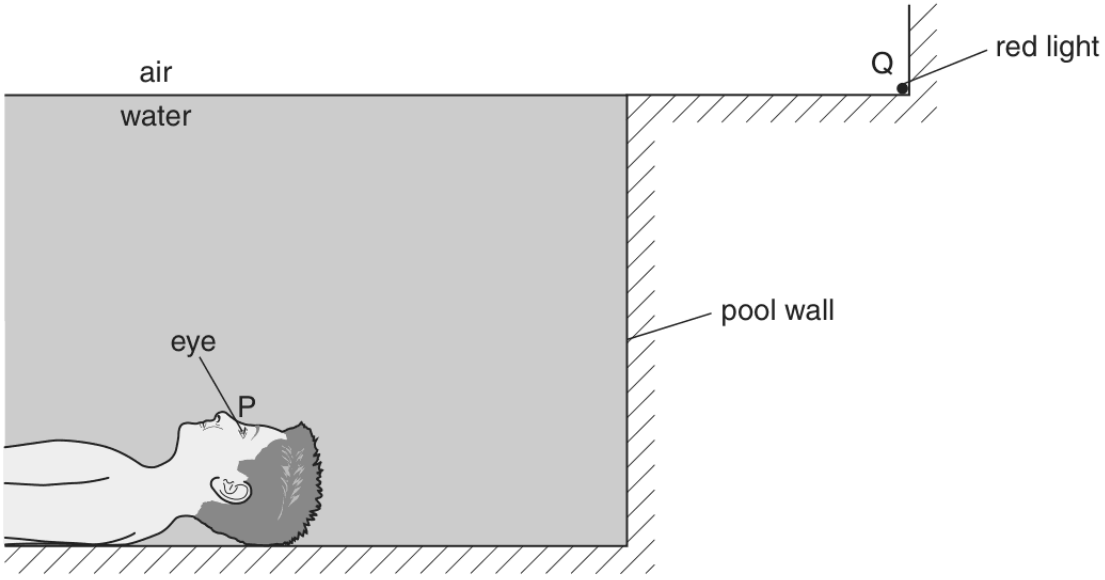


Fig. 9.1

(i) On Fig. 9.1, draw the ray of light travelling from Q to P. Mark the critical angle for light in water and label it C. [2]

(ii) Explain why the boy is unable to see the red light at Q when his eye is closer to the pool wall than P.

.....

.....

..... [2]

(iii) The critical angle is 49°. Calculate the refractive index of water.

refractive index = ... [2]

(iv) The red light is now replaced with a blue light. State the effect of this on the wavelength of the light in the air.

..... [1]

For Examiner's Use

(b) A small, very brightly illuminated display is located at the back of a projector. The projector lens produces an inverted and magnified image of the display on a white classroom wall.

Fig. 9.2 is a scale diagram showing the position and size of both the display and the image on the wall. R is a point on the display.

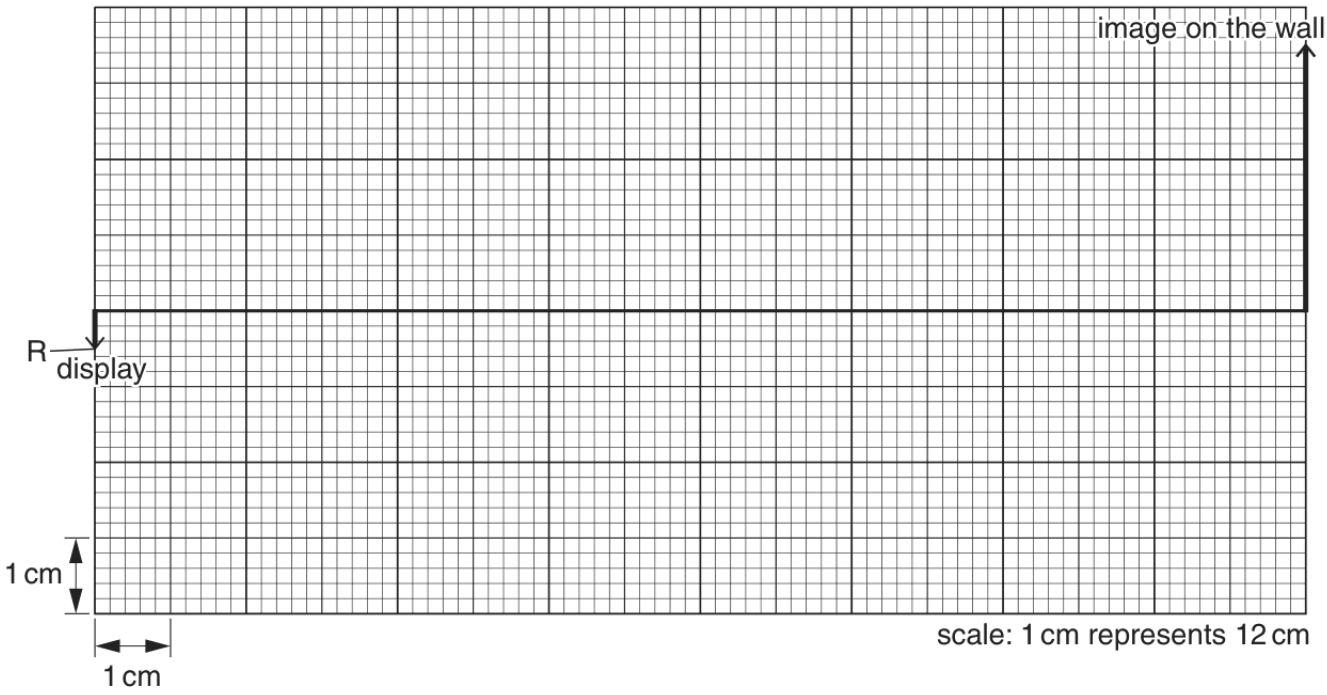


Fig. 9.2

(i) The image is inverted and magnified. State two other properties of the image.

1. .....

2. .....

[2]

(ii) On Fig. 9.2, draw the straight-line ray from R to the image. [1]

(iii) On Fig. 9.2, draw a vertical line representing the lens and label it L. [1]

(iv) A second ray from R to the image passes through a focal point (principal focus) of the lens. On Fig. 9.2, draw this ray and use it to mark this focal point. Label this focal point F. [2]

(v) Determine the focal length of the lens.

focal length = ... [2]

9	**(a)**	**(i)**	horizontal ray from Q to pool edge **and** on to P from corner	B1	
			critical angle marked *C* **or** obvious	B1	[2]
		(ii)	for $i = 90°$ **or** horizontal ray	B1	
			angle(in water) equals/cannot be less than critical/*C*	B1	[2]
		(iii)	(n =) sin *i*/sin *r* **or** 1/sin *C* **or** 1/*n* = sin *C* **or** sin 90°/sin 49° **or** 1/sin 49°	B1	
			1.3(2501)	B1	[2]
		(iv)	decreases	B1	[1]
	(b)	**(i)**	any **two** of: real less bright further from lens beyond 2*f*	B2	[2]
		(ii)	straight ray from R to **top** of image	B1	[1]
		(iii)	where ray crosses principal axis, vertical line (L **or** drawn lens)	B1	[1]
		(iv)	paraxial ray from R to lens refracted to top of image **or** paraxial ray from lens to top of image, traced back to R	M1	
			F marked	A1	[2]
		(v)	1.6 – 1.9 cm **or** attempt to use 1/*u* + 1/*v*	C1	
			19 – 23 cm (2 sig. fig. only)	A1	[2]

QUESTION · 5054_w10_qp_12.pdf · source page 9 of 16

24 An object is viewed through a concave (diverging) lens.

What is the correct description of the image formed?

A real, inverted, magnified

B real, upright, diminished

C virtual, inverted, magnified

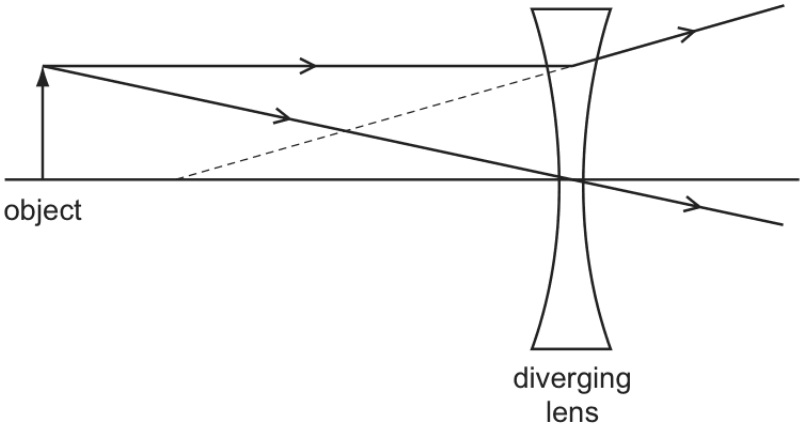
D virtual, upright, diminished

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 24

24	**D**

QUESTION · 5054_s11_qp_11.pdf · source page 10 of 16

23 The ray diagram shows two rays from a point on an object placed in front of a diverging (concave) lens.



What are the properties of the image produced?

A real and larger than the object

B real and smaller than the object

C virtual and larger than the object

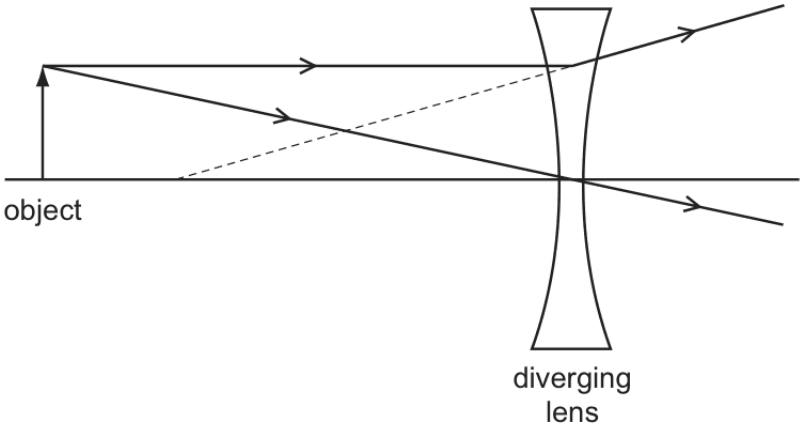
D virtual and smaller than the object

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 23

23	**D**

QUESTION · 5054_s11_qp_12.pdf · source page 10 of 16

24 The ray diagram shows two rays from a point on an object placed in front of a diverging (concave) lens.



What are the properties of the image produced?

A real and larger than the object

B real and smaller than the object

C virtual and larger than the object

D virtual and smaller than the object

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 24

24	**D**

10 (a) Describe an experiment to measure the critical angle for light in glass or perspex.

Your answer should include a labelled diagram.

.....

.....

.....

.....

.....

.....

.....

.....[5]

(b) Fig. 10.1 represents a simple camera.

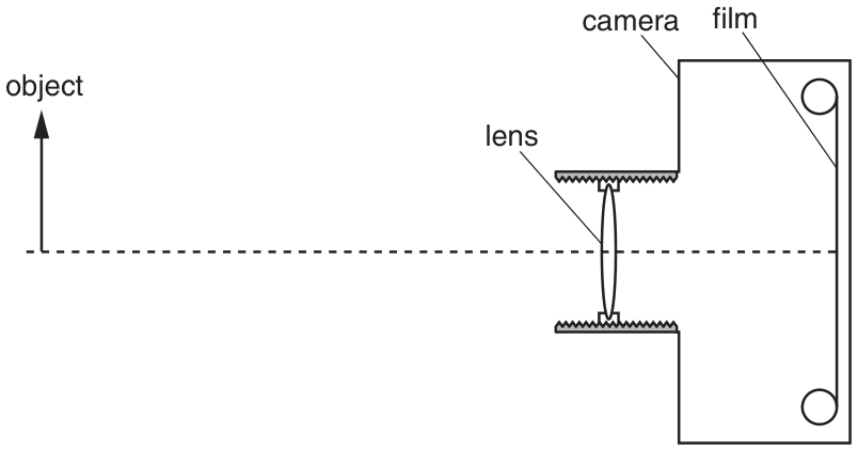


Fig. 10.1 (to scale)

(i) State the type of lens used in this simple camera.

.....[1]

(ii) Draw two rays from the top of the object to show how the image is formed on the film. Mark and label the image on the film. [3]

(iii) Define the term *linear magnification*.

.....

.....[1]

(iv) Fig. 10.1 is drawn to scale. Determine the linear magnification of the object shown in Fig. 10.1.

magnification = ...[1]

(v) Apart from its size, state one other property of the image formed by the lens.

.....[1]

(vi) Explain why, when taking photographs of other objects, it may be necessary to move the lens towards the film.

.....

.....

.....

.....[3]

10 (a) suitable block (semicircular/rectangular/prism) B1
suitable source of rays (e.g. ray box; pins on incident ray; laser **not** torch)
must be labelled on diagram or clear in text B1
diagram showing incident ray in glass/perspex (no arrow needed)
and correct refraction out into air B1
adjustment of (angle of incidence of) ray until along surface/just no longer emerges B1
(measure) correct angle marked or described clearly or C marked on diagram B1

(b) (i) converging or convex B1

(ii) ray from top through middle of lens undeviated B1
other ray from top of object to same position on film M1
correct image labelled/drawn/marked A1

(iii) ratio of size/height/length/distance of image to size/height/length/distance of object B1

(iv) 0.4(0) (±0.05) no ecf **(iii)** B1

(v) upside down // inverted // real // other side of lens to object // nearer lens than object B1

(vi) (otherwise) not focussed // to/adjust focus // to produce a clear/sharp image // B1
(otherwise) rays do not converge on film // to converge rays onto film //
image on film // object at different distance C1
(otherwise) image formed in front of film // object now further A1 [15]

4 Fig. 4.1 shows a glass lens in air and its two focal points F_1 and F_2.

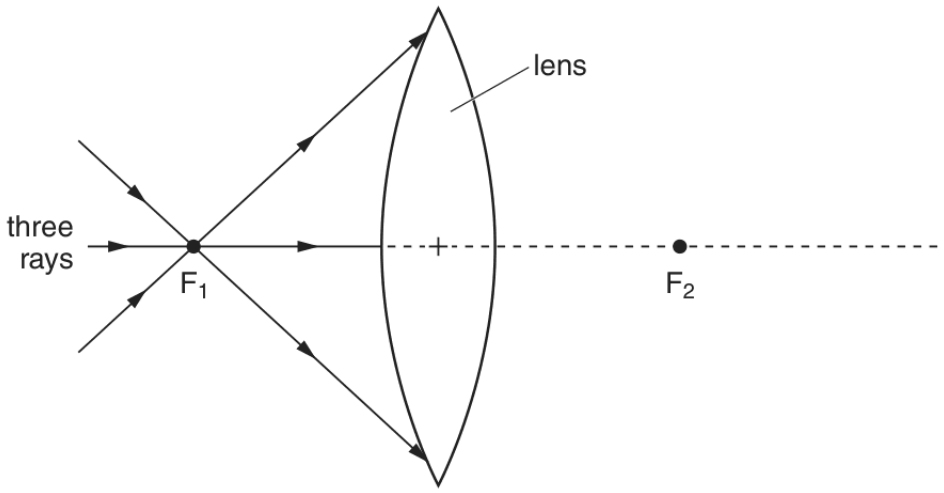


Fig. 4.1

Three rays of light pass through F_1 to the lens.

(a) On Fig. 4.1, continue the three rays through the lens and into the air. [2]

(b) State what happens to the speed of light on

(i) entering the glass lens from air,

.....[1]

(ii) leaving the lens and returning to the air.

.....[1]

(c) Light of wavelength 6.0×10^{-7} m travels in air at a speed of 3.0×10^{8} m/s.

(i) Calculate the frequency of this light.

frequency = ...[2]

(ii) State the effect, if any, on the frequency as the light enters the glass from air.

.....[1]

4	**(a)**	one outer ray parallel to principal axis three rays parallel to the principal axis	C1 A1	
	(b) **(i)**	(speed) reduced **or** slows down	B1	
	(ii)	(speed) returns to original value/$3.0 \times 10^{8}\,\text{m/s}$	B1	
	(c) **(i)**	$(f =)\ c/\lambda$ **or** $3.0 \times 10^{8} / 6.0 \times 10^{-7}$ $5(.0) \times 10^{14}\,\text{Hz}$	C1 A1	
	(ii)	no effect/unchanged/$(f =)\ 5(.0) \times 10^{14}\,\text{Hz}$	B1	[7]

8 Fig. 8.1 shows a short-sighted eye.

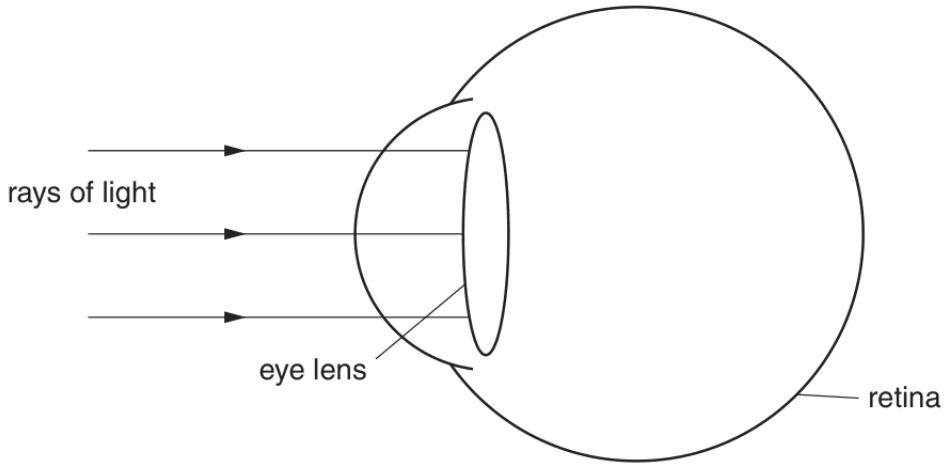


Fig. 8.1

Rays of light from a distant star are parallel as they reach the lens of the eye. Refraction of light as it enters the eye has been ignored in Fig. 8.1.

(a) (i) On Fig. 8.1, continue the rays to show their paths inside the short-sighted eye until they strike the retina. [2]

(ii) Explain how your diagram shows that the image of the star seen by the observer is blurred.

.....

.....[1]

(b) Fig. 8.2 shows three parallel rays of light.

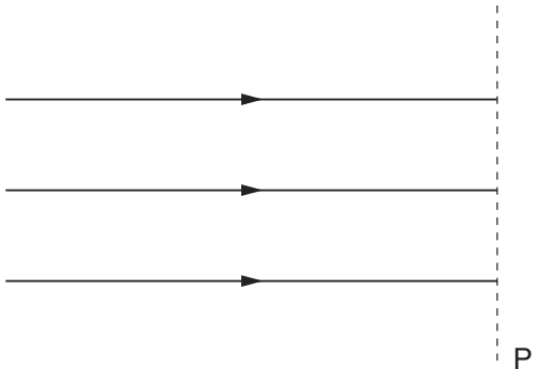


Fig. 8.2

(i) On the line P in Fig. 8.2, draw the shape of a lens that is used to correct short sight. [1]

(ii) On Fig. 8.2, continue the three rays through the lens that you have drawn. [1]

8	**(a)**	**(i)**	central ray undeviated emerging from lens	M1	
			two outer rays meet the central ray at a point inside the eye **and** carry on to strike the retina	A1	

		(ii)	light (from a single point) is spread over an area (on the retina) **or** rays do not meet at a point on the retina **or** image formed/rays meet/principal focus off retina	B1	
	(b)	**(i)**	**any** diverging lens: biconcave, planoconcave, convexoconcave – i.e. lens **clearly** thinner at the centre	B1	
		(ii)	**all** rays diverge	B1	[5]

QUESTION · 5054_s12_qp_11.pdf · source page 9 of 16

24 In a short-sighted eye, light from distant objects is not focused on the retina.

Where is this light focused and what type of lens is needed to correct the problem?

	where focused	lens needed
A	behind the retina	converging lens
B	behind the retina	diverging lens
C	in front of the retina	converging lens
D	in front of the retina	diverging lens

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 24

24 **D**

10 (a) Fig. 10.1 shows the path of a ray of blue light as it passes through a glass prism.

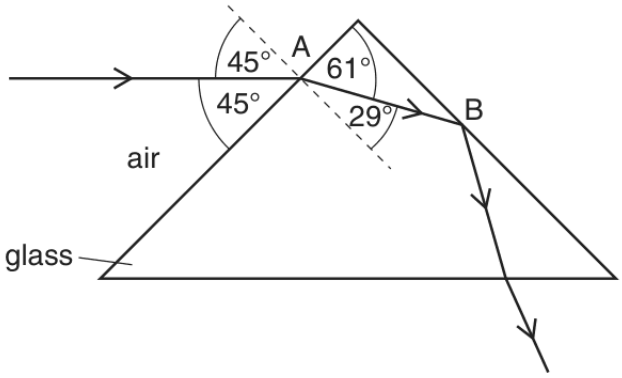


Fig. 10.1

(i) State the wave term used to describe what happens to the ray of light at A.

.....[1]

(ii) Using angles from Fig. 10.1, calculate the refractive index of the glass.

refractive index = ...[3]

(iii) Explain why the ray does not emerge from the prism at B.

.....

.....[2]

(iv) Fig. 10.2 shows a second, horizontal, ray of blue light striking the prism at point C.

On Fig. 10.2, continue the path of the second ray through and out of the glass prism. [2]

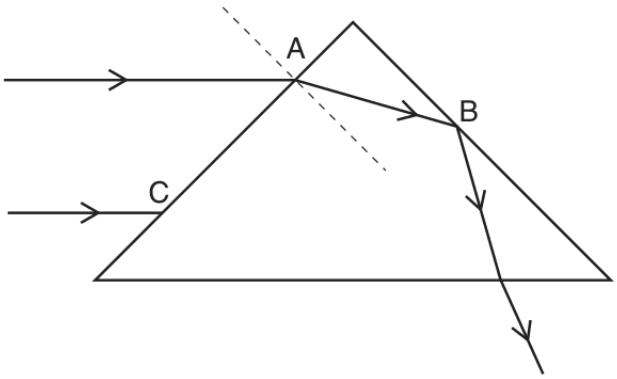


Fig. 10.2

(b) The camera lens shown in Fig. 10.3 is used to photograph the object O.

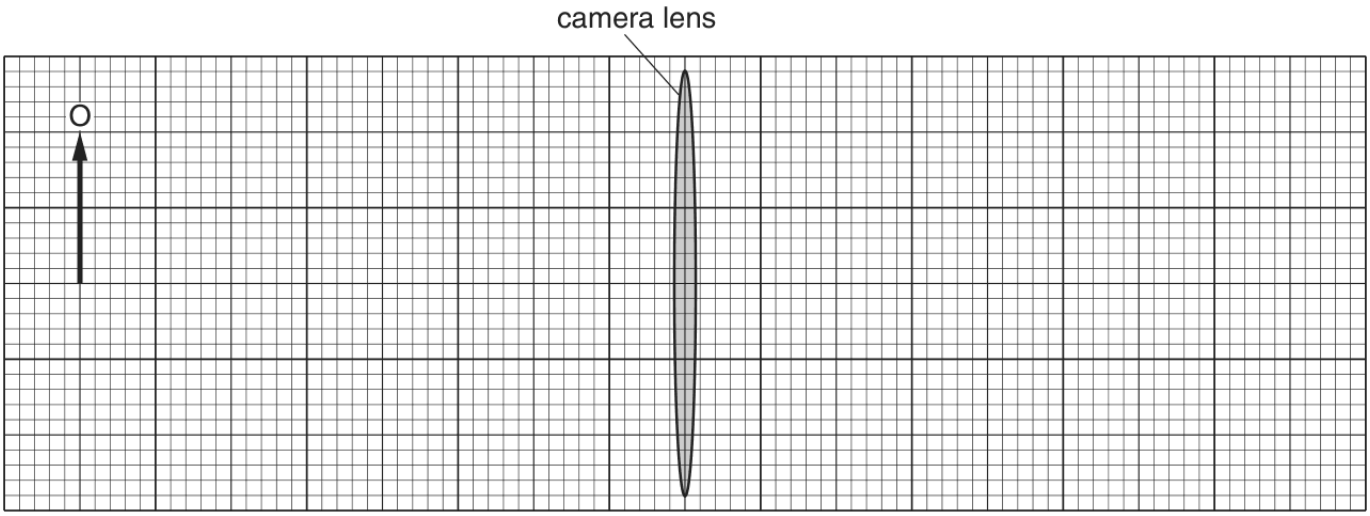


Fig. 10.3 (full scale)

The object O is 2.0 cm high and is placed 8.0 cm from the centre of the lens. The lens has a focal length of 3.0 cm.

(i) Draw rays on Fig. 10.3 to find the position and height of the image formed by the lens. Label the image I. [3]

(ii) Determine the height of the image.

.....[1]

(iii) The image formed by the lens is a real image.

1. Explain the difference between a real image and a virtual image.

.....

.....

.....[1]

2. Explain how a converging lens is used to produce and view a virtual image.

.....

.....

.....

.....[2]

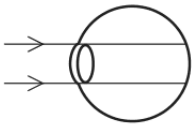
10	**(a)**	**(i)**	refraction	B1
		(ii)	(n =) sin i/sin r	C1
			sin 45°/sin 29°	C1
			1.4585 to more than 1 sig. fig.	A1
		(iii)	the angle of incidence/incident angle is greater than the critical angle	B1
			total internal reflection occurs	B1
		(iv)	correct refraction at C with ray parallel to AB	B1
			correct reflection (and correct refraction on other face i.e. downwards)	B1
	(b)	**(i)**	Any TWO of: undeviated ray through centre of lens ray parallel to axis through point 3 cm from lens on right after lens ray through point 3 cm to left of lens parallel to axis after lens	M2
			rays converge and vertical image drawn and labelled I	A1
		(ii)	1.2 ± 0.2 cm	B1
		(iii)	**1.** real image (can be) formed on screen; virtual image not found on screen; rays converge on real image; rays do not converge on virtual image; rays only appear/seem to come from a point on virtual image	B1
			2. place object within focal length; between lens and focal point/principal focus	B1
			view from other side of lens; look through lens; image same side as/behind object	B1 [15]

QUESTION · 5054_s12_qp_12.pdf · source page 11 of 16

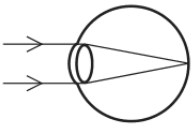
24 A man is short-sighted.

Which ray diagram shows what happens in his eye when he looks at a distant object?

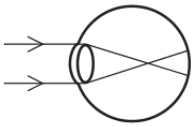
A



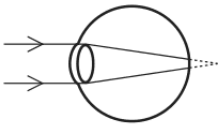
B



C



D



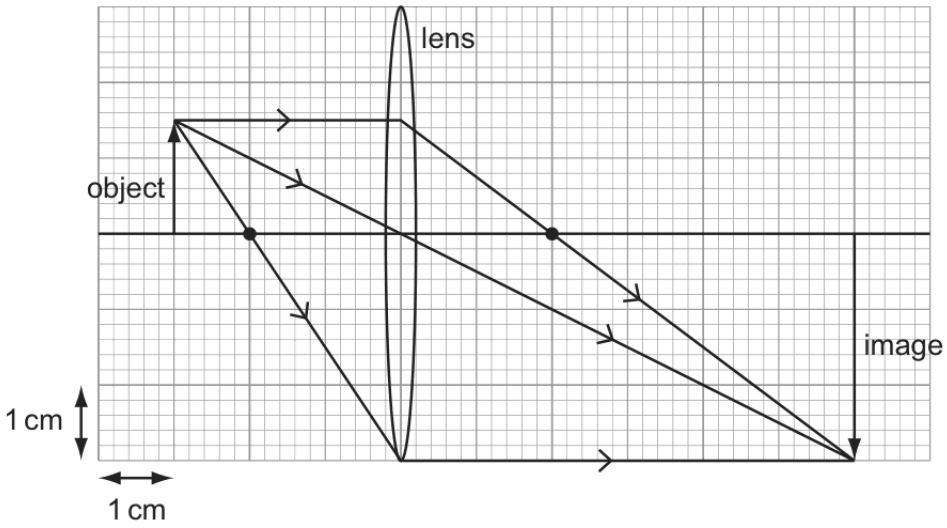
MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 24

24	**C**

QUESTION · 5054_w12_qp_11.pdf · source page 10 of 16

24 An object of height 1.5 cm is placed in front of a converging lens of focal length 2.0 cm.

The arrangement is shown on the full-scale ray diagram.



What is the linear magnification produced by the lens?

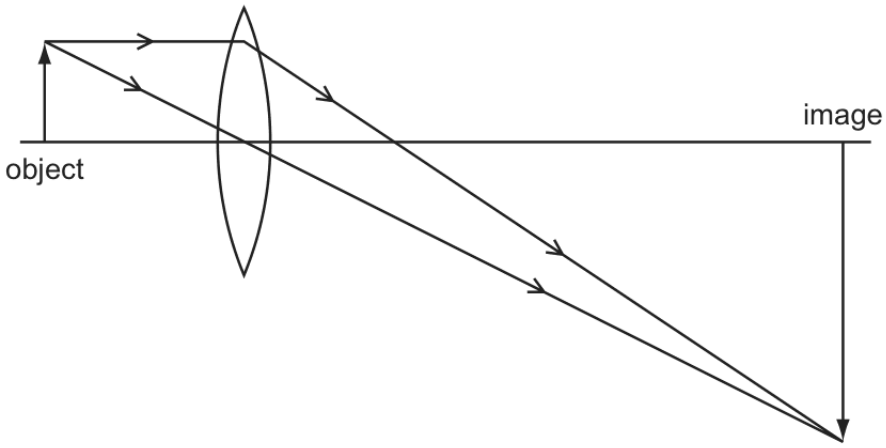
A 2.0 **B** 3.0 **C** 4.0 **D** 6.0

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 24

24	**A**

QUESTION · 5054_w12_qp_12.pdf · source page 9 of 16

24 A lens is used to produce a magnified image, as shown in the scale diagram.



What is the linear magnification of the object?

A 0.33 **B** 3.0 **C** 4.0 **D** 6.0

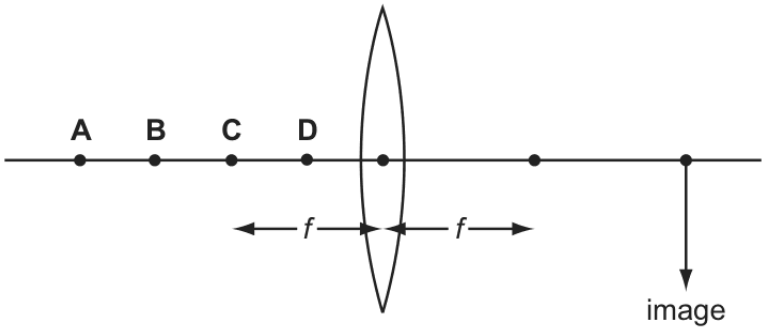
MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 24

24	**B**

QUESTION · 5054_s13_qp_11.pdf · source page 9 of 20

21 The diagram shows a thin converging lens of focal length *f*.

Where must an object be placed to produce a real image in the position shown?

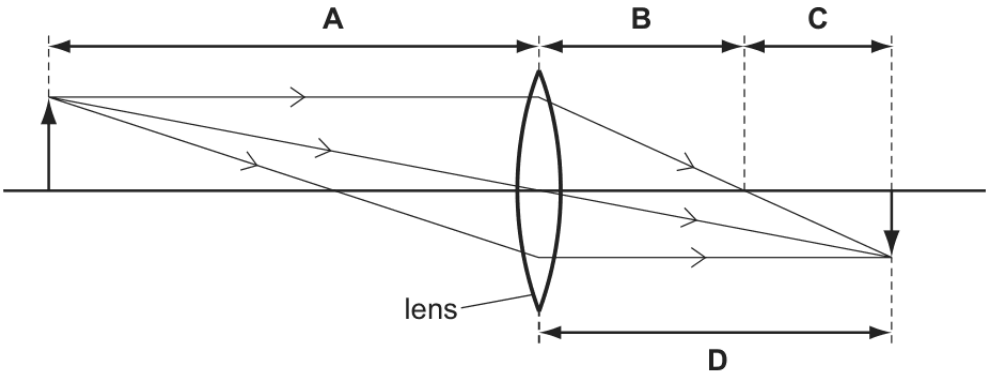


MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 21

21	**A**

QUESTION · 5054_s13_qp_12.pdf · source page 10 of 16

26 Which length is the focal length of the lens?

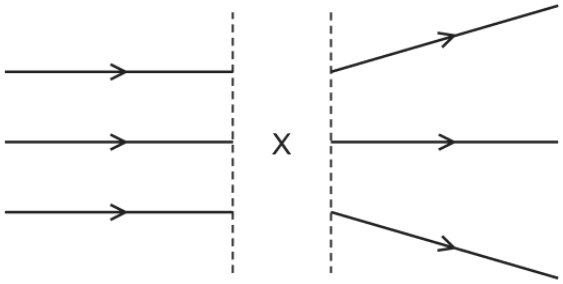


MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 26

26	**B**

QUESTION · 5054_w13_qp_11.pdf · source page 10 of 20

21 The diagram shows rays of light.



What is in the space labelled X?

A a converging lens

B a diverging lens

C a plane mirror

D a rectangular glass block

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 21

21	**B**

QUESTION · 5054_s14_qp_11.pdf · source page 10 of 16

27 Which row applies to a short-sighted eye viewing a distant object?

	position of the image	lens needed for correction
A	behind the retina	converging lens
B	behind the retina	diverging lens
C	in front of the retina	converging lens
D	in front of the retina	diverging lens

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 27

27 **D**

3 A collector views a postage stamp of height 1.5 cm through a lens. The lens is 2.0 cm from the stamp. The image has a linear magnification of 3.0.

The stamp, the image of the stamp and the position of the lens are shown full scale in Fig. 3.1.

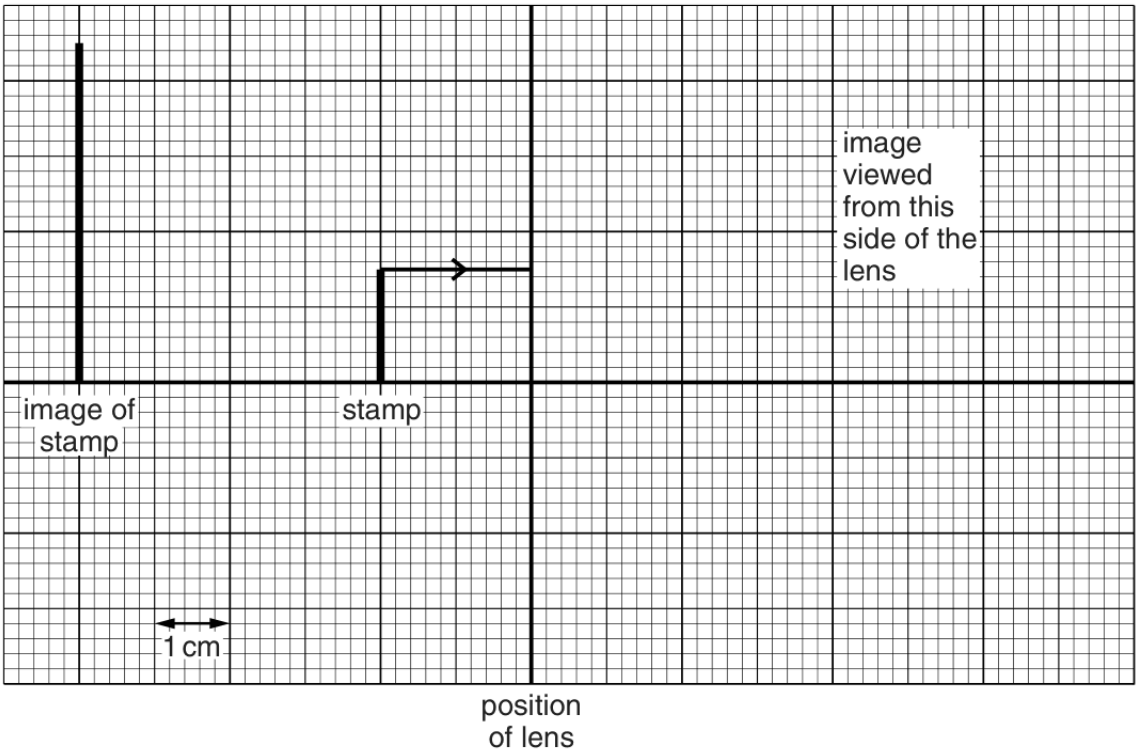


Fig. 3.1 (full scale)

A ray of light from the top of the stamp to the lens is shown on Fig. 3.1.

(a) State the type of lens used.

..... [1]

(b) State what is meant by *linear magnification*.

.....

..... [1]

(c) **(i)** On Fig. 3.1, complete the path of the ray from the top of the stamp after it passes through the lens. [1]

(ii) Use your drawing to determine the focal length of the lens.

..... [1]

(iii) On Fig. 3.1, draw two additional rays from the top of the stamp to show how the image is formed. [1]

3 **(a)** converging **or** convex [B1]

(b) image height ÷ object height [B1]

(c) **(i)** line when extended back joins top of image with intersection of ray and lens [B1]

(ii) 3.0 ± 0.1 cm ecf from diagram [B1]

(iii) any two further lines from top of stamp that appear to come from the top of the image [B1]

[5]

6 **(a)** A beam of parallel light strikes a converging lens of focal length 2.8 cm.

The width of the beam before it reaches the lens is 1.0 cm. The width changes on the other side of the lens.

State a distance from the lens where the width of the beam is

(i) less than 1.0 cm,

..... [1]

(ii) more than 1.0 cm.

..... [1]

(b) An object is placed 3.0 cm from a converging lens of focal length of 2.8 cm. Fig. 6.1 is an incomplete, full-scale ray diagram for this arrangement.

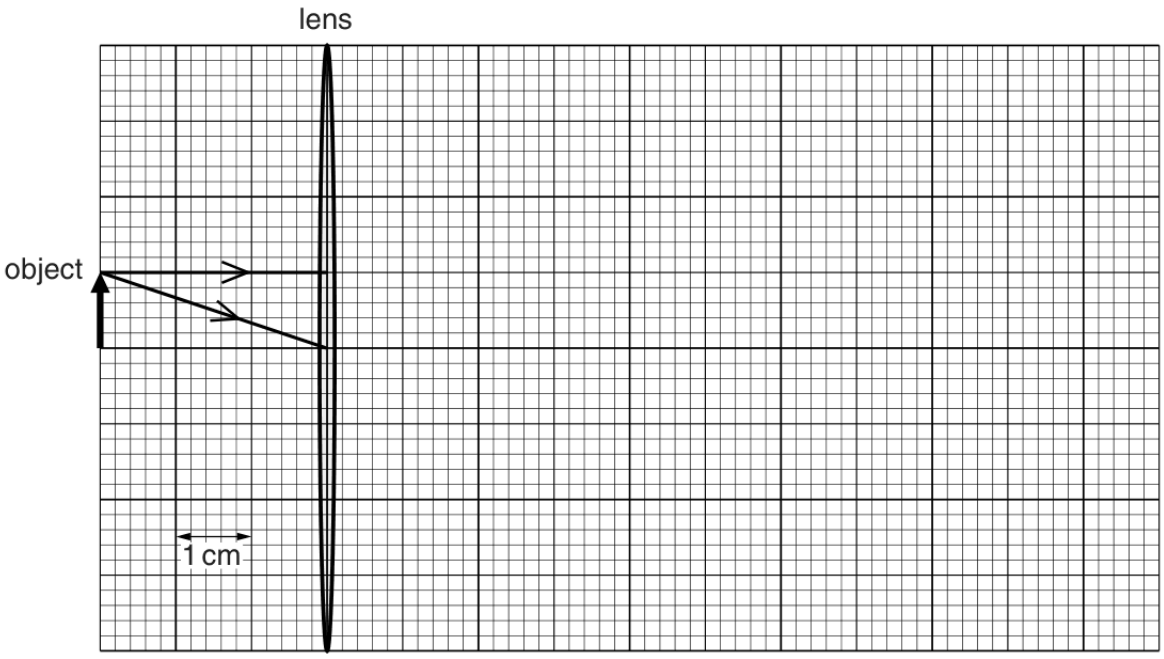


Fig. 6.1 (full scale)

(i) On Fig. 6.1, draw the paths of the two rays after they pass through the lens. [2]

(ii) Explain how your ray diagram shows that the image is more than 11 cm from the lens.

.....

..... [1]

(iii) Underline **three** of the following words which describe the image.

diminished inverted magnified real upright virtual [1]

6 **(a)** **(i)** any single value between 0 and 5.6 cm or a range all of whose values are correct [B1]

(ii) any value beyond 5.6 cm [B1]

(b) **(i)** ray through optical centre undeviated [B1]
other ray correct through or to axis 2.8 cm (± ½ small square) from lens [B1]

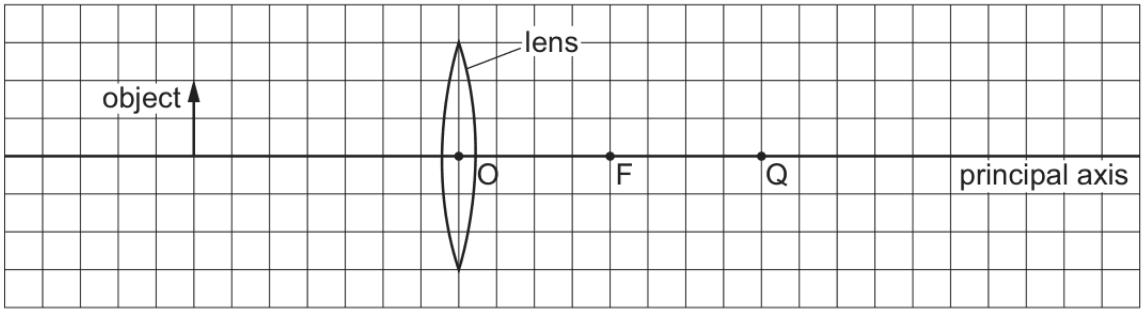
(ii) lines drawn meet after 11 cm or rays do not meet (on page) or rays almost parallel [B1]

(iii) inverted, magnified, real all 3 needed and none wrong [B1]

[6]

QUESTION · 5054_w14_qp_11.pdf · source page 12 of 20

22 The diagram shows an object on the principal axis of a converging (convex) lens. A principal focus of the lens is at F.



Where is the image formed by the lens?

A between O and F

B between F and Q

C at Q

D to the right of Q

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 22

22 **D**

QUESTION · 5054_w14_qp_11.pdf · source page 12 of 20

23 A digital camera uses a lens to produce a diminished (reduced in size) image on a light sensor.

Which row shows the correct type of lens and the nature of the image?

	type of lens	nature of image
A	converging	inverted
B	converging	upright
C	diverging	inverted
D	diverging	upright

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 23

23 **A**

QUESTION · 5054_s15_qp_11.pdf · source page 8 of 16

23 An image is formed by a thin converging lens when it is used as a magnifying glass.

What is the correct description of the image?

A real and erect

B real and inverted

C virtual and erect

D virtual and inverted

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 23

23	**C**

QUESTION · 5054_s15_qp_12.pdf · source page 10 of 16

27 An image is formed by a thin converging lens when it is used as a magnifying glass.

What is the correct description of the image?

A real and erect

B real and inverted

C virtual and erect

D virtual and inverted

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 27

27	**C**

6 A burning candle is placed close to a thin converging lens. The candle acts as the object.

A white, vertical screen is moved to a position on the other side of the lens from the candle.

Fig. 6.1 is a full-scale diagram, on graph paper, of the lens and the screen.

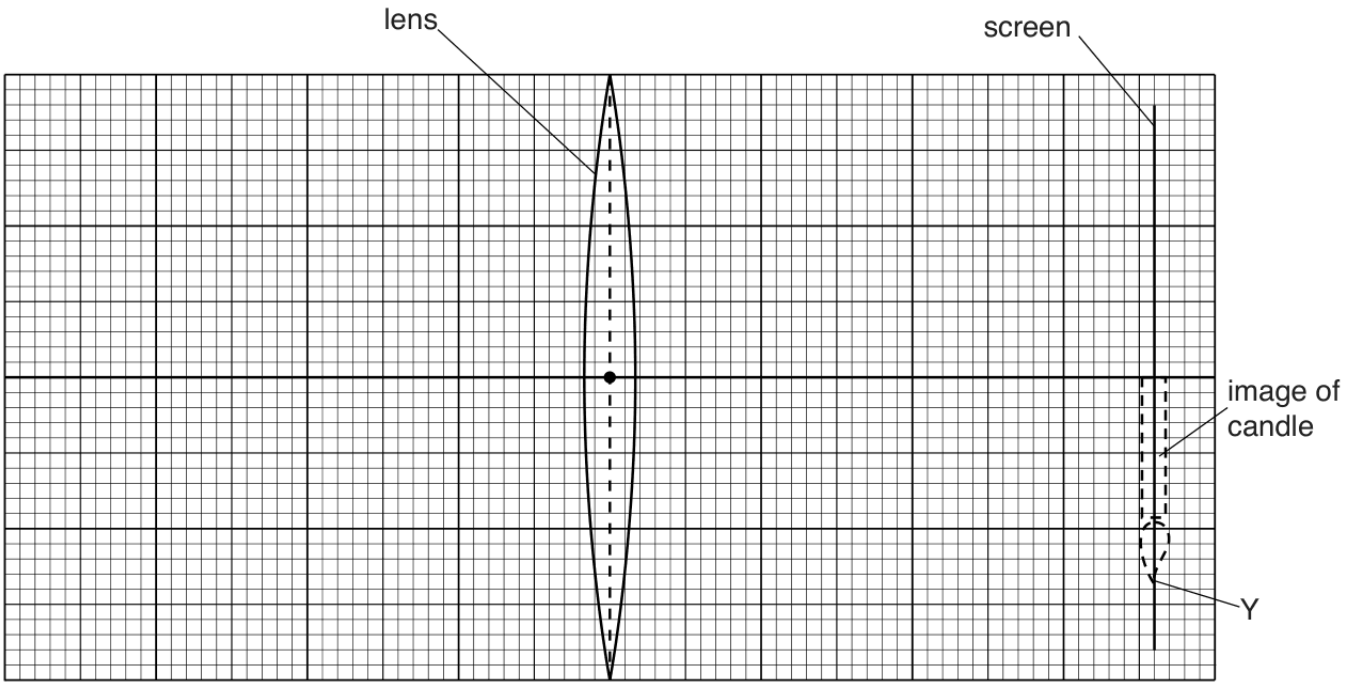


Fig. 6.1

The focal length of the lens is 2.4 cm. The screen is 7.2 cm from the centre of the lens. A sharply focused, inverted image of the candle is produced on the screen, as shown in Fig. 6.1.

(a) Define the term *focal length*.

.....

.....

..... [1]

(b) **(i)** On Fig. 6.1, mark and label with an F, each of the two focal points (principal foci) of the lens. [1]

(ii) The point Y is the tip of the image.

On Fig. 6.1, draw a ray diagram to locate the position of the top of the object. Label this point X. [3]

(iii) Using Fig. 6.1, determine the distance of the candle from the centre of the lens.

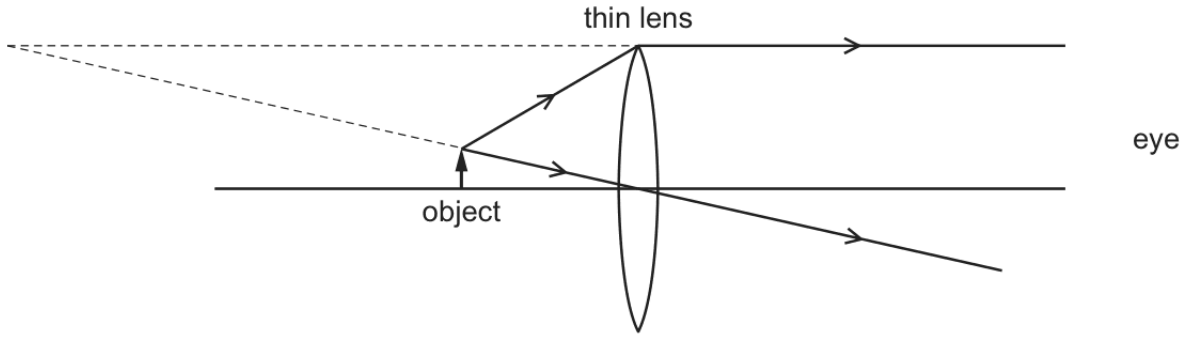
distance = ... [1]

6	**(a)**	**distance** from (optical) centre to focal point (principal focus)	B1	
	(b) **(i)**	**both** Fs correctly positioned at ±1 mm	B1	
	(ii)	**two** of: paraxial ray to lens through focal point to image ray through optical centre ray through focal point and then paraxial to image (**ign.** arrows)	M2	
		X at crossing point of rays	A1	
	(iii)	3.4–3.8 cm	B1	[6]

QUESTION · 5054_w15_qp_12.pdf · source page 10 of 16

22 An object is viewed through a converging lens.

The diagram shows the paths of two rays from the top of the object to an eye.



How does the image compare with the object?

A It is larger and inverted.

B It is larger and upright.

C It is smaller and inverted.

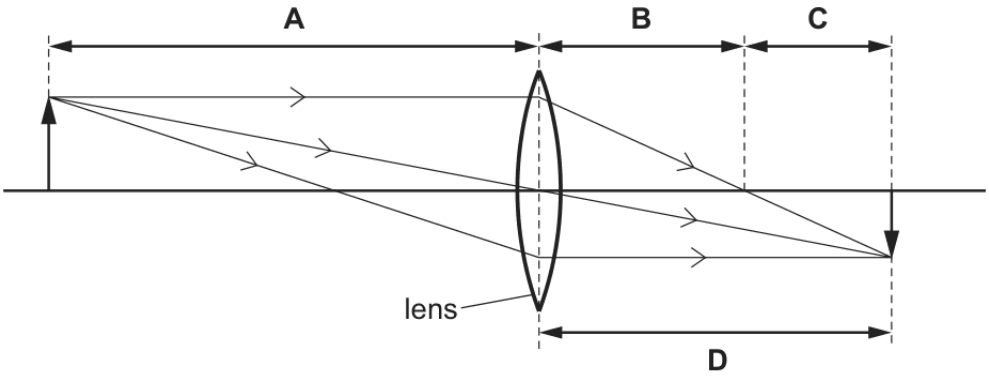
D It is smaller and upright.

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 22

22	**B**

QUESTION · 5054_s16_qp_11.pdf · source page 9 of 16

22 Which length is the focal length of the lens shown in the diagram?

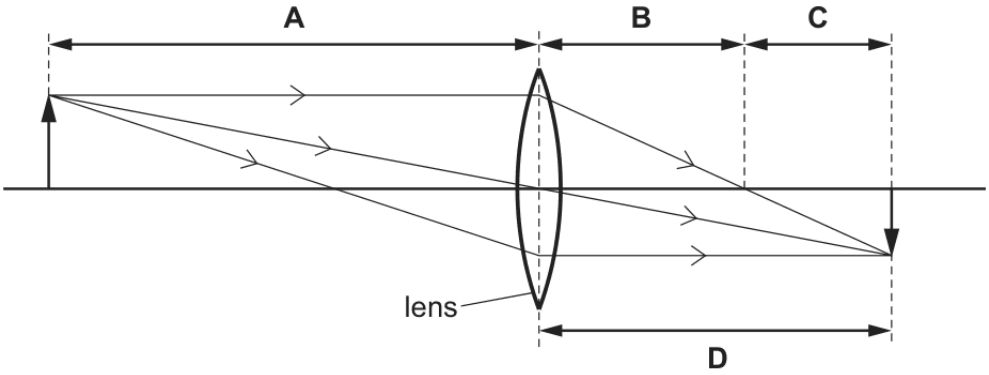


MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 22

22	**B**

QUESTION · 5054_s16_qp_12.pdf · source page 7 of 16

19 Which length is the focal length of the lens shown in the diagram?

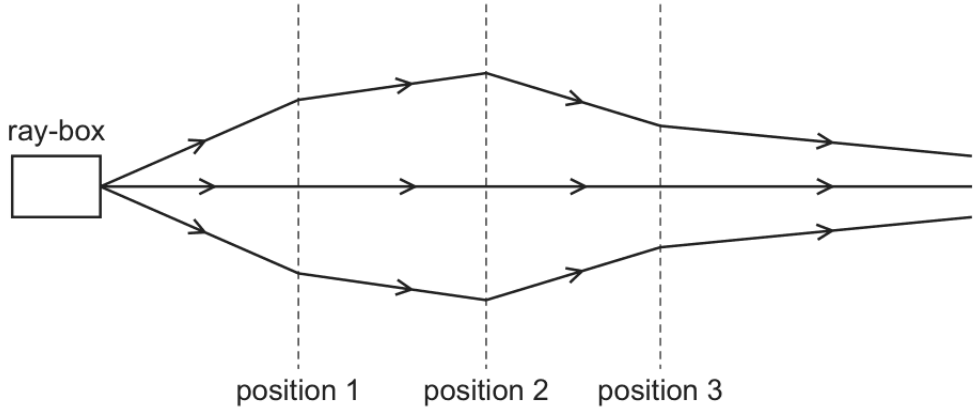


MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 19

19	**B**

QUESTION · 5054_s16_qp_12.pdf · source page 8 of 16

20 The rays of light from a ray-box pass through three lenses placed at positions 1, 2 and 3.



What type of lens is used at each position?

	position 1	position 2	position 3
A	converging	converging	converging
B	converging	converging	diverging
C	diverging	converging	diverging
D	diverging	diverging	converging

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 20

20 **B**

5 A person can focus clearly on objects that are far away, but near objects appear blurred.

Fig. 5.1 shows three rays from a point on a near object as they pass through the person's eye. The refraction of the light as it enters the eye is ignored.

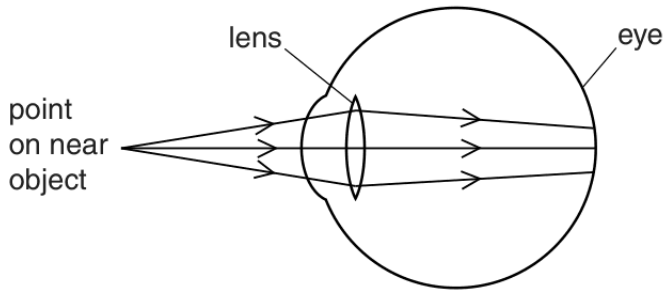


Fig. 5.1

(a) (i) State the name of the eye defect shown in Fig. 5.1.

.....[1]

(ii) Explain what causes the image of the near object to appear blurred.

.....

.....

.....[2]

(b) A lens is used to correct the defect.

(i) On Fig. 5.2, draw a suitable lens placed in front of the eye and continue the path of the three rays to the back of the eye.

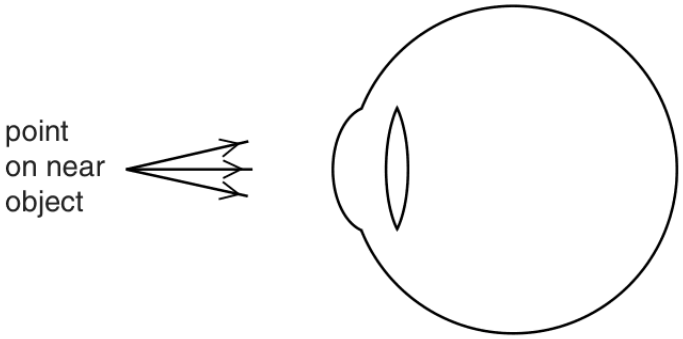


Fig. 5.2 [2]

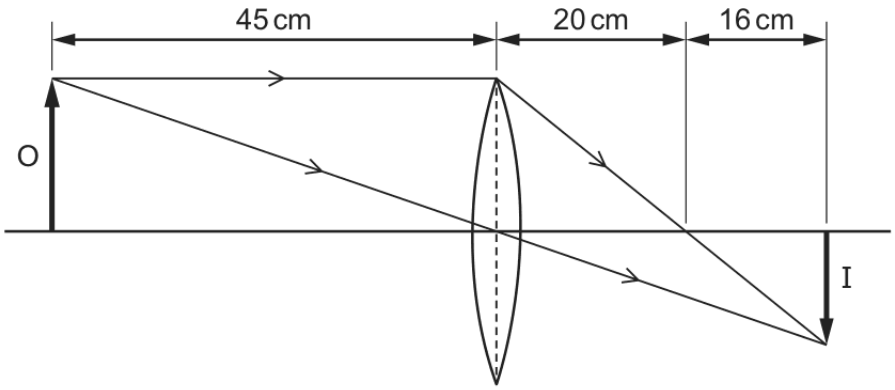
(ii) State what type of lens is used.

.....[1]

5	**(a)**	**(i)**	long-sight **or** far-sight **or** hypermetropia	B1
		(ii)	rays do not come together (on back of eye) **or** rays do not converge (on retina) **or** it / the image is not formed on retina / back of eye **or** it / the image is formed behind retina / back of eye	B1
	(b)	**(i)**	lens between rays and eyeball **and** a converging lens shown	B1
		(ii)	converging or convex	B1

QUESTION · 5054_w16_qp_11.pdf · source page 11 of 20

25 In the diagram, a convex lens forms an image I of an object O. The diagram is not drawn to scale.



What is the focal length of the lens?

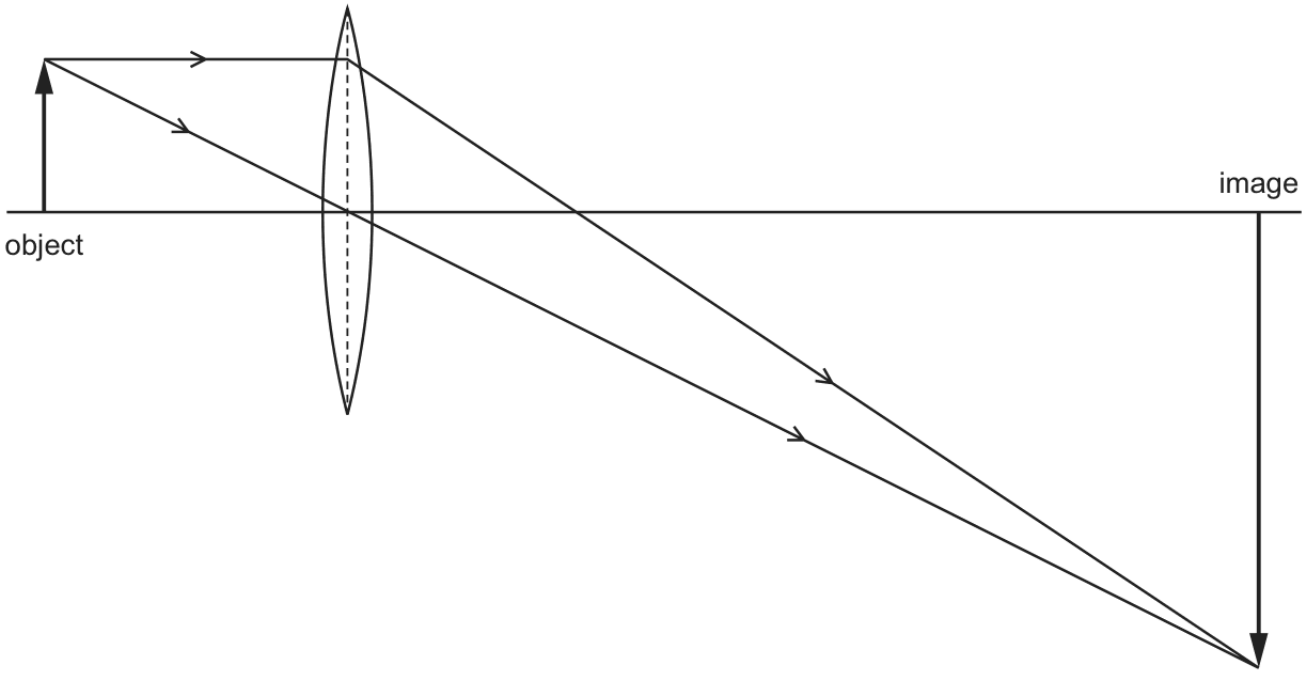
A 16 cm **B** 20 cm **C** 36 cm **D** 45 cm

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 25

25 **B**

QUESTION · 5054_w16_qp_11.pdf · source page 11 of 20

26 A lens is used to produce a magnified image, as shown in the scale diagram.



What is the linear magnification produced by the lens?

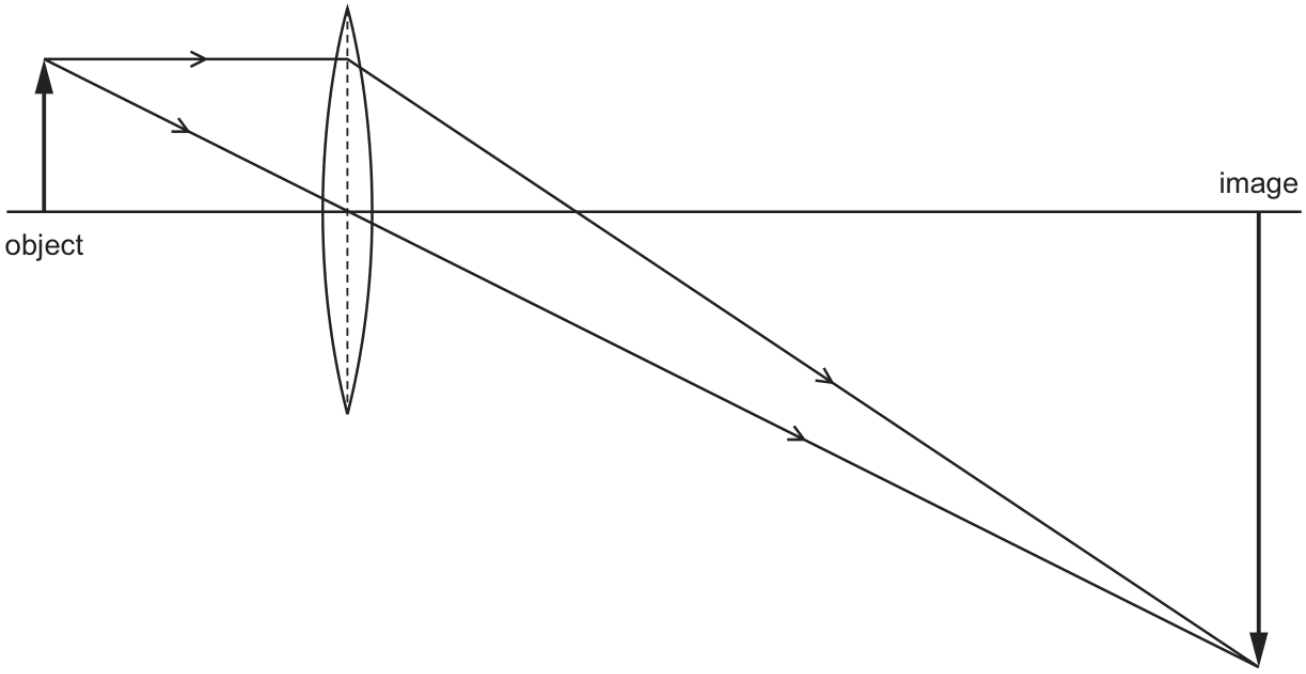
A 0.33 **B** 3.0 **C** 4.0 **D** 6.0

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 26

26	**B**

QUESTION · 5054_w16_qp_12.pdf · source page 10 of 16

27 A lens is used to produce a magnified image, as shown in the scale diagram.



What is the linear magnification produced by the lens?

A 0.33 **B** 3.0 **C** 4.0 **D** 6.0

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 27

27	**B**

QUESTION · 5054_w16_qp_12.pdf · source page 11 of 16

28 A girl is long-sighted.

Which statement is correct?

A She sees close objects less clearly than a person with normal vision.

B She sees distant objects more clearly than a person with normal vision.

C The fault is corrected with a diverging lens.

D The image of a close object is formed in front of her retina.

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 28

28	**A**

4 Fig. 4.1 shows a converging lens, an object O and the image I produced by the lens.

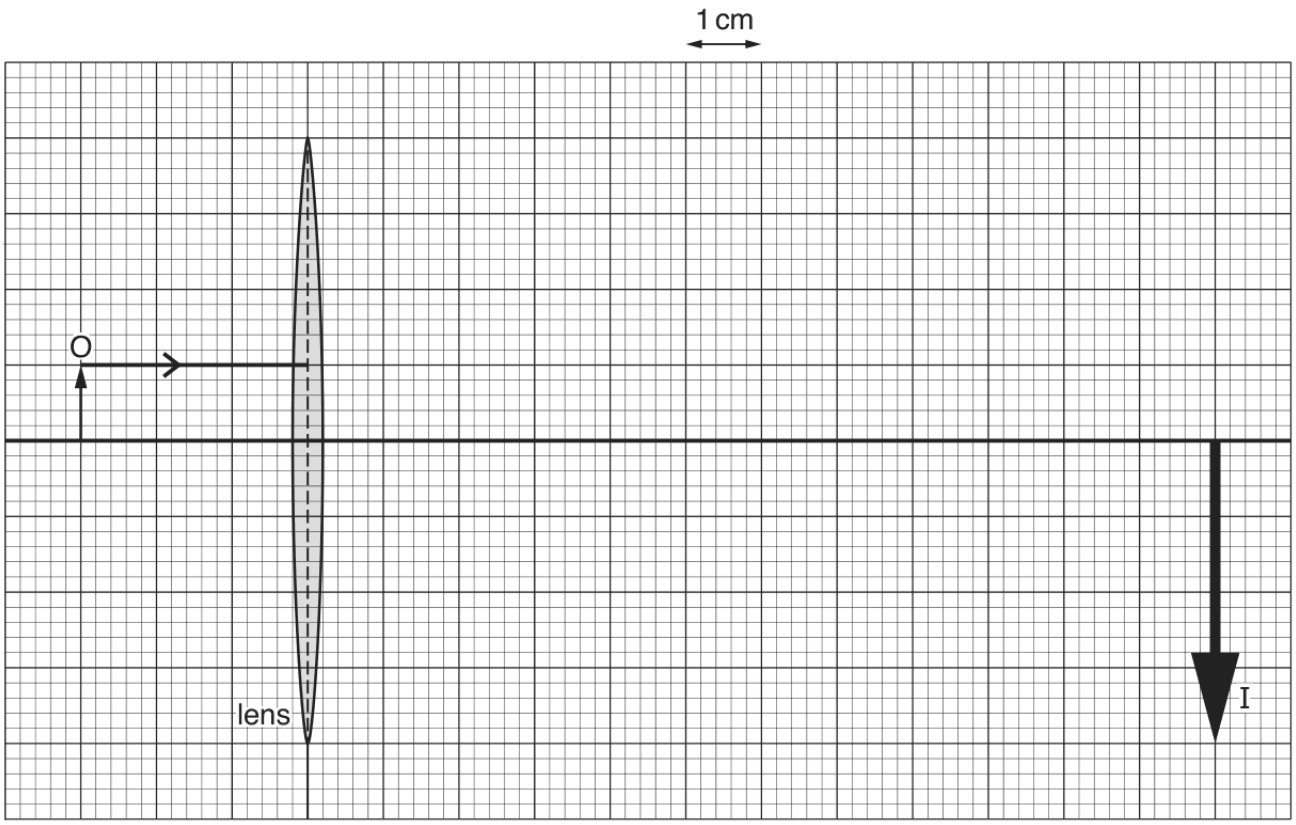


Fig. 4.1

(a) The image formed is real and has a linear magnification of 4.0.

(i) Define the term *linear magnification*.

.....

.....[1]

(ii) Explain what is meant by *real image*.

.....

.....[1]

(b) On Fig. 4.1 a horizontal ray has been drawn from the top of the object to the lens.

(i) Continue this ray until it meets the image.

(ii) Using Fig. 4.1, determine the focal length of the lens.

focal length = ...

(iii) Draw another two rays from the top of the object to show how the image is formed.

[3]

Question	Answer	Marks
4(a)(i)	image height / object height **or** image distance / object distance	**B1**
4(a)(ii)	rays go to / meet at image **or** can be formed on a screen	**B1**
4(b)(i)	horizontal ray continued to bottom of I	**B1**
4(b)(ii)	2.4 cm	**B1**
4(b)(iii)	any two other rays drawn, e.g. through centre of lens	**B1**

QUESTION · 5054_w17_qp_11.pdf · source page 10 of 20

21 What is the name and shape of the lens used to correct short sight?

	name of lens	shape of lens
A	converging	
B	converging	
C	diverging	
D	diverging	

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 21

21	C	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 11 of 20

22 An object is placed in front of a converging lens. The lens forms a magnified image of the object on a screen.

Which statement is correct?

A The distance between the object and the lens is greater than the focal length.

B The image formed is a virtual image.

C The image is the right way up.

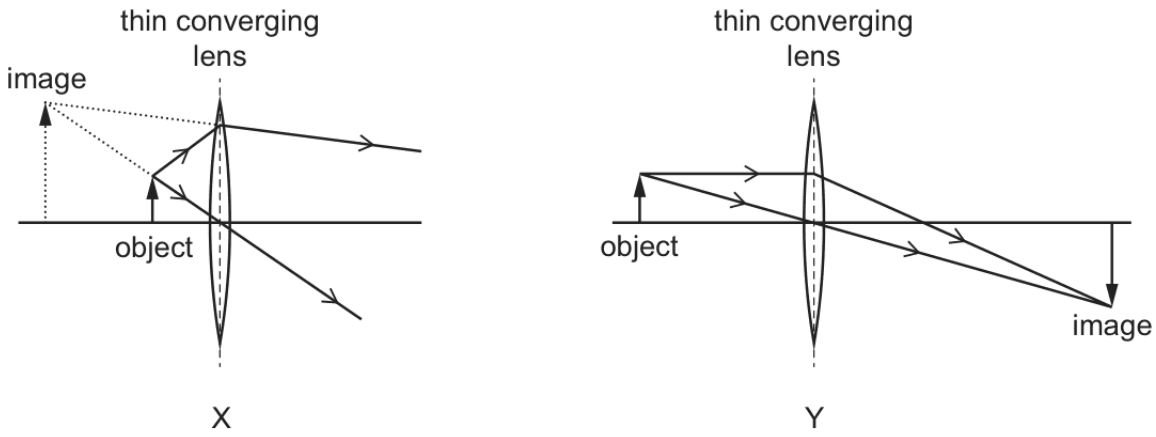
D The lens is acting as a magnifying glass.

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 22

22	A	**1**

QUESTION · 5054_w17_qp_12.pdf · source page 10 of 16

23 The ray diagrams, X and Y, show two ways in which a thin converging lens produces an image that is larger than the object.



Which devices use a lens as shown in diagram X and in diagram Y?

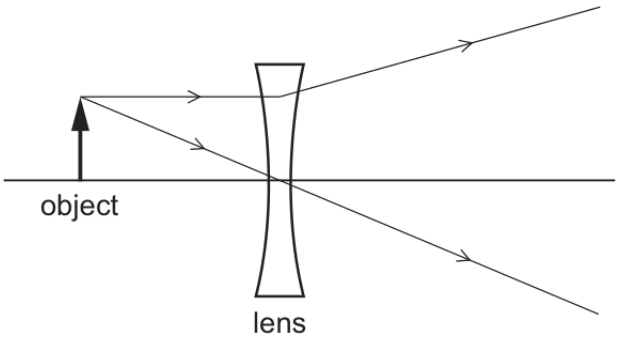
	X	Y
A	camera	magnifying glass
B	magnifying glass	projector
C	photographic enlarger	camera
D	photographic enlarger	projector

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 23

23	B	**1**

QUESTION · 5054_s18_qp_11.pdf · source page 10 of 20

24 The diagram shows the paths of two rays from the top of an object through a lens.



The object is viewed from the opposite side of the lens to the object.

How does the image compare with the object?

A larger and inverted

B larger and the same way up

C smaller and inverted

D smaller and the same way up

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 24

24	D	**1**

6 A converging lens has a focal length of 3.0 cm. An object of height 2.0 cm is placed 5.0 cm from the centre of the lens. Fig. 6.1 shows the arrangement of the object and the lens.

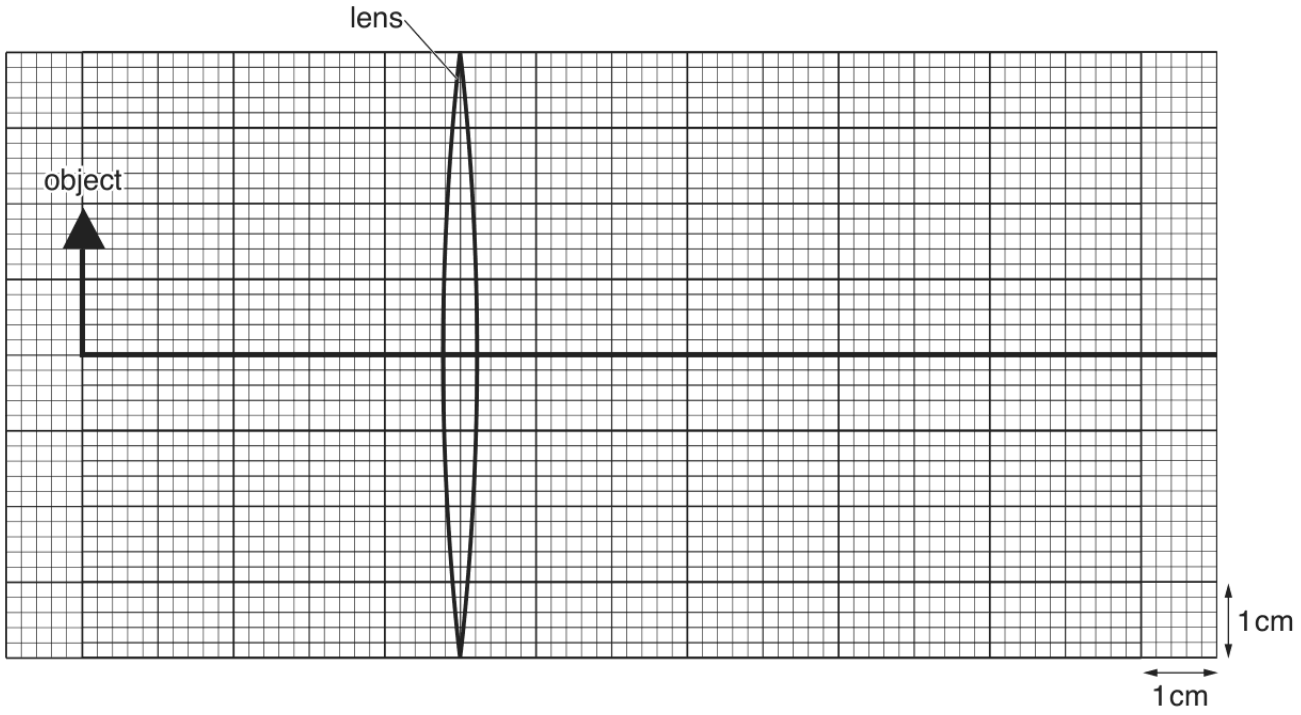


Fig. 6.1 (to scale)

(a) On Fig. 6.1, draw rays from the top of the object to show how the lens forms an image of the object. Mark the image clearly. [2]

(b) The image is magnified. State **one** other feature of the image.

.....[1]

(c) Calculate the linear magnification produced by the lens in this case.

magnification = ... [2]

(d) State the name of **one** optical device that produces a magnified image as shown by Fig. 6.1.

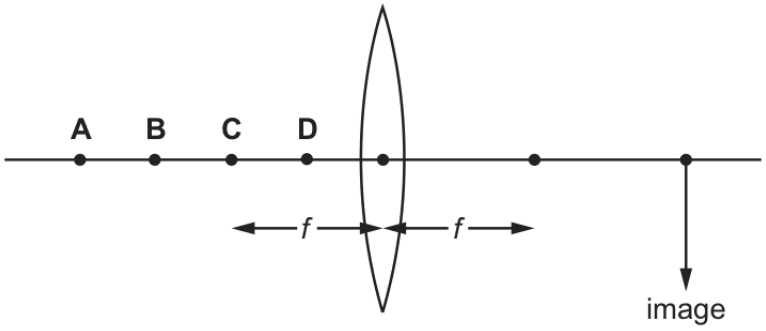
.....[1]

Question	Answer	Marks
6(a)	**any** two of • ray through middle of lens undeviated • ray parallel to axis passes through focus, 3 cm from lens • ray through focus on left of lens parallel to axis after lens	**B2**
6(b)	inverted **or** real	**B1**
6(c)	image size / object size **or** image distance / object distance numerical or algebraic	**C1**
	rays from lens intersect on diagram (for image) **and** 1.3–1.7	**A1**
6(d)	projector **or** photographic enlarger	**B1**

QUESTION · 5054_w18_qp_11.pdf · source page 7 of 16

19 The diagram shows a thin converging lens of focal length *f*.

Where must an object be placed to produce a real image in the position shown?



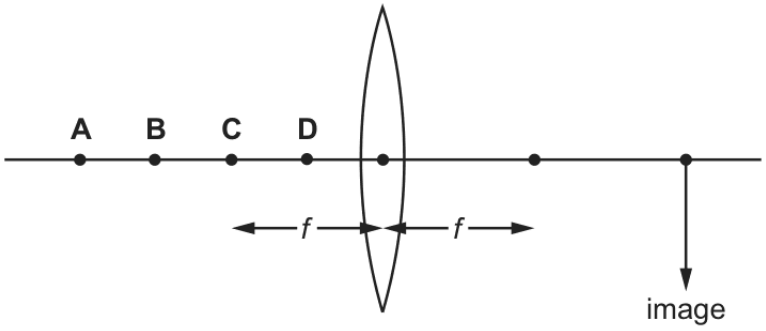
MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 19

19	A	**1**

QUESTION · 5054_w18_qp_12.pdf · source page 10 of 16

23 The diagram shows a thin converging lens of focal length *f*.

Where must an object be placed to produce a real image in the position shown?



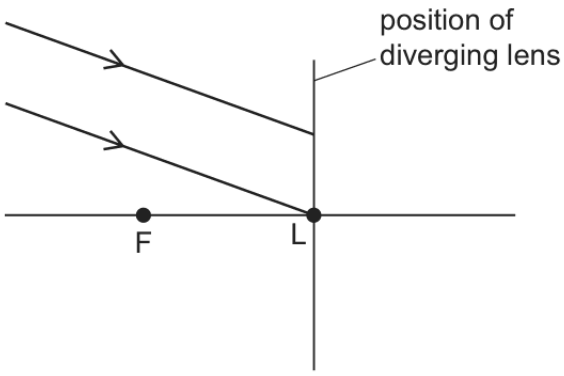
MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 23

23	A	**1**

QUESTION · 5054_s19_qp_11.pdf · source page 11 of 20

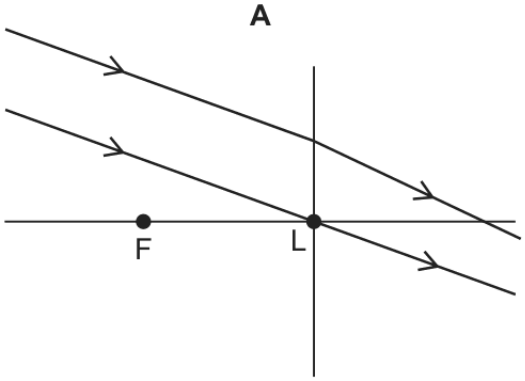
25 A parallel beam of light is incident on a thin diverging lens.

The focal length of the lens is FL, as shown in the diagram.

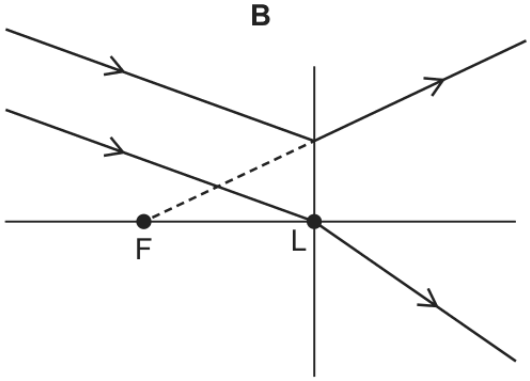


Which ray diagram shows the beam after it has passed through the lens?

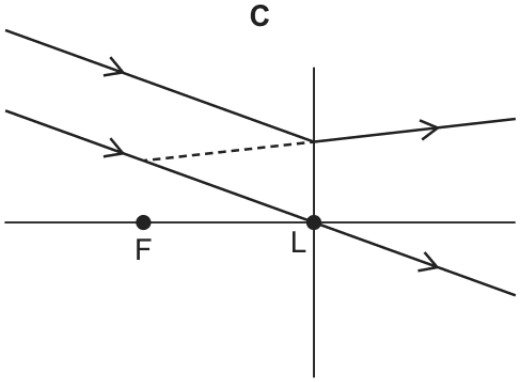
A



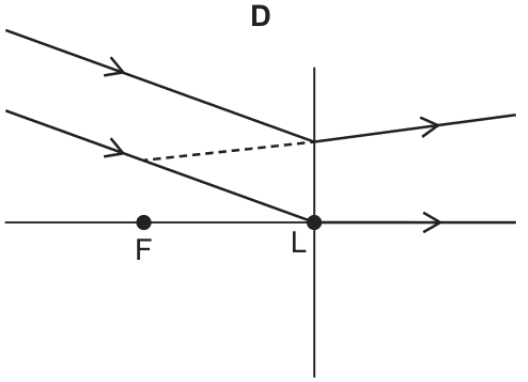
B



C



D



MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 25

25	C	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 9 of 16

27 An object is placed at a distance from a converging lens that is equal to twice the focal length of the lens.

Which statement about the image is correct?

A It is enlarged.

B It is inverted.

C It is on the same side of the lens as the object.

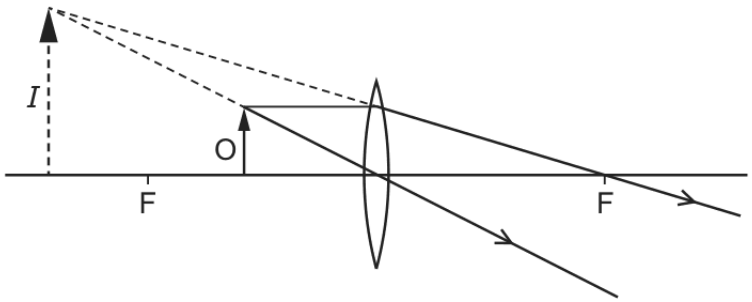
D It is virtual.

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 27

27	B	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 10 of 16

28 The lens in the diagram produces an image I of the object O.



Why is this **not** the ray diagram for a photographic enlarger?

A The image is magnified.

B The image is virtual.

C The lens is a converging lens.

D The lens is too thin.

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 28

28	B	**1**

10 Visible light is one component of the electromagnetic spectrum.

(a) State the speed of light in a vacuum.

..... [1]

(b) State the colours of the visible spectrum.

..... [1]

(c) An object O of height 3.0 cm is placed 4.0 cm from the centre of a diverging lens.

Fig. 10.1 shows the object O, the diverging lens and the two focal points (principal focuses), F_1 and F_2, of the lens.

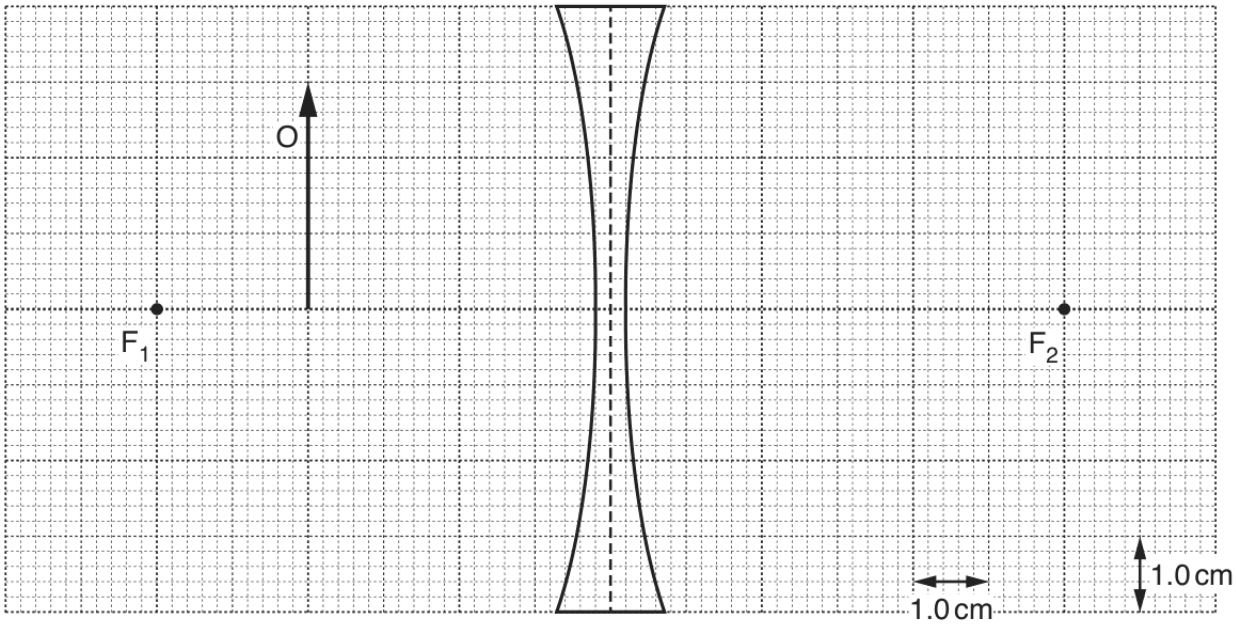


Fig. 10.1

(i) Determine the focal length of this lens.

focal length = ... [1]

(ii) The diverging lens produces an image of object O.

On Fig. 10.1, by drawing rays from the tip of O, locate this image and label it I. [4]

(iii) Determine the height of the image.

height = ... [1]

(iv) Determine the magnification of the image.

magnification = ... [1]

(v) State **two** ways in which a virtual image differs from a real image.

1. .....

.....

2. .....

..... [2]

(vi) State one use for a diverging lens.

.....

..... [1]

(d) Light passing from air into glass refracts in a similar way to a water wave passing from deep water into shallow water.

Fig. 10.2 represents light passing from air into glass at an angle to the surface.

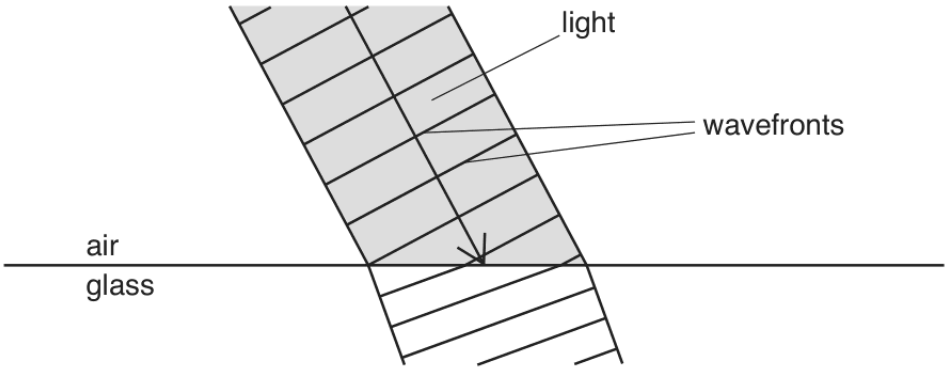


Fig. 10.2

One side of a wavefront strikes the glass before the other side.

Explain why the wavefronts change direction as the light enters the glass.

.....

.....

.....

..... [3]

[Total: 15]

MARK SCHEME · 5054_w19_ms_21.pdf · source page 9 of 10 · Question 10

Question	Answer	Marks
10(a)	3.0×10^8 m / s	**B1**
10(b)	red orange yellow green blue indigo violet (any order)	**B1**
10(c)(i)	(–)6.(0) cm	**B1**

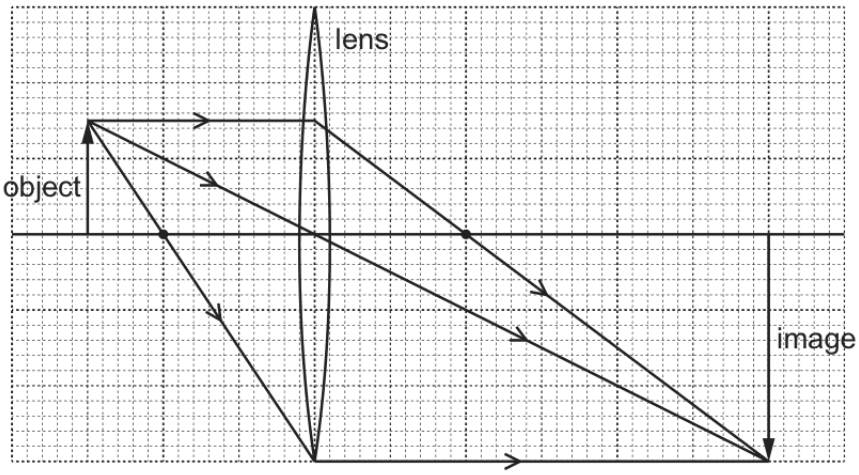
MARK SCHEME · 5054_w19_ms_21.pdf · source page 10 of 10 · Question 10

Question	Answer	Marks
10(c)(ii)	any **two** rays drawn from: paraxial ray that refracts and seems to come from F_1 ray through the optical centre of lens ray that aims for F_2 but refracts and emerges paraxially	**B2**
	rays traced back to point	**B1**
	(point) labelled I **and** rest of image drawn down to the principal axis	**B1**
10(c)(iii)	1.7–1.9 cm	**B1**
10(c)(iv)	candidate's **10(c)(iii)** / 3.0 evaluated	**B1**
10(c)(v)	any **two** from: a real image can be projected on to a screen light actually passes through a real image on same side (of lens) as object **or** on opposite side of mirror to object	**B2**
10(c)(vi)	correction of short-sight / myopia	**B1**
10(d)	light travels more slowly in glass **or** light changes speed	**B1**
	one side / left-hand side of wavefront slows down first	**B1**
	wavelength decreases **or** wavefront travels a shorter distance in the same time	**B1**

QUESTION · 5054_w19_qp_12.pdf · source page 11 of 16

26 An object is placed in front of a converging lens of focal length 4.0 cm. The height of the image is 6.0 cm.

The arrangement is shown on the scale diagram.



What is the linear magnification produced by the lens?

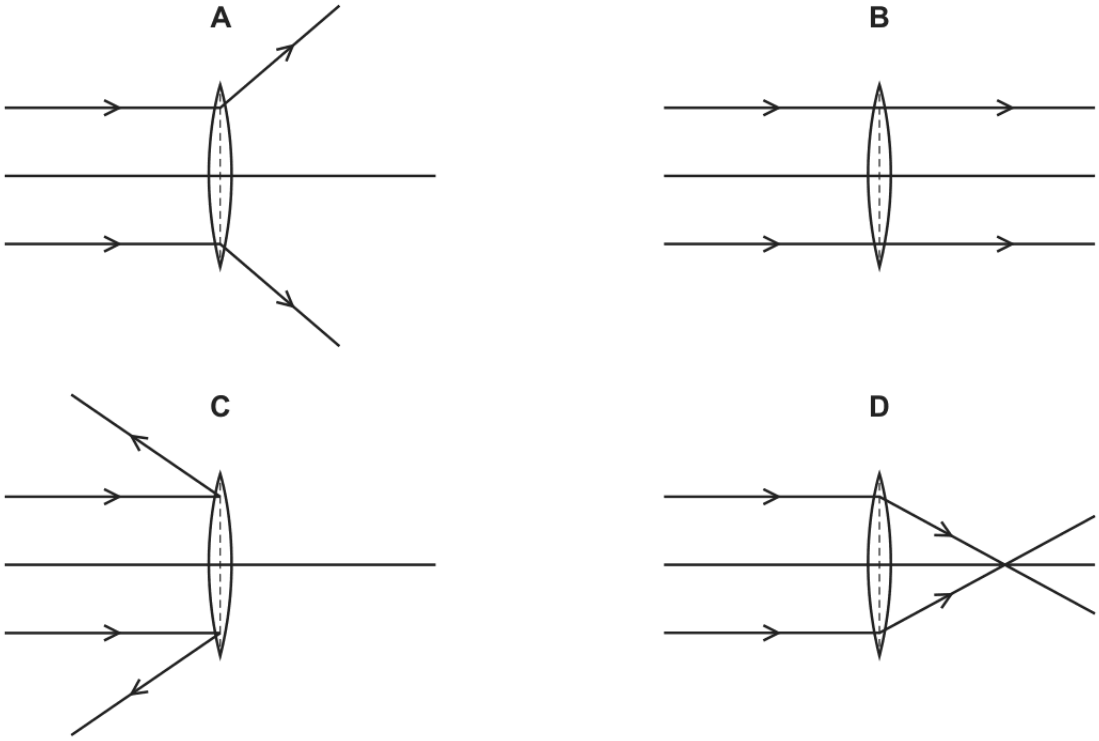
A 0.50 **B** 1.5 **C** 2.0 **D** 6.0

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 26

26	C	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 10 of 16

28 Which diagram shows the action of a converging lens on a parallel beam of light?



MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 2 of 3 · row for Question 28

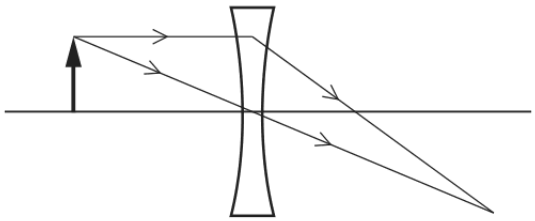
28	D	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 12 of 16

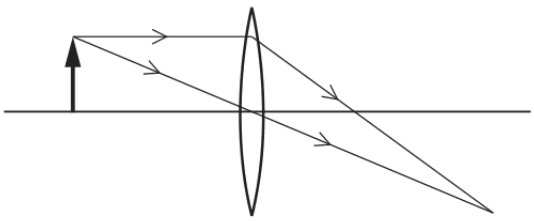
33 A converging glass lens is used to produce a virtual, magnified image.

Which ray diagram shows the rays passing through the converging lens?

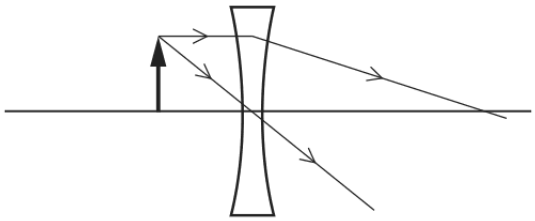
A



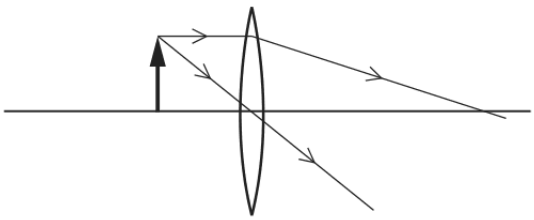
B



C



D

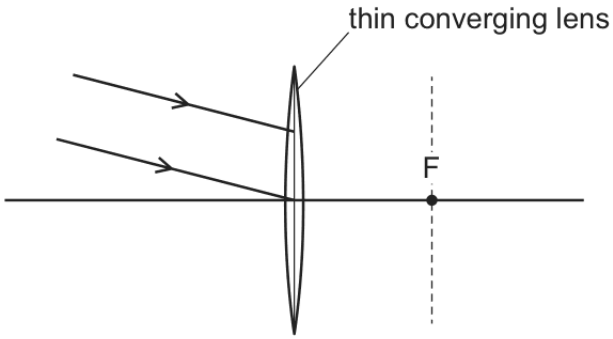


MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 3 of 3 · row for Question 33

33	D	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 12 of 16

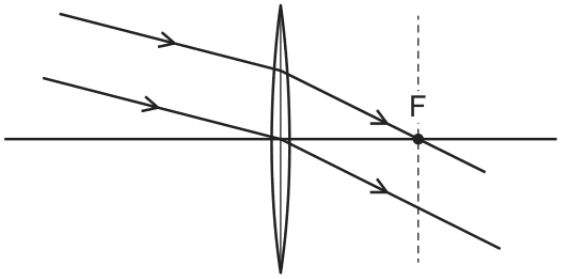
29 A parallel beam of light is incident on a thin converging lens.



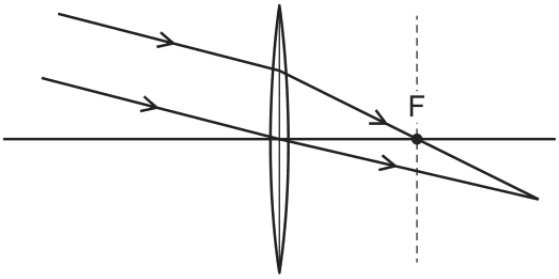
F is one focal point of the lens.

Which ray diagram shows the light after it has passed through the lens?

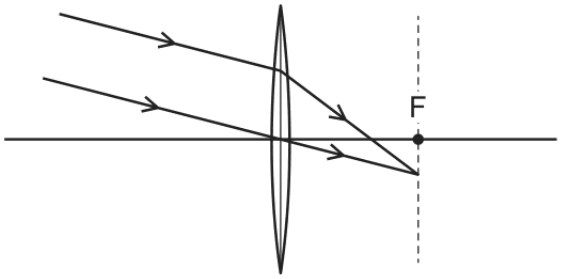
A



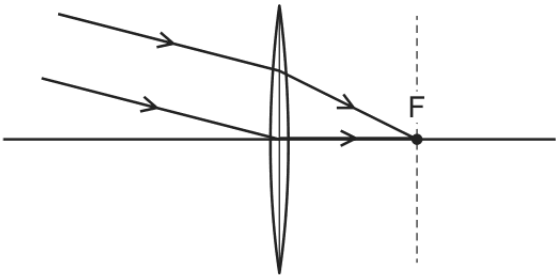
B



C



D



MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 3 of 3 · row for Question 29

29	C	**1**

8 A thin converging lens is made of a transparent material of refractive index 1.4.

(a) A ray of light travelling in air strikes the surface of the lens at an angle of incidence of 55°.

(i) Calculate the angle of refraction.

angle of refraction = ... [2]

(ii) Place a tick (✓) in **one** of the boxes in the third column of Table 8.1 to indicate how the light ray deviates and what happens to the speed of the light in the ray as it enters the lens.

Table 8.1

direction of deviation	speed of light	
away from the normal	decreases	
away from the normal	does not change	
away from the normal	increases	
towards the normal	decreases	
towards the normal	does not change	
towards the normal	increases	

[1]

(iii) State what happens to the frequency of the light in the ray as it enters the glass.

..... [1]

(b) The focal length of the lens is 2.5 cm.

(i) State what is meant by *focal length*.

.....

..... [1]

(ii) Fig. 8.1 is a full-scale diagram that shows an object O of height 3.0 cm and the lens.

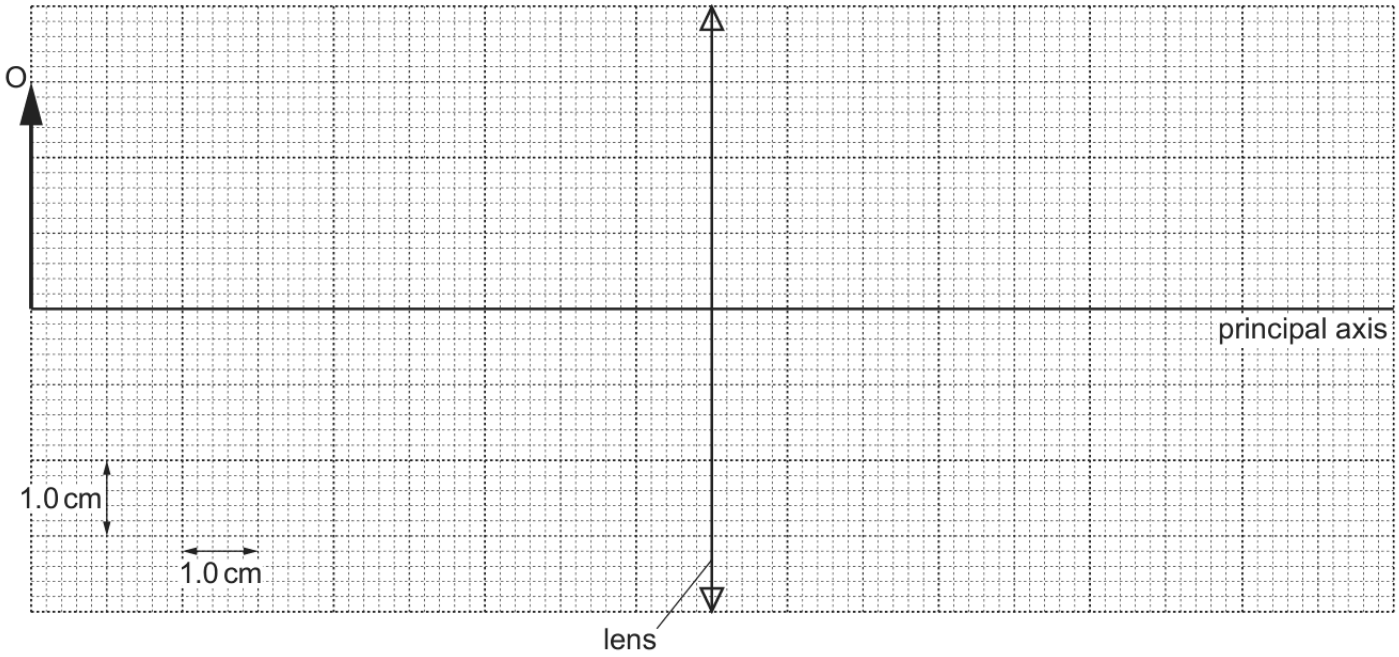


Fig. 8.1 (full scale)

By drawing on Fig. 8.1, locate and mark the image I of O. [4]

(iii) Determine the distance of I from the lens and calculate the magnification of O produced by the lens.

distance = ...

magnification = ...
[3]

(c) Describe how a converging lens is used in a camera.

.....

.....

.....

.....

..... [3]

[Total: 15]

Question	Answer	Marks
8(a)(i)	n = sin i / sin r **or** 1.4 = sin 55° / sin r **or** (r =) $\sin^{-1}$(sin i / n) **or** (r =) $\sin^{-1}$(sin 55°/ 1.4)	**C1**
	36°	**A1**
8(a)(ii)	towards (the normal) **and** speed decreases (i.e. fourth box **only**)	**B1**
8(a)(ii)	stays constant	**B1**
8(b)(i)	distance between (one) focal point / principal focus and (the centre of) lens	**B1**
8(b)(ii)	at least one correct focal point labelled / clear / used	**B1**
	any **two** from drawn: paraxial ray from tip of O to lens and refracted through correct far F ray from tip of O through optical centre and undeviated ray from tip of O through correct near F to lens and refracted paraxially	**B2**
	I marked at intersection of correct rays	**B1**
8(b)(ii)	3.4 cm ⩽ distance ⩽ 3.5 cm	**B1**
	(magnification =) candidate's distance / 9.0 **or** candidate's height / 3.0	**C1**
	0.37 ⩽ magnification ⩽ 0.40 **c.a.o.**	**A1**
8(c)	on next page	
8(c)	any **three** from: light (from subject / object) focussed / refracted by lens by moving / sliding / adjusting lens real image (of subject / object) formed on digital light detector / film	**B3**

QUESTION · 5054_s21_qp_11.pdf · source page 9 of 16

23 Which statement about human vision is correct?

A In a normal eye, the image on the retina is magnified and upright.

B In a long-sighted eye, distant objects form images in front of the retina.

C Short-sighted eyes produce only virtual images.

D Short-sight is corrected by the use of a diverging lens.

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 23

23	D	**1**

5 Fig. 5.1 shows part of the ray diagram of a lens being used as a magnifying glass.

Three rays are shown coming from the top of an object O.

The points labelled F are one focal length from the lens.

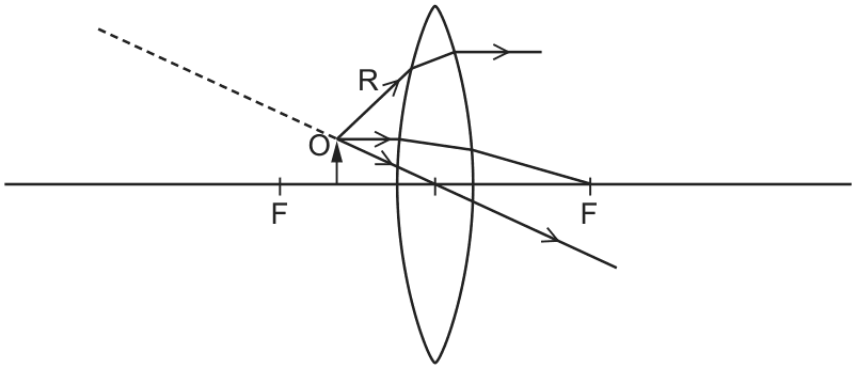


Fig. 5.1

(a) State the name of the type of lens shown in Fig. 5.1.

..... [1]

(b) Describe what happens to the ray of light R:

• as it enters the lens .....

.....

• as it leaves the lens. .....

.....

[2]

(c) Using all three rays from O, complete Fig. 5.1 to show the image formed. [2]

(d) Underline **all** of the words in the list that describe the image formed in **(c)**.

inverted upright real virtual

[1]

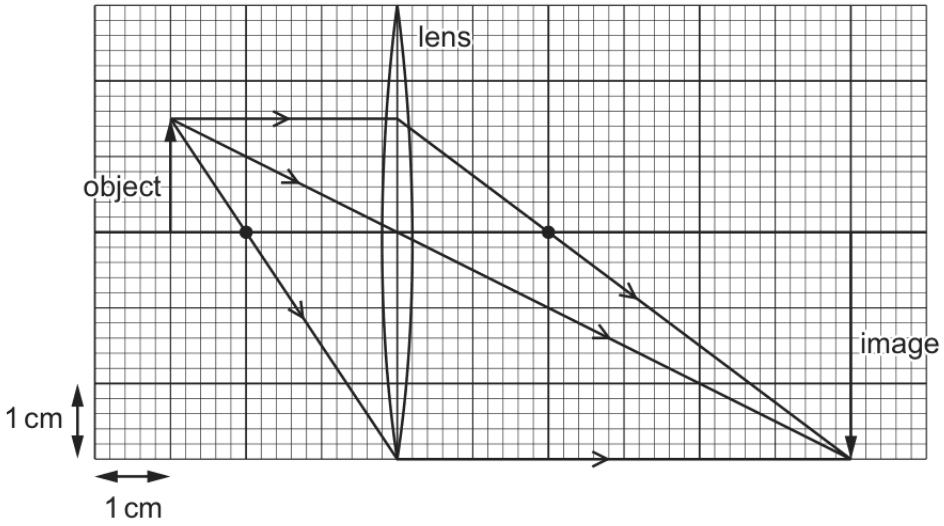
[Total: 6]

Question	Answer	Marks
5(a)	convex **or** converging	**B1**
5(b)	as it enters – refracts (towards normal) **or** changes speed	**B1**
	as it leaves, one of: • bends / refracts away from normal • speed increases • wavelength increases • emerges parallel to the (principal) axis	**B1**
5(c)	either emergent ray R or the ray through F extended to the left of the lens	**B1**
	both ray through F and R shown and image marked at intersection of extended rays	**B1**
5(d)	upright **and** virtual	**B1**

QUESTION · 5054_s21_qp_12.pdf · source page 10 of 16

25 An object of height 1.5 cm is placed in front of a converging lens of focal length 2.0 cm.

The arrangement is shown on the full-scale ray diagram.



What is the linear magnification produced by the lens?

A 2.0 **B** 3.0 **C** 4.0 **D** 6.0

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 25

25	A	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 10 of 16

26 In which optical instrument is the distance between the object and the lens less than the focal length of the lens?

A camera

B magnifying glass

C photographic enlarger

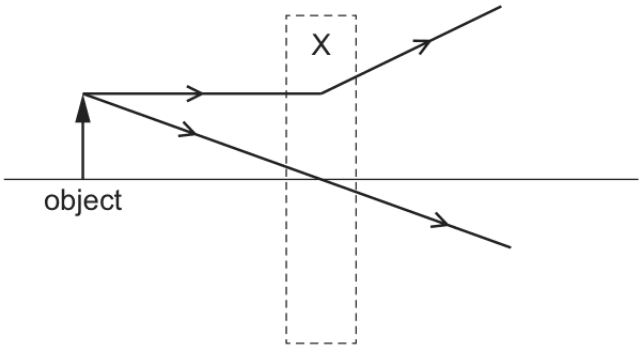
D projector

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 26

26	B	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 10 of 20

21 Two rays of light pass through a lens in region X.



Which type of lens is in region X and which type of image is formed?

	type of lens	type of image
A	converging	real
B	converging	virtual
C	diverging	real
D	diverging	virtual

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 21

21	D	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 10 of 20

22 A girl is long-sighted.

Which statement is correct?

A She sees close objects less clearly than a person with normal vision.

B She sees distant objects more clearly than a person with normal vision.

C The fault is corrected with a diverging lens.

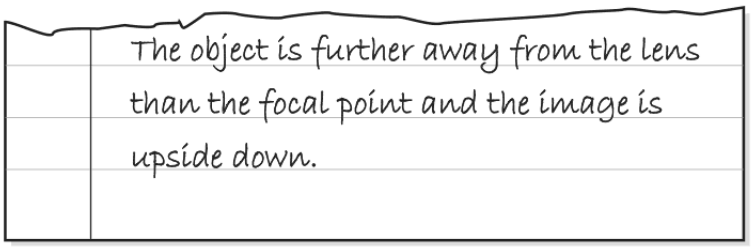
D The image of a close object is formed in front of her retina.

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 22

22	A	**1**

QUESTION · 5054_w21_qp_12.pdf · source page 10 of 16

24 A piece of paper torn from an exercise book is shown.



Which process is being described?

A the formation of a virtual image by a diverging lens

B the formation of a virtual image by a converging lens

C the formation of a real image by a diverging lens

D the formation of a real image by a converging lens

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 24

24	D	**1**

QUESTION · 5054_w21_qp_12.pdf · source page 10 of 16

25 A girl is long-sighted.

Which statement is correct?

A She sees close objects less clearly than a person with normal vision.

B She sees distant objects more clearly than a person with normal vision.

C The fault is corrected with a diverging lens.

D The image of a close object is formed in front of her retina.

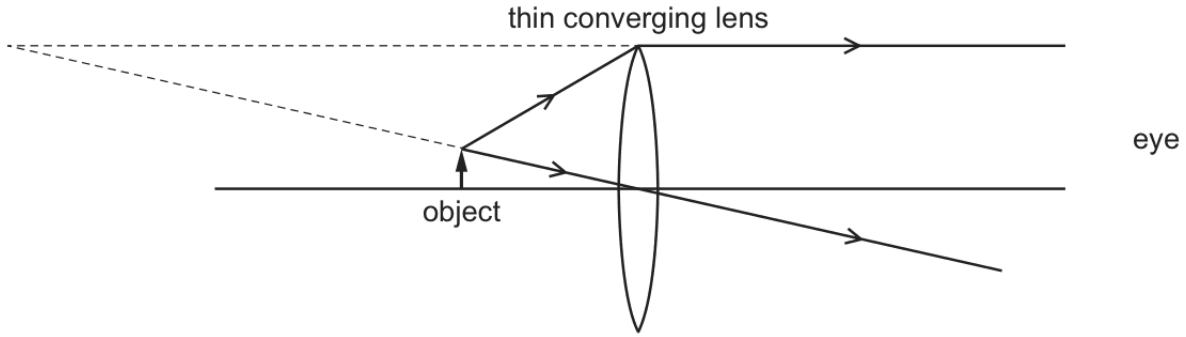
MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 25

25	A	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 8 of 16

21 An object is viewed through a thin converging lens.

The diagram shows the paths of two rays from the top of the object to an eye.



How does the image compare with the object?

A It is larger and inverted.

B It is larger and upright.

C It is smaller and inverted.

D It is smaller and upright.

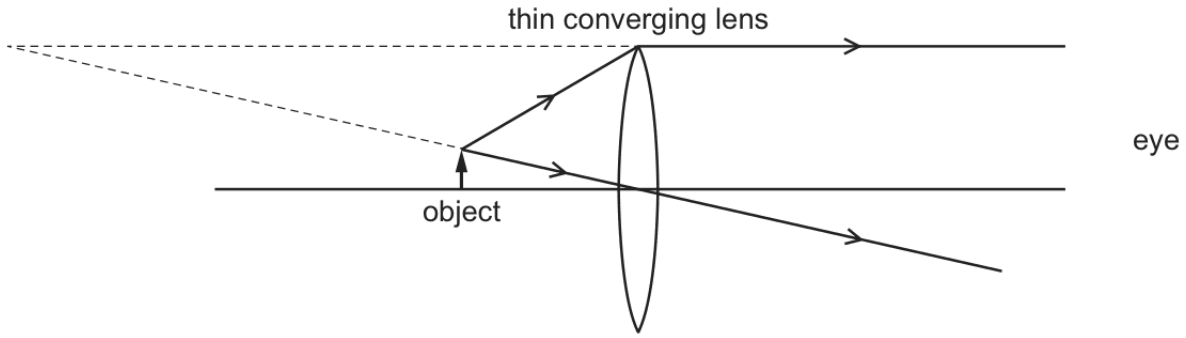
MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 21

21	B	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 12 of 20

25 An object is viewed through a thin converging lens.

The diagram shows the paths of two rays from the top of the object to an eye.



How does the image compare with the object?

A It is larger and inverted.

B It is larger and upright.

C It is smaller and inverted.

D It is smaller and upright.

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 25

25	B	**1**

8 A parallel beam of light travelling in air is incident on a glass lens.

The beam is perpendicular to the lens as shown in Fig. 8.1.

The dashed line P indicates the position of the lens. The centre of the lens is dot C.

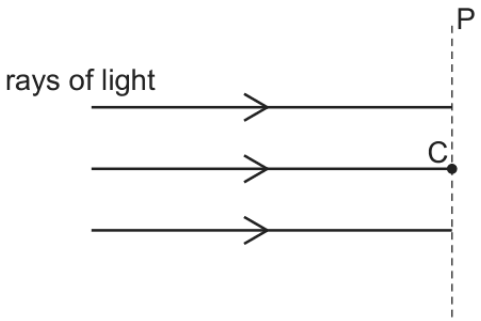


Fig. 8.1

(a) The lens is a diverging lens.

On Fig. 8.1:

(i) indicate the shape of the lens by drawing the outline of the lens around dashed line P [1]

(ii) draw the path taken by each ray of light after it passes through the lens. [2]

(b) Diverging lenses are used to correct short-sight.

(i) Fig. 8.2 is a simplified diagram of a short-sighted eye. Light from a distant object strikes the eye lens and enters the eye.

On Fig. 8.2, continue the three rays in the eye until they reach the back of the eye.

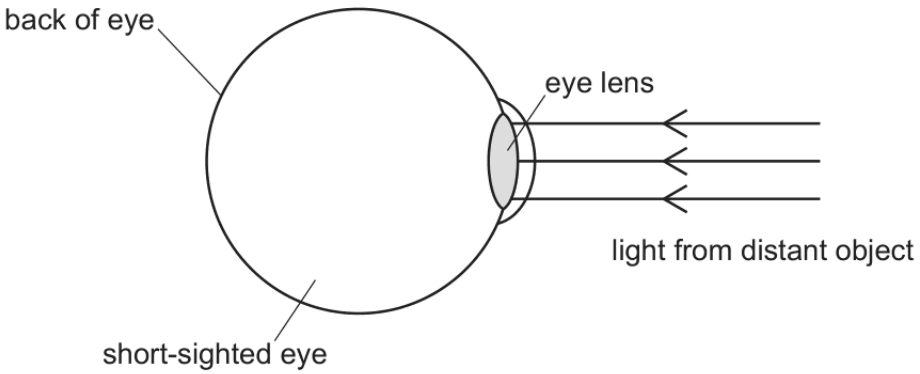


Fig. 8.2

[2]

(ii) State how the image of a distant object detected by a normal eye differs from the image detected by the short-sighted eye.

.....

..... [1]

(iii) Explain how a diverging lens corrects the sight of a short-sighted eye viewing a distant object.

.....

..... [1]

(c) The focal length of the diverging lens is 4.0 cm.

An object of height 3.5 cm is placed 6.0 cm from the centre of the lens.

(i) Fig. 8.3 is a full-scale diagram drawn on a grid, on which the dashed line L represents the lens and the arrow O the object.

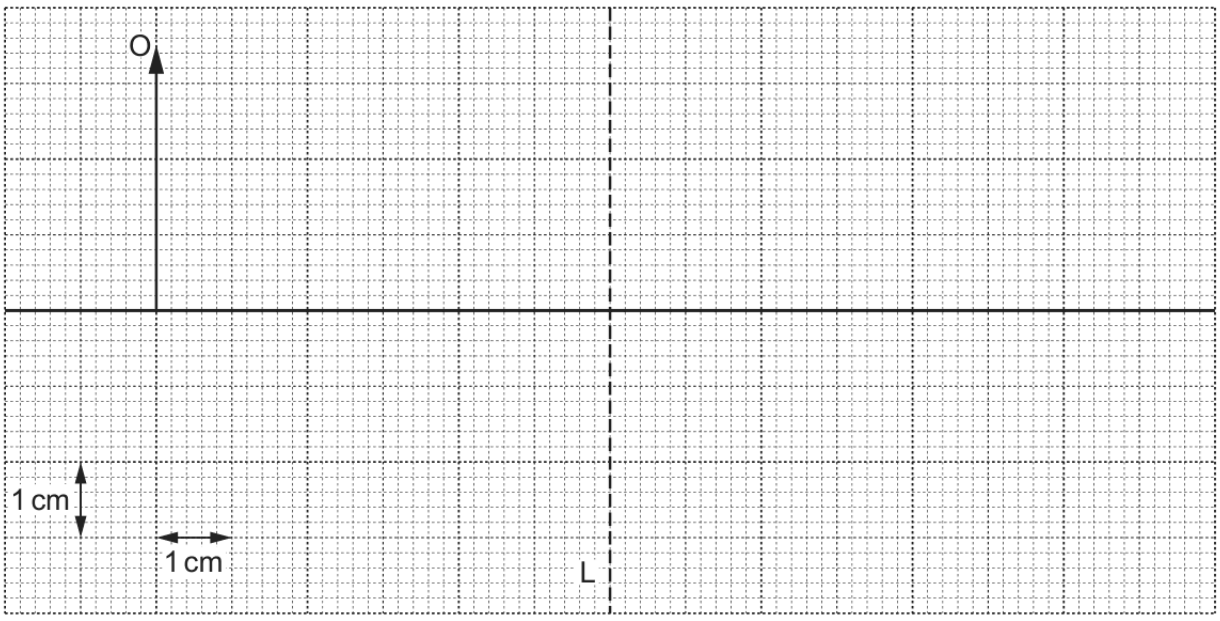


Fig. 8.3

By drawing on Fig. 8.3, find the position of the image I of object O. Draw image I and label it I. [4]

(ii) Explain whether the image produced is real or virtual.

.....

..... [1]

(iii) On the grid in Fig. 8.3, write an E in a position from which an eye can see the image. [1]

(iv) Determine the linear magnification produced.

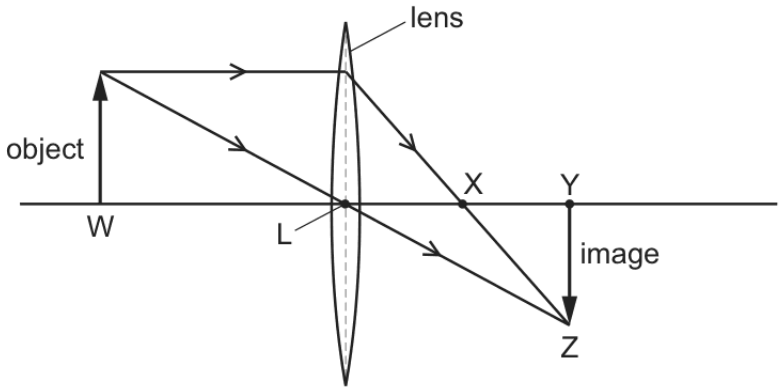
linear magnification = ... [2]

[Total: 15]

Question	Answer	Marks
8(a)(i)	vertical biconcave lens shape along P	**B1**
8(a)(ii)	on right of P, middle ray continues in straight line	**B1**
	other two rays diverge from middle ray	**B1**
8(b)(i)	continuations of the three rays meet at a point **and** on continuation of middle ray	**B1**
	meeting point within eye **and** rays strike back of eye at different places	**B1**
8(b)(ii)	image is sharp / focused / not blurred	**B1**
8(b)(iii)	light diverges before entering eye **or** meeting point now on the back of the eye	**B1**
8(c)(i)	**one** point on principal axis 4.0 cm from the centre of the lens indicated	**B1**
	any **two** rays from: paraxial ray from tip of object to lens **and** seems to come from left-hand principal focus after refraction ray from tip of object towards right-hand principal focus (until lens) **and** paraxial after refraction ray from tip of object reaching the centre of the lens	**B2**
	image marked at correct intersection	**B1**
8(c)(ii)	virtual **and** light only seems to come from the image **or** light does not pass through image	**B1**
8(c)(iii)	E marked on the right of the dashed line	**B1**
8(c)(iv)	height of candidate's I / 3.5	**C1**
	0.30 ≤ magnification ≤ 0.50	**A1**

QUESTION · 5054_w22_qp_12.pdf · source page 10 of 16

24 A thin converging lens forms a real, focused image of an object, as shown.



Which distance is equal to the focal length of the lens?

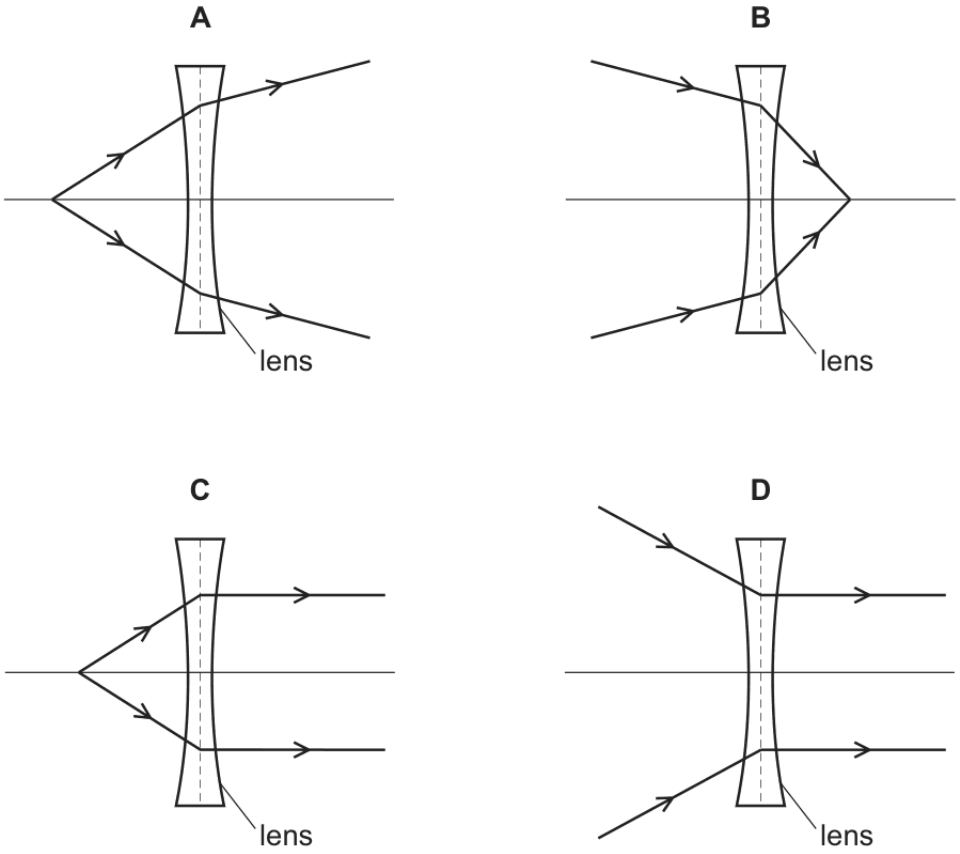
A LW **B** LX **C** LY **D** LZ

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 24

24	B	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 11 of 16

25 Which ray diagram shows the action of a diverging lens?



MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 25

25	D	**1**

6 The virtual reality headset in Fig. 6.1 contains a display and a lens, as shown in Fig. 6.2.

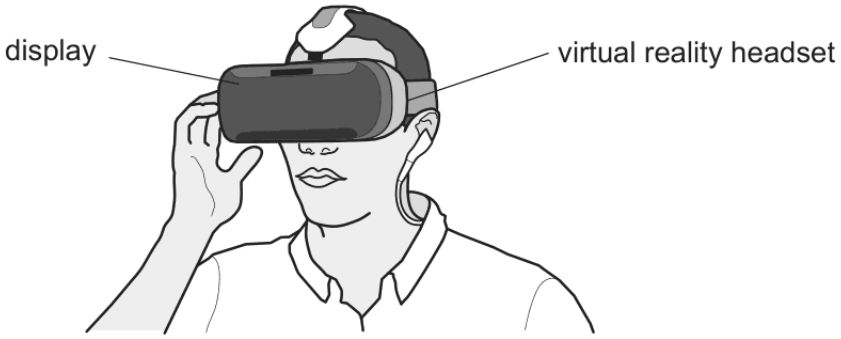


Fig. 6.1

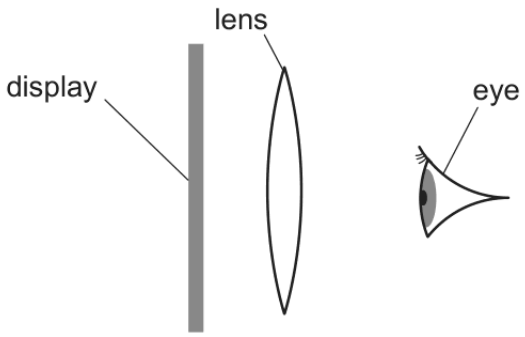


Fig. 6.2 (not to scale)

The display is the object for the lens.

The lens acts as a magnifying glass and forms a virtual image of the display.

(a) (i) Describe where the display must be positioned relative to the focal length of the lens for the lens to act as a magnifying glass for the image on the display.

.....

..... [1]

(ii) Explain how a virtual image is formed.

.....

.....

..... [1]

(b) An arrow on the display is 3.4 cm from the lens. The virtual image of the arrow is 22 cm from the lens.

Fig. 6.3 shows the arrow O, the lens L and the virtual image of the arrow I, drawn on a grid with a scale of 1:2.

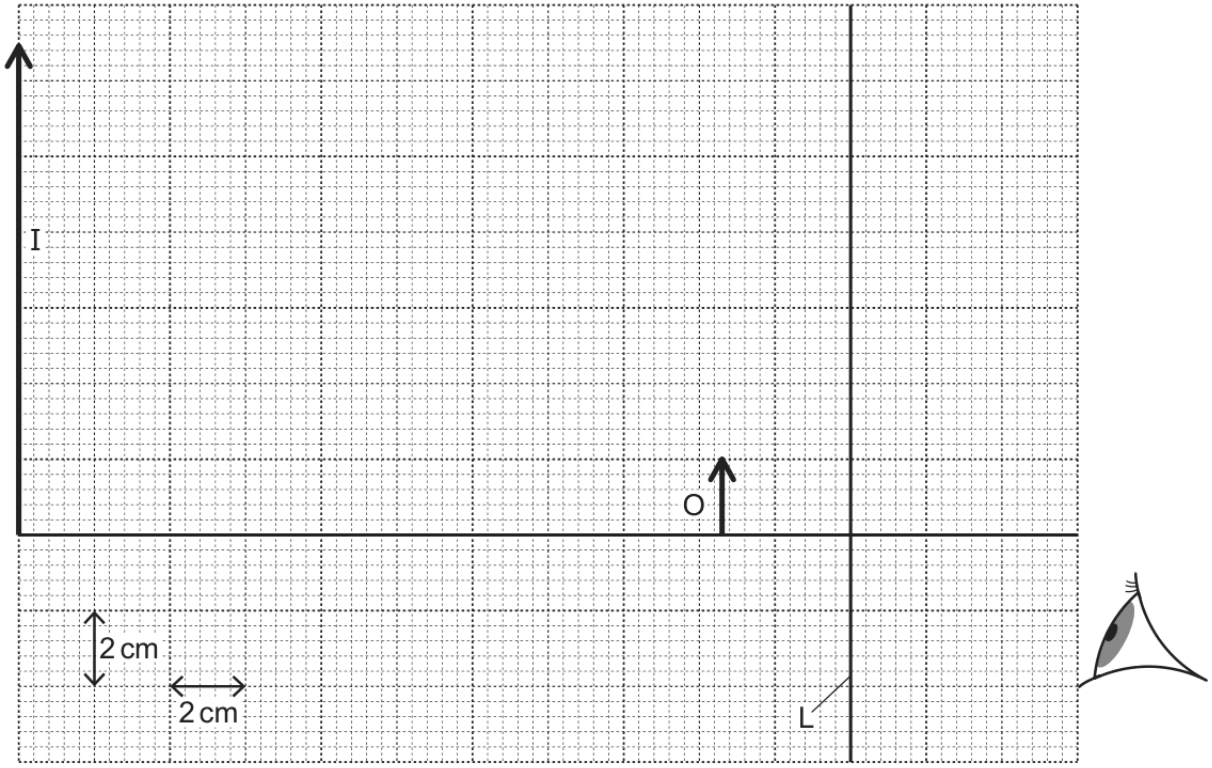


Fig. 6.3 (scale: 1 cm represents 2 cm)

(i) On Fig. 6.3, draw a ray diagram to show the formation of the virtual image I. [3]

(ii) Determine the focal length of the lens.

focal length = ... cm [1]

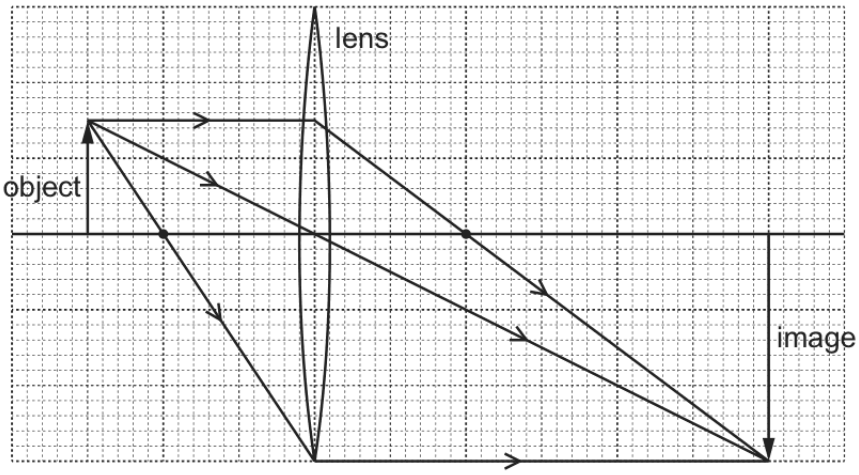
[Total: 6]

Question	Answer	Marks
6(a)(i)	object within focal length / same distance from lens as the focal length	**B1**
6(a)(ii)	from divergent / parallel rays **or** rays followed back (behind lens meet at point on image **or** rays do not meet on a screen / do not converge **or** rays appear to come from image	**B1**
6(b)(i)	ray through optical centre from top of object	**B1**
	ray parallel to axis from top of object to lens **and** line from junction with lens to top of image	**B1**
	both rays extended back to image **and** ray parallel to axis correctly refracted through lens to appear to come from top of image	**B1**
6(b)(ii)	4(.0 cm)	**B1**

QUESTION · 5054_w23_qp_11.pdf · source page 9 of 16

21 An object is placed in front of a converging lens of focal length 4.0 cm. The height of the image is 6.0 cm.

The arrangement is shown on the scale diagram.



What is the linear magnification produced by the lens?

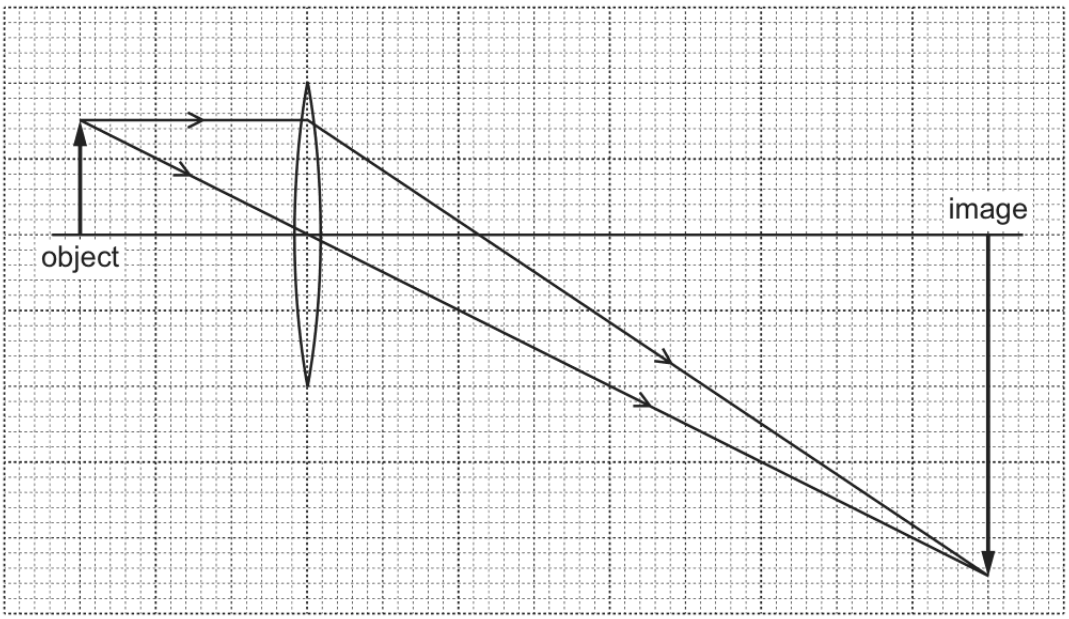
A 0.50 **B** 1.5 **C** 2.0 **D** 6.0

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 21

21	C	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 12 of 20

29 A lens is used to produce a magnified image, as shown in the scale diagram.



What is the linear magnification produced by the lens?

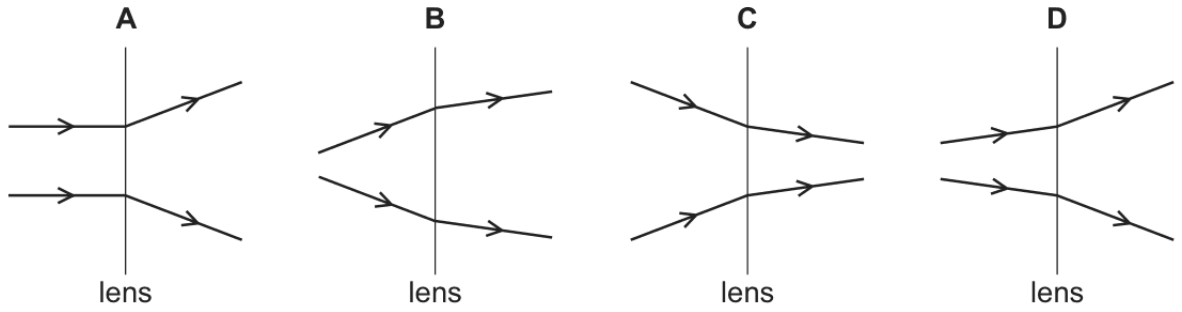
A 0.33 **B** 3.0 **C** 4.0 **D** 6.0

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 3 of 3 · row for Question 29

29	B	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 9 of 16

25 Which diagram shows rays of light passing through a converging lens?



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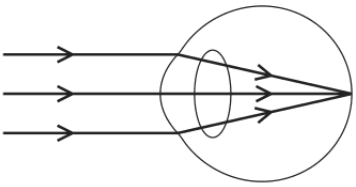
25	B	**1**

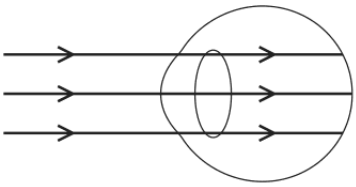
QUESTION · 5054_s24_qp_12.pdf · source page 9 of 16

24 Which diagram shows how light from a distant object forms an image in a normal eye?

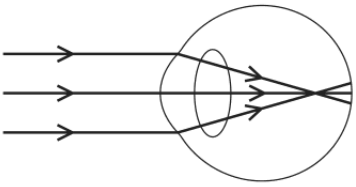
A



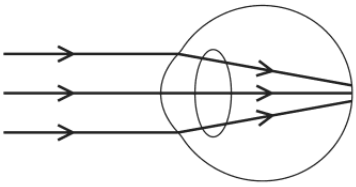
B



C



D



MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 24

24	A	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 10 of 16

26 A thin converging lens is used as a magnifying glass.

Which row gives the nature of the image produced and an expression that is the linear magnification?

	nature of image	expression for linear magnification
A	real	$\frac{\text{image length}}{\text{object length}}$
B	real	$\frac{\text{object length}}{\text{image length}}$
C	virtual	$\frac{\text{image length}}{\text{object length}}$
D	virtual	$\frac{\text{object length}}{\text{image length}}$

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 26

26	C	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 11 of 16

26 A short-sighted student looks at a distant object.

Which row shows the position of an image in a short-sighted eye and the type of lens used to correct short-sightedness?

	position of image in short-sighted eye	lens used to correct short-sightedness
A	in front of retina	diverging
B	behind retina	converging
C	on retina	converging
D	behind retina	diverging

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 26

26	A	**1**

5 Some glass lenses are converging lenses, and others are diverging lenses.

(a) Draw the cross-section of a diverging lens.

[1]

(b) Fig. 5.1 shows the cross-section of a converging lens, the principal axis and the two principal focuses (focal points) F_1 and F_2 on a full-scale grid.

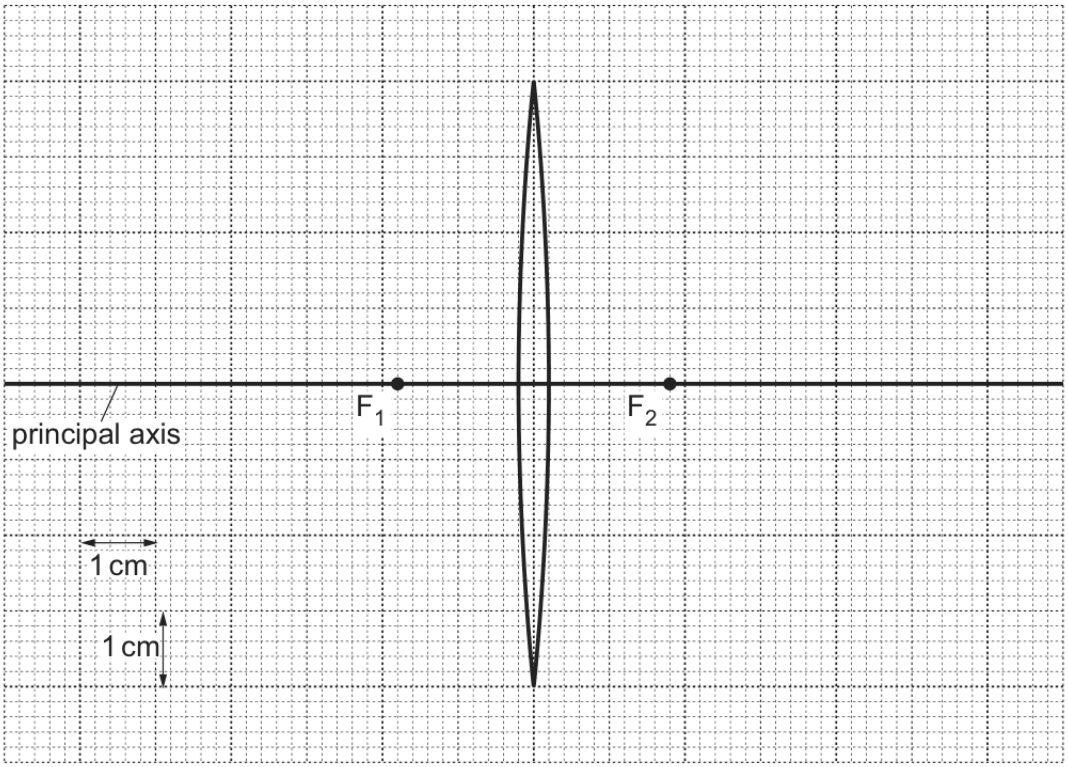


Fig. 5.1

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A student places an object of height 2.1 cm at a distance of 3.0 cm from the centre of the lens.

(i) Using Fig. 5.1, determine the focal length of the lens.

focal length = ... cm [1]

(ii) On Fig. 5.1, draw a vertical arrow of height 2.1 cm that is 3.0 cm from the centre of the lens and label the arrow **O**. The arrow is the object. [1]

(iii) On Fig. 5.1, draw **two** rays from the tip of the object arrow to find the tip of the image.

Draw another arrow to show the image. [3]

(iv) Using the image marked on Fig. 5.1 in **(b)(iii)**, determine the linear magnification produced.

magnification = ... [2]

(v) Explain whether the image of the object is real or virtual.

.....

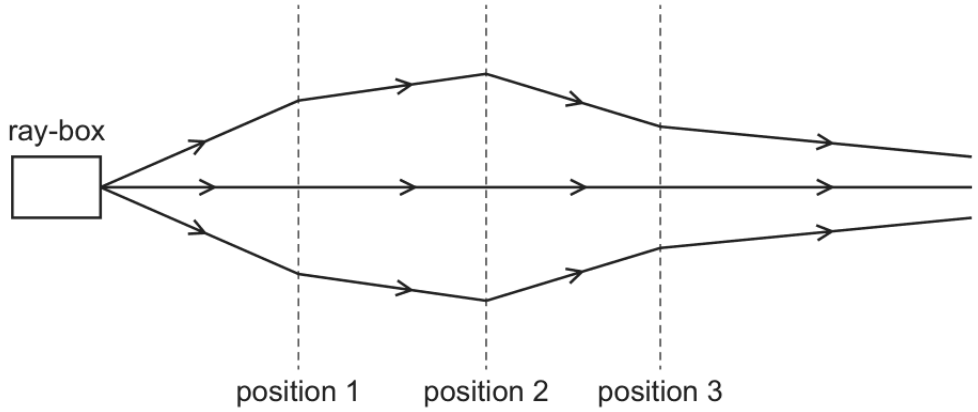
..... [1]

[Total: 9]

Question	Answer	Marks
5(a)	a drawn lens that is thinner at the middle (than at the edges)	**B1**
5(b)(i)	1.8 (cm)	**B1**
5(b)(ii)	vertical arrow of height 2.1 cm drawn 3.0 cm from lens (and labelled O)	**B1**
5(b)(iii)	any **two** from: paraxial ray from tip of object to lens that then refracts through F_2 ray from tip of object, straight through and past optical centre ray from tip of object through F_1 to lens that then emerges paraxially	**B2**
	image drawn from (sensible) intersection to principal axis	**B1**
5(b)(iv)	(linear magnification =) h_I / h_o **or** height of candidate's image / 2.1 (cm)	**C1**
	$1.2 \leqslant$ magnification $\leqslant 1.8$	**A1**
5(b)(v)	real **and** formed by converging light / rays	**B1**

QUESTION · 5054_s25_qp_11.pdf · source page 10 of 20

22 The rays of light from a ray-box pass through three lenses placed at positions 1, 2 and 3.



Which type of lens is used at each position?

	position 1	position 2	position 3
A	converging	converging	converging
B	converging	converging	diverging
C	diverging	converging	diverging
D	diverging	diverging	converging

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 22

22	B	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 11 of 20

24 When a converging lens is used as a magnifying glass, what is the nature of the image?

A real and inverted

B real and upright

C virtual and inverted

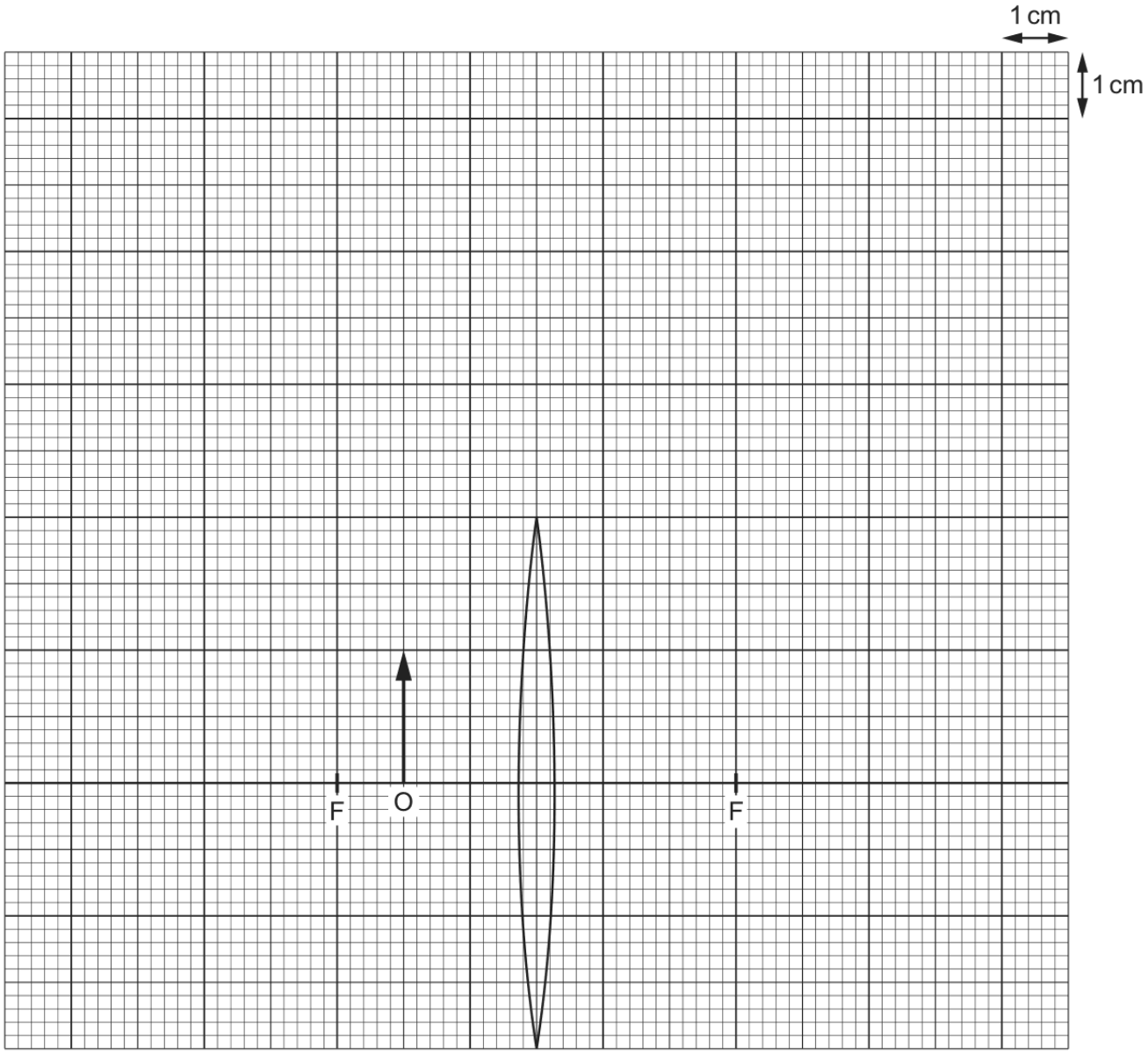
D virtual and upright

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 24

24	D	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 10 of 16

23 An object O is placed between a converging lens and one principal focus F of the lens.



What is the magnification of the image formed?

A 1.0 **B** 2.0 **C** 3.0 **D** 6.0

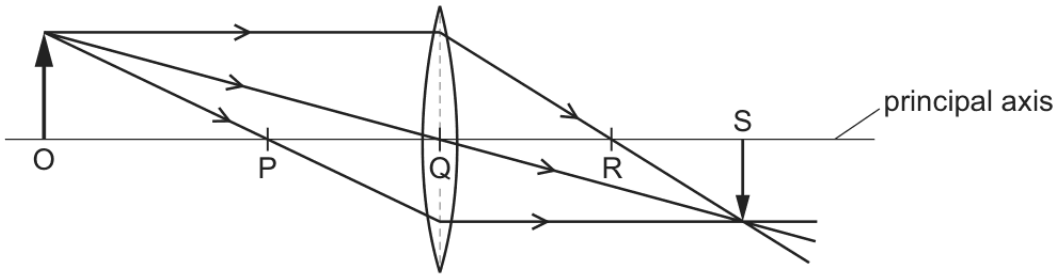
MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 23

23	C	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 10 of 16

21 An object O is placed near to a thin converging lens.

The diagram shows three rays of light from the object passing through the lens.



Which row describes the image formed and gives the position of a principal focus (focal point) of the lens?

	image formed	position of principal focus
A	real	R
B	real	Q
C	virtual	P
D	virtual	S

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 21

21	A	**1**

4 Fig. 4.1 shows a ray of light striking the left-hand surface of a thin glass converging lens.

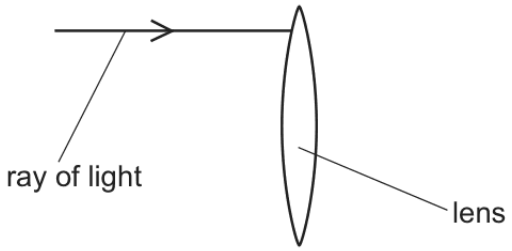


Fig. 4.1

(a) Explain why the ray refracts and changes direction as it enters the glass.

.....

.....

.....

..... [2]

(b) Explain what is meant by:

(i) the principal focus (focal point) of a lens

.....

.....

.....

..... [2]

(ii) the focal length of a lens.

.....

..... [1]

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(c) An object O of height 1.8 cm is placed at a point 2.2 cm to the left of the centre of the lens.

Fig. 4.2 is a full-scale diagram that shows the converging lens, its principal axis and its two principal focuses, F_1 and F_2.

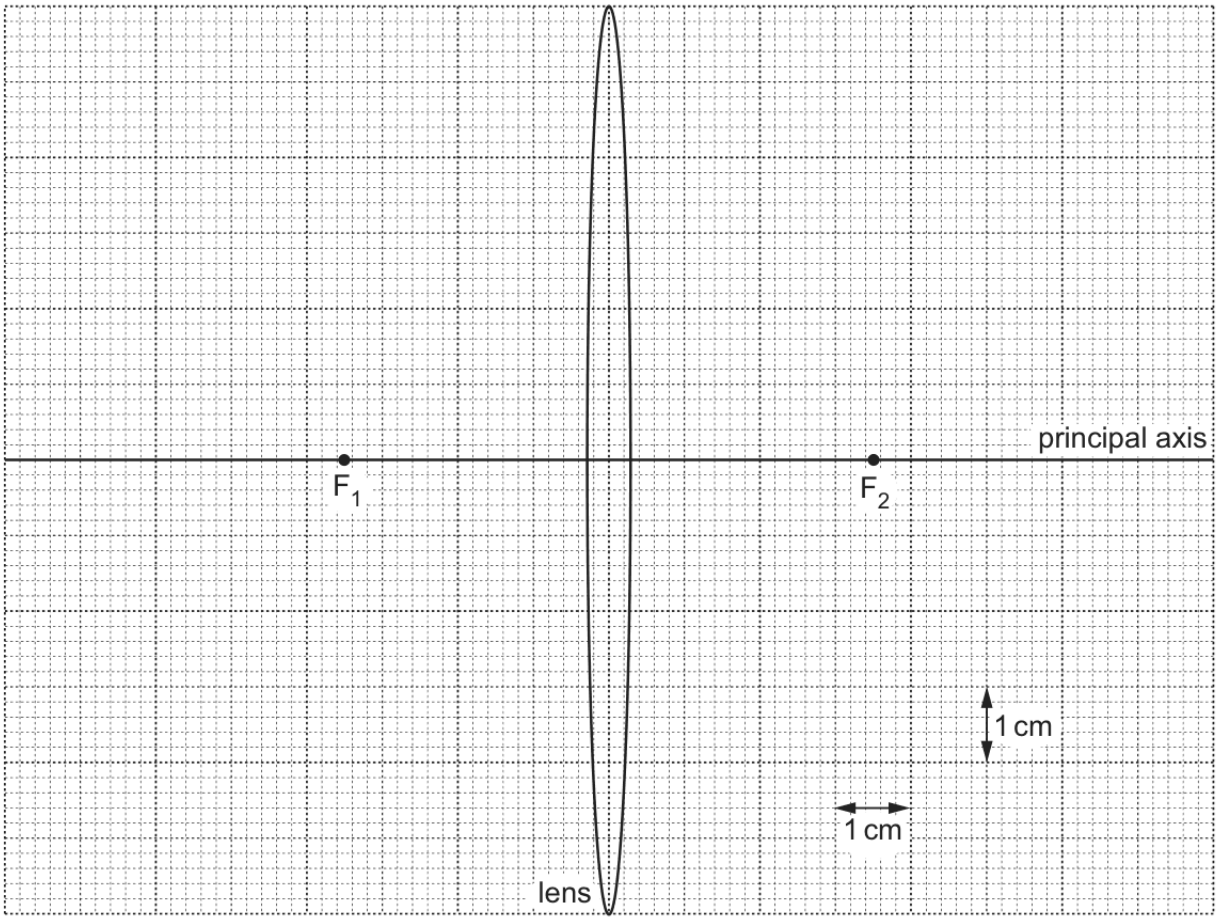


Fig. 4.2

(i) On Fig. 4.2, draw an arrow to represent the height and position of object O. The arrow should start on the principal axis. [1]

(ii) On Fig. 4.2, draw **two** rays from the tip of the arrow drawn in **(c)(i)** to locate the position of the tip of the image I of object O.

Draw an arrow to represent the image I of the object O. Label the image I. [3]

(iii) State whether the image I of the object O is a real or a virtual image, and explain your answer.

.....

.....

..... [1]

[Total: 10]

MARK SCHEME · 5054_w25_ms_21.pdf · source page 9 of 14 · Question 4

Question	Answer	Marks
4(a)	any **two** from: (in glass) light / it travels more slowly / at a different speed enters (an optically) denser medium / medium with different refractive index wavelength changes (and frequency stays the same) angle of incidence is greater than 0	**B2**

MARK SCHEME · 5054_w25_ms_21.pdf · source page 10 of 14 · Question 4

Question	Answer	Marks
4(b)(i)	(point) where parallel rays meet (after refraction)	**C1**
	(point) where rays parallel to principal axis meet (after refraction) **or** (point) where paraxial rays meet (after refraction)	**A1**
4(b)(ii)	distance between principal focus / focal point and (centre of) lens	**B1**
4(c)(i)	arrow of height 1.8 cm **and** 2.2 cm **and** on left of the centre of the lens	**B1**
4(c)(ii)	any **two** from: paraxial ray from tip of O to lens **and** from lens to F_2 straight ray through centre of lens ray that seems to come from F_1 from tip of O to lens **and** emerges paraxially from lens	**M2**
	two rays traced back to image (labelled I)	**A1**
4(c)(iii)	virtual **and** rays (merely) seem to come from I / do not pass through the image / object distance less than focal length / the image is upright	**B1**

QUESTION · 5054_w25_qp_12.pdf · source page 12 of 16

31 A student does an investigation using four thin lenses. The lenses are used to produce focused images of four different objects.

The student measures the length of the image and the length of the object. The results are shown in the table.

In which case is the magnification produced the greatest?

	image length / cm	object length / cm
A	3.5	0.80
B	4.7	0.60
C	4.8	0.70
D	5.2	0.90

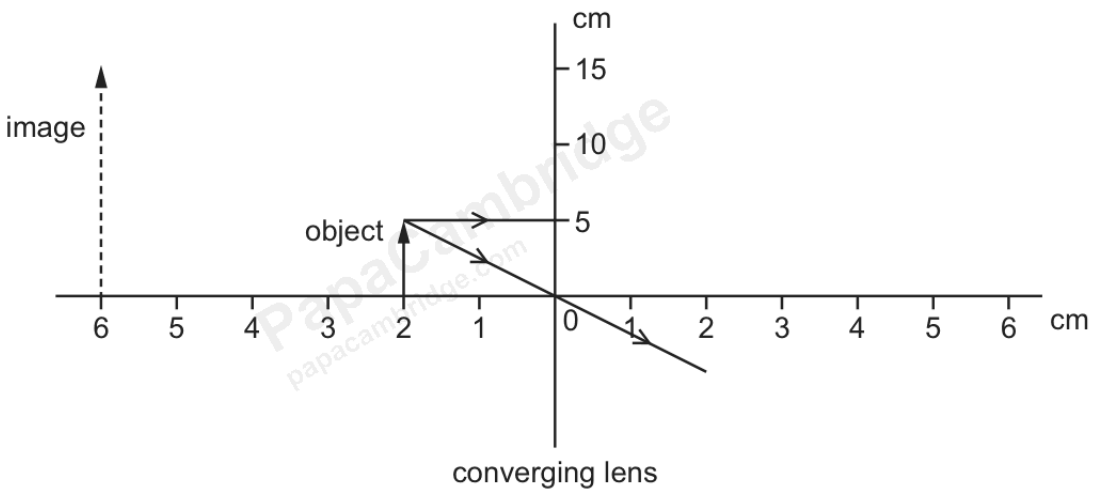
MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 3 of 3 · row for Question 31

31	B	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 11 of 16

26 An object of height 5.0 cm is placed 2.0 cm from a converging (convex) lens which is being used as a magnifying glass.

The image produced is 6.0 cm from the lens and has height 15.0 cm.



What is the focal length of the lens?

A 2.0 cm **B** 3.0 cm **C** 4.0 cm **D** 6.0 cm

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 26

26	B	**1**

5 A student uses a glass block and two identical glass prisms to demonstrate the action of a converging lens.

Figure 5.1 shows three parallel rays of red light incident on the glass block and the two prisms.

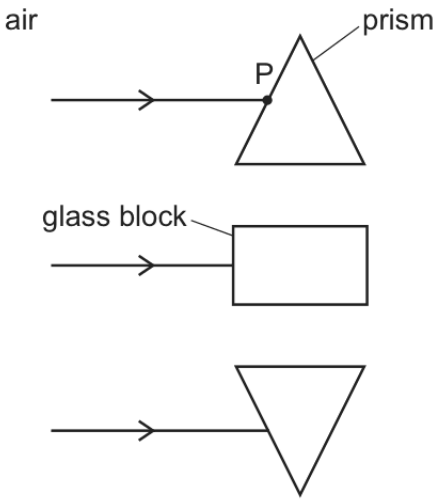


Figure 5.1

(a) (i) On Figure 5.1, draw and label the normal to the prism at point P. [1]

(ii) Complete Figure 5.1 to show how the rays of light travel through and out of both prisms and the glass block. [2]

(iii) The action of a diverging lens is also demonstrated using the two identical glass prisms and the glass block.

On Figure 5.2, draw the prisms arranged so that they demonstrate the action of a diverging lens.

Show the rays of light travelling through the prisms and the glass block.

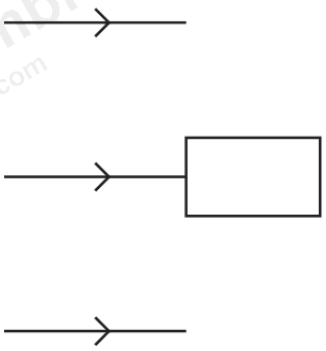


Figure 5.2

[2]

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(b) Figure 5.3 shows a converging lens, an object and two rays of light from the top of the object. The principal focus (focal point) on each side of the lens is labelled F.

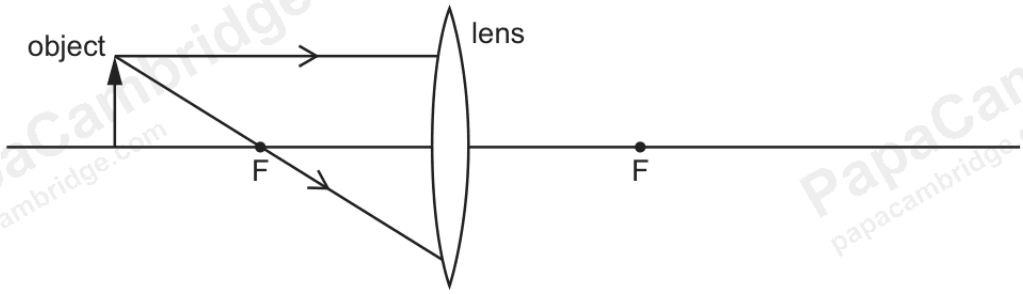


Figure 5.3

Continue the paths of both rays to show where the image of the object is formed. Draw and label the image. [2]

(c) Figure 5.4 shows the electromagnetic spectrum.

radio waves	microwaves	infrared	visible light	ultraviolet	X-rays	gamma rays

Figure 5.4

State **two** properties of electromagnetic waves that change from one end of the spectrum to the other end, and describe how these properties change from one end to the other.

1 .....

.....

2 .....

.....

[2]

[Total: 9]

5(a)(i)	correct normal	**B1**	by eye; either one side or the other of boundary or both

bod

Question	Answer	Marks	Guidance
5(a)(ii)	three correct refractions on entering and leaving the prisms	**B1**	**I** central block **con** dispersion, e.g. ray splits on refraction **allow** one wrong refraction of a ray at a boundary
	horizontal ray through and emerging from central glass block	**B1**	
5(a)(iii)	correct arrangement of prisms	**B1**	i.e. Intention that prisms inverted vertically, e.g. apex of prisms face central block
	rays diverge after prisms (top diverges up and bottom diverges down)	**B1**	**I** refraction in block and prisms**; I** ray leaving the block **not** if rays split on refraction; no prisms shown
5(b)	one correct ray; top ray passes through / to F on right of lens **or** bottom ray horizontal on right of lens	**C1**	
	both rays correct **and** image shown	**A1**	**A** point of intersection marked as I / image **A** arrow either way up as image at image position without a label
5(c)	ANY 2 from • wavelength decreases (towards the right) • frequency increases (towards the right) • damage / ionisation / energy increases (towards the right)	**B2**	**A** reverse arguments **con** speed changes assume left to right unless stated otherwise **A** just 'radio waves largest / large(r) λ'; 'γ small(est) λ' 'gamma have high(est) / high(er) frequency'; 'radio low(est) frequency' **SC** allow 1 mark for *wavelength increases and freq. decreases;* radio has largest λ **and** smallest frequency ORA for gamma

Chapter 3 Waves

3.2 Light

3.2.4 Dispersion of light

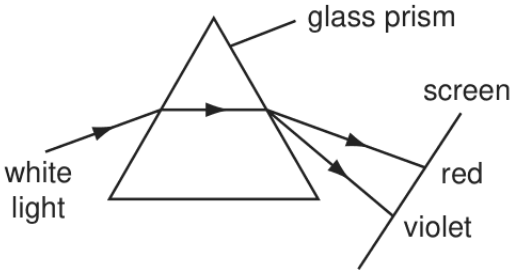
42 entries: 29 Paper 1 questions, 13 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

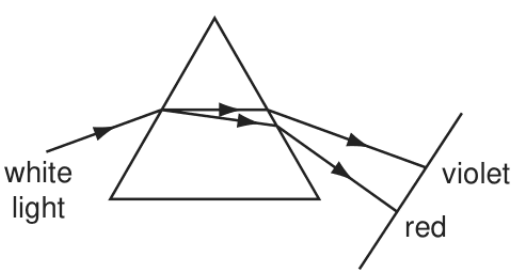
QUESTION · 5054_s04_qp_1.pdf · source page 11 of 20

23 Which diagram correctly shows the dispersion of white light by a glass prism?

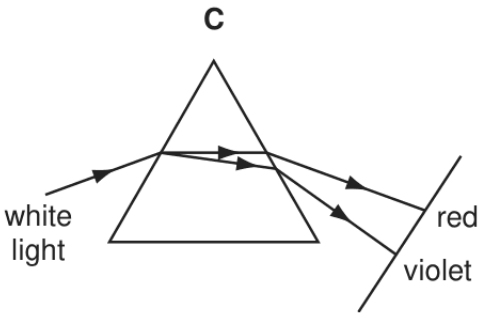
A



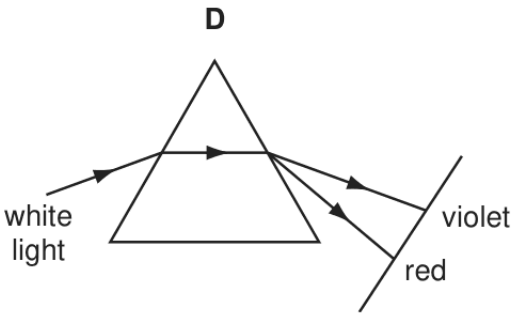
B



C



D

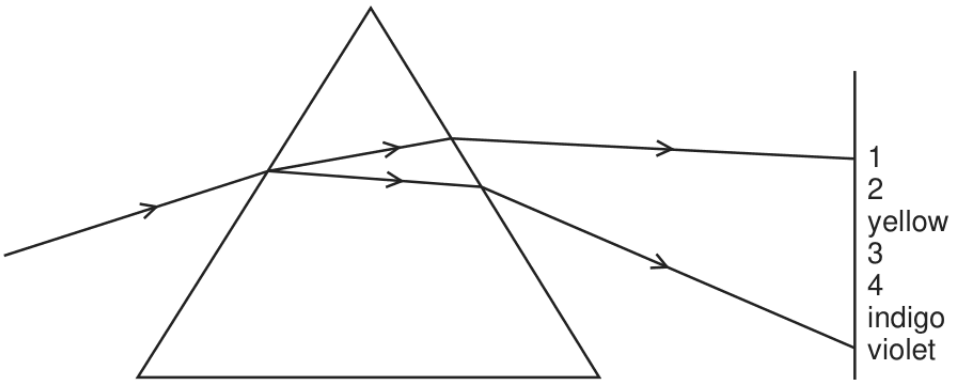


MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 23

23	**C**

QUESTION · 5054_w04_qp_1.pdf · source page 12 of 20

26 The diagram shows the spectrum produced when white light is dispersed by a glass prism.



What are the numbered visible colours?

	1	2	3	4
A	infra-red	red	green	ultra-violet
B	red	green	orange	blue
C	red	orange	green	blue
D	red	orange	green	ultra-violet

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 26

26 **C**

For Examiner's Use

2 Fig. 2.1 shows the lens of a simple camera being used to photograph an object.

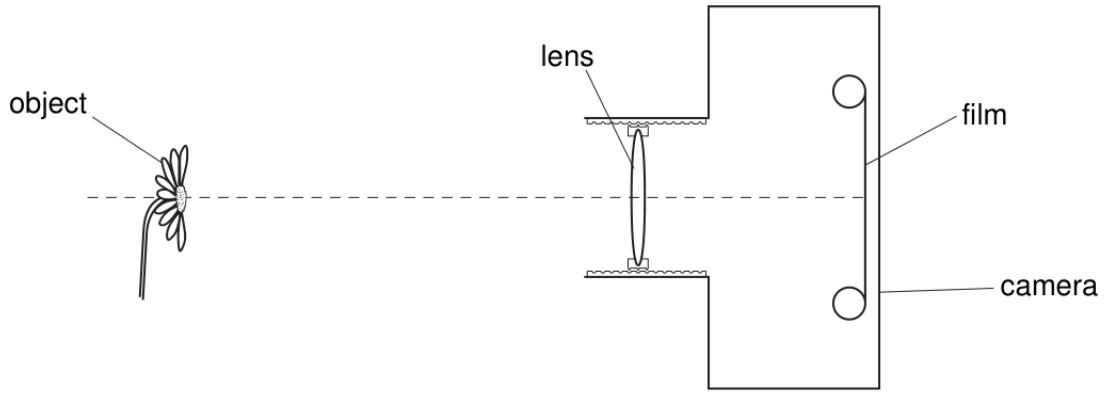


Fig. 2.1

The lens forms a focused image of the object on the film.

(a) Draw two rays from the top of the object to show how the lens forms the image. [2]

(b) The object moves closer to the camera. State how the lens is adjusted to keep the image in focus.

.....

.....

..... [1]

(c) Complete Fig. 2.2 to show how white light is split into a spectrum when it passes through a glass prism. [3]

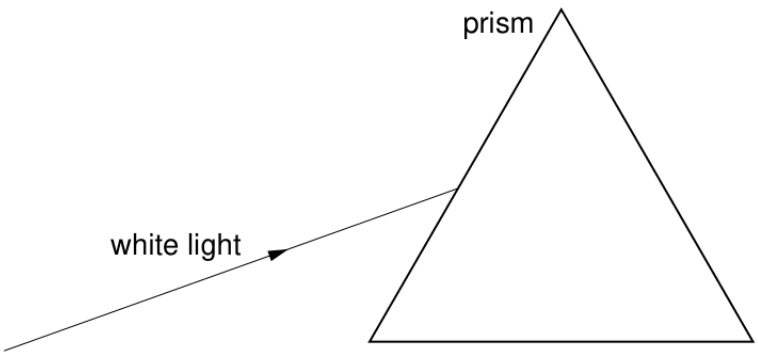
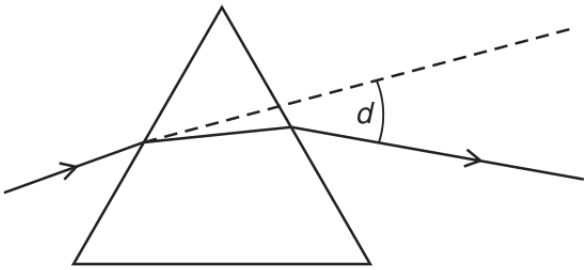


Fig. 2.2

2	**(a)**	straight line through optical centre by eye	**M1**	
		one other line from same point on object correctly to image on film	**A1**	
	(b)	move lens towards object/to left/away from film	**B1**	
	(c)	**1st and 2nd face** correct refraction for all rays shown	**B1**	
		dispersion into at least two rays at first face only	**B1**	
		colours marked on diverging rays outside prism (any 2 visible colours from spectrum, any order, accept letters)	**B1**	**6**

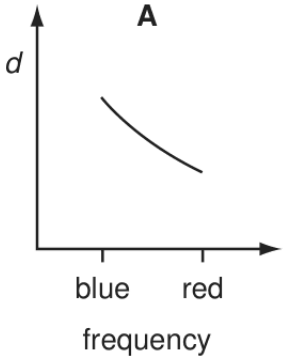
QUESTION · 5054_s06_qp_1.pdf · source page 9 of 20

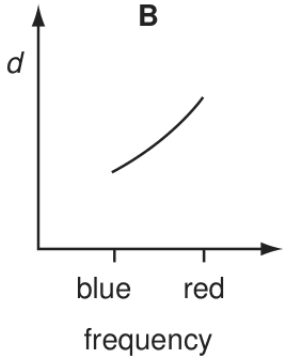
24 Light rays are deviated by a prism.

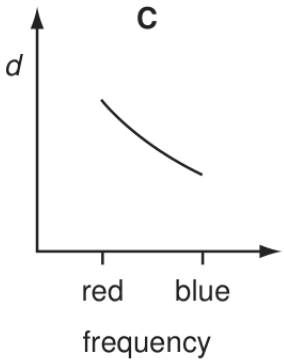


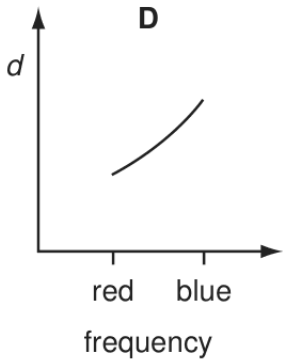
The deviation angle *d* is measured for light rays of different frequency, including blue light and red light.

Which graph is correct?









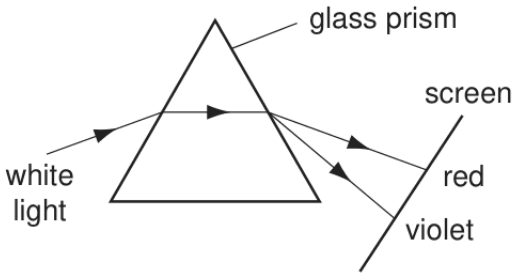
MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 24

24	**D**

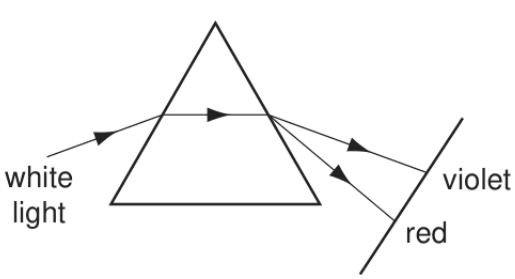
QUESTION · 5054_w07_qp_1.pdf · source page 10 of 20

21 Which diagram correctly shows the dispersion of white light by a glass prism?

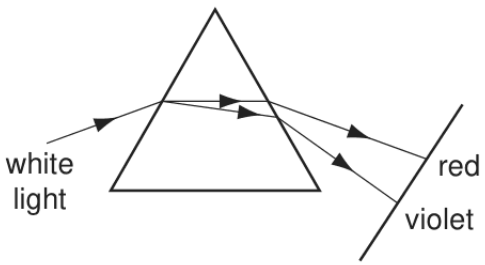
A



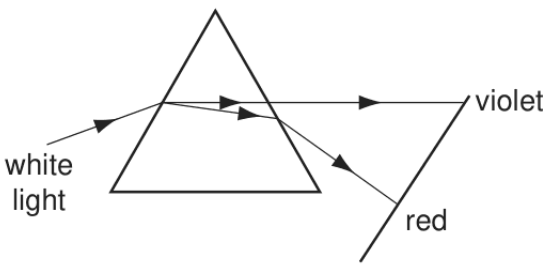
B



C



D



MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 21

21	C

QUESTION · 5054_s08_qp_1.pdf · source page 11 of 20

25 Which colour, red or blue, has the higher frequency and which has the longer wavelength?

	higher frequency	longer wavelength
A	blue	blue
B	blue	red
C	red	blue
D	red	red

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 25

25 **B**

For Examiner's Use

6 Fig. 6.1 shows a ray of white light from a ray-box passing into a glass prism. A spectrum is formed between P and Q on the screen.

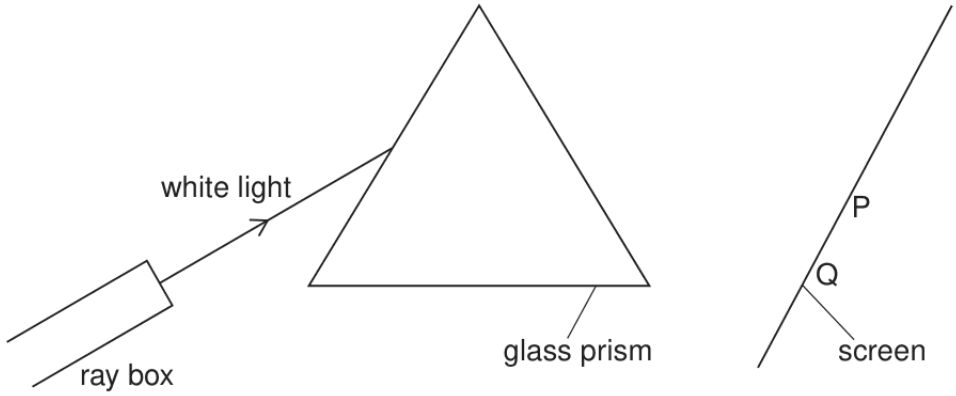


Fig. 6.1

(a) State the colour of the light at end P of the spectrum.

..... [1]

(b) State whether the value of each of these properties for blue light is greater than, equal to or less than the value for red light.

(i) speed in a vacuum ..... [1]

(ii) wavelength ..... [1]

(c) Fig. 6.2 shows the ray passing through a red filter before it reaches the prism.

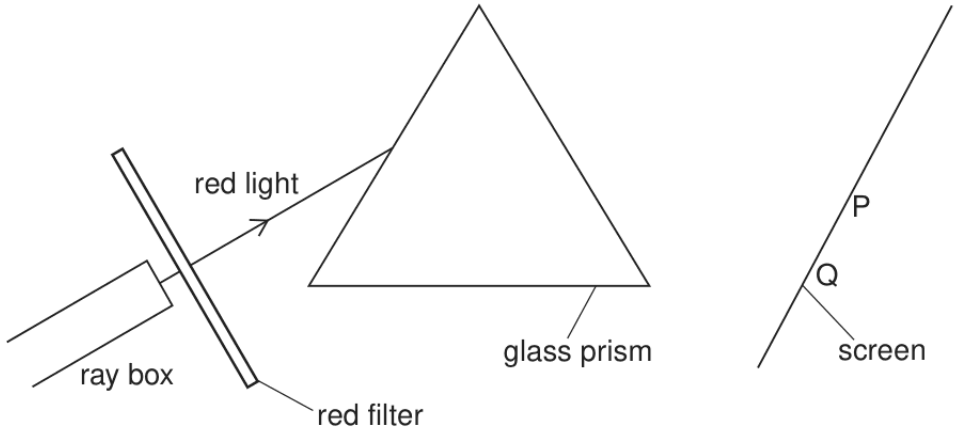


Fig. 6.2

Complete Fig. 6.2 to show the ray of red light passing through and emerging from the prism. [2]

6	**(a)**	red	B1	
	(b)	**(i)** equal to	B1	
		(ii) less than	B1	
	(c)	two correct refractions on Fig. 6.2	M1	
		no dispersion **and** ray ends close to P	A1	[5]

QUESTION · 5054_s09_qp_1.pdf · source page 10 of 20

22 When white light is dispersed by a prism, compared with blue light, the **red** light is

A slowed down less and refracted less.

B slowed down less and refracted more.

C slowed down more and refracted less.

D slowed down more and refracted more.

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 22

22	**A**

For Examiner's Use

5 Fig. 5.1 shows a ray of white light incident on a glass prism.

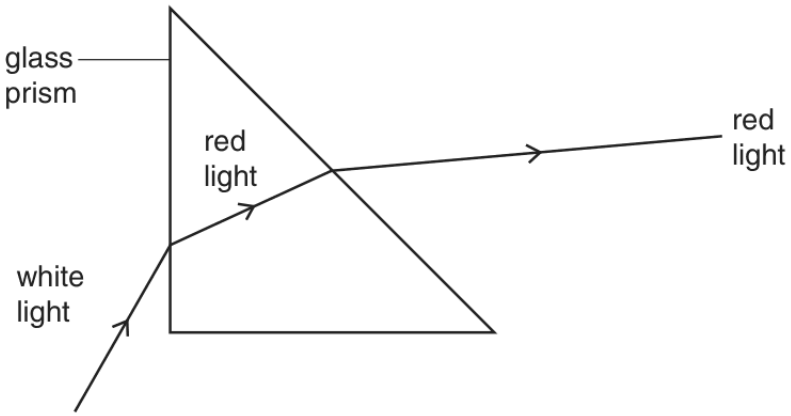


Fig. 5.1

Fig. 5.1 is not complete; it shows only the ray of red light produced from the white light.

(a) (i) On Fig. 5.1, draw the ray of blue light produced in, and beyond, the prism. [2]

(ii) State **two** colours of the spectrum found between the red and blue rays.

.....[1]

(b) Fig. 5.2 shows the same prism with the ray of white light incident at a different angle.

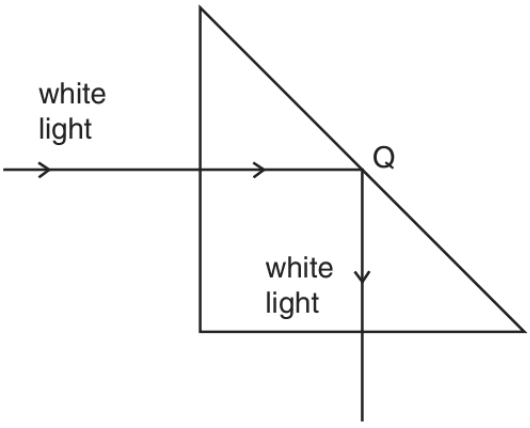


Fig. 5.2

Fig. 5.2 is complete; no spectrum is produced.

(i) Explain why no light emerges from the prism at Q.

.....

.....[1]

(ii) Explain why no spectrum is produced.

.....

.....[1]

5	**(a)**	**(i)**	correct direction of refraction at **both** faces (**not** along normal)	B1	
			blue below red **and** blue and red diverge	B1	[2]
		(ii)	any **two** from orange, yellow, green	B1	[1]
	(b)	**(i)**	**total internal reflection or** angle of incidence greater than critical angle	B1	[1]
		(ii)	all colours reflected at same angle **or** all have i = r	B1	[1]

QUESTION · 5054_w11_qp_11.pdf · source page 8 of 20

20 How do the speed and the wavelength of red light in air compare with the speed and the wavelength of violet light in air?

	speed of red light	wavelength of red light
A	greater	greater
B	greater	less
C	same	greater
D	same	less

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 20

20 **C**

QUESTION · 5054_w11_qp_12.pdf · source page 10 of 20

22 How do the speed and the wavelength of red light in air compare with the speed and the wavelength of violet light in air?

	speed of red light	wavelength of red light
A	greater	greater
B	greater	less
C	same	greater
D	same	less

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 22

22	**C**

QUESTION · 5054_w12_qp_11.pdf · source page 10 of 16

25 Red and violet are the colours at the ends of the visible spectrum.

How do the frequencies and the wavelengths of these colours compare?

	higher frequency	longer wavelength
A	red	red
B	red	violet
C	violet	red
D	violet	violet

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 25

25 **C**

QUESTION · 5054_w12_qp_12.pdf · source page 9 of 16

25 Which statement about red light and blue light is correct?

A Red light has a higher frequency than blue light.

B Red light has a longer wavelength than blue light.

C Red light has the same speed in glass as blue light.

D Red light is refracted by a glass prism more than blue light.

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 25

25	**B**

For Examiner's Use

10 A student traces the path of a ray of blue light as it enters and as it leaves a glass prism. Fig. 10.1 shows the trace obtained by the student.

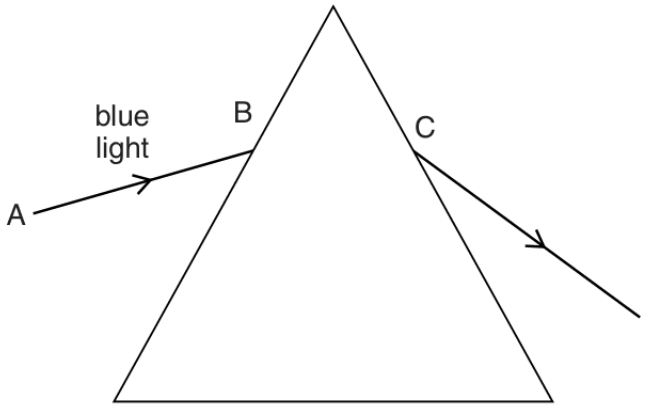


Fig. 10.1

(a) On Fig. 10.1, draw and label, at the point B, the normal, the angle of incidence *i* and the angle of refraction *r*. [3]

(b) State, in terms of the properties of light waves, why the light refracts at B.

.....

..... [1]

(c) The angle of incidence for the ray of blue light at B is 45°. The refractive index of the glass is 1.5. Calculate the angle of refraction at B.

angle of refraction = ... [3]

(d) The student performs another experiment with a ray of red light along the line AB.

On Fig. 10.1, show the path taken by this ray of light as it passes through and leaves the prism. [2]

For Examiner's Use

(e) The student performs another experiment with a semicircular glass block and a ray of white light. Fig.10.2 shows the path taken by this ray of light as it enters the glass at P until it hits the straight edge at Q.

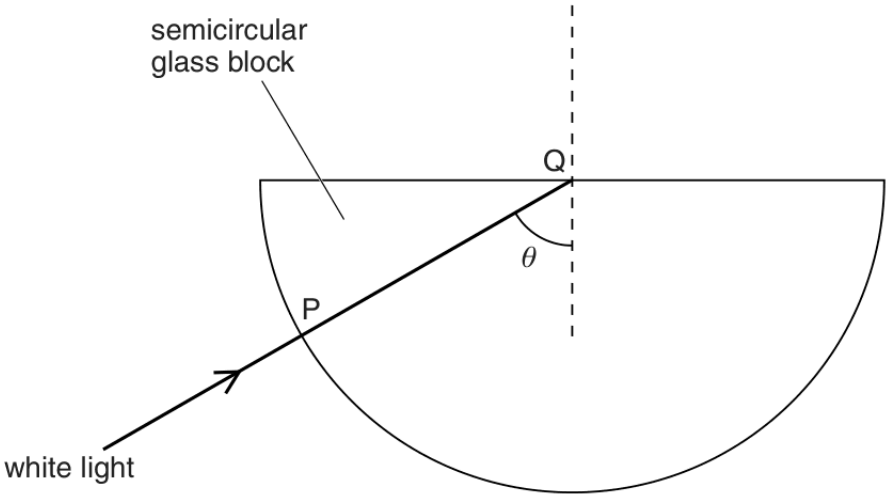


Fig. 10.2

The student finds that there is no change in direction as the ray enters the glass at P and that no light passes out of the glass at Q.

(i) Explain why the ray does not change direction at P.

.....

..... [1]

(ii) Explain why no light passes out of the glass at Q.

.....

.....

..... [2]

(iii) On Fig. 10.2, draw the complete path followed by this ray. [1]

(iv) The student directs the ray of white light into the glass along different paths, so that the angle θ is slowly reduced.

Describe what happens to the ray at Q.

.....

.....

..... [2]

MARK SCHEME · 5054_s13_ms_22.pdf · source page 5 of 7 · Question 10

10 (a) correct normal by eye B1
correct angle of incidence between candidate's normal and incident ray B1
correct angle of refraction marked between candidate's normal and BC B1 [3]

(b) decrease / change in speed / wavelength B1 [1]

MARK SCHEME · 5054_s13_ms_22.pdf · source page 6 of 7 · Question 10

(c) n = sin i/sin r seen in any form C1

(sin r =) sin 45°/1.5 C1
or 0.47(14) seen

28(.1)° C1 [3]

(d) refracts less at first face and on correct side of normal B1

refraction at second face away from normal so that red ray and blue ray are diverging B1 [2]

(e) (i) angle of incidence is 0 B1
or ray along normal/perpendicular to glass

(ii) angle of incidence/θ is larger than critical angle B1
total internal reflection occurs B1

(iii) reflected ray drawn correctly **and** emerging without refraction from block B1

(iv) (eventually) light emerges (into air at Q) B1
or light refracts (out at Q)
or (weak) refracted ray appears

light emerging at Q coloured in some way B1 [6]
or correct description of movement of reflected ray (as θ decreases)

[Total: 15]

For Examiner's Use

10 A student goes for a walk in the mountains. During a storm, she sees lightning strike a hillside in the distance. Several seconds later, she hears the thunder caused by the lightning.

(a) (i) Explain why she hears the thunder several seconds after she sees the lightning.

.....

.....[1]

(ii) The student knows the distance to the hillside. She waits for lightning to strike the hillside again. Describe how she can determine a value for the speed of sound in air.

.....

.....

.....

.....[2]

(b) After the storm, the student sees a rainbow.

(i) State the speed of light in air.

speed = ...[1]

(ii) Calculate the wavelength in air of light of frequency 7.5×10^{14} Hz.

wavelength = ...[2]

(iii) State the colours of the spectrum in order of increasing wavelength.

.....

.....[2]

For Examiner's Use

(c) In the laboratory, the student sends blue light into a glass prism placed on a sheet of paper. The arrangement is shown in Fig. 10.1.

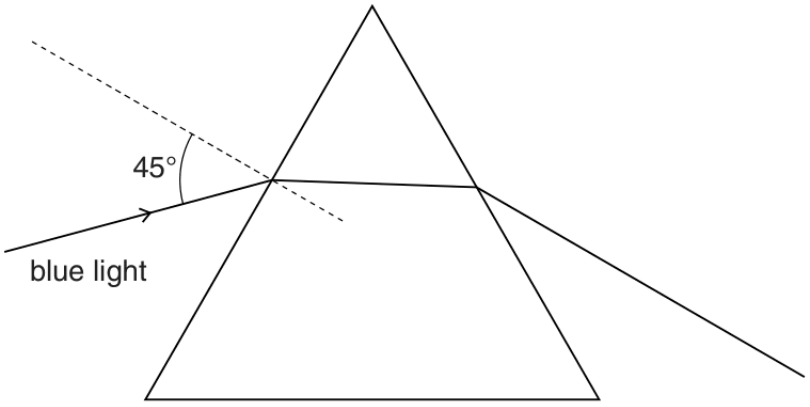


Fig. 10.1

The blue light enters the prism with an angle of incidence of 45°.

(i) On Fig. 10.1, mark the angle of refraction in the glass and label it *r*. [1]

(ii) The student wishes to determine the angle of refraction. On the sheet of paper, she draws the path the ray takes in the glass. Describe how she could do this.

.....

.....

.....[2]

(iii) **1.** State the formula that relates the angle of incidence *i*, the angle of refraction *r* and the refractive index *n*.

.....

.....[1]

2. The angle of refraction in the glass is 28°.

Calculate the refractive index of the glass.

refractive index = ...[1]

(iv) The student sends a ray of red light to the glass prism along the same path as the blue light.

On Fig. 10.1, mark the path taken by the red light after it enters the prism. [2]

MARK SCHEME · 5054_w13_ms_21.pdf · source page 4 of 6 · Question 10

10	**(a)**	**(i)**	speed of sound is (much) less than the speed of light (accept quoted values)	B1	
		(ii)	**measure** the time delay (between the lightning and thunder)	B1	
			divide distance by time/delay	B1	[3]

MARK SCHEME · 5054_w13_ms_21.pdf · source page 5 of 6 · Question 10

	(b)	**(i)**	3.0×10^8 m / s	B1	
		(ii)	(λ =) c/f **or** $3.0 \times 10^8 / 7.5 \times 10^{14}$	C1	
			4.0×10^{-7} m	A1	
		(iii)	(in any order) blue, green, orange, red, yellow, (indigo), (violet) **or** VIBGYOR	C1	
			violet, indigo, blue, green, yellow, orange, red	A1	[5]
	(c)	**(i)**	correct angle clear/labelled r	B1	
		(ii)	mark/determine entrance and exit points (e.g. trace rays back to glass)	B1	
			join/draw line between entrance and exit points	B1	
		(iii)	**1.** n = sin i/sin r	B1	
			2. 1.5/1.51/1.506176 with no unit (**not** just 1.5 without working out)	B1	
		(iv)	correct direction of refraction at **both** faces	M1	
			completely correct (above blue)	A1	[7]

[Total: 15]

5 Visible light, radio waves, X-rays, gamma rays and microwaves are some of the components of the electromagnetic spectrum.

(a) State two other components of the electromagnetic spectrum.

1. .....

2. .....
[1]

(b) White light is a mixture of different colours.

Fig. 5.1 shows a ray of white light entering a glass prism.

The white light separates into a number of colours. Only the blue light and the red light are shown.

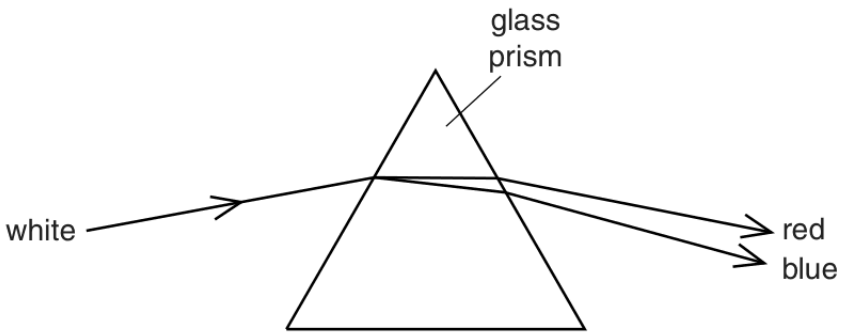


Fig. 5.1

Explain why the blue light and the red light separate as shown.

Use the term *refractive index* in your answer.

.....

.....

.....

.....

..... [3]

5 **(a)** ultra violet **and** infra-red [B1]

(b) blue refracts / bends / deviates more [B1]
blue slows more (than red when entering glass) or blue and red have different speeds (from each other in glass) [B1]
blue and red have different refractive indices [B1]

[4]

QUESTION · 5054_w15_qp_11.pdf · source page 8 of 16

20 Which statement about red light and blue light is correct?

A Red light has a higher frequency than blue light.

B Red light has a longer wavelength than blue light.

C Red light has the same speed in glass as blue light.

D Red light is refracted by a glass prism more than blue light.

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 20

20	**B**

QUESTION · 5054_w15_qp_12.pdf · source page 10 of 16

24 Which statement about red light and blue light is correct?

A Red light has a higher frequency than blue light.

B Red light has a longer wavelength than blue light.

C Red light has the same speed in glass as blue light.

D Red light is refracted by a glass prism more than blue light.

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 24

24	**B**

QUESTION · 5054_w16_qp_11.pdf · source page 12 of 20

27 Blue and yellow are colours in the visible spectrum.

Which colour has the lower frequency and which colour has the longer wavelength?

	lower frequency	longer wavelength
A	blue	blue
B	blue	yellow
C	yellow	blue
D	yellow	yellow

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 27

27 **D**

10 A laser produces a beam of red light.

(a) The red light from the laser has a frequency of 4.3×10^{14} Hz.

(i) State the speed of light in air.

.....[1]

(ii) Calculate the wavelength of this red light in air.

wavelength = ...[2]

(b) Red light from the laser strikes one side of a glass prism at an angle of incidence *i*. The light refracts towards the normal as it enters the prism.

Fig. 10.1 shows the prism, the light and a screen.

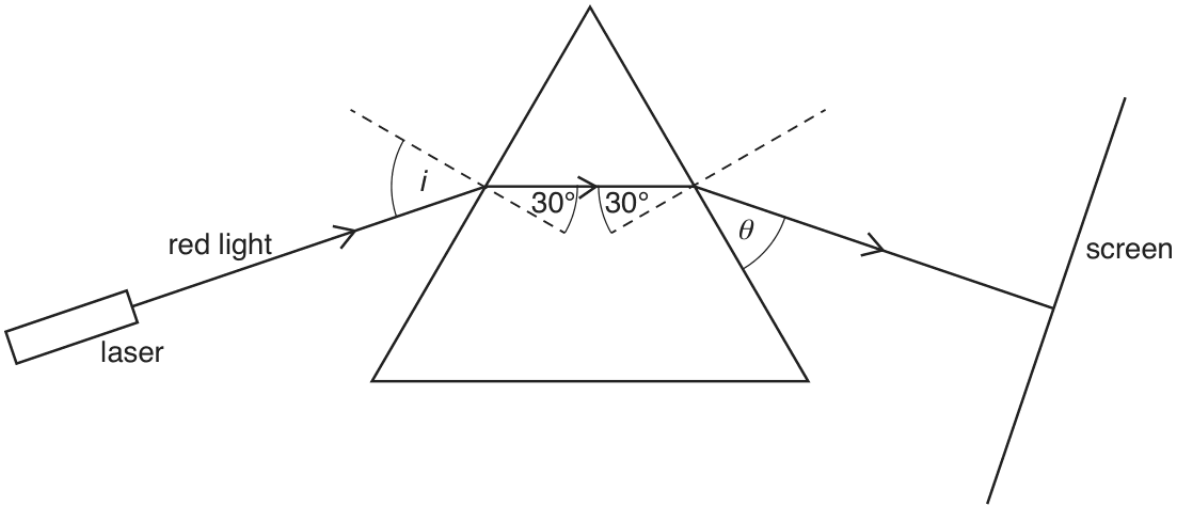


Fig. 10.1 (not to scale)

(i) State what happens to the speed of light as it enters the glass.

.....[1]

(ii) The refractive index of the glass is 1.5. The angle of refraction in the glass, where the light enters the prism, is 30°.

Calculate the angle of incidence *i*.

i = ...[2]

(iii) The light then passes back into the air and strikes the front of the white screen, as shown in Fig. 10.1.

Calculate θ, the angle between the ray in air and the side of the prism.

$\theta =$ .....[1]

(c) The laser in **(b)** is replaced with a filament lamp and a slit, as shown in Fig. 10.2.

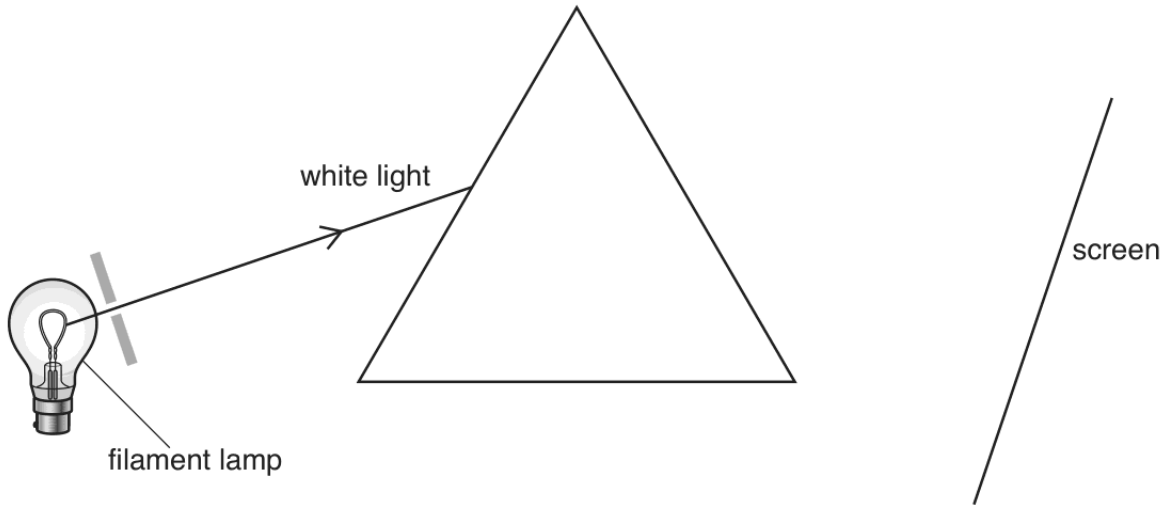


Fig. 10.2 (not to scale)

A ray of white light now strikes the prism.

(i) On Fig. 10.2, draw what happens to the light as it passes through the prism and strikes the screen. [2]

(ii) Describe what is seen on the screen.

.....

.....

.....

.....[2]

(iii) In addition to visible light, the filament lamp also emits some infra-red radiation. This infra-red radiation is able to pass through glass.

1. On the screen in Fig. 10.2, mark an X to indicate a place where infra-red radiation strikes the screen. [1]

2. Infra-red radiation is often detected by using a sensitive thermometer with a bulb that has been painted black.

Explain why the blackened bulb makes the thermometer a good detector of infra-red radiation.

.....

.....[1]

(d) Explain the role played by infra-red radiation in intruder alarms.

.....

.....

.....

.....

.....[2]

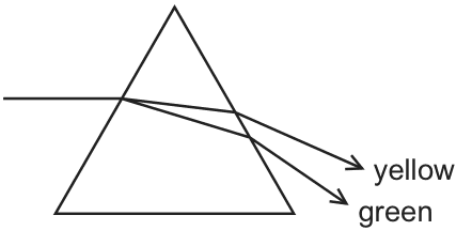
10	**(a)**	**(i)**	3.0×10^8 m / s			B1	
		(ii)	(λ =)c / f **or** $3.0 \times 10^8 / 4.3 \times 10^{14}$			C1	
			7.0×10^{-7} m			A1	[3]
	(b)	**(i)**	decreases			B1	
		(ii)	sin(i) = n × sin(r) **or** 1.5 × sin(30°) **or** 0.75			C1	
			49°			A1	
		(iii)	41°			B1	[4]
	(c)	**(i)**	dispersion at **both** surfaces **and** refractions in correct direction			B1	
			violet / blue light below the red light shown			B1	
		(ii)	spectrum **or** band of (continuous) colours **or** colours of rainbow			B1	
			red, orange, yellow, green, blue, (indigo, violet)			B1	
		(iii)	**1** X marked above red			B1	
			2 it is / black surfaces are good absorbers (of IR radiation)			B1	[6]
	(d)	intruder / human being emits IR		IR beam broken		IR reflected	B1
			or		**or**		
		intruder warm **or** IR detected		does not reach detector		change detected	B1	[2]
							[15]

QUESTION · 5054_s17_qp_12.pdf · source page 14 of 20

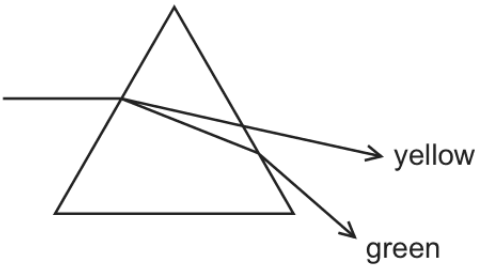
28 A narrow beam of yellow and green light is separated as it passes through a prism.

Which ray diagram is correct?

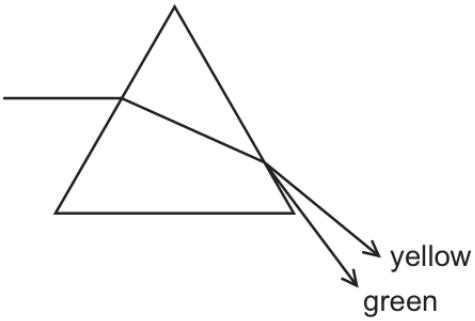
A



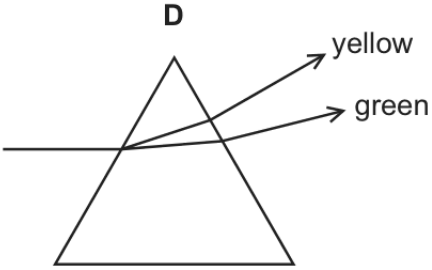
B



C



D



MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 2 of 3 · row for Question 28

28	A	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Electromagnetic radiation is produced by the Sun and travels from the Sun to the Earth.

(a) Some of the statements below are true and some of them are false.

Put a tick (✓) in the box after each statement to show whether it is true or false.

	true	false
Gamma rays are used to kill cancerous cells but can also cause cancer.	☐	☐
Infra-red is used in sun-beds.	☐	☐
Radio waves have the highest frequency in the electromagnetic spectrum.	☐	☐
The higher the frequency of the radiation the smaller is the wavelength in air.	☐	☐

[2]

(b) The glass prism in Fig. 9.1 is used to split white light from the Sun into different colours.

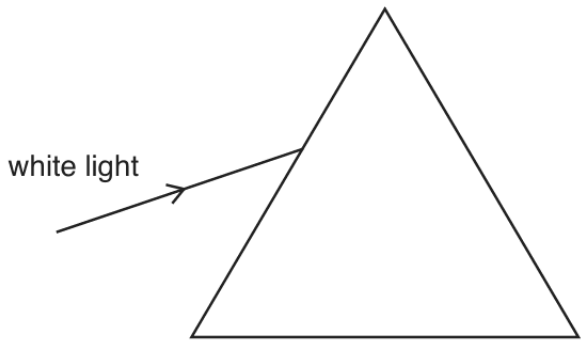


Fig. 9.1

(i) Draw, on Fig. 9.1, to show the action of the prism on the light as it passes through the prism. [3]

(ii) Explain why the prism splits white light into different colours.

.....

.....

.....

.....[2]

(c) The Earth is 1.5×10^8 km from the Sun.

(i) State the speed of light in a vacuum.

.....[1]

(ii) Calculate the time taken for light to travel from the Sun to the Earth.

time = ...[2]

(d) The Sun also produces a solar wind, which is a stream of charged particles.

In one part of space, the solar wind is due to the movement of protons and alpha-particles away from the Sun. A charge of 180 C passes through a small volume of space in 2.0 minutes.

(i) Calculate the electric current in this volume of space.

current = ...[2]

(ii) Complete the table in Fig. 9.2 for an alpha-particle, a proton and a gamma ray.

	nature	charge	stopped by
alpha-particle			
proton	a proton	positive	thin metal
gamma ray	electromagnetic radiation		

Fig. 9.2

[3]

Question	Answer	Marks
9(a)	**true false false true** all 4 correct 2 marks	**B1**
	3 correct 1 mark	**B1**
9(b)(i)	correct refraction towards normal on entering prism	**B1**
	splitting into colours occurs at first surface	**B1**
	red at top and blue / violet at bottom <u>inside</u> and outside prism if shown	**B1**
9(b)(ii)	(different colours have) different speeds in the glass **or** different refractive indices	**B1**
	different (angles of) <u>refraction</u> **or** <u>refract</u> at different angles	**B1**
9(c)(i)	$3(.0) \times 10^8$ m / s	**B1**
9(c)(ii)	(t=) d / s in any form algebraic or numerical, e.g.	**C1**
	500 s	**A1**
9(d)(i)	(I=) Q / t in any form algebraic or numerical, e.g. 180 / 2	**C1**
	1.5 A	**A1**
9(d)(ii)	1st column helium nucleus **or** 2 protons and 2 neutrons	**B1**
	2nd column positive **and** neutral / no / charge / none	**B1**
	3rd column paper / <u>thin</u> metal / skin / few cm air etc. **and** (thick) lead / thick metal / concrete / stone / earth OR never completely stopped / nothing	**B1**

QUESTION · 5054_s18_qp_12.pdf · source page 11 of 16

28 Which statement about blue light is correct?

A Blue light has a smaller frequency than red light.

B Blue light has a longer wavelength than red light.

C Blue light has the same speed in glass as red light.

D Blue light is refracted more by a glass prism than red light.

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 2 of 3 · row for Question 28

28	D	**1**

QUESTION · 5054_w18_qp_11.pdf · source page 8 of 16

20 White light is dispersed by a prism. Compared with blue light, the **red** light is

A slowed down less and refracted less.

B slowed down less and refracted more.

C slowed down more and refracted less.

D slowed down more and refracted more.

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 20

20	A	**1**

QUESTION · 5054_w18_qp_12.pdf · source page 10 of 16

24 White light is dispersed by a prism. Compared with blue light, the **red** light is

A slowed down less and refracted less.

B slowed down less and refracted more.

C slowed down more and refracted less.

D slowed down more and refracted more.

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 24

24	A	**1**

QUESTION · 5054_s19_qp_11.pdf · source page 11 of 20

26 The following lists show colours of the spectrum.

Which list shows these colours in order of increasing frequency?

A blue, violet, red, orange, yellow, green

B green, blue, violet, red, orange, yellow

C red, orange, violet, yellow, green, blue

D red, orange, yellow, green, blue, violet

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 26

26	D	**1**

10 Visible light is one component of the electromagnetic spectrum.

(a) State the speed of light in a vacuum.

..... [1]

(b) State the colours of the visible spectrum.

..... [1]

(c) An object O of height 3.0 cm is placed 4.0 cm from the centre of a diverging lens.

Fig. 10.1 shows the object O, the diverging lens and the two focal points (principal focuses), F_1 and F_2, of the lens.

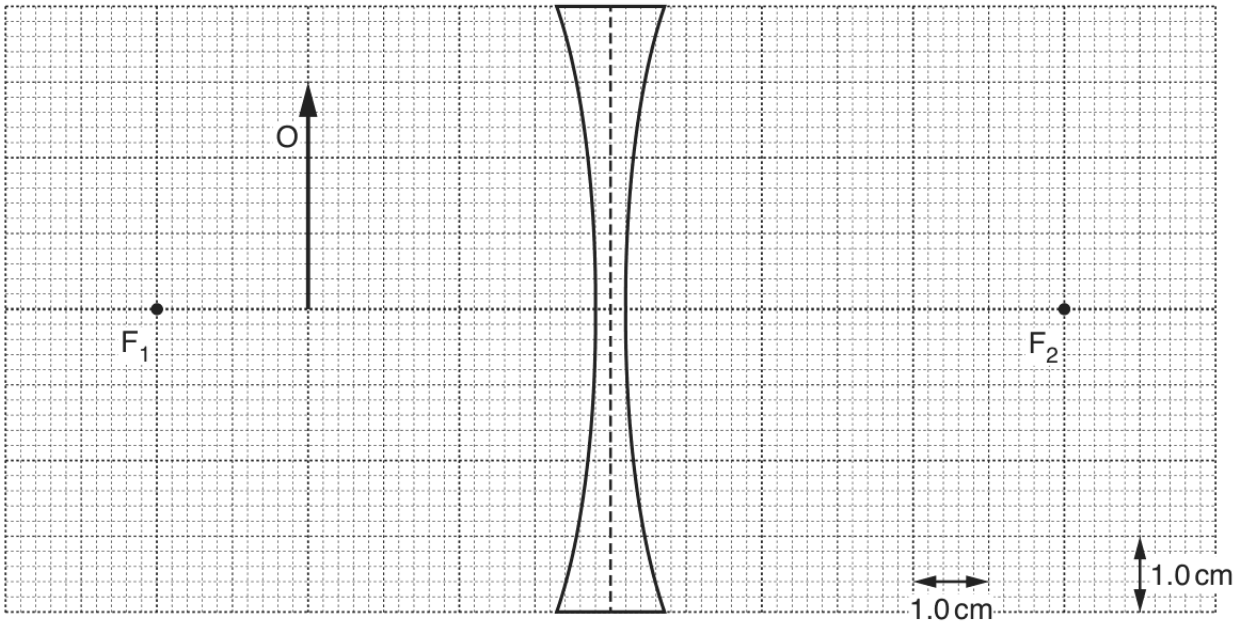


Fig. 10.1

(i) Determine the focal length of this lens.

focal length = ... [1]

(ii) The diverging lens produces an image of object O.

On Fig. 10.1, by drawing rays from the tip of O, locate this image and label it I. [4]

(iii) Determine the height of the image.

height = ... [1]

(iv) Determine the magnification of the image.

magnification = ... [1]

(v) State **two** ways in which a virtual image differs from a real image.

1. .....

.....

2. .....

..... [2]

(vi) State one use for a diverging lens.

.....

..... [1]

(d) Light passing from air into glass refracts in a similar way to a water wave passing from deep water into shallow water.

Fig. 10.2 represents light passing from air into glass at an angle to the surface.

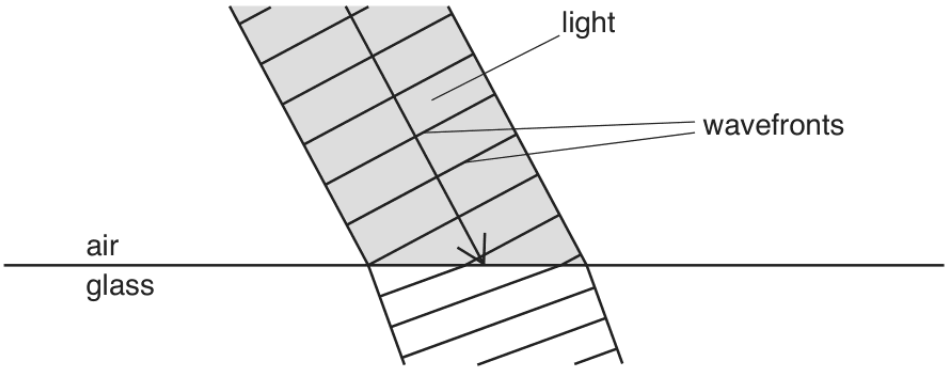


Fig. 10.2

One side of a wavefront strikes the glass before the other side.

Explain why the wavefronts change direction as the light enters the glass.

.....

.....

.....

..... [3]

[Total: 15]

MARK SCHEME · 5054_w19_ms_21.pdf · source page 9 of 10 · Question 10

Question	Answer	Marks
10(a)	3.0×10^8 m / s	**B1**
10(b)	red orange yellow green blue indigo violet (any order)	**B1**
10(c)(i)	(–)6.(0) cm	**B1**

MARK SCHEME · 5054_w19_ms_21.pdf · source page 10 of 10 · Question 10

Question	Answer	Marks
10(c)(ii)	any **two** rays drawn from: paraxial ray that refracts and seems to come from F_1 ray through the optical centre of lens ray that aims for F_2 but refracts and emerges paraxially	**B2**
	rays traced back to point	**B1**
	(point) labelled I **and** rest of image drawn down to the principal axis	**B1**
10(c)(iii)	1.7–1.9 cm	**B1**
10(c)(iv)	candidate's **10(c)(iii)** / 3.0 evaluated	**B1**
10(c)(v)	any **two** from: a real image can be projected on to a screen light actually passes through a real image on same side (of lens) as object **or** on opposite side of mirror to object	**B2**
10(c)(vi)	correction of short-sight / myopia	**B1**
10(d)	light travels more slowly in glass **or** light changes speed	**B1**
	one side / left-hand side of wavefront slows down first	**B1**
	wavelength decreases **or** wavefront travels a shorter distance in the same time	**B1**

QUESTION · 5054_s20_qp_11.pdf · source page 10 of 16

29 What is a feature of red light compared with that of violet light?

A A prism deviates red light more.

B Red light has a lower frequency.

C Red light has a shorter wavelength.

D The speed of red light in a vacuum is smaller.

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 3 of 3 · row for Question 29

29	B	**1**

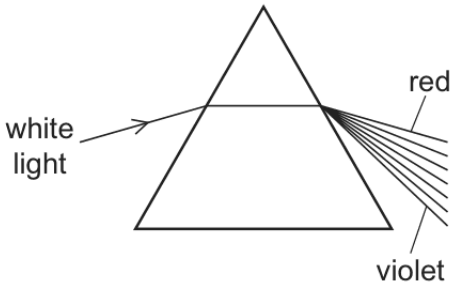
QUESTION · 5054_s21_qp_11.pdf · source page 10 of 16

24 White light enters a prism and forms a spectrum.

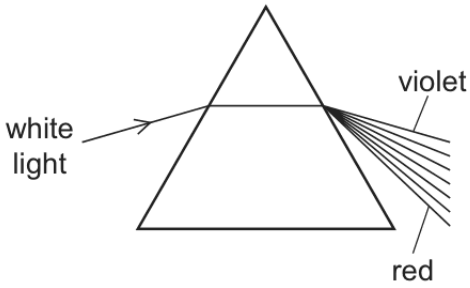
The rays in the air are labelled.

Which diagram shows how the white light is dispersed by the prism?

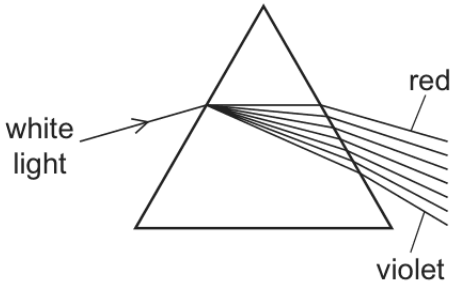
A



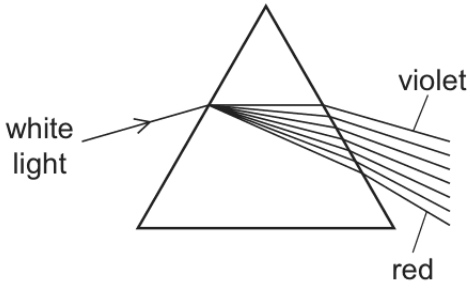
B



C



D



MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 24

24	C	**1**

10 **(a)** Fig. 10.1 shows a water wave moving from shallow into deep water. The wavefronts shown represent the crests of the wave.

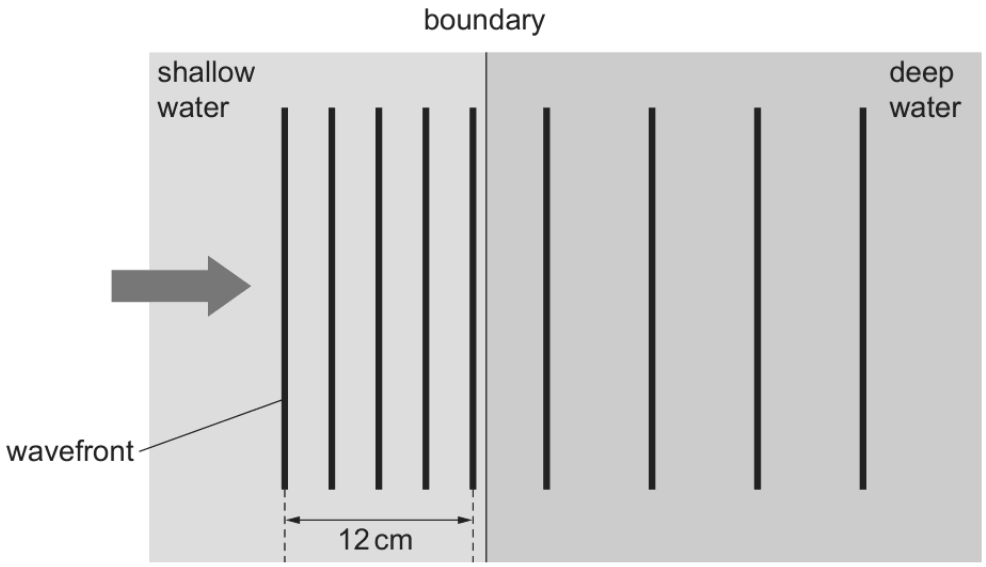


Fig. 10.1 (not to scale)

The water wave is made by dipping a wooden bar up and down in the water. The bar makes 10 complete up and down movements in 5.0 s.

(i) Determine the wavelength of the wave in the shallow water.

wavelength = ... [1]

(ii) Calculate the frequency of the wave produced.

frequency = ... [2]

(iii) Determine the speed of the water wave in the shallow water.

speed = ... [2]

(iv) On Fig. 10.2, draw lines to show what happens to the wavelength, the frequency and the speed of the water wave as it enters the deep water.

One line has been drawn for you.

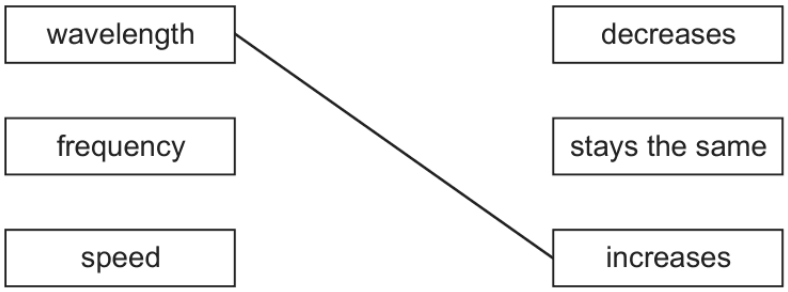


Fig. 10.2

[1]

(v) The direction of the boundary is now altered. The water wave now enters the deep water from the shallow water at an angle, as shown in Fig. 10.3.

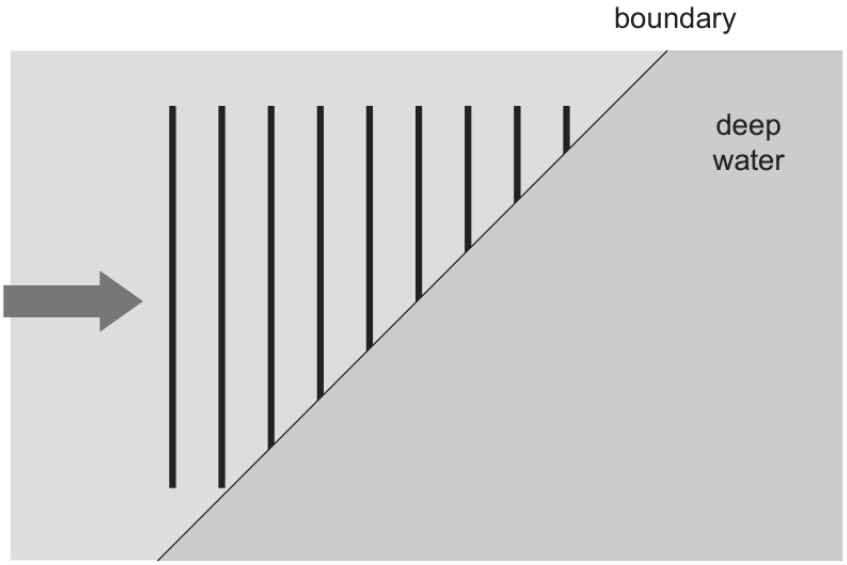


Fig. 10.3

On Fig. 10.3, draw the wavefronts in the deep water. [3]

(b) White light is made up of different colours.

(i) State the name of four of the colours in the visible spectrum and place them in order from the smallest wavelength to the largest wavelength.

smallest wavelength .....

.....

.....

largest wavelength .....

[2]

(ii) A narrow beam of white light can be split into different colours.

Fig. 10.4 shows rays of white light emitted from a lamp.

Complete Fig. 10.4 to show how a narrow beam is produced from these rays and how a spectrum is shown on the screen. Label your diagram.

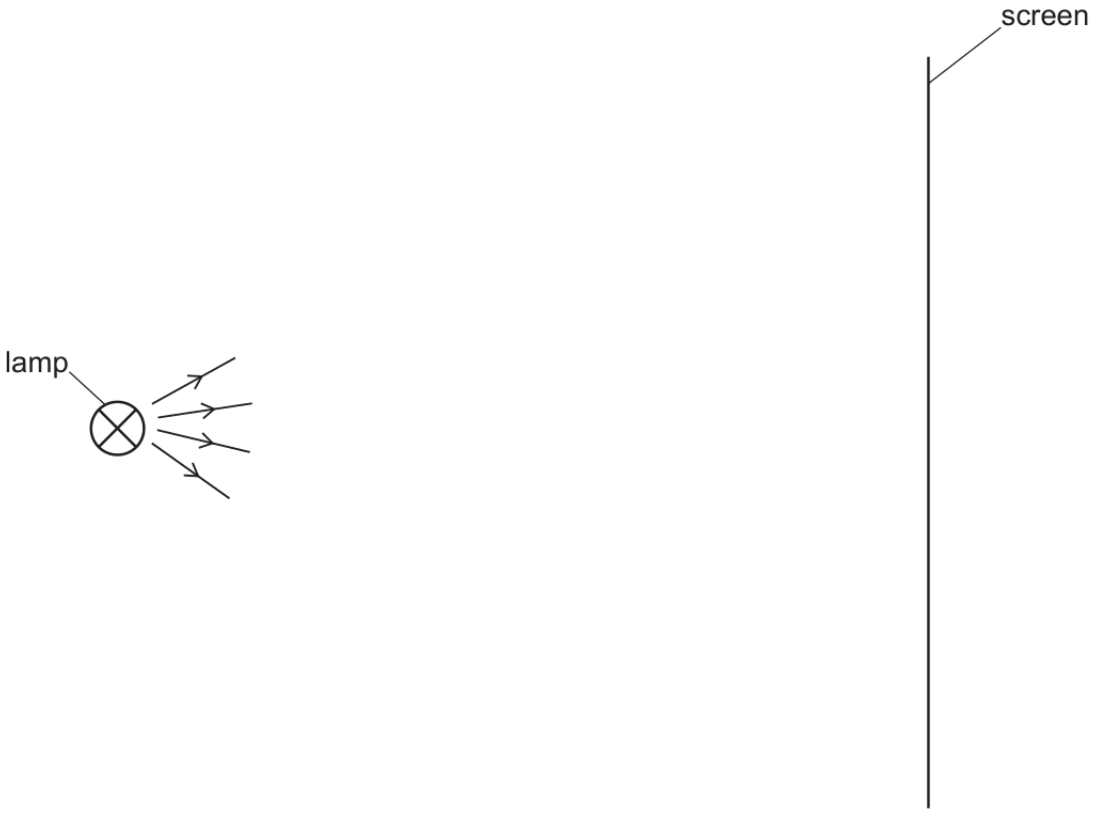


Fig. 10.4

[4]

[Total: 15]

MARK SCHEME · 5054_s21_ms_22.pdf · source page 14 of 16 · Question 10

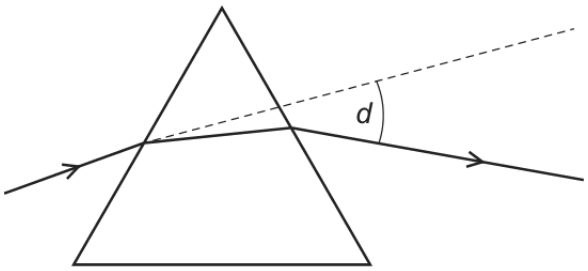
Question	Answer	Marks
10(a)(i)	3 (.0) cm	**B1**
10(a)(ii)	number of waves per second **or** 5 / 10 **or** $f = 1 / T$	**C1**
	2(.0) Hz	**A1**
10(a)(iii)	($v =$) $f \lambda$ **or** s = d / t numerical or algebraic in any form	**C1**
	6(.0) cm / s	**A1**
10(a)(iv)	frequency stays the same **and** speed increases	**B1**

MARK SCHEME · 5054_s21_ms_22.pdf · source page 15 of 16 · Question 10

Question	Answer	Marks
10(a)(v)	⩾ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in-between	**B1**
	constant wavelength shown between all refracted wavefronts	**B1**
	all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal	**B1**
10(b)(i)	four colours from violet, indigo, blue, green, yellow, orange, red	**B1**
	correct order of four of the above colours	**B1**
10(b)(ii)	prism	**B1**
	slit (to produce narrow beam)	**B1**
	dispersion shown at first or second face, i.e. a ray becoming at least two rays at least one face	**B1**
	correct refraction on first and second faces for all rays shown	**B1**

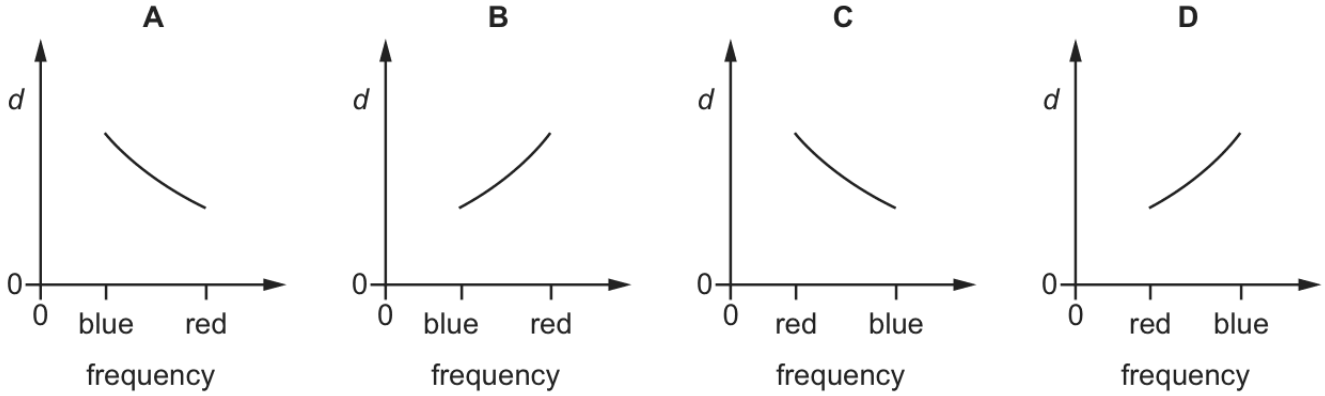
QUESTION · 5054_w21_qp_11.pdf · source page 11 of 20

23 Light rays are deviated by a prism.



The deviation angle *d* is measured for light rays of different frequency, including blue light and red light.

Which graph of *d* against frequency is correct?

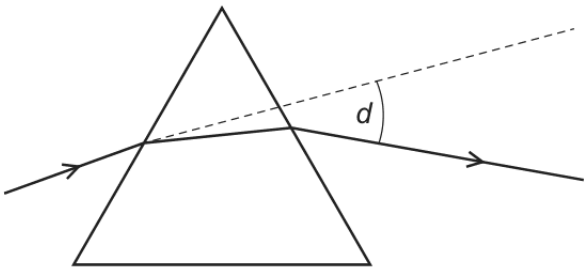


MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 23

23	D	**1**

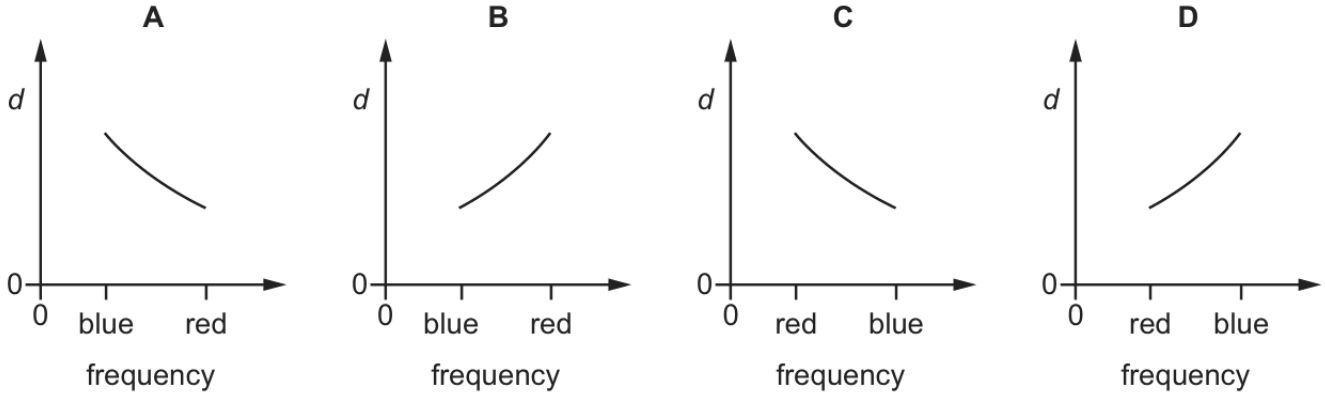
QUESTION · 5054_w21_qp_12.pdf · source page 11 of 16

26 Light rays are deviated by a prism.



The deviation angle *d* is measured for light rays of different frequency, including blue light and red light.

Which graph of *d* against frequency is correct?

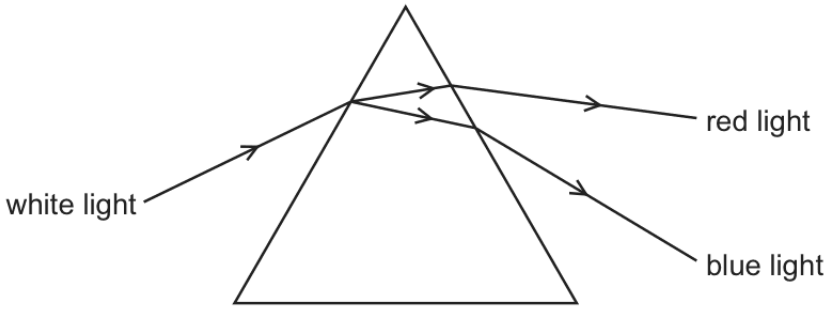


MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 26

26	D	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 9 of 16

22 A ray of white light enters a prism as shown.



Which row is correct?

	wave properties observed	frequency of red light compared with frequency of blue light
A	dispersion only	smaller
B	refraction only	greater
C	dispersion and refraction	smaller
D	dispersion and refraction	greater

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 22

22	C	**1**

4 Two parallel rays of light, one red and one blue, enter a glass prism.

Fig. 4.1 shows both rays of light before they enter the prism. The blue ray is also shown incident on a different side of the prism after passing through the prism.

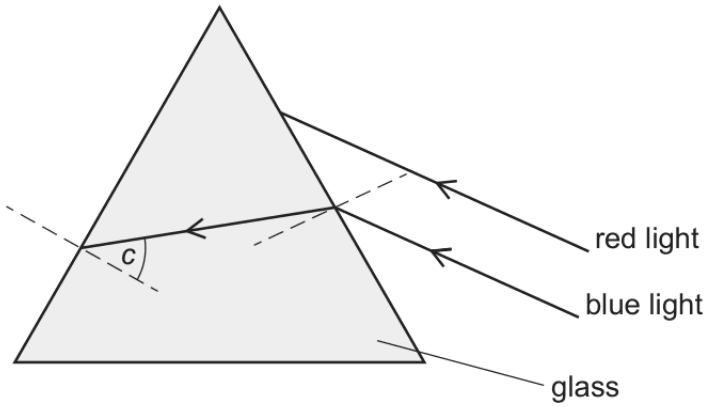


Fig. 4.1 (not to scale)

The ray of blue light strikes the left side of the prism at an angle equal to its critical angle *c*.

(a) (i) On Fig. 4.1, mark and label the angle of incidence *i* and the angle of refraction *r* for the blue light as it enters the prism. [1]

(ii) On Fig. 4.1, continue the path of the blue light after it strikes the left side of the prism. [1]

(iii) The refractive index of glass for red light is smaller than the refractive index for blue light.

On Fig. 4.1, draw the path of the red light as it travels in the prism and after it strikes the left side of the prism. [2]

(b) (i) State what is meant by the critical angle.

.....

..... [2]

(ii) The refractive index of glass for blue light is 1.5.

Calculate the critical angle *c* for blue light in glass. Show your working.

c = ... [2]

[Total: 8]

MARK SCHEME · 5054_s22_ms_21.pdf · source page 6 of 11 · Question 4

Question	Answer	Marks
4(a)(i)	correct *i* and *r* marked	**B1**
4(a)(ii)	reflection shown at approximately the correct angle **or** ray along surface	**B1**
4(a)(iii)	correct refraction as red light enters prism showing less refraction than blue light	**B1**
	correct refraction as red light emerges from left side of prism	**B1**

MARK SCHEME · 5054_s22_ms_21.pdf · source page 7 of 11 · Question 4

Question	Answer	Marks
4(b)(i)	angle of incidence	**B1**
	smallest angle when ray totally internally reflects **or** (when ray is) just internally reflected **or** when ray (refracted/emerges) along surface / at 90° to normal / angle of refraction is 90° **OR** angle of refraction when the angle of incidence is 90° B2	**B1**
4(b)(ii)	$\sin c = 1 / n$ in any form numerical or algebraic	**B1**
	42°	**B1**

QUESTION · 5054_s22_qp_12.pdf · source page 12 of 20

26 Violet and indigo light have the shortest wavelengths in the spectrum of visible light.

Which three colours, in order of increasing wavelength, immediately follow indigo?

A blue → green → orange

B blue → green → yellow

C green → blue → yellow

D yellow → green → orange

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 26

26	B	**1**

4 Fig. 4.1 shows a ray of white light incident on a glass prism.

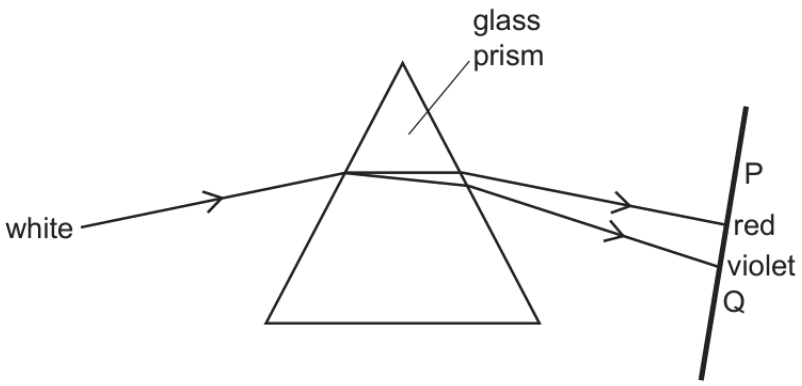


Fig. 4.1 (not to scale)

Refraction causes the white light to separate into different colours.

(a) Define the term ‘angle of refraction’.

.....

..... [2]

(b) The angle of incidence of the white light as it enters the prism is 40° and the angle of refraction for the red light is 25°.

Calculate the refractive index of the glass for red light. Show your working.

refractive index = ... [2]

(c) Using Fig. 4.1, state and explain how the refractive index for red light differs from the refractive index for violet light.

.....

.....

.....

..... [2]

(d) The source of white light used in Fig. 4.1 produces other types of electromagnetic radiation as well as visible light.

State the name of the invisible radiation found at P and the invisible radiation found at Q.

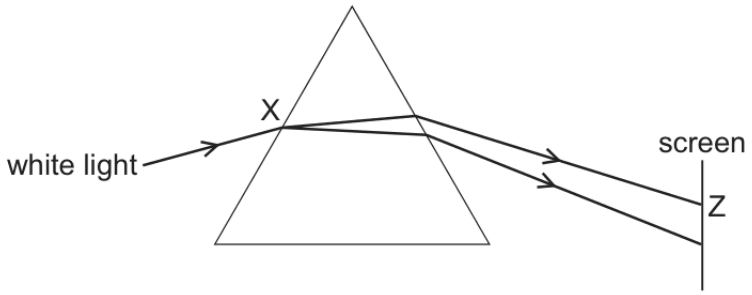
at P ..... at Q ..... [1]

[Total: 7]

Question	Answer	Marks
4(a)	clear angle is to / from the normal **or** 90°– angle between ray and surface	**C1**
	angle between/with the refracted ray / beam and the normal or vv **or** angle from the normal to the refracted ray / beam or vv	**A1**
4(b)	(refractive index =) sin*i* / sin*r* algebraic or numerical in any form	**M1**
	1.5	**A1**
4(c)	refractive index smaller (for red)	**B1**
	larger angle of refraction (for red) **or** (red) bends / refracts less (towards normal) assume comparison is for light **entering** glass at equal incident angles unless clear otherwise	**B1**
4(d)	P infrared **and Q** ultraviolet	**B1**

QUESTION · 5054_s23_qp_11.pdf · source page 10 of 20

22 The diagram shows white light entering a prism.



What happens at point X and what is the colour of the light that strikes the screen at point Z?

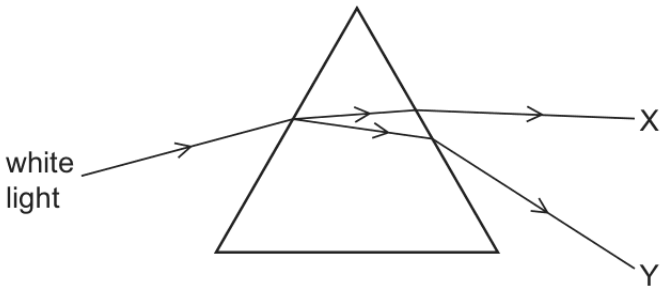
	at point X	at point Z
A	dispersion only	blue light
B	dispersion and refraction	blue light
C	dispersion and refraction	red light
D	refraction only	red light

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 22

22	C	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 10 of 16

26 A beam of white light is incident on a glass prism.



X and Y are the two ends of the visible spectrum.

Which statement is correct?

A All the light travels at the same speed in the prism.

B The light at X has a lower frequency than the light at Y.

C The light at X refracts more as it leaves the prism than the light at Y.

D The white light refracts away from the normal as it enters the prism.

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 26

26	B	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 9 of 16

25 The colour of visible light is related to the wavelength of the light.

Which list of colours is in order of increasing wavelength?

A blue → green → yellow → red

B blue → green → red → yellow

C green → red → yellow → blue

D red → yellow → green → blue

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 25

25	A	**1**

4 A student directs a beam of white light from a filament lamp towards a glass prism in a dark room.

Fig. 4.1 shows the beam of white light incident on the left-hand side of the glass prism.

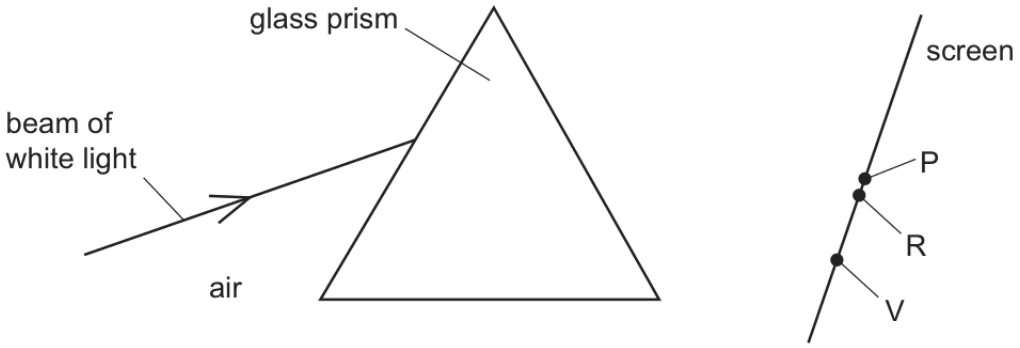


Fig. 4.1

A screen is placed to the right of the glass prism. Red light is observed at point R on the screen and violet light is observed at point V.

(a) Draw on Fig. 4.1 to show the paths of the red light and the violet light between the left-hand side of the prism and the screen at point R and point V. [2]

(b) State what these observations show about the change in speed of red light and change in speed of violet light as each enters the glass from the air.

.....

.....

..... [2]

(c) Five other colours of light appear on the screen between the red light at point R and the violet light at point V.

(i) State the names of the five colours and list them in the correct order from red to violet.

red ..... **violet** [2]

(ii) State how the frequency and the wavelength of the coloured light change going from red light to violet light.

frequency .....

wavelength .....
[1]

(d) On Fig. 4.1, point P is shown immediately next to point R.

No light is observed at point P, but a detector at point P registers radiation reaching the detector.

State which region of the electromagnetic spectrum is detected at point P.

..... [1]

[Total: 8]

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MARK SCHEME · 5054_w24_ms_21.pdf · source page 6 of 10 · Question 4

Question	Answer	Marks
4(a)	at least **one** ray from left-hand side showing correct refraction at both faces **and** no incorrect rays	**B1**
	at least **two** correct rays (i.e. two rays within glass) **and** one ending at R and the other at V	**B1**
4(b)	both (red and violet light) travel more slowly in glass than in air	**B1**
	red light travels more quickly (than violet light) in glass	**B1**
4(c)(i)	any **four** (of the five) colours in order **or** five colours in any order	**C1**
	red orange yellow green blue indigo violet	**B1**

MARK SCHEME · 5054_w24_ms_21.pdf · source page 7 of 10 · Question 4

Question	Answer	Marks
4(c)(ii)	frequency increases **and** wavelength decreases	**B1**
4(d)	infrared (radiation)	**B1**

QUESTION · 5054_s25_qp_12.pdf · source page 10 of 16

24 Which statement about red light and violet light is correct?

A A prism deviates red light more than it deviates violet light.

B Red light has a lower frequency than violet light.

C Red light has a shorter wavelength than violet light.

D The speed of red light in a vacuum is less than the speed of violet light in a vacuum.

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 24

24	B	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 10 of 16

20 The following lists show colours of the spectrum.

Which list shows these colours in order of increasing frequency?

A blue → violet → red → orange → yellow → green

B green → blue → violet → red → orange → yellow

C red → orange → violet → yellow → green → blue

D red → orange → yellow → green → blue → violet

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 20

20	D	**1**

Chapter 3 Waves

3.3 Electromagnetic spectrum

99 entries: 53 Paper 1 questions, 46 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

Section B

Answer **two** questions from this section.

Use the separate answer paper provided.

9 Fig. 9.1 shows the main parts of an electric grill. An electric current in the heating element causes it to become red hot and to emit visible light and infra-red radiation. The infra-red radiation is used to cook food.

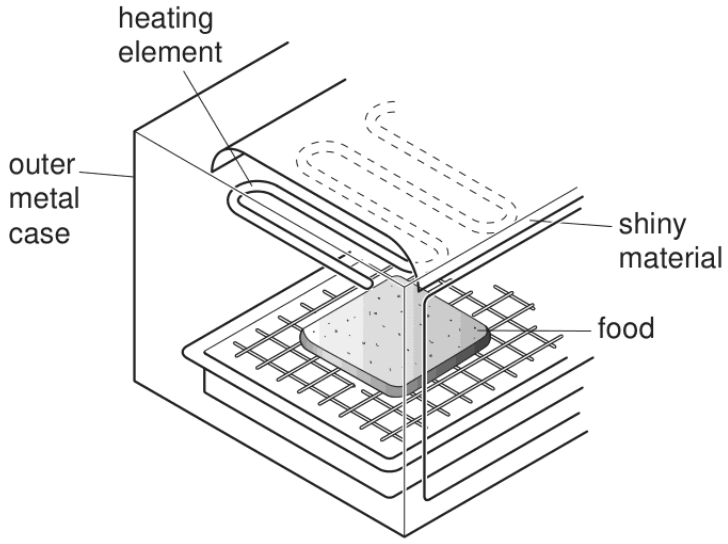


Fig. 9.1

(a) Infra-red radiation has a longer wavelength than visible light. Both are parts of the electromagnetic spectrum.

(i) Draw a diagram of the electromagnetic spectrum that shows visible light, infra-red and **three** other parts of the spectrum. Your diagram should put the parts of the spectrum in order of increasing wavelength.

(ii) Explain why the grill has a layer of shiny material between the heating element and the outer case.

[7]

(b) For safety, the electric grill is fitted with an earth wire. State where the earth wire is connected to the electric grill and explain how it acts as a safety device. [4]

(c) The current in the heating element is 8.3 A when it is connected to the 230 V mains supply.

(i) State the equation that links current, power and voltage.

(ii) Calculate the power rating of the heating element when it is working correctly. Give your answer to an appropriate number of significant figures.

(iii) A student thinks that if the heating element is connected to a mains supply of 115 V then the power produced is halved. Explain why the student is wrong.

[4]

SECTION B

9	**(a)**	**(i)**	Any three other parts of spectrum radio, microwaves, u.v., X, γ (-1 any wrong if>3 **ignore** t.v.)		M3
			correct order for all including visible (accept colours) and I.R.		A1
		(ii)	reflection **of infra-red or radiation** (from shiny material) more energy hits food **or** reflection towards food cooks food faster avoids wasting heat / energy **or** more efficient avoids heating outer case or burning hand	ANY 2	B1 B2
	(b)		connected to (outer metal) case		B1
			if live touches case **or** case becomes live		B1
			allows current / charge to earth / ground blows fuse (and disconnects circuit)		B1
			or no current through person **or** no electrocution / electric shock		B1
	(c)	**(i)**	P = V I in any algebraic form		B1
		(ii)	230 * 8.3		C1
			1900 W (accept 1910 W but **not** power 1/4)		A1
		(iii)	current decreases (halves) **or** power 1/4		B1
				Total	**[15]**

QUESTION · 5054_s05_qp_1.pdf · source page 9 of 20

23 Which statement is true for all electromagnetic waves?

A They are longitudinal.

B They can be seen.

C They have the same frequency in air.

D They travel at the same speed in a vacuum.

MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 23

23	**D**

For Examiner's Use

4 X-rays, microwaves, ultra-violet rays and infra-red rays are different types of radiation in the electromagnetic spectrum.

(a) Write the name of one of these types of radiation in each of the boxes, placing them in order of increasing wavelength.

shortest wavelength ⟶ longest wavelength

[2]

(b) State one use of ultra-violet radiation.

.....

.....

..... [1]

(c) State two properties that are common to all types of radiation in the electromagnetic spectrum.

1. .....

.....

2. .....

..... [2]

4	**(a)**	X-rays, ultra-violet, infra-red, microwaves in each box allow one mark if moving one box gives correct order	**B2**	
	(b)	sun-beds (accept tanning), fluorescent tubes, sterilisation, illuminating marks on property (phosphors) **not** just marking property	**B1**	
	(c)	transverse, same speed, will diffract, reflect, refract etc. (allow only 1) travel in a vacuum (accept need no medium) any 2	**B2**	**5**

QUESTION · 5054_s06_qp_1.pdf · source page 9 of 20

25 Which wave is part of the electromagnetic spectrum?

	$\frac{\text{speed}}{\text{m/s}}$	type
A	330	longitudinal
B	330	transverse
C	3×10^8	longitudinal
D	3×10^8	transverse

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 25

25 **D**

11 Stars that are being formed emit infra-red radiation. Some of this radiation is received by a telescope that orbits the Earth. Microwave signals from the telescope are sent to the Earth's surface, as shown in Fig. 11.1.

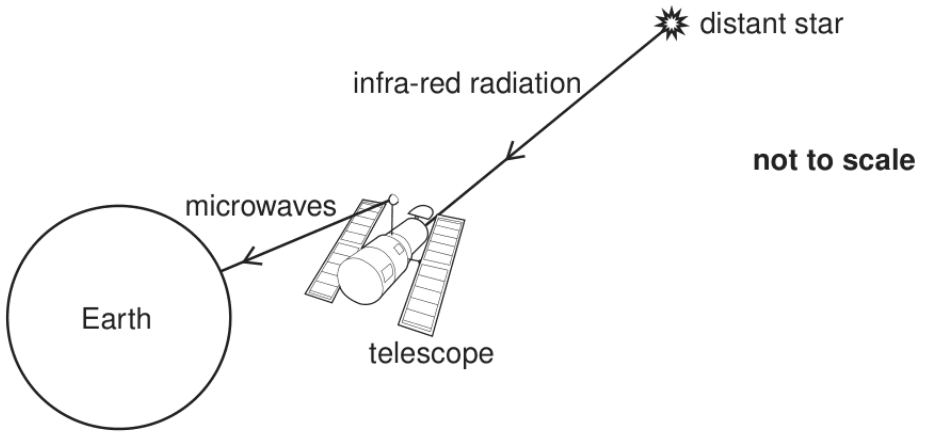


Fig. 11.1

(a) **(i)** Microwaves travel at a speed of 300 000 km/s. The telescope is 600 km above the Earth's surface. Calculate the time for a microwave signal from the telescope to reach the Earth's surface. [3]

(ii) Infra-red and microwave radiation are both part of the electromagnetic spectrum.

State **two** other similarities and give **one** difference between infra-red and microwave radiation. [3]

(b) As clouds of gas and dust come together to form a star, there is an energy change similar to the energy change as a ball falls to the ground.

The temperature in the clouds of dust and gas becomes so high that nuclear fusion occurs.

In one such reaction, two isotopes of hydrogen, ${}^{2}_{1}\mathrm{H}$ and ${}^{3}_{1}\mathrm{H}$, fuse together.

(i) Explain why the gas and dust come together. [1]

(ii) State the energy change that causes the rise in temperature as the gas and dust come together. [2]

(iii) Explain why high temperatures are needed for nuclear fusion to occur. [2]

(iv) Describe the structure of a nucleus of ${}^{3}_{1}\mathrm{H}$. [2]

(v) State which element is created by the fusion of two hydrogen nuclei. [1]

(vi) State **one** effect of nuclear fusion on a star. [1]

11 (a)	**(i)**	d = speed × time in any format		C1
		600/300 000 **or** 600 000/300 000 000		C1
		0.002 s		A1
	(ii)	**similarities:**		
		same speed (in vacuum)		
		travel in a vacuum		
		travel in straight lines		
		refract/reflect/diffract/interfere		
		carry energy		
		transverse/polarisable ANY 2 (-1 each clearly wrong answer beyond 2)		B2
		(NOT both obey c = fλ/waves/invisible/undeflected by magnetic/electric field)		
		differences:		
		wavelength		
		frequency		
		microwave received by aerials ANY 1 line		B1
		(wavelength of IR different YES; wavelength of IR longer NO)		
(b)	**(i)**	gravity		B1
	b	potential energy to kinetic energy		B1
		kinetic energy to heat/thermal energy		B1
		OR potential energy to heat/thermal energy	**OR**	B2
		-1 each clearly wrong answer beyond 2		
	(iii)	nuclei repel **or** nuclei are positive		B1
		nuclei need high speed/ K.E. (so high temperature)		B1
	(iv)	1 proton **or** proton number = 1		B1
		2 neutrons **or** neutron number = 2 (electron(s) **max 1**)		B1
	(v)	He **or** helium		B1
	(vi)	energy/heat produced **or** raises temperature **or** becomes hot **or** causes star to expand **or** counters gravitational collapse **or** loses mass		B1 15

QUESTION · 5054_s07_qp_1.pdf · source page 11 of 20

22 Which type of wave is used to send telephone signals to and from a satellite?

A infra-red waves

B light waves

C microwaves

D sound waves

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 22

22	**C**

QUESTION · 5054_w07_qp_1.pdf · source page 10 of 20

22 Which does **not** normally use infra-red radiation?

A electric grill

B intruder alarm

C television remote controller

D sunbed

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 22

22	**D**

For Examiner's Use

5 **(a)** Each object in the table below emits one main type of electromagnetic wave. Complete the table by writing in the name of the type of wave. One line has been written for you.

Object	Main type of electromagnetic wave emitted
radio transmitter	radio wave
remote control for a television	
radioactive source	

[2]

(b) X-rays are used in hospitals to produce images of bones and to show whether bones are broken.

(i) State what is used to detect X-rays.

.....

..... [1]

(ii) Explain the properties of X-rays that enable an image of a bone to be produced.

.....

.....

.....

.....

.....

..... [2]

5 **(a)** infrared B1
gamma (rays/waves) B1

(b) **(i)** fluorescent (screen), photographic (plate), CCD/semiconductor/photoelectric/GM tube B1

(ii) (X-rays) absorbed/stopped by bone **or** do not penetrate bone (**not** reflected by bone) B1
less absorption/pass **through** flesh/skin/body, etc. **or** travel in straight lines
or effect on detector, e.g. ionisation, photo black (on development), light emitted B1

[Total: 5]

For Examiner's Use

6 Fig. 6.1 shows a ray of white light from a ray-box passing into a glass prism. A spectrum is formed between P and Q on the screen.

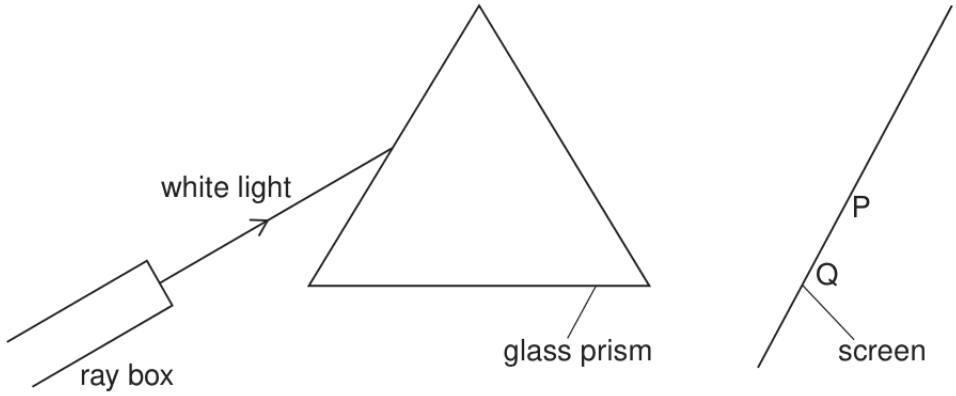


Fig. 6.1

(a) State the colour of the light at end P of the spectrum.

..... [1]

(b) State whether the value of each of these properties for blue light is greater than, equal to or less than the value for red light.

(i) speed in a vacuum ..... [1]

(ii) wavelength ..... [1]

(c) Fig. 6.2 shows the ray passing through a red filter before it reaches the prism.

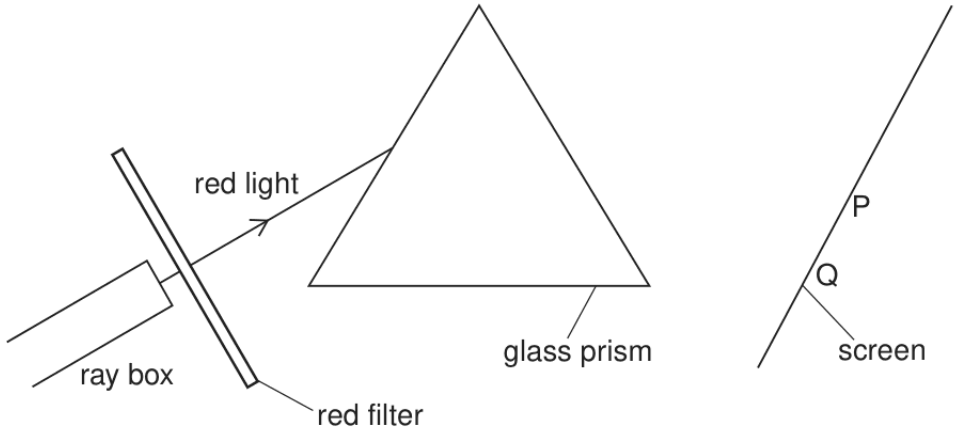


Fig. 6.2

Complete Fig. 6.2 to show the ray of red light passing through and emerging from the prism. [2]

6	**(a)**	red	B1	
	(b)	**(i)** equal to	B1	
		(ii) less than	B1	
	(c)	two correct refractions on Fig. 6.2	M1	
		no dispersion **and** ray ends close to P	A1	[5]

QUESTION · 5054_s10_qp_11.pdf · source page 10 of 20

22 A hospital needs to sterilise medical equipment.

Which electromagnetic waves could be used?

A infra-red

B microwaves

C radiowaves

D ultraviolet

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 22

22	**D**

QUESTION · 5054_s10_qp_12.pdf · source page 9 of 20

21 A hospital needs to sterilise medical equipment.

Which electromagnetic waves could be used?

A infra-red

B microwaves

C radiowaves

D ultraviolet

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 21

21	**D**

For Examiner's Use

3 Fig. 3.1 shows a metal roof. One side is facing the Sun.

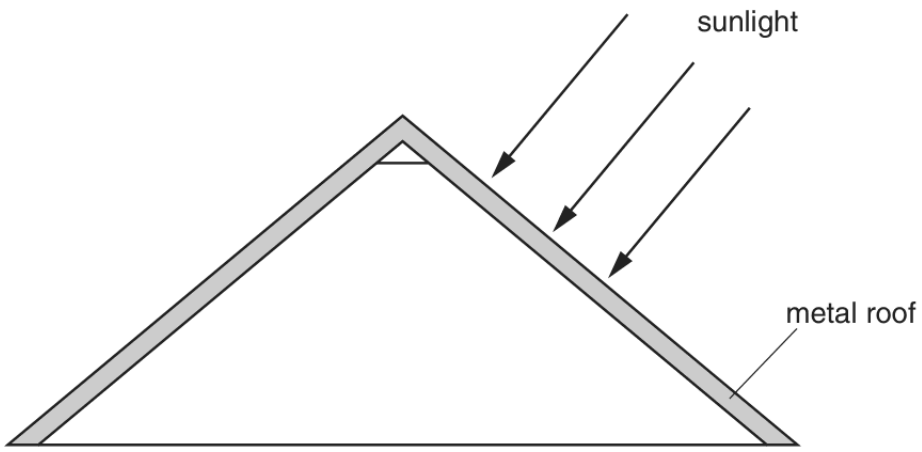


Fig. 3.1

(a) State the means by which thermal energy (heat) is transferred from the Sun to the Earth and explain why other means of thermal energy transfer are not involved.

.....

.....[2]

(b) Describe how thermal energy is transferred through the metal roof from the heated surface.

.....

.....

.....[2]

(c) During the night, the metal roof loses 1.2×10^6 J of thermal energy and its temperature falls by 20 °C.

The specific heat capacity of the metal in the roof is 400 J/(kg °C).

Calculate the mass of metal in the roof.

mass = ...[2]

3	**(a)**	radiation or infra-red or electromagnetic waves	B1	
		travels through space/vacuum or does not require medium/molecules/particles or medium required for conduction and/or convection or for other methods	B1	[2]
	(b)	conduction occurs or atoms/particles/molecules vibrate or electrons given energy	B1	
		heat/energy/vibration passed on from one particle to another or electrons move to other parts/diffuse/hit atoms	B1	[2]
	(c)	($Q =$) mcT algebraic or numerical in any form (e.g. $1.2 \times 10^6 = m \times 400 \times 20$)	C1	
		150 kg	A1	[2]

[Total: 6]

QUESTION · 5054_w10_qp_11.pdf · source page 11 of 20

24 Which pair of emissions travels with the same speed in air?

A alpha-particles and gamma-rays

B gamma-rays and infra-red waves

C infra-red waves and sound waves

D sound waves and alpha-particles

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 24

24	**B**

For Examiner's Use

4 A large number of television sets in a shop are tuned to the same station. The signal travels to some of the sets via a satellite in orbit above the Earth.

(a) (i) Explain why only electromagnetic waves are used to transmit the television signal.

.....

..... [1]

(ii) State the region of the electromagnetic spectrum used to transmit the television signal.

.....

..... [1]

(iii) Suggest why satellites are sometimes used to transmit television signals.

.....

..... [1]

(b) The other television sets receive the signal directly from a ground-based transmitter very near to the shop. A salesman notices that the signal sent via the satellite arrives 0.24 s later than the signal sent directly to the other television sets.

The speed of electromagnetic waves is 3.0×10^8 m/s.

Calculate the distance travelled by the television signal in 0.24 s.

distance = ... [2]

4	**(a)**	**(i)**	e/m waves can travel/satellite in a **vacuum/space**	B1	[1]
		(ii)	microwave/radio wave (region)	B1	[1]
		(iii)	greater coverage/less ground-based infrastructure/less obstruction	B1	[1]
	(b)		(x =) vt **or** $3.0 \times 10^8 \times 0.24$	C1	
			7.2×10^7 m **or** 72 000 km	A1	[2]

For Examiner's Use

9 **(a)** Fig. 9.1 shows a young boy lying on his back on the bottom of a swimming pool. He is holding his breath and his eyes are open. A red light is positioned on the ground at Q.

At first the boy's head is touching the pool wall. He notices that, as he slides away from the pool wall, his eye reaches a point P where he first sees the light at Q. Fig. 9.1 shows the boy in this position.

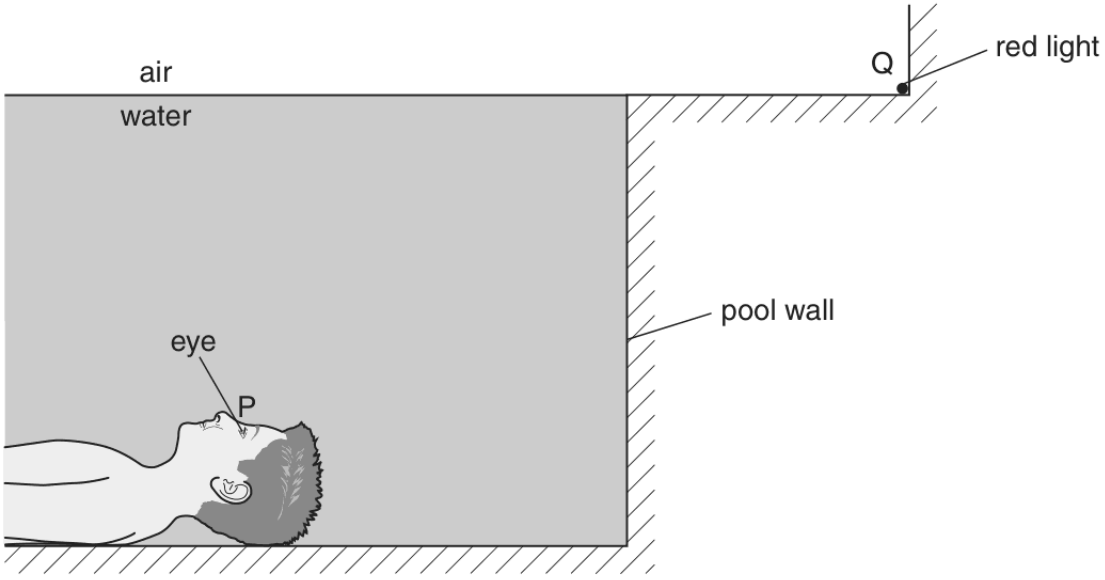


Fig. 9.1

(i) On Fig. 9.1, draw the ray of light travelling from Q to P. Mark the critical angle for light in water and label it C. [2]

(ii) Explain why the boy is unable to see the red light at Q when his eye is closer to the pool wall than P.

.....

.....

..... [2]

(iii) The critical angle is 49°. Calculate the refractive index of water.

refractive index = ... [2]

(iv) The red light is now replaced with a blue light. State the effect of this on the wavelength of the light in the air.

..... [1]

For Examiner's Use

(b) A small, very brightly illuminated display is located at the back of a projector. The projector lens produces an inverted and magnified image of the display on a white classroom wall.

Fig. 9.2 is a scale diagram showing the position and size of both the display and the image on the wall. R is a point on the display.

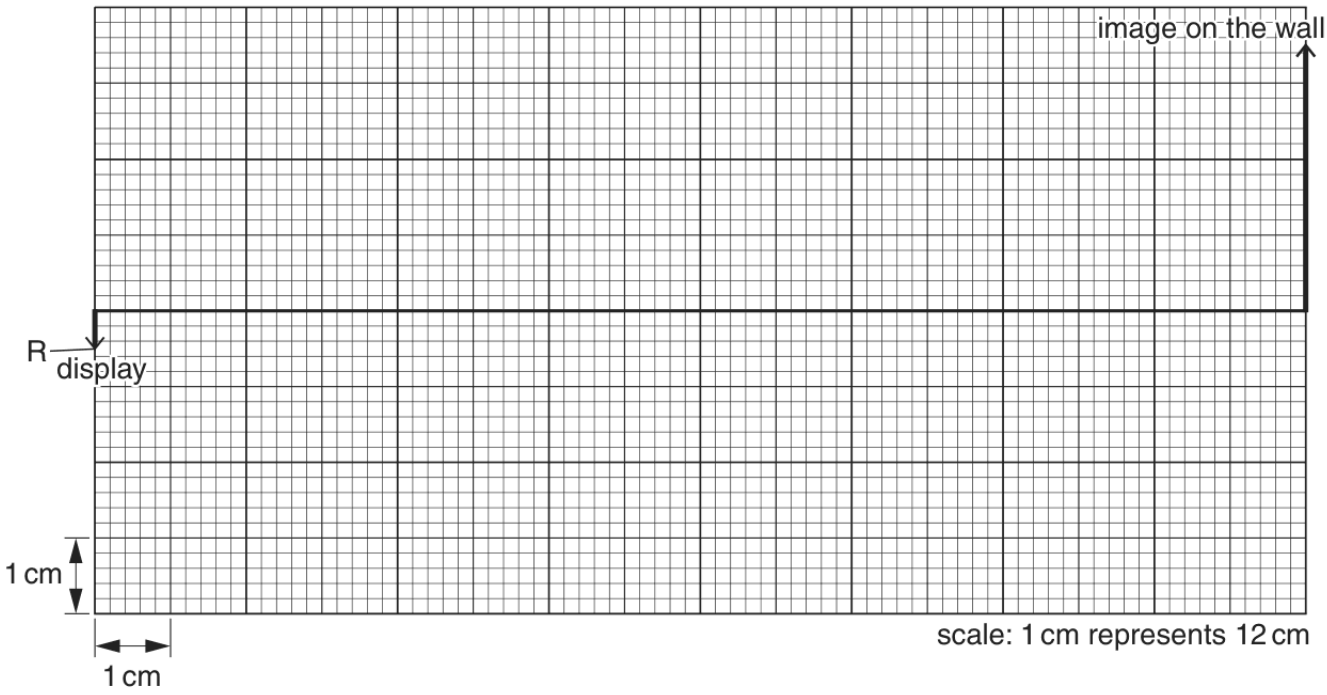


Fig. 9.2

(i) The image is inverted and magnified. State two other properties of the image.

1. .....

2. .....
[2]

(ii) On Fig. 9.2, draw the straight-line ray from R to the image. [1]

(iii) On Fig. 9.2, draw a vertical line representing the lens and label it L. [1]

(iv) A second ray from R to the image passes through a focal point (principal focus) of the lens. On Fig. 9.2, draw this ray and use it to mark this focal point. Label this focal point F. [2]

(v) Determine the focal length of the lens.

focal length = ... [2]

9	**(a)**	**(i)**	horizontal ray from Q to pool edge **and** on to P from corner	B1	
			critical angle marked *C* **or** obvious	B1	[2]
		(ii)	for $i = 90°$ **or** horizontal ray	B1	
			angle(in water) equals/cannot be less than critical/*C*	B1	[2]
		(iii)	(n =) sin *i*/sin *r* **or** 1/sin *C* **or** 1/n = sin *C* **or** sin 90°/sin 49° **or** 1/sin 49°	B1	
			1.3(2501)	B1	[2]
		(iv)	decreases	B1	[1]
	(b)	**(i)**	any **two** of: real less bright further from lens beyond 2*f*	B2	[2]
		(ii)	straight ray from R to **top** of image	B1	[1]
		(iii)	where ray crosses principal axis, vertical line (L **or** drawn lens)	B1	[1]
		(iv)	paraxial ray from R to lens refracted to top of image **or** paraxial ray from lens to top of image, traced back to R	M1	
			F marked	A1	[2]
		(v)	1.6 – 1.9 cm **or** attempt to use 1/u + 1/v	C1	
			19 – 23 cm (2 sig. fig. only)	A1	[2]

QUESTION · 5054_w10_qp_12.pdf · source page 8 of 16

22 Which pair of emissions travels with the same speed in air?

A alpha-particles and gamma-rays

B gamma-rays and infra-red waves

C infra-red waves and sound waves

D sound waves and alpha-particles

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 22

22	**B**

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a very large plane mirror, inclined at 45° to the horizontal, beneath a pattern on the high ceiling of a hall.

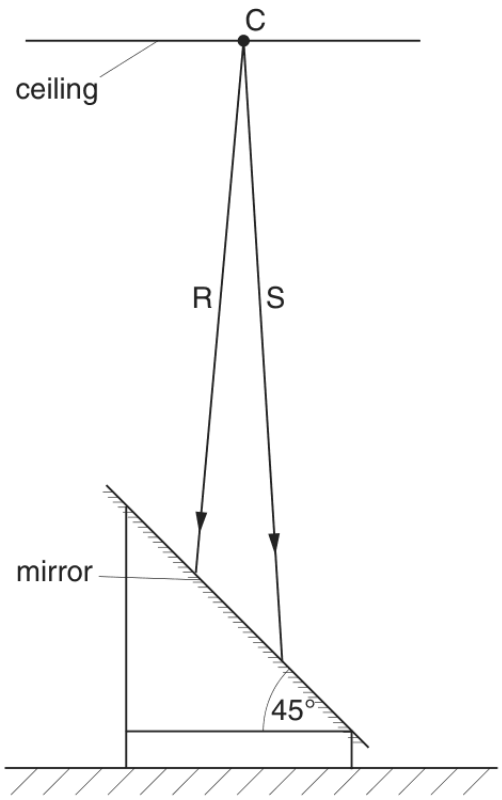


Fig. 9.1

The mirror is set on a stand at head-height immediately below the centre C of the pattern. R and S are two rays of light from C that strike the mirror.

(a) (i) On Fig. 9.1, continue the rays R and S after they strike the mirror. [1]

(ii) On Fig. 9.1, show how these rays are used to locate the image of C and mark and label the position of this image with the letter I. [2]

(iii) State two characteristics of this image.

1. .....

2. .....
[2]

(iv) Suggest how the mirror helps visitors to the hall to see the pattern on the ceiling.

.....

.....[1]

(b) Violet light from C has a wavelength of 4.0×10^{-7} m.

For Examiner's Use

(i) Calculate the frequency of this light, clearly stating the value of any constant used in the calculation.

frequency = ...[3]

(ii) State two different components of the electromagnetic spectrum that have wavelengths smaller than the wavelength of violet light.

1. .....

2. .....
[2]

(iii) **1.** Discuss a medical application of **one** of these components of the electromagnetic spectrum.

.....

.....

.....

.....

.....[3]

2. State a health risk associated with this component of the electromagnetic spectrum.

.....

.....[1]

Section B

9 (a) (i) **one** correctly reflected ray (by eye) B1

(ii) **two** reflected rays traced back to an image B1
(labelled) image in correct position (by eye) B1

(iii) any **two** of:
virtual
full size/mag = 1 **or** same distance from mirror as C
laterally inverted (**ign** upright)
dimmer B2

(iv) more comfortable/no neck strain/no need to look up/reflects to eyes B1 [6]

(b) (i) (c =) 3(.00) × 10^8 (m/s) **or** 3(.00) × 10^5 (km/s) **or** used in equation B1
(f =) c/λ **or** (3.0 × 10^8/their **stated** value/330)/4.0 × 10^{-7} C1
7.5 × 10^{14} Hz **or** correct answer from **stated** value (incl. unit)
or 8.2/8.25/8.3 × 10^8 Hz A1

(ii) any **two**:
UV(radiation); X(radiation); γ(radiation) B2

(iii) **1.**

UV absorbed by skin	psoriasis destroyed	cells multiply less rapidly
X-rays absorbed by bones/not absorbed by flesh	shadow/image of bones	on film/CCD
γ-rays emitted by absorbed isotope	position/shape of organ etc. revealed	on film/CCD
tumour/cancer absorbs X/γ-ray	tumour destroyed	photons/energy/stops cells multiplying
bacteria absorb UV/X/γ-ray	Bacteria killed	sterilisation/stops bacteria multiplying

2.

UV:	X-rays:	γ-ray:	
damages eyes/skin cancer	cancer/hair loss/ radiation sickness	cancer/hair loss/ radiation sickness	B1 [9]

QUESTION · 5054_s11_qp_11.pdf · source page 10 of 16

24 Which application may use the part of the electromagnetic spectrum called microwaves?

A cooking vegetables

B detecting small cracks in metals

C gaining a sun-tan

D lighting a fluorescent tube

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 24

24	**A**

5 The fluorescent tube shown in Fig. 5.1 converts electrical energy into light more efficiently than a filament lamp.

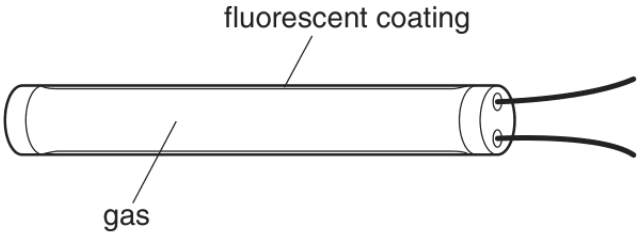


Fig. 5.1

In the gas inside the tube, both light and ultra-violet radiation are produced.

There is a fluorescent coating on the inside surface of the tube.

(a) Explain the purpose of the fluorescent coating.

.....

.....

..... [2]

(b) State the name of one region of the electromagnetic spectrum with wavelengths shorter than ultra-violet radiation.

..... [1]

(c) The ultra-violet radiation inside the tube has a wavelength of 3.6×10^{-7} m. Calculate the frequency of the ultra-violet radiation.
(speed of light = 3.0×10^{8} m/s)

frequency = ... [2]

5 **(a)** gives out (visible) light // glows B1
when hit by uv/electrons **or** spark/discharge/current in **tube** B1

(b) X-rays or gamma B1

(c) (f =) v/λ numerical or any algebraic form, e.g. $v = f\lambda$ C1
8.3×10^{14} Hz A1 [5]

QUESTION · 5054_s11_qp_12.pdf · source page 8 of 16

20 Which application may use the part of the electromagnetic spectrum called microwaves?

A cooking vegetables

B detecting small cracks in metals

C gaining a sun-tan

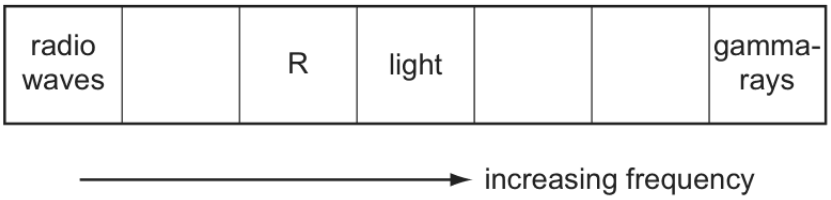
D lighting a fluorescent tube

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 20

20	**A**

QUESTION · 5054_w11_qp_11.pdf · source page 10 of 20

24 The diagram shows the main sections of the electromagnetic spectrum in order of increasing frequency. Some of the sections are labelled.



The section R has a frequency just below that of light.

Which application uses the section R?

A killing cancerous cells

B satellite television

C sterilisation

D television remote controller

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 24

24	**D**

4 Fig. 4.1 shows a glass lens in air and its two focal points F_1 and F_2.

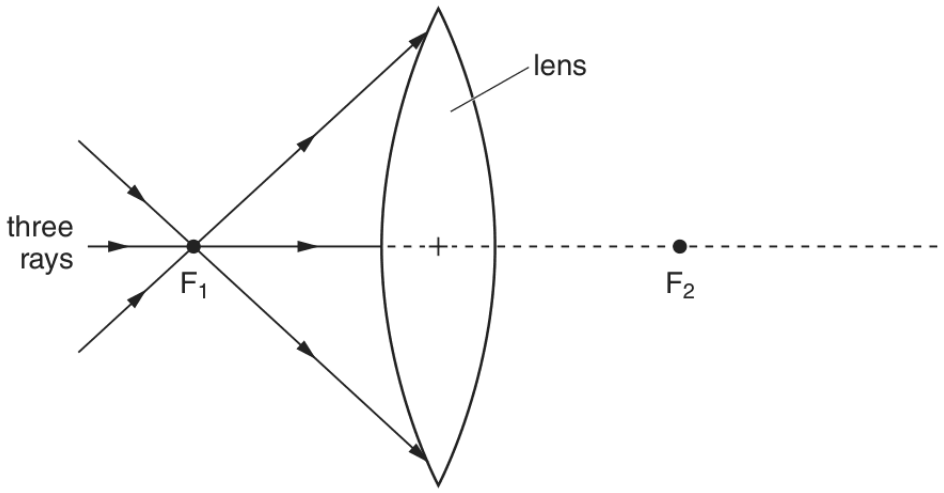


Fig. 4.1

Three rays of light pass through F_1 to the lens.

(a) On Fig. 4.1, continue the three rays through the lens and into the air. [2]

(b) State what happens to the speed of light on

(i) entering the glass lens from air,

.....[1]

(ii) leaving the lens and returning to the air.

.....[1]

(c) Light of wavelength 6.0×10^{-7} m travels in air at a speed of 3.0×10^{8} m/s.

(i) Calculate the frequency of this light.

frequency = ...[2]

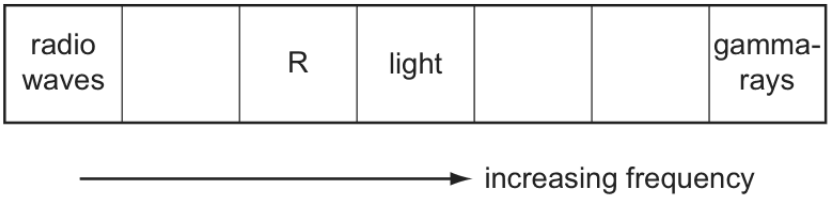
(ii) State the effect, if any, on the frequency as the light enters the glass from air.

.....[1]

4	**(a)**	one outer ray parallel to principal axis three rays parallel to the principal axis	C1 A1	
	(b) **(i)**	(speed) reduced **or** slows down	B1	
	(ii)	(speed) returns to original value/$3.0 \times 10^{8}\,\text{m/s}$	B1	
	(c) **(i)**	$(f =)\ c/\lambda$ **or** $3.0 \times 10^{8} / 6.0 \times 10^{-7}$ $5(.0) \times 10^{14}\,\text{Hz}$	C1 A1	
	(ii)	no effect/unchanged/$(f =)\ 5(.0) \times 10^{14}\,\text{Hz}$	B1	[7]

QUESTION · 5054_w11_qp_12.pdf · source page 10 of 20

23 The diagram shows the main sections of the electromagnetic spectrum in order of increasing frequency. Some of the sections are labelled.



The section R has a frequency just below that of light.

Which application uses the section R?

A killing cancerous cells

B satellite television

C sterilisation

D television remote controller

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 23

23	**D**

4 Fig. 4.1 represents a microwave travelling in air through points A and B.

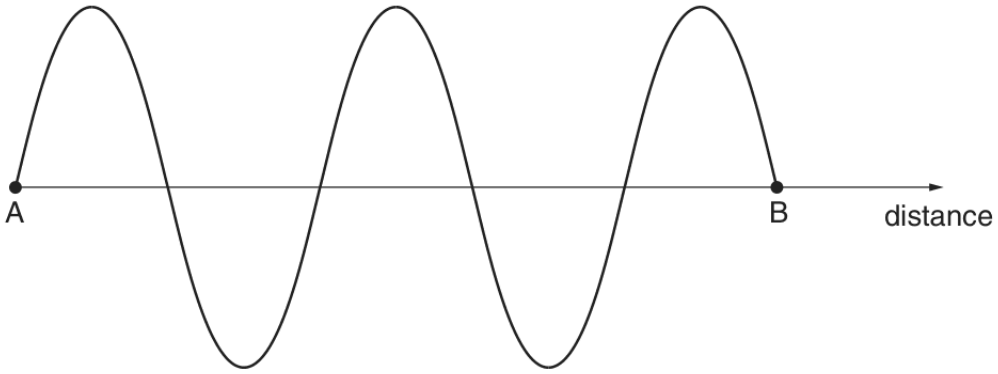


Fig. 4.1 (not to scale)

The distance between A and B is 40 cm.

(a) State the speed of microwaves in air.

speed = ...[1]

(b) Determine the wavelength of the microwave shown in Fig. 4.1.

wavelength = ...[1]

(c) Describe how microwaves are used in the transmission of television signals by satellite.

.....

.....

.....

.....[3]

(d) State two properties common to all electromagnetic waves.

1. .....

2. .....
[2]

4 **(a)** $3(.00) \times 10^8\,m/s$ B1

(b) 0.16 m **or** 16 cm B1

(c) any **three** of:
travel through space/vacuum
pass through the atmosphere/not reflected by ionosphere
encoded (with the signal)
(satellite) amplifies/boosts signal
sent to/received by satellite
transmitted/sent by satellite
transmitted/received by a (satellite) **dish** (on Earth) B3

(d) any **two** of:
same (high) speed (in air) **or** travel at speed of light
travel in vacuum/space **or** no medium needed
transfer/transmit energy
transverse (stated **or** explained)
(oscillating) magnetic **and** electric fields/waves
reflection/refraction/diffraction/interference/polarisation B2 [7]

QUESTION · 5054_s12_qp_11.pdf · source page 10 of 16

25 Below are four statements about the uses of electromagnetic radiation.

Gamma rays are used in medical treatment.

Infra-red waves are used in sunbeds.

Microwaves are used in satellite television.

X-rays are used in intruder alarms.

How many of these statements are correct?

A 1 **B** 2 **C** 3 **D** 4

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 25

25 **B**

5 Fig. 5.1 shows the electromagnetic spectrum.

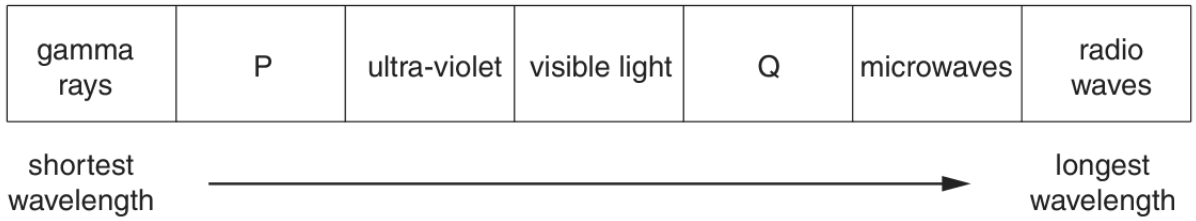


Fig. 5.1

Two components of the spectrum, P and Q, have not been named.

(a) State the name of

(i) component P,

.....[1]

(ii) component Q.

.....[1]

(b) All electromagnetic waves have a wavelength and a frequency. They all have changing magnetic and electric fields. State two other properties of all electromagnetic waves.

1. .....

.....

2. .....

.....

[2]

(c) State the component of the electromagnetic spectrum used for satellite communication.

.....[1]

5	**(a)**	**(i)**	X-ray(s)	B1	
		(ii)	infra-red	B1	
	(b)		any TWO lines: same speed (in vacuo) travel in a vacuum; need no medium carry energy transverse can reflect/refract/diffract/interfere/polarise	B2	
	(c)		microwaves	B1	[5]

QUESTION · 5054_s12_qp_12.pdf · source page 11 of 16

25 Which device uses ultra-violet radiation?

A electric grill

B intruder alarm

C television remote controller

D sunbed

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 25

25	**D**

For Examiner's Use

6 **(a)** Fig. 6.1 shows part of a page from a pupil's notebook.

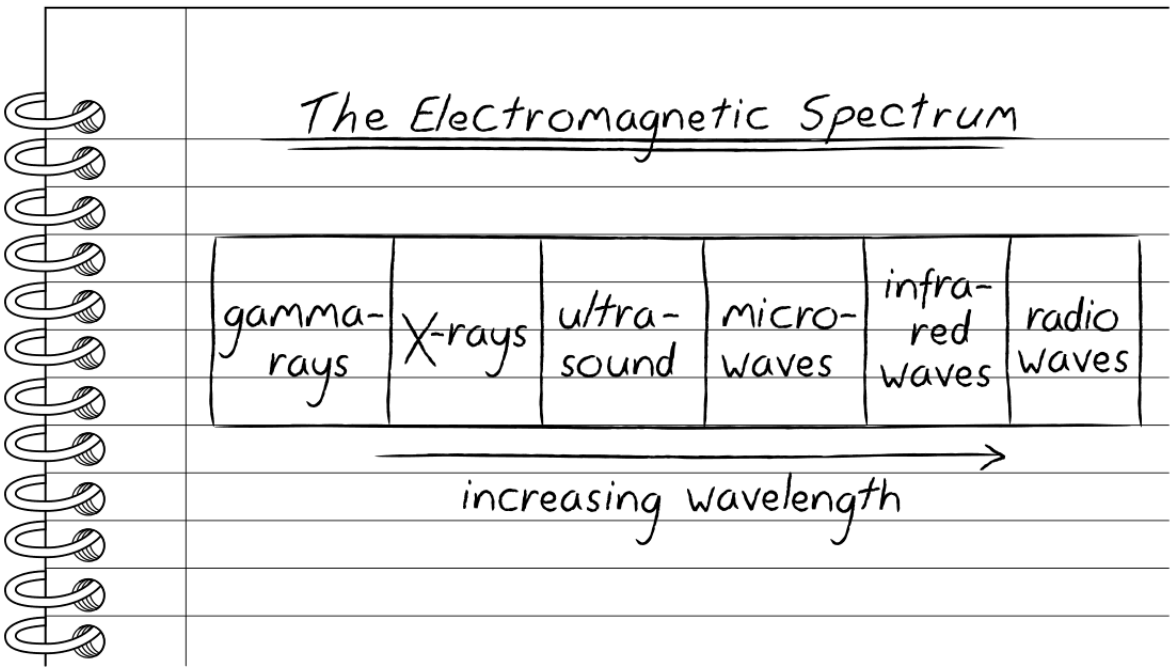


Fig. 6.1

List three errors in the pupil's notes.

1. .....

2. .....

3. .....

[3]

(b) State and explain one application of X-rays in engineering.

.....

.....

..... [2]

6 **(a)** any **three** of:
infra-red and microwaves reversed
visible light is omitted
ultrasound is not e.m./should not be included
ultraviolet is missing ('ultrasound instead of light' scores 2) B3

(b)

engineering use	M1	detail/explanation	A1
detecting cracks in metal		(more) X-rays pass through crack/poor weld	
or		**or**	
checking welds		image of crack on film/screen	
astronomy		hot stars emit X-rays	
crystallography		diffraction reveals pattern of atoms	
fluorescence		substances re-emit different energies	
(airport/border) security		contents of luggage/lorries revealed	
paintings investigated		underpainting revealed	

(**not** medical use) [5]

For Examiner's Use

10 A laser produces red light of frequency 4.7×10^{14} Hz. The speed of light in glass is 2.0×10^{8} m/s.

(a) Calculate the wavelength in glass of light from this laser.

wavelength = ...[2]

(b) Describe an experiment to verify *the law of reflection* for light. You may include a diagram in your answer.

.....

.....

.....

.....

.....

.....

.....[5]

(c) Fig. 10.1 shows a ray of light travelling in an optical fibre. The ray strikes the side of the fibre at P.

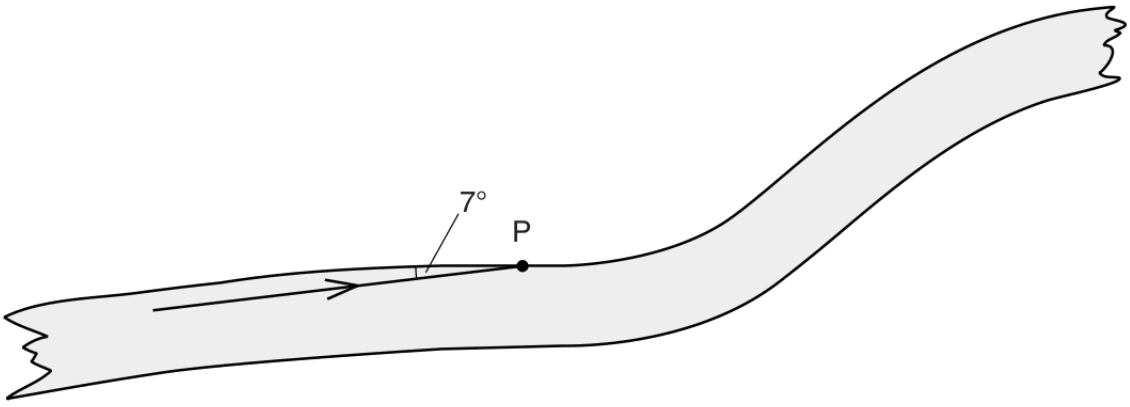


Fig. 10.1

The angle between the ray and the side of the fibre is 7°.

For Examiner's Use

(i) Determine the angle of incidence of the ray at P.

angle = ...[1]

(ii) State and explain what happens to the ray at P.

.....

.....

.....[2]

(d) A room is illuminated by wall lamps. Fig. 10.2 shows a mirror on the wall behind one of the lamps.

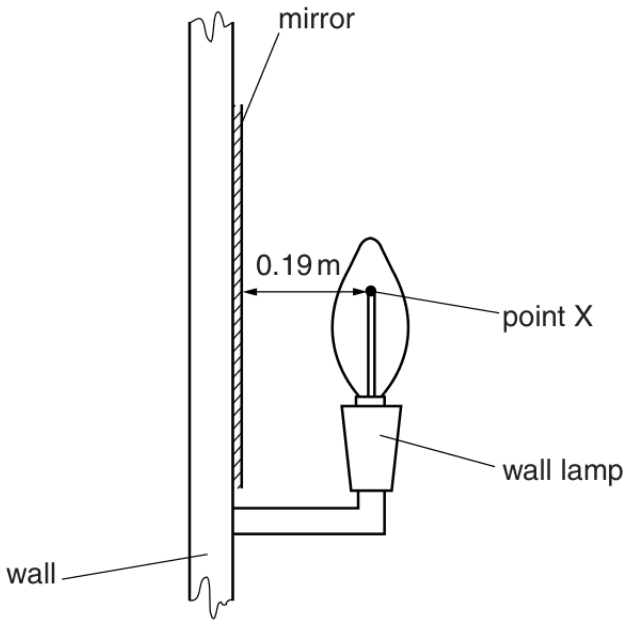


Fig. 10.2 (not to scale)

X is a point on the filament of the lamp. It is 0.19 m in front of the mirror.

(i) On Fig. 10.2, draw rays from X and locate the image of X. Label the image I. [3]

(ii) State the distance between I and the mirror.

distance = ...[1]

(iii) Suggest one advantage of placing a mirror behind the lamp in the room.

.....

.....[1]

10 (a) $(\lambda =)\, v/f$ **or** $2 \times 10^8 / 4.7 \times 10^{14}$ C1
4.3×10^{-7} m A1 [2]

(b)

raybox/light source/ laser **and** mirror		pin(s) **and** mirror	B1
shine ray at mirror		place two pins	B1
mark rays	**or**	two more pins in line with image	B1
measure *i* and *r* **and** equal		measure *i* and *r* **and** equal	B1
repeat		repeat	B1 [5]

(c) (i) 83° B1

(ii) total internal reflection **or** TIR **cao** B1
angle of incidence exceeds critical angle B1 [3]

(d) (i) (at least) one ray from X to mirror M1
(at least) **two** rays from X to mirror **and** correct reflections A1
rays traced back to marked I **or** I marked in correct place (by eye) B1

(ii) 0.19 m B1

(iii) less/no light wasted **or** hall brighter B1 [5]

[Total: 15]

QUESTION · 5054_s13_qp_11.pdf · source page 10 of 20

22 Which wave property has the same value for all X-rays travelling in air?

A amplitude

B frequency

C speed

D wavelength

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 22

22	**C**

For Examiner's Use

4 **(a)** The list below contains three components of the electromagnetic spectrum.

infra-red **gamma rays** **visible light**

Arrange the components in order of increasing wavelength.

..... [1]

(b) Satellites are used in the transmission of some television signals.

Fig. 4.1 shows a satellite above the television station where a television signal is generated.

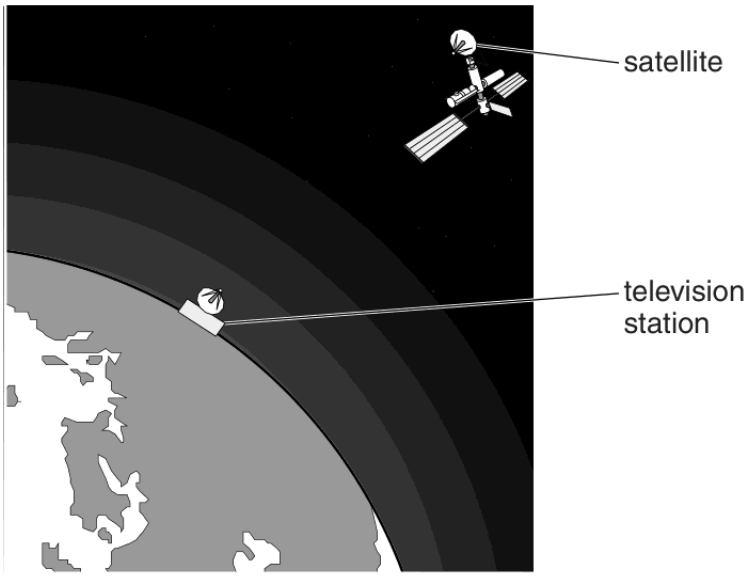


Fig. 4.1 (not to scale)

(i) State which component of the electromagnetic spectrum is used to transmit the television signal to the satellite.

..... [1]

(ii) Explain how the satellite is used.

.....

..... [1]

(iii) Suggest one advantage of using a satellite to transmit television signals.

.....

..... [1]

4	**(a)**	gamma rays, visible light, infra-red	B1	
	(b) **(i)**	microwaves	B1	
	(ii)	satellite (receives and) sends/transmits/emits/boosts/amplifies signal	B1	
	(iii)	cover a large area over the horizon / only one (transmitter/station) needed etc. **or** unaffected by tall buildings/hills **or** no obstructions	B1	[4]

QUESTION · 5054_w13_qp_11.pdf · source page 10 of 20

22 The table lists the main components of the electromagnetic spectrum and their approximate frequency range.

	frequency / Hz
gamma rays	10^{22} to 10^{19}
X-rays	10^{21} to 10^{18}
ultra-violet	10^{18} to 10^{15}
visible light	10^{15} to 10^{14}
infra-red	10^{14} to 10^{12}
microwaves	10^{12} to 10^{9}
radiowaves	10^{9} to 10^{3}

Which range of frequencies can be used to detect cracks inside a block of metal?

A 10^{3} Hz to 10^{12} Hz

B 10^{12} Hz to 10^{15} Hz

C 10^{15} Hz to 10^{18} Hz

D 10^{18} Hz to 10^{22} Hz

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 22

22 **D**

For Examiner's Use

4 Fig. 4.1 shows food being cooked in an electric grill.

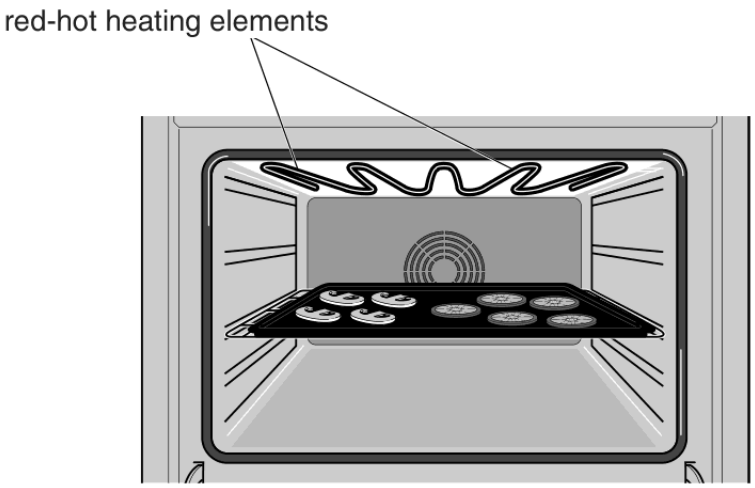


Fig. 4.1

There are red-hot heating elements above the food and thermal energy (heat) is transmitted to the food by radiation.

(a) Explain what is meant, in this case, by *radiation*.

.....

.....

.....[2]

(b) Explain why very little thermal energy is transmitted to the food by

(i) conduction,

.....

.....[1]

(ii) convection.

.....

.....[1]

4	**(a)**	(energy transmitted) by electromagnetic/infra-red (wave)/can travel through a vacuum	B1	
		infra-red **or** visible < λ < microwaves **or** λ just longer than visible (i.e. infra-red scores 2/2)	B1	
	(b) **(i)**	air is a poor conductor	B1	
	(ii)	convection occurs (primarily) upwards/hot air rises (**not** heat rises)	B1	[4]

For Examiner's Use

10 A student goes for a walk in the mountains. During a storm, she sees lightning strike a hillside in the distance. Several seconds later, she hears the thunder caused by the lightning.

(a) (i) Explain why she hears the thunder several seconds after she sees the lightning.

.....

.....[1]

(ii) The student knows the distance to the hillside. She waits for lightning to strike the hillside again. Describe how she can determine a value for the speed of sound in air.

.....

.....

.....

.....[2]

(b) After the storm, the student sees a rainbow.

(i) State the speed of light in air.

speed = ...[1]

(ii) Calculate the wavelength in air of light of frequency 7.5×10^{14} Hz.

wavelength = ...[2]

(iii) State the colours of the spectrum in order of increasing wavelength.

.....

.....[2]

For Examiner's Use

(c) In the laboratory, the student sends blue light into a glass prism placed on a sheet of paper. The arrangement is shown in Fig. 10.1.

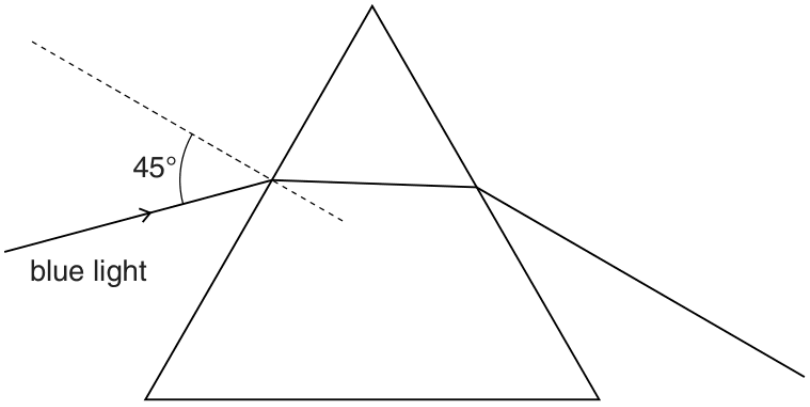


Fig. 10.1

The blue light enters the prism with an angle of incidence of 45°.

(i) On Fig. 10.1, mark the angle of refraction in the glass and label it *r*. [1]

(ii) The student wishes to determine the angle of refraction. On the sheet of paper, she draws the path the ray takes in the glass. Describe how she could do this.

.....

.....

.....[2]

(iii) **1.** State the formula that relates the angle of incidence *i*, the angle of refraction *r* and the refractive index *n*.

.....

.....[1]

2. The angle of refraction in the glass is 28°.

Calculate the refractive index of the glass.

refractive index = ...[1]

(iv) The student sends a ray of red light to the glass prism along the same path as the blue light.

On Fig. 10.1, mark the path taken by the red light after it enters the prism. [2]

MARK SCHEME · 5054_w13_ms_21.pdf · source page 4 of 6 · Question 10

10	**(a)**	**(i)**	speed of sound is (much) less than the speed of light (accept quoted values)	B1	
		(ii)	**measure** the time delay (between the lightning and thunder)	B1	
			divide distance by time/delay	B1	[3]

MARK SCHEME · 5054_w13_ms_21.pdf · source page 5 of 6 · Question 10

	(b)	**(i)**	3.0×10^8 m / s	B1	
		(ii)	(λ =) c/f **or** $3.0 \times 10^8 / 7.5 \times 10^{14}$	C1	
			4.0×10^{-7} m	A1	
		(iii)	(in any order) blue, green, orange, red, yellow, (indigo), (violet) **or** VIBGYOR	C1	
			violet, indigo, blue, green, yellow, orange, red	A1	[5]
	(c)	**(i)**	correct angle clear/labelled r	B1	
		(ii)	mark/determine entrance and exit points (e.g. trace rays back to glass)	B1	
			join/draw line between entrance and exit points	B1	
		(iii)	**1.** n = sin i/sin r	B1	
			2. 1.5/1.51/1.506176 with no unit (**not** just 1.5 without working out)	B1	
		(iv)	correct direction of refraction at **both** faces	M1	
			completely correct (above blue)	A1	[7]

[Total: 15]

QUESTION · 5054_s14_qp_11.pdf · source page 10 of 16

26 Which statement is correct?

A Gamma rays have a longer wavelength than ultra-violet waves.

B Infra-red waves have a lower frequency than radio waves.

C Microwaves have a longer wavelength than visible light.

D X-rays have a higher speed in air than visible light.

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 26

26	**C**

QUESTION · 5054_s14_qp_12.pdf · source page 10 of 16

27 Which statement is correct for all electromagnetic waves?

A They are transverse.

B They cannot travel in a vacuum.

C They have the same frequency.

D They travel through lead.

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 27

27	**A**

5 Visible light, radio waves, X-rays, gamma rays and microwaves are some of the components of the electromagnetic spectrum.

(a) State two other components of the electromagnetic spectrum.

1. .....

2. .....
[1]

(b) White light is a mixture of different colours.

Fig. 5.1 shows a ray of white light entering a glass prism.

The white light separates into a number of colours. Only the blue light and the red light are shown.

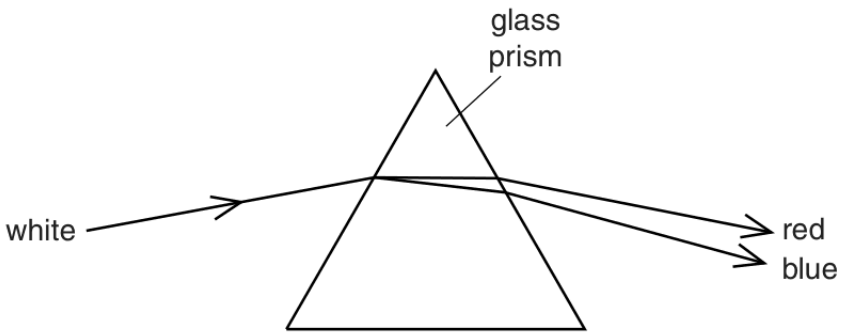


Fig. 5.1

Explain why the blue light and the red light separate as shown.

Use the term *refractive index* in your answer.

.....

.....

.....

.....

..... [3]

5 **(a)** ultra violet **and** infra-red [B1]

(b) blue refracts / bends / deviates more [B1]
blue slows more (than red when entering glass) or blue and red have different speeds (from each other in glass) [B1]
blue and red have different refractive indices [B1]

[4]

QUESTION · 5054_w14_qp_11.pdf · source page 12 of 20

24 Which application uses microwaves?

A detecting small cracks in metals

B gaining a sun-tan

C lighting a fluorescent tube

D satellite television

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 24

24	**D**

QUESTION · 5054_w14_qp_12.pdf · source page 7 of 16

17 Which application uses microwaves?

A detecting small cracks in metals

B gaining a sun-tan

C lighting a fluorescent tube

D satellite television

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 17

17 **D**

10 **(a)** State the speed of light in air.

.....[1]

(b) Fig. 10.1 shows a ray of blue light passing from air into a glass block and refracting at the surface.

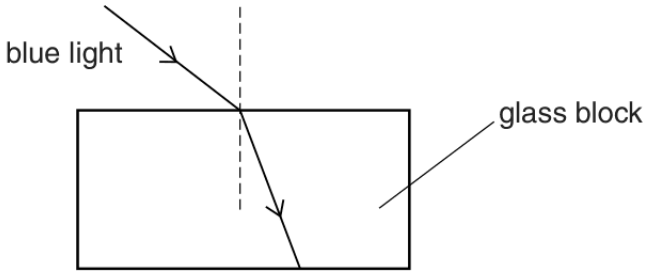


Fig. 10.1 (not to scale)

(i) As the light enters the glass, state what happens to

1. the speed of the light,

.....[1]

2. the frequency of the light,

.....[1]

3. the wavelength of the light.

.....[1]

(ii) On Fig. 10.1, mark and label the angle of incidence *i* and the angle of refraction *r*. [2]

(c) The refractive index of glass for blue light is 1.5.

(i) A ray of blue light strikes the surface of a glass block at an angle of incidence of 89°.

Calculate the angle of refraction of the light in the block.

angle of refraction = ...[3]

(ii) Explain why the angle of refraction of blue light in glass is always less than 45°.

.....

.....

.....

.....[2]

(d) Blue light, travelling in air, strikes the side of a different glass block and continues in the same direction as it enters the glass block. Fig. 10.2 shows the ray of light and the shape of the glass block. The critical angle for this glass is 42°.

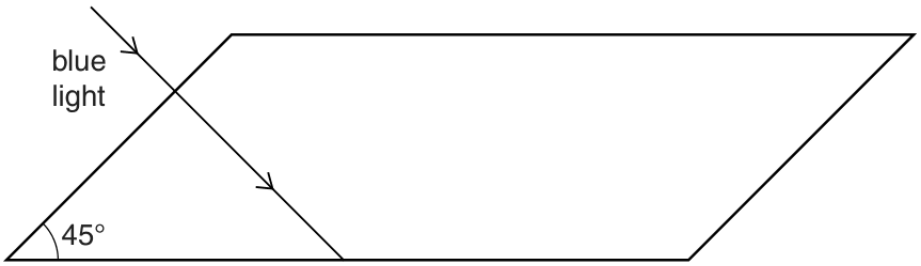


Fig. 10.2

(i) Explain why the light continues in the same direction as it enters the glass block.

.....

.....

.....[2]

(ii) On Fig. 10.2, complete the path of the light until it leaves the glass. [2]

10 (a) 3.0×10^8 m / s — B1 **[1]**

(b) (i)
1. decreases **cao** — B1
2. no change **cao** — B1
3. decreases **cao** — B1

(ii)
1. *i* correctly marked (to normal) — B1
2. *r* correctly marked (to normal) — B1 **[5]**

(c) (i)
sin *i* / sin *r* = *n* **or** sin *i* / sin *r* = 1.5 — C1
sin 89 / sin *r* = 1.5 **or** sin 89 / 1.5 **or** 0.67(0.666565) — C1
42° **or** 41.8025° — A1

(ii)

i equal to / close to 90°	sin *i* / sin 45	$\sin^{-1}(1/n) / \sin^{-1}(1/1.5)$	
and *r* less than 45°	= 1.5	**and** 41.8°	B1
	or	**or**	
i never bigger than	sin *i* >1	*r* not be more than *c*	
89° / 90°			B1 **[5]**

(d) (i)

(sin) *i* = 0 **or** ray enters directly /		wavefront / light hits surface	
along normal / perpendicular	**or**	all together	B1
(sin) *r* = 0 **or** no refraction		all slows down together	B1

(ii)
correct reflection at bottom surface (by eye) — M1
second correct reflection at top and no refraction at either point — A1 **[4]**

[Total: 15]

QUESTION · 5054_s15_qp_12.pdf · source page 11 of 16

28 Which component of the electromagnetic spectrum is used for television transmission from satellites?

A microwaves

B radio waves

C ultra-violet

D X-rays

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 28

28	**A**

QUESTION · 5054_w15_qp_12.pdf · source page 9 of 16

20 Which statement is correct?

A Infra-red radiation cannot travel in a vacuum.

B Infra-red radiation cannot travel in solids or in gases.

C Infra-red radiation can only travel in a vacuum.

D Infra-red radiation can travel in a vacuum and in gases.

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 20

20	**D**

QUESTION · 5054_s16_qp_11.pdf · source page 8 of 16

18 Which statement about thermal radiation is correct?

A In a vacuum, thermal radiation travels at the speed of light.

B Thermal radiation is a longitudinal wave.

C Thermal radiation travels as an ultra-violet wave.

D White surfaces are better emitters of thermal radiation than black surfaces.

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 18

18	**A**

QUESTION · 5054_s16_qp_11.pdf · source page 9 of 16

24 Microwaves are used to transmit television signals to and from a satellite.

Which statement about microwaves is correct?

A They have a longer wavelength than radio waves.

B They penetrate the atmosphere without significant loss of energy.

C They travel much faster than radio waves in a vacuum.

D They warm the satellite and stop it freezing.

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 24

24	**B**

QUESTION · 5054_s16_qp_11.pdf · source page 10 of 16

25 Where are gamma-rays used?

A in fluorescent tubes

B in killing cancerous cells

C in pre-natal scanning

D in sunbeds

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 25

25	**B**

6 Fig. 6.1 shows some of the colours in the visible part of the electromagnetic spectrum.

red	orange	yellow	green	blue

Fig. 6.1

(a) State which colour in Fig. 6.1 has

(i) the largest wavelength, ... [1]

(ii) the highest frequency. ..... [1]

(b) Visible light, infra-red, ultra-violet and radio waves are four components of the electromagnetic spectrum.

State two other components of the electromagnetic spectrum and describe a different use that is made of each component.

1. component .....

description of use .....

.....

.....

2. component .....

description of use .....

.....

.....

[4]

6 **(a)** **(i)** red B1

(ii) blue B1

(b) ANY 2 from (the use must agree with the type) B4

Microwaves B1
use – satellite television, telephone, mobile / cell phones; B1
cooking, heating in a microwave oven, television
remote, radar, communication

X(-rays) B1
use – hospital use in medical imaging or security imaging, killing cancerous cells, B1
and engineering applications such as detecting cracks in metal, crystallography

gamma (rays) B1
use – medical treatment in killing cancerous cells, and engineering applications B1
such as detecting cracks in metal, sterilisation, tracer applications, radiotherapy

QUESTION · 5054_w16_qp_11.pdf · source page 12 of 20

28 Which two types of electromagnetic radiation are both used to kill cancerous cells and are both used to detect cracks in metals?

A gamma-rays and microwaves

B gamma-rays and X-rays

C microwaves and ultra-violet radiation

D ultra-violet radiation and X-rays

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 28

28	**B**

10 A laser produces a beam of red light.

(a) The red light from the laser has a frequency of 4.3×10^{14} Hz.

(i) State the speed of light in air.

.....[1]

(ii) Calculate the wavelength of this red light in air.

wavelength = ...[2]

(b) Red light from the laser strikes one side of a glass prism at an angle of incidence *i*. The light refracts towards the normal as it enters the prism.

Fig. 10.1 shows the prism, the light and a screen.

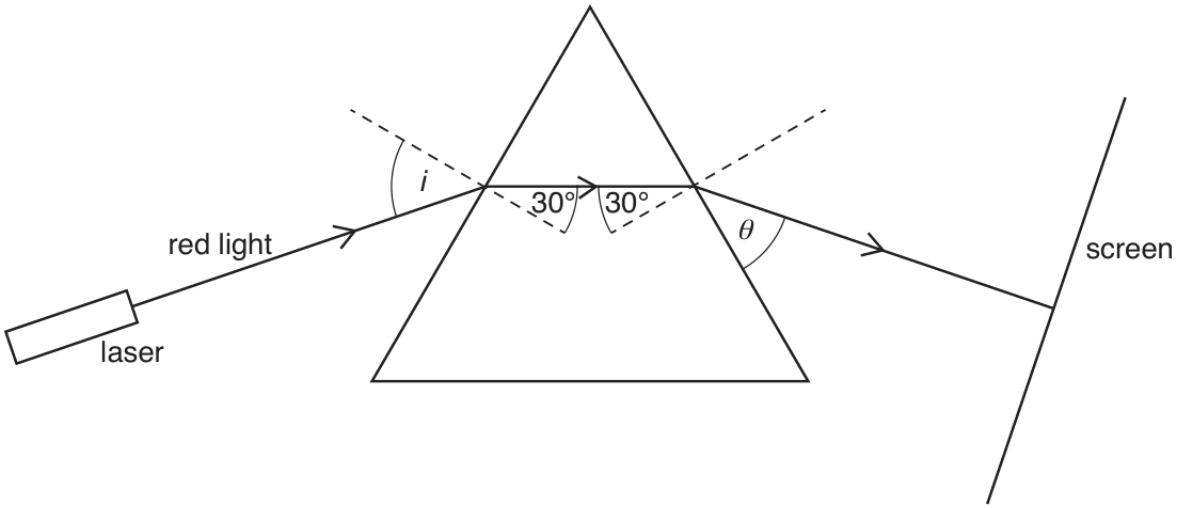


Fig. 10.1 (not to scale)

(i) State what happens to the speed of light as it enters the glass.

.....[1]

(ii) The refractive index of the glass is 1.5. The angle of refraction in the glass, where the light enters the prism, is 30°.

Calculate the angle of incidence *i*.

i = ...[2]

(iii) The light then passes back into the air and strikes the front of the white screen, as shown in Fig. 10.1.

Calculate θ, the angle between the ray in air and the side of the prism.

$\theta =$ .....[1]

(c) The laser in **(b)** is replaced with a filament lamp and a slit, as shown in Fig. 10.2.

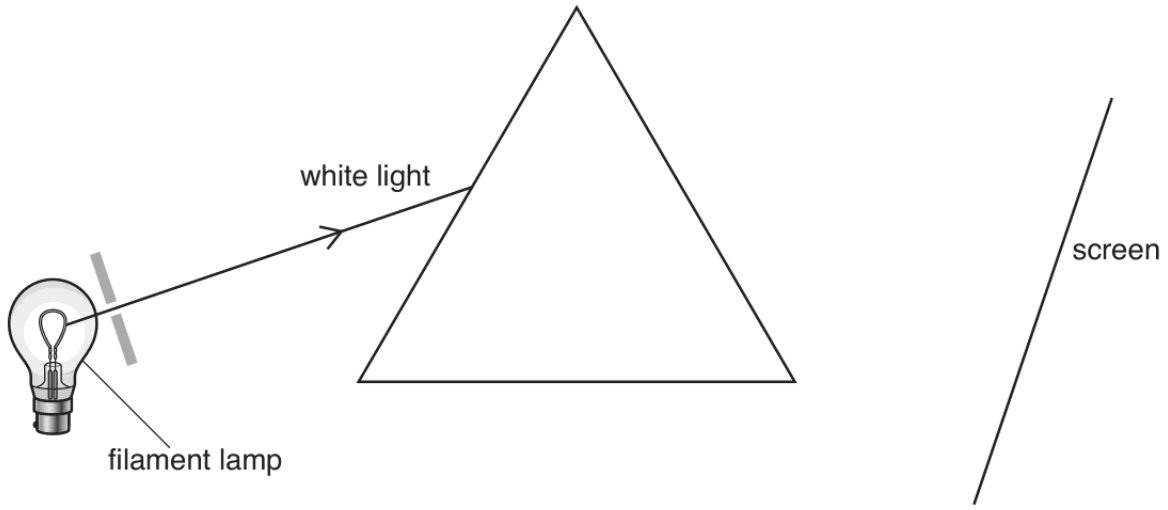


Fig. 10.2 (not to scale)

A ray of white light now strikes the prism.

(i) On Fig. 10.2, draw what happens to the light as it passes through the prism and strikes the screen. [2]

(ii) Describe what is seen on the screen.

.....

.....

.....

.....[2]

(iii) In addition to visible light, the filament lamp also emits some infra-red radiation. This infra-red radiation is able to pass through glass.

1. On the screen in Fig. 10.2, mark an X to indicate a place where infra-red radiation strikes the screen. [1]

2. Infra-red radiation is often detected by using a sensitive thermometer with a bulb that has been painted black.

Explain why the blackened bulb makes the thermometer a good detector of infra-red radiation.

.....

.....[1]

(d) Explain the role played by infra-red radiation in intruder alarms.

.....

.....

.....

.....

.....[2]

10	**(a)**	**(i)**	3.0×10^8 m / s			B1	
		(ii)	$(\lambda =) c / f$ **or** $3.0 \times 10^8 / 4.3 \times 10^{14}$			C1	
			7.0×10^{-7} m			A1	[3]
	(b)	**(i)**	decreases			B1	
		(ii)	$\sin(i) = n \times \sin(r)$ **or** $1.5 \times \sin(30°)$ **or** 0.75			C1	
			49°			A1	
		(iii)	41°			B1	[4]
	(c)	**(i)**	dispersion at **both** surfaces **and** refractions in correct direction			B1	
			violet / blue light below the red light shown			B1	
		(ii)	spectrum **or** band of (continuous) colours **or** colours of rainbow			B1	
			red, orange, yellow, green, blue, (indigo, violet)			B1	
		(iii)	**1** X marked above red			B1	
			2 it is / black surfaces are good absorbers (of IR radiation)			B1	[6]
	(d)	intruder / human being emits IR		IR beam broken		IR reflected	B1	
			or		**or**			
		intruder warm **or** IR detected		does not reach detector		change detected	B1	[2]
								[15]

QUESTION · 5054_s17_qp_11.pdf · source page 12 of 16

29 A wall poster showing the electromagnetic spectrum is displayed in a laboratory.

A section of the electromagnetic spectrum has been accidentally ripped from this wall poster.

Which piece is missing?

A infra-red visible light microwave

B microwave infra-red visible light

C visible light infra-red microwave

D visible light microwave infra-red

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 3 of 3 · row for Question 29

29	C	**1**

QUESTION · 5054_s17_qp_11.pdf · source page 12 of 16

30 Which type of wave is used to send television signals to a satellite?

A infra-red waves

B light waves

C microwaves

D sound waves

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 3 of 3 · row for Question 30

30	C	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 14 of 20

29 Which type of wave is used to send television signals to a satellite?

A infra-red waves

B light waves

C microwaves

D sound waves

MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 3 of 3 · row for Question 29

29	C	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 11 of 20

23 A type of electromagnetic radiation possesses the following properties.

- It is ionising.
- Its frequency is higher than the frequency of microwaves.
- It is not detected by the human eye.

What is this radiation?

A gamma rays

B infra-red

C light

D radio waves

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 23

23	A	**1**

10 Fig. 10.1 represents the electromagnetic spectrum.

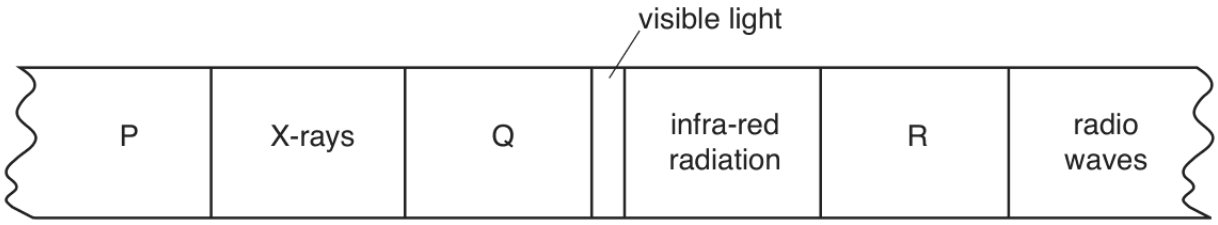


Fig. 10.1

(a) State the name of the component of the electromagnetic spectrum that is found in region

(i) P, ...

(ii) Q, ...

(iii) R. .....
[2]

(b) On Fig. 10.1, mark a tick (✓) in all of the boxes that represent the components with frequencies greater than the frequencies of visible light. [1]

(c) The speed of electromagnetic radiation in a vacuum is 3.0×10^8 m/s.

A television remote controller uses infra-red radiation of wavelength 9.4×10^{-7} m.

(i) Calculate the frequency of this radiation.

frequency = ... [3]

(ii) Explain how infra-red radiation is used in television remote controllers.

.....

.....

.....

..... [2]

(d) Fig. 10.2 shows a ray of green light in air striking the side of a glass prism.

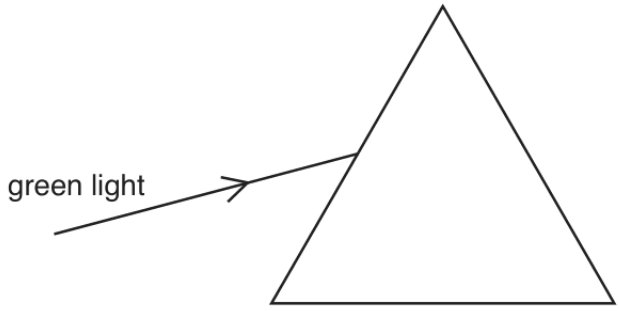


Fig. 10.2 (not to scale)

The refractive index of the glass for green light is 1.5.

(i) On Fig. 10.2,

1. draw the normal for this ray,

2. mark the angle of incidence with a letter *i*.

[1]

(ii) This angle of incidence is 57°.

Calculate the angle of refraction in the glass.

angle of refraction = ... [2]

(iii) State what happens, as the light enters the glass, to the light's

1. frequency,

.....

2. speed,

.....

3. wavelength.

.....

[2]

(iv) On Fig. 10.2, draw the path taken by the light as it passes through the glass and into the air. [2]

Question	Answer	Marks
10(a)	P - gamma(-rays) **or** γ(-rays)	
	Q - ultraviolet (radiation)	
	R - microwaves	
	any one correct	**C1**
	all three correct	**A1**
10(b)	P **and** X-rays **and** Q ticked	**B1**
10(c)(i)	(f =)c / λ **or** 3.0×10^{8} / 9.4×10^{-7}	**C1**
	3.2×10^{N}	**C1**
	3.2×10^{14} Hz	**A1**
10(c)(ii)	infra-red / radiation / signal / wave emitted by control **and** received at set	**B1**
	infra-red / radiation / signal / wave is encoded **or** is decoded	**B1**
10(d)(i)	normal indicated **and** angle of incidence indicated	**B1**
10(d)(ii)	n = sin i / sin r **or** 1.5 = sin 57(°) / sin r **or** (r =)$\sin^{-1}$(sin 57(°) / n) **or** $\sin^{-1}$(sin 57(°) / 1.5)	**C1**
	34°	**A1**
10(d)(iii)	**1** no change	**B1**
	2 **3** decreases **and** decreases	**B1**

Question	Answer	Marks
10(d)(iv)	ray in glass between normal and continuation of the incident ray	**B1**
	ray in air between continuation of the refracted ray and side of prism	**B1**

QUESTION · 5054_w17_qp_12.pdf · source page 10 of 16

24 The diagram shows the electromagnetic spectrum with three components named. The spectrum is in order from long wavelength to short wavelength.

Which component of the spectrum is used in a sunbed to produce a suntan?

long wavelength short wavelength

radio waves	**A**	**B**	visible light	**C**	**D**	gamma rays

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 24

24	C	**1**

5 A process called *nuclear fusion* takes place in the Sun.

(a) Discuss the process of nuclear fusion taking place in the Sun.

.....

.....

.....

.....

.....

.....

.....[3]

(b) On Earth, solar panels and solar cells are powered by energy from the Sun.

Describe how this energy from the Sun travels to Earth.

.....

.....

.....

.....[2]

Question	Answer	Marks
5(a)	joining together of (small) nuclei (to make bigger nuclei)	**B1**
	energy released	**B1**
	hydrogen (used) **or** helium (produced)	**B1**
5(b)	electromagnetic radiation / infra-red / light / ultraviolet	**B1**
	travels through vacuum **or** no medium needed	**B1**

QUESTION · 5054_s18_qp_11.pdf · source page 2 of 20

1 Which statement about electromagnetic waves is correct?

A All electromagnetic waves have speeds in air of approximately $3 \times 10^8\,\text{m/s}$.

B In air, some electromagnetic waves travel faster than light.

C The electromagnetic waves with the largest wavelength are in the infra-red region.

D The electromagnetic waves with the smallest wavelength are in the X-ray region.

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 1

1	A	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Electromagnetic radiation is produced by the Sun and travels from the Sun to the Earth.

(a) Some of the statements below are true and some of them are false.

Put a tick (✓) in the box after each statement to show whether it is true or false.

	true	false
Gamma rays are used to kill cancerous cells but can also cause cancer.	☐	☐
Infra-red is used in sun-beds.	☐	☐
Radio waves have the highest frequency in the electromagnetic spectrum.	☐	☐
The higher the frequency of the radiation the smaller is the wavelength in air.	☐	☐

[2]

(b) The glass prism in Fig. 9.1 is used to split white light from the Sun into different colours.

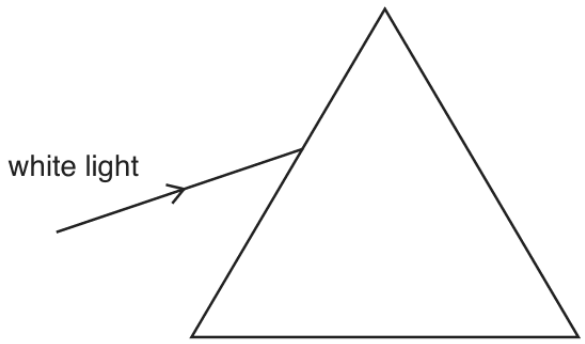


Fig. 9.1

(i) Draw, on Fig. 9.1, to show the action of the prism on the light as it passes through the prism. [3]

(ii) Explain why the prism splits white light into different colours.

.....

.....

.....

.....[2]

(c) The Earth is 1.5×10^8 km from the Sun.

(i) State the speed of light in a vacuum.

.....[1]

(ii) Calculate the time taken for light to travel from the Sun to the Earth.

time = ...[2]

(d) The Sun also produces a solar wind, which is a stream of charged particles.

In one part of space, the solar wind is due to the movement of protons and alpha-particles away from the Sun. A charge of 180 C passes through a small volume of space in 2.0 minutes.

(i) Calculate the electric current in this volume of space.

current = ...[2]

(ii) Complete the table in Fig. 9.2 for an alpha-particle, a proton and a gamma ray.

	nature	charge	stopped by
alpha-particle			
proton	a proton	positive	thin metal
gamma ray	electromagnetic radiation		

Fig. 9.2

[3]

Question	Answer	Marks
9(a)	**true false false true** all 4 correct 2 marks	**B1**
	3 correct 1 mark	**B1**
9(b)(i)	correct refraction towards normal on entering prism	**B1**
	splitting into colours occurs at first surface	**B1**
	red at top and blue / violet at bottom <u>inside</u> and outside prism if shown	**B1**
9(b)(ii)	(different colours have) different speeds in the glass **or** different refractive indices	**B1**
	different (angles of) <u>refraction</u> **or** <u>refract</u> at different angles	**B1**
9(c)(i)	$3(.0) \times 10^{8}$ m / s	**B1**
9(c)(ii)	(t=) d / s in any form algebraic or numerical, e.g.	**C1**
	500 s	**A1**
9(d)(i)	(I=) Q / t in any form algebraic or numerical, e.g. 180 / 2	**C1**
	1.5 A	**A1**
9(d)(ii)	1st column helium nucleus **or** 2 protons and 2 neutrons	**B1**
	2nd column positive **and** neutral / no / charge / none	**B1**
	3rd column paper / <u>thin</u> metal / skin / few cm air etc. **and** (thick) lead / thick metal / concrete / stone / earth OR never completely stopped / nothing	**B1**

QUESTION · 5054_s18_qp_12.pdf · source page 11 of 16

30 A television controller emits an infra-red beam.

Which statement about infra-red radiation is correct?

A It causes ionisation.

B It consists of longitudinal waves.

C It has a higher frequency than ultra-violet light.

D It travels at the speed of light.

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 3 of 3 · row for Question 30

30	D	**1**

5 A microwave signal from a transmitter on the Earth's surface is sent up to a satellite in orbit and is then immediately transmitted back to Earth.

(a) A pulse of microwaves from the transmitter is received back at the transmitter 0.24 s later.

The speed of microwaves is 3.0×10^8 m / s.

Calculate the distance from the transmitter on the Earth to the satellite.

distance = ... [2]

(b) The microwave signal is used in a telephone system that produces sound.

Waves are described as longitudinal or transverse. Some waves are also electromagnetic.

(i) Place ticks (✓) in the table in Fig. 5.1 to show whether microwaves and sound waves are longitudinal or transverse and to show whether they are electromagnetic.

	longitudinal	transverse	electromagnetic
microwaves			
sound			

[2]

Fig. 5.1

(ii) A longitudinal wave contains a compression.

Describe what is meant by a *compression*.

.....

.....

.....[1]

Question	Answer	Marks
5(a)	(d=) s × t in any form algebraic or numerical,	**C1**
	3.6×10^7 m	**A1**
5(b)(i)	1st two columns correct (sound longitudinal and microwaves transverse)	**B1**
	3rd column correct microwaves electromagnetic	**B1**
5(b)(ii)	Layers / molecules / particles / spring coils close together **or** high(er) pressure (than atmospheric)	**B1**

QUESTION · 5054_w18_qp_11.pdf · source page 8 of 16

21 What is the approximate value for

$$\frac{\text{speed of light in air}}{\text{speed of sound in air}}?$$

A 10^2 **B** 10^4 **C** 10^6 **D** 10^8

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 21

21	C	**1**

9 Some waves are longitudinal and some waves are transverse.

(a) State how a longitudinal wave differs from a transverse wave.

.....

..... [1]

(b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

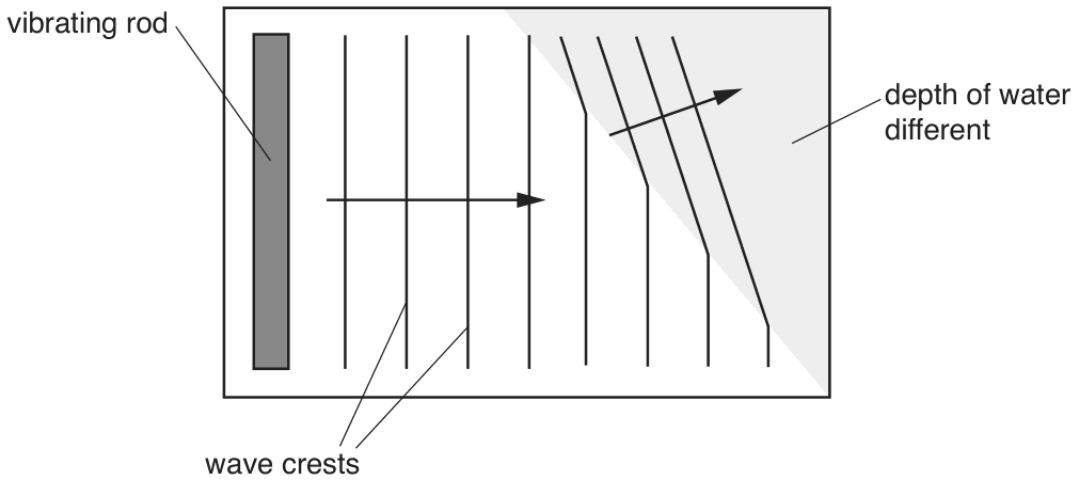


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

(i) In the left-hand section of the tank, the wavelength of the wave is 0.019 m and it travels at 0.17 m/s.

Calculate the frequency of the wave.

frequency = ... [2]

(ii) State what happens to the frequency of the wave, as it passes into the right-hand section of the tank.

..... [1]

(iii) Using Fig. 9.1, state and explain what happens to the speed of the wave as it passes into the right-hand section of the tank.

.....

.....

..... [2]

(c) Fig. 9.2 shows light, in air, striking the vertical side of a rectangular glass block at an angle of incidence of 60°.

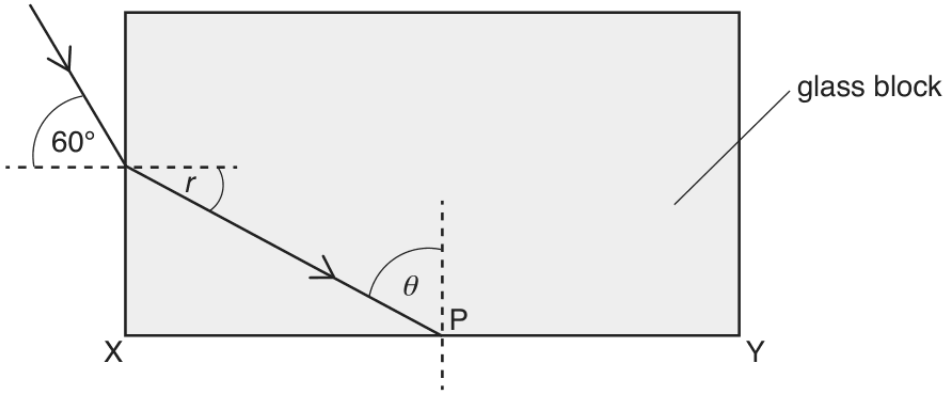


Fig. 9.2

The refractive index of the glass is 1.6. The light travels in the glass and strikes side XY at P.

(i) Underline all the terms that describe a light wave.

electromagnetic **longitudinal** **transverse**

[1]

(ii) At the point where the light enters the glass, the angle of refraction is *r*.

Calculate angle *r*.

r = ... [2]

(iii) **1.** Calculate the critical angle *c* for light travelling in the block.

c = ... [2]

2. At P, the angle θ between the ray and the normal is given by $\theta = 90° - r$.

State and explain what happens to the light when it strikes side XY.

.....

.....

..... [2]

3. On Fig. 9.2, draw the path of the light after it strikes side XY at P and the path of the light when it is again travelling in the air. [2]

Question	Answer	Marks
9(a)	the vibration direction is parallel to the wave / energy travel direction **or** longitudinal waves have compressions and rarefactions **or** transverse waves have crests and troughs **or** longitudinal waves cannot be polarised	**B1**
9(b)(i)	(f =) v / λ **or** 0.17 / 0.019	**C1**
	8.9 Hz	**A1**
9(b)(ii)	(frequency) stays constant **or** does not change	**B1**
9(b)(iii)	(speed) decreases	**B1**
	(Fig. 9.1 shows that) wavelength decreases **or** refraction is towards normal **or** $i > r$ **or** top of wave (in shaded region) lags behind bottom of wave	**B1**
9(c)(i)	electromagnetic **and** transverse underlined	**B1**
9(c)(ii)	$n = \sin(i) / \sin(r)$ **or** (r =) $\sin^{-1}(\sin(i) / n)$ **or** (r =) $\sin^{-1}(\sin(60°) / 1.6)$	**C1**
	33(°)	**A1**

Question	Answer	Marks
9(c)(iii)	1 $n = 1 / \sin(c)$ **or** (c =) $\sin^{-1}(1 / n)$ **or** $\sin^{-1}(1 / 1.6)$	**C1**
	39°	**A1**
	2 total internal reflection	**B1**
	$\theta > c$ **or** 57(°) > 39(°) **or** angle of incidence greater than the critical angle	**B1**
	3 reflected ray at P (correct angle by eye)	**B1**
	light in air at 30° (by eye) to vertical surface	**B1**

QUESTION · 5054_w18_qp_12.pdf · source page 11 of 16

26 Which statement about radio waves is correct?

A Radio waves are sound waves.

B Radio waves are used to kill cancerous cells.

C Radio waves are used in television communications.

D Radio waves have frequencies higher than those of visible light.

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 26

26	C	**1**

4 A house has several solar panels on the roof.

These panels use energy from the Sun both to generate electricity and to raise the temperature of water that passes through tubes inside the panels.

(a) The panels on the roof of the house have a black surface.

(i) State how energy from the Sun travels through space before it reaches the Earth.

.....

..... [1]

(ii) Explain the advantage of using panels that have a black surface.

.....

.....

.....

..... [2]

(b) On one occasion, the panels are supplying an electric current of 15 A at a voltage of 24 V.

(i) Calculate the electrical energy generated by the panels in one hour.

electrical energy = ... [2]

(ii) In the same time, 51 kg of cold water is pumped through the panels. The temperature of the water increases from 16 °C to 45 °C.

The specific heat capacity of water is 4200 J / (kg °C).

Calculate the increase in thermal energy of the water.

thermal energy increase = ... [3]

Question	Answer	Marks
4(a)(i)	radiation **or** infrared (radiation / waves) **or** light	**B1**
4(a)(ii)	it / a black surface is a good absorber / poor reflector of radiation	**B1**
	more energy / power output **or** more electricity produced	**B1**
4(b)(i)	(P =) VIt **or** 24 × 15 × 3600 **or** 24 × 15 × 60 **or** 22 000 (J)	**C1**
	1.3×10^6 J	**A1**
4(b)(ii)	(ΔQ =) $mc\Delta T$ **or** 29 (°C) **or** 45 – 16 (°C)	**C1**
	51 × 4200 × 29 **or** 51 × 4200 × (45 – 16)	**C1**
	6.2×10^6 J	**A1**

QUESTION · 5054_s19_qp_11.pdf · source page 6 of 20

16 Which statement is correct?

A Infrared radiation cannot travel in a vacuum.

B Infrared radiation cannot travel in solids or in gases.

C Infrared radiation can only travel in a vacuum.

D Infrared radiation can travel in a vacuum and in gases.

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 16

16	D	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 8 of 16

21 Which statement is correct?

A Infrared radiation cannot travel in a vacuum.

B Infrared radiation cannot travel in solids or in gases.

C Infrared radiation can only travel in a vacuum.

D Infrared radiation can travel in a vacuum and in gases.

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 2 of 3 · row for Question 21

21	D	**1**

4 A star emits electromagnetic radiation over a range of wavelengths.

Fig. 4.1 shows the brightness of the radiation from the star at different wavelengths.

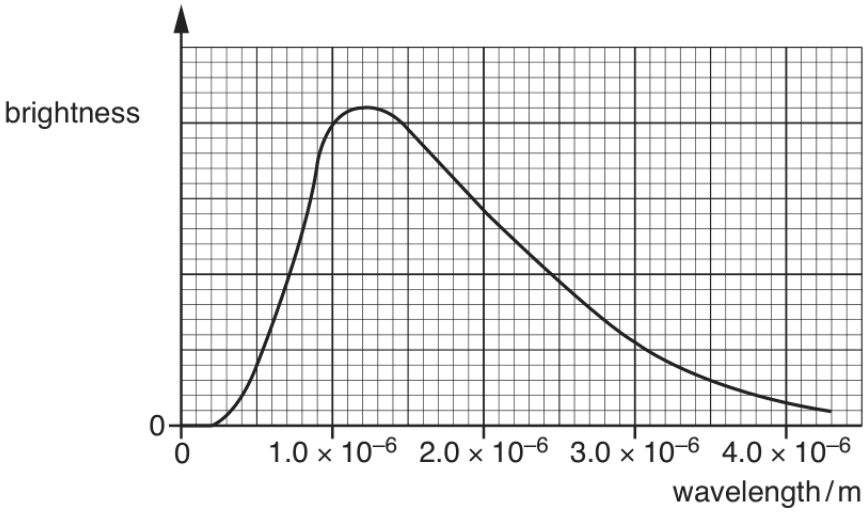


Fig. 4.1

Radiation from the star is brightest at one wavelength.

(a) Determine the wavelength where the radiation is brightest.

wavelength = ... [1]

(b) Visible light has a wavelength between 4.0×10^{-7} m and 7.0×10^{-7} m.

The radiation in **(a)** lies just outside the visible part of the electromagnetic spectrum.

State the name of the region of the spectrum that contains the radiation in **(a)**.

..... [1]

(c) Determine the frequency of the radiation in **(a)**.

The speed of light is 3.0×10^{8} m/s.

frequency = ... [3]

[Total: 5]

Question	Answer	Marks
4(a)	1.15–1.25 × 10^{-6} (m)	**B1**
4(b)	infra-red	**B1**
4(c)	($f = v / \lambda$) algebraic or numerical	**C1**
	3 × 10^{8} / 1.2 × 10^{-6}	**C1**
	2.5 × 10^{14} Hz	**A1**

QUESTION · 5054_w19_qp_11.pdf · source page 12 of 20

25 The frequency of microwaves used in an experiment is 3.0×10^9 Hz. Their speed in air is 3.0×10^8 m/s.

What is a possible frequency and a possible speed in air of a radio wave?

	frequency / Hz	speed in air / m/s
A	3.0×10^5	1500
B	3.0×10^5	3.0×10^8
C	3.0×10^{14}	1500
D	3.0×10^{14}	3.0×10^8

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 25

25	B	**1**

4 Fig. 4.1 represents the electromagnetic spectrum.

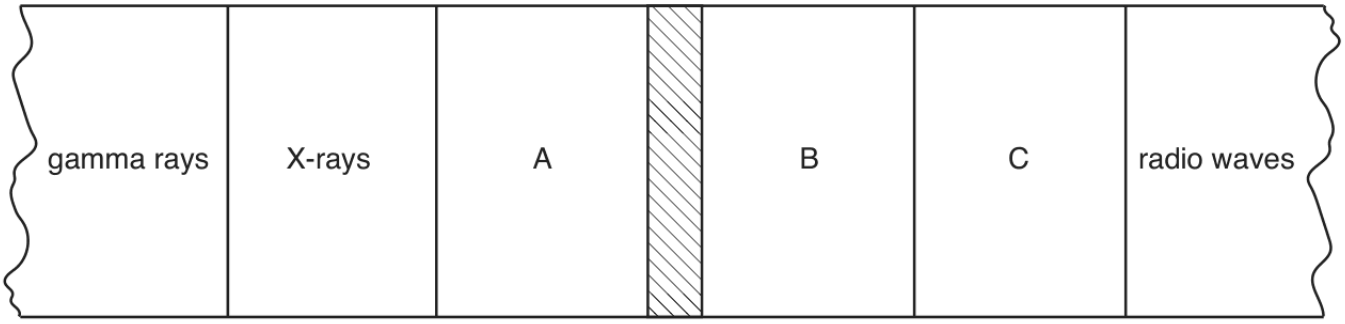


Fig. 4.1

The shaded region in the middle is the visible light spectrum.

(a) State the name of the component labelled:

- A .....
- B .....
- C .....

[2]

(b) State the name of the component of the electromagnetic spectrum that has:

the greatest frequency .....

the smallest wavelength .....

[1]

(c) Explain how X-rays are used to produce an image of a broken bone in a patient's leg.

.....

.....

.....

.....

..... [3]

[Total: 6]

Question	Answer	Marks
4(a)	(A:) ultraviolet (radiation) (B:) infra-red (radiation) (C:) microwaves any **two** correct	**C1**
	all **three** correct	**A1**
4(b)	gamma-rays **and** gamma-rays	**B1**
4(c)	X-rays pass through flesh **and** not (to the same extent) through bone	**B1**
	X-rays detected photographically **or** (digital) detector behind bone	**B1**
	no / less exposure / detection reveals bone **or** exposure / detection reveals break	**B1**

10 Visible light is one component of the electromagnetic spectrum.

(a) State the speed of light in a vacuum.

..... [1]

(b) State the colours of the visible spectrum.

..... [1]

(c) An object O of height 3.0 cm is placed 4.0 cm from the centre of a diverging lens.

Fig. 10.1 shows the object O, the diverging lens and the two focal points (principal focuses), F_1 and F_2, of the lens.

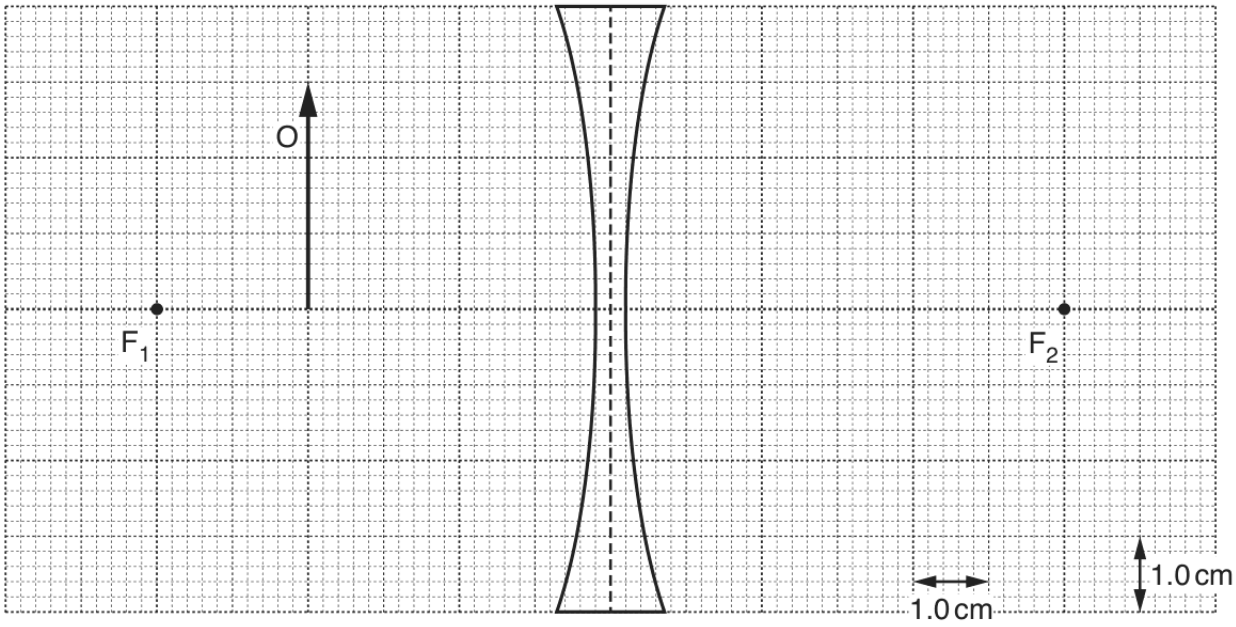


Fig. 10.1

(i) Determine the focal length of this lens.

focal length = ... [1]

(ii) The diverging lens produces an image of object O.

On Fig. 10.1, by drawing rays from the tip of O, locate this image and label it I. [4]

(iii) Determine the height of the image.

height = ... [1]

(iv) Determine the magnification of the image.

magnification = ... [1]

(v) State **two** ways in which a virtual image differs from a real image.

1. .....

.....

2. .....

.....

[2]

(vi) State one use for a diverging lens.

.....

..... [1]

(d) Light passing from air into glass refracts in a similar way to a water wave passing from deep water into shallow water.

Fig. 10.2 represents light passing from air into glass at an angle to the surface.

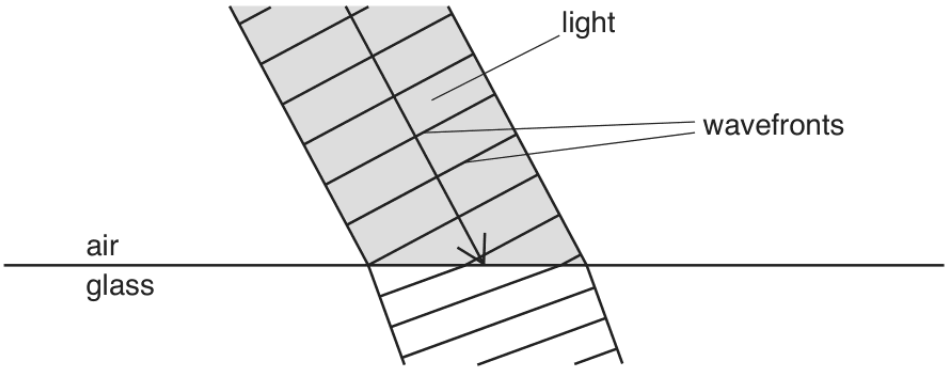


Fig. 10.2

One side of a wavefront strikes the glass before the other side.

Explain why the wavefronts change direction as the light enters the glass.

.....

.....

.....

..... [3]

[Total: 15]

MARK SCHEME · 5054_w19_ms_21.pdf · source page 9 of 10 · Question 10

Question	Answer	Marks
10(a)	3.0×10^8 m / s	**B1**
10(b)	red orange yellow green blue indigo violet (any order)	**B1**
10(c)(i)	(–)6.(0) cm	**B1**

MARK SCHEME · 5054_w19_ms_21.pdf · source page 10 of 10 · Question 10

Question	Answer	Marks
10(c)(ii)	any **two** rays drawn from: paraxial ray that refracts and seems to come from F_1 ray through the optical centre of lens ray that aims for F_2 but refracts and emerges paraxially	**B2**
	rays traced back to point	**B1**
	(point) labelled I **and** rest of image drawn down to the principal axis	**B1**
10(c)(iii)	1.7–1.9 cm	**B1**
10(c)(iv)	candidate's **10(c)(iii)** / 3.0 evaluated	**B1**
10(c)(v)	any **two** from: a real image can be projected on to a screen light actually passes through a real image on same side (of lens) as object **or** on opposite side of mirror to object	**B2**
10(c)(vi)	correction of short-sight / myopia	**B1**
10(d)	light travels more slowly in glass **or** light changes speed	**B1**
	one side / left-hand side of wavefront slows down first	**B1**
	wavelength decreases **or** wavefront travels a shorter distance in the same time	**B1**

3 Solar panels are used to heat water.

(a) The energy absorbed by solar panels comes from the Sun.

(i) Describe how this energy is produced inside the Sun.

.....

.....

..... [2]

(ii) State how this energy is transferred to Earth.

..... [1]

(b) Explain why solar panels are usually black.

.....

.....

.....

..... [2]

[Total: 5]

MARK SCHEME · 5054_w19_ms_22.pdf · source page 5 of 10 · Question 3

Question	Answer	Marks
3(a)(i)	(nuclear) fusion (reactions) **or** nuclei fuse	**B1**
	small nuclei / hydrogen nuclei joining together (and release energy) **or** larger nucleus produced	**B1**
3(a)(ii)	infra-red (radiation) **or** ultraviolet (radiation) **or** (visible) light	**B1**
3(b)	black surfaces are (good) absorbers	**M1**
	of (infra-red) radiation / heat / (visible) light / thermal energy	**A1**

9 Ultrasound and X-rays are both used in medical imaging.

(a) (i) Define what is meant by *ultrasound*.

.....

..... [2]

(ii) Describe what happens to ultrasound waves as they meet the boundary between two different materials.

.....

..... [2]

(iii) To produce the image of an unborn child, an ultrasound emitter and receiver are placed close together on the mother's skin.

Fig. 9.1 shows pulses detected by the receiver.

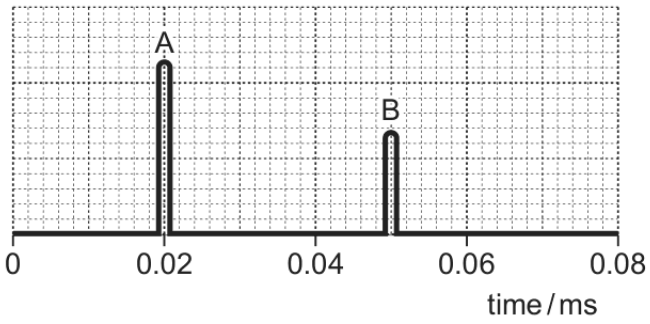


Fig. 9.1

Pulse A is the emitted pulse and pulse B is the first pulse that returns from the unborn child.

The average speed of ultrasound in human tissue is 1500 m / s.

Calculate the distance between the emitter and the child.

distance = ... [3]

(iv) The speed of ultrasound in human tissue is close to the speed of sound in water.

Suggest approximate values for the speed of sound in gases and solids.

speed in gases .....

speed in solids .....
[2]

(b) Fig. 9.2 shows an X-ray image of a hand. An X-ray detector is placed just below the hand. An image of the bones and human tissue around the bones is formed on a screen by the detector.

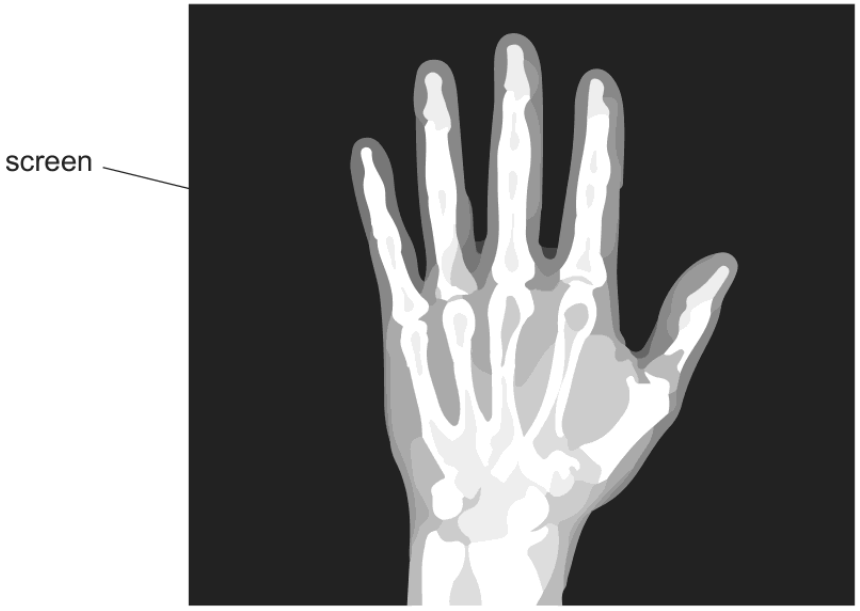


Fig. 9.2

(i) Describe what happens to the X-rays to produce the image.

.....

.....

.....

.....

..... [3]

(ii) The wavelength of the X-rays used is 2.0×10^{-9} m. The speed of electromagnetic waves is 3.0×10^{8} m/s.

Calculate the frequency of the X-rays.

frequency = ... [2]

(iii) Suggest one reason why X-rays are **not** used to form an image of an unborn child.

.....

..... [1]

[Total: 15]

MARK SCHEME · 5054_s20_ms_21.pdf · source page 9 of 11 · Question 9

Question	Answer	Marks
9(a)(i)	(sound that) cannot be heard	**C1**
	(sound with) frequency above 20 kHz	**A1**
9(a)(ii)	(some) reflect	**B1**
	(some) pass into new material **or** (some) absorbed by material	**B1**
9(a)(iii)	(d =) s × t in any form numerical or algebraic	**C1**
	1500×0.03 ($\times 10^{-3}$ / 2)	**C1**
	0.022 m **or** 0.023 m	**A1**
9(a)(iv)	***gases*** 150 – 1000 m/s	**B1**
	solids 2000 – 8000 m/s	**B1**

MARK SCHEME · 5054_s20_ms_21.pdf · source page 10 of 11 · Question 9

Question	Answer	Marks
9(b)(i)	(more) X-rays make the image black(er)	**B1**
	X-rays absorbed by bone **or** do not reach detector through bone	**B1**
	some X-rays pass through tissue **or** tissue absorbs some X-rays	**B1**
9(b)(ii)	(f =) v / λ in any form numerical or algebraic	**C1**
	1.5×10^{17} Hz	**A1**
9(b)(iii)	(X-rays) cause mutation / cancer / kill cells / ionisation	**B1**

5 The components of the electromagnetic spectrum have different uses.

Microwaves are used in cooking and ultraviolet rays are used in sterilisation.

(a) Draw **one** line from each component of the spectrum to another suitable use for that component.

component	use
microwaves	sunbeds
	television controller
ultraviolet	satellite television
	airport security check of cases

[2]

(b) Fig. 5.1 shows a microwave oven used to heat soup. The container for the soup is a glass bowl.

Microwaves created inside the oven are reflected by the metal walls.

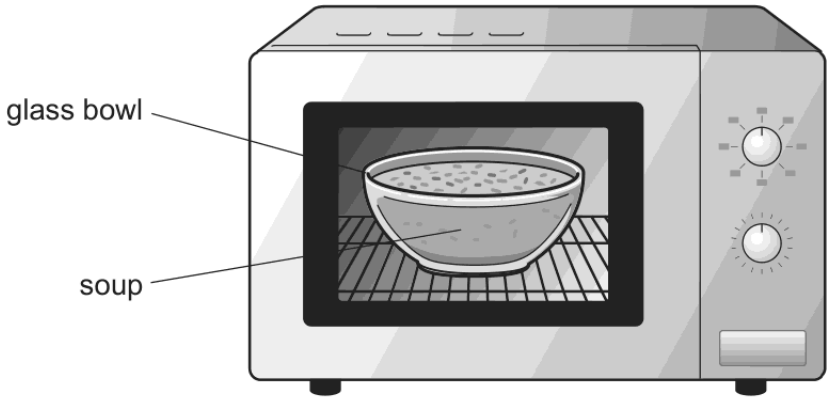


Fig. 5.1

(i) Explain why the choice of material for the container is important in microwave cooking.

.....

.....

..... [1]

(ii) The soup is mostly water.

Microwaves are completely absorbed by a few centimetres of water.

As a result, microwaves do not reach the centre of the soup.

The instructions suggest that, after the microwave oven is turned off, the soup:

- is not stirred
- is left for some minutes so that the centre becomes hot.

State the name of and describe each of the **two** processes by which thermal energy transfers throughout the soup after the microwave oven is turned off.

name of first process .....

description .....

.....

.....

name of second process .....

description .....

.....

..... [3]

[Total: 6]

MARK SCHEME · 5054_s20_ms_22.pdf · source page 8 of 13 · Question 5

Question	Answer	Marks
5(a)	microwaves → satellite television	**B1**
	ultraviolet → sunbeds	**B1**
5(b)(i)	(microwaves) must pass through **or** not be absorbed (by container) **or** not reflected	**B1**

MARK SCHEME · 5054_s20_ms_22.pdf · source page 9 of 13 · Question 5

Question	Answer	Marks
5(b)(ii)	convection **and/or** conduction named	**B1**
	(convection involves) hot liquid rises (cold liquid falls)	**B1**
	(conduction involves) energy passes from molecule to molecule	**B1**

QUESTION · 5054_w20_qp_12.pdf · source page 10 of 16

25 Applications use different components of the electromagnetic spectrum.

Which shows correct applications for X-rays, ultraviolet light and microwaves?

	X-rays	ultraviolet light	microwaves
A	mobile phone	fluorescent tube	intruder alarm
B	killing cancerous cells	sterilising surgical instruments	satellite television
C	medical imaging	television controller	sunbed
D	sterilising surgical instruments	television controller	detecting cracks in metal

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 25

25	B	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 10 of 16

27 Which two waves are components of the electromagnetic spectrum?

A light and sound

B water waves and infrared

C ultrasound and ultraviolet

D X-rays and microwaves

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 27

27	D	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 11 of 20

24 Which component of the electromagnetic spectrum is used for the remote control of a television?

A gamma rays

B infrared rays

C radio waves

D ultraviolet rays

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 24

24	B	**1**

3 A type of nuclear reaction takes place at the centre of the Sun. The reaction releases thermal energy.

(a) State the name of the type of nuclear reaction that takes place at the centre of the Sun.

..... [1]

(b) Thermal energy is emitted from the surface of the Sun into space. The energy is transferred through the vacuum of space by thermal radiation.

(i) Describe the radiation that is emitted from the surface of the Sun.

.....

.....

..... [2]

(ii) State the speed of this radiation in a vacuum.

speed = ... [1]

(iii) The Earth is 1.5×10^{11} m from the surface of the Sun.

Calculate the time taken for this radiation to travel to Earth.

time = ... [2]

(c) Explain one advantage of wearing white clothes on sunny days.

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
3(a)	(nuclear) fusion	**B1**
3(b)(i)	electromagnetic (waves)	**B1**
	produced by hot objects **or** infrared (radiation)	**B1**
3(b)(ii)	3.0×10^8 m / s	**B1**
3(b)(iii)	(t =) x / v **or** $1.5 \times 10^{11} / 3.0 \times 10^8$	**C1**
	500 s	**A1**
3(c)	white surfaces are (good) reflectors / poor absorbers of radiation	**B1**
	less thermal energy absorbed / wearer stays cooler	**B1**

QUESTION · 5054_w21_qp_12.pdf · source page 11 of 16

27 A satellite orbits the Earth at a height of 300 km above the surface. The speed of electromagnetic waves is 3.0×10^8 m/s.

What is the minimum time taken for a radio signal transmitted from the Earth to reach the satellite?

A 1.0×10^{-6} s **B** 2.0×10^{-6} s **C** 1.0×10^{-3} s **D** 2.0×10^{-3} s

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 27

27	C	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 9 of 16

23 Which waves are **not** part of the electromagnetic spectrum?

A radio waves

B television signals

C ultrasound

D ultraviolet

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 23

23	C	**1**

4 Fig. 4.1 shows a ray of white light incident on a glass prism.

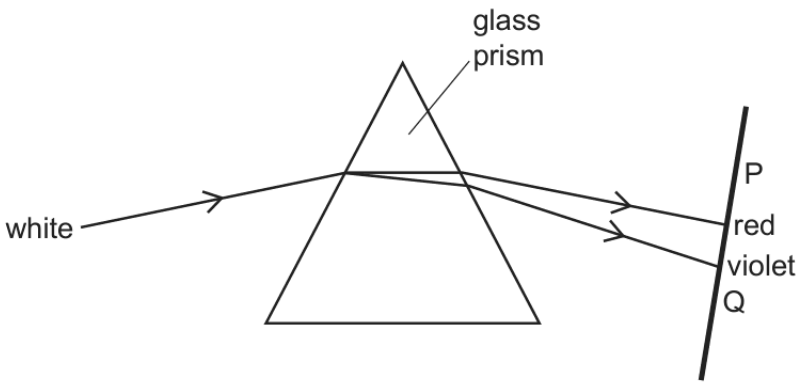


Fig. 4.1 (not to scale)

Refraction causes the white light to separate into different colours.

(a) Define the term 'angle of refraction'.

.....

..... [2]

(b) The angle of incidence of the white light as it enters the prism is 40° and the angle of refraction for the red light is 25°.

Calculate the refractive index of the glass for red light. Show your working.

refractive index = ... [2]

(c) Using Fig. 4.1, state and explain how the refractive index for red light differs from the refractive index for violet light.

.....

.....

.....

..... [2]

(d) The source of white light used in Fig. 4.1 produces other types of electromagnetic radiation as well as visible light.

State the name of the invisible radiation found at P and the invisible radiation found at Q.

at P ..... at Q ..... [1]

[Total: 7]

Question	Answer	Marks
4(a)	clear angle is to / from the normal **or** 90°– angle between ray and surface	**C1**
	angle between/with the refracted ray / beam and the normal or vv **or** angle from the normal to the refracted ray / beam or vv	**A1**
4(b)	(refractive index =) sin*i* / sin*r* algebraic or numerical in any form	**M1**
	1.5	**A1**
4(c)	refractive index smaller (for red)	**B1**
	larger angle of refraction (for red) **or** (red) bends / refracts less (towards normal) assume comparison is for light **entering** glass at equal incident angles unless clear otherwise	**B1**
4(d)	P infrared **and Q** ultraviolet	**B1**

8 **(a)** Fig. 8.1 shows a ripple tank and the crests of the water wave that is produced in it.

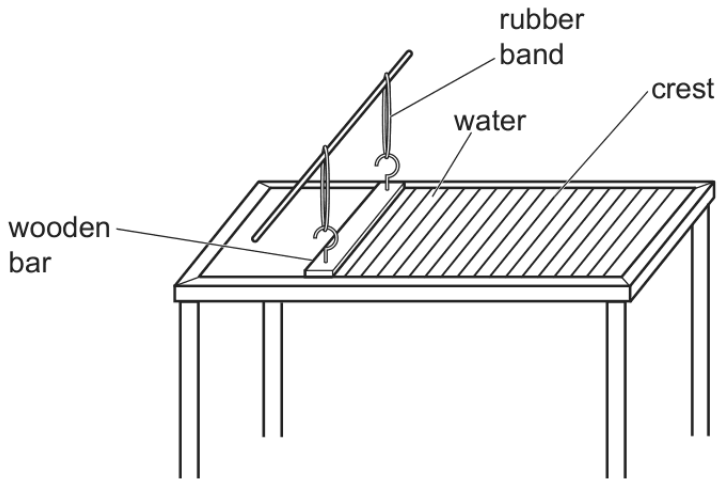


Fig. 8.1

The frequency of the water wave is 2.0 Hz and its amplitude is 3.0 mm.

(i) Calculate the number of crests produced in 1.5 s.

number = ... [1]

(ii) The height of the wave is measured from the level of the undisturbed surface of the water. The height of the wave at one point is 0 at time = 0.

On Fig. 8.2, draw a graph to show how the height of the wave at this point varies with time.

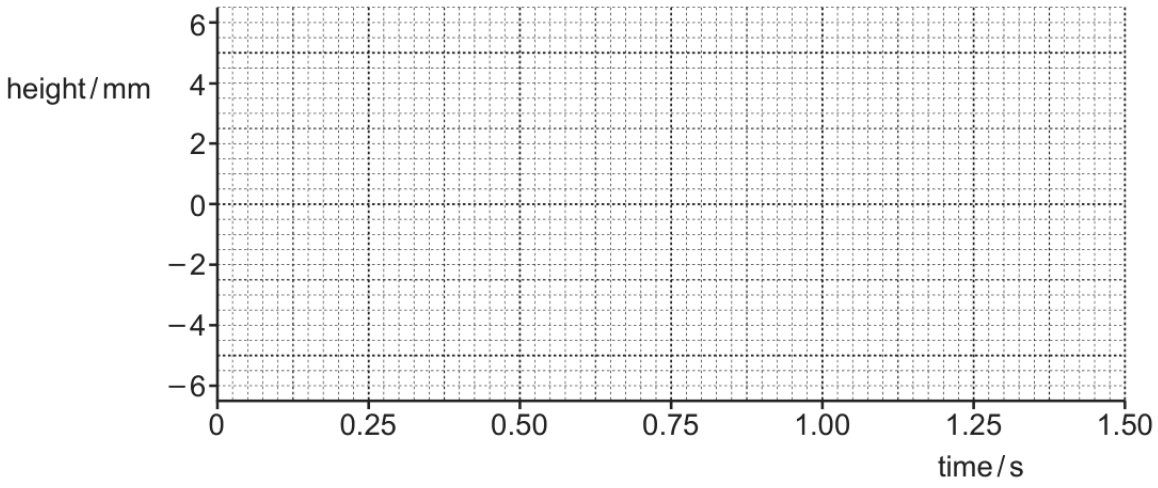


Fig. 8.2

[2]

(b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....

..... [1]

(ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed .....

wavelength .....
[2]

(c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

(i) State the additional apparatus needed to demonstrate refraction.

..... [1]

(ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.

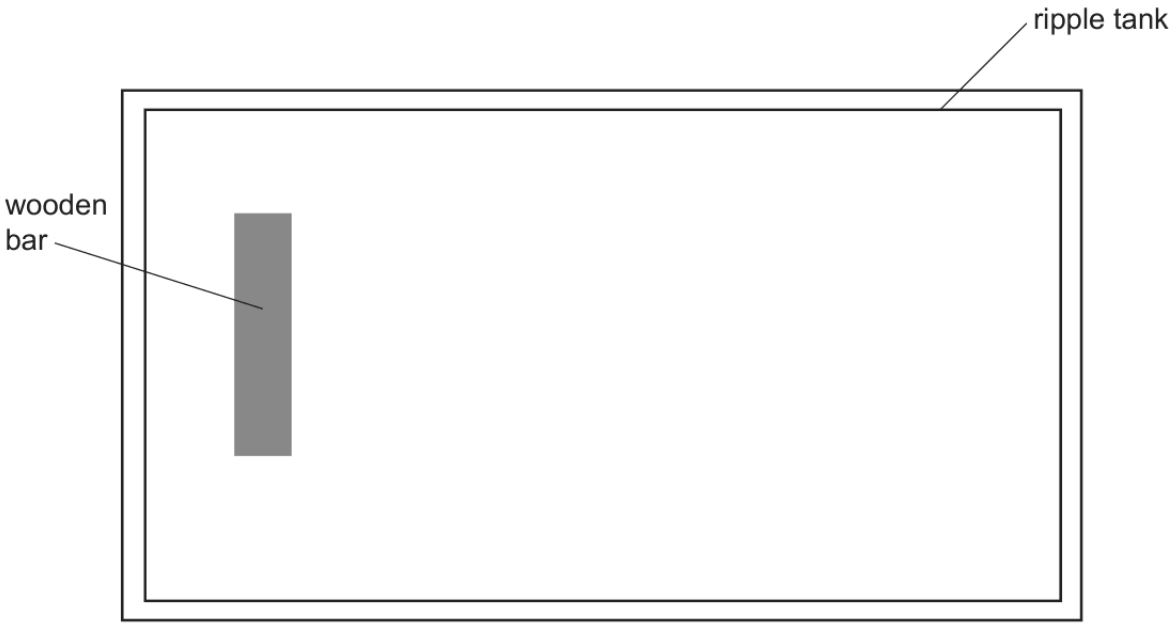


Fig. 8.3

[3]

(d) Fig. 8.4 shows a connection to the internet made from a remote station A using a satellite above the Earth's surface.

Data is sent between stations A and B using microwaves which travel to and from the satellite.

Station B is connected directly to the internet using optical fibre.

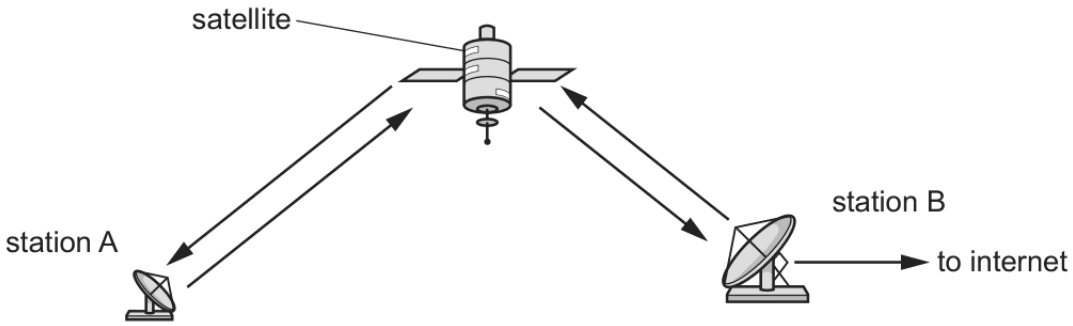


Fig. 8.4 (not to scale)

The speed of microwaves is 3.0×10^8 m/s.

Stations A and B are each 560 km from the satellite.

(i) Calculate the time taken for data to travel from A to B using microwaves.

time = ... [3]

(ii) The same data can be sent from A to B in a shorter time along the surface of the Earth using infrared waves to carry the information in optical fibres.

State **one** property that infrared waves and microwaves have in common.

.....

..... [1]

(iii) Suggest **one** other advantage of using optical fibres to connect stations A and B directly.

.....

..... [1]

[Total: 15]

Question 9 begins over the page.

Question	Answer	Marks
8(a)(i)	3	**B1**
8(a)(ii)	shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change)	**B1**
	1 wave taking 0.5 s for first wave shown within half a square **or** 1.5 s shown for the number of waves in **(a)(i)**	**B1**
8(b)(i)	move bar up and down more often (per second) / faster **or** make rubber band(s) strong(er) / short(er) / tight(er) / more in number **or** use bar of less mass / weight	**B1**
8(b)(ii)	*speed* remains the same	**B1**
	wavelength decreases	**B1**
8(c)(i)	(piece of) plastic / (piece of) glass / block / plate **or** a shallow(er) / deep(er) area / boundary	**B1**
8(c)(ii)	diagram showing within the ripple tank: • crests parallel to bar hitting boundary with a boundary labelled	**B1**
	• correct refraction of plane crests either towards or away from normal on the same side of the normal **or** no direction change if crests are initially parallel to boundary but with a wavelength change	**B1**
	• different wavelength shown before and after refraction and constant before and after (by eye)	**B1**
8(d)(i)	(t =) distance / speed algebraic or numerical seen in any form	**C1**
	2 × 560(000) seen or used **or** 0.0019 (s)	**C1**
	0.0037 s	**A1**

Question	Answer	Marks
8(d)(ii)	One of (both): • reflect / refract • have same speed (in a vacuum) • travel in a vacuum / don't need a medium • are electromagnetic • are transverse • carry energy / cause heating • have larger wavelength / smaller frequency than visible light	**B1**
8(d)(iii)	less likely to be hacked / intercepted / greater security **or** more channels / better bandwidth possible / more data **or** less likely for transmission to be interrupted **or** cost of satellites saved / satellites expensive	**B1**

4 X-rays are transverse waves that are part of a spectrum of waves.

(a) State the name of the spectrum of waves that includes X-rays.

..... [1]

(b) **(i)** Explain what is meant by 'frequency'.

.....

.....

..... [2]

(ii) State the name of the waves in this spectrum that have the greatest frequency.

..... [1]

(c) X-rays are used in hospitals to produce an image of broken bones.

(i) Explain how this is done. You may draw a diagram if you wish.

.....

.....

.....

..... [3]

(ii) Explain why X-rays are **not** used for pre-natal scanning.

.....

..... [1]

[Total: 8]

Question	Answer	Marks
4(a)	electromagnetic (waves) **or** components of electromagnetic spectrum	**B1**
4(b)(i)	number of oscillations / cycles / wavelengths	**B1**
	per unit time	**B1**
4(b)(ii)	gamma rays	**B1**
4(c)(i)	X-rays directed towards body part with (suspected) broken bone	**B1**
	X-rays pass through flesh but not (to same extent) through bones	**B1**
	image produced by digital / electronic / CCD / CMOS detector	**B1**
4(c)(ii)	X-rays (may) cause cancer / cell mutation / birth defect / miscarriage **or** X-rays show the bone structure rather than foetal development **or** image would not be clear	**B1**

QUESTION · 5054_w22_qp_12.pdf · source page 11 of 16

26 Many devices produce electromagnetic waves when operating.

Which device produces electromagnetic waves of the highest frequency?

A mobile phone

B sunbed

C television controller

D toaster

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 26

26	B	**1**

QUESTION · 5054_s23_qp_11.pdf · source page 11 of 20

23 Which row names a region of the electromagnetic spectrum with a wavelength longer than that of visible light and a region with a frequency that is greater than that of visible light?

	wavelength longer than visible light	frequency greater than visible light
A	infrared	ultraviolet
B	microwave	radio
C	radio	microwave
D	ultraviolet	infrared

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 23

23	A	**1**

7 **(a)** Ultraviolet radiation is one component of the electromagnetic spectrum.

(i) State the name of **two** components of the electromagnetic spectrum that have a smaller wavelength than ultraviolet radiation.

1 .....

2 .....
[1]

(ii) State **one** useful application of ultraviolet radiation.

..... [1]

(iii) Exposure to ultraviolet radiation from the Sun damages the eyes.

State **one** type of damage to the eye caused by ultraviolet radiation.

..... [1]

(b) Fig. 7.1 shows a ray of light. The ray passes into a semi-circular block of glass at A and leaves the glass at B, travelling along the surface to C.

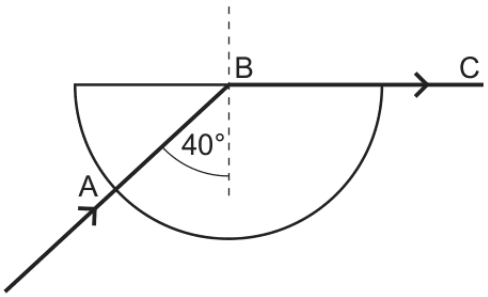


Fig. 7.1

(i) State the name given to the angle of incidence marked as 40°.

..... [1]

(ii) Calculate the refractive index of the glass.

refractive index = ... [2]

[Total: 6]

MARK SCHEME · 5054_s23_ms_22.pdf · source page 9 of 12 · Question 7

Question	Answer	Marks
7(a)(i)	X-rays **and** gamma (radiation / rays)	**B1**
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	**B1**
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	**B1**

MARK SCHEME · 5054_s23_ms_22.pdf · source page 10 of 12 · Question 7

Question	Answer	Marks
7(b)(i)	critical angle	**B1**
7(b)(ii)	(n =) sin$i$ / sin$r$ **or** 1 / sin$c$	**C1**
	1.6	**A1**

5 Most of the energy emitted by the Sun is in three regions of the electromagnetic spectrum.

(a) The name of one of these three regions is the ultraviolet region.

(i) State the names of all three regions in order of increasing frequency.

smallest frequency .....

intermediate frequency .....

largest frequency .....

[2]

(ii) Some of the ultraviolet radiation emitted by the Sun reaches the surface of the Earth.

Give one possible damaging effect of ultraviolet radiation on the human body and the property of ultraviolet radiation that causes this damage.

damaging effect .....

.....

property .....

.....

[2]

(b) Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

.....

.....

..... [2]

[Total: 6]

Question	Answer	Marks
5(a)(i)	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation)	**B1**
	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation) **and** in this order	**B1**
5(a)(ii)	damaging effect: (skin) cancer / cataracts	**B1**
	property: it / ultraviolet radiation is ionising	**B1**

Question	Answer	Marks
5(b)	in a transverse wave, the vibration direction is perpendicular / at right angles to the propagation direction	**B1**
	in a longitudinal wave, the vibration direction is parallel to the propagation direction	**B1**

QUESTION · 5054_w23_qp_12.pdf · source page 12 of 20

30 Ultraviolet radiation is a component of the electromagnetic spectrum.

Which application uses ultraviolet radiation?

A Bluetooth technology

B prenatal scanning

C sterilising water

D thermal imaging

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 3 of 3 · row for Question 30

30	C	**1**

10 The isotope iodine-131 emits two types of nuclear radiation. Gamma radiation is one of the types emitted.

A sample of iodine-131 produces a narrow beam of radiation which passes into a magnetic field.

Fig. 10.1 shows some of the paths taken in the magnetic field by the radiation emitted from the sample.

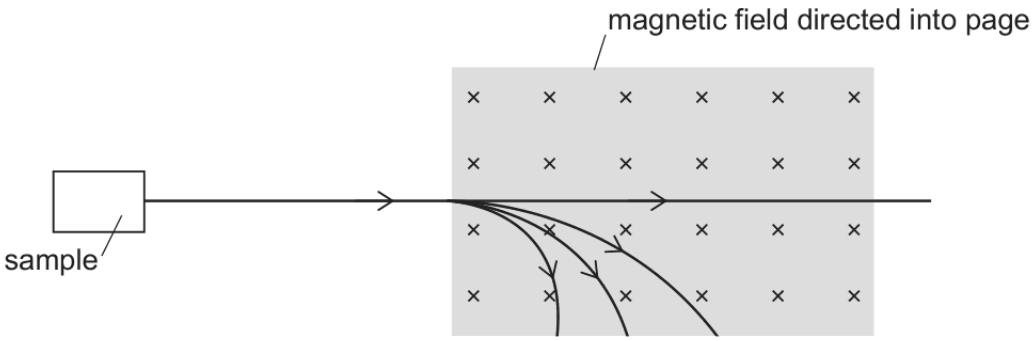


Fig. 10.1

(a) The direction of the magnetic field is into the page.

(i) On Fig. 10.1, mark with a gamma symbol (γ) any of the paths taken by the gamma radiation. [1]

(ii) Determine the other type of radiation emitted by iodine-131.

Explain how you obtain your answer.

type of radiation .....

explanation .....

.....

.....

.....
[4]

(iii) In another experiment, the beam of radiation from the sample is incident on a 5 mm thick sheet of lead.

State what happens to the two types of radiation in the beam.

.....

.....

..... [2]

Question 10 continues over the page.

(b) The gamma radiation produced has a frequency of 8.8×10^{19} Hz.

The speed of electromagnetic radiation in a vacuum is 3.0×10^{8} m/s.

Calculate the wavelength of this gamma radiation in a vacuum.

wavelength = ... m [2]

[Total: 9]

Question	Answer	Marks
10(a)(i)	horizontal path marked with γ **and** no other path indicated as γ	**B1**
10(a)(ii)	beta (β-radiation)	**B1**
	left-hand rule (mentioned)	**B1**
	first finger into page **and** thumb downwards	**B1**
	current opposite to motion **or** particles negative (and so are beta)	**B1**
10(a)(iii)	beta (β-radiation) completely absorbed	**B1**
	some gamma rays absorbed **or** (some) gamma passes through	**B1**
10(b)	($\lambda =$) $v / f = 3.0 \times 10^{8} / 8.8 \times 10^{19}$	**C1**
	3.4×10^{-12} (m)	**A1**

QUESTION · 5054_s24_qp_11.pdf · source page 10 of 16

27 Which components of the electromagnetic spectrum are used to communicate from a remote controller to a television set and from the Earth to a satellite?

	remote controller to TV set	the Earth to a satellite
A	infrared	microwave
B	infrared	ultraviolet
C	light	microwave
D	light	ultraviolet

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 27

27	A	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 8 of 16

22 Which statement about infrared radiation is correct?

A In a vacuum, infrared radiation travels at the speed of light.

B Infrared radiation is a longitudinal wave.

C Infrared radiation has a higher frequency than ultraviolet radiation.

D White surfaces are better emitters of infrared radiation than black surfaces.

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 22

22	A	**1**

4 **(a)** Fig. 4.1 shows light passing through a triangular glass prism.

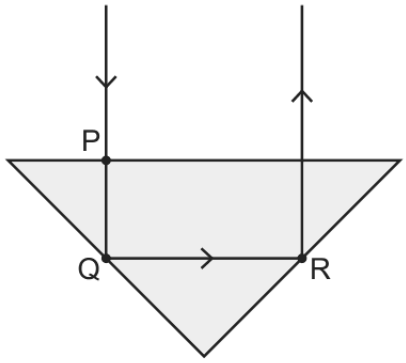


Fig. 4.1

(i) State the value of the angle of incidence at point P.

angle of incidence = ... ° [1]

(ii) Draw the normal and the angle of incidence at point R.

Label the angle of incidence. [2]

(iii) State **two** conditions needed so that no light refracts from the glass into the air at point Q.

1 .....

.....

2 .....

.....
[2]

(b) Information is sent across the internet using pulses of visible light through long, thin glass fibres and electrical signals through copper wires.

(i) State the name of **one** other type of electromagnetic radiation used to transmit information through long, thin glass fibres.

..... [1]

(ii) Suggest **two** advantages of using glass fibres rather than copper wires to transmit information from the internet.

1 .....

.....

2 .....

.....
[2]

[Total: 8]

Question	Answer	Marks
4(a)(i)	0	**B1**
4(a)(ii)	normal correct by eye	**B1**
	angle of incidence correct and labelled	**B1**
4(a)(ii)	angle of incidence greater than critical angle	**B1**
	from more dense to less dense (medium) / from high to low refractive index	**B1**
4(b)(i)	infrared	**B1**
4(b)(ii)	ANY 2 from • more calls per second **or** greater bandwidth **or** faster speed • less chance of hacking • less interference • less (information) noise • longer transmission distances (before losing signal) / less attenuation • thinner **or** lighter **or** less mass/weight	**B2**

QUESTION · 5054_w24_qp_11.pdf · source page 11 of 16

27 A student makes four statements about infrared waves.

1 They can be seen by the human eye.

2 They travel at 330 m / s in air.

3 They are transverse waves.

4 They are used by intruder alarms.

Which statements are correct for infrared waves?

A 1 and 3 **B** 2, 3 and 4 **C** 2 only **D** 3 and 4 only

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 2 of 3 · row for Question 27

27	D	**1**

4 A student directs a beam of white light from a filament lamp towards a glass prism in a dark room.

Fig. 4.1 shows the beam of white light incident on the left-hand side of the glass prism.

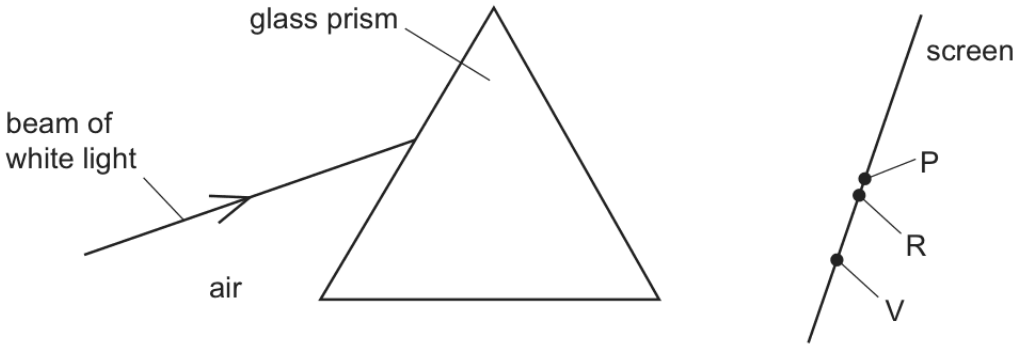


Fig. 4.1

A screen is placed to the right of the glass prism. Red light is observed at point R on the screen and violet light is observed at point V.

(a) Draw on Fig. 4.1 to show the paths of the red light and the violet light between the left-hand side of the prism and the screen at point R and point V. [2]

(b) State what these observations show about the change in speed of red light and change in speed of violet light as each enters the glass from the air.

.....

.....

..... [2]

(c) Five other colours of light appear on the screen between the red light at point R and the violet light at point V.

(i) State the names of the five colours and list them in the correct order from red to violet.

red ..... **violet** [2]

(ii) State how the frequency and the wavelength of the coloured light change going from red light to violet light.

frequency .....

wavelength .....
[1]

(d) On Fig. 4.1, point P is shown immediately next to point R.

No light is observed at point P, but a detector at point P registers radiation reaching the detector.

State which region of the electromagnetic spectrum is detected at point P.

..... [1]

[Total: 8]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

MARK SCHEME · 5054_w24_ms_21.pdf · source page 6 of 10 · Question 4

Question	Answer	Marks
4(a)	at least **one** ray from left-hand side showing correct refraction at both faces **and** no incorrect rays	**B1**
	at least **two** correct rays (i.e. two rays within glass) **and** one ending at R and the other at V	**B1**
4(b)	both (red and violet light) travel more slowly in glass than in air	**B1**
	red light travels more quickly (than violet light) in glass	**B1**
4(c)(i)	any **four** (of the five) colours in order **or** five colours in any order	**C1**
	red orange yellow green blue indigo violet	**B1**

MARK SCHEME · 5054_w24_ms_21.pdf · source page 7 of 10 · Question 4

Question	Answer	Marks
4(c)(ii)	frequency increases **and** wavelength decreases	**B1**
4(d)	infrared (radiation)	**B1**

QUESTION · 5054_w24_qp_12.pdf · source page 9 of 16

23 Which two waves are components of the electromagnetic spectrum?

A light and sound

B ultrasound and ultraviolet

C water waves and infrared

D X-rays and microwaves

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 23

23	D	**1**

5 The microwave region is one region of the electromagnetic spectrum.

(a) Fig. 5.1 represents the electromagnetic spectrum divided into the seven main regions.

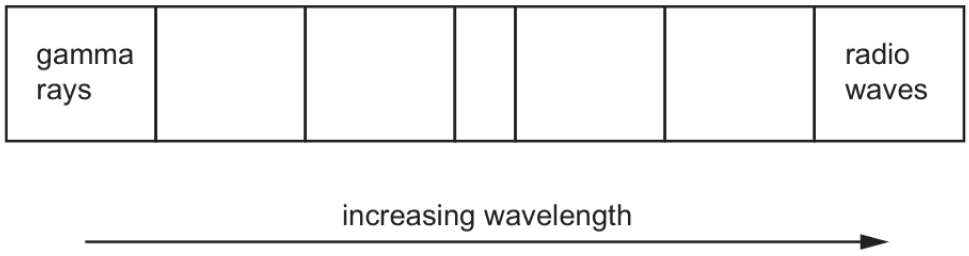


Fig. 5.1

The regions are arranged from left to right in order of increasing wavelength.

On Fig. 5.1:

- indicate the microwave region by marking it with the letter M
- indicate the ultraviolet region by marking it with the letters UV.

[1]

(b) The frequency of the microwaves used by some satellite television systems is 12 GHz.

(i) The speed of microwaves in a vacuum is 3.0×10^8 m/s.

Calculate the wavelength of these microwaves in a vacuum.

wavelength = ... m [3]

(ii) Describe how microwaves are used in satellite television systems.

.....

.....

.....

..... [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

(c) All microwave ovens have a switch that turns the oven off when the door is opened.

Suggest why it is important for a microwave oven to switch off when the door is opened.

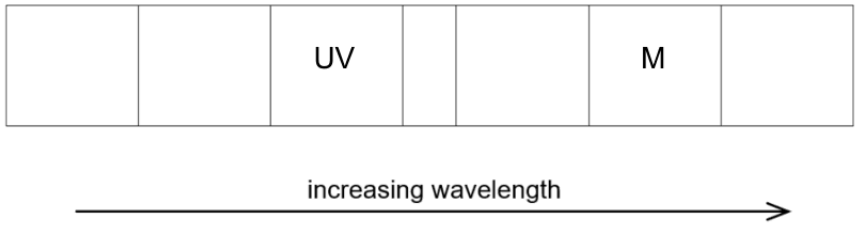
.....

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
5(a)		**B1**
5(b)(i)	$(\lambda =) \frac{c}{f}$ **or** $\frac{3.0 \times 10^{8}}{1.2 \times 10^{N}}$	**C1**
	2.5×10^{X} (m)	**C1**
	0.025 (m)	**A1**
5(b)(ii)	encoded microwave / microwave signal / microwave carrying data / microwave transmitting data (to satellite from Earth)	**B1**
	satellite transmits microwave to Earth / dish / antenna / subscriber / television	**B1**
5(c)	to prevent microwaves escaping / leaving the oven / spreading out of the device	**B1**
	microwaves / they can cause burns (in living tissue / people) / damage to tissue through heating / blisters	**B1**

QUESTION · 5054_s26_qp_11.pdf · source page 12 of 16

28 Which statement is correct?

A Gamma rays have a longer wavelength than ultraviolet waves.

B Infrared waves have a lower frequency than radio waves.

C Microwaves have a longer wavelength than visible light.

D X-rays have a higher speed in air than visible light.

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 28

28	C	**1**

4 **(a)** All electromagnetic waves transfer energy.

State **two** other properties that are the same for all electromagnetic waves.

1 .....

.....

2 .....

.....

[2]

(b) Figure 4.1 shows an X-ray image of healthy teeth.

The detector used to create the image is placed inside the mouth, behind the teeth.

A source of X-rays is placed outside the mouth, in front of the teeth.

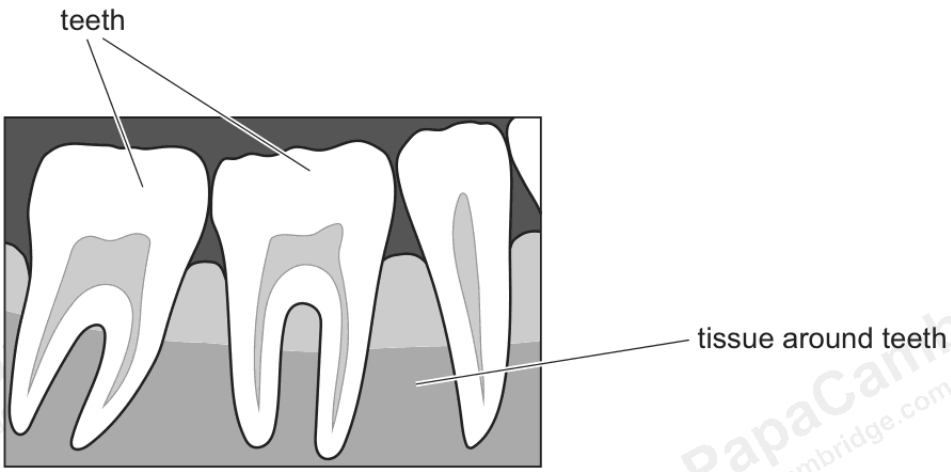


Figure 4.1

Explain why the X-rays produce an image that shows white teeth, grey tissue around the teeth and a black background.

.....

.....

.....

.....

..... [2]

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(c) Ultrasound is another type of wave used in hospitals to produce medical images.

A scanner placed on the skin of a patient emits an ultrasound wave.

(i) The ultrasound wave causes a motion of the particles in the tissue of the patient.

Describe this motion.

.....

.....

..... [2]

(ii) The ultrasound wave is reflected from a bone in the patient and arrives back at the scanner 8.0×10^{-5} s after it was emitted.

The speed of ultrasound in the patient is 1500 m / s.

Calculate the distance between the scanner and the bone.

distance = ... m [2]

(iii) Some of the ultrasound passes through the patient and emerges into the air.

The speed of ultrasound in air is smaller than the speed of ultrasound in the patient.

State and explain the change in wavelength as ultrasound travels from the patient into the air.

change in wavelength .....

explanation .....

.....

.....

.....

[2]

[Total: 10]

Question	Answer	Marks	Guidance
4(a)	Any 2 from: • can reflect / diffract / refract • travel through a vacuum / do not require a medium • transverse • are oscillating electric and magnetic fields • travel at the speed of light / same speed (in vacuum/air)	**2**	**I** no mass, no charge
4(b)	Any two from: • X rays pass through the spaces **or** X-rays make the image darker / go black • (most / all) X-rays absorbed by / cannot penetrate the teeth • X-rays partially absorbed by the tissue / some pass through tissue	**2**	X-rays reflected max 1 mark
4(c)(i)	oscillation / vibration longitudinal **or** backwards and forwards **or** in the direction of the sound / parallel to the direction of travel of the wave **or** formation of compressions and rarefactions	**1**	**A** backwards and forwards or any clear oscillation, eg up and down; left and right; to and fro **A** ultrasound longitudinal **I** left and right, up and down
4(c)(ii)	(d =) vt algebraic or numerical 0.06 (0) m	**1**	**A** ½ vt; ½ × 8.0 × 10^{-5} × 1500; 0.12 m; 12 cm
4(c)(iii)	(wavelength) decreases (in air) $v = f\lambda$ **and** frequency the same **or** each (complete) wave travels less distance (before the next arrives)	**1**	**A** frequency the same **and** $v \propto \lambda$

QUESTION · 5054_s26_qp_12.pdf · source page 11 of 16

26 Which statement is correct?

A Infrared radiation **cannot** travel in a vacuum.

B Infrared radiation **cannot** travel in solids or in gases.

C Infrared radiation can only travel in a vacuum.

D Infrared radiation can travel in a vacuum and in gases.

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 26

26	D	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 11 of 16

27 Ultraviolet waves and radio waves are both components of the electromagnetic spectrum.

Why do ultraviolet waves cause skin cancer and cataracts while radio waves do **not**?

A Ultraviolet waves have a greater speed in air than radio waves.

B Ultraviolet waves have longer wavelengths than radio waves.

C Ultraviolet waves have lower frequencies than radio waves.

D Ultraviolet waves are ionising, but radio waves are **not** ionising.

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MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 27

27	D	**1**

5 A student uses a glass block and two identical glass prisms to demonstrate the action of a converging lens.

Figure 5.1 shows three parallel rays of red light incident on the glass block and the two prisms.

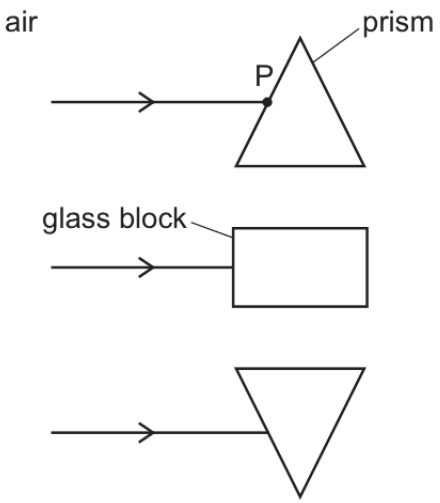

Figure 5.1

(a) (i) On Figure 5.1, draw and label the normal to the prism at point P. [1]

(ii) Complete Figure 5.1 to show how the rays of light travel through and out of both prisms and the glass block. [2]

(iii) The action of a diverging lens is also demonstrated using the two identical glass prisms and the glass block.

On Figure 5.2, draw the prisms arranged so that they demonstrate the action of a diverging lens.

Show the rays of light travelling through the prisms and the glass block.

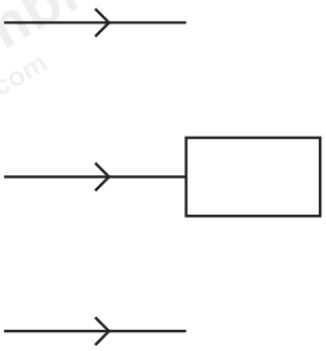

Figure 5.2

[2]

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(b) Figure 5.3 shows a converging lens, an object and two rays of light from the top of the object. The principal focus (focal point) on each side of the lens is labelled F.

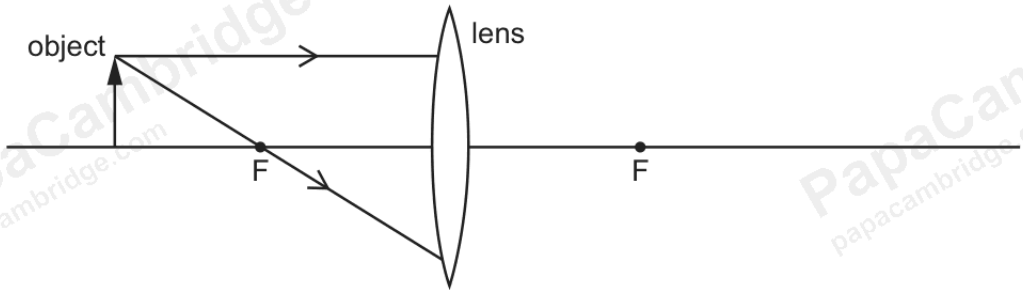


Figure 5.3

Continue the paths of both rays to show where the image of the object is formed. Draw and label the image. [2]

(c) Figure 5.4 shows the electromagnetic spectrum.

radio waves	microwaves	infrared	visible light	ultraviolet	X-rays	gamma rays

Figure 5.4

State **two** properties of electromagnetic waves that change from one end of the spectrum to the other end, and describe how these properties change from one end to the other.

1 .....

.....

2 .....

.....

[2]

[Total: 9]

5(a)(i)	correct normal	**B1**	by eye; either one side or the other of boundary or both

bod

Question	Answer	Marks	Guidance
5(a)(ii)	three correct refractions on entering and leaving the prisms	**B1**	**I** central block **con** dispersion, e.g. ray splits on refraction **allow** one wrong refraction of a ray at a boundary
	horizontal ray through and emerging from central glass block	**B1**	
5(a)(iii)	correct arrangement of prisms	**B1**	i.e. Intention that prisms inverted vertically, e.g. apex of prisms face central block
	rays diverge after prisms (top diverges up and bottom diverges down)	**B1**	**I** refraction in block and prisms**; I** ray leaving the block **not** if rays split on refraction; no prisms shown
5(b)	one correct ray; top ray passes through / to F on right of lens **or** bottom ray horizontal on right of lens	**C1**	
	both rays correct **and** image shown	**A1**	**A** point of intersection marked as I / image **A** arrow either way up as image at image position without a label
5(c)	ANY 2 from • wavelength decreases (towards the right) • frequency increases (towards the right) • damage / ionisation / energy increases (towards the right)	**B2**	**A** reverse arguments **con** speed changes assume left to right unless stated otherwise **A** just 'radio waves largest / large(r) λ'; 'γ small(est) λ' 'gamma have high(est) / high(er) frequency'; 'radio low(est) frequency' **SC** allow 1 mark for *wavelength increases and freq. decreases;* radio has largest λ **and** smallest frequency ORA for gamma

Chapter 3 Waves

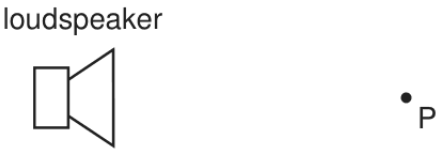
3.4 Sound

153 entries: 115 Paper 1 questions, 38 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

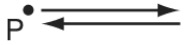
QUESTION · 5054_s04_qp_1.pdf · source page 11 of 20

24 The diagram shows a loudspeaker that is producing a continuous sound wave of frequency 200 Hz in air.



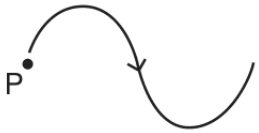
Which diagram best shows how the sound causes a molecule at P to move during $\frac{1}{200}$ s ?

A



B

C



D

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 24

24	**A**

For Examiner's Use

4 Fig. 4.1 shows a student clapping in front of a vertical wall. The wall reflects the sound.

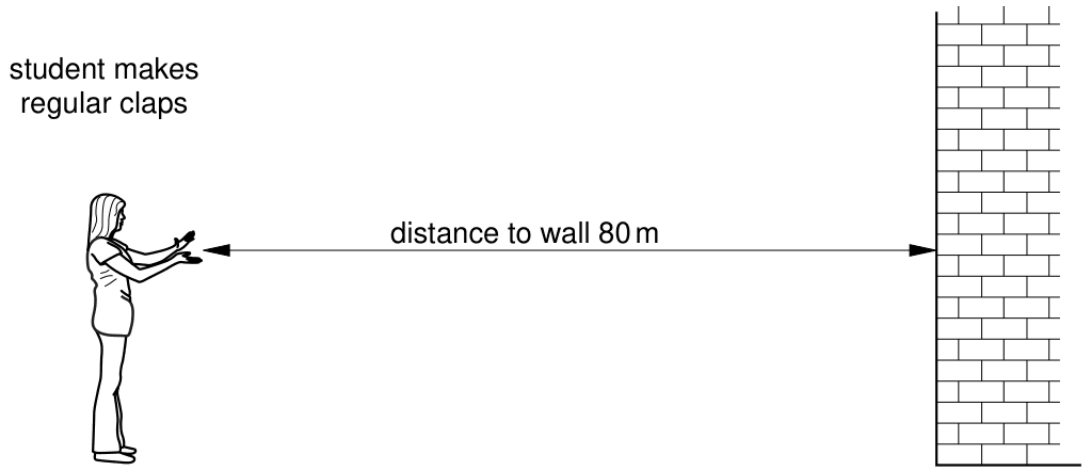


Fig. 4.1

The student changes the number of claps made in 1 minute until the reflection of each clap returns to her at exactly the same time as she makes the next clap.

The speed of sound in air is 330 m/s.

(a) Explain what is meant by *speed*.

.....

.....[1]

(b) Calculate the time between claps.

time = …….. [3]

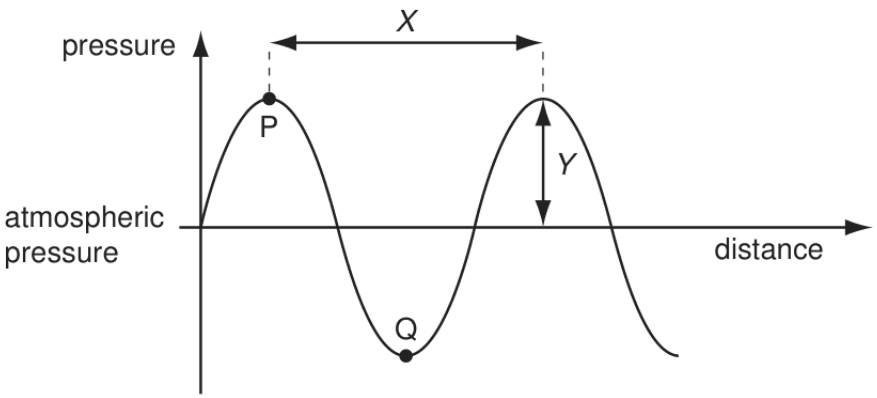
(c) Calculate the number of claps in 1 minute.

number of claps = …….. [2]

4	**(a)**	distance traveled per unit time **or** in one second / distance ÷ time		
		or rate of change of distance		B1
	(b)	s = d/t in any algebraic or numerical form		C1
		any doubling of distance or final time		C1
		0.48 s	(allow 0.24s 2/3 accept 0.5s)	A1
	(c)	60/0.48 (5)		C1
		123.75 accept 120, 123, 124	(ecf **(b)**)	A1
			Total	**[6]**

QUESTION · 5054_s05_qp_1.pdf · source page 10 of 20

24 The graph shows, at one instant, the pressure variation along a sound wave.



Which point on the diagram represents a rarefaction and what is the wavelength of the sound wave?

	rarefaction at	wavelength is
A	P	*X*
B	P	*Y*
C	Q	*X*
D	Q	*Y*

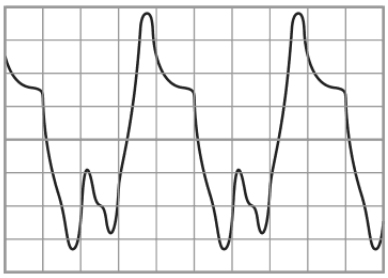
MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 24

24 **C**

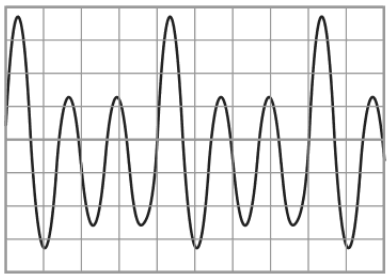
QUESTION · 5054_w05_qp_1.pdf · source page 13 of 20

24 Wave forms are shown on an oscilloscope for a flute and a bassoon playing the same note. The oscilloscope settings are the same for both wave forms.

flute



bassoon



What is the difference between the two sounds?

A the amplitude

B the frequency

C the quality (timbre)

D the wavelength

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 24

24	**C**

For Examiner's Use

5 This question is about the sound produced by a mobile phone and the energy changes while it is operating.

(a) The sound produced by a ringing phone consists of two notes, one after the other.

Fig. 5.1 shows the trace on an oscilloscope screen produced by the first of the notes.

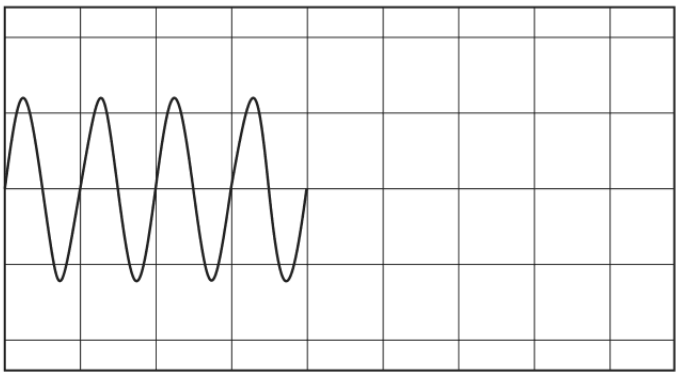


Fig. 5.1

The second note is louder and has a higher pitch.

(i) On Fig. 5.1, continue the trace to show what happens when the second note is sounding. [1]

(ii) Explain the differences between the two traces.

.....

.....

.....

.....

..... [2]

(b) The mobile phone contains a rechargeable battery.

Energy is stored inside the battery as it is charged.

Describe the useful energy change that takes place inside the battery as it is charged from the mains.

.....

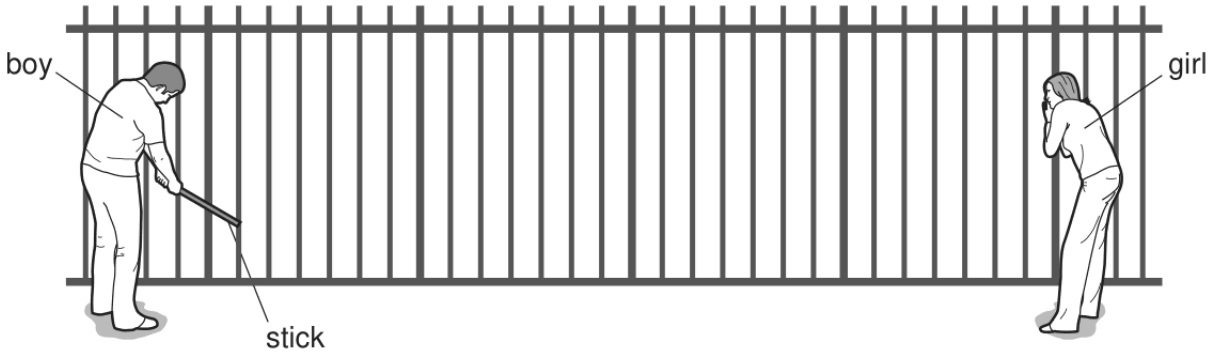
.....

..... [2]

5	**(a)**	**(i)**	diagram with larger amplitude and shorter “wavelength”	**B1**	
		(ii)	louder means larger amplitude/height	**B1**	
			higher pitch means higher frequency/more waves on screen/shorter wavelength	**B1**	
	(b)		electrical at start	**B1**	
			chemical at end and a clear transformation without errors (allow 1 mark for chemical to electrical)	**B1**	**5**

QUESTION · 5054_s06_qp_1.pdf · source page 10 of 20

26 A boy strikes a rigid metal fence with a stick to create a sound along the fence. A girl listens with her ear against the fence. One second after the fence is struck, the girl hears a sound through the air.



How long will it take for the sound to reach the girl through the fence?

A 0 second

B less than 1 second

C 1 second

D more than 1 second

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 26

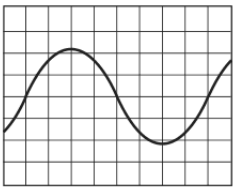
26	**B**

QUESTION · 5054_w06_qp_1.pdf · source page 9 of 16

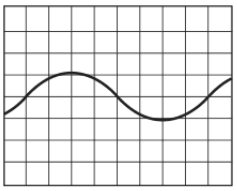
23 The diagrams show oscilloscope traces of sounds picked up by microphones. The oscilloscope controls are set in the same position for all the traces.

Which trace shows the sound that is both loud and low-pitched?

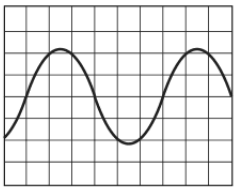
A



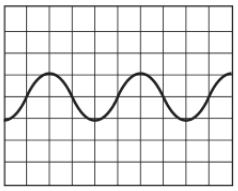
B



C



D

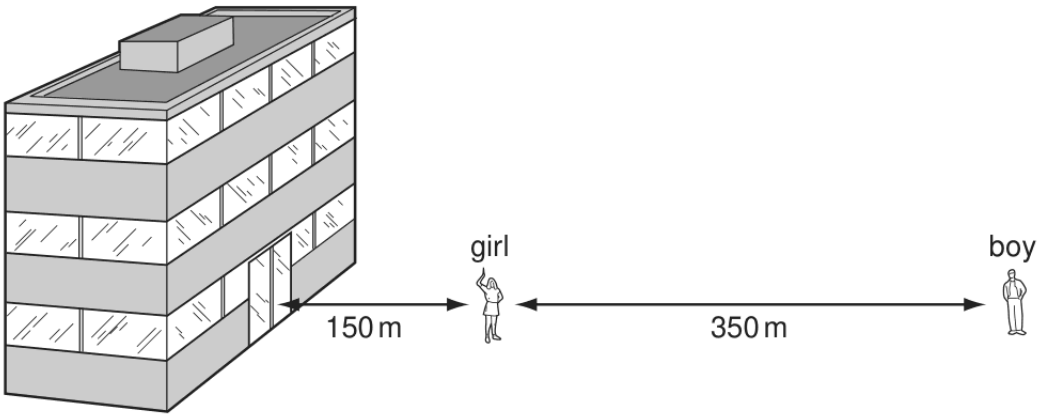


MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 23

23	**A**

QUESTION · 5054_s07_qp_1.pdf · source page 11 of 20

23 A girl, standing 150 m in front of a tall building, fires a shot using a starting pistol. A boy, standing 350 m from the girl, hears two bangs 1 s apart.



From this information, what is the speed of sound in air?

A 300 m / s **B** 350 m / s **C** 500 m / s **D** 650 m / s

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 23

23	**A**

QUESTION · 5054_w07_qp_1.pdf · source page 10 of 20

23 A flash of lightning and the corresponding thunder clap are detected 6 s apart. It is calculated that the lightning struck about 1800 m away.

On which assumption is the calculation based?

A Light reaches us almost instantaneously, but sound travels at 300 m / s.

B Light travels 300 m / s faster than sound.

C Sound reaches us almost instantaneously, but light travels at 300 m / s.

D The sound of the thunder was emitted 6 s after the flash.

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 23

23	**A**

For Examiner's Use

6 Fig. 6.1 shows the cone of a loudspeaker.

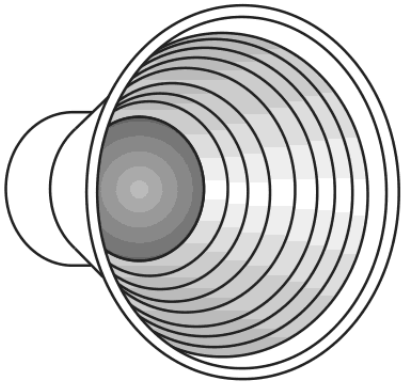


Fig. 6.1

(a) Sound is being produced. Describe in detail the behaviour of the cone and the air near to it.

.....

.....

.....

.....

..... [2]

(b) The lowest frequency that a human can hear is 20 Hz.

(i) State the highest frequency that a human with normal hearing can hear.

..... [1]

(ii) Calculate the longest wavelength of sound that a human can hear. The speed of sound in air is 340 m/s.

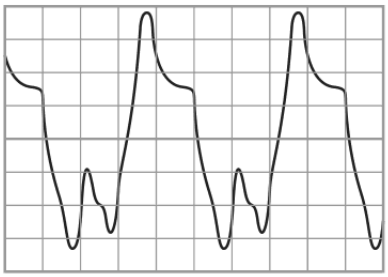
wavelength = ... [2]

6	**(a)**	cone/molecules vibrate	B1	
		molecules (vibrate) longitudinally/back and forward (in direction of sound) **or** compressions **and** rarefactions mentioned (**e.g.** longitudinal waves)	B1	
	(b)	**(i)** a number from 18,000 to 22,000 Hz	B1	
		(ii) (v =) f λ algebraic **or** numerical using 20 Hz **or** candidate's **(i)**	C1	
		17 m	A1	[5]

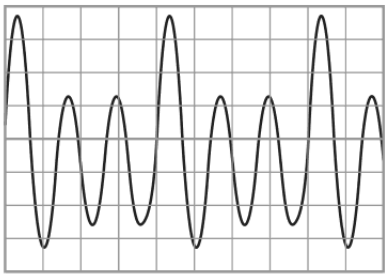
QUESTION · 5054_s08_qp_1.pdf · source page 11 of 20

26 Waveforms are shown on a cathode-ray oscilloscope for a flute and a guitar playing the same note. The oscilloscope settings are the same for both waveforms.

flute



guitar



What is the difference between the two sounds?

A the amplitude

B the frequency

C the quality (timbre)

D the wavelength

MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 26

26	**C**

Section B

Answer **two** questions from this section.

Use the lined pages provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 Fig. 9.1 shows one swimmer in a race starting before the signal.

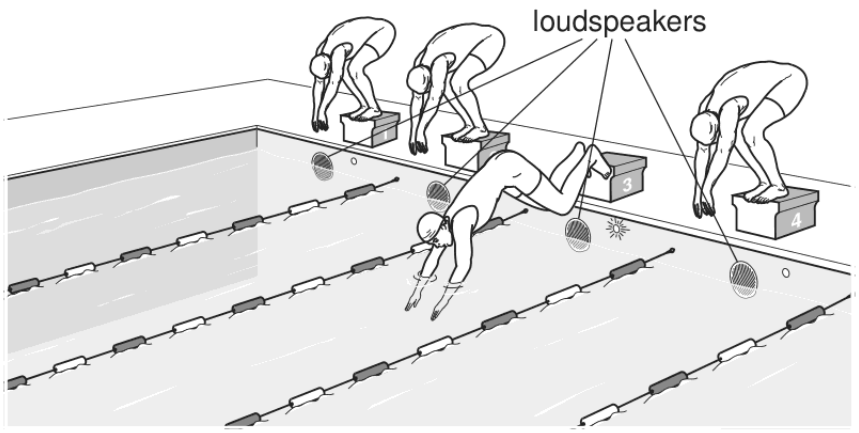


Fig. 9.1

The swimmer is called back by a loud, low-pitched sound from a loudspeaker positioned just at water level. The speed of sound in air is 330 m/s.

(a) (i) Describe how the loudspeaker causes sound to travel through the air. [3]

(ii) Explain, in terms of wave properties, what is meant by *loud* and *low-pitched*. [3]

(iii) The swimmer is 0.57 m from the loudspeaker when he hears the sound. Calculate the time taken for the sound to reach him through the air. [2]

(iv) Explain how the time taken differs when sound travels the same distance through air and through water. [2]

(b) The loudspeaker produces sound of frequency 0.20 kHz.

(i) Calculate the wavelength of this sound. [3]

(ii) Draw a diagram to show what is meant by the term *wavelength* when applied to a **longitudinal** wave such as sound. [2]

Section B

9	**(a)**	**(i)**	Any **three** lines: vibration of **cone/loudspeaker** vibration of **air/particles** (molecules) **particles/molecules** pass on vibrations/energy (to neighbours) compressions **and** rarefactions **or** longitudinal wave/movement of particle (max 3)	 B1 B1 B1 B1	
		(ii)	loud – large amplitude/max displacement low-pitched – frequency/no. of waves per sec low frequency, small frequency, etc. (long wavelength 1/2)	B1 M1 A1	
		(iii)	(t =) d/s **or** 0.57/330 0.0017 s	C1 A1	
		(iv)	speed of sound greater in water/liquid **or** v.v. less time taken in water/liquid **or** heard sooner/faster	B1 B1	[10]
	(b)	**(i)**	$v = f\lambda$ **or** 200 seen (λ =) v/f **or** 330/200 **or** 330/0.2 **or** 1650 (m) 1.6/1.65/1.7 m	C1 C1 A1	
		(ii)	attempt at compressions and rarefactions/longitudinal wave correct wavelength marked	M1 A1	[5]

[Total: 15]

QUESTION · 5054_s09_qp_1.pdf · source page 10 of 20

23 An ultrasonic tape-measure is used to find the distance to a wall. It sends out an ultrasonic pulse and times how long it takes for the reflected pulse to return from the wall.

The ultrasound has a frequency, a wavelength and a speed.

Which pair of values is needed to find the distance to the wall?

A frequency and wavelength

B frequency and time taken for the pulse to return

C speed and time taken for the pulse to return

D wavelength and time taken for the pulse to return

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 23

23	**C**

For Examiner's Use

6 Fig. 6.1 shows how ultrasound is used to produce an image of the heart.

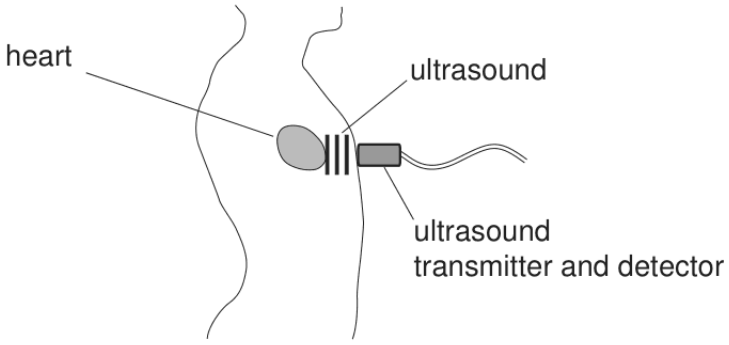


Fig. 6.1

(a) Define *ultrasound*.

.....

..... [1]

(b) The ultrasound has a wavelength of 1.2×10^{-3} m. The speed of the ultrasound in the human body is 1500 m/s.

Calculate the frequency of the ultrasound.

frequency = ... [2]

(c) Ultrasound is a longitudinal wave.
Describe how particles in the body move as the ultrasound passes.

You may draw a diagram if you wish.

.....

.....

.....[2]

(d) There are small bubbles of gas in the body.

Explain why these bubbles expand and contract as the ultrasound passes.

.....

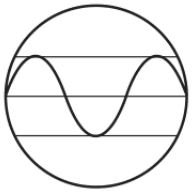
.....

..... [1]

6	**(a)**	(sound) too high a frequency to be heard or (frequency) above 20 kHz	B1	
	(b)	(f =) v/λ or $v = f\lambda$ algebraic or numerical	C1	
		1 250 000 Hz	A1	
	(c)	vibrate/oscillate	C1	
		vibration etc. in same direction as/parallel to wave/energy or horizontally	A1	
	(d)	pressure increases and decreases or compressions and rarefactions mentioned in **(d)** or particles come together and move apart	B1	[6]

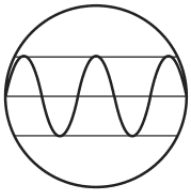
QUESTION · 5054_w09_qp_1.pdf · source page 12 of 20

23 The diagram shows the trace produced on a cathode-ray oscilloscope (c.r.o.) by a sound.

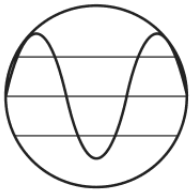


Which trace is produced when both the loudness and the pitch of the sound are increased?

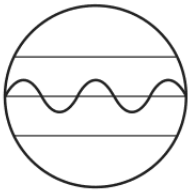
A



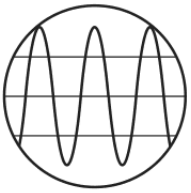
B



C



D



MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 23

23	**D**

QUESTION · 5054_s10_qp_11.pdf · source page 10 of 20

23 A flash of lightning and the corresponding sound of the thunder are detected 6 s apart. A student calculates that the lightning struck about 1800 m away.

On which assumption is the calculation based?

A Light reaches us almost instantaneously, but sound travels at 300 m / s.

B Light travels 300 m / s faster than sound.

C Light travels 300 times faster than sound.

D The sound of the thunder was emitted 6 s after the flash.

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 23

23	**A**

QUESTION · 5054_s10_qp_12.pdf · source page 9 of 20

20 A flash of lightning and the corresponding sound of the thunder are detected 6 s apart. A student calculates that the lightning struck about 1800 m away.

On which assumption is the calculation based?

A Light reaches us almost instantaneously, but sound travels at 300 m / s.

B Light travels 300 m / s faster than sound.

C Light travels 300 times faster than sound.

D The sound of the thunder was emitted 6 s after the flash.

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 20

20	**A**

For Examiner's Use

10 (a) Describe a method for measuring the speed of sound in air. In your account, state clearly

- how the sound is made,
- what measurements are taken,
- how the result is calculated,
- one precaution to produce an accurate result.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[5]

(b) Ultrasound is used in quality control to detect cracks in metal.

Pulses of ultrasound are sent into the metal from a transmitter. A detector is placed next to the transmitter on the front surface of the metal.

Fig. 10.1 shows the oscilloscope trace of the ultrasound pulses produced if the metal contains no cracks.

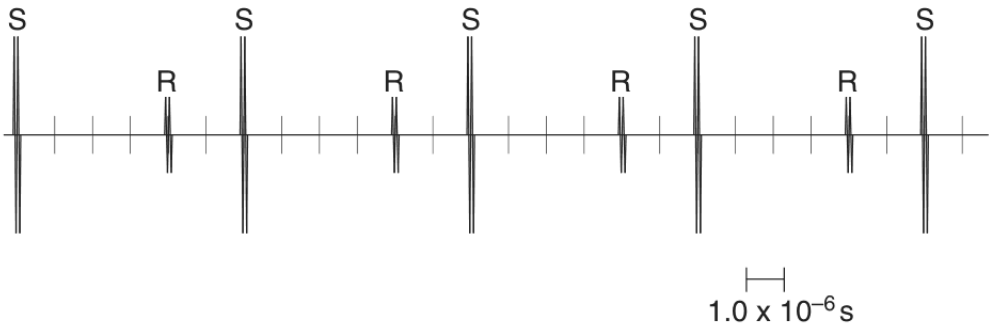


Fig. 10.1

One division along the *x*-axis represents 1.0×10^{-6} s.

Pulses labelled S are the pulses initially sent out from the transmitter. Each pulse labelled R is the reflection from the back surface of the metal of the previous pulse S.

For Examiner's Use

(i) State what is meant by *ultrasound*.

.....

.....[2]

(ii) Use Fig. 10.1 to calculate the number of pulses sent out by the source in one second.

number = ...[2]

(iii) Suggest two reasons why the amplitude of R is less than the amplitude of S.

1. .....

.....

2. .....

.....
[2]

(iv) Some time later, the piece of metal is tested again. It now has a small crack half-way between the front surface and the back surface.

On Fig. 10.1, draw the position and size of the pulses produced by this crack.

Label each of these pulses C. [2]

(v) A second beam of ultrasound has a frequency of 8.0×10^6 Hz and a speed of 4000 m/s in the metal.

Calculate the wavelength of this ultrasound in the metal.

wavelength = ...[2]

10	**(a)**	how sound is made e.g. gun, clap hands, hit metal	B1	
		correct measurement of time, e.g. from seeing flash to hearing sound, clap-echo	B1	
		correct measurement of distance, e.g. gun to observer, observer to wall	B1	
		correct calculation for measurements, e.g. d/t or $2d/t$	B1	
		precaution e.g. time clap on echo and time 10; ensure no wind; repeat in opposite direction; repeat and average; use large distance; use more than 200 m	B1	[5]
	(b) **(i)**	(sound/wave/vibration) of high frequency or (sound that) cannot be heard	C1	
		(frequency) above 15–20 kHz	A1	[2]
	(ii)	$f = 1/T$ or 6×10^{-6} (s) seen or 2,3,4 pulses in 12,18,24 μs	C1	
		1.7×10^{5} allow 166667	A1	[2]

	(iii)	not all sound is reflected (from back surface) or some passes through the back (some energy/sound) absorbed (by metal) (sound/energy) spreads out/scattered/reflected in other directions/dispersed/ travels a (greater) distance any 2 lines	B2	[2]
	(iv)	at least one pulse half way between S and R in the long gap	B1	
		total height of pulse smaller than S and 3 or more drawn and labelled C	B1	[2]
	(v)	$v = f\lambda$ in any algebraic or numerical form e.g. $4000/8 \times 10^{6}$	C1	
		$5(.0) \times 10^{-4}$ m	A1	[2]

[Total: 15]

QUESTION · 5054_w10_qp_11.pdf · source page 11 of 20

25 During a thunderstorm, there is an interval of 1.70 s between an observer seeing the lightning and hearing the thunder. The speed of sound is 340 m / s.

What is the distance between the observer and the storm?

A 100 m **B** 200 m **C** 578 m **D** 1160 m

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 25

25	**C**

For Examiner's Use

6 Fig. 6.1 shows the coil of a loudspeaker attached to a cardboard cone. One pole of a stationary cylindrical magnet lies near to the coil.

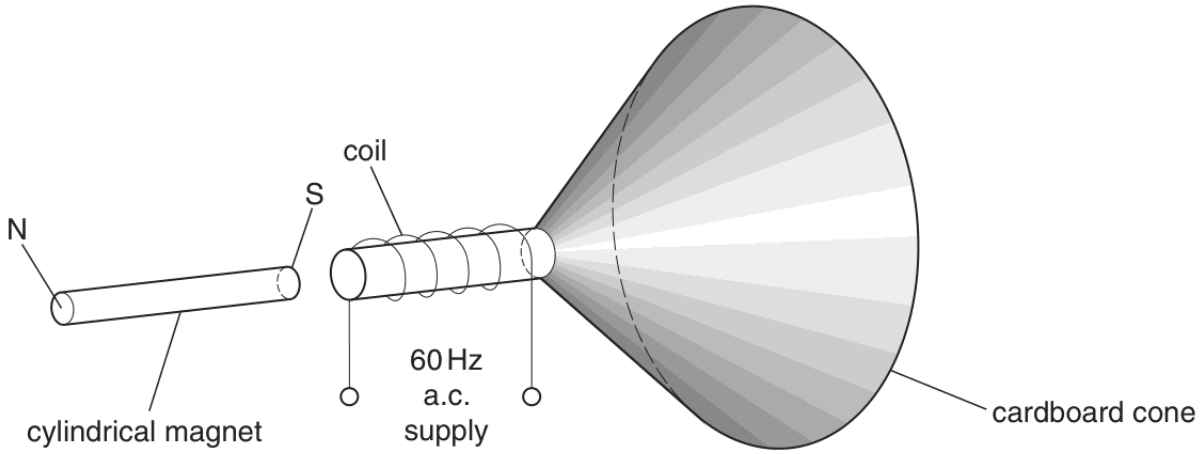


Fig. 6.1

There is an alternating current in the coil of the loudspeaker. A student hears the note produced.

(a) (i) Explain why the cone of the loudspeaker vibrates.

.....

.....

.....

..... [3]

(ii) Explain how the vibrating cone produces sound waves in the air.

.....

.....

..... [2]

(b) A stronger cylindrical magnet is now used. State the difference in the note heard.

.....

..... [1]

6	**(a)**	**(i)**	current in magnetic field **or** motor effect/LH rule **or** coil is magnet	B1	
			(produces) force **or** current **direction** changes **or** coil moves **or** repulsion **and** attraction	B1	
			force changes direction/backwards and forwards	B1	[3]
		(ii)	air (atoms/molecules/particles) (next to cone) vibrates	B1	
			compressions **and** rarefactions **or** high **and** low pressure **or** vibrations passed on **or** longitudinal	B1	[2]
	(b)		the note is louder/has greater intensity (**not** changed frequency)	B1	[1]

QUESTION · 5054_w10_qp_12.pdf · source page 8 of 16

21 During a thunderstorm, there is an interval of 1.70 s between an observer seeing the lightning and hearing the thunder. The speed of sound is 340 m / s.

What is the distance between the observer and the storm?

A 100 m **B** 200 m **C** 578 m **D** 1160 m

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 21

21	**C**

For Examiner's Use

5 A microphone is connected to a cathode-ray oscilloscope (c.r.o.). A note produced by a musical instrument causes a trace on the screen. The trace is stored electronically in the c.r.o.

Fig. 5.1 shows the stored trace displayed on the screen.

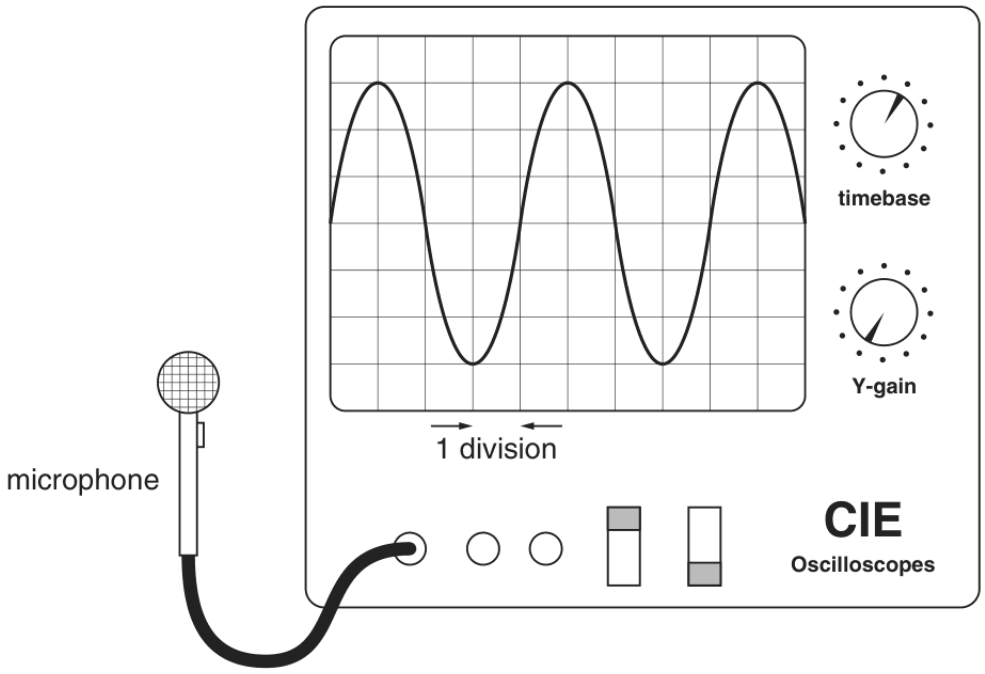


Fig. 5.1

The timebase setting on the oscilloscope is 0.20 ms / division.

(a) Determine the frequency of this note.

frequency = ...[3]

For Examiner's Use

(b) A note from a second musical instrument is played and the trace S produced by the new note is also displayed on the screen. Fig. 5.2 shows the screen displaying both the original trace and S.

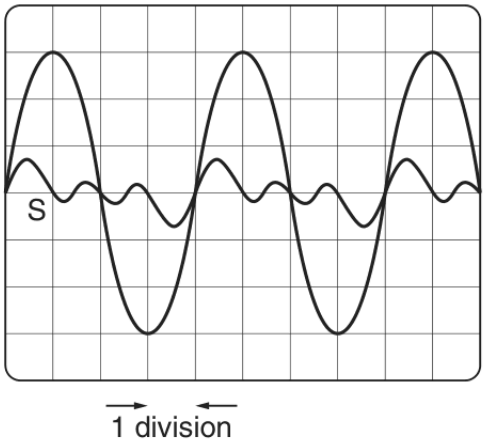


Fig. 5.2

In each of the three spaces below, describe either a similarity or a difference between the **sounds** made by the two notes.

1. .....

2. .....

3. .....

[3]

5 **(a)** 0.80 **or** 0.0008 **or** 4 × 0.20 **or** 4 × 0.0002 **or** 4 divisions — C1
(f =) 1/T **or** 1.2/1.25/1.3 (Hz) — C1
1200/1250/1300 Hz — A1

(b) any **three** of:
equal/same pitch/frequency (**ign** wavelength)
original note louder/ S quieter/softer (**ign** amplitude)
{ different qualities/timbres/
{ more frequencies/overtones/harmonics in S — B3 [6]

QUESTION · 5054_s11_qp_11.pdf · source page 10 of 16

25 Sound travels at different speeds in air, water and steel.

For these materials, which row is correct?

	sound travels slowest in	sound travels fastest in
A	air	steel
B	air	water
C	steel	air
D	water	air

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 25

25 **A**

6 Fig. 6.1 shows a bat.

Fig. 6.1

Bats emit short bursts of ultrasound. The echoes of the ultrasound help the bat find insects and prevent the bat flying into objects.

(a) State what is meant by an *echo*.

.....

..... [1]

(b) Fig. 6.2 shows the variation with time of air pressure caused by a burst of ultrasound.

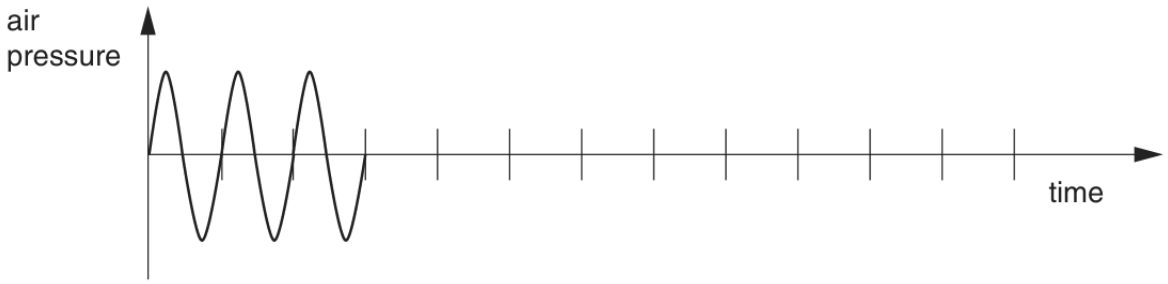


Fig. 6.2

On Fig. 6.2, draw a possible echo formed by this burst of ultrasound. [2]

(c) State

(i) the approximate range of frequencies of sound audible to humans,

..... [1]

(ii) how the frequency of ultrasound differs from frequencies in this audible range.

.....

..... [1]

6	**(a)**	reflection (of sound/ultrasound)	B1	
	(b)	waves of same period (by eye)	B1	
		smaller amplitude (by eye)	B1	
	(c)	**(i)** 20–20 000 Hz	B1	
		(ii) higher than **(i)**	B1	[5]

QUESTION · 5054_s11_qp_12.pdf · source page 11 of 16

26 Sound travels at different speeds in air, water and steel.

For these materials, which row is correct?

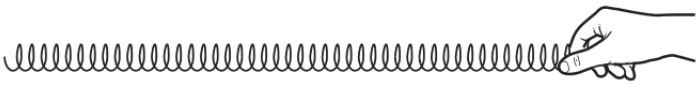
	sound travels slowest in	sound travels fastest in
A	air	steel
B	air	water
C	steel	air
D	water	air

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 26

26	**A**

QUESTION · 5054_w11_qp_11.pdf · source page 10 of 20

23 A student uses a spring to demonstrate waves. He moves the spring with his hand.

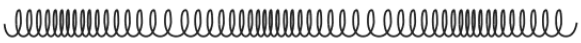


spring placed on bench

Which diagram demonstrates the type of wave produced by a source of sound?

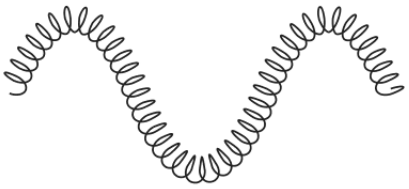
A

B



C

D



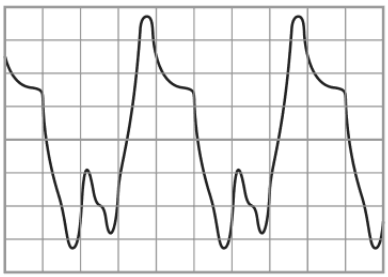
MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 23

23	**B**

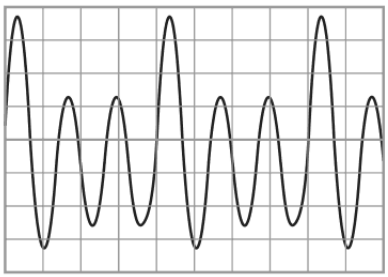
QUESTION · 5054_w11_qp_11.pdf · source page 11 of 20

26 Waveforms are shown on a cathode-ray oscilloscope for a flute and for a guitar, each playing the same note. The oscilloscope settings are the same for both waveforms.

flute



guitar



What is the difference between the two sounds?

A the amplitude

B the frequency

C the quality (timbre)

D the wavelength

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 26

26	**C**

7 Fig. 7.1 is the circuit diagram for the volume control of a loudspeaker.

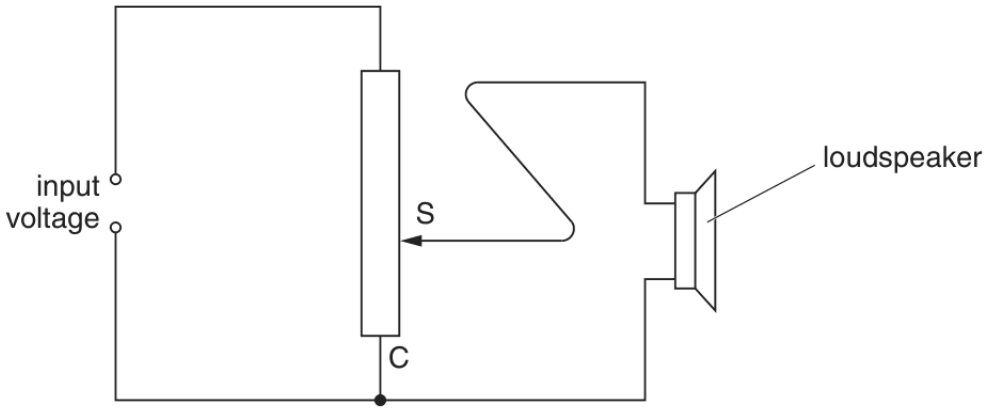


Fig. 7.1

The circuit includes a variable potential divider (potentiometer). The input voltage is connected across the ends of the potential divider. The loudspeaker is connected between the sliding contact S and the terminal C.

(a) The sliding contact S is moved towards C. State and explain the effect on the sound produced.

.....

.....

.....[2]

(b) The loudspeaker emits a sound of constant pitch. Over a period of time, this sound becomes louder and louder. State the effect, if any, on the amplitude and on the frequency of the sound wave during this time.

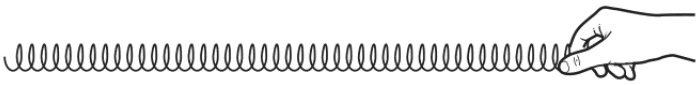
the amplitude .....

the frequency .....
[2]

7	**(a)**	volume decreases/quieter/less sound	B1	
		(in some way) resistance between S and C decreases **or** (in some way) voltage (to loudspeaker) reduced	B1	
	(b)	(the amplitude) increases	B1	
		(the frequency) remains constant	B1	[4]

QUESTION · 5054_w11_qp_12.pdf · source page 11 of 20

24 A student uses a spring to demonstrate waves. He moves the spring with his hand.

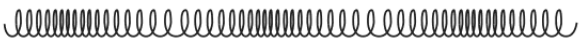


spring placed on bench

Which diagram demonstrates the type of wave produced by a source of sound?

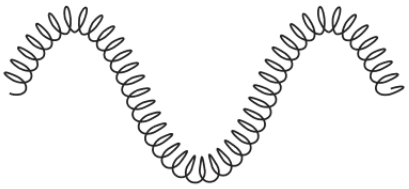
A

B



C

D



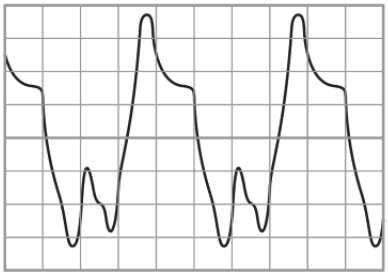
MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 24

24	**B**

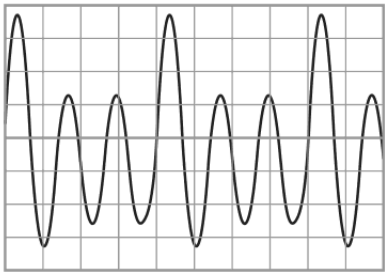
QUESTION · 5054_w11_qp_12.pdf · source page 11 of 20

25 Waveforms are shown on a cathode-ray oscilloscope for a flute and for a guitar, each playing the same note. The oscilloscope settings are the same for both waveforms.

flute



guitar



What is the difference between the two sounds?

A the amplitude

B the frequency

C the quality (timbre)

D the wavelength

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 25

25	**C**

6 A sound from an electronic organ is played into a microphone which is connected to a cathode-ray oscilloscope (c.r.o.). Fig. 6.1 shows the trace on the c.r.o. produced by the sound.

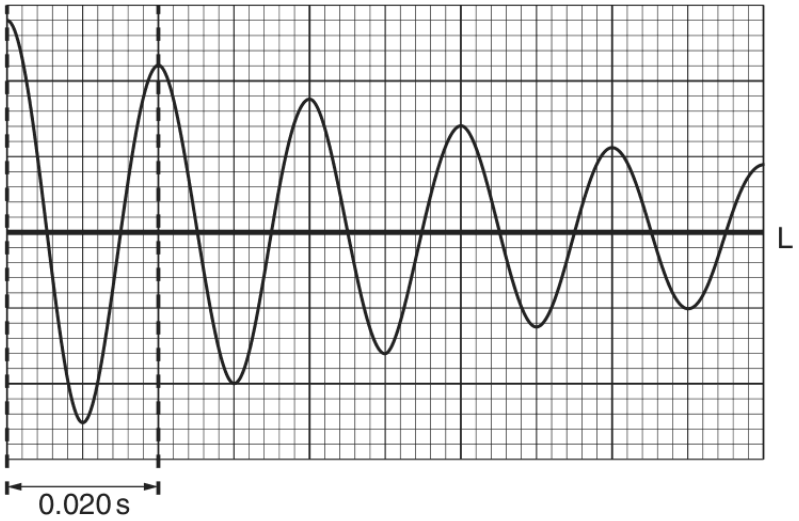


Fig. 6.1

The horizontal line marked L is the trace seen when there is no sound.

(a) (i) State how Fig. 6.1 shows that the loudness of the sound decreases with time.

.....[1]

(ii) State how Fig. 6.1 shows that the pitch of the sound is constant.

.....

.....[1]

(b) (i) State what is meant by the *frequency* of a wave.

.....

.....[1]

(ii) The time for one complete cycle of the wave in Fig. 6.1 is 0.020 s.
Calculate the frequency of the wave.

frequency = ...[2]

(iii) Determine the time taken for the amplitude of the trace in Fig. 6.1 to decrease to half its initial value.

time = ...[1]

6	**(a)**	**(i)**	amplitude decreases	B1	
		(ii)	constant frequency/time for one wave/wavelength/period	B1	
	(b)	**(i)**	number of (complete) cycles in one second	B1	
		(ii)	(f =) 1/T in any form numerical or algebraic; 1/0.02 50 Hz	C1 A1	
		(iii)	0.06 s	B1	[6]

Section B

For Examiner's Use

Answer **two** questions from this section. Answer in the spaces provided.

9 A submarine is at rest deep beneath the surface of the sea. The gravitational field strength is 10 N/kg, the atmospheric pressure is 1.0×10^5 Pa and the density of sea water is 1.0×10^3 kg/m^3.

(a) Calculate

(i) the pressure due to the sea water at a depth of 120 m,

pressure = ... [2]

(ii) the total pressure at a depth of 120 m.

total pressure = ... [1]

(b) Fig. 9.1 shows a horizontal hatch of surface area 0.45 m^2 on the top of the submarine. The hatch is at a depth of 120 m beneath the surface of the sea.

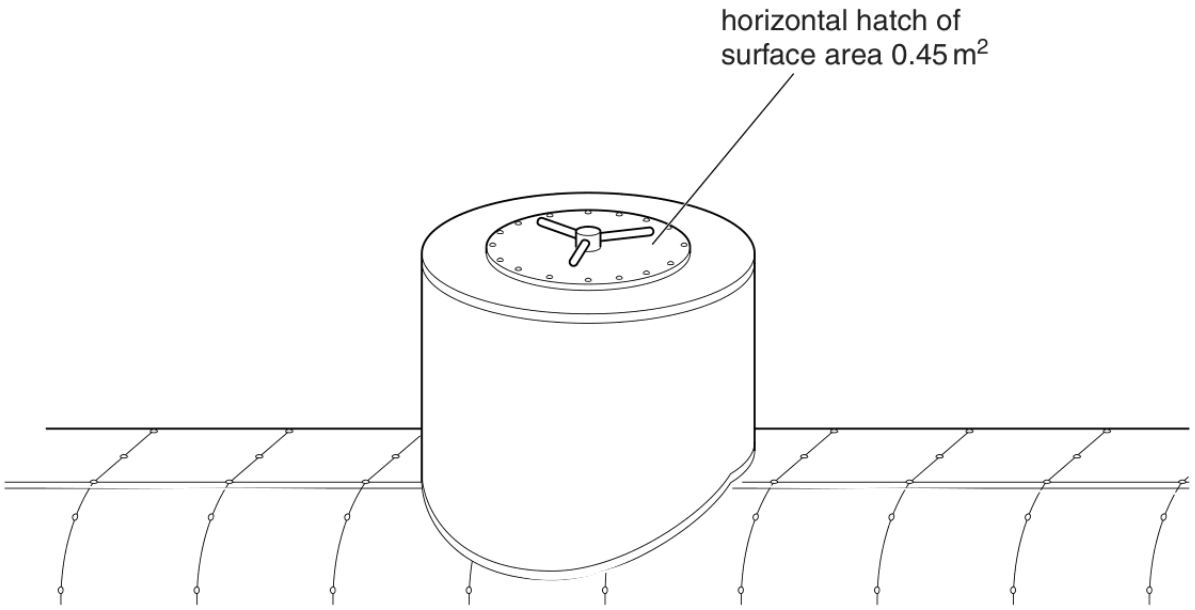


Fig. 9.1

For Examiner's Use

(i) Calculate the downward force acting on the surface of the hatch.

force = ... [2]

(ii) State two reasons why the force needed to lift the hatch is different from the value calculated in **(i)**.

1. .....

2. .....
[2]

(c) The submarine uses ultrasound to detect obstacles.

(i) State what is meant by *ultrasound*.

.....

..... [2]

(ii) The ultrasound is produced by a vibrating source.

Describe how the ultrasound is transmitted through the water.

.....

.....

.....

..... [3]

(iii) An ultrasound pulse is emitted. At a later time, the reflected pulse is detected. The time between the emission and the detection of the pulse is *t*.

1. State the other quantity that must be known in order to determine the distance of the obstacle from the submarine.

..... [1]

2. State how the distance of the obstacle from the submarine is calculated.

..... [1]

(iv) State one other use of ultrasound.

.....

..... [1]

Section B

9	**(a)**	**(i)**	($\Delta P =$) ρgh **or** 1000 × 10 × 120	C1	
			1.2×10^6 Pa	A1	
		(ii)	1.3×10^6 Pa	B1	[3]
	(b)	**(i)**	($F =$)PA **or** $1.2 \times 10^6 \times 0.45$ **or** $1.3 \times 10^6 \times 0.45$ **or** 5.4×10^5 (N)	C1	
			$5.8/5.85/5.9 \times 10^5$ N	A1	
		(ii)	any **two** of: weight of hatch pressure inside submarine friction at seal/hinge/water resistance lever effect	B2	[4]
	(c)	**(i)**	sound or pressure wave	B1	
			frequency > 20 kHz/**frequency** beyond human hearing/inaudible	B1	
		(ii)	(water) molecules/particles vibrate/oscillate	B1	
			molecules collide with other molecules/neighbours	B1	
			pass on vibration/energy (to neighbours) **or** longitudinal (vibration/wave) or compressions and rarefactions	B1	
		(iii)	**1.** speed of sound/ultrasound (in water/sea water)	B1	
			2. speed × $t \div 2$	B1	
		(iv)	cleaning/quality control/detecting cracks/prenatal screening/ kidney stones/detecting shoals of fish/(used by dolphins/bats)	B1	[8]

[Total: 15]

QUESTION · 5054_s13_qp_11.pdf · source page 10 of 20

23 Which range is the approximate range of audible frequencies for a human?

A $0.2\,\text{Hz} \rightarrow 200\,\text{Hz}$

B $2\,\text{Hz} \rightarrow 2000\,\text{Hz}$

C $20\,\text{Hz} \rightarrow 20\,000\,\text{Hz}$

D $200\,\text{Hz} \rightarrow 2\,000\,000\,\text{Hz}$

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 23

23	**C**

QUESTION · 5054_s13_qp_11.pdf · source page 10 of 20

24 Which row correctly compares the speeds of sound in air, liquid and solid?

	highest→ lowest		
A	air	liquid	solid
B	air	solid	liquid
C	liquid	air	solid
D	solid	liquid	air

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 24

24 **D**

QUESTION · 5054_s13_qp_11.pdf · source page 10 of 20

25 A guitar string is made to vibrate.

What makes the pitch of the note rise?

A a decrease in the amplitude of vibration

B a decrease in the frequency of vibration

C an increase in the amplitude of vibration

D an increase in the frequency of vibration

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 25

25	**D**

QUESTION · 5054_s13_qp_12.pdf · source page 10 of 16

27 What is a possible frequency of an ultrasound wave?

A 0.1 kHz **B** 3 kHz **C** 10 kHz **D** 30 kHz

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 27

27	**D**

QUESTION · 5054_s13_qp_12.pdf · source page 10 of 16

28 A guitar string is made to vibrate.

What makes the pitch of the note rise?

A a decrease in the amplitude of vibration

B a decrease in the frequency of vibration

C an increase in the amplitude of vibration

D an increase in the frequency of vibration

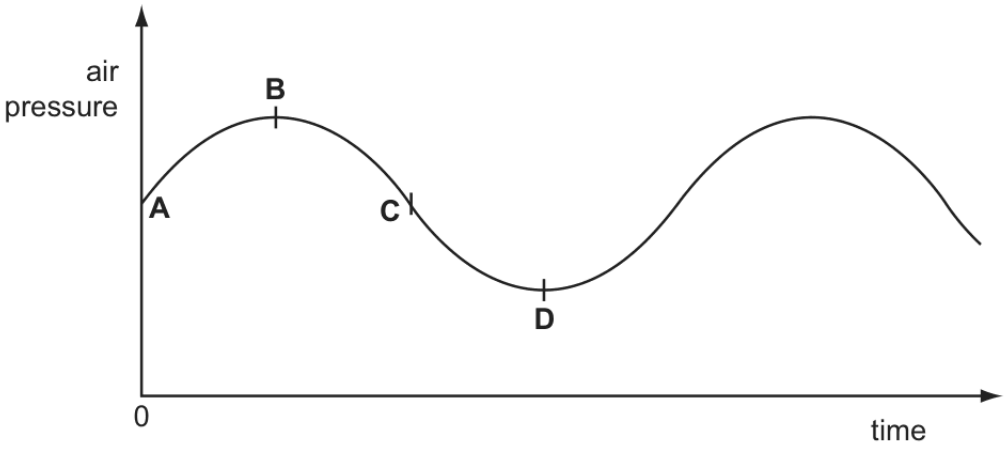
MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 28

28	**D**

QUESTION · 5054_w13_qp_11.pdf · source page 11 of 20

23 The graph shows how the pressure varies as a sound wave passes through air.

Which point represents a compression?

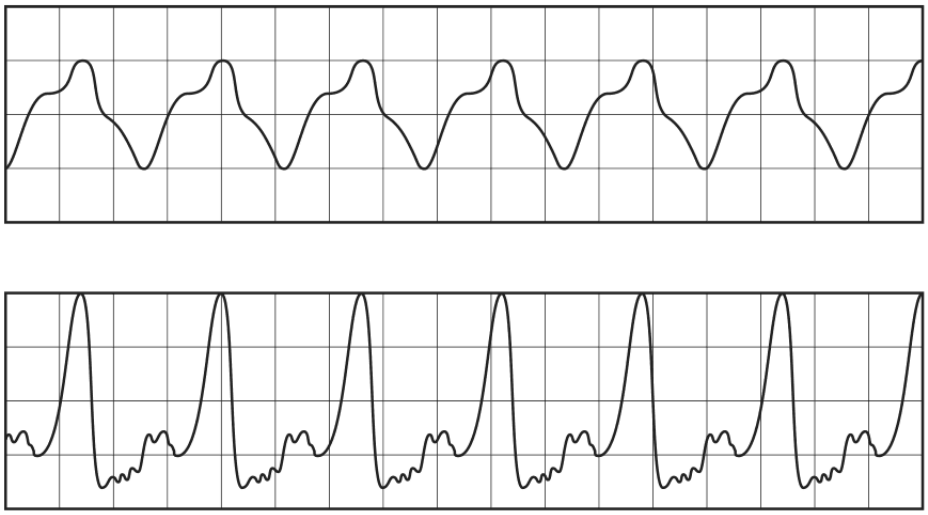


MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 23

23	**B**

QUESTION · 5054_w13_qp_11.pdf · source page 11 of 20

24 The sounds produced by two musical instruments are directed towards a microphone connected to an oscilloscope (c.r.o.). The waveforms produced on the screen are shown.



The waveforms show that the sounds produced have a different property.

What is the property?

A frequency

B speed

C timbre (quality)

D wavelength

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 24

24	**C**

For Examiner's Use

10 A student goes for a walk in the mountains. During a storm, she sees lightning strike a hillside in the distance. Several seconds later, she hears the thunder caused by the lightning.

(a) (i) Explain why she hears the thunder several seconds after she sees the lightning.

.....

.....[1]

(ii) The student knows the distance to the hillside. She waits for lightning to strike the hillside again. Describe how she can determine a value for the speed of sound in air.

.....

.....

.....

.....[2]

(b) After the storm, the student sees a rainbow.

(i) State the speed of light in air.

speed = ...[1]

(ii) Calculate the wavelength in air of light of frequency 7.5×10^{14} Hz.

wavelength = ...[2]

(iii) State the colours of the spectrum in order of increasing wavelength.

.....

.....[2]

For Examiner's Use

(c) In the laboratory, the student sends blue light into a glass prism placed on a sheet of paper. The arrangement is shown in Fig. 10.1.

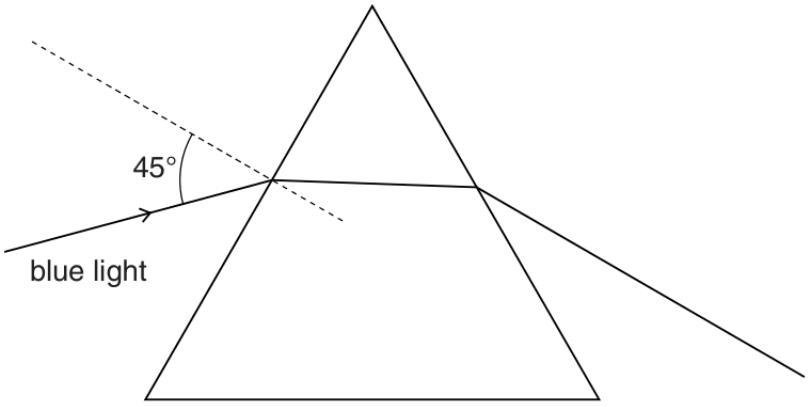


Fig. 10.1

The blue light enters the prism with an angle of incidence of 45°.

(i) On Fig. 10.1, mark the angle of refraction in the glass and label it *r*. [1]

(ii) The student wishes to determine the angle of refraction. On the sheet of paper, she draws the path the ray takes in the glass. Describe how she could do this.

.....

.....

.....[2]

(iii) **1.** State the formula that relates the angle of incidence *i*, the angle of refraction *r* and the refractive index *n*.

.....

.....[1]

2. The angle of refraction in the glass is 28°.

Calculate the refractive index of the glass.

refractive index = ...[1]

(iv) The student sends a ray of red light to the glass prism along the same path as the blue light.

On Fig. 10.1, mark the path taken by the red light after it enters the prism. [2]

MARK SCHEME · 5054_w13_ms_21.pdf · source page 4 of 6 · Question 10

10	**(a)**	**(i)**	speed of sound is (much) less than the speed of light (accept quoted values)	B1	
		(ii)	**measure** the time delay (between the lightning and thunder)	B1	
			divide distance by time/delay	B1	[3]

MARK SCHEME · 5054_w13_ms_21.pdf · source page 5 of 6 · Question 10

	(b)	**(i)**	3.0×10^8 m / s	B1	
		(ii)	(λ =) c/f **or** $3.0 \times 10^8 / 7.5 \times 10^{14}$	C1	
			4.0×10^{-7} m	A1	
		(iii)	(in any order) blue, green, orange, red, yellow, (indigo), (violet) **or** VIBGYOR	C1	
			violet, indigo, blue, green, yellow, orange, red	A1	[5]
	(c)	**(i)**	correct angle clear/labelled r	B1	
		(ii)	mark/determine entrance and exit points (e.g. trace rays back to glass)	B1	
			join/draw line between entrance and exit points	B1	
		(iii)	**1.** n = sin i/sin r	B1	
			2. 1.5/1.51/1.506176 with no unit (**not** just 1.5 without working out)	B1	
		(iv)	correct direction of refraction at **both** faces	M1	
			completely correct (above blue)	A1	[7]

[Total: 15]

QUESTION · 5054_w13_qp_12.pdf · source page 12 of 20

25 Two campers are woken up in the middle of the night by a thunderstorm. Their tent is lit up by a flash of lightning and they hear the thunderclap 5.0 s later. The speed of sound is 340 m / s.

How far away from the tent is the lightning?

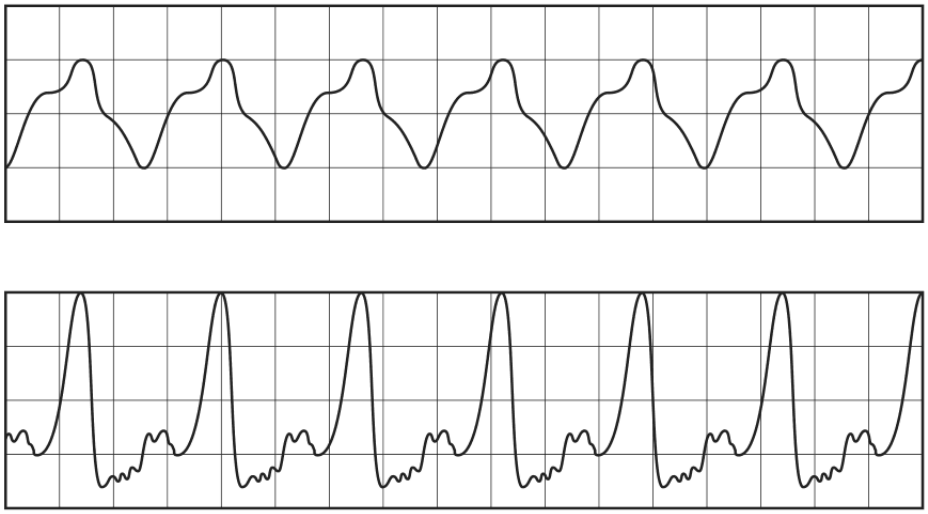
A 68 m **B** 850 m **C** 1700 m **D** 3400 m

MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 25

25	**C**

QUESTION · 5054_w13_qp_12.pdf · source page 12 of 20

26 The sounds produced by two musical instruments are directed towards a microphone connected to an oscilloscope (c.r.o.). The waveforms produced on the screen are shown.



The waveforms show that the sounds produced have a different property.

What is the property?

A frequency

B speed

C timbre (quality)

D wavelength

MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 26

26	**C**

For Examiner's Use

5 A physics textbook states that sound is a longitudinal pressure wave with a frequency within the audible range.

(a) Explain what is meant by a *longitudinal wave*.

.....

.....

.....[2]

(b) **(i)** State the approximate range of audible frequencies.

highest frequency: .....

lowest frequency: .....
[2]

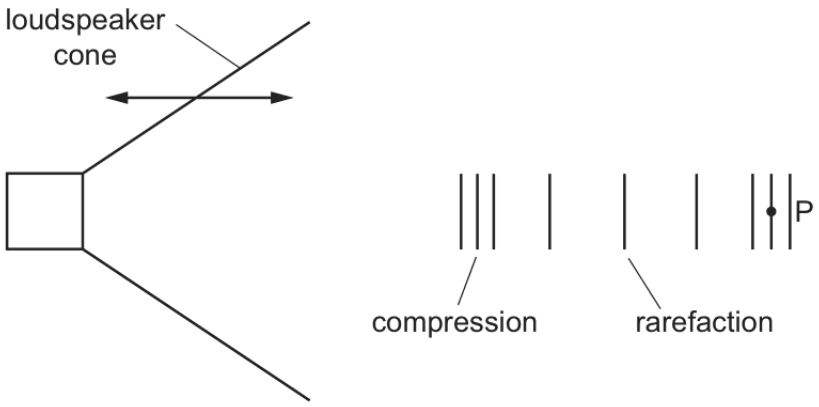
(ii) The speed of sound in air is 330 m / s. Using your answer in **(i)**, calculate the shortest wavelength in air of sound in the audible range.

wavelength = ...[2]

5	**(a)**	transmission of energy through a medium **or** vibration **or** oscillation **or** two opposite motions (e.g. up and down) **or** compressions and rarefactions	C1	
		vibration direction parallel to energy travel/wave direction **or** similar	A1	
	(b) **(i)**	$1.5–2.5 \times 10^4$ Hz **or** 15–25 kHz cao	B1	
		15–25 Hz cao	B1	
	(ii)	(λ =) c/f **or** 330/either of candidate's frequencies	C1	
		330/candidate's higher frequency and correctly calculated with unit (candidate's higher frequency is either the one stated as the highest **or** the one that is in fact the higher)	A1	[6]

QUESTION · 5054_s14_qp_11.pdf · source page 9 of 16

22 Compressions and rarefactions are sent out from a loudspeaker cone as it vibrates backwards and forwards. The frequency of vibration is 50 Hz.



A compression is at point P. How much time elapses before the next rarefaction arrives at P?

A 0.010 s **B** 0.020 s **C** 25 s **D** 50 s

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 22

22	**A**

QUESTION · 5054_s14_qp_11.pdf · source page 10 of 16

28 Which statement about the speed of sound is correct?

A sound travels fastest in a vacuum

B sound travels fastest in gases

C sound travels fastest in liquids

D sound travels fastest in solids

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 28

28	**D**

10 A sound wave travels through air.

(a) Explain what is meant by the *wavelength* of a wave.

.....

..... [1]

(b) Fig. 10.1 represents air molecules in the sound wave at one instant.

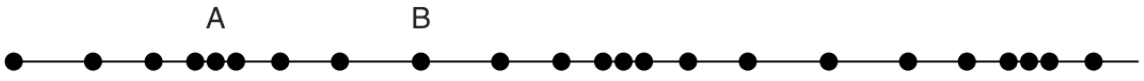


Fig. 10.1

(i) Describe the motion of a molecule of air due to the sound.

.....

..... [2]

(ii) State one difference between the motion of molecule A and the motion of molecule B.

.....

..... [1]

(c) Describe an experiment that shows that a medium is needed to transmit sound waves. Draw a labelled diagram of the apparatus.

.....

.....

.....

.....

.....

..... [4]

(d) A short pulse of sound waves produces an echo from a wall 20 m away. The echo arrives back at the source of the sound 0.12 s after the pulse is produced.

The wavelength of the sound wave is 30 mm.

(i) State what is meant by an *echo*.

.....

..... [1]

(ii) Calculate the speed of the sound.

speed = ... [2]

(iii) Calculate the frequency of the sound.

frequency = ... [3]

(iv) The sound wave produces compressions and rarefactions in the air as it passes.

Calculate the distance between a compression and the nearest rarefaction.

distance = ... [1]

10 (a) (shortest) distance between two points in phase [B1]

(b) (i) oscillate **or** vibrate **or** move closer and further [B1]
backwards and forwards **or** in direction of wave (energy) **or** longitudinal mentioned [B1]

(ii) move in opposite directions **or** when A is on right B is on left (of mean)
or A molecules next move apart **and** B next move together [B1]

(c) diagram containing sound source (bell / tuning fork, etc.) in container [B1]
vacuum / pump connected to container [B1]
remove air [M1]
sound decreases [A1]

(d) (i) sound (that returns) after a reflection [B1]

(ii) (s=) d / t in any algebraic or numerical form e.g. 20 (or 40) / 0.12 (or 0.06) [C1]
330 **or** 333 m / s [A1]

(iii) (f=) v / λ in any algebraic or numerical form e.g. 330 / 30 (or 0.03) [C1]
correct conversion of 30 mm to 0.03 m [C1]
11 100 Hz **or** 11 000 Hz [A1]

(iv) 0.015 m **or** 15 mm [B1]

[15]

QUESTION · 5054_s14_qp_12.pdf · source page 10 of 16

28 Which frequency is in the ultrasound range?

A 35 Hz **B** 350 Hz **C** 3500 Hz **D** 35 000 Hz

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 28

28	**D**

QUESTION · 5054_s14_qp_12.pdf · source page 11 of 16

29 In an experiment to determine the speed of sound in air, a student stands 200 m away from a cliff and claps two pieces of wood together.

His class-mates standing next to him start stopwatches when the two pieces of wood meet and stop the stopwatches when they hear the echo.

Their times are:

1.44 s 1.70 s 1.58 s 1.76 s

Which value for the speed of sound do they obtain?

A 62 m/s **B** 123 m/s **C** 247 m/s **D** 340 m/s

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 29

29	**C**

QUESTION · 5054_w14_qp_11.pdf · source page 13 of 20

25 A star explodes in outer space.

Which waves from the exploding star do **not** reach the Earth?

A infra-red

B light

C radio

D sound

MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 25

25	**D**

5 The device shown in Fig. 5.1 uses the reflection of ultrasound to measure distances.

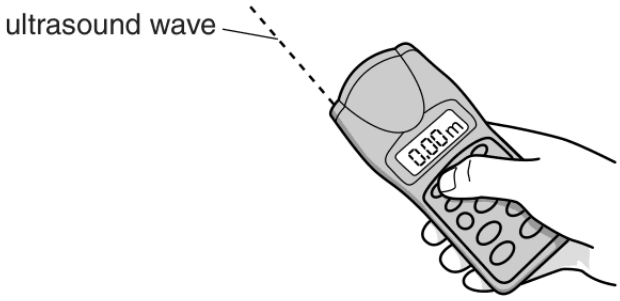


Fig. 5.1

(a) State what is meant by *ultrasound*.

.....

.....

.....[2]

(b) Fig. 5.2 shows a builder using the ultrasound device to measure the width of a room.

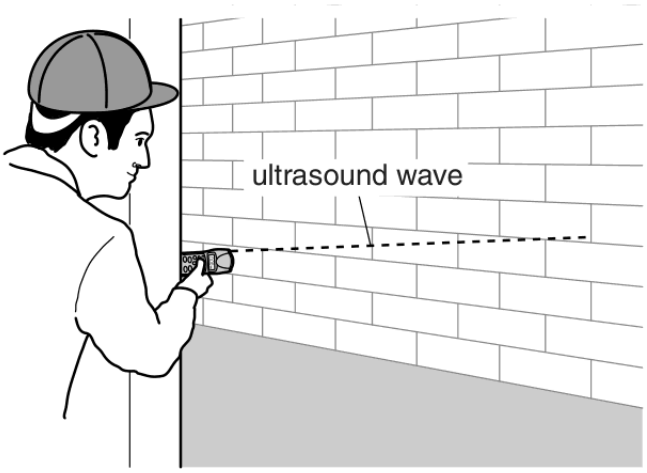


Fig. 5.2

The ultrasound device is placed against one wall and it emits an ultrasound wave that reflects back from the opposite wall.

The time between sending out the ultrasound wave and receiving the reflection is 0.030 s. The speed of ultrasound in air is 340 m / s.

Calculate the distance between the device and the opposite wall.

distance = ...[2]

5	**(a)**	longitudinal / pressure / sound (wave) **or** compressions **and** rarefactions	B1	
		(frequency) greater than 15 – 25 kHz / above limits of audibility	B1	
	(b)	(x =) vt / 2 **or** 340 × 0.030 / 2 **or** 340 × 0.015 **or** 10.2	C1	
		5.1 m	A1	**[4]**

QUESTION · 5054_w14_qp_12.pdf · source page 7 of 16

18 A student stands at a distance *d* from the base of a tall cliff.

He claps together two pieces of wood and measures the time that elapses before he hears the echo. He conducts the experiment five times and obtains these results.

0.72 s 0.80 s 0.71 s 0.81 s 0.71 s

The speed of sound is 320 m / s.

What is the distance *d*?

A 120 m **B** 240 m **C** 480 m **D** 600 m

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 18

18 **A**

QUESTION · 5054_w14_qp_12.pdf · source page 8 of 16

19 Ultrasound has many uses.

For what are ultrasound waves used?

A killing cancerous cells

B pre-natal scanning

C sunbeds

D telephones

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 19

19	**B**

QUESTION · 5054_s15_qp_11.pdf · source page 8 of 16

20 In which situation do sound waves **not** travel?

A from a satellite in space to Earth

B from a ship to a submarine

C from an explosion underground to the surface

D through a balloon filled with helium gas

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 20

20	**A**

QUESTION · 5054_s15_qp_12.pdf · source page 11 of 16

29 In which situation do sound waves **not** travel?

A from a satellite in space to Earth

B from a ship to a submarine

C from an explosion underground to the surface

D through a balloon filled with helium gas

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 29

29	**A**

QUESTION · 5054_s15_qp_12.pdf · source page 11 of 16

30 Two sound waves X and Y are compared.

X has the greater frequency.

Y has the greater amplitude.

How do the loudness and pitch of sound wave Y compare to those of X?

A Y is louder and higher pitch.

B Y is louder and lower pitch.

C Y is quieter and higher pitch.

D Y is quieter and lower pitch.

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 30

30	**B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows the wavefronts of a water wave in deep water in a ripple tank.

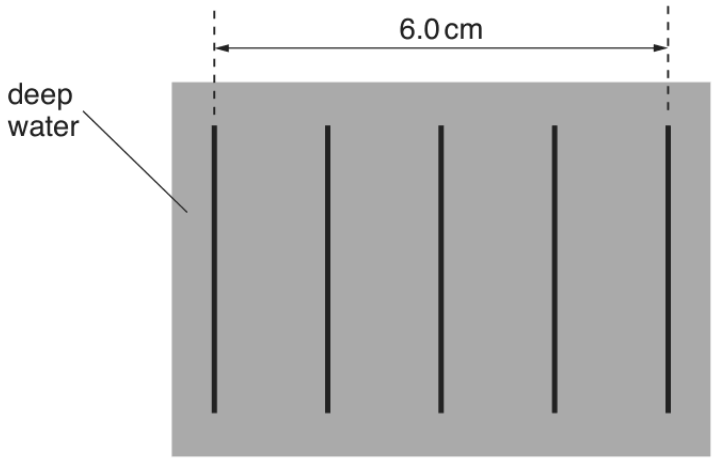


Fig. 9.1

The frequency of the wave in deep water is 5.0 Hz.

(a) Explain what is meant by the *frequency* of a wave.

.....

.....

..... [2]

(b) **(i)** Determine the wavelength of the wave in deep water.

..... [1]

(ii) Calculate the speed of the wave in deep water.

speed = ... [2]

(c) The wave passes from deep water into shallow water. The speed of the wave is less in shallower water.

(i) State and explain how this affects the wavelength of the wave.

.....

.....

..... [2]

(ii) The wave in deep water shown in Fig. 9.2 travels towards the right and enters the shallow water at an angle. The wave refracts.

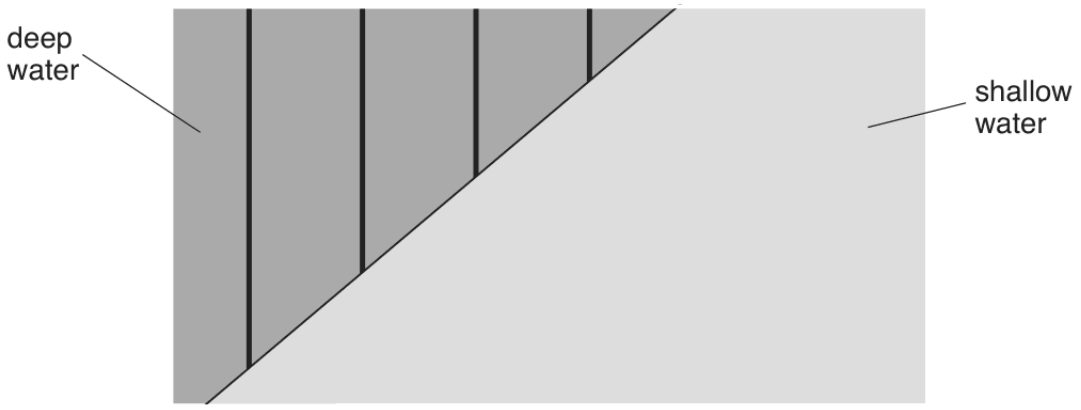


Fig. 9.2

On Fig. 9.2, draw the wavefronts in the shallow water. [3]

(d) Sound is also a wave.

(i) Describe one difference between a sound wave and a water wave.

.....

.....

.....

..... [2]

(ii) The speed of sound in carbon dioxide gas is less than the speed of sound in air. Using this information, or otherwise, describe an experiment to show the refraction of sound waves. You may include a diagram of your apparatus.

.....

.....

.....

.....

.....

..... [3]

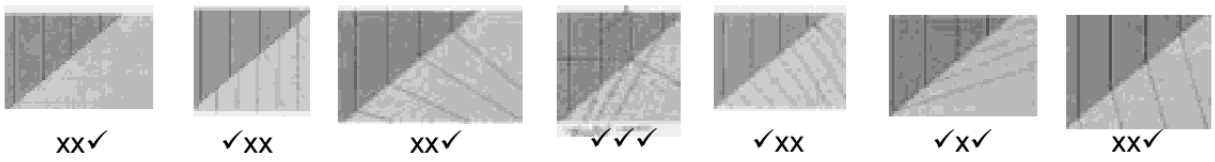
9 **(a)** number of waves (that pass a point) M1
or number of oscillations (passing a point)
in unit time **or** per second **or** in 1 second A1

(b) **(i)** 1.5 cm B1

(ii) (v =)$f\lambda$ **or** 5 × 1.5 seen C1
or (s=)d / t **and** f = 1 / t
7.5 cm / s A1

(c) **(i)** wavelength decreases B1
travels a shorter distance in the same time B1
or frequency stays the same (and $v = f\lambda$)

(ii) wavefronts with smaller wavelength (by eye) B1
smaller angle to surface (by eye) **and** slanted down B1
wavefronts join those in shallow water B1



xx✓ ✓xx xx✓ ✓✓✓ ✓xx ✓x✓ xx✓

(d) **(i)**

sound	water
particles / wave / source • vibrate / oscillate / move in direction of (travel of) wave / along wave • move backwards and forwards	particles / wave / source • vibrate / oscillate / move at 90° to direction of (travel of) wave • move up and down
(contains) compressions and rarefactions **or** particles come closer / further apart	(contains) crests and troughs
speed 300–330 m / s	wave slower (than sound)

B1
B1
one row only

(ii) method of generating sound, e.g. (loud) speaker (and signal generator) B1
apparatus that enables refraction clear, e.g. carbon dioxide in balloon or any shape where refraction is possible B1
method of detecting refraction, e.g. microphone and how it is used to show refraction B1

QUESTION · 5054_w15_qp_11.pdf · source page 9 of 16

22 A musical note is produced by two sources.

The traces produced by each source on the screen of a cathode-ray oscilloscope (c.r.o.) are shown below.

What is the difference between the sounds?

A the amplitude

B the frequency

C the pitch

D the quality

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 22

22	**D**

7 Audible sound waves and ultrasound waves both consist of compressions and rarefactions.

(a) Describe one difference between a compression and a rarefaction.

.....

.....

..... [1]

(b) In a particular medium, the speed of both audible sound and ultrasound waves is *v*.

(i) State the equation that relates the wavelength λ of a sound wave to its frequency *f*.

.....

..... [1]

(ii) Explain how, in this medium, the wavelength of an audible sound wave compares with the wavelength of an ultrasound wave.

.....

.....

.....

..... [1]

(c) Describe one use of ultrasound waves.

.....

.....

.....

.....

..... [2]

7	**(a)**	at compression: molecules closer together **or** pressure higher **or** vice versa for rarefaction		B1	
	(b)	**(i)** $v = f\lambda$ **or** in words		B1	
		(ii) larger **and** because the frequency is lower		B1	
	(c)	states one use	(e.g. prenatal scanning)	B1	
		basic idea	(e.g. ultrasound reflects off foetus)	B1	[5]

QUESTION · 5054_w15_qp_12.pdf · source page 11 of 16

25 A loudspeaker produces a sound wave of frequency 50 Hz.
The amplitude of the sound wave is increased.

What is heard?

A a louder sound of a higher pitch

B a louder sound of the same pitch

C a sound of higher pitch but the same loudness

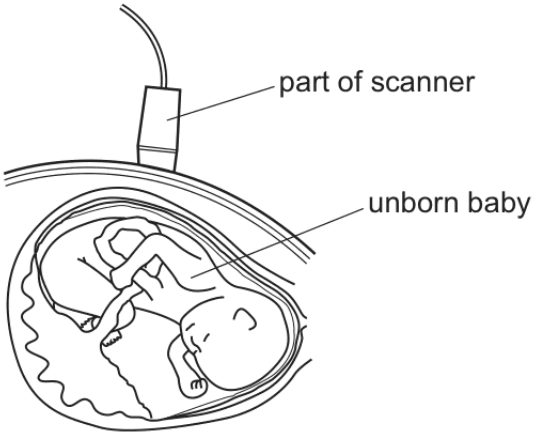
D a sound of the same pitch and loudness as before

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 25

25	**B**

QUESTION · 5054_w15_qp_12.pdf · source page 11 of 16

26 An ultrasound scanner produces an image of an unborn baby.



How does the scanner form an image?

A from ultrasound absorbed by the baby

B from ultrasound emitted by the baby

C from ultrasound reflected by the baby

D from ultrasound refracted by the baby

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 26

26	**C**

QUESTION · 5054_w15_qp_12.pdf · source page 11 of 16

27 The sound from the siren of a ship is reflected by a cliff. An echo is heard by a sailor on the deck of the ship, 4.0 s after the siren is sounded. The speed of sound in air is 320 m / s.

How far from the cliff is the ship?

A 80 m **B** 160 m **C** 640 m **D** 1280 m

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 27

27	**C**

QUESTION · 5054_s16_qp_11.pdf · source page 10 of 16

26 An intruder alarm is adjusted to give a quieter sound without affecting the pitch of the note.

How are the amplitude and the frequency of the sound affected?

	amplitude	frequency
A	lower	lower
B	lower	same
C	same	lower
D	same	same

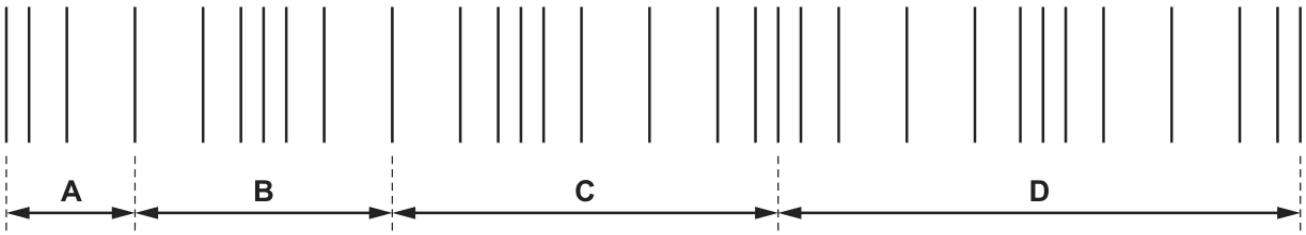
MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 26

26	**B**

QUESTION · 5054_s16_qp_12.pdf · source page 9 of 16

22 A sound wave travels through air. The lines in the diagram show the positions of layers of air at one particular time.

Which distance shows the wavelength of the wave?



MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 22

22	**B**

QUESTION · 5054_w16_qp_11.pdf · source page 12 of 20

29 Which property of a sound wave increases as the loudness of the sound increases?

A amplitude

B frequency

C speed

D wavelength

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 29

29	**A**

5 Fig. 5.1 shows a loudspeaker inside a glass bell-jar that contains air.

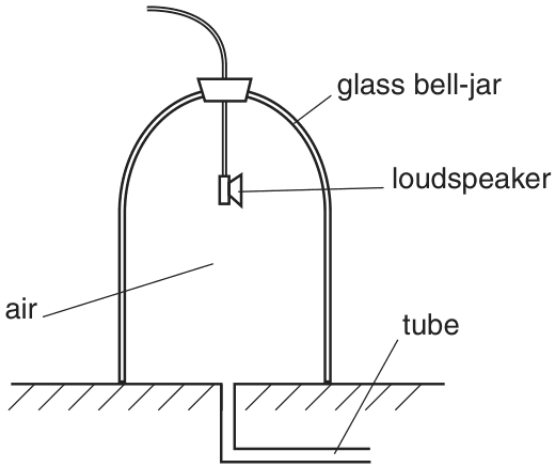


Fig. 5.1

The loudspeaker produces a sound of frequency 2200 Hz.

(a) State what is meant by the *frequency* of a sound.

.....

.....

.....[1]

(b) **(i)** The speed of sound in air is 330 m / s.

Calculate the wavelength of the sound in the air.

wavelength = ...[2]

(ii) The sound passes from the air into the glass.

State what happens to

1. the frequency of the sound,

.....

2. the speed of the sound.

.....
[1]

(c) (i) Describe

1. how sound is produced by the loudspeaker,

.....

.....

.....[1]

2. how sound is transmitted through the air.

.....

.....

.....[2]

(ii) A pump is connected to the tube at the bottom of the glass bell-jar. The pump is switched on and it removes the air from the glass bell-jar.

Explain why the volume of the sound heard outside the glass bell-jar decreases.

.....

.....

.....[1]

5	**(a)**	number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time		B1	
	(b)	**(i)**	(λ =)c / f **or** 330 / 2200	C1	
			0.15 m	A1	
		(ii)	**1** no change **and**		
			2 increases	B1	
	(c)	**(i)**	**1** loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules)	B1	
			2 compressions **and** rarefactions / molecules vibrate / longitudinal wave vibration / oscillation / energy passed on	B1 B1	
		(ii)	fewer / no molecules / particles **and** less / no energy / vibration transferred	B1	[8]

QUESTION · 5054_w16_qp_12.pdf · source page 11 of 16

29 Sound travels at different speeds in solids, liquids and gases.

Which list shows these states of matter in order of increasing speed of sound (slowest to fastest)?

A gas → liquid → solid

B gas → solid → liquid

C liquid → solid → gas

D solid → liquid → gas

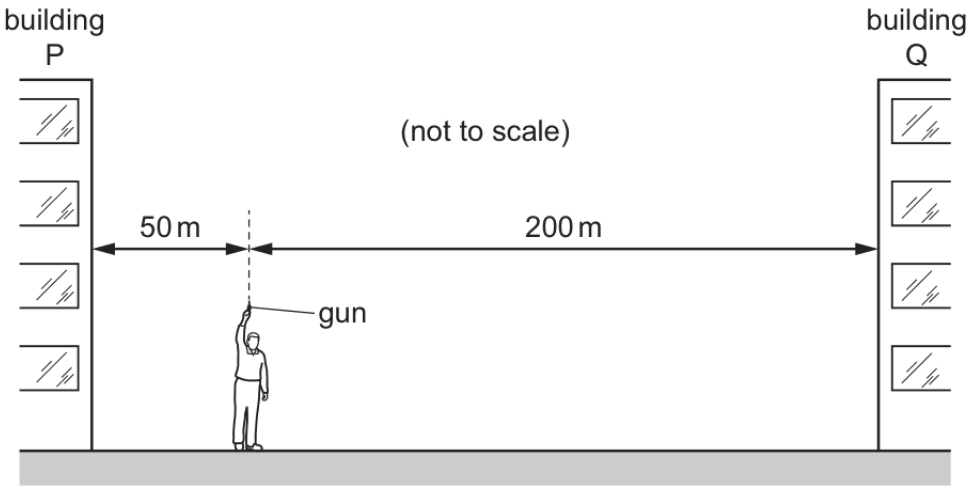
MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 29

29	**A**

QUESTION · 5054_w16_qp_12.pdf · source page 11 of 16

30 A man stands between two tall buildings, P and Q. The diagram is not drawn to scale.

He is 50 m from P and 200 m from Q.



He fires a gun and the first two echoes he hears are 1.0 second apart.

What is the speed of sound calculated from this observation?

A 250 m / s **B** 300 m / s **C** 330 m / s **D** 500 m / s

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 30

30	**B**

QUESTION · 5054_s17_qp_11.pdf · source page 13 of 16

31 A ship that is stationary on the surface of the sea sends pulses of sound vertically downwards towards the sea bed.

Each pulse that reflects from the sea bed is received 1.0 s after it is sent out.

A whale swims under the boat and a pulse is received 0.60 s after it is sent out.

The speed of sound in sea water is 1500 m / s.

What is the distance of the whale above the sea bed?

A 300 m **B** 450 m **C** 600 m **D** 750 m

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 3 of 3 · row for Question 31

31	A	**1**

10 **(a)** Fig. 10.1 shows two rays from an object that is placed in front of a plane mirror.

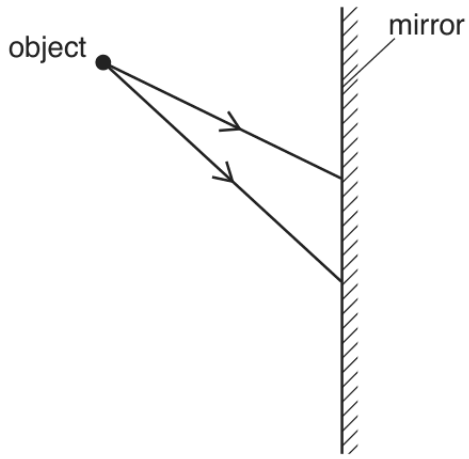


Fig. 10.1

(i) On Fig. 10.1, draw the two reflected rays and locate the position of the image. [2]

(ii) Apart from its position, state one characteristic of the image.

.....[1]

(b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

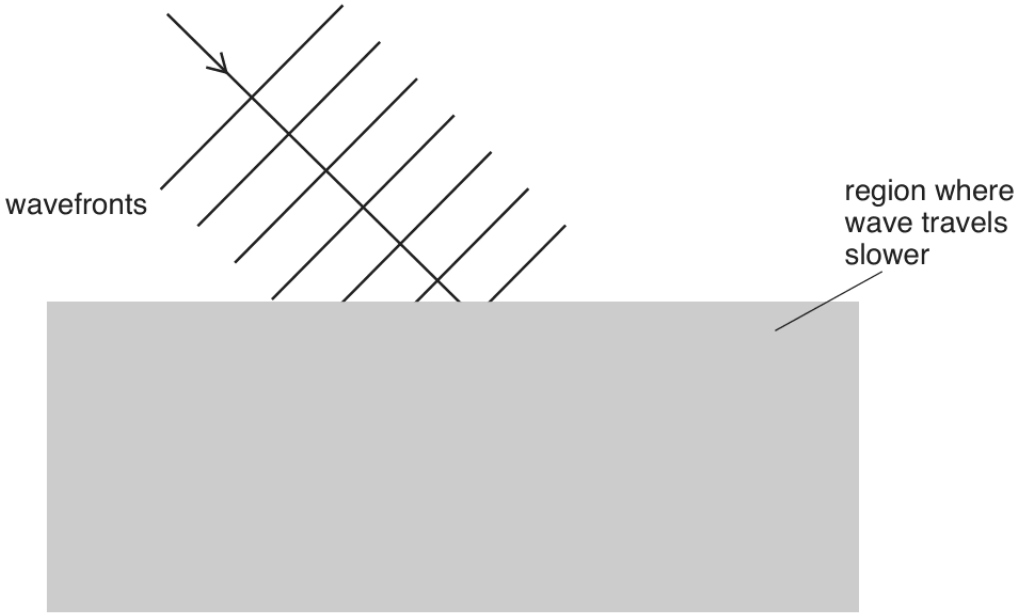


Fig. 10.2

(i) State what is meant by *wavefront*.

.....

.....[2]

(ii) Both reflection and refraction occur at the boundary.

1. On Fig. 10.2, draw the reflected wavefronts. [1]

2. On Fig. 10.2, draw the refracted wavefronts. [2]

(c) A loudspeaker produces a sound of frequency 2.0 kHz. The wavelength of this sound in air is 16 cm.

(i) Calculate the speed of sound in air.

speed = ...[2]

(ii) **1.** State the range of frequencies that can be heard by a healthy human ear.

.....[1]

2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength = ...[1]

(iii) Describe a simple experiment to show that sound waves obey the law of reflection. You may draw a diagram if you wish.

.....

.....

.....

.....[3]

Question	Answer	Marks
10(a)(i)	both reflected rays correct by eye	**B1**
	image in correct position shown by continuation of rays behind mirror	**B1**
10(a)(ii)	virtual **or** upright **or** laterally inverted **or** same size (as object)	**B1**
10(b)(i)	line (joining points)	**C1**
	line joining points / particles on wave with same phase **or** line joining points along a crest etc.	**A1**
10(b)(ii)1	correct angle to surface ± 7° with correct orientation **and** similar wavelength (by eye)	**B1**
10(b)(ii)2	at least two lines at smaller angle to surface with correct orientation in slower medium	**B1**
	showing smaller and constant wavelength with wavefronts deviated in correct direction	**B1**
10(c)(i)	(v =) fλ **or** 2(000) × 16	**C1**
	32 000 cm / s **or** 320 m / s	**A1**
10(c)(ii)1	a range with 15–25 Hz as the lowest frequency **and** 15–30 kHz as the highest	**B1**
10(c)(ii)2	1.6 cm (if highest frequency is 20 000 Hz)	**B1**
10(c)(iii)	tube or other method to produce (narrow) beam of sound on source and/or on detector	**B1**
	stated detector **or** stated reflector	**B1**
	detector moved to find maximum loudness **or** angles *i* and *r* measured with suitable experiment	**B1**

6 **(a)** Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

(b) A sound wave has a frequency of 3.8 kHz and a speed of 330 m/s.

(i) Calculate the wavelength of the sound.

wavelength = ...[2]

(ii) A different sound has a frequency of 3.8 Hz.

State and explain whether a human with normal hearing is able to hear this sound.

.....

.....

.....[1]

Question	Answer	Marks
6(a)	longitudinal - vibration / oscillation / movement to and fro **and** in direction of wave **or** has compressions and rarefactions	**B1**
	transverse – vibration / oscillation / movement up and down **and** at right angles to wave **or** has crests and troughs	**B1**
6(b)(i)	(λ =) v / f or 330 / 3800	**B1**
	0.087 m **or** 8.7 cm	**B1**
6(b)(ii)	not heard **and** because below the range of audible frequencies **or** audible range is 20 – 20 000 Hz **or** too low a pitch / frequency	**B1**

QUESTION · 5054_w17_qp_11.pdf · source page 11 of 20

24 Which row gives the speed of sound in air, in water and in steel?

	speed in air m/s	speed in water m/s	speed in steel m/s
A	330	1500	6000
B	330	6000	1500
C	6000	330	1500
D	6000	1500	330

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 24

24	A	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 12 of 20

25 As a sound wave travels from one medium to another, its wavelength increases.

What happens to the frequency and to the speed of the sound?

	frequency	speed
A	decreases	decreases
B	decreases	increases
C	stays constant	decreases
D	stays constant	increases

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 25

25	D	**1**

5 Some medical processes involve the use of ultrasound.

(a) Explain what is meant by *ultrasound*.

.....

.....

.....

..... [2]

(b) Explain briefly how ultrasound is used in pre-natal scanning.

.....

.....

.....

.....

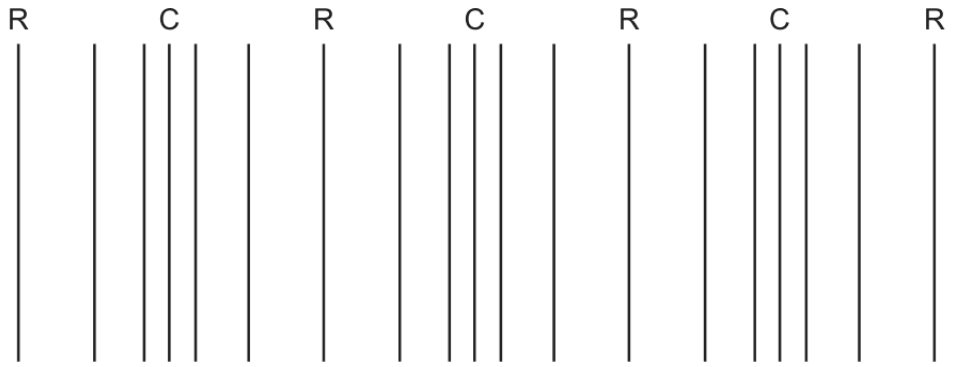
.....

..... [3]

Question	Answer	Marks
5(a)	pressure / vibrational / longitudinal (wave) **or** (wave that consists of) compressions **and** rarefactions **or** sound wave **or** inaudible sound	**B1**
	frequency greater than 15 kHz–25 kHz	**B1**
5(b)	(ultrasound) transmitted into body	**B1**
	echo / reflection from (baby / fetus)	**B1**
	image produced	**B1**

QUESTION · 5054_w17_qp_12.pdf · source page 8 of 16

19 A sound wave in a solid is represented by a series of lines. The diagram shows compressions C and rarefactions R at one instant in time.



Which statement is correct?

A All particles on one line move in the same direction at the same time.

B The direction of travel of the sound wave is parallel to the lines.

C The distance between a compression and its adjacent rarefaction is a wavelength.

D The pattern of lines represents a transverse wave.

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 19

19	A	**1**

QUESTION · 5054_w17_qp_12.pdf · source page 10 of 16

25 Which row gives the speed of sound in air, in water and in steel?

	speed in air m/s	speed in water m/s	speed in steel m/s
A	330	1500	6000
B	330	6000	1500
C	6000	330	1500
D	6000	1500	330

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 25

25	A	**1**

6 Fig. 6.1 shows a girl standing a few hundred metres in front of a large building.

Fig. 6.1 (not to scale)

The girl uses a signal generator and a loudspeaker to send a short pulse of sound towards the building. The sound has a frequency of 3700 Hz.

A short time later, the girl hears an echo.

(a) State what is meant by the term *echo*.

.....

.....[1]

(b) The pitch of the echo is the same as that of the original sound but the echo is not as loud.

State what has happened to

(i) the amplitude of the sound wave,

.....[1]

(ii) the frequency of the sound wave.

.....[1]

(c) The speed of sound in air is 330 m/s.

Calculate the wavelength of this sound.

wavelength = ...[2]

Question	Answer	Marks
6(a)	reflection of sound	**B1**
6(b)(i)	decreases	**B1**
6(b)(ii)	does not change	**B1**
6(c)	(λ =) c / f **or** 330 / 3700	**C1**
	0.089 m **or** 8.9 cm **or** 89 mm	**A1**

QUESTION · 5054_s18_qp_11.pdf · source page 11 of 20

25 A young person with healthy ears can hear a range of frequencies.

What is the approximate range of frequencies?

A 2 Hz to 2000 Hz only

B 2 Hz to 20 000 Hz only

C 20 Hz to 2000 Hz only

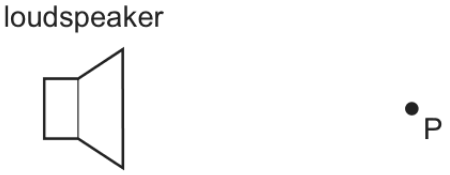
D 20 Hz to 20 000 Hz only

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 25

25	D	**1**

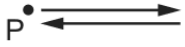
QUESTION · 5054_s18_qp_11.pdf · source page 11 of 20

26 The diagram shows a loudspeaker that is producing a continuous sound wave of frequency 200 Hz in air.



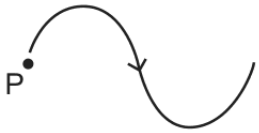
Which diagram best shows how the sound wave causes a molecule at P to move during $\frac{1}{200}$ s ?

A



B

C



D

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 2 of 3 · row for Question 26

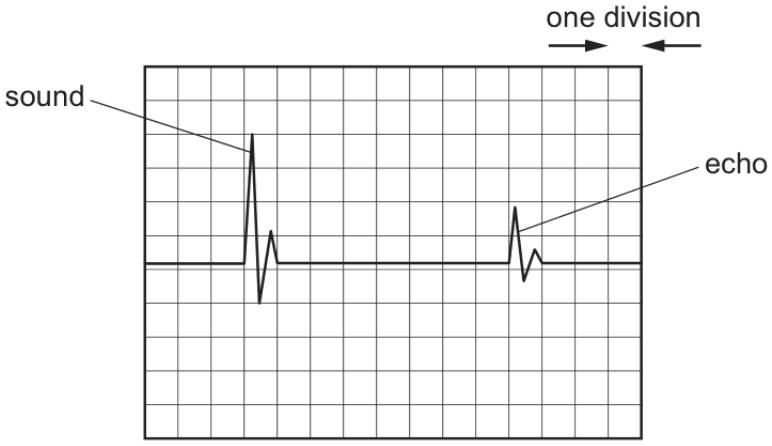
26	A	**1**

QUESTION · 5054_s18_qp_11.pdf · source page 16 of 20

34 A loudspeaker and a microphone are placed in front of a wall.

The loudspeaker makes a sound which is detected by the microphone.

The microphone is connected to an oscilloscope which is set so that each division on the screen represents 0.01 s. The microphone detects the original sound and the echo.



display on oscilloscope

The speed of sound in air is 300 m / s.

What is the distance between the loudspeaker and the wall?

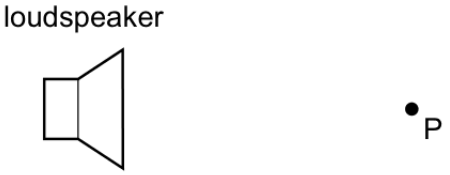
A 6.0 m **B** 12 m **C** 24 m **D** 48 m

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 3 of 3 · row for Question 34

34	B	**1**

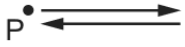
QUESTION · 5054_s18_qp_12.pdf · source page 12 of 16

31 The diagram shows a loudspeaker that is producing a continuous sound wave of frequency 200 Hz in air.

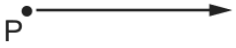


Which diagram best shows how the sound wave causes a molecule at P to move during $\frac{1}{200}$ s ?

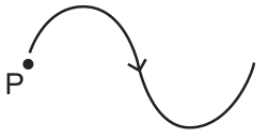
A



B



C



D

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 3 of 3 · row for Question 31

31	A	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 12 of 16

32 Which frequency is in the range heard by a healthy human ear?

A 12 mHz **B** 12 Hz **C** 12 kHz **D** 12 MHz

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 3 of 3 · row for Question 32

32	C	**1**

QUESTION · 5054_s18_qp_12.pdf · source page 12 of 16

33 What is one of the uses of ultrasound?

A cleaning jewellery

B satellite communication

C fluorescent tubes

D optical fibres

MARK SCHEME (official answer key) · 5054_s18_ms_12.pdf · source page 3 of 3 · row for Question 33

33	A	**1**

QUESTION · 5054_w18_qp_11.pdf · source page 8 of 16

22 What are the lowest frequency and the highest frequency of the sound that a human being with normal hearing can hear?

	lowest frequency	highest frequency
A	2.0 Hz	2.0 kHz
B	2.0 Hz	20 000 Hz
C	20 Hz	20 kHz
D	20 kHz	20 000 kHz

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 22

22	C	**1**

QUESTION · 5054_w18_qp_11.pdf · source page 8 of 16

23 Ultrasound has a number of uses in engineering and medicine.

What is ultrasound **not** used for?

A cleaning

B pre-natal scanning

C quality control

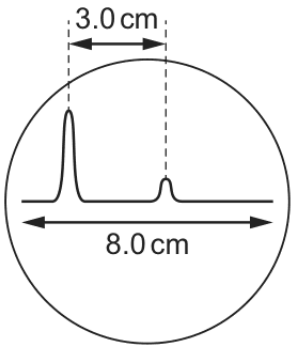
D sterilising medical equipment

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 23

23	D	**1**

QUESTION · 5054_w18_qp_11.pdf · source page 13 of 16

36 The diagram shows the trace on an oscilloscope screen. The first peak occurs when a pulse of sound is emitted. The second, smaller peak is the echo received from a wall.



The trace moves across the screen in a time of 24 ms.

How long does it take for the sound to travel to the wall?

A 4.5 ms **B** 9.0 ms **C** 12 ms **D** 32 ms

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 3 of 3 · row for Question 36

36	A	**1**

QUESTION · 5054_w18_qp_12.pdf · source page 11 of 16

27 Ultrasound is used to map the ocean floor.

During one survey, the depth of water is 1200 m. An ultrasound pulse is sent from the surface and when it returns to the ship, another pulse is sent immediately. In any period of 8.0 s, five pulses are sent down from the surface.

What is the speed of the ultrasound in water?

A 150 m / s **B** 300 m / s **C** 750 m / s **D** 1500 m / s

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 2 of 3 · row for Question 27

27	D	**1**

9 Both sound and ultrasound are waves that travel in solids, liquids and gases.

(a) State how sound differs from ultrasound.

.....

..... [1]

(b) Sound and ultrasound are longitudinal waves that consist of compressions and rarefactions.

(i) Explain what is meant by a wave and describe how a longitudinal wave differs from a transverse wave.

.....

.....

.....

.....

..... [3]

(ii) Fig. 9.1 represents a longitudinal wave of frequency 25 000 Hz travelling in oil.

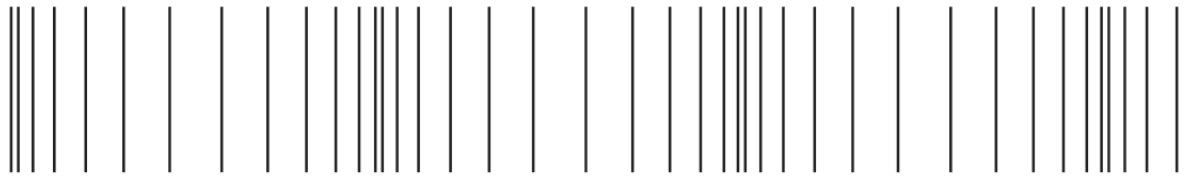


Fig. 9.1 (full scale)

1. On Fig. 9.1, mark two points at the centre of two different rarefactions and label each one R. [1]

2. On Fig. 9.1, draw a double-headed arrow to indicate a distance that is equal to one wavelength of the wave. [1]

3. Measure the length of the arrow drawn on Fig. 9.1 and use it to determine the speed of the wave in oil.

speed = ... [2]

(c) A sound wave travelling in a liquid, passes into air.

(i) State what happens to the speed of the sound wave as it enters the air.

..... [1]

(ii) The wave in the liquid travels towards the surface at an angle. Fig. 9.2 shows the centres of the compressions of the sound wave in the liquid.

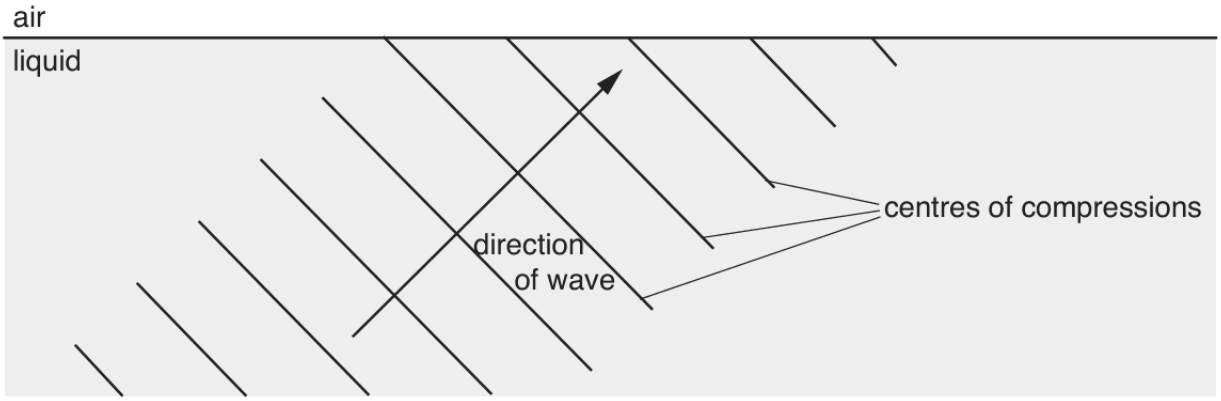


Fig. 9.2

Some compressions shown have reached the liquid-air boundary. The parts of these compressions in the air are not shown on Fig. 9.2.

On Fig. 9.2, draw the parts of these compressions that are in the air. [3]

(d) Describe, in outline, how ultrasound is used in cleaning.

.....

.....

.....

.....

..... [3]

Question	Answer	Marks
9(a)	frequency of sound wave small(er) **or** its frequency is less than 20 000 Hz	**B1**
9(b)(i)	transmission of energy	**B1**
	(through a medium) with no net movement of medium **or** by vibrating particles	**B1**
	vibrations parallel (and antiparallel) to wave / energy travel direction **or** cannot be polarised	**B1**
9(b)(ii)	1 two centres of rarefactions labelled *R*	**B1**
	2 distance from one point to adjacent identical point indicated (with double-headed arrow)	**B1**
	3. (v =) $f\lambda$ **or** 25 000 × 0.047 / 0.048 / 0.049 **or** 25 000 × 4.7 / 4.8 / 4.9 **or** 25 000 × 47 / 48 / 49	**C1**
	1200 m / s	**A1**
9(c)(i)	decreases	**B1**
9(c)(ii)	four / five straight lines in air that touch the compressions still in the liquid **and** no intermediate / extra lines between the correct lines	**B1**
	at least four compressions in air parallel to each other	**B1**
	at least four straight lines at shallower angle from horizontal **and** slope correct	**B1**

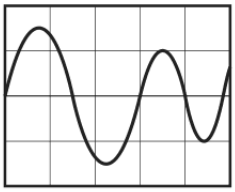
Question	Answer	Marks
9(d)	object (to be cleaned) immersed in liquid / solvent	**B1**
	object / liquid agitated / vibrated by ultrasound	**B1**
	dirt (particles) shaken off **or** dislodges / removes dirt	**B1**

QUESTION · 5054_s19_qp_11.pdf · source page 12 of 20

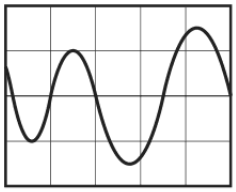
27 Four sound waves are displayed on the screen of a cathode-ray oscilloscope.

Which sound wave gets louder and has a pitch that decreases?

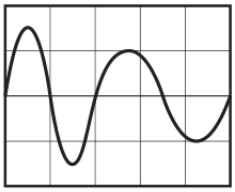
A



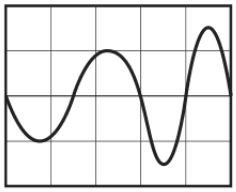
B



C



D



MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 2 of 3 · row for Question 27

27	B	**1**

10 In Fig. 10.1 the vertical lines represent the crests of a wave in a ripple tank.

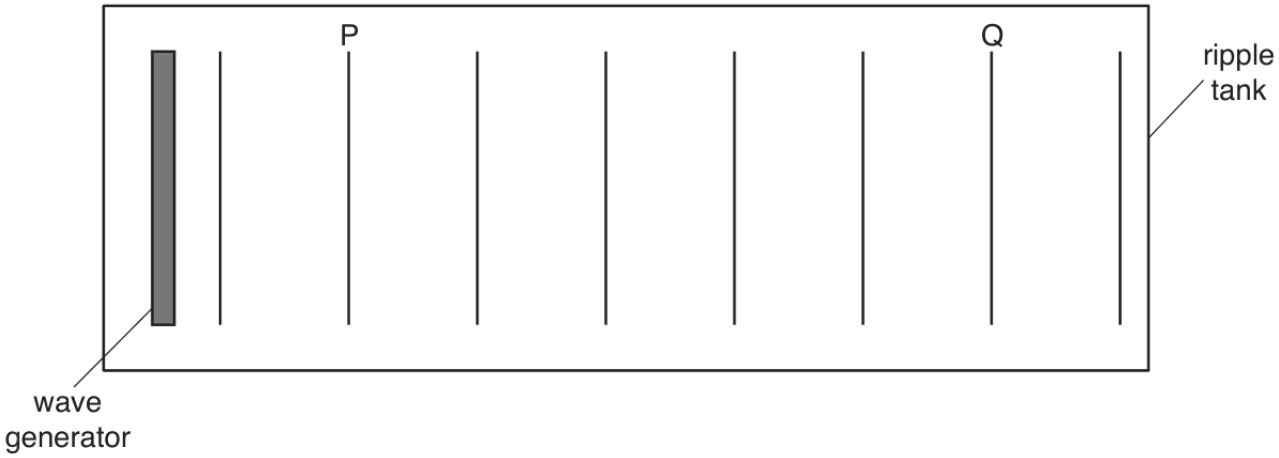


Fig. 10.1

Fig. 10.1 is drawn to scale, with 1.0 cm on the diagram representing 4.0 cm in the tank.

A crest takes 2.0 s to travel from P to Q.

(a) (i) Determine the wavelength of the wave in the ripple tank.

wavelength = ... [2]

(ii) Define the term *frequency*.

.....

..... [2]

(iii) Determine the frequency of the wave.

frequency = ... [1]

(iv) The wave generator is a bar of wood.

Describe how it is used to produce the wave shown.

.....

.....

..... [2]

(v) Some water in the ripple tank is removed. The water becomes less deep. No other change is made.

State how the change in depth affects the wavelength and the speed of the wave.

speed .....

wavelength .....
[1]

(b) A student reads in a textbook that the speed of sound in air is about 300 m / s.

He plans an experiment to check this value.

He asks another student to stand a long distance away in a large open space.

One student has a starting pistol. The pistol, when fired, produces a loud sound and a puff of smoke at the same instant.

(i) Other pieces of apparatus are needed for this experiment in order to measure the speed of sound.

Name two other pieces of apparatus that are needed.

1. .....

2. .....
[2]

(ii) Describe how all the measurements are made.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(iii) State typical values for the speed of sound in:

1. a liquid

.....

2. a solid.

.....

[2]

[Total: 15]

Question	Answer	Marks
10(a)(i)	1.7 (cm) seen or any multiple of wavelength measured on diagram	**C1**
	6.4–7.2 cm	**A1**
10(a)(ii)	number of waves/cycles/oscillations in one second / unit time	**C1**
	number of wavelengths/crests/troughs/ wave cycles/wave oscillations generated/made/pass a point in one second/unit time **or** number of oscillations of a source/particle in one second / unit time	**A1**
10(a)(iii)	2.5 Hz	**B1**
10(a)(iv)	moved up and down	**B1**
	regularly **or** at constant frequency / 2.5 times a second	**B1**
10(a)(v)	*speed* smaller / decreases **and** *wavelength* smaller / decreases	**B1**
10(b)(i)	tape measure **or** trundle wheel **or** microphones	**B1**
	stopwatch **or** timer	**B1**
10(b)(ii)	start timer on seeing the smoke from gun / sound picked up by microphone	**B1**
	stop timer on hearing sound	**B1**
	measure (large) distance between students	**B1**
10(b)(iii)	both between and including 1000 and 10 000 m/s	**B1**
	solid speed > liquid speed	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 10 of 16

29 What is the approximate range of audible frequencies for a healthy human ear?

A 2.0 Hz to 2.0 kHz

B 20 Hz to 2.0 kHz

C 20 Hz to 20 kHz

D 200 Hz to 200 kHz

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 3 of 3 · row for Question 29

29	C	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 10 of 16

30 The speed of sound in air is c_a, the speed of sound in water is c_w and the speed of sound in ice is c_i.

Which relationship is correct?

A $c_a > c_i$ **B** $c_a > c_w$ **C** $c_i > c_w$ **D** $c_w = c_a$

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 3 of 3 · row for Question 30

30	C	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 13 of 20

28 A person hears an echo after a sound hits a solid cliff face.

What causes the echo?

A absorption

B dispersion

C reflection

D refraction

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 2 of 3 · row for Question 28

28	C	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 13 of 20

29 What is a typical value for the speed of sound in air?

A $3.3 \times 10^4\,\text{cm/s}$

B $3.3 \times 10^4\,\text{m/s}$

C $3.3 \times 10^2\,\text{km/s}$

D $3.3 \times 10^8\,\text{m/s}$

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 3 of 3 · row for Question 29

29	A	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 12 of 16

27 Which row shows the speed of sound in air (c_a), liquid (c_L) and solid (c_s) in order from slowest to fastest?

	slowest →		fastest
A	c_L	c_s	c_a
B	c_L	c_a	c_s
C	c_a	c_L	c_s
D	c_s	c_L	c_a

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 27

27	C	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 12 of 16

28 A man stands 80 m in front of a cliff face. He makes a loud bang and listens for the echo.

He makes a loud bang once every second. He hears an echo exactly halfway between the bang that caused it and the next bang.

What is the speed of sound?

A 40 m/s **B** 80 m/s **C** 160 m/s **D** 320 m/s

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 2 of 3 · row for Question 28

28	D	**1**

5 Both sound and ultrasound are longitudinal waves.

(a) Describe what is meant by *longitudinal*.

.....

.....

..... [2]

(b) Fig. 5.1 shows an ultrasound cleaner being used to clean jewellery.

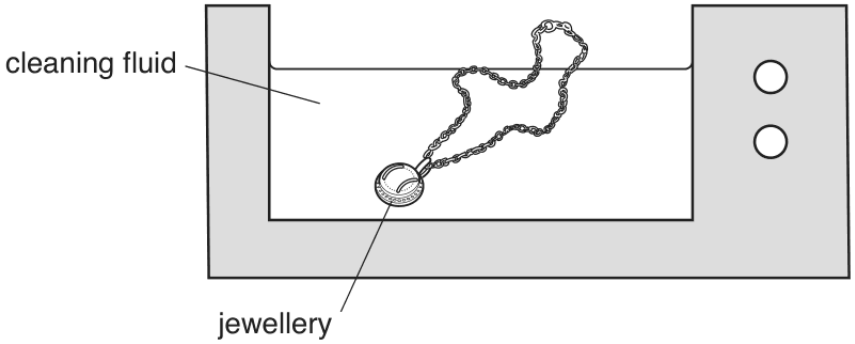


Fig. 5.1

The jewellery is lowered into the cleaning fluid and ultrasound waves of frequency 42 000 Hz are produced in the fluid.

(i) The speed of ultrasound in the fluid is 1500 m / s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = ... [2]

(ii) Describe how the ultrasound cleans the jewellery.

.....

.....

..... [2]

[Total: 6]

MARK SCHEME · 5054_w19_ms_22.pdf · source page 5 of 10 · Question 5

Question	Answer	Marks
5(a)	vibration / oscillation of <u>particles</u> **or** energy transfer **or** compressions and rarefactions	**B1**
	vibration / oscillation (of particles) parallel to energy travel direction / propagation direction	**B1**

MARK SCHEME · 5054_w19_ms_22.pdf · source page 6 of 10 · Question 5

Question	Answer	Marks
5(b)(i)	(λ =) v / f **or** 1500 / 42 000	**C1**
	0.036 m **or** 3.6 cm	**A1**
5(b)(ii)	(ultrasound) vibrates (the cleaning fluid / jewellery)	**B1**
	dirt shaken off	**B1**

QUESTION · 5054_s20_qp_11.pdf · source page 11 of 16

30 What is the ratio of the longest sound wavelength audible to a healthy human ear to the shortest?

A 20 : 1 **B** 1000 : 1 **C** 10 000 : 1 **D** 20 000 : 1

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 3 of 3 · row for Question 30

30	B	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 11 of 16

31 Ultrasound is used to clean jewellery in a liquid.

What is another use of ultrasound?

A optical fibre communication

B pre-natal scanning

C sunbeds

D telephone communications

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 3 of 3 · row for Question 31

31	B	**1**

9 Ultrasound and X-rays are both used in medical imaging.

(a) (i) Define what is meant by *ultrasound*.

.....

..... [2]

(ii) Describe what happens to ultrasound waves as they meet the boundary between two different materials.

.....

..... [2]

(iii) To produce the image of an unborn child, an ultrasound emitter and receiver are placed close together on the mother's skin.

Fig. 9.1 shows pulses detected by the receiver.

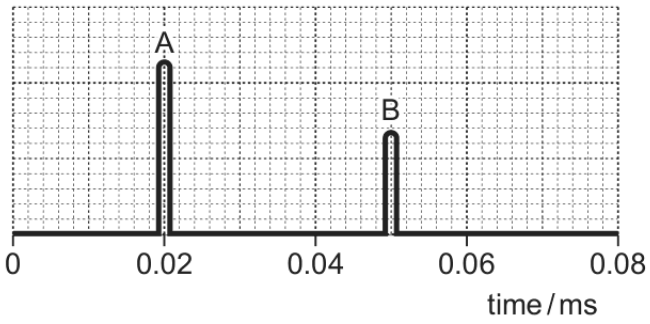


Fig. 9.1

Pulse A is the emitted pulse and pulse B is the first pulse that returns from the unborn child.

The average speed of ultrasound in human tissue is 1500 m / s.

Calculate the distance between the emitter and the child.

distance = ... [3]

(iv) The speed of ultrasound in human tissue is close to the speed of sound in water.

Suggest approximate values for the speed of sound in gases and solids.

speed in gases .....

speed in solids .....
[2]

(b) Fig. 9.2 shows an X-ray image of a hand. An X-ray detector is placed just below the hand. An image of the bones and human tissue around the bones is formed on a screen by the detector.

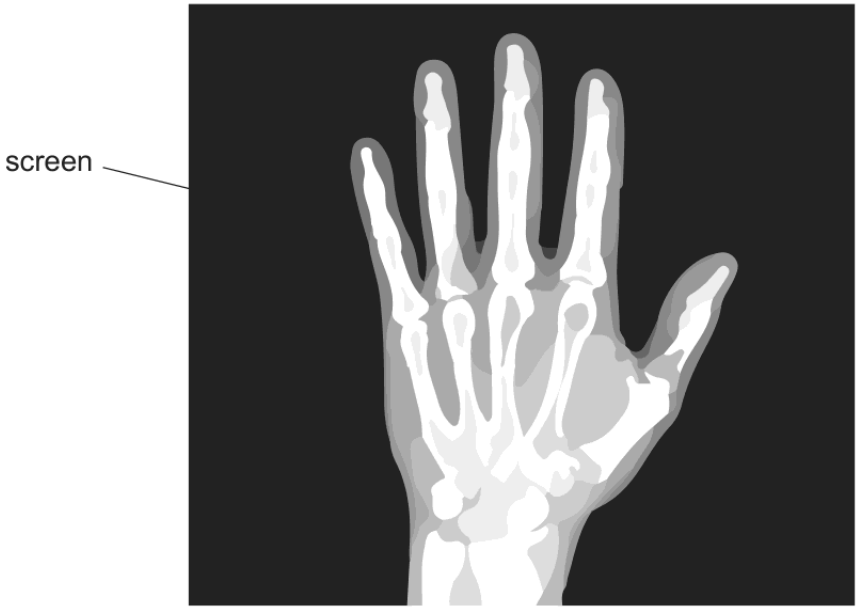


Fig. 9.2

(i) Describe what happens to the X-rays to produce the image.

.....

.....

.....

.....

..... [3]

(ii) The wavelength of the X-rays used is 2.0×10^{-9} m. The speed of electromagnetic waves is 3.0×10^{8} m/s.

Calculate the frequency of the X-rays.

frequency = ... [2]

(iii) Suggest one reason why X-rays are **not** used to form an image of an unborn child.

.....

..... [1]

[Total: 15]

MARK SCHEME · 5054_s20_ms_21.pdf · source page 9 of 11 · Question 9

Question	Answer	Marks
9(a)(i)	(sound that) cannot be heard	**C1**
	(sound with) frequency above 20 kHz	**A1**
9(a)(ii)	(some) reflect	**B1**
	(some) pass into new material **or** (some) absorbed by material	**B1**
9(a)(iii)	(d =) s × t in any form numerical or algebraic	**C1**
	1500×0.03 ($\times 10^{-3}$ / 2)	**C1**
	0.022 m **or** 0.023 m	**A1**
9(a)(iv)	***gases*** 150 – 1000 m/s	**B1**
	solids 2000 – 8000 m/s	**B1**

MARK SCHEME · 5054_s20_ms_21.pdf · source page 10 of 11 · Question 9

Question	Answer	Marks
9(b)(i)	(more) X-rays make the image black(er)	**B1**
	X-rays absorbed by bone **or** do not reach detector through bone	**B1**
	some X-rays pass through tissue **or** tissue absorbs some X-rays	**B1**
9(b)(ii)	(f =) v / λ in any form numerical or algebraic	**C1**
	1.5×10^{17} Hz	**A1**
9(b)(iii)	(X-rays) cause mutation / cancer / kill cells / ionisation	**B1**

10 An oscilloscope is a device used to display waveforms.

(a) Inside the oscilloscope, a beam of electrons is emitted from a metal filament by thermionic emission.

The emitted electrons are accelerated away from the filament by a potential difference of 2000 V.

(i) State what must happen to the metal filament for thermionic emission to occur.

.....

..... [1]

(ii) Explain why the electrons accelerate away from the filament.

.....

.....

..... [1]

(iii) The charge on one electron is 1.6×10^{-19} C.

Calculate the maximum kinetic energy of one electron after it has been accelerated through 2000 V.

kinetic energy = ... [2]

(b) Fig. 10.1 is the trace on the screen of the oscilloscope.

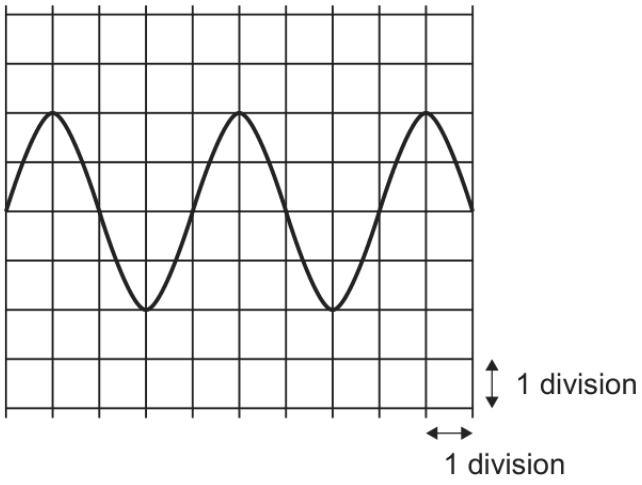


Fig. 10.1

The settings on the oscilloscope are 10 ms / division for the *x*-axis and 3.0 V / division for the *y*-axis.

(i) Calculate the amplitude of the trace shown in Fig. 10.1.

amplitude = ... [2]

(ii) Calculate the time for one oscillation (complete wavelength) of the trace shown in Fig. 10.1.

time = ... [1]

(c) The trace shown in Fig. 10.1 is caused by a sound.

The sound travels through the air to a microphone from the place that it is made. The microphone is connected to the oscilloscope which displays the waveform shown.

(i) Sound is a type of wave.

State which type.

..... [1]

(ii) Describe the motion of the air molecules as the sound passes through the air to the microphone.

.....

.....

..... [2]

(iii) Describe and explain how the trace on the screen changes as the pitch of the sound becomes higher. The settings on the oscilloscope are unchanged.

.....

.....

..... [2]

(d) A student investigates what happens when a diode is connected between the microphone and the oscilloscope.

Fig. 10.2 shows the circuit that he uses.

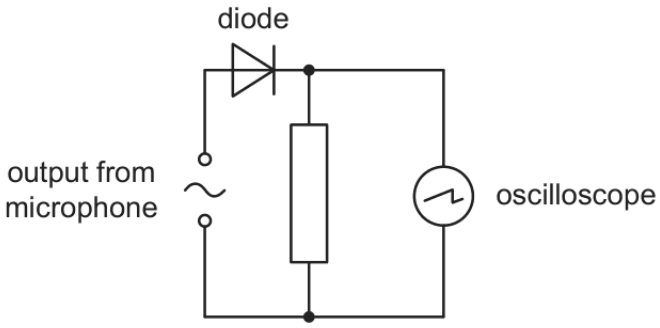


Fig. 10.2

Without the diode, the output from the microphone is an alternating current and the trace is as shown in Fig. 10.1 on page 18.

(i) Describe the action of the diode on the current from the microphone.

.....

..... [1]

(ii) On the grid in Fig. 10.3, sketch the trace seen on the screen when the diode is used.

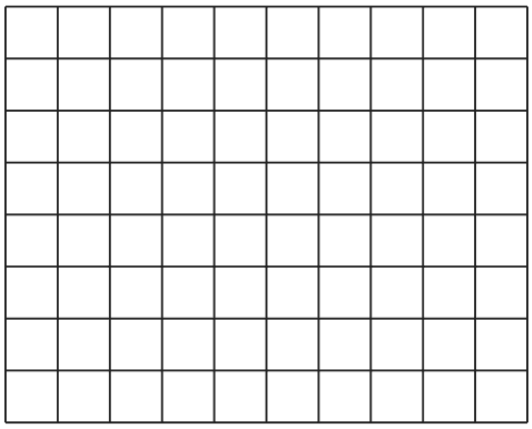


Fig. 10.3

[2]

[Total: 15]

MARK SCHEME · 5054_s20_ms_22.pdf · source page 12 of 13 · Question 10

Question	Answer	Marks
10(a)(i)	filament is hot / heated	**B1**
10(a)(ii)	attracted by a positive potential **or** repelled by filament **or** filament is negative	**B1**
10(a)(iii)	$2000 \times 1.6 \times 10^{-19}$	**C1**
	3.2×10^{-16} J	**A1**
10(b)(i)	two divisions	**C1**
	6(.0) V	**A1**
10(b)(ii)	40 ms	**B1**
10(c)(i)	longitudinal	**B1**

MARK SCHEME · 5054_s20_ms_22.pdf · source page 13 of 13 · Question 10

Question	Answer	Marks
10(c)(ii)	oscillation described, e.g. back and forth	**B1**
	in direction of travel (of wave / energy) **or** compressions and rarefactions stated or described	**B1**
10(c)(iii)	more waves seen on screen	**B1**
	higher frequency (of sound) **or** more waves per second **or** time for one wave less	**B1**
10(d)(i)	diode passes current in only one direction **or** current only flows for half the cycle / oscillation **or** current is d.c. / direct	**B1**
10(d)(ii)	at least one top half and no bottom half **or** one bottom half and no top half	**C1**
	three top halves of the wave and flat sections along mid line **or** two bottom halves of the wave and flat sections along mid line	**A1**

QUESTION · 5054_w20_qp_11.pdf · source page 12 of 16

30 Which description of a sound wave is correct?

A a longitudinal electromagnetic wave

B a transverse electromagnetic wave

C an oscillation of particles parallel to the direction of travel of the wave energy

D an oscillation of particles perpendicular to the direction of travel of the wave energy

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 3 of 3 · row for Question 30

30	C	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 13 of 16

31 Where does sound travel the fastest?

A in a vacuum

B in gases

C in liquids

D in solids

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 3 of 3 · row for Question 31

31	D	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 10 of 16

26 Where does sound travel the fastest?

A in a vacuum

B in gases

C in liquids

D in solids

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 26

26	D	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 11 of 16

27 A scientific research boat uses pulses of ultrasound to detect fish underneath it.

A pulse of ultrasound is bounced from a group of fish 6000 m directly below the boat.

The next pulse is sent out 2.0 s after the echo from the previous pulse has been received.

The speed of ultrasound in water is 1500 m / s.

What is the interval between the two pulses?

A 4.0 s **B** 6.0 s **C** 8.0 s **D** 10 s

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 27

27	D	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 11 of 16

28 Which row correctly describes ultrasound?

	type of wave	frequency
A	longitudinal	less than 20 kHz
B	longitudinal	greater than 20 kHz
C	transverse	less than 20 kHz
D	transverse	greater than 20 kHz

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 3 of 3 · row for Question 28

28	B	**1**

QUESTION · 5054_s21_qp_11.pdf · source page 10 of 16

25 The sound from a ship is reflected by a cliff. An echo is heard by a sailor on the ship 4.0 s after the sound is made. The speed of sound in air is 320 m / s.

How far from the cliff is the ship?

A 160 m **B** 640 m **C** 1280 m **D** 2560 m

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 25

25	B	**1**

4 **(a)** Fig. 4.1 shows a wave on a rope and Fig. 4.2 shows a wave on a spring. Both waves are moving in the direction shown by the arrows.

Fig. 4.1

Fig. 4.2

(i) State the name of each of the two types of wave shown.

Fig. 4.1 .....

Fig. 4.2 .....
[1]

(ii) On Fig. 4.1, draw a wave which has a smaller frequency than the wave shown. [1]

(iii) Describe the movement of point X on the spring in Fig. 4.2 as the wave passes.

.....

..... [1]

(b) Table 4.1 shows some sentences that a student writes about waves.

His teacher places a ring around each mistake.

Write a suitable correction for each mistake in Table 4.1. One has been done for you.

Table 4.1

sentence	correction
Sound travels at 3.0×10^8 m/s in air.	330
Sound with a higher pitch has a larger amplitude.	
X-rays are used for pre-natal scanning.	
In the electromagnetic spectrum, microwaves have the highest frequency.	

[3]

[Total: 6]

Question	Answer	Marks
4(a)(i)	Fig.4.1 transverse **and** Fig.4.2 longitudinal	**B1**
4(a)(ii)	wave drawn with larger wavelength	**B1**
4(a)(iii)	backwards and forwards, left and right, to and fro, vibration parallel to arrow / direction of wave (movement)	**B1**
4(b)	frequency	**B1**
	ultrasound	**B1**
	gamma	**B1**

QUESTION · 5054_s21_qp_12.pdf · source page 10 of 16

28 The sound from a ship is reflected by a cliff. An echo is heard by a sailor on the ship 4.0 s after the sound is made. The speed of sound in air is 320 m / s.

How far from the cliff is the ship?

A 160 m **B** 640 m **C** 1280 m **D** 2560 m

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 2 of 3 · row for Question 28

28	B	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 11 of 16

29 Students are asked for the uses of ultrasound.

Three suggested uses for ultrasound are:

P cleaning jewellery

Q tanning on a sunbed

R obtaining an image of an unborn baby.

Which suggestions are correct?

A P and Q only **B** P and R only **C** Q and R only **D** P, Q and R

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 3 of 3 · row for Question 29

29	B	**1**

4 **(a)** In a demonstration, a teacher uses a loud ticking clock, two hollow tubes A and B, a barrier and a smooth surface. The clock is used as a source of sound.

Fig. 4.1 shows tube A and the clock, both fixed in position on the left of the barrier.

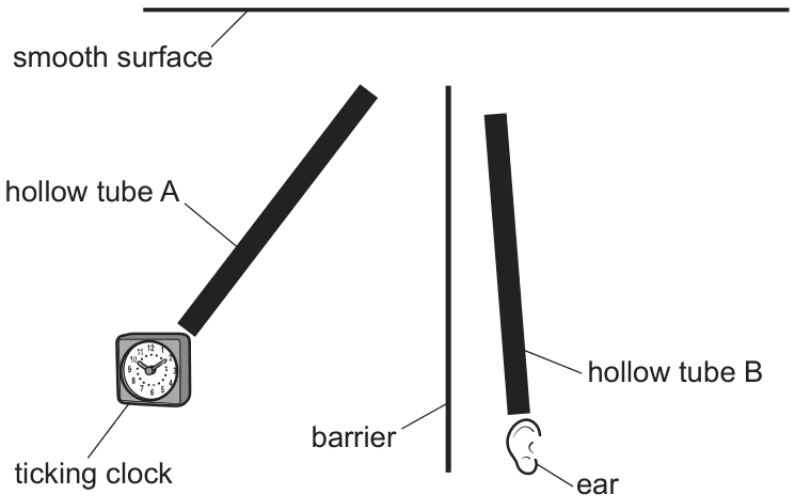


Fig. 4.1

Sound from the ticking clock passes along tube A and is incident on the smooth surface.

A student listens to the sound passing along tube B, which is on the right of the barrier.

(i) On Fig. 4.1, draw tube B in the position where the sound heard by the student is loudest. [1]

(ii) Explain your answer to **(i)**.

.....

.....

..... [2]

(b) The speed of sound in air is 330 m / s.

State a typical value for the speed of sound in a solid.

..... [1]

[Total: 4]

Question 5 begins over the page

Question	Answer	Marks
4(a)(i)	tube in correct position and direction	**B1**
4(a)(ii)	reflection (of sound waves) mentioned	**B1**
	angle of incidence = angle of reflection **or** angle between A and surface = angle between B and surface	**B1**
4(b)	between 1000 m / s and 10 000 m / s	**B1**

QUESTION · 5054_w21_qp_11.pdf · source page 11 of 20

25 A student stands a distance x in front of a large wall.

He claps his hands at a regular rate so that each clap coincides with the echo from the previous clap.

In a time t, he claps his hands N times.

Which expression is used to calculate the speed of sound in air?

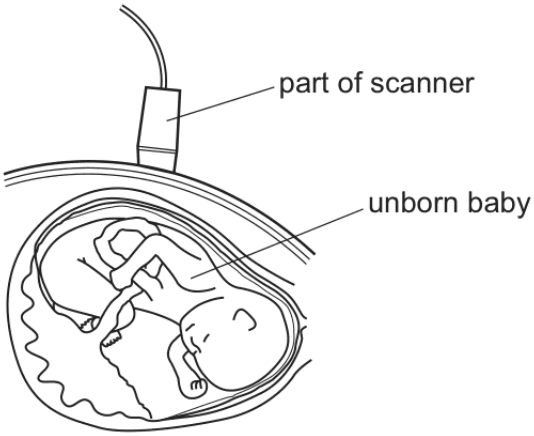
A $\frac{x}{Nt}$ **B** $\frac{2x}{Nt}$ **C** $\frac{Nx}{t}$ **D** $\frac{2Nx}{t}$

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 25

25	D	**1**

QUESTION · 5054_w21_qp_11.pdf · source page 12 of 20

26 An ultrasound scanner produces an image of an unborn baby.



What does the scanner use to form an image of the baby?

A ultrasound absorbed by the baby

B ultrasound emitted by the baby

C ultrasound reflected by the baby

D ultrasound refracted by the baby

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 26

26	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 A loudspeaker is made from a coil of wire fixed to a cardboard tube. The tube is attached to a cardboard cone.

Fig. 8.1 shows part of the arrangement of the loudspeaker.

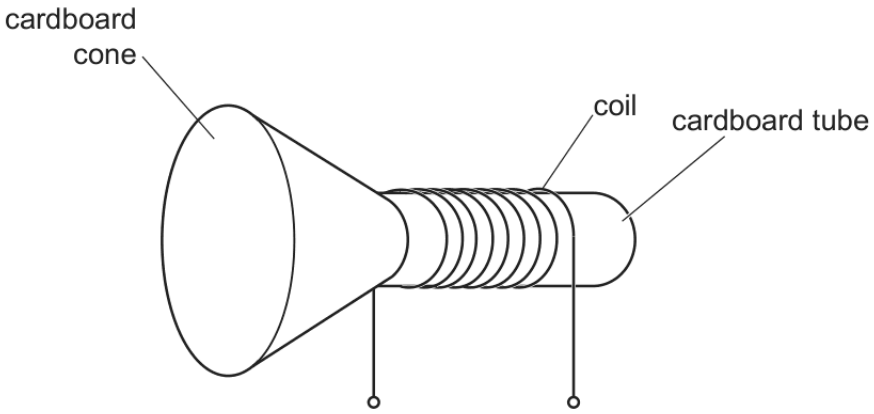


Fig. 8.1

When there is a current in the coil, the coil experiences a force.

(a) State what else is needed in a loudspeaker to make a current-carrying wire experience a force.

..... [1]

(b) A student connects the coil to the output of an alternating current (a.c.) generator. Fig. 8.2 shows how the electromotive force (e.m.f.) produced by the generator varies with time.

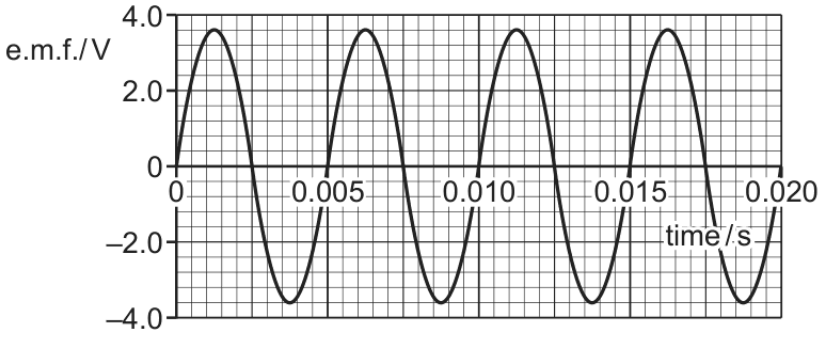


Fig. 8.2

The coil, tube and cone vibrate backwards and forwards.

(i) Explain why the e.m.f. shown in Fig. 8.2 makes the coil vibrate.

.....

.....

..... [2]

(ii) The vibrating cone produces sound in the surrounding air.

Explain, in terms of molecules, how the cone produces a sound wave that travels through the air.

.....

.....

.....

.....

..... [3]

(iii) Sound is a longitudinal wave.

Describe the difference between a longitudinal wave and a transverse wave.

.....

.....

.....

..... [2]

(iv) Using Fig. 8.2, determine the number of times that the cone reverses its direction of motion in 1.0 s.

number = ... [2]

(v) The speed of sound in air is 340 m / s.

Calculate the wavelength of the sound.

wavelength = ... [2]

(c) The student adjusts the generator so that the maximum voltage of the output is now 3.0 V. Everything else stays the same as the output shown on the graph in Fig. 8.2.

(i) Explain any effect on the loudness of the sound.

.....

..... [2]

(ii) Explain any effect on the pitch of the sound.

.....

..... [1]

[Total: 15]

Question 9 starts over the page

MARK SCHEME · 5054_w21_ms_21.pdf · source page 8 of 11 · Question 8

Question	Answer	Marks
8(a)(i)	magnetic field **or** magnet	**B1**
8(b)(i)	the current (in coil continually) reverses (direction) **or** the force (on the coil continually) reverses	**B1**
	the current (in the coil continually) reverses (direction) **and** the force (on the coil continually) reverses	**B1**

MARK SCHEME · 5054_w21_ms_21.pdf · source page 9 of 11 · Question 8

Question	Answer	Marks
8(b)(ii)	cone moves forwards **and** air is compressed / hits (air) molecules	**B1**
	cone moves backwards **and** rarefaction produced / molecules sucked back	**B1**
	(air) molecules vibrate **or** production of compressions and rarefactions continues	**B1**
8(b)(iii)	vibration direction is parallel to energy travel / propagation direction	**B1**
	in transverse wave, vibration direction is perpendicular to propagation / energy travel direction	**B1**
8(b)(iv)	(time period =) 0.0050 (s) **or** 1 / 0.0050 **or** 200 (times per second)	**C1**
	400 (times per second)	**A1**
8(b)(v)	(λ =) c / v **or** 340 / 200	**C1**
	1.7 m	**A1**
8(c)(i)	decreases / less loud / quieter	**B1**
	amplitude of sound / vibration decreases **or** (maximum) force on coil decreases	**B1**
8(c)(ii)	(pitch) stays the same / unchanged **and** no change to frequency / time period	**B1**

QUESTION · 5054_w21_qp_12.pdf · source page 11 of 16

28 Which of the following is most closely associated with echoes?

A focussing

B reflection

C refraction

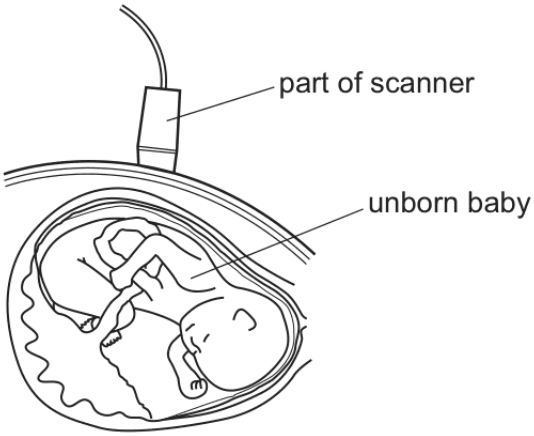
D total internal reflection

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 2 of 3 · row for Question 28

28	B	**1**

QUESTION · 5054_w21_qp_12.pdf · source page 12 of 16

29 An ultrasound scanner produces an image of an unborn baby.



What does the scanner use to form an image of the baby?

A ultrasound absorbed by the baby

B ultrasound emitted by the baby

C ultrasound reflected by the baby

D ultrasound refracted by the baby

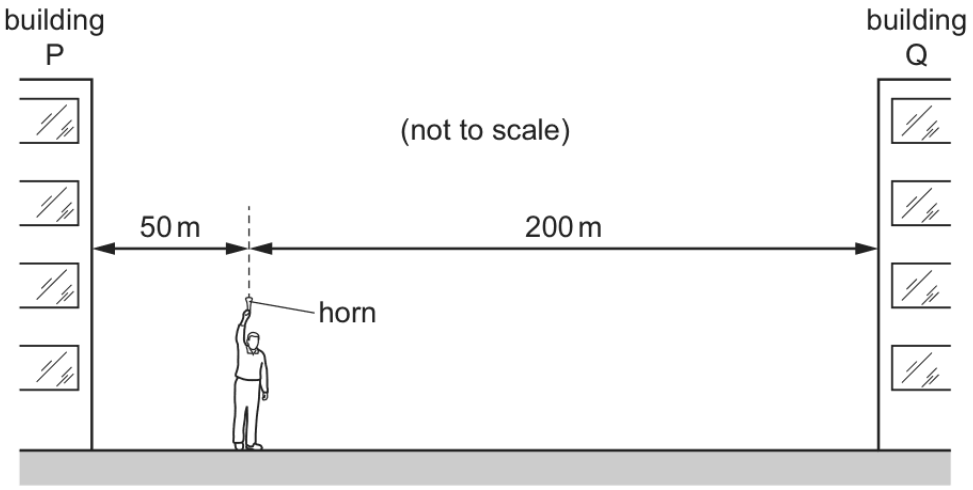
MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 3 of 3 · row for Question 29

29	C	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 10 of 16

24 A man stands between two tall buildings, P and Q.

He is 50 m from P and 200 m from Q.



He sounds a horn.

He hears the first echo from building P and 1.0 s later he hears the first echo from building Q.

What is the speed of sound calculated using this information?

A 250 m / s **B** 300 m / s **C** 330 m / s **D** 500 m / s

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 24

24	B	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 10 of 16

25 Sound travels at different speeds in each of the three states of matter, solid, liquid and gas.

Which row shows where sound travels fastest and where sound travels slowest?

	state where sound travels fastest	state where sound travels slowest
A	gas	liquid
B	gas	solid
C	liquid	gas
D	solid	gas

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 25

25	D	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 10 of 16

26 What is one of the uses of ultrasound?

A cleaning jewellery

B fluorescent tubes

C optical fibres

D sunbeds

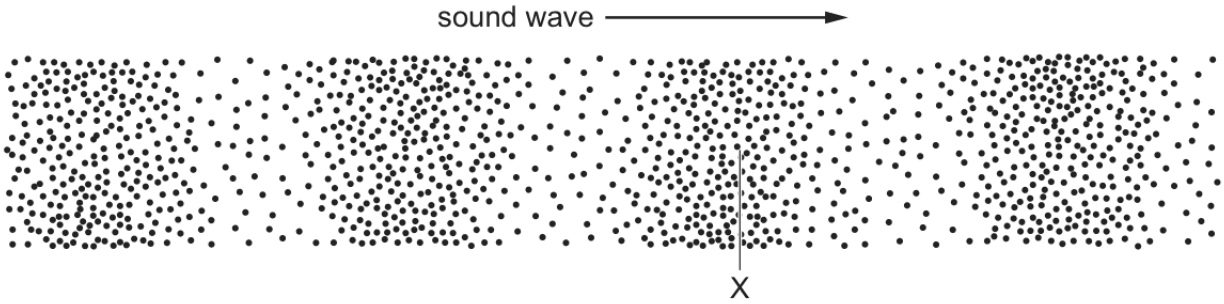
MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 26

26	A	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 13 of 20

27 A sound wave passes through a substance.

The diagram shows the position of molecules of the substance at one point in time.



Which type of wave is a sound wave and what is at X?

	type of wave	at X
A	longitudinal	compression
B	longitudinal	rarefaction
C	transverse	compression
D	transverse	rarefaction

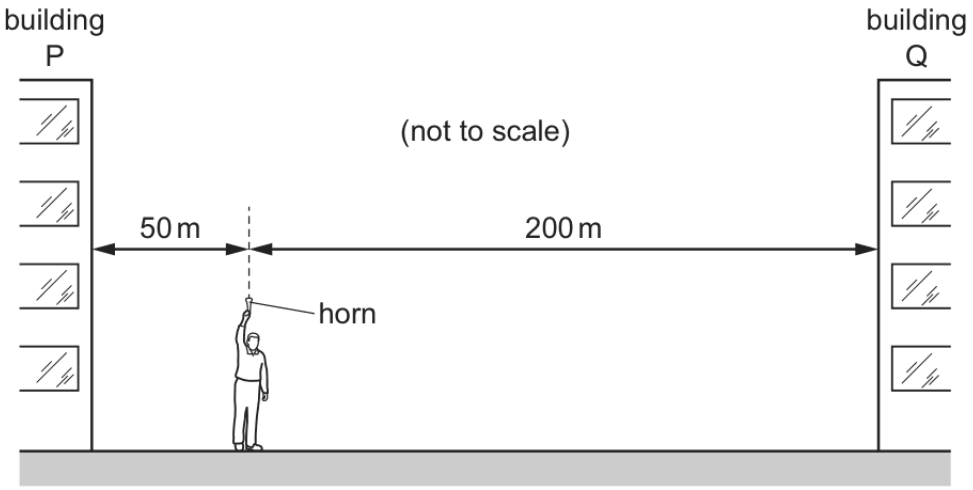
MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 27

27	A	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 13 of 20

28 A man stands between two tall buildings, P and Q.

He is 50 m from P and 200 m from Q.



He sounds a horn.

He hears the first echo from building P and 1.0 s later he hears the first echo from building Q.

What is the speed of sound calculated using this information?

A 250 m / s **B** 300 m / s **C** 330 m / s **D** 500 m / s

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 2 of 3 · row for Question 28

28	B	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 14 of 20

29 What is one of the uses of ultrasound?

A cleaning jewellery

B fluorescent tubes

C optical fibres

D sunbeds

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 3 of 3 · row for Question 29

29	A	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 10 of 16

25 An oscilloscope is used to demonstrate a sound.

Which feature of the oscilloscope trace is affected by the quality (timbre) of the sound?

A amplitude

B frequency

C shape

D wavelength

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 25

25	C	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 11 of 16

27 Which two frequencies are both outside the range of audible frequencies for a healthy human ear?

A 2.0 Hz and 500 Hz

B 2.0 Hz and 25 kHz

C 25 Hz and 25 kHz

D 500 Hz and 25 kHz

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 27

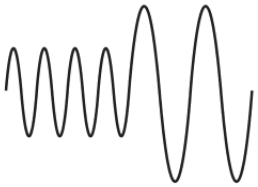
27	B	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 12 of 16

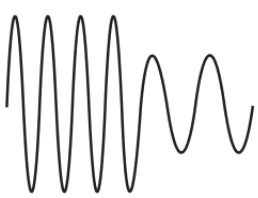
28 Sound waves are displayed as a trace on the screen of an oscilloscope.

Which trace shows a sound that becomes quieter with a higher pitch?

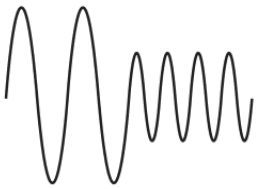
A



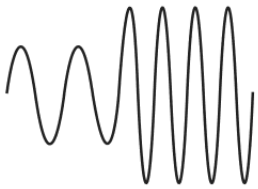
B



C



D



MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 2 of 3 · row for Question 28

28	C	**1**

4 A sound wave, travelling in air or water, contains compressions and rarefactions.

(a) Describe an experiment to show that a medium is required to transmit sound waves. You may draw a labelled diagram if you wish.

.....

.....

.....

.....

..... [3]

(b) When a sound wave passes through air or water, the molecules of the air or water move.

Describe the motion of the molecules due to the sound wave.

.....

.....

..... [2]

(c) Describe what is meant by ‘a rarefaction’.

.....

..... [1]

(d) Fig. 4.1 shows a dolphin.

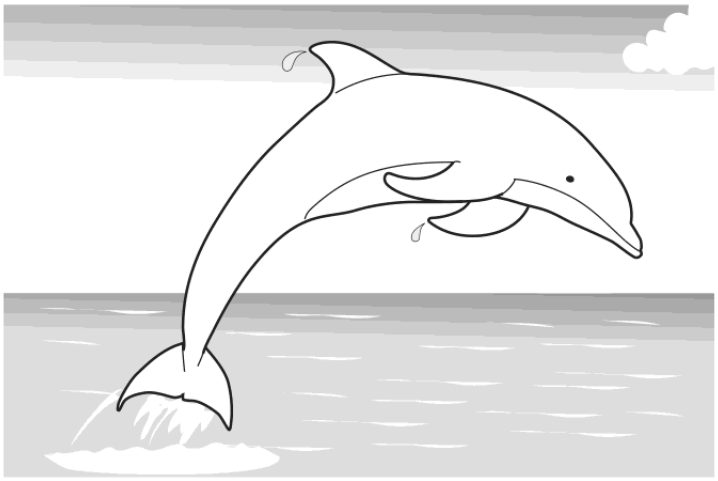


Fig. 4.1

The range of audible sound frequencies for the dolphin is 75 Hz to 140 000 Hz.

In water, the speed of sound is 1500 m / s.

Calculate the smallest wavelength of sound in water that the dolphin can hear.

wavelength = ... [2]

[Total: 8]

MARK SCHEME · 5054_w22_ms_22.pdf · source page 6 of 11 · Question 4

Question	Answer	Marks
4(a)	equipment e.g. (evacuable) container **and** (suspended electric) bell / source of sound **and** (air) pump	**B1**
	action e.g. ringing / sound (produced) / sound heard **and** evacuate air / vacuum (produced)	**B1**
	observation e.g. transmission / hearing of sound ceases / no sound	**B1**

MARK SCHEME · 5054_w22_ms_22.pdf · source page 7 of 11 · Question 4

Question	Answer	Marks
4(b)	(they) vibrate	**M1**
	(vibration) parallel to / in the direction of propagation / of energy travel	**A1**
4(c)	(point where) molecules are further apart (than average) **or** (point where) pressure / density is lower (than average)	**B1**
4(d)	(λ =) v / f **or** 1500 / 140 000	**C1**
	1.1×10^{-2} m **or** 1.1 cm **or** 11 mm	**A1**

QUESTION · 5054_s23_qp_11.pdf · source page 11 of 20

24 Sound travels at different speeds in different states of matter.

Which list gives the three states of matter arranged in order of the speed of sound from slowest to fastest?

A gas → liquid → solid

B liquid → solid → gas

C solid → gas → liquid

D solid → liquid → gas

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 24

24	A	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 10 of 16

23 Ultrasound is a type of sound wave that is not audible to a healthy human ear.

Which range of frequencies contains **only** ultrasound?

A 30 Hz–3.0 kHz

B 300 Hz–30 kHz

C 3.0 kHz–3.0 MHz

D 30 kHz–30 MHz

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 23

23	D	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 9 of 16

22 Which range of sound frequencies includes only frequencies of sound that can be heard by a healthy human ear?

A 0.5–50 Hz **B** 5–500 Hz **C** 50–5000 Hz **D** 500–50 000 Hz

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 22

22	C	**1**

6 The loudspeaker shown in Fig. 6.1 produces a sound of frequency 800 Hz.

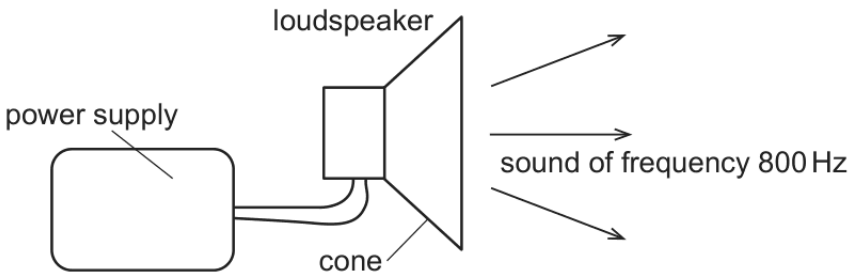


Fig. 6.1

(a) State what is meant by 'frequency'.

.....

..... [2]

(b) Explain whether the sound produced by the speaker is audible to a human with healthy hearing.

.....

..... [1]

(c) Describe the motion of the cone in Fig. 6.1 and explain how this causes the production of sound.

.....

.....

.....

.....

..... [3]

(d) The Sun is a very large and extremely violent ball of gas. Electromagnetic radiation from explosions on the surface of the Sun reaches the Earth.

Explain why sound from explosions on the surface of the Sun does **not** reach the Earth.

.....

.....

..... [2]

[Total: 8]

MARK SCHEME · 5054_w23_ms_22.pdf · source page 8 of 11 · Question 6

Question	Answer	Marks
6(a)	(frequency =) the number of wavelengths	**B1**
	(passing a point) per unit time	**B1**
6(b)	yes **and** within audible range (of human with healthy hearing)	**B1**

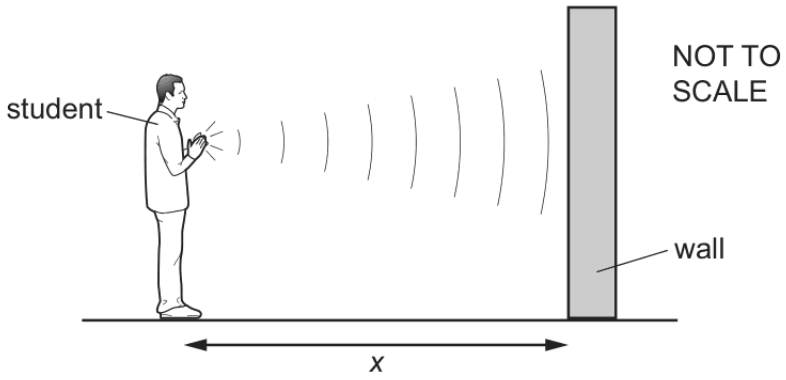
MARK SCHEME · 5054_w23_ms_22.pdf · source page 9 of 11 · Question 6

Question	Answer	Marks
6(c)	it / the loudspeaker / cone / coil is vibrating	**B1**
	when it / cone moves (forwards) it produces a compression / compresses (air) **or** when it moves (backwards) it produces a rarefaction / expands (air)	**B1**
	vibrations / compressions / rarefactions / energy / change in pressure / motion passed on (to neighbouring air)	**B1**
6(d)	space is a vacuum **or** no medium / air in space	**C1**
	sound cannot travel in a vacuum / space / without a medium	**A1**

QUESTION · 5054_s24_qp_11.pdf · source page 11 of 16

28 A student claps his hands in front of a wall and hears the echoes. He claps at the same moment as he hears the echo of the previous clap.

Another student starts a stop-watch on the first clap and stops it on the eleventh clap. The reading on the stop-watch is 9.4 s and the speed of sound is 320 m / s.



What is the distance *x* between the student and the wall?

A 140 m **B** 150 m **C** 270 m **D** 300 m

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 2 of 3 · row for Question 28

28	B	**1**

4 **(a)** Ultrasound of frequency 30 kHz has a wavelength of 0.011 m in air.

(i) State what is meant by 'frequency'.

.....

..... [1]

(ii) Calculate the speed of ultrasound in air.

Show your working.

speed = ... m/s [2]

(iii) State **one** use of ultrasound.

..... [1]

(b) Sound waves are longitudinal waves. Water waves are transverse waves.

(i) Describe, by referring to the movement of the particles in the wave, the difference between a longitudinal wave and a transverse wave.

You may include a labelled diagram to help your description.

.....

.....

.....

.....

..... [3]

(ii) State **one** source of waves that generates both longitudinal and transverse waves.

..... [1]

(c) Fig. 4.1 shows radio waves of long and short wavelength passing over the same hill.

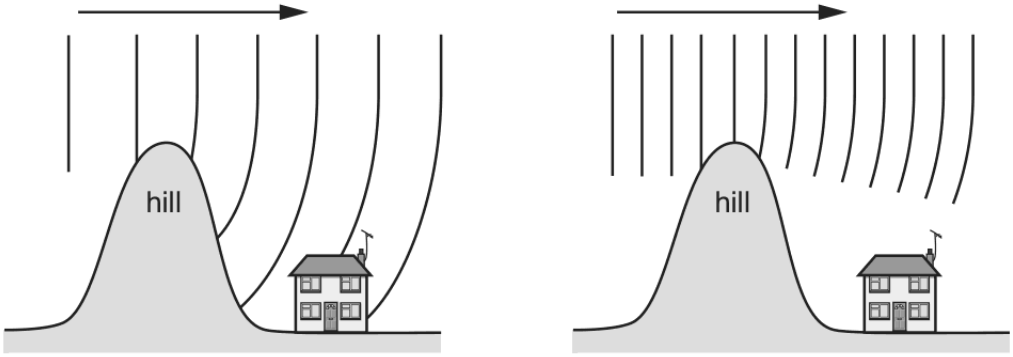


diagram A diagram B

Fig. 4.1 (not to scale)

Explain why the radio waves in diagram A reach the house but the radio waves in diagram B do not reach the house.

.....

.....

.....

..... [2]

[Total: 10]

Question	Answer	Marks
4(a)(i)	number of (complete) waves (passing a point) per second	**B1**
4(a)(ii)	(V =) $f \lambda$ in any form	**C1**
	330 m / s	**A1**
4(a)(iii)	ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance	**B1**
4(b)(i)	longitudinal wave - oscillation backwards and forwards / in direction of travel of wave	**B1**
	transverse wave - oscillation at right angles to (direction of travel of) wave	**B1**
	diagram of each type	**B1**
4(b)(ii)	earthquake (waves) or other correct source which is a single source that generates both types of wave, e.g. lightning	**B1**
4(c)	long wavelengths bend around hill **and** short wavelengths do not bend	**B1**
	(due to) diffraction	**B1**

QUESTION · 5054_s24_qp_12.pdf · source page 9 of 16

23 The speed of sound in air is 330 m / s.

Which sound is classed as ultrasound?

A a sound with a wavelength of 250 cm

B a sound with a wavelength of 25 cm

C a sound with a wavelength of 2.5 cm

D none of the above

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 23

23	D	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 10 of 16

27 Which application is **not** a use for ultrasound?

A cleaning jewellery

B scanning an unborn baby

C sonar

D sterilising water

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 27

27	D	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 12 of 16

29 A person hears an echo after a sound hits a solid cliff face.

What causes the echo?

A absorption

B dispersion

C reflection

D refraction

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 3 of 3 · row for Question 29

29	C	**1**

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

5 Ultrasound is a longitudinal wave which cannot be heard by humans as the frequency of ultrasound is too high.

(a) Ultrasound waves **cannot** travel in a vacuum.

Explain why.

.....

.....

..... [2]

(b) Describe how a longitudinal wave differs from a transverse wave.

.....

.....

.....

..... [2]

(c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m / s.

(i) Calculate the wavelength of this ultrasound wave in soft human tissue.

wavelength = ... m [3]

(ii) The ultrasound wave passes from the soft tissue into bone where the speed of the ultrasound wave is greater than 1500 m / s.

State what happens to the frequency and to the wavelength of the ultrasound wave as it passes into the bone.

frequency .....

wavelength .....
[2]

[Total: 9]

Question	Answer	Marks
5(a)	ultrasound / sound / it is transmitted by vibrating matter / particles	**B1**
	no matter / particles in a vacuum / no medium	**B1**
5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	**B1**
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	**B1**
5(c)(i)	($\lambda =$) v / f **or** $1500 / 8.4 \times 10^{6}$	**C1**
	1.8×10^{N}	**C1**
	1.8×10^{-4} (m)	**A1**
5(c)(ii)	frequency: does not change	**B1**
	wavelength: increases	**B1**

QUESTION · 5054_w24_qp_12.pdf · source page 8 of 16

19 Which row gives the velocity and wavelength of a wave that is an ultrasound wave?

	velocity / m/s	wavelength / cm
A	330	6.6
B	890	89
C	1500	6.0
D	4000	25

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 19

19	C	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 11 of 20

25 A sound wave consists of compressions and rarefactions.

Which type of wave is sound and how does the pressure in a rarefaction compare with the pressure in a compression?

	type of wave	pressure in a rarefaction
A	longitudinal	smaller
B	longitudinal	greater
C	transverse	smaller
D	transverse	greater

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 25

25	A	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 11 of 20

26 What happens to a sound wave when the note heard gets louder?

A Its amplitude increases.

B Its frequency increases.

C Its speed increases.

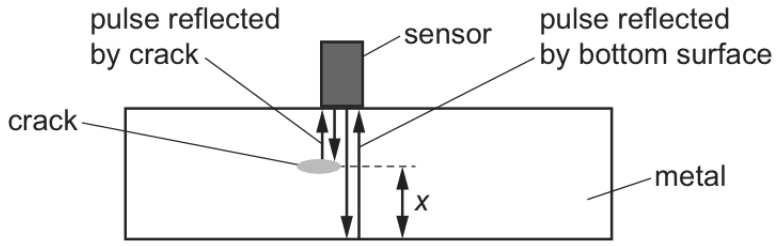
D Its wavelength increases.

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 26

26	A	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 12 of 20

27 A pulse of ultrasound from a sensor is reflected back to the sensor by a crack in a piece of metal and by the bottom surface of the metal.



The reflection from the crack is received back at the sensor 1.0×10^{-5} s after the pulse is sent out.

The reflection from the bottom surface is received back at the sensor 1.5×10^{-5} s after the pulse is sent out.

The speed of ultrasound in the metal is 5200 m / s.

What is the distance *x* between the crack and the bottom surface of the metal?

A 0.013 m **B** 0.026 m **C** 0.052 m **D** 0.078 m

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 27

27	A	**1**

5 **(a)** Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.

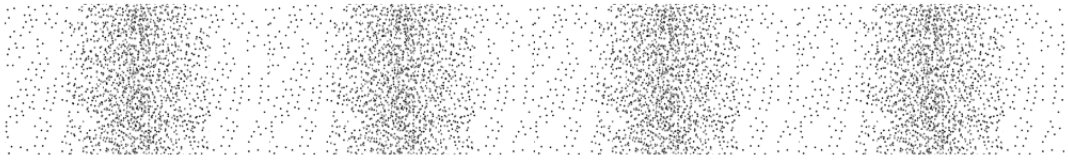


Fig. 5.1

(i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]

(ii) Describe the difference between a compression and a rarefaction.

.....

.....

..... [1]

(b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.

(i) The wooden bar makes 45 complete oscillations in 1.0 minute.

Calculate the frequency of the wave produced.

frequency = ...Hz [1]

(ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed .....

wavelength .....
[2]

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(iii) The crests of the water wave move into the shallow region shown in Fig. 5.2.

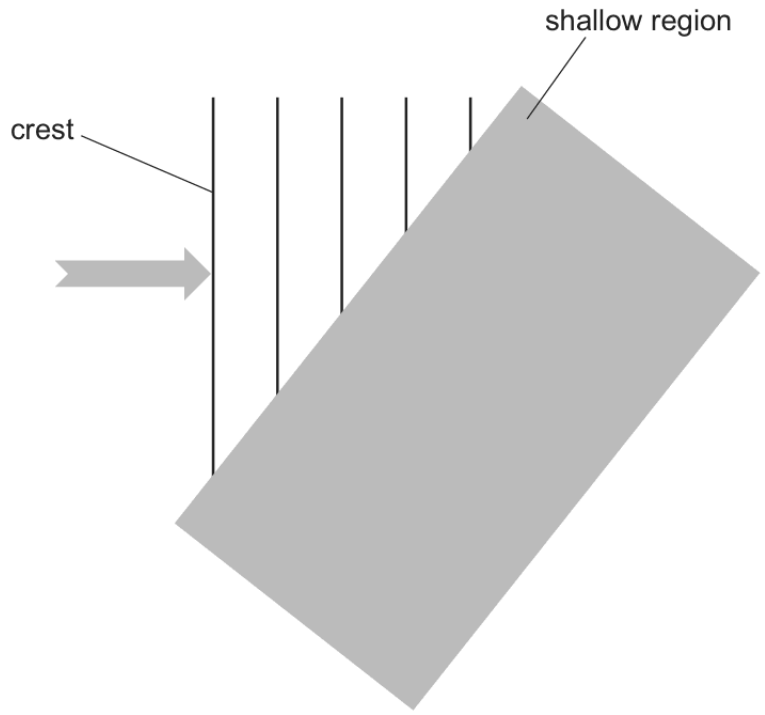


Fig. 5.2

On Fig. 5.2, draw the crests in the shallow region. [2]

(c) Describe what is meant by the diffraction of a water wave.

.....

.....

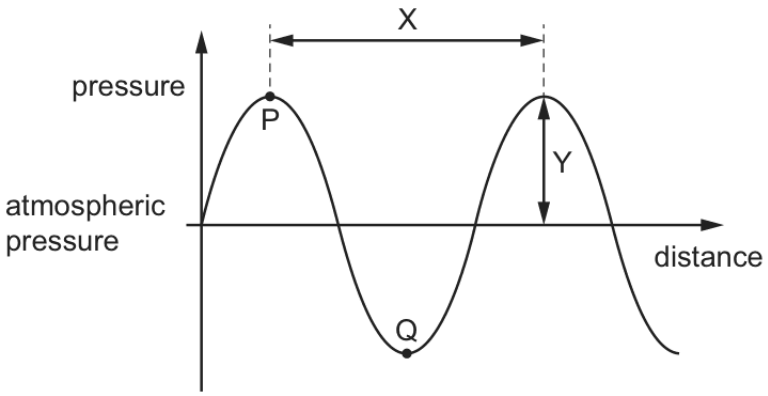
..... [1]

[Total: 8]

Question	Answer	Marks
5(a)(i)	C and R marked correctly	**B1**
5(a)(ii)	compressions are where pressure is high / particles close together / density is high	**B1**
5(b)(i)	0.75 (Hz)	**B1**
5(b)(ii)	*speed:* no change	**B1**
	wavelength: decreases	**B1**
5(b)(iii)	at least 3 wavefronts in the shallow region parallel to each other **and** on correct side of normal	**M1**
	refracted towards the surface on correct side of surface **and** join wavefronts shown	**A1**
5(c)	spreading out / bending at an edge / obstacle **or** spreading out after passing through a (small) gap / hole	**B1**

QUESTION · 5054_s25_qp_12.pdf · source page 8 of 16

18 The graph shows, at one instant, the pressure variation along a sound wave.



Which point on the diagram represents a rarefaction and what is the wavelength of the sound wave?

	rarefaction	wavelength
A	P	X
B	P	Y
C	Q	X
D	Q	Y

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 18

18	C	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 11 of 16

25 Ultrasound is used to clean jewellery in a liquid.

What is another use of ultrasound?

A optical fibre communication

B prenatal scanning

C photography

D telephone communications

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 25

25	B	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 11 of 16

26 A ship that is stationary on the surface of the sea sends pulses of sound vertically downwards towards the sea bed.

Each pulse that reflects from the sea bed is received 1.0 s after it is sent out.

A whale swims under the boat and a pulse is received 0.60 s after it is sent out.

The speed of sound in sea water is 1500 m / s.

What is the distance of the whale above the sea bed?

A 300 m **B** 450 m **C** 600 m **D** 750 m

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 26

26	A	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 11 of 16

27 Which row describes ultrasound waves?

	type of wave	frequency
A	longitudinal	smaller than 20 kHz
B	longitudinal	greater than 20 kHz
C	transverse	smaller than 20 kHz
D	transverse	greater than 20 kHz

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 27

27	B	**1**

4 A sound wave from a vibrating tuning fork is displayed on the oscilloscope shown in Fig. 4.1. The sound wave is detected using a microphone.

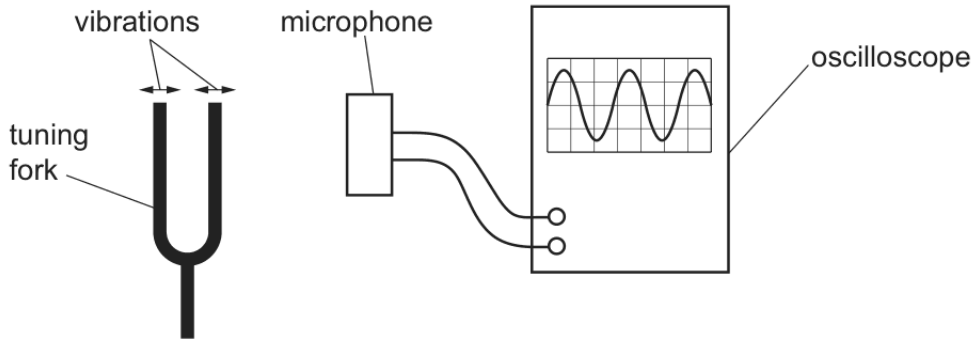


Fig. 4.1

(a) Describe the motion of the air particles as the sound wave passes from the tuning fork to the microphone.

.....

.....

..... [2]

(b) Fig. 4.2 shows the trace seen on the oscilloscope when the tuning fork vibrates.

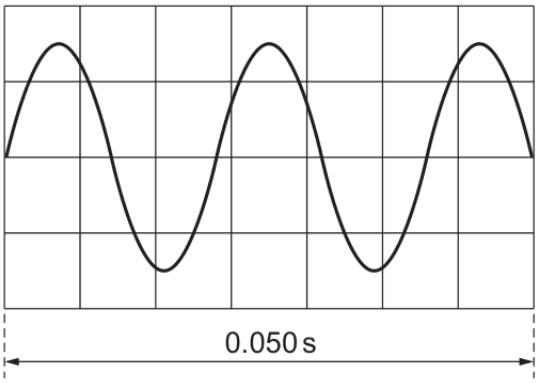


Fig. 4.2

The frequency of the sound is constant.

(i) State what is meant by 'frequency'.

.....

..... [1]

(ii) The trace shown on Fig. 4.2 is produced in 0.050 s.

Calculate the frequency of the sound wave.

frequency = ... Hz [2]

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(iii) A student near the tuning fork observes the trace and listens to the sound. He notices that the amplitude of the trace decreases.

State how the sound heard by the student changes.

..... [1]

(iv) The tuning fork is replaced by a new tuning fork that produces a sound wave of half the frequency. The controls on the oscilloscope are not changed.

On Fig. 4.3, draw the trace obtained with the new tuning fork. The trace obtained with the original tuning fork is shown.

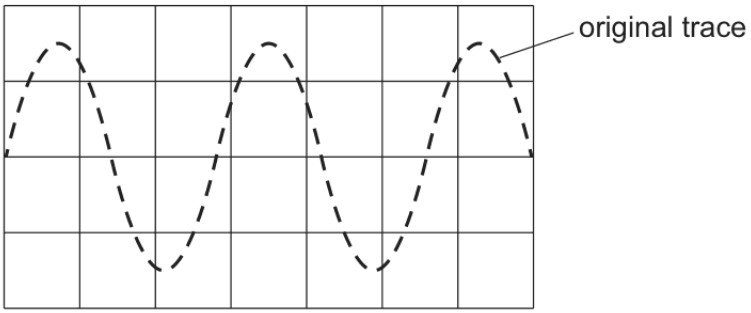

Fig. 4.3

[1]

(c) Fig. 4.4 shows three adjacent wavefronts of the sound wave as the wavefronts move towards a gap in a wall created by an open door.

Another student, standing at P on the other side of the wall, hears the sound.

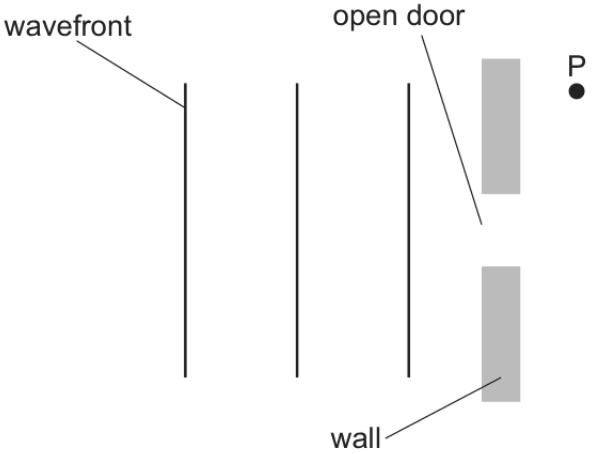

Fig. 4.4

(i) On Fig. 4.4, draw the three wavefronts after they pass through the gap. [2]

(ii) State the name of the process involved as the wavefronts pass through the gap and reach P.

..... [1]

[Total: 10]

Question	Answer	Marks
4(a)	*motion of particles:* backwards and forwards / to and fro / closer and further apart	**B1**
	(oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork **or** mention of compressions and rarefactions	**B1**
4(b)(i)	number of oscillations per second **or** number of wave(length)s (passing a point) per second	**B1**
4(b)(ii)	2.5 (waves) seen **or** (1 wave takes) 0.02 (s) **or** (f =) 1 / T or (number of) waves / time	**C1**
	50 (Hz)	**A1**
4(b)(iii)	quieter / less volume	**B1**
4(b)(iv)	half frequency shown **and** at least one complete oscillation seen	**B1**
4(c)(i)	(at least) three correctly curved wavefronts centered on gap	**B1**
	same wavelength as on left **and** some curving in correct direction	**B1**
4(c)(ii)	diffraction	**B1**

QUESTION · 5054_w25_qp_11.pdf · source page 9 of 16

18 The sentences describe sound waves. Three words have been omitted.

Sound waves are longitudinal waves that consist of regions of higher pressure called .....1..... and regions of lower pressure called .....2..... .

An echo occurs when a sound wave is .....3..... .

Which words complete gaps 1, 2 and 3?

	1	2	3
A	compressions	rarefactions	reflected
B	compressions	rarefactions	refracted
C	rarefactions	compressions	refracted
D	rarefactions	compressions	reflected

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 18

18	A	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 11 of 16

24 A firework is launched vertically from the ground.

When the firework reaches a height of 360 m, the firework explodes, producing coloured light.

A student on the ground below the firework incorrectly measures the time between seeing the coloured light and hearing the explosion. The student obtains a value of 1.8 s.

What is the speed of sound that the student's value suggests?

A 200 m/s **B** 360 m/s **C** 400 m/s **D** 650 m/s

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 24

24	A	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 12 of 16

25 A loudspeaker produces a sound wave of frequency 50 Hz.

The amplitude of the sound wave is increased.

What is heard?

A a louder sound of a higher pitch

B a louder sound of the same pitch

C a sound of higher pitch but the same loudness

D a sound of the same pitch and same loudness

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 25

25	B	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 3 of 16

5 A ship uses ultrasound to measure the depth of the sea.

There is a time interval of 0.52 s between transmitting and detecting a pulse of ultrasound.

The speed of ultrasound in sea water is 1500 m / s.

What is the depth of the sea?

A 390 m **B** 780 m **C** 1600 m **D** 5800 m

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 5

5	A	**1**

QUESTION · 5054_w25_qp_12.pdf · source page 11 of 16

26 The table shows the speed and the wavelength for different sound waves in four different materials.

wave	speed of sound / m/s	wavelength / m
1	340	0.19
2	1500	0.050
3	4000	0.23
4	6700	0.31

Which waves are ultrasound waves?

A 1 and 3 **B** 1 only **C** 2 and 4 **D** 4 only

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 2 of 3 · row for Question 26

26	C	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 11 of 16

27 The approximate range of audible frequencies for a sheep is 100 Hz to 30 000 Hz.

Which statement is correct?

A A human can hear sounds of higher frequencies than a sheep can hear.

B A sheep can hear sounds of lower frequencies than a human can hear.

C The highest frequency of sound that a human can hear is approximately 10 000 Hz lower than the highest frequency of sound a sheep can hear.

D The lowest frequency of sound that a human can hear is approximately 1900 Hz higher than the lowest frequency of sound a sheep can hear.

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 2 of 3 · row for Question 27

27	C	**1**

4 **(a)** All electromagnetic waves transfer energy.

State **two** other properties that are the same for all electromagnetic waves.

1 .....

.....

2 .....

.....

[2]

(b) Figure 4.1 shows an X-ray image of healthy teeth.

The detector used to create the image is placed inside the mouth, behind the teeth.

A source of X-rays is placed outside the mouth, in front of the teeth.

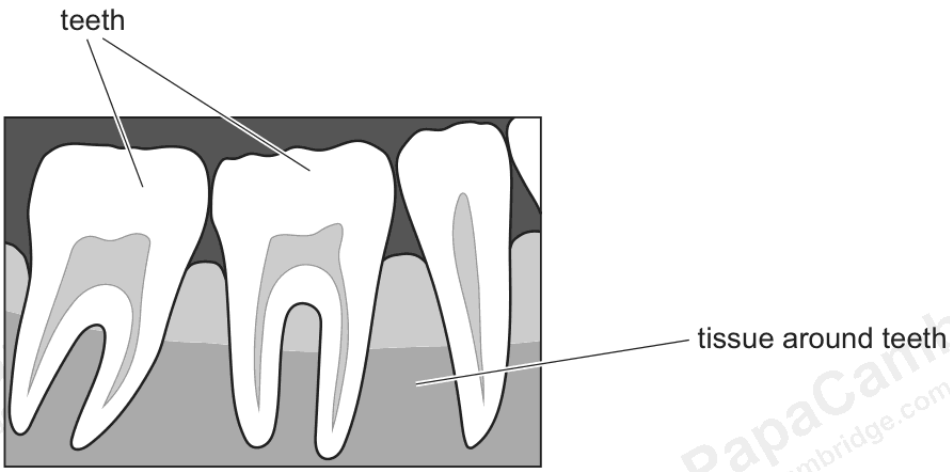

Figure 4.1

Explain why the X-rays produce an image that shows white teeth, grey tissue around the teeth and a black background.

.....

.....

.....

.....

..... [2]

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(c) Ultrasound is another type of wave used in hospitals to produce medical images.

A scanner placed on the skin of a patient emits an ultrasound wave.

(i) The ultrasound wave causes a motion of the particles in the tissue of the patient.

Describe this motion.

.....

.....

..... [2]

(ii) The ultrasound wave is reflected from a bone in the patient and arrives back at the scanner 8.0×10^{-5} s after it was emitted.

The speed of ultrasound in the patient is 1500 m / s.

Calculate the distance between the scanner and the bone.

distance = ... m [2]

(iii) Some of the ultrasound passes through the patient and emerges into the air.

The speed of ultrasound in air is smaller than the speed of ultrasound in the patient.

State and explain the change in wavelength as ultrasound travels from the patient into the air.

change in wavelength .....

explanation .....

.....

.....

.....

[2]

[Total: 10]

Question	Answer	Marks	Guidance
4(a)	Any 2 from: • can reflect / diffract / refract • travel through a vacuum / do not require a medium • transverse • are oscillating electric and magnetic fields • travel at the speed of light / same speed (in vacuum/air)	**2**	**I** no mass, no charge
4(b)	Any two from: • X rays pass through the spaces **or** X-rays make the image darker / go black • (most / all) X-rays absorbed by / cannot penetrate the teeth • X-rays partially absorbed by the tissue / some pass through tissue	**2**	X-rays reflected max 1 mark
4(c)(i)	oscillation / vibration longitudinal **or** backwards and forwards **or** in the direction of the sound / parallel to the direction of travel of the wave **or** formation of compressions and rarefactions	**1**	**A** backwards and forwards or any clear oscillation, eg up and down; left and right; to and fro **A** ultrasound longitudinal **I** left and right, up and down
4(c)(ii)	(d =) vt algebraic or numerical 0.06 (0) m	**1**	**A** ½ vt; ½ × 8.0 × 10^{-5} × 1500; 0.12 m; 12 cm
4(c)(iii)	(wavelength) decreases (in air) $v = f\lambda$ **and** frequency the same **or** each (complete) wave travels less distance (before the next arrives)	**1**	**A** frequency the same **and** $v \propto \lambda$

QUESTION · 5054_s26_qp_12.pdf · source page 11 of 16

28 A fishing boat emits a pulse of ultrasound. The pulse is reflected from some fish underneath the boat.

The reflected pulse reaches the boat 0.32 s after it is emitted.

The speed of ultrasound in water is 1500 m / s.

What is the distance between the boat and the fish?

A 240 m **B** 480 m **C** 4700 m **D** 9400 m

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 2 of 3 · row for Question 28

28	A	**1**

Chapter 4 Electricity and magnetism

4.2 Electrical quantities

4.2.1 Electrical charge

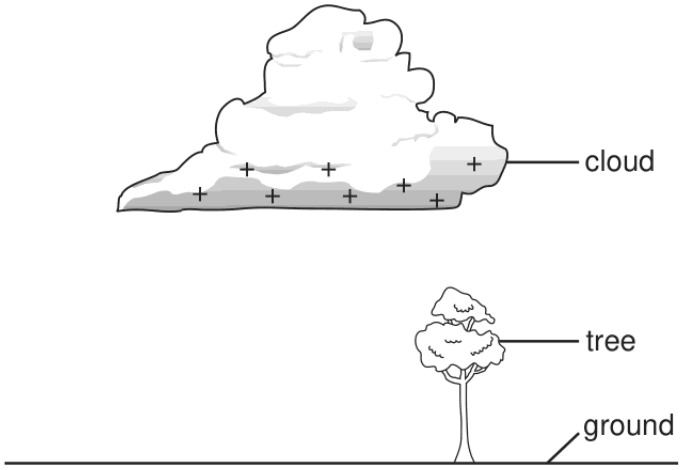
119 entries: 90 Paper 1 questions, 29 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 13 of 20

27 On a stormy day, a large, positively-charged cloud is above a tree.

An electrical charge is induced on the tree as charged particles flow through it.



What is the charge induced on the tree and how do the charged particles move?

	charge on tree	movement of charged particles through tree
A	negative	negatively charged particles move down the tree
B	negative	negatively charged particles move up the tree
C	positive	positively charged particles move down the tree
D	positive	positively charged particles move up the tree

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 27

27 **B**

For Examiner's Use

6 An experiment to show charging by induction uses a metal sphere mounted on an insulated support. The sphere is initially uncharged and is shown in Fig. 6.1.

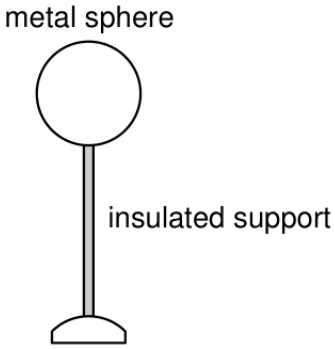


Fig. 6.1

(a) A negatively charged rod is brought near the sphere, as shown in Fig. 6.2.

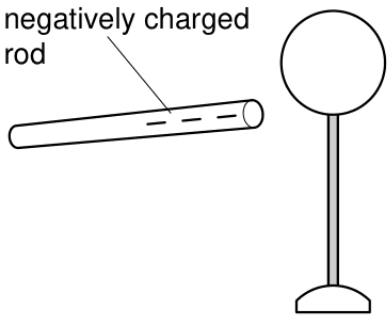


Fig. 6.2

(i) State and explain the movement of electrons in the sphere that occurs as the rod is brought near.

.....

.....

.....

.....

(ii) On Fig. 6.2, draw the charges on the metal sphere.

[3]

For Examiner's Use

(b) The metal sphere is now touched at point A by a wire connected to earth, as shown in Fig. 6.3.

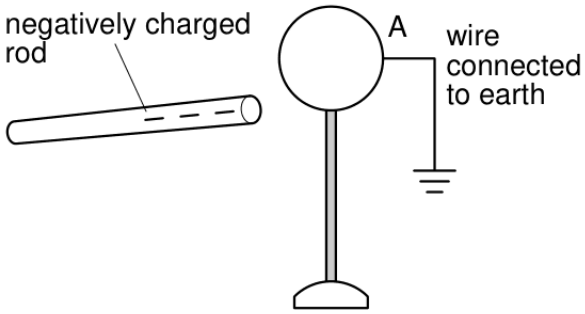


Fig. 6.3

On Fig. 6.3, draw the charges on the metal sphere. [1]

(c) The wire connected to earth is removed. Then the negatively charged rod is also removed, as shown in Fig. 6.4.

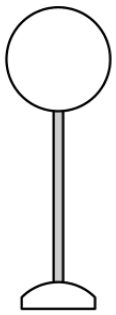


Fig. 6.4

On Fig. 6.4, draw the charges on the metal sphere. [1]

(d) The support is made from an insulator.
State one material that may be used to make the support.

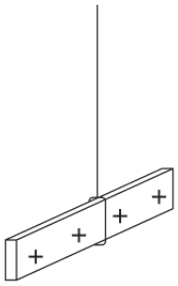
.....[1]

6	**(a)**	**(i)**	**electrons/they** move (on sphere) away from rod/to right	**B1**	
			negative or electrons repelled by (negative on) rod **or** like charges repel	**B1**	
			(actual movement of positive charge max 1 positive electrons max 1)		
		(ii)	positive nearest rod **and** negative on side furthest from rod	**B1**	
			(allow charges just outside sphere no need for same no. of +ve and –ve charges)		
	(b)		only positive on side near rod		
			no e.c.f. **a (ii)**	**B1**	

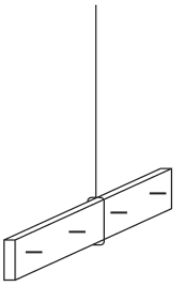
	(c)		>1 positive charge distributed over sphere		
			e.c.f.	**B1**	
	(d)		plastic/perspex/polythene/rubber/ebonite/glass/wood etc.	**B1**	**6**

QUESTION · 5054_s05_qp_1.pdf · source page 11 of 20

26 The diagram shows a positively charged acetate strip and a negatively charged polythene strip that are freely suspended.



acetate strip



polythene strip

Two rods **X** and **Y** are brought up in turn to these two strips. Rod **X** attracts the acetate strip but repels the polythene strip. Rod **Y** does not repel either the acetate strip or the polythene strip.

Which type of charge is on each rod?

	rod **X**	rod **Y**
A	negative	positive
B	negative	uncharged
C	positive	negative
D	positive	uncharged

MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 26

26 **B**

QUESTION · 5054_w05_qp_1.pdf · source page 13 of 20

26 Four processes are used to charge an isolated metal sphere.

P The sphere is earthed by touching it.

Q The earth connection is removed from the sphere.

R A charged rod is brought close to the sphere.

S The charged rod is removed.

In which order should these processes be carried out to charge the sphere?

	first → last			
A	P	Q	R	S
B	P	R	S	Q
C	R	P	Q	S
D	R	S	P	Q

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 26

26 **C**

For Examiner's Use

6 An electrostatic generator is used to produce sparks, as shown in Fig. 6.1.

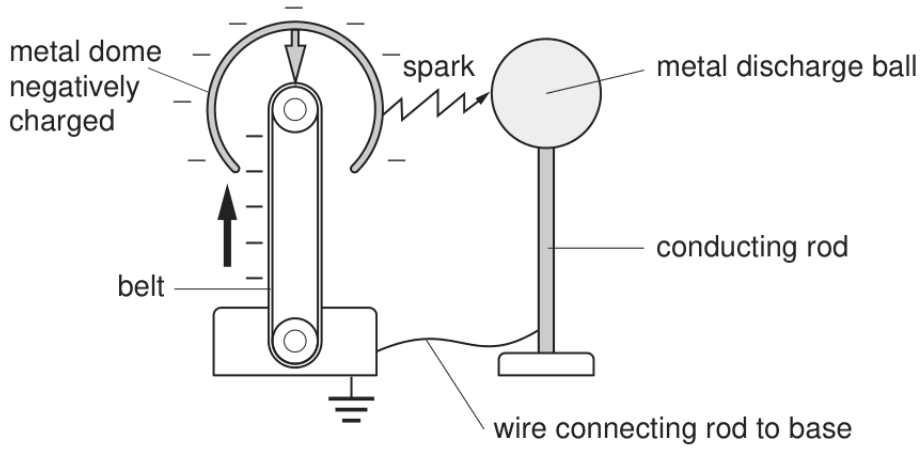


Fig. 6.1

The belt carries negative charge to the dome, making the dome negatively charged.

(a) Before a spark is produced, the discharge ball becomes positively charged.

(i) Describe and explain the movement of electrons in the discharge ball and in the rod as the ball becomes charged.

.....

.....

.....

..... [2]

(ii) On Fig. 6.1, mark with an X where there is the most positive charge on the discharge ball. [1]

(b) A spark jumps between the dome and the discharge ball when there is enough negative charge on the dome.

A charge of 0.00016 C flows in the spark in a time of 0.012 s.

Calculate the average current in the spark. State the equation that you use.

current = ... [3]

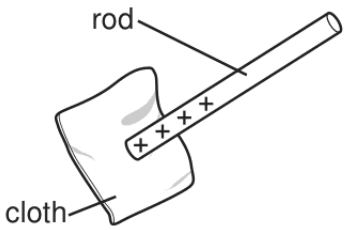
6	**(a)**	**(i)**	electrons move down rod **o**r away from ball	**B1**	
			like charges repel **or** electrons repelled by (charge on) dome	**B1**	
		(ii)	X on left side of ball	**B1**	
	(b)		Q = It formula seen in any algebraic form	**B1**	
			0.00016/0.012	**C1**	
			0.0133 A	**A1**	**6**

QUESTION · 5054_s06_qp_1.pdf · source page 11 of 20

29 In an electrostatics experiment, a plastic rod is rubbed with a cloth. The cloth becomes negatively charged.

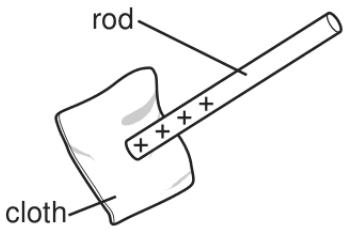
Which diagram shows the charge on the rod, and describes the movement of charge?

A



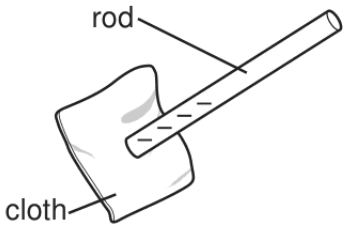
electrons move from the rod onto the cloth

B



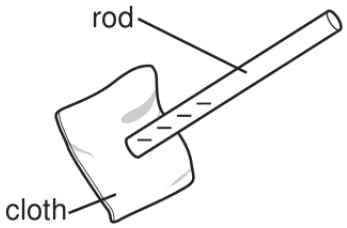
protons move from the cloth onto the rod

C



electrons move from the cloth onto the rod

D



protons move from the rod onto the cloth

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 29

29 **A**

For Examiner's Use

5 Two conducting spheres are rubbed and become charged, as shown in Fig. 5.1.

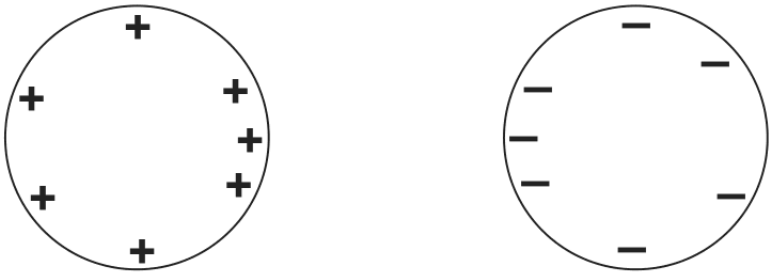


Fig. 5.1

(a) The spheres are uncharged before they are rubbed.

Explain, in terms of the movement of electrons, how they become charged.

.....

.....

.....

..... [1]

(b) On Fig. 5.1, draw the electric field pattern between the spheres. Show the direction of the field lines. [2]

(c) A metal wire is used to connect the two spheres together. In a time of 2.0×10^{-3} s, a charge of 1.8×10^{-13} C passes through the wire.

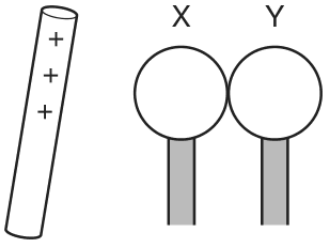
Calculate the current in the wire.

current = ... [2]

5	**(a)**	(electrons) move onto negative/right sphere **and** off positive/left sphere	B1	
	(b)	3 or more (approx. correct none wrong) lines from one sphere to the other	C1	
		arrow on at least 3 lines from + to – sphere	A1	
	(c)	Q = It in any format algebraic or numerical	C1	
		9×10^{-11} A	A1	**[5]**

QUESTION · 5054_w06_qp_1.pdf · source page 11 of 16

26 Two insulated and uncharged metal spheres X and Y are touching. While a positively charged rod is near X, the spheres are moved apart. After this action, X has a negative charge.



What will be the charge on Y?

A negative and smaller than that on X

B negative and the same size as that on X

C positive and smaller than that on X

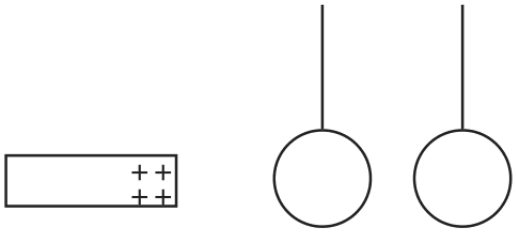
D positive and the same size as that on X

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 26

26	**D**

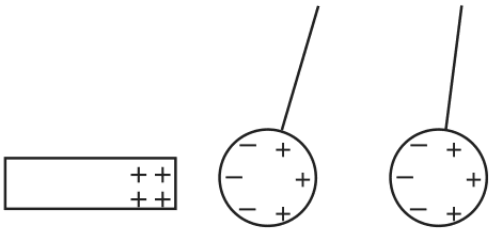
QUESTION · 5054_s07_qp_1.pdf · source page 13 of 20

26 Two uncharged metal spheres, not touching one another, are suspended by means of cotton thread. A positively charged rod is brought near.

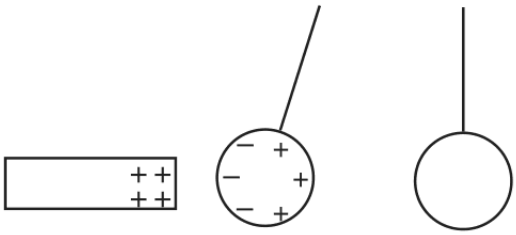


Which diagram shows what happens to the spheres?

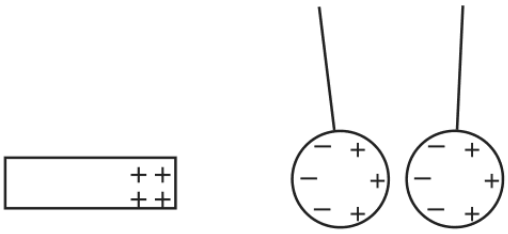
A



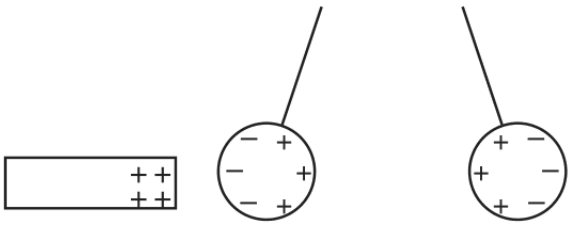
B



C



D



MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 26

26	**A**

11 A plastic rod is rubbed with a cloth and becomes negatively charged.

(a) Explain how the rod becomes negatively charged when rubbed with a cloth. [2]

(b) An uncharged metal-coated sphere hangs from an insulating thread. The sphere is brought near to the rod. The sphere is attracted to the rod, as shown in Fig. 11.1.

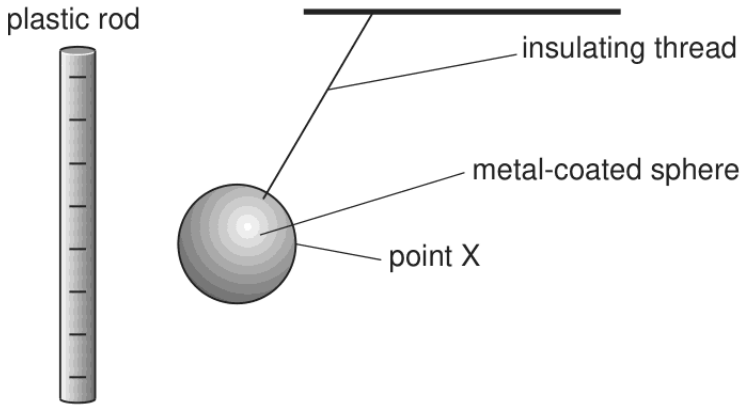


Fig. 11.1

(i) Describe and explain what happens to the free electrons in the metal-coated sphere as it approaches the rod. [2]

(ii) Draw a diagram to show how charge is distributed on the sphere. [1]

(iii) Explain why the uncharged sphere is attracted to the negatively-charged rod. [2]

(c) With the charged rod still close, point X on the metal-coated sphere is earthed.

(i) State what is meant by *earthing* the sphere. [1]

(ii) Describe and explain what happens to the free electrons in the metal-coated sphere as it is earthed. [2]

(iii) Draw a diagram to show how the charge is now distributed on the sphere. [1]

(d) Describe **one** device where electrostatic charging is used. In your answer include a diagram and explain how and why the charge is produced. [4]

11 (a) –ve charge/electrons moves or rod gains electrons C1
–ve charge/electrons move from cloth to rod A1
positive electrons scores 0/2 in **(a)** and **(b)(i)** +ve moves max 1 mark

(b) (i) (electrons) move to right/to X/to opposite side (to rod) / B1
(electrons or –ve) repelled (by rod) or like charges repel B1

(ii) +ve on left and –ve on right, inside or outside sphere B1

(iii) +ve attracted to rod or unlike charges or +ve and –ve attract B1
repulsion of –ve on sphere (by rod) weaker (than attraction) B1 [7]

(c) (i) connection of sphere to earth/ground/0 V B1

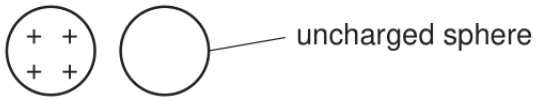
(ii) move down to the ground/earth **or** electrons on right/at X removed B1
repelled (by –ve on rod) **or** move from –ve to 0 potential B1

(iii) only +ve on sphere at left or clearly more positive on left than on right B1 [4]

(d) Sensible example of a **use** of charging,
e.g. precipitator, photocopier, spray painting, gold leaf electroscope, plates in CRO M1
ink jet printer, Van de Graff generator, piezoelectric devices, capacitor, lightning conductor
simple diagram showing effect A1
a correctly charged object clear A1
description of the function that the charge performs A1 [4]

QUESTION · 5054_w07_qp_1.pdf · source page 11 of 20

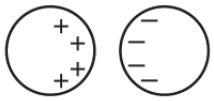
26 A positively charged insulated metal sphere is brought close to, but not touching, a similar uncharged metal sphere.



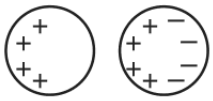
Which diagram shows the charge distribution on the spheres?

A

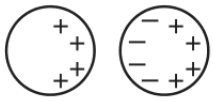
B



C



D



MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 26

26	**D**

10 **(a)** Describe an experiment to show the difference between an electrical insulator and an electrical conductor. Name one example of each. [4]

(b) Fig. 10.1 is a sketch graph of the current in a component P against the potential difference (p.d.) across it.

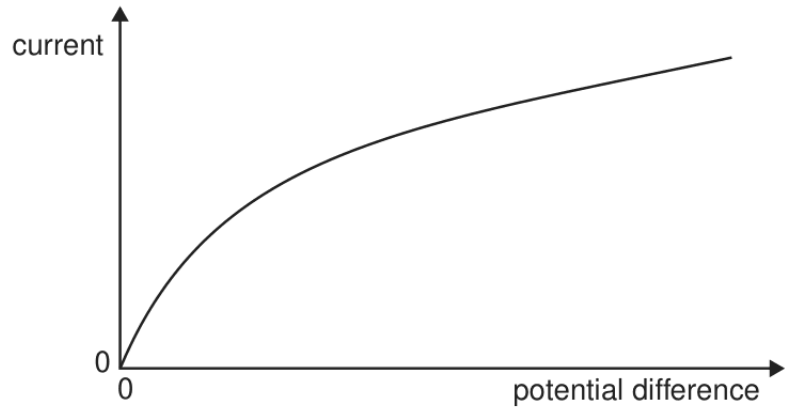


Fig. 10.1

(i) Define *resistance*. [1]

(ii) State how the resistance of P varies with the p.d. across it. [1]

(iii) Suggest what component P is. [1]

(iv) Explain why the resistance of P varies with the p.d. across it. [2]

(c) Component P is used in the electrical circuit shown in Fig. 10.2.

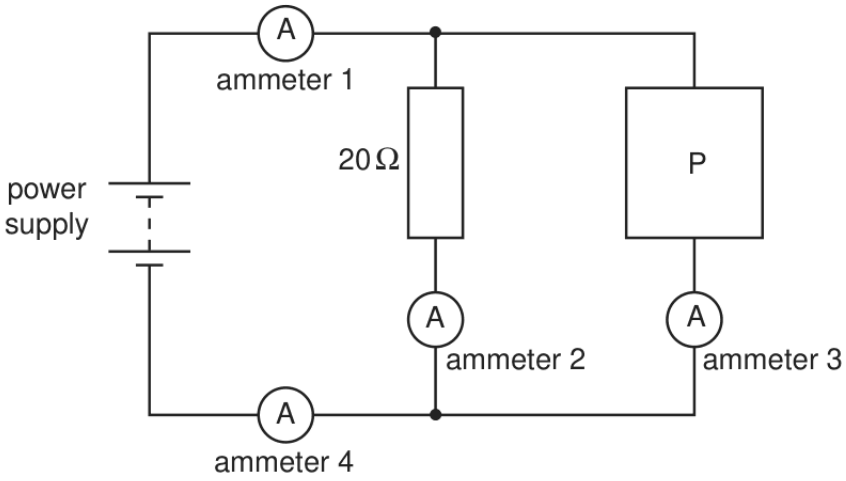


Fig. 10.2

The current in ammeter 2 is 0.40 A and the current in ammeter 3 is 0.60 A.

(i) Determine the readings of ammeters 1 and 4. [1]

(ii) Calculate the p.d. across the 20 Ω resistor. [2]

(iii) State the p.d. across the power supply. [1]

(iv) Calculate the resistance of P in this circuit. [2]

10	**(a)**	electrical circuit containing cell/source, ammeter/lamp/bell and component under test		
		or charged gold-leaf electroscope and component **or** other sensible apparatus	M1	
		one correct observation	A1	
		named conductor (any metal/carbon/graphite **accept** water)	B1	
		named insulator (e.g. plastic **accept** paper **and** wood)	B1	[4]
	(b)	**(i)** voltage/current **or** V/I **not** volts/amps	B1	
		(ii) resistance increases at higher p.d. (**not** resistance increases)	B1	
		(iii) (filament) lamp/bulb or **PTC** thermisitor (**not** metal conductor)	B1	
		(iv) temperature changes	C1	
		higher current/voltage produces higher temperatures	A1	[5]
	(c)	**(i)** 1.0A **both** for A_1 and A_4	B1	
		(ii) (V=) IR in any form or 20 x 0.4	C1	
		8(.0) V	A1	
		(iii) 8 V or same as **(ii)**	B1	
		(iv) **(ii)** / 0.6	C1	
		13 Ω (accept 2 or more sig figs **or** recurring decimal **not** fractions)	A1	[6]

QUESTION · 5054_s08_qp_1.pdf · source page 12 of 20

28 A perspex rod can be charged positively by rubbing it with a woollen cloth.

How does the rod gain its charge?

A The rod gains electrons.

B The rod gains protons.

C The rod loses electrons.

D The rod loses protons.

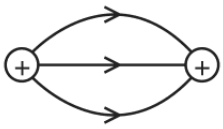
MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 28

28	**C**

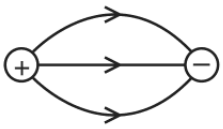
QUESTION · 5054_s08_qp_1.pdf · source page 12 of 20

29 Which diagram correctly shows the electric field lines between two point charges?

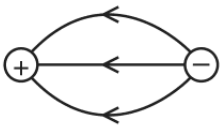
A



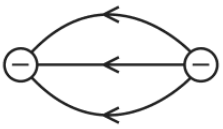
B



C



D



MARK SCHEME (official answer key) · 5054_s08_ms_1.pdf · source page 2 of 2 · row for Question 29

29	**B**

QUESTION · 5054_w08_qp_1.pdf · source page 13 of 20

27 A piece of polythene is rubbed with a cloth duster. The polythene becomes negatively charged and the cloth becomes positively charged.

What happens to the polythene and to the cloth to cause this?

	polythene	cloth
A	gains electrons	gains protons
B	gains electrons	loses electrons
C	loses protons	gains protons
D	loses protons	loses electrons

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 27

27 **B**

QUESTION · 5054_s09_qp_1.pdf · source page 11 of 20

25 A negatively-charged balloon is brought towards a wall.

Which statement explains what happens?

A Negative charges on the balloon attract the negative charges in the wall.

B Negative charges on the balloon have no effect on the charges in the wall.

C Negative charges on the balloon repel the negative charges in the wall.

D Negative charges on the balloon repel the positive charges in the wall.

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 25

25	**C**

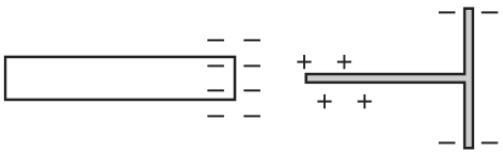
QUESTION · 5054_w09_qp_1.pdf · source page 12 of 20

25 A negatively-charged rod is brought close to an isolated T-shaped piece of metal.

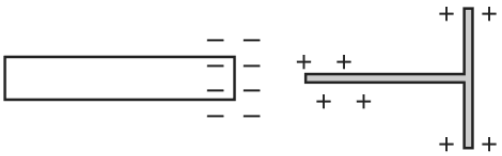
Initially, the metal is uncharged.

Which diagram shows the induced charge on the metal?

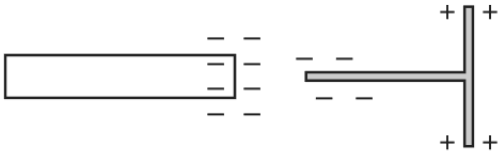
A



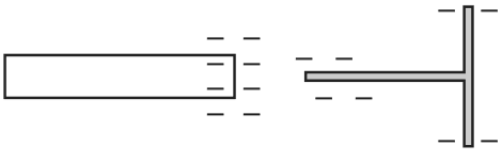
B



C



D



MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 25

25	**A**

For Examiner's Use

3 A hockey player trains on a nylon-fibre surface. As he runs around, his shoes rub against the surface and he becomes positively charged.

(a) (i) Explain, in terms of the particles involved, how he becomes positively charged.

.....

.....

.....[2]

(ii) State what happens to the nylon-fibre surface as he becomes positively charged.

.....

.....[1]

(b) At the end of the training session, the hockey player touches a metal gate and feels an electric shock.

(i) State how this shock is produced.

.....

.....[1]

(ii) The shock lasts for 0.15 ms. During this time, the current has an average value of 1.6 mA. Calculate the size of the charge on the hockey player just before he touches the gate.

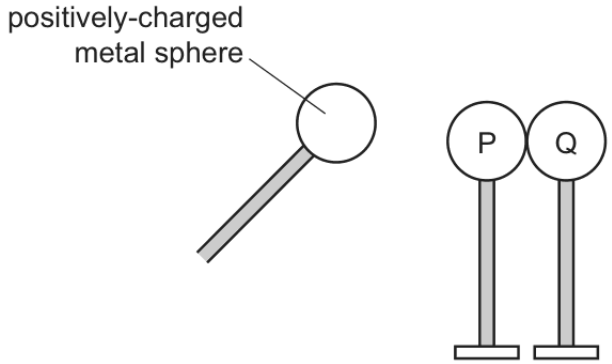
charge = ...[3]

3	**(a)**	**(i)**	(he) loses –ve charge		C1	
			electrons lost (to surface)	(positive electrons 0/2)	A1	
		(ii)	(becomes) negative/gains electrons		B1	
	(b)	**(i)**	(he) discharges/(re)gains electrons/–ve charge (**not** current flow)		B1	
		(ii)	(Q =)It **or** 1.6 × 0.15 **or** 0.0016 × 0.00015		C1	
			2.4×10^{n}		C1	
			2.4×10^{-7} C		A1	[7]

QUESTION · 5054_s10_qp_11.pdf · source page 11 of 20

25 Two metal spheres P and Q are mounted on insulating stands and are touching each other. They are uncharged.

A positively-charged metal sphere on an insulating handle is brought close to P but does not touch it. This induces charges on P and Q.



The positively-charged metal sphere is held in this position and sphere Q is moved to the right, away from sphere P.

What are the signs of the induced charges on P and Q and how do the sizes of these charges compare?

	charge on P	charge on Q	sizes of the charges
A	negative	positive	equal
B	negative	positive	unequal
C	positive	negative	equal
D	positive	negative	unequal

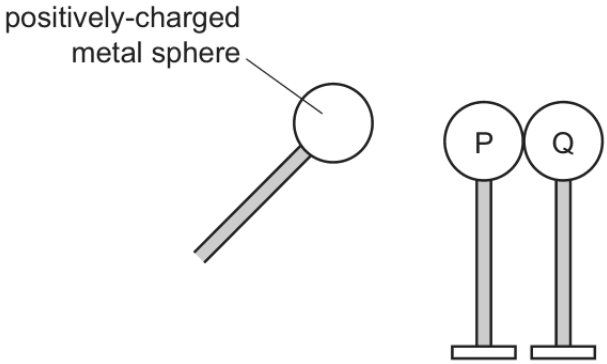
MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 25

25 **A**

QUESTION · 5054_s10_qp_12.pdf · source page 11 of 20

24 Two metal spheres P and Q are mounted on insulating stands and are touching each other. They are uncharged.

A positively-charged metal sphere on an insulating handle is brought close to P but does not touch it. This induces charges on P and Q.



The positively-charged metal sphere is held in this position and sphere Q is moved to the right, away from sphere P.

What are the signs of the induced charges on P and Q and how do the sizes of these charges compare?

	charge on P	charge on Q	sizes of the charges
A	negative	positive	equal
B	negative	positive	unequal
C	positive	negative	equal
D	positive	negative	unequal

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 24

24 **A**

For Examiner's Use

6 **(a)** An aeroplane becomes positively-charged as it flies through the air, because it loses particles from its metal surface.

(i) State the name of the particles lost from the metal surface.

.....[1]

(ii) The tyres of the aeroplane are made from an electrical conductor.

Explain what happens to the charge on the aeroplane when it lands.

.....

.....[2]

(b) Suggest why it is necessary to keep an aeroplane connected electrically to earth during refuelling.

.....

.....[2]

6 **(a)** **(i)** electrons B1 [1]

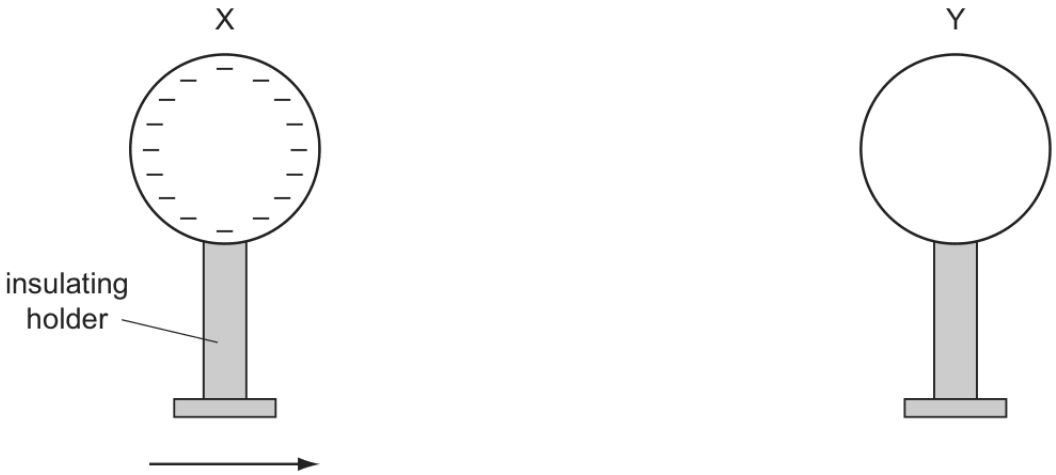
(ii) neutralised/charge becomes zero/loses all charge/charge goes to earth B1
electrons move to plane/tyres from ground/earth/zero potential/surface/land B1 [2]

(b) (otherwise) plane/tank/fuel becomes or is charged or charge builds up in some way
or stays neutral/uncharged or (earthing) conducts charge away (to ground) B1
avoids sparks or prevents explosion/fire/fuel igniting/blast
or sparks/fires, etc. may be produced B1 [2]

[Total: 5]

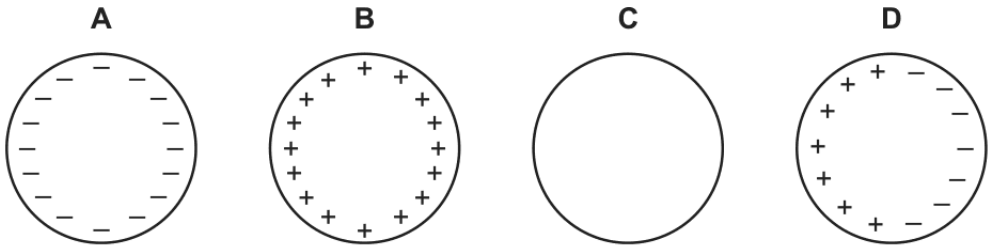
QUESTION · 5054_w10_qp_11.pdf · source page 12 of 20

27 A negatively-charged sphere X is brought up to an identical uncharged sphere Y. The spheres do not touch.



Sphere Y is 'earthed' by touching it with a finger, which is then removed. Sphere X is then moved away from sphere Y.

What is the final charge, if any, on sphere Y?

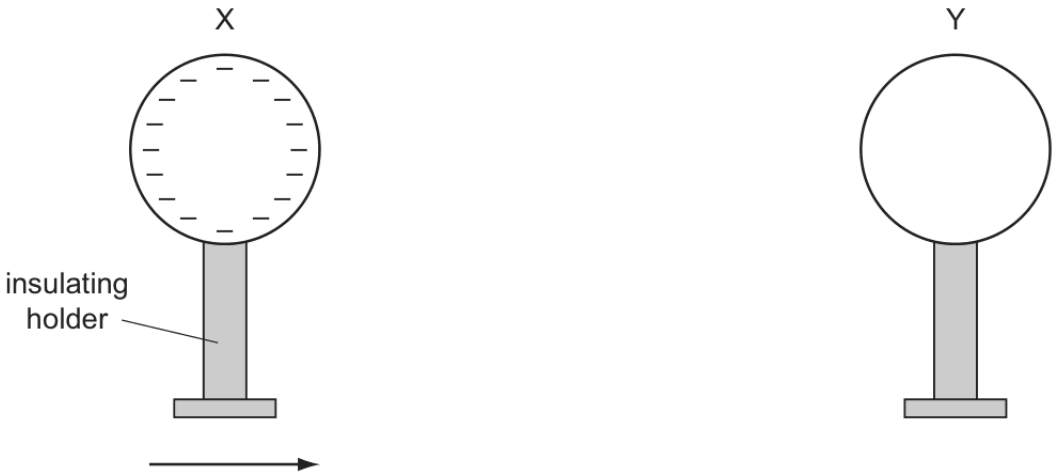


MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 27

27	**B**

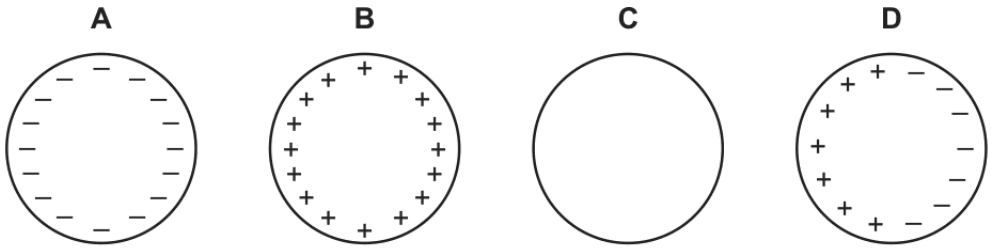
QUESTION · 5054_w10_qp_12.pdf · source page 11 of 16

29 A negatively-charged sphere X is brought up to an identical uncharged sphere Y. The spheres do not touch.



Sphere Y is 'earthed' by touching it with a finger, which is then removed. Sphere X is then moved away from sphere Y.

What is the final charge, if any, on sphere Y?



MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 29

29	**B**

QUESTION · 5054_s11_qp_11.pdf · source page 11 of 16

27 To charge an isolated metal sphere by induction, the following four processes are required.

P The sphere is earthed by touching it.

Q The earth connection is removed from the sphere.

R A charged rod is brought close to the sphere.

S The charged rod is removed.

In which order can these stages be carried out to charge the isolated metal sphere?

A P → Q → R → S

B P → R → S → Q

C R → P → Q → S

D R → P → S → Q

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 27

27	**C**

QUESTION · 5054_s11_qp_12.pdf · source page 11 of 16

29 To charge an isolated metal sphere by induction, the following four processes are required.

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MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 29

29	**C**

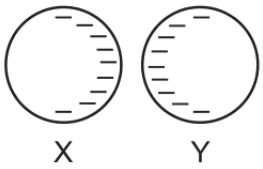
QUESTION · 5054_w11_qp_11.pdf · source page 12 of 20

27 Two metal spheres X and Y are on insulating bases. Both spheres are negatively charged.

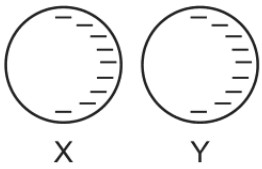
Sphere X is moved towards sphere Y until they almost touch.

Which diagram shows the final pattern of charges?

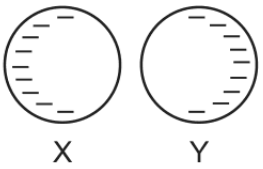
A



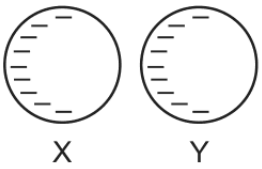
B



C



D



MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 27

27	**C**

5 Fig. 5.1 shows a nylon rod being rubbed with a woollen cloth.

Fig. 5.1

The rod becomes negatively charged.

(a) (i) Explain how rubbing the nylon rod makes it negatively charged.

.....

.....[1]

(ii) State what happens to the cloth as the rod becomes negatively charged.

.....

.....[1]

(b) Fig. 5.2 shows the negatively-charged rod placed near to a conducting sphere. The sphere is suspended by an insulating thread.

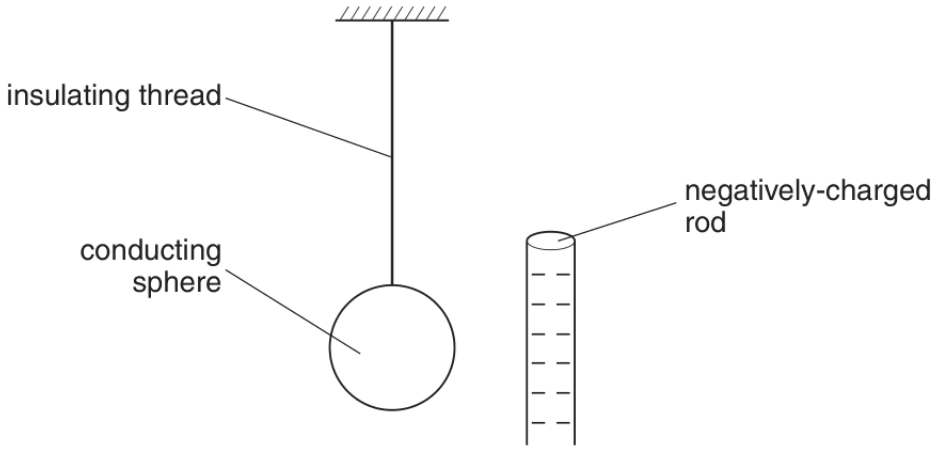


Fig. 5.2

(i) On Fig. 5.2, draw the distribution of positive and negative charges on the sphere. [2]

(ii) Explain why the sphere is attracted to the rod.

.....

.....

.....[2]

5	**(a)**	**(i)**	**electrons** move to the rod	B1	
		(ii)	becomes positively-charged/loses electrons	B1	
	(b)	**(i)**	positives on right **and** negatives on left	M1	
			equal numbers(at least 2) **and** roughly symmetrical	A1	
		(ii)	positive charges attracted	B1	
			attraction larger than repulsion **or** positives closer (than negatives to rod)	B1	[6]

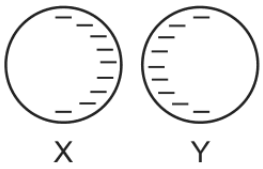
QUESTION · 5054_w11_qp_12.pdf · source page 15 of 20

33 Two metal spheres X and Y are on insulating bases. Both spheres are negatively charged.

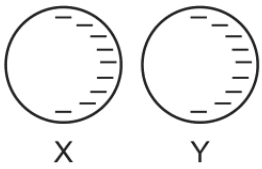
Sphere X is moved towards sphere Y until they almost touch.

Which diagram shows the final pattern of charges?

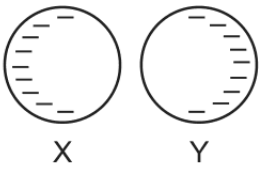
A



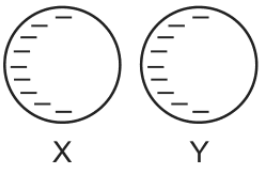
B



C



D



MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 33

33	**C**

QUESTION · 5054_s12_qp_11.pdf · source page 10 of 16

27 A polythene rod is rubbed with a duster. The duster then attracts small pieces of paper.

Are the rod and the duster charged or uncharged?

	rod	duster
A	charged	charged
B	charged	not charged
C	not charged	charged
D	not charged	not charged

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 27

27 **A**

11 Fig. 11.1 shows part of a smoke detector.

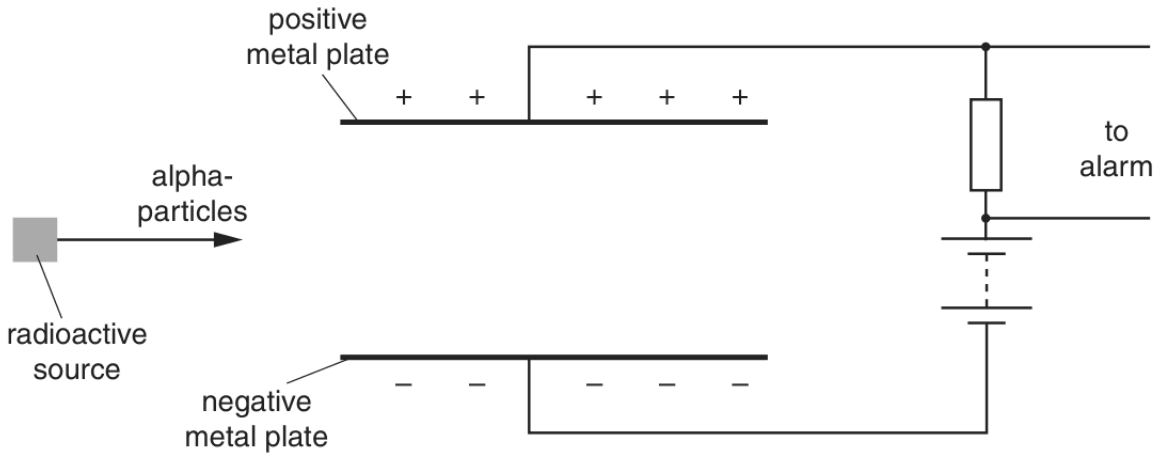


Fig. 11.1 (not to scale)

Alpha-particles from the radioactive source ionise the air between the metal plates. There is then a small current in the resistor. When smoke enters the detector, the current changes and the alarm sounds.

(a) Explain why a source emitting only gamma-rays is unsuitable for use in the smoke detector.

.....

.....[1]

(b) There is an electric field between the metal plates.

(i) On Fig. 11.1, draw the electric field between the plates. [3]

(ii) The alpha-particles in Fig. 11.1 are deflected slightly as they pass through the electric field.

Explain why the alpha-particles are deflected and state the direction of the deflection.

.....

.....

.....[2]

(c) State and explain how the path of gamma-rays differs from the path of alpha-particles as they pass through an electric field.

.....

.....

.....

.....[2]

(d) The source used is the radioactive isotope americium-241, which has a half-life of 500 years.

(i) A sample of americium-241 is left for 1500 years. Determine the fraction of the original number of atoms of americium-241 left in the sample after this time.

fraction = ...[2]

(ii) Another isotope of the same element, americium-242, has a half-life of 16 hours. Explain why the half-life of americium-242 makes this isotope unsuitable for use in a smoke detector.

.....

.....

.....[1]

(iii) Atoms of americium-241 and americium-242 contain protons, neutrons and electrons.

1. State **two** similarities in the structure of an atom of americium-241 and an atom of americium-242.

.....

.....

.....[2]

2. Describe the difference between an atom of americium-241 and an atom of americium-242.

.....

.....

.....[2]

11 **(a)** (gamma) produces little or no ionisation; passes out of detector; requires shielding; reaches people B1

(b) **(i)** at least 3 lines between plates and in middle at least one **straight**, **vertical** line B1
correct curvature at edges B1
at least one arrow down and no arrows wrong B1

(ii) alpha charged (positively); alpha repelled by positive/attracted by negative B1
deflected down/towards positive (plate) B1

(c) (gamma) undeviated; straight line B1
(gamma) uncharged B1

(d) **(i)** any attempt at halving or 3 **half-lives** seen C1
1/8; 0.125; 12.5% A1

(ii) decays too fast; have to replace source often; current falls too quickly; detector only works for a short time B1

(iii) **1.** any TWO of:
number of protons
number of electrons
charge on nucleus B2

2. number of neutrons B1
nucleon or mass number or mass B1 [15]
(americium-242 has **one** extra neutron gets 2 marks)

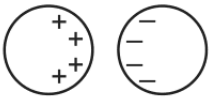
QUESTION · 5054_s12_qp_12.pdf · source page 12 of 16

27 A positively-charged insulated metal sphere is brought close to, but does not touch, a similar uncharged metal sphere.

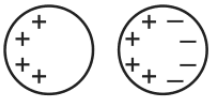
Which diagram shows the charge distribution on the spheres?

A

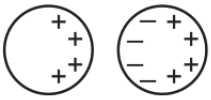
B



C



D



MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 27

27	**D**

7 Fig. 7.1 shows two charged metal spheres.

One sphere has a positive charge and the other sphere has a negative charge.

Fig. 7.1

(a) On Fig. 7.1, draw the electric field between the two spheres. [2]

(b) The negative charge on the sphere is removed and a wire is used to connect the positive sphere to earth. The charge on the positive sphere decreases from 4.8×10^{-9} C to zero in a time of 2.0×10^{-6} s.

Calculate the average current in the wire.

current = ...[2]

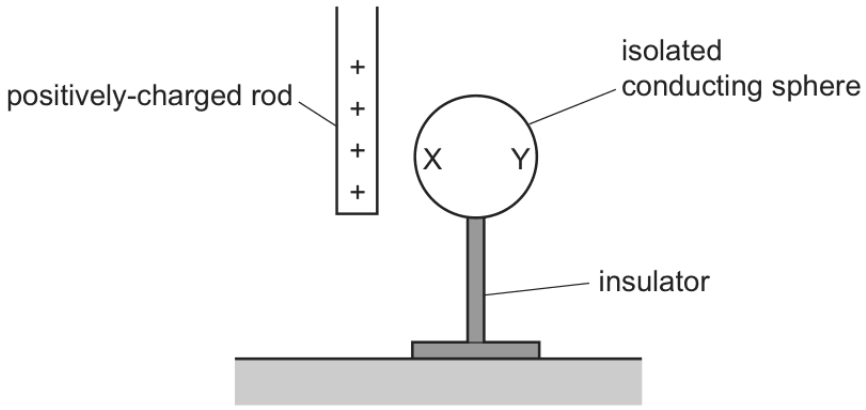
7 **(a)** three lines from one sphere to the other **and** some lines should spread out as they leave one sphere **and** come together nearing the other B1

correct direction on at least one line and none wrong B1

(b) (I =) Q/t in any form numerical or algebraic C1
2.4×10^{-3} A A1 [4]

QUESTION · 5054_w12_qp_11.pdf · source page 11 of 16

27 A positively-charged rod is brought near to an isolated uncharged conducting sphere.



What are the charges on sides X and Y of the sphere?

A Both X and Y are positively charged.

B Both X and Y are negatively charged.

C X is positively charged and Y is negatively charged.

D X is negatively charged and Y is positively charged.

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 27

27	**D**

For Examiner's Use

11 **(a)** Fig. 11.1 shows a thunder cloud with a flat, positively charged base. It passes over a tall tree growing in a region of flat, open land.

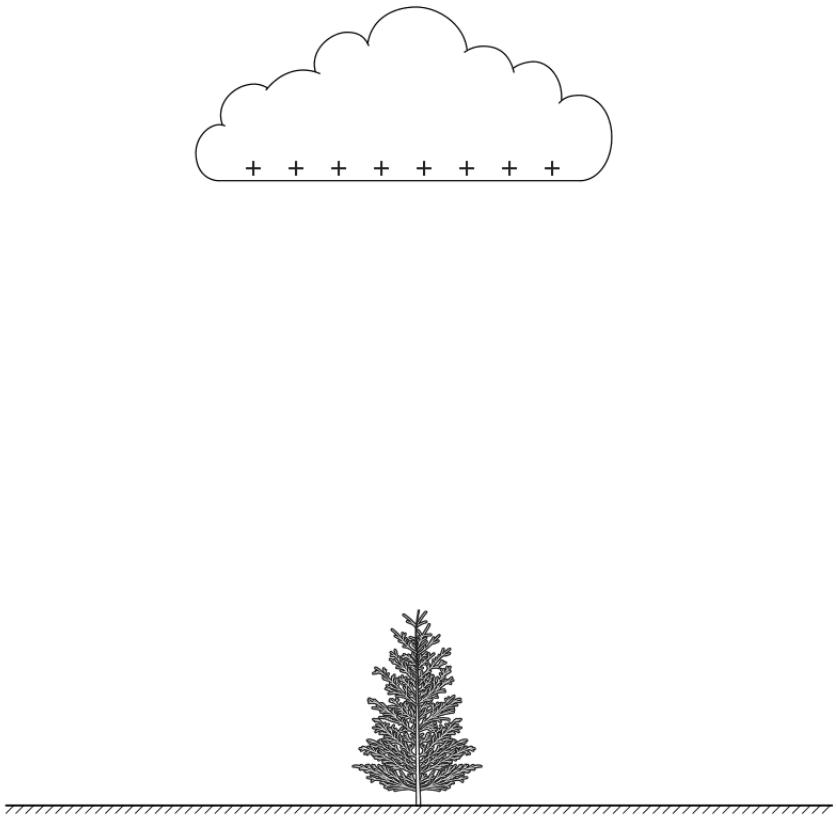


Fig. 11.1 (not to scale)

(i) On Fig. 11.1, mark the charge on the tree. [1]

(ii) Explain how the tree becomes charged.

.....

.....

..... [2]

(iii) A lightning strike occurs and, in 2.0×10^{-4} s, a charge of 560 C passes from the cloud to the tree. The size of the charge on an electron is 1.6×10^{-19} C.

1. Calculate the number of electrons that pass between the tree and the cloud.

number = ... [2]

For Examiner's Use

2. Calculate the average current in the lightning strike.

current = ... [2]

(b) Two flat metal plates are positioned horizontally, one above the other. Fig. 11.2 shows the positive terminal of a high-voltage supply unit connected to the bottom plate and the negative terminal connected to the top plate.

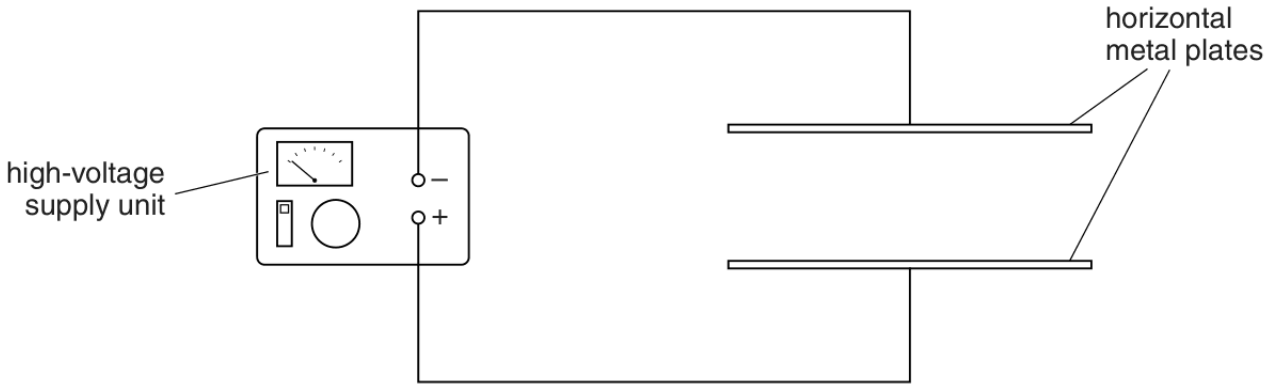


Fig. 11.2

The high-voltage supply is switched on.

(i) On Fig. 11.2, draw the shape and the direction of the electric field produced. [3]

(ii) Suggest why a small, charged oil droplet placed between the two metal plates accelerates upwards.

.....

.....

.....

..... [3]

(iii) The oil droplet touches the top metal plate. State and explain what happens to the charge on the oil droplet.

.....

.....

..... [2]

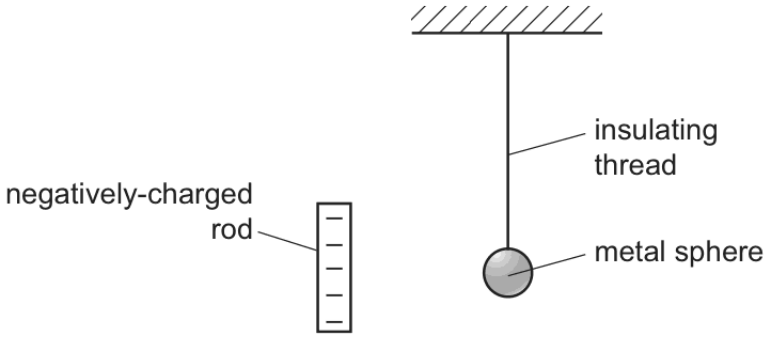
11	**(a)**	**(i)**	correct negative charges on tree.	B1	
		(ii)	electrons/-ve charges attracted by cloud/+ve charges	B1	
			electrons from ground or correct induction mentioned	B1	
		(iii)	**1.** $560/1.6 \times 10^{-19}$	C1	
			3.5×10^{21}	A1	
			2. $(I =)Q/t$ or $560/2 \times 10^{-4}$	C1	
			2.8×10^{6} A	A1	[7]
	(b)	**(i)**	at least 4 vertical lines between plates	B1	
			equally spaced or curved at edges	B1	
			arrows +ve to –ve/upwards	B1	
		(ii)	oil droplet positively charged	B1	
			attraction/force on (droplet) and in direction of field/upwards	B1	
			force greater than weight (of droplet) or resultant force	B1	
		(iii)	(droplet becomes) negative	C1	
			(droplet) gains electrons	A1	[8]

[Total: 15]

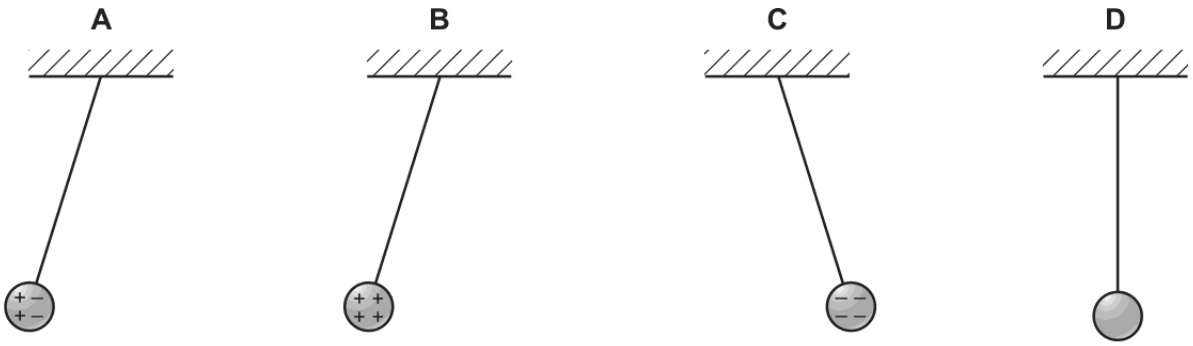
QUESTION · 5054_w12_qp_12.pdf · source page 10 of 16

27 A small uncharged metal sphere hangs from an insulating thread.

A negatively-charged rod moves close to the sphere.



Which diagram shows the charges on the sphere and its final position?

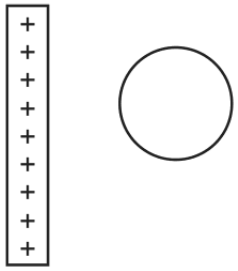


MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 27

27 **A**

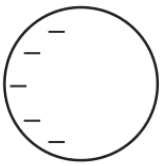
QUESTION · 5054_s13_qp_11.pdf · source page 12 of 20

28 A positively-charged rod is held close to, but not touching, an isolated metal sphere.

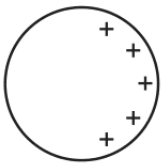


Which diagram shows the charges that are induced on the sphere?

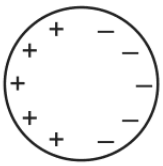
A



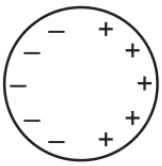
B



C



D



MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 28

28	**D**

QUESTION · 5054_s13_qp_12.pdf · source page 11 of 16

31 A student rubs a rod held in his hand.

Which action causes the rod to gain a large electrostatic charge?

A rubbing an iron rod with a steel magnet

B rubbing an iron rod with a woollen duster

C rubbing a polythene rod with a steel magnet

D rubbing a polythene rod with a woollen duster

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 31

31	**D**

For Examiner's Use

5 A student rubs a polythene rod with a dry cloth. The polythene rod becomes negatively charged and the cloth becomes positively charged.

(a) Describe, in terms of the movement of electrons, what happens during the charging process.

.....

..... [2]

(b) There is an electric field around the charged rod.

Explain what is meant by an *electric field*.

.....

..... [1]

(c) The charged rod is placed near a stream of water from a tap, as shown in Fig. 5.1.

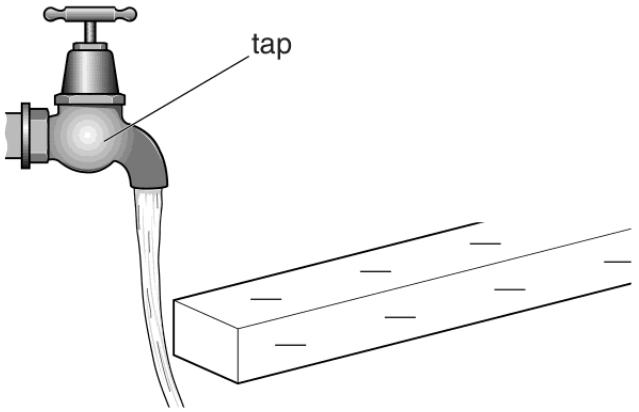


Fig. 5.1

The stream of water bends because the rod is charged.

(i) Explain how the charge on the rod causes positive and negative particles in the water to move.

.....

.....

..... [2]

(ii) On Fig. 5.1, draw the charge present on the stream of water. [1]

5 **(a)** electrons move onto polythene / rod B1

electrons/negative charge move off cloth B1

(b) (region of space) where force is exerted on a charge B1

(c) **(i)** unlike charges attract B1
or (rod) attracts +ve charge/ions/particles

repels like charge B1
or (rod) repels –ve charge/ions/electrons/particles

(ii) (net) positive charge on water near rod B1 [6]

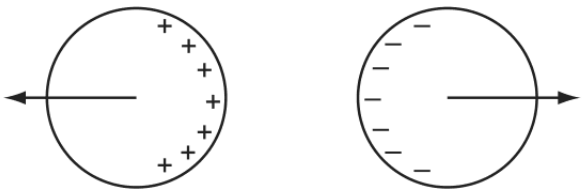
QUESTION · 5054_w13_qp_11.pdf · source page 12 of 20

26 Two charged conducting spheres are placed close to one another.

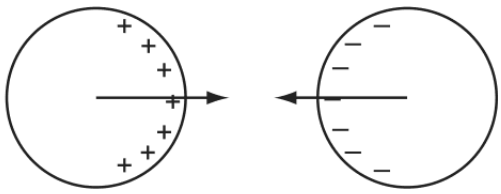
One sphere is positively charged and the other is negatively charged.

Which diagram shows the distribution of charges and the forces acting on the spheres?

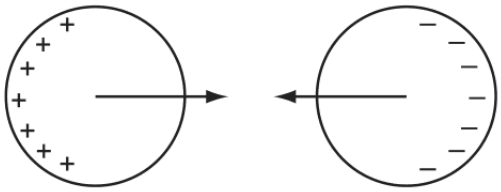
A



B



C



D

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 26

26	**B**

QUESTION · 5054_w13_qp_11.pdf · source page 13 of 20

27 Which row shows a conductor and an insulator?

	conductor	insulator
A	aluminium	copper
B	aluminium	glass
C	plastic	copper
D	plastic	glass

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 27

27 **B**

For Examiner's Use

6 At an airport, fuel is pumped through a pipe from a tanker to an aeroplane, as shown in Fig. 6.1.

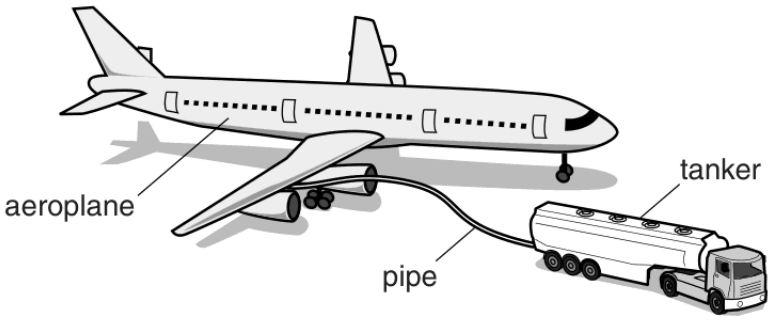


Fig. 6.1

As it rubs against the pipe, the fuel becomes negatively charged and this charges the aeroplane.

(a) Explain, in terms of the particles involved, how the fuel becomes negatively charged.

.....

.....

.....[2]

(b) Suggest and explain one problem that can arise when an aeroplane becomes charged.

.....

.....

.....[2]

(c) To prevent an aeroplane becoming charged, a metal cable connects the aeroplane to the ground.

Explain

(i) why the cable is made of metal,

.....

.....[1]

(ii) how the cable prevents the aeroplane becoming charged.

.....

.....[1]

6	**(a)**	electrons (move)	M1	
		to the fuel **or** from the pipe **or** pipe becomes positively charged (**not** moving protons/+ve charges)	A1	
	(b)	spark (jumps from the plane)	B1	
		ignite the fuel/explosion/blast	B1	
		or		
		current from ground	(B1)	
		shock (to worker/passenger)	(B1)	
	(c) **(i)**	(metal an electrical) conductor **or** has low resistance **or allows/lets** charges/ electrons to flow through it (this is general: about the conduction property of metals)	B1	
	(ii)	charge/electrons flow along the cable **or** (plane/charges) earthed (this is specific: about the conduction in this case)	B1	[6]

5 A positively charged sphere C is brought close to a small, uncharged metal sphere U. Sphere U is suspended from an insulating thread, as shown in Fig. 5.1.

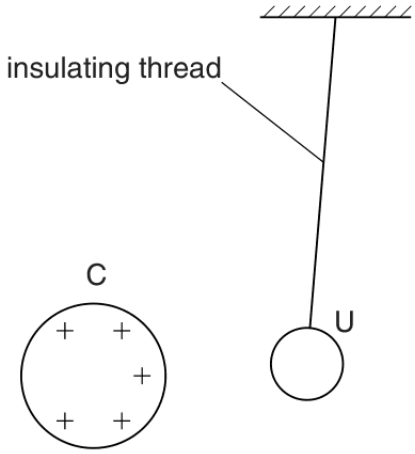


Fig. 5.1

(a) On Fig. 5.1, draw the induced charges on sphere U. [1]

(b) Sphere C is moved towards sphere U until the spheres touch. Sphere U is then repelled by sphere C, as shown in Fig. 5.2. The charges on C and U are not shown.

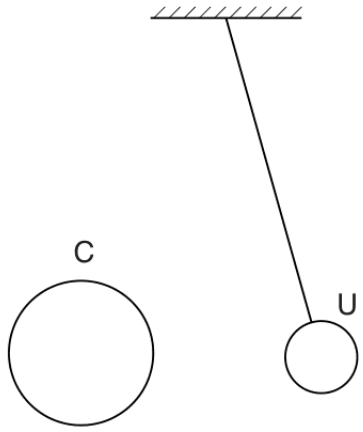


Fig. 5.2

(i) State and explain what happens to the charge on the two spheres as they touch.

charge on C: .....

.....

charge on U: .....

.....
[3]

(ii) Explain why U is repelled by C.

.....

..... [1]

5 **(a)** **equal numbers** (5 or less) of negative charges on left **and** positive charges on right [B1]

(b) **(i)** C becomes less positive / less charged [B1]
U becomes (completely) positive [B1]
electrons / negative charge flows from U to C **or** + (on C) and – (on U) cancel / neutralise [B1]

(ii) like charges repel **or** both have same charge **or** both positive [B1]

[5]

QUESTION · 5054_s14_qp_12.pdf · source page 11 of 16

31 Which row shows an electrical conductor and an insulator?

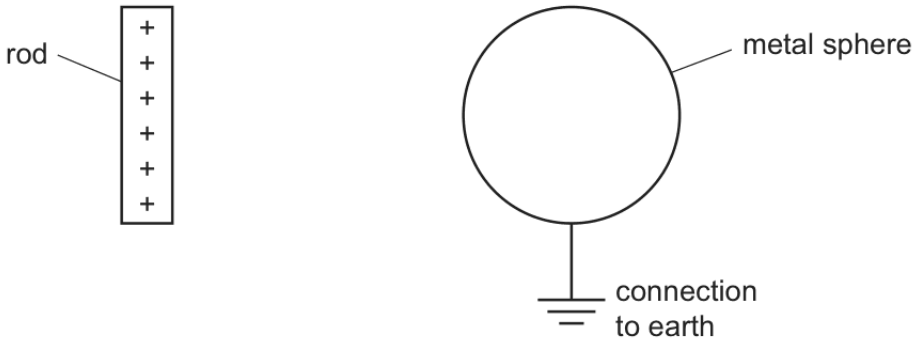
	electrical conductor	insulator
A	aluminium	rubber
B	copper	aluminium
C	plastic	copper
D	rubber	plastic

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 31

31 **A**

QUESTION · 5054_s14_qp_12.pdf · source page 12 of 16

32 A metal sphere is connected to earth. A positively charged rod approaches the sphere and stops before touching it.



What is the movement of charge on the sphere and what is the final charge on the sphere?

	movement of charge	final charge on sphere
A	negative charge moves from earth to the sphere	negative
B	negative charge moves from earth to the sphere	neutral
C	positive charge moves from the sphere to earth	negative
D	positive charge moves from the sphere to earth	neutral

MARK SCHEME (official answer key) · 5054_s14_ms_12.pdf · source page 2 of 2 · row for Question 32

32 **A**

6 The base of a storm cloud is negatively charged. Fig. 6.1 shows the cloud above flat ground.

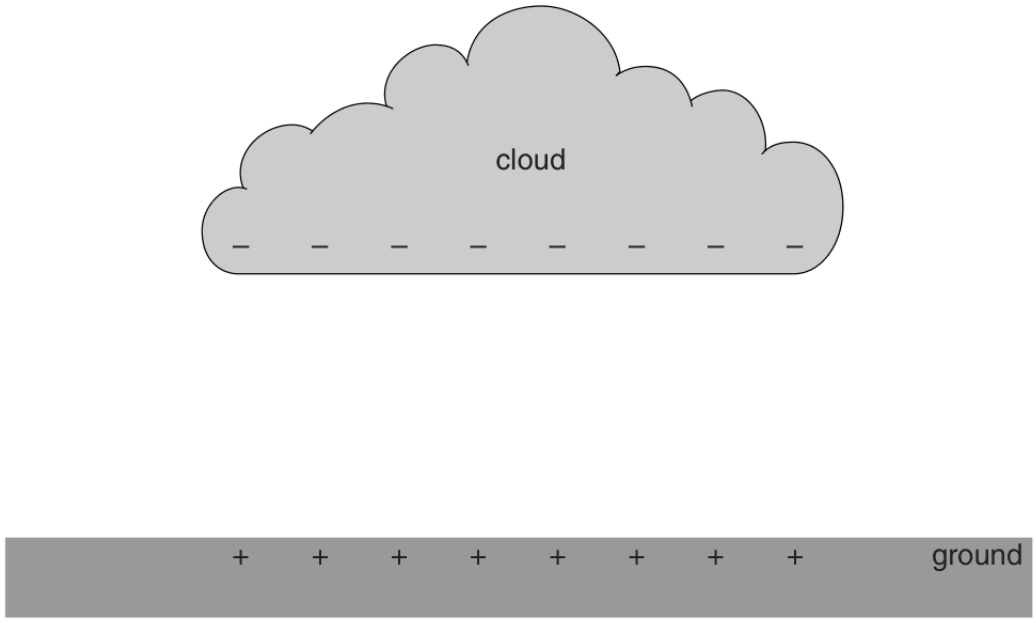


Fig. 6.1

(a) The cloud causes the ground beneath it to become positively charged.

Explain, in terms of the particles involved, how the ground becomes positively charged.

.....

.....

.....[2]

(b) In the space between the negative charge on the cloud and the positive charge on the ground, there is an electric field.

State what is meant by an *electric field*.

.....

.....[1]

(c) A lightning strike takes place. In 0.0015 s, a charge of 180 C passes between the cloud and the ground.

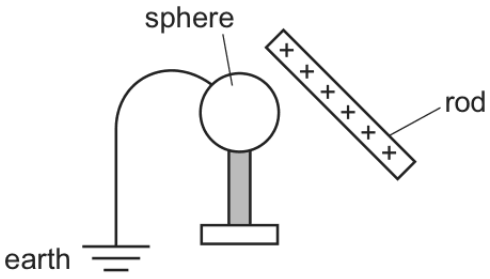
Calculate the average current in the lightning strike.

current = ...[2]

6	**(a)**	**electrons** repelled by cloud (leaving ground positive) **not** positive charge / protons move	B1	
		like charges repel **or** electrons negative	B1	
	(b)	(region) where (electric) charge experiences a force	B1	
	(c)	(I =) Q/t **or** 180 / 0.0015	C1	
		1.2×10^5 A	A1	**[5]**

QUESTION · 5054_w14_qp_12.pdf · source page 9 of 16

22 A positively charged rod is held close to an earthed metal sphere.



What describes the charge on the metal sphere?

A It is negative because electrons are attracted towards the rod.

B It is neutral because electrons are attracted towards the rod and protons are repelled.

C It is neutral because it is earthed.

D It is positive because protons are repelled by the rod.

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 22

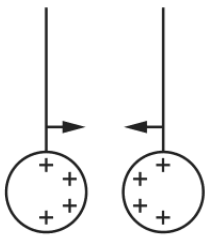
22	**A**

QUESTION · 5054_w14_qp_12.pdf · source page 9 of 16

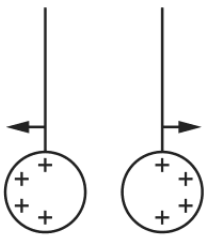
23 Two charged metal spheres are suspended close to each other.

Which diagram shows the charge distribution on the spheres and the directions of the forces on the spheres?

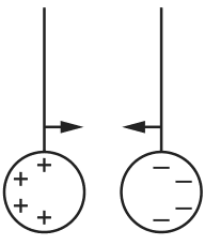
A



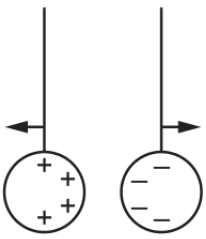
B



C



D



MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 23

23	**B**

QUESTION · 5054_w14_qp_12.pdf · source page 9 of 16

24 Which list contains only electrical insulators?

A glass, plastic, rubber

B glass, plastic, steel

C glass, rubber, steel

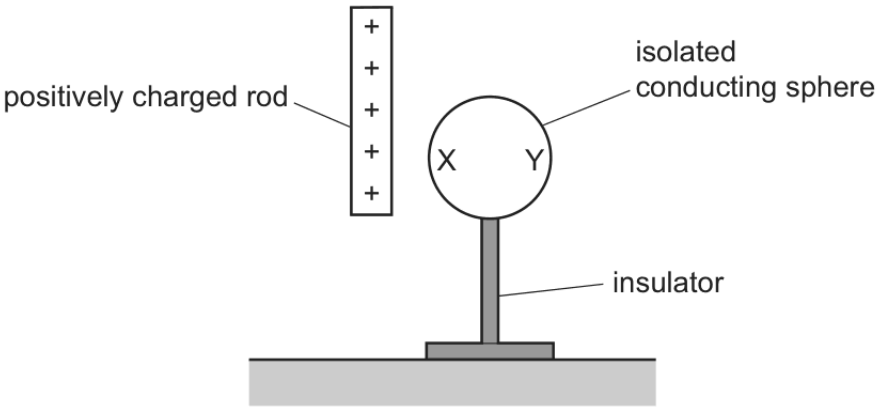
D plastic, rubber, steel

MARK SCHEME (official answer key) · 5054_w14_ms_12.pdf · source page 2 of 2 · row for Question 24

24	**A**

QUESTION · 5054_s15_qp_12.pdf · source page 12 of 16

32 A positively charged rod is brought near to an isolated uncharged conducting sphere.



What are the charges on sides X and Y of the sphere?

A Both X and Y are positively charged.

B Both X and Y are negatively charged.

C X is positively charged and Y is negatively charged.

D X is negatively charged and Y is positively charged.

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 32

32	**D**

5 When a balloon is rubbed on hair, the balloon becomes negatively charged. The balloon is shown in Fig. 5.1.

Fig. 5.1

(a) Explain how rubbing causes the balloon to become negatively charged.

.....

.....

..... [2]

(b) Explain why the hair is pulled towards the balloon.

.....

.....

.....

..... [2]

(c) Explain why it is important that the balloon is made from an electrical insulator.

.....

..... [1]

(d) State one example where static electricity is useful.

.....

..... [1]

5 **(a)** negative charge moves from hair / person / head to balloon C1
electrons move from hair / person / head to the balloon A1

(b) hair is positive (at end) B1
opposite charges attract B1
or positive and negative attract

(c) **charges / electrons** B1
- don't flow away
- aren't conducted (to earth / person)

stay on balloon / on insulator

(d) any sensible example e.g. photocopier, electrostatic precipitator, flu ash removal, spray painting, printing, crop spraying, lightning fixes nitrogen in atmosphere etc. B1

QUESTION · 5054_w15_qp_11.pdf · source page 10 of 16

24 A polythene rod becomes charged when rubbed with a dry woollen cloth.

The charge caused is a result of

A friction producing a movement of electrons.

B friction producing a movement of protons.

C magnetism producing a movement of electrons.

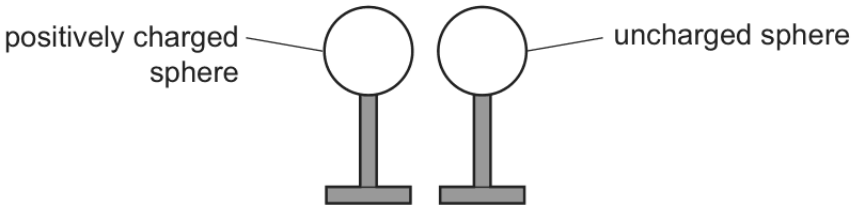
D magnetism producing a movement of protons.

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 24

24	**A**

QUESTION · 5054_w15_qp_11.pdf · source page 10 of 16

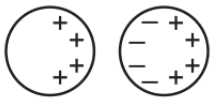
25 A positively charged insulated metal sphere is brought close to an uncharged insulated metal sphere.



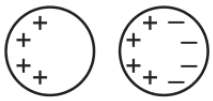
Which diagram shows the charge distribution on the spheres?

A

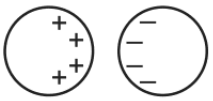
B



C



D



MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 25

25	**B**

QUESTION · 5054_w15_qp_12.pdf · source page 12 of 16

29 Which row correctly names an electrical conductor and an insulator?

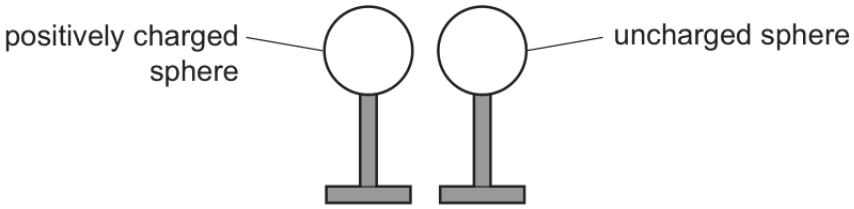
	electrical conductor	insulator
A	aluminium	iron
B	iron	rubber
C	plastic	aluminium
D	rubber	plastic

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 29

29 **B**

QUESTION · 5054_w15_qp_12.pdf · source page 12 of 16

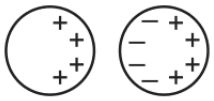
30 A positively charged insulated metal sphere is brought close to an uncharged insulated metal sphere.



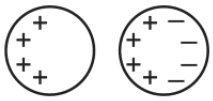
Which diagram shows the charge distribution on the spheres?

A

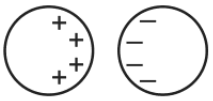
B



C



D



MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 30

30	**B**

10 Thin wire, covered in plastic insulation, is used to make a solenoid (long coil). The solenoid is connected to a sensitive ammeter. Fig. 10.1 shows the N-pole of a steel magnet placed next to the solenoid.

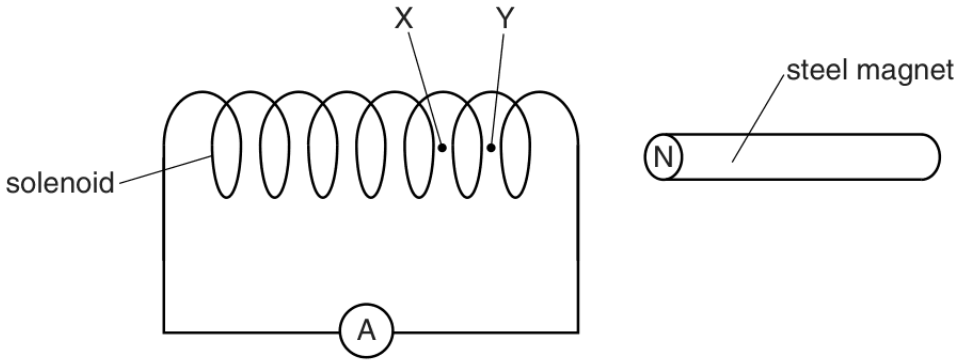


Fig. 10.1

Point X and point Y are on the axis of the solenoid.

(a) **(i)** Explain why plastic is an electrical insulator.

.....

.....

..... [1]

(ii) Explain why the magnet is **not** made from

1. aluminium,

..... [1]

2. iron.

..... [1]

(b) In one experiment, the magnet in Fig. 10.1 is moved to the left and passes into the solenoid. The N-pole of the magnet travels from Y to X at a constant speed. As it moves, the ammeter shows a small current.

(i) Explain why there is a current in the solenoid when the magnet is moving.

.....

.....

.....

.....

.....

..... [3]

(ii) The N-pole travels from Y to X in 0.14 s. As it moves, the current shown on the ammeter is 0.045 mA. The resistance of the solenoid is 1.2 Ω.

Calculate

1. the potential difference (p.d.) across the solenoid,

potential difference = ... [2]

2. the charge that passes through the solenoid as the N-pole moves from Y to X.

charge = ... [2]

(c) In a second experiment, the speed of the N-pole is greater than its speed in the first experiment. It now takes only 0.070 s to travel from Y to X.

A current in the same direction is shown on the ammeter.

(i) State and explain how the size of this current compares with the size of the current in the first experiment.

.....

.....

..... [2]

(ii) The same quantity of charge passes through the coil in both the first and second experiments.

Explain why this is the case.

.....

.....

..... [1]

(d) State two ways in which the equipment shown in Fig. 10.1 can be used to produce a current in the solenoid that is in the opposite direction.

1. .....

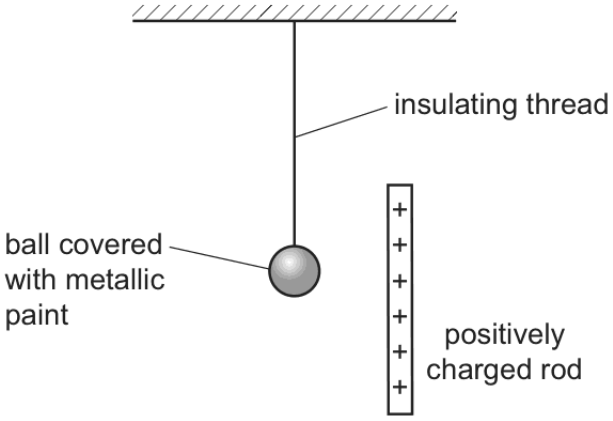
2. .....
[2]

10	**(a)**	**(i)**	no free electrons (in plastic) **or** all electrons are bound / structural	B1	
		(ii)	(aluminium) is not magnet(ic) **or** cannot be magnetised	B1	
			(iron) is a temporary / soft magnetic material **or** is not a permanent magnet	B1	[3]
	(b)	**(i)**	magnetic field / flux (mentioned)	B1	
			(magnetic) field lines cut wire / solenoid / circuit **or** changing magnetic field / flux	B1	
			voltage/e.m.f. **induced**	B1	
		(ii)	(V =)IR **or** 0.045×1.2 **or** 0.000045×1.2	C1	
			5.4×10^{-5} V **or** 0.054 mV	A1	
			(Q =)It **or** 0.045×0.14 **or** 0.000045×0.14	C1	
			6.3×10^{-6} C **or** 0.0063 mC	A1	[7]
	(c)	**(i)**	larger **or** twice the current	B1	
			(magnetic) field lines cut faster **or** (magnetic) field changes faster **or** twice the current	B1	

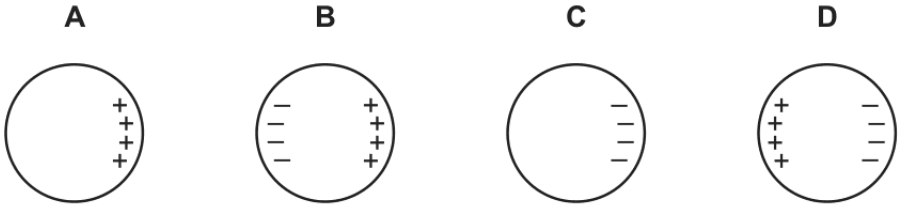
		(ii)	double the current for half the time **or** larger current for less time **or** product $I \times t$ is the same	B1	[3]
	(d)	any **two** from: insert S-pole (at same end) insert (N-pole) at other end **or** from other direction withdraw N-pole (from same end implied) withdraw S-pole from other end **or** pass through completely		B2	[2]
					[15]

QUESTION · 5054_s16_qp_11.pdf · source page 11 of 16

28 The diagram shows an uncharged ball coated with metallic paint. The ball is suspended from an insulating thread. It is placed near a positively charged rod.



Which diagram shows the charge distribution on the ball?



MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 28

28	**D**

QUESTION · 5054_s16_qp_12.pdf · source page 9 of 16

24 A polythene rod is rubbed with a cloth and the cloth becomes positively charged.

Which statement describes the transfer of charge?

A Negative charges are transferred from the cloth to the polythene.

B Negative charges are transferred from the polythene to the cloth.

C Positive charges are transferred from the cloth to the polythene.

D Positive charges are transferred from the polythene to the cloth.

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 24

24	**A**

7 An uncharged piece of metal P rests on an insulator. A positively charged rod is placed close to P, as shown in Fig. 7.1.

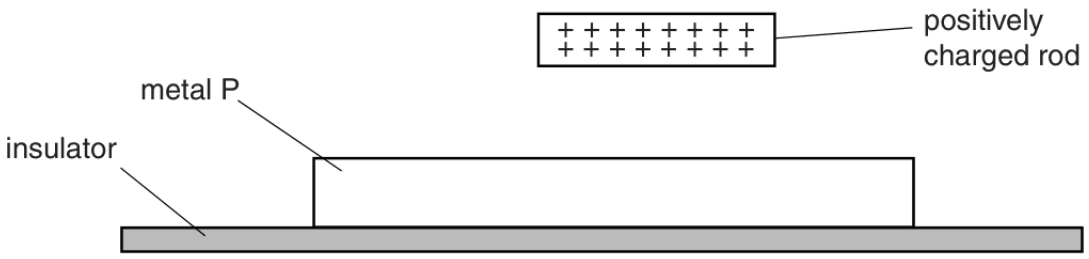


Fig. 7.1

(a) State a material that is an electrical insulator.

.....[1]

(b) On Fig. 7.1, draw the distribution of charges on P. [2]

(c) P is then connected to earth by a wire, as shown in Fig. 7.2.

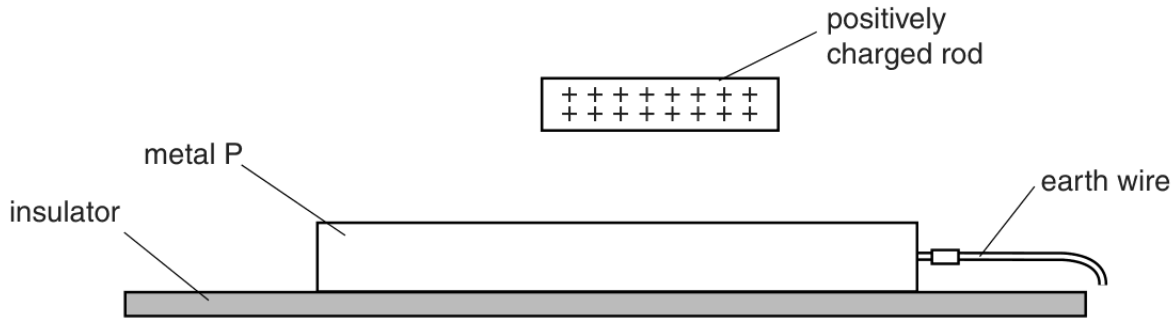


Fig. 7.2

(i) On Fig. 7.2, show the distribution of charges on P with the earth wire connected. [1]

(ii) State what happens to the charges on P if the positively charged rod is removed

1. with the earth wire still connected to P,

.....

.....[1]

2. after the earth wire is disconnected from P.

.....

.....[1]

7 **(a)** any insulator, e.g. perspex, plastic, nylon, rubber B1

(b) top of P shows a net negative charge with some negative charges under rod B1
bottom of P has equal number of positive charges B1

(c) **(i)** clear net negative charge on P B1
and (net) negative charges above or at middle line

(ii) **1** negative (charges)/electrons flow to earth B1
or (P) <u>becomes</u> neutral

2 charges spread over P B1

QUESTION · 5054_w16_qp_11.pdf · source page 13 of 20

31 A positively charged metal sphere is placed midway between two previously uncharged metal rods, one of which is connected to earth.

Which diagram shows the charges on the rods?

A

B

C

D

MARK SCHEME (official answer key) · 5054_w16_ms_11.pdf · source page 2 of 2 · row for Question 31

31	**B**

QUESTION · 5054_w16_qp_12.pdf · source page 12 of 16

32 A positively charged metal sphere is placed midway between two previously uncharged metal rods, one of which is connected to earth.

Which diagram shows the charges on the rods?

A

B

C

D

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 32

32	**B**

QUESTION · 5054_s17_qp_11.pdf · source page 13 of 16

32 A charged plastic ball is at rest.

Which fields are found in the region surrounding the ball?
(Ignore the magnetic field of the Earth.)

A electric, gravitational and magnetic

B electric and gravitational only

C electric and magnetic only

D gravitational and magnetic only

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 3 of 3 · row for Question 32

32	B	**1**

QUESTION · 5054_s17_qp_11.pdf · source page 14 of 16

35 A shoe becomes positively charged by friction when it rubs against a carpet.

What happens as the shoe becomes charged?

A Negative electrons are transferred to the carpet.

B Negative electrons are transferred to the shoe.

C Positive electrons are transferred to the carpet.

D Positive electrons are transferred to the shoe.

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 3 of 3 · row for Question 35

35	A	**1**

8 Fig. 8.1 shows charges on a rod and on a cloth.

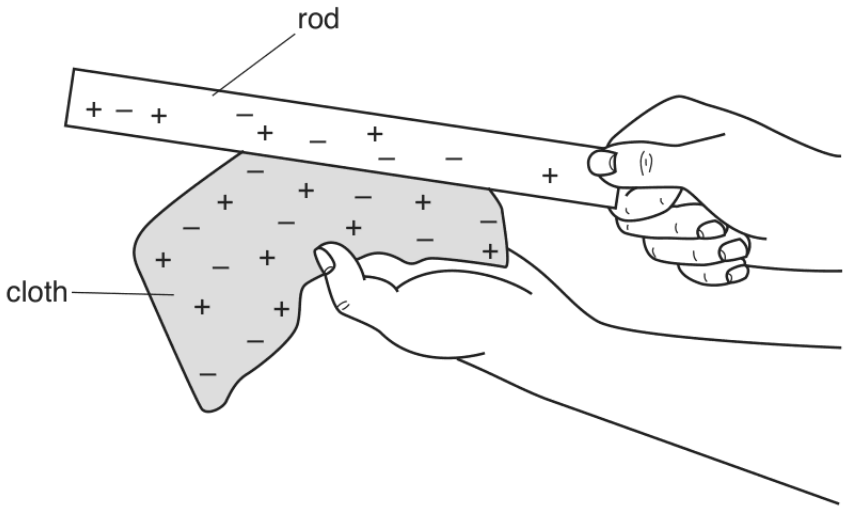


Fig. 8.1

(a) State how Fig. 8.1 shows that the rod and the cloth are uncharged.

.....

.....[1]

(b) The plastic rod becomes negatively charged when rubbed by the cloth.

(i) Describe what happens to the charges on the rod and on the cloth during the charging process.

.....

.....

.....

.....[2]

(ii) Describe a simple experiment to show that the plastic rod is charged.

You may draw a diagram if you wish.

.....

.....

.....

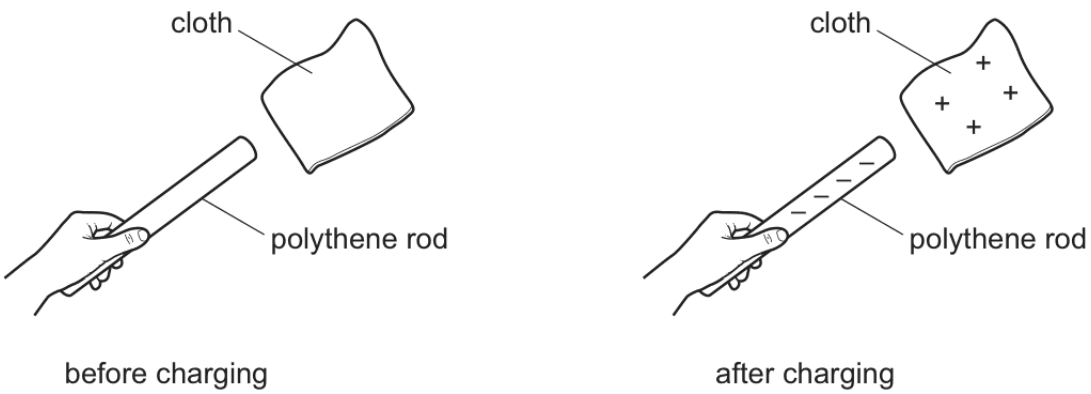
.....

.....[2]

Question	Answer	Marks
8(a)	equal (numbers of) positive and negative charges	**B1**
8(b)(i)	negative charge moves from cloth to rod	**C1**
	electrons move from cloth to rod	**A1**
8(b)(ii)	apparatus needed, e.g. (small) pieces of paper / water stream / (gold leaf) electroscope / suspended or pivoted other charged rod / charged object / conducting object	**B1**
	correct statement of what is seen / felt with apparatus	**B1**

QUESTION · 5054_w17_qp_11.pdf · source page 13 of 20

28 A polythene rod is charged by rubbing it with a cloth.



Which statement explains why the cloth is charged?

A Electrons move from the cloth to the rod.

B Electrons move from the rod to the cloth.

C Protons move from the cloth to the rod.

D Protons move from the rod to the cloth.

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 2 of 3 · row for Question 28

28	A	**1**

11 **(a)** Fig. 11.1 shows a negatively charged rod close to an uncharged metal sphere that is mounted on an insulating stand.

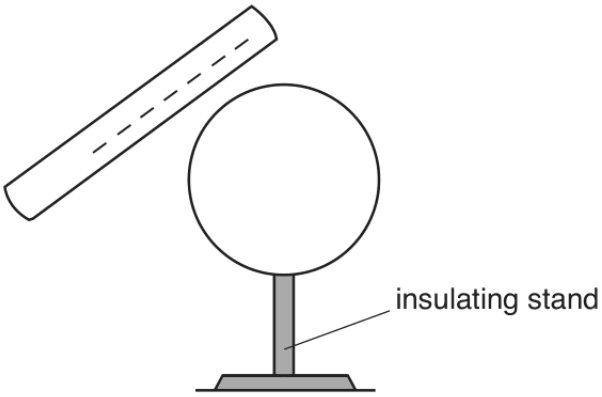


Fig. 11.1

(i) Suggest a material for the insulating stand.

..... [1]

(ii) On Fig. 11.1, draw the distribution of charges on the sphere. [2]

(iii) A metal wire connected to earth is touched against the sphere.

1. Explain what happens to the charge on the sphere.

.....

.....

.....

.....

.....

..... [3]

2. The earth wire remains in contact with the sphere.

Describe what happens in the wire as the rod is moved away.

.....

.....

..... [1]

(b) A circuit contains a 7.5 V d.c. power supply and three resistors.

Fig. 11.2 is the circuit diagram.

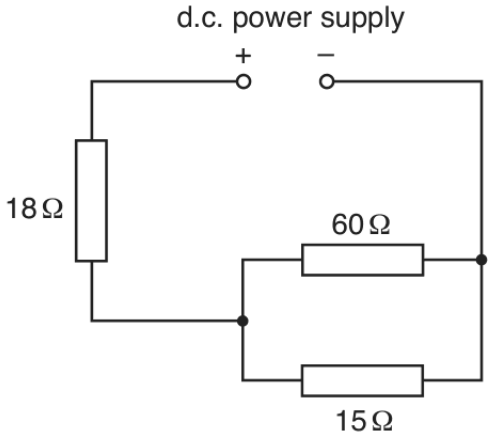


Fig. 11.2

(i) Determine the total resistance of the circuit.

resistance = ... [3]

(ii) Calculate the current in the power supply.

current = ... [2]

(iii) There is a different current in each of the three resistors in the circuit in Fig. 11.2.

Underline the resistor in which the current is

1. the largest,

15 Ω resistor **18 Ω resistor** **60 Ω resistor**

2. the smallest.

15 Ω resistor **18 Ω resistor** **60 Ω resistor**

[1]

(iv) The 7.5 V d.c. power supply is made from five 1.5 V cells.

In the space below, draw a diagram that shows how the cells are arranged.

[1]

(v) The 18 Ω resistor is made of metal.

State and explain what happens to the current in this resistor as its temperature increases.

.....

.....

..... [1]

Question	Answer	Marks
11(a)(i)	any suitable solid insulator (e.g. nylon, plastic, glass, rubber, polystyrene)	**B1**
11(a)(ii)	positive charges near to rod	**B1**
	negative charges opposite rod **and** equal in number **and** 7 or fewer	**B1**
11(a)(iii)	**1** electrons / negative charges flow towards earth	**B1**
	repelled (by negative charge on rod)	**B1**
	(sphere) becomes positive	**B1**
	2 flow of electrons / negative charge **and** (in direction) earth to sphere	**B1**
11(b)(i)	$1 / R = 1 / R_1 + 1 / R_2$ **or** $R_1 R_2 / (R_1 + R_2)$ **or** 1 / R = 1 / 15 + 1 / 60 **or** $15 \times 60 / 75$ **or** $15 \times 60 / (15 + 60)$	**C1**
	12 (Ω) **or** 0.083 (Ω)	**C1**
	30 Ω	**A1**
11(b)(ii)	(I =)V / R **or** 7.5 / 30	**C1**
	0.25 A	**A1**

Question	Answer	Marks
11(b)(iii)	**18 Ω resistor** underlined **2** **and 60 Ω resistor** underlined	**B1**
11(b)(iv)	five cells in series **and** all in same direction	**B1**
11(b)(v)	resistance increases **and** current decreases	**B1**

QUESTION · 5054_w17_qp_12.pdf · source page 11 of 16

27 A plastic block is rubbed with a silk handkerchief and the block becomes charged.

What causes the block to become charged?

A Negative electrons flow from the handkerchief and positive electrons flow from the block.

B Negative electrons flow from the handkerchief to the block.

C Positive electrons flow from the block to the handkerchief.

D Positive electrons flow to the handkerchief and negative electrons flow to the block.

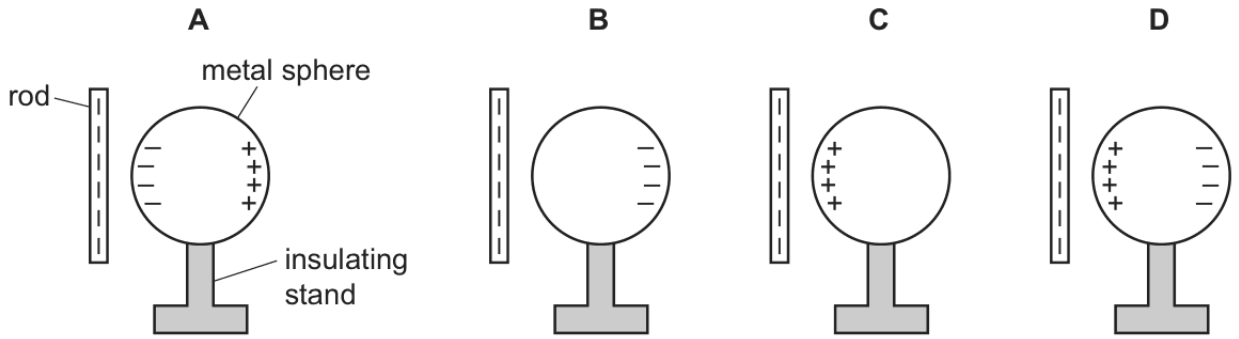
MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 27

27	B	**1**

QUESTION · 5054_w17_qp_12.pdf · source page 11 of 16

28 A negatively-charged rod is brought near to an uncharged metal sphere on an insulating stand.

Which diagram shows the charge distribution on the sphere?



MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 2 of 3 · row for Question 28

28	D	**1**

7 Electrostatics can be a nuisance but is also useful.

(a) A person walks across a carpet. As he touches a door handle he receives an electric shock.

Suggest why this happens.

.....

.....

.....[2]

(b) In spray painting, small drops of paint with a positive charge emerge from a nozzle. The drops are used to paint the leg of a metal chair, which has a negative charge, as shown in Fig. 7. 1.

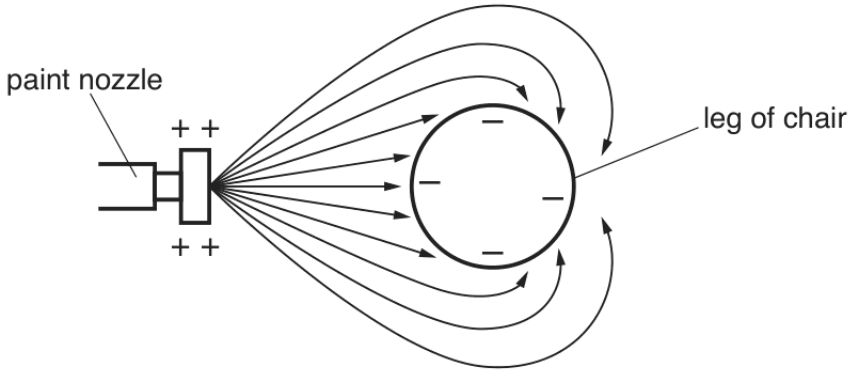


Fig. 7.1 (not to scale)

Fig. 7.2 shows what happens when there is no charge on the nozzle, or on the drops or on the leg of the chair.

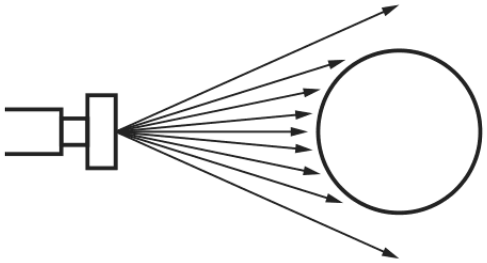


Fig. 7.2 (not to scale)

(i) The drops in Fig. 7.1 spread out more as they leave the nozzle than those in Fig. 7.2.

Explain why.

.....

.....[2]

(ii) Explain why the paint reaches the back of the leg in Fig. 7.1.

.....

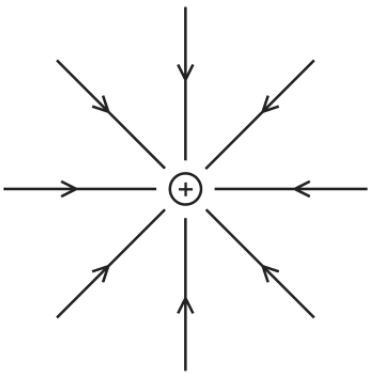
.....[1]

Question	Answer	Marks
7(a)	friction (between shoes and carpet) causes charging **or** <u>electrons</u> move from foot to carpet or carpet to foot	**B1**
	charge / electrons (on person) flows (to handle causing shock) or v.v.	**B1**
7(b)(i)	they / drops repel (each other)	**B1**
	they / drops have same charge (as each other)	**B1**
7(b)(ii)	(positive) drops attracted by negative (leg)	**B1**

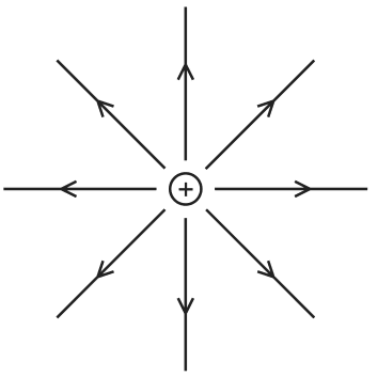
QUESTION · 5054_w18_qp_11.pdf · source page 9 of 16

24 Which diagram shows the electric field around an isolated point positive charge?

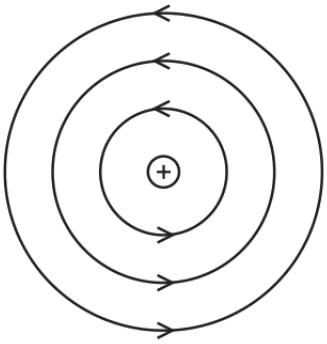
A



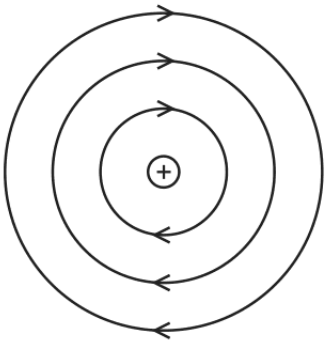
B



C



D

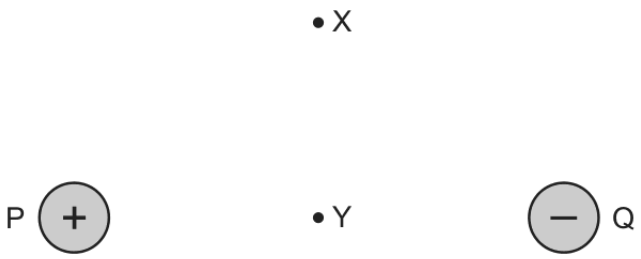


MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 24

24	B	**1**

QUESTION · 5054_w18_qp_11.pdf · source page 9 of 16

25 An electric field is formed by two isolated, equal and opposite charges P and Q.



How does the field at X compare with the field at Y?

A It is stronger at X than at Y and is in the opposite direction.

B It is stronger at X than at Y and is in the same direction.

C It is weaker at X than at Y and is in the opposite direction.

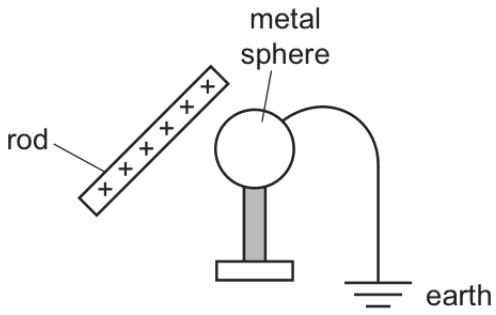
D It is weaker at X than at Y and is in the same direction.

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 25

25	D	**1**

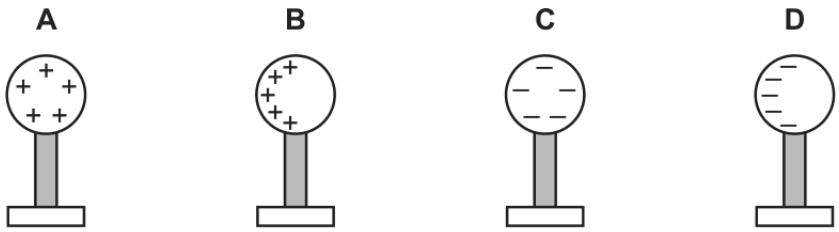
QUESTION · 5054_w18_qp_11.pdf · source page 10 of 16

26 A positively charged rod is held close to an insulated metal sphere. The sphere is earthed as shown.



The earth connection is removed and then the rod is removed.

Which diagram shows the charges on the sphere after the rod is removed?



MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 26

26	C	**1**

5 Two uncharged conducting spheres, K and L, are mounted on insulating stands.

(a) State how the structure of an insulator differs from that of a conductor and give one example of an insulator.

.....

.....

..... [2]

(b) The two spheres are in contact.

Fig. 5.1 shows a positively-charged insulating rod held next to K.

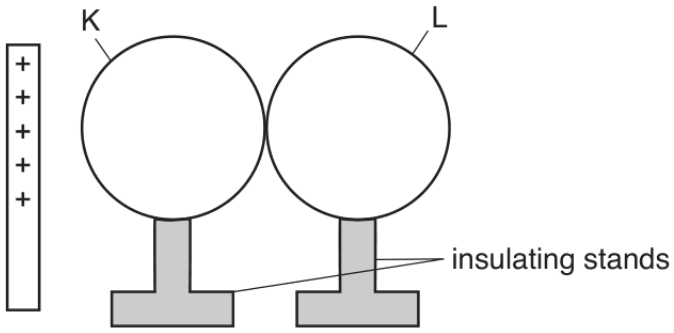


Fig. 5.1

(i) On Fig. 5.1, draw the charges on K and on L. [2]

(ii) Sphere L is moved to the right, a long way from sphere K.

1. The positively charged rod is moved away.

State what happens to the charge on K.

.....

..... [1]

2. An earthing wire is connected to sphere L.

State what happens to the charge on L.

.....

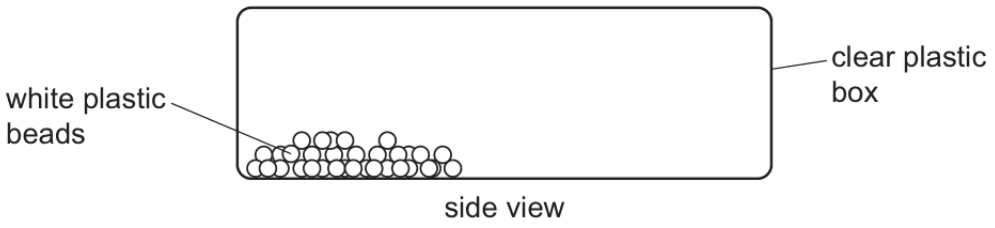
..... [1]

Question	Answer	Marks
5(a)	charges / electrons are not free to move (in an insulator)	**B1**
	any sensible example e.g. plastic / nylon / glass / rubber	**B1**
5(b)(i)	negative charge on left of K **and** positive charge on right of L	**M1**
	equal numbers of charges **and** number ⩽ 5 **and** no charges on right of K **and** no charges on left of L	**A1**

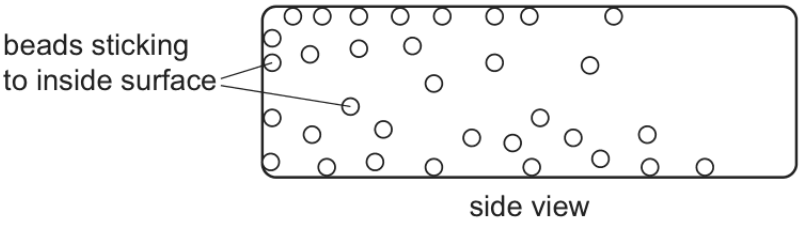
Question	Answer	Marks
5(b)(ii)	1 (negative) charge spreads throughout (surface of) sphere	**B1**
	2 (positive) charge on L disappears **or** L becomes neutral **or** equal number of positive and negative charges	**B1**

QUESTION · 5054_s19_qp_11.pdf · source page 15 of 20

31 The diagram shows some white plastic beads in a clear plastic box.



The box is shaken and the beads rub against the box. The beads stick to the inside surface of the box.



Which row is a possible explanation for this?

	box	beads
A	gains electrons	no change of electrons
B	loses electrons	gain electrons
C	no change of electrons	lose electrons
D	no change of electrons	gain electrons

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 3 of 3 · row for Question 31

31	B	**1**

5 **(a)** Fig. 5.1 shows part of a machine used to investigate electrostatic charging.

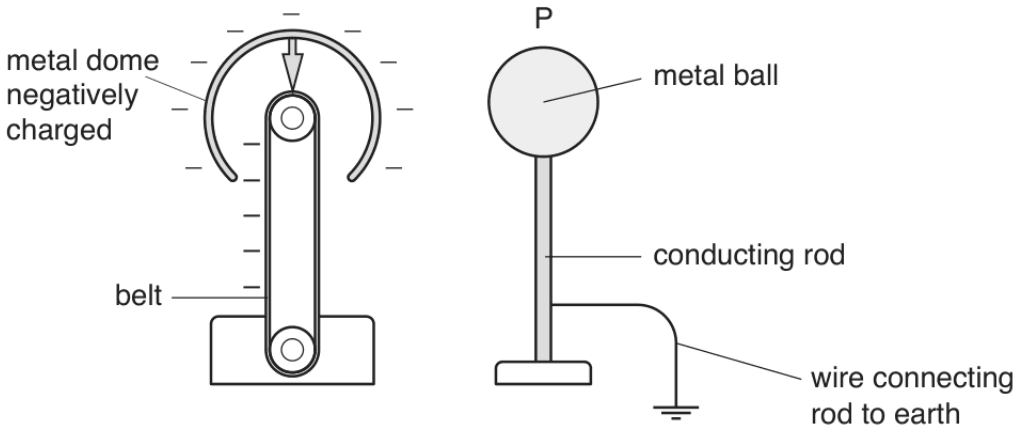


Fig. 5.1

Before the machine is switched on the metal dome and the ball are uncharged.

When the machine is switched on, the metal dome becomes negatively charged.

Explain how charging the dome causes the metal ball P to become positively charged.

.....

.....

.....

..... [2]

(b) A photocopier works using electrostatic charging. It contains a drum whose surface conducts charge where it is exposed to light.

Fig. 5.2 shows a piece of paper, the drum and the heater of a photocopier.

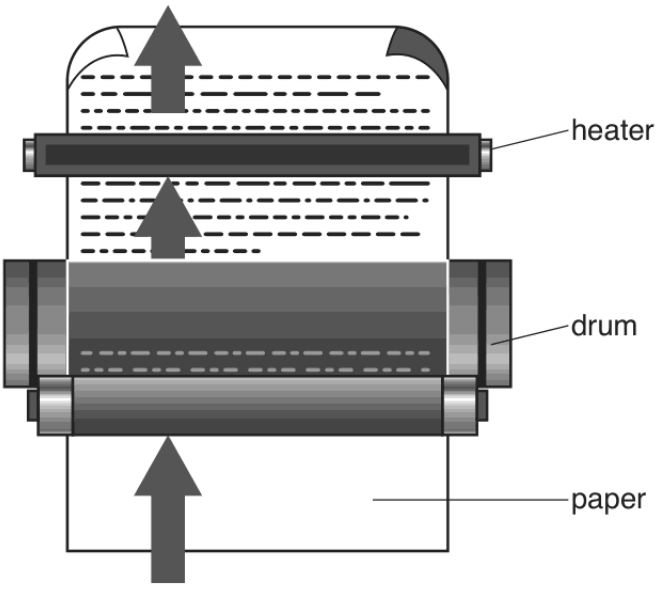


Fig. 5.2

The photocopier produces a copy of an original page on a piece of paper.

The sentences below describe the process of making the copy but the sentences are in the wrong order.

A The drum is given a positive charge.

B The positively charged parts of the drum attract a black powder.

C Where light hits the drum, the positive charge leaks away.

D An image of the original page is projected onto the drum.

E The drum rolls against a piece of paper, transferring powder to the paper.

F The paper is heated which makes the black powder stick to it.

Arrange the sentences in the correct order. The first and last sentences are already in the correct boxes.

A	→		→		→		→		→	**F**

[2]

[Total: 4]

Question	Answer	Marks
5(a)	negative charge / electrons / experience repulsive force (from dome)	**B1**
	negative charge / electrons are earthed / move to earth	**B1**
5(b)	D at start or E at end	**C1**
	D C B E	**A1**

8 Fig. 8.1 shows a device that is used to show the force on an electron in an electric field.

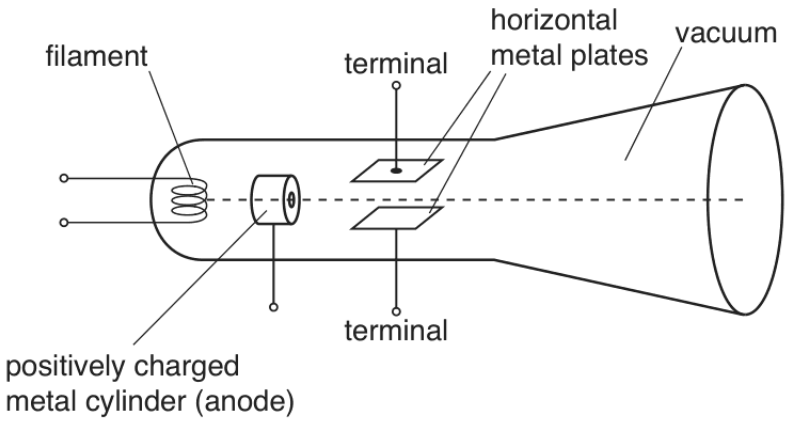


Fig. 8.1

(a) Explain how a continuous flow of electrons from the filament is produced in this apparatus.

.....

.....

.....

.....

..... [2]

(b) When an electron is in an electric field there is a force on the electron.

The direction of this force is in the opposite direction to the electric field.

Describe how the metal plates in Fig. 8.1 can be used to show this.

.....

.....

.....

.....

..... [2]

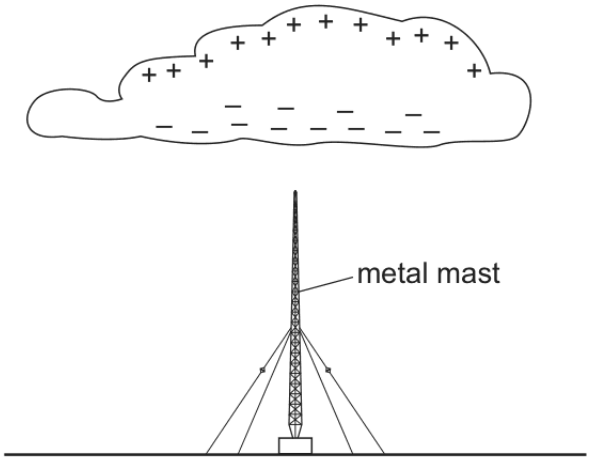
[Total: 4]

Question	Answer	Marks
8(a)	filament is heated / hot (emits electrons) **or** thermionic emission occurs at the filament	**B1**
	cylinder / anode attracts / accelerates electrons	**B1**
8(b)	connect a battery / p.d. / voltage / power supply across plates	**B1**
	electrons deflect / attract towards positive / away from negative **or** field is from positive to negative	**B1**

QUESTION · 5054_w19_qp_11.pdf · source page 15 of 20

31 The base of a cloud is negatively charged.

The cloud is over a metal mast as shown.



Which row describes the charge induced at the top of the mast and how it is produced?

	charge at the top of the mast	charge at the top of mast is produced by
A	negative	electrons moving to the top of the mast
B	negative	protons moving to the bottom of the mast
C	positive	electrons moving to the bottom of the mast
D	positive	protons moving to the top of the mast

MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 3 of 3 · row for Question 31

31	C	**1**

QUESTION · 5054_w19_qp_11.pdf · source page 15 of 20

32 An aircraft flying through the air becomes electrically charged.

What causes this to happen?

A background radiation in the aircraft

B friction of the aircraft with the air

C heating of the aircraft surface

D radio waves emitted by the aircraft

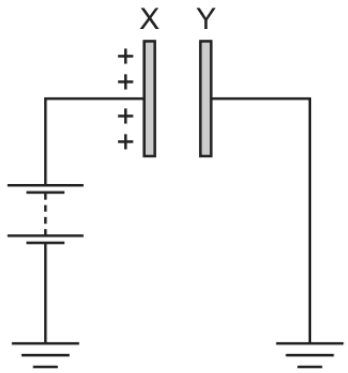
MARK SCHEME (official answer key) · 5054_w19_ms_11.pdf · source page 3 of 3 · row for Question 32

32	B	**1**

QUESTION · 5054_w19_qp_12.pdf · source page 13 of 16

30 A metal plate X is attached to earth by a battery. The plate has a positive charge.

A second metal plate Y is initially uncharged and connected to earth. This plate is positioned parallel to X.



What happens as Y is brought closer to X?

A Electrons flow from Y to earth only.

B Electrons flow to Y from earth only.

C Protons flow from Y to earth and electrons flow to Y from earth.

D Protons flow to Y from earth and electrons flow from Y to earth.

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 3 of 3 · row for Question 30

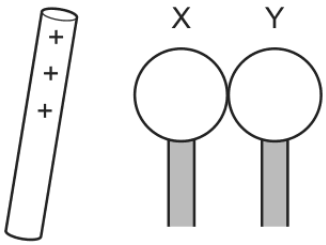
30	B	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 12 of 16

34 Two insulated and uncharged metal spheres X and Y are touching.

A positively charged rod is held near X and then the spheres are moved apart.

X now has a negative charge.



What is the charge on Y?

A negative and smaller than that on X

B negative and the same size as that on X

C positive and smaller than that on X

D positive and the same size as that on X

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 3 of 3 · row for Question 34

34	D	**1**

QUESTION · 5054_s20_qp_11.pdf · source page 13 of 16

35 A stationary negative charge in an electric field experiences an electric force in the direction shown.

What is the direction of the electric field?

A to the left

B to the right

C down the page

D up the page

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 3 of 3 · row for Question 35

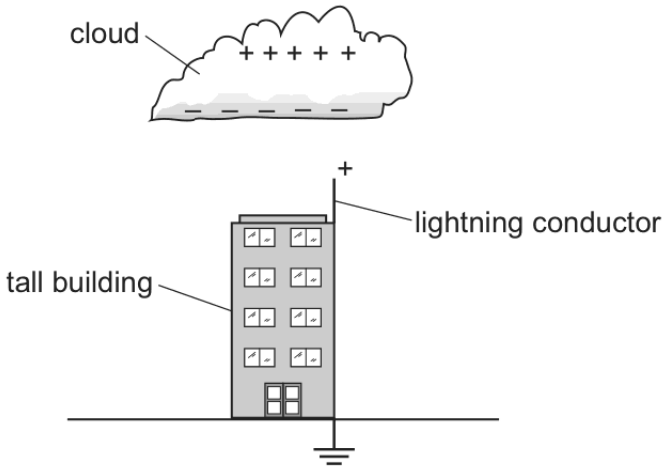
35	A	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 12 of 16

34 A metal lightning conductor is placed on the top of a tall building.

The conductor is connected to Earth.

When a charged cloud passes over the building, the metal lightning conductor becomes positively charged.



What happens to the lightning conductor to produce a positive charge at its top?

A Electrons move downwards and protons move upwards in the conductor.

B Only electrons move in the conductor.

C Only protons move in the conductor.

D Protons in the air attach themselves to the conductor.

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 3 of 3 · row for Question 34

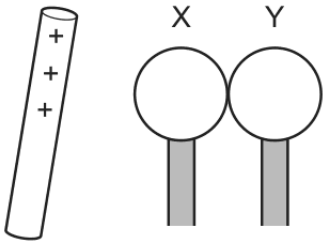
34	B	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 13 of 16

35 Two insulated and uncharged metal spheres X and Y are touching.

A positively charged rod is held near X and then the spheres are moved apart.

X now has a negative charge.



What is the charge on Y?

A negative and smaller than that on X

B negative and the same size as that on X

C positive and smaller than that on X

D positive and the same size as that on X

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 3 of 3 · row for Question 35

35	D	**1**

QUESTION · 5054_s20_qp_12.pdf · source page 13 of 16

36 A stationary negative charge in an electric field experiences an electric force in the direction shown.

What is the direction of the electric field?

A to the left

B to the right

C down the page

D up the page

MARK SCHEME (official answer key) · 5054_s20_ms_12.pdf · source page 3 of 3 · row for Question 36

36	A	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 4 of 16

7 An electrically charged plastic ball is at rest.

Which types of field are caused by the ball?

A electric, gravitational and magnetic

B electric and gravitational only

C electric and magnetic only

D gravitational and magnetic only

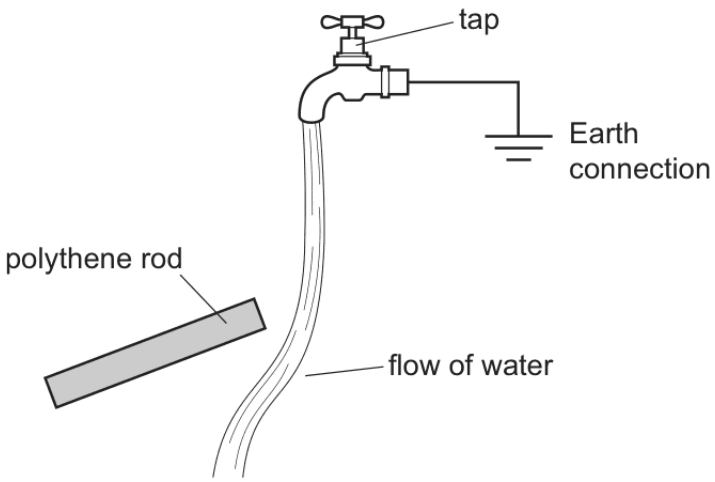
MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 2 of 3 · row for Question 7

7	B	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 14 of 16

33 A polythene rod is brought near to a stream of water from a metal tap which is earthed. Tap water is an electrical conductor.

The water moves towards the rod, as shown.



Which description of the rod and water is correct?

	rod	water
A	charged	charged
B	charged	uncharged
C	uncharged	charged
D	uncharged	uncharged

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 3 of 3 · row for Question 33

33	A	**1**

QUESTION · 5054_w20_qp_11.pdf · source page 14 of 16

34 A small negative charge is placed at a point where an electric field is vertically upwards.

There is a force on the charge due to the field.

In which direction does it act?

A vertically upwards

B vertically downwards

C horizontally to the right

D horizontally to the left

MARK SCHEME (official answer key) · 5054_w20_ms_11.pdf · source page 3 of 3 · row for Question 34

34	B	**1**

QUESTION · 5054_w20_qp_12.pdf · source page 5 of 16

9 An electrically charged plastic ball is at rest.

Which types of field are caused by the ball?

A electric, gravitational and magnetic

B electric and gravitational only

C electric and magnetic only

D gravitational and magnetic only

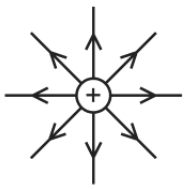
MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 2 of 3 · row for Question 9

9	B	**1**

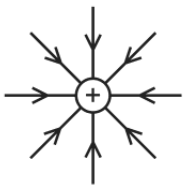
QUESTION · 5054_s21_qp_11.pdf · source page 11 of 16

27 Which diagram shows the pattern and direction of the electric field lines near a positive charge?

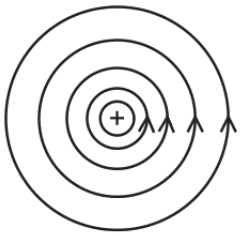
A



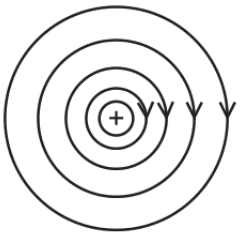
B



C



D

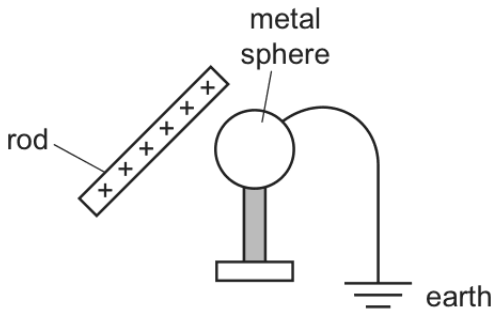


MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 27

27	A	**1**

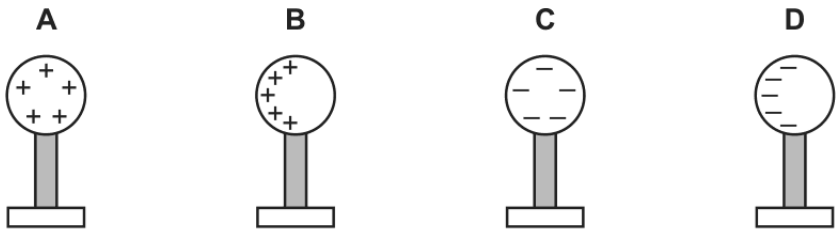
QUESTION · 5054_s21_qp_11.pdf · source page 11 of 16

28 A positively charged rod is held close to an insulated metal sphere. The sphere is earthed as shown.



The earth connection is removed and then the rod is removed.

Which diagram shows the charges on the sphere after the rod is removed?

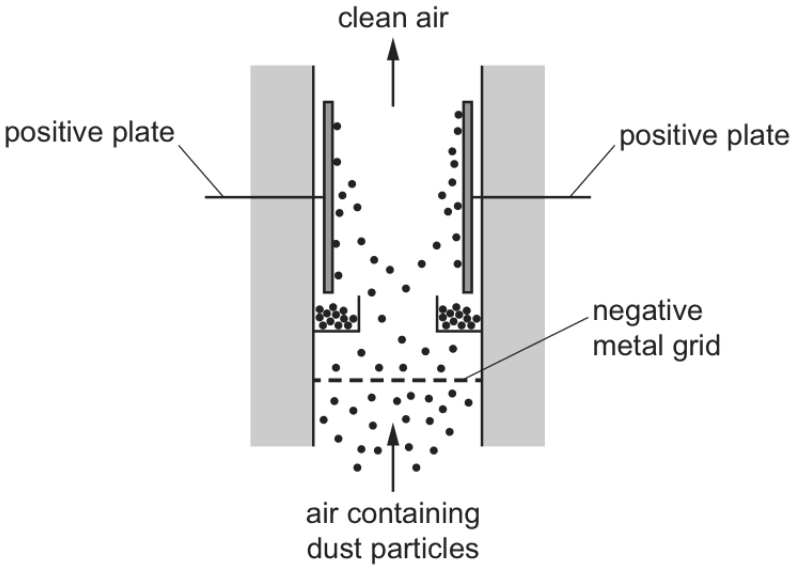


MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 2 of 3 · row for Question 28

28	C	**1**

QUESTION · 5054_s21_qp_11.pdf · source page 12 of 16

29 The diagram shows an electrostatic precipitator. It can be used to remove dust from air.



What happens at the negative grid?

A Dust particles gain electrons.

B Dust particles gain protons.

C Dust particles lose electrons.

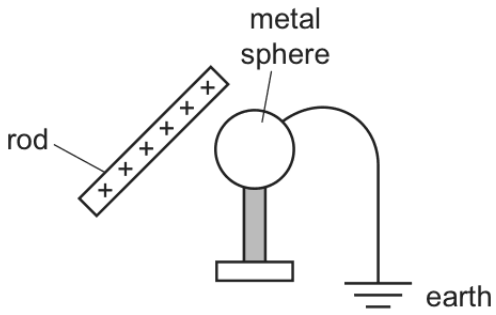
D Dust particles lose protons.

MARK SCHEME (official answer key) · 5054_s21_ms_11.pdf · source page 3 of 3 · row for Question 29

29	A	**1**

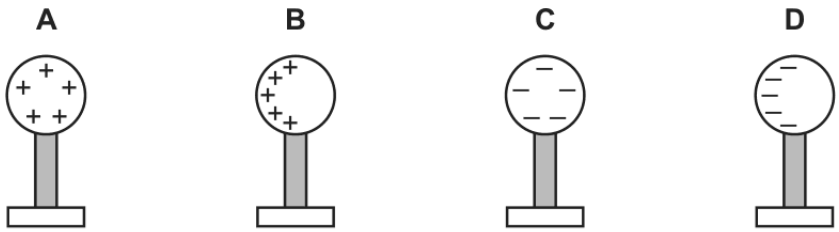
QUESTION · 5054_s21_qp_12.pdf · source page 11 of 16

31 A positively charged rod is held close to an insulated metal sphere. The sphere is earthed as shown.



The earth connection is removed and then the rod is removed.

Which diagram shows the charges on the sphere after the rod is removed?



MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 3 of 3 · row for Question 31

31	C	**1**

5 Fig. 5.1 shows a method of producing sandpaper using static electricity.

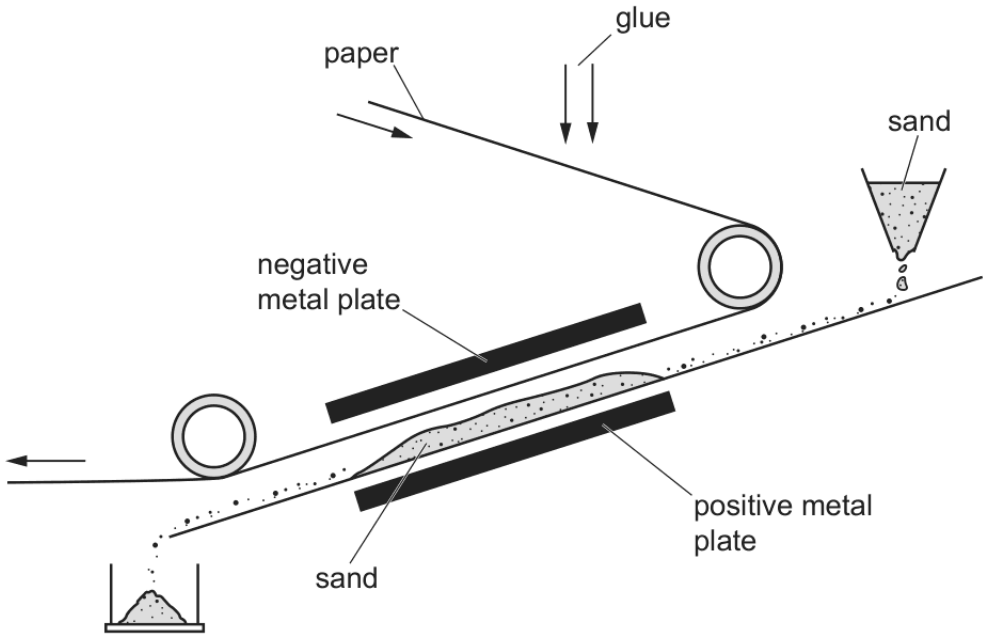


Fig. 5.1

Glue is sprayed on to moving paper. The sticky paper then passes between two metal plates.

One of the plates is positive and the other plate is negative.

There is an electric field between the plates.

(a) On Fig. 5.2, draw the electric field between the two metal plates. Show the direction of the electric field.

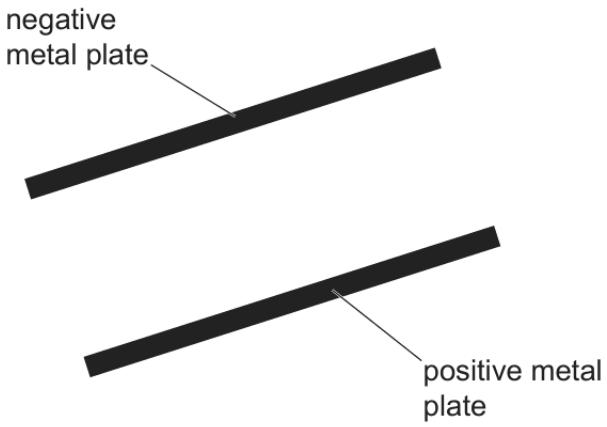


Fig. 5.2

[2]

(b) Grains of sand are present just below the sticky paper.

Each grain of sand is a conductor of electricity.

Fig. 5.3 shows two layers of sand grains in the space between the two plates.

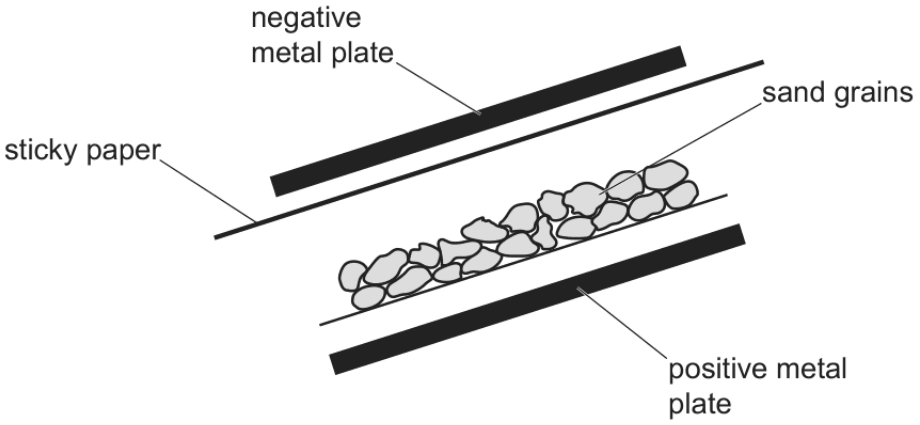


Fig. 5.3

(i) Explain, using ideas about the movement of charge, why the top layer of sand becomes positively charged.

.....

..... [1]

(ii) State why the top layer of sand moves towards the sticky paper.

.....

..... [1]

(c) A student receives an electric shock by walking across a certain type of carpet and then nearly touching a piece of earthed metal with his hand.

The potential difference between the hand and the metal is 2000 V and the charge transferred is 4.0×10^{-4} C.

Calculate the energy transferred when the spark jumps across the air gap from the metal onto his hand.

energy = ... [2]

[Total: 6]

Question	Answer	Marks
5(a)	at least three straight lines perpendicular to plates	**B1**
	arrow on at least one line from positive to negative plate	**B1**
5(b)(i)	electrons / negative charge (on sand / grains) move to(wards) the positive plate / downwards / to(wards) or into the bottom layer	**B1**
5(b)(ii)	positive (sand grain) attracted to or by negative (plate)	**B1**

Question	Answer	Marks
5(c)	(E) = QV numerical or algebraic	**C1**
	0.8(0) J	**A1**

QUESTION · 5054_w21_qp_11.pdf · source page 3 of 20

5 An electron has mass.

Where does a stationary electron experience a force?

A in an electric field only

B in a gravitational field only

C in a gravitational field and an electric field only

D in a gravitational, electric and magnetic field

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 2 of 3 · row for Question 5

5	C	**1**

5 A plastic rod that is initially neutral is rubbed with a woollen cloth. After the rod has been rubbed, it is positively charged.

(a) (i) Explain, in terms of particles, why the rod is now positively charged.

.....

..... [1]

(ii) State what happens to the woollen cloth.

..... [1]

(b) An uncharged, conducting sphere is suspended from an insulating thread. The positively charged rod is placed near to the sphere, as shown in Fig. 5.1.

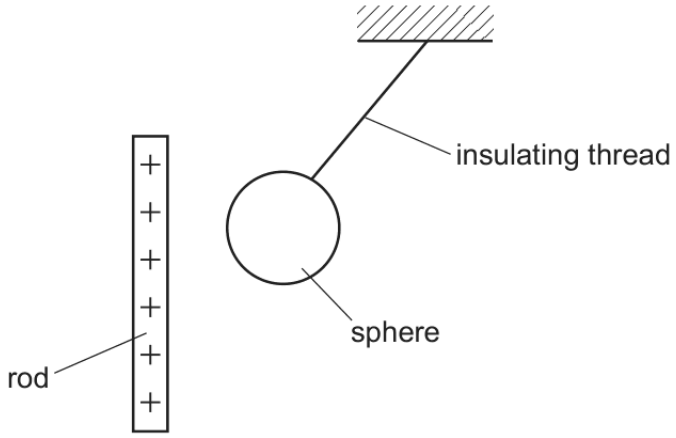


Fig. 5.1

(i) On Fig. 5.1, draw the distribution of charge on the sphere. [2]

(ii) Explain why the sphere is pulled towards the rod.

.....

.....

.....

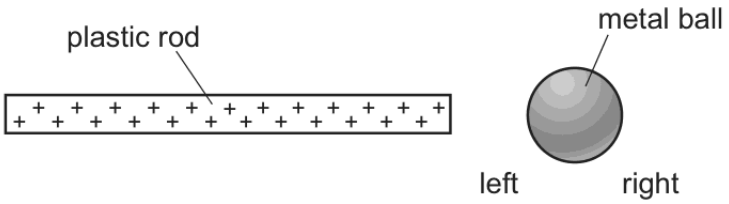
..... [2]

[Total: 6]

Question	Answer	Marks
5(a)(i)	electrons are transferred from rod (on to cloth)	**B1**
5(a)(ii)	it / the cloth becomes negatively charged	**B1**
5(b)(i)	more negative (than positive) signs on left of sphere **and** more positive (than negative) signs on right	**M1**
	equal numbers of positive and negative signs **and** six or fewer	**A1**
5(b)(ii)	negative charges/electrons (on sphere) attracted by positive charges (on rod)	**B1**
	force of attraction greater than repulsion (of positive charges) **or** distance between negative (on sphere) and positive charges (on rod) less than distance between positive and positive	**B1**

QUESTION · 5054_w21_qp_12.pdf · source page 13 of 16

31 A positively charged plastic rod is held near but not touching an insulated and previously uncharged metal ball. No charge leaves the rod.



What are possible charges on the left-hand side and right-hand side of the ball?

	charge on left-hand side of ball	charge on right-hand side of ball
A	-2.0×10^{-9} A	-1.0×10^{-9} A
B	-2.0×10^{-9} A	$+2.0 \times 10^{-9}$ A
C	-2.0×10^{-9} C	$+2.0 \times 10^{-9}$ C
D	$+2.0 \times 10^{-9}$ C	-2.0×10^{-9} C

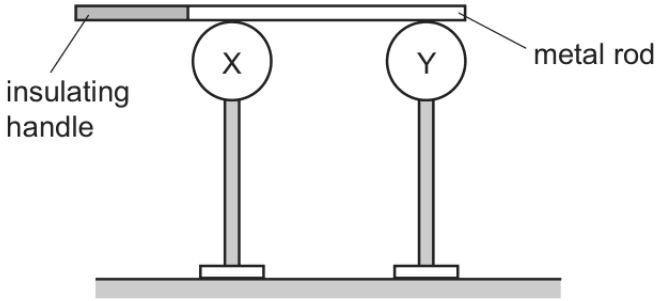
MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 3 of 3 · row for Question 31

31	C	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 11 of 16

28 Two metal spheres are mounted on insulating stands. Sphere X is initially uncharged and sphere Y is initially positively charged.

A metal rod, held by an insulating handle, is placed in contact with X and Y as shown.



What are the charges on X and on Y after the rod is placed in contact with them?

	charge on X	charge on Y
A	positive	positive
B	positive	uncharged
C	uncharged	positive
D	uncharged	uncharged

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 2 of 3 · row for Question 28

28	A	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 11 of 16

29 Which device involves the use of static electrical charges?

A a computer hard disk drive

B a motor

C a photocopier

D a transformer

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 3 of 3 · row for Question 29

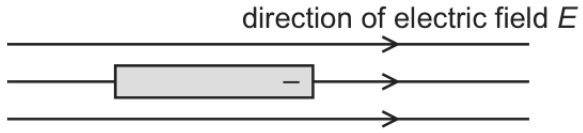
29	C	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 12 of 16

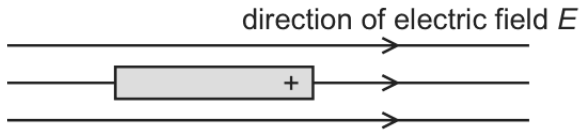
30 An initially uncharged copper rod is placed in a uniform electric field *E*. The rod is parallel to the field.

Which diagram shows the charges induced on the rod?

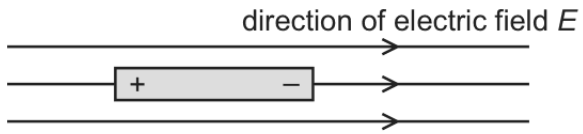
A



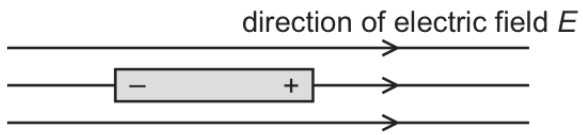
B



C



D

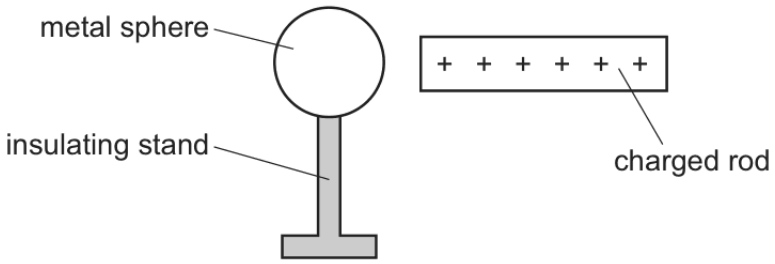


MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 3 of 3 · row for Question 30

30	D	**1**

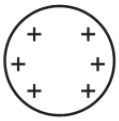
QUESTION · 5054_s22_qp_12.pdf · source page 14 of 20

30 A positively charged rod is held near an uncharged metal sphere on an insulating stand.

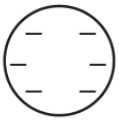


What is the distribution of charge on the sphere?

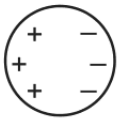
A



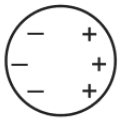
B



C



D



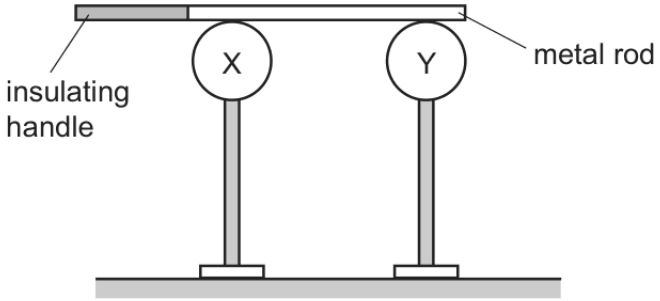
MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 3 of 3 · row for Question 30

30	C	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 14 of 20

31 Two metal spheres are mounted on insulating stands. Sphere X is initially uncharged and sphere Y is initially positively charged.

A metal rod, held by an insulating handle, is placed in contact with X and Y as shown.



What are the charges on X and on Y after the rod is placed in contact with them?

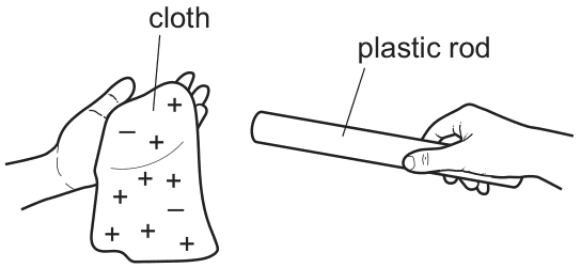
	charge on X	charge on Y
A	positive	positive
B	positive	uncharged
C	uncharged	positive
D	uncharged	uncharged

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 3 of 3 · row for Question 31

31	A	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 11 of 16

27 A plastic rod and a cloth are initially uncharged. The rod is then rubbed with the cloth and the cloth becomes positively charged.



What is the sign of the charge on the plastic rod and why does it become charged?

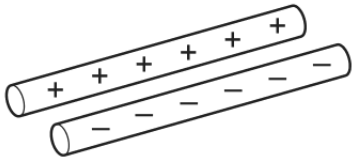
	charge on rod	reason why rod becomes charged
A	negative	electrons move from the cloth to the rod
B	negative	protons move from the rod to the cloth
C	positive	electrons move from the rod to the cloth
D	positive	protons move from the cloth to the rod

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 27

27	A	**1**

QUESTION · 5054_w22_qp_12.pdf · source page 13 of 16

31 Two oppositely charged plastic rods are placed next to each other.



Which row describes and explains how the rods move?

	movement	explanation
A	away from each other	opposite charges attract
B	away from each other	opposite charges repel
C	towards each other	opposite charges attract
D	towards each other	opposite charges repel

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 3 of 3 · row for Question 31

31	C	**1**

8 An uncharged plastic rod is rubbed with an uncharged woollen cloth.

As a result, the rod becomes negatively charged.

(a) Explain, in terms of particles, how the rod becomes charged and state what happens to the cloth.

.....

.....

.....

..... [2]

(b) Fig. 8.1 shows a metal sphere X on a plastic stand.

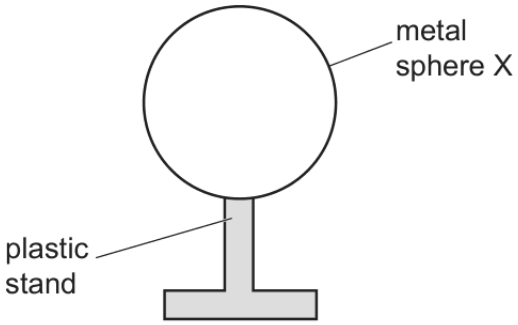


Fig. 8.1

Explain why metals are electrical conductors and why plastics are electrical insulators.

.....

.....

.....

..... [2]

(c) The negatively charged plastic rod is held above sphere X.

Fig. 8.2 shows that it does not touch the sphere.

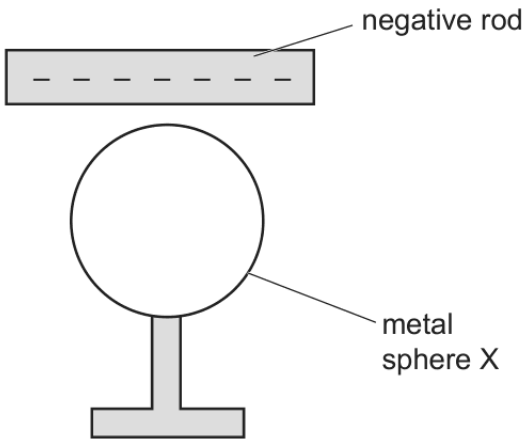


Fig. 8.2

(i) By drawing on Fig. 8.2, show the distribution of charge on X. [2]

(ii) An earth wire is now connected to the metal sphere as shown in Fig. 8.3.

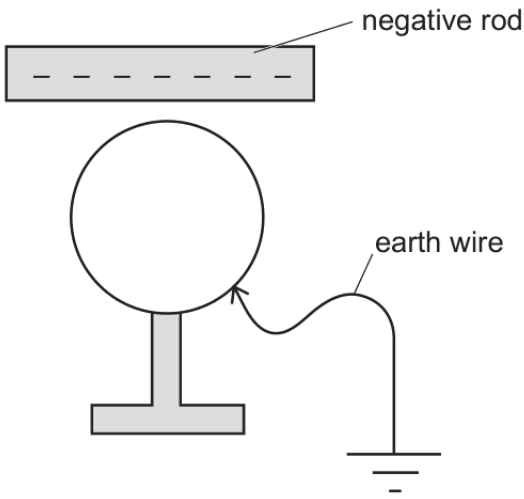


Fig. 8.3

State what happens to the charges on the sphere as the earth wire is connected.

.....

..... [1]

(iii) The earth wire in **(ii)** stays connected to the metal sphere.

Describe the distribution of charge on the sphere.

.....

.....

..... [2]

(iv) The earth wire is disconnected from the sphere and then the negatively charged rod is moved away from the sphere.

Explain what happens to the charges on the sphere as the charged rod is moved away.

.....

.....

..... [2]

(d) A second metal sphere Y is negatively charged.

There is an electric field surrounding Y.

(i) State what is meant by 'electric field'.

.....

..... [1]

(ii) Fig. 8.4 shows the negatively charged sphere Y. The negative charges are not shown.

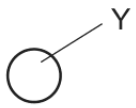


Fig. 8.4

On Fig. 8.4, draw the pattern and mark the direction of the electric field lines surrounding sphere Y. [3]

[Total: 15]

MARK SCHEME · 5054_w22_ms_22.pdf · source page 9 of 11 · Question 8

Question	Answer	Marks
8(a)	electrons move from the cloth / to the rod	**B1**
	cloth becomes positively charged	**B1**
8(b)	mention of free / mobile / delocalised / sea of electrons	**B1**
	(free) electrons present in metal **and** not present in plastic	**B1**
8(c)(i)	positive signs at top (half) of sphere **and** negative signs at the bottom	**B1**
	equal number of each **and** number ⩽ 7	**B1**
8(c)(ii)	negative charges / electrons flow to earth (and positive charges do not move	**B1**
8(c)(iii)	(only) top of sphere is / stays positive	**B1**
	bottom of sphere is uncharged / neutral	**B1**

MARK SCHEME · 5054_w22_ms_22.pdf · source page 10 of 11 · Question 8

Question	Answer	Marks
8(c)(iv)	charges repel each other **or** as earth wire is disconnected, charge distribution does not change	**B1**
	(positive) charge is spread (uniformly) over (the surface of) the sphere	**B1**
8(d)(i)	(region of space) where an (electrically) charged particle / charge experiences an (extra) force	**B1**
8(d)(ii)	at least **five** straight lines outside Y **and** none curved **and** perpendicular to the surface	**B1**
	at least one arrowhead pointing inwards **and** none wrong	**B1**
	field lines uniformly spaced at surface of sphere **and** no crossing / touching anywhere	**B1**

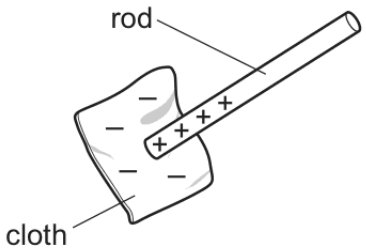
QUESTION · 5054_s23_qp_11.pdf · source page 13 of 20

27 In an electrostatics experiment, a plastic rod is rubbed with a cloth.

The **cloth** becomes negatively charged.

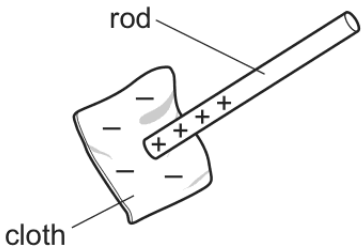
Which diagram shows the charge on the rod and describes the movement of charge?

A



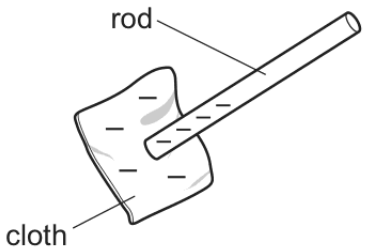
electrons move from the rod on to the cloth

B



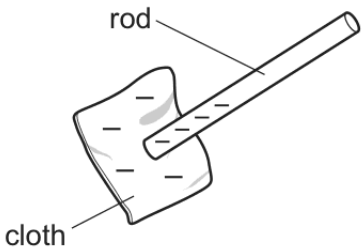
protons move from the cloth on to the rod

C



electrons move from the cloth on to the rod

D



protons move from the rod on to the cloth

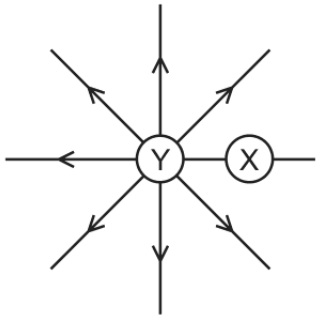
MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 27

27	A	**1**

QUESTION · 5054_s23_qp_11.pdf · source page 13 of 20

28 Object X is stationary and positively charged. It experiences a force due to the field produced by object Y.

The arrows show the direction of the field produced by Y.



Which statement about the direction of the force on X is correct?

A It is towards the right because it is in an electric field.

B It is towards the left because it is in an electric field.

C It is towards the right because it is in a magnetic field.

D It is towards the left because it is in a magnetic field.

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 2 of 3 · row for Question 28

28	A	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 10 of 16

26 A shoe becomes positively charged by friction when it rubs against a carpet.

What happens as the shoe becomes charged?

A Negative electrons are transferred to the carpet.

B Negative electrons are transferred to the shoe.

C Positive electrons are transferred to the carpet.

D Positive electrons are transferred to the shoe.

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 26

26	A	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 9 of 16

23 An aircraft flying through the air becomes electrically charged.

What causes this to happen?

A background radiation in the aircraft

B friction of the aircraft with the air

C heating of the aircraft surface

D radio waves emitted by the aircraft

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 23

23	B	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 9 of 16

24 Two metal spheres are brought close to each other. The two spheres repel each other.

Which statement explains this?

A Both spheres are charged and both are positive.

B Both spheres are charged but one is positive and the other is negative.

C The spheres have the same number of electrons.

D The spheres have the same number of protons.

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 24

24	A	**1**

QUESTION · 5054_w23_qp_12.pdf · source page 13 of 20

31 A student rubs different rods held in his hand.

Which rod gains a large electrostatic charge?

A a polythene rod rubbed with a steel magnet

B a polythene rod rubbed with a woollen duster

C an iron rod rubbed with a steel magnet

D an iron rod rubbed with a woollen duster

MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 3 of 3 · row for Question 31

31	B	**1**

QUESTION · 5054_s24_qp_11.pdf · source page 12 of 16

30 Plastic and wool are insulating materials.

Samples of plastic and wool are rubbed together. This causes a transfer of electrons.

Which row shows the direction of electron transfer and the final charge on both materials?

	plastic	wool	final charge on plastic	final charge on wool
A	gains electrons	loses electrons	neutral	positive
B	gains electrons	loses electrons	negative	positive
C	loses electrons	gains electrons	negative	positive
D	loses electrons	gains electrons	positive	neutral

MARK SCHEME (official answer key) · 5054_s24_ms_11.pdf · source page 3 of 3 · row for Question 30

30	B	**1**

5 An initially uncharged rubber balloon is rubbed with a woollen cloth as shown in Fig. 5.1.

Rubbing the balloon causes the balloon to have a negative charge.

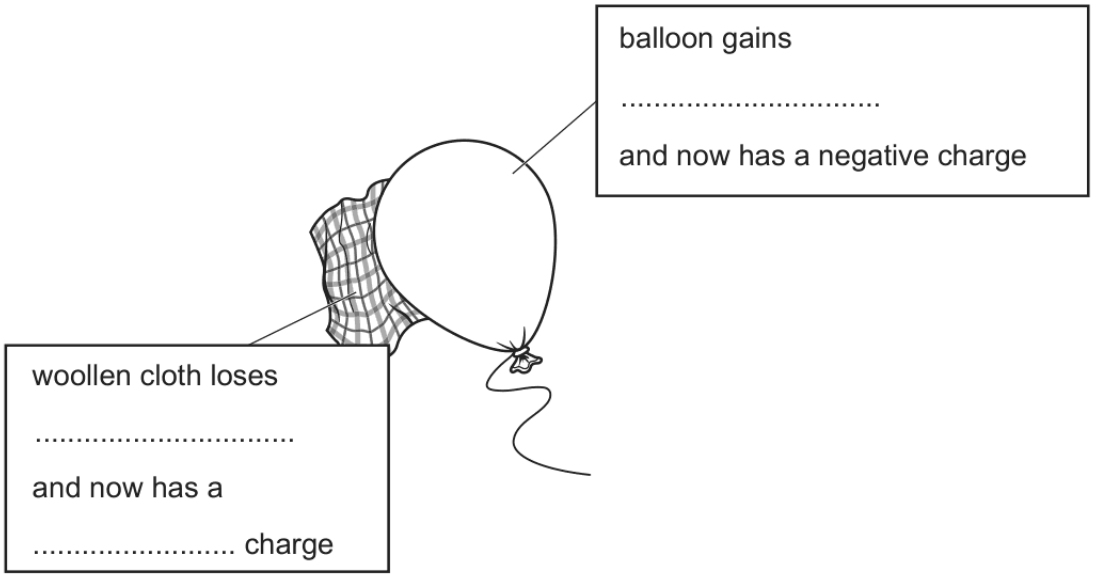


Fig. 5.1

(a) (i) On Fig. 5.1, complete the labels on the diagram. [2]

(ii) Explain why the balloon stays negatively charged for a long time.

.....

.....

.....

..... [2]

(b) Rubbing the balloon causes the temperature of the air inside it to rise.

Explain, in terms of the particles of air, why the volume of the balloon increases when the temperature of the air rises.

.....

.....

.....

.....

.....

..... [3]

[Total: 7]

MARK SCHEME · 5054_s24_ms_22.pdf · source page 7 of 11 · Question 5

Question	Answer	Marks
5(a)(i)	electrons on both top lines	**B1**
	positive (charge on cloth)	**B1**
5(a)(ii)	balloon / rubber / air is an insulator	**C1**
	electrons / charge unable to move through rubber / balloon / air **or** little ionisation in air (caused by background radiation)	**A1**

MARK SCHEME · 5054_s24_ms_22.pdf · source page 8 of 11 · Question 5

Question	Answer	Marks
5(b)	particles move faster	**B1**
	particles hit sides	**C1**
	particles hit sides more often / more frequently **or** with greater force	**A1**

QUESTION · 5054_w24_qp_11.pdf · source page 12 of 16

30 A plastic strip is rubbed with a cloth and the plastic becomes negatively charged.

Which statement explains this?

A Electrons are transferred from earth to the plastic.

B Electrons are transferred from the cloth to the plastic.

C Electrons move from one side of the plastic to the other side of the plastic.

D Protons are transferred from the plastic to the cloth.

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 3 of 3 · row for Question 30

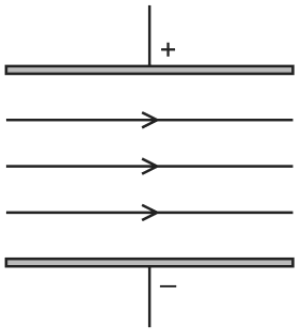
30	B	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 13 of 16

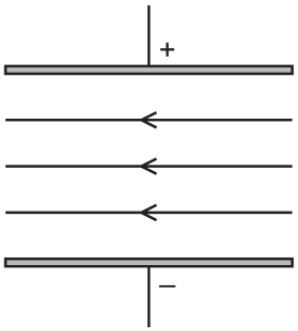
31 Two parallel, conducting plates are charged. The charge on the upper plate is positive. The charge on the lower plate is equal in size but negative.

Which diagram shows the pattern and the direction of the electric field between the two plates?

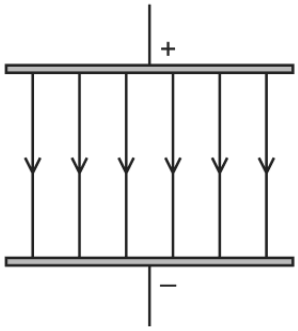
A



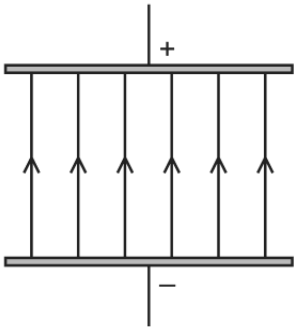
B



C



D



MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 3 of 3 · row for Question 31

31	C	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 10 of 16

27 A plastic rod becomes charged when it is rubbed with a cloth.

Which statement gives the reason for the charge on the rod?

A The plastic rod loses protons and becomes negatively charged.

B The plastic rod loses electrons and becomes negatively charged.

C The plastic rod gains protons and becomes positively charged.

D The plastic rod loses electrons and becomes positively charged.

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 27

27	D	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 12 of 16

27 Which material is an electrical conductor at room temperature?

A aluminium

B glass

C plastic

D rubber

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 27

27	A	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 13 of 16

28 A student investigates electrostatic charge using a balloon and a cloth. The balloon and cloth are initially uncharged.

The student rubs the balloon with the cloth. The balloon gains a negative charge of magnitude Q.

The cloth becomes charged and a force acts between the balloon and the cloth.

Which row correctly describes the magnitude and sign of the charge on the cloth and the type of force between the balloon and the cloth?

	magnitude of charge on cloth	sign of charge on cloth	type of force
A	equal to Q	positive	attraction
B	equal to Q	negative	repulsion
C	greater than Q	positive	repulsion
D	greater than Q	negative	attraction

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 2 of 3 · row for Question 28

28	A	**1**

7 An insulating rod is made of plastic and is initially uncharged. The rod is rubbed with a cloth and becomes negatively charged.

(a) Describe, in terms of particles, what happens as the rod becomes negatively charged.

.....

.....

..... [2]

(b) A light ball has a thin conducting surface. The ball is suspended from the ceiling by an insulating thread. The ball is not charged.

The negatively charged rod is brought close to the ball.

Fig. 7.1 shows that the ball experiences a force attracting it towards the negatively charged rod.

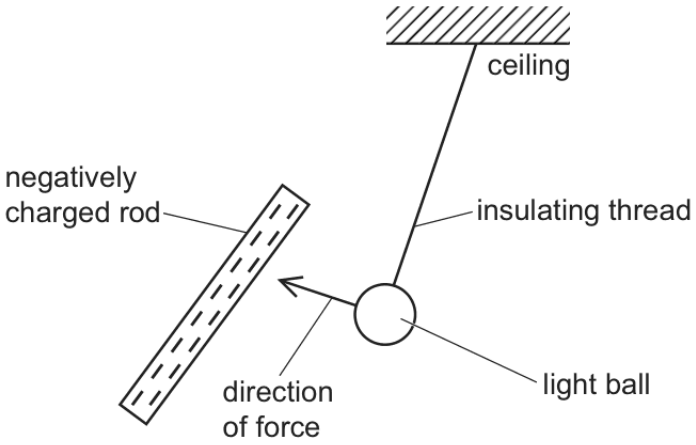


Fig. 7.1

(i) Explain, in terms of particles, how an electrical conductor differs from an electrical insulator.

.....

.....

..... [1]

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(ii) The conducting surface of the ball contains both negatively charged particles and positively charged particles.

Explain what happens to the negatively charged particles and to the positively charged particles as the rod is brought close to the ball.

.....

.....

.....

..... [3]

(iii) Explain why the resultant force on the ball is towards the rod.

.....

.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
7(a)	electrons mentioned **or** movement of negative charge mentioned	**C1**
	electrons move (from cloth) to rod	**A1**
7(b)(i)	(in conductors) electrons are free to move	**B1**
7(b)(ii)	like charges repel	**B1**
	negative charges / electrons move away from the rod / to the right	**B1**
	positive charges / protons do not move **or** left-hand side (of ball) becomes positively charged	**B1**
7(b)(iii)	unlike / opposite charges attract	**B1**
	attractive force stronger (than repulsive force on negative charges) / positive charges closer to the rod.	**B1**

QUESTION · 5054_s26_qp_11.pdf · source page 12 of 16

30 An uncharged plastic rod is rubbed with a cloth.

The plastic rod gains a negative charge.

Which statement is correct?

A Electrons transfer from the cloth to the rod.

B Electrons transfer from the rod to the cloth.

C Protons transfer from the cloth to the rod.

D Protons transfer from the rod to the cloth.

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 3 of 3 · row for Question 30

30	A	**1**

QUESTION · 5054_s26_qp_12.pdf · source page 13 of 16

31 Why are metals good electrical conductors?

A Metals have electrons that are free to move.

B Metals have isotopes that are radioactive.

C Metals have nuclei that have a positive charge.

D Metals have protons that are fixed in their nuclei.

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MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 3 of 3 · row for Question 31

31	A	**1**

Chapter 4 Electricity and magnetism

4.2 Electrical quantities

4.2.2 Electrical current

75 entries: 35 Paper 1 questions, 40 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

For Examiner's Use

7 Fig. 7.1 shows an electrical circuit.

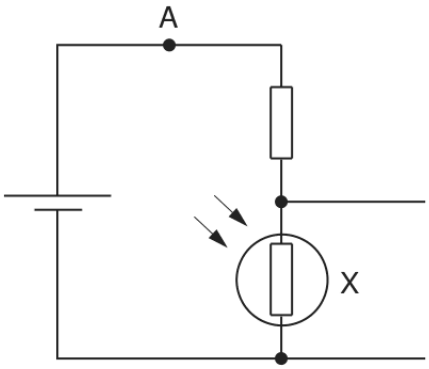


Fig. 7.1

(a) On Fig. 7.1, draw an arrow at A to show the direction of flow of the electrons in the wire. [1]

(b) What is the name of component X?

..... [1]

(c) State and explain how the potential difference across X varies as the light shining on it becomes brighter.

.....

.....

.....

.....

..... [2]

7	**(a)** arrow anticlockwise anywhere near top line of circuit	**B1**	
	(b) LDR **or** light dependent resistor	**B1**	
	(c) less resistance of X	**B1**	
	same change in voltage as resistance (voltage decreases alone B1)	**B1**	**4**

For Examiner's Use

6 An electrostatic generator is used to produce sparks, as shown in Fig. 6.1.

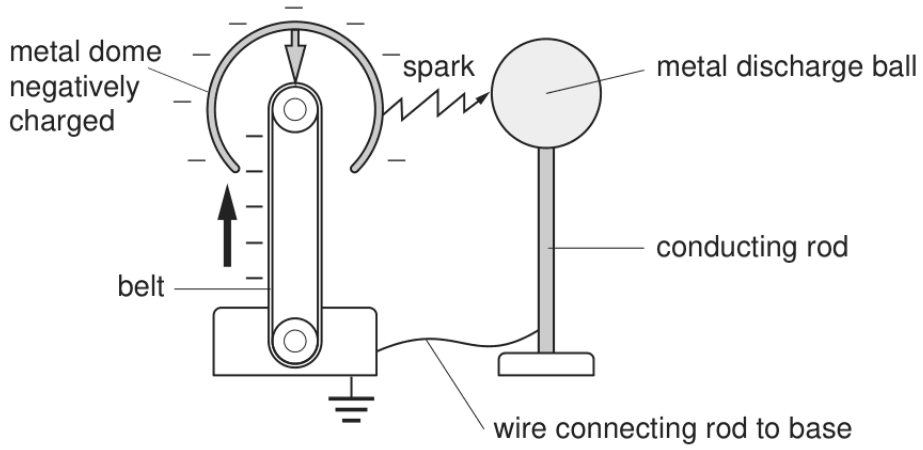


Fig. 6.1

The belt carries negative charge to the dome, making the dome negatively charged.

(a) Before a spark is produced, the discharge ball becomes positively charged.

(i) Describe and explain the movement of electrons in the discharge ball and in the rod as the ball becomes charged.

.....

.....

.....

..... [2]

(ii) On Fig. 6.1, mark with an X where there is the most positive charge on the discharge ball. [1]

(b) A spark jumps between the dome and the discharge ball when there is enough negative charge on the dome.

A charge of 0.00016 C flows in the spark in a time of 0.012 s.

Calculate the average current in the spark. State the equation that you use.

current = ... [3]

6	**(a)**	**(i)**	electrons move down rod **o**r away from ball	**B1**	
			like charges repel **or** electrons repelled by (charge on) dome	**B1**	
		(ii)	X on left side of ball	**B1**	
	(b)		Q = It formula seen in any algebraic form	**B1**	
			0.00016/0.012	**C1**	
			0.0133 A	**A1**	**6**

For Examiner's Use

5 Two conducting spheres are rubbed and become charged, as shown in Fig. 5.1.

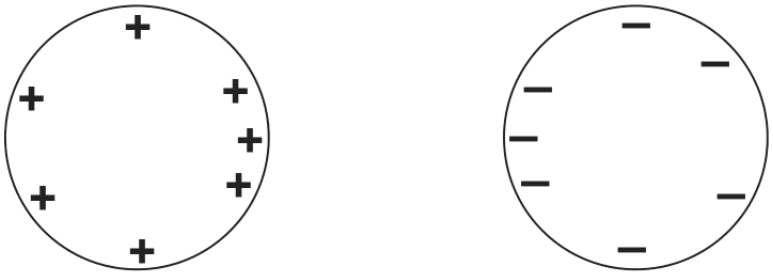


Fig. 5.1

(a) The spheres are uncharged before they are rubbed.

Explain, in terms of the movement of electrons, how they become charged.

.....

.....

.....

..... [1]

(b) On Fig. 5.1, draw the electric field pattern between the spheres. Show the direction of the field lines. [2]

(c) A metal wire is used to connect the two spheres together. In a time of 2.0×10^{-3} s, a charge of 1.8×10^{-13} C passes through the wire.

Calculate the current in the wire.

current = ... [2]

5	**(a)**	(electrons) move onto negative/right sphere **and** off positive/left sphere	B1	
	(b)	3 or more (approx. correct none wrong) lines from one sphere to the other	C1	
		arrow on at least 3 lines from + to – sphere	A1	
	(c)	Q = It in any format algebraic or numerical	C1	
		9×10^{-11} A	A1	**[5]**

For Examiner's Use

8 Fig. 8.1 shows the part of a cathode-ray oscilloscope in which an electron beam is produced.

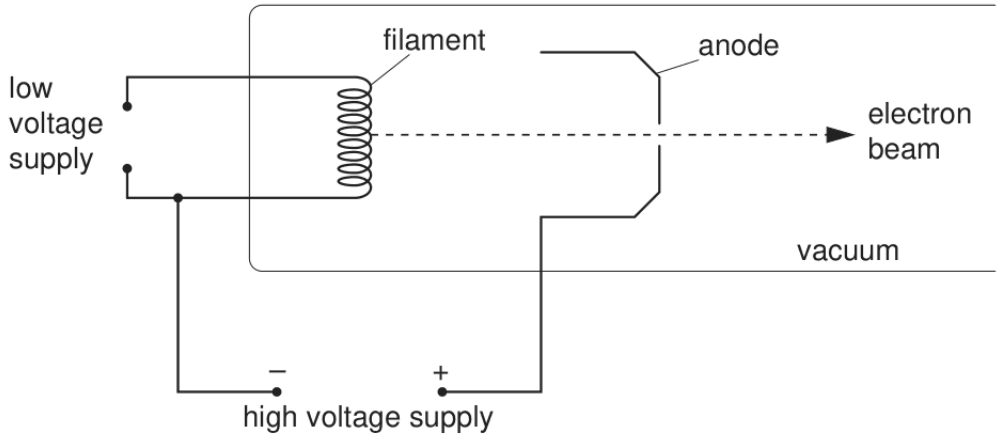


Fig. 8.1

(a) State why the filament emits electrons.

..... [1]

(b) State

(i) how the electrons are made to move at high speed,

.....

.....

..... [1]

(ii) why a vacuum is needed inside the oscilloscope.

.....

.....

..... [1]

(c) In each second, 8.0×10^{14} electrons pass through the hole in the anode.
The charge on an electron is 1.6×10^{-19} C. Calculate the current in the electron beam.

current = ... [2]

8 **(a)** thermionic emission **or** hot (filament/metal) B1

(b) **(i)** attracted by anode/+ve **or** repelled by filament/–ve B1

(ii) no obstruction/interference **or** electrons reach screen/travel through CRO **or** otherwise electrons collide (with atoms)/lose energy/deflected B1

(c) $8.0 \times 10^{14} \times 1.6 \times 10^{-19}$ C1
1.3×10^{-4} **or** 1.28×10^{-4} A A1

[Total: 5]

QUESTION · 5054_w08_qp_1.pdf · source page 13 of 20

28 The current in an electric heater is 10 A. It is switched on for five minutes.

How much charge flows through the heater?

A 0.5 C **B** 2 C **C** 50 C **D** 3000 C

MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 28

28	**D**

10 Fig. 10.1 shows a wire passing through a hole in a horizontal, plastic board.

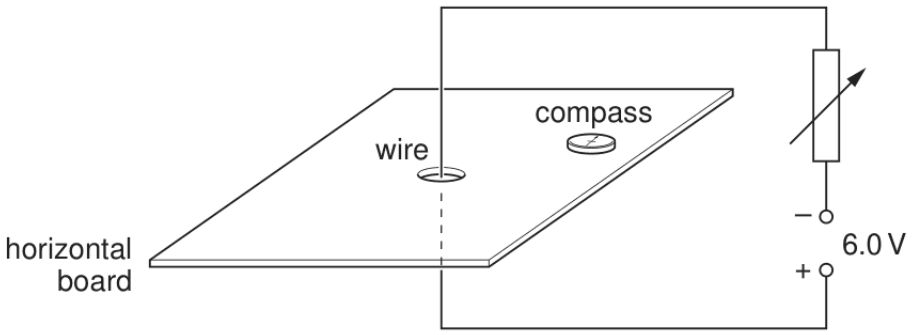


Fig. 10.1

The wire carries a current vertically upwards. A student moves a small compass around the board and plots the magnetic field lines due to the current.

(a) **(i)** Draw a diagram of the board as seen from above and mark on it the magnetic field lines due to the current. [3]

(ii) The current is increased. Describe how the magnetic field changes. [1]

(b) A 6.0 V power supply produces a current of 8.0 A in the wire. Calculate

(i) the total resistance of the circuit, [2]

(ii) the charge that flows through the wire in 2.0 minutes. [2]

(c) The north pole of a bar magnet is held on the left of the wire and the south pole of another bar magnet is held on the right, as shown in Fig. 10.2.

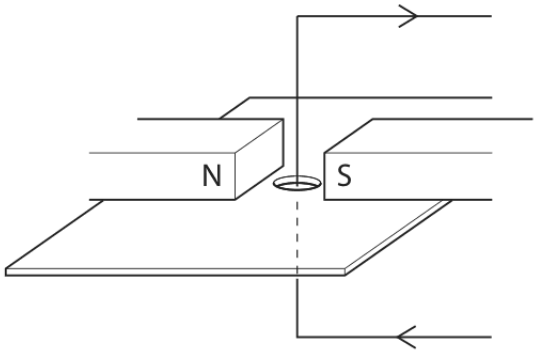


Fig. 10.2

(i) State the direction of the magnetic field at the wire due to these two poles. [1]

(ii) The wire is flexible. Describe the effect of this magnetic field on the current-carrying wire. [2]

(iii) The current in the wire is now reversed. State the effect of this on the wire. [1]

(iv) Describe how this effect is put to use in a d.c. motor. [3]

10	**(a)**	**(i)**	at least 2 concentric, complete circles	B1	
			increasing gap	B1	
			at least 1 anticlockwise arrow **and** none incorrect	B1	
		(ii)	stronger **or** more lines **or** lines closer together **or** extends further	B1	[4]
	(b)	**(i)**	(R =) V/I **or** 6.0/8.0	C1	
			0.75 Ω	A1	
		(ii)	(Q =) It **or** 8.0 × 120 **or** 8.0 × 2	C1	
			960 C (16 C scores 1/2)	A1	[4]
	(c)	**(i)**	L→R **or** N→S	B1	
		(ii)	force (on wire) **or** wire bends/moves	M1	
			into page/perpendicular to field/away (from us)/LH rule quoted	A1	
		(iii)	force reverses **or** out of page **or** bends the other way e.c.f.	B1	[4]
		(iv)	accept first two marks on unlabelled diagram		
			(wire becomes) coil / armature /solenoid	B1	
			force/movement opposite on sides of coil **or** moment	B1	
			current reverses during rotation/due to commutator or split ring	B1	[3]

[Total: 15]

For Examiner's Use

3 A hockey player trains on a nylon-fibre surface. As he runs around, his shoes rub against the surface and he becomes positively charged.

(a) **(i)** Explain, in terms of the particles involved, how he becomes positively charged.

.....

.....

.....[2]

(ii) State what happens to the nylon-fibre surface as he becomes positively charged.

.....

.....[1]

(b) At the end of the training session, the hockey player touches a metal gate and feels an electric shock.

(i) State how this shock is produced.

.....

.....[1]

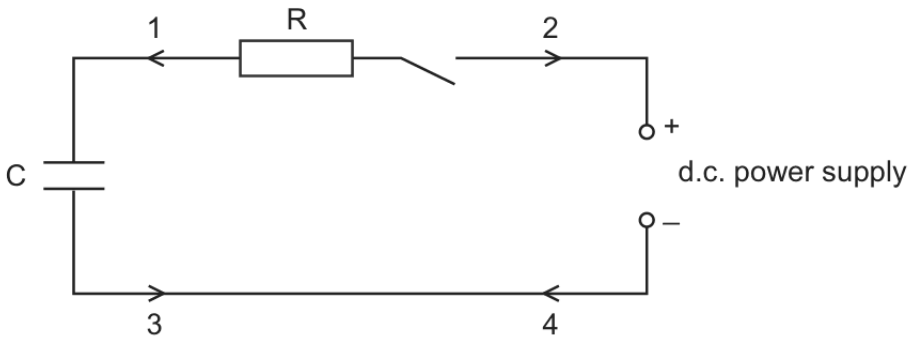
(ii) The shock lasts for 0.15 ms. During this time, the current has an average value of 1.6 mA. Calculate the size of the charge on the hockey player just before he touches the gate.

charge = ...[3]

3	**(a)**	**(i)**	(he) loses –ve charge		C1	
			electrons lost (to surface)	(positive electrons 0/2)	A1	
		(ii)	(becomes) negative/gains electrons		B1	
	(b)	**(i)**	(he) discharges/(re)gains electrons/–ve charge (**not** current flow)		B1	
		(ii)	(Q =)It **or** 1.6 × 0.15 **or** 0.0016 × 0.00015		C1	
			2.4×10^{n}		C1	
			2.4×10^{-7} C		A1	[7]

QUESTION · 5054_s10_qp_11.pdf · source page 16 of 20

36 A capacitor C charges when it is connected to a d.c. power supply.



Which arrows show the direction of the conventional current when the capacitor is charging?

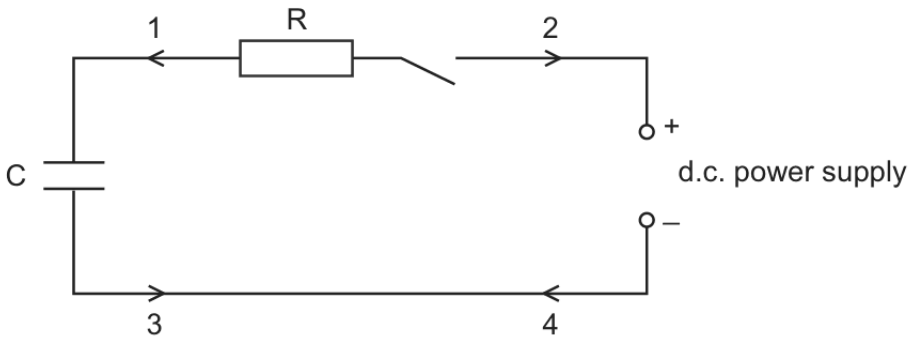
A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 36

36	**A**

QUESTION · 5054_s10_qp_12.pdf · source page 15 of 20

34 A capacitor C charges when it is connected to a d.c. power supply.



Which arrows show the direction of the conventional current when the capacitor is charging?

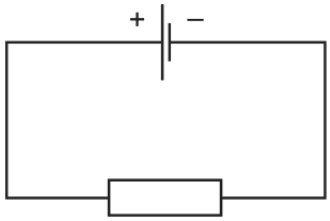
A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 34

34	**A**

QUESTION · 5054_w10_qp_11.pdf · source page 13 of 20

29 The diagram shows a simple electric circuit.



Which row describes the charge on an electron and the direction of electron flow through the resistor?

	charge on an electron	direction of electron flow
A	negative	– to +
B	negative	+ to –
C	positive	– to +
D	positive	+ to –

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 29

29 **A**

For Examiner's Use

7 A householder installs a burglar alarm. The alarm does not operate for the first 200 s after it is switched on. This time delay lets the householder leave the house and lock up without setting off the alarm.

Fig. 7.1 is a circuit inside the burglar alarm.

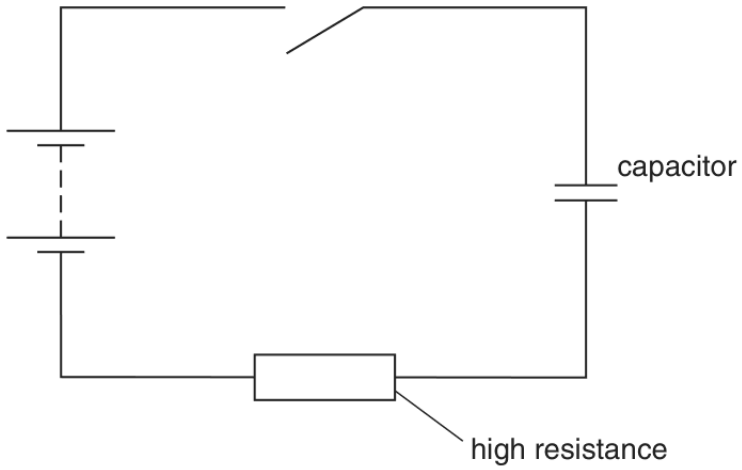


Fig. 7.1

Initially, the switch is open.

(a) (i) The switch is now closed. Explain what happens to the potential difference across the capacitor as time passes.

.....

.....

..... [2]

(ii) Suggest how this circuit may be used for the time delay of the burglar alarm.

.....

..... [1]

(b) The capacitor in the circuit is uncharged when the burglar alarm is first switched on. A current then starts to charge the capacitor and after 200 s, the charge stored is 5.4×10^{-7} C.

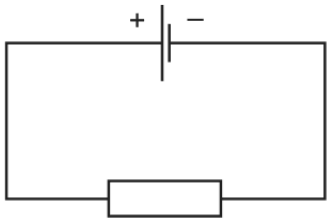
Calculate the average value of the current during this time.

current = ... [2]

7	**(a)**	**(i)**	p.d. rises	B1	
			the capacitor charges/at a decreasing rate/to a maximum value	B1	[2]
		(ii)	it takes a certain time/200 s to reach certain charge/p.d. **or** certain charge/p.d. activates alarm	B1	[1]
	(b)		(I =) Q/t **or** $5.4 \times 10^{-7}/200$	C1	
			$2.7(0) \times 10^{-9}$ A	A1	[2]

QUESTION · 5054_w10_qp_12.pdf · source page 11 of 16

28 The diagram shows a simple electric circuit.



Which row describes the charge on an electron and the direction of electron flow through the resistor?

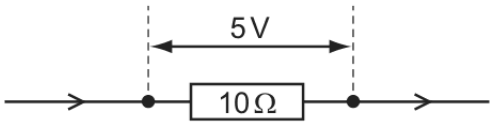
	charge on an electron	direction of electron flow
A	negative	– to +
B	negative	+ to –
C	positive	– to +
D	positive	+ to –

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 28

28 **A**

QUESTION · 5054_s11_qp_11.pdf · source page 11 of 16

28 The potential difference (p.d.) across a 10 Ω resistor is 5 V.



How much charge passes through the 10 Ω resistor in 30 seconds?

A 2 C **B** 15 C **C** 60 C **D** 1500 C

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 28

28	**B**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 shows a rotating magnet in an alternating current generator that is used to power a lamp.

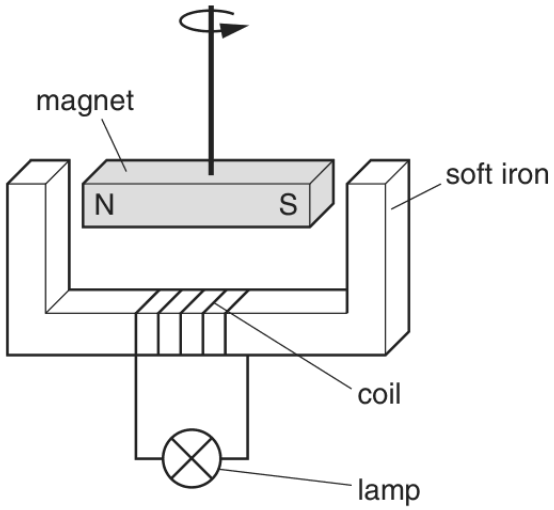


Fig. 9.1

(a) (i) State how an *alternating* current differs from a *direct* current.

.....

.....

..... [1]

(ii) Explain, in detail, how **alternating** current is produced by the apparatus shown in Fig. 9.1.

.....

.....

.....

.....

.....

..... [4]

(iii) State two ways in which the current in the lamp may be increased.

1. .....

2. .....

[2]

(b) The generators at a power station produce a voltage of 25 000 V. This voltage is stepped up to 400 000 V by a transformer for long-distance transmission on overhead power lines. The voltage is later stepped down to 240 V.

(i) State and explain why the voltage is stepped up for long-distance transmission.

.....

.....

..... [2]

(ii) Calculate the ratio of the number of turns in the primary coil of the step-up transformer to the number of turns in its secondary coil.

ratio = ... [1]

(iii) State one advantage and one disadvantage of using thicker wire in the overhead power lines.

advantage: .....

.....

disadvantage: .....

.....
[2]

(iv) An electric drill of power 1000 W is used in a country where the mains voltage is 240 V.
State and explain the most appropriate fuse to use with this drill.
You should select a fuse from the following values: 1 A, 3 A, 4 A, 13 A.

.....

.....

.....

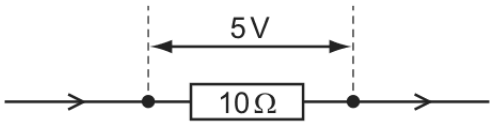
..... [3]

Section B

9	**(a)**	**(i)**	d.c. current flows in one direction **or** a.c. current flows in one direction then the other	B1	
		(ii)	mention of magnetic field/flux	B1	
			(magnetic) field lines // flux cuts coil // flux changes in coil	B1	
			induction of voltage/current	B1	
			something **relevant** reverses (e.g. field/flux cuts in one direction then the other // N pole approaches then leaves // N pole approaches and S pole approaches) **and** link to a.c.	B1	
		(iii)	**two** of: thicker wires; more turns of coil; stronger magnet; faster rotation; lower resistance (of lamp)	B2	
	(b)	**(i)**	to reduce heat/energy/power loss (on the power lines)	B1	
			(higher voltage means) lower current	B1	
		(ii)	25:400 // 1:16 // 0.0625	B1	
		(iii)	reduces resistance // less power loss // costs less to run // more current // more power	B1	
			increases weight // more support needed // more wind resistance // more ice forms // costs more to install	B1	
		(iv)	13 A	B1	
			(I =) P/V // 4.2 A // 4.17 etc A	B1	
			must choose higher value to avoid fuse blowing // other fuses melt	B1	[15]

QUESTION · 5054_s11_qp_12.pdf · source page 11 of 16

27 The potential difference (p.d.) across a 10 Ω resistor is 5 V.



How much charge passes through the 10 Ω resistor in 30 seconds?

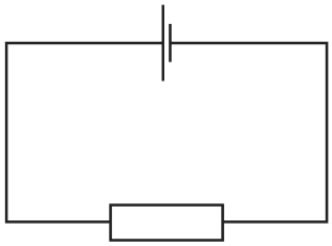
A 2 C **B** 15 C **C** 60 C **D** 1500 C

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 27

27	**B**

QUESTION · 5054_s12_qp_12.pdf · source page 15 of 16

37 The electrical circuit shown consists of a cell connected to a resistor.



What are the directions of the electron flow and of the conventional current in the **resistor**?

	electron flow	conventional current
A	→	→
B	→	←
C	←	←
D	←	→

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 37

37 **D**

7 Fig. 7.1 shows two charged metal spheres.

One sphere has a positive charge and the other sphere has a negative charge.

Fig. 7.1

(a) On Fig. 7.1, draw the electric field between the two spheres. [2]

(b) The negative charge on the sphere is removed and a wire is used to connect the positive sphere to earth. The charge on the positive sphere decreases from 4.8×10^{-9} C to zero in a time of 2.0×10^{-6} s.

Calculate the average current in the wire.

current = ...[2]

7 **(a)** three lines from one sphere to the other **and** some lines should spread out as they leave one sphere **and** come together nearing the other B1

correct direction on at least one line and none wrong B1

(b) (I =) Q/t in any form numerical or algebraic C1
2.4×10^{-3} A A1 [4]

3 Fig. 3.1 shows an electric motor being used on a building site to lift bricks.

For Examiner's Use

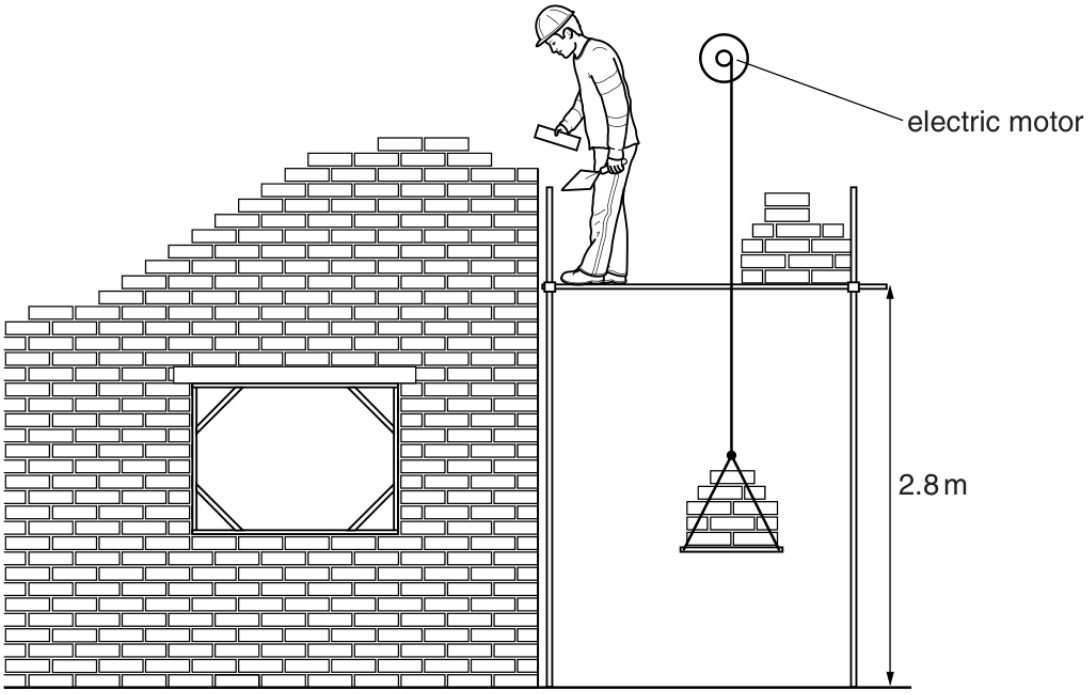


Fig. 3.1

The gravitational field strength is 10 N / kg. The motor lifts bricks of total mass 54 kg through a height of 2.8 m in 3.0 s.

(a) Calculate

(i) the work done in lifting the bricks,

work done = ... [2]

(ii) the useful output power of the motor.

power = ... [2]

(b) Suggest two reasons why the electrical power supplied to the motor is greater than its useful output power.

1. .....

2. .....
[2]

For Examiner's Use

(c) (i) Fig. 3.2 shows a battery and an electric motor. The battery is used to power the motor. Complete Fig. 3.2 to show the circuit diagram of a circuit that is used to determine the power supplied to the electric motor.

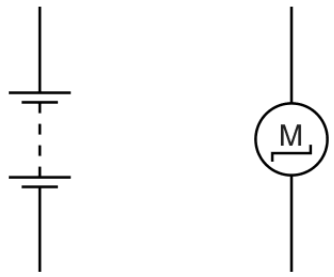


Fig. 3.2 [2]

(ii) State how the power supplied is calculated.

.....

..... [1]

3	**(a)**	**(i)**	(WD =)*mgh* **or** 54 × 10 × 2.8	C1	
			1500/1510/1512 J	A1	
		(ii)	(*P* =) WD/*t* or *E*/*t* **or** 1500/3 **or** 1510/3 **or** 1512/3	C1	
			500/503/504 W	A1	
	(b)		any **two** of: also lifting board/rope heat in motor/wires/cable friction with something named e.g. axle/spindle/air	B2	
	(c)	**(i)**	power supply, motor and ammeter in series (**ignore** series voltmeter and other components)	B1	
			voltmeter to measure voltage across motor	B1	
		(ii)	current (reading) × voltage (reading) or VI	B1	[9]

For Examiner's Use

7 **(a)** Fig. 7.1 shows a solenoid made from wire wound around a plastic cylinder.

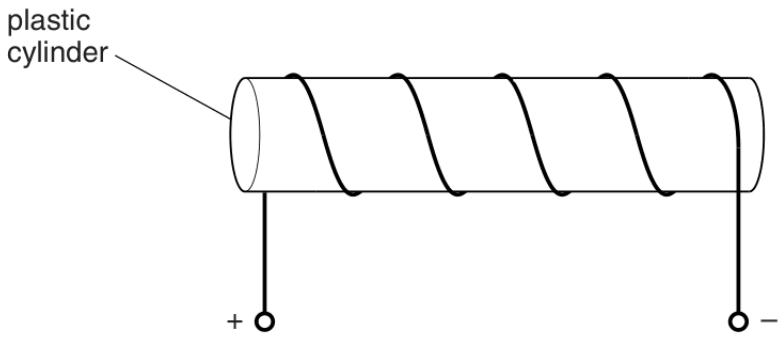


Fig. 7.1

A current in the solenoid produces a magnetic field.

On Fig. 7.1, draw the pattern of the magnetic field lines inside and outside the cylinder. [2]

(b) Fig. 7.2 shows a beam of beta-particles, in a vacuum, passing into a magnetic field.

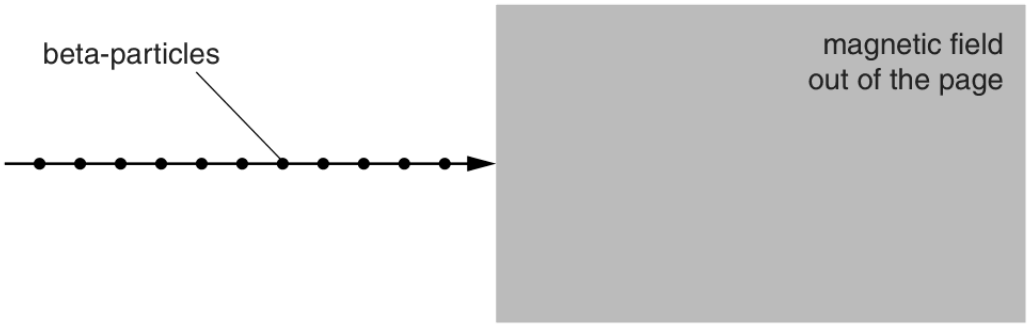


Fig. 7.2

The movement of the beta-particles from left to right is an electric current.

(i) On Fig. 7.2, draw an arrow to show the direction of the conventional current. [1]

(ii) A solenoid is used to produce the magnetic field that lies within the shaded region of Fig. 7.2. The direction of the field is out of the page.

1. On Fig. 7.2, draw the path followed by one of the beta-particles in the shaded region. [2]

2. The direction of the current in the solenoid is reversed. State what happens to the path of the beta-particle.

.....

..... [1]

7	**(a)**	(at least) **two** parallel horizontal lines within the cylinder	B1	
		(at least) **two** correctly shaped lines outside the cylinder	B1	
	(b)	**(i)** ← (right to left) **and** on diagram (somewhere)	B1	
		(ii) **1.** path continuously curving in same direction	M1	
		upwards (**ignore** lines outside the shaded area)	A1	
		2. (changes to) downwards (curve) **not** reverses/opposite direction	B1	[6]

For Examiner's Use

11 **(a)** Fig. 11.1 shows a thunder cloud with a flat, positively charged base. It passes over a tall tree growing in a region of flat, open land.

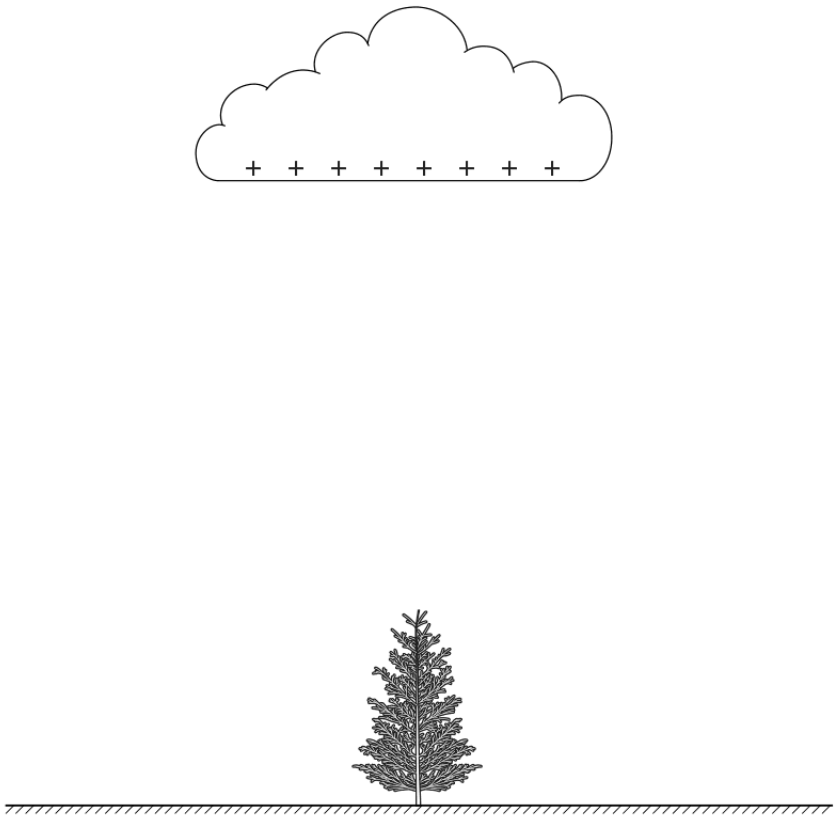


Fig. 11.1 (not to scale)

(i) On Fig. 11.1, mark the charge on the tree. [1]

(ii) Explain how the tree becomes charged.

.....

.....

..... [2]

(iii) A lightning strike occurs and, in 2.0×10^{-4} s, a charge of 560 C passes from the cloud to the tree. The size of the charge on an electron is 1.6×10^{-19} C.

1. Calculate the number of electrons that pass between the tree and the cloud.

number = ... [2]

2. Calculate the average current in the lightning strike.

For Examiner's Use

current = ... [2]

(b) Two flat metal plates are positioned horizontally, one above the other. Fig. 11.2 shows the positive terminal of a high-voltage supply unit connected to the bottom plate and the negative terminal connected to the top plate.

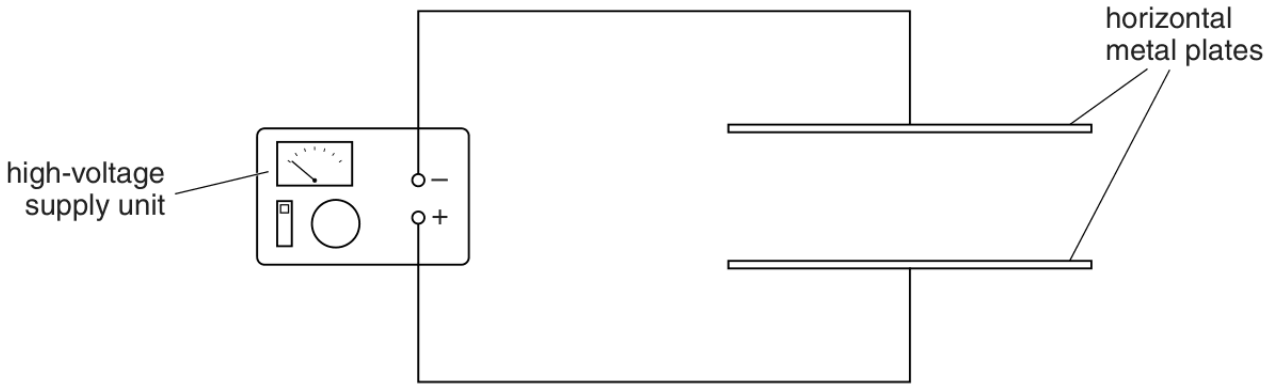


Fig. 11.2

The high-voltage supply is switched on.

(i) On Fig. 11.2, draw the shape and the direction of the electric field produced. [3]

(ii) Suggest why a small, charged oil droplet placed between the two metal plates accelerates upwards.

.....

.....

.....

..... [3]

(iii) The oil droplet touches the top metal plate. State and explain what happens to the charge on the oil droplet.

.....

.....

..... [2]

11	**(a)**	**(i)**	correct negative charges on tree.	B1	
		(ii)	electrons/-ve charges attracted by cloud/+ve charges	B1	
			electrons from ground or correct induction mentioned	B1	
		(iii)	**1.** $560/1.6 \times 10^{-19}$	C1	
			3.5×10^{21}	A1	
			2. $(I =)Q/t$ or $560/2 \times 10^{-4}$	C1	
			2.8×10^{6} A	A1	[7]
	(b)	**(i)**	at least 4 vertical lines between plates	B1	
			equally spaced or curved at edges	B1	
			arrows +ve to –ve/upwards	B1	
		(ii)	oil droplet positively charged	B1	
			attraction/force on (droplet) and in direction of field/upwards	B1	
			force greater than weight (of droplet) or resultant force	B1	
		(iii)	(droplet becomes) negative	C1	
			(droplet) gains electrons	A1	[8]

[Total: 15]

For Examiner's Use

6 The filament in a cathode-ray oscilloscope (c.r.o.) emits charged particles by thermionic emission.

(a) (i) State the name of the charged particles that are emitted.

.....[1]

(ii) Explain what causes the particles to be emitted.

.....

.....[1]

(iii) State why there must be a vacuum in the c.r.o. tube.

.....

.....[1]

(b) The particles emitted have a charge of size 1.6×10^{-19} C and carry a current of 5.6×10^{-3} A between the filament and the screen of the c.r.o.

Calculate the number of particles that strike the screen every second.

number = ...[2]

6	**(a)**	**(i)**	electrons **cao** (**not** positive electrons)	B1	
		(ii)	(from) heated (filament) **or** heat **or** boiled off (from filament) **or** knocked out by energetic/fast-moving atoms	B1	
		(iii)	to allow electrons to reach the screen **or** no collisions with (air) atoms/molecules/particles	B1	
	(b)		$(1/t =)\ I/Q$ **or** $1.6 \times 10^{-19}/5.6 \times 10^{-3}$ **or** $5.6 \times 10^{-3}/1.6 \times 10^{-19}$ **or** $2.86/2.9 \times 10^{-17}$	C1	
			3.5×10^{16}	A1	[5]

QUESTION · 5054_s13_qp_11.pdf · source page 12 of 20

29 A lightning flash carries 40 C of charge and lasts for 5.0 ms.

What is the average current in the flash?

A 0.20 A **B** 8.0 A **C** 200 A **D** 8000 A

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 29

29	**D**

QUESTION · 5054_s13_qp_12.pdf · source page 11 of 16

32 A lightning flash carries 40 C of charge and lasts for 5.0 ms.

What is the average current in the flash?

A 0.20 A **B** 8.0 A **C** 200 A **D** 8000 A

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 32

32	**D**

6 Fig. 6.1 shows the current-voltage graph for a filament lamp.

For Examiner's Use

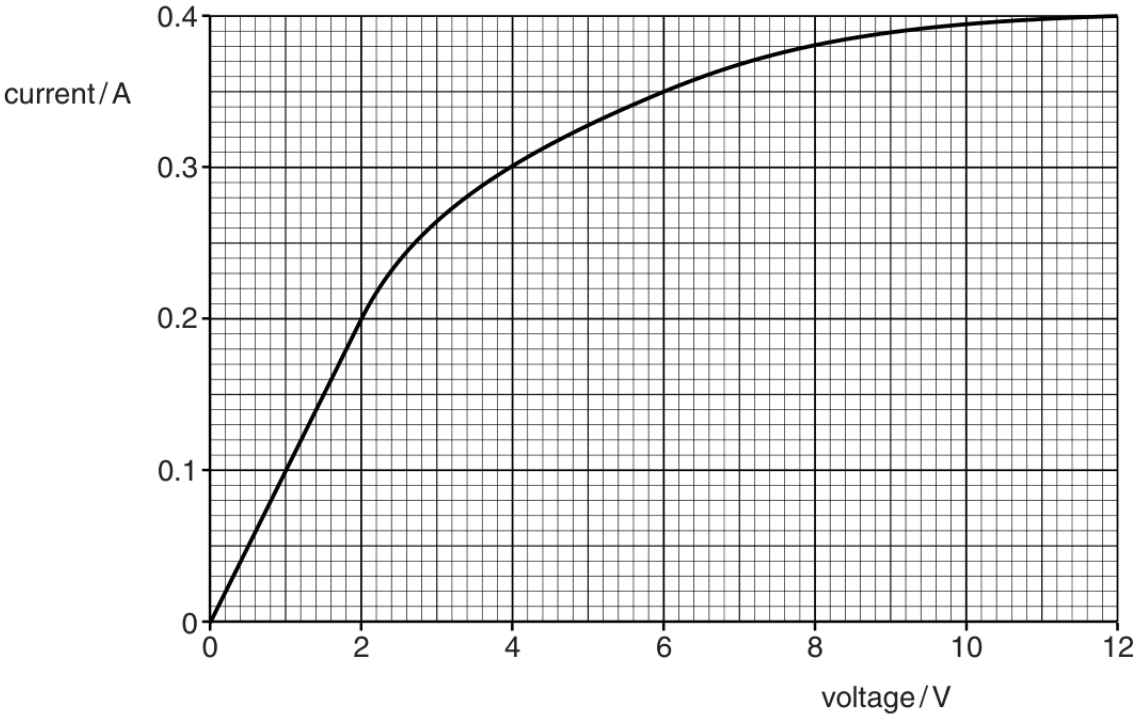


Fig. 6.1

(a) (i) State the range of voltages where the resistance increases.

..... [1]

(ii) State why the resistance of the lamp increases as the voltage increases.

.....

..... [1]

For Examiner's Use

(b) The filament lamp is connected in the circuit shown in Fig. 6.2.

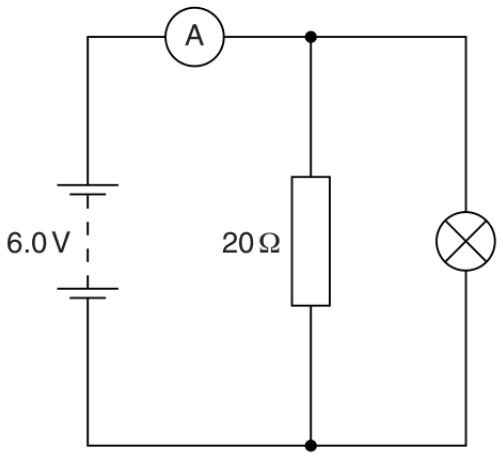


Fig. 6.2

There are currents in the lamp, the 20 Ω resistor and the ammeter.

(i) State what is meant by an *electric current*.

.....

..... [1]

(ii) Use Fig. 6.1 to determine the current in the filament lamp.

current = ... [1]

(iii) Determine the current in the ammeter.

current = ... [3]

6	**(a)**	**(i)**	2 to 2.1 (V) to any value between 11 and 12 (V) **or** above 2/2.1(V)	B1	
		(ii)	temperature increases / gets hotter	B1	
	(b)	**(i)**	(rate of) flow of charge/electrons	B1	
		(ii)	0.35 A cao	B1	
		(iii)	(I=) V/R algebraic **or** 6/20 **or** 6/0.35	C1	
			0.3(0) (A) **or** $1/R_T = 1/20 + 1/17.1$ **or** (R_T =) 9.2 (Ω) seen	C1	
			0.65 A	A1	[7]

For Examiner's Use

11 A student places a small electrical heater inside a cup of water, as shown in Fig. 11.1.

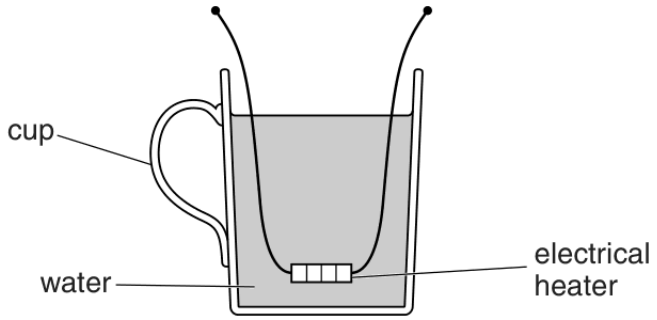


Fig. 11.1

The student determines the electrical power of the heater.

(a) In the space above the cup on Fig. 11.1, draw the electrical circuit that the student uses. Include an ammeter, a voltmeter and a power supply. [2]

(b) The voltage of the power supply is 12 V and the current is 4.2 A.

(i) Calculate the electrical power input to the heater.

power = ... [2]

(ii) Calculate the energy input to the heater in 8.0 minutes. Give your answer in kW h.

energy = ... kW h [3]

For Examiner's Use

(c) During heating, the student notices that some of the water evaporates from the cup.

(i) Describe, using ideas about molecules of water, what happens during evaporation.

.....

.....

..... [2]

(ii) The student finds that the amount of evaporation increases when the temperature of the water is higher.

State and explain one other change that increases the amount of evaporation.

.....

.....

..... [2]

(iii) State two differences between evaporation and boiling.

1. .....

.....

2. .....

.....
[2]

(d) The student turns off the power supply and the water cools.

Describe and explain how convection in the air causes the water to cool.

.....

.....

.....

..... [2]

MARK SCHEME · 5054_s13_ms_22.pdf · source page 6 of 7 · Question 11

11 (a) power supply with ammeter and heater in series B1
voltmeter in parallel with heater/ power supply B1 [2]

(b) (i) (*P*=)*VI* in any form C1
or 4.2 × 12

50(.4) W A1

(ii) (*E*=) *Pt* i.e. any power × any time e.g. 50(.4) × 8 C1

8/60 C1
or 0.13(3) seen
or division by 1000 seen anywhere

0.0067(2) (kWh) A1 [5]

MARK SCHEME · 5054_s13_ms_22.pdf · source page 7 of 7 · Question 11

(c) (i) molecules escape (from surface/leave water) / become gas or vapour / break bonds C1

fast(er) moving / high energy/ energetic molecules escape A1

(ii)

change **M1**	**explanation** **A1**
wind / draught / breeze	wind knocks molecules away
or larger surface area	more chance/possibility of escape/more space to escape **or** more molecules come to/near/at surface
or decrease humidity / drier air	fewer molecules return/from air
or decrease atmospheric pressure	fewer air molecules to hit during escape

(iii) evaporation occurs at surface **and** boiling inside liquid/bubbles
evaporation occurs at any temperature (accept room temperature) **and** boiling occurs at boiling point/100°C/ fixed / specific temperature
evaporation increased by draughts/higher temp/more area **and** boiling is not
OR increase in pressure stops boiling but only reduces evaporation
any two B2 [6]

(d) water heats air (by conduction) B1
or water loses heat/energy (to cup or air)
or air gains heat/energy (from water)

hot / heated air / particles rise B1 [2]
or cold air / particles sink
or hot air is less dense
or cold air is more dense

[Total: 15]

QUESTION · 5054_w13_qp_11.pdf · source page 13 of 20

28 Which of the following is equivalent to one coulomb?

A one ampere second

B one ampere per volt

C one volt ampere

D one volt per ampere

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 28

28	**A**

For Examiner's Use

11 An ammeter, a variable resistor and a 4.0Ω resistor are connected in series with a 6.0V battery. A cathode-ray oscilloscope (c.r.o.) is connected in parallel with the 4.0Ω resistor through a switch. Fig. 11.1 is the circuit diagram.

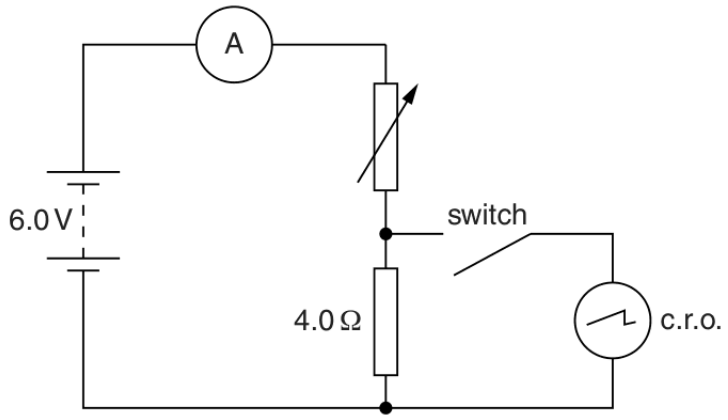


Fig. 11.1

The switch is open.

(a) The variable resistor is adjusted so that it has a resistance of 8.0Ω.

Calculate

(i) the current measured by the ammeter,

current = ...[2]

(ii) the potential difference (p.d.) across the 4.0Ω resistor.

p.d. = ...[1]

(b) The resistance of the variable resistor is adjusted until the p.d. across the 4.0Ω resistor is 5.0V.

State the effect that this adjustment has on

(i) the current measured by the ammeter,

.....[1]

(ii) the resistance of the variable resistor.

.....[1]

For Examiner's Use

(c) The trace on the c.r.o. is a horizontal line across the middle of the screen and the p.d. across the 4.0Ω resistor is still 5.0V.

The Y-gain control is set at 2.0V/cm and the switch is closed.

State the effect on the trace of closing the switch.

.....

.....

.....[2]

(d) Fig. 11.2 shows, in outline, the basic structure of a c.r.o. viewed from the side.

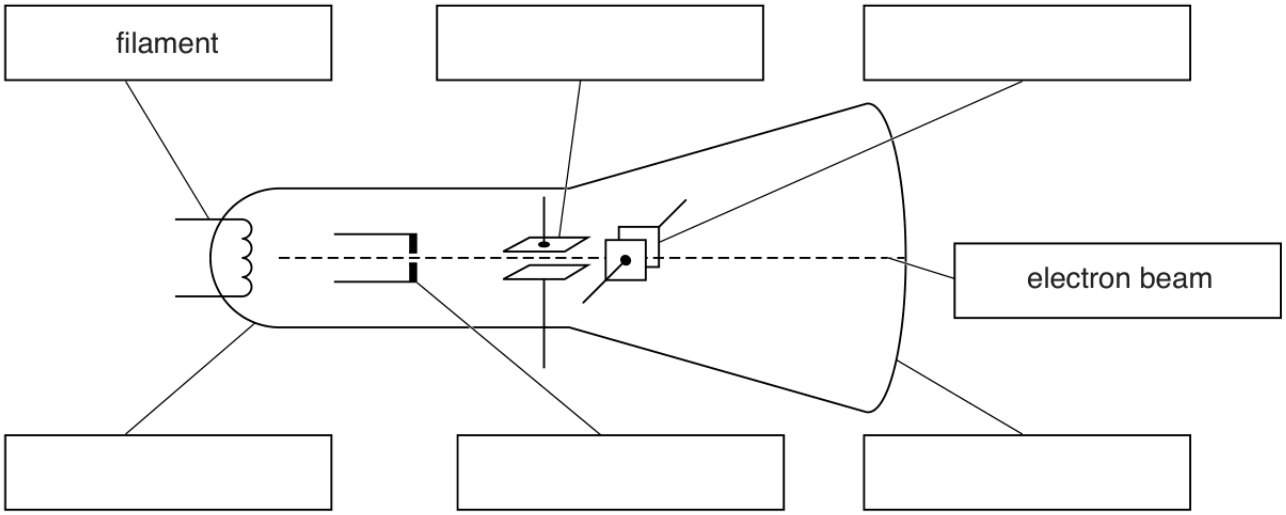


Fig. 11.2

(i) On Fig. 11.2 write, in the **five** empty boxes, the names of the parts of the c.r.o. indicated. [3]

(ii) Explain how the electron beam is produced.

.....

.....

.....[2]

(iii) Explain why the tube of the c.r.o. must be evacuated.

.....

.....[1]

Question 11 continues on page 16.

(iv) In the c.r.o. shown in Fig. 11.2, the electrons travel from left to right.

For Examiner's Use

1. Explain why the flow of electrons is an electric current.

.....

.....[1]

2. State the direction of the conventional current due to the electrons.

.....

.....[1]

11 (a) (i) (I =)V/R **or** 6.0/12.0 **or** 6.0/(4.0+8.0) **or** (in **(ii)**) (V =)IR **or** 0.50 × 4.0 — C1
0.50 A — A1

(ii) 2.0 V (scores C1 in **(a)(i)** if not already scored) — A1 [3]

(b) (i) increased **or** becomes 1.25 A — B1

(ii) decreases **or** becomes 0.8 Ω — B1 [2]

(c) moves up or down **or** 5.0/2.0 — C1
moves up or down by 2.5 cm — A1 [2]

(d) (i)

	Y-plates	X-plates
(glass) tube	anode	ZnS/screen

(5 correct 3 marks, 4 correct 2 marks, 3 correct 1 mark
X and Y plates reversed –1; **allow** focussing anode) — B3

(ii) filament heated/thermionic emission — B1
(thermionic) electrons attracted by anode **or** repelled by cathode — B1

(iii) to prevent/otherwise collisions with air molecules/to allow to reach the screen/to avoid deflection — B1

(iv) **1.** electrons are charged — B1

2. backwards **or** towards the back **or** opposite to electron motion **or** to the left **or** from the right — B1 [8]

[Total: 15]

For Examiner's Use

11 A metal filament lamp is connected to a power supply. The electromotive force (e.m.f.) produced by the supply can be varied. Fig. 11.1 is the circuit diagram.

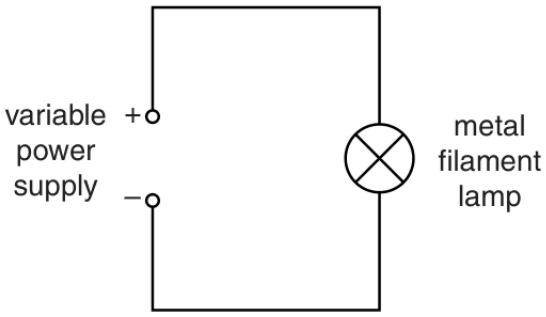


Fig. 11.1

(a) State what is meant by *electromotive force (e.m.f.)*.

.....

.....

.....[2]

(b) Add appropriate circuit symbols to Fig. 11.1 to show the position of

(i) an ammeter that measures the current in the circuit, [1]

(ii) a voltmeter that measures the voltage across the lamp. [1]

(c) A student adds meters to the circuit and makes measurements of the current and voltage. He then plots the current/voltage graph shown in Fig. 11.2.

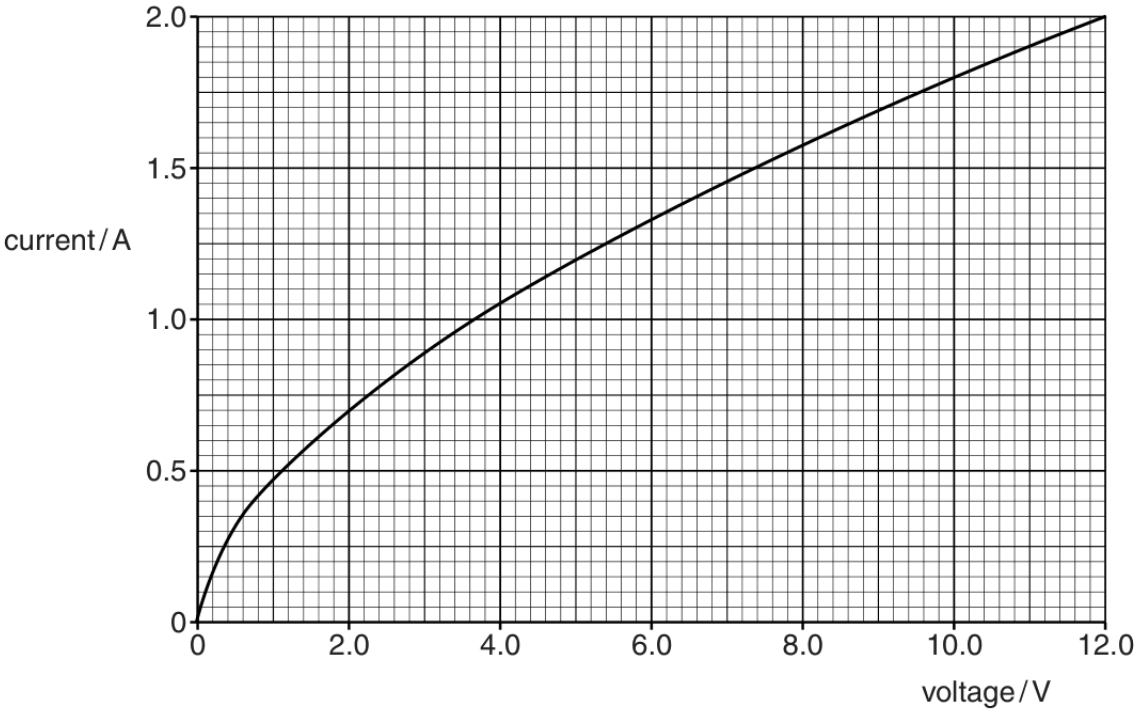


Fig. 11.2

For Examiner's Use

(i) Using values from Fig. 11.2, calculate the resistance of the lamp when the current is 0.70 A. Give your answer to an appropriate number of significant figures.

resistance = ...[3]

(ii) State what, if anything, happens to the resistance of the lamp as the voltage increases.

.....

.....[1]

(d) In normal use, the lamp is connected to a 12 V supply.

Use Fig. 11.2 to determine the power of the lamp when it is used in this way.

power = ...[2]

(e) When connected to a 12 V supply, thermionic emission occurs in the lamp.

(i) Describe what is meant by *thermionic emission*.

.....

.....

.....[2]

(ii) Thermionic emission is used in a cathode-ray oscilloscope (c.r.o.) tube.

1. State why air must be removed from the tube of the c.r.o.

.....

.....[1]

2. A voltage is connected across the Y-plates in the c.r.o.

State and explain the effect on the trace on the screen.

.....

.....

.....[2]

11	**(a)**	work done per (unit) charge/coulomb/C **or** energy transformed per (unit) charge/ coulomb/C	B1	
		property of a source (of electricity) **or** energy transformed to electrical energy per (unit) charge/coulomb/C	B1	[2]
	(b) **(i)**	ammeter in series	B1	
	(ii)	voltmeter in parallel with lamp **or** lamp and ammeter	B1	[2]
	(c) **(i)**	(V =) 2.0 (V)	C1	
		(R =) V/I **or** 2.0/0.70	C1	
		2.9/2.86 Ω (i.e. 2 **or** 3 s.f. only)	A1	
	(ii)	(resistance) increases	B1	[4]
	(d)	(P =) VI **or** (P =) V^2/R **or** I^2R **or** 12 × 2.0 **or** 12 × 0.70	C1	
		24 W	A1	[2]
	(e) **(i)**	emission of electrons	M1	
		from heated metal/named metal/filament/wire	A1	
	(ii) **1.**	prevents collision with air (molecules) **or** prevents deflection **or** lets electrons/particles reach screen/travel unimpeded	B1	
	2.	moves vertically (e.g. up/down/above/below **or** vertical line) **not** with horizontal movement due to this voltage	B1	
		attracted by positive **or** repelled by negative **or** attracted by one plate and repelled by the other **or** electric field (acts on charge)	B1	[5]

[Total: 15]

2 Fig. 2.1 shows apparatus used when determining the specific heat capacity of aluminium.

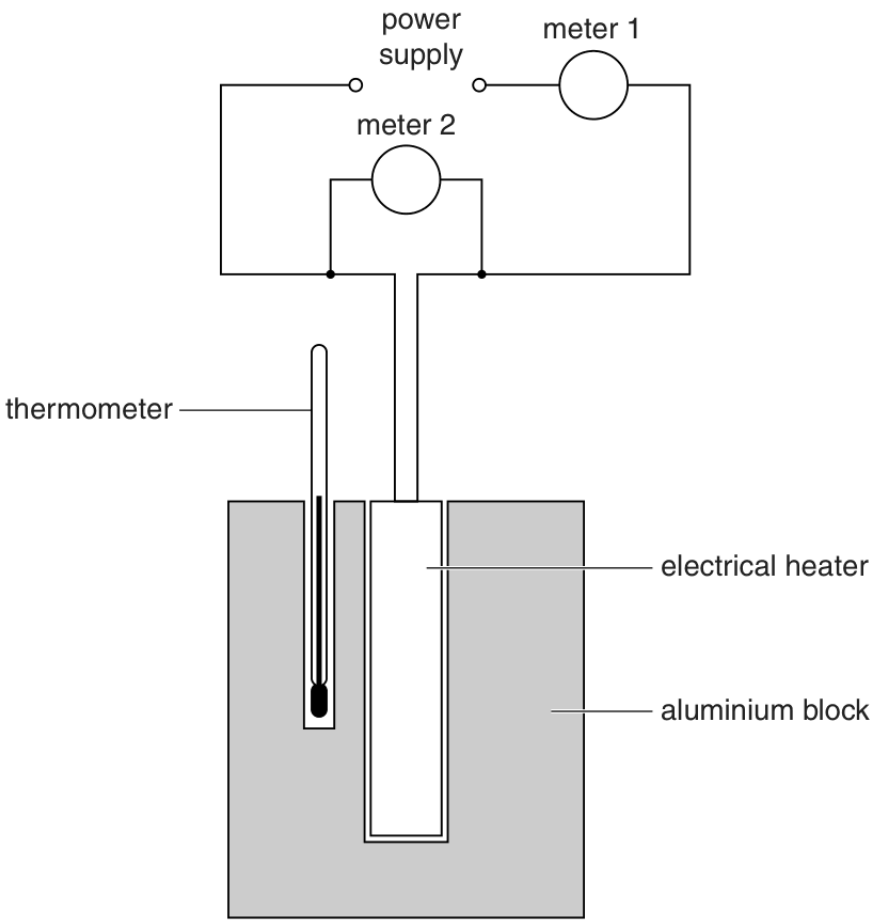


Fig. 2.1

Meter 1 and meter 2 are electrical meters.

(a) State the quantities measured by meter 1 and by meter 2.

meter 1: .....

meter 2: .....

[1]

(b) The following results are obtained in the experiment:

initial temperature of block	= 18 °C
final temperature of block	= 40 °C
mass of block	= 0.85 kg
energy supplied to heater	= 17 000 J

Assuming that no heat is lost from the block during the experiment, calculate the specific heat capacity of aluminium.

specific heat capacity = ... [2]

(c) In practice, some heat is lost from the block and the air above it becomes hot.

(i) Describe and explain the process by which the hot air moves away from the block.

.....

.....

.....

..... [2]

(ii) Suggest how to reduce the loss of heat from the block in the experiment.

.....

..... [1]

2 **(a)** current **and** voltage / p.d. / e.m.f. in correct order [B1]

(b) (c=) E / mT in symbols or numbers e.g. 17 000 = 0.85 × c × 22 [C1]
910 J / (kg °C) [A1]

(c) **(i)** (hot air) rises **or** convection mentioned [B1]
(hot) air less dense [B1]

(ii) lag **or** cover with insulating material **or** warmer room **or** start with colder block [B1]

[6]

11 **(a)** A torch contains a filament lamp connected to a battery. The battery contains four 1.5 V cells connected in series.

Fig. 11.1 shows the current/potential difference (voltage) graph for the filament lamp.

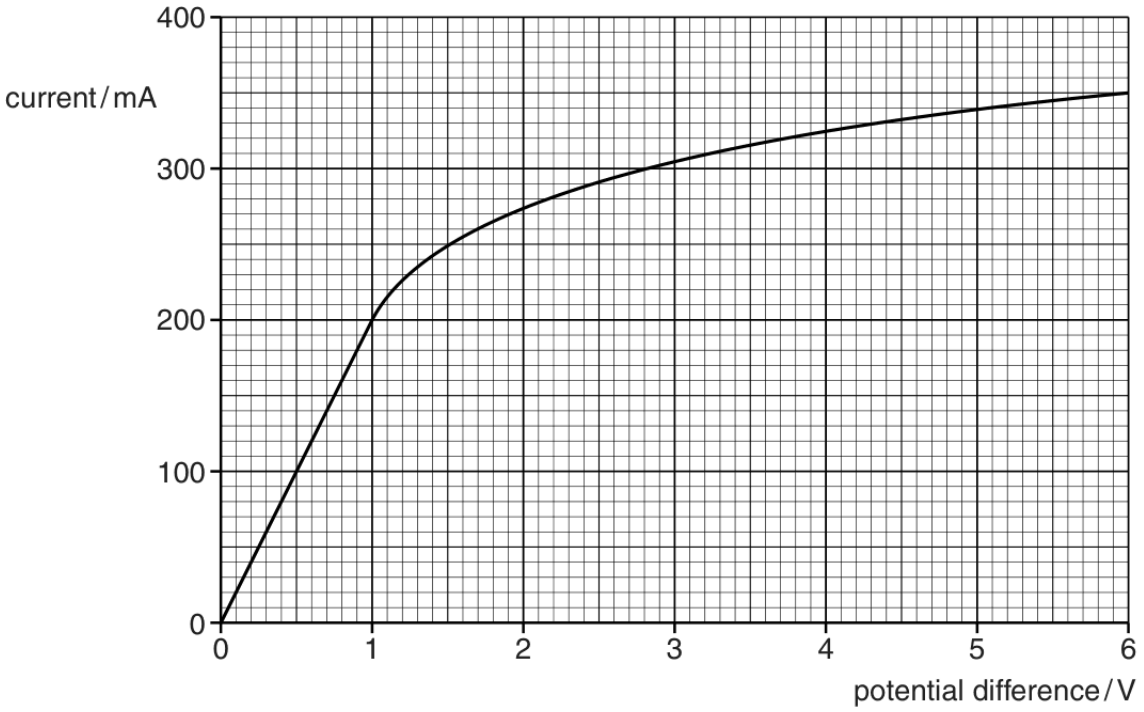


Fig. 11.1

(i) State what is meant by the *potential difference* (p.d.) across the lamp.

.....

..... [2]

(ii) Fig. 11.1 shows that the current in the lamp is not proportional to the p.d.

1. State how Fig. 11.1 shows this.

.....

..... [1]

2. Suggest why the current is not proportional to the p.d.

.....

..... [1]

(iii) The lamp is at normal brightness and there is a p.d. of 6.0V across the lamp.

1. Use Fig. 11.1 to find the current in the lamp.

current = ... [1]

2. Calculate the charge that passes through the lamp in 2.0 hours.

charge = ... [2]

3. The battery in the torch is able to keep the lamp lit for 2.0 hours.
Estimate the initial energy stored in the battery.

energy = ... [2]

4. Explain why your value is only an estimate.

.....

..... [1]

(b) Suggest and explain an advantage of using two cells in parallel rather than a single cell.

.....

.....

..... [2]

(c) Another torch contains a light-emitting diode (LED) instead of a filament lamp.
The battery in the torch uses four 1.5V cells, some in series and some in parallel.
The p.d. across the LED when in operation is 3.0V.

In the space below, draw a circuit diagram showing how the four cells, a switch and the LED are connected.

[3]

11 (a) (i) work done **or** energy produced / needed [M1]
per unit charge **or** per coulomb (passing through lamp) [A1]

(ii) **1**. not straight **or** curves **or** gradient changes **or** data used correctly [B1]
2. gets hotter **or** temperature changes **or** resistance increases [B1]

(iii) **1**. 350 mA **or** 0.35 A [B1]
2. Q = It in any form algebraic or numerical **or** 2 × 60 × 60 **or** 7200 (s) seen [C1]
2520 C **or** 2500 C [A1]
3. (E=) QV or V*I*t in any algebraic or numerical form, e.g. 0.35 × 6 × 2 [C1]
15100 J **or** 15120 J **or** 15000 J **or** 4.2 kWh [A1]
4. current and / or voltage falls / varies
or some energy remains (in cell)
or some energy / heat produced **in cell**
or correct argument involving internal resistance of cell [B1]

(b) lasts longer [M1]
(because) larger energy (initially) **or** smaller current (in each cell) [A1]
or
avoids failure if one cell fails (M1)
(because) other (parallel) cell takes over (A1)
or
larger current / power (in external resistor / lamp) (M1)
(because) smaller (internal) resistance of combined cells (A1)

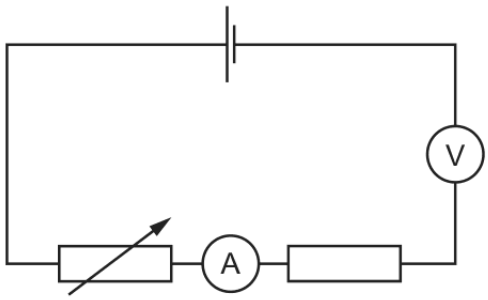
(c) symbol for LED clear and in correct direction for cells shown [B1]
4 cells (not all in parallel or in series), switch and LED (labelled or shown) that switches on and off properly [B1]
cells connected correctly to give 3.0 V total e.m.f., e.g. two sets of parallel cells connected in series or three in parallel and one in series [B1]

[15]

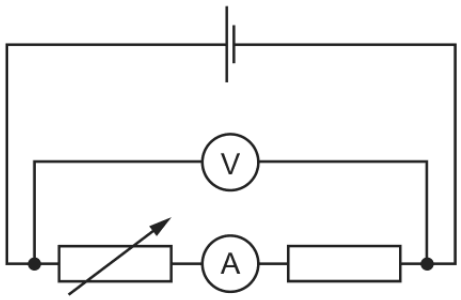
QUESTION · 5054_w14_qp_11.pdf · source page 14 of 20

29 Which circuit is connected correctly to measure the current in a fixed resistor and the potential difference (p.d.) across the same resistor?

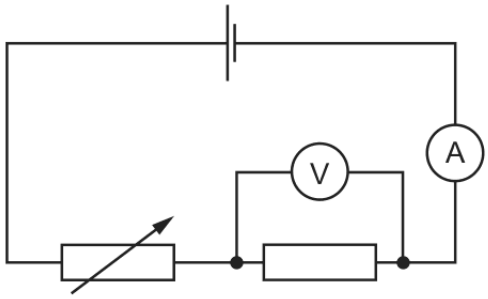
A



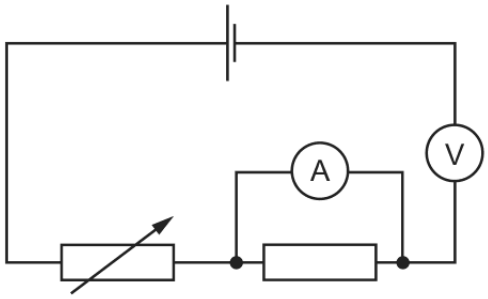
B



C



D



MARK SCHEME (official answer key) · 5054_w14_ms_11.pdf · source page 2 of 2 · row for Question 29

29	**C**

6 The base of a storm cloud is negatively charged. Fig. 6.1 shows the cloud above flat ground.

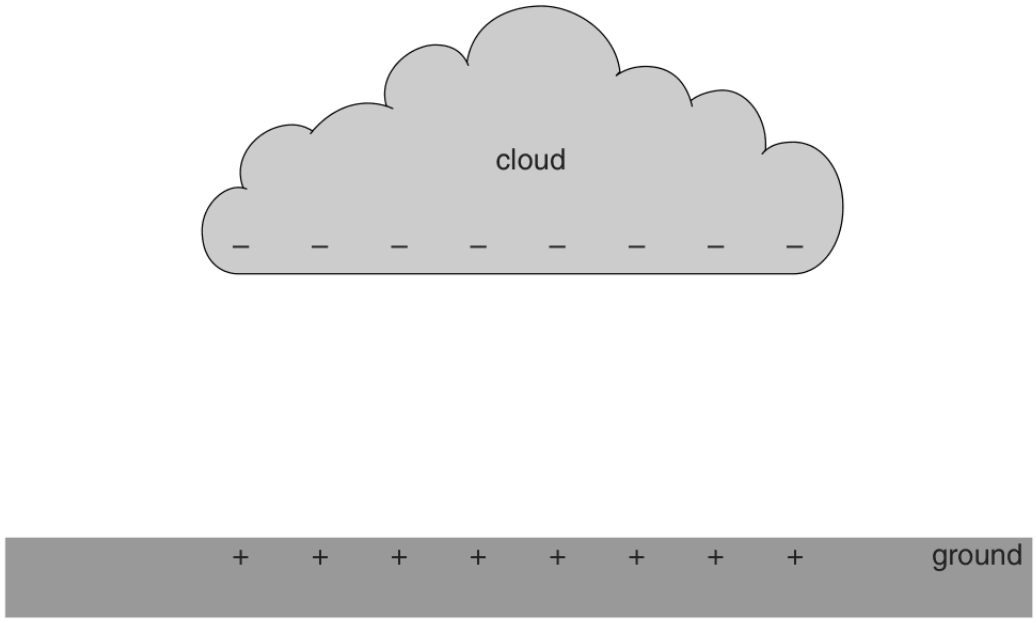


Fig. 6.1

(a) The cloud causes the ground beneath it to become positively charged.

Explain, in terms of the particles involved, how the ground becomes positively charged.

.....

.....

.....[2]

(b) In the space between the negative charge on the cloud and the positive charge on the ground, there is an electric field.

State what is meant by an *electric field*.

.....

.....[1]

(c) A lightning strike takes place. In 0.0015 s, a charge of 180 C passes between the cloud and the ground.

Calculate the average current in the lightning strike.

current = ...[2]

6	**(a)**	**electrons** repelled by cloud (leaving ground positive) **not** positive charge / protons move	B1	
		like charges repel **or** electrons negative	B1	
	(b)	(region) where (electric) charge experiences a force	B1	
	(c)	(I =) Q/t **or** 180 / 0.0015	C1	
		1.2×10^5 A	A1	**[5]**

11 A student makes a 2.0V battery by connecting two cells of electromotive force (e.m.f.) 2.0V in parallel. The battery, an ammeter with different ranges and three different resistors are used to set up the circuit shown in Fig. 11.1.

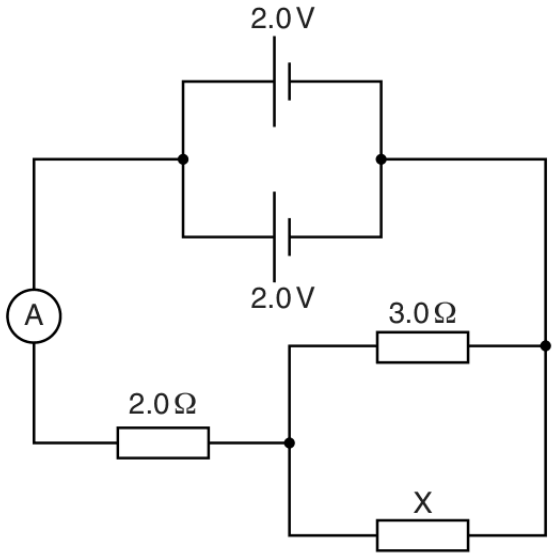


Fig. 11.1

(a) (i) Explain what is meant by *electromotive force*.

.....

.....

.....[2]

(ii) State one advantage of using two cells in parallel rather than using a single 2.0V cell.

.....

.....[1]

(b) Resistor X and the 3.0 Ω resistor have a combined resistance that is equal to 2.0 Ω.

Calculate

(i) the total resistance of the circuit,

total resistance = ...[1]

(ii) the resistance of X.

resistance of X = ...[2]

(c) **(i)** Determine the reading of the ammeter.

reading = ...[2]

(ii) Suggest a suitable range for the ammeter.

.....[1]

(d) The current in the 2.0 Ω resistor is I_2.
The current in the 3.0 Ω resistor is I_3.
The current in X is I_X.

State the equation that relates I_2, I_3 and I_X.

.....[1]

(e) State the potential difference (p.d.) across

(i) the 2.0 Ω resistor,

p.d. = ...[1]

(ii) the 3.0 Ω resistor.

p.d. = ...[1]

Question 11 continues on page 18.

(f) The student sets up a second circuit using a variable d.c. power supply, an ammeter and a 12V metal filament lamp. The circuit is shown in Fig. 11. 2.

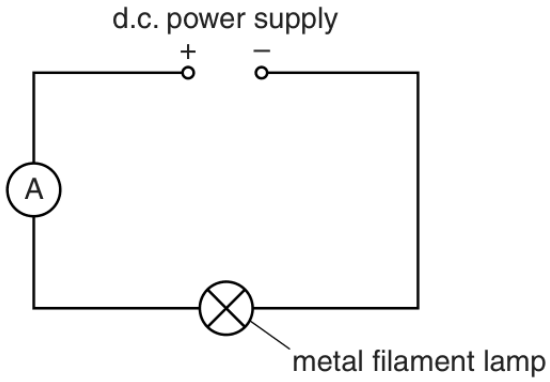


Fig. 11.2

The d.c. power supply is set to 12V and the ammeter reading is 1.5 A.

The student changes the e.m.f. of the d.c. power supply to 6.0V. The lamp dims and the ammeter reading changes.

(i) State and explain what happens to the resistance of the metal filament of the lamp.

.....

.....

.....[2]

(ii) State whether the new ammeter reading is less than, equal to or greater than 0.75 A.

.....[1]

11	**(a)**	**(i)**	energy to drive charge around a circuit **or** terminal p.d. on open circuit	B1	
			energy to drive **unit** charge around a circuit **or** energy / charge	B1	
		(ii)	lasts longer **or** lower internal resistance **or** can replace a cell without switching off **or** continues to work if one cell is flat **ignore** more current (**not** greater e.m.f. / voltage)	B1	**[3]**
	(b)	**(i)**	4.0 Ω	B1	
		(ii)	$(1/R_{tot} =)1/R_1 + 1/R_2$ **or** $1/3 + 1/X$ **or** product / sum **or** $(3 \times X)/(3 + X)$ **or** $\frac{1}{X} = \frac{1}{2} - \frac{1}{3}$	C1	
			6.0 Ω	A1	**[3]**
	(c)	**(i)**	$(I =)\ V/R$ **or** 2.0 / 4.0	C1	
			0.50 A	A1	
		(ii)	(from) 0 **and** (to) 0.50 to 5.0 A	B1	**[3]**
	(d)		$I_2 = I_3 + I_X$	B1	**[1]**
	(e)	**(i)**	1.0 V	B1	
		(ii)	1.0 V	B1	**[2]**
	(f)	**(i)**	temperature decreases	B1	
			resistance decreases	B1	
		(ii)	greater than 0.75 A (e.c.f. resistance increases in **(f)(i)**)	B1	**[3]**
					[Total: 15]

QUESTION · 5054_s15_qp_11.pdf · source page 11 of 16

29 A charge of 7.5 C flows through a resistor in 5.0 s. A student has ammeters with different ranges that he can use to measure the current in the resistor.

Which ammeter range is the most appropriate?

A 0-1 A **B** 0-2 A **C** 0-5 A **D** 0-40 A

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 29

29	**B**

QUESTION · 5054_w15_qp_11.pdf · source page 10 of 16

26 The flash on a camera operates when a capacitor discharges through a flash-tube.
In 3.2 ms, a charge of 6.0 C passes through the flash-tube.

What is the average current in the flash-tube?

A 0.019 A

B 1.9 A

C 19 A

D 1900 A

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 26

26	**D**

10 Thin wire, covered in plastic insulation, is used to make a solenoid (long coil). The solenoid is connected to a sensitive ammeter. Fig. 10.1 shows the N-pole of a steel magnet placed next to the solenoid.

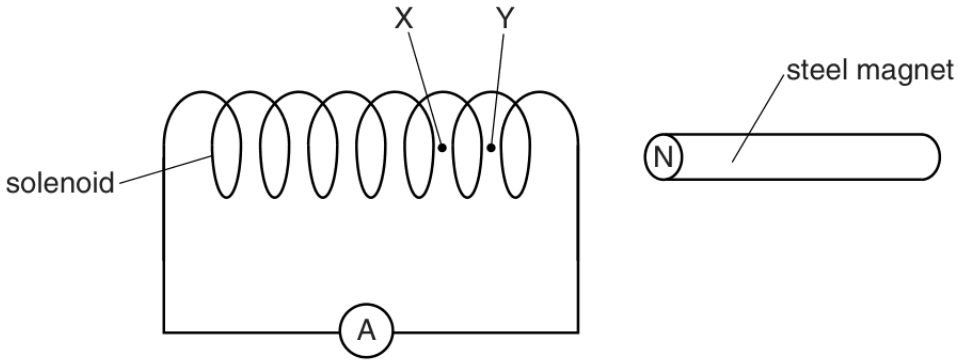


Fig. 10.1

Point X and point Y are on the axis of the solenoid.

(a) (i) Explain why plastic is an electrical insulator.

.....

.....

..... [1]

(ii) Explain why the magnet is **not** made from

1. aluminium,

..... [1]

2. iron.

..... [1]

(b) In one experiment, the magnet in Fig. 10.1 is moved to the left and passes into the solenoid. The N-pole of the magnet travels from Y to X at a constant speed. As it moves, the ammeter shows a small current.

(i) Explain why there is a current in the solenoid when the magnet is moving.

.....

.....

.....

.....

.....

..... [3]

(ii) The N-pole travels from Y to X in 0.14 s. As it moves, the current shown on the ammeter is 0.045 mA. The resistance of the solenoid is 1.2 Ω.

Calculate

1. the potential difference (p.d.) across the solenoid,

potential difference = ... [2]

2. the charge that passes through the solenoid as the N-pole moves from Y to X.

charge = ... [2]

(c) In a second experiment, the speed of the N-pole is greater than its speed in the first experiment. It now takes only 0.070 s to travel from Y to X.

A current in the same direction is shown on the ammeter.

(i) State and explain how the size of this current compares with the size of the current in the first experiment.

.....

.....

..... [2]

(ii) The same quantity of charge passes through the coil in both the first and second experiments.

Explain why this is the case.

.....

.....

..... [1]

(d) State two ways in which the equipment shown in Fig. 10.1 can be used to produce a current in the solenoid that is in the opposite direction.

1. .....

2. .....
[2]

10	**(a)**	**(i)**	no free electrons (in plastic) **or** all electrons are bound / structural	B1	
		(ii)	(aluminium) is not magnet(ic) **or** cannot be magnetised	B1	
			(iron) is a temporary / soft magnetic material **or** is not a permanent magnet	B1	[3]
	(b)	**(i)**	magnetic field / flux (mentioned)	B1	
			(magnetic) field lines cut wire / solenoid / circuit **or** changing magnetic field / flux	B1	
			voltage/e.m.f. **induced**	B1	
		(ii)	(V =)IR **or** 0.045×1.2 **or** 0.000045×1.2	C1	
			5.4×10^{-5} V **or** 0.054 mV	A1	
			(Q =)It **or** 0.045×0.14 **or** 0.000045×0.14	C1	
			6.3×10^{-6} C **or** 0.0063 mC	A1	[7]
	(c)	**(i)**	larger **or** twice the current	B1	
			(magnetic) field lines cut faster **or** (magnetic) field changes faster **or** twice the current	B1	

		(ii)	double the current for half the time **or** larger current for less time **or** product $I \times t$ is the same	B1	[3]
	(d)	any **two** from: insert S-pole (at same end) insert (N-pole) at other end **or** from other direction withdraw N-pole (from same end implied) withdraw S-pole from other end **or** pass through completely		B2	[2]
					[15]

QUESTION · 5054_s16_qp_11.pdf · source page 11 of 16

29 The current in a car headlamp is 2.0 A. The headlamp is switched on for 4.0 minutes.

How much charge passes through the headlamp?

A 8.0 C **B** 30 C **C** 120 C **D** 480 C

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 29

29	**D**

QUESTION · 5054_s16_qp_12.pdf · source page 9 of 16

25 A charge of 45 C flows through an electric appliance in 3.0 minutes.

What is the average current in the appliance?

A 0.25 A **B** 4.0 A **C** 15 A **D** 135 A

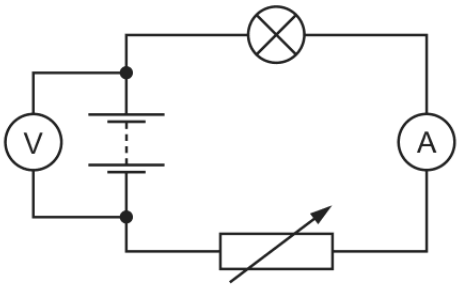
MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 25

25	**A**

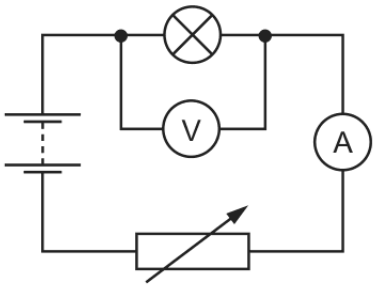
QUESTION · 5054_s16_qp_12.pdf · source page 11 of 16

28 Which diagram shows the arrangement of the ammeter and voltmeter to obtain readings to find the power of a lamp?

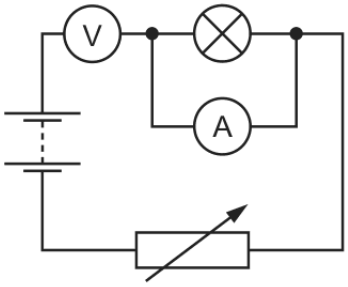
A



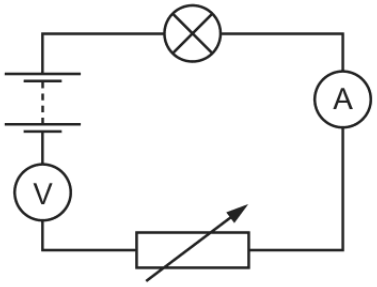
B



C



D



MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 28

28 **B**

9 **(a)** The primary coil of a transformer is connected to the 230 V a.c. mains supply.

(i) Describe how the transformer operates.

.....

.....

.....

.....

.....

.....[3]

(ii) The output voltage of the transformer is alternating and cannot be used directly as a d.c. power supply.

In the space below, draw the symbol of a circuit component that allows the current to pass in only one direction and state the name of the component.

circuit symbol

name of component: .....
[2]

(b) Fig. 9.1 shows a d.c. power supply being used to charge a rechargeable battery.

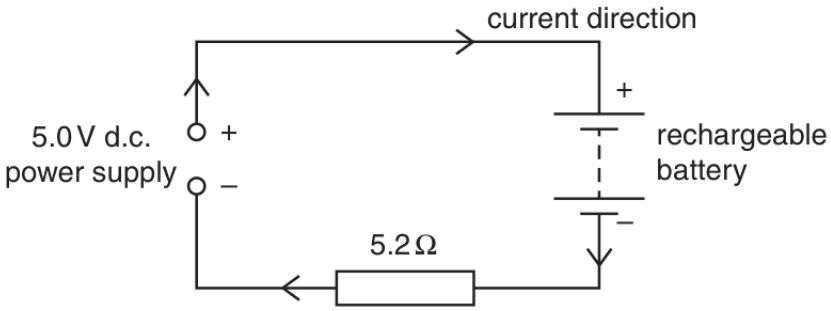


Fig. 9.1

The rechargeable battery is in series with the d.c. power supply and a 5.2 Ω resistor. The e.m.f. of the power supply is 5.0 V.

(i) Explain what is meant by *e.m.f.*

.....

.....

.....[2]

(ii) The e.m.f. of the power supply and the e.m.f. of the battery act against each other in opposite directions in the circuit.

The e.m.f. of the battery is 3.7 V.

1. Calculate the resultant e.m.f. in the circuit.

resultant e.m.f. = ...[1]

2. Calculate the current in the circuit.

current = ...[2]

3. The current in the circuit remains constant as the battery is charged.

Calculate the charge that passes through the resistor in 1.5 hours.

charge = ...[3]

(c) A battery charger has a plastic casing and double insulation. The battery charger is not earthed.

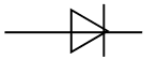
Explain why it is not necessary for the battery charger to be earthed.

.....

.....

.....

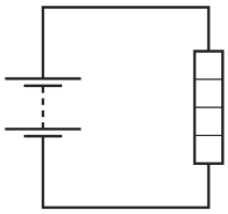
.....[2]

9	**(a)**	**(i)**	magnetic field mentioned	B1	
			alternating / changing magnetic field	B1	
			current / voltage / e.m.f. induced (in secondary coil)	B1	
		(ii)		B1	
			diode	B1	[5]
	(b)	**(i)**	work done / energy transferred per unit charge	M1	
			electrical energy to other forms **or** for whole circuit **or** property of supply	A1	

	(ii)	**1**	1.3 V	B1	
		2	(I =)V/R **or** 1.3/5.2	C1	
			0.25 A	A1	
		3	(Q =)It **or** 0.25 × 1.5 × 3600 **or** 0.25 × 1.5	C1	
			0.25 × 1.5 × 3600 **or** 0.37/0.375/0.38	C1	
			1300 / 1350 / 1400 C	A1	[8]
	(c)		plastic / casing is an (electrical) insulator	M1	
			no shock possible	A1	[2]
					[15]

QUESTION · 5054_w16_qp_12.pdf · source page 12 of 16

33 An ammeter and a voltmeter are connected into the circuit shown when determining the power produced by a heater.



How are these meters connected, in relation to the heater?

	ammeter	voltmeter
A	in parallel	in parallel
B	in parallel	in series
C	in series	in parallel
D	in series	in series

MARK SCHEME (official answer key) · 5054_w16_ms_12.pdf · source page 2 of 2 · row for Question 33

33 **C**

9 A vertical solenoid (long coil) with an iron core is held in a wooden clamp above a laboratory bench.

The solenoid is connected in series with a battery, a switch S, an ammeter and a variable resistor. There is a voltmeter in parallel with the solenoid.

Fig. 9.1 represents this apparatus.

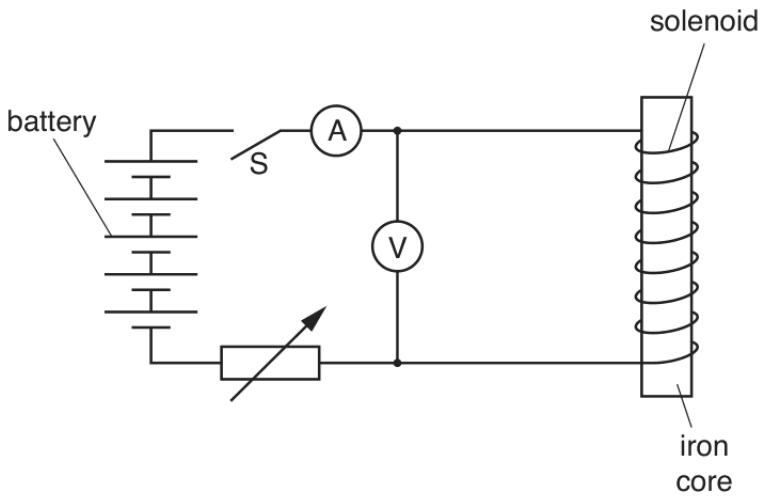


Fig. 9.1

A student closes switch S and a current in the circuit produces a reading on the ammeter.

(a) State what is meant by *current*.

.....

.....[1]

(b) The battery consists of five 1.5 V cells in series. The reading on the ammeter is 4.0 A.

(i) State the size of the electromotive force (e.m.f.) of the battery.

.....[1]

(ii) Calculate the total resistance of the series circuit.

resistance = ...[2]

(iii) The reading on the voltmeter is 6.5 V.

Calculate the power dissipated in the solenoid.

power = ...[2]

(iv) The solenoid is made of copper and the student notices that, as time passes, the solenoid becomes extremely warm.

State and explain the effect of this temperature increase on the ammeter reading.

.....

.....

.....[2]

(c) (i) The current in the solenoid magnetises the iron core so that the lower end of the core is an N-pole.

On Fig. 9.1, draw the pattern and indicate the direction of the magnetic field in the region surrounding the iron core. [3]

(ii) The student holds an iron cylinder against the bottom surface of the iron core in the solenoid. When he releases the iron cylinder, it stays in contact with the iron core.

1. Explain why the iron cylinder does not fall.

.....

.....

.....

.....

.....[3]

2. The switch S is opened.

State and explain whether the iron cylinder remains in contact with the iron core.

.....

.....

.....[1]

9	**(a)**	rate of flow of charge **or** charge flowing per unit time	B1	[1]
	(b) **(i)**	7.5 V	B1	
	(ii)	(R =)V / I **or** 7.5 / 4.0	C1	
		1.9 Ω	A1	
	(iii)	(P =) VI **or** 6.5 × 4.0	C1	
		26 W	A1	
	(iv)	resistance increases	M1	
		(reading of ammeter) decreases	A1	[7]
	(c) **(i)**	at least two lines on left **and** two lines on right of core **and** correct shape (by eye)	B1	
		good shape (by eye) **and** into poles **and** no straight sections **and** at least one line on each side	B1	
		at least one arrow N to S (primarily upwards) **and** none wrong	B1	[3]

	(ii) **1**	cylinder is magnetised (by induction)	B1	
		top (of cylinder) is an S-pole	B1	
		unlike poles attract **or** S-pole attracts N-pole	B1	
	2	it does not (remain in contact**) and** iron is temporary / soft magnetic material / core (and cylinder) lose magnetisation	B1	[4]
				[15]

QUESTION · 5054_s17_qp_11.pdf · source page 14 of 16

36 During a thunderstorm, a lightning flash occurs when charge flows between a cloud and the Earth. On average, 20 C of charge flows during each flash and there are three flashes every minute.

What is the average current between the cloud and the Earth during the thunderstorm?

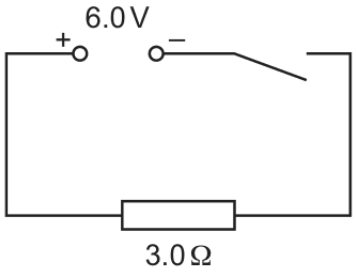
A 1.0 A **B** 36 A **C** 60 A **D** 1800 A

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 3 of 3 · row for Question 36

36	A	**1**

QUESTION · 5054_s17_qp_12.pdf · source page 15 of 20

33 The circuit shown is switched on for 1.0 minute.



How much charge passes through the 3.0 Ω resistor?

A 2.0 C **B** 12 C **C** 120 C **D** 720 C

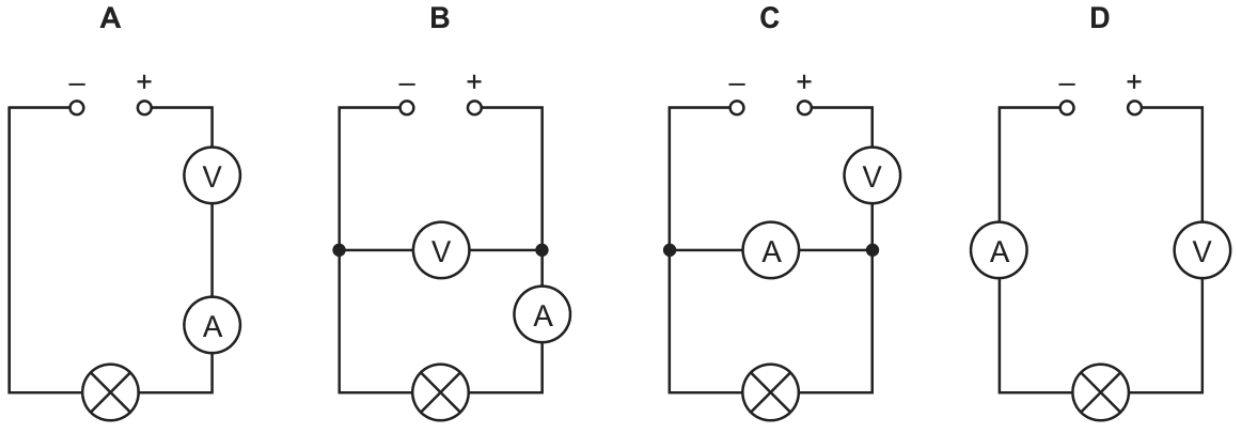
MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 3 of 3 · row for Question 33

33	C	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 14 of 20

30 In a circuit, a voltmeter is used to measure the potential difference across a lamp. An ammeter is used to measure the current in the lamp.

Which diagram shows the circuit?



MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 3 of 3 · row for Question 30

30	B	**1**

8 In the engine compartment of a car, the battery supplies a current of 120 A to the starter motor.

(a) The starter motor is switched on for 3.5 s.

Calculate the charge that passes through the starter motor in this time.

charge = ...[2]

(b) The starter motor circuit is switched on and off using a relay.

The relay is controlled using a second circuit. A switch in the second circuit is used by the driver to operate the relay and turn the starter motor on and off.

Explain how the current in the second circuit controls the relay.

.....

.....

.....

.....

.....[2]

Question	Answer	Marks
8(a)	(Q =) It **or** 120×3.5	**C1**
	420 C	**A1**
8(b)	(small current) magnetises something physical **or** produces a magnetic field **or** produces an electromagnet	**M1**
	(terminals of motor) switch / contacts / terminals (in first circuit) attract / close	**A1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Electromagnetic radiation is produced by the Sun and travels from the Sun to the Earth.

(a) Some of the statements below are true and some of them are false.

Put a tick (✓) in the box after each statement to show whether it is true or false.

	true	false
Gamma rays are used to kill cancerous cells but can also cause cancer.	☐	☐
Infra-red is used in sun-beds.	☐	☐
Radio waves have the highest frequency in the electromagnetic spectrum.	☐	☐
The higher the frequency of the radiation the smaller is the wavelength in air.	☐	☐

[2]

(b) The glass prism in Fig. 9.1 is used to split white light from the Sun into different colours.

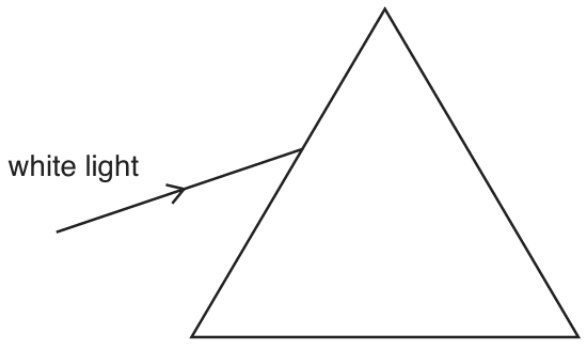


Fig. 9.1

(i) Draw, on Fig. 9.1, to show the action of the prism on the light as it passes through the prism. [3]

(ii) Explain why the prism splits white light into different colours.

.....

.....

.....

.....[2]

(c) The Earth is 1.5×10^8 km from the Sun.

(i) State the speed of light in a vacuum.

.....[1]

(ii) Calculate the time taken for light to travel from the Sun to the Earth.

time = ...[2]

(d) The Sun also produces a solar wind, which is a stream of charged particles.

In one part of space, the solar wind is due to the movement of protons and alpha-particles away from the Sun. A charge of 180 C passes through a small volume of space in 2.0 minutes.

(i) Calculate the electric current in this volume of space.

current = ...[2]

(ii) Complete the table in Fig. 9.2 for an alpha-particle, a proton and a gamma ray.

	nature	charge	stopped by
alpha-particle			
proton	a proton	positive	thin metal
gamma ray	electromagnetic radiation		

Fig. 9.2

[3]

Question	Answer	Marks
9(a)	**true false false true** all 4 correct 2 marks	**B1**
	3 correct 1 mark	**B1**
9(b)(i)	correct refraction towards normal on entering prism	**B1**
	splitting into colours occurs at first surface	**B1**
	red at top and blue / violet at bottom inside and outside prism if shown	**B1**
9(b)(ii)	(different colours have) different speeds in the glass **or** different refractive indices	**B1**
	different (angles of) refraction **or** refract at different angles	**B1**
9(c)(i)	3(.0) $\times$ 10^8 m / s	**B1**
9(c)(ii)	(t=) d / s in any form algebraic or numerical, e.g.	**C1**
	500 s	**A1**
9(d)(i)	(I=) Q / t in any form algebraic or numerical, e.g. 180 / 2	**C1**
	1.5 A	**A1**
9(d)(ii)	1st column helium nucleus **or** 2 protons and 2 neutrons	**B1**
	2nd column positive **and** neutral / no / charge / none	**B1**
	3rd column paper / thin metal / skin / few cm air etc. **and** (thick) lead / thick metal / concrete / stone / earth OR never completely stopped / nothing	**B1**

7 Fig. 7.1 shows the structure of a transformer.

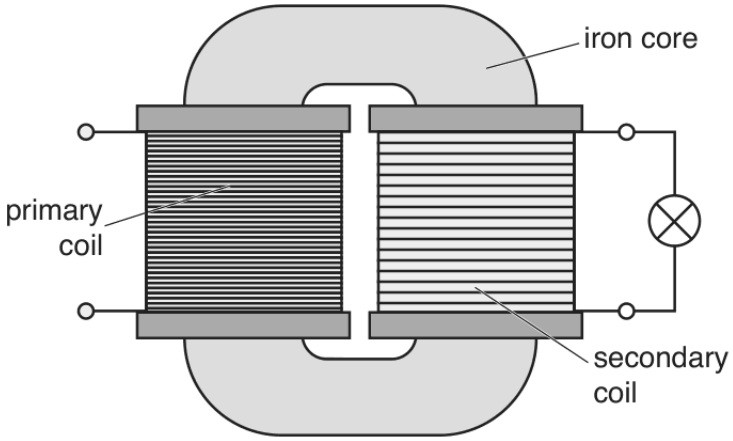


Fig. 7.1

A lamp is connected to the secondary coil.

(a) Explain why the core of the transformer is made from iron.

.....

.....

..... [2]

(b) When there is an alternating current (a.c.) in the primary coil, the lamp is lit.

When there is a direct current (d.c.) in the primary coil, the lamp is **not** lit.

(i) State two ways in which an alternating current differs from a direct current.

1. .....

.....

2. .....

.....
[2]

(ii) Explain why the lamp is **not** lit when there is a direct current in the primary coil.

.....

.....

.....

..... [2]

MARK SCHEME · 5054_w18_ms_22.pdf · source page 6 of 10 · Question 7

Question	Answer	Marks
7(a)	magnetic (material)	**B1**
	temporary / soft magnetic (material)	**B1**
7(b)(i)	it / a.c. changes direction **or** changes polarity / from positive to negative (continually)	**B1**
	it / a.c. has varying size **or** is sinusoidal / like a sine wave	**B1**

MARK SCHEME · 5054_w18_ms_22.pdf · source page 7 of 10 · Question 7

Question	Answer	Marks
7(b)(ii)	magnetic field (in core / secondary coil) is not changing / remains constant	**B1**
	no (electromagnetic) induction	**B1**

10 A student sets up the circuit shown in Fig. 10.1 in a laboratory at room temperature.

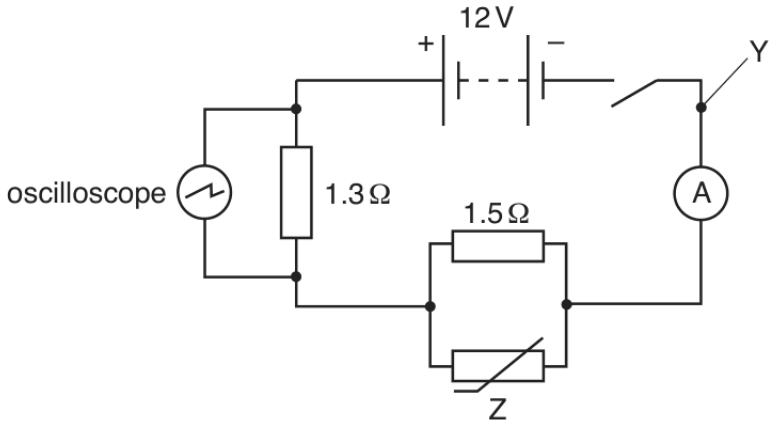


Fig. 10.1

The electromotive force (e.m.f.) of the battery is 12 V.

The switch is closed.

(a) The connecting wires in the circuit are made from copper covered by insulating plastic.

State the name of the particles that flow in the copper and state the direction in which they are flowing at point Y in the circuit.

.....

.....

..... [2]

(b) At room temperature, the resistance of component Z is 6.0 Ω.

(i) State the name of component Z.

..... [1]

(ii) Calculate the resistance of the whole circuit.

resistance = ... [3]

(iii) Calculate the current measured by the ammeter.

current = ... [2]

(iv) The current in the ammeter is I_A, the current in the 1.5 Ω resistor is I_R and the current in component Z is I_Z.

Write down an equation that relates I_A, I_R and I_Z.

..... [1]

(c) Fig. 10.2 shows the screen of the oscilloscope.

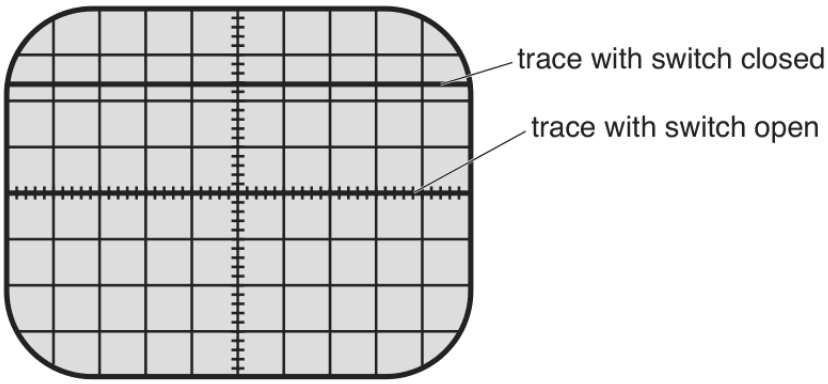


Fig. 10.2

Before the switch is closed, the trace is a horizontal line across the middle of the screen, as shown in Fig. 10.2. When the switch is closed, the trace remains horizontal and moves up the screen.

Component Z is heated.

State and explain what is observed on the oscilloscope screen as the temperature of Z increases.

.....

.....

.....

.....

.....

..... [4]

(d) The 12 V battery is made from cells of electromotive force (e.m.f.) 1.5 V which are all in series.

(i) Calculate the number of 1.5 V cells that make up the battery.

number = ... [1]

(ii) State the e.m.f. of a battery made by connecting all of these cells in parallel.

e.m.f. = ... [1]

MARK SCHEME · 5054_w18_ms_22.pdf · source page 9 of 10 · Question 10

Question	Answer	Marks
10(a)	electrons **c.a.o.**	**M1**
	towards the ammeter **or** away from the negative terminal **or** towards the positive terminal	**A1**
10(b)(i)	thermistor **c.a.o.**	**B1**
10(b)(ii)	$1 / R_T = 1 / R_1 + 1 / R_2$ **or** $1 / R_T = 1 / 1.5 + 1 / 6.0$ **or** $(R_T =) R_1R_2 / (R_1 + R_2)$ **or** $1.5 \times 6.0 / (1.5 + 6.0)$	**C1**
	1.2 (Ω)	**C1**
	2.5 Ω	**A1**
10(b)(iii)	$(I =) V / R$ **or** $12 / 2.5$	**C1**
	4.8 (A)	**A1**
10(b)(iv)	$I_A = I_R + I_Z$	**B1**
10(c)	resistance of Z / thermistor decreases	**B1**
	resistance of parallel combination decreases **or** total resistance (of circuit) decreases **or** current increases	**B1**
	voltage (across 1.3 Ω) increases	**B1**
	trace moves towards top of screen / upwards	**B1**

MARK SCHEME · 5054_w18_ms_22.pdf · source page 10 of 10 · Question 10

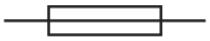
Question	Answer	Marks
10(d)(i)	8	**B1**
10(d)(ii)	1.5 V	**B1**

QUESTION · 5054_s19_qp_11.pdf · source page 16 of 20

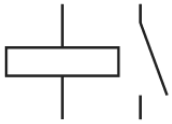
34 What is the symbol for a device that measures current?

A

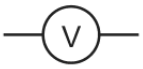
B



C



D



MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 3 of 3 · row for Question 34

34	A	**1**

5 A student uses the circuit shown in Fig. 5.1 to investigate a resistor R.

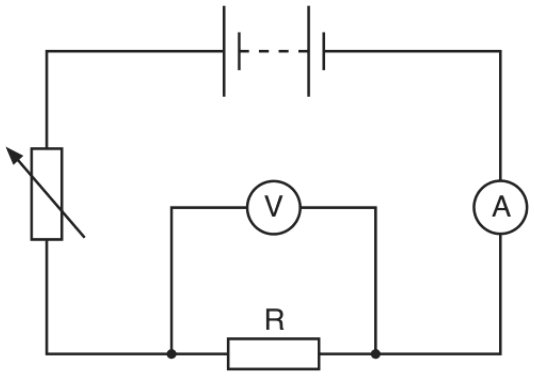


Fig. 5.1

(a) Describe how the student uses the apparatus in Fig. 5.1 to obtain a range of ammeter and voltmeter readings.

.....

..... [1]

(b) The readings obtained are shown in Table 5.1.

Table 5.1

voltmeter reading / V	ammeter reading / mA
7.6	320
5.2	220
2.4	100

(i) State Ohm's law.

.....

.....

..... [2]

(ii) Using the results in Table 5.1, show that the resistor obeys Ohm's law.

.....

.....

.....

..... [1]

(iii) The student has available an ammeter with five ranges marked:

- 0 – 10 A
- 0 – 200 mA
- 0 – 20 mA
- 0 – 2 mA
- 0 – 0.2 mA

Suggest how the student makes best use of the different ranges during the investigation.

.....

.....

.....

..... [1]

[Total: 5]

Question	Answer	Marks
5(a)	adjusts the variable resistor / rheostat **or** change the number of cells	**B1**
5(b)(i)	current proportional to voltage **or** V = kI where k is a constant	**M1**
	provided temperature / physical conditions unchanged	**A1**
5(b)(ii)	at least two values of V/I or I/V shown to be the same;	**B1**
5(b)(iii)	use 0–10 A range for 220 and 320 mA readings **and** 0–200 mA range for 100 mA reading **or** start with 10 A / highest range **and** either • reduce to 200 mA for last reading • reduce range **and** keeping high value on scale / not over scale	**B1**

9 Fig. 9.1 shows a boiling liquid at its boiling point, trapped in a cylinder by a piston.

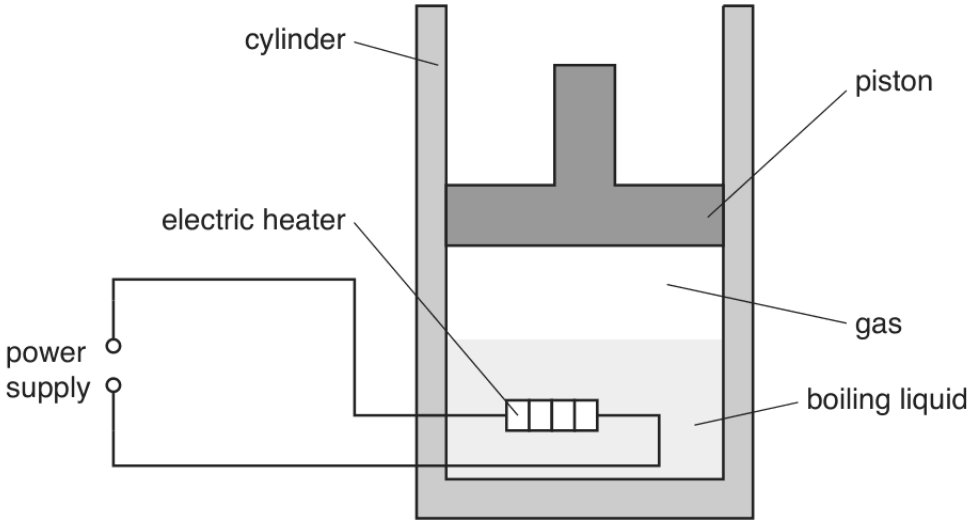


Fig. 9.1

There is an electric heater in the liquid which is connected to a power supply. The thermal energy produced by the heater gradually turns the boiling liquid into a gas.

(a) State what is meant by *boiling point*.

.....

..... [2]

(b) Explain, in terms of molecules, why it is necessary to supply energy in order to turn a liquid at its boiling point into a gas.

.....

.....

.....

..... [3]

(c) In order to determine the power of the electric heater, two meters are connected into the circuit. One meter measures the current in the heater and the other meter measures the potential difference (p.d.) across it.

On the circuit in Fig. 9.1, draw symbols to show the two meters used and where they are connected. [2]

(d) The current in the heater is 2.0 A and the p.d. across it is 6.0 V.

(i) Calculate the power of the heater.

power = ... [2]

(ii) Calculate the thermal energy supplied to the liquid by the heater in 1.0 minute.

energy = ... [2]

(iii) The specific latent heat of vaporisation of the liquid is 9.0×10^5 J/kg.

Calculate the mass of liquid that vaporises every minute.

mass = ... [2]

(e) The piston is free to move in the cylinder. As the liquid boils, the piston is pushed upwards in the cylinder at a constant speed. The volume occupied by the gas just above the liquid increases.

Discuss whether the upward force on the piston changes as the piston moves upwards at constant speed.

.....

.....

.....

..... [2]

[Total: 15]

Question	Answer	Marks
9(a)	temperature	**B1**
	at which a liquid becomes a gas / vapour	**B1**
9(b)	forces (between the molecules of the liquid) / bonds mentioned	**B1**
	move molecules apart **or** overcome forces (between the molecules)	**B1**
	work done against forces **or** gains (molecular) potential energy	**B1**
9(c)	ammeter symbol in series	**B1**
	voltmeter symbol across the heater / power supply	**B1**
9(d)(i)	(P =) VI **or** 6.0×2.0	**C1**
	12 W	**A1**
9(d)(ii)	(E =) Pt **or** 12×60 **or** (E =) VIt **or** $6.0 \times 2.0 \times 60$	**C1**
	720 J	**A1**
9(d)(iii)	(m =) Q / l_v **or** $720 / 9.0 \times 10^5$	**C1**
	8.0×10^{-4} kg **or** 0.80 g	**A1**
9(e)	(it / upward force) does not change	**B1**
	always equal to weight of piston + force due to atmospheric pressure **or** no resultant force **or** piston is not accelerating	**B1**

10 Fig. 10.1 shows a motor lifting a mass. Fig. 10.2 shows part of the circuit diagram of the connections to the motor.

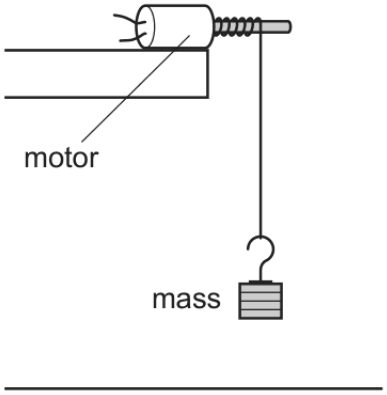


Fig. 10.1

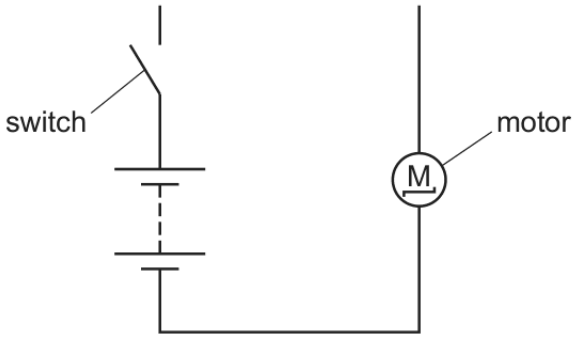


Fig. 10.2

(a) The current in the motor is 1.5A and the voltage supplied by the battery is 8.0 V.

(i) Complete the circuit diagram in Fig. 10.2 to show an ammeter and a voltmeter in the correct positions to take these measurements while the motor is working. [2]

(ii) The motor takes 4.0 s to lift the mass.

Calculate the electrical energy transferred to the motor in this time.

energy = ... [2]

(iii) The motor lifts the 150 g mass through a height of 80 cm in the 4.0 s.

Calculate the gravitational potential energy gained by the mass.

The gravitational field strength g = 10 N / kg.

gravitational potential energy = ... [3]

(iv) State **two** reasons why the gravitational potential energy gained by the mass is less than the electrical energy supplied to the motor.

1. .....

2. .....

[2]

(b) Fig. 10.3 shows the structure of the motor.

When the mass reaches the top of its motion, the switch is opened. This disconnects the battery and causes the mass to fall. The coil turns as the mass falls.

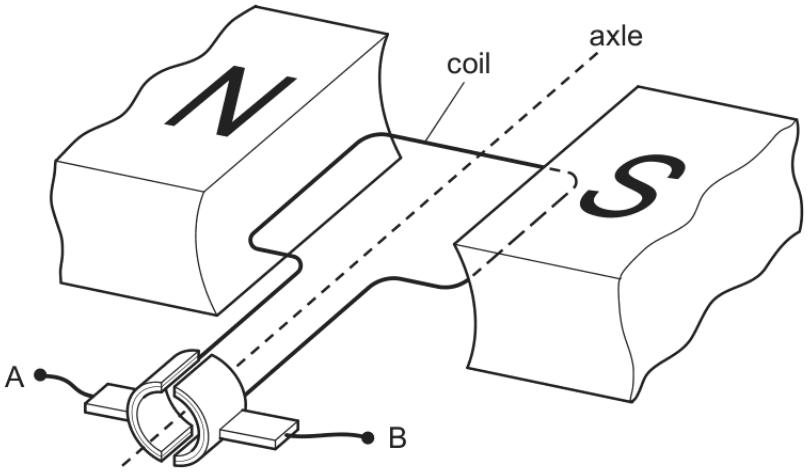


Fig. 10.3

As the coil turns, a small voltage is produced.

(i) Explain why a voltage is produced as the coil turns.

.....

.....

..... [3]

(ii) As the mass falls, a student connects a wire between the points A and B shown in Fig. 10.3.

He notices that the mass takes a longer time to fall when the wire is connected.

The student suggests that this is an example of Lenz's law.

State Lenz's law and suggest why the mass takes longer to fall.

.....

.....

.....

.....

.....

..... [3]

[Total: 15]

MARK SCHEME · 5054_s20_ms_21.pdf · source page 10 of 11 · Question 10

Question	Answer	Marks
10(a)(i)	ammeter in series connecting switch with motor	**B1**
	voltmeter across motor or battery	**B1**
10(a)(ii)	(E =) VIt in any form numerical or algebraic **or** P =VI and P = E / t in any form	**C1**
	48 J	**A1**
10(a)(iii)	(E = mgh) in any form numerical or algebraic	**C1**
	$0.15 \times 0.8(0) \times 10$	**C1**
	1.2 J	**A1**
10(a)(iv)	ANY two of • motor / mass has kinetic energy • heat generated in motor / wires • heat generated in air due to drag / resistance • work done to lift string	**B2**

MARK SCHEME · 5054_s20_ms_21.pdf · source page 11 of 11 · Question 10

Question	Answer	Marks
10(b)(i)	mention of magnetic field (between N and S poles)	**B1**
	coil cuts lines of magnetic flux / magnetic field lines **or** flux / field in coil changes	**B1**
	induction	**B1**
10(b)(ii)	***statement of Lenz's law***, e.g. (induced) current flows in a direction to oppose the change producing it	**B1**
	coil becomes a magnet **or** current flowing in coil produces force in field	**M1**
	poles of coil attracted to opposite poles on magnet **or** force up on a side as side moves down	**A1**

QUESTION · 5054_w20_qp_12.pdf · source page 11 of 16

30 A car battery delivers 150 kC to a circuit before it needs recharging.

Two headlamps connected in parallel to the battery are switched on. There is a current of 4.0 A in each lamp.

How much time passes before the battery needs recharging?

A 2.6 hours **B** 5.2 hours **C** 7.8 hours **D** 10.4 hours

MARK SCHEME (official answer key) · 5054_w20_ms_12.pdf · source page 3 of 3 · row for Question 30

30	B	**1**

9 Fig. 9.1 shows a permanent magnet lying on a piece of paper.

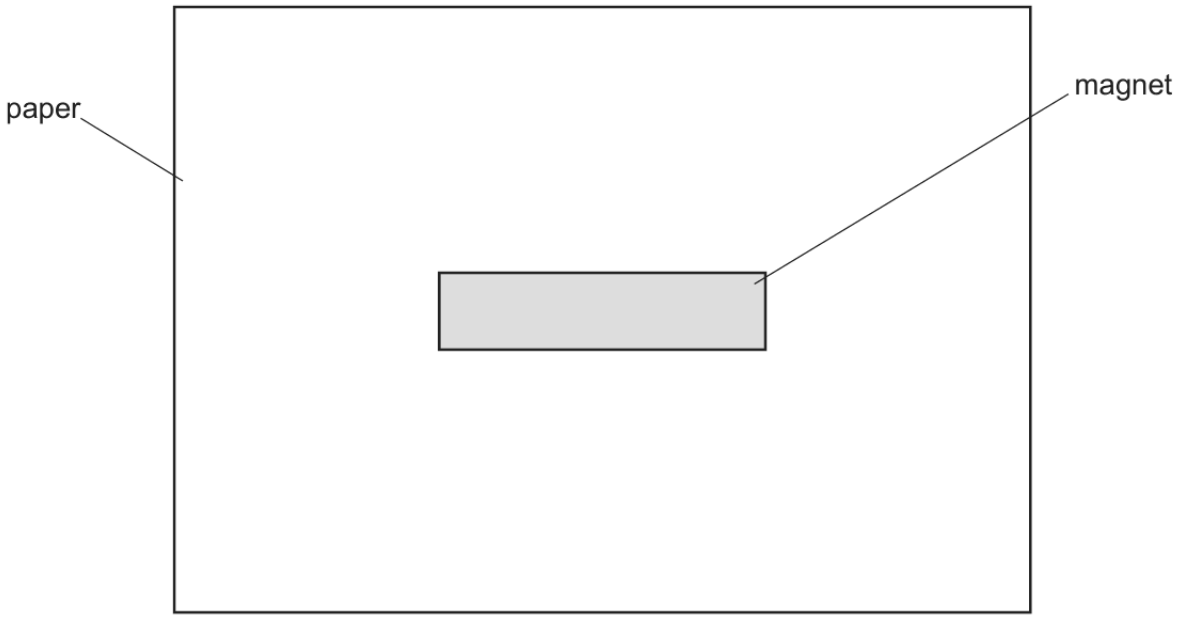


Fig. 9.1

(a) Underline the material in the list from which it is possible to make a strong, permanent magnet.

aluminium copper glass iron mercury plastic steel

[1]

(b) Describe an experiment to plot the pattern and the direction of the magnetic field surrounding the magnet. You may draw on Fig. 9.1, if you wish.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(c) Fig. 9.2 shows the N-pole of a magnet placed in front of the S-pole of a second magnet.

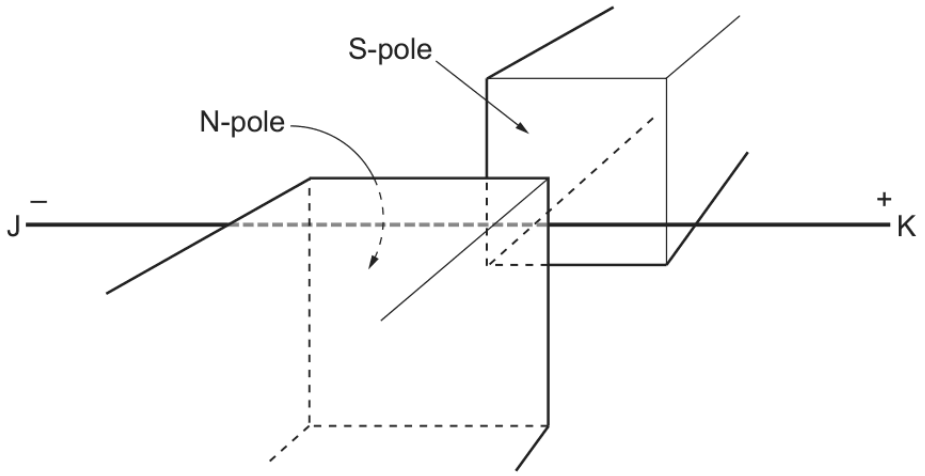


Fig. 9.2

A section of a horizontal, metal wire JK lies in the magnetic field between the two magnetic poles. End K of the metal wire is connected to the positive terminal of a battery and end J is connected to the negative terminal.

(i) Explain in terms of electrons, why there is a current in the wire and state the direction of the conventional current.

.....

.....

..... [2]

(ii) The part of JK that is in the magnetic field experiences a force F.

State the direction of F and describe how this direction is deduced.

.....

.....

.....

.....

..... [4]

(iii) The equipment in Fig. 9.2 is used in a similar experiment.

The part of JK that lies between the poles of the magnets, now passes through a long iron tube that is fixed in position.

The tube is shown in Fig. 9.3.

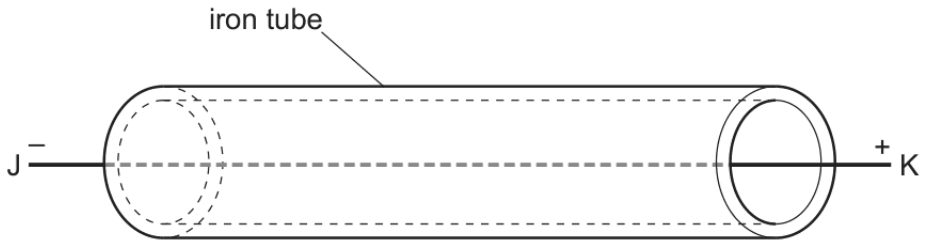


Fig. 9.3

JK is connected to the battery in the same way as before.

State what happens to force F on the wire.

Explain your answer.

.....

.....

..... [2]

(d) The iron tube and the wire JK are removed.

A square, vertical coil is placed between the poles so that the plane of the coil lies in the magnetic field as shown in Fig. 9.4.

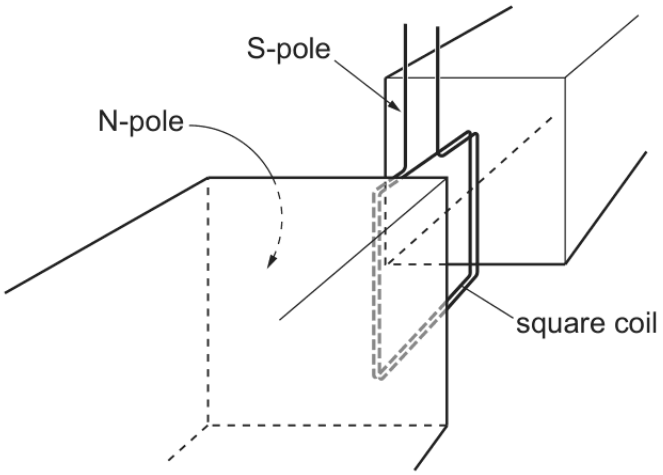


Fig. 9.4

Explain why the coil tries to rotate when there is a current in the coil.

.....

.....

.....

..... [2]

[Total: 15]

MARK SCHEME · 5054_w20_ms_22.pdf · source page 10 of 11 · Question 9

Question	Answer	Marks
9(a)	**steel** underlined **and** no other material indicated	**B1**
9(b)	use of (plotting) compass	**B1**
	place the (plotting) compass (on the paper next to the magnet) **and** mark a dot where the compass needle points	**B1**
	move compass / other end of needle next to the dot **and** mark a dot where the compass needle points	**B1**
	continue **and** needle points in the direction of the magnetic field	**B1**
9(c)(i)	(free) electrons attracted to K / positive terminal **or** repelled by J / negative terminal	**B1**
	(conventional current) from K to J **or** from positive **or** to negative **or** from right **or** to left	**B1**
9(c)(ii)	(Fleming) left-hand rule mentioned **or** equivalent rule mentioned	**B1**
	magnetic field related to index / pointer finger / equivalent	**B1**
	current related to middle finger / equivalent	**B1**
	thumb points down the page / equivalent **and** force down (the page)	**B1**
	or catapult field explanation: field lines clockwise when viewed from K **or** anticlockwise when viewed from J	**B1**
	extra field lines / stronger field above wire	**B1**
	fewer field lines / weaker field below wire	**B1**
	force towards weaker field	**B1**
9(c)(ii)	there is no force F **or** it disappears / becomes equal to zero	**B1**
	wire is magnetically screened / shielded by the iron tube	**B1**

MARK SCHEME · 5054_w20_ms_22.pdf · source page 11 of 11 · Question 9

Question	Answer	Marks
9(d)	moment acts (on coil) **or** forces (on vertical sides) in opposite directions	**B1**
	currents (in vertical sides) **or** in opposite directions	**B1**
	or coil becomes / acts as a magnet **or** coil produces magnetic field	**(B1)**
	N face (of coil) attracted to S pole (of magnet) **or** S face (of coil) attracted to N pole (of magnet) **or** N face (of coil) repelled by N pole (of magnet) **or** S face (of coil) repelled by N pole (of magnet))	**(B1)**

9 Fig. 9.1 is the current–voltage graph for a filament lamp.

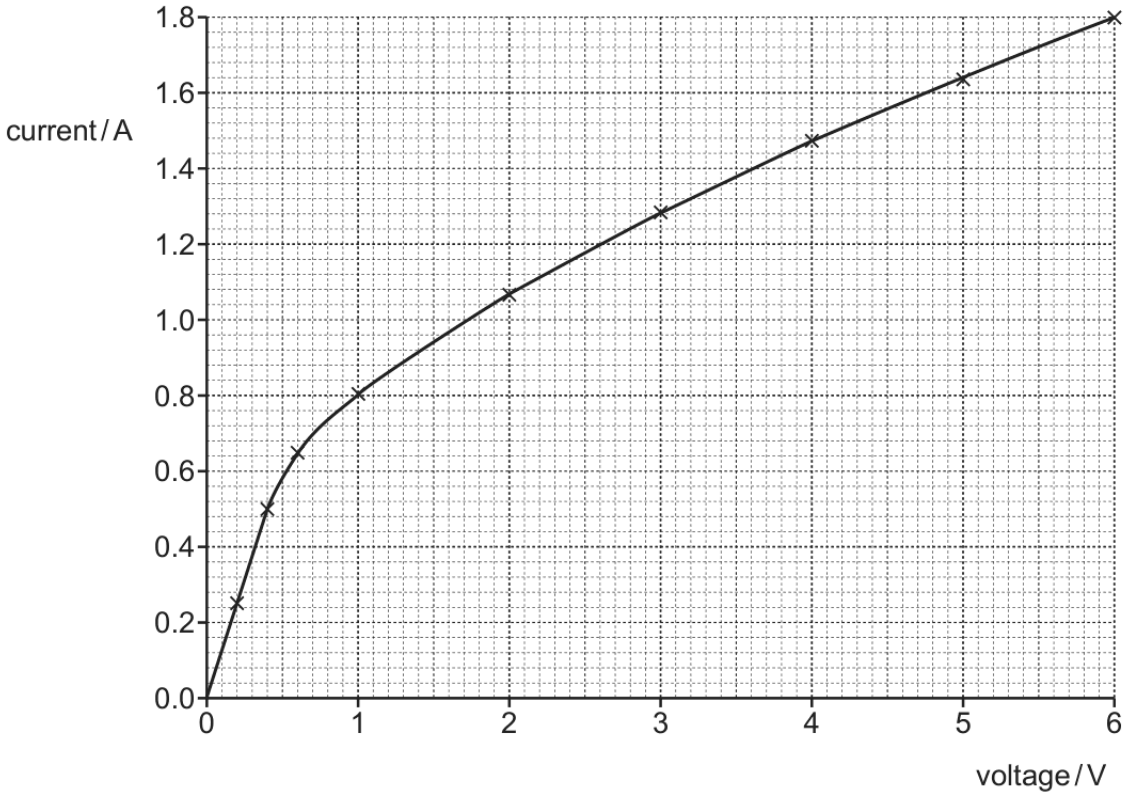


Fig. 9.1

(a) Fig. 9.2 shows an incomplete circuit diagram of the circuit that a student uses to obtain the readings for the graph.

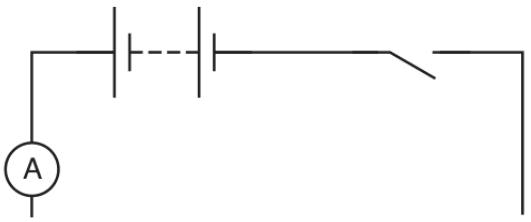


Fig. 9.2

Complete the circuit diagram using suitable circuit symbols for the lamp, a voltmeter and a variable resistor. [3]

(b) The ammeter used by the student has ranges 0−10 A, 0−1 A, 0−100 mA and 0−10 mA.

Describe how the student uses the different ranges to obtain readings which allow all nine points to be plotted precisely on the graph while using the ammeter safely.

.....

.....

.....

..... [2]

(c) State how Fig. 9.1 shows that the lamp does **not** obey Ohm's law.

.....

..... [1]

(d) Using Fig. 9.1, determine the resistance of the lamp when:

(i) the voltmeter reading is 0.40 V

resistance = ...

(ii) the voltmeter reading is 6.0 V.

resistance = ...
[3]

(e) (i) Explain why the resistance of the lamp changes as the voltage changes.

.....

.....

..... [2]

(ii) State how the resistance of a wire depends upon its length and how it depends on its cross-sectional area.

length .....

.....

cross-sectional area .....

.....
[2]

(iii) The student obtains a long sample of the same type of wire as the filament in the lamp.

The wire he obtains has the same cross-sectional area as the filament.

He cuts a 1.0 m length of the wire and measures its resistance at room temperature.

He obtains the value 16 Ω.

Using the appropriate resistance value from **(d)**, estimate the length of the filament wire in the lamp.

length = ... [2]

[Total: 15]

Question 10 begins over the page

MARK SCHEME · 5054_s21_ms_21.pdf · source page 10 of 12 · Question 9

Question	Answer	Marks
9(a)	correct symbol for lamp	**B1**
	correct symbol for voltmeter placed across lamp	**B1**
	correct symbol for variable resistor in series with lamp and battery **or** used as a potential divider	**B1**
9(b)	start on 10 A scale (and lower if reading small) **or** uses 10 A scale for points from 1 to 1.8 A **or** uses 10 A scale for voltages above 2 V (and for 2 V)	**B1**
	for lower readings switch to (0-) 1 A scale	**B1**

MARK SCHEME · 5054_s21_ms_21.pdf · source page 11 of 12 · Question 9

Question	Answer	Marks
9(c)	graph curved **or** not a straight line (through origin) **or** current not (directly) proportional to voltage	**B1**
9(d)(i)	(R =) V / I in any form algebraic or numerical	**C1**
	at 0.40 V, resistance = 0.80 (± 0.03) Ω	**A1**
9(d)(ii)	at 6.0 V, resistance = 3.3–3.4 Ω	**B1**
9(e)(i)	temperature not constant / increases	**C1**
	resistance increases as temperature increases	**A1**
9(e)(ii)	(resistance is directly) proportional to length	**B1**
	(resistance is) inversely proportional to area **or** proportional to 1 / area	**B1**
9(e)(iii)	any clear use of proportionality using 0.80 Ω	**B1**
	0.050 m	**B1**

QUESTION · 5054_s21_qp_12.pdf · source page 12 of 16

33 In which example is the current 1.0 A?

A a charge of 0.10 C passes through a resistor in 100 ms

B a 10 V lamp with input power 0.10 W

C a resistor of resistance 0.50 Ω with a potential difference (p.d.) of 2.0 V across it

D two resistors with a current of 0.50 A connected in series

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 3 of 3 · row for Question 33

33	A	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 **(a)** Fig. 9.1 represents the conversion of energy in a mobile phone.

Fig. 9.1

(i) State the form of energy stored in the battery.

..... [1]

(ii) State the useful form of output energy from the battery.

..... [1]

(iii) State **two** useful forms of output energy from the mobile phone.

..... [2]

(b) When the battery is in use, the average current is 1.3 mA.

Determine the charge that passes through the battery in a time of 2.0 minutes.

charge = ... [2]

(c) A student estimates the thermal energy produced in the battery when the mobile phone is used and uses this value to estimate the efficiency of the battery.

He measures a temperature rise of 5.0 °C within the battery in a period of time when the useful energy output from the battery is 5200 J.

The mass of the battery is 110 g and its specific heat capacity is 830 J / (kg °C).

(i) Calculate the thermal energy (heat) needed to raise the temperature of the battery by 5.0 °C.

thermal energy = ... [3]

(ii) State what is meant by *efficiency*.

.....

..... [2]

(iii) Calculate the efficiency of the battery.

efficiency = ... [2]

(iv) Suggest and explain **one** reason why the value for the efficiency calculated in **(iii)** is larger than the actual efficiency of the battery.

.....

.....

.....

..... [2]

[Total: 15]

MARK SCHEME · 5054_s21_ms_22.pdf · source page 13 of 16 · Question 9

Question	Answer	Marks
9(a)(i)	chemical	**B1**
9(a)(ii)	electrical	**B1**
9(a)(iii)	ANY 2 from • light • sound • radio / microwave / electromagnetic (waves)	**B2**
9(b)	(*Q* =) *It* in any form	**C1**
	0.16 C	**A1**

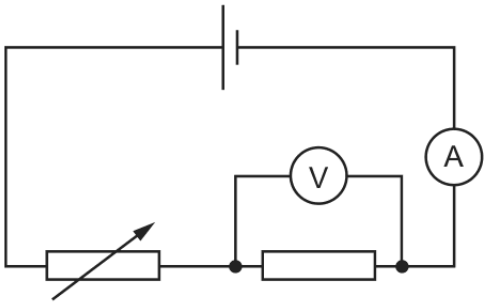
MARK SCHEME · 5054_s21_ms_22.pdf · source page 14 of 16 · Question 9

Question	Answer	Marks
9(c)(i)	(*E* =) *mcT* numerical or algebraic in any form	**C1**
	$0.11 \times 830 \times 5$	**C1**
	460 J	**A1**
9(c)(ii)	energy output / energy input	**C1**
	useful energy output / (total) energy input	**A1**
9(c)(iii)	(energy input =) 460 + 5200 **or** 5660 (J) seen **or** 5200 ÷ (energy in **9(c)(i)** + 5200)	**C1**
	92% **or** 0.92	**A1**
9(c)(iv)	(thermal) energy lost / heat lost (from battery to environment)	**B1**
	energy input larger than calculated / 5660 (J) **or** temperature rise should be larger / is smaller than it should be **or** thermal energy / heat produced larger than 460 J / calculated	**B1**

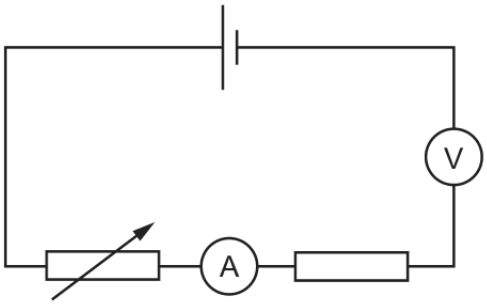
QUESTION · 5054_w21_qp_11.pdf · source page 14 of 20

30 Which circuit is connected to measure the current in the fixed resistor and the potential difference (p.d.) across the same resistor?

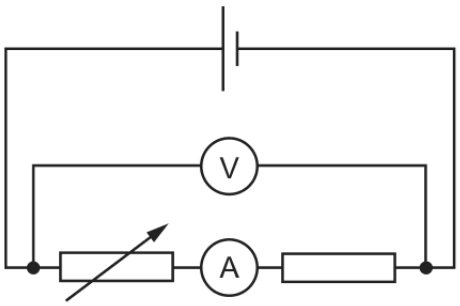
A



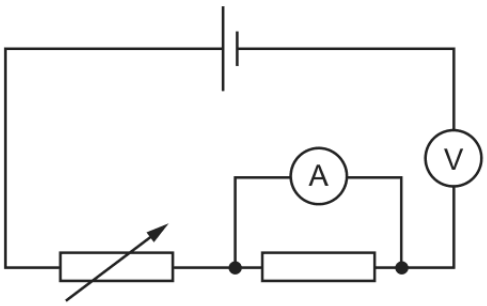
B



C



D



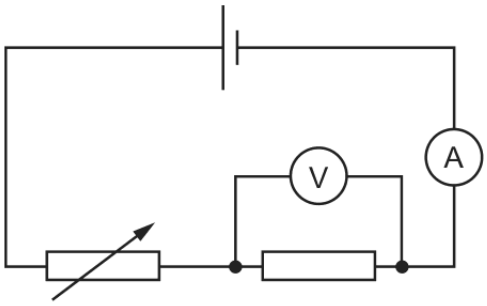
MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 3 of 3 · row for Question 30

30	A	**1**

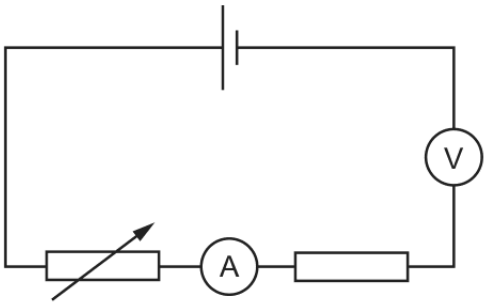
QUESTION · 5054_w21_qp_12.pdf · source page 13 of 16

32 Which circuit is connected to measure the current in the fixed resistor and the potential difference (p.d.) across the same resistor?

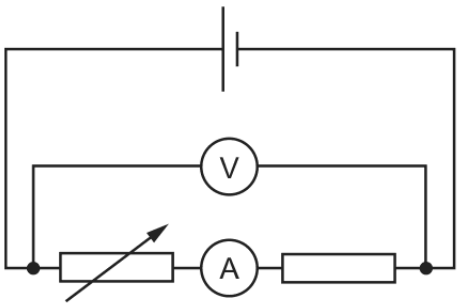
A



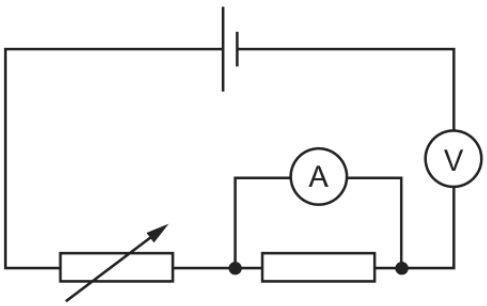
B



C



D



MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 3 of 3 · row for Question 32

32	A	**1**

5 EITHER

Fig. 5.1 shows a circuit that contains a direct current (d.c.) power supply, a light-dependent resistor (LDR), two fixed resistors, M and N, and two ammeters.

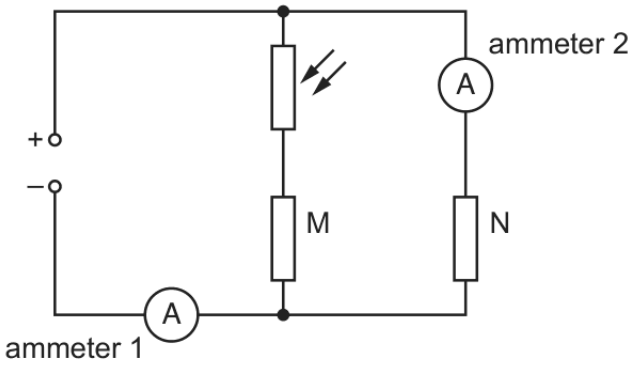


Fig. 5.1

(a) The reading on ammeter 1 is 1.6 mA.

Calculate the time taken for a charge of 7.2 C to pass through ammeter 1.

time = ... [2]

(b) Some time later, the brightness of the light incident on the LDR decreases.

Explain what happens to:

(i) the reading on ammeter 2

.....

..... [1]

(ii) the time taken for a charge of 7.2 C to pass through ammeter 1.

.....

.....

..... [2]

[Total: 5]

OR

Fig. 5.2 shows the symbol for an electronic circuit component.

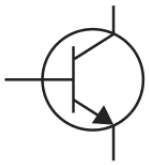


Fig. 5.2

(a) State the name of the component shown in Fig. 5.2.

..... [1]

(b) Fig. 5.3 is the diagram of a circuit that includes this component, a light-dependent resistor (LDR) and a relay.

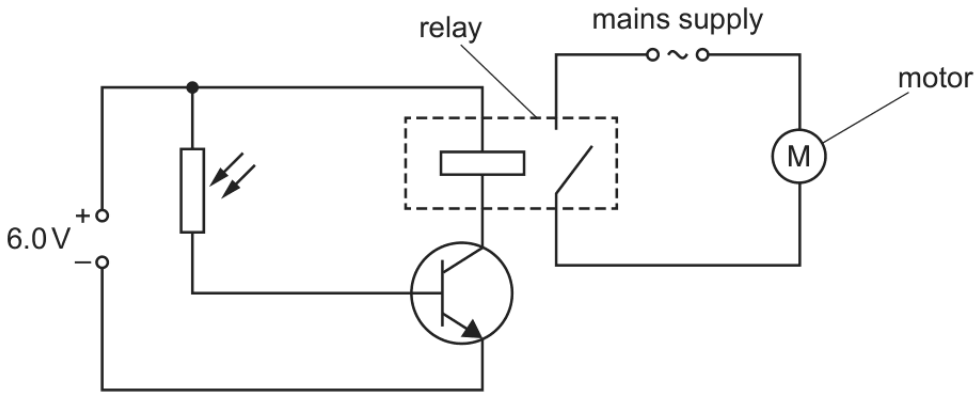


Fig. 5.3

The motor in Fig. 5.3 is part of the pump in a garden fountain. In the dark, the pump does not work.

In the morning, the brightness of the light incident on the LDR increases.

(i) State what happens to the LDR.

..... [1]

(ii) Explain why the motor switches on.

.....

.....

.....

..... [3]

[Total: 5]

Question	Answer	Marks
5E(a)	(t =) Q / I **or** 7.2 / 0.0016 **or** 7.2 / 1.6 **or** 4.5×10^{N}	**C1**
	4500 s	**A1**
5E(b)(i)	does not change **and** p.d. / voltage across resistor N does not change / ammeter 2 is not in the branch that contains the LDR	**B1**
5E(b)(ii)	resistance of LDR increases **or** current (in M / in ammeter 1) decreases	**B1**
	time taken increases	**B1**
5O(a)	(bipolar npn) transistor	**B1**
5O(b)(i)	its resistance decreases **or** current in the LDR increases	**B1**
5O(b)(ii)	base current **or** base-emitter voltage increases	**B1**
	there is now a current in the relay coil **or** transistor switches (circuit) on **or** coil / core magnetised	**B1**
	relay coil attracts the relay switch / closes switch (in the mains circuit)	**B1**

9 A filament lamp is connected to a 240 V alternating current (a.c.) mains supply.

(a) Describe how the output of an a.c. supply differs from the output of a direct current (d.c.) supply.

Sketch **two** voltage–time graphs in the blank space to help your explanation.

.....

..... [3]

(b) The lamp is rated at 60 W and is designed to be used with a 240 V supply.

(i) Calculate the current in the lamp.

current = ... [2]

(ii) Calculate the resistance of the lamp.

resistance = ... [2]

(iii) A room is lit by five of these filament lamps connected in parallel.

State **two** advantages of connecting the lamps in parallel rather than in series.

1. .....

.....

2. .....

.....

[2]

(iv) The five lamps in **(b)(iii)** are lit for an average time of 5.5 hours a day for a year. Electricity costs \$0.15 / kW h.

Calculate the cost of using these lamps for a year.

cost = ... [2]

(c) A student takes the lamp in **(b)** to school and connects it in a circuit using a 1.5 V cell, an ammeter and a voltmeter. The circuit is used to determine the resistance of the lamp.

(i) In the blank space, draw the circuit diagram of the circuit used to determine the resistance of the lamp.

[2]

(ii) The value of the resistance of the filament lamp in this circuit differs greatly from the value calculated in **(b)(ii)**.

State how the resistance value in this circuit differs and explain why it differs.

.....

.....

..... [2]

[Total: 15]

MARK SCHEME · 5054_w21_ms_22.pdf · source page 10 of 11 · Question 9

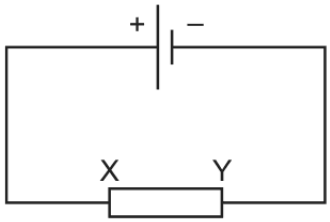
Question	Answer	Marks
9(a)	sinusoidal voltage-time graph labelled a.c. drawn	**B1**
	voltage-time labelled d.c. graph with horizontal, non-zero line	**B1**
	an a.c. varies with time **or** an a.c. changes direction / polarity (at regular time intervals) **or** a d.c. is constant with time **or** d.c. is never negative	**B1**
9(b)(i)	(I =) P / V **or** 60 / 240	**C1**
	0.25 A	**A1**
9(b)(ii)	(R =) V / I **or** 240 / 0.25	**C1**
	960 Ω	**A1**
9(b)(iii)	any **two** from: if one lamp blows the others stay lit can be operated separately each lamp has the correct voltage across it **or** lamps glow brightly / normally	**B2**

MARK SCHEME · 5054_w21_ms_22.pdf · source page 11 of 11 · Question 9

Question	Answer	Marks
9(b)(iv)	(number of units =) $5 \times 60 \times 5.5 \times 365 / 1000$ **or** 600 **or** 602.25 **or** 18(.07)	**C1**
	\$90 **or** \$90.34	**A1**
9(c)(i)	cell, lamp and ammeter in series **and** no other components	**B1**
	voltmeter in parallel with lamp	**B1**
9(c)(ii)	(in this circuit) the resistance is (very much) smaller	**B1**
	(metal) filament is cooler	**B1**

QUESTION · 5054_s22_qp_11.pdf · source page 12 of 16

31 The diagram shows a simple electric circuit.



Which row describes the charge on an electron and the direction of electron flow through the resistor?

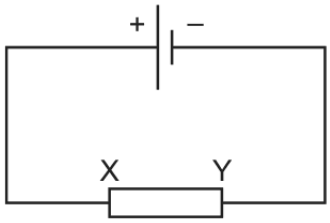
	charge on an electron	direction of electron flow
A	negative	X to Y
B	negative	Y to X
C	positive	X to Y
D	positive	Y to X

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 3 of 3 · row for Question 31

31	B	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 15 of 20

32 The diagram shows a simple electric circuit.



Which row describes the charge on an electron and the direction of electron flow through the resistor?

	charge on an electron	direction of electron flow
A	negative	X to Y
B	negative	Y to X
C	positive	X to Y
D	positive	Y to X

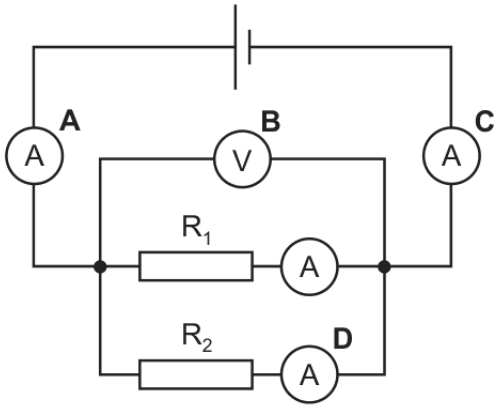
MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 3 of 3 · row for Question 32

32	B	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 15 of 20

34 The diagram shows a simple electric circuit.

Which meter measures the current in resistor R_2?



MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 3 of 3 · row for Question 34

34	D	**1**

6 An electromagnet is used to separate objects that are magnetic from objects that are non-magnetic.

Fig. 6.1 shows the electromagnet suspended from the arm of a crane.

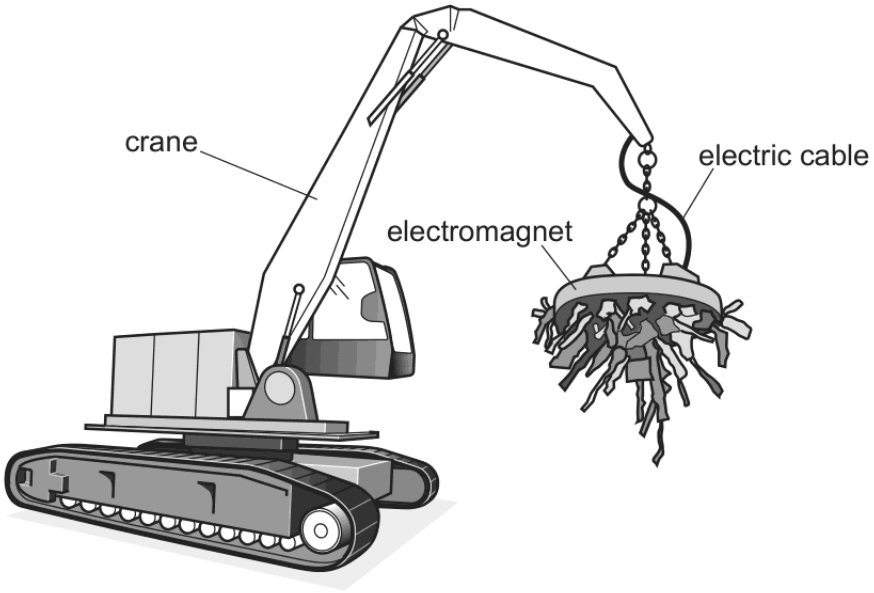


Fig. 6.1

(a) State the name of the metal from which the core of the electromagnet is made.

..... [1]

(b) The electromagnet is powered by a 220 V d.c. (direct current) supply.

The electromagnet is switched on and the current in the circuit is 39 A.

(i) Calculate the power transferred to the electromagnet.

power = ... [2]

(ii) The d.c. supply is a set of batteries.

The batteries power the magnet for 4.5 hours before they need to be replaced.

Calculate the charge driven around the complete circuit in this time.

charge = ... [3]

(c) On a very cold morning, when the electromagnet is first switched on, the current is greater than 39A before decreasing to the usual value.

Explain why the current is initially greater than 39A.

.....

..... [1]

[Total: 7]

Question	Answer	Marks
6(a)	soft iron	**B1**
6(b)(i)	(P =) VI **or** 220 × 39	**C1**
	8.6×10^3 W	**A1**
6(b)(ii)	(Q =) It **or** 39 × 4.5 **or** 39 × 4.5 × 60 **or** 39 × 4.5 × 60 × 60 **or** 39 × 4.5 × 3600	**C1**
	180 **or** 11 000 **or** 39 × 4.5 × 60 × 60 **or** 39 × 4.5 × 3600	**C1**
	6.3×10^5 C	**A1**
6(c)	resistance increases with temperature **or** resistance lower (because of (low / ambient) temperature)	**B1**

9 The power supply in an electric circuit is a battery of electromotive force (e.m.f.) 12 V.

(a) State **two** ways in which the e.m.f. of a battery differs from that of an alternating current (a.c.) power supply.

1 .....

2 .....

[2]

(b) The circuit includes three resistors and two open switches, S_1 and S_2.

Fig. 9.1 shows the circuit.

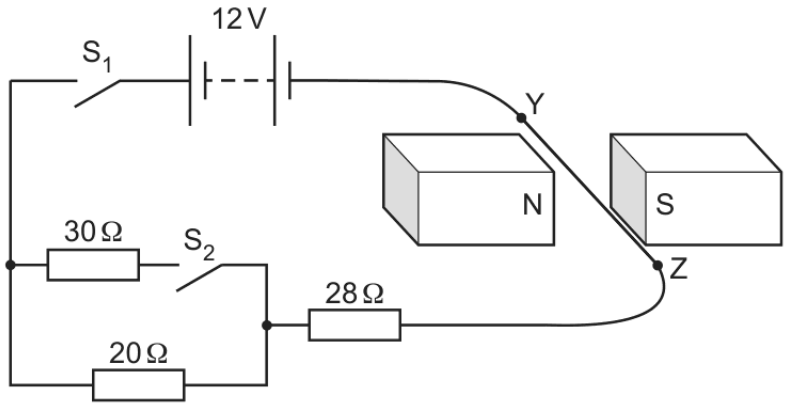


Fig. 9.1

YZ is a straight, horizontal section of connecting wire that lies between two magnets.

S_1 is now closed.

(i) Calculate the current in YZ.

current = ... [2]

(ii) Explain why YZ experiences a force.

.....

.....

..... [2]

(iii) Tick the box which describes the direction of the force on YZ.

towards N	☐	towards Z	☐
towards S	☐	downwards	☐
towards Y	☐	upwards	☐

[1]

(iv) Explain how the direction of the force on YZ is determined.

.....

.....

..... [2]

(c) Switch S_2 in the circuit in Fig. 9.1 is now also closed.

(i) Calculate the total resistance of the circuit.

resistance = ... [3]

(ii) Explain what happens to the force on YZ as switch S_2 is closed.

.....

.....

..... [2]

(iii) The current in the 20 Ω resistor is I_{20}. The current in the 30 Ω resistor is I_{30}.

State a value for the ratio I_{20}/I_{30}.

ratio I_{20}/I_{30} = ... [1]

[Total: 15]

Question	Answer	Marks
9(a)	always positive **or** never negative **or** direction does not change	**B1**
	constant value **or** size does not change	**B1**
9(b)(i)	(I =) V / R **or** 12 / (20 + 28) **or** 12 / 48	**C1**
	0.25 A	**A1**
9(b)(ii)	YZ in magnetic <u>field</u> **or** magnetic <u>field</u> between poles (of magnet)	**M1**
	(electric) current in YZ **or** interacts with field of current / YZ	**A1**
9(b)(iii)	downwards; middle, right-hand box ticked **i.e.** towards N ☐ towards Z ☐ towards S ☐ downwards ✓ towards Y ☐ upwards ☐	**B1**
9(b)(iv)	current from Z to Y page / into page **and** magnetic field left to right	**B1**
	(Fleming's) left-hand rule / other appropriate rule mentioned / described	**B1**

Question	Answer	Marks
9(c)(i)	$1 / R_T = 1 / R_1 + 1 / R_2$ **or** $(R_T =)\ R_1 R_2 / (R_1 + R_2)$ **or** $1 / R_T = 1 / 20 + 1 / 30$ **or** $1 / R_T = 5 / 60$ **or** $1 / R_T = 1 / 12$ **or** $(R_T =)\ 20 \times 30 / (20 + 30)$	**C1**
	12	**C1**
	40 W	**A1**
9(c)(ii)	resistance decreases **or** resistance is less (with S_2 closed)	**B1**
	current increases **and** force increases	**B1**
9(c)(iii)	1.5	**B1**

QUESTION · 5054_w22_qp_12.pdf · source page 13 of 16

32 A meter that measures the rate of flow of charge in a resistor is placed in a circuit.

What is the name of this meter and how is it connected to the resistor?

	name	connection
A	ammeter	in parallel
B	ammeter	in series
C	voltmeter	in parallel
D	voltmeter	in series

MARK SCHEME (official answer key) · 5054_w22_ms_12.pdf · source page 3 of 3 · row for Question 32

32	B	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 11 of 16

27 A metal conductor is connected between the positive and negative terminals of a battery.

Which row gives the direction of movement of the particles that flow in the conductor and their name?

	direction of movement	name of particles
A	from positive terminal to negative terminal	protons
B	from positive terminal to negative terminal	electrons
C	from negative terminal to positive terminal	protons
D	from negative terminal to positive terminal	electrons

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 27

27	D	**1**

3 Fig. 3.1 shows a solar-powered charger connected to a cell phone (mobile phone).

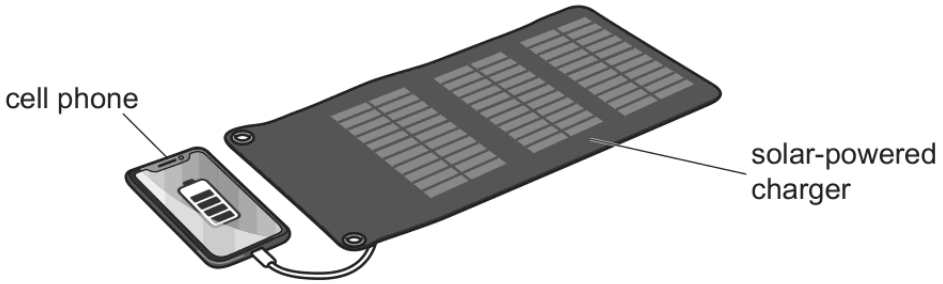


Fig. 3.1

The battery inside the cell phone is charged by the solar-powered charger.

(a) **(i)** Complete Fig. 3.2 to show the useful transfer of energy from the Sun to the battery.

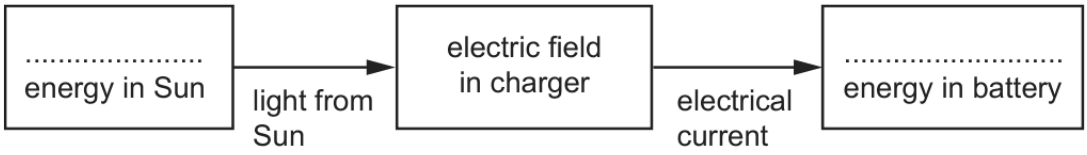


Fig. 3.2

[2]

(ii) Explain why the battery takes a long time to charge on a cloudy day.

.....

..... [1]

(b) After use, the outside surface of the cell phone is warm. When switched off, the cell phone cools down.

Name and describe the **three** processes by which thermal energy is transferred as the cell phone cools down.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) It takes 4.5 hours to charge the battery with an average current of 300 mA.

Calculate the quantity of charge that enters the battery. Give the unit of your answer.

charge = ... unit ..... [3]

[Total: 9]

Question	Answer	Marks
3(a)(i)	nuclear (energy in Sun)	**B1**
	chemical (energy in battery)	**B1**
3(a)(ii)	less / no light / energy (reaching the charger) **or** less / no electrical current (produced)	**B1**
3(b)	conduction **and** energy passing from particle to particle in solid / from outside particles to air particles	**B1**
	convection **and** in air / hot air rises	**B1**
	radiation **and** emitted from surface	**B1**
3(c)	(Q =) It in any form	**C1**
	4 900	**A1**
	C **or** coulomb	**B1**

QUESTION · 5054_s25_qp_11.pdf · source page 12 of 20

28 A 12 V battery is connected to a resistor of resistance 100 Ω.

How much charge passes through the resistor in 30 minutes?

A 3.6 C **B** 220 C **C** 250 C **D** 15 000 C

MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 2 of 3 · row for Question 28

28	B	**1**

QUESTION · 5054_s25_qp_11.pdf · source page 12 of 20

29 A circuit consists of a resistor connected between the positive terminal and the negative terminal of a power supply.

Which row describes the direction of conventional current and the direction of flow of free electrons in the resistor?

	direction of conventional current	direction of flow of free electrons
A	from negative to positive	from negative to positive
B	from negative to positive	from positive to negative
C	from positive to negative	from negative to positive
D	from positive to negative	from positive to negative

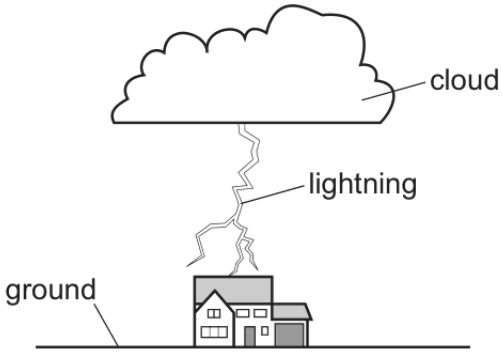
MARK SCHEME (official answer key) · 5054_s25_ms_11.pdf · source page 3 of 3 · row for Question 29

29	C	**1**

QUESTION · 5054_w25_qp_11.pdf · source page 14 of 16

30 The diagram shows a lightning strike.

Charge flows between the cloud and objects on the ground.



not to scale

During the lightning strike, there is a current of 2.8×10^4 A between the cloud and the house for 52 ms.

How much charge flows during the lightning strike?

A 1500 C **B** 15 000 C **C** 24 000 C **D** 54 000 C

MARK SCHEME (official answer key) · 5054_w25_ms_11.pdf · source page 3 of 3 · row for Question 30

30	A	**1**

6 An electric car is driven by a direct current (d.c.) motor that is powered by a direct current power supply.

(a) Describe how a direct current differs from an alternating current (a.c.).

.....

.....

.....

..... [1]

(b) Fig. 6.1 represents the motor circuit. Point P is a point in the circuit between the motor and the positive terminal of the power supply.

Point P is shown to the left of the motor in the circuit diagram.

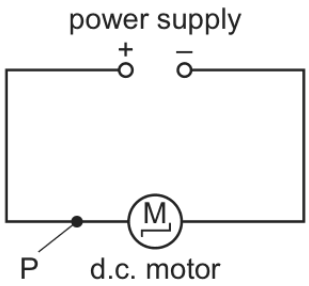


Fig. 6.1

During a time of 120 s, the current in the circuit at point P is 41 A.

(i) Calculate the charge that flows through point P in the time of 120 s.

charge = ... C [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

(ii) The current in the circuit is due to the flow of electrons. Each electron carries a charge of magnitude 1.6×10^{-19} C.

State the direction which the electrons move past point P and determine the number of electrons that pass point P in 1.0 s.

direction .....

number of electrons = ...
[2]

(c) The power supply for the electric car is a number of batteries, each made from a large number of identical cells.

The e.m.f. (electromotive force) of each cell is 3.7 V.

(i) Define the term 'electromotive force'.

.....

.....

..... [2]

(ii) Each battery consists of 92 cells in series.

Calculate the e.m.f. of one battery.

e.m.f. = ... V [2]

(iii) The power supply is made by connecting 85 of these batteries in parallel.

State the e.m.f. of the power supply.

e.m.f. = ... V [1]

[Total: 10]

MARK SCHEME · 5054_w25_ms_22.pdf · source page 11 of 14 · Question 6

Question	Answer	Marks
6(a)	a d.c. / it has one direction **or** does not (repeatedly) reverse direction **or** polarity / direction does not change	**B1**
6(b)(i)	(Q =) It **or** 41 × 120	**C1**
	4.9×10^3 (C)	**A1**

MARK SCHEME · 5054_w25_ms_22.pdf · source page 12 of 14 · Question 6

Question	Answer	Marks
6(b)(ii)	(number =) $\frac{41}{1.6 \times 10–19}$ **or** $\frac{4.9 \times 103}{120 \times 1.6 \times 10–19}$ **or** 2.6×10^N	**C1**
	2.6×10^{20} **and** towards the positive terminal / away from the motor	**A1**
6(c)(i)	(electrical) work done (by a source) in moving charge around a (complete) circuit;	**B1**
	work done (by a source) per <u>unit</u> charge	**B1**
6(c)(ii)	(e.m.f. =) 92 × 3.7	**C1**
	340 (V)	**A1**
6(c)(iii)	same answer as candidate's **6(c)(ii) or** 340 (V)	**B1**

QUESTION · 5054_s26_qp_11.pdf · source page 13 of 16

31 What is a possible unit for electric current?

A ampere / second

B coulomb / second

C joule / second

D volt / second

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 3 of 3 · row for Question 31

31	B	**1**

Chapter 4 Electricity and magnetism

4.2 Electrical quantities

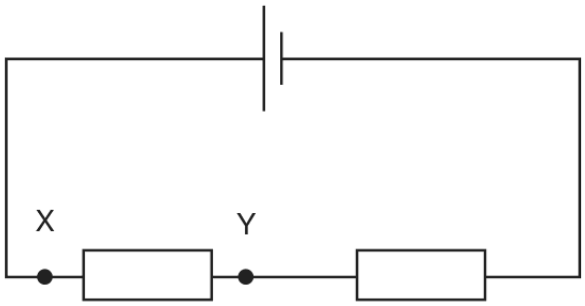
4.2.3 Electromotive force and potential difference

73 entries: 44 Paper 1 questions, 29 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

QUESTION · 5054_s04_qp_1.pdf · source page 13 of 20

28 The diagram shows two resistors connected in series with a cell.



Which statement defines the potential difference across XY?

A the power needed to drive a unit charge through the cell

B the power needed to drive a unit charge between X and Y

C the work done in driving a unit charge through the cell

D the work done in driving a unit charge between X and Y

MARK SCHEME (official answer key) · 5054_s04_ms.pdf · source page 3 of 16 · row for Question 28

28	**D**

For Examiner's Use

6 Fig. 6.1 shows a circuit in which a voltmeter is placed across a resistor.

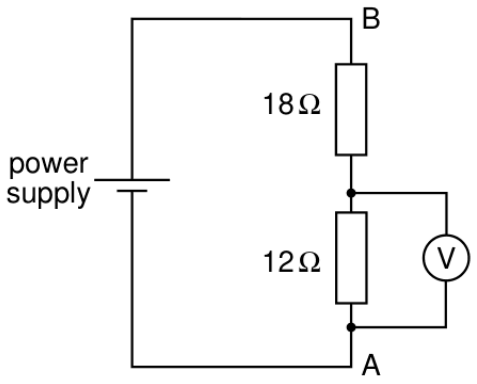


Fig. 6.1

The potential difference across the 12 Ω resistor is 4.0 V.

The voltmeter has three different ranges: 0 to 3.0 V, 0 to 6.0 V and 0 to 30 V. The best range for use in this circuit is 0 to 6.0 V.

(a) Explain why

(i) using the voltmeter on the range 0 to 3.0 V is unsuitable,

.....

.....

(ii) using the voltmeter on the range 0 to 30 V is unsuitable.

.....

.....

[2]

(b) (i) Calculate the current in the 12 Ω resistor. State the formula that you use.

current = ...

(ii) Calculate the p.d. between A and B in Fig. 6.1.

p.d. = ...

[5]

6	**(a)**	**(i)**	voltage past maximum or 3V / off scale / outside range	B1
		(ii)	reading less accurate or sensitive / not far up scale or smaller deflection	
	(b)	**(i)**	V = I R in any algebraic format	B1
			4/12	C1
			0.33 A (accept 1/3 A)	A1
		(ii)	**(i)** * 30 **or (i)** * 18 + 4 **or** 30*4/12	C1
			9.9 – 10 V (e.c.f **(i)**, e.g. if **(i)** = 0.3, 0.3*30 = 9V or 0.3*18+4 = 9.4 V)	A1
			only 1 unit error in this question	
			Total	**[7]**

QUESTION · 5054_w04_qp_1.pdf · source page 13 of 20

28 The potential difference between the ends of a conductor is 12 V.

How much electrical energy is converted to other forms of energy in the conductor when 100 C of charge flows through it?

A 0.12 J **B** 8.3 J **C** 88 J **D** 1200 J

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 28

28	**D**

QUESTION · 5054_s05_qp_1.pdf · source page 11 of 20

27 A cell is connected to a resistor.

What is the e.m.f. of the cell equal to?

A The potential difference across the resistor for each unit of current.

B The power produced in the circuit for each unit of charge that passes.

C The work done in the circuit for each unit of charge that passes.

D The work done in the circuit for each unit of current.

MARK SCHEME (official answer key) · 5054_s05_ms_1.pdf · source page 3 of 3 · row for Question 27

27	**C**

For Examiner's Use

8 Three cells are connected in series making a battery, as shown in Fig. 8.1. The e.m.f. of each cell is 1.5 V. A resistance of 15 Ω is connected to the battery.

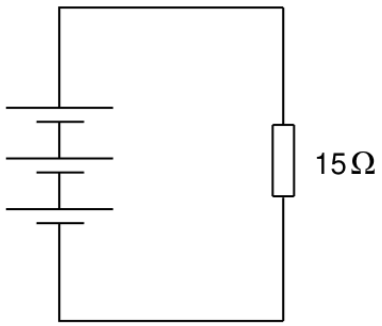


Fig. 8.1

(a) What is the total e.m.f. of the battery?

..... [1]

(b) Calculate the current in the circuit.

State the equation that you use.

current = ... [3]

(c) A battery can be made from cells connected in parallel, as shown in Fig. 8.2.

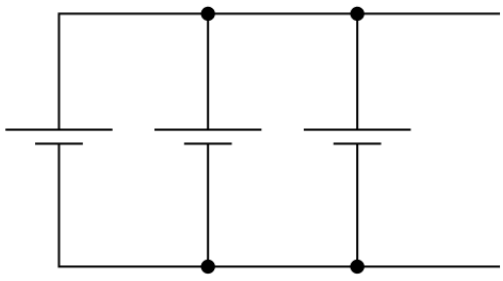


Fig. 8.2

State one advantage of connecting the cells in parallel.

.....

.....

.....

..... [1]

8	**(a)** 4.5 V	**B1**	
	(b) I =V/R in any form using symbols or words	**B1**	
	4.5/15	**C1**	
	0.3 A	**A1**	
	(c) provides smaller (internal) resistance **or** lasts longer **or** less lost voltage **or** one (cell) fails others work **or** less heat/energy lost	**B1**	**5**

QUESTION · 5054_w05_qp_1.pdf · source page 14 of 20

27 Why can birds stand on an overhead high voltage transmission line without suffering any harm?

A Their bodies have a very high resistance.

B Their feet are very good insulators.

C The spaces between their feathers act as insulators.

D They are not connected to earth.

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 27

27	**D**

QUESTION · 5054_s06_qp_1.pdf · source page 12 of 20

30 An electrical quantity is defined as 'the energy converted by a source in driving a unit charge round a complete circuit.'

What is this quantity called?

A current

B electromotive force

C potential difference

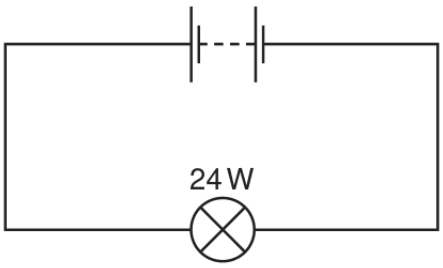
D power

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 30

30	**B**

QUESTION · 5054_w07_qp_1.pdf · source page 12 of 20

28 A battery is used to light a 24 W electric lamp. The battery provides a charge of 120 C in 60 s.



What is the potential difference across the bulb?

A 5 V **B** 12 V **C** 24 V **D** 120 V

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 28

28	**B**

For Examiner's Use

7 A motorcycle battery consists of six 2.0 V cells in series. The battery supplies energy to the headlight.

(a) State the total electromotive force (e.m.f.) of the battery.

..... [1]

(b) The motorcycle headlight contains two identical filament lamps F and G. Filament lamp F is always lit but filament lamp G is turned on and off by switch D.
Fig. 7.1 is the circuit diagram.

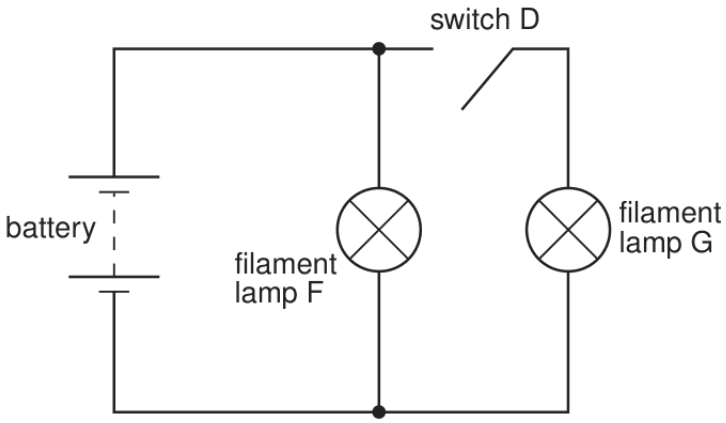


Fig. 7.1

When switch D is open, the battery supplies a current of 4.6 A.

Complete the table of Fig. 7.2.

	current supplied by battery / A	current in filament lamp F / A	current in filament lamp G / A
switch D open	4.6		
switch D closed			

Fig. 7.2 [3]

(c) Calculate the energy supplied by the battery as an electric charge of 200 C moves through the circuit.

energy supplied = ... [2]

7	**(a)**	12(.0) V	B1	
	(b)	top row: 4.6 **and** 0	B1	
		bottom row: square 1 = square 2 + square 3 **or** 9.2	B1	
		bottom row: 4.6 in squares 2 **and** 3 **cao**	B1	
	(c)	(*E*=) *QV* **or** *VIt* **or** 200 × 12	C1	
		2400 J **accept** 2370–2410 J e.c.f.	A1	[6]

QUESTION · 5054_s09_qp_1.pdf · source page 12 of 20

27 One volt can also be written as

A one coulomb per ampere.

B one coulomb per joule.

C one joule per ampere.

D one joule per coulomb.

MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 27

27	**D**

11 Fig. 11.1 shows a 9.0V battery connected in series with a 16.0Ω resistor and a small metal conductor X at room temperature.

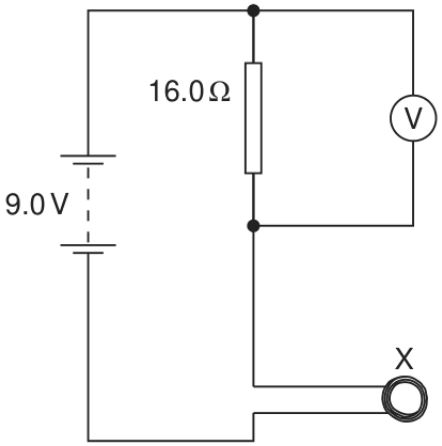


Fig. 11.1

A voltmeter measures the potential difference (p.d.) across the 16.0Ω resistor. At room temperature the resistance of X is 4.0Ω.

(a) (i) Calculate the current supplied by the battery. [2]

(ii) Calculate the p.d. across the 16.0Ω resistor. [1]

(b) X is heated slowly to a very high temperature.

(i) Sketch a graph to show how the resistance of a metal conductor depends on its temperature. [2]

(ii) State and explain how the voltmeter reading changes as the temperature of X rises. [2]

(iii) Suggest a suitable range for the voltmeter. [2]

There is a choice of part (c) on the following page.

EITHER (c) (i) Describe in outline how the circuit in Fig. 11.1 can be used as a thermometer. [2]

(ii) State two advantages of a thermometer such as this over a liquid-in-glass thermometer. [2]

(iii) The change in voltmeter reading with temperature is non-linear. Explain what is meant by *non-linear*. [2]

OR **(c)** The voltmeter is removed from the circuit in Fig. 11.1 and the negative terminal of the battery is earthed. A connection is made from the circuit to the inputs A and B of a NAND gate. The output of the NAND gate is connected to an LED (light-emitting diode) and a resistor. This is shown in Fig. 11.2.

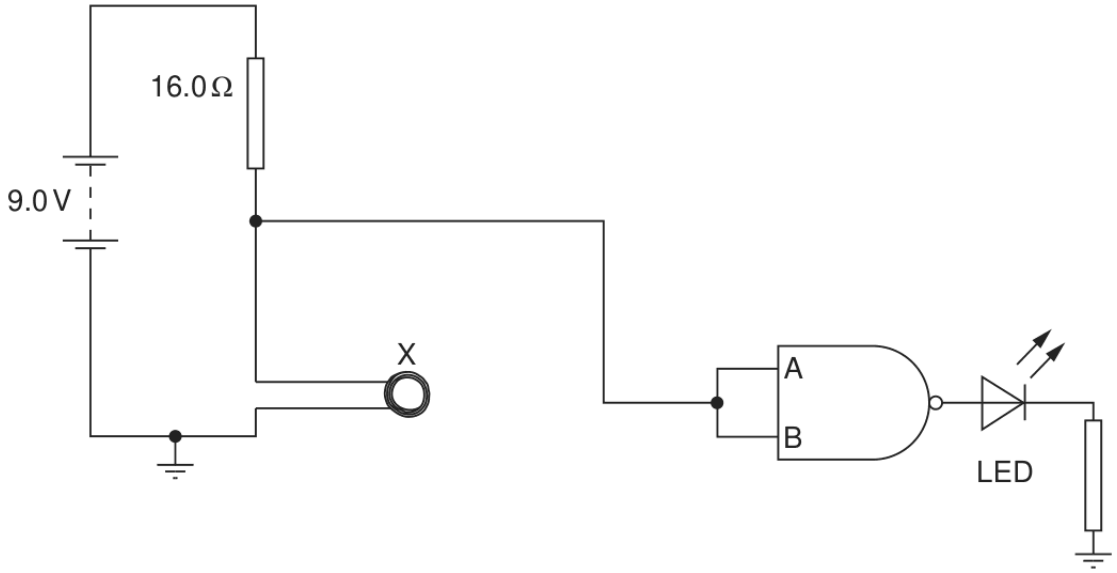


Fig. 11.2

(i) Draw the truth table for a NAND gate. [2]

(ii) The LED is lit. State the inputs at A and B and explain why these inputs cause the LED to light. [2]

(iii) X is slowly heated to a very high temperature. At a certain temperature, the LED switches off. Explain why this happens. [2]

11 (a) (i) (I =)V/R **or** V = IR (in **(i)**/**(ii)**) **or** 9.0/20 (in **(i)**) **or** 0.45 × 16 (in **(ii)**) C1
0.45 A A1

(ii) 7.2 V (max 3 for **(i)** and **(ii)** together) A1 [3]
C1 may be awarded for **either** A mark

(b) (i) R → T **and** line of positive slope throughout B1
straight line, positive intercept on R-axis **and** slope/0 on **kelvin** scale B1

(ii) voltmeter reading falls B1
current (supplied by battery) falls **or** X takes greater proportion of p.d. **or** 16 Ω takes smaller proportion of p.d. B1

(iii) 0 **and** to/→/- B1
8/9/10/whole number not greater than 20 V (usual unit penalty) B1 [6]

EITHER:

(c) (i) use small, metal conductor as probe/sensor **or** calibrate V reading (with known T) B1
the voltmeter reading is used to find T B1

(ii) any **two** from: high temperatures /remote reading/robust/quick acting/direct input to computer/low heat capacity B2

(iii) equal changes in one/T do **not** produce a equal changes in the other/V **or** graph with axes labelled **not** straight **or not** proportional to B2 [6]
not a straight line **or** not same change **or** change in V is not uniform 1/2

OR:

(c) (i)

In		Out	
0	0	1	all correct B1
0	1	1	
1	0	1	
1	1	0	correct B1

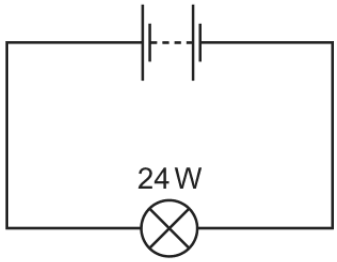
(ii) both inputs = 0 B1
output = 1 B1

(iii) A **and** B inputs = 1 B1
output = 0 B1 [6]

[Total: 15]

QUESTION · 5054_w10_qp_11.pdf · source page 12 of 20

28 A battery is used to light a 24 W electric lamp. The battery provides a charge of 120 C in 60 s.



What is the potential difference across the lamp?

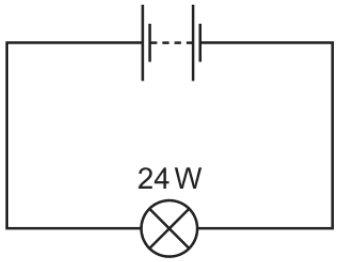
A 5 V **B** 12 V **C** 24 V **D** 120 V

MARK SCHEME (official answer key) · 5054_w10_ms_11.pdf · source page 2 of 2 · row for Question 28

28	**B**

QUESTION · 5054_w10_qp_12.pdf · source page 12 of 16

30 A battery is used to light a 24 W electric lamp. The battery provides a charge of 120 C in 60 s.



What is the potential difference across the lamp?

A 5 V **B** 12 V **C** 24 V **D** 120 V

MARK SCHEME (official answer key) · 5054_w10_ms_12.pdf · source page 2 of 2 · row for Question 30

30	**B**

11 Fig. 11.1 shows a workman using a cordless electric drill.

For Examiner's Use

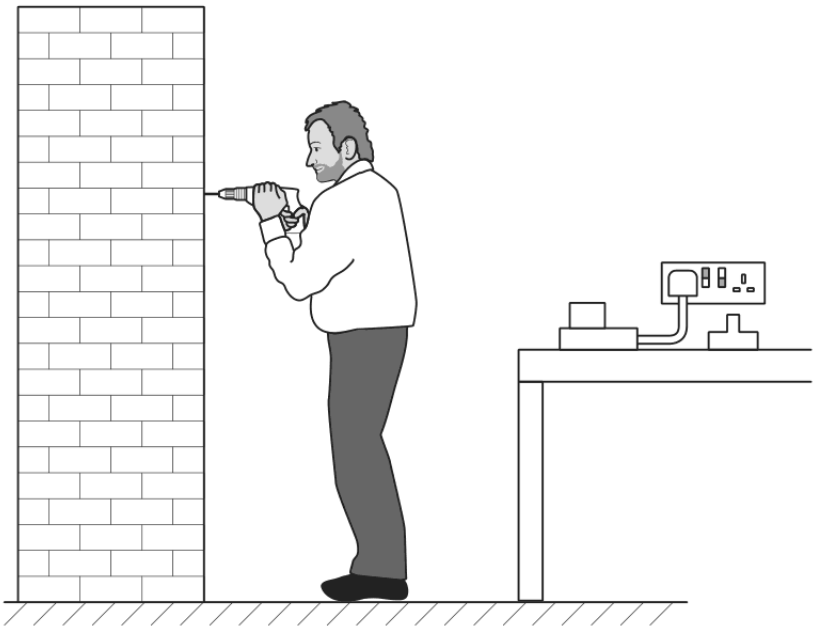


Fig. 11.1

The motor of the drill is powered by a rechargeable battery with an electromotive force (e.m.f.) of 18 V. When the drill is used, the power supplied to the motor is 450 W.

(a) Explain what is meant by an *e.m.f. of 18 V*.

.....

.....

.....[2]

(b) The workman uses the drill for 90 minutes.

Calculate

(i) the electrical energy supplied to the motor,

energy = ...[3]

(ii) the charge that the battery supplies.

charge = ...[2]

For Examiner's Use

(c) After 90 minutes of use, the battery is flat. It is connected to a charger and is recharged. The charger includes a transformer that produces a 23V a.c. output from a 230V a.c. mains supply.

(i) Draw a labelled diagram to show the structure of the transformer.

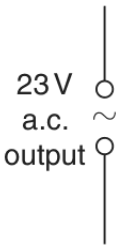
[2]

(ii) State how the transformer ensures that the a.c. output has a value of 23V when the input is the 230V a.c. mains supply.

.....

.....[1]

(iii) The 23V a.c. output of the transformer is used in a rectifying circuit to produce a direct current that recharges the battery. Complete the circuit diagram of the circuit.



[2]

(d) State and explain **one** advantage of using an alternating current (a.c.) for long-distance transmission of electrical power.

.....

.....

.....

.....[3]

11	**(a)**	energy released/unit charge **or** power released/unit current	C1	
		18 J/C **or 18** W/A	A1	[2]
	(b)	**(i)** (*t* =) 5400 **or** 60 × 90 **or** 1.5 **or** 90/60 **or** (*E* =) *Pt* **or** 450 × 90	B1	
		450 × 60 × 90 **or** 450 × 5400 **or** 4.0/4.05/4.1 × 10^4 **or** 0.45 × 1.5		
		or 0.45 × 90/60 **or** 450 × 1.5 **or** 450 × 90/60	C1	
		2.4(3) × 10^6 J **or** 0.675 kWh	A1	
		(ii) (*Q* =) *E*/emf (**ign.** emf = *E*/*Q*) **OR** (*I* =) 25 (A) **or** 25 × 5400		
		or 2.4(3) × 10^6/18 **or** 25 × 60 × 90	C1	
		1.3/1.35/1.4 × 10^5 C	A1	[5]
	(c)	**(i)** laminated/iron core	B1	
		two coils on core	B1	
		(ii) turns ratio = 10:1 (may be shown on diagram)	B1	
		(iii) diode symbol	B1	
		symbol for battery/cell (**allow** either polarity w.r.t. diode) **and** complete circuit	B1	[5]

	(d)	can be transformed/operate transformer/voltage can be changed	B1	
		high voltage/low current transmission (possible)		
		or changing magnetic field	B1	
		less energy/power loss **or** less heating (in wires) **or** thinner wires	B1	[3]

MARKING SCHEME CODE:

10 (a) A 1500 Ω resistor is used on an electronic circuit board in a television.

Fig. 10.1 shows the bands on a resistor.

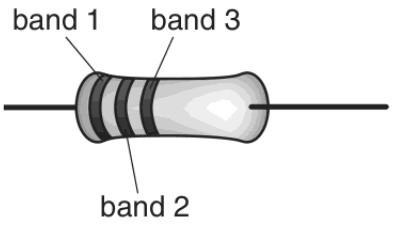


Fig. 10.1

digit or multiplier	colour
0	black
1	brown
2	red
3	orange
4	yellow
5	green
6	blue
7	violet
8	grey
9	white

Fig. 10.2

The table of Fig. 10.2 shows the colour code for the bands.

(i) State the colours of the bands on a 1500 Ω resistor.

band 1 .....

band 2 .....

band 3 .....

[2]

(ii) Determine the maximum value of resistance that can be shown with this three-band colour code.

resistance = ... [1]

(iii) The 1500 Ω resistor on the electronic circuit board has a correct power rating of 0.25 W.

State a disadvantage of using a different 1500 Ω resistor with a power rating that is

1. much less than 0.25 W,

.....

..... [1]

2. much greater than 0.25 W.

.....

..... [1]

(b) Fig. 10.3 shows part of a circuit that includes a variable resistor R and a battery of e.m.f. 9.0V.

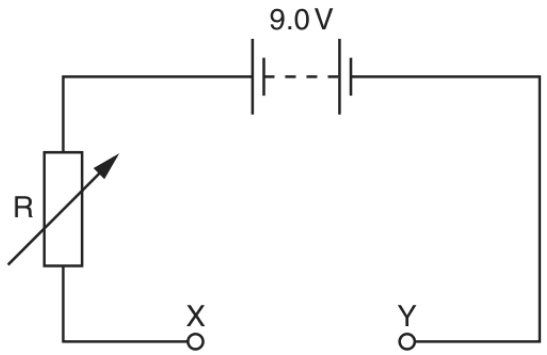


Fig. 10.3

(i) State one similarity and one difference between electromotive force (e.m.f.) and potential difference (p.d.).

similarity: .....

.....

difference: .....

.....
[3]

(ii) A light-emitting diode (LED) is connected between points X and Y, so that it emits light.

On Fig. 10.3, draw the symbol for the LED connected between points X and Y. [2]

(iii) The resistance of R is increased. State what happens to the p.d. across the LED and to the current in the LED.

p.d.: .....

current: .....
[1]

(iv) The LED is marked "maximum current 25 mA when the p.d. is 1.7 V".

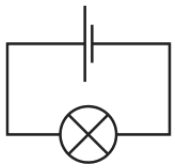
Calculate the minimum value of the resistance of R.

resistance = ... [4]

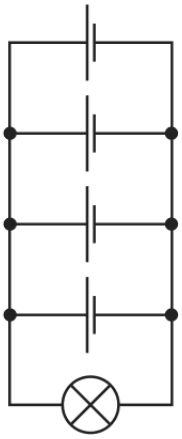
10	**(a)**	**(i)**	brown and green	B1	
			red	B1	
		(ii)	99×10^9 or 9.9×10^{10} Ω	B1	
		(iii)	*less*: (likely to) burn out/blow // become too hot	B1	
			greater: (likely to) be large (in size)	B1	
	(b)	**(i)**	both involve energy and charge // measured in J/C/volts/by voltmeter	B1	
			energy change is from other forms (accept chemical) to electrical in e.m.f.	B1	
			energy change is from electrical to other forms (accept heat/light) in p.d.	B1	
			(**or** e.m.f. is property of source and p.d. is property of (part of) circuit B2)		
		(ii)	correct symbol	B1	
			correct direction	B1	
		(iii)	p.d. reduces/(approximately) constant **and** current reduces	B1	
		(iv)	$R = V/I$ in any form, e.g. 1.7/0.025, 1.7/25	C1	
			correct conversion to mA, e.g. 0.025/7.3 seen	C1	
			68 or 360 seen	C1	
			290 Ω, 292 Ω	A1	[15]

QUESTION · 5054_w11_qp_11.pdf · source page 13 of 20

29 A lamp, designed to work at 1.5 V, is connected to a cell of electromotive force (e.m.f.) 1.5 V. The lamp lights at normal brightness.



The lamp is now connected to four similar cells, each of e.m.f. 1.5 V, arranged in parallel.



What is the effect of connecting the extra cells in this way?

A The lamp burns out.

B The lamp is dimmer.

C The lamp produces light for a longer time.

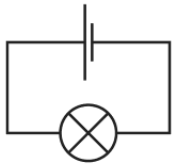
D The lamp produces light for a shorter time.

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 29

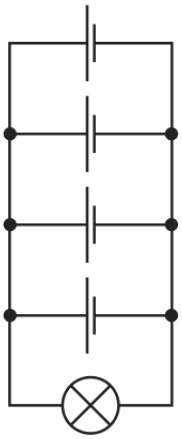
29	**C**

QUESTION · 5054_w11_qp_12.pdf · source page 14 of 20

32 A lamp, designed to work at 1.5 V, is connected to a cell of electromotive force (e.m.f.) 1.5 V. The lamp lights at normal brightness.



The lamp is now connected to four similar cells, each of e.m.f. 1.5 V, arranged in parallel.



What is the effect of connecting the extra cells in this way?

A The lamp burns out.

B The lamp is dimmer.

C The lamp produces light for a longer time.

D The lamp produces light for a shorter time.

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 32

32	**C**

7 Fig. 7.1 is the circuit diagram of a circuit set up by a student.

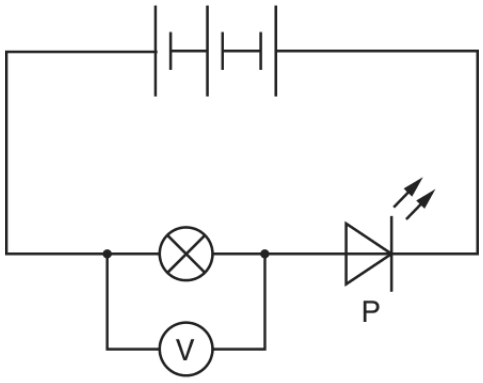


Fig. 7.1

The student uses three cells, each of electromotive force (e.m.f.) 1.5V. The cells are connected in series, as shown.

(a) State the name of component P.

.....[1]

(b) State what is meant by the *electromotive force* of a cell.

.....

.....[1]

(c) The student expects the reading on the voltmeter to be 4.5V. The actual reading is 0.2V.

Suggest **two** reasons why the voltmeter reading is much less than 4.5V.

.....

.....

.....[2]

(d) **(i)** In the space below, draw three cells connected in parallel.

[1]

(ii) State one advantage of connecting cells in parallel.

.....

.....[1]

7	**(a)**	LED; light-emitting diode	B1	
	(b)	energy/work done per unit charge/coulomb	B1	
	(c)	voltage/p.d. across P	B1	
		one cell connected the wrong way/acting against the others	B1	
	(d)	**(i)** correct arrangement	B1	
		(ii) cells last longer; cells run down slower; one cells fails the others still work; reduces (internal) resistance; if cell removed circuit not broken	B1	[6]

QUESTION · 5054_w12_qp_11.pdf · source page 12 of 16

28 Three identical cells are connected in parallel to a resistor.

What is the advantage of using three cells in parallel, rather than a single cell?

A Each cell produces more energy.

B Each cell supplies more charge.

C Each cell takes longer to run down.

D The total electromotive force (e.m.f.) is larger.

MARK SCHEME (official answer key) · 5054_w12_ms_11.pdf · source page 2 of 2 · row for Question 28

28	**C**

QUESTION · 5054_s13_qp_11.pdf · source page 12 of 20

30 Which of the following is equivalent to 1 V?

A 1 J / A

B 1 J / C

C 1 J / s

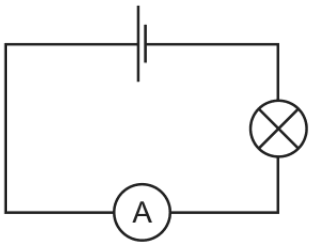
D 1 J / W

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 30

30	**B**

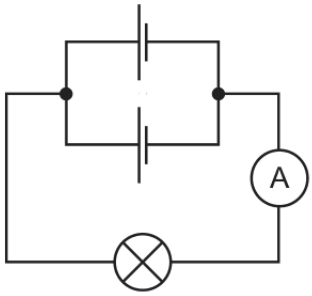
QUESTION · 5054_s13_qp_11.pdf · source page 14 of 20

32 A cell is connected in series with an ammeter and a lamp. The reading on the ammeter is 1 A.

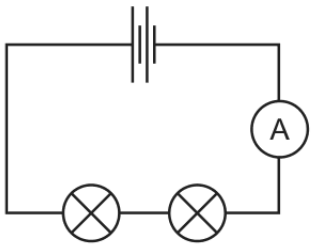


In which circuit, using identical lamps, ammeters and cells, is the reading on the ammeter 2 A?

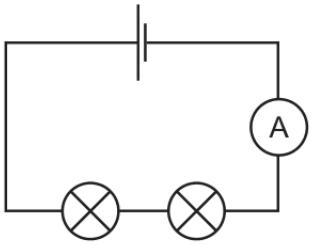
A



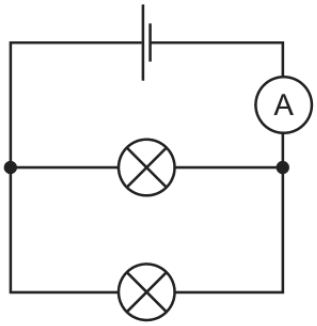
B



C



D



MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 32

32 **D**

For Examiner's Use

11 A metal filament lamp is connected to a power supply. The electromotive force (e.m.f.) produced by the supply can be varied. Fig. 11.1 is the circuit diagram.

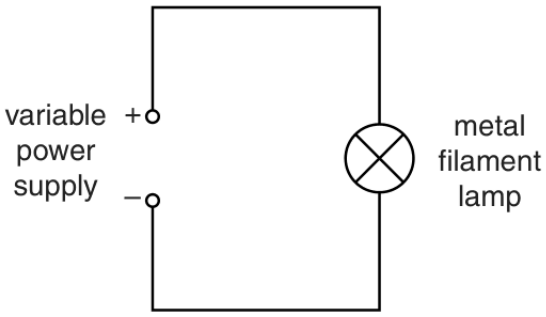


Fig. 11.1

(a) State what is meant by *electromotive force (e.m.f.)*.

.....

.....

.....[2]

(b) Add appropriate circuit symbols to Fig. 11.1 to show the position of

(i) an ammeter that measures the current in the circuit, [1]

(ii) a voltmeter that measures the voltage across the lamp. [1]

(c) A student adds meters to the circuit and makes measurements of the current and voltage. He then plots the current/voltage graph shown in Fig. 11.2.

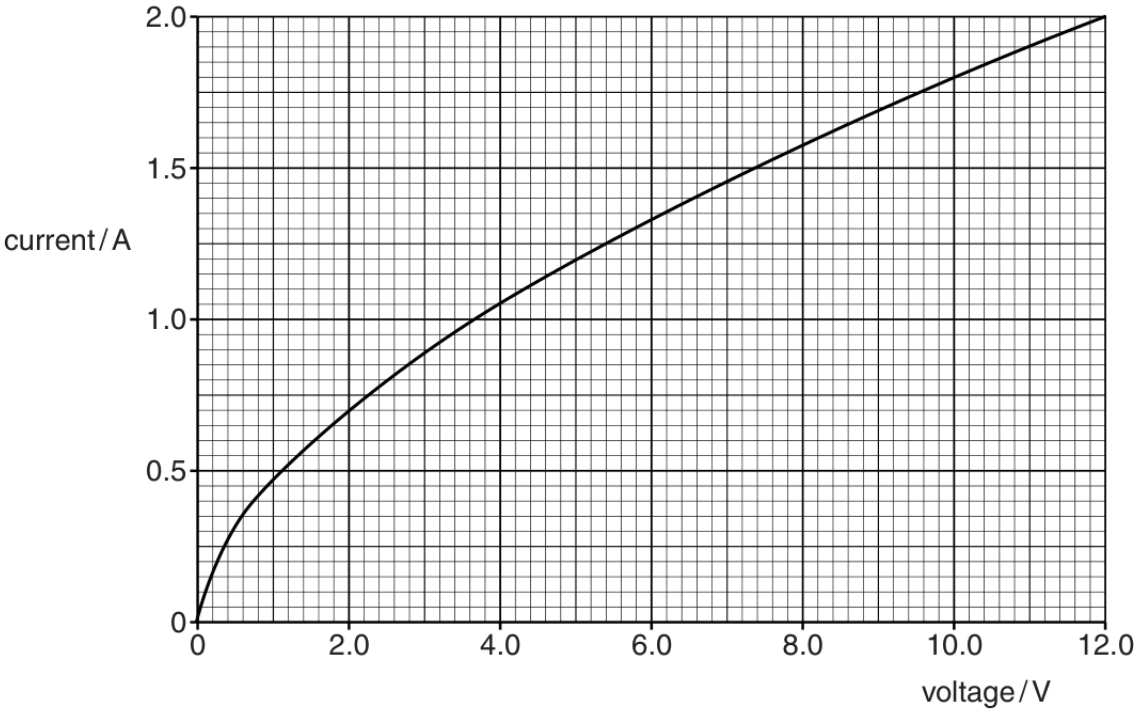


Fig. 11.2

For Examiner's Use

(i) Using values from Fig. 11.2, calculate the resistance of the lamp when the current is 0.70 A. Give your answer to an appropriate number of significant figures.

resistance = ...[3]

(ii) State what, if anything, happens to the resistance of the lamp as the voltage increases.

.....

.....[1]

(d) In normal use, the lamp is connected to a 12 V supply.

Use Fig. 11.2 to determine the power of the lamp when it is used in this way.

power = ...[2]

(e) When connected to a 12 V supply, thermionic emission occurs in the lamp.

(i) Describe what is meant by *thermionic emission*.

.....

.....

.....[2]

(ii) Thermionic emission is used in a cathode-ray oscilloscope (c.r.o.) tube.

1. State why air must be removed from the tube of the c.r.o.

.....

.....[1]

2. A voltage is connected across the Y-plates in the c.r.o.

State and explain the effect on the trace on the screen.

.....

.....

.....[2]

11 (a) work done per (unit) charge/coulomb/C **or** energy transformed per (unit) charge/coulomb/C — B1
property of a source (of electricity) **or** energy transformed to electrical energy per (unit) charge/coulomb/C — B1 [2]

(b) (i) ammeter in series — B1

(ii) voltmeter in parallel with lamp **or** lamp and ammeter — B1 [2]

(c) (i) (V =) 2.0 (V) — C1
(R =) V/I **or** 2.0/0.70 — C1
2.9/2.86 Ω (i.e. 2 **or** 3 s.f. only) — A1

(ii) (resistance) increases — B1 [4]

(d) (P =) VI **or** (P =) V^2/R **or** I^2R **or** 12 × 2.0 **or** 12 × 0.70 — C1
24 W — A1 [2]

(e) (i) emission of electrons — M1
from heated metal/named metal/filament/wire — A1

(ii) 1. prevents collision with air (molecules) **or** prevents deflection **or** lets electrons/particles reach screen/travel unimpeded — B1

2. moves vertically (e.g. up/down/above/below **or** vertical line) **not** with horizontal movement due to this voltage — B1
attracted by positive **or** repelled by negative **or** attracted by one plate and repelled by the other **or** electric field (acts on charge) — B1 [5]

[Total: 15]

11 **(a)** A torch contains a filament lamp connected to a battery. The battery contains four 1.5 V cells connected in series.

Fig. 11.1 shows the current/potential difference (voltage) graph for the filament lamp.

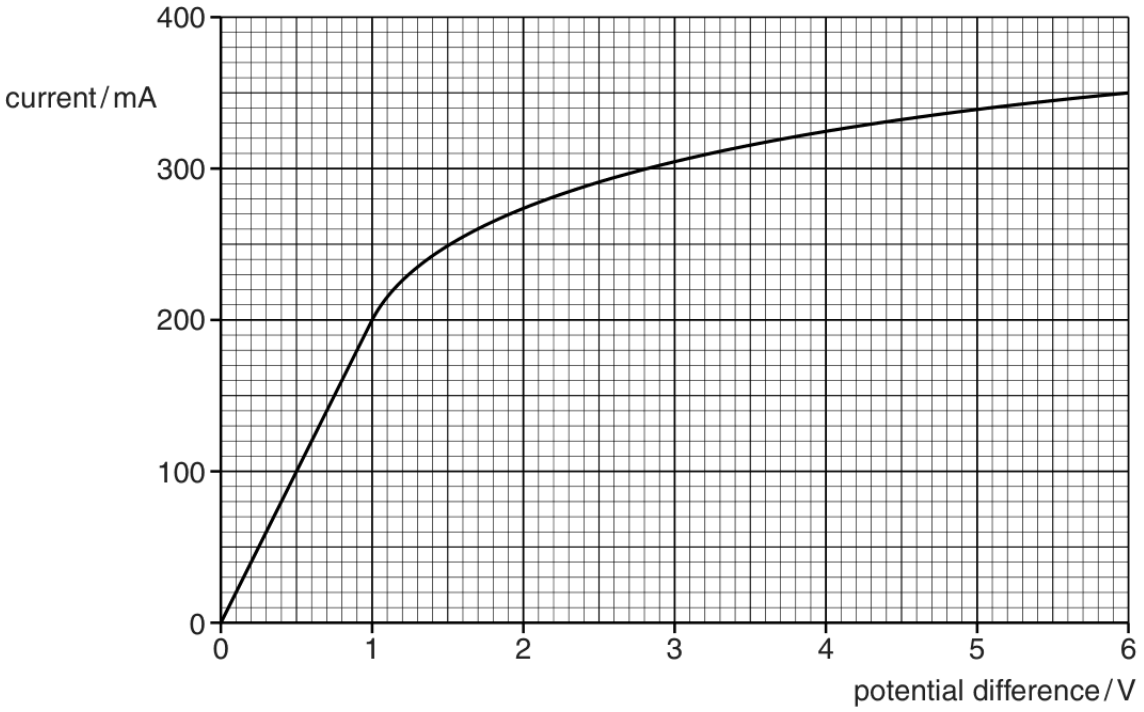


Fig. 11.1

(i) State what is meant by the *potential difference* (p.d.) across the lamp.

.....

..... [2]

(ii) Fig. 11.1 shows that the current in the lamp is not proportional to the p.d.

1. State how Fig. 11.1 shows this.

.....

..... [1]

2. Suggest why the current is not proportional to the p.d.

.....

..... [1]

(iii) The lamp is at normal brightness and there is a p.d. of 6.0V across the lamp.

1. Use Fig. 11.1 to find the current in the lamp.

current = ... [1]

2. Calculate the charge that passes through the lamp in 2.0 hours.

charge = ... [2]

3. The battery in the torch is able to keep the lamp lit for 2.0 hours.
Estimate the initial energy stored in the battery.

energy = ... [2]

4. Explain why your value is only an estimate.

.....

..... [1]

(b) Suggest and explain an advantage of using two cells in parallel rather than a single cell.

.....

.....

..... [2]

(c) Another torch contains a light-emitting diode (LED) instead of a filament lamp.
The battery in the torch uses four 1.5V cells, some in series and some in parallel.
The p.d. across the LED when in operation is 3.0V.

In the space below, draw a circuit diagram showing how the four cells, a switch and the LED are connected.

[3]

11 (a) (i) work done **or** energy produced / needed [M1]
per unit charge **or** per coulomb (passing through lamp) [A1]

(ii) **1**. not straight **or** curves **or** gradient changes **or** data used correctly [B1]
2. gets hotter **or** temperature changes **or** resistance increases [B1]

(iii) **1**. 350 mA **or** 0.35 A [B1]
2. Q = It in any form algebraic or numerical **or** 2 × 60 × 60 **or** 7200 (s) seen [C1]
2520 C **or** 2500 C [A1]
3. (E=) QV or V*I*t in any algebraic or numerical form, e.g. 0.35 × 6 × 2 [C1]
15100 J **or** 15120 J **or** 15000 J **or** 4.2 kWh [A1]
4. current and / or voltage falls / varies
or some energy remains (in cell)
or some energy / heat produced **in cell**
or correct argument involving internal resistance of cell [B1]

(b) lasts longer [M1]
(because) larger energy (initially) **or** smaller current (in each cell) [A1]
or
avoids failure if one cell fails (M1)
(because) other (parallel) cell takes over (A1)
or
larger current / power (in external resistor / lamp) (M1)
(because) smaller (internal) resistance of combined cells (A1)

(c) symbol for LED clear and in correct direction for cells shown [B1]
4 cells (not all in parallel or in series), switch and LED (labelled or shown) that switches on and off properly [B1]
cells connected correctly to give 3.0 V total e.m.f., e.g. two sets of parallel cells connected in series or three in parallel and one in series [B1]

[15]

11 A student makes a 2.0V battery by connecting two cells of electromotive force (e.m.f.) 2.0V in parallel. The battery, an ammeter with different ranges and three different resistors are used to set up the circuit shown in Fig. 11.1.

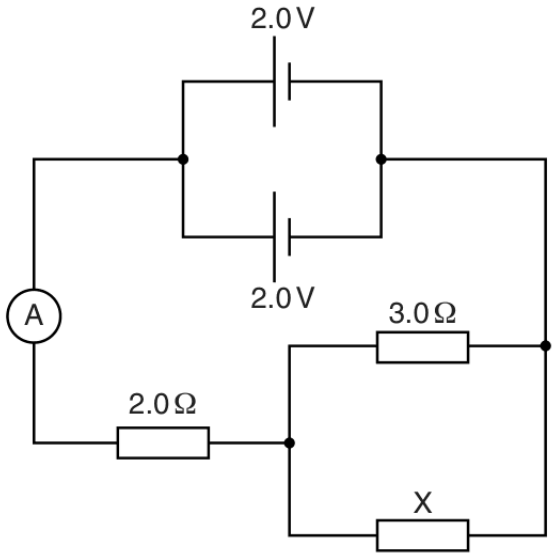


Fig. 11.1

(a) (i) Explain what is meant by *electromotive force*.

.....

.....

.....[2]

(ii) State one advantage of using two cells in parallel rather than using a single 2.0V cell.

.....

.....[1]

(b) Resistor X and the 3.0 Ω resistor have a combined resistance that is equal to 2.0 Ω.

Calculate

(i) the total resistance of the circuit,

total resistance = ...[1]

(ii) the resistance of X.

resistance of X = ...[2]

(c) **(i)** Determine the reading of the ammeter.

reading = ...[2]

(ii) Suggest a suitable range for the ammeter.

.....[1]

(d) The current in the 2.0 Ω resistor is I_2.
The current in the 3.0 Ω resistor is I_3.
The current in X is I_X.

State the equation that relates I_2, I_3 and I_X.

.....[1]

(e) State the potential difference (p.d.) across

(i) the 2.0 Ω resistor,

p.d. = ...[1]

(ii) the 3.0 Ω resistor.

p.d. = ...[1]

Question 11 continues on page 18.

(f) The student sets up a second circuit using a variable d.c. power supply, an ammeter and a 12V metal filament lamp. The circuit is shown in Fig. 11. 2.

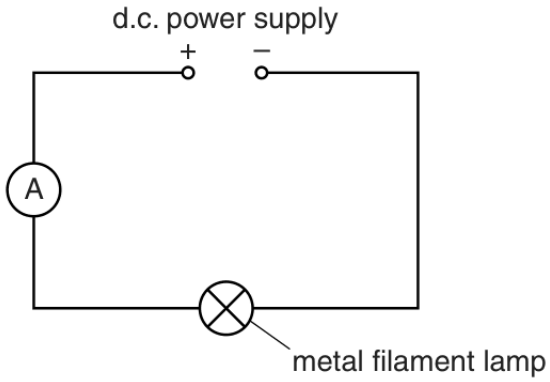


Fig. 11.2

The d.c. power supply is set to 12V and the ammeter reading is 1.5 A.

The student changes the e.m.f. of the d.c. power supply to 6.0V. The lamp dims and the ammeter reading changes.

(i) State and explain what happens to the resistance of the metal filament of the lamp.

.....

.....

.....[2]

(ii) State whether the new ammeter reading is less than, equal to or greater than 0.75 A.

.....[1]

11	**(a)**	**(i)**	energy to drive charge around a circuit **or** terminal p.d. on open circuit	B1	
			energy to drive **unit** charge around a circuit **or** energy / charge	B1	
		(ii)	lasts longer **or** lower internal resistance **or** can replace a cell without switching off **or** continues to work if one cell is flat **ignore** more current (**not** greater e.m.f. / voltage)	B1	**[3]**
	(b)	**(i)**	4.0 Ω	B1	
		(ii)	$(1/R_{tot} =)1/R_1 + 1/R_2$ **or** $1/3 + 1/X$ **or** product / sum **or** $(3 \times X)/(3 + X)$ **or** $\frac{1}{X} = \frac{1}{2} - \frac{1}{3}$	C1	
			6.0 Ω	A1	**[3]**
	(c)	**(i)**	$(I =)\ V/R$ **or** 2.0 / 4.0	C1	
			0.50 A	A1	
		(ii)	(from) 0 **and** (to) 0.50 to 5.0 A	B1	**[3]**
	(d)		$I_2 = I_3 + I_X$	B1	**[1]**
	(e)	**(i)**	1.0 V	B1	
		(ii)	1.0 V	B1	**[2]**
	(f)	**(i)**	temperature decreases	B1	
			resistance decreases	B1	
		(ii)	greater than 0.75 A (e.c.f. resistance increases in **(f)(i)**)	B1	**[3]**
					[Total: 15]

7 The circuit of Fig. 7.1 includes an ammeter, a 6.0 Ω resistor, a length of metal resistance wire and three 1.5 V cells connected in parallel.

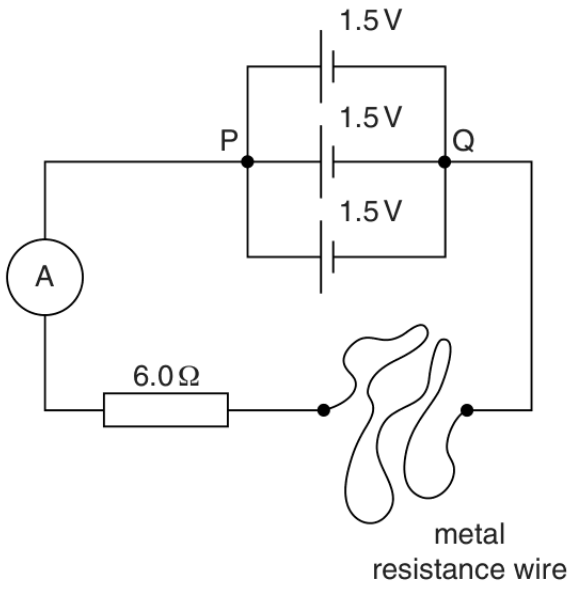


Fig. 7.1

(a) State

(i) one advantage of using cells in parallel rather than using a single cell,

.....

.....[1]

(ii) the potential difference (p.d.) between points P and Q in the circuit of Fig. 7.1.

p.d. = ...[1]

(b) **(i)** The ammeter in Fig. 7.1 reads 0.075 A.

Calculate the resistance of the resistance wire.

resistance = ...[3]

(ii) The temperature of the metal resistance wire increases.

State and explain the effect of this temperature increase on the ammeter reading.

.....

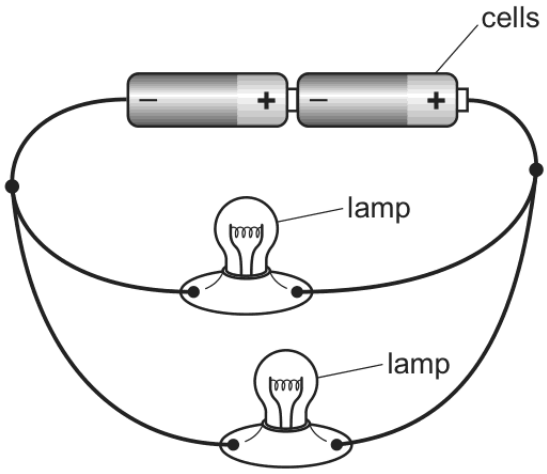
.....

.....[2]

7	**(a)**	**(i)**	lasts longer **or** one cell can be replaced without switching off the circuit **or** less (internal) resistance **or** if one fails the others still work	B1	
		(ii)	1.5 V	B1	
	(b)	**(i)**	(R =)V / I **or** 1.5 / 0.075	C1	
			20 (Ω) **or** 1.5 / (0.075 – 6.0)	C1	
			14 Ω	A1	
		(ii)	decreases	B1	
			resistance of wire increases	B1	**[7]**

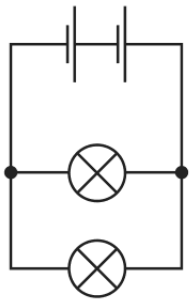
QUESTION · 5054_s15_qp_11.pdf · source page 11 of 16

28 The diagram shows a circuit.

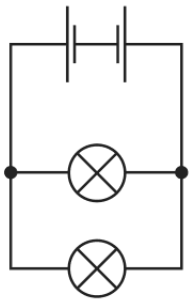


Which circuit diagram shows this circuit?

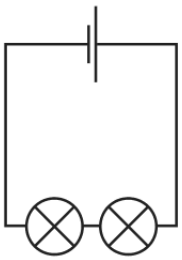
A



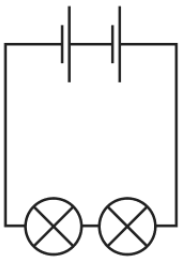
B



C



D



MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 28

28	**A**

QUESTION · 5054_s15_qp_11.pdf · source page 11 of 16

30 Three identical cells are connected in parallel to a resistor.

What is the advantage of using three cells in parallel, rather than using a single cell?

A Each cell produces more energy.

B Each cell supplies more charge.

C Each cell takes longer to run down.

D The total electromotive force (e.m.f.) is larger.

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 30

30	**C**

QUESTION · 5054_s15_qp_11.pdf · source page 12 of 16

31 When the flash on a camera is used, a charge of 1.5 C flows through the flash-tube in 0.0030 s.

The average voltage across the flash-tube is 3600 V.

What is the electrical energy supplied to the flash-tube and what is the average power supplied?

	energy / J	power / W
A	2400	7.2
B	2400	800 000
C	5400	16.2
D	5400	1.8×10^6

MARK SCHEME (official answer key) · 5054_s15_ms_11.pdf · source page 2 of 2 · row for Question 31

31	**D**

QUESTION · 5054_w15_qp_11.pdf · source page 10 of 16

27 Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d.)?

A current

B electromotive force (e.m.f.)

C energy

D power

MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 27

27	**C**

QUESTION · 5054_s16_qp_11.pdf · source page 11 of 16

30 How can *one volt* also be expressed?

A one coulomb per ampere

B one coulomb per joule

C one joule per ampere

D one joule per coulomb

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 30

30	**D**

QUESTION · 5054_s16_qp_12.pdf · source page 10 of 16

26 How can *one volt* also be expressed?

A one coulomb per ampere

B one coulomb per joule

C one joule per ampere

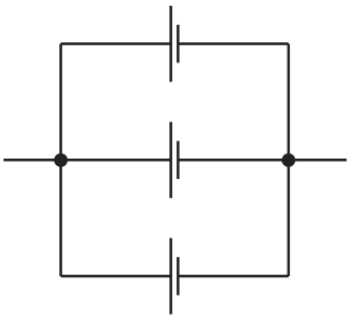
D one joule per coulomb

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 26

26	**D**

QUESTION · 5054_s16_qp_12.pdf · source page 10 of 16

27 The diagram shows a battery of three 1.5 V cells.



What is an advantage of this arrangement of cells?

A The battery can supply a current for a longer time than a single 1.5 V cell.

B The battery can supply any e.m.f. between 0 V and 4.5 V.

C The battery supplies more energy to each coulomb of charge than a single 1.5 V cell.

D The e.m.f. of the battery is 4.5 V.

MARK SCHEME (official answer key) · 5054_s16_ms_12.pdf · source page 2 of 2 · row for Question 27

27	**A**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 is a circuit diagram.

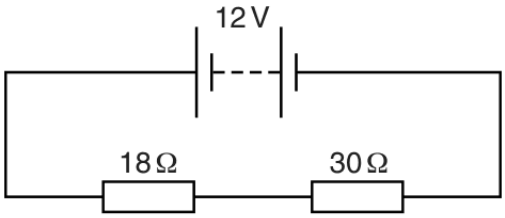


Fig. 9.1

The electromotive force (e.m.f.) of the battery is 12V.

(a) Explain what is meant by *electromotive force*.

.....

.....[2]

(b) An 18 Ω resistor and a 30 Ω resistor are connected in series with the battery.

(i) **1.** State the relationship between the current I_B in the battery, the current I_1 in the 18 Ω resistor and the current I_2 in the 30 Ω resistor.

.....[1]

2. State an equation that relates the e.m.f. E of the battery, the potential difference V_1 across the 18 Ω resistor and the potential difference V_2 across the 30 Ω resistor.

.....[1]

(ii) Calculate the current in the battery.

current = ...[2]

(iii) Calculate the potential difference (p.d.) across the 18 Ω resistor.

p.d. = ...[1]

(iv) Calculate the power produced in the 18 Ω resistor.

power = ...[2]

(c) The resistors obey Ohm's law.

State Ohm's law and describe one limitation on this law.

.....

.....

.....[2]

(d) The resistors are made from the same material, but have different cross-sectional areas and different lengths.

State the relationship between

(i) the resistance R and the length l of a wire of constant cross-sectional area,

.....

(ii) the resistance R and the cross-sectional area A of a wire of constant length.

.....
[2]

(e) A resistor is shown in Fig. 9.2 and part of the resistor colour code is shown in Fig. 9.3.

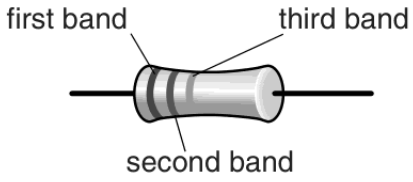


Fig. 9.2

digit or multiplier	colour
0	black
1	brown
2	red
3	orange

Fig. 9.3

State the colour of the bands on the 30 Ω resistor.

first band = ...

second band = ...

third band = ...
[2]

9	**(a)**		(amount of) energy / work (dissipated by source) by unit charge (around a circuit)	M1 A1
	(b)	**(i)**	**1** they are the same **or** $I_B = I_1 = I_2$	B1
			2 $E = V_1 + V_2$	B1
		(ii)	(I=) V / R in any form algebraic or numerical	C1
			0.25 A	A1
		(iii)	4.5 V	B1
		(iv)	(P=) VI **or** (P=) I^2R **or** (P=) V^2 / R in any form algebraic or numerical	C1
			1.1(25) W	A1

	(c)		current is (directly) proportional to voltage **or** voltage / current is a constant	B1
			law holds for constant physical conditions / constant temperature / constant pressure / for metals	B1
	(d)	**(i)**	(directly) proportional **or** (R) ∝ 1	B1
		(ii)	inversely proportional **or** (R) ∝ 1 / A	B1
	(e)		1st band orange	B1
			2nd and 3rd bands both black	B1

9 **(a)** The primary coil of a transformer is connected to the 230 V a.c. mains supply.

(i) Describe how the transformer operates.

.....

.....

.....

.....

.....

.....[3]

(ii) The output voltage of the transformer is alternating and cannot be used directly as a d.c. power supply.

In the space below, draw the symbol of a circuit component that allows the current to pass in only one direction and state the name of the component.

circuit symbol

name of component: .....
[2]

(b) Fig. 9.1 shows a d.c. power supply being used to charge a rechargeable battery.

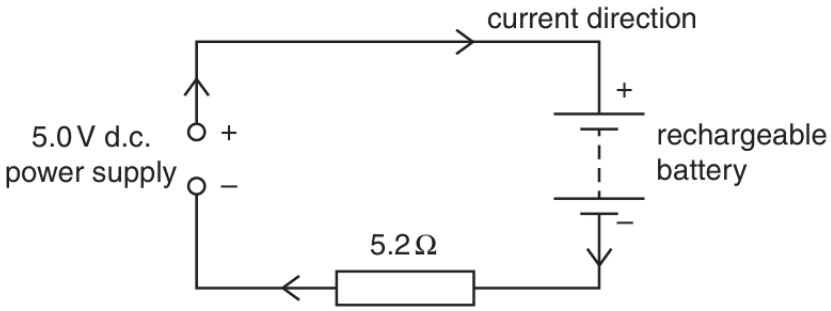


Fig. 9.1

The rechargeable battery is in series with the d.c. power supply and a 5.2 Ω resistor. The e.m.f. of the power supply is 5.0 V.

(i) Explain what is meant by *e.m.f.*

.....

.....

.....[2]

(ii) The e.m.f. of the power supply and the e.m.f. of the battery act against each other in opposite directions in the circuit.

The e.m.f. of the battery is 3.7 V.

1. Calculate the resultant e.m.f. in the circuit.

resultant e.m.f. = ...[1]

2. Calculate the current in the circuit.

current = ...[2]

3. The current in the circuit remains constant as the battery is charged.

Calculate the charge that passes through the resistor in 1.5 hours.

charge = ...[3]

(c) A battery charger has a plastic casing and double insulation. The battery charger is not earthed.

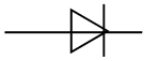
Explain why it is not necessary for the battery charger to be earthed.

.....

.....

.....

.....[2]

9	**(a)**	**(i)**	magnetic field mentioned	B1	
			alternating / changing magnetic field	B1	
			current / voltage / e.m.f. induced (in secondary coil)	B1	
		(ii)		B1	
			diode	B1	[5]
	(b)	**(i)**	work done / energy transferred per unit charge	M1	
			electrical energy to other forms **or** for whole circuit **or** property of supply	A1	

	(ii)	**1**	1.3 V	B1	
		2	(I =)V/R **or** 1.3/5.2	C1	
			0.25 A	A1	
		3	(Q =)It **or** 0.25 × 1.5 × 3600 **or** 0.25 × 1.5	C1	
			0.25 × 1.5 × 3600 **or** 0.37/0.375/0.38	C1	
			1300 / 1350 / 1400 C	A1	[8]
	(c)		plastic / casing is an (electrical) insulator	M1	
			no shock possible	A1	[2]
					[15]

9 A vertical solenoid (long coil) with an iron core is held in a wooden clamp above a laboratory bench.

The solenoid is connected in series with a battery, a switch S, an ammeter and a variable resistor. There is a voltmeter in parallel with the solenoid.

Fig. 9.1 represents this apparatus.

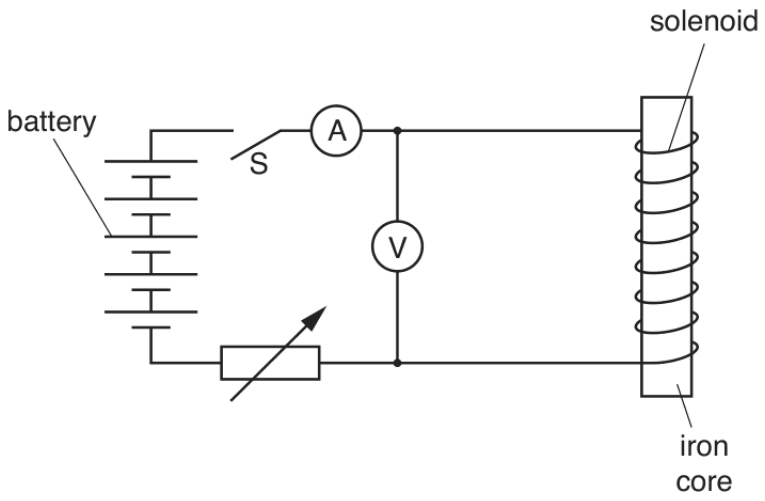


Fig. 9.1

A student closes switch S and a current in the circuit produces a reading on the ammeter.

(a) State what is meant by *current*.

.....

.....[1]

(b) The battery consists of five 1.5 V cells in series. The reading on the ammeter is 4.0 A.

(i) State the size of the electromotive force (e.m.f.) of the battery.

.....[1]

(ii) Calculate the total resistance of the series circuit.

resistance = ...[2]

(iii) The reading on the voltmeter is 6.5 V.

Calculate the power dissipated in the solenoid.

power = ...[2]

(iv) The solenoid is made of copper and the student notices that, as time passes, the solenoid becomes extremely warm.

State and explain the effect of this temperature increase on the ammeter reading.

.....

.....

.....[2]

(c) (i) The current in the solenoid magnetises the iron core so that the lower end of the core is an N-pole.

On Fig. 9.1, draw the pattern and indicate the direction of the magnetic field in the region surrounding the iron core. [3]

(ii) The student holds an iron cylinder against the bottom surface of the iron core in the solenoid. When he releases the iron cylinder, it stays in contact with the iron core.

1. Explain why the iron cylinder does not fall.

.....

.....

.....

.....

.....[3]

2. The switch S is opened.

State and explain whether the iron cylinder remains in contact with the iron core.

.....

.....

.....[1]

9	**(a)**	rate of flow of charge **or** charge flowing per unit time	B1	[1]
	(b) **(i)**	7.5 V	B1	
	(ii)	(R =)V / I **or** 7.5 / 4.0	C1	
		1.9 Ω	A1	
	(iii)	(P =) VI **or** 6.5 × 4.0	C1	
		26 W	A1	
	(iv)	resistance increases	M1	
		(reading of ammeter) decreases	A1	[7]
	(c) **(i)**	at least two lines on left **and** two lines on right of core **and** correct shape (by eye)	B1	
		good shape (by eye) **and** into poles **and** no straight sections **and** at least one line on each side	B1	
		at least one arrow N to S (primarily upwards) **and** none wrong	B1	[3]

	(ii) **1**	cylinder is magnetised (by induction)	B1	
		top (of cylinder) is an S-pole	B1	
		unlike poles attract **or** S-pole attracts N-pole	B1	
	2	it does not (remain in contact**) and** iron is temporary / soft magnetic material / core (and cylinder) lose magnetisation	B1	[4]
				[15]

QUESTION · 5054_s17_qp_11.pdf · source page 14 of 16

37 Why are batteries sometimes made from several identical cells arranged in parallel?

A The battery has a greater e.m.f. than a single cell on its own.

B The battery has a lower e.m.f. than a single cell on its own.

C The battery lasts longer than a single cell on its own.

D The battery supplies a smaller current than a single cell on its own.

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 3 of 3 · row for Question 37

37	C	**1**

10 (a) Measurements are made of the current in a metal wire and the potential difference (p.d.) across it. Fig. 10.1 shows a graph of the current against the p.d. for the wire.

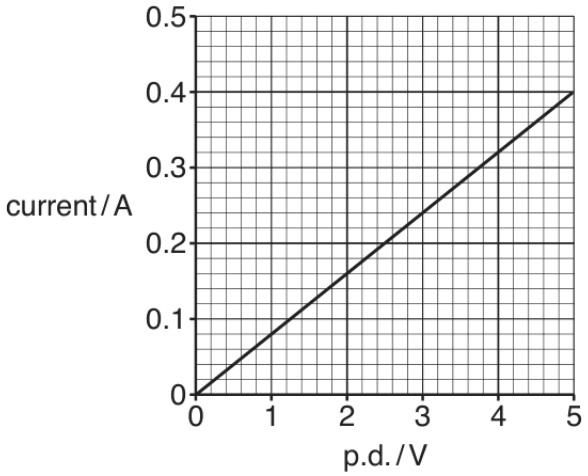


Fig. 10.1

(i) State the relationship between the current and the p.d. for the metal wire.

.....

.....[1]

(ii) State how the graph shows that the temperature of the wire does not change in the experiment.

.....

.....[1]

(iii) Another wire of the same material has the same length as the original wire but has only half the cross-sectional area.

1. Compare the resistance of the new wire and the original wire.

.....

.....[1]

2. On Fig. 10.1, draw the graph for the new wire. [1]

(b) Three resistors are connected to a d.c. power supply, as shown in Fig. 10.2. The electromotive force (e.m.f.) of the power supply is 6.0 V.

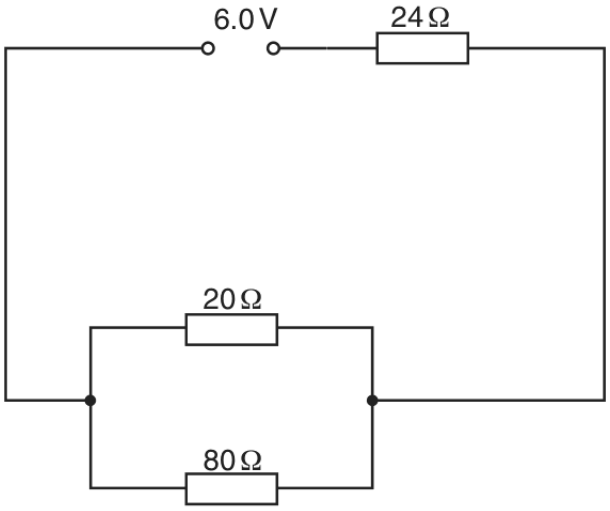


Fig. 10.2

(i) Calculate

1. the total resistance of the circuit,

resistance = ...[2]

2. the current supplied by the power supply,

current = ...[2]

3. the p.d. across the 20 Ω resistor.

p.d. = ...[2]

(ii) The d.c. power supply is a battery that contains a number of cells. Each cell has an electromotive force (e.m.f.) of 1.5 V.

1. State what is meant by the *electromotive force (e.m.f.)* of a cell.

.....

.....

.....[1]

2. In the space below, draw a diagram to show how four of these cells are connected to make a battery of e.m.f. 6.0 V. Mark the positive terminal of the battery.

[2]

3. In another arrangement to produce a battery of e.m.f. 6.0 V, eight cells of e.m.f. 1.5 V are connected. In this arrangement, some cells are connected in series and some in parallel.

In the space below, draw a diagram to show how the eight cells are connected to make the battery.

[1]

(c) State one advantage of using a battery made from cells in parallel rather than a single cell.

.....

.....[1]

Question	Answer	Marks
10(a)(i)	directly proportional	**B1**
10(a)(ii)	straight line **or** does not curve **or** constant gradient	**B1**
10(a)(iii)1	greater **or** twice as large	**B1**
10(a)(iii)2	straight line with half the gradient	**B1**
10(b)(i)1	$1 / R_t = 1 / R_1 + 1 / R_2$ in any form e.g. $1 / R = 1 / 20 + 1 / 80$ **or** 16 (Ω) seen	**C1**
	40 Ω	**A1**
10(b)(i)2	(I=)V / R in any form e.g. 6 / 40	**C1**
	0.15 A	**A1**
10(b)(i)3	0.15 × 16 **or** 0.15 × 24 **or** 3.6 (V) seen **or** current split in ratio 1:4, e.g. 0.03 A and 0.12 A seen **or** clear attempt at potential divider formula	**C1**
	2.4 V	**A1**
10(b)(ii)1	work done ÷ charge	**B1**
10(b)(ii)2	correct circuit symbol for a cell and positive correct	**B1**
	four cells, correct symbol, correctly in series	**B1**
10(b)(ii)3	four cells in series and another four in parallel **or** any other series and parallel arrangement of 8 cells with connections to and from battery	**B1**
10(c)	lasts longer **or** if one cell fails it still works **or** contains more energy	**B1**

QUESTION · 5054_w17_qp_11.pdf · source page 13 of 20

29 What is *electromotive force (e.m.f.)*?

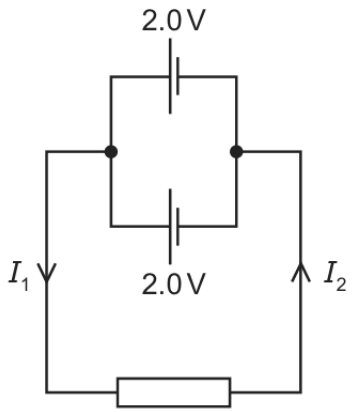
A $\frac{\text{charge}}{\text{work done}}$ **B** $\frac{\text{charge}}{\text{time}}$ **C** $\frac{\text{work done}}{\text{charge}}$ **D** $\frac{\text{work done}}{\text{time}}$

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 3 of 3 · row for Question 29

29	C	**1**

QUESTION · 5054_w17_qp_11.pdf · source page 14 of 20

31 The battery in a circuit contains two identical cells connected in parallel.



Which row shows the relationship between currents I_1 and I_2 and gives the electromotive force (e.m.f.) of the battery?

	relationship	e.m.f. / V
A	$I_1 > I_2$	2.0
B	$I_1 > I_2$	4.0
C	$I_1 = I_2$	2.0
D	$I_1 = I_2$	4.0

MARK SCHEME (official answer key) · 5054_w17_ms_11.pdf · source page 3 of 3 · row for Question 31

31	C	**1**

11 **(a)** Fig. 11.1 shows a negatively charged rod close to an uncharged metal sphere that is mounted on an insulating stand.

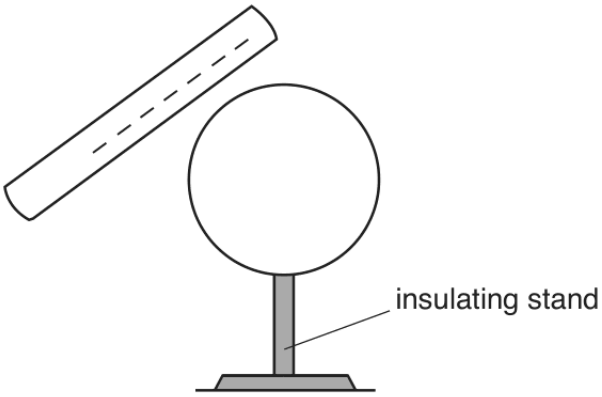


Fig. 11.1

(i) Suggest a material for the insulating stand.

..... [1]

(ii) On Fig. 11.1, draw the distribution of charges on the sphere. [2]

(iii) A metal wire connected to earth is touched against the sphere.

1. Explain what happens to the charge on the sphere.

.....

.....

.....

.....

.....

..... [3]

2. The earth wire remains in contact with the sphere.

Describe what happens in the wire as the rod is moved away.

.....

.....

..... [1]

(b) A circuit contains a 7.5 V d.c. power supply and three resistors.

Fig. 11.2 is the circuit diagram.

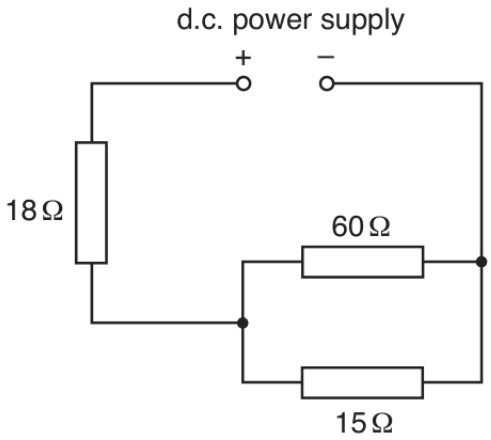


Fig. 11.2

(i) Determine the total resistance of the circuit.

resistance = ... [3]

(ii) Calculate the current in the power supply.

current = ... [2]

(iii) There is a different current in each of the three resistors in the circuit in Fig. 11.2.

Underline the resistor in which the current is

1. the largest,

15 Ω resistor **18 Ω resistor** **60 Ω resistor**

2. the smallest.

15 Ω resistor **18 Ω resistor** **60 Ω resistor**

[1]

(iv) The 7.5 V d.c. power supply is made from five 1.5 V cells.

In the space below, draw a diagram that shows how the cells are arranged.

[1]

(v) The 18 Ω resistor is made of metal.

State and explain what happens to the current in this resistor as its temperature increases.

.....

.....

..... [1]

MARK SCHEME · 5054_w17_ms_21.pdf · source page 9 of 10 · Question 11

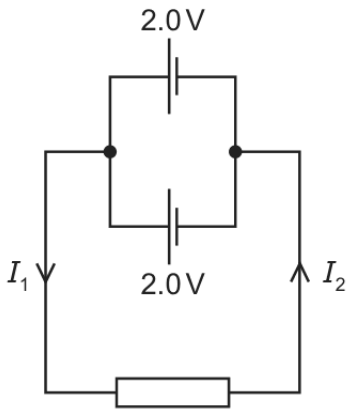
Question	Answer	Marks
11(a)(i)	any suitable solid insulator (e.g. nylon, plastic, glass, rubber, polystyrene)	**B1**
11(a)(ii)	positive charges near to rod	**B1**
	negative charges opposite rod **and** equal in number **and** 7 or fewer	**B1**
11(a)(iii)	**1** electrons / negative charges flow towards earth	**B1**
	repelled (by negative charge on rod)	**B1**
	(sphere) becomes positive	**B1**
	2 flow of electrons / negative charge **and** (in direction) earth to sphere	**B1**
11(b)(i)	$1 / R = 1 / R_1 + 1 / R_2$ **or** $R_1 R_2 / (R_1 + R_2)$ **or** 1 / R = 1 / 15 + 1 / 60 **or** $15 \times 60 / 75$ **or** $15 \times 60 / (15 + 60)$	**C1**
	12 (Ω) **or** 0.083 (Ω)	**C1**
	30 Ω	**A1**
11(b)(ii)	(I =)V / R **or** 7.5 / 30	**C1**
	0.25 A	**A1**

MARK SCHEME · 5054_w17_ms_21.pdf · source page 10 of 10 · Question 11

Question	Answer	Marks
11(b)(iii)	**18 Ω resistor** underlined **2** **and 60 Ω resistor** underlined	**B1**
11(b)(iv)	five cells in series **and** all in same direction	**B1**
11(b)(v)	resistance increases **and** current decreases	**B1**

QUESTION · 5054_w17_qp_12.pdf · source page 13 of 16

31 The battery in a circuit contains two identical cells connected in parallel.



Which row shows the relationship between currents I_1 and I_2 and gives the electromotive force (e.m.f.) of the battery?

	relationship	e.m.f. / V
A	$I_1 > I_2$	2.0
B	$I_1 > I_2$	4.0
C	$I_1 = I_2$	2.0
D	$I_1 = I_2$	4.0

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 3 of 3 · row for Question 31

31	C	**1**

10 A student connects a battery to two resistors. The circuit diagram is shown in Fig. 10.1.

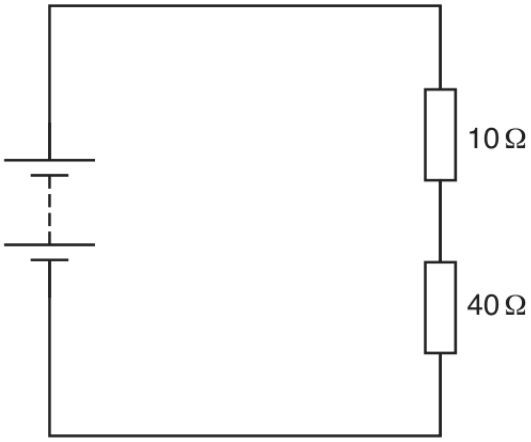


Fig. 10.1

The potential difference (p.d.) across the 40 Ω resistor is 9.6 V.

(a) State what is meant by the *potential difference* across a resistor.

.....

.....[2]

(b) **(i)** Calculate the current in the 40 Ω resistor.

current = ...[2]

(ii) Calculate the electromotive force (e.m.f.) of the battery.

e.m.f. = ...[2]

(c) The student has three different voltmeters to measure the p.d. across the 40 Ω resistor. These are labelled 0–2 V, 0–20 V and 0–200 V. Each has a pointer that shows the p.d. on a scale.

State and explain which of the three voltmeters is best to measure this p.d.

.....

.....

.....

.....[2]

(d) (i) Calculate the power P produced in the 10 Ω resistor.

P = ...[2]

(ii) The student has available two 10 Ω resistors, with power ratings of $\frac{1}{2}P$ and $2P$.

Suggest why a resistor with a power rating of $\frac{1}{2}P$ is not suitable for the circuit in Fig. 10.1.

.....

.....[1]

(e) The student adds a resistor R to the circuit, to make the circuit shown in Fig. 10.2.

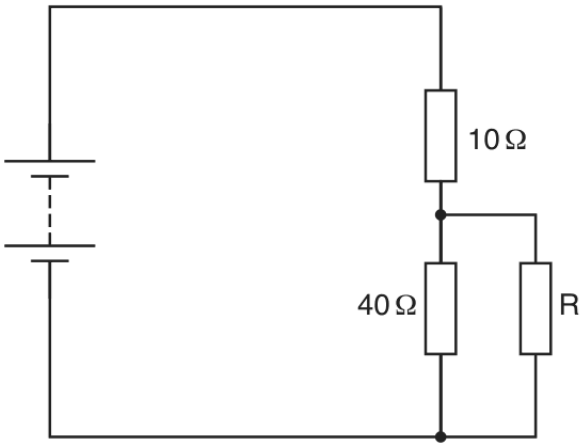


Fig. 10.2

Complete the table in Fig. 10.3 to show what happens as resistor R is connected.

You should state whether each quantity increases, decreases or stays the same and give a brief explanation of why any change occurs. Calculations are not required.

quantity	increases, decreases or stays the same as resistor R is added	brief explanation of why the change occurs
current in 10 Ω resistor	increases	
p.d. across 10 Ω resistor		
p.d. across 40 Ω resistor		

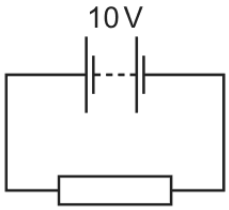
[4]

Fig. 10.3

Question	Answer	Marks
10(a)	work done / energy transferred	**B1**
	work done / energy transferred in taking unit charge (through resistor)	**B1**
10(b)(i)	(I=) V / R in any form algebraic or numerical	**C1**
	0.24 A	**A1**
10(b)(ii)	(total resistance of) 50 (Ω) seen or 2.4 (V) p.d. across 10 Ω resistor	**C1**
	12 V	**A1**
10(c)	voltmeter chosen is 0–20 V	**B1**
	correct explanation of why 0–200 V meter is unsuitable **OR** correct explanation of why 0–2 V meter is unsuitable	**B1**
10(d)(i)	(P)= VI or I^2R or V^2 / R in any form	**C1**
	0.58 W	**A1**
10(d)(ii)	½P rated resistor blows / too hot / melts	**B1**
10(e)	current in 10 Ω | increases | resistance decreases (of parallel combination or total resistance)	**B1**
	p.d. across 10 Ω | increases | p.d.is proportional to current **or** (p.d.) increases as current increases **or** potential divider argument	**M1+A1**
	p.d. across 40 Ω | decreases | sum of p.d.s constant / 12 V / value in **(b)(ii)** **or** p.d. across 10 Ω increases **or** resistance (40 Ω and R) a smaller fraction of total	**B1**

QUESTION · 5054_w18_qp_11.pdf · source page 11 of 16

29 A battery of electromotive force (e.m.f.) 10 V is connected to a resistor.



A charge of 2.0 coulombs passes through the resistor.

How much work is done as the charge passes through the resistor?

A 0.20 J **B** 2.0 J **C** 5.0 J **D** 20 J

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 3 of 3 · row for Question 29

29	D	**1**

7 A student sets up the circuit shown in Fig. 7.1.

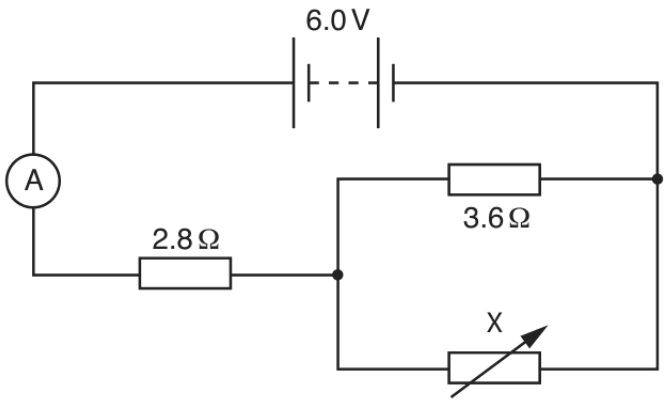


Fig. 7.1

The electromotive force (e.m.f.) of the battery is 6.0 V.

(a) The resistance of the variable resistor X is set to 1.8 Ω.

Determine

(i) the total resistance of the circuit,

resistance = ... [3]

(ii) the current measured by the ammeter.

current = ... [2]

(b) **(i)** State what is meant by the potential difference (p.d.) across a component in a circuit.

.....

..... [1]

(ii) Add to Fig. 7.1 the symbol for a component that is measuring the p.d. across the 2.8 Ω resistor. [1]

(iii) The resistance of X is increased.

Explain why the p.d. across the 2.8 Ω resistor decreases.

.....

.....

..... [2]

Question	Answer	Marks
7(a)(i)	$1 / R_T = 1 / R_1 + 1 / R_2$ **or** $1 / R_T = 1 / 3.6 + 1 / 1.8$ **or** (R_T =) $R_1 R_2 / (R_1 + R_2)$ **or** $1.8 \times 3.6 / (1.8 + 3.6)$	**C1**
	1.2 (Ω)	**C1**
	4.0 Ω	**A1**
7(a)(ii)	(I =) V / R **or** 6.0 / 4.0	**C1**
	1.5 (A)	**A1**
7(b)(i)	work done / energy (released) per unit charge (passed through component)	**B1**
7(b)(ii)	voltmeter symbol **and** across 2.8 Ω resistor	**B1**

Question	Answer	Marks
7(b)(iii)	total resistance increases **or** resistance of circuit / parallel combination increases	**B1**
	current (in 2.8 Ω resistor) decreases	**B1**

QUESTION · 5054_w18_qp_12.pdf · source page 13 of 16

32 A defibrillator is a device that is used to give an electric shock to a patient's heart.

It supplies an electric shock with energy 240 J at an average voltage of 2000 V for 10 ms.

What is the average current it supplies?

A 0.012 A **B** 1.2 A **C** 12 A **D** 120 A

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32	C	**1**

10 A student sets up the circuit shown in Fig. 10.1 in a laboratory at room temperature.

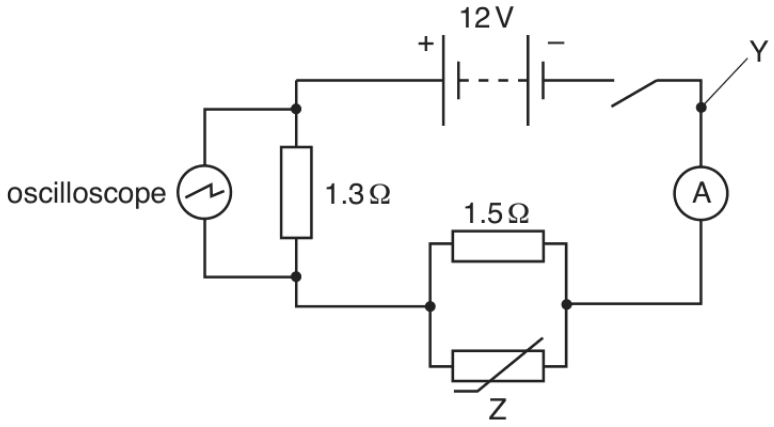


Fig. 10.1

The electromotive force (e.m.f.) of the battery is 12 V.

The switch is closed.

(a) The connecting wires in the circuit are made from copper covered by insulating plastic.

State the name of the particles that flow in the copper and state the direction in which they are flowing at point Y in the circuit.

.....

.....

..... [2]

(b) At room temperature, the resistance of component Z is 6.0 Ω.

(i) State the name of component Z.

..... [1]

(ii) Calculate the resistance of the whole circuit.

resistance = ... [3]

(iii) Calculate the current measured by the ammeter.

current = ... [2]

(iv) The current in the ammeter is I_A, the current in the 1.5 Ω resistor is I_R and the current in component Z is I_Z.

Write down an equation that relates I_A, I_R and I_Z.

..... [1]

(c) Fig. 10.2 shows the screen of the oscilloscope.

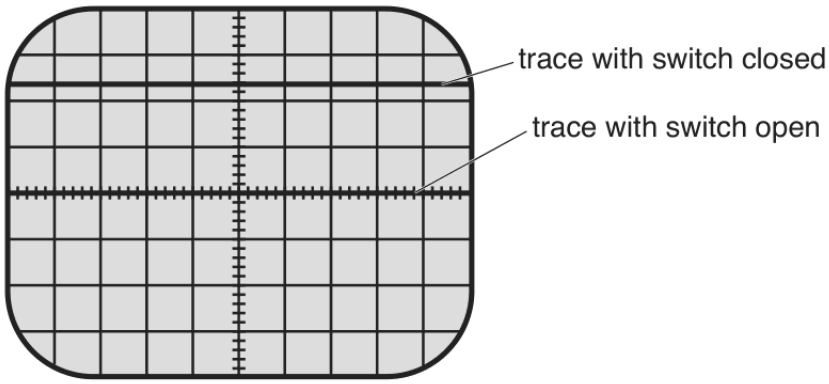


Fig. 10.2

Before the switch is closed, the trace is a horizontal line across the middle of the screen, as shown in Fig. 10.2. When the switch is closed, the trace remains horizontal and moves up the screen.

Component Z is heated.

State and explain what is observed on the oscilloscope screen as the temperature of Z increases.

.....

.....

.....

.....

.....

..... [4]

(d) The 12 V battery is made from cells of electromotive force (e.m.f.) 1.5 V which are all in series.

(i) Calculate the number of 1.5 V cells that make up the battery.

number = ... [1]

(ii) State the e.m.f. of a battery made by connecting all of these cells in parallel.

e.m.f. = ... [1]

MARK SCHEME · 5054_w18_ms_22.pdf · source page 9 of 10 · Question 10

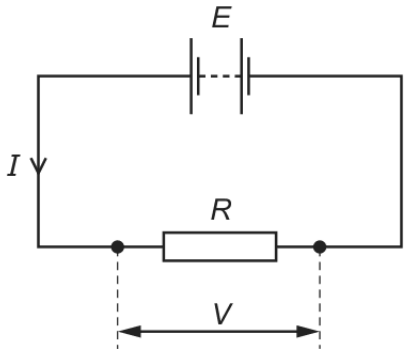
Question	Answer	Marks
10(a)	electrons **c.a.o.**	**M1**
	towards the ammeter **or** away from the negative terminal **or** towards the positive terminal	**A1**
10(b)(i)	thermistor **c.a.o.**	**B1**
10(b)(ii)	$1 / R_T = 1 / R_1 + 1 / R_2$ **or** $1 / R_T = 1 / 1.5 + 1 / 6.0$ **or** $(R_T =) R_1R_2 / (R_1 + R_2)$ **or** $1.5 \times 6.0 / (1.5 + 6.0)$	**C1**
	1.2 (Ω)	**C1**
	2.5 Ω	**A1**
10(b)(iii)	$(I =) V / R$ **or** $12 / 2.5$	**C1**
	4.8 (A)	**A1**
10(b)(iv)	$I_A = I_R + I_Z$	**B1**
10(c)	resistance of Z / thermistor decreases	**B1**
	resistance of parallel combination decreases **or** total resistance (of circuit) decreases **or** current increases	**B1**
	voltage (across 1.3 Ω) increases	**B1**
	trace moves towards top of screen / upwards	**B1**

MARK SCHEME · 5054_w18_ms_22.pdf · source page 10 of 10 · Question 10

Question	Answer	Marks
10(d)(i)	8	**B1**
10(d)(ii)	1.5 V	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 13 of 16

34 The diagram shows a simple electrical circuit.



Which statement **cannot** be correct?

A The current I is 2 A.

B The e.m.f. E is 12 J.

C The p.d. V (across the resistor) is 10 V.

D The resistance R is 5 Ω.

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 3 of 3 · row for Question 34

34	B	**1**

6 A 60 W filament lamp is powered by the 240 V mains electricity supply.

The lamp is switched on.

(a) Calculate:

(i) the current in the lamp

current = ... [2]

(ii) the resistance of the lamp.

resistance = ... [2]

(b) The lamp is now unplugged from the mains supply and connected in series with two 1.5 V cells. Fig. 6.1 is the circuit diagram.

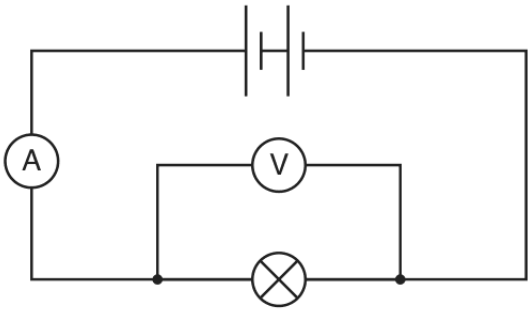


Fig. 6.1

(i) State the reading on the voltmeter.

reading = ... [1]

(ii) State and explain how the resistance of the lamp now differs from the value calculated in **(a)(ii)**.

.....

.....

..... [2]

(iii) When a filament lamp blows, it very often does so immediately after being switched on.

Suggest why this is the case.

.....

..... [1]

[Total: 8]

Question	Answer	Marks
6(a)(i)	(*I* =) *P* / *V* **or** 60 / 240	**C1**
	0.25 A	**A1**
6(a)(ii)	(*R* =) *V* / *I* **or** 240 / 0.25	**C1**
	960 Ω	**A1**
6(b)(i)	3.0 V	**B1**
6(b)(ii)	resistance is smaller	**B1**
	(filament at a) lower temperature	**B1**
6(b)(iii)	current surge (due to lower resistance when cold)	**B1**

QUESTION · 5054_s20_qp_11.pdf · source page 13 of 16

36 A battery consists of three identical cells in parallel.

What is the unit of electromotive force (e.m.f.) and to what is the e.m.f. of the battery equal?

	unit	e.m.f. of the battery is equal to
A	J / C	the sum of the e.m.f.s of the three cells
B	J / C	the e.m.f. of one of the cells
C	N / V	the sum of the e.m.f.s of the three cells
D	N / V	the e.m.f. of one of the cells

MARK SCHEME (official answer key) · 5054_s20_ms_11.pdf · source page 3 of 3 · row for Question 36

36	B	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 Fig. 8.1 shows the circuit diagram of a temperature gauge. It contains an ammeter, a thermistor, a fixed resistor R and a battery.

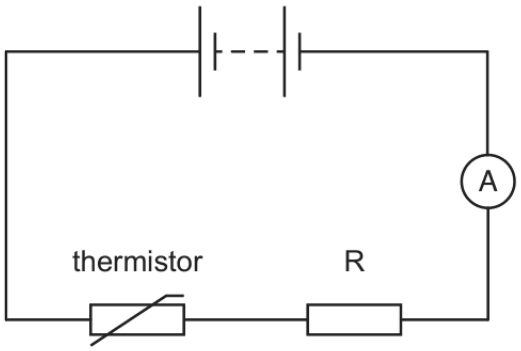


Fig. 8.1

The current is measured at different temperatures and a graph of the results is shown in Fig. 8.2.

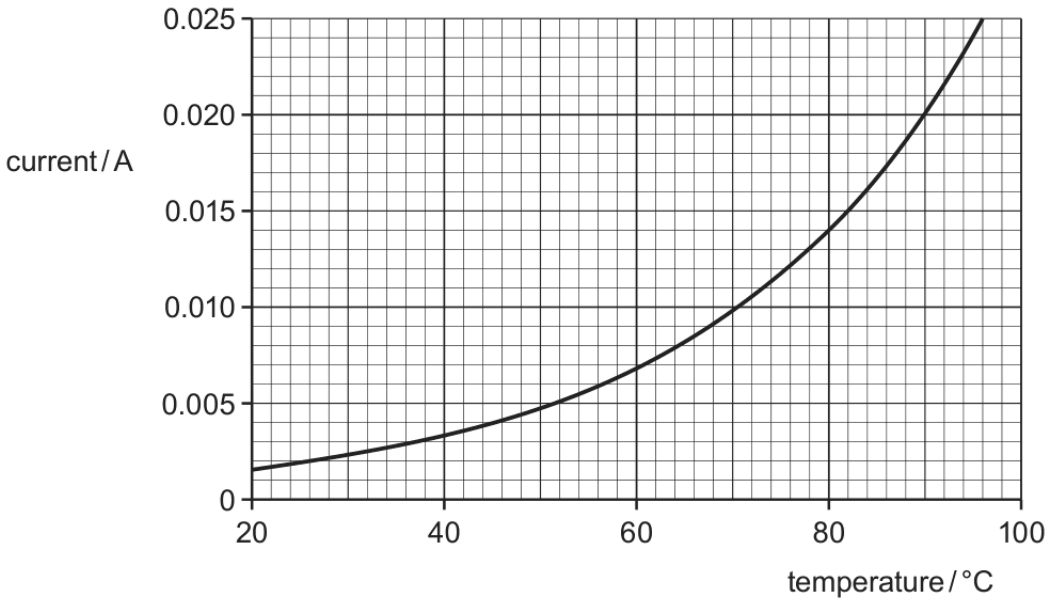


Fig. 8.2

(a) (i) State how the resistance of the thermistor changes with temperature and explain how Fig. 8.2 shows this change.

.....

.....

..... [2]

(ii) Resistance and current are used for the measurement of temperature.

State **one** other physical property that is used for the measurement of temperature.

.....

..... [1]

(b) A temperature scale involves the use of fixed points.

State what is meant by:

(i) *the ice point*

.....

..... [1]

(ii) *the steam point*.

.....

..... [1]

(c) At 80 °C, the potential difference (p.d.) across the thermistor is 3.6 V.

(i) State what is meant by *potential difference*.

.....

..... [1]

(ii) Calculate the resistance of the thermistor when the temperature is 80 °C.

resistance = ... [3]

(iii) The electromotive force (e.m.f.) of the battery is 25 V.

Calculate the resistance of the resistor R.

resistance = ... [2]

(d) The current scale on the ammeter is linear.

Using values from Fig. 8.2, a student marks the scale on the ammeter with temperature values that correspond to the values of the current.

The temperature is then read directly from the temperature scale on the ammeter.

(i) State what is meant by a *sensitive* thermometer.

.....

.....

..... [1]

(ii) State and explain at which temperatures this thermometer is most sensitive.

.....

.....

.....

..... [2]

(e) The temperature gauge is used to measure the temperature of a room.

Suggest why the temperature obtained using the gauge is slightly higher than the actual temperature of the room.

.....

..... [1]

[Total: 15]

MARK SCHEME · 5054_s20_ms_22.pdf · source page 10 of 13 · Question 8

Question	Answer	Marks
8(a)(i)	resistance decreases as temperature increases	**B1**
	current increases (and voltage constant) as temperature increases	**B1**
8(a)(ii)	volume (of liquid/gas), expansion (of solid/liquid or gas), length (of solid), pressure (of gas) length (of bimetallic strip), e.m.f. (of thermocouple), colour (of liquid crystals or very hot objects)	**B1**
8(b)(i)	temperature of melting ice **or** freezing water	**B1**
8(b)(ii)	temperature of boiling water (at 1 atmosphere)	**B1**
8(c)(i)	work done / energy change ÷ charge (passed through a component)	**B1**
8(c)(ii)	(R =) V / I in any form	**C1**
	(I =) 0.014(0) (A)	**C1**
	260 Ω	**A1**

MARK SCHEME · 5054_s20_ms_22.pdf · source page 11 of 13 · Question 8

Question	Answer	Marks
8(c)(iii)	(V across R =) 25-3.6 **or** 21(.4 V) **OR** (total resistance =) 25 / 0.014	**C1**
	1500 Ω	**A1**
8(d)(i)	(a thermometer where the quantity) changes a lot per unit temperature rise / °C / same temperature rise	**B1**
8(d)(ii)	at high temperatures	**B1**
	large(r) change in current / meter deflection for unit temperature rise / °C / same temperature rise **or** large(r) gradient	**B1**
8(e)	heat (thermal energy) produced in the thermistor / resistor	**B1**

10 An oscilloscope is a device used to display waveforms.

(a) Inside the oscilloscope, a beam of electrons is emitted from a metal filament by thermionic emission.

The emitted electrons are accelerated away from the filament by a potential difference of 2000 V.

(i) State what must happen to the metal filament for thermionic emission to occur.

.....

..... [1]

(ii) Explain why the electrons accelerate away from the filament.

.....

.....

..... [1]

(iii) The charge on one electron is 1.6×10^{-19} C.

Calculate the maximum kinetic energy of one electron after it has been accelerated through 2000 V.

kinetic energy = ... [2]

(b) Fig. 10.1 is the trace on the screen of the oscilloscope.

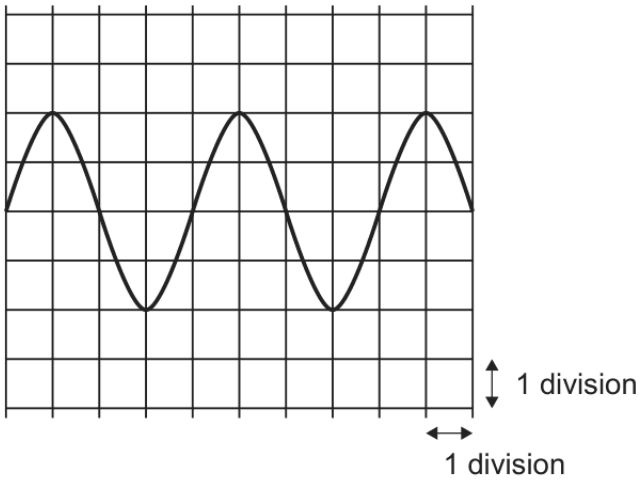


Fig. 10.1

The settings on the oscilloscope are 10 ms / division for the *x*-axis and 3.0 V / division for the *y*-axis.

(i) Calculate the amplitude of the trace shown in Fig. 10.1.

amplitude = ... [2]

(ii) Calculate the time for one oscillation (complete wavelength) of the trace shown in Fig. 10.1.

time = ... [1]

(c) The trace shown in Fig. 10.1 is caused by a sound.

The sound travels through the air to a microphone from the place that it is made. The microphone is connected to the oscilloscope which displays the waveform shown.

(i) Sound is a type of wave.

State which type.

..... [1]

(ii) Describe the motion of the air molecules as the sound passes through the air to the microphone.

.....

.....

..... [2]

(iii) Describe and explain how the trace on the screen changes as the pitch of the sound becomes higher. The settings on the oscilloscope are unchanged.

.....

.....

..... [2]

(d) A student investigates what happens when a diode is connected between the microphone and the oscilloscope.

Fig. 10.2 shows the circuit that he uses.

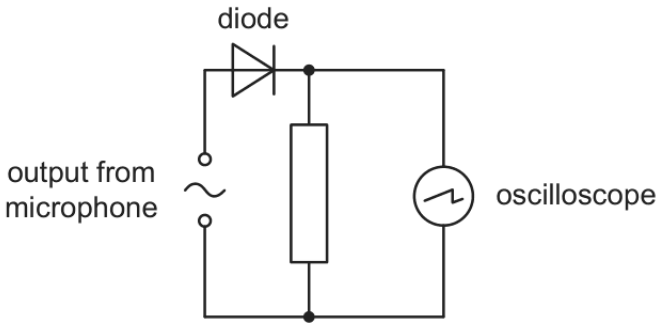


Fig. 10.2

Without the diode, the output from the microphone is an alternating current and the trace is as shown in Fig. 10.1 on page 18.

(i) Describe the action of the diode on the current from the microphone.

.....

..... [1]

(ii) On the grid in Fig. 10.3, sketch the trace seen on the screen when the diode is used.

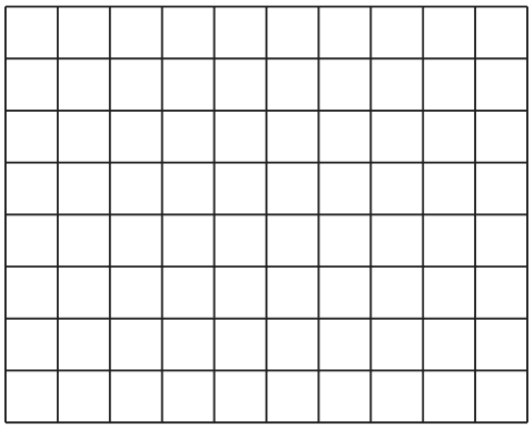


Fig. 10.3

[2]

[Total: 15]

MARK SCHEME · 5054_s20_ms_22.pdf · source page 12 of 13 · Question 10

Question	Answer	Marks
10(a)(i)	filament is hot / heated	**B1**
10(a)(ii)	attracted by a positive potential **or** repelled by filament **or** filament is negative	**B1**
10(a)(iii)	$2000 \times 1.6 \times 10^{-19}$	**C1**
	3.2×10^{-16} J	**A1**
10(b)(i)	two divisions	**C1**
	6(.0) V	**A1**
10(b)(ii)	40 ms	**B1**
10(c)(i)	longitudinal	**B1**

MARK SCHEME · 5054_s20_ms_22.pdf · source page 13 of 13 · Question 10

Question	Answer	Marks
10(c)(ii)	oscillation described, e.g. back and forth	**B1**
	in direction of travel (of wave / energy) **or** compressions and rarefactions stated or described	**B1**
10(c)(iii)	more waves seen on screen	**B1**
	higher frequency (of sound) **or** more waves per second **or** time for one wave less	**B1**
10(d)(i)	diode passes current in only one direction **or** current only flows for half the cycle / oscillation **or** current is d.c. / direct	**B1**
10(d)(ii)	at least one top half and no bottom half **or** one bottom half and no top half	**C1**
	three top halves of the wave and flat sections along mid line **or** two bottom halves of the wave and flat sections along mid line	**A1**

10 Fig. 10.1 shows a wire of length l and cross-sectional area A.

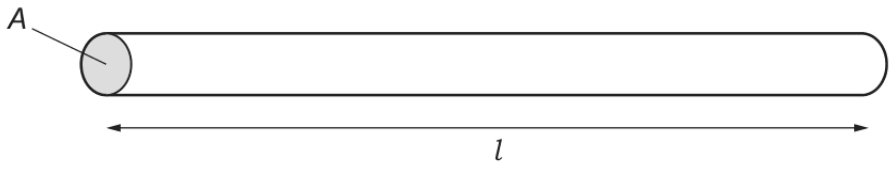


Fig. 10.1

(a) State how the resistance of the wire in Fig. 10.1 depends on:

(i) l

..... [1]

(ii) A.

..... [1]

(b) The cross-sectional area of a piece of metal wire is $7.5 \times 10^{-4}\,\text{cm}^2$. The resistance of a 1.0 m length of the same wire is 6.4 Ω.

The wire is made from metal W.

Fig. 10.2 shows a solid cube of side 1.0 cm. It is also made from metal W.

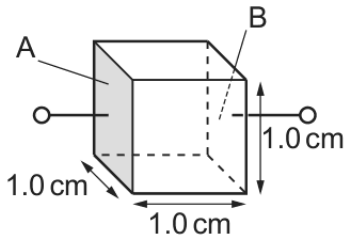


Fig. 10.2

Calculate the resistance between the two opposite faces A and B of the cube.

resistance = ... [2]

(c) The wire in part **(b)** is taped to a metre rule.

Fig. 10.3 shows that a 1.0 m length of the wire (resistance 6.4 Ω) is connected in series with a switch, a cell of electromotive force (e.m.f.) 1.2 V and a resistor of resistance 9.6 Ω.

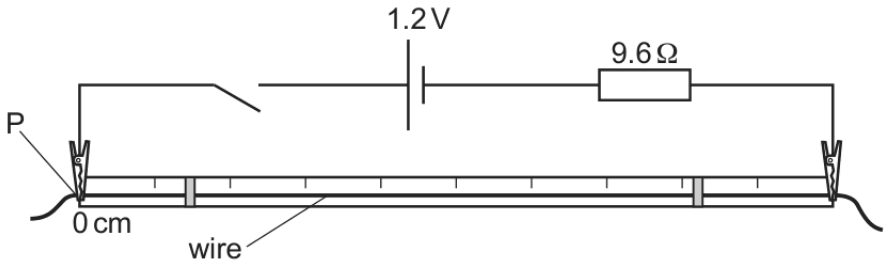


Fig. 10.3

The switch is closed.

(i) Explain what is meant by *electromotive force* (*e.m.f.*).

.....

.....

..... [2]

(ii) Calculate the potential difference (p.d.) across the 1.0 m length of the wire.

p.d. = ... [3]

(iii) One input terminal of an oscilloscope is connected to the wire at point P, the 0 cm mark of the metre rule.

The other terminal of the oscilloscope is connected to a sliding contact. Initially, this contact touches the wire at point P.

The Y-gain setting on the oscilloscope is 0.20 V / cm.

Fig. 10.4 shows the screen of the oscilloscope with a horizontal trace across the middle of the screen.

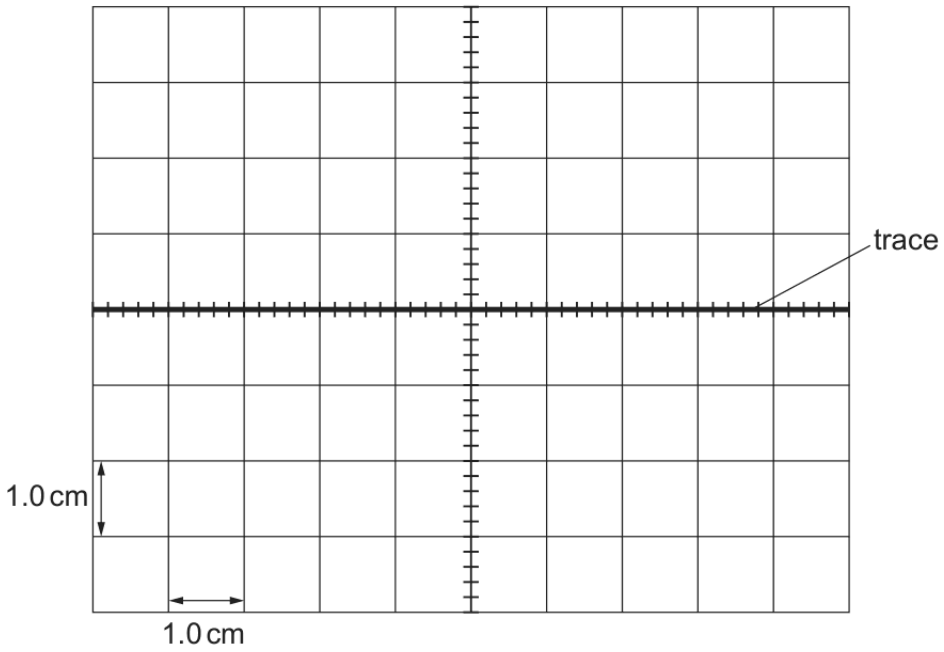


Fig. 10.4

The sliding contact is slowly moved along the wire until it reaches the other end of the metre rule.

Describe and explain what happens to the trace on the screen.

.....

.....

.....

..... [3]

(d) A second, identical 1.2 V cell is connected in parallel with the cell in the circuit in Fig. 10.3.

(i) State **one** advantage of using two cells in parallel rather than a single cell.

.....

..... [1]

(ii) State and explain the effect on the trace in **(c)(iii)** of adding the second cell in parallel.

.....

.....

..... [2]

[Total: 15]

Question	Answer	Marks
10(a)(i)	(resistance) is directly proportional to length	**B1**
10(a)(ii)	(resistance) is inversely proportional to (cross-sectional) area	**B1**
10(b)	(R =) $6.4 \times 7.5 \times 10^{-4} / 100$ **or** 4.8×10^{-n} (n is an integer)	**C1**
	$4.8 \times 10^{-5}\ \Omega$	**A1**
10(c)(i)	energy / work done in driving a charge round a circuit **or** energy transferred to electrical energy	**B1**
	energy / work done per unit charge **or** energy / charge	**B1**
10(c)(ii)	(R =) 6.4 + 9.6 **or** 16	**C1**
	(V =) $1.2 \times 6.4 / 16$ **or** current = 1.2 / 16 **or** current = 0.075	**C1**
	0.48 V	**A1**
10(c)(iii)	trace moves vertically away from centre line	**B1**
	moves distance of 2.4 cm	**B1**
	remains horizontal **or** distance moved by trace directly proportional to distance moved by jockey **or** comment about tape	**B1**
10(d)(i)	the battery lasts twice as long as the single cell	**B1**
10(d)(ii)	no effect **or** trace moves in an identical manner	**B1**
	e.m.f. of identical cells in parallel equals the e.m.f. of a single cell	**B1**

7 **(a)** Fig. 7.1 shows two circuit components.

Fig. 7.1

For each of these components state what causes an **increase** in its resistance.

thermistor .....

.....

LDR .....

.....
[2]

(b) Fig. 7.2 shows a battery connected to a diode, an ammeter and a resistor.

Fig. 7.3 shows the current–voltage graph for the diode.

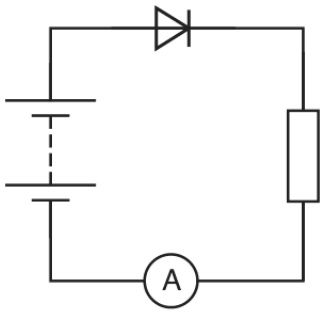


Fig. 7.2

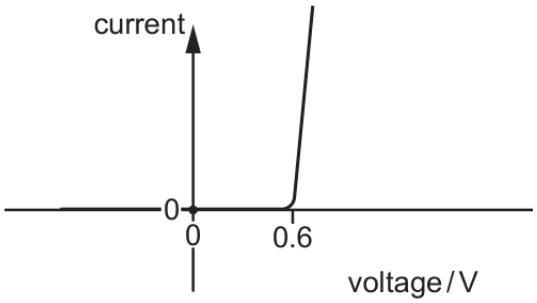


Fig. 7.3

The current in the resistor is 2.0 A and the potential difference across the resistor is 5.4 V.

(i) State what is meant by the *electromotive force* (e.m.f.) of the battery.

.....

..... [1]

(ii) Explain why the e.m.f. of the battery must be greater than 5.4 V.

.....

.....

..... [2]

(iii) Calculate the electrical energy input to the resistor in 5.0 minutes.

energy input = ... [2]

(iv) The circuit is reconnected with the diode in the opposite direction, as shown in Fig. 7.4.

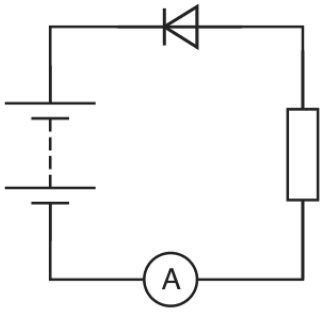


Fig. 7.4

Explain what is observed.

.....

.....

..... [1]

[Total: 8]

Question	Answer	Marks
7(a)	lower temperature / cooling of thermistor	**B1**
	less light on LDR	**B1**
7(b)(i)	energy / work per unit charge	**B1**
7(b)(ii)	Any 2 from: • voltage across diode • voltage across ammeter • voltage across internal resistance (of battery) • (e.m.f. / voltage of battery =) 5.4 + 0.6 (or larger)	**B2**
7(b)(iii)	(*E* =) *VQ* **or** *VIt* in any form algebraic or numerical	**C1**
	3200 J	**A1**
7(b)(iv)	no current / no ammeter reading **and** diode has been reversed / diode does not conduct / has (very) high resistance / blocks current / reference to current / voltage graph	**B1**

5 Fig. 5.1 shows a method of producing sandpaper using static electricity.

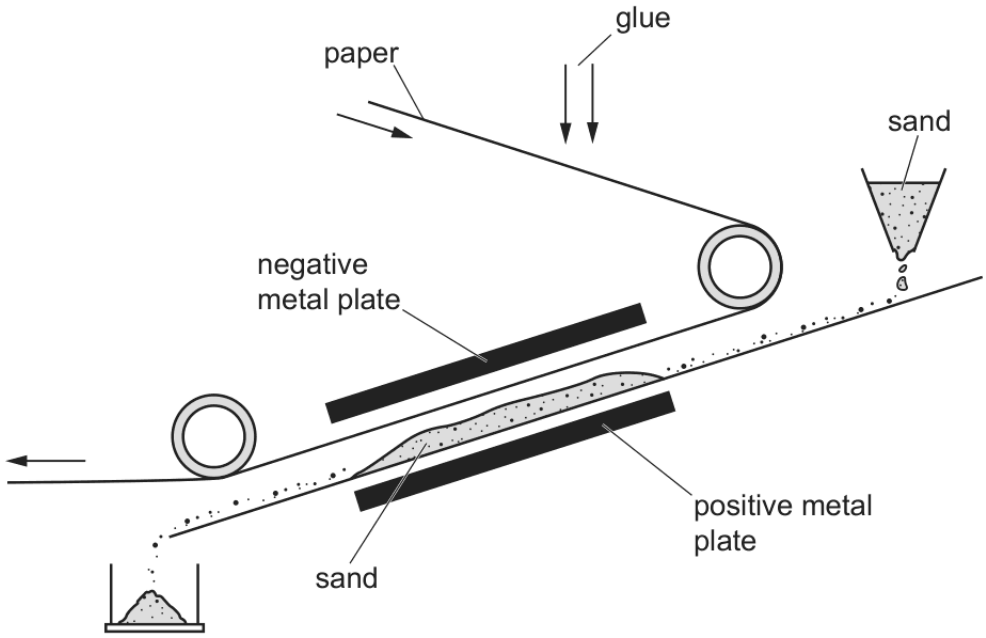


Fig. 5.1

Glue is sprayed on to moving paper. The sticky paper then passes between two metal plates.

One of the plates is positive and the other plate is negative.

There is an electric field between the plates.

(a) On Fig. 5.2, draw the electric field between the two metal plates. Show the direction of the electric field.

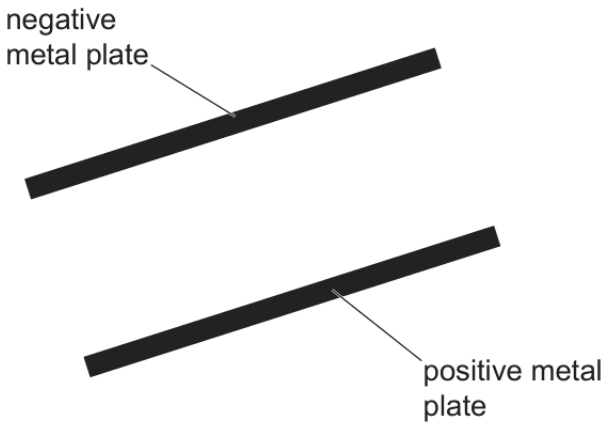


Fig. 5.2

[2]

(b) Grains of sand are present just below the sticky paper.

Each grain of sand is a conductor of electricity.

Fig. 5.3 shows two layers of sand grains in the space between the two plates.

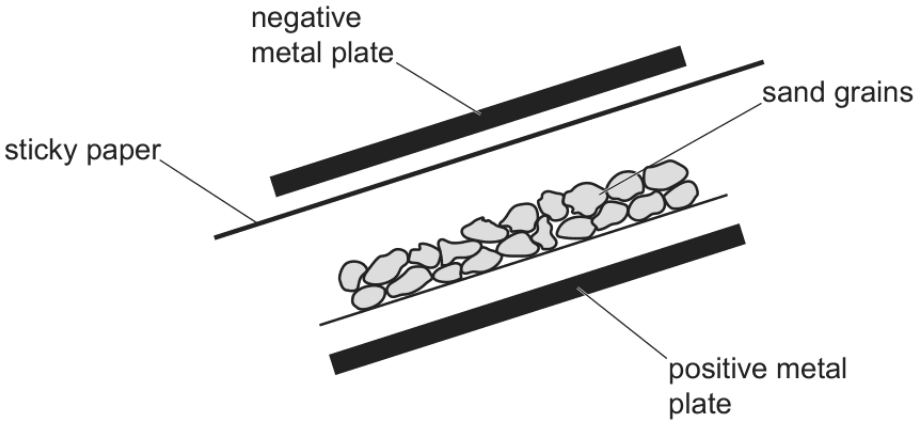


Fig. 5.3

(i) Explain, using ideas about the movement of charge, why the top layer of sand becomes positively charged.

.....

..... [1]

(ii) State why the top layer of sand moves towards the sticky paper.

.....

..... [1]

(c) A student receives an electric shock by walking across a certain type of carpet and then nearly touching a piece of earthed metal with his hand.

The potential difference between the hand and the metal is 2000 V and the charge transferred is 4.0×10^{-4} C.

Calculate the energy transferred when the spark jumps across the air gap from the metal onto his hand.

energy = ... [2]

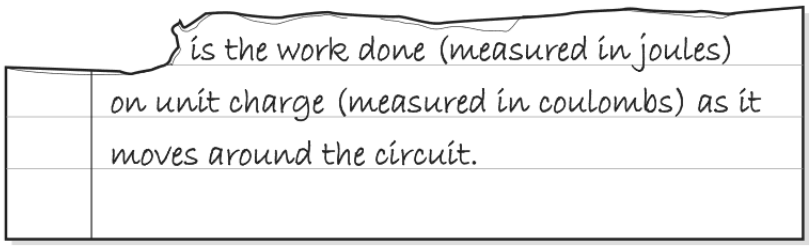
[Total: 6]

Question	Answer	Marks
5(a)	at least three straight lines perpendicular to plates	**B1**
	arrow on at least one line from positive to negative plate	**B1**
5(b)(i)	electrons / negative charge (on sand / grains) move to(wards) the positive plate / downwards / to(wards) or into the bottom layer	**B1**
5(b)(ii)	positive (sand grain) attracted to or by negative (plate)	**B1**

Question	Answer	Marks
5(c)	(E) = QV numerical or algebraic	**C1**
	0.8(0) J	**A1**

QUESTION · 5054_w21_qp_11.pdf · source page 13 of 20

29 A piece of paper torn out of an exercise book is shown.



What is being defined?

A electrical resistance

B electric current

C electrical power

D electromotive force

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 3 of 3 · row for Question 29

29	D	**1**

6 A circuit contains a battery of electromotive force (e.m.f.) 4.5 V, a 28 Ω resistor and a variable resistor. Fig. 6.1 is the circuit diagram.

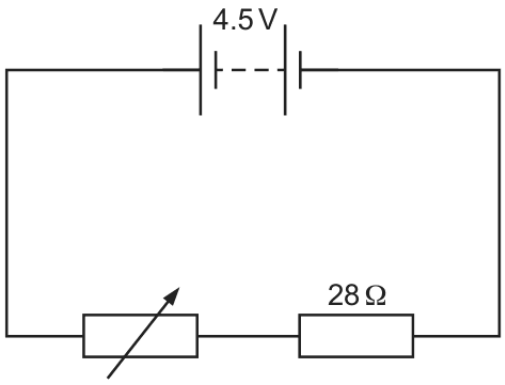


Fig. 6.1

(a) The battery consists of three identical cells in series.

Determine the e.m.f. of each cell.

e.m.f. of each cell = ... [1]

(b) The potential difference (p.d.) across the variable resistor is 2.5 V.

(i) State what is meant by *potential difference*.

.....

.....

.....

..... [2]

(ii) Calculate the resistance of the variable resistor.

resistance = ... [3]

[Total: 6]

Question	Answer	Marks
6(a)	1.5 V	**B1**
6(b)(i)	work done / energy transferred when charge passes through a component	**C1**
	$\frac{\text{work done / energy transferred}}{\text{charge passed}}$	**A1**
6(b)(ii)	4.5 – 2.5 **or** 2.0 (V) seen	**C1**
	(R =) V / I **or** 2.5 / (2.0 / 28) **or** R / 28 = 2.5 / 2.0	**C1**
	35 Ω	**A1**

QUESTION · 5054_w21_qp_12.pdf · source page 14 of 16

33 Quantity X and quantity Y have the same unit Z.

X is the energy dissipated by a battery in driving unit charge round the complete circuit.

Y is the work done in a component in driving unit charge through the component.

What are X, Y and Z?

	X	Y	Z
A	electromotive force	potential difference	coulomb
B	electromotive force	potential difference	volt
C	potential difference	electromotive force	coulomb
D	potential difference	electromotive force	volt

MARK SCHEME (official answer key) · 5054_w21_ms_12.pdf · source page 3 of 3 · row for Question 33

33	B	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 13 of 16

32 When the flash on a camera is used, a charge of 1.5 C flows for 0.0030 s through the flash lamp.

The average voltage across the flash lamp is 3600 V.

What is the electrical energy supplied to the flash lamp and what is the average power supplied?

	energy / J	power / W
A	2400	7.2
B	2400	800 000
C	5400	16.2
D	5400	1.8×10^6

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 3 of 3 · row for Question 32

32	D	**1**

QUESTION · 5054_s22_qp_11.pdf · source page 13 of 16

33 What is the unit of potential difference (p.d.) and which measuring instrument is used to measure p.d.?

	unit	instrument
A	ampere	ammeter
B	ampere	voltmeter
C	volt	ammeter
D	volt	voltmeter

MARK SCHEME (official answer key) · 5054_s22_ms_11.pdf · source page 3 of 3 · row for Question 33

33	D	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 15 of 20

33 Charge of 0.40 C passes through a resistor in 1.0 s.

In 2.0 s, 20 J of thermal energy is produced in the resistor.

What is the potential difference across the resistor?

A 0.020 V **B** 0.040 V **C** 25 V **D** 50 V

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 3 of 3 · row for Question 33

33	C	**1**

QUESTION · 5054_s22_qp_12.pdf · source page 16 of 20

35 When the flash on a camera is used, a charge of 1.5 C flows for 0.0030 s through the flash lamp.

The average voltage across the flash lamp is 3600 V.

What is the electrical energy supplied to the flash lamp and what is the average power supplied?

	energy / J	power / W
A	2400	7.2
B	2400	800 000
C	5400	16.2
D	5400	1.8×10^6

MARK SCHEME (official answer key) · 5054_s22_ms_12.pdf · source page 3 of 3 · row for Question 35

35	D	**1**

QUESTION · 5054_w22_qp_11.pdf · source page 11 of 16

28 Which expression gives the electromotive force (e.m.f.) of a battery?

A work done × time

B work done × charge

C $\frac{\text{work done}}{\text{time}}$

D $\frac{\text{work done}}{\text{charge}}$

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 2 of 3 · row for Question 28

28	D	**1**

5 The power supply in a circuit is made from several cells, each with the same electromotive force (e.m.f.). The cells are connected in parallel.

(a) Explain what is meant by 'electromotive force'.

.....

.....

..... [2]

(b) State **one** advantage of using several cells in parallel rather than a single cell as the power supply.

.....

..... [1]

(c) The power supply is connected in series to a resistor of resistance 4000 Ω and a thermistor. There is a voltmeter in parallel with the resistor.

Fig. 5.1 is the circuit diagram.

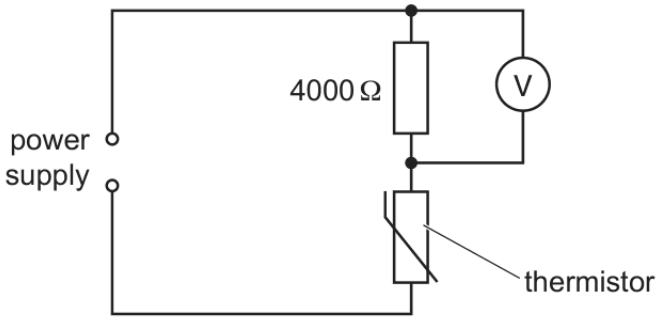


Fig. 5.1

(i) The temperature of the thermistor increases.

Explain what happens to the reading on the voltmeter.

.....

.....

.....

..... [3]

(ii) The e.m.f. of the power supply is 1.5 V.

Calculate the reading on the voltmeter when the resistance of the thermistor is 8000 Ω.

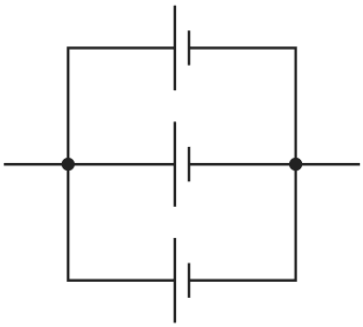
reading = ... [2]

[Total: 8]

Question	Answer	Marks
5(a)	energy / work done per unit charge **or** $\frac{\text{energy}}{\text{charge}}$ **or** $\frac{\text{work done}}{\text{charge}}$	**B1**
	property of a source / cell / battery / power supply **or** energy used driving charge around a (complete) circuit **or** energy transferred to electrical energy	**B1**
5(b)	it lasts for a longer time **or** more (electrical) energy available **or** power supply still functions if one cell fails / runs flat **or** can replace one cell without stopping circuit / stopping current	**B1**
5(c)(i)	resistance (of thermistor / circuit) decreases **c.a.o.**	**B1**
	current change **or** voltage change across thermistor matching the change in the resistance of the resistor	**B1**
	reading of voltmeter increases **c.a.o.**	**B1**
5(c)(ii)	(V_1 =) $V_0 \times R_1 / R_0$ **or** $1.5 \times 4000 / (4000 + 8000)$ **or** $1.5 \times 4000 / 12\,000$ **or** $I = V / R$ **or** $1.5 \times 8000 / (4000 + 8000)$ **or** I =1.5 / 12 000 **or** 0.125 mA	**C1**
	0.50 V	**A1**

QUESTION · 5054_s23_qp_11.pdf · source page 14 of 20

29 The diagram shows a battery of three 1.5 V cells.



What is an advantage of this arrangement of cells?

A The battery can supply a current for a longer time than a single 1.5 V cell.

B The battery can supply any e.m.f. between 0 V and 4.5 V.

C The battery supplies more energy to each coulomb of charge than a single 1.5 V cell.

D The e.m.f. of the battery is 4.5 V.

MARK SCHEME (official answer key) · 5054_s23_ms_11.pdf · source page 3 of 3 · row for Question 29

29	A	**1**

QUESTION · 5054_s23_qp_12.pdf · source page 11 of 16

28 Three identical cells are connected in parallel to a resistor.

What is the advantage of using three cells in parallel rather than using a single cell?

A Each cell produces more energy.

B Each cell supplies more charge.

C Each cell takes longer to discharge.

D The e.m.f. is greater than that of a single cell.

MARK SCHEME (official answer key) · 5054_s23_ms_12.pdf · source page 2 of 3 · row for Question 28

28	C	**1**

QUESTION · 5054_w23_qp_11.pdf · source page 10 of 16

27 The current in an electric motor is 2.0 A.

It transfers 20 J of energy in 5.0 s.

What is the potential difference (p.d.) across the motor?

A 0.50 V **B** 2.0 V **C** 8.0 V **D** 50 V

MARK SCHEME (official answer key) · 5054_w23_ms_11.pdf · source page 2 of 3 · row for Question 27

27	B	**1**

QUESTION · 5054_s24_qp_12.pdf · source page 10 of 16

28 Which equation is correct for potential difference (p.d.)?

A p.d. = voltage × current

B $\text{p.d.} = \frac{\text{energy}}{\text{time}}$

C $\text{p.d.} = \frac{\text{work done}}{\text{charge}}$

D $\text{p.d.} = \frac{\text{current}}{\text{resistance}}$

MARK SCHEME (official answer key) · 5054_s24_ms_12.pdf · source page 2 of 3 · row for Question 28

28	C	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 13 of 16

32 Which quantity is defined as the electrical work done by a source in moving a unit charge around a complete circuit?

A power

B potential difference

C electromotive force

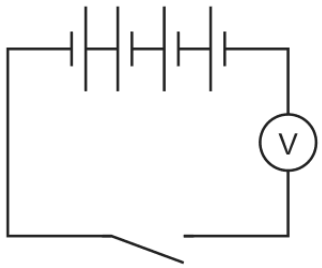
D energy

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 3 of 3 · row for Question 32

32	C	**1**

QUESTION · 5054_w24_qp_11.pdf · source page 14 of 16

33 Four 1.5 V cells are connected in a circuit as shown.



The switch is closed.

What is the reading on the voltmeter?

A 0 V **B** 1.5 V **C** 3.0 V **D** 4.5 V

MARK SCHEME (official answer key) · 5054_w24_ms_11.pdf · source page 3 of 3 · row for Question 33

33	C	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 11 of 16

29 How can one volt also be expressed?

A one coulomb per ampere

B one coulomb per joule

C one joule per ampere

D one joule per coulomb

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 3 of 3 · row for Question 29

29	D	**1**

QUESTION · 5054_s25_qp_12.pdf · source page 11 of 16

28 A quantity is defined as the work done by a source in moving unit charge around a complete circuit.

What is the quantity and what is its unit?

	quantity	unit
A	electromotive force	N
B	electromotive force	V
C	potential difference	N
D	potential difference	V

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 2 of 3 · row for Question 28

28	B	**1**

6 An electric car is driven by a direct current (d.c.) motor that is powered by a direct current power supply.

(a) Describe how a direct current differs from an alternating current (a.c.).

.....

.....

.....

..... [1]

(b) Fig. 6.1 represents the motor circuit. Point P is a point in the circuit between the motor and the positive terminal of the power supply.

Point P is shown to the left of the motor in the circuit diagram.

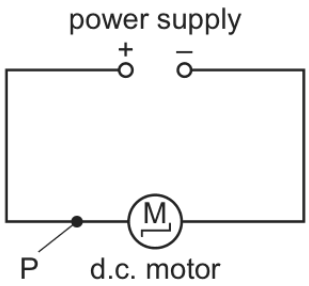


Fig. 6.1

During a time of 120 s, the current in the circuit at point P is 41 A.

(i) Calculate the charge that flows through point P in the time of 120 s.

charge = ... C [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

(ii) The current in the circuit is due to the flow of electrons. Each electron carries a charge of magnitude 1.6×10^{-19} C.

State the direction which the electrons move past point P and determine the number of electrons that pass point P in 1.0 s.

direction .....

number of electrons = ...
[2]

(c) The power supply for the electric car is a number of batteries, each made from a large number of identical cells.

The e.m.f. (electromotive force) of each cell is 3.7 V.

(i) Define the term 'electromotive force'.

.....

.....

..... [2]

(ii) Each battery consists of 92 cells in series.

Calculate the e.m.f. of one battery.

e.m.f. = ... V [2]

(iii) The power supply is made by connecting 85 of these batteries in parallel.

State the e.m.f. of the power supply.

e.m.f. = ... V [1]

[Total: 10]

MARK SCHEME · 5054_w25_ms_22.pdf · source page 11 of 14 · Question 6

Question	Answer	Marks
6(a)	a d.c. / it has one direction **or** does not (repeatedly) reverse direction **or** polarity / direction does not change	**B1**
6(b)(i)	(Q =) It **or** 41 × 120	**C1**
	4.9×10^3 (C)	**A1**

MARK SCHEME · 5054_w25_ms_22.pdf · source page 12 of 14 · Question 6

Question	Answer	Marks
6(b)(ii)	(number =) $\frac{41}{1.6 \times 10{-}19}$ **or** $\frac{4.9 \times 103}{120 \times 1.6 \times 10{-}19}$ **or** 2.6×10^N	**C1**
	2.6×10^{20} **and** towards the positive terminal / away from the motor	**A1**
6(c)(i)	(electrical) work done (by a source) in moving charge around a (complete) circuit;	**B1**
	work done (by a source) per unit charge	**B1**
6(c)(ii)	(e.m.f. =) 92 × 3.7	**C1**
	340 (V)	**A1**
6(c)(iii)	same answer as candidate's **6(c)(ii) or** 340 (V)	**B1**

6 Figure 6.1 shows a light-dependent resistor (LDR) connected in series with a fixed resistor of resistance 200 Ω and a battery.

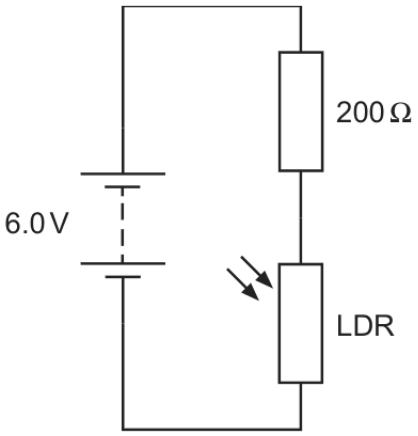


Figure 6.1

The electromotive force (e.m.f.) of the battery is 6.0 V.

The potential difference (p.d.) across the LDR is 2.4 V.

(a) Both e.m.f. and p.d. are measured in volts.

The definitions of e.m.f. and p.d both involve the work done per unit charge.

State how the definitions of e.m.f. and p.d are **different** from each other.

.....

.....

.....

..... [2]

(b) **(i)** State the equation that relates resistance, current and voltage.

..... [1]

(ii) Compare the current in the fixed resistor with the current in the LDR.

..... [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

(c) Calculate:

(i) the p.d. across the fixed resistor

p.d. = ...V [1]

(ii) the resistance of the LDR.

resistance = ... Ω [2]

(d) The brightness of the light shining on the LDR increases.

Using words or phrases from the list below, complete Table 6.1 to show the effect of this increase in brightness.

You may use each word or phrase once, more than once or not all.

increases **stays the same** **decreases**

Table 6.1

	effect of increase in brightness
resistance of LDR	.....
current in LDR	.....
current in fixed resistor	.....
p.d. across the fixed resistor	.....
p.d. across the LDR	.....

[3]

[Total: 10]

Question	Answer	Marks	Guidance
6(a)	e.m.f. (work done on unit charge or voltage) in complete circuit or by the source / battery	**B1**	**e.m.f:** energy converted to electrical energy; / **p.d:** electrical energy converted to other types
	p.d. (work done on unit charge or voltage) in / on a component / conductor / resistor / equipment / between two points	**B1**	**not** emf is a force / **ignore** (work for) part / section of a circuit; force in p.d explanation i.e. do not accept 'force across a component'
6(b)(i)	resistance = voltage / current	**B1**	**A** $R = V / I$ in any form symbol or words
6(b)(ii)	same / equal	**B1**	
6(c)(i)	3.6 (V) seen	**B1**	
6(c)(ii)	(current =) 3.6 / 200 or 0.018 (A) seen	**C1**	**ecf** as usual **but not** for 6 / 200 **A 6(c)(i)** / 200 **A** potential divider calculation ((R_{LDR} =) $R_{FR} \times V_{LDR} / V_{FR}$ or numerical equivalent)
	130 Ω	**A1**	133.3
6(d)	*1st two rows* decreases **and** increases	**B1**	if 4th row is 'stay the same' then accept 5th row 'stay the same'
	2nd, 3rd and 4th rows same as each other	**B1**	
	5th row opposite of 4th row	**B1**	

Chapter 4 Electricity and magnetism

4.2 Electrical quantities

4.2.4 Resistance

154 entries: 73 Paper 1 questions, 81 Paper 2 questions

Order: year, session (May/June, then October/November), variant, paper, question number.
Each entry: the whole question, followed at once by the whole official mark scheme of that question.

For Examiner's Use

6 Fig. 6.1 shows a circuit in which a voltmeter is placed across a resistor.

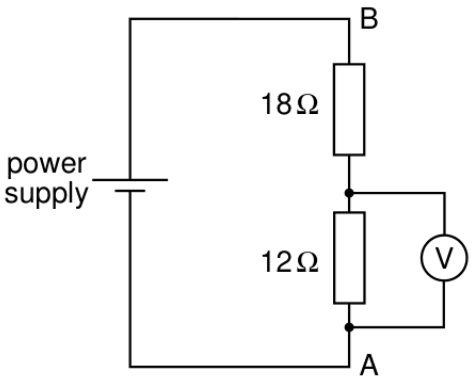


Fig. 6.1

The potential difference across the 12 Ω resistor is 4.0 V.

The voltmeter has three different ranges: 0 to 3.0 V, 0 to 6.0 V and 0 to 30 V. The best range for use in this circuit is 0 to 6.0 V.

(a) Explain why

(i) using the voltmeter on the range 0 to 3.0 V is unsuitable,

.....

.....

(ii) using the voltmeter on the range 0 to 30 V is unsuitable.

.....

.....

[2]

(b) (i) Calculate the current in the 12 Ω resistor. State the formula that you use.

current = ...

(ii) Calculate the p.d. between A and B in Fig. 6.1.

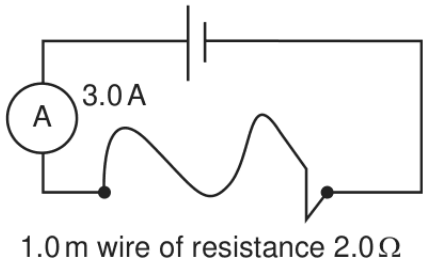
p.d. = ...

[5]

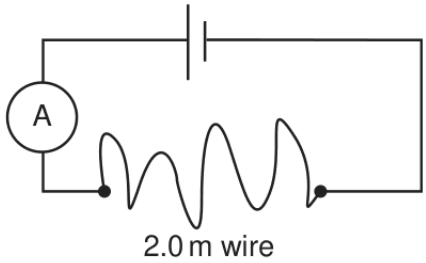
6	**(a)**	**(i)**	voltage past maximum or 3V / off scale / outside range	B1
		(ii)	reading less accurate or sensitive / not far up scale or smaller deflection	
	(b)	**(i)**	V = I R in any algebraic format	B1
			4/12	C1
			0.33 A (accept 1/3 A)	A1
		(ii)	**(i)** * 30 **or (i)** * 18 + 4 **or** 30*4/12	C1
			9.9 – 10 V (e.c.f **(i)**, e.g. if **(i)** = 0.3, 0.3*30 = 9V or 0.3*18+4 = 9.4 V)	A1
			only 1 unit error in this question	
			Total	**[7]**

QUESTION · 5054_w04_qp_1.pdf · source page 14 of 20

29 A wire of length 1.0 m has a resistance of 2.0 Ω. When connected to a cell, the current in the wire is 3.0 A.



A 2.0 m length of the same wire is connected to the same cell.



What is the resistance of the wire and the current in it?

	resistance	current
A	1.0 Ω	1.5 A
B	1.0 Ω	3.0 A
C	4.0 Ω	1.5 A
D	4.0 Ω	3.0 A

MARK SCHEME (official answer key) · 5054_w04_ms_1.pdf · source page 3 of 3 · row for Question 29

29 **C**

For Examiner's Use

7 Fig. 7.1 shows high voltage cables used to transmit electrical energy.

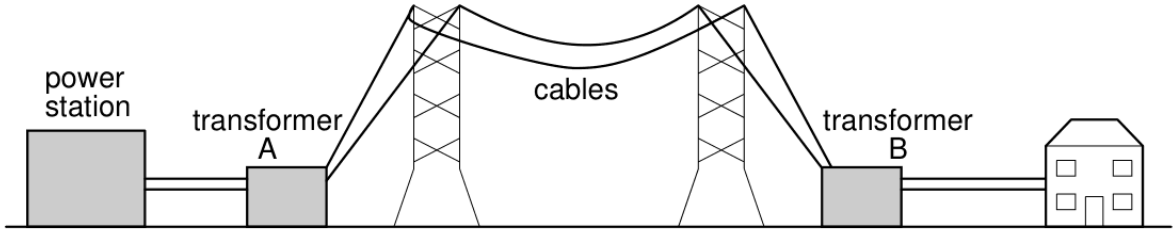


Fig. 7.1

(a) State the purpose of transformer B.

.....[1]

(b) In the space below, draw a labelled diagram to show the structure of transformer B.

[3]

(c) **(i)** Explain why high voltages are used to transmit electrical power.

.....

.....

.....

.....

.....

.....

For Examiner's Use

(ii) Fig. 7.2 shows how the loss of thermal energy from a cable varies with the thickness of the cable.

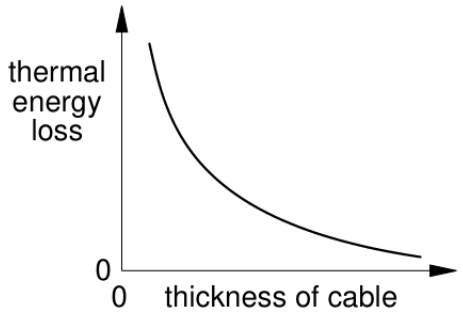


Fig. 7.2

Explain why the loss of thermal energy is less if the cable is thicker.

.....

.....

.....

.....

.....

[4]

7 **(a)** to step down/reduce the **voltage**

(**ignore** just step down transformer) **B1**

(b) two coils (no label needed)

(**not** if primary connects secondary) **B1**

output/secondary has fewer turns than input/primary clear; coils labelled

or right-hand coil has fewer turns **B1**

complete (soft) iron (core) **labelled**

(**ignore** circuit symbol) **B1**

(c) **(i)** less energy/power/heat loss/heating

(**ignore** just more efficient) **B1**

current is reduced/low

(**not** if resistance changes) **B1**

(ii) resistance is decrease

(resitivity is not resistance) **B1**

electrical power/energy related to resistance

e.g. $P = I^2R$, P prop to R (**not** V^2/R alone)

or resistance α 1/area

(**accept** power related to R etc. given in **(i)**) **B1** **8**

For Examiner's Use

8 Three cells are connected in series making a battery, as shown in Fig. 8.1. The e.m.f. of each cell is 1.5 V. A resistance of 15 Ω is connected to the battery.

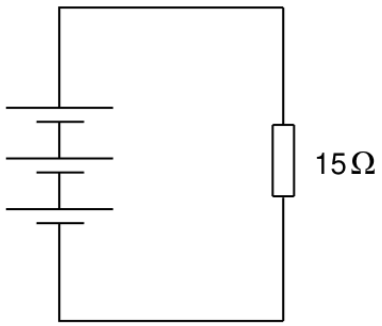


Fig. 8.1

(a) What is the total e.m.f. of the battery?

..... [1]

(b) Calculate the current in the circuit.

State the equation that you use.

current = ... [3]

(c) A battery can be made from cells connected in parallel, as shown in Fig. 8.2.

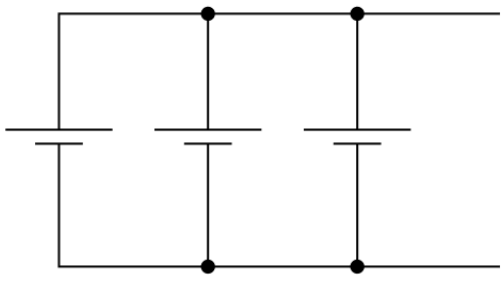


Fig. 8.2

State one advantage of connecting the cells in parallel.

.....

.....

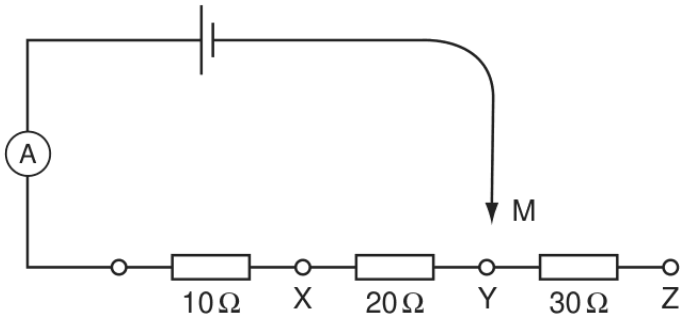
.....

..... [1]

8	**(a)** 4.5 V	**B1**	
	(b) I =V/R in any form using symbols or words	**B1**	
	4.5/15	**C1**	
	0.3 A	**A1**	
	(c) provides smaller (internal) resistance **or** lasts longer **or** less lost voltage **or** one (cell) fails others work **or** less heat/energy lost	**B1**	**5**

QUESTION · 5054_w05_qp_1.pdf · source page 14 of 20

28 The diagram shows a cell connected in series with an ammeter and three resistors (10 Ω, 20 Ω, 30 Ω). The circuit can be completed by a moveable contact M.



When M is connected to X, the ammeter reads 0.6 A.

What is the ammeter reading when M is connected to Y?

A 0.1 A **B** 0.2 A **C** 0.3 A **D** 0.6 A

MARK SCHEME (official answer key) · 5054_w05_ms_1.pdf · source page 2 of 2 · row for Question 28

28 **B**

11 Fig. 11.1 shows how the currents in a lamp L and in a wire W vary with the potential difference (p.d.) applied.

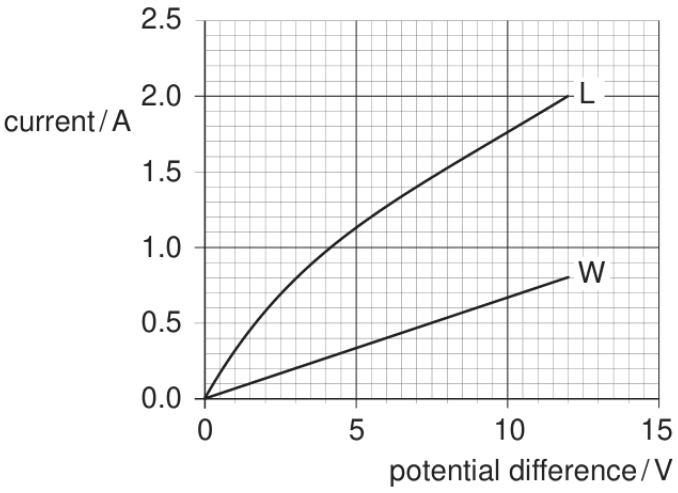


Fig. 11.1

(a) (i) Draw a diagram of the circuit that you would use to take the readings of current and p.d. for the lamp. State the range you would choose for the voltmeter. [5]

(ii) Describe how the resistance of the lamp L varies as the p.d. increases. [1]

(b) (i) Fig. 11.2 shows the lamp L and the wire W connected in series. The current in the circuit is 0.8 A.

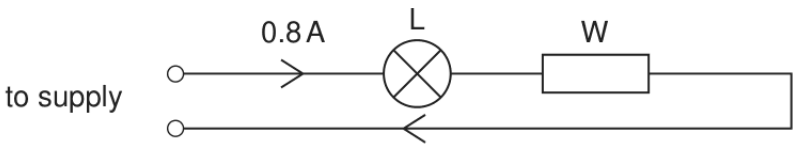


Fig. 11.2

Determine, using the graph or by calculation,

1. the p.d. across L,

2. the p.d. across W,

3. the p.d. of the supply,

4. the resistance of the lamp L in Fig. 11.2. [5]

(ii) Fig. 11.3 shows the lamp and wire connected in parallel.

A p.d. of 12 V is connected across them.

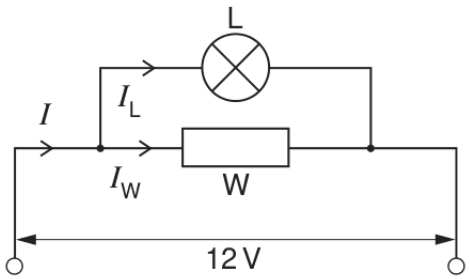


Fig. 11.3

Determine, using the graph or by calculation,

1. the current I_W in the wire W,

2. the current I_L in the lamp L,

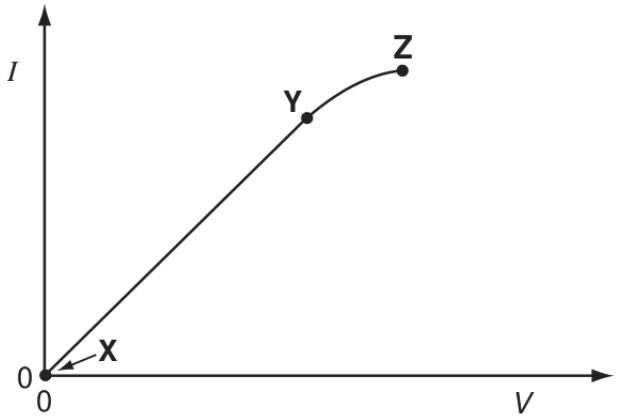
3. the total current I in the circuit,

4. the resistance of the wire W. [4]

11	**(a)**	**(i)**		correct symbol for supply, lamp	**B1**
				voltmeter across lamp or resistor	**B1**
				ammeter in series with lamp or resistor	**B1**
				power supply and variable resistor or variable power supply stated **and** no errors	**B1**
				voltmeter range (0 to) any value between 12 and 20 V	**B1**
		(ii)		resistance increases (at higher p.d./higher temperature)	**B1**
	(b)	**(i)**	1	3(.0) V	**B1**
			2	12 V	**B1**
			3	15 V **or 1 + 2**	**B1**
			4	R = 3/0.8 or V/I seen anywhere in **(b)** formula or numerical values clear	**C1**
				3.75 Ω (accept 3.7 or 3.8 but not 4)	**A1**
		(ii)	1	0.8 A	**B1**
			2	2(.0) A	**B1**
			3	2.8 A **or 1 + 2**	**B1**
			4	15 Ω **or** ecf 12/(ii **1**)	**B1**

QUESTION · 5054_s06_qp_1.pdf · source page 12 of 20

31 The diagram shows the current I / voltage V graph for a length of resistance wire.



Where can Ohm's law be applied to the wire?

A at **Y** only

B at **Z** only

C from **X** to **Y**

D from **X** to **Z**

MARK SCHEME (official answer key) · 5054_s06_ms_1.pdf · source page 2 of 2 · row for Question 31

31	**C**

11 Fig.11.1 shows a wiring diagram for two different lamps A and B.

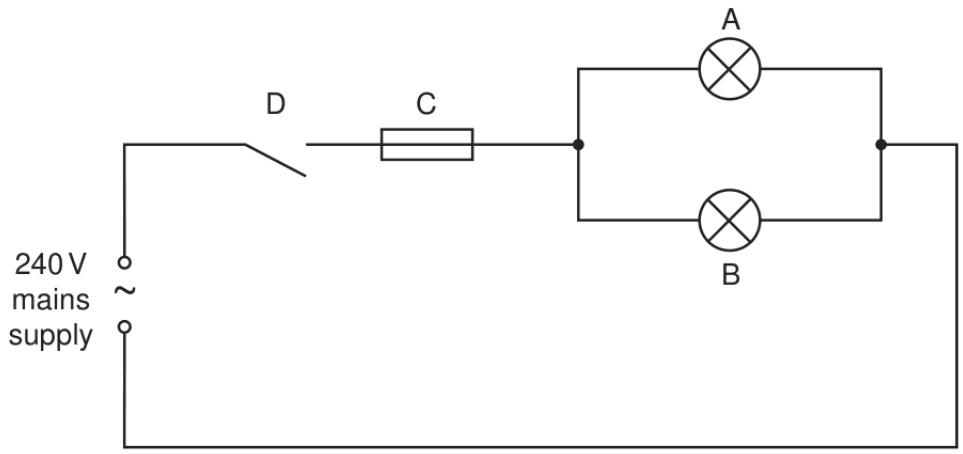


Fig. 11.1

(a) State the names of components C and D and explain the purpose, in this circuit, of each of these components. [4]

(b) When both lamps are working correctly, the current in C is 0.42 A.

Lamp A is marked 240 V, 60 W. Calculate

(i) the current in lamp A, [2]

(ii) the current in lamp B, [2]

(iii) the resistance of lamp A. [3]

(c) The two lamps in Fig. 11.1 are connected in parallel. When wiring the circuit in a house, an electrician makes a mistake and connects the two lamps in series to the mains supply. He switches them on.

(i) Draw a circuit diagram showing the two lamps connected in series to the mains supply. [1]

(ii) State whether the current in the two lamps is larger than, the same as, or smaller than the currents you have calculated in **(b)**. Explain your answer. [2]

(d) Another lamp is made using the same material for the filament as lamp A.

The filament in this new lamp has the same length as the filament in lamp A but has half the cross-sectional area.

State the resistance of the new lamp. [1]

11	**(a)**	fuse	B1	
		limits the current (not controls current)	B1	
		switch	B1	
		turns current/bulbs/circuit on **and** off	B1	**[4]**
	(b) **(i)**	I = P/V or 60/240	C1	
		0.25 (A)	A1	
	(ii)	0.42-**(i)**	C1	
		0.17 (A)	A1	
	(iii)	R = V/I or 240/**(i)**	C1	
		960 (Ω)	A1	
		units correct in **(i)**, **(ii)** and **(iii)**	B1	**[7]**
	(c) **(i)**	supply and two lamps in series	B1	
	(ii)	smaller than	B1	
		higher resistance in series **or** lamps have less than 240V across them	B1	
	(d)	2 x **(iii)** **or** twice as large **or** 1920 Ω	B1	**[4] (15)**

QUESTION · 5054_w06_qp_1.pdf · source page 11 of 16

27 A wire has a resistance of 8 Ω. A second wire, made of the same material, has half the length and twice the cross-sectional area.

What is the resistance of the second wire?

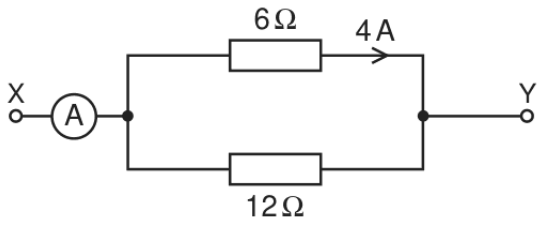
A 1 Ω **B** 2 Ω **C** 8 Ω **D** 16 Ω

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 27

27	**B**

QUESTION · 5054_w06_qp_1.pdf · source page 11 of 16

28 Two resistors of 6 Ω and 12 Ω are arranged in parallel. A p.d. is connected across the terminals X and Y. The current through the 6 Ω resistor is 4 A.



What is the current in the ammeter?

A 4 A **B** 6 A **C** 8 A **D** 12 A

MARK SCHEME (official answer key) · 5054_w06_ms_1.pdf · source page 2 of 2 · row for Question 28

28 **B**

For Examiner's Use

8 Fig. 8.1 shows an electrical circuit using two resistors.

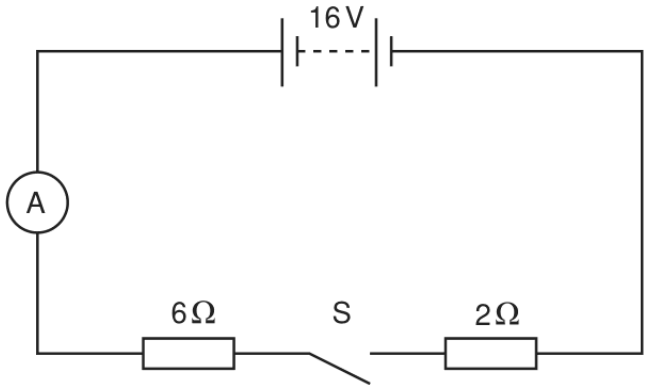


Fig. 8.1

(a) The switch S is open and the ammeter reading is zero.

State the value of the potential difference across the 6 Ω resistor.

potential difference = ... [1]

(b) Switch S is now closed.

(i) State the value of the total resistance of the circuit.

resistance = ...[1]

(ii) Calculate the current in the ammeter. State clearly the formula that you use.

current = ...[2]

(iii) Calculate the potential difference across the 6 Ω resistor.

p.d. = ...[2]

8	**(a)**		0 (V)	B1
	(b)	**(i)**	8 Ω (i.e. accept 1 sig.fig.)	B1
		(ii)	R = V/I any algebraic form in (ii) or (iii)	B1
			2 A (i.e. accept 1 sig.fig.) ecf (i)	B1
		(iii)	16/8 in (ii) **or** (ii) × 6	C1
			12 V ecf (ii)	A1 6

QUESTION · 5054_s07_qp_1.pdf · source page 13 of 20

27 Which factors will both increase the resistance of a wire in a circuit?

	size of wire	temperature of wire
A	longer	lower
B	shorter	lower
C	thicker	higher
D	thinner	higher

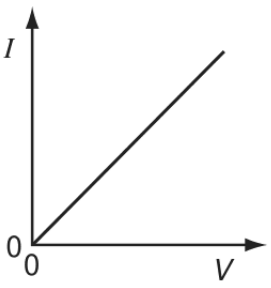
MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 27

27	**D**

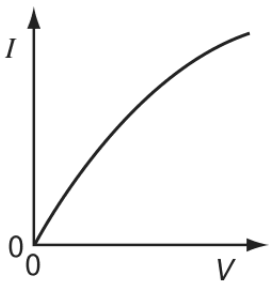
QUESTION · 5054_s07_qp_1.pdf · source page 14 of 20

28 Which graph best represents how current I varies with voltage V in a component in which the resistance **increases** as the current **increases**.

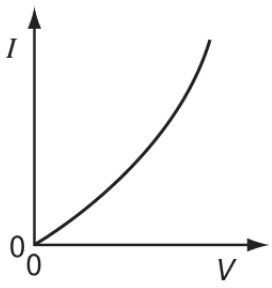
A



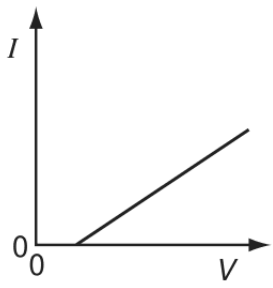
B



C



D

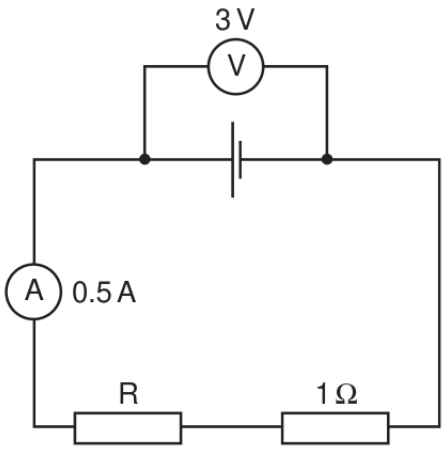


MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 28

28	**B**

QUESTION · 5054_s07_qp_1.pdf · source page 14 of 20

30 The diagram shows a circuit.



The ammeter has negligible resistance.

What is the resistance of the resistor R?

A 0.5 Ω **B** 1.5 Ω **C** 5 Ω **D** 6 Ω

MARK SCHEME (official answer key) · 5054_s07_ms_1.pdf · source page 2 of 2 · row for Question 30

30 **C**

For Examiner's Use

7 Fig. 7.1 shows an electrical circuit containing a 12V power supply and a number of resistors.

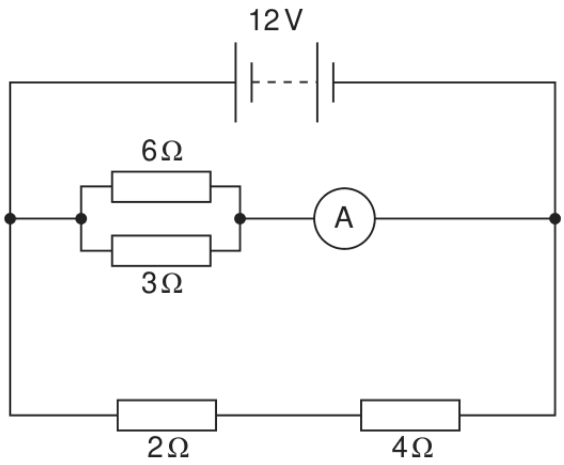


Fig. 7.1

(a) Calculate the combined resistance of

(i) the 2Ω and 4Ω resistors in series,

resistance = ... [1]

(ii) the 3Ω and 6Ω resistors in parallel.

resistance = ... [2]

(b) Calculate the reading of the ammeter in Fig. 7.1.

ammeter reading = ... [2]

(c) Determine the potential difference across the 4Ω resistor.

p.d. = ... [2]

7	**(a)**	**(i)**	6 Ω	B1	
		(ii)	$1/R = 1/R_1 + 1/R_2$ algebraic or numerical	C1	
			2 Ω	A1	
	(b)		I = V/R algebraic or numerical	C1	
			6 A ecf (ii)	A1	
	(c)		(I =) 2 (A) **or** proportionality idea/potential divider idea seen	C1	
			8 V	A1	[7]

QUESTION · 5054_w07_qp_1.pdf · source page 11 of 20

27 Under which condition does Ohm's law apply?

A The current must be constant.

B The power must be constant.

C The temperature must be constant.

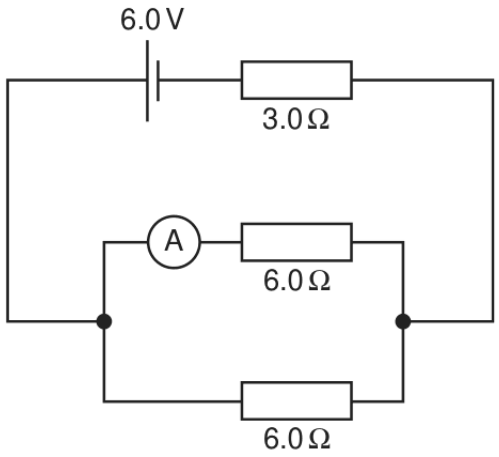
D The voltage must be constant.

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 27

27	**C**

QUESTION · 5054_w07_qp_1.pdf · source page 13 of 20

30 The following circuit is set up.



What is the reading on the ammeter?

A 0.33 A **B** 0.50 A **C** 0.67 A **D** 1.0 A

MARK SCHEME (official answer key) · 5054_w07_ms_1.pdf · source page 2 of 2 · row for Question 30

30	**B**

10 **(a)** Describe an experiment to show the difference between an electrical insulator and an electrical conductor. Name one example of each. [4]

(b) Fig. 10.1 is a sketch graph of the current in a component P against the potential difference (p.d.) across it.

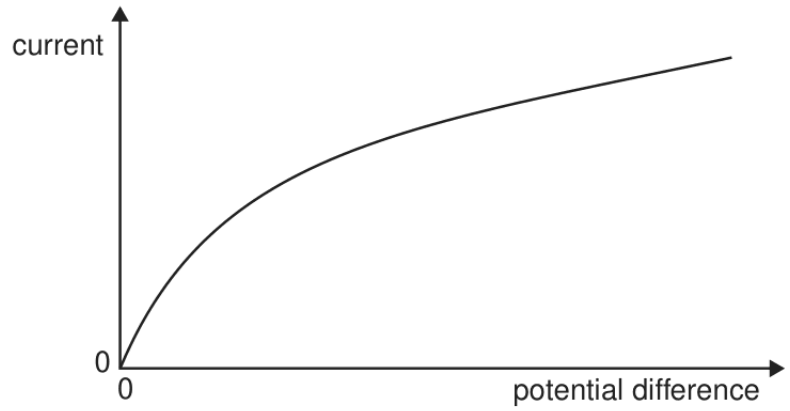


Fig. 10.1

(i) Define *resistance*. [1]

(ii) State how the resistance of P varies with the p.d. across it. [1]

(iii) Suggest what component P is. [1]

(iv) Explain why the resistance of P varies with the p.d. across it. [2]

(c) Component P is used in the electrical circuit shown in Fig. 10.2.

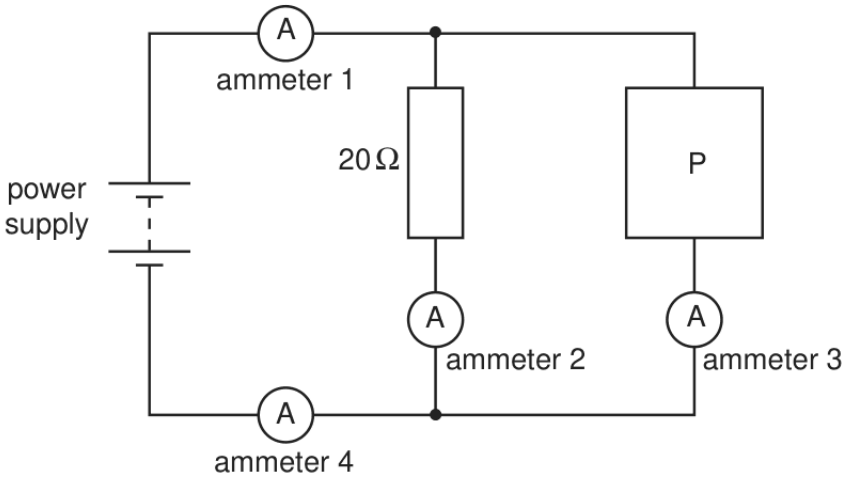


Fig. 10.2

The current in ammeter 2 is 0.40 A and the current in ammeter 3 is 0.60 A.

(i) Determine the readings of ammeters 1 and 4. [1]

(ii) Calculate the p.d. across the 20 Ω resistor. [2]

(iii) State the p.d. across the power supply. [1]

(iv) Calculate the resistance of P in this circuit. [2]

10 (a) electrical circuit containing cell/source, ammeter/lamp/bell and component under test
or charged gold-leaf electroscope and component **or** other sensible apparatus M1
one correct observation A1
named conductor (any metal/carbon/graphite **accept** water) B1
named insulator (e.g. plastic **accept** paper **and** wood) B1 [4]

(b) (i) voltage/current **or** V/I **not** volts/amps B1

(ii) resistance increases at higher p.d. (**not** resistance increases) B1

(iii) (filament) lamp/bulb or **PTC** thermisitor (**not** metal conductor) B1

(iv) temperature changes C1
higher current/voltage produces higher temperatures A1 [5]

(c) (i) 1.0A **both** for A_1 and A_4 B1

(ii) (V=) IR in any form or 20 x 0.4 C1
8(.0) V A1

(iii) 8 V or same as **(ii)** B1

(iv) **(ii)** / 0.6 C1
13 Ω (accept 2 or more sig figs **or** recurring decimal **not** fractions) A1 [6]

For Examiner's Use

6 A filament lamp is labelled 240 V, 0.20 A.

(a) Calculate the resistance of the lamp at normal brightness.

resistance = ... [2]

(b) The lamp is connected to a 240 V supply and switched on at time $t = 0$.

Fig. 6.1 shows the variation with t of the resistance of the lamp.

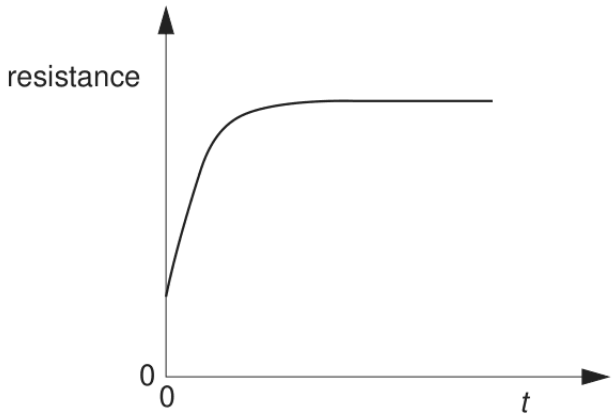


Fig. 6.1

Describe how the current in the lamp varies with t.

.....

.....

.....

.....

..... [2]

(c) A second filament lamp has higher resistance.

State **two** differences between the two filaments that may cause the change in resistance.

1 The filament in the second lamp is .....

.....

2 The filament in the second lamp is .....

..... [2]

6 **(a)** $R = V/I$ in any algebraic (e.g. $V = IR$) or numerical form C1
1200 Ω A1

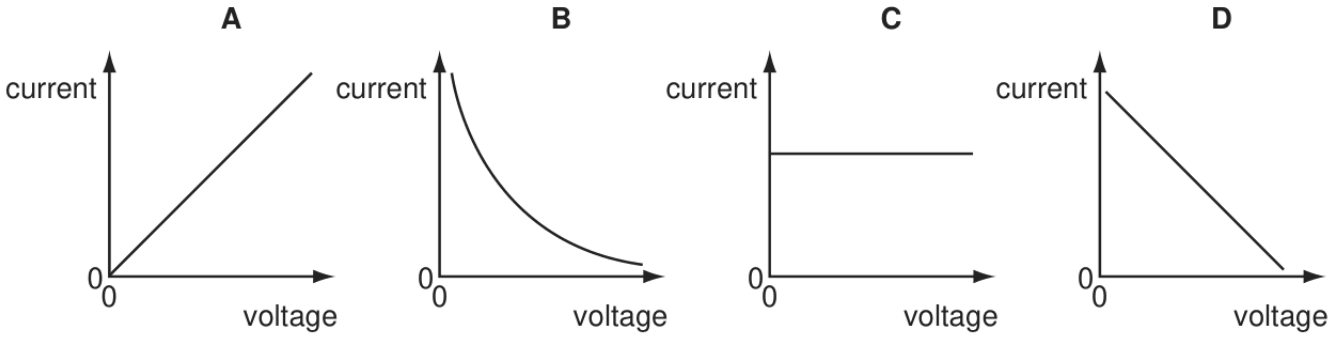
(b) decreases M1
to constant value/to 0.2 A A1

(c) longer **or** thinner **or** hotter **or** material/made of poorer conductor (higher resistivity) B2

[Total: 6]

QUESTION · 5054_w08_qp_1.pdf · source page 14 of 20

29 Which graph shows how the current changes when the voltage across a fixed resistance is varied?



MARK SCHEME (official answer key) · 5054_w08_ms_1.pdf · source page 2 of 2 · row for Question 29

29 **A**

10 Fig. 10.1 shows a wire passing through a hole in a horizontal, plastic board.

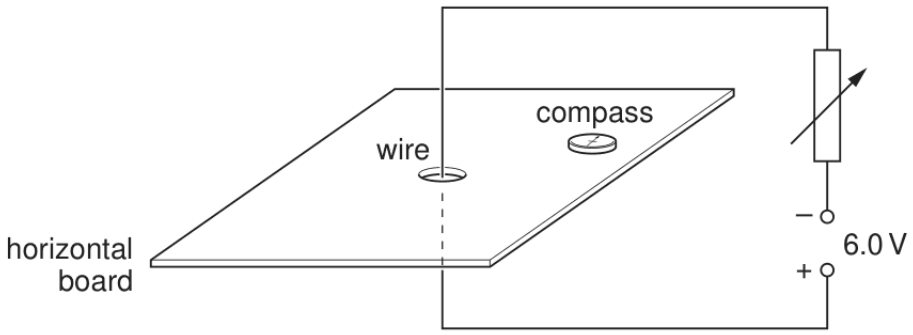


Fig. 10.1

The wire carries a current vertically upwards. A student moves a small compass around the board and plots the magnetic field lines due to the current.

(a) **(i)** Draw a diagram of the board as seen from above and mark on it the magnetic field lines due to the current. [3]

(ii) The current is increased. Describe how the magnetic field changes. [1]

(b) A 6.0 V power supply produces a current of 8.0 A in the wire. Calculate

(i) the total resistance of the circuit, [2]

(ii) the charge that flows through the wire in 2.0 minutes. [2]

(c) The north pole of a bar magnet is held on the left of the wire and the south pole of another bar magnet is held on the right, as shown in Fig. 10.2.

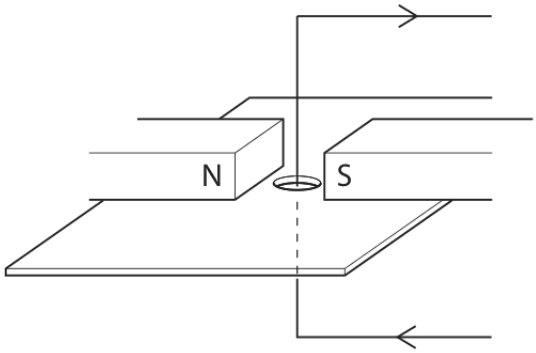


Fig. 10.2

(i) State the direction of the magnetic field at the wire due to these two poles. [1]

(ii) The wire is flexible. Describe the effect of this magnetic field on the current-carrying wire. [2]

(iii) The current in the wire is now reversed. State the effect of this on the wire. [1]

(iv) Describe how this effect is put to use in a d.c. motor. [3]

10	**(a)**	**(i)**	at least 2 concentric, complete circles	B1	
			increasing gap	B1	
			at least 1 anticlockwise arrow **and** none incorrect	B1	
		(ii)	stronger **or** more lines **or** lines closer together **or** extends further	B1	[4]
	(b)	**(i)**	(R =) V/I **or** 6.0/8.0	C1	
			0.75 Ω	A1	
		(ii)	(Q =) It **or** 8.0 × 120 **or** 8.0 × 2	C1	
			960 C (16 C scores 1/2)	A1	[4]
	(c)	**(i)**	L→R **or** N→S	B1	
		(ii)	force (on wire) **or** wire bends/moves	M1	
			into page/perpendicular to field/away (from us)/LH rule quoted	A1	
		(iii)	force reverses **or** out of page **or** bends the other way e.c.f.	B1	[4]
		(iv)	accept first two marks on unlabelled diagram		
			(wire becomes) coil / armature /solenoid	B1	
			force/movement opposite on sides of coil **or** moment	B1	
			current reverses during rotation/due to commutator or split ring	B1	[3]

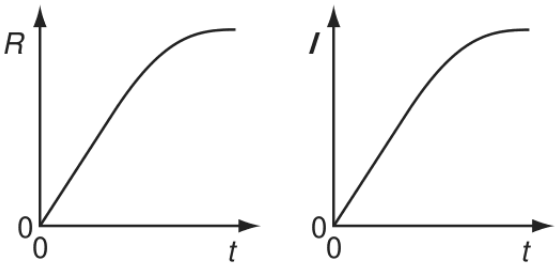
[Total: 15]

QUESTION · 5054_s09_qp_1.pdf · source page 12 of 20

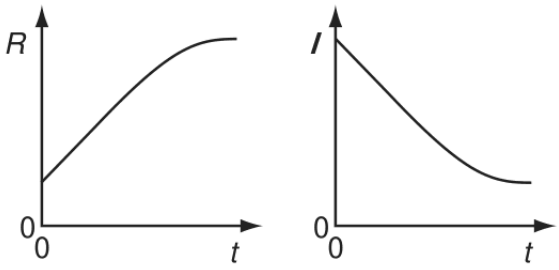
26 When a filament lamp is switched on, there is a current in the lamp. As the temperature of the filament rises, its resistance changes.

Which pair of graphs shows how the resistance *R* of the filament and the current *I* vary with time after the lamp is switched on?

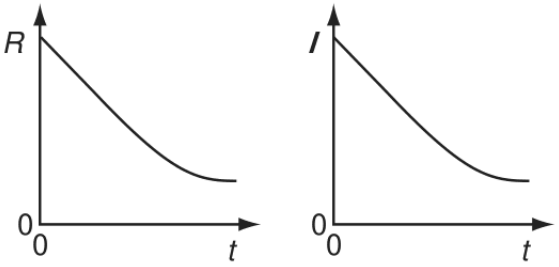
A



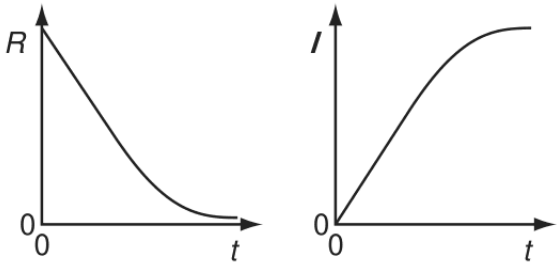
B



C



D



MARK SCHEME (official answer key) · 5054_s09_ms_1.pdf · source page 2 of 2 · row for Question 26

26	**B**

Section B

Answer **two** questions from this section.

Use the lined pages provided and, if necessary,
continue on the separate sheets available from the Supervisor.

9 **(a)** A lamp is marked 24 V, 100 W. Describe an experiment to check that the electrical power supplied to the lamp is 100 W when the potential difference (p.d.) across it is 24 V.
In your account you should

- include a circuit diagram,
- state the readings that are taken,
- show how the result is calculated from the readings. [4]

(b) Two lamps are connected in parallel to a 240 V mains supply, as shown in Fig. 9.1.

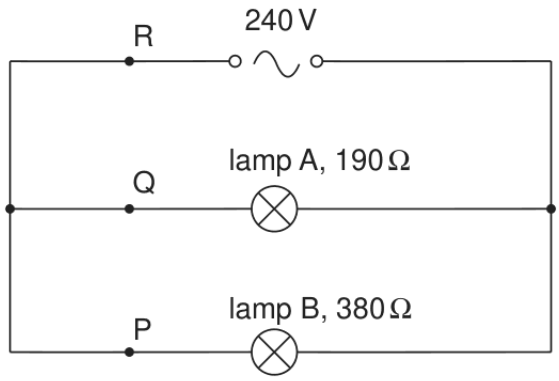


Fig. 9.1

Lamp A has a resistance of 190 Ω and lamp B has a resistance of 380 Ω.

(i) Calculate the current at points P, Q and R. [3]

(ii) Calculate the total resistance of the circuit. [2]

(c) Fig. 9.2 shows the same lamps connected in series to the mains supply.

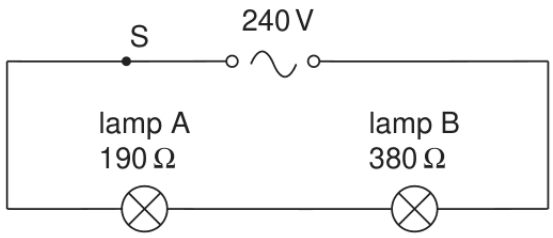


Fig. 9.2

(i) Calculate the current at point S. [2]

(ii) Calculate the p.d. across lamp A. [1]

(d) In a house, all lamps are connected in parallel to the mains supply, not in series.

State and explain two reasons for this.

In your explanation you may refer to the results of your calculations in **(b)** and **(c)**. [3]

Section B

9	**(a)**	circuit diagram showing power supply, lamp and ammeter in series	B1	
		voltmeter across lamp	B1	
		ensure voltage is 24 V in some way e.g. power supply 24 V	B1	
		$V \times I$ or voltmeter × ammeter readings	B1	
	(b) **(i)**	P 0.63(2) A	B1	
		Q 1.26(3) A	B1	
		R 1.89(5), 1.9 A or sum of candidate's P and Q	B1	
	(ii)	240/current at R or $1/R = 1/R_1 + 1/R_2$	C1	
		127, 130, 126.7 Ω ecf **(i)**	A1	
	(c) **(i)**	(I =) V/R numerical or algebraic	C1	
		0.42 A	A1	
	(ii)	80 V or 79.8 V ecf **(i)**	B1	
	(d)	one lamp goes out/blows/fuses/switched off they do not all go out/others stay on	B1	
		lamps are working at correct/more brightness/voltage/current power	B1	
		reference to voltage is 240 V across each lamp or voltage shared in series/<240 V or current value(s) quoted	B1	[15]

QUESTION · 5054_w09_qp_1.pdf · source page 13 of 20

27 Ohm's law states that the current in a conductor is proportional to the potential difference across it, provided that a certain quantity remains constant.

What is this quantity?

A length

B pressure

C temperature

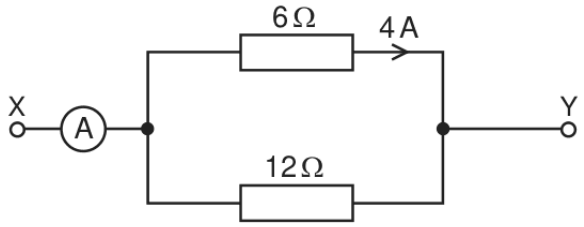
D thickness

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 27

27	**C**

QUESTION · 5054_w09_qp_1.pdf · source page 13 of 20

28 Two resistors of 6 Ω and 12 Ω are arranged in parallel. A potential difference is connected across the terminals X and Y. The current in the 6 Ω resistor is 4 A.



What is the current in the ammeter?

A 4 A **B** 6 A **C** 8 A **D** 12 A

MARK SCHEME (official answer key) · 5054_w09_ms_1.pdf · source page 2 of 2 · row for Question 28

28	**B**

11 Fig. 11.1 shows a 9.0V battery connected in series with a 16.0Ω resistor and a small metal conductor X at room temperature.

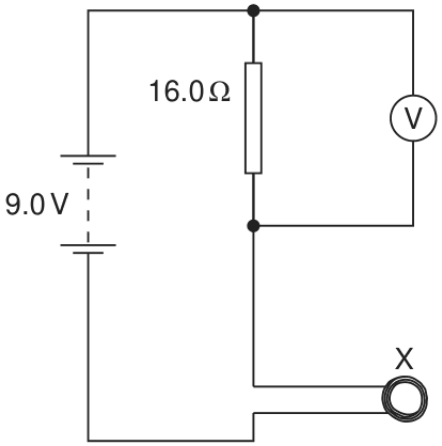


Fig. 11.1

A voltmeter measures the potential difference (p.d.) across the 16.0Ω resistor. At room temperature the resistance of X is 4.0Ω.

(a) **(i)** Calculate the current supplied by the battery. [2]

(ii) Calculate the p.d. across the 16.0Ω resistor. [1]

(b) X is heated slowly to a very high temperature.

(i) Sketch a graph to show how the resistance of a metal conductor depends on its temperature. [2]

(ii) State and explain how the voltmeter reading changes as the temperature of X rises. [2]

(iii) Suggest a suitable range for the voltmeter. [2]

There is a choice of part (c) on the following page.

EITHER (c) (i) Describe in outline how the circuit in Fig. 11.1 can be used as a thermometer. [2]

(ii) State two advantages of a thermometer such as this over a liquid-in-glass thermometer. [2]

(iii) The change in voltmeter reading with temperature is non-linear. Explain what is meant by *non-linear*. [2]

OR **(c)** The voltmeter is removed from the circuit in Fig. 11.1 and the negative terminal of the battery is earthed. A connection is made from the circuit to the inputs A and B of a NAND gate. The output of the NAND gate is connected to an LED (light-emitting diode) and a resistor. This is shown in Fig. 11.2.

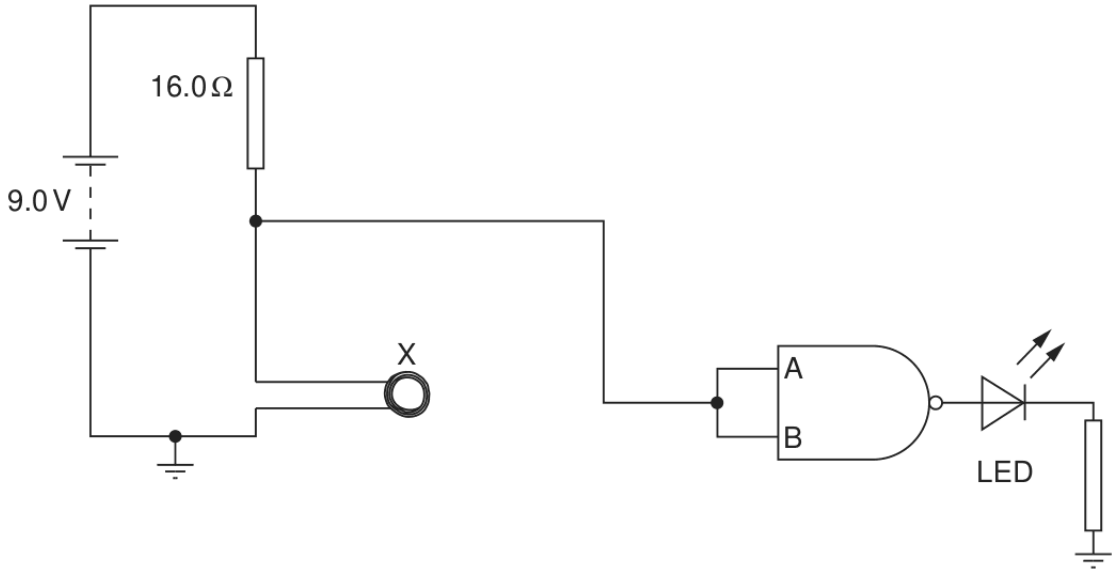


Fig. 11.2

(i) Draw the truth table for a NAND gate. [2]

(ii) The LED is lit. State the inputs at A and B and explain why these inputs cause the LED to light. [2]

(iii) X is slowly heated to a very high temperature. At a certain temperature, the LED switches off. Explain why this happens. [2]

11 (a) (i) (I =)V/R **or** V = IR (in **(i)**/**(ii)**) **or** 9.0/20 (in **(i)**) **or** 0.45 × 16 (in **(ii)**) C1
0.45 A A1

(ii) 7.2 V (max 3 for **(i)** and **(ii)** together) A1 [3]
C1 may be awarded for **either** A mark

(b) (i) R → T **and** line of positive slope throughout B1
straight line, positive intercept on R-axis **and** slope/0 on **kelvin** scale B1

(ii) voltmeter reading falls B1
current (supplied by battery) falls **or** X takes greater proportion of p.d. **or** 16 Ω takes smaller proportion of p.d. B1

(iii) 0 **and** to/→/- B1
8/9/10/whole number not greater than 20 V (usual unit penalty) B1 [6]

EITHER:

(c) (i) use small, metal conductor as probe/sensor **or** calibrate V reading (with known T) B1
the voltmeter reading is used to find T B1

(ii) any **two** from: high temperatures /remote reading/robust/quick acting/direct input to computer/low heat capacity B2

(iii) equal changes in one/T do **not** produce a equal changes in the other/V **or** graph with axes labelled **not** straight **or not** proportional to B2 [6]
not a straight line **or** not same change **or** change in V is not uniform 1/2

OR:

(c) (i)

In		Out	
0	0	1	all correct B1
0	1	1	
1	0	1	
1	1	0	correct B1

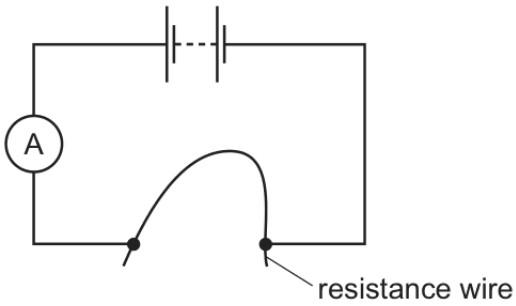
(ii) both inputs = 0 B1
output = 1 B1

(iii) A **and** B inputs = 1 B1
output = 0 B1 [6]

[Total: 15]

QUESTION · 5054_s10_qp_11.pdf · source page 12 of 20

26 A length of resistance wire is used as a resistor in a simple circuit.



Four separate changes are made to the wire.

Which change will **not** reduce the value of the resistance of the wire?

A It is covered in an insulating sleeve.

B Its cross-sectional area is increased.

C Its length is decreased.

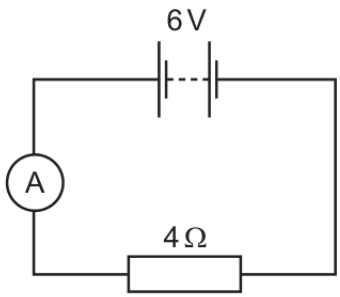
D Its temperature is decreased.

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 26

26	**A**

QUESTION · 5054_s10_qp_11.pdf · source page 12 of 20

27 A 6 V supply is connected in series with an ammeter and a 4 Ω resistor.



What is the reading on the ammeter?

A 0.67 A **B** 1.5 A **C** 10 A **D** 24 A

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 27

27 **B**

QUESTION · 5054_s10_qp_11.pdf · source page 12 of 20

28 Sets of voltage-current readings are obtained for different electrical components.

Which set of readings is for a 100 Ω resistor?

A	voltage / V	–3	–2	–1	0	+1	+2	+3
	current / mA	–30	–15	–5	0	+5	+15	+30
B	voltage / V	–3	–2	–1	0	+1	+2	+3
	current / mA	–30	–20	–10	0	+10	+20	+30
C	voltage / V	–3	–2	–1	0	+1	+2	+3
	current / mA	–60	–40	–20	0	+20	+40	+60
D	voltage / V	–3	–2	–1	0	+1	+2	+3
	current / mA	–60	–45	–30	0	+30	+45	+60

MARK SCHEME (official answer key) · 5054_s10_ms_11.pdf · source page 2 of 2 · row for Question 28

28 **B**

For Examiner's Use

6 Fig. 6.1 shows a circuit used in an experiment to measure the current in a lamp for different values of the potential difference (p.d.) across the lamp.

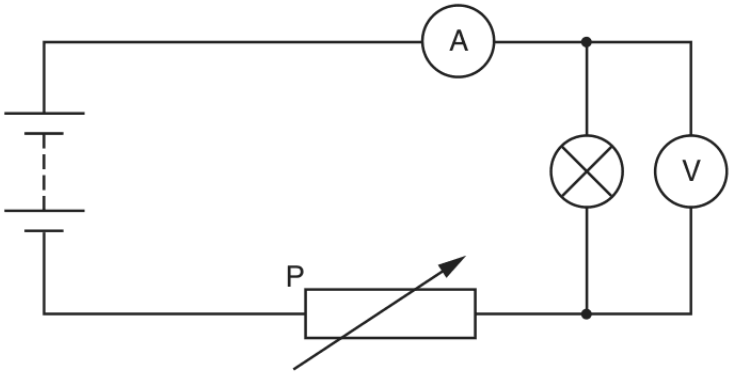


Fig. 6.1

(a) State the name of component P.

.....[1]

(b) On the axes below, sketch a graph of current against p.d. for a filament lamp.

[2]

(c) **(i)** State how the resistance of the lamp changes as the p.d. increases.

.....[1]

(ii) Explain how the graph in **(b)** shows the change you have described in **(c)(i)**.

.....

.....

.....[1]

6	**(a)**	variable resistor **or** rheostat	B1	[1]
	(b)	curved line starting at origin **allow** straight at first **not** two straight lines	C1	
		correct curvature **from origin** with decreasing gradient	A1	[2]
	(c) **(i)**	(resistance) increases (as p.d. increases)	B1	[1]
	(ii)	explanation, e.g. lower current than expected for given p.d. **or** given current needs larger p.d. **or** correct explanation involving $R = V/I$ **accept** gradient decreases	B1	[1]

For Examiner's Use

7 **(a)** Fig. 7.1 shows a circuit with two resistors in series.

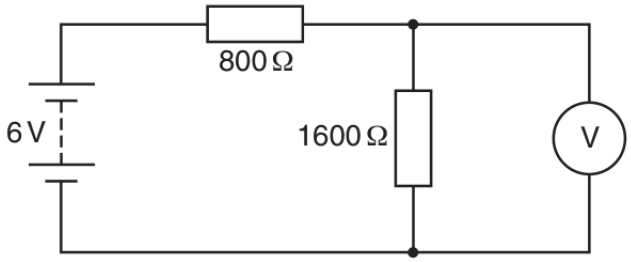


Fig. 7.1

(i) Calculate the current in the circuit.

current = ……..... [2]

(ii) Calculate the reading on the voltmeter.

voltmeter reading = ……..... [1]

(b) Fig. 7.2 and Fig. 7.3 show circuits containing a capacitor and a transistor respectively.

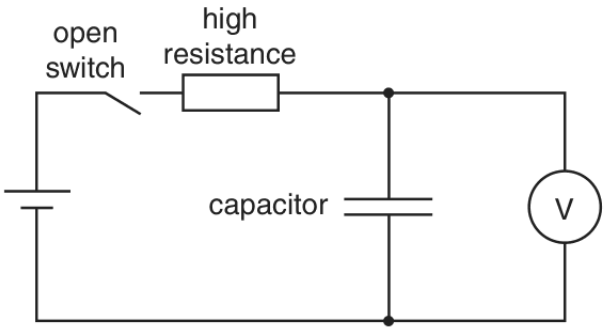


Fig. 7.2

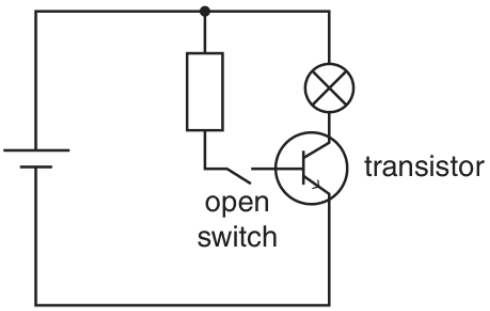


Fig. 7.3

EITHER Explain the action of the capacitor shown in Fig. 7.2 when the switch is closed.

.....

.....

..... [2]

OR Explain the action of the transistor shown in Fig. 7.3 when the switch is closed.

.....

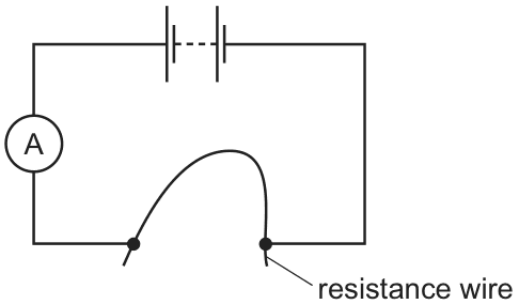
.....

..... [2]

7	**(a)**	**(i)**	$I = V/R$ **or** 2400 seen	C1	
			0.0025 A (2.5 mA)	A1	[2]
		(ii)	4 V **or** 1600 × **(i)** ecf	B1	[1]
	(b)	**EITHER**			
			capacitor stores charge/charges up/stores energy	B1	
			takes time/delay e.g. voltmeter reading rises slowly/capacitor charges up slowly or to a maximum	B1	
			OR		
			(small) current into transistor/base **or** large base/emitter voltage (>0.6 V)	B1	
			switches (transistor) on **or** large current collector/emitter or resistance of transistor reduced	B1	[2]

QUESTION · 5054_s10_qp_12.pdf · source page 12 of 20

27 A length of resistance wire is used as a resistor in a simple circuit.



Four separate changes are made to the wire.

Which change will **not** reduce the value of the resistance of the wire?

A It is covered in an insulating sleeve.

B Its cross-sectional area is increased.

C Its length is decreased.

D Its temperature is decreased.

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 27

27 **A**

QUESTION · 5054_s10_qp_12.pdf · source page 13 of 20

30 Sets of voltage-current readings are obtained for different electrical components.

Which set of readings is for a 100 Ω resistor?

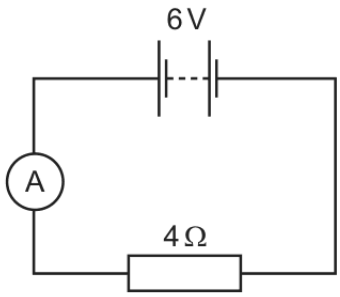
A	voltage / V	–3	–2	–1	0	+1	+2	+3
	current / mA	–30	–15	–5	0	+5	+15	+30
B	voltage / V	–3	–2	–1	0	+1	+2	+3
	current / mA	–30	–20	–10	0	+10	+20	+30
C	voltage / V	–3	–2	–1	0	+1	+2	+3
	current / mA	–60	–40	–20	0	+20	+40	+60
D	voltage / V	–3	–2	–1	0	+1	+2	+3
	current / mA	–60	–45	–30	0	+30	+45	+60

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 30

30 **B**

QUESTION · 5054_s10_qp_12.pdf · source page 13 of 20

31 A 6 V supply is connected in series with an ammeter and a 4 Ω resistor.



What is the reading on the ammeter?

A 0.67 A **B** 1.5 A **C** 10 A **D** 24 A

MARK SCHEME (official answer key) · 5054_s10_ms_12.pdf · source page 2 of 2 · row for Question 31

31	**B**

For Examiner's Use

11 **(a)** Describe an experiment, including an ammeter and a voltmeter, to show how the resistance of a wire varies with temperature. Draw a circuit diagram.

.....

.....

.....

.....

.....

.....[4]

(b) The filament in a lamp is made from metal wire.

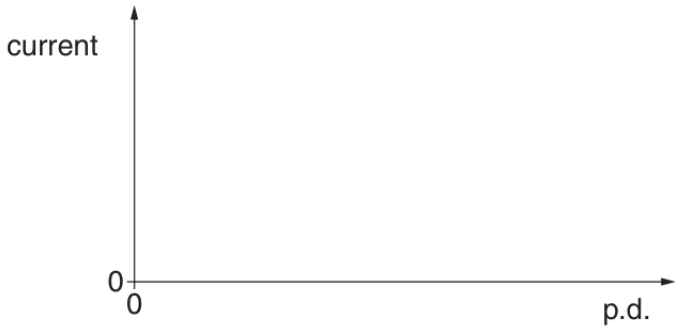
(i) Describe the variation with temperature of the resistance of the filament lamp.

.....

.....

.....[2]

(ii) On Fig. 11.1, sketch a graph of the variation of the current in the filament against the potential difference (p.d.) across it.



[2]

Fig. 11.1

(c) Fig. 11.2 is a circuit used to monitor changes in room temperature.

For Examiner's Use

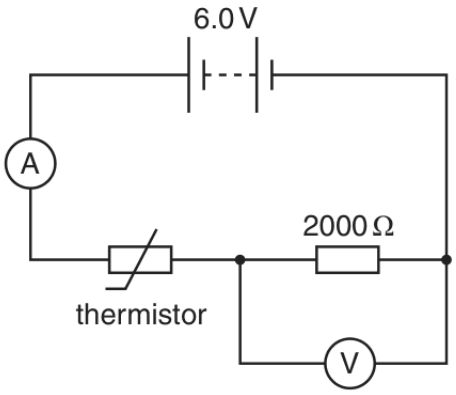


Fig. 11.2

A thermistor is connected in series with a 6.0V battery and a 2000 Ω resistor.

(i) The temperature of the room increases. State and explain what happens to

1. the reading on the ammeter,

.....

.....[2]

2. the reading on the voltmeter.

.....

.....[2]

(ii) At a certain temperature, the reading on the voltmeter is 3.8V. Calculate the resistance of the thermistor at this temperature.

resistance = ...[3]

11 (a) circuit diagram with cell and ammeter in series with resistor/wire/lamp and voltmeter across resistor/wire/lamp B1
R = V/I in any form or gradient of V, I graph B1
thermometer/thermocouple used or shown B1
statement of how different temperatures obtained,
e.g. water bath/oven/heat room/change supply voltage or current or series resistor to change temperature B1 [4]

(b) (i) resistance increases with temperature M1
(approximately) linear, proportional, straight line increase A1 [2]

(ii) curved line starting at origin C1
correct curvature from origin with decreasing gradient
allow zero gradient not negative gradient A1 [2]

(c) (i) 1 (current) increases B1
thermistor resistance decreases B1 [2]

2 (voltmeter reading) increases B1
greater fraction of voltage across resistor or potential divider equation explained
or greater current through fixed/constant/2000 Ω resistor B1 [2]

(ii) (voltage across thermistor) 2.2 (V) or attempt to use potential divider formula C1
(current) 3.8 / 2000 or 1.9×10^{-3} (A)
or 3.8 = 6 × 2000/(R+2000) or other correct potential divider equation C1
1200 Ω allow 1157 – 1160 A1 [3]

[Total: 15]

For Examiner's Use

4 At night, a bright floodlight is used to illuminate a building. The floodlight is a metal filament lamp. It is connected in series with a variable resistor and a 230 V mains power supply.

When the floodlight is first switched on, the resistance of the variable resistor is at its maximum value. In the following few minutes the variable resistor is adjusted so that its resistance decreases slowly to zero. This causes the brightness of the lamp to increase until it reaches a maximum. At maximum brightness, the current in the lamp is 12 A.

(a) Calculate the resistance of the lamp at maximum brightness.

resistance = ...[2]

(b) State and explain what happens to the resistance of the lamp as its brightness increases.

.....

.....

.....[2]

(c) Suggest how, by using the variable resistor in this way, the filament lamp is protected from damage.

.....

.....

.....[2]

4	**(a)**	(R =) V/I **or** 230/12	C1	
		19/19.2/19.1 Ω etc.	A1	
	(b)	(resistance) increases	B1	
		as the temperature increases/gets hotter/gets heated	B1	
	(c)	(if switched on suddenly) low R → high/excess current **or** it prevents high/excess current	B1	
		bulb/filament/fuse blown/damaged **or** wires damaged (**ign** lamp/filament lamp damaged)	B1	[6]

QUESTION · 5054_s11_qp_11.pdf · source page 11 of 16

29 Which changes both cause a decrease in the resistance of a copper wire?

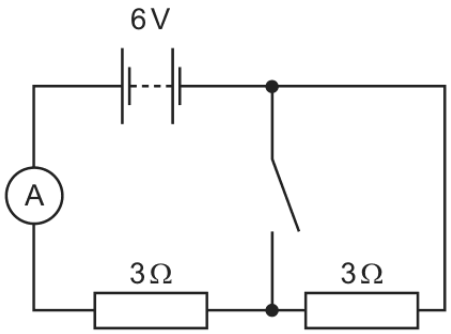
	size of wire	temperature of wire
A	decrease in length	lower
B	increase in length	lower
C	decrease in thickness	higher
D	increase in thickness	higher

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 29

29 **A**

QUESTION · 5054_s11_qp_11.pdf · source page 12 of 16

30 The diagram shows a circuit.



What is the reading on the ammeter when the switch is open, and the reading when it is closed?

	ammeter reading when open / A	ammeter reading when closed / A
A	1	1
B	1	2
C	2	1
D	2	2

MARK SCHEME (official answer key) · 5054_s11_ms_11.pdf · source page 2 of 2 · row for Question 30

30 **B**

10 (a) A 1500 Ω resistor is used on an electronic circuit board in a television.

Fig. 10.1 shows the bands on a resistor.

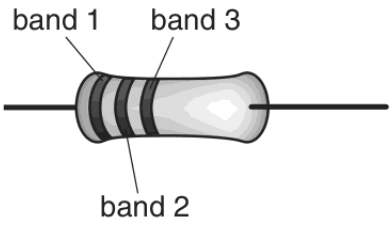


Fig. 10.1

digit or multiplier	colour
0	black
1	brown
2	red
3	orange
4	yellow
5	green
6	blue
7	violet
8	grey
9	white

Fig. 10.2

The table of Fig. 10.2 shows the colour code for the bands.

(i) State the colours of the bands on a 1500 Ω resistor.

band 1 .....

band 2 .....

band 3 .....

[2]

(ii) Determine the maximum value of resistance that can be shown with this three-band colour code.

resistance = ... [1]

(iii) The 1500 Ω resistor on the electronic circuit board has a correct power rating of 0.25 W.

State a disadvantage of using a different 1500 Ω resistor with a power rating that is

1. much less than 0.25 W,

.....

..... [1]

2. much greater than 0.25 W.

.....

..... [1]

(b) Fig. 10.3 shows part of a circuit that includes a variable resistor R and a battery of e.m.f. 9.0V.

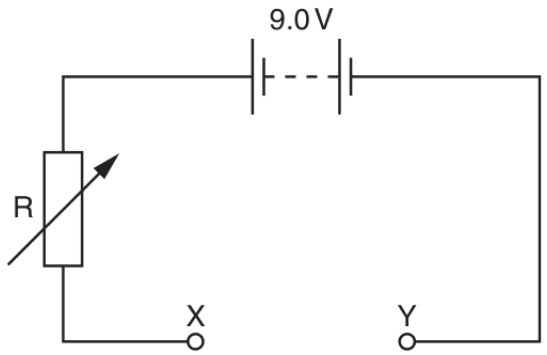


Fig. 10.3

(i) State one similarity and one difference between electromotive force (e.m.f.) and potential difference (p.d.).

similarity: .....

.....

difference: .....

.....
[3]

(ii) A light-emitting diode (LED) is connected between points X and Y, so that it emits light.

On Fig. 10.3, draw the symbol for the LED connected between points X and Y. [2]

(iii) The resistance of R is increased. State what happens to the p.d. across the LED and to the current in the LED.

p.d.: .....

current: .....
[1]

(iv) The LED is marked "maximum current 25 mA when the p.d. is 1.7 V".

Calculate the minimum value of the resistance of R.

resistance = ... [4]

10	**(a)**	**(i)**	brown and green	B1	
			red	B1	
		(ii)	99×10^9 or 9.9×10^{10} Ω	B1	
		(iii)	*less*: (likely to) burn out/blow // become too hot	B1	
			greater: (likely to) be large (in size)	B1	
	(b)	**(i)**	both involve energy and charge // measured in J/C/volts/by voltmeter	B1	
			energy change is from other forms (accept chemical) to electrical in e.m.f.	B1	
			energy change is from electrical to other forms (accept heat/light) in p.d. (**or** e.m.f. is property of source and p.d. is property of (part of) circuit B2)	B1	
		(ii)	correct symbol	B1	
			correct direction	B1	
		(iii)	p.d. reduces/(approximately) constant **and** current reduces	B1	
		(iv)	$R = V/I$ in any form, e.g. 1.7/0.025, 1.7/25	C1	
			correct conversion to mA, e.g. 0.025/7.3 seen	C1	
			68 or 360 seen	C1	
			290 Ω, 292 Ω	A1	[15]

QUESTION · 5054_s11_qp_12.pdf · source page 11 of 16

28 Which changes both cause a decrease in the resistance of a copper wire?

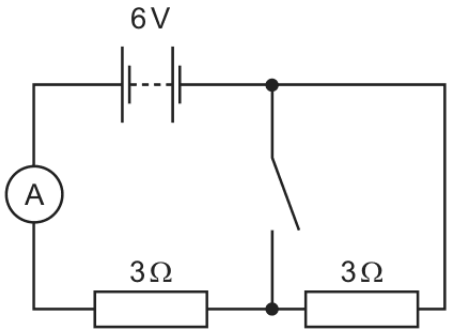
	size of wire	temperature of wire
A	decrease in length	lower
B	increase in length	lower
C	decrease in thickness	higher
D	increase in thickness	higher

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 28

28 **A**

QUESTION · 5054_s11_qp_12.pdf · source page 12 of 16

31 The diagram shows a circuit.



What is the reading on the ammeter when the switch is open, and the reading when it is closed?

	ammeter reading when open / A	ammeter reading when closed / A
A	1	1
B	1	2
C	2	1
D	2	2

MARK SCHEME (official answer key) · 5054_s11_ms_12.pdf · source page 2 of 2 · row for Question 31

31 **B**

5 Fig. 5.1 shows part of a low-voltage lighting circuit.

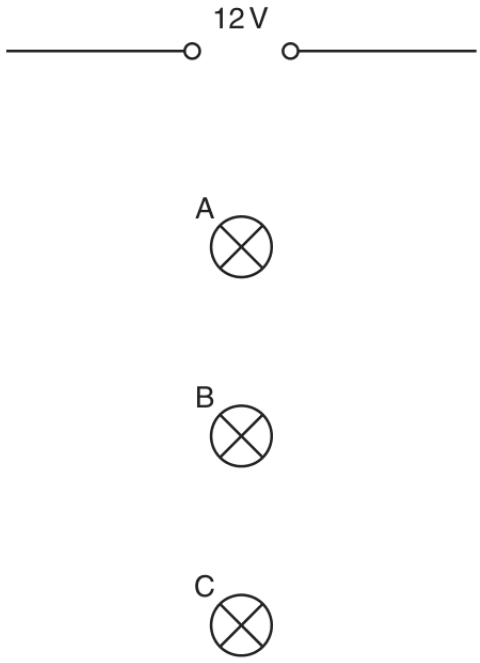


Fig. 5.1

The power supply voltage is 12 V.

(a) On Fig. 5.1, complete the circuit, adding components as necessary, so that:

- the total current in the circuit can be measured,
- lamp A is on all the time,
- lamps B and C are in series with each other and are switched on or off together.

[3]

(b) All the lamps are on. The potential difference (p.d.) across lamp B is 8.0 V and the current in lamp B is 50 mA.

(i) Calculate the resistance of lamp B.

resistance = ... [2]

(ii) State the current in lamp C.

current = ... [1]

5 **(a)** ammeter in series with supply // ammeters in series with A **and** in series with B & C B1
A across cell with no switch (condone closed switch) **not** B1
B and C in series with switch (closed or open) and cell B1

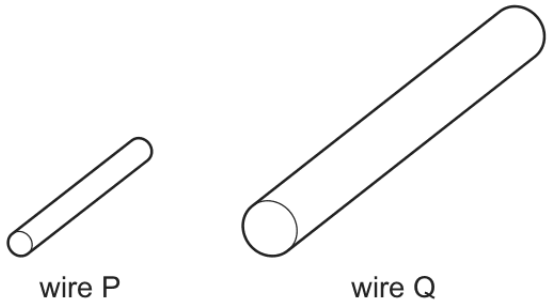
(b) **(i)** (R =) V/I in any form numerical or algebraic, e.g. 8/50, 8/0.05 C1
160 Ω A1

(ii) 50 mA // 0.05(0) A B1 [6]

QUESTION · 5054_w11_qp_11.pdf · source page 12 of 20

28 The resistance of a cylindrical wire P is 80 Ω. A second wire Q is made from the same material.

The cross-sectional area of Q is four times that of P. The length of Q is twice the length of P.



What is the resistance of Q?

A 10 Ω **B** 40 Ω **C** 160 Ω **D** 640 Ω

MARK SCHEME (official answer key) · 5054_w11_ms_11.pdf · source page 2 of 2 · row for Question 28

28	**B**

6 When a car is moving, its electrical equipment is powered by an a.c. generator.

(a) The coil of the a.c. generator is rotated by the car engine.

(i) On the axes in Fig. 6.1, sketch a graph of the output voltage of the coil against time for two rotations of the coil of the generator.

[1]

Fig. 6.1

(ii) The speed of the car increases and so does the speed of rotation of the coil.

State two changes that this causes to the output voltage.

change 1 .....

.....

change 2 .....

.....

[2]

(b) When the car engine is off, the current in a lamp from a 12 V battery is 0.50 A. Calculate the resistance of the lamp.

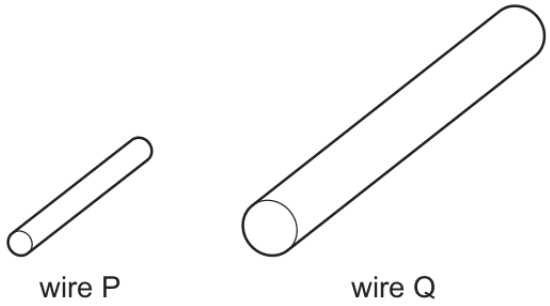
resistance = ...[2]

6	**(a)**	**(i)**	recognisable sine/cosine curve (≥ 2.0 cycles)	B1	
		(ii)	larger (peak)(voltage)	B1	
			higher frequency/shorter period/described in words (**allow** shorter wavelength)	B1	
	(b)		(R =) V/I **or** 12/0.50	C1	
			24 Ω	A1	[5]

QUESTION · 5054_w11_qp_12.pdf · source page 12 of 20

27 The resistance of a cylindrical wire P is 80 Ω. A second wire Q is made from the same material.

The cross-sectional area of Q is four times that of P. The length of Q is twice the length of P.



What is the resistance of Q?

A 10 Ω **B** 40 Ω **C** 160 Ω **D** 640 Ω

MARK SCHEME (official answer key) · 5054_w11_ms_12.pdf · source page 2 of 2 · row for Question 27

27	**B**

6 An electrical engineer measures the potential difference across a length of metal wire and the current in the wire. He does this for different values of the current.

(a) (i) Draw a labelled circuit diagram of a circuit that enables the engineer to do this.

[3]

(ii) Describe how the circuit is used.

.....

.....

.....

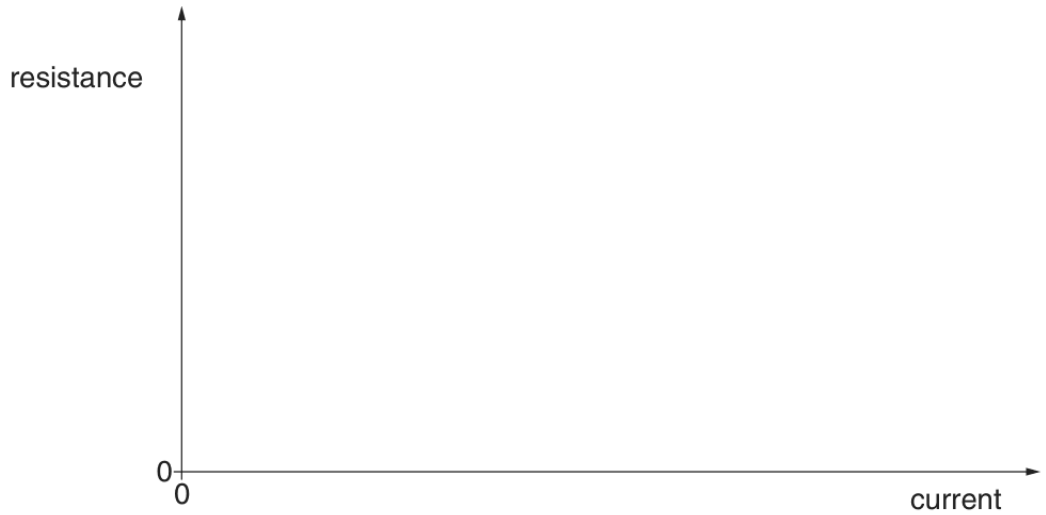
.....[2]

(iii) State the equation that defines resistance.

.....[1]

(b) The engineer performs the experiment and notices that the potential difference across the wire is directly proportional to the current in it. He calculates the resistance of the wire and plots a graph of the resistance against the current.

On Fig. 6.1, sketch the shape of this graph.



[1]

Fig. 6.1

6	**(a)**	**(i)**	power supply, (wire/resistor/bulb) and ammeter in series	B1	
			voltmeter across wire/resistor/bulb **labelled/clear**	B1	
			variable power supply **or** rheostat in series **or** potentiometer	B1	
			correct symbols **or** labelled throughout		
		(ii)	read ammeter **and** voltmeter / measure voltage **and** current	B1	
			vary power supply/rheostat/current	B1	
		(iii)	(R =) V/I (**ign.** V/A)	B1	
	(b)		horizontal line **and** above axis	B1	[7]

7 The current in the transmission line used to supply electrical power to a village is 65 A. The power is transmitted at a voltage of 23 000 V.

(a) Calculate the power supplied to the village.

power = ...[2]

(b) The transmission line has a resistance of 3.0 Ω. Calculate

(i) the potential difference across a 3.0 Ω resistor that carries a current of 65 A,

potential difference = ...[2]

(ii) the thermal energy (heat) produced in 1.0 s in a 3.0 Ω resistor that carries a current of 65 A.

heat = ...[1]

(c) **(i)** State one advantage of transmitting electrical power at a high voltage.

.....

.....[1]

(ii) Suggest one reason why the mains power supply to the houses in the village is at a voltage much lower than 23 000 V.

.....

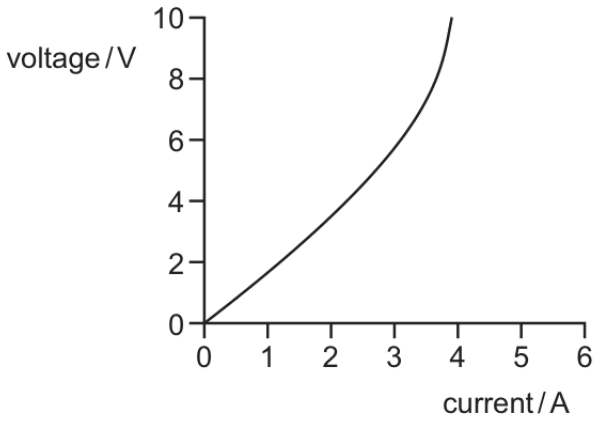
.....

.....[1]

7	**(a)**	(P =) VI **or** 23 000 × 65	C1	
		1.49/1.5/1.50/1.495 × 10^6 W	A1	
	(b)	**(i)** (V =) IR **or** 65 × 3	C1	
		190/195/200 V	A1	
		(ii) 1.3(1.27 etc.) × 10^4 J	B1	
	(c)	**(i)** low current/less energy/power wasted/less heat generated/less voltage **loss**/ more efficient/thinner wires	B1	
		(ii) **step-down** transformer between them **or** less insulation needed **or** less dangerous **or** less chance of electric shock **or** less danger of sparking /fire	B1	[7]

QUESTION · 5054_s12_qp_11.pdf · source page 11 of 16

28 The voltage / current graph for a filament lamp is shown.



The voltage across the lamp increases. What happens to the temperature of the lamp and the resistance of the lamp?

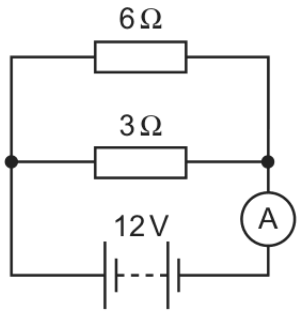
	temperature	resistance
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 28

28 **D**

QUESTION · 5054_s12_qp_11.pdf · source page 11 of 16

29 A 12 V battery is connected across a parallel arrangement of two resistors.



What is the reading on the ammeter?

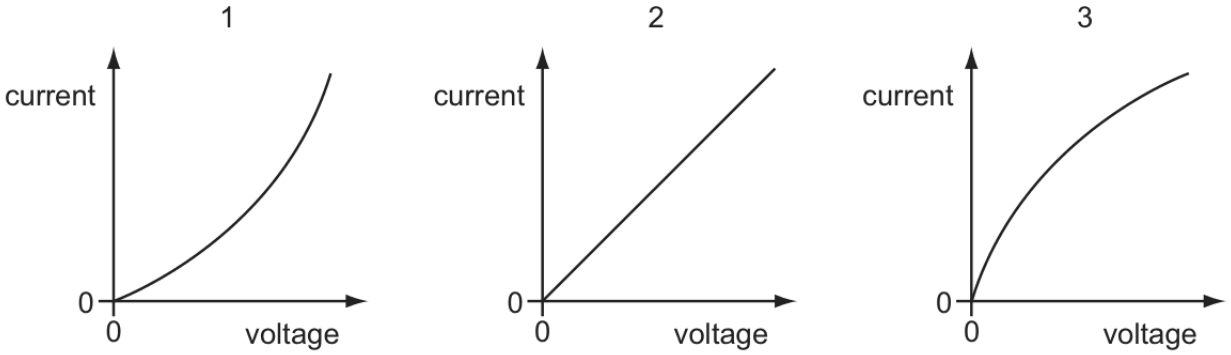
A 1.3 A **B** 2.0 A **C** 4.0 A **D** 6.0 A

MARK SCHEME (official answer key) · 5054_s12_ms_11.pdf · source page 2 of 2 · row for Question 29

29	**D**

QUESTION · 5054_s12_qp_12.pdf · source page 12 of 16

28 The current / voltage graphs are for different electrical components.



Which graph is for a resistor at constant temperature and which is for a filament lamp?

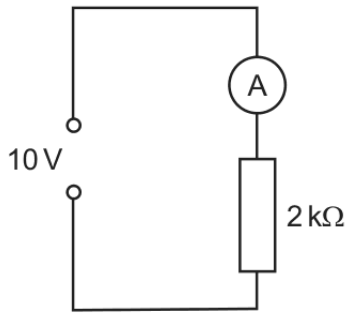
	resistor	lamp
A	1	2
B	2	1
C	2	3
D	3	2

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 28

28 **C**

QUESTION · 5054_s12_qp_12.pdf · source page 12 of 16

29 The diagram shows an ammeter connected in a circuit.



What is the current in the ammeter?

A 5 mA **B** 20 mA **C** 0.2 A **D** 5 A

MARK SCHEME (official answer key) · 5054_s12_ms_12.pdf · source page 2 of 2 · row for Question 29

29	**A**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 A student investigates how the resistance of a wire depends upon its length. The student uses an ammeter, a voltmeter, a battery, a fixed resistor and the wire under test, all connected in an electrical circuit.

(a) (i) In the space below, draw a circuit diagram of the apparatus. Label the wire under test with the letter W.

[3]

(ii) Describe how the student obtains one complete set of results.

.....

.....

.....[2]

(iii) During the experiment, the student keeps the temperature of the wire constant.

1. Suggest why it is sensible to keep the temperature of the wire constant.

.....

.....[1]

2. Suggest how the student keeps the temperature of the wire constant.

.....

.....[1]

(b) Fig. 8.1 shows part of a circuit containing three resistors X, Y and Z.

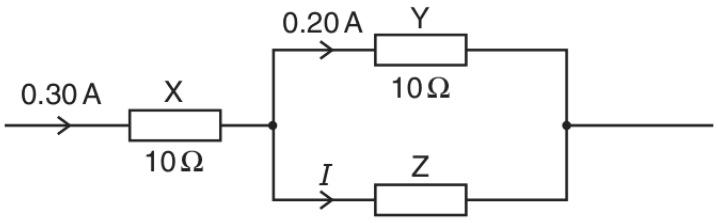


Fig. 8.1

The resistance of X and the resistance of Y are each 10 Ω. The current through X is 0.30 A. The current through Y is 0.20 A.

(i) Calculate the potential difference (p.d.) across Y.

p.d. = ...[2]

(ii) Calculate the current I in resistor Z.

I = ...[1]

(iii) Explain in words, rather than by calculation, how Fig. 8.1 shows that the resistance of Z is larger than the resistance of Y. Use ideas about p.d. and current in your answer.

.....

.....

.....[2]

(iv) Calculate the total resistance of the resistors X, Y and Z in this circuit.

resistance = ...[3]

Section B

8 (a) (i) **correct circuit symbols** containing, in any circuit, a
- battery/cell/d.c. power supply
- ammeter
- voltmeter
- fixed resistor B1

ammeter clearly measures current through W B1

voltmeter clearly across W if W shown or a resistor if not B1

(ii) Any 2 from
- resistance (calculated from) V/I or $V = IR$ seen
- length (of wire), V and I **all three** measured
- change length **and** V and I measured B2

(iii) **1.** resistance/resistivity changes (with temperature)
OR wire gets hot **and** melts/burns/catches fire/dangerous
OR V only proportional to I at constant temperature B1

2. use of a water bath/heat sink
OR use small currents
OR take reading (quickly) and switch off B1

(b) (i) (V =) IR in any form numerical or algebraic C1
2(.0) V A1

(ii) 0.1(0) A B1

(iii) (Z) has the same potential difference/voltage B1

(Z) has less/small(er) current (thus larger resistance) B1

(iv) (p.d. across X =) 0.3 × 10 (V)
OR (R_Z =) 2/0.1 OR 20 (Ω) seen C1

(total p.d.) 5 (V)
OR 6.7 (Ω) seen
OR $1/R_T = 1/R_1 + 1/R_2$ in any form numerical or algebraic OR 20/3 seen C1

16.7 Ω ; 17 Ω ; 16.67 Ω; 16.66 Ω A1 [15]

For Examiner's Use

8 A 600 Ω resistor and a thermistor are connected in series with an ammeter and a 20 V d.c. (direct current) power supply. A voltmeter is in parallel with the resistor.

Fig. 8.1 is the circuit diagram.

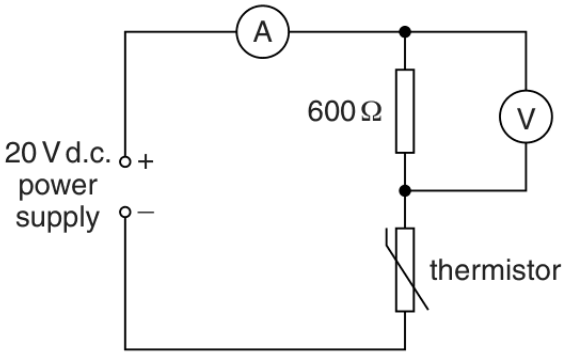


Fig. 8.1

The ammeter reads 0.025 A.

(a) Calculate

(i) the reading on the voltmeter,

reading = ... [2]

(ii) the resistance of the thermistor.

resistance = ... [2]

(b) The temperature of the thermistor increases.

State what, if anything, happens to

(i) the resistance of the thermistor,

..... [1]

(ii) the ammeter reading and to the voltmeter reading.

ammeter reading: .....

voltmeter reading: .....

[1]

8	**(a)**	**(i)**	(V =)IR or 0.025 × 600	C1	
			15 V	A1	
		(ii)	5(V) **or** 5/0.025 **or** 800 **or** 800–600	C1	
			200 Ω	A1	
	(b)	**(i)**	decreases	B1	
		(ii)	ammeter: opposite to **(i)**		
			voltmeter: same as ammeter (both changes correct)	B1	[6]

[Total: 45]

QUESTION · 5054_w12_qp_12.pdf · source page 10 of 16

28 A metal wire has length l and cross-sectional area A.

What is the resistance proportional to?

A $A + l$ **B** $\frac{A}{l}$ **C** $\frac{l}{A}$ **D** $l \times A$

MARK SCHEME (official answer key) · 5054_w12_ms_12.pdf · source page 2 of 2 · row for Question 28

28	**C**

For Examiner's Use

11 (a) A wire is wound around a soft-iron core forming a solenoid, as shown in Fig. 11.1.

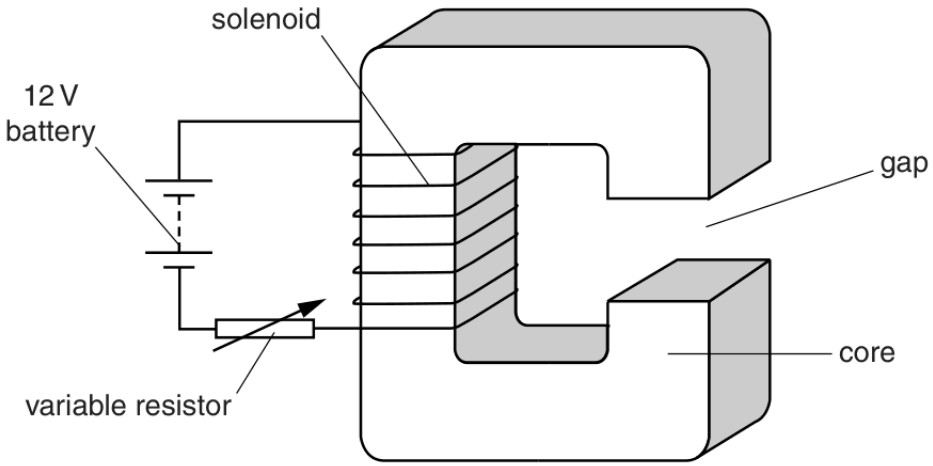


Fig. 11.1

There is a gap in the core. The solenoid is connected in series with a 12V battery and a variable resistor (rheostat). The resistance of the solenoid is 0.30 Ω and the variable resistor is set so that it has a resistance of 4.5 Ω.

(i) Calculate the current in the solenoid.

current = ...[3]

(ii) The current in the solenoid magnetises the soft-iron core.

Explain how the electric circuit is used to increase the strength of the magnetic field.

.....

.....

.....[2]

For Examiner's Use

(iii) Fig. 11.2 shows a horizontal, current-carrying wire PQ in the gap.

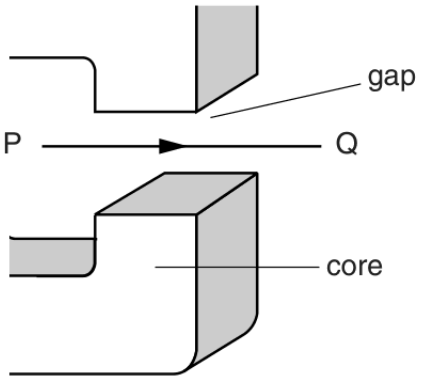


Fig. 11.2

1. The magnetic field in the gap is uniform and vertically upwards. The current in PQ is from left to right. Describe the effect of the magnetic field on PQ.

.....

.....[2]

2. State the effect on PQ of increasing the strength of the magnetic field in the gap.

.....

.....[1]

(b) The starter motor in a car is powered by a 12V battery that is positioned next to the motor. The current in the motor is 75A.

(i) Calculate the power supplied by the battery.

power = ...[2]

(ii) Suggest and explain why the wires that connect the motor to the battery are very thick.

.....

.....

.....[2]

(c) A relay is used to switch on a starter motor in a car. Describe how the relay works. You may include a diagram in your answer.

For Examiner's Use

.....

.....

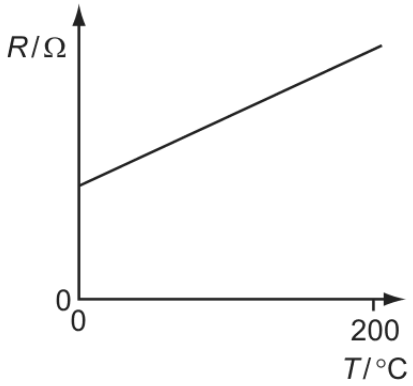
.....

.....[3]

11	**(a)**	**(i)**	4.5 + 0.3 **or** 4.8	C1	
			(*I* =)*V*/*R* **or** 12/4.8 **or** 12/4.5 **or** 12/0.3 **or** 12/0.28125	C1	
			2.5 A	A1	
		(ii)	decrease resistance (of variable resistor)	B1	
			increase current (in solenoid)	B1	
		(iii)	**1.** force on PQ/wire **or** PQ/wire moves	M1	
			force/movement out of page/outwards/towards observer (**not** upwards)	A1	
			2. force/speed/acceleration larger	B1	[8]
	(b)	**(i)**	(*P* =)*VI* **or** 75 × 12	C1	
			900 W	A1	
		(ii)	(thick wires) have low resistance	B1	
			(thick wires) not as hot/do not melt	B1	
	(c)		current to relay/coil/solenoid/electromagnet	B1	
			core/relay/coil/solenoid/electromagnet magnetised	B1	
			connections made (in motor circuit)	B1	[7]
					[Total: 15]

QUESTION · 5054_s13_qp_11.pdf · source page 13 of 20

31 The electrical resistance of a metal wire varies with temperature as shown.



The resistance of the wire at 0 °C is 1.00 Ω and at 100 °C is 1.40 Ω.

What is the resistance of the wire at 200 °C?

A 1.80 Ω **B** 1.96 Ω **C** 2.00 Ω **D** 2.80 Ω

MARK SCHEME (official answer key) · 5054_s13_ms_11.pdf · source page 2 of 2 · row for Question 31

31	**A**

For Examiner's Use

6 Fig. 6.1 is a graph of current against potential difference (p.d.) for a length of metal wire.

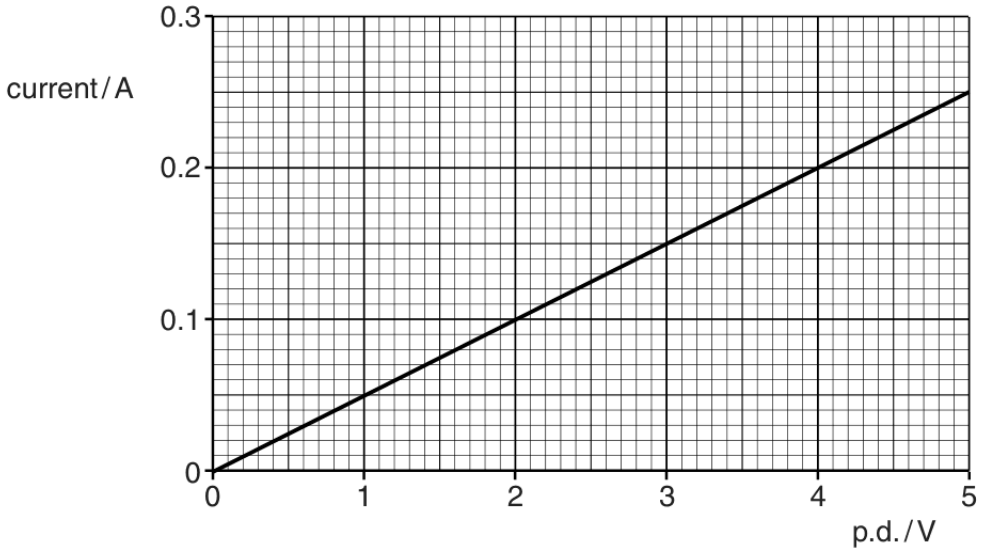


Fig. 6.1

(a) The metal wire obeys Ohm's law. State Ohm's law in words.

.....

.....

.....[2]

(b) Calculate the resistance of the metal wire.

resistance = ...[2]

(c) A new wire is made from the same metal as the original wire. The two wires have the same length. The cross-sectional area of the new wire is half that of the original wire.

(i) Calculate the resistance of the new wire.

resistance = ...[1]

(ii) On Fig. 6.1, draw a line to show how current varies with p.d. for the new wire. [2]

6 **(a)** current is directly proportional to voltage (**accept** voltage/current = constant, but **not** just = R) B1
if temperature/physical conditions constant B1

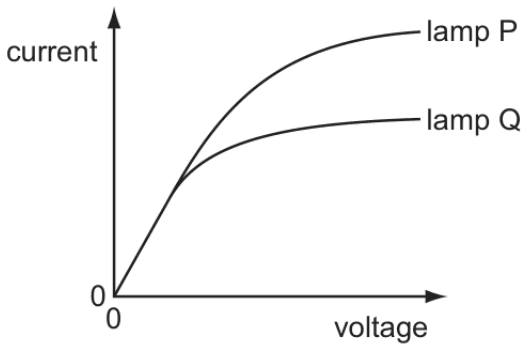
(b) (R=) V/I in any form algebraic or using any value of V and I from graph C1
20 Ω A1

(c) **(i)** 40 Ω **or** 2 × **(b)** B1

(ii) straight line graph through origin below given line ecf **(b)** (e.g. **accept** above line if R < 20) M1
goes through 0.1 A at 4 V ecf **(b)** (e.g. allow through 0.2 A at 2 V if R = 10 Ω) A1 [7]

QUESTION · 5054_s13_qp_12.pdf · source page 11 of 16

33 The diagram shows the current-voltage graph of two filament lamps.



Which statement is correct?

A Lamp P has a lower resistance than lamp Q at all currents.

B Lamp P has the same resistance as lamp Q at low currents.

C Lamp P has the same resistance as lamp Q at high currents.

D Lamp P has a higher resistance than lamp Q at all currents.

MARK SCHEME (official answer key) · 5054_s13_ms_12.pdf · source page 2 of 2 · row for Question 33

33	**B**

6 Fig. 6.1 shows the current-voltage graph for a filament lamp.

For Examiner's Use

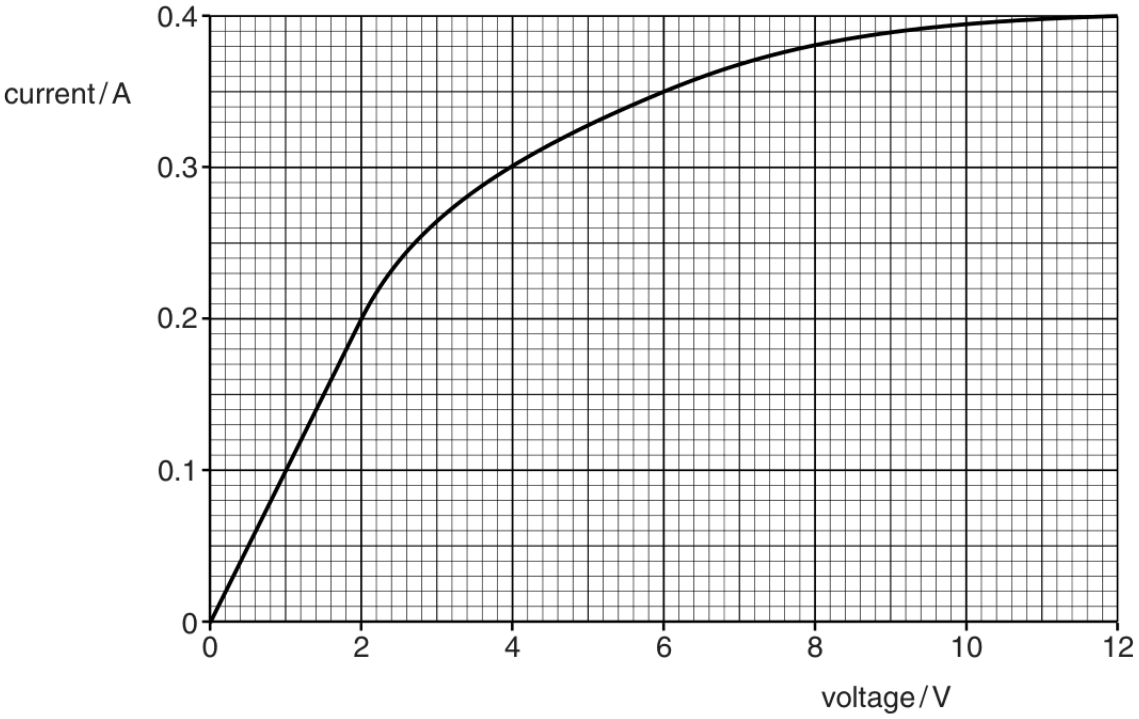


Fig. 6.1

(a) (i) State the range of voltages where the resistance increases.

..... [1]

(ii) State why the resistance of the lamp increases as the voltage increases.

.....

..... [1]

For Examiner's Use

(b) The filament lamp is connected in the circuit shown in Fig. 6.2.

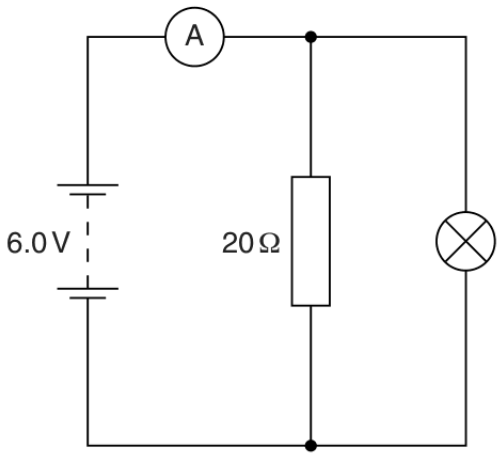


Fig. 6.2

There are currents in the lamp, the 20 Ω resistor and the ammeter.

(i) State what is meant by an *electric current*.

.....

..... [1]

(ii) Use Fig. 6.1 to determine the current in the filament lamp.

current = ... [1]

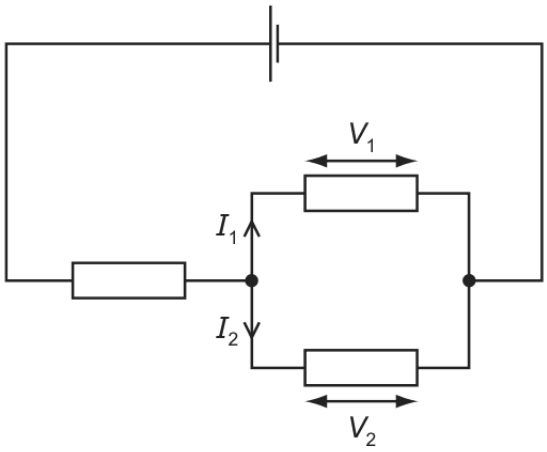
(iii) Determine the current in the ammeter.

current = ... [3]

6	**(a)**	**(i)**	2 to 2.1 (V) to any value between 11 and 12 (V) **or** above 2/2.1(V)	B1	
		(ii)	temperature increases / gets hotter	B1	
	(b)	**(i)**	(rate of) flow of charge/electrons	B1	
		(ii)	0.35 A cao	B1	
		(iii)	(I=) V/R algebraic **or** 6/20 **or** 6/0.35	C1	
			0.3(0) (A) **or** $1/R_T = 1/20 + 1/17.1$ **or** (R_T =) 9.2 (Ω) seen	C1	
			0.65 A	A1	[7]

QUESTION · 5054_w13_qp_11.pdf · source page 14 of 20

30 The circuit shows three resistors connected to a cell.



The resistors have different values of resistance.

V_1 and V_2 are potential differences and I_1 and I_2 are currents as shown.

Which pair of relationships is correct?

A $I_1 = I_2$ and $V_1 = V_2$

B $I_1 = I_2$ and $V_1 \neq V_2$

C $I_1 \neq I_2$ and $V_1 = V_2$

D $I_1 \neq I_2$ and $V_1 \neq V_2$

MARK SCHEME (official answer key) · 5054_w13_ms_11.pdf · source page 2 of 2 · row for Question 30

30	**C**

For Examiner's Use

11 An ammeter, a variable resistor and a 4.0Ω resistor are connected in series with a 6.0V battery. A cathode-ray oscilloscope (c.r.o.) is connected in parallel with the 4.0Ω resistor through a switch. Fig. 11.1 is the circuit diagram.

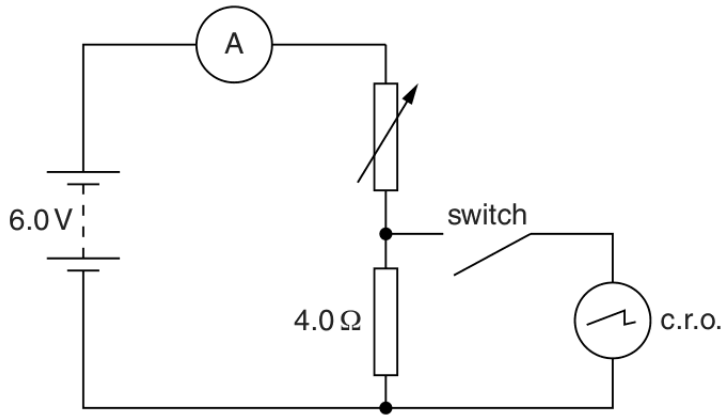


Fig. 11.1

The switch is open.

(a) The variable resistor is adjusted so that it has a resistance of 8.0Ω.

Calculate

(i) the current measured by the ammeter,

current = ...[2]

(ii) the potential difference (p.d.) across the 4.0Ω resistor.

p.d. = ...[1]

(b) The resistance of the variable resistor is adjusted until the p.d. across the 4.0Ω resistor is 5.0V.

State the effect that this adjustment has on

(i) the current measured by the ammeter,

.....[1]

(ii) the resistance of the variable resistor.

.....[1]

For Examiner's Use

(c) The trace on the c.r.o. is a horizontal line across the middle of the screen and the p.d. across the 4.0Ω resistor is still 5.0V.

The Y-gain control is set at 2.0V/cm and the switch is closed.

State the effect on the trace of closing the switch.

.....

.....

.....[2]

(d) Fig. 11.2 shows, in outline, the basic structure of a c.r.o. viewed from the side.

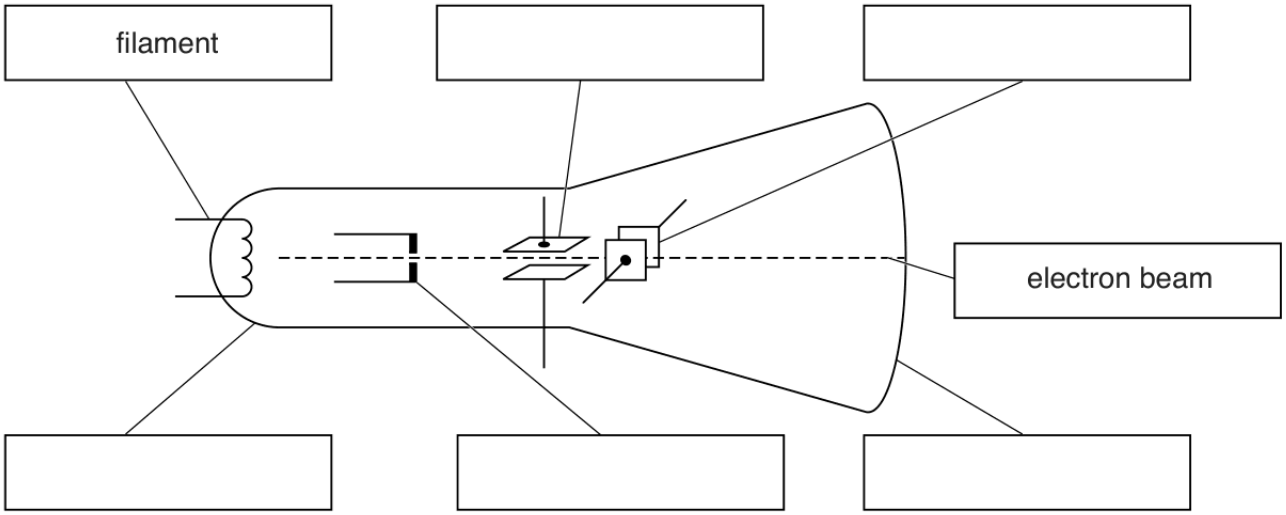


Fig. 11.2

(i) On Fig. 11.2 write, in the **five** empty boxes, the names of the parts of the c.r.o. indicated. [3]

(ii) Explain how the electron beam is produced.

.....

.....

.....[2]

(iii) Explain why the tube of the c.r.o. must be evacuated.

.....

.....[1]

Question 11 continues on page 16.

For Examiner's Use

(iv) In the c.r.o. shown in Fig. 11.2, the electrons travel from left to right.

1. Explain why the flow of electrons is an electric current.

.....

.....[1]

2. State the direction of the conventional current due to the electrons.

.....

.....[1]

11 (a) (i) (I =)V/R **or** 6.0/12.0 **or** 6.0/(4.0+8.0) **or** (in **(ii)**) (V =)IR **or** 0.50 × 4.0 — C1
0.50 A — A1

(ii) 2.0 V (scores C1 in **(a)(i)** if not already scored) — A1 [3]

(b) (i) increased **or** becomes 1.25 A — B1

(ii) decreases **or** becomes 0.8 Ω — B1 [2]

(c) moves up or down **or** 5.0/2.0 — C1
moves up or down by 2.5 cm — A1 [2]

(d) (i)

	Y-plates	X-plates
(glass) tube	anode	ZnS/screen

(5 correct 3 marks, 4 correct 2 marks, 3 correct 1 mark
X and Y plates reversed –1; **allow** focussing anode) — B3

(ii) filament heated/thermionic emission — B1
(thermionic) electrons attracted by anode **or** repelled by cathode — B1

(iii) to prevent/otherwise collisions with air molecules/to allow to reach the screen/to avoid deflection — B1

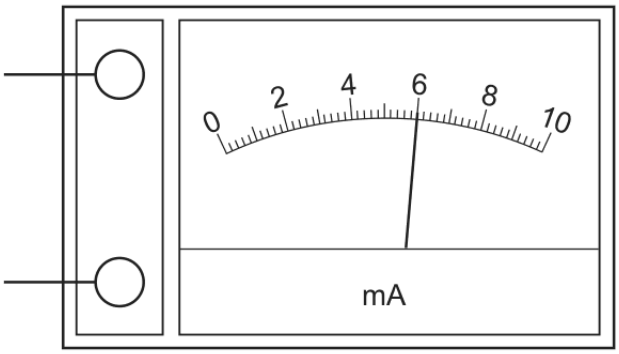
(iv) **1.** electrons are charged — B1

2. backwards **or** towards the back **or** opposite to electron motion **or** to the left **or** from the right — B1 [8]

[Total: 15]

QUESTION · 5054_w13_qp_12.pdf · source page 13 of 20

28 An ammeter is used to measure the current in a 300 Ω resistor. The ammeter is shown below.



What is the potential difference across the resistor?

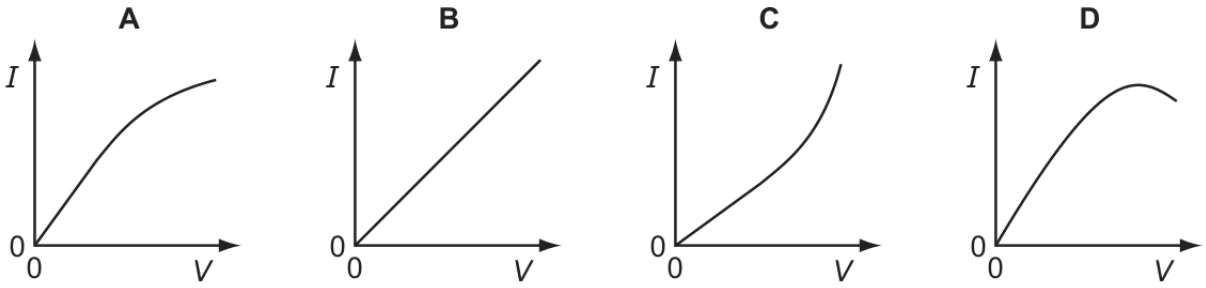
A 0.050 V **B** 1.8 V **C** 50 V **D** 1800 V

MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 28

28	**B**

QUESTION · 5054_w13_qp_12.pdf · source page 14 of 20

30 Which is the current / voltage (I / V) graph of a filament lamp?



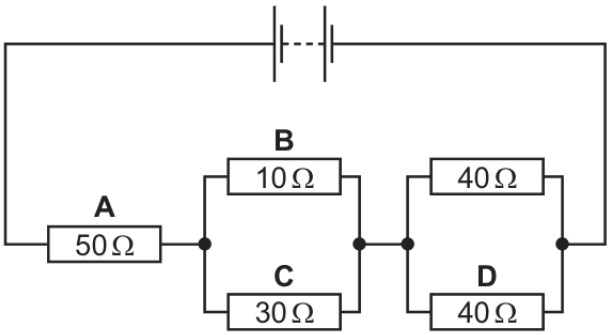
MARK SCHEME (official answer key) · 5054_w13_ms_12.pdf · source page 2 of 2 · row for Question 30

30	**A**

QUESTION · 5054_w13_qp_12.pdf · source page 14 of 20

31 The diagram shows a circuit containing five resistors connected to a battery.

In which resistor is the current the smallest?



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31	**C**

For Examiner's Use

11 A metal filament lamp is connected to a power supply. The electromotive force (e.m.f.) produced by the supply can be varied. Fig. 11.1 is the circuit diagram.

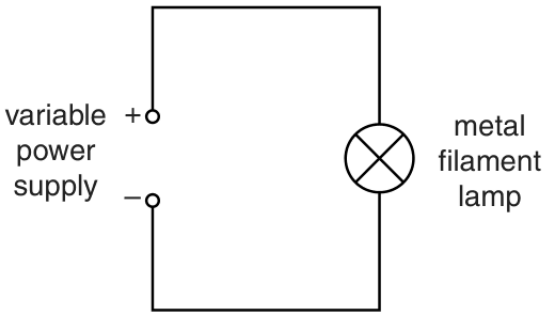


Fig. 11.1

(a) State what is meant by *electromotive force (e.m.f.)*.

.....

.....

.....[2]

(b) Add appropriate circuit symbols to Fig. 11.1 to show the position of

(i) an ammeter that measures the current in the circuit, [1]

(ii) a voltmeter that measures the voltage across the lamp. [1]

(c) A student adds meters to the circuit and makes measurements of the current and voltage. He then plots the current/voltage graph shown in Fig. 11.2.

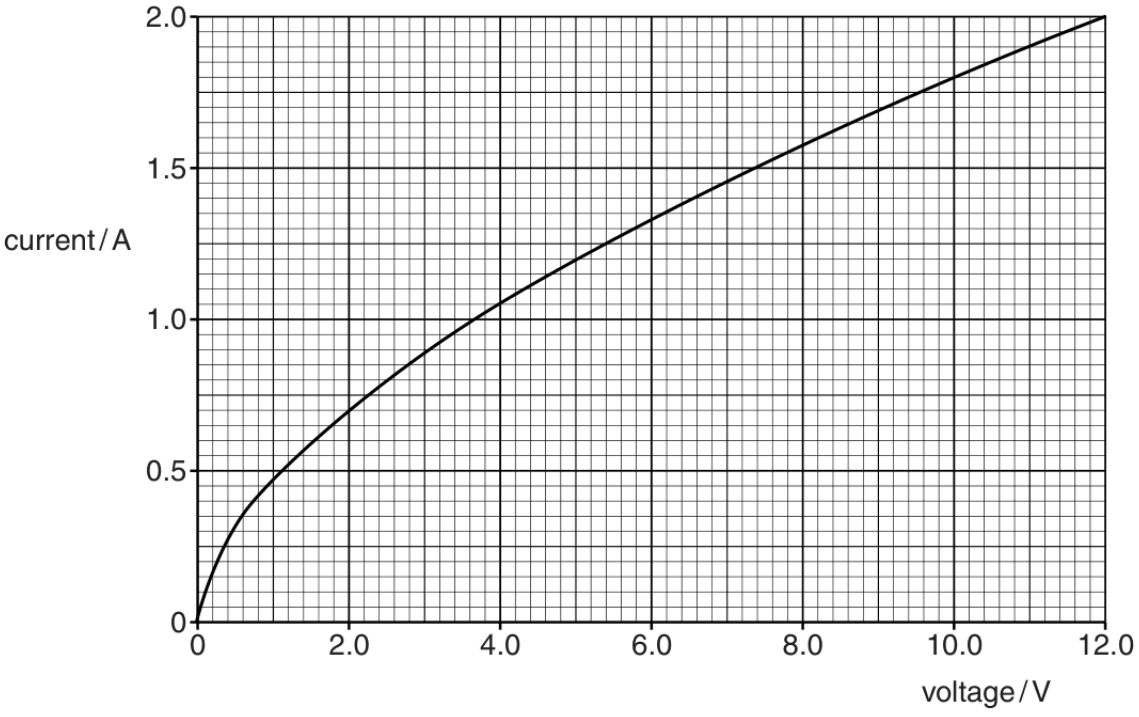


Fig. 11.2

For Examiner's Use

(i) Using values from Fig. 11.2, calculate the resistance of the lamp when the current is 0.70 A. Give your answer to an appropriate number of significant figures.

resistance = ...[3]

(ii) State what, if anything, happens to the resistance of the lamp as the voltage increases.

.....

.....[1]

(d) In normal use, the lamp is connected to a 12 V supply.

Use Fig. 11.2 to determine the power of the lamp when it is used in this way.

power = ...[2]

(e) When connected to a 12 V supply, thermionic emission occurs in the lamp.

(i) Describe what is meant by *thermionic emission*.

.....

.....

.....[2]

(ii) Thermionic emission is used in a cathode-ray oscilloscope (c.r.o.) tube.

1. State why air must be removed from the tube of the c.r.o.

.....

.....[1]

2. A voltage is connected across the Y-plates in the c.r.o.

State and explain the effect on the trace on the screen.

.....

.....

.....[2]

11	**(a)**	work done per (unit) charge/coulomb/C **or** energy transformed per (unit) charge/coulomb/C	B1	
		property of a source (of electricity) **or** energy transformed to electrical energy per (unit) charge/coulomb/C	B1	[2]
	(b) **(i)**	ammeter in series	B1	
	(ii)	voltmeter in parallel with lamp **or** lamp and ammeter	B1	[2]
	(c) **(i)**	(V =) 2.0 (V)	C1	
		(R =) V/I **or** 2.0/0.70	C1	
		2.9/2.86 Ω (i.e. 2 **or** 3 s.f. only)	A1	
	(ii)	(resistance) increases	B1	[4]
	(d)	(P =) VI **or** (P =) V^2/R **or** I^2R **or** 12 × 2.0 **or** 12 × 0.70	C1	
		24 W	A1	[2]
	(e) **(i)**	emission of electrons	M1	
		from heated metal/named metal/filament/wire	A1	
	(ii) **1.**	prevents collision with air (molecules) **or** prevents deflection **or** lets electrons/particles reach screen/travel unimpeded	B1	
	2.	moves vertically (e.g. up/down/above/below **or** vertical line) **not** with horizontal movement due to this voltage	B1	
		attracted by positive **or** repelled by negative **or** attracted by one plate and repelled by the other **or** electric field (acts on charge)	B1	[5]

[Total: 15]

QUESTION · 5054_s14_qp_11.pdf · source page 11 of 16

30 A piece of wire has a resistance of 16 Ω.

Another wire made from the same metal has four times the length and twice the cross-sectional area.

What is the resistance of the wire?

A 8 Ω **B** 32 Ω **C** 96 Ω **D** 128 Ω

MARK SCHEME (official answer key) · 5054_s14_ms_11.pdf · source page 2 of 2 · row for Question 30

30	**B**

7 **(a)** Fig. 7.1 shows a variable potential divider (potentiometer) connected to a 6.0 V battery.

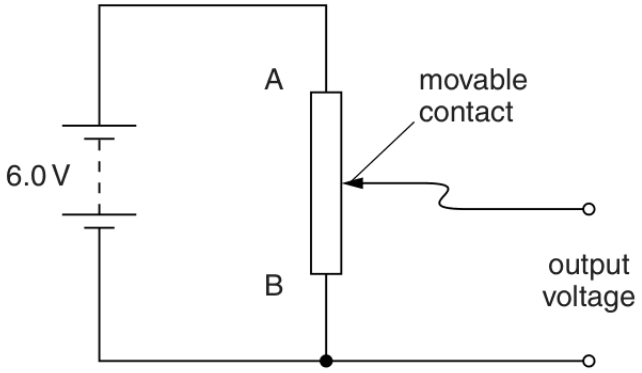


Fig. 7.1

The movable contact can slide between A and B. As it moves, the output voltage changes.

Complete the table below to show the output voltage when the contact is at A and at B.

position of movable contact	A	B
output voltage		

[2]

(b) Fig. 7.2 shows a thermistor in a potential divider circuit that is used to monitor temperature. The fixed resistor has a resistance of 2000 Ω.

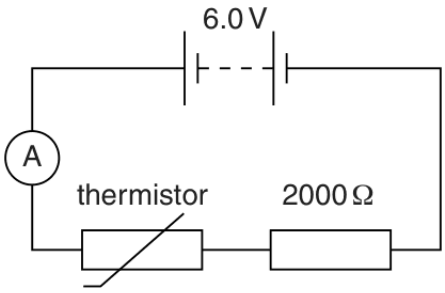


Fig. 7.2

(i) Explain why the voltage across the thermistor decreases as its temperature rises.

.....

..... [2]

(ii) At one temperature, the thermistor has a resistance of 1000 Ω. Calculate the current in the thermistor.

current = ... [2]

7 (a)

A	B
6.(0) V	0 (V)

[B1, B1]

(b) (i) resistance (of thermistor) decreases [B1]
current increases **and** larger voltage across 2000 Ω / fixed resistor
or smaller **fraction** of voltage across thermistor **or** potential divider explained [B1]

(ii) (I =) V / R in symbols or numbers [A1]
0.002(0) A; 2 mA [C1]

[6]

11 (a) A torch contains a filament lamp connected to a battery. The battery contains four 1.5 V cells connected in series.

Fig. 11.1 shows the current/potential difference (voltage) graph for the filament lamp.

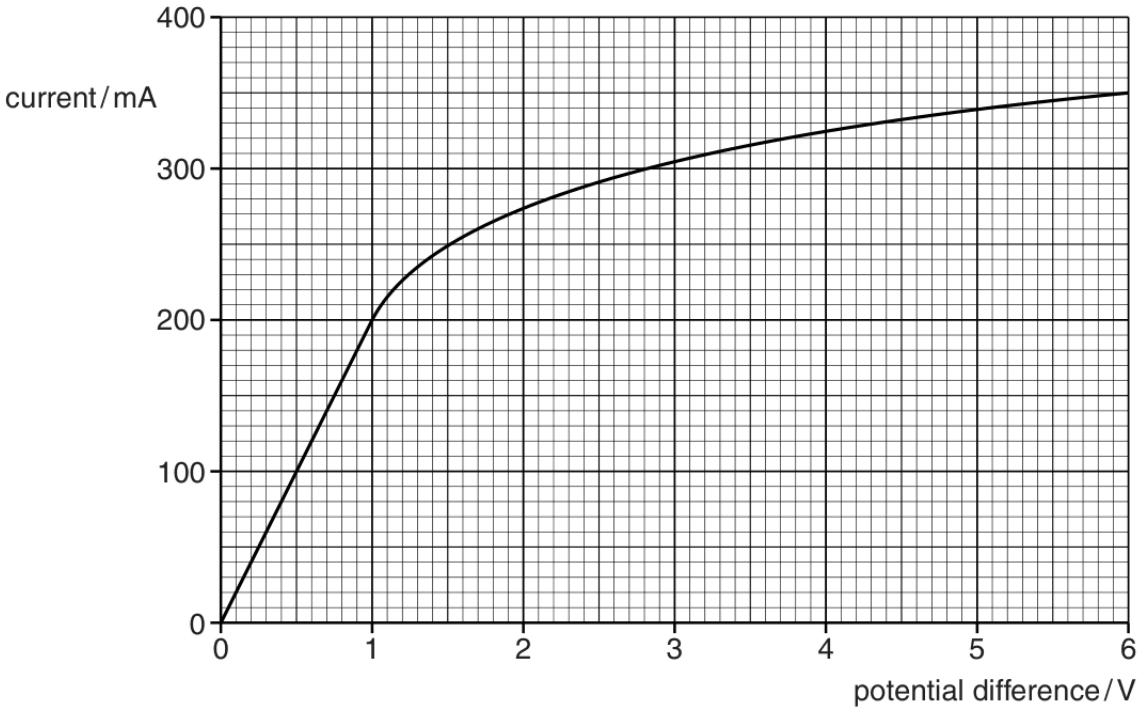


Fig. 11.1

(i) State what is meant by the *potential difference* (p.d.) across the lamp.

.....

..... [2]

(ii) Fig. 11.1 shows that the current in the lamp is not proportional to the p.d.

1. State how Fig. 11.1 shows this.

.....

..... [1]

2. Suggest why the current is not proportional to the p.d.

.....

..... [1]

(iii) The lamp is at normal brightness and there is a p.d. of 6.0V across the lamp.

1. Use Fig. 11.1 to find the current in the lamp.

current = ... [1]

2. Calculate the charge that passes through the lamp in 2.0 hours.

charge = ... [2]

3. The battery in the torch is able to keep the lamp lit for 2.0 hours.
Estimate the initial energy stored in the battery.

energy = ... [2]

4. Explain why your value is only an estimate.

.....

..... [1]

(b) Suggest and explain an advantage of using two cells in parallel rather than a single cell.

.....

.....

..... [2]

(c) Another torch contains a light-emitting diode (LED) instead of a filament lamp.
The battery in the torch uses four 1.5V cells, some in series and some in parallel.
The p.d. across the LED when in operation is 3.0V.

In the space below, draw a circuit diagram showing how the four cells, a switch and the LED are connected.

[3]

11 (a) (i) work done **or** energy produced / needed [M1]
per unit charge **or** per coulomb (passing through lamp) [A1]

(ii) **1**. not straight **or** curves **or** gradient changes **or** data used correctly [B1]
2. gets hotter **or** temperature changes **or** resistance increases [B1]

(iii) **1**. 350 mA **or** 0.35 A [B1]
2. Q = It in any form algebraic or numerical **or** 2 × 60 × 60 **or** 7200 (s) seen [C1]
2520 C **or** 2500 C [A1]
3. (E=) QV or V*I*t in any algebraic or numerical form, e.g. 0.35 × 6 × 2 [C1]
15100 J **or** 15120 J **or** 15000 J **or** 4.2 kWh [A1]
4. current and / or voltage falls / varies
or some energy remains (in cell)
or some energy / heat produced **in cell**
or correct argument involving internal resistance of cell [B1]

(b) lasts longer [M1]
(because) larger energy (initially) **or** smaller current (in each cell) [A1]
or
avoids failure if one cell fails (M1)
(because) other (parallel) cell takes over (A1)
or
larger current / power (in external resistor / lamp) (M1)
(because) smaller (internal) resistance of combined cells (A1)

(c) symbol for LED clear and in correct direction for cells shown [B1]
4 cells (not all in parallel or in series), switch and LED (labelled or shown) that switches on and off properly [B1]
cells connected correctly to give 3.0 V total e.m.f., e.g. two sets of parallel cells connected in series or three in parallel and one in series [B1]

[15]

11 A student sets up the circuit shown in Fig. 11.1.

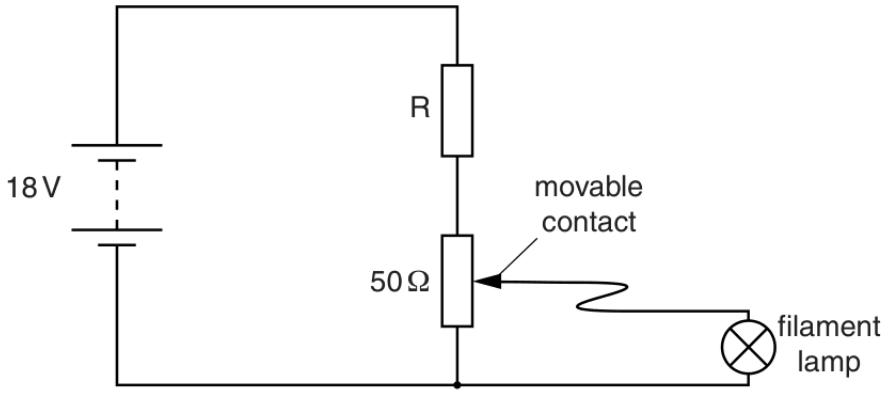


Fig. 11.1

R is a fixed resistor in the circuit. The filament lamp is marked 12 V, 0.25 A.

The circuit is used to produce a current/voltage graph for the filament lamp. The ammeter and voltmeter needed are not shown.

To obtain different readings, the student changes the position of the movable contact.

(a) On Fig. 11.1, draw the symbols for an ammeter and a voltmeter in the correct positions. [3]

(b) Explain why it is sensible to include the resistor R in this circuit.

.....

.....

..... [2]

(c) **(i)** On Fig. 11.2, sketch a current/voltage graph for the lamp.

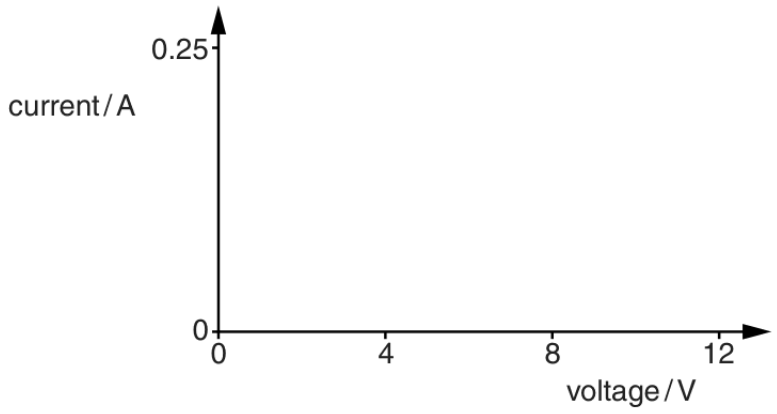


Fig. 11.2 [3]

(ii) State and explain how a current/voltage graph for a fixed resistor is different from the graph for a filament lamp.

.....

.....

..... [2]

(d) Fig. 11.3 shows the position of the movable contact when the voltage across the lamp is 12 V and the current in the lamp is 0.25 A.

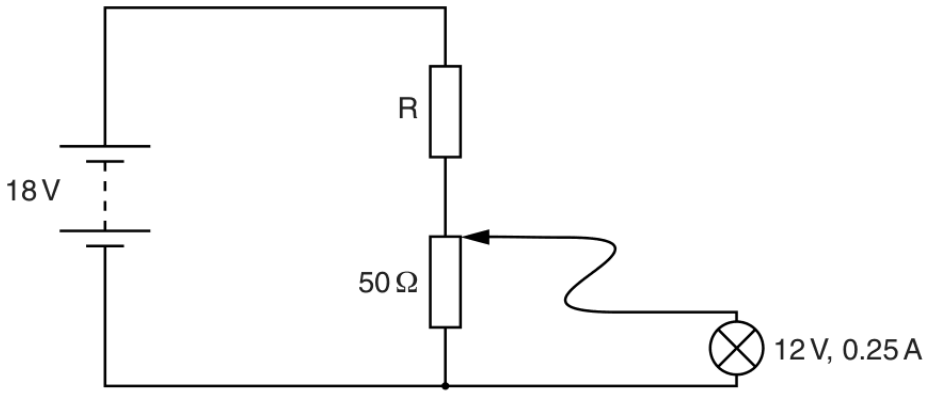


Fig. 11.3

Determine

(i) the current in the 50 Ω resistor,

current = ... [2]

(ii) the current in R,

current = ... [1]

(iii) the potential difference (p.d.) across R,

p.d. = ... [1]

(iv) the resistance of R.

resistance = ... [1]

11 **(a)** ammeter and voltmeter correct symbols [B1]
ammeter in series with lamp [B1]
voltmeter in parallel with lamp [B1]

(b) R limits or reduces the current / voltage [B1]
otherwise lamp blows
or more of the 50 Ω can be used to adjust voltage / current [B1]

(c) **(i)** 12 V, 0.25 A correctly plotted (by eye) [B1]
curved line from origin [B1]
correct curvature – decreasing slope [B1]

(ii) straight line (for fixed resistor) [B1]
lamp has changing temperature or changing resistance
or fixed resistor has constant temperature or constant **resistance** [B1]

(d) **(i)** (I =) V / R in any algebraic or numerical form, e.g. 12 / 50 [C1]
0.24 A [A1]

(ii) 0.49 A [B1]

(iii) 6(.0) V [B1]

(iv) 12(.24) Ω [B1]

[15]

11 A student makes a 2.0V battery by connecting two cells of electromotive force (e.m.f.) 2.0V in parallel. The battery, an ammeter with different ranges and three different resistors are used to set up the circuit shown in Fig. 11.1.

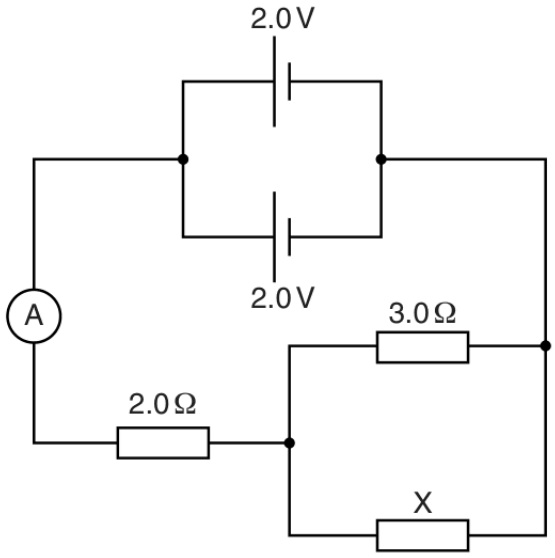


Fig. 11.1

(a) (i) Explain what is meant by *electromotive force*.

.....

.....

.....[2]

(ii) State one advantage of using two cells in parallel rather than using a single 2.0V cell.

.....

.....[1]

(b) Resistor X and the 3.0 Ω resistor have a combined resistance that is equal to 2.0 Ω.

Calculate

(i) the total resistance of the circuit,

total resistance = ...[1]

(ii) the resistance of X.

resistance of X = ...[2]

(c) **(i)** Determine the reading of the ammeter.

reading = ...[2]

(ii) Suggest a suitable range for the ammeter.

.....[1]

(d) The current in the 2.0 Ω resistor is I_2.
The current in the 3.0 Ω resistor is I_3.
The current in X is I_X.

State the equation that relates I_2, I_3 and I_X.

.....[1]

(e) State the potential difference (p.d.) across

(i) the 2.0 Ω resistor,

p.d. = ...[1]

(ii) the 3.0 Ω resistor.

p.d. = ...[1]

Question 11 continues on page 18.

(f) The student sets up a second circuit using a variable d.c. power supply, an ammeter and a 12V metal filament lamp. The circuit is shown in Fig. 11. 2.

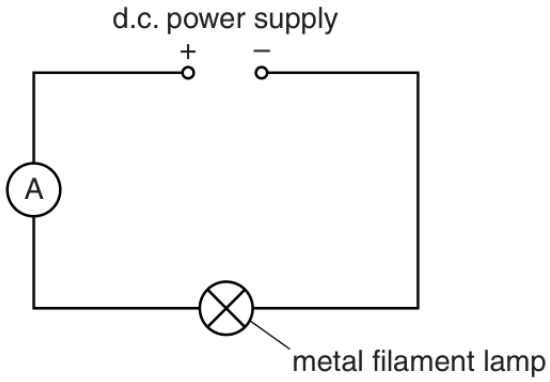


Fig. 11.2

The d.c. power supply is set to 12V and the ammeter reading is 1.5 A.

The student changes the e.m.f. of the d.c. power supply to 6.0V. The lamp dims and the ammeter reading changes.

(i) State and explain what happens to the resistance of the metal filament of the lamp.

.....

.....

.....[2]

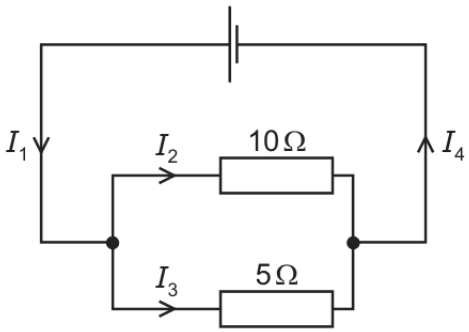
(ii) State whether the new ammeter reading is less than, equal to or greater than 0.75 A.

.....[1]

11	**(a)**	**(i)**	energy to drive charge around a circuit **or** terminal p.d. on open circuit	B1	
			energy to drive **unit** charge around a circuit **or** energy / charge	B1	
		(ii)	lasts longer **or** lower internal resistance **or** can replace a cell without switching off **or** continues to work if one cell is flat **ignore** more current (**not** greater e.m.f. / voltage)	B1	**[3]**
	(b)	**(i)**	4.0 Ω	B1	
		(ii)	$(1/R_{tot} =)1/R_1 + 1/R_2$ **or** $1/3 + 1/X$ **or** product / sum **or** $(3 \times X)/(3 + X)$ **or** $\frac{1}{X} = \frac{1}{2} - \frac{1}{3}$	C1	
			6.0 Ω	A1	**[3]**
	(c)	**(i)**	$(I =)\ V/R$ **or** 2.0 / 4.0	C1	
			0.50 A	A1	
		(ii)	(from) 0 **and** (to) 0.50 to 5.0 A	B1	**[3]**
	(d)		$I_2 = I_3 + I_X$	B1	**[1]**
	(e)	**(i)**	1.0 V	B1	
		(ii)	1.0 V	B1	**[2]**
	(f)	**(i)**	temperature decreases	B1	
			resistance decreases	B1	
		(ii)	greater than 0.75 A (e.c.f. resistance increases in **(f)(i)**)	B1	**[3]**
					[Total: 15]

QUESTION · 5054_w14_qp_12.pdf · source page 10 of 16

25 The currents in different parts of the circuit are I_1, I_2, I_3 and I_4.



Which statement is correct?

A $I_1 = I_4$ and I_2 is greater than I_3.

B $I_1 = I_4$ and I_3 is greater than I_2.

C I_2 is greater than I_1 and less than I_3.

D I_2 is greater than I_1 and greater than I_3.

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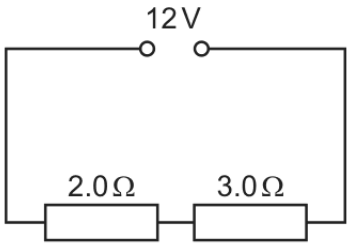
25	**B**

QUESTION · 5054_w14_qp_12.pdf · source page 10 of 16

26 The diagram shows a circuit that has two resistors in series with a 12 V supply.



What is the current in the circuit?

A 2.4 A **B** 10 A **C** 14 A **D** 60 A

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26	**A**

7 The circuit of Fig. 7.1 includes an ammeter, a 6.0 Ω resistor, a length of metal resistance wire and three 1.5 V cells connected in parallel.

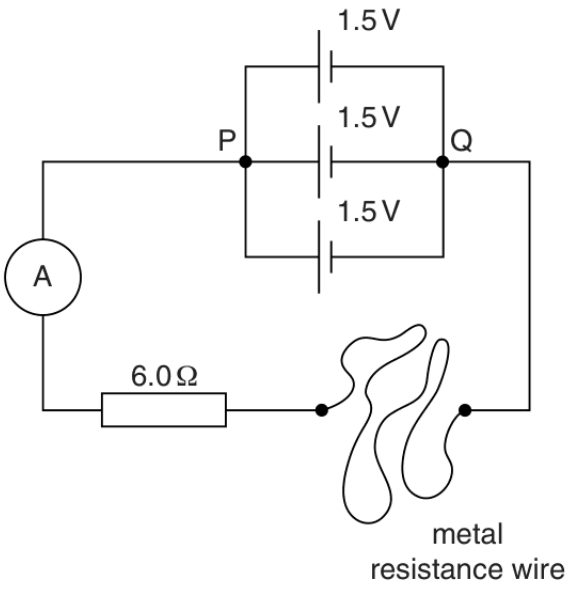

Fig. 7.1

(a) State

(i) one advantage of using cells in parallel rather than using a single cell,

.....

.....[1]

(ii) the potential difference (p.d.) between points P and Q in the circuit of Fig. 7.1.

p.d. = ...[1]

(b) **(i)** The ammeter in Fig. 7.1 reads 0.075 A.

Calculate the resistance of the resistance wire.

resistance = ...[3]

(ii) The temperature of the metal resistance wire increases.

State and explain the effect of this temperature increase on the ammeter reading.

.....

.....

.....[2]

7	(a)	(i)	lasts longer **or** one cell can be replaced without switching off the circuit **or** less (internal) resistance **or** if one fails the others still work	B1	
		(ii)	1.5 V	B1	
	(b)	(i)	(R =)V / I **or** 1.5 / 0.075	C1	
			20 (Ω) **or** 1.5 / (0.075 – 6.0)	C1	
			14 Ω	A1	
		(ii)	decreases	B1	
			resistance of wire increases	B1	**[7]**

QUESTION · 5054_s15_qp_11.pdf · source page 12 of 16

32 A 1.0 m length of wire of cross-sectional area 0.40 mm^2 has a resistance of 2.0 Ω.

What is the resistance of a 0.50 m length of wire, of the same material, with a cross-sectional area of 0.80 mm^2?

A 0.5 Ω **B** 1.0 Ω **C** 4.0 Ω **D** 8.0 Ω

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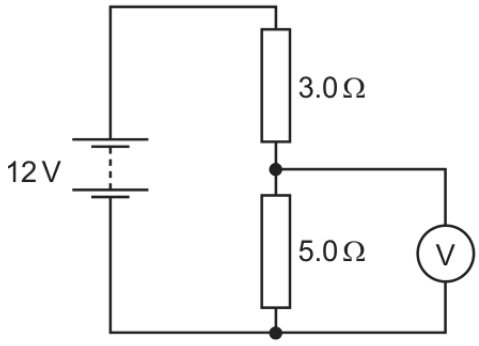
32	**A**

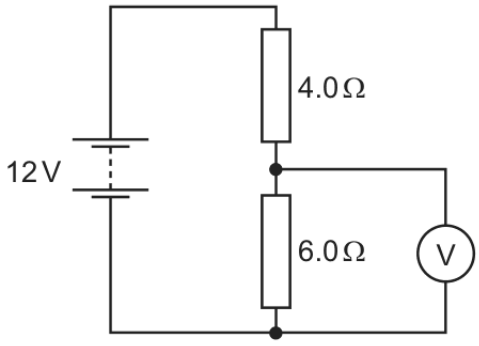
QUESTION · 5054_s15_qp_11.pdf · source page 13 of 16

34 In which circuit is the voltmeter reading 7.2 V?

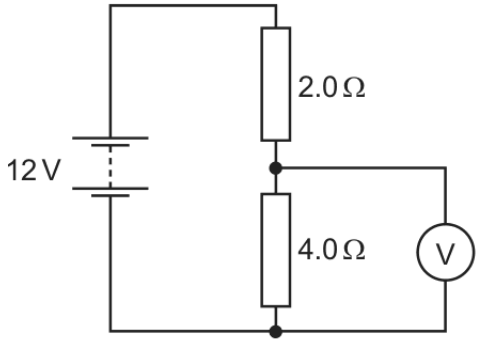
A



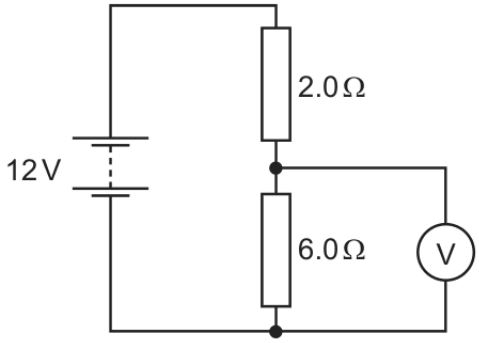
B



C



D



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34	**B**

6 Fig. 6.1 shows a circuit that contains a resistor connected to a power supply of 6.0 V and a lamp L.

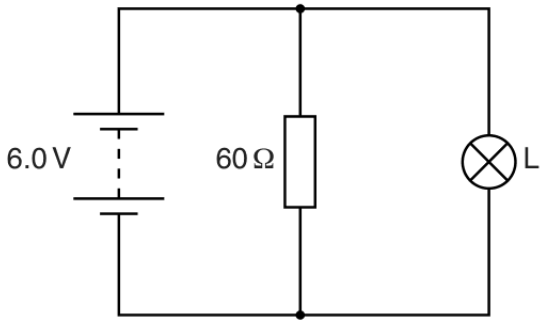


Fig. 6.1

The resistor has a resistance of 60 Ω. The lamp is marked 6.0 V, 0.90 W.

(a) Calculate

(i) the current in the resistor,

current = ...[2]

(ii) the current in the power supply.

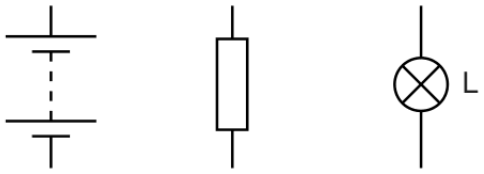
current = ...[2]

(b) A second lamp is added to the circuit shown in Fig. 6.1.

The second lamp is in series with the 60 Ω resistor, but is **not** in series with lamp L.

(i) In the space below draw a circuit diagram of this new circuit.

The power supply, 60 Ω resistor and lamp L have been drawn for you.



[1]

(ii) The two lamps are identical.

Explain why the second lamp is dimmer than lamp L.

A calculation is not required.

.....

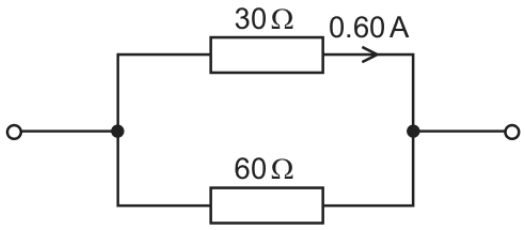
.....

.....[1]

6	**(a)**	**(i)**	(I=) V / R or 6/60	C1
			0.1(0) A	A1
		(ii)	(I=) P / V or 0.9 / 6	C1
			or 0.15 (A) seen	
			0.25 A	A1
	(b)	**(i)**	lamp correctly drawn in series with resistor but not the lamp	B1
		(ii)	less voltage (across lamp) **because** some voltage across resistor / shares voltage with resistor	B1
			or less current **because** of effect of resistor	

QUESTION · 5054_s15_qp_12.pdf · source page 13 of 16

34 Two resistors of resistances 30 Ω and 60 Ω are arranged in parallel. The current in the 30 Ω resistor is 0.60 A.



What is the potential difference across the 60 Ω resistor?

A 9.0 V **B** 18 V **C** 36 V **D** 54 V

MARK SCHEME (official answer key) · 5054_s15_ms_12.pdf · source page 2 of 2 · row for Question 34

34	**B**

10 Fig. 10.1 shows a relay connected to a cell and a switch.

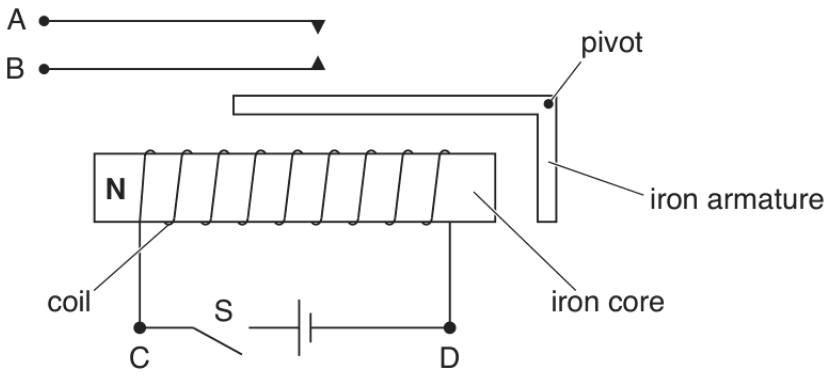


Fig. 10.1

(a) When switch S is closed, the iron core is magnetised. The left side of the core is an N-pole, as shown in Fig. 10.1. The iron armature is attracted to the core.

(i) On Fig. 10.1, mark

1. the S-pole of the iron core

2. the N-pole and the S-pole of the iron armature.

[2]

(ii) The cell is reversed.

State what happens to the poles and to the armature.

.....

..... [2]

(iii) Explain why the core is made of iron and not steel.

.....

.....

..... [2]

(b) Fig. 10.2 shows the relay connected in a circuit to a 12V battery. The bell is not ringing.

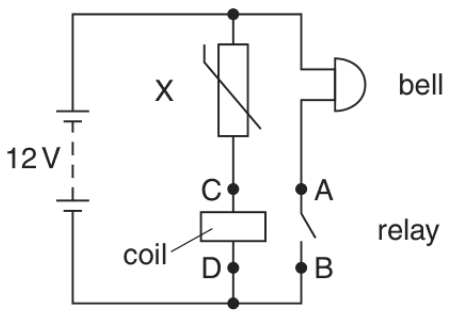


Fig. 10.2

(i) State the name of component X.

..... [1]

(ii) Explain why the bell rings when the temperature of X rises.

.....

.....

..... [2]

(iii) When the resistance of X is 2000 Ω, the current in the coil is 1.5 mA. This causes the contacts in the relay to close. The resistance of the bell is 200 Ω.

Calculate

1. the potential difference (p.d.) across X,

p.d. = ... [2]

2. the p.d. across the coil,

p.d. = ... [1]

3. the current in the battery.

current = ... [2]

(iv) Component X is removed from the circuit and replaced by a different component Y. The bell now rings when bright light shines on Y.

State the name of component Y.

..... [1]

MARK SCHEME · 5054_s15_ms_22.pdf · source page 5 of 6 · Question 10

10	**(a)**	**(i)**	**1**	S-pole on right of core	B1
			2	N-pole anywhere on vertical section of armature **and** S-pole anywhere on horizontal section of armature **or** N-pole on left of vertical section of armature **and** S-pole on right	B1
		(ii)		poles (on core) reverse / change positions	B1
				(armature still) attracted (to core)	B1
		(iii)		(iron is a) temporary magnet	B1
				or (iron) easily demagnetised **or** steel retains magnetism	
				when current off / no battery / switch off / circuit open **and** armature released / does not stay attracted / opens connections (at AB)	B1

MARK SCHEME · 5054_s15_ms_22.pdf · source page 6 of 6 · Question 10

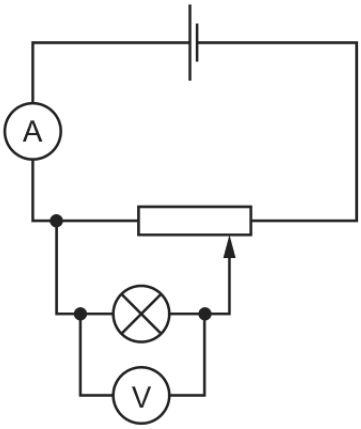
	(b)	**(i)**		thermistor	B1
		(ii)		resistance (of X) decreases	B1
				current (in coil) increases **or** more voltage across coil **and** **either** relay switch closes **or** circuit (to bell) complete	B1
		(iii)	**1**	(V=) IR **or** 1.5 ($\times 10^{-3}$) $\times$ 2000	C1
				3(.0) V	A1
			2	9(.0) V	B1
			3	12 / 200 **or** 0.06 (A) **or** 60 (mA) seen **or** (R_T =) 195(.12 Ω)	C1
				61(.5) mA **or** 0.061(5) A **or** 62 mA **or** 0.062 A	A1
		(iv)		light dependent resistor **or** LDR	B1

QUESTION · 5054_w15_qp_11.pdf · source page 14 of 16

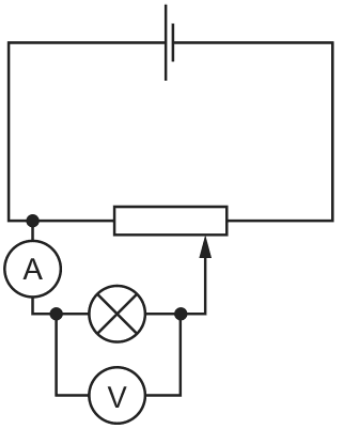
34 An experiment is set up to investigate how the current in a filament lamp changes with the potential difference across it.

Which circuit is correct?

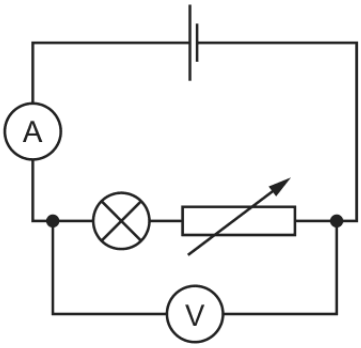
A



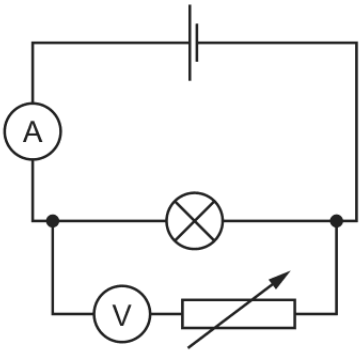
B



C



D



MARK SCHEME (official answer key) · 5054_w15_ms_11.pdf · source page 2 of 2 · row for Question 34

34	**B**

8 A potential divider is made from a light-dependent resistor (LDR) and a 6.0 kΩ fixed resistor. The potential divider is connected in series with a 12 V d.c. power supply and a voltmeter is connected across the 6.0 kΩ resistor. Fig. 8.1 is the circuit diagram.

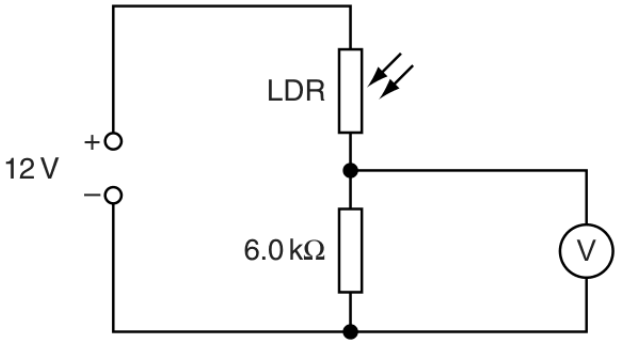


Fig. 8.1

A light shines on the LDR. The resistance of the LDR is 2.0 kΩ.

(a) Calculate

(i) the current in the LDR,

current = ... [2]

(ii) the reading on the voltmeter.

reading = ... [1]

(b) The brightness of the light on the LDR is gradually increased.

State and explain what happens to the reading on the voltmeter.

.....

.....

..... [2]

(c) Suggest a use for a potential divider made from an LDR and a fixed resistor.

.....

..... [1]

8 **(a)** **(i)** $(I =)V/R$ **or** 12/(6000 + 2000) **or** 12/8000 **or** 12/2000 **or** 12/6000 **or** in **(ii)** $(V =)IR$ **or** 0.0015×6000 C1

1.5 mA A1

(ii) 9.0 V B1

(b) (reading/it) increases B1

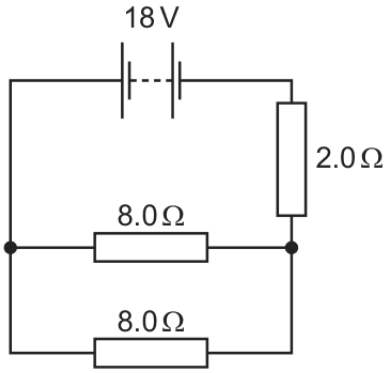
resistance of LDR falls B1

(c) light meter/sensor **or** automatic light switch **or** something sensible B1 [6]

[45]

QUESTION · 5054_w15_qp_12.pdf · source page 13 of 16

33 A power supply of 18 V is connected to three resistors, as shown.



What is the potential difference across the 2.0 Ω resistor?

A 2.0 V **B** 3.6 V **C** 6.0 V **D** 12 V

MARK SCHEME (official answer key) · 5054_w15_ms_12.pdf · source page 2 of 2 · row for Question 33

33	**C**

10 Thin wire, covered in plastic insulation, is used to make a solenoid (long coil). The solenoid is connected to a sensitive ammeter. Fig. 10.1 shows the N-pole of a steel magnet placed next to the solenoid.

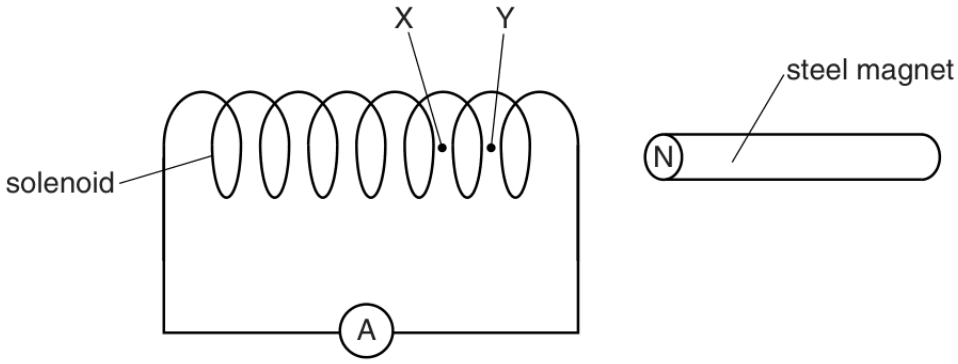


Fig. 10.1

Point X and point Y are on the axis of the solenoid.

(a) **(i)** Explain why plastic is an electrical insulator.

.....

.....

..... [1]

(ii) Explain why the magnet is **not** made from

1. aluminium,

..... [1]

2. iron.

..... [1]

(b) In one experiment, the magnet in Fig. 10.1 is moved to the left and passes into the solenoid. The N-pole of the magnet travels from Y to X at a constant speed. As it moves, the ammeter shows a small current.

(i) Explain why there is a current in the solenoid when the magnet is moving.

.....

.....

.....

.....

.....

..... [3]

(ii) The N-pole travels from Y to X in 0.14 s. As it moves, the current shown on the ammeter is 0.045 mA. The resistance of the solenoid is 1.2 Ω.

Calculate

1. the potential difference (p.d.) across the solenoid,

potential difference = ... [2]

2. the charge that passes through the solenoid as the N-pole moves from Y to X.

charge = ... [2]

(c) In a second experiment, the speed of the N-pole is greater than its speed in the first experiment. It now takes only 0.070 s to travel from Y to X.

A current in the same direction is shown on the ammeter.

(i) State and explain how the size of this current compares with the size of the current in the first experiment.

.....

.....

..... [2]

(ii) The same quantity of charge passes through the coil in both the first and second experiments.

Explain why this is the case.

.....

.....

..... [1]

(d) State two ways in which the equipment shown in Fig. 10.1 can be used to produce a current in the solenoid that is in the opposite direction.

1. .....

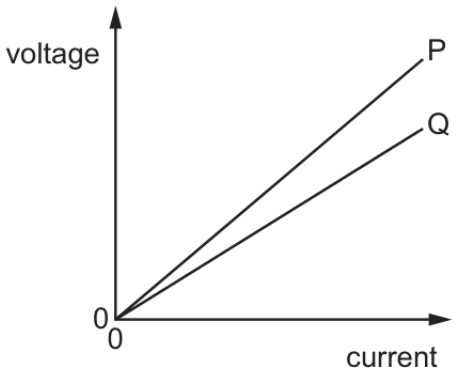
2. .....
[2]

10	**(a)**	**(i)**	no free electrons (in plastic) **or** all electrons are bound / structural	B1	
		(ii)	(aluminium) is not magnet(ic) **or** cannot be magnetised	B1	
			(iron) is a temporary / soft magnetic material **or** is not a permanent magnet	B1	[3]
	(b)	**(i)**	magnetic field / flux (mentioned)	B1	
			(magnetic) field lines cut wire / solenoid / circuit **or** changing magnetic field / flux	B1	
			voltage/e.m.f. **induced**	B1	
		(ii)	(V =)IR **or** 0.045×1.2 **or** 0.000045×1.2	C1	
			5.4×10^{-5} V **or** 0.054 mV	A1	
			(Q =)It **or** 0.045×0.14 **or** 0.000045×0.14	C1	
			6.3×10^{-6} C **or** 0.0063 mC	A1	[7]
	(c)	**(i)**	larger **or** twice the current	B1	
			(magnetic) field lines cut faster **or** (magnetic) field changes faster **or** twice the current	B1	

		(ii)	double the current for half the time **or** larger current for less time **or** product $I \times t$ is the same	B1	[3]
	(d)	any **two** from: insert S-pole (at same end) insert (N-pole) at other end **or** from other direction withdraw N-pole (from same end implied) withdraw S-pole from other end **or** pass through completely		B2	[2]
					[15]

QUESTION · 5054_s16_qp_11.pdf · source page 12 of 16

32 The graph is the voltage-current graph for two resistance wires P and Q.



The wires are made from the same material and have equal lengths.

The resistances of the wires and their cross-sectional areas are different.

Which wire has the greater resistance and which wire has the larger cross-sectional area?

	greater resistance	larger cross-sectional area
A	P	P
B	P	Q
C	Q	P
D	Q	Q

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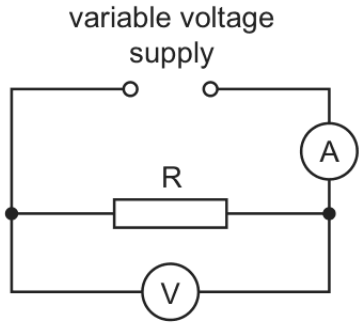
32 **B**

QUESTION · 5054_s16_qp_11.pdf · source page 14 of 16

34 A student is investigating resistance using the circuit shown.



The resistance of R is approximately 5 Ω.

What are the most suitable ranges for the voltmeter and for the ammeter?

	voltmeter range	ammeter range
A	0–2 V	0–0.5 A
B	0–2 V	0–2 A
C	0–5 V	0–5 A
D	0–10 V	0–5 A

MARK SCHEME (official answer key) · 5054_s16_ms_11.pdf · source page 2 of 2 · row for Question 34

34 **A**

8 Fig. 8.1 shows a circuit that contains a battery, a fixed resistor of resistance 1000 Ω and a variable resistor R.

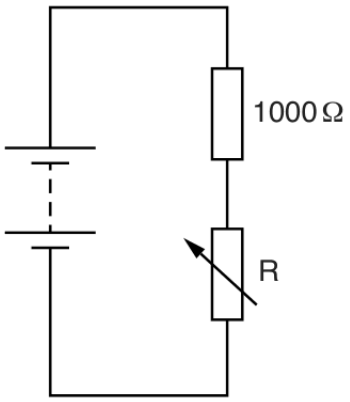


Fig. 8.1

(a) (i) State the relationship between the current I_B in the battery, the current I_1 in the fixed resistor and the current I_2 in the variable resistor.

.....

.....[1]

(ii) State an equation that relates the potential difference (p.d.) V_B across the battery, the p.d. V_1 across the fixed resistor and the p.d. V_2 across the variable resistor.

.....

.....[1]

(b) The p.d. across the battery is 9.0 V. The resistance of R is set at 500 Ω.

Calculate the current in R.

current = ...[2]

8	**(a)**	**(i)**	same / equal **or** $I_B = I_1 = I_2$	B1
		(ii)	(p.d. of) battery is sum of (p.d. across) fixed resistor and (p.d. across) the variable resistor **or** $V_B = V_1 + V_2$	B1
	(b)		(I=) V / R numerical or algebraic in any form	C1
			0.006(0) A	A1

8 A transformer and a diode are used to charge a battery.

Fig. 8.1 shows the transformer, which contains a soft-iron core and two coils.

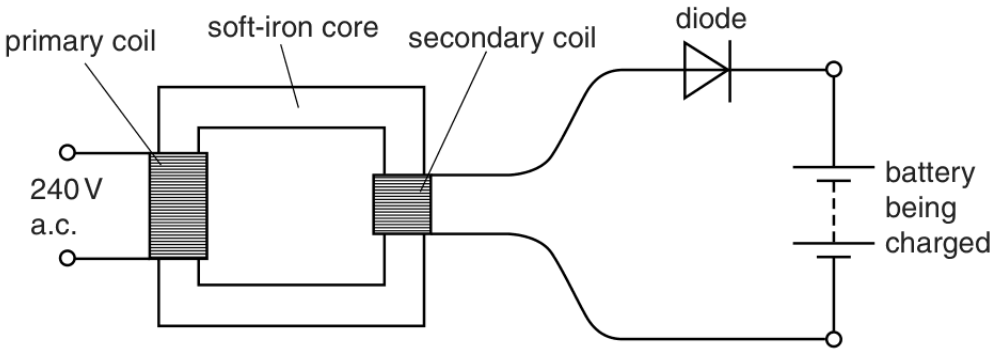


Fig. 8.1

The primary coil is connected to the 240V a.c. mains supply. The secondary coil is connected in series with the diode and the battery.

(a) Explain why an electromotive force (e.m.f.) is induced in the secondary coil.

.....

.....

.....

.....

.....[2]

(b) The e.m.f. induced in the secondary coil is less than 240V.

Suggest why.

.....

.....[1]

(c) Suggest why steel is not used as the core of a transformer.

.....

.....[1]

(d) Describe the action of the diode.

.....

.....[1]

8	**(a)**	current / a.c (in primary coil) creates magnetic field **or** current / a.c magnetises iron	B1
		changing magnetic field (in secondary)	B1
	(b)	it / secondary has less turns (than primary) **or** primary has more turns (than secondary) **or** (some) flux escapes	B1
	(c)	(steel is) a permanent magnet **or** weaker fields produced **or** (steel) difficult to magnetise / demagnetise **or** (steel) is a hard magnetic material	B1
	(d)	passes current / charge in one direction **or** has high resistance / is an insulator when current in one direction / reverse biased	B1

Section B

Answer **two** questions from this section. Answer in the spaces provided.

9 Fig. 9.1 is a circuit diagram.

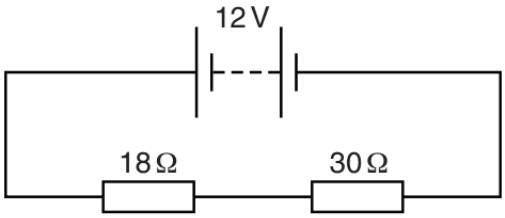


Fig. 9.1

The electromotive force (e.m.f.) of the battery is 12V.

(a) Explain what is meant by *electromotive force*.

.....

.....[2]

(b) An 18 Ω resistor and a 30 Ω resistor are connected in series with the battery.

(i) **1.** State the relationship between the current I_B in the battery, the current I_1 in the 18 Ω resistor and the current I_2 in the 30 Ω resistor.

.....[1]

2. State an equation that relates the e.m.f. E of the battery, the potential difference V_1 across the 18 Ω resistor and the potential difference V_2 across the 30 Ω resistor.

.....[1]

(ii) Calculate the current in the battery.

current = ...[2]

(iii) Calculate the potential difference (p.d.) across the 18 Ω resistor.

p.d. = ...[1]

(iv) Calculate the power produced in the 18 Ω resistor.

power = ...[2]

(c) The resistors obey Ohm's law.

State Ohm's law and describe one limitation on this law.

.....

.....

.....[2]

(d) The resistors are made from the same material, but have different cross-sectional areas and different lengths.

State the relationship between

(i) the resistance R and the length l of a wire of constant cross-sectional area,

.....

(ii) the resistance R and the cross-sectional area A of a wire of constant length.

.....
[2]

(e) A resistor is shown in Fig. 9.2 and part of the resistor colour code is shown in Fig. 9.3.

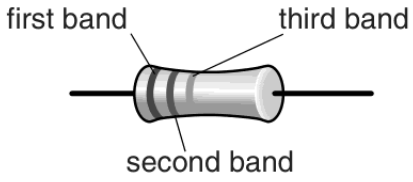


Fig. 9.2

digit or multiplier	colour
0	black
1	brown
2	red
3	orange

Fig. 9.3

State the colour of the bands on the 30 Ω resistor.

first band = ...

second band = ...

third band = ...
[2]

9	**(a)**		(amount of) energy / work (dissipated by source) by unit charge (around a circuit)	M1 A1
	(b)	**(i)**	**1** they are the same **or** $I_B = I_1 = I_2$	B1
			2 $E = V_1 + V_2$	B1
		(ii)	(I=) V / R in any form algebraic or numerical 0.25 A	C1 A1
		(iii)	4.5 V	B1
		(iv)	(P=) VI **or** (P=) I^2R **or** (P=) V^2 / R in any form algebraic or numerical 1.1(25) W	C1 A1

	(c)		current is (directly) proportional to voltage **or** voltage / current is a constant law holds for constant physical conditions / constant temperature / constant pressure / for metals	B1 B1
	(d)	**(i)**	(directly) proportional **or** (R) ∝ 1	B1
		(ii)	inversely proportional **or** (R) ∝ 1 / A	B1
	(e)		1st band orange 2nd and 3rd bands both black	B1 B1

9 A vertical solenoid (long coil) with an iron core is held in a wooden clamp above a laboratory bench.

The solenoid is connected in series with a battery, a switch S, an ammeter and a variable resistor. There is a voltmeter in parallel with the solenoid.

Fig. 9.1 represents this apparatus.

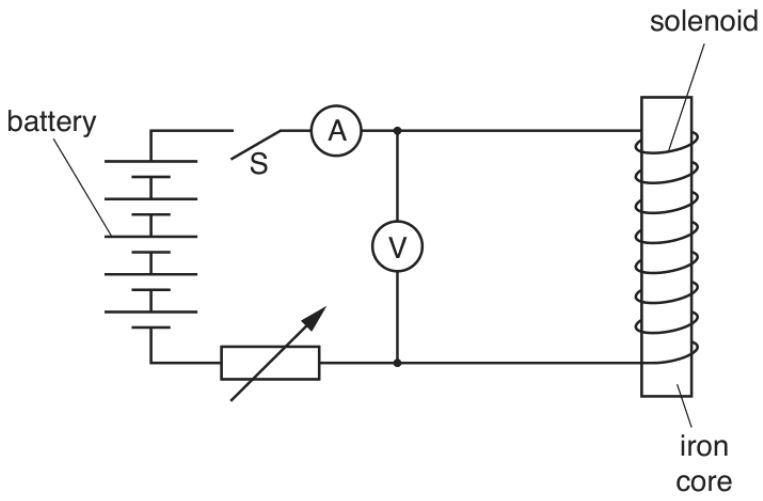


Fig. 9.1

A student closes switch S and a current in the circuit produces a reading on the ammeter.

(a) State what is meant by *current*.

.....

.....[1]

(b) The battery consists of five 1.5 V cells in series. The reading on the ammeter is 4.0 A.

(i) State the size of the electromotive force (e.m.f.) of the battery.

.....[1]

(ii) Calculate the total resistance of the series circuit.

resistance = ...[2]

(iii) The reading on the voltmeter is 6.5 V.

Calculate the power dissipated in the solenoid.

power = ...[2]

(iv) The solenoid is made of copper and the student notices that, as time passes, the solenoid becomes extremely warm.

State and explain the effect of this temperature increase on the ammeter reading.

.....

.....

.....[2]

(c) (i) The current in the solenoid magnetises the iron core so that the lower end of the core is an N-pole.

On Fig. 9.1, draw the pattern and indicate the direction of the magnetic field in the region surrounding the iron core. [3]

(ii) The student holds an iron cylinder against the bottom surface of the iron core in the solenoid. When he releases the iron cylinder, it stays in contact with the iron core.

1. Explain why the iron cylinder does not fall.

.....

.....

.....

.....

.....[3]

2. The switch S is opened.

State and explain whether the iron cylinder remains in contact with the iron core.

.....

.....

.....[1]

9	**(a)**	rate of flow of charge **or** charge flowing per unit time	B1	[1]
	(b) **(i)**	7.5 V	B1	
	(ii)	(R =)V / I **or** 7.5 / 4.0	C1	
		1.9 Ω	A1	
	(iii)	(P =) VI **or** 6.5 × 4.0	C1	
		26 W	A1	
	(iv)	resistance increases	M1	
		(reading of ammeter) decreases	A1	[7]
	(c) **(i)**	at least two lines on left **and** two lines on right of core **and** correct shape (by eye)	B1	
		good shape (by eye) **and** into poles **and** no straight sections **and** at least one line on each side	B1	
		at least one arrow N to S (primarily upwards) **and** none wrong	B1	[3]

	(ii) **1**	cylinder is magnetised (by induction)	B1	
		top (of cylinder) is an S-pole	B1	
		unlike poles attract **or** S-pole attracts N-pole	B1	
	2	it does not (remain in contact**) and** iron is temporary / soft magnetic material / core (and cylinder) lose magnetisation	B1	[4]
				[15]

QUESTION · 5054_s17_qp_11.pdf · source page 14 of 16

38 A cylinder of conducting material has resistance R.

A second cylinder of the same material is twice as long but has half the cross-sectional area.

What is the resistance of the second cylinder?

A $\frac{R}{2}$ **B** R **C** $2R$ **D** $4R$

MARK SCHEME (official answer key) · 5054_s17_ms_11.pdf · source page 3 of 3 · row for Question 38

38	D	**1**

11 A student connects a thermistor in series with a battery and an ammeter. He uses the resistance of the thermistor to determine temperature.

(a) In the space below, draw a circuit diagram showing the thermistor, the battery and the ammeter.

[3]

(b) The student places the thermistor in a water bath at various temperatures and records the reading on the ammeter.

Fig. 11.1 shows how the current varies with temperature.

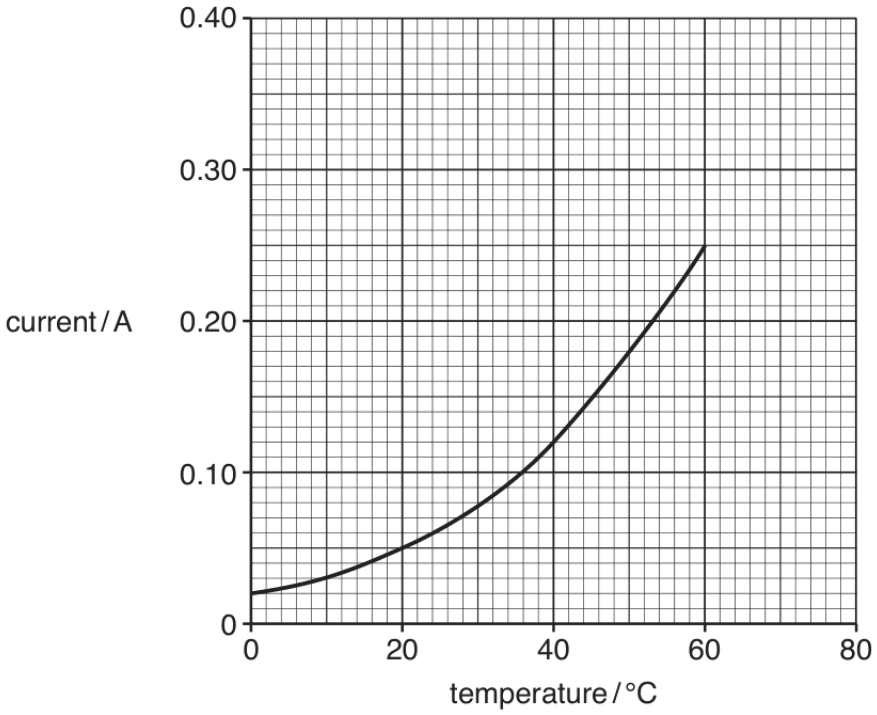


Fig. 11.1

The potential difference (p.d.) across the thermistor **remains constant**.

The resistance of the thermistor is 240 Ω at a temperature of 20 °C.

(i) Using Fig. 11.1, calculate

1. the potential difference (p.d.) across the thermistor,

p.d. = ...[3]

2. the resistance of the thermistor at a temperature of 40 °C.

resistance = ...[2]

(ii) The resistance of the thermistor is 240 Ω at a temperature of 20 °C and 48 Ω at a temperature of 60 °C.

The resistance of the thermistor does not change linearly with temperature.

Describe how the resistances at 20 °C, 40 °C and 60 °C show this is true.

.....

.....

.....

.....

.....[2]

(c) A thermocouple thermometer is another type of thermometer.

(i) Describe the structure of a thermocouple thermometer. You may draw a diagram if you wish.

.....

.....

.....

.....

.....[3]

(ii) State two advantages of a thermocouple thermometer over a liquid-in-glass thermometer.

1. .....

.....

2. .....

.....
[2]

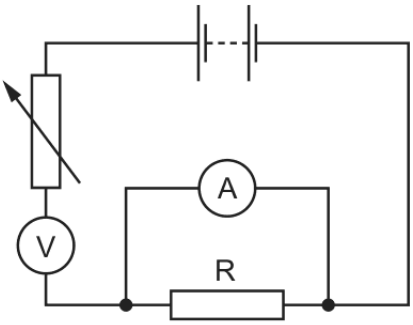
Question	Answer	Marks
11(a)	battery and ammeter symbols correct	**B1**
	thermistor symbol correct	**B1**
	all connected in series	**B1**
11(b)(i)1	0.05 (A) seen	**C1**
	(V =) IR **or** 0.05 × 240	**C1**
	12 V	**A1**
11(b)(i)2	(R =) V / I **or** 0.12 (A) seen	**C1**
	100 Ω	**A1**
11(b)(ii)	description of change in resistance in two equal changes in temperature, e.g. 240–100 and 100–48	**C1**
	difference values obtained are not the same, with one value shown e.g. 140 and 52 (Ω)	**A1**
11(c)(i)	two different metals connected together	**B1**
	voltmeter, ammeter or galvanometer in series	**B1**
	clear junction of two different metals used as the place to measure temperature	**B1**
11(c)(ii)	ANY 2 lines from measures high(er) temperatures more sensitive to changes in temperature measures fast(er) changing temperatures / reading can be taken more rapidly produce electronic output / output may be stored or processed / may be read remotely	**B2**

QUESTION · 5054_s17_qp_12.pdf · source page 16 of 20

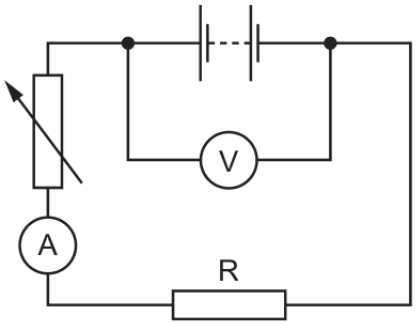
34 The readings of an ammeter and a voltmeter are used to calculate the resistance of resistor R.

Which circuit shows the ammeter and voltmeter in the correct positions?

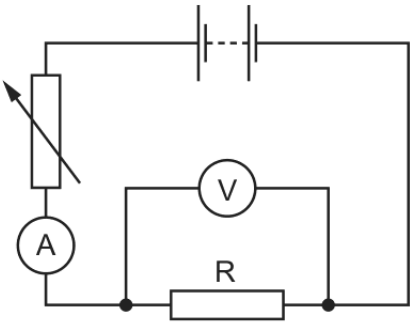
A



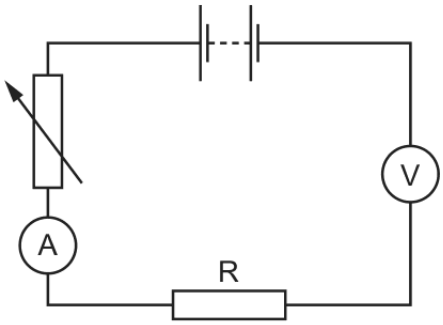
B



C



D



MARK SCHEME (official answer key) · 5054_s17_ms_12.pdf · source page 3 of 3 · row for Question 34

34	C	**1**

10 (a) Measurements are made of the current in a metal wire and the potential difference (p.d.) across it. Fig. 10.1 shows a graph of the current against the p.d. for the wire.

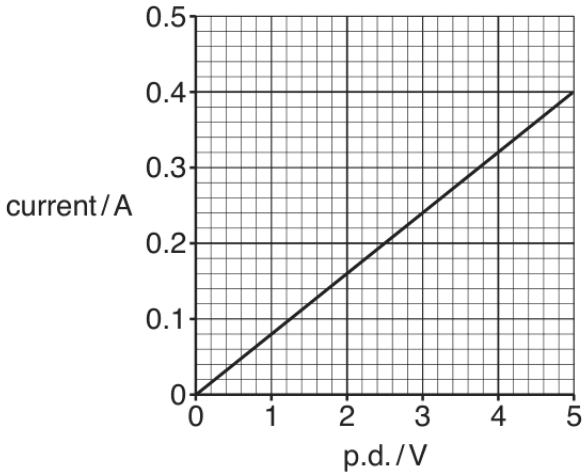


Fig. 10.1

(i) State the relationship between the current and the p.d. for the metal wire.

.....

.....[1]

(ii) State how the graph shows that the temperature of the wire does not change in the experiment.

.....

.....[1]

(iii) Another wire of the same material has the same length as the original wire but has only half the cross-sectional area.

1. Compare the resistance of the new wire and the original wire.

.....

.....[1]

2. On Fig. 10.1, draw the graph for the new wire. [1]

(b) Three resistors are connected to a d.c. power supply, as shown in Fig. 10.2. The electromotive force (e.m.f.) of the power supply is 6.0 V.

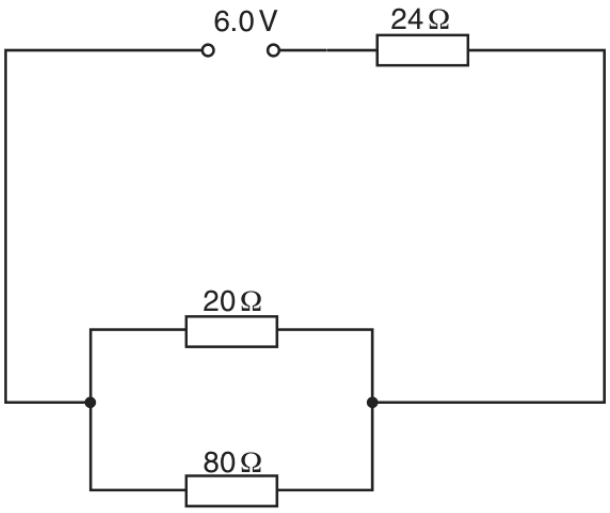


Fig. 10.2

(i) Calculate

1. the total resistance of the circuit,

resistance = ...[2]

2. the current supplied by the power supply,

current = ...[2]

3. the p.d. across the 20 Ω resistor.

p.d. = ...[2]

(ii) The d.c. power supply is a battery that contains a number of cells. Each cell has an electromotive force (e.m.f.) of 1.5 V.

1. State what is meant by the *electromotive force (e.m.f.)* of a cell.

.....

.....

.....[1]

2. In the space below, draw a diagram to show how four of these cells are connected to make a battery of e.m.f. 6.0 V. Mark the positive terminal of the battery.

[2]

3. In another arrangement to produce a battery of e.m.f. 6.0 V, eight cells of e.m.f. 1.5 V are connected. In this arrangement, some cells are connected in series and some in parallel.

In the space below, draw a diagram to show how the eight cells are connected to make the battery.

[1]

(c) State one advantage of using a battery made from cells in parallel rather than a single cell.

.....

.....[1]

Question	Answer	Marks
10(a)(i)	directly proportional	**B1**
10(a)(ii)	straight line **or** does not curve **or** constant gradient	**B1**
10(a)(iii)1	greater **or** twice as large	**B1**
10(a)(iii)2	straight line with half the gradient	**B1**
10(b)(i)1	$1 / R_t = 1 / R_1 + 1 / R_2$ in any form e.g. $1 / R = 1 / 20 + 1 / 80$ **or** 16 (Ω) seen	**C1**
	40 Ω	**A1**
10(b)(i)2	(I=)V / R in any form e.g. 6 / 40	**C1**
	0.15 A	**A1**
10(b)(i)3	0.15×16 **or** 0.15×24 **or** 3.6 (V) seen **or** current split in ratio 1:4, e.g. 0.03 A and 0.12 A seen **or** clear attempt at potential divider formula	**C1**
	2.4 V	**A1**
10(b)(ii)1	work done ÷ charge	**B1**
10(b)(ii)2	correct circuit symbol for a cell and positive correct	**B1**
	four cells, correct symbol, correctly in series	**B1**
10(b)(ii)3	four cells in series and another four in parallel **or** any other series and parallel arrangement of 8 cells with connections to and from battery	**B1**
10(c)	lasts longer **or** if one cell fails it still works **or** contains more energy	**B1**

11 **(a)** Fig. 11.1 shows a negatively charged rod close to an uncharged metal sphere that is mounted on an insulating stand.

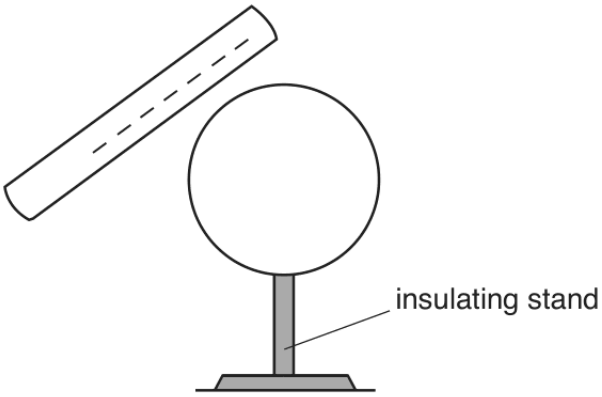


Fig. 11.1

(i) Suggest a material for the insulating stand.

..... [1]

(ii) On Fig. 11.1, draw the distribution of charges on the sphere. [2]

(iii) A metal wire connected to earth is touched against the sphere.

1. Explain what happens to the charge on the sphere.

.....

.....

.....

.....

.....

..... [3]

2. The earth wire remains in contact with the sphere.

Describe what happens in the wire as the rod is moved away.

.....

.....

..... [1]

(b) A circuit contains a 7.5 V d.c. power supply and three resistors.

Fig. 11.2 is the circuit diagram.

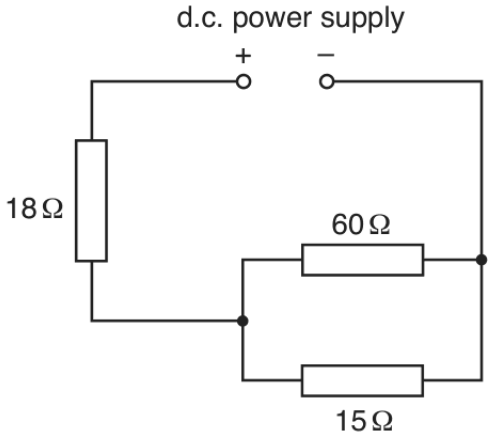


Fig. 11.2

(i) Determine the total resistance of the circuit.

resistance = ... [3]

(ii) Calculate the current in the power supply.

current = ... [2]

(iii) There is a different current in each of the three resistors in the circuit in Fig. 11.2.

Underline the resistor in which the current is

1. the largest,

15 Ω resistor **18 Ω resistor** **60 Ω resistor**

2. the smallest.

15 Ω resistor **18 Ω resistor** **60 Ω resistor**

[1]

(iv) The 7.5 V d.c. power supply is made from five 1.5 V cells.

In the space below, draw a diagram that shows how the cells are arranged.

[1]

(v) The 18 Ω resistor is made of metal.

State and explain what happens to the current in this resistor as its temperature increases.

.....

.....

..... [1]

Question	Answer	Marks
11(a)(i)	any suitable solid insulator (e.g. nylon, plastic, glass, rubber, polystyrene)	**B1**
11(a)(ii)	positive charges near to rod	**B1**
	negative charges opposite rod **and** equal in number **and** 7 or fewer	**B1**
11(a)(iii)	**1** electrons / negative charges flow towards earth	**B1**
	repelled (by negative charge on rod)	**B1**
	(sphere) becomes positive	**B1**
	2 flow of electrons / negative charge **and** (in direction) earth to sphere	**B1**
11(b)(i)	$1 / R = 1 / R_1 + 1 / R_2$ **or** $R_1 R_2 / (R_1 + R_2)$ **or** 1 / R = 1 / 15 + 1 / 60 **or** $15 \times 60 / 75$ **or** $15 \times 60 / (15 + 60)$	**C1**
	12 (Ω) **or** 0.083 (Ω)	**C1**
	30 Ω	**A1**
11(b)(ii)	(I =)V / R **or** 7.5 / 30	**C1**
	0.25 A	**A1**

Question	Answer	Marks
11(b)(iii)	**18 Ω resistor** underlined **2** **and 60 Ω resistor** underlined	**B1**
11(b)(iv)	five cells in series **and** all in same direction	**B1**
11(b)(v)	resistance increases **and** current decreases	**B1**

QUESTION · 5054_w17_qp_12.pdf · source page 12 of 16

30 A piece of wire has a resistance of 16 Ω.

The wire is 20 cm long and has a cross-sectional area of 2.0 mm^2.

Which wire of the same material has a resistance of 8.0 Ω?

	length / cm	cross-sectional area / mm^2
A	10	1.0
B	10	4.0
C	20	1.0
D	20	4.0

MARK SCHEME (official answer key) · 5054_w17_ms_12.pdf · source page 3 of 3 · row for Question 30

30	D	**1**

7 A 4.5 V battery is connected in a circuit with an ammeter, a light-dependent resistor (LDR) and a 1800 Ω fixed resistor. Fig. 7.1 is the circuit diagram.

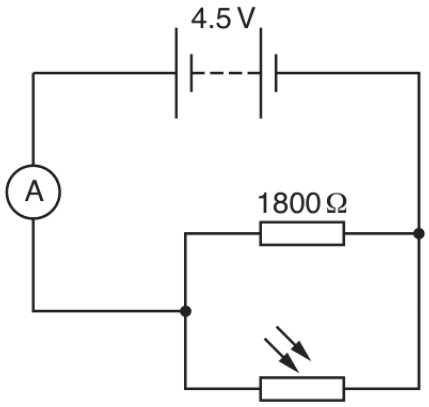


Fig. 7.1

(a) The resistance of the LDR is 9000 Ω.

Calculate

(i) the total resistance of the circuit,

resistance = ...[2]

(ii) the reading on the ammeter.

reading = ...[2]

(b) A very bright lamp is switched on and the light intensity on the LDR increases.

State and explain what happens to

(i) the current in the LDR,

.....

.....[1]

(ii) the current in the 1800 Ω resistor.

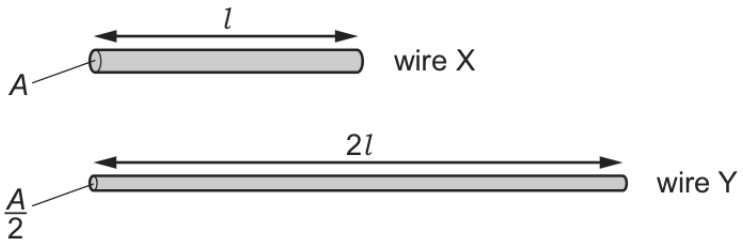
.....

.....[1]

Question	Answer	Marks
7(a)(i)	$1/R = 1/R_1 + 1/R_2$ **or** $R_1R_2/(R_1 + R_2)$ **or** $1/R = 1/1800 + 1/9000$ **or** $9000 \times 1800/10\,800$ **or** 0.00066667	**C1**
	1500 Ω	**A1**
7(a)(ii)	(I =) V/R **or** $4.5/1500$	**C1**
	0.0030 A **or** 3.0×10^{-3} A **or** 3.0 mA	**A1**
7(b)(i)	increases **and** resistance of LDR decreases	**B1**
7(b)(ii)	does not change **and** resistance / e.m.f. does not change / not affected by LDR	**B1**

QUESTION · 5054_s18_qp_11.pdf · source page 13 of 20

29 A copper wire X has resistance *R*. Another copper wire Y has twice the length and half the cross-sectional area of X.



What is the resistance of Y?

A $\frac{R}{2}$ **B** R **C** $2R$ **D** $4R$

MARK SCHEME (official answer key) · 5054_s18_ms_11.pdf · source page 3 of 3 · row for Question 29

29	D	**1**

5 A student has three resistors; each resistor is labelled 3.9 Ω, 3 W.

She checks the value of the resistance of **one** of the resistors.

(a) In the space below, draw a circuit diagram of a circuit that she can use to find this resistance. Include a battery, a variable resistor, a voltmeter and an ammeter in the circuit.

[2]

(b) When the circuit is turned on, the potential difference (p.d.) across the resistor is 4.2 V and the current in the resistor is 1.2 A.

(i) Calculate the resistance of the resistor.

resistance = ...[2]

(ii) Shortly after the circuit is turned on, the resistor overheats and gives off smoke.

Explain why it overheats. Include a calculation in your answer.

.....

.....

.....

.....

.....[2]

(iii) The other two resistors are also labelled 3.9 Ω, 3 W. The first resistor is now removed and the student connects these other two resistors in series to the p.d. of 4.2 V.

State and explain whether these resistors overheat.

.....

.....

.....

.....[1]

Question	Answer	Marks
5(a)	correct circuit symbols for battery, resistor, variable resistor, ammeter and voltmeter	**B1**
	ammeter in series with resistor **and** voltmeter in parallel in correct series circuit	**B1**
5(b)(i)	(R=) V / I in any form numerical or algebraic	**C1**
	3.5 Ω	**A1**
5(b)(ii)	voltage **or** current **or** power too large	**B1**
	calculation of power as 5(.04) W or 5.6 W **or** max voltage as 3.4(2) V or 3.2(4) V **or** max current as 0.88 A or 0.93 A	**B1**
5(b)(iii)	no **and** half voltage (2.1 V across each) **or** half current (0.54 A through each) **or** quarter power (1.13 W in each)	**B1**

10 A student connects a battery to two resistors. The circuit diagram is shown in Fig. 10.1.

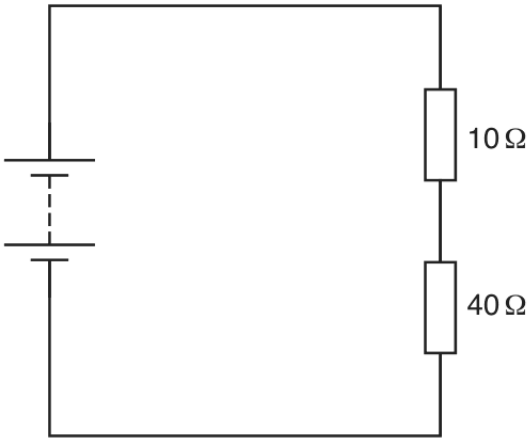


Fig. 10.1

The potential difference (p.d.) across the 40 Ω resistor is 9.6 V.

(a) State what is meant by the *potential difference* across a resistor.

.....

.....[2]

(b) **(i)** Calculate the current in the 40 Ω resistor.

current = ...[2]

(ii) Calculate the electromotive force (e.m.f.) of the battery.

e.m.f. = ...[2]

(c) The student has three different voltmeters to measure the p.d. across the 40 Ω resistor. These are labelled 0–2 V, 0–20 V and 0–200 V. Each has a pointer that shows the p.d. on a scale.

State and explain which of the three voltmeters is best to measure this p.d.

.....

.....

.....

.....[2]

(d) (i) Calculate the power P produced in the 10 Ω resistor.

$P =$ .....[2]

(ii) The student has available two 10 Ω resistors, with power ratings of $\frac{1}{2}P$ and $2P$.

Suggest why a resistor with a power rating of $\frac{1}{2}P$ is not suitable for the circuit in Fig. 10.1.

.....

.....[1]

(e) The student adds a resistor R to the circuit, to make the circuit shown in Fig. 10.2.

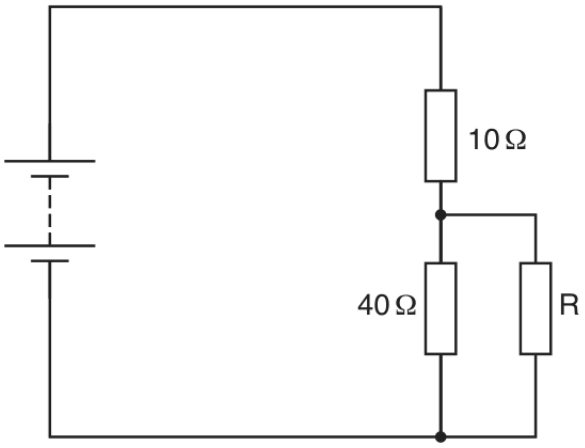


Fig. 10.2

Complete the table in Fig. 10.3 to show what happens as resistor R is connected.

You should state whether each quantity increases, decreases or stays the same and give a brief explanation of why any change occurs. Calculations are not required.

quantity	increases, decreases or stays the same as resistor R is added	brief explanation of why the change occurs
current in 10 Ω resistor	increases	
p.d. across 10 Ω resistor		
p.d. across 40 Ω resistor		

[4]

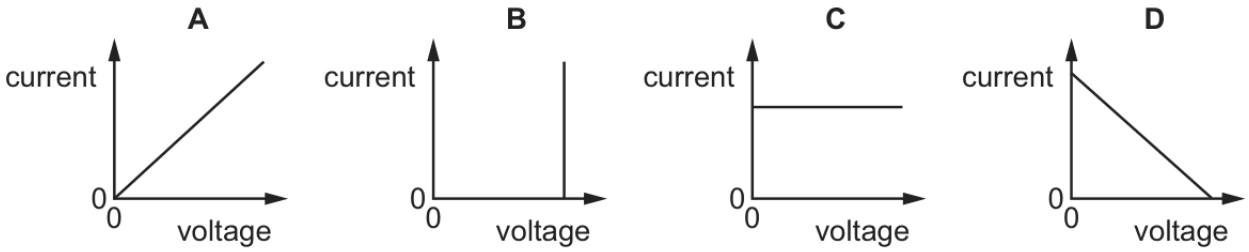
Fig. 10.3

Question	Answer	Marks
10(a)	work done / energy transferred	**B1**
	work done / energy transferred in taking unit charge (through resistor)	**B1**
10(b)(i)	(I=) V / R in any form algebraic or numerical	**C1**
	0.24 A	**A1**
10(b)(ii)	(total resistance of) 50 (Ω) seen or 2.4 (V) p.d. across 10 Ω resistor	**C1**
	12 V	**A1**
10(c)	voltmeter chosen is 0–20 V	**B1**
	correct explanation of why 0–200 V meter is unsuitable **OR** correct explanation of why 0–2 V meter is unsuitable	**B1**
10(d)(i)	(P)= VI or I^2R or V^2 / R in any form	**C1**
	0.58 W	**A1**
10(d)(ii)	½P rated resistor blows / too hot / melts	**B1**
10(e)	current in 10 Ω | increases | <u>resistance</u> decreases (of parallel combination or total resistance)	**B1**
	p.d. across 10 Ω | increases | p.d.is proportional to current **or (**p.d.) increases as current increases **or** potential divider argument	**M1+A1**
	p.d. across 40 Ω | decreases | sum of p.d.s constant / 12 V / value in **(b)(ii)** **or** p.d. across 10 Ω increases **or** resistance (40 Ω and R) a smaller fraction of total	**B1**

QUESTION · 5054_w18_qp_11.pdf · source page 10 of 16

27 The voltage across a resistor of fixed resistance changes.

Which graph shows how the current in the resistor changes with the voltage?



MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 27

27	A	**1**

QUESTION · 5054_w18_qp_11.pdf · source page 10 of 16

28 A wire of length 0.50 m and cross-sectional area $1.0 \times 10^{-6}\,m^2$ has a resistance of 0.75 Ω.

Another wire of the same material has a length of 2.0 m and a cross-sectional area of $0.50 \times 10^{-6}\,m^2$.

What is the resistance of the longer wire?

A 0.094 Ω **B** 0.38 Ω **C** 1.5 Ω **D** 6.0 Ω

MARK SCHEME (official answer key) · 5054_w18_ms_11.pdf · source page 2 of 3 · row for Question 28

28	D	**1**

7 A student sets up the circuit shown in Fig. 7.1.

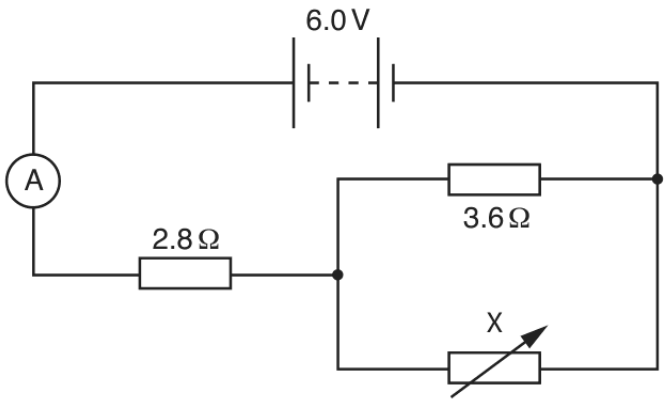


Fig. 7.1

The electromotive force (e.m.f.) of the battery is 6.0 V.

(a) The resistance of the variable resistor X is set to 1.8 Ω.

Determine

(i) the total resistance of the circuit,

resistance = ... [3]

(ii) the current measured by the ammeter.

current = ... [2]

(b) **(i)** State what is meant by the potential difference (p.d.) across a component in a circuit.

.....

..... [1]

(ii) Add to Fig. 7.1 the symbol for a component that is measuring the p.d. across the 2.8 Ω resistor. [1]

(iii) The resistance of X is increased.

Explain why the p.d. across the 2.8 Ω resistor decreases.

.....

.....

..... [2]

MARK SCHEME · 5054_w18_ms_21.pdf · source page 6 of 10 · Question 7

Question	Answer	Marks
7(a)(i)	$1 / R_T = 1 / R_1 + 1 / R_2$ **or** $1 / R_T = 1 / 3.6 + 1 / 1.8$ **or** (R_T =) $R_1 R_2 / (R_1 + R_2)$ **or** $1.8 \times 3.6 / (1.8 + 3.6)$	**C1**
	1.2 (Ω)	**C1**
	4.0 Ω	**A1**
7(a)(ii)	(I =) V / R **or** 6.0 / 4.0	**C1**
	1.5 (A)	**A1**
7(b)(i)	work done / energy (released) per unit charge (passed through component)	**B1**
7(b)(ii)	voltmeter symbol **and** across 2.8 Ω resistor	**B1**

MARK SCHEME · 5054_w18_ms_21.pdf · source page 7 of 10 · Question 7

Question	Answer	Marks
7(b)(iii)	total resistance increases **or** resistance of circuit / parallel combination increases	**B1**
	current (in 2.8 Ω resistor) decreases	**B1**

QUESTION · 5054_w18_qp_12.pdf · source page 12 of 16

29 Students are asked to describe an experiment to measure the resistance of a metallic conductor.

Which description is correct?

A Connect an ammeter in parallel and a voltmeter in series with the conductor then use $R = \frac{I}{V}$.

B Connect an ammeter in parallel and a voltmeter in series with the conductor then use $R = \frac{V}{I}$.

C Connect a voltmeter in parallel and an ammeter in series with the conductor then use $R = \frac{I}{V}$.

D Connect a voltmeter in parallel and an ammeter in series with the conductor then use $R = \frac{V}{I}$.

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 3 of 3 · row for Question 29

29	D	**1**

QUESTION · 5054_w18_qp_12.pdf · source page 12 of 16

31 A wire of length 0.50 m and cross-sectional area $1.0 \times 10^{-6}\,m^2$ has a resistance of 0.75 Ω.

Another wire of the same material has a length of 2.0 m and a cross-sectional area of $0.50 \times 10^{-6}\,m^2$.

What is the resistance of the longer wire?

A 0.094 Ω **B** 0.38 Ω **C** 1.5 Ω **D** 6.0 Ω

MARK SCHEME (official answer key) · 5054_w18_ms_12.pdf · source page 3 of 3 · row for Question 31

31	D	**1**

10 A student sets up the circuit shown in Fig. 10.1 in a laboratory at room temperature.

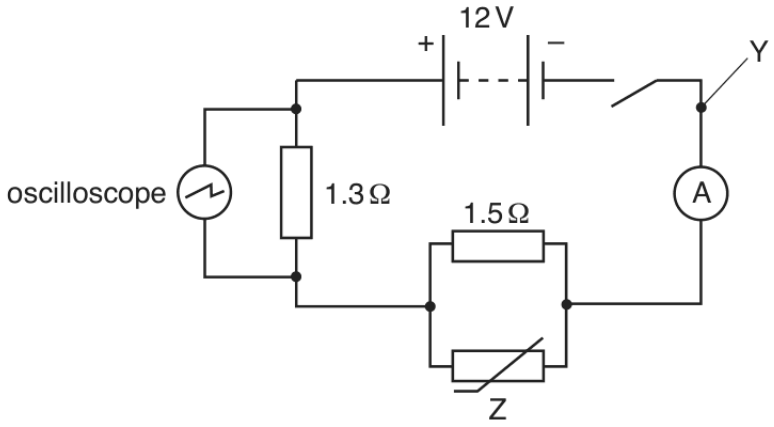


Fig. 10.1

The electromotive force (e.m.f.) of the battery is 12 V.

The switch is closed.

(a) The connecting wires in the circuit are made from copper covered by insulating plastic.

State the name of the particles that flow in the copper and state the direction in which they are flowing at point Y in the circuit.

.....

.....

..... [2]

(b) At room temperature, the resistance of component Z is 6.0 Ω.

(i) State the name of component Z.

..... [1]

(ii) Calculate the resistance of the whole circuit.

resistance = ... [3]

(iii) Calculate the current measured by the ammeter.

current = ... [2]

(iv) The current in the ammeter is I_A, the current in the 1.5 Ω resistor is I_R and the current in component Z is I_Z.

Write down an equation that relates I_A, I_R and I_Z.

..... [1]

(c) Fig. 10.2 shows the screen of the oscilloscope.

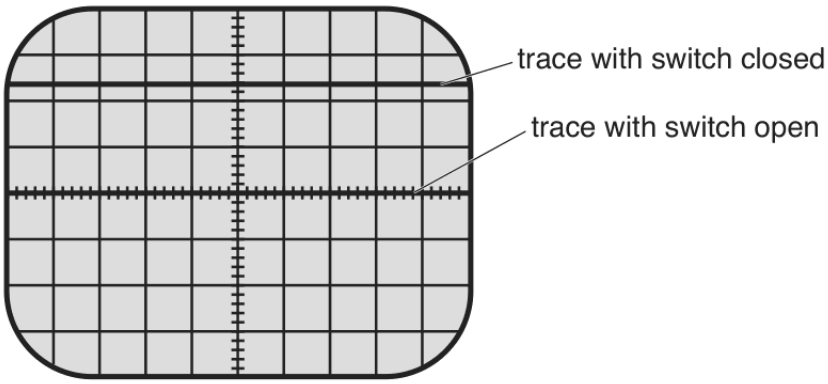


Fig. 10.2

Before the switch is closed, the trace is a horizontal line across the middle of the screen, as shown in Fig. 10.2. When the switch is closed, the trace remains horizontal and moves up the screen.

Component Z is heated.

State and explain what is observed on the oscilloscope screen as the temperature of Z increases.

.....

.....

.....

.....

.....

..... [4]

(d) The 12 V battery is made from cells of electromotive force (e.m.f.) 1.5 V which are all in series.

(i) Calculate the number of 1.5 V cells that make up the battery.

number = ... [1]

(ii) State the e.m.f. of a battery made by connecting all of these cells in parallel.

e.m.f. = ... [1]

MARK SCHEME · 5054_w18_ms_22.pdf · source page 9 of 10 · Question 10

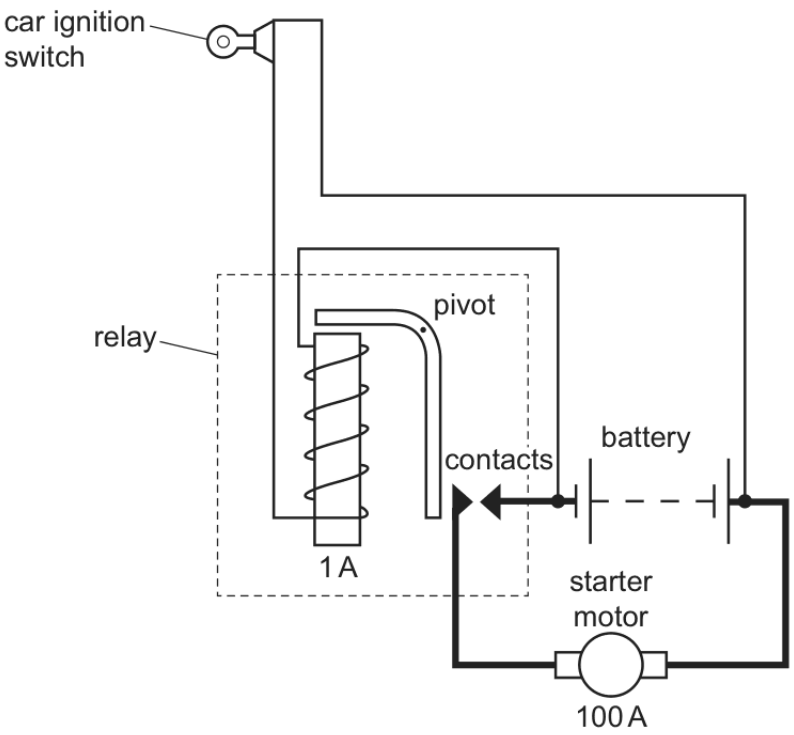
Question	Answer	Marks
10(a)	electrons **c.a.o.**	**M1**
	towards the ammeter **or** away from the negative terminal **or** towards the positive terminal	**A1**
10(b)(i)	thermistor **c.a.o.**	**B1**
10(b)(ii)	$1 / R_T = 1 / R_1 + 1 / R_2$ **or** $1 / R_T = 1 / 1.5 + 1 / 6.0$ **or** $(R_T =) R_1R_2 / (R_1 + R_2)$ **or** $1.5 \times 6.0 / (1.5 + 6.0)$	**C1**
	1.2 (Ω)	**C1**
	2.5 Ω	**A1**
10(b)(iii)	$(I =) V / R$ **or** $12 / 2.5$	**C1**
	4.8 (A)	**A1**
10(b)(iv)	$I_A = I_R + I_Z$	**B1**
10(c)	resistance of Z / thermistor decreases	**B1**
	resistance of parallel combination decreases **or** total resistance (of circuit) decreases **or** current increases	**B1**
	voltage (across 1.3 Ω) increases	**B1**
	trace moves towards top of screen / upwards	**B1**

MARK SCHEME · 5054_w18_ms_22.pdf · source page 10 of 10 · Question 10

Question	Answer	Marks
10(d)(i)	8	**B1**
10(d)(ii)	1.5 V	**B1**

QUESTION · 5054_s19_qp_11.pdf · source page 14 of 20

30 The diagram shows a car ignition switch and starter motor.



The ignition switch is in a circuit with long, thin wires. The starter motor is in a circuit with short, thick wires.

What is the explanation for the choice of wires?

A Each circuit needs to contain the same total mass of wire.

B Thicker wires heat up more quickly when the relay is switched on.

C Thin wires have lower resistances.

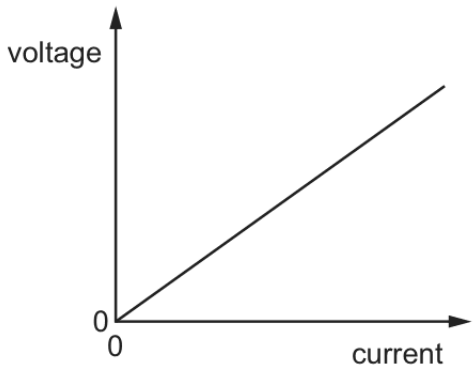
D The ignition switch circuit carries a smaller current than the starter motor circuit.

MARK SCHEME (official answer key) · 5054_s19_ms_11.pdf · source page 3 of 3 · row for Question 30

30	D	**1**

QUESTION · 5054_s19_qp_11.pdf · source page 16 of 20

33 The voltage / current graph for a metal wire is shown.



What does the gradient of this graph represent?

A the charge passing through the wire

B the e.m.f. of the battery connected to the wire

C the energy produced in the wire

D the resistance of the wire

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33	D	**1**

5 A student uses the circuit shown in Fig. 5.1 to investigate a resistor R.

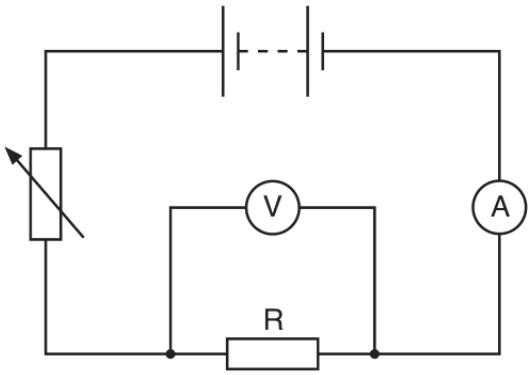


Fig. 5.1

(a) Describe how the student uses the apparatus in Fig. 5.1 to obtain a range of ammeter and voltmeter readings.

.....

..... [1]

(b) The readings obtained are shown in Table 5.1.

Table 5.1

voltmeter reading / V	ammeter reading / mA
7.6	320
5.2	220
2.4	100

(i) State Ohm's law.

.....

.....

..... [2]

(ii) Using the results in Table 5.1, show that the resistor obeys Ohm's law.

.....

.....

.....

..... [1]

(iii) The student has available an ammeter with five ranges marked:

- 0 – 10 A
- 0 – 200 mA
- 0 – 20 mA
- 0 – 2 mA
- 0 – 0.2 mA

Suggest how the student makes best use of the different ranges during the investigation.

.....

.....

.....

..... [1]

[Total: 5]

Question	Answer	Marks
5(a)	adjusts the variable resistor / rheostat **or** change the number of cells	**B1**
5(b)(i)	current proportional to voltage **or** V = kI where k is a constant	**M1**
	provided temperature / physical conditions unchanged	**A1**
5(b)(ii)	at least two values of V/I or I/V shown to be the same;	**B1**
5(b)(iii)	use 0–10 A range for 220 and 320 mA readings **and** 0–200 mA range for 100 mA reading **or** start with 10 A / highest range **and** either • reduce to 200 mA for last reading • reduce range **and** keeping high value on scale / not over scale	**B1**

8 Fig. 8.1 shows a circuit diagram containing a 2000 Ω resistor of constant resistance and a thermistor.

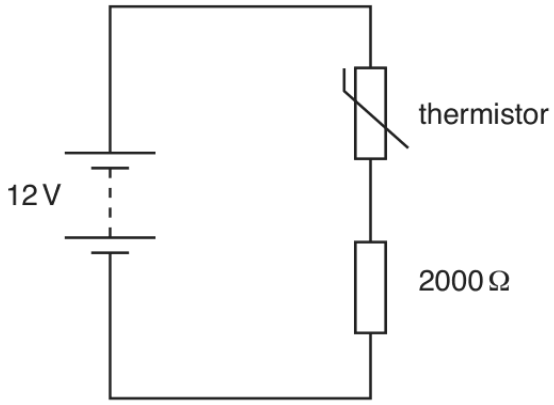


Fig. 8.1

(a) (i) At one temperature, the resistance of the thermistor is 1200 Ω.

Calculate the potential difference (p.d.) across the 2000 Ω resistor at this temperature.

p.d. = ... [2]

(ii) Explain why the p.d. across the 2000 Ω resistor increases when the temperature increases.

.....

.....

.....

..... [2]

(b) The circuit in Fig. 8.1 can be adapted to enable a lamp to switch on and off automatically.

Fig. 8.2 shows the circuit adapted using a relay and Fig. 8.3 shows the circuit adapted using a transistor.

In both cases, the lamp switches on when the temperature of the thermistor increases.

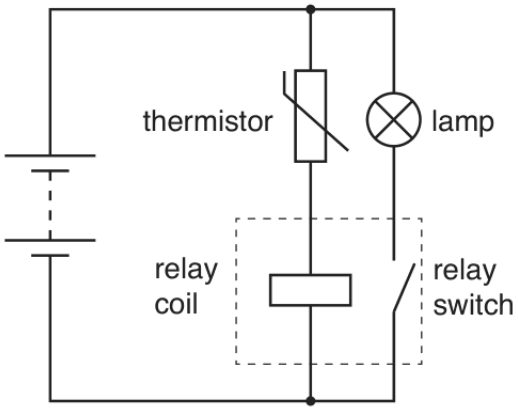


Fig. 8.2

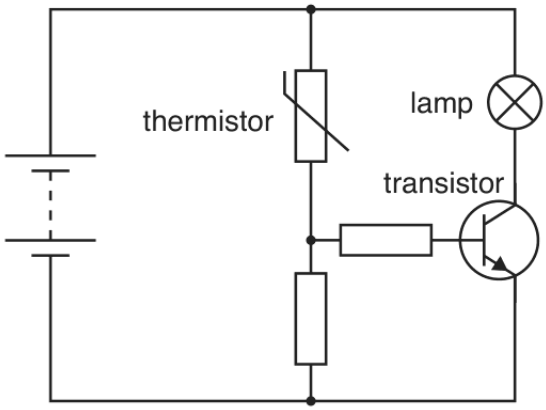


Fig. 8.3

EITHER

Describe what happens inside the relay in Fig. 8.2 that causes the lamp to switch on.

.....

.....

.....

.....

OR

Describe how the circuit and transistor in Fig. 8.3 cause the lamp to switch on.

.....

.....

.....

.....

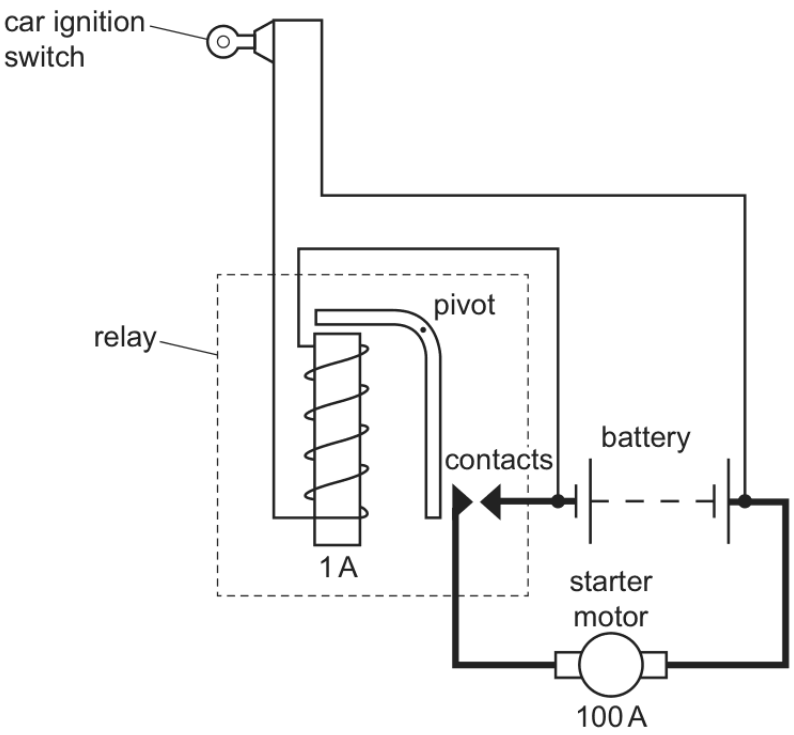
[2]

[Total: 6]

Question	Answer	Marks
8(a)(i)	attempt to use potential divider formula or split 12 V in ratio 200:1200 **or** I = 12 / (2000 + 1200) numerical or algebraic with total resistance	**C1**
	7.5 V	**A1**
8(a)(ii)	resistance (of thermistor / circuit) falls	**B1**
	fixed resistor has larger fraction / share of total voltage **or** voltage across thermistor falls **and** sum of voltages the same **or** current in circuit increases (and V = IR explained for fixed resistor)	**B1**
	EITHER	
8(b)	(increased current in coil causes) coil / core / relay to become magnetised	**B1**
	coil / core attracts (iron / magnetic) switch	**B1**
	OR	
8(b)	transistor switches on / (gives) high (collector) current	**B1**
	occurs when base and emitter voltage is high (> 0.6 V) **or** small current into base gives larger output / collector current	**B1**

QUESTION · 5054_s19_qp_12.pdf · source page 12 of 16

32 The diagram shows a car ignition switch and starter motor.



The ignition switch is in a circuit with long, thin wires. The starter motor is in a circuit with short, thick wires.

What is the explanation for the choice of wires?

A Each circuit needs to contain the same total mass of wire.

B Thicker wires heat up more quickly when the relay is switched on.

C Thin wires have lower resistances.

D The ignition switch circuit carries a smaller current than the starter motor circuit.

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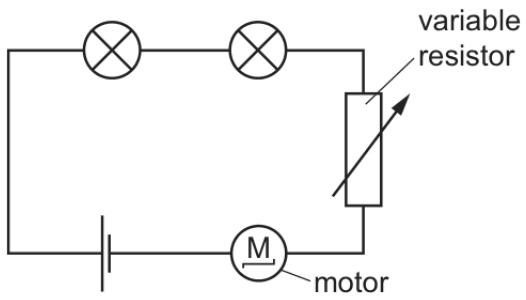
32	D	**1**

QUESTION · 5054_s19_qp_12.pdf · source page 13 of 16

33 The diagram shows a simple circuit.



What happens when the resistance of the variable resistor is increased?

A The lamps are dimmer and the motor rotates more quickly.

B The lamps are dimmer and the motor rotates more slowly.

C The lamps have the same brightness and the motor rotates more quickly.

D The lamps have the same brightness and the motor rotates more slowly.

MARK SCHEME (official answer key) · 5054_s19_ms_12.pdf · source page 3 of 3 · row for Question 33

33	B	**1**

6 Two lamps, P and Q, are connected to a battery of electromotive force (e.m.f.) 6.0 V and an ammeter, as shown in Fig. 6.1.

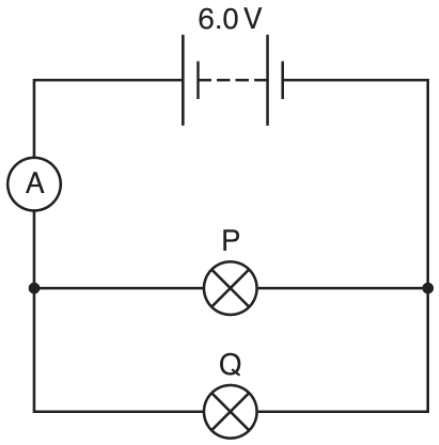


Fig. 6.1

Lamp P has a resistance of 15 Ω. The ammeter reading is 0.65 A.

(a) Calculate the current in lamp P.

current = ... [2]

(b) Calculate the resistance of lamp Q.

resistance = ... [2]

(c) The two lamps are now connected in series to the ammeter and the same battery.

(i) In the space below draw the circuit diagram.

[1]

(ii) Explain why the ammeter reading is less than 0.65 A when the lamps are connected in series.

.....

..... [1]

[Total: 6]

Question	Answer	Marks
6(a)	(I=)V / R algebraic or numerical	**C1**
	0.4(0) A	**A1**
6(b)	current in Q is 0.25 (A) **or** (total resistance) 6 / 0.25 **or** 6 / 0.65 **or** 9.2(3 Ω) seen	**C1**
	24 Ω	**A1**
6(c)(i)	correct circuit with two lamps ammeter and battery in series	**B1**
6(c)(ii)	(total) resistance increases **or** (each) lamp has lower p.d. across it	**B1**

6 A 60 W filament lamp is powered by the 240 V mains electricity supply.

The lamp is switched on.

(a) Calculate:

(i) the current in the lamp

current = ... [2]

(ii) the resistance of the lamp.

resistance = ... [2]

(b) The lamp is now unplugged from the mains supply and connected in series with two 1.5 V cells. Fig. 6.1 is the circuit diagram.

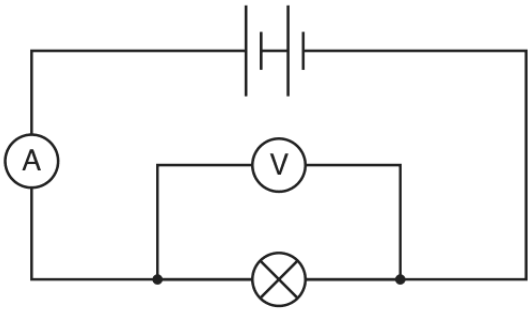


Fig. 6.1

(i) State the reading on the voltmeter.

reading = ... [1]

(ii) State and explain how the resistance of the lamp now differs from the value calculated in **(a)(ii)**.

.....

.....

..... [2]

(iii) When a filament lamp blows, it very often does so immediately after being switched on.

Suggest why this is the case.

.....

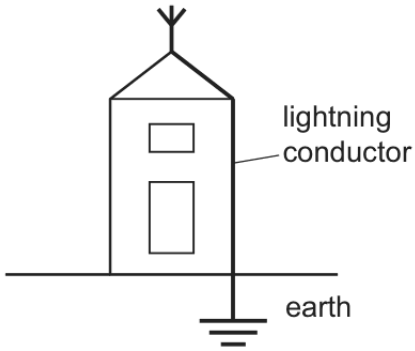
..... [1]

[Total: 8]

Question	Answer	Marks
6(a)(i)	(*I* =) *P* / *V* **or** 60 / 240	**C1**
	0.25 A	**A1**
6(a)(ii)	(*R* =) *V* / *I* **or** 240 / 0.25	**C1**
	960 Ω	**A1**
6(b)(i)	3.0 V	**B1**
6(b)(ii)	resistance is smaller	**B1**
	(filament at a) lower temperature	**B1**
6(b)(iii)	current surge (due to lower resistance when cold)	**B1**

QUESTION · 5054_w19_qp_12.pdf · source page 13 of 16

31 The diagram shows a metal lightning conductor attached to a building.



Why is the lightning conductor made of copper?

A Both positive and negative charges can flow through copper.

B Copper can be electrostatically charged.

C So the lightning conductor has a low electrical resistance.

D So the heat is conducted away quickly.

MARK SCHEME (official answer key) · 5054_w19_ms_12.pdf · source page 3 of 3 · row for Question 31

31	C	**1**

6 An electric circuit contains a 700 Ω resistor and a light-dependent resistor (LDR). Fig. 6.1 is the circuit diagram.

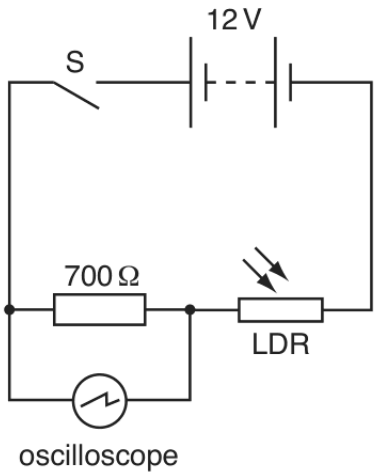


Fig. 6.1

The electromotive force (e.m.f.) of the battery is 12 V.

An oscilloscope is connected across the fixed resistor. Fig. 6.2 shows the oscilloscope, including the settings of the timebase and the Y-gain controls. Line Q shows the position of the trace on the oscilloscope when the switch S is open.

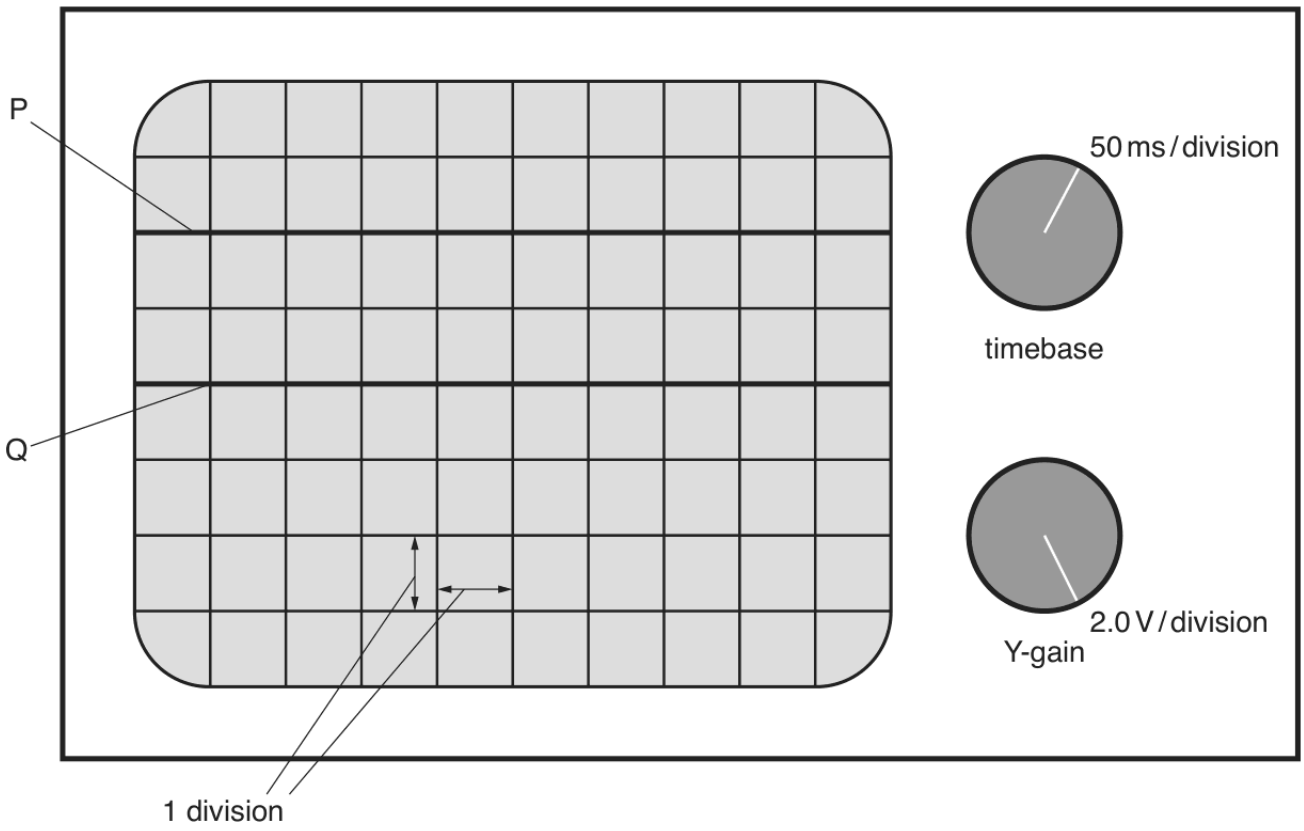


Fig. 6.2

The switch S is closed and the trace on the oscilloscope moves to the position shown by line P in Fig. 6.2.

(a) (i) Determine the potential difference (p.d.) across the 700 Ω resistor.

p.d. = ... [1]

(ii) Determine the resistance of the LDR.

resistance = ... [3]

(b) The intensity of the light incident on the LDR gradually increases.

State and explain how the trace on the oscilloscope screen moves.

.....

.....

.....

..... [3]

[Total: 7]

Question	Answer	Marks
6(a)(i)	4.0 V **c.a.o.**	**B1**
6(a)(ii)	(V =) 12 – 4.0 **or** 8.0 **or** 4.0 / 700 **or** 5.7×10^{-3}	**C1**
	(R =) V / I **or** 8.0 × 1 / (4.0 / 700) **or** 700 × (8.0 / 4.0) **or** 12 / (4.0 / 700) **or** 2100 (Ω)	**C1**
	1400 Ω	**A1**
6(b)	resistance of LDR decreases	**B1**
	current increases **or** p.d. across 700 Ω resistor / oscilloscope increases **or** p.d. across LDR decreases	**M1**
	trace moves up the screen	**A1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 Fig. 8.1 shows a lamp from a car. It contains two metal filaments.

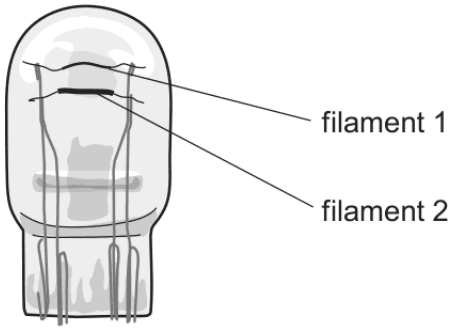


Fig. 8.1

(a) (i) Complete the boxes to describe the transfer of energy that takes place when the lamp is switched on.

..... energy	→	..... and ..... energy

[3]

(ii) The efficiency of the metal filament lamp is less than 10%.

State what is meant by *efficiency*.

.....

..... [2]

(b) The two filaments are usually connected in parallel to a car battery.

A student investigates what happens when the filaments are connected in series, rather than in parallel. He uses the same battery for the investigation.

State whether the current, the voltage across each filament and the total power produced *increases*, *decreases* or *stays the same* when the two filaments are connected in series.

current .....

voltage .....

power .....

[2]

(c) Fig. 8.2 shows the current–voltage graph for the two filaments.

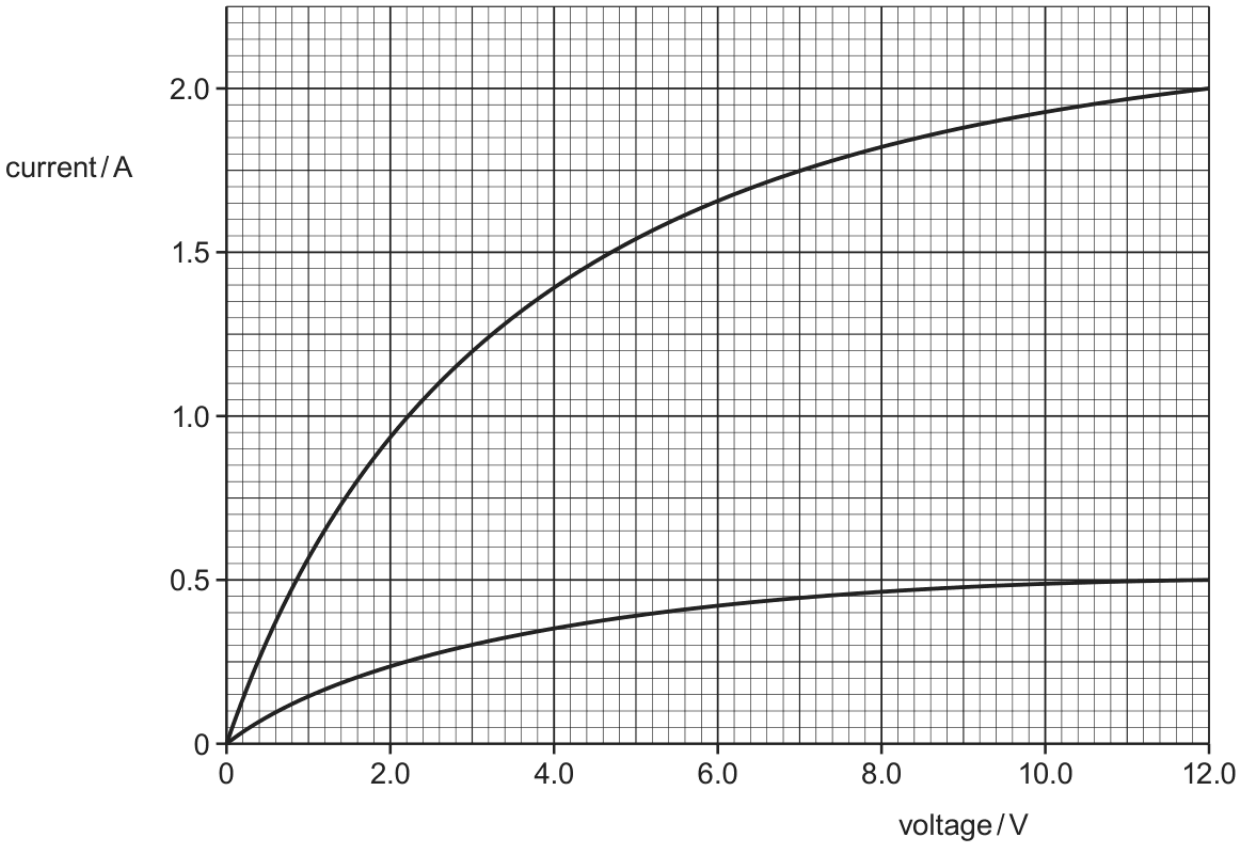


Fig. 8.2

(i) Calculate the total resistance of the two filaments when they are connected in parallel to a voltage of 12 V.

resistance = ... [3]

(ii) The two filaments are made from the same type of metal and have the same length, when uncoiled. They both operate at the same temperature.

Suggest why one filament has a resistance that is greater than that of the other filament.

.....

..... [1]

(d) Fig. 8.3 shows a relay used to switch on a car headlamp.

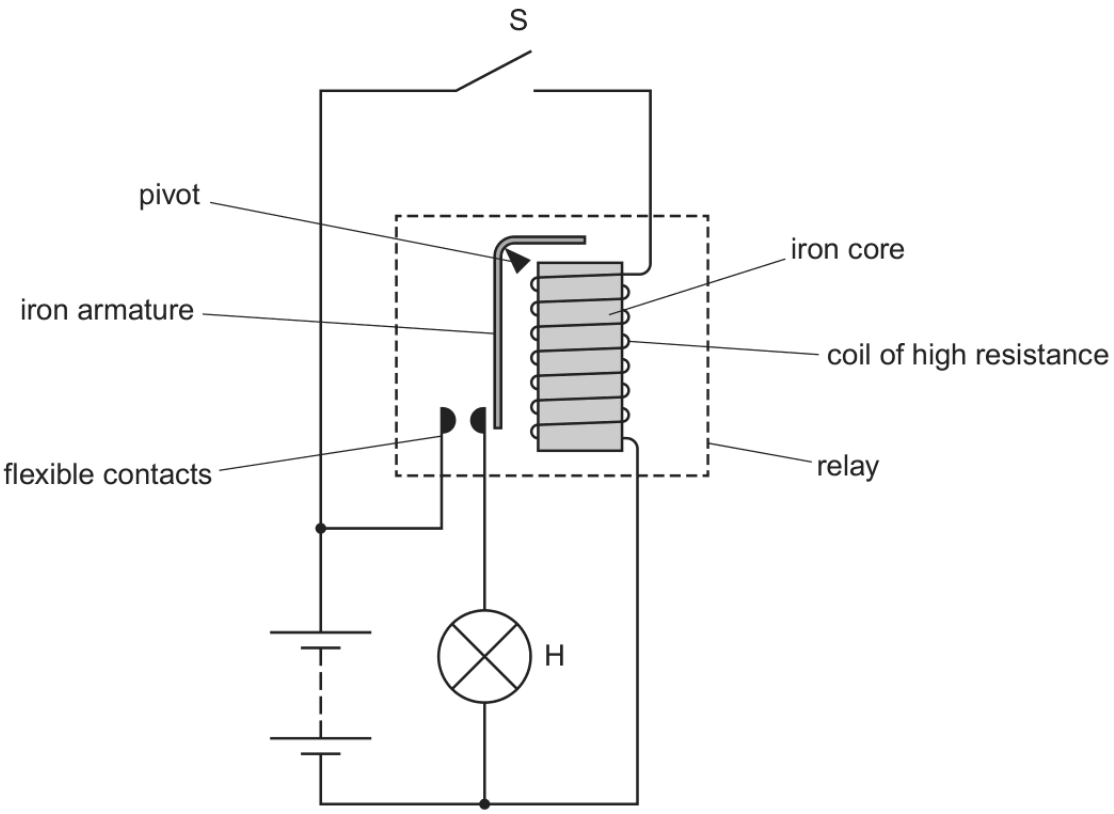


Fig. 8.3

Explain why headlamp H lights up when switch S is closed.

.....

.....

.....

.....

.....

..... [4]

[Total: 15]

MARK SCHEME · 5054_s20_ms_21.pdf · source page 8 of 11 · Question 8

Question	Answer	Marks
8(a)(i)	electrical in first box	**B1**
	heat	**B1**
	light	**B1**
8(a)(ii)	energy output / energy input	**C1**
	useful output energy / total energy input	**A1**
8(b)	two decrease	**C1**
	all three decrease	**A1**
8(c)(i)	(R =) V / I in any form numerical or algebraic	**C1**
	total current = 2.5 A **or** both resistances 24 (Ω) and 6(.0 Ω)	**C1**
	4.8 Ω	**A1**

MARK SCHEME · 5054_s20_ms_21.pdf · source page 9 of 11 · Question 8

Question	Answer	Marks
8(c)(ii)	smaller area / diameter / radius	**B1**
8(d)	current in the coil	**B1**
	coil / core becomes magnetised	**B1**
	attracts iron armature	**B1**
	armature turns **and** contacts close	**B1**

QUESTION · 5054_s20_qp_12.pdf · source page 13 of 16

37 A metal wire of length l and cross-sectional area A has resistance R.

A second wire is made from the same metal. It has a length $2l$ and a cross-sectional area $4A$.

What is the resistance of the second wire?

A $8R$ **B** $2R$ **C** $\frac{R}{2}$ **D** $\frac{R}{8}$

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37	C	**1**

Section B

Answer **two** questions from this section. Answer in the spaces provided.

8 Fig. 8.1 shows the circuit diagram of a temperature gauge. It contains an ammeter, a thermistor, a fixed resistor R and a battery.

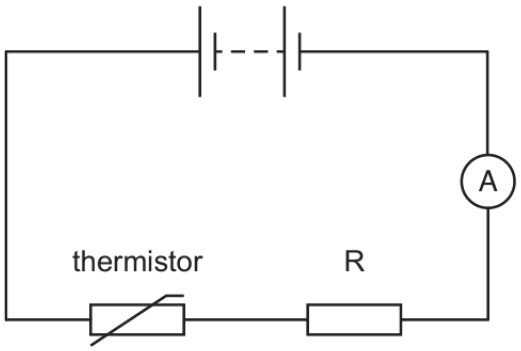


Fig. 8.1

The current is measured at different temperatures and a graph of the results is shown in Fig. 8.2.

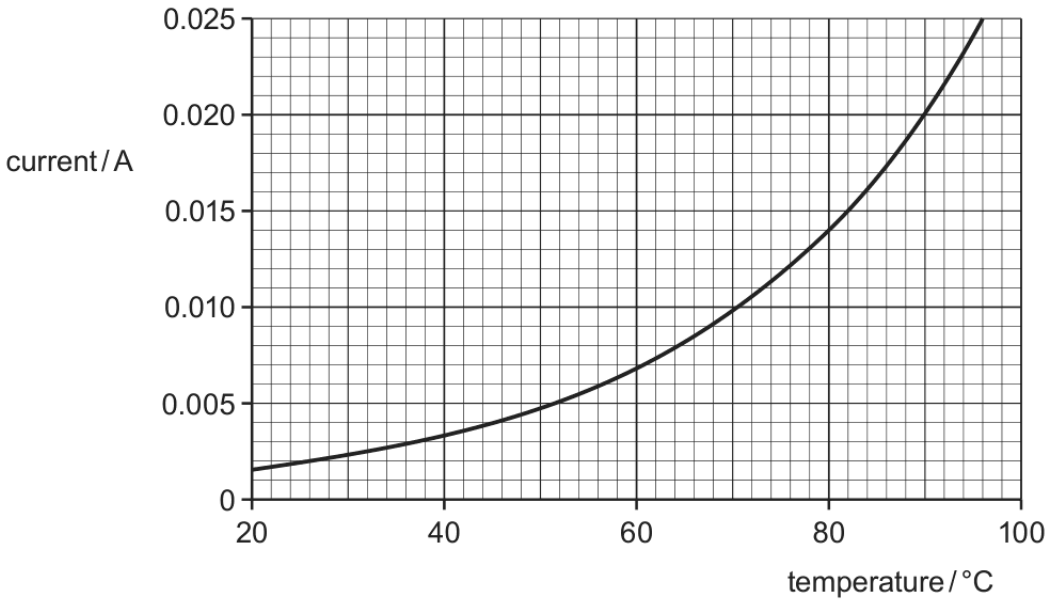


Fig. 8.2

(a) (i) State how the resistance of the thermistor changes with temperature and explain how Fig. 8.2 shows this change.

.....

.....

..... [2]

(ii) Resistance and current are used for the measurement of temperature.

State **one** other physical property that is used for the measurement of temperature.

.....

..... [1]

(b) A temperature scale involves the use of fixed points.

State what is meant by:

(i) *the ice point*

.....

..... [1]

(ii) *the steam point*.

.....

..... [1]

(c) At 80 °C, the potential difference (p.d.) across the thermistor is 3.6 V.

(i) State what is meant by *potential difference*.

.....

..... [1]

(ii) Calculate the resistance of the thermistor when the temperature is 80 °C.

resistance = ... [3]

(iii) The electromotive force (e.m.f.) of the battery is 25 V.

Calculate the resistance of the resistor R.

resistance = ... [2]

(d) The current scale on the ammeter is linear.

Using values from Fig. 8.2, a student marks the scale on the ammeter with temperature values that correspond to the values of the current.

The temperature is then read directly from the temperature scale on the ammeter.

(i) State what is meant by a *sensitive* thermometer.

.....

.....

..... [1]

(ii) State and explain at which temperatures this thermometer is most sensitive.

.....

.....

.....

..... [2]

(e) The temperature gauge is used to measure the temperature of a room.

Suggest why the temperature obtained using the gauge is slightly higher than the actual temperature of the room.

.....

..... [1]

[Total: 15]

MARK SCHEME · 5054_s20_ms_22.pdf · source page 10 of 13 · Question 8

Question	Answer	Marks
8(a)(i)	resistance decreases as temperature increases	**B1**
	current increases (and voltage constant) as temperature increases	**B1**
8(a)(ii)	volume (of liquid/gas), expansion (of solid/liquid or gas), length (of solid), pressure (of gas) length (of bimetallic strip), e.m.f. (of thermocouple), colour (of liquid crystals or very hot objects)	**B1**
8(b)(i)	temperature of melting ice **or** freezing water	**B1**
8(b)(ii)	temperature of boiling water (at 1 atmosphere)	**B1**
8(c)(i)	work done / energy change ÷ charge (passed through a component)	**B1**
8(c)(ii)	(R =) V / I in any form	**C1**
	(I =) 0.014(0) (A)	**C1**
	260 Ω	**A1**

MARK SCHEME · 5054_s20_ms_22.pdf · source page 11 of 13 · Question 8

Question	Answer	Marks
8(c)(iii)	(V across R =) 25-3.6 **or** 21(.4 V) **OR** (total resistance =) 25 / 0.014	**C1**
	1500 Ω	**A1**
8(d)(i)	(a thermometer where the quantity) changes a lot per unit temperature rise / °C / same temperature rise	**B1**
8(d)(ii)	at high temperatures	**B1**
	large(r) change in current / meter deflection for unit temperature rise / °C / same temperature rise **or** large(r) gradient	**B1**
8(e)	heat (thermal energy) produced in the thermistor / resistor	**B1**

10 An oscilloscope is a device used to display waveforms.

(a) Inside the oscilloscope, a beam of electrons is emitted from a metal filament by thermionic emission.

The emitted electrons are accelerated away from the filament by a potential difference of 2000 V.

(i) State what must happen to the metal filament for thermionic emission to occur.

.....

..... [1]

(ii) Explain why the electrons accelerate away from the filament.

.....

.....

..... [1]

(iii) The charge on one electron is 1.6×10^{-19} C.

Calculate the maximum kinetic energy of one electron after it has been accelerated through 2000 V.

kinetic energy = ... [2]

(b) Fig. 10.1 is the trace on the screen of the oscilloscope.

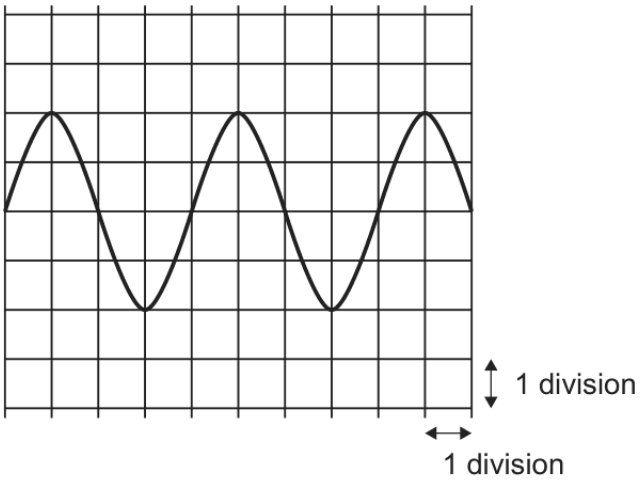


Fig. 10.1

The settings on the oscilloscope are 10 ms / division for the *x*-axis and 3.0 V / division for the *y*-axis.

(i) Calculate the amplitude of the trace shown in Fig. 10.1.

amplitude = ... [2]

(ii) Calculate the time for one oscillation (complete wavelength) of the trace shown in Fig. 10.1.

time = ... [1]

(c) The trace shown in Fig. 10.1 is caused by a sound.

The sound travels through the air to a microphone from the place that it is made. The microphone is connected to the oscilloscope which displays the waveform shown.

(i) Sound is a type of wave.

State which type.

..... [1]

(ii) Describe the motion of the air molecules as the sound passes through the air to the microphone.

.....

.....

..... [2]

(iii) Describe and explain how the trace on the screen changes as the pitch of the sound becomes higher. The settings on the oscilloscope are unchanged.

.....

.....

..... [2]

(d) A student investigates what happens when a diode is connected between the microphone and the oscilloscope.

Fig. 10.2 shows the circuit that he uses.

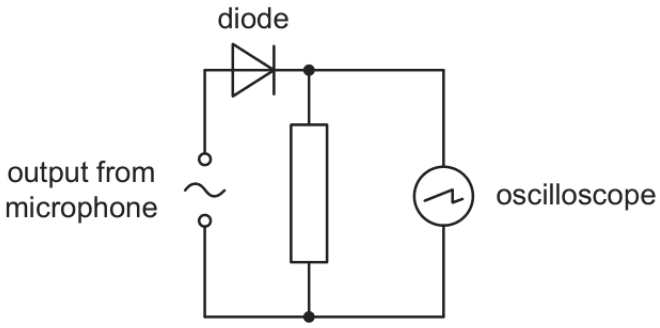


Fig. 10.2

Without the diode, the output from the microphone is an alternating current and the trace is as shown in Fig. 10.1 on page 18.

(i) Describe the action of the diode on the current from the microphone.

.....

..... [1]

(ii) On the grid in Fig. 10.3, sketch the trace seen on the screen when the diode is used.

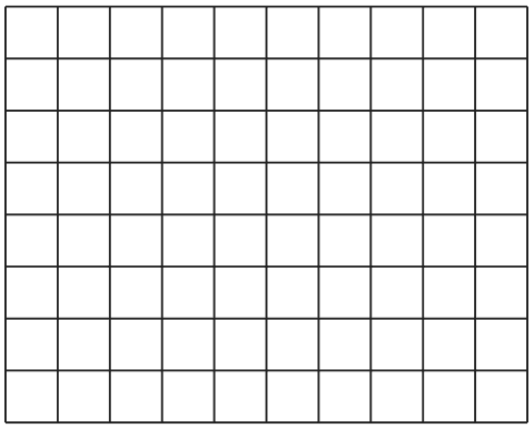


Fig. 10.3

[2]

[Total: 15]

MARK SCHEME · 5054_s20_ms_22.pdf · source page 12 of 13 · Question 10

Question	Answer	Marks
10(a)(i)	filament is hot / heated	**B1**
10(a)(ii)	attracted by a positive potential **or** repelled by filament **or** filament is negative	**B1**
10(a)(iii)	$2000 \times 1.6 \times 10^{-19}$	**C1**
	3.2×10^{-16} J	**A1**
10(b)(i)	two divisions	**C1**
	6(.0) V	**A1**
10(b)(ii)	40 ms	**B1**
10(c)(i)	longitudinal	**B1**

MARK SCHEME · 5054_s20_ms_22.pdf · source page 13 of 13 · Question 10

Question	Answer	Marks
10(c)(ii)	oscillation described, e.g. back and forth	**B1**
	in direction of travel (of wave / energy) **or** compressions and rarefactions stated or described	**B1**
10(c)(iii)	more waves seen on screen	**B1**
	higher frequency (of sound) **or** more waves per second **or** time for one wave less	**B1**
10(d)(i)	diode passes current in only one direction **or** current only flows for half the cycle / oscillation **or** current is d.c. / direct	**B1**
10(d)(ii)	at least one top half and no bottom half **or** one bottom half and no top half	**C1**
	three top halves of the wave and flat sections along mid line **or** two bottom halves of the wave and flat sections along mid line	**A1**

10 Fig. 10.1 shows a wire of length l and cross-sectional area A.

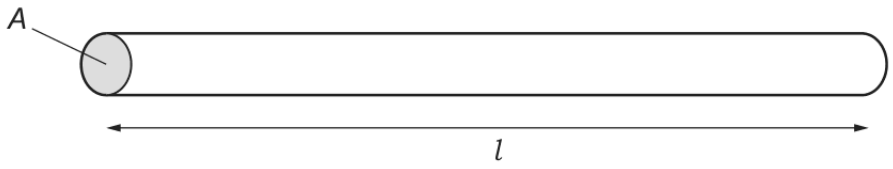


Fig. 10.1

(a) State how the resistance of the wire in Fig. 10.1 depends on:

(i) l

..... [1]

(ii) A.

..... [1]

(b) The cross-sectional area of a piece of metal wire is $7.5 \times 10^{-4}\,\text{cm}^2$. The resistance of a 1.0 m length of the same wire is 6.4 Ω.

The wire is made from metal W.

Fig. 10.2 shows a solid cube of side 1.0 cm. It is also made from metal W.

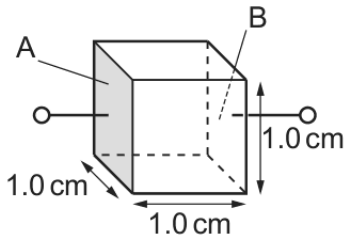


Fig. 10.2

Calculate the resistance between the two opposite faces A and B of the cube.

resistance = ... [2]

(c) The wire in part **(b)** is taped to a metre rule.

Fig. 10.3 shows that a 1.0 m length of the wire (resistance 6.4 Ω) is connected in series with a switch, a cell of electromotive force (e.m.f.) 1.2 V and a resistor of resistance 9.6 Ω.

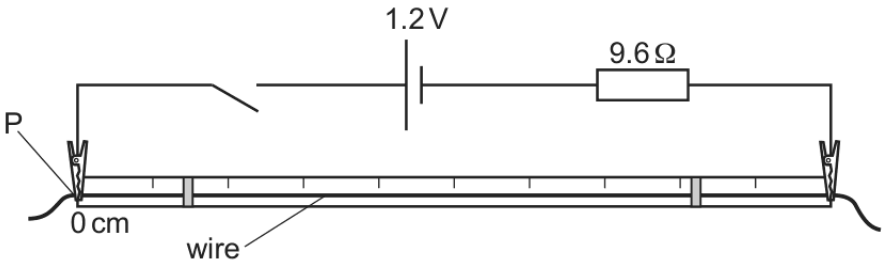


Fig. 10.3

The switch is closed.

(i) Explain what is meant by *electromotive force* (*e.m.f.*).

.....

.....

..... [2]

(ii) Calculate the potential difference (p.d.) across the 1.0 m length of the wire.

p.d. = ... [3]

(iii) One input terminal of an oscilloscope is connected to the wire at point P, the 0 cm mark of the metre rule.

The other terminal of the oscilloscope is connected to a sliding contact. Initially, this contact touches the wire at point P.

The Y-gain setting on the oscilloscope is 0.20 V / cm.

Fig. 10.4 shows the screen of the oscilloscope with a horizontal trace across the middle of the screen.

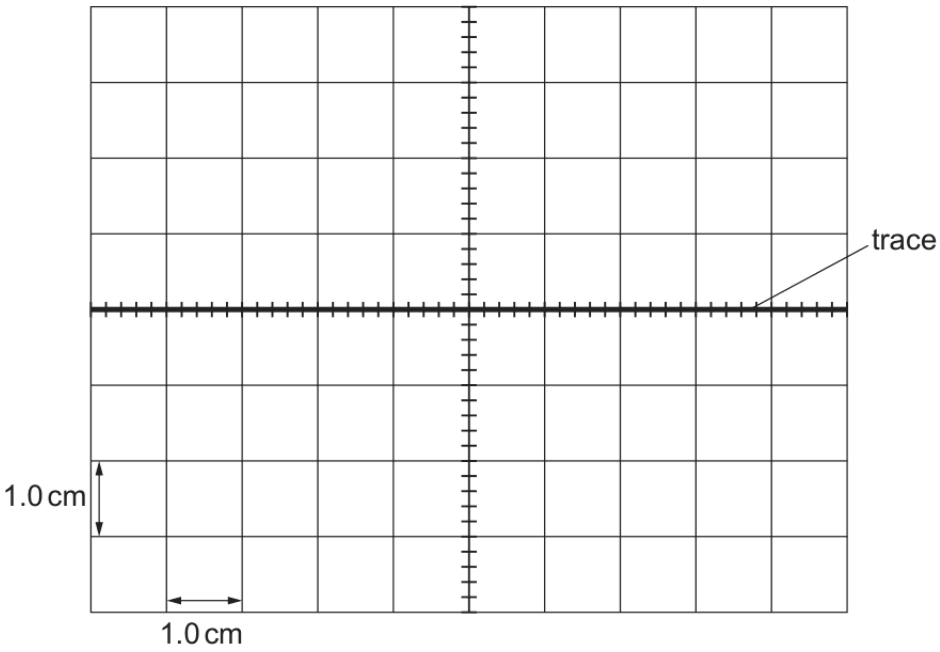


Fig. 10.4

The sliding contact is slowly moved along the wire until it reaches the other end of the metre rule.

Describe and explain what happens to the trace on the screen.

.....

.....

.....

..... [3]

(d) A second, identical 1.2 V cell is connected in parallel with the cell in the circuit in Fig. 10.3.

(i) State **one** advantage of using two cells in parallel rather than a single cell.

.....

..... [1]

(ii) State and explain the effect on the trace in **(c)(iii)** of adding the second cell in parallel.

.....

.....

..... [2]

[Total: 15]

Question	Answer	Marks
10(a)(i)	(resistance) is directly proportional to length	**B1**
10(a)(ii)	(resistance) is inversely proportional to (cross-sectional) area	**B1**
10(b)	(R =) $6.4 \times 7.5 \times 10^{-4} / 100$ **or** 4.8×10^{-n} (n is an integer)	**C1**
	$4.8 \times 10^{-5}\ \Omega$	**A1**
10(c)(i)	energy / work done in driving a charge round a circuit **or** energy transferred to electrical energy	**B1**
	energy / work done per unit charge **or** energy / charge	**B1**
10(c)(ii)	(R =) 6.4 + 9.6 **or** 16	**C1**
	(V =) $1.2 \times 6.4 / 16$ **or** current = 1.2 / 16 **or** current = 0.075	**C1**
	0.48 V	**A1**
10(c)(iii)	trace moves vertically away from centre line	**B1**
	moves distance of 2.4 cm	**B1**
	remains horizontal **or** distance moved by trace directly proportional to distance moved by jockey **or** comment about tape	**B1**
10(d)(i)	the battery lasts twice as long as the single cell	**B1**
10(d)(ii)	no effect **or** trace moves in an identical manner	**B1**
	e.m.f. of identical cells in parallel equals the e.m.f. of a single cell	**B1**

7 **(a)** Fig. 7.1 shows two circuit components.

Fig. 7.1

For each of these components state what causes an **increase** in its resistance.

thermistor .....

.....

LDR .....

.....

[2]

(b) Fig. 7.2 shows a battery connected to a diode, an ammeter and a resistor.

Fig. 7.3 shows the current–voltage graph for the diode.

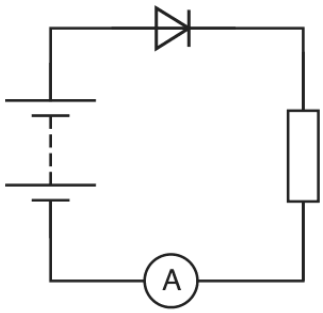


Fig. 7.2

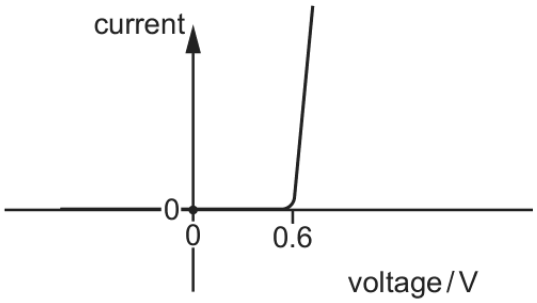


Fig. 7.3

The current in the resistor is 2.0 A and the potential difference across the resistor is 5.4 V.

(i) State what is meant by the *electromotive force* (e.m.f.) of the battery.

.....

..... [1]

(ii) Explain why the e.m.f. of the battery must be greater than 5.4 V.

.....

.....

..... [2]

(iii) Calculate the electrical energy input to the resistor in 5.0 minutes.

energy input = ... [2]

(iv) The circuit is reconnected with the diode in the opposite direction, as shown in Fig. 7.4.

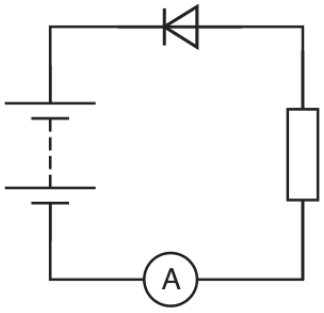


Fig. 7.4

Explain what is observed.

.....

.....

..... [1]

[Total: 8]

Question	Answer	Marks
7(a)	lower temperature / cooling of thermistor	**B1**
	less light on LDR	**B1**
7(b)(i)	energy / work per unit charge	**B1**
7(b)(ii)	Any 2 from: • voltage across diode • voltage across ammeter • voltage across internal resistance (of battery) • (e.m.f. / voltage of battery =) 5.4 + 0.6 (or larger)	**B2**
7(b)(iii)	(*E* =) *VQ* **or** *VIt* in any form algebraic or numerical	**C1**
	3200 J	**A1**
7(b)(iv)	no current / no ammeter reading **and** diode has been reversed / diode does not conduct / has (very) high resistance / blocks current / reference to current / voltage graph	**B1**

9 Fig. 9.1 is the current–voltage graph for a filament lamp.

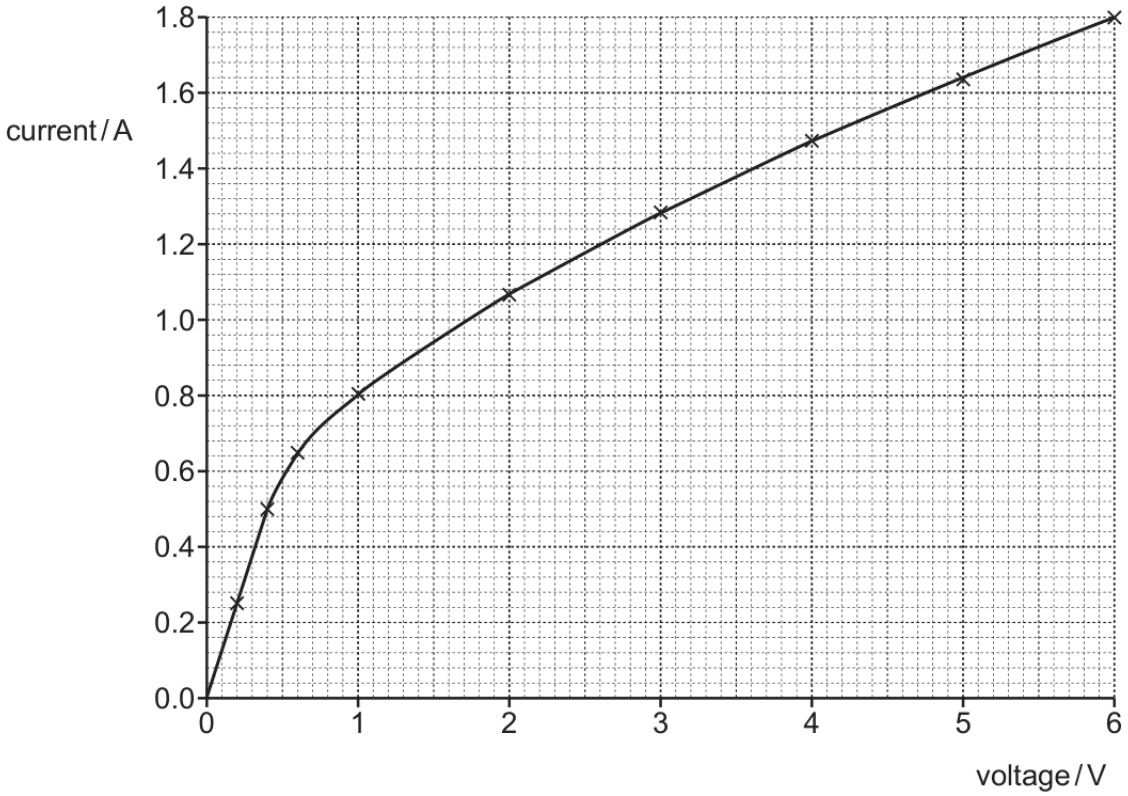


Fig. 9.1

(a) Fig. 9.2 shows an incomplete circuit diagram of the circuit that a student uses to obtain the readings for the graph.

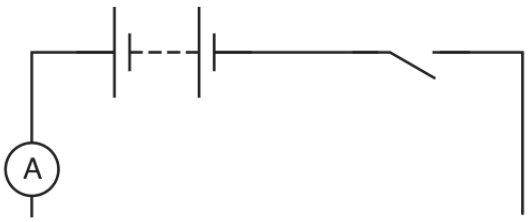


Fig. 9.2

Complete the circuit diagram using suitable circuit symbols for the lamp, a voltmeter and a variable resistor. [3]

(b) The ammeter used by the student has ranges 0−10 A, 0−1 A, 0−100 mA and 0−10 mA.

Describe how the student uses the different ranges to obtain readings which allow all nine points to be plotted precisely on the graph while using the ammeter safely.

.....

.....

.....

..... [2]

(c) State how Fig. 9.1 shows that the lamp does **not** obey Ohm's law.

.....

..... [1]

(d) Using Fig. 9.1, determine the resistance of the lamp when:

(i) the voltmeter reading is 0.40 V

resistance = ...

(ii) the voltmeter reading is 6.0 V.

resistance = ...
[3]

(e) (i) Explain why the resistance of the lamp changes as the voltage changes.

.....

.....

..... [2]

(ii) State how the resistance of a wire depends upon its length and how it depends on its cross-sectional area.

length .....

.....

cross-sectional area .....

.....
[2]

(iii) The student obtains a long sample of the same type of wire as the filament in the lamp.

The wire he obtains has the same cross-sectional area as the filament.

He cuts a 1.0 m length of the wire and measures its resistance at room temperature.

He obtains the value 16 Ω.

Using the appropriate resistance value from **(d)**, estimate the length of the filament wire in the lamp.

length = ... [2]

[Total: 15]

Question 10 begins over the page

MARK SCHEME · 5054_s21_ms_21.pdf · source page 10 of 12 · Question 9

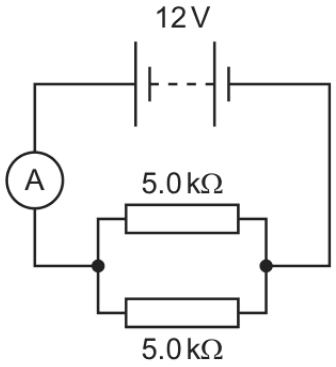
Question	Answer	Marks
9(a)	correct symbol for lamp	**B1**
	correct symbol for voltmeter placed across lamp	**B1**
	correct symbol for variable resistor in series with lamp and battery **or** used as a potential divider	**B1**
9(b)	start on 10 A scale (and lower if reading small) **or** uses 10 A scale for points from 1 to 1.8 A **or** uses 10 A scale for voltages above 2 V (and for 2 V)	**B1**
	for lower readings switch to (0-) 1 A scale	**B1**

MARK SCHEME · 5054_s21_ms_21.pdf · source page 11 of 12 · Question 9

Question	Answer	Marks
9(c)	graph curved **or** not a straight line (through origin) **or** current not (directly) proportional to voltage	**B1**
9(d)(i)	(R =) V / I in any form algebraic or numerical	**C1**
	at 0.40 V, resistance = 0.80 (± 0.03) Ω	**A1**
9(d)(ii)	at 6.0 V, resistance = 3.3–3.4 Ω	**B1**
9(e)(i)	temperature not constant / increases	**C1**
	resistance increases as temperature increases	**A1**
9(e)(ii)	(resistance is directly) proportional to length	**B1**
	(resistance is) inversely proportional to area **or** proportional to 1 / area	**B1**
9(e)(iii)	any clear use of proportionality using 0.80 Ω	**B1**
	0.050 m	**B1**

QUESTION · 5054_s21_qp_12.pdf · source page 12 of 16

32 The diagram shows a circuit.



The ammeter has four different range settings.

Which range setting gives a reading near the mid-point of the range?

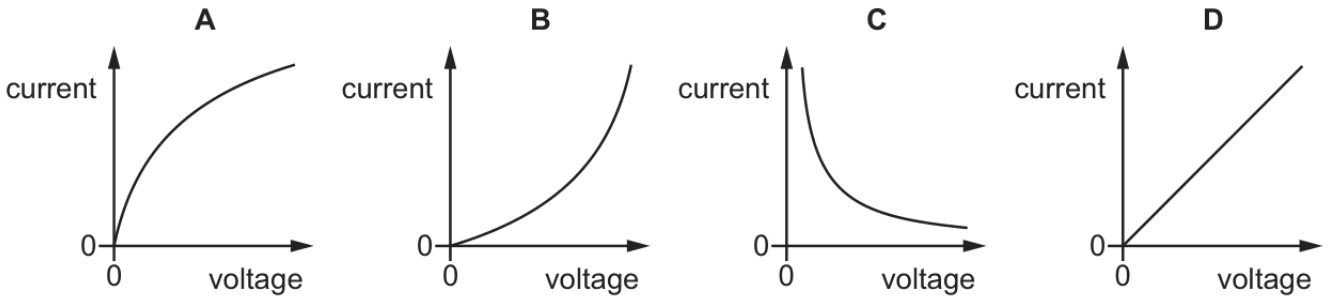
A 0–1 mA **B** 0–10 mA **C** 0–1.0 A **D** 0–10 A

MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 3 of 3 · row for Question 32

32	B	**1**

QUESTION · 5054_s21_qp_12.pdf · source page 12 of 16

34 Which graph shows the relationship between current and voltage for a filament lamp?



MARK SCHEME (official answer key) · 5054_s21_ms_12.pdf · source page 3 of 3 · row for Question 34

34	A	**1**

6 The power input to a filament lamp when connected to a 12 V supply is 20 W.

(a) State what is meant by *power*.

.....

..... [1]

(b) Calculate:

(i) the current in the filament

current = ... [2]

(ii) the resistance of the filament.

resistance = ... [1]

(c) The filament in another lamp is made from thinner wire of the same length and made from the same material. Both lamps are connected to a 12 V supply.

State and explain how using thinner wire in the filament affects the power input to the lamp.

.....

.....

..... [2]

[Total: 6]

Question	Answer	Marks
6(a)	work done / energy (transfer) per unit time	**B1**
6(b)(i)	(I =) P / V in any form numerical or algebraic	**C1**
	1.7 A	**A1**
6(b)(ii)	6.9−7.2 Ω	**B1**
6(c)	resistance (of wire) is larger	**B1**
	power is less **and** **either** current is smaller **or** $P = V^2 / R$ ***or*** $P = VI$ **and** $V = IR$	**B1**

8 Fig. 8.1 shows a potential divider circuit containing two resistors R_1 and R_2.

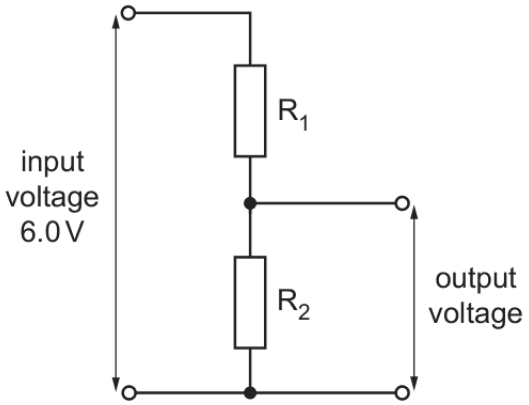


Fig. 8.1

(a) There is an input voltage of 6.0 V.

Complete Table 8.1 to show the output voltage for different values of the resistances of the two resistors.

Table 8.1

resistance of R_1 / Ω	resistance of R_2 / Ω	output voltage / V
8.0	8.0	.....
800	1200	.....

[3]

EITHER

(b) In the circuit shown in Fig. 8.1, resistor R_1 is replaced by a light dependent resistor (LDR).

(i) In the space below, draw the circuit symbol for an LDR.

[1]

(ii) The LDR is moved from darkness into daylight.

State what happens to the resistance of the LDR and the output voltage.

resistance .....

output voltage .....
[1]

(iii) A potential divider circuit can also be used to produce an output voltage that changes with temperature.

State the name of the component that is used instead of the LDR.

..... [1]

OR

(b) A simple bistable circuit contains NOR gates.

(i) In the space below, draw the circuit symbol of a NOR gate.

[1]

(ii) Describe what is meant by a *bistable circuit*.

.....

.....

.....

..... [2]

[Total: 6]

MARK SCHEME · 5054_s21_ms_22.pdf · source page 12 of 16 · Question 8

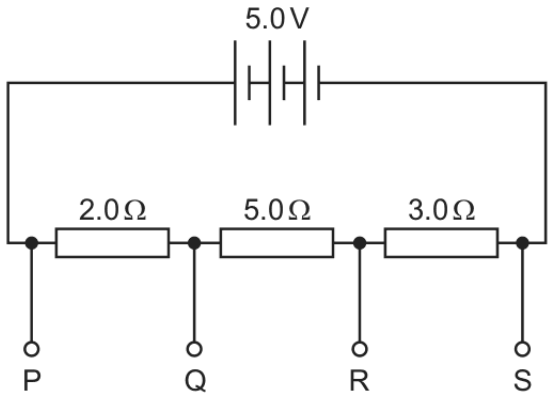
Question	Answer	Marks
8(a)	top row 3(.0) V	**B1**
	potential divider formula / working shown e.g. 1200 / 2000 (× 6) or 800 / 2000 (× 6) **or** split 6 (V) in ratio 1200:800 **or** I = 6 / 2000	**C1**
	second row 3.6 (V)	**A1**

MARK SCHEME · 5054_s21_ms_22.pdf · source page 13 of 16 · Question 8

Question	Answer	Marks
	EITHER	
8(b)(i)	correct symbol with arrows towards resistor	**B1**
8(b)(ii)	resistance decreases **and** output voltage increases	**B1**
8(b)(iii)	thermistor	**B1**
	OR	
8(b)(i)	correct symbol	**B1**
8(b)(ii)	two (stable) states	**B1**
	switches (between states) by an applied signal **or** exhibits memory	**B1**

QUESTION · 5054_w21_qp_11.pdf · source page 14 of 20

31 The diagram shows a circuit with a 5.0 V power supply, three resistors and four output terminals P, Q, R and S.



Between which pair of terminals is there a potential difference of 1.5 V?

A PQ **B** PR **C** QS **D** RS

MARK SCHEME (official answer key) · 5054_w21_ms_11.pdf · source page 3 of 3 · row for Question 31

31	D	**1**

6 A circuit contains a battery of electromotive force (e.m.f.) 4.5 V, a 28 Ω resistor and a variable resistor. Fig. 6.1 is the circuit diagram.

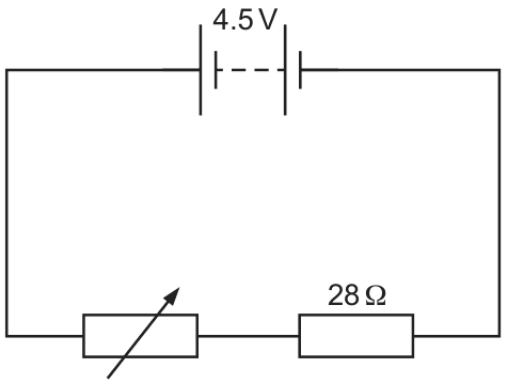


Fig. 6.1

(a) The battery consists of three identical cells in series.

Determine the e.m.f. of each cell.

e.m.f. of each cell = ... [1]

(b) The potential difference (p.d.) across the variable resistor is 2.5 V.

(i) State what is meant by *potential difference*.

.....

.....

.....

..... [2]

(ii) Calculate the resistance of the variable resistor.

resistance = ... [3]

[Total: 6]

Question	Answer	Marks
6(a)	1.5 V	**B1**
6(b)(i)	work done / energy transferred when charge passes through a component	**C1**
	$\frac{\text{work done / energy transferred}}{\text{charge passed}}$	**A1**
6(b)(ii)	4.5 – 2.5 **or** 2.0 (V) seen	**C1**
	(R =) V / I **or** 2.5 / (2.0 / 28) **or** R / 28 = 2.5 / 2.0	**C1**
	35 Ω	**A1**

9 A filament lamp is connected to a 240 V alternating current (a.c.) mains supply.

(a) Describe how the output of an a.c. supply differs from the output of a direct current (d.c.) supply.

Sketch **two** voltage–time graphs in the blank space to help your explanation.

.....

..... [3]

(b) The lamp is rated at 60 W and is designed to be used with a 240 V supply.

(i) Calculate the current in the lamp.

current = ... [2]

(ii) Calculate the resistance of the lamp.

resistance = ... [2]

(iii) A room is lit by five of these filament lamps connected in parallel.

State **two** advantages of connecting the lamps in parallel rather than in series.

1. .....

.....

2. .....

.....

[2]

(iv) The five lamps in **(b)(iii)** are lit for an average time of 5.5 hours a day for a year. Electricity costs $0.15/kWh.

Calculate the cost of using these lamps for a year.

cost = ... [2]

(c) A student takes the lamp in **(b)** to school and connects it in a circuit using a 1.5 V cell, an ammeter and a voltmeter. The circuit is used to determine the resistance of the lamp.

(i) In the blank space, draw the circuit diagram of the circuit used to determine the resistance of the lamp.

[2]

(ii) The value of the resistance of the filament lamp in this circuit differs greatly from the value calculated in **(b)(ii)**.

State how the resistance value in this circuit differs and explain why it differs.

.....

.....

..... [2]

[Total: 15]

MARK SCHEME · 5054_w21_ms_22.pdf · source page 10 of 11 · Question 9

Question	Answer	Marks
9(a)	sinusoidal voltage-time graph labelled a.c. drawn	**B1**
	voltage-time labelled d.c. graph with horizontal, non-zero line	**B1**
	an a.c. varies with time **or** an a.c. changes direction / polarity (at regular time intervals) **or** a d.c. is constant with time **or** d.c. is never negative	**B1**
9(b)(i)	(I =) P / V **or** 60 / 240	**C1**
	0.25 A	**A1**
9(b)(ii)	(R =) V / I **or** 240 / 0.25	**C1**
	960 Ω	**A1**
9(b)(iii)	any **two** from: if one lamp blows the others stay lit can be operated separately each lamp has the correct voltage across it **or** lamps glow brightly / normally	**B2**

MARK SCHEME · 5054_w21_ms_22.pdf · source page 11 of 11 · Question 9

Question	Answer	Marks
9(b)(iv)	(number of units =) $5 \times 60 \times 5.5 \times 365 / 1000$ **or** 600 **or** 602.25 **or** 18(.07)	**C1**
	\$90 **or** \$90.34	**A1**
9(c)(i)	cell, lamp and ammeter in series **and** no other components	**B1**
	voltmeter in parallel with lamp	**B1**
9(c)(ii)	(in this circuit) the resistance is (very much) smaller	**B1**
	(metal) filament is cooler	**B1**

6 **(a)** A light-dependent resistor (LDR) is used to sense the amount of light in a room.

Fig. 6.1 shows part of the circuit used. The LDR is not shown.

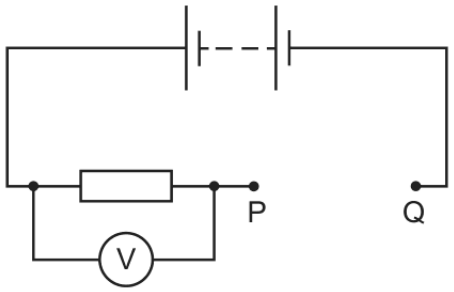


Fig. 6.1

(i) Complete Fig. 6.1 by drawing the circuit symbol for an LDR between points P and Q. [1]

(ii) Explain how the voltmeter reading changes as the amount of light falling on the LDR increases.

.....

.....

..... [2]

(b) Fig. 6.2 shows part of a circuit containing three resistors.

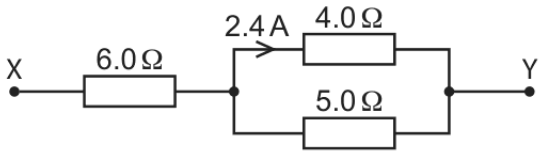


Fig. 6.2

The current in the 4.0 Ω resistor is 2.4 A.

(i) Show that the current in the 5.0 Ω resistor is 1.9 A.

[2]

(ii) Calculate the potential difference (p.d.) between points X and Y.

p.d. = ... [2]

[Total: 7]

Section B begins over the page

Question	Answer	Marks
6(a)(i)	correct symbol 2 or 3 arrows onto resistor symbol	**B1**
6(a)(ii)	resistance (of LDR) decreases	**B1**
	(voltmeter reading) increases **and** more current in circuit **or** potential divider argument comparing voltage across resistor / LDR / voltage of cell	**B1**
6(b)(i)	$V=IR$ seen in any form **or** (voltage across 4 or 5 Ω) 2.4×4.0 seen	**B1**
	(I =) 9.6 / 5 = 1.9(2 A)	**B1**
6(b)(ii)	(current in 6.0 Ω =) 4.3(2) (A) **or** (voltage across 6 Ω=) 25.8 or 25.9(2) (V) seen **or** resistance of parallel resistors = 20 / 9 (Ω) **or** total resistance = $8\frac{2}{9}$ Ω)	**C1**
	35 V	**A1**

5 **(a)** A student connects a battery, thermistor X and resistor Y in parallel, as shown in Fig. 5.1.

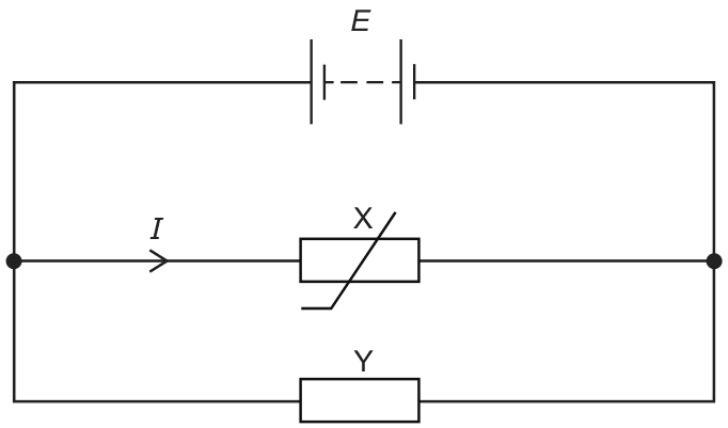


Fig. 5.1

- The electromotive force (e.m.f.) of the battery is E.
- The current in X is I.
- The resistance of Y is greater than the resistance of X.

Tick **two** boxes, one to show the correct statement about the potential difference across Y and the other to show the correct statement about the current in Y.

The potential difference across Y:

is less than E ☐

is equal to E ☐

is greater than E. ☐

The current in Y:

is less than I ☐

is equal to I ☐

is greater than I. ☐

[1]

(b) The student connects thermistor X in series with the resistor Y and a battery of e.m.f. 6.0 V, as shown in Fig. 5.2.

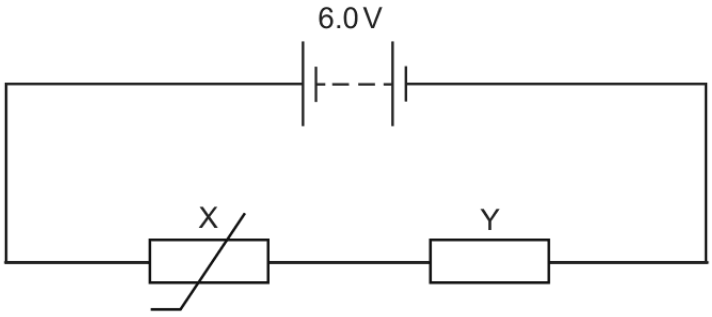


Fig. 5.2

In this circuit, at room temperature, the resistance of thermistor X is 600 Ω and the current in thermistor X is 0.0020 A.

(i) Calculate the power produced in the thermistor.

power = ... [2]

(ii) Calculate the resistance of Y.

resistance = ... [3]

(iii) The thermistor is cooled.

Explain why this causes the potential difference across Y to decrease.

.....

.....

..... [2]

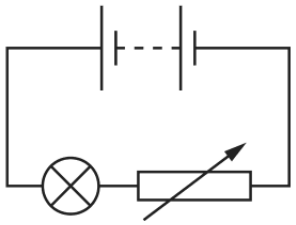
[Total: 8]

Question	Answer	Marks
5(a)	**middle** box of 1st three boxes <u>and</u> **top** box of 2nd three boxes	**B1**
5(b)(i)	(P =) VI **or** I^2R algebraic or numerical seen in any form	**C1**
	0.0024 W	**A1**
5(b)(ii)	$V = IR$ algebraic or numerical in any form **or** potential divider formula seen	**C1**
	4.8 (V) seen **or** 6.0 – 1.2 / 0.002 **or** 3000 − 600 **or** $6 = 0.002 \times (600+R_Y)$ **or** $1.2 = 6 \times 600 / (R_Y + 600)$	**C1**
	2400 Ω	**A1**

Question	Answer	Marks
5(b)(iii)	resistance of thermistor / X increases (on cooling)	**B1**
	current (in thermistor) decreases **or** potential divider argument used to explain decrease in pd across Y **or** pd across <u>X increases</u>	**B1**

QUESTION · 5054_w22_qp_11.pdf · source page 12 of 16

30 A lamp is connected in series with a battery and a variable resistor. The resistance of the variable resistor is decreased gradually.



Which statement is correct?

A The current in the lamp increases.

B The current in the variable resistor decreases.

C The e.m.f. of the battery decreases.

D The total resistance in the circuit increases.

MARK SCHEME (official answer key) · 5054_w22_ms_11.pdf · source page 3 of 3 · row for Question 30

30	A	**1**

6 An electromagnet is used to separate objects that are magnetic from objects that are non-magnetic.

Fig. 6.1 shows the electromagnet suspended from the arm of a crane.

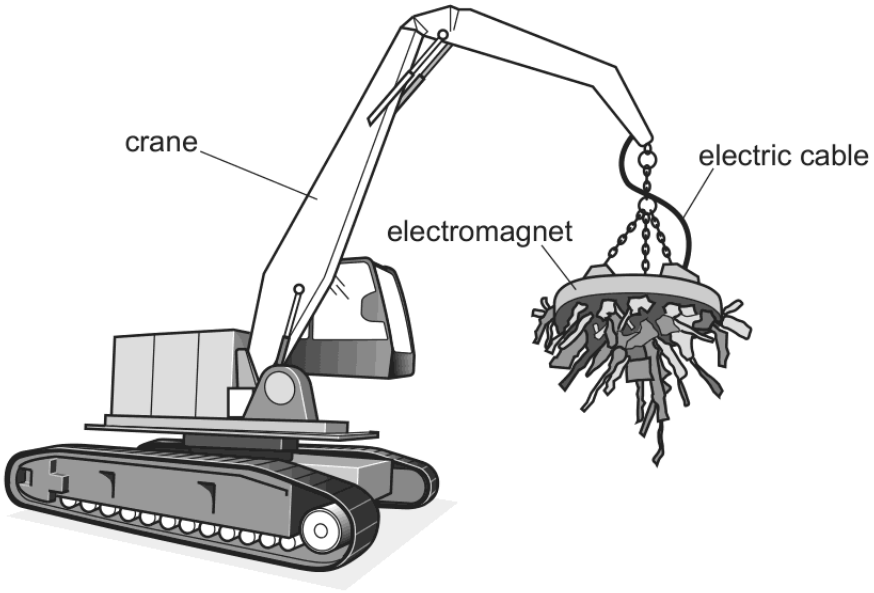


Fig. 6.1

(a) State the name of the metal from which the core of the electromagnet is made.

..... [1]

(b) The electromagnet is powered by a 220 V d.c. (direct current) supply.

The electromagnet is switched on and the current in the circuit is 39 A.

(i) Calculate the power transferred to the electromagnet.

power = ... [2]

(ii) The d.c. supply is a set of batteries.

The batteries power the magnet for 4.5 hours before they need to be replaced.

Calculate the charge driven around the complete circuit in this time.

charge = ... [3]

(c) On a very cold morning, when the electromagnet is first switched on, the current is greater than 39A before decreasing to the usual value.

Explain why the current is initially greater than 39A.

.....

..... [1]

[Total: 7]

Question	Answer	Marks
6(a)	soft iron	**B1**
6(b)(i)	($P =$) VI **or** 220×39	**C1**
	8.6×10^3 W	**A1**
6(b)(ii)	($Q =$) It **or** 39×4.5 **or** $39 \times 4.5 \times 60$ **or** $39 \times 4.5 \times 60 \times 60$ **or** $39 \times 4.5 \times 3600$	**C1**
	180 **or** 11 000 **or** $39 \times 4.5 \times 60 \times 60$ **or** $39 \times 4.5 \times 3600$	**C1**
	6.3×10^5 C	**A1**
6(c)	resistance increases with temperature **or** resistance lower (because of (low / ambient) temperature)	**B1**

9 The power supply in an electric circuit is a battery of electromotive force (e.m.f.) 12 V.

(a) State **two** ways in which the e.m.f. of a battery differs from that of an alternating current (a.c.) power supply.

1 .....

2 .....
[2]

(b) The circuit includes three resistors and two open switches, S_1 and S_2.

Fig. 9.1 shows the circuit.

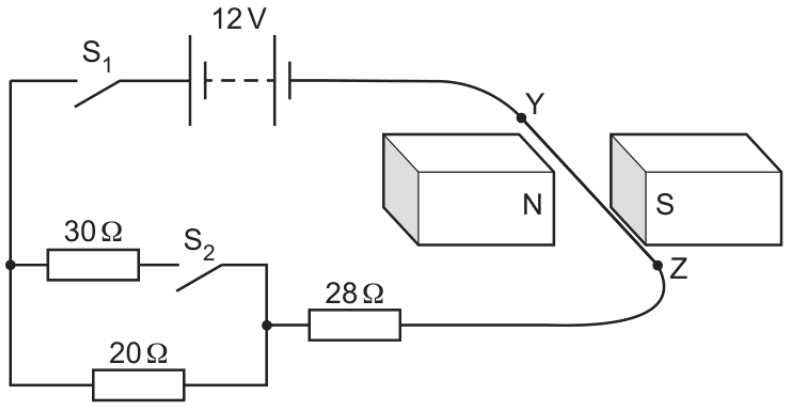


Fig. 9.1

YZ is a straight, horizontal section of connecting wire that lies between two magnets.

S_1 is now closed.

(i) Calculate the current in YZ.

current = ... [2]

(ii) Explain why YZ experiences a force.

.....

.....

..... [2]

(iii) Tick the box which describes the direction of the force on YZ.

towards N	☐	towards Z	☐
towards S	☐	downwards	☐
towards Y	☐	upwards	☐

[1]

(iv) Explain how the direction of the force on YZ is determined.

.....

.....

..... [2]

(c) Switch S_2 in the circuit in Fig. 9.1 is now also closed.

(i) Calculate the total resistance of the circuit.

resistance = ... [3]

(ii) Explain what happens to the force on YZ as switch S_2 is closed.

.....

.....

..... [2]

(iii) The current in the 20 Ω resistor is I_{20}. The current in the 30 Ω resistor is I_{30}.

State a value for the ratio I_{20}/I_{30}.

ratio I_{20}/I_{30} = ... [1]

[Total: 15]

Question	Answer	Marks
9(a)	always positive **or** never negative **or** direction does not change	**B1**
	constant value **or** size does not change	**B1**
9(b)(i)	(I =) V / R **or** 12 / (20 + 28) **or** 12 / 48	**C1**
	0.25 A	**A1**
9(b)(ii)	YZ in magnetic <u>field</u> **or** magnetic <u>field</u> between poles (of magnet)	**M1**
	(electric) current in YZ **or** interacts with field of current / YZ	**A1**
9(b)(iii)	downwards; middle, right-hand box ticked **i.e.** towards N ☐ towards Z ☐ towards S ☐ downwards ☑ towards Y ☐ upwards ☐	**B1**
9(b)(iv)	current from Z to Y page / into page **and** magnetic field left to right	**B1**
	(Fleming's) left-hand rule / other appropriate rule mentioned / described	**B1**

Question	Answer	Marks
9(c)(i)	$1 / R_T = 1 / R_1 + 1 / R_2$ **or** $(R_T =)\ R_1 R_2 / (R_1 + R_2)$ **or** $1 / R_T = 1 / 20 + 1 / 30$ **or** $1 / R_T = 5 / 60$ **or** $1 / R_T = 1 / 12$ **or** $(R_T =)\ 20 \times 30 / (20 + 30)$	**C1**
	12	**C1**
	40 W	**A1**
9(c)(ii)	resistance decreases **or** resistance is less (with S_2 closed)	**B1**
	current increases **and** force increases	**B1**
9(c)(iii)	1.5	**B1**

3 Fig. 3.1 shows a circuit containing three resistors, a 12 V power supply and an ammeter.

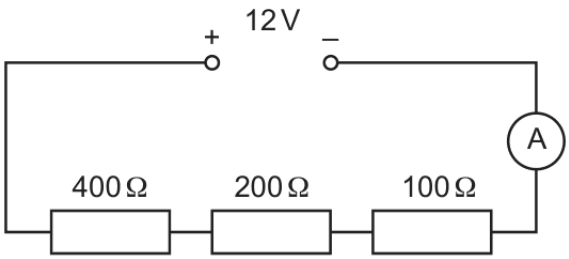


Fig. 3.1

(a) Calculate the current in the 400 Ω resistor.

current = ... A [2]

(b) Calculate the power produced in the 400 Ω resistor.

power = ... W [2]

(c) A student uses all of the components shown in Fig. 3.1 in another circuit.

She connects them so that there is the largest possible current in each resistor.

The current in one of the resistors is larger than the current in the other two resistors. The student connects the ammeter into the circuit to measure the current in this resistor.

Draw the circuit diagram of the arrangement. Label each resistor with the value of its resistance.

[2]

[Total: 6]

Question	Answer	Marks
3(a)	(I =) V/R in any form **or** (total resistance =) 700 (Ω)	**B1**
	0.017 (A)	**B1**
3(b)	(P =) I^2 R **or** VI **and** either V^2/R **or** $V=IR$ numerical or algebraic in any form	**C1**
	or voltage across 400 Ω resistor = **(a)** × 400 (6.857 V) 0.12 (W)	**A1**
3(c)	circuit drawn with all 3 resistors in parallel with cell / power supply	**B1**
	ammeter in series with 100 Ω resistor **or** in series with resistor with largest current in arrangement drawn	**B1**

6 Fig. 6.1 shows an electric circuit containing a filament lamp, a resistor R, a 12 V battery and five meters.

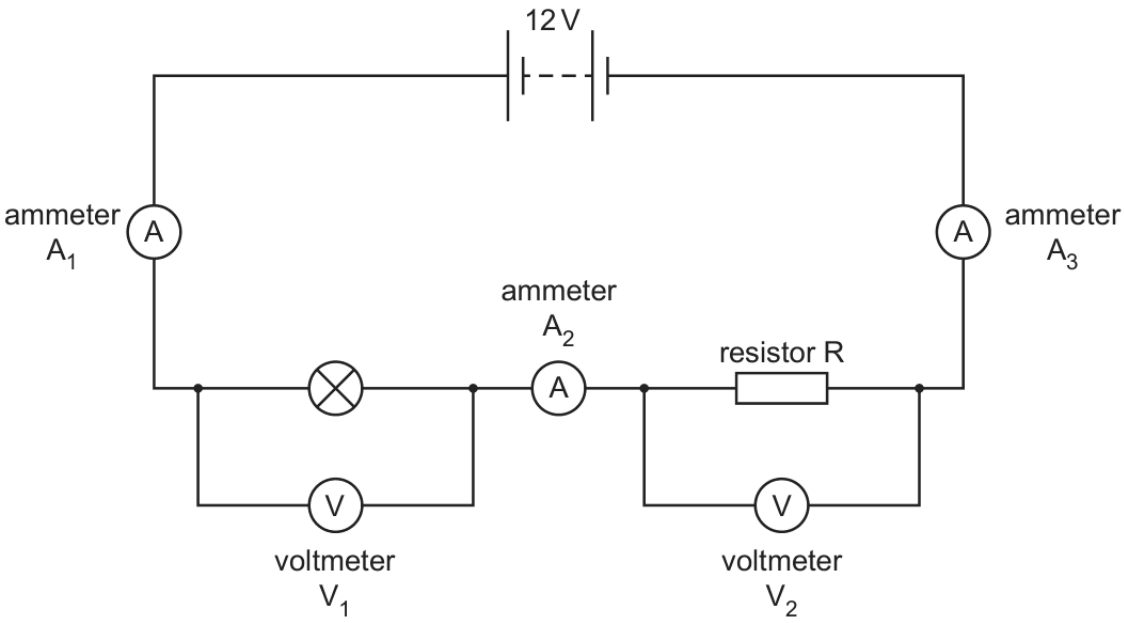


Fig. 6.1

(a) (i) The reading on ammeter A_1 is 0.25 A.

The reading on voltmeter V_1 is 3.0 V.

Determine the readings on the other meters.

reading on ammeter A_2 = ... A

reading on ammeter A_3 = ... A

reading on voltmeter V_2 = ... V
[2]

(ii) Calculate the resistance of resistor R.

resistance of resistor R = ... Ω [2]

(iii) The resistor obeys Ohm's law.

State Ohm's law.

.....

.....

..... [2]

(b) Fig. 6.2 shows the current–voltage graph for the filament lamp.

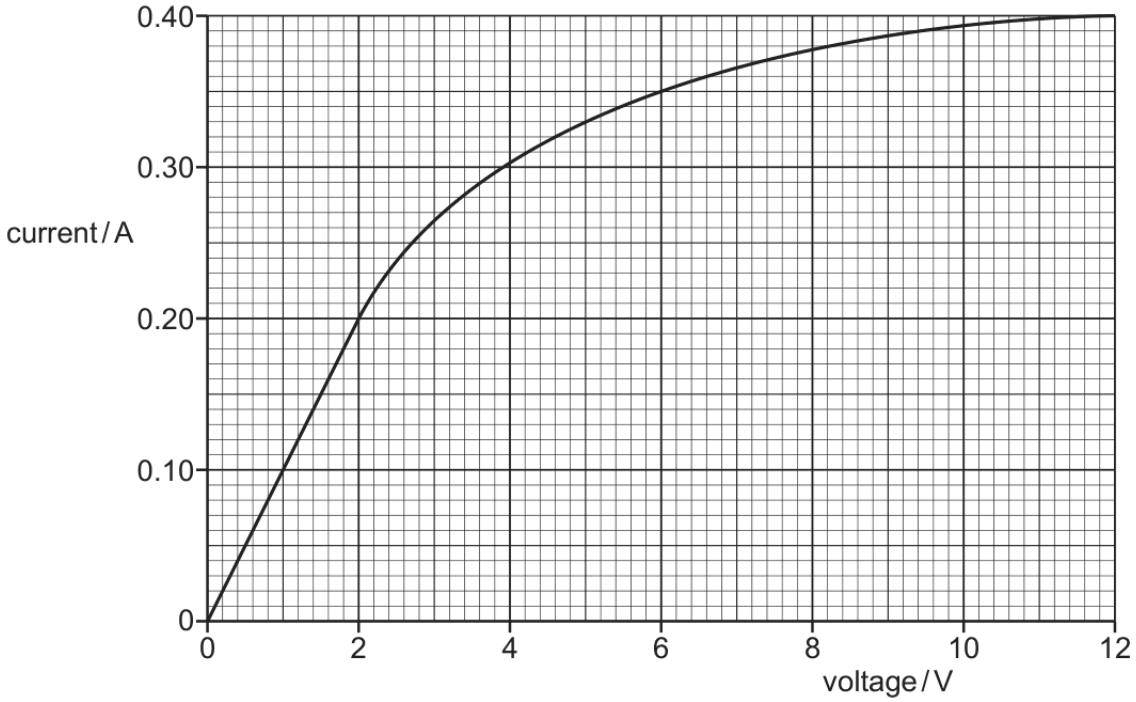


Fig. 6.2

The battery in Fig. 6.1 is replaced with a different battery which has a different e.m.f. (electromotive force).

The voltage across the lamp increases to 6.0 V.

Use data from the graph to determine the e.m.f. of the second battery.

Show your working.

e.m.f. = ... V [3]

[Total: 9]

Question	Answer	Marks
6(a)(i)	ammeter A_2 and A_3 both 0.25 (A)	**B1**
	voltmeter V_2 9(.0 V)	**B1**
6(a)(ii)	(R =) V / I in any form	**C1**
	36 (Ω)	**A1**
6(a)(iii)	current is (directly) proportional to voltage / potential difference	**B1**
	provided the temperature is constant	**B1**
6(b)	0.35 (A) seen **or** total resistance 36 + 6 / 0.35; 36 + 17(.1); 53(.1) Ω	**B1**
	voltage (across R) 0.35 × 36 **or** 12.6 (V) seen	**B1**
	19 V	**B1**

6 Fig. 6.1 shows a circuit that contains two identical cells, each of e.m.f. 1.5 V, that are connected in parallel.

The circuit also contains two ammeters, a thermistor T and two resistors R and S.

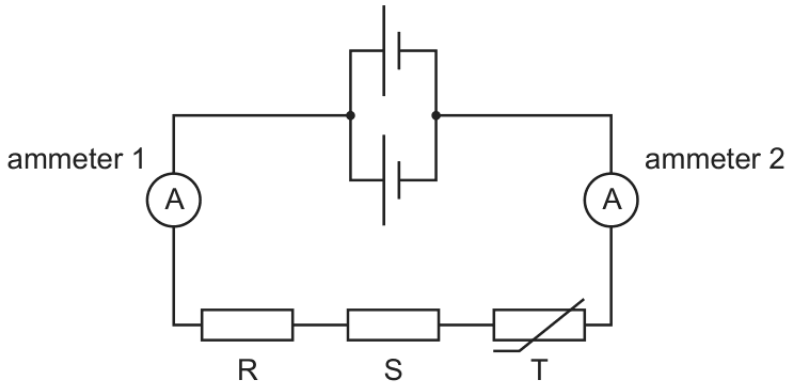


Fig. 6.1

The reading on ammeter 1 is 0.20 A.

(a) State why the reading on ammeter 2 is also 0.20 A.

.....

..... [1]

(b) The resistance of R is 1.5 Ω and the resistance of S is 3.5 Ω.

(i) Calculate the resistance of T.

resistance = ... Ω [3]

(ii) The temperature of T increases.

Explain why the potential difference (p.d.) across T decreases.

.....

.....

.....

.....

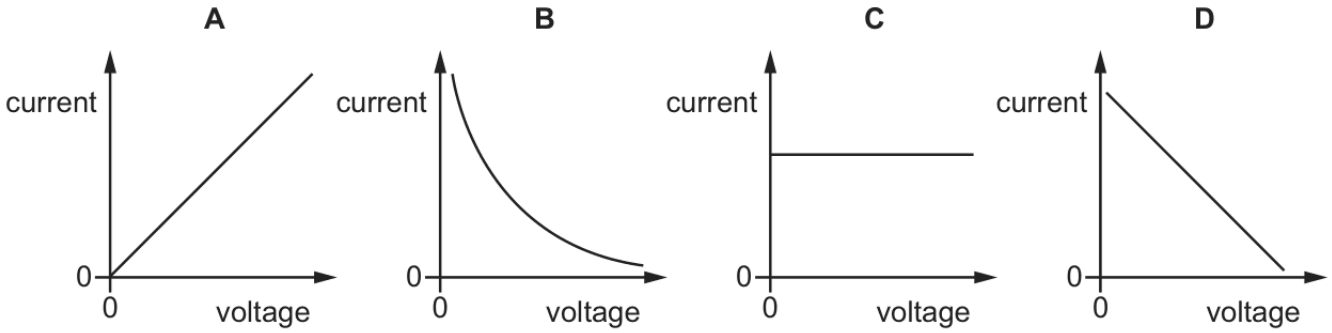
..... [3]

[Total: 7]

Question	Answer	Marks
6(a)	(because) the current at every point in a series circuit is the same	**B1**
6(b)(i)	(R =) V / I **or** 1.5 / 0.20 **or** 7.5 (Ω)	**C1**
	(R_T =) 7.5 – 1.5 –3.5	**C1**
	2.5 (Ω)	**A1**
6(b)(ii)	resistance (of T / thermistor) decreases (as the temperature increases)	**B1**
	current in circuit increases **or** smaller fraction of total resistance	**B1**
	p.d. across R and S increases **or** (p.d. is) smaller fraction of e.m.f.	**B1**

QUESTION · 5054_w23_qp_12.pdf · source page 13 of 20

33 Which graph shows how the current changes when the voltage across a resistor of constant resistance is varied?



MARK SCHEME (official answer key) · 5054_w23_ms_12.pdf · source page 3 of 3 · row for Question 33

33	A	**1**

7 An electric circuit contains a battery, a fixed resistor R, a voltmeter and a thermistor.

Fig. 7.1 is an incomplete circuit diagram of the arrangement.

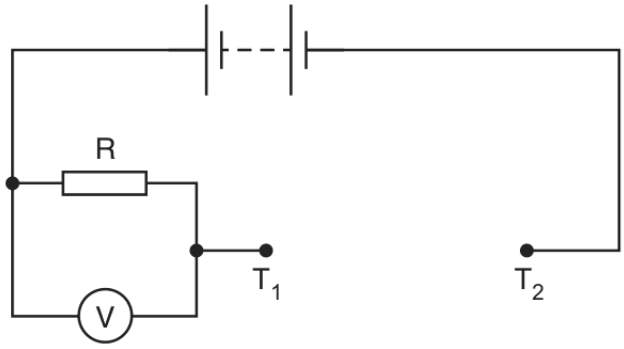


Fig. 7.1

The thermistor is connected between terminals T_1 and T_2.

(a) Complete Fig. 7.1 by drawing the circuit symbol for a thermistor connected between T_1 and T_2. [1]

(b) The resistance of R is 2.5 Ω. The reading on the voltmeter is 0.40 V.

Calculate:

(i) the current in R

current = ... A [2]

(ii) the power transferred in R.

power = ... W [2]

(c) The temperature of the thermistor increases.

Explain what happens to the reading on the voltmeter.

.....

.....

..... [2]

[Total: 7]

Question	Answer	Marks
7(a)	**and** drawn between T_1 and T_2	**B1**
7(b)(i)	(I =) V / R **or** 0.40 / 2.5	**C1**
	0.16 (A)	**A1**
7(b)(ii)	(P =) VI **or** 0.40 × 0.16 **or** (P =) I^2R **or** $0.16^2 \times 2.5$	**C1**
	0.064 (W)	**A1**
7(c)	resistance (of the thermistor / circuit) decreases	**B1**
	increase in current in circuit / R takes a larger proportion of the total resistance / voltage **and** voltmeter reading / it increases**.**	**B1**

5 Fig. 5.1 shows part of a circuit containing a 240 V mains supply connected to a lamp and two heaters of resistance 40 Ω and 60 Ω.

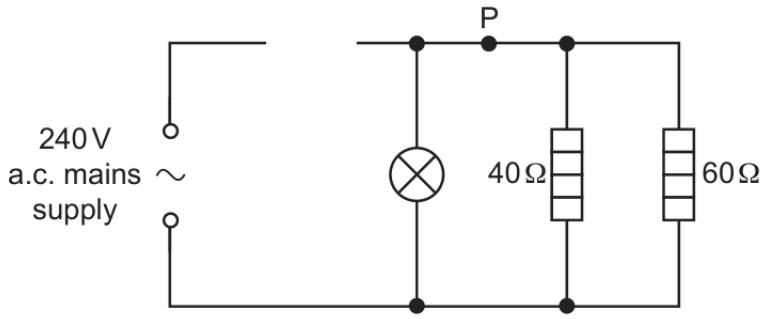


Fig. 5.1

(a) Complete Fig. 5.1 by drawing the symbol for a fuse in the gap in the circuit. [1]

(b) **(i)** On Fig. 5.2 sketch a current−voltage graph for a filament lamp.

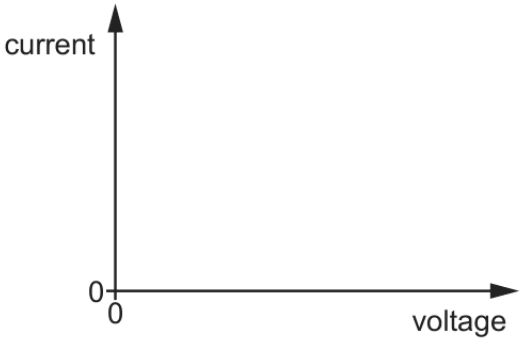


Fig. 5.2

[1]

(ii) Explain the shape of the current−voltage graph drawn in **(i)**.

.....

.....

..... [2]

(c) Calculate the current at point P in the circuit shown in Fig. 5.1.

current = ... A [3]

(d) The current in the lamp is 1.5A.

(i) Suggest a suitable fuse rating for the fuse.

..... [1]

(ii) Explain why a fuse rating much larger than the value in **(i)** is **not** suitable.

.....

.....

.....

..... [2]

(iii) Explain why it is necessary to connect a fuse in the live wire rather than the neutral wire or earth wire.

.....

..... [1]

[Total: 11]

MARK SCHEME · 5054_s24_ms_21.pdf · source page 7 of 10 · Question 5

Question	Answer	Marks
5(a)	correct symbol;	**B1**
5(b)(i)	correct curvature	**B1**
5(b)(ii)	(at higher current / voltage) temperature increases	**B1**
	(at higher current / voltage) resistance increases	**B1**

MARK SCHEME · 5054_s24_ms_21.pdf · source page 8 of 10 · Question 5

Question	Answer	Marks
5(c)	(I =) V / R in any form	**C1**
	4 (A) or 6 (A) seen **or** R_{total} = 24 (Ω)	**C1**
	10 (A)	**A1**
5(d)(i)	any integer between and including 12 and 19 (A)	**B1**
5(d)(ii)	fuse does not melt / blow or circuit does not shut down **aaw**	**B1**
	quickly enough or when there is a fault **or** when the current is large / dangerous	**B1**
5(d)(iii)	<u>when fuse melts</u> nothing is live / electrocution not possible OR supply of voltage is cut OR voltage is supplied by live wire.	**B1**

6 Fig. 6.1 shows a circuit diagram containing a battery, a light-dependent resistor (LDR) and a fixed resistor of resistance 240 Ω connected in series.

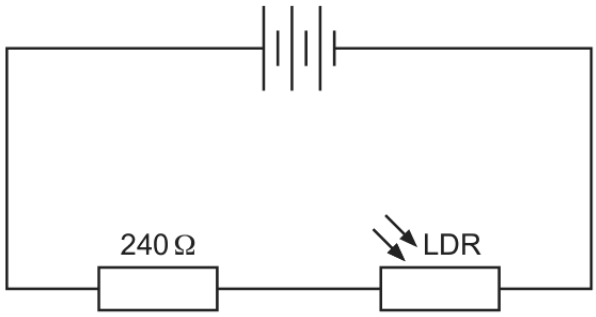


Fig. 6.1

There is a lamp near the circuit. Light from the lamp is incident on the LDR when the lamp is switched on.

Fig. 6.2 shows the current−voltage graph for the LDR with the lamp switched on and with the lamp switched off.

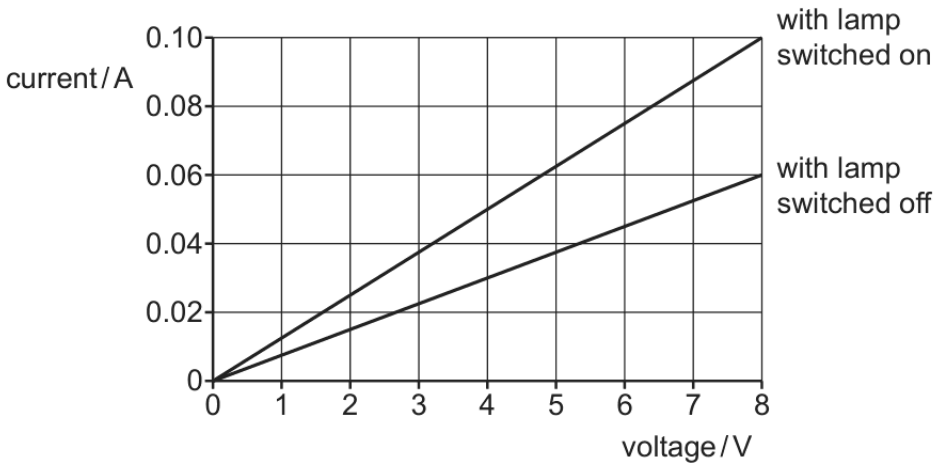


Fig. 6.2

(a) State Ohm's law.

.....

.....

..... [2]

(b) Explain how the graph lines in Fig. 6.2 show that Ohm's law applies to the LDR.

.....

.....

..... [1]

(c) Use values from Fig. 6.2 to explain the effect of light on the resistance of the LDR.

.....

.....

..... [2]

(d) With the lamp switched on, the current in the LDR is 0.050A.

(i) Determine the current in the fixed resistor.

current in fixed resistor = ... A [1]

(ii) Calculate the electromotive force (e.m.f.) of the cell.

e.m.f. = ... V [3]

[Total: 9]

Question	Answer	Marks
6(a)	current is (directly) proportional to potential difference (p.d.)	**B1**
	at constant temperature	**B1**
6(b)	graph is a straight line **and** through the origin (for any one intensity of illumination).	**B1**
6(c)	resistance decreases in bright(er) light	**B1**
	(use of Fig. 6.2 e.g.) resistance values calculated as 80 and 130 (Ω) **or** at same voltage with light has larger current **or** at same current with light needs less voltage	**B1**
6(d)(i)	0.05(0 A)	**B1**
6(d)(ii)	(V =) IR in any form	**C1**
	(voltage across fixed resistor) 12 (V) **or** (total resistance =) 320 (Ω)	**C1**
	16 (V)	**A1**

7 A filament lamp connected to a 12 V power supply transfers 24 W of power.

(a) Calculate:

(i) the current in the lamp

current = ... A [2]

(ii) the resistance of the lamp in this circuit.

resistance = ... Ω [2]

(b) A student has a battery of electromotive force (e.m.f.) 12 V.

The student uses the battery in a circuit with the 12 V filament lamp to obtain a range of suitable readings and plots the current–voltage graph for the lamp.

(i) Fig. 7.1 shows the battery and the filament lamp.

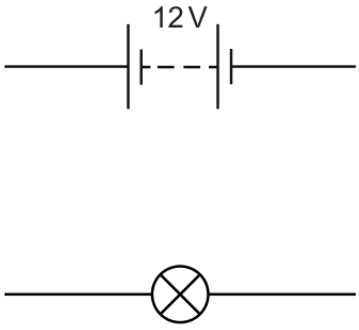


Fig. 7.1

On Fig. 7.1, complete the circuit diagram of a suitable circuit.

You will need to add additional components. [2]

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(ii) On Fig. 7.2, draw the shape of the current–voltage graph for the filament lamp.

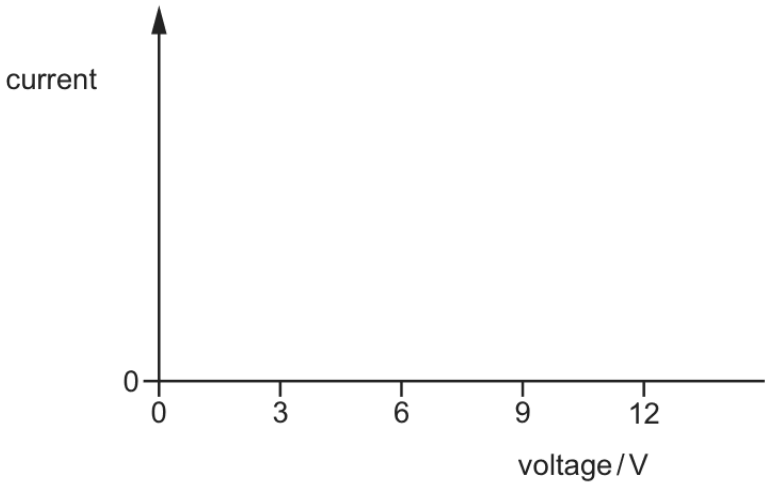


Fig. 7.2

[2]

(iii) State what happens to the resistance of a filament lamp as the applied voltage increases and explain **one** reason for this happening.

.....

.....

..... [2]

[Total: 10]

MARK SCHEME · 5054_w24_ms_21.pdf · source page 8 of 10 · Question 7

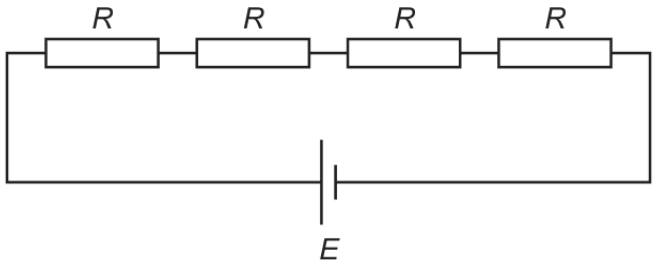
Question	Answer	Marks
7(a)(i)	(I =) P / V or 24 / 12	**C1**
	2.0 (A)	**A1**
7(a)(ii)	(R =) V / I **or** 12 / 2.0	**C1**
	6.0 (Ω)	**A1**

MARK SCHEME · 5054_w24_ms_21.pdf · source page 9 of 10 · Question 7

Question	Answer	Marks
7(b)(i)	variable resistor in series with battery and lamp	**B1**
	ammeter in series with battery and lamp **and** voltmeter across lamp	**B1**
	or	
	potential divider circuit correctly connected	**B1**
	ammeter in series with lamp and voltmeter across lamp	**B1**
7(b)(ii)	curved line with gradient ⩾ 0 **and** starting at origin	**M1**
	curve of decreasing gradient and gradient always > 0	**A1**
7(b)(iii)	resistance increases	**B1**
	temperature increases/filament gets hot **or** particles / ions vibrate more vigorously (and impede electron flow)	**B1**

QUESTION · 5054_w24_qp_12.pdf · source page 11 of 16

28 Four resistors, each of resistance R, are connected in series with a cell of e.m.f. E.



From left to right, what is the current in each of the resistors?

A $\frac{E}{4R}$ $\frac{E}{4R}$ $\frac{E}{4R}$ $\frac{E}{4R}$

B $\frac{E}{4R}$ $\frac{E}{3R}$ $\frac{E}{2R}$ $\frac{E}{R}$

C $\frac{E}{R}$ $\frac{E}{2R}$ $\frac{E}{3R}$ $\frac{E}{4R}$

D $\frac{E}{R}$ $\frac{E}{R}$ $\frac{E}{R}$ $\frac{E}{R}$

MARK SCHEME (official answer key) · 5054_w24_ms_12.pdf · source page 2 of 3 · row for Question 28

28	A	**1**

QUESTION · 5054_w24_qp_12.pdf · source page 11 of 16

30 X is a piece of wire with length l, area a and resistance R.

Wire Y is made from the same material and is connected to wire X to produce a component with a total resistance of $\frac{R}{2}$.

What are the dimensions of Y and how is it connected to X?

	length	area	connection
A	$2l$	$0.25a$	in parallel
B	$2l$	$0.25a$	in series
C	$4l$	$4a$	in parallel
D	$4l$	$4a$	in series

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30	C	**1**

6 Fig. 6.1 shows the circuit symbol for a relay and two labelled terminals.

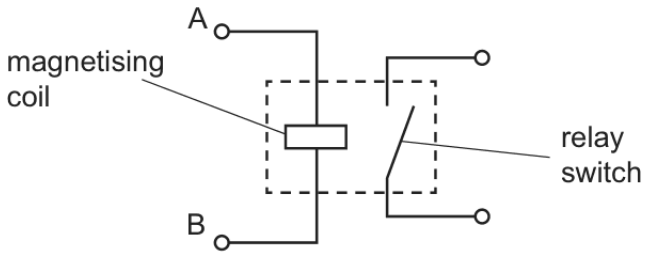

Fig. 6.1

(a) By referring to Fig. 6.1, explain the effect of a current passing from A to B.

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..... [2]

(b) Fig. 6.2 shows two circuits linked by a relay. The heater transfers energy at a rate of 40 W when it is connected to a 12 V supply.

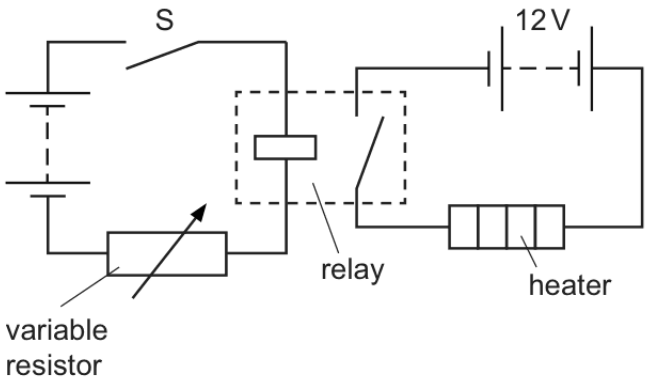

Fig. 6.2

Switch S is closed. The variable resistor is adjusted so that its resistance *R* increases from 0.

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Fig. 6.3 shows how the power transferred in the 40 W heater varies as R increases from 0. The graph shows a sharp decrease when the resistance R reaches R_H.

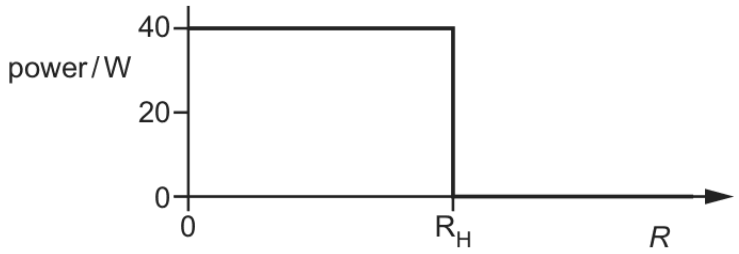

Fig. 6.3

Explain the shape of the graph in Fig. 6.3.

.....

.....

.....

.....

.....

..... [3]

(c) The 40 W heater connected to the 12 V supply is switched on.

Calculate:

(i) the energy transferred in the heater in 2.0 minutes

energy = ... J [2]

(ii) the resistance of the heater.

resistance = ... Ω [3]

[Total: 10]

MARK SCHEME · 5054_w24_ms_22.pdf · source page 7 of 10 · Question 6

Question	Answer	Marks
6(a)	current / it in AB / the (magnetising) coil produces magnetic field	**B1**
	(relay) switch (between X and Y) closes **or** switch attracted by magnetic field / coil **or** (relay) switch made of iron	**B1**

MARK SCHEME · 5054_w24_ms_22.pdf · source page 8 of 10 · Question 6

Question	Answer	Marks
6(b)	(initially) the current in the relay coil is large **or** relay switch (in relay circuit / relay) is closed **or** heater is connected to a 12 V supply **or** voltage of supply does not change **or** heater is switched on	**B1**
	(as R increases) current / magnetic field (in relay coil) decreases	**B1**
	(at $R = R_H$) relay switch (in relay coil) opens **or** magnetic field not strong enough to hold switch closed	**B1**
6(c)(i)	($E =$) Pt **or** 40×2.0 **or** $40 \times 2.0 \times 60$ **or** 80	**C1**
	4800 (J)	**A1**
6(c)(ii)	$I = P / V$ **or** $I = 40 / 12$ **or** $I = 3.3$ (A)	**C1**
	($R =$) V / I **or** $12 / 3.3$	**C1**
	3.6 (Ω)	**A1**

6 A student sets up a circuit to determine the resistance of a length of wire.

The circuit contains a battery of unknown e.m.f., a length of a wire used to make a resistor X, an ammeter, a voltmeter and a variable resistor R.

(a) Fig. 6.1 shows part of the circuit diagram.

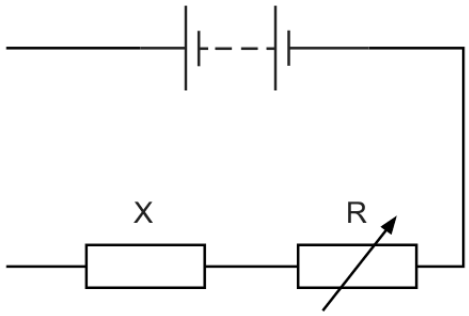

Fig. 6.1

(i) On Fig. 6.1, complete the circuit diagram by adding **one** voltmeter and **one** ammeter in suitable places to allow the determination of the resistance of X. [1]

(ii) Explain how X and R act as a variable potential divider.

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..... [2]

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(b) The student determines the resistance of the resistor X for five different lengths of the wire making it. The lengths of wire range from 20 cm to 60 cm. The type of wire and the cross-sectional area of the wire are kept constant.

Fig. 6.2 shows a graph of the results.

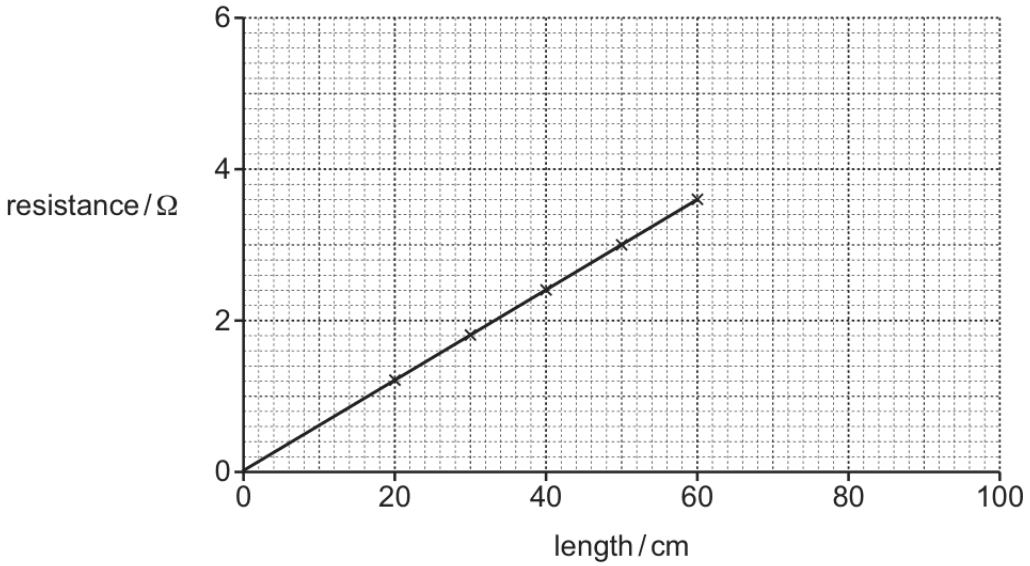


Fig. 6.2

(i) State the relationship between the resistance of the wire and the length of the wire.

..... [1]

(ii) Calculate the current in a 90 cm length of the wire when there is a potential difference (p.d.) of 9.0 V across it.

Show your working.

current = ...A [3]

(iii) The p.d. across the wire making the resistor X is kept constant for all the measurements of resistance.

Describe the relationship between the current in the wire and the length of the wire.

.....

..... [2]

(c) State how the resistance of a wire depends upon the cross-sectional area of the wire.

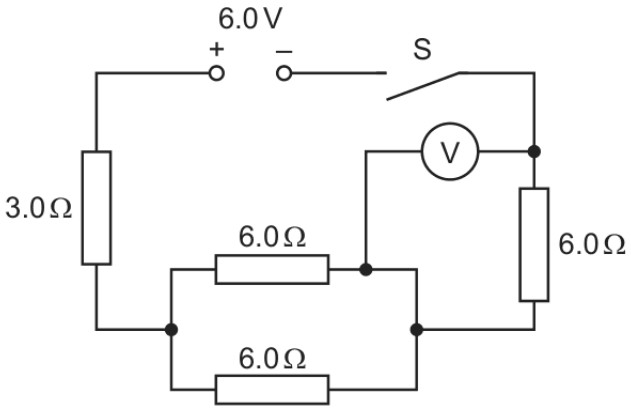
..... [1]

[Total: 10]

Question	Answer	Marks
6(a)(i)	voltmeter across X and ammeter in series anywhere in circuit	**B1**
6(a)(ii)	*Any two from*: • X and R share voltage / p.d. of battery **or** e.m.f. of battery = p.d. across X + p.d. across R • as R varies, p.d. across X **and** R vary • if R increases then p.d. across R increases • If R increases then p.d, across X decreases	**B2**
6(b)(i)	directly proportional	**B1**
6(b)(ii)	resistance = 90 × 3.6 / 60 **or** 5.4 Ω seen	**B1**
	(I =) V / R in any form or with any resistance	**C1**
	1.7 A	**A1**
6(b)(iii)	as length increases current decreases	**C1**
	(current) inversely proportional (to length)	**A1**
6(c)	inversely proportional	**B1**

QUESTION · 5054_s25_qp_12.pdf · source page 12 of 16

29 A student connects the circuit shown.



The switch S is closed.

What is the reading on the voltmeter?

A 1.0 V **B** 3.0 V **C** 4.5 V **D** 6.0 V

MARK SCHEME (official answer key) · 5054_s25_ms_12.pdf · source page 3 of 3 · row for Question 29

29	B	**1**

5 A student sets up the circuit shown in Fig. 5.1 to measure the resistance of a length of wire. The circuit contains four identical cells.

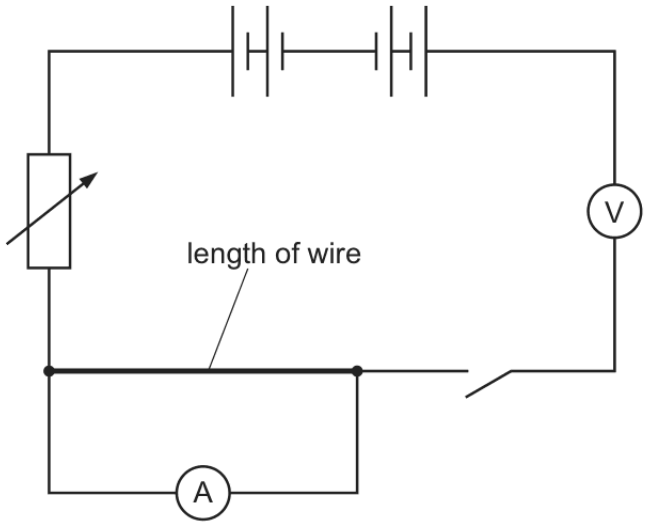


Fig. 5.1

The student finds that both meters read zero when the switch is closed. This is because the circuit shown in Fig. 5.1 is not suitable for the measurement.

(a) Describe **all** the changes to the circuit in Fig. 5.1 that are needed so that the meters produce readings that can be used to calculate the resistance of the length of wire.

.....

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..... [2]

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(b) Fig. 5.2 shows the current–voltage graph for a 9.0 cm length of wire (line P) and for a different length of the same wire with the same cross-sectional area (line Q).

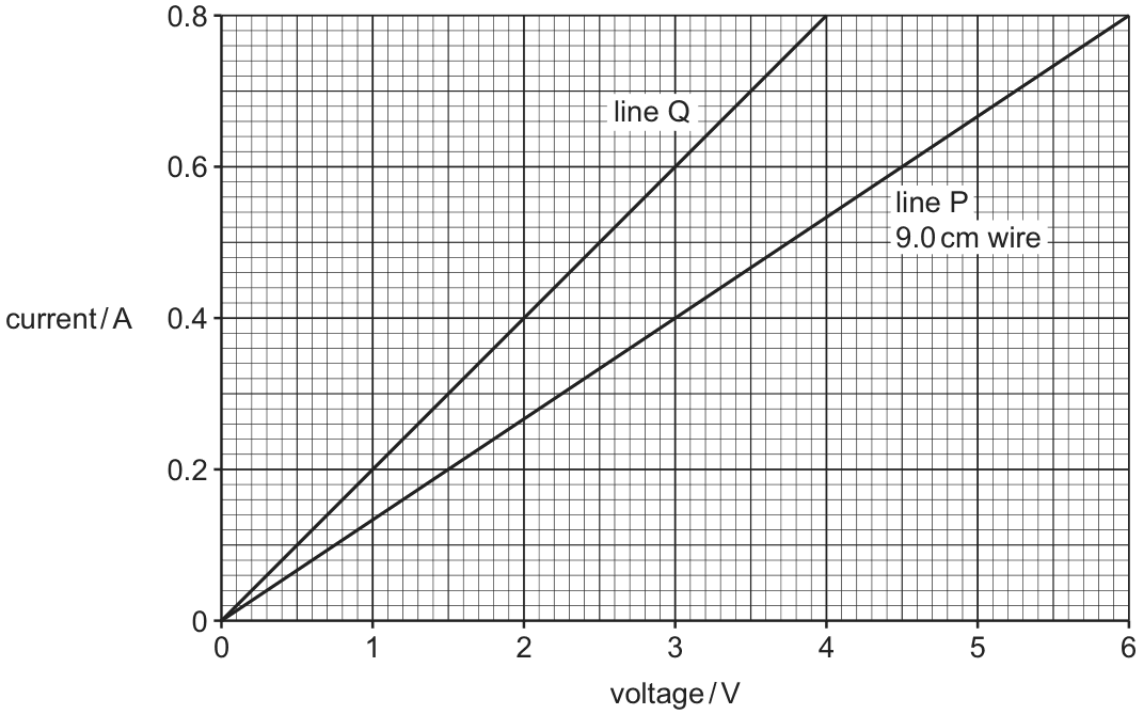

Fig. 5.2

(i) Calculate the resistance of the 9.0 cm length of wire.

resistance = ... Ω [2]

(ii) Explain whether line Q is obtained with a length of wire that is longer or shorter than 9.0 cm.

.....

..... [1]

(iii) Calculate the length of the wire used to obtain line Q.

length = ... cm [1]

[Total: 6]

Question	Answer	Marks
5(a)	swap voltmeter for ammeter	**B1**
	have cells all facing the same way, e.g. reverse left two cells	**B1**
5(b)(i)	(R =) V / I in any form, e.g. 6 / 0.8	**C1**
	7.5 (Ω)	**A1**
5(b)(ii)	shorter **and** *one of*: • resistance smaller **or** resistance proportional to length • larger current (for same voltage) • smaller voltage to produce (an equal) current	**B1**
5(b)(iii)	6 (.0 cm)	**B1**

6 A circuit consists of a battery of e.m.f. 9.0 V, a switch, a resistor P, an ammeter, a 4.0 Ω resistor and a voltmeter.

Fig. 6.1 is the circuit diagram of the circuit.

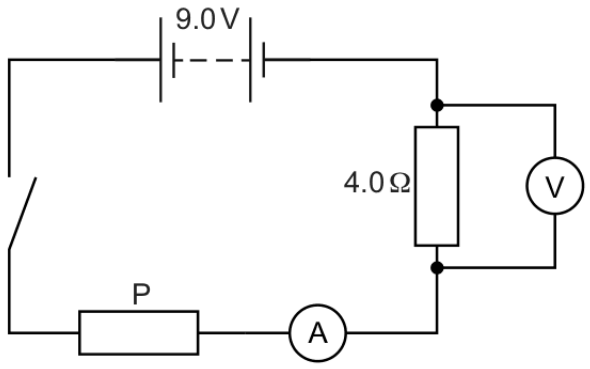

Fig. 6.1

(a) State Ohm's law.

.....

.....

..... [2]

(b) Resistor P is a cylinder of length l, made from conducting modelling clay.

Fig. 6.2 shows resistor P.

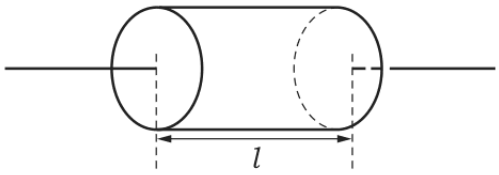

Fig. 6.2

P has a resistance of 8.0 Ω.

(i) The switch in the circuit in Fig. 6.1 is closed.

Calculate the reading on the ammeter.

ammeter reading = ...A [2]

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(ii) Calculate the power transferred electrically to resistor P.

power transferred = ... W [2]

(c) Resistor P is removed from the circuit.

The conducting modelling clay from resistor P is reshaped to make a second cylindrical resistor that has the same volume as resistor P but has a length $\frac{l}{2}$.

Fig. 6.3 shows the reshaped resistor.

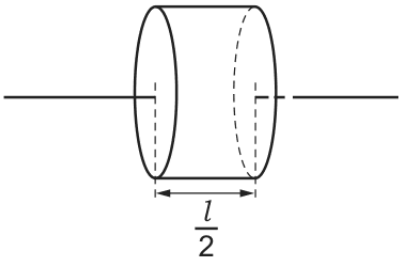


Fig. 6.3

(i) Calculate the resistance of the reshaped resistor.

resistance of reshaped resistor = ... Ω [2]

(ii) The reshaped resistor is connected into the circuit instead of resistor P.

Calculate the new reading on the voltmeter. Show your working.

voltmeter reading = ... V [2]

[Total: 10]

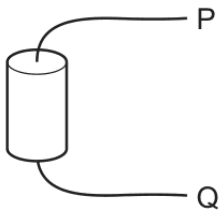
Question	Answer	Marks
6(a)	potential difference / p.d. (across a component) is directly proportional to the current (in the component)	**B1**
	at constant <u>temperature</u> / <u>physical conditions</u>	**B1**
6(b)(i)	(ammeter reading =) V/R **or** $\frac{E}{R_P + R_Q}$ **or** $\frac{9.0}{8.0 + 4.0}$	**C1**
	0.75 (A)	**A1**
6(b)(ii)	(power transferred =) IV **or** $0.75 \times (0.75 \times 8.0)$ **or** 0.75×6.0	**C1**
	4.5 (W)	**A1**
6(c)(i)	(length halves and so cross-sectional) area doubles **or** (new resistance =) $\frac{8.0 \times 0.50}{2.0}$	**C1**
	2.0 (Ω)	**A1**

Question	Answer	Marks
6(c)(ii)	(voltmeter reading =) $\frac{E \times R_Q}{R_{new} + R_Q}$ **or** $\frac{9.0 \times 4.0}{2.0 + 4.0}$ **or** (I =) 1.5 A	**B1**
	6.0 (V)	**B1**

QUESTION · 5054_w25_qp_12.pdf · source page 13 of 16

33 A cardboard tube has two wires, P and Q, connected to its ends. Inside the tube, wire P and wire Q are connected to two resistors.

The potential difference (p.d.) between wire P and wire Q is 9.0 V and the current in the wires is 0.75 A.



Which combination of resistors is inside the tube?

A a 0.75 Ω resistor and a 6.0 Ω resistor in parallel

B a 0.75 Ω resistor and a 6.0 Ω resistor in series

C a 20 Ω resistor and a 30 Ω resistor in parallel

D a 20 Ω resistor and a 30 Ω resistor in series

MARK SCHEME (official answer key) · 5054_w25_ms_12.pdf · source page 3 of 3 · row for Question 33

33	C	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 13 of 16

32 The resistance of a metal wire depends on its length and on its cross-sectional area.

What are the relationships for the resistance of the wire?

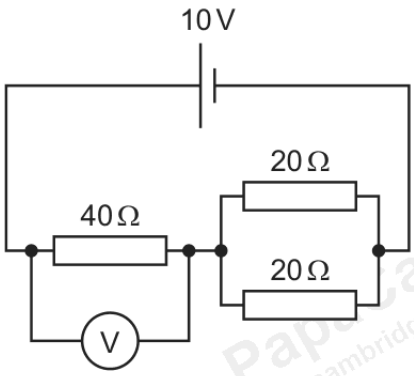
	resistance and length	**resistance and cross-sectional area**
A	direct proportion	direct proportion
B	direct proportion	inverse proportion
C	inverse proportion	direct proportion
D	inverse proportion	inverse proportion

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 3 of 3 · row for Question 32

32	B	**1**

QUESTION · 5054_s26_qp_11.pdf · source page 14 of 16

34 The diagram shows a circuit that contains three resistors, a cell and a voltmeter.



What is the reading on the voltmeter?

A 2.0 V **B** 3.3 V **C** 5.0 V **D** 8.0 V

MARK SCHEME (official answer key) · 5054_s26_ms_11.pdf · source page 3 of 3 · row for Question 34

34	D	**1**

6 Figure 6.1 shows two identical car headlamps connected in parallel to the 12 V battery of a car. The lamps are filament lamps.

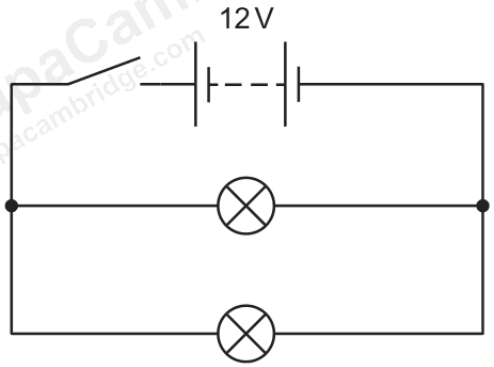


Figure 6.1

The switch is closed, and both lamps operate at normal brightness.

(a) The power of each lamp is 40 W.

(i) Calculate the current in each lamp and the current in the battery.

current in each lamp = ... A

current in battery = ... A
[3]

(ii) Calculate the resistance of the circuit when the lamps are operating at normal brightness.

resistance = ... Ω [2]

(iii) When the circuit is first switched on, the current in each filament lamp is large and then quickly decreases.

Explain why the current in each lamp decreases.

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..... [2]

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(b) During use, some metal atoms evaporate from the filament of a filament lamp and condense on the inside surface of the lamp. The inside surface of a lamp darkens where the metal atoms condense.

When there is a gas inside the lamp, the metal atoms condense mainly at the top of the lamp.

When there is a vacuum inside the lamp, the metal atoms condense uniformly over the inside surface.

A filament lamp with gas inside and a filament lamp with a vacuum inside are shown in Figure 6.2.

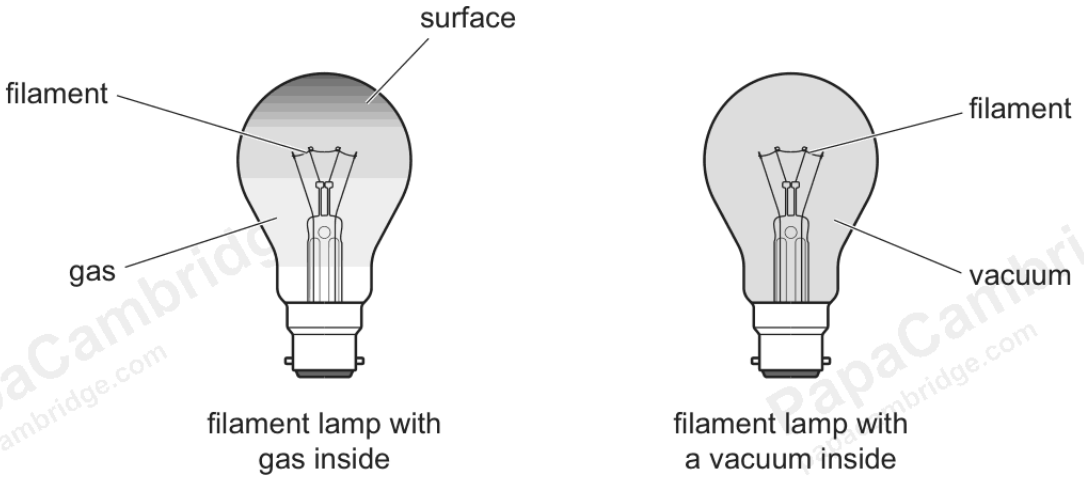


Figure 6.2

Explain why the metal atoms condense mainly at the top of the lamp when there is a gas inside the lamp.

.....

.....

.....

.....

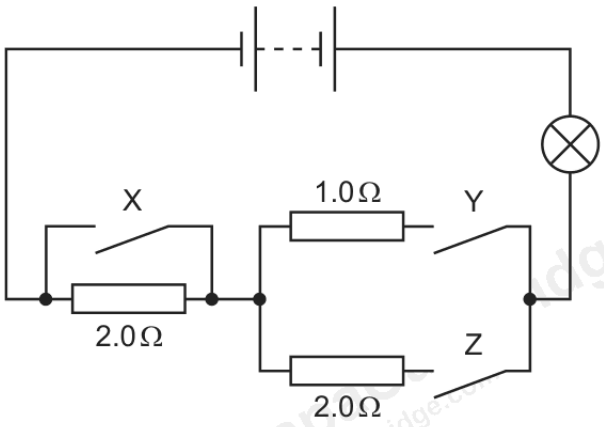
..... [2]

[Total: 9]

Question	Answer	Marks	Guidance
6(a)(i)	(I =) P/V in any form current in lamp = 3.3 (A) current in battery twice that in lamp / 6.7 **or** 6.6 (A)	**1**	3.33 penalise s.f. / fractions only once
6(a)(ii)	(R =) V/I in any form **or** (R =) P/I^2 1.8 (Ω)	**1**	**ecf** 12 / current in battery but accept current in lamp for 1st mark **accept** (R =) V^2/P
6(a)(iii)	As the temperature (of the filament) rise(s) / lamp heats up the resistance (of the filament/lamp) increases	**1**	**A** lamp releases heat **I** heat lost / wires get hot **I** resistance is high (must be an increase)
6(b)	hot gas (near filament) / less dense particles rise **or** convection occurs hot gas less dense	**1**	**A** hot particles rise / hot metal atoms rise **I** hot particles / metal atoms less dense

QUESTION · 5054_s26_qp_12.pdf · source page 13 of 16

32 The diagram shows a lamp, three resistors and three switches, X, Y and Z, connected to a battery.



For which combination of switch settings does the lamp shine the brightest?

	X	Y	Z
A	closed	closed	closed
B	closed	closed	open
C	open	closed	closed
D	open	open	closed

MARK SCHEME (official answer key) · 5054_s26_ms_12.pdf · source page 3 of 3 · row for Question 32

32	A	**1**

6 Figure 6.1 shows a light-dependent resistor (LDR) connected in series with a fixed resistor of resistance 200 Ω and a battery.

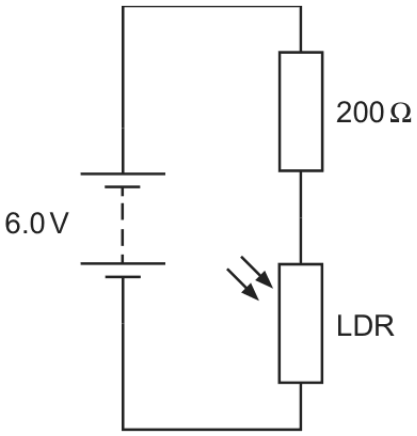


Figure 6.1

The electromotive force (e.m.f.) of the battery is 6.0 V.

The potential difference (p.d.) across the LDR is 2.4 V.

(a) Both e.m.f. and p.d. are measured in volts.

The definitions of e.m.f. and p.d both involve the work done per unit charge.

State how the definitions of e.m.f. and p.d are **different** from each other.

.....

.....

.....

..... [2]

(b) **(i)** State the equation that relates resistance, current and voltage.

..... [1]

(ii) Compare the current in the fixed resistor with the current in the LDR.

..... [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

(c) Calculate:

(i) the p.d. across the fixed resistor

p.d. = ...V [1]

(ii) the resistance of the LDR.

resistance = ... Ω [2]

(d) The brightness of the light shining on the LDR increases.

Using words or phrases from the list below, complete Table 6.1 to show the effect of this increase in brightness.

You may use each word or phrase once, more than once or not all.

increases **stays the same** **decreases**

Table 6.1

	effect of increase in brightness
resistance of LDR	.....
current in LDR	.....
current in fixed resistor	.....
p.d. across the fixed resistor	.....
p.d. across the LDR	.....

[3]

[Total: 10]

Question	Answer	Marks	Guidance
6(a)	e.m.f. (work done on unit charge or voltage) in complete circuit or by the source / battery	**B1**	**e.m.f:** energy converted to electrical energy; / **p.d:** electrical energy converted to other types
	p.d. (work done on unit charge or voltage) in / on a component / conductor / resistor / equipment / between two points	**B1**	**not** emf is a force / **ignore** (work for) part / section of a circuit; force in p.d explanation i.e. do not accept 'force across a component'
6(b)(i)	resistance = voltage / current	**B1**	**A** $R = V / I$ in any form symbol or words
6(b)(ii)	same / equal	**B1**	
6(c)(i)	3.6 (V) seen	**B1**	
6(c)(ii)	(current =) 3.6 / 200 or 0.018 (A) seen	**C1**	**ecf** as usual **but not** for 6 / 200 **A 6(c)(i)** / 200 **A** potential divider calculation ($(R_{LDR} =) R_{FR} \times V_{LDR} / V_{FR}$ or numerical equivalent)
	130 Ω	**A1**	133.3
6(d)	*1st two rows* decreases **and** increases	**B1**	if 4th row is 'stay the same' then accept 5th row 'stay the same'
	2nd, 3rd and 4th rows same as each other	**B1**	
	5th row opposite of 4th row	**B1**	